UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2024

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from ______ to ______.

Commission file number: **001-37580**

Alphabet Inc.

(Exact name of registrant as specified in its charter)

Delaware	**61-1767919**
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)

1600 Amphitheatre Parkway
Mountain View, CA 94043
(Address of principal executive offices, including zip code)
(650) 253-0000
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Class A Common Stock, $0.001 par value	**GOOGL**	**Nasdaq Stock Market LLC (Nasdaq Global Select Market)**
Class C Capital Stock, $0.001 par value	**GOOG**	**Nasdaq Stock Market LLC (Nasdaq Global Select Market)**

Securities registered pursuant to Section 12(g) of the Act:

Title of each class
None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “smaller reporting company,” and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐		

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C.7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of June 28, 2024, the aggregate market value of shares held by non-affiliates of the registrant (based upon the closing sale prices of such shares on the Nasdaq Global Select Market on June 28, 2024) was approximately $2.0 trillion. For purposes of calculating the aggregate market value of shares held by non-affiliates, we have assumed that all outstanding shares are held by non-affiliates, except for shares held by each of our executive officers, directors, and 5% or greater stockholders. In the case of 5% or greater stockholders, we have not deemed such stockholders to be affiliates unless there are facts and circumstances which would indicate that such stockholders exercise any control over our company, or unless they hold 10% or more of our outstanding common stock. These assumptions should not be deemed to constitute an admission that all executive officers, directors, and 5% or greater stockholders are, in fact, affiliates of our company, or that there are not other persons who may be deemed to be affiliates of our company. Further information concerning shareholdings of our officers, directors, and principal stockholders is included or incorporated by reference in Part III, Item 12 of this Annual Report on Form 10-K.

As of January 28, 2025, there were 5,833 million shares of Alphabet’s Class A stock outstanding, 860 million shares of Alphabet’s Class B stock outstanding, and 5,497 million shares of the Alphabet’s Class C stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant’s Proxy Statement for the 2025 Annual Meeting of Stockholders are incorporated herein by reference in Part III of this Annual Report on Form 10-K to the extent stated herein. Such proxy statement will be filed with the Securities and Exchange Commission within 120 days of the registrant’s fiscal year ended December 31, 2024.

Alphabet Inc.
Form 10-K
For the Fiscal Year Ended December 31, 2024

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Note About Forward-Looking Statements

This Annual Report on Form 10-K contains forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. These include, among other things, statements regarding:

- the growth of our business and revenues and our expectations about the factors that influence our success and trends in our business;
- fluctuations in our revenues and margins and various factors contributing to such fluctuations;
- our expectation that the continuing shift to an online world as the digital economy evolves will continue to benefit our business;
- our expectation that the revenues that we derive beyond advertising will continue to increase and may affect our margins;
- our expectation that our traffic acquisition costs (TAC) and the associated TAC rate will fluctuate, which could affect our overall margins;
- our expectation that our monetization trends will fluctuate, which could affect our revenues and margins;
- fluctuations in paid clicks and cost-per-click as well as impressions and cost-per-impression, and various factors contributing to such fluctuations;
- our expectation that we will continue to periodically review, refine, and update our methodologies for monitoring, gathering, and counting the number of paid clicks and impressions, and for identifying the revenues generated by the corresponding click and impression activity;
- our expectation that our results will be affected by our performance in international markets as users in developing economies increasingly come online;
- our expectation that our foreign exchange risk management program will not fully offset our net exposure to fluctuations in foreign currency exchange rates;
- the expected variability of gains and losses related to hedging activities under our foreign exchange risk management program;
- the amount and timing of revenue recognition from customer contracts with commitments for performance obligations, including our estimate of the remaining amount of commitments and when we expect to recognize revenue;
- our expectation that our capital expenditures will increase, including our expected spend and the expected increase in our technical infrastructure investment to support the growth of our business and our long-term initiatives, in particular in support of artificial intelligence (AI) products and services;
- our plans to continue to invest in new businesses, products, services and technologies, and systems, as well as to continue to invest in acquisitions and strategic investments;
- our pace of hiring and our plans to provide competitive compensation programs;
- our expectation that our cost of revenues, research and development (R&D) expenses, sales and marketing expenses, and general and administrative expenses may increase in amount and/or may increase as a percentage of revenues and may be affected by a number of factors;
- estimates of our future employee compensation expenses;
- our expectation that our other income (expense), net (OI&E), will fluctuate in the future, as it is largely driven by market dynamics;
- our expectation that our effective tax rate and cash tax payments could increase in future years;
- seasonal fluctuations in internet usage, advertising expenditures, and underlying business trends such as traditional retail seasonality, which are likely to cause fluctuations in our quarterly results;
- the sufficiency of our sources of funding;
- our potential exposure in connection with new and pending investigations, proceedings, and other contingencies, including the possibility that certain legal proceedings to which we are a party could harm our business, financial condition, and operating results;

- our expectation that we will continue to face heightened regulatory scrutiny, and changes in regulatory conditions, laws, and public policies, which could affect our business practices and financial results;
- the expected timing, amount, and effect of Alphabet Inc.'s share repurchases and dividends;
- our long-term sustainability goals;

as well as other statements regarding our future operations, financial condition and prospects, and business strategies. Forward-looking statements may appear throughout this report and other documents we file with the Securities and Exchange Commission (SEC), including without limitation, the following sections: Part I, Item 1 "Business;" Part I, Item 1A "Risk Factors;" and Part II, Item 7 "Management's Discussion and Analysis of Financial Condition and Results of Operations." Forward-looking statements generally can be identified by words such as "anticipates," "believes," "could," "estimates," "expects," "intends," "may," "plans," "predicts," "projects," "will be," "will continue," "will likely result," and similar expressions. These forward-looking statements are based on current expectations and assumptions that are subject to risks and uncertainties, which could cause our actual results to differ materially from those reflected in the forward-looking statements. Factors that could cause or contribute to such differences include, but are not limited to, those discussed in this Annual Report on Form 10-K, including the risks discussed in Part I, Item 1A "Risk Factors" and the trends discussed in Part II, Item 7 "Management's Discussion and Analysis of Financial Condition and Results of Operations," and those discussed in other documents we file with the SEC. We undertake no obligation to revise or publicly release the results of any revision to these forward-looking statements, except as required by law. Given these risks and uncertainties, readers are cautioned not to place undue reliance on such forward-looking statements.

As used herein, "Alphabet," "the company," "we," "us," "our," and similar terms include Alphabet Inc. and its subsidiaries, unless the context indicates otherwise.

"Alphabet," "Google," and other trademarks of ours appearing in this report are our property. We do not intend our use or display of other companies' trade names or trademarks to imply an endorsement or sponsorship of us by such companies, or any relationship with any of these companies.

PART I

ITEM 1. BUSINESS

Overview

As our founders Larry and Sergey wrote in the original founders' letter, "Google is not a conventional company. We do not intend to become one." That unconventional spirit has been a driving force throughout our history, inspiring us to tackle big problems and invest in moonshots. It led us to be a pioneer in the development of AI and, since 2016, an AI-first company. We continue this work under the leadership of Alphabet and Google CEO, Sundar Pichai.

Alphabet is a collection of businesses — the largest of which is Google. We report Google in two segments, Google Services and Google Cloud, and all non-Google businesses collectively as Other Bets. Supporting these businesses, we have centralized certain AI-related research and development which is reported in Alphabet-level activities. Alphabet's structure is about helping each of our businesses prosper through strong leaders and independence.

Access and Technology for Everyone

The Internet is one of the world's most powerful equalizers; it propels ideas, people, and businesses large and small. Our mission to organize the world's information and make it universally accessible and useful is as relevant today as it was when we were founded in 1998. Since then, we have evolved from a company that helps people find answers to a company that also helps people get things done.

We are focused on building an even more helpful Google for everyone, and we aspire to give everyone the tools they need to increase their knowledge, health, happiness, and success. Google Search helps people find information and make sense of the world in more natural and intuitive ways, with trillions of searches on Google every year. YouTube provides people with entertainment, information, and opportunities to learn something new and helps support the creator economy through the YouTube Partner Program. Google Cloud helps customers solve today's business challenges, improve productivity, reduce costs, and unlock new growth engines. We are continually innovating and building new products and features to help our users, partners, customers, and communities and have invested more than $150 billion in research and development in the last five years in support of these efforts.

Making AI Helpful for Everyone

We believe AI is a profound platform shift, one that can bring meaningful and positive change to people and societies across the world, and to our business. At Google, we have been bringing AI into our products and services for more than a decade and making them available to our users.

In 2023, we took a significant step on our journey to make AI more helpful for everyone with the introduction of Gemini, our natively multimodal AI model. In 2024, we launched Gemini 2.0, our most capable model yet. Gemini can generalize and seamlessly understand, operate across, and combine different types of information including text, code, audio, image, and video. Gemini is powering AI features across our products and services that are helping people everyday. Today, all seven of our two billion-user products — Android, Chrome, Gmail, Maps, Play Store, Search, and YouTube — are using Gemini.

Our Google Cloud products, including Google Cloud Platform and Google Workspace, help organizations stay at the forefront of innovation with our AI-optimized infrastructure, AI development platform, world-class models, as well as assistive agents and applications.

We believe AI has the potential to solve important societal, scientific, and engineering challenges. For example, in 2020, Google DeepMind's AlphaFold system solved a 50-year-old protein folding challenge, and in 2024, we introduced AlphaFold 3, built on the previous models, to predict the structure and interactions of all the molecules in life's processes.

We are focused on transforming our products to provide the most helpful tools to our users and customers as we deliver on our mission to organize the world's information and make it universally accessible and useful.

Deliver the Most Advanced, Safe, and Responsible AI

We aim to build the most advanced, safe, and responsible AI through a full stack of robust AI-optimized infrastructure, including data centers, chips, and a global fiber network; world class research teams; and a broad global reach through products and platforms that touch billions of people and customers around the world.

We are driving efficiencies in our data centers, while making significant hardware and model improvements. For example, since we started serving AI Overviews to our users, we have significantly lowered machine costs and latency through hardware, engineering, and technical breakthroughs. Our AI-optimized infrastructure allows us to use, and offer our customers, a range of AI accelerator options, including our own custom-built Tensor Processing Units (TPUs).

Our teams across Alphabet leverage Gemini, as well as other AI models we have previously developed and announced, to deliver the best product and service experiences for our users, advertisers, partners, customers, and developers. We are using Gemini 2.0 in new research prototypes, including Project Astra, which explores the future capabilities of a universal AI assistant and Project Mariner, an early prototype capable of taking actions in Chrome as an experimental extension.

We believe our approach to AI must be both bold and responsible. That means developing AI in a way that maximizes the positive benefits to society while addressing its potential challenges, guided by our AI Principles.

Enable Organizations and Developers to Innovate on Google Cloud

AI is a major technology shift for enterprises. Globally, businesses from startups to large enterprises, and the public sector are thinking about how to drive transformation. That is why we are focused on making it easy and scalable for others to innovate, and grow, with AI. That means providing advanced computing infrastructure and expanding access to Google's latest AI models. Our Vertex AI platform gives developers the ability to train, tune, augment, test, and deploy applications using Gemini, Imagen, Veo, and other generative AI models. Gemini for Google Cloud provides pre-packaged AI agents that can assist developers to write, document, test, and operate software as well as assist cybersecurity teams to analyze, detect, protect, and respond to threats.

Improve Knowledge, Learning, Creativity, and Productivity

As AI continues to improve rapidly, we are focused on transforming our products to provide our users the most helpful tools. Examples include AI Overviews, which allow users to ask their most complex questions, break down concepts in a format that is useful for them, or take a video and ask a question out loud, and NotebookLM, which helps users understand complex information by summarizing sources and providing relevant quotes. We are embedding the power of generative AI to continue helping our users express themselves and get things done. For example, the Gemini app allows users to collaborate with new AI features that include image generation capabilities, coding support, and app integration. Gemini for Google Workspace helps users write, organize, visualize, accelerate workflows, and have richer meetings. Additionally, we are using Gemini to improve recommendations on YouTube. We also know businesses of all sizes around the world rely on Google Ads to find customers and grow their businesses — and we make that even easier with AI. For example, advertisers are able to use our updated image generation in Google Ads to produce high-quality imagery for their campaigns, and Demand Gen helps them reach their target audiences.

Build the Most Helpful Personal Computing Platforms and Devices

Over the years, our Pixel phones have incorporated AI compute directly into the device and built experiences on top of it. Our latest Pixel devices were built with Gemini Nano, bringing the best AI-assistive experiences to our users, such as Gemini on Pixel, Pixel Screenshots, and Pixel Studio. Our Android and Chrome operating systems enable users to access Gemini, and other Google AI features throughout their day. For example, with Circle to Search built directly into the Android experience, users can search anything they see on their phone using a simple gesture.

Moonshots

Many companies get comfortable doing what they have always done, making only incremental changes. This incrementalism leads to irrelevance over time, especially in technology, where change tends to be revolutionary, not evolutionary. Our acquisitions of YouTube and Android and subsequent launch of Chrome have matured into major platforms for digital video and mobile devices and a safer, popular browser. In Other Bets, our fully autonomous driving technology company, Waymo, is now providing fully autonomous, paid ride-hailing services to customers in multiple cities.

Our early investments in AI started out as moonshots but are now incorporated into our core products and central to future developments. We continue to look toward the future and to invest for the long term, most notably for the application of AI to our products and services, as well as other frontier technologies such as quantum computing. As we said in the original founders' letter, we will not shy away from high-risk, high-reward projects that we believe in, as they are the key to our long-term success.

Privacy and Security

We make it a priority to protect the privacy and security of our products, users, and customers, even if there are near-term financial consequences. We do this by continuously investing in building products that are secure by default; strictly upholding responsible data practices that emphasize privacy by design; and building easy-to-use settings that put people in control. We are continually enhancing these efforts over time, whether by enabling users to auto-delete their data, applying privacy technologies like on-device processing, giving people tools to control their experience, or advancing anti-malware, anti-phishing, and password security features.

Google

For reporting purposes Google comprises two segments: Google Services and Google Cloud.

Google Services

Serving Our Users

We have always been committed to building helpful products that can improve the lives of millions of people worldwide. Our product innovations are what make our services widely used, and our brand one of the most recognized in the world. Google Services' core products and platforms include ads, Android, Chrome, devices, Gmail, Google Drive, Google Maps, Google Photos, Google Play, Search, and YouTube, with broad and growing adoption by users around the world.

Our products and services have come a long way since the company was founded more than 25 years ago. While Google Search started as a way to find web pages, organized into ten blue links, we have driven technical advancements and product innovations that have transformed Google Search into a dynamic, multimodal experience. We first expanded from traditional desktop browsers into mobile web search, making it easier to navigate on smaller screens. As new types of content surfaced on the internet, Universal Search made it possible to search multiple content types, like news, images, videos, and more, to deliver rich, relevant results. The introduction of new search modalities, like voice and visual search, made it easier for people to express their curiosity in natural and intuitive ways. We took that a step further with multisearch, which lets people search with text and images at the same time. Large language models have made it possible to express more natural language queries, vastly improving the types of questions users can ask, and the quality of results. Each advancement has made it easier and more natural for people to find what they are looking for. We are now using Gemini customized for Google Search to provide our users an improved Search experience, and AI Overviews has been released in more than one hundred countries, reaching more than one billion users.

This drive to make information more accessible and helpful has led us over the years to improve the discovery and creation of digital content both on the web and through platforms like Google Play and YouTube. People are consuming many forms of digital content, including watching videos, streaming TV, playing games, listening to music, reading books, and using apps. Working with content creators and partners, we continue to build new ways for people around the world to create and find great digital content.

Fueling all of these great digital experiences are extraordinary platforms and devices. That is why we continue to invest in platforms like our Android mobile operating system, Chrome browser, and Chrome operating system, as well as our family of devices. We see tremendous potential for devices to be helpful and make people's lives easier by combining the best of our AI, software, and hardware. This potential is reflected in our latest generation of devices, such as the new Pixel 9 series and the Pixel Watch 3. Creating products and services that people rely on every day is a journey that we are investing in for the long-term.

How We Make Money

We have built world-class advertising technologies for advertisers, agencies, and publishers to power their digital marketing businesses. Our advertising solutions help millions of companies grow their businesses through our wide range of products across devices and formats, and we aim to ensure positive user experiences by serving the right ads at the right time and by building deep partnerships with brands and agencies. AI has been foundational to our advertising business for more than a decade. Products like Demand Gen, Performance Max, and Product Studio use the full power of our AI to help advertisers find untapped and incremental conversion opportunities.

Google Services generates revenues primarily by delivering both performance and brand advertising that appears on Google Search & other properties, YouTube, and Google Network partners' properties ("Google Network properties"). We continue to invest in both performance and brand advertising and seek to improve the measurability of advertising so advertisers understand the effectiveness of their campaigns.

- **Performance advertising** creates and delivers relevant ads that users will click on leading to direct engagement with advertisers. Performance advertising lets our advertisers connect with users while driving measurable results. Our ads tools allow performance advertisers to create simple text-based ads.
- **Brand advertising** helps enhance users' awareness of and affinity for advertisers' products and services, through videos, text, images, and other interactive ads that run across various devices. We help brand advertisers deliver digital videos and other types of ads to specific audiences for their brand-building marketing campaigns.

We have allocated substantial resources to stopping bad advertising practices and protecting users on the web. We focus on creating the best advertising experiences for our users and advertisers in many ways, including filtering out invalid traffic, removing billions of bad ads from our systems every year, and closely monitoring the sites, apps, and videos where ads appear and blocklisting them when necessary to ensure that ads do not fund bad content.

In addition, Google Services increasingly generates revenues from products and services beyond advertising, including:

- **consumer subscriptions,** which primarily include revenues from YouTube services, such as YouTube TV, YouTube Music and Premium, and NFL Sunday Ticket, as well as Google One;
- **platforms**, which primarily include revenues from Google Play sales of apps and in-app purchases; and
- **devices,** which primarily include sales of the Pixel family of devices.

Google Cloud

Through our Google Cloud Platform and Google Workspace offerings, Google Cloud generates revenues primarily from consumption-based fees and subscriptions for infrastructure, platform, applications, and other cloud services. Customers use Google Cloud in multiple ways such as:

- **AI-optimized Infrastructure:** provides open, reliable, and scalable compute, networking, and storage to enable customers to run workloads anywhere — on our Cloud, at the edge, or in their data centers. It can be used to migrate and modernize IT systems and to train and serve various types of AI models.
- **Developer Platform:** provides developers, through the Vertex AI platform, the ability to train, tune, augment, test, and deploy applications using Gemini and other leading generative AI models. We offer widely used first-party, third-party, and open models along with services such as vector search, grounding, and distillation to further improve the cost and quality of models.
- **Cybersecurity:** provides AI powered cybersecurity solutions to help customers analyze, detect, protect, and respond to a broad range of cybersecurity threats, to further strengthen security outcomes, prioritize which threats to investigate, and identify attack paths, as well as accelerate resolution of cybersecurity threats.

- **Data and Analytics:** provides a variety of different types of databases — relational, key-value, and in-memory — to store and manage data for different types of applications and deliver data to AI agents. Our data platform also unifies data lakes, data warehouses, data governance, and advanced machine learning into a single platform that helps users analyze data using AI models across any cloud.
- **Applications:** offers a broad applications portfolio, including Gemini for Google Cloud and Gemini for Google Workspace, as well as purpose-built agents for Search and our Customer Engagement Suite. AI has been used in Google Workspace for years to improve grammar, efficiency, security, and more with features like Smart Reply, Smart Compose, and malware and phishing protection in Gmail. Now, Gemini for Google Workspace brings our AI-powered agents into Gmail, Docs, Sheets and more to help users write, organize, visualize, accelerate workflows, and have richer meetings.

Other Bets

Across Alphabet, we are also using technology to try to solve big problems that affect a wide variety of industries, including transportation and health technology. Alphabet's investment in the portfolio of Other Bets includes businesses that are at various stages of development, ranging from those in the R&D phase to those that are in the beginning stages of commercialization. Our goal is for them to become thriving, successful businesses. Other Bets operate as independent companies and some of them have their own boards with independent members and outside investors. While these early-stage businesses naturally come with considerable uncertainty, some of them are already generating revenue and making important strides in their industries. Revenues from Other Bets are generated primarily from the sale of healthcare-related services, and internet services.

Competition

Our business is characterized by rapid change as well as new and disruptive technologies. We face formidable competition in every aspect of our business, including but not limited to, from:

- general purpose search engines and information services;
- vertical search engines and e-commerce providers for queries on topics such as those related to travel, jobs, and health, which users may navigate directly to rather than go through Google;
- online advertising platforms and networks, including online shopping and streaming services;
- other forms of advertising, such as billboards, magazines, newspapers, radio, and television as our advertisers typically advertise in multiple media, both online and offline;
- digital content and application platform providers;
- providers of enterprise cloud services;
- developers and providers of AI products and services;
- companies that design, manufacture, and market consumer hardware products, including businesses that have developed proprietary platforms;
- providers of digital video services;
- social networks, which users may rely on for product or service referrals, rather than seeking information through traditional search engines;
- providers of workspace communication and connectivity products; and
- digital assistant providers.

Competing successfully depends heavily on our ability to develop and distribute innovative products and technologies to the marketplace across our businesses. For example, for advertising, competing successfully depends on attracting and retaining:

- users, for whom other products and services are literally one click away, largely on the basis of the relevance of our advertising, as well as the general usefulness, security, and availability of our products and services;
- advertisers, primarily based on our ability to generate sales leads, and ultimately customers, and to deliver their advertisements in an efficient and effective manner across a variety of distribution channels; and
- content providers, primarily based on the quality of our advertiser base, our ability to help these partners generate revenues from advertising, and the terms of our agreements with them.

For additional information about competition, see Item 1A Risk Factors of this Annual Report on Form 10-K.

Ongoing Commitment to Sustainability

Our environmental strategy has two key pillars, supported by our dedication to accessible information and technological innovation:

- Our products: We are empowering people with information about the environmental impacts of their choices.
- Our operations: We are working to drive sustainability and efficiency across our operations and value chain.

Through our products, we have an aspiration to help individuals, cities, and other partners collectively reduce one gigaton of their carbon equivalent emissions annually by 2030.

In 2021, we set an ambitious goal to reach net-zero emissions across all of our operations and value chain by 2030. To make progress toward this effort, we aim to reduce 50% of our combined Scope 1, Scope 2 (market-based), and Scope 3 absolute emissions (compared to our 2019 base year) by 2030, and we plan to invest in nature-based and technology-based carbon removal solutions to neutralize our remaining emissions.

Our primary approach to reducing our Scope 2 emissions is through the procurement of carbon-free energy (CFE). In 2020, we set a goal to run on 24/7 CFE every hour of every day on every grid where we operate by 2030. In 2023, we began implementing our carbon removals strategy, and we have begun establishing impactful partnerships and have started contracting for carbon removal credits.

Achieving net-zero emissions and 24/7 CFE by 2030 are extremely ambitious goals. We also know that our path to net-zero emissions will not be easy or linear. Some of our plans may take years to deliver results, particularly where they involve building new large-scale infrastructure with long lead times. Our approach will continue to evolve and will require us to navigate significant uncertainty, including the uncertainty around the future environmental impact of AI, which is complex and difficult to predict. In addition, solutions for some key global challenges do not currently exist, and will depend heavily on the development and improvement of new technologies by us and by the energy sector. As our business and industry continue to evolve, we expect our total GHG emissions to rise before dropping toward our absolute emissions reduction target.

For additional information about risks and uncertainties applicable to our work on sustainability and efficiency, see Item 1A Risk Factors of this Annual Report on Form 10-K.

Culture and Workforce

We are a company of curious, talented, and passionate people. We embrace collaboration and creativity, and encourage the iteration of ideas to address complex challenges in technology and society.

Our people are critical for our continued success, so we work hard to create an environment where employees can have fulfilling careers, and be happy, healthy, and perform at a high level. We offer industry-leading benefits and programs to take care of the diverse needs of our employees and their families, including opportunities for career growth and development, resources to support their financial health, and access to excellent healthcare choices. Our competitive compensation programs help us to attract and retain key talent, and we will continue to invest in recruiting talented people to technical and non-technical roles, and rewarding them well. We provide a variety of high quality training and support to managers to build and strengthen their capabilities-–ranging from courses for new managers, to learning resources that help them provide feedback and manage performance, to coaching and individual support.

As of December 31, 2024, Alphabet had 183,323 employees. We have work councils and statutory employee representation obligations in certain countries, and we are committed to supporting protected labor rights, maintaining an open culture, and listening to all employees. Supporting healthy and open dialogue is central to how we work, and we communicate information about the company through multiple internal channels to our employees.

When appropriate we partner with outside companies on a contractual basis to provide a specialized service or to temporarily cover a short-term need. The employees of our suppliers and staffing partners — vendors and temporary staff, respectively — and independent contractors who are self-employed, make up our extended workforce. We choose our partners and staffing agencies carefully, and review their compliance with Google’s Supplier Code of Conduct.

Government Regulation

We are subject to numerous United States (U.S.) federal, state, and local, as well as foreign laws and regulations covering a wide variety of subjects, and the scope of this coverage continues to broaden with continuing new legal and regulatory developments in the U.S. and internationally. Like other companies in the technology industry, we face increasingly heightened scrutiny from both U.S. and foreign governments with respect to our compliance with laws and regulations. Many of these laws and regulations are evolving and their applicability and scope, as interpreted by the courts, remain uncertain. Particularly with regard to AI; competition; consumer protection; content moderation; data privacy and security; news publications; and sustainability and other social matters, we have seen an increase in new and evolving laws and regulations, as well as related enforcement actions and investigations, being proposed and implemented in recent years by legislative and regulatory bodies around the world. As we have seen in recent years, different laws and regulations on the same topic may not always have the same requirements, and even when requirements overlap, the rules are not always consistently implemented, interpreted, and enforced from jurisdiction to jurisdiction.

Our compliance with these laws and regulations may be onerous and could, individually or in the aggregate, increase our cost of doing business, make our products and services less useful, limit our ability to pursue certain business practices or offer certain products and services, cause us to change our business models and operations, affect our competitive position relative to our peers, and/or otherwise harm our business, reputation, financial condition, and operating results.

For additional information about government regulation applicable to our business, see Item 1A Risk Factors; Trends in Our Business and Financial Effect in Part II, Item 7; and Legal Matters in Note 10 of the Notes to Consolidated Financial Statements included in Part II, Item 8 of this Annual Report on Form 10-K.

Intellectual Property

We rely on various intellectual property laws, confidentiality procedures and contractual provisions to protect our proprietary technology and our brand. We have registered, and applied for the registration of, U.S. and international trademarks, service marks, domain names, and copyrights. We have also filed patent applications in the U.S. and foreign countries covering certain of our technology, and acquired patent assets to supplement our portfolio. We have licensed in the past, and expect that we may license in the future, certain of our rights to other parties. For additional information, see Item 1A Risk Factors of this Annual Report on Form 10-K.

Available Information

Our website is located at www.abc.xyz, and our investor relations website is located at www.abc.xyz/investor. Access to our Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, and our Proxy Statements, and any amendments to these reports, is available on our investor relations website, free of charge, after we file or furnish them with the SEC and they are available on the SEC's website at www.sec.gov.

We webcast via our investor relations YouTube channel and website our earnings calls and certain events we participate in or host with members of the investment community. Our investor relations website also provides notifications of news or announcements regarding our financial performance and other items that may be material or of interest to our investors, including SEC filings, investor events, press and earnings releases, and blogs. We also share Google news and product updates on Google's Keyword blog at https://www.blog.google/ and News From Google page on X at x.com/NewsFromGoogle, and our executive officers may also use certain social media channels, such as X and LinkedIn, to communicate information about earnings results and company updates, which may be of interest or material to our investors. Further, corporate governance information, including our certificate of incorporation, bylaws, governance guidelines, board committee charters, and code of conduct, is also available on our investor relations website under the heading "Governance." The information contained on, or that may be accessed through our websites or our executive officers' social media channels, is not incorporated by reference into this Annual Report on Form 10-K or in any other report or document we file with the SEC, and any references to our websites are intended to be inactive textual references only.

ITEM 1A. RISK FACTORS

Our operations and financial results are subject to various risks and uncertainties, including but not limited to those described below, which could harm our business, reputation, financial condition, and operating results, and affect the trading price of our Class A and Class C stock.

Risks Specific to our Company

We generate a significant portion of our revenues from advertising. Reduced spending by advertisers, a loss of partners, or new and existing technologies that block ads online and/or affect our ability to personalize ads could harm our business.

We generated more than 75% of total revenues from online advertising in 2024. Many of our advertisers, companies that distribute our products and services, digital publishers, and content providers can terminate their contracts with us at any time. These partners may not continue to do business with us if we do not create more value (such as increased numbers of users or customers, new sales leads, increased brand awareness, or more effective monetization) than their available alternatives. Changes to our advertising policies and data privacy practices, such as our initiatives related to third-party cookies, including our announcement in July 2024 to move from phasing out all third-party cookies to a proposed user choice model (which remains subject to continuing discussions with regulators), as well as changes to other companies' advertising and/or data privacy practices have in the past, and may in the future, affect the advertising services that we are able to provide. In addition, technologies have been developed that make personalized ads more difficult, or that block the display of ads altogether, and some providers of online services have integrated technologies that could potentially impair the availability and functionality of third-party digital advertising. Failing to provide superior value or deliver advertisements effectively and competitively could harm our business, reputation, financial condition, and operating results.

In addition, expenditures by advertisers tend to correlate with overall economic conditions. Adverse macroeconomic conditions have affected, and may in the future affect, the demand for advertising, resulting in fluctuations in the amounts our advertisers spend on advertising, which could harm our financial condition and operating results.

We face intense competition. If we do not continue to innovate and provide products and services that are useful to users, customers, and other partners, we may not remain competitive, which could harm our business, financial condition, and operating results.

Our business environment is rapidly evolving and intensely competitive. Our businesses face changing technologies, shifting user needs, and frequent introductions of rival products and services. To compete successfully, we must accurately anticipate technology developments and deliver innovative, relevant and useful products, services, and technologies in a timely manner. As our businesses evolve, the competitive pressure to innovate will encompass a wider range of products and services. We must continue to invest significant resources in technical infrastructure and R&D, including through acquisitions, in order to enhance our technology, products, and services.

We have many competitors in different industries. Our current and potential domestic and international competitors range from large and established companies to emerging start-ups. Some competitors have longer operating histories and well-established relationships in various sectors. They can use their experience and resources in ways that could affect our competitive position, including by making acquisitions and entering into other strategic arrangements; continuing to invest heavily in technical infrastructure, R&D, and in talent; initiating intellectual property and competition claims (whether or not meritorious); and continuing to compete for users, advertisers, customers, and content providers. Further, discrepancies in enforcement of existing laws may enable our lesser known competitors to aggressively interpret those laws without commensurate scrutiny, thereby affording them competitive advantages. Our competitors may also be able to innovate and provide products and services faster or more cost effectively than we can or may foresee the need for products and services before we do.

We are expanding our investment in AI across the entire company. This includes generative AI and continuing to integrate AI capabilities into our products and services. AI technology and services are highly competitive, rapidly evolving, and require significant investment, including technical infrastructure, development and operational costs, to meet the changing needs and expectations of our existing users and attract new users. Our ability to deploy certain AI technologies critical for our products and services and for our business strategy may depend on the availability and pricing of third-party equipment and other technical infrastructure operations costs, including network capacity, energy, and equipment costs. Additionally, other companies may develop AI products and technologies that are similar or superior to our technologies or more cost-effective to develop and/or deploy. Other companies may also have (or in the future may obtain) patents or other proprietary rights that would prevent, limit, or interfere with our ability to make, use, or sell our own AI products and services.

Our financial condition and operating results may also suffer if our products and services are not responsive in a timely manner to the evolving needs and desires of our users, advertisers, publishers, customers, and content providers, or if we miscalculate those needs and desires and invest significantly in areas that fail to gain sufficient market traction. As new and existing technologies continue to develop, competitors and new entrants may be able to offer experiences that are, or that are perceived to be, substantially similar to or better than ours. These technologies could reduce usage of our products and services, and force us to compete in different ways and expend significant resources to develop and operate equal or better products and services. Competitors' success in providing compelling products and services or in attracting and retaining users, advertisers, publishers, customers, and content providers could harm our financial condition and operating results.

Our ongoing investment in new businesses, products, services, and technologies is inherently risky, and could divert management attention and harm our business, financial condition, and operating results.

We have invested and expect to continue to invest in new businesses, products, services, and technologies in a wide range of industries beyond online advertising. The investments that we are making across our businesses, such as building AI capabilities into new and existing products and services, reflect our ongoing efforts to innovate and provide products and services that are helpful to users, advertisers, publishers, customers, and content providers. Our investments ultimately may not be commercially viable or may not result in an adequate return of capital and, in pursuing new strategies, we may incur unanticipated liabilities.

We have invested significantly and expect to continue to invest significantly in our property and equipment, including our technical infrastructure, and we expect these assets to benefit our business over their estimated useful lives. Changes in facts and circumstances such as changes to ongoing business operations, changes in the planned use and utilization of assets, and/or technological advancements, could indicate a change in the period over which we expect to benefit from the asset and impact our financial condition and operating results.

Innovations in our products and services could also result in changes to user behavior and affect our revenue trends. These endeavors involve significant risks and uncertainties, including diversion of resources and management attention from current operations, different monetization models, and the use of alternative investment, governance, or compensation structures that may fail to adequately align incentives across the company or otherwise accomplish their objectives.

Within Google Services, we continue to invest heavily in devices, including our smartphones, home devices, and wearables, which is a highly competitive market with frequent introduction of new products and services, rapid adoption of technological advancements by competitors, increased market saturation in developed countries, short product life cycles, evolving industry standards, continual improvement in performance characteristics, and price and feature sensitivity on the part of consumers and businesses. There can be no assurance we will be able to provide devices that compete effectively.

Within Google Cloud, we devote significant resources to develop and deploy our enterprise-ready cloud services, including Google Cloud Platform and Google Workspace, and we are advancing our AI platforms and models to support these tools and technologies. We are incurring costs to build and maintain infrastructure to support cloud computing services, invest in cybersecurity, and hire talent, particularly to support and scale our sales force. At the same time, our competitors are rapidly developing and deploying cloud-based services. Pricing, including platform switching costs, are challenging across the industry, and delivery models are competitive and constantly evolving, and we may therefore not achieve our business objectives. Further, our business with financial services, healthcare, and public sector customers may present additional risks, including regulatory compliance risks. For instance, we may be subject to government audits and cost reviews, and any failure to comply or any deficiencies found may expose us to legal, financial, and/or reputational risks. Evolving laws and regulations may require us to make new capital investments, build new products, and seek partners to deliver localized services in other countries, and we may not be able to meet sovereign operating requirements.

Within Other Bets, we are investing significantly in areas such as health, life sciences, and transportation, among others. These investment areas face intense competition from large, experienced, and well-funded competitors, and our offerings, many of which involve the development of new and emerging technologies, may not be successful, or be able to compete effectively or operate at sufficient levels of profitability.

In addition, new and evolving products and services, including those that use AI, raise ethical, technological, legal, regulatory, and other challenges, which may negatively affect our brands and demand for our products and services. Because all of these investment areas are inherently risky, no assurance can be given that such strategies and offerings will be successful or will not harm our reputation, financial condition, and operating results.

Our revenue growth rate could decline over time, and we may experience downward pressure on our operating margin in the future.

Our revenue growth rate could decline over time as a result of a number of factors, including changes in the devices and modalities used to access our products and services; changes in geographic mix; deceleration or declines in advertiser spending; competition; changes in customer usage and demand for our existing products and increasing demand for new technologies; decreases in the pricing of our products and services; ongoing product and policy changes; and shifts to lower priced products and services.

In addition, we may experience downward pressure on our operating margin resulting from a variety of factors, such as an increase in the mix of lower-margin products and services, in particular from the continued expansion of our business into new fields, including products and services such as our devices, Google Cloud, and consumer subscription products, as well as significant investments in Other Bets, all of which may have margins lower than those we generate from advertising. In particular, margins on our devices have had, and may continue to have, an adverse effect on our consolidated margins due to pressures on pricing and higher cost of sales. We may also experience downward pressure on our operating margins from increasing regulations, increasing competition, and increasing costs for many aspects of our business, including higher level of investment in technical infrastructure. Further, certain of our costs and expenses are generally less variable in nature and may not correlate to changes in revenue. We may also not be able to continue to execute our efficiency efforts successfully or in a timely manner. Due to these factors and the evolving nature of our business, our historical revenue growth rate and historical operating margin may not be indicative of our future performance. For additional information, see Trends in Our Business and Financial Effect and Revenues and Monetization Metrics in Part II, Item 7 of this Annual Report on Form 10-K.

Our intellectual property rights are valuable, and any inability to protect them could reduce the value of our products, services, and brands as well as affect our ability to compete.

Our patents, trademarks, trade secrets, copyrights, and other intellectual property rights are important assets for us. Various events outside of our control pose a threat to our intellectual property rights, as well as to our products, services, and technologies. For example, effective intellectual property protection may not be available in every country in which our products and services are distributed or made available through the Internet. Also, the efforts we have taken to protect our proprietary rights may not be sufficient or effective. Although we seek to obtain patent protection for our innovations, it is possible we may not be able to protect some of these innovations. Moreover, we may not have adequate patent or copyright protection for certain innovations that later turn out to be important. There is always the possibility that the scope of the protection gained will be insufficient or that an issued patent may be deemed invalid or unenforceable.

We also seek to maintain certain intellectual property as trade secrets. The secrecy of such trade secrets and other sensitive information could be compromised, which could cause us to lose the competitive advantage resulting from these trade secrets. We also face risks associated with our trademarks. For example, there is a risk that the word “Google” could become so commonly used that it becomes synonymous with the word “search.” Some courts have ruled that "Google" is a protectable trademark, but it is possible that other courts, particularly those outside of the U.S., may reach a different determination. If this happens, we could lose protection for this trademark, which could result in other people using the word “Google” to refer to their own products, thus diminishing our brand.

Any significant impairment of our intellectual property rights could harm our business and our ability to compete. Also, protecting our intellectual property rights is costly and time consuming. Any increase in the unauthorized use of our intellectual property could make it more expensive to do business and harm our financial condition and operating results.

Our business depends on strong brands, and failing to maintain and enhance our brands would hurt our ability to expand our base of users, advertisers, customers, content providers, and other partners.

Our strong brands have significantly contributed to the success of our business. Maintaining and enhancing the brands within Google Services, Google Cloud, and Other Bets increases our ability to enter new categories and launch new and innovative products and services that better serve the needs of our users, advertisers, customers, content providers, and other partners. Our brands have been, and may in the future be, negatively affected by a number of factors, including, among others, reputational issues, third-party content shared on our platforms, data privacy and security issues and developments, and product or technical performance failures. For example, if we fail to respond appropriately to the sharing of misinformation or objectionable content on our services and/or products or objectionable practices by advertisers, or otherwise to adequately address user concerns, our users may lose confidence in our brands.

Furthermore, failure to maintain and enhance our brands could harm our business, reputation, financial condition, and operating results. Our success will depend largely on our ability to remain a technology leader and continue to provide high-quality, trustworthy, innovative products and services that are truly useful and play a valuable role in a range of settings.

We face a number of manufacturing and supply chain risks that could harm our business, financial condition, and operating results.

We face a number of risks related to manufacturing and supply chain management, which could affect our ability to supply both our products and our services.

We rely on contract manufacturers to manufacture or assemble our devices as well as servers and networking equipment used in our technical infrastructure, certain components of which we may supply. We rely on third parties to supply components and distribute our products and services. Our business could be negatively affected if we are not able to engage these companies with the necessary capabilities or capacity on reasonable terms, or if those we engage fail to meet their obligations (whether due to financial difficulties or other reasons), or make adverse changes in the pricing or other material terms of our arrangements with them.

We have experienced and/or may in the future experience supply shortages, price increases, quality issues, and/or longer lead times that could negatively affect our operations, driven by raw material and/or component availability, manufacturing capacity, labor shortages, industry allocations, logistics capacity, inflation, foreign currency exchange rates, tariffs, sanctions and export controls, trade disputes and barriers, forced labor concerns, sustainability sourcing requirements, geopolitical tensions, armed conflicts, natural disasters or pandemics, the effects of climate change (such as sea level rise, drought, flooding, heat waves, wildfires and resultant air quality effects and power shutdowns associated with wildfire prevention, and increased storm severity), power and transmission availability, and significant changes in the financial or business condition of our suppliers. Some of the components we use in our technical infrastructure and our devices are available from only one or limited sources, and we may not be able to find replacement vendors on favorable terms in the event of a supply chain disruption. A significant supply interruption that affects us or our vendors could delay critical data center upgrades or expansions and delay consumer product availability.

We may enter into long-term contracts for materials and products that commit us to significant terms and conditions. We may face costs for materials and products that are not consumed due to market demand, technological change, changed consumer preferences, quality, product recalls, and warranty issues. For instance, because certain of our hardware supply contracts have volume-based pricing or minimum purchase requirements, if the volume of sales of our devices decreases or does not reach projected targets, we could face increased materials and manufacturing costs or other financial liabilities that could make our products more costly per unit to manufacture and harm our financial condition and operating results. Furthermore, certain of our competitors may negotiate more favorable contractual terms based on volume and other commitments that may provide them with competitive advantages and may affect our supply. For example, industry supply capacity for AI accelerators, including Graphics Processing Units, or GPUs, as well as our custom-built TPUs, is highly competitive and rapidly evolving. If we are unable to negotiate favorable contractual terms or our competitors claim the supply or capacity first, we may face supply constraints.

Our devices have had, and in the future may have, quality issues resulting from design, manufacturing, or operations. Sometimes, these issues may be caused by components we purchase from other manufacturers or suppliers. If the quality of our products and services does not meet expectations or our products or services are defective or require a recall, it could harm our business, reputation, financial condition, and operating results.

We require our suppliers and business partners to comply with laws and, where applicable, our company policies and practices, such as the Google Supplier Code of Conduct, regarding workplace and employment practices, data security, environmental compliance, and intellectual property licensing, but we do not control them or their practices. Violations of law or unethical business practices could result in supply chain disruptions, canceled orders, harm to key relationships, and damage to our reputation. Their failure to procure necessary license rights to intellectual property could affect our ability to sell our products or services and expose us to litigation or financial claims.

Interruption to, interference with, or failure of our complex information technology and communications systems could hurt our ability to effectively provide our products and services, which could harm our reputation, financial condition, and operating results.

The availability of our products and services and fulfillment of our customer contracts depend on the continuing operation of our information technology and communications systems. Our systems are vulnerable to damage, interference, or interruption from modifications or upgrades, terrorist attacks, state-sponsored attacks, natural disasters or pandemics, geopolitical tensions or armed conflicts, export controls and sanctions, the effects of climate change (such as sea level rise, drought, flooding, heat waves, wildfires and resultant air quality effects and power shutdowns associated with wildfire prevention, and increased storm severity), power and transmission availability challenges, utility outages, telecommunications failures, computer viruses, software bugs, cyber attacks, supply-chain attacks, computer denial of service attacks, phishing schemes, or other attempts to harm or access our systems. Some of our data centers are located in areas with a high risk of major earthquakes or other natural disasters. Our data centers are also subject to break-ins, sabotage, and intentional acts of vandalism, and, in some cases, to potential disruptions resulting from problems experienced by facility operators or disruptions as a result of geopolitical tensions and conflicts happening in the area. Some of our systems are not fully redundant, and disaster recovery planning cannot account for all eventualities. The occurrence of a natural disaster or pandemic, closure of a facility, or other unanticipated problems affecting our data centers could result in lengthy interruptions in our service. In addition, our products and services are highly technical and complex and have contained in the past, and may contain in the future, errors or vulnerabilities, which could result in interruptions in or failure of our services or systems. Any of these incidents could impede or prevent us from effectively offering products and providing services, which could harm our business, reputation, financial condition, and operating results.

Our international operations expose us to additional risks that could harm our business, reputation, financial condition, and operating results.

Our international operations are significant to our revenues and net income, and we plan to continue to grow internationally. International revenues accounted for approximately 51% of our consolidated revenues in 2024. In addition to risks described elsewhere in this section, our international operations expose us to other risks, including the following:

- restrictions on foreign ownership and investments, and stringent foreign exchange controls that might prevent us from repatriating cash earned in countries outside the U.S.;
- sanctions, import and export controls, other market access barriers, political unrest, geopolitical tensions, changes in regimes, or armed conflict (such as ongoing conflicts in the Middle East and Ukraine), any of which may affect our business continuity, increase our operating costs, limit demand for our products and services, limit our ability to source components or final products, or prevent or impede us from operating in certain jurisdictions, complying with local laws, or offering products or services;
- longer payment cycles in some countries, increased credit risk, and higher levels of payment fraud;
- an evolving foreign policy landscape that could harm our revenues and could subject us to litigation, new regulatory costs and challenges (including new customer requirements), uncertainty regarding regulatory outcomes, and other liabilities under local laws that may not offer due process or clear legal precedent;
- anti-corruption laws, such as the U.S. Foreign Corrupt Practices Act, and other local laws prohibiting certain payments to government officials, violations of which could result in civil and criminal penalties; and
- different employee/employer relationships; different labor laws, regulations, and labor practices; and other challenges caused by distance, language, local expertise, and cultural differences, increasing the complexity of doing business in multiple jurisdictions.

Because we conduct business in currencies other than U.S. dollars but report our financial results in U.S. dollars, we have faced, and will continue to face, exposure to fluctuations in foreign currency exchange rates. Although we hedge a portion of our international currency exposure, significant fluctuations in exchange rates between the U.S. dollar and foreign currencies have in the past and may in the future adversely affect our revenues and earnings. Hedging programs are also inherently risky and could expose us to additional risks that could harm our financial condition and operating results.

We are exposed to fluctuations in the fair values of our investments and, in some instances, our financial statements incorporate inherently subjective valuation methodologies.

The fair value of our debt and equity investments may in the future be, and certain investments have been in the past, negatively affected by liquidity, credit deterioration or losses, performance and financial results of the underlying entities, foreign exchange rates, changes in interest rates, the effect of new or changing regulations, the stock market in general, or other factors.

We measure certain of our non-marketable equity and debt securities, certain other instruments including stock-based compensation awards settled in the stock of Other Bet companies, and certain assets and liabilities acquired in a business combination, at fair value on a nonrecurring basis, which is inherently subjective and requires management judgment and estimation. All gains and losses on non-marketable equity securities are recognized in OI&E, which increases the volatility of our OI&E. The unrealized gains and losses or impairments we record from fair value remeasurements in any particular period may differ significantly from the gains and losses we ultimately realize on such investments.

As a result of these factors, the value of our investments could decline, which could harm our financial condition and operating results.

Risks Related to our Industry

People access our products and services through a variety of platforms and devices that continue to evolve with the advancement of technology and user preferences. If manufacturers and users do not widely adopt versions of our products and services developed for these interfaces, our business could be harmed.

People access our products and services through a growing variety of devices such as desktop computers, phones, laptops and tablets, video game consoles, voice-activated speakers, wearables (including virtual reality and augmented reality devices), automobiles, and television-streaming devices. Our products and services may be less popular on some interfaces. Each manufacturer or distributor may establish unique technical standards for its devices, and our products and services may not be available or may only be available with limited functionality for our users or our advertisers on these devices as a result. Some manufacturers may also elect not to include our products on their devices.

It is hard to predict the challenges we may encounter in adapting our products and services and developing competitive new products and services. We expect to continue to devote significant resources to creating and supporting products and services across multiple platforms and devices. Failing to attract and retain a substantial number of device manufacturers, suppliers, distributors, developers, and users, or failing to develop products and technologies that work well on new devices and platforms, could harm our business, financial condition, and operating results and ability to capture future business opportunities.

Issues in the development and use of AI may result in reputational harm and increased liability exposure.

Our evolving AI-related efforts may give rise to risks related to harmful content, inaccuracies, discrimination, intellectual property infringement or misappropriation, violation of rights of publicity, defamation, data privacy, cybersecurity, and other issues. As a result of these and other challenges associated with innovative technologies, our implementation of AI systems could subject us to competitive harm, regulatory action, legal liability (including under new and proposed legislation and regulations), new applications of existing data protection, privacy, intellectual property, and other laws, and brand or reputational harm.

Some uses of AI will present ethical issues and may have broad effects on society. In order to implement AI responsibly and minimize unintended harmful effects, we have already devoted and will continue to invest significant resources to develop, test, and maintain our products and services, but we may not be able to identify or resolve all AI-related issues, deficiencies, and/or failures before they arise. Unintended consequences, uses, or customization of our AI tools and systems may negatively affect human rights, privacy, employment, or other social concerns, which may result in claims, lawsuits, brand or reputational harm, and increased regulatory scrutiny, any of which could harm our business, financial condition, and operating results.

Data privacy and security concerns relating to our technology and our practices could harm our reputation, cause us to incur significant liability, and deter current and potential users or customers from using our products and services. Computer viruses, software bugs or defects, security breaches, and attacks on our systems could result in the improper disclosure and use of user data and interference with our users' and customers' ability to use our products and services, harming our business and reputation.

Concerns about, including the adequacy of, our practices with regard to the collection, use, governance, disclosure, or security of personal data or other data-privacy-related matters, even if unfounded, could harm our business, reputation, financial condition, and operating results. Our policies and practices may change over time as expectations and regulations regarding privacy and data change.

Our products and services involve the storage, handling, and transmission of proprietary and other sensitive information. Malicious software such as viruses, software bugs, theft, misuse, defects, vulnerabilities in our products and services, as well as cyber attacks, phishing schemes, and other types of security breaches expose us to a risk of loss or improper use and disclosure of such information, which could result in litigation and other potential liabilities, including regulatory fines and penalties, as well as reputational harm. Additionally, our products incorporate highly technical and complex technologies, and thus our technologies and software have contained, and are likely in the future to contain, undetected errors, bugs, and/or vulnerabilities. We continue to add new features involving AI to our offerings and internal systems, and features that rely on AI may be susceptible to unanticipated security threats as our and the market's understanding of AI-centric security risks and protection methods continue to develop. We have in the past discovered, and may in the future discover, some errors in our software code only after we have released the code. Systems and control failures, security breaches, failure to comply with our privacy policies, and/or inadvertent disclosure of user data could result in regulatory and legal exposure, seriously harm our reputation, brand, and business, and impair our ability to attract and retain users or customers. Such incidents have occurred in the past and may continue to occur due to the scale and nature of our products and services. While there is no guarantee that such incidents will not cause significant damage, we expect to continue to expend significant resources to maintain security protections that limit the effect of bugs, theft, misuse, and security vulnerabilities or breaches.

We experience cyber attacks and other attempts to gain unauthorized access to our systems on a regular basis. Cyber attacks continue to evolve in sophistication and volume, and inherently may be difficult to detect for long periods of time. For example, the development and implementation of AI technologies may further increase our exposure to or exacerbate the risks of cyber attacks or other security incidents, particularly where such technologies are exploited by third parties to breach our or other parties' systems, including when such technologies are used to target our employees or impersonate members of senior management in order to gain unauthorized access to our systems. We have also seen, and will continue to see, industry-wide software supply chain vulnerabilities, which could affect our or other parties' systems. We expect to continue to experience such incidents or vulnerabilities in the future. Our efforts to prevent security incidents and address undesirable activity on our platform may require us to spend additional resources to prepare and defend against such threats, and could also increase the risk of retaliatory attack. In addition, we face the risk of cyber attacks by nation-states and state-sponsored actors. These attacks may target us or our customers, particularly our public sector customers (including federal, state, and local governments). Geopolitical tensions or armed conflicts, such as the ongoing conflict in the Middle East and Ukraine, may increase these risks.

We may experience security and/or privacy issues, whether due to employee or insider error or malfeasance, system errors, or vulnerabilities in our or other parties' systems. While we may not determine some of these issues to be material at the time they occur and may remedy them quickly, there is no guarantee that these issues will not ultimately result in significant legal, financial, and reputational harm, including government inquiries, enforcement actions, litigation, and negative publicity. There is also no guarantee that a series of related issues may not be determined to be material at a later date in the aggregate, even if they may not be material individually at the time of their occurrence. Because the techniques used to obtain unauthorized access to, disable or degrade service provided by or otherwise sabotage systems change frequently and often are recognized only after being launched against a target, even taking all reasonable precautions, including those required by law, we have been unable in the past and may continue to be unable to anticipate or detect attacks or vulnerabilities or implement adequate preventative measures.

Further, if any partners with whom we share user or other customer information fail to implement adequate data-security practices, fail to comply with our terms and policies, or otherwise suffer a network or other security breach, our users' data may be improperly accessed, used, or disclosed. If an actual or perceived breach of our or our business partners' or service providers' security occurs, the market perception of the effectiveness of our security measures would be harmed, we could lose users and customers, our trade secrets or those of our business partners may be compromised, and we may be exposed to significant legal and financial risks, including legal claims (which may include class-action litigation) and regulatory actions, fines, and penalties. Any of the foregoing consequences could harm our business, reputation, financial condition, and operating results.

While we have dedicated significant resources to privacy and security incident response capabilities, including dedicated worldwide incident response teams, our response process, particularly during times of a natural disaster or pandemic, may not be adequate, may fail to accurately assess the severity of an incident, may not be fast enough to prevent or limit harm, or may fail to sufficiently remediate an incident. As a result, we may suffer significant legal, reputational, or financial exposure, which could harm our business, financial condition, and operating results.

For additional information, see also our risk factor on privacy and data protection regulations under 'Risks Related to Laws, Regulations, and Policies' below.

Our ongoing investments in safety, security, and content review will likely continue to identify abuse of our platforms and misuse of user data.

In addition to our efforts to prevent and mitigate cyber attacks, we are making significant investments in safety, security, and review efforts to combat misuse of our services and unauthorized access to user data by third parties, including investigation and review of platform applications that could access the information of users of our services. As a result of these efforts, we have in the past discovered, and may in the future discover, incidents of unnecessary access to or misuse of user data or other undesirable activity by third parties. However, we may not have discovered, and may in the future not discover, all such incidents or activity, whether as a result of our data limitations, including our lack of visibility over our encrypted services, the scale of activity on our platform, or other factors, including factors outside of our control such as a natural disaster or pandemic, and we may learn of such incidents or activity via third parties. Such incidents and activities may include the use of user data or our systems in a manner inconsistent with our terms, contracts or policies, the existence of false or undesirable user accounts, election interference, improper ad purchases, activities that threaten people's safety on- or off-line, or instances of spamming, scraping, or spreading disinformation. While we may not determine some of these incidents to be material at the time they occurred and we may remedy them quickly, there is no guarantee that these issues will not ultimately result in significant legal, financial, and reputational harm, including government inquiries and enforcement actions, litigation, and negative publicity. There is also no guarantee that a series of related issues may not be determined to be material at a later date in the aggregate, even if they may not be material individually at the time of their occurrence.

We may also be unsuccessful in our efforts to enforce our policies or otherwise prevent or remediate any such incidents. Any of the foregoing developments may negatively affect user trust and engagement, harm our reputation and brands, require us to change our business practices in ways that harm our business operations, and adversely affect our business and financial results. Any such developments may also subject us to additional litigation and regulatory inquiries, which could result in monetary penalties and damages, divert management's time and attention, and lead to enhanced regulatory oversight.

Problematic content on our platforms, including low-quality user-generated content, web spam, content farms, and other violations of our guidelines could affect the quality of our services, which could harm our reputation and deter our current and potential users from using our products and services.

We, like others in the industry, face violations of our content guidelines across our platforms, including sophisticated attempts by bad actors to manipulate our hosting and advertising systems to fraudulently generate revenues, or to otherwise generate traffic that does not represent genuine user interest or intent. While we invest significantly in efforts to promote high-quality and relevant results and to detect and prevent low-quality content and invalid traffic, we have been unable and may continue to be unable to detect and prevent all such abuses or promote uniformly high-quality content. Increased use of AI in our offerings and internal systems may create new avenues of abuse for bad actors.

Many websites violate or attempt to violate our guidelines, including by seeking to inappropriately rank higher in search results than our search engine's assessment of their relevance and utility would rank them. Such efforts have affected, and may continue to affect, the quality of content on our platforms and lead them to display false, misleading, or undesirable content. Although English-language web spam in our search results has been reduced, and web spam in most other languages is limited, we expect web spammers will continue to seek inappropriate ways to improve their rankings. Although we continue to invest in and deploy proprietary technology to detect and prevent web spam on our platforms, there is no guarantee that our technology will always be successful, and our users may have negative experiences on our platforms if our technology fails to work as intended, which may affect our users' decisions in continuing to use our platforms. We also face other challenges from low-quality and irrelevant content websites, including content farms, which are websites that generate large quantities of low-quality content to help them improve their search rankings. We are continually launching algorithmic changes designed to detect and prevent abuse from low-quality websites, but we may not always be successful. We also face other challenges on our platforms, including violations of our content guidelines involving incidents such as attempted election interference; activities that threaten the safety and/or well-being of our users, including minors, on- or off-line; and the spreading of misinformation or disinformation.

If we fail to either detect and prevent an increase in problematic content or effectively promote high-quality content, it could hurt our reputation for delivering relevant information or reduce use of our platforms, harming our financial condition and operating results. It may also subject us to litigation and regulatory actions, which could result in monetary penalties and damages and divert management’s time and attention.

Our business depends on continued and unimpeded access to the Internet by us and our users. Internet access providers may be able to restrict, block, degrade, or charge for access to certain of our products and services, which could lead to additional expenses and the loss of users and advertisers.

Our products and services depend on the ability of our users to access the Internet, and certain of our products require significant network capacity to work effectively. Currently, this access is provided by companies that have significant market power in the broadband and internet access marketplace, including incumbent telephone companies, cable companies, mobile communications companies, and government-owned service providers. Some of these providers have taken, or have stated that they may take, measures that could degrade, disrupt, or increase the cost of user access to certain of our products by restricting or prohibiting the use of their infrastructure to support or facilitate our offerings, by charging increased fees to us or our users to provide our offerings, or by providing our competitors preferential access. Some jurisdictions have adopted regulations prohibiting certain forms of discrimination by internet access providers; however, substantial uncertainty exists in the U.S. and elsewhere regarding such protections. In addition, in some jurisdictions, our products and services have been subject to government-initiated restrictions or blockages. These could harm existing key relationships, including with our users, customers, advertisers, and/or content providers, and impair our ability to attract new ones; harm our reputation; and increase costs, thereby negatively affecting our business.

Risks Related to Laws, Regulations, and Policies

We are subject to a variety of new, existing, and changing laws and regulations worldwide that could harm our business, and will likely be subject to an even broader scope of laws and regulations as we continue to expand our business.

We are subject to numerous U.S. federal and state as well as foreign laws and regulations covering a wide variety of subjects, and our introduction of new businesses, products, services, and technologies will likely continue to subject us to additional laws and regulations. In recent years, governments around the world have proposed and adopted a large number of new laws and regulations relevant to the digital economy, particularly in the areas of data privacy and security, competition, AI, and online content. The costs of compliance with these measures are high and are likely to increase in the future, including as a result of differing, and sometimes conflicting, laws and regulations.

New or changing laws and regulations, or interpretations or applications of existing laws and regulations in a manner inconsistent with our interpretations of such laws and regulations and/or our practices, have resulted in, and may continue to result in, less useful products and services, altered business models and operations, limited ability to pursue certain business practices or offer certain products and services, substantial costs, and civil or criminal liability. Examples include laws and regulations regarding:

- **Competition and technology platforms' business practices:** Laws and regulations focused on large technology platforms, including the Digital Markets Act in the European Union (EU); regulations and legal settlements in the U.S., South Korea, and elsewhere that affect Google Play's billing policies, fees, and business model; as well as litigation and new and expected regulations in a range of jurisdictions, including, among others, in the UK, Japan, and India.
- **AI:** Laws and regulations focused on the development, use, and provision of AI technologies and other digital products and services, which could result in monetary penalties or other regulatory actions. For example, the EU AI Act came into force on August 1, 2024, and will generally become fully applicable after a two-year transitional period (although certain obligations will take effect at an earlier or later time). The EU AI Act introduces various requirements for AI systems and models placed on the market or put into service in the EU, including specific transparency and other requirements for general purpose AI systems and the models on which those systems are based. In the U.S., there is increasing uncertainty as to the federal government's approach to AI regulation going forward, as the continued applicability of the White House's 2023 Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence, which lays out a framework for the U.S. government, among other things, to monitor private sector development of certain foundation models, remains subject to regulatory development. Several states are considering enacting or have already enacted regulations concerning the use of AI technologies, including those focused on consumer protection, and depending on the scope of AI regulation at the federal level, some states may move to regulate AI model development and deployment. Further, at the federal and state level, there have been various proposals (and in some cases laws enacted) addressing "deepfakes" and other AI-generated synthetic media.
- **Data privacy, collection, processing, and portability:** Laws and regulations further restricting the collection, processing, and/or sharing of user or advertising-related data, including privacy and data protection laws; laws affecting the processing of children's data (as discussed further below), data breach notification laws; laws limiting data transfers (including data localization laws); and laws requiring data portability (including the EU Data Act, as discussed further below).
- **Copyright and other intellectual property:** Copyright and related laws, including the EU Directive on Copyright in the Digital Single Market and European Economic Area transpositions, which have introduced new licensing regimes, increase liability with respect to content uploaded by users or linked to from our platforms, or create property rights in news publications that could require payments to news agencies and publishers, which may result in other regulatory actions.
- **Content moderation:** Various laws covering content moderation and removal, and related disclosure obligations, such as the EU's Digital Services Act, Florida's Senate Bill 7072 and Texas' House Bill 20, and laws and proposed legislation in Singapore, Australia, and the United Kingdom that impose penalties for failure to remove certain types of content or require disclosure of information about the operation of our services and algorithms, which may make it harder for services like Google Search and YouTube to detect and deal with low-quality, deceptive, or harmful content, or on the other hand, may impinge on the rights of free expression, which, in turn, could impact how our platforms are viewed by users. Additionally, legislators are increasingly focused on regulating online child safety, including content protections for minors under eighteen years of age. These regulations could result in our having to modify our products and services and incur additional costs to operate and monitor minors' experiences on our products and services.
- **Consumer protection:** Consumer protection laws, including the EU's New Deal for Consumers, which could result in monetary penalties and create a range of new compliance obligations.

In addition, the applicability and scope of these and other laws and regulations, as interpreted by courts, regulators, or administrative bodies, remain uncertain and could be interpreted in ways that harm our business. For example, we rely on statutory safe harbors, like those set forth in the Digital Millennium Copyright Act and Section 230 of the Communications Decency Act in the U.S. and the Digital Services Act in Europe, to protect against liability for various linking, caching, ranking, recommending, and hosting activities. Legislation or court rulings affecting these safe harbors may harm us and may impose significant operational challenges. There are legislative proposals and pending litigation in the U.S., EU, and around the world that could diminish or eliminate safe harbor protection for websites and online platforms. Our development, use, and commercialization of AI products and services (including our implementation of AI in our offerings and internal systems) could subject us to regulatory action and legal liability, including under specific legislation regulating AI, as well as new applications of existing data protection, cybersecurity, privacy, intellectual property, and other laws.

Further, we are subject to evolving laws, regulations, policies, and international accords relating to matters beyond our core products and services, including environmental sustainability, climate change, human capital, and employment matters. In response, we have implemented robust programs, adopted reporting frameworks and principles, and announced a number of complex and ambitious goals and initiatives that may require considerable investments. We cannot guarantee that our goals and initiatives will be fully realized on the timelines we expect or at all, and projects that are completed as planned may not achieve the results we anticipate.

We are and may continue to be subject to claims, lawsuits, regulatory and government inquiries and investigations, enforcement actions, consent orders, and other forms of regulatory scrutiny and legal liability, including competition matters, that could harm our business, reputation, financial condition, and operating results.

We are subject to claims, lawsuits, regulatory and government inquiries and investigations, other proceedings, and orders involving competition, intellectual property, data privacy and security, tax and related compliance, labor and employment, commercial disputes, content generated by our users, goods and services offered by advertisers or publishers using our platforms, personal injury, and other matters.

The U.S. Department of Justice (DOJ), various U.S. states, and other plaintiffs have filed, and may continue to file in the future, several antitrust lawsuits about various aspects of our business, including our advertising technologies and practices, the operation and distribution of Google Search, and the operation and distribution of the Android operating system and Play Store.

For example, the DOJ and a number of state Attorneys General filed a lawsuit alleging that Google violated antitrust laws relating to Search and Search advertising, and in August 2024, the U.S. District Court for the District of Columbia ruled that Google violated such antitrust laws. The Court is holding a separate proceeding to determine remedies, which could include alterations to our products and services and our business models and operations, including structural remedies, and/or our distribution arrangements, among other changes. While we plan to appeal, there can be no assurance that our appeal will succeed, or that we will be able to change or decrease the severity of any remedies that may be ordered, and any or all of these potential remedies could harm our business, reputation, financial condition, and operating results.

Also, in December 2023, a California jury delivered a verdict in *Epic Games v. Google* finding that Google violated antitrust laws relating to Google Play's business. The presiding judge issued a remedies decision in October 2024 that ordered a variety of alterations to our business models and operations and contractual agreements for Android and Google Play. We are appealing the verdict, but if we are unsuccessful, we could face significant expenses to implement the remedies, and such costs and alterations could harm our business, reputation, financial condition, and operating results.

Furthermore, in December 2020, several State Attorneys General, led by the Texas Attorney General, filed an antitrust lawsuit in the U.S. District Court for the Eastern District of Texas alleging that Google violated antitrust and other laws relating to its advertising technology. This case is set for trial in March 2025, and we could face significant civil penalties. In January 2023, the DOJ and several Attorneys General sued in the Eastern District of Virginia alleging similar antitrust violations relating to Google’s advertising technology. Trial in the DOJ matter concluded in September 2024 with a decision expected in early 2025. If we are unsuccessful, we could face an order on remedies that could harm our business, reputation, financial condition, and operating results.

In addition to these regulatory proceedings, private individual and collective actions that overlap with claims pursued by regulators are pending in the U.S. and in several other jurisdictions. Adverse results in these or similar future lawsuits may include awards of monetary damages and remedies that could harm our business, reputation, financial condition, and operating results.

Other regulatory agencies in the U.S. and around the world, including competition enforcers, consumer protection agencies, and data protection authorities, have challenged and may continue to challenge our business practices and compliance with laws and regulations. We are cooperating with these investigations and defending litigation or appealing decisions where appropriate.

We are also subject to a variety of claims including product warranty, product liability, and consumer protection claims related to product defects, among other litigation, and we may also be subject to claims involving health and safety, hazardous materials usage, other environmental effects, AI training, development, and commercialization, or service disruptions or failures. Claims have been brought, and we expect will continue to be brought, against us for defamation, negligence, breaches of contract, patent, copyright and trademark infringement, unfair competition, unlawful activity, torts, privacy rights violations, fraud, or other legal theories based on the nature and content of information available on or via our services, the design and effect of our products and services, or due to our involvement in hosting, transmitting, marketing, branding, or providing access to content created by third parties.

Various laws, regulations, investigations, enforcement lawsuits, and regulatory actions have involved in the past, and may in the future result in substantial fines and penalties, injunctive relief, ongoing monitoring and auditing obligations, changes to our products and services, alterations to our business models and operations, including divestiture, and collateral related civil litigation or other adverse consequences. Any of these legal proceedings could also result in legal costs, diversion of management resources, and negative publicity, all of which could harm our business, reputation, financial condition, and operating results.

Estimating liabilities for our pending proceedings is a complex, fact-specific, and speculative process that requires significant judgment, and the amounts we are ultimately liable for may be less than or exceed our estimates. The resolution of one or more such proceedings has resulted in, and may in the future result in, additional substantial fines, penalties, injunctions, and other sanctions that could harm our business, reputation, financial condition, and operating results.

For additional information about the ongoing material legal proceedings to which we are subject, see Legal Proceedings in Item 3 of this Annual Report on Form 10-K.

Privacy, data protection, data usage, and portability regulations are complex and rapidly evolving areas. Any failure or alleged failure to comply with these laws could harm our business, reputation, financial condition, and operating results.

Authorities around the world have adopted and are considering a number of legislative and regulatory proposals concerning data protection, data usage and portability, and encryption of user data. Additionally, the increasing adoption of AI technologies, which rely on the collection of large amounts of data and use of such data for training purposes, has led data protection authorities around the world to consider and adopt new and evolving interpretations of data protection laws, imposing specific obligations with respect to the processing of personal data, including required notices, consents, and opt-outs. Further, the increased risk of inadvertent disclosure of confidential information or personal data in connection with the utilization of AI technologies may result in stronger regulatory scrutiny, leading to legal and regulatory investigations and enforcement actions that may negatively affect our business, even if unfounded. Adverse legal rulings, legislation, or regulation have resulted in, and may continue to result in, fines and orders requiring that we change our practices, which have had and could continue to have an adverse effect on how we provide services, harming our business, reputation, financial condition, and operating results. These laws and regulations are evolving and subject to interpretation, and compliance obligations could cause us to incur substantial costs or harm the quality and operations of our products and services in ways that harm our business. Examples of these laws include:

- The EU General Data Protection Regulation and the United Kingdom General Data Protection Regulations, which apply to all of our activities conducted from an establishment in the EU or the United Kingdom, respectively, or related to products and services that we offer to EU or the United Kingdom users or customers, respectively, or the monitoring of their behavior in the EU or the UK, respectively.
- Various U.S. federal, U.S. state, and foreign privacy laws related to the processing and security of personal data, including (1) comprehensive privacy laws that provide data privacy rights (including, in California, a private right of action in the event of a data breach resulting from our failure to implement and maintain reasonable security procedures and practices) and impose significant obligations on controllers and processors of consumer data; (2) laws imposing obligations on businesses that collect or disclose biometric information (including, in Illinois, Texas, and Washington); (3) laws governing the collection and processing of children and minor’s data and how companies provide age appropriate online experiences (including, in the U.S., the Children’s Online Privacy Protection Act of 1998, the pending Children and Teens' Online Privacy Protection Act (COPPA 2.0), which passed the U.S. Senate in 2024, and similar U.S. state laws related to children’s privacy, such as the New York Child Data Protection Act, and the United Kingdom’s Age-Appropriate Design Code); and (4) laws regulating internet-connected devices (such as, in California, the Internet of Things Security Law).
- The EU’s Digital Markets Act, which will require in-scope companies to obtain user consent for combining data across certain products and require search engines to share anonymized data with rival companies, among other changes; and the EU Data Act, which introduces new data portability requirements with respect to connected products (i.e., ‘internet of things’ products) and related services, as well as interoperability obligations on data processing services.

Further, we are subject to evolving laws and regulations that dictate whether, how, and under what circumstances we can transfer, process and/or receive personal data, as well as ongoing enforcement actions from supervisory authorities related to cross-border transfers of personal data. The validity of various data transfer mechanisms we currently rely upon remains subject to legal, regulatory, and political developments globally, which may require us to adapt our existing arrangements.

We face, and may continue to face, intellectual property infringement or misappropriation, violation of rights of publicity, and other claims that could be costly to defend, result in significant damage awards or other costs (including indemnification awards), and limit our ability to use certain technologies.

We, like other internet, technology, and media companies, are frequently subject to litigation based on allegations of infringement or other violations of intellectual property rights, including patent, copyright, trade secrets, and trademarks. Parties have also sought broad injunctive relief against us by filing claims in U.S. and international courts and the U.S. International Trade Commission (ITC) for exclusion and cease-and-desist orders. In addition, patent-holding companies may frequently seek to generate income from patents they have obtained by bringing claims against us. As we continue to expand our business, the number of intellectual property claims against us has increased and may continue to increase as we develop and acquire new products, services, and technologies, including AI technologies.

Adverse results in any of these lawsuits may include awards of monetary damages, costly royalty or licensing agreements (if licenses are available at all), or orders limiting our ability to sell our products and services in the U.S. or elsewhere, including by preventing us from offering certain features, functionalities, products, or services in certain jurisdictions. They may also cause us to change our business practices in ways that could result in a loss of revenues for us and otherwise harm our business.

Many of our agreements with our customers and partners, including certain suppliers, require us to defend against certain intellectual property infringement claims and in some cases indemnify them for certain intellectual property infringement claims against them, which could result in increased costs for defending such claims or significant damages if there were an adverse ruling in any such claims. Such customers and partners may also discontinue the use of our products, services, and technologies, as a result of injunctions or otherwise, which could result in loss of revenues and harm our business. Moreover, intellectual property indemnities provided to us by our suppliers, when obtainable, may not cover all damages and losses suffered by us and our customers arising from intellectual property infringement claims. Furthermore, in connection with our divestitures, we have agreed, and may in the future agree, to provide indemnification for certain potential liabilities, including those associated with intellectual property claims. Regardless of their merits, intellectual property claims are often time consuming and expensive to litigate or settle. To the extent such claims are successful, they could harm our business, including our product and service offerings, financial condition, and operating results.

We could be subject to changes in tax rates, the adoption of new U.S. or international tax legislation, or exposure to additional tax liabilities.

We are subject to a variety of taxes and tax collection obligations in the U.S. and numerous foreign jurisdictions. Our effective tax rates are affected by a variety of factors, including changes in the mix of earnings in jurisdictions with different statutory tax rates, net gains and losses on hedges and related transactions under our foreign exchange risk management program, changes in our stock price for shares issued as employee compensation, changes in the valuation of our deferred tax assets or liabilities, and the application of different provisions of tax laws or changes in tax laws, regulations, or accounting principles (including changes in the interpretation of existing laws). Further, if we are unable or fail to collect taxes on behalf of customers, employees and partners as the withholding agent, we could become liable for taxes that are levied against third parties.

We are subject to regular review and audit by both domestic and foreign tax authorities. As a result, we have received, and may in the future receive, assessments in multiple jurisdictions, on various tax-related assertions, such as transfer-pricing adjustments or permanent-establishment claims. Any adverse outcome of such a review or audit could harm our financial condition and operating results, require adverse changes to our business practices, or subject us to additional litigation and regulatory inquiries. In addition, the determination of our worldwide provision for income taxes and other tax liabilities requires significant judgment and often involves uncertainty. Although we believe our estimates are reasonable, the ultimate tax outcome may differ from the amounts recorded in our financial statements and may affect our financial results in the period or periods for which such determination is made.

Furthermore, due to shifting economic and political conditions, tax policies, laws, or rates in various jurisdictions may be subject to significant changes in ways that could harm our financial condition and operating results. For example, various jurisdictions around the world have enacted or are considering revenue-based taxes such as digital services taxes and other targeted taxes, which could lead to inconsistent and potentially overlapping international tax regimes. The U.S. has imposed a 1% excise tax on the fair market value of shares repurchased, which could potentially increase in the future. The Organization for Economic Cooperation and Development (OECD) is coordinating negotiations among more than 140 countries with the goal of achieving consensus around substantial changes to international tax policies, including the implementation of a minimum global effective tax rate of 15%, on which various jurisdictions around the world have either introduced draft legislation or adopted final legislation. Our effective tax rate and cash tax payments could increase as a result of these changes.

Risks Related to Ownership of our Stock

We cannot guarantee that any share repurchase program or dividend program will be continuously active or fully consummated or will enhance long-term stockholder value, and share repurchases or dividends could increase the volatility of our stock prices and could diminish our cash reserves.

We engage in share repurchases of our Class A and Class C stock from time to time in accordance with authorizations from the Board of Directors of Alphabet. Our repurchase program does not have an expiration date and does not obligate Alphabet to repurchase any specific dollar amount or to acquire any specific number of shares. In April 2024, we announced the approval of our cash dividend program and began paying regular cash dividends to our Class A, Class B, and Class C stockholders. Any and all future cash dividends are subject to declaration by our Board of Directors in its sole discretion, and in accordance with the requirements of any applicable laws, rules, and regulations, including the Delaware General Corporation Law. Our cash dividend program does not require, and our Board of Directors may decide not to declare, a cash dividend each quarter, and does not obligate our Board of Directors to declare a dividend at any specific dollar amount per share. Any such decision by our Board of Directors may depend on a variety of factors that it may deem relevant, including but not limited to our earnings, liquidity, financial condition, other capital deployment opportunities, level of indebtedness, and general market conditions. Our share repurchases or dividends could affect our share trading prices, increase their volatility, reduce our cash reserves and may be suspended or terminated at any time, which may result in a decrease in the trading prices of our stock.

The concentration of our stock ownership limits our stockholders' ability to influence corporate matters.

Our Class B stock has 10 votes per share, our Class A stock has one vote per share, and our Class C stock has no voting rights. As of December 31, 2024, Larry Page and Sergey Brin beneficially owned approximately 87.4% of our outstanding Class B stock, which represented approximately 52.1% of the voting power of our outstanding common stock. Through their stock ownership, Larry and Sergey have significant influence over all matters requiring stockholder approval, including the election of directors and significant corporate transactions, such as a merger or other sale of our company or our assets, for the foreseeable future. In addition, because our Class C stock carries no voting rights (except as required by applicable law), the issuance of the Class C stock, including in future stock-based acquisition transactions and to fund employee equity incentive programs, could continue Larry and Sergey's current relative voting power and their ability to elect all of our directors and to determine the outcome of most matters submitted to a vote of our stockholders. The share repurchases made pursuant to our repurchase program may also affect Larry and Sergey's relative voting power. This concentrated control limits or severely restricts other stockholders' ability to influence corporate matters and we may take actions that some of our stockholders do not view as beneficial, which could reduce the market price of our Class A stock and our Class C stock.

Provisions in our charter documents and under Delaware law could discourage a takeover that stockholders may consider favorable.

Provisions in Alphabet's certificate of incorporation and bylaws may have the effect of delaying or preventing a change of control or changes in our management. These provisions include the following:

- Our Board of Directors has the right to elect directors to fill a vacancy created by the expansion of the Board of Directors or the resignation, death, or removal of a director.
- Our stockholders may not act by written consent, which makes it difficult to take certain actions without holding a stockholders' meeting.
- Our certificate of incorporation prohibits cumulative voting in the election of directors. This limits the ability of minority stockholders to elect director candidates.
- Stockholders must provide advance notice to nominate individuals for election to the Board of Directors or to propose matters that can be acted upon at a stockholders' meeting. These provisions may discourage or deter a potential acquirer from conducting a solicitation of proxies to elect the acquirer's own slate of directors or otherwise attempting to obtain control of our company.

- Our Board of Directors may issue, without stockholder approval, shares of undesignated preferred stock, which makes it possible for our Board of Directors to issue preferred stock with voting or other rights or preferences that could impede the success of any attempt to acquire us.

As a Delaware corporation, we are also subject to certain Delaware anti-takeover provisions. Under Delaware law, a corporation may not engage in a business combination with any holder of 15% or more of its outstanding voting stock unless the holder has held the stock for three years or, among other things, the Board of Directors has approved the transaction. Our Board of Directors could rely on Delaware law to prevent or delay an acquisition of us.

The trading price for our Class A stock and non-voting Class C stock may continue to be volatile.

The trading price of our stock has at times experienced significant volatility and may continue to be volatile. In addition to the factors discussed in this report, the trading prices of our Class A stock and Class C stock have fluctuated, and may continue to fluctuate widely, in response to various factors, including, among others, the size or continuity of either our share repurchase or dividend programs, the activities of our peers and changes in broader economic and political conditions around the world. These broad market and industry factors could harm the market price of our Class A stock and our Class C stock, regardless of our actual operating performance.

General Risks

Our operating results may fluctuate, which makes our results difficult to predict and could cause our results to fall short of expectations.

Our operating results have fluctuated, and may in the future fluctuate, as a result of a number of factors, many outside of our control, including the cyclical nature and seasonality in our business and geopolitical events. As a result, comparing our operating results (including our expenses as a percentage of our revenues) on a period-to-period basis may not be meaningful, and our past results should not be relied on as an indication of our future performance. Consequently, our operating results in future quarters may fall below expectations.

Acquisitions, joint ventures, investments, and divestitures could result in operating difficulties, dilution, and other consequences that could harm our business, financial condition, and operating results.

Acquisitions, joint ventures, investments, and divestitures are important elements of our overall corporate strategy and use of capital, and these transactions could be material to our financial condition and operating results. We expect to continue to evaluate and enter into discussions regarding a wide array of such potential strategic arrangements, which could create unforeseen operating difficulties and expenditures. Some of the areas where we face risks include:

- diversion of management time and focus from operating our business to challenges related to acquisitions and other strategic arrangements;
- failure to obtain required approvals on a timely basis, if at all, from governmental authorities; conditions placed upon approval that could, among other things, delay or prevent us from completing a transaction, or otherwise restrict our ability to realize the expected financial or strategic goals of a transaction; or investigations or litigation by governmental authorities related to our acquisitions and other strategic arrangements;
- failure to successfully integrate the acquired operations, technologies, services, and personnel (including cultural integration and retention of employees) and further develop the acquired business or technology;
- implementation of controls (or remediation of control deficiencies), procedures, and policies at the acquired company;
- integration of the acquired company's accounting and other administrative systems, and the coordination of product, engineering, and sales and marketing functions;
- transition of operations, users, and customers onto our existing platforms;
- in the case of foreign acquisitions, the need to integrate operations across different cultures and languages and to address the particular economic, currency, political, and regulatory risks associated with specific countries;
- failure to accomplish commercial, strategic or financial objectives with respect to investments, joint ventures, and other strategic arrangements;
- failure to realize the value of investments and joint ventures due to a lack of liquidity;
- liability for activities of the acquired company before the acquisition, including patent and trademark infringement claims, data privacy and security issues, violations of laws, commercial disputes, tax liabilities, warranty claims, product liabilities, and other known and unknown liabilities; and

- litigation or other claims in connection with the acquired company, including claims from terminated employees, customers, former stockholders, or other third parties.

Our failure to address these risks or other problems encountered in connection with our past or future acquisitions and other strategic arrangements could cause us to fail to realize their anticipated benefits, incur unanticipated liabilities, and harm our business generally.

Our acquisitions and other strategic arrangements could also result in dilutive issuances of our equity securities, the incurrence of debt, contingent liabilities, or amortization expenses, or impairment of goodwill and/or purchased long-lived assets, and restructuring charges, any of which could harm our financial condition and operating results. Also, the anticipated benefits or value of our acquisitions and other strategic arrangements may not materialize. In connection with our divestitures and certain other strategic arrangements, we have agreed, and may in the future agree, to provide indemnification for certain potential liabilities, which could harm our financial condition and operating results.

We rely on highly skilled personnel and, if we are unable to retain or motivate key personnel, hire qualified personnel, or maintain and continue to adapt our corporate culture, we may not be able to grow or operate effectively.

Our performance and future success depends in large part upon the continued service of key technical leads as well as members of our senior management team. For instance, Sundar Pichai is critical to the overall management of Alphabet and its subsidiaries and plays an important role in the development of our technology, maintaining our culture, and setting our strategic direction.

Our ability to compete effectively and our future success depend on our continuing to identify, hire, develop, motivate, and retain highly skilled personnel for all areas of our organization. Competition in our industry for qualified employees is intense, and certain of our competitors have directly targeted, and may continue to target, our employees. In addition, our compensation arrangements, such as our equity award programs, may not always be successful in attracting new employees and retaining and motivating our existing employees. Immigration policy and regulatory changes may also affect our ability to hire, mobilize, or retain some of our global talent. All of our executive officers and key employees are at-will employees, and we do not maintain any key-person life insurance policies.

In addition, we believe that our corporate culture fosters innovation, creativity, and teamwork. As our organization grows and evolves, we may need to adapt our corporate culture and work environments to ever-changing circumstances, such as during times of a natural disaster or pandemic, and these changes could affect our ability to compete effectively or have an adverse effect on our corporate culture. Under our hybrid work models, we may experience increased costs and/or disruption, in addition to potential effects on our ability to operate effectively and maintain our corporate culture.

ITEM 1B. UNRESOLVED STAFF COMMENTS

Not applicable.

ITEM 1C. CYBERSECURITY

We maintain a comprehensive process for identifying, assessing, and managing material risks from cybersecurity threats as part of our broader risk management system and processes. For example, some risks include our software supply chain and other third-party dependencies, vulnerabilities in our products and services, theft of our intellectual property, and attempts to compromise our infrastructure. We obtain input, as appropriate, for our cybersecurity risk management program on the security industry and threat trends from multiple external experts and internal threat intelligence teams. Teams of dedicated privacy, safety, and security professionals oversee cybersecurity risk management and mitigation, incident prevention, detection, and remediation. These teams comprise professionals with deep cybersecurity expertise across multiple industries and are led by our Vice President of Privacy, Safety, and Security Engineering, who has 20 years of experience, including roles in technology infrastructure for two other large public companies. Our executive leadership team, along with input from the above teams, are responsible for our overall enterprise risk management system and processes and regularly consider cybersecurity risks in the context of other material risks to the company.

As part of our cybersecurity risk management system, our incident management teams track and log privacy and security incidents across Alphabet, our vendors, and other third-party service providers to remediate and resolve any such incidents. Significant incidents are reviewed regularly by a cross-functional working group to determine whether further escalation is appropriate. Any incident assessed as potentially being or potentially becoming material is promptly escalated for further assessment, and then reported to designated members of our senior management. We consult with outside counsel as appropriate, including on materiality analysis and disclosure matters, and our senior management makes the final materiality determinations and disclosure and other compliance decisions. Our management apprises Alphabet's independent public accounting firm of matters and any relevant developments.

The Audit and Compliance Committee has oversight responsibility for risks and incidents relating to cybersecurity threats, including compliance with disclosure requirements, cooperation with law enforcement, and related effects on financial and other risks, and it reports any findings and recommendations, as appropriate, to the full Board for consideration. Senior management regularly discusses cyber risks and trends and, should they arise, any material incidents with the Audit and Compliance Committee. Internal Audit maintains a dedicated cybersecurity auditing team that independently tests our cybersecurity controls.

Our business strategy, results of operations and financial condition have not been materially affected by risks from cybersecurity threats, including as a result of previously identified cybersecurity incidents, but we cannot provide assurance that they will not be materially affected in the future by such risks or any future material incidents. For more information on our cybersecurity related risks, see Item 1A Risk Factors of this Annual Report on Form 10-K.

ITEM 2. PROPERTIES

Our headquarters are located in Mountain View, California. We own and lease office facilities and data centers around the world, primarily in Asia, Europe, and North America. We believe our existing facilities are in good condition and suitable for the conduct of our business.

ITEM 3. LEGAL PROCEEDINGS

For a description of our material pending legal proceedings, see Legal Matters in Note 10 of the Notes to Consolidated Financial Statements included in Part II, Item 8 of this Annual Report on Form 10-K, which is incorporated herein by reference.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT’S COMMON EQUITY, RELATED STOCKHOLDER MATTERS, AND ISSUER PURCHASES OF EQUITY SECURITIES

As of October 2, 2015, Alphabet Inc. became the successor issuer of Google Inc. pursuant to Rule 12g-3(a) under the Exchange Act. Our Class A stock has been listed on the Nasdaq Global Select Market under the symbol “GOOG” since August 19, 2004, and under the symbol "GOOGL" since April 3, 2014. Prior to August 19, 2004, there was no public market for our stock. Our Class B stock is neither listed nor traded. Our Class C stock has been listed on the Nasdaq Global Select Market under the symbol “GOOG” since April 3, 2014.

Holders of Record

As of December 31, 2024, there were approximately 7,743 and 1,671 stockholders of record of our Class A stock and Class C stock, respectively. Because many of our shares of Class A stock and Class C stock are held by brokers and other institutions on behalf of stockholders, we are unable to estimate the total number of stockholders represented by these record holders. As of December 31, 2024, there were approximately 57 stockholders of record of our Class B stock.

Dividend Program

Dividend payments to stockholders of Class A, Class B, and Class C shares were $3.5 billion, $519 million, and $3.3 billion, respectively, for the year ended December 31, 2024. The company intends to pay quarterly cash dividends in the future, subject to review and approval by the company’s Board of Directors in its sole discretion. We regularly evaluate our cash and capital structure, including the size, pace, and form of capital return to stockholders.

Issuer Purchases of Equity Securities

The following table presents information with respect to Alphabet's repurchases of Class A and Class C stock during the quarter ended December 31, 2024:

Period	Total Number of Class A Shares Purchased (in thousands)(1)	Total Number of Class C Shares Purchased (in thousands)(1)	Average Price Paid per Class A Share(2)	Average Price Paid per Class C Share(2)	Total Number of Shares Purchased as Part of Publicly Announced Programs (in thousands)(1)	Approximate Dollar Value of Shares that May Yet Be Purchased Under the Program (in millions)
October 1 - 31	5,792	30,080	$ 167.66	$ 168.84	35,872	$ 53,699
November 1 - 30	4,325	22,450	$ 174.97	$ 175.98	26,775	$ 49,023
December 1 - 31	3,559	19,572	$ 187.07	$ 187.52	23,131	$ 44,704
Total	13,676	72,102			85,778	

(1) Repurchases are being executed from time to time, subject to general business and market conditions and other investment opportunities, through open market purchases or privately negotiated transactions, including through Rule 10b5-1 plans. The repurchase program does not have an expiration date. For additional information related to share repurchases, see Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

(2) Average price paid per share includes costs associated with the repurchases.

Stock Performance Graphs

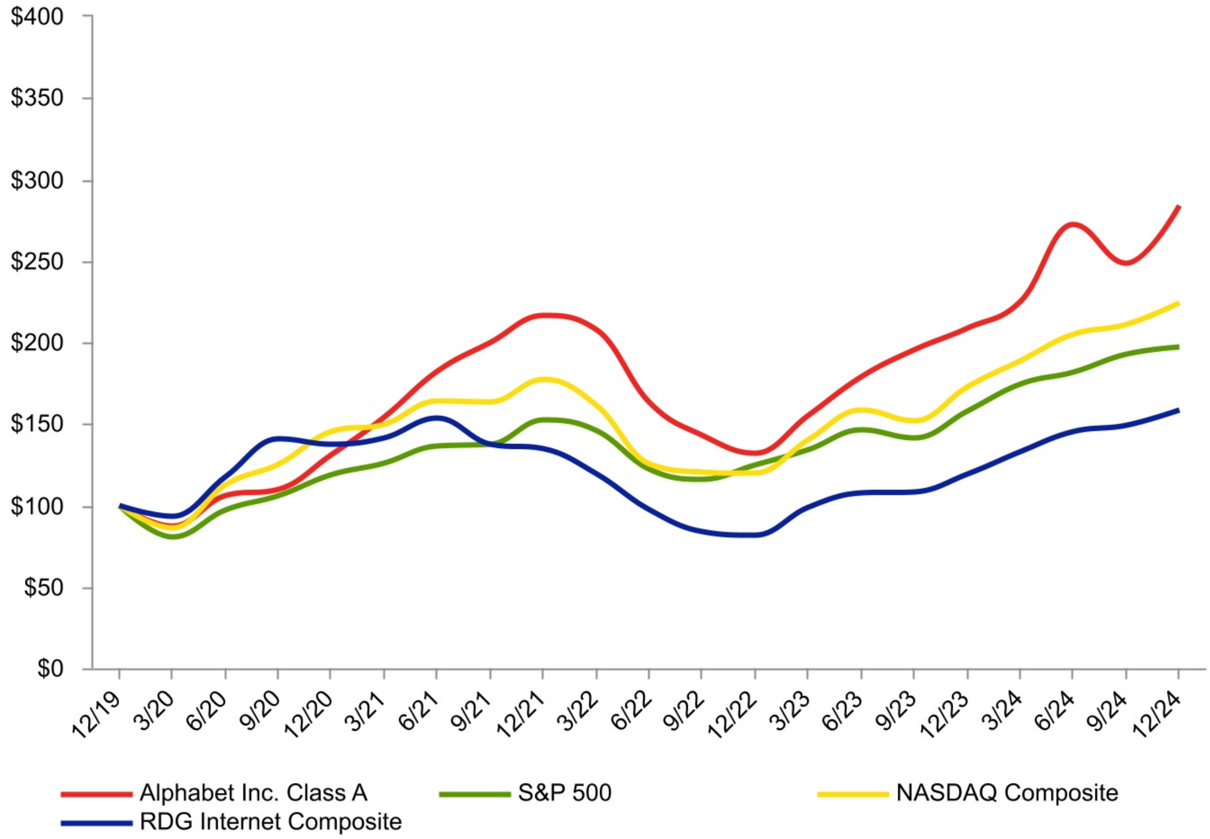
The graph below matches Alphabet Inc. Class A's cumulative five-year total stockholder return on common stock with the cumulative total returns of the S&P 500 index, the NASDAQ Composite index, and the RDG Internet Composite index. The graph tracks the performance of a $100 investment in our common stock and in each index (with the reinvestment of all dividends) from December 31, 2019, to December 31, 2024. The returns shown are based on historical results and are not intended to suggest future performance.

COMPARISON OF CUMULATIVE 5-YEAR TOTAL RETURN*

ALPHABET INC. CLASS A COMMON STOCK

Among Alphabet Inc., the S&P 500 Index, the

NASDAQ Composite Index, and the RDG Internet Composite Index



*$100 invested on December 31, 2019, in stock or index, including reinvestment of dividends.

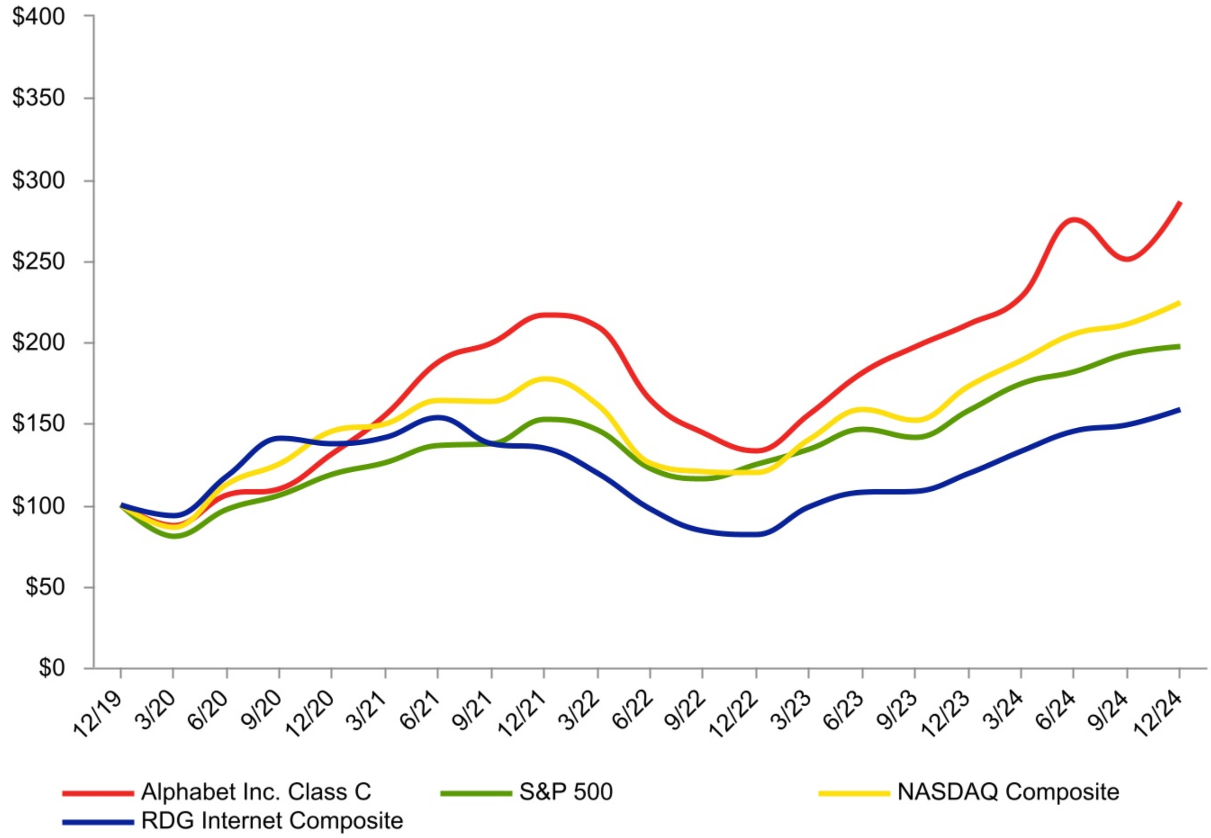
The graph below matches Alphabet Inc. Class C's cumulative five-year total stockholder return on capital stock with the cumulative total returns of the S&P 500 index, the NASDAQ Composite index, and the RDG Internet Composite index. The graph tracks the performance of a $100 investment in our Class C capital stock and in each index (with the reinvestment of all dividends) from December 31, 2019, to December 31, 2024. The returns shown are based on historical results and are not intended to suggest future performance.

COMPARISON OF CUMULATIVE 5-YEAR TOTAL RETURN*

ALPHABET INC. CLASS C CAPITAL STOCK

Among Alphabet Inc., the S&P 500 Index, the

NASDAQ Composite Index, and the RDG Internet Composite Index



*$100 invested on December 31, 2019, in stock or in index, including reinvestment of dividends.

ITEM 6. [Reserved]

ITEM 7. MANAGEMENT’S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

Please read the following discussion and analysis of our financial condition and results of operations together with “Note about Forward-Looking Statements,” Part I, Item 1 "Business," Part I, Item 1A "Risk Factors," and our consolidated financial statements and related notes included under Item 8 of this Annual Report on Form 10-K.

The following section generally discusses 2024 results compared to 2023 results. Discussion of 2023 results compared to 2022 results to the extent not included in this report can be found in Item 7 of our 2023 Annual Report on Form 10-K.

Understanding Alphabet’s Financial Results

Alphabet is a collection of businesses — the largest of which is Google. We report Google in two segments, Google Services and Google Cloud; we also report all non-Google businesses collectively as Other Bets. For further details on our segments, see Part I, Item 1 Business and Note 15 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Trends in Our Business and Financial Effect

The following long-term trends have contributed to the results of our consolidated operations, and we anticipate that they will continue to affect our future results:

- **Users' behaviors and advertising continue to shift online as the digital economy evolves.**

The continuing evolution of the online world has contributed to the growth of our business and our revenues since inception. We expect that this evolution will continue to benefit our business and our revenues, although at a slower pace than we have experienced historically. In addition, we face increasing competition for user engagement and advertisers, including from other developers and providers of AI products and services, which may affect our revenues.

- **Users continue to access our products and services using diverse devices and modalities, which allows for new advertising formats that may benefit our revenues but adversely affect our margins.**

Our users are accessing our products and services via diverse devices and modalities beyond traditional desktop, such as smartphones, wearables, connected TVs, and smart home devices, and want to be able to be connected no matter where they are or what they are doing. We are focused on expanding our products and services to stay in front of these trends in order to maintain and grow our business.

We benefit from advertising revenues generated from different channels, including mobile, and newer advertising formats. The margins from these channels and newer products have generally been lower than those from traditional desktop search. Additionally, as the market for a particular device type or modality matures, our advertising revenues may be affected. For example, changing dynamics within the global smartphone market, such as increased market saturation in developed countries, can affect our mobile advertising revenues.

We expect TAC paid to our distribution partners and Google Network partners to increase as our revenues grow and TAC as a percentage of our advertising revenues ("TAC rate") to be affected by changes in device mix; geographic mix; partner agreement terms; partner mix; the percentage of queries channeled through paid access points; product mix; the relative revenue growth rates of advertising revenues from different channels; and revenue share terms.

We expect these trends to continue to affect our revenues and put pressure on our margins.

- **As online advertising evolves, we continue to expand our product offerings, which may affect our monetization.**

As interactions between users and advertisers change, and as online user behavior evolves, for example with AI, we continue to expand our product offerings to serve these changing needs, which may affect monetization of our products and services. We expect to continue to incorporate AI innovations into our products, such as AI in Search, that could affect our monetization trends. When developing new products and services we generally focus first on user experience and then on monetization.

- **As users in developing economies increasingly come online, our revenues from international markets continue to increase, and may require continued investments. In addition, movements in foreign exchange rates affect such revenues.**

The shift to online, as well as the advent of the multi-device world, has brought opportunities outside of the U.S., including in emerging markets, such as India. We continue to invest heavily and develop localized versions of our products and advertising programs relevant to our users in these markets. This has led to a trend of increased revenues from emerging markets. We expect that our results will continue to be affected by our performance in these markets, particularly as low-cost mobile devices become more available. This trend could affect our revenues as developing markets initially monetize at a lower rate than more mature markets.

International revenues represent a significant portion of our revenues and are subject to fluctuations in foreign currency exchange rates relative to the U.S. dollar. While we have a foreign exchange risk management program designed to reduce our exposure to these fluctuations, this program does not fully offset their effect on our revenues and earnings.

- **The revenues that we derive beyond advertising are increasing and may adversely affect our margins.**

Revenues from cloud, consumer subscriptions, platforms, and devices, which may have differing characteristics than our advertising revenues, have grown over time, and we expect this trend to continue as we focus on expanding our products and services. The margins on these revenues vary significantly and are generally lower than the margins on our advertising revenues. For example, sales of our devices adversely affect our consolidated margins due to pressures on pricing and higher cost of sales.

- **As we continue to serve our users and expand our businesses, we will invest heavily in operating and capital expenditures.**

We continue to make significant research and development investments in areas of strategic focus as we seek to develop new, innovative offerings, improve our existing offerings, and rapidly and responsibly deploy AI across our businesses. We also expect to increase, relative to 2024, our investment in our technical infrastructure, including servers, network equipment, and data centers, to support the growth of our business and our long-term initiatives, in particular in support of AI products and services. In addition, acquisitions and strategic investments remain important elements in our use of capital and contribute to the breadth and depth of our offerings, expand our expertise in engineering and other functional areas, and build strong partnerships around strategic initiatives.

- **We continue to face an evolving regulatory environment, and we are subject to claims, lawsuits, investigations, and other forms of potential legal liability, which could affect our business practices and financial results.**

Changes in social, political, economic, tax, and regulatory conditions or in laws and policies governing a wide range of topics and related legal matters, including investigations, lawsuits, and regulatory actions, have resulted in fines and caused us to change our business practices. As these global trends continue, our cost of doing business may increase, our products and services may become less useful, our ability to pursue certain business practices or offer certain products or services may be limited, and we may need to change our business models and operations to comply with evolving regulatory and legal matters. For additional information, see Part I, Item 1A Risk Factors and Legal Matters in Note 10 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

- **Our employees are critical to our success and we expect to continue investing in them.**

Our employees are among our best assets and are critical for our continued success. We expect to continue hiring talented employees around the globe and to provide competitive compensation programs. For additional information, see Culture and Workforce in Part I, Item 1 Business of this Annual Report on Form 10-K.

Revenues and Monetization Metrics

We generate revenues by delivering relevant, cost-effective online advertising; cloud-based solutions that provide enterprise customers of all sizes with infrastructure, platform services, and applications; sales of other products and services, such as fees received for subscription-based products, apps and in-app purchases, and devices. For additional information on how we recognize revenue, see Note 1 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

In addition to the long-term trends and their financial effect on our business discussed above, fluctuations in our revenues have been and may continue to be affected by a combination of factors, including:

- changes in foreign currency exchange rates;
- changes in pricing, such as those resulting from changes in fee structures, discounts, and customer incentives;
- general economic conditions and various external dynamics, including geopolitical events, regulations, and other measures and their effect on advertiser, consumer, and enterprise spending;
- new product, service, and market launches; and

- seasonality.

Additionally, fluctuations in our revenues generated from advertising ("Google advertising"), other sources ("Google subscriptions, platforms, and devices"), Google Cloud, and Other Bets have been, and may continue to be, affected by other factors unique to each set of revenues, as described below.

Google Services

Google Services revenues consist of Google advertising as well as Google subscriptions, platforms, and devices revenues.

Google Advertising

Google advertising revenues are comprised of the following:

- Google Search & other, which includes revenues generated on Google search properties (including revenues from traffic generated by search distribution partners who use Google.com as their default search in browsers, toolbars, etc.), and other Google owned and operated properties like Gmail, Google Maps, and Google Play;
- YouTube ads, which includes revenues generated on YouTube properties; and
- Google Network, which includes revenues generated on Google Network properties participating in AdMob, AdSense, and Google Ad Manager.

We use certain metrics to track how well traffic across various properties is monetized as it relates to our advertising revenues: paid clicks and cost-per-click pertain to traffic on Google Search & other properties, while impressions and cost-per-impression pertain to traffic on our Google Network properties.

Paid clicks represent engagement by users and include clicks on advertisements by end-users on Google search properties and other Google owned and operated properties including Gmail, Google Maps, and Google Play. Cost-per-click is defined as click-driven revenues divided by our total number of paid clicks and represents the average amount we charge advertisers for each engagement by users.

Impressions include impressions displayed to users on Google Network properties participating primarily in AdMob, AdSense, and Google Ad Manager. Cost-per-impression is defined as impression-based and click-based revenues divided by our total number of impressions, and represents the average amount we charge advertisers for each impression displayed to users.

As our business evolves, we periodically review, refine, and update our methodologies for monitoring, gathering, and counting the number of paid clicks and the number of impressions, and for identifying the revenues generated by the corresponding click and impression activity.

Fluctuations in our advertising revenues, as well as the change in paid clicks and cost-per-click on Google Search & other properties and the change in impressions and cost-per-impression on Google Network properties and the correlation between these items have been, and may continue to be, affected by factors in addition to the general factors described above, such as:

- advertiser competition for keywords;
- changes in advertising quality, formats, delivery or policy;
- changes in device mix;
- seasonal fluctuations in internet usage, advertising expenditures, and underlying business trends, such as traditional retail seasonality; and
- traffic growth in emerging markets compared to more mature markets and across various verticals and channels.

Google subscriptions, platforms, and devices

Google subscriptions, platforms, and devices revenues are comprised of the following:

- consumer subscriptions, which primarily include revenues from YouTube services, such as YouTube TV, YouTube Music and Premium, and NFL Sunday Ticket, as well as Google One;
- platforms, which primarily include revenues from Google Play sales of apps and in-app purchases;
- devices, which primarily include sales of the Pixel family of devices; and
- other products and services.

Fluctuations in our Google subscriptions, platforms, and devices revenues have been, and may continue to be, affected by factors in addition to the general factors described above, such as changes in customer usage and demand, number of subscribers, and the timing of product launches.

Google Cloud

Google Cloud revenues are comprised of the following:

- Google Cloud Platform, which generates consumption-based fees and subscriptions for infrastructure, platform, and other services. These services provide access to solutions such as AI offerings including our AI infrastructure, Vertex AI platform, and Gemini for Google Cloud; cybersecurity; and data and analytics;
- Google Workspace, which includes subscriptions for cloud-based communication and collaboration tools for enterprises, such as Calendar, Gmail, Docs, Drive, and Meet, with integrated features like Gemini for Google Workspace; and
- other enterprise services.

Fluctuations in our Google Cloud revenues have been, and may continue to be, affected by factors in addition to the general factors described above, such as changes in customer usage and demand.

Other Bets

Revenues from Other Bets are generated primarily from the sale of healthcare-related services, and internet services.

Costs and Expenses

Our cost structure has two components: cost of revenues and operating expenses. Our operating expenses include costs related to R&D, sales and marketing, and general and administrative functions. Certain of our costs and expenses, including those associated with the operation of our technical infrastructure as well as components of our operating expenses, are generally less variable in nature and may not correlate to changes in revenue. Additionally, fluctuations in employee compensation expenses may not directly correlate with changes in headcount, due to factors such as annual stock-based compensation (SBC) awards that generally vest over four years.

Cost of Revenues

Cost of revenues is comprised of TAC and other costs of revenues.

- TAC includes:
    - amounts paid to our distribution partners who make available our search access points and services. Our distribution partners include browser providers, mobile carriers, original equipment manufacturers, and software developers; and
    - amounts paid to Google Network partners primarily for ads displayed on their properties.
- Other cost of revenues primarily includes:
    - content acquisition costs, which are payments to content providers from whom we license video and other content for distribution, primarily related to YouTube (we pay fees to these content providers based on revenues generated, subscriber counts, or a flat fee);
    - depreciation expense related to our technical infrastructure;
    - employee compensation expenses related to our technical infrastructure and other operations such as content review and customer and product support;
    - inventory and other costs related to the devices we sell; and
    - other technical infrastructure operations costs, including network capacity, energy, and equipment costs.

TAC as a percentage of revenues generated from ads placed on Google Network properties are significantly higher than TAC as a percentage of revenues generated from ads placed on Google Search & other properties, because most of the advertiser revenues from ads served on Google Network properties are paid as TAC to our Google Network partners.

Operating Expenses

Operating expenses are generally incurred during our normal course of business, which we categorize as either R&D, sales and marketing, or general and administrative.

The main components of our R&D expenses are:

- depreciation;
- employee compensation expenses for engineering and technical employees responsible for R&D related to our existing and new products and services; and
- third-party services fees primarily relating to consulting and outsourced services in support of our engineering and product development efforts.

The main components of our sales and marketing expenses are:

- employee compensation expenses for employees engaged in sales and marketing, sales support, and certain customer service functions; and
- spend relating to our advertising and promotional activities in support of our products and services.

The main components of our general and administrative expenses are:

- employee compensation expenses for employees in finance, human resources, information technology, legal, and other administrative support functions;
- expenses relating to legal and other matters, including certain fines and settlements; and
- third-party services fees, including audit, consulting, outside legal, and other outsourced administrative services.

Other Income (Expense), Net

OI&E, net primarily consists of interest income (expense), the effect of foreign currency exchange gains (losses), net gains (losses) and impairment on our marketable and non-marketable securities, performance fees, and income (loss) and impairment from our equity method investments.

For additional information, including how we account for our investments and factors that can drive fluctuations in the value of our investments, see Note 1 and Note 3 of the Notes to Consolidated Financial Statements included in Item 8 as well as Item 7A Quantitative and Qualitative Disclosures About Market Risk of this Annual Report on Form 10-K.

Provision for Income Taxes

Provision for income taxes represents the estimated amount of federal, state, and foreign income taxes incurred in the U.S. and the many jurisdictions in which we operate. The provision includes the effect of reserve provisions and changes to reserves that are considered appropriate as well as the related net interest and penalties.

For additional information, including a reconciliation of the U.S. federal statutory rate to our effective tax rate, see Note 14 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Executive Overview

The following table summarizes our consolidated financial results (in millions, except for per share information and percentages):

	Year Ended December 31, 2023	Year Ended December 31, 2024	$ Change	% Change
Consolidated revenues	$ 307,394	$ 350,018	$ 42,624	14 %
Change in consolidated constant currency revenues[(1)]				15 %
Cost of revenues	$ 133,332	$ 146,306	$ 12,974	10 %
Operating expenses	$ 89,769	$ 91,322	$ 1,553	2 %
Operating income	$ 84,293	$ 112,390	$ 28,097	33 %
Operating margin	27 %	32 %		5 %
Other income (expense), net	$ 1,424	$ 7,425	$ 6,001	421 %
Net income	$ 73,795	$ 100,118	$ 26,323	36 %
Diluted EPS[(2)]	$ 5.80	$ 8.04	$ 2.24	39 %

(1) See "Use of Non-GAAP Constant Currency Information" below for details relating to our use of constant currency information.

(2) For additional information on the calculation of diluted EPS, see Note 12 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

- Revenues were $350.0 billion, an increase of 14% year over year, primarily driven by an increase in Google Services revenues of $32.4 billion, or 12%, and an increase in Google Cloud revenues of $10.1 billion, or 31%.
- Total constant currency revenues, which exclude the effect of hedging, increased 15% year over year.
- Cost of revenues was $146.3 billion, an increase of 10% year over year, primarily driven by increases in content acquisition costs, TAC, and depreciation expense.
- Operating expenses were $91.3 billion, an increase of 2% year over year, primarily driven by increases in depreciation expense, employee compensation expenses, and third-party services fees. These increases were partially offset by reductions in charges related to legal and other matters and charges related to our office space optimization efforts. The overall increase in employee compensation expenses was partially offset by a reduction in employee severance and related charges.

Other Information:

- Dividend payments to stockholders of Class A, Class B, and Class C shares, which were first paid in June 2024, were $3.5 billion, $519 million, and $3.3 billion, respectively, totaling $7.4 billion for the year ended December 31, 2024. For additional information, see Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.
- Repurchases of Class A and Class C shares were $11.9 billion and $50.2 billion, respectively, totaling $62.0 billion for the year ended December 31, 2024. For additional information, see Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.
- Employee severance and related charges for the year ended December 31, 2024 were $1.0 billion, a decrease of $1.1 billion as compared to the year ended December 31, 2023. Office space charges, including accelerated rent and accelerated depreciation, for the year ended December 31, 2024 were $796 million, a decrease of $1.3 billion as compared to the year ended December 31, 2023. Substantially all of these charges were included in Alphabet-level activities.
- Operating cash flow was $125.3 billion for the year ended December 31, 2024.
- Capital expenditures, which primarily reflected investments in technical infrastructure, were $52.5 billion for the year ended December 31, 2024.
- As of December 31, 2024, we had 183,323 employees.

Financial Results

Revenues

The following table presents revenues by type (in millions):

	Year Ended December 31,	
	2023	2024
Google Search & other	$ 175,033	$ 198,084
YouTube ads	31,510	36,147
Google Network	31,312	30,359
Google advertising	237,855	264,590
Google subscriptions, platforms, and devices	34,688	40,340
Google Services total	272,543	304,930
Google Cloud	33,088	43,229
Other Bets	1,527	1,648
Hedging gains (losses)	236	211
Total revenues	$ 307,394	$ 350,018

Google Services

Google advertising revenues

Google Search & other

Google Search & other revenues increased $23.1 billion from 2023 to 2024. The overall growth was driven by interrelated factors including increases in search queries resulting from growth in user adoption and usage on mobile devices; growth in advertiser spending; and improvements we have made in ad formats and delivery.

YouTube ads

YouTube ads revenues increased $4.6 billion from 2023 to 2024. The growth was driven by our brand advertising products followed by our direct response advertising products, both of which benefited from increased spending by our advertisers.

Google Network

Google Network revenues decreased $953 million from 2023 to 2024, primarily driven by a decrease in Google Ad Manager and AdMob revenues. Additionally, Google Network revenues were adversely affected by changes in foreign currency exchange rates.

Monetization Metrics

The following table presents changes in monetization metrics for Google Search & other revenues (paid clicks and cost-per-click) and Google Network revenues (impressions and cost-per-impression), expressed as a percentage, from 2023 to 2024:

Google Search & other	
Paid clicks change	5 %
Cost-per-click change	7 %
Google Network	
Impressions change	(11)%
Cost-per-impression change	10 %

Changes in paid clicks and impressions are driven by a number of interrelated factors, including changes in advertiser spending; ongoing product and policy changes; and, as it relates to paid clicks, fluctuations in search queries resulting from changes in user adoption and usage, primarily on mobile devices.

Changes in cost-per-click and cost-per-impression are driven by a number of interrelated factors including changes in device mix, geographic mix, advertiser spending, ongoing product and policy changes, product mix, property mix, and changes in foreign currency exchange rates.

Google subscriptions, platforms, and devices

Google subscriptions, platforms, and devices revenues increased $5.7 billion from 2023 to 2024. The growth was primarily driven by an increase in subscription revenues, largely from growth in the number of paid subscribers for YouTube services followed by Google One.

Google Cloud

Google Cloud revenues increased $10.1 billion from 2023 to 2024 primarily driven by growth in Google Cloud Platform largely from infrastructure services.

Revenues by Geography

The following table presents revenues by geography as a percentage of revenues, determined based on the addresses of our customers:

	Year Ended December 31,	
	2023	2024
United States	47 %	49 %
EMEA	30 %	29 %
APAC	17 %	16 %
Other Americas	6 %	6 %
Hedging gains (losses)	0 %	0 %

For additional information, see Note 2 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Use of Non-GAAP Constant Currency Information

International revenues, which represent a significant portion of our revenues, are generally transacted in multiple currencies and therefore are affected by fluctuations in foreign currency exchange rates.

The effect of currency exchange rates on our business is an important factor in understanding period-to-period comparisons. We use non-GAAP constant currency revenues ("constant currency revenues") and non-GAAP percentage change in constant currency revenues ("percentage change in constant currency revenues") for financial and operational decision-making and as a means to evaluate period-to-period comparisons. We believe the presentation of results on a constant currency basis in addition to U.S. Generally Accepted Accounting Principles (GAAP) results helps improve the ability to understand our performance, because it excludes the effects of foreign currency volatility that are not indicative of our core operating results.

Constant currency information compares results between periods as if exchange rates had remained constant period over period. We define constant currency revenues as revenues excluding the effect of foreign currency exchange rate movements ("FX Effect") as well as hedging activities, which are recognized at the consolidated level. We use constant currency revenues to determine the constant currency revenue percentage change on a year-on-year basis. Constant currency revenues are calculated by translating current period revenues using prior year comparable period exchange rates, as well as excluding any hedging effects realized in the current period.

Constant currency revenue percentage change is calculated by determining the change in current period revenues over prior year comparable period revenues where current period foreign currency revenues are translated using prior year comparable period exchange rates and hedging effects are excluded from revenues of both periods.

These results should be considered in addition to, not as a substitute for, results reported in accordance with GAAP. Results on a constant currency basis, as we present them, may not be comparable to similarly titled measures used by other companies and are not a measure of performance presented in accordance with GAAP.

The following table presents the foreign currency exchange effect on international revenues and total revenues (in millions, except percentages):

	Year Ended December 31,		Year Ended December 31, 2024					
					% Change from Prior Period			
	2023	2024	Less FX Effect	Constant Currency Revenues	As Reported	Less Hedging Effect	Less FX Effect	Constant Currency Revenues
United States	$ 146,286	$ 170,447	$ 0	$ 170,447	17 %		0 %	17 %
EMEA	91,038	102,127	41	102,086	12 %		0 %	12 %
APAC	51,514	56,815	(1,369)	58,184	10 %		(3)%	13 %
Other Americas	18,320	20,418	(1,608)	22,026	11 %		(9)%	20 %
Revenues, excluding hedging effect	307,158	349,807	(2,936)	352,743	14 %		(1)%	15 %
Hedging gains (losses)	236	211						
Total revenues(1)	$ 307,394	$ 350,018		$ 352,743	14 %	0 %	(1)%	15 %

(1) Total constant currency revenues of $352.7 billion for 2024 increased $45.6 billion compared to $307.2 billion in revenues, excluding hedging effect, for 2023.

EMEA revenue growth was not materially affected by changes in foreign currency exchange rates, as the effect of the U.S. dollar strengthening relative to the Turkish lira was offset by the U.S. dollar weakening relative to the British pound and the euro.

APAC revenue growth was unfavorably affected by changes in foreign currency exchange rates, primarily due to the U.S. dollar strengthening relative to the Japanese yen.

Other Americas revenue growth was unfavorably affected by changes in foreign currency exchange rates, primarily due to the U.S. dollar strengthening relative to the Argentine peso and the Brazilian real.

Costs and Expenses

Cost of Revenues

The following table presents cost of revenues, including TAC (in millions, except percentages):

	Year Ended December 31,	
	2023	2024
TAC	$ 50,886	$ 54,900
Other cost of revenues	82,446	91,406
Total cost of revenues	$ 133,332	$ 146,306
Total cost of revenues as a percentage of revenues	43 %	42 %

Cost of revenues increased $13.0 billion from 2023 to 2024 due to an increase in other cost of revenues and TAC of $9.0 billion and $4.0 billion, respectively.

The increase in TAC from 2023 to 2024 was largely due to an increase in TAC paid to distribution partners, primarily driven by growth in revenues subject to TAC. The TAC rate decreased from 21.4% to 20.7% from 2023 to 2024 primarily due to a revenue mix shift from Google Network properties to Google Search & other properties. The TAC rate on Google Search & other revenues increased from 2023 to 2024 primarily due to increases related to mobile searches, which carries higher TAC because more mobile searches are channeled through paid access points. The TAC rate on Google Network revenues was substantially consistent from 2023 to 2024.

The increase in other cost of revenues from 2023 to 2024 was primarily due to increases in content acquisition costs, largely for YouTube, depreciation expense, and other technical infrastructure operations costs.

Research and Development

The following table presents R&D expenses (in millions, except percentages):

	Year Ended December 31,	
	2023	2024
Research and development expenses	$ 45,427	$ 49,326
Research and development expenses as a percentage of revenues	15 %	14 %

R&D expenses increased $3.9 billion from 2023 to 2024, primarily driven by increases in employee compensation expenses of $1.5 billion, depreciation expense of $1.4 billion, and third-party services fees of $698 million, partially offset by a reduction in charges related to our office space optimization efforts of $640 million. The increase in employee compensation expenses was primarily driven by a $1.3 billion increase in SBC expenses, which includes the reduction in valuation-based compensation liabilities related to certain Other Bets recognized in the prior year comparable period, partially offset by a $537 million decrease in severance and related charges.

Sales and Marketing

The following table presents sales and marketing expenses (in millions, except percentages):

	Year Ended December 31,	
	2023	2024
Sales and marketing expenses	$ 27,917	$ 27,808
Sales and marketing expenses as a percentage of revenues	9 %	8 %

Sales and marketing expenses decreased $109 million from 2023 to 2024, due to a combination of factors, none of which were individually significant.

General and Administrative

The following table presents general and administrative expenses (in millions, except percentages):

	Year Ended December 31,	
	2023	2024
General and administrative expenses	$ 16,425	$ 14,188
General and administrative expenses as a percentage of revenues	5 %	4 %

General and administrative expenses decreased $2.2 billion from 2023 to 2024, primarily driven by a reduction in charges related to legal and other matters of $1.3 billion and a decrease in employee compensation expenses of $285 million, primarily due to a decrease in average headcount, in addition to a combination of factors, none of which were individually significant.

Segment Profitability

We report our segment results as Google Services, Google Cloud, and Other Bets. Additionally, certain costs are not allocated to our segments because they represent Alphabet-level activities. For further details on our segments, see Part I, Item 1 Business and Note 15 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

The following table presents segment operating income (loss) (in millions).

	Year Ended December 31,	
	2023	2024
Operating income (loss):		
Google Services	$ 95,858	$ 121,263
Google Cloud	1,716	6,112
Other Bets	(4,095)	(4,444)
Alphabet-level activities[(1)]	(9,186)	(10,541)
Total income from operations	$ 84,293	$ 112,390

(1) In addition to the costs included in Alphabet-level activities, hedging gains (losses) related to revenue were $236 million and $211 million in 2023 and 2024, respectively. For the years ended December 31, 2023 and 2024, Alphabet-level activities included substantially all of the charges related to employee severance and our office space charges.

Google Services

Google Services operating income increased $25.4 billion from 2023 to 2024. The increase in operating income was primarily driven by an increase in revenues, partially offset by increases in content acquisition costs and TAC. Additionally, a reduction in employee compensation expenses contributed to the increase in operating income.

Google Cloud

Google Cloud operating income increased $4.4 billion from 2023 to 2024. The increase in operating income was primarily driven by an increase in revenues, partially offset by increases in usage costs for technical infrastructure as well as employee compensation expenses, largely driven by headcount growth.

Other Bets

Other Bets operating loss increased $349 million from 2023 to 2024. The increase in operating loss was primarily due to an increase in expenses, largely driven by employee compensation expenses in addition to a combination of factors, none of which were individually significant. The increase in employee compensation expenses was primarily as a result of the reduction in valuation-based compensation liabilities related to certain Other Bets recognized in the prior year comparable period.

Other Income (Expense), Net

The following table presents OI&E, (in millions):

	Year Ended December 31,	
	2023	2024
Interest income	$ 3,865	$ 4,482
Interest expense	(308)	(268)
Foreign currency exchange gain (loss), net	(1,238)	(409)
Gain (loss) on debt securities, net	(1,215)	(1,043)
Gain (loss) on equity securities, net	392	3,714
Performance fees	257	218
Income (loss) and impairment from equity method investments, net	(628)	(188)
Other	299	919
Other income (expense), net	$ 1,424	$ 7,425

OI&E, net increased $6.0 billion from 2023 to 2024 primarily due to an increase in net gains on equity securities and a decrease in net losses on foreign currency exchange. The net gains on equity securities were primarily due to net unrealized gains on non-marketable equity securities driven by fair value adjustments related to observable transactions, partially offset by a decrease in net unrealized gains on marketable equity securities due to market-driven changes. Foreign currency exchange net losses decreased compared to the prior year primarily due to prior year losses in unhedged currencies.

In January 2025, we recognized an $8.0 billion unrealized gain on our non-marketable equity securities related to our investment in a private company. The unrealized gain reflects an increase in the fair value measurement of our investment following an observable transaction in January 2025.

For additional information, see Note 3 and Note 7 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Provision for Income Taxes

The following table presents provision for income taxes (in millions, except effective tax rate):

	Year Ended December 31,	
	2023	2024
Income before provision for income taxes	$ 85,717	$ 119,815
Provision for income taxes	$ 11,922	$ 19,697
Effective tax rate	13.9 %	16.4 %

The effective tax rate increased from 2023 to 2024. This increase was primarily due to a one-time adjustment for tax rule changes issued by the Internal Revenue Service (IRS) that affected the 2023 rate related to U.S. federal foreign tax credits, as well as a separate rule change with guidance on the capitalization and amortization of R&D

expenses. Additionally, a decrease in the 2024 U.S. federal Foreign Derived Intangible Income tax deduction contributed to an increase in the effective tax rate. These factors were partially offset by an increase in stock-based compensation-related tax benefits in 2024.

The OECD is coordinating negotiations among more than 140 countries with the goal of achieving consensus around substantial changes to international tax policies, including the implementation of a minimum global effective tax rate of 15%. Some countries have already implemented the legislation effective January 1, 2024, and we expect others to follow, however this did not have a material effect on our income tax provision for the 2024 fiscal year.

Financial Condition

Cash, Cash Equivalents, and Marketable Securities

As of December 31, 2024, we had $95.7 billion in cash, cash equivalents, and short-term marketable securities. Cash equivalents and marketable securities are comprised of time deposits, money market funds, highly liquid government bonds, corporate debt securities, mortgage-backed and asset-backed securities, and marketable equity securities.

Sources, Uses of Cash and Related Trends

Our principal sources of liquidity are cash, cash equivalents, and marketable securities, as well as the cash flow that we generate from operations. The primary use of capital continues to be to invest for the long-term growth of the business. We regularly evaluate our cash and capital structure, including the size, pace, and form of capital return to stockholders.

The following table presents our cash flows (in millions):

	Year Ended December 31,	
	2023	2024
Net cash provided by operating activities	$ 101,746	$ 125,299
Net cash used in investing activities	$ (27,063)	$ (45,536)
Net cash used in financing activities	$ (72,093)	$ (79,733)

Cash Provided by Operating Activities

Our largest source of cash provided by operations are advertising revenues generated by Google Search & other properties, Google Network properties, and YouTube properties. In Google Services, we also generate cash through consumer subscriptions, the sale of apps and in-app purchases, and devices. In Google Cloud, we generate cash through consumption-based fees and subscriptions for infrastructure, platform, applications, and other cloud services.

Our primary uses of cash from operating activities include payments to distribution and Google Network partners, to employees for compensation, and to content providers. Other uses of cash from operating activities include payments to suppliers for devices, to tax authorities for income taxes, and other general corporate expenditures.

Net cash provided by operating activities increased from 2023 to 2024 due to an increase in cash received from customers, partially offset by an increase in cash payments for cost of revenues and operating expenses.

Cash Used in Investing Activities

Cash provided by investing activities consists primarily of maturities and sales of investments in marketable and non-marketable securities. Cash used in investing activities consists primarily of purchases of marketable and non-marketable securities, purchases of property and equipment, and payments for acquisitions.

Net cash used in investing activities increased from 2023 to 2024 primarily due to an increase in purchases of property and equipment and purchases of marketable securities, partially offset by increases in maturities and sales of marketable securities. The increase in purchases of property and equipment is primarily driven by investments in technical infrastructure.

Cash Used in Financing Activities

Cash provided by financing activities consists primarily of proceeds from issuance of debt and proceeds from the sale of interests in consolidated entities. Cash used in financing activities consists primarily of repurchases of stock, net payments related to stock-based award activities, payment of dividends, and repayments of debt.

Net cash used in financing activities increased from 2023 to 2024 due to dividend payments and net payments related to stock-based award activities, partially offset by an increase in proceeds from issuance of debt, net of repayments.

.iquidity and Material Cash Requirements

We expect existing cash, cash equivalents, short-term marketable securities, and cash flows from operations and financing activities to continue to be sufficient to fund our operating activities and cash commitments for investing and financing activities for at least the next 12 months, and thereafter for the foreseeable future.

Capital Expenditures and Leases

We make investments in land, buildings, and servers and network equipment through purchases of property and equipment and lease arrangements to provide capacity for the growth of our services and products.

Capital Expenditures

Our capital investments in property and equipment consist primarily of the following major categories:

- technical infrastructure, which consists of our investments in servers and network equipment for computing, storage, and networking requirements for ongoing business activities, including AI, and data center land and building construction; and
- office facilities, ground-up development projects, and building improvements (also referred to as "fit-outs").

Assets not yet in service are those that are not ready for our intended use, including assets in the process of construction or assembly, and consists primarily of technical infrastructure. The time frame from date of purchase to placement in service of these assets may extend from months to years. For example, our data center construction projects are generally multi-year projects with multiple phases, where we acquire land and buildings, construct buildings, and secure and install servers and network equipment.

During the years ended December 31, 2023 and 2024, we spent $32.3 billion and $52.5 billion on capital expenditures, respectively. We expect to increase, relative to 2024, our investment in our technical infrastructure, including servers, network equipment, and data centers, to support the growth of our business and our long-term initiatives, in particular in support of AI products and services. Depreciation of our property and equipment commences when the deployment of such assets are completed and are ready for our intended use. For the years ended December 31, 2023 and 2024, our depreciation on property and equipment was $11.9 billion and $15.3 billion, respectively.

Leases

For the years ended December 31, 2023 and 2024, we recognized additional operating lease assets of $2.9 billion and $2.5 billion, and additional finance lease assets of $564 million and $313 million, respectively. As of December 31, 2024, the amount of total future lease payments under operating leases, which had a weighted average remaining lease term of 7.8 years, was $17.0 billion, of which $3.2 billion is short-term, and total future lease payments under finance leases, which had a weighted average remaining lease term of 10.4 years, was $1.9 billion, of which $257 million is short-term. As of December 31, 2024, we have entered into leases that have not yet commenced with future short-term and long-term lease payments of $773 million and $6.5 billion, respectively, that are not yet recorded on our Consolidated Balance Sheets. These leases will commence between 2025 and 2028 with non-cancelable lease terms of one to 25 years.

For the years ended December 31, 2023 and 2024, our operating lease expenses (including variable lease costs) were $4.5 billion and $4.7 billion, respectively. Finance lease costs were $504 million and $444 million for the years ended December 31, 2023 and 2024, respectively. For additional information, see Note 4 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Financing

We have a short-term debt financing program of up to $10.0 billion through the issuance of commercial paper. Net proceeds from this program are used for general corporate purposes. As of December 31, 2024, we had $2.3 billion of short-term commercial paper outstanding.

As of December 31, 2024, we had senior unsecured notes outstanding with a total carrying value of $11.9 billion with short-term and long-term future interest payments of $197 million and $3.4 billion, respectively.

As of December 31, 2024, we had $10.0 billion of revolving credit facilities, $4.0 billion expiring in April 2025 and $6.0 billion expiring in April 2028. The interest rates for all credit facilities are determined based on a formula using certain market rates, as well as our progress toward the achievement of certain sustainability goals. No amounts have been borrowed under the credit facilities. For additional information, see Note 6 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

We primarily utilize contract manufacturers for the assembly of our servers used in our technical infrastructure and devices we sell. We have agreements where we may purchase components directly from suppliers and then supply these components to contract manufacturers for use in the assembly of the servers and devices. Certain of these arrangements result in a portion of the cash received from and paid to the contract manufacturers to be presented as financing activities in the Consolidated Statements of Cash Flows included in Item 8 of this Annual Report on Form 10-K.

Share Repurchase Program

During 2024, we repurchased and subsequently retired 379 million shares for $62.0 billion.

In April 2024, the Board of Directors of Alphabet authorized the company to repurchase up to an additional $70.0 billion of its Class A and Class C shares. As of December 31, 2024, $44.7 billion remained available for Class A and Class C share repurchases.

The following table presents Class A and Class C shares repurchased and subsequently retired (in millions):

	Year Ended December 31, 2023		Year Ended December 31, 2024	
	Shares	Amount	Shares	Amount
Class A share repurchases	78	$ 9,316	73	$ 11,855
Class C share repurchases	450	52,868	306	50,192
Total share repurchases[1]	528	$ 62,184	379	$ 62,047

[1] Shares repurchased include unsettled repurchases.

For additional information, see Note 11 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Dividend Program

During the year ended December 31, 2024 total cash dividends, which were first paid in June 2024, were $3.5 billion, $519 million, and $3.3 billion for Class A, Class B, and Class C shares, respectively.

The company intends to pay quarterly cash dividends in the future, subject to review and approval by the company's Board of Directors in its sole discretion.

European Commission Fines

In 2017, 2018, and 2019, the European Commission (EC) announced decisions that certain actions taken by Google infringed European competition law and imposed fines of €2.4 billion ($2.7 billion as of June 27, 2017), €4.3 billion ($5.1 billion as of June 30, 2018), and €1.5 billion ($1.7 billion as of March 20, 2019), respectively.

In September 2022, the General Court affirmed the EC decision but reduced the 2018 fine from €4.3 billion to €4.1 billion. We subsequently appealed the General Court's affirmation of the EC decision with the European Court of Justice, which remains pending.

In September 2024, the European Court of Justice rejected our appeal of the 2017 decision and upheld the €2.4 billion fine. In the third quarter of 2024, we made a cash payment of $3.0 billion for the 2017 shopping fine.

In September 2024, the EU's General Court overturned the 2019 decision and annulled the €1.5 billion fine. The EC has appealed the General Court's decision to the European Court of Justice.

We included the outstanding EC fines, including any under appeal, in accrued expenses and other current liabilities on our Consolidated Balance Sheets. For additional information, see Note 10 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Taxes

As of December 31, 2024, we had income taxes payable of $2.7 billion related to a one-time transition tax payable incurred as a result of the U.S. Tax Cuts and Jobs Act, which is due in 2025. We also had long-term taxes payable of $8.8 billion primarily related to uncertain tax positions as of December 31, 2024.

Purchase Commitments and Other Contractual Obligations

As of December 31, 2024, we had material purchase commitments and other contractual obligations of $55.4 billion, of which $32.5 billion was short-term. These amounts primarily consist of purchase orders for certain technical infrastructure as well as the non-cancelable portion or the minimum cancellation fee in certain agreements related to commitments to purchase licenses, including content licenses, inventory, and network capacity. For those agreements with variable terms, we do not estimate the non-cancelable obligation beyond any minimum quantities and/or pricing as of December 31, 2024. In certain instances, the amount of our contractual obligations may change based on the expected timing of order fulfillment from our suppliers. For more information related to our content licenses, see Note 10 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

In addition, we regularly enter into multi-year, non-cancellable agreements to purchase renewable energy and energy attributes, such as renewable energy certificates. These agreements do not include a minimum dollar commitment. The amounts to be paid under these agreements are based on the actual volumes to be generated and are not readily determinable.

Critical Accounting Estimates

We prepare our consolidated financial statements in accordance with GAAP. In doing so, we have to make estimates and assumptions. Our critical accounting estimates are those estimates that involve a significant level of uncertainty at the time the estimate was made, and changes in them have had or are reasonably likely to have a material effect on our financial condition or results of operations. Accordingly, actual results could differ materially from our estimates. We base our estimates on past experience and other assumptions that we believe are reasonable under the circumstances, and we evaluate these estimates on an ongoing basis. We have reviewed our critical accounting estimates with the Audit and Compliance Committee of our Board of Directors.

For a summary of significant accounting policies and the effect on our financial statements, see Note 1 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

Fair Value Measurements of Non-Marketable Equity Securities

We measure certain financial instruments at fair value on a nonrecurring basis, consisting primarily of our non-marketable equity securities. These investments are accounted for under the measurement alternative method ("the measurement alternative") and are measured at cost, less impairment, subject to upward and downward adjustments resulting from observable price changes for identical or similar investments of the same issuer. These adjustments require quantitative assessments of the fair value of our securities, which may require the use of unobservable inputs. Adjustments are determined primarily based on a market approach as of the transaction date and involve the use of estimates using the best information available, which may include cash flow projections or other available market data.

Non-marketable equity securities are also evaluated for impairment, based on qualitative factors including the companies' financial and liquidity position and access to capital resources, among others. When indicators of impairment exist, we prepare quantitative measurements of the fair value of our equity investments using a market approach or an income approach, which requires judgment and the use of unobservable inputs, including discount rates, investee revenues and costs, and comparable market data of private and public companies, among others. When the quantitative remeasurements of fair value indicate an impairment exists, we write down the investment to its current fair value.

We also have compensation arrangements with payouts based on realized returns from certain investments, i.e. performance fees. We record compensation expense based on the estimated payouts on an ongoing basis, which may result in expense recognized before investment returns are realized and compensation is paid and may require the use of unobservable inputs.

Property and Equipment

We assess the reasonableness of the useful lives of our property and equipment periodically as well as when other changes occur, such as when there are changes to ongoing business operations, changes in the planned use and utilization of assets, or technological advancements, that could indicate a change in the period over which we expect to benefit from the asset.

Income Taxes

We are subject to income taxes in the U.S. and foreign jurisdictions. Significant judgment is required in evaluating our uncertain tax positions and determining our provision for income taxes.

Recording an uncertain tax position involves various qualitative considerations, including evaluation of comparable and resolved tax exposures, applicability of tax laws, and likelihood of settlement. We evaluate uncertain tax positions periodically, considering changes in facts and circumstances, such as new regulations or recent judicial opinions, as well as the status of audit activities by taxing authorities. Although we believe we have adequately reserved for our uncertain tax positions, no assurance can be given that the final tax outcome of these matters will not be different. To the extent that the final tax outcome of these matters is different than the amounts recorded, such differences will affect the provision for income taxes and the effective tax rate in the period in which such determination is made.

The provision for income taxes includes the effect of reserve provisions and changes to reserves as well as the related net interest and penalties. In addition, we are subject to the continuous examination of our income tax returns by the IRS and other tax authorities which may assert assessments against us. We regularly assess the likelihood of adverse outcomes resulting from these examinations and assessments to determine the adequacy of our provision for income taxes.

Loss Contingencies

We are subject to claims, lawsuits, regulatory and government inquiries and investigations, other proceedings, and consent orders involving competition, intellectual property, data privacy and security, tax and related compliance, labor and employment, commercial disputes, content generated by our users, goods and services offered by advertisers or publishers using our platforms, personal injury, consumer protection, and other matters. Certain of these matters include speculative claims for substantial or indeterminate amounts of damages. We record a liability when we believe that it is probable that a loss has been incurred and the amount can be reasonably estimated. If we determine that a loss is reasonably possible and the loss or range of loss can be estimated, we disclose the possible loss in Note 10 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

We evaluate, on a regular basis, developments in our legal matters that could affect the amount of liability that has been previously accrued, and the matters and related reasonably possible losses disclosed, and make adjustments and changes to our disclosures. Significant judgment is required to determine both the likelihood and the estimated amount of a loss related to such matters. Until the final resolution of such matters, there may be an exposure to loss in excess of the amount recorded, and such amounts could be material.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

We are exposed to financial market risks, including changes in foreign currency exchange rates, interest rates, and equity investment risks.

Foreign Currency Exchange Risk

We transact business globally in multiple currencies. International revenues, as well as costs and expenses denominated in foreign currencies, expose us to the risk of fluctuations in foreign currency exchange rates against the U.S. dollar. As discussed below, we enter into derivative instruments to hedge foreign currency risk. Principal currencies hedged included the Australian dollar, British pound, Canadian dollar, Euro, and Japanese yen. For the purpose of analyzing foreign currency exchange risk, we considered the historical trends in foreign currency exchange rates and determined that it was reasonably possible that adverse changes in exchange rates of 10% could be experienced.

We use foreign currency forward and option contracts to offset the foreign exchange risk on monetary assets and liabilities denominated in currencies other than the functional currency of the subsidiary. These forward and option contracts reduce, but do not entirely eliminate, the effect of foreign currency exchange rate movements on our assets and liabilities. The foreign currency gains and losses on these assets and liabilities are recorded in OI&E, which are offset by the gains and losses on the forward and option contracts.

If an adverse 10% foreign currency exchange rate change was applied to net monetary assets, liabilities, and commitments denominated in currencies other than the functional currencies at the balance sheet date, it would have resulted in an adverse effect on income before income taxes of approximately $503 million and $135 million as of December 31, 2023 and 2024, respectively, after consideration of the effect of foreign exchange contracts in place for the years ended December 31, 2023 and 2024.

We use foreign currency forward and option contracts, including collars (an option strategy comprised of a combination of purchased and written options) to protect forecasted U.S. dollar-equivalent earnings from changes in foreign currency exchange rates. When the U.S. dollar strengthens, gains from foreign currency forward and option contacts reduce the foreign currency losses related to our earnings. When the U.S. dollar weakens, losses from foreign currency forward and option contracts offset the foreign currency gains related to our earnings. These hedging contracts reduce, but do not entirely eliminate, the effect of foreign currency exchange rate movements. We designate these contracts as cash flow hedges for accounting purposes. We reflect the gains and losses of foreign currency spot rate changes as a component of accumulated other comprehensive income (AOCI) and subsequently reclassify them into revenues to offset the hedged exposures as they occur.

If the U.S. dollar weakened by 10% as of December 31, 2023 and 2024, the amount recorded in AOCI related to our cash flow hedges before tax effect would have been approximately $1.5 billion and $1.6 billion lower as of

December 31, 2023 and 2024, respectively. The change in the value recorded in AOCI would be expected to offset a corresponding foreign currency change in forecasted hedged revenues when recognized.

We use foreign exchange forward contracts designated as net investment hedges to hedge the foreign currency risks related to investment in foreign subsidiaries. These forward contracts serve to offset the foreign currency translation risk from our foreign operations.

If the U.S. dollar weakened by 10%, the amount recorded in cumulative translation adjustment (CTA) within AOCI related to our net investment hedges before tax effect would have been approximately $946 million and $660 million lower as of December 31, 2023 and 2024, respectively. The change in value recorded in CTA would be expected to offset a corresponding foreign currency translation gain or loss from our investment in foreign subsidiaries.

Interest Rate Risk

Our Corporate Treasury investment strategy is to achieve a return that will allow us to preserve capital and maintain liquidity. We invest primarily in debt securities, including government bonds, corporate debt securities, mortgage-backed and asset-backed securities, money market and other funds, time deposits, and interest rate derivatives. By policy, we limit the amount of credit exposure to any one issuer. Our investments in both fixed rate and floating rate interest earning securities carry a degree of interest rate risk. Fixed rate securities may have their fair market value adversely affected due to a rise in interest rates, while floating rate securities may produce less income than predicted if interest rates fall. Unrealized gains and losses on our marketable debt securities are primarily due to interest rate fluctuations as compared to interest rates at the time of purchase. For certain fixed and variable rate debt securities, we have elected the fair value option for which changes in fair value are recorded in OI&E. We measure securities for which we have not elected the fair value option at fair value with gains and losses recorded in AOCI until the securities are sold, less any expected credit losses.

We use value-at-risk (VaR) analysis to determine the potential effect of fluctuations in interest rates on the value of our investment portfolio. The VaR is the expected loss in fair value, for a given confidence interval, for our investment portfolio due to adverse movements in interest rates. We use a variance/covariance VaR model with 95% confidence interval. The estimated one-day loss in fair value of our investment portfolio as of December 31, 2023 and 2024 are shown below (in millions):

	As of December 31,		12-Month Average As of December 31,	
	2023	2024	2023	2024
Risk category - interest rate	$ 296	$ 208	$ 271	$ 230

Actual future gains and losses associated with our investment portfolio may differ materially from the sensitivity analyses performed as of December 31, 2023 and 2024 due to the inherent limitations associated with predicting the timing and amount of changes in interest rates and our actual exposures and positions. VaR analysis is not intended to represent actual losses but is used as a risk estimation.

Equity Investment Risk

Our marketable and non-marketable equity securities are subject to a wide variety of market-related risks that could substantially reduce or increase the fair value of our holdings.

Our marketable equity securities are publicly traded stocks or funds and our non-marketable equity securities are investments in privately held companies, some of which are in the startup or development stages.

We record marketable equity securities not accounted for under the equity method at fair value based on readily determinable market values, of which publicly traded stocks and mutual funds are subject to market price volatility, and represent $6.0 billion and $5.1 billion of our investments as of December 31, 2023 and 2024, respectively. A hypothetical adverse price change of 10% on our December 31, 2024 balance would decrease the fair value of marketable equity securities by $508 million. From time to time, we may enter into derivatives to hedge the market price risk on certain of our marketable equity securities.

Our non-marketable equity securities not accounted for under the equity method are primarily adjusted to fair value for observable transactions for identical or similar investments of the same issuer or impairment (referred to as the measurement alternative). The fair value measured at the time of the observable transaction is not necessarily an indication of the current fair value as of the balance sheet date. These investments, especially those that are in the early stages, are inherently risky because the technologies or products these companies have under development are typically in the early phases and may never materialize, and they may experience a decline in financial condition, which could result in a loss of a substantial part of our investment in these companies. Valuations of our equity investments in private companies are inherently more complex due to the lack of readily available market data and observable transactions at lower valuations could result in significant losses. In addition, global economic conditions could result in additional volatility. The success of our investment in any private company is also typically dependent on the likelihood of our ability to realize appreciation in the value of investments through liquidity events such as public offerings, acquisitions, private sales or other market events. Changes in the valuation of non-marketable equity securities may not directly correlate with changes in valuation of marketable equity securities. As of December 31, 2023 and 2024, the carrying value of our non-marketable equity securities, which were accounted for under the measurement alternative, was $28.8 billion and $35.5 billion, respectively.

The carrying values of our equity method investments, which totaled approximately $1.7 billion and $2.0 billion as of December 31, 2023 and 2024, respectively, generally do not fluctuate based on market price changes. However, these investments could be impaired if the carrying value exceeds the fair value and is not expected to recover.

For additional information about our equity investments, see Note 1 and Note 3 of the Notes to Consolidated Financial Statements included in Item 8 of this Annual Report on Form 10-K.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

Alphabet Inc.

INDEX TO CONSOLIDATED FINANCIAL STATEMENTS

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of Alphabet Inc.

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Alphabet Inc. (the Company) as of December 31, 2023 and 2024, the related consolidated statements of income, comprehensive income, stockholders' equity and cash flows for each of the three years in the period ended December 31, 2024, and the related notes and financial statement schedule listed in the Index at Item 15 (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2023 and 2024, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2024, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2024, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework), and our report dated February 4, 2025 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the account or disclosure to which it relates.

Loss Contingencies

Description of the Matter	The Company is subject to claims, lawsuits, regulatory and government inquiries and investigations, other proceedings, and orders involving competition, intellectual property, data privacy and security, tax and related compliance, labor and employment, commercial disputes, content generated by its users, goods and services offered by advertisers or publishers using their platforms, personal injury, and other matters. As described in Note 10 to the consolidated financial statements “Commitments and Contingencies” such claims, lawsuits, regulatory and government inquiries and investigations, other proceedings, and orders could result in adverse consequences. Significant judgment is required to determine both the likelihood and the estimated amount of a loss related to such matters. Auditing management’s accounting for and disclosure of loss contingencies from these matters involved challenging and subjective auditor judgment in assessing the Company’s evaluation of the probability of a loss, and the estimated amount or range of loss.
How We Addressed the Matter in Our Audit	We tested relevant controls over the identified risks associated with management’s accounting for and disclosure of these matters. This included controls over management’s assessment of the probability of incurrence of a loss and whether the loss or range of loss was reasonably estimable and the development of related disclosures. Our audit procedures included gaining an understanding of previous rulings and the status of ongoing lawsuits, reviewing letters from internal and external legal counsel addressing the matters, meeting with internal legal counsel to discuss the allegations, and obtaining a representation letter from management on these matters. We also evaluated the Company’s disclosures in relation to these matters.

/s/ Ernst & Young LLP

We have served as the Company's auditor since 1999.

San Jose, California
February 4, 2025

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of Alphabet Inc.

Opinion on Internal Control Over Financial Reporting

We have audited Alphabet Inc.'s internal control over financial reporting as of December 31, 2024, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, Alphabet Inc. (the Company) maintained, in all material respects, effective internal control over financial reporting as of December 31, 2024, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the 2024 consolidated financial statements of the Company and our report dated February 4, 2025 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP

San Jose, California
February 4, 2025

Alphabet Inc.
CONSOLIDATED BALANCE SHEETS
(in millions, except par value per share amounts)

	As of December 31, 2023	As of December 31, 2024
Assets		
Current assets:		
Cash and cash equivalents	$ 24,048	$ 23,466
Marketable securities	86,868	72,191
Total cash, cash equivalents, and marketable securities	110,916	95,657
Accounts receivable, net	47,964	52,340
Other current assets	12,650	15,714
Total current assets	171,530	163,711
Non-marketable securities	31,008	37,982
Deferred income taxes	12,169	17,180
Property and equipment, net	134,345	171,036
Operating lease assets	14,091	13,588
Goodwill	29,198	31,885
Other non-current assets	10,051	14,874
Total assets	$ 402,392	$ 450,256
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	$ 7,493	$ 7,987
Accrued compensation and benefits	15,140	15,069
Accrued expenses and other current liabilities	46,168	51,228
Accrued revenue share	8,876	9,802
Deferred revenue	4,137	5,036
Total current liabilities	81,814	89,122
Long-term debt	11,870	10,883
Income taxes payable, non-current	8,474	8,782
Operating lease liabilities	12,460	11,691
Other long-term liabilities	4,395	4,694
Total liabilities	119,013	125,172
Commitments and Contingencies (Note 10)		
Stockholders' equity:		
Preferred stock, $0.001 par value per share, 100 shares authorized; no shares issued and outstanding	0	0
Class A, Class B, and Class C stock and additional paid-in capital, $0.001 par value per share: 300,000 shares authorized (Class A 180,000, Class B 60,000, Class C 60,000); 12,460 (Class A 5,899, Class B 870, Class C 5,691) and 12,211 (Class A 5,835, Class B 861, Class C 5,515) shares issued and outstanding	76,534	84,800
Accumulated other comprehensive income (loss)	(4,402)	(4,800)
Retained earnings	211,247	245,084
Total stockholders' equity	283,379	325,084
Total liabilities and stockholders' equity	$ 402,392	$ 450,256

See accompanying notes.

Alphabet Inc.
CONSOLIDATED STATEMENTS OF INCOME
(in millions, except per share amounts)

	Year Ended December 31,		
	2022	2023	2024
Revenues	$ 282,836	$ 307,394	$ 350,018
Costs and expenses:			
Cost of revenues	126,203	133,332	146,306
Research and development	39,500	45,427	49,326
Sales and marketing	26,567	27,917	27,808
General and administrative	15,724	16,425	14,188
Total costs and expenses	207,994	223,101	237,628
Income from operations	74,842	84,293	112,390
Other income (expense), net	(3,514)	1,424	7,425
Income before income taxes	71,328	85,717	119,815
Provision for income taxes	11,356	11,922	19,697
Net income	$ 59,972	$ 73,795	$ 100,118
Basic net income per share (Note 12)	$ 4.59	$ 5.84	$ 8.13
Diluted net income per share (Note 12)	$ 4.56	$ 5.80	$ 8.04

See accompanying notes.

Alphabet Inc.
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME
(in millions)

	Year Ended December 31,		
	2022	2023	2024
Net income	$ 59,972	$ 73,795	$ 100,118
Other comprehensive income (loss):			
Change in foreign currency translation adjustment, net of income tax benefit (expense) of $(134), $63 and $(49)	(1,836)	735	(1,673)
Available-for-sale investments:			
Change in net unrealized gains (losses)	(4,720)	1,344	(116)
Less: reclassification adjustment for net (gains) losses included in net income	1,007	1,168	782
Net change, net of income tax benefit (expense) of $1,056, $(698), and $(190)	(3,713)	2,512	666
Cash flow hedges:			
Change in net unrealized gains (losses)	1,275	168	775
Less: reclassification adjustment for net (gains) losses included in net income	(1,706)	(214)	(166)
Net change, net of income tax benefit (expense) of $110, $2, and $(151)	(431)	(46)	609
Other comprehensive income (loss)	(5,980)	3,201	(398)
Comprehensive income	$ 53,992	$ 76,996	$ 99,720

See accompanying notes.

Alphabet Inc.
CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY
(in millions)

	Class A, Class B, Class C Stock and Additional Paid-In Capital		Accumulated Other Comprehensive Income (Loss)	Retained Earnings	Total Stockholders' Equity
	Shares	Amount			
Balance as of December 31, 2021	13,242	$ 61,774	$ (1,623)	$ 191,484	$ 251,635
Stock issued	137	8	0	0	8
Stock-based compensation	0	19,525	0	0	19,525
Tax withholding related to vesting of restricted stock units and other	0	(9,754)	0	(1)	(9,755)
Repurchases of stock	(530)	(3,404)	0	(55,892)	(59,296)
Sale of interest in consolidated entities	0	35	0	0	35
Net income	0	0	0	59,972	59,972
Other comprehensive income (loss)	0	0	(5,980)	0	(5,980)
Balance as of December 31, 2022	12,849	68,184	(7,603)	195,563	256,144
Stock issued	139	0	0	0	0
Stock-based compensation	0	22,578	0	0	22,578
Tax withholding related to vesting of restricted stock units and other	0	(10,164)	0	9	(10,155)
Repurchases of stock	(528)	(4,064)	0	(58,120)	(62,184)
Net income	0	0	0	73,795	73,795
Other comprehensive income (loss)	0	0	3,201	0	3,201
Balance as of December 31, 2023	12,460	76,534	(4,402)	211,247	283,379
Stock issued	130	0	0	0	0
Stock-based compensation	0	22,937	0	0	22,937
Tax withholding related to vesting of restricted stock units and other	0	(12,507)	0	(16)	(12,523)
Repurchases of stock	(379)	(3,359)	0	(58,688)	(62,047)
Dividends and dividend equivalents declared ($0.60 per share)	0	41	0	(7,577)	(7,536)
Sale of interest in consolidated entities	0	1,154	0	0	1,154
Net income	0	0	0	100,118	100,118
Other comprehensive income (loss)	0	0	(398)	0	(398)
Balance as of December 31, 2024	12,211	$ 84,800	$ (4,800)	$ 245,084	$ 325,084

See accompanying notes.

Alphabet Inc.
CONSOLIDATED STATEMENTS OF CASH FLOWS
(in millions)

	Year Ended December 31,		
	2022	2023	2024
Operating activities			
Net income	$ 59,972	$ 73,795	$ 100,118
Adjustments:			
Depreciation of property and equipment	13,475	11,946	15,311
Stock-based compensation expense	19,362	22,460	22,785
Deferred income taxes	(8,081)	(7,763)	(5,257)
Loss (gain) on debt and equity securities, net	5,519	823	(2,671)
Other	3,483	4,330	3,419
Changes in assets and liabilities, net of effects of acquisitions:			
Accounts receivable, net	(2,317)	(7,833)	(5,891)
Income taxes, net	584	523	(2,418)
Other assets	(5,046)	(2,143)	(1,397)
Accounts payable	707	664	359
Accrued expenses and other liabilities	3,915	3,937	(1,161)
Accrued revenue share	(445)	482	1,059
Deferred revenue	367	525	1,043
Net cash provided by operating activities	91,495	101,746	125,299
Investing activities			
Purchases of property and equipment	(31,485)	(32,251)	(52,535)
Purchases of marketable securities	(78,874)	(77,858)	(86,679)
Maturities and sales of marketable securities	97,822	86,672	103,428
Purchases of non-marketable securities	(2,531)	(3,027)	(5,034)
Maturities and sales of non-marketable securities	150	947	882
Acquisitions, net of cash acquired, and purchases of intangible assets	(6,969)	(495)	(2,931)
Other investing activities	1,589	(1,051)	(2,667)
Net cash used in investing activities	(20,298)	(27,063)	(45,536)
Financing activities			
Net payments related to stock-based award activities	(9,300)	(9,837)	(12,190)
Repurchases of stock	(59,296)	(61,504)	(62,222)
Dividend payments	0	0	(7,363)
Proceeds from issuance of debt, net of costs	52,872	10,790	13,589
Repayments of debt	(54,068)	(11,550)	(12,701)
Proceeds from sale of interest in consolidated entities, net	35	8	1,154
Net cash used in financing activities	(69,757)	(72,093)	(79,733)
Effect of exchange rate changes on cash and cash equivalents	(506)	(421)	(612)
Net increase (decrease) in cash and cash equivalents	934	2,169	(582)
Cash and cash equivalents at beginning of period	20,945	21,879	24,048
Cash and cash equivalents at end of period	$ 21,879	$ 24,048	$ 23,466

See accompanying notes.

Alphabet Inc.
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

Note 1. Summary of Significant Accounting Policies

Nature of Operations

Google was incorporated in California in September 1998 and re-incorporated in the State of Delaware in August 2003. In 2015, we implemented a holding company reorganization, and as a result, Alphabet Inc. ("Alphabet") became the successor issuer to Google.

We generate revenues by delivering relevant, cost-effective online advertising; cloud-based solutions that provide enterprise customers of all sizes with infrastructure, platform services, and applications; sales of other products and services, such as fees received for subscription-based products, apps and in-app purchases, and devices.

Basis of Consolidation

The consolidated financial statements of Alphabet include the accounts of Alphabet and entities consolidated under the variable interest and voting models. Intercompany balances and transactions have been eliminated.

Use of Estimates

Preparation of consolidated financial statements in conformity with GAAP requires us to make estimates and assumptions that affect the amounts reported and disclosed in the financial statements and the accompanying notes. Actual results could differ materially from these estimates due to uncertainties. On an ongoing basis, we evaluate our estimates, including those related to the allowance for credit losses; contingent liabilities; fair values of financial instruments and goodwill; income taxes; inventory; and useful lives of property and equipment, among others. We base our estimates on assumptions, both historical and forward looking, that are believed to be reasonable, and the results of which form the basis for making judgments about the carrying values of assets and liabilities.

Revenue Recognition

Revenues are recognized when control of the promised goods or services is transferred to our customers, and the collectibility of an amount that we expect in exchange for those goods or services is probable. Sales and other similar taxes are excluded from revenues.

Advertising Revenues

We generate advertising revenues primarily by delivering advertising on:

- Google Search and other properties, including revenues from traffic generated by search distribution partners who use Google.com as their default search in browsers, toolbars, etc. and other Google owned and operated properties like Gmail, Google Maps, and Google Play;
- YouTube properties; and
- Google Network properties, including revenues from Google Network properties participating in AdMob, AdSense, and Google Ad Manager.

Our customers generally purchase advertising inventory through Google Ads, Google Ad Manager, Google Display & Video 360, and Google Marketing Platform, among others.

We offer advertising by delivering both performance and brand advertising. We recognize revenues for performance advertising when a user engages with the advertisement. For brand advertising, we recognize revenues when the ad is displayed, or a user views the ad.

For ads placed on Google Network properties, we evaluate whether we are the principal (i.e., report revenues on a gross basis) or agent (i.e., report revenues on a net basis). Generally, we report advertising revenues for ads placed on Google Network properties on a gross basis, that is, the amounts billed to our customers are recorded as revenues, and amounts paid to Google Network partners are recorded as cost of revenues. Where we are the principal, we control the advertising inventory before it is transferred to our customers. Our control is evidenced by our sole ability to monetize the advertising inventory before it is transferred to our customers and is further supported by us being primarily responsible to our customers and having a level of discretion in establishing pricing.

Google Subscriptions, Platforms, and Devices

Google subscriptions, platforms, and devices revenues consist of revenues from:

- consumer subscriptions, which primarily include revenues from YouTube services, such as YouTube TV, YouTube Music and Premium, and NFL Sunday Ticket, as well as Google One;
- platforms, which primarily include revenues from Google Play sales of apps and in-app purchases;
- devices, which primarily include sales of the Pixel family of devices; and
- other products and services.

Subscription revenues are recognized ratably over the period of the subscription, primarily monthly. We report revenues from Google Play sales of apps and in-app purchases on a net basis because our performance obligation is to facilitate a transaction between app developers and end users for which we earn a service fee.

Google Cloud Revenues

Google Cloud revenues consist of revenues from:

- Google Cloud Platform, which generates consumption-based fees and subscriptions for infrastructure, platform, and other services. These services provide access to solutions such as AI offerings including our AI infrastructure, Vertex AI platform, and Gemini for Google Cloud; cybersecurity; and data and analytics;
- Google Workspace, which includes subscriptions for cloud-based communication and collaboration tools for enterprises, such as Calendar, Gmail, Docs, Drive, and Meet, with integrated features like Gemini for Google Workspace; and
- other enterprise services.

Our cloud services are generally provided on either a consumption or subscription basis and may have contract terms longer than a year. Revenues related to cloud services provided on a consumption basis are recognized when the customer utilizes the services, based on the quantity of services consumed. Revenues related to cloud services provided on a subscription basis are recognized ratably over the contract term as the customer receives and consumes the benefits of the cloud services.

Arrangements with Multiple Performance Obligations

Our contracts with customers may include multiple performance obligations. For such arrangements, we allocate revenues to each performance obligation based on its relative standalone selling price. We generally determine standalone selling prices based on the prices charged to customers.

Customer Incentives and Credits

Certain customers receive cash-based incentives or credits, which are accounted for as variable consideration. We estimate these amounts based on the expected amount to be provided to customers and reduce revenues. We believe that there will not be significant changes to our estimates of variable consideration related to customer incentives and credits.

Sales Commissions

We expense sales commissions when incurred when the period of the expected benefit is one year or less. We recognize an asset for certain sales commissions and amortize if the expected benefit period is greater than one year. These costs are recorded within sales and marketing expenses.

Cost of Revenues

Cost of revenues consists of TAC and other costs of revenues.

- TAC includes:
  - amounts paid to our distribution partners who make available our search access points and services. Our distribution partners include browser providers, mobile carriers, original equipment manufacturers, and software developers; and
  - amounts paid to Google Network partners primarily for ads displayed on their properties.
- Other cost of revenues includes:
  - content acquisition costs, which are payments to content providers from whom we license video and other content for distribution, primarily related to YouTube (we pay fees to these content providers based on revenues generated, subscriber counts, or a flat fee);
  - depreciation expense related to our technical infrastructure;

- ◦ employee compensation expenses related to our technical infrastructure and other operations such as content review and customer and product support;
- ◦ inventory and other costs related to the devices we sell; and
- ◦ other technical infrastructure operations costs, including network capacity, energy, and equipment costs.

Software Development Costs

We expense software development costs, including costs to develop software products or the software component of products to be sold, leased, or marketed to external users, before technological feasibility is reached. Technological feasibility is typically reached shortly before the release of such products. As a result, development costs that meet the criteria for capitalization were not material for the periods presented.

Software development costs also include costs to develop software to be used solely to meet internal needs and cloud-based applications used to deliver our services. We capitalize development costs related to these software applications once the preliminary project stage is complete and it is probable that the project will be completed and the software will be used to perform the function intended. Costs capitalized for developing such software applications were not material for the periods presented.

Stock-based Compensation

Stock-based compensation (SBC) primarily consists of Alphabet restricted stock units (RSUs). RSUs are equity classified and measured at the fair market value of the underlying stock at the grant date. We recognize RSU expense using the straight-line attribution method over the requisite service period and account for forfeitures as they occur. RSUs are awarded dividend equivalents, which are subject to the same vesting conditions as the underlying award, and settled in Class C shares.

For RSUs, shares are issued on the vesting dates net of the applicable statutory income tax withholding to be paid by us on behalf of our employees. As a result, fewer shares are issued than the number of RSUs outstanding, and the income tax withholding is recorded as a reduction to additional paid-in capital.

Additionally, SBC includes other stock-based awards, such as performance stock units (PSUs) that include market conditions and awards that may be settled in cash or the stock of certain Other Bet companies. PSUs and certain awards granted by Other Bet companies are equity classified and expense is recognized over the requisite service period. Certain awards granted by Other Bet companies are liability classified and remeasured at fair value through settlement. The fair value of awards granted by Other Bet companies is based on the equity valuation of the respective Other Bet company.

Advertising and Promotional Expenses

We expense advertising and promotional costs in the period in which they are incurred. For the years ended December 31, 2022, 2023, and 2024, advertising and promotional expenses totaled approximately $9.2 billion, $8.7 billion, and $8.7 billion, respectively.

Performance Fees

Performance fees refer to compensation arrangements with payouts based on realized returns from certain investments. We record compensation expense based on the estimated payouts on an ongoing basis, which may result in expense recognized before investment returns are realized and compensation is paid and may require the use of unobservable inputs. Performance fees are recorded as a component of OI&E.

Fair Value Measurements

Fair value is an exit price, representing the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants. Assets and liabilities recorded at fair value are measured and classified in accordance with a three-tier fair value hierarchy based on the observability of the inputs available in the market used to measure fair value:

Level 1 - Observable inputs that reflect quoted prices (unadjusted) for identical assets or liabilities in active markets.

Level 2 - Inputs that are based upon quoted prices for similar instruments in active markets, quoted prices for identical or similar instruments in markets that are not active, and model-based valuation techniques for which all significant inputs are observable in the market or can be derived from observable market data. Where applicable, these models project future cash flows and discount the future amounts to a present value using market-based observable inputs including interest rate curves, foreign exchange rates, and credit ratings.

Level 3 - Unobservable inputs that are supported by little or no market activities.

The fair value hierarchy requires an entity to maximize the use of observable inputs and minimize the use of unobservable inputs when measuring fair value. The determination of fair value involves the use of appropriate valuation methods and relevant inputs into valuation models.

Our financial assets and liabilities that are measured at fair value on a recurring basis include cash equivalents, marketable securities, derivative financial instruments, and certain non-marketable debt securities. Our financial assets measured at fair value on a nonrecurring basis include non-marketable equity securities. Other financial assets and liabilities are carried at cost with fair value disclosed, if required.

We measure certain other instruments, and certain assets and liabilities acquired in a business combination, also at fair value on a nonrecurring basis.

Financial Instruments

Our financial instruments include cash, cash equivalents, marketable and non-marketable securities, derivative financial instruments, accounts receivable, and convertible notes.

Credit Risks

We are subject to credit risk primarily from cash equivalents, marketable debt securities, derivative financial instruments, including foreign exchange contracts, accounts receivable, and convertible notes. We manage our credit risk exposure through timely assessment of our counterparty creditworthiness, credit limits, and use of collateral management. Foreign exchange contracts are transacted with various financial institutions with high credit standing. Accounts receivable are typically unsecured and are derived from revenues earned from customers located around the world. We manage our credit risk exposure by performing ongoing evaluations to determine customer credit and we limit the amount of credit we extend. We generally do not require collateral from our customers.

Cash Equivalents

We invest excess cash primarily in asset-backed and mortgage-backed securities, corporate debt securities, government bonds, money market funds, and time deposits.

Marketable Securities

We classify all marketable debt securities that have effective maturities of three months or less from the date of purchase as cash equivalents and those with effective maturities of greater than three months as marketable securities on our Consolidated Balance Sheets. We determine the appropriate classification of our investments in marketable debt securities at the time of purchase and reevaluate such designation at each balance sheet date. We have classified and accounted for our marketable debt securities as available-for-sale. After consideration of our risk versus reward objectives, as well as our liquidity requirements, we may sell these debt securities prior to their effective maturities. As we view these securities as available to support current operations, we classify highly liquid securities with maturities beyond 12 months as current assets under the caption marketable securities on the Consolidated Balance Sheets. We carry these securities at fair value, and report the unrealized gains and losses, net of taxes, as a component of stockholders' equity, except for the changes in allowance for expected credit losses, which are recorded in OI&E. For certain marketable debt securities we have elected the fair value option, for which changes in fair value are recorded in OI&E. We determine any realized gains and losses on the sale of marketable debt securities on a specific identification method, and we record such gains and losses as a component of OI&E.

Our investments in marketable equity securities are measured at fair value with the related gains and losses, including unrealized, recognized in OI&E. We classify our marketable equity securities subject to long-term lock-up restrictions beyond 12 months as other non-current assets on the Consolidated Balance Sheets.

Non-Marketable Securities

Non-marketable securities primarily consist of equity securities. We account for non-marketable equity securities through which we exercise significant influence but do not have control over the investee under the equity method. Other non-marketable equity securities that we hold are primarily accounted for under the measurement alternative. Under the measurement alternative, the carrying value is measured at cost, less any impairment, plus or minus changes resulting from observable price changes in orderly transactions for identical or similar investments of the same issuer. Adjustments are determined primarily based on a market approach as of the transaction date and are recorded as a component of OI&E.

Non-marketable securities that do not have effective contractual maturity dates are classified as other non-current assets on the Consolidated Balance Sheets.

Derivative Financial Instruments

See Note 3 for the accounting policy pertaining to derivative financial instruments.

Accounts Receivable

Our payment terms for accounts receivable vary by the types and locations of our customers and the products or services offered. The term between invoicing and when payment is due is not significant. For certain products or services and customers, we require payment before the products or services are delivered to the customer. Additionally, accounts receivable includes amounts for services performed in advance of the right to invoice the customer.

We maintain an allowance for credit losses for accounts receivable, which is recorded as an offset to accounts receivable, and changes in such are classified as general and administrative expense in the Consolidated Statements of Income. We assess collectibility by reviewing accounts receivable on a collective basis where similar characteristics exist and on an individual basis when we identify specific customers with known disputes or collectibility issues. In determining the amount of the allowance for credit losses, we consider historical collectibility based on past due status and make judgments about the creditworthiness of customers based on ongoing credit evaluations. We also consider customer-specific information, current market conditions, and reasonable and supportable forecasts of future economic conditions.

Convertible Notes

Our investments in convertible notes are primarily recorded at amortized cost which includes unpaid principal balances, deferred origination costs, and any related discount or premium, net of allowances for credit losses, and are included within other non-current assets on our Consolidated Balance Sheets.

Other

Our financial instruments also include debt and equity investments in companies with which we also entered into commercial arrangements at or near the same time. For these transactions, judgment is required in assessing the substance of the arrangements, including assessing whether the components of the arrangements should be accounted for as separate transactions under the applicable GAAP, and determining the value of the components of the arrangements, including the fair value of the investments. Additionally, if our investment in such companies becomes impaired, we may need to re-evaluate the accounting for the commercial arrangement, including reducing any remaining performance obligations.

Impairment of Investments

We periodically review our debt and non-marketable equity securities for impairment.

For debt securities in an unrealized loss position, we determine whether a credit loss exists. The credit loss is estimated by considering available information relevant to the collectibility of the security and information about past events, current conditions, and reasonable and supportable forecasts. Any credit loss is recorded as a charge to OI&E, not to exceed the amount of the unrealized loss. Unrealized losses other than the credit loss are recognized in AOCI. If we have an intent to sell, or if it is more likely than not that we will be required to sell a debt security in an unrealized loss position before recovery of its amortized cost basis, we will write down the security to its fair value and record the corresponding charge as a component of OI&E.

For non-marketable equity securities, including equity method investments, we consider whether impairment indicators exist by evaluating the companies' financial and liquidity position and access to capital resources, among other indicators. If the assessment indicates that the investment is impaired, we write down the investment to its fair value by recording the corresponding charge as a component of OI&E. We prepare quantitative measurements of the fair value of our equity investments using a market approach or an income approach.

Inventory

Inventory consists primarily of finished goods and is stated at the lower of cost and net realizable value. Cost is computed using the first-in, first-out method.

Variable Interest Entities

We determine at the inception of each arrangement whether an entity in which we have made an investment or in which we have other variable interests is considered a variable interest entity (VIE). We consolidate VIEs when we are the primary beneficiary. We are the primary beneficiary of a VIE when we have the power to direct activities that most significantly affect the economic performance of the VIE and have the obligation to absorb the majority of their losses or benefits. If we are not the primary beneficiary in a VIE, we account for the investment or other variable interests in a VIE in accordance with applicable GAAP.

Periodically, we assess whether any changes in our interest or relationship with the entity affect our determination of whether the entity is a VIE and, if so, whether we are the primary beneficiary.

Property and Equipment

Property and equipment is comprised of technical infrastructure, office space, corporate and other assets currently in service, and assets not yet in service. Technical infrastructure includes data center land, buildings and leasehold improvements, and servers and network equipment. Office space includes office land, buildings and leasehold improvements. Assets not yet in service are those that are not ready for our intended use, including data center buildings and servers in the process of construction or assembly.

Property and equipment are stated at cost less accumulated depreciation. Depreciation commences once assets are ready for our intended use and is recorded using the straight-line method over the estimated useful lives of the assets, which we regularly evaluate for factors such as technological obsolescence and our planned use and utilization. We depreciate data center and office buildings over periods of seven to 40 years. We depreciate servers and network equipment generally over a period of six years. We depreciate corporate and other assets over periods of two to 25 years. We depreciate leasehold improvements over the shorter of the remaining lease term or the estimated useful lives of the assets. Land is not depreciated.

Goodwill

We allocate goodwill to reporting units based on the expected benefit from the business combination. We evaluate our reporting units periodically, as well as when changes in our operating segments occur. For changes in reporting units, we reassign goodwill using a relative fair value allocation approach. We test our goodwill for impairment at least annually, or more frequently if events or changes in circumstances indicate that the asset may be impaired. Goodwill impairments were not material for the periods presented.

Leases

We determine if an arrangement is a lease at inception. Our lease agreements generally contain lease and non-lease components. Payments under our lease arrangements are primarily fixed. Non-lease components primarily include payments for maintenance and utilities. We combine fixed payments for non-lease components with lease payments and account for them together as a single lease component which increases the amount of our lease assets and liabilities.

Certain lease agreements contain variable payments, which are expensed as incurred and not included in the lease assets and liabilities. These amounts primarily include payments affected by the Consumer Price Index, and payments for maintenance and utilities.

Lease assets and liabilities are recognized at the present value of the future lease payments at the lease commencement date. The interest rate used to determine the present value of the future lease payments is our incremental borrowing rate, because the interest rate implicit in our leases is not readily determinable. Our incremental borrowing rate is estimated to approximate the interest rate on a collateralized basis with similar terms and payments, and in economic environments where the leased asset is located. Our lease terms include periods under options to extend or terminate the lease when it is reasonably certain that we will exercise that option. We generally use the base, non-cancelable, lease term when determining the lease assets and liabilities. Lease assets also include any prepaid lease payments and lease incentives.

Lease assets and liabilities are included on our Consolidated Balance Sheets. The current portion of our operating lease liabilities is included in accrued expenses and other current liabilities, and the long-term portion is included in operating lease liabilities. Finance lease assets are included in property and equipment, net. Finance lease liabilities are included in accrued expenses and other current liabilities or other long-term liabilities.

Operating lease expense (excluding variable lease costs) is recognized on a straight-line basis over the lease term. Finance lease expense is recognized on a straight-line basis over the shorter of the lease term or the useful life of the asset, and interest expense is recognized based on the incremental borrowing rate.

Impairment of Long-Lived Assets

We review leases, property and equipment, and intangible assets, excluding goodwill, for impairment when events or changes in circumstances indicate the carrying amount may not be recoverable. The evaluation is performed at the lowest level of identifiable cash flows independent of other assets. We measure recoverability of these assets by comparing the carrying amounts to the future undiscounted cash flows that the assets or the asset group are expected to generate. If the carrying value of the assets or asset group is not recoverable, the impairment recognized is measured as the amount by which the carrying value exceeds its fair value.

Income Taxes

We account for income taxes using the asset and liability method, under which we recognize the amount of taxes payable or refundable for the current year and deferred tax assets and liabilities for the future tax consequences of events that have been recognized in our financial statements or tax returns. We measure current and deferred tax assets and liabilities based on provisions of enacted tax law. We evaluate the likelihood of future realization of our deferred tax assets based on all available evidence and establish a valuation allowance to reduce deferred tax assets when it is more likely than not that they will not be realized or release a valuation allowance to increase deferred tax assets when it is more likely than not that they will be realized. We have elected to account for the tax effects of the global intangible low tax income provision as a current period expense.

We recognize the financial statement effects of a tax position when it is more likely than not that, based on technical merits, the position will be sustained upon examination. The tax benefits of the position recognized in the financial statements are then measured based on the largest amount of benefit that is greater than 50% likely to be realized upon settlement with a taxing authority. In addition, we recognize interest and penalties related to unrecognized tax benefits as a component of the income tax provision.

Business Combinations

We include the results of operations of the businesses that we acquire as of the acquisition date. We allocate the purchase price of the acquisitions to the assets acquired and liabilities assumed based on their estimated fair values, except for revenue contracts acquired, which are recognized in accordance with our revenue recognition policy. The excess of the purchase price over the fair values of identifiable assets and liabilities is recorded as goodwill. Acquisition-related expenses are recognized separately from the business combination and are expensed as incurred.

Foreign Currency

We translate the financial statements of our international subsidiaries to U.S. dollars using month-end exchange rates for assets and liabilities, and average rates for the period derived from month-end exchange rates for revenues, costs, and expenses. We record translation gains and losses in AOCI as a component of stockholders’ equity. We reflect net foreign exchange transaction gains and losses resulting from the conversion of the transaction currency to functional currency as a component of foreign currency exchange gain (loss) in OI&E.

Recently Issued Accounting Pronouncements Not Yet Adopted

In December 2023, the Financial Accounting Standards Board (FASB) issued Accounting Standards Update (ASU) 2023-09 "Income Taxes (Topics 740): Improvements to Income Tax Disclosures" to expand the disclosure requirements for income taxes. Upon adoption we will be required to disclose additional specified categories in the rate reconciliation in both percentage and dollar amounts. We will also be required to disclose the amount of income taxes paid disaggregated by jurisdiction, among other disclosure requirements. The standard can be applied either prospectively or retrospectively. We will adopt the standard in our 2025 annual period and are currently assessing the effect that the updated standard will have on our financial statement disclosures.

In November 2024, the FASB issued ASU 2024-03 "Income Statement: Reporting Comprehensive Income-Expense Disaggregation Disclosures (Subtopic 220-40)" to improve the disclosures about an entity’s expenses. Upon adoption, we will be required to disclose in the notes to the financial statements a disaggregation of certain expense categories included within the expense captions on the face of the income statement. The standard is effective for our 2027 annual period, and our interim periods beginning in 2028, with early adoption permitted. The standard can be applied either prospectively or retrospectively. We are currently assessing adoption timing and the effect that the updated standard will have on our financial statement disclosures.

Recently Adopted Accounting Pronouncements

In November 2023, the FASB issued ASU No. 2023-07 “Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures” which expands annual and interim disclosure requirements for reportable segments, primarily through enhanced disclosures about significant segment expenses. We adopted this ASU for our 2024 annual period with the comparative periods updated to reflect additional disclosures. See Note 15 for further details.

Prior Period Reclassifications

Certain amounts in prior periods have been reclassified to conform with current period presentation.

Note 2. Revenues

Disaggregated Revenues

The following table presents revenues disaggregated by type (in millions):

	Year Ended December 31,		
	2022	2023	2024
Google Search & other	$ 162,450	$ 175,033	$ 198,084
YouTube ads	29,243	31,510	36,147
Google Network	32,780	31,312	30,359
Google advertising	224,473	237,855	264,590
Google subscriptions, platforms, and devices	29,055	34,688	40,340
Google Services total	253,528	272,543	304,930
Google Cloud	26,280	33,088	43,229
Other Bets	1,068	1,527	1,648
Hedging gains (losses)	1,960	236	211
Total revenues	$ 282,836	$ 307,394	$ 350,018

No individual customer or groups of affiliated customers represented more than 10% of our revenues in 2022, 2023, or 2024.

The following table presents revenues disaggregated by geography, based on the addresses of our customers (in millions):

	Year Ended December 31,					
	2022		2023		2024	
United States	$ 134,814	48 %	$ 146,286	47 %	$ 170,447	49 %
EMEA[(1)]	82,062	29	91,038	30	102,127	29
APAC[(1)]	47,024	16	51,514	17	56,815	16
Other Americas[(1)]	16,976	6	18,320	6	20,418	6
Hedging gains (losses)	1,960	1	236	0	211	0
Total revenues	$ 282,836	100 %	$ 307,394	100 %	$ 350,018	100 %

(1) Regions represent Europe, the Middle East, and Africa (EMEA); Asia-Pacific (APAC); and Canada and Latin America ("Other Americas").

Revenue Backlog

As of December 31, 2024, we had $93.2 billion of remaining performance obligations (“revenue backlog”), primarily related to Google Cloud. Revenue backlog represents commitments in customer contracts for future services that have not yet been recognized as revenue. We expect to recognize approximately half of the revenue backlog as revenues over the next 24 months with the remainder to be recognized thereafter. The estimated revenue backlog and timing of revenue recognition for these commitments is largely driven by our ability to deliver in accordance with relevant contract terms and when our customers utilize services. Revenue backlog includes related deferred revenue currently recorded as well as amounts that will be invoiced in future periods, and excludes contracts with an original expected term of one year or less and cancellable contracts.

Deferred Revenues

We record deferred revenues when cash payments are received or due in advance of our performance, including amounts which are refundable. Deferred revenues primarily relate to Google Cloud and Google subscriptions, platforms, and devices. Total deferred revenue as of December 31, 2023 was $5.0 billion, of which $3.9 billion was recognized as revenues for the year ended December 31, 2024. Total deferred revenue as of December 31, 2024 was $6.0 billion.

Note 3. Financial Instruments

Fair Value Measurements

Investments Measured at Fair Value on a Recurring Basis

Cash, cash equivalents, and marketable equity securities are measured at fair value and classified within Level 1 and Level 2 in the fair value hierarchy, because we use quoted prices for identical assets in active markets or inputs that are based upon quoted prices for similar instruments in active markets.

Debt securities are measured at fair value and classified within Level 2 in the fair value hierarchy, because we use quoted market prices to the extent available or alternative pricing sources and models utilizing market observable inputs to determine fair value. For certain marketable debt securities, we have elected the fair value option for which changes in fair value are recorded in OI&E. The fair value option was elected for these securities to align with the unrealized gains and losses from related derivative contracts.

The following tables summarize our cash, cash equivalents, and marketable securities measured at fair value on a recurring basis (in millions):

		As of December 31, 2023					
	Fair Value Hierarchy	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Marketable Securities
Fair value changes recorded in other comprehensive income							
Time deposits	Level 2	$ 2,628	$ 0	$ 0	$ 2,628	$ 2,628	$ 0
Government bonds	Level 2	38,106	233	(679)	37,660	1,993	35,667
Corporate debt securities	Level 2	22,457	112	(637)	21,932	0	21,932
Mortgage-backed and asset-backed securities	Level 2	17,243	88	(634)	16,697	0	16,697
Total investments with fair value change reflected in other comprehensive income[1]		80,434	433	(1,950)	78,917	4,621	74,296
Fair value adjustments recorded in net income							
Money market funds	Level 1				6,480	6,480	0
Current marketable equity securities[2]	Level 1				4,282	0	4,282
Mutual funds	Level 2				311	0	311
Government bonds	Level 2				1,952	347	1,605
Corporate debt securities	Level 2				3,782	91	3,691
Mortgage-backed and asset-backed securities	Level 2				2,683	0	2,683
Total investments with fair value change recorded in net income					19,490	6,918	12,572
Cash					0	12,509	0
Total		$ 80,434	$ 433	$ (1,950)	$ 98,407	$ 24,048	$ 86,868

(1) Represents gross unrealized gains and losses for debt securities recorded to AOCI.

(2) The long-term portion of marketable equity securities (subject to long-term lock-up restrictions) of $1.4 billion as of December 31, 2023 is included within other non-current assets.

	Fair Value Hierarchy	As of December 31, 2024					
		Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Marketable Securities
Fair value changes recorded in other comprehensive income							
Time deposits	Level 2	$ 2,217	$ 0	$ 0	$ 2,217	$ 2,081	$ 136
Government bonds	Level 2	27,551	83	(214)	27,420	50	27,370
Corporate debt securities	Level 2	18,300	79	(222)	18,157	0	18,157
Mortgage-backed and asset-backed securities	Level 2	14,437	63	(385)	14,115	0	14,115
Total investments with fair value change reflected in other comprehensive income(1)		62,505	225	(821)	61,909	2,131	59,778
Fair value adjustments recorded in net income							
Money market funds	Level 1				$ 8,154	$ 8,154	$ 0
Current marketable equity securities(2)	Level 1				4,708	0	4,708
Mutual funds	Level 2				105	0	105
Government bonds	Level 2				2,035	696	1,339
Corporate debt securities	Level 2				3,037	78	2,959
Mortgage-backed and asset-backed securities	Level 2				3,302	0	3,302
Total investments with fair value change recorded in net income					21,341	8,928	12,413
Cash					0	12,407	0
Total		$ 62,505	$ 225	$ (821)	$ 83,250	$ 23,466	$ 72,191

(1) Represents gross unrealized gains and losses for debt securities recorded to AOCI.

(2) The long-term portion of marketable equity securities (subject to long-term lock-up restrictions) of $266 million as of December 31, 2024 is included within other non-current assets.

Investments Measured at Fair Value on a Nonrecurring Basis

Our non-marketable equity securities are investments in privately held companies without readily determinable market values. The carrying value of our non-marketable equity securities is adjusted to fair value upon observable transactions for identical or similar investments of the same issuer or impairment. Non-marketable equity securities that have been remeasured during the period based on observable transactions are classified within Level 2 or Level 3 in the fair value hierarchy. Non-marketable equity securities that have been remeasured due to impairment are classified within Level 3. Our valuation methods include option pricing models, market comparable approach, and common stock equivalent method, which may include a combination of the observable transaction price at the transaction date and other unobservable inputs including volatility, expected time to exit, risk free rate, and the rights and obligations of the securities we hold. These inputs vary significantly based on investment type.

As of December 31, 2024, the carrying value of our non-marketable equity securities was $35.5 billion, of which $19.9 billion were remeasured at fair value during the year ended December 31, 2024, and were primarily classified within Level 2 of the fair value hierarchy at the time of measurement.

Debt Securities

The following table summarizes the estimated fair value of investments in available-for-sale marketable debt securities by effective contractual maturity dates (in millions):

	As of December 31, 2024
Due in 1 year or less	$ 6,341
Due in 1 year through 5 years	37,221
Due in 5 years through 10 years	10,920
Due after 10 years	12,896
Total	$ 67,378

The following tables present fair values and gross unrealized losses recorded to AOCI, aggregated by investment category and the length of time that individual securities have been in a continuous loss position (in millions):

	As of December 31, 2023					
	Less than 12 Months		12 Months or Greater		Total	
	Fair Value	Unrealized Loss	Fair Value	Unrealized Loss	Fair Value	Unrealized Loss
Government bonds	$ 1,456	$ (22)	$ 13,897	$ (657)	$ 15,353	$ (679)
Corporate debt securities	827	(5)	15,367	(592)	16,194	(597)
Mortgage-backed and asset-backed securities	2,945	(26)	7,916	(608)	10,861	(634)
Total	$ 5,228	$ (53)	$ 37,180	$ (1,857)	$ 42,408	$ (1,910)

	As of December 31, 2024					
	Less than 12 Months		12 Months or Greater		Total	
	Fair Value	Unrealized Loss	Fair Value	Unrealized Loss	Fair Value	Unrealized Loss
Government bonds	$ 11,119	$ (126)	$ 2,576	$ (88)	$ 13,695	$ (214)
Corporate debt securities	4,228	(17)	6,838	(168)	11,066	(185)
Mortgage-backed and asset-backed securities	5,222	(106)	3,813	(279)	9,035	(385)
Total	$ 20,569	$ (249)	$ 13,227	$ (535)	$ 33,796	$ (784)

We determine realized gains or losses on the sale or extinguishment of debt securities on a specific identification method. The following table summarizes gains and losses for debt securities, reflected as a component of OI&E (in millions):

	Year Ended December 31,		
	2022	2023	2024
Unrealized gain (loss) on fair value option debt securities	$ (557)	$ 386	$ 30
Gross realized gain on debt securities	103	182	482
Gross realized loss on debt securities	(1,588)	(1,833)	(1,553)
(Increase) decrease in allowance for credit losses	(22)	50	(2)
Total gain (loss) on debt securities recognized in other income (expense), net	$ (2,064)	$ (1,215)	$ (1,043)

Equity Investments

The carrying value of equity securities is measured as the total initial cost plus the cumulative net gain (loss). Gains and losses, including impairments, are included as a component of OI&E in the Consolidated Statements of Income. See Note 7 for further details on OI&E. Certain of our non-marketable equity securities include our investments in VIE where we are not the primary beneficiary. See Note 5 for further details on VIE.

The carrying values for marketable and non-marketable equity securities are summarized below (in millions):

	As of December 31, 2023			As of December 31, 2024		
	Marketable Equity Securities	Non-Marketable Equity Securities	Total	Marketable Equity Securities	Non-Marketable Equity Securities	Total
Total initial cost	$ 5,418	$ 17,616	$ 23,034	$ 4,767	$ 21,240	$ 26,007
Cumulative net gain (loss)[(1)]	555	11,150	11,705	312	14,291	14,603
Carrying value	$ 5,973	$ 28,766	$ 34,739	$ 5,079	$ 35,531	$ 40,610

(1) Non-marketable equity securities cumulative net gain (loss) is comprised of $18.1 billion and $22.7 billion of gains and $6.9 billion and $8.4 billion of losses (including impairments) as of December 31, 2023 and 2024, respectively.

Gains and Losses on Marketable and Non-marketable Equity Securities

Gains and losses (including impairments), net, for marketable and non-marketable equity securities included in OI&E are summarized below (in millions):

	Year Ended December 31,		
	2022	2023	2024
Realized net gain (loss) on equity securities sold during the period	$ (442)	$ 690	$ 186
Unrealized net gain (loss) on marketable equity securities	(3,242)	790	156
Unrealized net gain (loss) on non-marketable equity securities[(1)]	229	(1,088)	3,372
Total gain (loss) on equity securities in other income (expense), net	$ (3,455)	$ 392	$ 3,714

(1) Unrealized gain (loss) on non-marketable equity securities accounted for under the measurement alternative is comprised of $3.3 billion, $1.8 billion, and $5.6 billion of upward adjustments and $3.0 billion, $2.9 billion, and $2.2 billion of downward adjustments (including impairments) for the years ended December 31, 2022, 2023, and 2024, respectively.

In the table above, realized net gain (loss) on equity securities sold during the period reflects the difference between the sale proceeds and the carrying value of the equity securities at the beginning of the period or the purchase date, if later.

Cumulative net gains (losses) on equity securities sold during the period, which is summarized in the following table (in millions), represents the total net gains (losses) recognized after the initial purchase date of the equity security sold during the period. While these net gains (losses) may have been reflected in periods prior to the period of sale, we believe they are important supplemental information as they reflect the economic net gains (losses) on the securities sold during the period. Cumulative net gains (losses) are calculated as the difference between the sale price and the initial purchase price for the equity security sold during the period.

	2023	2024
Total sale price	$ 1,981	$ 2,827
Total initial cost	1,512	2,079
Cumulative net gains (losses)	$ 469	$ 748

Equity Securities Accounted for Under the Equity Method

As of December 31, 2023 and 2024, equity securities accounted for under the equity method had a carrying value of approximately $1.7 billion and $2.0 billion, respectively. Our share of gains and losses, including impairments, are included as a component of OI&E, in the Consolidated Statements of Income. See Note 7 for further details on OI&E. Certain of our equity method securities include our investments in VIEs where we are not the primary beneficiary. See Note 4 for further details on VIEs.

Convertible Notes

As of December 31, 2023 and December 31, 2024, we had investments in convertible notes of $921 million and $2.9 billion, respectively, the majority of which are convertible notes held for investment. Our convertible notes held for investment are recorded at amortized cost which includes unpaid principal balances, deferred origination costs, and any related discount or premium, net of allowances for credit losses, and are included within other non-current assets on our Consolidated Balance Sheets.

Derivative Financial Instruments

We use derivative instruments to manage risks relating to our ongoing business operations. The primary risk managed is foreign exchange risk. We use foreign currency contracts to reduce the risk that our cash flows, earnings, and investment in foreign subsidiaries will be adversely affected by foreign currency exchange rate fluctuations. We also enter into derivative instruments to partially offset our exposure to other risks and enhance investment returns.

We recognize derivative instruments in the Consolidated Balance Sheets at fair value and classify the derivatives primarily within Level 2 in the fair value hierarchy. We present our collar contracts (an option strategy comprised of a combination of purchased and written options) at net fair values and present all other derivatives at gross fair values. The accounting treatment for derivatives is based on the intended use and hedge designation.

Cash Flow Hedges

We designate foreign currency forward and option contracts (including collars) as cash flow hedges to hedge certain forecasted revenue transactions denominated in currencies other than the U.S. dollar. These contracts have maturities of 24 months or less.

Cash flow hedge amounts included in the assessment of hedge effectiveness are deferred in AOCI and subsequently reclassified to revenue when the hedged item is recognized in earnings. We exclude forward points and time value from our assessment of hedge effectiveness and amortize them on a straight-line basis over the life of the hedging instrument in revenues. The difference between fair value changes of the excluded component and the amount amortized to revenues is recorded in AOCI.

As of December 31, 2024, the net accumulated gain on our foreign currency cash flow hedges before tax effect was $731 million, which is expected to be reclassified from AOCI into revenues within the next 12 months.

Fair Value Hedges

We designate foreign currency forward contracts as fair value hedges to hedge foreign currency risks for our marketable securities denominated in currencies other than the U.S. dollar. Fair value hedge amounts included in the assessment of hedge effectiveness are recognized in OI&E, along with the offsetting gains and losses of the related hedged items. We exclude forward points from the assessment of hedge effectiveness and recognize changes in the excluded component in OI&E.

Net Investment Hedges

We designate foreign currency forward contracts as net investment hedges to hedge the foreign currency risks related to our investment in foreign subsidiaries. Net investment hedge amounts included in the assessment of hedge effectiveness are recognized in AOCI along with the foreign currency translation adjustment. We exclude forward points from the assessment of hedge effectiveness and recognize changes in the excluded component in OI&E.

Other Derivatives

We enter into foreign currency forward and option contracts that are not designated as hedging instruments to hedge intercompany transactions and other monetary assets or liabilities denominated in currencies other than the functional currency of a subsidiary. Gains and losses on these derivatives that are not designated as accounting hedges are primarily recorded in OI&E along with the foreign currency gains and losses on monetary assets and liabilities.

We also use derivatives not designated as hedging instruments to manage risks relating to interest rates, commodity prices, and credit exposures, and to enhance investment returns. From time to time, we enter into derivatives to hedge the market price risk on certain of our marketable equity securities. Gains and losses arising from other derivatives are primarily reflected within the “other” component of OI&E. See Note 7 for further details.

The gross notional amounts of outstanding derivative instruments were as follows (in millions):

	As of December 31,	
	2023	2024
Derivatives designated as hedging instruments:		
Foreign exchange contracts		
Cash flow hedges	$ 18,039	$ 20,315
Fair value hedges	$ 2,065	$ 1,562
Net investment hedges	$ 9,472	$ 6,986
Derivatives not designated as hedging instruments:		
Foreign exchange contracts	$ 39,722	$ 44,227
Other contracts	$ 10,818	$ 15,082

The fair values of outstanding derivative instruments were as follows (in millions):

	As of December 31, 2023		As of December 31, 2024	
	Assets(1)	Liabilities(2)	Assets(1)	Liabilities(2)
Derivatives designated as hedging instruments:				
Foreign exchange contracts	$ 205	$ 242	$ 1,054	$ 0
Derivatives not designated as hedging instruments:				
Foreign exchange contracts	134	156	200	593
Other contracts	114	47	474	19
Total derivatives not designated as hedging instruments	248	203	674	612
Total	$ 453	$ 445	$ 1,728	$ 612

(1) Derivative assets are recorded as other current and non-current assets in the Consolidated Balance Sheets.

(2) Derivative liabilities are recorded as accrued expenses and other liabilities, current and non-current in the Consolidated Balance Sheets.

The gains (losses) on derivatives in cash flow hedging and net investment hedging relationships recognized in other comprehensive income (OCI) are summarized below (in millions):

	Year Ended December 31,		
	2022	2023	2024
Derivatives in cash flow hedging relationship:			
Foreign exchange contracts			
Amount included in the assessment of effectiveness	$ 1,699	$ 90	$ 857
Amount excluded from the assessment of effectiveness	(188)	84	77
Derivatives in net investment hedging relationship:			
Foreign exchange contracts			
Amount included in the assessment of effectiveness	608	(287)	223
Total	$ 2,119	$ (113)	$ 1,157

The table below presents the gains (losses) of our derivatives included in the Consolidated Statements of Income: (in millions):

	Year Ended December 31,					
	2022		2023		2024	
	Revenues	Other income (expense), net	Revenues	Other income (expense), net	Revenues	Other income (expense), net
Total amounts included in the Consolidated Statements of Income	$ 282,836	$ (3,514)	$ 307,394	$ 1,424	$ 350,018	$ 7,425
Effect of cash flow hedges:						
Foreign exchange contracts						
Amount reclassified from AOCI to income	$ 2,046	$ 0	$ 213	$ 0	$ 174	$ 0
Amount excluded from the assessment of effectiveness (amortized)	(85)	0	24	0	37	0
Effect of fair value hedges:						
Foreign exchange contracts						
Hedged items	0	(162)	0	59	0	(59)
Derivatives designated as hedging instruments	0	163	0	(59)	0	58
Amount excluded from the assessment of effectiveness	0	16	0	15	0	13
Effect of net investment hedges:						
Foreign exchange contracts						
Amount excluded from the assessment of effectiveness	0	171	0	187	0	137
Effect of non designated hedges:						
Foreign exchange contracts	0	(395)	0	7	0	335
Other contracts	0	144	0	53	0	174
Total gains (losses)	$ 1,961	$ (63)	$ 237	$ 262	$ 211	$ 658

Offsetting of Derivatives

We enter into master netting arrangements and collateral security arrangements to reduce credit risk. Cash collateral received related to derivative instruments under our collateral security arrangements are included in other current assets with a corresponding liability. Cash and non-cash collateral pledged related to derivative instruments under our collateral security arrangements are included in other current assets.

The gross amounts of derivative instruments subject to master netting arrangements with various counterparties, and cash and non-cash collateral received and pledged under such agreements were as follows (in millions):

	As of December 31, 2023					
				Gross Amounts Not Offset in the Consolidated Balance Sheets, but Have Legal Rights to Offset		
	Gross Amounts Recognized	Gross Amounts Offset in the Consolidated Balance Sheets	Net Amounts Presented in the Consolidated Balance Sheets	Financial Instruments[1]	Cash and Non-Cash Collateral Received or Pledged	Net Amounts
Derivatives assets	$ 535	$ (82)	$ 453	$ (213)	$ (75)	$ 165
Derivatives liabilities	$ 527	$ (82)	$ 445	$ (213)	$ (16)	$ 216

	As of December 31, 2024					
				Gross Amounts Not Offset in the Consolidated Balance Sheets, but Have Legal Rights to Offset		
	Gross Amounts Recognized	Gross Amounts Offset in the Consolidated Balance Sheets	Net Amounts Presented in the Consolidated Balance Sheets	Financial Instruments[1]	Cash and Non-Cash Collateral Received or Pledged	Net Amounts
Derivatives assets	$ 1,776	$ (48)	$ 1,728	$ (516)	$ (721)	$ 491
Derivatives liabilities	$ 660	$ (48)	$ 612	$ (516)	$ (9)	$ 87

(1) The balances as of December 31, 2023 and 2024 were related to derivatives allowed to be net settled in accordance with our master netting agreements.

Note 4. Leases

We have entered into operating and finance lease agreements primarily for data centers, land, and offices throughout the world with varying lease terms.

Components of lease costs were as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
Operating lease cost	$ 2,900	$ 3,362	$ 3,304
Finance lease cost:			
Amortization of lease assets	474	469	413
Interest on lease liabilities	34	35	31
Finance lease cost	508	504	444
Variable lease cost	838	1,182	1,425
Total lease cost	$ 4,246	$ 5,048	$ 5,173

Supplemental balance sheet information related to leases was as follows (in millions):

	December 31,	
	2023	2024
Weighted average remaining lease term		
Operating leases	8.1 years	7.8 years
Finance leases	10.7 years	10.4 years
Weighted average discount rate		
Operating leases	3.1 %	3.4 %
Finance leases	2.0 %	2.8 %

	December 31, 2023	December 31, 2024
Operating leases:		
Operating lease assets	$ 14,091	$ 13,588
Accrued expenses and other liabilities	$ 2,791	$ 2,887
Operating lease liabilities	12,460	11,691
Total operating lease liabilities	$ 15,251	$ 14,578
Finance Leases:		
Property and equipment, at cost	$ 4,403	$ 4,622
Accumulated depreciation	(1,652)	(2,037)
Property and equipment, net	$ 2,751	$ 2,585
Accrued expenses and other liabilities	$ 283	$ 235
Other long-term liabilities	1,383	1,442
Total finance lease liabilities	$ 1,666	$ 1,677

Supplemental cash flow information related to leases was as follows (in millions):

	Year Ended December 31, 2022	2023	2024
Cash payments for lease liabilities:			
Operating cash flows used for operating leases	$ 2,722	$ 3,173	$ 3,425
Operating cash flows used for finance leases	$ 34	$ 35	$ 31
Financing cash flows used for finance leases[1]	$ 586	$ 705	$ 405
Assets obtained in exchange for lease liabilities:			
Operating leases	$ 4,383	$ 2,877	$ 2,510
Finance leases	$ 577	$ 564	$ 313

(1) Financing cash flows used for financing leases are included within financing activities of the Consolidated Statements of Cash Flows as repayments of debt.

Future lease payments as of December 31, 2024 were as follows (in millions):

	Operating Leases	Finance Leases
2025	$ 3,162	$ 257
2026	2,824	208
2027	2,311	208
2028	1,838	197
2029	1,448	166
Thereafter	5,455	852
Total future lease payments	17,038	1,888
Less imputed interest	(2,460)	(211)
Total lease liability balance	$ 14,578	$ 1,677

As of December 31, 2024, we have entered into leases that have not yet commenced with short-term and long-term future lease payments of $773 million and $6.5 billion, respectively, that are not yet recorded on our Consolidated Balance Sheets. These leases will commence between 2025 and 2028 with non-cancelable lease terms between one and 25 years.

Note 5. Variable Interest Entities

Consolidated VIEs

We consolidate VIEs in which we hold a variable interest and are the primary beneficiary. The results of operations and financial position of these VIEs are included in our consolidated financial statements.

For certain consolidated VIEs, their assets are not available to us, and their creditors do not have recourse to us. As of December 31, 2023 and 2024, assets that can only be used to settle obligations of these VIEs were $4.9 billion and $8.7 billion, respectively and are primarily included in cash and cash equivalents on our Consolidated Balance Sheets. As of December 31, 2023 and 2024, liabilities for which creditors only have recourse to the VIEs were $2.5 billion and $2.3 billion, respectively. We may continue to fund ongoing operations of certain VIEs that are included within Other Bets.

Waymo, a fully autonomous driving technology company and a consolidated VIE, received $5.6 billion in funding during the year ended December 31, 2024, the majority of which was funded by Alphabet. Investments from external parties were accounted for as equity transactions and resulted in recognition of noncontrolling interests.

As of December 31, 2023 and 2024, total noncontrolling interests (NCI) in our consolidated subsidiaries were $3.4 billion and $4.2 billion, respectively, of which $1.1 billion was redeemable noncontrolling interests (RNCI) for both periods. NCI and RNCI are included within additional paid-in capital. Net loss attributable to noncontrolling interests was not material for any period presented and is included within the "other" component of OI&E. See Note 7 for further details on OI&E.

Unconsolidated VIEs

We have investments in VIEs in which we are not the primary beneficiary. These VIEs include private companies that are primarily early stage companies and certain renewable energy entities in which activities involve power generation using renewable sources.

We have determined that the governance structures of these entities do not allow us to direct the activities that would significantly affect their economic performance. Therefore, we are not the primary beneficiary, and the results of operations and financial position of these VIEs are not included in our consolidated financial statements. We account for these investments primarily as non-marketable equity securities or equity method investments, which are included within non-marketable securities on our Consolidated Balance Sheets. The maximum exposure of these unconsolidated VIEs is generally based on the current carrying value of the investments and any future funding commitments. As of December 31, 2023 and 2024, our future funding commitments related to unconsolidated VIE investments were $1.7 billion and $1.5 billion, respectively.

Note 6. Debt

Short-Term Debt

We have a debt financing program of up to $10.0 billion through the issuance of commercial paper. Net proceeds from this program are used for general corporate purposes. We had no commercial paper outstanding as of December 31, 2023 and $2.3 billion of commercial paper outstanding with a weighted-average effective interest rate of 4.4% as of December 31, 2024.The estimated fair value of the commercial paper approximated its carrying value as of December 31, 2024

Our short-term debt balance also includes the current portion of certain long-term debt.

Long-Term Debt

Total outstanding long-term debt is summarized below (in millions, except percentages):

	Maturity	Coupon Rate	Effective Interest Rate	As of December 31, 2023	As of December 31, 2024
Debt					
2014 Notes issuance	2024	3.38%	3.38%	$ 1,000	$ 0
2016 Notes issuance	2026	2.00%	2.23%	2,000	2,000
2020 Notes issuance	2025 - 2060	0.45% - 2.25%	0.57% - 2.33%	10,000	10,000
Total face value of long-term debt				13,000	12,000
Unamortized discount and debt issuance costs				(130)	(118)
Less: Current portion of long-term notes[(1)]				(1,000)	(999)
Total long-term debt				**$ 11,870**	$ 10,883

(1) Total current portion of long-term debt is included within accrued expenses and other current liabilities. See Note 7 for further details.

The notes in the table above are fixed-rate senior unsecured obligations and rank equally with each other. We may redeem the notes at any time in whole or in part at specified redemption prices. The effective interest rates are based on proceeds received with interest payable semi-annually.

The total estimated fair value of the outstanding notes was approximately $10.3 billion and $9.0 billion as of December 31, 2023 and December 31, 2024, respectively. The fair value was determined based on observable market prices of identical instruments in less active markets and is categorized accordingly as Level 2 in the fair value hierarchy.

As of December 31, 2024, the future principal payments for long-term debt were as follows (in millions):

2025	$ 1,000
2026	2,000
2027	1,000
2028	0
2029	0
Thereafter	8,000
Total	$ 12,000

Credit Facility

As of December 31, 2024, we had $10.0 billion of revolving credit facilities, of which $4.0 billion expires in April 2025 and $6.0 billion expires in April 2028. The interest rates for all credit facilities are determined based on a formula using certain market rates, as well as our progress toward the achievement of certain sustainability goals. No amounts were outstanding under the credit facilities as of December 31, 2023 and 2024.

Note 7. Supplemental Financial Statement Information

Accounts Receivable

The allowance for credit losses on accounts receivable was $771 million and $879 million as of December 31, 2023 and 2024, respectively.

Property and Equipment, Net

Property and equipment, net, consisted of the following (in millions):

	As of December 31,	
	2023	2024
Technical infrastructure	$ 112,504	$ 139,596
Office space	40,435	43,714
Corporate and other assets	13,728	16,519
Property and equipment, in service	166,667	199,829
Less: accumulated depreciation	(67,458)	(79,390)
Add: assets not yet in service	35,136	50,597
Property and equipment, net	$ 134,345	$ 171,036

Accrued Expenses and Other Current Liabilities

Accrued expenses and other current liabilities consisted of the following (in millions):

	As of December 31,	
	2023	2024
European Commission fines[(1)]	$ 9,525	$ 6,322
Accrued purchases of property and equipment[(2)]	4,679	7,104
Accrued customer liabilities	4,140	4,304
Current operating lease liabilities	2,791	2,887
Income taxes payable, net	2,748	2,905
Other accrued expenses and current liabilities	22,285	27,706
Accrued expenses and other current liabilities	$ 46,168	$ 51,228

(1) The amounts related to the EC fines, including any under appeal, are included in accrued expenses and other current liabilities on our Consolidated Balance Sheets. Amounts include the effects of foreign exchange and interest. In the third quarter of 2024 we made a cash payment of $3.0 billion for the 2017 EC shopping fine. See Note 10 for further details.

(2) Additional property and equipment purchases of $2.8 billion and $3.2 billion as of December 31, 2023 and 2024, respectively, were included in accounts payable.

Accumulated Other Comprehensive Income (Loss)

Components of AOCI, net of income tax, were as follows (in millions):

	Foreign Currency Translation Adjustments	Unrealized Gains (Losses) on Available-for-Sale Investments	Unrealized Gains (Losses) on Cash Flow Hedges	Total
Balance as of December 31, 2021	$ (2,306)	$ 236	$ 447	$ (1,623)
Other comprehensive income (loss) before reclassifications	(1,836)	(4,720)	1,463	(5,093)
Amounts excluded from the assessment of hedge effectiveness recorded in AOCI	0	0	(188)	(188)
Amounts reclassified from AOCI	0	1,007	(1,706)	(699)
Other comprehensive income (loss)	(1,836)	(3,713)	(431)	(5,980)
Balance as of December 31, 2022	(4,142)	(3,477)	16	(7,603)
Other comprehensive income (loss) before reclassifications	735	1,344	84	2,163
Amounts excluded from the assessment of hedge effectiveness recorded in AOCI	0	0	84	84
Amounts reclassified from AOCI	0	1,168	(214)	954
Other comprehensive income (loss)	735	2,512	(46)	3,201
Balance as of December 31, 2023	(3,407)	(965)	(30)	(4,402)
Other comprehensive income (loss) before reclassifications	(1,673)	(116)	698	(1,091)
Amounts excluded from the assessment of hedge effectiveness recorded in AOCI	0	0	77	77
Amounts reclassified from AOCI	0	782	(166)	616
Other comprehensive income (loss)	(1,673)	666	609	(398)
Balance as of December 31, 2024	$ (5,080)	$ (299)	$ 579	$ (4,800)

The effects on net income of amounts reclassified from AOCI were as follows (in millions):

AOCI Components	Location	Year Ended December 31, 2022	2023	2024
Unrealized gains (losses) on available-for-sale investments				
	Other income (expense), net	$ (1,291)	$ (1,497)	$ (1,008)
	Benefit (provision) for income taxes	284	329	226
	Net of income tax	(1,007)	(1,168)	(782)
Unrealized gains (losses) on cash flow hedges				
Foreign exchange contracts	Revenue	2,046	213	174
Interest rate contracts	Other income (expense), net	6	6	1
	Benefit (provision) for income taxes	(346)	(5)	(9)
	Net of income tax	1,706	214	166
Total amount reclassified, net of income tax		$ 699	$ (954)	$ (616)

Other Income (Expense), Net

Components of OI&E were as follows (in millions):

	Year Ended December 31, 2022	2023	2024
Interest income	$ 2,174	$ 3,865	$ 4,482
Interest expense[(1)]	(357)	(308)	(268)
Foreign currency exchange gain (loss), net	(654)	(1,238)	(409)
Gain (loss) on debt securities, net	(2,064)	(1,215)	(1,043)
Gain (loss) on equity securities, net	(3,455)	392	3,714
Performance fees	798	257	218
Income (loss) and impairment from equity method investments, net	(337)	(628)	(188)
Other	381	299	919
Other income (expense), net	$ (3,514)	$ 1,424	$ 7,425

(1) Interest expense is net of interest capitalized of $128 million, $181 million, and $194 million for the years ended December 31, 2022, 2023, and 2024, respectively.

Note 8. Business Combinations

character.ai

In accordance with the accounting requirements under Accounting Standards Codification Topic 805, for the year ended December 31, 2024, we recorded $2.7 billion of goodwill and $413 million of intangible assets resulting from a transaction with character.ai (“Character”). In August 2024, we entered into a license agreement with Character pursuant to which we obtained a non-exclusive license to its then current large language model technology. We paid Character $2.7 billion in cash and canceled our convertible instruments. We also hired certain employees of Character. Goodwill was recorded in Google Services and Google Cloud and is deductible for tax purposes.

Note 9. Goodwill

Changes in the carrying amount of goodwill for the years ended December 31, 2023 and 2024 were as follows (in millions):

	Google Services	Google Cloud	Other Bets	Total
Balance as of December 31, 2022	$ 20,847	$ 7,205	$ 908	$ 28,960
Additions	240	3	0	243
Foreign currency translation and other adjustments	31	(9)	(27)	(5)
Balance as of December 31, 2023	21,118	7,199	881	29,198
Additions	2,441	295	0	2,736
Foreign currency translation and other adjustments	(38)	(4)	(7)	(49)
Balance as of December 31, 2024	$ 23,521	$ 7,490	$ 874	$ 31,885

Note 10. Commitments and Contingencies

Commitments

We have content licensing agreements with future fixed or minimum guaranteed commitments of $8.8 billion as of December 31, 2024, of which the majority is paid quarterly through the first quarter of 2030.

Indemnifications

In the normal course of business, including to facilitate transactions in our services and products and corporate activities, we indemnify certain parties, including advertisers, Google Network partners, distribution partners, customers of Google Cloud offerings, lessors, and service providers with respect to certain matters. We have agreed to defend and/or hold certain parties harmless against losses arising from a breach of representations or covenants, or out of intellectual property infringement or other claims made against certain parties. Several of these agreements limit the time within which an indemnification claim can be made and the amount of the claim. In addition, we have entered into indemnification agreements with our officers and directors, and our bylaws contain similar indemnification obligations to our agents.

It is not possible to make a reasonable estimate of the maximum potential amount under these indemnification agreements due to the unique facts and circumstances involved in each particular agreement. Additionally, the payments we have made under such agreements have not had a material adverse effect on our results of operations, cash flows, or financial position. However, to the extent that valid indemnification claims arise in the future, future payments by us could be significant and could have a material adverse effect on our results of operations or cash flows in a particular period.

As of December 31, 2024, we did not have any material indemnification claims that were probable or reasonably possible.

Legal Matters

We record a liability when we believe that it is probable that a loss has been incurred, and the amount can be reasonably estimated. If we determine that a loss is reasonably possible and the loss or range of loss can be estimated, we disclose the reasonably possible loss. We evaluate developments in our legal matters that could affect the amount of liability that has been previously accrued, and the matters and related reasonably possible losses disclosed, and make adjustments as appropriate.

Certain outstanding matters seek speculative, substantial or indeterminate monetary amounts, substantial changes to our business practices and products, or structural remedies. Significant judgment is required to determine both the likelihood of there being a loss and the estimated amount of a loss related to such matters, and we may be unable to estimate the reasonably possible loss or range of losses. The outcomes of outstanding legal matters are inherently unpredictable and subject to significant uncertainties, and could, either individually or in aggregate, have a material adverse effect.

We expense legal fees in the period in which they are incurred.

Antitrust Matters

We are subject to formal and informal inquiries and investigations as well as litigation on various competition matters by regulatory authorities and private parties in the U.S., Europe, and other jurisdictions globally, including the following:

- *Shopping:* In June 2017, the EC announced its decision that certain actions taken by Google relating to its display and ranking of shopping search results and ads infringed European antitrust laws and imposed a €2.4 billion fine. We appealed the EC decision and implemented product changes to bring shopping ads into compliance with the EC's decision. In September 2024, the European Court of Justice rejected our appeal and upheld the €2.4 billion fine. In the third quarter of 2024, we made a cash payment of $3.0 billion for the fine.

- *Android:* In July 2018, the EC announced its decision that certain provisions in Google’s Android-related distribution agreements infringed European antitrust laws, imposed a €4.3 billion fine, and directed the termination of the conduct at issue. We appealed the EC decision and implemented changes to certain of our Android distribution practices. In September 2022, the General Court affirmed the EC decision but reduced the fine from €4.3 billion to €4.1 billion. We subsequently appealed the General Court's affirmation of the EC decision with the European Court of Justice, which remains pending. In 2018, we recognized a charge of $5.1 billion for the fine, which we reduced by $217 million in 2022.
- *AdSense for Search:* In March 2019, the EC announced its decision that certain provisions in Google's agreements with AdSense for Search partners infringed European antitrust laws, imposed a fine of €1.5 billion, and directed actions related to AdSense for Search partners' agreements, which we implemented prior to the decision. In 2019, we recognized a charge of $1.7 billion for the fine and appealed the EC decision. In September 2024, the General Court overturned the EC decision and annulled the €1.5 billion fine. The EC has appealed the General Court's decision with the European Court of Justice.
- *Search:* In October 2020, the DOJ and a number of state Attorneys General filed a lawsuit in the U.S. District Court for the District of Columbia alleging that Google violated U.S. antitrust laws relating to Search and Search advertising. In August 2024, the U.S. District Court for the District of Columbia ruled that Google violated such U.S. antitrust laws. A separate proceeding is being held to determine remedies, the range of which vary widely. The DOJ has proposed a high level remedy framework, which includes alterations to our products and services and our business models and operations, including structural remedies, and/or our distribution arrangements, among other changes, some of which could have a material adverse effect on our business. We have filed our own remedies proposal ahead of a hearing on remedies in April 2025. We expect a decision likely in the second half of 2025, after which we intend to appeal.

  Further, in June 2022, the Australian Competition and Consumer Commission and in October 2023, the Japanese Fair Trade Commission each opened an investigation into Search distribution practices.

  Given the nature of these matters, we cannot estimate a possible loss.

- *Advertising Technology:* In December 2020, a number of state Attorneys General filed a lawsuit in the U.S. District Court for the Eastern District of Texas alleging that Google violated U.S. antitrust laws as well as state deceptive trade laws relating to its advertising technology, and a trial is scheduled for March 2025. Additionally, in January 2023, the DOJ, along with a number of state Attorneys General, filed a lawsuit in the U.S. District Court for the Eastern District of Virginia alleging that Google violated U.S. antitrust laws relating to its advertising technology, and a number of additional state Attorneys General subsequently joined the lawsuit. The trial ended in September 2024, and we expect a decision in early 2025.

  Further, in June 2023, the EC issued a Statement of Objections informing Google of its preliminary view that Google violated European antitrust laws relating to its advertising technology, to which we responded.

  Given the nature of these matters, we cannot estimate a possible loss.

- *Google Play:* In July 2021, a number of state Attorneys General filed a lawsuit in the U.S. District Court for the Northern District of California alleging that Google’s operation of Android and Google Play violated U.S. antitrust laws and state antitrust and consumer protection laws. In September 2023, we reached a settlement in principle with 50 state Attorneys General and three territories and recognized a charge. Final approval of the settlement remains pending before the court. In May 2024, we funded the settlement amount to an escrow agent.

  In December 2023, a California jury delivered a verdict in *Epic Games v. Google* finding that Google violated U.S. antitrust laws related to Google Play's business. Epic did not seek monetary damages. The presiding judge issued a remedies decision in October 2024, ordering a variety of alterations to our business models and operations and contractual agreements for Android and Google Play. We are appealing the verdict and the trial court judge temporarily paused the implementation of the remedies while the Court of Appeals considers our request to pause implementation of the remedies pending the duration of the appeal. Given the nature of this matter, we cannot estimate a possible loss.

- *European Digital Markets Act:* In March 2024, the EC opened two investigations regarding Google's compliance with certain provisions of EU's Digital Markets Act relating to Google Play and Search. Given the preliminary stages of this matter, we cannot estimate a possible loss.

In addition to these proceedings, private individual and collective actions that overlap with claims pursued by regulatory authorities are pending in the U.S. and in several other jurisdictions.

We believe we have strong arguments against these open claims and will defend ourselves vigorously. We continue to cooperate with federal and state regulators in the U.S., the EC, and other regulators around the world.

Privacy Matters

We are subject to a number of privacy-related laws and regulations, and we currently are party to a number of privacy investigations and lawsuits ongoing in multiple jurisdictions. For example, there are ongoing investigations and litigation in the U.S. and the EU, including those relating to our collection and use of location information, alleged violations of state biometric statutes, the choices we offer users, and advertising practices, which could result in significant fines, judgments, and product changes.

Patent and Intellectual Property Claims

We have had patent, copyright, trade secret, and trademark infringement lawsuits filed against us claiming that certain of our products, services, and technologies infringe others' intellectual property rights. Adverse results in these lawsuits may include awards of substantial monetary damages, costly royalty or licensing agreements, or orders preventing us from offering certain features, functionalities, products, or services. As a result, we may have to change our business practices and develop non-infringing products or technologies, which could result in a loss of revenues for us and otherwise harm our business. In addition, the U.S. International Trade Commission (ITC) has increasingly become an important forum to litigate intellectual property disputes because an ultimate loss in an ITC action can result in a prohibition on importing infringing products into the U.S. Because the U.S. is an important market, a prohibition on importation could have an adverse effect on us, including preventing us from importing many important products into the U.S. or necessitating workarounds that may limit certain features of our products. Further, our customers and partners may discontinue the use of our products, services, and technologies, as a result of injunctions or otherwise, which could result in loss of revenues and adversely affect our business.

Other

We are subject to claims, lawsuits, regulatory and government inquiries and investigations, other proceedings, and consent orders involving competition, intellectual property, data privacy and security, tax and related compliance, labor and employment, commercial disputes, content generated by our users, goods and services offered by advertisers or publishers using our platforms, personal injury, consumer protection, and other matters. For example, we periodically have data incidents that we report to relevant regulators as required by law. Such claims, consent orders, lawsuits, regulatory and government investigations, and other proceedings could result in substantial fines and penalties, injunctive relief, ongoing monitoring and auditing obligations, changes to our products and services, alterations to our business models and operations, and collateral related civil litigation or other adverse consequences, all of which could harm our business, reputation, financial condition, and operating results.

We have ongoing legal matters relating to Russia. For example, some matters concern civil judgments that include compounding penalties imposed upon us in connection with disputes regarding the termination of accounts, including those of sanctioned parties. We do not expect these ongoing legal matters will have a material adverse effect.

Non-Income Taxes

We are under audit by various domestic and foreign tax authorities with regards to non-income tax matters. The subject matter of non-income tax audits primarily arises from disputes on the tax treatment and tax rate applied to the sale of our products and services in these jurisdictions and the tax treatment of certain employee benefits. We accrue non-income taxes that may result from examinations by, or any negotiated agreements with, these tax authorities when a loss is probable and reasonably estimable. If we determine that a loss is reasonably possible and the loss or range of loss can be estimated, we disclose the reasonably possible loss. Due to the inherent complexity and uncertainty of these matters and judicial process in certain jurisdictions, the final outcome may be materially different from our expectations.

See Note 14 for information regarding income tax contingencies.

Note 11. Stockholders' Equity

Class A and Class B Common Stock and Class C Capital Stock

Our Board of Directors has authorized three classes of stock, Class A and Class B common stock, and Class C capital stock. The rights of the holders of each class of our common and capital stock are identical, except with respect to voting. Each share of Class A common stock is entitled to one vote per share. Each share of Class B common stock is entitled to 10 votes per share. Class C capital stock has no voting rights, except as required by applicable law. Shares of Class B common stock may be converted at any time at the option of the stockholder and automatically convert upon sale or transfer to Class A common stock.

Share Repurchases

In the years ended December 31, 2022, 2023, and 2024, we continued to repurchase both Class A and Class C shares in a manner deemed in the best interest of the company and its stockholders, taking into account the economic cost and prevailing market conditions, including the relative trading prices and volumes of the Class A and Class C shares. During the years ended December 31, 2022, 2023, and 2024, we repurchased $59.3 billion, $62.2 billion, and $62.0 billion, respectively, of Alphabet's Class A and Class C shares.

In April 2024, the Board of Directors of Alphabet authorized the company to repurchase up to an additional $70.0 billion of its Class A and Class C shares. As of December 31, 2024, $44.7 billion remained available for Class A and Class C share repurchases.

The following table presents Class A and Class C shares repurchased and subsequently retired (in millions):

	Year Ended December 31, 2023		Year Ended December 31, 2024	
	Shares	Amount	Shares	Amount
Class A share repurchases	78	$ 9,316	73	$ 11,855
Class C share repurchases	450	52,868	306	50,192
Total share repurchases[1]	528	$ 62,184	379	$ 62,047

[1] Shares repurchased include unsettled repurchases.

Repurchases are executed from time to time, subject to general business and market conditions and other investment opportunities, through open market purchases or privately negotiated transactions, including through Rule 10b5-1 plans. The repurchase program does not have an expiration date.

Dividends

During the year ended December 31, 2024, total cash dividends, which were first paid in June 2024, were $3.5 billion, $519 million, and $3.3 billion for Class A, Class B, and Class C shares, respectively.

The company intends to pay quarterly cash dividends in the future, subject to review and approval by the company's Board of Directors in its sole discretion.

Note 12. Net Income Per Share

We compute net income per share of Class A, Class B, and Class C stock using the two-class method. Basic net income per share is computed using the weighted-average number of shares outstanding during the period. Diluted net income per share is computed using the weighted-average number of shares and the effect of potentially dilutive securities outstanding during the period. Potentially dilutive securities consist of RSUs and other contingently issuable shares. The dilutive effect of outstanding RSUs and other contingently issuable shares is reflected in diluted earnings per share by application of the treasury stock method. The computation of the diluted net income per share of Class A stock assumes the conversion of Class B stock, while the diluted net income per share of Class B stock does not assume the conversion of those shares.

In accordance with our certificate of incorporation, the rights, including the liquidation and dividend rights, of the holders of our Class A, Class B, and Class C stock are identical, except with respect to voting. Furthermore, there are a number of safeguards built into our certificate of incorporation, as well as Delaware law, which preclude our Board of Directors from declaring or paying unequal per share dividends on our Class A, Class B, and Class C stock. Specifically, Delaware law provides that amendments to our certificate of incorporation which would have the effect of adversely altering the rights, powers, or preferences of a given class of stock must be approved by the class of stock adversely affected by the proposed amendment. In addition, our certificate of incorporation provides that before any such amendment may be put to a stockholder vote, it must be approved by the unanimous consent of our Board of Directors.

Immaterial differences in net income per share across our Class A, Class B, and Class C shares may arise due to the allocation of distributed earnings, which is based on the holders as of the record date, compared with the allocation of undistributed earnings and number of shares, which is based on the weighted average shares outstanding over the periods.

The following tables set forth the computation of basic and diluted net income per share of Class A, Class B, and Class C stock (in millions, except per share amounts):

	Year Ended December 31, 2022			
	Class A	Class B	Class C	Consolidated
Basic net income per share:				
Numerator				
Allocation of distributed earnings (cash dividends paid)	$ 0	$ 0	$ 0	$ 0
Allocation of undistributed earnings	27,518	4,072	28,382	59,972
Net income	$ 27,518	$ 4,072	$ 28,382	$ 59,972
Denominator				
Number of shares used in per share computation	5,994	887	6,182	13,063
Basic net income per share	$ 4.59	$ 4.59	$ 4.59	$ 4.59
Diluted net income per share:				
Numerator				
Allocation of total earnings for basic computation	$ 27,518	$ 4,072	$ 28,382	$ 59,972
Reallocation of total earnings as a result of conversion of Class B to Class A shares	4,072	0	0	_[1]
Reallocation of undistributed earnings	(230)	(30)	230	_[1]
Net income	$ 31,360	$ 4,042	$ 28,612	$ 59,972
Denominator				
Number of shares used in basic computation	5,994	887	6,182	13,063
Weighted-average effect of dilutive securities				
Add:				
Conversion of Class B to Class A shares outstanding	887	0	0	_[1]
Restricted stock units and other contingently issuable shares	0	0	96	96
Number of shares used in per share computation	6,881	887	6,278	13,159
Diluted net income per share	$ 4.56	$ 4.56	$ 4.56	$ 4.56

[1] Not applicable for consolidated net income per share.

	Year Ended December 31, 2023			
	Class A	Class B	Class C	Consolidated
Basic net income per share:				
Numerator				
Allocation of distributed earnings (cash dividends paid)	$ 0	$ 0	$ 0	$ 0
Allocation of undistributed earnings	34,601	5,124	34,070	73,795
Net income	$ 34,601	$ 5,124	$ 34,070	$ 73,795
Denominator				
Number of shares used in per share computation	5,922	877	5,831	12,630
Basic net income per share	$ 5.84	$ 5.84	$ 5.84	$ 5.84
Diluted net income per share:				
Numerator				
Allocation of total earnings for basic computation	$ 34,601	$ 5,124	$ 34,070	$ 73,795
Reallocation of total earnings as a result of conversion of Class B to Class A shares	5,124	0	0	_[(1)]
Reallocation of undistributed earnings	(287)	(37)	287	_[(1)]
Net income	$ 39,438	$ 5,087	$ 34,357	$ 73,795
Denominator				
Number of shares used in basic computation	5,922	877	5,831	12,630
Weighted-average effect of dilutive securities				
Add:				
Conversion of Class B to Class A shares outstanding	877	0	0	_[(1)]
Restricted stock units and other contingently issuable shares	0	0	92	92
Number of shares used in per share computation	6,799	877	5,923	12,722
Diluted net income per share	$ 5.80	$ 5.80	$ 5.80	$ 5.80

(1) Not applicable for consolidated net income per share.

	Year Ended December 31, 2024			
	Class A	Class B	Class C	Consolidated
Basic net income per share:				
Numerator				
Allocation of distributed earnings (cash dividends paid)	$ 3,509	$ 519	$ 3,335	$ 7,363
Allocation of undistributed earnings	44,085	6,520	42,150	92,755
Net income	$ 47,594	$ 7,039	$ 45,485	$ 100,118
Denominator				
Number of shares used in per share computation	5,855	866	5,598	12,319
Basic net income per share	$ 8.13	$ 8.13	$ 8.13	$ 8.13
Diluted net income per share:				
Numerator				
Allocation of total earnings for basic computation	$ 47,594	$ 7,039	$ 45,485	$ 100,118
Reallocation of total earnings as a result of conversion of Class B to Class A shares	7,039	0	0	_[(1)]
Reallocation of undistributed earnings	(520)	(67)	520	_[(1)]
Net income	$ 54,113	$ 6,972	$ 46,005	$ 100,118
Denominator				
Number of shares used in basic computation	5,855	866	5,598	12,319
Weighted-average effect of dilutive securities				
Add:				
Conversion of Class B to Class A shares outstanding	866	0	0	_[(1)]
Restricted stock units and other contingently issuable shares	0	0	128	128
Number of shares used in per share computation	6,721	866	5,726	12,447
Diluted net income per share	$ 8.05	$ 8.05	$ 8.03	$ 8.04

(1) Not applicable for consolidated net income per share.

Note 13. Compensation Plans

Stock Plans

Our stock plans include the Alphabet Amended and Restated 2021 Stock Plan ("Alphabet 2021 Stock Plan") and Other Bets stock-based plans. Under our stock plans, RSUs and other types of awards may be granted. Under the Alphabet 2021 Stock Plan, an RSU award is an agreement to issue shares of our Class C stock at the time the award vests. RSUs generally vest over four years contingent upon employment on the vesting date. RSUs are awarded dividend equivalents, which are subject to the same vesting conditions as the underlying award, and settled in Class C shares.

As of December 31, 2024, there were 633 million shares of Class C stock reserved for future issuance under the Alphabet 2021 Stock Plan.

Stock-Based Compensation

For the years ended December 31, 2022, 2023, and 2024, total SBC expense was $19.5 billion, $22.1 billion, and $22.8 billion, including amounts associated with awards we expect to settle in Alphabet stock of $18.8 billion, $21.7 billion, and $22.0 billion, respectively.

For the years ended December 31, 2022, 2023, and 2024, we recognized tax benefits on total SBC expense, which are reflected in the provision for income taxes in the Consolidated Statements of Income, of $3.9 billion, $4.5 billion, and $4.6 billion, respectively.

For the years ended December 31, 2022, 2023, and 2024, tax benefit realized related to awards vested or exercised during the period was $4.7 billion, $5.6 billion, and $6.8 billion, respectively. These amounts do not include the indirect effects of stock-based awards, which primarily relate to the R&D tax credit.

Stock-Based Award Activities

The following table summarizes the activities for unvested Alphabet RSUs, which include dividend equivalents awarded to holders of unvested stock, for the year ended December 31, 2024 (in millions, except per share amounts):

	Number of Shares	Weighted-Average Grant-Date Fair Value
Unvested as of December 31, 2023	338	$ 104.93
Granted	195	$ 140.04
Vested	(199)	$ 110.89
Forfeited/canceled	(35)	$ 113.52
Unvested as of December 31, 2024	299	$ 122.77

The weighted-average grant-date fair value of RSUs granted during the years ended December 31, 2022 and 2023 was $127.22 and $97.59, respectively. Total fair value of RSUs, as of their respective vesting dates, during the years ended December 31, 2022, 2023, and 2024, were $23.9 billion, $26.6 billion, and $33.3 billion, respectively.

As of December 31, 2024, there was $34.8 billion of unrecognized compensation cost related to unvested RSUs. This amount is expected to be recognized over a weighted-average period of 2.5 years.

Note 14. Income Taxes

Income from continuing operations before income taxes consisted of the following (in millions):

	Year Ended December 31,		
	2022	2023	2024
Domestic operations	$ 61,307	$ 73,600	$ 108,076
Foreign operations	10,021	12,117	11,739
Total	$ 71,328	$ 85,717	$ 119,815

Provision for income taxes consisted of the following (in millions):

	Year Ended December 31,		
	2022	2023	2024
Current:			
Federal and state	$ 17,120	$ 17,125	$ 22,485
Foreign	2,434	2,526	2,468
Total	19,554	19,651	24,953
Deferred:			
Federal and state	(8,052)	(7,482)	(5,350)
Foreign	(146)	(247)	94
Total	(8,198)	(7,729)	(5,256)
Provision for income taxes	$ 11,356	$ 11,922	$ 19,697

The reconciliation of federal statutory income tax rate to our effective income tax rate was as follows:

	Year Ended December 31,		
	2022	2023	2024
U.S. federal statutory tax rate	21.0 %	21.0 %	21.0 %
Foreign income taxed at different rates	3.0	0.3	0.5
Foreign-derived intangible income deduction	(5.4)	(4.6)	(3.8)
Stock-based compensation expense	(1.2)	(0.8)	(1.5)
Federal research credit	(2.2)	(1.8)	(1.5)
Deferred tax asset valuation allowance	0.9	0.6	0.4
State and local income taxes	0.8	1.0	1.1
Effect of tax law change	0.0	(1.4)	0.0
Other	(1.0)	(0.4)	0.2
Effective tax rate	15.9 %	13.9 %	16.4 %

In 2023, the IRS issued a rule change allowing taxpayers to temporarily apply the regulations in effect prior to 2022 related to U.S. federal foreign tax credits as well as a separate rule change with guidance on the capitalization and amortization of R&D expenses. A cumulative one-time adjustment for these tax rule changes was recorded in 2023.

Deferred Income Taxes

Deferred income taxes reflect the net effects of temporary differences between the carrying amounts of assets and liabilities for financial reporting purposes and the amounts used for income tax purposes. Significant components of our deferred tax assets and liabilities were as follows (in millions):

	As of December 31,	
	2023	2024
Deferred tax assets:		
Accrued employee benefits	$ 1,855	$ 1,834
Accruals and reserves not currently deductible	2,481	2,552
Tax credits	6,609	6,384
Net operating losses	2,965	3,472
Operating leases	3,526	3,336
Capitalized research and development	17,757	25,903
Other	1,951	1,376
Total deferred tax assets	37,144	44,857
Valuation allowance	(10,999)	(11,493)
Total deferred tax assets net of valuation allowance	26,145	33,364
Deferred tax liabilities:		
Property and equipment, net	(8,189)	(9,932)
Net investment gains	(2,405)	(2,978)
Operating leases	(2,965)	(2,986)
Other	(902)	(1,008)
Total deferred tax liabilities	(14,461)	(16,904)
Net deferred tax assets (liabilities)	$ 11,684	$ 16,460

As of December 31, 2024, our federal, state, and foreign net operating loss carryforwards for income tax purposes were approximately $8.4 billion, $19.4 billion, and $2.5 billion respectively. If not utilized, the federal net operating loss carryforwards will begin to expire in 2025, foreign net operating loss carryforwards will begin to expire in 2025 and the state net operating loss carryforwards will begin to expire in 2033. It is more likely than not that the majority of the net operating loss carryforwards will not be realized. The net operating loss carryforwards are subject to various annual limitations under the tax laws of the different jurisdictions.

As of December 31, 2024, our Federal and California research and development credit carryforwards for income tax purposes were approximately $700 million and $6.4 billion, respectively. If not utilized, the Federal R&D credit will begin to expire in 2037 and the California R&D credit can be carried over indefinitely. We believe the majority of the federal tax credit and state tax credit is not likely to be realized.

As of December 31, 2024, our investment tax credit carryforwards for state income tax purposes were approximately $1.2 billion and will begin to expire in 2030. We use the flow-through method of accounting for investment tax credits. We believe this tax credit is not likely to be realized.

As of December 31, 2024, we maintained a valuation allowance with respect to California deferred tax assets, certain federal net operating losses, certain state net operating losses and tax credits, net deferred tax assets relating to Other Bet companies, and certain foreign net operating losses that we believe are not likely to be realized. We continue to reassess the remaining valuation allowance quarterly, and if future evidence allows for a partial or full release of the valuation allowance, a tax benefit will be recorded accordingly.

Cash paid for income taxes, net of refunds, was $18.9 billion, $19.2 billion, and $27.4 billion as of December 31, 2022, 2023, and 2024, respectively.

Uncertain Tax Positions

The following table summarizes the activity related to our gross unrecognized tax benefits (in millions):

	Year Ended December 31,		
	2022	2023	2024
Beginning gross unrecognized tax benefits	$ 5,158	$ 7,055	$ 9,438
Increases related to prior year tax positions	253	740	896
Decreases related to prior year tax positions	(437)	(682)	(83)
Decreases related to settlement with tax authorities	(140)	(21)	(311)
Increases related to current year tax positions	2,221	2,346	2,679
Ending gross unrecognized tax benefits	$ 7,055	$ 9,438	$ 12,619

We are subject to income taxes in the U.S. and foreign jurisdictions. Significant judgment is required in evaluating our uncertain tax positions and determining our provision for income taxes. The total amount of gross unrecognized tax benefits was $7.1 billion, $9.4 billion, and $12.6 billion as of December 31, 2022, 2023, and 2024, respectively, of which $5.3 billion, $7.4 billion, and $10.0 billion, if recognized, would affect our effective tax rate, respectively.

As of December 31, 2023 and 2024, we accrued $622 million and $1.1 billion in interest and penalties in provision for income taxes, respectively.

We file income tax returns in the U.S. federal jurisdiction and in many state and foreign jurisdictions. Our two major tax jurisdictions are the U.S. federal and Ireland. We are subject to the continuous examination of our income tax returns by the IRS and other tax authorities. The IRS is currently examining our 2016 through 2021 tax returns. We have also received tax assessments in multiple foreign jurisdictions asserting transfer pricing adjustments or permanent establishment. We continue to defend such claims as presented.

The tax years 2016 through 2023 remain subject to examination by the appropriate governmental agencies for Irish tax purposes. There are other ongoing audits in various other jurisdictions that are not material to our financial statements.

We regularly assess the likelihood of adverse outcomes resulting from these examinations to determine the adequacy of our provision for income taxes. We continue to monitor the progress of ongoing discussions with tax authorities and the effect, if any, of the expected expiration of the statute of limitations in various taxing jurisdictions.

We believe that an adequate provision has been made for any adjustments that may result from tax examinations. However, the outcome of tax audits cannot be predicted with certainty. If any issues addressed in our tax audits are resolved in a manner not consistent with management's expectations, we could be required to adjust our provision for income taxes in the period such resolutions occur. Although the timing of resolution, settlement, and closure of audits is not certain, we do not believe it is reasonably possible that our unrecognized tax benefits from certain U.S. federal, state, and non U.S. tax positions will materially change in the next 12 months.

Note 15. Information about Segments and Geographic Areas

We report our segment results as Google Services, Google Cloud, and Other Bets:

- Google Services includes products and services such as ads, Android, Chrome, devices, Google Maps, Google Play, Search, and YouTube. Google Services generates revenues primarily from advertising; fees received for consumer subscription-based products such as YouTube TV, YouTube Music and Premium, and NFL Sunday Ticket, as well as Google One; the sale of apps and in-app purchases; and devices.
- Google Cloud includes infrastructure and platform services, applications, and other services for enterprise customers. Google Cloud generates revenues primarily from consumption-based fees and subscriptions received for Google Cloud Platform services, Google Workspace communication and collaboration tools, and other enterprise services.
- Other Bets is a combination of multiple operating segments that are not individually material. Revenues from Other Bets are generated primarily from the sale of healthcare-related services and internet services.

Revenues, certain costs, such as costs associated with content and traffic acquisition, certain engineering activities, and devices, as well as certain operating expenses are directly attributable to our segments. Due to the integrated nature of Alphabet, other costs and expenses, such as technical infrastructure and office facilities, are managed centrally at a consolidated level. These costs, including the associated depreciation, are allocated to operating segments as a service cost generally based on usage, headcount, or revenue.

As announced in April 2024, we consolidated teams that focus on building general AI models across Google Research and Google DeepMind to further accelerate our progress in AI. General AI model development teams previously under Google Research in our Google Services segment are reported within Alphabet-level activities prospectively beginning in the second quarter of 2024. As further announced, in October 2024, the Gemini app team that is developing the direct consumer interface to our Gemini models joined Google DeepMind. The costs associated with the Gemini app team continue to be reported within our Google Services segment.

Certain costs are not allocated to our segments because they represent Alphabet-level activities. These costs primarily include certain AI-focused shared R&D activities, including development costs of our general AI models; corporate initiatives such as our philanthropic activities; corporate shared costs such as certain finance, human resource, and legal costs, including certain fines and settlements. Charges associated with employee severance and office space reductions during 2023 and 2024 were also not allocated to our segments. Additionally, hedging gains (losses) related to revenue are not allocated to our segments.

Our Chief Operating Decision Maker (CODM) is our Chief Executive Officer, Sundar Pichai. Our CODM uses segment operating income (loss) to allocate resources to our segments in our annual planning process and to assess the performance of our segments, primarily by monitoring actual results versus the annual plan. Our operating segments are not evaluated using asset information.

The following table presents revenue, profitability, and expense information about our segments (in millions):

	Year Ended December 31,		
	2022	2023	2024
Revenues:			
Google Services	$ 253,528	$ 272,543	$ 304,930
Google Cloud	26,280	33,088	43,229
Other Bets	1,068	1,527	1,648
Hedging gains (losses)	1,960	236	211
Total revenues	$ 282,836	$ 307,394	$ 350,018
Operating income (loss):			
Google Services	$ 82,699	$ 95,858	$ 121,263
Google Cloud	(1,922)	1,716	6,112
Other Bets	(4,636)	(4,095)	(4,444)
Alphabet-level activities	(1,299)	(9,186)	(10,541)
Total income from operations	$ 74,842	$ 84,293	$ 112,390
Supplemental information about our segment expenses:			
Google Services:			
Employee compensation expenses	$ 43,529	$ 46,224	$ 44,560
Other costs and expenses	127,300	130,461	139,107
Total Google Services costs and expenses	$ 170,829	$ 176,685	$ 183,667
Google Cloud:			
Employee compensation expenses	$ 16,132	$ 19,054	$ 20,519
Other costs and expenses	12,070	12,318	16,598
Total Google Cloud costs and expenses	$ 28,202	$ 31,372	$ 37,117

Google Services and Google Cloud employee compensation expenses include the costs associated with direct and allocated employees. Google Services and Google Cloud other costs and expenses primarily include direct costs, such as advertising and promotional activities and third party services fees as well as allocated costs, such as technical infrastructure and office facilities usage costs. Additionally, Google Services other costs and expenses include content and traffic acquisition costs and device costs.

See Note 2 for information relating to revenues by geography.

The following table presents long-lived assets by geographic area, which includes property and equipment, net and operating lease assets (in millions):

	As of December 31,	
	2023	2024
Long-lived assets:		
United States	$ 110,053	$ 138,993
International	38,383	45,631
Total long-lived assets	$ 148,436	$ 184,624

Note 16. Subsequent Event

In January 2025, we recognized an $8.0 billion unrealized gain on our non-marketable equity securities related to our investment in a private company. The unrealized gain reflects an increase in the fair value measurement of our investment following an observable transaction in January 2025. See Note 3 and Note 7 for further details on equity investments and OI&E.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our chief executive officer and chief financial officer, evaluated the effectiveness of our disclosure controls and procedures pursuant to Rule 13a-15 under the Exchange Act, as of the end of the period covered by this Annual Report on Form 10-K.

Based on this evaluation, our chief executive officer and chief financial officer concluded that, as of December 31, 2024, our disclosure controls and procedures are designed at a reasonable assurance level and are effective to provide reasonable assurance that information we are required to disclose in reports that we file or submit under the Exchange Act is recorded, processed, summarized, and reported within the time periods specified in the SEC's rules and forms, and that such information is accumulated and communicated to our management, including our chief executive officer and chief financial officer, as appropriate, to allow timely decisions regarding required disclosure.

Changes in Internal Control over Financial Reporting

There have been no changes in our internal control over financial reporting that occurred during the quarter ended December 31, 2024 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Management's Report on Internal Control over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting, as defined in Rule 13a-15(f) of the Exchange Act. Our management conducted an evaluation of the effectiveness of our internal control over financial reporting based on the framework in *Internal Control—Integrated Framework* issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework). Based on this evaluation, management concluded that our internal control over financial reporting was effective as of December 31, 2024. Management reviewed the results of its assessment with our Audit and Compliance Committee. The effectiveness of our internal control over financial reporting as of December 31, 2024 has been audited by Ernst & Young LLP, an independent registered public accounting firm, as stated in its report which is included in Item 8 of this Annual Report on Form 10-K.

Limitations on Effectiveness of Controls and Procedures

In designing and evaluating the disclosure controls and procedures, management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving the desired control objectives. In addition, the design of disclosure controls and procedures must reflect the fact that there are resource constraints and that management is required to apply its judgment in evaluating the benefits of possible controls and procedures relative to their costs.

ITEM 9B. OTHER INFORMATION

10b5-1 Trading Plans

During the quarter ended December 31, 2024, each of the following Section 16 director and officer adopted, modified, or terminated a "Rule 10b5-1 trading arrangement" (as defined in Item 408 of Regulation S-K of the Exchange Act):

- John Hennessy, Chair of the Board of Directors of Alphabet, through the John L. Hennessy & Andrea J. Hennessy Revocable Trust adopted a trading plan on November 5, 2024 (with the first trade under the plan scheduled for March 13, 2025). The trading plan will be effective until March 13, 2026 to sell 6,000 shares of Class C Capital Stock and 1,200 shares of Class A Common Stock.
- Sundar Pichai, Chief Executive Officer, together with the Pichai Family Foundation, adopted a trading plan on December 2, 2024 (with the first trade under the plan scheduled for April 2, 2025). The trading plan will be effective until April 18, 2026 to sell an aggregate of 876,000 shares of Class C Capital Stock.

There were no "non-Rule 10b5-1 trading arrangements" (as defined in Item 408 of Regulation S-K of the Exchange Act) adopted, modified, or terminated during the quarter ended December 31, 2024 by our directors and Section 16 officers. Each of the Rule 10b5-1 trading arrangements are in accordance with our Policy Against Insider

Trading and actual sale transactions made pursuant to such trading arrangements will be disclosed publicly in Section 16 filings with the SEC in accordance with applicable securities laws, rules, and regulations.

Compensatory Arrangements of Certain Officers

On April 16, 2024, the Leadership Development, Inclusion and Compensation Committee of the Board of Directors of Alphabet approved the accrual of dividend equivalent units to current and future holders of all unvested stock units in connection with any regular cash dividend that may be declared by the Board of Directors of Alphabet from time to time. As stock units are not outstanding shares of stock and thus would not otherwise be entitled to participate in any dividends, the crediting of dividend equivalent units is intended to preserve the equity-based incentives intended by the company when the stock units were granted and to treat the holders of unvested stock units consistently with all stockholders.

Required Disclosure Pursuant to Section 13(r) of the Exchange Act

As previously disclosed, Google LLC, a subsidiary of Alphabet, filed notifications with the Russian Federal Security Service (FSB) pursuant to Russian encryption control requirements, which must be complied with prior to the import of covered items. The information provided pursuant to Section 13(r) of the Exchange Act in Part II, Item 5 of our Quarterly Report on Form 10-Q for the quarter ended September 30, 2024 is incorporated herein by reference.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS, AND CORPORATE GOVERNANCE

The information required by this item will be included under the caption “Directors, Executive Officers, and Corporate Governance” in our Proxy Statement for the 2025 Annual Meeting of Stockholders to be filed with the SEC within 120 days of the fiscal year ended December 31, 2024 (2025 Proxy Statement) and is incorporated herein by reference. The information required by this item regarding delinquent filers pursuant to Item 405 of Regulation S-K will be included under the caption “Delinquent Section 16(a) Reports” in the 2025 Proxy Statement and is incorporated herein by reference.

We have adopted insider trading policies and procedures governing the purchase, sale, and other dispositions of securities of Alphabet by directors, officers, and employees that we believe are reasonably designed to promote compliance with insider trading laws, rules and regulations, and applicable Nasdaq listing standards. Our insider trading policy states, among other things, that our directors, officers, and employees are prohibited from trading in such securities while in possession of material, nonpublic information. The foregoing summary of our insider trading policies and procedures does not purport to be complete and is qualified by reference to our Policy Against Insider Trading filed as an exhibit to this Annual Report on Form 10-K.

ITEM 11. EXECUTIVE COMPENSATION

The information required by this item will be included under the captions “Director Compensation,” “Executive Compensation” and “Directors, Executive Officers, and Corporate Governance—Corporate Governance and Board Matters—Compensation Committee Interlocks and Insider Participation” in the 2025 Proxy Statement and is incorporated herein by reference, except as to information disclosed therein pursuant to Item 402(v) of Regulation S-K relating to pay versus performance.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information required by this item will be included under the captions “Common Stock Ownership of Certain Beneficial Owners and Management” and “Equity Compensation Plan Information” in the 2025 Proxy Statement and is incorporated herein by reference.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

The information required by this item will be included under the captions “Certain Relationships and Related Transactions” and “Directors, Executive Officers, and Corporate Governance—Corporate Governance and Board Matters—Director Independence” in the 2025 Proxy Statement and is incorporated herein by reference.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

The information required by this item will be included under the caption “Independent Registered Public Accounting Firm” in the 2025 Proxy Statement and is incorporated herein by reference.

PART IV

ITEM 15. EXHIBITS, FINANCIAL STATEMENT SCHEDULES

We have filed the following documents as part of this Annual Report on Form 10-K:

1. Consolidated Financial Statements

2. Financial Statement Schedules

Schedule II: Valuation and Qualifying Accounts

The table below details the activity of the allowance for credit losses and sales credits for the years ended December 31, 2022, 2023, and 2024 (in millions):

	Balance at Beginning of Year	Additions	Usage	Balance at End of Year
Year ended December 31, 2022	$ 1,389	$ 2,125	$ (2,301)	$ 1,213
Year ended December 31, 2023	$ 1,213	$ 3,115	$ (2,737)	$ 1,591
Year ended December 31, 2024	$ 1,591	$ 2,895	$ (2,850)	$ 1,636

Note: Additions to the allowance for credit losses are charged to expense. Additions to the allowance for sales credits are charged against revenues.

All other schedules have been omitted because they are not required, not applicable, or the required information is otherwise included.

3. Exhibits

Exhibit Number		Description	Incorporated by reference herein: Form	Date
2.01		Agreement and Plan of Merger, dated October 2, 2015, by and among Google Inc., the Registrant and Maple Technologies Inc.	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
3.01		Amended and Restated Certificate of Incorporation of the Registrant	Current Report on Form 8-K (File No. 001-37580)	June 3, 2022
3.02		Amended and Restated Bylaws of the Registrant, dated October 19, 2022	Current Report on Form 8-K (File No. 001-37580)	October 25, 2022
4.01		Specimen Class A Common Stock certificate	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.02		Specimen Class C Capital Stock certificate	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.03	◆	Alphabet Inc. Deferred Compensation Plan	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.04		Transfer Restriction Agreement, dated October 2, 2015, between the Registrant and Larry Page and certain of his affiliates	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.05		Transfer Restriction Agreement, dated October 2, 2015, between the Registrant and Sergey Brin and certain of his affiliates	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.06		Joinder Agreement, dated December 31, 2021, among the Registrant, Sergey Brin and certain of his affiliates	Annual Report on Form 10-K (File No. 001-37580)	February 2, 2022

Exhibit Number		Description	Incorporated by reference herein Form	Date
4.07		Transfer Restriction Agreement, dated October 2, 2015, between the Registrant and Eric E. Schmidt and certain of its affiliates	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.08		Class C Undertaking, dated October 2, 2015, executed by the Registrant	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
4.09		Indenture, dated February 12, 2016, between the Registrant and The Bank of New York Mellon Trust Company, N.A., as Trustee	Registration Statement on Form S-3 (File No. 333-209510)	February 12, 2016
4.10		Registrant Registration Rights Agreement dated December 14, 2015	Registration Statement on Form S-3 (File No. 333-209518)	February 12, 2016
4.11		First Supplemental Indenture, dated April 27, 2016, between the Registrant and The Bank of New York Mellon Trust Company, N.A., as trustee	Current Report on Form 8-K (File No. 001-37580)	April 27, 2016
4.12		Form of the Registrant's 1.998% Note due 2026	Current Report on Form 8-K (File No. 001-37580)	August 9, 2016
4.13		Form of Global Note representing the Registrant's 0.450% notes due 2025	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.14		Form of Global Note representing the Registrant's 0.800% notes due 2027	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.15		Form of Global Note representing the Registrant's 1.100% notes due 2030	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.16		Form of Global Note representing the Registrant's 1.900% notes due 2040	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.17		Form of Global Note representing the Registrant's 2.050% notes due 2050	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.18		Form of Global Note representing the Registrant's 2.250% notes due 2060	Current Report on Form 8-K (File No. 001-37580)	August 5, 2020
4.19		Description of Registrant's Securities	Annual Report on Form 10-K (File No. 001-37580)	February 3, 2023
10.01	◆	Form of Indemnification Agreement entered into between the Registrant, its affiliates and its directors and officers	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
10.02	◆	Form of Offer Letter for Directors	Annual Report on Form 10-K (File No. 001-37580)	January 31, 2024
10.03	◆	Letter Agreement dated June 3, 2024, between Anat Ashkenazi and Alphabet	Current Report on Form 8-K (File No. 001-36380)	June 7, 2024
10.04	◆	Compensation Plan Agreement, dated October 2, 2015, between Google Inc. and the Registrant	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
10.05	◆	Director Arrangements Agreement, dated October 2, 2015, between Google Inc. and the Registrant	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
10.06	◆	Alphabet Inc. Deferred Compensation Plan	Current Report on Form 8-K (File No. 001-37580)	October 2, 2015
10.07	◆	Alphabet Inc. Amended and Restated 2012 Stock Plan	Current Report on Form 8-K (File No. 001-37580)	June 5, 2020
10.07.1	◆	Alphabet Inc. Amended and Restated 2012 Stock Plan - Form of Alphabet Restricted Stock Unit Agreement	Annual Report on Form 10-K (File No. 001-37580)	February 4, 2020
10.08	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan	Current Report on Form 8-K (file No. 001-37580)	June 8, 2023
10.08.1	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet Restricted Stock Unit Agreement	Quarterly Report on Form 10-Q (file No. 001-37580)	July 28, 2021

Exhibit Number		Description	Incorporated by reference herein: Form	Incorporated by reference herein: Date
10.08.2	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet Restricted Stock Unit Agreement	Quarterly Report on Form 10-Q (File No. 001-37580)	July 26, 2023
10.08.3	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan – Form of Alphabet Restricted Stock Unit Agreement	Quarterly Report on Form 10-Q (File No. 001-37580)	April 26, 2024
10.08.4	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet 2022 CEO Performance Stock Unit Agreement	Annual Report on Form 10-K (File No. 001-37580)	February 3, 2023
10.08.5	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan – Form of Alphabet CEO Performance Stock Unit Agreement	Quarterly Report on Form 10-Q (File No. 001-37580)	April 26, 2024
10.08.6	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet 2022 Non-CEO Performance Stock Unit Agreement	Annual Report on Form 10-K (File No. 001-37580)	February 2, 2022
10.08.7	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan - Form of Alphabet 2023 Non-CEO Performance Stock Unit Agreement	Quarterly Report on Form 10-Q (File No. 001-37580)	July 26, 2023
10.08.8	◆	Alphabet Inc. Amended and Restated 2021 Stock Plan – Form of Alphabet Non-CEO Performance Stock Unit Agreement	Quarterly Report on Form 10-Q (File No. 001-37580)	April 26, 2024
10.09	◆	Alphabet Inc. Company Bonus Plan, as amended	Annual Report on Form 10-K (File No. 001-37350)	February 2, 2023
19.01	*	Alphabet Inc. Policy Against Insider Trading		
21.01	*	Subsidiaries of the Registrant		
23.01	*	Consent of Independent Registered Public Accounting Firm		
24.01	*	Power of Attorney (incorporated by reference to the signature page of this Annual Report on Form 10-K)		
31.01	*	Certification of Chief Executive Officer pursuant to Exchange Act Rules 13a-14(a) and 15d-14(a), as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002		
31.02	*	Certification of Chief Financial Officer pursuant to Exchange Act Rules 13a-14(a) and 15d-14(a), as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002		
32.01	‡	Certifications of Chief Executive Officer and Chief Financial Officer pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002		
97.01	◆	Clawback Policy	Annual Report on Form 10-K (File No. 001-37350)	January 31, 2024
101.INS	*	Inline XBRL Instance Document - the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document		
101.SCH	*	Inline XBRL Taxonomy Extension Schema Document		
101.CAL	*	Inline XBRL Taxonomy Extension Calculation Linkbase Document		
101.DEF	*	Inline XBRL Taxonomy Extension Definition Linkbase Document		

Exhibit Number		Description	Incorporated by reference herein	
			Form	Date
101.LAB	*	Inline XBRL Taxonomy Extension Label Linkbase Document		
101.PRE	*	Inline XBRL Taxonomy Extension Presentation Linkbase Document		
104	*	Cover Page Interactive Data File (embedded within the Inline XBRL document and contained in Exhibit 101)		

◆ Indicates management compensatory plan, contract, or arrangement.
* Filed herewith.
‡ Furnished herewith.

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this Annual Report on Form 10-K to be signed on its behalf by the undersigned, thereunto duly authorized.

Date: February 4, 2025

ALPHABET INC.

By: /S/ SUNDAR PICHAI

Sundar Pichai
Chief Executive Officer
(Principal Executive Officer of the Registrant)

POWER OF ATTORNEY

KNOW ALL PERSONS BY THESE PRESENTS, that each person whose signature appears below constitutes and appoints Sundar Pichai and Anat Ashkenazi, jointly and severally, his or her attorney-in-fact, with the power of substitution, for him or her in any and all capacities, to sign any amendments to this Annual Report on Form 10-K and to file the same, with exhibits thereto and other documents in connection therewith, with the Securities and Exchange Commission, hereby ratifying and confirming all that each of said attorneys-in-fact, or his or her substitute or substitutes, may do or cause to be done by virtue hereof.

Pursuant to the requirements of the Securities Exchange Act of 1934, this Annual Report on Form 10-K has been signed below by the following persons on behalf of the Registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Sundar Pichai **Sundar Pichai**	Chief Executive Officer and Director (Principal Executive Officer)	February 4, 2025
/s/ Anat Ashkenazi **Anat Ashkenazi**	Senior Vice President and Chief Financial Officer (Principal Financial Officer)	February 4, 2025
/s/ Amie Thuener O'Toole **Amie Thuener O'Toole**	Vice President, Corporate Controller and Principal Accounting Officer	February 4, 2025
/s/ Frances H. Arnold **Frances H. Arnold**	Director	February 4, 2025
/s/ Sergey Brin **Sergey Brin**	Co-Founder and Director	February 4, 2025
/s/ R. Martin Chavez **R. Martin Chávez**	Director	February 4, 2025
/s/ L. John Doerr **L. John Doerr**	Director	February 4, 2025
/s/ Roger W. Ferguson Jr. **Roger W. Ferguson Jr.**	Director	February 4, 2025
/s/ John L. Hennessy **John L. Hennessy**	Director, Chair	February 4, 2025
/s/ Larry Page **Larry Page**	Co-Founder and Director	February 4, 2025
/s/ K. Ram Shriram **K. Ram Shriram**	Director	February 4, 2025
/s/ Robin L. Washington **Robin L. Washington**	Director	February 4, 2025

Exhibit 19.01

Alphabet's Policy Against Insider Trading

Employees, officers, and directors of Alphabet or its subsidiaries ("**Alphabet**"), their immediate family members (e.g., spouses, domestic partners, children, and parents) or other people living in the same household, and their controlled entities (trusts that they control or benefit from, or partnerships or other entities they control) (collectively, "**Covered Persons**") may not do any of the following:

- trade securities of Alphabet while having material nonpublic information related to Alphabet, unless the trade is made through Alphabet's Employee Trading Plan program [link] or a plan approved under Alphabet's 10b5-1 Trading Plan Guidelines [link],
- trade securities of any other company while having material nonpublic information about or that may affect that company acquired through an Alphabet role or relationship,
- "tip" others about material nonpublic information that involves Alphabet or other companies and that was gained through an Alphabet role or relationship, or
- otherwise violate insider trading laws.

This Policy applies until a Covered Person is no longer providing services to Alphabet and no longer has any material nonpublic information subject to this Policy.
Consultants and contractors working with Alphabet are not directly subject to this Policy, but should comply with applicable securities laws against insider trading, as well as contractual obligations regarding the handling of Alphabet confidential information.

Definitions

"**Material**" Information is information a reasonable investor would consider important in deciding whether to buy or sell securities. Either positive or negative information may be material. As a general rule, information likely to affect the market price of a security is likely material.

Some examples of the types of information that may be material:

- Financial results
- News of a pending or proposed merger, acquisition or disposition
- Significant developments involving major corporate relationships, such as joint ventures and other strategic partnerships
- Changes in dividend policy
- Proposed stock splits
- New product announcements of a significant nature
- Significant cybersecurity incidents
- New equity or debt offerings

- Positive or negative developments in litigation or investigations
- Major changes to senior management

Financial information is particularly likely to be material. For example, nonpublic information about the results of Alphabet's operations for even part of a quarter might be material in helping to predict Alphabet's results for the quarter.

"**Nonpublic**" information is information that has not been widely disseminated to the public market long enough for the public to absorb and evaluate it.

"**Securities**" of Alphabet or other companies include equity securities (e.g., Class A common stock, Class B common stock, Class C capital stock), restricted stock units, performance stock units, stock options, preferred stock, debt securities, warrants, convertible debentures, and derivative securities (including exchange-traded options).

"**Tipping**" is sharing material nonpublic information with another person or making recommendations or expressing opinions about trading in the impacted securities on the basis of such information (even if the information itself is not shared).

"**Trading**" includes directly or indirectly buying, selling, pledging, or otherwise acquiring or disposing of securities, as well as agreeing to do any of these things. While bona fide gifts and donations are generally permitted during open trading windows, a gift of securities can constitute trading if there is an arrangement (written or otherwise) with the recipient to sell such securities.

Other Restrictions

<u>Trading Windows</u>

Employees, officers, and directors of Alphabet or its subsidiaries and their controlled entities may not trade, give, or donate Alphabet securities when their applicable trading window is closed unless they do so through Alphabet's Employee Trading Plan program [link] or under a plan approved under Alphabet's 10b5-1 Trading Plan Guidelines [link]

Trading windows for Alphabet employees depend on their level:

- The quarterly trading window for individuals at Employee Level 8 (Directors) and above (including officers and directors) and their controlled entities opens on the second trading day following the public disclosure of Alphabet's financial results for the prior fiscal quarter and ends at the close of market on the first trading day of the third calendar month of the current quarter.
- The quarterly trading window for individuals at Employee Level 7 and below and their controlled entities opens on the second trading day following the public disclosure of Alphabet's financial results for the previous fiscal quarter and ends at the close of market on the first trading day of the next fiscal quarter.

Even during an open trading window, employees, officers, and directors of Alphabet or its subsidiaries (and their controlled entities) with material nonpublic information relating to Alphabet may not trade Alphabet securities until at least the second trading day after such information becomes public.

Alphabet's Insider Trading Compliance Officer may designate special trading restriction periods for individuals having material nonpublic information, such as news of a pending material acquisition. Alphabet will notify individuals subject to a special trading restriction. Those persons may not inform others of the notification or any material inside information, and they (and their controlled entities) may not engage in any transaction involving Alphabet's securities until instructed otherwise by the Insider Trading Compliance Officer.

Alphabet's quarterly trading windows and special trading restriction periods do not apply to family members or other persons living in the same household as the employees, officers, and directors of Alphabet or its subsidiaries. However, any family members and other persons living in the same household as the employees, officers, and directors of Alphabet or its subsidiaries are strongly advised to adhere to the applicable trading window as well as any special trading restriction periods that may apply to the employees, officers, and directors of Alphabet or its subsidiaries.

Alphabet's Insider Trading Compliance Officer has the authority to determine who will be subject to Alphabet's trading windows, and may modify the list of Alphabet persons subject to the quarterly trading windows and/or special trading restriction periods from time to time.

Other Prohibited Transactions

Covered Persons may not engage in short-term ("in and out") trading, short sales, hedging, or other derivative transactions involving Alphabet securities, pledge Alphabet securities as collateral, hold Alphabet securities in margin accounts, or place limit orders or stop orders for Alphabet securities that are likely to remain open during a closed trading window or other restricted period.

clearance

Outside of an approved 10b5-1 trading program, members of Alphabet's Board of Directors and all of Alphabet's Section 16 officers may not trade Alphabet's securities, even during an open trading window, without first having the transaction pre-cleared by Alphabet's Insider Trading Compliance Officer.

Exemptions

These restrictions do not apply to the: (i) exercise of Alphabet stock options when the exercise price is paid in cash ("cashless" exercises of stock options are covered by this Policy); (ii) transactions involving only a change in the form of ownership (e.g., a person transferring shares to a trust of which he or she is the sole beneficiary); (iii) mandatory sales of shares under non-US stock award agreements or under 409A exercise elections; or (iv) grants of stock, options, or restricted stock units from Alphabet.

Policy Violations

Individuals who trade on material nonpublic information (or tip such information to others) are subject to civil and criminal penalties. Violations of this Policy by Alphabet employees may also result in demotion, termination, or other disciplinary actions. This Policy goes beyond legal requirements, and Alphabet may determine that specific conduct violates this Policy, even if it does not violate the law, and may take disciplinary action prior to the filing or resolution of any civil or criminal action against the alleged violator.

Questions?

If you have questions about this Policy or how it applies, please visit the **insider trading compliance resource center** [link] for additional support and resources.

Exhibit 21.01

SUBSIDIARIES OF THE REGISTRANT

The following is a list of subsidiaries of Alphabet Inc., omitting subsidiaries which, considered in the aggregate as a single subsidiary, would not constitute a significant subsidiary as of December 31, 2024:

Name of Subsidiary	Jurisdiction of Incorporation or Organization
Google LLC	Delaware
XXVI Holdings Inc.	Delaware
Alphabet Capital US LLC	Delaware

Exhibit 23.01

CONSENT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

We consent to the incorporation by reference in the following Registration Statements:

(1) Registration Statement (Form S-8 No. 333-207254) pertaining to the Google Inc. 2004 Stock Plan, Alphabet Inc. 2012 Stock Plan, AdMob, Inc. 2006 Stock Plan, UK Sub-Plan of the AdMob, Inc. 2006 Stock Plan, Motorola Mobility Holdings, Inc. 2011 Incentive Compensation Plan and Alphabet Inc. Deferred Compensation Plan,

(2) Registration Statement (Form S-8 No. 333-212914) pertaining to the Alphabet Inc. 2012 Stock Plan,

(3) Registration Statement (Form S-8 No. 333-215873) pertaining to the Alphabet Inc. Deferred Compensation Plan,

(4) Registration Statement (Form S-8 No. 333-219435) pertaining to the Alphabet Inc. 2012 Stock Plan,

(5) Registration Statement (Form S-8 No. 333-226309) pertaining to the Alphabet Inc. 2012 Stock Plan,

(6) Registration Statement (Form S-8 No. 333-232836) pertaining to the Alphabet Inc. Amended and Restated 2012 Stock Plan,

(7) Registration Statement (Form S-8 No. 333-236248) pertaining to the Alphabet Inc. Deferred Compensation Plan,

(8) Registration Statement (Form S-8 No. 333-240221) pertaining to the Alphabet Inc. Amended and Restated 2012 Stock Plan,

(9) Registration Statement (Form S-8 No. 333-256731) pertaining to the Alphabet Inc. 2021 Stock Plan,

(10) Post-effective Amendment (Form S-8 No. 333-256731) pertaining to the Alphabet Inc. Amended and Restated 2012 Stock Plan and the Alphabet Inc. 2021 Stock Plan,

(11) Registration Statement (Form S-3 No. 333-262473) of Alphabet Inc.,

(12) Registration Statement (Form S-8 No. 333-266339) pertaining to the Alphabet Inc. Amended and Restated 2021 Stock Plan; and

(13) Registration Statement (Form S-8 No. 333-273428) pertaining to the Alphabet Inc. Amended and Restated 2021 Stock Plan;

of our reports dated February 4, 2025, with respect to the consolidated financial statements of Alphabet Inc. and the effectiveness of internal control over financial reporting of Alphabet Inc. included in this Annual Report (Form 10-K) of Alphabet Inc. for the year ended December 31, 2024.

/s/ Ernst & Young LLP

San Jose, California

February 4, 2025

Exhibit 31.01

CERTIFICATION OF CHIEF EXECUTIVE OFFICER
PURSUANT TO
EXCHANGE ACT RULES 13a-14(a) AND 15d-14(a),
AS ADOPTED PURSUANT TO
SECTION 302 OF THE SARBANES-OXLEY ACT OF 2002

I, Sundar Pichai, certify that:

1. I have reviewed this Annual Report on Form 10-K of Alphabet Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
   (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
   (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: February 4, 2025

/S/ SUNDAR PICHAI

Sundar Pichai
Chief Executive Officer
(Principal Executive Officer)

Exhibit 31.02

CERTIFICATION OF CHIEF FINANCIAL OFFICER
PURSUANT TO
EXCHANGE ACT RULES 13a-14(a) AND 15d-14(a),
AS ADOPTED PURSUANT TO
SECTION 302 OF THE SARBANES-OXLEY ACT OF 2002

I, Anat Ashkenazi, certify that:

1. I have reviewed this Annual Report on Form 10-K of Alphabet Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant’s other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
   (c) Evaluated the effectiveness of the registrant’s disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
   (d) Disclosed in this report any change in the registrant’s internal control over financial reporting that occurred during the registrant’s most recent fiscal quarter (the registrant’s fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant’s internal control over financial reporting; and
5. The registrant’s other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant’s auditors and the audit committee of the registrant’s board of directors (or persons performing the equivalent functions):
   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant’s ability to record, process, summarize and report financial information; and
   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant’s internal control over financial reporting.

Date: February 4, 2025

/s/ ANAT ASHKENAZI
Anat Ashkenazi
Senior Vice President, Chief Financial Officer

Exhibit 32.01

CERTIFICATIONS OF CHIEF EXECUTIVE OFFICER AND CHIEF FINANCIAL OFFICER
PURSUANT TO
18 U.S.C. SECTION 1350,
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002

I, Sundar Pichai, certify pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that the Annual Report on Form 10-K of Alphabet Inc. for the fiscal year ended December 31, 2024, fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and that information contained in such Annual Report on Form 10-K fairly presents, in all material respects, the financial condition and results of operations of Alphabet Inc.

Date: February 4, 2025	By:	/s/ SUNDAR PICHAI
	Name:	Sundar Pichai
	Title:	Chief Executive Officer (Principal Executive Officer)

I, Anat Ashkenazi, certify pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that the Annual Report on Form 10-K of Alphabet Inc. for the fiscal year ended December 31, 2024, fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and that information contained in such Annual Report on Form 10-K fairly presents, in all material respects, the financial condition and results of operations of Alphabet Inc.

Date: February 4, 2025	By:	/s/ ANAT ASHKENAZI
	Name:	Anat Ashkenazi
	Title:	Senior Vice President, Chief Financial Officer

The foregoing certifications are not deemed filed with the Securities and Exchange Commission for purposes of Section 18 of the Securities Exchange Act of 1934, as amended (Exchange Act), and are not to be incorporated by reference into any filing of Alphabet Inc. under the Securities Act of 1933, as amended, or the Exchange Act, whether made before or after the date hereof, regardless of any general incorporation language in such filing.

2 0 2 4

amazon

ANNUAL REPORT

amazon

Dear Shareholders:

2024 was a strong year for Amazon.

Our total revenue grew 11% year-over-year ("YoY") from $575B to $638B. By segment, North America revenue increased 10% YoY from $353B to $387B, International revenue grew 9% YoY from $131B to $143B, and AWS revenue increased 19% YoY, from $91B to $108B. For perspective, just 10 years ago, AWS revenue was $4.6B; and in that same year, Amazon's total revenue was $89B.

Amazon's operating income in 2024 improved 86% YoY, from $36.9B (an operating margin of 6.4%) to $68.6B (an operating margin of 10.8%). Free Cash Flow, adjusted for equipment finance leases improved from $35.5B in 2023 to $36.2B.

Apart from the financial results, we made our customers' lives meaningfully better and easier. In our Stores business, we substantially expanded selection, continued lowering prices (independent research firm, Profitero, found Amazon the lowest-priced online U.S. Retailer for the eighth year in a row), and for the second year in a row, we shipped at record speed to our Prime members. AWS launched a slew of new infrastructure and AI services that make it even easier to build remarkable customer experiences, including our latest custom AI silicon (Trainium2), a new set of frontier foundation models in Amazon Nova, and significant expansion of available models and features in our leading Generative AI ("GenAI") services Amazon SageMaker and Amazon Bedrock. Prime Video continued to offer compelling original shows, including new seasons for *Fallout, Reacher, The Boys,* and *The Lord of the Rings: Rings of Power*, movies like *Road House, The Idea of You,* and *Red One*, live sports like *Thursday Night Football* and *UEFA Champions League* in Europe (with the NBA and NASCAR coming in 2025), and new selection, highlighted by Apple TV+ joining Prime Video Channels. We launched a series of new Kindle devices that included a new color version, a larger Scribe option, and our fastest Paperwhites ever (the collection of which drove the highest Kindle unit sales for a single quarter in over a decade). And, we continued to add more selection, price transparency, and same day shipping for Amazon Pharmacy.

These accomplishments are a subset of what the team launched in 2024, but represent a lot of invention, hard work, and thoughtful execution across Amazon. I'm thankful for my teammates and their delivery this past year (some of which you can see in our 2024 results, others of which won't be visible for the next few years).

A Why Culture

Every year in my annual letter, I try to share insight into what makes Amazon tick. At the highest level, we're aiming to be Earth's most customer-centric company, making customers' lives better and easier every day. This is not easy to do in general, let alone year after year. In fact, it's actually quite hard, especially with the rapid rate of change in technology, customer habits, and new products from large and small companies alike. If we want to have a chance at succeeding in our mission, we have to constantly question everything around us.

We've had this long-held philosophy at Amazon about two-way and one-way door decisions. A two-way door decision is one where if you get the decision wrong, you can walk back through that door, revert to where you were, and there are few (if any) ramifications. You can make these decisions quickly and locally. A one-way door decision is one where it's quite difficult (if not impossible) to walk back through that door if you get the decision wrong, so these decisions are made more methodically. But, both of these constructs assume the door is unlocked. A lot of invention is about trying to open doors that have historically seemed bolted shut. And, over the past 30 years, we've found one of the most important keys to unlock these doors has been a simple question: "Why?"

"Why does this customer experience have to be this way?" "Why can't it be better?" "What are the constraints—why must we accept them?" "Why can't we invent around that?" "Why will it take so long to get to customers?" Why?

My Dad has told me that I was the kind of kid who kept asking why, perhaps to an annoying extent. He's also reminded me how shortly after I joined Amazon in 1997, he tried to persuade me to work somewhere more traditional (and on the east coast closer to family)—only to realize that I'd already found the perfect fit.

That's because Amazon is a Why company. We ask why, and why not, constantly. It helps us deconstruct problems, get to root causes, understand blockers, and unlock doors that might have previously seemed impenetrable. Amazon has an unusually high quotient of this WhyQ (let's call it "YQ"), and it frames the way we think about everything that we do.

Starting in 1995, we asked why can't we offer customers every in-print book?

Then, we asked, why limit ourselves to in-print—why can't we also offer every out-of-print book?

Why not offer every book, ever written, in any language—all available within 60 seconds on a device that's light and fits in the palm of your hand (Kindle)?

When we offer reviews, why must they all come from professional "experts?" Customers are great resources and will be brutally honest. Why not include customer reviews even if they sometimes dissuade a purchase?

Why not offer more than Books? What about Music, Video, Electronics, Tools, Kitchen, Apparel, Home Furnishings?

Why not practically everything?

Why should we be the only sellers of these items? Millions of third-party merchants and small sellers offer similar or unique items. Why not let customers choose the selection, price, and delivery speed they prefer from among these millions of sellers?

After struggling for a couple years to create awareness for sellers' selection, we asked ourselves why not show their selection on the same product detail pages as our first-party selection (where all the traffic was)?

Why not allow our sellers to also store items in our fulfillment network, enable those items to have fast, Prime delivery, and fulfill those items for sellers (a program called Fulfillment by Amazon)?

Why not experiment with relevant advertisements in our store to expose customers to new sellers and items (versus only what our algorithms might surface based on past purchases)?

Why does every company need their own capital-intensive datacenters and infrastructure? Why should every development team keep reinventing services like compute, storage, database, and analytics? Why should builders spend 80% of their time on the undifferentiated heavy lifting vs. their unique customer experience? Why not build a set of services (AWS) to solve that for internal and external builders?

Why do I have to buy a physical video to watch a movie? Why do I need cable or linear TV to watch amazing TV shows (Prime Video)?

Why can't I get my Prime shipping benefits on other websites than just Amazon (Buy with Prime)?

I can go on. But, you get the idea. Every one of these Whys have led to significant invention, and every one of them have made customers' lives better and easier. Some of these seem obvious now. But at the time, these were provocative questions that required curiosity, risk-taking, experimentation, and persistence to make these into success stories.

Enabling a Why Culture

If you believe having high YQ is critical to inventing for customers, how do you enable it? In my opinion, it's not solved with one mechanism. It needs to be built deeply into your culture and leadership team, and has to be fiercely protected over time if you're lucky enough to be successful. Here are a few of the strategies we employ.

Create leadership principles that set the tone. We have 16 Leadership Principles that guide our behavior. They're all integral underpinnings to our YQ, but I'll touch on three in particular:

<u>Are Right a Lot</u>

"Leaders are right a lot. They have strong judgment and good instincts. They seek diverse perspectives and work to disconfirm their beliefs."

When we first instituted this leadership principle, some people incorrectly assumed it meant that the best leaders were the ones whose ideas were chosen (i.e. they were right, a lot). It led to some people overly digging in and fighting for their ideas. There's nothing wrong with pushing for what you believe. But, in my experience, the best leaders want to hear others' views. They don't wilt or bristle when challenged; they're intrigued. Effective leaders change their minds when presented with new compelling information (which makes it ironic how people dismiss politicians as "flip-floppers" when they change their position). Ultimately, leaders are responsible for getting to the best answer for customers, regardless of whose original idea is chosen.

<u>Learn and be Curious</u>

"Leaders are never done learning and always seek to improve themselves. They are curious about new possibilities and act to explore them."

In the nearly 28 years I've been at Amazon, the biggest difference in the relative growth of companies and individuals has been their aptitude to learn. At a certain point, some leaders seem to lose their thirst to learn. It's hard to know the reason in each case, but it's as if some people find it too exhausting, too time-consuming, or too threatening to not have all the answers. Regardless, the day we stop learning at Amazon is the day we risk undermining what we're capable of building in the future. People with high YQ are always curious how they can get better, become wiser, and incorporate their new knowledge into better customer experiences.

<u>Have Backbone; Disagree and Commit</u>

"Leaders are obligated to respectfully challenge decisions when they disagree, even when doing so is uncomfortable or exhausting. Leaders have conviction and are tenacious. They do not compromise for the sake of social cohesion. Once a decision is determined, they commit wholly."

We don't just empower people to challenge one another, we obligate them to do so if they disagree. Questioning, asking the hard questions, forcing the discussion (versus silently thinking a mistake is being made) is necessary to getting to better answers for customers. "I told you so" has no currency at Amazon. It's also important to focus on the second part of this leadership principle: *disagree and commit*. While constructive debate is useful; at some point, teams need to make a decision and take action. From that point on, everybody—even those who advocated for a different solution than the one chosen—must commit to making that decision a success. That means the team goes all in—no pocket-vetoing nor hedging between other options. That's the only way we can preserve speed and confidence that if an issue is heavily debated, the team will ultimately pull together.

Create norms that support the Why. Similar to how our Leadership Principles guide our behavior, we've built norms over the years that guide how we work. Here are a few examples:

<u>Narratives</u>. We stopped presenting information to each other inside the company via powerpoint in 2004. Given how high level powerpoints are, we found that powerpoint was easy for the presenter to prepare, but harder for the audience to understand the substantive issues. Instead, we moved to writing narratives with a maximum of six pages in the body. Narratives are harder for the presenter (it's hard to write a thoughtful six-page document that highlights the key issues in enough detail to be crisp and clear), but much easier for the audience to engage with and ask the right Why questions.

<u>Working backwards documents</u>. When we build services or features, before we start coding, we write Press Release and Frequently Asked Questions ("FAQ") documents. The Press Release is intended to ensure that what we're proposing building is remarkable to customers (so we don't get to launch and ask "wait, why did we think customers would find this interesting?"). And, the FAQ is designed to force ourselves to ask the

hard questions about which customers will use this capability, what they'll like most, what they'll be most disappointed with, why are we drawing the launch line where we are, why is it better than current alternatives, how should we think about pricing, what pricing dimensions we recommend, and why have we made the architectural decisions we have. The Press Release and FAQ are how we work backwards from customers, and how we push ourselves to ask questions customers would if they were in these meetings.

<u>Be together whenever possible</u>. There are many paths that can lead to breakthrough innovation. Occasionally, a lone genius comes up with a brilliant idea, and everyone else simply executes it. While that can work, it's not how we typically operate. Amazon invention is deeply collaborative. It starts with a seed of an idea, then a group of smart, mission-driven people refine, challenge, and build on it together. And, we've found that this process is far more effective in person than remote. Of course, you can invent with everybody remote (and some cultures seem to prefer that). However, in my experience, it doesn't compare to being in the same room. The energy, the pace, the spontaneous brainstorming, the willingness for people to jump in, the way ideas evolve in real time, and the post-meeting iteration is much better when in the same room—and yields better outcomes for our customers and teams. With what's happening in AI right now, and the likelihood that every customer experience we've ever known will be reinvented, there has never been a more important time, in my opinion, to optimize to invent well.

<u>Tolerating messy meetings</u>. It's hard to "schedule" innovation. You can't book 60 minutes to invent Amazon Prime, or AWS, or Alexa+, or Fulfillment by Amazon, or Regionalization in our Fulfillment Network, or Project Kuiper. These inventions are borne out of somebody asking why we can't change what's possible for customers, and then they take on a life of their own, often meandering down multiple dead ends before getting to a final destination. This might bother some regimented folks. But, when we're inventing, we accept the process being beautifully imperfect.

Operate like a startup (in our case, the world's largest startup). We strive to operate like the world's largest startup. What does that mean?

First, whatever we're contemplating building has to be focused on solving a real customer problem or meaningfully improving a customer experience. Companies can get off track prioritizing technology because they're excited about the technology. Great startups are on a mission to change what's possible for customers.

Second, we have a disproportionate need for builders. These are inventors. They're people constantly dissecting customer experiences, even ones that seem pretty good today, and asking why they can't be better. They're divinely discontent (maybe annoyingly so for team members proud of what they've previously built), and never feel like the job is done.

Third, we want owners. One of the strengths of Amazon over the first 30 years is that we've hired really smart, motivated, inventive, ambitious people who have been great owners. And, that means that our teammates are constantly asking themselves, "What would I do if this was my own money?" "What would I do if I started this company and I was the majority owner?" "Hey, I know I've only been asked to own a part of this project, but I'm not sure if the other parts are being driven well—should I stick my nose into this and make sure or just trust somebody's got it?" Owners feel accountable. They care deeply about the quality and effectiveness of what they own, and view the company's mission as their mission (we want missionaries, not mercenaries). That's part of what our effort to increase the ratio of individual contributors versus managers is about. We want flatter organizations where our owners doing the work feel like they own the two-way door decisions (which are the vast majority), can move rapidly, and are fully accountable for solving the Whys of their customer experiences.

Fourth, speed disproportionately matters for every business, in every industry, at all times. It's a false binary to argue that you can move fast <u>or</u> deliver high standards. If you want to be fast, you can be fast, and still be high quality. We've done it for many years (though we can still be faster). Speed is a leadership decision. The leadership team has to believe it's a priority, reinforce it constantly, organize and remove structural barriers, and build in modular ways that enable pace. But, speed does not happen unless the entire company and culture embrace it. We have this persistent feeling, throughout the company and in every business in which we operate, that there are closing windows all around us. We operate in fiercely competitive market segments, with highly talented, well-funded, ambitious companies at every turn. Customers are always looking for something better. We spend a lot of time identifying how to unlock these experiences for them as quickly as possible, and know if we don't, somebody else will.

Another way to gain speed is to eliminate bureaucracy. There is a difference between process and bureaucracy. When you're running something at scale, you need mechanisms to deliver the right experience and constant improvement for customers. However, as companies grow and add more managers, unneeded processes get layered on that add little value. Last fall, I asked teammates across the company to send me bureaucracy examples that they were experiencing. I've received almost 1,000 of these emails, and read every single one. Builders hate bureaucracy. It slows them down, frustrates them, and keeps them from doing what they came here to do. As leaders, we don't always see the red tape buried deep in our organizations, but we can sure as heck eliminate it when we do. We've already made over 375 changes based on this feedback. We need to move fast, and we are committed to rooting out bureaucracy that ties up time and dispirits our teammates.

Fifth, you have to be scrappy. As businesses succeed and get larger, they sometimes forget how things got started. We built Amazon Simple Storage Service (S3) with 13 people; Amazon Elastic Compute Cloud (EC2) with 11 people. Managers can confuse themselves that the way to grow and get ahead is to accumulate large teams. Historically, we've had periods where we've allowed this thinking to hold sway. But, it's not the way we fundamentally think about building teams and products, and have adjusted to reflect that again. Our best leaders get the most done with the least number of resources required to do the job. They pride themselves on being lean.

Sixth, you have to be willing to take risks. This sounds easier than it is. You need clever enough people to identify worthwhile bets. And if you have these inventive, ambitious builders with high standards, they're not used to failure. They suspect external (and maybe internal) ridicule awaits them if they try something very different that doesn't work out. So, people often play it safe. But, you can't achieve something extraordinary for customers by playing "not to lose." If your Whys take you down an invention path that delivers an experience that doesn't look like what's been done before, let customer obsession be your compass. You rarely, if ever, change the world by doing the same thing as everybody else.

And finally, you have to care most about delivering compelling results for customers. It's not how charismatic you are. It's not whether you're really good at managing up or sideways. What matters is what we actually get done for customers. That's what we want to reward.

Next generation Whys

While the team and I feel quite optimistic about the progress and potential of our existing businesses, we have plenty of new Whys we're asking. Below are a few of them and some quick thoughts.

Why is AI so important? Will it really have as much impact as some claim and when?

Generative AI is going to reinvent virtually every customer experience we know, and enable altogether new ones about which we've only fantasized. The early AI workloads being deployed focus on productivity and cost avoidance (e.g. customer service, business process orchestration, workflow, translation, etc.). This is saving companies a lot of money. Increasingly, you'll see AI change the norms in coding, search, shopping, personal assistants, primary care, cancer and drug research, biology, robotics, space, financial services, neighborhood networks—everything. Some of these areas are already seeing rapid progress; others are still in their infancy. But, if your customer experiences aren't planning to leverage these intelligent models, their ability to query giant corpuses of data and quickly find your needle in the haystack, their ability to keep getting smarter with more feedback and data, and their future agentic capabilities, you will not be competitive. How soon? It won't all happen in a year or two, but, it won't take ten either. It's moving faster than almost anything technology has ever seen.

OK, I buy AI is big; but why invest this much this quickly?

Fundamentally, if your mission is to make customers' lives better and easier every day, and you believe every customer experience will be reinvented by AI, you're going to invest deeply and broadly in AI. That's why there are more than 1,000 GenAI applications being built across Amazon, aiming to meaningfully change customer experiences in shopping, coding, personal assistants, streaming video and music, advertising, healthcare, reading, and home devices, to name a few. It's also why AWS is quickly developing the key primitives (or building blocks) for AI development, such as custom silicon AI chips in Amazon Trainium to provide better price-performance on training and inference, highly flexible model-building and inference services in Amazon SageMaker and Amazon Bedrock, our own frontier models in Amazon Nova to provide lower cost and latency for customers' applications, and agent creation and management capabilities.

There is also substantial capital investment required. In AWS, the faster demand grows, the more datacenters, chips, and hardware we need to procure (and AI chips are much more expensive than CPU chips). We spend this capital upfront, even though these assets are useful for many years (in the case of datacenters, for at least 15-20 years). We only start monetizing this capital investment many months after we spend the capital, and over many years—which leads to attractive long-term FCF and ROIC (as people have seen in AWS over the last several years). But in periods, like now, of unusually high demand (our AI revenue is growing at triple digit YoY percentages and represents a multi-billion-dollar annual revenue run rate), you're deploying a lot of capital. We continue to believe AI is a once-in-a-lifetime reinvention of everything we know, the demand is unlike anything we've seen before, and our customers, shareholders, and business will be well-served by our investing aggressively now.

Why do chips and AI have to be this expensive for customers?

AI does not have to be as expensive as it is today, and it won't be in the future. Chips are the biggest culprit. Most AI to date has been built on one chip provider. It's pricey. Trainium should help, as our new Trainium2 chips offer 30-40% better price-performance than the current GPU-powered compute instances generally available today. While model training still accounts for a large amount of the total AI spend, inference (which are the predictions or outputs of the models) will represent the overwhelming majority of future AI cost because customers train their models periodically, but produce inferences constantly in large-scale AI applications. Inference will become another building block service, along with compute, storage, database, and others. We feel strong urgency to make inference less expensive for customers. More price-performant chips will help. But, inference will also get meaningfully more efficient in the next couple of years with improvements in model distillation, prompt caching, computing infrastructure, and model architectures. Reducing the cost per unit in AI will unleash AI being used as expansively as customers desire, and also lead to more overall AI spending. It's like what happened with AWS. Revolutionizing the cost of compute and storage happily led to lower cost per unit, and more invention, better customer experiences, and more absolute infrastructure spend.

Why have personal assistants not yet taken off? How can Alexa help?

A great personal assistant can answer virtually any question <u>and</u> get things done on your behalf. There have been no digital solutions that can do both yet. That is, until Alexa+ arrived. Alexa+ is not only comparably intelligent to the leading chatbots, but can take a plethora of real actions for you. She can play music, play video, move media from one of your devices to another, set alarms and timers, control your smart home, order across hundreds of millions of ecommerce items, make reservations for restaurants or Ubers, order concert tickets, alert you when your favorite artist announces a tour, find a plumber to fix your sink, and memorize whatever you've done on Amazon. This is pretty game-changing for consumers, and just the start of what Alexa+ will do. We have over 600 million Alexa devices out there today, and expect Alexa+ to play an even more vital role in the lives of these hundreds of millions of customers in the future.

Why can't we get items to customers even faster? Does it matter?

Every year, people ask whether we've reached the law of diminishing returns on speed of delivery. Our data shows this not to be the case. When we promise faster delivery times, customers complete purchases at a meaningfully higher rate and shop with us more frequently. Amazon Prime started with unlimited, free, two-day delivery for a million products; it's now grown to over 300 million items, with tens of millions available in one day (or better). An increasing number of deliveries happen same day. This speed improvement is primarily due to our regionalization redesign of our fulfillment network, our new placement algorithms, and the introduction of our innovative same-day fulfillment centers. Although we've set speed records for two consecutive years, we're still honing these innovations, and have others planned. And, don't forget Prime Air, our drones that will get items to customers inside an hour. We are not done improving speed.

Why can't people in small towns enjoy the same fast delivery speeds as people in cities?

As some other companies are abandoning small-town customers due to cost to serve, we're going the other way—we're investing to serve our rural customers even better. We've already expanded Same-Day and Overnight Delivery to dozens of smaller cities and towns across the U.S., with more coming. This expansion will provide even faster Amazon delivery speeds for many millions of customers, particularly in less densely populated areas, enabling us to deliver over a billion packages each year to customers living in 13,000 zip codes spanning 1.2 million square miles.

Related, why can't we help the hundreds of millions of people without broadband connectivity?

There are about 400-500 million households around the world, most in small, rural towns that don't have access to broadband connectivity. They can't leverage the Internet to learn, shop, do business, access entertainment, and communicate the same way people take for granted in bigger cities. This digital divide is what Project Kuiper, our low Earth orbit satellite network, aims to solve. We're just launching our first production satellites, and will ultimately have over 3,200 in orbit over the next few years. While capital-intensive to launch, we believe Kuiper will be a meaningful operating income and ROIC business for us.

Why does healthcare have to be so stressful?

Healthcare, especially in the U.S., is quite frustrating. It's hard to get fast appointments with primary care physicians, often harder with specialists. There's a lot of waiting around. Physicians spend only a few minutes with patients. Then, patients have to drive somewhere (often not close) to get their medications. And, when they get to the pharmacy, they're often surprised by the pricing, what's covered by their insurance, and what you can easily access that's not behind a locked shelf. Customers deserve better. It's why you see such positive customer sentiment and growth for Amazon Pharmacy and Amazon One Medical, and we continue to iterate quickly on selection and transparency for Amazon Pharmacy, and physical clinic capacity for One Medical.

These are some of the Why questions we're asking ourselves right now, and I'm excited about the future inventions to come. We're not going to be bored any time soon.

When I first started working, I thought it was unfathomable that my Dad worked at the same place for 45 years. How could that be? That's so long. I used to tell my friends that would never be me. Now, with almost 28 years and counting at Amazon, I have to answer those same friends with their own Why question.

After all these years, why are you still at Amazon?

I'm obviously a Superfan, but there are several compelling parts to working at Amazon. First, I'm not sure that any company prioritizes customers as relentlessly as we do. Lots of companies say they will; few follow through. Second, it's challenging to find a company where you can make a bigger impact on the world than you can at Amazon. Third, we make significant long-term investments and bets in both inventions and people. This allows our teams to iterate on ideas, and make the right long-term decisions for customers and the company. And, I've never encountered a more intelligent, creative, ambitious, hungry, hard-working, and missionary group of teammates than we have at Amazon. In my opinion, this is a remarkable set of qualities to have at a company. And, for builders who want to change the world, and who have fire in their belly, there's no better place to be than Amazon.

We operate like the world's largest startup in large part because of our culture of Why. We don't always get everything right, and we learn and iterate like crazy. But, we're constantly choosing to prioritize customers, delivery, invention, ownership, speed, scrappiness, curiosity, and building a company that outlasts us all. It remains Day One.

Sincerely,

Andy Jassy
President and Chief Executive Officer
Amazon.com, Inc.

P.S. As we have always done, our original 1997 Shareholder Letter follows. What's written there is as true today as it was in 1997.

amazon.com

1997 LETTER TO SHAREHOLDERS

(Reprinted from the 1997 Annual Report)

To our shareholders:

Amazon.com passed many milestones in 1997: by year-end, we had served more than 1.5 million customers, yielding 838% revenue growth to $147.8 million, and extended our market leadership despite aggressive competitive entry.

But this is Day 1 for the Internet and, if we execute well, for Amazon.com. Today, online commerce saves customers money and precious time. Tomorrow, through personalization, online commerce will accelerate the very process of discovery. Amazon.com uses the Internet to create real value for its customers and, by doing so, hopes to create an enduring franchise, even in established and large markets.

We have a window of opportunity as larger players marshal the resources to pursue the online opportunity and as customers, new to purchasing online, are receptive to forming new relationships. The competitive landscape has continued to evolve at a fast pace. Many large players have moved online with credible offerings and have devoted substantial energy and resources to building awareness, traffic, and sales. Our goal is to move quickly to solidify and extend our current position while we begin to pursue the online commerce opportunities in other areas. We see substantial opportunity in the large markets we are targeting. This strategy is not without risk: it requires serious investment and crisp execution against established franchise leaders.

It's All About the Long Term

We believe that a fundamental measure of our success will be the shareholder value we create over the *long term*. This value will be a direct result of our ability to extend and solidify our current market leadership position. The stronger our market leadership, the more powerful our economic model. Market leadership can translate directly to higher revenue, higher profitability, greater capital velocity, and correspondingly stronger returns on invested capital.

Our decisions have consistently reflected this focus. We first measure ourselves in terms of the metrics most indicative of our market leadership: customer and revenue growth, the degree to which our customers continue to purchase from us on a repeat basis, and the strength of our brand. We have invested and will continue to invest aggressively to expand and leverage our customer base, brand, and infrastructure as we move to establish an enduring franchise.

Because of our emphasis on the long term, we may make decisions and weigh tradeoffs differently than some companies. Accordingly, we want to share with you our fundamental management and decision-making approach so that you, our shareholders, may confirm that it is consistent with your investment philosophy:

- We will continue to focus relentlessly on our customers.
- We will continue to make investment decisions in light of long-term market leadership considerations rather than short-term profitability considerations or short-term Wall Street reactions.
- We will continue to measure our programs and the effectiveness of our investments analytically, to jettison those that do not provide acceptable returns, and to step up our investment in those that work best. We will continue to learn from both our successes and our failures.

- We will make bold rather than timid investment decisions where we see a sufficient probability of gaining market leadership advantages. Some of these investments will pay off, others will not, and we will have learned another valuable lesson in either case.
- When forced to choose between optimizing the appearance of our GAAP accounting and maximizing the present value of future cash flows, we'll take the cash flows.
- We will share our strategic thought processes with you when we make bold choices (to the extent competitive pressures allow), so that you may evaluate for yourselves whether we are making rational long-term leadership investments.
- We will work hard to spend wisely and maintain our lean culture. We understand the importance of continually reinforcing a cost-conscious culture, particularly in a business incurring net losses.
- We will balance our focus on growth with emphasis on long-term profitability and capital management. At this stage, we choose to prioritize growth because we believe that scale is central to achieving the potential of our business model.
- We will continue to focus on hiring and retaining versatile and talented employees, and continue to weight their compensation to stock options rather than cash. We know our success will be largely affected by our ability to attract and retain a motivated employee base, each of whom must think like, and therefore must actually be, an owner.

We aren't so bold as to claim that the above is the "right" investment philosophy, but it's ours, and we would be remiss if we weren't clear in the approach we have taken and will continue to take.

With this foundation, we would like to turn to a review of our business focus, our progress in 1997, and our outlook for the future.

Obsess Over Customers

From the beginning, our focus has been on offering our customers compelling value. We realized that the Web was, and still is, the World Wide Wait. Therefore, we set out to offer customers something they simply could not get any other way, and began serving them with books. We brought them much more selection than was possible in a physical store (our store would now occupy 6 football fields), and presented it in a useful, easy-to-search, and easy-to-browse format in a store open 365 days a year, 24 hours a day. We maintained a dogged focus on improving the shopping experience, and in 1997 substantially enhanced our store. We now offer customers gift certificates, 1-Click^{SM} shopping, and vastly more reviews, content, browsing options, and recommendation features. We dramatically lowered prices, further increasing customer value. Word of mouth remains the most powerful customer acquisition tool we have, and we are grateful for the trust our customers have placed in us. Repeat purchases and word of mouth have combined to make Amazon.com the market leader in online bookselling.

By many measures, Amazon.com came a long way in 1997:

- Sales grew from $15.7 million in 1996 to $147.8 million – an 838% increase.
- Cumulative customer accounts grew from 180,000 to 1,510,000 – a 738% increase.
- The percentage of orders from repeat customers grew from over 46% in the fourth quarter of 1996 to over 58% in the same period in 1997.
- In terms of audience reach, per Media Metrix, our Web site went from a rank of 90th to within the top 20.
- We established long-term relationships with many important strategic partners, including America Online, Yahoo!, Excite, Netscape, GeoCities, AltaVista, @Home, and Prodigy.

Infrastructure

During 1997, we worked hard to expand our business infrastructure to support these greatly increased traffic, sales, and service levels:

- Amazon.com's employee base grew from 158 to 614, and we significantly strengthened our management team.
- Distribution center capacity grew from 50,000 to 285,000 square feet, including a 70% expansion of our Seattle facilities and the launch of our second distribution center in Delaware in November.
- Inventories rose to over 200,000 titles at year-end, enabling us to improve availability for our customers.
- Our cash and investment balances at year-end were $125 million, thanks to our initial public offering in May 1997 and our $75 million loan, affording us substantial strategic flexibility.

Our Employees

The past year's success is the product of a talented, smart, hard-working group, and I take great pride in being a part of this team. Setting the bar high in our approach to hiring has been, and will continue to be, the single most important element of Amazon.com's success.

It's not easy to work here (when I interview people I tell them, "You can work long, hard, or smart, but at Amazon.com you can't choose two out of three"), but we are working to build something important, something that matters to our customers, something that we can all tell our grandchildren about. Such things aren't meant to be easy. We are incredibly fortunate to have this group of dedicated employees whose sacrifices and passion build Amazon.com.

Goals for 1998

We are still in the early stages of learning how to bring new value to our customers through Internet commerce and merchandising. Our goal remains to continue to solidify and extend our brand and customer base. This requires sustained investment in systems and infrastructure to support outstanding customer convenience, selection, and service while we grow. We are planning to add music to our product offering, and over time we believe that other products may be prudent investments. We also believe there are significant opportunities to better serve our customers overseas, such as reducing delivery times and better tailoring the customer experience. To be certain, a big part of the challenge for us will lie not in finding new ways to expand our business, but in prioritizing our investments.

We now know vastly more about online commerce than when Amazon.com was founded, but we still have so much to learn. Though we are optimistic, we must remain vigilant and maintain a sense of urgency. The challenges and hurdles we will face to make our long-term vision for Amazon.com a reality are several: aggressive, capable, well-funded competition; considerable growth challenges and execution risk; the risks of product and geographic expansion; and the need for large continuing investments to meet an expanding market opportunity. However, as we've long said, online bookselling, and online commerce in general, should prove to be a very large market, and it's likely that a number of companies will see significant benefit. We feel good about what we've done, and even more excited about what we want to do.

1997 was indeed an incredible year. We at Amazon.com are grateful to our customers for their business and trust, to each other for our hard work, and to our shareholders for their support and encouragement.

Jeff P Bezos

Jeffrey P. Bezos
Founder and Chief Executive Officer
Amazon.com, Inc.

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2024

or

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to .

Commission File No. 000-22513

AMAZON.COM, INC.

(Exact name of registrant as specified in its charter)

Delaware	**91-1646860**
(State or other jurisdiction of incorporation or organization)	**(I.R.S. Employer Identification No.)**

410 Terry Avenue North
Seattle, Washington 98109-5210
(206) 266-1000
(Address and telephone number, including area code, of registrant's principal executive offices)

Securities registered pursuant to Section 12(b) of the Act:

Title of Each Class	Trading Symbol(s)	Name of Each Exchange on Which Registered
Common Stock, par value $.01 per share	AMZN	Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act:
None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Exchange Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Exchange Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

Aggregate market value of voting stock held by non-affiliates of the registrant as of June 30, 2024	$	1,815,014,489,485
Number of shares of common stock outstanding as of January 29, 2025		10,597,729,352

DOCUMENTS INCORPORATED BY REFERENCE

The information required by Part III of this Report, to the extent not set forth herein, is incorporated herein by reference from the registrant's definitive proxy statement relating to the Annual Meeting of Shareholders to be held in 2025, which definitive proxy statement shall be filed with the Securities and Exchange Commission within 120 days after the end of the fiscal year to which this Report relates.

AMAZON.COM, INC.
FORM 10-K
For the Fiscal Year Ended December 31, 2024

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AMAZON.COM, INC.

PART I

Item 1. *Business*

This Annual Report on Form 10-K and the documents incorporated herein by reference contain forward-looking statements based on expectations, estimates, and projections as of the date of this filing. Actual results and outcomes may differ materially from those expressed in forward-looking statements. See Item 1A of Part I — "Risk Factors." As used herein, "Amazon.com," "we," "our," and similar terms include Amazon.com, Inc. and its subsidiaries, unless the context indicates otherwise.

General

We seek to be Earth's most customer-centric company. We are guided by four principles: customer obsession rather than competitor focus, passion for invention, commitment to operational excellence, and long-term thinking. In each of our segments, we serve our primary customer sets, consisting of consumers, sellers, developers, enterprises, content creators, advertisers, and employees.

We have organized our operations into three segments: North America, International, and Amazon Web Services ("AWS"). These segments reflect the way the Company evaluates its business performance and manages its operations. Information on our net sales is contained in Item 8 of Part II, "Financial Statements and Supplementary Data — Note 10 — Segment Information."

Consumers

We serve consumers through our online and physical stores and focus on selection, price, and convenience. We design our stores to enable hundreds of millions of unique products to be sold by us and by third parties across dozens of product categories. Customers access our offerings through our websites, mobile apps, Alexa, devices, streaming, and physically visiting our stores. We also manufacture and sell electronic devices, including Kindle, Fire tablet, Fire TV, Echo, Ring, Blink, and eero, and we develop and produce media content. We seek to offer our customers low prices, fast and free delivery, easy-to-use functionality, and timely customer service. In addition, we offer subscription services such as Amazon Prime, a membership program that includes fast, free shipping on tens of millions of items, access to award-winning movies and series, live sports, and other benefits.

We fulfill customer orders in a number of ways, including through: North America and International fulfillment networks that we operate; co-sourced and outsourced arrangements in certain countries; digital delivery; and through our physical stores. We operate customer service centers globally, which are supplemented by co-sourced arrangements. See Item 2 of Part I, "Properties."

Sellers

We offer programs that enable sellers to grow their businesses, sell their products in our stores, and fulfill orders using our services. We are not the seller of record in these transactions. We earn fixed fees, a percentage of sales, per-unit activity fees, interest, or some combination thereof, for our seller programs.

Developers and Enterprises

We serve developers and enterprises of all sizes, including start-ups, government agencies, and academic institutions, through AWS, which offers a broad set of on-demand technology services, including compute, storage, database, analytics, and machine learning, and other services.

Content Creators

We offer programs that allow authors, independent publishers, musicians, filmmakers, Twitch streamers, skill and app developers, and others to publish and sell content.

Advertisers

We provide advertising services to sellers, vendors, publishers, authors, and others, through programs such as sponsored ads, display, and video advertising.

Competition

Our businesses encompass a large variety of product types, service offerings, and delivery channels. The worldwide marketplace in which we compete is evolving rapidly and intensely competitive, and we face a broad array of competitors from many different industry sectors around the world. Our current and potential competitors include: (1) physical, e-commerce, and omnichannel retailers, publishers, vendors, distributors, manufacturers, and producers of the products we offer and sell to consumers and businesses; (2) publishers, producers, and distributors of physical, digital, and interactive media of all types and all distribution channels; (3) web search engines, comparison shopping websites, social networks, web portals, virtual assistants, and other online and app-based means of discovering, using, or acquiring goods and services, either directly or in collaboration with other retailers; (4) companies that provide e-commerce services, including website development and hosting, omnichannel sales, inventory and supply chain management, advertising, fulfillment, customer service, and payment processing; (5) companies that provide fulfillment and logistics services for themselves or for third parties, whether online or offline; (6) companies that provide information technology services or products, including on-premises or cloud-based infrastructure, tools and services relating to artificial intelligence, and other services; (7) companies that design, manufacture, market, or sell consumer electronics, communications, and other electronic devices and services; (8) companies that sell grocery products online and in physical stores; (9) companies that provide advertising services, whether in digital or other formats; and (10) providers of virtual or in-person healthcare services. We believe that the principal competitive factors in our retail businesses include selection, price, and convenience, including fast and reliable fulfillment. Additional competitive factors for our seller and enterprise services include the quality, speed, and reliability of our services and tools, as well as customers' ability and willingness to change business practices. Some of our current and potential competitors have greater resources, longer histories, more customers, greater brand recognition, and greater control over inputs critical to our various businesses. They may secure better terms from suppliers, adopt more aggressive pricing, pursue restrictive distribution agreements that restrict our access to supply, direct consumers to their own offerings instead of ours, lock-in potential customers with restrictive terms, and devote more resources to technology, infrastructure, fulfillment, and marketing. The internet facilitates competitive entry and comparison shopping, which enhances the ability of new, smaller, or lesser-known businesses to compete against us. Each of our businesses is also subject to rapid change and the development of new business models and the entry of new and well-funded competitors. Other companies also may enter into business combinations or alliances that strengthen their competitive positions.

Intellectual Property

We regard our trademarks, service marks, copyrights, patents, domain names, trade dress, trade secrets, proprietary technologies, and similar intellectual property as critical to our success, and we rely on trademark, copyright, and patent law, trade-secret protection, and confidentiality and/or license agreements with our employees, customers, partners, and others to protect our proprietary rights. We have registered, or applied for the registration of, a number of U.S. and international domain names, trademarks, service marks, and copyrights. Additionally, we have filed U.S. and international patent applications covering certain of our proprietary technology.

Seasonality

Our business is affected by seasonality, which historically has resulted in higher sales volume during our fourth quarter, which ends December 31.

Human Capital

Our employees are critical to our mission of being Earth's most customer-centric company. As of December 31, 2024, we employed approximately 1,556,000 full-time and part-time employees. Additionally, we use independent contractors and temporary personnel to supplement our workforce. Competition for qualified personnel is intense, particularly for software engineers, computer scientists, and other technical staff, and constrained labor markets have increased competition for personnel across other parts of our business.

We strive to be Earth's best employer. We rely on numerous and evolving initiatives to implement this objective and invent mechanisms for talent development, including competitive pay and benefits, flexible work arrangements, and skills training and educational programs such as Amazon Career Choice (education funding for eligible employees). Over 240,000 Amazon employees around the world have participated in Career Choice. We also continue to inspect and refine the mechanisms we use to hire, develop, evaluate, and retain our employees. In addition, safety is integral to everything we do at Amazon and we continue to invest in safety improvements such as capital improvements, new safety technology, vehicle safety controls, and engineering ergonomic solutions. Our safety team is dedicated to using the science of safety to solve complex problems and establish new industry best practices. We also provide mentorship and support resources to our employees, and have deployed numerous programs that advance employee engagement, communication, and feedback.

Available Information

Our investor relations website is amazon.com/ir and we encourage investors to use it as a way of easily finding information about us. We promptly make available on this website, free of charge, the reports that we file or furnish with the Securities and Exchange Commission ("SEC"), corporate governance information (including our Code of Business Conduct and Ethics), and select press releases.

Executive Officers and Directors

The following tables set forth certain information regarding our Executive Officers and Directors as of January 29, 2025:

Information About Our Executive Officers

Name	Age	Position
Jeffrey P. Bezos	61	Executive Chair
Andrew R. Jassy	57	President and Chief Executive Officer
Matthew S. Garman	48	CEO Amazon Web Services
Douglas J. Herrington	58	CEO Worldwide Amazon Stores
Brian T. Olsavsky	61	Senior Vice President and Chief Financial Officer
Shelley L. Reynolds	60	Vice President, Worldwide Controller, and Principal Accounting Officer
David A. Zapolsky	61	Senior Vice President, Global Public Policy and General Counsel

Jeffrey P. Bezos. Mr. Bezos founded Amazon.com in 1994 and has served as Executive Chair since July 2021. He has served as Chair of the Board since 1994 and served as Chief Executive Officer from May 1996 until July 2021, and as President from 1994 until June 1999 and again from October 2000 to July 2021.

Andrew R. Jassy. Mr. Jassy has served as President and Chief Executive Officer since July 2021, CEO Amazon Web Services from April 2016 until July 2021, and Senior Vice President, Amazon Web Services, from April 2006 until April 2016.

Matthew S. Garman. Mr. Garman has served as CEO Amazon Web Services since June 2024, Senior Vice President, Amazon Web Services from February 2021 until June 2024, Vice President, Marketing, Sales and Support of Amazon Web Services from January 2020 to February 2021, Vice President, AWS Compute Services from September 2018 to January 2020, and Vice President, EC2 from December 2012 to September 2018.

Douglas J. Herrington. Mr. Herrington has served as CEO Worldwide Amazon Stores since July 2022, Senior Vice President, North America Consumer from January 2015 to July 2022, Senior Vice President, Consumables from May 2014 to December 2014, and Vice President, Consumables from May 2005 to April 2014.

Brian T. Olsavsky. Mr. Olsavsky has served as Senior Vice President and Chief Financial Officer since June 2015, Vice President, Finance for the Global Consumer Business from December 2011 to June 2015, and numerous financial leadership roles across Amazon with global responsibility since April 2002.

Shelley L. Reynolds. Ms. Reynolds has served as Vice President, Worldwide Controller, and Principal Accounting Officer since April 2007.

David A. Zapolsky. Mr. Zapolsky has served as Senior Vice President, Global Public Policy and General Counsel since May 2023. He served as our Secretary from September 2012 to January 2024, Senior Vice President and General Counsel from May 2014 to May 2023, Vice President and General Counsel from September 2012 to May 2014, and as Vice President and Associate General Counsel for Litigation and Regulatory matters from April 2002 until September 2012.

Board of Directors

Name	Age	Position
Jeffrey P. Bezos	61	Executive Chair
Andrew R. Jassy	57	President and Chief Executive Officer
Keith B. Alexander	73	Former Chair and CEO of IronNet, Inc.
Edith W. Cooper	63	Former Executive Vice President, Goldman Sachs Group, Inc.
Jamie S. Gorelick	74	Partner, Wilmer Cutler Pickering Hale and Dorr LLP
Daniel P. Huttenlocher	66	Dean, MIT Schwarzman College of Computing
Andrew Y. Ng	48	Managing General Partner, AI Fund, L.P.
Indra K. Nooyi	69	Former Chair and CEO, PepsiCo, Inc.
Jonathan J. Rubinstein	68	Former co-CEO, Bridgewater Associates, LP
Brad D. Smith	60	President, Marshall University
Patricia Q. Stonesifer	68	Former President and Chief Executive Officer, Martha's Table
Wendell P. Weeks	65	Chairman and CEO, Corning Incorporated

Item 1A. *Risk Factors*

Please carefully consider the following discussion of significant factors, events, and uncertainties that make an investment in our securities risky. The events and consequences discussed in these risk factors could, in circumstances we may or may not be able to accurately predict, recognize, or control, have a material adverse effect on our business, growth, reputation, prospects, financial condition, operating results (including components of our financial results), cash flows, liquidity, and stock price. These risk factors do not identify all risks that we face; our operations could also be affected by factors, events, or uncertainties that are not presently known to us or that we currently do not consider to present significant risks to our operations. In addition to the factors discussed in Item 7 of Part II, "Management's Discussion and Analysis of Financial Condition and Results of Operations," and in the risk factors below, global economic and geopolitical conditions and additional or unforeseen circumstances, developments, or events may give rise to or amplify many of the risks discussed below. Many of the risks discussed below also impact our customers, including third-party sellers, which could indirectly have a material adverse effect on us.

Business and Industry Risks

We Face Intense Competition

Our businesses are rapidly evolving and intensely competitive, and we have many competitors across geographies, including cross-border competition, and in different industries, including physical, e-commerce, and omnichannel retail, e-commerce services, web and infrastructure computing services, electronic devices, digital content, advertising, grocery, healthcare, communications, and transportation and logistics services. Some of our current and potential competitors have greater resources, longer histories, more customers, and/or greater brand recognition, particularly with our newly-launched products and services and in our newer geographic regions. They may secure better terms from vendors, adopt more aggressive pricing, and devote more resources to technology, infrastructure, fulfillment, and marketing.

Competition continues to intensify, including with the development of new business models and the entry of new and well-funded competitors, and as our competitors enter into business combinations or alliances and established companies in other market segments expand to become competitive with our business. In addition, new and enhanced technologies, including search, web and infrastructure computing services, practical applications of artificial intelligence and machine learning, digital content, satellites, and electronic devices continue to increase our competition. The internet facilitates competitive entry and comparison shopping, which enhances the ability of new, smaller, or lesser known businesses to compete against us. As a result of competition, our product and service offerings may not be successful, we may fail to gain or may lose business, and we may be required to increase our spending or lower prices, any of which could materially reduce our sales and profits.

Our Expansion into New Products, Services, Technologies, and Geographic Regions Subjects Us to Additional Risks

We may have limited or no experience in our newer market segments, and our customers may not adopt our product or service offerings. These offerings, which can present new and difficult technology challenges, may subject us to claims if customers of these offerings experience, or are otherwise impacted by, service disruptions, delays, setbacks, or failures or quality issues. In addition, profitability or other intended benefits, if any, in our newer activities may not meet our expectations, and we may not be successful enough in these newer activities to recoup our investments in them, which investments are often significant. Failure to realize the benefits of amounts we invest in new technologies, products, or services could result in the value of those investments being written down or written off. In addition, our sustainability initiatives may be unsuccessful for a variety of reasons, including if we are unable to realize the expected benefits of new technologies or if we do not successfully plan or execute new strategies, which could harm our business or damage our reputation.

Our International Operations Expose Us to a Number of Risks

Our international activities are significant to our revenues and profits, and we plan to further expand internationally. In certain international market segments, we have relatively little operating experience and may not benefit from any first-to-market advantages or otherwise succeed. It is costly to establish, develop, and maintain international operations and stores, and promote our brand internationally. Our international operations may not become profitable on a sustained basis.

In addition to risks described elsewhere in this section, our international sales and operations are subject to a number of risks, including:

- local economic and political conditions;
- government regulation (such as regulation of our product and service offerings and of competition); restrictive governmental actions (such as trade protection measures, including export duties and quotas and custom duties and tariffs, and restrictions around the import and export of certain products, technologies, and components); nationalization; and restrictions on foreign ownership;
- restrictions on sales or distribution of certain products or services and uncertainty regarding liability for products, services, and content, including uncertainty as a result of less internet-friendly legal systems, local laws, lack of legal precedent, and varying rules, regulations, and practices regarding the physical and digital distribution of media products and enforcement of intellectual property rights;
- business licensing or certification requirements, such as for imports, exports, web services, electronic devices, and communications services;
- limitations on the repatriation and investment of funds and foreign currency exchange restrictions;
- limited fulfillment and technology infrastructure;
- shorter payable and longer receivable cycles and the resultant negative impact on cash flow;
- laws and regulations regarding privacy, data use, data protection, data security, data localization, network security, consumer protection, payments, advertising, and restrictions on pricing or discounts;
- lower levels of use of the internet;
- lower levels of consumer spending and fewer opportunities for growth compared to the U.S.;
- lower levels of credit card usage and increased payment risk;
- difficulty in staffing, developing, and managing foreign operations as a result of distance, language, and cultural differences;
- different employee/employer relationships and the existence of works councils and labor unions;
- compliance with the U.S. Foreign Corrupt Practices Act and other applicable U.S. and foreign laws prohibiting corrupt payments to government officials and other third parties;
- laws and policies of the U.S. and other jurisdictions affecting trade, foreign investment, loans, and taxes; and
- geopolitical events, including war and terrorism.

As international physical, e-commerce, and omnichannel retail, cloud services, and other services grow, competition will intensify, including through adoption of evolving business models. Local companies may have a substantial competitive advantage because of their greater understanding of, and focus on, the local customer, as well as their more established local brand names. The inability to hire, train, retain, and manage sufficient required personnel may limit our international growth.

The People's Republic of China ("PRC") and India regulate Amazon's and its affiliates' businesses and operations in country through regulations and license requirements that may restrict (i) foreign investment in and operation of the internet, IT infrastructure, data centers, retail, delivery, and other sectors, (ii) internet content, and (iii) the sale of media and other products and services. For example, in order to meet local ownership, regulatory licensing, and cybersecurity requirements, we provide certain technology services in China through contractual relationships with third parties that hold PRC licenses to provide services. In India, the government restricts the ownership or control of Indian companies by foreign entities involved in online multi-brand retail trading activities. For www.amazon.in, we provide certain marketing tools and logistics services to third-party sellers to enable them to sell online and deliver to customers, and we hold an indirect minority interest in an entity that is a third-party seller on the www.amazon.in marketplace. Although we believe these structures and activities comply with existing laws, they involve unique risks, and the PRC and India may from time to time consider and implement additional changes in their regulatory, licensing, or other requirements that could impact these structures and activities. There are substantial uncertainties regarding the interpretation of PRC and Indian laws and regulations, and it is possible that these governments will ultimately take a view contrary to ours. In addition, our Chinese and Indian businesses and operations may be unable to continue to operate if we or our affiliates are unable to access sufficient funding or, in China, enforce contractual relationships we or our affiliates have in place. Violation of any existing or future PRC, Indian, or other laws or regulations or changes in the interpretations of those laws and regulations could result in our businesses in those countries being subject to fines and other financial penalties, having licenses revoked, or being forced to restructure our operations or shut down entirely.

In addition, because China-based sellers account for significant portions of our third-party seller services and advertising revenues, and China-based suppliers provide significant portions of our components and finished goods, regulatory and trade restrictions, data protection and cybersecurity laws, economic factors, geopolitical events, security issues, or other factors negatively impacting China-based sellers and suppliers could adversely affect our operating results.

The Variability in Our Retail Business Places Increased Strain on Our Operations

Demand for our products and services can fluctuate significantly for many reasons, including as a result of seasonality, promotions, product launches, or unforeseeable events, such as in response to global economic conditions such as recessionary fears or rising inflation, natural or human-caused disasters (including public health crises) or extreme weather (including as a result of climate change), or geopolitical events. For example, we expect a disproportionate amount of our retail sales to occur during our fourth quarter. Our failure to stock or restock popular products in sufficient amounts such that we fail to meet customer demand could significantly affect our revenue and our future growth. When we overstock products, we may be required to take significant inventory markdowns or write-offs and incur commitment costs, which could materially reduce profitability. We regularly experience increases in our net shipping cost due to complimentary upgrades, split-shipments, and additional long-zone shipments necessary to ensure timely delivery for the holiday season. If too many customers access our websites within a short period of time due to increased demand, we may experience system interruptions that make our websites unavailable or prevent us from efficiently fulfilling orders, which may reduce the volume of goods we offer or sell and the attractiveness of our products and services. In addition, we may be unable to adequately staff our fulfillment network and customer service centers during these peak periods and delivery and other fulfillment companies and customer service co-sourcers may be unable to meet the seasonal demand. Risks described elsewhere in this Item 1A relating to fulfillment network optimization and inventory are magnified during periods of high demand.

As a result of holiday sales, as of December 31 of each year, our cash, cash equivalents, and marketable securities balances typically reach their highest level (other than as a result of cash flows provided by or used in investing and financing activities) because consumers primarily use credit cards in our stores and the related receivables settle quickly. Typically, there is also a corresponding increase in accounts payable as of December 31 due to inventory purchases and third-party seller sales. Our accounts payable balance generally declines during the first three months of the year as vendors and sellers are paid, resulting in a corresponding decline in our cash, cash equivalents, and marketable securities balances.

We Are Impacted by Fraudulent or Unlawful Activities of Sellers

The law relating to the liability of online service providers is currently unsettled. In addition, governmental agencies have in the past and could in the future require changes in the way this business is conducted. Under our seller programs, we maintain policies and processes designed to prevent sellers from collecting payments, fraudulently or otherwise, when buyers never receive the products they ordered or when the products received are materially different from the sellers' descriptions, and to prevent sellers in our stores or through other stores from selling unlawful, counterfeit, pirated, or stolen goods, selling goods in an unlawful or unethical manner, violating the proprietary rights of others, or otherwise violating our policies. When these policies and processes are circumvented or fail to operate sufficiently, it can harm our business or damage our reputation and we could face civil or criminal liability for unlawful activities by our sellers. Under our A-to-z Guarantee, we may reimburse customers for payments up to certain limits in these situations, and as our third-party seller sales grow, the cost of this program will increase and could negatively affect our operating results.

We Face Risks Related to Adequately Protecting Our Intellectual Property Rights and Being Accused of Infringing Intellectual Property Rights of Third Parties

We regard our trademarks, service marks, copyrights, patents, trade dress, trade secrets, proprietary technology, and similar intellectual property as critical to our success, and we rely on trademark, copyright, and patent law, trade secret protection, and confidentiality and/or license agreements with our employees, customers, and others to protect our proprietary rights. Effective intellectual property protection is not available in every country in which our products and services are made available. We also may not be able to acquire or maintain appropriate domain names in all countries in which we do business. Furthermore, regulations governing domain names may not protect our trademarks and similar proprietary rights. We may be unable to prevent third parties from acquiring domain names that are similar to, infringe upon, or diminish the value of our trademarks and other proprietary rights.

We are not always able to discover or determine the extent of any unauthorized use of our proprietary rights. Actions taken by third parties that license our proprietary rights may materially diminish the value of our proprietary rights or reputation. The protection of our intellectual property requires the expenditure of significant financial and managerial resources. Moreover, the steps we take to protect our intellectual property do not always adequately protect our rights or prevent third parties from infringing or misappropriating our proprietary rights. We also cannot be certain that others will not independently develop or otherwise acquire equivalent or superior technology or other intellectual property rights.

We have been subject to, and expect to continue to be subject to, claims and legal proceedings regarding alleged infringement by us of the intellectual property rights of third parties. Such claims, whether or not meritorious, have in the past, and may in the future, result in the expenditure of significant financial and managerial resources, injunctions against us, or significant payments for damages, including to satisfy indemnification obligations or to obtain licenses from third parties who allege that we have infringed their rights. Such licenses may not be available on terms acceptable to us or at all. These risks have been amplified by the increase in third parties whose sole or primary business is to assert such claims. In addition, our and our customers' development and use of artificial intelligence may result in increased claims of infringement or other claims, including those based on unauthorized use of third-party technology or content.

Our digital content offerings depend in part on effective digital rights management technology to control access to digital content. Breach or malfunctioning of the digital rights management technology that we use could subject us to claims, and content providers may be unwilling to include their content in our service.

We Have Foreign Exchange Risk

The results of operations of, and certain of our intercompany balances associated with, our international stores and product and service offerings are exposed to foreign exchange rate fluctuations. Due to these fluctuations, operating results may differ materially from expectations, and we may record significant gains or losses on the remeasurement of intercompany balances. As we have expanded our international operations, our exposure to exchange rate fluctuations has increased. We also hold cash equivalents and/or marketable securities in foreign currencies such as British Pounds, Canadian Dollars, Euros, and Japanese Yen. When the U.S. Dollar strengthens compared to these currencies, cash equivalents, and marketable securities balances, when translated, may be materially less than expected and vice versa.

Operating Risks

Our Expansion Places a Significant Strain on our Management, Operational, Financial, and Other Resources

We are continuing to rapidly and significantly expand our global operations, including increasing our product and service offerings and scaling our infrastructure to support our retail and services businesses. The complexity of the current scale of our business can place significant strain on our management, personnel, operations, systems, technical performance, financial resources, and internal financial control and reporting functions, and our expansion increases these factors. Failure to manage growth effectively could damage our reputation, limit our growth, and negatively affect our operating results.

We Experience Significant Fluctuations in Our Operating Results and Growth Rate

We are not always able to accurately forecast our growth rate. We base our expense levels and investment plans on sales estimates. A significant portion of our expenses and investments is fixed, and we are not always able to adjust our spending quickly enough if our sales are less than expected.

Our revenue growth may not be sustainable, and our percentage growth rates may decrease. Our revenue and operating profit growth depends on the continued growth of demand for the products and services offered by us or our sellers, and our business is affected by, among other things, general economic, business, and geopolitical conditions worldwide. A softening of demand, whether caused by changes in customer preferences or a weakening of the U.S. or global economies, may result in decreased revenue or growth.

Our sales and operating results will also fluctuate for many other reasons, including due to factors described elsewhere in this section and the following:

- our ability to retain and increase sales to existing customers, attract new customers, and satisfy our customers' demands;
- our ability to retain and expand our network of sellers;
- our ability to offer products on favorable terms, manage inventory, and fulfill orders;
- the introduction of competitive stores, websites, products, services, price decreases, or improvements;

- changes in usage or adoption rates of the internet, e-commerce, electronic devices, web services, satellite communications services, and artificial intelligence and machine learning technologies, products, and services, including outside the U.S.;
- timing, effectiveness, and costs of expansion and upgrades of our systems and infrastructure;
- the success of our geographic, service, and product line expansions;
- the extent to which we finance, and the terms of any such financing for, our current operations and future growth;
- the outcomes of legal proceedings and claims, which may include significant monetary damages or injunctive relief and could have a material adverse impact on our operating results;
- variations in the mix of products and services we sell;
- variations in our level of merchandise and vendor returns;
- the extent to which we offer fast and free delivery, continue to reduce prices worldwide, and provide additional benefits to our customers;
- factors affecting our reputation or brand image (including any actual or perceived inability to achieve our goals or commitments, whether related to sustainability, customers, employees, or other topics), and public perceptions regarding our positions on social or ethical issues and our development and use of artificial intelligence, machine learning, and automation technologies, products, and services;
- the extent to which we invest in technology and infrastructure, fulfillment, and other expense categories;
- availability of and increases in the prices of transportation (including fuel), resources such as land, water, and energy, commodities like paper and packing supplies and hardware products, and technology infrastructure products, including as a result of inflationary pressures;
- constrained labor markets, which increase our payroll costs;
- the extent to which operators of the networks between our customers and our stores successfully charge fees to grant our customers unimpaired and unconstrained access to our online services;
- our ability to collect amounts owed to us when they become due;
- the extent to which new and existing technologies, or industry trends, restrict online advertising or affect our ability to customize advertising or otherwise tailor our product and service offerings;
- the extent to which use of our services is affected by spyware, viruses, phishing and other spam emails, denial of service attacks, data theft, computer intrusions, outages, and similar events;
- the extent to which we fail to maintain our unique culture of innovation, customer obsession, and long-term thinking, which has been critical to our growth and success;
- disruptions from natural or human-caused disasters (including public health crises) or extreme weather (including as a result of climate change), geopolitical events and security issues (including terrorist attacks, armed hostilities, and political conflicts, including those involving China), labor or trade disputes (including restrictive governmental actions impacting us, our customers, and our third-party sellers and suppliers in China or other foreign countries), tariff policy changes, and similar events; and
- potential negative impacts of climate change, including: increased operating costs due to more frequent extreme weather events or climate-related changes, such as rising temperatures and water scarcity; increased investment requirements associated with the transition to a low-carbon economy; decreased demand for our products and services as a result of changes in customer behavior; increased compliance costs due to more extensive and global regulations and third-party requirements; and reputational damage resulting from perceptions of our environmental impact.

We Face Risks Related to Successfully Optimizing and Operating Our Fulfillment Network and Data Centers

Failures to adequately predict customer demand and consumer spending patterns or otherwise optimize and operate our fulfillment network and data centers successfully from time to time result in excess or insufficient fulfillment or data center capacity, service interruptions, increased costs, and impairment charges, any of which could materially harm our business. As we continue to add fulfillment and data center capability or add new businesses with different requirements, our fulfillment and data center networks become increasingly complex and operating them becomes more challenging. There can be no assurance that we will be able to operate our networks effectively.

In addition, failure to optimize inventory management or staffing in our fulfillment network increases our net shipping cost by increasing the distance products are shipped and reducing the number of units per shipment or delivery. We and our co-sourcers may be unable to adequately staff our fulfillment network and customer service centers. For example, productivity across our fulfillment network is affected by regional labor market constraints, which increase payroll costs and make it difficult to hire, train, and deploy a sufficient number of people to operate our fulfillment network as efficiently as we would like.

Under some of our commercial agreements, we maintain the inventory of other companies, thereby increasing the complexity of tracking inventory and operating our fulfillment network. Our failure to adequately predict seller demand for storage or to properly handle such inventory or the inability of the other businesses on whose behalf we perform inventory fulfillment services to accurately forecast product demand may result in us being unable to secure sufficient storage space or to optimize our fulfillment network or cause other unexpected costs and other harm to our business and reputation.

We rely on a limited number of shipping companies to deliver inventory to us and completed orders to our customers. An inability to negotiate acceptable terms with these companies or performance problems, staffing limitations, or other difficulties experienced by these companies or by our own transportation systems, including as a result of labor market constraints and related costs, could negatively impact our operating results and customer experience. In addition, our ability to receive inbound inventory efficiently and ship completed orders to customers also may be negatively affected by natural or human-caused disasters (including public health crises) or extreme weather (including as a result of climate change), geopolitical events and security issues, labor or trade disputes, and similar events.

We Could Be Harmed by Data Loss or Other Security Incidents

Because we collect, process, store, and transmit large amounts of data, including confidential, classified, sensitive, proprietary, and business and personal information, failure to prevent, detect, or mitigate data loss, theft, misuse, unauthorized access, or other security incidents or vulnerabilities affecting our or our vendors' or customers' technology, products, and systems, could: expose us or our customers to a risk of loss, disclosure, or misuse of such information; adversely affect our operating results; result in litigation, liability, or regulatory action (including under laws related to privacy, data use, data protection, data security, network security, and consumer protection); deter customers or sellers from using our stores, products, and services; and otherwise harm our business and reputation. We use third-party technology and systems for a variety of reasons, including, without limitation, encryption and authentication technology, employee email, content delivery to customers, back-office support, and other functions. Some of our systems have experienced past security incidents, and, although they did not have a material adverse effect on our operating results, there can be no assurance that future incidents will not have material adverse effects on our operations or financial results. Although we have developed systems and processes that are designed to protect customer data and prevent, detect, or mitigate such incidents, including systems and processes designed to reduce the impact of a security incident at a third-party vendor or customer, such measures cannot provide absolute security and may fail to operate as intended or be circumvented.

We Face Risks Related to System Interruption and Lack of Redundancy

We experience occasional system interruptions and delays that make our websites and services unavailable or slow to respond and prevent us from efficiently accepting or fulfilling orders or providing services to customers and third parties, which may reduce our net sales and the attractiveness of our products and services. Steps we take to add software and hardware, upgrade our systems and network infrastructure, and improve the stability and efficiency of our systems may not be sufficient to avoid system interruptions or delays that could adversely affect our operating results.

Our computer and communications systems and operations in the past have been, or in the future could be, damaged or interrupted due to events such as natural or human-caused disasters (including public health crises) or extreme weather (including as a result of climate change), geopolitical events and security issues (including terrorist attacks and armed hostilities), computer viruses, physical or electronic break-ins, operational failures (including from energy shortages), and similar events or disruptions. Any of these events could cause system interruption, delays, and loss of critical data, and could prevent us from accepting and fulfilling customer orders and providing services, which could make our product and service offerings less attractive and subject us to liability. Our systems are not fully redundant and our disaster recovery planning may not be sufficient. In addition, our insurance may not provide sufficient coverage to compensate for related losses. Any of these events could damage our reputation and be expensive to remedy.

The Loss of Key Senior Management Personnel or the Failure to Hire and Retain Highly Skilled and Other Personnel Could Negatively Affect Our Business

We depend on our senior management and other key personnel, including our President and CEO. We do not have "key person" life insurance policies. We also rely on other highly skilled personnel. Competition for qualified personnel in the industries in which we operate, as well as senior management, has historically been intense. For example, we experience significant competition in the technology industry, particularly for software engineers, computer scientists, and other technical staff. In addition, changes we make to our current and future work environments may not meet the needs or expectations of our employees or may be perceived as less favorable compared to other companies' policies, which could negatively impact our ability to hire and retain qualified personnel. The loss of any of our executive officers or other key employees, the failure to successfully transition key roles, or the inability to hire, train, retain, and manage qualified personnel, could harm our business.

We also rely on a significant number of personnel to operate our stores, fulfillment network, and data centers and carry out our other operations. Failure to successfully hire, train, manage, and retain sufficient personnel to meet our needs can strain our operations, increase payroll and other costs, and harm our business and reputation. In addition, changes in laws and regulations applicable to employees, independent contractors, and temporary personnel could increase our payroll costs, decrease our operational flexibility, and negatively impact how we are able to staff our operations and supplement our workforce.

We are also subject to labor union efforts to organize groups of our employees from time to time. These organizational efforts, if successful, decrease our operational flexibility, which could adversely affect our operating efficiency. In addition, our response to any organizational efforts could be perceived negatively and harm our business and reputation.

Our Supplier Relationships Subject Us to a Number of Risks

We have significant suppliers, including content and technology licensors, and in some cases, limited or single-sources of supply, that are important to our sourcing, services, manufacturing, and any related ongoing servicing of merchandise and content. We do not have long-term arrangements with most of our suppliers to guarantee availability of merchandise, content, components, or services, particular payment terms, or the extension of credit limits. Decisions by our current suppliers to limit or stop selling or licensing merchandise, content, components, or services to us on acceptable terms, or delay delivery, including as a result of one or more supplier bankruptcies due to poor economic conditions, as a result of natural or human-caused disasters (including public health crises) or geopolitical events, or for other reasons, may result in our being unable to procure alternatives from other suppliers in a timely and efficient manner and on acceptable terms, or at all. For example, we rely on a limited group of suppliers for semiconductor products, including products related to artificial intelligence infrastructure such as graphics processing units. Constraints on the availability of these products could adversely affect our ability to develop and operate artificial intelligence technologies, products, or services. In addition, violations by our suppliers or other vendors of applicable laws, regulations, contractual terms, intellectual property rights of others, or our Supply Chain Standards, as well as products or practices regarded as unethical, unsafe, or hazardous, could expose us to claims, damage our reputation, limit our growth, and negatively affect our operating results.

Our Commercial Agreements, Strategic Alliances, and Other Business Relationships Expose Us to Risks

We provide physical, e-commerce, and omnichannel retail, cloud services, and other services to businesses through commercial agreements, strategic alliances, and business relationships. Under these agreements, we provide web services, technology, fulfillment, computing, digital storage, and other services, as well as enable sellers to offer products or services through our stores. These arrangements are complex and require substantial infrastructure capacity, personnel, and other resource commitments, which may limit the amount of business we can service. We may not be able to implement, maintain, and develop the components of these commercial relationships, which may include web services, fulfillment, customer service, inventory management, tax collection, payment processing, hardware, content, and third-party software, and engaging third parties to perform services. The amount of compensation we receive under certain of our commercial agreements is partially dependent on the volume of the other company's sales. Therefore, when the other company's offerings are not successful, the compensation we receive may be lower than expected or the agreement may be terminated. Moreover, we may not be able to enter into additional or alternative commercial relationships and strategic alliances on favorable terms. We also may be subject to claims from businesses to which we provide these services if we are unsuccessful in implementing, maintaining, or developing these services.

As our agreements terminate, we may be unable to renew or replace these agreements on comparable terms, or at all. We may in the future enter into amendments on less favorable terms or encounter parties that have difficulty meeting their contractual obligations to us, which could adversely affect our operating results.

Our present and future commercial agreements, strategic alliances, and business relationships create additional risks such as:

- disruption of our ongoing business, including loss of management focus on existing businesses;
- impairment of other relationships;
- variability in revenue and income from entering into, amending, or terminating such agreements or relationships; and
- difficulty integrating under the commercial agreements.

Our Business Suffers When We Are Unsuccessful in Making, Integrating, and Maintaining Acquisitions and Investments

We have acquired and invested in a number of companies, and we may in the future acquire or invest in or enter into joint ventures with additional companies. These transactions involve risks such as:

- disruption of our ongoing business, including loss of management focus on existing businesses;
- problems retaining key personnel;
- additional operating losses and expenses of the businesses we acquired or in which we invested;
- the potential impairment of tangible and intangible assets and goodwill, including as a result of acquisitions;
- the potential impairment of customer and other relationships of the company we acquired or in which we invested or our own customers as a result of any integration of operations;
- the difficulty of completing such transactions, including obtaining regulatory approvals or satisfying other closing conditions, and achieving anticipated benefits within expected timeframes, or at all;
- the difficulty of incorporating acquired operations, technology, and rights into our offerings, and unanticipated expenses related to such integration;
- the difficulty of integrating a new company's accounting, financial and sustainability reporting, management, information and data security, human resource, and other administrative systems to permit effective management, and the lack of control if such integration is delayed or not successfully implemented;
- losses we may incur as a result of declines in the value of an investment or as a result of incorporating an investee's financial performance into our financial results;
- for investments in which an investee's financial performance is incorporated into our financial results, either in full or in part, or investments for which we are required to file financial statements or provide financial information, the dependence on the investee's accounting, financial and sustainability reporting, and similar systems, controls, and processes;
- the difficulty of implementing at companies we acquire the controls, procedures, and policies appropriate for a larger public company;
- the risks associated with businesses we acquire or invest in, which may differ from or be more significant than the risks our other businesses face;
- potential unknown liabilities associated with a company we acquire or in which we invest; and
- for foreign transactions, additional risks related to the integration of operations across different cultures and languages, and the economic, political, and regulatory risks associated with specific countries.

As a result of future acquisitions or mergers, we might need to issue additional equity securities, spend our cash, or incur debt, contingent liabilities, or amortization expenses related to intangible assets, any of which could reduce our profitability and harm our business or only be available on unfavorable terms, if at all. In addition, valuations supporting our acquisitions and strategic investments could change rapidly. We could determine that such valuations have experienced impairments or other-than-temporary declines in fair value which could adversely impact our financial results.

We Face Significant Inventory Risk

In addition to risks described elsewhere in this Item 1A relating to fulfillment network and inventory optimization by us and third parties, we are exposed to significant inventory risks that may adversely affect our operating results as a result of seasonality, new product launches, rapid changes in product cycles and pricing, defective merchandise, changes in customer demand and consumer spending patterns, changes in consumer tastes with respect to our products, spoilage, and other factors. We endeavor to accurately predict these trends and avoid overstocking or understocking products we manufacture and/or sell. Demand for products, however, can change significantly between the time inventory or components are ordered and the date of sale. In addition, when we begin selling or manufacturing a new product or offering a new service, it may be difficult to establish vendor relationships, determine appropriate product or component selection, and accurately forecast demand. The acquisition of certain types of inventory or components requires significant lead-time and prepayment and they may not be returnable. We carry a broad selection and significant inventory levels of certain products, such as consumer electronics, and at times we are unable to sell products in sufficient quantities or to meet demand during the relevant selling seasons. Any one of the inventory risk factors set forth above may adversely affect our operating results.

We Are Subject to Payments-Related Risks

We accept payments using a variety of methods, including credit card, debit card, credit accounts (including promotional financing), gift cards, direct debit from a customer's bank account, consumer invoicing, checks, and payment upon delivery. For existing and future payment options we offer to our customers, we currently are subject to, and may become subject to additional, regulations and compliance requirements (including obligations to implement enhanced authentication processes that could result in significant costs and reduce the ease of use of our payments products), as well as fraud. For certain payment methods, including credit and debit cards, we pay interchange and other fees, which may increase over time and raise our operating costs and lower profitability. We rely on third parties to provide certain Amazon-branded payment methods and payment processing services, including the processing of credit cards, debit cards, electronic checks, and promotional financing. In each case, it could disrupt our business if these companies become unwilling or unable to provide these services to us. We also offer co-branded credit card programs, which could adversely affect our operating results if renewed on less favorable terms or terminated. We are also subject to payment card association operating rules, including data security rules, certification requirements, and rules governing electronic funds transfers, which could change or be reinterpreted to make it difficult or impossible for us to comply. Failure to comply with these rules or requirements, as well as any breach, compromise, or failure to otherwise detect or prevent fraudulent activity involving our data security systems, could result in our being liable for card issuing banks' costs, subject to fines and higher transaction fees, and loss of our ability to accept credit and debit card payments from our customers, process electronic funds transfers, or facilitate other types of online payments, and our business and operating results could be adversely affected.

In addition, we provide regulated services in certain jurisdictions because we enable customers to keep account balances with us and transfer money to third parties, and because we provide services to third parties to facilitate payments on their behalf. Jurisdictions subject us to requirements for licensing, regulatory inspection, bonding and capital maintenance, the use, handling, and segregation of transferred funds, consumer disclosures, maintaining or processing data, and authentication. We are also subject to or voluntarily comply with a number of other laws and regulations relating to payments, money laundering, international money transfers, privacy, data use, data protection, data security, data localization, network security, consumer protection, and electronic fund transfers. If we were found to be in violation of applicable laws or regulations, we could be subject to additional requirements and civil and criminal penalties, or forced to cease providing certain services.

We Have a Rapidly Evolving Business Model and Our Stock Price Is Highly Volatile

We have a rapidly evolving business model. The trading price of our common stock fluctuates significantly in response to, among other risks, the risks described elsewhere in this Item 1A, as well as:

- changes in interest rates;
- conditions or trends in the internet and the industry segments we operate in;
- quarterly variations in operating results;
- fluctuations in the stock market in general and market prices for internet-related companies in particular;
- changes in financial estimates by us or decisions to increase or decrease future spending or investment levels;
- changes in financial estimates and recommendations by securities analysts;
- changes in our capital structure, including issuance of additional debt or equity to the public;
- changes in the valuation methodology of, or performance by, other e-commerce or technology companies; and
- transactions in our common stock by major investors and certain analyst reports, news, social media activity, and speculation.

Volatility in our stock price could adversely affect our business and financing opportunities and force us to increase our cash compensation to employees or grant larger stock awards than we have historically, which could hurt our operating results or reduce the percentage ownership of our existing stockholders, or both.

Legal and Regulatory Risks

Government Regulation Is Evolving and Unfavorable Changes Could Harm Our Business

We are subject to general business regulations and laws, as well as regulations and laws specifically governing the internet, physical, e-commerce, and omnichannel retail, digital content, web services, electronic devices, advertising, artificial intelligence technologies and services, satellite communications services, healthcare, and other products and services that we offer or sell. These regulations and laws cover taxation, privacy, data use, data protection, data security, data localization, network security, consumer protection, pricing, content, copyrights, distribution, transportation, communications, electronic device certification, electronic waste, energy consumption, environmental and climate-related regulation, electronic contracts and other communications, competition, employment, trade and protectionist measures, web services, the provision of online payment services, registration, licensing, and information reporting requirements, unencumbered internet access to our services or access to our facilities, the design and operation of websites, health, safety, and sanitation standards, the characteristics, legality, and quality of products and services, product labeling, the commercial operation of unmanned aircraft systems, healthcare, and other matters. It is not clear how existing laws governing issues such as property ownership, libel, privacy, data use, data protection, data security, data localization, network security, and consumer protection apply to aspects of our operations such as the internet, e-commerce, digital content, web services, electronic devices, advertising, artificial intelligence technologies and services, satellite communications services, and healthcare. A large number of jurisdictions regulate our operations, and the extent, nature, and scope of such regulations is evolving and expanding as the scope of our businesses expand. We are regularly subject to formal and informal reviews, investigations, and other proceedings by governments and regulatory authorities under existing laws, regulations, or interpretations or pursuing new and novel approaches to regulate our operations. For example, we face a number of open investigations based on claims that aspects of our operations infringe competition or consumer protection rules, including aspects of Amazon's operation of its stores, including its fulfillment network and Prime, and certain aspects of AWS's offering of cloud services. We strongly dispute these claims and intend to defend ourselves vigorously in these investigations. Similarly, we face investigations under a growing patchwork of laws and regulations governing the collection, use, and disclosure of data, the interpretation of which continues to evolve, leading to uncertainty about how regulators will view our privacy practices. In addition, regulators and lawmakers are increasingly focused on controlling additional aspects of the operations of technology companies and companies they have characterized to be online "gatekeepers" through the application of existing regulations and laws and the adoption of new regulations and laws, which increases our compliance costs and limits the operation of our business. Unfavorable regulations, laws, decisions, or interpretations by government or regulatory authorities applying those laws and regulations, or inquiries, investigations, or enforcement actions threatened or initiated by them, could cause us to incur substantial costs, expose us to unanticipated civil and criminal liability or penalties (including substantial monetary fines), diminish the demand for, or availability of, our products and services, increase our cost of doing business, require us to change our business practices in a manner materially adverse to our business, damage our reputation, impede our growth, or otherwise have a material effect on our operations. The media, political, and regulatory scrutiny we face, which may continue to increase, amplifies these risks.

Claims, Litigation, Government Investigations, and Other Proceedings May Adversely Affect Our Business and Results of Operations

As an innovative company offering a wide range of consumer and business products and services around the world, we are regularly subject to actual and threatened claims, litigation, reviews, investigations, and other proceedings, including proceedings by governments and regulatory authorities, involving a wide range of issues, including patent and other intellectual property matters, taxes, labor and employment (including the characterization of delivery drivers), competition and antitrust, privacy, data use, data protection, data security, data localization, network security, consumer protection, commercial disputes, goods and services offered by us and by third parties (including artificial intelligence technologies and services), healthcare, and other matters. The number and scale of these proceedings have increased over time as our businesses have expanded in scope and geographic reach, as our products, services, and operations have become more complex and available to, and used by, more people, and as governments and regulatory authorities seek to regulate us on a pre-emptive basis. For example, we are litigating a number of matters alleging price fixing, monopolization, and consumer protection claims, including those brought by state attorneys general and the Federal Trade Commission. Any of these types of proceedings can have an adverse effect on us because of legal costs, disruption of our operations, diversion of management resources, negative publicity, and other factors. The outcomes of these matters are inherently unpredictable and subject to significant uncertainties. Determining legal reserves or possible losses from such matters involves judgment and may not reflect the full range of uncertainties and unpredictable outcomes. Until the final resolution of such matters, we may be exposed to losses in excess of the amount recorded, and such amounts could be material. Should any of our estimates and assumptions change or prove to have been incorrect, it could have a material effect on our business, consolidated financial position, results of operations, or cash flows. In addition, it is possible that a resolution of one or more such proceedings, including as a result of a settlement, could involve licenses, sanctions, consent decrees, or orders requiring us to make substantial future payments, preventing us from offering certain products or services, requiring us to change our business practices in a manner materially adverse to our business, requiring development of non-infringing or otherwise altered products or technologies, damaging our reputation, or otherwise having a material effect on our operations.

We Are Subject to Product Liability Claims When People or Property Are Harmed by the Products We Sell or Manufacture

Some of the products we sell or manufacture expose us to product liability or food safety claims relating to personal injury or illness, death, or environmental or property damage, and can require product recalls or other actions. Third parties who sell products using our services and stores also expose us to product liability claims. Additionally, under our A-to-z Guarantee, we may reimburse customers for certain product liability claims up to certain limits in these situations, and as our third-party seller sales grow, the cost of this program will increase and could negatively affect our operating results. Although we maintain liability insurance, we cannot be certain that our coverage will be adequate for liabilities actually incurred or that insurance will continue to be available to us on economically reasonable terms, or at all. Although we impose contractual terms on sellers that are intended to prohibit sales of certain type of products, we may not be able to detect, enforce, or collect sufficient damages for breaches of such agreements. In addition, some of our agreements with our vendors and sellers do not indemnify us from product liability.

We Face Additional Tax Liabilities and Collection Obligations

We are subject to a variety of taxes and tax collection obligations in the U.S. (federal and state) and numerous foreign jurisdictions. We may recognize additional tax expense and be subject to additional tax liabilities, including other liabilities for tax collection obligations due to changes in laws, regulations, administrative practices, principles, and interpretations related to tax, including changes to the global tax framework, competition, and other laws and accounting rules in various jurisdictions. Such changes could come about as a result of economic, political, and other conditions. An increasing number of jurisdictions are considering or have adopted laws or administrative practices that impose new tax measures, including revenue-based taxes, targeting online commerce and the remote selling of goods and services. These include new obligations to withhold or collect sales, consumption, value added, or other taxes on online marketplaces and remote sellers, or other requirements that may result in liability for third party obligations. For example, non-U.S. jurisdictions have proposed or enacted taxes on online advertising and marketplace service revenues. Proliferation of these or similar unilateral tax measures may continue unless broader international tax reform is implemented. In addition, the European Union and other countries (including those in which we operate) have enacted or have committed to enact global minimum taxes, which may increase our tax expense.

Our results of operations and cash flows could be adversely affected by additional taxes imposed on us prospectively or retroactively or additional taxes or penalties resulting from the failure to comply with any collection obligations or failure to provide information about our customers, suppliers, and other third parties for tax reporting purposes to various government agencies. In some cases we also may not have sufficient notice to enable us to build systems and adopt processes to properly comply with new reporting or collection obligations by the effective date.

Our tax expense and liabilities are also affected by other factors, such as changes in our business operations, acquisitions, investments, entry into new businesses and geographies, intercompany transactions, the relative amount of our foreign earnings, losses incurred in jurisdictions for which we are not able to realize related tax benefits, the applicability of special or extraterritorial tax regimes, changes in foreign exchange rates, changes in our stock price, changes to our forecasts of income and loss and the mix of jurisdictions to which they relate, and changes in our tax assets and liabilities and their valuation. In the ordinary course of our business, there are many transactions and calculations for which the ultimate tax determination is uncertain. Significant judgment is required in evaluating and estimating our tax expense, assets, and liabilities.

We are also subject to tax controversies in various jurisdictions that can result in tax assessments against us. Developments in an audit, investigation, or other tax controversy can have a material effect on our operating results or cash flows in the period or periods for which that development occurs, as well as for prior and subsequent periods. Due to the inherent complexity and uncertainty of these matters, interpretations of certain tax laws by authorities, and judicial, administrative, and regulatory processes in certain jurisdictions, the final outcome of any such controversy may be materially different from our expectations. For example, the Indian tax authority has asserted that tax applies to cloud services fees paid to Amazon in the U.S. We are contesting this position; however, if this matter is adversely resolved, we may be required to pay additional amounts with respect to current and prior periods and our taxes in the future could increase. We regularly assess the likelihood of an adverse outcome resulting from these proceedings to determine the adequacy of our tax accruals. Although we believe our tax estimates are reasonable, the final outcome of audits, investigations, and any other tax controversies could be materially different from our historical tax accruals.

We Are Subject to Risks Related to Government Contracts and Related Procurement Regulations

Our contracts with U.S., as well as state, local, and foreign, government entities are subject to various procurement regulations and other requirements relating to their formation, administration, and performance. We are subject to audits and investigations relating to our government contracts, and any violations could result in various civil and criminal penalties and administrative sanctions, including termination of contract, refunding or suspending of payments, forfeiture of profits, payment of fines, and suspension or debarment from future government business. In addition, some of these contracts are subject to periodic funding approval and/or provide for termination by the government at any time, without cause.

Item 1B. *Unresolved Staff Comments*

None.

Item 1C. *Cybersecurity*

We have processes in place for assessing, identifying, and managing material risks from potential unauthorized occurrences on or through our electronic information systems that could adversely affect the confidentiality, integrity, or availability of our information systems or the information residing on those systems. These include a wide variety of mechanisms, controls, technologies, methods, systems, and other processes that are designed to prevent, detect, or mitigate data loss, theft, misuse, unauthorized access, or other security incidents or vulnerabilities affecting the data. The data include confidential, proprietary, and business and personal information that we collect, process, store, and transmit as part of our business, including on behalf of third parties. We also use systems and processes designed to reduce the impact of a security incident at a third-party vendor or customer. Additionally, we use processes to oversee and identify material risks from cybersecurity threats associated with our use of third-party technology and systems, including: technology and systems we use for encryption and authentication; employee email; content delivery to customers; back-office support; and other functions.

As part of our risk management process, we conduct application security assessments, vulnerability management, penetration testing, security audits, and ongoing risk assessments. We also maintain a variety of incident response plans that are utilized when incidents are detected. We require employees with access to information systems, including all corporate employees, to undertake data protection and cybersecurity training and compliance programs annually.

We have a unified and centrally-coordinated team, led by our chief security officer, that is responsible for implementing and maintaining centralized cybersecurity and data protection practices at Amazon in close coordination with senior leadership and other teams across Amazon. Reporting to our chief security officer are a number of experienced chief information security officers responsible for various parts of our business, including AWS, each of whom is supported by a team of trained cybersecurity professionals. In addition to our extensive in-house cybersecurity capabilities, at times we also engage assessors, consultants, auditors, or other third parties to assist with assessing, identifying, and managing cybersecurity risks.

Our cybersecurity risks and associated mitigations are evaluated by senior leadership, including as part of our enterprise risk assessments that are reviewed by the Audit Committee and our Board of Directors. Such risks and mitigations are also subject to oversight by the Security Committee of our Board of Directors. Additional information about cybersecurity risks we face is discussed in Item 1A of Part I, “Risk Factors,” under the heading “We Could Be Harmed by Data Loss or Other Security Incidents,” which should be read in conjunction with the information above.

The Security Committee, which is comprised of independent directors, oversees our policies and procedures for protecting our cybersecurity infrastructure and for compliance with applicable data protection and security regulations, and related risks. The Security Committee receives reports regarding such risks from management, including our chief security officer, and reports to the Board at least annually. The Security Committee also oversees the Board’s response to any significant cybersecurity incidents.

Our chief security officer, who has extensive cybersecurity knowledge and skills gained from over 15 years of work experience on the security team at Amazon and an extensive career in the technology and cybersecurity industries as a senior executive in the federal government, heads the team responsible for implementing and maintaining cybersecurity and data protection practices at Amazon and reports directly to the Chief Executive Officer.

Item 2. *Properties*

As of December 31, 2024, we operated the following facilities (in thousands):

Description of Use	Leased Square Footage (1)	Owned Square Footage	Location
Office space	29,551	9,104	North America
Office space	23,771	1,802	International
Physical stores (2)	23,975	707	North America
Physical stores (2)	222	—	International
Fulfillment, data centers, and other	448,168	36,869	North America
Fulfillment, data centers, and other	176,287	18,958	International
Total	701,974	67,440	

(1) For leased properties, represents the total leased space excluding sub-leased space.
(2) This includes 618 North America and 27 International stores as of December 31, 2024.

Segment	Leased Square Footage (1)	Owned Square Footage (1)
North America	457,104	20,741
International	166,673	11,741
AWS	24,875	24,052
Total	648,652	56,534

(1) Segment amounts exclude corporate facilities. Shared facilities are allocated among the segments based on usage and primarily relate to facilities that hold our technology infrastructure. See Item 8 of Part II, "Financial Statements and Supplementary Data — Note 10 — Segment Information."

We own and lease our corporate headquarters in Washington's Puget Sound region and Arlington, Virginia.

Item 3. *Legal Proceedings*

See Item 8 of Part II, "Financial Statements and Supplementary Data — Note 7 — Commitments and Contingencies — Legal Proceedings."

Item 4. *Mine Safety Disclosures*

Not applicable.

PART II

Item 5. ***Market for the Registrant's Common Stock, Related Shareholder Matters, and Issuer Purchases of Equity Securities***

Market Information

Our common stock is traded on the Nasdaq Global Select Market under the symbol "AMZN."

Holders

As of January 29, 2025, there were 12,135 shareholders of record of our common stock, although there is a much larger number of beneficial owners.

Recent Sales of Unregistered Securities

None.

Issuer Purchases of Equity Securities

None.

Item 6. ***Reserved***

Item 7. *Management's Discussion and Analysis of Financial Condition and Results of Operations*

Forward-Looking Statements

This Annual Report on Form 10-K includes forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. All statements other than statements of historical fact, including statements regarding guidance, industry prospects, or future results of operations or financial position, made in this Annual Report on Form 10-K are forward-looking. We use words such as anticipates, believes, expects, future, intends, and similar expressions to identify forward-looking statements. Forward-looking statements reflect management's current expectations and are inherently uncertain. Actual results and outcomes could differ materially for a variety of reasons, including, among others, fluctuations in foreign exchange rates, changes in global economic conditions and customer demand and spending, inflation, interest rates, regional labor market constraints, world events, the rate of growth of the internet, online commerce, cloud services, and new and emerging technologies, the amount that Amazon.com invests in new business opportunities and the timing of those investments, the mix of products and services sold to customers, the mix of net sales derived from products as compared with services, the extent to which we owe income or other taxes, competition, management of growth, potential fluctuations in operating results, international growth and expansion, the outcomes of claims, litigation, government investigations, and other proceedings, fulfillment, sortation, delivery, and data center optimization, risks of inventory management, variability in demand, the degree to which we enter into, maintain, and develop commercial agreements, proposed and completed acquisitions and strategic transactions, payments risks, and risks of fulfillment throughput and productivity. In addition, global economic and geopolitical conditions and additional or unforeseen circumstances, developments, or events may give rise to or amplify many of these risks. These risks and uncertainties, as well as other risks and uncertainties that could cause our actual results or outcomes to differ significantly from management's expectations, are described in greater detail in Item 1A of Part I, "Risk Factors."

Overview

Our primary source of revenue is the sale of a wide range of products and services to customers. The products offered through our stores include merchandise and content we have purchased for resale and products offered by third-party sellers, and we also manufacture and sell electronic devices and produce media content. Generally, we recognize gross revenue from items we sell from our inventory as product sales and recognize our net share of revenue of items sold by third-party sellers as service sales. We seek to increase unit sales across our stores, through increased product selection, across numerous product categories. We also offer other services such as compute, storage, and database offerings, fulfillment, advertising, publishing, and digital content subscriptions.

Our financial focus is on long-term, sustainable growth in free cash flows. Free cash flows are driven primarily by increasing operating income and efficiently managing accounts receivable, inventory, accounts payable, and cash capital expenditures, including our decision to purchase or lease property and equipment. Increases in operating income primarily result from increases in sales of products and services and efficiently managing our operating costs, partially offset by investments we make in longer-term strategic initiatives, including capital expenditures focused on improving the customer experience. To increase sales of products and services, we focus on improving all aspects of the customer experience, including lowering prices, improving availability, offering faster delivery and performance times, increasing selection, producing original content, increasing product categories and service offerings, expanding product information, improving ease of use, improving reliability, and earning customer trust. See "Results of Operations — Non-GAAP Financial Measures" below for additional information on our non-GAAP free cash flows financial measures.

We seek to reduce our variable costs per unit and work to leverage our fixed costs. Our variable costs include product and content costs, payment processing and related transaction costs, picking, packaging, and preparing orders for shipment, transportation, customer service support, costs necessary to run AWS, and a portion of our marketing costs. Our fixed costs include the costs necessary to build and run our technology infrastructure; to build, enhance, and add features to our online stores, web services, electronic devices, and digital offerings; and to build and optimize our fulfillment network. Variable costs generally change directly with sales volume, while fixed costs generally are dependent on the timing of capacity needs, geographic expansion, category expansion, and other factors. To decrease our variable costs on a per unit basis and enable us to lower prices for customers, we seek to increase our direct sourcing, increase discounts from suppliers, and reduce defects in our processes. To minimize unnecessary growth in fixed costs, we seek to improve process efficiencies and maintain a lean culture.

We seek to turn inventory quickly and collect from consumers before our payments to vendors and sellers become due. Because consumers primarily use credit cards in our stores, our receivables from consumers settle quickly. We expect variability in inventory turnover over time since it is affected by numerous factors, including our product mix, the mix of sales by us and by third-party sellers, our continuing focus on in-stock inventory availability and selection of product offerings, supply chain disruptions and resulting vendor lead times, our investment in new geographies and product lines, and the extent to which we choose to utilize third-party fulfillment providers. We also expect some variability in accounts payable days over time since they are affected by several factors, including the mix of product sales, the mix of sales by third-party sellers, the mix of suppliers, seasonality, and changes in payment and other terms over time, including the effect of balancing pricing and timing of payment terms with suppliers.

We expect spending in technology and infrastructure will increase over time as we add computer scientists, designers, software and hardware engineers, and merchandising employees. Our technology and infrastructure investment and capital spending projects often support a variety of product and service offerings due to geographic expansion and the cross-functionality of our systems and operations. We seek to invest efficiently in several areas of technology and infrastructure, including AWS, and expansion of new and existing product categories and service offerings, as well as in infrastructure to enhance the customer experience and improve our process efficiencies. We believe that advances in technology, specifically the speed and reduced cost of processing power, data storage and analytics, improved wireless connectivity, and the practical applications of artificial intelligence and machine learning, will continue to improve users' experience on the internet and increase its ubiquity in people's lives. To best take advantage of these continued advances in technology, we are investing in AWS, which offers a broad set of on-demand technology services, including compute, storage, database, analytics, and machine learning, and other services to developers and enterprises of all sizes. We are also investing in initiatives to build and deploy innovative and efficient software and electronic devices as well as other initiatives including the development of a satellite network for global broadband service and autonomous vehicles for ride-hailing services.

We seek to efficiently manage shareholder dilution while maintaining the flexibility to issue shares for strategic purposes, such as financings, acquisitions, and aligning employee compensation with shareholders' interests. We utilize restricted stock units as our primary vehicle for equity compensation because we believe this compensation model aligns the long-term interests of our shareholders and employees. In measuring shareholder dilution, we include all vested and unvested stock awards outstanding, without regard to estimated forfeitures. Total shares outstanding plus outstanding stock awards were 10.8 billion and 10.9 billion as of December 31, 2023 and 2024.

Our financial reporting currency is the U.S. Dollar and changes in foreign exchange rates significantly affect our reported results and consolidated trends. For example, if the U.S. Dollar weakens year-over-year relative to currencies in our international locations, our consolidated net sales and operating expenses will be higher than if currencies had remained constant. Likewise, if the U.S. Dollar strengthens year-over-year relative to currencies in our international locations, our consolidated net sales and operating expenses will be lower than if currencies had remained constant. We believe that our increasing diversification beyond the U.S. economy through our growing international businesses benefits our shareholders over the long-term. We also believe it is useful to evaluate our operating results and growth rates before and after the effect of currency changes.

In addition, the remeasurement of our intercompany balances can result in significant gains and losses associated with the effect of movements in foreign exchange rates. Currency volatilities may continue, which may significantly impact (either positively or negatively) our reported results and consolidated trends and comparisons.

For additional information about each line item addressed above, refer to Item 8 of Part II, "Financial Statements and Supplementary Data — Note 1 — Description of Business, Accounting Policies, and Supplemental Disclosures."

Our Annual Report on Form 10-K for the year ended December 31, 2023 includes a discussion and analysis of our financial condition and results of operations for the year ended December 31, 2022 in Item 7 of Part II, "Management's Discussion and Analysis of Financial Condition and Results of Operations."

Critical Accounting Estimates

The preparation of financial statements in conformity with generally accepted accounting principles of the United States ("GAAP") requires estimates and assumptions that affect the reported amounts of assets and liabilities, revenues and expenses, and related disclosures of contingent liabilities in the consolidated financial statements and accompanying notes. Critical accounting estimates are those estimates made in accordance with GAAP that involve a significant level of estimation uncertainty and have had or are reasonably likely to have a material impact on the financial condition or results of operations of the Company. Based on this definition, we have identified the critical accounting estimates addressed below. We also have other key accounting policies, which involve the use of estimates, judgments, and assumptions that are significant to understanding our results. For additional information, see Item 8 of Part II, "Financial Statements and Supplementary Data — Note 1 — Description of Business, Accounting Policies, and Supplemental Disclosures." Although we believe that our estimates, assumptions, and judgments are reasonable, they are based upon information presently available. Actual results may differ significantly from these estimates under different assumptions, judgments, or conditions.

Inventories

Inventories, consisting of products available for sale, are primarily accounted for using the first-in first-out method, and are valued at the lower of cost and net realizable value. This valuation requires us to make judgments, based on currently available information, about the likely method of disposition, such as through sales to individual customers, returns to product vendors, or liquidations, and expected recoverable values of each disposition category. These assumptions about future disposition of inventory are inherently uncertain and changes in our estimates and assumptions may cause us to realize material write-downs in the future. As a measure of sensitivity, for every 1% of additional inventory valuation allowance as of December 31, 2024, we would have recorded an additional cost of sales of approximately $365 million.

In addition, we enter into supplier commitments for certain electronic device components and certain products. These commitments are based on forecasted customer demand. If we reduce these commitments, we may incur additional costs.

Income Taxes

We are subject to income taxes in the U.S. (federal and state) and numerous foreign jurisdictions. Tax laws, regulations, administrative practices, principles, and interpretations in various jurisdictions may be subject to significant change, with or without notice, due to economic, political, and other conditions, and significant judgment is required in evaluating and estimating our provision and accruals for these taxes. There are many transactions that occur during the ordinary course of business for which the ultimate tax determination is uncertain. In addition, our actual and forecasted earnings are subject to change due to economic, political, and other conditions and significant judgment is required in determining our ability to use our deferred tax assets.

Our effective tax rates could be affected by numerous factors, such as changes in our business operations, acquisitions, investments, entry into new businesses and geographies, intercompany transactions, the relative amount of our foreign earnings, including earnings being lower than anticipated in jurisdictions where we have lower statutory rates and higher than anticipated in jurisdictions where we have higher statutory rates, losses incurred in jurisdictions for which we are not able to realize related tax benefits, the applicability of special tax regimes, changes in foreign exchange rates, changes in our stock price, changes to our forecasts of income and loss and the mix of jurisdictions to which they relate, changes in our deferred tax assets and liabilities and their valuation, changes in the laws, regulations, administrative practices, principles, and interpretations related to tax, including changes to the global tax framework, competition, and other laws and accounting rules in various jurisdictions. In addition, a number of countries have enacted or are actively pursuing changes to their tax laws applicable to corporate multinationals.

We are also currently subject to tax controversies in various jurisdictions, and these jurisdictions may assess additional income tax liabilities against us. Developments in an audit, investigation, or other tax controversy could have a material effect on our operating results or cash flows in the period or periods for which that development occurs, as well as for prior and subsequent periods. We regularly assess the likelihood of an adverse outcome resulting from these proceedings to determine the adequacy of our tax accruals. Although we believe our tax estimates are reasonable, the final outcome of audits, investigations, and any other tax controversies could be materially different from our historical income tax provisions and accruals.

Liquidity and Capital Resources

Cash flow information is as follows (in millions):

	Year Ended December 31,	
	2023	2024
Cash provided by (used in):		
Operating activities	$ 84,946	$ 115,877
Investing activities	(49,833)	(94,342)
Financing activities	(15,879)	(11,812)

Our principal sources of liquidity are cash flows generated from operations and our cash, cash equivalents, and marketable securities balances, which, at fair value, were $86.8 billion and $101.2 billion as of December 31, 2023 and 2024. Amounts held in foreign currencies were $23.5 billion and $25.5 billion as of December 31, 2023 and 2024. Our foreign currency balances include British Pounds, Canadian Dollars, Euros, Indian Rupees, and Japanese Yen.

Cash provided by (used in) operating activities was $84.9 billion and $115.9 billion in 2023 and 2024. Our operating cash flows result primarily from cash received from our consumer, seller, developer, enterprise, and content creator customers, and advertisers, offset by cash payments we make for products and services, employee compensation, payment processing and related transaction costs, operating leases, and interest payments. Cash received from our customers and other activities generally corresponds to our net sales. The increase in operating cash flow in 2024, compared to the prior year, was due to an increase in net income (loss), excluding non-cash expenses, and changes in working capital. Working capital at any specific point in time is subject to many variables, including variability in demand, inventory management and category expansion, the timing of cash receipts and payments, customer and vendor payment terms, and fluctuations in foreign exchange rates.

Cash provided by (used in) investing activities corresponds with cash capital expenditures, including leasehold improvements, incentives received from property and equipment vendors, proceeds from asset sales, cash outlays for acquisitions, investments in other companies and intellectual property rights, and purchases, sales, and maturities of marketable securities. Cash provided by (used in) investing activities was $(49.8) billion and $(94.3) billion in 2023 and 2024, with the variability caused primarily by purchases, sales, and maturities of marketable securities and cash capital expenditures. Cash capital expenditures were $48.1 billion, and $77.7 billion in 2023 and 2024, which primarily reflect investments in technology infrastructure (the majority of which is to support AWS business growth) and in additional capacity to support our fulfillment network. We expect cash capital expenditures to increase in 2025, primarily driven by investments in technology infrastructure. We made cash payments, net of acquired cash, related to acquisition and other investment activity of $5.8 billion and $7.1 billion in 2023 and 2024. We funded the acquisition of 1Life Healthcare, Inc. (One Medical) in 2023 with cash on hand. In Q3 2023, we invested $1.25 billion in a convertible note from Anthropic, PBC. In Q1 2024, we invested $2.75 billion in a second convertible note. In Q4 2024, we entered into an agreement and invested $1.3 billion in a third convertible note, and will invest an additional $2.7 billion by Q4 2025.

Cash provided by (used in) financing activities was $(15.9) billion and $(11.8) billion in 2023 and 2024. Cash inflows from financing activities resulted from proceeds from short-term debt, and other and long-term-debt of $18.1 billion and $5.1 billion in 2023 and 2024. Cash outflows from financing activities resulted from payments of short-term debt, and other, long-term debt, finance leases, and financing obligations of $34.0 billion and $17.0 billion in 2023 and 2024. Property and equipment acquired under finance leases was $642 million and $854 million in 2023 and 2024.

We had no borrowings outstanding under the two unsecured revolving credit facilities or the commercial paper programs as of December 31, 2024. See Item 8 of Part II, "Financial Statements and Supplementary Data — Note 6 — Debt" for additional information.

As of December 31, 2024, cash, cash equivalents, and marketable securities held by foreign subsidiaries were $6.3 billion. We intend to invest substantially all of our foreign subsidiary earnings, as well as our capital in our foreign subsidiaries, indefinitely outside of the U.S. in those jurisdictions in which we would incur significant, additional costs upon repatriation of such amounts.

Our U.S. taxable income is reduced by accelerated depreciation deductions and increased by the impact of capitalized research and development expenses. U.S. tax rules provide for enhanced accelerated depreciation deductions by allowing us to expense a portion of qualified property, primarily equipment. These enhanced deductions are scheduled to phase out annually from 2023 through 2026. Our federal tax provision included accelerated depreciation deductions for 2022, 2023, and 2024. Additionally, effective January 1, 2022, research and development expenses are required to be capitalized and amortized for U.S. tax purposes, which delays the deductibility of these expenses. Cash paid for U.S. (federal and state) and foreign income taxes (net of refunds) totaled $11.2 billion and $12.3 billion for 2023 and 2024.

As of December 31, 2023 and 2024, restricted cash, cash equivalents, and marketable securities were $503 million and $3.5 billion. See Item 8 of Part II, "Financial Statements and Supplementary Data — Note 6 — Debt" and "Financial Statements and Supplementary Data — Note 7 — Commitments and Contingencies" for additional discussion of our principal contractual commitments, as well as our pledged assets. Additionally, we have purchase obligations and open purchase orders, including for inventory and capital expenditures, that support normal operations and are primarily due in the next twelve months. These purchase obligations and open purchase orders are generally cancellable in full or in part through the contractual provisions.

We believe that cash flows generated from operations and our cash, cash equivalents, and marketable securities balances, as well as our borrowing arrangements, will be sufficient to meet our anticipated operating cash needs for at least the next twelve months. However, any projections of future cash needs and cash flows are subject to substantial uncertainty. See Item 1A of Part I, "Risk Factors." We continually evaluate opportunities to sell additional equity or debt securities, obtain credit facilities, obtain finance and operating lease arrangements, enter into financing obligations, repurchase common stock, pay dividends, or repurchase, refinance, or otherwise restructure our debt for strategic reasons or to further strengthen our financial position.

The sale of additional equity or convertible debt securities would be dilutive to our shareholders. In addition, we will, from time to time, consider the acquisition of, or investment in, complementary businesses, products, services, capital infrastructure, and technologies, which might affect our liquidity requirements or cause us to secure additional financing, or issue additional equity or debt securities. There can be no assurance that additional credit lines or financing instruments will be available in amounts or on terms acceptable to us, if at all. In addition, economic conditions and actions by policymaking bodies are contributing to changing interest rates and significant capital market volatility, which, along with any increases in our borrowing levels, could increase our future borrowing costs.

Results of Operations

We have organized our operations into three segments: North America, International, and AWS. These segments reflect the way the Company evaluates its business performance and manages its operations. See Item 8 of Part II, “Financial Statements and Supplementary Data — Note 10 — Segment Information.”

Overview

Macroeconomic factors, including changes in inflation and interest rates, global economic and geopolitical developments, and the development and adoption of technologies and services, including artificial intelligence, have direct and indirect impacts on our results of operations that are difficult to isolate and quantify. These could affect customer demand for our products and services, our ability to predict growth needs, expenses, and the benefits we gain from new technologies. We expect some or all of them to continue to impact our operations into Q1 2025.

Net Sales

Net sales include product and service sales. Product sales represent revenue from the sale of products and related shipping fees and digital media content where we record revenue gross. Service sales primarily represent third-party seller fees, which includes commissions and any related fulfillment and shipping fees, AWS sales, advertising services, Amazon Prime membership fees, and certain digital media content subscriptions. Net sales information is as follows (in millions):

	Year Ended December 31,	
	2023	2024
Net Sales:		
North America	$ 352,828	$ 387,497
International	131,200	142,906
AWS	90,757	107,556
Consolidated	$ 574,785	$ 637,959
Year-over-year Percentage Growth:		
North America	12 %	10 %
International	11	9
AWS	13	19
Consolidated	12	11
Year-over-year Percentage Growth, excluding the effect of foreign exchange rates:		
North America	12 %	10 %
International	11	10
AWS	13	19
Consolidated	12	11
Net Sales Mix:		
North America	61 %	61 %
International	23	22
AWS	16	17
Consolidated	100 %	100 %

Sales increased 11% in 2024, compared to the prior year. Changes in foreign exchange rates reduced net sales by $2.3 billion in 2024. For a discussion of the effect of foreign exchange rates on sales growth, see “Effect of Foreign Exchange Rates” below.

North America sales increased 10% in 2024, compared to the prior year. The sales growth primarily reflects increased unit sales, including sales by third-party sellers, advertising sales, and subscription services. Increased unit sales were driven largely by our continued focus on price, selection, and convenience for our customers, including from our fast shipping offers. Changes in foreign exchange rates reduced North America net sales by $462 million in 2024.

International sales increased 9% in 2024, compared to the prior year. The sales growth primarily reflects increased unit sales, including sales by third-party sellers, advertising sales, and subscription services. Increased unit sales were driven largely by our continued focus on price, selection, and convenience for our customers, including from our fast shipping offers. Changes in foreign exchange rates reduced International net sales by $1.8 billion in 2024.

AWS sales increased 19% in 2024, compared to the prior year. The sales growth primarily reflects increased customer usage, partially offset by pricing changes primarily driven by long-term customer contracts.

Operating Expenses

Information about operating expenses is as follows (in millions):

	Year Ended December 31,	
	2023	2024
Operating Expenses:		
Cost of sales	$ 304,739	$ 326,288
Fulfillment	90,619	98,505
Technology and infrastructure	85,622	88,544
Sales and marketing	44,370	43,907
General and administrative	11,816	11,359
Other operating expense (income), net	767	763
Total operating expenses	$ 537,933	$ 569,366
Year-over-year Percentage Growth (Decline):		
Cost of sales	6 %	7 %
Fulfillment	7	9
Technology and infrastructure	17	3
Sales and marketing	5	(1)
General and administrative	(1)	(4)
Other operating expense (income), net	(39)	(1)
Percent of Net Sales:		
Cost of sales	53.0 %	51.1 %
Fulfillment	15.8	15.4
Technology and infrastructure	14.9	13.9
Sales and marketing	7.7	6.9
General and administrative	2.1	1.8
Other operating expense (income), net	0.1	0.1

Cost of Sales

Cost of sales primarily consists of the purchase price of consumer products, inbound and outbound shipping costs, including costs related to sortation and delivery centers and where we are the transportation service provider, and digital media content costs where we record revenue gross, including video and music.

The increase in cost of sales in 2024, compared to the prior year, is primarily due to increased product and shipping costs resulting from increased sales, partially offset by fulfillment network efficiencies, including lower transportation costs. Changes in foreign exchange rates reduced cost of sales by $1.7 billion in 2024.

Shipping costs were $89.5 billion and $95.8 billion in 2023 and 2024. Shipping costs to receive products from our suppliers are included in our inventory and recognized as cost of sales upon sale of products to our customers. We expect our cost of shipping to continue to increase to the extent our customers accept and use our shipping offers at an increasing rate, we use more expensive shipping methods, and we offer additional services. We seek to mitigate costs of shipping over time in part through achieving higher sales volumes, optimizing our fulfillment network, negotiating better terms with our suppliers, and achieving better operating efficiencies. We believe that offering low prices to our customers is fundamental to our future success, and one way we offer lower prices is through shipping offers.

Costs to operate our AWS segment are primarily classified as "Technology and infrastructure" as we leverage a shared infrastructure that supports both our internal technology requirements and external sales to AWS customers.

Fulfillment

Fulfillment costs primarily consist of those costs incurred in operating and staffing our North America and International fulfillment centers, physical stores, and customer service centers and payment processing costs. While AWS payment processing and related transaction costs are included in "Fulfillment," AWS costs are primarily classified as "Technology and infrastructure." Fulfillment costs as a percentage of net sales may vary due to several factors, such as payment processing and related transaction costs, our level of productivity and accuracy, changes in volume, size, and weight of units received and fulfilled, the extent to which third-party sellers utilize Fulfillment by Amazon services, timing of fulfillment network and physical store expansion, the extent we utilize fulfillment services provided by third parties, mix of products and services sold, and our ability to affect customer service contacts per unit by implementing improvements in our operations and enhancements to our customer self-service features. Additionally, sales by our sellers have higher payment processing and related transaction costs as a percentage of net sales compared to our retail sales because payment processing costs are based on the gross purchase price of underlying transactions.

The increase in fulfillment costs in 2024, compared to the prior year, is primarily due to increased sales and investments in our fulfillment network, partially offset by fulfillment network efficiencies. Changes in foreign exchange rates reduced fulfillment costs by $223 million in 2024.

We seek to expand our fulfillment network to accommodate a greater selection and in-stock inventory levels and to meet anticipated shipment volumes from sales of our own products as well as sales by third parties for which we provide the fulfillment services. We regularly evaluate our facility requirements.

Technology and Infrastructure

Technology and infrastructure costs include payroll and related expenses for employees involved in the research and development of new and existing products and services, development, design, and maintenance of our stores, curation and display of products and services made available in our online stores, and infrastructure costs. Infrastructure costs include servers, networking equipment, and data center related depreciation and amortization, rent, utilities, and other expenses necessary to support AWS and other Amazon businesses. Collectively, these costs reflect the investments we make in order to offer a wide variety of products and services to our customers, including expenditures related to initiatives to build and deploy innovative and efficient software and electronic devices and the development of a satellite network for global broadband service and autonomous vehicles for ride-hailing services.

We seek to invest efficiently in numerous areas of technology and infrastructure so we may continue to enhance the customer experience and improve our process efficiency through rapid technology developments, while operating at an ever increasing scale. Our technology and infrastructure investment and capital spending projects often support a variety of product and service offerings due to geographic expansion and the cross-functionality of our systems and operations. We expect spending in technology and infrastructure to increase over time as we continue to add employees and infrastructure. These costs are allocated to segments based on usage. The increase in technology and infrastructure costs in 2024, compared to the prior year, is primarily due to an increase in spending on infrastructure, partially offset by decreased payroll and related costs associated with technical teams responsible for expanding our existing products and services and initiatives to introduce new products and service offerings and a reduction in depreciation and amortization expense from our change in the estimated useful life of our servers. Changes in foreign exchange rates reduced technology and infrastructure costs by $244 million in 2024. We currently expense the majority of the costs associated with the development of our satellite network for global broadband service (including production, launch, and payroll costs, and launch services deposits upon launch). We will capitalize certain of these costs once the service achieves commercial viability, including sales to customers.

Sales and Marketing

Sales and marketing costs include advertising and payroll and related expenses for personnel engaged in marketing and selling activities, including sales commissions related to AWS. We direct customers to our stores primarily through a number of marketing channels, such as our sponsored search, third-party customer referrals, social and online advertising, television advertising, and other initiatives. Our marketing costs are largely variable, based on growth in sales and changes in rates. To the extent there is increased or decreased competition for these traffic sources, or to the extent our mix of these channels shifts, we would expect to see a corresponding change in our marketing costs.

The decrease in sales and marketing costs in 2024, compared to the prior year, is primarily due to decreased payroll and related expenses for personnel engaged in marketing and selling activities, partially offset by increased advertising expenses. Changes in foreign exchange rates reduced sales and marketing costs by $263 million in 2024.

While costs associated with Amazon Prime membership benefits and other shipping offers are not included in sales and marketing expense, we view these offers as effective worldwide marketing tools, and intend to continue offering them indefinitely.

General and Administrative

The decrease in general and administrative costs in 2024, compared to the prior year, is primarily due to a decrease in payroll and related expenses.

Other Operating Expense (Income), Net

Other operating expense (income), net was $767 million and $763 million during 2023 and 2024, and was primarily related to asset impairments and the amortization of intangible assets.

Operating Income (Loss)

Operating income (loss) by segment is as follows (in millions):

	Year Ended December 31,	
	2023	**2024**
Operating Income (Loss)		
North America	$ 14,877	$ 24,967
International	(2,656)	3,792
AWS	24,631	39,834
Consolidated	$ 36,852	$ 68,593

Operating income was $36.9 billion and $68.6 billion for 2023 and 2024. We believe that operating income is a more meaningful measure than gross profit and gross margin due to the diversity of our product categories and services. For more information on the operating expenses that impact segment operating income, see “Operating Expenses” and the descriptions of operating expense line item changes on pages 25 to 27, and “Note 10 — Segment Information” on page 65.

The increase in North America operating income in 2024, compared to the prior year, is primarily due to increased unit sales and increased advertising sales, partially offset by increased fulfillment and shipping costs.

The International operating income in 2024, as compared to the operating loss in the prior year, is primarily due to increased unit sales and increased advertising sales, partially offset by increased shipping and fulfillment costs. Changes in foreign exchange rates did not significantly impact operating income in 2024.

The increase in AWS operating income in 2024, compared to the prior year, is primarily due to increased sales, decreased payroll and related expenses, and a reduction in depreciation and amortization expense from our change in the estimated useful lives of our servers, partially offset by spending on technology infrastructure that was primarily driven by additional investments to support AWS business growth. Changes in foreign exchange rates positively impacted operating income by $240 million in 2024.

Interest Income and Expense

Our interest income was $2.9 billion and $4.7 billion during 2023 and 2024, primarily due to a higher average balance of invested funds at prevailing rates. We generally invest our excess cash in investment grade short- to intermediate-term marketable debt securities and AAA-rated money market funds. Our interest income corresponds with the average balance of invested funds based on the prevailing rates, which vary depending on the geographies and currencies in which they are invested.

Interest expense was $3.2 billion and $2.4 billion in 2023 and 2024 and was primarily related to debt and finance leases. See Item 8 of Part II, “Financial Statements and Supplementary Data — Note 4 — Leases and Note 6 — Debt” for additional information.

Our long-term lease liabilities were $77.3 billion and $78.3 billion as of December 31, 2023 and 2024. Our long-term debt was $58.3 billion and $52.6 billion as of December 31, 2023 and 2024. See Item 8 of Part II, “Financial Statements and Supplementary Data — Note 4 — Leases and Note 6 — Debt” for additional information.

Other Income (Expense), Net

Other income (expense), net was $938 million and $(2.3) billion during 2023 and 2024. The primary components of other income (expense), net are related to equity securities valuations and adjustments, equity warrant valuations, and foreign currency. Included in other income (expense), net in 2023 and 2024 is a marketable equity securities valuation gain (loss) of $797 million and $(1.6) billion from our equity investment in Rivian.

Income Taxes

Our effective tax rate is subject to significant variation due to several factors, including variability in our pre-tax and taxable income and loss and the mix of jurisdictions to which they relate, intercompany transactions, the applicability of special tax regimes, changes in how we do business, acquisitions, investments, developments in tax controversies, changes in our stock price, changes in our deferred tax assets and liabilities and their valuation, foreign currency gains (losses), changes in statutes, regulations, case law, and administrative practices, principles, and interpretations related to tax, including changes to the global tax framework, competition, and other laws and accounting rules in various jurisdictions, and relative changes of expenses or losses for which tax benefits are not recognized. Our effective tax rate can be more or less volatile based on the amount of pre-tax income or loss. For example, the impact of discrete items and non-deductible expenses on our effective tax rate is greater when our pre-tax income is lower. In addition, we record valuation allowances against deferred tax assets when there is uncertainty about our ability to generate future income in relevant jurisdictions.

We recorded a provision (benefit) for income taxes of $7.1 billion and $9.3 billion in 2023 and 2024. See Item 8 of Part II, "Financial Statements and Supplementary Data — Note 9 — Income Taxes" for additional information.

Non-GAAP Financial Measures

Regulation G, Conditions for Use of Non-GAAP Financial Measures, and other SEC regulations define and prescribe the conditions for use of certain non-GAAP financial information. Our measures of free cash flows and the effect of foreign exchange rates on our consolidated statements of operations meet the definition of non-GAAP financial measures.

We provide multiple measures of free cash flows because we believe these measures provide additional perspective on the impact of acquiring property and equipment with cash and through finance leases and financing obligations.

Free Cash Flow

Free cash flow is cash flow from operations reduced by "Purchases of property and equipment, net of proceeds from sales and incentives." The following is a reconciliation of free cash flow to the most comparable GAAP cash flow measure, "Net cash provided by (used in) operating activities," for 2023 and 2024 (in millions):

	Year Ended December 31,	
	2023	2024
Net cash provided by (used in) operating activities	$ 84,946	$ 115,877
Purchases of property and equipment, net of proceeds from sales and incentives	(48,133)	(77,658)
Free cash flow	$ 36,813	$ 38,219
Net cash provided by (used in) investing activities	$ (49,833)	$ (94,342)
Net cash provided by (used in) financing activities	$ (15,879)	$ (11,812)

Free Cash Flow Less Principal Repayments of Finance Leases and Financing Obligations

Free cash flow less principal repayments of finance leases and financing obligations is free cash flow reduced by "Principal repayments of finance leases" and "Principal repayments of financing obligations." Principal repayments of finance leases and financing obligations approximates the actual payments of cash for our finance leases and financing obligations. The following is a reconciliation of free cash flow less principal repayments of finance leases and financing obligations to the most comparable GAAP cash flow measure, "Net cash provided by (used in) operating activities," for 2023 and 2024 (in millions):

	Year Ended December 31,	
	2023	2024
Net cash provided by (used in) operating activities	$ 84,946	$ 115,877
Purchases of property and equipment, net of proceeds from sales and incentives	(48,133)	(77,658)
Free cash flow	36,813	38,219
Principal repayments of finance leases	(4,384)	(2,043)
Principal repayments of financing obligations	(271)	(669)
Free cash flow less principal repayments of finance leases and financing obligations	$ 32,158	$ 35,507
Net cash provided by (used in) investing activities	$ (49,833)	$ (94,342)
Net cash provided by (used in) financing activities	$ (15,879)	$ (11,812)

Free Cash Flow Less Equipment Finance Leases and Principal Repayments of All Other Finance Leases and Financing Obligations

Free cash flow less equipment finance leases and principal repayments of all other finance leases and financing obligations is free cash flow reduced by equipment acquired under finance leases, which is included in "Property and equipment acquired under finance leases, net of remeasurements and modifications," principal repayments of all other finance lease liabilities, which is included in "Principal repayments of finance leases," and "Principal repayments of financing obligations." All other finance lease liabilities and financing obligations consists of property. In this measure, equipment acquired under finance leases is reflected as if these assets had been purchased with cash, which is not the case as these assets have been leased. The following is a reconciliation of free cash flow less equipment finance leases and principal repayments of all other finance leases and financing obligations to the most comparable GAAP cash flow measure, "Net cash provided by (used in) operating activities," for 2023 and 2024 (in millions):

	Year Ended December 31,	
	2023	2024
Net cash provided by (used in) operating activities	$ 84,946	$ 115,877
Purchases of property and equipment, net of proceeds from sales and incentives	(48,133)	(77,658)
Free cash flow	36,813	38,219
Equipment acquired under finance leases (1)	(310)	(572)
Principal repayments of all other finance leases (2)	(683)	(767)
Principal repayments of financing obligations	(271)	(669)
Free cash flow less equipment finance leases and principal repayments of all other finance leases and financing obligations	$ 35,549	$ 36,211
Net cash provided by (used in) investing activities	$ (49,833)	$ (94,342)
Net cash provided by (used in) financing activities	$ (15,879)	$ (11,812)

(1) For the year ended December 31, 2023 and 2024, this amount relates to equipment included in "Property and equipment acquired under finance leases, net of remeasurements and modifications" of $642 million and $854 million.
(2) For the year ended December 31, 2023 and 2024, this amount relates to property included in "Principal repayments of finance leases" of $4,384 million and $2,043 million.

All of these free cash flows measures have limitations as they omit certain components of the overall cash flow statement and do not represent the residual cash flow available for discretionary expenditures. For example, these measures of free cash flows do not incorporate the portion of payments representing principal reductions of debt or cash payments for business acquisitions. Additionally, our mix of property and equipment acquisitions with cash or other financing options may change over time. Therefore, we believe it is important to view free cash flows measures only as a complement to our entire consolidated statements of cash flows.

Effect of Foreign Exchange Rates

Information regarding the effect of foreign exchange rates, versus the U.S. Dollar, on our net sales, operating expenses, and operating income is provided to show reported period operating results had the foreign exchange rates remained the same as those in effect in the comparable prior year period. The effect on our net sales, operating expenses, and operating income from changes in our foreign exchange rates versus the U.S. Dollar is as follows (in millions):

	Year Ended December 31, 2023			Year Ended December 31, 2024		
	As Reported	Exchange Rate Effect (1)	At Prior Year Rates (2)	As Reported	Exchange Rate Effect (1)	At Prior Year Rates (2)
Net sales	$574,785	$ 71	$574,856	$637,959	$ 2,335	$640,294
Operating expenses	537,933	531	538,464	569,366	2,466	571,832
Operating income	36,852	(460)	36,392	68,593	(131)	68,462

(1) Represents the change in reported amounts resulting from changes in foreign exchange rates from those in effect in the comparable prior year period for operating results.
(2) Represents the outcome that would have resulted had foreign exchange rates in the reported period been the same as those in effect in the comparable prior year period for operating results.

Guidance

We provided guidance on February 6, 2025, in our earnings release furnished on Form 8-K as set forth below. These forward-looking statements reflect Amazon.com's expectations as of February 6, 2025, and are subject to substantial uncertainty. Our results are inherently unpredictable and may be materially affected by many factors, such as fluctuations in foreign exchange rates, changes in global economic and geopolitical conditions and customer demand and spending (including the impact of recessionary fears), inflation, interest rates, regional labor market constraints, world events, the rate of growth of the internet, online commerce, cloud services, and new and emerging technologies, as well as those outlined in Item 1A of Part I, "Risk Factors."

First Quarter 2025 Guidance

- Net sales are expected to be between $151.0 billion and $155.5 billion, or to grow between 5% and 9% compared with first quarter 2024. This guidance anticipates an unusually large, unfavorable impact of approximately $2.1 billion, or 150 basis points, from foreign exchange rates. Also, as a reminder, in first quarter 2024 the impact from Leap Year added approximately $1.5 billion in net sales.
- Operating income is expected to be between $14.0 billion and $18.0 billion, compared with $15.3 billion in first quarter 2024.
- This guidance assumes, among other things, that no additional business acquisitions, restructurings, or legal settlements are concluded.

Item 7A. ***Quantitative and Qualitative Disclosures About Market Risk***

We are exposed to market risk for the effect of interest rate changes, foreign currency fluctuations, and changes in the market values of our investments. Information relating to quantitative and qualitative disclosures about market risk is set forth below and in Item 7 of Part II, “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Liquidity and Capital Resources.”

Interest Rate Risk

Our exposure to market risk for changes in interest rates relates primarily to our investment portfolio and our debt. Our long-term debt is carried at amortized cost and fluctuations in interest rates do not impact our consolidated financial statements. However, the fair value of our long-term debt, which pays interest at a fixed rate, will generally fluctuate with movements of interest rates, increasing in periods of declining rates of interest and declining in periods of increasing rates of interest.

We generally invest our excess cash in investment grade short- to intermediate-term marketable debt securities and AAA-rated money market funds. Marketable debt securities with fixed interest rates may have their fair market value adversely affected due to a rise in interest rates, and we may suffer losses in principal if forced to sell securities that have declined in market value due to changes in interest rates. The following table provides information about our cash equivalents and marketable debt securities, including principal cash flows by expected maturity and the related weighted-average interest rates as of December 31, 2024 (in millions, except percentages):

	2025	2026	2027	2028	2029	Thereafter	Total	Estimated Fair Value as of December 31, 2024
Money market funds	$ 28,282	$ —	$ —	$ —	$ —	$ —	$28,282	$ 28,282
Weighted average interest rate	4.42 %	— %	— %	— %	— %	— %	4.42 %	
Corporate debt securities	47,908	1,779	1,086	311	55	—	51,139	50,912
Weighted average interest rate	4.65 %	3.48 %	4.49 %	4.65 %	4.83 %	— %	4.60 %	
U.S. government and agency securities	1,986	928	215	65	50	213	3,457	3,401
Weighted average interest rate	3.95 %	2.92 %	3.83 %	2.72 %	2.57 %	2.04 %	3.51 %	
Asset-backed securities	450	262	195	226	188	220	1,541	1,523
Weighted average interest rate	3.19 %	4.74 %	4.59 %	4.84 %	3.29 %	3.40 %	3.92 %	
Foreign government and agency securities	151	2	27	—	—	—	180	177
Weighted average interest rate	4.50 %	4.60 %	4.31 %	— %	— %	— %	4.48 %	
Other debt securities	44	8	8	8	—	—	68	67
Weighted average interest rate	0.99 %	4.49 %	4.11 %	4.29 %	— %	— %	2.13 %	
	$ 78,821	$ 2,979	$ 1,531	$ 610	$ 293	$ 433	$84,667	
Cash equivalents and marketable debt securities								$ 84,362

As of December 31, 2024, we had long-term debt with a face value of $58.0 billion, including the current portion, primarily consisting of fixed rate unsecured senior notes. See Item 8 of Part II, “Financial Statements and Supplementary Data — Note 6 — Debt” for additional information.

Foreign Exchange Risk

During 2024, net sales from our International segment accounted for 22% of our consolidated revenues. Net sales and related expenses generated from our internationally-focused stores, including within Canada and Mexico (which are included in our North America segment), are primarily denominated in the functional currencies of the corresponding stores and primarily include Euros, British Pounds, and Japanese Yen. The results of operations of, and certain of our intercompany balances associated with, our internationally-focused stores and AWS are exposed to foreign exchange rate fluctuations. Upon consolidation, as foreign exchange rates vary, net sales and other operating results may differ materially from expectations, and we may record significant gains or losses on the remeasurement of intercompany balances. For example, as a result of fluctuations in foreign exchange rates throughout the year compared to rates in effect the prior year, International segment net sales decreased by $1.8 billion in comparison with the prior year.

We have foreign exchange risk related to foreign-denominated cash, cash equivalents, and marketable securities ("foreign funds"). Based on the balance of foreign funds as of December 31, 2024, of $25.5 billion, an assumed 5%, 10%, and 20% adverse change to foreign exchange would result in declines of $1.3 billion, $2.6 billion, and $5.1 billion.

We also have foreign exchange risk related to our intercompany balances denominated in various currencies. Based on the intercompany balances as of December 31, 2024, an assumed 5%, 10%, and 20% adverse change to foreign exchange rates would result in losses of $305 million, $605 million, and $1.2 billion, recorded to "Other income (expense), net."

See Item 7 of Part II, "Management's Discussion and Analysis of Financial Condition and Results of Operations — Results of Operations — Effect of Foreign Exchange Rates" for additional information on the effect on reported results of changes in foreign exchange rates.

Equity Investment Risk

As of December 31, 2024, our recorded value in equity, equity warrant, and convertible debt investments in public and private companies was $22.1 billion. Our equity and equity warrant investments in publicly traded companies, which include our equity investment in Rivian, represent $4.6 billion of our investments as of December 31, 2024, and are recorded at fair value, which is subject to market price volatility. We record our equity warrant investments in private companies at fair value and adjust our equity investments in private companies for observable price changes or impairments. We record our available-for-sale convertible debt investments in private companies at fair value, which primarily relate to Anthropic, PBC. Valuations of private companies are inherently more complex due to the lack of readily available market data. The current global economic conditions provide additional uncertainty. As such, we believe that market sensitivities are not practicable. See Item 8 of Part II, "Financial Statements and Supplementary Data — Note 1 — Description of Business, Accounting Policies, and Supplemental Disclosures" for additional information.

Item 8. ***Financial Statements and Supplementary Data***

INDEX TO CONSOLIDATED FINANCIAL STATEMENTS

Report of Independent Registered Public Accounting Firm

The Board of Directors and Shareholders
Amazon.com, Inc.

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Amazon.com, Inc. (the Company) as of December 31, 2024 and 2023, the related consolidated statements of operations, comprehensive income (loss), stockholders' equity, and cash flows for each of the three years in the period ended December 31, 2024, and the related notes (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2024 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2024, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2024, based on criteria established in Internal Control-Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) and our report dated February 6, 2025 expressed an unqualified opinion thereon.

Basis for Opinion

These consolidated financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's consolidated financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the consolidated financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

	Uncertain Tax Positions
Description of the Matter	As discussed in Notes 1 and 9 of the consolidated financial statements, the Company is subject to income taxes in the U.S. and numerous foreign jurisdictions and during the ordinary course of business, there are many tax positions for which the ultimate tax determination is uncertain. As a result, significant judgment is required in evaluating the Company's tax positions and determining its provision for income taxes. The Company uses significant judgment in (1) determining whether a tax position's technical merits are more likely than not to be sustained and (2) measuring the amount of tax benefit that qualifies for recognition. As of December 31, 2024, the Company reported accrued liabilities of \$6.5 billion for various tax contingencies. Auditing the recognition and measurement of the Company's tax contingencies was challenging because the evaluation of whether a tax position is more likely than not to be sustained and the measurement of the benefit of various tax positions can be complex and involves significant auditor judgment. Management's evaluation of tax positions is based on interpretations of tax laws and legal rulings, and may be impacted by regulatory changes and judicial and examination activity.
How We Addressed the Matter in Our Audit	We tested controls over the Company's process to assess the technical merits of its tax contingencies, including controls over: the assessment as to whether a tax position is more likely than not to be sustained; the measurement of the benefit of its tax positions, both initially and on an ongoing basis; and the development of the related disclosures. We involved our international tax, transfer pricing, and research and development tax professionals in assessing the technical merits of certain of the Company's tax positions. Depending on the nature of the specific tax position and, as applicable, developments with the relevant tax authorities relating thereto, our procedures included obtaining and examining the Company's analysis including the Company's correspondence with such tax authorities and evaluating the underlying facts upon which the tax positions are based. We used our knowledge of and experience with international, transfer pricing, and other income tax laws of the relevant taxing jurisdictions to evaluate the Company's accounting for its tax contingencies. We evaluated developments in the applicable regulatory environments to assess potential effects on the Company's positions, including recent decisions in relevant court cases. We analyzed the appropriateness of the Company's assumptions and the accuracy of the Company's calculations and data used to determine the amount of tax benefits to recognize. We evaluated the Company's income tax disclosures in relation to these matters.

/s/ Ernst & Young LLP

We have served as the Company's auditor since 1996.
Seattle, Washington
February 6, 2025

AMAZON.COM, INC.
CONSOLIDATED STATEMENTS OF CASH FLOWS
(in millions)

	Year Ended December 31,		
	2022	2023	2024
CASH, CASH EQUIVALENTS, AND RESTRICTED CASH, BEGINNING OF PERIOD	$ 36,477	$ 54,253	$ 73,890
OPERATING ACTIVITIES:			
Net income (loss)	(2,722)	30,425	59,248
Adjustments to reconcile net income (loss) to net cash from operating activities:			
Depreciation and amortization of property and equipment and capitalized content costs, operating lease assets, and other	41,921	48,663	52,795
Stock-based compensation	19,621	24,023	22,011
Non-operating expense (income), net	16,966	(748)	2,012
Deferred income taxes	(8,148)	(5,876)	(4,648)
Changes in operating assets and liabilities:			
Inventories	(2,592)	1,449	(1,884)
Accounts receivable, net and other	(8,622)	(8,348)	(3,249)
Other assets	(13,275)	(12,265)	(14,483)
Accounts payable	2,945	5,473	2,972
Accrued expenses and other	(1,558)	(2,428)	(2,904)
Unearned revenue	2,216	4,578	4,007
Net cash provided by (used in) operating activities	46,752	84,946	115,877
INVESTING ACTIVITIES:			
Purchases of property and equipment	(63,645)	(52,729)	(82,999)
Proceeds from property and equipment sales and incentives	5,324	4,596	5,341
Acquisitions, net of cash acquired, non-marketable investments, and other	(8,316)	(5,839)	(7,082)
Sales and maturities of marketable securities	31,601	5,627	16,403
Purchases of marketable securities	(2,565)	(1,488)	(26,005)
Net cash provided by (used in) investing activities	(37,601)	(49,833)	(94,342)
FINANCING ACTIVITIES:			
Common stock repurchased	(6,000)	—	—
Proceeds from short-term debt, and other	41,553	18,129	5,142
Repayments of short-term debt, and other	(37,554)	(25,677)	(5,060)
Proceeds from long-term debt	21,166	—	—
Repayments of long-term debt	(1,258)	(3,676)	(9,182)
Principal repayments of finance leases	(7,941)	(4,384)	(2,043)
Principal repayments of financing obligations	(248)	(271)	(669)
Net cash provided by (used in) financing activities	9,718	(15,879)	(11,812)
Foreign currency effect on cash, cash equivalents, and restricted cash	(1,093)	403	(1,301)
Net increase (decrease) in cash, cash equivalents, and restricted cash	17,776	19,637	8,422
CASH, CASH EQUIVALENTS, AND RESTRICTED CASH, END OF PERIOD	$ 54,253	$ 73,890	$ 82,312

See accompanying notes to consolidated financial statements.

AMAZON.COM, INC.
CONSOLIDATED STATEMENTS OF OPERATIONS
(in millions, except per share data)

	Year Ended December 31,		
	2022	2023	2024
Net product sales	$ 242,901	$ 255,887	$ 272,311
Net service sales	271,082	318,898	365,648
Total net sales	513,983	574,785	637,959
Operating expenses:			
Cost of sales	288,831	304,739	326,288
Fulfillment	84,299	90,619	98,505
Technology and infrastructure	73,213	85,622	88,544
Sales and marketing	42,238	44,370	43,907
General and administrative	11,891	11,816	11,359
Other operating expense (income), net	1,263	767	763
Total operating expenses	501,735	537,933	569,366
Operating income	12,248	36,852	68,593
Interest income	989	2,949	4,677
Interest expense	(2,367)	(3,182)	(2,406)
Other income (expense), net	(16,806)	938	(2,250)
Total non-operating income (expense)	(18,184)	705	21
Income (loss) before income taxes	(5,936)	37,557	68,614
Benefit (provision) for income taxes	3,217	(7,120)	(9,265)
Equity-method investment activity, net of tax	(3)	(12)	(101)
Net income (loss)	$ (2,722)	$ 30,425	$ 59,248
Basic earnings per share	$ (0.27)	$ 2.95	$ 5.66
Diluted earnings per share	$ (0.27)	$ 2.90	$ 5.53
Weighted-average shares used in computation of earnings per share:			
Basic	10,189	10,304	10,473
Diluted	10,189	10,492	10,721

See accompanying notes to consolidated financial statements.

AMAZON.COM, INC.
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME (LOSS)
(in millions)

	Year Ended December 31,		
	2022	2023	2024
Net income (loss)	$ (2,722)	$ 30,425	$ 59,248
Other comprehensive income (loss):			
Foreign currency translation adjustments, net of tax of $100, $(55), and $226	(2,586)	1,027	(3,333)
Available-for-sale debt securities:			
Change in net unrealized gains (losses), net of tax of $159, $(110), and $(2,086)	(823)	366	6,339
Less: reclassification adjustment for losses (gains) included in “Other income (expense), net,” net of tax of $0, $(15), and $(2)	298	50	5
Net change	(525)	416	6,344
Other, net of tax of $0, $(1), and $1	—	4	(5)
Total other comprehensive income (loss)	(3,111)	1,447	3,006
Comprehensive income (loss)	$ (5,833)	$ 31,872	$ 62,254

See accompanying notes to consolidated financial statements.

AMAZON.COM, INC.
CONSOLIDATED BALANCE SHEETS
(in millions, except per share data)

	December 31, 2023	December 31, 2024
ASSETS		
Current assets:		
Cash and cash equivalents	$ 73,387	$ 78,779
Marketable securities	13,393	22,423
Inventories	33,318	34,214
Accounts receivable, net and other	52,253	55,451
Total current assets	172,351	190,867
Property and equipment, net	204,177	252,665
Operating leases	72,513	76,141
Goodwill	22,789	23,074
Other assets	56,024	82,147
Total assets	$ 527,854	$ 624,894
LIABILITIES AND STOCKHOLDERS' EQUITY		
Current liabilities:		
Accounts payable	$ 84,981	$ 94,363
Accrued expenses and other	64,709	66,965
Unearned revenue	15,227	18,103
Total current liabilities	164,917	179,431
Long-term lease liabilities	77,297	78,277
Long-term debt	58,314	52,623
Other long-term liabilities	25,451	28,593
Commitments and contingencies (Note 7)		
Stockholders' equity:		
Preferred stock ($0.01 par value; 500 shares authorized; no shares issued or outstanding)	—	—
Common stock ($0.01 par value; 100,000 shares authorized; 10,898 and 11,108 shares issued; 10,383 and 10,593 shares outstanding)	109	111
Treasury stock, at cost	(7,837)	(7,837)
Additional paid-in capital	99,025	120,864
Accumulated other comprehensive income (loss)	(3,040)	(34)
Retained earnings	113,618	172,866
Total stockholders' equity	201,875	285,970
Total liabilities and stockholders' equity	$ 527,854	$ 624,894

See accompanying notes to consolidated financial statements.

AMAZON.COM, INC.
CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY
(in millions)

	Common Stock						
	Shares	Amount	Treasury Stock	Additional Paid-In Capital	Accumulated Other Comprehensive Income (Loss)	Retained Earnings	Total Stockholders' Equity
Balance as of January 1, 2022	10,175	$ 106	$ (1,837)	$ 55,437	$ (1,376)	$ 85,915	$ 138,245
Net loss	—	—	—	—	—	(2,722)	(2,722)
Other comprehensive income (loss)	—	—	—	—	(3,111)	—	(3,111)
Stock-based compensation and issuance of employee benefit plan stock	113	2	—	19,629	—	—	19,631
Common stock repurchased	(46)	—	(6,000)	—	—	—	(6,000)
Balance as of December 31, 2022	10,242	108	(7,837)	75,066	(4,487)	83,193	146,043
Net income	—	—	—	—	—	30,425	30,425
Other comprehensive income (loss)	—	—	—	—	1,447	—	1,447
Stock-based compensation and issuance of employee benefit plan stock	141	1	—	23,959	—	—	23,960
Balance as of December 31, 2023	10,383	109	(7,837)	99,025	(3,040)	113,618	201,875
Net income	—	—	—	—	—	59,248	59,248
Other comprehensive income (loss)	—	—	—	—	3,006	—	3,006
Stock-based compensation and issuance of employee benefit plan stock	210	2	—	21,839	—	—	21,841
Balance as of December 31, 2024	10,593	$ 111	$ (7,837)	$ 120,864	$ (34)	$ 172,866	$ 285,970

See accompanying notes to consolidated financial statements.

Note 1 — DESCRIPTION OF BUSINESS, ACCOUNTING POLICIES, AND SUPPLEMENTAL DISCLOSURES

Description of Business

We seek to be Earth’s most customer-centric company. In each of our segments, we serve our primary customer sets, consisting of consumers, sellers, developers, enterprises, content creators, advertisers, and employees. We serve consumers through our online and physical stores and focus on selection, price, and convenience. We offer programs that enable sellers to grow their businesses, sell their products in our stores, and fulfill orders using our services, and programs that allow authors, independent publishers, musicians, filmmakers, Twitch streamers, skill and app developers, and others to publish and sell content. We serve developers and enterprises of all sizes through AWS, which offers a broad set of on-demand technology services, including compute, storage, database, analytics, and machine learning, and other services. We also manufacture and sell electronic devices. In addition, we provide advertising services to sellers, vendors, publishers, authors, and others, through programs such as sponsored ads, display, and video advertising.

We have organized our operations into three segments: North America, International, and AWS. See “Note 10 — Segment Information.”

Common Stock Split

On May 27, 2022, we effected a 20-for-1 stock split of our common stock and proportionately increased the number of authorized shares of common stock. All share, restricted stock unit (“RSU”), and per share or per RSU information throughout this Annual Report on Form 10-K has been retroactively adjusted to reflect the stock split. The shares of common stock retain a par value of $0.01 per share. Accordingly, an amount equal to the par value of the increased shares resulting from the stock split was reclassified from “Additional paid-in capital” to “Common stock.”

Principles of Consolidation

The consolidated financial statements include the accounts of Amazon.com, Inc. and its consolidated entities (collectively, the “Company”), consisting of its wholly-owned subsidiaries and those entities in which we have a variable interest and of which we are the primary beneficiary, including certain entities in India and certain entities that support our healthcare services. Intercompany balances and transactions between consolidated entities are eliminated.

Use of Estimates

The preparation of financial statements in conformity with GAAP requires estimates and assumptions that affect the reported amounts of assets and liabilities, revenues and expenses, and related disclosures of contingent liabilities in the consolidated financial statements and accompanying notes. Estimates are used for, but not limited to, income taxes, useful lives of equipment, commitments and contingencies, valuation of acquired intangibles and goodwill, stock-based compensation forfeiture rates, vendor funding, inventory valuation, collectability of receivables, impairment of property and equipment and operating leases, valuation and impairment of investments, self-insurance liabilities, and viewing patterns of capitalized video content. Actual results could differ materially from these estimates. We review the useful lives of equipment on an ongoing basis.

In Q4 2024, we completed a useful life study for certain types of heavy equipment and are increasing the useful life from ten years to thirteen years for such equipment effective January 1, 2025. Based on heavy equipment included in “Property and equipment, net” as of December 31, 2024, we estimate an increase in 2025 operating income of approximately $0.9 billion, which will be recorded primarily in “Fulfillment” and impact our North America and International segments.

We completed our most recent servers and networking equipment useful life study in Q4 2024, and are changing the useful lives of a subset of our servers and networking equipment, effective January 1, 2025, from six years to five years. For those assets included in “Property and equipment, net” as of December 31, 2024, whose useful life will change from six years to five years, we anticipate a decrease in 2025 operating income of approximately $0.7 billion. We expect to continue to acquire more of these server and networking assets in 2025. In 2024, we also determined, primarily in the fourth quarter, to retire early certain of our servers and networking equipment. We recorded approximately $920 million of accelerated depreciation and related charges for the quarter ended December 31, 2024 related to these decisions. The accelerated depreciation will continue into 2025 and decrease operating income by approximately $0.6 billion in 2025. These two changes above are due to an increased pace of technology development, particularly in the area of artificial intelligence and machine learning.

We had previously increased the useful life of our servers from five years to six years effective January 1, 2024. The effect of this change for the year ended December 31, 2024, based on servers that were included in “Property and equipment, net" as of December 31, 2023 and those acquired during the year ended December 31, 2024, was a reduction in depreciation and amortization expense of $3.2 billion and a benefit to net income of $2.5 billion, or $0.23 per basic share and $0.23 per diluted share.

These server and networking equipment useful life changes primarily impact our AWS segment.

For the year ended December 31, 2022, we recorded approximately $1.1 billion, of which $720 million was recorded in the fourth quarter, of impairments of property and equipment and operating leases primarily related to physical stores. These charges were recorded in "Other operating expense (income), net" and primarily impacted our North America segment. For the year ended December 31, 2022, we also recorded expenses of approximately $480 million, primarily in "Fulfillment," primarily relating to terminating contracts for certain leases not yet commenced as well as other purchase commitments, which primarily impacted our North America segment.

For the year ended December 31, 2022, we recorded approximately $720 million, of which $640 million was recorded in the fourth quarter, of estimated severance costs primarily related to planned role eliminations. These charges were recorded primarily in "Technology and infrastructure," "Fulfillment," and "General and administrative" and primarily impacted our North America segment.

Charges for impairment, expenses for terminating contracts and other commitments, and severance costs were not material to our consolidated results of operations for the years ended December 31, 2023 and 2024.

Supplemental Cash Flow Information

The following table shows supplemental cash flow information (in millions):

	Year Ended December 31,		
	2022	2023	2024
SUPPLEMENTAL CASH FLOW INFORMATION:			
Cash paid for interest on debt, net of capitalized interest	$ 1,561	$ 2,608	$ 1,858
Cash paid for operating leases	$ 8,633	$ 10,453	$ 12,341
Cash paid for interest on finance leases	$ 374	$ 308	$ 287
Cash paid for interest on financing obligations	$ 207	$ 196	$ 219
Cash paid for income taxes, net of refunds	$ 6,035	$ 11,179	$ 12,308
Assets acquired under operating leases	$ 18,800	$ 14,052	$ 15,424
Property and equipment acquired under finance leases, net of remeasurements and modifications	$ 675	$ 642	$ 854
Property and equipment recognized during the construction period of build-to-suit lease arrangements	$ 3,187	$ 357	$ 97
Property and equipment derecognized after the construction period of build-to-suit lease arrangements, with the associated leases recognized as operating	$ 5,158	$ 1,374	$ —

Earnings Per Share

Basic earnings per share is calculated using our weighted-average outstanding common shares. Diluted earnings per share is calculated using our weighted-average outstanding common shares including the dilutive effect of stock awards as determined under the treasury stock method. In periods when we have a net loss, stock awards are excluded from our calculation of earnings per share as their inclusion would have an antidilutive effect.

The following table shows the calculation of diluted shares (in millions):

	Year Ended December 31,		
	2022	2023	2024
Shares used in computation of basic earnings per share	10,189	10,304	10,473
Total dilutive effect of outstanding stock awards	—	188	248
Shares used in computation of diluted earnings per share	10,189	10,492	10,721

Revenue

Revenue is measured based on the amount of consideration that we expect to receive, reduced by estimates for return allowances, promotional discounts, and rebates. Revenue also excludes any amounts collected on behalf of third parties, including sales and indirect taxes. In arrangements where we have multiple performance obligations, the transaction price is allocated to each performance obligation using the relative stand-alone selling price. We generally determine stand-alone selling prices based on the prices charged to customers or using expected cost plus a margin.

A description of our principal revenue generating activities is as follows:

Retail sales - We offer consumer products through our online and physical stores. Revenue is recognized when control of the goods is transferred to the customer, which generally occurs upon our delivery to a third-party carrier or, in the case of an Amazon delivery, to the customer.

Third-party seller services - We offer programs that enable sellers to sell their products in our stores, and fulfill orders using our services. We are not the seller of record in these transactions. The commissions and any related fulfillment and shipping fees we earn from these arrangements are recognized when the services are rendered, which generally occurs upon delivery of the related products to a third-party carrier or, in the case of an Amazon delivery, to the customer.

Advertising services - We provide advertising services to sellers, vendors, publishers, authors, and others, through programs such as sponsored ads, display, and video advertising. Revenue is recognized as ads are delivered based on the number of clicks or impressions.

Subscription services - Our subscription sales include fees associated with Amazon Prime memberships and access to content including digital video, audiobooks, digital music, e-books, and other non-AWS subscription services. Prime memberships provide our customers with access to an evolving suite of benefits that represent a single stand-ready obligation. Subscriptions are paid for at the time of or in advance of delivering the services. Revenue from such arrangements is recognized over the subscription period.

AWS - Our AWS arrangements include global sales of compute, storage, database, and other services. Revenue is allocated to services using stand-alone selling prices and is primarily recognized when the customer uses these services, based on the quantity of services rendered, such as compute or storage capacity delivered on-demand. Certain services, including compute and database, are also offered as a fixed quantity over a specified term, for which revenue is recognized ratably. Sales commissions we pay in connection with contracts that exceed one year are capitalized and amortized over the contract term.

Other - Other revenue includes sales related to various other offerings, such as healthcare services, certain licensing and distribution of video content, and shipping services, and our co-branded credit card agreements. Revenue is recognized when content is licensed or distributed and as or when services are performed.

Return Allowances

Return allowances, which reduce revenue and cost of sales, are estimated using historical experience. Liabilities for return allowances are included in “Accrued expenses and other” and were $1.3 billion, $1.4 billion, and $1.4 billion as of December 31, 2022, 2023, and 2024. Additions to the allowance were $5.5 billion, $5.2 billion, and $5.5 billion and deductions from the allowance were $5.2 billion, $5.1 billion, and $5.5 billion in 2022, 2023, and 2024. Included in “Inventories” on our consolidated balance sheets are assets totaling $948 million, $992 million, and $998 million as of December 31, 2022, 2023, and 2024, for the rights to recover products from customers associated with our liabilities for return allowances.

Cost of Sales

Cost of sales primarily consists of the purchase price of consumer products, inbound and outbound shipping costs, including costs related to sortation and delivery centers and where we are the transportation service provider, and digital media content costs where we record revenue gross, including video and music. Shipping costs to receive products from our suppliers are included in our inventory, and recognized as cost of sales upon sale of products to our customers. Payment processing and related transaction costs, including those associated with seller transactions, are classified in “Fulfillment” on our consolidated statements of operations.

Vendor Agreements

We have agreements with our vendors to receive consideration primarily for cooperative marketing efforts, promotions, incentives, and volume rebates. We generally consider these amounts received from vendors to be a reduction of the prices we pay for their goods, including property and equipment, or services, and are recorded as a reduction of the cost of inventory, cost of services, or cost of property and equipment. Volume rebates typically depend on reaching minimum purchase thresholds. We evaluate the likelihood of reaching purchase thresholds using past experience and current year forecasts. When volume rebates can be reasonably estimated, we record a portion of the rebate as we make progress towards the purchase threshold.

Fulfillment

Fulfillment costs primarily consist of those costs incurred in operating and staffing our North America and International segments' fulfillment centers, physical stores, and customer service centers, including facilities and equipment expenses, such as depreciation and amortization, and rent; costs attributable to buying, receiving, inspecting, and warehousing inventories; picking, packaging, and preparing customer orders for shipment; payment processing and related transaction costs, including costs associated with our guarantee for certain seller transactions; responding to inquiries from customers; and supply chain management for our manufactured electronic devices. Fulfillment costs also include amounts paid to third parties that assist us in fulfillment and customer service operations.

Technology and Infrastructure

Technology and infrastructure costs include payroll and related expenses for employees involved in the research and development of new and existing products and services, development, design, and maintenance of our stores, curation and display of products and services made available in our online stores, and infrastructure costs. Infrastructure costs include servers, networking equipment, and data center related depreciation and amortization, rent, utilities, and other expenses necessary to support AWS and other Amazon businesses. Collectively, these costs reflect the investments we make in order to offer a wide variety of products and services to our customers, including expenditures related to initiatives to build and deploy innovative and efficient software and electronic devices and the development of a satellite network for global broadband service and autonomous vehicles for ride-hailing services. Technology and infrastructure costs are generally expensed as incurred.

Sales and Marketing

Sales and marketing costs include advertising and payroll and related expenses for personnel engaged in marketing and selling activities, including sales commissions related to AWS. We pay commissions to third parties when their customer referrals result in sales. We also participate in cooperative advertising arrangements with certain of our vendors, and other third parties.

Advertising and other promotional costs to market our products and services are expensed as incurred and were $20.6 billion, $20.3 billion, and $21.4 billion in 2022, 2023, and 2024.

General and Administrative

General and administrative expenses primarily consist of costs for corporate functions, including payroll and related expenses; facilities and equipment expenses, such as depreciation and amortization expense and rent; and professional fees.

Stock-Based Compensation

Compensation cost for all equity-classified stock awards expected to vest is measured at fair value on the date of grant and recognized over the service period. The fair value of restricted stock units is determined based on the number of shares granted and the quoted price of our common stock. Such value is recognized as expense over the service period, net of estimated forfeitures, using the accelerated method. Under this method, approximately 50% of the grant date fair value is recognized as expense in the first year of grant for the majority of our stock-based compensation awards. The accelerated method also adds a higher level of sensitivity and complexity in estimating forfeitures. If an award is forfeited early in its life, the adjustment to compensation expense is much greater under an accelerated method than under a straight-line method. The estimated number of stock awards that will ultimately vest requires judgment, and to the extent actual results or updated estimates differ from our current estimates, such amounts will be recorded as a cumulative adjustment in the period estimates are revised. We consider many factors when estimating expected forfeitures, including historical forfeiture experience. Additionally, we have stock-based compensation awards that are expected to settle in cash. These liability-classified awards are remeasured to fair value at the end of each reporting period until settlement or expiration.

Other Operating Expense (Income), Net

Other operating expense (income), net, consists primarily of the amortization of intangible assets and asset impairments.

Other Income (Expense), Net

Other income (expense), net, is as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
Marketable equity securities valuation gains (losses)	$ (13,870)	$ 984	$ (1,278)
Equity warrant valuation gains (losses)	(2,132)	26	(192)
Upward adjustments relating to equity investments in private companies	76	40	49
Foreign currency gains (losses)	(340)	65	(408)
Other, net	(540)	(177)	(421)
Total other income (expense), net	$ (16,806)	$ 938	$ (2,250)

Included in other income (expense), net in 2022, 2023, and 2024 is a marketable equity securities valuation gain (loss) of $(12.7) billion, $797 million, and $(1.6) billion from our equity investment in Rivian Automotive, Inc. ("Rivian"). Our investment in Rivian's preferred stock was accounted for at cost, with adjustments for observable changes in prices or impairments, prior to Rivian's initial public offering in November 2021, which resulted in the conversion of our preferred stock to Class A common stock. As of December 31, 2024, we held 158 million shares of Rivian's Class A common stock, representing an approximate 14% ownership interest, and an approximate 13% voting interest. We determined that we have the ability to exercise significant influence over Rivian through our equity investment, our commercial arrangement for the purchase of electric vehicles and jointly-owned intellectual property, and one of our employees serving on Rivian's board of directors. We elected the fair value option to account for our equity investment in Rivian, which is included in "Marketable securities" on our consolidated balance sheets, and had a fair value of $3.7 billion and $2.1 billion as of December 31, 2023 and December 31, 2024. The investment was subject to regulatory sales restrictions resulting in a discount for lack of marketability of approximately $800 million as of December 31, 2021, which expired in Q1 2022.

Income Taxes

Income tax expense includes U.S. (federal and state) and foreign income taxes. Certain foreign subsidiary earnings and losses are subject to current U.S. taxation and the subsequent repatriation of those earnings is not subject to tax in the U.S. We intend to invest substantially all of our foreign subsidiary earnings, as well as our capital in our foreign subsidiaries, indefinitely outside of the U.S. in those jurisdictions in which we would incur significant, additional costs upon repatriation of such amounts.

Deferred income tax balances reflect the effects of temporary differences between the carrying amounts of assets and liabilities and their tax bases, as well as net operating loss and tax credit carryforwards, and are stated at enacted tax rates expected to be in effect when taxes are actually paid or recovered.

Deferred tax assets represent amounts available to reduce income taxes payable in future periods. Deferred tax assets are evaluated for future realization and reduced by a valuation allowance to the extent we believe they will not be realized. We consider many factors when assessing the likelihood of future realization of our deferred tax assets, including recent cumulative loss experience and expectations of future earnings, capital gains and investment in such jurisdiction, the carry-forward periods available to us for tax reporting purposes, and other relevant factors.

We utilize a two-step approach to recognizing and measuring uncertain income tax positions (income tax contingencies). The first step is to evaluate the tax position for recognition by determining if the weight of available evidence indicates it is more likely than not the position will be sustained on audit, including resolution of related appeals or litigation processes. The second step is to measure the tax benefit as the largest amount which is more than 50% likely of being realized upon ultimate settlement. We consider many factors when evaluating our tax positions and estimating our tax benefits, which may require periodic adjustments and which may not accurately forecast actual outcomes. We include interest and penalties related to our income tax contingencies in income tax expense.

Fair Value of Financial Instruments

Fair value is defined as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. To increase the comparability of fair value measures, the following hierarchy prioritizes the inputs to valuation methodologies used to measure fair value:

Level 1 — Valuations based on quoted prices for identical assets and liabilities in active markets.

Level 2 — Valuations based on observable inputs other than quoted prices included in Level 1, such as quoted prices for similar assets and liabilities in active markets, quoted prices for identical or similar assets and liabilities in markets that are not active, or other inputs that are observable or can be corroborated by observable market data.

Level 3 — Valuations based on unobservable inputs reflecting our own assumptions, consistent with reasonably available assumptions made by other market participants. These valuations require significant judgment.

We measure the fair value of money market funds and certain marketable equity securities based on quoted prices in active markets for identical assets or liabilities. Other marketable securities were valued either based on recent trades of securities in inactive markets or based on quoted market prices of similar instruments and other significant inputs derived from or corroborated by observable market data.

Cash and Cash Equivalents

We classify all highly liquid instruments with an original maturity of three months or less as cash equivalents.

Inventories

Inventories, consisting of products available for sale, are primarily accounted for using the first-in, first-out method, and are valued at the lower of cost and net realizable value. This valuation requires us to make judgments, based on currently available information, about the likely method of disposition, such as through sales to individual customers, returns to product vendors, or liquidations, and expected recoverable values of each disposition category. The inventory valuation allowance, representing a write-down of inventory, was $3.0 billion as of December 31, 2023 and 2024.

We provide Fulfillment by Amazon services in connection with certain of our sellers' programs. Third-party sellers maintain ownership of their inventory, regardless of whether fulfillment is provided by us or the third-party sellers, and therefore these products are not included in our inventories.

We also purchase electronic device components from a variety of suppliers and use several contract manufacturers to provide manufacturing services for our products. During the normal course of business, in order to manage manufacturing lead times and help ensure adequate supply, we enter into agreements with contract manufacturers and suppliers for certain electronic device components. We have certain non-cancellable purchase commitments arising from these agreements. These commitments are based on forecasted customer demand. If we reduce these commitments, we may incur additional costs.

Accounts Receivable, Net and Other

Included in "Accounts receivable, net and other" on our consolidated balance sheets are receivables primarily related to customers, vendors, and prepaid expenses and other current assets. As of December 31, 2023 and 2024, customer receivables, net, were $34.1 billion and $34.3 billion, vendor receivables, net, were $8.5 billion and $11.6 billion, and other receivables, net, were $4.3 billion and $3.4 billion. Prepaid expenses and other current assets, which include amounts related to non-income taxes and satellite network launch services deposits, were $5.4 billion and $6.3 billion as of December 31, 2023 and December 31, 2024. We currently expense satellite network launch services deposits upon launch to "Technology and infrastructure."

We estimate losses on receivables based on expected losses, including our historical experience of actual losses. Receivables are considered impaired and written-off when it is probable that all contractual payments due will not be collected in accordance with the terms of the agreement. The allowance for doubtful accounts was $1.4 billion, $1.7 billion, and $2.0 billion as of December 31, 2022, 2023, and 2024. Additions to the allowance were $1.6 billion, $1.9 billion, and $1.9 billion, and deductions to the allowance were $1.3 billion, $1.6 billion, and $1.6 billion in 2022, 2023, and 2024.

Software Development Costs

We incur software development costs related to products to be sold, leased, or marketed to external users, internal-use software, and our websites. Software development costs capitalized were not significant for the years presented. All other costs, including those related to design or maintenance, are expensed as incurred.

Property and Equipment, Net

Property and equipment are stated at cost less accumulated depreciation and amortization. Incentives that we receive from property and equipment vendors are recorded as a reduction to our costs. Property includes buildings and land that we own, along with property we have acquired under build-to-suit lease arrangements when we have control over the building during the construction period and finance lease arrangements. Equipment includes assets such as servers and networking equipment, heavy equipment, and other fulfillment equipment. Depreciation and amortization is recorded on a straight-line basis over the estimated useful lives of the assets (generally the lesser of 40 years or the remaining life of the underlying building, five years prior to January 1, 2024 and six years subsequent to January 1, 2024 for our servers, six years for our networking equipment, ten years for heavy equipment, and three to ten years for other fulfillment equipment). Depreciation and amortization expense is classified within the corresponding operating expense categories on our consolidated statements of operations.

Leases

We categorize leases with contractual terms longer than twelve months as either operating or finance. Finance leases are generally those leases that allow us to substantially utilize or pay for the entire asset over its estimated life. Assets acquired under finance leases are recorded in "Property and equipment, net." All other leases are categorized as operating leases. Our leases generally have terms that range from one to ten years for equipment and one to twenty years for property.

Certain lease contracts include obligations to pay for other services, such as operations and maintenance. For leases of property, we account for these other services as a component of the lease. For substantially all other leases, the services are accounted for separately and we allocate payments to the lease and other services components based on estimated stand-alone prices.

Lease liabilities are recognized at the present value of the fixed lease payments, reduced by landlord incentives using a discount rate based on similarly secured borrowings available to us. Lease assets are recognized based on the initial present value of the fixed lease payments, reduced by landlord incentives, plus any direct costs from executing the leases or lease prepayments reclassified from "Other assets" upon lease commencement. Leasehold improvements are capitalized at cost and amortized over the lesser of their expected useful life or the lease term.

When we have the option to extend the lease term, terminate the lease before the contractual expiration date, or purchase the leased asset, and it is reasonably certain that we will exercise the option, we consider the option in determining the classification and measurement of the lease. Our leases may include variable payments based on measures that include changes in price indices, market interest rates, or the level of sales at a physical store, which are expensed as incurred.

Costs associated with operating lease assets are recognized on a straight-line basis within operating expenses over the term of the lease. Finance lease assets are amortized within operating expenses on a straight-line basis over the shorter of the estimated useful lives of the assets or, in the instance where title does not transfer at the end of the lease term, the lease term. The interest component of a finance lease is included in interest expense and recognized using the effective interest method over the lease term.

We establish assets and liabilities for the present value of estimated future costs to retire long-lived assets at the termination or expiration of a lease. Such assets are amortized over the lease period into operating expense, and the recorded liabilities are accreted to the future value of the estimated retirement costs.

Financing Obligations

We record assets and liabilities for estimated construction costs under build-to-suit lease arrangements when we have control over the building during the construction period. If we continue to control the building after the construction period, the arrangement is classified as a financing obligation instead of a lease. The building is depreciated over the shorter of its useful life or the term of the obligation.

If we do not control the building after the construction period ends, the assets and liabilities for construction costs are derecognized, and we classify the lease as operating.

Goodwill and Indefinite-Lived Intangible Assets

We evaluate goodwill and indefinite-lived intangible assets for impairment annually or more frequently when an event occurs or circumstances change that indicate the carrying value may not be recoverable. We may elect to utilize a qualitative assessment to evaluate whether it is more likely than not that the fair value of a reporting unit or indefinite-lived intangible asset is less than its carrying value and if so, we perform a quantitative test. We compare the carrying value of each reporting unit and indefinite-lived intangible asset to its estimated fair value and if the fair value is determined to be less than the carrying value, we recognize an impairment loss for the difference. We estimate the fair value of the reporting units using discounted cash flows. Forecasts of future cash flows are based on our best estimate of future net sales and operating expenses, based primarily on expected category expansion, pricing, market segment share, and general economic conditions.

We completed the required annual impairment test of goodwill for all reporting units and indefinite-lived intangible assets as of April 1, 2024, resulting in no impairments. The fair value of our reporting units substantially exceeded their carrying value. There were no events that caused us to update our annual impairment test. See "Note 5 — Acquisitions, Goodwill, and Acquired Intangible Assets."

Other Assets

Included in “Other assets” on our consolidated balance sheets are amounts primarily related to video and music content, net of accumulated amortization; long-term deferred tax assets; acquired intangible assets, net of accumulated amortization; convertible notes and certain equity investments; satellite network launch services deposits; and affordable housing loans. We recognize certain transactions with governments when there is reasonable assurance that incentives included in the agreements, such as cash or certain tax credits, will be received and we are able to comply with any related conditions. These incentives are recorded as reductions to the cost of related assets or expenses.

Digital Video and Music Content

We obtain video content, inclusive of episodic television and movies, and music content for customers through licensing agreements that have a wide range of licensing provisions including both fixed and variable payment schedules. When the license fee for a specific video or music title is determinable or reasonably estimable and the content is available to us, we recognize an asset and a corresponding liability for the amounts owed. We reduce the liability as payments are made and we amortize the asset to “Cost of sales” on an accelerated basis, based on estimated usage or viewing patterns, or on a straight-line basis. If the licensing fee is not determinable or reasonably estimable, no asset or liability is recorded and licensing costs are expensed as incurred. We also develop original video content for which the production costs are capitalized and amortized to “Cost of sales” predominantly on an accelerated basis that follows the estimated viewing patterns associated with the content. The weighted average remaining life of our capitalized video content is 3.1 years. We review usage and viewing patterns impacting the amortization of capitalized video content on an ongoing basis and reflect any changes prospectively.

Our produced and licensed video content is primarily monetized together as a unit, referred to as a film group, in each major geography where we offer Amazon Prime memberships. These film groups are evaluated for impairment whenever an event occurs or circumstances change indicating the fair value is less than the carrying value. The total capitalized costs of video, which is primarily released content, and music as of December 31, 2023 and 2024 were $17.4 billion and $19.6 billion. Total video and music expense was $18.9 billion and $20.4 billion for the year ended December 31, 2023 and 2024. Total video and music expense includes licensing and production costs associated with content offered within Amazon Prime memberships, and costs associated with digital subscriptions and sold or rented content.

Cash Equivalents and Marketable Securities

We generally invest our excess cash in investment grade short- to intermediate-term marketable debt securities and AAA-rated money market funds. Such investments are included in “Cash and cash equivalents” or “Marketable securities” on the accompanying consolidated balance sheets.

Marketable debt securities are classified as available-for-sale and reported at fair value with unrealized gains and losses included in “Accumulated other comprehensive income (loss).” Each reporting period, we evaluate whether declines in fair value below carrying value are due to expected credit losses, as well as our ability and intent to hold the investment until a forecasted recovery occurs. Expected credit losses are recorded as an allowance through “Other income (expense), net” on our consolidated statements of operations.

Equity investments that have readily determinable fair values, including investments for which we have elected the fair value option, are included in “Marketable securities” on our consolidated balance sheets and measured at fair value with changes recognized in “Other income (expense), net” on our consolidated statements of operations.

Non-Marketable Investments

Notes that are convertible to equity classified as available-for-sale are reported at fair value with unrealized gains and losses included in “Accumulated other comprehensive income (loss).” Credit losses, if any, are recorded as an allowance through “Other income (expense), net” on our consolidated statements of operations. Upon conversion, the amount of the notes reported at fair value are reclassified generally from available-for-sale to equity investments accounted for at cost, with any associated unrealized gain or loss reclassified from “Accumulated other comprehensive income (loss)” to “Other income (expense), net” on our consolidated statements of operations.

In Q3 2023, we invested in a $1.25 billion note from Anthropic, PBC, which is convertible to equity. In Q1 2024, we invested $2.75 billion in a second convertible note. In Q4 2024, we entered into an agreement and invested $1.3 billion in a third convertible note, and will invest an additional $2.7 billion by Q4 2025. The notes are classified as available-for-sale and are classified as Level 3 assets, and as of December 31, 2024 had an estimated fair value of approximately $13.8 billion. In making these estimates, we utilized valuation methods based on information available, including the rights and obligations of the convertible notes, other outstanding classes of securities, observable transactions such as new securities offerings, estimates of expected time to and type of liquidity events and anticipated securities offerings, and discounts for lack of marketability. Subsequent to December 31, 2024, a portion of the notes were converted to nonvoting preferred stock. As a result of this conversion, a significant portion of the unrealized gain associated with the notes as of December 31, 2024 was reclassified and a gain will be recorded in “Other income (expense), net” in our Q1 2025 consolidated statement of operations. We also have a commercial arrangement primarily for the provision of AWS cloud services, which includes the use of AWS chips.

Equity investments in private companies for which we do not have the ability to exercise significant influence are accounted for at cost, with adjustments for observable changes in prices or impairments recognized in “Other income (expense), net” on our consolidated statements of operations. Each reporting period, we perform a qualitative assessment to evaluate whether the investment is impaired. Our assessment includes a review of recent operating results and trends, recent sales/acquisitions of the investee securities, and other publicly available data. If the investment is impaired, we write it down to its estimated fair value. As of December 31, 2023 and 2024, these investments had a carrying value of $754 million and $989 million.

Equity investments where we can exercise significant influence, but not control, over an investee are accounted for using the equity method of accounting, or at fair value if we elect the fair value option. Our share of the earnings or losses as reported by equity-method investees, amortization of basis differences, related gains or losses, and impairments, if any, are recognized in “Equity-method investment activity, net of tax” on our consolidated statements of operations. Each reporting period, we evaluate whether declines in fair value below carrying value are other-than-temporary and if so, we write down the investment to its estimated fair value. As of December 31, 2023 and 2024, these investments had a carrying value of $614 million and $1.2 billion.

As of December 31, 2023 and 2024, equity warrants measured at fair value were $2.2 billion and $2.7 billion, with changes recognized in “Other income (expense), net” on our consolidated statements of operations. These warrants are classified as Level 2 and 3 assets.

These non-marketable investments are included within “Other assets” on our consolidated balance sheets.

Certain of our investments represent a variable interest in an entity. We do not consolidate the entities in which we hold these investments because we are not the primary beneficiary.

Long-Lived Assets

Long-lived assets, other than goodwill and indefinite-lived intangible assets, are reviewed for impairment whenever events or changes in circumstances indicate that the carrying amount of the assets might not be recoverable. Conditions that would necessitate an impairment assessment include a significant decline in the observable market value of an asset, a significant change in the extent or manner in which an asset is used, or any other significant adverse change that would indicate that the carrying amount of an asset or group of assets may not be recoverable.

For long-lived assets used in operations, including lease assets, impairment losses are only recorded if the asset’s carrying amount is not recoverable through its undiscounted, probability-weighted future cash flows. We measure the impairment loss based on the difference between the carrying amount and estimated fair value. Long-lived assets are considered held for sale when certain criteria are met, including when management has committed to a plan to sell the asset, the asset is available for sale in its immediate condition, and the sale is probable within one year of the reporting date. Assets held for sale are reported at the lower of cost or fair value less costs to sell. Assets held for sale were not significant as of December 31, 2023 and 2024.

Accrued Expenses and Other

Included in “Accrued expenses and other” on our consolidated balance sheets are liabilities primarily related to tax-related liabilities, leases and asset retirement obligations, self-insurance liabilities, payroll and related expenses, current debt, unredeemed gift cards, customer liabilities, marketing liabilities, acquired digital media content, and other operating expenses.

As of December 31, 2023 and 2024, our liabilities for payroll related expenses were $7.7 billion and $7.5 billion and our liabilities for unredeemed gift cards were $5.3 billion and $5.4 billion. We reduce the liability for a gift card when redeemed by a customer. The portion of gift cards that we do not expect to be redeemed is recognized based on customer usage patterns.

Self-Insurance Liabilities

Although we maintain certain high-deductible, third-party insurance coverage for catastrophic losses, we effectively self-insure for exposure primarily related to workers’ compensation, employee healthcare benefits, general and product liability, and automobile liability, including liability resulting from third-party transportation service providers. We estimate self-insurance liabilities by considering historical claims experience, frequency and costs of claims, projected claims development, inflation, and other actuarial assumptions. Changes in the number or costs of claims, healthcare costs, judgment and settlement amounts, associated legal expenses, and other factors could cause actual results to differ materially from these estimates. In the fourth quarter of 2022, we increased our reserves for general, product, and automobile liabilities by $1.3 billion primarily driven by changes in our estimates about the costs of asserted and unasserted claims, which was primarily recorded in “Cost of sales” on our consolidated statements of operations and impacted our North America segment. Increases to our reserves driven by changes in estimates were not material to our consolidated results of operations for the years ended December 31, 2023 and 2024. As of December 31, 2023 and 2024, our total self-insurance liabilities were $6.3 billion and $8.5 billion and are included in "Accrued expenses and other" on our consolidated balance sheets.

Unearned Revenue

Unearned revenue is recorded when payments are received or due in advance of performing our service obligations and is recognized over the service period. Unearned revenue primarily relates to prepayments of AWS services and Amazon Prime memberships. Our total unearned revenue as of December 31, 2023 was $20.9 billion, of which $14.2 billion was recognized as revenue during the year ended December 31, 2024 and our total unearned revenue as of December 31, 2024 was $24.6 billion. Included in "Other long-term liabilities" on our consolidated balance sheets was $5.7 billion and $6.5 billion of unearned revenue as of December 31, 2023 and 2024.

Additionally, we have performance obligations, primarily related to AWS, associated with commitments in customer contracts for future services that have not yet been recognized in our financial statements. For contracts with original terms that exceed one year, those commitments not yet recognized were approximately $177 billion as of December 31, 2024. The weighted average remaining life of our long-term contracts is 4.1 years. However, the amount and timing of revenue recognition is largely driven by customer usage, which can extend beyond the original contractual term.

Other Long-Term Liabilities

Included in "Other long-term liabilities" on our consolidated balance sheets are liabilities primarily related to financing obligations, unearned revenue, asset retirement obligations, tax contingencies, deferred tax liabilities, and digital video and music content.

Foreign Currency

We have internationally-focused stores for which the net sales generated, as well as most of the related expenses directly incurred from those operations, are denominated in local functional currencies. The functional currency of our subsidiaries that either operate or support these stores is generally the same as the local currency. Assets and liabilities of these subsidiaries are translated into U.S. Dollars at period-end foreign exchange rates, and revenues and expenses are translated at average rates prevailing throughout the period. Translation adjustments are included in "Accumulated other comprehensive income (loss)," a separate component of stockholders' equity. Transaction gains and losses including intercompany transactions denominated in a currency other than the functional currency of the entity involved are included in "Other income (expense), net" on our consolidated statements of operations. In connection with the settlement and remeasurement of intercompany balances, we recorded gains (losses) of $386 million, $(329) million, and $413 million in 2022, 2023, and 2024.

Accounting Pronouncements Not Yet Adopted

In December 2023, the Financial Accounting Standards Board ("FASB") issued an Accounting Standards Update ("ASU") amending existing income tax disclosure guidance, primarily requiring more detailed disclosure for income taxes paid and the effective tax rate reconciliation. The ASU is effective for annual reporting periods beginning after December 15, 2024, with early adoption permitted and can be applied on either a prospective or retroactive basis. We expect to adopt the ASU on a retroactive basis.

In November 2024, the FASB issued an ASU amending existing income statement disclosure guidance, primarily requiring more detailed disclosure for expenses. The ASU is effective for annual reporting periods beginning after December 15, 2026, and interim periods within fiscal years beginning after December 15, 2027, with early adoption permitted. The amendments can be applied on either a prospective or retroactive basis. We are currently evaluating the ASU to determine its impact on our disclosures.

Note 2 — FINANCIAL INSTRUMENTS

Cash, Cash Equivalents, Restricted Cash, and Marketable Securities

As of December 31, 2023 and 2024, our cash, cash equivalents, restricted cash, and marketable securities primarily consisted of cash, AAA-rated money market funds, U.S. and foreign government and agency securities, other investment grade securities, and marketable equity securities. Cash equivalents and marketable securities are recorded at fair value. The following table summarizes, by major security type, our cash, cash equivalents, restricted cash, and marketable securities that are measured at fair value on a recurring basis and are categorized using the fair value hierarchy (in millions):

	December 31, 2023			
	Cost or Amortized Cost	Gross Unrealized Gains	Gross Unrealized Losses	Total Estimated Fair Value
Cash	$ 11,706	$ —	$ —	$ 11,706
Level 1 securities:				
Money market funds	39,160	—	—	39,160
Equity securities (1)				4,658
Level 2 securities:				
Foreign government and agency securities	505	—	—	505
U.S. government and agency securities	1,789	1	(91)	1,699
Corporate debt securities	27,996	—	(191)	27,805
Asset-backed securities	1,707	—	(61)	1,646
Other debt securities	108	—	(4)	104
	$ 82,971	$ 1	$ (347)	$ 87,283
Less: Restricted cash, cash equivalents, and marketable securities (2)				(503)
Total cash, cash equivalents, and marketable securities				$ 86,780

	December 31, 2024			
	Cost or Amortized Cost	Gross Unrealized Gains	Gross Unrealized Losses	Total Estimated Fair Value
Cash	$ 17,055	$ —	$ —	$ 17,055
Level 1 securities:				
Money market funds	28,282	—	—	28,282
Equity securities (1)				3,318
Level 2 securities:				
Foreign government and agency securities	177	—	—	177
U.S. government and agency securities	3,452	1	(52)	3,401
Corporate debt securities	50,959	3	(50)	50,912
Asset-backed securities	1,539	2	(18)	1,523
Other debt securities	68	—	(1)	67
	$ 101,532	$ 6	$ (121)	$ 104,735
Less: Restricted cash, cash equivalents, and marketable securities (2)				(3,533)
Total cash, cash equivalents, and marketable securities				$ 101,202

(1) The related unrealized gain (loss) recorded in "Other income (expense), net" was $(13.6) billion, $1.0 billion, and $(1.3) billion for the years ended December 31, 2022, 2023, and 2024.

(2) We are required to pledge or otherwise restrict a portion of our cash, cash equivalents, and marketable debt securities primarily as collateral for real estate, amounts due to third-party sellers in certain jurisdictions, debt, standby and trade letters of credit, and licenses of digital media content. We classify cash, cash equivalents, and marketable debt securities with use restrictions of less than twelve months as "Accounts receivable, net and other" and of twelve months or longer as non-current "Other assets" on our consolidated balance sheets. See "Note 7 — Commitments and Contingencies."

The following table summarizes gross gains and gross losses realized on sales of marketable debt securities (in millions):

	Year Ended December 31,		
	2022	2023	2024
Realized gains	$ 43	$ 2	$ 3
Realized losses	341	67	10

The following table summarizes the remaining contractual maturities of our cash equivalents and marketable debt securities as of December 31, 2024 (in millions):

	Amortized Cost	Estimated Fair Value
Due within one year	$ 78,065	$ 78,046
Due after one year through five years	5,023	4,969
Due after five years through ten years	560	553
Due after ten years	829	794
Total	$ 84,477	$ 84,362

Actual maturities may differ from the contractual maturities because borrowers may have certain prepayment conditions.

Consolidated Statements of Cash Flows Reconciliation

The following table provides a reconciliation of the amount of cash, cash equivalents, and restricted cash reported within the consolidated balance sheets to the total of the same such amounts shown in the consolidated statements of cash flows (in millions):

	December 31, 2023	December 31, 2024
Cash and cash equivalents	$ 73,387	$ 78,779
Restricted cash included in accounts receivable, net and other	497	247
Restricted cash included in other assets	6	3,286
Total cash, cash equivalents, and restricted cash shown in the consolidated statements of cash flows	$ 73,890	$ 82,312

Note 3 — PROPERTY AND EQUIPMENT

Property and equipment, at cost, consisted of the following (in millions):

	December 31,	
	2023	2024
Gross property and equipment (1):		
Land and buildings	$ 105,293	$ 123,039
Equipment	185,039	218,893
Other assets	5,116	5,487
Construction in progress	28,840	46,636
Gross property and equipment	324,288	394,055
Total accumulated depreciation and amortization (1)	120,111	141,390
Total property and equipment, net	$ 204,177	$ 252,665

(1) Includes the original cost and accumulated depreciation of fully-depreciated assets.

Depreciation and amortization expense on property and equipment was $24.9 billion, $30.2 billion, and $32.1 billion which includes amortization of property and equipment acquired under finance leases of $6.1 billion, $5.9 billion, and $3.9 billion for 2022, 2023, and 2024.

Note 4 — LEASES

We have entered into non-cancellable operating and finance leases for fulfillment network, data center, office, and physical store facilities as well as server and networking equipment, aircraft, and vehicles. Gross assets acquired under finance leases, including those where title transfers at the end of the lease, are recorded in "Property and equipment, net" and were \$62.5 billion and \$56.5 billion as of December 31, 2023 and 2024. Accumulated amortization associated with finance leases was \$44.7 billion and \$41.8 billion as of December 31, 2023 and 2024.

Lease cost recognized in our consolidated statements of operations is summarized as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
Operating lease cost	\$ 8,847	\$ 10,550	\$ 11,961
Finance lease cost:			
Amortization of lease assets	6,097	5,899	3,866
Interest on lease liabilities	361	304	285
Finance lease cost	6,458	6,203	4,151
Variable lease cost	1,852	2,165	2,465
Total lease cost	\$ 17,157	\$ 18,918	\$ 18,577

Other information about lease amounts recognized in our consolidated financial statements is as follows:

	December 31, 2023	December 31, 2024
Weighted-average remaining lease term – operating leases	11.3 years	10.6 years
Weighted-average remaining lease term – finance leases	11.9 years	11.9 years
Weighted-average discount rate – operating leases	3.3 %	3.5 %
Weighted-average discount rate – finance leases	2.7 %	3.0 %

Our lease liabilities were as follows (in millions):

	December 31, 2023		
	Operating Leases	Finance Leases	Total
Gross lease liabilities	\$ 90,777	\$ 14,106	\$ 104,883
Less: imputed interest	(15,138)	(1,997)	(17,135)
Present value of lease liabilities	75,639	12,109	87,748
Less: current portion of lease liabilities	(8,419)	(2,032)	(10,451)
Total long-term lease liabilities	\$ 67,220	\$ 10,077	\$ 77,297

	December 31, 2024		
	Operating Leases	Finance Leases	Total
Gross lease liabilities	\$ 95,294	\$ 12,520	\$ 107,814
Less: imputed interest	(15,698)	(1,918)	(17,616)
Present value of lease liabilities	79,596	10,602	90,198
Less: current portion of lease liabilities	(10,546)	(1,375)	(11,921)
Total long-term lease liabilities	\$ 69,050	\$ 9,227	\$ 78,277

Note 5 — ACQUISITIONS, GOODWILL, AND ACQUIRED INTANGIBLE ASSETS

2022 Acquisition Activity

On March 17, 2022, we acquired MGM Holdings Inc., for cash consideration of approximately $6.1 billion, net of cash acquired, to provide more digital media content options for customers. We also assumed $2.5 billion of debt, which we repaid immediately after closing. The acquired assets primarily consist of $3.4 billion of video content and $4.9 billion of goodwill.

During 2022, we also completed acquisition activity for aggregate cash consideration of $141 million, net of cash acquired.

2023 Acquisition Activity

On February 22, 2023, we acquired 1Life Healthcare, Inc. (One Medical), for cash consideration of approximately $3.5 billion, net of cash acquired, to provide healthcare options for customers. The acquired assets primarily consist of $1.3 billion of intangible assets and $2.5 billion of goodwill, which is allocated to our North America segment.

During 2023, we also completed acquisition activity for immaterial aggregate cash consideration, net of cash acquired.

2024 Acquisition Activity

During 2024, we completed acquisition activity for aggregate cash consideration of $780 million, net of cash acquired.

The primary reasons for these transactions were to acquire technologies and know-how to enable Amazon to serve customers more effectively or to expand our customer base.

Pro forma results of operations have not been presented because the effects of the 2024 transactions, individually and in the aggregate, were not material to our consolidated results of operations. Transaction-related costs were expensed as incurred and were not significant.

Goodwill

The goodwill resulting from the acquisition activity is primarily related to expected improvements in technology performance and functionality, as well as sales growth from future product and service offerings and new customers, together with certain intangible assets that do not qualify for separate recognition. The goodwill resulting from the acquisition activity is generally not deductible for tax purposes. The following summarizes our goodwill activity in 2023 and 2024 by segment (in millions):

	North America	International	AWS	Consolidated
Goodwill - January 1, 2023	$ 16,621	$ 2,411	$ 1,256	$ 20,288
Acquisition activity	2,494	—	—	2,494
Other adjustments (1)	11	1	(5)	7
Goodwill - December 31, 2023	19,126	2,412	1,251	22,789
Acquisition activity	191	77	52	320
Other adjustments (1)	(28)	(4)	(3)	(35)
Goodwill - December 31, 2024	$ 19,289	$ 2,485	$ 1,300	$ 23,074

(1) Primarily includes changes in foreign exchange rates.

Intangible Assets

Acquired identifiable intangible assets are valued primarily by using discounted cash flows. These assets are included within "Other assets" on our consolidated balance sheets and consist of the following (in millions):

	December 31,						
	2023			2024			
	Acquired Intangibles, Gross (1)	Accumulated Amortization (1)	Acquired Intangibles, Net	Acquired Intangibles, Gross (1)	Accumulated Amortization (1)	Acquired Intangibles, Net	Weighted Average Life Remaining
Finite-lived intangible assets (2):							
Marketing-related	$ 2,643	$ (738)	$ 1,905	$ 2,629	$ (860)	$ 1,769	16.8
Contract-based	4,800	(1,129)	3,671	5,767	(1,541)	4,226	10.9
Technology- and content-based	743	(340)	403	1,246	(284)	962	3.2
Customer-related	749	(188)	561	764	(282)	482	5.7
Total finite-lived intangible assets	$ 8,935	$ (2,395)	$ 6,540	$ 10,406	$ (2,967)	$ 7,439	11.0
IPR&D and other (3)	$ 1,147		$ 1,147	$ 1,163		$ 1,163	
Total acquired intangibles	$ 10,082	$ (2,395)	$ 7,687	$ 11,569	$ (2,967)	$ 8,602	

(1) Excludes the original cost and accumulated amortization of fully-amortized intangibles.
(2) Finite-lived intangible assets, excluding acquired video content, have estimated useful lives of between one and twenty-five years, and are being amortized to operating expenses on a straight-line basis.
(3) Intangible assets acquired in a business combination that are in-process and used in research and development activities are considered indefinite-lived until the completion or abandonment of the research and development efforts. Once the research and development efforts are completed, we determine the useful life and begin amortizing the assets.

Amortization expense for acquired finite-lived intangibles was $604 million, $706 million, and $838 million in 2022, 2023, and 2024. Expected future amortization expense of acquired finite-lived intangible assets as of December 31, 2024 is as follows (in millions):

Year Ended December 31,	
2025	$ 994
2026	918
2027	796
2028	632
2029	615
Thereafter	3,484
	$ 7,439

Note 6 — DEBT

As of December 31, 2024, we had $58.0 billion of unsecured senior notes outstanding (the "Notes"). Our total long-term debt obligations are as follows (in millions):

	Maturities (1)	Stated Interest Rates	Effective Interest Rates	December 31, 2023	December 31, 2024
2014 Notes issuance of $6.0 billion	2034 - 2044	4.80% - 4.95%	4.93% - 5.12%	4,000	2,750
2017 Notes issuance of $17.0 billion	2025 - 2057	3.15% - 5.20%	3.02% - 4.33%	15,000	13,000
2020 Notes issuance of $10.0 billion	2025 - 2060	0.80% - 2.70%	0.88% - 2.77%	9,000	9,000
2021 Notes issuance of $18.5 billion	2026 - 2061	1.00% - 3.25%	1.14% - 3.31%	17,500	15,000
April 2022 Notes issuance of $12.8 billion	2025 - 2062	3.00% - 4.10%	3.13% - 4.15%	12,750	11,250
December 2022 Notes issuance of $8.3 billion	2025 - 2032	4.55% - 4.70%	4.61% - 4.74%	8,250	7,000
Credit Facility				682	—
Total face value of long-term debt				67,182	58,000
Unamortized discount and issuance costs, net				(374)	(360)
Less: current portion of long-term debt				(8,494)	(5,017)
Long-term debt				$ 58,314	$ 52,623

(1) The weighted-average remaining lives of the 2014, 2017, 2020, 2021, April 2022, and December 2022 Notes were 15.4, 15.2, 16.5, 14.2, 12.8, and 4.6 years as of December 31, 2024. The combined weighted-average remaining life of the Notes was 13.4 years as of December 31, 2024.

Interest on the Notes is payable semi-annually in arrears. We may redeem the Notes at any time in whole, or from time to time, in part at specified redemption prices. We are not subject to any financial covenants under the Notes. The estimated fair value of the Notes was approximately $60.6 billion and $50.2 billion as of December 31, 2023 and 2024, which is based on quoted prices for our debt as of those dates.

As of September 30, 2024, we had repaid outstanding borrowings and terminated the secured revolving credit facility with a lender that was secured by certain seller receivables (the "Credit Facility"). The Credit Facility bore interest based on the daily Secured Overnight Financing Rate plus 1.25%, and had a commitment fee of up to 0.45% on the undrawn portion. There were $682 million of borrowings outstanding under the Credit Facility as of December 31, 2023, which had an interest rate of 6.6%. As of December 31, 2023, we had pledged $806 million of our cash and seller receivables as collateral for debt related to our Credit Facility. The estimated fair value of the Credit Facility, which was based on Level 2 inputs, approximated its carrying value as of December 31, 2023.

As of December 31, 2024, future principal payments for our total long-term debt were as follows (in millions):

Year Ended December 31,	
2025	$ 5,000
2026	2,750
2027	8,750
2028	2,250
2029	3,000
Thereafter	36,250
	$ 58,000

In January 2023, we entered into an $8.0 billion unsecured 364-day term loan with a syndicate of lenders (the "Term Loan"), maturing in January 2024 and bearing interest at the Secured Overnight Financing Rate specified in the Term Loan plus 0.75%. The Term Loan was classified as short-term debt and included within "Accrued expenses and other" on our consolidated balance sheets. As of December 31, 2023, the entire amount of the Term Loan had been repaid.

We have U.S. Dollar and Euro commercial paper programs (the "Commercial Paper Programs") under which we may from time to time issue unsecured commercial paper up to a total of $20.0 billion (including up to €3.0 billion) at the date of issue, with individual maturities that may vary but will not exceed 397 days from the date of issue. There were no borrowings outstanding under the Commercial Paper Programs as of December 31, 2023 and 2024. We use the net proceeds from the issuance of commercial paper for general corporate purposes.

We have a $15.0 billion unsecured revolving credit facility with a syndicate of lenders (the “Credit Agreement”), with a term that extends to November 2028 and may be extended for one or more additional one-year terms subject to approval by the lenders. The interest rate applicable to outstanding balances under the Credit Agreement is the applicable benchmark rate specified in the Credit Agreement plus 0.45%, with a commitment fee of 0.03% on the undrawn portion of the credit facility. There were no borrowings outstanding under the Credit Agreement as of December 31, 2023 and 2024.

In October 2024, we entered into a $5.0 billion unsecured 364-day revolving credit facility with a syndicate of lenders (the “Short-Term Credit Agreement”), which replaced the prior 364-day revolving credit agreement entered into in November 2023. The Short-Term Credit Agreement matures in October 2025 and may be extended for one additional period of 364 days subject to approval by the lenders. The interest rate applicable to outstanding balances under the Short-Term Credit Agreement is the Secured Overnight Financing Rate specified in the Short-Term Credit Agreement plus 0.45%, with a commitment fee of 0.03% on the undrawn portion. There were no borrowings outstanding under the Short-Term Credit Agreement or the prior 364-day revolving credit agreement as of December 31, 2023 and 2024.

We also utilize other short-term credit facilities for working capital purposes. There were $147 million and $151 million of borrowings outstanding under these facilities as of December 31, 2023 and 2024, which were included in “Accrued expenses and other” on our consolidated balance sheets. In addition, we had $8.3 billion of unused letters of credit as of December 31, 2024.

Note 7 — COMMITMENTS AND CONTINGENCIES

Commitments

The following summarizes our principal contractual commitments, excluding open orders for purchases that support normal operations and are generally cancellable, as of December 31, 2024 (in millions):

	Year Ended December 31,						
	2025	2026	2027	2028	2029	Thereafter	Total
Long-term debt principal and interest	$ 6,858	$ 4,458	$10,404	$ 3,644	$ 4,344	$ 55,831	$ 85,539
Operating lease liabilities	12,002	11,023	10,087	9,205	8,534	44,443	95,294
Finance lease liabilities, including interest	1,491	1,495	1,317	1,114	1,022	6,081	12,520
Financing obligations, including interest (1)	511	515	523	531	540	6,397	9,017
Leases not yet commenced	2,695	3,691	5,011	4,253	4,286	41,691	61,627
Unconditional purchase obligations (2)	8,536	7,941	5,413	4,321	3,868	20,296	50,375
Other commitments (3)	2,739	1,470	1,016	923	1,025	11,456	18,629
Total commitments	$34,832	$30,593	$33,771	$23,991	$23,619	$186,195	$333,001

(1) Includes non-cancellable financing obligations for fulfillment network and data center facilities. Excluding interest, current financing obligations of $271 million and $312 million are recorded within “Accrued expenses and other” and $6.6 billion and $7.1 billion are recorded within “Other long-term liabilities” as of December 31, 2023 and 2024. The weighted-average remaining term of the financing obligations was 17.0 years and 16.1 years and the weighted-average imputed interest rate was 3.1% as of December 31, 2023 and 2024.

(2) Includes unconditional purchase obligations related to long-term agreements to acquire and license digital media content, procure energy, and license software that are not reflected on the consolidated balance sheets. For those agreements with variable terms, we do not estimate the total obligation beyond any minimum quantities and/or pricing as of the reporting date. Purchase obligations associated with renewal provisions solely at the option of the content provider are included to the extent such commitments are fixed or a minimum amount is specified. Energy agreements based on actual generation without a fixed or minimum volume commitment are not included. Our energy agreements generally provide the right to receive energy certificates for no additional consideration.

(3) Includes asset retirement obligations, the estimated timing and amounts of payments for rent and tenant improvements associated with build-to-suit lease arrangements that are under construction, and liabilities associated with digital media content agreements with initial terms greater than one year. Excludes approximately $6.5 billion of income tax contingencies for which we cannot make a reasonably reliable estimate of the amount and period of payment, if any.

Suppliers

During 2024, no vendor accounted for 10% or more of our purchases. We generally do not have long-term contracts or arrangements with our vendors to guarantee the availability of merchandise, particular payment terms, or the extension of credit limits.

Other Contingencies

We are disputing claims and denials of refunds or credits, and monitoring or evaluating potential claims, related to various non-income taxes (such as sales, value added, consumption, service, and similar taxes), including in jurisdictions in which we already collect and remit these taxes. These non-income tax controversies typically include (i) the taxability of products and services, including cross-border intercompany transactions, (ii) collection and withholding on transactions with third parties, including as a result of evolving requirements imposed on marketplaces with respect to third-party sellers, and (iii) the adequacy of compliance with reporting obligations, including evolving documentation requirements. Due to the inherent complexity and uncertainty of these matters and the judicial and regulatory processes in certain jurisdictions, the final outcome of any such controversies may be materially different from our expectations.

Legal Proceedings

The Company is involved from time to time in claims, proceedings, and litigation, including the following:

In May 2018, Rensselaer Polytechnic Institute and CF Dynamic Advances LLC filed a complaint against Amazon.com, Inc. in the United States District Court for the Northern District of New York. The complaint alleged, among other things, that "Alexa Voice Software and Alexa enabled devices" infringe U.S. Patent No. 7,177,798, entitled "Natural Language Interface Using Constrained Intermediate Dictionary of Results." The complaint sought an injunction, an unspecified amount of damages, enhanced damages, an ongoing royalty, interest, attorneys' fees, and costs. In March 2023, the plaintiffs alleged in their damages report that in the event of a finding of liability Amazon could be subject to $140 million to $267 million in damages. In March 2024, the district court granted summary judgment ruling that the patent is invalid and dismissed the case. In April 2024, the plaintiffs filed a notice of appeal. We dispute the allegations of wrongdoing and will continue to defend ourselves vigorously in this matter.

In December 2018, Kove IO, Inc. filed a complaint against Amazon Web Services, Inc. in the United States District Court for the Northern District of Illinois. The complaint alleged, among other things, that Amazon S3 and DynamoDB infringe U.S. Patent Nos. 7,814,170 and 7,103,640, each entitled "Network Distributed Tracking Wire Transfer Protocol"; and 7,233,978, entitled "Method and Apparatus for Managing Location Information in a Network Separate from the Data to Which the Location Information Pertains." The complaint sought an unspecified amount of damages, enhanced damages, attorneys' fees, costs, interest, and injunctive relief. In April 2024, a jury found that Amazon infringed the asserted patents and awarded Kove $525 million in damages. In August 2024, the court awarded Kove $148 million in pre-judgment interest. In September 2024, we filed a notice of appeal. We disagree with the jury's findings and will continue to defend ourselves vigorously in this matter.

Beginning in June 2019 with Wilcosky v. Amazon.com, Inc., now pending in the United States District Court for the Northern District of Illinois ("N.D. Ill."), private litigants have filed a number of cases in U.S. federal and state courts, including Hogan v. Amazon.com, Inc. (N.D. Ill.), alleging, among other things, that Amazon's collection, storage, use, retention, and protection of biometric identifiers violated the Illinois Biometric Information Privacy Act. The complaints allege purported classes of Illinois residents who had biometric identifiers collected through Amazon products or services, including Amazon Photos, Alexa, AWS cloud services, Ring, Amazon Connect, Amazon's Flex driver app, and Amazon's virtual try-on technology. The complaints seek certification as class actions, unspecified amounts of damages, injunctive relief, attorneys' fees, costs, and interest. We dispute the allegations of wrongdoing and intend to defend ourselves vigorously in these matters.

Beginning in March 2020 with Frame-Wilson v. Amazon.com, Inc. filed in the United States District Court for the Western District of Washington ("W.D. Wash."), private litigants have filed a number of cases in the U.S. and Canada alleging, among other things, price fixing arrangements between Amazon.com, Inc. and vendors and third-party sellers in Amazon's stores, monopolization and attempted monopolization, and consumer protection and unjust enrichment claims. Attorneys General for the District of Columbia and California brought similar suits in May 2021 and September 2022 in the Superior Court of the District of Columbia and the California Superior Court for the County of San Francisco, respectively. Some of the private cases include allegations of several distinct purported classes, including consumers who purchased a product through Amazon's stores and consumers who purchased a product offered by Amazon through another e-commerce retailer. The complaints seek billions of dollars of alleged damages, treble damages, punitive damages, injunctive relief, civil penalties, attorneys' fees, and costs. The Federal Trade Commission and a number of state Attorneys General filed a similar lawsuit in September 2023 in the W.D. Wash. alleging violations of federal antitrust and state antitrust and consumer protection laws. That complaint alleges, among other things, that Amazon has a monopoly in markets for online superstores and marketplace services, and unlawfully maintains those monopolies through anticompetitive practices relating to our pricing policies, advertising practices, the structure of Prime, and promotion of our own products on our website. The complaint seeks injunctive and structural relief, an unspecified amount of damages, and costs. In May 2024, the Attorney General of Arizona filed a complaint in the Superior Court of Arizona in Maricopa County alleging that Amazon's practices related to pricing and the Featured Offers in its stores violate state antitrust and consumer protection laws. That complaint also seeks injunctive relief, an unspecified amount of damages, civil penalties, and costs. Amazon's motions to dismiss were granted in part and denied in part in Frame-Wilson in March 2022 and March 2023, De Coster v. Amazon.com, Inc. (W.D. Wash.) in January 2023, and the California Attorney General's lawsuit in March 2023. All three courts dismissed claims alleging that Amazon's pricing policies are inherently illegal and denied dismissal of claims alleging that Amazon's pricing policies are an unlawful restraint of trade. In August 2024, the DC Court of Appeals overturned a prior decision by the DC Superior Court dismissing the DC Attorney General's lawsuit and that case is now proceeding. In September 2024, the United States District Court for the W.D. Wash. granted in part Amazon's motion to dismiss the suit brought by the FTC and certain state Attorneys General with respect to five state law claims and denied the motion with respect to the remaining claims. We dispute the allegations of wrongdoing and intend to defend ourselves vigorously in these matters.

In October 2020, Broadband iTV, Inc. filed a complaint against Amazon.com, Inc., Amazon.com Services LLC, and Amazon Web Services, Inc. in the United States District Court for the Western District of Texas. The complaint alleges, among other things, that certain Amazon Prime Video features and services infringe U.S. Patent Nos. 9,648,388, 10,546,750, and 10,536,751, each entitled "Video-On-Demand Content Delivery System for Providing Video-On-Demand Services to TV Services Subscribers"; 10,028,026, entitled "System for Addressing On-Demand TV Program Content on TV Services Platform of a Digital TV Services Provider"; and 9,973,825, entitled "Dynamic Adjustment of Electronic Program Guide Displays Based on Viewer Preferences for Minimizing Navigation in VOD Program Selection." The complaint seeks an unspecified amount of damages. In April 2022, Broadband iTV alleged in its damages report that in the event of a finding of liability Amazon could be subject to $166 million to $986 million in damages. In September 2022, the district court granted summary judgment, holding that the patents are invalid. In October 2022, Broadband iTV filed a notice of appeal. In September 2024, the United States Court of Appeals for the Federal Circuit affirmed the district court's judgment. This decision is subject to appeal. We dispute the allegations of wrongdoing and will continue to defend ourselves vigorously in this matter.

In July 2021, the Luxembourg National Commission for Data Protection (the "CNPD") issued a decision against Amazon Europe Core S.à r.l. claiming that Amazon's processing of personal data did not comply with the EU General Data Protection Regulation. The decision imposes a fine of €746 million and corresponding practice revisions. We believe the CNPD's decision to be without merit and intend to defend ourselves vigorously in this matter.

In December 2021, the Italian Competition Authority (the "ICA") issued a decision against Amazon Services Europe S.à r.l., Amazon Europe Core S.à r.l., Amazon EU S.à r.l., Amazon Italia Services S.r.l., and Amazon Italia Logistica S.r.l. claiming that certain of our marketplace and logistics practices in Italy infringe EU competition rules. The decision imposes remedial actions and a fine of €1.13 billion, which we have paid and will seek to recover pending conclusion of all appeals. We believe the ICA's decision to be without merit and intend to defend ourselves vigorously in this matter.

Beginning in October 2023, Nokia Technologies Oy and related entities filed complaints alleging infringement of patents related to video-related technologies against Amazon.com, Inc. and related entities in multiple courts in the United States, India, the United Kingdom, Germany, and Brazil, the Unified Patent Court of the European Union, and the United States International Trade Commission. The complaints allege, among other things, that certain Amazon Prime Video services and features of Amazon devices carrying the Prime Video app infringe Nokia's patents; some of the complaints additionally allege infringement by Freevee, Twitch, and Amazon voice assistants. The complaints seek, among other things, injunctive relief and, in some cases, unspecified money damages, enhanced damages, attorneys' fees, costs, interest, and declaratory relief. These matters are at various procedural stages, with preliminary and final injunctions issued in certain instances. We dispute the allegations of wrongdoing and intend to defend ourselves vigorously in these matters.

In addition, we are regularly subject to claims, litigation, and other proceedings, including potential regulatory proceedings, involving patent and other intellectual property matters, taxes, labor and employment, competition and antitrust, privacy and data protection, consumer protection, commercial disputes, goods and services offered by us and by third parties, and other matters.

The outcomes of our legal proceedings and other contingencies are inherently unpredictable, subject to significant uncertainties, and could be material to our operating results and cash flows for a particular period. We evaluate, on a regular basis, developments in our legal proceedings and other contingencies that could affect the amount of liability, including amounts in excess of any previous accruals and reasonably possible losses disclosed, and make adjustments and changes to our accruals and disclosures as appropriate. For the matters we disclose that do not include an estimate of the amount of loss or range of losses, such an estimate is not possible or is immaterial, and we may be unable to estimate the possible loss or range of losses that could potentially result from the application of non-monetary remedies. Until the final resolution of such matters, if any of our estimates and assumptions change or prove to have been incorrect, we may experience losses in excess of the amounts recorded, which could have a material effect on our business, consolidated financial position, results of operations, or cash flows.

See also “Note 9 — Income Taxes.”

Note 8 — STOCKHOLDERS’ EQUITY

Preferred Stock

We have authorized 500 million shares of $0.01 par value preferred stock. No preferred stock was outstanding for any year presented.

Common Stock

Common shares outstanding plus shares underlying outstanding stock awards totaled 10.6 billion, 10.8 billion, and 10.9 billion, as of December 31, 2022, 2023, and 2024. These totals include all vested and unvested stock awards outstanding, including those awards we estimate will be forfeited.

Stock Repurchase Activity

In March 2022, the Board of Directors authorized a program to repurchase up to $10.0 billion of our common stock, with no fixed expiration. We repurchased 46.2 million shares of our common stock for $6.0 billion in 2022. There were no repurchases of common stock in 2023 or 2024. As of December 31, 2024, we have $6.1 billion remaining under the repurchase program.

Stock Award Plans

Employees vest in restricted stock unit awards over the corresponding service term, generally between two and five years. The majority of restricted stock unit awards are granted at the date of hire or in Q2 as part of the annual compensation review and primarily vest semi-annually in Q2 and Q4 of the relevant compensation year.

Stock Award Activity

Stock-based compensation expense is as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
Cost of sales	$ 757	$ 836	$ 838
Fulfillment	2,745	3,090	2,973
Technology and infrastructure	10,621	13,434	12,150
Sales and marketing	3,875	4,623	4,084
General and administrative	1,623	2,040	1,966
Total stock-based compensation expense (1)	$ 19,621	$ 24,023	$ 22,011

(1) The related tax benefits were $4.3 billion, $5.4 billion, and $5.0 billion for 2022, 2023, and 2024.

The following table summarizes our restricted stock unit activity (in millions):

	Number of Units	Weighted Average Grant-Date Fair Value
Outstanding as of January 1, 2022	279.9	$ 134
Units granted	262.8	142
Units vested	(113.3)	114
Units forfeited	(45.0)	143
Outstanding as of December 31, 2022	384.4	144
Units granted	218.1	106
Units vested	(139.9)	143
Units forfeited	(56.8)	135
Outstanding as of December 31, 2023	405.8	125
Units granted	126.9	183
Units vested	(209.7)	132
Units forfeited	(39.9)	133
Outstanding as of December 31, 2024	283.1	145

Scheduled vesting for outstanding restricted stock units as of December 31, 2024, is as follows (in millions):

	Year Ended						
	2025	2026	2027	2028	2029	Thereafter	Total
Scheduled vesting — restricted stock units	146.0	88.5	34.3	11.5	1.4	1.4	283.1

As of December 31, 2024, there was $16.2 billion of net unrecognized compensation cost related to unvested stock-based compensation arrangements. This compensation is recognized on an accelerated basis with more than half of the compensation expected to be expensed in the next twelve months, and has a remaining weighted-average recognition period of 1.0 year. The estimated forfeiture rate as of December 31, 2022, 2023, and 2024 was 26.5%, 26.1%, and 25.6%.

During 2022, 2023, and 2024, the fair value of restricted stock units that vested was $12.8 billion, $17.6 billion, and $39.6 billion.

Common Stock Available for Future Issuance

As of December 31, 2024, common stock available for future issuance to employees is 1.5 billion shares.

Note 9 — INCOME TAXES

In 2022, 2023, and 2024, we recorded a net tax provision (benefit) of $(3.2) billion, $7.1 billion, and $9.3 billion. Our U.S. taxable income is reduced by accelerated depreciation deductions and increased by the impact of capitalized research and development expenses. Cash paid for income taxes, net of refunds, was $6.0 billion, $11.2 billion, and $12.3 billion for 2022, 2023, and 2024.

Certain foreign subsidiary earnings and losses are subject to current U.S. taxation and the subsequent repatriation of those earnings is not subject to tax in the U.S. The U.S. tax rules also provide for enhanced accelerated depreciation deductions by allowing us to expense a portion of qualified property, primarily equipment. These enhanced deductions are scheduled to phase out annually from 2023 through 2026. Our federal tax provision included accelerated depreciation deductions for 2022, 2023, and 2024. Effective January 1, 2022, research and development expenses are required to be capitalized and amortized for U.S. tax purposes.

The components of the provision (benefit) for income taxes, net are as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
U.S. Federal:			
Current	$ 2,175	$ 8,652	$ 9,039
Deferred	(6,686)	(5,505)	(4,101)
Total	(4,511)	3,147	4,938
U.S. State:			
Current	1,074	2,158	2,109
Deferred	(1,302)	(498)	(453)
Total	(228)	1,660	1,656
International:			
Current	1,682	2,186	2,765
Deferred	(160)	127	(94)
Total	1,522	2,313	2,671
Provision (benefit) for income taxes, net	$ (3,217)	$ 7,120	$ 9,265

U.S. and international components of income (loss) before income taxes are as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
U.S.	$ (8,225)	$ 32,328	$ 61,947
International	2,289	5,229	6,667
Income (loss) before income taxes	$ (5,936)	$ 37,557	$ 68,614

The items accounting for differences between income taxes computed at the federal statutory rate and the provision (benefit) recorded for income taxes are as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
Income taxes computed at the federal statutory rate	$ (1,246)	$ 7,887	$ 14,409
Effect of:			
Tax impact of foreign earnings and losses	(370)	594	199
State taxes, net of federal benefits	(173)	1,307	1,306
Tax credits	(1,006)	(2,362)	(2,805)
Stock-based compensation (1)	612	1,047	(1,688)
Foreign income deduction (2)	(1,258)	(1,429)	(2,379)
Other, net	224	76	223
Total	$ (3,217)	$ 7,120	$ 9,265

(1) Includes non-deductible stock-based compensation and excess tax benefits or shortfalls from stock-based compensation. Our tax provision includes $33 million and $519 million of tax shortfalls from stock-based compensation for 2022 and 2023, and $2.8 billion of excess tax benefits from stock-based compensation for 2024.

(2) U.S. companies are eligible for a deduction that lowers the effective tax rate on certain foreign income. This regime is referred to as the Foreign-Derived Intangible Income deduction and is dependent on the amount of our U.S. taxable income.

We recorded a provision for income taxes in 2023 as compared to an income tax benefit in 2022 primarily due to an increase in pretax income, a decrease in the tax impact of foreign earnings and losses driven by a decline in the favorable effects of corporate restructuring transactions, and an increase in tax shortfalls from stock-based compensation. This was partially offset by an increase in federal research and development credits, which included approximately $600 million of tax benefit recorded in 2023 related to a change in the estimated qualifying expenditures associated with our 2022 U.S. federal R&D credit.

Our provision for income taxes in 2024 was higher than in 2023 primarily due to an increase in pretax income, partially offset by an increase in excess tax benefits from stock-based compensation and an increase in our foreign income deduction.

We intend to invest substantially all of our foreign subsidiary earnings, as well as our capital in our foreign subsidiaries, indefinitely outside of the U.S. in those jurisdictions in which we would incur significant, additional costs upon repatriation of such amounts.

Deferred income tax assets and liabilities are as follows (in millions):

	December 31, 2023	December 31, 2024
Deferred tax assets (1):		
Loss carryforwards U.S. - Federal/States	$ 610	$ 692
Loss carryforwards - Foreign	2,796	2,687
Accrued liabilities, reserves, and other expenses	3,751	4,254
Stock-based compensation	5,279	4,089
Depreciation and amortization	1,114	1,133
Operating lease liabilities	19,922	20,921
Capitalized research and development	14,800	22,701
Other items	745	1,688
Tax credits	1,582	1,773
Total gross deferred tax assets	50,599	59,938
Less valuation allowances (2)	(4,811)	(4,893)
Deferred tax assets, net of valuation allowances	45,788	55,045
Deferred tax liabilities:		
Depreciation and amortization	(12,454)	(16,240)
Operating lease assets	(18,648)	(19,517)
Other items	(1,489)	(3,323)
Net deferred tax assets (liabilities), net of valuation allowances	$ 13,197	$ 15,965

(1) Deferred tax assets are presented after tax effects and net of tax contingencies.

(2) Relates primarily to deferred tax assets that would only be realizable upon the generation of net income in certain foreign taxing jurisdictions or future capital gains, as well as tax credits.

Our valuation allowances primarily relate to foreign deferred tax assets, including substantially all of our foreign net operating loss carryforwards as of December 31, 2024. Our foreign net operating loss carryforwards for income tax purposes as of December 31, 2024 were approximately $9.9 billion before tax effects and certain of these amounts are subject to annual limitations under applicable tax law. If not utilized, a portion of these losses will begin to expire in 2025.

Income Tax Contingencies

We are subject to income taxes in the U.S. (federal and state) and numerous foreign jurisdictions. Significant judgment is required in evaluating our tax positions and determining our provision for income taxes. During the ordinary course of business, there are many transactions and calculations for which the ultimate tax determination is uncertain. We establish reserves for tax-related uncertainties based on estimates of whether, and the extent to which, additional taxes will be due. These reserves are established when we believe that certain positions might be challenged despite our belief that our tax return positions are fully supportable. We adjust these reserves in light of changing facts and circumstances, such as the outcome of tax audits. The provision for income taxes includes the impact of reserve provisions and changes to reserves that are considered appropriate.

The reconciliation of our income tax contingencies is as follows (in millions):

	December 31,		
	2022	2023	2024
Gross tax contingencies – January 1	$ 3,242	$ 4,002	$ 5,228
Gross increases to tax positions in prior periods	274	440	154
Gross decreases to tax positions in prior periods	(172)	(38)	(129)
Gross increases to current period tax positions	706	1,009	1,392
Settlements with tax authorities	(20)	(106)	(9)
Lapse of statute of limitations	(28)	(79)	(151)
Gross tax contingencies – December 31 (1)	$ 4,002	$ 5,228	$ 6,485

(1) As of December 31, 2024, we had approximately $6.5 billion of income tax contingencies of which $4.4 billion, if fully recognized, would decrease our effective tax rate.

As of December 31, 2023 and 2024, we had accrued interest and penalties, net of federal income tax benefit, related to tax contingencies of $194 million and $316 million. Interest and penalties, net of federal income tax benefit, recognized for the years ended December 31, 2022, 2023, and 2024 were $(7) million, $91 million, and $121 million.

We are under examination, or may be subject to examination, by the Internal Revenue Service for the calendar year 2016 and thereafter. These examinations may lead to ordinary course adjustments or proposed adjustments to our taxes or our net operating losses with respect to years under examination as well as subsequent periods.

We are also subject to taxation in various states and foreign jurisdictions including China, France, Germany, India, Japan, Luxembourg, and the United Kingdom. We are under, or may be subject to, audit or examination and additional assessments by the relevant authorities in respect of these particular jurisdictions primarily for 2011 and thereafter. We are currently disputing tax assessments in multiple jurisdictions, including with respect to the allocation and characterization of income.

In September 2022, the Luxembourg tax authority (“LTA”) denied the tax basis of certain intangible assets that we distributed from Luxembourg to the U.S. in 2021. When we are assessed by the LTA, we will need to remit taxes related to this matter. We believe the LTA’s position is without merit, we intend to defend ourselves vigorously in this matter, and we expect to recoup taxes paid.

The Indian tax authority (“ITA”) has asserted that tax applies to cloud services fees paid to Amazon in the U.S. We will need to remit taxes related to this matter until it is resolved, which payments could be significant in the aggregate. We believe the ITA’s position is without merit, we are defending our position vigorously, and we expect to recoup taxes paid. If this matter is adversely resolved, we could recognize significant additional tax expense, including for taxes previously paid.

Changes in tax laws, regulations, administrative practices, principles, and interpretations may impact our tax contingencies. Due to various factors, including the inherent complexities and uncertainties of the judicial, administrative, and regulatory processes in certain jurisdictions, the timing of the resolution of income tax controversies is highly uncertain, and the amounts ultimately paid, if any, upon resolution of the issues raised by the taxing authorities may differ from the amounts accrued. It is reasonably possible that within the next twelve months we will receive additional assessments by various tax authorities or possibly reach resolution of income tax controversies in one or more jurisdictions. These assessments or settlements could result in changes to our contingencies related to positions on prior years’ tax filings. The actual amount of any change could vary significantly depending on the ultimate timing and nature of any settlements. We cannot currently provide an estimate of the range of possible outcomes.

Note 10 — SEGMENT INFORMATION

We have organized our operations into three segments: North America, International, and AWS. We allocate to segment results the operating expenses “Fulfillment,” “Technology and infrastructure,” “Sales and marketing,” and “General and administrative” based on usage, which is generally reflected in the segment in which the costs are incurred. The majority of technology costs recorded in “Technology and infrastructure” are incurred in the U.S. and are included in our North America and AWS segments. The majority of infrastructure costs recorded in “Technology and infrastructure” are allocated to the AWS segment based on usage. There are no internal revenue transactions between our reportable segments. Our chief operating decision maker (“CODM”) is our President and Chief Executive Officer. Our CODM regularly reviews consolidated net sales, consolidated operating expenses, and consolidated operating income (loss) by segment. Amounts included in consolidated operating expenses include “Cost of sales,” “Fulfillment,” “Technology and infrastructure,” “Sales and marketing,” “General and administrative,” and “Other operating expense (income), net.” Our CODM manages our business primarily by reviewing consolidated results by segment on a quarterly basis, and using those results along with forecasts and other non-financial information in our annual budgeting process.

North America

The North America segment primarily consists of amounts earned from retail sales of consumer products (including from sellers) and advertising and subscription services through North America-focused online and physical stores. This segment includes export sales from these online stores.

International

The International segment primarily consists of amounts earned from retail sales of consumer products (including from sellers) and advertising and subscription services through internationally-focused online stores. This segment includes export sales from these internationally-focused online stores (including export sales from these online stores to customers in the U.S., Mexico, and Canada), but excludes export sales from our North America-focused online stores.

AWS

The AWS segment consists of amounts earned from global sales of compute, storage, database, and other services for start-ups, enterprises, government agencies, and academic institutions.

Information on reportable segments and reconciliation to consolidated net income (loss) is as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
North America			
Net sales	$ 315,880	$ 352,828	$ 387,497
Operating expenses	318,727	337,951	362,530
Operating income (loss)	$ (2,847)	$ 14,877	$ 24,967
International			
Net sales	$ 118,007	$ 131,200	$ 142,906
Operating expenses	125,753	133,856	139,114
Operating income (loss)	$ (7,746)	$ (2,656)	$ 3,792
AWS			
Net sales	$ 80,096	$ 90,757	$ 107,556
Operating expenses	57,255	66,126	67,722
Operating income	$ 22,841	$ 24,631	$ 39,834
Consolidated			
Net sales	$ 513,983	$ 574,785	$ 637,959
Operating expenses	501,735	537,933	569,366
Operating income	12,248	36,852	68,593
Total non-operating income (expense)	(18,184)	705	21
Benefit (provision) for income taxes	3,217	(7,120)	(9,265)
Equity-method investment activity, net of tax	(3)	(12)	(101)
Net income (loss)	$ (2,722)	$ 30,425	$ 59,248

Net sales by groups of similar products and services, which also have similar economic characteristics, is as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
Net Sales:			
Online stores (1)	$ 220,004	$ 231,872	$ 247,029
Physical stores (2)	18,963	20,030	21,215
Third-party seller services (3)	117,716	140,053	156,146
Advertising services (4)	37,739	46,906	56,214
Subscription services (5)	35,218	40,209	44,374
AWS	80,096	90,757	107,556
Other (6)	4,247	4,958	5,425
Consolidated	$ 513,983	$ 574,785	$ 637,959

(1) Includes product sales and digital media content where we record revenue gross. We leverage our retail infrastructure to offer a wide selection of consumable and durable goods that includes media products available in both a physical and digital format, such as books, videos, games, music, and software. These product sales include digital products sold on a transactional basis. Digital media content subscriptions that provide unlimited viewing or usage rights are included in "Subscription services."
(2) Includes product sales where our customers physically select items in a store. Sales to customers who order goods online for delivery or pickup at our physical stores are included in "Online stores."
(3) Includes commissions and any related fulfillment and shipping fees, and other third-party seller services.
(4) Includes sales of advertising services to sellers, vendors, publishers, authors, and others, through programs such as sponsored ads, display, and video advertising.
(5) Includes annual and monthly fees associated with Amazon Prime memberships, as well as digital video, audiobook, digital music, e-book, and other non-AWS subscription services.
(6) Includes sales related to various other offerings, such as healthcare services, certain licensing and distribution of video content, and shipping services, and our co-branded credit card agreements.

Net sales are attributed to countries primarily based on country-focused online and physical stores or, for AWS purposes, the selling entity. Net sales attributed to countries that represent a significant portion of consolidated net sales are as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
United States	$ 356,113	$ 395,637	$ 438,015
Germany	33,598	37,588	40,856
United Kingdom	30,074	33,591	37,855
Japan	24,396	26,002	27,401
Rest of world	69,802	81,967	93,832
Consolidated	$ 513,983	$ 574,785	$ 637,959

Total segment assets exclude corporate assets, such as cash and cash equivalents, marketable securities, other long-term investments, corporate facilities, goodwill and other acquired intangible assets, and tax assets. Technology infrastructure assets are allocated among the segments based on usage, with the majority allocated to the AWS segment. Total segment assets reconciled to consolidated amounts are as follows (in millions):

	December 31,		
	2022	2023	2024
North America (1)	$ 185,268	$ 196,029	$ 210,120
International (1)	64,666	69,718	69,487
AWS (2)	88,491	108,533	155,953
Corporate	124,250	153,574	189,334
Consolidated	$ 462,675	$ 527,854	$ 624,894

(1) North America and International segment assets primarily consist of property and equipment, operating leases, inventory, accounts receivable, and digital video and music content.
(2) AWS segment assets primarily consist of property and equipment, accounts receivable, and operating leases.

Property and equipment, net by segment is as follows (in millions):

	December 31,		
	2022	2023	2024
North America	$ 90,076	$ 93,632	$ 103,041
International	23,347	24,357	25,618
AWS	60,324	72,701	110,683
Corporate	12,968	13,487	13,323
Consolidated	$ 186,715	$ 204,177	$ 252,665

Total net additions to property and equipment include technology infrastructure assets, which are allocated among the segments based on usage, with the majority allocated to the AWS segment. Total net additions to property and equipment include the effect of non-cash activity such as property and equipment acquired but not yet paid for. Total net additions to property and equipment are as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
North America (1)	$ 23,682	$ 17,529	$ 24,348
International (1)	6,711	4,144	6,643
AWS (2)	27,755	24,843	53,267
Corporate	2,688	1,828	1,494
Consolidated	$ 60,836	$ 48,344	$ 85,752

(1) Includes property and equipment added under finance leases of $422 million, $525 million, and $616 million in 2022, 2023, and 2024, and under build-to-suit lease arrangements of $3.2 billion, $356 million, and $89 million in 2022, 2023, and 2024.
(2) Includes property and equipment added under finance leases of $253 million, $117 million, and $238 million in 2022, 2023, and 2024, and under build-to-suit lease arrangements of $20 million, $1 million, and $8 million in 2022, 2023, and 2024.

U.S. property and equipment, net and operating leases were $180.0 billion, $196.0 billion, and $241.6 billion, as of December 31, 2022, 2023, and 2024, and non-U.S. property and equipment, net and operating leases were $72.9 billion, $80.7 billion, and $87.2 billion as of December 31, 2022, 2023, and 2024. Except for the U.S., property and equipment, net and operating leases in any single country were less than 10% of consolidated property and equipment, net and operating leases.

Depreciation and amortization expense on property and equipment, including corporate property and equipment, are allocated to all segments based on usage. Total depreciation and amortization expense, by segment, is as follows (in millions):

	Year Ended December 31,		
	2022	2023	2024
North America	$ 11,565	$ 13,678	$ 14,285
International	3,483	4,016	4,462
AWS	9,876	12,531	13,320
Consolidated	$ 24,924	$ 30,225	$ 32,067

Item 9. *Changes in and Disagreements with Accountants On Accounting and Financial Disclosure*

None.

Item 9A. *Controls and Procedures*

Evaluation of Disclosure Controls and Procedures

We carried out an evaluation required by the Securities Exchange Act of 1934 (the “1934 Act”), under the supervision and with the participation of our principal executive officer and principal financial officer, of the effectiveness of the design and operation of our disclosure controls and procedures, as defined in Rule 13a-15(e) of the 1934 Act, as of December 31, 2024. Based on this evaluation, our principal executive officer and principal financial officer concluded that, as of December 31, 2024, our disclosure controls and procedures were effective to provide reasonable assurance that information required to be disclosed by us in the reports that we file or submit under the 1934 Act is recorded, processed, summarized, and reported within the time periods specified in the SEC’s rules and forms and to provide reasonable assurance that such information is accumulated and communicated to our management, including our principal executive officer and principal financial officer, as appropriate to allow timely decisions regarding required disclosure.

Management’s Report on Internal Control over Financial Reporting

Management is responsible for establishing and maintaining adequate internal control over financial reporting, as defined in Rule 13a-15(f) of the 1934 Act. Management has assessed the effectiveness of our internal control over financial reporting as of December 31, 2024 based on criteria established in Internal Control — Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission. As a result of this assessment, management concluded that, as of December 31, 2024, our internal control over financial reporting was effective in providing reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. Ernst & Young has independently assessed the effectiveness of our internal control over financial reporting and its report is included below.

Changes in Internal Control Over Financial Reporting

There were no changes in our internal control over financial reporting during the quarter ended December 31, 2024 that materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Limitations on Controls

Our disclosure controls and procedures and internal control over financial reporting are designed to provide reasonable assurance of achieving their objectives as specified above. Management does not expect, however, that our disclosure controls and procedures or our internal control over financial reporting will prevent or detect all error and fraud. Any control system, no matter how well designed and operated, is based upon certain assumptions and can provide only reasonable, not absolute, assurance that its objectives will be met. Further, no evaluation of controls can provide absolute assurance that misstatements due to error or fraud will not occur or that all control issues and instances of fraud, if any, within the Company have been detected.

Report of Independent Registered Public Accounting Firm

The Board of Directors and Shareholders
Amazon.com, Inc.

Opinion on Internal Control Over Financial Reporting

We have audited Amazon.com, Inc.'s internal control over financial reporting as of December 31, 2024, based on criteria established in Internal Control-Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, Amazon.com, Inc. (the Company) maintained, in all material respects, effective internal control over financial reporting as of December 31, 2024, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2024 and 2023, the related consolidated statements of operations, comprehensive income (loss), stockholders' equity, and cash flows for each of the three years in the period ended December 31, 2024, and the related notes and our report dated February 6, 2025 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP

Seattle, Washington
February 6, 2025

Item 9B. *Other Information*

On November 7, 2024, Douglas Herrington, CEO Worldwide Amazon Stores, adopted a trading plan intended to satisfy Rule 10b5-1(c) to sell up to 158,970 shares of Amazon.com, Inc. common stock over a period ending on December 31, 2025, subject to certain conditions.

On November 7, 2024, Shelley Reynolds, Vice President, Worldwide Controller, adopted a trading plan intended to satisfy Rule 10b5-1(c) to sell up to 16,938 shares of Amazon.com, Inc. common stock over a period ending on November 29, 2025, subject to certain conditions.

On November 11, 2024, Brian Olsavsky, Senior Vice President and Chief Financial Officer, adopted a trading plan intended to satisfy Rule 10b5-1(c) to sell up to 32,370 shares of Amazon.com, Inc. common stock over a period ending on May 30, 2025, subject to certain conditions.

On November 12, 2024, David Zapolsky, Senior Vice President, Global Public Policy and General Counsel, adopted a trading plan intended to satisfy Rule 10b5-1(c) to sell up to 87,070 shares of Amazon.com, Inc. common stock over a period ending on December 31, 2025, subject to certain conditions.

On November 18, 2024, Andrew Jassy, President and Chief Executive Officer, adopted a trading plan intended to satisfy Rule 10b5-1(c) to sell up to 80,400 shares of Amazon.com, Inc. common stock over a period ending on December 31, 2025, subject to certain conditions.

Item 9C. *Disclosure Regarding Foreign Jurisdictions that Prevent Inspections*

Not applicable.

PART III

Item 10. *Directors, Executive Officers, and Corporate Governance*

Information regarding our Executive Officers required by Item 10 of Part III is set forth in Item 1 of Part I “Business — Information About Our Executive Officers.” Information required by Item 10 of Part III regarding our Directors and any material changes to the process by which security holders may recommend nominees to the Board of Directors is included in our Proxy Statement relating to our 2025 Annual Meeting of Shareholders, and is incorporated herein by reference. Information relating to our Code of Business Conduct and Ethics, insider trading policies and procedures, and, to the extent applicable, compliance with Section 16(a) of the 1934 Act is set forth in our Proxy Statement relating to our 2025 Annual Meeting of Shareholders and is incorporated herein by reference. To the extent permissible under Nasdaq rules, we intend to disclose amendments to our Code of Business Conduct and Ethics, as well as waivers of the provisions thereof, on our investor relations website under the heading “Corporate Governance” at amazon.com/ir.

Item 11. *Executive Compensation*

Information required by Item 11 of Part III is included in our Proxy Statement relating to our 2025 Annual Meeting of Shareholders and is incorporated herein by reference.

Item 12. *Security Ownership of Certain Beneficial Owners and Management and Related Shareholder Matters*

Information required by Item 12 of Part III is included in our Proxy Statement relating to our 2025 Annual Meeting of Shareholders and is incorporated herein by reference.

Item 13. *Certain Relationships and Related Transactions, and Director Independence*

Information required by Item 13 of Part III is included in our Proxy Statement relating to our 2025 Annual Meeting of Shareholders and is incorporated herein by reference.

Item 14. ***Principal Accountant Fees and Services***

Information required by Item 14 of Part III is included in our Proxy Statement relating to our 2025 Annual Meeting of Shareholders and is incorporated herein by reference.

PART IV

Item 15. ***Exhibits, Financial Statement Schedules***

(a) *List of Documents Filed as a Part of This Report:*

(1) *Index to Consolidated Financial Statements:*

Report of Ernst & Young LLP, Independent Registered Public Accounting Firm

Consolidated Statements of Cash Flows for each of the three years ended December 31, 2024

Consolidated Statements of Operations for each of the three years ended December 31, 2024

Consolidated Statements of Comprehensive Income (Loss) for each of the three years ended December 31, 2024

Consolidated Balance Sheets as of December 31, 2023 and 2024

Consolidated Statements of Stockholders' Equity for each of the three years ended December 31, 2024

Notes to Consolidated Financial Statements

Report of Ernst & Young LLP, Independent Registered Public Accounting Firm

(2) *Index to Financial Statement Schedules:*

All schedules have been omitted because the required information is included in the consolidated financial statements or the notes thereto, or because it is not required.

(3) *Index to Exhibits*

See exhibits listed under Part (b) below.

(b) *Exhibits:*

Exhibit Number	Description
3.1	Amended and Restated Certificate of Incorporation of Amazon.com, Inc. (incorporated by reference to the Company's Current Report on Form 8-K, filed May 27, 2022).
3.2	Amended and Restated Bylaws of Amazon.com, Inc. (incorporated by reference to the Company's Current Report on Form 8-K, filed May 3, 2024).
4.1	Indenture, dated as of November 29, 2012, between Amazon.com, Inc. and Wells Fargo Bank, National Association, as trustee (incorporated by reference to the Company's Current Report on Form 8-K, filed November 29, 2012).
4.2	Supplemental Indenture, dated as of April 13, 2022, among Amazon.com, Inc., Wells Fargo Bank, National Association, as prior trustee, and Computershare Trust Company, National Association, as successor trustee, containing Form of 2.730% Note due 2024, Form of 3.000% Note due 2025, Form of 3.300% Note due 2027, Form of 3.450% Note due 2029, Form of 3.600% Note due 2032, Form of 3.950% Note due 2052, and Form of 4.100% Note due 2062 (incorporated by reference to the Company's Current Report on Form 8-K, filed April 13, 2022).
4.3	Officers' Certificate of Amazon.com, Inc., dated as of December 5, 2014, containing Form of 2.600% Note due 2019, Form of 3.300% Note due 2021, Form of 3.800% Note due 2024, Form of 4.800% Note due 2034, and Form of 4.950% Note due 2044 (incorporated by reference to the Company's Current Report on Form 8-K, filed December 5, 2014).
4.4	Officers' Certificate of Amazon.com, Inc., dated as of August 22, 2017, containing Form of 1.900% Note due 2020, Form of 2.400% Note due 2023, Form of 2.800% Note due 2024, Form of 3.150% Note due 2027, Form of 3.875% Note due 2037, Form of 4.050% Note due 2047, and Form of 4.250% Note due 2057 (incorporated by reference to the Company's Current Report on Form 8-K, filed August 22, 2017).
4.5	Officers' Certificate of Amazon.com, Inc., dated as of December 20, 2017, containing Form of 5.200% Note due 2025 (incorporated by reference to the Company's Current Report on Form 8-K, filed December 20, 2017).

4.6	Officers’ Certificate of Amazon.com, Inc., dated as of June 3, 2020, containing Form of 0.400% Note due 2023, Form of 0.800% Note due 2025, Form of 1.200% Note due 2027, Form of 1.500% Note due 2030, Form of 2.500% Note due 2050, and Form of 2.700% Note due 2060 (incorporated by reference to the Company’s Current Report on Form 8-K, filed June 3, 2020).
4.7	Officers’ Certificate of Amazon.com, Inc., dated as of May 12, 2021, containing Form of 0.250% Note due 2023, Form of 0.450% Note due 2024, Form of 1.000% Note due 2026, Form of 1.650% Note due 2028, Form of 2.100% Note due 2031, Form of 2.875% Note due 2041, Form of 3.100% Note due 2051, and Form of 3.250% Note due 2061 (incorporated by reference to the Company’s Current Report on Form 8-K, filed May 12, 2021).
4.8	Officers’ Certificate of Amazon.com, Inc., dated as of December 1, 2022, containing Form of 4.700% Note due 2024, Form of 4.600% Note due 2025, Form of 4.550% Note due 2027, Form of 4.650% Note due 2029, and Form of 4.700% Note due 2032 (incorporated by reference to the Company’s Current Report on Form 8-K, filed December 1, 2022).
4.9	Description of Securities (incorporated by reference to the Company’s Annual Report on Form 10-K for the Year ended December 31, 2019).
10.1†	1997 Stock Incentive Plan (amended and restated) (incorporated by reference to the Company’s Quarterly Report on Form 10-Q for the Quarter ended June 30, 2022).
10.2†	1999 Nonofficer Employee Stock Option Plan (amended and restated) (incorporated by reference to the Company’s Quarterly Report on Form 10-Q for the Quarter ended June 30, 2022).
10.3†	Form of Indemnification Agreement between Amazon.com, Inc. and each of its Directors (incorporated by reference to Exhibit 10.1 to the Company’s Registration Statement on Form S-1 (Registration No. 333-23795) filed March 24, 1997, as amended on April 21, 1997).
10.4†	Form of Restricted Stock Unit Agreement for Officers and Employees (incorporated by reference to the Company’s Annual Report on Form 10-K for the Year ended December 31, 2002).
10.5†	Form of Restricted Stock Unit Agreement for Directors (incorporated by reference to the Company’s Annual Report on Form 10-K for the Year ended December 31, 2002).
10.6†	Form of Restricted Stock Agreement (incorporated by reference to the Company’s Annual Report on Form 10-K for the Year ended December 31, 2001).
10.7†	Form of Global Restricted Stock Unit Award Agreement for Executive Officers (incorporated by reference to the Company's Annual Report on Form 10-K for the Year ended December 31, 2023).
10.8	Five-Year Revolving Credit Agreement, dated as of November 1, 2023, among Amazon.com, Inc., Citibank N.A., as administrative agent, and the lenders party thereto (incorporated by reference to the Company’s Current Report on Form 8-K, filed November 1, 2023).
10.9	364-Day Revolving Credit Agreement, dated as of October 30, 2024, among Amazon.com, Inc., Citibank N.A., as administrative agent, and the lenders party thereto (incorporated by reference to the Company’s Quarterly Report on Form 10-Q for the Quarter ended September 30, 2024).
19.1	Amazon.com, Inc. Insider Trading Policy.
21.1	List of Significant Subsidiaries.
23.1	Consent of Independent Registered Public Accounting Firm.
31.1	Certification of Andrew R. Jassy, President and Chief Executive Officer of Amazon.com, Inc., pursuant to Rule 13a-14(a) under the Securities Exchange Act of 1934.
31.2	Certification of Brian T. Olsavsky, Senior Vice President and Chief Financial Officer of Amazon.com, Inc., pursuant to Rule 13a-14(a) under the Securities Exchange Act of 1934.
32.1	Certification of Andrew R. Jassy, President and Chief Executive Officer of Amazon.com, Inc., pursuant to 18 U.S.C. Section 1350.

32.2 Certification of Brian T. Olsavsky, Senior Vice President and Chief Financial Officer of Amazon.com, Inc., pursuant to 18 U.S.C. Section 1350.

97.1 Amazon.com, Inc. Clawback Policy (incorporated by reference to the Company's Annual Report on Form 10-K for the Year ended December 31, 2023).

101 The following financial statements from the Company’s Annual Report on Form 10-K for the year ended December 31, 2024, formatted in Inline XBRL: (i) Consolidated Statements of Cash Flows, (ii) Consolidated Statements of Operations, (iii) Consolidated Statements of Comprehensive Income (Loss), (iv) Consolidated Balance Sheets, (v) Consolidated Statements of Stockholders’ Equity, and (vi) Notes to Consolidated Financial Statements, tagged as blocks of text and including detailed tags.

As permitted by Item 601(b)(4)(iii)(A) of Regulation S-K, the Company has not filed with this Annual Report on Form 10-K certain instruments defining the rights of holders of long-term debt of the Company and its subsidiaries because the total amount of securities authorized thereunder does not exceed 10 percent of the total assets of the Company and its subsidiaries on a consolidated basis. The Company agrees to furnish a copy of such agreements to the Commission upon request.

104 The cover page from the Company’s Annual Report on Form 10-K for the year ended December 31, 2024, formatted in Inline XBRL (included as Exhibit 101).

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† Executive Compensation Plan or Agreement.

Item 16. ***Form 10-K Summary***

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this Report to be signed on its behalf by the undersigned, thereunto duly authorized, as of February 6, 2025.

AMAZON.COM, INC.

By: /s/ Andrew R. Jassy

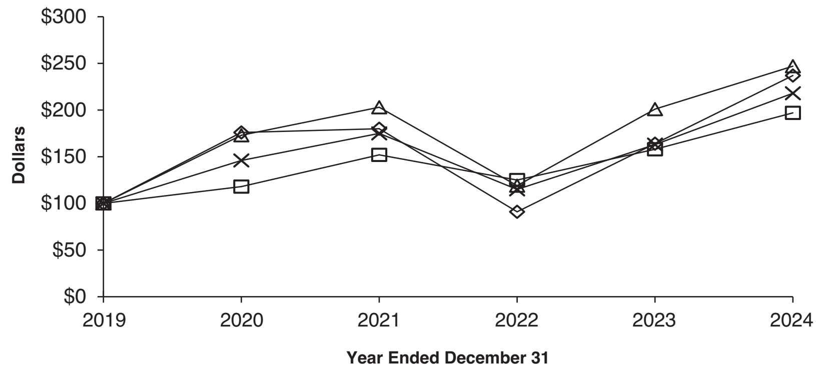
Andrew R. Jassy
President and Chief Executive Officer

Pursuant to the requirements of the Securities Exchange Act of 1934, this Report has been signed below by the following persons on behalf of the registrant and in the capacities indicated as of February 6, 2025.

Signature	Title
/s/ Andrew R. Jassy **Andrew R. Jassy**	President and Chief Executive Officer (Principal Executive Officer) and Director
/s/ Brian T. Olsavsky **Brian T. Olsavsky**	Senior Vice President and Chief Financial Officer (Principal Financial Officer)
/s/ Shelley L. Reynolds **Shelley L. Reynolds**	Vice President, Worldwide Controller (Principal Accounting Officer)
/s/ Jeffrey P. Bezos **Jeffrey P. Bezos**	Executive Chair
/s/ Keith B. Alexander **Keith B. Alexander**	Director
/s/ Edith W. Cooper **Edith W. Cooper**	Director
/s/ Jamie S. Gorelick **Jamie S. Gorelick**	Director
/s/ Daniel P. Huttenlocher **Daniel P. Huttenlocher**	Director
/s/ Andrew Y. Ng **Andrew Y. Ng**	Director
/s/ Indra K. Nooyi **Indra K. Nooyi**	Director
/s/ Jonathan J. Rubinstein **Jonathan J. Rubinstein**	Director
/s/ Brad D. Smith **Brad D. Smith**	Director
/s/ Patricia Q. Stonesifer **Patricia Q. Stonesifer**	Director
/s/ Wendell P. Weeks **Wendell P. Weeks**	Director

Stock Price Performance Graph

The graph set forth below compares cumulative total return on the common stock with the cumulative total return of the NYSE Technology Index, the S&P 500 Index, and the S&P 500 Consumer Discretionary Distribution & Retail Index, resulting from an initial investment of $100 in each and, except in the case of the NYSE Technology Index, assuming the reinvestment of any dividends, based on closing prices. Measurement points are the last trading day of each of Amazon's fiscal years ended December 31, 2019, 2020, 2021, 2022, 2023, and 2024.



	Legend	Cumulative Total Return Year Ended December 31, 2019	2020	2021	2022	2023	2024
Amazon.com, Inc.	—◇—	$100	$176	$180	$91	$164	$237
NYSE Technology Index	—△—	100	173	203	119	201	247
S&P 500 Index	—□—	100	118	152	125	158	197
S&P 500 Consumer Discretionary Distribution & Retail Index	—×—	100	146	175	115	163	218

Note: Stock price performance shown in the Stock Price Performance Graph for the common stock is historical and not necessarily indicative of future price performance.

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-Q

(Mark One)

☒ **QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the quarterly period ended December 30, 2023

or

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from ______ to ______.

Commission File Number: **001-36743**

Apple Inc.

(Exact name of Registrant as specified in its charter)

California	**94-2404110**
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)
One Apple Park Way Cupertino, California	**95014**
(Address of principal executive offices)	(Zip Code)

(408) 996-1010

(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading symbol(s)	Name of each exchange on which registered
Common Stock, $0.00001 par value per share	**AAPL**	**The Nasdaq Stock Market LLC**
0.000% Notes due 2025	—	**The Nasdaq Stock Market LLC**
0.875% Notes due 2025	—	**The Nasdaq Stock Market LLC**
1.625% Notes due 2026	—	**The Nasdaq Stock Market LLC**
2.000% Notes due 2027	—	**The Nasdaq Stock Market LLC**
1.375% Notes due 2029	—	**The Nasdaq Stock Market LLC**
3.050% Notes due 2029	—	**The Nasdaq Stock Market LLC**
0.500% Notes due 2031	—	**The Nasdaq Stock Market LLC**
3.600% Notes due 2042	—	**The Nasdaq Stock Market LLC**

Indicate by check mark whether the Registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the Registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days.

Yes ☒ No ☐

Indicate by check mark whether the Registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the Registrant was required to submit such files).

Yes ☒ No ☐

Indicate by check mark whether the Registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “smaller reporting company,” and “emerging growth company” in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the Registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the Registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act).

Yes ☐ No ☒

15,441,881,000 shares of common stock were issued and outstanding as of January 19, 2024.

Apple Inc.

Form 10-Q

For the Fiscal Quarter Ended December 30, 2023

TABLE OF CONTENTS

PART I — FINANCIAL INFORMATION

Item 1. Financial Statements

Apple Inc.

CONDENSED CONSOLIDATED STATEMENTS OF OPERATIONS (Unaudited)
(In millions, except number of shares, which are reflected in thousands, and per-share amounts)

	Three Months Ended	
	December 30, 2023	December 31, 2022
Net sales:		
Products	$ 96,458	$ 96,388
Services	23,117	20,766
Total net sales	119,575	117,154
Cost of sales:		
Products	58,440	60,765
Services	6,280	6,057
Total cost of sales	64,720	66,822
Gross margin	54,855	50,332
Operating expenses:		
Research and development	7,696	7,709
Selling, general and administrative	6,786	6,607
Total operating expenses	14,482	14,316
Operating income	40,373	36,016
Other income/(expense), net	(50)	(393)
Income before provision for income taxes	40,323	35,623
Provision for income taxes	6,407	5,625
Net income	$ 33,916	$ 29,998
Earnings per share:		
Basic	$ 2.19	$ 1.89
Diluted	$ 2.18	$ 1.88
Shares used in computing earnings per share:		
Basic	15,509,763	15,892,723
Diluted	15,576,641	15,955,718

See accompanying Notes to Condensed Consolidated Financial Statements.

Apple Inc.

CONDENSED CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME (Unaudited)
(In millions)

	Three Months Ended	
	December 30, 2023	December 31, 2022
Net income	$ 33,916	$ 29,998
Other comprehensive income/(loss):		
Change in foreign currency translation, net of tax	308	(14)
Change in unrealized gains/losses on derivative instruments, net of tax:		
Change in fair value of derivative instruments	(531)	(988)
Adjustment for net (gains)/losses realized and included in net income	(823)	(1,766)
Total change in unrealized gains/losses on derivative instruments	(1,354)	(2,754)
Change in unrealized gains/losses on marketable debt securities, net of tax:		
Change in fair value of marketable debt securities	3,045	900
Adjustment for net (gains)/losses realized and included in net income	75	65
Total change in unrealized gains/losses on marketable debt securities	3,120	965
Total other comprehensive income/(loss)	2,074	(1,803)
Total comprehensive income	$ 35,990	$ 28,195

See accompanying Notes to Condensed Consolidated Financial Statements.

Apple Inc.

CONDENSED CONSOLIDATED BALANCE SHEETS (Unaudited)

(In millions, except number of shares, which are reflected in thousands, and par value)

	December 30, 2023	September 30, 2023
ASSETS:		
Current assets:		
Cash and cash equivalents	$ 40,760	$ 29,965
Marketable securities	32,340	31,590
Accounts receivable, net	23,194	29,508
Vendor non-trade receivables	26,908	31,477
Inventories	6,511	6,331
Other current assets	13,979	14,695
Total current assets	143,692	143,566
Non-current assets:		
Marketable securities	99,475	100,544
Property, plant and equipment, net	43,666	43,715
Other non-current assets	66,681	64,758
Total non-current assets	209,822	209,017
Total assets	$ 353,514	$ 352,583
LIABILITIES AND SHAREHOLDERS' EQUITY:		
Current liabilities:		
Accounts payable	$ 58,146	$ 62,611
Other current liabilities	54,611	58,829
Deferred revenue	8,264	8,061
Commercial paper	1,998	5,985
Term debt	10,954	9,822
Total current liabilities	133,973	145,308
Non-current liabilities:		
Term debt	95,088	95,281
Other non-current liabilities	50,353	49,848
Total non-current liabilities	145,441	145,129
Total liabilities	279,414	290,437
Commitments and contingencies		
Shareholders' equity:		
Common stock and additional paid-in capital, $0.00001 par value: 50,400,000 shares authorized; 15,460,223 and 15,550,061 shares issued and outstanding, respectively	75,236	73,812
Retained earnings/(Accumulated deficit)	8,242	(214)
Accumulated other comprehensive loss	(9,378)	(11,452)
Total shareholders' equity	74,100	62,146
Total liabilities and shareholders' equity	$ 353,514	$ 352,583

See accompanying Notes to Condensed Consolidated Financial Statements.

Apple Inc.

CONDENSED CONSOLIDATED STATEMENTS OF SHAREHOLDERS' EQUITY (Unaudited)
(In millions, except per-share amounts)

	Three Months Ended	
	December 30, 2023	December 31, 2022
Total shareholders' equity, beginning balances	$ 62,146	$ 50,672
Common stock and additional paid-in capital:		
Beginning balances	73,812	64,849
Common stock withheld related to net share settlement of equity awards	(1,660)	(1,434)
Share-based compensation	3,084	2,984
Ending balances	75,236	66,399
Retained earnings/(Accumulated deficit):		
Beginning balances	(214)	(3,068)
Net income	33,916	29,998
Dividends and dividend equivalents declared	(3,774)	(3,712)
Common stock withheld related to net share settlement of equity awards	(1,018)	(978)
Common stock repurchased	(20,668)	(19,000)
Ending balances	8,242	3,240
Accumulated other comprehensive income/(loss):		
Beginning balances	(11,452)	(11,109)
Other comprehensive income/(loss)	2,074	(1,803)
Ending balances	(9,378)	(12,912)
Total shareholders' equity, ending balances	$ 74,100	$ 56,727
Dividends and dividend equivalents declared per share or RSU	$ 0.24	$ 0.23

See accompanying Notes to Condensed Consolidated Financial Statements.

Apple Inc.

CONDENSED CONSOLIDATED STATEMENTS OF CASH FLOWS (Unaudited)
(In millions)

	Three Months Ended	
	December 30, 2023	December 31, 2022
Cash, cash equivalents and restricted cash, beginning balances	$ 30,737	$ 24,977
Operating activities:		
Net income	33,916	29,998
Adjustments to reconcile net income to cash generated by operating activities:		
Depreciation and amortization	2,848	2,916
Share-based compensation expense	2,997	2,905
Other	(989)	(317)
Changes in operating assets and liabilities:		
Accounts receivable, net	6,555	4,275
Vendor non-trade receivables	4,569	2,320
Inventories	(137)	(1,807)
Other current and non-current assets	(1,457)	(4,099)
Accounts payable	(4,542)	(6,075)
Other current and non-current liabilities	(3,865)	3,889
Cash generated by operating activities	39,895	34,005
Investing activities:		
Purchases of marketable securities	(9,780)	(5,153)
Proceeds from maturities of marketable securities	13,046	7,127
Proceeds from sales of marketable securities	1,337	509
Payments for acquisition of property, plant and equipment	(2,392)	(3,787)
Other	(284)	(141)
Cash generated by/(used in) investing activities	1,927	(1,445)
Financing activities:		
Payments for taxes related to net share settlement of equity awards	(2,591)	(2,316)
Payments for dividends and dividend equivalents	(3,825)	(3,768)
Repurchases of common stock	(20,139)	(19,475)
Repayments of term debt	—	(1,401)
Repayments of commercial paper, net	(3,984)	(8,214)
Other	(46)	(389)
Cash used in financing activities	(30,585)	(35,563)
Increase/(Decrease) in cash, cash equivalents and restricted cash	11,237	(3,003)
Cash, cash equivalents and restricted cash, ending balances	$ 41,974	$ 21,974
Supplemental cash flow disclosure:		
Cash paid for income taxes, net	$ 7,255	$ 828

See accompanying Notes to Condensed Consolidated Financial Statements.

Apple Inc.

Notes to Condensed Consolidated Financial Statements (Unaudited)

Note 1 – Summary of Significant Accounting Policies

Basis of Presentation and Preparation

The condensed consolidated financial statements include the accounts of Apple Inc. and its wholly owned subsidiaries (collectively "Apple" or the "Company"). In the opinion of the Company's management, the condensed consolidated financial statements reflect all adjustments, which are normal and recurring in nature, necessary for fair financial statement presentation. The preparation of these condensed consolidated financial statements and accompanying notes in conformity with U.S. generally accepted accounting principles ("GAAP") requires the use of management estimates. Certain prior period amounts in the condensed consolidated financial statements and accompanying notes have been reclassified to conform to the current period's presentation. These condensed consolidated financial statements and accompanying notes should be read in conjunction with the Company's annual consolidated financial statements and accompanying notes included in its Annual Report on Form 10-K for the fiscal year ended September 30, 2023 (the "2023 Form 10-K").

The Company's fiscal year is the 52- or 53-week period that ends on the last Saturday of September. An additional week is included in the first fiscal quarter every five or six years to realign the Company's fiscal quarters with calendar quarters, which occurred in the first fiscal quarter of 2023. The Company's fiscal years 2024 and 2023 span 52 and 53 weeks, respectively. Unless otherwise stated, references to particular years, quarters, months and periods refer to the Company's fiscal years ended in September and the associated quarters, months and periods of those fiscal years.

Note 2 – Revenue

Net sales disaggregated by significant products and services for the three months ended December 30, 2023 and December 31, 2022 were as follows (in millions):

	Three Months Ended	
	December 30, 2023	December 31, 2022
iPhone®	$ 69,702	$ 65,775
Mac®	7,780	7,735
iPad®	7,023	9,396
Wearables, Home and Accessories	11,953	13,482
Services	23,117	20,766
Total net sales	$ 119,575	$ 117,154

Total net sales include $3.5 billion of revenue recognized in the three months ended December 30, 2023 that was included in deferred revenue as of September 30, 2023 and $3.4 billion of revenue recognized in the three months ended December 31, 2022 that was included in deferred revenue as of September 24, 2022.

The Company's proportion of net sales by disaggregated revenue source was generally consistent for each reportable segment in Note 10, "Segment Information and Geographic Data" for the three months ended December 30, 2023 and December 31, 2022, except in Greater China, where iPhone revenue represented a moderately higher proportion of net sales.

As of December 30, 2023 and September 30, 2023, the Company had total deferred revenue of $12.5 billion and $12.1 billion, respectively. As of December 30, 2023, the Company expects 66% of total deferred revenue to be realized in less than a year, 26% within one-to-two years, 7% within two-to-three years and 1% in greater than three years.

Note 3 – Earnings Per Share

The following table shows the computation of basic and diluted earnings per share for the three months ended December 30, 2023 and December 31, 2022 (net income in millions and shares in thousands):

	Three Months Ended	
	December 30, 2023	December 31, 2022
Numerator:		
Net income	$ 33,916	$ 29,998
Denominator:		
Weighted-average basic shares outstanding	15,509,763	15,892,723
Effect of dilutive share-based awards	66,878	62,995
Weighted-average diluted shares	15,576,641	15,955,718
Basic earnings per share	$ 2.19	$ 1.89
Diluted earnings per share	$ 2.18	$ 1.88

Approximately 89 million restricted stock units ("RSUs") were excluded from the computation of diluted earnings per share for the three months ended December 31, 2022 because their effect would have been antidilutive.

Note 4 – Financial Instruments

Cash, Cash Equivalents and Marketable Securities

The following tables show the Company's cash, cash equivalents and marketable securities by significant investment category as of December 30, 2023 and September 30, 2023 (in millions):

	December 30, 2023						
	Adjusted Cost	Unrealized Gains	Unrealized Losses	Fair Value	Cash and Cash Equivalents	Current Marketable Securities	Non-Current Marketable Securities
Cash	$ 29,542	$ —	$ —	$ 29,542	$ 29,542	$ —	$ —
Level 1:							
Money market funds	2,000	—	—	2,000	2,000	—	—
Mutual funds	448	35	(11)	472	—	472	—
Subtotal	2,448	35	(11)	2,472	2,000	472	—
Level 2 (1):							
U.S. Treasury securities	24,041	12	(920)	23,133	7,303	4,858	10,972
U.S. agency securities	5,791	—	(448)	5,343	243	98	5,002
Non-U.S. government securities	17,326	54	(675)	16,705	—	11,175	5,530
Certificates of deposit and time deposits	1,448	—	—	1,448	1,119	329	—
Commercial paper	1,361	—	—	1,361	472	889	—
Corporate debt securities	75,360	112	(3,964)	71,508	81	13,909	57,518
Municipal securities	562	—	(14)	548	—	185	363
Mortgage- and asset-backed securities	22,369	53	(1,907)	20,515	—	425	20,090
Subtotal	148,258	231	(7,928)	140,561	9,218	31,868	99,475
Total (2)	$ 180,248	$ 266	$ (7,939)	$ 172,575	$ 40,760	$ 32,340	$ 99,475

	September 30, 2023						
	Adjusted Cost	Unrealized Gains	Unrealized Losses	Fair Value	Cash and Cash Equivalents	Current Marketable Securities	Non-Current Marketable Securities
Cash	$ 28,359	$ —	$ —	$ 28,359	$ 28,359	$ —	$ —
Level 1:							
Money market funds	481	—	—	481	481	—	—
Mutual funds and equity securities	442	12	(26)	428	—	428	—
Subtotal	923	12	(26)	909	481	428	—
Level 2 (1):							
U.S. Treasury securities	19,406	—	(1,292)	18,114	35	5,468	12,611
U.S. agency securities	5,736	—	(600)	5,136	36	271	4,829
Non-U.S. government securities	17,533	6	(1,048)	16,491	—	11,332	5,159
Certificates of deposit and time deposits	1,354	—	—	1,354	1,034	320	—
Commercial paper	608	—	—	608	—	608	—
Corporate debt securities	76,840	6	(5,956)	70,890	20	12,627	58,243
Municipal securities	628	—	(26)	602	—	192	410
Mortgage- and asset-backed securities	22,365	6	(2,735)	19,636	—	344	19,292
Subtotal	144,470	18	(11,657)	132,831	1,125	31,162	100,544
Total (2)	$ 173,752	$ 30	$ (11,683)	$ 162,099	$ 29,965	$ 31,590	$ 100,544

(1) The valuation techniques used to measure the fair values of the Company's Level 2 financial instruments, which generally have counterparties with high credit ratings, are based on quoted market prices or model-driven valuations using significant inputs derived from or corroborated by observable market data.

(2) As of December 30, 2023 and September 30, 2023, total marketable securities included $13.9 billion and $13.8 billion, respectively, that were restricted from general use, related to the European Commission decision finding that Ireland granted state aid to the Company, and other agreements.

The following table shows the fair value of the Company's non-current marketable debt securities, by contractual maturity, as of December 30, 2023 (in millions):

Due after 1 year through 5 years	$ 72,994
Due after 5 years through 10 years	9,368
Due after 10 years	17,113
Total fair value	$ 99,475

Derivative Instruments and Hedging

The Company may use derivative instruments to partially offset its business exposure to foreign exchange and interest rate risk. However, the Company may choose not to hedge certain exposures for a variety of reasons, including accounting considerations or the prohibitive economic cost of hedging particular exposures. There can be no assurance the hedges will offset more than a portion of the financial impact resulting from movements in foreign exchange or interest rates.

Foreign Exchange Rate Risk

To protect gross margins from fluctuations in foreign exchange rates, the Company may use forwards, options or other instruments, and may designate these instruments as cash flow hedges. The Company generally hedges portions of its forecasted foreign currency exposure associated with revenue and inventory purchases, typically for up to 12 months.

To protect the Company's foreign currency–denominated term debt or marketable securities from fluctuations in foreign exchange rates, the Company may use forwards, cross-currency swaps or other instruments. The Company designates these instruments as either cash flow or fair value hedges. As of December 30, 2023, the maximum length of time over which the Company is hedging its exposure to the variability in future cash flows for term debt–related foreign currency transactions is 19 years.

The Company may also use derivative instruments that are not designated as accounting hedges to protect gross margins from certain fluctuations in foreign exchange rates, as well as to offset a portion of the foreign currency gains and losses generated by the remeasurement of certain assets and liabilities denominated in non-functional currencies.

Interest Rate Risk

To protect the Company's term debt or marketable securities from fluctuations in interest rates, the Company may use interest rate swaps, options or other instruments. The Company designates these instruments as either cash flow or fair value hedges.

The notional amounts of the Company's outstanding derivative instruments as of December 30, 2023 and September 30, 2023 were as follows (in millions):

	December 30, 2023	September 30, 2023
Derivative instruments designated as accounting hedges:		
Foreign exchange contracts	$ 66,735	$ 74,730
Interest rate contracts	$ 19,375	$ 19,375
Derivative instruments not designated as accounting hedges:		
Foreign exchange contracts	$ 102,108	$ 104,777

The carrying amounts of the Company's hedged items in fair value hedges as of December 30, 2023 and September 30, 2023 were as follows (in millions):

	December 30, 2023	September 30, 2023
Hedged assets/(liabilities):		
Current and non-current marketable securities	$ 15,102	$ 14,433
Current and non-current term debt	$ (18,661)	$ (18,247)

Accounts Receivable

Trade Receivables

The Company's third-party cellular network carriers accounted for 34% and 41% of total trade receivables as of December 30, 2023 and September 30, 2023, respectively. The Company requires third-party credit support or collateral from certain customers to limit credit risk.

Vendor Non-Trade Receivables

The Company has non-trade receivables from certain of its manufacturing vendors resulting from the sale of components to these vendors who manufacture subassemblies or assemble final products for the Company. The Company purchases these components directly from suppliers. The Company does not reflect the sale of these components in products net sales. Rather, the Company recognizes any gain on these sales as a reduction of products cost of sales when the related final products are sold by the Company. As of December 30, 2023, the Company had two vendors that individually represented 10% or more of total vendor non-trade receivables, which accounted for 50% and 20%. As of September 30, 2023, the Company had two vendors that individually represented 10% or more of total vendor non-trade receivables, which accounted for 48% and 23%.

Note 5 – Condensed Consolidated Financial Statement Details

The following table shows the Company's condensed consolidated financial statement details as of December 30, 2023 and September 30, 2023 (in millions):

Property, Plant and Equipment, Net

	December 30, 2023	September 30, 2023
Gross property, plant and equipment	$ 116,176	$ 114,599
Accumulated depreciation	(72,510)	(70,884)
Total property, plant and equipment, net	$ 43,666	$ 43,715

Note 6 – Debt

Commercial Paper

The Company issues unsecured short-term promissory notes pursuant to a commercial paper program. The Company uses net proceeds from the commercial paper program for general corporate purposes, including dividends and share repurchases. As of December 30, 2023 and September 30, 2023, the Company had $2.0 billion and $6.0 billion of commercial paper outstanding, respectively. The following table provides a summary of cash flows associated with the issuance and maturities of commercial paper for the three months ended December 30, 2023 and December 31, 2022 (in millions):

	Three Months Ended	
	December 30, 2023	December 31, 2022
Maturities 90 days or less:		
Repayments of commercial paper, net	$ (3,984)	$ (5,569)
Maturities greater than 90 days:		
Repayments of commercial paper	—	(2,645)
Total repayments of commercial paper, net	$ (3,984)	$ (8,214)

Term Debt

As of December 30, 2023 and September 30, 2023, the Company had outstanding fixed-rate notes with varying maturities for an aggregate carrying amount of $106.0 billion and $105.1 billion, respectively (collectively the “Notes”). As of December 30, 2023 and September 30, 2023, the fair value of the Company’s Notes, based on Level 2 inputs, was $96.7 billion and $90.8 billion, respectively.

Note 7 – Shareholders’ Equity

Share Repurchase Program

During the three months ended December 30, 2023, the Company repurchased 118 million shares of its common stock for $20.5 billion. The Company’s share repurchase program does not obligate the Company to acquire a minimum amount of shares. Under the program, shares may be repurchased in privately negotiated or open market transactions, including under plans complying with Rule 10b5-1 under the Securities Exchange Act of 1934, as amended (the “Exchange Act”).

Note 8 – Share-Based Compensation

Restricted Stock Units

A summary of the Company’s RSU activity and related information for the three months ended December 30, 2023 is as follows:

	Number of RSUs (in thousands)	Weighted-Average Grant Date Fair Value Per RSU	Aggregate Fair Value (in millions)
Balance as of September 30, 2023	180,247	$ 135.91	
RSUs granted	74,241	$ 171.58	
RSUs vested	(42,490)	$ 110.75	
RSUs canceled	(3,026)	$ 109.05	
Balance as of December 30, 2023	208,972	$ 154.09	$ 40,233

The fair value as of the respective vesting dates of RSUs was $7.7 billion and $6.8 billion for the three months ended December 30, 2023 and December 31, 2022, respectively.

Share-Based Compensation

The following table shows share-based compensation expense and the related income tax benefit included in the Condensed Consolidated Statements of Operations for the three months ended December 30, 2023 and December 31, 2022 (in millions):

	Three Months Ended	
	December 30, 2023	December 31, 2022
Share-based compensation expense	$ 2,997	$ 2,905
Income tax benefit related to share-based compensation expense	$ (1,235)	$ (1,178)

As of December 30, 2023, the total unrecognized compensation cost related to outstanding RSUs was $27.4 billion, which the Company expects to recognize over a weighted-average period of 2.9 years.

Note 9 – Contingencies

The Company is subject to various legal proceedings and claims that have arisen in the ordinary course of business and that have not been fully resolved. The outcome of litigation is inherently uncertain. In the opinion of management, there was not at least a reasonable possibility the Company may have incurred a material loss, or a material loss greater than a recorded accrual, concerning loss contingencies for asserted legal and other claims.

Note 10 – Segment Information and Geographic Data

The following table shows information by reportable segment for the three months ended December 30, 2023 and December 31, 2022 (in millions):

	Three Months Ended	
	December 30, 2023	December 31, 2022
Americas:		
Net sales	$ 50,430	$ 49,278
Operating income	$ 20,357	$ 17,864
Europe:		
Net sales	$ 30,397	$ 27,681
Operating income	$ 12,711	$ 10,017
Greater China:		
Net sales	$ 20,819	$ 23,905
Operating income	$ 8,622	$ 10,437
Japan:		
Net sales	$ 7,767	$ 6,755
Operating income	$ 3,819	$ 3,236
Rest of Asia Pacific:		
Net sales	$ 10,162	$ 9,535
Operating income	$ 4,579	$ 3,851

A reconciliation of the Company's segment operating income to the Condensed Consolidated Statements of Operations for the three months ended December 30, 2023 and December 31, 2022 is as follows (in millions):

	Three Months Ended	
	December 30, 2023	December 31, 2022
Segment operating income	$ 50,088	$ 45,405
Research and development expense	(7,696)	(7,709)
Other corporate expenses, net	(2,019)	(1,680)
Total operating income	$ 40,373	$ 36,016

Item 2. Management's Discussion and Analysis of Financial Condition and Results of Operations

This Item and other sections of this Quarterly Report on Form 10-Q ("Form 10-Q") contain forward-looking statements, within the meaning of the Private Securities Litigation Reform Act of 1995, that involve risks and uncertainties. Forward-looking statements provide current expectations of future events based on certain assumptions and include any statement that does not directly relate to any historical or current fact. For example, statements in this Form 10-Q regarding the potential future impact of macroeconomic conditions on the Company's business and results of operations are forward-looking statements. Forward-looking statements can also be identified by words such as "future," "anticipates," "believes," "estimates," "expects," "intends," "plans," "predicts," "will," "would," "could," "can," "may," and similar terms. Forward-looking statements are not guarantees of future performance and the Company's actual results may differ significantly from the results discussed in the forward-looking statements. Factors that might cause such differences include, but are not limited to, those discussed in Part I, Item 1A of the 2023 Form 10-K under the heading "Risk Factors." The Company assumes no obligation to revise or update any forward-looking statements for any reason, except as required by law.

Unless otherwise stated, all information presented herein is based on the Company's fiscal calendar, and references to particular years, quarters, months or periods refer to the Company's fiscal years ended in September and the associated quarters, months and periods of those fiscal years.

The following discussion should be read in conjunction with the 2023 Form 10-K filed with the U.S. Securities and Exchange Commission (the "SEC") and the condensed consolidated financial statements and accompanying notes included in Part I, Item 1 of this Form 10-Q.

Available Information

The Company periodically provides certain information for investors on its corporate website, www.apple.com, and its investor relations website, investor.apple.com. This includes press releases and other information about financial performance, information on environmental, social and governance matters, and details related to the Company's annual meeting of shareholders. The information contained on the websites referenced in this Form 10-Q is not incorporated by reference into this filing. Further, the Company's references to website URLs are intended to be inactive textual references only.

Business Seasonality and Product Introductions

The Company has historically experienced higher net sales in its first quarter compared to other quarters in its fiscal year due in part to seasonal holiday demand. Additionally, new product and service introductions can significantly impact net sales, cost of sales and operating expenses. The timing of product introductions can also impact the Company's net sales to its indirect distribution channels as these channels are filled with new inventory following a product launch, and channel inventory of an older product often declines as the launch of a newer product approaches. Net sales can also be affected when consumers and distributors anticipate a product introduction.

Fiscal Period

The Company's fiscal year is the 52- or 53-week period that ends on the last Saturday of September. An additional week is included in the first fiscal quarter every five or six years to realign the Company's fiscal quarters with calendar quarters, which occurred in the first quarter of 2023. The Company's fiscal years 2024 and 2023 span 52 and 53 weeks, respectively.

Quarterly Highlights

The Company's first quarter of 2024 included 13 weeks, compared to 14 weeks during the first quarter of 2023.

The Company's total net sales increased 2% or $2.4 billion during the first quarter of 2024 compared to the same quarter in 2023, driven primarily by higher net sales of iPhone and Services, partially offset by lower net sales of iPad and Wearables, Home and Accessories.

During the first quarter of 2024, the Company announced an updated MacBook Pro® 14-in., MacBook Pro 16-in. and iMac®.

The Company repurchased $20.5 billion of its common stock and paid dividends and dividend equivalents of $3.8 billion during the first quarter of 2024.

Macroeconomic Conditions

Macroeconomic conditions, including inflation, changes in interest rates, and currency fluctuations, have directly and indirectly impacted, and could in the future materially impact, the Company's results of operations and financial condition.

Segment Operating Performance

The following table shows net sales by reportable segment for the three months ended December 30, 2023 and December 31, 2022 (dollars in millions):

	Three Months Ended		
	December 30, 2023	December 31, 2022	Change
Net sales by reportable segment:			
Americas	$ 50,430	$ 49,278	2 %
Europe	30,397	27,681	10 %
Greater China	20,819	23,905	(13)%
Japan	7,767	6,755	15 %
Rest of Asia Pacific	10,162	9,535	7 %
Total net sales	$ 119,575	$ 117,154	2 %

Americas

Americas net sales increased 2% or $1.2 billion during the first quarter of 2024 compared to the same quarter in 2023 due primarily to higher net sales of Services and iPhone, partially offset by lower net sales of iPad. The strength in foreign currencies relative to the U.S. dollar had a net favorable year-over-year impact on Americas net sales during the first quarter of 2024.

Europe

Europe net sales increased 10% or $2.7 billion during the first quarter of 2024 compared to the same quarter in 2023 due primarily to higher net sales of iPhone. The strength in foreign currencies relative to the U.S. dollar had a net favorable year-over-year impact on Europe net sales during the first quarter of 2024.

Greater China

Greater China net sales decreased 13% or $3.1 billion during the first quarter of 2024 compared to the same quarter in 2023 due primarily to lower net sales of iPhone, iPad and Wearables, Home and Accessories. The weakness in the renminbi relative to the U.S. dollar had an unfavorable year-over-year impact on Greater China net sales during the first quarter of 2024.

Japan

Japan net sales increased 15% or $1.0 billion during the first quarter of 2024 compared to the same quarter in 2023 due primarily to higher net sales of iPhone. The weakness in the yen relative to the U.S. dollar had an unfavorable year-over-year impact on Japan net sales during the first quarter of 2024.

Rest of Asia Pacific

Rest of Asia Pacific net sales increased 7% or $627 million during the first quarter of 2024 compared to the same quarter in 2023 due primarily to higher net sales of iPhone, partially offset by lower net sales of Wearables, Home and Accessories.

Products and Services Performance

The following table shows net sales by category for the three months ended December 30, 2023 and December 31, 2022 (dollars in millions):

	Three Months Ended		
	December 30, 2023	December 31, 2022	Change
Net sales by category:			
iPhone	$ 69,702	$ 65,775	6 %
Mac	7,780	7,735	1 %
iPad	7,023	9,396	(25)%
Wearables, Home and Accessories	11,953	13,482	(11)%
Services	23,117	20,766	11 %
Total net sales	$ 119,575	$ 117,154	2 %

iPhone

iPhone net sales increased 6% or $3.9 billion during the first quarter of 2024 compared to the same quarter in 2023 due primarily to higher net sales of Pro models, partially offset by lower net sales of other models.

Mac

Mac net sales were relatively flat during the first quarter of 2024 compared to the same quarter in 2023.

iPad

iPad net sales decreased 25% or $2.4 billion during the first quarter of 2024 compared to the same quarter in 2023 due primarily to lower net sales of iPad Pro, iPad 9th generation and iPad Air.

Wearables, Home and Accessories

Wearables, Home and Accessories net sales decreased 11% or $1.5 billion during the first quarter of 2024 compared to the same quarter in 2023 due primarily to lower net sales of Wearables and Accessories.

Services

Services net sales increased 11% or $2.4 billion during the first quarter of 2024 compared to the same quarter in 2023 due primarily to higher net sales from advertising, video and cloud services.

Gross Margin

Products and Services gross margin and gross margin percentage for the three months ended December 30, 2023 and December 31, 2022 were as follows (dollars in millions):

	Three Months Ended	
	December 30, 2023	December 31, 2022
Gross margin:		
Products	$ 38,018	$ 35,623
Services	16,837	14,709
Total gross margin	$ 54,855	$ 50,332
Gross margin percentage:		
Products	39.4%	37.0%
Services	72.8%	70.8%
Total gross margin percentage	45.9%	43.0%

Products Gross Margin

Products gross margin increased during the first quarter of 2024 compared to the same quarter in 2023 due primarily to cost savings and a different Products mix, partially offset by the weakness in foreign currencies relative to the U.S. dollar and lower Products volume.

Products gross margin percentage increased during the first quarter of 2024 compared to the same quarter in 2023 due primarily to cost savings and a different Products mix, partially offset by the weakness in foreign currencies relative to the U.S. dollar.

Services Gross Margin

Services gross margin increased during the first quarter of 2024 compared to the same quarter in 2023 due primarily to higher Services net sales and a different Services mix.

Services gross margin percentage increased during the first quarter of 2024 compared to the same quarter in 2023 due primarily to a different Services mix.

The Company’s future gross margins can be impacted by a variety of factors, as discussed in Part I, Item 1A of the 2023 Form 10-K under the heading “Risk Factors.” As a result, the Company believes, in general, gross margins will be subject to volatility and downward pressure.

Operating Expenses

Operating expenses for the three months ended December 30, 2023 and December 31, 2022 were as follows (dollars in millions):

	Three Months Ended	
	December 30, 2023	December 31, 2022
Research and development	$ 7,696	$ 7,709
Percentage of total net sales	6%	7%
Selling, general and administrative	$ 6,786	$ 6,607
Percentage of total net sales	6%	6%
Total operating expenses	$ 14,482	$ 14,316
Percentage of total net sales	12%	12%

Research and Development

Research and development ("R&D") expense was relatively flat during the first quarter of 2024 compared to the same quarter in 2023.

Selling, General and Administrative

Selling, general and administrative expense increased 3% or $179 million during the first quarter of 2024 compared to the same quarter in 2023.

Provision for Income Taxes

Provision for income taxes, effective tax rate and statutory federal income tax rate for the three months ended December 30, 2023 and December 31, 2022 were as follows (dollars in millions):

	Three Months Ended	
	December 30, 2023	December 31, 2022
Provision for income taxes	$ 6,407	$ 5,625
Effective tax rate	15.9%	15.8%
Statutory federal income tax rate	21%	21%

The Company's effective tax rate for the first quarter of 2024 was lower than the statutory federal income tax rate due primarily to a lower effective tax rate on foreign earnings, tax benefits from share-based compensation, and the impact of the U.S. federal R&D credit, partially offset by state income taxes.

The Company's effective tax rate for the first quarter of 2024 was relatively flat compared to the same quarter in 2023.

Liquidity and Capital Resources

The Company believes its balances of cash, cash equivalents and unrestricted marketable securities, along with cash generated by ongoing operations and continued access to debt markets, will be sufficient to satisfy its cash requirements and capital return program over the next 12 months and beyond.

The Company's contractual cash requirements have not changed materially since the 2023 Form 10-K, except for manufacturing purchase obligations.

Manufacturing Purchase Obligations

The Company utilizes several outsourcing partners to manufacture subassemblies for the Company's products and to perform final assembly and testing of finished products. The Company also obtains individual components for its products from a wide variety of individual suppliers. As of December 30, 2023, the Company had manufacturing purchase obligations of $38.0 billion, with $37.9 billion payable within 12 months.

Capital Return Program

In addition to its contractual cash requirements, the Company has an authorized share repurchase program. The program does not obligate the Company to acquire a minimum amount of shares. As of December 30, 2023, the Company's quarterly cash dividend was $0.24 per share. The Company intends to increase its dividend on an annual basis, subject to declaration by the Board of Directors.

Recent Accounting Pronouncements

Income Taxes

In December 2023, the Financial Accounting Standards Board (the "FASB") issued Accounting Standards Update ("ASU") No. 2023-09, *Income Taxes (Topic 740): Improvements to Income Tax Disclosures* ("ASU 2023-09"), which will require the Company to disclose specified additional information in its income tax rate reconciliation and provide additional information for reconciling items that meet a quantitative threshold. ASU 2023-09 will also require the Company to disaggregate its income taxes paid disclosure by federal, state and foreign taxes, with further disaggregation required for significant individual jurisdictions. The Company will adopt ASU 2023-09 in its fourth quarter of 2026. ASU 2023-09 allows for adoption using either a prospective or retrospective transition method.

Segment Reporting

In November 2023, the FASB issued ASU No. 2023-07, *Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures* ("ASU 2023-07"), which will require the Company to disclose segment expenses that are significant and regularly provided to the Company's chief operating decision maker ("CODM"). In addition, ASU 2023-07 will require the Company to disclose the title and position of its CODM and how the CODM uses segment profit or loss information in assessing segment performance and deciding how to allocate resources. The Company will adopt ASU 2023-07 in its fourth quarter of 2025 using a retrospective transition method.

Critical Accounting Estimates

The preparation of financial statements and related disclosures in conformity with GAAP and the Company's discussion and analysis of its financial condition and operating results require the Company's management to make judgments, assumptions and estimates that affect the amounts reported. Note 1, "Summary of Significant Accounting Policies" of the Notes to Condensed Consolidated Financial Statements in Part I, Item 1 of this Form 10-Q and in the Notes to Consolidated Financial Statements in Part II, Item 8 of the 2023 Form 10-K describe the significant accounting policies and methods used in the preparation of the Company's condensed consolidated financial statements. There have been no material changes to the Company's critical accounting estimates since the 2023 Form 10-K.

Item 3. Quantitative and Qualitative Disclosures About Market Risk

There have been no material changes to the Company's market risk during the first three months of 2024. For a discussion of the Company's exposure to market risk, refer to the Company's market risk disclosures set forth in Part II, Item 7A, "Quantitative and Qualitative Disclosures About Market Risk" of the 2023 Form 10-K.

Item 4. Controls and Procedures

Evaluation of Disclosure Controls and Procedures

Based on an evaluation under the supervision and with the participation of the Company's management, the Company's principal executive officer and principal financial officer have concluded that the Company's disclosure controls and procedures as defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act were effective as of December 30, 2023 to provide reasonable assurance that information required to be disclosed by the Company in reports that it files or submits under the Exchange Act is (i) recorded, processed, summarized and reported within the time periods specified in the SEC rules and forms and (ii) accumulated and communicated to the Company's management, including its principal executive officer and principal financial officer, as appropriate to allow timely decisions regarding required disclosure.

Changes in Internal Control over Financial Reporting

There were no changes in the Company's internal control over financial reporting during the first quarter of 2024, which were identified in connection with management's evaluation required by paragraph (d) of Rules 13a-15 and 15d-15 under the Exchange Act, that have materially affected, or are reasonably likely to materially affect, the Company's internal control over financial reporting.

PART II — OTHER INFORMATION

Item 1. Legal Proceedings

Epic Games

Epic Games, Inc. ("Epic") filed a lawsuit in the U.S. District Court for the Northern District of California (the "District Court") against the Company alleging violations of federal and state antitrust laws and California's unfair competition law based upon the Company's operation of its App Store®. On September 10, 2021, the District Court ruled in favor of the Company with respect to nine out of the ten counts included in Epic's claim. The District Court found that certain provisions of the Company's App Store Review Guidelines violate California's unfair competition law and issued an injunction enjoining the Company from prohibiting developers from including in their apps external links that direct customers to purchasing mechanisms other than Apple in-app purchasing. The injunction applies to apps on the U.S. storefront of the iOS and iPadOS® App Store. On April 24, 2023, the U.S. Court of Appeals for the Ninth Circuit (the "Circuit Court") affirmed the District Court's ruling. On June 7, 2023, the Company and Epic filed petitions with the Circuit Court requesting further review of the decision. On June 30, 2023, the Circuit Court denied both petitions. On July 17, 2023, the Circuit Court granted Apple's motion to stay enforcement of the injunction pending appeal to the U.S. Supreme Court (the "Supreme Court"). On January 16, 2024, the Supreme Court denied both the Company's and Epic's petitions and the stay terminated. The Supreme Court's denial of Epic's petition confirms the District Court's ruling in favor of the Company with respect to all of the antitrust claims. Following termination of the stay, the Company implemented a plan to comply with the injunction and filed a statement of compliance with the District Court. On January 31, 2024, Epic filed a notice with the District Court indicating its intent to dispute the Company's compliance plan.

Masimo

Masimo Corporation and Cercacor Laboratories, Inc. (together, "Masimo") filed a complaint before the U.S. International Trade Commission (the "ITC") alleging infringement by the Company of five patents relating to the functionality of the blood oxygen feature in Apple Watch® Series 6 and 7. In its complaint, Masimo sought a permanent exclusion order prohibiting importation to the U.S. of certain Apple Watch models that include blood oxygen sensing functionality. On October 26, 2023, the ITC entered a limited exclusion order (the "Order") prohibiting importation and sales in the U.S. of Apple Watch models with blood oxygen sensing functionality, which includes Apple Watch Series 9 and Apple Watch Ultra™ 2. The Company subsequently proposed a redesign of Apple Watch Series 9 and Apple Watch Ultra 2 to the U.S. Customs and Border Protection (the "CBP") and appealed the Order. On January 12, 2024, the CBP found that the Company's proposed redesign of Apple Watch Series 9 and Apple Watch Ultra 2 falls outside the scope of the Order, permitting the Company to import and sell the models in the U.S.

Other Legal Proceedings

The Company is subject to other legal proceedings and claims that have not been fully resolved and that have arisen in the ordinary course of business. The Company settled certain matters during the first quarter of 2024 that did not individually or in the aggregate have a material impact on the Company's financial condition or operating results. The outcome of litigation is inherently uncertain. If one or more legal matters were resolved against the Company in a reporting period for amounts above management's expectations, the Company's financial condition and operating results for that reporting period could be materially adversely affected.

Item 1A. Risk Factors

The Company's business, reputation, results of operations, financial condition and stock price can be affected by a number of factors, whether currently known or unknown, including those described in Part I, Item 1A of the 2023 Form 10-K under the heading "Risk Factors." When any one or more of these risks materialize from time to time, the Company's business, reputation, results of operations, financial condition and stock price can be materially and adversely affected. Except as set forth below, there have been no material changes to the Company's risk factors since the 2023 Form 10-K.

The technology industry, including, in some instances, the Company, is subject to intense media, political and regulatory scrutiny, which exposes the Company to increasing regulation, government investigations, legal actions and penalties.

From time to time, the Company has made changes to its App Store, including actions taken in response to litigation, competition, market conditions and legal and regulatory requirements. The Company expects to make further business changes in the future. For example, in the U.S. the Company has implemented changes to how developers communicate with consumers within apps on the U.S. storefront of the iOS and iPadOS App Store regarding alternative purchasing mechanisms.

In January 2024, the Company announced changes to iOS, the App Store and Safari® in the European Union to comply with the Digital Markets Act (the "DMA"), including new business terms and alternative fee structures for iOS apps, alternative methods of distribution for iOS apps, alternative payment processing for apps across the Company's operating systems, and additional tools and application programming interfaces ("APIs") for developers. Although the Company's compliance plan is intended to address the DMA's obligations, it is still subject to potential challenge by the European Commission or private litigants. In addition, other jurisdictions may seek to require the Company to make changes to its business. While the changes introduced by the Company in the European Union are intended to reduce new privacy and security risks the DMA poses to European Union users, many risks will remain.

The Company is also currently subject to antitrust investigations in various jurisdictions around the world, which can result in legal proceedings and claims against the Company that could, individually or in the aggregate, have a materially adverse impact on the Company's business, results of operations and financial condition. For example, the Company is the subject of investigations in Europe and other jurisdictions relating to App Store terms and conditions. If such investigations result in adverse findings against the Company, the Company could be exposed to significant fines and may be required to make further changes to its App Store business, all of which could materially adversely affect the Company's business, results of operations and financial condition.

Further, the Company has commercial relationships with other companies in the technology industry that are or may become subject to investigations and litigation that, if resolved against those other companies, could materially adversely affect the Company's commercial relationships with those business partners and materially adversely affect the Company's business, results of operations and financial condition. For example, the Company earns revenue from licensing arrangements with other companies to offer their search services on the Company's platforms and applications, and certain of these arrangements are currently subject to government investigations and legal proceedings.

There can be no assurance the Company's business will not be materially adversely affected, individually or in the aggregate, by the outcomes of such investigations, litigation or changes to laws and regulations in the future. Changes to the Company's business practices to comply with new laws and regulations or in connection with other legal proceedings can negatively impact the reputation of the Company's products for privacy and security and otherwise adversely affect the experience for users of the Company's products and services, and result in harm to the Company's reputation, loss of competitive advantage, poor market acceptance, reduced demand for products and services, and lost sales.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds

Purchases of Equity Securities by the Issuer and Affiliated Purchasers

Share repurchase activity during the three months ended December 30, 2023 was as follows (in millions, except number of shares, which are reflected in thousands, and per-share amounts):

Periods	Total Number of Shares Purchased	Average Price Paid Per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs	Approximate Dollar Value of Shares That May Yet Be Purchased Under the Plans or Programs (1)
October 1, 2023 to November 4, 2023:				
August 2023 ASRs	6,498	(2)	6,498	
Open market and privately negotiated purchases	45,970	$ 174.03	45,970	
November 5, 2023 to December 2, 2023:				
Open market and privately negotiated purchases	33,797	$ 187.14	33,797	
December 3, 2023 to December 30, 2023:				
Open market and privately negotiated purchases	31,782	$ 194.29	31,782	
Total	118,047			$ 53,569

(1) As of December 30, 2023, the Company was authorized by the Board of Directors to purchase up to $90 billion of the Company's common stock under a share repurchase program announced on May 4, 2023, of which $36.4 billion had been utilized. The program does not obligate the Company to acquire a minimum amount of shares. Under the program, shares may be repurchased in privately negotiated or open market transactions, including under plans complying with Rule 10b5-1 under the Exchange Act.

(2) In August 2023, the Company entered into accelerated share repurchase agreements ("ASRs") to purchase up to a total of $5.0 billion of the Company's common stock. In October 2023, the purchase periods for these ASRs ended and an additional 6 million shares were delivered and retired. In total, 29 million shares were delivered under these ASRs at an average repurchase price of $174.93 per share.

Item 3. Defaults Upon Senior Securities

None.

Item 4. Mine Safety Disclosures

Not applicable.

Item 5. Other Information

Insider Trading Arrangements

On November 11, 2023 and November 27, 2023, respectively, Luca Maestri, the Company's Senior Vice President and Chief Financial Officer, and Katherine L. Adams, the Company's Senior Vice President and General Counsel, each entered into a trading plan intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) under the Exchange Act. The plans provide for the sale of all shares vested during the duration of the plans pursuant to certain equity awards granted to Mr. Maestri and Ms. Adams, respectively, excluding any shares withheld by the Company to satisfy income tax withholding and remittance obligations. Mr. Maestri's plan will expire on December 31, 2024, and Ms. Adams's plan will expire on November 1, 2024, subject to early termination for certain specified events set forth in the plans.

Item 6. Exhibits

		Incorporated by Reference		
Exhibit Number	Exhibit Description	Form	Exhibit	Filing Date/ Period End Date
31.1*	Rule 13a-14(a) / 15d-14(a) Certification of Chief Executive Officer.			
31.2*	Rule 13a-14(a) / 15d-14(a) Certification of Chief Financial Officer.			
32.1**	Section 1350 Certifications of Chief Executive Officer and Chief Financial Officer.			
101*	Inline XBRL Document Set for the condensed consolidated financial statements and accompanying notes in Part I, Item 1, "Financial Statements" of this Quarterly Report on Form 10-Q.			
104*	Inline XBRL for the cover page of this Quarterly Report on Form 10-Q, included in the Exhibit 101 Inline XBRL Document Set.			

* Filed herewith.

** Furnished herewith.

SIGNATURE

Pursuant to the requirements of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned thereunto duly authorized.

Date: February 1, 2024	Apple Inc.
	By: /s/ Luca Maestri
	Luca Maestri
	Senior Vice President, Chief Financial Officer

Exhibit 31.1

CERTIFICATION

I, Timothy D. Cook, certify that:

1. I have reviewed this quarterly report on Form 10-Q of Apple Inc.;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the Registrant as of, and for, the periods presented in this report;

4. The Registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the Registrant and have:

   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the Registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

   (c) Evaluated the effectiveness of the Registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   (d) Disclosed in this report any change in the Registrant's internal control over financial reporting that occurred during the Registrant's most recent fiscal quarter (the Registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the Registrant's internal control over financial reporting; and

5. The Registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the Registrant's auditors and the audit committee of the Registrant's board of directors (or persons performing the equivalent functions):

   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the Registrant's ability to record, process, summarize and report financial information; and

   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the Registrant's internal control over financial reporting.

Date: February 1, 2024

By: /s/ Timothy D. Cook

Timothy D. Cook

Chief Executive Officer

Exhibit 31.2

CERTIFICATION

I, Luca Maestri, certify that:

1. I have reviewed this quarterly report on Form 10-Q of Apple Inc.;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the Registrant as of, and for, the periods presented in this report;

4. The Registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the Registrant and have:

   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the Registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

   (c) Evaluated the effectiveness of the Registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   (d) Disclosed in this report any change in the Registrant's internal control over financial reporting that occurred during the Registrant's most recent fiscal quarter (the Registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the Registrant's internal control over financial reporting; and

5. The Registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the Registrant's auditors and the audit committee of the Registrant's board of directors (or persons performing the equivalent functions):

   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the Registrant's ability to record, process, summarize and report financial information; and

   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the Registrant's internal control over financial reporting.

Date: February 1, 2024

By: /s/ Luca Maestri

Luca Maestri

Senior Vice President,
Chief Financial Officer

Exhibit 32.1

**CERTIFICATIONS OF CHIEF EXECUTIVE OFFICER AND CHIEF FINANCIAL OFFICER
PURSUANT TO
18 U.S.C. SECTION 1350,
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002**

I, Timothy D. Cook, certify, as of the date hereof, pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that the Quarterly Report of Apple Inc. on Form 10-Q for the period ended December 30, 2023 fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and that information contained in such Form 10-Q fairly presents in all material respects the financial condition and results of operations of Apple Inc. at the dates and for the periods indicated.

Date: February 1, 2024

By: /s/ Timothy D. Cook
Timothy D. Cook
Chief Executive Officer

I, Luca Maestri, certify, as of the date hereof, pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that the Quarterly Report of Apple Inc. on Form 10-Q for the period ended December 30, 2023 fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and that information contained in such Form 10-Q fairly presents in all material respects the financial condition and results of operations of Apple Inc. at the dates and for the periods indicated.

Date: February 1, 2024

By: /s/ Luca Maestri
Luca Maestri
Senior Vice President,
Chief Financial Officer

A signed original of this written statement required by Section 906 has been provided to Apple Inc. and will be retained by Apple Inc. and furnished to the Securities and Exchange Commission or its staff upon request.

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION

Washington, D.C. 20549

FORM 10-K

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the Fiscal Year Ended June 30, 2024

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the Transition Period From to

Commission File Number 001-37845

MICROSOFT CORPORATION

WASHINGTON	**91-1144442**
(STATE OF INCORPORATION)	**(I.R.S. ID)**

ONE MICROSOFT WAY, REDMOND, WASHINGTON 98052-6399

(425) 882-8080

www.microsoft.com/investor

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Name of exchange on which registered
Common stock, $0.00000625 par value per share	**MSFT**	**NASDAQ**
3.125% Notes due 2028	**MSFT**	**NASDAQ**
2.625% Notes due 2033	**MSFT**	**NASDAQ**

Securities registered pursuant to Section 12(g) of the Act:

None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large Accelerated Filer ☒	Accelerated Filer ☐
Non-accelerated Filer ☐	Smaller Reporting Company ☐
	Emerging Growth Company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☒

As of December 31, 2023, the aggregate market value of the registrant's common stock held by non-affiliates of the registrant was $2.8 trillion based on the closing sale price as reported on the NASDAQ National Market System. As of July 25, 2024, there were 7,433,038,381 shares of common stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the definitive Proxy Statement to be delivered to shareholders in connection with the Annual Meeting of Shareholders to be held on December 10, 2024 are incorporated by reference into Part III.

MICROSOFT CORPORATION
FORM 10-K
For the Fiscal Year Ended June 30, 2024
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Note About Forward-Looking Statements

This report includes estimates, projections, statements relating to our business plans, objectives, and expected operating results that are "forward-looking statements" within the meaning of the Private Securities Litigation Reform Act of 1995, Section 27A of the Securities Act of 1933, and Section 21E of the Securities Exchange Act of 1934. Forward-looking statements may appear throughout this report, including the following sections: "Business" (Part I, Item 1 of this Form 10-K), "Risk Factors" (Part I, Item 1A of this Form 10-K), and "Management's Discussion and Analysis of Financial Condition and Results of Operations" (Part II, Item 7 of this Form 10-K). These forward-looking statements generally are identified by the words "believe," "project," "expect," "anticipate," "estimate," "intend," "strategy," "future," "opportunity," "plan," "may," "should," "will," "would," "will be," "will continue," "will likely result," and similar expressions. Forward-looking statements are based on current expectations and assumptions that are subject to risks and uncertainties that may cause actual results to differ materially. We describe risks and uncertainties that could cause actual results and events to differ materially in "Risk Factors," "Management's Discussion and Analysis of Financial Condition and Results of Operations," and "Quantitative and Qualitative Disclosures About Market Risk" (Part II, Item 7A of this Form 10-K). Readers are cautioned not to place undue reliance on forward-looking statements, which speak only as of the date they are made. We undertake no obligation to update or revise publicly any forward-looking statements, whether because of new information, future events, or otherwise.

PART I

ITEM 1. BUSINESS

GENERAL

Embracing Our Future

Microsoft is a technology company committed to making digital technology and artificial intelligence ("AI") available broadly and doing so responsibly, with a mission to empower every person and every organization on the planet to achieve more. We create platforms and tools, powered by AI, that deliver innovative solutions that meet the evolving needs of our customers. From infrastructure and data, to business applications and collaboration, we provide unique, differentiated value to customers. We strive to create local opportunity, growth, and impact in every country around the world.

We have entered a new age of AI that will fundamentally transform productivity for every individual, organization, and industry on earth, while helping us address some of our most pressing challenges. Microsoft's AI offerings, including Copilot and our Copilot stack, are already orchestrating a new era of AI transformation, driving better business outcomes across every role and industry. As a company, we believe we can be the democratizing force for this new generation of technology and the opportunity it will help unlock for every country, community, and individual.

We believe AI should be as empowering across communities as it is powerful, and we're committed to ensuring it is responsibly designed and built with safety and security from the outset.

What We Offer

Founded in 1975, we develop and support software, services, devices, and solutions that deliver new value for customers and help people and businesses realize their full potential.

We offer an array of services, including cloud-based solutions that provide customers with software, services, platforms, and content, and we provide solution support and consulting services. We also deliver relevant online advertising to a global audience.

Our products include operating systems, cross-device productivity and collaboration applications, server applications, business solution applications, desktop and server management tools, software development tools, and video games. We also design and sell devices, including PCs, tablets, gaming and entertainment consoles, other intelligent devices, and related accessories.

The Ambitions That Drive Us

To achieve our vision, our research and development efforts focus on three interconnected ambitions:

- Reinvent productivity and business processes.
- Build the intelligent cloud and intelligent edge platform.
- Create more personal computing.

Reinvent Productivity and Business Processes

At Microsoft, we provide technology and resources to help our customers create a secure, productive work environment. Our family of products plays a key role in the ways the world works, learns, and connects.

Our growth depends on securely delivering continuous innovation and advancing our leading productivity and collaboration tools and services, including Microsoft 365, LinkedIn, and Dynamics 365. Microsoft 365 is an AI first platform that brings together Office, Windows, Copilot, and Enterprise Mobility + Security to help organizations empower their employees. Copilot for Microsoft 365 combines AI with business data in the Microsoft Graph and Microsoft 365 applications. Microsoft Teams is a comprehensive platform for communication and collaboration, with meetings, calling, chat, file collaboration, and the ability to bring all of the applications teams use into a single place. Microsoft Viva is an employee experience platform that brings together communications, knowledge, learning, resources, and insights.

Together, the Microsoft Cloud, Dynamics 365, Microsoft Teams, and our AI offerings bring a new era of collaborative applications for every role and business function to get insights and business impact faster. Dynamics 365 is a portfolio of intelligent business applications that delivers operational efficiency and breakthrough customer experiences. Our role-based extensions of Microsoft Copilot – Copilot for Sales, Copilot for Service, and Copilot for Finance – bring together the power of Copilot for Microsoft 365 with role-specific insights and workflow assistance to streamline business processes. Copilot Studio allows customers to customize Copilot for Microsoft 365 or build their own Copilot. Microsoft Power Platform helps domain experts drive productivity gains with low-code/no-code tools, robotic process automation, virtual agents, and business intelligence. Copilot Pro is a consumer subscription service that offers faster and more powerful AI assistance in Microsoft 365 apps and on the web. LinkedIn combines our unique data with this new generation of AI to transform the way professionals learn, sell, market, and get hired.

Build the Intelligent Cloud and Intelligent Edge Platform

Digital transformation and adoption of AI continues to revolutionize more business workstreams for organizations in every sector across the globe. For enterprises, digital technology empowers employees, optimizes operations, engages customers, and in some cases, changes the very core of products and services. We continue to invest in high performance and sustainable computing to meet the growing demand for fast access to Microsoft services provided by our network of cloud computing and AI infrastructure and datacenters.

Our cloud business benefits from three economies of scale: datacenters that deploy computational resources at significantly lower cost per unit than smaller ones; datacenters that coordinate and aggregate diverse customer, geographic, and application demand patterns, improving the utilization of computing, storage, and network resources; and multi-tenancy locations that lower application maintenance labor costs.

The Microsoft Cloud provides the best integration across the technology stack while offering openness, improving time to value, reducing costs, and increasing agility. As the foundation of the Microsoft Cloud, Azure uniquely offers hybrid consistency, developer productivity, data and AI capabilities, and trusted security and compliance.

We offer supercomputing power for AI at scale to run large workloads, complemented by our rapidly expanding portfolio of AI cloud services and hardware, which includes custom-built silicon and strong partnerships with chip manufacturers. We have introduced purpose-built cloud infrastructure for AI workloads including a custom AI accelerator, Azure Maia, and a custom in-house central processing unit, Azure Cobalt.

Our AI platform, Azure AI, is helping organizations transform, bringing intelligence and insights to the hands of their employees and customers to solve their most pressing challenges. We offer a wide selection of industry-leading frontier and open models, including from partners, as well as state-of-the-art tooling, and AI-optimized infrastructure, delivering the Copilot stack for Microsoft, enterprises, and developers. Organizations large and small are deploying Azure AI solutions to achieve more at scale, more easily, with the proper enterprise-level responsible AI and safety and security protections. Azure AI Studio provides a full lifecycle toolchain customers can use to ground these models on their own data, create prompt workflows, and help ensure they are deployed and used safely.

GitHub Copilot is at the forefront of AI-powered software development, giving developers a tool to write code easier and faster. From GitHub to Visual Studio, we provide a developer tool chain for everyone, no matter the technical experience, across all platforms.

We have a long-term partnership with OpenAI, a leading AI research and deployment company. We deploy OpenAI’s models across our consumer and enterprise products. As OpenAI’s exclusive cloud provider, Azure powers all of OpenAI's workloads. We have also increased our investments in the development and deployment of specialized supercomputing systems to accelerate OpenAI’s research.

Our hybrid infrastructure offers integrated, end-to-end security, compliance, identity, and management capabilities to support the real-world needs and evolving regulatory requirements of commercial customers and enterprises. Our industry clouds bring together capabilities across the entire Microsoft Cloud, along with industry-specific customizations. Azure Arc simplifies governance and management by delivering a consistent multi-cloud and on-premises management platform.

The Microsoft Intelligent Data Platform fully integrates databases, analytics, and governance. Microsoft Fabric is an end-to-end, unified analytics platform that brings together all the data and analytics tools that organizations need.

Nuance is a leader in conversational AI and ambient intelligence across industries, including healthcare, financial services, retail, and telecommunications. Microsoft and Nuance enable organizations to accelerate their business goals with security-focused, cloud-based solutions infused with AI.

As the rate and pace of cyberthreats continue to accelerate, security is a top priority for every organization. Microsoft offers customers integrated products addressing security, compliance, identity, management, and privacy across customers’ multi-cloud, application, and device assets. With Copilot for Security, Microsoft offers an AI cybersecurity product that enables security professionals to respond to cyberthreats quickly.

Windows 365 enables users to stream a full Windows experience from the Microsoft Cloud to any device.

Create More Personal Computing

We strive to make computing more personal, enabling users to interact with technology in more intuitive, engaging, and dynamic ways.

Windows 11 offers innovations focused on performance, productivity, and creativity, including Copilot in Windows. Windows 11 security and privacy features include operating system security, application security, and user and identity security. Dev Home is an open-source experience in Windows to help developer productivity. We are committed to designing and marketing first-party devices to help drive innovation, create new device categories, and stimulate demand in the Windows ecosystem. The Surface family includes Surface Pro, Surface Laptop, and other Surface products. Copilot+ PCs are a new class of Windows 11 PCs that are powered by a neural processing unit. These PCs use on-device AI for enhanced performance and features.

Copilot is an AI assistant that helps users navigate the web, answer questions, and create content. Microsoft Edge is our fast and secure browser that helps protect users’ data and offers enhanced browsing capabilities including quick access to AI-powered tools, apps, and more. The AI-powered Bing search engine with Copilot delivers better search, more complete answers, and the ability to generate content.

Microsoft is expanding how billions of people globally access and play video games on PC, console, mobile, and cloud. We put game development front and center, backed by innovative hardware, experiences, and a subscription service, Xbox Game Pass, that allows those games to reach more players across more devices. Activision Blizzard, Inc. (“Activision Blizzard”), a leader in game development and an interactive entertainment content publisher, joined Microsoft in October 2023.

Our Future Opportunity

We are focused on helping customers use the breadth and depth of the Microsoft Cloud to get the most value out of their digital spend while leading the AI platform wave across our solution areas. We continue to develop complete, intelligent solutions for our customers that empower people to be productive and collaborate, while safeguarding businesses and simplifying IT management. Our goal is to lead the industry in several distinct areas of technology over the long term, which we expect will translate to sustained growth. We are investing significant resources in:

- Transforming the workplace to deliver new modern, modular business applications, drive deeper insights, and improve how people communicate, collaborate, learn, work, and interact with one another.
- Building and running cloud-based services in ways that utilize ubiquitous computing to unleash new experiences and opportunities for businesses and individuals.
- Applying AI and ambient intelligence to drive insights, revolutionize many types of work and business processes, and provide substantive productivity gains using natural methods of communication.
- Tackling security from all angles with our integrated, end-to-end solutions spanning security, compliance, identity, and management, across all clouds and platforms.
- Inventing new gaming experiences that bring people together around their shared love for games on any devices and pushing the boundaries of innovation with console and PC gaming.
- Using Windows to fuel our cloud business, grow our share of the PC market, and drive increased engagement with our services like Microsoft Edge, Bing, Microsoft Teams, Microsoft 365 Consumer, Xbox Game Pass, and more.

Our future growth depends on our ability to transcend current product category definitions, business models, and sales motions.

Corporate Social Responsibility

Commitment to Sustainability

Microsoft’s approach to addressing climate change starts with the sustainability of our own business. In 2020, we committed to being a carbon negative, water positive, and zero waste company by 2030.

Since announcing that commitment, we have seen major changes both in the technology sector and in our understanding of what it will take to meet our climate goals. New technologies, including generative AI, hold promise for new innovations that can help address the climate crisis. At the same time, the infrastructure and electricity needed for these technologies create new challenges for meeting sustainability commitments across the tech sector.

In May 2024, we released our Environmental Sustainability Report which looked back at our progress in several areas during fiscal year 2023. In four areas we are on track, and in each of these we see progress that has the potential to have global impact beyond our own sustainability work. These are:

- Reducing our direct operational emissions (Scope 1 and 2).
- Accelerating carbon removal.
- Designing for circularity to minimize waste and reusing cloud hardware.
- Improving biodiversity and protecting more land than we use.

At the same time, there are two areas where we’re not yet on track, and in each of these we are intensively engaged in work to identify and pursue additional breakthroughs. These are:

- Reducing our indirect emissions (Scope 3).
- Reducing our water use and replenishing more water than we consume in our datacenter operations.

Even amid the challenges, we remain optimistic. We're encouraged by ongoing progress across our campuses and datacenters, and throughout our value chain.

Addressing Racial Injustice and Inequity

In June 2020, we outlined a series of multi-year commitments designed to address the racial injustice and inequity experienced by racial and ethnic minorities in the United States, including Black and African American communities. We remain committed to addressing racial injustice and inequity and helping improve lived experiences at Microsoft, in employees' communities, and beyond.

In fiscal year 2024, we continued to collaborate with partners and worked within neighborhoods and communities to advance projects and programs. We grew our Nonprofit Tech Acceleration for Black and African American Communities program, to help more than 3,000 local organizations in nearly 1,900 Black and African American communities use technical solutions to modernize and streamline operations. We also expanded our Technology Education and Learning Support ("TEALS") program to reach nearly 550 high schools across 21 racial equity expansion regions with the support of nearly 1,500 volunteers, 12% of whom identify as Black or African American.

We have committed $150 million in Minority Depository Institutions and funds supporting Black and African American-owned small businesses. These commitments drive sustained impact by directly enabling an increase of funds into local communities, improving diverse, small-business access to capital, and increasing skill development. We continue to partner with diverse-owned banking partners and asset managers to catalyze growth and industry participation. Additionally, we enriched our supplier pipeline, achieving our goal to spend $500 million with double the number of Black- and African American-owned suppliers. We have also provided 162 low- or no-interest loans to our small to medium-sized partners through our Partner Capital Fund.

We also continue to make progress toward our overall commitment to double the number of Black and African American and Hispanic and Latinx leaders in the U.S. by 2025.

Investing in Digital Skills

Microsoft's Skills for Jobs initiative aims to support a more skills-based labor market, with greater flexibility and accessible learning paths to develop the right skills needed for the most in-demand jobs. This initiative brings together classes, Career Essentials Certificates, and other resources from LinkedIn, GitHub, and Microsoft Learn, and is built on data insights drawn from LinkedIn's Economic Graph. Our goal was to train and certify 10 million learners by 2025. As of May 2024, we have surpassed that goal, training and certifying 12.6 million learners. We also launched a campaign in the United States in 2021 to help skill and recruit 250,000 people into the nation's cybersecurity workforce by 2025, representing half of the country's workforce shortage. To that end, we are making curriculum available free of charge to all of the nation's higher education institutions, providing training for new and existing faculty, and providing scholarships and supplemental resources to 25,000 students. The cyber skills initiative has expanded to 27 additional countries that show elevated cyberthreat risks coupled with significant gaps in their cybersecurity workforces, where we've partnered with nonprofits and other educational institutions to train the next generation of cybersecurity workers.

Generative AI is creating unparalleled opportunities to empower workers globally, but only if everyone has the skills to use it. In June 2023, we launched an AI Skills initiative to help everyone learn how to harness the power of AI. This includes a new LinkedIn learning pathway offering new coursework on learning the foundations of generative AI. We also launched a new global grant challenge to uncover new ways of training workers on generative AI and provide greater access to digital learning events and resources. Additionally, we extended our reach in rural communities, including through our TechSpark initiative in the United States. As of June 2024, we've helped more than 2.5 million people in 92% of the world's countries learn how to use AI.

HUMAN CAPITAL RESOURCES

Microsoft aims to recruit, develop, and retain world-changing talent from a diversity of backgrounds. To foster their and our success, we seek to create an environment where people can thrive and do their best work. We strive to maximize the potential of our human capital resources by creating a respectful, rewarding, and inclusive work environment that enables our global employees to create products and services that further our mission. Microsoft's culture is grounded in growth mindset. This means everyone is on a continuous journey to learn and grow, operating as one company instead of multiple siloed businesses. Our culture also embeds the security of customers and Microsoft as a priority for every employee and across all of our organizations.

As of June 30, 2024, we employed approximately 228,000 people on a full-time basis, 126,000 in the U.S. and 102,000 internationally. Of the total employed people, 86,000 were in operations, including product support and consulting services, datacenter operations, and manufacturing and distribution; 81,000 were in product research and development; 45,000 were in sales and marketing; and 16,000 were in general and administration. Certain employees are subject to collective bargaining agreements.

We design our programs to attract, reward, and retain top talent, enable our employees' continual growth, and reinforce our culture and values. Our total compensation opportunity is highly differentiated and market competitive. Our intended result is a global performance and development approach that fosters our culture, drives company performance, and competitive compensation that ensures equitable pay by role while supporting pay for performance.

Diversity and inclusion are core to our business. As reported in our Global Diversity and Inclusion Reports, we monitor pay equity and career progress across multiple dimensions. We encourage every person at Microsoft to play an active role in creating an inclusive environment.

We have invested significantly in employee wellbeing and offer a differentiated benefits package which includes many physical, emotional, and financial wellness programs. Our Occupational Health and Safety program helps to protect employees' safety while they are working. We also have introduced Hybrid Workplace Flexibility guidance to better support leaders, managers, and employees in hybrid work scenarios.

We believe providing employees with access to continual learning enables them to drive impact for the company. We provide individuals and teams with access to first and third-party content resources across professions, disciplines, and roles, and offer skilling opportunities to support employees' growth while driving organizations' needs.

Our employee listening systems enable us to gather feedback directly from our workforce to inform our programs and employee needs globally, giving us real-time insights into ways we can support our employees. As a company, we will continue to leverage data and research to inform decision making, balancing the needs of the business, team, and individual.

OPERATING SEGMENTS

We operate our business and report our financial performance using three segments: Productivity and Business Processes, Intelligent Cloud, and More Personal Computing. Our segments provide management with a comprehensive financial view of our key businesses. The segments enable the alignment of strategies and objectives across the development, sales, marketing, and services organizations, and they provide a framework for timely and rational allocation of resources within businesses.

Additional information on our operating segments and geographic and product information is contained in Note 19 – Segment Information and Geographic Data of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K).

Our reportable segments are described below.

Productivity and Business Processes

Our Productivity and Business Processes segment consists of products and services in our portfolio of productivity, communication, and information services, spanning a variety of devices and platforms. This segment primarily comprises:

- Office Commercial (Office 365 subscriptions, the Office 365 portion of Microsoft 365 Commercial subscriptions, and Office licensed on-premises), comprising Office, Exchange, SharePoint, Microsoft Teams, Office 365 Security and Compliance, Microsoft Viva, and Copilot for Microsoft 365.
- Office Consumer, including Microsoft 365 Consumer and Copilot Pro subscriptions, Office licensed on-premises, and other Office services.
- LinkedIn, including Talent Solutions, Marketing Solutions, Premium Subscriptions, and Sales Solutions.
- Dynamics business solutions, including Dynamics 365, comprising a set of intelligent, cloud-based applications across ERP, CRM, Power Apps, and Power Automate; and on-premises ERP and CRM applications.

Office Commercial

Office Commercial is designed to increase personal, team, and organizational productivity through a range of products and services. Growth depends on our ability to reach new users in new markets such as frontline workers, small and medium businesses, and growth markets, as well as add value to our core product and service offerings to span AI and productivity categories such as communication, collaboration, analytics, security, and compliance. Office Commercial revenue is mainly affected by a combination of continued installed base growth and average revenue per user expansion, as well as the continued shift from Office licensed on-premises to Office 365.

Office Consumer

Office Consumer is designed to increase personal productivity and creativity through a range of products and services. Growth depends on our ability to reach new users, add value to our core product set with new features including AI tools, and continue to expand our product and service offerings into new markets. Office Consumer revenue is mainly affected by the percentage of customers that buy Office with their new devices and the continued shift from Office licensed on-premises to Microsoft 365 Consumer subscriptions. Office Consumer Services revenue is mainly affected by the demand for communication and storage through Skype, Outlook.com, and OneDrive, which is largely driven by subscriptions, advertising, and the sale of minutes.

LinkedIn

LinkedIn connects the world’s professionals to make them more productive and successful and transforms the way companies hire, market, sell, and learn. Our vision is to create economic opportunity for every member of the global workforce through the ongoing development of the world’s first Economic Graph, a digital representation of the global economy. In addition to LinkedIn’s free services, LinkedIn offers monetized solutions designed to offer AI-enabled insights and productivity: Talent Solutions, Marketing Solutions, Premium Subscriptions, and Sales Solutions. Talent Solutions provide insights for workforce planning and tools to hire, nurture, and develop talent. Talent Solutions also includes Learning Solutions, which help businesses close critical skills gaps in times where companies are having to do more with existing talent. Marketing Solutions help companies reach, engage, and convert their audiences at scale. Premium Subscriptions enable professionals to manage their professional identity, grow their network, find jobs, access knowledge, and connect with talent through additional services like premium search. Sales Solutions help companies strengthen customer relationships, empower teams with digital selling tools, and acquire new opportunities. Growth will depend on our ability to increase the number of LinkedIn members and our ability to continue offering insight and AI-enabled services that provide value for our members and increase their engagement. LinkedIn revenue is mainly affected by demand from enterprises and professionals for subscriptions to Talent Solutions, Sales Solutions, and Premium Subscriptions offerings, as well as member engagement and the quality of the sponsored content delivered to those members to drive Marketing Solutions.

Dynamics

Dynamics provides cloud-based and on-premises business solutions for financial management, enterprise resource planning (“ERP”), customer relationship management (“CRM”), and supply chain management, as well as other low code application development platforms and AI offerings, for small and medium businesses, large organizations, and divisions of global enterprises. Dynamics revenue is driven by the number of users licensed and applications consumed, expansion of average revenue per user, and the continued shift to Dynamics 365, a unified set of cloud-based intelligent business applications, including our low code development platforms, such as Power Apps and Power Automate.

Competition

Competitors to Office include software and global application vendors, such as Apple, Cisco Systems, Google, Meta, Proofpoint, Slack, Symantec, Zoom, and numerous web-based and mobile application competitors as well as local application developers. Apple distributes versions of its pre-installed application software, such as email and calendar products, through its PCs, tablets, and phones. Cisco Systems is using its position in enterprise communications equipment to grow its unified communications business. Google provides a hosted messaging and productivity suite. Meta offers communication tools to enable productivity and engagement within organizations. Proofpoint and Symantec provide security solutions across email security, information protection, and governance. Slack provides teamwork and collaboration software. Zoom offers videoconferencing and cloud phone solutions. Web-based offerings competing with individual applications have also positioned themselves as alternatives to our products and services. We compete by providing powerful, flexible, secure, integrated industry-specific, and easy-to-use productivity and collaboration tools and services that create comprehensive solutions and work well with technologies our customers already have both on-premises or in the cloud.

LinkedIn faces competition from online professional networks, recruiting companies, talent management companies, and larger companies that are focusing on talent management and human resource services; job boards; traditional recruiting firms; and companies that provide learning and development products and services. Marketing Solutions competes with online and offline outlets that generate revenue from advertisers and marketers, and Sales Solutions competes with online and offline outlets for companies with lead generation and customer intelligence and insights.

Dynamics competes with cloud-based and on-premises business solution providers such as Oracle, Salesforce, SAP, Service Now, UI Path, and WorkDay.

Intelligent Cloud

Our Intelligent Cloud segment consists of our public, private, and hybrid server products and cloud services that can power modern business and developers. This segment primarily comprises:

- Server products and cloud services, including Azure and other cloud services; SQL Server, Windows Server, Visual Studio, System Center, and related Client Access Licenses ("CALs"); and Nuance and GitHub.
- Enterprise and partner services, including Enterprise Support Services, Industry Solutions, Nuance professional services, Microsoft Partner Network, and Learning Experience.

Server Products and Cloud Services

Azure is a comprehensive set of cloud services that offer developers, IT professionals, and enterprises freedom to build, deploy, and manage applications on any platform or device. Customers can use Azure through our global network of datacenters for computing, networking, storage, mobile and web application services, AI, Internet of Things ("IoT"), cognitive services, and machine learning. Azure enables customers to devote more resources to development and use of applications that benefit their organizations, rather than managing on-premises hardware and software. Azure revenue is mainly affected by infrastructure-as-a-service and platform-as-a-service consumption-based services, and per user-based services such as Enterprise Mobility + Security.

Azure AI offerings provide a competitive advantage as companies seek ways to optimize and scale their business with machine learning. With Azure's purpose-built, AI-optimized infrastructure, customers can use a variety of large language models and developer tools to create the next generation of AI apps and services.

Our server products are designed to make IT professionals, developers, and their systems more productive and efficient. Server software is integrated server infrastructure and middleware designed to support software applications built on the Windows Server operating system. This includes the server platform, database, business intelligence, storage, management and operations, virtualization, service-oriented architecture platform, security, and identity software. We also license standalone and software development lifecycle tools for software architects, developers, testers, and project managers. Server products revenue is mainly affected by purchases through volume licensing programs, licenses sold to original equipment manufacturers ("OEM"), and retail packaged products. CALs provide access rights to certain server products, including SQL Server and Windows Server, and revenue is reported along with the associated server product.

Nuance and GitHub include both cloud and on-premises offerings. Nuance provides healthcare and enterprise AI solutions. GitHub provides a collaboration platform and code hosting service for developers.

Enterprise and Partner Services

Enterprise and Partner Services, including Enterprise Support Services, Industry Solutions, Nuance professional services, Microsoft Partner Network, and Learning Experience, assist customers in developing, deploying, and managing Microsoft server solutions, Microsoft desktop solutions, and Nuance conversational AI and ambient intelligent solutions, along with providing training and certification to developers and IT professionals on various Microsoft products.

Competition

Azure faces diverse competition from companies such as Amazon, Broadcom, Google, IBM, Oracle, and open source offerings. Azure's competitive advantage includes enabling a hybrid cloud, allowing deployment of existing datacenters with our public cloud into a single, cohesive infrastructure, and the ability to run at a scale that meets the needs of businesses of all sizes and complexities. Our AI offerings compete with AI products from hyperscalers such as Amazon and Google, as well as products from other emerging competitors, including Anthropic, OpenAI, Meta, and other open source offerings, many of which are also current or potential partners. Our Azure Security offerings include our cloud security solution and security information and event management solution, which compete with companies such as Palo Alto Networks and Cisco. Our Enterprise Mobility + Security offerings also compete with products from a range of competitors including identity vendors, security solution vendors, and numerous other security point solution vendors. We believe our cloud's global scale, coupled with our broad portfolio of identity and security solutions, allows us to effectively solve complex cybersecurity challenges for our customers and differentiates us from the competition.

Our server products face competition from a wide variety of server operating systems and applications offered by companies with a range of market approaches. Vertically integrated computer manufacturers such as Hewlett-Packard, IBM, and Oracle offer their own versions of the Unix operating system preinstalled on server hardware and nearly all computer manufacturers offer server hardware for the Linux operating system.

We compete to provide enterprise-wide computing and point solutions with numerous commercial software vendors that offer solutions and middleware technology platforms, software applications for connectivity, security, hosting, database, and e-business servers. IBM and Oracle lead a group of companies that compete with our enterprise-wide computing solutions. Commercial competitors for our server applications for PC-based distributed client-server environments include Broadcom, IBM, and Oracle. Our web application platform software competes with open source software such as Apache, Linux, MySQL, and PHP. In middleware, we compete against Java vendors.

Our database, business intelligence, and data warehousing solutions offerings compete with products from Databricks, IBM, Oracle, SAP, Snowflake, and other companies. Our system management solutions compete with server management and server virtualization platform providers, such as BMC, Broadcom, Hewlett-Packard, and IBM. Our products for software developers compete against offerings from Adobe, IBM, Oracle, and other companies, and also against open source projects, including Eclipse (sponsored by IBM, Oracle, and SAP), PHP, and Ruby on Rails.

We believe our server products provide customers with advantages in performance, total costs of ownership, and productivity by delivering superior applications, development tools, compatibility with a broad base of hardware and software applications, security, and manageability.

Our Enterprise and Partner Services business competes with a wide range of companies that provide strategy and business planning, application development, and infrastructure services, including multinational consulting firms and small niche businesses focused on specific technologies.

More Personal Computing

Our More Personal Computing segment consists of products and services that put customers at the center of the experience with our technology. This segment primarily comprises:

- Windows, including Windows OEM licensing and other non-volume licensing of the Windows operating system; Windows Commercial, comprising volume licensing of the Windows operating system, Windows cloud services, and other Windows commercial offerings; patent licensing; and Windows Internet of Things.
- Devices, including Surface, HoloLens, and PC accessories.
- Gaming, including Xbox hardware and Xbox content and services, comprising first-party content (such as Activision Blizzard) and third-party content, including games and in-game content; Xbox Game Pass and other subscriptions; Xbox Cloud Gaming; advertising; third-party disc royalties; and other cloud services.
- Search and news advertising, comprising Bing (including Copilot), Microsoft News, Microsoft Edge, and third-party affiliates.

Windows

The Windows operating system is designed to deliver a more personal computing experience for users by enabling consistency of experience, applications, and information across their devices. Windows OEM revenue is impacted significantly by the number of Windows operating system licenses purchased by OEMs, which they pre-install on the devices they sell. In addition to computing device market volume, Windows OEM revenue is impacted by:

- The mix of computing devices based on form factor and screen size.
- Differences in device market demand between developed markets and growth markets.
- Growth of the AI PC category
- Attachment of Windows to devices shipped.
- Customer mix between consumer, small and medium businesses, and large enterprises.
- Changes in inventory levels in the OEM channel.
- Pricing changes and promotions, pricing variation that occurs when the mix of devices manufactured shifts from local and regional system builders to large multinational OEMs, and different pricing of Windows versions licensed.
- Constraints in the supply chain of device components.
- Piracy.

Windows Commercial revenue, which includes volume licensing of the Windows operating system and Windows cloud services such as Microsoft Defender for Endpoint, is affected mainly by the demand from commercial customers for Microsoft 365 and our advanced security offerings. Windows Commercial revenue often reflects the number of information workers in a licensed enterprise and is relatively independent of the number of PCs sold in a given year.

Patent licensing includes our programs to license patents we own for use across a broad array of technology areas, including mobile devices and cloud offerings.

Windows IoT extends the power of Windows and the cloud to intelligent systems by delivering specialized operating systems, tools, and services for use in embedded devices.

Devices

We design and sell devices, such as Surface (including Copilot+ PCs), HoloLens, and PC accessories. Our devices are designed to enable people and organizations to connect to the people and content that matter most using Windows and integrated Microsoft products and services. Surface is designed to help organizations, students, and consumers be more productive. Growth in Devices is dependent on total PC shipments, the ability to attract new customers, our product roadmap, and expanding into new categories.

Gaming

Our gaming platform is designed to provide a variety of entertainment through a unique combination of content, community, and cloud services. Our game content is developed through a collection of first-party studios creating iconic and differentiated gaming experiences. We continue to invest in new gaming studios and content to expand our intellectual property roadmap and leverage new content creators. These unique gaming experiences are the cornerstone of Xbox Game Pass, a subscription service and gaming community with access to a curated library of over 400 first- and third-party console and PC titles.

The gamer remains at the heart of the Xbox ecosystem. We are identifying new opportunities to attract gamers across a variety of different end points through our first- and third-party content and business diversification across subscriptions, ads, and digital stores. We've seen new devices from third-party manufacturers along with key PC and mobile end points that help us empower gamers to play in a way that is most convenient to them. We are focused on growing the platform and expanding to new ecosystems to engage as many gamers as possible.

Xbox enables people to connect and share online gaming experiences that are accessible on Xbox consoles, Windows-enabled devices, and other devices. Xbox is designed to benefit users by providing access to a network of certified applications and services and to benefit our developer and partner ecosystems by providing access to a large customer base. Xbox revenue is mainly affected by subscriptions and sales of first- and third-party content, as well as advertising. Growth of our Gaming business is determined by the overall active user base through Xbox enabled content, availability of games, providing exclusive game content that gamers seek, the computational power and reliability of the devices used to access our content and services, and the ability to create new experiences.

Search and News Advertising

Our Search and news advertising business is designed to deliver relevant search, native, and display advertising to a global audience. Our Microsoft Edge browser and Bing search engine with Copilot are key tools to enable user acquisition and engagement, while our technology platform enables accelerated delivery of digital advertising solutions. In addition to first-party tools, we have several partnerships with companies, such as Yahoo, through which we provide and monetize search offerings. Growth depends on our ability to attract new users, understand intent, and match intent with relevant content on advertising offerings.

Competition

Windows faces competition from various software products and from alternative platforms and devices, mainly from Apple and Google, and Microsoft Defender for Endpoint competes with CrowdStrike on endpoint security solutions. We believe Windows competes effectively by giving customers choice, value, flexibility, security, an easy-to-use interface, and compatibility with a broad range of hardware and software applications, including those that enable productivity.

Devices face competition from various computer, tablet, and hardware manufacturers who offer a unique combination of high-quality industrial design and innovative technologies across various price points. These manufacturers, many of which are also current or potential partners and customers, include Apple and our Windows OEMs.

Xbox and our cloud gaming services face competition from various online gaming ecosystems and game streaming services, including those operated by Amazon, Apple, Meta, and Tencent. We also compete with other providers of entertainment services such as video streaming platforms. Our gaming platform competes with console platforms from Nintendo and Sony, both of which have a large, established base of customers. We believe our gaming platform is effectively positioned against, and uniquely differentiated from, competitive products and services based on significant innovation in hardware architecture, user interface, developer tools, online gaming and entertainment services, and continued strong content from our own first-party game franchises as well as other digital content offerings.

Our Search and news advertising business competes with Google, OpenAI, and a wide array of websites, social platforms like Meta, and portals that provide content and online offerings to end users.

OPERATIONS

We have regional operations service centers that support our operations, including customer contract and order processing, billing, credit and collections, customer lifecycle operations, information processing, and vendor management and logistics. The centers in Ireland and Romania support the African, European, and Middle East regions; the centers in India and Ireland support the Asia-Pacific region; and the centers in Arlington, Virginia, Atlanta, Georgia, Charlotte, North Carolina, Fargo, North Dakota, Fort Lauderdale, Florida, Redmond, Washington, Reno, Nevada, and San Jose, Costa Rica support the Americas regions.

In addition to our operations centers, we also operate datacenters throughout each of these regions. We continue to identify and evaluate opportunities to expand our datacenter locations and increase our server capacity to meet the evolving needs of our customers, particularly given the growing demand for AI services. Our datacenters depend on the availability of permitted and buildable land, predictable energy, networking supplies, and servers, including graphics processing units ("GPUs") and other components.

Our devices are primarily manufactured by third-party contract manufacturers. For the majority of our products, we have the ability to use other manufacturers if a current vendor becomes unavailable or unable to meet our requirements. However, some of our products contain certain components for which there are very few qualified suppliers. Extended disruptions at these suppliers could impact our ability to manufacture devices on time to meet consumer demand.

RESEARCH AND DEVELOPMENT

Product and Service Development, and Intellectual Property

We develop most of our products and services internally through the following engineering groups.

- *Cloud and AI* – focuses on making IT professionals, developers, partners, independent software vendors, and their systems more productive and efficient through development of Azure AI platform and cloud infrastructure, server, database, CRM, ERP, software development tools and services, AI cognitive services, and other business process applications and services for enterprises.
- *Strategic Missions and Technologies* – focuses on incubating technical products and support solutions with transformative potential for the future of cloud computing and continued company growth, such as quantum computing and advanced AI for science.
- *Experiences and Devices* – focuses on delivering high value end-user experiences across our products, services, and devices, including Microsoft 365, Windows, Microsoft Teams, and the Surface line of devices.
- *Microsoft AI* – focuses on delivering online experiences targeted at consumers (including Bing, Copilot, Start/MSN, and other advertising-based services) and developing advanced AI models.
- *Microsoft Security* – focuses on delivering a comprehensive portfolio of services that protect our customers' digital infrastructure through cloud platform and application security, data protection and governance, identity and network access, and device management.
- *Technology and Research* – focuses on fundamental research, product and business incubations, and forward-looking AI innovations that span infrastructure, services, and applications. This engineering group includes Microsoft Research, one of the world's largest corporate research organizations, which focuses on fundamental research in AI, computer science, and a broad range of other disciplines.
- *LinkedIn* – focuses on our services that transform the way professionals grow their network and find jobs and the way businesses hire, market, sell, and learn.
- *Gaming* – focuses on developing hardware, content, and services across a large range of platforms to help grow our user base through game experiences and social interaction.

Internal development allows us to maintain competitive advantages that come from product differentiation and closer technical control over our products and services. It also gives us the freedom to decide which modifications and enhancements are most important and when they should be implemented. We strive to obtain information as early as possible about changing usage patterns and hardware advances that may affect software and hardware design. Before releasing new software platforms, and as we make significant modifications to existing platforms, we provide application vendors with a range of resources and guidelines for development, training, and testing. Generally, we also create product documentation internally.

We protect our intellectual property investments in a variety of ways. We work actively in the U.S. and internationally to ensure the enforcement of copyright, trademark, trade secret, and other protections that apply to our software and hardware products, services, business plans, and branding. We are a leader among technology companies in pursuing patents and currently have a portfolio of over 63,000 U.S. and international patents issued and over 23,000 pending worldwide. While we employ much of our internally-developed intellectual property in our products and services, we also engage in outbound licensing of specific patented technologies that are incorporated into licensees' products. From time to time, we enter into broader cross-license agreements with other technology companies covering entire groups of patents. We may also purchase or license technology that we incorporate into our products and services. At times, we make select intellectual property broadly available at no or low cost to achieve a strategic objective, such as promoting industry standards, advancing interoperability, supporting societal and/or environmental efforts, or attracting and enabling our external development community. Our engagement with open source software also causes us to license our intellectual property rights broadly in certain situations.

While it may be necessary in the future to seek or renew licenses relating to various aspects of our products and services, we believe, based upon past experience and industry practice, such licenses generally can be obtained on commercially reasonable terms. We believe our continuing research and product development are not materially dependent on any single license or other agreement with a third party relating to the development of our products.

Investing in the Future

Our success is based on our ability to create new and compelling products, services, and experiences for our users, to initiate and embrace disruptive technology trends, to enter new geographic and product markets, and to drive broad adoption of our products and services. We invest in a range of emerging technology trends and breakthroughs that we believe offer significant opportunities to deliver value to our customers and growth for the company. Based on our assessment of key technology trends, we maintain our long-term commitment to research and development across a wide spectrum of technologies, tools, and platforms spanning digital work and life experiences, cloud computing, AI, devices, and operating systems.

While our main product research and development facilities are located in Redmond, Washington, we also operate research and development facilities in other parts of the U.S. and around the world. This global approach helps us remain competitive in local markets and enables us to continue to attract top talent from across the world.

We plan to continue to make significant investments in a broad range of product research and development activities, and as appropriate we will coordinate our research and development across operating segments and leverage the results across the company. This includes continuing to support fundamental research, which provides us with a unique perspective on future trends and contributes to our innovation.

DISTRIBUTION, SALES, AND MARKETING

We market and distribute our products and services through the following channels: OEMs, direct, and distributors and resellers. Our sales organization performs a variety of functions, including working directly with commercial enterprises and public-sector organizations worldwide to identify and meet their technology and digital transformation requirements; managing OEM relationships; and supporting system integrators, independent software vendors, and other partners who engage directly with our customers to perform sales, consulting, and fulfillment functions for our products and services.

OEMs

We distribute our products and services through OEMs that pre-install our software on new devices and servers they sell. The largest component of the OEM business is the Windows operating system pre-installed on devices. OEMs also sell devices pre-installed with other Microsoft products and services, including applications such as Office and the capability to subscribe to Microsoft 365 Consumer.

There are two broad categories of OEMs. The largest category of OEMs are direct OEMs as our relationship with them is managed through a direct agreement between Microsoft and the OEM. We have distribution agreements covering one or more of our products with virtually all the multinational OEMs, including Dell, Hewlett-Packard, Lenovo, and with many regional and local OEMs. The second broad category of OEMs are system builders consisting of lower-volume PC manufacturers, which source Microsoft software for pre-installation and local redistribution primarily through the Microsoft distributor channel rather than through a direct agreement or relationship with Microsoft.

Direct

Many organizations that license our products and services transact directly with us through Enterprise Agreements and Enterprise Services contracts, with sales support from system integrators, independent software vendors, web agencies, and partners that advise organizations on licensing our products and services ("Enterprise Agreement Software Advisors" or "ESA"). Microsoft offers direct sales programs targeted to reach small, medium, and corporate customers, in addition to those offered through the reseller channel. A large network of partner advisors support many of these sales.

We also sell commercial and consumer products and services directly to customers, such as cloud services, search, and gaming, through our digital marketplaces and online stores. Additionally, our Microsoft Experience Centers are designed to facilitate deeper engagement with our partners and customers across industries.

Distributors and Resellers

Organizations also license our products and services indirectly, primarily through licensing solution partners ("LSP"), distributors, value-added resellers ("VAR"), and retailers. Although each type of reselling partner may reach organizations of all sizes, LSPs are primarily engaged with large organizations, distributors resell primarily to VARs, and VARs typically reach small and medium organizations. ESAs are also typically authorized as LSPs and operate as resellers for our other volume licensing programs. Microsoft Cloud Solution Provider is our main partner program for reselling cloud services.

We distribute our retail packaged products primarily through independent non-exclusive distributors, authorized replicators, resellers, and retail outlets. Individual consumers obtain these products primarily through retail outlets. We distribute our devices through third-party retailers. We have a network of field sales representatives and field support personnel that solicit orders from distributors and resellers and provide product training and sales support.

Our Dynamics business solutions are also licensed to enterprises through a global network of channel partners providing vertical solutions and specialized services.

LICENSING OPTIONS

We offer options for organizations of varying sizes that want to purchase our cloud services and on-premise software. We license these organizations under volume licensing agreements to allow the customer to acquire multiple licenses of products and services instead of having to acquire separate licenses through retail channels. These volume licensing programs have varying programmatic requirements and benefits to best meet the needs of our customers.

Software Assurance ("SA") conveys rights to new software and upgrades for perpetual licenses released over the contract period. It also provides support, tools, training, and other licensing benefits to help customers deploy and use software efficiently. SA is required to be purchased with certain volume licensing agreements and is an optional purchase with others.

Volume Licensing Programs

Enterprise Agreement

Enterprise Agreements offer large organizations a manageable volume licensing program that gives them the flexibility to buy cloud services and software licenses under one agreement. Enterprise Agreements are designed for medium or large organizations that want to license Microsoft products and services organization-wide over a three-year period. Organizations can elect to purchase perpetual licenses (covered with SA) and/or subscribe to cloud services.

Microsoft Customer Agreement

Microsoft Customer Agreements are simplified purchase agreements presented, accepted, and stored through a digital experience. Microsoft Customer Agreements are non-expiring agreements that are designed to support all customers over time, whether purchasing through a partner or directly from Microsoft.

Microsoft Online Subscription Agreement

Microsoft Online Subscription Agreements are designed for small and medium organizations that want to subscribe to, activate, provision, and maintain cloud services seamlessly and directly via the web. These agreements allow customers to acquire monthly or annual subscriptions for cloud-based services.

Microsoft Products and Services Agreement

Microsoft Products and Services Agreements are designed for medium and large organizations that want to license cloud services and on-premises software as needed, with no organization-wide commitment, under a single, non-expiring agreement. Organizations purchase perpetual licenses or subscribe to licenses. SA is optional for customers that purchase perpetual licenses.

Open Value

Open Value agreements are a simple, cost-effective way to acquire the latest Microsoft technology. These agreements are designed for small and medium organizations that want to license cloud services and on-premises software over a three-year period. Under Open Value agreements, organizations can elect to purchase perpetual licenses or subscribe to licenses and SA is included.

Select Plus

A Select Plus agreement is designed for government and academic organizations to acquire on-premises licenses at any affiliate or department level, while realizing advantages as one organization. Organizations purchase perpetual licenses and SA is optional.

Partner Programs

The Microsoft Cloud Solution Provider Program offers customers an easy way to license the cloud services they need in combination with the value-added services offered by their systems integrator, managed services provider, or cloud reseller partner. Partners in this program can easily package their own products and services to directly provision, manage, and support their customer subscriptions.

The Microsoft Services Provider License Agreement allows hosting service providers and independent software vendors who want to license eligible Microsoft software products to provide hosted applications and software services to their end customers. Partners license software over a three-year period and are billed monthly based on units licensed.

The Independent Software Vendor Royalty Program enables partners to integrate Microsoft products into other applications and then license the unified business solution to their end users.

CUSTOMERS

Our customers include individual consumers, small and medium organizations, large global enterprises, public-sector institutions, Internet service providers, application developers, and OEMs. Our practice is to ship our products promptly upon receipt of purchase orders from customers; consequently, backlog is not significant.

GOVERNMENT REGULATION

We are subject to a wide range of laws, regulations, and legal requirements in the U.S. and globally, including those that may apply to our products and online services offerings, and those that impose requirements related to user privacy, telecommunications, data storage and protection, advertising, and online content. How these laws and regulations apply to our business is often unclear, subject to change over time, and sometimes may be inconsistent from jurisdiction to jurisdiction. To comply with the accelerating global regulatory obligations, we established a regulatory governance framework and to create a repeatable system-focused approach to regulatory governance with an initial focus on four domains: Responsible AI, Privacy, Digital Safety, and Cybersecurity. The framework is designed to help us maintain customer trust and confidence in our products, remain in compliance with regulators around the globe, and effectively scale our capability to address the growing number of complex regulations. Through the framework, our legal and regulatory subject matter experts ingest regulations, develop standards and implementation guidance, and, when appropriate, work with our engineers to develop and implement products to monitor compliance. Our business teams, with legal support, manage the compliance programs and prepare external regulatory and commercial reporting, and our internal audit teams conduct reviews of our programs and processes. While we intended to create a unified approach to regulatory compliance, some of the programs and processes established pursuant to the framework are tailored to meet specific regulatory obligations, such as with the creation of independent compliance functions required by the European Union (“EU”) Digital Markets Act and the EU Digital Services Act, which oversee, monitor, and assess the company’s compliance with these acts.

For a description of the risks we face related to regulatory matters, refer to Risk Factors (Part I, Item 1A of this Form 10-K).

INFORMATION ABOUT OUR EXECUTIVE OFFICERS

Our executive officers as of July 30, 2024 were as follows:

Name	Age	Position with the Company
Satya Nadella	56	Chairman and Chief Executive Officer
Judson B. Althoff	51	Executive Vice President and Chief Commercial Officer
Kathleen T. Hogan	58	Executive Vice President and Chief Human Resources Officer
Amy E. Hood	52	Executive Vice President and Chief Financial Officer
Takeshi Numoto	53	Executive Vice President and Chief Marketing Officer
Bradford L. Smith	65	Vice Chair and President
Christopher D. Young	52	Executive Vice President, Business Development, Strategy, and Ventures

Mr. Nadella was appointed Chairman of the Board in June 2021 and Chief Executive Officer in February 2014. He served as Executive Vice President, Cloud and Enterprise from July 2013 until that time. From 2011 to 2013, Mr. Nadella served as President, Server and Tools. From 2009 to 2011, he was Senior Vice President, Online Services Division. From 2008 to 2009, he was Senior Vice President, Search, Portal, and Advertising. Since joining Microsoft in 1992, Mr. Nadella’s roles also included Vice President of the Business Division.

Mr. Althoff was appointed Executive Vice President and Chief Commercial Officer in July 2021. He served as Executive Vice President, Worldwide Commercial Business from July 2017 until that time. Prior to that, Mr. Althoff served as the President of Microsoft North America. Mr. Althoff joined Microsoft in March 2013 as President of Microsoft North America. Mr. Althoff also serves on the Board of Directors of Ecolab Inc.

Ms. Hogan was appointed Executive Vice President and Chief Human Resources Officer in June 2023. Ms. Hogan had been Executive Vice President, Human Resources since November 2014. Prior to that Ms. Hogan was Corporate Vice President of Microsoft Services. She also served as Corporate Vice President of Customer Service and Support. Ms. Hogan joined Microsoft in 2003. Ms. Hogan also serves on the Board of Directors of Alaska Air Group, Inc.

Ms. Hood was appointed Executive Vice President and Chief Financial Officer in July 2013, subsequent to her appointment as Chief Financial Officer in May 2013. From 2010 to 2013, Ms. Hood was Chief Financial Officer of the Microsoft Business Division. Since joining Microsoft in 2002, Ms. Hood has also held finance-related positions in the Server and Tools Business and the corporate finance organization. Ms. Hood also serves on the Board of Directors of 3M Corporation.

Mr. Numoto was appointed Executive Vice President and Chief Marketing Officer in October 2023. He served as Executive Vice President and Commercial Chief Marketing Officer from March 2020. Mr. Numoto served as a Corporate Vice President, Cloud Marketing from January 2012. Prior to that, Mr. Numoto served as a Corporate Vice President for Office 365 Marketing from 2004, where he led the transformation from traditional on-premises packaged software to the introduction of Office 365. Since joining Microsoft in 1997, Mr. Numoto has held multiple roles in Windows Program Management and Office Marketing.

Mr. Smith was appointed Vice Chair and President in September 2021. Prior to that, he served as President and Chief Legal Officer since September 2015. He served as Executive Vice President, General Counsel, and Secretary from 2011 to 2015, and served as Senior Vice President, General Counsel, and Secretary from 2001 to 2011. Mr. Smith was also named Chief Compliance Officer in 2002. Since joining Microsoft in 1993, he was Deputy General Counsel for Worldwide Sales and previously was responsible for managing the European Law and Corporate Affairs Group, based in Paris. Mr. Smith also serves on the Board of Directors of Netflix, Inc.

Mr. Young has served as Executive Vice President, Business Development, Strategy, and Ventures since joining Microsoft in November 2020. Prior to Microsoft, he served as the Chief Executive Officer of McAfee, LLC from 2017 to 2020, and served as a Senior Vice President and General Manager of Intel Security Group from 2014 until 2017, when he led the initiative to spin out McAfee into a standalone company. Mr. Young also serves on the Board of Directors of American Express Company.

AVAILABLE INFORMATION

Our Internet address is www.microsoft.com. At our Investor Relations website, www.microsoft.com/investor, we make available free of charge a variety of information for investors. Our goal is to maintain the Investor Relations website as a portal through which investors can easily find or navigate to pertinent information about us, including:

- Our annual report on Form 10-K, quarterly reports on Form 10-Q, current reports on Form 8-K, and any amendments to those reports, as soon as reasonably practicable after we electronically file that material with or furnish it to the Securities and Exchange Commission ("SEC") at www.sec.gov.
- Information on our business strategies, financial results, and metrics for investors.
- Announcements of investor conferences, speeches, and events at which our executives talk about our product, service, and competitive strategies. Archives of these events are also available.
- Press releases on quarterly earnings, product and service announcements, legal developments, and international news.
- Corporate governance information including our articles of incorporation, bylaws, governance guidelines, committee charters, codes of conduct and ethics, global corporate social responsibility initiatives, and other governance-related policies.
- Other news and announcements that we may post from time to time that investors might find useful or interesting.
- Opportunities to sign up for email alerts to have information pushed in real time.

We publish a variety of reports and resources related to our Corporate Social Responsibility programs and progress on our Reports Hub website, www.microsoft.com/corporate-responsibility/reports-hub, including reports on sustainability, responsible sourcing, accessibility, digital trust, and public policy engagement.

The information found on these websites is not part of, or incorporated by reference into, this or any other report we file with, or furnish to, the SEC. In addition to these channels, we use social media to communicate to the public. It is possible that the information we post on social media could be deemed to be material to investors. We encourage investors, the media, and others interested in Microsoft to review the information we post on the social media channels listed on our Investor Relations website.

ITEM 1A. RISK FACTORS

Our operations and financial results are subject to various risks and uncertainties, including those described below, that could adversely affect our business, operations, financial condition, results of operations, liquidity, and the trading price of our common stock.

STRATEGIC AND COMPETITIVE RISKS

We face intense competition across all markets for our products and services, which may adversely affect our results of operations.

Competition in the technology sector

Our competitors range in size from diversified global companies with significant research and development resources to small, specialized firms whose narrower product lines may let them be more effective in deploying technical, marketing, and financial resources. Barriers to entry in many of our businesses are low and many of the areas in which we compete evolve rapidly with changing and disruptive technologies, shifting user needs, and frequent introductions of new products and services. If we do not continue to innovate and provide products, devices, and services that appeal to businesses and consumers, we may not remain competitive, which may adversely affect our business, financial condition, and results of operations.

Competition among platform-based ecosystems

An important element of our business model has been to create platform-based ecosystems on which many participants can build diverse solutions. A well-established ecosystem creates beneficial network effects among users, application developers, and the platform provider that can accelerate growth. Establishing significant scale in the marketplace is necessary to achieve and maintain attractive margins. We face significant competition from firms that provide competing platforms.

- A competing vertically-integrated model, in which a single firm controls the software and hardware elements of a product and related services, has succeeded with some consumer products such as PCs, tablets, smartphones, gaming consoles, wearables, and other endpoint devices. Competitors pursuing this model also earn revenue from services integrated with the hardware and software platform, including applications and content sold through their integrated marketplaces. They may also be able to claim security and performance benefits from their vertically integrated offer. We also offer some vertically-integrated hardware and software products and services. Shifting a portion of our business to a vertically integrated model may increase our cost of revenue and reduce our operating margins.
- We derive substantial revenue from licenses of Windows operating systems on PCs. We face significant competition from competing platforms developed for new devices and form factors such as smartphones and tablets. These devices compete on multiple bases including price and the perceived utility of the device and its platform. Users continue to turn to these devices to perform functions that in the past were performed by PCs. Even if many users view these devices as complementary to a PC, the prevalence of these devices may make it more difficult to attract application developers to our PC operating system platforms. Competing with operating systems licensed at low or no cost may decrease our PC operating system margins. Popular products or services offered on competing platforms could increase their competitive strength. In addition, some of our devices compete with products made by our original equipment manufacturer (“OEM”) partners, which may affect their commitment to our platform.
- Competing platforms have content and application marketplaces with scale and significant installed bases. The variety and utility of content and applications available on a platform are important to device purchasing decisions. Users may incur costs to move data and buy new content and applications when switching platforms. To compete, we must successfully enlist developers to write applications for our platform and ensure that these applications have high quality, security, customer appeal, and value. Efforts to compete with competitors’ content and application marketplaces may increase our cost of revenue and lower our operating margins. Competitors’ rules governing their content and applications marketplaces may restrict our ability to distribute products and services through them in accordance with our technical and business model objectives.

For all of these reasons, we may not be able to compete successfully against our current and future competitors, which may adversely affect our business, operations, financial condition, and results of operations.

Business model competition

Companies compete with us based on a growing variety of business models.

- A material part of our business involves cloud-based services available across the spectrum of computing devices. Our competitors continue to develop and deploy cloud-based services for consumers and business customers, and pricing and delivery models are evolving. We and our competitors are devoting significant resources to develop and deploy our cloud-based strategies.
- We are investing in artificial intelligence ("AI") across the entire company and infusing generative AI capabilities into our consumer and commercial offerings. We expect AI technology and services to be a highly competitive and rapidly evolving market, and new competitors continue to enter the market. We will bear significant development and operational costs to build and support the AI models, services, platforms, and infrastructure necessary to meet the needs of our customers. To compete effectively we must also be responsive to technological change, new and potential regulatory developments, and public scrutiny.
- Even as we transition more of our business to infrastructure-, platform-, and software-as-a-service business model, the license-based proprietary software model generates a substantial portion of our software revenue. We bear the costs of converting original ideas into software products through investments in research and development, offsetting these costs with the revenue received from licensing our products. Many of our competitors also develop and sell software to businesses and consumers under this model.
- Other competitors develop and offer free applications, online services, and content, and make money by selling third-party advertising. Advertising revenue funds development of products and services these competitors provide to users at little or no cost, competing directly with our revenue-generating products.
- Some companies compete with us by modifying and then distributing open source software at little or no cost to end users, using open source AI models, and earning revenue on advertising or integrated products and services. These firms do not bear the full costs of research and development for the open source products. Some open source products mimic the features and functionality of our products.

The competitive pressures described above may cause decreased sales volumes, price reductions, and/or increased operating costs, such as for research and development, marketing, and sales incentives, which may adversely affect our financial condition and results of operations.

Our focus on cloud-based and AI services presents execution and competitive risks. We are incurring significant costs to build and maintain infrastructure to support cloud computing and AI services. These costs will reduce the operating margins. Whether we succeed in cloud-based and AI services depends on our execution in several areas, including:

- Continuing to bring to market compelling cloud-based and AI experiences and products that generate increasing traffic and market share.
- Maintaining the utility, compatibility, and performance of our cloud-based and AI services on the growing array of computing devices, including PCs, smartphones, tablets, gaming consoles, and other devices.
- Continuing to enhance the attractiveness of our cloud platforms to third-party developers.
- Ensuring our cloud-based services meet the reliability expectations and specific requirements of our customers and maintain the security of their data as well as help them meet their own compliance needs.
- Making our suite of cloud-based services platform-agnostic, available on a wide range of devices and ecosystems, including those of our competitors.

It is uncertain whether our strategies will continue to attract users or generate the revenue required to succeed. If we are not effective in executing organizational and technical changes to increase efficiency and accelerate innovation, or if we fail to generate sufficient usage of our new products and services, we may not grow revenue in line with the infrastructure and development investments described above. This may adversely affect our operations, financial condition, and results of operations.

Our AI systems offer users powerful tools and capabilities. However, there may be instances where these systems are used in ways that are unintended or inappropriate. In addition, some users may also engage in fraudulent or abusive activities through our cloud-based services, such as unauthorized account access, payment fraud, or terms of service violations including cryptocurrency mining or launching cyberattacks. While are committed to detecting and controlling such misuse of our cloud-based and AI services, our efforts may not be effective, and we may incur reputational damage or experience adverse impacts to our business and results of operations.

RISKS RELATING TO THE EVOLUTION OF OUR BUSINESS

We make significant investments in products and services that may not achieve expected returns. We will continue to make significant investments in research, development, and marketing for existing products, services, and technologies. In addition, we are focused on developing new AI platform services and incorporating AI into existing products and services. We also invest in the development and acquisition of a variety of hardware for productivity, communication, and entertainment, including PCs, tablets, and gaming devices. Investments in new technology are speculative. Commercial success depends on many factors, including innovation, developer support, and effective distribution and marketing. If customers do not perceive our latest offerings as providing significant new functionality or other value, they may reduce their purchases of new software and hardware products or upgrades, unfavorably affecting revenue. We may not achieve significant revenue from new product, service, and distribution channel investments for several years, if at all. New products and services may not be profitable or may not achieve operating margins as high as we have experienced historically. We may not get engagement in certain features that drive post-sale monetization opportunities. Our data-handling practices across our products and services will continue to be under scrutiny. Perceptions of mismanagement, driven by regulatory activity or negative public reaction to our practices or product experiences, could negatively impact product and feature adoption. Developing new technologies is complex. It can require long development and testing periods. We could experience significant delays in new releases or significant problems in creating new products or services. These factors could adversely affect our business, financial condition, and results of operations.

Acquisitions, joint ventures, and strategic alliances may have an adverse effect on our business. We expect to continue making acquisitions and entering into joint ventures and strategic alliances as part of our long-term business strategy. For example, in March 2022 we completed our acquisition of Nuance Communications, Inc., and in October 2023 we completed our acquisition of Activision Blizzard, Inc. (“Activision Blizzard”). In January 2023 we announced the third phase of our OpenAI strategic partnership. Acquisitions and other transactions and arrangements involve significant challenges and risks, including that they do not advance our business strategy, that we get an unsatisfactory return on our investment, that they raise new compliance-related obligations and challenges, that we have difficulty integrating and retaining new employees, business systems, and technology, that they distract management from our other businesses, or that announced transactions may not be completed. If an arrangement fails to adequately anticipate changing circumstances and interests of a party, it may result in early termination or renegotiation of the arrangement. We also have limited ability to control or influence third parties with whom we have arrangements, which may impact our ability to realize the anticipated benefits. The success of these transactions and arrangements depend in part on our ability to leverage them to enhance our existing products and services or develop compelling new ones, as well as the acquired companies’ ability to meet our policies and processes in areas such as data governance, privacy, and cybersecurity. It may take longer than expected to realize the full benefits from these transactions and arrangements, such as increased revenue or enhanced efficiencies, or the benefits may ultimately be smaller than we expected. In addition, an acquisition may be subject to challenge even after it has been completed. For example, the Federal Trade Commission continues to challenge our Activision Blizzard acquisition and could, if successful, alter or unwind the transaction. These events could adversely affect our business, operations, financial condition, and results of operations.

If our goodwill or amortizable intangible assets become impaired, we may be required to record a significant charge to earnings. We acquire other companies and intangible assets and may not realize all the economic benefit from those acquisitions, which could cause an impairment of goodwill or intangibles. We review our amortizable intangible assets for impairment when events or changes in circumstances indicate the carrying value may not be recoverable. We test goodwill for impairment at least annually. Factors that may be a change in circumstances, indicating that the carrying value of our goodwill or amortizable intangible assets may not be recoverable, include a decline in our stock price and market capitalization, reduced future cash flow estimates, and slower growth rates in industry segments in which we participate. We have in the past recorded, and may in the future be required to record, a significant charge in our consolidated financial statements during the period in which any impairment of our goodwill or amortizable intangible assets is determined, negatively affecting our results of operations.

CYBERSECURITY, DATA PRIVACY, AND PLATFORM ABUSE RISKS

Cyberattacks and security vulnerabilities could lead to reduced revenue, increased costs, liability claims, or harm to our reputation or competitive position.

Security of our information technology

Threats to IT security can take a variety of forms. Individual and groups of hackers and sophisticated organizations, including state-sponsored organizations or nation-states, continuously undertake attacks that pose threats to our customers and our IT, and we have experienced cybersecurity incidents in which such actors have gained unauthorized access to our IT systems and data, including customer systems and data. These actors use a wide variety of methods, which include developing and deploying malicious software; exploiting known and potential vulnerabilities or intentionally designed processes in hardware, software, or other infrastructure to attack our products and services or gain access to our networks and datacenters; using social engineering techniques to induce our employees, users, partners, or customers to disclose sensitive information, such as passwords, or take other actions to gain access to our data or our users' or customers' data; or acting in a coordinated manner or conducting coordinated attacks. For example, as previously disclosed in our Form 8-K filed with the Securities and Exchange Commission on January 19, 2024 and amended on March 8, 2024, beginning in late November 2023, a nation-state associated threat actor used a password spray attack to compromise a legacy test account and, in turn, gain access to Microsoft email accounts. The threat actor used and may continue to use information it obtained to gain, or attempt to gain, unauthorized access to some of our source code repositories and internal systems, and the threat actor may utilize this information to otherwise adversely affect our business and results of operations. This incident has and may continue to result in harm to our reputation and customer relationships. Additionally, we may discover additional impacts of this or other incidents as part of our ongoing examination of this incident. Nation-state and state-sponsored actors can sustain malicious activities for extended periods and deploy significant resources to plan and carry out attacks. Nation-state attacks against us, our customers, or our partners have and may continue to intensify during periods of intense diplomatic or armed conflict, such as the ongoing conflict in Ukraine. Cyber incidents and attacks, individually or in the aggregate, could adversely affect our financial condition, results of operations, competitive position, and reputation, or expose us to legal or regulatory risk.

Inadequate account security or organizational security practices, including those of companies we have acquired or those of the third parties we utilize, have resulted and may result in unauthorized access to our IT systems and data, including customer systems and data, in the future. For example, system administrators may fail to timely remove employee account access when no longer appropriate. Employees or third parties may intentionally compromise our or our users' security or systems or reveal confidential information. Malicious actors may employ the IT supply chain to introduce malware through software updates or compromised supplier accounts or hardware.

Cyberthreats are constantly evolving and becoming increasingly sophisticated and complex, increasing the difficulty of detecting and successfully defending against them. Threat actors may also utilize emerging technologies, such as AI and machine learning. We may have no current capability to detect certain vulnerabilities or new attack methods, which may allow them to persist in the environment over long periods of time. It may be difficult to determine the best way to investigate, mitigate, contain, and remediate the harm caused by a cyber incident. Such efforts may not be successful, and we may make errors or fail to take necessary actions. It is possible that threat actors may gain undetected access to other networks and systems after establishing a foothold on an internal system. Cyber incidents and attacks can have cascading impacts that unfold with increasing speed across our internal networks and systems, as well as those of our partners and customers. In addition, it may take considerable time for us to investigate and evaluate the full impact of incidents, particularly for sophisticated attacks. These factors may inhibit our ability to provide prompt, full, and reliable information about the incident to our customers, partners, regulators, and the public. Breaches of our facilities, network, or data security can disrupt the security of our systems and business applications, impair our ability to provide services to our customers and protect the privacy of their data, result in product development delays, compromise confidential or technical business information, result in theft or misuse of our intellectual property or other assets, subject us to ransomware attacks, require us to allocate more resources to improve technologies or remediate the impacts of attacks, or otherwise adversely affect our business. In addition, actions taken to remediate an incident could result in outages, data losses, and disruptions of our services.

Our internal IT environment continues to evolve. Often, we are early adopters of new devices and technologies. We embrace new ways of sharing data and communicating internally and with partners and customers using methods such as social networking and other consumer-oriented technologies. Increasing use of generative AI models in our internal systems may create new attack methods for adversaries. Our business policies and internal security controls may not keep pace with these changes as new threats emerge or the emerging cybersecurity regulations in jurisdictions worldwide.

Security of our products, services, devices, and customers' data

The security of our products and services is important in our customers' decisions to purchase or use our products or services across cloud and on-premises environments. Security threats are a significant challenge to companies like us, whose business is providing technology products and services to others. Threats to or attacks on our own IT infrastructure, such as the nation-state attack described in the prior risk factor, have also affected our customers and may do so in the future. Customers using our cloud-based services rely on the security of our infrastructure, including hardware and other elements provided by third parties, to ensure the reliability of our services and the protection of their data. Adversaries tend to focus their efforts on the most popular operating systems, programs, and services, including many of ours, and we expect that to continue. In addition, adversaries can attack our customers' on-premises or cloud environments, sometimes exploiting previously unknown ("zero-day") vulnerabilities, such as the attack in early calendar year 2021 with several of our Exchange Server on-premises products. Vulnerabilities in these or any product can persist even after we have issued security patches if customers have not installed the most recent updates, or if the attackers exploited the vulnerabilities before patching to install additional malware to further compromise customers' systems. Adversaries will continue to attack customers using our cloud services as customers embrace digital transformation. Adversaries that acquire user account information can use that information to compromise our users' accounts, including where accounts share the same attributes such as passwords. Inadequate account security practices may also result in unauthorized access, and user activity may result in ransomware or other malicious software impacting a customer's use of our products or services. There may be vulnerabilities in open source software that may make our products susceptible to cyberattacks as we increasingly incorporate open source software into our products. Additionally, features that rely on generative AI may be susceptible to unanticipated security threats from adversaries as we add new generative AI features to our services while continuously developing our understanding of security risks and protection methods in the new field of generative AI.

Our customers operate complex IT systems with third-party hardware and software from multiple vendors that may include systems acquired over many years. They expect our products and services to support all these systems and products, including those that no longer incorporate the strongest current security advances or standards. As a result, we may not be able to discontinue support in our services for a product, service, standard, or feature solely because a more secure alternative is available. Failure to utilize the most current security advances and standards can increase our customers' vulnerability to attack. Further, customers of widely varied sizes and technical sophistication use our technology, and consequently may still have limited capabilities and resources to help them adopt and implement state-of-the-art cybersecurity practices and technologies. In addition, we must account for this wide variation of technical sophistication when defining default settings for our products and services, including security default settings, as these settings may limit or otherwise impact other aspects of IT operations and some customers may have limited capability to review and reset these defaults.

Cyberattacks may adversely impact our customers even if our production services are not directly compromised. We are committed to notifying our customers whose systems have been impacted as we become aware and have actionable information for customers to help protect themselves. We are also committed to providing guidance and support on detection, tracking, and remediation. We may not be able to detect the existence or extent of these attacks for all of our customers or have information on how to detect or track an attack, especially where an attack involves on-premises software such as Exchange Server where we may have no or limited visibility into our customers' computing environments.

Any of the foregoing events could result in reputational harm, loss of revenue, increased costs, or otherwise adversely affect our business, financial condition, and results of operations.

Development and deployment of defensive measures

To defend against security threats to our internal IT systems, our cloud-based services, and our customers' systems, we must continuously engineer more secure products and services, enhance security, threat detection, and reliability features, escalate and improve the deployment of software updates to address security vulnerabilities in our own products as well as those provided by others in a timely manner, develop mitigation technologies that help to secure customers from attacks even when software updates are not deployed, maintain the digital security infrastructure that protects the integrity of our network, products, and services, and provide security tools such as firewalls, anti-virus software, and advanced security and information about the need to deploy security measures and the impact of doing so.

The cost of measures to protect products and customer-facing services could reduce our operating margins. If we fail to do these things well, actual or perceived security vulnerabilities in our products and services, data corruption issues, or reduced performance could harm our reputation and lead customers to reduce or delay future purchases of products or subscriptions to services, or to use competing products or services. Customers may also spend more on protecting their existing computer systems from attack, which could delay adoption of additional products or services. Customers in certain industries such as financial services, health care, and government may have enhanced or specialized expectations and requirements to which we must engineer our products and services. Customers and third parties granted access to their systems may fail to update their systems, continue to run software or operating systems we no longer support, or may fail timely to install or enable security patches, or may otherwise fail to adopt adequate security practices Any of these could adversely affect our reputation and results of operations. Actual or perceived vulnerabilities may lead to claims against us. Our license agreements typically contain provisions that eliminate or limit our exposure to liability, but there is no assurance these provisions will withstand legal challenges. At times, to achieve commercial objectives, we may enter into agreements with larger liability exposure to customers.

Our products operate in conjunction with and are dependent on products and components across a broad ecosystem of third parties. If there is a security vulnerability in one of these components, and if there is a security exploit targeting it, we may experience adverse impacts to our results of operations, reputation, or competitive position.

Disclosure and misuse of personal data could result in liability and harm our reputation. As we continue to grow the number, breadth, and scale of our cloud-based offerings, we store and process increasingly large amounts of personal data of our customers and users. The continued occurrence of high-profile data breaches provides evidence of an external environment increasingly hostile to information security. Despite our efforts to improve the security controls across our business groups and geographies, it is possible our security controls over personal data, our training of employees and third parties on data security, and other practices we follow may not prevent the improper disclosure or misuse of customer or user data we or our vendors store and manage. Relatedly, despite our efforts to continuously improve security controls, it is possible that we may fail to identify or mitigate insider threat activities that could lead to the misuse of our systems or customer and user data. In addition, third parties who have limited access to our customer or user data may use this data in unauthorized ways. Improper disclosure or misuse could harm our reputation, lead to legal exposure to customers or users, or subject us to liability under laws that protect personal data, resulting in increased costs or loss of revenue. Our software products and services also enable our customers and users to store and process personal data on-premises or in a cloud-based environment we host. Government authorities can sometimes require us to produce customer or user data in response to valid legal orders. In the U.S. and elsewhere, we advocate for transparency concerning these requests and appropriate limitations on government authority to compel disclosure. Despite our efforts to protect customer and user data, perceptions that the collection, use, and retention of personal information is not satisfactorily protected could inhibit sales of our products or services and could limit adoption of our cloud-based solutions by consumers, businesses, and government entities. Additional security measures we may take to address customer or user concerns, or constraints on our flexibility to determine where and how to operate datacenters in response to customer or user expectations or governmental rules or actions, may increase costs or hinder sales of our products and services.

We may not be able to protect information in our products and services from use by others. LinkedIn and other Microsoft products and services contain valuable information and content protected by contractual restrictions or technical measures. In certain cases, we have made commitments to our members and users to limit access to or use of this information. Changes in the law or interpretations of the law may weaken our ability to prevent third parties from scraping or gathering information or content through use of bots or other measures and using it for their own benefit which could adversely affect our business, financial condition, and results of operations.

Abuse of our platforms may harm our reputation or user engagement.

Advertising, professional, marketplace, and gaming platform abuses

For platform products and services that provide content or host ads that come from or can be influenced by third parties, our reputation or user engagement may be negatively affected by activity that is hostile or inappropriate. This activity may come from users impersonating other people or organizations, including through the use of AI technologies, dissemination of information that may be viewed as misleading or intended to manipulate the opinions of our users, or the use of our products or services that violates our terms of service or otherwise for objectionable or illegal ends. Preventing or responding to these actions may require us to make substantial investments in people and technology and these investments may not be successful, adversely affecting our business, financial condition, and results of operations.

Other digital safety abuses

Our hosted consumer services as well as our enterprise services may be used to generate or disseminate harmful or illegal content in violation of our terms or applicable law. We may not proactively discover such content due to scale, the limitations of existing technologies, and conflicting legal frameworks. When discovered by users and others, such content may negatively affect our reputation, our brands, and user engagement. Regulations and other initiatives to make platforms responsible for preventing or eliminating harmful content online have been enacted, and we expect this to continue. We may be subject to enhanced regulatory oversight, civil or criminal liability, or reputational damage if we fail to comply with content moderation regulations, adversely affecting our business, financial condition, and results of operations.

Our products and services, how they are used by customers, and how third-party products and services interact with them, may present security, privacy, and execution risks. Our products and services may contain defects in design, manufacture, or operation that make them insecure or ineffective for their intended purposes. For example, an Internet of Things solution may have multiple layers of hardware, sensors, processors, software, and firmware, several of which we may not develop or control, and may have limited ability to be updated or patched. Further, customers control our products and services, including our AI products, within their environments, and may deploy them in high-risk scenarios or utilize them inappropriately. As a result, our products and services may increasingly affect personal health and safety. Our products may also collect large amounts of data in manners which may not satisfy customers or regulatory requirements. Our customers also operate complex IT systems with third-party hardware and software from multiple vendors whose products or personnel may take or fail to take actions which impact the reliability or security of our products and services. If our products and services do not work as intended, are utilized in methods not intended, violate the law, or harm individuals or businesses, we may be subject to legal claims or enforcement actions. These risks, if realized, may increase our costs, damage our reputation, or adversely affect our results of operations.

Issues in the development and use of AI may result in reputational or competitive harm or liability. We are building AI into many of our offerings, including our productivity services, and we are also making AI available for our customers to use in solutions that they build. This AI may be developed by Microsoft or others, including our strategic partner, OpenAI. We expect these elements of our business to grow. We envision a future in which AI operating in devices, applications, and the cloud helps our customers be more productive in their work and personal lives. As with many innovations, AI presents risks and challenges that could affect its adoption, and therefore our business. AI algorithms or training methodologies may be flawed. Datasets may be overbroad, insufficient, or contain biased information. Content generated by AI systems may be offensive, illegal, inaccurate, or otherwise harmful. Ineffective or inadequate AI development or deployment practices by Microsoft or others could result in incidents that impair the acceptance of AI solutions, cause harm to individuals, customers, or society, or result in our products and services not working as intended. Human review of certain outputs may be required. Our implementation of AI systems could result in legal liability, regulatory action, brand, reputational, or competitive harm, or other adverse impacts. These risks may arise from current copyright infringement and other claims related to AI training and output, new and proposed legislation and regulations, such as the European Union's ("EU") AI Act and the U.S.'s AI Executive Order, and new applications of data protection, privacy, consumer protection, intellectual property, and other laws. Some AI scenarios present ethical issues or may have broad impacts on society. If we enable or offer AI solutions that have unintended consequences, unintended usage or customization by our customers and partners, are contrary to our responsible AI policies and practices, or are otherwise controversial because of their impact on human rights, privacy, employment, or other social, economic, or political issues, our reputation, competitive position, business, financial condition, and results of operations may be adversely affected.

OPERATIONAL RISKS

We may have excessive outages, data losses, and disruptions of our online services if we fail to maintain an adequate operations infrastructure. Our increasing user traffic, growth in services, and the complexity of our products and services demand more computing power. We spend substantial amounts to build, purchase, or lease datacenters and equipment and to upgrade our technology and network infrastructure to handle more traffic on our websites and in our datacenters. Our datacenters depend on the availability of permitted and buildable land, predictable energy, networking supplies, and servers, including graphics processing units and other components. The cost or availability of these dependencies could be adversely affected by a variety of factors, including the transition to a clean energy economy, local and regional environmental regulations, and geopolitical disruptions. These demands continue to increase as we introduce new products and services and support the growth and the augmentation of existing services, including through the incorporation of AI features and/or functionality. We are rapidly growing our business of providing a platform and back-end hosting for services provided by third parties to their end users. Maintaining, securing, and expanding this infrastructure is expensive and complex, and requires development of principles for datacenter builds in geographies with higher safety and reliability risks. It requires that we maintain an Internet connectivity infrastructure and storage and compute capacity that is robust and reliable within competitive and regulatory constraints that continue to evolve. Inefficiencies or operational failures, including temporary or permanent loss of customer data, outages, insufficient Internet connectivity, insufficient or unavailable power or water supply, or inadequate storage and compute capacity could diminish the quality of our products, services, and user experience, resulting in contractual liability, claims by customers and other third parties, regulatory actions, damage to our reputation, and loss of current and potential users, subscribers, and advertisers, each of which may adversely affect our business, operations, financial condition, and results of operations.

We may experience quality or supply problems. There are limited suppliers for certain device and datacenter components. We continue to identify and evaluate opportunities to expand our datacenter locations and increase our server capacity to meet the evolving needs of our customers, particularly given the growing demand for AI services. Capacity available to us may be affected as competitors use some of the same suppliers and materials for hardware components. If components are delayed or become unavailable, whether because of supplier capacity constraint, industry shortages, legal or regulatory changes that restrict supply sources, or other reasons, we may not obtain timely replacement supplies, resulting in reduced sales or inadequate datacenter capacity to support the delivery and continued development of our products and services. Component shortages, excess or obsolete inventory, or price reductions resulting in inventory adjustments may increase our cost of revenue. Datacenter servers, Xbox consoles, Surface devices, and other hardware are assembled in Asia and other geographies that may be subject to disruptions in the supply chain, resulting in shortages which may adversely affect our business, operations, financial condition, and results of operations.

Our software products and services also may experience quality or reliability problems. The highly sophisticated software we develop may contain bugs and other defects that interfere with their intended operation. Our customers increasingly rely on us for critical business functions and multiple workloads. Many of our products and services are interdependent on one another. Our products and services may be impacted by interaction with third-party products and services. Our customers may also utilize their own or third-party products and services whose reliability is dependent on interaction with our products and services. Each of these circumstances potentially magnifies the impact of quality or reliability issues. Any defects we do not detect and fix in pre-release testing could cause reduced sales, damage to our reputation, repair or remediation costs, delays in the release of new products or versions, or legal liability, which could adversely affect our business, financial condition, and results of operations. Although our license agreements typically contain provisions that eliminate or limit our exposure to liability, there is no assurance these provisions will withstand legal challenge.

Our hardware products such as Xbox consoles, Surface devices, and other devices we design and market are highly complex. Failure to prevent, detect, or address defects in design, manufacture, or associated software could result in recalls, safety alerts, or product liability claims, which could adversely affect our business and results of operations.

LEGAL, REGULATORY, AND LITIGATION RISKS

Government enforcement under competition laws and new market regulation may limit how we design and market our products. Government agencies closely scrutinize us under U.S. and foreign competition laws. Governments are actively enforcing competition laws and regulations and enacting new regulations to intervene in digital markets, and this includes markets such as the EU, the United Kingdom, the U.S., and China. Some jurisdictions also allow competitors or consumers to assert claims of anti-competitive conduct. U.S. and foreign antitrust authorities have previously brought enforcement actions and continue to scrutinize our business.

For example, the European Commission ("the Commission") has designated Windows and LinkedIn as core platform services subject to obligations under the EU Digital Markets Act, which prohibits certain self-preferencing behaviors and places limitations on certain data use among other obligations. The Commission also continues to closely scrutinize the design of high-volume Microsoft products and the terms on which we make certain technologies used in these products, such as file formats, programming interfaces, and protocols, available to other companies. Flagship product releases such as Microsoft 365 and Windows can receive significant scrutiny under EU or other competition laws.

Our portfolio of first-party devices continues to grow; at the same time, our OEM partners offer a large variety of devices for our platforms. As a result, we increasingly both cooperate and compete with our OEM partners, creating a risk that we fail to do so in compliance with competition rules. Regulatory scrutiny in this area may increase. Certain foreign governments, particularly in China and other countries in Asia, have advanced arguments under their competition laws that exert downward pressure on royalties for our intellectual property.

Competition law enforcement actions and court decisions along with new market regulations may result in fines or hinder our ability to provide the benefits of our software to consumers and businesses, reducing the attractiveness of our products and the revenue that comes from them. New competition law actions or obligations under market regulation schemes could be initiated, potentially using previous actions as precedent. The outcome of such actions, or steps taken to avoid them, could adversely affect us in a variety of ways, including causing us to withdraw products from or modify products for certain markets, decreasing the value of our assets, adversely affecting our ability to monetize our products, or inhibiting our ability to consummate acquisition or impose conditions on acquisitions that may reduce their value, which may adversely affect our business, financial condition, and results of operations.

Laws and regulations relating to anti-corruption and trade could result in increased costs, fines, criminal penalties, or reputational damage. The Foreign Corrupt Practices Act ("FCPA") and other anti-corruption laws and regulations ("Anti-Corruption Laws") prohibit corrupt payments by our employees, vendors, or agents, and the accounting provisions of the FCPA require us to maintain accurate books and records and adequate internal controls. From time to time, we receive inquiries from authorities in the U.S. and elsewhere which may be based on reports from employees and others about our business activities outside the U.S. and our compliance with Anti-Corruption Laws. Periodically, we receive such reports directly and investigate them, and also cooperate with investigations by U.S. and foreign law enforcement authorities. An example of increasing international regulatory complexity is the EU Whistleblower Directive, initiated in 2021, which presents compliance challenges as it is implemented in different forms by EU member states. Most countries in which we operate also have competition laws that prohibit competitors from colluding or otherwise attempting to reduce competition between themselves. While we devote substantial resources to our U.S. and international compliance programs and have implemented policies, training, and internal controls designed to reduce the risk of corrupt payments and collusive activity, our employees, partners, vendors, or agents may violate our policies. Our failure to comply with Anti-Corruption Laws or competition laws could result in significant fines and penalties, criminal sanctions against us, our officers, or our employees, prohibitions on the conduct of our business, and damage to our reputation, which could adversely affect our business, financial condition, and results of operations.

Increasing trade laws, policies, sanctions, and other regulatory requirements also affect our operations in and outside the U.S. relating to trade and investment. Economic sanctions in the U.S., the EU, and other countries prohibit most business with restricted entities or countries. U.S. export controls restrict Microsoft from offering many of its products and services to, or making investments in, certain entities in specified countries. U.S. import controls restrict us from integrating certain information and communication technologies into our supply chain and allow for government review of transactions involving information and communications technology from countries determined to be foreign adversaries. Supply chain regulations may impact the availability of goods or result in additional regulatory scrutiny. Periods of intense diplomatic or armed conflict, such as the ongoing conflict in Ukraine, may result in (1) new and rapidly evolving sanctions and trade restrictions, which may impair trade with sanctioned individuals and countries, and (2) negative impacts to regional trade ecosystems among our customers, partners, and us. Non-compliance with sanctions as well as general ecosystem disruptions could result in reputational harm, operational delays, monetary fines, loss of revenue, increased costs, loss of export privileges, or criminal sanctions, which could adversely affect our business, financial condition, and results of operations.

Laws and regulations relating to the handling of personal data may impede the adoption of our services or result in increased costs, legal claims, fines against us, or reputational damage. The growth of our Internet- and cloud-based services internationally relies increasingly on the movement of data across national boundaries. Legal requirements relating to the collection, storage, handling, and transfer of personal data continue to evolve. For example, while the EU-U.S. Data Privacy Framework ("DPF") has been recognized as adequate under EU law to allow transfers of personal data from the EU to certified companies in the U.S., the DPF is subject to further legal challenge which could cause the legal requirements for data transfers from the EU to be uncertain. EU data protection authorities have and may again block the use of certain U.S.-based services that involve the transfer of data to the U.S. In the EU and other markets, potential new rules and restrictions on the flow of data across borders could increase the cost and complexity of delivering our products and services. In addition, the EU General Data Protection Regulation ("GDPR"), which applies to all of our activities conducted from an establishment in the EU or related to products and services offered in the EU, imposes a range of compliance obligations regarding the handling of personal data. More recently, the EU has been developing new requirements related to the use of data, including in the Digital Markets Act, the Digital Services Act, and the Data Act, that add additional rules and restriction on the use of data in our products and services. Engineering efforts to build and maintain capabilities to facilitate compliance with these laws involve substantial expense and the diversion of engineering resources from other projects. We might experience reduced demand for our offerings if we are unable to engineer products that meet our legal duties or help our customers meet their obligations under these and other data regulations, or if our implementation to comply makes our offerings less attractive. Compliance with these obligations depends in part on how particular regulators interpret and apply them. If we fail to comply, or if regulators assert we have failed to comply (including in response to complaints made by customers), it may lead to regulatory enforcement actions, which can result in significant monetary penalties, private lawsuits, reputational damage, blockage of product offerings or of international data transfers, and loss of customers. The highest fines assessed under GDPR have recently been increasing, especially against large technology companies, and European data protection authorities have taken action to block or remove services from their markets. Jurisdictions around the world, such as China, India, and states in the U.S. have adopted, or are considering adopting or expanding, laws and regulations imposing obligations regarding the collection, handling, and transfer of personal data.

Our investment in gaining insights from data is becoming central to the value of the services we deliver to customers, including AI services, to operational efficiency and key opportunities in monetization, and to customer perceptions of quality. Our ability to use data in this way may be constrained by regulatory developments that impede realizing the expected return from this investment. Ongoing legal analyses, reviews, and inquiries by regulators of Microsoft practices, or relevant practices of other organizations, may result in burdensome or inconsistent requirements, including data sovereignty and localization requirements, affecting the location, movement, collection, and use of our customer and internal employee data as well as the management of that data. Compliance with applicable laws and regulations regarding personal data may require changes in services, business practices, or internal systems that result in increased costs, lower revenue, reduced efficiency, or greater difficulty in competing with foreign-based firms. Compliance with data regulations might limit our ability to innovate or offer certain features and functionality in some jurisdictions where we operate. Failure to comply with existing or new rules may result in significant penalties or orders to stop the alleged noncompliant activity, negative publicity, and diversion of management time and effort.

Existing and increasing legal and regulatory requirements could adversely affect our results of operations. We are subject to a wide range of laws, regulations, and legal requirements in the U.S. and globally, including those that may apply to our products and online services offerings, and those that impose requirements related to user privacy, telecommunications, data storage and protection, digital accessibility, advertising, and online content. Laws in several jurisdictions, including EU Member State laws under the European Electronic Communications Code, increasingly define certain of our services as regulated telecommunications services. This trend may continue and will result in these offerings being subject to additional data protection, security, law enforcement surveillance, and other obligations. Regulators and private litigants may assert that our collection, use, and management of customer data and other information is inconsistent with their laws and regulations, including laws that apply to the tracking of users via technology such as cookies. In addition, laws requiring us to retrieve and produce customer data in response to compulsory legal demands from law enforcement and governmental authorities are expanding and the requests we are experiencing are increasing in volume and complexity. New environmental, social, and governance laws and regulations are expanding mandatory disclosure, reporting, and diligence requirements. Legislative or regulatory action relating to cybersecurity requirements may increase the costs to develop, implement, or secure our products and services. Legislative and regulatory action is emerging in the areas of AI and content moderation, which could increase costs or restrict opportunity. For example, the EU's AI Act may increase costs or impact the provision or operation of our AI models and services in the European market.

How these laws and regulations apply to our business is often unclear, subject to change over time, and sometimes may be inconsistent from jurisdiction to jurisdiction. In addition, governments' approach to enforcement, and our products and services, are continuing to evolve. Compliance with existing, expanding, or new laws and regulations may involve significant costs or require changes in products or business practices that could adversely affect our results of operations. Noncompliance could result in the imposition of penalties, criminal sanctions, or orders we cease the alleged noncompliant activity. In addition, there is increasing pressure from advocacy groups, regulators, competitors, customers, and other stakeholders across many of these areas. If our products do not meet customer expectations or legal requirements, we could face regulatory or legal actions, and our business, operations, financial condition, and results of operations could be adversely affected.

We have claims and lawsuits against us that may result in adverse outcomes. We are subject to a variety of claims and lawsuits. These claims may arise from a wide variety of business practices and initiatives, including major new product releases, AI services, significant business transactions, warranty or product claims, employment practices, and regulation. As we continue to expand our business and offerings, we may experience new and novel legal claims. Adverse outcomes in some or all of these claims may result in significant monetary damages or injunctive relief that could adversely affect our ability to conduct our business. Litigation and other claims are subject to inherent uncertainties and management's view of these matters may change in the future. A material adverse impact to our financial condition and results of operations could occur for the period in which the effect of an unfavorable outcome becomes probable and reasonably estimable.

Our business with government customers may present additional uncertainties. We derive substantial revenue from government contracts. Government contracts generally can present risks and challenges not present in private commercial agreements. For instance, we may be subject to government audits and investigations relating to these contracts, we could be suspended or debarred as a governmental contractor, we could incur civil and criminal fines and penalties, and under certain circumstances contracts may be rescinded. Some agreements may allow a government to terminate without cause and provide for higher liability limits for certain losses. Some contracts may be subject to periodic funding approval, reductions, cancellations, or delays which could adversely impact public-sector demand for our products and services. These events could negatively impact our financial condition, results of operations, and reputation.

We may have additional tax liabilities. We are subject to income taxes in the U.S. and many foreign jurisdictions. Significant judgment is required in determining our worldwide provision for income taxes. In the course of our business, there are many transactions and calculations where the ultimate tax determination is uncertain. We may recognize additional tax expense and be subject to additional tax liabilities due to changes in tax laws, regulations, and administrative practices and principles, including changes to the global tax framework, in various jurisdictions. In recent years, multiple domestic and international tax proposals were proposed to impose greater tax burdens on large multinational enterprises. For example, the Organisation for Economic Co-operation and Development continues to advance proposals or guidance in international taxation, including the establishment of a global minimum tax.

We are regularly under audit by tax authorities in different jurisdictions. Although we believe that our provision for income taxes and our tax estimates are reasonable, tax authorities may disagree with certain positions we have taken. In addition, economic and political pressures to increase tax revenue in various jurisdictions may make resolving tax disputes favorably more difficult. We are currently under Internal Revenue Service ("IRS") audit for prior tax years and have received Notices of Proposed Adjustment ("NOPAs") from the IRS for the tax years 2004 to 2013. The primary issues in the NOPAs relate to intercompany transfer pricing. In the NOPAs, the IRS is seeking an additional tax payment of $28.9 billion plus penalties and interest. The final resolution of the proposed adjustments, and other audits or litigation, may differ from the amounts recorded in our consolidated financial statements and adversely affect our results of operations in the period or periods in which that determination is made.

We earn a significant amount of our operating income outside the U.S. A change in the mix of earnings and losses in countries with differing statutory tax rates, changes in our business or structure, or the expiration of or disputes about certain tax agreements in a particular country may result in higher effective tax rates for the company. In addition, changes in U.S. federal and state or international tax laws applicable to corporate multinationals, other global fundamental law changes currently being considered by many countries, including in the U.S., and changes in taxing jurisdictions' administrative interpretations, decisions, policies, and positions may materially adversely affect our financial condition and results of operations.

We are subject to evolving sustainability regulatory requirements and expectations, which exposes us to increased costs and legal and reputational risks. Laws, regulations, and policies relating to environmental, social, and governance matters are being developed and formalized in Europe, the U.S., and elsewhere, which may include specific, target-driven frameworks and disclosure requirements. In addition, we have established and publicly announced goals and commitments to become carbon negative, water positive, zero waste, and protect more land than we use. Any failure or perceived failure to pursue or fulfill our sustainability goals and commitments or to satisfy various sustainability reporting standards or regulatory requirements within the timelines we announce, or at all, could result in claims and lawsuits, regulatory actions, or damage to our reputation, each of which may adversely affect our business, operations, financial condition, and results of operations.

INTELLECTUAL PROPERTY RISKS

We face risks related to the protection and utilization of our intellectual property that may result in our business and operating results being harmed. Protecting our intellectual property rights and combating unlicensed copying and use of our software, source code, and other intellectual property on a global basis is difficult. Similarly, the absence of harmonized patent laws makes it more difficult to ensure consistent respect for patent rights.

Changes in the law may continue to weaken our ability to prevent the use of patented technology. Our increasing engagement with open source software will also cause us to license our intellectual property rights broadly in certain situations. If we are unable to protect our intellectual property, our results of operations may be adversely affected.

Source code, the detailed program commands for our operating systems and other software programs, is critical to our business. If our source code leaks, we might lose future trade secret protection for that code. It may then become easier for third parties to compete with our products by copying functionality, which could adversely affect our results of operations. Unauthorized access to or disclosure of source code or other intellectual property also could increase the security risks described elsewhere in these risk factors.

Third parties may claim that we infringe their intellectual property. From time to time, others claim we infringe their intellectual property rights, including current copyright infringement and other claims arising from AI training and output. To resolve these claims, we may enter into royalty-bearing data access or licensing agreements on terms that are less favorable than currently available, stop selling or redesign affected products or services, or pay damages to satisfy indemnification commitments with our customers. Adverse outcomes could also include monetary damages or injunctive relief that may limit or prevent importing, marketing, and selling our products or services that have infringing technologies. We have paid significant amounts to settle claims related to the use of technology and intellectual property rights and to procure intellectual property rights as part of our strategy to manage this risk, and may continue to do so, which could adversely affect our results of operations.

GENERAL RISKS

If our reputation or our brands are damaged, our business and results of operations may be harmed. Our reputation and brands are globally recognized and are important to our business. Our reputation and brands affect our ability to attract and retain consumer, business, and public-sector customers. There are numerous ways our reputation or brands could be damaged. These include product safety or quality issues, our environmental impact and sustainability, supply chain practices, or human rights record. We may experience backlash from customers, government entities, advocacy groups, employees, and other stakeholders that disagree with our product offering decisions, public policy positions, or corporate philanthropic initiatives. Damage to our reputation or our brands may occur from, among other things:

- The introduction of new features, products, services, or terms of service that customers, users, or partners do not like.
- Public scrutiny of our decisions regarding user privacy, data practices, content, or development and deployment of AI.
- Data security breaches, cybersecurity incidents, responsible AI failures, compliance failures, or actions of partners or individual employees.

Social media may increase the likelihood, speed, and magnitude of negative brand events. If our brands or reputation are damaged, it could adversely affect our business, results of operations, or ability to attract the most highly qualified employees.

Adverse economic or market conditions may harm our business. Worsening economic conditions, including inflation, recession, pandemic, or other changes in economic conditions, may cause lower IT spending and adversely affect our results of operations. If demand for PCs, servers, and other computing devices declines, or consumer or business spending for those products declines, our results of operations may be adversely affected.

Our product distribution system relies on an extensive partner and retail network. OEMs building devices that run our software have also been a significant means of distribution. The impact of economic conditions on our partners, such as the bankruptcy of a major distributor, OEM, or retailer, could cause sales channel disruption.

Challenging economic conditions also may impair the ability of our customers to pay for products and services they have purchased. As a result, allowances for doubtful accounts and write-offs of accounts receivable may increase.

We maintain an investment portfolio of various holdings, types, and maturities. These investments are subject to general credit, liquidity, market, and interest rate risks, which may be exacerbated by market downturns or events that affect global financial markets. A significant part of our investment portfolio comprises U.S. government securities. If global financial markets decline for long periods, or if there is a downgrade of the U.S. government credit rating due to an actual or threatened default on government debt, our investment portfolio may be adversely affected and we could determine that more of our investments have experienced a decline in fair value, requiring impairment charges that could adversely affect our financial condition and results of operations.

Catastrophic events or geopolitical conditions may disrupt our business. A disruption or failure of our systems, operations, or supply chain because of a major earthquake, weather event, cyberattack, terrorist attack, pandemic, or other catastrophic event could cause delays in completing sales, providing services, or performing other critical functions. Our corporate headquarters, a significant portion of our research and development activities, and certain other essential business operations are in the Seattle, Washington area, and we have other business operations in the Silicon Valley area of California, both of which are seismically active regions. A catastrophic event that results in the destruction or disruption of any of our critical business or IT systems, or the infrastructure or systems they rely on, such as power grids, could harm our ability to conduct normal business operations or adversely affect our results of operations. Providing our customers with more services and solutions in the cloud puts a premium on the resilience of our systems and strength of our business continuity management plans and magnifies the potential negative consequences of prolonged service outages.

Abrupt political change, terrorist activity, and armed conflict, such as the ongoing conflict in Ukraine, pose economic and other risks, which may negatively impact our ability to sell to and collect from customers, increase our operating costs, or otherwise disrupt our operations in markets both directly and indirectly impacted by such events. These conditions also may add uncertainty to the timing and budget for technology investment decisions by our customers and may cause supply chain disruptions for hardware manufacturers. Geopolitical change may result in changing regulatory systems and requirements and market interventions that could impact our operating strategies, access to national, regional, and global markets, hiring, and profitability. Geopolitical instability may lead to sanctions and impact our ability to do business in some markets or with some public-sector customers. Any of these changes may negatively affect our results of operations.

The occurrence of regional epidemics or a global pandemic, such as COVID-19, may adversely affect our business, operations, financial condition, and results of operations. The extent to which global pandemics impact our business going forward will depend on factors such as the duration and scope of the pandemic; governmental, business, and individuals' actions in response to the pandemic; and the impact on economic activity, including the possibility of recession or financial market instability. Measures to contain a global pandemic may intensify other risks described in these Risk Factors.

The long-term effects of climate change on the global economy and the IT industry in particular are unclear. Environmental regulations or changes in the supply, demand, or available sources of energy or other resources may affect the availability or cost of goods and services, including natural resources, necessary to run our business. Changes in climate where we operate may increase the costs of powering and cooling computer hardware we use to develop software and provide cloud-based services.

Our global business exposes us to operational and economic risks. Our customers are located throughout the world and a significant part of our revenue comes from international sales. The global nature of our business creates operational, economic, and geopolitical risks. Global, regional, and local economic developments, monetary policy, inflation, and recession, as well as political and military disputes, may adversely affect our results of operations. In addition, our international growth strategy includes certain markets, the developing nature of which presents several risks, including deterioration of social, political, labor, or economic conditions in a country or region, and difficulties in staffing and managing foreign operations. Emerging nationalist and protectionist trends and concerns about human rights, the environment, and political expression in specific countries may significantly alter the trade and commercial environments. Changes to trade policy or agreements as a result of populism, protectionism, or economic nationalism may result in higher tariffs, local sourcing initiatives, and non-local sourcing restrictions, export controls, investment restrictions, or other developments that make it more difficult to sell our products in foreign countries. Disruptions of these kinds in developed or emerging markets could negatively impact demand for our products and services, impair our ability to operate in certain regions, or increase operating costs. Although we hedge a portion of our international currency exposure, significant fluctuations in foreign exchange rates between the U.S. dollar and foreign currencies may adversely affect our results of operations.

Our business depends on our ability to attract and retain talented employees. Our business is based on successfully attracting, training, and retaining talented employees representing diverse backgrounds, experiences, and skill sets. The market for highly skilled workers and leaders in our industry is extremely competitive. Maintaining our brand and reputation, as well as a diverse and inclusive work environment that enables all our employees to thrive, are important to our ability to recruit and retain employees. We are also limited in our ability to recruit internationally by restrictive domestic immigration laws. Restraints on the flow of technical and professional talent, including as a result of changes to U.S. immigration policies or laws, may inhibit our ability to adequately staff our research and development efforts. If we are less successful in our recruiting efforts, or if we cannot retain highly skilled workers and key leaders, our ability to develop and deliver successful products and services may be adversely affected. Effective succession planning is also important to our long-term success. Failure to ensure effective transfer of knowledge and smooth transitions involving key employees could hinder our strategic planning and execution. How employment-related laws are interpreted and applied to our workforce practices may result in increased operating costs and less flexibility in how we meet our workforce needs. Our global workforce is predominantly non-unionized, although we do have some employees in the U.S. and internationally who are represented by unions or works councils. In the U.S., there has been a general increase in workers exercising their right to form or join a union. The unionization of significant employee populations could result in higher costs and other operational changes necessary to respond to changing conditions and to establish new relationships with worker representatives.

ITEM 1B. UNRESOLVED STAFF COMMENTS

We have received no written comments regarding our periodic or current reports from the staff of the Securities and Exchange Commission that were issued 180 days or more preceding the end of our fiscal year 2024 that remain unresolved.

ITEM 1C. CYBERSECURITY

RISK MANAGEMENT AND STRATEGY

Microsoft plays a central role in the world's digital ecosystem. We have made it the top corporate priority to protect the computing environment used by our customers and employees and to support the resiliency of our cloud infrastructure and services, products, devices, and our internal corporate resources from determined adversaries. In response to the evolving cybersecurity threat landscape, we launched the Secure Future Initiative ("SFI") in November 2023 and expanded the scope of SFI in May 2024. The SFI focuses our business strategy and efforts on continual improvement in cybersecurity protection, and is aligned around three security principles:

- **Secure by Design:** Security comes first when designing any product or service.
- **Secure by Default:** Security protections are enabled and enforced by default, require no extra effort, and are not optional.
- **Secure Operations:** Security controls and monitoring will continuously be improved to meet current and future threats.

We operate a cybersecurity program and governance framework designed to protect our computing environments against cybersecurity threats, and we have controls, policies, and procedures to identify, manage, and mitigate cybersecurity threats. Annually, we assess our cybersecurity program's alignment with the National Institute of Standards & Technology's Cyber Security Framework ("NIST") and other applicable industry standards. We also undertake integrated planning and preparedness activities to support business continuity and operational resiliency. We assess our program's effectiveness through various exercises, including tabletop simulations and production environment tests, penetration and vulnerability tests, red team exercises, and other related activities. We conduct mandatory cybersecurity training, provide employees with tools to report suspected incidents and assess their own security posture, and conduct real-time simulated employee education exercises, such as phishing email campaigns designed to emulate real-world attacks. We also engage in robust cybersecurity assessments and remediation efforts for acquired companies.

Our computing environments, products, and services are reviewed by our internal audit teams as well as independent third-party assessors. We are committed to managing the most significant risks to our strategies and ambitions, including cybersecurity risks. The Enterprise Risk Management ("ERM") organization supports management in this commitment by facilitating the semiannual risk assessment, which documents the priority and status of these risks and aligns them with our strategic mitigation efforts. ERM is structured using a framework based on the Committee of Sponsoring Organization ("COSO") guidance on Enterprise Risk Management Integrating Strategy with Performance and it also aligns with the International Organization for Standardization 31000:2018 Risk Management Standard.

We continuously monitor our computing environments, products, and services for vulnerabilities and signs of compromise, and we utilize our own security products to combat cybersecurity threats. We integrate security into our computing environments, products, and services through our Security Development Lifecycle ("SDL"). Our SDL introduces security and privacy considerations throughout all phases of our development process and through the adoption of zero-trust end-to-end architecture. We utilize machine learning and AI-powered security tools to gain insights from over 78 trillion signals per day and over 135 million managed devices. We track over 300 unique threat actors, including 160 nation-state actors and 50 ransomware groups. To support our efforts, we operate a Cyber Defense Operations Center connected to over 10,000 security and threat intelligence experts, including engineers, researchers, data scientists, cybersecurity experts, threat hunters, geopolitical analysts, investigators, and frontline responders across the globe.

When appropriate, we utilize external service providers to assess, test, or otherwise assist our program. We also leverage third parties by working with external researchers, operating bug bounty programs, and managing coordinated vulnerability disclosure programs with security organizations. We maintain a systematic approach to assessing and controlling the cybersecurity risks presented by third-party service providers. We require third-party service providers to manage their cybersecurity risks in defined ways, undergo cybersecurity reviews, notify us of cyber events, and satisfy additional contractual requirements.

We seek to improve the entire cybersecurity ecosystem through multistakeholder diplomacy to set and uphold expectations for state behavior, advancement of government policy that strengthens cybersecurity and resiliency, disruption and deterrence of cybercrime, protection of national security interests, and disruption of digital threats to democracies. We also establish processes and innovate solutions for us and our customers to address the growing number and complexity of cybersecurity regulations.

When we experience a cybersecurity incident, we utilize our well-established incident response plans that operate both across the company and at the product and services level. Incidents are first triaged for severity, and then more deeply assessed to establish a plan of record and activate internal and external notification, disclosure, and communication plans, as applicable. Engineering and development resources are mobilized to resolve or remediate the incident. After the incident is resolved, a comprehensive post-incident review process is conducted.

We describe the risks from cybersecurity threats, including previous cybersecurity incidents, in section “Risk Factors” (Part I, Item 1A of this Form 10-K). As of the date of this Form 10-K, we do not believe any risks from cybersecurity threats have materially affected or are reasonably likely to materially affect us, including our results of operations or financial condition. However, the cybersecurity threat environment is increasingly challenging, and we, along with the entire digital ecosystem, are under constant and increasing threat. As discussed above, our business strategy is tied to the SFI and we are committed to continuously monitoring cybersecurity threats, enhancing the security of our products, investing in our cybersecurity infrastructure, and collaborating with peers, customers, service providers, regulators, and governments to advance our and the entire digital ecosystem’s cybersecurity defenses and resiliency.

GOVERNANCE

Our Board of Directors oversees cybersecurity risk. Cybersecurity reviews by the Board are scheduled to occur at least quarterly, or more frequently as determined to be necessary or advisable. Presentations to the Board of Directors are made by senior management, including our Chief Information Security Officer (“CISO”), our EVP of Microsoft Security, and the head of our Customer Security and Trust organization. The presentations address topics such as cybersecurity threats, incidents, top risks and related remediation efforts, results from internal and third-party assessments, progress towards risk-mitigation goals, the functioning of our incident response program, regulatory developments, and digital diplomacy efforts. In addition, we have an escalation process in place to inform senior management and the Board of significant issues. Cybersecurity issues are also considered during separate Board meeting discussions regarding important matters like ERM, audit issues, operational budgeting, business continuity planning, mergers and acquisitions, brand management, and other relevant matters.

Our CISO leads the strategy, engineering, and operations of cybersecurity across the company, and reports to the EVP of Microsoft Security. Our CISO has extensive experience assessing and managing cybersecurity programs and cybersecurity risk. Before joining Microsoft, our CISO served in a prior Chief Technology Officer role as well as in senior leadership, engineering, and operational roles within multiple organizations. In addition to the Board’s oversight of cybersecurity risk, to support the CISO, we have established a Cybersecurity Governance Council (“CGC”) charged with overseeing initiatives that safeguard Microsoft’s infrastructure. The CGC is comprised of an executive-level team of Deputy CISOs with cybersecurity backgrounds and expertise relevant to their roles. The CGC responsibilities include approving our enterprise security risk assessment process and results, determining the appropriate cybersecurity risk level and mitigations, reviewing the NIST CSF alignment, and supporting compliance with cybersecurity regulations. Our cybersecurity efforts are supported directly by Microsoft’s security and threat intelligence experts and our employees across the company, all of whom receive cybersecurity awareness training and education and are expected to support our efforts.

ITEM 2. PROPERTIES

Our corporate headquarters are located in Redmond, Washington. We have approximately 15 million square feet of space located in King County, Washington that is used for engineering, sales, marketing, and operations, among other general and administrative purposes. These facilities include approximately 12 million square feet of owned space situated on approximately 530 acres of land we own at our corporate headquarters, and approximately 3 million square feet of space we lease.

We own and lease other facilities domestically and internationally, primarily for offices, datacenters, and research and development. The largest owned international properties include space in the following locations: China, India, Ireland, and the Netherlands. The largest leased international properties include space in the following locations: Australia, Canada, China, France, Germany, India, Ireland, Israel, Japan, the Netherlands, and the United Kingdom. Refer to Research and Development (Part I, Item 1 of this Form 10-K) for further discussion of our research and development facilities.

The table below shows a summary of the square footage of our properties owned and leased domestically and internationally as of June 30, 2024:

(Square feet in millions)

Location	**Owned**	**Leased**	**Total**
U.S.	**30**	**20**	**50**
International	**10**	**25**	**35**
Total	**40**	**45**	**85**

ITEM 3. LEGAL PROCEEDINGS

Refer to Note 15 – Contingencies of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for information regarding legal proceedings in which we are involved.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS, AND ISSUER PURCHASES OF EQUITY SECURITIES

MARKET AND STOCKHOLDERS

Our common stock is traded on the NASDAQ Stock Market under the symbol MSFT. On July 25, 2024, there were 81,346 registered holders of record of our common stock.

SHARE REPURCHASES AND DIVIDENDS

Following are our monthly share repurchases for the fourth quarter of fiscal year 2024:

Period	Total Number of Shares Purchased	Average Price Paid Per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs	Approximate Dollar Value of Shares That May Yet Be Purchased Under the Plans or Programs
				(In millions)
April 1, 2024 – April 30, 2024	**2,444,905**	**$ 413.75**	**2,444,905**	**$ 12,138**
May 1, 2024 – May 31, 2024	**2,233,450**	**416.85**	**2,233,450**	**11,207**
June 1, 2024 – June 30, 2024	**1,963,873**	**436.58**	**1,963,873**	**10,349**
	6,642,228		**6,642,228**	

All share repurchases were made using cash resources. Our share repurchases may occur through open market purchases or pursuant to a Rule 10b5-1 trading plan. The above table excludes shares repurchased to settle employee tax withholding related to the vesting of stock awards.

Our Board of Directors declared the following dividends during the fourth quarter of fiscal year 2024:

Declaration Date	Record Date	Payment Date	Dividend Per Share	Amount
				(In millions)
June 12, 2024	**August 15, 2024**	**September 12, 2024**	**$ 0.75**	**$ 5,575**

We returned $8.4 billion to shareholders in the form of share repurchases and dividends in the fourth quarter of fiscal year 2024. Refer to Note 16 – Stockholders' Equity of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for further discussion regarding share repurchases and dividends.

ITEM 6. [RESERVED]

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

The following Management's Discussion and Analysis of Financial Condition and Results of Operations ("MD&A") is intended to help the reader understand the results of operations and financial condition of Microsoft Corporation. MD&A is provided as a supplement to, and should be read in conjunction with, our consolidated financial statements and the accompanying Notes to Financial Statements (Part II, Item 8 of this Form 10-K). This section generally discusses the results of our operations for the year ended June 30, 2024 compared to the year ended June 30, 2023. For a discussion of the year ended June 30, 2023 compared to the year ended June 30, 2022, please refer to Part II, Item 7, "Management's Discussion and Analysis of Financial Condition and Results of Operations" in our Annual Report on Form 10-K for the year ended June 30, 2023.

OVERVIEW

Microsoft is a technology company committed to making digital technology and artificial intelligence ("AI") available broadly and doing so responsibly, with a mission to empower every person and every organization on the planet to achieve more. We create platforms and tools, powered by AI, that deliver innovative solutions that meet the evolving needs of our customers.

We generate revenue by offering a wide range of cloud-based solutions, content, and other services to people and businesses; licensing and supporting an array of software products; delivering relevant online advertising to a global audience; and designing and selling devices. Our most significant expenses are related to compensating employees; supporting and investing in our cloud-based services, including datacenter operations; designing, manufacturing, marketing, and selling our other products and services; and income taxes.

Highlights from fiscal year 2024 compared with fiscal year 2023 included:

- Microsoft Cloud revenue increased 23% to $137.4 billion.
- Office Commercial products and cloud services revenue increased 14% driven by Office 365 Commercial growth of 16%.
- Office Consumer products and cloud services revenue increased 4% and Microsoft 365 Consumer subscribers grew to 82.5 million.
- LinkedIn revenue increased 9%.
- Dynamics products and cloud services revenue increased 19% driven by Dynamics 365 growth of 24%.
- Server products and cloud services revenue increased 22% driven by Azure and other cloud services growth of 30%.
- Windows revenue increased 8% with Windows original equipment manufacturer licensing ("Windows OEM") revenue growth of 7% and Windows Commercial products and cloud services revenue growth of 11%.
- Devices revenue decreased 15%.
- Xbox content and services revenue increased 50% driven by 44 points of net impact from the Activision Blizzard Inc. ("Activision Blizzard") acquisition. The net impact reflects the change of Activision Blizzard content from third-party to first-party.
- Search and news advertising revenue excluding traffic acquisition costs increased 12%.

On October 13, 2023, we completed our acquisition of Activision Blizzard for a total purchase price of $75.4 billion, consisting primarily of cash. The financial results of Activision Blizzard have been included in our consolidated financial statements since the date of the acquisition. Activision Blizzard is reported as part of our More Personal Computing segment. Refer to Note 8 – Business Combinations of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for further discussion.

Industry Trends

Our industry is dynamic and highly competitive, with frequent changes in both technologies and business models. Each industry shift is an opportunity to conceive new products, new technologies, or new ideas that can further transform the industry and our business. At Microsoft, we push the boundaries of what is possible through a broad range of research and development activities that seek to identify and address the changing demands of customers and users, industry trends, and competitive forces.

Economic Conditions, Challenges, and Risks

The markets for software, devices, and cloud-based services are dynamic and highly competitive. Our competitors are developing new software and devices, while also deploying competing cloud-based services for consumers and businesses. The devices and form factors customers prefer evolve rapidly, influencing how users access services in the cloud and, in some cases, the user’s choice of which suite of cloud-based services to use. Aggregate demand for our software, services, and devices is also correlated to global macroeconomic and geopolitical factors, which remain dynamic. We must continue to evolve and adapt over an extended time in pace with this changing environment.

The investments we are making in cloud and AI infrastructure and devices will continue to increase our operating costs and may decrease our operating margins. We continue to identify and evaluate opportunities to expand our datacenter locations and increase our server capacity to meet the evolving needs of our customers, particularly given the growing demand for AI services. Our datacenters depend on the availability of permitted and buildable land, predictable energy, networking supplies, and servers, including graphics processing units (“GPUs”) and other components. Our devices are primarily manufactured by third-party contract manufacturers. For the majority of our products, we have the ability to use other manufacturers if a current vendor becomes unavailable or unable to meet our requirements. However, some of our products contain certain components for which there are very few qualified suppliers. Extended disruptions at these suppliers could impact our ability to manufacture devices on time to meet consumer demand.

Our success is highly dependent on our ability to attract and retain qualified employees. We hire a mix of university and industry talent worldwide. We compete for talented individuals globally by offering an exceptional working environment, broad customer reach, scale in resources, the ability to grow one’s career across many different products and businesses, and competitive compensation and benefits.

Our international operations provide a significant portion of our total revenue and expenses. Many of these revenue and expenses are denominated in currencies other than the U.S. dollar. As a result, changes in foreign exchange rates may significantly affect revenue and expenses. Fluctuations in the U.S. dollar relative to certain foreign currencies did not have a material impact on reported revenue and expenses from our international operations in fiscal year 2024.

Refer to Risk Factors (Part I, Item 1A of this Form 10-K) for a discussion of these factors and other risks.

Seasonality

Our revenue fluctuates quarterly and is generally higher in the fourth quarter of our fiscal year. Fourth quarter revenue is driven by a higher volume of multi-year contracts executed during the period.

Change in Accounting Estimate

In July 2022, we completed an assessment of the useful lives of our server and network equipment. Due to investments in software that increased efficiencies in how we operate our server and network equipment, as well as advances in technology, we determined we should increase the estimated useful lives of both server and network equipment from four years to six years. This change in accounting estimate was effective beginning fiscal year 2023.

Reportable Segments

We report our financial performance based on the following segments: Productivity and Business Processes, Intelligent Cloud, and More Personal Computing. The segment amounts included in MD&A are presented on a basis consistent with our internal management reporting.

Additional information on our reportable segments is contained in Note 19 – Segment Information and Geographic Data of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K).

Metrics

We use metrics in assessing the performance of our business and to make informed decisions regarding the allocation of resources. We disclose metrics to enable investors to evaluate progress against our ambitions, provide transparency into performance trends, and reflect the continued evolution of our products and services. Our commercial and other business metrics are fundamentally connected based on how customers use our products and services. The metrics are disclosed in the MD&A or the Notes to Financial Statements (Part II, Item 8 of this Form 10-K). Financial metrics are calculated based on financial results prepared in accordance with accounting principles generally accepted in the United States of America ("GAAP"), and growth comparisons relate to the corresponding period of last fiscal year.

In the first quarter of fiscal year 2024, we made updates to the presentation and method of calculation for certain metrics, revising our Microsoft Cloud revenue metric to include revenue growth and expanding our Microsoft 365 Consumer subscribers metric to include Microsoft 365 Basic subscribers, aligning with how we manage our business.

Commercial

Our commercial business primarily consists of Server products and cloud services, Office Commercial, Windows Commercial, the commercial portion of LinkedIn, Enterprise and partner services, and Dynamics. Our commercial metrics allow management and investors to assess the overall health of our commercial business and include leading indicators of future performance.

Commercial remaining performance obligation	Commercial portion of revenue allocated to remaining performance obligations, which includes unearned revenue and amounts that will be invoiced and recognized as revenue in future periods
Microsoft Cloud revenue and revenue growth	Revenue from Azure and other cloud services, Office 365 Commercial, the commercial portion of LinkedIn, Dynamics 365, and other commercial cloud properties
Microsoft Cloud gross margin percentage	Gross margin percentage for our Microsoft Cloud business

Productivity and Business Processes and Intelligent Cloud

Metrics related to our Productivity and Business Processes and Intelligent Cloud segments assess the health of our core businesses within these segments. The metrics reflect our cloud and on-premises product strategies and trends.

Office Commercial products and cloud services revenue growth	Revenue from Office Commercial products and cloud services (Office 365 subscriptions, the Office 365 portion of Microsoft 365 Commercial subscriptions, and Office licensed on-premises), comprising Office, Exchange, SharePoint, Microsoft Teams, Office 365 Security and Compliance, Microsoft Viva, and Copilot for Microsoft 365
Office Consumer products and cloud services revenue growth	Revenue from Office Consumer products and cloud services, including Microsoft 365 Consumer and Copilot Pro subscriptions, Office licensed on-premises, and other Office services
Office 365 Commercial seat growth	The number of Office 365 Commercial seats at end of period where seats are paid users covered by an Office 365 Commercial subscription
Microsoft 365 Consumer subscribers	The number of Microsoft 365 Consumer and Copilot Pro subscribers at end of period
Dynamics products and cloud services revenue growth	Revenue from Dynamics products and cloud services, including Dynamics 365, comprising a set of intelligent, cloud-based applications across ERP, CRM, Power Apps, and Power Automate; and on-premises ERP and CRM applications
LinkedIn revenue growth	Revenue from LinkedIn, including Talent Solutions, Marketing Solutions, Premium Subscriptions, and Sales Solutions
Server products and cloud services revenue growth	Revenue from Server products and cloud services, including Azure and other cloud services; SQL Server, Windows Server, Visual Studio, System Center, and related Client Access Licenses (“CALs”); and Nuance and GitHub

More Personal Computing

Metrics related to our More Personal Computing segment assess the performance of key lines of business within this segment. These metrics provide strategic product insights which allow us to assess the performance across our commercial and consumer businesses. As we have diversity of target audiences and sales motions within the Windows business, we monitor metrics that are reflective of those varying motions.

Windows OEM revenue growth	Revenue from sales of Windows Pro and non-Pro licenses sold through the OEM channel
Windows Commercial products and cloud services revenue growth	Revenue from Windows Commercial products and cloud services, comprising volume licensing of the Windows operating system, Windows cloud services, and other Windows commercial offerings
Devices revenue growth	Revenue from Devices, including Surface, HoloLens, and PC accessories
Xbox content and services revenue growth	Revenue from Xbox content and services, comprising first-party content (such as Activision Blizzard) and third-party content, including games and in-game content; Xbox Game Pass and other subscriptions; Xbox Cloud Gaming; advertising; third-party disc royalties; and other cloud services
Search and news advertising revenue (ex TAC) growth	Revenue from search and news advertising excluding traffic acquisition costs (“TAC”) paid to Bing Ads network publishers and news partners

SUMMARY RESULTS OF OPERATIONS

(In millions, except percentages and per share amounts)	2024	2023	Percentage Change
Revenue	**$ 245,122**	$ 211,915	16%
Gross margin	**171,008**	146,052	17%
Operating income	**109,433**	88,523	24%
Net income	**88,136**	72,361	22%
Diluted earnings per share	**11.80**	9.68	22%
Adjusted gross margin (non-GAAP)	**171,008**	146,204	17%
Adjusted operating income (non-GAAP)	**109,433**	89,694	22%
Adjusted net income (non-GAAP)	**88,136**	73,307	20%
Adjusted diluted earnings per share (non-GAAP)	**11.80**	9.81	20%

Adjusted gross margin, operating income, net income, and diluted earnings per share (“EPS”) are non-GAAP financial measures. Prior year non-GAAP financial measures exclude the impact of a $1.2 billion charge in the second quarter of fiscal year 2023 (“Q2 charge”), which included employee severance expenses, impairment charges resulting from changes to our hardware portfolio, and costs related to lease consolidation activities. Refer to the Non-GAAP Financial Measures section below for a reconciliation of our financial results reported in accordance with GAAP to non-GAAP financial results.

Fiscal Year 2024 Compared with Fiscal Year 2023

Revenue increased $33.2 billion or 16% driven by growth across each of our segments. Intelligent Cloud revenue increased driven by Azure. Productivity and Business Processes revenue increased driven by Office 365 Commercial. More Personal Computing revenue increased driven by Gaming.

Cost of revenue increased $8.3 billion or 13% driven by growth in Microsoft Cloud and Gaming, offset in part by a decline in Devices.

Gross margin increased $25.0 billion or 17% driven by growth across each of our segments.

- Gross margin percentage increased slightly. Excluding the impact of the change in accounting estimate for the useful lives of our server and network equipment, gross margin percentage increased 2 points driven by improvement in More Personal Computing.
- Microsoft Cloud gross margin percentage decreased slightly to 71%. Excluding the impact of the change in accounting estimate, Microsoft Cloud gross margin percentage increased slightly driven by improvements in Azure and Office 365 Commercial, inclusive of scaling our AI infrastructure, offset in part by sales mix shift to Azure.

Operating expenses increased $4.0 billion or 7% driven by Gaming, with 7 points of growth from the Activision Blizzard acquisition, and investments in cloud engineering, offset in part by the prior year Q2 charge.

Operating income increased $20.9 billion or 24% driven by growth across each of our segments.

Prior year gross margin, operating income, net income, and diluted EPS were negatively impacted by the Q2 charge, which resulted in decreases of $152 million, $1.2 billion, $946 million, and $0.13, respectively.

SEGMENT RESULTS OF OPERATIONS

(In millions, except percentages)	2024	2023	Percentage Change
Revenue			
Productivity and Business Processes	**$ 77,728**	$ 69,274	12%
Intelligent Cloud	**105,362**	87,907	20%
More Personal Computing	**62,032**	54,734	13%
Total	**$ 245,122**	$ 211,915	16%
Operating Income			
Productivity and Business Processes	**$ 40,540**	$ 34,189	19%
Intelligent Cloud	**49,584**	37,884	31%
More Personal Computing	**19,309**	16,450	17%
Total	**$ 109,433**	$ 88,523	24%

Reportable Segments

Fiscal Year 2024 Compared with Fiscal Year 2023

Productivity and Business Processes

Revenue increased $8.5 billion or 12%.

- Office Commercial products and cloud services revenue increased $5.8 billion or 14%. Office 365 Commercial revenue grew 16% with seat growth of 7%, driven by small and medium business and frontline worker offerings, as well as growth in revenue per user. Office Commercial products revenue declined 16% driven by continued customer shift to cloud offerings.
- Office Consumer products and cloud services revenue increased $237 million or 4%. Microsoft 365 Consumer subscribers grew 10% to 82.5 million.
- LinkedIn revenue increased $1.4 billion or 9% driven by growth across all lines of business – Talent Solutions, Premium Subscriptions, Marketing Solutions, and Sales Solutions.
- Dynamics products and cloud services revenue increased $1.0 billion or 19% driven by Dynamics 365. Dynamics 365 revenue grew 24% driven by growth across all workloads.

Operating income increased $6.4 billion or 19%.

- Gross margin increased $6.5 billion or 12% driven by growth in Office 365 Commercial. Gross margin percentage decreased slightly. Excluding the impact of the change in accounting estimate, gross margin percentage increased slightly driven by improvement in Office 365 Commercial.
- Operating expenses increased $159 million or 1%.

Intelligent Cloud

Revenue increased $17.5 billion or 20%.

- Server products and cloud services revenue increased $17.8 billion or 22% driven by Azure and other cloud services. Azure and other cloud services revenue grew 30% driven by growth in our consumption-based services. Server products revenue increased 3% driven by continued demand for our hybrid solutions, including Windows Server and SQL Server running in multi-cloud environments.
- Enterprise and partner services revenue decreased $306 million or 4% driven by declines in Enterprise Support Services and Industry Solutions.

Operating income increased $11.7 billion or 31%.

- Gross margin increased $11.6 billion or 19% driven by growth in Azure. Gross margin percentage decreased slightly. Excluding the impact of the change in accounting estimate, gross margin percentage increased slightly primarily driven by improvement in Azure, inclusive of scaling our AI infrastructure, offset in part by sales mix shift to Azure.
- Operating expenses decreased slightly primarily driven by the prior year Q2 charge, offset in part by investments in Azure.

More Personal Computing

Revenue increased $7.3 billion or 13%.

- Windows revenue increased $1.7 billion or 8% driven by growth in Windows Commercial and Windows OEM. Windows Commercial products and cloud services revenue increased 11% driven by demand for Microsoft 365. Windows OEM revenue increased 7%.
- Gaming revenue increased $6.0 billion or 39% driven by growth in Xbox content and services. Xbox content and services revenue increased 50% driven by 44 points of net impact from the Activision Blizzard acquisition. Xbox hardware revenue decreased 13% driven by lower volume of consoles sold.
- Search and news advertising revenue increased $418 million or 3%. Search and news advertising revenue excluding traffic acquisition costs increased 12% driven by higher search volume.
- Devices revenue decreased $815 million or 15%.

Operating income increased $2.9 billion or 17%.

- Gross margin increased $6.8 billion or 23% driven by growth in Gaming, with 10 points of net impact from the Activision Blizzard acquisition, as well as growth in Windows. Gross margin percentage increased driven by sales mix shift to higher margin businesses and improvement in Devices.
- Operating expenses increased $3.9 billion or 31% driven by Gaming, with 34 points of growth from the Activision Blizzard acquisition.

OPERATING EXPENSES

Research and Development

(In millions, except percentages)		2024		2023	Percentage Change
Research and development	**$**	**29,510**	$	27,195	9%
As a percent of revenue		**12%**		13%	(1)ppt

Research and development expenses include payroll, employee benefits, stock-based compensation expense, and other headcount-related expenses associated with product development. Research and development expenses also include third-party development and programming costs and the amortization of purchased software code and services content.

Fiscal Year 2024 Compared with Fiscal Year 2023

Research and development expenses increased $2.3 billion or 9% driven by Gaming, with 7 points of growth from the Activision Blizzard acquisition, and investments in cloud engineering.

Sales and Marketing

(In millions, except percentages)		2024		2023	Percentage Change
Sales and marketing	**$**	**24,456**	$	22,759	7%
As a percent of revenue		**10%**		11%	(1)ppt

Sales and marketing expenses include payroll, employee benefits, stock-based compensation expense, and other headcount-related expenses associated with sales and marketing personnel, and the costs of advertising, promotions, trade shows, seminars, and other programs.

Fiscal Year 2024 Compared with Fiscal Year 2023

Sales and marketing expenses increased $1.7 billion or 7% driven by Gaming, with 6 points of growth from the Activision Blizzard acquisition.

General and Administrative

(In millions, except percentages)	2024	2023	Percentage Change
General and administrative	**$ 7,609**	$ 7,575	0%
As a percent of revenue	**3%**	4%	(1)ppt

General and administrative expenses include payroll, employee benefits, stock-based compensation expense, employee severance expense incurred as part of a corporate program, and other headcount-related expenses associated with finance, legal, facilities, certain human resources and other administrative personnel, certain taxes, and legal and other administrative fees.

Fiscal Year 2024 Compared with Fiscal Year 2023

General and administrative expenses increased slightly as growth from the Activision Blizzard acquisition was offset in part by the prior year Q2 charge.

OTHER INCOME (EXPENSE), NET

The components of other income (expense), net were as follows:

(In millions)		
Year Ended June 30,	2024	2023
Interest and dividends income	**$ 3,157**	$ 2,994
Interest expense	**(2,935)**	(1,968)
Net recognized gains (losses) on investments	**(118)**	260
Net losses on derivatives	**(187)**	(456)
Net gains (losses) on foreign currency remeasurements	**(244)**	181
Other, net	**(1,319)**	(223)
Total	**$ (1,646)**	$ 788

We use derivative instruments to manage risks related to foreign currencies, interest rates, equity prices, and credit; to enhance investment returns; and to facilitate portfolio diversification. Gains and losses from changes in fair values of derivatives that are not designated as hedging instruments are primarily recognized in other income (expense), net.

Fiscal Year 2024 Compared with Fiscal Year 2023

Interest and dividends income increased due to higher yields. Interest expense increased due to the issuance of commercial paper. Net recognized losses on investments increased primarily due to higher equity impairments and lower gains on equity investments. Net losses on derivatives decreased primarily due to lower losses on equity derivatives. Other, net primarily reflects net recognized losses on equity method investments.

INCOME TAXES

Effective Tax Rate

Our effective tax rate for fiscal years 2024 and 2023 was 18% and 19%, respectively. The decrease in our effective tax rate was primarily due to tax benefits from tax law changes, including the impact from the issuance of Notice 2023-55 and Notice 2023-80 by the Internal Revenue Service ("IRS") and U.S. Treasury Department. Notice 2023-55, issued in the first quarter of fiscal year 2024, delayed the effective date of final foreign tax credit regulations to fiscal year 2024 for Microsoft. Notice 2023-80, issued in the second quarter of fiscal year 2024, further delayed the effective date of final foreign tax credit regulations indefinitely.

Our effective tax rate was lower than the U.S. federal statutory rate, primarily due to earnings taxed at lower rates in foreign jurisdictions resulting from producing and distributing our products and services through our foreign regional operations center in Ireland.

The mix of income before income taxes between the U.S. and foreign countries impacted our effective tax rate as a result of the geographic distribution of, and customer demand for, our products and services. In fiscal year 2024, our U.S. income before income taxes was $62.9 billion and our foreign income before income taxes was $44.9 billion. In fiscal year 2023, our U.S. income before income taxes was $52.9 billion and our foreign income before income taxes was $36.4 billion.

The Organisation for Economic Co-operation and Development ("OECD") published its model rules "Tax Challenges Arising From the Digitalisation of the Economy - Global Anti-Base Erosion Model Rules (Pillar Two)" which established a global minimum corporate tax rate of 15% for certain multinational enterprises. Many countries have implemented or are in the process of implementing the Pillar Two legislation, which will apply to Microsoft beginning in fiscal year 2025. While we do not currently estimate a material impact to our consolidated financial statements, we continue to monitor the impact as countries implement legislation and the OECD provides additional guidance.

Uncertain Tax Positions

We remain under audit by the IRS for tax years 2014 to 2017. With respect to the audit for tax years 2004 to 2013, on September 26, 2023, we received Notices of Proposed Adjustment ("NOPAs") from the IRS. The primary issues in the NOPAs relate to intercompany transfer pricing. In the NOPAs, the IRS is seeking an additional tax payment of $28.9 billion plus penalties and interest. As of June 30, 2024, we believe our allowances for income tax contingencies are adequate. We disagree with the proposed adjustments and will vigorously contest the NOPAs through the IRS's administrative appeals office and, if necessary, judicial proceedings. We do not expect a final resolution of these issues in the next 12 months. Based on the information currently available, we do not anticipate a significant increase or decrease to our income tax contingencies for these issues within the next 12 months.

We are subject to income tax in many jurisdictions outside the U.S. Our operations in certain jurisdictions remain subject to examination for tax years 1996 to 2023, some of which are currently under audit by local tax authorities. The resolution of each of these audits is not expected to be material to our consolidated financial statements.

NON-GAAP FINANCIAL MEASURES

Adjusted gross margin, operating income, net income, and diluted EPS are non-GAAP financial measures. Prior year non-GAAP financial measures exclude the impact of the Q2 charge, which includes employee severance expenses, impairment charges resulting from changes to our hardware portfolio, and costs related to lease consolidation activities. We believe these non-GAAP measures aid investors by providing additional insight into our operational performance and help clarify trends affecting our business. For comparability of reporting, management considers non-GAAP measures in conjunction with GAAP financial results in evaluating business performance. These non-GAAP financial measures presented should not be considered a substitute for, or superior to, the measures of financial performance prepared in accordance with GAAP.

The following table reconciles our financial results reported in accordance with GAAP to non-GAAP financial results:

(In millions, except percentages and per share amounts)	2024	2023	Percentage Change
Gross margin	**$ 171,008**	$ 146,052	17%
Severance, hardware-related impairment, and lease consolidation costs	**0**	152	*
Adjusted gross margin (non-GAAP)	**$ 171,008**	$ 146,204	17%
Operating income	**$ 109,433**	$ 88,523	24%
Severance, hardware-related impairment, and lease consolidation costs	**0**	1,171	*
Adjusted operating income (non-GAAP)	**$ 109,433**	$ 89,694	22%
Net income	**$ 88,136**	$ 72,361	22%
Severance, hardware-related impairment, and lease consolidation costs	**0**	946	*
Adjusted net income (non-GAAP)	**$ 88,136**	$ 73,307	20%
Diluted earnings per share	**$ 11.80**	$ 9.68	22%
Severance, hardware-related impairment, and lease consolidation costs	**0**	0.13	*
Adjusted diluted earnings per share (non-GAAP)	**$ 11.80**	$ 9.81	20%

* *Not meaningful.*

LIQUIDITY AND CAPITAL RESOURCES

We expect existing cash, cash equivalents, short-term investments, cash flows from operations, and access to capital markets to continue to be sufficient to fund our operating activities and cash commitments for investing and financing activities, such as dividends, share repurchases, debt maturities, material capital expenditures, and the transition tax related to the Tax Cuts and Jobs Act (“TCJA”), for at least the next 12 months and thereafter for the foreseeable future.

Cash, Cash Equivalents, and Investments

Cash, cash equivalents, and short-term investments totaled $75.5 billion and $111.3 billion as of June 30, 2024 and 2023, respectively. Equity and other investments were $14.6 billion and $9.9 billion as of June 30, 2024 and 2023, respectively. Our short-term investments are primarily intended to facilitate liquidity and capital preservation. They consist predominantly of highly liquid investment-grade fixed-income securities, diversified among industries and individual issuers. The investments are predominantly U.S. dollar-denominated securities, but also include foreign currency-denominated securities to diversify risk. Our fixed-income investments are exposed to interest rate risk and credit risk. The credit risk and average maturity of our fixed-income portfolio are managed to achieve economic returns that correlate to certain fixed-income indices. The settlement risk related to these investments is insignificant given that the short-term investments held are primarily highly liquid investment-grade fixed-income securities.

Valuation

In general, and where applicable, we use quoted prices in active markets for identical assets or liabilities to determine the fair value of our financial instruments. This pricing methodology applies to our Level 1 investments, such as U.S. government securities, common and preferred stock, and mutual funds. If quoted prices in active markets for identical assets or liabilities are not available to determine fair value, then we use quoted prices for similar assets and liabilities or inputs other than the quoted prices that are observable either directly or indirectly. This pricing methodology applies to our Level 2 investments, such as commercial paper, certificates of deposit, U.S. agency securities, foreign government bonds, mortgage- and asset-backed securities, corporate notes and bonds, and municipal securities. Level 3 investments are valued using internally-developed models with unobservable inputs. Assets and liabilities measured at fair value on a recurring basis using unobservable inputs are an immaterial portion of our portfolio.

A majority of our investments are priced by pricing vendors and are generally Level 1 or Level 2 investments as these vendors either provide a quoted market price in an active market or use observable inputs for their pricing without applying significant adjustments. Broker pricing is used mainly when a quoted price is not available, the investment is not priced by our pricing vendors, or when a broker price is more reflective of fair values in the market in which the investment trades. Our broker-priced investments are generally classified as Level 2 investments because the broker prices these investments based on similar assets without applying significant adjustments. In addition, all our broker-priced investments have a sufficient level of trading volume to demonstrate that the fair values used are appropriate for these investments. Our fair value processes include controls that are designed to ensure appropriate fair values are recorded. These controls include model validation, review of key model inputs, analysis of period-over-period fluctuations, and independent recalculation of prices where appropriate.

Cash Flows

Cash from operations increased $31.0 billion to $118.5 billion for fiscal year 2024, primarily due to an increase in cash received from customers. Cash used in financing decreased $6.2 billion to $37.8 billion for fiscal year 2024, primarily due to a $5.0 billion decrease in common stock repurchases and a $3.3 billion increase in proceeds from issuance of debt, net of repayments, offset in part by a $2.0 billion increase in dividends paid. Cash used in investing increased $74.3 billion to $97.0 billion for fiscal year 2024, primarily due to a $67.5 billion increase in cash used for acquisitions of companies, net of cash acquired, and purchases of intangible and other assets and a $16.4 billion increase in additions to property and equipment.

Debt Proceeds

We issue debt to take advantage of favorable pricing and liquidity in the debt markets, reflecting our credit rating. The proceeds of these issuances were or will be used for general corporate purposes, which may include, among other things, funding for working capital, capital expenditures, repurchases of capital stock, acquisitions, and repayment of existing debt. Refer to Note 11 – Debt of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for further discussion.

Unearned Revenue

Unearned revenue comprises mainly unearned revenue related to volume licensing programs, which may include Software Assurance (“SA”) and cloud services. Unearned revenue is generally invoiced annually at the beginning of each contract period for multi-year agreements and recognized ratably over the coverage period. Unearned revenue also includes payments for other offerings for which we have been paid in advance and earn the revenue when we transfer control of the product or service. Refer to Note 1 – Accounting Policies of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for further discussion.

The following table outlines the expected future recognition of unearned revenue as of June 30, 2024:

(In millions)

Three Months Ending		
September 30, 2024	**$**	**22,529**
December 31, 2024		**17,664**
March 31, 2025		**12,076**
June 30, 2025		**5,313**
Thereafter		**2,602**
Total	**$**	**60,184**

If our customers choose to license cloud-based versions of our products and services rather than licensing transaction-based products and services, the associated revenue will shift from being recognized at the time of the transaction to being recognized over the subscription period or upon consumption, as applicable. Refer to Note 13 – Unearned Revenue of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for further discussion.

Material Cash Requirements and Other Obligations

Contractual Obligations

The following table summarizes the payments due by fiscal year for our outstanding contractual obligations as of June 30, 2024:

(In millions)	2025	Thereafter	Total
Long-term debt: [(a)]			
Principal payments	$ 2,250	$ 48,971	$ 51,221
Interest payments	1,618	27,041	28,659
Construction commitments [(b)]	29,892	5,499	35,391
Operating and finance leases, including imputed interest [(c)]	12,250	160,475	172,725
Purchase commitments [(d)]	68,280	3,742	72,022
Total	$ 114,290	$ 245,728	$ 360,018

(a) *Refer to Note 11 – Debt of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K).*
(b) *Refer to Note 7 – Property and Equipment of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K).*
(c) *Refer to Note 14 – Leases of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K).*
(d) *Purchase commitments primarily relate to datacenters and include open purchase orders and take-or-pay contracts that are not presented as construction commitments above.*

Income Taxes

As a result of the TCJA, we are required to pay a one-time transition tax on deferred foreign income not previously subject to U.S. income tax. Under the TCJA, the transition tax is payable in interest-free installments over eight years, with 8% due in each of the first five years, 15% in year six, 20% in year seven, and 25% in year eight. As of June 30, 2024, we had a remaining transition tax liability of $7.6 billion, of which $3.8 billion is short-term and payable in the first quarter of fiscal year 2025.

Share Repurchases

During fiscal years 2024 and 2023, we repurchased 32 million shares and 69 million shares of our common stock for $12.0 billion and $18.4 billion, respectively, through our share repurchase program. All repurchases were made using cash resources. As of June 30, 2024, $10.3 billion remained of our $60 billion share repurchase program. Refer to Note 16 – Stockholders' Equity of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for further discussion.

Dividends

During fiscal years 2024 and 2023, our Board of Directors declared dividends totaling $22.3 billion and $20.2 billion, respectively. We intend to continue returning capital to shareholders in the form of dividends, subject to declaration by our Board of Directors. Refer to Note 16 – Stockholders' Equity of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for further discussion.

Other Planned Uses of Capital

We will continue to invest in sales, marketing, product support infrastructure, and existing and advanced areas of technology, as well as acquisitions that align with our business strategy. Additions to property and equipment will continue, including new facilities, datacenters, and computer systems for research and development, sales and marketing, support, and administrative staff. We expect capital expenditures to increase in coming years to support growth in our cloud offerings and our investments in AI infrastructure and training. We have operating and finance leases for datacenters, corporate offices, research and development facilities, Microsoft Experience Centers, and certain equipment. We have not engaged in any related party transactions or arrangements with unconsolidated entities or other persons that are reasonably likely to materially affect liquidity or the availability of capital resources.

RECENT ACCOUNTING GUIDANCE

Refer to Note 1 – Accounting Policies of the Notes to Financial Statements (Part II, Item 8 of this Form 10-K) for further discussion.

CRITICAL ACCOUNTING ESTIMATES

Our consolidated financial statements and accompanying notes are prepared in accordance with GAAP. Preparing consolidated financial statements requires management to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenue, and expenses. Critical accounting estimates are those estimates that involve a significant level of estimation uncertainty and could have a material impact on our financial condition or results of operations. We have critical accounting estimates in the areas of revenue recognition, impairment of investment securities, goodwill, research and development costs, legal and other contingencies, income taxes, and business combinations – valuation of intangible assets.

Revenue Recognition

Our contracts with customers often include promises to transfer multiple products and services to a customer. Determining whether products and services are considered distinct performance obligations that should be accounted for separately versus together may require significant judgment. When a cloud-based service includes both on-premises software licenses and cloud services, judgment is required to determine whether the software license is considered distinct and accounted for separately, or not distinct and accounted for together with the cloud service and recognized over time. Certain cloud services, primarily Office 365, depend on a significant level of integration, interdependency, and interrelation between the desktop applications and cloud services, and are accounted for together as one performance obligation. Revenue from Office 365 is recognized ratably over the period in which the cloud services are provided.

Judgment is required to determine the standalone selling price (“SSP") for each distinct performance obligation. We use a single amount to estimate SSP for items that are not sold separately, including on-premises licenses sold with SA or software updates provided at no additional charge. We use a range of amounts to estimate SSP when we sell each of the products and services separately and need to determine whether there is a discount to be allocated based on the relative SSP of the various products and services.

In instances where SSP is not directly observable, such as when we do not sell the product or service separately, we determine the SSP using information that may include market conditions and other observable inputs. We typically have more than one SSP for individual products and services due to the stratification of those products and services by customers and circumstances. In these instances, we may use information such as the size of the customer and geographic region in determining the SSP.

Due to the various benefits from and the nature of our SA program, judgment is required to assess the pattern of delivery, including the exercise pattern of certain benefits across our portfolio of customers.

Our products are generally sold with a right of return, we may provide other credits or incentives, and in certain instances we estimate customer usage of our products and services, which are accounted for as variable consideration when determining the amount of revenue to recognize. Returns and credits are estimated at contract inception and updated at the end of each reporting period if additional information becomes available. Changes to our estimated variable consideration were not material for the periods presented.

Impairment of Investment Securities

We review debt investments quarterly for credit losses and impairment. If the cost of an investment exceeds its fair value, we evaluate, among other factors, general market conditions, credit quality of debt instrument issuers, and the extent to which the fair value is less than cost. This determination requires significant judgment. In making this judgment, we employ a systematic methodology that considers available quantitative and qualitative evidence in evaluating potential impairment of our investments. In addition, we consider specific adverse conditions related to the financial health of, and business outlook for, the investee. If we have plans to sell the security or it is more likely than not that we will be required to sell the security before recovery, then a decline in fair value below cost is recorded as an impairment charge in other income (expense), net and a new cost basis in the investment is established. If market, industry, and/or investee conditions deteriorate, we may incur future impairments.

Equity investments without readily determinable fair values are written down to fair value if a qualitative assessment indicates that the investment is impaired and the fair value of the investment is less than carrying value. We perform a qualitative assessment on a periodic basis. We are required to estimate the fair value of the investment to determine the amount of the impairment loss. Once an investment is determined to be impaired, an impairment charge is recorded in other income (expense), net.

Goodwill

We allocate goodwill to reporting units based on the reporting unit expected to benefit from the business combination. We evaluate our reporting units on an annual basis and, if necessary, reassign goodwill using a relative fair value allocation approach. Goodwill is tested for impairment at the reporting unit level (operating segment or one level below an operating segment) on an annual basis (May 1) and between annual tests if an event occurs or circumstances change that would more likely than not reduce the fair value of a reporting unit below its carrying value. These events or circumstances could include a significant change in the business climate, legal factors, operating performance indicators, competition, or sale or disposition of a significant portion of a reporting unit.

Application of the goodwill impairment test requires judgment, including the identification of reporting units, assignment of assets and liabilities to reporting units, assignment of goodwill to reporting units, and determination of the fair value of each reporting unit. The fair value of each reporting unit is estimated primarily through the use of a discounted cash flow methodology. This analysis requires significant judgments, including estimation of future cash flows, which is dependent on internal forecasts, estimation of the long-term rate of growth for our business, estimation of the useful life over which cash flows will occur, and determination of our weighted average cost of capital.

The estimates used to calculate the fair value of a reporting unit change from year to year based on operating results, market conditions, and other factors. Changes in these estimates and assumptions could materially affect the determination of fair value and goodwill impairment for each reporting unit.

Research and Development Costs

Costs incurred internally in researching and developing a computer software product are charged to expense until technological feasibility has been established for the product. Once technological feasibility is established, software costs are capitalized until the product is available for general release to customers. Judgment is required in determining when technological feasibility of a product is established. We have determined that technological feasibility for our software products is reached after all high-risk development issues have been resolved through coding and testing. Generally, this occurs shortly before the products are released to production. The amortization of these costs is included in cost of revenue over the estimated life of the products.

Legal and Other Contingencies

The outcomes of legal proceedings and claims brought against us are subject to significant uncertainty. An estimated loss from a loss contingency such as a legal proceeding or claim is accrued by a charge to income if it is probable that an asset has been impaired or a liability has been incurred and the amount of the loss can be reasonably estimated. In determining whether a loss should be accrued we evaluate, among other factors, the degree of probability of an unfavorable outcome and the ability to make a reasonable estimate of the amount of loss. Changes in these factors could materially impact our consolidated financial statements.

Income Taxes

The objectives of accounting for income taxes are to recognize the amount of taxes payable or refundable for the current year, and deferred tax liabilities and assets for the future tax consequences of events that have been recognized in an entity's financial statements or tax returns. We recognize the tax benefit from an uncertain tax position only if it is more likely than not that the tax position will be sustained on examination by the taxing authorities, based on the technical merits of the position. The tax benefits recognized in the financial statements from such a position are measured based on the largest benefit that has a greater than 50% likelihood of being realized upon ultimate settlement. Accounting literature also provides guidance on derecognition of income tax assets and liabilities, classification of deferred income tax assets and liabilities, accounting for interest and penalties associated with tax positions, and income tax disclosures. Judgment is required in assessing the future tax consequences of events that have been recognized in our consolidated financial statements or tax returns. Variations in the actual outcome of these future tax consequences could materially impact our consolidated financial statements.

Business Combinations – Valuation of Intangible Assets

Accounting for business combinations requires significant judgments when allocating the purchase price to the estimated fair values of assets acquired and liabilities assumed at the acquisition date. Determination of fair value involves estimates and assumptions which can be complex, most notably with respect to intangible assets. Critical estimates used in the valuation of intangible assets include, but are not limited to, the amount and timing of projected cash flows, useful lives, and discount rates. While management's estimates of fair value are based on assumptions that are believed to be reasonable, these assumptions are inherently uncertain as they pertain to forward-looking views of our business and market conditions. The judgments made in this valuation process could materially impact our consolidated financial statements.

STATEMENT OF MANAGEMENT'S RESPONSIBILITY FOR FINANCIAL STATEMENTS

Management is responsible for the preparation of the consolidated financial statements and related information that are presented in this report. The consolidated financial statements, which include amounts based on management's estimates and judgments, have been prepared in conformity with accounting principles generally accepted in the United States of America.

The Company designs and maintains accounting and internal control systems to provide reasonable assurance at reasonable cost that assets are safeguarded against loss from unauthorized use or disposition, and that the financial records are reliable for preparing consolidated financial statements and maintaining accountability for assets. These systems are augmented by written policies, an organizational structure providing division of responsibilities, careful selection and training of qualified personnel, and a program of internal audits.

The Company engaged Deloitte & Touche LLP, an independent registered public accounting firm, to audit and render an opinion on the consolidated financial statements and internal control over financial reporting in accordance with the standards of the Public Company Accounting Oversight Board (United States).

The Board of Directors, through its Audit Committee, consisting solely of independent directors of the Company, meets periodically with management, internal auditors, and our independent registered public accounting firm to ensure that each is meeting its responsibilities and to discuss matters concerning internal controls and financial reporting. Deloitte & Touche LLP and the internal auditors each have full and free access to the Audit Committee.

Satya Nadella
Chief Executive Officer

Amy E. Hood
Executive Vice President and Chief Financial Officer

Alice L. Jolla
Corporate Vice President and Chief Accounting Officer

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

RISKS

We are exposed to economic risk from foreign exchange rates, interest rates, credit risk, and equity prices. We use derivatives instruments to manage these risks, however, they may still impact our consolidated financial statements.

Foreign Currencies

Certain forecasted transactions, assets, and liabilities are exposed to foreign currency risk. We monitor our foreign currency exposures daily to maximize the economic effectiveness of our foreign currency positions, including hedges. Principal currency exposures include the Euro, Japanese yen, British pound, Canadian dollar, and Australian dollar.

Interest Rate

Securities held in our fixed-income portfolio are subject to different interest rate risks based on their maturities. We manage the average maturity of the fixed-income portfolio to achieve economic returns that correlate to certain global fixed-income indices.

Credit

Our fixed-income portfolio is diversified and consists primarily of investment-grade securities. We manage credit exposures relative to broad-based indices to facilitate portfolio diversification.

Equity

Securities held in our equity investments portfolio are subject to price risk.

SENSITIVITY ANALYSIS

The following table sets forth the potential loss in future earnings or fair values, including associated derivatives, resulting from hypothetical changes in relevant market rates or prices:

(In millions)

Risk Categories	Hypothetical Change	June 30, 2024	Impact
Foreign currency – Revenue	10% decrease in foreign exchange rates	$ (9,605)	Earnings
Foreign currency – Investments	10% decrease in foreign exchange rates	(38)	Fair Value
Interest rate	100 basis point increase in U.S. treasury interest rates	(1,343)	Fair Value
Credit	100 basis point increase in credit spreads	(318)	Fair Value
Equity	10% decrease in equity market prices	(1,078)	Earnings

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

INCOME STATEMENTS

(In millions, except per share amounts)

Year Ended June 30,	**2024**	**2023**	**2022**
Revenue:			
Product	**$ 64,773**	$ 64,699	$ 72,732
Service and other	**180,349**	147,216	125,538
Total revenue	**245,122**	211,915	198,270
Cost of revenue:			
Product	**15,272**	17,804	19,064
Service and other	**58,842**	48,059	43,586
Total cost of revenue	**74,114**	65,863	62,650
Gross margin	**171,008**	146,052	135,620
Research and development	**29,510**	27,195	24,512
Sales and marketing	**24,456**	22,759	21,825
General and administrative	**7,609**	7,575	5,900
Operating income	**109,433**	88,523	83,383
Other income (expense), net	**(1,646)**	788	333
Income before income taxes	**107,787**	89,311	83,716
Provision for income taxes	**19,651**	16,950	10,978
Net income	**$ 88,136**	$ 72,361	$ 72,738
Earnings per share:			
Basic	**$ 11.86**	$ 9.72	$ 9.70
Diluted	**$ 11.80**	$ 9.68	$ 9.65
Weighted average shares outstanding:			
Basic	**7,431**	7,446	7,496
Diluted	**7,469**	7,472	7,540

Refer to accompanying notes.

COMPREHENSIVE INCOME STATEMENTS

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Net income	**$ 88,136**	$ 72,361	$ 72,738
Other comprehensive income (loss), net of tax:			
Net change related to derivatives	**24**	(14)	6
Net change related to investments	**957**	(1,444)	(5,360)
Translation adjustments and other	**(228)**	(207)	(1,146)
Other comprehensive income (loss)	**753**	(1,665)	(6,500)
Comprehensive income	**$ 88,889**	$ 70,696	$ 66,238

Refer to accompanying notes.

BALANCE SHEETS

(In millions)

June 30,	**2024**	**2023**
Assets		
Current assets:		
Cash and cash equivalents	**$ 18,315**	$ 34,704
Short-term investments	**57,228**	76,558
Total cash, cash equivalents, and short-term investments	**75,543**	111,262
Accounts receivable, net of allowance for doubtful accounts of **$830** and $650	**56,924**	48,688
Inventories	**1,246**	2,500
Other current assets	**26,021**	21,807
Total current assets	**159,734**	184,257
Property and equipment, net of accumulated depreciation of **$76,421** and $68,251	**135,591**	95,641
Operating lease right-of-use assets	**18,961**	14,346
Equity and other investments	**14,600**	9,879
Goodwill	**119,220**	67,886
Intangible assets, net	**27,597**	9,366
Other long-term assets	**36,460**	30,601
Total assets	**$ 512,163**	$ 411,976
Liabilities and stockholders' equity		
Current liabilities:		
Accounts payable	**$ 21,996**	$ 18,095
Short-term debt	**6,693**	0
Current portion of long-term debt	**2,249**	5,247
Accrued compensation	**12,564**	11,009
Short-term income taxes	**5,017**	4,152
Short-term unearned revenue	**57,582**	50,901
Other current liabilities	**19,185**	14,745
Total current liabilities	**125,286**	104,149
Long-term debt	**42,688**	41,990
Long-term income taxes	**27,931**	25,560
Long-term unearned revenue	**2,602**	2,912
Deferred income taxes	**2,618**	433
Operating lease liabilities	**15,497**	12,728
Other long-term liabilities	**27,064**	17,981
Total liabilities	**243,686**	205,753
Commitments and contingencies		
Stockholders' equity:		
Common stock and paid-in capital – shares authorized 24,000; outstanding **7,434** and 7,432	**100,923**	93,718
Retained earnings	**173,144**	118,848
Accumulated other comprehensive loss	**(5,590)**	(6,343)
Total stockholders' equity	**268,477**	206,223
Total liabilities and stockholders' equity	**$ 512,163**	$ 411,976

Refer to accompanying notes.

CASH FLOWS STATEMENTS

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Operations			
Net income	**$ 88,136**	$ 72,361	$ 72,738
Adjustments to reconcile net income to net cash from operations:			
Depreciation, amortization, and other	**22,287**	13,861	14,460
Stock-based compensation expense	**10,734**	9,611	7,502
Net recognized losses (gains) on investments and derivatives	**305**	196	(409)
Deferred income taxes	**(4,738)**	(6,059)	(5,702)
Changes in operating assets and liabilities:			
Accounts receivable	**(7,191)**	(4,087)	(6,834)
Inventories	**1,284**	1,242	(1,123)
Other current assets	**(1,648)**	(1,991)	(709)
Other long-term assets	**(6,817)**	(2,833)	(2,805)
Accounts payable	**3,545**	(2,721)	2,943
Unearned revenue	**5,348**	5,535	5,109
Income taxes	**1,687**	(358)	696
Other current liabilities	**4,867**	2,272	2,344
Other long-term liabilities	**749**	553	825
Net cash from operations	**118,548**	87,582	89,035
Financing			
Proceeds from issuance of debt, maturities of 90 days or less, net	**5,250**	0	0
Proceeds from issuance of debt	**24,395**	0	0
Repayments of debt	**(29,070)**	(2,750)	(9,023)
Common stock issued	**2,002**	1,866	1,841
Common stock repurchased	**(17,254)**	(22,245)	(32,696)
Common stock cash dividends paid	**(21,771)**	(19,800)	(18,135)
Other, net	**(1,309)**	(1,006)	(863)
Net cash used in financing	**(37,757)**	(43,935)	(58,876)
Investing			
Additions to property and equipment	**(44,477)**	(28,107)	(23,886)
Acquisition of companies, net of cash acquired, and purchases of intangible and other assets	**(69,132)**	(1,670)	(22,038)
Purchases of investments	**(17,732)**	(37,651)	(26,456)
Maturities of investments	**24,775**	33,510	16,451
Sales of investments	**10,894**	14,354	28,443
Other, net	**(1,298)**	(3,116)	(2,825)
Net cash used in investing	**(96,970)**	(22,680)	(30,311)
Effect of foreign exchange rates on cash and cash equivalents	**(210)**	(194)	(141)
Net change in cash and cash equivalents	**(16,389)**	20,773	(293)
Cash and cash equivalents, beginning of period	**34,704**	13,931	14,224
Cash and cash equivalents, end of period	**$ 18,315**	$ 34,704	$ 13,931

Refer to accompanying notes.

STOCKHOLDERS' EQUITY STATEMENTS

(In millions, except per share amounts)

Year Ended June 30,	2024	2023	2022
Common stock and paid-in capital			
Balance, beginning of period	**$ 93,718**	$ 86,939	$ 83,111
Common stock issued	**2,002**	1,866	1,841
Common stock repurchased	**(5,712)**	(4,696)	(5,688)
Stock-based compensation expense	**10,734**	9,611	7,502
Other, net	**181**	(2)	173
Balance, end of period	**100,923**	93,718	86,939
Retained earnings			
Balance, beginning of period	**118,848**	84,281	57,055
Net income	**88,136**	72,361	72,738
Common stock cash dividends	**(22,293)**	(20,226)	(18,552)
Common stock repurchased	**(11,547)**	(17,568)	(26,960)
Balance, end of period	**173,144**	118,848	84,281
Accumulated other comprehensive loss			
Balance, beginning of period	**(6,343)**	(4,678)	1,822
Other comprehensive income (loss)	**753**	(1,665)	(6,500)
Balance, end of period	**(5,590)**	(6,343)	(4,678)
Total stockholders' equity	**$ 268,477**	$ 206,223	$ 166,542
Cash dividends declared per common share	**$ 3.00**	$ 2.72	$ 2.48

Refer to accompanying notes.

NOTES TO FINANCIAL STATEMENTS

NOTE 1 — ACCOUNTING POLICIES

Accounting Principles

Our consolidated financial statements and accompanying notes are prepared in accordance with accounting principles generally accepted in the United States of America ("GAAP").

We have recast certain prior period amounts to conform to the current period presentation. The recast of these prior period amounts had no impact on our consolidated balance sheets, consolidated income statements, or consolidated cash flows statements.

Principles of Consolidation

The consolidated financial statements include the accounts of Microsoft Corporation and its subsidiaries. Intercompany transactions and balances have been eliminated.

Estimates and Assumptions

Preparing financial statements requires management to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenue, and expenses. Examples of estimates and assumptions include: for revenue recognition, determining the nature and timing of satisfaction of performance obligations, and determining the standalone selling price ("SSP") of performance obligations, variable consideration, and other obligations such as product returns and refunds; loss contingencies; product warranties; the fair value of and/or potential impairment of goodwill and intangible assets for our reporting units; product life cycles; useful lives of our tangible and intangible assets; allowances for doubtful accounts; the market value of, and demand for, our inventory; stock-based compensation forfeiture rates; when technological feasibility is achieved for our products; the potential outcome of uncertain tax positions that have been recognized in our consolidated financial statements or tax returns; and determining the timing and amount of impairments for investments. Actual results and outcomes may differ from management's estimates and assumptions due to risks and uncertainties.

In July 2022, we completed an assessment of the useful lives of our server and network equipment. Due to investments in software that increased efficiencies in how we operate our server and network equipment, as well as advances in technology, we determined we should increase the estimated useful lives of both server and network equipment from four years to six years. This change in accounting estimate was effective beginning fiscal year 2023.

Foreign Currencies

Assets and liabilities recorded in foreign currencies are translated at the exchange rate on the balance sheet date. Revenue and expenses are translated at average rates of exchange prevailing during the year. Translation adjustments resulting from this process are recorded to other comprehensive income.

Revenue

Product Revenue and Service and Other Revenue

Product revenue includes sales from operating systems, cross-device productivity and collaboration applications, server applications, business solution applications, desktop and server management tools, software development tools, video games, and hardware such as PCs, tablets, gaming and entertainment consoles, other intelligent devices, and related accessories.

Service and other revenue includes sales from cloud-based solutions that provide customers with software, services, platforms, and content such as Office 365, Azure, Dynamics 365, and gaming; solution support; and consulting services. Service and other revenue also includes sales from online advertising and LinkedIn.

Revenue Recognition

Revenue is recognized upon transfer of control of promised products or services to customers in an amount that reflects the consideration we expect to receive in exchange for those products or services. We enter into contracts that can include various combinations of products and services, which are generally capable of being distinct and accounted for as separate performance obligations. Revenue is recognized net of allowances for returns and any taxes collected from customers, which are subsequently remitted to governmental authorities.

Nature of Products and Services

Licenses for on-premises software provide the customer with a right to use the software as it exists when made available to the customer. Customers may purchase perpetual licenses or subscribe to licenses, which provide customers with the same functionality and differ mainly in the duration over which the customer benefits from the software. Revenue from distinct on-premises licenses is recognized upfront at the point in time when the software is made available to the customer. In cases where we allocate revenue to software updates, primarily because the updates are provided at no additional charge, revenue is recognized as the updates are provided, which is generally ratably over the estimated life of the related device or license.

Certain volume licensing programs, including Enterprise Agreements, include on-premises licenses combined with Software Assurance (“SA”). SA conveys rights to new software and upgrades released over the contract period and provides support, tools, and training to help customers deploy and use products more efficiently. On-premises licenses are considered distinct performance obligations when sold with SA. Revenue allocated to SA is generally recognized ratably over the contract period as customers simultaneously consume and receive benefits, given that SA comprises distinct performance obligations that are satisfied over time.

Cloud services, which allow customers to use hosted software over the contract period without taking possession of the software, are provided on either a subscription or consumption basis. Revenue related to cloud services provided on a subscription basis is recognized ratably over the contract period. Revenue related to cloud services provided on a consumption basis, such as the amount of storage used in a period, is recognized based on the customer utilization of such resources. When cloud services require a significant level of integration and interdependency with software and the individual components are not considered distinct, all revenue is recognized over the period in which the cloud services are provided.

Revenue from search advertising is recognized when the advertisement appears in the search results or when the action necessary to earn the revenue has been completed. Revenue from consulting services is recognized as services are provided.

Our hardware is generally highly dependent on, and interrelated with, the underlying operating system and cannot function without the operating system. In these cases, the hardware and software license are accounted for as a single performance obligation and revenue is recognized at the point in time when ownership is transferred to resellers or directly to end customers through retail stores and online marketplaces.

Refer to Note 19 – Segment Information and Geographic Data for further information, including revenue by significant product and service offering.

Significant Judgments

Our contracts with customers often include promises to transfer multiple products and services to a customer. Determining whether products and services are considered distinct performance obligations that should be accounted for separately versus together may require significant judgment. When a cloud-based service includes both on-premises software licenses and cloud services, judgment is required to determine whether the software license is considered distinct and accounted for separately, or not distinct and accounted for together with the cloud service and recognized over time. Certain cloud services, primarily Office 365, depend on a significant level of integration, interdependency, and interrelation between the desktop applications and cloud services, and are accounted for together as one performance obligation. Revenue from Office 365 is recognized ratably over the period in which the cloud services are provided.

Judgment is required to determine the SSP for each distinct performance obligation. We use a single amount to estimate SSP for items that are not sold separately, including on-premises licenses sold with SA or software updates provided at no additional charge. We use a range of amounts to estimate SSP when we sell each of the products and services separately and need to determine whether there is a discount to be allocated based on the relative SSP of the various products and services.

In instances where SSP is not directly observable, such as when we do not sell the product or service separately, we determine the SSP using information that may include market conditions and other observable inputs. We typically have more than one SSP for individual products and services due to the stratification of those products and services by customers and circumstances. In these instances, we may use information such as the size of the customer and geographic region in determining the SSP.

Due to the various benefits from and the nature of our SA program, judgment is required to assess the pattern of delivery, including the exercise pattern of certain benefits across our portfolio of customers.

Our products are generally sold with a right of return, we may provide other credits or incentives, and in certain instances we estimate customer usage of our products and services, which are accounted for as variable consideration when determining the amount of revenue to recognize. Returns and credits are estimated at contract inception and updated at the end of each reporting period if additional information becomes available. Changes to our estimated variable consideration were not material for the periods presented.

Contract Balances and Other Receivables

Timing of revenue recognition may differ from the timing of invoicing to customers. We record a receivable when revenue is recognized prior to invoicing, or unearned revenue when revenue is recognized subsequent to invoicing. For multi-year agreements, we generally invoice customers annually at the beginning of each annual coverage period. We record a receivable related to revenue recognized for multi-year on-premises licenses as we have an unconditional right to invoice and receive payment in the future related to those licenses.

Unearned revenue comprises mainly unearned revenue related to volume licensing programs, which may include SA and cloud services. Unearned revenue is generally invoiced annually at the beginning of each contract period for multi-year agreements and recognized ratably over the coverage period. Unearned revenue also includes payments for consulting services to be performed in the future, LinkedIn subscriptions, Office 365 subscriptions, Xbox subscriptions, Windows post-delivery support, Dynamics business solutions, and other offerings for which we have been paid in advance and earn the revenue when we transfer control of the product or service.

Refer to Note 13 – Unearned Revenue for further information, including unearned revenue by segment and changes in unearned revenue during the period.

Payment terms and conditions vary by contract type, although terms generally include a requirement of payment within 30 to 60 days. In instances where the timing of revenue recognition differs from the timing of invoicing, we have determined our contracts generally do not include a significant financing component. The primary purpose of our invoicing terms is to provide customers with simplified and predictable ways of purchasing our products and services, not to receive financing from our customers or to provide customers with financing. Examples include invoicing at the beginning of a subscription term with revenue recognized ratably over the contract period, and multi-year on-premises licenses that are invoiced annually with revenue recognized upfront.

As of June 30, 2024 and 2023, long-term accounts receivable, net of allowance for doubtful accounts, was $4.9 billion and $4.5 billion, respectively, and is included in other long-term assets in our consolidated balance sheets.

The allowance for doubtful accounts reflects our best estimate of probable losses inherent in the accounts receivable balance. We determine the allowance based on known troubled accounts, historical experience, and other currently available evidence.

Activity in the allowance for doubtful accounts was as follows:

(In millions) Year Ended June 30,	2024	2023	2022
Balance, beginning of period	**$ 716**	$ 710	$ 798
Charged to costs and other	**386**	258	157
Write-offs	**(218)**	(252)	(245)
Balance, end of period	**$ 884**	$ 716	$ 710

Allowance for doubtful accounts included in our consolidated balance sheets:

(In millions) June 30,	2024	2023	2022
Accounts receivable, net of allowance for doubtful accounts	**$ 830**	$ 650	$ 633
Other long-term assets	**54**	66	77
Total	**$ 884**	$ 716	$ 710

As of June 30, 2024 and 2023, other receivables related to activities to facilitate the purchase of server components were $10.5 billion and $9.2 billion, respectively, and are included in other current assets in our consolidated balance sheets.

We record financing receivables when we offer certain customers the option to acquire our software products and services offerings through a financing program in a limited number of countries. As of June 30, 2024 and 2023, our financing receivables, net were $4.5 billion and $5.3 billion, respectively, for short-term and long-term financing receivables, which are included in other current assets and other long-term assets in our consolidated balance sheets. We record an allowance to cover expected losses based on troubled accounts, historical experience, and other currently available evidence.

Assets Recognized from Costs to Obtain a Contract with a Customer

We recognize an asset for the incremental costs of obtaining a contract with a customer if we expect the benefit of those costs to be longer than one year. We have determined that certain sales incentive programs meet the requirements to be capitalized. Total capitalized costs to obtain a contract were immaterial during the periods presented and are included in other current and long-term assets in our consolidated balance sheets.

We apply a practical expedient to expense costs as incurred for costs to obtain a contract with a customer when the amortization period would have been one year or less. These costs include our internal sales organization compensation program and certain partner sales incentive programs as we have determined annual compensation is commensurate with annual sales activities.

Cost of Revenue

Cost of revenue includes: manufacturing and distribution costs for products sold and programs licensed; operating costs related to product support service centers and product distribution centers; costs incurred to include software on PCs sold by original equipment manufacturers (“OEM”), to drive traffic to our websites, and to acquire online advertising space; costs incurred to support and maintain cloud-based and other online products and services, including datacenter costs and royalties; warranty costs; inventory valuation adjustments; costs associated with the delivery of consulting services; and the amortization of capitalized software development costs. Capitalized software development costs are amortized over the estimated lives of the products.

Product Warranty

We provide for the estimated costs of fulfilling our obligations under hardware and software warranties at the time the related revenue is recognized. For hardware warranties, we estimate the costs based on historical and projected product failure rates, historical and projected repair costs, and knowledge of specific product failures (if any). The specific hardware warranty terms and conditions vary depending upon the product sold and the country in which we do business, but generally include parts and labor over a period generally ranging from 90 days to three years. For software warranties, we estimate the costs to provide bug fixes, such as security patches, over the estimated life of the software. We regularly reevaluate our estimates to assess the adequacy of the recorded warranty liabilities and adjust the amounts as necessary.

Research and Development

Research and development expenses include payroll, employee benefits, stock-based compensation expense, and other headcount-related expenses associated with product development. Research and development expenses also include third-party development and programming costs and the amortization of purchased software code and services content. Such costs related to software development are included in research and development expense until the point that technological feasibility is reached, which for our software products, is generally shortly before the products are released to production. Once technological feasibility is reached, such costs are capitalized and amortized to cost of revenue over the estimated lives of the products.

Sales and Marketing

Sales and marketing expenses include payroll, employee benefits, stock-based compensation expense, and other headcount-related expenses associated with sales and marketing personnel, and the costs of advertising, promotions, trade shows, seminars, and other programs. Advertising costs are expensed as incurred. Advertising expense was $1.7 billion, $904 million, and $1.5 billion in fiscal years 2024, 2023, and 2022, respectively.

Stock-Based Compensation

Compensation cost for stock awards, which include restricted stock units (“RSUs”) and performance stock units (“PSUs”), is measured at the fair value on the grant date and recognized as expense, net of estimated forfeitures, over the related service or performance period. The fair value of stock awards is based on the quoted price of our common stock on the grant date less the present value of expected dividends not received during the vesting period. We measure the fair value of PSUs using a Monte Carlo valuation model. Compensation cost for RSUs is recognized using the straight-line method and for PSUs is recognized using the accelerated method.

Compensation expense for the employee stock purchase plan (“ESPP”) is measured as the discount the employee is entitled to upon purchase and is recognized in the period of purchase.

Income Taxes

Income tax expense includes U.S. and international income taxes, and interest and penalties on uncertain tax positions. Certain income and expenses are not reported in tax returns and financial statements in the same year. The tax effect of such temporary differences is reported as deferred income taxes. Deferred tax assets are reported net of a valuation allowance when it is more likely than not that a tax benefit will not be realized. All deferred income taxes are classified as long-term in our consolidated balance sheets.

Financial Instruments

Investments

We consider all highly liquid interest-earning investments with a maturity of three months or less at the date of purchase to be cash equivalents. The fair values of these investments approximate their carrying values. In general, investments with original maturities of greater than three months and remaining maturities of less than one year are classified as short-term investments. Investments with maturities beyond one year may be classified as short-term based on their highly liquid nature and because such marketable securities represent the investment of cash that is available for current operations.

Debt investments are classified as available-for-sale and realized gains and losses are recorded using the specific identification method. Changes in fair value, excluding credit losses and impairments, are recorded in other comprehensive income. Fair value is calculated based on publicly available market information or other estimates determined by management. If the cost of an investment exceeds its fair value, we evaluate, among other factors, general market conditions, credit quality of debt instrument issuers, and the extent to which the fair value is less than cost. To determine credit losses, we employ a systematic methodology that considers available quantitative and qualitative evidence. In addition, we consider specific adverse conditions related to the financial health of, and business outlook for, the investee. If we have plans to sell the security or it is more likely than not that we will be required to sell the security before recovery, then a decline in fair value below cost is recorded as an impairment charge in other income (expense), net and a new cost basis in the investment is established. If market, industry, and/or investee conditions deteriorate, we may incur future impairments.

Equity investments with readily determinable fair values are measured at fair value. Equity investments without readily determinable fair values are measured using the equity method or measured at cost with adjustments for observable changes in price or impairments (referred to as the measurement alternative). We perform a qualitative assessment on a periodic basis and recognize an impairment if there are sufficient indicators that the fair value of the investment is less than carrying value. Changes in value are recorded in other income (expense), net.

Investments that are considered variable interest entities (“VIEs”) are evaluated to determine whether we are the primary beneficiary of the VIE, in which case we would be required to consolidate the entity. We evaluate whether we have (1) the power to direct the activities that most significantly impact the VIE’s economic performance, and (2) the obligation to absorb losses or the right to receive benefits from the VIE that could potentially be significant to the VIE. We have determined we are not the primary beneficiary of any of our VIE investments. Therefore, our VIE investments are not consolidated and the majority are accounted for under the equity method of accounting.

Derivatives

Derivative instruments are recognized as either assets or liabilities and measured at fair value. The accounting for changes in the fair value of a derivative depends on the intended use of the derivative and the resulting designation.

For derivative instruments designated as fair value hedges, gains and losses are recognized in other income (expense), net with offsetting gains and losses on the hedged items. Gains and losses representing hedge components excluded from the assessment of effectiveness are recognized in other income (expense), net.

For derivative instruments designated as cash flow hedges, gains and losses are initially reported as a component of other comprehensive income and subsequently recognized in other income (expense), net with the corresponding hedged item. Gains and losses representing hedge components excluded from the assessment of effectiveness are recognized in other income (expense), net.

For derivative instruments that are not designated as hedges, gains and losses from changes in fair values are primarily recognized in other income (expense), net.

Fair Value Measurements

We account for certain assets and liabilities at fair value. The hierarchy below lists three levels of fair value based on the extent to which inputs used in measuring fair value are observable in the market. We categorize each of our fair value measurements in one of these three levels based on the lowest level input that is significant to the fair value measurement in its entirety. These levels are:

- *Level 1* – inputs are based upon unadjusted quoted prices for identical instruments in active markets. Our Level 1 investments include U.S. government securities, common and preferred stock, and mutual funds. Our Level 1 derivative assets and liabilities include those actively traded on exchanges.

- *Level 2* – inputs are based upon quoted prices for similar instruments in active markets, quoted prices for identical or similar instruments in markets that are not active, and model-based valuation techniques (e.g. the Black-Scholes model) for which all significant inputs are observable in the market or can be corroborated by observable market data for substantially the full term of the assets or liabilities. Where applicable, these models project future cash flows and discount the future amounts to a present value using market-based observable inputs including interest rate curves, credit spreads, foreign exchange rates, and forward and spot prices for currencies. Our Level 2 investments include commercial paper, certificates of deposit, U.S. agency securities, foreign government bonds, mortgage- and asset-backed securities, corporate notes and bonds, and municipal securities. Our Level 2 derivative assets and liabilities include certain cleared swap contracts and over-the-counter forward, option, and swap contracts.
- *Level 3* – inputs are generally unobservable and typically reflect management's estimates of assumptions that market participants would use in pricing the asset or liability. The fair values are therefore determined using model-based techniques, including option pricing models and discounted cash flow models. Our Level 3 assets and liabilities include investments in corporate notes and bonds, municipal securities, and goodwill and intangible assets, when they are recorded at fair value due to an impairment charge. Unobservable inputs used in the models are significant to the fair values of the assets and liabilities.

We measure equity investments without readily determinable fair values on a nonrecurring basis. The fair values of these investments are determined based on valuation techniques using the best information available, and may include quoted market prices, market comparables, and discounted cash flow projections.

Our other current financial assets and current financial liabilities have fair values that approximate their carrying values.

Inventories

Inventories are stated at average cost, subject to the lower of cost or net realizable value. Cost includes materials, labor, and manufacturing overhead related to the purchase and production of inventories. Net realizable value is the estimated selling price less estimated costs of completion, disposal, and transportation. We regularly review inventory quantities on hand, future purchase commitments with our suppliers, and the estimated utility of our inventory. If our review indicates a reduction in utility below carrying value, we reduce our inventory to a new cost basis through a charge to cost of revenue.

Property and Equipment

Property and equipment is stated at cost less accumulated depreciation, and depreciated using the straight-line method over the shorter of the estimated useful life of the asset or the lease term. The estimated useful lives of our property and equipment are generally as follows: computer software developed or acquired for internal use, three years; computer equipment, two to six years; buildings and improvements, five to 15 years; leasehold improvements, three to 20 years; and furniture and equipment, one to 10 years. Land is not depreciated.

Leases

We determine if an arrangement is a lease at inception. Operating leases are included in operating lease right-of-use ("ROU") assets, other current liabilities, and operating lease liabilities in our consolidated balance sheets. Finance leases are included in property and equipment, other current liabilities, and other long-term liabilities in our consolidated balance sheets.

ROU assets represent our right to use an underlying asset for the lease term and lease liabilities represent our obligation to make lease payments arising from the lease. Operating lease ROU assets and liabilities are recognized at commencement date based on the present value of lease payments over the lease term. As most of our leases do not provide an implicit rate, we generally use our incremental borrowing rate based on the estimated rate of interest for collateralized borrowing over a similar term of the lease payments at commencement date. The operating lease ROU asset also includes any lease payments made and excludes lease incentives. Our lease terms may include options to extend or terminate the lease when it is reasonably certain that we will exercise that option. Lease expense for lease payments is recognized on a straight-line basis over the lease term.

We have lease agreements with lease and non-lease components, which are generally accounted for separately. For certain equipment leases, such as vehicles, we account for the lease and non-lease components as a single lease component. Additionally, for certain equipment leases, we apply a portfolio approach to effectively account for the operating lease ROU assets and liabilities.

Goodwill

Goodwill is tested for impairment at the reporting unit level (operating segment or one level below an operating segment) on an annual basis (May 1) and between annual tests if an event occurs or circumstances change that would more likely than not reduce the fair value of a reporting unit below its carrying value.

Intangible Assets

Our intangible assets are subject to amortization and are amortized over the estimated useful life in proportion to the economic benefits received. We evaluate the recoverability of intangible assets periodically by taking into account events or circumstances that may warrant revised estimates of useful lives or that indicate the asset may be impaired.

Related Party Transactions

In March 2024, we entered into an agreement with Inflection AI, Inc. (“Inflection”), pursuant to which we obtained a non-exclusive license to Inflection’s intellectual property. Reid Hoffman, a member of our Board of Directors, is a co-founder of and serves on the board of directors of Inflection. As of the date of the agreement with Inflection, Reprogrammed Interchange LLC (“Reprogrammed”) and entities affiliated with Greylock Ventures (“Greylock”) each held less than a 10% equity interest in Inflection. Mr. Hoffman may be deemed to beneficially own the shares held by Reprogrammed and Greylock by virtue of his relationship with such entities. Mr. Hoffman did not participate in any portions of the meetings of our Board of Directors or any committee thereof to review and approve the transaction with Inflection.

Recent Accounting Guidance

Segment Reporting – Improvements to Reportable Segment Disclosures

In November 2023, the Financial Accounting Standards Board (“FASB”) issued a new standard to improve reportable segment disclosures. The guidance expands the disclosures required for reportable segments in our annual and interim consolidated financial statements, primarily through enhanced disclosures about significant segment expenses. The standard will be effective for us beginning with our annual reporting for fiscal year 2025 and interim periods thereafter, with early adoption permitted. We are currently evaluating the impact of this standard on our segment disclosures.

Income Taxes – Improvements to Income Tax Disclosures

In December 2023, the FASB issued a new standard to improve income tax disclosures. The guidance requires disclosure of disaggregated income taxes paid, prescribes standardized categories for the components of the effective tax rate reconciliation, and modifies other income tax-related disclosures. The standard will be effective for us beginning with our annual reporting for fiscal year 2026, with early adoption permitted. We are currently evaluating the impact of this standard on our income tax disclosures.

NOTE 2 — EARNINGS PER SHARE

Basic earnings per share (“EPS”) is computed based on the weighted average number of shares of common stock outstanding during the period. Diluted EPS is computed based on the weighted average number of shares of common stock plus the effect of dilutive potential common shares outstanding during the period using the treasury stock method. Dilutive potential common shares include outstanding stock options and stock awards.

The components of basic and diluted EPS were as follows:

(In millions, except per share amounts)

Year Ended June 30,	2024	2023	2022
Net income available for common shareholders (A)	**$ 88,136**	$ 72,361	$ 72,738
Weighted average outstanding shares of common stock (B)	**7,431**	7,446	7,496
Dilutive effect of stock-based awards	**38**	26	44
Common stock and common stock equivalents (C)	**7,469**	7,472	7,540
Earnings Per Share			
Basic (A/B)	**$ 11.86**	$ 9.72	$ 9.70
Diluted (A/C)	**$ 11.80**	$ 9.68	$ 9.65

Anti-dilutive stock-based awards excluded from the calculations of diluted EPS were immaterial during the periods presented.

NOTE 3 — OTHER INCOME (EXPENSE), NET

The components of other income (expense), net were as follows:

(In millions)

Year Ended June 30,	2024	2023	2022
Interest and dividends income	**$ 3,157**	$ 2,994	$ 2,094
Interest expense	**(2,935)**	(1,968)	(2,063)
Net recognized gains (losses) on investments	**(118)**	260	461
Net losses on derivatives	**(187)**	(456)	(52)
Net gains (losses) on foreign currency remeasurements	**(244)**	181	(75)
Other, net	**(1,319)**	(223)	(32)
Total	**$ (1,646)**	$ 788	$ 333

Other, net primarily reflects net recognized losses on equity method investments.

Net Recognized Gains (Losses) on Investments

Net recognized gains (losses) on debt investments were as follows:

(In millions)

Year Ended June 30,	2024	2023	2022
Realized gains from sales of available-for-sale securities	**$ 22**	$ 36	$ 162
Realized losses from sales of available-for-sale securities	**(98)**	(124)	(138)
Impairments and allowance for credit losses	**23**	(10)	(81)
Total	**$ (53)**	$ (98)	$ (57)

Net recognized gains (losses) on equity investments were as follows:

(In millions)

Year Ended June 30,	2024	2023	2022
Net realized gains on investments sold	**$ 18**	$ 75	$ 29
Net unrealized gains on investments still held	**146**	303	509
Impairments of investments	**(229)**	(20)	(20)
Total	**$ (65)**	$ 358	$ 518

NOTE 4 — INVESTMENTS

Investment Components

The components of investments were as follows:

(In millions)	Fair Value Level	Adjusted Cost Basis	Unrealized Gains	Unrealized Losses	Recorded Basis	Cash and Cash Equivalents	Short-term Investments	Equity and Other Investments
June 30, 2024								
Changes in Fair Value Recorded in Other Comprehensive Income								
Commercial paper	Level 2	$ 4,666	$ 0	$ 0	$ 4,666	$ 4,666	$ 0	$ 0
Certificates of deposit	Level 2	1,547	0	0	1,547	1,503	44	0
U.S. government securities	Level 1	49,603	4	(2,948)	46,659	14	46,645	0
U.S. agency securities	Level 2	17	0	0	17	0	17	0
Foreign government bonds	Level 2	319	3	(16)	306	0	306	0
Mortgage- and asset-backed securities	Level 2	944	3	(35)	912	0	912	0
Corporate notes and bonds	Level 2	9,106	28	(318)	8,816	0	8,816	0
Corporate notes and bonds	Level 3	1,641	0	(1)	1,640	0	140	1,500
Municipal securities	Level 2	262	0	(13)	249	0	249	0
Municipal securities	Level 3	104	0	(17)	87	0	87	0
Total debt investments		$ 68,209	$ 38	$ (3,348)	$ 64,899	$ 6,183	$ 57,216	$ 1,500
Changes in Fair Value Recorded in Net Income								
Equity investments	Level 1				$ 3,547	$ 561	$ 0	$ 2,986
Equity investments	Other				10,114	0	0	10,114
Total equity investments					$ 13,661	$ 561	$ 0	$ 13,100
Cash					$ 11,571	$ 11,571	$ 0	$ 0
Derivatives, net [(a)]					12	0	12	0
Total					$ 90,143	$ 18,315	$ 57,228	$ 14,600

(In millions)	Fair Value Level	Adjusted Cost Basis	Unrealized Gains	Unrealized Losses	Recorded Basis	Cash and Cash Equivalents	Short-term Investments	Equity and Other Investments
June 30, 2023								
Changes in Fair Value Recorded in Other Comprehensive Income								
Commercial paper	Level 2	$ 16,589	$ 0	$ 0	$ 16,589	$ 12,231	$ 4,358	$ 0
Certificates of deposit	Level 2	2,701	0	0	2,701	2,657	44	0
U.S. government securities	Level 1	65,237	2	(3,870)	61,369	2,991	58,378	0
U.S. agency securities	Level 2	2,703	0	0	2,703	894	1,809	0
Foreign government bonds	Level 2	498	1	(24)	475	0	475	0
Mortgage- and asset-backed securities	Level 2	824	1	(39)	786	0	786	0
Corporate notes and bonds	Level 2	10,809	8	(583)	10,234	0	10,234	0
Corporate notes and bonds	Level 3	120	0	0	120	0	120	0
Municipal securities	Level 2	285	1	(18)	268	7	261	0
Municipal securities	Level 3	103	0	(16)	87	0	87	0
Total debt investments		$ 99,869	$ 13	$ (4,550)	$ 95,332	$ 18,780	$ 76,552	$ 0
Changes in Fair Value Recorded in Net Income								
Equity investments	Level 1				$ 10,138	$ 7,446	$ 0	$ 2,692
Equity investments	Other				7,187	0	0	7,187
Total equity investments					$ 17,325	$ 7,446	$ 0	$ 9,879
Cash					$ 8,478	$ 8,478	$ 0	$ 0
Derivatives, net [(a)]					6	0	6	0
Total					$ 121,141	$ 34,704	$ 76,558	$ 9,879

(a) *Refer to Note 5 – Derivatives for further information on the fair value of our derivative instruments.*

Equity investments presented as “Other” in the tables above include investments without readily determinable fair values measured using the equity method or measured at cost with adjustments for observable changes in price or impairments, and investments measured at fair value using net asset value as a practical expedient which are not categorized in the fair value hierarchy. As of June 30, 2024 and 2023, equity investments without readily determinable fair values measured at cost with adjustments for observable changes in price or impairments were $3.9 billion and $4.2 billion, respectively.

Unrealized Losses on Debt Investments

Debt investments with continuous unrealized losses for less than 12 months and 12 months or greater and their related fair values were as follows:

	Less than 12 Months		12 Months or Greater			
(In millions)	Fair Value	Unrealized Losses	Fair Value	Unrealized Losses	Total Fair Value	Total Unrealized Losses
June 30, 2024						
U.S. government and agency securities	**$ 529**	**$ (12)**	**$ 45,821**	**$ (2,936)**	**$ 46,350**	**$ (2,948)**
Foreign government bonds	**79**	**(2)**	**180**	**(14)**	**259**	**(16)**
Mortgage- and asset-backed securities	**201**	**(1)**	**409**	**(34)**	**610**	**(35)**
Corporate notes and bonds	**1,310**	**(9)**	**5,779**	**(310)**	**7,089**	**(319)**
Municipal securities	**38**	**(1)**	**243**	**(29)**	**281**	**(30)**
Total	**$ 2,157**	**$ (25)**	**$ 52,432**	**$ (3,323)**	**$ 54,589**	**$ (3,348)**

	Less than 12 Months		12 Months or Greater			
(In millions)	Fair Value	Unrealized Losses	Fair Value	Unrealized Losses	Total Fair Value	Total Unrealized Losses
June 30, 2023						
U.S. government and agency securities	$ 7,950	$ (336)	$ 45,273	$ (3,534)	$ 53,223	$ (3,870)
Foreign government bonds	77	(5)	391	(19)	468	(24)
Mortgage- and asset-backed securities	257	(5)	412	(34)	669	(39)
Corporate notes and bonds	2,326	(49)	7,336	(534)	9,662	(583)
Municipal securities	111	(3)	186	(31)	297	(34)
Total	$ 10,721	$ (398)	$ 53,598	$ (4,152)	$ 64,319	$ (4,550)

Unrealized losses from fixed-income securities are primarily attributable to changes in interest rates. Management does not believe any remaining unrealized losses represent impairments based on our evaluation of available evidence.

Debt Investment Maturities

The following table outlines maturities of our debt investments as of June 30, 2024:

(In millions)	Adjusted Cost Basis	Estimated Fair Value
June 30, 2024		
Due in one year or less	**$ 19,815**	**$ 19,596**
Due after one year through five years	**38,954**	**36,779**
Due after five years through 10 years	**8,028**	**7,242**
Due after 10 years	**1,412**	**1,282**
Total	**$ 68,209**	**$ 64,899**

NOTE 5 — DERIVATIVES

We use derivative instruments to manage risks related to foreign currencies, interest rates, equity prices, and credit; to enhance investment returns; and to facilitate portfolio diversification. Our objectives for holding derivatives include reducing, eliminating, and efficiently managing the economic impact of these exposures as effectively as possible. Our derivative programs include strategies that both qualify and do not qualify for hedge accounting treatment.

Foreign Currencies

Certain forecasted transactions, assets, and liabilities are exposed to foreign currency risk. We monitor our foreign currency exposures daily to maximize the economic effectiveness of our foreign currency hedge positions.

Foreign currency risks related to certain non-U.S. dollar-denominated investments are hedged using foreign exchange forward contracts that are designated as fair value hedging instruments. Foreign currency risks related to certain Euro-denominated debt are hedged using foreign exchange forward contracts that are designated as cash flow hedging instruments.

Certain options and forwards not designated as hedging instruments are also used to manage the variability in foreign exchange rates on certain balance sheet amounts and to manage other foreign currency exposures.

Interest Rate

Interest rate risks related to certain fixed-rate debt are hedged using interest rate swaps that are designated as fair value hedging instruments to effectively convert the fixed interest rates to floating interest rates.

Securities held in our fixed-income portfolio are subject to different interest rate risks based on their maturities. We manage the average maturity of our fixed-income portfolio to achieve economic returns that correlate to certain broad-based fixed-income indices using option, futures, and swap contracts. These contracts are not designated as hedging instruments and are included in “Other contracts” in the tables below.

Equity

Securities held in our equity investments portfolio are subject to market price risk. At times, we may hold options, futures, and swap contracts. These contracts are not designated as hedging instruments.

Credit

Our fixed-income portfolio is diversified and consists primarily of investment-grade securities. We use credit default swap contracts to manage credit exposures relative to broad-based indices and to facilitate portfolio diversification. These contracts are not designated as hedging instruments and are included in “Other contracts” in the tables below.

Credit-Risk-Related Contingent Features

Certain counterparty agreements for derivative instruments contain provisions that require our issued and outstanding long-term unsecured debt to maintain an investment grade credit rating and require us to maintain minimum liquidity of $1.0 billion. To the extent we fail to meet these requirements, we will be required to post collateral, similar to the standard convention related to over-the-counter derivatives. As of June 30, 2024, our long-term unsecured debt rating was AAA, and cash investments were in excess of $1.0 billion. As a result, no collateral was required to be posted.

The following table presents the notional amounts of our outstanding derivative instruments measured in U.S. dollar equivalents:

(In millions)	June 30, 2024	June 30, 2023
Designated as Hedging Instruments		
Foreign exchange contracts purchased	**$ 1,492**	$ 1,492
Interest rate contracts purchased	**1,100**	1,078
Not Designated as Hedging Instruments		
Foreign exchange contracts purchased	**7,167**	7,874
Foreign exchange contracts sold	**31,793**	25,159
Equity contracts purchased	**4,016**	3,867
Equity contracts sold	**2,165**	2,154
Other contracts purchased	**2,113**	1,224
Other contracts sold	**811**	581

Fair Values of Derivative Instruments

The following table presents our derivative instruments:

(In millions)	Derivative Assets	Derivative Liabilities	Derivative Assets	Derivative Liabilities
		June 30, 2024		June 30, 2023
Designated as Hedging Instruments				
Foreign exchange contracts	**$ 24**	**$ (76)**	$ 34	$ (67)
Interest rate contracts	**19**	**0**	16	0
Not Designated as Hedging Instruments				
Foreign exchange contracts	**213**	**(230)**	249	(332)
Equity contracts	**63**	**(491)**	165	(400)
Other contracts	**12**	**(3)**	5	(6)
Gross amounts of derivatives	**331**	**(800)**	469	(805)
Gross amounts of derivatives offset in the balance sheets	**(151)**	**152**	(202)	206
Cash collateral received	**0**	**(104)**	0	(125)
Net amounts of derivatives	**$ 180**	**$ (752)**	$ 267	$ (724)
Reported as				
Short-term investments	**$ 12**	**$ 0**	$ 6	$ 0
Other current assets	**149**	**0**	245	0
Other long-term assets	**19**	**0**	16	0
Other current liabilities	**0**	**(401)**	0	(341)
Other long-term liabilities	**0**	**(351)**	0	(383)
Total	**$ 180**	**$ (752)**	$ 267	$ (724)

Gross derivative assets and liabilities subject to legally enforceable master netting agreements for which we have elected to offset were $304 million and $800 million, respectively, as of June 30, 2024, and $442 million and $804 million, respectively, as of June 30, 2023.

The following table presents the fair value of our derivatives instruments on a gross basis:

(In millions)	Level 1	Level 2	Level 3	Total
June 30, 2024				
Derivative assets	**$ 0**	**$ 327**	**$ 4**	**$ 331**
Derivative liabilities	**(1)**	**(799)**	**0**	**(800)**
June 30, 2023				
Derivative assets	0	462	7	469
Derivative liabilities	0	(805)	0	(805)

Gains (losses) on derivative instruments recognized in other income (expense), net were as follows:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Designated as Fair Value Hedging Instruments			
Foreign exchange contracts			
Derivatives	**$ 0**	$ 0	$ 49
Hedged items	**0**	0	(50)
Excluded from effectiveness assessment	**0**	0	4
Interest rate contracts			
Derivatives	**(23)**	(65)	(92)
Hedged items	**(25)**	38	108
Designated as Cash Flow Hedging Instruments			
Foreign exchange contracts			
Amount reclassified from accumulated other comprehensive loss	**(48)**	61	(79)
Not Designated as Hedging Instruments			
Foreign exchange contracts	**367**	(73)	383
Equity contracts	**(177)**	(420)	13
Other contracts	**(15)**	(41)	(85)

Gains (losses), net of tax, on derivative instruments recognized in our consolidated comprehensive income statements were as follows:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Designated as Cash Flow Hedging Instruments			
Foreign exchange contracts			
Included in effectiveness assessment	**$ (14)**	$ 34	$ (57)

NOTE 6 — INVENTORIES

The components of inventories were as follows:

(In millions)

June 30,	**2024**	**2023**
Raw materials	**$ 394**	$ 709
Work in process	**7**	23
Finished goods	**845**	1,768
Total	**$ 1,246**	$ 2,500

NOTE 7 — PROPERTY AND EQUIPMENT

The components of property and equipment were as follows:

(In millions)

June 30,	2024	2023
Land	**$ 8,163**	$ 5,683
Buildings and improvements	**93,943**	68,465
Leasehold improvements	**9,594**	8,537
Computer equipment and software	**93,780**	74,961
Furniture and equipment	**6,532**	6,246
Total, at cost	**212,012**	163,892
Accumulated depreciation	**(76,421)**	(68,251)
Total, net	**$ 135,591**	$ 95,641

During fiscal years 2024, 2023, and 2022, depreciation expense was $15.2 billion, $11.0 billion, and $12.6 billion, respectively.

As of June 30, 2024, we have committed $35.4 billion for the construction of new buildings, building improvements, and leasehold improvements, primarily related to datacenters.

NOTE 8 — BUSINESS COMBINATIONS

Activision Blizzard, Inc.

On October 13, 2023, we completed our acquisition of Activision Blizzard, Inc. (“Activision Blizzard”) for a total purchase price of $75.4 billion, consisting primarily of cash. Activision Blizzard is a leader in game development and an interactive entertainment content publisher. The acquisition will accelerate the growth in our gaming business across mobile, PC, console, and cloud gaming. The financial results of Activision Blizzard have been included in our consolidated financial statements since the date of the acquisition. Activision Blizzard is reported as part of our More Personal Computing segment.

The purchase price allocation as of the date of acquisition was based on a preliminary valuation and is subject to revision as more detailed analyses are completed and additional information about the fair value of assets acquired and liabilities assumed becomes available. The primary areas that remain preliminary relate to the fair values of goodwill and income taxes.

The major classes of assets and liabilities to which we have preliminarily allocated the purchase price were as follows:

(In millions)

Cash and cash equivalents	$ 12,976
Goodwill	50,969
Intangible assets	21,969
Other assets	2,501
Long-term debt	(2,799)
Long-term income taxes	(1,914)
Deferred income taxes	(4,677)
Other liabilities	(3,617)
Total purchase price	$ 75,408

Goodwill was assigned to our More Personal Computing segment. The goodwill was primarily attributed to increased synergies that are expected to be achieved from the integration of Activision Blizzard. Substantially all of the goodwill is expected to be non-deductible for income tax purposes.

Following are the details of the purchase price allocated to the intangible assets acquired:

(In millions, except average life)		Amount	Weighted Average Life
Marketing-related	$	11,619	24 years
Technology-based		9,689	4 years
Customer-related		661	4 years
Fair value of intangible assets acquired	$	21,969	15 years

Following is the net impact of the Activision Blizzard acquisition on our consolidated income statements since the date of acquisition:

(In millions) Year Ended June 30,		2024
Revenue	**$**	**5,729**
Operating loss		**(1,362)**

The change of Activision Blizzard content from third-party to first-party is reflected in the net impact.

Following are the supplemental consolidated financial results of Microsoft Corporation on an unaudited pro forma basis, as if the acquisition had been consummated on July 1, 2022:

(In millions, except per share amounts) Year Ended June 30,		2024		2023
Revenue	**$**	**247,442**	$	219,790
Net income		**88,308**		71,383
Diluted earnings per share		**11.82**		9.55

These pro forma results were based on estimates and assumptions, which we believe are reasonable. They are not the results that would have been realized had we been a combined company during the periods presented and are not necessarily indicative of our consolidated results of operations in future periods. The pro forma results include adjustments related to purchase accounting, primarily amortization of intangible assets. Acquisition costs and other nonrecurring charges were immaterial and are included in the earliest period presented.

Nuance Communications, Inc.

On March 4, 2022, we completed our acquisition of Nuance Communications, Inc. (“Nuance”) for a total purchase price of $18.8 billion, consisting primarily of cash. Nuance is a cloud and artificial intelligence (“AI”) software provider with healthcare and enterprise AI experience, and the acquisition will build on our industry-specific cloud offerings. The financial results of Nuance have been included in our consolidated financial statements since the date of the acquisition. Nuance is reported as part of our Intelligent Cloud segment.

The allocation of the purchase price to goodwill was completed as of December 31, 2022. The major classes of assets and liabilities to which we have allocated the purchase price were as follows:

(In millions)		
Goodwill [(a)]	$	16,326
Intangible assets		4,365
Other assets		42
Other liabilities [(b)]		(1,972)
Total	$	18,761

(a) *Goodwill was assigned to our Intelligent Cloud segment and was primarily attributed to increased synergies that are expected to be achieved from the integration of Nuance. None of the goodwill is expected to be deductible for income tax purposes.*
(b) *Includes $986 million of convertible senior notes issued by Nuance in 2015 and 2017, substantially all of which have been redeemed.*

Following are the details of the purchase price allocated to the intangible assets acquired:

(In millions, except average life)		Amount	Weighted Average Life
Customer-related	$	2,610	9 years
Technology-based		1,540	5 years
Marketing-related		215	4 years
Total	$	4,365	7 years

NOTE 9 — GOODWILL

Changes in the carrying amount of goodwill were as follows:

(In millions)		June 30, 2022		Acquisitions		Other		June 30, 2023		Acquisitions		Other		June 30, 2024
Productivity and Business Processes	$	24,811	$	11	$	(47)	**$**	**24,775**	**$**	**0**	**$**	**2**	**$**	**24,777**
Intelligent Cloud		30,182		223		64		**30,469**		**0**		**(28)**		**30,441**
More Personal Computing		12,531		0		111		**12,642**		**51,235** [(a)]		**125** [(a)]		**64,002**
Total	$	67,524	$	234	$	128	**$**	**67,886**	**$**	**51,235**	**$**	**99**	**$**	**119,220**

(a) *Includes goodwill of $51.0 billion related to Activision Blizzard. See Note 8 – Business Combinations for further information.*

The measurement periods for the valuation of assets acquired and liabilities assumed end as soon as information on the facts and circumstances that existed as of the acquisition dates becomes available, but do not exceed 12 months. Adjustments in purchase price allocations may require a change in the amounts allocated to goodwill during the periods in which the adjustments are determined.

Any change in the goodwill amounts resulting from foreign currency translations and purchase accounting adjustments are presented as “Other” in the table above. Also included in “Other” are business dispositions and transfers between segments due to reorganizations, as applicable.

Goodwill Impairment

We test goodwill for impairment annually on May 1 at the reporting unit level, primarily using a discounted cash flow methodology with a peer-based, risk-adjusted weighted average cost of capital. We believe use of a discounted cash flow approach is the most reliable indicator of the fair values of the businesses.

No instances of impairment were identified in our May 1, 2024, May 1, 2023, or May 1, 2022 tests. As of June 30, 2024 and 2023, accumulated goodwill impairment was $11.3 billion.

NOTE 10 — INTANGIBLE ASSETS

The components of intangible assets, all of which are finite-lived, were as follows:

(In millions)	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount
June 30,			2024			2023
Marketing-related	**$ 16,500**	**$ (3,101)**	**$ 13,399**	$ 4,935	$ (2,473)	$ 2,462
Technology-based	**21,913**	**(10,741)**	**11,172**	11,245	(7,589)	3,656
Customer-related	**6,038**	**(3,051)**	**2,987**	7,281	(4,047)	3,234
Contract-based	**58**	**(19)**	**39**	29	(15)	14
Total	**$ 44,509** [(a)]	**$ (16,912)**	**$ 27,597**	$ 23,490	$ (14,124)	$ 9,366

(a) *Includes intangible assets of $22.0 billion related to Activision Blizzard. See Note 8 – Business Combinations for further information.*

No material impairments of intangible assets were identified during fiscal years 2024, 2023, or 2022. We estimate that we have no significant residual value related to our intangible assets.

The components of intangible assets acquired during the periods presented were as follows:

(In millions)	Amount	Weighted Average Life	Amount	Weighted Average Life
Year Ended June 30,	2024		2023	
Marketing-related	**$ 11,619**	**24 years**	$ 7	5 years
Technology-based	**10,947**	**4 years**	522	7 years
Customer-related	**660**	**4 years**	0	0 years
Contract-based	**38**	**4 years**	12	3 years
Total	**$ 23,264**	**14 years**	$ 541	6 years

Intangible assets amortization expense was $4.8 billion, $2.5 billion, and $2.0 billion for fiscal years 2024, 2023, and 2022, respectively.

The following table outlines the estimated future amortization expense related to intangible assets held as of June 30, 2024:

(In millions)	
Year Ending June 30,	
2025	**$ 5,892**
2026	**4,471**
2027	**2,793**
2028	**1,909**
2029	**1,728**
Thereafter	**10,804**
Total	**$ 27,597**

NOTE 11 — DEBT

Short-term Debt

As of June 30, 2024, we had $6.7 billion of commercial paper issued and outstanding, with a weighted average interest rate of 5.4% and maturities ranging from 28 days to 152 days. The estimated fair value of this commercial paper approximates its carrying value. As of June 30, 2023, we had no commercial paper issued or outstanding.

Long-term Debt

The components of long-term debt were as follows:

(In millions, issuance by calendar year)	Maturities (calendar year)	Stated Interest Rate	Effective Interest Rate	June 30, 2024	June 30, 2023
2009 issuance of $3.8 billion	**2039**	**5.20%**	**5.24%**	**$ 520**	$ 520
2010 issuance of $4.8 billion	**2040**	**4.50%**	**4.57%**	**486**	486
2011 issuance of $2.3 billion	**2041**	**5.30%**	**5.36%**	**718**	718
2012 issuance of $2.3 billion	**2042**	**3.50%**	**3.57%**	**454**	454
2013 issuance of $5.2 billion	**2043**	**3.75%–4.88%**	**3.83%–4.92%**	**314**	1,814
2013 issuance of €4.1 billion	**2028–2033**	**2.63%–3.13%**	**2.69%–3.22%**	**2,465**	2,509
2015 issuance of $23.8 billion	**2025–2055**	**2.70%–4.75%**	**2.77%–4.78%**	**9,805**	9,805
2016 issuance of $19.8 billion	**2026–2056**	**2.40%–3.95%**	**2.46%–4.03%**	**7,930**	9,430
2017 issuance of $17.1 billion (a)	**2026–2057**	**3.30%–4.50%**	**3.38%–5.49%**	**6,833**	8,945
2020 issuance of $10.1 billion (a)	**2030–2060**	**1.35%–2.68%**	**2.53%–5.43%**	**10,111**	10,000
2021 issuance of $8.2 billion	**2052–2062**	**2.92%–3.04%**	**2.92%–3.04%**	**8,185**	8,185
2023 issuance of $0.1 billion (a)	**2026–2050**	**1.35%–4.50%**	**5.16%–5.49%**	**56**	0
2024 issuance of $3.3 billion (a)	**2026–2050**	**1.35%–4.50%**	**5.16%–5.49%**	**3,344**	0
Total face value				**51,221**	52,866
Unamortized discount and issuance costs				**(1,227)**	(438)
Hedge fair value adjustments (b)				**(81)**	(106)
Premium on debt exchange				**(4,976)**	(5,085)
Total debt				**44,937**	47,237
Current portion of long-term debt				**(2,249)**	(5,247)
Long-term debt				**$ 42,688**	$ 41,990

(a) *Includes $3.6 billion of debt at face value related to the Activision Blizzard acquisition, the majority of which was exchanged for Microsoft registered securities in June 2024. See Note 8 – Business Combinations for further information.*
(b) *Refer to Note 5 – Derivatives for further information on the interest rate swaps related to fixed-rate debt.*

As of June 30, 2024 and 2023, the estimated fair value of long-term debt, including the current portion, was $42.3 billion and $46.2 billion, respectively. The estimated fair values are based on Level 2 inputs.

Debt in the table above is comprised of senior unsecured obligations and ranks equally with our other outstanding obligations. Interest is paid semi-annually, except for the Euro-denominated debt, which is paid annually. Cash paid for interest on our debt for fiscal years 2024, 2023, and 2022 was $1.7 billion, $1.7 billion, and $1.9 billion, respectively.

The following table outlines maturities of our long-term debt, including the current portion, as of June 30, 2024:

(In millions)		
Year Ending June 30,		
2025	**$**	**2,250**
2026		**3,000**
2027		**9,250**
2028		**0**
2029		**1,876**
Thereafter		**34,845**
Total	**$**	**51,221**

NOTE 12 — INCOME TAXES

Provision for Income Taxes

The components of the provision for income taxes were as follows:

(In millions)						
Year Ended June 30,		**2024**		**2023**		**2022**
Current Taxes						
U.S. federal	**$**	**12,165**	$	14,009	$	8,329
U.S. state and local		**2,366**		2,322		1,679
Foreign		**9,858**		6,678		6,672
Current taxes	**$**	**24,389**	$	23,009	$	16,680
Deferred Taxes						
U.S. federal	**$**	**(4,791)**	$	(6,146)	$	(4,815)
U.S. state and local		**(379)**		(477)		(1,062)
Foreign		**432**		564		175
Deferred taxes	**$**	**(4,738)**	$	(6,059)	$	(5,702)
Provision for income taxes	**$**	**19,651**	$	16,950	$	10,978

U.S. and foreign components of income before income taxes were as follows:

(In millions)						
Year Ended June 30,		**2024**		**2023**		**2022**
U.S.	**$**	**62,886**	$	52,917	$	47,837
Foreign		**44,901**		36,394		35,879
Income before income taxes	**$**	**107,787**	$	89,311	$	83,716

Effective Tax Rate

The items accounting for the difference between income taxes computed at the U.S. federal statutory rate and our effective rate were as follows:

Year Ended June 30,	2024	2023	2022
Federal statutory rate	**21.0%**	21.0%	21.0%
Effect of:			
Foreign earnings taxed at lower rates	**(1.4)%**	(1.8)%	(1.3)%
Impact of intangible property transfers	**0%**	0%	(3.9)%
Foreign-derived intangible income deduction	**(1.1)%**	(1.3)%	(1.1)%
State income taxes, net of federal benefit	**1.5%**	1.6%	1.4%
Research and development credit	**(1.1)%**	(1.1)%	(0.9)%
Excess tax benefits relating to stock-based compensation	**(1.1)%**	(0.7)%	(1.9)%
Interest, net	**1.1%**	0.8%	0.5%
Other reconciling items, net	**(0.7)%**	0.5%	(0.7)%
Effective rate	**18.2%**	19.0%	13.1%

In the first quarter of fiscal year 2022, we transferred certain intangible properties from our Puerto Rico subsidiary to the U.S. The transfer of intangible properties resulted in a $3.3 billion net income tax benefit in the first quarter of fiscal year 2022, as the value of future U.S. tax deductions exceeded the current tax liability from the U.S. global intangible low-taxed income (“GILTI”) tax.

The decrease from the federal statutory rate in fiscal year 2024 and 2023 is primarily due to earnings taxed at lower rates in foreign jurisdictions resulting from producing and distributing our products and services through our foreign regional operations center in Ireland. The decrease from the federal statutory rate in fiscal year 2022 is primarily due to the net income tax benefit related to the transfer of intangible properties, earnings taxed at lower rates in foreign jurisdictions resulting from producing and distributing our products and services through our foreign regional operations center in Ireland, and tax benefits relating to stock-based compensation. In fiscal years 2024 and 2023, our foreign regional operating center in Ireland, which is taxed at a rate lower than the U.S. rate, generated 83% and 81% of our foreign income before tax. In fiscal year 2022, our foreign regional operating centers in Ireland and Puerto Rico, which are taxed at rates lower than the U.S. rate, generated 71% of our foreign income before tax. Other reconciling items, net consists primarily of tax credits and GILTI tax, and in fiscal year 2024, includes tax benefits from tax law changes. In fiscal year 2024, tax benefits from tax law changes primarily relates to the issuance of Notice 2023-55 and Notice 2023-80 by the Internal Revenue Service (“IRS”) and U.S. Treasury Department. Notice 2023-55, issued in the first quarter of fiscal year 2024, delayed the effective date of final foreign tax credit regulations to fiscal year 2024 for Microsoft. Notice 2023-80, issued in the second quarter of fiscal year 2024, further delayed the effective date of final foreign tax credit regulations indefinitely. In fiscal years 2024, 2023, and 2022, there were no individually significant other reconciling items.

The decrease in our effective tax rate for fiscal year 2024 compared to fiscal year 2023 was primarily due to tax benefits from tax law changes, including the delay of the effective date of final foreign tax credit regulations. The increase in our effective tax rate for fiscal year 2023 compared to fiscal year 2022 was primarily due to a $3.3 billion net income tax benefit in the first quarter of fiscal year 2022 related to the transfer of intangible properties and a decrease in tax benefits relating to stock-based compensation.

The components of the deferred income tax assets and liabilities were as follows:

(In millions)

June 30,	2024	2023
Deferred Income Tax Assets		
Stock-based compensation expense	**$ 765**	$ 681
Accruals, reserves, and other expenses	**4,381**	3,131
Loss and credit carryforwards	**1,741**	1,441
Amortization	**4,159**	9,440
Leasing liabilities	**6,504**	5,041
Unearned revenue	**3,717**	3,296
Book/tax basis differences in investments and debt	**9**	373
Capitalized research and development	**11,442**	6,958
Other	**426**	489
Deferred income tax assets	**33,144**	30,850
Less valuation allowance	**(1,045)**	(939)
Deferred income tax assets, net of valuation allowance	**$ 32,099**	$ 29,911
Deferred Income Tax Liabilities		
Leasing assets	**$ (6,503)**	$ (4,680)
Depreciation	**(3,940)**	(2,674)
Deferred tax on foreign earnings	**(1,837)**	(2,738)
Other	**(167)**	(89)
Deferred income tax liabilities	**$ (12,447)**	$ (10,181)
Net deferred income tax assets	**$ 19,652**	$ 19,730
Reported As		
Other long-term assets	**$ 22,270**	$ 20,163
Long-term deferred income tax liabilities	**(2,618)**	(433)
Net deferred income tax assets	**$ 19,652**	$ 19,730

Deferred income tax balances reflect the effects of temporary differences between the carrying amounts of assets and liabilities and their tax bases and are stated at enacted tax rates expected to be in effect when the taxes are paid or recovered.

As of June 30, 2024, we had federal, state, and foreign net operating loss carryforwards of $476 million, $899 million, and $2.6 billion, respectively. The federal and state net operating loss carryforwards have varying expiration dates ranging from fiscal year 2025 to 2044 or indefinite carryforward periods, if not utilized. The majority of our foreign net operating loss carryforwards do not expire. Certain acquired net operating loss carryforwards are subject to an annual limitation but are expected to be realized with the exception of those which have a valuation allowance. As of June 30, 2024, we had $456 million federal capital loss carryforwards for U.S. tax purposes from our acquisition of Nuance. The federal capital loss carryforwards are subject to an annual limitation and will expire in fiscal year 2025.

The valuation allowance disclosed in the table above relates to the foreign net operating loss carryforwards, federal capital loss carryforwards, and other net deferred tax assets that may not be realized.

Income taxes paid, net of refunds, were $23.4 billion, $23.1 billion, and $16.0 billion in fiscal years 2024, 2023, and 2022, respectively.

Uncertain Tax Positions

Gross unrecognized tax benefits related to uncertain tax positions as of June 30, 2024, 2023, and 2022, were $22.8 billion, $17.1 billion, and $15.6 billion, respectively, which were primarily included in long-term income taxes in our consolidated balance sheets. If recognized, the resulting tax benefit would affect our effective tax rates for fiscal years 2024, 2023, and 2022 by $19.6 billion, $14.4 billion, and $13.3 billion, respectively.

As of June 30, 2024, 2023, and 2022, we had accrued interest expense related to uncertain tax positions of $6.8 billion, $5.2 billion, and $4.3 billion, respectively, net of income tax benefits. The provision for income taxes for fiscal years 2024, 2023, and 2022 included interest expense related to uncertain tax positions of $1.5 billion, $918 million, and $36 million, respectively, net of income tax benefits.

The aggregate changes in the gross unrecognized tax benefits related to uncertain tax positions were as follows:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Beginning unrecognized tax benefits	**$ 17,120**	$ 15,593	$ 14,550
Decreases related to settlements	**(76)**	(329)	(317)
Increases for tax positions related to the current year	**1,903**	1,051	1,145
Increases for tax positions related to prior years [(a)]	**4,289**	870	461
Decreases for tax positions related to prior years	**(464)**	(60)	(246)
Decreases due to lapsed statutes of limitations	**(12)**	(5)	0
Ending unrecognized tax benefits	**$ 22,760**	$ 17,120	$ 15,593

(a) *Fiscal year 2024 includes unrecognized tax benefits of $3.4 billion related to the acquisition of Activision Blizzard. See Note 8 – Business Combinations for further information.*

We remain under audit by the IRS for tax years 2014 to 2017. With respect to the audit for tax years 2004 to 2013, on September 26, 2023, we received Notices of Proposed Adjustment ("NOPAs") from the IRS. The primary issues in the NOPAs relate to intercompany transfer pricing. In the NOPAs, the IRS is seeking an additional tax payment of $28.9 billion plus penalties and interest. As of June 30, 2024, we believe our allowances for income tax contingencies are adequate. We disagree with the proposed adjustments and will vigorously contest the NOPAs through the IRS's administrative appeals office and, if necessary, judicial proceedings. We do not expect a final resolution of these issues in the next 12 months. Based on the information currently available, we do not anticipate a significant increase or decrease to our income tax contingencies for these issues within the next 12 months.

We are subject to income tax in many jurisdictions outside the U.S. Our operations in certain jurisdictions remain subject to examination for tax years 1996 to 2023, some of which are currently under audit by local tax authorities. The resolution of each of these audits is not expected to be material to our consolidated financial statements.

NOTE 13 — UNEARNED REVENUE

Unearned revenue by segment was as follows:

(In millions)

June 30,	**2024**	**2023**
Productivity and Business Processes	**$ 30,879**	$ 27,572
Intelligent Cloud	**23,117**	21,563
More Personal Computing	**6,188**	4,678
Total	**$ 60,184**	$ 53,813

Changes in unearned revenue were as follows:

(In millions)

Year Ended June 30, 2024	
Balance, beginning of period	**$ 53,813**
Deferral of revenue	**148,701**
Recognition of unearned revenue	**(142,330)**
Balance, end of period	**$ 60,184**

Revenue allocated to remaining performance obligations, which includes unearned revenue and amounts that will be invoiced and recognized as revenue in future periods, was $275 billion as of June 30, 2024, of which $269 billion is related to the commercial portion of revenue. We expect to recognize approximately 45% of our total company remaining performance obligation revenue over the next 12 months and the remainder thereafter.

NOTE 14 — LEASES

We have operating and finance leases for datacenters, corporate offices, research and development facilities, Microsoft Experience Centers, and certain equipment. Our leases have remaining lease terms of less than 1 year to 17 years, some of which include options to extend the leases for up to 5 years, and some of which include options to terminate the leases within 1 year.

The components of lease expense were as follows:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Operating lease cost	**$ 3,555**	$ 2,875	$ 2,461
Finance lease cost:			
Amortization of right-of-use assets	**$ 1,800**	$ 1,352	$ 980
Interest on lease liabilities	**734**	501	429
Total finance lease cost	**$ 2,534**	$ 1,853	$ 1,409

Supplemental cash flow information related to leases was as follows:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash flows from operating leases	**$ 3,550**	$ 2,706	$ 2,368
Operating cash flows from finance leases	**734**	501	429
Financing cash flows from finance leases	**1,286**	1,056	896
Right-of-use assets obtained in exchange for lease obligations:			
Operating leases	**6,703**	3,514	5,268
Finance leases	**11,633**	3,128	4,234

Supplemental balance sheet information related to leases was as follows:

(In millions, except lease term and discount rate)

June 30,	**2024**	**2023**
Operating Leases		
Operating lease right-of-use assets	**$ 18,961**	$ 14,346
Other current liabilities	**$ 3,580**	$ 2,409
Operating lease liabilities	**15,497**	12,728
Total operating lease liabilities	**$ 19,077**	$ 15,137
Finance Leases		
Property and equipment, at cost	**$ 32,248**	$ 20,538
Accumulated depreciation	**(6,386)**	(4,647)
Property and equipment, net	**$ 25,862**	$ 15,891
Other current liabilities	**$ 2,349**	$ 1,197
Other long-term liabilities	**24,796**	15,870
Total finance lease liabilities	**$ 27,145**	$ 17,067
Weighted Average Remaining Lease Term		
Operating leases	**7 years**	8 years
Finance leases	**12 years**	11 years
Weighted Average Discount Rate		
Operating leases	**3.3%**	2.9%
Finance leases	**3.9%**	3.4%

The following table outlines maturities of our lease liabilities as of June 30, 2024:

(In millions)

Year Ending June 30,	**Operating Leases**	**Finance Leases**
2025	$ 4,124	$ 3,311
2026	3,549	3,021
2027	2,981	3,037
2028	2,405	3,026
2029	1,924	2,638
Thereafter	6,587	19,116
Total lease payments	21,570	34,149
Less imputed interest	(2,493)	(7,004)
Total	$ 19,077	$ 27,145

As of June 30, 2024, we had additional operating and finance leases, primarily for datacenters, that had not yet commenced of $8.6 billion and $108.4 billion, respectively. These operating and finance leases will commence between fiscal year 2025 and fiscal year 2030 with lease terms of 1 year to 20 years.

NOTE 15 — CONTINGENCIES

U.S. Cell Phone Litigation

Microsoft Mobile Oy, a subsidiary of Microsoft, along with other handset manufacturers and network operators, is a defendant in 45 lawsuits filed in the Superior Court for the District of Columbia by individual plaintiffs who allege that radio emissions from cellular handsets caused their brain tumors and other adverse health effects. We assumed responsibility for these claims in our agreement to acquire Nokia's Devices and Services business and have been substituted for the Nokia defendants. Twelve of these cases were consolidated for certain pre-trial proceedings; the remaining cases are stayed. In a separate 2009 decision, the Court of Appeals for the District of Columbia held that adverse health effect claims arising from the use of cellular handsets that operate within the U.S. Federal Communications Commission radio frequency emission guidelines ("FCC Guidelines") are pre-empted by federal law. The plaintiffs allege that their handsets either operated outside the FCC Guidelines or were manufactured before the FCC Guidelines went into effect. The lawsuits also allege an industry-wide conspiracy to manipulate the science and testing around emission guidelines.

In 2013, the defendants in the consolidated cases moved to exclude the plaintiffs' expert evidence of general causation on the basis of flawed scientific methodologies. In 2014, the trial court granted in part and denied in part the defendants' motion to exclude the plaintiffs' general causation experts. The defendants filed an interlocutory appeal to the District of Columbia Court of Appeals challenging the standard for evaluating expert scientific evidence. In October 2016, the Court of Appeals issued its decision adopting the standard advocated by the defendants and remanding the cases to the trial court for further proceedings under that standard. The plaintiffs have filed supplemental expert evidence, portions of which were stricken by the court. A hearing on general causation took place in September of 2022. In April of 2023, the court granted defendants' motion to strike the testimony of plaintiffs' experts that cell phones cause brain cancer and entered an order excluding all of plaintiffs' experts from testifying. The parties agreed to a stipulated dismissal of the consolidated cases to allow plaintiffs to appeal the expert testimony order. Plaintiffs appealed the court's order in August of 2023, and the parties have filed their briefs on the appeal. A hearing on the status of the stayed cases occurred in December of 2023. In July 2024, the court entered summary judgment in nine of the stayed cases on the grounds that plaintiffs had agreed to be bound by the general causation outcome in the consolidated cases.

Irish Data Protection Commission Matter

In 2018, the Irish Data Protection Commission ("IDPC") began investigating a complaint against LinkedIn as to whether LinkedIn's targeted advertising practices violated the recently implemented European Union General Data Protection Regulation ("GDPR"). Microsoft cooperated throughout the period of inquiry. In April 2023, the IDPC provided LinkedIn with a non-public preliminary draft decision alleging GDPR violations and proposing a fine. In July 2024, the IDPC provided LinkedIn with a revised non-public draft decision. There is no set timeline for the IDPC to issue a final decision, at which time Microsoft will consider its options to appeal.

Other Contingencies

We also are subject to a variety of other claims and suits that arise from time to time in the ordinary course of our business. Although management currently believes that resolving claims against us, individually or in aggregate, will not have a material adverse impact in our consolidated financial statements, these matters are subject to inherent uncertainties and management's view of these matters may change in the future.

As of June 30, 2024, we accrued aggregate legal liabilities of $641 million. While we intend to defend these matters vigorously, adverse outcomes that we estimate could reach approximately $600 million in aggregate beyond recorded amounts are reasonably possible. Were unfavorable final outcomes to occur, there exists the possibility of a material adverse impact in our consolidated financial statements for the period in which the effects become reasonably estimable.

NOTE 16 — STOCKHOLDERS' EQUITY

Shares Outstanding

Shares of common stock outstanding were as follows:

(In millions)			
Year Ended June 30,	**2024**	**2023**	**2022**
Balance, beginning of year	**7,432**	7,464	7,519
Issued	**34**	37	40
Repurchased	**(32)**	(69)	(95)
Balance, end of year	**7,434**	7,432	7,464

Share Repurchases

On September 18, 2019, our Board of Directors approved a share repurchase program authorizing up to $40.0 billion in share repurchases. This share repurchase program commenced in February 2020 and was completed in November 2021.

On September 14, 2021, our Board of Directors approved a share repurchase program authorizing up to $60.0 billion in share repurchases. This share repurchase program commenced in November 2021, following completion of the program approved on September 18, 2019, has no expiration date, and may be terminated at any time. As of June 30, 2024, $10.3 billion remained of this $60.0 billion share repurchase program.

We repurchased the following shares of common stock under the share repurchase programs:

(In millions)	Shares	Amount	Shares	Amount	Shares	Amount
Year Ended June 30,		**2024**		**2023**		**2022**
First Quarter	**11**	**$ 3,560**	17	$ 4,600	21	$ 6,200
Second Quarter	**7**	**2,800**	20	4,600	20	6,233
Third Quarter	**7**	**2,800**	18	4,600	26	7,800
Fourth Quarter	**7**	**2,800**	14	4,600	28	7,800
Total	**32**	**$ 11,960**	69	$ 18,400	95	$ 28,033

All repurchases were made using cash resources. Shares repurchased during the first quarter of fiscal year 2022 were under the share repurchase program approved on September 18, 2019. Shares repurchased during the second quarter of fiscal year 2022 were under the share repurchase programs approved on September 18, 2019 and September 14, 2021. All other shares repurchased were under the share repurchase program approved on September 14, 2021. The above table excludes shares repurchased to settle employee tax withholding related to the vesting of stock awards of $5.3 billion, $3.8 billion, and $4.7 billion for fiscal years 2024, 2023, and 2022, respectively.

Dividends

Our Board of Directors declared the following dividends:

Declaration Date	Record Date	Payment Date	Dividend Per Share	Amount
Fiscal Year 2024				**(In millions)**
September 19, 2023	**November 16, 2023**	**December 14, 2023**	**$ 0.75**	**$ 5,574**
November 28, 2023	**February 15, 2024**	**March 14, 2024**	**0.75**	**5,573**
March 12, 2024	**May 16, 2024**	**June 13, 2024**	**0.75**	**5,574**
June 12, 2024	**August 15, 2024**	**September 12, 2024**	**0.75**	**5,575**
Total			**$ 3.00**	**$ 22,296**
Fiscal Year 2023				
September 20, 2022	November 17, 2022	December 8, 2022	$ 0.68	$ 5,066
November 29, 2022	February 16, 2023	March 9, 2023	0.68	5,059
March 14, 2023	May 18, 2023	June 8, 2023	0.68	5,054
June 13, 2023	August 17, 2023	September 14, 2023	0.68	5,051
Total			$ 2.72	$ 20,230

The dividend declared on June 12, 2024 was included in other current liabilities as of June 30, 2024.

NOTE 17 — ACCUMULATED OTHER COMPREHENSIVE INCOME (LOSS)

The following table summarizes the changes in accumulated other comprehensive income (loss) by component:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Derivatives			
Balance, beginning of period	**$ (27)**	$ (13)	$ (19)
Unrealized gains (losses), net of tax of **$(4)**, $9, and $(15)	**(14)**	34	(57)
Reclassification adjustments for (gains) losses included in other income (expense), net	**48**	(61)	79
Tax expense (benefit) included in provision for income taxes	**(10)**	13	(16)
Amounts reclassified from accumulated other comprehensive loss	**38**	(48)	63
Net change related to derivatives, net of tax of **$6**, $(4), and $1	**24**	(14)	6
Balance, end of period	**$ (3)**	$ (27)	$ (13)
Investments			
Balance, beginning of period	**$ (3,582)**	$ (2,138)	$ 3,222
Unrealized gains (losses), net of tax of **$247**, $(393), and $(1,440)	**915**	(1,523)	(5,405)
Reclassification adjustments for losses included in other income (expense), net	**53**	99	57
Tax benefit included in provision for income taxes	**(11)**	(20)	(12)
Amounts reclassified from accumulated other comprehensive loss	**42**	79	45
Net change related to investments, net of tax of **$258**, $(373), and $(1,428)	**957**	(1,444)	(5,360)
Balance, end of period	**$ (2,625)**	$ (3,582)	$ (2,138)
Translation Adjustments and Other			
Balance, beginning of period	**$ (2,734)**	$ (2,527)	$ (1,381)
Translation adjustments and other, net of tax of **$0**, $0, and $0	**(228)**	(207)	(1,146)
Balance, end of period	**$ (2,962)**	$ (2,734)	$ (2,527)
Accumulated other comprehensive loss, end of period	**$ (5,590)**	$ (6,343)	$ (4,678)

NOTE 18 — EMPLOYEE STOCK AND SAVINGS PLANS

We grant stock-based compensation to employees and directors. Awards that expire or are canceled without delivery of shares generally become available for issuance under the plans. We issue new shares of Microsoft common stock to satisfy vesting of awards granted under our stock plans. We also have an ESPP for all eligible employees.

Stock-based compensation expense and related income tax benefits were as follows:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Stock-based compensation expense	**$ 10,734**	$ 9,611	$ 7,502
Income tax benefits related to stock-based compensation	**1,826**	1,651	1,293

Stock Plans

Stock awards entitle the holder to receive shares of Microsoft common stock as the award vests. Stock awards generally vest over a service period of four years or five years.

Executive Incentive Plan

Under the Executive Incentive Plan, the Compensation Committee approves stock awards to executive officers and certain senior executives. RSUs generally vest ratably over a service period of four years. PSUs generally vest over a performance period of three years. The number of shares the PSU holder receives is based on the extent to which the corresponding performance goals have been achieved.

Activity for All Stock Plans

The fair value of stock awards was estimated on the date of grant using the following assumptions:

Year ended June 30,	2024	2023	2022
Dividends per share (quarterly amounts)	**$ 0.68 – 0.75**	$ 0.62 – 0.68	$ 0.56 – 0.62
Interest rates	**3.8% – 5.6%**	2.0% – 5.4%	0.03% – 3.6%

During fiscal year 2024, the following activity occurred under our stock plans:

	Shares	Weighted Average Grant-Date Fair Value
	(In millions)	
Stock Awards		
Nonvested balance, beginning of year	**96**	**$ 250.37**
Granted [(a)]	**41**	**339.46**
Vested	**(42)**	**246.71**
Forfeited	**(7)**	**270.59**
Nonvested balance, end of year	**88**	**$ 292.28**

(a) *Includes 1 million of PSUs granted at target and performance adjustments above target levels for each of the fiscal years 2024, 2023, and 2022.*

As of June 30, 2024, total unrecognized compensation costs related to stock awards were $20.3 billion. These costs are expected to be recognized over a weighted average period of three years. The weighted average grant-date fair value of stock awards granted was $339.46, $252.59, and $291.22 for fiscal years 2024, 2023, and 2022, respectively. The fair value of stock awards vested was $16.0 billion, $11.9 billion, and $14.1 billion, for fiscal years 2024, 2023, and 2022, respectively. As of June 30, 2024, an aggregate of 129 million shares were authorized for future grant under our stock plans.

Employee Stock Purchase Plan

We have an ESPP for all eligible employees. Shares of our common stock may be purchased by employees at three-month intervals at 90% of the fair market value on the last trading day of each three-month period. Employees may purchase shares having a value not exceeding 15% of their gross compensation during an offering period.

Employees purchased the following shares during the periods presented:

(Shares in millions)

Year Ended June 30,	2024	2023	2022
Shares purchased	**6**	7	7
Average price per share	**$ 339.46**	$ 245.59	$ 259.55

As of June 30, 2024, 68 million shares of our common stock were reserved for future issuance through the ESPP.

Savings Plans

We have savings plans in the U.S. that qualify under Section 401(k) of the Internal Revenue Code, and a number of savings plans in international locations. Eligible U.S. employees may contribute a portion of their salary into the savings plans, subject to certain limitations. We match a portion of each dollar a participant contributes into the plans. Employer-funded retirement benefits for all plans were $1.7 billion, $1.6 billion, and $1.4 billion in fiscal years 2024, 2023, and 2022, respectively, and were expensed as contributed.

NOTE 19 — SEGMENT INFORMATION AND GEOGRAPHIC DATA

In its operation of the business, management, including our chief operating decision maker, who is also our Chief Executive Officer, reviews certain financial information, including segmented internal profit and loss statements prepared on a basis not consistent with GAAP. During the periods presented, we reported our financial performance based on the following segments: Productivity and Business Processes, Intelligent Cloud, and More Personal Computing.

Our reportable segments are described below.

Productivity and Business Processes

Our Productivity and Business Processes segment consists of products and services in our portfolio of productivity, communication, and information services, spanning a variety of devices and platforms. This segment primarily comprises:

- Office Commercial (Office 365 subscriptions, the Office 365 portion of Microsoft 365 Commercial subscriptions, and Office licensed on-premises), comprising Office, Exchange, SharePoint, Microsoft Teams, Office 365 Security and Compliance, Microsoft Viva, and Copilot for Microsoft 365.
- Office Consumer, including Microsoft 365 Consumer and Copilot Pro subscriptions, Office licensed on-premises, and other Office services.
- LinkedIn, including Talent Solutions, Marketing Solutions, Premium Subscriptions, and Sales Solutions.
- Dynamics business solutions, including Dynamics 365, comprising a set of intelligent, cloud-based applications across ERP, CRM, Power Apps, and Power Automate; and on-premises ERP and CRM applications.

Intelligent Cloud

Our Intelligent Cloud segment consists of our public, private, and hybrid server products and cloud services that can power modern business and developers. This segment primarily comprises:

- Server products and cloud services, including Azure and other cloud services; SQL Server, Windows Server, Visual Studio, System Center, and related Client Access Licenses ("CALs"); and Nuance and GitHub.
- Enterprise and partner services, including Enterprise Support Services, Industry Solutions, Nuance professional services, Microsoft Partner Network, and Learning Experience.

More Personal Computing

Our More Personal Computing segment consists of products and services that put customers at the center of the experience with our technology. This segment primarily comprises:

- Windows, including Windows OEM licensing and other non-volume licensing of the Windows operating system; Windows Commercial, comprising volume licensing of the Windows operating system, Windows cloud services, and other Windows commercial offerings; patent licensing; and Windows Internet of Things.
- Devices, including Surface, HoloLens, and PC accessories.

- Gaming, including Xbox hardware and Xbox content and services, comprising first-party content (such as Activision Blizzard) and third-party content, including games and in-game content; Xbox Game Pass and other subscriptions; Xbox Cloud Gaming; advertising; third-party disc royalties; and other cloud services.
- Search and news advertising, comprising Bing (including Copilot), Microsoft News, Microsoft Edge, and third-party affiliates.

Revenue and costs are generally directly attributed to our segments. However, due to the integrated structure of our business, certain revenue recognized and costs incurred by one segment may benefit other segments. Revenue from certain contracts is allocated among the segments based on the relative value of the underlying products and services, which can include allocation based on actual prices charged, prices when sold separately, or estimated costs plus a profit margin. Cost of revenue is allocated in certain cases based on a relative revenue methodology. Operating expenses that are allocated primarily include those relating to marketing of products and services from which multiple segments benefit and are generally allocated based on relative gross margin.

In addition, certain costs are incurred at a corporate level and allocated to our segments. These allocated costs generally include legal, including settlements and fines, information technology, human resources, finance, excise taxes, field selling, shared facilities services, customer service and support, and severance incurred as part of a corporate program. Each allocation is measured differently based on the specific facts and circumstances of the costs being allocated and is generally based on relative gross margin or relative headcount.

Segment revenue and operating income were as follows during the periods presented:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
Revenue			
Productivity and Business Processes	**$ 77,728**	$ 69,274	$ 63,364
Intelligent Cloud	**105,362**	87,907	74,965
More Personal Computing	**62,032**	54,734	59,941
Total	**$ 245,122**	$ 211,915	$ 198,270
Operating Income			
Productivity and Business Processes	**$ 40,540**	$ 34,189	$ 29,690
Intelligent Cloud	**49,584**	37,884	33,203
More Personal Computing	**19,309**	16,450	20,490
Total	**$ 109,433**	$ 88,523	$ 83,383

No sales to an individual customer or country other than the United States accounted for more than 10% of revenue for fiscal years 2024, 2023, or 2022. Revenue, classified by the major geographic areas in which our customers were located, was as follows:

(In millions)

Year Ended June 30,	**2024**	**2023**	**2022**
United States [(a)]	**$ 124,704**	$ 106,744	$ 100,218
Other countries	**120,418**	105,171	98,052
Total	**$ 245,122**	$ 211,915	$ 198,270

(a) *Includes billings to OEMs and certain multinational organizations because of the nature of these businesses and the impracticability of determining the geographic source of the revenue.*

Revenue, classified by significant product and service offerings, was as follows:

(In millions)			
Year Ended June 30,	2024	2023	2022
Server products and cloud services	**$ 97,726**	$ 79,970	$ 67,350
Office products and cloud services	**54,875**	48,848	44,970
Windows	**23,244**	21,507	24,732
Gaming	**21,503**	15,466	16,230
LinkedIn	**16,372**	14,989	13,631
Search and news advertising	**12,576**	12,158	11,526
Enterprise and partner services	**7,594**	7,900	7,605
Dynamics products and cloud services	**6,481**	5,437	4,687
Devices	**4,706**	5,521	7,306
Other	**45**	119	233
Total	**$ 245,122**	$ 211,915	$ 198,270

We have recast certain prior period amounts to conform to the way we internally manage and monitor our business.

Our Microsoft Cloud revenue, which includes Azure and other cloud services, Office 365 Commercial, the commercial portion of LinkedIn, Dynamics 365, and other commercial cloud properties, was $137.4 billion, $111.6 billion, and $91.4 billion in fiscal years 2024, 2023, and 2022, respectively. These amounts are primarily included in Server products and cloud services, Office products and cloud services, LinkedIn, and Dynamics products and cloud services in the table above.

Assets are not allocated to segments for internal reporting presentations. A portion of amortization and depreciation is included with various other costs in an overhead allocation to each segment. It is impracticable for us to separately identify the amount of amortization and depreciation by segment that is included in the measure of segment profit or loss.

Long-lived assets, excluding financial instruments and tax assets, classified by the location of the controlling statutory company and with countries over 10% of the total shown separately, were as follows:

(In millions)			
June 30,	2024	2023	2022
United States	**$ 186,106**	$ 114,380	$ 106,430
Other countries	**115,263**	72,859	59,938
Total	**$ 301,369**	$ 187,239	$ 166,368

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of Microsoft Corporation

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Microsoft Corporation and subsidiaries (the "Company") as of June 30, 2024 and 2023, the related consolidated statements of income, comprehensive income, cash flows, and stockholders' equity, for each of the three years in the period ended June 30, 2024, and the related notes (collectively referred to as the "financial statements"). In our opinion, the financial statements present fairly, in all material respects, the financial position of the Company as of June 30, 2024 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended June 30, 2024, in conformity with accounting principles generally accepted in the United States of America.

We have also audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of June 30, 2024, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission and our report dated July 30, 2024, expressed an unqualified opinion on the Company's internal control over financial reporting.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current-period audit of the financial statements that were communicated or required to be communicated to the audit committee and that (1) relate to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Revenue Recognition – Refer to Note 1 to the financial statements

Critical Audit Matter Description

The Company recognizes revenue upon transfer of control of promised products or services to customers in an amount that reflects the consideration the Company expects to receive in exchange for those products or services. The Company offers customers the ability to acquire multiple licenses of software products and services, including cloud-based services, in its customer agreements through its volume licensing programs.

Significant judgment is exercised by the Company in determining revenue recognition for certain customer agreements, and includes the following:

- Determination of whether products and services are considered distinct performance obligations that should be accounted for separately versus together, such as software licenses and related services that are sold with cloud-based services.
- The pattern of delivery (i.e., timing of when revenue is recognized) for each distinct performance obligation.
- Identification and treatment of contract terms that may impact the timing and amount of revenue recognized (e.g., variable consideration, optional purchases, and free services).
- Determination of stand-alone selling prices for each distinct performance obligation and for products and services that are not sold separately.

Given these factors and due to the volume of transactions, the related audit effort in evaluating management's judgments in determining revenue recognition for certain customer agreements was extensive and required a high degree of auditor judgment.

How the Critical Audit Matter Was Addressed in the Audit

Our principal audit procedures related to the Company's revenue recognition for certain customer agreements included the following:

- We tested the effectiveness of controls related to the identification of distinct performance obligations, the determination of the timing of revenue recognition, and the estimation of variable consideration.
- We evaluated management's significant accounting policies related to certain customer agreements for reasonableness.
- We selected a sample of customer agreements and performed the following procedures:

- Obtained and read contract source documents for each selection, including master agreements, and other documents that were part of the agreement.
- Tested management's identification and treatment of contract terms.
- Assessed the terms in the customer agreement and evaluated the appropriateness of management's application of their accounting policies, along with their use of estimates, in the determination of revenue recognition conclusions.

- We evaluated the reasonableness of management's estimate of stand-alone selling prices for products and services that are not sold separately.
- We tested the mathematical accuracy of management's calculations of revenue and the associated timing of revenue recognized in the financial statements.

Income Taxes – Uncertain Tax Positions – Refer to Note 12 to the financial statements

Critical Audit Matter Description

The Company's long-term income taxes liability includes uncertain tax positions related to transfer pricing issues that remain unresolved with the Internal Revenue Service ("IRS"). The Company remains under IRS audit, or subject to IRS audit, for tax years subsequent to 2003. In the current fiscal year, the Company received Notices of Proposed Adjustments (“NOPAs”) for the tax years 2004 to 2013, primarily related to intercompany transfer pricing. While the Company has settled a portion of the IRS audits, resolution of the remaining matters could have a material impact on the Company's financial statements.

Conclusions on recognizing and measuring uncertain tax positions involve significant estimates and management judgment and include complex considerations of the Internal Revenue Code, related regulations, tax case laws, and prior-year audit settlements. Given the complexity and the subjective nature of certain transfer pricing issues that remain unresolved with the IRS, evaluating management's estimates relating to their determination of uncertain tax positions required extensive audit effort and a high degree of auditor judgment, including involvement of our tax specialists.

How the Critical Audit Matter Was Addressed in the Audit

Our principal audit procedures to evaluate management's estimates of uncertain tax positions related to unresolved transfer pricing issues included the following:

- We evaluated the appropriateness and consistency of management's methods and assumptions used in the identification, recognition, measurement, and disclosure of uncertain tax positions, which included testing the effectiveness of the related internal controls.
- We read and evaluated management's documentation, including relevant accounting policies and information obtained by management from outside tax specialists, that detailed the basis of the uncertain tax positions.
- We tested the reasonableness of management's judgments regarding the future resolution of the uncertain tax positions, including an evaluation of the technical merits of the uncertain tax positions.
- For those uncertain tax positions that had not been effectively settled, we evaluated whether management had appropriately considered new information, including the NOPAs received in the current fiscal year, that could significantly change the recognition, measurement, or disclosure of the uncertain tax positions.
- We evaluated the reasonableness of management's estimates by considering how tax law, including statutes, regulations, and case law, impacted management's judgments.

Business Combinations – Estimate for Valuation of Acquired Intangible Assets – Refer to Note 8 to the financial statements

Critical Audit Matter Description

On October 13, 2023, the Company completed the acquisition of Activision Blizzard, Inc. The Company accounted for the Activision Blizzard, Inc., acquisition as a business combination and, accordingly, allocated the purchase price to the assets acquired and liabilities assumed based on their respective estimated fair values as of the date of acquisition. Identifiable intangible assets acquired included marketing-related intangible assets, technology-based intangible assets, and customer-related intangible assets. The excess of the purchase consideration over the fair value of identifiable assets acquired and liabilities assumed was recorded as goodwill.

We identified the fair value determination of certain marketing-related and technology-based intangible assets for the business combination as a critical audit matter due to the significant judgment required in determining their estimated fair values. Management's estimates of fair value included assumptions for revenue and expense forecasts and the selection of appropriate discount rates. There was a high degree of auditor judgment and subjectivity in applying audit procedures and evaluating the significant assumptions relating to the estimates, including involvement of our fair value specialists.

How the Critical Audit Matter Was Addressed in the Audit

Our audit procedures related to management's estimates of the fair value of certain marketing-related and technology-based intangible assets acquired included the following, among others:

- We tested the operating effectiveness of internal controls over the business combination, including internal controls over the revenue and expense forecasts and the selection of appropriate discount rates.
- We assessed the knowledge, skills, abilities, and objectivity of management's valuation specialist and evaluated the work performed.

- When assessing the reasonableness of assumptions related to forecasted revenue and expenses, we evaluated whether the assumptions used were reasonable considering historical financial information of Activision Blizzard, Inc., and the Company’s forecasted financial information.

- With the assistance of our fair value specialists, we evaluated the reasonableness of the discount rates by:

- Testing the source information underlying the discount rates and testing the mathematical accuracy of the calculations.

- Developing a range of independent estimates and comparing those to the discount rates selected by management.

/s/ DELOITTE & TOUCHE LLP

Seattle, Washington
July 30, 2024

We have served as the Company's auditor since 1983.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

Not applicable.

ITEM 9A. CONTROLS AND PROCEDURES

Under the supervision and with the participation of our management, including the Chief Executive Officer and Chief Financial Officer, we have evaluated the effectiveness of our disclosure controls and procedures as required by Exchange Act Rule 13a-15(b) as of the end of the period covered by this report. Based on that evaluation, the Chief Executive Officer and Chief Financial Officer have concluded that these disclosure controls and procedures are effective.

REPORT OF MANAGEMENT ON INTERNAL CONTROL OVER FINANCIAL REPORTING

Our management is responsible for establishing and maintaining adequate internal control over financial reporting for the Company. Internal control over financial reporting is a process to provide reasonable assurance regarding the reliability of our financial reporting for external purposes in accordance with accounting principles generally accepted in the United States of America. Internal control over financial reporting includes maintaining records that in reasonable detail accurately and fairly reflect our transactions; providing reasonable assurance that transactions are recorded as necessary for preparation of our consolidated financial statements; providing reasonable assurance that receipts and expenditures of company assets are made in accordance with management authorization; and providing reasonable assurance that unauthorized acquisition, use, or disposition of company assets that could have a material effect on our consolidated financial statements would be prevented or detected on a timely basis. Because of its inherent limitations, internal control over financial reporting is not intended to provide absolute assurance that a misstatement of our consolidated financial statements would be prevented or detected.

Management conducted an evaluation of the effectiveness of our internal control over financial reporting based on the framework in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission. Our assessment of, and conclusion on, the effectiveness of internal control over financial reporting did not include the internal controls of Activision Blizzard, Inc., acquired on October 13, 2023, which is included in our consolidated financial statements since the date of acquisition and represented less than 1% of our total assets as of June 30, 2024 after excluding goodwill and intangible assets acquired, and 2% of our total revenues for the year ended June 30, 2024. Based on this evaluation, management concluded that the Company’s internal control over financial reporting was effective as of June 30, 2024. There were no changes in our internal control over financial reporting during the quarter ended June 30, 2024 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting. Deloitte & Touche LLP has audited our internal control over financial reporting as of June 30, 2024; their report is included in Item 9A.

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of Microsoft Corporation

Opinion on Internal Control over Financial Reporting

We have audited the internal control over financial reporting of Microsoft Corporation and subsidiaries (the "Company") as of June 30, 2024, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO). In our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of June 30, 2024, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by COSO.

We have also audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated financial statements as of and for the year ended June 30, 2024, of the Company and our report dated July 30, 2024, expressed an unqualified opinion on those financial statements.

As described in the Report of Management on Internal Control over Financial Reporting, management excluded from its assessment the internal control over financial reporting at Activision Blizzard, Inc., which was acquired on October 13, 2023, and whose financial statements constitute less than 1 percent of total assets as of June 30, 2024 after excluding goodwill and intangible assets acquired, and 2 percent of total revenues for the year ended June 30, 2024. Accordingly, our audit did not include the internal control over financial reporting at Activision Blizzard, Inc.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Report of Management on Internal Control over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects. Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ DELOITTE & TOUCHE LLP

Seattle, Washington
July 30, 2024

ITEM 9B. OTHER INFORMATION

Insider Trading Arrangements

None of our officers or directors, as defined in Rule 16a-1(f) of the Securities Exchange Act of 1934, adopted, modified, or terminated a “Rule 10b5-1 trading arrangement” or a “non-Rule 10b5-1 trading arrangement,” as defined in Item 408 of Regulation S-K, during the three months ended June 30, 2024.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS, AND CORPORATE GOVERNANCE

A list of our executive officers and biographical information appears in Part I, Item 1 of this Form 10-K. Information about our directors may be found under the caption “Our Director Nominees” in our Proxy Statement for the Annual Meeting of Shareholders to be held December 10, 2024 (the “Proxy Statement”). Information about our Audit Committee may be found under the caption “Board Committees” in the Proxy Statement. That information is incorporated herein by reference.

We have adopted the Microsoft Finance Code of Professional Conduct (the “finance code of ethics”), a code of ethics that applies to our Chief Executive Officer, Chief Financial Officer, Chief Accounting Officer, and other finance organization employees. The finance code of ethics is publicly available on our website at https://aka.ms/FinanceCodeProfessionalConduct. If we make any substantive amendments to the finance code of ethics or grant any waiver, including any implicit waiver, from a provision of the code to our Chief Executive Officer, Chief Financial Officer, or Chief Accounting Officer, we will disclose the nature of the amendment or waiver on that website or in a report on Form 8-K.

We have adopted insider trading policies and procedures applicable to our directors, officers, and employees, and have implemented processes for the company, that we believe are reasonably designed to promote compliance with insider trading laws, rules, and regulations, and the Nasdaq Stock Market LLC listing standards.

Our General Insider Trading Policy prohibits our employees and related persons and entities from trading in securities of Microsoft and other companies while in possession of material, nonpublic information. Our General Insider Trading Policy also prohibits our employees from disclosing material, nonpublic information Microsoft, or another publicly traded company, to others who may trade on the basis of that information. A copy of our General Insider Trading Policy is filed as Exhibit 19.1 to this Form 10-K.

Our Restricted Trading Window Policy requires that certain officers of the company (corporate vice presidents and above) and other designated employees only transact in Microsoft securities during an open window period, subject to limited exceptions. In addition, certain officers of the company are required to obtain approval in advance of transactions in Microsoft securities. A copy of our Restricted Trading Window Policy is filed as Exhibit 19.2 to this Form 10-K.

Our executive officers and directors must also comply with additional trading restrictions. A copy of our Insider Trading Compliance and Preclearance Policies for Section 16 Officers and Directors of Microsoft is filed as Exhibit 19.3 to this Form 10-K.

ITEM 11. EXECUTIVE COMPENSATION

The information in the Proxy Statement set forth under the captions “Director Compensation,” “Named Executive Officer Compensation,” “Compensation Committee Report,” and, if required, “Compensation Committee Interlocks and Insider Participation,” is incorporated herein by reference.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information in the Proxy Statement set forth under the captions “Stock Ownership Information,” “Principal Shareholders,” and “Equity Compensation Plan Information” is incorporated herein by reference.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

The information set forth in the Proxy Statement under the captions “Director Independence Guidelines” and “Certain Relationships and Related Transactions” is incorporated herein by reference.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

Information concerning fees and services provided by our principal accountant, Deloitte & Touche LLP (PCAOB ID No. 34), appears in the Proxy Statement under the headings “Fees Billed by Deloitte & Touche” and “Policy on Audit Committee Pre-Approval of Audit and Permissible Non-Audit Services of Independent Auditor” and is incorporated herein by reference.

PART IV

ITEM 15. EXHIBIT AND FINANCIAL STATEMENT SCHEDULES

(a) Financial Statements and Schedules

The financial statements are set forth under Part II, Item 8 of this Form 10-K, as indexed below. Financial statement schedules have been omitted since they either are not required, not applicable, or the information is otherwise included.

(b) Exhibit Listing

			Incorporated by Reference			
Exhibit Number	Exhibit Description	Filed Herewith	Form	Period Ending	Exhibit	Filing Date
3.1	Amended and Restated Articles of Incorporation of Microsoft Corporation		8-K		3.1	12/1/2016
3.2	Bylaws of Microsoft Corporation		8-K		3.2	7/3/2023
4.1	Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee ("Base Indenture")		S-3ASR		4.1	10/29/2015
4.2	Form of First Supplemental Indenture for 2.95% Notes due 2014, 4.20% Notes due 2019, and 5.20% Notes due 2039, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Base Indenture		8-K		4.2	5/15/2009
4.5	Form of Second Supplemental Indenture for 0.875% Notes due 2013, 1.625% Notes due 2015, 3.00% Notes due 2020, and 4.50% Notes due 2040, dated as of September 27, 2010, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee		8-K		4.2	9/27/2010

Exhibit Number	Exhibit Description	Filed Herewith	Incorporated by Reference Form	Period Ending	Exhibit	Filing Date
4.6	Third Supplemental Indenture for 2.500% Notes due 2016, 4.000% Notes due 2021, and 5.300% Notes due 2041, dated as of February 8, 2011, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee		8-K		4.2	2/8/2011
4.7	Fourth Supplemental Indenture for 0.875% Notes due 2017, 2.125% Notes due 2022, and 3.500% Notes due 2042, dated as of November 7, 2012, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee		8-K		4.1	11/7/2012
4.8	Fifth Supplemental Indenture for 2.625% Notes due 2033, dated as of May 2, 2013, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee		8-K		4.1	5/1/2013
4.9	Sixth Supplemental Indenture for 1.000% Notes due 2018, 2.375% Notes due 2023, and 3.750% Notes due 2043, dated as of May 2, 2013, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee		8-K		4.2	5/1/2013
4.10	Seventh Supplemental Indenture for 2.125% Notes due 2021 and 3.125% Notes due 2028, dated as of December 6, 2013, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee		8-K		4.1	12/6/2013

Exhibit Number	Exhibit Description	Filed Herewith	Incorporated by Reference Form	Period Ending	Exhibit	Filing Date
4.11	Eighth Supplemental Indenture for 1.625% Notes due 2018, 3.625% Notes due 2023, and 4.875% Notes due 2043, dated as of December 6, 2013, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee		8-K		4.2	12/6/2013
4.12	Ninth Supplemental Indenture for 1.850% Notes due 2020, 2.375% Notes due 2022, 2.700% Notes due 2025, 3.500% Notes due 2035, 3.750% Notes due 2045, and 4.000% Notes due 2055, dated as of February 12, 2015, between Microsoft Corporation and U.S. Bank National Association, as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee		8-K		4.1	2/12/2015
4.13	Tenth Supplemental Indenture for 1.300% Notes due 2018, 2.000% Notes due 2020, 2.650% Notes due 2022, 3.125% Notes due 2025, 4.200% Notes due 2035, 4.450% Notes due 2045, and 4.750% Notes due 2055, dated as of November 3, 2015, between Microsoft Corporation and U.S. Bank National Association, as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee		8-K		4.1	11/3/2015
4.14	Eleventh Supplemental Indenture for 1.100% Notes due 2019, 1.550% Notes due 2021, 2.000% Notes due 2023, 2.400% Notes due 2026, 3.450% Notes due 2036, 3.700% Notes due 2046, and 3.950% Notes due 2056, dated as of August 8, 2016, between Microsoft Corporation and U.S. Bank, National Association, as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee		8-K		4.1	8/5/2016

Exhibit Number	Exhibit Description	Filed Herewith	Incorporated by Reference			
			Form	Period Ending	Exhibit	Filing Date
4.15	Twelfth Supplemental Indenture for 1.850% Notes due 2020, 2.400% Notes due 2022, 2.875% Notes due 2024, 3.300% Notes due 2027, 4.100% Notes due 2037, 4.250% Notes due 2047, and 4.500% Notes due 2057, dated as of February 6, 2017, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee		8-K		4.1	2/3/2017
4.16	Thirteenth Supplemental Indenture for 2.525% Notes due 2050 and 2.675% Notes due 2060, dated as of June 1, 2020, between Microsoft Corporation and U.S. Bank National Association, as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee		8-K		4.1	6/1/2020
4.17	Fourteenth Supplemental Indenture for 2.921% Notes due 2052 and 3.041% Notes due 2062, dated as of March 17, 2021, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee, to the Indenture, dated as of May 18, 2009, between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee		8-K		4.1	3/17/2021
4.18	Fifteenth Supplemental Indenture, dated as of November 6, 2023, by and between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as Trustee		8-K		4.2	11/6/2023
4.19	Indenture, dated as of September 19, 2016, by and between Activision Blizzard, Inc. and Wells Fargo Bank, National Association, as Trustee, with respect to Activision Blizzard, Inc.'s 3.400% Senior Notes due 2026		8-K		4.9	11/6/2023

Exhibit Number	Exhibit Description	Filed Herewith	Incorporated by Reference Form	Period Ending	Exhibit	Filing Date
4.20	Base Indenture, dated as of May 26, 2017, by and between Activision Blizzard, Inc. and Wells Fargo Bank, National Association, as Trustee, with respect to Activision Blizzard, Inc.'s 3.400% Senior Notes due 2027, 1.350% Senior Notes due 2030, 4.500% Senior Notes due 2047 and 2.500% Senior Notes due 2050		8-K		4.10	11/6/2023
4.21	First Supplemental Indenture, dated as of May 26, 2017, by and between Activision Blizzard, Inc. and Wells Fargo Bank, National Association, as Trustee, with respect to Activision Blizzard, Inc.'s 3.400% Senior Notes due 2027 and 4.500% Senior Notes due 2047		8-K		4.11	11/6/2023
4.22	Second Supplemental Indenture, dated as of August 10, 2020, by and between Activision Blizzard, Inc. and Wells Fargo Bank, National Association, as Trustee, with respect to Activision Blizzard, Inc.'s 1.350% Senior Notes due 2030 and 2.500% Senior Notes due 2050		8-K		4.12	11/6/2023
4.23	First Supplemental Indenture, dated as of October 27, 2023, by and between Activision Blizzard, Inc. and Computershare Trust Company, N.A., with respect to Activision Blizzard, Inc.'s 3.400% Senior Notes due 2026		8-K		4.13	11/6/2023
4.24	Third Supplemental Indenture, dated as of October 27, 2023, by and between Activision Blizzard, Inc. and Computershare Trust Company, N.A., with respect to Activision Blizzard, Inc.'s 3.400% Senior Notes due 2027 and 4.500% Senior Notes due 2047		8-K		4.14	11/6/2023
4.25	Fourth Supplemental Indenture, dated as of October 27, 2023, by and between Activision Blizzard, Inc. and Computershare Trust Company, N.A., with respect to Activision Blizzard, Inc.'s 1.350% Senior Notes due 2030 and 2.500% Senior Notes due 2050		8-K		4.15	11/6/2023
4.26	Description of Securities	X				
10.1*	Microsoft Corporation 2001 Stock Plan		10-Q	9/30/2016	10.1	10/20/2016
10.4*	Microsoft Corporation Employee Stock Purchase Plan		10-K	6/30/2012	10.4	7/26/2012

Exhibit Number	Exhibit Description	Filed Herewith	Incorporated by Reference Form	Period Ending	Exhibit	Filing Date
10.5*	Microsoft Corporation Deferred Compensation Plan	X				
10.6*	Microsoft Corporation 2017 Stock Plan		DEF14A		Annex C	10/16/2017
10.7*	Form of Stock Award Agreement Under the Microsoft Corporation 2017 Stock Plan		10-Q	3/31/2018	10.26	4/26/2018
10.8*	Form of Performance Stock Award Agreement Under the Microsoft Corporation 2017 Stock Plan		10-Q	3/31/2018	10.27	4/26/2018
10.9	Amended and Restated Officers' Indemnification Trust Agreement between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee		10-Q	9/30/2016	10.12	10/20/2016
10.10	Assumption of Beneficiaries' Representative Obligations Under Amended and Restated Officers' Indemnification Trust Agreement		10-K	6/30/2020	10.25	7/30/2020
10.11	Form of Indemnification Agreement and Amended and Restated Directors' Indemnification Trust Agreement between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee		10-K	6/30/2019	10.13	8/1/2019
10.12	Assumption of Beneficiaries' Representative Obligations Under Amended and Restated Directors' Indemnification Trust Agreement		10-K	6/30/2020	10.26	7/30/2020
10.14*	Microsoft Corporation Deferred Compensation Plan for Non-Employee Directors		10-Q	12/31/2017	10.14	1/31/2018
10.15*	Microsoft Corporation Executive Incentive Plan		8-K		10.1	9/19/2018
10.19*	Microsoft Corporation Executive Incentive Plan		10-Q	9/30/2016	10.17	10/20/2016
10.20*	Form of Executive Incentive Plan (Executive Officer SAs) Stock Award Agreement under the Microsoft Corporation 2001 Stock Plan		10-Q	9/30/2016	10.18	10/20/2016
10.21*	Form of Executive Incentive Plan Performance Stock Award Agreement under the Microsoft Corporation 2001 Stock Plan		10-Q	9/30/2016	10.25	10/20/2016
10.22*	Senior Executive Severance Benefit Plan		10-Q	9/30/2016	10.22	10/20/2016
10.23*	Offer Letter, dated February 3, 2014, between Microsoft Corporation and Satya Nadella		8-K		10.1	2/4/2014

Exhibit Number	Exhibit Description	Filed Herewith	Incorporated by Reference			
			Form	Period Ending	Exhibit	Filing Date
10.24*	Long-Term Performance Stock Award Agreement between Microsoft Corporation and Satya Nadella		10-Q	12/31/2014	10.24	1/26/2015
10.25*	Offer Letter, dated October 25, 2020, between Microsoft Corporation and Christopher Young		10-Q	9/30/2021	10.27	10/26/2021
19.1	General Insider Trading Policy	X				
19.2	Restricted Trading Window Policy	X				
19.3	Insider Trading Compliance and Preclearance Policies for Section 16 Officers and Directors of Microsoft	X				
21	Subsidiaries of Registrant	X				
23.1	Consent of Independent Registered Public Accounting Firm	X				
31.1	Certification of Chief Executive Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002	X				
31.2	Certification of Chief Financial Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002	X				
32.1**	Certification of Chief Executive Officer Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002	X				
32.2**	Certification of Chief Financial Officer Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002	X				
97.1*	Microsoft Corporation Executive Compensation Recovery Policy	X				
101.INS	Inline XBRL Instance Document—the instance document does not appear in the Interactive Data File as its XBRL tags are embedded within the Inline XBRL document	X				
101.SCH	Inline XBRL Taxonomy Extension Schema With Embedded Linkbase Documents	X				
104	Cover page formatted as Inline XBRL and contained in Exhibit 101	X				

* *Indicates a management contract or compensatory plan or arrangement.*
** *Furnished, not filed.*

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned; thereunto duly authorized, in the City of Redmond, State of Washington, on July 30, 2024.

MICROSOFT CORPORATION

/s/ ALICE L. JOLLA
Alice L. Jolla
Corporate Vice President and Chief Accounting Officer (Principal Accounting Officer)

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of Registrant and in the capacities indicated on July 30, 2024.

Signature	Title
/s/ SATYA NADELLA Satya Nadella	Chairman and Chief Executive Officer (Principal Executive Officer)
/s/ REID HOFFMAN Reid Hoffman	Director
/s/ HUGH F. JOHNSTON Hugh F. Johnston	Director
/s/ TERI L. LIST Teri L. List	Director
/s/ CATHERINE MACGREGOR Catherine MacGregor	Director
/s/ MARK A. L. MASON Mark A. L. Mason	Director
/s/ SANDRA E. PETERSON Sandra E. Peterson	Lead Independent Director
/s/ PENNY S. PRITZKER Penny S. Pritzker	Director
/s/ CARLOS A. RODRIGUEZ Carlos A. Rodriguez	Director
/s/ CHARLES W. SCHARF Charles W. Scharf	Director
/s/ JOHN W. STANTON John W. Stanton	Director
/s/ EMMA N. WALMSLEY Emma N. Walmsley	Director
/s/ AMY E. HOOD Amy E. Hood	Executive Vice President and Chief Financial Officer (Principal Financial Officer)
/s/ ALICE L. JOLLA Alice L. Jolla	Corporate Vice President and Chief Accounting Officer (Principal Accounting Officer)

Exhibit 4.26

**DESCRIPTION OF THE REGISTRANT'S SECURITIES
REGISTERED PURSUANT TO SECTION 12 OF THE
SECURITIES EXCHANGE ACT OF 1934**

As of July 30, 2024, Microsoft Corporation has three classes of securities registered under Section 12 of the Securities Exchange Act of 1934, as amended (the “Exchange Act”): (1) our Common Stock; (2) our 3.125% Notes due 2028; and (3) our 2.625% Notes due 2033.

Description of Common Stock

The following description of our Common Stock is a summary and does not purport to be complete. It is subject to and qualified in its entirety by reference to our Amended and Restated Articles of Incorporation (the “Articles of Incorporation”) and our Amended and Restated Bylaws (the “Bylaws”), each of which are incorporated by reference as an exhibit to the Annual Report on Form 10-K of which this Exhibit 4.1 is a part. We encourage you to read our Articles of Incorporation, our Bylaws and the applicable provisions of Washington Business Corporation Act, Title 23B of the Revised Code of Washington, for additional information.

Authorized Capital Shares

Our authorized capital shares consist of 24,000,000,000 shares of common stock, $0.00000625 par value per share (“Common Stock”), and 100,000,000 shares of series preferred stock, $0.01 par value per share (“Preferred Stock”). The outstanding shares of our Common Stock are fully paid and nonassessable.

Voting Rights

Holders of Common Stock are entitled to one vote per share on all matters voted on by the stockholders, including the election of directors. Our Common Stock does not have cumulative voting rights.

Dividend Rights

Subject to the rights of holders of outstanding shares of Preferred Stock, if any, the holders of Common Stock are entitled to receive dividends, if any, as may be declared from time to time by the Board of Directors in its discretion out of funds legally available for the payment of dividends.

Liquidation Rights

Subject to any preferential rights of outstanding shares of Preferred Stock, holders of Common Stock will share ratably in all assets legally available for distribution to our stockholders in the event of dissolution.

Other Rights and Preferences

Our Common Stock has no sinking fund or redemption provisions or preemptive, conversion or exchange rights. Holders of Common Stock may act by unanimous written consent.

Listing

The Common Stock is traded on The Nasdaq Stock Market LLC under the trading symbol “MSFT.”

Description of the Notes

The following description of our 3.125% Notes due 2028 (the “2028 Notes”) and our 2.625% Notes due 2033 (the “2033 Notes” and, together with the 2028 Notes, the “notes”), is a summary and does not purport to be complete. It is subject to and qualified in its entirety by reference to the indenture, dated as of May 18, 2009 (the “Base Indenture”), between Microsoft Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee, as supplemented in the case of the 2028 Notes, by the seventh supplemental indenture, dated as of December 6, 2013, and, as supplemented in the case of the 2033

Notes, by the fifth supplemental indenture dated as of May 1, 2013 (the Base Indenture, as supplemented by the fifth and seventh supplemental indentures, the "indenture"), which are incorporated by reference as exhibits to the Annual Report on Form 10-K of which this Exhibit 4.1 is a part. The 2028 Notes and the 2033 Notes are each traded on The New York Stock Exchange under the bond trading symbols of "MSFT28" and "MSFT33," respectively.

We encourage you to read the above referenced indenture, as supplemented, for additional information.

General

The following is a description of certain of the specific terms and conditions of the indenture supplements with respect to each of the notes.

The 2033 Notes were initially issued in an €550,000,000 aggregate principal amount. The 2028 Notes were initially issued in an €1,750,000,000 aggregate principal amount. We are permitted to issue additional notes of each series of notes without the consent of the holders of that series of notes, but we will not issue such additional notes unless they are fungible for U.S. federal income tax purposes with the relevant series of notes offered. As of July 30, 2024 no such additional notes have been issued.

The notes are senior unsecured obligations and rank equally with our other unsecured and unsubordinated debt from time to time outstanding.

The maturity date of the 2028 Notes is December 6, 2028. The maturity date of the 2033 Notes is May 2, 2033.

The notes will be subject to legal defeasance and covenant defeasance as provided under the "Discharge, Defeasance and Covenant Defeasance" section set forth below.

The notes were issued in a form of one or more fully registered global securities, without coupons, in denominations of €100,000 in principal amount and integral multiples of €1,000 in excess thereof.

The notes are not redeemable at the option of the holder prior to maturity and will not benefit from any sinking fund.

Interest and Principal

The 2028 Notes bear interest from December 6, 2013 at a fixed interest rate of 3.125% per annum. The 2033 Notes bear interest from May 2, 2013 at a fixed interest rate of 2.625% per annum. Interest is paid annually on December 6 for the 2028 Notes and on May 2 for the 2033 Notes, and on the maturity date of each series of notes (the "interest payment date"). We will pay interest on the notes to the persons in whose names the notes are registered at the close of business on the May 1 or December 5 or May 1, as applicable (in each case, whether or not a business day) immediately preceding the related interest payment date. Interest on the notes will be computed on the basis of the actual number of days in the period for which interest is being calculated and the actual number of days from and including the last date on which interest was paid on the notes (or the original issue date if no interest has been paid or duly provided for on the notes), to but excluding the next date on which interest is paid or duly provided for. This payment convention is referred to as Actual/Actual (ICMA) as defined in the rulebook of the International Capital Market Association.

We will pay the principal of and interest on each note to the registered holder in euro in immediately available funds; *provided* that for the 2028 Notes, if the euro is unavailable to us due to the imposition of exchange controls or other circumstances beyond our control or if the euro is no longer being used by the then member states of the European Monetary Union that have adopted the euro as their currency or for the settlement of transactions by public institutions of or within the international banking community, then all payments in respect of the notes will be made in U.S. dollars until the euro is again available to us or so used. The amount payable on any date in euro will be converted into U.S. dollars at the rate mandated by the U.S. Federal Reserve Board as of the close of business on the second business day prior to the relevant payment date or, in the event the U.S. Federal Reserve Board has not mandated a rate of conversion, on the basis of the most recent U.S. dollar/euro exchange rate published in The Wall Street Journal on or prior to the second business day prior to the relevant payment date. Any payment in respect of the notes so made in U.S. dollars will not constitute an event of default under the notes or the indenture governing the notes. So long as the notes are in book-entry form, we will make payments of principal and interest through the London paying agent described below.

Interest payable on any interest payment date for a series of notes or the maturity date for that series of notes will be the amount of interest accrued from, and including, the next preceding interest payment date for such notes in respect of which interest has been paid or duly provided for (or from and including the original issue date, if no interest has been paid or duly provided for with respect to the notes of that series) to, but excluding, such interest payment date or maturity date, as the case may be. If any interest payment date falls on a day that is not a business day, the interest payment will be made on the next succeeding business day, and we will not be liable for any additional interest as a result of the delay in payment. If a maturity date falls on a day that is not a business day, the related payment of principal and interest will be made on the next succeeding business day, and no interest will accrue on the amounts so payable for the period from and after such date to the next succeeding business day. The term “business day” means any day, other than a Saturday or a Sunday, (1) which is not a day on which banking institutions are authorized or obligated by law or executive order to close in New York City or London and (2) on which the Trans-European Automated Real-time Gross Settlement Express Transfer system (the TARGET2 system), or any successor thereto, is open.

Optional Redemption

At any time prior to September 6, 2028, we will have the right at our option to redeem the 2028 Notes, in whole or in part, at any time or from time to time, on at least 30 days’ but not more than 60 days’ prior notice mailed to the registered address of each holder of the 2028 Notes, at a redemption price, calculated by us, equal to the greater of (1) 100% of the principal amount of the 2028 Notes to be redeemed and (2) the sum of the present values of each remaining scheduled payment of principal and interest on the 2028 Notes to be redeemed (exclusive of interest accrued to the date of redemption) discounted to the redemption date on an annual basis (Actual/Actual (ICMA)) at the applicable Bond Rate plus 20 basis points.

At any time prior to February 2, 2033, we will have the right at our option to redeem the notes, in whole or in part, at any time or from time to time, on at least 30 days’ but not more than 60 days’ prior notice mailed to the registered address of each holder of the notes, at a redemption price equal to the greater of (1) 100% of the principal amount of the notes to be redeemed and (2) the sum of the present values of each remaining scheduled payment of principal and interest on the notes to be redeemed (exclusive of interest accrued to the date of redemption) discounted to the redemption date on an annual basis (Actual/Actual (ICMA)) at the applicable Bond Rate plus 12.5 basis points.

At any time on or after September 6, 2028, we will have the right at our option to redeem the 2028 Notes, in whole or in part, on at least 30 days’ but not more than 60 days’ notice, at any time at a redemption price equal to 100% of the principal amount of the 2028 Notes to be redeemed.

At any time on or after February 2, 2033, we will have the right at our option to redeem the 2033 Notes, in whole or in part, on at least 30 days’ but not more than 60 days’ notice, at any time at a redemption price equal to 100% of the principal amount of the notes to be redeemed.

The redemption price for the notes will include, in each case, accrued and unpaid interest on the principal amount of the notes to be redeemed to the redemption date. The redemption price paid for the notes upon any such redemption will be paid in euro.

“Bond Rate” means, with respect to any redemption date, the rate per annum equal to the annual equivalent yield to maturity or interpolated maturity (on a day count basis) of the applicable Comparable Government Issue (computed as of the third business day immediately preceding the redemption date), assuming a price for such Comparable Government Issue (expressed as a percentage of its principal amount) equal to the applicable Comparable Price for such redemption date.

“Comparable Government Issue” means the euro-denominated security issued by the German government selected by an Independent Investment Banker as having an actual or interpolated maturity comparable to the remaining term of the series of notes to be redeemed that would be utilized, at the time of selection and in accordance with customary financial practice, in pricing new issues of corporate debt securities of a comparable maturity to the remaining term of the notes of such series.

“Comparable Price” means, with respect to any redemption date (1) the arithmetic average of the Reference Dealer Quotations for such redemption date, after excluding the highest and lowest such Reference Dealer Quotations or (2) if we obtain fewer than four such Reference Dealer Quotations, the arithmetic average of all such quotations for such redemption date.

“Independent Investment Banker” means an investment bank of international standing appointed by us.

“Reference Dealer” means a broker of, or a market maker in, the Comparable Government Issue selected by the Independent Investment Banker.

“Reference Dealer Quotation” means, with respect to each Reference Dealer and any redemption date, the arithmetic average, as determined by us, of the bid and asked prices for the applicable Comparable Government Issue (expressed in each case as a percentage of its principal amount) quoted in writing to us by such Reference Dealer at 11:00 a.m. (London time) on the third business day preceding such redemption date.

On and after a redemption date, interest will cease to accrue on the notes called for redemption or any portion of any series of the notes called for redemption (unless we default in the payment of the redemption price and accrued and unpaid interest). On or before the redemption date, we will deposit with the London paying agent money sufficient to pay the redemption price of and (unless the redemption date shall be an interest payment date) accrued and unpaid interest to the redemption date on the notes to be redeemed on such date. If less than all of the notes of a series are to be redeemed, the notes of such series to be redeemed will be selected by the trustee by such method as the trustee will deem fair and appropriate; *provided*, *however*, that no notes of a principal amount of €100,000 or less shall be redeemed in part.

Redemption Upon Tax Event

We may redeem the notes of any series at our option in whole, but not in part, on at least 15 days’ but not more than 60 days’ notice, at a redemption price equal to 100% of their principal amount (plus any accrued interest and additional amounts then payable with respect to the notes of that series), if we determine that (A) as a result of any change or amendment to the laws, treaties, regulations or rulings of the United States or any political subdivision or taxing authority thereof, which change or amendment is announced and becomes effective after the date of the applicable prospectus supplement, we have become obligated to pay additional amounts as described under “-Payment of Additional Amounts” on any notes of such series or (B) after the date of the applicable prospectus supplement, any change in the official application, enforcement or interpretation of those laws, treaties, regulations or rulings, including a holding by a court of competent jurisdiction in the United States or any other action, taken by any taxing authority or a court of competent jurisdiction in the United States, whether or not such action was taken or made with respect to us, results in a material probability that we have or will become obligated to pay additional amounts as described under “-Payment of Additional Amounts” on any notes of such series; *provided* that we determine, in our business judgment, that the obligation to pay such additional amounts cannot be avoided by use of reasonable measures available to us, not including substitution of the obligor under the notes of such series. Prior to the mailing of any notice of such a redemption, we will deliver to the trustee (1) an officer’s certificate stating that we are entitled to effect such a redemption and setting forth a statement of facts showing that the conditions precedent to the right of our company to so redeem have occurred and (2) an opinion of counsel to that effect based on that statement of facts.

Payment of Additional Amounts

All payments of principal and interest in respect of the notes will be made free and clear of, and without deduction or withholding for or on account of any present or future taxes, duties, assessments or other governmental charges of whatsoever nature imposed, levied, collected, withheld or assessed by the United States or any political subdivision or taxing authority of or in the United States, unless such withholding or deduction is required by law.

We will pay to the beneficial owner of notes who is a Non-U.S. Person (as defined below) additional amounts as may be necessary so that every net payment of the principal of and premium, if any, and interest on such holder's note, after deduction or withholding for or on account of any present or future tax, assessment or other governmental charge imposed upon that beneficial owner by the United States or any taxing authority thereof or therein, will not be less than the amount provided in such holder's notes to be then due and payable. We will not be required, however, to make any payment of additional amounts for or on account of:

(a) any tax, assessment or other governmental charge that would not have been imposed but for (1) the existence of any present or former connection (other than a connection arising solely from the ownership of those notes or the receipt of payments in respect of those notes) between that beneficial owner, or between a fiduciary, settlor, beneficiary of, member or shareholder of, or possessor of a power over, that beneficial owner, if that beneficial owner is an estate, trust, partnership or corporation, and the United States, including that beneficial owner, or that fiduciary, settlor, beneficiary, member, shareholder or possessor, being or having been a citizen or resident or treated as a resident of the United States or being or having been engaged in trade or business or present in the United States or having had a permanent establishment in the United States or (2) the presentation of a debt security for payment on a date more than 30 days after the later of the date on which that payment becomes due and payable and the date on which payment is duly provided for;

(b) any estate, inheritance, gift, sales, transfer, excise, personal property, wealth, interest equalization or similar tax, assessment or other governmental charge;

(c) any tax, assessment or other governmental charge imposed on foreign personal holding company income or by reason of that beneficial owner's past or present status as a passive foreign investment company, a controlled foreign corporation, a foreign tax exempt organization or a personal holding company with respect to the United States or as a corporation that accumulates earnings to avoid U.S. federal income tax;

(d) any tax, assessment or other governmental charge which is payable otherwise than by withholding from payment of principal of or premium, if any, or interest on such holder's notes;

(e) any tax, assessment or other governmental charge required to be withheld by any paying agent from any payment of principal of and premium, if any, or interest on any note if that payment can be made without withholding by any other paying agent;

(f) any tax, assessment or other governmental charge which would not have been imposed but for the failure of a beneficial owner or any holder of notes to comply with our request to comply with certification, information, documentation or other reporting requirements concerning the nationality, residence, identity or connections with the United States of the beneficial owner or any holder of the notes (including, but not limited to, the requirement to provide Internal Revenue Service Forms W-8BEN, Forms W-8ECI, or any subsequent versions thereof or successor thereto, and including, without limitation, any documentation requirement under an applicable income tax treaty);

(g) any tax, assessment or other governmental charge imposed on interest received by (1) a 10% shareholder (as defined in Section 871(h)(3)(B) of the U.S. Internal Revenue Code of 1986, as amended (the "Code"), and the regulations that may be promulgated thereunder) of our company or (2) a controlled foreign corporation that is related to us within the meaning of Section 864(d)(4) of the Code, or (3) a bank receiving interest described in Section 881(c)(3)(A) of the Code;

(h) any withholding or deduction that is imposed on a payment to an individual and is required to be made pursuant to European Council Directive 2003/48/EC relating to the taxation of savings, or any law implementing or complying with, or introduced in order to conform to, such Directive;

(i) with respect to the 2028 Notes, any taxes payable under Sections 1471 through 1474 of the Code (or any amended or successor version of such Sections), any regulations or other guidance thereunder, or any agreement (including any intergovernmental agreement) entered into in connection therewith; or

(j) any combination of items (a), (b), (c), (d), (e), (f), (g), (h) and (i);

nor will we pay any additional amounts to any beneficial owner or holder of notes who is a fiduciary or partnership to the extent that a beneficiary or settlor with respect to that fiduciary or a member of that partnership or a beneficial owner thereof would not have been entitled to the payment of those additional amounts had that beneficiary, settlor, member or beneficial owner been the beneficial owner of those notes.

As used in the preceding paragraph, “Non-U.S. Person” means any corporation, partnership, individual or fiduciary that is, for United States federal income tax purposes, a foreign corporation, a non-resident alien individual who has not made a valid election to be treated as a United States resident, a non-resident fiduciary of a foreign estate or trust or a foreign partnership, one or more of the members of which is, as to the United States, a foreign corporation, a non-resident alien individual or a non-resident fiduciary of a foreign estate or trust.

Book-Entry and Settlement

Each series of notes was issued in the form of one or more global securities, in definitive, fully registered form without interest coupons, each of which we refer to as a “global security.” Each such global security was deposited with The Bank of New York Mellon, as common depositary (the “Common Depositary”) and registered in the name of the Common Depositary or its nominee.

Beneficial interests in the global securities are represented, and transfers of such beneficial interest was effected, through accounts of financial institutions acting on behalf of beneficial owners as direct or indirect participants in Clearstream Banking, *société anonyme*, which we refer to as “Clearstream,” or Euroclear Bank SA/ NV, as operator of the Euroclear System, which we refer to as “Euroclear.” Investors may hold notes directly through Clearstream or Euroclear, if they are participants in such systems, or indirectly through organizations that are participants in such systems.

Beneficial interests in the global securities will be shown on, and transfers of beneficial interests in the global securities are made only through, records maintained by Clearstream or Euroclear and their participants. The London paying agent will wire payments on the notes to the Common Depositary as the holder of the global securities. The trustee, the London paying agent and we will treat the Common Depositary or any successor nominee to the Common Depositary as the owner of the global securities for all purposes. Accordingly, the trustee, the London paying agent and we will have no direct responsibility or liability to pay amounts due with respect to the global securities to you or any other beneficial owners in the global securities. Any redemption or other notices with respect to the notes will be sent by us directly to Clearstream or Euroclear, which will, in turn, inform the direct participants (or the indirect participants), which will then contact you as a beneficial holder, all in accordance with the rules of Clearstream or Euroclear, as the case may be, and the internal procedures of the direct participant (or the indirect participant) through which beneficial interest in the notes are held.

Certificated Notes

Subject to certain conditions, the notes represented by the global securities are exchangeable for certificated notes in definitive form of like tenor in minimum denominations of €100,000 principal amount and integral multiples of €1,000 in excess thereof if:

(1) the Common Depositary notifies us that it is unwilling or unable to continue as depositary or if the Common Depositary ceases to be eligible under the indenture and we do not appoint a successor depository within 90 days;

(2) we determine that the notes will no longer be represented by global securities and execute and deliver to the trustee an order to that effect; or

(3) an event of default with respect to the notes will have occurred and be continuing.

Any note that is exchangeable as above is exchangeable for certificated notes issuable in authorized denominations and registered in such names as the Common Depositary shall direct. Subject to the foregoing, a global security is not exchangeable, except for a global security of the same aggregate denomination to be registered in the name of the Common Depositary or its nominee.

Trustee, Paying Agents and Security Registrar

The Bank of New York Mellon Trust Company, N.A. is the trustee under the indenture governing the notes. The Bank of New York Mellon Trust Company, N.A. is a national banking association organized under and governed by the laws of the United States of America, and provides trust services and acts as indenture trustee for numerous corporate securities issuances, including for other series of debt securities of which we are the issuer. The Bank of New York Mellon, London Branch, is the paying agent for the notes in London.

Base Indenture Provisions:

Governing Law

The indenture and the notes are governed by, and construed in accordance with, the laws of the State of New York.

Consolidation, Merger and Sale of Assets

The indenture provides that we may consolidate with or merge with or into any other person, and may sell, transfer, or lease or convey all or substantially all of our properties and assets to another person; provided that the following conditions are satisfied:

- we are the continuing entity, or the resulting, surviving or transferee person (the “Successor”) is a person organized and existing under the laws of the United States of America, any state thereof or the District of Columbia and the Successor (if not us) will expressly assume, by supplemental indenture, all of our obligations under the debt securities and the indenture and, for each security that by its terms provides for conversion, provide for the right to convert such security in accordance with its terms;

- immediately after giving effect to such transaction, no default or event of default under the indenture has occurred and is continuing; and

- if requested, the trustee receives from us, an officers’ certificate and an opinion of counsel that the merger, consolidation or transfer and such supplemental indenture, as the case may be, complies with the applicable provisions of the indenture.

If we consolidate or merge with or into any other person or sell, transfer, lease or convey all or substantially all of our properties and assets in accordance with the indenture, the Successor will be substituted for us in the indenture, with the same effect as if it had been an original party to the indenture.

As a result, the Successor may exercise our rights and powers under the indenture, and we will be released from all our liabilities and obligations under the indenture and under the debt securities.

Any substitution of the Successor for us might be deemed for federal income tax purposes to be an exchange of the debt securities for "new" debt securities, resulting in recognition of gain or loss for such purposes and possibly certain other adverse tax consequences to beneficial owners of the debt securities. Holders should consult their own tax advisors regarding the tax consequences of any such substitution.

For purposes of this covenant, "person" means any individual, corporation, partnership, limited liability company, joint venture, association, joint-stock company, trust, unincorporated organization or government or any agency or political subdivision thereof or any other entity.

Events of Default

Each of the following events are defined in the indenture as an "event of default" (whatever the reason for such event of default and whether or not it will be voluntary or involuntary or be effected by operation of law or pursuant to any judgment, decree or order of any court or any order, rule or regulation of any administrative or governmental body) with respect to the debt securities of any series:

(1) default in the payment of any installment of interest on any debt securities of that series for 30 days after becoming due;

(2) default in the payment of principal of or premium, if any, on any debt securities of that series when it becomes due and payable at its stated maturity, upon optional redemption, upon declaration or otherwise;

(3) default in the deposit of any sinking fund payment, when and as due by the terms of any debt securities of that series;

(4) default in the performance, or breach, of any covenant or agreement of ours in the indenture with respect to the debt securities of that series (other than as referred to in clause (1), (2) or (3) above), which continues for a period of 90 days after written notice to us by the trustee or to us and the trustee by the holders of at least 25% in aggregate principal amount of the outstanding debt securities of that series;

(5) we pursuant to or within the meaning of the Bankruptcy Law:

- commence a voluntary case or proceeding;
- consent to the entry of an order for relief against us in an involuntary case or proceeding;
- consent to the appointment of a Custodian of us or for all or substantially all of our property;
- make a general assignment for the benefit of our creditors;
- file a petition in bankruptcy or answer or consent seeking reorganization or relief;
- consent to the filing of such petition or the appointment of or taking possession by a Custodian; or
- take any comparable action under any foreign laws relating to insolvency;

(6) a court of competent jurisdiction enters an order or decree under any Bankruptcy Law that:

- is for relief against us in an involuntary case, or adjudicates us insolvent or bankrupt;

- appoints a Custodian of us or for all or substantially all of our property; or
- orders the winding-up or liquidation of us (or any similar relief is granted under any foreign laws);

and the order or decree remains unstayed and in effect for 90 days; or

(7) any other event of default provided with respect to debt securities of that series occurs.

“Bankruptcy Law” means Title 11, United States Code or any similar federal or state or foreign law for the relief of debtors.

“Custodian” means any custodian, receiver, trustee, assignee, liquidator or other similar official under any Bankruptcy Law.

If an event of default with respect to debt securities of any series (other than an event of default relating to certain events of bankruptcy, insolvency, or reorganization of us) occurs and is continuing, the trustee by notice to us, or the holders of at least 25% in aggregate principal amount of the outstanding debt securities of that series by notice to us and the trustee, may, and the trustee at the request of these holders will, declare the principal of and premium, if any, and accrued and unpaid interest on all the debt securities of that series to be due and payable. Upon such a declaration, such principal, premium and accrued and unpaid interest will be due and payable immediately. If an event of default relating to certain events of bankruptcy, insolvency, or reorganization of us occurs and is continuing, the principal of and premium, if any, and accrued and unpaid interest on the debt securities of that series will become and be immediately due and payable without any declaration or other act on the part of the trustee or any holders.

The holders of not less than a majority in aggregate principal amount of the outstanding debt securities of any series may rescind a declaration of acceleration and its consequences, if we have deposited certain sums with the trustee and all events of default with respect to the debt securities of that series, other than the non-payment of the principal or interest which have become due solely by such acceleration, have been cured or waived, as provided in the indenture.

An event of default for a particular series of debt securities does not necessarily constitute an event of default for any other series of debt securities issued under the indenture.

We are required to furnish the trustee annually a statement by certain of our officers to the effect that, to the best of their knowledge, we are not in default in the fulfillment of any of our obligations under the indenture or, if there has been a default in the fulfillment of any such obligation, specifying each such default.

No holder of any debt securities of any series will have any right to institute any judicial or other proceeding with respect to the indenture, or for the appointment of a receiver or trustee, or for any other remedy unless:

(1) an event of default has occurred and is continuing and such holder has given the trustee prior written notice of such continuing event of default with respect to the debt securities of that series;

(2) the holders of not less than 25% of the aggregate principal amount of the outstanding debt securities of that series have requested the trustee to institute proceedings in respect of such event of default;

(3) the trustee has been offered indemnity reasonably satisfactory to it against its costs, expenses and liabilities in complying with such request;

(4) the trustee has failed to institute proceedings 60 days after the receipt of such notice, request and offer of indemnity; and

(5) no direction inconsistent with such written request has been given for 60 days by the holders of a majority in aggregate principal amount of the outstanding debt securities of that series.

The holders of a majority in aggregate principal amount of outstanding debt securities of a series will have the right, subject to certain limitations, to direct the time, method and place of conducting any proceeding for any remedy available to the trustee with respect to the debt securities of that series or exercising any trust or power conferred to the trustee, and to waive certain defaults. The indenture provides that if an event of default occurs and is continuing, the trustee will exercise such of its rights and powers under the indenture, and use the same degree of care and skill in their exercise, as a prudent person would exercise or use under the circumstances in the conduct of such person's own affairs. Subject to such provisions, the trustee will be under no obligation to exercise any of its rights or powers under the indenture at the request of any of the holders of the debt securities of a series unless they will have offered to the trustee security or indemnity satisfactory to the trustee against the costs, expenses and liabilities which might be incurred by it in compliance with such request.

Notwithstanding the foregoing, the holder of any debt security will have an absolute and unconditional right to receive payment of the principal of and premium, if any, and interest on that debt security on or after the due dates expressed in that debt security and to institute suit for the enforcement of payment.

Discharge, Defeasance and Covenant Defeasance

We may discharge certain obligations to holders of the debt securities of a series that have not already been delivered to the trustee for cancellation and that either have become due and payable or will become due and payable within one year (or scheduled for redemption within one year) by depositing with the trustee, in trust, money in an amount sufficient to pay the entire indebtedness including the principal and premium, if any, and interest to the date of such deposit (if the debt securities have become due and payable) or to the maturity thereof or the redemption date of the debt securities of that series, as the case may be. We may direct the trustee to invest such funds in U.S. Treasury securities with a maturity of one year or less or in a money market fund that invests solely in short-term U.S. Treasury securities.

The indenture provides that we may elect either (1) to defease and be discharged from any and all obligations with respect to the debt securities of a series (except for, among other things, obligations to register the transfer or exchange of the debt securities, to replace temporary or mutilated, destroyed, lost or stolen debt securities, to maintain an office or agency with respect to the debt securities and to hold moneys for payment in trust) ("legal defeasance") or (2) to be released from our obligations to comply with the restrictive covenants under the indenture, and any omission to comply with such obligations will not constitute a default or an event of default with respect to the debt securities of a series and clauses (4) and (7) under "-Events of Default" will no longer be applied ("covenant defeasance"). Legal defeasance or covenant defeasance, as the case may be, will be conditioned upon, among other things, the irrevocable deposit by us with the trustee, in trust, of an amount, or U.S. government obligations, or both, applicable to the debt securities of that series which through the scheduled payment of principal and interest in accordance with their terms will provide money in an amount sufficient to pay the principal or premium, if any, and interest on the debt securities on the scheduled due dates therefor.

If we effect covenant defeasance with respect to the debt securities of any series, the amount, or U.S. government obligations, or both, on deposit with the trustee will be sufficient, in the opinion of a nationally recognized firm of independent accountants, to pay amounts due on the debt securities of that series at the time of the stated maturity but may not be sufficient to pay amounts due on the debt securities of that series at the time of the acceleration resulting from such event of default. However, we would remain liable to make payment of such amounts due at the time of acceleration.

We have elected pursuant to Section 1301 of the indenture, to have both Section 1302 (legal defeasance) and Section 1303 (covenant defeasance) of the indenture apply to the all of the notes.

With respect to the 2028 Notes, "U.S. government obligation" means (I)(x) any security which is (i) a direct obligation of the United States of America for the payment of which the full faith and credit of the United States of America is pledged or (ii) an obligation of a Person (as defined in the indenture) controlled or supervised by and acting as an agency or instrumentality of the United States of America the payment of which is unconditionally guaranteed as a full faith and credit obligation by the United States of

America, which, in either case (I)(x)(i) or (ii), is not callable or redeemable at the option of the issuer thereof, and (y) any depositary receipt issued by a bank (as defined in Section 3(a)(2) of the Securities Act) as custodian with respect to any U.S. government obligation which is specified in clause (x) above and held by such bank for the account of the holder of such depositary receipt, or with respect to any specific payment of principal of or interest on any U.S. government obligation which is so specified and held; *provided* that (except as required by law) such custodian is not authorized to make any deduction from the amount payable to the holder of such depositary receipt from any amount received by the custodian in respect of the U.S. government obligation or the specific payment of principal or interest evidenced by such depositary receipt or (II)(x) any security which is (i) a direct obligation of the German Government (as defined in the indenture) or (ii) an obligation of a Person (as defined in the indenture) controlled or supervised by and acting as an agency or instrumentality of the German Government (as defined in the indenture) the payment of which is fully and unconditionally guaranteed by the German Government (as defined in the indenture), the central bank of the German Government (as defined in the indenture) or a governmental agency of the German Government (as defined in the indenture), which, in either case (II)(x)(i) or (ii), is not callable or redeemable at the option of the issuer thereof, and (y) certificates, depositary receipts or other instruments which evidence a direct ownership interest in obligations described in clause (II)(x)(i) or (ii) above or in any specific principal or interest payments due in respect thereof.

With respect to the 2033 Notes, “U.S. government obligation” means (x) any security which is (i) a direct obligation of the United States of America for the payment of which the full faith and credit of the United States of America is pledged or (ii) an obligation of a Person (as defined in the indenture) controlled or supervised by and acting as an agency or instrumentality of the United States of America the payment of which is unconditionally guaranteed as a full faith and credit obligation by the United States of America, which, in either case (i) or (ii), is not callable or redeemable at the option of the issuer thereof, and (y) any depositary receipt issued by a bank (as defined in Section 3(a)(2) of the Securities Act of 1933, as amended (the “Securities Act”)) as custodian with respect to any U.S. government obligation which is specified in clause (x) above and held by such bank for the account of the holder of such depositary receipt, or with respect to any specific payment of principal of or interest on any U.S. government obligation which is so specified and held; provided that (except as required by law) such custodian is not authorized to make any deduction from the amount payable to the holder of such depositary receipt from any amount received by the custodian in respect of the U.S. government obligation or the specific payment of principal or interest evidenced by such depositary receipt.

We will be required to deliver to the trustee an opinion of counsel that the deposit and related defeasance will not cause the holders and beneficial owners of the debt securities of that series to recognize income, gain or loss for federal income tax purposes. If we elect legal defeasance, that opinion of counsel must be based upon a ruling from the U.S. Internal Revenue Service or a change in law to that effect.

We may exercise our legal defeasance option notwithstanding our prior exercise of our covenant defeasance option.

Exhibit 10.5

MICROSOFT CORPORATION DEFERRED COMPENSATION PLAN
(Amended and Restated Effective as of April 1, 2024)

1. Purpose.

The purpose of the Microsoft Corporation Deferred Compensation Plan (the "Plan") is to further the long-term growth of Microsoft Corporation (the "Company") by allowing selected Company executives and other senior management or highly compensated employees to defer receipt of certain compensation in order to keep such employees' financial interests aligned with the Company and provide them with a long-term incentive to continue employment with the Company.

The Plan was formerly known as the 1998 Microsoft Corporation Stock Option Gain and Bonus Deferral Program. The name of the Plan was changed pursuant to a restatement effective January 1, 2006.

This Plan is intended (1) to comply with section 409A of the Internal Revenue Code, as amended (the "Code") and official guidance issued thereunder (except with respect to amounts covered by Appendix B), and (2) to be "a plan which is unfunded and is maintained by an employer primarily for the purpose of providing deferred compensation for a select group of management or highly compensated employees" within the meaning of sections 201(2), 301(a)(3) and 401(a)(1) of the Employee Retirement Income Security Act of 1974. Notwithstanding any other provision of this Plan, this Plan shall be interpreted, operated and administered in a manner consistent with these intentions.

2. Effective Date.

The Plan was originally effective November 18, 1998. Except as specifically set forth below, this restatement of the Plan is effective as of April 1, 2024.

3. Definitions.

Account - means a bookkeeping account established by the Company for each Participant electing to defer Eligible Income under the Plan, which may include sub-accounts for different types of Eligible Income deferred and for amounts payable at different times and/or payable in different forms.

Acquisition Retention Bonus - means a bonus provided to a Newly Hired Eligible Employee who continues employment with the Company or a Designated Subsidiary after the acquisition of a business by the Company or a Designated Subsidiary or who begins employment with the Company or a Designated Subsidiary as part of a strategic alliance.

Affiliate - means any corporation or other entity that is treated as a single employer with the Company under Code section 414.

Annual Base Salary - means the regular annual base salary paid to an Eligible Employee.

Annual Bonus - means the amount payable to an Eligible Employee as an annual bonus that is awarded in connection with the Company's annual process under the Annual Bonus Plan or the cash portion of awards under the Executive Incentive Plan.

Board - means the Board of Directors of Microsoft Corporation.

Code - means the Internal Revenue Code of 1986, as amended.

Company - means Microsoft Corporation.

Date of Hire - means the date of a Participant's first day of active employment with the Company and its Affiliates.

Designated Subsidiary - means a subsidiary of the Company that has been approved for participation in the Plan by the Senior HR Officer. A listing of the Designated Subsidiaries is in Appendix A.

Disabled - means:

(a) A Participant (1) is unable to engage in any substantial gainful activity by reason of any medically determinable physical or mental impairment which can be expected to result in death or can be expected to last for a continuous period of not less than 12 months, or (2) is, by reason of any medically determinable physical or mental impairment which can be expected to result in death or can be expected to last for a continuous period of not less than 12 months, receiving income replacement benefits for a period of not less than 3 months under an accident and health plan covering employees of the participant's employer.

(b) The Plan Administrator, in its complete and sole discretion, shall determine whether a Participant is Disabled. The Plan Administrator may require that the Participant submit to an examination on an annual basis, at the expense of the Company, by a competent physician or medical clinic selected by the Plan Administrator to assist in determining whether the Participant is Disabled. On the basis of such medical evidence, the determination of the Plan Administrator as to whether or not the Participant is Disabled (or whether such Participant continues to be Disabled) shall be conclusive.

Eligible Employee - means:

(a)An Employee of the Company or a Designated Subsidiary working in the U.S. at the Company's stock level 68 or above and, effective with respect to Eligible Income earned for periods beginning on or after January 1, 2012, an Employee of the Company or a Designated Subsidiary working in the U.S. at the Company's stock level 67 or above.

(b)An Employee meeting the criteria of subsection (a) will not fail to be considered an Eligible Employee solely as a result of being on paid or unpaid leave.

Eligible Income - means compensation which may be deferred under the Plan, as from time to time determined by the Plan Administrator, including without limitation (1) Regular Enrollment Compensation and (2) New Hire Enrollment Compensation.
Amounts will qualify as "Eligible Income" only if the Participant is on the U.S. payroll of the Company or its Affiliates at the time the amount is payable to the Participant absent deferral. For

the avoidance of doubt, “Eligible Income” does not include severance pay, revenue-based incentives, commitment-based incentives, SCA bonuses, or any other pay that the Plan Administrator does not classify as Regular Enrollment Compensation or New Hire Enrollment Compensation.

Employee - means an individual who is a regular employee on the U.S. payroll of the Company or its Affiliates. The term "Employee" shall not include a person hired as an independent contractor, leased employee, consultant, or a person otherwise designated by the Company or an Affiliate as not eligible to participate in the Plan, even if such person is determined to be a common law employee of the Company or an Affiliate, retroactively or prospectively, by any governmental or judicial authority.

ERISA - means the Employee Retirement Income Security Act of 1974, as amended.

Fiscal Year Compensation - means "fiscal year compensation" as defined under Treas. Reg. § 1.409A-2(a)(6) or any successor thereto.

Hire Date - means the date an Employee becomes employed by the Company or a Designated Subsidiary. In the case of an individual who becomes an Employee upon the acquisition of a business by the Company or a Designated Subsidiary, the Employee's "Hire Date" shall be such Employee’s transfer date.

Investment Options - means a set of investment options, which may include investment options offered under the 401(k) Plan, and which are from time to time determined by the Plan Administrator and used to credit earnings, gains, and losses on Account balances.

Key Employee - means an employee treated as a "specified employee" under Code section 409A(a)(2)(B)(i) as of such employee’s Separation from Service (i.e., a key employee (as defined under Code section 416(i) without regard to paragraph (5) thereof) of a corporation any stock of which is publicly traded on an established securities market or otherwise). Key Employees shall be determined in accordance with Code section 409A, using a December 31 identification date. A listing of Key Employees as of an identification date shall be effective for the 12-month period beginning on the April 1 following the identification date.

New Hire Enrollment Compensation - means compensation for a Newly Hired Eligible Employee which is from time to time determined by the Plan Administrator, including without limitation a (1) Signing Bonus and (2) Acquisition Retention Bonus.

Newly Hired Eligible Employee - means an individual hired by the Company or a Designated Subsidiary who meets the criteria for an Eligible Employee on such individual’s Hire Date, provided that an individual who has previously worked for the Company or an Affiliate will only qualify as a "Newly Hired Eligible Employee" if such individual meets the requirements of Treas. Reg. § 1.409A-2(a)(7) or any successor thereto. Generally, a re-hired individual will meet these requirements if (1) such individual has been paid any and all amounts due under the Plan (and any plans required to be aggregated with the Plan under Code section 409A) prior to re-hire, or (2) such individual has not been eligible to participate, other than the accrual of earnings, in the Plan (or any other plan required to be aggregated with the Plan under Code section 409A) for at least 24 months.

Open Enrollment - means the period or periods during each Plan Year when Eligible Employees may elect to defer amounts under the Plan. Open Enrollment shall be held at the time or times designated by the Plan Administrator.

Participant - means an Eligible Employee who elects to defer Eligible Income under the Plan.

Performance-Based Compensation - means "performance-based compensation" as defined under Code section 409A.

Plan - means the Microsoft Corporation Deferred Compensation Plan, as amended from time to time.

Plan Administrator - means the Senior HR Officer or, with respect to the eligibility of executive officers of the Company to participate in the Plan, the Compensation Committee of the Board.

Plan Year - means the 12-month period from January 1 to December 31.

Pre-2014 Election Amount - means an amount deferred under the Plan (and earnings thereon) pursuant to a deferral election that (1) becomes irrevocable during an Open Enrollment period occurring before July 1, 2013, or (2) is made by a Newly Hired Eligible Employee under Section 5.1(b)(ii) and becomes irrevocable on or before December 31, 2013. Thus, Annual Base Salary amounts deferred after 2013 and Annual Bonuses deferred based on elections made after 2013 will not be considered Pre-2014 Election Amounts.

Regular Enrollment Compensation - means compensation which is from time to time determined by the Plan Administrator, including without limitation (1) Annual Base Salary and (2) Annual Bonus.

Retirement - means a Separation from Service after attaining Retirement Age.

Retirement Age - means one specified date for each Participant occurring on the earlier of: (1) Participant's attainment of age sixty-five (65), or (2) the later of Participant's attainment of age fifty-five (55) or the tenth (10th) anniversary of such Participant’s Date of Hire. When an Employee becomes eligible to participate in the Plan, the Plan Administrator shall determine the Retirement Age for the Employee as one specified date in accordance with the foregoing.

Senior HR Officer - means the senior officer in charge of the Human Resources department.

Separation from Service or Separates from Service - means a "separation from service" with the Company and its Affiliates within the meaning of Code section 409A.

Signing Bonus - means a bonus provided to a Newly Hired Eligible Employee upon acceptance of an offer of employment with the Company or a Designated Subsidiary, including bonuses provided to a Newly Hired Eligible Employee who accepted an offer to continue employment with the Company or a Designated Subsidiary after the acquisition of a business by the Company or a Designated Subsidiary or to begin employment with the Company or Designated Subsidiary as part of a strategic alliance.

401(k) Plan - means the Microsoft Corporation Savings Plus 401(k) Plan.

4. Participation.

4.1.An Eligible Employee becomes an active Participant in the Plan on the date the Eligible Employee first enrolls in the Plan by electing to defer all or any portion of such Eligible Employee’s Eligible Income. An Eligible Employee may enroll in the Plan during Open Enrollment in accordance with Section 5.1(b)(i) or pursuant to Section 5.1(c). A Newly Hired Eligible Employee may enroll before such Newly Hired Eligible Employee’s Hire Date in accordance with 5.1(b)(ii).

4.2.An Eligible Employee who has been an active Participant under the Plan will cease to be a Participant on the date such Eligible Employee’s Account is fully distributed.

5. Participant Accounts.

5.1. Elections to Defer Eligible Income.

(a) Initial Deferral Election. An Eligible Employee may make an irrevocable election to defer the following types of Eligible Income in one (1) percent increments up to the specified maximum percentages:

(i) An Eligible Employee may elect to defer up to 75% of such Eligible Employee’s Annual Base Salary.

(ii) An Eligible Employee may elect to defer up to 100% of an Annual Bonus.

(iii) An Eligible Employee may elect to defer up to 90% of New Hire Enrollment Compensation.

Eligible Employees are not permitted to defer gains on the exercise of a stock option under the Plan after December 31, 2004.

(b) Time and Manner of Making an Initial Election.

(i) An Eligible Employee may make an election to defer one or more types of Regular Enrollment Compensation during an Open Enrollment period that occurs in the Plan Year preceding the Plan Year in which the Regular Enrollment Compensation begins to be earned. A deferral election shall be made in accordance with procedures established by the Plan Administrator. An Employee's election during such an Open Enrollment period will not be given effect if the Employee ceases to be an Eligible Employee by the last day of the month in which the Open Enrollment period occurs.

(ii) A Newly Hired Eligible Employee may make an election to defer one or more types of New Hire Enrollment Compensation in accordance with procedures established by the Plan Administrator, provided such election occurs before such Newly Hired Eligible Employee’s Hire Date and such election shall only apply to amounts earned after the election is filed. A Newly Hired Eligible Employee may make an election to defer Regular Enrollment Compensation during an Open Enrollment period that follows or coincides with such Newly Hired Eligible Employee’s Hire Date.

(c) Alternative Election Deadlines. Notwithstanding the rules in subsection (b), if the Plan Administrator, in its sole discretion, determines that:

(i) Eligible Income constitutes Performance-Based Compensation that is based on services performed over a performance period of at least twelve (12) months, the Plan Administrator may establish procedures, including an Open Enrollment period, under which an Eligible Employee may elect to defer such Performance-Based Compensation, but such election must be made no later than six (6) months before the end of the performance period; or

(ii) Eligible Income constitutes Fiscal Year Compensation, the Plan Administrator may establish procedures, including an Open Enrollment period, under which an Eligible Employee may elect to defer such Fiscal Year Compensation, but such election must be made no later than the last day of the Company's fiscal year immediately preceding the first fiscal year in which services are performed related to such Eligible Income.

An Employee's election under this Section will not be given effect if the Employee ceases to be an Eligible Employee by the deadline stated above for making such an election.

(d) Cancellation of Election. If a Participant becomes Disabled, receives a hardship withdrawal under the 401(k) Plan, or the Plan Administrator determines the Participant meets the requirements to obtain a distribution under Section 6.6 on account of an unforeseeable emergency (regardless of whether such distribution is actually made) during a Plan Year, such Participant's deferrals for such Plan Year shall be cancelled. For the avoidance of doubt, this Section 5.1(d) shall not affect such Participant's deferral election made during Open Enrollment in such Plan Year with respect to amounts earned or payable in a subsequent Plan Year.

5.2. Crediting of Deferrals. Eligible Income deferred by a Participant under the Plan shall be credited to the Participant's Account as soon as practicable after the amounts would have otherwise been paid to the Participant.

5.3. Vesting. A Participant shall at all times be one-hundred (100) percent vested in any amounts credited to such Participant's Account.

5.4. Investments and Earnings. The Company shall periodically credit gains, losses and earnings to a Participant's Account, until the full balance of the Account has been distributed. Amounts shall be credited to a Participant's Account under this Section based on the results that would have been achieved had amounts credited to the Account been invested as soon as practicable after crediting into the Investment Options selected by the Participant. The Plan Administrator shall specify procedures to allow Participants to make elections as to the deemed investment of amounts newly credited to such Participants' Accounts, as well as the deemed investment of amounts previously credited to such Participants' Accounts. Nothing in this Section or otherwise in the Plan, however, will require the Company to actually invest any amounts in such Investment Options or otherwise.

5.5. Employment, State and Local Taxes. The Participant's share of FICA and FUTA taxes, or any state or local taxes, owed on Eligible Income the Participant elects to defer shall be deducted from the amount deferred or from other compensation payable to the Participant, at the election of the Company in accordance with Code section 409A.

6. Distribution of Account Balances.

6.1. Distribution Form.

(a) A Participant may elect to have amounts deferred under the Plan (and earnings thereon) distributed in a lump sum payment or in annual installments over a period ranging from three (3) to fifteen (15) years.

(b) A Participant must specify the form in which a deferred amount (and earnings thereon) will be distributed at the time of making the initial deferral election under Section 5.1.

(c) Notwithstanding the distribution form elected under subsection (a), if at the time a portion of a Participant's Account is to be distributed, the portion of the balance to be distributed is less than $50,000, that portion shall be distributed in a lump sum payment at such time, provided that this subsection (c) shall not apply to any amounts deferred under the Plan pursuant to a deferral election that becomes irrevocable on or after June 30, 2011 (and earnings thereon).

(d) Distribution of a Participant's Account balance shall be made in cash.

6.2. Distribution Time of Post-2013 Election Amounts. The rules in this Section 6.2 shall apply to amounts deferred under the Plan, excluding any portion of a Participant's Account attributable to Pre-2014 Election Amounts.

(a) A Participant may elect to have distribution of a deferred amount (and earnings thereon) commence as of the following dates:

(i) A specified time (a particular month and year); or

(ii) Upon the Participant's Separation from Service (in which case distributions will commence in the month following Separation from Service).

(b) A Participant must specify such Participant’s distribution commencement election at the time of making the initial deferral election under Section 5.1.

(c) If a Participant elects to have a deferred amount distributed as of a specified time, the specified time must be at least twelve (12) months after the date on which the final payment of the deferred amount would have been made to the Participant absent deferral.

6.3. Distribution of Pre-2014 Election Amounts Upon Separation from Service. The rules in this Section 6.3 shall only apply to Pre-2014 Election Amounts.

(a) If a Participant has reached Retirement Age at the time of Separation from Service, the distribution of a Pre-2014 Election Amount will commence as follows:

(i) If the Participant elected commencement upon Retirement, the distribution will commence in the month following Retirement.

(ii) If the Participant elected commencement upon a specified time, the distribution will commence in the specified month and year.

(b) Notwithstanding a Participant's elections with respect to Pre-2014 Election Amounts, (i) if a Participant Separates from Service prior to reaching Retirement Age, the portion of such Participant's Account balance attributable to Pre-2014 Election Amounts shall be distributed in an immediate lump sum payment in the month following the Separation from Service, and (ii) if a Participant becomes Disabled prior to attaining Retirement Age while employed with the Company or an Affiliate, the portion of such Participant's Account balance attributable to Pre-2014 Election Amounts shall be distributed in an immediate lump sum payment in the month following the date the Participant becomes Disabled.

6.4. Key Employee Delay. Except as otherwise permitted under IRS guidance, if a distribution is to be made upon the Separation from Service of a Key Employee, distribution may not be made before the date which is six months after the date of the Key Employee's Separation from Service (or, if earlier, before the date of death of the Key Employee). Any payments that would otherwise be made during this period of delay shall be paid in accordance with the elected distribution method and the terms of the Plan in the seventh month following Separation from Service (or, if earlier, following the Key Employee's death in accordance with Sections 6.5(a) and (b) below).

6.5. Distributions Upon Death.

(a) Notwithstanding a Participant's elections with respect to Pre-2014 Election Amounts, if a Participant dies prior to attaining Retirement Age while employed with the Company or an Affiliate, the portion of such Participant's Account balance attributable to the Pre-2014 Election Amounts shall be distributed to the Participant's beneficiary in an immediate single lump sum payment as soon as administratively feasible following the Participant's death. For the avoidance of doubt, if a Participant attains Retirement Age while employed with the Company or an Affiliate, and dies thereafter (whether employed by the Company or an Affiliate at the time of death or not), the portion of such Participant's Account balance attributable to the Pre-2014 Election Amounts shall be distributed to the Participant's beneficiary in the time and form of payment elected by the Participant under Section 6.1 and 6.3.

(b) Notwithstanding a Participant's elections under Sections 6.1 and 6.2, if a Participant dies, such Participant's Account balance, excluding any portion attributable to Pre-2014 Election Amounts, shall be distributed to the Participant's beneficiary in a single lump sum payment as soon as administratively feasible following the Participant's death.

(c) A Participant shall designate a beneficiary prior to death in accordance with procedures established by the Plan Administrator. If a Participant has not properly designated a beneficiary or if no designated beneficiary is living on the date of distribution, the Participant's Account shall be distributed to the Participant's beneficiary designated under the 401(k) Plan, or if no designated beneficiary under the 401(k) Plan is living, in accordance with the default provisions under the 401(k) Plan.

(d) For purposes of determining the proper death beneficiary under this Plan, this Plan shall not be interpreted as preempting applicable state law regarding the ownership rights of Accounts upon a Participant's death. For example, although this Plan states that upon a Participant's death, Account balances will be paid to such Participant's beneficiary, the personal representative may be obligated to pay any benefits owed to a spouse or otherwise as a result of any applicable community property laws.

6.6. Withdrawals for Unforeseeable Emergency. A Participant may withdraw all or any portion of such Participant's Account balance for an Unforeseeable Emergency. The amounts distributed with respect to an Unforeseeable Emergency may not exceed the amounts necessary to satisfy such Unforeseeable Emergency plus amounts necessary to pay taxes reasonably anticipated as a result of the distribution, after taking into account the extent to which such hardship is or may be relieved through reimbursement or compensation by insurance or otherwise or by liquidation of the Participant's assets (to the extent the liquidation of such assets would not itself cause severe financial hardship) or by cessation of deferrals under the Plan. "Unforeseeable Emergency" means for this purpose a severe financial hardship to a Participant resulting from an illness or accident of the Participant, the Participant's spouse, or a dependent of the Participant, loss of the Participant's property due to casualty, or other similar extraordinary and unforeseeable circumstances arising as a result of events beyond the control of the Participant.

Unless otherwise required by IRS guidance, a Participant shall not be required to take any available hardship withdrawals from the 401(k) Plan before being eligible to receive a withdrawal under this section.

6.7. Changes in Time or Form of Distribution. To the extent permitted in accordance with procedures established by the Plan Administrator, including (without limitation) restrictions with respect to the frequency with which participants can make such elections, a Participant may make one or more subsequent elections to change the time or form of a distribution to be made as of a specified time or upon the occurrence of a distributable event for a deferred amount, but such an election will be effective only if the following conditions are satisfied:

(a) The election may not take effect until at least twelve (12) months after the date on which the election is made;

(b) A distribution may not be made earlier than at least five (5) years from the date the distribution would have otherwise been made;

(c) In the case of an election to change the time or form of a distribution payable as of a specified time, the election must be made at least twelve (12) months before the date of the first scheduled distribution; and

(d) The election may not result in an impermissible acceleration of payment prohibited under Code section 409A.

6.8. Effect of Taxation. If a portion of the Participant's Account balance is includible in income under Code section 409A, such portion shall be distributed immediately to the Participant.

6.9. Payment of Taxes. If state, local, or foreign tax obligations arise from participation in the Plan that apply to an amount deferred under the Plan before such amount is paid or made available to the Participant (the "Taxes"), the Company shall pay a portion of such deferred amount by distribution (a) to the Participant in the form of withholding pursuant to provisions of applicable state, local, or foreign law; or (b) directly to the Participant. In no event shall the total payment under this Section 6.9 exceed the aggregate amount of the Taxes, and the income tax withholding related to such Taxes.

6.10. Settlement of Bona Fide Dispute. Subject to certain presumptions under Code section 409A, if an arm's length, bona fide dispute between a Participant and the Company arises as to the Participant's right to an amount deferred under the Plan, the payment of the deferred amount as part of a settlement of such dispute shall be distributed immediately to the Participant.

6.11. Offset for Obligations to Company. If the Participant has any debt, obligation or other liability representing an amount owing to the Company (the "Debt"), incurred in the ordinary course of such Participant's employment relationship, the Company shall offset the Debt against the Participant's Account balance. The Company shall reduce the Participant's Account balance in satisfaction of the Debt at the same time and in the same amount as the Debt otherwise would have been due and collected from the Participant; provided however, in no event shall the amount of such offset in any of the Company's taxable years exceed $5,000.

6.12. 2005 Deferred Compensation. Except as provided in Appendix C, Sections 6.1 - 6.11 shall govern the distribution of compensation earned and deferred under the Plan during the 2005 Plan Year.

6.13. Pre-2005 Deferrals. Notwithstanding the foregoing, Appendix B governs the distribution of amounts that were earned and vested (within the meaning of Code section 409A and regulations thereunder) under the Plan prior to 2005 (and earnings thereon) and are exempt from the requirements of Code section 409A.

7. Administration.

7.1. General Administration. The Plan Administrator shall be responsible for the operation and administration of the Plan and for carrying out the provisions hereof. The Plan Administrator shall have the full authority and discretion to make, amend, interpret, and enforce all appropriate rules and regulations for the administration of this Plan and decide or resolve any and all questions, including interpretations of this Plan, as may arise in connection with this Plan. Except as otherwise provided in Section 7.2, any such action taken by the Plan Administrator shall be final and conclusive on any party. To the extent the Plan Administrator has been granted discretionary authority under the Plan, the Plan Administrator's prior exercise of such authority shall not obligate it to exercise its authority in a like fashion thereafter. The Plan Administrator shall be entitled to rely conclusively upon all tables, valuations, certificates, opinions and reports furnished by any actuary, accountant, controller, counsel or other person employed or engaged by the Company with respect to the Plan. The Plan Administrator may, from time to time, employ agents and delegate to such agents, including other employees of the Company, such administrative duties as it sees fit.

7.2. Claims for Benefits.

(a) Filing a Claim. A Participant or the Participant's authorized representative may file a claim for benefits under the Plan. Any claim must be in writing and submitted to the Senior HR Officer at such address as may be specified from time to time. Claimants will be notified in writing of approved claims. A claim is considered approved only if its approval is communicated in writing to a claimant.

(b) Denial of Claim. If a Participant's claim is denied, in whole or in part, a written notice will be furnished to the claimant within 90 days of the date on which the claim is received by the Senior HR Officer. If special circumstances (such as for a hearing) require a longer period, the claimant will be notified in writing, prior to the expiration of the 90-day period, of the reasons for an extension of time; provided, however, that no extensions will be permitted beyond 90 days after the expiration of the initial 90-day period.

(c) Reasons for Denial. A denial or partial denial of a claim will be dated and signed by the Senior HR Officer and will clearly set forth:

(i) the specific reason or reasons for the denial;

(ii) specific reference to pertinent Plan provisions on which the denial is based;

(iii) a description of any additional material or information necessary for the claimant to perfect the claim and an explanation of why such material or information is necessary; and

(iv) an explanation of the procedure for review of the denied or partially denied claim set forth below, including the claimant's right to bring a civil action under ERISA section 502(a) following an adverse benefit determination on review.

(d) Review of Denial. Upon denial of a claim, in whole or in part, a claimant or such claimant's duly authorized representative will have the right to submit a written request to the Senior HR Officer for a full and fair review of the denied claim by filing a written notice of appeal with the Senior HR Officer within 60 days of the receipt by the claimant of written notice of the denial of the claim. A claimant or the claimant's authorized representative will have, upon request and free of charge, reasonable access to, and copies of, all documents, records, and other information relevant to the claimant's claim for benefits and may submit issues and comments in writing. The review will take into account all comments, documents, records, and other information submitted by the claimant relating to the claim, without regard to whether such information was submitted or considered in the initial benefit determination.

If the claimant fails to file a request for review within 60 days of the claim denial notification, the claim will be deemed abandoned and the claimant precluded from reasserting it. If the claimant does file a request for review, the claimant's request must include a description of the issues and evidence the claimant deems relevant. Failure to raise issues or present evidence on review will preclude those issues or evidence from being presented in any subsequent proceeding or judicial review of the claim.

(e) Decision Upon Review. The Senior HR Officer will provide a prompt written decision on review. If the claim is denied on review, the decision shall set forth:

(i) the specific reason or reasons for the adverse determination;

(ii) specific reference to pertinent Plan provisions on which the adverse determination is based;

(iii) a statement that the claimant is entitled to receive, upon request and free of charge, reasonable access to, and copies of, all documents, records, and other information relevant to the claimant's claim for benefits; and

(iv) a statement describing any voluntary appeal procedures offered by the Plan and the claimant's right to obtain the information about such procedures, as well as a statement of the claimant's right to bring an action under ERISA section 502(a).

A decision will be rendered no more than 60 days after the Senior HR Officer's receipt of the request for review, except that such period may be extended for an additional 60 days if the Senior HR Officer determines that special circumstances (such as the need for a hearing) require such extension. If an extension of time is required, written notice of the extension will be furnished to the claimant before the end of the initial 60-day period.

(f) Finality of Determinations; Exhaustion of Remedies. To the extent permitted by law, decisions reached under the claims procedures set forth in this Section shall be final and binding on all parties. No legal action for benefits under the Plan shall be brought unless and until the claimant has exhausted such claimant's remedies under this Section. In any such legal action, the claimant may only present evidence and theories which the claimant presented during the internal claims and appeals procedure. Any claims which the claimant does not in good faith pursue through the review stage of the procedure shall be treated as having been irrevocably waived. Judicial review of a claimant's denied claim shall be limited to a determination of whether the denial was an abuse of discretion based on the evidence and theories the claimant presented during the internal claims and appeals procedure. Any suit or legal action initiated by a claimant under the Plan must be brought by the claimant no later than one year following a final decision on appeal by the Senior HR Officer. The one-year limitation on suits for benefits will apply in any forum where a claimant initiates such suit or legal action.

(g) Disability Claims. Claims for disability benefits shall be determined under the DOL Regulation section 2560.503-1 which is hereby incorporated by reference.

8. Amendment and Termination.

8.1. Amendment or Termination. The Company reserves the right to amend or terminate the Plan when, in the sole discretion of the Company, such amendment or termination is advisable, pursuant to a resolution or other action taken by the Plan Administrator.

Notwithstanding the foregoing, no amendment of the Plan shall apply to amounts that were earned and vested (within the meaning of Code section 409A and regulations thereunder) under the Plan prior to 2005, unless the amendment specifically provides that it applies to such amounts. The purpose of this restriction is to prevent a Plan amendment from resulting in an inadvertent "material modification" to amounts that are "grandfathered" and exempt from the requirements of Code section 409A.

8.2. Effect of Amendment or Termination. No amendment or termination of the Plan shall decrease the amounts credited to a Participant's Account as of such amendment or termination. Upon termination of the Plan, Participants' Account balances shall be distributed in accordance with the terms of Section 6, unless the Company determines in its sole discretion that all such amounts shall be distributed upon termination in accordance with the requirements under Code section 409A.

9. General Provisions.

9.1. Rights Unsecured. The right of a Participant or the Participant's beneficiary to receive a distribution hereunder shall be an unsecured claim against the general assets of the Company, and neither the Participant nor the Participant's beneficiary shall have any rights in or against any amount credited to any Account or any other assets of the Company. The Plan at all times shall be considered entirely unfunded for tax purposes. Any funds set aside by the Company for the purpose of meeting its obligations under the Plan, including any amounts held by a trustee, shall continue for all purposes to be part of the general assets of the Company and shall be available to its general creditors in the event of the Company's bankruptcy or insolvency. The Company's obligation under this Plan shall be that of an unfunded and unsecured promise to pay money in the future.

9.2. No Right to Eligible Income. Nothing in this Plan shall be construed to give any Eligible Employee any right to be granted Eligible Income or any other type of compensation.

9.3. No Enlargement of Rights. No Participant or beneficiary shall have any right to receive a distribution under the Plan except in accordance with the terms of the Plan. Establishment of the Plan shall not be construed to give any Participant the right to continue to be employed by or provide services to the Company or its affiliates or to employment that is not terminable at will.

9.4. No Guarantee of Benefits. Nothing contained in the Plan shall constitute a guarantee by the Company or any other person or entity that the assets of the Company will be sufficient to pay any benefits hereunder.

9.5. Nonalienation of Benefits. This Plan inures to the benefit of and is binding upon the parties hereto and their successors, heirs and assigns; provided, however, that the amounts credited to a Participant's Account are not, except as provided in Sections 9.6 and 6.11, subject in any manner to anticipation, alienation, sale, transfer, assignment, pledge, encumbrance, charge, garnishment, execution or levy of any kind, either voluntary or involuntary, and any attempt to anticipate, alienate, sell, transfer, assign, pledge, encumber, charge or otherwise dispose of any right to any benefits payable hereunder, will be null and void and not binding on the Plan or the Company.

9.6. Taxes. In addition to its rights under Section 5.5, the Company or other payor may withhold from a benefit payment under the Plan or a Participant's wages any federal, state, or local taxes required by law to be withheld with respect to a payment or accrual under the Plan, and shall report such payments and other Plan-related information to the appropriate governmental agencies as required under applicable law.

9.7. Participant's Cooperation. The Participant shall cooperate with the Company by furnishing any and all information requested by the Plan Administrator in order to facilitate the payment of benefits hereunder, taking such physical examinations as the Plan Administrator may deem necessary and taking such other actions as may be requested by the Plan Administrator. If the Participant refuses to cooperate, the Company shall have no further obligation to the Participant under the Plan.

9.8. Incapacity of Recipient. If any person entitled to a distribution under the Plan is deemed in the Plan Administrator’s sole discretion to be incapacitated or otherwise incapable of personally receiving such payment, then, unless and until a claim for such payment is made by a duly appointed guardian or other legal representative of such person, the Plan Administrator may provide for such payment or any part thereof to be made to any other person or institution then contributing toward or providing for the care and maintenance of such person. Any such payment shall be a payment for the account of such person and a complete discharge of any liability of the Company and the Plan with respect to the payment.

9.9. Legally Binding. In the event of any consolidation, merger, acquisition or reorganization, the obligations of the Company under this Plan shall continue and be binding on the Company and its successors or assigns. The rights, privileges, benefits and obligations under the Plan are intended to be legal obligations of the Company and binding upon the Company, its successors and assigns.

9.10. Unclaimed Benefits. Each Participant shall keep the Plan Administrator informed of the Participant’s current address and the current address of the Participant’s designated beneficiary. The Plan Administrator shall not be obligated to search for the whereabouts of any person if the location of a person is not made known to the Plan Administrator.

9.11. Severability. In the event any provision of the Plan shall be held invalid or illegal for any reason, any illegality or invalidity shall not affect the remaining parts of the Plan, but the Plan shall be construed and enforced as if the illegal or invalid provision had never been inserted.

9.12. Words and Headings. Words in any gender shall include all genders and the singular shall include the plural, and vice versa, unless qualified by the context. Any headings used herein are included for ease of reference only, and are not to be construed so as to alter the terms hereof.

9.13. Applicable law and Venue. To the extent not preempted by federal law, the Plan shall be governed by the laws of the State of Washington. In the event the Company or any Participant (or beneficiary) initiates litigation related to this Plan, the venue for such action will be in King County, Washington.

9.14. Waiver of Breach. The waiver by the Company of any breach of any provision of the Plan by the Participant shall not operate or be construed as a waiver of any subsequent breach by the Participant.

9.15. Notice. Any notice or filing required or permitted to be given to the Plan Administrator under the Plan shall be sufficient if in writing and hand-delivered, or sent by first class mail to the principal office of the Company, directed to the attention of the Plan Administrator. Such notice shall be deemed given as of the date of delivery, or, if delivery is made by mail, as of the date shown on the postmark.

9.16. Attorneys' Fees and Costs. In the event that a dispute regarding benefits arises between the Company or Plan Administrator and a Participant (or beneficiary) and such dispute is resolved through arbitration or litigation in court, the prevailing party(ies) shall be entitled to their reasonable attorneys' fees and costs incurred in such action.

The Company has caused this restated Plan to be duly adopted and executed on ______________________.

[Kathleen Hogan signature block]

APPENDIX A

DESIGNATED SUBSIDIARIES

(As of April 1, 2024)

1654: MOL Corporation
1693: Vexcel Corporation
1548: Microsoft Online, Inc.

APPENDIX B

GRANDFATHERED AMOUNTS

Distribution of amounts that were earned and vested (within the meaning of Code section 409A and regulations thereunder) under the Plan prior to 2005 (and earnings thereon) and are exempt from the requirements of Code section 409A shall be made in accordance with the Plan terms as in effect on December 31, 2004 and as summarized in this Appendix B.

B.1 Timing. As soon as practicable following the final day of the Deferral Period for a specific deferral, the Company will distribute to the Participant (or in the case of the Participant's death, the Participant's estate), all proceeds in the Participant's Deferred Bonus Account and will issue to the Participant (or in the event of the Participant's death, the personal representative or beneficiaries of the Participant's estate) shares of Stock credited to the Participant's Deferred Stock Option Gain Account, that are attributed to that deferral. With respect to a specific deferral, the final day of the Deferral Period shall be the earliest of the last day of the Deferral Period selected by the Participant or the date such Participant has a Termination of Employment. Upon Termination of Employment, a Participant will have the same rights with respect to an unexercised Option that such Participant would have if the Participant had not elected to defer the Stock Option Gain relating to that Option. The portion of a Participant's Accounts that can be attributed to a specific deferral shall be determined in the sole discretion of the Plan Administrator.

B.2 Extension of Deferral Period. On a one time basis with respect to each deferral, a Participant may elect in accordance with procedures established by the Plan Administrator to extend the Deferral Period for a Bonus or Stock Option Gain for an additional five (5), seven (7), or ten (10) years, provided that such extension is elected in the calendar year prior, and at least six (6) months prior, to the expiration of the initial Deferral Period and the Participant is an Eligible Executive at the time such Participant makes the election to extend the Deferral Period.

B.3 Disability. In the event of a Participant's Disability and upon application by such Participant, the Plan Administrator may determine that payment of all, or part, of such Participant's Accounts shall be made in a different manner, or on an earlier date than the time or times specified in Section 8.1 above, but only to the extent determined by the Plan Administrator to be reasonably required to satisfy the Participant's need.

B.4 Investment of Accounts. Notwithstanding Section 5.4, a Participant shall not have the right to select among Investment Options for amounts credited to the Participant's Deferred Stock Option Gain Account. Such amounts shall be treated as if invested in Stock at all times.

B.5 Definitions. For purposes of this Appendix B, the following terms shall have the meanings indicated below:

Bonus means the amount payable by the Company to an Eligible Employee as an individual performance bonus, executive bonus or any other bonus/incentive award that is approved by the Plan Administrator for deferral under the Plan.

Deferral Period means with respect to a specific deferral of a Bonus or Stock Option Gain, the period of five (5), seven (7), or ten (10) years from the date on which the corresponding Bonus would otherwise have been paid or the date the Option was scheduled to expire had it not been exercised; provided that, in the event of the Participant's Termination of Employment, the Deferral Period shall end on the date of Termination of Employment.

Deferred Bonus Account means a bookkeeping account established for Bonuses deferred under the Plan.

Deferred Stock Option Gain Account means a bookkeeping account established for Stock Option Gains deferred under the Plan.

Disability means any long-term disability as defined under the Company's long-term disability plan. The Plan Administrator, in its complete and sole discretion, shall determine a Participant's Disability. The Plan Administrator may require that the Participant submit to an examination on an annual basis, at the expense of the Company, by a competent physician or medical clinic selected by the Plan Administrator to assist in the determination of Disability. On the basis of such medical evidence, the determination of the Plan Administrator as to whether or not a condition of Disability exists or continues shall be conclusive.

Eligible Executive means a full-time employee of the Company who is (i) an elected officer of the Company, (ii) at the level of Vice President or above, (iii) at Level 16 or above on the Company's salary range, and (iv) working within the United States of America. In addition, the Plan Administrator may, in the Plan Administrator's discretion, extend coverage to persons who are selected by the Plan Administrator and who either (y) meet all of the foregoing requirements except that such person works outside of the United States of America, or (z) is an officer of a subsidiary of the Company.

Mature Shares means shares of the Company's Stock delivered by a Participant in payment of the exercise price of an Option; provided that Mature Shares shall not include any shares of the Company's Stock that may be received upon exercise of such Option, nor Stock that the Participant purchased pursuant to a prior stock option exercise which occurred less than six months prior to the exercise of such Option.

Option shall mean one or more non-qualified stock options, issued to a Participant under any stock option plan of the Company, with respect to which the Participant has elected to defer the Stock Option Gain. Option shall not include any rights under the Company's Employee Stock Purchase Plan.

Stock means Microsoft Corporation common stock.

Stock Option Gain means the number of shares underlying an Option minus the number of Mature Shares required to pay the exercise price for those shares. For example, if a Participant elects to defer the gain on 100 shares and is required to deliver 10 shares of Stock as payment for the exercise price on the 100 shares, the Stock Option Gain will be 90 shares.

Termination of Employment means the termination of the Participant's employment relationship with the Company for any reason including, without limitation, involuntary termination with or without cause, voluntary termination, disability, death, or retirement.

APPENDIX C

2005 DEFERRED COMPENSATION

This Appendix C sets forth the special rules applicable to compensation eligible for deferral under the Plan from January 1, 2005 through December 31, 2005. Unless otherwise defined herein, capitalized terms used but not otherwise defined herein shall have the meanings given to them in the Plan and Appendix B.

C-1. 2005 Initial Deferral Elections. Notwithstanding anything in Section 5.1 of the Plan to the contrary and only with respect to compensation earned during the 2005 Plan Year ("2005 Income"), an Eligible Employee may make an irrevocable election to defer up to 100% of a Bonus in ten (10) percent increments. Eligible Employees are not permitted to defer gains on the exercise of a stock option under the Plan after December 31, 2004.

C-2. Time of Distribution. The Company will distribute to the Participant (or in the case of the Participant's death, the Participant’s estate) all proceeds in the Participant's Deferred Bonus Account that are attributed to a specific deferral upon the earlier of: (1) the last day of the Deferral Period elected by the Participant; or (2) the date of the Participant's Separation from Service; provided that, if a distribution is to be made upon the Separation from Service of a Key Employee, such distribution is subject to the six month delay set forth in Section 6.4 of the Plan.

For purposes of this Appendix C, "Deferral Period" means with respect to a specific deferral of a Bonus, the period, as elected by the Participant at the time of the deferral election, of five (5), seven (7), or ten (10) years from the date on which the corresponding Bonus would otherwise have been paid.

C-3. Changes in Time or Form of Distribution. To the extent the Company allows a Participant to make a subsequent election to change the time or form of distribution of 2005 Income deferred under the Plan, such election will be effective only if the conditions set forth in Section 6.7 of the Plan are satisfied.

C-4. General Application of the Plan. Other than as set forth above, the terms of the Plan in all other respects and in compliance with Code section 409A shall govern the distribution of 2005 Income deferred under the Plan from January 1, 2005 through December 31, 2005.

Exhibit 19.1

General Insider Trading Policy

1. PURPOSE

To prevent the misuse of material, nonpublic information about Microsoft or other publicly traded companies obtained by virtue of your position at Microsoft.

2. SUMMARY

This policy and Federal and State securities laws prohibit a person from trading in securities of a company when that person has material, nonpublic information relating to that company.

As an employee of Microsoft, you may have access to financial, business, or other information about Microsoft, or other companies, that is both material and not available to the public. If you are in possession of material, nonpublic information about any company, including Microsoft, the securities laws and Microsoft prohibit you from trading in (or gifting) the securities of that company, and you may not disclose the information to anyone else, except as specifically authorized in the performance of your job responsibilities.

This policy also applies to:

- immediate family members who live with you or use your address as their regular address (e.g., spouse, domestic partner, minor children, college students, parents, grandparents, grandchildren, siblings and in-laws);
- trusts, if you are: (1) a trustee who has or shares investment control of the trust securities and you or a member of your immediate family (whether or not they live with you) has an economic interest in the trust securities; (2) a beneficiary who has or shares investment control; or (3) a person who has the power to revoke the trust; and
- partnerships or other entities over which you have a controlling influence.

You are responsible for assuring that individuals or entities listed above comply with this policy.

3. REQUIREMENTS

A. Do Not Trade While in Possession of Material, Nonpublic Information

As an employee of Microsoft, you may have access to financial, business and other information about Microsoft or other companies. When you possess information about Microsoft that is both material and nonpublic, the securities laws and Microsoft prohibit you from buying or selling (or gifting) Microsoft securities.

Similarly, when you possess material, nonpublic information about any publicly traded company that you acquired through your role at Microsoft, including but not limited to a customer or partner of Microsoft or an economically-linked company such as a competitor of Microsoft, the securities laws and Microsoft prohibit you from buying or selling securities in that company.

- Information is 'material' if there is a substantial likelihood that a reasonable investor would consider it important in deciding whether to buy or sell that company's securities.
- Information is 'nonpublic' if it has not been broadly communicated to the marketplace in a manner making it generally available to the investing public. Rumors, even if true and widely reported in the media, do not constitute public disclosure unless publicly confirmed by the company to which it relates.

  Common examples of material information may include:

- Financial results
- Internal projections that significantly differ from external expectations
- Significant proposed or pending acquisitions, dispositions, mergers or joint ventures
- Changes in key executives
- Significant plans, developments or problems related to important products and services
- Changes in the terms of a significant contract with a major customer or partner, or the acquisition or loss of such a contract
- Cybersecurity risks and incidents (including vulnerabilities and breaches)
- Significant litigation or regulatory actions
- Other notable transactions, events or developments outside the ordinary course of business

Material information can be good news or bad news, and can relate to any aspect of a company's business. Either way, the law prohibits making profits and avoiding losses by using material, nonpublic information. It makes no difference whether you receive the information during the course of your job responsibilities or overhear it in the hallway or lunchroom, or anywhere at or away from your normal workplace.

B. Do Not Disclose Material, Nonpublic Information to Others

Do not disclose material, nonpublic information about Microsoft, or another publicly traded company, to anyone else. This is considered illegal 'tipping' and you could be held criminally liable under federal law even if you yourself did not make a trade.

Do not recommend or suggest that anyone else trade the securities of any company, including Microsoft, even if you do not disclose the material, nonpublic information.

C. Limited Exceptions to These Restrictions

There are a few situations where transactions in securities are not prohibited even if you have material, nonpublic information.

- Purchases of Microsoft stock through an existing purchase election under the Employee Stock Purchase Plan (ESPP). However, you may sell your ESPP shares only at a time when you do not have material, nonpublic information.

- Acquiring Microsoft stock through the vesting of employee Stock Awards. You may sell your Stock Award shares only at a time when you do not have material, nonpublic information.
- Exercising options, if you pay the exercise price and withholding taxes in cash. Again, you may sell your shares only at a time when you do not have material, nonpublic information.
- Trades made pursuant to a predetermined, written Rule 10b5-1 Trading Plan. For information on Trading Plans and the specific policies and procedures associated with them, please visit the Rule 10b5-1 Trading plans page.

D. Restrictions on Microsoft Officers and Other Designated Employees

Microsoft has a policy requiring certain officers of the company (corporate vice presidents and above) and other designated employees to only transact in Microsoft securities during specific times throughout the year, commonly referred to as 'trading windows'. Those officers and employees will be notified if they are subject to the Restricted Trading Window Policy.

E. Ask Questions

If you have any questions about this policy or how it applies under certain circumstances, please visit the Insider trading compliance site or contact (**)@microsoft.com.

4. PROCEDURE

Any person who knows of, or believes there is, a violation of this policy must report the violation immediately to (**)@microsoft.com.

5. EXCEPTIONS

There are no exceptions to this policy.

6. ENFORCEMENT

Violation of this policy may result in disciplinary action, up to and including immediate termination of employment. Employees who engage in insider trading or tipping may also be subject to civil and criminal fines, as well as imprisonment.

7. APPLICATION

This policy applies to all employees.

8. COUNTRY OR BUSINESS UNIT SUPPLEMENT

9. RELATED DOCUMENTS

Restricted Trading Window Policy

10. TRANSLATIONS

11. DOCUMENT HISTORY

Exhibit 19.2

Restricted Trading Window Policy

1. PURPOSE

This policy describes the restrictions on trading Microsoft securities that apply to Microsoft officers and other designated persons due to their regular access to material, nonpublic information about Microsoft.

Members of the Senior Leadership Team (“SLT”) who are direct reports to the CEO (“SLT CEO Direct Reports”) are subject to an additional notification requirement described below.

2. SUMMARY

Microsoft's officers and other designated employees are subject to additional trading restrictions for Microsoft securities to those set forth in the General Insider Trading Policy. In general, if you fall in this group you may only trade (or gift) Microsoft securities during an open window period when your trading status is designated as Trading Allowed in the Insider Trading Compliance Tool.

3. REQUIREMENTS

A. Application of the Policy

This policy applies to all transactions in Microsoft securities by Microsoft's officers and other employees with regular access to material, nonpublic information (collectively 'Insiders'). In general, you will know you are an Insider and subject to this policy because you will be notified. The requirements in this policy are in addition to the General Insider Trading Policy, which applies to all employees.

This policy applies to transactions involving Microsoft securities that are beneficially owned by Insiders. 'Beneficial ownership' is based on the Insider's direct or economic interest in the securities. Some common examples of relationships that may give rise to beneficial ownership include, but are not limited to:

- Microsoft shares owned by immediate family members who live with you or use your address as their regular address (e.g., spouse, domestic partner, minor children, college students, parents, grandparents, grandchildren, siblings and in-laws).
- Microsoft shares held in a trust, if the Insider is: (1) a trustee who has or shares investment control of the trust securities and the Insider or a member of his or her immediate family (whether or not they live with you) has an economic interest in the trust securities; (2) a beneficiary who has or shares investment control; or (3) a person who has the power to revoke the trust.

- Microsoft shares held in the name of a partnership or other entity over which the Insider has a controlling influence.

Insiders are responsible for assuring that individuals or entities who may have an interest in securities that are beneficially owned by the Insider ('Covered Persons') comply with this policy. Accordingly, any reference below to 'Insiders' should be read to include 'Covered Persons.'

B. Open Trading Window

If you are an Insider, you may only trade (or gift) Microsoft securities during an Open Window Period and when your status is Trading Allowed in the Insider Trading Compliance Tool. However, even during an Open Window Period, you may not trade Microsoft securities if you possess material, nonpublic information about Microsoft. For more information about material, nonpublic information, see the General Insider Trading Policy.

An Open Window Period means a defined period of time following Microsoft's quarterly earnings release when the public has been apprised of all relevant material information about Microsoft's recent business and financial performance.

- Open Window Periods typically begin with the opening of trading on the second full trading day following Microsoft's public release of quarterly earnings, and end at the close of trading 15 business days before the end of the fiscal quarter in which the earnings were released.
- For example, if Microsoft made an earnings announcement on a Thursday, the Open Window Period would open the following Monday, and would end at the close of the trading 15 business days before the end of that fiscal quarter.
- After you become subject to this Restricted Trading Window Policy, and as long as you have an active restriction reason, you will receive three emails each quarter from the insider trading compliance tool. (1) You will receive an email on the first business day after the public release of quarterly earnings to confirm the date the trading window will open and that your trading status will be Trading Allowed. (2) You will receive an email 14 calendar days before the trading window closes to let you know the date when your trading status will change to No Trading Allowed. (3) You will receive an email on the date the window closes to confirm that your trading status is No Trading Allowed.
- Occasionally, a quarterly window may be delayed, closed early, or skipped. You will be notified if this happens.
- Microsoft may also institute additional 'blackout periods' during which no trades may be made, at which time you should receive notification of such restriction.

Do not make a trade in Microsoft securities until you receive notice that the Open Window Period has begun and you have confirmed that your status in the Insider Trading Compliance Tool is Trading Allowed.

You may not disclose to any outside third party, including brokers or financial advisers, the terms of Microsoft's Open Window Period, the date on which an Open Window Period has begun or ended, or that a blackout period has been designated. You may disclose your status in the Insider Trading Compliance Tool as Trading Allowed or No Trading Allowed.

If your employment with Microsoft ends during a blackout period, you will be subject to the remainder of that blackout period. If your employment with Microsoft ends during an Open Window Period, you will not be subject to the next blackout period. However, even if you are not subject to our blackout period after you leave Microsoft, you should not trade in Microsoft securities if you are aware of material, nonpublic information. That restriction stays with you as long as the information you possess is material and not publicly disseminated within the meaning of the General Insider Trading Policy.

C. Application of Trading Restrictions to Particular Situations

1. Stock Awards

Transactions involving Microsoft stock received by an Insider pursuant to a Stock Award may be made only during an Open Window Period and your status in the Insider Trading Compliance Tool is Trading Allowed.

2. Gifts

Gifts of Microsoft stock may be made only during an Open Window Period and your status in the Insider Trading Compliance Tool is Trading Allowed.

3. Employee Stock Purchase Plan

You may purchase Microsoft stock under Microsoft's Employee Stock Purchase Plan ('**ESPP**') at any time. However, you may only sell Microsoft stock purchased pursuant to the ESPP during an Open Window Period and your status in the Insider Trading Compliance Tool is Trading Allowed.

4. Stock Options

You may exercise stock options outside an Open Window Period only if you pay the exercise price and withholding taxes in cash, and then hold the shares (known as an 'exercise and hold'). Any sale of stock as part of a broker-assisted cashless exercise of an option, any other market sale for the purpose of generating the cash needed to pay the exercise price of an option, or any delivery of shares to pay the exercise price (if permissible under the applicable stock plan) may only be done during an Open Window Period.

5. Transactions Pursuant to Rule 10b5-1 Plans

Sales of Microsoft stock outside of the Open Window Period or during a blackout period can only be made pursuant to an approved written trading plan (known as a '10b5-1 Plan') that you entered into at a time that sales could be made in accordance with the restrictions set forth in Section B above. To learn more, please visit the Rule 10b5-1 Trading plans page.

D. Senior Leadership Team Advance Notification Requirement

In addition to the requirements above, SLT CEO Direct Reports are also required to notify CELA in advance of transactions in Microsoft securities, including, but not limited to, option exercises, gifts, purchases, sales, and pledges of Microsoft stock or debt. It does not apply to purchases made pursuant to the ESPP or sales made pursuant to a 10b5-1 Plan, but does apply to the other scenarios described in Section C above.

The notification must include a brief description of the nature of the transaction (e.g., exercise of stock option, purchase, sale, gift, pledge of stock for loan, etc.), the type of Microsoft securities involved, and the proposed date on which the Insider will enter into the transaction. Notifications should be given by sending email to (**)@microsoft.com, or by phoning any of the following individuals ("Contacts"):

(**)

Advance notification can be provided by the Insider or his or her broker.

As soon as possible following notification, one of the Contacts will review the transaction and advise the Insider whether he or she may proceed with the transaction. During this process, the Insider or his or her broker will be asked to answer some preclearance questions from a Preclearance Checklist. A copy of the Preclearance Checklist may be requested from (**)@microsoft.com. If advance notification of a transaction is provided via e-mail, the Insider may wish to attach a completed Preclearance Checklist for the proposed transaction.

E. Questions

If you have a question about this policy, its application in relation to a proposed transaction, or reporting responsibilities, you should email (**)@microsoft.com.

4. PROCEDURE

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5. EXCEPTIONS

There are no exceptions to this policy.

6. ENFORCEMENT

Violation of this policy may result in disciplinary action, up to and including immediate termination of employment.

Insiders who engage in insider trading or tipping may also be subject to civil and criminal fines, as well as imprisonment.

7. APPLICATION

This policy applies to Microsoft officers and other designated persons, and their Covered Persons.

8. COUNTRY OR BUSINESS UNIT SUPPLEMENT

9. RELATED DOCUMENTS

Corporate Governance Guidelines
General Insider Trading Policy

10. TRANSLATIONS

11. DOCUMENT HISTORY

Exhibit 19.3

Insider Trading Compliance and Preclearance Policies
for Section 16 Officers and Directors of Microsoft

This document sets forth the insider trading compliance policies that apply to transactions in Microsoft securities by Microsoft officers and directors who are subject to Section 16 of the Securities Exchange Act of 1934 (“**Insiders**”).

The policies contained in this document apply to all securities “beneficially owned” by Insiders. Insiders are responsible for assuring that other individuals or entities (“**Covered Persons**”) who may have an interest in securities that are “beneficially owned” by an Insider comply with these policies. For more information regarding beneficial ownership of securities and Covered Persons, please see Section 6 below.

Covered Persons must comply with the policies below. Accordingly, any reference below to “**Insiders**” should be read to include Covered Persons. In addition to the policies below, all Directors must comply with the Director stock ownership requirements located in the *Corporate Governance Guidelines* and Executive Officers must comply with the *Stock Ownership Requirements for Microsoft Corporation Executives*.

1. <u>Restrictions on Trading</u>

A. <u>Prohibition on Trading While in Possession of Material, Nonpublic Information</u>. Under Federal securities law and regulations, no Insider may trade in Microsoft securities while possessing material, nonpublic information concerning the Company, even if a quarterly window period described in Section 2 is open. The window period is not a safe harbor that eliminates the need for caution about whether an Insider should refrain from trading because he or she possesses material, nonpublic information about Microsoft. For more information, see the General Insider Trading Policy.

B. <u>Trading Restricted to Open Window Periods</u>. No Insider may trade in Microsoft securities outside of an applicable Open Window Period described below in Section 2, or during any trading blackout periods designated by Microsoft, except pursuant to the terms of a Rule 10b5-1 Plan as outlined in Section 4.E below. For more information, see the Restricted Trading Window Policy.

C. <u>Prohibition on Trading in Derivatives</u>. Trading in derivative securities (other than derivative securities issued by Microsoft as compensation) related to Microsoft stock or debt such as options, puts, calls, warrants or similar financial instruments may create the appearance among regulators and Microsoft shareholders that an Insider is engaging in short-term, speculative transactions or inappropriately hedging the risk of declines in Microsoft stock. For this reason, Insiders may not trade in options, puts, calls or other derivative instruments related to Microsoft stock or debt.

D. <u>Prohibition on Pledging</u>. No Insider may purchase Microsoft stock on margin, borrow against Microsoft stock held in a margin account, or pledge Microsoft stock as collateral for a loan.

2. Trading Window and Blackout Periods

A. An Insider may trade in Microsoft securities only during quarterly window periods that follow each quarterly earnings announcement. This policy is intended to help prevent accusations of trading on material, nonpublic information. The theory underlying the policy is that the period of time following an earnings release is a time when the public has been apprised of all relevant material information about the Company's recent business and financial performance. The window period generally begins on the commencement of trading on the second full trading day following Microsoft's public release of quarterly earnings, and ends at the close of trading 15 business days prior to the end of the third month of the fiscal quarter in which the earnings are released (the "**Open Window Period**"). If, for example, Microsoft made an earnings announcement on Thursday, the trading window would be expected to open the following Monday. There are rare situations when a quarterly window period is delayed, closed early or skipped. Microsoft employees and directors who are subject to the trading window policy are sent email each quarter advising them of the first and last day of that quarter's Open Window Period. If an Open Window Period is delayed, closed early or skipped, Insiders will be sent email alerting them to this development. An Insider should not trade unless he or she has received notice about the opening of the trading window.

B. Insiders who have material, nonpublic information concerning Microsoft may not trade in Microsoft securities even during Open Window Periods.

C. Insiders may not trade in Microsoft securities outside of an Open Window Period or during any blackout period that Microsoft may designate. Insiders may not disclose to any outside third party, including brokers or financial advisers, the terms of the trading window policy, the date on which an Open Window Period has begun or ended, or that a blackout period has been designated.

3. Pre-Clearance Policies and Procedures

A. Potential Section 16(b) Liability. Insiders are subject to the short-swing profit recapture rules of Section 16 of the Securities Exchange Act of 1934. Compliance with the law includes prompt reporting of an Insider's transactions involving Microsoft securities. The Section 16 reporting requirements also apply to transactions in Microsoft securities owned by certain other individuals and entities related to an Insider, as explained in greater detail in Section 5 below. Any late or delinquent filings are required to be reported in Microsoft's annual proxy statement. In order to assist Insiders in complying with these reporting requirements, the procedures described below have been adopted so that CELA, Corporate Legal Group may prepare and file with the SEC on an Insider's behalf the appropriate form reporting an Insider's transactions.

The purpose of these filings with the SEC is to give any interested party the ability to examine the filings to determine whether an Insider has violated the so-called "short-swing profit" rules, which prohibit a specified officer or director of a public company from profiting on sales and purchases of stock that occur within 6 months of each other. Neither lack of familiarity with the rules nor inadvertent non-compliance will excuse a violation of the short-swing profit rules. Generally, this law should not present Insiders at Microsoft with a problem because Microsoft stock purchases that an Insider makes from the Company (e.g. stock acquired on exercise of stock options and stock purchased under the Employee Stock Purchase Plan) are all exempt from short swing profit liability under SEC rules, and therefore any sales of stock an Insider undertakes typically would not have any non-exempt purchases of stock with which they could be matched. However, an Insider may potentially have a problem if the Insider purchases Microsoft shares in the open market or changes an investment election under the Microsoft 401(k) plan with respect to investing in Microsoft stock. The advance notification procedures described below are intended to help Insiders avoid transactions that would violate Section 16(b), as well as meet the 2-day filing deadline for reporting most Insider transactions in Microsoft securities. Insiders are expected to cooperate with the Company fully and promptly in disclosing their transactions involving Company securities.

B. Preclearance Requirement. All Insiders are required to obtain approval from the CELA Corporate Legal Group before entering into any transaction involving Microsoft securities, including, but not limited to, option exercises, gifts, purchases, and sales, of Microsoft stock or debt. Some very limited exceptions apply as described in Section 4 below. The Insider or his or her broker will be asked to answer some preclearance questions in a Preclearance Checklist, which includes a brief description of the nature of the transaction (e.g., exercise of stock option, purchase, sale, gift, etc.), the type and number of Microsoft securities involved, and the proposed date on which the Insider will enter into the transaction. A copy of the Preclearance Checklist may be requested from (**)@microsoft.com.

Preclearance requests should be sent by email to (**)@microsoft.com. For questions, please call any of the following individuals ("**Contacts**"):

(**)

As soon as possible following the preclearance request, one of the Contacts will review the transaction and advise the Insider whether he or she may proceed with the transaction.

C. Confirmation of Transaction. Immediately after the transaction is executed, the Insider or his or her broker must provide the details of the transaction to one or more of the Contacts, either by phone or e-mail to (**)@microsoft.com. This information must include the following:

i. type of Microsoft security involved;

ii. a brief description of the nature of the transaction (e.g., purchase, sale, exercise, etc.);

iii. number of shares involved;

iv. price per share;

v. the total aggregate price/value; and

vi. the transaction date.

Please note that this information must be provided on the date the transaction is executed (commonly referred to as the "trade date"), as opposed to on the date the transaction is settled. In nearly all cases, the trade date is the date that gives rise to the Section 16 reporting obligation.

4. **Application of the Trading Restrictions and Preclearance Policies to Particular Situations**

A. Stock Awards.

i. Trading Restrictions. Transactions involving Microsoft stock received by an Insider pursuant to a Stock Award may be made only in accordance with, and at times permitted by, the trading restrictions contained in Sections 1 and 2 above.

ii. Preclearance and Confirmation. Insiders do not need to request preclearance or provide confirmation of receipt of a Stock Award. Insiders must request preclearance and provide confirmation of a transaction as outlined in Section 3 above for any sales of Microsoft stock received under a Stock Award.

B. Gifts.

i. Trading Restrictions. Gifts of Microsoft stock may be made only in accordance with, and at times permitted by, the trading restrictions contained in Sections 1 and 2 above.

ii. Preclearance and Confirmation. Any gift is subject to the preclearance requirement and confirmation of the transaction as outlined in Section 3 above.

C. Employee Stock Purchase Plan.

i. Trading Restrictions. There is no restriction on purchases of Microsoft stock under Microsoft's Employee Stock Purchase Plan ("**ESPP**"). However, an Insider's sale of Microsoft stock purchased pursuant to the ESPP may be made only in accordance with, and at times permitted by, the trading restrictions contained in Sections 1 and 2 above.

ii. Preclearance and Confirmation. Insiders do not need to request preclearance or provide confirmation of purchases of Microsoft stock pursuant to an election under the ESPP. However, Insiders must request preclearance and provide confirmation of the transaction as outlined in Section 3 above for any sales of Microsoft stock purchased pursuant to the ESPP.

D. Stock Options.

i. Trading Restrictions. Stock options may be exercised outside an Open Window Period only if the Insider pays the exercise price and withholding taxes in cash, and then holds the shares (also known as an "exercise and hold"). Any sale of stock as part of a broker-assisted cashless exercise of an option, any other market sale for the purpose of generating the cash needed to pay the exercise price of an option, or any delivery of shares to pay the exercise price (if permissible under the applicable stock plan) may only be done in accordance with, and at times permitted by, the trading restrictions described in Section 1 and 2 above.

ii. Preclearance and Confirmation. Any exercise of an employee stock option is subject to the preclearance requirement and confirmation of transaction process as outlined in Section 3 above.

E. Transactions in Microsoft Stock Pursuant to Rule 10b5-1 Plans.

i. Trading Restrictions. SEC Rule 10b5-1 allows trading under certain circumstances by Insiders during periods when the Insider would otherwise be prohibited from trading (for example, during periods when the trading window is closed or the insider possesses material, nonpublic information). In order to qualify for the affirmative defense offered by this Rule, the plan must be contained in an agreement that meets certain requirements, and the Insider must not exercise any discretion regarding the transactions under the plan during a period when trading would otherwise be prohibited (typically referred to as a "**10b5-1 Plan**"). An Insider may make sales of Microsoft stock outside of the Open Window Period or during a blackout period only if such sales are made pursuant to a 10b5-1 Plan that consists of an agreement with the selling broker that has been prepared by a broker-dealer or a lawyer representing the Insider, reviewed and approved by the CELA Corporate Legal Group, and entered into by the Insider at a time that sales could be made in accordance with the restrictions set forth in Sections 1 and 2 above. For additional information regarding 10b5-1 Plans and the policies associated with them, please send email to (**)@microsoft.com.

ii. Preclearance and Confirmation. Insiders engaging in transactions under an approved 10b5-1 Plan generally will not have an obligation to request preclearance of such transactions. However, Insiders (or their brokers) must immediately confirm a transaction as provided in Section 3 above upon execution of a transaction under a 10b5-1 Plan.

5. Tax Services from Microsoft Independent Auditors

Microsoft's independent auditor (currently Deloitte & Touche LLP) is prohibited from providing tax services to individuals in a financial reporting oversight role at Microsoft, or immediate family members of such persons, who do not meet the exceptions provided by PCAOB Rule 3523.

Employee Insiders not in a financial reporting oversight role and directors may use Microsoft's independent auditor for individual tax services, after providing advance notice to and receiving approval from CELA (via the *audsvc* alias).

If you are uncertain whether you a person in a financial reporting oversight role covered by this policy or are otherwise prohibited from using Microsoft's independent auditor for tax services, you may contact the*(**)@microsoft.com* alias.

6. Covered Persons

Insiders are subject to the Section 16 reporting requirements with respect to all Microsoft securities that they beneficially own. The determination of "beneficial ownership" is based on the Insider's direct or economic interest in the securities. Although not exhaustive, we have listed below the primary types of relationships that may give rise to beneficial ownership, thus triggering the Section 16 reporting requirement.

- Microsoft shares owned by immediate family members who live with the Insider or use the Insider's address as their regular address (e.g., spouse, domestic partner, minor children, college students, parents, grandparents, grandchildren, siblings and in-laws).
- Microsoft shares held in a trust, if the Insider is: (1) a trustee who has or shares investment control of the trust securities and the Insider or a member of his or her immediate family (whether or not they share the same household) has an economic interest in the trust securities; (2) a beneficiary and has or shares investment control; or (3) a person who has the power to revoke the trust.
- Microsoft shares held in the name of a partnership or other entity over which the Insider has a controlling influence.

Insiders may wish to consult with a Contact or send e-mail to (**)@microsoft.com to clarify reporting responsibilities with respect to any Microsoft securities over which the Insider has voting control or in which the Insider has an economic interest.

7. Sanctions

Persons who are found to have traded on material, nonpublic information can face serious criminal and civil penalties. Insiders that violate the prohibition on short-swing profit transactions will be required to disgorge the deemed profit from the transaction to the Company. Any late or delinquent filings are required to be reported in Microsoft's annual proxy statement. In addition, the SEC has broad authority to seek sanctions from late filers; these sanctions may include significant fines and other penalties. Violation of these policies may result in disciplinary action up to and including immediate termination of employment. Violation of these policies also may be reported in Company filings with the SEC.

8. Questions

Anyone with questions about these policies and procedures, or the application of them to a proposed transaction, may obtain additional guidance by sending e-mail to (**)@microsoft.com. The ultimate responsibility for adhering to these policies and avoiding unlawful transactions rests with the Insider.

Exhibit 21

SUBSIDIARIES OF REGISTRANT

The following is a list of subsidiaries of Microsoft Corporation as of June 30, 2024, omitting subsidiaries which, considered in the aggregate, would not constitute a significant subsidiary.

Name	Where Incorporated
Microsoft Ireland Research	Ireland
Microsoft Global Finance	Ireland
Microsoft Ireland Operations Limited	Ireland
Microsoft Online, Inc.	United States
LinkedIn Corporation	United States
LinkedIn Ireland Unlimited Company	Ireland
Nuance Communications, Inc.	United States
Activision Blizzard, Inc.	United States
Activision Publishing, Inc.	United States

Exhibit 23.1

CONSENT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

We consent to the incorporation by reference in Registration Statement Nos. 333-109185, 333-118764, 333-52852, 333-132100, 333-161516, 333-75243, 333-185757, and 333-221833 on Form S-8 and Registration Statement No. 333-261590 on Form S-3 of our reports dated July 30, 2024, relating to the financial statements of Microsoft Corporation and the effectiveness of Microsoft Corporation's internal control over financial reporting appearing in this Annual Report on Form 10-K of Microsoft Corporation for the year ended June 30, 2024.

/s/ Deloitte & Touche LLP

Seattle, Washington
July 30, 2024

Exhibit 31.1

CERTIFICATION

I, Satya Nadella, certify that:

1. I have reviewed this annual report on Form 10-K of Microsoft Corporation;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:

a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and

5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of registrant's Board of Directors (or persons performing the equivalent functions):

a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

/s/ SATYA NADELLA

Satya Nadella
Chief Executive Officer

July 30, 2024

Exhibit 31.2

CERTIFICATION

I, Amy E. Hood, certify that:

1. I have reviewed this annual report on Form 10-K of Microsoft Corporation;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:

a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and

5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of registrant's Board of Directors (or persons performing the equivalent functions):

a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

/s/ AMY E. HOOD

Amy E. Hood
Executive Vice President and
Chief Financial Officer

July 30, 2024

Exhibit 32.1

CERTIFICATION PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002
(18 U.S.C. SECTION 1350)

In connection with the Annual Report of Microsoft Corporation, a Washington corporation (the "Company"), on Form 10-K for the year ended June 30, 2024, as filed with the Securities and Exchange Commission (the "Report"), Satya Nadella, Chief Executive Officer of the Company, does hereby certify, pursuant to § 906 of the Sarbanes-Oxley Act of 2002 (18 U.S.C. § 1350), that to his knowledge:

(1) The Report fully complies with the requirements of section 13(a) or 15(d) of the Securities Exchange Act of 1934; and

(2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

/s/ SATYA NADELLA
Satya Nadella
Chief Executive Officer

July 30, 2024

[A signed original of this written statement required by Section 906 has been provided to Microsoft Corporation and will be retained by Microsoft Corporation and furnished to the Securities and Exchange Commission or its staff upon request.]

Exhibit 32.2

CERTIFICATION PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002
(18 U.S.C. SECTION 1350)

In connection with the Annual Report of Microsoft Corporation, a Washington corporation (the “Company”), on Form 10-K for the year ended June 30, 2024, as filed with the Securities and Exchange Commission (the “Report”), Amy E. Hood, Chief Financial Officer of the Company, does hereby certify, pursuant to § 906 of the Sarbanes-Oxley Act of 2002 (18 U.S.C. § 1350), that to her knowledge:

(1) The Report fully complies with the requirements of section 13(a) or 15(d) of the Securities Exchange Act of 1934; and

(2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

/s/ Amy E. Hood

Amy E. Hood
Executive Vice President and
Chief Financial Officer

July 30, 2024

[A signed original of this written statement required by Section 906 has been provided to Microsoft Corporation and will be retained by Microsoft Corporation and furnished to the Securities and Exchange Commission or its staff upon request.]

Microsoft Corporation
Executive Compensation Recovery Policy

This policy covers Microsoft's Covered Officers and explains when Microsoft will be required or authorized, as applicable, to seek recovery of Incentive Compensation awarded or paid to Covered Officers. Please refer to Exhibit A attached hereto (the "Definitions Exhibit") for the definitions of capitalized terms used throughout this Policy.

1. **Miscalculation of Financial Performance Measure Results.** In the event of a Restatement, Microsoft will seek to recover, reasonably promptly, all Recoverable Incentive Compensation from a Covered Officer during the Applicable Period. Such recovery, in the case of a Restatement, will be made without regard to any individual knowledge or responsibility related to the Restatement or the Recoverable Incentive Compensation. Notwithstanding the foregoing, if Microsoft is required to undertake a Restatement, Microsoft will not be required to recover the Recoverable Incentive Compensation if the Compensation Committee determines it Impracticable to do so, after exercising a normal due process review of all the relevant facts and circumstances.

   Microsoft will seek to recover all Recoverable Incentive Compensation that was awarded or paid in accordance with the definition of "Recoverable Incentive Compensation" set forth on the Definitions Exhibit. If such Recoverable Incentive Compensation was not awarded or paid on a formulaic basis, Microsoft will seek to recover the amount that the Compensation Committee determines in good faith should be recouped.

2. **Legal and Compliance Violations**. Compliance with the law and Microsoft's Standards of Business Conduct and other corporate policies is a pre-condition to earning Incentive Compensation. If Microsoft in its sole discretion concludes that a Covered Officer (1) committed a significant legal or compliance violation in connection with the Covered Officer's employment, including a violation of Microsoft's corporate policies or Microsoft's Standards of Business Conduct (each, "Misconduct"), or (2) was aware of or willfully blind to Misconduct that occurred in an area over which the Covered Officer had supervisory authority, Microsoft may, at the direction of the Compensation Committee, seek recovery of all or a portion of the Recoverable Incentive Compensation awarded or paid to the Covered Officer for the Applicable Period in which the violation occurred. In addition, Microsoft may, at the direction of the Compensation Committee, conclude that any unpaid or unvested Incentive Compensation has not been earned and must be forfeited.

   In the event of Misconduct, Microsoft may seek recovery of Recoverable Incentive Compensation even if the Misconduct did not result in an award or payment greater than would have been awarded or paid absent the Misconduct.

   In the event of Misconduct, in determining whether to seek recovery and the amount, if any, by which the payment or award should be reduced, the Compensation Committee may consider—among other things— the seriousness of the Misconduct, whether the Covered Officer was unjustly enriched, whether seeking the recovery would prejudice Microsoft's interests in any way, including in a proceeding or investigation, and any other factors it deems relevant to the determination.

3. **Other Actions**. The Compensation Committee may, subject to applicable law, seek recovery in the manner it chooses, including by seeking reimbursement from the Covered Officer of all or part of the compensation awarded or paid, by electing to withhold unpaid compensation, by set-off, or by rescinding or canceling unvested stock.

   In the reasonable exercise of its business judgment under this Policy, the Compensation Committee may in its sole discretion determine whether and to what extent additional action is appropriate to address the circumstances surrounding a Restatement or Misconduct to minimize the likelihood of any recurrence and to impose such other discipline as it deems appropriate.

4. **No Indemnification or Reimbursement**. Notwithstanding the terms of any other policy, program, agreement or arrangement, in no event will Microsoft or any of its affiliates indemnify or reimburse a Covered Officer for any loss under this Policy and in no event will Microsoft or any of its affiliates pay premiums on any insurance policy that would cover a Covered Officer's potential obligations with respect to Recoverable Incentive Compensation under this Policy.

5. **Administration of Policy**. The Compensation Committee will have full authority to administer this Policy. Actions of the Compensation Committee pursuant to this Policy will be taken by the vote of a majority of its members. The Compensation Committee will, subject to the provisions of this Policy and Rule 10D-1 of the Securities Exchange Act of 1934, as amended (the "Exchange Act"), and Microsoft's applicable exchange listing standards, make such determinations and interpretations and take such actions in connection with this Policy as it deems necessary, appropriate or advisable. All determinations and interpretations made by the Compensation Committee will be final, binding and conclusive.

6. **Other Claims and Rights**. The remedies under this Policy are in addition to, and not in lieu of, any legal and equitable claims Microsoft or any of its affiliates may have or any actions that may be imposed by law enforcement agencies, regulators, administrative bodies, or other authorities. Further, the exercise by the Compensation Committee of any rights pursuant to this Policy will not impact any other rights that Microsoft or any of its affiliates may have with respect to any Covered Officer subject to this Policy.

7. **Condition to Eligibility for Incentive Compensation**. All Incentive Compensation subject to this Policy will not be earned, even if already paid, until the Policy ceases to apply to such Incentive Compensation and any other vesting conditions applicable to such Incentive Compensation are satisfied.

8. **Amendment; Termination**. The Board or the Compensation Committee may amend or terminate this Policy at any time.

9. **Effectiveness**. Except as otherwise determined in writing by the Compensation Committee, this Policy will apply to any Incentive Compensation that (a) in the case of any Restatement, is Received by Covered Officers prior to, on or following the Effective Date, and (b) in the case of Misconduct, is awarded or paid to a Covered Officer on or after the Effective Date. This Policy will survive and continue notwithstanding any termination of a Covered Officer's employment with Microsoft and its affiliates.

10. **Successors**. This Policy shall be binding and enforceable against all Covered Officers and their successors, beneficiaries, heirs, executors, administrators, or other legal representatives.

11. **Governing Law**. To the extent not preempted by U.S. federal law, this Policy will be governed by and construed in accordance with the laws of the State of Washington, without reference to principles of conflict of laws.

EXHIBIT A

DEFINITIONS

"Applicable Period" means (a) in the case of any Restatement, the three completed fiscal years of Microsoft immediately preceding the earlier of (i) the date the Board, a committee of the Board, or the officer or officers of Microsoft authorized to take such action if Board action is not required, concludes (or reasonably should have concluded) that a Restatement is required or (ii) the date a regulator, court or other legally authorized entity directs Microsoft to undertake a Restatement, and (b) in the case of any Misconduct, such period as the Compensation Committee or Board determines to be appropriate in light of the scope and nature of the Misconduct. The "Applicable Period" also includes any transition period (that results from a change in Microsoft's fiscal year) within or immediately following the three completed fiscal years identified in the preceding sentence.

"Board" means the Board of Directors of Microsoft.

"Compensation Committee" means Microsoft's committee of independent directors responsible for executive compensation decisions, or in the absence of such a committee, a majority of the independent directors serving on the Board.

"Covered Officer" means (a) in the case of any Restatement, any person who is, or was at any time, during the Applicable Period, an Executive Officer of Microsoft or an SLT Member, and (b) in the case of any Misconduct, any person who was an Executive Officer or an SLT Member at the time of the Misconduct. For the avoidance of doubt, a Covered Officer may include a former Executive Officer or SLT Member that left Microsoft, retired, or transitioned to an employee role (including after serving as an Executive Officer in an interim capacity) during the Applicable Period.

"Effective Date" means July 1, 2023.

"Executive Officer" means Microsoft's president, principal financial officer, principal accounting officer (or if there is no such accounting officer, the controller), any vice-president in charge of a principal business unit, division, or function (such as sales, administration, or finance), any other officer who performs a policy-making function, or any other person (including an officer of Microsoft's parent(s) or subsidiaries) who performs similar policy-making functions for Microsoft.

"Financial Performance Measure" means a measure that is determined and presented in accordance with the accounting principles used in preparing Microsoft's financial statements (including "non-GAAP" financial measures, such as those appearing in Microsoft's earnings releases or Management Discussion and Analysis), and any measure that is derived wholly or in part from such measure. Stock price and total shareholder return (and any measures derived wholly or in part therefrom) shall be considered Financial Performance Measures.

"Impracticable." The Compensation Committee may determine in good faith that recovery of Recoverable Incentive Compensation is "Impracticable" (a) in the case of any Restatement, if: (i) pursuing such recovery would violate home country law of the jurisdiction of incorporation of the Company where that law was adopted prior to November 28, 2022 and Microsoft provides an opinion of counsel to that effect acceptable to Microsoft's listing exchange; (ii) the direct expense paid to a third party to assist in enforcing this Policy would exceed the Recoverable Incentive Compensation and Microsoft has (A) made a reasonable attempt to recover such amounts and (B) provided documentation of such attempts to recover to Microsoft's applicable listing exchange; or (iii) recovery would likely cause an otherwise tax-qualified retirement plan, under which benefits are broadly available to employees of Microsoft, to fail to meet the requirements of the Internal Revenue Code of 1986, as amended, and (b) in the case of any Misconduct, in its sole discretion, in light of the scope and nature of the Misconduct.

"Incentive Compensation" means any compensation that is granted, earned, or vested based wholly or in part upon the attainment of a Financial Performance Measure. Incentive Compensation does not include any base salaries (except with respect to any salary increases earned wholly or in part based on the attainment of a Financial Performance Measure performance goal); bonuses paid solely at the discretion of the Compensation Committee or Board that are not paid from a "bonus pool" that is determined by satisfying a Financial Performance Measure performance goal; bonuses paid solely upon satisfying one or more subjective standards and/or completion of a specified employment period; non-equity incentive plan awards earned solely upon satisfying one or more strategic measures or operational measures; and equity awards that vest solely based on the passage of time and/or attaining one or more non-Financial Performance Measures. Notwithstanding the foregoing, in the case of any Misconduct, Incentive Compensation will include all forms of cash and equity incentive compensation, including, without limitation, cash bonuses and equity awards that are received or vest solely based on the passage of time and/or attaining one or more non-Financial Performance Measures.

"Received." Incentive Compensation is deemed "Received" in Microsoft's fiscal period during which the Financial Performance Measure specified in the Incentive Compensation award is attained, even if the payment or grant of the Incentive Compensation occurs after the end of that period.

"Recoverable Incentive Compensation" means (a) in the case of any Restatement, the amount of any Incentive Compensation (calculated on a pre-tax basis) Received by a Covered Officer during the Applicable Period that is in excess of the amount that otherwise would have been Received if the calculation were based on the Restatement, and (b) in the case of any Misconduct, the amount of any Incentive Compensation (calculated on a pre-tax basis) awarded or paid to a Covered Officer during the Applicable Period that the Compensation Committee determines, in its sole discretion, to be appropriate in light of the scope and nature of the Misconduct. For the avoidance of doubt, in the case of any Restatement, Recoverable Incentive Compensation does not include any Incentive Compensation Received by a person (i) before such person began service as a Covered Officer and (ii) who did not serve as a Covered Officer at any time during the performance period for that Incentive Compensation. For the avoidance of doubt, in the case of any Restatement, Recoverable Incentive Compensation may include Incentive Compensation Received by a person while serving as an employee if such person previously served as a Covered Officer and then transitioned to an employee role. For Incentive Compensation based on (or derived from) stock price or total shareholder return where the amount of Recoverable Incentive Compensation is not subject to mathematical recalculation directly from the information in the applicable Restatement, the amount will be determined by the Compensation Committee based on a reasonable estimate of the effect of the Restatement on the stock price or total shareholder return upon which the Incentive Compensation was Received (in which case, Microsoft will maintain documentation of such determination of that reasonable estimate and provide such documentation to Microsoft's applicable listing exchange).

"Restatement" means an accounting restatement of any of Microsoft's financial statements filed with the Securities and Exchange Commission under the Exchange Act, or the Securities Act of 1933, as amended, due to Microsoft's material noncompliance with any financial reporting requirement under U.S. securities laws, regardless of whether Microsoft or Covered Officer misconduct was the cause for such restatement. "Restatement" includes any required accounting restatement to correct an error in previously issued financial statements that is material to the previously issued financial statements (commonly referred to as “Big R” restatements), or that would result in a material misstatement if the error were corrected in the current period or left uncorrected in the current period (commonly referred to as “little r” restatements).

“SLT Member” means any person who has been designated as a member of Microsoft’s Senior Leadership Team, other than an Executive Officer.

Updated July 1, 2023

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-Q

☒ **QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the Quarterly Period Ended March 31, 2025

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the Transition Period From to

Commission File Number 001-37845

MICROSOFT CORPORATION

WASHINGTON	**91-1144442**
(STATE OF INCORPORATION)	**(I.R.S. ID)**

ONE MICROSOFT WAY, REDMOND, WASHINGTON 98052-6399
(425) 882-8080
www.microsoft.com/investor

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Name of exchange on which registered
Common stock, $0.00000625 par value per share	**MSFT**	**NASDAQ**
3.125% Notes due 2028	**MSFT**	**NASDAQ**
2.625% Notes due 2033	**MSFT**	**NASDAQ**

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large Accelerated Filer ☒	Accelerated Filer ☐
Non-accelerated Filer ☐	Smaller Reporting Company ☐
	Emerging Growth Company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

Indicate the number of shares outstanding of each of the issuer's classes of common stock, as of the latest practicable date.

Class	**Outstanding as of April 24, 2025**
Common Stock, $0.00000625 par value per share	7,432,543,865 shares

MICROSOFT CORPORATION

FORM 10-Q

For the Quarter Ended March 31, 2025

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PART I. FINANCIAL INFORMATION

ITEM 1. FINANCIAL STATEMENTS

INCOME STATEMENTS

(In millions, except per share amounts) (Unaudited)	Three Months Ended March 31,		Nine Months Ended March 31,	
	2025	2024	**2025**	2024
Revenue:				
Product	**$ 15,319**	$ 17,080	**$ 46,810**	$ 51,556
Service and other	**54,747**	44,778	**158,473**	128,839
Total revenue	**70,066**	61,858	**205,283**	180,395
Cost of revenue:				
Product	**3,037**	4,339	**10,187**	13,834
Service and other	**18,882**	14,166	**53,630**	40,596
Total cost of revenue	**21,919**	18,505	**63,817**	54,430
Gross margin	**48,147**	43,353	**141,466**	125,965
Research and development	**8,198**	7,653	**23,659**	21,454
Sales and marketing	**6,212**	6,207	**18,369**	17,640
General and administrative	**1,737**	1,912	**5,233**	5,363
Operating income	**32,000**	27,581	**94,205**	81,508
Other expense, net	**(623)**	(854)	**(3,194)**	(971)
Income before income taxes	**31,377**	26,727	**91,011**	80,537
Provision for income taxes	**5,553**	4,788	**16,412**	14,437
Net income	**$ 25,824**	$ 21,939	**$ 74,599**	$ 66,100
Earnings per share:				
Basic	**$ 3.47**	$ 2.95	**$ 10.03**	$ 8.90
Diluted	**$ 3.46**	$ 2.94	**$ 9.99**	$ 8.85
Weighted average shares outstanding:				
Basic	**7,434**	7,431	**7,434**	7,431
Diluted	**7,461**	7,472	**7,466**	7,467

Refer to accompanying notes.

COMPREHENSIVE INCOME STATEMENTS

(In millions) (Unaudited)	Three Months Ended March 31,		Nine Months Ended March 31,	
	2025	2024	**2025**	2024
Net income	**$ 25,824**	$ 21,939	**$ 74,599**	$ 66,100
Other comprehensive income (loss), net of tax:				
Net change related to derivatives	**(20)**	10	**4**	28
Net change related to investments	**450**	(202)	**1,130**	869
Translation adjustments and other	**353**	(294)	**(377)**	11
Other comprehensive income (loss)	**783**	(486)	**757**	908
Comprehensive income	**$ 26,607**	$ 21,453	**$ 75,356**	$ 67,008

Refer to accompanying notes.

BALANCE SHEETS

(In millions) (Unaudited)

	March 31, 2025	June 30, 2024
Assets		
Current assets:		
Cash and cash equivalents	**$ 28,828**	$ 18,315
Short-term investments	**50,790**	57,228
Total cash, cash equivalents, and short-term investments	**79,618**	75,543
Accounts receivable, net of allowance for doubtful accounts of **$695** and $830	**51,700**	56,924
Inventories	**848**	1,246
Other current assets	**24,478**	26,021
Total current assets	**156,644**	159,734
Property and equipment, net of accumulated depreciation of **$87,074** and $76,421	**183,939**	135,591
Operating lease right-of-use assets	**24,475**	18,961
Equity and other investments	**16,035**	14,600
Goodwill	**119,329**	119,220
Intangible assets, net	**23,968**	27,597
Other long-term assets	**38,234**	36,460
Total assets	**$ 562,624**	$ 512,163
Liabilities and stockholders' equity		
Current liabilities:		
Accounts payable	**$ 26,250**	$ 21,996
Short-term debt	**0**	6,693
Current portion of long-term debt	**2,999**	2,249
Accrued compensation	**10,579**	12,564
Short-term income taxes	**6,805**	5,017
Short-term unearned revenue	**44,636**	57,582
Other current liabilities	**22,937**	19,185
Total current liabilities	**114,206**	125,286
Long-term debt	**39,882**	42,688
Long-term income taxes	**25,061**	27,931
Long-term unearned revenue	**2,840**	2,602
Deferred income taxes	**2,522**	2,618
Operating lease liabilities	**17,686**	15,497
Other long-term liabilities	**38,536**	27,064
Total liabilities	**240,733**	243,686
Commitments and contingencies		
Stockholders' equity:		
Common stock and paid-in capital – shares authorized 24,000; outstanding **7,434** and 7,434	**106,965**	100,923
Retained earnings	**219,759**	173,144
Accumulated other comprehensive loss	**(4,833)**	(5,590)
Total stockholders' equity	**321,891**	268,477
Total liabilities and stockholders' equity	**$ 562,624**	$ 512,163

Refer to accompanying notes.

CASH FLOWS STATEMENTS

(In millions) (Unaudited)	Three Months Ended March 31, 2025	Three Months Ended March 31, 2024	Nine Months Ended March 31, 2025	Nine Months Ended March 31, 2024
Operations				
Net income	**$ 25,824**	$ 21,939	**$ 74,599**	$ 66,100
Adjustments to reconcile net income to net cash from operations:				
Depreciation, amortization, and other	**8,740**	6,027	**22,950**	15,907
Stock-based compensation expense	**2,980**	2,703	**8,901**	8,038
Net recognized losses (gains) on investments and derivatives	**(298)**	49	**553**	261
Deferred income taxes	**(2,244)**	(1,323)	**(4,835)**	(3,593)
Changes in operating assets and liabilities:				
Accounts receivable	**(2,461)**	(2,028)	**5,598**	6,055
Inventories	**52**	260	**390**	1,229
Other current assets	**1,076**	951	**642**	880
Other long-term assets	**(518)**	(2,137)	**(3,368)**	(5,577)
Accounts payable	**1,179**	648	**1,221**	(659)
Unearned revenue	**(1,032)**	(645)	**(12,923)**	(10,309)
Income taxes	**1,298**	2,622	**(1,081)**	2,493
Other current liabilities	**2,839**	2,803	**576**	215
Other long-term liabilities	**(391)**	48	**292**	313
Net cash from operations	**37,044**	31,917	**93,515**	81,353
Financing				
Proceeds from issuance (repayments) of debt, maturities of 90 days or less, net	**0**	(3,810)	**(5,746)**	6,392
Proceeds from issuance of debt	**0**	6,352	**0**	24,198
Repayments of debt	**(2,250)**	(11,589)	**(3,216)**	(16,005)
Common stock issued	**546**	522	**1,508**	1,468
Common stock repurchased	**(4,781)**	(4,213)	**(13,874)**	(13,044)
Common stock cash dividends paid	**(6,169)**	(5,572)	**(17,913)**	(16,197)
Other, net	**(382)**	(498)	**(1,614)**	(1,006)
Net cash used in financing	**(13,036)**	(18,808)	**(40,855)**	(14,194)
Investing				
Additions to property and equipment	**(16,745)**	(10,952)	**(47,472)**	(30,604)
Acquisition of companies, net of cash acquired and divestitures, and purchases of intangible and other assets	**(981)**	(1,575)	**(4,235)**	(67,790)
Purchases of investments	**(4,474)**	(2,183)	**(8,144)**	(14,901)
Maturities of investments	**6,721**	3,350	**11,461**	23,218
Sales of investments	**2,161**	1,941	**6,688**	8,871
Other, net	**604**	(1,281)	**(325)**	(916)
Net cash used in investing	**(12,714)**	(10,700)	**(42,027)**	(82,122)
Effect of foreign exchange rates on cash and cash equivalents	**52**	(80)	**(120)**	(107)
Net change in cash and cash equivalents	**11,346**	2,329	**10,513**	(15,070)
Cash and cash equivalents, beginning of period	**17,482**	17,305	**18,315**	34,704
Cash and cash equivalents, end of period	**$ 28,828**	$ 19,634	**$ 28,828**	$ 19,634

Refer to accompanying notes.

STOCKHOLDERS' EQUITY STATEMENTS

(In millions, except per share amounts) (Unaudited)	Three Months Ended March 31,		Nine Months Ended March 31,	
	2025	2024	**2025**	2024
Common stock and paid-in capital				
Balance, beginning of period	**$ 104,829**	$ 97,480	**$ 100,923**	$ 93,718
Common stock issued	**546**	522	**1,508**	1,468
Common stock repurchased	**(1,390)**	(1,512)	**(4,366)**	(4,213)
Stock-based compensation expense	**2,980**	2,703	**8,901**	8,038
Other, net	**0**	0	**(1)**	182
Balance, end of period	**106,965**	99,193	**106,965**	99,193
Retained earnings				
Balance, beginning of period	**203,482**	145,737	**173,144**	118,848
Net income	**25,824**	21,939	**74,599**	66,100
Common stock cash dividends	**(6,168)**	(5,573)	**(18,508)**	(16,718)
Common stock repurchased	**(3,379)**	(2,709)	**(9,476)**	(8,836)
Balance, end of period	**219,759**	159,394	**219,759**	159,394
Accumulated other comprehensive loss				
Balance, beginning of period	**(5,616)**	(4,949)	**(5,590)**	(6,343)
Other comprehensive income (loss)	**783**	(486)	**757**	908
Balance, end of period	**(4,833)**	(5,435)	**(4,833)**	(5,435)
Total stockholders' equity	**$ 321,891**	$ 253,152	**$ 321,891**	$ 253,152
Cash dividends declared per common share	**$ 0.83**	$ 0.75	**$ 2.49**	$ 2.25

Refer to accompanying notes.

NOTES TO FINANCIAL STATEMENTS

(Unaudited)

NOTE 1 — ACCOUNTING POLICIES

Accounting Principles

Our unaudited interim consolidated financial statements and accompanying notes are prepared in accordance with accounting principles generally accepted in the United States of America (“GAAP”). In the opinion of management, the unaudited interim consolidated financial statements reflect all adjustments of a normal recurring nature that are necessary for a fair presentation of the results for the interim periods presented. Interim results are not necessarily indicative of results for a full year. The information included in this Form 10-Q should be read in conjunction with information included in the Microsoft Corporation fiscal year 2024 Form 10-K and Form 8-K filed with the U.S. Securities and Exchange Commission on July 30, 2024 and December 3, 2024, respectively.

Principles of Consolidation

The consolidated financial statements include the accounts of Microsoft Corporation and its subsidiaries. Intercompany transactions and balances have been eliminated.

Recast of Certain Prior Period Information

In August 2024, we announced changes to the composition of our segments. These changes align our segments with how we currently manage our business, most notably bringing the commercial components of Microsoft 365 together in the Productivity and Business Processes segment. Beginning in fiscal year 2025, the information that our chief operating decision maker is regularly provided and reviews for purposes of allocating resources and assessing performance reflects these segment changes. Prior period segment information has been recast to conform to the way we internally manage and monitor our business during fiscal year 2025. These changes impacted Note 8 – Goodwill, Note 12 – Unearned Revenue, and Note 17 – Segment Information and Geographic Data.

The recast of prior period information had no impact on our consolidated balance sheets, consolidated income statements, or consolidated cash flows statements.

Estimates and Assumptions

Preparing financial statements requires management to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenue, and expenses. Examples of estimates and assumptions include: for revenue recognition, determining the nature and timing of satisfaction of performance obligations, and determining the standalone selling price of performance obligations, variable consideration, and other obligations such as product returns and refunds; loss contingencies; product warranties; the fair value of and/or potential impairment of goodwill and intangible assets for our reporting units; product life cycles; useful lives of our tangible and intangible assets; allowances for doubtful accounts; the market value of, and demand for, our inventory; stock-based compensation forfeiture rates; when technological feasibility is achieved for our products; the potential outcome of uncertain tax positions that have been recognized in our consolidated financial statements or tax returns; and determining the timing and amount of impairments for investments. Actual results and outcomes may differ from management’s estimates and assumptions due to risks and uncertainties.

Financial Instruments

Investments

We consider all highly liquid interest-earning investments with a maturity of three months or less at the date of purchase to be cash equivalents. The fair values of these investments approximate their carrying values. In general, investments with original maturities of greater than three months and remaining maturities of less than one year are classified as short-term investments. Investments with maturities beyond one year may be classified as short-term based on their highly liquid nature and because such marketable securities represent the investment of cash that is available for current operations.

Debt investments are classified as available-for-sale and realized gains and losses are recorded using the specific identification method. Changes in fair value, excluding credit losses and impairments, are recorded in other comprehensive income. Fair value is calculated based on publicly available market information or other estimates determined by management. If the cost of an investment exceeds its fair value, we evaluate, among other factors, general market conditions, credit quality of debt instrument issuers, and the extent to which the fair value is less than cost. To determine credit losses, we employ a systematic methodology that considers available quantitative and qualitative evidence. In addition, we consider specific adverse conditions related to the financial health of, and business outlook for, the investee. If we have plans to sell the security or it is more likely than not that we will be required to sell the security before recovery, then a decline in fair value below cost is recorded as an impairment charge in other income (expense), net and a new cost basis in the investment is established. If market, industry, and/or investee conditions deteriorate, we may incur future impairments.

Equity investments with readily determinable fair values are measured at fair value. Equity investments without readily determinable fair values are measured using the equity method or measured at cost with adjustments for observable changes in price or impairments (referred to as the measurement alternative). We perform a qualitative assessment on a periodic basis and recognize an impairment if there are sufficient indicators that the fair value of the investment is less than carrying value. Changes in value are recorded in other income (expense), net.

Investments that are considered variable interest entities ("VIEs") are evaluated to determine whether we are the primary beneficiary of the VIE, in which case we would be required to consolidate the entity. We evaluate whether we have (1) the power to direct the activities that most significantly impact the VIE's economic performance, and (2) the obligation to absorb losses or the right to receive benefits from the VIE that could potentially be significant to the VIE. We have determined we are not the primary beneficiary of any of our VIE investments. Therefore, our VIE investments are not consolidated and the majority are accounted for under the equity method of accounting. We have an investment in OpenAI Global, LLC ("OpenAI") and have made total funding commitments of $13 billion. The investment is accounted for under the equity method of accounting.

Derivatives

Derivative instruments are recognized as either assets or liabilities and measured at fair value. The accounting for changes in the fair value of a derivative depends on the intended use of the derivative and the resulting designation.

For derivative instruments designated as fair value hedges, gains and losses are recognized in other income (expense), net with offsetting gains and losses on the hedged items. Gains and losses representing hedge components excluded from the assessment of effectiveness are recognized in other income (expense), net.

For derivative instruments designated as cash flow hedges, gains and losses are initially reported as a component of other comprehensive income and subsequently recognized in other income (expense), net with the corresponding hedged item. Gains and losses representing hedge components excluded from the assessment of effectiveness are recognized in other income (expense), net.

For derivative instruments that are not designated as hedges, gains and losses from changes in fair values are primarily recognized in other income (expense), net.

Fair Value Measurements

We account for certain assets and liabilities at fair value. The hierarchy below lists three levels of fair value based on the extent to which inputs used in measuring fair value are observable in the market. We categorize each of our fair value measurements in one of these three levels based on the lowest level input that is significant to the fair value measurement in its entirety. These levels are:

- *Level 1* – inputs are based upon unadjusted quoted prices for identical instruments in active markets. Our Level 1 investments include U.S. government securities, common and preferred stock, and mutual funds. Our Level 1 derivative assets and liabilities include those actively traded on exchanges.

- *Level 2* – inputs are based upon quoted prices for similar instruments in active markets, quoted prices for identical or similar instruments in markets that are not active, and model-based valuation techniques (e.g. the Black-Scholes model) for which all significant inputs are observable in the market or can be corroborated by observable market data for substantially the full term of the assets or liabilities. Where applicable, these models project future cash flows and discount the future amounts to a present value using market-based observable inputs including interest rate curves, credit spreads, foreign exchange rates, and forward and spot prices for currencies. Our Level 2 investments include commercial paper, certificates of deposit, U.S. agency securities, foreign government bonds, mortgage- and asset-backed securities, corporate notes and bonds, and municipal securities. Our Level 2 derivative assets and liabilities include certain cleared swap contracts and over-the-counter forward, option, and swap contracts.
- *Level 3* – inputs are generally unobservable and typically reflect management's estimates of assumptions that market participants would use in pricing the asset or liability. The fair values are therefore determined using model-based techniques, including option pricing models and discounted cash flow models. Our Level 3 assets and liabilities include investments in corporate notes and bonds, municipal securities, and goodwill and intangible assets, when they are recorded at fair value due to an impairment charge. Unobservable inputs used in the models are significant to the fair values of the assets and liabilities.

We measure equity investments without readily determinable fair values on a nonrecurring basis. The fair values of these investments are determined based on valuation techniques using the best information available, and may include quoted market prices, market comparables, and discounted cash flow projections.

Our other current financial assets and current financial liabilities have fair values that approximate their carrying values.

Contract Balances and Other Receivables

As of both March 31, 2025 and June 30, 2024, long-term accounts receivable, net of allowance for doubtful accounts, was $4.9 billion and is included in other long-term assets in our consolidated balance sheets.

As of March 31, 2025 and June 30, 2024, other receivables related to activities to facilitate the purchase of server components were $10.8 billion and $10.5 billion, respectively, and are included in other current assets in our consolidated balance sheets.

We record financing receivables when we offer certain customers the option to acquire our software products and services offerings through a financing program in a limited number of countries. As of March 31, 2025 and June 30, 2024, our financing receivables, net were $2.6 billion and $4.5 billion, respectively, for short-term and long-term financing receivables, which are included in other current assets and other long-term assets in our consolidated balance sheets. We record an allowance to cover expected losses based on troubled accounts, historical experience, and other currently available evidence.

Related Party Transactions

In March 2024, we entered into an agreement with Inflection AI, Inc. ("Inflection"), pursuant to which we obtained a non-exclusive license to Inflection's intellectual property. Reid Hoffman, a member of our Board of Directors, is a co-founder of and serves on the board of directors of Inflection. As of the date of the agreement with Inflection, Reprogrammed Interchange LLC ("Reprogrammed") and entities affiliated with Greylock Ventures ("Greylock") each held less than a 10% equity interest in Inflection. Mr. Hoffman may be deemed to beneficially own the shares held by Reprogrammed and Greylock by virtue of his relationship with such entities. Mr. Hoffman did not participate in any portions of the meetings of our Board of Directors or any committee thereof to review and approve the transaction with Inflection.

Recent Accounting Guidance

Segment Reporting – Improvements to Reportable Segment Disclosures

In November 2023, the Financial Accounting Standards Board ("FASB") issued a new standard to improve reportable segment disclosures. The guidance expands the disclosures required for reportable segments in our annual and interim consolidated financial statements, primarily through enhanced disclosures about significant segment expenses. The standard will be effective for us beginning with our annual reporting for fiscal year 2025 and interim periods thereafter, with early adoption permitted. We are currently evaluating the impact of this standard on our segment disclosures.

Income Taxes – Improvements to Income Tax Disclosures

In December 2023, the FASB issued a new standard to improve income tax disclosures. The guidance requires disclosure of disaggregated income taxes paid, prescribes standardized categories for the components of the effective tax rate reconciliation, and modifies other income tax-related disclosures. The standard will be effective for us beginning with our annual reporting for fiscal year 2026, with early adoption permitted. We are currently evaluating the impact of this standard on our income tax disclosures.

Income Statement – Disaggregation of Income Statement Expenses

In November 2024, the FASB issued a new standard to expand disclosures about income statement expenses. The guidance requires disaggregation of certain costs and expenses included in each relevant expense caption on our consolidated income statements in a separate note to the financial statements at each interim and annual reporting period, including amounts of purchases of inventory, employee compensation, depreciation, and intangible asset amortization. The standard will be effective for us beginning with our annual reporting for fiscal year 2028 and interim periods thereafter, with early adoption permitted. We are currently evaluating the impact of this standard on our disclosures.

NOTE 2 — EARNINGS PER SHARE

Basic earnings per share ("EPS") is computed based on the weighted average number of shares of common stock outstanding during the period. Diluted EPS is computed based on the weighted average number of shares of common stock plus the effect of dilutive potential common shares outstanding during the period using the treasury stock method. Dilutive potential common shares include outstanding stock options and stock awards.

The components of basic and diluted EPS were as follows:

(In millions, except per share amounts)	Three Months Ended March 31,		Nine Months Ended March 31,	
	2025	**2024**	**2025**	**2024**
Net income available for common shareholders (A)	**$ 25,824**	$ 21,939	**$ 74,599**	$ 66,100
Weighted average outstanding shares of common stock (B)	**7,434**	7,431	**7,434**	7,431
Dilutive effect of stock-based awards	**27**	41	**32**	36
Common stock and common stock equivalents (C)	**7,461**	7,472	**7,466**	7,467
Earnings Per Share				
Basic (A/B)	**$ 3.47**	$ 2.95	**$ 10.03**	$ 8.90
Diluted (A/C)	**$ 3.46**	$ 2.94	**$ 9.99**	$ 8.85

Anti-dilutive stock-based awards excluded from the calculations of diluted EPS were immaterial during the periods presented.

NOTE 3 — OTHER INCOME (EXPENSE), NET

The components of other income (expense), net were as follows:

(In millions)	Three Months Ended March 31, 2025	2024	Nine Months Ended March 31, 2025	2024
Interest and dividends income	**$ 597**	$ 619	**$ 1,878**	$ 2,519
Interest expense	**(594)**	(800)	**(1,770)**	(2,234)
Net recognized gains (losses) on investments	**111**	(25)	**(286)**	(63)
Net gains (losses) on derivatives	**187**	(24)	**(267)**	(198)
Net gains (losses) on foreign currency remeasurements	**89**	(138)	**112**	(203)
Other, net	**(1,013)**	(486)	**(2,861)**	(792)
Total	**$ (623)**	$ (854)	**$ (3,194)**	$ (971)

Other, net primarily reflects net recognized losses on equity method investments, including OpenAI.

Net Recognized Gains (Losses) on Investments

Net recognized gains (losses) on debt investments were as follows:

(In millions)	Three Months Ended March 31, 2025	2024	Nine Months Ended March 31, 2025	2024
Realized gains from sales of available-for-sale securities	**$ 8**	$ 8	**$ 25**	$ 14
Realized losses from sales of available-for-sale securities	**(17)**	(24)	**(51)**	(78)
Impairments and allowance for credit losses	**3**	3	**1**	15
Total	**$ (6)**	$ (13)	**$ (25)**	$ (49)

Net recognized gains (losses) on equity investments were as follows:

(In millions)	Three Months Ended March 31, 2025	2024	Nine Months Ended March 31, 2025	2024
Net realized gains on investments sold	**$ 9**	$ 15	**$ 66**	$ 29
Net unrealized gains (losses) on investments still held	**135**	(7)	**572**	156
Impairments of investments	**(27)**	(20)	**(899)**	(199)
Total	**$ 117**	$ (12)	**$ (261)**	$ (14)

NOTE 4 — INVESTMENTS

Investment Components

The components of investments were as follows:

(In millions)	Fair Value Level	Adjusted Cost Basis	Unrealized Gains	Unrealized Losses	Recorded Basis	Cash and Cash Equivalents	Short-term Investments	Equity and Other Investments
March 31, 2025								
Changes in Fair Value Recorded in Other Comprehensive Income								
Commercial paper	Level 2	$ 11,101	$ 0	$ 0	$ 11,101	$ 11,002	$ 99	$ 0
Certificates of deposit	Level 2	3,517	0	0	3,517	3,473	44	0
U.S. government securities	Level 1	42,349	7	(1,747)	40,609	858	39,751	0
U.S. agency securities	Level 2	1,953	0	0	1,953	1,944	9	0
Foreign government bonds	Level 2	321	6	(13)	314	1	313	0
Mortgage- and asset-backed securities	Level 2	1,610	6	(27)	1,589	0	1,589	0
Corporate notes and bonds	Level 2	8,637	62	(139)	8,560	0	8,560	0
Corporate notes and bonds	Level 3	2,410	0	(12)	2,398	0	123	2,275
Municipal securities	Level 2	216	1	(9)	208	0	208	0
Municipal securities	Level 3	104	0	(16)	88	0	88	0
Total debt investments		$ 72,218	$ 82	$ (1,963)	$ 70,337	$ 17,278	$ 50,784	$ 2,275
Changes in Fair Value Recorded in Net Income								
Equity investments	Level 1				$ 4,427	$ 870	$ 0	$ 3,557
Equity investments	Other				9,930	0	0	9,930
Total equity investments					$ 14,357	$ 870	$ 0	$ 13,487
Cash					$ 10,680	$ 10,680	$ 0	$ 0
Derivatives, net [(a)]					279	0	6	273
Total					$ 95,653	$ 28,828	$ 50,790	$ 16,035

(In millions)	Fair Value Level	Adjusted Cost Basis	Unrealized Gains	Unrealized Losses	Recorded Basis	Cash and Cash Equivalents	Short-term Investments	Equity and Other Investments
June 30, 2024								
Changes in Fair Value Recorded in Other Comprehensive Income								
Commercial paper	Level 2	$ 4,666	$ 0	$ 0	$ 4,666	$ 4,666	$ 0	$ 0
Certificates of deposit	Level 2	1,547	0	0	1,547	1,503	44	0
U.S. government securities	Level 1	49,603	4	(2,948)	46,659	14	46,645	0
U.S. agency securities	Level 2	17	0	0	17	0	17	0
Foreign government bonds	Level 2	319	3	(16)	306	0	306	0
Mortgage- and asset-backed securities	Level 2	944	3	(35)	912	0	912	0
Corporate notes and bonds	Level 2	9,106	28	(318)	8,816	0	8,816	0
Corporate notes and bonds	Level 3	1,641	0	(1)	1,640	0	140	1,500
Municipal securities	Level 2	262	0	(13)	249	0	249	0
Municipal securities	Level 3	104	0	(17)	87	0	87	0
Total debt investments		$ 68,209	$ 38	$ (3,348)	$ 64,899	$ 6,183	$ 57,216	$ 1,500
Changes in Fair Value Recorded in Net Income								
Equity investments	Level 1				$ 3,547	$ 561	$ 0	$ 2,986
Equity investments	Other				10,114	0	0	10,114
Total equity investments					$ 13,661	$ 561	$ 0	$ 13,100
Cash					$ 11,571	$ 11,571	$ 0	$ 0
Derivatives, net [(a)]					12	0	12	0
Total					$ 90,143	$ 18,315	$ 57,228	$ 14,600

(a) *Refer to Note 5 – Derivatives for further information on the fair value of our derivative instruments.*

Equity investments presented as “Other” in the tables above include investments without readily determinable fair values measured at cost with adjustments for observable changes in price or impairments, measured using the equity method, or measured at fair value using net asset value as a practical expedient which are not categorized in the fair value hierarchy. As of March 31, 2025 and June 30, 2024, equity investments without readily determinable fair values measured at cost with adjustments for observable changes in price or impairments were $2.9 billion and $3.9 billion, respectively, and equity investments measured using the equity method were $6.8 billion and $6.0 billion, respectively.

Unrealized Losses on Debt Investments

Debt investments with continuous unrealized losses for less than 12 months and 12 months or greater and their related fair values were as follows:

(In millions)	Less than 12 Months Fair Value	Less than 12 Months Unrealized Losses	12 Months or Greater Fair Value	12 Months or Greater Unrealized Losses	Total Fair Value	Total Unrealized Losses
March 31, 2025						
U.S. government and agency securities	**$ 492**	**$ (50)**	**$ 38,754**	**$ (1,697)**	**$ 39,246**	**$ (1,747)**
Foreign government bonds	**66**	**(3)**	**127**	**(10)**	**193**	**(13)**
Mortgage- and asset-backed securities	**649**	**(3)**	**227**	**(24)**	**876**	**(27)**
Corporate notes and bonds	**1,720**	**(14)**	**3,629**	**(137)**	**5,349**	**(151)**
Municipal securities	**0**	**0**	**212**	**(25)**	**212**	**(25)**
Total	**$ 2,927**	**$ (70)**	**$ 42,949**	**$ (1,893)**	**$ 45,876**	**$ (1,963)**

(In millions)	Less than 12 Months Fair Value	Less than 12 Months Unrealized Losses	12 Months or Greater Fair Value	12 Months or Greater Unrealized Losses	Total Fair Value	Total Unrealized Losses
June 30, 2024						
U.S. government and agency securities	$ 529	$ (12)	$ 45,821	$ (2,936)	$ 46,350	$ (2,948)
Foreign government bonds	79	(2)	180	(14)	259	(16)
Mortgage- and asset-backed securities	201	(1)	409	(34)	610	(35)
Corporate notes and bonds	1,310	(9)	5,779	(310)	7,089	(319)
Municipal securities	38	(1)	243	(29)	281	(30)
Total	$ 2,157	$ (25)	$ 52,432	$ (3,323)	$ 54,589	$ (3,348)

Unrealized losses from fixed-income securities are primarily attributable to changes in interest rates. Management does not believe any remaining unrealized losses represent impairments based on our evaluation of available evidence.

Debt Investment Maturities

The following table outlines maturities of our debt investments as of March 31, 2025:

(In millions)	Adjusted Cost Basis	Estimated Fair Value
March 31, 2025		
Due in one year or less	**$ 31,746**	**$ 31,609**
Due after one year through five years	**32,127**	**30,870**
Due after five years through 10 years	**6,919**	**6,530**
Due after 10 years	**1,426**	**1,328**
Total	**$ 72,218**	**$ 70,337**

NOTE 5 — DERIVATIVES

We use derivative instruments to manage risks related to foreign currencies, interest rates, equity prices, and credit; to enhance investment returns; and to facilitate portfolio diversification. Our objectives for holding derivatives include reducing, eliminating, and efficiently managing the economic impact of these exposures as effectively as possible. Our derivative programs include strategies that both qualify and do not qualify for hedge accounting treatment.

Foreign Currencies

Certain forecasted transactions, assets, and liabilities are exposed to foreign currency risk. We monitor our foreign currency exposures daily to maximize the economic effectiveness of our foreign currency hedge positions.

Foreign currency risks related to certain Euro-denominated debt are hedged using foreign exchange forward contracts that are designated as cash flow hedging instruments.

Certain options and forwards not designated as hedging instruments are also used to manage the variability in foreign exchange rates on certain balance sheet amounts and to manage other foreign currency exposures.

Interest Rate

Interest rate risks related to certain fixed-rate debt are hedged using interest rate swaps that are designated as fair value hedging instruments to effectively convert the fixed interest rates to floating interest rates.

Securities held in our fixed-income portfolio are subject to different interest rate risks based on their maturities. We manage the average maturity of our fixed-income portfolio to achieve economic returns that correlate to certain broad-based fixed-income indices using option, futures, and swap contracts. These contracts are not designated as hedging instruments and are included in “Other contracts” in the tables below.

Equity

Securities held in our equity investments portfolio are subject to market price risk. At times, we may hold options, futures, and swap contracts. These contracts are not designated as hedging instruments.

Credit

Our fixed-income portfolio is diversified and consists primarily of investment-grade securities. We use credit default swap contracts to manage credit exposures relative to broad-based indices and to facilitate portfolio diversification. These contracts are not designated as hedging instruments and are included in “Other contracts” in the tables below.

Credit-Risk-Related Contingent Features

Certain counterparty agreements for derivative instruments contain provisions that require our issued and outstanding long-term unsecured debt to maintain an investment grade credit rating and require us to maintain minimum liquidity of $1.0 billion. To the extent we fail to meet these requirements, we will be required to post collateral, similar to the standard convention related to over-the-counter derivatives. As of March 31, 2025, our long-term unsecured debt rating was AAA, and cash investments were in excess of $1.0 billion. As a result, no collateral was required to be posted.

The following table presents the notional amounts of our outstanding derivative instruments measured in U.S. dollar equivalents:

(In millions)	March 31, 2025	June 30, 2024
Designated as Hedging Instruments		
Foreign exchange contracts purchased	**$ 1,492**	$ 1,492
Interest rate contracts purchased	**1,151**	1,100
Not Designated as Hedging Instruments		
Foreign exchange contracts purchased	**9,641**	7,167
Foreign exchange contracts sold	**34,881**	31,793
Equity contracts purchased	**5,100**	4,016
Equity contracts sold	**2,170**	2,165
Other contracts purchased	**2,693**	2,113
Other contracts sold	**1,195**	811

Fair Values of Derivative Instruments

The following table presents our derivative instruments:

(In millions)	Derivative Assets	Derivative Liabilities	Derivative Assets	Derivative Liabilities
		March 31, 2025		June 30, 2024
Designated as Hedging Instruments				
Foreign exchange contracts	$ 26	$ (80)	$ 24	$ (76)
Interest rate contracts	5	0	19	0
Not Designated as Hedging Instruments				
Foreign exchange contracts	224	(751)	213	(230)
Equity contracts	282	(1,085)	63	(491)
Other contracts	13	(4)	12	(3)
Gross amounts of derivatives	550	(1,920)	331	(800)
Gross amounts of derivatives offset in the balance sheets	(142)	145	(151)	152
Cash collateral received	0	(126)	0	(104)
Net amounts of derivatives	$ 408	$ (1,901)	$ 180	$ (752)
Reported as				
Short-term investments	$ 6	$ 0	$ 12	$ 0
Other current assets	124	0	149	0
Equity and other investments	273	0	0	0
Other long-term assets	5	0	19	0
Other current liabilities	0	(1,730)	0	(401)
Other long-term liabilities	0	(171)	0	(351)
Total	$ 408	$ (1,901)	$ 180	$ (752)

Gross derivative assets and liabilities subject to legally enforceable master netting agreements for which we have elected to offset were $259 million and $1.9 billion, respectively, as of March 31, 2025, and $304 million and $800 million, respectively, as of June 30, 2024.

The following table presents the fair value of our derivatives instruments on a gross basis:

(In millions)	Level 1	Level 2	Level 3	Total
March 31, 2025				
Derivative assets	$ 0	$ 267	$ 283	$ 550
Derivative liabilities	0	(1,920)	0	(1,920)
June 30, 2024				
Derivative assets	0	327	4	331
Derivative liabilities	(1)	(799)	0	(800)

Gains (losses) on derivative instruments recognized in other income (expense), net were as follows:

(In millions)	Three Months Ended March 31, 2025	2024	Nine Months Ended March 31, 2025	2024
Designated as Fair Value Hedging Instruments				
Interest rate contracts				
Derivatives	**$ 3**	$ (21)	**$ 5**	$ (15)
Hedged items	**(12)**	10	**(36)**	(21)
Designated as Cash Flow Hedging Instruments				
Foreign exchange contracts				
Amount reclassified from accumulated other comprehensive loss	**50**	(37)	**(7)**	(32)
Not Designated as Hedging Instruments				
Foreign exchange contracts	**(372)**	299	**383**	171
Equity contracts	**176**	(22)	**(283)**	(196)
Other contracts	**18**	(8)	**16**	(5)

Gains (losses), net of tax, on derivative instruments recognized in our consolidated comprehensive income statements were as follows:

(In millions)	Three Months Ended March 31, 2025	2024	Nine Months Ended March 31, 2025	2024
Designated as Cash Flow Hedging Instruments				
Foreign exchange contracts				
Included in effectiveness assessment	**$ 20**	$ (19)	**$ (1)**	$ 3

NOTE 6 — INVENTORIES

The components of inventories were as follows:

(In millions)	March 31, 2025	June 30, 2024
Raw materials	**$ 327**	$ 394
Work in process	**13**	7
Finished goods	**508**	845
Total	**$ 848**	$ 1,246

NOTE 7 — BUSINESS COMBINATIONS

Activision Blizzard, Inc.

On October 13, 2023, we completed our acquisition of Activision Blizzard, Inc. (“Activision Blizzard”) for a total purchase price of $75.4 billion, consisting primarily of cash. Activision Blizzard is a leader in game development and an interactive entertainment content publisher. The acquisition will accelerate the growth in our gaming business across mobile, PC, console, and cloud gaming. The financial results of Activision Blizzard have been included in our consolidated financial statements since the date of the acquisition. Activision Blizzard is reported as part of our More Personal Computing segment.

The allocation of the purchase price to the assets acquired and liabilities assumed was completed as of September 30, 2024. The major classes of assets and liabilities to which we have allocated the purchase price were as follows:

(In millions)		
Cash and cash equivalents	$	12,976
Goodwill		51,001
Intangible assets		21,969
Other assets		2,503
Long-term debt		(2,799)
Long-term income taxes		(1,946)
Deferred income taxes		(4,676)
Other liabilities		(3,620)
Total purchase price	$	75,408

Goodwill was assigned to our More Personal Computing segment. The goodwill was primarily attributed to increased synergies that are expected to be achieved from the integration of Activision Blizzard. Substantially all of the goodwill is expected to be non-deductible for income tax purposes.

Following are the details of the purchase price allocated to the intangible assets acquired:

(In millions, except average life)		Amount	Weighted Average Life
Marketing-related	$	11,619	24 years
Technology-based		9,689	4 years
Customer-related		661	4 years
Fair value of intangible assets acquired	$	21,969	15 years

Following are the supplemental consolidated financial results of Microsoft Corporation on an unaudited pro forma basis, as if the acquisition had been consummated on July 1, 2022:

(In millions, except per share amounts)		Three Months Ended March 31,		Nine Months Ended March 31,
		2024		2024
Revenue	$	61,856	$	182,717
Net income		21,931		66,278
Diluted earnings per share		2.94		8.88

These pro forma results were based on estimates and assumptions, which we believe are reasonable. They are not the results that would have been realized had we been a combined company during the periods presented and are not necessarily indicative of our consolidated results of operations in future periods. The pro forma results include adjustments related to purchase accounting, primarily amortization of intangible assets. Acquisition costs and other nonrecurring charges were immaterial and are included in the earliest period presented.

NOTE 8 — GOODWILL

Changes in the carrying amount of goodwill were as follows:

(In millions)		June 30, 2024		Acquisitions		Other		March 31, 2025
Productivity and Business Processes	$	31,361	**$**	**0**	**$**	**20**	**$**	**31,381**
Intelligent Cloud		25,648		**0**		**7**		**25,655**
More Personal Computing		62,211		**0**		**82**		**62,293**
Total	$	119,220	**$**	**0**	**$**	**109**	**$**	**119,329**

We have recast certain prior period amounts to conform to the way we internally manage and monitor our business. Refer to Note 1 – Accounting Policies for further information.

The measurement periods for the valuation of assets acquired and liabilities assumed end as soon as information on the facts and circumstances that existed as of the acquisition dates becomes available, but do not exceed 12 months. Adjustments in purchase price allocations may require a change in the amounts allocated to goodwill during the periods in which the adjustments are determined.

Any change in the goodwill amounts resulting from foreign currency translations and purchase accounting adjustments are presented as “Other” in the table above. Also included in “Other” are business dispositions and transfers between segments due to reorganizations, as applicable.

As discussed in Note 1 – Accounting Policies, during the first quarter of fiscal year 2025 we made changes to our segments. These segment changes also resulted in changes to our reporting units. We reallocated goodwill across impacted reporting units using a relative fair value approach. In addition, we completed an assessment of any potential goodwill impairment for all reporting units immediately prior to the reallocation and determined that no impairment existed.

NOTE 9 — INTANGIBLE ASSETS

The components of intangible assets, all of which are finite-lived, were as follows:

(In millions)	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount
			March 31, 2025			June 30, 2024
Marketing-related	**$ 16,503**	**$ (3,703)**	**$ 12,800**	$ 16,500	$ (3,101)	$ 13,399
Technology-based	**22,437**	**(13,813)**	**8,624**	21,913	(10,741)	11,172
Customer-related	**4,382**	**(2,001)**	**2,381**	6,038	(3,051)	2,987
Contract-based	**193**	**(30)**	**163**	58	(19)	39
Total	**$ 43,515**	**$ (19,547)**	**$ 23,968**	$ 44,509	$ (16,912)	$ 27,597

Intangible assets amortization expense was $1.5 billion and $4.5 billion for the three and nine months ended March 31, 2025, respectively, and $1.4 billion and $3.4 billion for the three and nine months ended March 31, 2024, respectively.

The following table outlines the estimated future amortization expense related to intangible assets held as of March 31, 2025:

(In millions)	
Year Ending June 30,	
2025 (excluding the nine months ended March 31, 2025)	**$ 1,525**
2026	**4,582**
2027	**2,866**
2028	**2,007**
2029	**1,834**
Thereafter	**11,154**
Total	**$ 23,968**

NOTE 10 — DEBT

Short-term Debt

As of March 31, 2025, we had no commercial paper issued or outstanding. As of June 30, 2024, we had $6.7 billion of commercial paper issued and outstanding, with a weighted average interest rate of 5.4% and maturities ranging from 28 days to 152 days. The estimated fair value of this commercial paper approximates its carrying value.

Long-term Debt

The components of long-term debt were as follows:

(In millions, issuance by calendar year)	Maturities (calendar year)	Stated Interest Rate	Effective Interest Rate	March 31, 2025	June 30, 2024
2009 issuance of $3.8 billion	**2039**	**5.20%**	**5.24%**	**$ 520**	$ 520
2010 issuance of $4.8 billion	**2040**	**4.50%**	**4.57%**	**486**	486
2011 issuance of $2.3 billion	**2041**	**5.30%**	**5.36%**	**718**	718
2012 issuance of $2.3 billion	**2042**	**3.50%**	**3.57%**	**454**	454
2013 issuance of $5.2 billion	**2043**	**3.75%–4.88%**	**3.83%–4.92%**	**314**	314
2013 issuance of €4.1 billion	**2028–2033**	**2.63%–3.13%**	**2.69%–3.22%**	**2,484**	2,465
2015 issuance of $23.8 billion	**2025–2055**	**3.13%–4.75%**	**3.18%–4.78%**	**7,555**	9,805
2016 issuance of $19.8 billion	**2026–2056**	**2.40%–3.95%**	**2.46%–4.03%**	**7,930**	7,930
2017 issuance of $17.1 billion	**2026–2057**	**3.30%–4.50%**	**3.38%–5.49%**	**6,833**	6,833
2020 issuance of $10.1 billion	**2030–2060**	**1.35%–2.68%**	**2.53%–5.43%**	**10,111**	10,111
2021 issuance of $8.2 billion	**2052–2062**	**2.92%–3.04%**	**2.92%–3.04%**	**8,185**	8,185
2023 issuance of $0.1 billion	**2026–2050**	**1.35%–4.50%**	**5.16%–5.49%**	**56**	56
2024 issuance of $3.3 billion	**2026–2050**	**1.35%–4.50%**	**5.16%–5.49%**	**3,344**	3,344
Total face value				**48,990**	51,221
Unamortized discount and issuance costs				**(1,171)**	(1,227)
Hedge fair value adjustments [(a)]				**(45)**	(81)
Premium on debt exchange				**(4,893)**	(4,976)
Total debt				**42,881**	44,937
Current portion of long-term debt				**(2,999)**	(2,249)
Long-term debt				**$ 39,882**	$ 42,688

(a) *Refer to Note 5 – Derivatives for further information on the interest rate swaps related to fixed-rate debt.*

As of March 31, 2025 and June 30, 2024, the estimated fair value of long-term debt, including the current portion, was $40.2 billion and $42.3 billion, respectively. The estimated fair values are based on Level 2 inputs.

Debt in the table above is comprised of senior unsecured obligations and ranks equally with our other outstanding obligations. Interest is paid semi-annually, except for the Euro-denominated debt, which is paid annually.

The following table outlines maturities of our long-term debt, including the current portion, as of March 31, 2025:

(In millions)	
Year Ending June 30,	
2025 (excluding the nine months ended March 31, 2025)	**$ 0**
2026	**3,000**
2027	**9,250**
2028	**0**
2029	**1,890**
Thereafter	**34,850**
Total	**$ 48,990**

NOTE 11 — INCOME TAXES

Effective Tax Rate

Our effective tax rate was 18% for both the three months ended March 31, 2025 and 2024, and 18% for both the nine months ended March 31, 2025 and 2024. Our effective tax rate for the three months ended March 31, 2025 was primarily impacted by changes in the mix of our earnings and tax expenses between the U.S. and foreign countries. Our effective tax rate for the nine months ended March 31, 2025 was primarily impacted by tax benefits from tax law changes in the prior fiscal year, including the delay of the effective date of final foreign tax credit regulations, and changes in the mix of our earnings and tax expenses between the U.S. and foreign countries.

Our effective tax rate was lower than the U.S. federal statutory rate for the three and nine months ended March 31, 2025, primarily due to earnings taxed at lower rates in foreign jurisdictions resulting from producing and distributing our products and services through our foreign regional operations center in Ireland.

Uncertain Tax Positions

As of March 31, 2025 and June 30, 2024, unrecognized tax benefits and other income tax liabilities were $26.4 billion and $24.9 billion, respectively, and are included in long-term income taxes in our consolidated balance sheets.

We remain under audit by the IRS for tax years 2014 to 2017. With respect to the audit for tax years 2004 to 2013, on September 26, 2023, we received Notices of Proposed Adjustment (“NOPAs”) from the IRS. The primary issues in the NOPAs relate to intercompany transfer pricing. In the NOPAs, the IRS is seeking an additional tax payment of $28.9 billion plus penalties and interest. As of March 31, 2025, we believe our allowances for income tax contingencies are adequate. We disagree with the proposed adjustments and will vigorously contest the NOPAs through the IRS’s administrative appeals office and, if necessary, judicial proceedings. We do not expect a final resolution of these issues in the next 12 months. Based on the information currently available, we do not anticipate a significant increase or decrease to our income tax contingencies for these issues within the next 12 months.

We are subject to income tax in many jurisdictions outside the U.S. Our operations in certain jurisdictions remain subject to examination for tax years 1996 to 2024, some of which are currently under audit by local tax authorities. The resolution of each of these audits is not expected to be material to our consolidated financial statements.

NOTE 12 — UNEARNED REVENUE

Unearned revenue by segment was as follows:

(In millions)

	March 31, 2025	June 30, 2024
Productivity and Business Processes	**$ 34,244**	$ 43,599
Intelligent Cloud	**10,240**	13,683
More Personal Computing	**2,992**	2,902
Total	**$ 47,476**	$ 60,184

We have recast certain prior period amounts to conform to the way we internally manage and monitor our business. Refer to Note 1 – Accounting Policies for further information.

Changes in unearned revenue were as follows:

(In millions)

Nine Months Ended March 31, 2025	
Balance, beginning of period	**$ 60,184**
Deferral of revenue	**118,709**
Recognition of unearned revenue	**(131,417)**
Balance, end of period	**$ 47,476**

Revenue allocated to remaining performance obligations, which includes unearned revenue and amounts that will be invoiced and recognized as revenue in future periods, was $321 billion as of March 31, 2025, of which $315 billion is related to the commercial portion of revenue. We expect to recognize approximately 40% of our total company remaining performance obligation revenue over the next 12 months and the remainder thereafter.

NOTE 13 — LEASES

We have operating and finance leases for datacenters, corporate offices, research and development facilities, Microsoft Experience Centers, and certain equipment. Our leases have remaining lease terms of less than 1 year to 20 years, some of which include options to extend the leases for up to 5 years, and some of which include options to terminate the leases within 1 year.

The components of lease expense were as follows:

(In millions)	Three Months Ended March 31,		Nine Months Ended March 31,	
	2025	**2024**	**2025**	**2024**
Operating lease cost	**$ 1,476**	$ 882	**$ 3,929**	$ 2,473
Finance lease cost:				
Amortization of right-of-use assets	**$ 900**	$ 453	**$ 2,426**	$ 1,241
Interest on lease liabilities	**374**	190	**986**	507
Total finance lease cost	**$ 1,274**	$ 643	**$ 3,412**	$ 1,748

Supplemental cash flow information related to leases was as follows:

(In millions)	Three Months Ended March 31,		Nine Months Ended March 31,	
	2025	**2024**	**2025**	**2024**
Cash paid for amounts included in the measurement of lease liabilities:				
Operating cash flows from operating leases	**$ 955**	$ 836	**$ 3,124**	$ 2,433
Operating cash flows from finance leases	**344**	190	**938**	507
Financing cash flows from finance leases	**352**	323	**1,634**	896
Right-of-use assets obtained in exchange for lease obligations:				
Operating leases	**1,918**	1,831	**6,909**	4,482
Finance leases	**3,241**	3,421	**14,008**	6,921

Supplemental balance sheet information related to leases was as follows:

(In millions, except lease term and discount rate)

	March 31, 2025	June 30, 2024
Operating Leases		
Operating lease right-of-use assets	**$ 24,475**	$ 18,961
Other current liabilities	**$ 5,238**	$ 3,580
Operating lease liabilities	**17,686**	15,497
Total operating lease liabilities	**$ 22,924**	$ 19,077
Finance Leases		
Property and equipment, at cost	**$ 46,275**	$ 32,248
Accumulated depreciation	**(8,650)**	(6,386)
Property and equipment, net	**$ 37,625**	$ 25,862
Other current liabilities	**$ 2,889**	$ 2,349
Other long-term liabilities	**36,325**	24,796
Total finance lease liabilities	**$ 39,214**	$ 27,145
Weighted Average Remaining Lease Term		
Operating leases	**6 years**	7 years
Finance leases	**12 years**	12 years
Weighted Average Discount Rate		
Operating leases	**3.5%**	3.3%
Finance leases	**4.1%**	3.9%

The following table outlines maturities of our lease liabilities as of March 31, 2025:

(In millions)

Year Ending June 30,	**Operating Leases**	**Finance Leases**
2025 (excluding the nine months ended March 31, 2025)	**$ 1,709**	**$ 1,065**
2026	**5,700**	**4,465**
2027	**4,823**	**4,507**
2028	**3,286**	**4,509**
2029	**2,287**	**3,866**
Thereafter	**7,728**	**32,307**
Total lease payments	**25,533**	**50,719**
Less imputed interest	**(2,609)**	**(11,505)**
Total	**$ 22,924**	**$ 39,214**

As of March 31, 2025, we had additional operating and finance leases, primarily for datacenters, that had not yet commenced of $4.4 billion and $94.8 billion, respectively. These operating and finance leases will commence between fiscal year 2025 and fiscal year 2030 with lease terms of 1 year to 20 years.

NOTE 14 — CONTINGENCIES

U.S. Cell Phone Litigation

Microsoft Mobile Oy, a subsidiary of Microsoft, along with other handset manufacturers and network operators, is a defendant in 45 lawsuits filed in the Superior Court for the District of Columbia by individual plaintiffs who allege that radio emissions from cellular handsets caused their brain tumors and other adverse health effects. We assumed responsibility for these claims in our agreement to acquire Nokia’s Devices and Services business and have been substituted for the Nokia defendants. Twelve of these cases were consolidated for certain pre-trial proceedings; the remaining cases are stayed. In a separate 2009 decision, the Court of Appeals for the District of Columbia held that adverse health effect claims arising from the use of cellular handsets that operate within the U.S. Federal Communications Commission radio frequency emission guidelines (“FCC Guidelines”) are pre-empted by federal law. The plaintiffs allege that their handsets either operated outside the FCC Guidelines or were manufactured before the FCC Guidelines went into effect. The lawsuits also allege an industry-wide conspiracy to manipulate the science and testing around emission guidelines.

In 2013, the defendants in the consolidated cases moved to exclude the plaintiffs’ expert evidence of general causation on the basis of flawed scientific methodologies. In 2014, the trial court granted in part and denied in part the defendants’ motion to exclude the plaintiffs’ general causation experts. The defendants filed an interlocutory appeal to the District of Columbia Court of Appeals challenging the standard for evaluating expert scientific evidence. In October 2016, the Court of Appeals issued its decision adopting the standard advocated by the defendants and remanding the cases to the trial court for further proceedings under that standard. The plaintiffs have filed supplemental expert evidence, portions of which were stricken by the court. A hearing on general causation took place in September of 2022. In April of 2023, the court granted defendants’ motion to strike the testimony of plaintiffs’ experts that cell phones cause brain cancer and entered an order excluding all of plaintiffs’ experts from testifying. The parties agreed to a stipulated dismissal of the consolidated cases to allow plaintiffs to appeal the expert testimony order. Plaintiffs appealed the court’s order in August of 2023, and the appeal was argued in January of 2025. A hearing on the status of the stayed cases occurred in December of 2023. In July 2024, the court entered summary judgment in nine of the stayed cases on the grounds that plaintiffs had agreed to be bound by the general causation outcome in the consolidated cases.

Irish Data Protection Commission Matter

In 2018, the Irish Data Protection Commission (“IDPC”) began investigating a complaint against LinkedIn as to whether LinkedIn’s targeted advertising practices violated the recently implemented European Union General Data Protection Regulation (“GDPR”). Microsoft cooperated throughout the period of inquiry. In October 2024, the IDPC provided LinkedIn with a final decision alleging GDPR violations and assessing a fine. In November 2024, LinkedIn appealed the final decision to the Irish courts, and the next hearing is scheduled for May 2025.

Other Contingencies

We also are subject to a variety of other claims and suits that arise from time to time in the ordinary course of our business. Although management currently believes that resolving claims against us, individually or in aggregate, will not have a material adverse impact in our consolidated financial statements, these matters are subject to inherent uncertainties and management’s view of these matters may change in the future.

As of March 31, 2025, we accrued aggregate legal liabilities of $530 million. While we intend to defend these matters vigorously, adverse outcomes that we estimate could reach approximately $800 million in aggregate beyond recorded amounts are reasonably possible. Were unfavorable final outcomes to occur, there exists the possibility of a material adverse impact in our consolidated financial statements for the period in which the effects become reasonably estimable.

NOTE 15 — STOCKHOLDERS' EQUITY

Share Repurchases

On September 14, 2021, our Board of Directors approved a share repurchase program authorizing up to $60.0 billion in share repurchases. This share repurchase program commenced in November 2021, has no expiration date, and may be terminated at any time. As of March 31, 2025, $549 million remained of this $60.0 billion share repurchase program.

On September 16, 2024, our Board of Directors approved a share repurchase program authorizing up to $60.0 billion in share repurchases. This share repurchase program will commence following completion of the program approved on September 14, 2021, has no expiration date, and may be terminated at any time.

We repurchased the following shares of common stock under the share repurchase program:

(In millions)	Shares	Amount	Shares	Amount
Fiscal Year		2025		2024
First Quarter	**7**	**$ 2,800**	11	$ 3,560
Second Quarter	**8**	**3,500**	7	2,800
Third Quarter	**8**	**3,500**	7	2,800
Total	**23**	**$ 9,800**	25	$ 9,160

All repurchases were made using cash resources. All shares repurchased were under the share repurchase program approved on September 14, 2021. The above table excludes shares repurchased to settle employee tax withholding related to the vesting of stock awards of $1.3 billion and $4.1 billion for the three and nine months ended March 31, 2025, respectively, and $1.4 billion and $3.9 billion for the three and nine months ended March 31, 2024, respectively.

Dividends

Our Board of Directors declared the following dividends:

Declaration Date	Record Date	Payment Date	Dividend Per Share	Amount
Fiscal Year 2025				(In millions)
September 16, 2024	**November 21, 2024**	**December 12, 2024**	**$ 0.83**	**$ 6,170**
December 3, 2024	**February 20, 2025**	**March 13, 2025**	**0.83**	**6,169**
March 11, 2025	**May 15, 2025**	**June 12, 2025**	**0.83**	**6,170**
Total			**$ 2.49**	**$ 18,509**
Fiscal Year 2024				
September 19, 2023	November 16, 2023	December 14, 2023	$ 0.75	$ 5,574
November 28, 2023	February 15, 2024	March 14, 2024	0.75	5,573
March 12, 2024	May 16, 2024	June 13, 2024	0.75	5,574
Total			$ 2.25	$ 16,721

The dividend declared on March 11, 2025 was included in other current liabilities as of March 31, 2025.

NOTE 16 — ACCUMULATED OTHER COMPREHENSIVE INCOME (LOSS)

The following table summarizes the changes in accumulated other comprehensive income (loss) by component:

(In millions)	Three Months Ended March 31, 2025	Three Months Ended March 31, 2024	Nine Months Ended March 31, 2025	Nine Months Ended March 31, 2024
Derivatives				
Balance, beginning of period	**$ 21**	$ (9)	**$ (3)**	$ (27)
Unrealized gains (losses), net of tax of **$5**, $(5), **$(1),** and $1	**20**	(19)	**(1)**	3
Reclassification adjustments for (gains) losses included in other income (expense), net	**(50)**	37	**7**	32
Tax expense (benefit) included in provision for income taxes	**10**	(8)	**(2)**	(7)
Amounts reclassified from accumulated other comprehensive loss	**(40)**	29	**5**	25
Net change related to derivatives, net of tax of **$(5)**, $3, **$1**, and $8	**(20)**	10	**4**	28
Balance, end of period	**$ 1**	$ 1	**$ 1**	$ 1
Investments				
Balance, beginning of period	**$ (1,945)**	$ (2,511)	**$ (2,625)**	$ (3,582)
Unrealized gains (losses), net of tax of **$118**, $(56), **$294**, and $221	**445**	(212)	**1,110**	830
Reclassification adjustments for losses included in other income (expense), net	**6**	13	**25**	49
Tax benefit included in provision for income taxes	**(1)**	(3)	**(5)**	(10)
Amounts reclassified from accumulated other comprehensive loss	**5**	10	**20**	39
Net change related to investments, net of tax of **$119,** $(53), **$299**, and $231	**450**	(202)	**1,130**	869
Balance, end of period	**$ (1,495)**	$ (2,713)	**$ (1,495)**	$ (2,713)
Translation Adjustments and Other				
Balance, beginning of period	**$ (3,692)**	$ (2,429)	**$ (2,962)**	$ (2,734)
Translation adjustments and other, net of tax of **$0**, $0, **$0**, and $0	**353**	(294)	**(377)**	11
Balance, end of period	**$ (3,339)**	$ (2,723)	**$ (3,339)**	$ (2,723)
Accumulated other comprehensive loss, end of period	**$ (4,833)**	$ (5,435)	**$ (4,833)**	$ (5,435)

NOTE 17 — SEGMENT INFORMATION AND GEOGRAPHIC DATA

In its operation of the business, management, including our chief operating decision maker, who is also our Chief Executive Officer, reviews certain financial information, including segmented internal profit and loss statements prepared on a basis not consistent with GAAP. During the periods presented, we reported our financial performance based on the following segments: Productivity and Business Processes, Intelligent Cloud, and More Personal Computing.

We have recast certain prior period amounts to conform to the way we internally manage and monitor our business. Refer to Note 1 – Accounting Policies for further information.

Our reportable segments are described below.

Productivity and Business Processes

Our Productivity and Business Processes segment consists of products and services in our portfolio of productivity, communication, and information services, spanning a variety of devices and platforms. This segment primarily comprises:

- Microsoft 365 Commercial products and cloud services, including Microsoft 365 Commercial cloud, comprising Microsoft 365 Commercial, Enterprise Mobility + Security, the cloud portion of Windows Commercial, the per-user portion of Power BI, Exchange, SharePoint, Microsoft Teams, Microsoft 365 Security and Compliance, Microsoft Viva, and Microsoft 365 Copilot; and Microsoft 365 Commercial products, comprising Windows Commercial on-premises and Office licensed on-premises.
- Microsoft 365 Consumer products and cloud services, including Microsoft 365 Consumer subscriptions, Office licensed on-premises, and other consumer services.
- LinkedIn, including Talent Solutions, Marketing Solutions, Premium Subscriptions, and Sales Solutions.
- Dynamics products and cloud services, including Dynamics 365, comprising a set of intelligent, cloud-based applications across ERP, CRM, Power Apps, and Power Automate; and on-premises ERP and CRM applications.

Intelligent Cloud

Our Intelligent Cloud segment consists of our public, private, and hybrid server products and cloud services that can power modern business and developers. This segment primarily comprises:

- Server products and cloud services, including Azure and other cloud services, comprising cloud and AI consumption-based services, GitHub cloud services, Nuance Healthcare cloud services, virtual desktop offerings, and other cloud services; and Server products, comprising SQL Server, Windows Server, Visual Studio, System Center, related Client Access Licenses (“CALs”), and other on-premises offerings.
- Enterprise and partner services, including Enterprise Support Services, Industry Solutions, Nuance professional services, Microsoft Partner Network, and Learning Experience.

More Personal Computing

Our More Personal Computing segment consists of products and services that put customers at the center of the experience with our technology. This segment primarily comprises:

- Windows and Devices, including Windows, comprising Windows OEM licensing (Windows Pro and non-Pro licenses sold through the OEM channel), Windows Internet of Things, and patent licensing; and Devices, comprising Surface, HoloLens, and PC accessories.
- Gaming, including Xbox hardware and Xbox content and services, comprising first- and third-party content (including games and in-game content), Xbox Game Pass and other subscriptions, Xbox Cloud Gaming, advertising, third-party disc royalties, and other cloud services.
- Search and news advertising, comprising Bing (including Copilot), Microsoft News, Microsoft Edge, and third-party affiliates.

Revenue and costs are generally directly attributed to our segments. However, due to the integrated structure of our business, certain revenue recognized and costs incurred by one segment may benefit other segments. Revenue from certain contracts is allocated among the segments based on the relative value of the underlying products and services, which can include allocation based on actual prices charged, prices when sold separately, or estimated costs plus a profit margin. Cost of revenue is allocated in certain cases based on a relative revenue methodology. Operating expenses that are allocated primarily include those relating to marketing of products and services from which multiple segments benefit and are generally allocated based on relative gross margin.

In addition, certain costs are incurred at a corporate level and allocated to our segments. These allocated costs generally include legal, including settlements and fines, information technology, human resources, finance, excise taxes, field selling, shared facilities services, customer service and support, and severance incurred as part of a corporate program. Each allocation is measured differently based on the specific facts and circumstances of the costs being allocated and is generally based on relative gross margin or relative headcount.

Segment revenue and operating income were as follows during the periods presented:

(In millions)	Three Months Ended March 31, 2025	Three Months Ended March 31, 2024	Nine Months Ended March 31, 2025	Nine Months Ended March 31, 2024
Revenue				
Productivity and Business Processes	**$ 29,944**	$ 27,113	**$ 87,698**	$ 78,193
Intelligent Cloud	**26,751**	22,141	**76,387**	63,679
More Personal Computing	**13,371**	12,604	**41,198**	38,523
Total	**$ 70,066**	$ 61,858	**$ 205,283**	$ 180,395
Operating Income				
Productivity and Business Processes	**$ 17,379**	$ 15,143	**$ 50,780**	$ 43,955
Intelligent Cloud	**11,095**	9,515	**32,449**	27,978
More Personal Computing	**3,526**	2,923	**10,976**	9,575
Total	**$ 32,000**	$ 27,581	**$ 94,205**	$ 81,508

No sales to an individual customer or country other than the United States accounted for more than 10% of revenue for the three or nine months ended March 31, 2025 or 2024. Revenue, classified by the major geographic areas in which our customers were located, was as follows:

(In millions)	Three Months Ended March 31, 2025	Three Months Ended March 31, 2024	Nine Months Ended March 31, 2025	Nine Months Ended March 31, 2024
United States [a]	**$ 36,084**	$ 31,437	**$ 105,534**	$ 92,544
Other countries	**33,982**	30,421	**99,749**	87,851
Total	**$ 70,066**	$ 61,858	**$ 205,283**	$ 180,395

(a) *Includes billings to OEMs and certain multinational organizations because of the nature of these businesses and the impracticability of determining the geographic source of the revenue.*

Revenue, classified by significant product and service offerings, was as follows:

(In millions)	Three Months Ended March 31, 2025	Three Months Ended March 31, 2024	Nine Months Ended March 31, 2025	Nine Months Ended March 31, 2024
Server products and cloud services	**$ 24,761**	$ 20,266	**$ 70,557**	$ 57,925
Microsoft 365 Commercial products and cloud services	**21,883**	19,712	**63,449**	56,077
Gaming	**5,721**	5,451	**17,923**	16,481
LinkedIn	**4,311**	4,013	**13,190**	12,121
Windows and Devices	**4,144**	4,098	**12,985**	12,801
Search and news advertising	**3,504**	3,055	**10,287**	9,241
Enterprise and partner services	**1,946**	1,861	**5,766**	5,722
Dynamics products and cloud services	**1,929**	1,740	**5,691**	5,025
Microsoft 365 Consumer products and cloud services	**1,821**	1,648	**5,369**	4,970
Other	**46**	14	**66**	32
Total	**$ 70,066**	$ 61,858	**$ 205,283**	$ 180,395

Our Microsoft Cloud revenue, which includes Microsoft 365 Commercial cloud, Azure and other cloud services, the commercial portion of LinkedIn, and Dynamics 365, was $42.4 billion and $122.2 billion for the three and nine months ended March 31, 2025, respectively, and $35.2 billion and $100.8 billion for the three and nine months ended March 31, 2024, respectively. These amounts are included in Microsoft 365 Commercial products and cloud services, Server products and cloud services, LinkedIn, and Dynamics products and cloud services in the table above.

Assets are not allocated to segments for internal reporting presentations. A portion of amortization and depreciation is included with various other costs in an overhead allocation to each segment. It is impracticable for us to separately identify the amount of amortization and depreciation by segment that is included in the measure of segment profit or loss.

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and the Board of Directors of Microsoft Corporation

Results of Review of Interim Financial Information

We have reviewed the accompanying consolidated balance sheet of Microsoft Corporation and subsidiaries (the "Company") as of March 31, 2025, the related consolidated statements of income, comprehensive income, cash flows, and stockholders’ equity for the three-month and nine-month periods ended March 31, 2025 and 2024, and the related notes (collectively referred to as the “interim financial information”). Based on our reviews, we are not aware of any material modifications that should be made to the accompanying interim financial information for it to be in conformity with accounting principles generally accepted in the United States of America.

We have previously audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheet of the Company as of June 30, 2024, and the related consolidated statements of income, comprehensive income, cash flows, and stockholders’ equity for the year then ended (not presented herein); and in our report dated July 30, 2024 (December 3, 2024 as to the effects of the retrospective adjustments in Notes 1, 8, 9, 13, and 19), we expressed an unqualified opinion on those consolidated financial statements. In our opinion, the information set forth in the accompanying consolidated balance sheet as of June 30, 2024, is fairly stated, in all material respects, in relation to the consolidated balance sheet from which it has been derived.

Basis for Review Results

This interim financial information is the responsibility of the Company's management. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our reviews in accordance with standards of the PCAOB. A review of interim financial information consists principally of applying analytical procedures and making inquiries of persons responsible for financial and accounting matters. It is substantially less in scope than an audit conducted in accordance with the standards of the PCAOB, the objective of which is the expression of an opinion regarding the financial statements taken as a whole. Accordingly, we do not express such an opinion.

/S/ DELOITTE & TOUCHE LLP

Seattle, Washington
April 30, 2025

ITEM 2. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

Note About Forward-Looking Statements

This report includes estimates, projections, statements relating to our business plans, objectives, and expected operating results that are "forward-looking statements" within the meaning of the Private Securities Litigation Reform Act of 1995, Section 27A of the Securities Act of 1933, and Section 21E of the Securities Exchange Act of 1934. Forward-looking statements may appear throughout this report, including the following sections: "Management's Discussion and Analysis of Financial Condition and Results of Operations" and "Risk Factors" (Part II, Item 1A of this Form 10-Q). These forward-looking statements generally are identified by the words "believe," "project," "expect," "anticipate," "estimate," "intend," "strategy," "future," "opportunity," "plan," "may," "should," "will," "would," "will be," "will continue," "will likely result," and similar expressions. Forward-looking statements are based on current expectations and assumptions that are subject to risks and uncertainties that may cause actual results to differ materially. We describe risks and uncertainties that could cause actual results and events to differ materially in "Management's Discussion and Analysis of Financial Condition and Results of Operations," "Quantitative and Qualitative Disclosures About Market Risk" (Part I, Item 3 of this Form 10-Q), and "Risk Factors". We undertake no obligation to update or revise publicly any forward-looking statements, whether because of new information, future events, or otherwise.

The following Management's Discussion and Analysis of Financial Condition and Results of Operations ("MD&A") is intended to help the reader understand the results of operations and financial condition of Microsoft Corporation. MD&A is provided as a supplement to, and should be read in conjunction with, our Annual Report on Form 10-K for the year ended June 30, 2024, our Form 8-K filed on December 3, 2024, and our consolidated financial statements and the accompanying Notes to Financial Statements (Part I, Item 1 of this Form 10-Q).

OVERVIEW

Microsoft is a technology company committed to making digital technology and artificial intelligence ("AI") available broadly and doing so responsibly, with a mission to empower every person and every organization on the planet to achieve more. We create platforms and tools, powered by AI, that deliver innovative solutions that meet the evolving needs of our customers.

We generate revenue by offering a wide range of cloud-based solutions, content, and other services to people and businesses; licensing and supporting an array of software products; delivering relevant online advertising to a global audience; and designing and selling devices. Our most significant expenses are related to compensating employees; supporting and investing in our cloud-based services, including datacenter operations; designing, manufacturing, marketing, and selling our other products and services; and income taxes.

Highlights from the third quarter of fiscal year 2025 compared with the third quarter of fiscal year 2024 included:

- Microsoft Cloud revenue increased 20% to $42.4 billion.
- Microsoft 365 Commercial products and cloud services revenue increased 11% driven by Microsoft 365 Commercial cloud revenue growth of 12%.
- Microsoft 365 Consumer products and cloud services revenue increased 10% driven by Microsoft 365 Consumer cloud revenue growth of 10%.
- LinkedIn revenue increased 7%.
- Dynamics products and cloud services revenue increased 11% driven by Dynamics 365 revenue growth of 16%.
- Server products and cloud services revenue increased 22% driven by Azure and other cloud services revenue growth of 33%.
- Windows OEM and Devices revenue increased 3%.
- Xbox content and services revenue increased 8%.
- Search and news advertising revenue excluding traffic acquisition costs increased 21%.

Industry Trends

Our industry is dynamic and highly competitive, with frequent changes in both technologies and business models. Each industry shift is an opportunity to conceive new products, new technologies, or new ideas that can further transform the industry and our business. At Microsoft, we push the boundaries of what is possible through a broad range of research and development activities that seek to identify and address the changing demands of customers and users, industry trends, and competitive forces.

Economic Conditions, Challenges, and Risks

The markets for software, devices, and cloud-based services are dynamic and highly competitive. Our competitors are developing new software and devices, while also deploying competing cloud-based services for consumers and businesses. The devices and form factors customers prefer evolve rapidly, influencing how users access services in the cloud and, in some cases, the user's choice of which suite of cloud-based services to use. Aggregate demand for our software, services, and devices is also correlated to global macroeconomic and geopolitical factors, which remain dynamic. We must continue to evolve and adapt over an extended time in pace with this changing environment.

The investments we are making in cloud and AI infrastructure and devices will continue to increase our operating costs and may decrease our operating margins. We continue to identify and evaluate opportunities to expand our datacenter locations and increase our server capacity to meet the evolving needs of our customers, particularly given the growing demand for AI services. Our datacenters depend on the availability of permitted and buildable land, predictable energy, networking supplies, and servers, including graphics processing units ("GPUs") and other components. Our devices are primarily manufactured by third-party contract manufacturers. For the majority of our products, we have the ability to use other manufacturers if a current vendor becomes unavailable or unable to meet our requirements. However, some of our products contain certain components for which there are very few qualified suppliers. Extended disruptions at these suppliers could impact our ability to manufacture devices on time to meet consumer demand.

Our success is highly dependent on our ability to attract and retain qualified employees. We hire a mix of university and industry talent worldwide. We compete for talented individuals globally by offering an exceptional working environment, broad customer reach, scale in resources, the ability to grow one's career across many different products and businesses, and competitive compensation and benefits.

Our international operations provide a significant portion of our total revenue and expenses. Many of these revenue and expenses are denominated in currencies other than the U.S. dollar. As a result, changes in foreign exchange rates may significantly affect revenue and expenses. Fluctuations in the U.S. dollar relative to certain foreign currencies decreased reported revenue and expenses from our international operations for the three months ended March 31, 2025, and did not have a material impact on reported revenue and expenses from our international operations for the nine months ended March 31, 2025.

Further, global, regional, and local economic developments and changes in global trade policies such as restrictions on international trade, including tariffs and other controls on imports or exports, could result in increased supply chain challenges, cost volatility, and consumer and economic uncertainty which may adversely affect our results of operations.

Refer to Risk Factors (Part II, Item 1A of this Form 10-Q) for a discussion of these factors and other risks.

Seasonality

Our revenue fluctuates quarterly and is generally higher in the fourth quarter of our fiscal year. Fourth quarter revenue is driven by a higher volume of multi-year contracts executed during the period.

Reportable Segments

We report our financial performance based on the following segments: Productivity and Business Processes, Intelligent Cloud, and More Personal Computing. The segment amounts included in MD&A are presented on a basis consistent with our internal management reporting.

In August 2024, we announced changes to the composition of our segments. These changes align our segments with how we currently manage our business, most notably bringing the commercial components of Microsoft 365 together in the Productivity and Business Processes segment. Beginning in fiscal year 2025, the information that our chief operating decision maker is regularly provided and reviews for purposes of allocating resources and assessing performance reflects these segment changes. Prior period segment information has been recast to conform to the way we internally manage and monitor our business during fiscal year 2025.

Additional information on our reportable segments is contained in Note 17 – Segment Information and Geographic Data of the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q).

Metrics

We use metrics in assessing the performance of our business and to make informed decisions regarding the allocation of resources. We disclose metrics to enable investors to evaluate progress against our ambitions, provide transparency into performance trends, and reflect the continued evolution of our products and services. Our commercial and other business metrics are fundamentally connected based on how customers use our products and services. The metrics are disclosed in the MD&A or the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q). Financial metrics are calculated based on financial results prepared in accordance with accounting principles generally accepted in the United States of America ("GAAP"), and growth comparisons relate to the corresponding period of last fiscal year.

In the first quarter of fiscal year 2025, we made updates to our metrics in connection with the segment changes described above. These changes align our metrics with how we manage and monitor certain businesses. The key change was bringing the commercial components of Microsoft 365 together and creating a new Microsoft 365 Commercial cloud revenue growth metric. Other changes include combining Windows OEM and Devices into a single revenue growth metric that brings revenue from PC market-driven businesses together, as well as elevating our cloud revenue growth metrics to align to our strategic focus on cloud growth.

Commercial

Our commercial business primarily consists of Server products and cloud services, Microsoft 365 Commercial products and cloud services, the commercial portion of LinkedIn, Enterprise and partner services, and Dynamics products and cloud services. Our commercial metrics allow management and investors to assess the overall health of our commercial business and include leading indicators of future performance.

Commercial remaining performance obligation	Commercial portion of revenue allocated to remaining performance obligations, which includes unearned revenue and amounts that will be invoiced and recognized as revenue in future periods
Microsoft Cloud revenue and revenue growth	Revenue from Microsoft 365 Commercial cloud, Azure and other cloud services, the commercial portion of LinkedIn, and Dynamics 365
Microsoft Cloud gross margin percentage	Gross margin percentage for our Microsoft Cloud business

Productivity and Business Processes and Intelligent Cloud

Metrics related to our Productivity and Business Processes and Intelligent Cloud segments assess the health of our core businesses within these segments. The metrics primarily reflect growth across our cloud services.

Microsoft 365 Commercial cloud revenue growth	Revenue from Microsoft 365 Commercial subscriptions, comprising Microsoft 365 Commercial, Enterprise Mobility + Security, the cloud portion of Windows Commercial, the per-user portion of Power BI, Exchange, SharePoint, Microsoft Teams, Microsoft 365 Security and Compliance, Microsoft Viva, and Microsoft 365 Copilot
Microsoft 365 Commercial seat growth	The number of Microsoft 365 Commercial seats at end of period where seats are paid users covered by a Microsoft 365 Commercial subscription
Microsoft 365 Consumer cloud revenue growth	Revenue from Microsoft 365 Consumer subscriptions and other consumer services
Microsoft 365 Consumer subscribers	The number of Microsoft 365 Consumer subscribers at end of period
LinkedIn revenue growth	Revenue from LinkedIn, including Talent Solutions, Marketing Solutions, Premium Subscriptions, and Sales Solutions
Dynamics 365 revenue growth	Revenue from Dynamics 365, including a set of intelligent, cloud-based applications across ERP, CRM, Power Apps, and Power Automate
Azure and other cloud services revenue growth	Revenue from Azure and other cloud services, including cloud and AI consumption-based services, GitHub cloud services, Nuance Healthcare cloud services, virtual desktop offerings, and other cloud services

More Personal Computing

Metrics related to our More Personal Computing segment assess the performance of our key consumer businesses.

Windows OEM and Devices revenue growth	Revenue from sales of Windows Pro and non-Pro licenses sold through the OEM channel and sales of first-party Devices, including Surface, HoloLens, and PC accessories
Xbox content and services revenue growth	Revenue from Xbox content and services, comprising first- and third-party content (including games and in-game content), Xbox Game Pass and other subscriptions, Xbox Cloud Gaming, advertising, third-party disc royalties, and other cloud services
Search and news advertising revenue (ex TAC) growth	Revenue from search and news advertising excluding traffic acquisition costs ("TAC") paid to Bing Ads network publishers and news partners

SUMMARY RESULTS OF OPERATIONS

(In millions, except percentages and per share amounts)	**Three Months Ended March 31,**		**Percentage Change**	**Nine Months Ended March 31,**		**Percentage Change**
	2025	**2024**		**2025**	**2024**	
Revenue	**$ 70,066**	$ 61,858	13%	**$ 205,283**	$ 180,395	14%
Gross margin	**48,147**	43,353	11%	**141,466**	125,965	12%
Operating income	**32,000**	27,581	16%	**94,205**	81,508	16%
Net income	**25,824**	21,939	18%	**74,599**	66,100	13%
Diluted earnings per share	**3.46**	2.94	18%	**9.99**	8.85	13%

Three Months Ended March 31, 2025 Compared with Three Months Ended March 31, 2024

Revenue increased $8.2 billion or 13% with growth across each of our segments. Intelligent Cloud revenue increased driven by Azure. Productivity and Business Processes revenue increased driven by Microsoft 365 Commercial cloud. More Personal Computing revenue increased driven by Search and news advertising.

Cost of revenue increased $3.4 billion or 18% driven by growth in Microsoft Cloud.

Gross margin increased $4.8 billion or 11% with growth across each of our segments.

- Gross margin percentage decreased driven by Intelligent Cloud.
- Microsoft Cloud gross margin percentage decreased to 69% driven by the impact of scaling our AI infrastructure.

Operating expenses increased $375 million or 2% driven by investments in cloud and AI engineering.

Operating income increased $4.4 billion or 16% with growth across each of our segments.

Revenue, gross margin, and operating income included an unfavorable foreign currency impact of 2%, 2%, and 3%, respectively. Cost of goods sold included a favorable foreign currency impact of 2%.

Nine Months Ended March 31, 2025 Compared with Nine Months Ended March 31, 2024

Revenue increased $24.9 billion or 14% with growth across each of our segments. Intelligent Cloud revenue increased driven by Azure. Productivity and Business Processes revenue increased driven by Microsoft 365 Commercial cloud. More Personal Computing revenue increased driven by Gaming and Search and news advertising.

Cost of revenue increased $9.4 billion or 17% driven by growth in Microsoft Cloud.

Gross margin increased $15.5 billion or 12% with growth across each of our segments.

- Gross margin percentage decreased slightly driven by Intelligent Cloud, offset in part by More Personal Computing.
- Microsoft Cloud gross margin percentage decreased to 70% driven by the impact of scaling our AI infrastructure.

Operating expenses increased $2.8 billion or 6% driven by investments in cloud and AI engineering and the impact of the Activision Blizzard acquisition.

Operating income increased $12.7 billion or 16% with growth across each of our segments.

SEGMENT RESULTS OF OPERATIONS

(In millions, except percentages)	Three Months Ended March 31, 2025	Three Months Ended March 31, 2024	Percentage Change	Nine Months Ended March 31, 2025	Nine Months Ended March 31, 2024	Percentage Change
Revenue						
Productivity and Business Processes	**$ 29,944**	$ 27,113	10%	**$ 87,698**	$ 78,193	12%
Intelligent Cloud	**26,751**	22,141	21%	**76,387**	63,679	20%
More Personal Computing	**13,371**	12,604	6%	**41,198**	38,523	7%
Total	**$ 70,066**	$ 61,858	13%	**$ 205,283**	$ 180,395	14%
Operating Income						
Productivity and Business Processes	**$ 17,379**	$ 15,143	15%	**$ 50,780**	$ 43,955	16%
Intelligent Cloud	**11,095**	9,515	17%	**32,449**	27,978	16%
More Personal Computing	**3,526**	2,923	21%	**10,976**	9,575	15%
Total	**$ 32,000**	$ 27,581	16%	**$ 94,205**	$ 81,508	16%

Reportable Segments

Three Months Ended March 31, 2025 Compared with Three Months Ended March 31, 2024

Productivity and Business Processes

Revenue increased $2.8 billion or 10%.

- Microsoft 365 Commercial products and cloud services revenue increased $2.2 billion or 11%. Microsoft 365 Commercial cloud revenue grew 12% with Microsoft 365 Commercial seat growth of 7% driven by small and medium business and frontline worker offerings, as well as growth in revenue per user. Microsoft 365 Commercial products revenue grew 5% driven by the Windows Commercial on-premises components of Microsoft 365 suite sales and an increase in Office transactional purchasing.
- Microsoft 365 Consumer products and cloud services revenue increased $173 million or 10%. Microsoft 365 Consumer cloud revenue grew 10% driven by Microsoft 365 Consumer subscriber growth of 9% to 87.7 million, as well as growth in revenue per user from the price increase announced in January 2025, offset in part by continued mix shift to Microsoft 365 Basic.
- LinkedIn revenue increased $298 million or 7% with growth across all lines of business.
- Dynamics products and cloud services revenue increased $189 million or 11% driven by growth in Dynamics 365, offset in part by a decline in Dynamics on-premises products. Dynamics 365 revenue grew 16% with growth across all workloads.

Operating income increased $2.2 billion or 15%.

- Gross margin increased $2.3 billion or 10% driven by growth in Microsoft 365 Commercial cloud. Gross margin percentage was relatively unchanged inclusive of the impact of scaling our AI infrastructure.
- Operating expenses increased $53 million or 1% primarily driven by investments in cloud and AI engineering.

Revenue, gross margin, and operating income each included an unfavorable foreign currency impact of 3%.

Intelligent Cloud

Revenue increased $4.6 billion or 21%.

- Server products and cloud services revenue increased $4.5 billion or 22% driven by Azure and other cloud services. Azure and other cloud services revenue grew 33% driven by demand for our portfolio of services, including 16 points from our AI services. Server products revenue decreased 6% primarily driven by a decrease in transactional purchasing with continued customer shift to cloud offerings.
- Enterprise and partner services revenue increased $85 million or 5% driven by growth in Enterprise Support Services, offset in part by a decline in Industry Solutions.

Operating income increased $1.6 billion or 17%.

- Gross margin increased $1.9 billion or 13% driven by growth in Azure. Gross margin percentage decreased driven by the impact of scaling our AI infrastructure.
- Operating expenses increased $302 million or 6% driven by investments in cloud and AI engineering.

More Personal Computing

Revenue increased $767 million or 6%.

- Windows and Devices revenue increased $46 million or 1%. Windows OEM and Devices revenue increased 3% driven by growth in Windows OEM, with inventory levels remaining elevated due to tariff uncertainty.
- Gaming revenue increased $270 million or 5% driven by growth in Xbox content and services. Xbox content and services revenue increased 8% driven by growth in Xbox Game Pass, Call of Duty, and Minecraft. Xbox hardware revenue decreased 6%.

- Search and news advertising revenue increased $449 million or 15%. Search and news advertising revenue excluding traffic acquisition costs increased 21% driven by higher revenue per search and higher search volume.

Operating income increased $603 million or 21%.

- Gross margin increased $623 million or 9% driven by growth in Search and news advertising and Gaming. Gross margin percentage increased driven by improvement in Search and news advertising and Gaming.
- Operating expenses increased $20 million or 1%.

Gross margin and operating income both included an unfavorable foreign currency impact of 2%.

Nine Months Ended March 31, 2025 Compared with Nine Months Ended March 31, 2024

Productivity and Business Processes

Revenue increased $9.5 billion or 12%.

- Microsoft 365 Commercial products and cloud services revenue increased $7.4 billion or 13%. Microsoft 365 Commercial cloud revenue grew 14% driven by seat growth and growth in revenue per user. Microsoft 365 Commercial products revenue grew 7% driven by the Windows Commercial on-premises components of Microsoft 365 suite sales and an increase in Office transactional purchasing.
- Microsoft 365 Consumer products and cloud services revenue increased $399 million or 8%. Microsoft 365 Consumer cloud revenue grew 8% with continued growth in Microsoft 365 Consumer subscribers.
- LinkedIn revenue increased $1.1 billion or 9% with growth across all lines of business.
- Dynamics products and cloud services revenue increased $666 million or 13% driven by growth in Dynamics 365, offset in part by a decline in Dynamics on-premises products. Dynamics 365 revenue grew 18% with growth across all workloads.

Operating income increased $6.8 billion or 16%.

- Gross margin increased $7.4 billion or 12% driven by growth in Microsoft 365 Commercial cloud. Gross margin percentage decreased slightly driven by the impact of scaling our AI infrastructure.
- Operating expenses increased $544 million or 3% driven by investments in cloud and AI engineering and commercial sales.

Intelligent Cloud

Revenue increased $12.7 billion or 20%.

- Server products and cloud services revenue increased $12.6 billion or 22% driven by Azure and other cloud services. Azure and other cloud services revenue grew 32% driven by demand for our portfolio of services, including 14 points from our AI services. Server products revenue decreased 4% driven by a decrease in transactional purchasing.
- Enterprise and partner services revenue increased slightly driven by growth in Enterprise Support Services, offset in part by a decline in Industry Solutions.

Operating income increased $4.5 billion or 16%.

- Gross margin increased $5.6 billion or 13% driven by growth in Azure. Gross margin percentage decreased driven by the impact of scaling our AI infrastructure.
- Operating expenses increased $1.2 billion or 8% driven by investments in cloud and AI engineering.

More Personal Computing

Revenue increased $2.7 billion or 7%.

- Windows and Devices revenue increased $184 million or 1%. Windows OEM and Devices revenue increased 3% driven by growth in Windows OEM, offset in part by a decline in Devices.

- Gaming revenue increased $1.4 billion or 9% driven by growth in Xbox content and services, offset in part by a decline in Xbox hardware. Xbox content and services revenue increased 18% driven by the impact of the Activision Blizzard acquisition and Xbox Game Pass. Xbox hardware revenue decreased 26% driven by lower volume of consoles sold.
- Search and news advertising revenue increased $1.0 billion or 11%. Search and news advertising revenue excluding traffic acquisition costs increased 20% driven by higher search volume and higher revenue per search.

Operating income increased $1.4 billion or 15%.

- Gross margin increased $2.5 billion or 13% driven by growth in Gaming, including the impact of the Activision Blizzard acquisition, and Search and news advertising. Gross margin percentage increased driven by sales mix shift to higher margin businesses and improvement in Search and news advertising and Gaming.
- Operating expenses increased $1.1 billion or 11% driven by the impact of the Activision Blizzard acquisition.

OPERATING EXPENSES

Research and Development

(In millions, except percentages)	Three Months Ended March 31, 2025	2024	Percentage Change	Nine Months Ended March 31, 2025	2024	Percentage Change
Research and development	**$ 8,198**	$ 7,653	7%	**$ 23,659**	$ 21,454	10%
As a percent of revenue	**12%**	12%	0ppt	**12%**	12%	0ppt

Research and development expenses include payroll, employee benefits, stock-based compensation expense, and other headcount-related expenses associated with product development. Research and development expenses also include technology development costs, including AI training and other infrastructure costs, third-party development and programming costs, and the amortization of purchased software code and services content.

Three Months Ended March 31, 2025 Compared with Three Months Ended March 31, 2024

Research and development expenses increased $545 million or 7% driven by investments in cloud and AI engineering.

Nine Months Ended March 31, 2025 Compared with Nine Months Ended March 31, 2024

Research and development expenses increased $2.2 billion or 10% driven by investments in cloud and AI engineering and the impact of the Activision Blizzard acquisition.

Sales and Marketing

(In millions, except percentages)	Three Months Ended March 31, 2025	2024	Percentage Change	Nine Months Ended March 31, 2025	2024	Percentage Change
Sales and marketing	**$ 6,212**	$ 6,207	0%	**$ 18,369**	$ 17,640	4%
As a percent of revenue	**9%**	10%	(1)ppt	**9%**	10%	(1)ppt

Sales and marketing expenses include payroll, employee benefits, stock-based compensation expense, and other headcount-related expenses associated with sales and marketing personnel, and the costs of advertising, promotions, trade shows, seminars, and other programs.

Three Months Ended March 31, 2025 Compared with Three Months Ended March 31, 2024

Sales and marketing expenses were relatively unchanged.

Nine Months Ended March 31, 2025 Compared with Nine Months Ended March 31, 2024

Sales and marketing expenses increased $729 million or 4% driven by the impact of the Activision Blizzard acquisition and investments in commercial sales.

General and Administrative

(In millions, except percentages)	Three Months Ended March 31, 2025	Three Months Ended March 31, 2024	Percentage Change	Nine Months Ended March 31, 2025	Nine Months Ended March 31, 2024	Percentage Change
General and administrative	**$ 1,737**	$ 1,912	(9)%	**$ 5,233**	$ 5,363	(2)%
As a percent of revenue	**2%**	3%	(1)ppt	**3%**	3%	0ppt

General and administrative expenses include payroll, employee benefits, stock-based compensation expense, employee severance expense incurred as part of a corporate program, and other headcount-related expenses associated with finance, legal, facilities, certain human resources and other administrative personnel, certain taxes, and legal and other administrative fees.

Three Months Ended March 31, 2025 Compared with Three Months Ended March 31, 2024

General and administrative expenses decreased $175 million or 9% primarily driven by Gaming.

Nine Months Ended March 31, 2025 Compared with Nine Months Ended March 31, 2024

General and administrative expenses decreased $130 million or 2% primarily driven by Gaming, offset in part by investments in corporate functions.

OTHER INCOME (EXPENSE), NET

The components of other income (expense), net were as follows:

(In millions)	Three Months Ended March 31, 2025	Three Months Ended March 31, 2024	Nine Months Ended March 31, 2025	Nine Months Ended March 31, 2024
Interest and dividends income	**$ 597**	$ 619	**$ 1,878**	$ 2,519
Interest expense	**(594)**	(800)	**(1,770)**	(2,234)
Net recognized gains (losses) on investments	**111**	(25)	**(286)**	(63)
Net gains (losses) on derivatives	**187**	(24)	**(267)**	(198)
Net gains (losses) on foreign currency remeasurements	**89**	(138)	**112**	(203)
Other, net	**(1,013)**	(486)	**(2,861)**	(792)
Total	**$ (623)**	$ (854)	**$ (3,194)**	$ (971)

We use derivative instruments to manage risks related to foreign currencies, interest rates, equity prices, and credit; to enhance investment returns; and to facilitate portfolio diversification. Gains and losses from changes in fair values of derivatives that are not designated as hedging instruments are primarily recognized in other income (expense), net.

Three Months Ended March 31, 2025 Compared with Three Months Ended March 31, 2024

Interest and dividends income decreased primarily due to lower portfolio balances. Interest expense decreased primarily due to maturities of commercial paper, offset in part by higher finance lease interest expense. Net recognized gains on investments increased primarily due to higher gains on equity investments in the current period. Net gains on derivatives increased primarily due to gains on equity derivatives in the current period as compared to losses in the prior period. Other, net primarily reflects net recognized losses on equity method investments, including OpenAI.

Nine Months Ended March 31, 2025 Compared with Nine Months Ended March 31, 2024

Interest and dividends income decreased primarily due to lower portfolio balances. Interest expense decreased primarily due to maturities of commercial paper, offset in part by higher finance lease interest expense. Net recognized losses on investments increased primarily due to higher impairments, offset in part by higher gains on equity investments in the current period. Net losses on derivatives increased primarily due to higher losses on equity derivatives in the current period. Other, net primarily reflects net recognized losses on equity method investments, including OpenAI.

INCOME TAXES

Effective Tax Rate

Our effective tax rate was 18% for both the three months ended March 31, 2025 and 2024, and 18% for both the nine months ended March 31, 2025 and 2024. Our effective tax rate for the three months ended March 31, 2025 was primarily impacted by changes in the mix of our earnings and tax expenses between the U.S. and foreign countries. Our effective tax rate for the nine months ended March 31, 2025 was primarily impacted by tax benefits from tax law changes in the prior fiscal year, including the delay of the effective date of final foreign tax credit regulations, and changes in the mix of our earnings and tax expenses between the U.S. and foreign countries.

Our effective tax rate was lower than the U.S. federal statutory rate for the three and nine months ended March 31, 2025, primarily due to earnings taxed at lower rates in foreign jurisdictions resulting from producing and distributing our products and services through our foreign regional operations center in Ireland.

The Organisation for Economic Co-operation and Development (“OECD”) published its model rules “Tax Challenges Arising From the Digitalisation of the Economy - Global Anti-Base Erosion Model Rules (Pillar Two)” which established a global minimum corporate tax rate of 15% for certain multinational enterprises. Many countries have implemented or are in the process of implementing the Pillar Two legislation, which applies to Microsoft beginning in fiscal year 2025. While we do not currently estimate a material impact to our consolidated financial statements, we continue to monitor the impact as countries implement legislation and the OECD provides additional guidance.

Uncertain Tax Positions

We remain under audit by the IRS for tax years 2014 to 2017. With respect to the audit for tax years 2004 to 2013, on September 26, 2023, we received Notices of Proposed Adjustment (“NOPAs”) from the IRS. The primary issues in the NOPAs relate to intercompany transfer pricing. In the NOPAs, the IRS is seeking an additional tax payment of $28.9 billion plus penalties and interest. As of March 31, 2025, we believe our allowances for income tax contingencies are adequate. We disagree with the proposed adjustments and will vigorously contest the NOPAs through the IRS’s administrative appeals office and, if necessary, judicial proceedings. We do not expect a final resolution of these issues in the next 12 months. Based on the information currently available, we do not anticipate a significant increase or decrease to our income tax contingencies for these issues within the next 12 months.

We are subject to income tax in many jurisdictions outside the U.S. Our operations in certain jurisdictions remain subject to examination for tax years 1996 to 2024, some of which are currently under audit by local tax authorities. The resolution of each of these audits is not expected to be material to our consolidated financial statements.

LIQUIDITY AND CAPITAL RESOURCES

We expect existing cash, cash equivalents, short-term investments, cash flows from operations, and access to capital markets to continue to be sufficient to fund our operating activities and cash commitments for investing and financing activities, such as dividends, share repurchases, debt maturities, material capital expenditures, and the transition tax related to the Tax Cuts and Jobs Act (“TCJA”), for at least the next 12 months and thereafter for the foreseeable future.

Cash, Cash Equivalents, and Investments

Cash, cash equivalents, and short-term investments totaled $79.6 billion and $75.5 billion as of March 31, 2025 and June 30, 2024, respectively. Equity and other investments were $16.0 billion and $14.6 billion as of March 31, 2025 and June 30, 2024, respectively. Our short-term investments are primarily intended to facilitate liquidity and capital preservation. They consist predominantly of highly liquid investment-grade fixed-income securities, diversified among industries and individual issuers. The investments are predominantly U.S. dollar-denominated securities, but also include foreign currency-denominated securities to diversify risk. Our fixed-income investments are exposed to interest rate risk and credit risk. The credit risk and average maturity of our fixed-income portfolio are managed to achieve economic returns that correlate to certain fixed-income indices. The settlement risk related to these investments is insignificant given that the short-term investments held are primarily highly liquid investment-grade fixed-income securities.

Valuation

In general, and where applicable, we use quoted prices in active markets for identical assets or liabilities to determine the fair value of our financial instruments. This pricing methodology applies to our Level 1 investments, such as U.S. government securities, common and preferred stock, and mutual funds. If quoted prices in active markets for identical assets or liabilities are not available to determine fair value, then we use quoted prices for similar assets and liabilities or inputs other than the quoted prices that are observable either directly or indirectly. This pricing methodology applies to our Level 2 investments, such as commercial paper, certificates of deposit, U.S. agency securities, foreign government bonds, mortgage- and asset-backed securities, corporate notes and bonds, and municipal securities. Level 3 investments are valued using internally-developed models with unobservable inputs. Assets and liabilities measured at fair value on a recurring basis using unobservable inputs are an immaterial portion of our portfolio.

A majority of our investments are priced by pricing vendors and are generally Level 1 or Level 2 investments as these vendors either provide a quoted market price in an active market or use observable inputs for their pricing without applying significant adjustments. Broker pricing is used mainly when a quoted price is not available, the investment is not priced by our pricing vendors, or when a broker price is more reflective of fair values in the market in which the investment trades. Our broker-priced investments are generally classified as Level 2 investments because the broker prices these investments based on similar assets without applying significant adjustments. In addition, all our broker-priced investments have a sufficient level of trading volume to demonstrate that the fair values used are appropriate for these investments. Our fair value processes include controls that are designed to ensure appropriate fair values are recorded. These controls include model validation, review of key model inputs, analysis of period-over-period fluctuations, and independent recalculation of prices where appropriate.

Cash Flows

Cash from operations increased $12.2 billion to $93.5 billion for the nine months ended March 31, 2025, primarily due to an increase in cash received from customers, offset in part by an increase in cash paid to suppliers and employees and cash used to pay income taxes. Cash used in financing increased $26.7 billion to $40.9 billion for the nine months ended March 31, 2025, primarily due to $9.0 billion in cash used for repayments of debt, net of proceeds in the current period compared to $14.6 billion in proceeds from the issuance of debt, net of repayments in the prior period. Cash used in investing decreased $40.1 billion to $42.0 billion for the nine months ended March 31, 2025, primarily due to a $63.6 billion decrease in cash used for acquisitions of companies, net of cash acquired and divestitures, and purchases of intangible and other assets, offset in part by a $16.9 billion increase in additions to property and equipment and a $7.2 billion decrease in cash from net investment purchases, sales, and maturities.

Debt Proceeds

We issue debt to take advantage of favorable pricing and liquidity in the debt markets, reflecting our credit rating. The proceeds of these issuances were or will be used for general corporate purposes, which may include, among other things, funding for working capital, capital expenditures, repurchases of capital stock, acquisitions, and repayment of existing debt. Refer to Note 10 – Debt of the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q) for further discussion.

Unearned Revenue

Unearned revenue comprises mainly unearned revenue related to volume licensing programs, which may include Software Assurance (“SA”) and cloud services. Unearned revenue is generally invoiced annually at the beginning of each contract period for multi-year agreements and recognized ratably over the coverage period. Unearned revenue also includes payments for other offerings for which we have been paid in advance and earn the revenue when we transfer control of the product or service.

The following table outlines the expected future recognition of unearned revenue as of March 31, 2025:

(In millions)	
Three Months Ending	
June 30, 2025	$ **21,566**
September 30, 2025	**11,722**
December 31, 2025	**8,394**
March 31, 2026	**2,954**
Thereafter	**2,840**
Total	$ **47,476**

If our customers choose to license cloud-based versions of our products and services rather than licensing transaction-based products and services, the associated revenue will shift from being recognized at the time of the transaction to being recognized over the subscription period or upon consumption, as applicable. Refer to Note 12 – Unearned Revenue of the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q) for further discussion.

Material Cash Requirements and Other Obligations

Income Taxes

As a result of the TCJA, we are required to pay a one-time transition tax on deferred foreign income not previously subject to U.S. income tax. Under the TCJA, the transition tax is payable in interest-free installments over eight years, with 8% due in each of the first five years, 15% in year six, 20% in year seven, and 25% in year eight. As of March 31, 2025, our eighth transition tax installment of $4.4 billion is short-term and payable in the first quarter of fiscal year 2026.

Share Repurchases

For the nine months ended March 31, 2025 and 2024, we repurchased 23 million shares and 25 million shares of our common stock for $9.8 billion and $9.2 billion, respectively, through our share repurchase program. All repurchases were made using cash resources. As of March 31, 2025, $549 million remained of our $60 billion share repurchase program. Refer to Note 15 – Stockholders’ Equity of the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q) for further discussion.

Dividends

For the nine months ended March 31, 2025 and 2024, our Board of Directors declared dividends totaling $18.5 billion and $16.7 billion, respectively. We intend to continue returning capital to shareholders in the form of dividends, subject to declaration by our Board of Directors. Refer to Note 15 – Stockholders’ Equity of the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q) for further discussion.

Other Planned Uses of Capital

We will continue to invest in sales, marketing, product support infrastructure, and existing and advanced areas of technology, as well as acquisitions that align with our business strategy. Additions to property and equipment will continue, including new facilities, datacenters, and computer systems for research and development, sales and marketing, support, and administrative staff. We expect capital expenditures to increase in coming years to support growth in our cloud offerings and our investments in AI infrastructure and training. We have operating and finance leases for datacenters, corporate offices, research and development facilities, Microsoft Experience Centers, and certain equipment. We have not engaged in any related party transactions or arrangements with unconsolidated entities or other persons that are reasonably likely to materially affect liquidity or the availability of capital resources.

RECENT ACCOUNTING GUIDANCE

Refer to Note 1 – Accounting Policies of the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q) for further discussion.

CRITICAL ACCOUNTING ESTIMATES

Our consolidated financial statements and accompanying notes are prepared in accordance with GAAP. Preparing consolidated financial statements requires management to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenue, and expenses. Critical accounting estimates are those estimates that involve a significant level of estimation uncertainty and could have a material impact on our financial condition or results of operations. We have critical accounting estimates in the areas of revenue recognition, impairment of investment securities, goodwill, research and development costs, legal and other contingencies, income taxes, and business combinations – valuation of intangible assets.

Revenue Recognition

Our contracts with customers often include promises to transfer multiple products and services to a customer. Determining whether products and services are considered distinct performance obligations that should be accounted for separately versus together may require significant judgment. When a cloud-based service includes both on-premises software licenses and cloud services, judgment is required to determine whether the software license is considered distinct and accounted for separately, or not distinct and accounted for together with the cloud service and recognized over time. Certain cloud services, primarily Office 365, depend on a significant level of integration, interdependency, and interrelation between the desktop applications and cloud services, and are accounted for together as one performance obligation. Revenue from Office 365 is recognized ratably over the period in which the cloud services are provided.

Judgment is required to determine the standalone selling price (“SSP") for each distinct performance obligation. We use a single amount to estimate SSP for items that are not sold separately, including on-premises licenses sold with SA or software updates provided at no additional charge. We use a range of amounts to estimate SSP when we sell each of the products and services separately and need to determine whether there is a discount to be allocated based on the relative SSP of the various products and services.

In instances where SSP is not directly observable, such as when we do not sell the product or service separately, we determine the SSP using information that may include market conditions and other observable inputs. We typically have more than one SSP for individual products and services due to the stratification of those products and services by customers and circumstances. In these instances, we may use information such as the size of the customer and geographic region in determining the SSP.

Due to the various benefits from and the nature of our SA program, judgment is required to assess the pattern of delivery, including the exercise pattern of certain benefits across our portfolio of customers.

Our products are generally sold with a right of return, we may provide other credits or incentives, and in certain instances we estimate customer usage of our products and services, which are accounted for as variable consideration when determining the amount of revenue to recognize. Returns and credits are estimated at contract inception and updated at the end of each reporting period if additional information becomes available. Changes to our estimated variable consideration were not material for the periods presented.

Impairment of Investment Securities

We review debt investments quarterly for credit losses and impairment. If the cost of an investment exceeds its fair value, we evaluate, among other factors, general market conditions, credit quality of debt instrument issuers, and the extent to which the fair value is less than cost. This determination requires significant judgment. In making this judgment, we employ a systematic methodology that considers available quantitative and qualitative evidence in evaluating potential impairment of our investments. In addition, we consider specific adverse conditions related to the financial health of, and business outlook for, the investee. If we have plans to sell the security or it is more likely than not that we will be required to sell the security before recovery, then a decline in fair value below cost is recorded as an impairment charge in other income (expense), net and a new cost basis in the investment is established. If market, industry, and/or investee conditions deteriorate, we may incur future impairments.

Equity investments without readily determinable fair values are written down to fair value if a qualitative assessment indicates that the investment is impaired and the fair value of the investment is less than carrying value. We perform a qualitative assessment on a periodic basis. We are required to estimate the fair value of the investment to determine the amount of the impairment loss. Once an investment is determined to be impaired, an impairment charge is recorded in other income (expense), net.

Goodwill

We allocate goodwill to reporting units based on the reporting unit expected to benefit from the business combination. We evaluate our reporting units on an annual basis and, if necessary, reassign goodwill using a relative fair value allocation approach. Goodwill is tested for impairment at the reporting unit level (operating segment or one level below an operating segment) on an annual basis (May 1) and between annual tests if an event occurs or circumstances change that would more likely than not reduce the fair value of a reporting unit below its carrying value. These events or circumstances could include a significant change in the business climate, legal factors, operating performance indicators, competition, or sale or disposition of a significant portion of a reporting unit.

Application of the goodwill impairment test requires judgment, including the identification of reporting units, assignment of assets and liabilities to reporting units, assignment of goodwill to reporting units, and determination of the fair value of each reporting unit. The fair value of each reporting unit is estimated primarily through the use of a discounted cash flow methodology. This analysis requires significant judgments, including estimation of future cash flows, which is dependent on internal forecasts, estimation of the long-term rate of growth for our business, estimation of the useful life over which cash flows will occur, and determination of our weighted average cost of capital.

The estimates used to calculate the fair value of a reporting unit change from year to year based on operating results, market conditions, and other factors. Changes in these estimates and assumptions could materially affect the determination of fair value and goodwill impairment for each reporting unit.

Research and Development Costs

Costs incurred internally in researching and developing a computer software product are charged to expense until technological feasibility has been established for the product. Once technological feasibility is established, software costs are capitalized until the product is available for general release to customers. Judgment is required in determining when technological feasibility of a product is established. We have determined that technological feasibility for our software products is reached after all high-risk development issues have been resolved through coding and testing. Generally, this occurs shortly before the products are released to production. The amortization of these costs is included in cost of revenue over the estimated life of the products.

Legal and Other Contingencies

The outcomes of legal proceedings and claims brought against us are subject to significant uncertainty. An estimated loss from a loss contingency such as a legal proceeding or claim is accrued by a charge to income if it is probable that an asset has been impaired or a liability has been incurred and the amount of the loss can be reasonably estimated. In determining whether a loss should be accrued we evaluate, among other factors, the degree of probability of an unfavorable outcome and the ability to make a reasonable estimate of the amount of loss. Changes in these factors could materially impact our consolidated financial statements.

Income Taxes

The objectives of accounting for income taxes are to recognize the amount of taxes payable or refundable for the current year, and deferred tax liabilities and assets for the future tax consequences of events that have been recognized in an entity's financial statements or tax returns. We recognize the tax benefit from an uncertain tax position only if it is more likely than not that the tax position will be sustained on examination by the taxing authorities, based on the technical merits of the position. The tax benefits recognized in the financial statements from such a position are measured based on the largest benefit that has a greater than 50% likelihood of being realized upon ultimate settlement. Accounting literature also provides guidance on derecognition of income tax assets and liabilities, classification of deferred income tax assets and liabilities, accounting for interest and penalties associated with tax positions, and income tax disclosures. Judgment is required in assessing the future tax consequences of events that have been recognized in our consolidated financial statements or tax returns. Variations in the actual outcome of these future tax consequences could materially impact our consolidated financial statements.

Business Combinations – Valuation of Intangible Assets

Accounting for business combinations requires significant judgments when allocating the purchase price to the estimated fair values of assets acquired and liabilities assumed at the acquisition date. Determination of fair value involves estimates and assumptions which can be complex, most notably with respect to intangible assets. Critical estimates used in the valuation of intangible assets include, but are not limited to, the amount and timing of projected cash flows, useful lives, and discount rates. While management's estimates of fair value are based on assumptions that are believed to be reasonable, these assumptions are inherently uncertain as they pertain to forward-looking views of our business and market conditions. The judgments made in this valuation process could materially impact our consolidated financial statements.

ITEM 3. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

RISKS

We are exposed to economic risk from foreign exchange rates, interest rates, credit risk, and equity prices. We use derivatives instruments to manage these risks, however, they may still impact our consolidated financial statements.

Foreign Currencies

Certain forecasted transactions, assets, and liabilities are exposed to foreign currency risk. We monitor our foreign currency exposures daily to maximize the economic effectiveness of our foreign currency positions, including hedges. Principal currency exposures include the Euro, Japanese yen, British pound, Canadian dollar, and Australian dollar.

Interest Rate

Securities held in our fixed-income portfolio are subject to different interest rate risks based on their maturities. We manage the average maturity of the fixed-income portfolio to achieve economic returns that correlate to certain global fixed-income indices.

Credit

Our fixed-income portfolio is diversified and consists primarily of investment-grade securities. We manage credit exposures relative to broad-based indices to facilitate portfolio diversification.

Equity

Securities held in our equity investments portfolio are subject to price risk.

SENSITIVITY ANALYSIS

The following table sets forth the potential loss in future earnings or fair values, including associated derivatives, resulting from hypothetical changes in relevant market rates or prices:

(In millions)

Risk Categories	Hypothetical Change		March 31, 2025	Impact
Foreign currency – Revenue	10% decrease in foreign exchange rates	$	**(10,372)**	Earnings
Foreign currency – Investments	10% decrease in foreign exchange rates		**(21)**	Fair Value
Interest rate	100 basis point increase in U.S. treasury interest rates		**(1,045)**	Fair Value
Credit	100 basis point increase in credit spreads		**(333)**	Fair Value
Equity	10% decrease in equity market prices		**(1,247)**	Earnings

ITEM 4. CONTROLS AND PROCEDURES

Under the supervision and with the participation of our management, including the Chief Executive Officer and Chief Financial Officer, we have evaluated the effectiveness of our disclosure controls and procedures as required by Exchange Act Rule 13a-15(b) as of the end of the period covered by this report. Based on that evaluation, the Chief Executive Officer and Chief Financial Officer have concluded that these disclosure controls and procedures are effective. There were no changes in our internal control over financial reporting during the quarter ended March 31, 2025 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

PART II. OTHER INFORMATION

ITEM 1. LEGAL PROCEEDINGS

Refer to Note 14 – Contingencies of the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q) for information regarding legal proceedings in which we are involved.

ITEM 1A. RISK FACTORS

Our operations and financial results are subject to various risks and uncertainties, including those described below, that could adversely affect our business, operations, financial condition, results of operations, liquidity, and the trading price of our common stock.

STRATEGIC AND COMPETITIVE RISKS

We face intense competition across all markets for our products and services, which may adversely affect our results of operations.

Competition in the technology sector

Our competitors range in size from diversified global companies with significant research and development resources to small, specialized firms whose narrower product lines may let them be more effective in deploying technical, marketing, and financial resources. Barriers to entry in many of our businesses are low and many of the areas in which we compete evolve rapidly with changing and disruptive technologies, shifting user needs, and frequent introductions of new products and services. If we do not continue to innovate and provide products, devices, and services that appeal to businesses and consumers, we may not remain competitive, which may adversely affect our business, financial condition, and results of operations.

Competition among platform-based ecosystems

An important element of our business model has been to create platform-based ecosystems on which many participants can build diverse solutions. A well-established ecosystem creates beneficial network effects among users, application developers, and the platform provider that can accelerate growth. Establishing significant scale in the marketplace is necessary to achieve and maintain attractive margins. We face significant competition from firms that provide competing platforms.

- A competing vertically-integrated model, in which a single firm controls the hardware and software elements of a product and related services, has succeeded with some consumer products such as PCs, tablets, smartphones, gaming consoles, wearables, and other endpoint devices. Competitors pursuing this model also earn revenue from services integrated with the hardware and software platform, including applications and content sold through their integrated marketplaces. They may also be able to claim security and performance benefits from their vertically-integrated offer. We also offer some vertically-integrated hardware and software products and services. Shifting a portion of our business to a vertically-integrated model may increase our cost of revenue and reduce our operating margins.
- We derive substantial revenue from licenses of Windows operating systems on PCs. We face significant competition from competing platforms developed for new devices and form factors such as smartphones and tablets. These devices compete on multiple bases including price and the perceived utility of the device and its platform. Users continue to turn to these devices to perform functions that in the past were performed by PCs. Even if many users view these devices as complementary to a PC, the prevalence of these devices may make it more difficult to attract application developers to our PC operating system platforms. Competing with operating systems licensed at low or no cost may decrease our PC operating system margins. Popular products or services offered on competing platforms could increase their competitive strength. In addition, some of our devices compete with products made by our OEM partners, which may affect their commitment to our platform.

- Competing platforms have content and application marketplaces with scale and significant installed bases. The variety and utility of content and applications available on a platform are important to device purchasing decisions. Users may incur costs to move data and buy new content and applications when switching platforms. To compete, we must successfully enlist developers to write applications for our platform and ensure that these applications have high quality, security, customer appeal, and value. Efforts to compete with competitors' content and application marketplaces may increase our cost of revenue and lower our operating margins. Competitors' rules governing their content and applications marketplaces may restrict our ability to distribute products and services through them in accordance with our technical and business model objectives.

For all of these reasons, we may not be able to compete successfully against our current and future competitors, which may adversely affect our business, operations, financial condition, and results of operations.

Business model competition

Companies compete with us based on a growing variety of business models.

- A material part of our business involves cloud-based services available across the spectrum of computing devices. Our competitors continue to develop and deploy cloud-based services for consumers and business customers, and pricing and delivery models are evolving. We and our competitors are devoting significant resources to develop and deploy our cloud-based strategies.
- We are investing in artificial intelligence ("AI") across the entire company and infusing generative AI capabilities into our consumer and commercial offerings. AI technology and services are a highly competitive and rapidly evolving market, and new competitors continue to enter the market. We will bear significant development and operational costs to build and support the AI models, services, platforms, and infrastructure necessary to meet the needs of our customers. To compete effectively we must also be responsive to technological change, new and potential regulatory developments, and public scrutiny.
- Even as we transition more of our business to infrastructure-, platform-, and software-as-a-service business models, the license-based proprietary software model generates a substantial portion of our software revenue. We bear the costs of converting original ideas into software products through investments in research and development, offsetting these costs with the revenue received from licensing our products. Many of our competitors also develop and sell software to businesses and consumers under this model.
- Other competitors develop and offer free applications, online services, and content, and make money by selling third-party advertising. Advertising revenue funds development of products and services these competitors provide to users at little or no cost, competing directly with our revenue-generating products.
- Some companies compete with us by modifying and then distributing open source software at little or no cost to end users, developing, making available, or using AI models that are open, and earning revenue on advertising or integrated products and services. These firms do not bear the full costs of research and development for the open source products. Some open source products mimic the features and functionality of our products.

The competitive pressures described above may cause decreased sales volumes, price reductions, and/or increased operating costs, such as for research and development, marketing, and sales incentives, which may adversely affect our financial condition and results of operations.

Our focus on cloud-based and AI services presents execution and competitive risks. We are incurring significant costs to build and maintain infrastructure to support cloud-based and AI services. These costs will reduce the operating margins. Whether we succeed in cloud-based and AI services depends on our execution in several areas, including:

- Continuing to bring to market compelling cloud-based and AI services and products that generate increasing traffic and market share.
- Maintaining the utility, compatibility, and performance of our cloud-based and AI services on the growing array of computing devices, including PCs, smartphones, tablets, gaming consoles, and other devices.
- Continuing to enhance the attractiveness of our cloud platforms to third-party developers.

- Ensuring our cloud-based services meet the reliability expectations and specific requirements of our customers and maintain the security of their data as well as help them meet their own compliance needs.
- Making our suite of cloud-based services platform-agnostic, available on a wide range of devices and ecosystems, including those of our competitors.

It is uncertain whether our strategies will continue to attract users or generate the revenue required to succeed. If we are not effective in executing organizational and technical changes to increase efficiency and accelerate innovation, or if we fail to generate sufficient usage of our new products and services, we may not grow revenue in line with the infrastructure and development investments described above. This may adversely affect our operations, financial condition, and results of operations.

Our AI systems offer users powerful tools and capabilities. However, there may be instances where these systems are used in ways that are unintended or inappropriate. In addition, some users may also engage in fraudulent or abusive activities through our cloud-based and AI services, such as unauthorized account access, payment fraud, or terms of service violations including cryptocurrency mining or launching cyberattacks. While we are committed to detecting and controlling such misuse of our cloud-based and AI services, our efforts may not be effective, and we may incur reputational damage or experience adverse impacts to our business and results of operations.

RISKS RELATING TO THE EVOLUTION OF OUR BUSINESS

We make significant investments in products and services that may not achieve expected returns. We will continue to make significant investments in research, development, and marketing for existing products, services, and technologies. In addition, we are focused on developing new AI platform services and incorporating AI into existing products and services. We also invest in the development and acquisition of a variety of hardware for productivity, communication, and entertainment, including PCs, tablets, and gaming devices. Investments in new technology are speculative. Commercial success depends on many factors, including innovation, developer support, and effective distribution and marketing. If customers do not perceive our latest offerings as providing significant new functionality or other value, they may reduce their purchases of new software and hardware products or upgrades, unfavorably affecting revenue. We may not achieve significant revenue from new product, service, and distribution channel investments for several years, if at all. New products and services may not be profitable or may not achieve operating margins as high as we have experienced historically. We may not get engagement in certain features that drive post-sale monetization opportunities. Our data-handling practices across our products and services will continue to be under scrutiny. Perceptions of mismanagement, driven by regulatory activity or negative public reaction to our practices or product experiences, could negatively impact product and feature adoption. Developing new technologies is complex. It can require long development and testing periods. We could experience significant delays in new releases or significant problems in creating new products or services. These factors could adversely affect our business, financial condition, and results of operations.

Acquisitions, joint ventures, and strategic alliances may have an adverse effect on our business. We expect to continue making acquisitions and entering into joint ventures and strategic alliances as part of our long-term business strategy. For example, in March 2022 we completed our acquisition of Nuance Communications, Inc., and in October 2023 we completed our acquisition of Activision Blizzard, Inc. (“Activision Blizzard”). In January 2023 we announced the third phase of our OpenAI strategic partnership. Acquisitions and other transactions and arrangements involve significant challenges and risks, including that they do not advance our business strategy, that we get an unsatisfactory return on our investment, that they raise new compliance-related obligations and challenges, that we have difficulty integrating and retaining new employees, business systems, and technology, that they distract management from our other businesses, or that announced transactions may not be completed. If an arrangement fails to adequately anticipate changing circumstances and interests of a party, it may result in early termination or renegotiation of the arrangement. We also have limited ability to control or influence third parties with whom we have arrangements, which may impact our ability to realize the anticipated benefits. The success of these transactions and arrangements depend in part on our ability to leverage them to enhance our existing products and services or develop compelling new ones, as well as the acquired companies’ ability to meet our policies and processes in areas such as data governance, privacy, and cybersecurity. It may take longer than expected to realize the full benefits from these transactions and arrangements, such as increased revenue or enhanced efficiencies, or the benefits may ultimately be smaller than we expected. In addition, an acquisition may be subject to challenge even after it has been completed. For example, the Federal Trade Commission continues to challenge our Activision Blizzard acquisition and could, if successful, alter or unwind the transaction. These events could adversely affect our business, operations, financial condition, and results of operations.

If our goodwill or amortizable intangible assets become impaired, we may be required to record a significant charge to earnings. We acquire other companies and intangible assets and may not realize all the economic benefit from those acquisitions, which could cause an impairment of goodwill or intangibles. We review our amortizable intangible assets for impairment when events or changes in circumstances indicate the carrying value may not be recoverable. We test goodwill for impairment at least annually. Factors that may be a change in circumstances, indicating that the carrying value of our goodwill or amortizable intangible assets may not be recoverable, include a decline in our stock price and market capitalization, reduced future cash flow estimates, and slower growth rates in industry segments in which we participate. We have recorded, and may in the future be required to record, a significant charge in our consolidated financial statements during the period in which any impairment of our goodwill or amortizable intangible assets is determined, negatively affecting our results of operations.

CYBERSECURITY, DATA PRIVACY, AND PLATFORM ABUSE RISKS

Cyberattacks and security vulnerabilities could lead to reduced revenue, increased costs, liability claims, or harm to our reputation or competitive position.

Security of our information technology

Threats to security can take a variety of forms. Threat actors, including individual and groups of hackers and sophisticated organizations, including nation-states or state-sponsored organizations, continuously undertake attacks that pose threats to our customers and our internal infrastructure, and we have experienced cybersecurity incidents in which such actors have gained unauthorized access to our systems and data, including customer systems and data. These actors use a wide variety of methods, which include developing and deploying malicious software; exploiting known and potential vulnerabilities or intentionally designed processes in hardware, software, or other infrastructure to attack our products and services or gain access to our networks and datacenters; using social engineering techniques to induce our employees, users, partners, or customers to disclose sensitive information, such as passwords, or take other actions to gain access to our data or our users' or customers' data; or acting in a coordinated manner or conducting coordinated attacks. For example, as previously disclosed in our Form 8-K filed with the Securities and Exchange Commission on January 19, 2024 and amended on March 8, 2024, beginning in late November 2023, a nation-state associated threat actor used a password spray attack to compromise a legacy test account and, in turn, gain access to Microsoft email accounts. The threat actor used information it obtained to gain unauthorized access to some of our source code repositories and internal systems, and the threat actor may continue to utilize this and other information to attempt to gain access to our systems or otherwise adversely affect our business and results of operations. This incident has and may continue to result in harm to our reputation and customer relationships. Nation-state and state-sponsored actors can sustain malicious activities for extended periods and deploy significant resources to plan and carry out attacks. Nation-state attacks against us, our customers, or our partners have and may continue to intensify due to our transparency to our customers, other stakeholders, and the public about cyberattacks, and during elections or periods of intense diplomatic or armed conflict. Challenges or failures in applying security patches to all hardware and devices connected to our systems, including end-of-life and end-of-support equipment, have and may continue to result in unauthorized access to our systems and data in the future. Cyber incidents and attacks, individually or in the aggregate, could adversely affect our financial condition, results of operations, competitive position, and reputation, or expose us to legal or regulatory risk.

Inadequate account security or organizational security practices, including those of companies we have acquired or those of the third parties we utilize, have resulted and may result in unauthorized access to our systems and data, including customer systems and data. For example, system administrators may fail to timely remove employee account access when no longer appropriate. Employees or third parties may intentionally compromise our or our users' security or systems or reveal confidential information, and laws in foreign jurisdictions may compel actions by such parties against our interests and could limit our recourse. Malicious actors may employ the supply chain to introduce malware through software updates or compromised supplier accounts or hardware.

Cyberthreats are constantly evolving and becoming increasingly sophisticated and complex, increasing the difficulty of detecting and successfully defending against them. Threat actors may also utilize emerging technologies, such as AI and machine learning. Our current capabilities may not detect certain vulnerabilities or new attack methods, which may allow them to persist in the environment over long periods of time. It may be difficult to determine the best way to investigate, mitigate, contain, and remediate the harm caused by a cyber incident. Such efforts may not be successful, and we may make errors or fail to take necessary actions. It is possible that threat actors may gain undetected access to other networks and systems after establishing a foothold on an internal system. Cyber incidents and attacks can have cascading impacts that unfold with increasing speed across our internal networks and systems, as well as those of our partners and customers. In addition, it may take considerable time for us to investigate and evaluate the full impact of incidents, particularly for sophisticated attacks. As a result of these and other factors, we may not be able to provide prompt, full, and reliable information about the incident to our customers, partners, regulators, and the public. Breaches of our facilities, network, or data security can disrupt the security of our systems and business applications, impair our ability to provide services to our customers and protect the privacy of their data, result in product development delays, compromise confidential or technical business information, result in theft or misuse of our intellectual property or other assets, subject us to ransomware attacks, require us to allocate more resources to improve technologies or remediate the impacts of attacks, or otherwise adversely affect our business. In addition, actions taken to remediate an incident could result in outages, data losses, and disruptions of our services.

Our internal environment continues to evolve. Often, we are early adopters of new devices and technologies. We embrace new ways of sharing data and communicating internally and with partners and customers using methods such as social networking and other consumer-oriented technologies. Increasing use of generative AI models in our internal systems may create new attack surfaces or methods for adversaries. Our business policies and internal security controls may not keep pace with these changes as new threats emerge or the emerging cybersecurity regulations in jurisdictions worldwide.

Security of our products, services, devices, and customers' data

The security of our products and services is important in our customers' decisions to purchase or use our products or services across cloud and on-premises environments. Security threats are a significant challenge to companies like us, whose business is providing technology products and services to others. Threats to or attacks on our own infrastructure, such as the nation-state attack described in the prior risk factor, have also affected our customers and may do so in the future. Customers using our cloud-based services rely on the security of our infrastructure, including hardware and other elements provided by third parties, to ensure the reliability of our services and the protection of their data. Adversaries tend to focus their efforts on the most popular operating systems, programs, and services, including many of ours, as well as customers with sensitive data, and we expect that to continue. In addition, adversaries can attack our customers' on-premises or cloud environments, sometimes exploiting previously unknown ("zero-day") vulnerabilities, such as the attack in early calendar year 2021 with several of our Exchange Server on-premises products. Vulnerabilities in these or any product can persist even after we have issued security patches if customers have not installed the most recent updates, or if the attackers exploited the vulnerabilities before patching to install additional malware to further compromise customers' systems. Adversaries will continue to attack customers using our cloud services as customers embrace digital transformation. Adversaries that acquire user account information can use that information to compromise our users' accounts, including where accounts share the same attributes such as passwords. Inadequate account security practices may also result in unauthorized access, and user activity may result in ransomware or other malicious software impacting a customer's use of our products or services. There may be vulnerabilities in open source software that may make our products susceptible to cyberattacks as we increasingly incorporate open source software into our products. Additionally, features that rely on generative AI may be susceptible to unanticipated security threats from adversaries as we add new generative AI features to our services while continuously developing our understanding of security risks and protection methods in the new field of generative AI.

Our customers operate complex systems with third-party hardware and software from multiple vendors that may include systems acquired over many years. They expect our products and services to support all these systems and products, including those that no longer incorporate the strongest current security advances or standards. As a result, we may not be able to discontinue support in our services for a product, service, standard, or feature solely because a more secure alternative is available. Failure to utilize the most current security advances and standards can increase our customers' vulnerability to attack. Further, customers of widely varied sizes and technical sophistication use our technology, and consequently may still have limited capabilities and resources to help them adopt and implement state-of-the-art cybersecurity practices and technologies. In addition, we must account for this wide variation of technical sophistication when defining default settings for our products and services, including security default settings, as these settings may limit or otherwise impact other aspects of operations and some customers may have limited capability to review and reset these defaults.

Cyberattacks may adversely impact our customers even if our production services are not directly compromised. We are committed to notifying our customers whose systems have been impacted as we become aware and have actionable information for customers to help protect themselves. We are also committed to providing guidance and support on detection, tracking, and remediation. We may not be able to detect the existence or extent of these attacks for all of our customers or have information on how to detect or track an attack, especially where an attack involves on-premises software such as Exchange Server where we may have no or limited visibility into our customers' computing environments.

Any of the foregoing events could result in reputational harm, loss of revenue, increased costs, or otherwise adversely affect our business, financial condition, and results of operations.

Development and deployment of defensive measures

To defend against security threats to our internal infrastructure, our cloud-based services, and our customers' systems, we must continuously engineer more secure products and services, enhance security, threat detection, and reliability features, escalate and improve the deployment of software updates to address security vulnerabilities in our own products as well as those provided by others in a timely manner, develop mitigation technologies that help to secure customers from attacks even when software updates are not deployed, maintain the digital security infrastructure that protects the integrity of our network, products, and services, and provide security tools such as firewalls, anti-virus software, and advanced security and information about the need to deploy security measures and the impact of doing so.

The cost of measures to protect products and customer-facing services could reduce our operating margins. If we fail to do these things well, actual or perceived security vulnerabilities in our products and services, data corruption issues, or reduced performance could harm our reputation and lead customers to reduce or delay future purchases of products or subscriptions to services, or to use competing products or services. Customers may also spend more on protecting their existing computer systems from attack, which could delay adoption of additional products or services. Customers in certain industries such as financial services, health care, and government may have enhanced or specialized expectations and requirements to which we must engineer our products and services. Customers and third parties granted access to their systems may fail to update their systems, continue to run software or operating systems we no longer support, or may fail to timely install or enable security patches, or may otherwise fail to adopt adequate security practices. Any of these could adversely affect our reputation and results of operations. Actual or perceived vulnerabilities may lead to claims against us. Our license agreements typically contain provisions that eliminate or limit our exposure to liability, but there is no assurance these provisions will withstand legal challenges. At times, to achieve commercial objectives, we may enter into agreements with larger liability exposure to customers.

Our products operate in conjunction with and are dependent on products and components across a broad ecosystem of third parties. If there is a security vulnerability in one of these components, and if there is a security exploit targeting it, we may experience adverse impacts to our results of operations, reputation, or competitive position.

Disclosure and misuse of personal data could result in liability and harm our reputation. As we continue to grow the number, breadth, and scale of our cloud-based offerings, we store and process increasingly large amounts of personal data of our customers and users. The continued occurrence of high-profile data breaches provides evidence of an external environment increasingly hostile to information security. Despite our efforts to improve the security controls across our business groups and geographies, it is possible our security controls over personal data, our training of employees and third parties on data security, and other practices we follow may not prevent the improper disclosure or misuse of customer or user data we or our vendors store and manage. Relatedly, despite our efforts to continuously improve security controls, it is possible that we may fail to identify or mitigate insider threat activities that could lead to the misuse of our systems or customer and user data. In addition, third parties who have limited access to our customer or user data may use this data in unauthorized ways. Improper disclosure or misuse could harm our reputation, lead to legal exposure to customers or users, or subject us to liability under laws that protect personal data, resulting in increased costs or loss of revenue. Our software products and services also enable our customers and users to store and process personal data on-premises or in a cloud-based environment we host. Government authorities can sometimes require us to produce customer or user data in response to valid legal orders. In the U.S. and elsewhere, we advocate for transparency concerning these requests and appropriate limitations on government authority to compel disclosure. Despite our efforts to protect customer and user data, perceptions that the collection, use, and retention of personal information is not satisfactorily protected could inhibit sales of our products or services and could limit adoption of our cloud-based solutions by consumers, businesses, and government entities. Additional security measures we take to address customer or user concerns, or constraints on our flexibility to determine where and how to operate datacenters in response to customer or user expectations or governmental rules or actions, may increase costs or hinder sales of our products and services.

We may not be able to protect information in our products and services from use by others. LinkedIn and other Microsoft products and services contain valuable information and content protected by contractual restrictions or technical measures. In certain cases, we have made commitments to our members and users to limit access to or use of this information. Changes in the law or interpretations of the law may weaken our ability to prevent third parties from scraping or gathering information or content through use of bots or other measures and using it for their own benefit which could adversely affect our business, financial condition, and results of operations.

Abuse of our platforms may harm our reputation or user engagement.

Advertising, professional, marketplace, and gaming platform abuses

For platform products and services that provide content or host ads that come from or can be influenced by third parties, our reputation or user engagement may be negatively affected by activity that is hostile or inappropriate. This activity may come from users impersonating other people or organizations, including through the use of AI technologies, dissemination of information that may be viewed as misleading or intended to manipulate the opinions of our users, or the use of our products or services that violates our terms of service or otherwise for objectionable or illegal ends. Preventing or responding to these actions may require us to make substantial investments in people and technology and these investments may not be successful, adversely affecting our business, financial condition, and results of operations.

Other digital safety abuses

Our consumer services as well as our enterprise services may be used to find, generate, store, or disseminate harmful or illegal content in violation of our terms or applicable law. We may not proactively discover such content due to scale, the limitations of existing technologies, and conflicting legal frameworks. When discovered by users and others, such content may negatively affect our reputation, our brands, and user engagement. Regulations and other initiatives have been enacted to make platforms responsible for preventing or eliminating harmful content online, and we expect this to continue. At the same time, regulations and other initiatives regarding freedom of expression may conflict with such content moderation regulations. Failure to comply with content requirements may subject us to enhanced regulatory oversight, civil or criminal liability, or reputational damage, which could adversely affect our business, financial condition, and results of operations.

Our products and services, how they are used by customers, and how third-party products and services interact with them, may present security, privacy, and execution risks. Our products and services may contain defects in design, manufacture, or operation that make them insecure or ineffective for their intended purposes. For example, an Internet of Things solution may have multiple layers of hardware, sensors, processors, software, and firmware, several of which we may not develop or control, and may have limited ability to be updated or patched. Further, customers control our products and services, including our AI products, within their environments, and may deploy them in high-risk scenarios or utilize them inappropriately. As a result, our products and services may increasingly affect personal health and safety. Our products may also collect large amounts of data in manners which may not satisfy customers or regulatory requirements. Our customers also operate complex systems with third-party hardware and software from multiple vendors whose products or personnel may take or fail to take actions which impact the reliability or security of our products and services. If our products and services do not work as intended, are utilized in methods not intended, violate the law, or harm individuals or businesses, we may be subject to legal claims or enforcement actions. These risks, if realized, may increase our costs, damage our reputation, or adversely affect our results of operations.

Issues in the development, deployment, and use of AI may result in reputational or competitive harm or liability. We are building AI into many of our offerings, including our productivity services, and we are also making AI available for our customers to use in solutions that they build. This AI may be developed by Microsoft or others, including our strategic partner, OpenAI. We expect these elements of our business to grow. We envision a future in which AI operating in devices, applications, and the cloud helps our customers be more productive in their work and personal lives. As with many innovations, AI presents risks and challenges that could affect its adoption, and therefore our business. AI algorithms or training methodologies may be flawed. Datasets may be overbroad, insufficient, or contain biased or inaccurate information. Content generated by AI systems may be offensive, illegal, inaccurate, or otherwise harmful. Ineffective or inadequate AI development or deployment practices by Microsoft or others could result in incidents that impair the acceptance of AI solutions, cause harm to individuals, customers, or society, or result in our products and services not working as intended. Human review of certain inputs and outputs may be required, including for agentic AI systems that can take actions autonomously. Our implementation of AI systems could result in legal liability, regulatory action, brand, reputational, or competitive harm, or other adverse impacts. These risks may arise from intellectual property, data privacy, and other claims related to AI training and output, new and proposed legislation and regulations, such as the European Union's ("EU") AI Act, and new applications of data protection, privacy, consumer protection, intellectual property, and other laws. Some AI scenarios present ethical issues or may have broad impacts on society. There is also rising divergence globally in how to address these issues and impacts, with the result that we will need to navigate a web of different obligations in different geographies. Finally, if we enable or offer AI solutions that have unintended consequences, unintended usage or customization by our customers and partners, are contrary to our responsible AI policies and practices, or are otherwise controversial because of their impact on human rights, privacy, employment, or other social, economic, or political issues, our reputation, competitive position, business, financial condition, and results of operations may be adversely affected.

OPERATIONAL RISKS

We may have excessive outages, data losses, and disruptions of our online services if we fail to maintain an adequate operations infrastructure. Our increasing user traffic, growth in services, and the complexity of our products and services demand more computing power. We spend substantial amounts to build, purchase, or lease datacenters and equipment and to upgrade our technology and network infrastructure to handle more traffic on our websites and in our datacenters. Our datacenters depend on the availability of permitted and buildable land, predictable energy, networking supplies, and servers, including graphics processing units and other components. The cost or availability of these dependencies could be adversely affected by a variety of factors, including the transition to a clean energy economy, local and regional environmental regulations, and geopolitical disruptions. These demands continue to increase as we introduce new products and services and support the growth and the augmentation of existing services, including through the incorporation of AI features and/or functionality. We are rapidly growing our business of providing a platform and back-end hosting for services provided by third parties to their end users. Maintaining, securing, and expanding this infrastructure is expensive and complex, and requires development of principles for datacenter builds in geographies with higher safety and reliability risks. It requires that we maintain an Internet connectivity infrastructure and storage and compute capacity that is robust and reliable within competitive and regulatory constraints that continue to evolve. Inefficiencies or operational failures, including temporary or permanent loss of customer data, outages, insufficient Internet connectivity, insufficient or unavailable power or water supply, or inadequate storage and compute capacity could diminish the quality of our products, services, and user experience, resulting in contractual liability, claims by customers and other third parties, regulatory actions, damage to our reputation, and loss of current and potential users, subscribers, and advertisers, each of which may adversely affect our business, operations, financial condition, and results of operations.

We may experience supply or quality problems. There are limited suppliers for certain device and datacenter components. We continue to identify and evaluate opportunities to expand our datacenter locations and increase our server capacity to meet the evolving needs of our customers, particularly given the growing demand for AI services. Capacity available to us may be affected as competitors use some of the same suppliers and materials for hardware components. If components are delayed or become unavailable, whether because of supplier capacity constraint, industry shortages, legal or regulatory changes that restrict supply sources, or other reasons, we may not obtain timely replacement supplies, resulting in reduced sales or inadequate datacenter capacity to support the delivery and continued development of our products and services. Component shortages, excess or obsolete inventory, or price reductions resulting in inventory adjustments may increase our cost of revenue. Datacenter servers, Xbox consoles, Surface devices, and other hardware are assembled in Asia and other geographies that may be subject to disruptions in the supply chain, resulting in shortages which may adversely affect our business, operations, financial condition, and results of operations.

Our software products and services also may experience quality or reliability problems. The highly sophisticated software we develop may contain bugs and other defects that interfere with their intended operation. Our customers increasingly rely on us for critical business functions and multiple workloads. Many of our products and services are interdependent on one another. Our products and services may be impacted by interaction with third-party products and services. Our customers may also utilize their own or third-party products and services whose reliability is dependent on interaction with our products and services. Each of these circumstances potentially magnifies the impact of quality or reliability issues. Any defects we do not detect and fix in pre-release testing could cause reduced sales, damage to our reputation, repair or remediation costs, delays in the release of new products or versions, or legal liability, which could adversely affect our business, financial condition, and results of operations. Although our license agreements typically contain provisions that eliminate or limit our exposure to liability, there is no assurance these provisions will withstand legal challenge.

Our hardware products such as Xbox consoles, Surface devices, and other devices we design and market are highly complex. Failure to prevent, detect, or address defects in design, manufacture, or associated software could result in recalls, safety alerts, or product liability claims, which could adversely affect our business and results of operations.

LEGAL, REGULATORY, AND LITIGATION RISKS

Government enforcement under competition laws and new market regulation may limit how we design and market our products. Government agencies closely scrutinize us under U.S. and foreign competition laws. Governments are actively enforcing competition laws and regulations and enacting new regulations to intervene in digital markets, and this includes markets such as the EU, the United Kingdom, the U.S., and China. Some jurisdictions also allow competitors or consumers to assert claims of anti-competitive conduct. U.S. and foreign antitrust authorities have previously brought enforcement actions and continue to scrutinize our business.

For example, the European Commission (“the Commission”) has designated Windows and LinkedIn as core platform services subject to obligations under the EU Digital Markets Act, which prohibits certain self-preferencing behaviors and places limitations on certain data use among other obligations. The Commission also continues to closely scrutinize the design of high-volume Microsoft products and the terms on which we make certain technologies used in these products, such as file formats, programming interfaces, and protocols, available to other companies. Flagship product releases such as Microsoft 365 and Windows can receive significant scrutiny under EU or other competition laws.

Our portfolio of first-party devices continues to grow; at the same time, our OEM partners offer a large variety of devices for our platforms. As a result, we increasingly both cooperate and compete with our OEM partners, creating a risk that we fail to do so in compliance with competition rules. Regulatory scrutiny in this area may increase. Certain foreign governments, particularly in China and other countries in Asia, have advanced arguments under their competition laws that exert downward pressure on royalties for our intellectual property.

Competition law enforcement actions and court decisions along with new market regulations may result in fines or hinder our ability to provide the benefits of our software to consumers and businesses, reducing the attractiveness of our products and the revenue that comes from them. New competition law actions or obligations under market regulation schemes could be initiated, potentially using previous actions as precedent. The outcome of such actions, or steps taken to avoid them, could adversely affect us in a variety of ways, including causing us to withdraw products from or modify products for certain markets, decreasing the value of our assets, adversely affecting our ability to monetize our products, or inhibiting our ability to consummate acquisition or impose conditions on acquisitions that may reduce their value, which may adversely affect our business, financial condition, and results of operations.

Laws and regulations relating to anti-corruption and trade could result in increased costs, fines, criminal penalties, or reputational damage. The Foreign Corrupt Practices Act (“FCPA”) and other anti-corruption laws and regulations (“Anti-Corruption Laws”) prohibit corrupt payments by our employees, vendors, or agents, and the accounting provisions of the FCPA require us to maintain accurate books and records and adequate internal controls. From time to time, we receive inquiries from authorities in the U.S. and elsewhere which may be based on reports from employees and others about our business activities outside the U.S. and our compliance with Anti-Corruption Laws. Periodically, we receive such reports directly and investigate them, and also cooperate with investigations by U.S. and foreign law enforcement authorities. Most countries in which we operate also have competition laws that prohibit competitors from colluding or otherwise attempting to reduce competition between themselves. While we devote substantial resources to our U.S. and international compliance programs and have implemented policies, training, and internal controls designed to reduce the risk of corrupt payments and collusive activity, our employees, partners, vendors, or agents may violate our policies. Our failure to comply with Anti-Corruption Laws or competition laws could result in significant fines and penalties, criminal sanctions against us, our officers, or our employees, prohibitions on the conduct of our business, and damage to our reputation, which could adversely affect our business, financial condition, and results of operations.

Increasing trade laws, policies, sanctions, and other regulatory requirements also affect our operations in and outside the U.S. relating to trade and investment. Economic sanctions in the U.S., the EU, and other countries prohibit most business with restricted entities or countries. U.S. export controls restrict Microsoft from offering many of its products and services to, or making investments in, certain entities in specified countries. U.S. import controls restrict us from integrating certain information and communication technologies into our supply chain and allow for government review of transactions involving information and communications technology from countries determined to be foreign adversaries. Supply chain regulations may impact the availability of goods or result in additional regulatory scrutiny. Periods of intense diplomatic or armed conflict, such as the ongoing conflict in Ukraine, may result in (1) new and rapidly evolving sanctions and trade restrictions, which may impair trade with sanctioned individuals and countries, and (2) negative impacts to regional trade ecosystems among our customers, partners, and us. Non-compliance with sanctions as well as general ecosystem disruptions could result in reputational harm, operational delays, monetary fines, loss of revenue, increased costs, loss of export privileges, or criminal sanctions, which could adversely affect our business, financial condition, and results of operations.

Laws and regulations relating to the handling of personal data may impede the adoption of our services or result in increased costs, legal claims, fines against us, or reputational damage. The growth of our Internet- and cloud-based services internationally relies increasingly on the movement of data across national boundaries. Legal requirements relating to the collection, storage, handling, and transfer of personal data continue to evolve. For example, while the EU-U.S. Data Privacy Framework ("DPF") has been recognized as adequate under EU law to allow transfers of personal data from the EU to certified companies in the U.S., the DPF is subject to further legal challenge which could cause the legal requirements for data transfers from the EU to be uncertain. EU data protection authorities have and may again block the use of certain U.S.-based services that involve the transfer of data to the U.S. In the EU and other markets, potential new rules and restrictions on the flow of data across borders could increase the cost and complexity of delivering our products and services.

In addition, the EU General Data Protection Regulation ("GDPR"), which applies to all of our activities conducted from an establishment in the EU or related to products and services offered in the EU, imposes a range of compliance obligations regarding the handling of personal data. More recently, the EU has been developing new requirements related to the use of data, including in the Digital Markets Act, the Digital Services Act, and the Data Act, that add additional rules and restriction on the use of data in our products and services. Engineering efforts to build and maintain capabilities to facilitate compliance with these laws involve substantial expense and the diversion of engineering resources from other projects. We might experience reduced demand for our offerings if we are unable to engineer products that meet our legal duties or help our customers meet their obligations under these and other data regulations, or if our implementation to comply makes our offerings less attractive. Compliance with these obligations depends in part on how particular regulators interpret and apply them. If we fail to comply, or if regulators assert we have failed to comply (including in response to complaints made by customers), it may lead to regulatory enforcement actions, which can result in significant monetary penalties, private lawsuits, reputational damage, blockage of product offerings or of international data transfers, and loss of customers. The highest fines assessed under GDPR have recently been increasing, especially against large technology companies, and European data protection authorities have taken action to block or remove services from their markets. Jurisdictions around the world, such as China, India, and states in the U.S. have adopted, or are considering adopting or expanding, laws and regulations imposing obligations regarding the collection, handling, and transfer of personal data.

Our investment in gaining insights from data is becoming central to the value of the services we deliver to customers, including AI services, to operational efficiency and key opportunities in monetization, and to customer perceptions of quality. Our ability to use data in this way may be constrained by regulatory developments that impede realizing the expected return from this investment. Ongoing legal analyses, reviews, and inquiries by regulators of Microsoft practices, or relevant practices of other organizations, may result in burdensome or inconsistent requirements, including data sovereignty and localization requirements, affecting the location, movement, collection, and use of our customer and internal employee data as well as the management of that data. Compliance with applicable laws and regulations regarding personal data may require changes in services, business practices, or internal systems that result in increased costs, lower revenue, reduced efficiency, or greater difficulty in competing with foreign-based firms. Compliance with data regulations might limit our ability to innovate or offer certain features and functionality in some jurisdictions where we operate. Failure to comply with existing or new rules may result in significant penalties or orders to stop the alleged noncompliant activity, negative publicity, and diversion of management time and effort.

Existing and increasing legal and regulatory requirements could adversely affect our results of operations. We are subject to a wide range of laws, regulations, and legal requirements in the U.S. and globally, including those that may apply to our products and online services offerings, and those that impose requirements related to user privacy, telecommunications, data storage and protection, digital accessibility, advertising, and online content. Laws in several jurisdictions, including EU Member State laws under the European Electronic Communications Code and the Digital Services Act, increasingly define certain of our services as regulated services. This trend may continue and will result in these offerings being subject to additional data protection, security, digital safety, law enforcement surveillance, and other obligations. Regulators and private litigants may assert that our collection, use, and management of customer data and other information is inconsistent with their laws and regulations, including laws that apply to the tracking of users via technology such as cookies. In addition, laws requiring us to retrieve and produce customer data in response to compulsory legal demands from law enforcement and governmental authorities are expanding and the requests we are experiencing are increasing in volume and complexity. New environmental, social, and governance laws and regulations are expanding mandatory disclosure, reporting, and diligence requirements. Legislative or regulatory action relating to cybersecurity requirements may increase the costs to develop, implement, or secure our products and services. Legislative and regulatory action is emerging in the areas of AI and content moderation, which could increase costs or restrict opportunity. For example, the EU's AI Act may increase costs or impact the provision or operation of our AI models and services in the European market.

How these laws and regulations apply to our business is often unclear, subject to change over time, and sometimes may be inconsistent from jurisdiction to jurisdiction. In addition, governments' approach to enforcement, and our products and services, are continuing to evolve. Compliance with existing, expanding, or new laws and regulations may involve significant costs or require changes in products or business practices that could adversely affect our results of operations. Noncompliance could result in the imposition of penalties, criminal sanctions, or orders to cease the alleged noncompliant activity. In addition, there is increasing pressure from advocacy groups, regulators, competitors, customers, and other stakeholders across many of these areas. If our products do not meet customer expectations or legal requirements, we could face regulatory or legal actions, and our business, operations, financial condition, and results of operations could be adversely affected.

We have claims and lawsuits against us that may result in adverse outcomes. We are subject to a variety of claims and lawsuits. These claims may arise from a wide variety of business practices and initiatives, including major new product releases, AI services, significant business transactions, warranty or product claims, employment practices, and regulation. As we continue to expand our business and offerings, we may experience new and novel legal claims. Adverse outcomes in some or all of these claims may result in significant monetary damages or injunctive relief that could adversely affect our ability to conduct our business. Litigation and other claims are subject to inherent uncertainties and management's view of these matters may change in the future. A material adverse impact to our financial condition and results of operations could occur for the period in which the effect of an unfavorable outcome becomes probable and reasonably estimable.

Our business with government customers may present additional uncertainties. We derive substantial revenue from government contracts. Government contracts generally can present risks and challenges not present in private commercial agreements. For instance, we may be subject to government audits and investigations relating to these contracts, we could be suspended or debarred as a governmental contractor, we could incur civil and criminal fines and penalties, and under certain circumstances contracts may be rescinded. Some agreements may allow a government to terminate without cause and provide for higher liability limits for certain losses. Some contracts may be subject to periodic funding approval, reductions, cancellations, or delays which could adversely impact public-sector demand for our products and services. These events could negatively impact our financial condition, results of operations, and reputation.

We may have additional tax liabilities. We are subject to income taxes in the U.S. and many foreign jurisdictions. Significant judgment is required in determining our worldwide provision for income taxes. In the course of our business, there are many transactions and calculations where the ultimate tax determination is uncertain. We may recognize additional tax expense and be subject to additional tax liabilities due to changes in tax laws, regulations, and administrative practices and principles, including changes to the global tax framework, in various jurisdictions. In recent years, multiple domestic and international tax proposals were proposed to impose greater tax burdens on large multinational enterprises. For example, the Organisation for Economic Co-operation and Development continues to advance proposals or guidance in international taxation, including the establishment of a global minimum tax.

We are regularly under audit by tax authorities in different jurisdictions. Although we believe that our provision for income taxes and our tax estimates are reasonable, tax authorities may disagree with certain positions we have taken. In addition, economic and political pressures to increase tax revenue in various jurisdictions may make resolving tax disputes favorably more difficult. We are currently under Internal Revenue Service ("IRS") audit for prior tax years and have received Notices of Proposed Adjustment ("NOPAs") from the IRS for the tax years 2004 to 2013. The primary issues in the NOPAs relate to intercompany transfer pricing. In the NOPAs, the IRS is seeking an additional tax payment of $28.9 billion plus penalties and interest. The final resolution of the proposed adjustments, and other audits or litigation, may differ from the amounts recorded in our consolidated financial statements and adversely affect our results of operations in the period or periods in which that determination is made.

We earn a significant amount of our operating income outside the U.S. A change in the mix of earnings and losses in countries with differing statutory tax rates, changes in our business or structure, or the expiration of or disputes about certain tax agreements in a particular country may result in higher effective tax rates for the company. In addition, changes in U.S. federal and state or international tax laws applicable to corporate multinationals, other global fundamental law changes currently being considered by many countries, including in the U.S., and changes in taxing jurisdictions' administrative interpretations, decisions, policies, and positions may materially adversely affect our financial condition and results of operations.

We are subject to evolving sustainability regulatory requirements and expectations, which exposes us to increased costs and legal and reputational risks. Laws, regulations, and policies relating to environmental, social, and governance matters are being developed and formalized in Europe, the U.S., and elsewhere, which may include specific, target-driven frameworks and disclosure requirements. In addition, we have established and publicly announced goals and commitments to become carbon negative, water positive, zero waste, and protect more land than we use. Any failure or perceived failure to pursue or fulfill our sustainability goals and commitments or to satisfy various sustainability reporting standards or regulatory requirements within the timelines we announce, or at all, could result in claims and lawsuits, regulatory actions, or damage to our reputation, each of which may adversely affect our business, operations, financial condition, and results of operations.

INTELLECTUAL PROPERTY RISKS

We face risks related to the protection and utilization of our intellectual property that may result in our business and operating results being harmed. Protecting our intellectual property rights and combating unlicensed copying and use of our software, source code, and other intellectual property on a global basis is difficult. Similarly, the absence of harmonized patent laws makes it more difficult to ensure consistent respect for patent rights.

Changes in the law may continue to weaken our ability to prevent the use of patented technology. Our increasing engagement with open source software will also cause us to license our intellectual property rights broadly in certain situations. If we are unable to protect our intellectual property, our results of operations may be adversely affected.

Source code, the detailed program commands for our operating systems and other software programs, is critical to our business. If our source code leaks, we might lose future trade secret protection for that code. It may then become easier for third parties to compete with our products by copying functionality, which could adversely affect our results of operations. Unauthorized access to or disclosure of source code or other intellectual property also increases the security risks described elsewhere in these risk factors.

Third parties may claim that we infringe their intellectual property. From time to time, others claim we infringe their intellectual property rights, including current copyright infringement and other claims arising from AI training and output. To resolve these claims, we may enter into royalty-bearing data access or licensing agreements on terms that are less favorable than currently available, stop selling or redesign affected products or services, or pay damages to satisfy indemnification commitments with our customers. Adverse outcomes could also include monetary damages or injunctive relief that may limit or prevent importing, marketing, and selling our products or services that have infringing technologies. We have paid significant amounts to settle claims related to the use of technology and intellectual property rights and to procure intellectual property rights as part of our strategy to manage this risk, and may continue to do so, which could adversely affect our results of operations.

GENERAL RISKS

If our reputation or our brands are damaged, our business and results of operations may be harmed. Our reputation and brands are globally recognized and are important to our business. Our reputation and brands affect our ability to attract and retain consumer, business, and public-sector customers. There are numerous ways our reputation or brands could be damaged. These include product safety or quality issues, our environmental impact and sustainability, supply chain practices, or human rights record. We may experience backlash from customers, government entities, advocacy groups, employees, and other stakeholders that disagree with our product offering decisions, public policy positions, or corporate philanthropic initiatives. Damage to our reputation or our brands may occur from, among other things:

- The introduction of new features, products, services, or terms of service that customers, users, or partners do not like.
- Public scrutiny of our decisions regarding user privacy, data practices, content, or development and deployment of AI.
- Data security breaches, cybersecurity incidents, responsible AI failures, compliance failures, or actions of partners or individual employees.

Social media may increase the likelihood, speed, and magnitude of negative brand events. If our brands or reputation are damaged, it could adversely affect our business, results of operations, or ability to attract the most highly qualified employees.

Adverse economic or market conditions may harm our business. Worsening economic conditions, including inflation, recession, pandemic, or other changes in economic conditions, may cause lower IT spending and adversely affect our results of operations. If demand for PCs, servers, and other computing devices declines, or consumer or business spending for those products declines, our results of operations may be adversely affected.

Our product distribution system relies on an extensive partner and retail network. OEMs building devices that run our software have also been a significant means of distribution. The impact of economic conditions on our partners, such as the bankruptcy of a major distributor, OEM, or retailer, could cause sales channel disruption.

Challenging economic conditions also may impair the ability of our customers to pay for products and services they have purchased. As a result, allowances for doubtful accounts and write-offs of accounts receivable may increase.

We maintain an investment portfolio of various holdings, types, and maturities. These investments are subject to general credit, liquidity, market, and interest rate risks, which may be exacerbated by market downturns or events that affect global financial markets. A significant part of our investment portfolio comprises U.S. government securities. If global financial markets decline for long periods, or if there is a downgrade of the U.S. government credit rating due to an actual or threatened default on government debt, our investment portfolio may be adversely affected and we could determine that more of our investments have experienced a decline in fair value, requiring impairment charges that could adversely affect our financial condition and results of operations.

Catastrophic events or geopolitical conditions may disrupt our business. A disruption or failure of our systems, operations, or supply chain because of a major earthquake, weather event, cyberattack, terrorist attack, pandemic, or other catastrophic event could cause delays in completing sales, providing services, or performing other critical functions. Our corporate headquarters, a significant portion of our research and development activities, and certain other essential business operations are in the Seattle, Washington area, and we have other business operations in the Silicon Valley area of California, both of which are seismically active regions. A catastrophic event that results in the destruction or disruption of any of our critical business or systems, or the infrastructure or systems they rely on, such as power grids, could harm our ability to conduct normal business operations or adversely affect our results of operations. Providing our customers with more services and solutions in the cloud puts a premium on the resilience of our systems and strength of our business continuity management plans and magnifies the potential negative consequences of prolonged service outages.

Abrupt political change, terrorist activity, and armed conflict, such as the ongoing conflict in Ukraine, pose economic and other risks, which may negatively impact our ability to sell to and collect from customers, increase our operating costs, or otherwise disrupt our operations in markets both directly and indirectly impacted by such events. These conditions also may add uncertainty to the timing and budget for technology investment decisions by our customers and may cause supply chain disruptions for hardware manufacturers. Geopolitical change may result in changing regulatory systems and requirements and market interventions that could impact our operating strategies, access to national, regional, and global markets, hiring, and profitability. Geopolitical instability may lead to sanctions and impact our ability to do business in some markets or with some public-sector customers. Any of these changes may negatively affect our results of operations. Changes in geopolitical conditions also increase the security risks described elsewhere in these risk factors.

The occurrence of regional epidemics or a global pandemic, such as COVID-19, may adversely affect our business, operations, financial condition, and results of operations. The extent to which global pandemics impact our business going forward will depend on factors such as the duration and scope of the pandemic; governmental, business, and individuals' actions in response to the pandemic; and the impact on economic activity, including the possibility of recession or financial market instability. Measures to contain a global pandemic may intensify other risks described in these Risk Factors.

The long-term effects of climate change on the global economy and the IT industry in particular are unclear. Environmental regulations or changes in the supply, demand, or available sources of energy or other resources may affect the availability or cost of goods and services, including natural resources, necessary to run our business. Changes in climate where we operate may increase the costs of powering and cooling computer hardware we use to develop software and provide cloud-based services.

Our global business exposes us to operational and economic risks. Our customers, employees, and infrastructure are located throughout the world and a significant part of our revenue comes from international sales. The global nature of our business creates operational, economic, and geopolitical risks. Global, regional, and local economic developments, monetary policy, restrictions on international trade, such as tariffs and other controls on imports or exports, inflation, and recession, as well as political and military disputes, may adversely affect our results of operations. In addition, our international growth strategy includes certain markets, the developing nature of which presents several risks, including deterioration of social, political, labor, or economic conditions in a country or region, and difficulties in staffing and managing foreign operations. Emerging nationalist and protectionist trends and concerns about human rights, the environment, and political expression in specific countries may significantly alter the trade and commercial environments. Changes to trade policy or agreements as a result of populism, protectionism, or economic nationalism may result in higher tariffs, local sourcing initiatives, and non-local sourcing restrictions, export controls, investment restrictions, or other developments that make it more difficult to operate and sell our products in foreign countries. Disruptions of these kinds in developed or emerging markets could negatively impact demand for our products and services, impair our ability to operate in certain regions, or increase operating costs. Although we hedge a portion of our international currency exposure, significant fluctuations in foreign exchange rates between the U.S. dollar and foreign currencies may adversely affect our results of operations.

Our business depends on our ability to attract and retain talented employees. Our business is based on successfully attracting, training, and retaining talented employees representing diverse backgrounds, experiences, and skill sets. The market for highly skilled workers and leaders in our industry is extremely competitive. Maintaining our brand and reputation, as well as a diverse and inclusive work environment that enables all our employees to thrive, are important to our ability to recruit and retain employees. We are also limited in our ability to recruit internationally by restrictive domestic immigration laws. Restraints on the flow of technical and professional talent, including as a result of changes to U.S. immigration policies or laws, may inhibit our ability to adequately staff our research and development efforts. If we are less successful in our recruiting efforts, or if we cannot retain highly skilled workers and key leaders, our ability to develop and deliver successful products and services may be adversely affected. Effective succession planning is also important to our long-term success. Failure to ensure effective transfer of knowledge and smooth transitions involving key employees could hinder our strategic planning and execution. How employment-related laws are interpreted and applied to our workforce practices may result in increased operating costs and less flexibility in how we meet our workforce needs. Our global workforce is predominantly non-unionized, although we do have some employees in the U.S. and internationally who are represented by unions or works councils. In the U.S., there has been a general increase in workers exercising their right to form or join a union. The unionization of significant employee populations could result in higher costs and other operational changes necessary to respond to changing conditions and to establish new relationships with worker representatives.

ITEM 2. UNREGISTERED SALES OF EQUITY SECURITIES AND USE OF PROCEEDS

SHARE REPURCHASES AND DIVIDENDS

Following are our monthly share repurchases for the third quarter of fiscal year 2025:

Period	Total Number of Shares Purchased	Average Price Paid Per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs	Approximate Dollar Value of Shares That May Yet Be Purchased Under the Plans or Programs
				(In millions)
January 1, 2025 – January 31, 2025	**2,712,050**	**$ 427.65**	**2,712,050**	**$ 2,889**
February 1, 2025 – February 28, 2025	**2,814,771**	**408.03**	**2,814,771**	**1,741**
March 1, 2025 – March 31, 2025	**3,072,544**	**387.84**	**3,072,544**	**549**
	8,599,365		**8,599,365**	

All share repurchases were made using cash resources. Our share repurchases may occur through open market purchases or pursuant to a Rule 10b5-1 trading plan. The above table excludes shares repurchased to settle employee tax withholding related to the vesting of stock awards.

Our Board of Directors declared the following dividends during the third quarter of fiscal year 2025:

Declaration Date	Record Date	Payment Date	Dividend Per Share	Amount
				(In millions)
March 11, 2025	**May 15, 2025**	**June 12, 2025**	**$ 0.83**	**$ 6,170**

We returned $9.7 billion to shareholders in the form of share repurchases and dividends in the third quarter of fiscal year 2025. Refer to Note 15 – Stockholders' Equity of the Notes to Financial Statements (Part I, Item 1 of this Form 10-Q) for further discussion regarding share repurchases and dividends.

ITEM 5. OTHER INFORMATION

Insider Trading Arrangements

Our Section 16 officers and directors, as defined in Rule 16a-1(f) of the Securities Exchange Act of 1934 (the “Exchange Act”), may from time to time enter into plans for the purchase or sale of our common stock that are intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) of the Exchange Act. During the quarter ended March 31, 2025, the following Section 16 officers and directors adopted, modified, or terminated “Rule 10b5-1 trading arrangements” (as defined in Item 408 of Regulation S-K of the Exchange Act):

- Satya Nadella, our Chief Executive Officer and Chairman of the Board of Directors, adopted a new written 10b5-1 trading plan on March 7, 2025, during an open trading window. The plan’s maximum duration is until September 5, 2025. The first trade will not occur until September 2, 2025, at the earliest. Under the trading plan, Mr. Nadella will sell 80% of the net vested shares of our common stock upon the August 31, 2025 vest of a Performance Stock Award. The actual number of shares sold under the trading plan will depend upon achievement of previously disclosed performance metrics and a relative total shareholder return modifier applicable to the Performance Stock Award.

No other officers or directors, as defined in Rule 16a-1(f), adopted, modified, or terminated a “Rule 10b5-1 trading arrangement” or a “non-Rule 10b5-1 trading arrangement,” as defined in Item 408 of Regulation S-K, during the three months ended March 31, 2025.

ITEM 6. EXHIBITS

Exhibit Number	Exhibit Description	Filed Herewith	Incorporated by Reference			
			Form	Period Ending	Exhibit	Filing Date
15.1	Letter regarding unaudited interim financial information	X				
31.1	Certification of Chief Executive Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002	X				
31.2	Certification of Chief Financial Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002	X				
32.1*	Certification of Chief Executive Officer Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002	X				
32.2*	Certification of Chief Financial Officer Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002	X				
101.INS	Inline XBRL Instance Document–the instance document does not appear in the Interactive Data File as its XBRL tags are embedded within the Inline XBRL document	X				
101.SCH	Inline XBRL Taxonomy Extension Schema with Embedded Linkbase Documents	X				
104	Cover page formatted as Inline XBRL and contained in Exhibit 101	X				

* *Furnished, not filed.*

Items 3 and 4 are not applicable and have been omitted.

SIGNATURE

Pursuant to the requirements of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned; thereunto duly authorized.

MICROSOFT CORPORATION

/s/ ALICE L. JOLLA

Alice L. Jolla
Corporate Vice President and Chief Accounting Officer (Duly Authorized Officer)

April 30, 2025

Exhibit 15.1

April 30, 2025

The Board of Directors and Stockholders of Microsoft Corporation
One Microsoft Way
Redmond, WA 98052-6399

We are aware that our report dated April 30, 2025, on our review of the interim financial information of Microsoft Corporation and subsidiaries ("Microsoft") appearing in Microsoft's Quarterly Report on Form 10-Q for the quarter ended March 31, 2025, is incorporated by reference in Registration Statement Nos. 333-109185, 333-118764, 333-52852, 333-132100, 333-161516, 333-75243, 333-185757, and 333-221833 on Form S-8 and Registration Statement No. 333-283760 on Form S-3.

/s/ DELOITTE & TOUCHE LLP

Seattle, Washington

Exhibit 31.1

CERTIFICATION

I, Satya Nadella, certify that:

1. I have reviewed this quarterly report on Form 10-Q of Microsoft Corporation;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:

a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and

5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of registrant's Board of Directors (or persons performing the equivalent functions):

a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

/s/ SATYA NADELLA

Satya Nadella
Chief Executive Officer

April 30, 2025

Exhibit 31.2

CERTIFICATION

I, Amy E. Hood, certify that:

1. I have reviewed this quarterly report on Form 10-Q of Microsoft Corporation;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:

a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and

5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of registrant's Board of Directors (or persons performing the equivalent functions):

a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

/s/ AMY E. HOOD

Amy E. Hood
Executive Vice President and
Chief Financial Officer

April 30, 2025

Exhibit 32.1

CERTIFICATION PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002
(18 U.S.C. SECTION 1350)

In connection with the Quarterly Report of Microsoft Corporation, a Washington corporation (the “Company”), on Form 10-Q for the quarter ended March 31, 2025, as filed with the Securities and Exchange Commission (the “Report”), Satya Nadella, Chief Executive Officer of the Company, does hereby certify, pursuant to § 906 of the Sarbanes-Oxley Act of 2002 (18 U.S.C. § 1350), that to his knowledge:

(1) The Report fully complies with the requirements of section 13(a) or 15(d) of the Securities Exchange Act of 1934; and

(2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

/s/ SATYA NADELLA
Satya Nadella
Chief Executive Officer

April 30, 2025

[A signed original of this written statement required by Section 906 has been provided to Microsoft Corporation and will be retained by Microsoft Corporation and furnished to the Securities and Exchange Commission or its staff upon request.]

Exhibit 32.2

CERTIFICATION PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002
(18 U.S.C. SECTION 1350)

In connection with the Quarterly Report of Microsoft Corporation, a Washington corporation (the "Company"), on Form 10-Q for the quarter ended March 31, 2025, as filed with the Securities and Exchange Commission (the "Report"), Amy E. Hood, Chief Financial Officer of the Company, does hereby certify, pursuant to § 906 of the Sarbanes-Oxley Act of 2002 (18 U.S.C. § 1350), that to her knowledge:

(1) The Report fully complies with the requirements of section 13(a) or 15(d) of the Securities Exchange Act of 1934; and

(2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

/s/ AMY E. HOOD
Amy E. Hood
Executive Vice President and
Chief Financial Officer

April 30, 2025

[A signed original of this written statement required by Section 906 has been provided to Microsoft Corporation and will be retained by Microsoft Corporation and furnished to the Securities and Exchange Commission or its staff upon request.]

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
WASHINGTON, D.C. 20549

FORM 10-K

Annual report pursuant to Section 13 or 15(d) of the Securities Exchange Act of 1934

For the fiscal year ended	Commission file
December 31, 2024	number 1-5805

JPMorgan Chase & Co.

(Exact name of registrant as specified in its charter)

Delaware	13-2624428
(State or other jurisdiction of incorporation or organization)	(I.R.S. employer identification no.)
383 Madison Avenue,	
New York, New York	10179
(Address of principal executive offices)	(Zip Code)

Registrant's telephone number, including area code: (212) 270-6000
Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock	JPM	The New York Stock Exchange
Depositary Shares, each representing a one-four hundredth interest in a share of 5.75% Non-Cumulative Preferred Stock, Series DD	JPM PR D	The New York Stock Exchange
Depositary Shares, each representing a one-four hundredth interest in a share of 6.00% Non-Cumulative Preferred Stock, Series EE	JPM PR C	The New York Stock Exchange
Depositary Shares, each representing a one-four hundredth interest in a share of 4.75% Non-Cumulative Preferred Stock, Series GG	JPM PR J	The New York Stock Exchange
Depositary Shares, each representing a one-four hundredth interest in a share of 4.55% Non-Cumulative Preferred Stock, Series JJ	JPM PR K	The New York Stock Exchange
Depositary Shares, each representing a one-four hundredth interest in a share of 4.625% Non-Cumulative Preferred Stock, Series LL	JPM PR L	The New York Stock Exchange
Depositary Shares, each representing a one-four hundredth interest in a share of 4.20% Non-Cumulative Preferred Stock, Series MM	JPM PR M	The New York Stock Exchange
Guarantee of Callable Fixed Rate Notes due June 10, 2032 of JPMorgan Chase Financial Company LLC	JPM/32	The New York Stock Exchange
Guarantee of Alerian MLP Index ETNs due January 28, 2044 of JPMorgan Chase Financial Company LLC	AMJB	NYSE Arca, Inc.

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. ☐ Yes ☒ No

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. ☐ Yes ☒ No

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. ☒ Yes ☐ No

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). ☒ Yes ☐ No

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

☒ Large accelerated filer ☐ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐ Emerging growth company

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒ Yes ☐ No

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1 (b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). ☐ Yes ☒ No

The aggregate market value of JPMorgan Chase & Co. common stock held by non-affiliates as of June 30, 2024: $573,443,601,053

Number of shares of common stock outstanding as of January 31, 2025: 2,796,106,099

Documents incorporated by reference: Portions of the registrant's Proxy Statement for the annual meeting of stockholders to be held on May 20, 2025, are incorporated by reference in this Form 10-K in response to Items 10, 11, 12, 13 and 14 of Part III.

Form 10-K Index

Item 1. Business.

Overview

JPMorgan Chase & Co. ("JPMorganChase" or the "Firm", NYSE: JPM), a financial holding company incorporated under Delaware law in 1968, is a leading financial services firm based in the United States of America ("U.S."), with operations worldwide. JPMorganChase had $4.0 trillion in assets and $344.8 billion in stockholders' equity as of December 31, 2024. The Firm is a leader in investment banking, financial services for consumers and small businesses, commercial banking, financial transaction processing and asset management. Under the J.P. Morgan and Chase brands, the Firm serves millions of customers, predominantly in the U.S., and many of the world's most prominent corporate, institutional and government clients globally.

JPMorganChase's principal bank subsidiary is JPMorgan Chase Bank, National Association ("JPMorgan Chase Bank, N.A."), a national banking association with U.S. branches in 48 states and Washington, D.C. JPMorganChase's principal non-bank subsidiary is J.P. Morgan Securities LLC ("J.P. Morgan Securities"), a U.S. broker-dealer. The bank and non-bank subsidiaries of JPMorganChase operate nationally as well as through overseas branches and subsidiaries, representative offices and subsidiary foreign banks. The Firm's principal operating subsidiaries outside the U.S. are J.P. Morgan Securities plc and J.P. Morgan SE ("JPMSE"), which are subsidiaries of JPMorgan Chase Bank, N.A. and are based in the United Kingdom ("U.K.") and Germany, respectively.

The Firm's website is www.jpmorganchase.com. JPMorganChase makes available on its website, free of charge, annual reports on Form 10-K, quarterly reports on Form 10-Q and current reports on Form 8-K pursuant to Section 13(a) or Section 15(d) of the Securities Exchange Act of 1934, as soon as reasonably practicable after it electronically files or furnishes such material to the U.S. Securities and Exchange Commission (the "SEC") at www.sec.gov. JPMorganChase makes new and important information about the Firm available on its website at https://www.jpmorganchase.com, including on the Investor Relations section of its website at https://www.jpmorganchase.com/ir. Information on the Firm's website, including documents on the website that are referenced in this Form 10-K, is not incorporated by reference into this Annual Report on Form 10-K for the year ended December 31, 2024 ("2024 Form 10-K" or "Form 10-K") or the Firm's other filings with the SEC.

Business segments & Corporate

Effective in the second quarter of 2024, JPMorganChase reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank. As a result of the reorganization, the Firm has three reportable business segments – Consumer & Community Banking ("CCB"), Commercial & Investment Bank ("CIB") and Asset & Wealth Management ("AWM") – with the remaining activities in Corporate. The Firm's consumer business segment is CCB, and the Firm's wholesale business segments are CIB and AWM.

A description of the Firm's reportable business segments and the products and services that they provide to their respective client bases, as well as a description of Corporate activities, is provided in the Management's discussion and analysis of financial condition and results of operations section of this Form 10-K ("Management's discussion and analysis" or "MD&A") under the heading "Business Segment & Corporate Results," which begins on page 52, and in Note 32.

Competition

JPMorganChase and its subsidiaries and affiliates operate in highly competitive environments. Competitors include other banks, brokerage firms, investment banking companies, merchant banks, hedge funds, commodity trading companies, private equity firms, insurance companies, mutual fund companies, investment managers, credit card companies, mortgage banking companies, trust companies, securities processing companies, automobile financing companies, leasing companies, e-commerce and other internet-based companies, financial technology companies, and other companies engaged in providing similar and new products and services. The Firm's businesses generally compete on the basis of the quality and variety of the Firm's products and services, transaction execution, innovation, reputation and price. Competition also varies based on the types of clients, customers, industries and geographies served. With respect to some of its geographies and products, JPMorganChase competes globally; with respect to others, the Firm competes on a national or regional basis. New competitors in the financial services industry continue to emerge, including firms that offer products and services solely through the internet and non-financial companies that offer products and services that disintermediate traditional banking products and services offered by financial services firms such as JPMorganChase.

Supervision and regulation

The Firm is subject to extensive and comprehensive regulation under U.S. federal and state laws, as well as the applicable laws of the jurisdictions outside the U.S. in which the Firm does business.

Financial holding company:

Consolidated supervision. JPMorgan Chase & Co. is a bank holding company ("BHC") and a financial holding company ("FHC") under U.S. federal law, and is subject to comprehensive consolidated supervision, regulation and examination by the Board of Governors of the Federal Reserve System (the "Federal Reserve"). The Federal Reserve acts as the supervisor of the consolidated operations of BHCs. Certain of JPMorganChase's subsidiaries are also regulated directly by additional authorities based on the activities or licenses of those subsidiaries.

JPMorganChase's national bank subsidiary, JPMorgan Chase Bank, N.A., is supervised and regulated by the Office of the Comptroller of the Currency ("OCC") and, with respect to certain matters, by the Federal Deposit Insurance Corporation (the "FDIC").

JPMorganChase's U.S. broker-dealers are supervised and regulated by the Securities and Exchange Commission ("SEC") and the Financial Industry Regulatory Authority ("FINRA"). Subsidiaries of the Firm that engage in certain futures-related and swaps-related activities are supervised and regulated by the Commodity Futures Trading Commission ("CFTC"). J.P. Morgan Securities plc holds a banking license in the U.K. and is regulated by the U.K. Prudential Regulation Authority (the "PRA") and the U.K. Financial Conduct Authority ("FCA").

JPMSE is a Germany-based credit institution jointly regulated by the European Central Bank ("ECB"), the German Financial Supervisory Authority and the German Central Bank, as well as the local regulators in each of the countries in which it operates. The Firm's other non-U.S. subsidiaries are regulated by the banking, securities, prudential, payments and conduct regulatory authorities, as applicable, in the countries in which they operate.

Permissible business activities. The Bank Holding Company Act restricts BHCs from engaging in business activities other than the business of banking and certain closely-related activities. FHCs are permitted to engage in a broader range of financial activities. The Federal Reserve has the authority to limit an FHC's ability to conduct otherwise permissible activities if the FHC or any of its depository institution subsidiaries ceases to meet applicable eligibility requirements. The Federal Reserve may also impose corrective capital and/or managerial requirements on the FHC, and if deficiencies are persistent, may require divestiture of the FHC's depository institutions. If any depository institution controlled by an FHC fails to maintain a satisfactory rating under the Community Reinvestment Act, the Federal Reserve must prohibit the FHC and its subsidiaries from engaging in any new activities other than those permissible for BHCs, or acquiring a company engaged in such activities.

Capital and liquidity requirements. The Federal Reserve establishes capital, liquidity and leverage requirements for JPMorganChase that are generally consistent with the international Basel III capital and liquidity framework and evaluates the Firm's compliance with those requirements. The OCC establishes similar requirements for JPMorgan Chase Bank, N.A. Certain of the Firm's non-U.S. subsidiaries and branches are also subject to local capital and liquidity requirements.

Banking supervisors globally continue to refine and enhance the Basel III capital framework for financial institutions. In July 2023, U.S. banking regulators released a proposal to amend the U.S. risk-based capital framework to incorporate certain elements of the revised international Basel III capital framework. The proposal would significantly revise risk-based capital requirements for banks with assets of $100 billion or more, including the Firm and other U.S. global systemically important banks ("GSIBs"). Finalization of the proposal, including the required implementation date, is uncertain. The Firm continues to monitor developments and potential impacts.

In the EU and U.K., regulators have finalized the rules implementing their Basel III frameworks. The new rules became effective in the EU beginning January 1, 2025, with market risk aspects delayed until January 1, 2026. In January 2025, the PRA announced that it intends to delay the implementation of the new rules in the U.K. to January 1, 2027. There are certain transitional arrangements applicable in both the EU and U.K. until 2032 and 2030, respectively.

Stress tests. As a large BHC, JPMorganChase is subject to supervisory stress testing administered by the Federal Reserve as part of the Federal Reserve's annual Comprehensive Capital Analysis and Review ("CCAR") framework. The Firm must conduct annual company-run stress tests and must also submit an annual capital plan to the Federal Reserve, taking into account the results of separate stress tests designed by each of the Firm and the Federal Reserve. The Federal Reserve uses the results under the severely adverse scenario from its supervisory stress test to determine the Firm's Stress Capital Buffer ("SCB") requirement for the coming year, which forms part of the Firm's applicable capital buffers. The Firm is required to file its annual CCAR submission on April 5, 2025. The Federal Reserve will notify the Firm of its indicative SCB requirement by June 30, 2025 and final SCB requirement by August 31, 2025. The Firm's final

SCB requirement will become effective on October 1, 2025. The OCC requires JPMorgan Chase Bank, N.A. to perform separate, similar stress tests annually. The Firm publishes each year the results of the annual stress tests for the Firm and JPMorgan Chase Bank, N.A. under the supervisory "severely adverse" scenarios provided by the Federal Reserve and the OCC. In December 2024, the Federal Reserve indicated in a press release that it intends to seek public comment on changes to its stress testing framework. Additionally, there is a pending legal challenge to the manner in which stress testing is administered. Refer to Litigation and regulatory challenges on pages 6-7 for further information.

Refer to Capital Risk Management on pages 97–107 and Liquidity Risk Management on pages 108–115 for more information.

Enhanced prudential standards. As part of its mandate to identify and monitor risks to the financial stability of the U.S. posed by large banking organizations, the Financial Stability Oversight Council ("FSOC") recommends prudential standards and reporting requirements to the Federal Reserve for systemically important financial institutions ("SIFIs"), such as JPMorganChase. The Federal Reserve has adopted several rules to implement those heightened prudential standards, including rules relating to risk management and corporate governance of subject BHCs. JPMorganChase is required under these rules to comply with enhanced liquidity and overall risk management standards, including oversight by the board of directors of risk management activities.

Holding company as a source of strength. JPMorgan Chase & Co. is required to serve as a source of financial strength for its depository institution subsidiaries and to commit resources to support those subsidiaries, including when directed to do so by the Federal Reserve.

Regulation of acquisitions. Acquisitions by BHCs and their banks are subject to requirements, limitations and prohibitions established by law and by the Federal Reserve and the OCC. For example, FHCs and BHCs are required to obtain the approval of the Federal Reserve before they acquire more than 5% of the voting shares of an unaffiliated bank. In addition, acquisitions by financial companies are generally prohibited if, as a result of the acquisition, the total liabilities of the financial company would exceed 10% of the total liabilities of all financial companies, as determined under Federal Reserve regulations. Furthermore, for certain acquisitions, the Firm must provide written notice to the Federal Reserve prior to acquiring direct or indirect ownership or control of any voting shares of any company with over $10 billion in assets that is engaged in activities that are "financial in nature." Moreover, while FHCs may engage in a broader range of activities (including acquisitions) than BHCs, the Federal Reserve has the authority to limit an FHC's ability to conduct otherwise permissible acquisitions if the FHC or any of its depository institution subsidiaries ceases to meet applicable eligibility requirements.

Ongoing obligations. The Firm is subject to a five-year cooperation obligation under an order issued by the CFTC on September 29, 2020, relating to precious metals and U.S. Treasuries markets investigations. The Firm also remains subject to consent orders entered into in March 2024 with the OCC and the Board of Governors of the Federal Reserve System, and a resolution entered into in May 2024 with the CFTC, which relate to the Firm's processes to inventory trading venues and confirm the completeness of certain data fed to trade surveillance platforms.

Subsidiary banks:

The activities of JPMorgan Chase Bank, N.A., the Firm's principal subsidiary bank, are limited to those specifically authorized under the National Bank Act and related interpretations of the OCC. The OCC has authority to bring an enforcement action against JPMorgan Chase Bank, N.A. for unsafe or unsound banking practices, which could include limiting JPMorgan Chase Bank, N.A.'s ability to conduct otherwise permissible activities, or imposing corrective capital or managerial requirements on the bank.

FDIC deposit insurance. The FDIC deposit insurance fund provides insurance coverage for certain deposits and is funded through assessments on banks, including JPMorgan Chase Bank, N.A. The FDIC is required to maintain a minimum reserve ratio, which measures the balance of reserves in the deposit insurance fund against an estimate of FDIC-insured deposits, of 1.35%. The reserve ratio is currently below the statutory minimum and, in October 2022, the FDIC adopted a final rule to raise bank assessments and accelerate the time by which the reserve ratio would meet the statutory minimum. As a result, the FDIC has adopted a restoration plan to bring the reserve ratio up to the required 1.35% by September 30, 2028, with a longer-term target of maintaining a reserve ratio of 2%.

FDIC powers upon a bank insolvency. Upon any insolvency of JPMorgan Chase Bank, N.A., the FDIC could be appointed as conservator or receiver under the Federal Deposit Insurance Act. The FDIC has broad powers to transfer assets and liabilities without the approval of the institution's creditors.

Prompt corrective action. The Federal Deposit Insurance Corporation Improvement Act of 1991 requires the relevant federal banking regulator to take "prompt corrective action" with respect to a depository institution if that institution does not meet certain capital adequacy standards. The Federal Reserve is also authorized to take appropriate action against the parent BHC, such as JPMorgan Chase & Co., based on the undercapitalized status of any bank subsidiary. In certain instances, the BHC would be required to guarantee the performance of the capital restoration plan for its undercapitalized subsidiary.

Heightened Supervisory Standards. In the U.S., the OCC has established guidelines setting forth heightened standards for large banks, including minimum standards for the design and implementation of a risk governance framework for banks. Under these standards, a bank's risk governance framework must ensure that the bank's risk profile is easily distinguished and separate from that of its parent BHC for risk management purposes. The bank's board or risk committee is responsible for approving the bank's risk governance framework, providing active oversight of the bank's risk-taking activities, and holding management accountable for adhering to the risk governance framework.

The Firm's banking entities in the EU and the U.K. are subject to supervisory expectations published by the ECB and the PRA, respectively, addressing bank strategy, governance and risk management in the areas of climate change, operational resilience, reliance on IT systems and third-party services, and resilience from macro-financial and geopolitical shocks.

Restrictions on transactions with affiliates. JPMorgan Chase Bank, N.A. and its subsidiaries are subject to restrictions imposed by federal law on extensions of credit to, investments in stock or securities of, and derivatives, securities lending and certain other transactions with, JPMorgan Chase & Co. and certain other affiliates. These restrictions prevent JPMorgan Chase & Co. and other affiliates from borrowing from JPMorgan Chase Bank, N.A. and its subsidiaries unless the loans are secured in specified amounts and comply with certain other requirements.

Dividend restrictions. Federal law imposes limitations on the payment of dividends by national banks, such as JPMorgan Chase Bank, N.A. Refer to Note 26 for the amount of dividends that JPMorgan Chase Bank, N.A. could pay, at January 1, 2025, to JPMorganChase without the approval of the banking regulators. The OCC and the Federal Reserve also have authority to prohibit or limit the payment of dividends of a bank subsidiary that they supervise if, in the banking regulator's opinion, payment of a dividend would constitute an unsafe or unsound practice in light of the financial condition of the bank.

Depositor preference. Under federal law, the claims of a receiver of an insured depositary institution ("IDI") for administrative expense and the claims of holders of U.S. deposit liabilities (including the FDIC and deposits in non-U.S. branches that are dually payable in the U.S. and in a non-U.S. branch) have priority over the claims of other unsecured creditors of the institution, including depositors in non-U.S. branches and public noteholders.

Consumer supervision and regulation. JPMorganChase and JPMorgan Chase Bank, N.A. are subject to supervision and regulation in the U.S. by the Consumer Financial Protection Bureau ("CFPB") with respect to federal consumer protection laws, including laws relating to fair lending and the prohibition of unfair, deceptive or abusive acts or practices in connection with the offer, sale or provision of consumer financial products and services. The CFPB also has jurisdiction over small business lending activities with respect to fair lending and the Equal Credit Opportunity Act. As part of its regulatory oversight, the CFPB has authority to take enforcement actions against firms that offer certain products and services to consumers using practices that are deemed to be unfair, deceptive or abusive. In March 2024, the CFPB released a final rule which significantly reduces the late payment fees that large credit card issuers, including the Firm, are permitted to charge to customers (the "CFPB Late Fee Rule"). The final rule is currently stayed pending resolution of an industry-led challenge in federal court. Refer to Litigation and regulatory challenges on pages 6-7 for further information.

In addition, in October 2024, the CFPB issued a final rule that requires data providers, including banks, to make certain consumer data available to consumers and authorized third parties in electronic form beginning in April 2026 (the "CFPB Data Sharing Rule"). Refer to Litigation and regulatory challenges on pages 6-7 for further information.

Separately, in December 2024, the CFPB announced a final rule that would significantly restrict overdraft fees for certain insured depository institutions, including the Firm (the "CFPB Overdraft Rule"). The rule imposes certain requirements on overdraft protections that are similar to those that apply to credit cards, unless the institution prices the overdraft fee at $5 per transaction or the institution's cost as outlined by the CFPB. Refer to Litigation and regulatory challenges on pages 6-7 for further information.

In October 2023, the Federal Reserve Board proposed to lower the maximum interchange fee that large debit card issuers, including the Firm, would be permitted to receive for a debit card transaction. The proposal would also establish a process for automatically publishing an updated maximum fee amount every other year going forward. The Firm's consumer activities are also subject to regulation under state statutes which are enforced by the Attorney General or empowered agency of each state.

In the U.K., the Firm operates a retail bank through J.P. Morgan Europe Limited ("JPMEL") and provides retail investment management services through Nutmeg Saving and Investment Limited ("Nutmeg"). JPMEL is regulated by the PRA, and both JPMEL and Nutmeg are regulated by the FCA with respect to their conduct of financial services in the U.K., including obligations relating to the fair treatment of customers. JPMEL is also regulated by the U.K. Payment Systems Regulator with respect to its operation and use of payment systems. In addition, the retail businesses of JPMEL and Nutmeg are subject to U.K. consumer-protection legislation.

Securities and broker-dealer regulation:
The Firm conducts securities underwriting, dealing and brokerage activities in the U.S. through J.P. Morgan Securities LLC and other non-bank broker-dealer subsidiaries, all of which are subject to regulations of the SEC, FINRA and the New York Stock Exchange, among others. The Firm conducts similar securities activities outside the U.S. subject to local regulatory requirements. In the U.K., those activities are primarily conducted by J.P. Morgan Securities plc and in the EU, those activities are primarily conducted by JPMSE. Broker-dealers are subject to laws and regulations covering all aspects of the securities business, including sales and trading practices, securities offerings, publication of research reports, use of customer funds, the financing of client purchases, capital structure, record-keeping and retention, and the conduct of their directors, officers and employees. Refer to Broker-dealer regulatory capital on page 107 for information concerning the capital of J.P. Morgan Securities LLC, J.P. Morgan Securities plc and JPMSE. In addition, the Firm's sales and trading activities, which are conducted through both bank and non-bank subsidiaries, are subject to laws and regulations relating to market conduct, including prohibitions on manipulative or anti-competitive practices.

Investment management regulation:
The Firm's asset and wealth management businesses are subject to significant regulation in jurisdictions around the world relating to, among other things, the safeguarding and management of client assets, offerings of funds and marketing activities. Certain of the Firm's subsidiaries are registered with, and subject to oversight by, the SEC as investment advisers and broker-dealers. The Firm's registered investment advisers in the U.S. are subject to the fiduciary and other obligations imposed under the Investment Advisers Act of 1940 and applicable state and federal law. The Firm's bank fiduciary activities are subject to supervision by the OCC.

The Firm's asset and wealth management businesses continue to be subject to ongoing rule-making and implementation of new regulations and other guidance, including by the SEC and certain U.S. states with respect to enhanced standards of conduct and conflicts of interest. In April 2024, the Department of Labor ("DOL") finalized a new "fiduciary" rule that would significantly expand the scope for defining who can be deemed investment advice fiduciaries for purposes of retirement plans and individual retirement accounts ("IRAs") under the Employee Retirement Income Security Act of 1974, as amended (the "Fiduciary Rule"). Among the most significant impacts of the rule and related amendments to prohibited transaction exemptions would be the impact on the fee and compensation practices at financial institutions that offer investment recommendations to retirement clients, including in the context of rollovers from an employer plan to an IRA. The effective date of the Fiduciary Rule has been stayed by two federal courts. Refer to Litigation and regulatory challenges on pages 6-7 for further information.

Derivatives regulation:
The Firm is subject to comprehensive regulation of its derivatives businesses. In the U.S., JPMorgan Chase Bank, N.A., J.P. Morgan Securities LLC and J.P. Morgan Securities plc are registered with the CFTC as "swap dealers". In addition, JPMorgan Chase Bank, N.A. and J.P. Morgan Securities LLC are registered with the SEC as "security-based swap dealers". As a result, these entities are subject to a comprehensive regulatory framework applicable to their swap or security-based swap activities, including capital requirements, rules requiring the collateralization of uncleared swaps and security-based swaps, rules regarding segregation of counterparty collateral, business conduct and documentation standards, rules requiring the central clearing of standardized over-the-counter ("OTC") derivatives, requirements that certain standardized OTC swaps be traded on regulated trading venues, record-keeping and reporting obligations, and anti-fraud and anti-manipulation requirements. Similar requirements have also been established in the European Union ("EU") under the European Market Infrastructure Regulation ("EMIR") and the Markets in Financial Instruments Directive ("MiFID II"), as well as in the U.K. and other jurisdictions around the world.

J.P. Morgan Securities LLC is also registered with the CFTC as a futures commission merchant and is a member of the National Futures Association.

Data, privacy, cybersecurity and artificial intelligence regulation:
The Firm and its subsidiaries are subject to laws, rules and regulations globally concerning data, including data protection, consumer protection, privacy, cybersecurity, artificial intelligence and related

matters. These laws, rules and regulations are constantly evolving, subject to interpretation, remain a focus of regulators globally, may be enforced by private parties or government bodies, and continue to have a significant impact on all of the Firm's businesses and operations.

For example, the Digital Operational Resilience Act (DORA) mandates that the Firm's financial services subsidiaries operating in the EU comply with requirements relating to information and communications technology ("ICT") risk management, reporting, security control testing and ICT third party risks beginning in January 2025. In addition, the EU Artificial Intelligence Act regulates the development and deployment of artificial intelligence systems within the EU, with phased-in requirements that began in February 2025.

The Bank Secrecy Act and Economic Sanctions:
The Bank Secrecy Act ("BSA") requires all financial institutions, including banks and securities broker-dealers, to establish a risk-based system of internal controls reasonably designed to prevent money laundering and the financing of terrorism. The BSA includes a variety of record-keeping and reporting requirements, as well as due diligence/know-your-customer documentation requirements. The Firm is also subject to the regulations and economic sanctions programs administered and enforced by the U.S. Treasury's Office of Foreign Assets Control ("OFAC") and EU and U.K. authorities which target entities or individuals that are, or are located in countries that are, involved in activities including terrorism, hostilities, embezzlement or human rights violations. The Firm is also subject to economic sanctions laws, rules and regulations in other jurisdictions in which it operates, including those that conflict with or prohibit a firm such as JPMorganChase from complying with certain laws, rules and regulations to which it is otherwise subject.

Anti-Corruption:
The Firm is subject to laws and regulations relating to corrupt and illegal payments to government officials and others in the jurisdictions in which it operates, including the U.S. Foreign Corrupt Practices Act and the U.K. Bribery Act.

Compensation practices:
The Firm's compensation practices are subject to oversight by the Federal Reserve, as well as other agencies. The Federal Reserve has jointly issued guidance with the FDIC and the OCC that is designed to ensure that incentive compensation paid by banking organizations does not encourage imprudent risk-taking that threatens the organizations' safety and soundness. The Financial Stability Board ("FSB") has also established standards covering compensation principles for banks. The Firm's compensation practices are also subject to regulation and oversight by regulators in other jurisdictions, notably the Fifth Capital Requirements Directive ("CRD V"), as implemented in the EU and as largely adopted in the U.K, which includes compensation-related provisions. The European Banking Authority has instituted guidelines on compensation policies including under CRD V which in certain countries (such as Germany) are implemented or supplemented by local regulations or guidelines. The U.K. regulators have also instituted regulations and guidelines on compensation policies, which diverge in certain areas from EU rules. The Firm expects that the implementation of regulatory guidelines regarding compensation in the U.S. and other countries will continue to evolve, and may affect the manner in which the Firm structures its compensation programs and practices.

Sustainability:
Policymakers in the U.K. and the EU have continued to implement and enhance sustainability-related initiatives and disclosure requirements. The Corporate Sustainability Reporting Directive ("CSRD") will replace and significantly expand the scope and content of certain EU ESG reporting requirements, with phased-in requirements expected to start with fiscal year 2024. The implementation of CSRD into local law has been delayed in a number of member states, including in Germany, and the Firm continues to monitor developments and potential impacts. In addition, in July 2024, the EU enacted the Corporate Sustainability Due Diligence Directive ("CSDDD"), which provides for phased-in requirements starting in 2027. The CSDDD sets mandatory due diligence obligations for companies to address actual and potential human rights violations and environmental adverse impacts stemming from their own operations and business relationships, including the activities of certain companies with which they have established business relationships and also requires the adoption of company-specific climate-related transition plans. Both the CSRD and CSDDD will impact certain of the Firm's EU and non-EU entities.

Litigation and regulatory challenges:
Trade organizations representing the financial services industry and others have filed lawsuits challenging various laws, rules and regulations that, if enacted, adopted or implemented, could have significant adverse impacts on the results of operations or compliance costs of financial institutions, including the Firm. These matters include:

- **Stress tests:** The Bank Policy Institute ("BPI"), the U.S. Chamber of Commerce and other trade organizations filed an action against the Federal Reserve in the United States District Court for the Southern District of Ohio in December 2024 challenging the manner in which the annual stress testing process is administered.

- **CFPB Late Fee Rule**: This rule has been stayed pending resolution of an action challenging the rule filed against the CFPB in the United States District Court for the Northern District of Texas in March 2024 by trade organizations including the American Bankers Association and the Consumer Bankers Association.
- **CFPB Data Sharing Rule**: The BPI, the Kentucky Bankers Association and other organizations filed an action against the CFPB in the United States District Court for the Eastern District of Kentucky in October 2024 challenging key aspects of this rule.
- **CFPB Overdraft Rule**: An action filed by trade organizations led by the Mississippi Bankers Association against the CFPB in the United States District Court for the Southern District of Mississippi in December 2024 seeks a preliminary injunction to stay the October 1, 2025 effective date of this rule. The CFPB has consented in part to stay the effective date of the rule by 90 days and to temporarily stay the litigation. The preliminary injunction and the stay of litigation are pending court approval.
- **Fiduciary Rule**: Trade organizations including the Federation of Americans for Consumer Choice and the American Council of Life Insurers filed actions against the DOL seeking to enjoin this rule, and in July 2024, the effective date of the rule was stayed by two United States District Courts.

Human capital

JPMorganChase believes that its long-term growth and success depend on its ability to attract, develop and retain talented employees and foster an inclusive work environment. The information provided below relates to JPMorganChase's full-time and part-time employees and does not include the Firm's contractors.

Global workforce

As of December 31, 2024, JPMorganChase had 317,233 employees globally, an increase of 7,307 employees from the prior year. The increase was primarily attributable to growth in the number of front office and technology employees. JPMorganChase's employees are located in 66 countries, with 59% of the Firm's employees located in the U.S. The following table presents the distribution of the Firm's global workforce by region and by line of business ("LOB") and Corporate as of December 31, 2024:

Employee Breakdown by Region	
Region	**Employees**
North America	187,179
Asia-Pacific	93,941
Europe/Middle East/Africa	30,729
Latin America/Caribbean	5,384
Total Firm	**317,233**

Employee Breakdown by LOB and Corporate	
LOB	**Employees**
CCB	144,989
CIB	93,231
AWM	29,403
Corporate	49,610
Total Firm	**317,233**

Workforce composition

The following table presents information based on voluntary self-identifications by the Firm's employees, including members of the Firm's Operating Committee and other senior level employees, as well as members of the Board of Directors, as of December 31, 2024. Information on race/ethnicity of employees is categorized based on Equal Employment Opportunity ("EEO") classifications and is presented for U.S. employees who self-identified, and information on gender is presented for global employees who self-identified. Information on race/ethnicity and gender for members of the Operating Committee and the Board of Directors reflects all such members. Information on LGBTQ+ and veteran statuses is based on all U.S. employees, and all members of the Operating Committee and the Board of Directors. Information on disability status is based on all U.S. employees and all members of the Operating Committee.

December 31, 2024	**Total employees**	**Senior level employees**[e]	**Operating Committee**	**Board of Directors**[f]
Race/Ethnicity[a]:				
White	43 %	74 %	86 %	80 %
Hispanic	21 %	6 %	7 %	—
Asian	20 %	14 %	7 %	—
Black	13 %	5 %	—	20 %
Other[b]	3 %	1 %	—	—
Gender[c]:				
Men	51 %	71 %	53 %	50 %
Women	49 %	29 %	47 %	50 %
LGBTQ+[d]	4 %	2 %	7 %	—
Military veterans[d]	3 %	2 %	—	10 %
People with disabilities[d]	5 %	3 %	—	— [g]

(a) Based on EEO metrics. Presented as a percentage of the respective populations who self-identified race/ethnicity, which was 97% and 95% of the Firm's total U.S.-based employees and U.S.-based senior level employees, respectively, and all members of the Operating Committee and the Board of Directors. Information for the Operating Committee includes one member who is based outside of the U.S.
(b) Other includes American Indian or Alaskan Native, Native Hawaiian or Other Pacific Islander, and two or more races/ethnicities.
(c) Presented as a percentage of the respective populations who self-identified gender, which was 99% of each of the Firm's total global employees and senior level employees, and all members of the Operating Committee and the Board of Directors.
(d) Presented as a percentage of total U.S.-based employees, total U.S.-based senior level employees, all members of the Operating Committee, and all members of the Board of Directors, respectively.
(e) Senior level employees represents employees with the titles of Managing Director and above.
(f) Excludes Brad D. Smith and Michele G. Buck, who were elected to the Firm's Board of Directors, effective January 21, 2025 and March 17, 2025, respectively.
(g) The Firm has not asked members of the Board of Directors to self-identify disability status.

Attracting and retaining employees
The goal of JPMorganChase's recruitment efforts is to attract and hire highly qualified candidates in all roles and at all career levels. The Firm's hiring practices focus on the skills and qualifications of a candidate relative to the job requirements.

The Firm strives to provide both external candidates and internal employees who are seeking a different role with challenging and stimulating career opportunities. These opportunities range from internship training programs for students to entry-level, management and executive careers. During 2024, approximately 56% of the Firm's employment opportunities were filled by external candidates, with the remainder filled by existing employees.

Developing employees
JPMorganChase supports the professional development and career growth of its employees. The Firm requires that its employees, including new hires, complete a training curriculum which covers, among other topics, compliance with the Firm's Code of Conduct and information concerning Firm policies and standards, including those relating to cybersecurity. In addition, the Firm offers extensive voluntary training programs and educational resources to all employees covering a broad variety of topics such as leadership and management, artificial intelligence, data literacy and operational and professional skills. Leadership Edge, the Firm's global leadership and management development center of excellence, is focused on creating one Firmwide leadership culture.

Compensation and benefits
The Firm provides market-competitive compensation and benefits programs. JPMorganChase's compensation philosophy includes guiding principles that drive compensation-related decisions across the Firm, and includes: pay-for-performance practices designed to attract and retain top talent; responsiveness and alignment with shareholder interests; and reinforcement of the Firm's culture. The Firm follows a disciplined and balanced compensation framework, including the integration of risk, controls and conduct considerations. The Firm's compensation review processes seek to ensure that the Firm's employees are paid fairly and competitively for the work they do.

JPMorganChase offers extensive benefits and wellness packages to support employees and their families, which vary depending on location and include healthcare coverage, retirement benefits, life and disability insurance, access to on-site health and wellness centers, counseling and resources related to mental health, time away policies, child care access and support, tuition assistance, and financial coaching.

Item 1A. Risk Factors.

The following discussion sets forth the material risk factors that could affect JPMorganChase's financial condition and operations. Readers should not consider any descriptions of these factors to be a complete set of all potential risks that could affect the Firm. Any of the risk factors discussed below could by itself, or combined with other factors, materially and adversely affect JPMorganChase's business, results of operations, financial condition, capital position, liquidity, competitive position or reputation, including by materially increasing expenses or decreasing revenues, which could result in material losses or a decrease in earnings.

Summary

The principal risk factors that could adversely affect JPMorganChase's business, results of operations, financial condition, capital position, liquidity, competitive position or reputation include:

- **Regulatory** risks, including the impact that applicable laws, rules and regulations in the highly-regulated and supervised financial services industry, as well as changes to or in the application, interpretation or enforcement of those laws, rules and regulations, can have on JPMorganChase's business and operations, including JPMorganChase incurring additional costs associated with assessments, levies or other governmental charges; the ways in which differences in financial services regulation and supervision in different jurisdictions or with respect to certain competitors can negatively impact JPMorganChase's business; the ways in which governmental policies that discourage or penalize business relationships with clients in certain industries, or require specific business practices, can negatively affect JPMorganChase's businesses; the penalties and collateral consequences, and higher compliance and operational costs, that JPMorganChase may incur when resolving a regulatory investigation; the ways in which less predictable legal and regulatory frameworks in certain jurisdictions can negatively impact JPMorganChase's operations and financial results; and the losses that security holders and other unsecured creditors will absorb if JPMorganChase were to enter into a resolution.
- **Political** risks, including the potential negative effects on JPMorganChase's businesses due to economic uncertainty or instability caused by political developments.
- **Market** risks, including the effects that economic and market events and conditions, political developments, changes in interest rates and credit spreads, and market fluctuations can have on JPMorganChase's consumer and wholesale businesses and its investment and market-making positions and on JPMorganChase's earnings and its liquidity and capital levels.
- **Credit** risks, including potential negative effects from adverse changes in the financial condition of clients, customers, counterparties, custodians and central counterparties; the potential for losses due to declines in the value of collateral in stressed market conditions; and potential negative impacts from concentrations of credit risk with respect to clients, customers, counterparties and other market participants.
- **Liquidity** risks, including the risk that JPMorganChase's liquidity could be impaired by market-wide illiquidity or disruption, unforeseen liquidity or capital requirements, the inability to sell assets, default by a significant market participant, unanticipated outflows of cash or collateral, or lack of market or customer confidence in JPMorganChase; the dependence of JPMorgan Chase & Co. on the cash flows of its subsidiaries; and the potential adverse effects that any downgrade in any of JPMorganChase's credit ratings may have on its liquidity and cost of funding.
- **Capital** risks, including the risk that any failure by or inability of JPMorganChase to maintain the required level and composition of capital, or unfavorable changes in applicable capital requirements, could limit JPMorganChase's ability to distribute capital to shareholders or to support its business activities.
- **Operational** risks, including risks associated with JPMorganChase's dependence on its operational systems and its employees, as well as the systems and employees of third parties, market participants and service providers; the potential negative effects of failing to identify and address operational risks related to the failure of internal or external operational systems, the introduction of or changes to products, services and delivery platforms or the adoption of new technologies; risks related to safeguarding personal information; the harm that could be caused by a successful cyber attack affecting JPMorganChase or by other extraordinary events; risks associated with JPMorganChase's risk management framework and control environment, its models and estimations and associated judgments used in its stress testing and financial statements, and controls over disclosure and financial reporting; and potential adverse effects of failing to comply with applicable standards for the oversight of vendors and other service providers.
- **Strategic** risks, including the damage to JPMorganChase's competitive standing and results that could occur if management fails to develop and execute effective business strategies; risks associated with the significant and increasing competition that JPMorganChase faces; and the potential adverse impacts of climate change on the

effectiveness of JPMorganChase's existing business strategies with respect to its operations, clients and customers.

- **Conduct** risks, including the negative impact that can result from the actions or misconduct of employees, including any failure of employees to conduct themselves in accordance with JPMorganChase's expectations, policies and practices.
- **Reputation** risks, including the potential adverse effects on JPMorganChase's relationships with its clients, customers, shareholders, regulators and other stakeholders that could arise from employee misconduct, security breaches, inadequate risk management, compliance or operational failures, litigation and regulatory investigations, failure to satisfy expectations concerning environmental, social and governance concerns, failure to effectively manage conflicts of interest or to satisfy fiduciary obligations, or other factors that could damage JPMorganChase's reputation.
- **Country** risks, including potential impacts on JPMorganChase's businesses from an outbreak or escalation of hostilities between countries or within a country or region; and the potential adverse effects of local economic, political, regulatory and social factors on JPMorganChase's business and revenues in certain countries in which it operates.
- **People** risks, including the criticality of attracting and retaining qualified employees; and the potential adverse effects of unfavorable changes in immigration or travel policies on JPMorganChase's workforce.
- **Legal** risks, including those relating to litigation and regulatory and government investigations.

The above summary is subject in its entirety to the discussion of the risk factors set forth below.

Regulatory

JPMorganChase's businesses are highly regulated, and the laws, rules and regulations that apply to JPMorganChase have a significant impact on its business and operations.

JPMorganChase is a financial services firm with operations worldwide. JPMorganChase must comply with the laws, rules and regulations that apply to its operations in all of the jurisdictions around the world in which it does business, and financial services firms such as JPMorganChase are subject to extensive and constantly-evolving regulation and supervision.

The regulation and supervision of JPMorganChase significantly affects the way that it conducts its business and structures its operations, and JPMorganChase could be required to make changes to its business and operations in response to supervisory expectations or decisions or to new or changed laws, rules and regulations. These types of developments could result in JPMorganChase incurring additional costs or experiencing a reduction in revenues to comply with applicable laws, rules and regulations, which could reduce its profitability. Furthermore, JPMorganChase's entry into or acquisition of a new business or an increase in its principal investments may require JPMorganChase to comply with additional laws, rules, and regulations.

Additionally, JPMorganChase's ability to execute certain business initiatives could become more challenging due to increased regulation in the financial services industry, such as limitations on late payment, overdraft and interchange fees. This could adversely affect JPMorganChase's earnings from its consumer businesses, prompting the reevaluation or adjustment of certain businesses or product offerings, as well as the reallocation of resources and incurrence of restructuring costs, which could impact revenue and profitability in the affected lines of business.

In response to new and existing laws, rules and regulations and expanded supervision, JPMorganChase has in the past been and could in the future be, required to:

- limit the products and services that it offers
- reduce the liquidity that it can provide through its market-making activities
- refrain from engaging in business opportunities that it might otherwise pursue
- pay higher taxes (including as part of any minimum global tax regime), assessments, levies or other governmental charges, including in connection with the resolution of tax examinations
- incur losses, including with respect to fraudulent transactions perpetrated against its customers
- dispose of certain assets, and do so at times or prices that are disadvantageous
- impose restrictions on certain business activities, or
- increase the prices that it charges for products and services, which could reduce the demand for them.

Any failure by JPMorganChase to comply with the laws, rules and regulations to which it is subject could result in:

- increased regulatory and supervisory scrutiny
- regulatory and governmental enforcement actions
- the imposition of fines, penalties or other sanctions
- increased exposure to litigation, or
- harm to its reputation.

Differences and inconsistencies in financial services regulation and supervision can negatively impact JPMorganChase's businesses, operations and financial results.

The content and application of laws, rules and regulations affecting financial services firms can vary according to factors such as the size of the firm, the jurisdiction in which it is organized or operates, and other criteria. For example:

- larger firms such as JPMorganChase are often subject to more stringent supervision, regulation and regulatory scrutiny
- financial technology companies and other non-traditional competitors may not be subject to banking regulation, or may be supervised by a national or state regulatory agency that does not have the same resources or regulatory priorities as the regulatory agencies that supervise more diversified financial services firms, or
- the financial services regulatory and supervisory framework in a particular jurisdiction may favor financial institutions that are based in that jurisdiction.

These types of differences in the regulatory and supervisory framework can result in JPMorganChase losing market share to competitors that are less regulated or not subject to regulation, especially with respect to unregulated financial products.

There can also be significant differences in the ways that similar regulatory initiatives affecting the financial services industry are implemented in the U.S. and in other countries and regions in which JPMorganChase does business. For example, when adopting rules that are intended to implement a global regulatory or supervisory standard, a national regulator may introduce additional or more restrictive requirements, which can create competitive disadvantages for financial services firms, such as JPMorganChase, that may be subject to those enhanced regulations.

In addition, certain national and multi-national bodies and governmental agencies outside the U.S. have adopted laws, rules or regulations that may conflict with or prohibit JPMorganChase from complying with laws, rules and regulations to which it is otherwise subject, creating conflict of law issues that also increase its risk of non-compliance in those jurisdictions.

Legislative and regulatory initiatives outside the U.S. have required and could in the future require JPMorganChase to make significant modifications to its operations and legal entity structure in the relevant countries or regions in order to comply with those requirements. These include laws, rules and regulations that have been adopted or proposed, as well as regulatory expectations, relating to:

- the establishment of locally-based intermediate holding companies or operating subsidiaries
- requirements to maintain minimum amounts of capital or liquidity in locally-based subsidiaries
- the implementation of processes within locally-based subsidiaries to comply with local regulatory requirements or expectations
- the separation (or "ring fencing") of core banking products and services from markets activities
- requirements for the orderly resolution of financial institutions
- requirements for executing or settling transactions on exchanges or through central counterparties ("CCPs"), or for depositing funds with other financial institutions or clearing and settlement systems
- position limits and reporting rules for derivatives
- governance and accountability regimes
- conduct of business and control requirements, and
- restrictions on compensation.

These types of differences, inconsistencies and conflicts in financial services regulation have required and could in the future require JPMorganChase to:

- divest assets or restructure its operations
- maintain higher levels of capital and liquidity, or absorb increased capital and liquidity costs
- incur higher operational and compliance costs
- change the prices that it charges for its products and services
- curtail the products and services that it offers to its customers and clients
- curtail other business opportunities, including acquisitions or principal investments, that it otherwise would have pursued
- become subject to regulatory fines, penalties or other sanctions, or
- incur higher costs for complying with different legal and regulatory frameworks.

Any or all of these factors could harm JPMorganChase's ability to compete against other firms that are not subject to the same laws, rules and regulations or supervisory oversight, or harm JPMorganChase's businesses, results of operations and profitability.

Resolving regulatory investigations can subject JPMorganChase to significant penalties and collateral consequences, and could result in higher compliance costs or restrictions on its operations.

JPMorganChase is subject to heightened oversight and scrutiny from regulatory authorities in many jurisdictions. JPMorganChase has paid significant fines, provided other monetary relief, incurred other penalties and experienced other repercussions in connection with resolving investigations and enforcement actions by governmental agencies. JPMorganChase could become subject to similar regulatory or governmental resolutions or other actions in the future, and addressing the requirements of any such resolutions or actions could result in JPMorganChase incurring higher operational and compliance costs, including devoting substantial resources to the required remediation or needing to comply with other restrictions.

In connection with resolving specific regulatory investigations or enforcement actions, certain regulators have required JPMorganChase and other financial institutions to admit wrongdoing with respect to the activities that gave rise to the resolution. These types of admissions can lead to:

- greater exposure in litigation
- damage to JPMorganChase's reputation
- disqualification from doing business with certain clients or customers, or in specific jurisdictions, or
- other direct and indirect adverse effects.

Furthermore, government officials in the U.S. and other countries have demonstrated a willingness to bring criminal actions against financial institutions and have required that institutions plead guilty to criminal offenses or admit other wrongdoing in connection with resolving regulatory investigations or enforcement actions. Resolutions of this type can have significant collateral consequences for the subject financial institution, including:

- loss of clients, customers and business
- restrictions on offering certain products or services, and
- losing permission to operate certain businesses, either temporarily or permanently.

JPMorganChase expects that:

- it and other financial services firms will continue to be subject to heightened regulatory scrutiny and governmental investigations and enforcement actions
- governmental authorities will continue to require that financial institutions be penalized for actual or deemed violations of law with formal and punitive enforcement actions, including the imposition of significant monetary and other sanctions, rather than resolving these matters through informal supervisory actions, and
- governmental authorities will be more likely to pursue formal enforcement actions and resolutions against JPMorganChase to the extent that it has previously been subject to other governmental investigations or enforcement actions.

If JPMorganChase fails to meet the requirements of any resolution of a governmental investigation or enforcement action, or to maintain risk and control processes that meet the heightened standards and expectations of its regulators, it could be required to, among other things:

- enter into further resolutions of investigations or enforcement actions
- pay additional regulatory penalties or enter into judgments, or
- accept material regulatory restrictions on, or changes in the management of, its businesses.

In these circumstances, JPMorganChase could also become subject to other sanctions, or to prosecution or civil litigation with respect to the conduct that gave rise to an investigation or enforcement action. In addition, JPMorganChase can be subject to higher costs or requests for additional capital in connection with the resolution of governmental investigations and enforcement actions involving newly-acquired businesses, companies in which JPMorganChase has made principal investments, parties to joint ventures with JPMorganChase, and vendors with which JPMorganChase does business.

JPMorganChase's operations and financial results can be negatively impacted in jurisdictions with less predictable legal and regulatory frameworks.

JPMorganChase conducts existing and new business in certain countries, states, municipalities, territories and other jurisdictions in which the application of the rule of law is inconsistent, extralegal or less predictable, including with respect to:

- the absence of a statutory or regulatory basis or guidance for engaging in specific types of business or transactions
- conflicting or ambiguous laws, rules, regulations and judicial orders, or the inconsistent application or interpretation of existing laws, rules, regulations and judicial precedents
- actions by or at the direction of government officials or agencies
- uncertainty concerning the enforceability of intellectual property rights or contractual or other obligations
- difficulty in competing in economies in which the government controls or protects all or a portion of the local economy or specific businesses, or where graft or corruption may be pervasive
- the threat of regulatory investigations, civil litigations or criminal prosecutions that are arbitrary or otherwise contrary to established legal principles in other parts of the world, and

- the termination of licenses required to operate in the local market or the suspension of business relationships with governmental bodies.

If the application of the laws, rules, regulations and judicial precedents in any jurisdiction is susceptible to producing outcomes that are inconsistent, unexpected or contrary to established legal principles, this can create a more difficult environment in which JPMorganChase conducts its business and could negatively affect JPMorganChase's operations and reduce its earnings with respect to that jurisdiction. For example, JPMorganChase has faced actual and threatened litigation in Russia with respect to payments that JPMorganChase cannot make under, and is contractually excused from paying as a result of, relevant economic sanctions laws. That litigation has also resulted in the seizure of assets. In addition, conducting business in jurisdictions with less predictable legal and regulatory frameworks could require JPMorganChase to devote significant additional resources to understanding local laws, rules and regulations, as well as structuring its operations to comply with local laws, rules and regulations and implementing and administering related internal policies and procedures.

There can be no assurance that JPMorganChase will always be successful in its efforts to fully understand and to conduct its business in compliance with the laws, rules and regulations of all of the jurisdictions in which it operates, and the risk of non-compliance, or of interference with JPMorganChase's businesses, can be greater in jurisdictions that have less predictable legal and regulatory frameworks.

JPMorganChase's businesses may be negatively impacted by governmental policies that either discourage or penalize business with certain industries or require specific business practices.

JPMorganChase's businesses and results of operations may be adversely affected by actions or initiatives by national, state or local governmental authorities that:

- seek to discourage financial institutions from doing business with companies engaged in certain industries, or conversely, to penalize financial institutions that elect not to do business with such companies, or
- mandate specific business practices that companies operating in the relevant jurisdiction must adopt.

Because governmental policies in one jurisdiction may differ or conflict with those in other jurisdictions, JPMorganChase may face negative consequences regardless of the course of action it takes or elects not to take, including:

- restrictions or prohibitions on doing business within a particular jurisdiction, or with governmental entities in a jurisdiction
- the threat of enforcement actions, including under antitrust or other anti-competition laws, rules and regulations, and
- harm to its reputation arising from public criticism, including from politicians, activists and other stakeholders.

JPMorganChase has been prohibited from engaging in certain business activities in specific jurisdictions as a result of these types of governmental actions, and there is no assurance that it will not face similar restrictions on its business and operations in the future.

Requirements for the orderly resolution of JPMorganChase could result in JPMorganChase having to restructure or reorganize its businesses and could increase its funding or operational costs or curtail its businesses.

JPMorganChase is required under Federal Reserve and FDIC rules to prepare and submit periodically to those agencies a detailed plan for rapid and orderly resolution in bankruptcy, without extraordinary government support, in the event of material financial distress or failure. The evaluation of JPMorganChase's resolution plan by these agencies may change, and the requirements for resolution plans may be modified from time to time. Any such determinations or modifications could result in JPMorganChase needing to make changes to its legal entity structure or to certain internal or external activities, which could increase its funding or operational costs, or hamper its ability to serve clients and customers.

If the Federal Reserve and the FDIC were both to determine that a resolution plan submitted by JPMorganChase has deficiencies, they could jointly impose more stringent capital, leverage or liquidity requirements or restrictions on JPMorganChase's growth, activities or operations. The agencies could also require that JPMorganChase restructure, reorganize or divest assets or businesses in ways that could materially and adversely affect JPMorganChase's operations and strategy.

Holders of JPMorgan Chase & Co.'s debt and equity securities will absorb losses if it were to enter into a resolution.

Federal Reserve rules require that JPMorgan Chase & Co. (the "Parent Company") maintain minimum levels of unsecured external long-term debt and other loss-absorbing capacity with specific terms ("eligible LTD") for purposes of recapitalizing JPMorganChase's operating subsidiaries if the Parent Company were to enter into a resolution either:

- in a bankruptcy proceeding under Chapter 11 of the U.S. Bankruptcy Code, or
- in a receivership administered by the FDIC under Title II of the Dodd-Frank Act ("Title II").

If the Parent Company were to enter into a resolution, holders of eligible LTD, other unsecured creditors and holders of equity securities of the Parent Company will absorb the losses of the Parent Company and its subsidiaries.

The preferred "single point of entry" strategy under JPMorganChase's resolution plan contemplates that the Parent Company would enter bankruptcy proceedings and JPMorganChase's material subsidiaries would be recapitalized, as needed, so that they could continue normal operations or subsequently be divested or wound down in an orderly manner. As a result, the Parent Company's losses and any losses incurred by its subsidiaries would be imposed first on holders of the Parent Company's equity securities and thereafter on its unsecured creditors, including holders of eligible LTD. Claims of the Parent Company's shareholders and unsecured creditors would have a junior position to the claims of creditors of JPMorganChase's subsidiaries and to the claims of priority (as determined by statute) and secured creditors of the Parent Company.

Accordingly, in a resolution of the Parent Company in bankruptcy, unsecured creditors of the Parent Company, including holders of eligible LTD of the Parent Company, would realize value only to the extent available to the Parent Company as a shareholder of JPMorgan Chase Bank, N.A. and its other subsidiaries, and only after any claims of priority and secured creditors of the Parent Company have been fully repaid.

The FDIC has similarly indicated that a single point of entry recapitalization model would be its expected strategy to resolve a systemically important financial institution, such as the Parent Company, under Title II. However, the FDIC has not formally adopted or committed to any specific resolution strategy.

If the Parent Company were to approach, or enter into, a resolution, none of the Parent Company, the Federal Reserve or the FDIC is obligated to follow JPMorganChase's preferred resolution strategy, and losses to unsecured creditors of the Parent Company, including holders of eligible LTD, and to holders of equity securities of the Parent Company, under whatever strategy is ultimately followed, could be greater than they might have been under JPMorganChase's preferred strategy.

Political

Economic uncertainty or instability caused by political and geopolitical developments can negatively impact JPMorganChase's businesses.

Political developments in the U.S. and other countries can cause uncertainty in the economic environment and market conditions in which JPMorganChase operates its businesses. Certain governmental policy initiatives, as well as heightened geopolitical tensions, could significantly affect U.S. and global economic growth and cause higher volatility in the financial markets, including:

- monetary policies and actions taken by the Federal Reserve and other central banks or governmental authorities, including changes in interest rate levels and any sustained large-scale asset purchases or any suspension or reversal of those actions
- fiscal policies, including with respect to taxation and spending
- isolationist foreign policies
- economic or financial sanctions
- the implementation of tariffs and other protectionist trade policies
- changes to immigration policies, or
- actions that the government takes or fails to take in response to the effects of health emergencies, the spread of infectious diseases, epidemics or pandemics.

These types of political developments, and uncertainty about the possible outcomes of these developments, could:

- erode investor or consumer confidence in the U.S. economy and financial markets, which could potentially undermine the status of the U.S. dollar as a safe haven currency
- provoke retaliatory countermeasures by other countries and otherwise heighten tensions in regulatory, enforcement or diplomatic relations
- increase the risk of targeted cyber attacks
- increase concerns about whether the U.S. government will be funded, and its outstanding debt serviced, at any particular time
- result in periodic shutdowns of the U.S. government
- influence investor perceptions concerning government support of certain sectors of the economy or the economy as a whole
- influence monetary policy actions of the Federal Reserve to moderate the economic impact of political developments, including decisions on interest rate levels and asset purchases and sales
- adversely affect the financial condition or credit ratings of clients and counterparties with which JPMorganChase does business, or
- cause JPMorganChase to refrain from engaging in business opportunities that it might otherwise pursue.

These factors could lead to:

- slower growth rates, rising inflation or recession
- disruptions in labor markets
- greater market volatility

- a contraction of available credit and the widening of credit spreads
- U.S. dollar currency fluctuations
- lower investments in a particular country or sector of the economy
- large-scale sales of government debt and other debt and equity securities
- reduced commercial activity among trading partners or disruptions to supply chains, or
- the possible departure of a country from, or the dissolution or formation of, a political or economic alliance or treaty.

Under certain circumstances, such as geopolitically challenging situations in regions like Russia, the Middle East and China, these various risks could become highly correlated or combine in unprecedented ways.

Any of these potential outcomes could cause JPMorganChase to suffer losses on its market-making positions or in its investment portfolio, reduce its liquidity and capital levels, increase the allowance for credit losses or lead to higher net charge-offs, hamper its ability to deliver products and services to its clients and customers, and weaken its results of operations and financial condition or credit rating.

Market

Economic and market events and conditions can materially affect JPMorganChase's businesses and investment and market-making positions.

JPMorganChase's results of operations can be negatively affected by adverse changes in any of the following:

- investor, consumer and business sentiment
- events that reduce confidence in the financial markets
- inflation, deflation or recession
- high unemployment or, conversely, a tightening labor market
- the availability and cost of capital, liquidity and credit
- levels and volatility of interest rates, credit spreads and market prices for currencies, debt and equity securities and commodities, as well as the duration of any such changes
- the economic effects of an outbreak or escalation of war, hostilities, terrorism or other geopolitical instabilities, cyber attacks, climate change, natural disasters, severe weather conditions, health emergencies, the spread of infectious diseases, epidemics or pandemics or other extraordinary events beyond JPMorganChase's control, and
- the state of the U.S. and global economies.

All of these are affected by global economic, market and political events and conditions, including monetary policies and actions taken by central banks or other governmental authorities, as well as by the regulatory environment.

In addition, JPMorganChase's investment portfolio and market-making businesses can suffer losses due to unanticipated market events, including:

- severe declines in asset values
- unexpected credit events
- unforeseen events or conditions that may cause previously uncorrelated factors to become correlated (and vice versa)
- the inability to effectively hedge risks related to market-making and investment portfolio positions, or
- other market risks that may not have been appropriately taken into account in the development, structuring or pricing of a financial instrument.

If JPMorganChase experiences significant losses in its investment portfolio or from market-making activities, this could reduce JPMorganChase's profitability and its liquidity and capital levels, and thereby constrain the growth of its businesses.

JPMorganChase's consumer businesses can be negatively affected by adverse economic conditions and governmental policies.

JPMorganChase's consumer businesses are particularly affected by U.S. and global economic conditions, including:

- personal and household income distribution
- unemployment or underemployment
- prolonged periods of exceptionally high or low interest rates, or significant changes to interest rates
- changes in the value of collateral such as residential real estate and vehicles
- changes in housing prices
- the level of inflation and its effect on prices for goods and services
- consumer and small business confidence levels, and
- changes in consumer spending or in the level of consumer debt.

Heightened levels of unemployment or underemployment that result in reduced personal and household income could negatively affect consumer credit performance to the extent that consumers are less able to service their debts. In addition, sustained low growth, low or negative interest rates, inflationary pressures or recessionary conditions could diminish customer demand for the products and services offered by JPMorganChase's consumer businesses.

Adverse economic conditions could also lead to an increase in delinquencies, additions to the allowance for credit losses and higher net charge-offs, which can reduce JPMorganChase's earnings. These consequences could be significantly worse in certain geographies, including where declining industrial or manufacturing activity has resulted in or could result in higher levels of unemployment, or where high levels of consumer debt, such as outstanding student loans, could impair the ability of customers to pay their other consumer loan obligations.

JPMorganChase's earnings from its consumer businesses could also be adversely affected by governmental policies and actions that affect consumers, including:

- policies and initiatives relating to medical insurance, education, immigration and housing, or that may impact employment status
- laws, rules and regulations relating specifically to the financial services industry, such as limitations on late payment, overdraft and interchange fees, and
- policies aimed at the economy more broadly, such as higher taxes and increased regulation, which could result in reductions in consumer disposable income.

Unfavorable market and economic conditions can have an adverse effect on JPMorganChase's wholesale businesses.

In JPMorganChase's wholesale businesses, market and economic factors can affect the volume of transactions that JPMorganChase executes for its clients or for which it advises clients, and, therefore, the revenue that JPMorganChase receives from those transactions. These factors can also influence the willingness of other financial institutions and investors to participate in capital markets transactions that JPMorganChase manages, such as loan syndications or securities underwriting. Furthermore, if a significant and sustained deterioration in market conditions were to occur, the profitability of JPMorganChase's businesses engaged in capital markets activities, including loan syndication, securities underwriting and leveraged lending activities, could be reduced to the extent that those businesses:

- earn less fee revenue due to lower transaction volumes, including when clients are unwilling or unable to refinance their outstanding debt obligations in unfavorable market conditions, or
- dispose of portions of credit commitments at a loss, or hold larger residual positions in credit commitments that cannot be sold at favorable prices.

The fees that JPMorganChase earns from managing client assets or holding assets under custody for clients could be diminished by declining asset values or other adverse macroeconomic conditions. For example, higher interest rates or a downturn in financial markets could affect the valuation of client assets that JPMorganChase manages or holds under custody, which, in turn, could affect JPMorganChase's revenue from fees that are based on the amount of assets under management or custody. Similarly, adverse macroeconomic or market conditions could prompt outflows from JPMorganChase funds or accounts, or cause clients to invest in products that generate lower revenue. Substantial and unexpected withdrawals from a JPMorganChase fund can also hamper the investment performance of the fund, particularly if the outflows create the need for the fund to dispose of fund assets at disadvantageous times or prices, and could lead to further withdrawals based on the weaker investment performance.

An adverse change in market conditions in particular segments of the economy, such as a sudden and severe downturn in oil and gas prices or an increase in commodity prices, severe declines in commercial real estate values, or sustained changes in consumer behavior that affect specific economic sectors, could have a material adverse effect on clients of JPMorganChase whose operations or financial condition are directly or indirectly dependent on the health or stability of those market segments or economic sectors, as well as clients that are engaged in related businesses. JPMorganChase could incur credit losses on its loans and other commitments to clients that operate in, or are dependent on, any sector of the economy that is or comes under stress.

An economic downturn or sustained changes in consumer behavior that results in shifts in consumer and business spending could also have a negative impact on certain of JPMorganChase's wholesale clients, and thereby diminish JPMorganChase's earnings from its wholesale operations. For example, the businesses of certain of JPMorganChase's wholesale clients are dependent on consistent streams of rental income from commercial real estate properties, including offices, which are owned or being built by those clients. Sustained adverse economic conditions or hybrid work models could result in reductions in the rental cash flows that owners or developers receive from their tenants which, in turn, could depress the values of the properties, impair the ability of borrowers to service or refinance their commercial real estate loans and lead to an increase in foreclosures. These consequences could result in JPMorganChase experiencing increases in the allowance for credit losses, higher delinquencies, defaults and charge-offs within its commercial real estate loan portfolio and incurring higher costs for servicing a larger volume of delinquent loans in that portfolio. An increase in foreclosures could result in higher operational risk associated with JPMorganChase owning and managing real property, and any inadequacy in governance or control over the foreclosed properties could result in regulatory scrutiny and reputational harm.

Changes in interest rates and credit spreads can adversely affect JPMorganChase's earnings, its liquidity or its capital levels.

When interest rates are high or increasing, JPMorganChase can generally be expected to earn higher net interest income. However, higher interest rates can also lead to:

- fewer originations of commercial and residential real estate loans
- losses on underwriting exposures or incremental client-specific downgrades, or increases in the allowance for credit losses and net charge-offs due to higher financing costs for clients
- the loss of deposits, particularly if customers withdraw deposits because they believe that interest rates offered by JPMorganChase are lower than those of competitors or if JPMorganChase makes incorrect assumptions about depositor behavior
- losses on available-for-sale ("AFS") securities held in the investment securities portfolio
- lower net interest income if central banks introduce interest rate increases more quickly than anticipated and this results in a misalignment in the pricing of short-term and long-term borrowings
- less liquidity in the financial markets, and
- higher funding costs.

All of these outcomes could adversely affect JPMorganChase's earnings or its liquidity and capital levels, and any negative outcomes could be more severe in a prolonged period of high interest rates. Higher interest rates can also negatively affect the payment performance on loans within JPMorganChase's consumer and wholesale loan portfolios that are linked to variable interest rates. If borrowers of variable rate loans are unable to afford higher interest payments, those borrowers may reduce or stop making payments, thereby causing JPMorganChase to incur losses and increased operational costs related to servicing a higher volume of delinquent loans.

On the other hand, a low or negative interest rate environment may cause:

- net interest margins to be compressed, which could reduce the amounts that JPMorganChase earns on its investment securities portfolio to the extent that it is unable to reinvest contemporaneously in higher-yielding instruments
- unanticipated or adverse changes in depositor behavior, which could negatively affect JPMorganChase's broader asset and liability management strategy, and
- a reduction in the value of JPMorganChase's mortgage servicing rights ("MSRs") asset, resulting in decreased revenues.

When credit spreads widen, it becomes more expensive for JPMorganChase to borrow. JPMorganChase's credit spreads may widen or narrow not only in response to events and circumstances that are specific to JPMorganChase but also as a result of general economic and geopolitical events and conditions. Changes in JPMorganChase's credit spreads will affect, positively or negatively, JPMorganChase's earnings on certain liabilities, such as derivatives, that are recorded at fair value.

JPMorganChase's results may be materially affected by market fluctuations and significant changes in the valuation of financial instruments.

The value of securities, derivatives and other financial instruments which JPMorganChase owns or in which it makes markets can be materially affected by market fluctuations. Market volatility, illiquid market conditions and other disruptions in the financial markets may make it extremely difficult to value certain financial instruments. Subsequent valuations of financial instruments in future periods, in light of factors then prevailing, may result in significant changes in the value of these instruments. In addition, at the time of any disposition of these financial instruments, the price that JPMorganChase ultimately realizes will depend on the demand and liquidity in the market at that time and may be materially lower than their current fair value. Any of these factors could cause a decline in the value of financial instruments that JPMorganChase owns or in which it makes markets, which may have an adverse effect on JPMorganChase's results of operations.

JPMorganChase's risk management and monitoring processes, including its stress testing framework, seek to quantify and manage JPMorganChase's exposure to more extreme market moves. However, JPMorganChase's hedging and other risk management strategies may not be effective, and it could incur significant losses, if extreme market events were to occur.

Credit

JPMorganChase can be negatively affected by adverse changes in the financial condition of clients, counterparties, custodians and CCPs.

JPMorganChase routinely executes transactions with clients and counterparties such as corporations, financial institutions, asset managers, hedge funds, securities exchanges and government entities within and outside the U.S. Many of these transactions expose JPMorganChase to the credit risk of its clients and counterparties, and can involve JPMorganChase in disputes and litigation if a client or counterparty defaults. JPMorganChase can also be subject to losses or liability where a financial institution that it has appointed to provide custodial services for client assets or funds becomes insolvent as a result of fraud or the failure to abide by existing laws and obligations, or where clients are unable to access assets held by JPMorganChase as custodian due to governmental actions or other factors.

A default by, or the financial or operational failure of, a CCP through which JPMorganChase executes contracts would require JPMorganChase to replace those contracts, thereby increasing its operational costs and potentially resulting in losses. In addition, JPMorganChase can be exposed to losses if a member of a CCP in which JPMorganChase is also a member defaults on its obligations to the CCP because of requirements that each member of the CCP absorb a portion of those losses. Furthermore, JPMorganChase can be subject to bearing its share of non-default losses incurred by a CCP, including losses from custodial, settlement or investment activities or due to cyber or other security breaches.

As part of its clearing services activities, JPMorganChase is exposed to the risk of nonperformance by its clients, which it seeks to mitigate by requiring clients to provide adequate collateral. JPMorganChase is also exposed to intra-day credit risk of its clients in connection with providing cash management, clearing, custodial and other transaction services to those clients. If a client for which JPMorganChase provides these services becomes bankrupt or insolvent, JPMorganChase may incur losses, become involved in disputes and litigation with one or more CCPs, the client's bankruptcy estate and other creditors, or be subject to regulatory investigations. All of the foregoing events can increase JPMorganChase's operational and litigation costs, and JPMorganChase may suffer losses to the extent that any collateral that it has received is insufficient to cover those losses.

Transactions with government entities, including national, state, provincial, municipal and local authorities, can expose JPMorganChase to enhanced sovereign, credit, operational and reputation risks. Government entities may, among other things, claim that actions taken by government officials were beyond the legal authority of those officials or repudiate transactions authorized by a previous incumbent government. These types of actions have in the past caused, and could in the future cause, JPMorganChase to suffer losses or hamper its ability to conduct business in the relevant jurisdiction.

In addition, local laws, rules and regulations could limit JPMorganChase's ability to resolve disputes and litigation in the event of a counterparty default or unwillingness to make previously agreed-upon payments, which could subject JPMorganChase to losses.

Disputes may arise with counterparties to derivatives contracts with regard to the terms, the settlement procedures or the value of underlying collateral. The disposition of those disputes could cause JPMorganChase to incur unexpected transaction, operational and legal costs, or result in credit losses. These consequences can also impair JPMorganChase's ability to effectively manage its credit risk exposure from its market activities, or cause harm to JPMorganChase's reputation.

The financial or operational failure of a significant market participant, such as a major financial institution or a CCP, or concerns about the creditworthiness of such a market participant or its ability to fulfill its obligations, can cause substantial and cascading disruption within the financial markets, including in circumstances where coordinated action by multiple other market participants is required to address the failure or disruption. JPMorganChase's businesses could be significantly disrupted by such an event, particularly if it leads to other market participants incurring significant losses, experiencing liquidity issues or defaulting, and JPMorganChase is likely to have significant interrelationships with, and credit exposure to, such a significant market participant.

JPMorganChase may suffer losses if the value of collateral declines in stressed market conditions.

During periods of market stress or illiquidity, JPMorganChase's credit risk may be further increased when:

- JPMorganChase fails to realize the estimated value of the collateral it holds
- collateral is liquidated at prices that are not sufficient to recover the full amount owed to it, or
- counterparties are unable to post collateral, whether for operational or other reasons.

Furthermore, disputes with counterparties concerning the valuation of collateral may increase in times of significant market stress, volatility or illiquidity, and JPMorganChase could suffer losses during these periods if it is unable to realize the fair value of collateral or to manage declines in the value of collateral.

JPMorganChase could incur significant losses arising from concentrations of credit and market risk.

JPMorganChase is exposed to greater credit and market risk to the extent that groupings of its clients or counterparties, or obligors on securities and other financial instruments:

- engage in similar or related businesses, or in businesses in related industries
- do business in the same geographic region, or

- have business profiles, models or strategies that could cause their ability to meet their obligations to be similarly affected by changes in economic conditions.

For example, a significant deterioration in the credit quality of a counterparty, borrower or other obligor could lead to concerns about the creditworthiness of other counterparties, borrowers or obligors in similar, related or dependent industries. This type of interrelationship could exacerbate JPMorganChase's credit, liquidity and market risk exposure and potentially cause it to incur losses, including fair value losses in its market-making businesses and investment portfolios. In addition, JPMorganChase may be required to increase the allowance for credit losses or establish other reserves with respect to certain clients, industries or country exposures in order to align with directives or expectations of its banking regulators.

Similarly, challenging economic conditions that affect a particular industry or geographic area could lead to concerns about the credit quality of counterparties, borrowers or other obligors not only in that particular industry or geography but in related or dependent industries, wherever located. These conditions could also heighten concerns about the ability of customers of JPMorganChase's consumer businesses who live in those areas or work in those affected industries or related or dependent industries to meet their obligations to JPMorganChase. JPMorganChase regularly monitors various segments of its credit and market risk exposures to assess the potential risks of concentration or contagion, but its ability to diversify or hedge its exposure against those risks may be limited.

JPMorganChase's consumer businesses can also be harmed by an excessive expansion of consumer credit by bank or non-bank competitors. Heightened competition for certain types of consumer loans could prompt industry-wide reactions such as significant reductions in the pricing or margins of those loans or the making of loans to less-creditworthy borrowers. If large numbers of consumers subsequently default on their loans, whether due to weak credit profiles, an economic downturn or other factors, this could impair their ability to repay obligations owed to JPMorganChase and result in higher charge-offs and other credit-related losses. More broadly, widespread defaults on consumer debt could lead to recessionary conditions in the U.S. economy, and JPMorganChase's consumer businesses may earn lower revenues in such an environment.

If JPMorganChase is unable to reduce positions effectively during a market dislocation, this can increase both the market and credit risks associated with those positions and the level of risk-weighted-assets ("RWA") that JPMorganChase holds on its balance sheet. These factors could adversely affect JPMorganChase's capital position, funding costs and the profitability of its businesses.

Liquidity

JPMorganChase's ability to operate its businesses could be impaired if its liquidity is constrained.

JPMorganChase's liquidity can be impacted at any given time as a result of factors such as:

- market-wide illiquidity or disruption
- changes in liquidity or capital requirements resulting from changes in laws, rules and regulations, including those in response to economic effects of systemic events
- actions taken by the U.S. government or by the Federal Reserve to reduce its balance sheet, which may reduce deposits held by JPMorganChase and other financial institutions
- inability to sell assets, or to sell assets at favorable times or prices
- default by a CCP or other significant market participant
- unanticipated outflows of cash or collateral
- unexpected loss of deposits or higher than anticipated draws on lending-related commitments, and
- lack of market or customer confidence in JPMorganChase or financial institutions in general.

A reduction in JPMorganChase's liquidity may be caused by events over which it has little or no control. For example, periods of market stress, low investor confidence and significant market illiquidity could result in higher funding costs for JPMorganChase and could limit its access to some of its traditional sources of liquidity.

JPMorganChase may need to raise funding from alternative sources if its access to stable and lower-cost sources of funding, such as deposits and borrowings from Federal Home Loan Banks, is reduced. Alternative sources of funding could be more expensive or limited in availability. JPMorganChase's funding costs could also be negatively affected by actions that JPMorganChase may take in order to:

- satisfy applicable liquidity coverage ratio and net stable funding ratio requirements
- address obligations under its resolution plan, or
- satisfy regulatory requirements in jurisdictions outside the U.S. relating to the pre-positioning of liquidity in subsidiaries that are material legal entities.

More generally, if JPMorganChase fails to effectively manage its liquidity, this could constrain its ability to fund or invest in its businesses and subsidiaries, and thereby adversely affect its results of operations.

JPMorgan Chase & Co. is a holding company and depends on the cash flows of its subsidiaries to make payments on its outstanding securities.

JPMorgan Chase & Co. is a holding company that holds the stock of JPMorgan Chase Bank, N.A. and an intermediate holding company, JPMorgan Chase Holdings LLC (the “IHC”). The IHC in turn generally holds the stock of JPMorganChase’s subsidiaries other than JPMorgan Chase Bank, N.A. and its subsidiaries. The IHC also owns other assets and provides intercompany lending to the Parent Company.

The Parent Company is obligated to contribute to the IHC substantially all the net proceeds received from securities issuances (including issuances of senior and subordinated debt securities and of preferred and common stock).

The ability of JPMorgan Chase Bank, N.A. and the IHC to make payments to the Parent Company is also limited. JPMorgan Chase Bank, N.A. is subject to regulatory restrictions on its dividend distributions, as well as capital adequacy requirements, such as the Supplementary Leverage Ratio (“SLR”), and liquidity requirements and other regulatory restrictions on its ability to make payments to the Parent Company. The IHC is prohibited from paying dividends or extending credit to the Parent Company if certain capital or liquidity thresholds are breached, or if limits are otherwise imposed by the Parent Company’s management or Board of Directors.

As a result of these arrangements, the ability of the Parent Company to make various payments is dependent on its receiving dividends from JPMorgan Chase Bank, N.A. and dividends and borrowings from the IHC. These limitations could affect the Parent Company’s ability to:

- pay interest on its debt securities
- pay dividends on its equity securities
- redeem or repurchase outstanding securities, and
- fulfill its other payment obligations.

These arrangements could also result in the Parent Company seeking protection under bankruptcy laws or otherwise entering into resolution proceedings at a time earlier than would have been the case absent the existence of the capital and liquidity thresholds to which JPMorgan Chase Bank, N.A. and the IHC are subject.

Reductions in JPMorganChase’s credit ratings may adversely affect its liquidity and cost of funding.

JPMorgan Chase & Co. and certain of its principal subsidiaries are rated by credit rating agencies. Rating agencies evaluate general, firm-specific and industry-specific factors when determining credit ratings for a particular financial institution, including:

- expected future profitability
- risk management practices
- legal expenses
- ratings differentials between bank holding companies and their bank and non-bank subsidiaries
- regulatory developments
- assumptions about government support, and
- economic and geopolitical developments.

JPMorganChase closely monitors and manages, to the extent that it is able, factors that could influence its credit ratings. However, there is no assurance that JPMorganChase’s credit ratings will not be downgraded in the future. Furthermore, any such downgrade could occur at times of broader market instability when JPMorganChase’s options for responding to events may be more limited and general investor confidence is low.

A reduction in JPMorganChase’s credit ratings could curtail JPMorganChase’s business activities and reduce its profitability in a number of ways, including:

- reducing its access to capital markets
- materially increasing its cost of issuing and servicing securities
- triggering additional collateral or funding requirements, and
- decreasing the number of investors and counterparties that are willing or permitted to do business with or lend to JPMorganChase.

Any rating reduction could also increase the credit spreads charged by the market for taking credit risk on JPMorgan Chase & Co. and its subsidiaries. This could, in turn, adversely affect the value of debt and other obligations of JPMorgan Chase & Co. and its subsidiaries.

Capital

Maintaining the required level and composition of capital may impact JPMorganChase’s ability to support business activities, meet evolving regulatory requirements and distribute capital to shareholders.

JPMorganChase is subject to various regulatory capital requirements, including leverage- and risk-based capital requirements. In addition, as a Global Systemically Important Bank (“GSIB”), JPMorganChase is required to hold additional capital buffers, including a GSIB surcharge, a Stress Capital Buffer (“SCB”), and a countercyclical buffer, each of which is reassessed at least annually. The amount of capital that JPMorganChase is required to hold in order to satisfy these leverage- and risk-based requirements could increase at any given time due to factors such as:

- actions by banking regulators, including changes in laws, rules and regulations

- changes in the composition of JPMorganChase's balance sheet or developments that could increase RWA, such as increased market risk, customer delinquencies, client credit rating downgrades or other factors, and
- increases in estimated stress losses as determined by the Federal Reserve under the Comprehensive Capital Analysis and Review, which could increase JPMorganChase's SCB.

Any failure by or inability of JPMorganChase to maintain the required level and composition of capital, or unfavorable changes in applicable capital requirements, could have an adverse impact on JPMorganChase's shareholders, such as:

- reducing the amount of common stock that JPMorganChase is permitted to repurchase
- requiring the issuance of, or prohibiting the redemption of, capital instruments in a manner inconsistent with JPMorganChase's capital management strategy
- constraining the amount of dividends that may be paid on common stock, or
- curtailing JPMorganChase's business activities or operations.

In 2023, U.S. banking regulators released a proposal to implement the final Basel III reforms which would have significantly revised the risk-based capital requirements for banks with assets of $100 billion or more, including JPMorganChase. In addition, in 2023 the Federal Reserve released a proposal to amend the calculation of the GSIB surcharge. Uncertainty remains regarding the content of the final versions of these rule proposals and how they might ultimately apply to JPMorganChase. However, it is possible that the final rules could impact JPMorganChase's decisions concerning the business activities in which it will engage and its levels of capital distributions to its shareholders.

Operational

JPMorganChase's businesses are dependent on the effectiveness of internal and external operational systems.

JPMorganChase's businesses rely on the ability of JPMorganChase's financial, accounting, transaction execution, data processing and other operational systems, including devices supporting those systems, to process, record, monitor and report a large number of transactions on a continuous basis, and to do so accurately, quickly and securely. In addition to proper design, installation, maintenance and training, the effective functioning of JPMorganChase's operational systems depends on:

- the quality of the information contained in those systems, as inaccurate, outdated, incomplete or corrupted data can significantly compromise the functionality or reliability of a particular system and other systems to which it transmits or from which it receives information, and
- JPMorganChase's ability to continue to maintain and upgrade its systems on a regular and timely basis in line with technological advancements and evolving security requirements, maintain security and operational continuity of its systems, including by carefully managing any changes introduced to its systems, prevent unauthorized access and the misuse of access to its systems, and adhere to all applicable legal and regulatory requirements, particularly in regions where JPMorganChase may face a heightened risk of malicious activity.

JPMorganChase has experienced and expects that it will continue to experience failures and disruptions in the stability of its operational systems, including degraded performance of data processing systems, data quality issues, disruptions of network connectivity and malfunctioning software, as well as disruptions in its ability to access and use the operational systems of third parties and interruptions in service from third-party service providers. These incidents have resulted in various negative effects for customers, including the inability to access account information or transact through ATM, internet or mobile channels, the exfiltration of customer personal data, the recording of duplicative transactions and extended delays for customers requiring services from call centers. There can be no assurance that these and other types of operational failures or disruptions will not occur in the future.

JPMorganChase's ability to effectively manage the stability of its operational systems and infrastructure could be hindered by many factors, any of which could have a negative impact on JPMorganChase and its clients, customers and counterparties, including:

- JPMorganChase's ability to effectively maintain and upgrade systems and infrastructure can become more challenging as the speed, frequency, volume, interconnectivity and complexity of transactions continue to increase
- attempts by third parties to defraud JPMorganChase or its clients and customers continue to increase, evolve and become more complex, and during periods of market disruption or economic uncertainty, these attempts can be expected to further increase in volume
- errors made by JPMorganChase or another market participant, whether inadvertent or malicious, could cause widespread system disruption
- failure to detect weaknesses or shortcomings in operational systems in a timely manner
- isolated or seemingly insignificant errors in operational systems could compound, or migrate to other systems over time, to become larger issues

- disruptions in operational systems or in the ability of systems to communicate with each other could be caused by failures in synchronization or encryption software, or degraded performance of microprocessors, and
- attempts by third parties to block the use of key technology solutions by claiming that the use infringes on their intellectual property rights.

JPMorganChase also depends on its ability to access and use the operational systems of third parties, including its custodians, vendors (such as those that provide data and cloud computing services, and security and technology services) and other market participants (such as clearing and payment systems, CCPs and securities exchanges). These external operational systems with which JPMorgan is connected, whether directly or indirectly, can be sources of operational risk to JPMorganChase. JPMorganChase may be exposed not only to a systems failure or cyber attack that may be experienced by a vendor or market infrastructure with which JPMorganChase is directly connected, but also to a systems breakdown or cyber attack involving another party to which such a vendor or infrastructure is connected. Similarly, retailers, payment systems and processors, data aggregators and other external parties with which JPMorganChase's customers do business can increase JPMorganChase's operational risk. This is particularly the case where activities of customers or other parties are beyond JPMorganChase's security and control systems, including through the use of the internet, cloud computing services, and personal smart phones and other mobile devices or services.

If an external party obtains access to customer account data on JPMorganChase's systems, whether authorized or unauthorized, and that party misappropriates that data, this could result in negative outcomes for JPMorganChase and its clients and customers, including a heightened risk of fraudulent transactions using JPMorganChase's systems, losses from fraudulent transactions and reputational harm arising from the perception that JPMorganChase's systems may not be secure.

As JPMorganChase's interconnectivity with clients, customers and other external parties continues to expand, JPMorganChase increasingly faces the risk of operational failure or cyber attacks with respect to the systems of those parties. Security breaches affecting JPMorganChase's clients or customers, or systems breakdowns or failures, security breaches or human error or misconduct affecting other external parties, may require JPMorganChase to take steps to protect the integrity of its own operational systems or to safeguard confidential information, including restricting the access of customers to their accounts. These actions can increase JPMorganChase's operational costs and potentially diminish customer satisfaction and confidence in JPMorganChase.

Furthermore, the widespread and expanding interconnectivity among financial institutions, clearing banks, CCPs, payments processors, financial technology companies, securities exchanges, clearing houses and other financial market infrastructures increases the risk that the disruption of an operational system involving one institution or entity, including due to a cyber attack, may cause industry-wide operational disruptions that could materially affect JPMorganChase's ability to conduct business. In addition, the risks associated with the disruption of an operational system of a third party could be exacerbated to the extent that the services provided by that system are used by a significant number or proportion of market participants.

The ineffectiveness, failure or other disruption of operational systems upon which JPMorganChase depends, including due to a systems malfunction, cyber incident or other systems failure, could result in unfavorable ripple effects in the financial markets and for JPMorganChase and its clients and customers, including:

- delays or other disruptions in providing services, including the provision of liquidity or information to clients and customers
- impairment of JPMorganChase's ability to execute transactions, including delays or failures in the confirmation or settlement of transactions or in obtaining access to funds or other assets required for settlement
- the possibility that funds transfers, capital markets trades or other transactions are executed erroneously
- financial losses, including due to loss-sharing requirements of CCPs, payment systems or other market infrastructures, or as possible restitution to clients and customers
- higher operational costs associated with replacing services provided by a system that has experienced a failure or other disruption
- limitations on JPMorganChase's ability to collect data needed for its business and operations
- loss of confidence in the ability of JPMorganChase, or financial institutions generally, to protect against and withstand operational disruptions
- dissatisfaction among JPMorganChase's clients or customers
- significant exposure to litigation and regulatory fines, penalties or other sanctions, and
- harm to JPMorganChase's reputation.

If JPMorganChase's operational systems, or those of acquired businesses or of external parties on which

JPMorganChase's businesses depend, are unable to meet the requirements of JPMorganChase's businesses and operations or bank regulatory standards, or if they fail or have other significant shortcomings, JPMorganChase could be materially and adversely affected.

A successful cyber attack affecting JPMorganChase could cause significant harm to JPMorganChase and its clients and customers.

JPMorganChase experiences numerous cyber attacks on its computer systems, software, networks and other technology assets on a daily basis from various actors, including groups acting on behalf of hostile countries, cyber-criminals, "hacktivists" (i.e., individuals or groups that use technology to promote a political agenda or social change) and others. These cyber attacks can take many forms, including attempts to introduce computer viruses or malicious code, which are commonly referred to as "malware," into JPMorganChase's systems. These attacks are often designed to:

- obtain unauthorized access to JPMorganChase's systems or to confidential information belonging to JPMorganChase or its clients, customers, counterparties or employees
- manipulate data
- destroy data or systems with the aim of rendering services unavailable
- disrupt, sabotage or degrade service on JPMorganChase's systems
- steal money, or
- extort money through the use of so-called "ransomware."

JPMorganChase also experiences:

- distributed denial-of-service attacks intended to disrupt JPMorganChase's websites, including those that provide online banking and other services,
- a higher volume and complexity of cyber attacks against the backdrop of heightened geopolitical tensions, and
- a high volume of disruptions to internet-based services used by JPMorganChase that are provided by third parties.

JPMorganChase has experienced security breaches due to cyber attacks in the past, and it is inevitable that additional breaches will occur in the future. Any such breach could result in serious and harmful consequences for JPMorganChase or its clients and customers.

A principal reason that JPMorganChase cannot provide absolute security against cyber attacks is that it may not always be possible to anticipate, detect or recognize threats to JPMorganChase's systems, or to implement effective preventive measures against all breaches due to evolving risks, including:

- the techniques used in cyber attacks evolve frequently and increase in sophistication, and therefore may not be recognized until launched or may go undetected for extended periods
- cyber attacks can originate from a wide variety of sources, including JPMorganChase's own employees, cyber-criminals, hacktivists, groups linked to terrorist organizations or hostile nation-states that can sustain malicious activities for extended periods, or third parties whose objective is to disrupt the operations of financial institutions more generally
- JPMorganChase does not have control over the cybersecurity of the systems of the large number of clients, customers, counterparties and third-party service providers with which it does business, and
- it is possible that a third party, after establishing a foothold on an internal network without being detected, may gain access to other networks and systems.

The risk of a security breach due to a cyber attack could increase in the future due to factors such as:

- JPMorganChase's ongoing expansion of its digital banking and other internet-based product offerings and its internal use of internet-based products and applications, including those that use cloud computing services
- advances in artificial intelligence, such as the use of machine learning, generative artificial intelligence and quantum computing by malicious actors to develop more advanced social engineering attacks, including targeted phishing attacks
- the inability to maintain the security of information transmitted by JPMorganChase due to advances in quantum computing that may counteract or nullify existing information protections, and
- the acquisition and integration of new businesses.

In addition, a third party could misappropriate confidential information obtained by intercepting signals or communications from mobile devices used by JPMorganChase's employees.

The dynamic nature of the cyber threat landscape, including the pace of innovation and increased threat of novel attack methods, necessitates ongoing investment in, as well as enhancement and adaptation of, cybersecurity controls, including the adoption of enhanced security measures in certain jurisdictions. Failure to discover or address emerging threats, known vulnerabilities or shortcomings in cybersecurity controls, or to prioritize or complete enhancements to address them, in each case in a timely manner, may leave JPMorganChase vulnerable to cyber attacks, potentially resulting in data breaches, financial losses, reputational damage and regulatory penalties, including the failure to prioritize or complete enhancements relating to:

- preventing unauthorized access and protecting against the misuse of access, including the maintenance and enhancement of controls related to secure software development practices and identity and access management, such as those relating to the management of administrative access to systems
- detecting, escalating and addressing effectively and in a timely manner any vulnerabilities that may be present either in internally-developed software or externally-provided software or services, including vulnerabilities that could allow attackers to exploit unknown security flaws in software and hardware ("zero-day vulnerabilities")
- oversight of third-party vendors and early detection of attacks against those vendors, including ransomware attacks and attacks targeting vulnerabilities in third-party open-source software, in support of the secure development and maintenance of internal systems
- maintaining and enhancing controls related to technology asset management and inventory systems to prevent the risk of undetected vulnerabilities that could undermine JPMorganChase's ability to operate an effective control process
- upgrading the coverage and capabilities of systems and controls to protect JPMorganChase and its clients and customers from the impact of distributed denial-of-service attacks, or to recover from outages that could be caused by a malware or ransomware attack
- the continuing migration of client-facing services to the cloud, and modernization of those services
- strengthening network security and managing outbound connections to reduce the risk of data loss
- identifying, assessing and mitigating insider threat activities that could lead to the misuse of JPMorganChase's systems or client and customer information, and
- integrating acquired businesses where system integration may be complex or may require extensive and lengthy remediation or enhancement of controls.

A successful penetration or circumvention of the security of JPMorganChase's systems or the systems of a vendor, governmental body or another market participant could cause serious negative consequences, including:

- significant disruption of JPMorganChase's operations and those of its clients, customers and counterparties, including losing access to operational systems
- misappropriation of confidential information of JPMorganChase or that of its clients, customers, counterparties, employees or regulators
- disruption of or damage to JPMorganChase's systems and those of its clients, customers and counterparties
- the inability, or extended delays in the ability, to fully recover and restore data that has been stolen, manipulated or destroyed, or the inability to prevent systems from processing fraudulent transactions
- demands that JPMorganChase pay a ransom to a malicious actor that has perpetrated a cybersecurity breach
- unintended violations by JPMorganChase of applicable privacy and other laws
- financial loss to JPMorganChase or to its clients, customers, counterparties or employees
- losses to JPMorganChase in excess of cyber insurance policy coverage
- loss of confidence in JPMorganChase's cybersecurity and business resiliency measures
- dissatisfaction among JPMorganChase's clients, customers or counterparties
- significant exposure to litigation and regulatory fines, penalties or other sanctions, and
- harm to JPMorganChase's reputation.

The extent of a particular cyber attack, the methods and tools used by various actors, and the steps that JPMorganChase may need to take to investigate the attack may not be immediately clear, and it may take a significant amount of time before such an investigation can be completed. While such an investigation is ongoing, JPMorganChase may not necessarily know the full extent of the harm caused by the cyber attack, and that damage may continue to spread. These factors may inhibit JPMorganChase's ability to provide rapid, full and reliable information about the cyber attack to its clients, customers, counterparties and regulators, as well as the public. Furthermore, it may not be clear how best to contain and remediate the harm caused by the cyber attack, and certain errors or actions could be repeated or compounded before they are discovered and remediated. Any or all of these factors could further increase the costs and consequences of a cyber attack.

JPMorganChase can be negatively affected if it fails to identify and address operational risks associated with the introduction of or changes to products, services and delivery platforms or the adoption of new technologies.

When JPMorganChase launches a new product or service, introduces a new platform for the delivery or distribution of products or services (including mobile connectivity, electronic trading and cloud computing), acquires or invests in a business, makes changes to an existing product, service or delivery platform, or adopts a new technology, it may not fully appreciate or identify new operational risks that may arise from those changes, including increased reliance on third party providers, or may fail to implement adequate controls to mitigate the risks associated with those changes. Any significant failure in this regard could diminish JPMorganChase's ability to operate one or more of its businesses or result in:

- potential liability to clients, counterparties and customers
- higher compliance, operational or integration costs
- higher litigation costs, including regulatory fines, penalties and other sanctions
- damage to JPMorganChase's reputation
- impairment of JPMorganChase's liquidity
- regulatory intervention, or
- weaker competitive standing.

Any of the foregoing consequences could materially and adversely affect JPMorganChase's businesses and results of operations.

JPMorganChase's business and operations rely on its ability, and the ability of key external parties, to maintain appropriately-staffed workforces, and on the competence, trustworthiness, health and safety of employees.

JPMorganChase's ability to operate its businesses efficiently and profitably, to offer products and services that meet the expectations of its clients and customers, and to maintain an effective risk management framework is highly dependent on its ability to staff its operations appropriately and on the competence, trustworthiness, health and safety of its employees. JPMorganChase's businesses and operations similarly rely on the workforces of third parties, including employees of vendors, custodians and financial markets infrastructures, and of businesses that it may seek to acquire. JPMorganChase's businesses could be materially and adversely affected by:

- the ineffective implementation of business decisions
- any failure to institute controls that appropriately address risks associated with business activities, or to appropriately train employees with respect to those risks and controls
- staffing shortages, particularly in tight labor markets
- the possibility that significant portions of JPMorganChase's workforce are unable to work effectively, including because of illness, quarantines, shelter-in-place arrangements, government actions or other restrictions in connection with health emergencies, the spread of infectious diseases, epidemics or pandemics, or due to extraordinary events beyond JPMorganChase's control such as natural disasters or an outbreak or escalation of hostilities
- a significant operational breakdown or failure, theft, fraud or other unlawful conduct, or
- other negative outcomes caused by human error or misconduct by an employee of JPMorganChase or of another party on which JPMorganChase's businesses or operations rely.

JPMorganChase's operations could also be impaired if the measures taken by it or by governmental authorities to protect the health and safety of its employees are ineffective, or if any external party on which JPMorganChase relies fails to take appropriate and effective actions to protect the health and safety of its employees.

JPMorganChase faces substantial legal and operational risks in the processing and safeguarding of personal information.

JPMorganChase's businesses and operations are subject to complex and evolving laws, rules and regulations, both within and outside the U.S., governing the privacy and protection of personal information of individuals. Governmental authorities around the world have adopted and are considering the adoption of numerous legislative and regulatory initiatives concerning privacy, data protection and security. Litigation or enforcement actions relating to these laws, rules and regulations could result in fines or orders requiring that JPMorganChase change its data-related practices, which could have an adverse effect on JPMorganChase's ability to provide products and otherwise harm its business operations.

Implementing processes relating to JPMorganChase's collection, use, sharing and storage of personal information to comply with all applicable laws, rules and regulations in all relevant jurisdictions, including where the laws of different jurisdictions are in conflict, can:

- increase JPMorganChase's compliance and operating costs
- hinder the development of new products or services, curtail the offering of existing products or services, or affect how products and services are offered to clients and customers
- demand significant oversight by JPMorganChase's management, and
- require JPMorganChase to structure its businesses, operations and systems in less efficient ways.

Not all of JPMorganChase's clients, customers, vendors, counterparties and other external parties may have appropriate controls in place to protect the confidentiality, integrity or availability of the information exchanged between them and JPMorganChase, particularly where information is transmitted by electronic means. JPMorganChase could be exposed to litigation or regulatory fines, penalties or other sanctions if personal information of clients, customers, employees or others were to be mishandled or misused, such as situations where such information is:

- erroneously provided to parties who are not permitted to have the information, or
- intercepted or otherwise compromised by unauthorized third parties.

The increasing sophistication of artificial intelligence technologies poses a greater risk of identity fraud, as malicious actors may exploit artificial intelligence to create convincing false identities or manipulate verification processes. This challenge necessitates ongoing enhancements to client verification systems and security protocols to prevent unauthorized access and protect sensitive client information. Failure to manage these risks or to implement effective countermeasures could lead to unauthorized transactions, financial losses, reputational damage and increased regulatory scrutiny.

Concerns regarding the effectiveness of JPMorganChase's measures to safeguard personal information, or the perception that those measures are inadequate, could cause JPMorganChase to lose existing or potential clients and customers or employees, and thereby reduce JPMorganChase's revenues. Furthermore, any failure or perceived failure by JPMorganChase to comply with applicable privacy or data protection laws, rules and regulations, or any failure to appropriately calibrate, manage and monitor access by employees or third parties to personal information, could subject JPMorganChase to inquiries, examinations and investigations that could result in requirements to modify or cease certain operations or practices, significant liabilities or regulatory fines, penalties or other sanctions. Any of these could damage JPMorganChase's reputation and otherwise adversely affect its businesses.

In recent years, well-publicized incidents involving the inappropriate collection, use, sharing or storage of personal information have led to expanded governmental scrutiny of practices relating to the processing or safeguarding of personal information by companies in the U.S. and other countries. That scrutiny has in some cases resulted in, and could in the future lead to, the adoption of stricter laws, rules and regulations relating to the collection, use, sharing and storage of personal information. These types of laws, rules and regulations can prohibit or significantly restrict financial services firms such as JPMorganChase from transferring information across national borders or sharing information among affiliates or with third parties such as vendors, thereby increase compliance costs and operational risk, or restrict JPMorganChase's use of personal information when developing or offering products or services to customers. Some countries are considering or have adopted legislation implementing data protection requirements or requiring local storage and processing of data which could increase the cost and complexity of JPMorganChase's delivery of products and services. These restrictions could also inhibit JPMorganChase's development or marketing of certain products or services, or increase the costs of offering them to customers.

JPMorganChase's operations, results and reputation could be harmed by occurrences of extraordinary events beyond its control.

JPMorganChase's business and operational systems could be seriously disrupted, and its reputation could be harmed, by events or contributing factors that are wholly or partially beyond its control, including material instances of:

- cyber attacks
- security breaches of its physical premises, including threats to health and safety
- power, telecommunications or internet outages, or shutdowns of mass transit
- failure of, or loss of access to, technology or operational systems, including any resulting loss of critical data
- interruption of service from third-party service providers
- damage to or loss of property or assets of JPMorganChase or third parties, and any consequent injuries, including in connection with any construction projects undertaken by JPMorganChase
- effects of climate change
- natural disasters or severe weather conditions
- accidents such as explosions or structural failures
- health emergencies, the spread of infectious diseases, epidemics or pandemics, or
- events arising from local or larger-scale civil or political unrest, any outbreak or escalation of hostilities, or terrorist acts.

JPMorganChase operates a Firmwide resiliency framework that is intended to enable it to prepare for and adapt to changing conditions and withstand and recover from, and address any adverse effects on its operations caused by, disruptions that may impact critical business functions and supporting assets, including its staff, technology, data and facilities and those of third-party service providers.

Although not every form of disruption can be anticipated or defended against, JPMorganChase strives for resiliency or recovery in a range of scenarios in the event of a disruption, including due to the occurrence of an extraordinary event beyond its control. There can be no assurance that JPMorganChase's Firmwide resiliency framework will fully mitigate all potential resiliency risks to JPMorganChase, its clients, and customers and third parties with which it does business, or that its resiliency framework will be adequate to address the effects of simultaneous occurrences of multiple or extended disruption events. In addition, JPMorganChase's ability to respond effectively to a disruption event could be hampered to the extent that the members of its workforce, physical assets, systems and other support infrastructure, or those of its third-party service providers, that are needed to address the event are geographically dispersed, or conversely, if such an event were to occur in an area in which they are concentrated. Further, should extraordinary events or the factors that cause or contribute to those events become more chronic, the disruptive effects of those events on JPMorganChase's business and operations, and on its clients, customers, counterparties and employees, could become more significant and long-lasting.

Any significant failure or disruption of JPMorganChase's operations or operational systems, or the occurrence of one or more extraordinary events that are beyond its control, could:

- hinder JPMorganChase's ability to provide services to its clients and customers or to transact with its counterparties
- require it to expend significant resources to correct the failure or disruption or to address the event
- cause it to incur losses or liabilities, including from loss of revenue, damage to or loss of property, or injuries
- disrupt market infrastructure systems on which JPMorganChase's businesses rely
- expose it to litigation or regulatory fines, penalties or other sanctions, and
- harm its reputation.

The occurrence of one or more extraordinary events could also negatively impact the financial condition or creditworthiness of JPMorganChase's clients and customers, and could lead to an increase in delinquencies, additions to the allowance for credit losses and higher net charge-offs, which can reduce JPMorganChase's earnings.

Data quality is essential to JPMorganChase's business and operations, and if JPMorganChase fails to maintain adequate data management processes, this could adversely affect its ability to effectively manage its businesses, comply with applicable laws, rules and regulations, or remain competitive.

JPMorganChase relies on accurate, timely and complete data to effectively operate its systems and processes, including:

- assessing risk exposures and limits
- monitoring and detecting fraudulent transactions and cyber threats
- developing or maintaining models and other analytical and judgment-based estimations, including those that use machine learning or artificial intelligence
- implementing and maintaining compliance programs, and
- preparing financial statements, disclosures and regulatory reports, as well as internal reporting

Any deficiencies in JPMorganChase's data management processes, including with respect to the accuracy or completeness of data, the timeliness of data collection, the analysis or validation of data, or the safeguarding of data could undermine the reliability and effectiveness of its operations, including:

- risk management practices, including inaccurate or untimely risk reporting
- delivery of regulatory reporting or internal or external financial reporting
- compliance practices, such as those relating to transaction monitoring, customer screening, blocking and rejecting transactions, recordkeeping or reporting
- business activities, such as those related to managing JPMorganChase's market-making positions and liquidity and capital levels, including reliance on timely data for informed decision-making
- providing services to clients and customers, including transaction processing, lending services, account management and customer support, or
- fraud detection and prevention processes.

Any or all of these factors could impair the ability of JPMorganChase to make sound business decisions, cause it to incur higher operational and compliance costs, result in operational breakdowns or failure to meet its regulatory requirements, negatively affect clients and customers, or lead to reputational harm.

Enhanced regulatory and other standards for the oversight of vendors and other service providers can result in higher costs and other potential exposures.

JPMorganChase must comply with enhanced regulatory and other standards associated with doing business with vendors and other service providers, including standards relating to the outsourcing of functions as well as the performance of significant banking and other functions by subsidiaries. JPMorganChase incurs significant costs and expenses in connection with its initiatives to address the risks associated with oversight of its internal and external service providers. JPMorganChase's failure to appropriately assess and manage these relationships, especially those involving significant banking functions, shared services or other critical activities, could materially adversely affect JPMorganChase. Specifically, any such failure could result in:

- potential harm to clients and customers, and any liability associated with that harm
- regulatory fines, penalties or other sanctions
- lower revenues, and the opportunity cost from lost revenues
- increased operational costs, or
- harm to JPMorganChase's reputation.

JPMorganChase's risk management framework and control environment will not be effective in identifying and mitigating every risk to JPMorganChase.

Any inadequacy or lapse in JPMorganChase's risk management framework, governance structure, practices, models or reporting systems, or in its control environment, could expose it to unexpected losses, and its financial condition or results of operations could be materially and adversely affected. Any such inadequacy or lapse could:

- hinder the timely escalation of material risk issues to JPMorganChase's senior management and Board of Directors
- lead to business decisions that have negative outcomes for JPMorganChase
- require significant resources and time to remediate
- lead to non-compliance with laws, rules and regulations
- attract heightened regulatory scrutiny
- expose JPMorganChase to litigation, regulatory investigations or regulatory fines, penalties or other sanctions
- lead to potential harm to customers and clients, and any liability associated with that harm
- harm its reputation, or
- otherwise diminish confidence in JPMorganChase.

Many of JPMorganChase's risk management strategies and techniques consider historical market behavior and to some degree are based on management's subjective judgment or assumptions. For example, many models used by JPMorganChase are based on assumptions regarding historical correlations among prices of various asset classes or other market indicators. In times of market stress, including difficult or less liquid market environments, or in the event of other unforeseen circumstances, previously uncorrelated indicators may become correlated. Conversely, previously-correlated indicators may become uncorrelated at those times. Sudden market movements and unanticipated market or economic events could, in some circumstances, limit the effectiveness of JPMorganChase's risk management strategies, causing it to incur losses.

JPMorganChase could recognize unexpected losses, its capital levels could be reduced and it could face greater regulatory scrutiny if its models, estimations or judgments, including those used in its financial statements, are inadequate or incorrect.

JPMorganChase has developed and uses a variety of models and other analytical and judgment-based estimations to measure, monitor and implement controls over its market, credit, capital, liquidity, operational and other risks. JPMorganChase also uses internal models and estimations as a basis for its stress testing and in connection with the preparation of its financial statements under U.S. generally accepted accounting principles ("U.S. GAAP").

These models and estimations are based on a variety of assumptions and historical trends, and are periodically reviewed and modified as necessary. The models and estimations that JPMorganChase uses, including those that use machine learning or artificial intelligence, may not be effective in all cases to identify, observe and mitigate risk due to a variety of factors, such as:

- reliance on historical trends that may not persist in the future, including assumptions underlying the models and estimations such as correlations among certain market indicators or asset prices
- inherent limitations associated with forecasting uncertain economic and financial outcomes
- historical trend information may be incomplete, or may not be indicative of severely negative market conditions such as extreme volatility, dislocation or lack of liquidity
- sudden illiquidity in markets or declines in prices of certain loans and securities may make it more difficult to value certain financial instruments
- technology that is introduced to run models or estimations may not perform as expected, or may not be well understood by the personnel using the technology
- models and estimations may contain erroneous data, valuations, formulas or algorithms
- review processes may fail to detect flaws in models and estimations, and

- models may inadvertently incorporate biases present in data used in the models.

JPMorganChase may experience unexpected losses if models, estimates or judgments used or applied in connection with its risk management activities or the preparation of its financial statements are inadequate or incorrect. For example, where quoted market prices are not available for certain financial instruments that require a determination of their fair value, JPMorganChase may make fair value determinations based on internally developed models or other means which ultimately rely to some degree on management estimates and judgment. In addition, JPMorganChase may experience increased uncertainty in its estimates if assets acquired differ from those used to develop those models, which may lead to unexpected losses.

Similarly, JPMorganChase establishes an allowance for expected credit losses related to its credit exposures which requires significant judgments, including forecasts of how macroeconomic conditions might impair the ability of JPMorganChase's clients and customers to repay their loans or other obligations. These types of estimates and judgments may not prove to be accurate due to a variety of factors, including when the current and forecasted environments are significantly different from the historical environments upon which the models were developed. The increased uncertainty may necessitate a greater degree of judgment and analytics to inform any adjustments that JPMorganChase may make to model outputs than would otherwise be the case.

Some of the models and other analytical and judgment-based estimations used by JPMorganChase in managing risks are subject to review by, and require the approval of, JPMorganChase's regulators. These reviews are required before JPMorganChase may use those models and estimations for calculating market risk RWA, credit risk RWA and operational risk RWA under Basel III. If JPMorganChase's models or estimations are not approved by its regulators, it may be subject to higher capital charges, which could adversely affect its financial results or limit the ability to expand its businesses.

Lapses, weaknesses or deficiencies in controls over disclosure or financial reporting could materially affect JPMorganChase's profitability or reputation.

JPMorganChase's businesses and operations are subject to complex and evolving laws, rules and regulations, both within and outside the U.S., requiring continuous enhancements to various disclosures in its financial statements and regulatory reports.

There can be no assurance that JPMorganChase's disclosure controls and procedures will be effective in every circumstance, or that a material weakness or significant deficiency in internal control over financial reporting will not occur. Any such lapse, weakness or deficiency could result in inaccurate financial reporting which, in turn, could:

- materially and adversely affect JPMorganChase's business and results of operations or financial condition
- restrict its ability to access the capital markets
- require it to expend significant resources to correct the lapse, weakness or deficiency
- expose it to litigation or regulatory fines, penalties or other sanctions
- harm its reputation, or
- otherwise diminish investor confidence in JPMorganChase.

Strategic

JPMorganChase's results or competitive standing could suffer if its management fails to develop and execute effective business strategies, and to anticipate changes affecting those strategies.

The development and execution of effective business strategies by JPMorganChase's management, along with the ability to anticipate and respond to shifts in the competitive environment, are critical to JPMorganChase's competitive standing and to achieving its strategic objectives. These strategies relate to:

- the products and services that JPMorganChase offers
- the geographies in which it operates
- the types of clients and customers that it serves
- the businesses that it acquires or in which it invests
- the counterparties with which it does business
- the technologies that it adopts or in which it invests, which may include new and currently unproven technologies, and
- the methods, distribution channels and third party service providers by or through which it offers products and services.

If management makes choices about these strategies and goals that prove to be incorrect, are based on incomplete, inaccurate or fraudulent information, do not accurately assess the competitive landscape and industry trends, or fail to address changing regulatory and market environments or the expectations of clients, customers, investors, employees and other stakeholders, then the franchise values and growth prospects of JPMorganChase's businesses may suffer and its earnings could decline.

JPMorganChase's growth prospects also depend on management's ability to develop and execute effective business plans to address these strategic priorities, both in the near term and over longer time horizons. Management's effectiveness in this regard will affect

JPMorganChase's ability to develop and enhance its resources, control expenses and return capital to shareholders. Each of these objectives could be adversely affected by any failure on the part of management to:

- devise effective business plans and strategies
- offer products and services that meet changing expectations of clients and customers
- allocate capital in a manner that promotes long-term stability to enable JPMorganChase to build and invest in market-leading businesses, even in a highly stressed environment
- allocate capital appropriately due to imprecise modeling or subjective judgments made in connection with those allocations
- appropriately assess and monitor principal investments made to enhance or accelerate JPMorganChase's business strategies
- conduct appropriate due diligence on prospective business acquisitions or investments, or effectively integrate newly-acquired businesses
- appropriately address concerns of clients, customers, investors, employees and other stakeholders, including with respect to climate and other ESG matters
- react quickly to changes in market conditions or market structures, or
- develop and enhance the operational, technology, risk, financial and managerial resources and capabilities necessary to grow and manage JPMorganChase's businesses.

Furthermore, JPMorganChase may incur costs in connection with disposing of excess properties, premises and facilities, and those costs could be material to its results of operations.

JPMorganChase faces significant and increasing competition in the rapidly evolving financial services industry.

JPMorganChase operates in a highly competitive environment in which it must evolve and adapt to changes in financial regulation, technological advances, increased public scrutiny and changes in economic conditions. JPMorganChase expects that competition in the U.S. and global financial services industry will continue to be intense. Competitors include:

- other banks and financial institutions
- trading, advisory and investment management firms
- finance companies
- technology companies, and
- other non-bank firms that are engaged in providing similar as well as new products and services.

JPMorganChase cannot provide assurance that the significant competition in the financial services industry will not materially and adversely affect its future results of operations. For example, aggressive or less disciplined lending practices by non-bank competitors could lead to a loss of market share for traditional banks, and in an economic downturn could result in instability in the financial services industry and adversely impact other market participants, including JPMorganChase.

New competitors in the financial services industry continue to emerge. For example, technological advances and the growth of e-commerce have made it possible for non-depository institutions to offer products and services that traditionally were banking products. These advances have also allowed financial institutions and other companies to provide electronic and internet-based financial solutions, including electronic securities and cryptocurrency trading, lending and other extensions of credit to consumers, payments processing and online automated algorithmic-based investment advice. Furthermore, both financial institutions and their non-banking competitors face the risk that payments processing and other products and services, including deposits and other traditional banking products, could be significantly disrupted by the use of new technologies, such as cryptocurrencies and other applications using secure distributed ledgers, that may not require intermediation. New technologies have required and could require JPMorganChase to spend more to modify or adapt its products to attract and retain clients and customers or to match products and services offered by its competitors, including technology companies. In addition, new technologies may be used by customers, or breached or infiltrated by third parties, in unexpected ways, which can increase JPMorganChase's costs for complying with laws, rules and regulations that apply to the offering of products and services through those technologies and reduce the income that JPMorganChase earns from providing products and services through those technologies.

Ongoing or increased competition may put pressure on the pricing for JPMorganChase's products and services or may cause JPMorganChase to lose market share, particularly with respect to traditional banking products. This competition may be based on quality and variety of products and services offered, transaction execution, innovation, reputation and price. The failure of any of JPMorganChase's businesses to meet the expectations of clients and customers, whether due to general market conditions, under-performance, a decision not to offer a particular product or service, changes in client and customer expectations or other factors, could affect JPMorganChase's ability to attract or retain clients and customers. Any such impact could, in turn, reduce

JPMorganChase's revenues. Increased competition also may require JPMorganChase to make additional capital investments in its businesses, or to extend more of its capital on behalf of its clients to remain competitive.

The effects of climate change could adversely affect JPMorganChase's business and operations, both directly and as a result of impacts on its clients and customers.

JPMorganChase operates in many regions, countries and communities around the world where its business, and the activities of its clients and customers, could be adversely affected by climate change. Climate change could manifest as a financial risk to JPMorganChase either through changes in the physical climate or from the process of transitioning to a lower-carbon economy. Both physical risks and transition risks associated with climate change could have negative impacts on the financial condition or creditworthiness of JPMorganChase's clients and customers, on JPMorganChase's exposure to affected companies and markets, and on the effectiveness of JPMorganChase's existing business strategy with respect to its operations, clients and customers.

Physical risks include the increased frequency or severity of acute weather events, such as floods, wildfires and tropical cyclones, and chronic shifts in the climate, such as rising sea levels, persistent changes in precipitation levels, or increases in average ambient temperatures. Potential adverse impacts of climate-related physical risks to JPMorganChase, its clients or customers include:

- declines in asset values, including due to the destruction or degradation of property
- reduced availability or increased cost of insurance for clients of JPMorganChase
- interruptions to business operations, including supply chain disruption, and
- population migration or unemployment in affected regions.

Transition risks arise from the financial and economic consequences of society's shift towards a lower-carbon economy, such as changes in public policy, adoption of new technologies or changes in consumer preferences towards low-carbon goods and services. These risks could also be influenced by changes in the physical climate. Potential adverse impacts of transition risks to JPMorganChase, its clients or customers include:

- sudden devaluation of assets, including unanticipated write-downs ("stranded assets")
- increased operational and compliance costs driven by changes in climate policy
- increased energy costs driven by governmental actions and initiatives such as emission pricing and accelerated decarbonization policies
- negative consequences to business models, and the need to make changes in response to those consequences, and
- damage to JPMorganChase's reputation, including due to any perception that its business practices are contrary to public policy or the preferences of different stakeholders.

Climate risks can also arise from inconsistencies and conflicts in the manner in which climate policy and financial regulations are implemented in the many regions where JPMorganChase operates, including initiatives to apply and enforce policy and regulation with extraterritorial effect. Additionally, internal models and estimations used in climate risk assessments have an increased level of uncertainty due to limited historical trend information and the absence of standardized, reliable and comprehensive greenhouse gas emissions data, which could lead to inaccurate disclosures or financial reporting.

Conduct

Conduct failure by JPMorganChase employees can harm clients and customers, impact market integrity, damage JPMorganChase's reputation and trigger litigation and regulatory action.

JPMorganChase's employees interact with clients, customers, counterparties and other market and industry participants, and with each other, every day. All employees are expected to demonstrate values and exhibit the behaviors that are an integral part of JPMorganChase's Code of Conduct and Business Principles. JPMorganChase endeavors to embed conduct risk management throughout an employee's life cycle, including recruiting, onboarding, training and development, and performance management. Conduct risk management is also an integral component of JPMorganChase's promotion and compensation processes.

Notwithstanding these expectations, policies and practices, certain employees have engaged in improper or illegal conduct in the past. These instances of misconduct have resulted in litigation, and resolutions of governmental investigations or enforcement actions involving consent orders, deferred prosecution agreements, non-prosecution agreements and other civil or criminal sanctions. There is no assurance that further inappropriate or unlawful actions by employees have not occurred or will not occur, lead to a violation of the terms of these resolutions (and associated consequences), or that any such actions will always be detected, deterred or prevented.

JPMorganChase's reputation could be harmed by, and collateral consequences could result from, a failure by one or more employees to conduct themselves in accordance with JPMorganChase's expectations, policies and practices, including by acting in ways that harm clients, customers, other market participants, employees or others. Some examples of this include:

- improperly selling and marketing JPMorganChase's products or services
- engaging in insider trading, market manipulation or unauthorized trading
- engaging in improper or fraudulent behavior in connection with government relief programs
- facilitating a transaction where a material objective is to achieve a particular tax, accounting or financial disclosure treatment that may be subject to scrutiny by governmental or regulatory authorities, or where the proposed treatment is unclear or may not reflect the economic substance of the transaction
- failing to fulfill fiduciary obligations or other duties owed to clients or customers
- violating antitrust or anti-competition laws by colluding with other market participants
- using electronic communications channels that have not been approved by JPMorganChase
- engaging in discriminatory behavior or harassment with respect to clients, customers or employees, or acting contrary to JPMorganChase's goal of fostering an inclusive workplace
- managing or reporting risks in ways that subordinate JPMorganChase's risk appetite to business performance goals or employee compensation objectives, and
- misappropriating property, confidential or proprietary information, or technology assets belonging to JPMorganChase, its clients and customers or third parties.

The consequences of any failure by one or more employees to conduct themselves in accordance with JPMorganChase's expectations, policies or practices could include litigation, or regulatory or other governmental investigations or enforcement actions. Any of these proceedings or actions could result in judgments, settlements, fines, penalties or other sanctions, or lead to:

- financial losses
- increased operational and compliance costs
- greater scrutiny by regulators and other parties
- regulatory actions that require JPMorganChase to restructure, curtail or cease certain of its activities
- the need for significant oversight by JPMorganChase's management
- loss of clients or customers, and
- harm to JPMorganChase's reputation.

The foregoing risks could be heightened with respect to newly-acquired businesses if JPMorganChase fails to successfully integrate employees of those businesses or any of those employees do not conduct themselves in accordance with JPMorganChase's expectations, policies and practices.

Reputation

Damage to JPMorganChase's reputation could harm its businesses.

Maintaining trust in JPMorganChase is critical to its ability to attract and retain clients, customers, investors and employees. Damage to JPMorganChase's reputation can therefore cause significant harm to JPMorganChase's business and prospects, and can arise from numerous sources, including:

- employee misconduct, including discriminatory behavior or harassment with respect to clients, customers or employees, or actions that are contrary to JPMorganChase's goal of fostering an inclusive workplace
- security breaches, including as a result of cyber attacks
- failure to safeguard client, customer or employee information
- failure to manage risks associated with its client relationships, or with transactions or business activities in which JPMorganChase or its clients engage, including transactions or activities that may be unpopular among one or more constituencies
- rapid and broad dissemination of misinformation and disinformation across the media landscape, including social networking sites
- incorrect, biased or misleading results or content generated by artificial intelligence, leading to harmful outcomes, including discrimination in lending practices against vulnerable populations, fraud, manipulation of customers, privacy breaches or intellectual property infringement
- deficiencies or perceived failures in managing ESG-related initiatives, including modifying or failing to meet publicly-announced targets
- operational failures
- litigation or regulatory fines, penalties or other sanctions
- actions taken in executing regulatory and governmental requirements during a global or regional health emergency, spread of infectious disease, epidemic or pandemic
- regulatory investigations or enforcement actions, or resolutions of these matters, and
- failure or perceived failure to comply with laws, rules or regulations by JPMorganChase or its clients,

customers, counterparties or other parties, including newly-acquired businesses, companies in which JPMorganChase has made principal investments, parties to joint ventures with JPMorganChase, and vendors with which JPMorganChase does business.

Social and environmental activists have been targeting JPMorganChase and other financial services firms with public criticism concerning their business practices, including business relationships with clients that are engaged in certain sensitive industries, such as companies:

- whose products are or are perceived to be harmful to human health, or
- whose activities negatively affect or are perceived to negatively affect the environment, workers' rights or communities.

Activists have also taken actions intended to change or influence JPMorganChase's business practices with respect to ESG matters, including public protests at JPMorganChase's headquarters and other properties, and submitting specific ESG-related proposals for a vote by JPMorganChase's shareholders.

In addition, JPMorganChase has been and expects that it will continue to be criticized by activists, politicians and other members of the public concerning business practices or positions taken by JPMorganChase with respect to matters of public policy (such as diversity, equity and inclusion initiatives) or regarding transactions or other business or interactions between JPMorganChase and governmental or regulatory bodies. Furthermore, JPMorganChase's relationships or ability to transact with clients and customers, and with governmental or regulatory bodies in jurisdictions in which JPMorganChase does business, could be adversely affected if its decisions with respect to doing business with companies in certain sensitive industries are perceived to harm those companies or to align with particular political viewpoints. The foregoing types of criticism can be more widespread during election years in various jurisdictions, and could have the effect of focusing attention on a company such as JPMorganChase as part of a wider public debate on public policy matters. Furthermore, JPMorganChase's participation in or association with certain environmental and social industry groups or initiatives could be viewed by activists or governmental authorities as boycotting or other discriminatory business behavior.

These and other types of criticism and actions directed at JPMorganChase could potentially engender dissatisfaction among clients, customers, investors, employees, government officials and other stakeholders. In all of these cases, JPMorganChase's reputation and its business and results of operations could be harmed by:

- greater scrutiny from governmental or regulatory bodies, or further criticism from politicians and other members of the public, including in the form of governmental or regulatory investigations or litigation
- unfavorable coverage or commentary in the media, including through social media campaigns
- certain clients and customers ceasing doing business with JPMorganChase, and encouraging others to do so
- impairment of JPMorganChase's ability to attract new clients and customers, to expand its relationships with existing clients and customers, or to hire or retain employees, or
- certain investors opting to divest from investments in securities of JPMorganChase.

Actions by the financial services industry generally or individuals in the industry can also affect JPMorganChase's reputation. For example, the reputation of the industry as a whole can be damaged by concerns that:

- consumers have been treated unfairly by a financial institution, or
- a financial institution has acted inappropriately with respect to the methods used to offer products to customers.

If JPMorganChase is perceived to have engaged in these types of behaviors, this could weaken its reputation among clients or customers, employees or other stakeholders.

Failure to effectively manage potential conflicts of interest or to satisfy fiduciary obligations can result in litigation and enforcement actions, as well as damage JPMorganChase's reputation.

JPMorganChase's ability to manage potential conflicts of interest is highly complex due to the broad range of its business activities which encompass a variety of transactions, obligations and interests with and among JPMorganChase's clients and customers. JPMorganChase can become subject to litigation, enforcement actions, and heightened regulatory scrutiny, and its reputation can be damaged, by the failure or perceived failure to:

- adequately address or appropriately disclose conflicts of interest, including potential conflicts of interest that may arise in connection with providing multiple products and services in, or having one or more investments related to, the same transaction
- identify and address any conflict of interest that a third party with which it is does business may have with respect to a transaction involving JPMorganChase
- deliver appropriate standards of service and quality

- treat clients and customers fairly and with the appropriate standard of care
- use client and customer data responsibly and in a manner that meets legal requirements and regulatory expectations
- provide fiduciary products or services in accordance with the applicable legal and regulatory standards, or
- handle or use confidential information of customers or clients appropriately and in compliance with applicable data protection and privacy laws, rules and regulations.

A failure or perceived failure to appropriately address conflicts of interest or fiduciary obligations could result in customer dissatisfaction, litigation and regulatory fines, penalties or other sanctions, and heightened regulatory scrutiny and enforcement actions, all of which can lead to lost revenue and higher operating costs and cause serious harm to JPMorganChase's reputation.

Country

An outbreak or escalation of hostilities between countries or within a country or region could have a material adverse effect on the global economy and on JPMorganChase's businesses within the affected region or globally.

Aggressive actions by hostile governments or groups, including armed conflict or intensified cyber attacks, could expand in unpredictable ways by drawing in other countries or escalating into full-scale war with potentially catastrophic consequences, particularly if one or more of the combatants possess nuclear weapons. Depending on the scope of the conflict, the hostilities could result in:

- worldwide economic disruption
- heightened volatility in financial markets
- severe declines in asset values, accompanied by widespread sell-offs of investments
- sudden increases in prices in the energy and commodity markets or for certain safe haven currencies
- substantial depreciation of local currencies, potentially leading to defaults by borrowers and counterparties in the affected region
- disruption of global trade
- diminished consumer, business and investor confidence
- refugee and humanitarian crises, and
- new economic sanctions or other regulatory requirements, including those that introduce exceptional compliance challenges for multinational companies such as JPMorganChase.

Any of the above consequences could have significant negative effects on JPMorganChase's operations and earnings, both in the countries or regions directly affected by the hostilities or globally. Further, if the U.S. were to become directly involved in such a conflict, this could lead to a curtailment of any operations that JPMorganChase may have in the affected countries or region, as well as in any nation that is aligned against the U.S. in the hostilities. JPMorganChase could also experience more numerous and aggressive cyber attacks launched by or under the sponsorship of one or more of the adversaries in such a conflict.

JPMorganChase's business and operations in certain countries can be adversely affected by local economic, political, regulatory and social factors.

Some of the countries in which JPMorganChase conducts business have economies or markets that are less developed and more volatile or may have political, legal and regulatory regimes that are less established or predictable than other countries in which JPMorganChase operates. In addition, in some jurisdictions in which JPMorganChase conducts business, the local economy and business activities are subject to substantial government influence or control. Some of these countries have in the past experienced economic disruptions, including:

- extreme currency fluctuations
- high inflation
- low or negative growth
- defaults or reduced ability to service sovereign debt and
- increased fraud or other misrepresentation of value.

The governments in these countries have sometimes reacted to these developments by imposing restrictive policies that adversely affect the local and regional business environment, such as:

- price, capital or exchange controls, including imposition of punitive transfer and convertibility restrictions or forced currency exchange
- expropriation or nationalization of assets, including client assets, or confiscation of property, including intellectual property, and
- changes in laws, rules and regulations.

The impact of these actions could be accentuated in trading markets that are smaller, less liquid and more volatile than more-developed markets. These types of government actions can negatively affect JPMorganChase's operations in the relevant country, either directly or by suppressing the business activities of local clients or multi-national clients that conduct business in the jurisdiction.

In addition, emerging markets countries, as well as more developed countries, have been susceptible to unfavorable social developments arising from poor economic conditions or governmental actions, including:

- widespread demonstrations, civil unrest or general strikes
- crime and corruption
- security and personal safety issues
- an outbreak or escalation of hostilities, or other geopolitical instabilities
- overthrow of incumbent governments
- terrorist attacks, and
- other forms of internal discord.

These economic, political, regulatory and social developments have in the past resulted in, and in the future could lead to, conditions that can adversely affect JPMorganChase's operations in those countries and impair the revenues, growth and profitability of those operations. In addition, any of these events or circumstances in one country can affect JPMorganChase's operations and investments in another country or countries, including in the U.S.

People

JPMorganChase's ability to attract and retain qualified employees is critical to its success.

JPMorganChase's employees are its most important resource, and in many areas of the financial services industry, competition for qualified personnel is intense. JPMorganChase endeavors to attract talented new employees from a variety of backgrounds and retain, develop and motivate its existing employees. JPMorganChase's efforts to hire and retain talented employees could be hindered by factors such as:

- the emerging need for more-skilled workers in an evolving labor and workplace environment, including due to changes in technology
- targeted recruitment of JPMorganChase employees by competitors, and
- modifications to or discontinuation of JPMorganChase's hybrid work models.

JPMorganChase's performance and competitive position could be materially and adversely affected if it is unable to attract or retain qualified employees for its workforce or to devise and execute effective succession planning for key leadership roles, such as the Chief Executive Officer, members of the Operating Committee and other senior leaders.

In addition, advances in technology, such as automation and artificial intelligence, may lead to workforce displacement. This could require JPMorganChase to invest in additional employee training, manage impacts on morale and retention, and compete for employment candidates who possess more advanced technological skills, all of which could have a negative impact on JPMorganChase's business and operations.

Unfavorable changes in immigration or travel policies could adversely affect JPMorganChase's businesses and operations.

JPMorganChase relies on the skills, knowledge and expertise of employees located throughout the world. Changes in immigration or travel policies in the U.S. and other countries that unduly restrict or otherwise make it more difficult for employees or their family members to work in, or travel to or transfer between, jurisdictions in which JPMorganChase has operations or conducts its business could inhibit JPMorganChase's ability to attract and retain qualified employees, and thereby dilute the quality of its workforce, or could prompt JPMorganChase to make structural changes to its worldwide or regional operating models that cause its operations to be less efficient or more costly.

Legal

JPMorganChase faces significant legal risks from litigation and formal and informal regulatory and government investigations.

JPMorganChase is named as a defendant or is otherwise involved in many legal proceedings, including class actions, derivative actions and other litigation or disputes with third parties, as well as criminal proceedings. Actions currently pending against JPMorganChase may result in judgments, settlements, fines, penalties or other sanctions adverse to JPMorganChase. Any of these matters could materially and adversely affect JPMorganChase's business, financial condition or results of operations, or cause serious reputational harm. As a participant in the financial services industry, it is likely that JPMorganChase will continue to experience a high level of litigation and regulatory and government investigations related to its businesses and operations.

Regulators and other government agencies conduct examinations of JPMorganChase and its subsidiaries both on a routine basis and in targeted exams, and JPMorganChase's businesses and operations are subject to heightened regulatory oversight. This heightened regulatory scrutiny, or the results of such an investigation or examination, may lead to additional regulatory investigations or enforcement actions. There is no assurance that those actions will not result in resolutions or other enforcement actions against JPMorganChase. Furthermore, a single event involving a potential violation of law or regulation may give rise to numerous and overlapping investigations and proceedings, either by multiple federal, state or local agencies and officials in the U.S. or, in some instances, regulators and other governmental officials in non-U.S. jurisdictions.

In addition, if another financial institution violates a law or regulation relating to a particular business activity or practice, this will often give rise to an investigation by regulators and other governmental agencies of the same or similar activity or practice by JPMorganChase.

JPMorganChase could become subject to a significant regulatory investigation and be unable to disclose specific information concerning that investigation to the public if such a disclosure would violate JPMorganChase's obligations under applicable rules and regulations to maintain the confidentiality of confidential supervisory information, even if the resolution of that investigation could have a material adverse effect on JPMorganChase's business, operations, results or financial condition.

Regulatory investigations, examinations or other initiatives by U.S. and non-U.S. governmental authorities may subject JPMorganChase to judgments, settlements, fines, penalties or other sanctions, and may require JPMorganChase to restructure its operations and activities or to cease offering certain products or services. All of these potential outcomes could harm JPMorganChase's reputation or lead to higher operational costs, thereby reducing JPMorganChase's profitability, or result in collateral consequences. In addition, the extent of JPMorganChase's exposure to legal and regulatory matters can be unpredictable and could, in some cases, exceed the amount of reserves that JPMorganChase has established for those matters.

Item 1B. Unresolved Staff Comments.

None.

Item 1C. Cybersecurity.

Refer to the Operational Risk Management section of Management's discussion and analysis on pages 153–156 for a discussion of cybersecurity risk.

Item 2. Properties.

JPMorganChase's headquarters is located in New York City at 383 Madison Avenue, a 47-story office building that it owns. The demolition of the Firm's former headquarters at 270 Park Avenue in New York City was completed in 2021, and construction of a new headquarters on the same site is nearing completion.

The Firm owned or leased facilities in the following locations at December 31, 2024.

December 31, 2024 (in millions)	Approximate square footage
United States[(a)]	
New York City, New York	
383 Madison Avenue, New York, New York	1.1
All other New York City locations	6.0
Total New York City, New York	**7.1**
Other U.S. locations	
Columbus/Westerville, Ohio	3.5
Chicago, Illinois	2.7
Dallas/Plano/Fort Worth, Texas	2.5
Wilmington/Newark, Delaware	2.1
Houston, Texas	1.6
Jersey City, New Jersey	1.4
Phoenix/Tempe, Arizona	1.3
All other U.S. locations	33.8
Total United States	**56.0**
Europe, the Middle East and Africa ("EMEA")	
25 Bank Street, London, U.K.	1.4
All other U.K. locations	2.3
All other EMEA locations	1.5
Total EMEA	**5.2**
Asia-Pacific, Latin America and Canada	
India	6.4
Philippines	1.8
All other locations	2.8
Total Asia-Pacific, Latin America and Canada	**11.0**
Total	**72.2**

(a) At December 31, 2024, the Firm owned or leased 4,966 branches in 48 states and Washington D.C.

The premises and facilities occupied by JPMorganChase are collectively used across all of the Firm's business segments and for corporate purposes. JPMorganChase continues to evaluate its current and projected space requirements and may determine from time to time that certain of its properties (including the premises and facilities noted above) are no longer necessary for its operations. There is no assurance that the Firm will be able to dispose of any such excess properties, premises or facilities, or that it will not incur costs in connection with such dispositions. Such disposition costs may be material to the Firm's results of operations in a given period. Refer to the Consolidated Results of Operations on pages 59–62 for information on occupancy expense.

Item 3. Legal Proceedings.

Refer to Note 30 for a description of the Firm's material legal proceedings.

Item 4. Mine Safety Disclosures.

Not applicable.

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities.

Market for registrant's common equity

JPMorganChase's common stock is listed and traded on the New York Stock Exchange. Refer to "Five-year stock performance," on page 51 for a comparison of the cumulative total return for JPMorganChase common stock with the comparable total return of the S&P 500 Index, the KBW Bank Index and the S&P Financials Index over the five-year period ended December 31, 2024.

Refer to Capital actions in the Capital Risk Management section of Management's discussion and analysis on page 105 for information on the common dividend payout ratio. Refer to Note 21 and Note 26 for discussions of restrictions on dividend payments. On January 31, 2025, there were 196,005 holders of record of JPMorganChase common stock. Refer to Part III, Item 12 on page 43 for information regarding securities authorized for issuance under the Firm's employee share-based incentive plans.

Repurchases under the common share repurchase program

Refer to Capital actions in the Capital Risk Management section of Management's discussion and analysis on page 105 for information regarding repurchases under the Firm's common share repurchase program.

On June 28, 2024, the Firm announced that its Board of Directors had authorized a new $30 billion common share repurchase program, effective July 1, 2024. Through June 30, 2024, the Firm was authorized to purchase up to $30 billion of common shares under its previously-approved common share repurchase program that was announced on April 13, 2022.

Shares repurchased pursuant to the common share repurchase programs during 2024 were as follows:

Year ended December 31, 2024	Total number of shares of common stock repurchased	Average price paid per share of common stock[(a)]	Aggregate purchase price of common stock repurchases (in millions)[(a)]	Dollar value of remaining authorized repurchase (in millions)[(a)(b)]
First quarter	15,869,936	$ 179.50	$ 2,849	$ 16,886
Second quarter	27,019,730	196.83	5,318	11,568 [(c)]
Third quarter	30,343,933	209.61	6,361	23,639
October	6,173,254	218.00	1,345	22,294
November	5,142,243	241.03	1,240	21,054
December	7,170,130	241.10	1,728	19,326
Fourth quarter	18,485,627	233.37	4,313	19,326
Full year	91,719,226	$ 205.43	$ 18,841	$ 19,326

(a) Excludes excise tax and commissions. As part of the Inflation Reduction Act of 2022, a 1% excise tax is imposed on net share repurchases commencing January 1, 2023.
(b) Represents the amount remaining under the $30 billion repurchase program.
(c) The remaining $11.6 billion of share repurchase capacity under the prior Board authorization was canceled when the new $30 billion repurchase program was authorized by the Board of Directors effective July 1, 2024.

Item 6. Reserved

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations.

Management's discussion and analysis of financial condition and results of operations, entitled "Management's discussion and analysis," appears on pages 52–167. Such information should be read in conjunction with the Consolidated Financial Statements and Notes thereto, which appear on pages 172–321.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk.

Refer to the Market Risk Management section of Management's discussion and analysis on pages 141–149 for a discussion of quantitative and qualitative disclosures about market risk.

Item 8. Financial Statements and Supplementary Data.

The Consolidated Financial Statements, together with the Notes thereto and the report thereon dated February 14, 2025, of PricewaterhouseCoopers LLP, the Firm's independent registered public accounting firm (PCAOB ID 238), appear on pages 169–321.

The "Glossary of Terms and Acronyms" is included on pages 327–333.

Item 9. Changes in and Disagreements With Accountants on Accounting and Financial Disclosure.

None.

Item 9A. Controls and Procedures.

The internal control framework promulgated by the Committee of Sponsoring Organizations of the Treadway Commission ("COSO"), "Internal Control — Integrated Framework" ("COSO 2013"), provides guidance for designing, implementing and conducting internal control and assessing its effectiveness. The Firm used the COSO 2013 framework to assess the effectiveness of the Firm's internal control over financial reporting as of December 31, 2024. Refer to "Management's report on internal control over financial reporting" on page 168.

As of the end of the period covered by this report, an evaluation was carried out under the supervision and with the participation of the Firm's management, including its Chairman and Chief Executive Officer and its Chief Financial Officer, of the effectiveness of its disclosure controls and procedures (as defined in Rule 13a-15(e) under the Securities Exchange Act of 1934). Based on that evaluation, the Chairman and Chief Executive Officer and the Chief Financial Officer concluded that these disclosure controls and procedures were effective. Refer to Exhibits 31.1 and 31.2 for the Certifications furnished by the Chairman and Chief Executive Officer and Chief Financial Officer, respectively.

The Firm is committed to maintaining high standards of internal control over financial reporting. Nevertheless, because of its inherent limitations, internal control over financial reporting may not prevent or detect all misstatements. Deficiencies or lapses in internal controls may occur from time to time, and there can be no assurance that any such deficiencies will not result in significant deficiencies or material weaknesses in internal control in the future and collateral consequences therefrom. Refer to "Management's report on internal control over financial reporting" on page 168 for further information. There was no change in the Firm's internal control over financial reporting (as defined in Rule 13a-15(f) under the Securities Exchange Act of 1934) that occurred during the three months ended December 31, 2024, that has materially affected, or is reasonably likely to materially affect, the Firm's internal control over financial reporting.

Item 9B. Other Information.

Director and executive officer trading arrangements

The following table provides information concerning Rule 10b5-1 trading arrangements (as defined in Item 408 of Regulation S-K under the Securities Exchange Act of 1934) adopted in the fourth quarter of 2024 by any director or any officer who is subject to the filing requirements of Section 16 of the Securities Exchange Act of 1934 ("Section 16 Director or Officer"). These trading arrangements are intended to satisfy the affirmative defense of Rule 10b5-1(c). Certain of the Firm's Section 16 Directors or Officers may participate in employee stock purchase plans, 401(k) plans or dividend reinvestment plans of the Firm that have been designed to comply with Rule 10b5-1(c). No non-Rule 10b5-1 trading arrangements (as defined in Item 408 of Regulation S-K under the Securities Exchange Act of 1934) were adopted by any Section 16 Director or Officer during the fourth quarter of 2024. Additionally, no Rule 10b5-1 or non-Rule 10b5-1 trading arrangements were terminated by any Section 16 Director or Officer in the fourth quarter of 2024.

Name	Title	Adoption date	Duration[c]	Aggregate number of shares to be sold
Lori Beer	Chief Information Officer	November 15, 2024	November 15, 2024 - March 31, 2025	4,105
James Dimon[a]	Chairman and CEO	November 7, 2024	November 7, 2024 - August 1, 2025	1,000,000
Robin Leopold	Head of Human Resources	November 4, 2024	November 4, 2024 - December 31, 2025	2,500
Jennifer Piepszak[b]	Co-CEO, CIB	October 30, 2024	October 30, 2024 - March 31, 2025	8,545
Troy Rohrbaugh	Co-CEO, CIB	November 15, 2024	November 15, 2024 - June 30, 2025	75,000

(a) Transaction by trusts and an entity of which Mr. Dimon has either a direct or indirect pecuniary interest.
(b) On January 14, 2025, JPMorganChase announced that Ms. Piepszak became a Chief Operating Officer of the Firm, effective January 14, 2025.
(c) Sales under the trading arrangement will not commence until completion of the required cooling off period under Rule 10b5-1. Subject to compliance with Rule 10b5-1, duration could cease earlier than the final date shown above to the extent that the aggregate number of shares to be sold under the trading arrangement have been sold.

Item 9C. Disclosure regarding Foreign Jurisdictions that Prevent Inspections.

Not applicable.

Item 10. Directors, Executive Officers and Corporate Governance.

Executive officers of the registrant

Name	Age (at December 31, 2024)	Positions and offices
James Dimon	68	Chairman of the Board since December 2006 and Chief Executive Officer since December 2005.
Ashley Bacon	55	Chief Risk Officer since June 2013.
Jeremy Barnum	52	Chief Financial Officer since May 2021, prior to which he was Head of Global Research for the former Corporate & Investment Bank since February 2021. He previously served as Chief Financial Officer of the former Corporate & Investment Bank from July 2013 until February 2021.
Lori A. Beer	57	Chief Information Officer since September 2017.
Mary Callahan Erdoes	57	Chief Executive Officer of Asset & Wealth Management since September 2009.
Stacey Friedman	56	General Counsel since January 2016.
Marianne Lake	55	Chief Executive Officer of Consumer & Community Banking since January 2024, having previously served as its Co-Chief Executive Officer since May 2021. She was Chief Executive Officer of Consumer Lending from May 2019 until May 2021.
Robin Leopold	60	Head of Human Resources since January 2018.
Jennifer A. Piepszak[(a)(b)]	54	Co-Chief Executive Officer of the Commercial & Investment Bank, having previously served as Co-Chief Executive Officer of Consumer & Community Banking since May 2021, prior to which she had been Chief Financial Officer since May 2019.
Daniel E. Pinto[(a)(b)]	62	President and Chief Operating Officer since January 2022, Co-President and Co-Chief Operating Officer since January 2018. He also served as Chief Executive Officer of the former Corporate & Investment Bank from March 2014 until January 2024.
Troy Rohrbaugh[(a)]	54	Co-Chief Executive Officer of the Commercial & Investment Bank since January 2024, prior to which he had been the Co-Head of Markets & Securities Services since June 2023. He was Head of Global Markets from January 2019 until June 2023.

Unless otherwise noted, during the five fiscal years ended December 31, 2024, all of JPMorganChase's above-named executive officers have continuously held senior-level positions with JPMorganChase. There are no family relationships among the foregoing executive officers. Information to be provided in Items 10, 11, 12, 13 and 14 of this 2024 Form 10-K and not otherwise included herein is incorporated by reference to the Firm's Definitive Proxy Statement for its 2025 Annual Meeting of Stockholders to be held on May 20, 2025, which will be filed with the SEC within 120 days of the end of the Firm's fiscal year ended December 31, 2024.

(a) Effective in the second quarter of 2024, JPMorganChase reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank. Refer to Business Segment & Corporate Results on page 70 for further information.

(b) On January 14, 2025, JPMorganChase announced new responsibilities for certain executives: Mr. Pinto will retire at the end of 2026, will relinquish his duties as President and Chief Operating Officer as of June 30, 2025, and will continue to serve as Vice Chairman through the end of 2026; Ms. Piepszak became a Chief Operating Officer of the Firm, effective January 14, 2025; and Doug Petno, Co-Head of Global Banking, succeeded Ms. Piepszak as co-Chief Executive Officer of the Commercial & Investment Bank. Refer to Recent events on page 57 of this 2024 Form 10-K for further information.

Code of Conduct and Code of Ethics

JPMorganChase has adopted, and posted on its website at https://www.jpmorganchase.com, a Code of Conduct for all employees of the Firm and a Code of Ethics for its Chairman and Chief Executive Officer, Chief Financial Officer, Principal Accounting Officer and all other professionals of the Firm worldwide serving in a finance, accounting, treasury, tax or investor relations role. The Code of Ethics is also available in print upon request to the Firm's Investor Relations team. Within the time period required by the SEC, JPMorganChase will post on its website any amendment to the Code of Ethics and any waiver applicable to a director or executive officer.

Insider Trading Policy

JPMorganChase has adopted an insider trading policy applicable to its directors, officers and employees, as well as to JPMorganChase itself, governing the purchase, sale and other dispositions of the Firm's securities (the "Insider Trading Policy"). The Firm believes that the Insider Trading Policy is reasonably designed to promote compliance with applicable U.S. federal securities laws and the listing standards of the New York Stock Exchange relating to insider trading. The Insider Trading Policy is filed as Exhibit 19 to this 2024 Form 10-K.

Item 11. Executive Compensation.

Refer to Item 10.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters.

Refer to Item 10 for security ownership of certain beneficial owners and management.

The following table sets forth the total number of shares available for issuance under JPMorganChase's employee share-based incentive plans (including shares available for issuance to non-employee directors). The Firm is not authorized to grant share-based incentive awards to non-employees, other than to non-employee directors.

December 31, 2024	Number of shares to be issued upon exercise of outstanding stock appreciation rights	Weighted-average exercise price of outstanding stock appreciation rights	Number of shares remaining available for future issuance under stock incentive plans
Plan category			
Employee share-based incentive plans approved by shareholders	2,250,000 (a)	$ 152.19	81,151,866 (b)
Total	2,250,000	$ 152.19	81,151,866

(a) Does not include restricted stock units or performance stock units granted under the shareholder-approved Long-Term Incentive Plan ("LTIP"), as amended and restated effective May 21, 2024. Refer to Note 9 for further information.
(b) Represents shares available for future issuance under the shareholder-approved LTIP.

All shares available for future issuance will be issued under the shareholder-approved LTIP. Refer to Note 9 for further discussion.

Item 13. Certain Relationships and Related Transactions, and Director Independence.

Refer to Item 10.

Item 14. Principal Accounting Fees and Services.

Refer to Item 10.

Item 15. Exhibits, Financial Statement Schedules.

1	Financial statements
	The Consolidated Financial Statements, the Notes thereto and the report of the Independent Registered Public Accounting Firm thereon listed in Item 8 are set forth commencing on page 169.
2	Financial statement schedules
3	Exhibits
3.1	Restated Certificate of Incorporation of JPMorgan Chase & Co., effective April 5, 2006 (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed April 7, 2006).
3.2	Amendment to the Restated Certificate of Incorporation of JPMorgan Chase & Co., effective June 7, 2013 (incorporated by reference to Appendix F to the Proxy Statement on Schedule 14A of JPMorgan Chase & Co. (File No. 1-5805) filed April 10, 2013).
3.3	Certificate of Designations for Fixed-to-Floating Rate Non-Cumulative Preferred Stock, Series CC (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed October 20, 2017).
3.4	Certificate of Designations for 5.75% Non-Cumulative Preferred Stock, Series DD (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed September 21, 2018).
3.5	Certificate of Designations for 6.00% Non-Cumulative Preferred Stock, Series EE (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed January 24, 2019).
3.6	Certificate of Designations for 4.75% Non-Cumulative Preferred Stock, Series GG (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed November 7, 2019).
3.7	Certificate of Designations for Fixed-to-Floating Rate Non-Cumulative Preferred Stock, Series II (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed February 24, 2020).
3.8	Certificate of Designations for 4.55% Non-Cumulative Preferred Stock, Series JJ (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed March 17, 2021).
3.90	Certificate of Designations for 3.65% Fixed-Rate Reset Non-Cumulative Preferred Stock, Series KK (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed May 12, 2021).
3.10	Certificate of Designations for 4.625% Non-Cumulative Preferred Stock, Series LL (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed May 20, 2021).
3.11	Certificate of Designations for 4.20% Non-Cumulative Preferred Stock, Series MM (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed July 29, 2021).
3.12	Certificate of Designations for 6.875% Non-Cumulative Preferred Stock, Series NN (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File 1-5805) filed March 12, 2024).
3.13	Certificate of Designations for 6.500% Non-Cumulative Preferred Stock, Series OO (incorporated by reference to Exhibit 3.1 to the Current Report on From 8-K of JPMorgan Chase & Co. (File 1-5808) filed February 4, 2025).
3.14	By-laws of JPMorgan Chase & Co., as amended, effective September 19, 2023 (incorporated by reference to Exhibit 3.2 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed September 22, 2023).
4.1(a)	Indenture, dated as of October 21, 2010, between JPMorgan Chase & Co. and Deutsche Bank Trust Company Americas, as Trustee (incorporated by reference to Exhibit 4.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No.1-5805) filed October 21, 2010).

4.1(b) First Supplemental Indenture, dated as of January 13, 2017, between JPMorgan Chase & Co. and Deutsche Bank Trust Company Americas, as Trustee, to the Indenture, dated as of October 21, 2010 (incorporated by reference to Exhibit 4.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed January 13, 2017).

4.2(a) Subordinated Indenture, dated as of March 14, 2014, between JPMorgan Chase & Co. and U.S. Bank Trust National Association, as Trustee (incorporated by reference to Exhibit 4.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No.1-5805) filed March 14, 2014).

4.2(b) First Supplemental Indenture, dated as of January 13, 2017, between JPMorgan Chase & Co. and U.S. Bank Trust National Association, as Trustee, to the Subordinated Indenture, dated as of March 14, 2014 (incorporated by reference to Exhibit 4.2 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed January 13, 2017).

4.3(a) Indenture, dated as of May 25, 2001, between JPMorgan Chase & Co. and Bankers Trust Company (succeeded by Deutsche Bank Trust Company Americas), as Trustee (incorporated by reference to Exhibit 4(a)(1) to the Registration Statement on Form S-3 of JPMorgan Chase & Co. (File No. 333-52826) filed June 13, 2001).

4.3(b) Sixth Supplemental Indenture, dated as of January 13, 2017, between JPMorgan Chase & Co. and Bankers Trust Company (succeeded by Deutsche Bank Trust Company Americas), as Trustee, to the Indenture, dated as of May 25, 2001 (incorporated by reference to Exhibit 4.3 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed January 13, 2017).

4.4 Indenture, dated as of February 19, 2016, among JPMorgan Chase Financial Company LLC, JPMorgan Chase & Co. and Deutsche Bank Trust Company Americas, as Trustee (incorporated by reference to Exhibit 4(a)(7) to the Registration Statement on Form S-3 of JPMorgan Chase & Co. and JPMorgan Chase Financial Company LLC (File No. 333-209682) filed February 24, 2016).

4.5 Form of Deposit Agreement (incorporated by reference to Exhibit 4.3 to the Registration Statement on Form S-3 of JPMorgan Chase & Co. (File No. 333-191692) filed October 11, 2013).

4.6 Description of Securities of JPMorgan Chase & Co. registered pursuant to Section 12 of the Securities Exchange Act of 1934. [(b)]

Other instruments defining the rights of holders of long-term debt securities of JPMorgan Chase & Co. and its subsidiaries are omitted pursuant to Section (b)(4)(iii)(A) of Item 601 of Regulation S-K. JPMorgan Chase & Co. agrees to furnish copies of these instruments to the SEC upon request.

10.1 Deferred Compensation Plan for Non-Employee Directors of JPMorgan Chase & Co., as amended and restated July 2001 and as of December 31, 2004 (incorporated by reference to Exhibit 10.1 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2007).[(a)]

10.2 2005 Deferred Compensation Plan for Non-Employee Directors of JPMorgan Chase & Co., effective as of January 1, 2005 (incorporated by reference to Exhibit 10.2 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2007).[(a)]

10.3 2005 Deferred Compensation Program of JPMorgan Chase & Co., restated effective as of December 31, 2008 (incorporated by reference to Exhibit 10.4 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2008).[(a)]

10.4 JPMorgan Chase & Co. Amended and Restated Long-Term Incentive Plan, effective May 21, 2024 (incorporated by reference to the Appendix of the Schedule 14A of JPMorgan Chase & Co. (File No. 1-5805) filed April 8, 2024).[(a)]

10.5 Key Executive Performance Plan of JPMorgan Chase & Co., as amended and restated effective January 1, 2014 (incorporated by reference to Appendix G of the Schedule 14A of JPMorgan Chase & Co. (File No. 1-5805) filed April 10, 2013).[(a)]

10.6 Excess Retirement Plan of JPMorgan Chase & Co., restated and amended as of December 31, 2008, as amended (incorporated by reference to Exhibit 10.7 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2009).[(a)]

10.7 Executive Retirement Plan of JPMorgan Chase & Co., as amended and restated December 31, 2008 (incorporated by reference to Exhibit 10.9 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2008).[(a)]

10.8 Bank One Corporation Supplemental Savings and Investment Plan, as amended and restated effective December 31, 2008 (incorporated by reference to Exhibit 10.13 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2008).[(a)]

10.9 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for performance share units and restricted stock units for Operating Committee members (U.S. and U.K.), dated as of January 17, 2017 (incorporated by reference to Exhibit 10.23 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2016).[(a)]

10.10 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for performance share units and restricted stock units for Operating Committee members (U.S. and U.K.), dated as of January 16, 2018 (incorporated by reference to Exhibit 10.19 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2017).[(a)]

10.11 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units for Operating Committee members (U.S. and U.K.), dated as of January 15, 2019 (incorporated by reference to Exhibit 10.18 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2018).[(a)]

10.12 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 21, 2020 (incorporated by reference to Exhibit 10.18 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2019).[(a)]

10.13 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 19, 2021(incorporated by reference to Exhibit 10.17 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2020).[(a)]

10.14 Form of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for stock appreciation rights for Chairman/Chief Executive Officer, dated July 20, 2021 (incorporated by reference to Exhibit 99 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed July 20, 2021).[(a)]

10.15 Form of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for stock appreciation rights for President and Chief Operating Officer, dated December 14, 2021 (incorporated by reference to Exhibit 99 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed December 15, 2021).[(a)]

10.16 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 18, 2022 (incorporated by reference to Exhibit 10.20 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5808) for the year ended December 31, 2021).[(a)]

10.17 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 17, 2023 (incorporated by reference to Exhibit 10.18 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5808) for the year ended December 31, 2022).[(a)]

10.18 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 16, 2024 (incorporated by reference to Exhibit 10.19 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5808) for the year ended December 31, 2023).[(a)]

10.19 Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 21, 2025.[(a)(b)]

10.20	Employee Stock Purchase Plan of JPMorgan Chase & Co., as amended and restated effective as of January 1, 2019 (incorporated by reference to Exhibit 10.1 to the Quarterly Report on Form 10-Q of JPMorgan Chase & Co. (File No. 1-5805) for the quarter ended September 30, 2019).
10.21	Form of JPMorgan Chase & Co. Performance-Based Incentive Compensation Plan, effective as of January 1, 2021, as amended (incorporated by reference to Exhibit 10.23 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5808) for the year ended December 31, 2021).[a]
19	Insider Trading Policy - Firmwide[b]
21	List of subsidiaries of JPMorgan Chase & Co.[b]
22.1	Annual Report on Form 11-K of The JPMorgan Chase 401(k) Savings Plan for the year ended December 31, 2024 (to be filed pursuant to Rule 15d-21 under the Securities Exchange Act of 1934).
22.2	Subsidiary Guarantors and Issuers of Guaranteed Securities.[b]
23	Consent of independent registered public accounting firm.[b]
31.1	Certification.[b]
31.2	Certification.[b]
32	Certification pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.[c]
97	Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation.[b]
101.INS	The instance document does not appear in the interactive data file because its XBRL tags are embedded within the Inline XBRL document.[d]
101.SCH	XBRL Taxonomy Extension Schema Document.[b]
101.CAL	XBRL Taxonomy Extension Calculation Linkbase Document.[b]
101.DEF	XBRL Taxonomy Extension Definition Linkbase Document.[b]
101.LAB	XBRL Taxonomy Extension Label Linkbase Document.[b]
101.PRE	XBRL Taxonomy Extension Presentation Linkbase Document.[b]
104	Cover Page Interactive Data File (embedded within the Inline XBRL document and included in Exhibit 101).

(a) This exhibit is a management contract or compensatory plan or arrangement.
(b) Filed herewith.
(c) Furnished herewith. This exhibit shall not be deemed “filed” for purposes of Section 18 of the Securities Exchange Act of 1934, or otherwise subject to the liability of that Section. Such exhibit shall not be deemed incorporated into any filing under the Securities Act of 1933 or the Securities Exchange Act of 1934.
(d) Pursuant to Rule 405 of Regulation S-T, includes the following financial information included in the Firm’s Form 10-K for the year ended December 31, 2024, formatted in XBRL (eXtensible Business Reporting Language) interactive data files: (i) the Consolidated statements of income for the years ended December 31, 2024, 2023 and 2022, (ii) the Consolidated statements of comprehensive income for the years ended December 31, 2024, 2023 and 2022, (iii) the Consolidated balance sheets as of December 31, 2024 and 2023, (iv) the Consolidated statements of changes in stockholders’ equity for the years ended December 31, 2024, 2023 and 2022, (v) the Consolidated statements of cash flows for the years ended December 31, 2024, 2023 and 2022, and (vi) the Notes to Consolidated Financial Statements.

page 48 not used

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THREE-YEAR SUMMARY OF CONSOLIDATED FINANCIAL HIGHLIGHTS (unaudited)

As of or for the year ended December 31, (in millions, except per share, ratio, employee data and where otherwise noted)	2024	2023	2022
Selected income statement data			
Total net revenue	$ 177,556 [e]	$ 158,104	$ 128,695
Total noninterest expense	91,797 [e]	87,172	76,140
Pre-provision profit[a]	85,759	70,932	52,555
Provision for credit losses	10,678	9,320	6,389
Income before income tax expense	75,081	61,612	46,166
Income tax expense	16,610	12,060	8,490
Net income	$ 58,471	$ 49,552	$ 37,676
Earnings per share data			
Net income: Basic	$ 19.79	$ 16.25	$ 12.10
Diluted	19.75	16.23	12.09
Average shares: Basic	2,873.9	2,938.6	2,965.8
Diluted	2,879.0	2,943.1	2,970.0
Market and per common share data			
Market capitalization	670,618	489,320	393,484
Common shares at period-end	2,797.6	2,876.6	2,934.2
Book value per share	116.07	104.45	90.29
Tangible book value per share (“TBVPS”)[a]	97.30	86.08	73.12
Cash dividends declared per share	4.80	4.10	4.00
Selected ratios and metrics			
Return on common equity (“ROE”)	18 %	17 %	14 %
Return on tangible common equity (“ROTCE”)[a]	22	21	18
Return on assets (“ROA”)	1.43	1.30	0.98
Overhead ratio	52	55	59
Loans-to-deposits ratio	56	55	49
Firm Liquidity coverage ratio (“LCR”) (average)[b]	113	113	112
JPMorgan Chase Bank, N.A. LCR (average)[b]	124	129	151
Common equity Tier 1 (“CET1”) capital ratio[c][d]	15.7	15.0	13.2
Tier 1 capital ratio[c][d]	16.8	16.6	14.9
Total capital ratio[c][d]	18.5	18.5	16.8
Tier 1 leverage ratio[b][c]	7.2	7.2	6.6
Supplementary leverage ratio (“SLR”)[b][c]	6.1	6.1	5.6
Selected balance sheet data (period-end)			
Trading assets	$ 637,784	$ 540,607	$ 453,799
Investment securities, net of allowance for credit losses	681,320	571,552	631,162
Loans	1,347,988	1,323,706	1,135,647
Total assets	4,002,814	3,875,393	3,665,743
Deposits	2,406,032	2,400,688	2,340,179
Long-term debt	401,418	391,825	295,865
Common stockholders’ equity	324,708	300,474	264,928
Total stockholders’ equity	344,758	327,878	292,332
Employees	317,233	309,926	293,723
Credit quality metrics			
Allowances for credit losses	$ 26,866	$ 24,765	$ 22,204
Allowance for loan losses to total retained loans	1.87 %	1.75 %	1.81 %
Nonperforming assets	$ 9,300	$ 7,597	$ 7,247
Net charge-offs	8,638	6,209	2,853
Net charge-off rate	0.68 %	0.52 %	0.27 %

(a) Pre-provision profit, TBVPS and ROTCE are each non-GAAP financial measures. Tangible common equity (“TCE”) is also a non-GAAP financial measure. Refer to Explanation and Reconciliation of the Firm’s Use of Non-GAAP Financial Measures on pages 67–69 for a discussion of these measures.
(b) For the years ended December 31, 2024, 2023 and 2022, the percentage represents average ratios for the three months ended December 31, 2024, 2023 and 2022.
(c) The ratios reflect the Current Expected Credit Losses (“CECL”) capital transition provisions. Refer to Note 27 for additional information.
(d) Reflects the Firm’s ratios under the Basel III Standardized approach. Refer to Capital Risk Management on pages 97–107 for additional information.
(e) Total net revenue included a $7.9 billion net gain related to Visa shares, and total noninterest expense included a $1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation, both recorded in the second quarter of 2024. Refer to Executive Overview on pages 54–58, and Notes 2 and 6 for additional information on the exchange offer for Visa Class B-1 common stock.

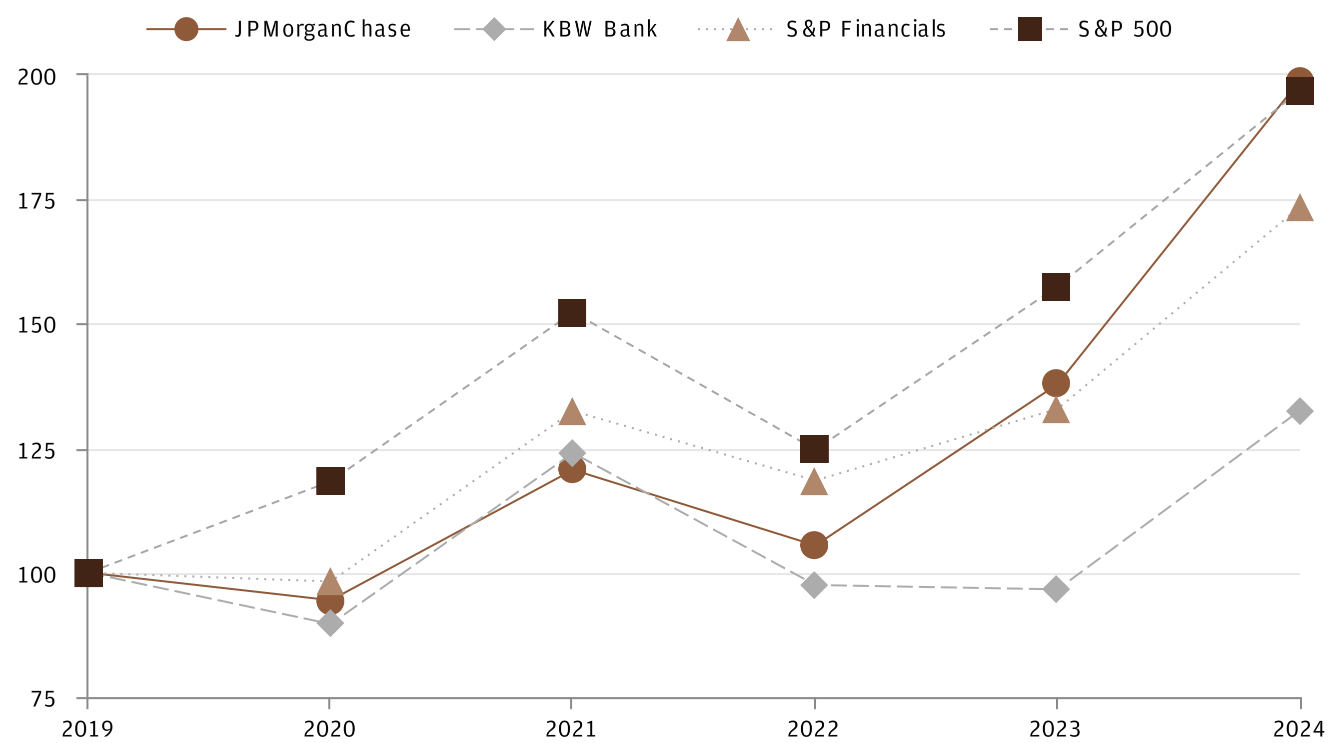
FIVE-YEAR STOCK PERFORMANCE

The following table and graph compare the five-year cumulative total return for JPMorgan Chase & Co. ("JPMorganChase" or the "Firm") common stock with the cumulative return of the S&P 500 Index, the KBW Bank Index and the S&P Financials Index. The S&P 500 Index is a commonly referenced equity benchmark in the United States of America ("U.S."), consisting of leading companies from different economic sectors. The KBW Bank Index seeks to reflect the performance of banks and thrifts that are publicly traded in the U.S. and is composed of leading national money center and regional banks and thrifts. The S&P Financials Index is an index of financial companies, all of which are components of the S&P 500. The Firm is a component of all three industry indices.

The following table and graph assume simultaneous investments of $100 on December 31, 2019, in JPMorganChase common stock and in each of the above indices. The comparison assumes that all dividends were reinvested.

December 31, (in dollars)	2019	2020	2021	2022	2023	**2024**
JPMorganChase	$ 100.00	$ 94.48	$ 120.68	$ 105.48	$ 137.91	**$ 198.96**
KBW Bank Index	100.00	89.69	124.08	97.53	96.66	**132.62**
S&P Financials Index	100.00	98.24	132.50	118.54	132.94	**173.57**
S&P 500 Index	100.00	118.39	152.34	124.75	157.54	**196.96**

December 31,
(in dollars)



The following is Management's discussion and analysis of the financial condition and results of operations ("MD&A") of JPMorganChase for the year ended December 31, 2024. The MD&A is included in both JPMorganChase's Annual Report for the year ended December 31, 2024 ("Annual Report") and its Annual Report on Form 10-K for the year ended December 31, 2024 ("2024 Form 10-K" or "Form 10-K") filed with the Securities and Exchange Commission ("SEC"). Refer to the Glossary of terms and acronyms on pages 327–333 for definitions of terms and acronyms used throughout the Annual Report and the 2024 Form 10-K.

This Form 10-K contains forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. These forward-looking statements are based on the current beliefs and expectations of JPMorganChase's management, speak only as of the date of this Form 10-K and are subject to significant risks and uncertainties. Refer to Forward-looking Statements on page 167 and Part 1, Item 1A: Risk Factors in this Form 10-K on pages 10-37 for a discussion of certain of those risks and uncertainties and the factors that could cause JPMorganChase's actual results to differ materially because of those risks and uncertainties. There is no assurance that actual results will be in line with any outlook information set forth herein, and the Firm does not undertake to update any forward-looking statements.

INTRODUCTION

JPMorgan Chase & Co. (NYSE: JPM), a financial holding company incorporated under Delaware law in 1968, is a leading financial services firm based in the United States of America ("U.S."), with operations worldwide. JPMorganChase had $4.0 trillion in assets and $344.8 billion in stockholders' equity as of December 31, 2024. The Firm is a leader in investment banking, financial services for consumers and small businesses, commercial banking, financial transaction processing and asset management. Under the J.P. Morgan and Chase brands, the Firm serves millions of customers, predominantly in the U.S., and many of the world's most prominent corporate, institutional and government clients globally.

JPMorganChase's principal bank subsidiary is JPMorgan Chase Bank, National Association ("JPMorgan Chase Bank, N.A."), a national banking association with U.S. branches in 48 states and Washington, D.C. JPMorganChase's principal non-bank subsidiary is J.P. Morgan Securities LLC ("J.P. Morgan Securities"), a U.S. broker-dealer. The bank and non-bank subsidiaries of JPMorganChase operate nationally as well as through overseas branches and subsidiaries, representative offices and subsidiary foreign banks. The Firm's principal operating subsidiaries outside the U.S. are J.P. Morgan Securities plc and J.P. Morgan SE ("JPMSE"), which are subsidiaries of JPMorgan Chase Bank, N.A. and are based in the United Kingdom ("U.K.") and Germany, respectively.

Business segments & Corporate: Effective in the second quarter of 2024, the Firm reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank. As a result of the reorganization, the Firm has three reportable business segments – Consumer & Community Banking ("CCB"), Commercial & Investment Bank ("CIB") and Asset & Wealth Management ("AWM") – with the remaining activities in Corporate. The Firm's consumer business segment is CCB, and the Firm's wholesale business segments are CIB and AWM.

A description of each of the Firm's reportable business segments, and the products and services that they provide to their respective client bases, as well as a description of Corporate activities, is provided in the Management's discussion and analysis of financial condition and results of operations section of this Form 10-K ("Management's discussion and analysis" or "MD&A") under the heading "Business Segment & Corporate Results," which begins on page 70, and in Note 32.

First Republic: On May 1, 2023, JPMorganChase acquired certain assets and assumed certain liabilities of First Republic Bank (the "First Republic acquisition") from the Federal Deposit Insurance Corporation ("FDIC"). References in this Form 10-K to "associated with First Republic," "impact of First Republic" or similar expressions refer to the relevant effects of the First Republic acquisition, as well as subsequent related business and activities, as applicable. Refer to Note 34 for additional information.

The Firm’s website is www.jpmorganchase.com. JPMorganChase makes available on its website, free of charge, annual reports on Form 10-K, quarterly reports on Form 10-Q and current reports on Form 8-K pursuant to Section 13(a) or Section 15(d) of the Securities Exchange Act of 1934, as soon as reasonably practicable after it electronically files or furnishes such material to the U.S. Securities and Exchange Commission (the “SEC”) at www.sec.gov. JPMorganChase makes new and important information about the Firm available on its website at https://www.jpmorganchase.com, including on the Investor Relations section of its website at https://www.jpmorganchase.com/ir. Information on the Firm's website, including documents on the website that are referenced in this Form 10-K, is not incorporated by reference into this 2024 Form 10-K or the Firm’s other filings with the SEC.

This executive overview of the MD&A highlights selected information and does not contain all of the information that is important to readers of the Firm's 2024 Form 10-K. For a complete description of the trends and uncertainties, as well as the risks and critical accounting estimates affecting the Firm, the 2024 Form 10-K should be read in its entirety.

Financial performance of JPMorganChase

Year ended December 31, (in millions, except per share data and ratios)	2024	2023	Change
Selected income statement data			
Noninterest revenue	**$ 84,973**	$ 68,837	23%
Net interest income	**92,583**	89,267	4
Total net revenue	**177,556**	158,104	12
Total noninterest expense	**91,797**	87,172	5
Pre-provision profit	**85,759**	70,932	21
Provision for credit losses	**10,678**	9,320	15
Net income	**58,471**	49,552	18
Diluted earnings per share	**19.75**	16.23	22
Selected ratios and metrics			
Return on common equity	**18 %**	17 %	
Return on tangible common equity	**22**	21	
Book value per share	**$ 116.07**	$ 104.45	11
Tangible book value per share	**97.3**	86.08	13
Capital ratios[(a)(b)]			
CET1 capital	**15.7 %**	15.0 %	
Tier 1 capital	**16.8**	16.6	
Total capital	**18.5**	18.5	
Memo:			
NII excluding Markets[(c)]	**$ 92,419**	$ 90,041	3
NIR excluding Markets[(c)]	**58,167**	44,361	31
Markets[(c)]	**30,007**	27,964	7
Total net revenue - managed basis	**$ 180,593**	$ 162,366	11

(a) The ratios reflect the CECL capital transition provisions. Refer to Note 27 for additional information.
(b) Reflects the Firm's ratios under the Basel III Standardized approach. Refer to Capital Risk Management on pages 97–107 for additional information.
(c) NII and NIR refer to net interest income and noninterest revenue, respectively. Markets consists of CIB's Fixed Income Markets and Equity Markets businesses.

Visa shares: On April 8, 2024, Visa Inc. commenced an initial exchange offer for its Class B-1 common shares. On May 6, 2024, the Firm announced that Visa had accepted the Firm's tender of its 37.2 million Visa Class B-1 common shares in exchange for a combination of Visa Class B-2 common shares and Visa Class C common shares ("Visa C shares"), resulting in a $7.9 billion net gain related to Visa shares recorded in the second quarter of 2024. As of September 30, 2024, the Firm had disposed of all of its Visa C shares through sales and through a contribution to the Firm's Foundation. Refer to Market Risk Management on pages 141–149, and Notes 2 and 6 for additional information.

First Republic: JPMorganChase acquired certain assets and assumed certain liabilities of First Republic Bank from the FDIC on May 1, 2023. As a result, the year-to-date results include the twelve-month impact of First Republic compared with eight months in the prior-year period. Where meaningful to the results, this is referred to in this Form 10-K as the "timing impact" of First Republic. Refer to Notes 6 and 34 for additional information.

Comparisons noted in the sections below are for the full year of 2024 versus the full year of 2023, unless otherwise specified.

Firmwide overview

JPMorganChase reported net income of $58.5 billion for 2024, up 18%, earnings per share of $19.75, ROE of 18% and ROTCE of 22%.

- **Total net revenue** was $177.6 billion, up 12%, reflecting:
  - **Net interest income** ("NII") of $92.6 billion, up 4%, driven by the impact of balance sheet actions, primarily reinvestments in the investment securities portfolio, higher revolving balances in Card Services, the timing impact of First Republic, higher wholesale deposit balances, and higher Markets net interest income, largely offset by deposit margin compression across the lines of business and lower average deposit balances in CCB. NII excluding Markets was $92.4 billion, up 3%.
  - **Noninterest revenue** ("NIR") was $85.0 billion, up 23%, predominantly driven by a $7.9 billion net gain related to Visa shares recorded in the second quarter of 2024, higher asset management fees in AWM and CCB, higher investment banking fees, and lower net investment securities losses in Treasury and CIO.

    The prior year included the estimated bargain purchase gain of $2.8 billion associated with First Republic.

- **Noninterest expense** was $91.8 billion, up 5%, driven by higher compensation expense, including higher revenue-related compensation and growth in the number of employees, as well as a $1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024, partially offset by lower FDIC-related expense, reflecting a $2.9 billion special assessment recognized in the fourth quarter of 2023, compared with a $725 million increase to the FDIC special assessment recognized in the first quarter of 2024.
- The **provision for credit losses** was $10.7 billion, reflecting $8.6 billion of net charge-offs and a net addition to the allowance for credit losses of $2.0 billion. Net charge-offs increased by $2.4 billion, driven by Card Services, reflecting the seasoning of vintages originated in recent years, credit normalization, and balance growth. The net addition to the allowance for credit losses included a net addition of $2.1 billion in **consumer**, driven by Card Services, and a net reduction of $19 million in **wholesale**.

  The provision in the prior year was $9.3 billion, reflecting $6.2 billion of net charge-offs and a $3.1 billion net addition to the allowance for credit losses.
- The total **allowance for credit losses** was $26.9 billion at December 31, 2024. The Firm had an allowance for loan losses to retained loans coverage ratio of 1.87%, compared with 1.75% in the prior year.
- The Firm's **nonperforming assets** totaled $9.3 billion at December 31, 2024, up 22%, driven by higher wholesale nonaccrual loans, which reflected downgrades in Real Estate, concentrated in Office, partially offset by lower consumer nonaccrual loans, which included loan sales. Refer to Wholesale Credit Portfolio and Consumer Credit Portfolio on pages 126–136 and pages 120–125, respectively, for additional information.
- Firmwide **average loans** of $1.3 trillion were up 6%, driven by higher loans across the lines of business.
- Firmwide **average deposits** of $2.4 trillion were up 1%, reflecting:
  - net inflows in Payments and net issuances of structured notes in Markets,
  - the timing impact of First Republic, and
  - growth in balances in new and existing client accounts in AWM,

  predominantly offset by
  - a decline in CCB primarily driven by a decrease in balances in existing accounts due to increased customer spending.

  Refer to Liquidity Risk Management on pages 108–115 for additional information.

Selected capital and other metrics

- **CET1 capital** was $275.5 billion, and the Standardized and Advanced CET1 ratios were 15.7% and 15.8%, respectively.
- **SLR** was 6.1%.
- **TBVPS** grew 13.0%, ending 2024 at $97.30.
- As of December 31, 2024, the Firm had eligible end-of-period **High Quality Liquid Assets** ("HQLA") of approximately $834 billion and **unencumbered marketable securities** with a fair value of approximately $594 billion, resulting in approximately $1.4 trillion of liquidity sources. Refer to Liquidity Risk Management on pages 108–115 for additional information.

Refer to Consolidated Results of Operations and Consolidated Balance Sheets Analysis on pages 59–62 and pages 63–65, respectively, for a further discussion of the Firm's results, including the provision for credit losses, and Note 34 for additional information on the First Republic acquisition.

Pre-provision profit, ROTCE, TCE, TBVPS, NII and NIR excluding Markets, and total net revenue on a managed basis are non-GAAP financial measures. Refer to Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 67–69 for a further discussion of each of these measures.

Business segment highlights

Selected business metrics for each of the Firm's lines of business ("LOB") are presented below for the full year of 2024.

CCB ROE 32%	• Average deposits down 6%; client investment assets up 14% • Average loans up 9%; Card Services net charge-off rate of 3.34% • Debit and credit card sales volume[(a)] up 8% • Active mobile customers[(b)] up 7%
CIB[(c)] ROE 18%	• Investment Banking fees up 37%; #1 ranking for Global Investment Banking fees with 9.3% wallet share for the year • Markets revenue up 7%, with Fixed Income Markets up 5% and Equity Markets up 13% • Average Banking & Payments loans up 2%; average client deposits[(d)] up 5%
AWM ROE 34%	• Assets under management ("AUM") of $4.0 trillion, up 18% • Average loans up 3%; average deposits up 9% including the transfer of First Republic deposits to AWM in 4Q23[(e)]

(a) Excludes Commercial Card.
(b) Users of all mobile platforms who have logged in within the past 90 days.
(c) Reflects the reorganization of the Firm's business segments. Refer to Business Segment & Corporate Results on pages 70–90 for additional information.
(d) Represents client deposits and other third-party liabilities pertaining to the Payments and Securities Services businesses.
(e) In the fourth quarter of 2023, CCB transferred certain deposits associated with First Republic to AWM and CIB.

Refer to the Business Segment & Corporate Results on pages 70–90 for a detailed discussion of results by business segment.

Credit provided and capital raised

JPMorganChase continues to support consumers, businesses and communities around the globe. The Firm provided new and renewed credit and raised capital for wholesale and consumer clients during 2024, consisting of approximately:

$2.8 trillion	Total credit provided and capital raised (including loans and commitments)
$250 billion	Credit for consumers
$40 billion	Credit for U.S. small businesses
$2.4 trillion	Credit and capital for corporations and non-U.S. government entities[(a)]
$65 billion	Credit and capital for nonprofit and U.S. government entities[(b)]

(a) Includes Individuals and Individual Entities primarily consisting of Global Private Bank clients within AWM.
(b) Includes states, municipalities, hospitals and universities.

Recent events

- On January 14, 2025, JPMorganChase announced new responsibilities for several of its senior executives:
  - Daniel Pinto, President and Chief Operating Officer ("COO"), will retire at the end of 2026. Mr. Pinto will relinquish his responsibilities as President and COO as of June 30, 2025. He will continue to serve the Firm as Vice Chairman through the end of 2026.
  - Jennifer Piepszak, Co-Chief Executive Officer of the Commercial & Investment Bank ("CIB"), was named a COO of the Firm, effective January 14, 2025.
  - Doug Petno, Co-head of Global Banking, succeeded Ms. Piepszak as Co-Chief Executive Officer of CIB.
- On December 12, 2024, the Firm announced that Michele G. Buck, 63, had been elected as a director of the Firm, effective March 17, 2025. Ms. Buck is Chairman of the Board, President and CEO of The Hershey Company.

Outlook

These current expectations are forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. Such forward-looking statements are based on the current beliefs and expectations of JPMorganChase's management, speak only as of the date of this Form 10-K, and are subject to significant risks and uncertainties. Refer to Forward-Looking Statements on page 167 and Part I, Item 1A: Risk Factors on pages 10-37 of this Form 10-K for a further discussion of certain of those risks and uncertainties and the other factors that could cause JPMorganChase's actual results to differ materially because of those risks and uncertainties. There is no assurance that actual results in 2025 will be in line with the outlook information set forth below, and the Firm does not undertake to update any forward-looking statements.

JPMorganChase's current outlook for full-year 2025 should be viewed against the backdrop of the global and U.S. economies, financial markets activity, the geopolitical environment, the competitive environment, client and customer activity levels, and regulatory and legislative developments in the U.S. and other countries where the Firm does business. Each of these factors will affect the performance of the Firm. The Firm will continue to make appropriate adjustments to its businesses and operations in response to ongoing developments in the business, economic, regulatory and legal environments in which it operates.

Full-year 2025

- Management expects net interest income to be approximately $94.0 billion and net interest income excluding Markets to be approximately $90.0 billion, market dependent.
- Management expects adjusted expense to be approximately $95.0 billion, market dependent.
- Management expects the net charge-off rate in Card Services to be approximately 3.60%.

Net interest income excluding Markets and adjusted expense are non-GAAP financial measures. Refer to Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 67–69.

Business Developments

First Republic acquisition

On May 1, 2023, JPMorganChase acquired certain assets and assumed certain liabilities of First Republic Bank (the "First Republic acquisition") from the Federal Deposit Insurance Corporation ("FDIC"), as receiver.

As of December 31, 2024, the Firm had substantially completed the conversion of operations, and the integration of clients, products and services, associated with the First Republic acquisition to align with the Firm's businesses and operations.

Refer to Note 34 for additional information on First Republic.

CONSOLIDATED RESULTS OF OPERATIONS

This section provides a comparative discussion of JPMorganChase's Consolidated Results of Operations on a reported basis for the two-year period ended December 31, 2024, unless otherwise specified. Refer to Consolidated Results of Operations on pages 54-57 of the Firm's Annual Report on Form 10-K for the year ended December 31, 2023 (the "2023 Form 10-K") for a discussion of the 2023 versus 2022 results. Factors that relate primarily to a single business segment or Corporate are discussed in more detail in the results of that segment or Corporate. Refer to pages 161–164 for a discussion of the Critical Accounting Estimates Used by the Firm that affect the Consolidated Results of Operations.

Revenue

Year ended December 31, (in millions)	2024	2023	2022
Investment banking fees	**$ 8,910**	$ 6,519	$ 6,686
Principal transactions	**24,787**	24,460	19,912
Lending- and deposit-related fees	**7,606**	7,413	7,098
Asset management fees	**17,801**	15,220	14,096
Commissions and other fees	**7,530**	6,836	6,581
Investment securities losses	**(1,021)**	(3,180)	(2,380)
Mortgage fees and related income	**1,401**	1,176	1,250
Card income	**5,497**	4,784	4,420
Other income[(a)(b)]	**12,462** [(c)]	5,609 [(d)]	4,322
Noninterest revenue	**84,973**	68,837	61,985
Net interest income	**92,583**	89,267	66,710
Total net revenue	**$177,556**	$ 158,104	$ 128,695

(a) Included operating lease income of $2.8 billion, $2.8 billion and $3.7 billion for the years ended December 31, 2024, 2023 and 2022, respectively. Refer to Note 6 for additional information.
(b) Effective January 1, 2024, as a result of adopting updates to the Accounting for Investments in Tax Credit Structures guidance, the amortization of certain of the Firm's alternative energy tax-oriented investments that was previously recognized in other income is now recognized in income tax expense. Refer to Notes 1, 6, 14 and 25 for additional information.
(c) Included the net gain related to Visa shares of $7.9 billion recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.
(d) Included the estimated bargain purchase gain of $2.8 billion for the year ended December 31, 2023 associated with the First Republic acquisition. Refer to Notes 6 and 34 for additional information.

2024 compared with 2023

Investment banking fees increased, reflecting in CIB the benefit of favorable market conditions, which resulted in:

- higher debt underwriting fees predominantly driven by higher industry-wide issuances in leveraged loans, and in high-grade and high-yield bonds,
- higher equity underwriting fees driven by higher industry-wide fees and wallet share gains in IPOs, and in follow-on and convertible securities offerings, and
- higher advisory fees driven by higher industry-wide mergers and acquisitions ("M&A") activity and wallet share gains.

Refer to CIB segment results on pages 77–83 and Note 6 for additional information.

Principal transactions revenue increased driven by CIB, reflecting:

- higher Equity Markets revenue in Prime Finance and Equity Derivatives,

predominantly offset by

- lower Fixed Income Markets revenue, reflecting the net impact of declines in revenue across macro businesses and higher revenue in Securitized Products.

Principal transactions revenue in CIB generally has offsets across other revenue lines, including net interest income. The Firm assesses the performance of its Markets business on a total net revenue basis.

The increase in CIB was partially offset by a net decrease in Corporate, reflecting lower revenue in Treasury and CIO, and gains compared with a net loss on certain legacy private equity investments in the prior year.

Refer to CIB segment and Corporate results on pages 77–83 and pages 88–90, respectively, and Note 6 for additional information.

Lending- and deposit-related fees increased, reflecting, in CIB, higher deposit-related fees, including cash management fees in Payments, on higher volume; and higher lending-related fees, including loan commitment fees. These factors were largely offset by a decline in the amortization of the fair value discount on certain acquired lending-related commitments associated with First Republic, primarily in AWM, as certain of the commitments have expired.

Refer to CIB and AWM segment results on pages 77–83 and pages 84–87, respectively, and Note 6 for additional information.

Asset management fees increased, reflecting, in AWM and CCB, higher average market levels and net inflows, as well as higher performance fees in AWM; and in CCB, the timing impact of First Republic. Refer to CCB and AWM segment results on pages 73–76 and pages 84–87, respectively, and Note 6 for additional information.

Commissions and other fees increased, predominantly due to higher brokerage commissions and fees on higher volume, and higher custody fees, in both CIB and AWM, as well as higher annuity sales commissions in CCB. Refer to CCB, CIB and AWM segment results on pages 73–76, pages 77–83 and pages 84–87, respectively, and Note 6 for additional information.

Investment securities losses decreased, reflecting lower losses on sales of securities, primarily U.S. Treasuries and U.S. GSE and government agency MBS, associated with repositioning the investment securities portfolio in Treasury and CIO. Refer to Corporate results on pages 88–90 and Note 10 for additional information.

Mortgage fees and related income increased in Home Lending, reflecting higher production revenue, which included the timing impact of First Republic. Refer to CCB segment results on pages 73–76, and Note 6 and 15 for additional information.

Card income increased, reflecting higher net interchange on increased debit and credit card sales volume, as well as higher annual fees in CCB, partially offset by an increase in amortization related to new account origination costs. Refer to CCB segment results on pages 73–76 and Note 6 for additional information.

Other income increased, reflecting:

- in Corporate:
  - the $7.9 billion net gain related to Visa shares recorded in the second quarter of 2024,

  partially offset by
  - the absence of the prior-year $2.8 billion estimated bargain purchase gain associated with the First Republic acquisition, and
- in CIB:
  - the impact of the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance effective January 1, 2024, resulting in the amortization of certain of the Firm's alternative energy tax-oriented investments previously recognized in other income which is now recognized in income tax expense.

  Both periods included impairment losses related to certain equity investments.

The prior year included a gain of $339 million on the original minority interest in China International Fund Management ("CIFM"), partially offset by net investment valuation losses, both in AWM. Refer to AWM segment results on pages 84–87 for additional information on CIFM; Notes 1, 6, 14 and 25 for additional information on the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance; Notes 2 and 6 for additional information on Visa shares; and Notes 6 and 34 for additional information on the First Republic acquisition.

Net interest income increased driven by the impact of balance sheet actions, primarily reinvestments in the investment securities portfolio, higher revolving balances in Card Services, the timing impact of First Republic, higher wholesale deposit balances and higher Markets net interest income. These factors were largely offset by deposit margin compression across the lines of business and lower average deposit balances in CCB.

The Firm’s average interest-earning assets were $3.5 trillion, up $212 billion, and the yield was 5.50%, up 36 bps. The net yield on these assets, on an FTE basis, was 2.63%, a decrease of 7 bps. The net yield excluding Markets was 3.84%, relatively flat when compared to the prior year.

Refer to the Consolidated average balance sheets, interest and rates schedule on pages 322–326 for additional information. Net yield excluding Markets is a non-GAAP financial measure. Refer to Explanation and Reconciliation of the Firm’s Use of Non-GAAP Financial Measures on pages 67–69 for an additional discussion of net yield excluding Markets.

Provision for credit losses

Year ended December 31, (in millions)	2024	2023	2022
Consumer, excluding credit card	$ 631	$ 935	$ 506
Credit card	9,292	6,048	3,353
Total consumer	**9,923**	6,983	3,859
Wholesale	731	2,299	2,476
Investment securities	24	38	54
Total provision for credit losses	**$10,678**	$ 9,320	$ 6,389

2024 compared with 2023

The **provision for credit losses** was $10.7 billion, reflecting $8.6 billion of net charge-offs and a $2.0 billion net addition to the allowance for credit losses.

Net charge-offs included $7.8 billion in **consumer**, predominantly driven by Card Services, reflecting the seasoning of vintages originated in recent years, credit normalization and balance growth, and $0.8 billion in **wholesale**, primarily in Real Estate, largely concentrated in Office.

The net addition to the allowance for credit losses consisted of:

- $2.1 billion in **consumer**, reflecting:
  - a $2.2 billion net addition in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years,

  partially offset by

  - a $125 million net reduction in Home Lending in the first quarter of 2024, and
- a net reduction of $19 million in **wholesale**, reflecting:
  - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in Markets, and a reduction due to charge-offs largely from collateral-dependent loans,

  predominantly offset by

  - net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024.

The provision in the prior year was $9.3 billion, reflecting net charge-offs of $6.2 billion and a $3.1 billion net addition to the allowance for credit losses, which included $1.2 billion to establish the allowance for the First Republic loans and lending-related commitments in the second quarter of 2023.

Refer to CCB, CIB and AWM segment and Corporate results on pages 73–76, pages 77–83, pages 84–87, and pages 88–90, respectively; Allowance for Credit Losses on pages 137–139; Critical Accounting Estimates Used by the Firm on pages 161–164; and Notes 12 and 13 for additional information on the credit portfolio and the allowance for credit losses.

Noninterest expense

Year ended December 31, (in millions)	2024		2023	2022
Compensation expense	**$ 51,357**		$46,465	$ 41,636
Noncompensation expense:				
Occupancy	**5,026**		4,590	4,696
Technology, communications and equipment[a]	**9,831**		9,246	9,358
Professional and outside services	**11,057**		10,235	10,174
Marketing	**4,974**		4,591	3,911
Other expense	**9,552**	(c)	12,045	6,365
Total noncompensation expense	**40,440**		40,707	34,504
Total noninterest expense	**$ 91,797**		$ 87,172	$ 76,140
Certain components of other expense[b]				
Legal expense	**$ 740**		$ 1,436	$ 266
FDIC-related expense	**1,893**		4,203	860
Operating losses	**1,417**		1,228	1,101

(a) Includes depreciation expense associated with auto operating lease assets. Refer to Note 18 for additional information.
(b) Refer to Note 6 for additional information.
(c) Included a $1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.

2024 compared with 2023

Compensation expense increased driven by:

- higher revenue-related compensation across the LOBs,
- growth in the number of employees across the LOBs and Corporate, primarily in front office and technology, and
- the impact of First Republic, predominantly in CCB, reflecting timing and the classification of the prior-year expense, which was recognized in other expense in Corporate in the second quarter of 2023 as the individuals associated with First Republic were not employees of the Firm until July 2023.

Noncompensation expense decreased as a result of:

- lower FDIC-related expense recognized in Corporate, which included the impact of a $2.9 billion special assessment recognized in the fourth quarter of 2023, compared with a $725 million increase to the FDIC special assessment recognized in the first quarter of 2024, and
- lower legal expense, reflecting the net impact of declines in CCB, CIB and Corporate, and an increase in AWM,

predominantly offset by

- a $1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024 in Corporate,
- higher investments in technology in the businesses, as well as marketing, predominantly in CCB,
- higher occupancy expense, which included the impact of net additions to the Firm's properties,
- higher distribution fees in AWM and brokerage expense in CIB, and
- the timing impact associated with First Republic, offset by the alignment of expense to compensation expense, as noted above.

Refer to Notes 2 and 6 for additional information on Visa shares; Note 6 for additional information on other expense; and Note 34 for additional information on the First Republic acquisition.

Income tax expense

Year ended December 31, (in millions, except rate)	2024		2023	2022
Income before income tax expense	**$ 75,081**		$ 61,612	$ 46,166
Income tax expense	**16,610**	(a)	12,060	8,490
Effective tax rate	**22.1 %**		19.6 %	18.4 %

(a) Effective January 1, 2024, as a result of adopting updates to the Accounting for Investments in Tax Credit Structures guidance, the amortization of certain of the Firm's alternative energy tax-oriented investments is now recognized in income tax expense. Refer to Notes 1, 6, 14 and 25 for additional information.

2024 compared with 2023

The **effective tax rate** increased predominantly driven by:

- the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance on January 1, 2024, and
- a higher level of pretax income and changes in the mix of income and expenses subject to U.S. federal, state and local taxes, including the impact of the net gain on Visa shares and the contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024.

The prior year included the impact to income tax expense associated with the First Republic acquisition that was reflected in the estimated bargain purchase gain, and an income tax benefit related to the finalization of certain income tax regulations, both of which resulted in a reduction in the Firm's effective tax rate.

Refer to Note 25 for additional information.

Consolidated balance sheets analysis

The following is a discussion of the significant changes between December 31, 2024 and 2023. Refer to pages 161–164 for a discussion of the Critical Accounting Estimates Used by the Firm that affect the Consolidated Balance Sheets.

Selected Consolidated balance sheets data

December 31, (in millions)	2024	2023	Change
Assets			
Cash and due from banks	**$ 23,372**	$ 29,066	(20)%
Deposits with banks	**445,945**	595,085	(25)
Federal funds sold and securities purchased under resale agreements	**295,001**	276,152	7
Securities borrowed	**219,546**	200,436	10
Trading assets	**637,784**	540,607	18
Available-for-sale securities	**406,852**	201,704	102
Held-to-maturity securities	**274,468**	369,848	(26)
Investment securities, net of allowance for credit losses	**681,320**	571,552	19
Loans	**1,347,988**	1,323,706	2
Allowance for loan losses	**(24,345)**	(22,420)	9
Loans, net of allowance for loan losses	**1,323,643**	1,301,286	2
Accrued interest and accounts receivable	**101,223**	107,363	(6)
Premises and equipment	**32,223**	30,157	7
Goodwill, MSRs and other intangible assets	**64,560**	64,381	—
Other assets	**178,197**	159,308	12
Total assets	**$ 4,002,814**	$ 3,875,393	3 %

Cash and due from banks and deposits with banks decreased driven by higher investment securities in Treasury and CIO, and Markets activities in CIB.

Federal funds sold and securities purchased under resale agreements increased driven by Markets, reflecting higher client-driven market-making activities.

Securities borrowed increased driven by Markets, reflecting a higher demand for securities to cover short positions.

Refer to Note 11 for additional information on securities purchased under resale agreements and securities borrowed.

Trading assets increased predominantly due to higher levels of debt and equity instruments in Markets related to client-driven market-making activities. Refer to Notes 2 and 5 for additional information.

Investment securities increased due to:

- higher available-for-sale ("AFS") securities, reflecting net purchases, primarily U.S. Treasuries and non-U.S. government debt securities, partially offset by maturities and paydowns, and
- lower held to-maturity ("HTM") securities driven by maturities and paydowns.

Refer to Corporate results on pages 88–90, Investment Portfolio Risk Management on page 140, and Notes 2 and 10 for additional information.

Loans increased, reflecting:

- higher outstanding balances in Card Services driven by growth in new accounts and normalization of revolving balances,
- higher wholesale loans in CIB, and
- higher securities-based lending in AWM due to higher client demand,

partially offset by

- a decline in Home Lending as paydowns and loan sales outpaced originations.

The **allowance for loan losses** increased, reflecting a net addition to the allowance for loan losses of $1.9 billion, consisting of:

- $2.1 billion net addition in **consumer**, primarily in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years, partially offset by a net reduction in Home Lending in the first quarter of 2024, and
- a net reduction of $176 million in **wholesale**, reflecting:
  - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in

- Markets, and a reduction due to charge-offs largely from collateral-dependent loans,

predominantly offset by

- net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024.

There was also a $128 million net addition to the allowance for lending-related commitments recognized in other liabilities on the Consolidated balance sheets.

Refer to Consolidated Results of Operations and Credit and Investment Risk Management on pages 59–62 and pages 117–140, respectively, Critical Accounting Estimates Used by the Firm on pages 161–164, and Notes 2, 3, 12 and 13 for additional information on loans and the total allowance for credit losses.

Accrued interest and accounts receivable decreased primarily driven by lower receivables in Payments related to the timing of processing payment activities, due to December 31, 2023 falling on a weekend, as well as lower client receivables related to client-driven activities in Markets.

Premises and equipment increased primarily as a result of the construction-in-process associated with the Firm's headquarters, and purchases of properties. Refer to Notes 16 and 18 for additional information.

Goodwill, MSRs and other intangibles: Refer to Note 15 for additional information.

Other assets increased primarily due to higher cash collateral placed with central counterparties ("CCP") in Markets, the impact of the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance effective January 1, 2024, and higher auto operating lease assets in CCB. Refer to Notes 1, 6, 14 and 25 for additional information on updates to the accounting guidance.

Selected Consolidated balance sheets data (continued)

December 31, (in millions)	2024	2023	Change
Liabilities			
Deposits	**$ 2,406,032**	$ 2,400,688	—
Federal funds purchased and securities loaned or sold under repurchase agreements	**296,835**	216,535	37
Short-term borrowings	**52,893**	44,712	18
Trading liabilities	**192,883**	180,428	7
Accounts payable and other liabilities	**280,672**	290,307	(3)
Beneficial interests issued by consolidated variable interest entities ("VIEs")	**27,323**	23,020	19
Long-term debt	**401,418**	391,825	2
Total liabilities	**3,658,056**	3,547,515	3
Stockholders' equity	**344,758**	327,878	5
Total liabilities and stockholders' equity	**$ 4,002,814**	$ 3,875,393	3 %

Deposits increased, reflecting:

- an increase in CIB due to net inflows predominantly in Payments, largely offset by net maturities of structured notes in Markets,
- an increase in AWM as a result of growth in balances in new and existing client accounts, reflecting the impact of higher-yielding product offerings, largely offset by continued migration into other investments, and
- a decline in CCB primarily driven by a decrease in balances in existing accounts due to increased customer spending and migration into higher-yielding investments, predominantly offset by new accounts.

Federal funds purchased and securities loaned or sold under repurchase agreements increased driven by Markets, reflecting higher client-driven market-making activities and higher secured financing of trading assets.

Short-term borrowings increased driven by Markets, reflecting net issuance of structured notes due to client demand, and higher financing requirements.

Refer to Liquidity Risk Management on pages 108–115 for additional information on deposits, federal funds purchased and securities loaned or sold under repurchase agreements, and short-term borrowings; Notes 2 and 17 for deposits; and Note 11 for federal funds purchased and securities loaned or sold under repurchase agreements.

Trading liabilities increased due to client-driven market-making activities primarily in Fixed Income Markets, which resulted in higher levels of short positions in debt instruments. Refer to Notes 2 and 5 for additional information.

Accounts payable and other liabilities decreased primarily driven by lower payables in Payments related to the timing of processing payment activities, due to December 31, 2023 falling on a weekend, as well as lower client payables related to client-driven activities in Markets, partially offset by the impact of the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance effective January 1, 2024. Refer to Note 19 for additional information on accounts payable; and Notes 1, 6, 14 and 25 for additional information on updates to the accounting guidance.

Beneficial interests issued by consolidated VIEs increased driven by the issuance of credit card securitizations in Treasury and CIO, and activity in municipal bond vehicles in CIB.

Refer to Liquidity Risk Management on pages 108–115; and Notes 14 and 28 for additional information on Firm-sponsored VIEs and loan securitization trusts.

Long-term debt increased, primarily driven by:

- net issuances of structured notes in Markets due to client demand,

partially offset by

- a decline in Treasury and CIO, reflecting the net impact of lower FHLB advances and higher long-term debt from net issuances.

Refer to Liquidity Risk Management on pages 108–115 and Note 34 for additional information on the First Republic acquisition.

Stockholders' equity increased, reflecting:

- net income,

largely offset by

- the impact of capital actions, including repurchases of common shares, the declaration of common and preferred stock dividends, and net redemption of preferred stock, and
- net unrealized losses in AOCI, including the impact of higher interest rates on cash flow hedges in Treasury and CIO.

Refer to Consolidated Statements of changes in stockholders' equity on page 175, Capital Actions on page 105, and Note 24 for additional information.

Consolidated cash flows analysis

The following is a discussion of cash flow activities during the years ended December 31, 2024 and 2023. Refer to Consolidated cash flows analysis on page 61 of the Firm's 2023 Form 10-K for a discussion of the 2022 activities.

	Year ended December 31,		
(in millions)	**2024**	2023	2022
Net cash provided by/ (used in)			
Operating activities	**$ (42,012)**	$ 12,974	$ 107,119
Investing activities	**(163,403)**	67,643	(137,819)
Financing activities	**63,447**	(25,571)	(126,257)
Effect of exchange rate changes on cash	**(12,866)**	1,871	(16,643)
Net increase/(decrease) in cash and due from banks and deposits with banks	**$(154,834)**	$ 56,917	$(173,600)

Operating activities

JPMorganChase's operating assets and liabilities primarily support the Firm's lending and capital markets activities. These assets and liabilities can vary significantly in the normal course of business due to the amount and timing of cash flows, which are affected by client-driven and risk management activities and market conditions. The Firm believes that cash flows from operations, available cash and other liquidity sources, and its capacity to generate cash through secured and unsecured sources, are sufficient to meet its operating liquidity needs.

- In 2024, cash used resulted from higher trading assets and higher securities borrowed, largely offset by net income.
- In 2023, cash provided primarily reflected net income, lower other assets, and accrued interest and accounts receivable, predominantly offset by higher trading assets, lower accounts payable and other liabilities, and higher securities borrowed.

Investing activities

The Firm's investing activities predominantly include originating held-for-investment loans, investing in the investment securities portfolio and other short-term instruments.

- In 2024, cash used resulted from net purchases of investment securities, net loan originations and higher securities purchased under resale agreements, partially offset by proceeds from sales and securitizations of loans held-for-investment.
- In 2023, cash provided resulted from net proceeds from investment securities, proceeds from sales and securitizations of loans held-for-investment, and lower securities purchased under resale agreements, largely offset by net originations of loans and net cash used in the First Republic Bank acquisition.

Financing activities

The Firm's financing activities include acquiring customer deposits and issuing long-term debt and preferred stock.

- In 2024, cash provided primarily reflected higher securities loaned or sold under repurchase agreements and net proceeds from long- and short-term borrowings, partially offset by net redemption of preferred stock.
- In 2023, cash used reflected lower deposits, which included the impact of the repayment of the deposits provided to First Republic Bank by the consortium of large U.S. banks that the Firm assumed as part of the First Republic acquisition, partially offset by higher securities loaned under repurchase agreements and net proceeds from long- and short-term borrowings.
- For both periods, cash was used for repurchases of common stock and cash dividends on common and preferred stock.

* * *

Refer to Consolidated Balance Sheets Analysis on pages 63–65, Capital Risk Management on pages 97–107, and Liquidity Risk Management on pages 108–115, and the Consolidated Statements of Cash Flows on page 176 for a further discussion of the activities affecting the Firm's cash flows.

Non-GAAP financial measures

The Firm prepares its Consolidated Financial Statements in accordance with U.S. GAAP; these financial statements appear on pages 172–176. That presentation, which is referred to as "reported" basis, provides the reader with an understanding of the Firm's results that can be tracked consistently from year-to-year and enables a comparison of the Firm's performance with the U.S. GAAP financial statements of other companies.

In addition to analyzing the Firm's results on a reported basis, management reviews Firmwide results, including the overhead ratio, on a "managed" basis; these Firmwide managed basis results are non-GAAP financial measures. The Firm also reviews the results of the lines of business on a managed basis. The Firm's definition of managed basis starts, in each case, with the reported U.S. GAAP results and includes certain reclassifications to present total net revenue for the Firm as a whole, and for each of the reportable business segments and Corporate, on an FTE basis. Accordingly, revenue from investments that receive tax credits and tax-exempt securities is presented in the managed results on a basis comparable to taxable investments and securities. These financial measures allow management to assess the comparability of revenue from year-to-year arising from both taxable and tax-exempt sources. The corresponding income tax impact related to tax-exempt items is recorded within income tax expense. These adjustments have no impact on net income as reported by the Firm as a whole or by each of the lines of business and Corporate.

Management also uses certain non-GAAP financial measures at the Firm and business-segment levels because these other non-GAAP financial measures provide information to investors about the underlying operational performance and trends of the Firm or of the particular business segment, as the case may be, and therefore facilitate a comparison of the Firm or the business segment with the performance of its relevant competitors. Refer to Business Segment & Corporate Results on pages 70–90 for additional information on these non-GAAP measures. Non-GAAP financial measures used by the Firm may not be comparable to similarly named non-GAAP financial measures used by other companies.

The following summary table provides a reconciliation from the Firm's reported U.S. GAAP results to managed basis.

Year ended December 31, (in millions, except ratios)	**2024** Reported	Fully taxable-equivalent adjustments[(b)]	Managed basis	2023 Reported	Fully taxable-equivalent adjustments[(b)]	Managed basis	2022 Reported	Fully taxable-equivalent adjustments[(b)]	Managed basis
Other income	**$12,462** [(a)]	**$ 2,560** [(a)]	**$15,022**	$ 5,609	$ 3,782	$ 9,391	$ 4,322	$ 3,148	$ 7,470
Total noninterest revenue	**84,973**	**2,560**	**87,533**	68,837	3,782	72,619	61,985	3,148	65,133
Net interest income	**92,583**	**477**	**93,060**	89,267	480	89,747	66,710	434	67,144
Total net revenue	**177,556**	**3,037**	**180,593**	158,104	4,262	162,366	128,695	3,582	132,277
Total noninterest expense	**91,797**	**NA**	**91,797**	87,172	NA	87,172	76,140	NA	76,140
Pre-provision profit	**85,759**	**3,037**	**88,796**	70,932	4,262	75,194	52,555	3,582	56,137
Provision for credit losses	**10,678**	**NA**	**10,678**	9,320	NA	9,320	6,389	NA	6,389
Income before income tax expense	**75,081**	**3,037**	**78,118**	61,612	4,262	65,874	46,166	3,582	49,748
Income tax expense	**16,610** [(a)]	**3,037** [(a)]	**19,647**	12,060	4,262	16,322	8,490	3,582	12,072
Net income	**$58,471**	**NA**	**$58,471**	$49,552	NA	$49,552	$37,676	NA	$37,676
Overhead ratio	**52 %**	**NM**	**51 %**	55 %	NM	54 %	59 %	NM	58 %

(a) Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures guidance, under the modified retrospective method. Refer to Notes 1, 6, 14 and 25 for additional information.

(b) Predominantly recognized in CIB and Corporate.

Net interest income, net yield, and noninterest revenue excluding Markets

In addition to reviewing net interest income, net yield, and noninterest revenue on a managed basis, management also reviews these metrics excluding Markets, as shown below. Markets consists of CIB's Fixed Income Markets and Equity Markets. These metrics, which exclude Markets, are non-GAAP financial measures. Management reviews these metrics to assess the performance of the Firm's lending, investing (including asset-liability management) and deposit-raising activities, apart from any volatility associated with Markets activities. In addition, management also assesses Markets business performance on a total revenue basis as offsets may occur across revenue lines. Management believes that these measures provide investors and analysts with alternative measures to analyze the revenue trends of the Firm.

Year ended December 31, (in millions, except rates)	2024	2023	2022
Net interest income – reported[a]	**$ 92,583**	$ 89,267	$ 66,710
Fully taxable-equivalent adjustments	**477**	480	434
Net interest income – managed basis	**$ 93,060**	$ 89,747	$ 67,144
Less: Markets net interest income[b]	**641**	(294)	4,789
Net interest income excluding Markets	**$ 92,419**	$ 90,041	$ 62,355
Average interest-earning assets[a]	**$3,537,567**	$3,325,708	$3,349,079
Less: Average Markets interest-earning assets[b]	**1,128,153**	985,777	953,195
Average interest-earning assets excluding Markets	**$2,409,414**	$2,339,931	$2,395,884
Net yield on average interest-earning assets – managed basis	**2.63 %**	2.70 %	2.00 %
Net yield on average Markets interest-earning assets[b]	**0.06**	(0.03)	0.50
Net yield on average interest-earning assets excluding Markets	**3.84 %**	3.85 %	2.60 %
Noninterest revenue – reported[c]	**$ 84,973**	$ 68,837	$ 61,985
Fully taxable-equivalent adjustments[c]	**2,560**	3,782	3,148
Noninterest revenue – managed basis	**$ 87,533**	$ 72,619	$ 65,133
Less: Markets noninterest revenue[b][d]	**29,366**	28,258	24,373
Noninterest revenue excluding Markets	**$ 58,167**	$ 44,361	$ 40,760
Memo: Total Markets net revenue[b]	**$ 30,007**	$ 27,964	$ 29,162

(a) Includes the effect of derivatives that qualify for hedge accounting. Taxable-equivalent amounts are used, also where applicable. Refer to Note 5 for additional information on hedge accounting.
(b) Refer to pages 81–82 for further information on Markets.
(c) Effective January 1, 2024, the Firm adopted updates to the Accounting for Investment in Tax Credit Stricture guidance, under the modified retrospective method. Refer to Notes 1, 6, 14 and 25 for additional information.
(d) Includes the market-related revenues of the former Commercial Banking business segment. Prior-period amounts have been revised to conform with the current presentation.

Calculation of certain U.S. GAAP and non-GAAP financial measures

Certain U.S. GAAP and non-GAAP financial measures are calculated as follows:

Book value per share ("BVPS")
Common stockholders' equity at period-end / Common shares at period-end

Overhead ratio
Total noninterest expense / Total net revenue

ROA
Reported net income / Total average assets

ROE
Net income* / Average common stockholders' equity

ROTCE
Net income* / Average tangible common equity

TBVPS
Tangible common equity at period-end / Common shares at period-end

* Represents net income applicable to common equity

In addition, the Firm reviews other non-GAAP measures such as:

- Adjusted expense, which represents noninterest expense excluding Firmwide legal expense, and
- Pre-provision profit, which represents total net revenue less total noninterest expense.

Management believes that these measures help investors to understand the effect of these items on reported results and provide an alternative presentation of the Firm's performance.

TCE, ROTCE and TBVPS

TCE, ROTCE and TBVPS are each non-GAAP financial measures. TCE represents the Firm's common stockholders' equity (i.e., total stockholders' equity less preferred stock) less goodwill and identifiable intangible assets (other than MSRs), net of related deferred tax liabilities. ROTCE measures the Firm's net income applicable to common equity as a percentage of average TCE. TBVPS represents the Firm's TCE at period-end divided by common shares at period-end. TCE, ROTCE and TBVPS are utilized by the Firm, as well as investors and analysts, in assessing the Firm's use of equity.

The following summary table provides a reconciliation from the Firm's common stockholders' equity to TCE.

	Period-end		Average		
			Year ended December 31,		
(in millions, except per share and ratio data)	**Dec 31, 2024**	Dec 31, 2023	**2024**	2023	2022
Common stockholders' equity	**$ 324,708**	$ 300,474	**$ 312,370**	$ 282,056	$ 253,068
Less: Goodwill	**52,565**	52,634	**52,627**	52,258	50,952
Less: Other intangible assets	**2,874**	3,225	**3,042**	2,572	1,112
Add: Certain deferred tax liabilities[a]	**2,943**	2,996	**2,970**	2,883	2,505
Tangible common equity	**$ 272,212**	$ 247,611	**$ 259,671**	$ 230,109	$ 203,509
Return on tangible common equity	**NA**	NA	**22 %**	21 %	18 %
Tangible book value per share	**$ 97.30**	$ 86.08	**NA**	NA	NA

(a) Represents deferred tax liabilities related to tax-deductible goodwill and to identifiable intangibles created in nontaxable transactions, which are netted against goodwill and other intangibles when calculating TCE.

BUSINESS SEGMENT & CORPORATE RESULTS

The Firm is managed on an LOB basis. Effective in the second quarter of 2024, the Firm reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank ("CIB"). As a result of the reorganization, the Firm has three reportable business segments – Consumer & Community Banking, Commercial & Investment Bank, and Asset & Wealth Management – with the remaining activities in Corporate.

The business segments are determined based on the products and services provided, or the type of customer served, and they reflect the manner in which financial information is evaluated by the Firm's Operating Committee, whose members act collectively as the Firm's chief operating decision maker. Segment results are presented on a managed basis. Refer to Explanation and Reconciliation of the Firm's use of Non-GAAP Financial Measures, on pages 67–69 for a definition of managed basis.

The following table depicts the Firm's reportable business segments.

JPMorganChase					
Consumer Business Segment			Wholesale Business Segments		
Consumer & Community Banking			Commercial & Investment Bank		Asset & Wealth Management
Banking & Wealth Management	Home Lending	Card Services & Auto	Banking & Payments	Markets & Securities Services	• Asset Management • Global Private Bank
• Consumer Banking • Business Banking • J.P. Morgan Wealth Management	• Home Lending Production • Home Lending Servicing • Real Estate Portfolios	• Card Services • Auto	• Investment Banking • Payments • Lending • Other	• Fixed Income Markets • Equity Markets • Securities Services • Credit Adjustments & Other	

Description of business segment reporting methodology

Results of the reportable business segments are intended to present each segment as if it were a stand-alone business. The management reporting process that derives business segment results includes the allocation of certain income and expense items. The Firm periodically assesses the assumptions, methodologies and reporting classifications used for segment reporting, and therefore further refinements may be implemented in future periods. The Firm also assesses the level of capital required for each LOB on at least an annual basis. The Firm's LOBs also provide various business metrics which are utilized by the Firm and its investors and analysts in assessing performance.

Revenue sharing

When business segments or businesses within each segment join efforts to sell products and services to the Firm's clients and customers, the participating businesses may agree to share revenue from those transactions. Revenue is generally recognized in the segment responsible for the related product or service, with allocations to the other segments or businesses involved in the transaction. The segment and business results reflect these revenue-sharing agreements.

Expense allocation
Where business segments use services provided by Corporate support units, or another business segment, the costs of those services are allocated to the respective business segments. The expense is generally allocated based on the actual cost and use of services provided. In contrast, certain costs and investments related to Corporate that are not currently utilized by any LOB are not allocated to the business segments and are retained in Corporate. Expense retained in Corporate generally includes costs that would not be incurred if the segments were stand-alone businesses, and other items not solely aligned with a particular reportable business segment.

Funds transfer pricing
Funds transfer pricing ("FTP") is the process by which the Firm allocates interest income and expense to the LOBs and Other Corporate and transfers the primary interest rate risk and liquidity risk to Treasury and CIO.

The funds transfer pricing process considers the interest rate and liquidity risk characteristics of assets and liabilities and off-balance sheet products. Periodically, the methodology and assumptions utilized in the FTP process are adjusted to reflect economic conditions and other factors, which may impact the allocation of net interest income to the segments. Effective in the fourth quarter of 2024, the Firm updated its FTP with respect to consumer deposits, which resulted in an increase in the funding benefit reflected within CCB's net interest income that is fully offset in Corporate, with no effect on the Firm's net interest income.

As a result of higher average interest rates, the cost of funds for assets and the FTP benefit earned for liabilities have generally increased in the current year, impacting the net interest income of the business segments. During the period ended December 31, 2024, this has resulted in a higher cost of funds for loans and Markets activities. In addition, rates paid to deposit holders increased more than the FTP benefit increase during the year, resulting in deposit margin compression.

Foreign exchange risk
Foreign exchange risk is transferred from the LOBs and Other Corporate to Treasury and CIO for certain revenues and expenses. Treasury and CIO manages these risks centrally and reports the impact of foreign exchange rate movements related to the transferred risk in its results. Refer to Market Risk Management on page 149 for additional information.

Debt expense and preferred stock dividend allocation
As part of the FTP process, almost all of the cost of the credit spread component of outstanding unsecured long-term debt and preferred stock dividends is allocated to the reportable business segments, while the balance of the cost is retained in Corporate. The methodology to allocate the cost of unsecured long-term debt and preferred stock dividends to the business segments is aligned with the relevant regulatory capital requirements and funding needs of the LOBs, as applicable. The allocated cost of unsecured long-term debt is included in a business segment's net interest income, and net income is reduced by preferred stock dividends, to arrive at a business segment's net income applicable to common equity. Refer to Capital Risk Management on pages 97–107 for additional information.

Capital allocation
The amount of capital assigned to each LOB and Corporate is referred to as equity. The Firm's current equity allocation methodology incorporates Basel III Standardized risk-weighted assets ("RWA") and the global systemically important banks ("GSIB") surcharge, both under rules currently in effect, as well as a simulation of capital depletion in a severe stress environment. At least annually, the assumptions, judgments and methodologies used to allocate capital are reassessed and, as a result, the capital allocated to the LOBs and Corporate may change. Refer to Line of business and Corporate equity on page 104 for additional information on capital allocation.

Segment & Corporate Results – Managed Basis

The following tables summarize the Firm's results by business segments and Corporate for the periods indicated.

Year ended December 31, (in millions, except ratios)	Consumer & Community Banking 2024	2023	2022	Commercial & Investment Bank 2024	2023	2022	Asset & Wealth Management 2024	2023	2022
Total net revenue	**$ 71,507**	$ 70,148	$ 54,814 (a)	**$ 70,114**	$ 64,353	$ 59,635 (a)	**$ 21,578**	$ 19,827	$ 17,748
Total noninterest expense	**38,036**	34,819	31,208 (a)	**35,353**	33,972	32,069 (a)	**14,414**	12,780	11,829
Pre-provision profit/(loss)	**33,471**	35,329	23,606	**34,761**	30,381	27,566	**7,164**	7,047	5,919
Provision for credit losses	**9,974**	6,899	3,813	**762**	2,091	2,426	**(68)**	159	128
Net income/(loss)	**17,603**	21,232	14,916 (a)	**24,846**	20,272	19,138 (a)	**5,421**	5,227	4,365
Return on equity ("ROE")	**32 %**	38 %	29 %	**18 %**	14 %	14 %	**34 %**	31 %	25 %

Year ended December 31, (in millions, except ratios)	Corporate 2024	2023	2022	Total 2024	2023	2022
Total net revenue	**$ 17,394** (b)	$ 8,038	$ 80	**$180,593** (b)	$162,366	$ 132,277
Total noninterest expense	**3,994** (c)	5,601	1,034	**91,797** (c)	87,172	76,140
Pre-provision profit/(loss)	**13,400**	2,437	(954)	**88,796**	75,194	56,137
Provision for credit losses	**10**	171	22	**10,678**	9,320	6,389
Net income/(loss)	**10,601**	2,821	(743)	**58,471**	49,552	37,676
Return on equity ("ROE")	**NM**	NM	NM	**18 %**	17 %	14 %

(a) In the first quarter of 2023, the allocations of revenue and expense to CCB associated with a Merchant Services revenue sharing agreement were discontinued and are now retained in Payments in CIB. Prior-period amounts have been revised to conform with the current presentation.
(b) Included a $7.9 billion net gain related to Visa shares recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.
(c) Included a $1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.

Refer to Note 32 for further details on total net revenue and total noninterest expense.

The following sections provide a comparative discussion of the Firm's results by business segments and Corporate as of or for the years ended December 31, 2024 and 2023, unless otherwise specified.

Consumer & Community Banking offers products and services to consumers and small businesses through bank branches, ATMs, digital (including mobile and online) and telephone banking. CCB is organized into Banking & Wealth Management (including Consumer Banking, Business Banking and J.P. Morgan Wealth Management), Home Lending (including Home Lending Production, Home Lending Servicing and Real Estate Portfolios) and Card Services & Auto. Banking & Wealth Management offers deposit, investment and lending products, cash management, payments and services. Home Lending includes mortgage origination and servicing activities, as well as portfolios consisting of residential mortgages and home equity loans. Card Services issues credit cards and offers travel services. Auto originates and services auto loans and leases.

Selected income statement data

Year ended December 31, (in millions, except ratios)	2024	2023	2022
Revenue			
Lending- and deposit-related fees	**$ 3,387**	$ 3,356	$ 3,316
Asset management fees	**4,014**	3,282	2,734
Mortgage fees and related income	**1,378**	1,175	1,236
Card income	**3,139**	2,532	2,469
All other income[a]	**4,731**	4,773	5,131
Noninterest revenue	**16,649**	15,118	14,886
Net interest income	**54,858**	55,030	39,928
Total net revenue	**71,507**	70,148	54,814
Provision for credit losses	**9,974**	6,899	3,813
Noninterest expense			
Compensation expense	**17,045**	15,171	13,092
Noncompensation expense[b]	**20,991**	19,648	18,116
Total noninterest expense	**38,036**	34,819 [d]	31,208
Income before income tax expense	**23,497**	28,430	19,793
Income tax expense	**5,894**	7,198	4,877
Net income	**$17,603**	$ 21,232	$ 14,916
Revenue by business			
Banking & Wealth Management	**$40,943**	$ 43,199	$ 30,059
Home Lending	**5,097**	4,140	3,674
Card Services & Auto	**25,467**	22,809	21,081
Mortgage fees and related income details:			
Production revenue	**627**	421	497
Net mortgage servicing revenue[c]	**751**	754	739
Mortgage fees and related income	**$ 1,378**	$ 1,175	$ 1,236
Financial ratios			
Return on equity	**32 %**	38 %	29 %
Overhead ratio	**53**	50	57

(a) Primarily includes operating lease income and commissions and other fees. Operating lease income was $2.8 billion, $2.8 billion and $3.6 billion for the years ended December 31, 2024, 2023 and 2022, respectively.
(b) Included depreciation expense on leased assets of $1.7 billion, $1.7 billion and $2.4 billion for the years ended December 31, 2024, 2023 and 2022, respectively.
(c) Included MSR risk management results of $159 million, $131 million and $93 million for the years ended December 31, 2024, 2023 and 2022, respectively.
(d) In the second quarter of 2023, substantially all of the expense associated with First Republic was reported in Corporate. Commencing in the third quarter of 2023, the expense was aligned to the appropriate LOB.

2024 compared with 2023

Net income was $17.6 billion, down 17%.

Net revenue was $71.5 billion, up 2%.

Net interest income was $54.9 billion, flat when compared with the prior year, reflecting:

- lower NII in Banking & Wealth Management ("BWM"), predominantly driven by deposit margin compression and lower average deposits,

offset by

- higher Card Services NII, predominantly driven by higher revolving balances, and
- the timing impact of First Republic in Home Lending.

Noninterest revenue was $16.6 billion, up 10%, predominantly driven by:

- higher asset management fees reflecting higher average market levels, including the timing impact of First Republic and, to a lesser extent, net inflows, as well as higher commissions from annuity sales in BWM,
- higher card income, driven by higher net interchange reflecting increased debit and credit card sales volume, and higher annual fees, partially offset by an increase in amortization related to new account origination costs, as well as
- higher production revenue in Home Lending, including the timing impact of First Republic.

Refer to Note 6 for additional information on card income, asset management fees, and commissions and other fees; and Critical Accounting Estimates on pages 161–164 for additional information on the credit card rewards liability.

Refer to Executive Overview on page 54 and Note 34 for additional information on First Republic.

Noninterest expense was $38.0 billion, up 9%, reflecting First Republic-related expense that was aligned to CCB from Corporate starting in the third quarter of 2023, impacting both compensation and noncompensation expense.

The increase in expense also reflected:

- higher compensation expense, largely driven by higher revenue-related compensation predominantly for advisors and bankers, and an increase in the number of employees, including in technology, and
- higher noncompensation expense, largely driven by continued investments in technology and marketing, as well as higher operating losses, partially offset by lower legal expense.

The provision for credit losses was $10.0 billion, reflecting:

- net charge-offs of $7.9 billion, up $2.6 billion, including $2.4 billion in Card Services, reflecting the seasoning of vintages originated in recent years, credit normalization and balance growth, and
- a $2.0 billion net addition to the allowance for credit losses, consisting of:
  - $2.2 billion in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years,

  partially offset by

  - a $125 million net reduction in Home Lending, primarily due to improvements in the outlook for home prices in the first quarter of 2024.

The provision in the prior year was $6.9 billion, reflecting net charge-offs of $5.3 billion, a $1.2 billion net addition to the allowance for credit losses, predominantly driven by Card Services, and a $408 million net addition to the allowance for credit losses to establish the allowance for the First Republic loans and lending-related commitments in the second quarter of 2023.

Refer to Credit and Investment Risk Management on pages 117–140 and Allowance for Credit Losses on pages 137–139 for a further discussion of the credit portfolios and the allowance for credit losses.

Selected metrics

As of or for the year ended December 31, (in millions, except employees)	2024	2023	2022
Selected balance sheet data (period-end)			
Total assets	**$650,268**	$ 642,951	$ 514,085
Loans:			
Banking & Wealth Management	**33,221**	31,142	29,008
Home Lending[(a)]	**246,498**	259,181	172,554
Card Services	**233,016**	211,175	185,175
Auto	**73,619**	77,705	68,191
Total loans	**586,354**	579,203	454,928
Deposits[(b)]	**1,056,652**	1,094,738	1,131,611
Equity	**54,500**	55,500	50,000
Selected balance sheet data (average)			
Total assets	**$ 631,648**	$ 584,367	$ 497,263
Loans:			
Banking & Wealth Management	**31,544**	30,142	31,545
Home Lending[(c)]	**252,542**	232,115	176,285
Card Services	**214,139**	191,424	163,335
Auto	**75,009**	72,674	68,098
Total loans	**573,234**	526,355	439,263
Deposits[(b)]	**1,064,215**	1,126,552	1,162,680
Equity	**54,500**	54,349	50,000
Employees	**144,989**	141,640	135,347

(a) At December 31, 2024, 2023 and 2022, Home Lending loans held-for-sale and loans at fair value were $8.1 billion, $3.4 billion and $3.0 billion, respectively.
(b) In the fourth quarter of 2023, CCB transferred approximately $18.8 billion of deposits associated with First Republic to AWM and CIB.
(c) Average Home Lending loans held-for-sale and loans at fair value were $7.1 billion, $4.8 billion and $7.3 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

Selected metrics

As of or for the year ended December 31, (in millions, except ratio data)	2024	2023	2022
Credit data and quality statistics			
Nonaccrual loans[(a)]	**$ 3,366**	$3,740	$ 3,899
Net charge-offs/(recoveries)			
Banking & Wealth Management	**442**	340	370
Home Lending	**(106)**	(56)	(229)
Card Services	**7,148**	4,699	2,403
Auto	**444**	357	144
Total net charge-offs/ (recoveries)	**$ 7,928**	$5,340	$ 2,688
Net charge-off/(recovery) rate			
Banking & Wealth Management	**1.40 %**	1.13 %	1.17 %
Home Lending	**(0.04)**	(0.02)	(0.14)
Card Services	**3.34**	2.45	1.47
Auto	**0.59**	0.49	0.21
Total net charge-off/ (recovery) rate	**1.40 %**	1.02 %	0.62 %
30+ day delinquency rate			
Home Lending[(b)]	**0.78 %**	0.66 %	0.83 %
Card Services	**2.17**	2.14	1.45
Auto	**1.43**	1.19	1.01
90+ day delinquency rate - Card Services	**1.14 %**	1.05 %	0.68 %
Allowance for loan losses			
Banking & Wealth Management	**$ 764**	$ 685	$ 722
Home Lending	**447**	578	867
Card Services	**14,608**	12,453	11,200
Auto	**692**	742	715
Total allowance for loan losses	**$ 16,511**	$14,458	$13,504

(a) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024, 2023 and 2022, mortgage loans 90 or more days past due and insured by U.S. government agencies were $84 million, $123 million and $187 million, respectively. In addition, the Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance
(b) At December 31, 2024, 2023 and 2022, excluded mortgage loans insured by U.S. government agencies of $122 million, $176 million and $258 million, respectively, that are 30 or more days past due. These amounts have been excluded based upon the government guarantee.

Selected metrics

As of or for the year ended December 31, (in billions, except ratios and where otherwise noted)	**2024**	2023	2022
Business Metrics			
CCB Consumer customers (in millions)	**84.4**	82.1	79.2
CCB Small business customers (in millions)	**7.0**	6.4	5.7
Number of branches	**4,966**	4,897	4,787
Active digital customers (in thousands)[(a)]	**70,813**	66,983	63,136
Active mobile customers (in thousands)[(b)]	**57,821**	53,828	49,710
Debit and credit card sales volume	**$1,805.4**	$1,678.6	$1,555.4
Total payments transaction volume (in trillions)[(c)]	**6.4**	5.9	5.6
Banking & Wealth Management			
Average deposits	**$1,049.3**	$ 1,111.7	$ 1,145.7
Deposit margin	**2.66 %**	2.84 %	1.71 %
Business Banking average loans	**$ 19.5**	$ 19.6	$ 22.3
Business Banking origination volume	**4.5**	4.8	4.3
Client investment assets[(d)]	**1,087.6**	951.1	647.1
Number of client advisors	**5,755**	5,456	5,029
Home Lending			
Mortgage origination volume by channel			
Retail	**$ 25.5**	$ 22.4	$ 38.5
Correspondent	**15.3**	12.7	26.9
Total mortgage origination volume[(e)]	**$ 40.8**	$ 35.1	$ 65.4
Third-party mortgage loans serviced (period-end)	**$ 648.0**	$ 631.2	$ 584.3
MSR carrying value (period-end)	**9.1**	8.5	8.0
Card Services			
Sales volume, excluding commercial card	**$1,259.3**	$ 1,163.6	$1,064.7
Net revenue rate	**10.03 %**	9.72 %	9.87 %
Net yield on average loans	**9.73**	9.61	9.77
New credit card accounts opened (in millions)	**10.0**	10.0	9.6
Auto			
Loan and lease origination volume	**$ 40.3**	$ 41.3	$ 30.4
Average auto operating lease assets	**11.1**	10.9	14.3

(a) Users of all web and/or mobile platforms who have logged in within the past 90 days.
(b) Users of all mobile platforms who have logged in within the past 90 days.
(c) Total payments transaction volume includes debit and credit card sales volume and gross outflows of ACH, ATM, teller, wires, BillPay, PayChase, Zelle, person-to-person and checks.
(d) Includes assets invested in managed accounts and J.P. Morgan mutual funds where AWM is the investment manager. Refer to AWM segment results on pages 84–87 for additional information.
(e) Firmwide mortgage origination volume was $47.4 billion, $41.4 billion and $81.8 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

The Commercial & Investment Bank is comprised of the Banking & Payments and Markets & Securities Services businesses. These businesses offer investment banking, lending, payments, market-making, financing, custody and securities products and services to a global base of corporate and institutional clients. Banking & Payments offers products and services in all major capital markets, including advising on corporate strategy and structure, capital-raising in equity and debt markets, and loan origination and syndication. Banking & Payments also provides services that enable clients to manage payments globally across liquidity and account solutions, commerce solutions, clearing, trade, and working capital. Markets & Securities Services includes Markets, which is a global market-maker across products, including cash and derivative instruments, and also offers sophisticated risk management solutions, lending, prime brokerage, clearing and research. Markets & Securities Services also includes Securities Services, a leading global custodian that provides custody, fund services, liquidity and trading services, and data solutions products.

(a) Reflects the reorganization of the Firm's business segments in the second quarter of 2024. Refer to Business Segment & Corporate Results on pages 70–90 for additional information.

Selected income statement data

Year ended December 31, (in millions)	**2024**	2023	2022
Revenue			
Investment banking fees	**$ 9,116**	$ 6,631	$ 6,977
Principal transactions	**24,382**	23,794	19,792
Lending- and deposit-related fees	**3,914**	3,423	3,662
Commissions and other fees	**5,278**	4,879	5,113
Card income	**2,310**	2,213	1,934
All other income	**3,253**	2,869	2,060
Noninterest revenue	**48,253**	43,809	39,538
Net interest income	**21,861**	20,544	20,097
Total net revenue[a]	**70,114**	64,353	59,635
Provision for credit losses	**762**	2,091	2,426
Noninterest expense			
Compensation expense	**18,191**	17,105	16,214
Noncompensation expense	**17,162**	16,867	15,855
Total noninterest expense	**35,353**	33,972	32,069
Income before income tax expense	**33,999**	28,290	25,140
Income tax expense	**9,153**	8,018	6,002
Net income	**$24,846**	$20,272	$ 19,138

(a) Included tax equivalent adjustments primarily from income tax credits from investments in alternative energy, affordable housing and new markets, income from tax-exempt securities and loans, and the related amortization and other tax benefits of the investments in alternative energy and affordable housing of $2.8 billion, $4.0 billion and $3.3 billion for the years ended December 31, 2024, 2023 and 2022, respectively. Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance, under the modified retrospective method. Refer to Notes 1, 6, 14 and 25 for additional information.

Selected income statement data

Year ended December 31, (in millions, except ratios)	**2024**	2023	2022
Financial ratios			
Return on equity	**18 %**	14 %	14 %
Overhead ratio	**50**	53	54
Compensation expense as percentage of total net revenue	**26**	27	27
Revenue by business			
Investment Banking	**$ 9,636**	$ 7,076	$ 7,205
Payments	**18,085**	17,818	13,490
Lending	**7,470**	6,896	5,882
Other	**76**	107	244
Total Banking & Payments	**35,267**	31,897	26,821
Fixed Income Markets[a]	**20,066**	19,180	19,074
Equity Markets[a]	**9,941**	8,784	10,088
Securities Services	**5,084**	4,772	4,488
Credit Adjustments & Other[b]	**(244)**	(280)	(836)
Total Markets & Securities Services	**34,847**	32,456	32,814
Total net revenue	**$70,114**	$64,353	$59,635

(a) In the fourth quarter of 2024, certain net funding costs that were previously allocated to Fixed Income Markets were reclassified to Equity Markets. Prior-period amounts have been revised to conform with the current presentation.
(b) Consists primarily of centrally-managed credit valuation adjustments ("CVA"), funding valuation adjustments ("FVA") on derivatives, other valuation adjustments, and certain components of fair value option elected liabilities, which are primarily reported in principal transactions revenue. Results are presented net of associated hedging activities and net of CVA and FVA amounts allocated to Fixed Income Markets and Equity Markets. Refer to Notes 2, 3 and 24 for additional information.

Banking & Payments Revenue by Client Coverage Segment: [(a)]

Global Corporate Banking & Global Investment Banking provides banking products and services generally to large corporations, financial institutions and merchants.

Commercial Banking provides banking products and services generally to middle market clients, including start-ups, small and mid-sized companies, local governments, municipalities, and nonprofits, as well as to commercial real estate clients.

Other includes amounts related to credit protection purchased against certain retained loans and lending-related commitments in Lending, the impact of equity investments in Payments and revenues not aligned with a primary client coverage segment.

(a) Global Banking is a client coverage view within the Banking & Payments business and is comprised of the Global Corporate Banking, Global Investment Banking and Commercial Banking client coverage segments.

Selected income statement data

Year ended December 31, (in millions)	**2024**	2023	2022
Banking & Payments revenue by client coverage segment			
Global Corporate Banking & Global Investment Banking	**$ 24,549**	$ 21,700	$ 19,325
Commercial Banking	**11,487**	11,050	7,906
Middle Market Banking	**7,759**	7,740	5,443
Commercial Real Estate Banking	**3,728**	3,310	2,463
Other	**(769)**	(853)	(410)
Total Banking & Payments revenue	**$ 35,267**	$ 31,897	$ 26,821

2024 compared with 2023

Net income was $24.8 billion, up 23%.

Net revenue was $70.1 billion, up 9%.

Banking & Payments revenue was $35.3 billion, up 11%.

- Investment Banking revenue was $9.6 billion, up 36%. Investment Banking fees were up 37%, driven by higher fees across products. The Firm ranked #1 for Global Investment Banking fees, according to Dealogic.
  - Debt underwriting fees were $4.1 billion, up 55%, predominantly driven by higher industry-wide issuances in leveraged loans, and in high-grade and high-yield bonds.
  - Equity underwriting fees were $1.7 billion, up 47%, driven by increased industry-wide fees and wallet share gains in IPOs, and in follow-on and convertible securities offerings.
  - Advisory fees were $3.3 billion, up 17%, driven by increased industry-wide M&A activity and wallet share gains.
- Payments revenue was $18.1 billion, up 1%, driven by fee growth on higher volumes as well as higher average deposits, predominantly offset by deposit margin compression, reflecting higher rates paid, and higher deposit-related client credits.
- Lending revenue was $7.5 billion, up 8%, predominantly driven by the impacts of higher rates and the First Republic acquisition.

Markets & Securities Services revenue was $34.8 billion, up 7%. Markets revenue was $30.0 billion, up 7%.

- Equity Markets revenue was $9.9 billion, up 13%, driven by higher revenue in Equity Derivatives and Prime Finance.
- Fixed Income Markets revenue was $20.1 billion, up 5%, driven by higher revenue in the Securitized Products Group, Currencies & Emerging Markets, and Credit, largely offset by lower revenue in Rates and Commodities.
- Securities Services revenue was $5.1 billion, up 7%, predominantly driven by fee growth on higher client activity and market levels.
- Credit Adjustments & Other was a loss of $244 million, compared with a loss of $280 million in the prior year.

Noninterest expense was $35.4 billion, up 4%, driven by higher compensation expense, including revenue-related compensation and an increase in the number of employees, as well as higher technology and brokerage expense partially offset by lower legal expense.

The provision for credit losses was $762 million, reflecting:

- net charge-offs of $617 million, primarily in Real Estate, largely concentrated in Office, and
- a $145 million net addition to the allowance for credit losses, driven by
  - net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024,

  predominantly offset by
  - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in Markets, and a reduction due to charge-offs predominantly from collateral-dependent loans.

The provision in the prior year was $2.1 billion, reflecting a $1.5 billion net addition to the allowance for credit losses, which included $608 million to establish the allowance for the First Republic loans and lending-related commitments in the second quarter of 2023, and net charge-offs of $588 million.

Selected metrics

As of or for the year ended December 31, (in millions, except employees)	2024	2023	2022
Selected balance sheet data (period-end)			
Total assets	**$1,773,194**	$1,638,493	$1,591,402
Loans:			
Loans retained	**483,043**	475,186	421,521
Loans held-for-sale and loans at fair value[(a)]	**40,324**	39,464	43,011
Total loans	**523,367**	514,650	464,532
Equity	**132,000**	138,000	128,000
Banking & Payments loans by client coverage segment (period-end)[(b)]			
Global Corporate Banking & Global Investment Banking	**$ 125,083**	$ 128,097	$ 128,165
Commercial Banking	**217,674**	221,550	180,624
Middle Market Banking	**72,814**	78,043	72,625
Commercial Real Estate Banking	**144,860**	143,507	107,999
Other	**187**	526	122
Total Banking & Payments loans	**342,944**	350,173	308,911
Selected balance sheet data (average)			
Total assets	**$1,912,466**	$ 1,716,755	$1,649,358
Trading assets-debt and equity instruments	**624,032**	508,792	405,948
Trading assets-derivative receivables	**57,028**	63,862	77,822
Loans:			
Loans retained	**$ 475,426**	$ 457,886	$ 395,015
Loans held-for-sale and loans at fair value[(a)]	**43,621**	40,891	48,196
Total loans	**$ 519,047**	$ 498,777	$ 443,211
Deposits[(c)]	**1,061,488**	996,295	1,033,880
Equity	**132,000**	137,507	128,000
Banking & Payments loans by client coverage segment (average)[(b)]			
Global Corporate Banking & Global Investment Banking	**$ 128,142**	$ 131,230	$ 122,174
Commercial Banking	**220,285**	209,244	173,289
Middle Market Banking	**75,605**	77,130	67,830
Commercial Real Estate Banking	**144,680**	132,114	105,459
Other	**354**	331	168
Total Banking & Payments loans	**$ 348,781**	$ 340,805	$ 295,631
Employees	**93,231**	92,271	88,139

(a) Loans held-for-sale and loans at fair value primarily reflect lending-related positions originated and purchased in Markets, including loans held for securitization.
(b) Refer to page 78 for a description of each of the client coverage segments.
(c) In the fourth quarter of 2023, certain deposits associated with First Republic were transferred to CIB from CCB.

Selected metrics

As of or for the year ended December 31, (in millions, except ratios)	2024	2023	2022
Credit data and quality statistics			
Net charge-offs/(recoveries)	**$ 689** [(d)]	$ 588	$ 166
Nonperforming assets:			
Nonaccrual loans:			
Nonaccrual loans retained[(a)]	**$3,258**	$ 1,675	$ 1,484
Nonaccrual loans held-for-sale and loans at fair value[(b)]	**1,502**	828	848
Total nonaccrual loans	**4,760**	2,503	2,332
Derivative receivables	**145**	364	296
Assets acquired in loan satisfactions	**213**	169	87
Total nonperforming assets	**$ 5,118**	$ 3,036	$ 2,715
Allowance for credit losses:			
Allowance for loan losses	**$7,294**	$ 7,326	$ 5,616
Allowance for lending-related commitments	**1,976**	1,849	2,278
Total allowance for credit losses	**$9,270**	$ 9,175	$ 7,894
Net charge-off/(recovery) rate[(c)]	**0.14 %**	0.13 %	0.04 %
Allowance for loan losses to period-end loans retained	**1.51**	1.54	1.33
Allowance for loan losses to nonaccrual loans retained[(a)]	**224**	437	378
Nonaccrual loans to total period-end loans	**0.91**	0.49	0.50

(a) Allowance for loan losses of $435 million, $251 million and $257 million were held against these nonaccrual loans at December 31, 2024, 2023 and 2022, respectively.
(b) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024, 2023 and 2022, mortgage loans 90 or more days past due and insured by U.S. government agencies were $37 million, $59 million and $115 million, respectively.
(c) Loans held-for-sale and loans at fair value were excluded when calculating the net charge-off/(recovery) rate.
(d) Includes $72 million related to a purchased credit deteriorated ("PCD") loan that was charged off in the fourth quarter of 2024.

Investment banking fees

Year ended December 31, (in millions)		2024		2023		2022
Advisory	**$**	**3,290**	$	2,814	$	3,051
Equity underwriting		**1,692**		1,151		1,034
Debt underwriting(a)		**4,134**		2,666		2,892
Total investment banking fees	**$**	**9,116**	$	6,631	$	6,977

(a) Represents long-term debt and loan syndications.

League table results – wallet share

		2024			2023			2022	
Year ended December 31,		Rank	Share		Rank	Share		Rank	Share
Based on fees(a)									
M&A(b)									
Global	**#**	**1**	**9.6 %**	#	2	9.0 %	#	2	7.9 %
U.S.		**1**	**11.4**		2	10.9		2	8.9
Equity and equity-related(c)									
Global		**1**	**11.0**		1	7.7		2	5.7
U.S.		**1**	**14.7**		1	14.4		1	14.0
Long-term debt(d)									
Global		**1**	**7.6**		1	7.0		1	6.9
U.S.		**1**	**11.4**		1	10.9		1	12.1
Loan syndications									
Global		**1**	**10.2**		1	11.9		1	11.0
U.S.		**1**	**11.8**		1	15.1		1	12.9
Global investment banking fees(e)	**#**	**1**	**9.3 %**	#	1	8.6 %	#	1	7.8 %

(a) Source: Dealogic as of January 2, 2025. Reflects the ranking of revenue wallet and market share.
(b) Global M&A excludes any withdrawn transactions. U.S. M&A revenue wallet represents wallet from client parents based in the U.S.
(c) Global equity and equity-related ranking includes rights offerings and Chinese A-Shares.
(d) Long-term debt rankings include investment-grade, high-yield, supranationals, sovereigns, agencies, covered bonds, asset-backed securities ("ABS") and mortgage-backed securities ("MBS"); and exclude money market, short-term debt and U.S. municipal securities.
(e) Global investment banking fees exclude money market, short-term debt and shelf securities.

Markets revenue

The following table summarizes selected income statement data for the Markets businesses. Markets includes both Fixed Income Markets and Equity Markets. Markets revenue consists of principal transactions, fees, commissions and other income, as well as net interest income. The Firm assesses its Markets business performance on a total revenue basis, as offsets generally occur across revenue line items. For example, securities that generate net interest income may be risk-managed by derivatives that are reflected at fair value in principal transactions revenue. Refer to Notes 6 and 7 for a description of the composition of these income statement line items.

Principal transactions reflects revenue on financial instruments and commodities transactions that arise from client-driven market-making activity. Principal transactions revenue includes amounts recognized upon executing new transactions with market participants, as well as "inventory-related revenue," which is revenue recognized from gains and losses on derivatives and other instruments that the Firm has been holding in anticipation of, or in response to, client demand, and changes in the fair value of instruments used by the Firm to actively manage the risk exposure arising from such inventory. Principal transactions revenue recognized upon executing new transactions with market participants is affected by many factors including the level of client activity, the bid-offer spread (which is the difference between the price at which a market participant is willing and able to sell an instrument to the Firm and the price at which another market participant is willing and able to buy it from the Firm, and vice versa), market liquidity and volatility. These factors are interrelated and sensitive to the same factors that drive inventory-related revenue, which include general market conditions, such as interest rates, foreign exchange rates, credit spreads, and equity and commodity prices, as well as other macroeconomic conditions.

For the periods presented below, the primary source of principal transactions revenue was the amount recognized upon executing new transactions.

Year ended December 31, (in millions, except where otherwise noted)	2024 Fixed Income Markets	2024 Equity Markets	2024 Total Markets	2023 Fixed Income Markets[c]	2023 Equity Markets[c]	2023 Total Markets	2022 Fixed Income Markets[c]	2022 Equity Markets[c]	2022 Total Markets
Principal transactions	**$ 10,603**	**$ 13,526**	**$ 24,129**	$ 13,198	$ 10,380	$ 23,578	$ 12,244	$ 8,284	$ 20,528
Lending- and deposit-related fees	**391**	**100**	**491**	307	40	347	303	22	325
Commissions and other fees	**605**	**2,086**	**2,691**	596	1,908	2,504	550	1,975	2,525
All other income	**2,120**	**(65)**	**2,055**	1,908	(79)	1,829	1,083	(88)	995
Noninterest revenue	**13,719**	**15,647**	**29,366**	16,009	12,249	28,258	14,180	10,193	24,373
Net interest income[a]	**6,347**	**(5,706)**	**641**	3,171	(3,465)	(294)	4,894	(105)	4,789
Total net revenue	**$ 20,066**	**$ 9,941**	**$ 30,007**	$ 19,180	$ 8,784	$ 27,964	$ 19,074	$ 10,088	$ 29,162
Loss days[b]			**1**			2			7

(a) The decline in Equity Markets net interest income was driven by higher funding costs.
(b) Markets consists of Fixed Income Markets and Equity Markets. Loss days represent the number of days for which Markets recorded losses in total net revenue, which includes revenue related to both trading and non-trading positions. The loss days determined under this measure differ from the measure used to determine backtesting gains and losses. Daily backtesting gains and losses include positions in the Firm's Risk Management value-at-risk ("VaR") measure and exclude certain components of total net revenue, which may more than offset backtesting gains or losses on a particular day. For more information on daily backtesting gains and losses, refer to the VaR discussion on pages 143–145.
(c) In the fourth quarter of 2024, certain net funding costs that were previously allocated to Fixed Income Markets were reclassified to Equity Markets. Prior-period amounts have been revised to conform with the current presentation.

Selected metrics

As of or for the year ended December 31, (in millions, except where otherwise noted)	2024	2023	2022
Assets under custody ("AUC") by asset class (period-end) (in billions):			
Fixed Income	**$ 16,409**	$ 15,543	$ 14,361
Equity	**14,848**	12,927	10,748
Other[a]	**4,023**	3,922	3,526
Total AUC	**$ 35,280**	$ 32,392	$ 28,635
Client deposits and other third-party liabilities (average)[b]	**$ 961,646**	$ 912,859	$ 981,653

(a) Consists of mutual funds, unit investment trusts, currencies, annuities, insurance contracts, options and other contracts.
(b) Client deposits and other third-party liabilities pertain to the Payments and Securities Services businesses.

International metrics

As of or for the year ended December 31, (in millions, except where otherwise noted)	2024	2023	2022
Total net revenue[(a)]			
Europe/Middle East/Africa	**$ 15,191**	$ 14,418	$ 15,716
Asia-Pacific	**8,867**	7,891	8,043
Latin America/Caribbean	**2,427**	2,161	2,288
Total international net revenue	**26,485**	24,470	26,047
North America	**43,629**	39,883	33,588
Total net revenue	**$ 70,114**	$ 64,353	$ 59,635
Loans retained (period-end)[(a)]			
Europe/Middle East/Africa	**$ 44,374**	$ 44,793	$ 40,715
Asia-Pacific	**16,107**	15,506	16,764
Latin America/Caribbean	**10,331**	8,610	8,866
Total international loans	**70,812**	68,909	66,345
North America	**412,231**	406,277	355,176
Total loans retained	**$ 483,043**	$ 475,186	$ 421,521
Client deposits and other third-party liabilities (average)[(b)]			
Europe/Middle East/Africa	**$ 264,227**	$ 247,804	$ 265,061
Asia-Pacific	**141,042**	135,388	136,539
Latin America/Caribbean	**42,716**	39,861	40,531
Total international	**$ 447,985**	$ 423,053	$ 442,131
North America	**513,661**	489,806	539,522
Total client deposits and other third-party liabilities	**$ 961,646**	$ 912,859	$ 981,653
AUC (period-end)[(b)] **(in billions)**			
North America	**$ 23,845**	$ 21,792	$ 19,219
All other regions	**11,435**	10,600	9,416
Total AUC	**$ 35,280**	$ 32,392	$ 28,635

(a) Total net revenue and loans retained (excluding loans held-for-sale and loans at fair value) are based on the location of the trading desk, booking location, or domicile of the client, as applicable.
(b) Client deposits and other third-party liabilities pertaining to the Payments and Securities Services businesses, and AUC, are based on the domicile of the client or booking location, as applicable.

Asset & Wealth Management, with client assets of $5.9 trillion, is a global leader in investment and wealth management.

Asset Management
Offers multi-asset investment management solutions across equities, fixed income, alternatives and money market funds to institutional and retail investors providing for a broad range of clients' investment needs.

Global Private Bank
Provides retirement products and services, brokerage, custody, estate planning, lending, deposits and investment management to high net worth clients.

The majority of AWM's client assets are in actively managed portfolios.

Selected income statement data

Year ended December 31, (in millions, except ratios)	2024	2023	2022
Revenue			
Asset management fees	**$13,693**	$11,826	$11,510
Commissions and other fees	**874**	697	$ 662
All other income	**456**	1,037 (a)	335 (a)(b)
Noninterest revenue	**15,023**	13,560	12,507
Net interest income	**6,555**	6,267	5,241
Total net revenue	**21,578**	19,827	17,748
Provision for credit losses	**(68)**	159	128
Noninterest expense			
Compensation expense	**7,984**	7,115	6,336
Noncompensation expense	**6,430**	5,665	5,493
Total noninterest expense	**14,414**	12,780	11,829
Income before income tax expense	**7,232**	6,888	5,791
Income tax expense	**1,811**	1,661	1,426
Net income	**$ 5,421**	$ 5,227	$4,365
Revenue by line of business			
Asset Management	**$10,175**	$ 9,129	$ 8,818
Global Private Bank	**11,403**	10,698	8,930
Total net revenue	**$21,578**	$19,827	$17,748
Financial ratios			
Return on equity	**34 %**	31 %	25 %
Overhead ratio	**67**	64	67
Pre-tax margin ratio:			
Asset Management	**31**	31	30
Global Private Bank	**35**	38	35
Asset & Wealth Management	**34**	35	33

(a) Includes the amortization of the fair value discount on certain acquired lending-related commitments associated with First Republic. The discount is deferred in other liabilities and recognized on a straight-line basis over the commitment period and was largely recognized in 2023 as the commitments were generally short term. Refer to Note 34 for additional information.
(b) Includes the gain on the original minority interest in CIFM upon the Firm's acquisition of the remaining 51% interest in the entity.

2024 compared with 2023

Net income was $5.4 billion, up 4%.

Net revenue was $21.6 billion, up 9%. Net interest income was $6.6 billion, up 5%. Noninterest revenue was $15.0 billion, up 11%.

Revenue from Asset Management was $10.2 billion, up 11%, driven by:

- higher asset management fees, reflecting higher average market levels and strong net inflows, as well as
- higher performance fees.

The prior year included a gain of $339 million on the original minority interest in CIFM upon the Firm's acquisition of the remaining 51% interest in the entity.

Revenue from Global Private Bank was $11.4 billion, up 7%, driven by:

- higher noninterest revenue, reflecting:
  - higher management fees on strong net inflows and higher average market levels, as well as higher brokerage fees,

  partially offset by
  - a decline in the amortization of the fair value discount on certain acquired lending-related commitments associated with First Republic that have expired, and
- higher net interest income driven by:
  - higher average deposits associated with First Republic, which were transferred to AWM from CCB in the fourth quarter of 2023, as well as wider spreads on loans and higher average loans,

  largely offset by
  - deposit margin compression reflecting higher rates paid.

The prior year included net investment valuation losses.

Noninterest expense was $14.4 billion, up 13%, predominantly driven by:

- higher compensation, including revenue-related compensation, and continued growth in private banking advisor teams, and
- higher distribution fees and legal expense,

The provision for credit losses was a net benefit of $68 million.

The provision in the prior year was $159 million, reflecting a $146 million addition to the allowance for credit losses to establish the allowance for the First Republic loans and lending-related commitments in the second quarter of 2023.

Asset Management has two high-level measures of its overall fund performance.

- **Percentage of active mutual fund and active ETF assets under management in funds rated 4- or 5-star:** Mutual fund rating services rank funds based on their risk adjusted performance over various periods. A 5-star rating is the best rating and represents the top 10% of industry-wide ranked funds. A 4-star rating represents the next 22.5% of industry-wide ranked funds. A 3-star rating represents the next 35% of industry-wide ranked funds. A 2-star rating represents the next 22.5% of industry-wide ranked funds. A 1-star rating is the worst rating and represents the bottom 10% of industry-wide ranked funds. An overall Morningstar rating is derived from a weighted average of the performance associated with a fund's three-, five and ten-year (if applicable) Morningstar Rating metrics. For U.S.-domiciled funds, separate star ratings are provided at the individual share class level. The Nomura "star rating" is based on three-year risk-adjusted performance only. Funds with fewer than three years of history are not rated and hence excluded from these rankings. All ratings, the assigned peer categories and the asset values used to derive these rankings are sourced from the applicable fund rating provider. Where applicable, the fund rating providers redenominate asset values into U.S. dollars. The percentage of AUM is based on star ratings at the share class level for U.S.-domiciled funds, and at a "primary share class" level to represent the star rating of all other funds, except for Japan, for which Nomura provides ratings at the fund level. The performance data may have been different if all share classes had been included. Past performance is not indicative of future results.
- **Percentage of active mutual fund and active ETF assets under management in funds ranked in the 1st or 2nd quartile (one, three and five years)**:All quartile rankings, the assigned peer categories and the asset values used to derive these rankings are sourced from the fund rating providers. Quartile rankings are based on the net-of-fee absolute return of each fund. Where applicable, the fund rating providers redenominate asset values into U.S. dollars. The percentage of AUM is based on fund performance and associated peer rankings at the share class level for U.S.-domiciled funds, at a "primary share class" level to represent the quartile ranking for U.K., Luxembourg and Hong Kong SAR funds and at the fund level for all other funds. The performance data may have been different if all share classes had been included. Past performance is not indicative of future results.

"**Primary share class**" means the C share class for European funds and Acc share class for Hong Kong SAR and Taiwan funds. If these share classes are not available, the oldest share class is used as the primary share class.

Selected metrics

As of or for the year ended December 31, (in millions, except ranking data, ratios and employees)	**2024**	2023	2022
% of JPM mutual fund assets and ETFs rated as 4- or 5-star[(a)]	**69 %**	69 %	73 %
% of JPM mutual fund assets and ETFs ranked in 1st or 2nd quartile:[(b)]			
1 year	**73**	40	68
3 years	**75**	67	76
5 years	**77**	71	81
Selected balance sheet data (period-end)[(c)]			
Total assets	**$255,385**	$245,512	$232,037
Loans	**236,303**	227,929	214,006
Deposits	**248,287**	233,232 [(d)]	233,130
Equity	**15,500**	17,000	17,000
Selected balance sheet data (average)[(c)]			
Total assets	**$246,254**	$240,222	$232,438
Loans	**227,676**	220,487	215,582
Deposits	**235,146**	216,178 [(d)]	261,489
Equity	**15,500**	16,671	17,000
Employees	**29,403**	28,485	26,041
Number of Global Private Bank client advisors	**3,775**	3,515	3,137
Credit data and quality statistics[(c)]			
Net charge-offs/(recoveries)	**$ 21**	$ 13	$ (7)
Nonaccrual loans	**700**	650	459
Allowance for credit losses:			
Allowance for loan losses	**$ 539**	$ 633	$ 494
Allowance for lending-related commitments	**35**	28	20
Total allowance for credit losses	**$ 574**	$ 661	$ 514
Net charge-off/(recovery) rate	**0.01 %**	0.01 %	— %
Allowance for loan losses to period-end loans	**0.23**	0.28	0.23
Allowance for loan losses to nonaccrual loans	**77**	97	108
Nonaccrual loans to period-end loans	**0.30**	0.29	0.21

(a) Represents the Morningstar Rating for all domiciled funds except for Japan domiciled funds which use Nomura. Includes only Asset Management retail active open-ended mutual funds and active ETFs that have a rating. Excludes money market funds, Undiscovered Managers Fund, and Brazil domiciled funds.

(b) Quartile ranking sourced from Morningstar, Lipper and Nomura based on country of domicile. Includes only Asset Management retail active open-ended mutual funds and active ETFs that are ranked by the aforementioned sources. Excludes money market funds, Undiscovered Managers Fund, and Brazil domiciled funds.
(c) Loans, deposits and related credit data and quality statistics relate to the Global Private Bank business.
(d) In the fourth quarter of 2023, certain deposits associated with First Republic were transferred to AWM from CCB.

Client assets

2024 compared with 2023

Assets under management were $4.0 trillion and client assets were $5.9 trillion, each up 18%, driven by continued net inflows and higher market levels.

Client assets

December 31, (in billions)	2024	2023	2022
Assets by asset class			
Liquidity	**$ 1,083**	$ 926	$ 654
Fixed income	**851**	751	638
Equity	**1,128**	868	670
Multi-asset	**764**	680	603
Alternatives	**219**	197	201
Total assets under management	**4,045**	3,422	2,766
Custody/brokerage/ administration/deposits	**1,887**	1,590	1,282
Total client assets[(a)]	**$ 5,932**	$ 5,012	$ 4,048
Assets by client segment			
Private Banking	**$ 1,234**	$ 974	$ 751
Global Institutional	**1,692**	1,488	1,252
Global Funds	**1,119**	960	763
Total assets under management	**$ 4,045**	$ 3,422	$ 2,766
Private Banking	**$ 2,974**	$ 2,452	$ 1,964
Global Institutional	**1,820**	1,594	1,314
Global Funds	**1,138**	966	770
Total client assets[(a)]	**$ 5,932**	$ 5,012	$ 4,048

(a) Includes CCB client investment assets invested in managed accounts and J.P. Morgan mutual funds where AWM is the investment manager.

Client assets (continued)

Year ended December 31, (in billions)	2024	2023	2022
Assets under management rollforward			
Beginning balance	**$ 3,422**	$ 2,766	$ 3,113
Net asset flows:			
Liquidity	**140**	242	(55)
Fixed income	**91**	70	13
Equity	**114**	70	35
Multi-asset	**19**	1	(9)
Alternatives	**10**	(1)	8
Market/performance/other impacts	**249**	274	(339)
Ending balance, December 31	**$ 4,045**	$ 3,422	$ 2,766
Client assets rollforward			
Beginning balance	**$ 5,012**	$ 4,048	$ 4,295
Net asset flows	**486**	490	49
Market/performance/other impacts	**434**	474	(296)
Ending balance, December 31	**$ 5,932**	$ 5,012	$ 4,048

Selected Metrics

	As of December 31,		
	2024	2023	Change
Firmwide Wealth Management			
Client assets (in billions)[(a)]	**$ 3,756**	$ 3,177	18 %
Number of client advisors	**9,530**	8,971	6
Stock Plan Administration[(b)]			
Number of stock plan participants (in thousands)	**1,327**	974	36
Client assets (in billions)	**$ 270**	$ 230	17 %

(a) Consists of Global Private Bank in AWM and client investment assets in J.P. Morgan Wealth Management in CCB.
(b) Relates to an equity plan administration business which was acquired in 2022 with the Firm's purchase of Global Shares. The increase in 2024 includes the impact of onboarding participants in the Firm's employee stock plans during the fourth quarter of 2024.

International metrics

Year ended December 31, (in billions, except where otherwise noted)	2024	2023	2022
Total net revenue (in millions)[(a)]			
Europe/Middle East/Africa	**$ 3,563**	$ 3,377	$ 3,240
Asia-Pacific	**2,023**	1,876	1,836
Latin America/Caribbean	**1,065**	985	967
Total international net revenue	**6,651**	6,238	6,043
North America	**14,927**	13,589	11,705
Total net revenue	**$ 21,578**	$ 19,827	$ 17,748
Assets under management			
Europe/Middle East/Africa	**$ 604**	$ 539	$ 487
Asia-Pacific	**302**	263	218
Latin America/Caribbean	**106**	86	69
Total international assets under management	**1,012**	888	774
North America	**3,033**	2,534	1,992
Total assets under management	**$ 4,045**	$ 3,422	$ 2,766
Client assets			
Europe/Middle East/Africa	**$ 841**	$ 740	$ 610
Asia-Pacific	**482**	406	331
Latin America/Caribbean	**254**	232	189
Total international client assets	**1,577**	1,378	1,130
North America	**4,355**	3,634	2,918
Total client assets	**$ 5,932**	$ 5,012	$ 4,048

(a) Regional revenue is based on the domicile of the client.

Corporate consists of Treasury and Chief Investment Office ("CIO") and Other Corporate. Treasury and CIO is predominantly responsible for measuring, monitoring, reporting and managing the Firm's liquidity, funding, capital, structural interest rate and foreign exchange risks.

Other Corporate includes staff functions and expense that is centrally managed as well as certain Firm initiatives and activities not solely aligned to a specific LOB. The major Other Corporate functions include Real Estate, Technology, Legal, Corporate Finance, Human Resources, Internal Audit, Risk Management, Compliance, Control Management, Corporate Responsibility and various Other Corporate groups.

Selected income statement and balance sheet data

Year ended December 31, (in millions, except employees)	2024		2023		2022
Revenue					
Principal transactions	**$ 152**		$ 302		$ (227)
Investment securities losses	**(1,020)**		(3,180)		(2,380)
All other income	**8,476**	(c)	3,010	(f)	809
Noninterest revenue	**7,608**		132		(1,798)
Net interest income	**9,786**		7,906		1,878
Total net revenue[(a)]	**17,394**		8,038		80
Provision for credit losses	**10**		171		22
Noninterest expense	**3,994**	(d)(e)	5,601	(e)(g)	1,034
Income/(loss) before income tax expense/(benefit)	**13,390**		2,266		(976)
Income tax expense/(benefit)	**2,789**		(555)	(h)	(233)
Net income/(loss)	**$ 10,601**		$ 2,821		$ (743)
Total net revenue					
Treasury and CIO	**9,638**		6,072		(439)
Other Corporate	**7,756**		1,966		519
Total net revenue	**$ 17,394**		$ 8,038		$ 80
Net income/(loss)					
Treasury and CIO	**7,013**		4,206		(197)
Other Corporate	**3,588**	(e)	(1,385)	(e)	(546)
Total net income/(loss)	**$ 10,601**		$ 2,821		$ (743)
Total assets (period-end)	**$1,323,967**		$1,348,437		$1,328,219
Loans (period-end)	**1,964**		1,924		2,181
Deposits[(b)]	**27,581**		21,826		14,203
Employees	**49,610**		47,530		44,196

(a) Included tax-equivalent adjustments, predominantly driven by tax-exempt income from municipal bonds, of $182 million, $211 million and $235 million for the years ended December 31, 2024, 2023 and 2022, respectively.
(b) Predominantly relates to the Firm's international consumer initiatives.
(c) Included the net gain related to Visa shares of $7.9 billion recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.
(d) Included a $1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.
(e) The first quarter of 2024 included an increase of $725 million to the FDIC special assessment reflecting the FDIC's revised estimate of Deposit Insurance Fund losses. The fourth quarter of 2023 included the $2.9 billion FDIC special assessment.
(f) Included the estimated bargain purchase gain of $2.8 billion for the year ended December 31, 2023 associated with the First Republic acquisition. Refer to Notes 6 and 34 for additional information.
(g) In the second quarter of 2023, substantially all of the expense associated with First Republic was reported in Corporate. Commencing in the third quarter of 2023, the expense was aligned to the appropriate LOBs.
(h) Income taxes associated with the First Republic acquisition were reflected in the estimated bargain purchase gain.

2024 compared with 2023

Net income was $10.6 billion, compared with $2.8 billion in the prior year.

Net revenue was $17.4 billion, compared with $8.0 billion in the prior year.

Net interest income was $9.8 billion, up 24%, driven by the impact of balance sheet actions, primarily reinvestments in the investment securities portfolio, partially offset by the net impact of rates.

Noninterest revenue was $7.6 billion, compared with $132 million in the prior year. Excluding the $7.9 billion net gain related to Visa shares recorded in the second quarter of 2024 and the prior-year $2.8 billion estimated bargain purchase gain associated with the First Republic acquisition, revenue was up $2.4 billion, predominantly driven by lower investment securities losses, primarily on sales of U.S. Treasuries and U.S. GSE and government agency MBS, associated with repositioning the investment securities portfolio in Treasury and CIO.

Noninterest expense was $4.0 billion, down 29%, driven by:

- a lower FDIC special assessment,
- lower expense associated with the First Republic acquisition as the prior year expense in Corporate included individuals associated with First Republic who were not employees of the Firm until July 2023, and this expense was subsequently aligned to the appropriate LOBs starting in the third quarter of 2023, and
- lower legal expense,

partially offset by

- a $1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024, and
- higher costs associated with the Firm's international consumer initiatives.

The provision for credit losses was $10 million.

The provision in the prior year was $171 million, reflecting a net addition to the allowance for credit losses related to a single name exposure, which was subsequently charged off upon the restructuring of a loan.

Refer to Note 10 and Note 13 for additional information on the investment securities portfolio and the allowance for credit losses.

The current period income tax expense was driven by changes in the level and mix of income and expenses subject to U.S. federal and state and local taxes, including the impact of the net gain on Visa shares and the contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024, partially offset by benefits related to tax audit settlements.

Other Corporate includes the Firm's international consumer initiatives, which primarily consists of Chase U.K., Nutmeg, and an ownership stake in C6 Bank.

The deposits within Corporate relate to the Firm’s international consumer initiatives and have increased as a result of growth in client accounts, reflecting the impact of additional product offerings.

Treasury and CIO overview

Treasury and CIO is predominantly responsible for measuring, monitoring, reporting and managing the Firm's liquidity, funding, capital, structural interest rate and foreign exchange risks. The risks managed by Treasury and CIO arise from the activities undertaken by the Firm's three reportable business segments to serve their respective client bases, which generate both on- and off-balance sheet assets and liabilities.

Treasury and CIO seeks to achieve the Firm's asset-liability management objectives generally by investing in high quality securities that are managed for the longer-term as part of the Firm's investment securities portfolio. Treasury and CIO also uses derivatives to meet the Firm's asset-liability management objectives. Refer to Note 5 for further information on derivatives. In addition, Treasury and CIO manages the Firm's cash position primarily through deposits at central banks and investments in short-term instruments. Refer to Liquidity Risk Management on pages 108–115 for further information on liquidity and funding risk. Refer to Market Risk Management on pages 141–149 for information on interest rate and foreign exchange risks.

The investment securities portfolio predominantly consists of U.S. and non-U.S. government securities, U.S. GSE and government agency and nonagency mortgage-backed securities, collateralized loan obligations, obligations of U.S. states and municipalities and other ABS. At December 31, 2024, the Treasury and CIO investment securities portfolio, net of the allowance for credit losses, was $678.3 billion, and the average credit rating of the securities comprising the portfolio was AA+ (based upon external ratings where available and, where not available, based primarily upon internal risk ratings). Refer to Note 10 for further information on the Firm's investment securities portfolio and internal risk ratings.

Selected income statement and balance sheet data

As of or for the year ended December 31, (in millions)	**2024**	2023	2022
Investment securities losses	**$ (1,020)**	$ (3,180)	$ (2,380)
Available-for-sale securities (average)	**$287,260**	$200,708	$239,924
Held-to-maturity securities (average)[(a)]	**321,384**	402,010	412,180
Investment securities portfolio (average)	**$608,644**	$ 602,718	$ 652,104
Available-for-sale securities (period-end)	**$403,796**	$ 199,354	$ 203,981
Held-to-maturity securities (period–end)[(a)]	**274,468**	369,848	425,305
Investment securities portfolio, net of allowance for credit losses (period–end)[(b)]	**$678,264**	$569,202	$629,286

(a) Effective January 1, 2023, the Firm adopted the portfolio layer method hedge accounting guidance. As permitted by the guidance, the Firm elected to transfer $7.1 billion of investment securities from HTM to AFS. During 2022, the Firm transferred $78.3 billion of investment securities from AFS to HTM for capital management purposes. Refer to Note 1 and Note 10 for additional information on the portfolio layer method hedge accounting guidance.

(b) As of December 31, 2024, 2023 and 2022, the allowance for credit losses on investment securities was $105 million, $94 million and $67 million, respectively.

Risk is an inherent part of JPMorganChase's business activities. When the Firm extends a consumer or wholesale loan, advises customers and clients on their investment decisions, makes markets in securities, or offers other products or services, the Firm takes on some degree of risk. The Firm's overall objective is to manage its business, and the associated risks, in a manner that balances serving the interests of its clients, customers and investors, and protecting the safety and soundness of the Firm.

The Firm believes that effective risk management requires, among other things:

- Acceptance of responsibility, including identification and escalation of risks by all individuals within the Firm;
- Ownership of risk identification, assessment, data and management within each of the LOBs and Corporate; and
- A Firmwide risk governance and oversight structure.

The Firm follows a disciplined and balanced compensation framework with strong internal governance and independent oversight by the Board of Directors (the "Board"). The impact of risk and control issues is carefully considered in the Firm's performance evaluation and incentive compensation processes.

Risk governance framework

The Firm's risk governance framework involves understanding drivers of risks, types of risks and impacts of risks.

Drivers of risks are factors that cause a risk to exist. Drivers of risks include the economic environment, regulatory or government policy, competitor or market evolution, business decisions, process or judgment error, deliberate wrongdoing, dysfunctional markets and natural disasters.

Types of risks are categories by which risks manifest themselves. The Firm's risks are generally categorized in the following four risk types:

- Strategic risk is the risk to earnings, capital, liquidity or reputation associated with poorly-designed or failed business plans or an inadequate response to changes in the operating environment.
- Credit and investment risk is the risk associated with the default or change in credit profile of a client, counterparty or customer; or loss of principal or a reduction in expected returns on investments, including consumer credit risk, wholesale credit risk and investment portfolio risk.
- Market risk is the risk associated with the effect of changes in market factors, such as interest and foreign exchange rates, equity and commodity prices, credit spreads or implied volatilities, on the value of assets and liabilities held for both the short and long term.
- Operational risk is the risk of an adverse outcome resulting from inadequate or failed internal processes or systems; human factors; or external events impacting the Firm's processes or systems. Operational risk includes cybersecurity, compliance, conduct, legal, and estimations and model risk.

Impacts of risks are consequences of risks, both quantitative and qualitative. There may be many consequences when risks manifest themselves, including quantitative impacts such as a reduction in earnings and capital, liquidity outflows, and fines or penalties, or qualitative impacts such as damage to the Firm's reputation, loss of clients and customers, and regulatory and enforcement actions.

The Firm's risk governance framework is managed on a Firmwide basis. The Firm has an Independent Risk Management ("IRM") function, which is comprised of Risk Management and Compliance. The Firm's Chief Executive Officer ("CEO") appoints, subject to approval by the Risk Committee of the Board of Directors (the "Board Risk Committee"), the Firm's Chief Risk Officer ("CRO") to lead the IRM function and maintain the risk governance framework of the Firm. The framework is subject to approval by the Board Risk Committee through its review and approval of the Risk Governance and Oversight Policy.

The Firm's CRO oversees and delegates authority to the Firmwide Risk Executives ("FREs"), the Chief Risk Officers of the LOBs and Corporate ("LOB CROs"), and the Firm's Chief Compliance Officer ("CCO"), who, in turn, establish Risk Management and Compliance organizations, develop the Firm's risk governance policies and standards, and define and oversee the implementation of the Firm's risk governance framework. The LOB CROs oversee risks that arise in their LOBs and Corporate, while FREs oversee risks that span across the LOBs and Corporate, as well as functions and regions. Each area of the Firm that gives rise to risk is expected to operate within the parameters identified by the IRM function, and within the risk and control standards established by its own management.

Three lines of defense

The Firm's "three lines of defense" are as follows:

The first line of defense consists of each LOB, Treasury and CIO, and certain Other Corporate initiatives, including their aligned Operations, Technology and Control Management. The first line of defense owns the risks, and identification of risks, associated with their respective activities and the design and execution of controls to manage those risks. Responsibilities also include adherence to applicable laws, rules and regulations and implementation of the risk governance framework established by IRM, which may include policies, standards, limits, thresholds and controls.

The second line of defense is the IRM function, which is separate from the first line of defense and is responsible for independently measuring risk, as well as assessing and challenging the risk management activities of the first line of defense. IRM is also responsible for the identification of risks within its organization, its own adherence to applicable laws, rules and regulations and for the development and implementation of policies and standards with respect to its own processes.

The third line of defense is Internal Audit, an independent function that provides objective assessment of the adequacy and effectiveness of Firmwide processes, controls, governance and risk management. The Internal Audit function is led by the General Auditor, who reports to the Audit Committee and administratively to the CEO.

In addition, there are other functions that contribute to the Firmwide control environment but are not considered part of a particular line of defense, including Corporate Finance, Human Resources and Legal. These other functions are responsible for the identification of risks within their respective organizations, adherence to applicable laws, rules and regulations and implementation of the risk governance framework established by IRM.

Risk identification and ownership

The LOBs and Corporate are responsible for the identification of risks within their respective organizations, as well as the design and execution of controls, including IRM-specified controls, to manage those risks. To support this activity, the Firm has a risk identification framework designed to facilitate the responsibility of each LOB and Corporate to identify material risks inherent to the Firm's businesses and operational activities, catalog them in a central repository and review material risks on a regular basis. The IRM function reviews and challenges the risks identified by each LOB and Corporate, maintains the central repository and provides the consolidated Firmwide results to the Firmwide Risk Committee ("FRC") and the Board Risk Committee.

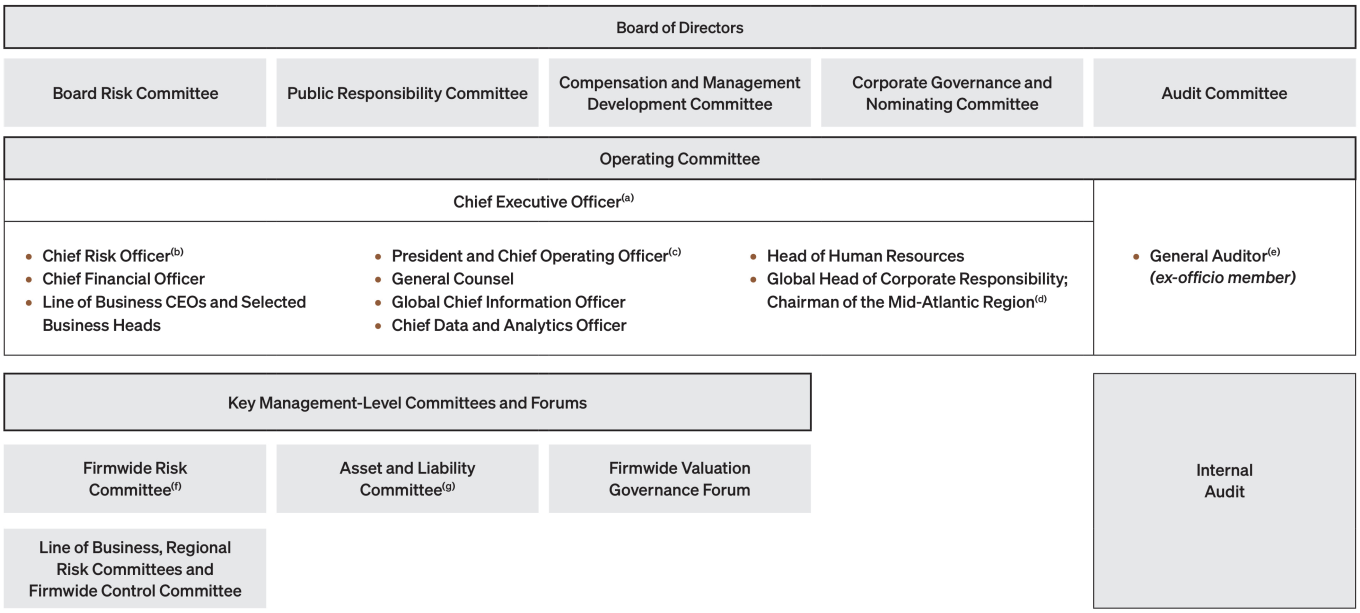
Risk appetite

The Firm's overall appetite for risk is governed by Risk Appetite frameworks for quantitative and qualitative risks. The Firm's risk appetite is periodically set and approved by senior management (including the CEO and CRO) and approved by the Board Risk Committee. Quantitative and qualitative risks are assessed to monitor and measure the Firm's capacity to take risk consistent with its stated risk appetite. Risk appetite results are reported to the Board Risk Committee.

Risk governance and oversight structure

The independent status of the IRM function is supported by a risk governance and oversight structure that provides channels for the escalation of risks and issues to senior management, the FRC and the Board of Directors, as appropriate.

The chart below illustrates the principal standing committees of the Board of Directors and key senior management-level committees in the Firm's risk governance and oversight structure. In addition, there are other committees, forums and channels of escalation that support the oversight of risk that are not shown in the chart below or described in this Form 10-K.



(a) The Firm's CEO is also the Chairman of the Board of Directors.
(b) The Firm's CRO reports to the Firm's CEO and the Board Risk Committee. The Firm's CRO may escalate directly to the Board of Directors (including its committees), as appropriate.
(c) Effective January 14, 2025, the co-CEO of the Commercial & Investment Bank became a Chief Operating Officer of the Firm. Refer to Recent events on page 57 for further information.
(d) Effective February 12, 2024, the Global Head of Corporate Responsibility and Chairman of the Mid-Atlantic Region became a member of the Operating Committee.
(e) The Firm's General Auditor reports to the Audit Committee and administratively to the Firm's CEO.
(f) The Firmwide Risk Committee escalates to the Board Risk Committee, as appropriate.
(g) The Asset and Liability Committee escalates to the Firm's CEO or the Board of Directors (including its committees), as appropriate.

The Firm's Operating Committee, which consists of the Firm's CEO, CRO, Chief Financial Officer ("CFO"), General Counsel, CEOs of the LOBs and other senior executives, is accountable to and may refer matters to the Firm's Board of Directors. The Operating Committee and certain other members of senior management are responsible for escalating to the Board the information necessary to facilitate the Board's exercise of its duties.

Board oversight

The Firm's Board of Directors actively oversees the business and affairs of the Firm. This includes monitoring the Firm's financial performance and condition and reviewing the strategic objectives and plans of the Firm. The Board carries out a significant portion of its oversight responsibilities through its principal standing committees, each of which consists solely of independent members of the Board. The Board Risk Committee is the principal committee that oversees risk matters. The Audit Committee oversees the control environment, and the Compensation & Management Development Committee oversees compensation and other management-related matters. Each committee of the Board oversees reputation risks, conduct risks, and environmental, social and governance ("ESG") matters within its scope of responsibility.

The JPMorgan Chase Bank, N.A. Board of Directors is responsible for the oversight of management of the bank, which it discharges both acting directly and through the principal standing committees of the Firm's Board of Directors. Risk and control oversight on behalf of JPMorgan Chase Bank N.A. is primarily the responsibility of the Board Risk Committee and the Audit Committee, respectively, and, with respect to compensation and other management-related matters, the Compensation & Management Development Committee.

The Board Risk Committee assists the Board in its oversight of management's responsibility to implement a global risk management framework reasonably designed to identify, assess and manage the Firm's risks. The Board Risk Committee's responsibilities include approval of applicable primary risk policies and review of certain associated frameworks, analysis and reporting established by management. Breaches in risk appetite and parameters, issues that may have a material adverse impact on the Firm, including capital and liquidity issues, and other significant risk-related matters are escalated to the Board Risk Committee, as appropriate.

The Audit Committee assists the Board in its oversight of management's responsibilities to ensure that there is an effective system of controls reasonably designed to safeguard the Firm's assets and income, ensure the integrity of the Firm's financial statements, and maintain compliance with the Firm's ethical standards, policies, plans and procedures, and with laws and regulations. It also assists the Board in its oversight of the qualifications, independence and performance of the Firm's independent registered public accounting firm, and of the performance of the Firm's Internal Audit function.

The Compensation & Management Development Committee ("CMDC") assists the Board in its oversight of the Firm's compensation principles and practices. The CMDC reviews and approves the Firm's compensation and qualified benefits programs. The Committee reviews the performance of Operating Committee members against their goals, and approves their compensation awards. In addition, the CEO's compensation award is subject to ratification by the independent directors of the Board. The CMDC also reviews the development of and succession for key executives. As part of the Board's role of reinforcing, demonstrating and communicating the "tone at the top," the CMDC oversees the Firm's culture, including reviewing updates from management regarding significant conduct issues and any related actions with respect to employees, including compensation actions.

The Public Responsibility Committee oversees and reviews the Firm's positions and practices on public responsibility matters such as community investment, fair lending, sustainability, consumer practices and other public policy issues that reflect the Firm's values and character and could impact the Firm's reputation among its stakeholders. The Committee also provides guidance on these matters to management and the Board, as appropriate.

The Corporate Governance & Nominating Committee exercises general oversight with respect to the governance of the Board of Directors. It reviews the qualifications of and recommends to the Board proposed nominees for election to the Board. The Committee evaluates and recommends to the Board corporate governance practices applicable to the Firm. It also reviews the framework for assessing the Board's performance and self-evaluation.

Management oversight

The Firm's senior management-level committees that are primarily responsible for key risk-related functions include:

The Firmwide Risk Committee ("FRC") is the Firm's highest management-level risk committee. It oversees the risks inherent in the Firm's business and provides a forum for discussion of risk-related and other topics and issues that are raised or escalated by its members and other committees.

The Firmwide Control Committee ("FCC") is an escalation committee for senior management to review and discuss the Firmwide compliance and operational risk environment, including identified issues, compliance and operational risk metrics and significant events that have been escalated.

Line of Business and Regional Risk Committees are responsible for overseeing the governance, limits and controls that have been established within the scope of their respective activities. These committees review the ways in which the particular LOB or the businesses operating in a particular region could be exposed to adverse outcomes, with a focus on identifying, accepting, escalating and/or requiring remediation of matters brought to these committees.

Line of Business and Corporate Function Control Committees oversee the risk and control environment of their respective business or function, inclusive of Operational Risk, Compliance and Conduct Risks. As part of that mandate, they are responsible for reviewing indicators of elevated or emerging risks and other data that may impact the level of compliance and operational risk in a business or function, addressing key compliance and operational risk issues, with an emphasis on processes with control concerns, and overseeing control remediation.

The Asset and Liability Committee ("ALCO") is responsible for overseeing the Firm's asset and liability management ("ALM"), including the activities and frameworks supporting management of the balance sheet, liquidity risk, interest rate risk and capital risk.

The Firmwide Valuation Governance Forum ("VGF") is composed of senior finance and risk executives and is responsible for overseeing the management of risks arising from valuation activities conducted across the Firm.

Risk governance and oversight functions

The Firm monitors and measures its risk through risk governance and oversight functions. The scope of a particular function or business activity may include one or more drivers, types and/or impacts of risk. For example, Country Risk Management oversees country risk which may be a driver of risk or an aggregation of exposures that could give rise to multiple risk types such as credit or market risk.

The following sections discuss the risk governance and oversight functions that have been established to oversee the risks inherent in the Firm's business activities.

Risk governance and oversight functions	Page
Strategic Risk	96
Capital Risk	97–107
Liquidity Risk	108-115
Reputation Risk	116
Consumer Credit Risk	120-125
Wholesale Credit Risk	126-136
Investment Portfolio Risk	140
Market Risk	141-149
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Climate Risk	152
Operational Risk	153-156
Compliance Risk	157
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STRATEGIC RISK MANAGEMENT

Strategic risk is the risk to earnings, capital, liquidity or reputation associated with poorly-designed or failed business plans or an inadequate response to changes in the operating environment.

Management and oversight

The Operating Committee, together with the senior leadership of each LOB and Corporate, are responsible for managing the Firm's most significant strategic risks. IRM engages regularly in strategic business discussions and decision-making, including participation in relevant business reviews and senior management meetings, risk and control committees and other relevant governance forums, and review of acquisitions and new business initiatives. The Board of Directors oversees management's strategic decisions, and the Board Risk Committee oversees IRM and the Firm's risk governance framework.

In the process of developing business plans and strategic initiatives, LOB and Corporate senior management identify the associated risks that are incorporated into the Firmwide Risk Identification framework and their impact on risk appetite.

In addition, IRM conducts a qualitative assessment of the LOB and Corporate strategic initiatives to assess their impact on the risk profile of the Firm.

The Firm's strategic planning process, which includes the development of the Firm's strategic plan and other strategic initiatives, is one component of managing the Firm's strategic risk. The strategic plan outlines the Firm's strategic framework and initiatives, and includes components such as budget, risk appetite, capital, earnings and asset-liability management objectives. Guided by the Firm's Business Principles, the Operating Committee and senior management teams in each LOB and Corporate review and update the strategic plan periodically, including evaluating the strategic framework and performance of strategic initiatives, assessing the operating environment, refining existing strategies and developing new strategies.

The Firm's strategic plan, together with IRM's assessment, are provided to the Board as part of its review and approval of the Firm's strategic plan, and the plan is also reflected in the Firm's budget.

The Firm's balance sheet strategy, which focuses on risk-adjusted returns, strong capital and robust liquidity, is also a component in the management of strategic risk. Refer to Capital Risk Management on pages 97–107 for further information on capital risk. Refer to Liquidity Risk Management on pages 108–115 for further information on liquidity risk. Refer to Reputation Risk Management on page 116 for further information on reputation risk.

Capital risk is the risk that the Firm has an insufficient level or composition of capital to support the Firm's business activities and associated risks during normal economic environments and under stressed conditions.

A strong capital position is essential to the Firm's business strategy and competitive position. Maintaining a strong balance sheet to manage through economic volatility is a strategic imperative of the Firm's Board of Directors, CEO and Operating Committee. The Firm's "fortress balance sheet" philosophy focuses on risk-adjusted returns, strong capital and robust liquidity. The Firm's capital risk management strategy focuses on maintaining long-term stability to enable the Firm to build and invest in market-leading businesses, including in highly stressed environments. Senior management considers the implications on the Firm's capital prior to making significant decisions that could impact future business activities. In addition to considering the Firm's earnings outlook, senior management evaluates all sources and uses of capital with a view to ensuring the Firm's capital strength.

Capital risk management

The Firm has a Capital Risk Management function whose primary objective is to provide independent oversight of capital risk across the Firm.

Capital Risk Management's responsibilities include:

- Defining, monitoring and reporting capital risk metrics;
- Establishing, calibrating and monitoring capital risk limits and indicators, including capital risk appetite;
- Developing processes to classify, monitor and report capital limit breaches;
- Performing assessments of the Firm's capital management activities, including changes made to the Contingency Capital Plan described below; and
- Conducting assessments of the Firm's regulatory capital framework intended to ensure compliance with applicable regulatory capital rules.

Capital management

Treasury and CIO is responsible for capital management.

The primary objectives of the Firm's capital management are to:

- Maintain sufficient capital in order to continue to build and invest in the Firm's businesses through normal economic cycles and in stressed environments;
- Retain flexibility to take advantage of future investment opportunities;
- Promote the Parent Company's ability to serve as a source of strength to its subsidiaries;
- Ensure the Firm operates above the minimum regulatory capital ratios as well as maintain "well-capitalized" status for the Firm and its principal insured depository institution ("IDI") subsidiary, JPMorgan Chase Bank, N.A., at all times under applicable regulatory capital requirements;
- Meet capital distribution objectives; and
- Maintain sufficient capital resources to operate throughout a resolution period in accordance with the Firm's preferred resolution strategy.

The Firm addresses these objectives through:

- Establishing internal minimum capital requirements and maintaining a strong capital governance framework. The internal minimum capital levels consider the Firm's regulatory capital requirements as well as an internal assessment of capital adequacy, in normal economic cycles and in stress events;
- Retaining flexibility in order to react to a range of potential events; and
- Regularly monitoring the Firm's capital position and following prescribed escalation protocols, both at the Firm and material legal entity levels.

Governance

Committees responsible for overseeing the Firm's capital management include the Capital Governance Committee, the Firmwide ALCO as well as regional ALCOs, and the CIO, Treasury and Corporate ("CTC") Risk Committee. In addition, the Board Risk Committee periodically reviews the Firm's capital risk tolerance. Refer to Firmwide Risk Management on pages 91–95 for additional discussion of the Firmwide ALCO and other risk-related committees.

Capital planning and stress testing

Comprehensive Capital Analysis and Review

The Federal Reserve requires the Firm, as a large Bank Holding Company ("BHC"), to submit at least annually a capital plan that has been reviewed and approved by the Board of Directors. The Federal Reserve uses Comprehensive Capital Analysis and Review ("CCAR") and other stress testing processes to assess whether large BHCs, such as the Firm, have sufficient capital during periods of economic and financial stress, and have robust, forward-looking capital assessment and planning processes in place that address each BHC's unique risks to enable it to absorb losses under certain stress scenarios. Through CCAR, the Federal Reserve evaluates each BHC's capital adequacy and internal capital adequacy assessment processes ("ICAAP"), as well as its plans to make capital distributions, such as dividend payments or stock repurchases. The Federal

Reserve uses results under the severely adverse scenario from its supervisory stress test to determine each firm's Stress Capital Buffer ("SCB") requirement for the coming year.

The Firm's current SCB requirement is 3.3% and will remain in effect until September 30, 2025. The Firm's Standardized CET1 capital ratio requirement, including regulatory buffers, was 12.3% as of December 31, 2024.

Refer to Capital actions on page 105 for information on actions taken by the Firm's Board of Directors.

Internal Capital Adequacy Assessment Process
Annually, the Firm prepares the ICAAP, which informs the Board of Directors of the ongoing assessment of the Firm's processes for managing the sources and uses of capital as well as compliance with supervisory expectations for capital planning and capital adequacy. The Firm's ICAAP integrates stress testing protocols with capital planning. The Firm's Audit Committee is responsible for reviewing and approving the capital planning framework.

Stress testing assesses the potential impact of alternative economic and business scenarios on the Firm's earnings and capital. Economic scenarios, and the parameters underlying those scenarios, are defined centrally and applied uniformly across the businesses. These scenarios are articulated in terms of macroeconomic factors, which are key drivers of business results; global market shocks, which generate short-term but severe trading losses; and idiosyncratic operational risk events. The scenarios are intended to capture and stress key vulnerabilities and idiosyncratic risks facing the Firm. In addition to CCAR and other periodic stress testing, management also considers tailored stress scenarios and sensitivity analyses, as necessary.

Contingency Capital Plan

The Firm's Contingency Capital Plan establishes the capital management framework for the Firm and specifies the principles underlying the Firm's approach towards capital management in normal economic conditions and in stressed environments. The Contingency Capital Plan defines how the Firm calibrates its targeted capital levels and meets minimum capital requirements, monitors the ongoing appropriateness of planned capital distributions, and sets out the capital contingency actions that are expected to be taken or considered at various levels of capital depletion during a period of stress.

Regulatory capital

The Federal Reserve establishes capital requirements, including well-capitalized standards, for the Firm as a consolidated financial holding company. The Office of the Comptroller of the Currency ("OCC") establishes similar minimum capital requirements and standards for the Firm's principal IDI subsidiary, JPMorgan Chase Bank, N.A. The U.S. capital requirements generally follow the Capital Accord of the Basel Committee, as amended from time to time.

Basel III Overview
The capital rules under Basel III establish minimum capital ratios and overall capital adequacy standards for large and internationally active U.S. BHCs and banks, including the Firm and JPMorgan Chase Bank, N.A. The minimum amount of regulatory capital that must be held by BHCs and banks is determined by calculating RWA, which are on-balance sheet assets and off-balance sheet exposures, weighted according to risk. Under the rules currently in effect, two comprehensive approaches are prescribed for calculating RWA: a standardized approach ("Basel III Standardized"), and an advanced approach ("Basel III Advanced").

For each of these risk-based capital ratios, the capital adequacy of the Firm is evaluated against the lower of the Standardized or Advanced approaches compared to their respective regulatory capital ratio requirements.

The current Basel III rules establish capital requirements for calculating credit risk RWA and market risk RWA, and in the case of Basel III Advanced, operational risk RWA. Key differences in the calculation of credit risk RWA between the Standardized and Advanced approaches are that for Basel III Advanced, credit risk RWA is based on risk-sensitive approaches which largely rely on the use of internal credit models and parameters, whereas for Basel III Standardized, credit risk RWA is generally based on supervisory risk-weightings which vary primarily by counterparty type and asset class. Market risk RWA is generally calculated consistently between Basel III Standardized and Basel III Advanced. In addition to the RWA calculated under these approaches, the Firm may supplement such amounts to incorporate management judgment and feedback from its regulators.

As of December 31, 2024, the Firm's Basel III Standardized ratios risk-based ratios were more binding than the Basel III Advanced risk-based ratios.

Additionally, Basel III requires that Advanced Approaches banking organizations, including the Firm, calculate their SLRs.

Refer to page 104 for additional information on SLR.

Key Regulatory Developments

U.S. Basel III Finalization

In July 2023, the Federal Reserve, the OCC and the FDIC released a proposal to amend the risk-based capital framework, entitled "Regulatory capital rule: Amendments applicable to large banking organizations and to banking organizations with significant trading activity", which is referred to in this Form 10-K as the "U.S. Basel III proposal". Under this proposal, changes to the framework would include replacement of the Advanced approach with an expanded risk-based approach for the calculation of RWA. In addition, the stress capital buffer requirement would be applicable to both the expanded risk-based approach and the Standardized approach.

GSIB Surcharge and TLAC and Eligible LTD Requirements

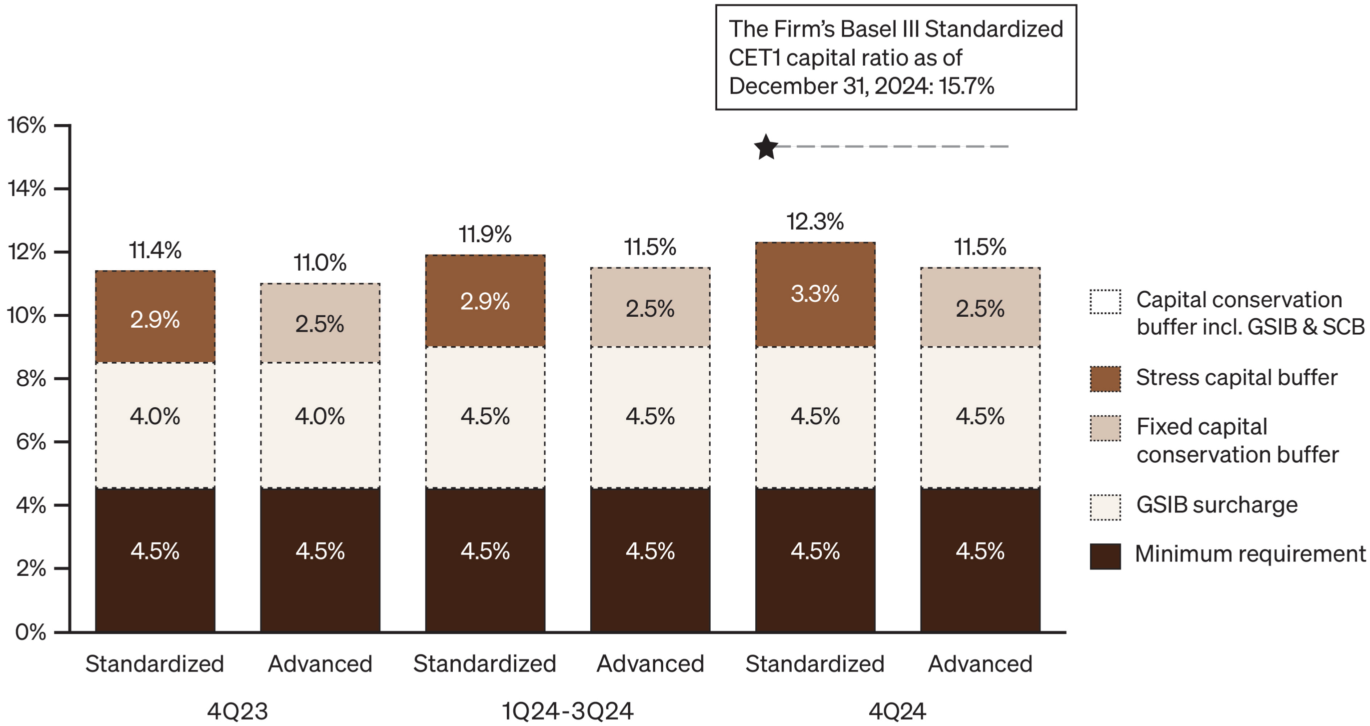
In July 2023, the Federal Reserve released a proposal to amend the calculation of the GSIB surcharge. Under the proposal, the annual GSIB surcharge would be based on an average of the quarterly surcharge calculations throughout the calendar year, with daily averaging required for certain measures. The proposal would also reduce surcharge increments from 50 bps to 10 bps and includes other technical amendments to the “Method 2” calculation. The proposed changes would revise risk-based capital requirements for the Firm and other U.S. GSIBs. Refer to Risk-based Capital Regulatory Requirements on page 100 for further information on the GSIB surcharge.

Additionally, in August 2023, the Federal Reserve, the FDIC and the OCC released a proposal to expand the eligible long-term debt ("eligible LTD") and clean holding company requirements under the existing total loss-absorbing capacity ("TLAC") rule to apply to non-GSIB banks with $100 billion or more in total consolidated assets. The proposal would also reduce the amount of LTD with remaining maturities of less than two years that count towards a U.S. GSIB's TLAC requirement and expand the existing capital deduction framework for LTD issued by GSIBs to include LTD issued by non-GSIB banks subject to the LTD requirements.

Finalization of the above proposals, including the required implementation dates, is uncertain. The Firm continues to monitor developments and potential impacts.

Risk-based Capital Regulatory Requirements

The following chart presents the Firm's Basel III CET1 capital ratio requirements under the Basel III rules currently in effect.



All banking institutions are currently required to have a minimum CET1 capital ratio of 4.5% of risk-weighted assets.

Certain banking organizations, including the Firm, are required to hold additional levels of capital to serve as a "capital conservation buffer". The capital conservation buffer incorporates a GSIB surcharge, a discretionary countercyclical capital buffer and a fixed capital conservation buffer of 2.5% for Advanced regulatory capital requirements, as well as a variable SCB requirement, floored at 2.5%, for Standardized regulatory capital requirements.

Under the Federal Reserve's GSIB rule, the Firm is required to assess its GSIB surcharge on an annual basis under two separately prescribed methods based on data for the previous fiscal year-end, and is subject to the higher of the two. "Method 1" reflects the GSIB surcharge as prescribed by the Basel Committee's assessment methodology, and is calculated across five criteria: size, cross-jurisdictional activity, interconnectedness, complexity and substitutability. "Method 2" modifies the Method 1 requirements to include a measure of short-term wholesale funding in place of substitutability, and introduces a GSIB score "multiplication factor".

The following table presents the Firm's effective GSIB surcharge for the years ended December 31, 2024 and 2023. For 2025, the Firm's effective regulatory minimum GSIB surcharge calculated under both Method 1 and Method 2 remains unchanged at 2.5% and 4.5%, respectively.

	2024	2023
Method 1	2.5 %	2.5 %
Method 2	4.5 %	4.0 %

The U.S. federal regulatory capital standards include a framework for setting a discretionary countercyclical capital buffer taking into account the macro financial environment in which large, internationally active banks function. As of December 31, 2024, the U.S. countercyclical capital buffer remained at 0%. The Federal Reserve will continue to review the buffer at least annually. The buffer can be increased if the Federal Reserve, the FDIC and the OCC determine that systemic risks are meaningfully above normal and can be calibrated up to an additional 2.5% of RWA subject to a 12-month implementation period.

Failure to maintain regulatory capital equal to or in excess of the risk-based regulatory capital minimum plus the capital conservation buffer (inclusive of the GSIB surcharge) and any countercyclical buffer will result in limitations to the amount of capital that the Firm may distribute, such as through dividends and common share repurchases, as well as on discretionary bonus payments for certain executive officers.

Total Loss-Absorbing Capacity

The Federal Reserve's TLAC rule requires the U.S. GSIB top-tier holding companies, including the Firm, to maintain minimum levels of external TLAC and eligible LTD. Refer to TLAC on page 106 for additional information.

Leverage-based Capital Regulatory Requirements
Supplementary leverage ratio

Banking organizations subject to the Basel III Advanced approach are currently required to have a minimum SLR of 3.0%. Certain banking organizations, including the Firm, are also required to hold an additional 2.0% leverage buffer. The SLR is defined as Tier 1 capital under Basel III divided by the Firm's total leverage exposure. Total leverage exposure is calculated by taking the Firm's total average on-balance sheet assets, less amounts permitted to be deducted for Tier 1 capital, and adding certain off-balance sheet exposures, as defined in regulatory capital rules. Refer to SLR on page 104 for additional information.

Failure to maintain an SLR equal to or greater than the regulatory requirement will result in limitations on the amount of capital that the Firm may distribute such as through dividends and common share repurchases, as well as on discretionary bonus payments for certain executive officers.

Other regulatory capital

In addition to meeting the capital ratio requirements of Basel III, the Firm and its principal IDI subsidiary, JPMorgan Chase Bank, N.A., must also maintain minimum capital and leverage ratios in order to be "well-capitalized" under the regulations issued by the Federal Reserve and the Prompt Corrective Action requirements of the FDIC Improvement Act, respectively. Refer to Note 27 for additional information.

Additional information regarding the Firm's capital ratios, as well as the U.S. federal regulatory capital standards to which the Firm is subject, is presented in Note 27. Refer to the Firm's Pillar 3 Regulatory Capital Disclosures reports, which are available on the Firm's website, for further information on the Firm's current capital measures.

Selected capital and RWA data

The following tables present the Firm's risk-based capital metrics under both the Basel III Standardized and Advanced approaches and leverage-based capital metrics. Refer to Note 27 for JPMorgan Chase Bank, N.A.'s risk-based and leverage-based capital metrics. First Republic Bank was not subject to Advanced approach regulatory capital requirements. As a result, for certain exposures associated with the First Republic acquisition, Advanced RWA and any impact on Advanced Total capital is calculated under the Standardized approach as permitted by the transition provisions in the U.S. capital rules. Refer to Note 34 for additional information on the First Republic acquisition.

	Standardized			Advanced		
(in millions, except ratios)	**December 31, 2024**	December 31, 2023	Capital ratio requirements[b]	**December 31, 2024**	December 31, 2023	Capital ratio requirements[b]
Risk-based capital metrics:[a]						
CET1 capital	**$ 275,513**	$ 250,585		**$ 275,513**	$ 250,585	
Tier 1 capital	**294,881**	277,306		**294,881**	277,306	
Total capital	**325,589**	308,497		**311,898** [c]	295,417 [c]	
Risk-weighted assets	**1,757,460**	1,671,995		**1,740,429** [c]	1,669,156 [c]	
CET1 capital ratio	**15.7 %**	15.0 %	**12.3 %**	**15.8 %**	15.0 %	**11.5 %**
Tier 1 capital ratio	**16.8**	16.6	**13.8**	**16.9**	16.6	**13.0**
Total capital ratio	**18.5**	18.5	**15.8**	**17.9**	17.7	**15.0**

(a) The capital metrics reflect the CECL capital transition provisions. As of December 31, 2024, CET1 capital reflected the remaining $720 million CECL benefit and were fully phased in as of January 1, 2025; as of December 31, 2023, CET1 capital reflected a $1.4 billion benefit. Refer to Note 27 for additional information.
(b) Represents minimum requirements and regulatory buffers applicable to the Firm for the period ended December 31, 2024. For the period ended December 31, 2023, the Basel III Standardized CET1, Tier 1, and Total capital ratio requirements applicable to the Firm were 11.4%, 12.9%, and 14.9%, respectively; the Basel III Advanced CET1, Tier 1, and Total capital ratio requirements applicable to the Firm were 11.0%, 12.5%, and 14.5%, respectively. Refer to Note 27 for additional information.
(c) Includes the impacts of certain assets associated with First Republic to which the Standardized approach has been applied as permitted by the transition provisions in the U.S. capital rules.

Three months ended (in millions, except ratios)	**December 31, 2024**	December 31, 2023	Capital ratio requirements[c]
Leverage-based capital metrics:[a]			
Adjusted average assets[b]	**$ 4,070,499**	$ 3,831,200	
Tier 1 leverage ratio	**7.2 %**	7.2 %	**4.0 %**
Total leverage exposure	**$ 4,837,568**	$ 4,540,465	
SLR	**6.1 %**	6.1 %	**5.0 %**

(a) The capital metrics reflect the CECL capital transition provisions. Refer to Note 27 for additional information.
(b) Adjusted average assets, for purposes of calculating the leverage ratios, includes quarterly average assets adjusted for on-balance sheet assets that are subject to deduction from Tier 1 capital, predominantly goodwill, inclusive of estimated equity method goodwill, and other intangible assets.
(c) Represents minimum requirements and regulatory buffers applicable to the Firm. Refer to Note 27 for additional information.

Capital components

The following table presents reconciliations of total stockholders' equity to Basel III CET1 capital, Tier 1 capital and Total capital as of December 31, 2024 and 2023.

(in millions)		**December 31, 2024**		December 31, 2023
Total stockholders' equity	**$**	**344,758**	$	327,878
Less: Preferred stock		20,050		27,404
Common stockholders' equity		**324,708**		300,474
Add:				
Certain deferred tax liabilities[a]		2,943		2,996
Other CET1 capital adjustments[b]		4,499		4,717
Less:				
Goodwill[c]		53,763		54,377
Other intangible assets		2,874		3,225
Standardized/Advanced CET1 capital		**275,513**		250,585
Add: Preferred stock		20,050		27,404
Less: Other Tier 1 adjustments		682		683
Standardized/Advanced Tier 1 capital	**$**	**294,881**	$	277,306
Long-term debt and other instruments qualifying as Tier 2 capital	$	10,312	$	11,779
Qualifying allowance for credit losses[d]		20,992		20,102
Other		(596)		(690)
Standardized Tier 2 capital	**$**	**30,708**	$	31,191
Standardized Total capital	**$**	**325,589**	$	308,497
Adjustment in qualifying allowance for credit losses for Advanced Tier 2 capital[e][f]		(13,691)		(13,080)
Advanced Tier 2 capital	**$**	**17,017**	$	18,111
Advanced Total capital	**$**	**311,898**	$	295,417

(a) Represents deferred tax liabilities related to tax-deductible goodwill and to identifiable intangibles created in nontaxable transactions, which are netted against goodwill and other intangibles when calculating CET1 capital.

(b) As of December 31, 2024 and 2023, included a net benefit associated with cash flow hedges and debit valuation adjustments ("DVA") related to structured notes recorded in AOCI of $5.2 billion and $4.3 billion and the benefit from the CECL capital transition provisions of $720 million and $1.4 billion, respectively.

(c) Goodwill deducted from capital includes goodwill associated with equity method investments in nonconsolidated financial institutions based on regulatory requirements. Refer to page 140 for additional information on principal investment risk.

(d) Represents the allowance for credit losses eligible for inclusion in Tier 2 capital up to 1.25% of credit risk RWA, including the impact of the CECL capital transition provision with any excess deducted from RWA. Refer to Note 27 for additional information on the CECL capital transition.

(e) Represents an adjustment to qualifying allowance for credit losses for the excess of eligible credit reserves over expected credit losses up to 0.6% of credit risk RWA, including the impact of the CECL capital transition provision with any excess deducted from RWA.

(f) As of December 31, 2024 and 2023, included an incremental $541 million and $655 million allowance for credit losses, respectively, on certain assets associated with First Republic to which the Standardized approach has been applied, as permitted by the transition provisions in the U.S. capital rules.

Capital rollforward

The following table presents the changes in Basel III CET1 capital, Tier 1 capital and Tier 2 capital for the year ended December 31, 2024.

Year ended December 31, (in millions)	**2024**
Standardized/Advanced CET1 capital at December 31, 2023	$ 250,585
Net income applicable to common equity	57,212
Dividends declared on common stock	(13,786)
Net purchase of treasury stock	(17,801)
Changes in additional paid-in capital	783
Changes related to AOCI applicable to capital:	
Unrealized gains/(losses) on investment securities	(87)
Translation adjustments, net of hedges[a]	(858)
Fair value hedges	(87)
Defined benefit pension and other postretirement employee benefit ("OPEB") plans	(63)
Changes related to other CET1 capital adjustments[b]	(385)
Change in Standardized/Advanced CET1 capital	24,928
Standardized/Advanced CET1 capital at December 31, 2024	**$ 275,513**
Standardized/Advanced Tier 1 capital at December 31, 2023	$ 277,306
Change in CET1 capital[b]	24,928
Net redemptions of noncumulative perpetual preferred stock	(7,354)
Other	1
Change in Standardized/Advanced Tier 1 capital	17,575
Standardized/Advanced Tier 1 capital at December 31, 2024	**$ 294,881**
Standardized Tier 2 capital at December 31, 2023	$ 31,191
Change in long-term debt and other instruments qualifying as Tier 2	(1,467)
Change in qualifying allowance for credit losses[b]	890
Other	94
Change in Standardized Tier 2 capital	(483)
Standardized Tier 2 capital at December 31, 2024	**$ 30,708**
Standardized Total capital at December 31, 2024	**$325,589**
Advanced Tier 2 capital at December 31, 2023	$ 18,111
Change in long-term debt and other instruments qualifying as Tier 2	(1,467)
Change in qualifying allowance for credit losses[b][c]	279
Other	94
Change in Advanced Tier 2 capital	(1,094)
Advanced Tier 2 capital at December 31, 2024	**$ 17,017**
Advanced Total capital at December 31, 2024	**$ 311,898**

(a) Includes foreign currency translation adjustments and the impact of related derivatives.

(b) Includes the impact of the CECL capital transition provisions and the cumulative effect of changes in accounting principles. Refer to Note 1 for additional information on changes in accounting principles and Note 27 for additional information on the CECL capital transition provisions.

(c) As of December 31, 2024 and 2023, included an incremental $541 million and $655 million allowance for credit losses, respectively, on certain assets associated with First Republic to which the Standardized approach has been applied, as permitted by the transition provisions in the U.S. capital rules.

RWA rollforward

The following table presents changes in the components of RWA under Basel III Standardized and Advanced approaches for the year ended December 31, 2024. The amounts in the rollforward categories are estimates, based on the predominant driver of the change.

	Standardized			Advanced			
Year ended December 31, 2024 (in millions)	Credit risk RWA[(c)]	Market risk RWA	**Total RWA**	Credit risk RWA[(c)(d)]	Market risk RWA	Operational risk RWA	**Total RWA**
December 31, 2023	$ 1,603,851	$ 68,144	**$ 1,671,995**	$ 1,155,261	$ 68,603	$ 445,292	**$ 1,669,156**
Model & data changes[(a)]	4,743	(366)	**4,377**	4,811	(366)	—	**4,445**
Movement in portfolio levels[(b)]	64,169	16,919	**81,088**	57,933	16,895	(8,000)	**66,828**
Changes in RWA	68,912	16,553	**85,465**	62,744	16,529	(8,000)	**71,273**
December 31, 2024	**$ 1,672,763**	**$ 84,697**	**$ 1,757,460**	**$ 1,218,005**	**$ 85,132**	**$ 437,292**	**$ 1,740,429**

(a) Model & data changes refer to material movements in levels of RWA as a result of revised methodologies and/or treatment per regulatory guidance (exclusive of rule changes).
(b) Movement in portfolio levels (inclusive of rule changes) refers to: for Credit risk RWA, changes in book size, changes in composition and credit quality, market movements, and deductions for excess eligible allowances for credit losses not eligible for inclusion in Tier 2 capital; for Market risk RWA, changes in position and market movements; and for Operational risk RWA, updates to cumulative losses and macroeconomic model inputs.
(c) As of December 31, 2024 and 2023, the Basel III Standardized Credit risk RWA included wholesale and retail off balance-sheet RWA of $208.0 billion and $208.5 billion, respectively; and the Basel III Advanced Credit risk RWA included wholesale and retail off balance-sheet RWA of $192.1 billion and $188.5 billion, respectively.
(d) As of December 31, 2024 and 2023, Credit risk RWA reflected approximately $43.3 billion and $52.4 billion, respectively, of RWA calculated under the Standardized approach for certain assets associated with First Republic as permitted by the transition provisions in the U.S. capital rules.

Refer to the Firm's Pillar 3 Regulatory Capital Disclosures reports, which are available on the Firm's website, for further information on Credit risk RWA, Market risk RWA and Operational risk RWA.

Supplementary leverage ratio

The following table presents the components of the Firm's SLR.

Three months ended (in millions, except ratio)	**December 31, 2024**	December 31, 2023
Tier 1 capital	**$ 294,881**	$ 277,306
Total average assets	4,125,167	3,885,632
Less: Regulatory capital adjustments[(a)]	54,668	54,432
Total adjusted average assets[(b)]	4,070,499	3,831,200
Add: Off-balance sheet exposures[(c)]	767,069	709,265
Total leverage exposure	**$ 4,837,568**	$4,540,465
SLR	**6.1 %**	6.1 %

(a) For purposes of calculating the SLR, includes quarterly average assets adjusted for on-balance sheet assets that are subject to deduction from Tier 1 capital, predominantly goodwill, inclusive of estimated equity method goodwill, other intangible assets and adjustments for the CECL capital transition provisions. Refer to Note 27 for additional information on the CECL capital transition.
(b) Adjusted average assets used for the calculation of Tier 1 leverage ratio.
(c) Off-balance sheet exposures are calculated as the average of the three month-end spot balances on applicable regulatory exposures during the reporting quarter. Refer to the Firm's Pillar 3 Regulatory Capital Disclosures reports for additional information.

Line of business and Corporate equity

Each LOB and Corporate is allocated capital by taking into consideration a variety of factors including capital levels of similarly rated peers and applicable regulatory capital requirements. ROE is measured and internal targets for expected returns are established as key measures of an LOB's performance.

The Firm's current equity allocation methodology incorporates Basel III Standardized RWA and the GSIB surcharge, both under rules currently in effect, as well as a simulation of capital depletion in a severe stress environment. At least annually, the assumptions, judgments and methodologies used to allocate capital are reassessed and, as a result, the capital allocated to the LOBs and Corporate may change. As of January 1, 2025, changes to the Firm's capital allocations are primarily a result of updates to the Firm's current capital requirements and changes in RWA for each LOB under rules currently in effect. Any capital that the Firm has accumulated in excess of these current requirements, including the capital required to meet the potential increased requirements of the U.S. Basel III proposal, has been retained in Corporate in addition to its allocated balance.

The following table presents the capital allocated to each LOB and Corporate.

		December 31,	
(in billions)	**January 1, 2025**	2024	2023
Consumer & Community Banking	**$ 56.0**	$ 54.5	$ 55.5
Commercial & Investment Bank	**149.5**	132.0	138.0
Asset & Wealth Management	**16.0**	15.5	17.0
Corporate	**103.2**	122.7	90.0
Total common stockholders' equity	**$ 324.7**	$ 324.7	$ 300.5

Capital actions

Common stock dividends

The Firm's common stock dividends are planned as part of the Capital Management governance framework in line with the Firm's capital management objectives.

On December 9, 2024, the Firm announced that its Board of Directors had declared a quarterly common stock dividend of $1.25 per share, payable on January 31, 2025. The Firm's dividends are subject to approval by the Board of Directors on a quarterly basis.

Refer to Note 21 and Note 26 for information regarding dividend restrictions.

The following table shows the common dividend payout ratio based on net income applicable to common equity.

Year ended December 31,	**2024**	2023	2022
Common dividend payout ratio	**24 %**	25 %	33 %

Common stock

On June 28, 2024, the Firm announced that its Board of Directors had authorized a new $30 billion common share repurchase program, effective July 1, 2024. Through June 30, 2024, the Firm was authorized to purchase up to $30 billion of common shares under its previously-approved common share repurchase program that was announced on April 13, 2022.

The following table sets forth the Firm's repurchases of common stock for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions)	**2024**	2023	2022[(b)]
Total number of shares of common stock repurchased	**91.7**	69.5	23.1
Aggregate purchase price of common stock repurchases[(a)]	**$ 18,841**	$ 9,898	$ 3,122

(a) Excludes excise tax and commissions. As part of the Inflation Reduction Act of 2022, a 1% excise tax is imposed on net share repurchases commencing January 1, 2023.

(b) In the second half of 2022, the Firm temporarily suspended share repurchases, which it resumed in the first quarter of 2023 under its common share repurchase program.

The Board of Directors' authorization to repurchase common shares is utilized at management's discretion. The $30 billion common share repurchase program approved by the Board of Directors does not establish specific price targets or timetables. Management determines the amount and timing of common share repurchases based on various factors, including market conditions; legal and regulatory considerations affecting the amount and timing of repurchase activity; the Firm's capital position (taking into account goodwill and intangibles); organic capital generation; current and proposed future capital requirements; and other investment opportunities. The amount of common shares that the Firm repurchases in any period may be substantially more or less than the amounts estimated or actually repurchased in prior periods, reflecting the dynamic nature of the decision-making process. The Firm's common share repurchases may be suspended by management at any time; and may be executed through open market purchases or privately negotiated transactions, or utilizing Rule 10b5-1 plans, which are written trading plans that the Firm may enter into from time to time under Rule 10b5-1 of the Securities Exchange Act of 1934 and which allow the Firm to repurchase its common shares during periods when it may otherwise not be repurchasing common shares — for example, during internal trading blackout periods.

Refer to capital planning and stress testing on pages 97–98 for additional information.

Refer to Part II, Item 5: Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities on page 39 of this 2024 Form 10-K for additional information regarding repurchases of the Firm's equity securities.

Preferred stock

Preferred stock dividends were $1.3 billion, $1.5 billion, and $1.6 billion for the years ended December 31, 2024, 2023, and 2022, respectively.

During the year ended and subsequent to December 31, 2024, the Firm issued and redeemed certain series of non-cumulative preferred stock. Refer to Note 21 for additional information on the Firm's preferred stock, including the issuance and redemption of preferred stock.

Subordinated Debt

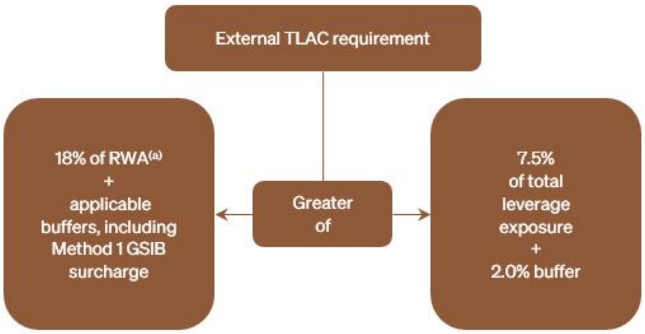
Refer to Long-term funding on page 114 and Note 20 for additional information on the Firm's subordinated debt.

Other capital requirements

Total Loss-Absorbing Capacity

The Federal Reserve's TLAC rule requires the U.S. GSIB top-tier holding companies, including the Firm, to maintain minimum levels of external TLAC and eligible long-term debt.

The external TLAC requirements and the minimum level of eligible long-term debt requirements are shown below:



(a) RWA is the greater of Standardized and Advanced compared to their respective regulatory capital ratio requirements.

Failure to maintain TLAC equal to or in excess of the regulatory minimum plus applicable buffers will result in limitations on the amount of capital that the Firm may distribute, such as through dividends and common share repurchases, as well as on discretionary bonus payments for certain executive officers.

The following table presents the eligible external TLAC and eligible LTD amounts, as well as a representation of these amounts as a percentage of the Firm's total RWA and total leverage exposure applying the impact of the CECL capital transition provisions as of December 31, 2024 and 2023.

	December 31, 2024		December 31, 2023	
(in billions, except ratio)	**External TLAC**	**LTD**	External TLAC	LTD
Total eligible amount	**$ 546.6**	**$ 236.8**	$ 513.8	$ 222.6
% of RWA	**31.1 %**	**13.5 %**	30.7 %	13.3 %
Regulatory requirements	**23.0**	**10.5**	23.0	10.0
Surplus/(shortfall)	**$ 142.3**	**$ 52.3**	$ 129.2	$ 55.4
% of total leverage exposure	**11.3 %**	**4.9 %**	11.3 %	4.9 %
Regulatory requirements	**9.5**	**4.5**	9.5	4.5
Surplus/(shortfall)	**$ 87.0**	**$ 19.2**	$ 82.5	$ 18.3

Effective January 1, 2024, the Firm's regulatory requirement for its eligible LTD to RWA ratio increased by 50 bps to 10.5%, due to the increase in the Firm's GSIB Method 2 requirements. The Firm's regulatory requirement for its TLAC to RWA ratio remained at 23.0%. Refer to Risk-based Capital Regulatory Requirements on pages 100–101 for further information on the GSIB surcharge.

Refer to Liquidity Risk Management on pages 108–115 for further information on long-term debt issued by the Parent Company.

Refer to Part I, Item 1A: Risk Factors on pages 10-37 of this 2024 Form 10-K for information on the financial consequences to holders of the Firm's debt and equity securities in a resolution scenario.

U.S. broker-dealer regulatory capital

J.P. Morgan Securities

JPMorganChase's principal U.S. broker-dealer subsidiary is J.P. Morgan Securities. J.P. Morgan Securities is subject to the regulatory capital requirements of Rule 15c3-1 under the Securities Exchange Act of 1934 (the "Net Capital Rule"). J.P. Morgan Securities is also registered as a futures commission merchant and is subject to regulatory capital requirements, including those imposed by the SEC, the Commodity Futures Trading Commission ("CFTC"), the Financial Industry Regulatory Authority ("FINRA") and the National Futures Association ("NFA").

J.P. Morgan Securities has elected to compute its minimum net capital requirements in accordance with the "Alternative Net Capital Requirements" of the Net Capital Rule.

The following table presents J.P. Morgan Securities' net capital.

December 31, 2024 (in millions)	Actual	Minimum
Net Capital	**$ 24,980**	**$ 5,999**

J.P. Morgan Securities is registered with the SEC as a security-based swap dealer and with the CFTC as a swap dealer. As a result of additional SEC and CFTC capital and financial reporting requirements for security-based swap dealers and swap dealers, J.P. Morgan Securities is subject to alternative minimum net capital requirements and required to hold "tentative net capital" in excess of $5.0 billion. J.P. Morgan Securities is also required to notify the SEC and CFTC in the event that its tentative net capital is less than $6.0 billion. Tentative net capital is net capital before deducting market and credit risk charges as defined by the Net Capital Rule. As of December 31, 2024, J.P. Morgan Securities maintained tentative net capital in excess of the minimum and notification requirements.

Non-U.S. subsidiary regulatory capital

J.P. Morgan Securities plc

J.P. Morgan Securities plc is a wholly-owned subsidiary of JPMorgan Chase Bank, N.A. and has authority to engage in banking, investment banking and broker-dealer activities. J.P. Morgan Securities plc is jointly regulated in the U.K. by the Prudential Regulation Authority ("PRA") and the Financial Conduct Authority ("FCA"). J.P. Morgan Securities plc is subject to the Capital Requirements Regulation ("CRR"), as adopted and amended in the U.K., and the capital rules in the PRA Rulebook. These requirements collectively represent the U.K.'s implementation of the Basel III standards. The PRA announced that it intends to delay the U.K.'s implementation of the final Basel III standards until January 1, 2027, with a three-year transitional period for certain aspects.

The Bank of England requires that U.K. banks, including U.K. regulated subsidiaries of overseas groups, maintain minimum requirements for own funds and eligible liabilities ("MREL"). As of December 31, 2024, J.P. Morgan Securities plc was compliant with its MREL requirements.

The following table presents J.P. Morgan Securities plc's risk-based and leverage-based capital metrics.

December 31, 2024 (in millions, except ratios)	Actual	Regulatory Minimum ratios[a]
Total capital	**$ 53,120**	
CET1 capital ratio	**17.0 %**	**4.5 %**
Tier 1 capital ratio	**22.1**	**6.0**
Total capital ratio	**27.1**	**8.0**
Tier 1 leverage ratio	**7.1**	**3.3** (b)

(a) Represents minimum Pillar 1 requirements specified by the PRA. J.P. Morgan Securities plc's capital ratios as of December 31, 2024 exceeded the minimum requirements, including the additional capital requirements specified by the PRA.

(b) At least 75% of the Tier 1 leverage ratio minimum must be met with CET1 capital.

J.P. Morgan SE

JPMSE is a wholly-owned subsidiary of JPMorgan Chase Bank, N.A. and has authority to engage in banking, investment banking and markets activities. JPMSE is regulated by the European Central Bank ("ECB"), the German Financial Supervisory Authority and the German Central Bank, as well as the local regulators in each of the countries in which it operates, and it is subject to EU capital requirements under Basel III. JPMSE is subject to the EU implementation of the final Basel III standards. Those standards became effective beginning on January 1, 2025, with the exception of market risk aspects for which the effective date is January 1, 2026.

JPMSE is required by the EU Single Resolution Board to maintain MREL. As of December 31, 2024, JPMSE was compliant with its MREL requirements.
The following table presents JPMSE's risk-based and leverage-based capital metrics.

December 31, 2024 (in millions, except ratios)	Actual	Regulatory Minimum ratios[a]
Total capital	**$ 43,298**	
CET1 capital ratio	**20.0 %**	**4.5 %**
Tier 1 capital ratio	**20.0**	**6.0**
Total capital ratio	**34.8**	**8.0**
Tier 1 leverage ratio	**6.1**	**3.0**

(a) Represents minimum Pillar 1 requirements specified by the EU CRR. J.P. Morgan SE's capital and leverage ratios as of December 31, 2024 exceeded the minimum requirements, including the additional capital requirements specified by EU regulators.

LIQUIDITY RISK MANAGEMENT

Liquidity risk is the risk that the Firm will be unable to meet its cash and collateral needs as they arise or that it does not have the appropriate amount, composition and tenor of funding and liquidity to support its assets and liabilities.

Liquidity risk management

The Firm has a Liquidity Risk Management ("LRM") function whose primary objective is to provide independent oversight of liquidity risk across the Firm. Liquidity Risk Management's responsibilities include:

- Defining, monitoring and reporting liquidity risk metrics;
- Independently establishing and monitoring limits and indicators, including liquidity risk appetite;
- Developing a process to classify, monitor and report limit breaches;
- Performing an independent review of liquidity risk management processes to evaluate their adequacy and effectiveness;
- Monitoring and reporting internal Firmwide and legal entity liquidity stress tests, regulatory defined metrics, as well as liquidity positions, balance sheet variances and funding activities; and
- Approving or escalating for review new or updated liquidity stress assumptions.

Liquidity management

Treasury and CIO is responsible for liquidity management.

The primary objectives of the Firm's liquidity management are to:

- Ensure that the Firm's core businesses and material legal entities are able to operate in support of client needs and meet contractual and contingent financial obligations through normal economic cycles as well as during stress events, and
- Manage an optimal funding mix and availability of liquidity sources.

The Firm addresses these objectives through:

- Analyzing and understanding the liquidity characteristics of the assets and liabilities of the Firm, LOBs, legal entities, as well as currencies, taking into account legal, regulatory, and operational restrictions;
- Developing and maintaining internal liquidity stress testing assumptions;
- Defining and monitoring Firmwide and legal entity-specific liquidity strategies, policies, reporting and contingency funding plans;
- Managing liquidity within the Firm's approved limits and indicators, including liquidity risk appetite tolerances;
- Managing compliance with regulatory requirements related to funding and liquidity risk; and
- Setting FTP in accordance with underlying liquidity characteristics of balance sheet assets and liabilities as well as certain off-balance sheet items.

As part of the Firm's overall liquidity management strategy, the Firm manages liquidity and funding using a centralized, global approach designed to:

- Optimize liquidity sources and uses;
- Monitor exposures;
- Identify constraints on the transfer of liquidity between the Firm's legal entities; and
- Maintain the appropriate amount of surplus liquidity at a Firmwide and legal entity level, where relevant.

Governance

Committees responsible for liquidity governance include the Firmwide ALCO, as well as regional ALCOs, the Treasurer Committee, and the CTC Risk Committee. In addition, the Board Risk Committee reviews and recommends to the Board of Directors, for approval, the Firm's liquidity risk tolerances, liquidity strategy, and liquidity policy. Refer to Firmwide Risk Management on pages 91–95 for further discussion of ALCO and other risk-related committees.

Internal stress testing

The Firm conducts internal liquidity stress testing to monitor liquidity positions at the Firm and its material legal entities under a variety of adverse scenarios, including scenarios analyzed as part of the Firm's resolution and recovery planning. Internal stress tests are produced on a daily basis, and other stress tests are performed in response to specific market events or concerns. Liquidity stress tests assume all of the Firm's contractual financial obligations are met and take into consideration:

- Varying levels of access to unsecured and secured funding markets;
- Estimated non-contractual and contingent cash outflows;
- Credit rating downgrades;
- Collateral haircuts; and
- Potential impediments to the availability and transferability of liquidity between jurisdictions and material legal entities such as regulatory, legal or other restrictions.

Liquidity outflows are modeled across a range of time horizons and currency dimensions and contemplate both market and idiosyncratic stresses.

Results of stress tests are considered in the formulation of the Firm's funding plan and assessment of its liquidity position. The Parent Company acts as a source of funding for the Firm through equity and long-term debt issuances, and its intermediate holding company, JPMorgan Chase Holdings LLC (the “IHC”), provides funding to support the ongoing operations of the Parent Company and its subsidiaries. The Firm manages liquidity at the Parent Company, the IHC, and operating subsidiaries at levels sufficient to comply with liquidity risk tolerances and minimum liquidity requirements, and to manage through periods of stress when access to normal funding sources may be disrupted.

Contingency funding plan

The Firm’s Contingency Funding Plan (“CFP”) sets out the strategies for addressing and managing liquidity resource needs during a liquidity stress event and incorporates liquidity risk limits, indicators and risk appetite tolerances. The CFP also identifies the alternative contingent funding and liquidity resources available to the Firm and its legal entities in a period of stress.

LCR and HQLA

The LCR rule requires that the Firm and JPMorgan Chase Bank, N.A. maintain an amount of eligible HQLA that is sufficient to meet their respective estimated total net cash outflows over a prospective 30 calendar-day period of significant stress. Eligible HQLA, for purposes of calculating the LCR, is the amount of unencumbered HQLA that satisfy certain operational considerations as defined in the LCR rule. HQLA primarily consist of cash and certain high-quality liquid securities as defined in the LCR rule.

Under the LCR rule, the amount of eligible HQLA held by JPMorgan Chase Bank, N.A. that is in excess of its stand-alone 100% minimum LCR requirement, and that is not transferable to non-bank affiliates, must be excluded from the Firm’s reported eligible HQLA.

Estimated net cash outflows are based on standardized stress outflow and inflow rates prescribed in the LCR rule, which are applied to the balances of the Firm’s assets, sources of funds, and obligations. The LCR for both the Firm and JPMorgan Chase Bank, N.A. is required to be a minimum of 100%.

The following table summarizes the Firm and JPMorgan Chase Bank, N.A.’s average LCR for the three months ended December 31, 2024, September 30, 2024 and December 31, 2023 based on the Firm’s interpretation of the LCR framework.

	Three months ended		
Average amount (in millions)	**December 31, 2024**	September 30, 2024	December 31, 2023
JPMorgan Chase & Co.:			
HQLA			
Eligible cash[(a)]	**$ 396,123**	$ 412,389	$ 485,263
Eligible securities[(b)(c)]	**464,877**	453,899	313,365
Total HQLA[(d)]	**$ 861,000**	$ 866,288	$ 798,628
Net cash outflows	**$ 763,648**	$ 762,072	$ 704,857
LCR	**113 %**	114 %	113 %
Net excess eligible HQLA[(d)]	**$ 97,352**	$ 104,216	$ 93,771
JPMorgan Chase Bank, N.A.:			
LCR	**124 %**	121 %	129 %
Net excess eligible HQLA	**$ 193,682**	$ 168,137	$ 215,190

(a) Represents cash on deposit at central banks, primarily the Federal Reserve Banks.
(b) Eligible HQLA securities may be reported in securities borrowed or purchased under resale agreements, trading assets, or investment securities on the Firm’s Consolidated balance sheets. For purposes of calculating the LCR, HQLA securities are included at fair value, which may differ from the accounting treatment under U.S. GAAP.
(c) Predominantly U.S. Treasuries, U.S. GSE and government agency MBS, and sovereign bonds net of regulatory haircuts under the LCR rule.
(d) Excludes average excess eligible HQLA at JPMorgan Chase Bank, N.A. that are not transferable to non-bank affiliates.

JPMorgan Chase Bank, N.A.'s average LCR increased during the three months ended December 31, 2024, compared with the three months ended September 30, 2024, driven by activities in CIB Markets, partially offset by lower market values of HQLA-eligible investment securities and funding maturities.

JPMorgan Chase Bank, N.A.’s average LCR for the three months ended December 31, 2024 decreased compared with the three months ended December 31, 2023, driven by dividend payments to the Parent Company and lending activity, largely offset by higher market values of HQLA-eligible investment securities, a reduction in unencumbered non-HQLA AFS securities, activities in CIB Markets, and long-term debt issuances.

Each of the Firm and JPMorgan Chase Bank, N.A.'s average LCR may fluctuate from period to period due to changes in their respective eligible HQLA and estimated net cash outflows as a result of ongoing business activity and from the impacts of Federal Reserve actions as well as other factors. Refer to the Firm's U.S. LCR Disclosure reports, which are available on the Firm's website, for a further discussion of the Firm's LCR.

Liquidity sources

In addition to the assets reported in the Firm's eligible HQLA discussed above, the Firm had unencumbered marketable securities, such as equity and debt securities, that the Firm believes would be available to raise liquidity. This includes excess eligible HQLA securities at JPMorgan Chase Bank, N.A. that are not transferable to non-bank affiliates. The fair value of these securities was approximately $594 billion and $649 billion as of December 31, 2024 and 2023, respectively, although the amount of liquidity that could be raised at any particular time would be dependent on prevailing market conditions. The decrease compared to December 31, 2023, was driven by reductions in unencumbered AFS securities in Treasury and CIO, excess eligible HQLA securities at JPMorgan Chase Bank, N.A., and unencumbered CIB trading assets.

The Firm had approximately $1.4 trillion of available cash and securities as of both December 31, 2024 and 2023. For each respective period, the amount was comprised of eligible end-of-period HQLA, excluding the impact of regulatory haircuts, of approximately $834 billion and $798 billion, and unencumbered marketable securities with a fair value of approximately $594 billion and $649 billion.

The Firm also had available borrowing capacity at the Federal Home Loan Banks ("FHLBs") and the discount window at the Federal Reserve Banks as a result of collateral pledged by the Firm to such banks of approximately $413 billion and $340 billion as of December 31, 2024 and 2023, respectively. This borrowing capacity excludes the benefit of cash and securities reported in the Firm's eligible HQLA or other unencumbered securities that are currently pledged at the Federal Reserve Banks discount window and other central banks. Available borrowing capacity increased from December 31, 2023 primarily due to a higher amount of commercial loans and credit card receivables pledged at the Federal Reserve Banks. Although available, the Firm does not view this borrowing capacity at the Federal Reserve Banks discount window and the other central banks as a primary source of liquidity.

NSFR

The net stable funding ratio ("NSFR") is a liquidity requirement for large banking organizations that is intended to measure the adequacy of "available" stable funding that is sufficient to meet their "required" amounts of stable funding over a one-year horizon.

For the three months ended December 31, 2024, both the Firm and JPMorgan Chase Bank, N.A. were compliant with the 100% minimum NSFR requirement, based on the Firm's interpretation of the final NSFR rule. Refer to the Firm's U.S. NSFR Disclosure report on the Firm's website for additional information.

Funding

Sources of funds

Management believes that the Firm's unsecured and secured funding capacity is sufficient to meet its on- and off-balance sheet obligations, which includes both short- and long-term cash requirements.

The Firm funds its global balance sheet through diverse sources of funding including deposits, secured and unsecured funding in the capital markets and stockholders' equity. Deposits are the primary funding source for JPMorgan Chase Bank, N.A. Additionally, JPMorgan Chase Bank, N.A. may access funding through short- or long-term secured borrowings, the issuance of unsecured long-term debt, or from borrowings from the IHC. The Firm's non-bank subsidiaries are primarily funded from long-term unsecured borrowings and short-term secured borrowings which are primarily securities loaned or sold under repurchase agreements. Excess funding is invested by Treasury and CIO in the Firm's investment securities portfolio or deployed in cash or other short-term liquid investments based on their interest rate and liquidity risk characteristics.

Refer to Note 28 for additional information on off–balance sheet obligations.

Deposits

The table below summarizes, by LOB and Corporate, the period-end and average deposit balances as of and for the years ended December 31, 2024 and 2023.

As of or for the year ended December 31,			Average	
(in millions)	**2024**	2023	**2024**	2023
Consumer & Community Banking[(a)]	**$ 1,056,652**	$ 1,094,738	**$ 1,064,215**	$ 1,126,552
Commercial & Investment Bank[(a)]	**1,073,512**	1,050,892	**1,061,488**	996,295
Asset & Wealth Management[(a)]	**248,287**	233,232	**235,146**	216,178
Corporate	**27,581**	21,826	**25,793**	20,042
Total Firm	**$ 2,406,032**	$ 2,400,688	**$ 2,386,642**	$ 2,359,067

(a) In the fourth quarter of 2023, CCB transferred certain deposits associated with First Republic to AWM and CIB.

The Firm believes that deposits provide a stable source of funding and reduce the Firm's reliance on the wholesale funding markets. A significant portion of the Firm's deposits are consumer deposits and wholesale operating deposits, which are both considered to be stable sources of liquidity. Wholesale operating deposits are generally considered to be stable sources of liquidity because they are generated from customers that maintain operating service relationships with the Firm.

The Firm believes that average deposit balances are generally more representative of deposit trends than period-end deposit balances. However, during periods of market disruption, average deposit trends may be impacted.

The following discussion excludes the impact of the transfer of certain First Republic deposits in the fourth quarter of 2023 from CCB to the other LOBs as the transfers had no net impact on Firmwide deposits.

Average deposits increased for the year ended December 31, 2024 compared to the year ended December 31, 2023, reflecting:

- an increase in CIB due to net inflows predominantly in Payments and net issuances of structured notes as a result of client demand in Markets, partially offset by deposit attrition, which included actions taken to reduce certain deposits,
- the timing impact of First Republic,
- an increase in AWM as a result of growth in balances in new and existing client accounts, reflecting the impact of higher-yielding product offerings, largely offset by continued migration into other investments, and
- a decline in CCB primarily driven by a decrease in balances in existing accounts due to increased customer spending, largely offset by new accounts.

Period-end deposits increased from December 31, 2023, reflecting:

- an increase in CIB due to net inflows predominantly in Payments, largely offset by net maturities of structured notes in Markets,
- an increase in AWM as a result of growth in balances in new and existing client accounts, reflecting the impact of higher-yielding product offerings, largely offset by continued migration into other investments, and
- a decline in CCB primarily driven by a decrease in balances in existing accounts due to increased customer spending and migration into higher-yielding investments, predominantly offset by new accounts.

Refer to the Firm's Consolidated Balance Sheets Analysis and the Business Segment & Corporate Results on pages 63–65 and pages 70–90, respectively, for further information on deposit and liability balance trends, as well as Executive Overview on pages 54–58 and Note 34 for additional information on the First Republic acquisition. Refer to Note 3 for further information on structured notes.

Certain deposits are covered by insurance protection that provides additional funding stability and results in a benefit to the LCR. Deposit insurance protection may be available to depositors in the countries in which the deposits are placed. For example, the FDIC provides deposit insurance protection for deposits placed in a U.S. depository institution. At December 31, 2024 and 2023[(a)], the Firmwide estimated uninsured deposits were $1,414.0 billion and $1,347.8 billion, respectively, primarily reflecting wholesale operating deposits.

Total uninsured deposits include time deposits. The table below presents an estimate of uninsured U.S. and non-U.S. time deposits, and their remaining maturities. The Firm's estimates of its uninsured U.S. time deposits are based on data that the Firm calculates periodically under applicable FDIC regulations. For purposes of this presentation, all non-U.S. time deposits are deemed to be uninsured.

	December 31, 2024		December 31, 2023	
(in millions)	U.S.	Non-U.S.	U.S.	Non-U.S.
Three months or less	**$ 119,333**	**$ 77,253**	$98,606 (a)	$ 77,466
Over three months but within 6 months	**11,040**	**12,229**	17,736	5,358
Over six months but within 12 months	**7,056**	**1,542**	10,294	4,820
Over 12 months	**823**	**1,924**	710	2,543
Total	**$138,252**	**$ 92,948**	$127,346 (a)	$ 90,187

(a) Prior-period amounts have been revised to include cash collateral for certain derivatives to align with a change in the methodology for calculating uninsured U.S. time deposits.

The table below shows the loan and deposit balances, the loans-to-deposits ratios, and deposits as a percentage of total liabilities, as of December 31, 2024 and 2023.

As of December 31, (in billions except ratios)	**2024**	2023
Deposits	**$ 2,406.0**	$ 2,400.7
Deposits as a % of total liabilities	**66 %**	68 %
Loans	**$ 1,348.0**	$ 1,323.7
Loans-to-deposits ratio	**56 %**	55 %

The following table provides a summary of the average balances and average interest rates of JPMorganChase's deposits for the years ended December 31, 2024, 2023, and 2022.

Year ended December 31,	Average balances			Average interest rates		
(in millions, except interest rates)	**2024**	2023	2022	**2024**	2023	2022
U.S. offices						
Noninterest-bearing	**$ 611,734**	$ 635,791	$ 691,206	**NA**	NA	NA
Interest-bearing						
Demand[(a)]	**282,533**	279,725	324,512	**3.90 %**	3.50 %	0.92 %
Savings[(b)]	**800,964**	864,558	971,788	**1.39**	1.10	0.28
Time	**223,503**	145,827	62,022	**4.93**	4.74	2.07
Total interest-bearing deposits	**1,307,000**	1,290,110	1,358,322	**2.54**	2.03	0.52
Total deposits in U.S. offices	**1,918,734**	1,925,901	2,049,528	**1.73**	1.36	0.34
Non-U.S. offices						
Noninterest-bearing	**26,858**	24,747	28,043	**NA**	NA	NA
Interest-bearing						
Demand	**346,179**	321,976	324,740	**3.13**	2.71	0.57
Time	**94,871**	86,443	65,604	**5.86**	5.82	1.85
Total interest-bearing deposits	**441,050**	408,419	390,344	**3.72**	3.37	0.78
Total deposits in non-U.S. offices	**467,908**	433,166	418,387	**3.50**	3.18	0.73
Total deposits	**$ 2,386,642**	$ 2,359,067	$ 2,467,915	**2.08 %**	1.70 %	0.41 %

(a) Includes Negotiable Order of Withdrawal accounts, and certain trust accounts.
(b) Includes Money Market Deposit Accounts.

Refer to Note 17 for additional information on deposits.

The following table summarizes short-term and long-term funding, excluding deposits, as of December 31, 2024 and 2023, and average balances for the years ended December 31, 2024 and 2023. Refer to the Consolidated Balance Sheets Analysis on pages 63–65 and Note 11 for additional information.

Sources of funds (excluding deposits)

As of or for the year ended December 31,			Average	
(in millions)	**2024**	2023	**2024**	2023
Commercial paper	**$ 14,932**	$ 14,737	**$ 11,398**	$ 12,675
Other borrowed funds	**13,018**	8,200	**12,040**	9,712
Federal funds purchased	**567**	787	**1,547**	1,754
Total short-term unsecured funding	**$ 28,517**	$ 23,724	**$ 24,985**	$ 24,141
Securities sold under agreements to repurchase[a]	**$ 291,500**	$ 212,804	**$ 357,144**	$ 249,661
Securities loaned[a]	**4,768**	2,944	**5,129**	4,671
Other borrowed funds	**24,943**	21,775	**25,504**	22,010
Obligations of Firm-administered multi-seller conduits[b]	**18,228**	17,781	**18,620**	14,918
Total short-term secured funding	**$ 339,439**	$ 255,304	**$ 406,397**	$ 291,260
Senior notes	**$ 203,639**	$ 191,202	**$ 199,908**	$ 181,803
Subordinated debt	**16,060**	19,708	**18,614**	20,374
Structured notes[c]	**98,792**	86,056	**93,483**	76,574
Total long-term unsecured funding	**$ 318,491**	$ 296,966	**$ 312,005**	$ 278,751
Credit card securitization[b]	**$ 5,312**	$ 2,998	**$ 5,138**	$ 1,634
FHLB advances	**29,257**	41,246	**35,040** [g]	28,865
Purchase Money Note[d]	**49,207**	$ 48,989	**$ 49,090**	$ 32,829
Other long-term secured funding[e]	**4,463**	4,624	**4,676**	4,513
Total long-term secured funding	**$ 88,239**	$ 97,857	**$ 93,944**	$ 67,841
Preferred stock[f]	**$ 20,050**	$ 27,404	**$ 24,054**	$ 27,404
Common stockholders' equity[f]	**$ 324,708**	$ 300,474	**$ 312,370**	$ 282,056

(a) Primarily consists of short-term securities loaned or sold under agreements to repurchase.
(b) Included in beneficial interests issued by consolidated variable interest entities on the Firm's Consolidated balance sheets.
(c) Includes certain TLAC-eligible long-term unsecured debt issued by the Parent Company.
(d) Reflects the Purchase Money Note associated with the First Republic acquisition on May 1, 2023. Refer to Note 34 for additional information.
(e) Includes long-term structured notes that are secured.
(f) Refer to Capital Risk Management on pages 97–107, Consolidated statements of changes in stockholders' equity on page 175, Note 21 and Note 22 for additional information on preferred stock and common stockholders' equity.
(g) Includes the timing impact of First Republic. Refer to the Executive Overview on pages 54–58 and Note 34 for additional information.

Short-term funding

The Firm's primary source of short-term secured funding is securities sold under agreements to repurchase. These instruments are secured predominantly by high-quality securities collateral, including government-issued debt and U.S. GSE and government agency MBS. Securities sold under agreements to repurchase increased at December 31, 2024, compared with December 31, 2023, driven by Markets, reflecting higher client-driven market-making activities and higher secured financing of trading assets.

The increase in secured other borrowed funds at December 31, 2024 from December 31, 2023, as well as the increase for the average year ended December 31, 2024, compared to the prior year period, were both due to higher financing requirements in Markets, partially offset by FHLB maturities in Treasury and CIO.

The balances associated with securities loaned or sold under agreements to repurchase fluctuate over time due to investment and financing activities of clients, the Firm's demand for financing, the ongoing management of the mix of the Firm's liabilities, including its secured and unsecured financing (for both the investment securities and market-making portfolios), and other market and portfolio factors.

The Firm's primary sources of short-term unsecured funding consist of issuances of wholesale commercial paper and other borrowed funds.

The decrease in average commercial paper for the year ended December 31, 2024 compared to the prior year period was due to lower issuances primarily as a result of short-term liquidity management.

The increase in unsecured other borrowed funds at December 31, 2024 from December 31, 2023, was predominantly driven by net issuances of structured notes in Markets.

Long-term funding

Long-term funding provides an additional source of stable funding and liquidity for the Firm. The Firm's long-term funding plan is driven primarily by expected client activity, liquidity considerations and regulatory requirements, including TLAC. Long-term funding objectives include maintaining diversification, maximizing market access and optimizing funding costs. The Firm evaluates various funding markets, tenors and currencies in creating its optimal long-term funding plan.

Unsecured funding and issuance

The significant majority of the Firm's total outstanding long-term debt has been issued by the Parent Company to provide flexibility in support of the funding needs of both bank and non-bank subsidiaries. The Parent Company advances substantially all net funding proceeds to its subsidiary, the IHC. The IHC does not issue debt to external counterparties. The increase in structured notes at December 31, 2024 from December 31, 2023, and for the average year ended December 31, 2024, compared to the prior year period, was primarily driven by net issuances of structured notes in Markets due to client demand.

The following table summarizes long-term unsecured issuance and maturities or redemptions for the years ended December 31, 2024 and 2023. Refer to Note 20 for additional information on the IHC and long-term debt.

Long-term unsecured funding

Year ended December 31,	2024	2023	2024	2023
(Notional in millions)	Parent Company		Subsidiaries	
Issuance				
Senior notes issued in the U.S. market	**$ 37,000**	$ 14,256	**$ —**	$ 3,750
Senior notes issued in non-U.S. markets	**4,079**	2,141	**—**	—
Total senior notes	**41,079**	16,397	**—**	3,750
Structured notes[(a)]	**3,944**	3,013	**54,993**	35,281
Total long-term unsecured funding – issuance	**$ 45,023**	$ 19,410	**$ 54,993**	$ 39,031
Maturities/redemptions				
Senior notes	**$ 25,765**	$ 21,483	**$ 65**	$ 67
Subordinated debt	**3,097**	2,090	**250**	—
Structured notes	**892**	1,532	**47,425**	28,777
Total long-term unsecured funding – maturities/redemptions	**$ 29,754**	$ 25,105	**$ 47,740**	$ 28,844

(a) Includes certain TLAC-eligible long-term unsecured debt issued by the Parent Company.

Secured funding and issuance

The Firm can also raise secured long-term funding through securitization of consumer credit card loans and FHLB advances. The following table summarizes the securitization issuance, the FHLB advances and their respective maturities or redemptions, as applicable for the years ended December 31, 2024 and 2023.

Long-term secured funding

Year ended December 31,	Issuance		Maturities/Redemptions	
(in millions)	**2024**	2023	**2024**	2023
Credit card securitization	**$ 2,348**	$ 1,998	**$ —**	$ 1,000
FHLB advances	**6,000**	39,775 [(c)]	**18,050**	9,485
Purchase Money Note[(a)]	**—**	50,000	**—**	$ —
Other long-term secured funding[(b)]	**1,578**	991	**1,049**	432
Total long-term secured funding	**$ 9,926**	$ 92,764	**$ 19,099**	$ 10,917

(a) Reflects the Purchase Money Note associated with the First Republic acquisition. Refer to Note 34 for additional information.
(b) Includes long-term structured notes that are secured.
(c) Includes FHLB advances associated with the First Republic acquisition on May 1, 2023. Refer to Note 34 for additional information.

The Firm's wholesale businesses also securitize loans for client-driven transactions; those client-driven loan securitizations are not considered to be a source of funding for the Firm and are not included in the table above. Refer to Note 14 for a further description of client-driven loan securitizations.

Credit ratings

The cost and availability of financing are influenced by credit ratings. Reductions in these ratings could have an adverse effect on the Firm's access to liquidity sources, increase the cost of funds, trigger additional collateral or funding requirements and decrease the number of investors and counterparties willing to lend to the Firm. The nature and magnitude of the impact of ratings downgrades depends on numerous contractual and behavioral factors, which the Firm believes are incorporated in its liquidity risk and stress testing metrics. The Firm believes that it maintains sufficient liquidity to withstand a potential decrease in funding capacity due to ratings downgrades.

Additionally, the Firm's funding requirements for VIEs and other third-party commitments may be adversely affected by a decline in credit ratings. Refer to Notes 5 and 14 for additional information.

The credit ratings of the Parent Company and certain of its principal subsidiaries as of December 31, 2024 were as follows:

	JPMorgan Chase & Co.			JPMorgan Chase Bank, N.A.			J.P. Morgan Securities LLC J.P. Morgan Securities plc J.P. Morgan SE[(a)]		
December 31, 2024	Long-term issuer	Short-term issuer	Outlook	Long-term issuer	Short-term issuer	Outlook	Long-term issuer	Short-term issuer	Outlook
Moody's Investors Service[(a)]	A1	P-1	Positive	Aa2	P-1	Developing	Aa3	P-1	Positive
Standard & Poor's[(b)]	A	A-1	Stable	AA-	A-1+	Stable	AA-	A-1+	Stable
Fitch Ratings	AA-	F1+	Stable	AA	F1+	Stable	AA	F1+	Stable

(a) On November 11, 2024, Moody's (i) affirmed the credit ratings of the Parent Company, JPMorgan Chase Bank, N.A. and the other subsidiaries listed above; (ii) revised its outlook for the Parent Company, J.P. Morgan Securities LLC and J.P. Morgan Securities plc from stable to positive; (iii) revised its outlook for JPMorgan Chase Bank, N.A. from negative to developing, reflecting its view with respect to possible support from the U.S. government; and (iv) assessed its outlook for J.P. Morgan SE as negative with an "(m)" modifier, reflecting a negative outlook for long-term bank deposits and a positive outlook for the long-term issuer rating.

(b) The credit ratings of the Parent Company, JPMorgan Chase Bank, N.A. and the other subsidiaries presented in the table reflect ratings upgrades by Standard & Poor's on November 15, 2024. Standard & Poor's also revised its outlook for the Parent Company and such subsidiaries from positive to stable.

JPMorganChase's unsecured debt does not contain requirements that would call for an acceleration of payments, maturities or changes in the structure of the existing debt, provide any limitations on future borrowings or require additional collateral, based on unfavorable changes in the Firm's credit ratings, financial ratios, earnings, or stock price.

Critical factors in maintaining high credit ratings include a stable and diverse earnings stream, strong capital and liquidity ratios, strong credit quality and risk management controls, and diverse funding sources. Rating agencies continue to evaluate economic and geopolitical trends, regulatory developments, future profitability, risk management practices, and litigation matters, as well as their broader ratings methodologies. Changes in any of these factors could lead to changes in the Firm's credit ratings.

Reputation risk is the risk that an action or inaction may negatively impact perception of the Firm's integrity and reduce confidence in the Firm's competence by various stakeholders, including clients, counterparties, customers, communities, investors, regulators, or employees.

The types of events that may result in reputation risk are wide-ranging and can be introduced by the Firm's employees, business strategies and activities, clients, customers and counterparties with which the Firm does business. These events could contribute to financial losses, litigation, regulatory enforcement actions, fines, penalties or other sanctions, as well as other harm to the Firm.

Organization and management

Reputation Risk Management is an independent risk management function that establishes the governance framework for managing reputation risk across the Firm's LOBs and Corporate. Reputation risk is inherently challenging to identify, manage, and quantify.

The Firm's reputation risk management function includes the following activities:

- Maintaining a Firmwide Reputation Risk Governance policy and a standard consistent with the reputation risk framework
- Providing oversight of the governance framework through processes and infrastructure to support consistent identification, escalation and monitoring of reputation risk issues Firmwide

Governance and oversight

The Reputation Risk Governance policy establishes the principles for managing reputation risk for the Firm. It is the responsibility of each LOB and Corporate, and the Firm's employees, to consider the reputation of the Firm when deciding whether to offer a new product, engage in a transaction or client relationship, enter a new jurisdiction, initiate a business process or consider any other activity. Environmental impacts and social concerns are important considerations in assessing the Firm's reputation risk, and are a component of the Firm's reputation risk governance.

Credit and investment risk is the risk associated with the default or change in credit profile of a client, counterparty or customer; or loss of principal or a reduction in expected returns on investments, including consumer credit risk, wholesale credit risk, and investment portfolio risk.

Credit risk management

Credit risk is the risk associated with the default or change in credit profile of a client, counterparty or customer. The Firm provides credit to a variety of clients and customers, ranging from large corporate and institutional clients to individual consumers and small businesses. In its consumer businesses, the Firm is exposed to credit risk primarily through its home lending, credit card, auto, and business banking businesses. In its wholesale businesses, the Firm is exposed to credit risk through its underwriting, lending, market-making, and hedging activities with and for clients and counterparties, as well as through its operating services activities (such as cash management and clearing activities), and securities financing activities. The Firm is also exposed to credit risk through its investment securities portfolio and cash placed with banks.

Credit Risk Management monitors and measures credit risk throughout the Firm, and defines credit risk policies, procedures and limits. The Firm's credit risk management governance includes the following activities:

- Maintaining a credit risk policy framework
- Monitoring and measuring credit risk across all portfolio segments, including transaction and exposure approval
- Setting industry and geographic concentration limits, as appropriate, and setting guidelines for credit review and analysis
- Assigning and maintaining credit approval authorities in connection with the approval of credit exposure
- Monitoring and independent assessment of criticized exposures and delinquent loans, and
- Estimating credit losses and supporting appropriate credit risk-based capital management

Risk identification and measurement

To measure credit risk, the Firm employs several methodologies for estimating the likelihood of obligor or counterparty default. Methodologies for measuring credit risk vary depending on several factors, including type of asset (e.g., consumer versus wholesale), risk measurement parameters (e.g., delinquency status and borrower's credit score versus wholesale risk-rating) and risk management and collection processes (e.g., retail collection center versus centrally managed workout groups). Credit risk measurement is based on the probability of default of an obligor or counterparty, the loss severity given a default event and the exposure at default.

Based on these factors and the methodology and estimates described in Note 13 and Note 10, the Firm estimates credit losses for its exposures. The allowance for loan losses reflects estimated credit losses related to the consumer and wholesale held-for-investment loan portfolios, the allowance for lending-related commitments reflects estimated credit losses related to the Firm's lending-related commitments and the allowance for investment securities reflects estimated credit losses related to the investment securities portfolio. Refer to Note 13, Note 10 and Critical Accounting Estimates used by the Firm on pages 161–164 for further information.

In addition, potential and unexpected credit losses are reflected in the allocation of credit risk capital and represent the potential volatility of actual losses relative to the established allowances for loan losses and lending-related commitments. The analyses for these losses include stress testing that considers alternative economic scenarios as described below.

Stress testing

Stress testing is important in measuring and managing credit risk in the Firm's credit portfolio. The stress testing process assesses the potential impact of alternative economic and business scenarios on estimated credit losses for the Firm. Economic scenarios and the underlying parameters are defined centrally, articulated in terms of macroeconomic factors and applied across the businesses. The stress test results may indicate credit migration, changes in delinquency trends and potential losses in the credit portfolio. In addition to the periodic stress testing processes, management also considers additional stresses outside these scenarios, including industry and country- specific stress scenarios, as appropriate. The Firm uses stress testing to inform decisions on setting risk appetite both at a Firm and LOB level, as well as to assess the impact of stress on individual counterparties.

Risk monitoring and management

The Firm has developed policies and practices that are designed to preserve the independence and integrity of the approval and decision-making process for extending credit so that credit risks are assessed accurately, approved properly, monitored regularly and managed actively at both the transaction and portfolio levels. The policy framework establishes credit approval authorities, concentration limits, risk-rating methodologies, portfolio review parameters and guidelines for management of distressed exposures. In addition, certain models, assumptions and inputs used in evaluating and monitoring credit risk are independently validated by groups that are separate from the LOBs.

Consumer credit risk is monitored for delinquency and other trends, including any concentrations at the portfolio level, as certain of these trends can be addressed through changes in underwriting policies and portfolio guidelines. Consumer Risk Management evaluates delinquency and other trends against business expectations, current and forecasted economic conditions, and industry benchmarks. Historical and forecasted economic performance and trends are incorporated into the modeling of estimated consumer credit losses and are part of the monitoring of the credit risk profile of the portfolio.

Wholesale credit risk is monitored regularly at an aggregate portfolio, industry, and individual client and counterparty level with established concentration limits that are reviewed and revised periodically as deemed appropriate by management. Industry and counterparty limits, as measured in terms of exposure and economic risk appetite, are subject to stress-based loss constraints.

Management of the Firm's wholesale credit risk exposure is accomplished through a number of means, including:

- Loan underwriting and credit approval processes
- Loan syndications and participations
- Loan sales and securitizations
- Credit derivatives
- Master netting agreements, and
- Collateral and other risk-reduction techniques

In addition to Credit Risk Management, an independent Credit Review function is responsible for:

- Independently assessing risk grades assigned to exposures in the Firm's wholesale credit portfolio and the timeliness of risk grade changes initiated by responsible business units; and
- Evaluating the effectiveness of the credit management processes of the LOBs and Corporate, including the adequacy of credit analyses and risk grading/loss given default ("LGD") rationales, proper monitoring and management of credit exposures, and compliance with applicable grading policies and underwriting guidelines.

Refer to Note 12 for further discussion of consumer and wholesale loans.

Risk reporting

To enable monitoring of credit risk and effective decision-making, aggregate credit exposure, credit quality forecasts, concentration levels and risk profile changes are reported regularly to senior members of Credit Risk Management. Detailed portfolio reporting of industry, clients, counterparties and customers, product and geography are prepared, and the appropriateness of the allowance for credit losses is reviewed by senior management at least on a quarterly basis. Through the risk reporting and governance structure, credit risk trends and limit exceptions are provided regularly to, and discussed with, risk committees, senior management and the Board of Directors.

Credit risk is the risk associated with the default or change in credit profile of a client, counterparty or customer.

In the following tables, total loans include loans retained (i.e., held-for-investment); loans held-for-sale; and certain loans accounted for at fair value. The following tables do not include loans which the Firm accounts for at fair value and classifies as trading assets; refer to Notes 2 and 3 for further information regarding these loans. Refer to Notes 12, 28, and 5 for additional information on the Firm's loans, lending-related commitments and derivative receivables, including the Firm's related accounting policies.

Refer to Note 10 for information regarding the credit risk inherent in the Firm's investment securities portfolio; and refer to Note 11 for information regarding credit risk inherent in the securities financing portfolio. Refer to Consumer Credit Portfolio on pages 120–125 and Note 12 for further discussions of the consumer credit environment, consumer loans and nonperforming exposure. Refer to Wholesale Credit Portfolio on pages 126–136 and Note 12 for further discussions of the wholesale credit environment and wholesale loans.

Total credit portfolio

December 31, (in millions)	Credit exposure		Nonperforming[c]	
	2024	2023	**2024**	2023
Loans retained	**$1,299,590**	$1,280,870	**$ 7,175**	$ 5,989
Loans held-for-sale	**7,048**	3,985	**160**	184
Loans at fair value	**41,350**	38,851	**1,502**	744
Total loans	**1,347,988**	1,323,706	**8,837**	6,917
Derivative receivables	**60,967**	54,864	**145**	364
Receivables from customers[a]	**51,929**	47,625	**—**	—
Total credit-related assets	**1,460,884**	1,426,195	**8,982**	7,281
Assets acquired in loan satisfactions				
Real estate owned	**NA**	NA	**284**	274
Other	**NA**	NA	**34**	42
Total assets acquired in loan satisfactions	**NA**	NA	**318**	316
Lending-related commitments	**1,577,622**	1,497,847	**737**	464
Total credit portfolio	**$3,038,506**	$2,924,042	**$10,037**	$ 8,061
Credit derivatives and credit-related notes used in credit portfolio management activities[b]	**$ (41,367)**	$ (37,779)	**$ —**	$ —
Liquid securities and other cash collateral held against derivatives	**(28,160)**	(22,461)	**NA**	NA

(a) Receivables from customers reflect held-for-investment margin loans to brokerage clients in CIB, CCB and AWM; these are reported within accrued interest and accounts receivable on the Consolidated balance sheets.

(b) Represents the net notional amount of protection purchased and sold through credit derivatives and credit-related notes used to manage credit exposures.

(c) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024 and 2023, mortgage loans 90 or more days past due and insured by U.S. government agencies were $121 million and $182 million, respectively. In addition, the Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance.

The following table provides information on Firmwide nonaccrual loans to total loans.

December 31, (in millions, except ratios)	**2024**	2023
Total nonaccrual loans	**$ 8,837**	$ 6,917
Total loans	**1,347,988**	1,323,706
Firmwide nonaccrual loans to total loans outstanding	**0.66 %**	0.52 %

The following table provides information about the Firm's net charge-offs and recoveries.

December 31, (in millions, except ratios)	**2024**	2023
Net charge-offs	**$ 8,638**	$ 6,209
Average retained loans	**1,271,344**	1,202,348
Net charge-off rates	**0.68 %**	0.52 %

CONSUMER CREDIT PORTFOLIO

The Firm's retained consumer portfolio consists primarily of loans and lending-related commitments for residential real estate, credit card, scored auto and business banking. The consumer credit portfolio also includes loans at fair value, predominantly in residential real estate. The Firm's focus is on serving primarily the prime segment of the consumer credit market. Originated mortgage loans are retained in the residential real estate portfolio, securitized or sold to U.S. government agencies and U.S. government-sponsored enterprises; other types of consumer loans are typically retained on the balance sheet. Refer to Note 12 for further information on the consumer loan portfolio. Refer to Note 28 for further information on lending-related commitments.

The following tables present consumer credit-related information with respect to the scored credit portfolio held in CCB, AWM, CIB and Corporate.

Consumer credit portfolio

December 31, (in millions)	Credit exposure		Nonaccrual loans[(i)]	
	2024	2023	**2024**	2023
Consumer, excluding credit card				
Residential real estate[(a)]	**$ 309,513**	$ 326,409	**$ 2,984**	$ 3,466
Auto and other[(b)(c)]	**66,821**	70,866	**249**	177
Total loans - retained	**376,334**	397,275	**3,233**	3,643
Loans held-for-sale	**945**	487	**155**	95
Loans at fair value[(d)]	**15,531**	12,331	**538**	465
Total consumer, excluding credit card loans	**392,810**	410,093	**3,926**	4,203
Lending-related commitments[(e)]	**44,844**	45,403		
Total consumer exposure, excluding credit card	**437,654**	455,496		
Credit card				
Loans retained[(f)]	**232,860**	211,123	**NA**	NA
Total credit card loans	**232,860**	211,123	**NA**	NA
Lending-related commitments[(e)(g)]	**1,001,311**	915,658		
Total credit card exposure	**1,234,171**	1,126,781		
Total consumer credit portfolio	**$ 1,671,825**	$ 1,582,277	**$ 3,926**	$ 4,203
Credit-related notes used in credit portfolio management activities[(h)]	**$ (479)**	$ (790)		

(in millions, except ratios)	Year ended December 31, Net charge-offs/ (recoveries)		Average loans - retained		Net charge-off/(recovery) rate[(j)]	
	2024	2023	**2024**	2023	**2024**	2023
Consumer, excluding credit card						
Residential real estate	**$ (101)**	$ (52)	**$ 316,042**	$ 296,515	**(0.03)%**	(0.02)%
Auto and other	**775**	684	**67,959**	67,546	**1.14**	1.01
Total consumer, excluding credit card - retained	**674**	632	**384,001**	364,061	**0.18**	0.17
Credit card - retained	**7,142**	4,698	**214,033**	191,412	**3.34**	2.45
Total consumer - retained	**$ 7,816**	$ 5,330	**$ 598,034**	$ 555,473	**1.31 %**	0.96 %

(a) Includes scored mortgage and home equity loans held in CCB and AWM.
(b) At December 31, 2024 and 2023, excluded operating lease assets of $12.8 billion and $10.4 billion, respectively. These operating lease assets are included in other assets on the Firm's Consolidated balance sheets. Refer to Note 18 for further information.
(c) Includes scored auto and business banking loans, and overdrafts.
(d) Includes scored mortgage loans held in CCB and CIB, and other consumer unsecured loans in CIB.
(e) Credit card, home equity and certain business banking lending-related commitments represent the total available lines of credit for these products. The Firm has not experienced, and does not anticipate, that all available lines of credit would be used at the same time. Refer to Note 28 for further information.
(f) Includes billed interest and fees.
(g) Also includes commercial card lending-related commitments primarily in CIB.
(h) Represents the notional amount of protection obtained through the issuance of credit-related notes that reference certain pools of residential real estate and auto loans in the retained consumer portfolio.
(i) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024 and 2023, mortgage loans 90 or more days past due and insured by U.S. government agencies were $121 million and $182 million, respectively. In addition, the Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status, as permitted by regulatory guidance.
(j) Average consumer loans held-for-sale and loans at fair value were $17.2 billion and $12.9 billion for the years ended December 31, 2024 and 2023, respectively. These amounts were excluded when calculating net charge-off/(recovery) rates.

Maturities and sensitivity to changes in interest rates

The table below sets forth loan maturities by scheduled repayments, by class of loan and the distribution between fixed and floating interest rates based on the stated terms of the loan agreements. The Firm estimated the principal repayment amounts for both the residential real estate and auto and other loan classes by calculating the weighted-average loan balance and interest rates for loan pools based on remaining loan term. Refer to Note 12 for further information on loan classes.

December 31, 2024 (in millions)	Within 1 year[(a)]	1-5 years	5-15 years	After 15 years	Total
Consumer, excluding credit card					
Residential real estate	**$ 21,442**	**$ 26,712**	**$109,608**	**$ 166,715**	**$324,477**
Auto and other	**19,404** [(b)]	**43,701**	**5,224**	**4**	**68,333**
Total consumer, excluding credit card loans	**$ 40,846**	**$ 70,413**	**$ 114,832**	**$ 166,719**	**$392,810**
Total credit card loans	**$231,799**	**$ 1,048**	**$ 13**	**$ —**	**$232,860**
Total consumer loans	**$272,645**	**$ 71,461**	**$ 114,845**	**$ 166,719**	**$625,670**
Loans due after one year at fixed interest rates					
Residential real estate		**$ 19,639**	**$ 57,351**	**$ 77,865**	
Auto and other		**43,565**	**2,957**	**4**	
Credit card		**1,048**	**13**	**—**	
Loans due after one year at variable interest rates					
Residential real estate		**$ 7,073**	**$ 52,257**	**$ 88,850**	
Auto and other		**136**	**2,267**	**—**	
Total consumer loans		**$ 71,461**	**$ 114,845**	**$ 166,719**	

(a) Includes loans held-for-sale and loans at fair value.
(b) Includes overdrafts.

Consumer, excluding credit card

Portfolio analysis

Loans decreased from December 31, 2023 driven by residential real estate loans and scored auto loans.

The following discussions provide information concerning individual loan products. Refer to Note 12 for further information about this portfolio, including information about delinquencies, loan modifications and other credit quality indicators.

Residential real estate: The residential real estate portfolio, including loans held-for-sale and loans at fair value, predominantly consists of prime mortgage loans and home equity lines of credit.

Retained loans decreased compared to December 31, 2023, predominantly driven by paydowns and loan sales, net of originations. Retained nonaccrual loans decreased compared to December 31, 2023, predominantly driven by loan sales. Net recoveries were higher for the year ended December 31, 2024 compared to the prior year, driven by loan sales.

Loans held-for-sale and nonaccrual loans held-for-sale increased from December 31, 2023, predominantly driven by transfers of certain retained loans in anticipation of securitization and loan sales, respectively.

Loans at fair value increased from December 31, 2023, predominantly driven by higher Home Lending loans, as originations outpaced warehouse loan sales. Nonaccrual loans at fair value increased compared to December 31, 2023, driven by CIB.

At December 31, 2024 and 2023, the carrying values of retained interest-only residential mortgage loans were $88.9 billion and $90.6 billion, respectively. These loans have an interest-only payment period generally followed by an adjustable-rate or fixed-rate fully amortizing payment period to maturity and are typically originated as higher-balance loans to higher-income borrowers. The credit performance of this portfolio is comparable to the performance of the broader prime mortgage portfolio.

The carrying value of retained home equity lines of credit outstanding was $14.5 billion at December 31, 2024, including $3.8 billion of HELOCs that have recast from interest-only to fully amortizing payments or have been modified, and $3.6 billion of interest-only balloon HELOCs, which primarily mature after 2030. The Firm manages the risk of HELOCs during their revolving period by reducing or canceling the undrawn line in accordance with the contract or to the extent otherwise permitted by law, including when there has been a demonstrable decline in the creditworthiness of the borrower or significant decrease in the value of the underlying property.

The following table provides a summary of the Firm's residential mortgage portfolio insured and/or guaranteed by U.S. government agencies, predominantly loans held-for-sale and loans at fair value. The Firm monitors its exposure to certain potential unrecoverable claim payments related to government-insured loans and considers this exposure in estimating the allowance for loan losses.

(in millions)	**December 31, 2024**	December 31, 2023
Current	**$ 462**	$ 446
30-89 days past due	**72**	102
90 or more days past due	**121**	182
Total government guaranteed loans	**$ 655**	$ 730

Geographic composition and current estimated loan-to-value ratio of residential real estate loans

At December 31, 2024, $217.7 billion, or 70% of the total retained residential real estate loan portfolio, was concentrated in California, New York, Florida, Texas and Massachusetts, compared to $228.4 billion, or 70% at December 31, 2023.

Average current estimated loan-to-value ("LTV") ratios have improved, reflecting an increase in home prices.

Refer to Note 12 for information on the geographic composition and current estimated LTVs of the Firm's residential real estate loans.

Auto and other: The auto and other loan portfolio, including loans at fair value, generally consists of prime-quality scored auto and business banking loans, other consumer unsecured loans, and overdrafts. The portfolio decreased when compared to December 31, 2023, predominantly due to loan securitizations. Net charge-offs increased compared to the prior year, predominantly due to net charge-offs of scored auto loans of $445 million compared to $357 million for the year ended December 31, 2023, reflecting a decline in used vehicle valuations. Refer to Note 14 for further information on securitization activity.

Nonperforming assets

The following table presents information as of December 31, 2024 and 2023, about consumer, excluding credit card, nonperforming assets.

Nonperforming assets[(a)]

December 31, (in millions)	2024	2023
Nonaccrual loans		
Residential real estate	$ 3,665	$ 4,015
Auto and other	261	188
Total nonaccrual loans	3,926	4,203
Assets acquired in loan satisfactions		
Real estate owned	78	120
Other	34	42
Total assets acquired in loan satisfactions	112	162
Total nonperforming assets	$ 4,038	$ 4,365

(a) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024 and 2023, mortgage loans 90 or more days past due and insured by U.S. government agencies were $121 million and $182 million, respectively.

Nonaccrual loans

The following table presents changes in consumer, excluding credit card, nonaccrual loans for the years ended December 31, 2024 and 2023.

Nonaccrual loan activity

Year ended December 31, (in millions)	2024	2023
Beginning balance	$ 4,203	$ 4,325
Additions:	3,225	2,894
Reductions:		
Principal payments and other	894	1,030
Sales	803	276
Charge-offs	665	472
Returned to performing status	963	1,052
Foreclosures and other liquidations	177	186
Total reductions	3,502	3,016
Net changes	(277)	(122)
Ending balance	$ 3,926	$ 4,203

Refer to Note 12 for further information about the consumer credit portfolio, including information about delinquencies, other credit quality indicators and loans that were in the process of active or suspended foreclosure.

Credit card

Total credit card loans increased from December 31, 2023 reflecting growth from new accounts and revolving balances. The December 31, 2024 30+ and 90+ day delinquency rates of 2.17% and 1.14%, respectively, increased compared to the December 31, 2023 30+ and 90+ day delinquency rates of 2.14% and 1.05%, respectively, in line with the Firm's expectations. Net charge-offs increased for the year ended December 31, 2024 compared to the prior year reflecting the seasoning of vintages originated in recent years, credit normalization and balance growth.

Consistent with the Firm's policy, all credit card loans typically remain on accrual status until charged off. However, the Firm's allowance for loan losses includes the estimated uncollectible portion of accrued and billed interest and fee income.

Geographic and FICO composition of credit card loans

At December 31, 2024, $109.0 billion, or 47% of the total retained credit card loan portfolio, was concentrated in California, Texas, New York, Florida and Illinois, compared to $98.1 billion, or 46%, at December 31, 2023.

Refer to Note 12 for further information about this portfolio, including information about delinquencies, geographic and FICO composition.

WHOLESALE CREDIT PORTFOLIO

In its wholesale businesses, the Firm is exposed to credit risk primarily through its underwriting, lending, market-making, and hedging activities with and for clients and counterparties, as well as through various operating services (such as cash management and clearing activities), securities financing activities and cash placed with banks. A portion of the loans originated or acquired by the Firm's wholesale businesses is generally retained on the balance sheet. The Firm distributes a significant percentage of the loans that it originates into the market as part of its syndicated loan business and to manage portfolio concentrations and credit risk. The wholesale portfolio is actively managed, in part by conducting ongoing, in-depth reviews of client credit quality and transaction structure, inclusive of collateral where applicable, and of industry, product and client concentrations. Refer to the industry discussion on pages 128–131 for further information.

The Firm's wholesale credit portfolio includes exposure held in CIB, AWM and Corporate, and risk-rated exposure held in CCB, for which the wholesale methodology is applied when determining the allowance for loan losses.

As of December 31, 2024, loans increased $19.8 billion, driven by higher loans in CIB and higher securities-based lending in AWM. Lending-related commitments decreased $5.3 billion, with decreases in AWM and CCB, largely offset by higher commitments in CIB.

As of December 31, 2024, nonperforming exposure increased by $2.3 billion, predominantly driven by Real Estate, concentrated in Office, Healthcare and Consumer & Retail, in each case resulting from downgrades.

For the year ended December 31, 2024, wholesale net charge-offs were $822 million, largely driven by Real Estate, concentrated in Office, and client-specific charge-offs across multiple industries including Consumer & Retail and Individuals.

Wholesale credit portfolio

December 31, (in millions)	Credit exposure		Nonperforming	
	2024	2023	**2024**	2023
Loans retained	**$ 690,396**	$ 672,472	**$ 3,942**	$ 2,346
Loans held-for-sale	**6,103**	3,498	**5**	89
Loans at fair value	**25,819**	26,520	**964**	279
Loans	**722,318**	702,490	**4,911**	2,714
Derivative receivables	**60,967**	54,864	**145**	364
Receivables from customers[(a)]	**51,929**	47,625	**—**	—
Total wholesale credit-related assets	**835,214**	804,979	**5,056**	3,078
Assets acquired in loan satisfactions				
Real estate owned	**NA**	NA	**206**	154
Other	**NA**	NA	**—**	—
Total assets acquired in loan satisfactions	**NA**	NA	**206**	154
Lending-related commitments	**531,467**	536,786	**737**	464
Total wholesale credit portfolio	**$1,366,681**	$1,341,765	**$ 5,999**	$ 3,696
Credit derivatives and credit-related notes used in credit portfolio management activities[(b)]	**$ (40,888)**	$ (36,989)	**$ —**	$ —
Liquid securities and other cash collateral held against derivatives	**(28,160)**	(22,461)	**NA**	NA

(a) Receivables from customers reflect held-for-investment margin loans to brokerage clients in CIB, CCB and AWM; these are reported within accrued interest and accounts receivable on the Consolidated balance sheets.

(b) Represents the net notional amount of protection purchased and sold through credit derivatives and credit-related notes used to manage both performing and nonperforming wholesale credit exposures; these derivatives do not qualify for hedge accounting under U.S. GAAP. Refer to Credit derivatives on page 136 and Note 5 for additional information.

Wholesale credit exposure – maturity and ratings profile

The following tables present the maturity and internal risk ratings profiles of the wholesale credit portfolio as of December 31, 2024 and 2023. The Firm generally considers internal ratings with qualitative characteristics equivalent to BBB-/Baa3 or higher as investment grade, and takes into consideration collateral and structural support when determining the internal risk rating for each credit facility. Refer to Note 12 for further information on internal risk ratings.

	Maturity profile[d]				Ratings profile			
December 31, 2024 (in millions, except ratios)	1 year or less	After 1 year through 5 years	After 5 years	Total	Investment-grade	Noninvestment-grade	Total	Total % of IG
Loans retained	**$ 225,982**	**$ 289,199**	**$ 175,215**	**$ 690,396**	**$ 471,670**	**$ 218,726**	**$ 690,396**	**68 %**
Derivative receivables				**60,967**			**60,967**	
Less: Liquid securities and other cash collateral held against derivatives				**(28,160)**			**(28,160)**	
Total derivative receivables, net of collateral	**11,515**	**7,418**	**13,874**	**32,807**	**24,707**	**8,100**	**32,807**	**75**
Lending-related commitments	**121,283**	**384,529**	**25,655**	**531,467**	**352,082**	**179,385**	**531,467**	**66**
Subtotal	**358,780**	**681,146**	**214,744**	**1,254,670**	**848,459**	**406,211**	**1,254,670**	**68**
Loans held-for-sale and loans at fair value[a]				**31,922**			**31,922**	
Receivables from customers				**51,929**			**51,929**	
Total exposure – net of liquid securities and other cash collateral held against derivatives				**$1,338,521**			**$ 1,338,521**	
Credit derivatives and credit-related notes used in credit portfolio management activities[b][c]	**$ (5,442)**	**$ (33,751)**	**$ (1,695)**	**$ (40,888)**	**$ (31,691)**	**$ (9,197)**	**$ (40,888)**	**78 %**

	Maturity profile[d]				Ratings profile			
December 31, 2023 (in millions, except ratios)	1 year or less	After 1 year through 5 years	After 5 years	Total	Investment-grade	Noninvestment-grade	Total	Total % of IG
Loans retained	$ 211,104	$ 280,821	$ 180,547	$ 672,472	$ 458,838	$ 213,634	$ 672,472	68 %
Derivative receivables				54,864			54,864	
Less: Liquid securities and other cash collateral held against derivatives				(22,461)			(22,461)	
Total derivative receivables, net of collateral	8,007	8,970	15,426	32,403	24,919	7,484	32,403	77
Lending-related commitments	143,337	368,646	24,803	536,786	341,611	195,175	536,786	64
Subtotal	362,448	658,437	220,776	1,241,661	825,368	416,293	1,241,661	66
Loans held-for-sale and loans at fair value[a]				30,018			30,018	
Receivables from customers				47,625			47,625	
Total exposure – net of liquid securities and other cash collateral held against derivatives				$1,319,304			$ 1,319,304	
Credit derivatives and credit-related notes used in credit portfolio management activities[b][c]	$ (3,311)	$ (28,353)	$ (5,325)	$ (36,989)	$ (28,869)	$ (8,120)	$ (36,989)	78 %

(a) Loans held-for-sale are primarily related to syndicated loans and loans transferred from the retained portfolio.
(b) These derivatives do not qualify for hedge accounting under U.S. GAAP.
(c) The notional amounts are presented on a net basis by underlying reference entity and the ratings profile shown is based on the ratings of the reference entity on which protection has been purchased. Predominantly all of the credit derivatives entered into by the Firm where it has purchased protection used in credit portfolio management activities are executed with investment-grade counterparties. In addition, the Firm obtains credit protection against certain loans in the retained loan portfolio through the issuance of credit-related notes.
(d) The maturity profile of retained loans, lending-related commitments and derivative receivables is generally based on remaining contractual maturity. Derivative contracts that are in a receivable position at December 31, 2024, may become payable prior to maturity based on their cash flow profile or changes in market conditions.

Wholesale credit exposure – industry exposures

The Firm focuses on the management and diversification of its industry exposures, and pays particular attention to industries with actual or potential credit concerns.

Exposures that are deemed to be criticized align with the U.S. banking regulators' definition of criticized exposures, which consist of the special mention, substandard and doubtful categories. Total criticized exposure, excluding loans held-for-sale and loans at fair value, was $44.7 billion and $41.4 billion at December 31, 2024 and 2023, representing approximately 3.5% and 3.3% of total wholesale credit exposure, respectively; of the $44.7 billion, $39.9 billion was performing. The increase in criticized exposure was driven by Real Estate resulting from downgrades, primarily in Multifamily and Office, and new commitments in Technology and Media, partially offset by Consumer & Retail resulting from net portfolio activity and upgrades.

The table below summarizes by industry the Firm's exposures as of December 31, 2024 and 2023. The industry of risk category is generally based on the client or counterparty's primary business activity. Refer to Note 4 for additional information on industry concentrations.

Wholesale credit exposure – industries[(a)]

						Selected metrics			
			Noninvestment-grade						
As of or for the year ended **December 31, 2024** (in millions)	Credit exposure[(f)(g)]	Investment-grade	Noncriticized	Criticized performing	Criticized nonperforming	30 days or more past due and accruing loans	Net charge-offs/ (recoveries)	Credit derivative and credit-related notes[(h)]	Liquid securities and other cash collateral held against derivative receivables
Real Estate	$ 207,050	$ 143,803	$ 50,865	$ 10,858	$ 1,524	$ 913	$ 345	$ (584)	$ —
Individuals and Individual Entities[(b)]	**144,145**	**118,650**	**24,831**	**217**	**447**	**831**	**122**	**—**	**—**
Asset Managers	**135,541**	**101,150**	**34,148**	**206**	**37**	**375**	**2**	**—**	**(9,194)**
Consumer & Retail	**129,815**	**62,800**	**60,141**	**6,055**	**819**	**252**	**123**	**(4,320)**	**—**
Technology, Media & Telecommunications	**84,716**	**45,021**	**28,629**	**10,592**	**474**	**79**	**94**	**(4,800)**	**—**
Industrials	**72,530**	**37,572**	**30,912**	**3,807**	**239**	**185**	**91**	**(2,312)**	**—**
Healthcare	**64,224**	**44,135**	**17,062**	**2,219**	**808**	**245**	**56**	**(3,286)**	**(34)**
Banks & Finance Companies	**61,287**	**36,884**	**24,119**	**257**	**27**	**36**	**—**	**(702)**	**(729)**
Utilities	**35,871**	**24,205**	**10,256**	**1,273**	**137**	**1**	**—**	**(2,700)**	**—**
State & Municipal Govt[(c)]	**35,039**	**33,303**	**1,711**	**9**	**16**	**90**	**—**	**(2)**	**(1)**
Automotive	**34,336**	**22,015**	**11,353**	**931**	**37**	**121**	**1**	**(997)**	**—**
Oil & Gas	**31,724**	**19,053**	**12,479**	**188**	**4**	**9**	**(3)**	**(1,711)**	**(2)**
Insurance	**24,267**	**17,847**	**6,198**	**222**	**—**	**2**	**—**	**(1,077)**	**(9,184)**
Chemicals & Plastics	**20,782**	**11,013**	**8,152**	**1,521**	**96**	**31**	**14**	**(1,164)**	**—**
Transportation	**17,019**	**9,462**	**7,135**	**391**	**31**	**17**	**(20)**	**(658)**	**—**
Metals & Mining	**15,860**	**7,373**	**7,860**	**590**	**37**	**9**	**—**	**(246)**	**(2)**
Central Govt	**13,862**	**13,580**	**157**	**125**	**—**	**4**	**—**	**(1,490)**	**(2,051)**
Securities Firms	**9,443**	**5,424**	**4,014**	**5**	**—**	**—**	**—**	**(13)**	**(2,635)**
Financial Markets Infrastructure	**4,446**	**4,201**	**245**	**—**	**—**	**—**	**—**	**(1)**	**—**
All other[(d)]	**140,873**	**117,986**	**22,398**	**398**	**91**	**10**	**(3)**	**(14,825)**	**(4,328)**
Subtotal	**$ 1,282,830**	**$ 875,477**	**$ 362,665**	**$ 39,864**	**$ 4,824**	**$ 3,210**	**$ 822**	**$ (40,888)**	**$ (28,160)**
Loans held-for-sale and loans at fair value	**31,922**								
Receivables from customers	**51,929**								
Total[(e)]	**$ 1,366,681**								

As of or for the year ended December 31, 2023 (in millions)	Credit exposure[(f)(g)]	Investment-grade	Noninvestment-grade Noncriticized	Criticized performing	Criticized nonperforming	Selected metrics: 30 days or more past due and accruing loans	Net charge-offs/ (recoveries)	Credit derivative and credit-related notes[(h)]	Liquid securities and other cash collateral held against derivative receivables
Real Estate	$ 208,261	$ 148,866	$ 50,190	$ 8,558	$ 647	$ 717	$ 275	$ (574)	$ —
Individuals and Individual Entities[(b)]	145,849	110,673	34,261	334	581	861	10	—	—
Asset Managers	129,574	83,857	45,623	90	4	201	1	—	(7,209)
Consumer & Retail	127,086	60,168	58,606	7,863	449	318	161	(4,204)	—
Technology, Media & Telecommunications	77,296	40,468	27,094	9,388	346	36	81	(4,287)	—
Industrials	75,092	40,951	30,586	3,419	136	213	31	(2,949)	—
Healthcare	65,025	43,163	18,396	3,005	461	130	17	(3,070)	—
Banks & Finance Companies	57,177	33,881	22,744	545	7	9	277	(511)	(412)
Utilities	36,061	25,242	9,929	765	125	1	(3)	(2,373)	—
State & Municipal Govt[(c)]	35,986	33,561	2,390	27	8	31	—	(4)	—
Automotive	33,977	23,152	10,060	640	125	59	—	(653)	—
Oil & Gas	34,475	18,276	16,076	111	12	45	11	(1,927)	(5)
Insurance	20,501	14,503	5,700	298	—	2	—	(961)	(6,898)
Chemicals & Plastics	20,773	11,353	8,352	916	152	106	2	(1,045)	—
Transportation	16,060	8,865	5,943	1,196	56	23	(26)	(574)	—
Metals & Mining	15,508	8,403	6,514	536	55	12	44	(229)	—
Central Govt	17,704	17,264	312	127	1	—	—	(3,490)	(2,085)
Securities Firms	8,689	4,570	4,118	1	—	—	—	(14)	(2,765)
Financial Markets Infrastructure	4,251	4,052	199	—	—	—	—	—	—
All other[(d)]	134,777	115,711	18,618	439	9	21	(2)	(10,124)	(3,087)
Subtotal	$ 1,264,122	$ 846,979	$ 375,711	$ 38,258	$ 3,174	$ 2,785	$ 879	$ (36,989)	$ (22,461)
Loans held-for-sale and loans at fair value	30,018								
Receivables from customers	47,625								
Total[(e)]	$ 1,341,765								

(a) The industry rankings presented in the table as of December 31, 2023, are based on the industry rankings of the corresponding exposures at December 31, 2024, not actual rankings of such exposures at December 31, 2023.
(b) Individuals and Individual Entities predominantly consists of Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB, and includes exposure to personal investment companies and personal and testamentary trusts.
(c) In addition to the credit risk exposure to states and municipal governments (both U.S. and non-U.S.) at December 31, 2024 and 2023, noted above, the Firm held: $6.1 billion and $5.9 billion, respectively, of trading assets; $17.9 billion and $21.4 billion, respectively, of AFS securities; and $9.3 billion and $9.9 billion, respectively, of HTM securities, issued by U.S. state and municipal governments. Refer to Note 2 and Note 10 for further information.
(d) All other includes: SPEs and Private education and civic organizations, representing approximately 94% and 6%, respectively, at both December 31, 2024 and 2023.
(e) Excludes cash placed with banks of $459.2 billion and $614.1 billion, at December 31, 2024 and 2023, respectively, which is predominantly placed with various central banks, primarily Federal Reserve Banks.
(f) Credit exposure is net of risk participations and excludes the benefit of credit derivatives and credit-related notes used in credit portfolio management activities held against derivative receivables or loans and liquid securities and other cash collateral held against derivative receivables.
(g) Credit exposure includes held-for-sale and fair value option elected lending-related commitments.
(h) Represents the net notional amounts of protection purchased and sold through credit derivatives and credit-related notes used to manage the credit exposures; these derivatives do not qualify for hedge accounting under U.S. GAAP. The All other category includes purchased credit protection on certain credit indices.

Presented below is additional detail on certain of the Firm's industry exposures.

Real Estate

Real Estate exposure was $207.1 billion as of December 31, 2024. Criticized exposure increased by $3.2 billion from $9.2 billion at December 31, 2023 to $12.4 billion at December 31, 2024, predominantly driven by downgrades, primarily in Multifamily and Office.

	December 31, 2024				
(in millions, except ratios)	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn[d]
Multifamily[a]	**$ 124,074**	**$ 7**	**$ 124,081**	**77 %**	**92 %**
Industrial	**19,092**	**17**	**19,109**	**65**	**72**
Other Income Producing Properties[b]	**16,411**	**158**	**16,569**	**50**	**63**
Office	**16,331**	**29**	**16,360**	**47**	**81**
Services and Non Income Producing	**14,047**	**57**	**14,104**	**62**	**46**
Retail	**12,230**	**23**	**12,253**	**77**	**75**
Lodging	**4,555**	**19**	**4,574**	**31**	**53**
Total Real Estate Exposure[c]	**$ 206,740**	**$ 310**	**$ 207,050**	**69 %**	**82 %**

	December 31, 2023				
(in millions, except ratios)	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn[d]
Multifamily[a]	$ 121,946	$ 21	$ 121,967	79 %	90 %
Industrial	20,254	18	20,272	70	72
Other Income Producing Properties[b]	15,542	208	15,750	55	63
Office	16,462	32	16,494	51	81
Services and Non Income Producing	16,145	74	16,219	62	46
Retail	12,763	48	12,811	75	73
Lodging	4,729	19	4,748	30	48
Total Real Estate Exposure	$ 207,841	$ 420	$ 208,261	71 %	80 %

(a) Total Multifamily exposure is approximately 99% performing. Multifamily exposure is largely in California.
(b) Other Income Producing Properties consists of clients with diversified property types or other property types outside of categories listed in the table above.
(c) Real Estate exposure is approximately 84% secured; unsecured exposure is largely investment-grade primarily to Real Estate Investment Trusts ("REITs") and Real Estate Operating Companies ("REOCs") whose underlying assets are generally diversified.
(d) Represents drawn exposure as a percentage of credit exposure.

Consumer & Retail

Consumer & Retail exposure was $129.8 billion as of December 31, 2024. Criticized exposure decreased by $1.4 billion from $8.3 billion at December 31, 2023 to $6.9 billion at December 31, 2024, driven by net portfolio activity and upgrades, largely offset by downgrades.

	December 31, 2024				
(in millions, except ratios)	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn[d]
Food and Beverage	**$ 34,774**	**$ 683**	**$ 35,457**	**61 %**	**34 %**
Retail	**34,917**	**261**	**35,178**	**51**	**31**
Business and Consumer Services[a]	**34,534**	**412**	**34,946**	**42**	**41**
Consumer Hard Goods	**13,796**	**208**	**14,004**	**43**	**35**
Leisure[b]	**10,186**	**44**	**10,230**	**26**	**43**
Total Consumer & Retail[c]	**$ 128,207**	**$ 1,608**	**$ 129,815**	**48 %**	**36 %**

	December 31, 2023				
(in millions, except ratios)	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn[d]
Food and Beverage	$ 32,256	$ 930	$ 33,186	57 %	36 %
Retail	36,042	334	36,376	51	30
Business and Consumer Services[a]	34,822	392	35,214	42	42
Consumer Hard Goods	13,169	197	13,366	43	33
Leisure[b]	8,784	160	8,944	25	47
Total Consumer & Retail	$ 125,073	$ 2,013	$ 127,086	47 %	36 %

(a) Retail consists of Home Improvement & Specialty Retailers, Restaurants, Discount & Drug Stores, Specialty Apparel, Department Stores and Supermarkets.
(b) Leisure consists of Arts & Culture, Travel Services, Gaming and Sports & Recreation. As of December 31, 2024, approximately 90% of the noninvestment-grade Leisure portfolio is secured.
(c) Consumer & Retail exposure is approximately 57% secured; unsecured exposure is approximately 80% investment-grade.
(d) Represents drawn exposure as a percent of credit exposure.

Oil & Gas

Oil & Gas exposure was $31.7 billion as of December 31, 2024. Criticized exposure was $192 million and $123 million at December 31, 2024 and 2023, respectively.

	December 31, 2024				
(in millions, except ratios)	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn[c]
Exploration & Production ("E&P") and Oil field Services	**$ 14,265**	**$ 848**	**$ 15,113**	**55 %**	**27 %**
Other Oil & Gas[a]	**16,306**	**305**	**16,611**	**65**	**19**
Total Oil & Gas[b]	**$ 30,571**	**$ 1,153**	**$ 31,724**	**60 %**	**23 %**

	December 31, 2023				
(in millions, except ratios)	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn[c]
Exploration & Production ("E&P") and Oil field Services	$ 18,121	$ 536	$ 18,657	51 %	26 %
Other Oil & Gas[a]	15,649	169	15,818	55	22
Total Oil & Gas	$ 33,770	$ 705	$ 34,475	53 %	25 %

(a) Other Oil & Gas includes Integrated Oil & Gas companies, Midstream/Oil Pipeline companies and refineries.
(b) Oil & Gas exposure is approximately 33% secured, and includes reserve-based lending to the Exploration & Production sub-sector; unsecured exposure is approximately 69% investment-grade.
(c) Represents drawn exposure as a percent of credit exposure.

Loans

In its wholesale businesses, the Firm provides loans to a variety of clients, ranging from large corporate and institutional clients to high-net-worth individuals. Refer to Note 12 for a further discussion on loans, including information about delinquencies, loan modifications and other credit quality indicators.

The following table presents the change in the nonaccrual loan portfolio for the years ended December 31, 2024 and 2023. Since December 31, 2023, nonaccrual loan exposure increased by $2.2 billion, predominantly driven by Real Estate, concentrated in Office, Healthcare and Consumer & Retail, in each case resulting from downgrades.

Wholesale nonaccrual loan activity

Year ended December 31, (in millions)	**2024**	2023
Beginning balance	**$ 2,714**	$ 2,395
Additions	**5,841**	3,543
Reductions:		
Paydowns and other	**2,387**	1,336
Gross charge-offs	**780**	965
Returned to performing status	**392**	616
Sales	**85**	307
Total reductions	**3,644**	3,224
Net changes	**2,197**	319
Ending balance	**$ 4,911**	$ 2,714

The following table presents net charge-offs/ recoveries, which are defined as gross charge-offs less recoveries, for the years ended December 31, 2024 and 2023. The amounts in the table below do not include gains or losses from sales of nonaccrual loans recognized in noninterest revenue.

Wholesale net charge-offs/(recoveries)

Year ended December 31, (in millions, except ratios)	**2024**	2023
Loans		
Average loans retained	**$ 673,310**	$ 646,875
Gross charge-offs	**1,022**	1,011
Gross recoveries collected	**(200)**	(132)
Net charge-offs/(recoveries)	**822**	879
Net charge-off/(recovery) rate	**0.12 %**	0.14 %

Maturities and sensitivity to changes in interest rates

The table below sets forth wholesale loan maturities and the distribution between fixed and floating interest rates based on the stated terms of the loan agreements by loan class. Refer to Note 12 for further information on loan classes.

December 31, 2024 (in millions, except ratios)	1 year or less[(b)]	After 1 year through 5 years	After 5 years through 15 years	After 15 years	Total
Wholesale loans:					
Secured by real estate	**$ 12,474**	**$ 57,125**	**$ 57,967**	**$ 42,597**	**$ 170,163**
Commercial and industrial	**55,731**	**109,839**	**8,587**	**94**	**174,251**
Other	**182,722**	**150,346**	**36,281**	**8,555**	**377,904**
Total wholesale loans	**$250,927**	**$ 317,310**	**$ 102,835**	**$ 51,246**	**$ 722,318**
Loans due after one year at fixed interest rates					
Secured by real estate		**$ 13,119**	**$ 17,943**	**$ 935**	
Commercial and industrial		**3,964**	**1,231**	**7**	
Other		**26,929**	**15,542**	**5,824**	
Loans due after one year at variable interest rates[(a)]					
Secured by real estate		**$ 44,006**	**$ 40,024**	**$ 41,662**	
Commercial and industrial		**105,875**	**7,356**	**87**	
Other		**123,417**	**20,739**	**2,731**	
Total wholesale loans		**$ 317,310**	**$ 102,835**	**$ 51,246**	

(a) Includes loans that have an initial fixed interest rate that resets to a variable rate as the variable rate will be the prevailing rate over the life of the loan.
(b) Includes loans held-for-sale, demand loans and overdrafts.

The following table presents net charge-offs/recoveries, average retained loans and net charge-off/recovery rate by loan class for the years ended December 31, 2024 and 2023.

	Year ended December 31,							
	Secured by real estate		Commercial and industrial		Other		Total	
(in millions, except ratios)	**2024**	2023	**2024**	2023	**2024**	2023	**2024**	2023
Net charge-offs/(recoveries)	**$ 313**	$ 178	**$ 381**	$ 370	**$ 128**	$ 331	**$ 822**	$ 879
Average retained loans	**162,653**	151,214	**169,363**	170,503	**341,294**	325,158	**673,310**	646,875
Net charge-off/(recovery) rate	**0.19 %**	0.12 %	**0.22 %**	0.22 %	**0.04 %**	0.10 %	**0.12 %**	0.14 %

Lending-related commitments

The Firm uses lending-related financial instruments, such as commitments (including revolving credit facilities) and guarantees, to address the financing needs of its clients. The contractual amounts of these financial instruments represent the maximum possible credit risk should the clients draw down on these commitments or when the Firm fulfills its obligations under these guarantees, and the clients subsequently fail to perform according to the terms of these contracts. Most of these commitments and guarantees have historically been refinanced, extended, cancelled, or expired without being drawn upon or a default occurring. As a result, the Firm does not believe that the total contractual amount of these wholesale lending-related commitments is representative of the Firm's expected future credit exposure or funding requirements. Refer to Note 28 for further information on wholesale lending-related commitments.

Receivables from customers

Receivables from customers reflect held-for-investment margin loans to brokerage clients in CIB, CCB and AWM that are collateralized by assets maintained in the clients' brokerage accounts (including cash on deposit, and primarily liquid and readily marketable debt or equity securities). To manage its credit risk, the Firm establishes margin requirements and monitors the required margin levels on an ongoing basis, and requires clients to deposit additional cash or other collateral, or to reduce positions, when appropriate. Credit risk arising from lending activities subject to collateral maintenance requirements is generally mitigated by factors such as the short-term nature of the activity, the fair value of collateral held and the Firm's right to call for, and the borrower's obligation to provide, additional margin when the fair value of the collateral declines. Because of these mitigating factors, these receivables generally do not require an allowance for credit losses. However, if in management's judgment, an allowance for credit losses is required, the Firm estimates expected credit losses based on the value of the collateral and probability of borrower default. These receivables are reported within accrued interest and accounts receivable on the Firm's Consolidated balance sheets.

Refer to Note 13 for further information on the Firm's accounting policies for the allowance for credit losses.

Derivative contracts

Derivatives enable clients and counterparties to manage risk, including credit risk and risks arising from fluctuations in interest rates, foreign exchange and equities and commodities prices. The Firm makes markets in derivatives in order to meet these needs and uses derivatives to manage certain risks associated with net open risk positions from its market-making activities, including the counterparty credit risk arising from derivative receivables. The Firm also uses derivative instruments to manage its own credit risk and other market risk exposure. The nature of the counterparty and the settlement mechanism of the derivative affect the credit risk to which the Firm is exposed. For over-the-counter ("OTC") derivatives, the Firm is exposed to the credit risk of the derivative counterparty. For exchange-traded derivatives ("ETD"), such as futures and options, and cleared over-the-counter ("OTC-cleared") derivatives, the Firm can also be exposed to the credit risk of the relevant CCP. Where possible, the Firm seeks to mitigate its credit risk exposures arising from derivative contracts through the use of legally enforceable master netting arrangements and collateral agreements. The percentage of the Firm's OTC derivative transactions subject to collateral agreements — excluding foreign exchange spot trades, which are not typically covered by collateral agreements due to their short maturity and centrally cleared trades that are settled daily — was approximately 86% and 87% at December 31, 2024 and 2023, respectively. Refer to Note 5 for additional information on the Firm's use of collateral agreements and for a further discussion of derivative contracts, counterparties and settlement types.

The fair value of derivative receivables reported on the Consolidated balance sheets was $61.0 billion and $54.9 billion at December 31, 2024 and 2023, respectively. The increase was primarily as a result of market movements. Derivative receivables represent the fair value of the derivative contracts after giving effect to legally enforceable master netting agreements and the related cash collateral held by the Firm.

In addition, the Firm holds liquid securities and other cash collateral that may be used as security when the fair value of the client's exposure is in the Firm's favor. For these purposes, the definition of liquid securities is consistent with the definition of high quality liquid assets as defined in the LCR rule.

In management's view, the appropriate measure of current credit risk should also take into consideration other collateral, which generally represents securities that do not qualify as high quality liquid assets under the LCR rule. The benefits of these additional collateral amounts for each counterparty are subject to a legally enforceable master netting agreement and limited to the net amount of the derivative receivables for each counterparty.

The Firm also holds additional collateral (primarily cash, G7 government securities, other liquid government agency and guaranteed securities, and corporate debt and equity securities) delivered by clients at the initiation of transactions, as well as collateral related to contracts that have a non-daily call frequency and collateral that the Firm has agreed to return but has not yet settled as of the reporting date. Although this collateral does not reduce the receivables balances and is not included in the tables below, it is available as security against potential exposure that could arise should the fair value of the client's derivative contracts move in the Firm's favor. Refer to Note 5 for additional information on the Firm's use of collateral agreements for derivative transactions.

The following tables summarize the net derivative receivables and the internal ratings profile for the periods presented.

Derivative receivables

December 31, (in millions)	2024	2023
Total, net of cash collateral	**$ 60,967**	$ 54,864
Liquid securities and other cash collateral held against derivative receivables	**(28,160)**	(22,461)
Total, net of liquid securities and other cash collateral	**$ 32,807**	$ 32,403
Other collateral held against derivative receivables	**(1,021)**	(993)
Total, net of collateral	**$ 31,786**	$ 31,410

Ratings profile of derivative receivables

	2024		2023	
December 31, (in millions, except ratios)	Exposure net of collateral	% of exposure net of collateral	Exposure net of collateral	% of exposure net of collateral
Investment-grade	**$ 23,783**	**75 %**	$ 24,004	76 %
Noninvestment-grade	**8,003**	**25**	7,406	24
Total	**$ 31,786**	**100 %**	$ 31,410	100 %

While useful as a current view of credit exposure, the net fair value of the derivative receivables does not capture the potential future variability of that credit exposure. To capture this variability, the Firm calculates, on a client-by-client basis, three measures of potential derivatives-related credit loss: Peak, Derivative Risk Equivalent ("DRE"), and Average exposure ("AVG"). These measures all incorporate netting and collateral benefits, where applicable.

Peak represents a conservative measure of potential derivative exposure, including the benefit of collateral, to a counterparty calculated in a manner that is broadly equivalent to a 97.5% confidence level over the life of the transaction. Peak is the primary measure used by the Firm for setting credit limits for derivative contracts, senior management reporting and derivatives exposure management.

DRE exposure is a measure that expresses the risk of derivative exposure, including the benefit of collateral, on a basis intended to be equivalent to the risk of loan exposures. DRE is a less extreme measure of potential credit loss than Peak.

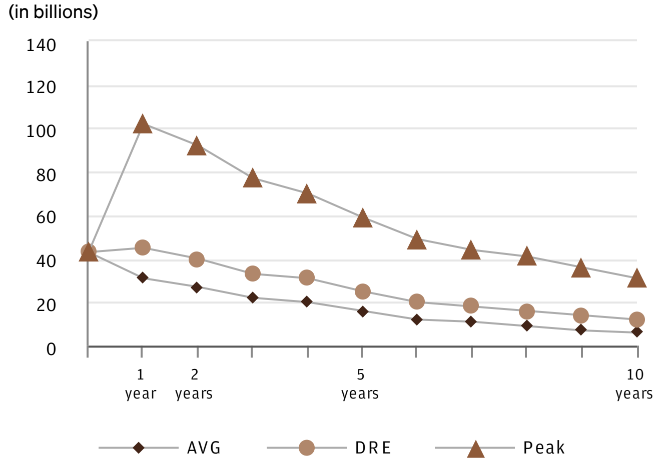
Finally, AVG is a measure of the expected fair value of the Firm's derivative exposures, including the benefit of collateral, at future time periods. AVG over the total life of the derivative contract is used as the primary metric for pricing purposes and is used to calculate credit risk capital and CVA, as further described below.

The fair value of the Firm's derivative receivables incorporates CVA to reflect the credit quality of counterparties. CVA is based on the Firm's AVG to a counterparty and the counterparty's credit spread in the credit derivatives market. The Firm believes that active risk management is essential to controlling the dynamic credit risk in the derivatives portfolio. In addition, the Firm's risk management process for derivatives exposures takes into consideration the potential impact of wrong-way risk, which is broadly defined as the risk that exposure to a counterparty is positively correlated with the impact of a default by the same counterparty, which could cause exposure to increase at the same time as the counterparty's capacity to meet its obligations is decreasing. Many factors may influence the nature and magnitude of these correlations over time. To the extent that these correlations are identified, the Firm may adjust the CVA associated with a particular counterparty's AVG. The Firm risk manages exposure to changes in CVA by entering into credit derivative contracts, as well as interest rate, foreign exchange, equity and commodity derivative contracts.

The below graph shows exposure profiles to the Firm's current derivatives portfolio over the next 10 years as calculated by the Peak, DRE and AVG metrics. The three measures generally show that exposure will decline after the first year, if no new trades are added to the portfolio.

Exposure profile of derivatives measures

December 31, 2024
(in billions)



Credit derivatives

The Firm uses credit derivatives for two primary purposes: first, in its capacity as a market-maker, and second, as an end-user to manage the Firm's own credit risk associated with various exposures.

Credit portfolio management activities

Included in the Firm's end-user activities are credit derivatives used to mitigate the credit risk associated with traditional lending activities (loans and lending-related commitments) and derivatives counterparty exposure in the Firm's wholesale businesses (collectively, "credit portfolio management activities"). Information on credit portfolio management activities is provided in the table below.

The Firm also uses credit derivatives as an end-user to manage other exposures, including credit risk arising from certain securities held in the Firm's market-making businesses. These credit derivatives are not included in credit portfolio management activities.

Credit derivatives and credit-related notes used in credit portfolio management activities

	Notional amount of protection purchased and sold[(a)]	
December 31, (in millions)	**2024**	2023
Credit derivatives and credit-related notes used to manage:		
Loans and lending-related commitments	**$ 25,216**	$ 24,157
Derivative receivables	**15,672**	12,832
Credit derivatives and credit-related notes used in credit portfolio management activities	**$ 40,888**	$ 36,989

(a) Amounts are presented net, considering the Firm's net protection purchased or sold with respect to each underlying reference entity or index.

The credit derivatives used in credit portfolio management activities do not qualify for hedge accounting under U.S. GAAP; these derivatives are reported at fair value, with gains and losses recognized in principal transactions revenue. In contrast, the loans and lending-related commitments being risk-managed are accounted for on an accrual basis. This asymmetry in accounting treatment, between loans and lending-related commitments and the credit derivatives used in credit portfolio management activities, causes earnings volatility that is not representative, in the Firm's view, of the true changes in value of the Firm's overall credit exposure.

The effectiveness of credit default swaps ("CDS") as a hedge against the Firm's exposures may vary depending on a number of factors, including the named reference entity (i.e., the Firm may experience losses on specific exposures that are different than the named reference entities in the purchased CDS); the contractual terms of the CDS (which may have a defined credit event that does not align with an actual loss realized by the Firm); and the maturity of the Firm's CDS protection (which in some cases may be shorter than the Firm's exposures). However, the Firm generally seeks to purchase credit protection with a maturity date that is the same or similar to the maturity date of the exposure for which the protection was purchased, and remaining differences in maturity are actively monitored and managed by the Firm. Refer to Credit derivatives in Note 5 for further information on credit derivatives and derivatives used in credit portfolio management activities.

ALLOWANCE FOR CREDIT LOSSES

The Firm’s allowance for credit losses represents management's estimate of expected credit losses over the remaining expected life of the Firm's financial assets measured at amortized cost and certain off-balance sheet lending-related commitments. The Firm's allowance for credit losses generally consists of:

- the allowance for loan losses, which covers the Firm’s retained loan portfolios (scored and risk-rated) and is presented separately on the Consolidated balance sheets,
- the allowance for lending-related commitments, which is reflected in accounts payable and other liabilities on the Consolidated balance sheets, and
- the allowance for credit losses on investment securities, which is reflected in investment securities on the Consolidated balance sheets.

Discussion of changes in the allowance

The allowance for credit losses as of December 31, 2024 was $26.9 billion, reflecting a net addition of $2.1 billion from December 31, 2023.

The net addition to the allowance for credit losses included:

- $2.1 billion in **consumer**, reflecting:
  - a $2.2 billion net addition in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years,

  partially offset by
  - a $125 million net reduction in Home Lending in the first quarter of 2024, and
- a net reduction of $30 million in **wholesale**, reflecting:
  - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in Markets, and a reduction due to charge-offs largely from collateral-dependent loans,

  predominantly offset by
  - net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm’s modeled credit loss estimates in the second quarter of 2024.

The Firm’s qualitative adjustments continued to include additional weight placed on the adverse scenarios to reflect ongoing uncertainties and downside risks related to the geopolitical and macroeconomic environment.

The Firm's allowance for credit losses is estimated using a weighted average of five internally developed macroeconomic scenarios. The adverse scenarios incorporate more punitive macroeconomic factors than the central case assumptions provided in the table below, resulting in:

- a weighted average U.S. unemployment rate peaking at 5.5% in the fourth quarter of 2025, and
- a weighted average U.S. real GDP level that is 1.9% lower than the central case at the end of the second quarter of 2026.

The following table presents the Firm’s central case assumptions for the periods presented:

	Central case assumptions at December 31, 2024		
	2Q25	**4Q25**	**2Q26**
U.S. unemployment rate(a)	**4.5 %**	**4.3 %**	**4.3 %**
YoY growth in U.S. real GDP(b)	**2.0 %**	**1.9 %**	**1.8 %**

	Central case assumptions at December 31, 2023		
	2Q24	4Q24	2Q25
U.S. unemployment rate(a)	4.1 %	4.4 %	4.1 %
YoY growth in U.S. real GDP(b)	1.8 %	0.7 %	1.0 %

(a) Reflects quarterly average of forecasted U.S. unemployment rate.
(b) The year over year growth in U.S. real GDP in the forecast horizon of the central scenario is calculated as the percentage change in U.S. real GDP levels from the prior year.

Subsequent changes to this forecast and related estimates will be reflected in the provision for credit losses in future periods.

Refer to Consumer Credit Portfolio on pages 120–125, Wholesale Credit Portfolio on pages 126–136 and Note 12 for additional information on the consumer and wholesale credit portfolios.

Refer to Critical Accounting Estimates Used by the Firm on pages 161–164 for further information on the allowance for credit losses and related management judgments.

Allowance for credit losses and related information

Year ended December 31, (in millions, except ratios)	2024 Consumer, excluding credit card	2024 Credit card	2024 Wholesale	2024 Total	2023 Consumer, excluding credit card	2023 Credit card	2023 Wholesale	2023 Total
Allowance for loan losses								
Beginning balance at January 1,	**$ 1,856**	**$ 12,450**	**$ 8,114**	**$ 22,420**	$ 2,040	$ 11,200	$ 6,486	$ 19,726
Cumulative effect of a change in accounting principle[a]	**NA**	**NA**	**NA**	**NA**	(489)	(100)	2	(587)
Gross charge-offs	**1,299**	**8,198**	**1,022**	**10,519**	1,151	5,491	1,011	7,653
Gross recoveries collected	**(625)**	**(1,056)**	**(200)**	**(1,881)**	(519)	(793)	(132)	(1,444)
Net charge-offs	**674**	**7,142**	**822**	**8,638**	632	4,698	879	6,209
Provision for loan losses	**624**	**9,292**	**578**	**10,494**	936	6,048	2,484	9,468
Other	**1**	**—**	**68**	**69**	1	—	21	22
Ending balance at December 31,	**$ 1,807**	**$ 14,600**	**$ 7,938**	**$ 24,345**	$ 1,856	$ 12,450	$ 8,114	$ 22,420
Allowance for lending-related commitments								
Beginning balance at January 1,	**$ 75**	**$ —**	**$ 1,899**	**$ 1,974**	$ 76	$ —	$ 2,306	$ 2,382
Provision for lending-related commitments	**7**	**—**	**121**	**128**	(1)	—	(407)	(408)
Other	**—**	**—**	**(1)**	**(1)**	—	—	—	—
Ending balance at December 31,	**$ 82**	**$ —**	**$ 2,019**	**$ 2,101**	$ 75	$ —	$ 1,899	$ 1,974
Impairment methodology								
Asset-specific[b]	**$ (728)**	**$ —**	**$ 526**	**$ (202)**	$ (876)	$ —	$ 392	$ (484)
Portfolio-based	**2,535**	**14,600**	**7,412**	**24,547**	2,732	12,450	7,722	22,904
Total allowance for loan losses	**$ 1,807**	**$ 14,600**	**$ 7,938**	**$ 24,345**	$ 1,856	$ 12,450	$ 8,114	$ 22,420
Impairment methodology								
Asset-specific	**$ —**	**$ —**	**$ 109**	**$ 109**	$ —	$ —	$ 89	$ 89
Portfolio-based	**82**	**—**	**1,910**	**1,992**	75	—	1,810	1,885
Total allowance for lending-related commitments	**$ 82**	**$ —**	**$ 2,019**	**$ 2,101**	$ 75	$ —	$ 1,899	$ 1,974
Total allowance for investment securities	**NA**	**NA**	**NA**	**$ 152**	NA	NA	NA	$ 128
Total allowance for credit losses[c]	**$ 1,889**	**$ 14,600**	**$ 9,957**	**$ 26,598**	$ 1,931	$ 12,450	$ 10,013	$ 24,522
Memo:								
Retained loans, end of period	**$376,334**	**$232,860**	**$690,396**	**$1,299,590**	$397,275	$ 211,123	$ 672,472	$1,280,870
Retained loans, average	**384,001**	**214,033**	**673,310**	**1,271,344**	364,061	191,412	646,875	1,202,348
Credit ratios								
Allowance for loan losses to retained loans	**0.48 %**	**6.27 %**	**1.15 %**	**1.87 %**	0.47 %	5.90 %	1.21 %	1.75 %
Allowance for loan losses to retained nonaccrual loans[d]	**56**	**NA**	**201**	**339**	51	NA	346	374
Allowance for loan losses to retained nonaccrual loans excluding credit card	**56**	**NA**	**201**	**136**	51	NA	346	166
Net charge-off rates	**0.18**	**3.34**	**0.12**	**0.68**	0.17	2.45	0.14	0.52

(a) Represents the impact to the allowance for loan losses upon the adoption of the Financial Instruments - Credit Losses: Troubled Debt Restructurings accounting guidance. Refer to Note 1 for further information.
(b) Includes collateral-dependent loans, including those for which foreclosure is deemed probable, and nonaccrual risk-rated loans.
(c) At December 31, 2024 and 2023, in addition to the allowance for credit losses in the table above, the Firm also had an allowance for credit losses of $268 million and $243 million, respectively, associated with certain accounts receivable in CIB.
(d) The Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance.

Allocation of allowance for loan losses

The table below presents a breakdown of the allowance for loan losses by loan class. Refer to Note 12 for further information on loan classes.

December 31, (in millions, except ratios)	2024 Allowance for loan losses	2024 Percent of retained loans to total retained loans	2023 Allowance for loan losses	2023 Percent of retained loans to total retained loans
Residential real estate	**$ 666**	**24 %**	$ 817	25 %
Auto and other	**1,141**	**5**	1,039	6
Consumer, excluding credit card	**1,807**	**29**	1,856	31
Credit card	**14,600**	**18**	12,450	16
Total consumer	**16,407**	**47**	14,306	47
Secured by real estate	**2,978**	**12**	2,997	13
Commercial and industrial	**3,350**	**13**	3,519	13
Other	**1,610**	**28**	1,598	27
Total wholesale	**7,938**	**53**	8,114	53
Total	**$ 24,345**	**100 %**	$ 22,420	100 %

INVESTMENT PORTFOLIO RISK MANAGEMENT

Investment portfolio risk is the risk associated with the loss of principal or a reduction in expected returns on investments arising from the investment securities portfolio or from principal investments. The investment securities portfolio is predominantly held by Treasury and CIO in connection with the Firm's balance sheet and asset-liability management objectives. Principal investments are predominantly privately-held financial instruments and are managed in the LOBs and Corporate. Investments are typically intended to be held over extended periods and, accordingly, the Firm has no expectation for short-term realized gains with respect to these investments.

Investment securities risk

Investment securities risk includes the exposure associated with a default in the payment of principal and interest. This risk is mitigated given that the investment securities portfolio held by Treasury and CIO predominantly consists of high-quality securities. At December 31, 2024, the Treasury and CIO investment securities portfolio, net of the allowance for credit losses, was $678.3 billion, and the average credit rating of the securities comprising the portfolio was AA+ (based upon external ratings where available, and where not available, based primarily upon internal risk ratings). Refer to Corporate results on pages 88–90 and Note 10 for further information on the investment securities portfolio and internal risk ratings. Refer to Liquidity Risk Management on pages 108–115 for further information on related liquidity risk. Refer to Market Risk Management on pages 141–149 for further information on the market risk inherent in the portfolio.

Governance and oversight

Investment securities risks are governed by the Firm's Risk Appetite framework, and reviewed at the CTC Risk Committee with regular updates provided to the Board Risk Committee.

The Firm's independent control functions are responsible for reviewing the appropriateness of the carrying value of investment securities in accordance with relevant policies. Approved levels for investment securities are established for each risk category, including capital and credit risks.

Principal investment risk

Principal investments are typically privately-held financial instruments representing ownership interests or other forms of junior capital. In general, principal investments include tax-oriented investments and investments made to enhance or accelerate the Firm's business strategies and exclude those that are consolidated on the Firm's balance sheets. These investments are made by dedicated investing businesses or as part of a broader business strategy. The Firm's principal investments are managed by the LOBs and Corporate and are reflected within their respective financial results. The Firm's investments will continue to evolve based on market circumstances and in line with its strategic initiatives.

The table below presents the aggregate carrying values of the principal investment portfolios as of December 31, 2024 and 2023.

(in billions)	**December 31, 2024**	December 31, 2023
Tax-oriented investments, primarily in alternative energy and affordable housing[a]	**$ 33.3**	$ 28.8
Private equity, various debt and equity instruments, and real assets	**9.1**	10.5
Total carrying value	**$ 42.4**	$ 39.3

(a) Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures guidance. Refer to Notes 1, 6, 14 and 25 for additional information.

Governance and oversight

The Firm's approach to managing principal investment risk is consistent with the Firm's risk governance structure. The Firm has established a Firmwide risk policy framework for all principal investing activities that includes approval by executives who are independent from the investing businesses, as appropriate.

The Firm's independent control functions are responsible for reviewing the appropriateness of the carrying value of investments in accordance with relevant policies. As part of the risk governance structure, approved levels for investments are established and monitored for each relevant business or segment in order to manage the overall size of the portfolios. The Firm also conducts stress testing on these portfolios using specific scenarios that estimate losses based on significant market moves and/or other risk events.

Market risk is the risk associated with the effect of changes in market factors such as interest and foreign exchange rates, equity and commodity prices, credit spreads or implied volatilities, on the value of assets and liabilities held for both the short and long term.

Market Risk Management

Market Risk Management monitors market risks throughout the Firm and defines market risk policies and procedures.

Market Risk Management seeks to facilitate efficient risk/return decisions, reduce volatility in operating performance and provide transparency into the Firm's market risk profile for senior management, the Board of Directors and regulators. Market Risk Management is responsible for the following functions:

- Maintaining a market risk policy framework
- Independently measuring and monitoring LOB, Corporate, and Firmwide market risk
- Defining, approving and monitoring limits
- Performing stress testing and qualitative risk assessments

Risk measurement

Measures used to capture market risk

There is no single measure to capture market risk and therefore Market Risk Management uses various metrics, both statistical and nonstatistical, to assess risk including:

- Value-at-risk
- Stress testing
- Profit and loss drawdowns
- Earnings-at-risk
- Economic Value Sensitivity
- Other sensitivity-based measures

Risk monitoring and control

Market risk exposure is managed primarily through a series of limits set in the context of the market environment and business strategy. In setting limits, Market Risk Management takes into consideration factors such as market volatility, product liquidity, accommodation of client business, and management judgment. Market Risk Management maintains different levels of limits. Firm level limits include VaR and stress limits. Similarly, LOB and Corporate limits include VaR and stress limits and may be supplemented by certain nonstatistical risk measures such as profit and loss drawdowns. Limits may also be set within the LOBs and Corporate, as well as at the legal entity level.

Market Risk Management sets limits and regularly reviews and updates them as appropriate. Senior management is responsible for reviewing and approving certain of these risk limits on an ongoing basis. Limits that have not been reviewed within specified time periods by Market Risk Management are reported to senior management. The LOBs and Corporate are responsible for adhering to established limits against which exposures are monitored and reported.

Limit breaches are required to be reported in a timely manner to limit approvers, which include Market Risk Management and senior management. In the event of a breach, Market Risk Management consults with senior members of appropriate groups within the Firm to determine the suitable course of action required to return the applicable positions to compliance, which may include a reduction in risk in order to remedy the breach or granting a temporary increase in limits to accommodate an expected increase in client activity and/or market volatility. Firm, Corporate or LOB-level limit breaches are escalated as appropriate.

Models used to measure market risk are inherently imprecise and are limited in their ability to measure certain risks or to predict losses. This imprecision may be heightened when sudden or severe shifts in market conditions occur. For additional discussion on model uncertainty refer to Estimations and Model Risk Management on page 160.

Market Risk Management periodically reviews the Firm's existing market risk measures to identify opportunities for enhancement, and to the extent appropriate, will calibrate those measures accordingly over time.

The following table summarizes the predominant business activities and related market risks, as well as positions which give rise to market risk and certain measures used to capture those risks, for each LOB and Corporate.

In addition to the predominant business activities, each LOB and Corporate may engage in principal investing activities. To the extent principal investments are deemed market risk sensitive, they are reflected in relevant risk measures and captured in the table below. Refer to Investment Portfolio Risk Management on page 140 for additional discussion on principal investments.

LOBs and Corporate	Predominant business activities	Related market risks	Positions included in Risk Management VaR	Positions included in earnings-at-risk	Positions included in other sensitivity-based measures
CCB	• Originates and services mortgage loans • Originates loans and takes deposits	• Risk from changes in the probability of newly originated mortgage commitments closing • Interest rate risk and prepayment risk	• Mortgage commitments, classified as derivatives • Warehouse loans that are fair value option elected, classified as loans – debt instruments • MSRs • Hedges of mortgage commitments, warehouse loans and MSRs, classified as derivatives • Interest-only and mortgage-backed securities, classified as trading assets debt instruments, and related hedges, classified as derivatives • Fair value option elected liabilities[(b)]	• Retained and held-for-sale loan portfolios • Deposits	• Fair value option elected liabilities DVA[(b)]
CIB[(a)]	• Makes markets and services clients across fixed income, foreign exchange, equities and commodities • Originates loans and takes deposits	• Risk of loss from adverse movements in market prices and implied volatilities across interest rate, foreign exchange, credit, commodity and equity instruments • Basis and correlation risk from changes in the way asset values move relative to one another • Interest rate risk and prepayment risk	• Trading assets/liabilities – debt and marketable equity instruments, and derivatives, including hedges of the retained loan portfolio • Certain securities purchased, loaned or sold under resale agreements and securities borrowed • Fair value option elected liabilities[(b)] • Certain fair value option elected loans • Derivative CVA and associated hedges • Marketable equity investments	• Retained and held-for-sale loan portfolios • Deposits	• Privately held equity and other investments measured at fair value; and certain real estate-related fair value option elected loans • Derivatives FVA and fair value option elected liabilities DVA[(b)] • Credit risk component of CVA and associated hedges for counterparties with credit spreads that have widened to elevated levels
AWM	• Provides initial capital investments in products such as mutual funds and capital invested alongside third-party investors • Originates loans and takes deposits	• Risk from adverse movements in market factors (e.g., market prices, rates and credit spreads) • Interest rate risk and prepayment risk	• Debt securities held in advance of distribution to clients, classified as trading assets - debt instruments • Trading assets/liabilities - derivatives that hedge the retained loan portfolio	• Retained and held-for-sale loan portfolios • Deposits	• Initial seed capital investments and related hedges, classified as derivatives • Certain deferred compensation and related hedges, classified as derivatives • Capital invested alongside third-party investors, typically in privately distributed collective vehicles managed by AWM (i.e., co-investments), as well as in third-party funds
Corporate	• Manages the Firm's liquidity, funding, capital, structural interest rate and foreign exchange risks	• Structural interest rate risk from the Firm's traditional banking activities • Structural non-USD foreign exchange risks	• Derivative positions measured through noninterest revenue in earnings • Marketable equity investments	• Deposits with banks and financing activities • Investment securities portfolio and related interest rate hedges • Cash flow hedges on retained loan portfolios in the LOBs • Long-term and short-term funding and related interest rate hedges • Deposits • Foreign exchange hedges of non-U.S. dollar capital investments	• Privately held equity and other investments measured at fair value • Foreign exchange exposure related to Firm-issued non-USD long-term debt ("LTD") and related hedges

(a) Effective in the second quarter of 2024, the Firm reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank ("CIB"). Refer to Business Segment & Corporate Results on pages 70–90 for additional information.
(b) Reflects structured notes in Risk Management VaR and the DVA on structured notes in other sensitivity-based measures.

Value-at-risk

JPMorganChase utilizes value-at-risk ("VaR"), a statistical risk measure, to estimate the potential loss from adverse market moves in the current market environment. The Firm has a single VaR framework used as a basis for calculating Risk Management VaR and Regulatory VaR.

The framework is employed across the Firm using historical simulation based on data for the previous 12 months. The framework's approach assumes that historical changes in market values are representative of the distribution of potential outcomes in the immediate future. The Firm believes the use of Risk Management VaR provides a daily measure of risk that is closely aligned to risk management decisions made by the LOBs and Corporate and, along with other market risk measures, provides the appropriate information needed to respond to risk events.

The Firm's Risk Management VaR is calculated assuming a one-day holding period and an expected tail-loss methodology which approximates a 95% confidence level. Risk Management VaR provides a consistent framework to measure risk profiles and levels of diversification across product types and is used for aggregating risks and monitoring limits across businesses. VaR results are reported as appropriate to various groups including senior management, the Board Risk Committee and regulators.

Underlying the overall VaR model framework are individual VaR models that simulate historical market returns for individual risk factors and/or product types. To capture material market risks as part of the Firm's risk management framework, comprehensive VaR model calculations are performed daily for businesses whose activities give rise to market risk. These VaR models are granular and incorporate numerous risk factors and inputs to simulate daily changes in market values over the historical period; inputs are selected based on the risk profile of each portfolio, as sensitivities and historical time series used to generate daily market values may be different across product types or risk management systems. The VaR model results across all portfolios are aggregated at the Firm level.

As VaR is based on historical data, it is an imperfect measure of market risk exposure and potential future losses. In addition, based on their reliance on available historical data, limited time horizons, and other factors, VaR measures are inherently limited in their ability to measure certain risks and to predict losses, particularly those associated with market illiquidity and sudden or severe shifts in market conditions.

For certain products, specific risk parameters are not captured in VaR due to the lack of liquidity and availability of appropriate historical data. The Firm uses proxies to estimate the VaR for these and other products when daily time series are not available. It is likely that using an actual price-based time series for these products, if available, would affect the VaR results presented. The Firm therefore considers other nonstatistical measures such as stress testing, in addition to VaR, to capture and manage its market risk positions.

As VaR model calculations require daily data and a consistent source for valuation, the daily market data used may be different than the independent third-party data collected for VCG price testing in its monthly valuation process. For example, in cases where market prices are not observable, or where proxies are used in VaR historical time series, the data sources may differ. Refer to Valuation process in Note 2 for further information on the Firm's valuation process.

The Firm's VaR model calculations are periodically evaluated and enhanced in response to changes in the composition of the Firm's portfolios, changes in market conditions, improvements in the Firm's modeling techniques and measurements, and other factors. Such changes may affect historical comparisons of VaR results. Refer to Estimations and Model Risk Management on page 160 for information regarding model reviews and approvals.

The Firm calculates separately a daily aggregated VaR in accordance with regulatory rules ("Regulatory VaR"), which is used to derive the Firm's regulatory VaR-based capital requirements under Basel III capital rules. This Regulatory VaR model framework currently assumes a ten business-day holding period and an expected tail-loss methodology which approximates a 99% confidence level. Regulatory VaR is applied to "covered" positions as defined by Basel III capital rules, which may be different than the positions included in the Firm's Risk Management VaR. For example, credit derivative hedges of accrual loans are included in the Firm's Risk Management VaR, while Regulatory VaR excludes these credit derivative hedges. In addition, in contrast to the Firm's Risk Management VaR, Regulatory VaR currently excludes the diversification benefit for certain VaR models.

Refer to JPMorganChase's Basel III Pillar 3 Regulatory Capital Disclosures reports, which are available on the Firm's website, for additional information on Regulatory VaR and the other components of market risk regulatory capital for the Firm (e.g., VaR-based measure, stressed VaR-based measure and the respective backtesting).

The table below shows the results of the Firm's Risk Management VaR measure using a 95% confidence level. VaR can vary significantly as positions change, market volatility fluctuates, and diversification benefits change.

Total VaR

As of or for the year ended December 31,	2024			2023		
(in millions)	Avg.	Min	Max	Avg.	Min	Max
CIB trading VaR by risk type[(a)]						
Fixed income	**$ 34**	**$ 26**	**$ 53**	$ 49	$ 31	$ 71
Foreign exchange	**15**	**7**	**23**	12	6	26
Equities	**8**	**4**	**15**	7	3	11
Commodities and other	**8**	**6**	**13**	11	6	19
Diversification benefit to CIB trading VaR [(b)]	**(32)**	**NM**	**NM**	(42)	NM	NM
CIB trading VaR	**33**	**27**	**42**	37	24	55
Credit Portfolio VaR[(c)]	**22**	**18**	**28**	14	8	26
Diversification benefit to CIB VaR[(b)]	**(16)**	**NM**	**NM**	(11)	NM	NM
CIB VaR	**39**	**27**	**52**	40	23	58
CCB VaR	**3**	**1**	**6**	7	1	15
AWM VaR[(d)]	**9**	**5**	**10**	1	—	10
Corporate VaR[(d)(e)]	**23**	**7**	**102**	12	9	17
Diversification benefit to other VaR[(b)]	**(10)**	**NM**	**NM**	(6)	NM	NM
Other VaR	**25**	**10**	**101**	14	9	22
Diversification benefit to CIB and other VaR[(b)]	**(17)**	**NM**	**NM**	(11)	NM	NM
Total VaR	**$ 47**	**$ 30**	**$ 91**	$ 43	$ 26	$ 57

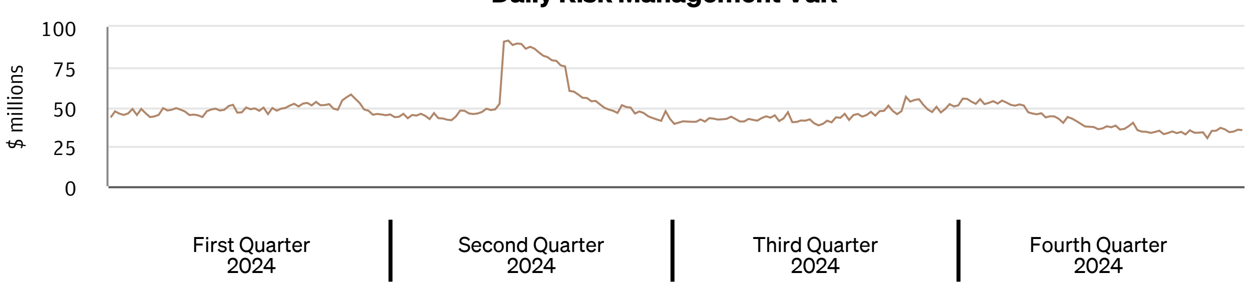
(a) The impact of the business segment reorganization in the second quarter of 2024 was not material to Total CIB VaR. Prior periods have not been revised. Refer to Business Segment & Corporate Results on pages 70–90 for additional information.
(b) Diversification benefit represents the difference between the portfolio VaR and the sum of its individual components. This reflects the non-additive nature of VaR due to imperfect correlation across LOBs, Corporate, and risk types. For maximum and minimum VaR, diversification benefit is not meaningful as the maximum and minimum VaR for each portfolio may have occurred on different trading days than the components.
(c) Includes the derivative CVA, hedges of the CVA and credit protection purchased against certain retained loans and lending-related commitments, which are reported in principal transactions revenue. This VaR does not include the retained loan portfolio, which is not reported at fair value. In line with the Firm's internal model governance, the credit risk component of CVA related to certain counterparties was removed from Credit Portfolio VaR due to the widening of the credit spreads for those counterparties to elevated levels. The related hedges were also removed to maintain consistency. This exposure is now reflected in other sensitivity-based measures.
(d) In the second quarter of 2024, the presentation of Corporate and other LOB VaR was updated to disaggregate AWM VaR due to the increase associated with credit protection purchased against certain retained loans and lending-related commitments. The VaR does not include the retained loan portfolio, which is not reported at fair value.
(e) Includes a legacy private equity position which is publicly traded, as well as Visa C shares which the Firm disposed of in the second and third quarters of 2024. The impact of Visa C shares resulted in elevated average and maximum Corporate VaR, Other VaR and Total VaR. Refer to Executive Overview on pages 54–58 for additional information.

2024 compared with 2023

Average Total VaR increased by $4 million for the year ended December 31, 2024 when compared with the prior year. The increase was predominantly driven by the impact of the Firm's receipt of Visa C shares on Corporate VaR and increases associated with credit protection purchased against certain retained loans and lending-related commitments within Credit Portfolio VaR and AWM VaR, largely offset by market volatility rolling out of the one-year historical look-back period impacting the Fixed income risk type.

The following graph presents daily Risk Management VaR for the four trailing quarters. The increase in VaR and subsequent decline observed in the second quarter of 2024 was primarily driven by changes in Visa C share exposure in the Firm's Corporate VaR.

Daily Risk Management VaR



VaR backtesting

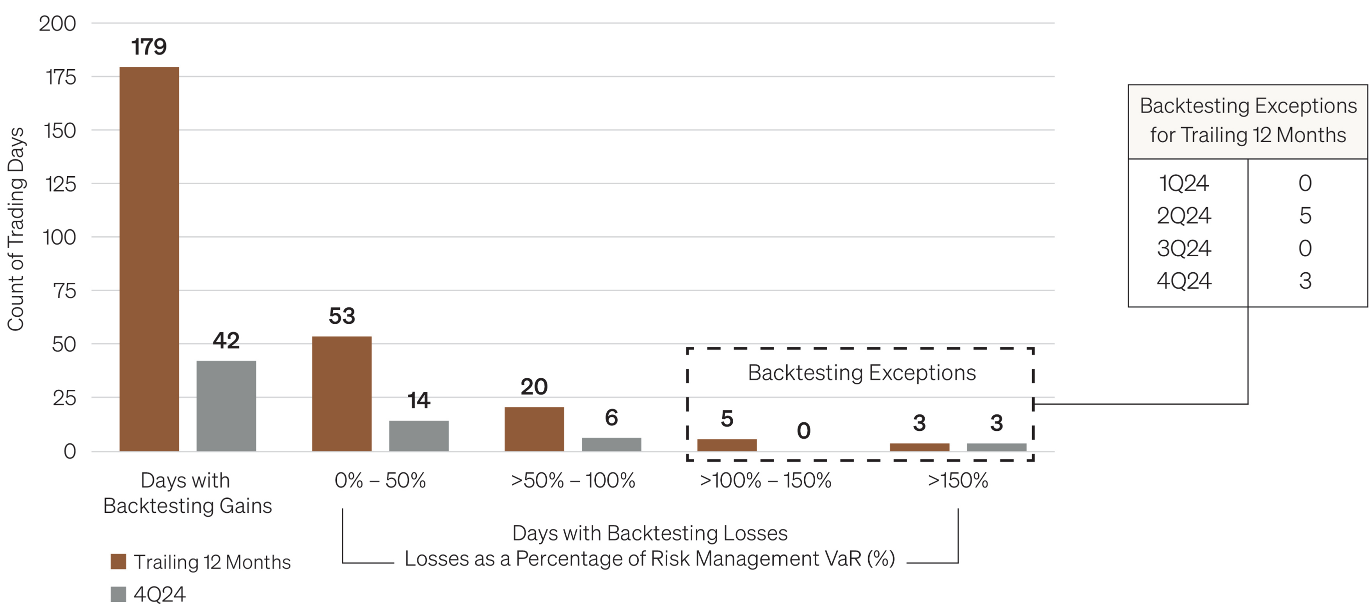
The Firm performs daily VaR model backtesting, which compares the daily Risk Management VaR results with the daily gains and losses that are utilized for VaR backtesting purposes. The gains and losses depicted in the chart below do not reflect the Firm's reported revenue as they exclude certain components of total net revenue, such as those associated with the execution of new transactions (i.e., intraday client-driven trading and intraday risk management activities), fees, commissions, other valuation adjustments and net interest income. These excluded components of total net revenue may more than offset the backtesting gain or loss on a particular day. The definition of backtesting gains and losses above is consistent with the requirements for backtesting under Basel III capital rules.

A backtesting exception occurs when the daily backtesting loss exceeds the daily Risk Management VaR for the prior day. Under the Firm's Risk Management VaR methodology, assuming current changes in market values are consistent with the historical changes used in the simulation, the Firm would expect to incur VaR backtesting exceptions five times every 100 trading days on average. The number of VaR backtesting exceptions observed can differ from the statistically expected number of backtesting exceptions if the current level of market volatility is materially different from the level of market volatility during the 12 months of historical data used in the VaR calculation.

For the 12 months ended December 31, 2024, the Firm posted backtesting gains on 179 of the 260 days, and observed eight VaR backtesting exceptions, of which three were in the three months ended December 31, 2024. Firmwide backtesting loss days can differ from the loss days for which Fixed Income Markets and Equity Markets posted losses, as disclosed in CIB Markets revenue, as the population of positions which comprise each metric are different and due to the exclusion of certain components of total net revenue in backtesting gains and losses as described above.

The following chart presents the distribution of Firmwide daily backtesting gains and losses for the trailing 12 months and three months ended December 31, 2024. The daily backtesting losses are displayed as a percentage of the corresponding daily Risk Management VaR. The count of days with backtesting losses are shown in aggregate, in fifty percentage point intervals. Backtesting exceptions are displayed within the intervals that are greater than one hundred percent. The results in the chart below differ from the results of backtesting disclosed in the Market Risk section of the Firm's Basel III Pillar 3 Regulatory Capital Disclosures reports, which are based on Regulatory VaR applied to the Firm's covered positions.

Distribution of Daily Backtesting Gains and Losses



Other risk measures

Stress testing

Along with VaR, stress testing is an important tool used to assess risk. While VaR reflects the risk of loss due to adverse changes in markets using recent historical market behavior, stress testing reflects the risk of loss from hypothetical changes in the value of market risk sensitive positions applied simultaneously. Stress testing measures the Firm's vulnerability to losses under a range of stressed but possible economic and market scenarios. The results are used to understand the exposures responsible for those potential losses and are measured against limits.

The Firm's stress framework covers market risk sensitive positions in the LOBs and Corporate. The framework is used to calculate multiple magnitudes of potential stress for both market rallies and market sell-offs, assuming significant changes in market factors such as credit spreads, equity prices, interest rates, currency rates and commodity prices, and combines them in multiple ways to capture an array of hypothetical economic and market scenarios.

The Firm generates a number of scenarios that focus on tail events in specific asset classes and geographies, including how the event may impact multiple market factors simultaneously. Scenarios also incorporate specific idiosyncratic risks and stress basis risk between different products. The flexibility in the stress framework allows the Firm to construct new scenarios that can test the outcomes against possible future stress events. Stress testing results are reported periodically to senior management of the Firm, as appropriate.

Stress methodologies are governed by the overall stress framework, under the oversight of Market Risk Management. The Firmwide Market Risk Stress Methodology Committee reviews and approves changes to stress testing methodology and scenarios across the Firm. Significant changes to the framework are escalated to senior management, as appropriate. In addition, stress methodology and the models to calculate the stress results are subject to the Firm's Estimations and Model Risk Management Policy

The Firm's stress testing framework is utilized in calculating the Firm's CCAR and other stress test results, which are reported periodically to the Board of Directors. In addition, stress testing results are incorporated into the Firm's Risk Appetite framework, and are reported periodically to the Board Risk Committee.

Profit and loss drawdowns

Profit and loss drawdowns are used to highlight trading losses above certain levels of risk tolerance. A profit and loss drawdown is a decline in revenue from its year-to-date peak level.

Structural interest rate risk management

The effect of interest rate exposure on the Firm's reported net income is important as interest rate risk represents one of the Firm's significant market risks. Interest rate risk arises not only from trading activities which are included in VaR, but also from the Firm's traditional banking activities, which include extension of loans and credit facilities, taking deposits, issuing debt, as well as the investment securities portfolio, and associated derivative instruments. Refer to the table on page 142 for a summary by LOB and Corporate identifying positions included in earnings-at-risk.

Governance

The CTC Risk Committee establishes the Firm's interest rate risk management policy and related limits, which are subject to approval by the Board Risk Committee. Treasury and CIO, working in partnership with the LOBs, calculates the Firm's structural interest rate risk profile and reviews it with senior management, including the CTC Risk Committee. In addition, oversight of structural interest rate risk is managed through a dedicated risk function reporting to the CTC CRO. This risk function is responsible for providing independent oversight and governance around assumptions and establishing and monitoring limits for structural interest rate risk, including limits related to Earnings-at-Risk and Economic Value Sensitivity. The Firm manages structural interest rate risk generally through its investment securities portfolio and interest rate derivatives.

Key risk drivers and risk management process

Structural interest rate risk can arise due to a variety of factors, including:

- Differences in timing among the maturity or repricing of assets, liabilities and off-balance sheet instruments
- Differences in the amounts of assets, liabilities and off-balance sheet instruments that are maturing or repricing at the same time
- Differences in the amounts by which short-term and long-term market interest rates change (for example, changes in the slope of the yield curve)
- The impact of changes in the maturity of various assets, liabilities or off-balance sheet instruments as interest rates change

The Firm manages interest rate exposure related to its assets and liabilities on a consolidated, Firmwide basis. Business units transfer their interest rate risk to Treasury and CIO through funds transfer pricing, which takes into account the elements of interest rate exposure that can be risk-managed in financial markets. These elements include asset and liability balances and contractual rates of interest, contractual principal payment schedules, expected prepayment experience, interest rate reset dates and maturities, rate indices used for repricing, and any interest rate ceilings or floors for adjustable rate products.

Earnings-at-Risk

One way that the Firm evaluates its structural interest rate risk is through earnings-at-risk. Earnings-at-risk estimates the Firm's interest rate exposure for a given interest rate scenario. It is presented as a sensitivity to a baseline, which includes net interest income and certain interest rate sensitive fees. The baseline uses market interest rates and, in the case of deposits, pricing assumptions. The Firm conducts simulations of changes to this baseline for interest rate-sensitive assets and liabilities denominated in U.S. dollars and other currencies ("non-U.S. dollar" currencies). These simulations primarily include retained and held-for-sale loans, deposits, deposits with banks and financing activities, investment securities, long-term debt, related interest rate hedges, and funds transfer pricing of other positions in risk management VaR and other sensitivity-based measures as described on page 142. Beginning in the fourth quarter of 2024, these simulations also include hedges of non-U.S. dollar foreign exchange exposures arising from capital investments. Refer to non-U.S. dollar foreign exchange risk on page 149 for more information.

Earnings-at-risk scenarios estimate the potential change to a baseline over the following 12 months utilizing multiple assumptions. These scenarios include a parallel shift involving changes to both short-term and long-term rates by an equal amount; a steeper yield curve involving holding short-term rates constant and increasing long-term rates; and a flatter yield curve involving increasing short-term rates and holding long-term rates constant or holding short-term rates constant and decreasing long-term rates. These scenarios consider many different factors, including:

- The impact on exposures as a result of instantaneous changes in interest rates from baseline rates.
- Forecasted balance sheet, as well as modeled prepayment and reinvestment behavior, but excluding assumptions about actions that could be taken by the Firm or its clients and customers in response to instantaneous rate changes. Mortgage prepayment assumptions are based on the interest rates used in the scenarios compared with underlying contractual rates, the time since origination, and other factors which are updated periodically based on historical experience. Deposit forecasts are a key assumption in the Firm's earnings-at-risk. The baseline reflects certain assumptions relating to the Federal Reserve's balance sheet policy (e.g., quantitative tightening and usage at the Reverse Repurchase Facility) that require management judgment. The amount of deposits that the Firm holds at any given time may be influenced by Federal Reserve actions, as well as broader monetary conditions and competition for deposits.
- The pricing sensitivity of deposits, known as deposit betas, represent the amount by which deposit rates paid could change upon a given change in market interest rates. Actual deposit rates paid may differ from the modeled assumptions, primarily due to customer behavior and competition for deposits.

The Firm performs sensitivity analyses of the assumptions used in earnings-at-risk scenarios, including with respect to deposit betas and forecasts of deposit balances, both of which are especially significant in the case of consumer deposits. The results of these sensitivity analyses are reported to the CTC Risk Committee and the Board Risk Committee.

The Firm's earnings-at-risk scenarios are periodically evaluated and enhanced in response to changes in the composition of the Firm's balance sheet, changes in market conditions, improvements in the Firm's simulation and other factors. In the second quarter of 2024, the Firm updated certain deposit rates paid assumptions which take into account observed pricing and client and customer behavior during the most recent economic cycle. These updated deposit rates paid assumptions impacted the U.S. dollar scenarios, resulting in an increase in positive sensitivity in higher interest rate scenarios, and an increase in negative sensitivity in lower interest rate scenarios.

The Firm's earnings-at-risk sensitivities are measures of the Firm's interest rate exposure. The Firm's actual net interest income for the rate changes presented may differ as the earnings-at-risk scenarios are modelled as instantaneous changes and exclude any actions that could be taken by the Firm or its clients or customers in response to rate changes. Other significant assumptions in the earnings-at-risk scenarios, including mortgage prepayments and deposit rates paid, may also differ from actual results. The Firm's forecast for net interest income is included in the Firm's outlook on page 57.

The Firm's sensitivities are presented in the table below.

December 31, (in billions)	2024[(a)]	2023[(b)]
Parallel shift:		
+100 bps shift in rates	**$ 2.3**	$ 3.1
-100 bps shift in rates	**(2.5)**	(2.8)
+200 bps shift in rates	**4.6**	6.2
-200 bps shift in rates	**(4.9)**	(6.1)
Steeper yield curve:		
+100 bps shift in long-term rates	**1.0**	0.6
-100 bps shift in short-term rates	**(1.4)**	(2.2)
Flatter yield curve:		
+100 bps shift in short-term rates	**1.2**	2.5
-100 bps shift in long-term rates	**(1.1)**	(0.6)

(a) Reflects the simultaneous shift of U.S. dollar and non-U.S. dollar rates, and the inclusion of the hedges of non-U.S. dollar capital investments. This inclusion had no impact on total sensitivities but increased U.S. dollar and decreased non-U.S. dollar sensitivities. Subsequent to this change, non-U.S. dollar sensitivities were insignificant.
(b) At December 31, 2023, represents the total of the Firm's U.S. dollar and non-U.S. dollar sensitivities as previously reported.

The change in the Firm's sensitivities as of December 31, 2024, compared to December 31, 2023, were primarily driven by Treasury and CIO balance sheet actions where the Firm added duration through investment securities activity, cash flow hedges of retained loans and fair value hedges of Firm debt. The impact on the sensitivities of the Treasury and CIO balance sheet actions were largely offset by the impact of deposits, primarily from the second quarter of 2024 update of the deposit rates paid assumptions for certain consumer and wholesale deposit products. Additionally, the results as of December 31, 2024 reflected the update to include hedges of the Firm's non-U.S. dollar capital investments. Although total results were not impacted, these hedges increase U.S. dollar sensitivities and decrease non-U.S. dollar sensitivities. In the absence of these updates the Firm's sensitivities as of December 31, 2024, would have been different by the amounts reported in the following table:

	Amounts by which reported sensitivities would have been different	
December 31, 2024 (in billions)	Impact from update in the second quarter of 2024	Impact from update in the fourth quarter of 2024
U.S. dollar:		
Parallel shift:		
+100 bps shift in rates	$ (1.0)	$ (0.6)
-100 bps shift in rates	0.9	0.6
+200 bps shift in rates	(1.9)	(1.3)
-200 bps shift in rates	1.5	1.3
Steeper yield curve:		
+100 bps shift in long-term rates	—	—
-100 bps shift in short-term rates	0.9	0.6
Flatter yield curve:		
+100 bps shift in short-term rates	(1.0)	(0.6)
-100 bps shift in long-term rates	—	—
Non-U.S. dollar:		
Parallel shift:		
+100 bps shift in rates	—	0.6
-100 bps shift in rates	—	(0.6)

Economic Value Sensitivity

In addition to earnings-at-risk, which is measured as a sensitivity to a baseline of earnings over the next 12 months, the Firm also measures Economic Value Sensitivity ("EVS"). EVS stress tests the longer-term economic value of equity by measuring the sensitivity of the Firm's current balance sheet, primarily retained loans, deposits, debt and investment securities as well as related hedges, under various interest rate scenarios. The Firm's pricing and cash flow assumptions associated with deposits, as well as prepayment assumptions for loans and securities, are significant factors in the EVS measure. In accordance with the CTC interest rate risk management policy, the Firm has established limits on EVS as a percentage of TCE.

Certain assumptions used in the EVS measure may differ from those required in the fair value measurement note to the Consolidated Financial Statements. For example, certain assets and liabilities with no stated maturity, such as credit card receivables and deposits, have longer assumed durations in the EVS measure. Additional information on long-term debt and held to maturity investment securities is disclosed on page 201 in Note 2.

Non-U.S. dollar foreign exchange risk

Non-U.S. dollar FX risk is the risk that changes in foreign exchange rates affect the value of the Firm's assets or liabilities or future results. The Firm has structural non-U.S. dollar FX exposures arising from capital investments, forecasted expense and revenue, the investment securities portfolio and non-U.S. dollar-denominated debt issuance. Treasury and CIO, working in partnership with the LOBs, primarily manage these risks on behalf of the Firm. Treasury and CIO may hedge certain of these risks using derivatives. Refer to Business Segment & Corporate Results on page 71 for additional information.

Other sensitivity-based measures

The Firm quantifies the market risk of certain debt and equity and credit and funding-related exposures by assessing the potential impact on net revenue, other comprehensive income ("OCI") and noninterest expense due to changes in relevant market variables. Refer to the predominant business activities that give rise to market risk on page 142 for additional information on the positions captured in other sensitivity-based measures.

The table below represents the potential impact to net revenue, OCI or noninterest expense for market risk sensitive instruments that are not included in VaR or earnings-at-risk. Where appropriate, instruments used for hedging purposes are reported net of the positions being hedged. The sensitivities disclosed in the table below may not be representative of the actual gain or loss that would have been realized at December 31, 2024 and 2023, as the movement in market parameters across maturities may vary and are not intended to imply management's expectation of future changes in these sensitivities.

Gain/(loss) (in millions) Activity	Description	Sensitivity measure	December 31, 2024	December 31, 2023
Debt and equity[(a)]				
Asset Management activities	Consists of seed capital and related hedges; fund co-investments[(c)]; and certain deferred compensation and related hedges[(d)]	10% decline in market value	**$ (53)**	$ (61)
Other debt and equity	Consists of certain real estate-related fair value option elected loans, privately held equity and other investments held at fair value[(c)]	10% decline in market value	**(1,030)**	(1,044)
Credit- and funding-related exposures				
Non-USD LTD cross-currency basis	Represents the basis risk on derivatives used to hedge the foreign exchange risk on the non-USD LTD[(e)]	1 basis point parallel tightening of cross currency basis	**(10)**	(12)
Non-USD LTD hedges foreign currency ("FX") exposure	Primarily represents the foreign exchange revaluation on the fair value of the derivative hedges[(e)]	10% depreciation of currency	**28**	16
Derivatives – funding spread risk	Impact of changes in the spread related to derivatives FVA[(c)]	1 basis point parallel increase in spread	**(2)**	(3)
CVA - counterparty credit risk[(b)]	Credit risk component of CVA and associated hedges	10% credit spread widening	**—**	—
Fair value option elected liabilities - funding spread risk	Impact of changes in the spread related to fair value option elected liabilities DVA[(e)]	1 basis point parallel increase in spread	**47**	46

(a) Excludes equity securities without readily determinable fair values that are measured under the measurement alternative. Refer to Note 2 for additional information.
(b) In line with the Firm's internal model governance, the credit risk component of CVA related to certain counterparties was removed from Credit Portfolio VaR due to the widening of the credit spreads for those counterparties to elevated levels. The related hedges were also removed to maintain consistency. This exposure is now reflected in the table above.
(c) Impact recognized through net revenue.
(d) Impact recognized through noninterest expense.
(e) Impact recognized through OCI.

COUNTRY RISK MANAGEMENT

The Firm, through its LOBs and Corporate, may be exposed to country risk resulting from financial, economic, political or other significant developments which adversely affect the value of the Firm's exposures related to a particular country or set of countries. The Country Risk Management group actively monitors the various portfolios which may be impacted by these developments and measures the extent to which the Firm's exposures are diversified given the Firm's strategy and risk tolerance relative to a country.

Organization and management

Country Risk Management is an independent risk management function that assesses and monitors exposure to country risk across the Firm.

The Firm's country risk management function includes the following activities:

- Maintaining policies, procedures and standards consistent with a comprehensive country risk framework
- Assigning sovereign ratings, assessing country risks and establishing risk tolerance relative to a country
- Measuring and monitoring country risk exposure and stress across the Firm
- Managing and approving country limits and reporting trends and limit breaches to senior management
- Developing surveillance tools, such as signaling models and ratings indicators, for early identification of potential country risk concerns
- Providing country risk scenario analysis

Sources and measurement

The Firm is exposed to country risk through its lending and deposits, investing, and market-making activities, whether cross-border or locally funded. Country exposure includes activity with both government and private-sector entities in a country.

Under the Firm's internal country risk management approach, attribution of exposure to an individual country is based on the country where the largest proportion of the assets of the counterparty, issuer, obligor or guarantor are located or where the largest proportion of its revenue is derived, which may be different than the domicile (i.e. legal residence) or country of incorporation.

Individual country exposures reflect an aggregation of the Firm's risk to an immediate default, with zero recovery, of the counterparties, issuers, obligors or guarantors attributed to that country. Activities which result in contingent or indirect exposure to a country are not included in the country exposure measure (for example, providing clearing services or secondary exposure to collateral on securities financing receivables).

Assumptions are sometimes required in determining the measurement and allocation of country exposure, particularly in the case of certain non-linear or index products, or where the nature of the counterparty, issuer, obligor or guarantor is not suitable for attribution to an individual country. The use of different measurement approaches or assumptions could affect the amount of reported country exposure.

Under the Firm's internal country risk measurement framework:

- Deposits with banks are measured as the cash balances placed with central banks, commercial banks, and other financial institutions
- Lending exposures are measured at the total committed amount (funded and unfunded), net of the allowance for credit losses and eligible cash and marketable securities collateral received
- Securities financing exposures are measured at their receivable balance, net of eligible collateral received
- Debt and equity securities are measured at the fair value of all positions, including both long and short positions
- Counterparty exposure on derivative receivables is measured at the derivative's fair value, net of the fair value of the eligible collateral received
- Credit derivatives exposure is measured at the net notional amount of protection purchased or sold for the same underlying reference entity, inclusive of the fair value of the derivative receivable or payable, reflecting the manner in which the Firm manages these exposures

The Firm's internal country risk reporting differs from the reporting provided under the FFIEC bank regulatory requirements.

Stress testing

Stress testing is an important component of the Firm's country risk management framework, which aims to estimate and limit losses arising from a country crisis by measuring the impact of adverse asset price movements to a country based on market shocks combined with counterparty specific assumptions. Country Risk Management periodically designs and runs tailored stress scenarios to test vulnerabilities to individual countries or sets of countries in response to specific or potential market events, sector performance concerns, sovereign actions and geopolitical risks. These tailored stress results are used to inform potential risk reduction across the Firm, as necessary.

Risk reporting

Country exposure and stress are measured and reported regularly, and used by Country Risk Management to identify trends and monitor high usages and breaches against limits.

For country risk management purposes, the Firm may report exposure to jurisdictions that are not fully autonomous, including dependent territories and Special Administrative Regions ("SAR") such as Hong Kong SAR, separately from the independent sovereign states with which they are associated.

The following table presents the Firm's top 20 exposures by country (excluding the U.S.) as of December 31, 2024, and their comparative exposures as of December 31, 2023. The top 20 country exposures represent the Firm's largest total exposures by individual country. Country exposures may fluctuate from period to period due to a variety of factors, including client activity, market flows and liquidity management activities undertaken by the Firm.

The increase in exposure to Germany when compared to December 31, 2023, was driven by an increase in cash placed with the central bank of Germany, predominantly due to client-driven market-making activities and higher client deposits.

The increase in exposure to Japan when compared to December 31, 2023, was driven by an increase in cash placed with the central bank of Japan as a result of client-driven market-making activities.

The Firm continues to monitor its exposure to Russia, which corresponds to cash placed with the central bank, but which excludes deposits placed on behalf of clients at the Deposit Insurance Agency of Russia. The Firm currently believes that its remaining exposure to Russia is not material. Refer to Note 30 on pages 310–311 for information concerning Russian litigation.

Top 20 country exposures (excluding the U.S.)[a]

	2024					2023[f]
December 31, (in billions)	Deposits with banks[b]	Lending[c]	Trading and investing[d]	Other[e]	Total exposure	Total exposure
Germany	$ 89.7	$ 12.6	$ 0.9	$0.7	$ 103.9	$ 84.8
United Kingdom	24.6	22.4	27.7	1.4	76.1	77.1
Japan	55.1	3.1	4.5	0.4	63.1	36.0
France	0.6	12.3	4.2	0.9	18.0	10.1
Canada	1.6	10.6	2.7	0.2	15.1	16.0
Brazil	3.5	4.2	7.0	—	14.7	16.7
Australia	5.0	7.4	1.9	—	14.3	18.3
Switzerland	4.7	4.2	1.4	3.3	13.6	10.9
Mainland China	3.1	6.2	4.1	—	13.4	14.0
India	1.1	5.2	4.1	0.9	11.3	9.7
Italy	0.1	8.2	1.8	0.3	10.4	6.0
South Korea	0.6	2.9	6.3	0.5	10.3	7.8
Saudi Arabia	0.8	5.7	2.9	—	9.4	7.7
Singapore	1.5	2.0	3.5	0.4	7.4	9.8
Mexico	1.3	4.4	1.5	—	7.2	8.2
Spain	0.2	4.6	1.2	0.1	6.1	6.3
Netherlands	—	6.6	(0.9)	0.2	5.9	5.6
Belgium	4.0	1.3	0.1	—	5.4	8.0
Malaysia	2.1	0.2	1.0	0.3	3.6	4.2
Luxembourg	0.9	1.7	1.0	—	3.6	4.0

(a) Country exposures presented in the table reflect 89% and 88% of total Firmwide non-U.S. exposure, where exposure is attributed to an individual country based on the Firm's internal country risk management approach, at December 31, 2024 and 2023, respectively.
(b) Predominantly represents cash placed with central banks.
(c) Includes loans and accrued interest receivable, lending-related commitments (net of eligible collateral and the allowance for credit losses). Excludes intra-day and operating exposures, such as those from settlement and clearing activities.
(d) Includes market-making positions and hedging, investment securities, and counterparty exposure on derivative and securities financings net of eligible collateral. Market-making positions and hedging includes exposure from single reference entity ("single-name"), index and other multiple reference entity transactions for which one or more of the underlying reference entities is in a country listed in the above table.
(e) Includes physical commodities inventory and clearing house guarantee funds.
(f) The country rankings presented in the table as of December 31, 2023, are based on the country rankings of the corresponding exposures at December 31, 2024, not actual rankings of such exposures at December 31, 2023.

CLIMATE RISK MANAGEMENT

Climate risk refers to the potential threats posed by climate change to the Firm and its clients, customers, operations and business strategy. Climate change is viewed as a driver of risk that may impact existing types of risks managed by the Firm. Climate risk is categorized into physical risk and transition risk.

Physical risk involves economic costs and financial losses due to a changing climate. Acute physical risk drivers include the increased frequency or severity of climate and weather events, such as floods, wildfires and tropical cyclones. Chronic physical risk drivers include more gradual shifts in the climate, such as sea level rise, persistent changes in precipitation levels and increases in average ambient temperatures.

Transition risk involves the financial and economic consequences of society's shift toward a lower-carbon economy. Transition risk drivers include possible changes in public policy, adoption of new technologies and shifts in consumer preferences. Transition risks may also be influenced by changes in the physical climate.

Organization and management

The Firm has a Climate Risk Management function that is responsible for establishing and maintaining the Firmwide framework and strategy for managing climate risks that may impact the Firm.

Other responsibilities of Climate Risk Management include:

- Setting policies, standards, procedures and processes to support identification, escalation, monitoring and management of climate risk across the Firm
- Developing metrics, scenarios and stress testing mechanisms designed to assess the range of potential climate-related financial and economic impacts to the Firm
- Establishing a Firmwide climate risk data strategy and the supporting climate risk technology infrastructure

The LOBs and Corporate are responsible for the identification, assessment and management of climate risks present in their business activities and for the adherence to applicable climate-related laws, rules and regulations.

Governance and oversight

The Firm's framework and strategy for managing climate risk is integrated into the Firm's risk governance structure. This framework allows for the escalation of significant climate risk-related issues to LOB Risk Committees. The Board Risk Committee also receives information on significant climate risks and climate-related initiatives, as appropriate.

Operational risk is the risk of an adverse outcome resulting from inadequate or failed internal processes or systems; human factors; or external events impacting the Firm's processes or systems. Operational Risk includes compliance, conduct, legal, and estimations and model risk. Operational risk is inherent in the Firm's activities and can manifest itself in various ways, including fraudulent acts, business disruptions (including those caused by extraordinary events beyond the Firm's control), cyber attacks, inappropriate employee behavior, failure to comply with applicable laws, rules and regulations or failure of vendors or other third party providers to perform in accordance with their agreements. Operational Risk Management attempts to manage operational risk at appropriate levels in light of the Firm's financial position, the characteristics of its businesses, and the markets and regulatory environments in which it operates.

Operational Risk Management Framework

The Firm's Compliance, Conduct, and Operational Risk ("CCOR") Management Framework is designed to enable the Firm to govern, identify, measure, monitor and test, manage and report on the Firm's operational risk.

Operational Risk Governance

The LOBs and Corporate are responsible for the management of operational risk. The Control Management Organization, which consists of control managers within each LOB and Corporate, is responsible for the day-to-day execution of the CCOR Framework.

The Firm's Global Chief Compliance Officer ("CCO") and FRE for Operational Risk and Qualitative Risk Appetite is responsible for defining the CCOR Management Framework and establishing the minimum standards for its execution. The LOB and Corporate aligned CCOR Lead Officers report to the Global CCO and FRE for Operational Risk and Qualitative Risk Appetite and are independent of the respective businesses or functions they oversee. The CCOR Management Framework is included in the Risk Governance and Oversight Policy that is reviewed and approved by the Board Risk Committee periodically.

Operational Risk Identification

The Firm utilizes a structured risk and control self-assessment process that is executed by the LOBs and Corporate. As part of this process, the LOBs and Corporate evaluate the effectiveness of their respective control environment to assess circumstances in which controls have failed, and to determine where remediation efforts may be required. The Firm's Operational Risk and Compliance organization ("Operational Risk and Compliance") provides oversight of and challenge to these evaluations and may also perform independent assessments of significant operational risk events and areas of concentrated or emerging risk.

Operational Risk Measurement

Operational Risk and Compliance performs an independent assessment of the operational risks inherent within the LOBs and Corporate, which includes evaluating the effectiveness of the control environments and reporting the results to senior management.

In addition, Operational Risk and Compliance assesses operational risks through quantitative means, including operational risk-based capital and estimation of operational risk losses under both baseline and stressed conditions.

The primary component of the operational risk-based capital estimate is the Loss Distribution Approach ("LDA") statistical model, which simulates the projected frequency and severity of operational risk losses based on historical data. The LDA model is used to estimate an aggregate operational risk loss over a one-year time horizon, at a 99.9% confidence level. The LDA model incorporates actual internal operational risk losses in the quarter following the period in which those losses were realized, and the calculation generally continues to reflect such losses even after the issues or business activities giving rise to the losses have been remediated or reduced.

As required under the Basel III capital framework, the Firm's operational risk capital methodology, which uses the Advanced Measurement Approach ("AMA"), incorporates internal and external losses as well as management's view of tail risk captured through operational risk scenario analysis, and evaluation of key business environment and internal control metrics. The Firm does not reflect the impact of insurance in its AMA estimate of operational risk capital.

The Firm considers the impact of stressed economic conditions on operational risk losses and develops a forward looking view of material operational risk events that may occur in a stressed environment. The Firm's operational risk stress testing framework is utilized in calculating results for the Firm's CCAR and other stress testing processes.

Refer to Capital Risk Management on pages 97–107 for information related to operational risk RWA, and CCAR.

Operational Risk Monitoring and Testing

The results of risk assessments performed by Operational Risk and Compliance are used in connection with their independent monitoring and testing compliance of the LOBs and Corporate with laws, rules and regulations. Through monitoring and testing, Operational Risk and Compliance independently identify areas of heightened operational risk and tests the effectiveness of controls within the LOBs and Corporate.

Management of Operational Risk
The operational risk areas or issues identified through monitoring and testing are escalated to the LOBs and Corporate to be remediated through action plans, as needed, to mitigate operational risk. Operational Risk and Compliance may advise the LOBs and Corporate in the development and implementation of action plans.

Operational Risk Reporting
All employees of the Firm are expected to escalate risks appropriately. Risks identified by Operational Risk and Compliance are escalated to the appropriate LOB and Corporate Control Committees, as needed. Operational Risk and Compliance has established standards designed to ensure that consistent operational risk reporting and operational risk reports are produced on a Firmwide basis as well as by the LOBs and Corporate. Reporting includes the evaluation of key risk and performance indicators against established thresholds as well as the assessment of different types of operational risk against stated risk appetite. The standards establish escalation protocols to senior management and to the Board of Directors.

Insurance
One of the ways in which operational risk may be mitigated is through insurance maintained by the Firm. The Firm purchases insurance from commercial insurers and maintains a wholly-owned captive insurer, Park Assurance Company. Insurance may also be required by third parties with whom the Firm does business.

Subcategories and examples of operational risks
Operational risk can manifest itself in various ways. Operational risk subcategories include Compliance risk, Conduct risk, Legal risk, and Estimations and Model risk. Refer to pages 157, 158, 159 and 160, respectively for more information on Compliance, Conduct, Legal, and Estimations and Model risk. Details on other select examples of operational risks such as business and technology resiliency, payment fraud and third-party outsourcing, as well as cybersecurity, are provided below.

Firmwide resiliency risk
Disruptions of the Firm's business and operations can occur due to forces beyond the Firm's control such as the spread of infectious diseases or pandemics, severe weather, natural disasters, the effects of climate change, power or telecommunications loss, failure of a third party to provide expected services, cyberattacks, civil or political unrest or terrorism. The Firm's resiliency framework is intended to enable the Firm to prepare for and adapt to changing conditions and withstand and recover from, and address adverse effects on its operations caused by, disruptions that may impact critical business functions and supporting assets, including its staff, technology, data and facilities, as well as those of third-party service providers. The framework includes governance, awareness training, planning and testing of recovery strategies, as well as strategic and tactical initiatives to identify, assess, and manage resiliency risks. The framework operates in accordance with the Firm's overall approach to Operational Risk Management, including alignment with technology, cybersecurity, data, physical security, crisis management, real estate and outsourcing programs.

Payment fraud risk
Payment fraud risk is the risk of external and internal parties unlawfully obtaining personal monetary benefit through misdirected or otherwise improper payment. The Firm employs various controls for managing payment fraud risk as well as providing employee and client education and awareness trainings.

Third-party outsourcing risk
The Firm's Third-Party Oversight ("TPO") and Inter-affiliates Oversight ("IAO") frameworks assist the LOBs and Corporate in selecting, documenting, onboarding, monitoring and managing their supplier relationships including services provided by affiliates. The objectives of the TPO framework are to hold suppliers and other third parties to an appropriate standard of operational performance and to mitigate key risks, including data loss and business disruptions. The Corporate Third-Party Oversight group is responsible for Firmwide training, monitoring, reporting and standards with respect to third-party outsourcing risks.

Cybersecurity risk
Cybersecurity risk is the risk of harm or loss resulting from misuse or abuse of technology or the unauthorized disclosure of data.

Overview

Cybersecurity risk is an important and continuously evolving focus for the Firm. Significant resources are devoted to protecting and enhancing the security of computer systems, software, networks, storage devices, and other technology. The Firm's security efforts are designed to protect against, among other things, cybersecurity attacks that can result in unauthorized access to confidential information, the destruction of data, disruptions to or degradations of service, the sabotaging of systems or other damage.

The Firm has experienced, and expects that it will continue to experience, a higher volume and complexity of cyber attacks against the backdrop of heightened geopolitical tensions. The Firm has implemented measures and controls reasonably designed to address this evolving environment, including enhanced threat monitoring. In addition, the Firm continues to review and enhance its capabilities to address associated risks, such as those relating to the management of administrative access to systems.

Third parties with which the Firm does business, that facilitate the Firm's business activities (e.g., vendors, supply chain, exchanges, clearing houses, central depositories, and financial intermediaries) or that the Firm has acquired are also sources of cybersecurity risk to the Firm. Third party incidents such as system breakdowns or failures, misconduct by the employees of such parties, or cyber attacks, including ransomware and supply-chain compromises, could have a material adverse effect on the Firm, including in circumstances in which an affected third party is unable to deliver a product or service to the Firm or where the incident delivers compromised software to the Firm or results in lost or compromised information of the Firm or its clients or customers.

Clients and customers are also sources of cybersecurity risk to the Firm and its information assets, particularly when their activities and systems are beyond the Firm's own security and control systems. The Firm engages in periodic discussions with its clients, customers and other external parties concerning cybersecurity risks including opportunities to improve cybersecurity.

Risks from cybersecurity threats, including any previous cybersecurity events, have not materially affected the Firm or its business strategy, results of operations or financial condition. Notwithstanding the comprehensive approach that the Firm takes to address cybersecurity risk, the Firm may not be successful in preventing or mitigating a future cybersecurity incident that could have a material adverse effect on the Firm or its business strategy, results of operations or financial condition.

Organization and management

The Global Chief Information Security Officer ("CISO") reports to the Global Chief Information Officer, and is a member of key cybersecurity governance forums. The CISO leads the Global Cybersecurity and Technology Controls organization, which is responsible for identifying technology and cybersecurity risks and for implementing and maintaining controls to manage cybersecurity threats. The CISO and the members of senior management within Global Technology and the Cybersecurity and Technology Controls organizations all have relevant expertise and experience in cybersecurity and information technology risk management, including relevant experience at the Firm, at other financial services companies or in other highly-regulated industries.

The CISO is responsible for the Firm's Information Security Program, which is designed to prevent, detect and respond to cyber attacks in order to help safeguard the confidentiality, integrity and availability of the Firm's infrastructure, resources and information. The program includes managing the Firm's global cybersecurity operations centers, providing training, conducting cybersecurity event simulation exercises, implementing the Firm's policies and standards relating to technology risk and cybersecurity management, and enhancing, as needed, the Firm's cybersecurity capabilities.

The Firm's Information Security Program includes the following functions:

Cyber Operations, which is responsible for implementing and maintaining controls designed to detect and defend the Firm against cyber attacks, and includes a dedicated function for incident response and ongoing monitoring for cybersecurity threats and vulnerabilities, including those among the Firm's third-party suppliers.

Technology Governance, Risk & Controls, which is responsible for operationalizing technology risk and control frameworks, analyzing regulatory developments that may impact the Firm, and developing control catalogs and assessments of controls, as well as overseeing governance and reporting of technology and cybersecurity risk.

Security Awareness, which provides awareness and training that reinforces information risk and security management practices and compliance with the Firm's policies, standards and practices. The training is mandatory for all employees globally on a periodic basis, and it is supplemented by Firmwide testing initiatives, including periodic phishing tests. The Firm also provides specialized security training to employees in specific roles, such as application developers. The Firm's Global Privacy Program requires all employees to take periodic training on data privacy that focuses on confidentiality and security, as well as responding to unauthorized access to or use of information.

Technology Resiliency, which establishes control requirements for planning and testing the prioritized recovery of technology services in the event of degradation or outage, including incident response planning, data backup and retention, and recovery readiness in support of the Firmwide Business Resiliency Program and operational risk management practices.

The Firm has a cybersecurity incident response plan designed to enable the Firm to respond to attempted cybersecurity incidents, coordinate as appropriate with law enforcement and other government agencies, notify clients and customers, as applicable, and recover from such incidents. In addition, the Firm actively partners with appropriate government and law enforcement agencies and peer industry forums, participating in discussions and simulations to assist in understanding the full spectrum of cybersecurity risks and in enhancing defenses and improving resiliency in the Firm's operating environment.

Governance and oversight

The governance structure for the Global Cybersecurity and Technology Controls organization is designed to appropriately identify, escalate and mitigate cybersecurity risks. Cybersecurity risk management and its governance and oversight are integrated into the Firm's operational risk management framework, including through the escalation of key risk and control issues to management and the development of risk mitigation plans for heightened risk and control issues. IRM independently assesses and challenges the activities and risk management practices of the Global Cybersecurity and Technology Controls organization related to the identification, assessment, measurement and mitigation of cybersecurity risk. As needed, the Firm engages third-party assessors or auditing firms with industry-recognized expertise on cybersecurity matters to review specific aspects of the Firm's cybersecurity risk management framework, processes and controls.

The governance and oversight for cybersecurity risk management includes governance forums that inform management of key areas of concern regarding the prevention, detection, mitigation and remediation of cybersecurity risks.

The Cybersecurity and Technology Controls Operating Committee ("CTOC") is the principal management committee that oversees the Firm's assessment and management of cybersecurity risk, including oversight of the implementation and maintenance of appropriate controls in support of the Firm's Information Security Program. The membership of the CTOC includes senior representatives from the Global Cybersecurity and Technology Controls organization and relevant corporate functions, including IRM and Internal Audit.

The CTOC escalates key operational risk and control issues, as appropriate, to the Global Technology Operating Committee ("GTOC") or its business control committee or to the appropriate LOB and Corporate Control Committees. The GTOC is responsible for the governance of the Firmwide Global Technology organization, including oversight of Firmwide technology strategies, the delivery of technology and technology operations, the effective use of information technology resources, and monitoring and resolving key operational risk and control matters arising in the Global Technology organization.

As part of its oversight of management's implementation and maintenance of the Firm's risk management framework, the Firm's Board of Directors receives periodic updates from the CIO, the CISO and senior members of the CTOC concerning cybersecurity matters. These updates generally include information regarding cybersecurity and technology developments, the Firm's Information Security Program and recommended changes to that program, cybersecurity policies and practices, and ongoing initiatives to improve information security, as well as any significant cybersecurity incidents and the Firm's efforts to address those incidents. The Audit Committee and the Risk Committee assist the Board in this oversight.

Compliance risk, a subcategory of operational risk, is the risk of failing to comply with laws, rules, regulations or codes of conduct and standards of self-regulatory organizations.

Overview

Each of the LOBs and Corporate hold primary ownership of and accountability for managing their compliance risk. The Firm's Operational Risk and Compliance Organization ("Operational Risk and Compliance"), which is independent of the LOBs and Corporate, provides independent review, monitoring and oversight of business operations with a focus on compliance with the laws, rules, and regulations applicable to the delivery of the Firm's products and services to clients and customers.

These compliance risks relate to a wide variety of laws, rules and regulations across the LOBs and Corporate, and jurisdictions, and include risks related to financial products and services, relationships and interactions with clients and customers, and employee activities. For example, compliance risks include those associated with anti-money laundering compliance, trading activities, market conduct, and complying with the laws, rules, and regulations related to the offering of products and services across jurisdictional borders. Compliance risk is also inherent in the Firm's fiduciary activities, including the failure to exercise the applicable standard of care to act in the best interest of fiduciary clients and customers or to treat fiduciary clients and customers fairly.

Other functions provide oversight of significant regulatory obligations that are specific to their respective areas of responsibility.

Operational Risk and Compliance implements policies and standards designed to govern, identify, measure, monitor and test, manage, and report on compliance risk.

Governance and oversight

Operational Risk and Compliance is led by the Firm's Global CCO and FRE for Operational Risk and Qualitative Risk Appetite.

The Firm maintains oversight and coordination of its compliance risk through the CCOR Management Framework. The Firm's Global CCO and FRE for Operational Risk and Qualitative Risk Appetite also provides regular updates to the Board Risk Committee and the Audit Committee on significant compliance risk issues, as appropriate.

Code of Conduct

The Firm has a Code of Conduct (the "Code") that sets forth the Firm's expectation that employees will conduct themselves with integrity, at all times. The Code provides the principles that help govern employee conduct with clients, customers, suppliers, vendors, shareholders, regulators, other employees, as well as with the markets and communities in which the Firm operates. The Code requires employees to promptly report any potential or actual violation of the Code, Firm policies, or laws, rules or regulations applicable to the Firm's business. It also requires employees to report any illegal or unethical conduct, or conduct that violates the underlying principles of the Code, by any of the Firm's employees, consultants, clients, customers, suppliers, contract or temporary workers, or business partners or agents. Training is assigned to newly hired employees after joining the Firm, and to current employees periodically thereafter. Employees are required to affirm their compliance with the Code annually.

Employees can report any potential or actual violations of the Code through the Firm's Conduct Hotline (the "Hotline") by phone, mobile device or the internet. The Hotline is anonymous, where permitted by law, is available at all times globally, has translation services, and is administered by an outside service provider. The Code prohibits retaliation against anyone who raises an issue or concern in good faith or assists with an inquiry or investigation. Periodically, the Audit Committee receives reports on the Code of Conduct program.

CONDUCT RISK MANAGEMENT

Conduct risk, a subcategory of operational risk, is the risk that any action or misconduct by an employee could lead to unfair client or customer outcomes, impact the integrity of the markets in which the Firm operates, harm employees or the Firm, or compromise the Firm's reputation.

Overview

Each LOB and Corporate is accountable for identifying and managing its conduct risk to provide appropriate engagement, ownership and sustainability of a culture consistent with the Firm's Business Principles. The Business Principles serve as a guide for how employees are expected to conduct themselves. With the Business Principles serving as a guide, the Firm's Code sets out the Firm's expectations for each employee and provides information and resources to help employees conduct business ethically and in compliance with applicable laws, rules and regulations everywhere the Firm operates. Refer to Compliance Risk Management on page 157 for further discussion of the Code.

Governance and oversight

The Firm's oversight and coordination of conduct risk is managed in the same manner as Compliance risk. Refer to Compliance Risk Management on page 157 for further information.

Conduct risk management encompasses various aspects of people management practices throughout the employee life cycle, including recruiting, onboarding, training and development, performance management, promotion and compensation processes. Each LOB, Treasury and CIO, and each designated corporate function completes an assessment of conduct risk periodically, reviews metrics and issues which may involve conduct risk, and provides conduct education as appropriate.

Legal risk, a subcategory of operational risk, is the risk of loss primarily caused by the actual or alleged failure to meet legal obligations that arise from the rule of law in jurisdictions in which the Firm operates, agreements with clients and customers, and products and services offered by the Firm.

Overview

The global Legal function ("Legal") provides legal services and advice to the Firm. Legal is responsible for managing the Firm's exposure to legal risk by:

- managing actual and potential litigation and enforcement matters, including internal reviews and investigations related to such matters
- advising on products and services, including contract negotiation and documentation
- advising on offering and marketing documents and new business initiatives
- managing dispute resolution
- interpreting existing laws, rules and regulations, and advising on changes to them
- advising on advocacy in connection with contemplated and proposed laws, rules and regulations, and
- providing legal advice to the LOBs, Corporate and the Board.

Legal selects, engages and manages outside counsel for the Firm on all matters in which outside counsel is engaged. In addition, Legal advises the Firm's Conflicts Office which reviews the Firm's wholesale transactions that may have the potential to create conflicts of interest for the Firm.

Governance and oversight

The Firm's General Counsel reports to the CEO and is a member of the Operating Committee, the Firmwide Risk Committee and the Firmwide Control Committee. The Firm's General Counsel and other members of Legal report on significant legal matters to the Firm's Board of Directors and to the Audit Committee.

Legal serves on and advises various committees and advises the Firm's LOBs and Corporate on potential reputation risk issues.

ESTIMATIONS AND MODEL RISK MANAGEMENT

Estimations and Model risk, a subcategory of operational risk, is the potential for adverse consequences from decisions based on incorrect or misused estimation outputs.

The Firm uses models and other analytical and judgment-based estimations, including those based upon machine learning or artificial intelligence techniques, across various businesses and functions. The estimation methods are of varying levels of sophistication and are used for many purposes, such as the valuation of positions and measurement of risk, assessing regulatory capital requirements, conducting stress testing, evaluating the allowance for credit losses and making business decisions. A dedicated independent function, Model Risk Governance and Review ("MRGR"), defines and governs the Firm's policies relating to the management of model risk and risks associated with certain analytical and judgment-based estimations, such as those used in risk management, budget forecasting and capital planning and analysis.

Model risks are owned by the users of the models within the LOBs and Corporate based on the specific purposes of such models. Users and developers of models are responsible for developing, implementing and testing their models, as well as referring models to MRGR for review and approval. Once models have been approved, model users and developers are responsible for maintaining a robust operating environment, and must monitor and evaluate the performance of the models on an ongoing basis. Model users and developers may seek to enhance models in response to changes in the relevant portfolios and in product and market developments, as well as to capture improvements in available modeling techniques and systems capabilities.

Models are tiered based on an internal standard according to their complexity, the exposure associated with the model and the Firm's reliance on the model. This tiering is subject to the approval of MRGR. In its review of a model, MRGR considers whether the model is suitable for the specific purposes for which it will be used. When reviewing a model, MRGR analyzes and challenges the model methodology and the reasonableness of model assumptions, and may perform or require additional testing, including back-testing of model outcomes. Model reviews are approved by the appropriate level of management within MRGR based on the relevant model tier.

Under the Firm's Estimations and Model Risk Management Policy, MRGR reviews and approves new models, as well as material changes to existing models, prior to their use. In certain circumstances, exceptions may be granted to the Firm's policy to allow a model to be used prior to review or approval. MRGR may also require the user to take appropriate actions to mitigate the model risk if it is to be used in the interim. These actions will depend on the model and may include, for example, limitation of trading activity.

While models are inherently imprecise, the degree of imprecision or uncertainty can be heightened by the market or economic environment. This is particularly true when the current and forecasted environments are significantly different from the historical environments upon which the models were developed. This increased uncertainty may necessitate a greater degree of judgment and analytics to inform any adjustments that the Firm may make to model outputs than would otherwise be the case. In addition, the Firm may experience increased uncertainty in its estimates if assets acquired differ from those used to develop the models.

Refer to Critical Accounting Estimates Used by the Firm on pages 161–164 and Note 2 for a summary of model-based valuations and other valuation techniques.

JPMorganChase's accounting policies and use of estimates are integral to understanding its reported results. The Firm's most complex accounting estimates require management's judgment to ascertain the appropriate carrying value of assets and liabilities. The Firm has established policies and control procedures intended to ensure that estimation methods, including any judgments made as part of such methods, are well-controlled, independently reviewed and applied consistently from period to period. The methods used and judgments made reflect, among other factors, the nature of the assets or liabilities and the related business and risk management strategies, which may vary across the Firm's businesses and portfolios. In addition, the policies and procedures are intended to ensure that the process for changing methodologies occurs in an appropriate manner. The Firm believes its estimates for determining the carrying value of its assets and liabilities are appropriate. The following is a brief description of the Firm's critical accounting estimates involving significant judgments.

Allowance for credit losses

The Firm's allowance for credit losses represents management's estimate of expected credit losses over the remaining expected life of the Firm's financial assets measured at amortized cost and certain off-balance sheet lending-related commitments. The allowance for credit losses generally comprises:

- The allowance for loan losses, which covers the Firm's retained loan portfolios (scored and risk-rated),
- The allowance for lending-related commitments, and
- The allowance for credit losses on investment securities.

The allowance for credit losses involves significant judgment on a number of matters including development and weighting of macroeconomic forecasts, incorporation of historical loss experience, assessment of risk characteristics, assignment of risk ratings, valuation of collateral, and the determination of remaining expected life. Refer to Note 10 and Note 13 for further information on these judgments as well as the Firm's policies and methodologies used to determine the Firm's allowance for credit losses.

One of the most significant judgments involved in estimating the Firm's allowance for credit losses relates to the macroeconomic forecasts used to estimate credit losses over the eight-quarter forecast period within the Firm's methodology. The eight-quarter forecast incorporates hundreds of macroeconomic variables ("MEVs") that are relevant for exposures across the Firm, with modeled credit losses being driven primarily by a subset of less than twenty variables. The specific variables that have the greatest effect on the modeled losses vary by portfolio and geography.

- Key MEVs for the consumer portfolio include regional U.S. unemployment rates and U.S. HPI.
- Key MEVs for the wholesale portfolio include U.S. unemployment, U.S. real GDP, U.S. equity prices, U.S. interest rates, U.S. corporate credit spreads, oil prices, U.S. commercial real estate prices and U.S. HPI.

Changes in the Firm's assumptions and forecasts of economic conditions could significantly affect its estimate of expected credit losses in the portfolio at the balance sheet date or lead to significant changes in the estimate from one reporting period to the next.

As a result of the First Republic acquisition, the Firm recorded an allowance for credit losses for the loans acquired and lending-related commitments assumed as of May 1, 2023. Due to differences in risk rating methodologies for the First Republic portfolio and the ongoing integration of products and systems, the allowance for credit losses for the acquired wholesale portfolio was initially measured based on similar risk characteristics from other facilities underwritten by the Firm. Starting in the second quarter of 2024, the acquired portfolio was incorporated into the Firm's modeled credit loss estimates and is now reflected in the wholesale sensitivity analysis below. Refer to Note 34 for additional information on the First Republic acquisition.

It is difficult to estimate how potential changes in any one factor or input might affect the overall allowance for credit losses because management considers a wide variety of factors and inputs in estimating the allowance for credit losses. Changes in the factors and inputs considered may not occur at the same rate and may not be consistent across all geographies or product types, and changes in factors and inputs may be directionally inconsistent, such that improvement in one factor or input may offset deterioration in others.

To consider the impact of a hypothetical alternate macroeconomic forecast, the Firm compared the modeled credit losses determined using its central and relative adverse macroeconomic scenarios, which are two of the five scenarios considered in estimating the allowances for loan losses and lending-related commitments. The central and relative adverse scenarios each included a full suite of MEVs, but differed in the levels, paths and peaks/troughs of those variables over the eight-quarter forecast period.

For example, compared to the Firm's central scenario shown on page 137 and in Note 13, the Firm's relative adverse scenario assumes an elevated U.S.

unemployment rate, averaging approximately 2.1% higher over the eight-quarter forecast, with a peak difference of approximately 3.0% in the fourth quarter of 2025.

This analysis is not intended to estimate expected future changes in the allowance for credit losses, for a number of reasons, including:

- The allowance as of December 31, 2024, reflects credit losses beyond those estimated under the central scenario due to the weight placed on the adverse scenarios.
- The impacts of changes in many MEVs are both interrelated and nonlinear, so the results of this analysis cannot be simply extrapolated for more severe changes in macroeconomic variables.
- Expectations of future changes in portfolio composition and borrower behavior can significantly affect the allowance for credit losses.

To demonstrate the sensitivity of credit loss estimates to macroeconomic forecasts as of December 31, 2024, the Firm compared the modeled estimates under its relative adverse scenario to its central scenario. Without considering offsetting or correlated effects in other qualitative components of the Firm's allowance for credit losses, the comparison between these two scenarios for the exposures below reflect the following differences:

- An increase of approximately $850 million for residential real estate loans and lending-related commitments
- An increase of approximately $3.7 billion for credit card loans
- An increase of approximately $4.1 billion for wholesale loans and lending-related commitments

This analysis relates only to the modeled credit loss estimates and is not intended to estimate changes in the overall allowance for credit losses as it does not reflect any potential changes in other adjustments to the quantitative calculation, which would also be influenced by the judgment management applies to the modeled lifetime loss estimates to reflect the uncertainty and imprecision of these modeled lifetime loss estimates based on then-current circumstances and conditions.

Recognizing that forecasts of macroeconomic conditions are inherently uncertain, the Firm believes that its process to consider the available information and associated risks and uncertainties is appropriately governed and that its estimates of expected credit losses were reasonable and appropriate for the period ended December 31, 2024.

Fair value

JPMorganChase carries a portion of its assets and liabilities at fair value. The majority of such assets and liabilities are measured at fair value on a recurring basis, including derivatives, structured note products and certain securities financing agreements. Certain assets and liabilities are measured at fair value on a nonrecurring basis, including certain mortgage, home equity and other loans, where the carrying value is based on the fair value of the underlying collateral.

Assets measured at fair value

The following table includes the Firm's assets measured at fair value and the portion of such assets that are classified within level 3 of the fair value hierarchy. Refer to Note 2 for further information.

December 31, 2024 (in millions, except ratios)	Total assets at fair value	Total level 3 assets
Federal funds sold and securities purchased under resale agreements	$ 286,771	$ —
Securities borrowed	83,962	—
Trading assets:		
Trading-debt and equity instruments	576,817	2,442
Derivative receivables[(a)]	60,967	8,452
Total trading assets	637,784	10,894
AFS securities	406,852	8
Loans	41,350	2,416
MSRs	9,121	9,121
Other	14,073	1,344
Total assets measured at fair value on a recurring basis	1,479,913	23,783
Total assets measured at fair value on a nonrecurring basis	2,489	1,742
Total assets measured at fair value	$ 1,482,402	$ 25,525
Total Firm assets	$ 4,002,814	
Level 3 assets at fair value as a percentage of total Firm assets[(a)]		1%
Level 3 assets at fair value as a percentage of total Firm assets at fair value[(a)]		2%

(a) For purposes of the table above, the derivative receivables total reflects the impact of netting adjustments; however, the $8.5 billion of derivative receivables classified as level 3 does not reflect the netting adjustment as such netting is not relevant to a presentation based on the transparency of inputs to the valuation of an asset. The level 3 balances would be reduced if netting were applied, including the netting benefit associated with cash collateral.

Valuation
Details of the Firm's processes for determining fair value are set out in Note 2. Estimating fair value requires the application of judgment. The type and level of judgment required is largely dependent on the amount of observable market information available to the Firm. For instruments valued using internally developed valuation models and other valuation techniques that use significant unobservable inputs and are therefore classified within level 3 of the fair value hierarchy, judgments used to estimate fair value are more significant than those required when estimating the fair value of instruments classified within levels 1 and 2.

In arriving at an estimate of fair value for an instrument within level 3, management must first determine the appropriate valuation model or other valuation technique to use. Second, the lack of observability of certain significant inputs requires management to assess relevant empirical data in deriving valuation inputs including, for example, transaction details, yield curves, interest rates, prepayment speeds, default rates, volatilities, correlations, prices (such as commodity, equity or debt prices), valuations of comparable instruments, foreign exchange rates and credit curves. Refer to Note 2 for a further discussion of the valuation of level 3 instruments, including unobservable inputs used.

For instruments classified in levels 2 and 3, management judgment must be applied to assess the appropriate level of valuation adjustments to reflect counterparty credit quality, the Firm's creditworthiness, market funding rates, liquidity considerations, unobservable parameters, and for portfolios that meet specified criteria, the size of the net open risk position. The judgments made are typically affected by the type of product and its specific contractual terms, and the level of liquidity for the product or within the market as a whole. In periods of heightened market volatility and uncertainty judgments are further affected by the wider variation of reasonable valuation estimates, particularly for positions that are less liquid. Refer to Note 2 for a further discussion of valuation adjustments applied by the Firm.

Imprecision in estimating unobservable market inputs or other factors can affect the amount of gain or loss recorded for a particular position. Furthermore, while the Firm believes its valuation methods are appropriate and consistent with those of other market participants, the methods and assumptions used reflect management judgment and may vary across the Firm's businesses and portfolios.

The Firm uses various methodologies and assumptions in the determination of fair value. The use of methodologies or assumptions different than those used by the Firm could result in a different estimate of fair value at the reporting date. Refer to Note 2 for a detailed discussion of the Firm's valuation process and hierarchy, and its determination of fair value for individual financial instruments.

Goodwill impairment

Under U.S. GAAP, goodwill must be allocated to reporting units and tested for impairment at least annually. The Firm's process and methodology used to conduct goodwill impairment testing is described in Note 15.

Management applies significant judgment when testing goodwill for impairment. The goodwill associated with each business combination is allocated to the related reporting units for goodwill impairment testing.

For the year ended December 31, 2024, the Firm reviewed current economic conditions, estimated market cost of equity, as well as actual business results and projections of business performance. Based on such reviews, the Firm has concluded that goodwill was not impaired as of December 31, 2024. For each of the reporting units, fair value exceeded carrying value by at least 10% and there was no indication of a significant risk of goodwill impairment based on current projections and valuations.

The projections for the Firm's reporting units are consistent with management's current business outlook assumptions in the short term, and the Firm's best estimates of long-term growth and return on equity in the longer term. Where possible, the Firm uses third-party and peer data to benchmark its assumptions and estimates.

Refer to Note 15 for additional information on goodwill, including the goodwill impairment assessment as of December 31, 2024.

Credit card rewards liability

JPMorganChase offers credit cards with various rewards programs which allow cardholders to earn rewards points based on their account activity and the terms and conditions of the rewards program. Generally, there are no limits on the points that an eligible cardholder can earn, nor do the points expire, and the points can be redeemed for a variety of rewards, including cash (predominantly in the form of account credits), gift cards and travel. The Firm maintains a rewards liability which represents the estimated cost of rewards points earned and expected to be redeemed by cardholders. The liability is accrued as the cardholder earns the benefit and is reduced when the cardholder redeems points. This liability was $14.4 billion and $13.2 billion at December 31, 2024 and 2023, respectively, and is recorded in accounts payable and other liabilities on the Consolidated balance sheets. The increase in the liability was driven by continued growth in rewards points earned on higher spend and promotional offers outpacing redemptions throughout 2024.

The rewards liability is sensitive to redemption rate ("RR") and cost per point ("CPP") assumptions. The RR assumption is used to estimate the number of points earned by customers that will be redeemed over the life of the account. The CPP assumption is used to estimate the cost of future point redemptions. These assumptions are evaluated periodically considering historical actuals, cardholder redemption behavior and management judgment. Updates to these assumptions will impact the rewards liability. As of December 31, 2024, a combined increase of 25 basis points in RR and 1 basis point in CPP would increase the rewards liability by approximately $442 million.

Income taxes

JPMorganChase is subject to the income tax laws of the various jurisdictions in which it operates, including U.S. federal, state and local, and non-U.S. jurisdictions. These laws are often complex and may be subject to different interpretations. To determine the financial statement impact of accounting for income taxes, including the provision for income tax expense and unrecognized tax benefits, JPMorganChase must make assumptions and judgments about how to interpret and apply these complex tax laws to numerous transactions and business events, as well as make judgments regarding the timing of when certain items may affect taxable income in the U.S. and non-U.S. tax jurisdictions.

JPMorganChase's interpretations of tax laws around the world are subject to review and examination by the various taxing authorities in the jurisdictions where the Firm operates, and disputes may occur regarding its view on a tax position. These disputes over interpretations with the various taxing authorities may be settled by audit, administrative appeals or adjudication in the court systems of the tax jurisdictions in which the Firm operates. JPMorganChase regularly reviews whether it may be assessed additional income taxes as a result of the resolution of these matters, and the Firm records additional unrecognized tax benefits, as appropriate. In addition, the Firm may revise its estimate of income taxes due to changes in income tax laws, legal interpretations, and business strategies. It is possible that revisions in the Firm's estimate of income taxes may materially affect the Firm's results of operations in any reporting period.

Deferred taxes arise from differences between assets and liabilities measured for financial reporting versus income tax return purposes. Deferred tax assets are recognized if, in management's judgment, their realizability is determined to be more likely than not. Deferred taxes are measured using enacted tax rates expected to apply to taxable income in the years in which those temporary differences are expected to be recovered or settled. The effect on deferred taxes of a change in tax rates is recognized within the provision for income taxes in the period enacted.

The Firm has also recognized deferred tax assets in connection with certain tax attributes, including net operating loss ("NOL") carryforwards and foreign tax credit ("FTC") carryforwards. The Firm performs regular reviews to ascertain whether its deferred tax assets are realizable. These reviews include management's estimates and assumptions regarding future taxable income, including foreign source income, and may incorporate various tax planning strategies, including strategies that may be available to utilize NOLs and FTCs before they expire. In connection with these reviews, if it is determined that a deferred tax asset is not realizable, a valuation allowance is established. The valuation allowance may be reversed in a subsequent reporting period if the Firm determines that, based on revised estimates of future taxable income or changes in tax planning strategies, it is more likely than not that all or part of the deferred tax asset will become realizable. As of December 31, 2024, management has determined it is more likely than not that the Firm will realize its deferred tax assets, net of the existing valuation allowance.

The Firm adjusts its unrecognized tax benefits as necessary when new information becomes available, including changes in tax law and regulations, and interactions with taxing authorities. Uncertain tax positions that meet the more-likely-than-not recognition threshold are measured to determine the amount of benefit to recognize. An uncertain tax position is measured at the largest amount of benefit that management believes is more likely than not to be realized upon settlement. It is possible that the reassessment of JPMorganChase's unrecognized tax benefits may have a material impact on its effective income tax rate in the period in which the reassessment occurs. Although the Firm believes that its estimates are reasonable, the final tax amount could be different from the amounts reflected in the Firm's income tax provisions and accruals. To the extent that the final outcome of these amounts is different than the amounts recorded, such differences will generally impact the Firm's provision for income taxes in the period in which such a determination is made.

The Firm's provision for income taxes is composed of current and deferred taxes. The current and deferred tax provisions are calculated based on estimates and assumptions that could differ from the actual results reflected in income tax returns filed during the subsequent year. Adjustments based on filed returns are generally recorded in the period when the tax returns are filed and the global tax implications are known, which could impact the Firm's effective tax rate.

Refer to Note 25 for additional information on income taxes.

Litigation reserves

Refer to Note 30 for a description of the significant estimates and judgments associated with establishing litigation reserves.

Financial Accounting Standards Board ("FASB") Standards Adopted since January 1, 2024

Standard	Summary of guidance	Effects on financial statements
Fair Value Measurement: Fair Value Measurement of Equity Securities Subject to Contractual Sale Restrictions *Issued June 2022*	• Clarifies that a contractual sale restriction is not considered part of the unit of account of the equity security and, therefore, is not considered in measuring fair value. • Requires disclosure for investments in equity securities subject to contractual sale restrictions, including: 1) fair value of these investments, 2) nature and remaining duration of the restriction(s) and 3) circumstances that could cause a lapse in the restriction(s).	• Adopted prospectively on January 1, 2024, with no impact to the Firm's Consolidated Financial Statements.
Investments - Equity Method and Joint Ventures: Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method *Issued March 2023*	• Expands the ability to elect proportional amortization on a program-by-program basis, for additional types of tax-oriented investments (beyond affordable housing tax credit investments). • May be adopted using a full retrospective method, or a modified retrospective method wherein the effect of adoption is reflected as an adjustment to retained earnings at the effective date.	• Adopted under the modified retrospective method on January 1, 2024. • Refer to Note 1 for further information.
Segment Reporting: Improvements to Reportable Segment Disclosures *Issued November 2023*	• Requires disclosure of significant segment expenses that are readily provided to the chief operating decision maker ("CODM") and included in segment profit or loss. • Requires disclosure of the composition and aggregate amount of other segment items, which represent the difference between profit or loss and segment revenues less significant segment expenses. • Requires disclosure of the title and position of the CODM and an explanation of how the CODM uses the reported segment measures in assessing segment performance and deciding how to allocate resources.	• Adopted retrospectively for the Firm's annual Consolidated Financial Statements for the year ended December 31, 2024.[a] • The adoption of this guidance resulted in additional reportable segment disclosures, primarily relating to significant segment expenses and the CODM. Refer to Note 32 for further information.

(a) The accounting standards update applies to the Firm's annual Consolidated Financial Statements for the year ended December 31, 2024, and interim financial statements thereafter.

FASB Standards Issued but not yet Adopted as of December 31, 2024

Standard	Summary of guidance	Effects on financial statements
Income Taxes: Improvements to Income Tax Disclosures *Issued December 2023*	• Requires disclosure of income taxes paid disaggregated by 1) federal, state, and foreign taxes and 2) individual jurisdiction on the basis of a quantitative threshold of equal to or greater than 5 percent of total income taxes paid (net of refunds received). • Requires disclosure of the effective tax rate reconciliation by specific categories, at a minimum, with accompanying qualitative disclosures, and separate disclosure of reconciling items based on quantitative thresholds. • Requires categories within the effective tax rate reconciliation to be further disaggregated if quantitative thresholds are met.	• Required effective date: Annual financial statements for the year ending December 31, 2025. • The guidance can be applied on a prospective basis with the option to apply the standard retrospectively. • The Firm is evaluating the potential impact on the Consolidated Financial Statements disclosures.
Income Statement - Reporting Comprehensive Income - Expense Disaggregation Disclosures: Disaggregation of Income Statement Expenses *Issued November 2024*	• Requires additional annual and interim disclosures about specific types of expenses presented in the Consolidated statements of income.	• Required effective date: Annual financial statements for the year ending December 31, 2027, and interim financial statements for the year ending December 31, 2028. (a) • The guidance can be applied on a prospective basis with the option to apply the standard retrospectively. • The Firm is evaluating the potential impact on the Consolidated Financial Statements disclosures, as well as the Firm's planned date of adoption.

(a) Early adoption is permitted.

FORWARD-LOOKING STATEMENTS

From time to time, the Firm has made and will make forward-looking statements. These statements can be identified by the fact that they do not relate strictly to historical or current facts. Forward-looking statements often use words such as “anticipate,” “target,” “expect,” “estimate,” “intend,” “plan,” “goal,” “believe,” or other words of similar meaning. Forward-looking statements provide JPMorganChase’s current expectations or forecasts of future events, circumstances, results or aspirations. JPMorganChase’s disclosures in this 2024 Form 10-K contain forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. The Firm also may make forward-looking statements in its other documents filed or furnished with the SEC. In addition, the Firm’s senior management may make forward-looking statements orally to investors, analysts, representatives of the media and others.

All forward-looking statements are, by their nature, subject to risks and uncertainties, many of which are beyond the Firm’s control. JPMorganChase’s actual future results may differ materially from those set forth in its forward-looking statements. While there is no assurance that any list of risks and uncertainties or risk factors is complete, below are certain factors which could cause actual results to differ from those in the forward-looking statements:

- Local, regional and global business, economic and political conditions and geopolitical events, including geopolitical tensions and hostilities;
- Changes in laws, rules and regulatory requirements, including capital and liquidity requirements affecting the Firm’s businesses, and the ability of the Firm to address those requirements;
- Heightened regulatory and governmental oversight and scrutiny of JPMorganChase’s business practices, including dealings with retail customers;
- Changes in trade, monetary and fiscal policies and laws;
- Changes in the level of inflation;
- Changes in income tax laws, rules, and regulations;
- Changes in FDIC assessments;
- Securities and capital markets behavior, including changes in market liquidity and volatility;
- Changes in investor sentiment or consumer spending or savings behavior;
- Ability of the Firm to manage effectively its capital and liquidity;
- Changes in credit ratings assigned to the Firm or its subsidiaries;
- Damage to the Firm’s reputation;
- Ability of the Firm to appropriately address social, environmental and sustainability concerns that may arise, including from its business activities;
- Ability of the Firm to deal effectively with an economic slowdown or other economic or market disruption, including, but not limited to, in the interest rate environment;
- Technology changes instituted by the Firm, its counterparties or competitors;
- The effectiveness of the Firm’s control agenda;
- Ability of the Firm to develop or discontinue products and services, and the extent to which products or services previously sold by the Firm require the Firm to incur liabilities or absorb losses not contemplated at their initiation or origination;
- Acceptance of the Firm’s new and existing products and services by the marketplace and the ability of the Firm to innovate and to increase market share;
- Ability of the Firm to attract and retain qualified employees;
- Ability of the Firm to control expenses;
- Competitive pressures;
- Changes in the credit quality of the Firm’s clients, customers and counterparties;
- Adequacy of the Firm’s risk management framework, disclosure controls and procedures and internal control over financial reporting;
- Adverse judicial or regulatory proceedings;
- Ability of the Firm to determine accurate values of certain assets and liabilities;
- Occurrence of natural or man-made disasters or calamities, including health emergencies, the spread of infectious diseases, epidemics or pandemics, an outbreak or escalation of hostilities or other geopolitical instabilities, the effects of climate change or extraordinary events beyond the Firm’s control, and the Firm’s ability to deal effectively with disruptions caused by the foregoing;
- Ability of the Firm to maintain the security of its financial, accounting, technology, data processing and other operational systems and facilities;
- Ability of the Firm to withstand disruptions that may be caused by any failure of its operational systems or those of third parties;
- Ability of the Firm to effectively defend itself against cyber attacks and other attempts by unauthorized parties to access information of the Firm or its customers or to disrupt the Firm’s systems; and
- The other risks and uncertainties detailed in Part I, Item 1A: Risk Factors in JPMorganChase’s 2024 Form 10-K.

Any forward-looking statements made by or on behalf of the Firm speak only as of the date they are made, and JPMorganChase does not undertake to update any forward-looking statements. The reader should, however, consult any further disclosures of a forward-looking nature the Firm may make in any subsequent Annual Reports on Form 10-Ks, Quarterly Reports on Form 10-Qs, or Current Reports on Form 8-K.

Management's report on internal control over financial reporting

Management of JPMorgan Chase & Co. ("JPMorganChase" or the "Firm") is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is a process designed by, or under the supervision of, the Firm's principal executive and principal financial officers, or persons performing similar functions, and effected by JPMorganChase's Board of Directors, management and other personnel, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with accounting principles generally accepted in the United States of America ("U.S. GAAP").

JPMorganChase's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records, that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the Firm's assets; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with U.S. GAAP, and that receipts and expenditures of the Firm are being made only in accordance with authorizations of JPMorganChase's management and directors; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of the Firm's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate. Management has completed an assessment of the effectiveness of the Firm's internal control over financial reporting as of December 31, 2024. In making the assessment, management used the "Internal Control — Integrated Framework" ("COSO 2013") promulgated by the Committee of Sponsoring Organizations of the Treadway Commission ("COSO").

Based upon the assessment performed, management concluded that as of December 31, 2024, JPMorganChase's internal control over financial reporting was effective based upon the COSO 2013 framework. Additionally, based upon management's assessment, the Firm determined that there were no material weaknesses in its internal control over financial reporting as of December 31, 2024.

The effectiveness of the Firm's internal control over financial reporting as of December 31, 2024, has been audited by PricewaterhouseCoopers LLP, an independent registered public accounting firm, as stated in their report which appears herein.

James Dimon
Chairman and Chief Executive Officer

Jeremy Barnum
Executive Vice President and Chief Financial Officer

February 14, 2025

pwc

To the Board of Directors and Shareholders of JPMorgan Chase & Co.:

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of JPMorgan Chase & Co. and its subsidiaries (the "Firm") as of December 31, 2024 and 2023, and the related consolidated statements of income, comprehensive income, changes in stockholders' equity and cash flows for each of the three years in the period ended December 31, 2024, including the related notes (collectively referred to as the "consolidated financial statements"). We also have audited the Firm's internal control over financial reporting as of December 31, 2024, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Firm as of December 31, 2024 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2024 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Firm maintained, in all material respects, effective internal control over financial reporting as of December 31, 2024, based on criteria established in *Internal Control – Integrated Framework* (2013) issued by the COSO.

Basis for Opinions

The Firm's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's report on internal control over financial reporting. Our responsibility is to express opinions on the Firm's consolidated financial statements and on the Firm's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Firm in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

PricewaterhouseCoopers LLP • 300 Madison Avenue • New York, NY 10017

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current period audit of the consolidated financial statements that were communicated or required to be communicated to the audit committee and that (i) relate to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Allowance for Loan Losses – Portfolio-based component of Wholesale Loan and Credit Card Loan Portfolios

As described in Note 13 to the consolidated financial statements, the allowance for loan losses for the portfolio-based component of the wholesale and credit card loan portfolios was $22.0 billion on total portfolio-based retained loans of $919.3 billion at December 31, 2024. The Firm's allowance for loan losses represents management's estimate of expected credit losses over the remaining expected life of the Firm's loan portfolios and considers expected future changes in macroeconomic conditions. The portfolio-based component of the Firm's allowance for loan losses for the wholesale and credit card retained loan portfolios begins with a quantitative calculation of expected credit losses over the expected life of the loan by applying credit loss factors to the estimated exposure at default. The credit loss factors applied are determined based on the weighted average of five internally developed macroeconomic scenarios that take into consideration the Firm's economic outlook as derived through forecast macroeconomic variables, the most significant of which are U.S. unemployment and U.S. real gross domestic product. This quantitative calculation is further adjusted to take into consideration model imprecision, emerging risk assessments, trends and other subjective factors that are not yet otherwise reflected in the credit loss estimate.

The principal considerations for our determination that performing procedures relating to the allowance for loan losses for the portfolio-based component of the wholesale and credit card loan portfolios is a critical audit matter are (i) the significant judgment and estimation by management in the forecast of macroeconomic variables, specifically U.S. unemployment and U.S. real gross domestic product, as the Firm's forecasts of economic conditions significantly affect its estimate of expected credit losses at the balance sheet date, (ii) the significant judgment and estimation by management in determining the quantitative calculation utilized in their credit loss estimates and the adjustments to take into consideration model imprecision, emerging risk assessments, trends and other subjective factors that are not yet otherwise reflected in the credit loss estimate, which in turn led to a high degree of auditor judgment, subjectivity, and effort in performing procedures and in evaluating audit evidence obtained relating to the credit loss estimates and the appropriateness of the adjustments to the credit loss estimates, and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the Firm's allowance for loan losses, including controls over model validation and generation of macroeconomic scenarios. These procedures also included, among others, testing management's process for estimating the allowance for loan losses, which involved (i) evaluating the appropriateness of the models and methodologies used in quantitative calculations; (ii) evaluating the reasonableness of forecasts of U.S. unemployment and U.S. real gross domestic product; (iii) testing the completeness and accuracy of data used in the estimate; and (iv) evaluating the reasonableness of management's adjustments to the quantitative output for the impacts of model imprecision, emerging risk assessments, trends and other subjective factors that are not yet otherwise reflected in the credit loss estimate. These procedures also included the use of professionals with specialized skill and knowledge to assist in evaluating the appropriateness of certain models, methodologies and macroeconomic variables.

Fair Value of Certain Level 3 Financial Instruments

As described in Notes 2 and 3 to the consolidated financial statements, the Firm carries $1.5 trillion of its assets and $586.2 billion of its liabilities at fair value on a recurring basis. Included in these balances are $10.9 billion of trading assets and $51.8 billion of liabilities measured at fair value on a recurring basis, collectively financial instruments, which are classified as level 3 as they contain one or more inputs to valuation which are unobservable and significant to their fair value measurement. The Firm utilized internally developed valuation models and unobservable inputs to estimate fair value of the level 3 financial instruments. The unobservable inputs used by management to estimate the fair value of certain of these financial instruments include interest rate volatility, equity volatility, Bermudan switch value and correlation relating to interest rates, interest rate-to-foreign exchange, equity prices, equity-to-foreign exchange and equity-to-interest rate.

The principal considerations for our determination that performing procedures relating to the fair value of certain level 3 financial instruments is a critical audit matter are (i) the significant judgment and estimation by management in determining the inputs to estimate fair value, which in turn led to a high degree of auditor judgment, subjectivity, and effort in performing procedures and in evaluating audit evidence obtained related to the fair value of these financial instruments, and (ii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the Firm's determination of the fair value, including controls over models, inputs, and data. These procedures also included, among others, the involvement of professionals with specialized skill and knowledge to assist in developing an independent estimate of fair value for a sample of these financial instruments and comparing management's estimate to the independently developed estimate of fair value. Developing the independent estimate involved testing the completeness and accuracy of data provided by management, developing independent inputs and, as appropriate, evaluating and utilizing management's aforementioned unobservable inputs.

PricewaterhouseCoopers LLP

New York, New York
February 14, 2025

We have served as the Firm's auditor since 1965.

JPMorgan Chase & Co.
Consolidated statements of income

Year ended December 31, (in millions, except per share data)	2024	2023	2022
Revenue			
Investment banking fees	**$ 8,910**	$ 6,519	$ 6,686
Principal transactions	**24,787**	24,460	19,912
Lending- and deposit-related fees	**7,606**	7,413	7,098
Asset management fees	**17,801**	15,220	14,096
Commissions and other fees	**7,530**	6,836	6,581
Investment securities losses	**(1,021)**	(3,180)	(2,380)
Mortgage fees and related income	**1,401**	1,176	1,250
Card income	**5,497**	4,784	4,420
Other income	**12,462**	5,609	4,322
Noninterest revenue	**84,973**	68,837	61,985
Interest income	**193,933**	170,588	92,807
Interest expense	**101,350**	81,321	26,097
Net interest income	**92,583**	89,267	66,710
Total net revenue	**177,556**	158,104	128,695
Provision for credit losses	**10,678**	9,320	6,389
Noninterest expense			
Compensation expense	**51,357**	46,465	41,636
Occupancy expense	**5,026**	4,590	4,696
Technology, communications and equipment expense	**9,831**	9,246	9,358
Professional and outside services	**11,057**	10,235	10,174
Marketing	**4,974**	4,591	3,911
Other expense	**9,552**	12,045	6,365
Total noninterest expense	**91,797**	87,172	76,140
Income before income tax expense	**75,081**	61,612	46,166
Income tax expense	**16,610**	12,060	8,490
Net income	**$ 58,471**	$ 49,552	$ 37,676
Net income applicable to common stockholders	**$ 56,868**	$ 47,760	$ 35,892
Net income per common share data			
Basic earnings per share	**$ 19.79**	$ 16.25	$ 12.10
Diluted earnings per share	**19.75**	16.23	12.09
Weighted-average basic shares	**2,873.9**	2,938.6	2,965.8
Weighted-average diluted shares	**2,879.0**	2,943.1	2,970.0

The Notes to Consolidated Financial Statements are an integral part of these statements.

JPMorgan Chase & Co.
Consolidated statements of comprehensive income

Year ended December 31, (in millions)	2024	2023	2022
Net income	**$ 58,471**	$ 49,552	$ 37,676
Other comprehensive income/(loss), after–tax			
Unrealized gains/(losses) on investment securities	**(87)**	5,381	(11,764)
Translation adjustments, net of hedges	**(858)**	329	(611)
Fair value hedges	**(87)**	(101)	98
Cash flow hedges	**(882)**	1,724	(5,360)
Defined benefit pension and OPEB plans	**(63)**	373	(1,241)
DVA on fair value option elected liabilities	**(36)**	(808)	1,621
Total other comprehensive income/(loss), after–tax	**(2,013)**	6,898	(17,257)
Comprehensive income	**$ 56,458**	$ 56,450	$ 20,419

The Notes to Consolidated Financial Statements are an integral part of these statements.

JPMorgan Chase & Co.
Consolidated balance sheets

December 31, (in millions, except share data)	2024	2023
Assets		
Cash and due from banks	**$ 23,372**	$ 29,066
Deposits with banks	**445,945**	595,085
Federal funds sold and securities purchased under resale agreements (included **$286,771** and $259,813 at fair value)	**295,001**	276,152
Securities borrowed (included **$83,962** and $70,086 at fair value)	**219,546**	200,436
Trading assets (included assets pledged of **$136,070** and $128,994)	**637,784**	540,607
Available-for-sale securities (amortized cost of **$411,045** and $205,456; included assets pledged of **$10,162** and $9,219)	**406,852**	201,704
Held-to-maturity securities	**274,468**	369,848
Investment securities, net of allowance for credit losses	**681,320**	571,552
Loans (included **$41,350** and $38,851 at fair value)	**1,347,988**	1,323,706
Allowance for loan losses	**(24,345)**	(22,420)
Loans, net of allowance for loan losses	**1,323,643**	1,301,286
Accrued interest and accounts receivable	**101,223**	107,363
Premises and equipment	**32,223**	30,157
Goodwill, MSRs and other intangible assets	**64,560**	64,381
Other assets (included **$15,122** and $12,306 at fair value and assets pledged of **$6,288** and $6,764)	**178,197**	159,308
Total assets[(a)]	**$ 4,002,814**	$ 3,875,393
Liabilities		
Deposits (included **$33,768** and $78,384 at fair value)	**$ 2,406,032**	$ 2,400,688
Federal funds purchased and securities loaned or sold under repurchase agreements (included **$226,329** and $169,003 at fair value)	**296,835**	216,535
Short-term borrowings (included **$26,521** and $20,042 at fair value)	**52,893**	44,712
Trading liabilities	**192,883**	180,428
Accounts payable and other liabilities (included **$5,893** and $5,637 at fair value)	**280,672**	290,307
Beneficial interests issued by consolidated VIEs (included **$1** and $1 at fair value)	**27,323**	23,020
Long-term debt (included **$100,780** and $87,924 at fair value)	**401,418**	391,825
Total liabilities[(a)]	**3,658,056**	3,547,515
Commitments and contingencies (refer to Notes 28, 29 and 30)		
Stockholders' equity		
Preferred stock ($1 par value; authorized 200,000,000 shares: issued **2,005,375** and 2,740,375 shares)	**20,050**	27,404
Common stock ($1 par value; authorized 9,000,000,000 shares; issued **4,104,933,895** shares)	**4,105**	4,105
Additional paid-in capital	**90,911**	90,128
Retained earnings	**376,166**	332,901
Accumulated other comprehensive losses	**(12,456)**	(10,443)
Treasury stock, at cost (**1,307,313,494** and 1,228,275,301 shares)	**(134,018)**	(116,217)
Total stockholders' equity	**344,758**	327,878
Total liabilities and stockholders' equity	**$ 4,002,814**	$ 3,875,393

(a) The following table presents information on assets and liabilities related to VIEs that are consolidated by the Firm at December 31, 2024 and 2023. The assets of the consolidated VIEs are used to settle the liabilities of those entities. The holders of the beneficial interests do not have recourse to the general credit of JPMorganChase. The assets and liabilities in the table below include third-party assets and liabilities of consolidated VIEs and exclude intercompany balances that eliminate in consolidation. Refer to Note 14 for a further discussion.

December 31, (in millions)	2024	2023
Assets		
Trading assets	**$ 3,885**	$ 2,170
Loans	**36,510**	37,611
All other assets	**681**	591
Total assets	**$ 41,076**	$ 40,372
Liabilities		
Beneficial interests issued by consolidated VIEs	**$ 27,323**	$ 23,020
All other liabilities	**454**	263
Total liabilities	**$ 27,777**	$ 23,283

The Notes to Consolidated Financial Statements are an integral part of these statements.

JPMorgan Chase & Co.
Consolidated statements of changes in stockholders' equity

Year ended December 31, (in millions, except per share data)	**2024**	2023	2022
Preferred stock			
Balance at January 1	**$ 27,404**	$ 27,404	$ 34,838
Issuance	**2,496**	—	—
Redemption	**(9,850)**	—	(7,434)
Balance at December 31	**20,050**	27,404	27,404
Common stock			
Balance at January 1 and December 31	**4,105**	4,105	4,105
Additional paid-in capital			
Balance at January 1	**90,128**	89,044	88,415
Shares issued and commitments to issue common stock for employee share-based compensation awards, and related tax effects	**768**	1,084	629
Other	**15**	—	—
Balance at December 31	**90,911**	90,128	89,044
Retained earnings			
Balance at January 1	**332,901**	296,456	272,268
Cumulative effect of change in accounting principles	**(161)**	449	—
Net income	**58,471**	49,552	37,676
Preferred stock dividends	**(1,259)**	(1,501)	(1,595)
Common stock dividends (**$4.80**, $4.10 and $4.00 per share for 2024, 2023 and 2022, respectively)	**(13,786)**	(12,055)	(11,893)
Balance at December 31	**376,166**	332,901	296,456
Accumulated other comprehensive income/(loss)			
Balance at January 1	**(10,443)**	(17,341)	(84)
Other comprehensive income/(loss), after-tax	**(2,013)**	6,898	(17,257)
Balance at December 31	**(12,456)**	(10,443)	(17,341)
Treasury stock, at cost			
Balance at January 1	**(116,217)**	(107,336)	(105,415)
Repurchase	**(19,007)**	(9,980)	(3,122)
Reissuance	**1,206**	1,099	1,201
Balance at December 31	**(134,018)**	(116,217)	(107,336)
Total stockholders' equity	**$ 344,758**	$ 327,878	$ 292,332

Effective January 1, 2024, the Firm adopted the Equity Method and Joint Ventures: Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method accounting guidance. Effective January 1, 2023, the Firm adopted the Financial Instruments – Credit Losses: Troubled Debt Restructurings, and Derivatives and Hedging: Fair Value Hedging – Portfolio Layer Method accounting guidance. Refer to Note 1 for further information.

The Notes to Consolidated Financial Statements are an integral part of these statements.

JPMorgan Chase & Co.
Consolidated statements of cash flows

Year ended December 31, (in millions)	2024	2023	2022
Operating activities			
Net income	$ 58,471	$ 49,552	$ 37,676
Adjustments to reconcile net income to net cash provided by operating activities:			
Provision for credit losses	10,678	9,320	6,389
Depreciation and amortization	7,938	7,512	7,051
Deferred tax (benefit)/expense	2,004	(4,534)	(2,738)
Estimated bargain purchase gain associated with the First Republic acquisition	(103)	(2,775)	—
Initial gain on the Visa share exchange	(7,990)	—	—
Other	1,985	4,301	5,174
Originations and purchases of loans held-for-sale	(212,238)	(115,245)	(149,167)
Proceeds from sales, securitizations and paydowns of loans held-for-sale	205,303	116,430	167,709
Net change in:			
Trading assets	(95,729)	(74,091)	(31,449)
Securities borrowed	(18,762)	(14,902)	20,203
Accrued interest and accounts receivable	5,735	19,928	(22,970)
Other assets	(7,650)	32,970	(2,882)
Trading liabilities	2,276	5,315	11,170
Accounts payable and other liabilities	(90)	(25,388)	58,614
Other operating adjustments	6,160	4,581	2,339
Net cash (used in)/provided by operating activities	**(42,012)**	12,974	107,119
Investing activities			
Net change in:			
Federal funds sold and securities purchased under resale agreements	(18,706)	39,740	(54,278)
Held-to-maturity securities:			
Proceeds from paydowns and maturities	99,363	53,056	48,626
Purchases	(4,709)	(4,141)	(33,676)
Available-for-sale securities:			
Proceeds from paydowns and maturities	38,499	53,744	39,159
Proceeds from sales	104,625	108,434	84,616
Purchases	(352,712)	(115,499)	(126,258)
Proceeds from sales and securitizations of loans held-for-investment	57,921	47,312	44,892
Other changes in loans, net	(83,176)	(88,343)	(128,968)
Net cash used in First Republic Acquisition	(2,362)	(9,920)	—
All other investing activities, net	(2,146)	(16,740)	(11,932)
Net cash (used in)/provided by investing activities	**(163,403)**	67,643	(137,819)
Financing activities			
Net change in:			
Deposits	3,299	(32,196)	(136,895)
Federal funds purchased and securities loaned or sold under repurchase agreements	80,288	13,801	8,455
Short-term borrowings	7,439	(1,934)	(8,984)
Beneficial interests issued by consolidated VIEs	1,543	9,029	2,205
Proceeds from long-term borrowings	109,915	75,417	78,442
Payments of long-term borrowings	(96,605)	(64,880)	(45,556)
Proceeds from issuance of preferred stock	2,500	—	—
Redemption of preferred stock	(9,850)	—	(7,434)
Treasury stock repurchased	(18,830)	(9,824)	(3,162)
Dividends paid	(14,783)	(13,463)	(13,562)
All other financing activities, net	(1,469)	(1,521)	234
Net cash provided by/(used in) financing activities	**63,447**	(25,571)	(126,257)
Effect of exchange rate changes on cash and due from banks and deposits with banks	(12,866)	1,871	(16,643)
Net increase/(decrease) in cash and due from banks and deposits with banks	(154,834)	56,917	(173,600)
Cash and due from banks and deposits with banks at the beginning of the period	624,151	567,234	740,834
Cash and due from banks and deposits with banks at the end of the period	**$ 469,317**	$ 624,151	$ 567,234
Cash interest paid	$ 99,642	$ 77,114	$ 23,143
Cash income taxes paid, net	11,715	9,908	4,355

The Notes to Consolidated Financial Statements are an integral part of these statements.

Note 1 – Basis of presentation

JPMorgan Chase & Co. ("JPMorganChase" or the "Firm"), a financial holding company incorporated under Delaware law in 1968, is a leading financial services firm based in the U.S., with operations worldwide. The Firm is a leader in investment banking, financial services for consumers and small businesses, commercial banking, financial transaction processing and asset management. Refer to Note 32 for further discussion of the Firm's reportable business segments.

The accounting and financial reporting policies of JPMorganChase and its subsidiaries conform to U.S. GAAP. Additionally, where applicable, the policies conform to the accounting and reporting guidelines prescribed by regulatory authorities.

Consolidation

The Consolidated Financial Statements include the accounts of JPMorganChase and other entities in which the Firm has a controlling financial interest. All material intercompany balances and transactions have been eliminated.

Assets held for clients in an agency or fiduciary capacity by the Firm are not assets of JPMorganChase and are not included on the Consolidated balance sheets.

The Firm determines whether it has a controlling financial interest in an entity by first evaluating whether the entity is a voting interest entity or a variable interest entity.

Voting interest entities

Voting interest entities are entities that have sufficient equity and provide the equity investors voting rights that enable them to make significant decisions relating to the entity's operations. For these types of entities, the Firm's determination of whether it has a controlling interest is primarily based on the amount of voting equity interests held. Entities in which the Firm has a controlling financial interest, through ownership of the majority of the entities' voting equity interests, or through other contractual rights that give the Firm control, are consolidated by the Firm.

Investments in companies in which the Firm has significant influence over operating and financing decisions (but does not own a majority of the voting equity interests) are accounted for (i) in accordance with the equity method of accounting, or (ii) at fair value if the fair value option was elected. These investments are generally included in other assets, with income or loss included in noninterest revenue.

Certain Firm-sponsored asset management funds are structured as limited partnerships or limited liability companies. For many of these entities, the Firm is the general partner or managing member, but the non-affiliated partners or members have the ability to remove the Firm as the general partner or managing member without cause (i.e., kick-out rights), based on a simple majority vote, or the non-affiliated partners or members have rights to participate in important decisions. Accordingly, the Firm does not consolidate these voting interest entities. However, in the limited cases where the non-managing partners or members do not have substantive kick-out or participating rights, the Firm evaluates the funds as VIEs and consolidates the funds if the Firm is the general partner or managing member and has both power and a potentially significant interest.

The Firm's investment companies and asset management funds have investments in both publicly-held and privately-held entities, including investments in buyouts, growth equity and venture opportunities. These investments are accounted for under investment company guidelines and, accordingly, irrespective of the percentage of equity ownership interests held, are carried on the Consolidated balance sheets at fair value, and are recorded in other assets, with income or loss included in noninterest revenue. If consolidated, the Firm retains the accounting under such specialized investment company guidelines.

Variable interest entities

VIEs are entities that, by design, either (1) lack sufficient equity to permit the entity to finance its activities without additional subordinated financial support from other parties, or (2) have equity investors that do not have the ability to make significant decisions relating to the entity's operations through voting rights, or do not have the obligation to absorb the expected losses, or do not have the right to receive the residual returns of the entity.

The most common type of VIE is an SPE. SPEs are commonly used in securitization transactions in order to isolate certain assets and distribute the cash flows from those assets to investors. The basic SPE structure involves a company selling assets to the SPE; the SPE funds the purchase of those assets by issuing securities to investors. The legal documents that govern the transaction specify how the cash earned on the assets must be allocated to the SPE's investors and other parties that have rights to those cash flows. SPEs are generally structured to insulate investors from claims on the SPE's assets by creditors of other entities, including the creditors of the seller of the assets.

The primary beneficiary of a VIE (i.e., the party that has a controlling financial interest) is required to consolidate the assets and liabilities of the VIE. The primary beneficiary is the party that has both (1) the power to direct the activities of the VIE that most significantly impact the VIE's economic performance; and (2) through its interests in the VIE, the obligation to absorb losses or the right to receive benefits from the VIE that could potentially be significant to the VIE.

To assess whether the Firm has the power to direct the activities of a VIE that most significantly impact the VIE's economic performance, the Firm considers all the facts and circumstances, including its role in establishing the VIE and its ongoing rights and responsibilities. This assessment includes, first, identifying the activities that most significantly impact the VIE's economic performance; and second, identifying which party, if any, has power over those activities. In general, the parties that make the most significant decisions affecting the VIE (such as asset managers, collateral managers, servicers, or owners of call options or liquidation rights over the VIE's assets) or have the right to unilaterally remove those decision-makers are deemed to have the power to direct the activities of a VIE.

To assess whether the Firm has the obligation to absorb losses of the VIE or the right to receive benefits from the VIE that could potentially be significant to the VIE, the Firm considers all of its economic interests, including debt and equity investments, servicing fees, and derivatives or other arrangements deemed to be variable interests in the VIE. This assessment requires that the Firm apply judgment in determining whether these interests, in the aggregate, are considered potentially significant to the VIE. Factors considered in assessing significance include: the design of the VIE, including its capitalization structure; subordination of interests; payment priority; relative share of interests held across various classes within the VIE's capital structure; and the reasons why the interests are held by the Firm.

The Firm performs on-going reassessments of: (1) whether entities previously evaluated under the majority voting-interest framework have become VIEs, based on certain events, and are therefore subject to the VIE consolidation framework; and (2) whether changes in the facts and circumstances regarding the Firm's involvement with a VIE cause the Firm's consolidation conclusion to change.

Refer to Note 14 for further discussion of Firm-sponsored VIEs.

Revenue recognition

Interest income
The Firm recognizes interest income on loans, debt securities, and other debt instruments, generally on a level-yield basis, based on the underlying contractual rate. Refer to Note 7 for further information.

Revenue from contracts with customers
JPMorganChase recognizes noninterest revenue from certain contracts with customers, in investment banking fees, deposit-related fees, asset management fees, commissions and other fees, and components of card income, when the Firm's related performance obligations are satisfied. Refer to Note 6 for further discussion of the Firm's revenue from contracts with customers.

Principal transactions revenue
JPMorganChase carries a portion of its assets and liabilities at fair value. Changes in fair value are reported primarily in principal transactions revenue. Refer to Notes 2 and 3 for further discussion of fair value measurement. Refer to Note 6 for further discussion of principal transactions revenue.

Use of estimates in the preparation of consolidated financial statements

The preparation of the Consolidated Financial Statements requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities, revenue and expense, and disclosures of contingent assets and liabilities. Actual results could be different from these estimates.

Foreign currency translation

JPMorganChase revalues assets, liabilities, revenue and expense denominated in non-U.S. currencies into U.S. dollars using applicable exchange rates.

Gains and losses relating to translating functional currency financial statements for U.S. reporting are included in the Consolidated statements of comprehensive income. Gains and losses relating to nonfunctional currency transactions, including non-U.S. operations where the functional currency is the U.S. dollar, are reported in the Consolidated statements of income.

Offsetting assets and liabilities

U.S. GAAP permits entities to present derivative receivables and derivative payables with the same counterparty and the related cash collateral receivables and payables on a net basis on the Consolidated balance sheets when a legally enforceable master netting agreement exists. U.S. GAAP also permits securities sold and purchased under repurchase agreements and securities borrowed or loaned under securities loan agreements to be presented net when specified conditions are met, including the existence of a legally enforceable master netting agreement. The Firm has elected to net such balances where it has determined that the specified conditions are met.

The Firm uses master netting agreements to mitigate counterparty credit risk in certain transactions, including derivative contracts, resale, repurchase, securities borrowed and securities loaned agreements. A master netting agreement is a single agreement with a counterparty that permits multiple transactions governed by that agreement to be terminated or accelerated and settled through a single payment in a single currency in the event of a default (e.g., bankruptcy, failure to make a required payment or securities transfer or deliver collateral or margin when due). Upon the exercise of derivatives termination rights by the non-defaulting party (i) all transactions are terminated, (ii) all transactions are valued and the positive values of "in the money" transactions are netted against the negative values of "out of the money" transactions and (iii) the only remaining payment obligation is of one of the parties to pay the netted termination amount. Upon exercise of default rights under repurchase agreements and securities loan agreements in general (i) all transactions are terminated and accelerated, (ii) all values of securities or cash held or to be delivered are calculated, and all such sums are netted against each other and (iii) the only remaining payment obligation is of one of the parties to pay the netted termination amount.

Typical master netting agreements for these types of transactions also often contain a collateral/margin agreement that provides for a security interest in, or title transfer of, securities or cash collateral/margin to the party that has the right to demand margin (the "demanding party"). The collateral/margin agreement typically requires a party to transfer collateral/margin to the demanding party with a value equal to the amount of the margin deficit on a net basis across all transactions governed by the master netting agreement, less any threshold. The collateral/margin agreement grants to the demanding party, upon default by the counterparty, the right to set-off any amounts payable by the counterparty against any posted collateral or the cash equivalent of any posted collateral/margin. It also grants to the demanding party the right to liquidate collateral/margin and to apply the proceeds to an amount payable by the counterparty.

Refer to Note 5 for further discussion of the Firm's derivative instruments. Refer to Note 11 for further discussion of the Firm's securities financing agreements.

Statements of cash flows

For JPMorganChase's Consolidated statements of cash flows, cash is defined as those amounts included in cash and due from banks and deposits with banks on the Consolidated balance sheets.

Accounting standard adopted January 1, 2024

Equity Method and Joint Ventures: Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method

The guidance expanded the types of tax-oriented investments, beyond affordable housing tax credit investments, that the Firm can elect on a program by program basis, to be accounted for using the proportional amortization method. This method requires the cost of eligible investments, within an elected program, to be amortized in proportion to the tax benefits received with the resulting amortization reported directly in income tax expense, which aligns with the associated tax credits and other tax benefits. Eligible investments must meet certain criteria, including that substantially all of the return is from income tax credits and other income tax benefits.

This guidance was adopted on January 1, 2024 under the modified retrospective method. The adoption of this guidance resulted in a change to the classification and timing of the amortization associated with certain of the Firm's alternative energy tax-oriented investments. As a result of the adoption, the amortization of these investments that was previously recognized in other income is now recognized in income tax expense. The change in accounting resulted in a decrease to retained earnings of $161 million and increased the Firm's income tax expense and the effective tax rate by approximately $450 million and two percentage points, respectively, in the first quarter of 2024, with no material impact to net income.

The guidance requires additional disclosure for all investments that generate income tax credits and other income tax benefits from a tax-oriented investment program for which the Firm has elected to apply the proportional amortization method. The guidance also requires a reevaluation of eligible investments when significant modifications or events occur that result in a change in the nature of the investment or a change in the Firm's relationship with the underlying project.

Refer to Notes 6, 14 and 25 for additional information.

Accounting standards adopted January 1, 2023

Derivatives and Hedging: Fair Value Hedging – Portfolio Layer Method

The adoption of this guidance expanded the ability to hedge a portfolio of fixed-rate assets in a qualifying hedge accounting relationship. As permitted by the guidance, the Firm elected to transfer HTM securities to AFS and designated those securities in a portfolio layer method hedge upon adoption. The adoption impact of the transfer on retained earnings was not material.

Financial Instruments – Credit Losses: Troubled Debt Restructurings ("TDRs")

The adoption of this guidance eliminated the requirement to measure the allowance for TDRs using a discounted cash flow ("DCF") methodology and allowed the option of a non-DCF portfolio-based approach for modified loans to troubled borrowers. The Firm elected this option for all portfolios of modified loans to troubled borrowers except collateral-dependent loans and nonaccrual risk-rated loans, for which the Firm elected to continue applying a DCF methodology. The adoption of this guidance under the modified retrospective method on January 1, 2023, resulted in a $446 million increase to retained earnings.

Significant accounting policies

The following table identifies JPMorganChase's other significant accounting policies and the Note and page where a detailed description of each policy can be found.

Fair value measurement	Note 2	page 181
Fair value option	Note 3	page 203
Derivative instruments	Note 5	page 209
Noninterest revenue and noninterest expense	Note 6	page 225
Interest income and Interest expense	Note 7	page 229
Pension and other postretirement employee benefit plans	Note 8	page 230
Employee share-based incentives	Note 9	page 233
Investment securities	Note 10	page 235
Securities financing activities	Note 11	page 240
Loans	Note 12	page 243
Allowance for credit losses	Note 13	page 266
Variable interest entities	Note 14	page 271
Goodwill, mortgage servicing rights, and other intangible assets	Note 15	page 280
Premises and equipment	Note 16	page 285
Leases	Note 18	page 286
Accounts payable & other liabilities	Note 19	page 288
Long-term debt	Note 20	page 289
Earnings per share	Note 23	page 294
Income taxes	Note 25	page 296
Off–balance sheet lending-related financial instruments, guarantees and other commitments	Note 28	page 302
Litigation	Note 30	page 309

Note 2 – Fair value measurement

JPMorganChase carries a portion of its assets and liabilities at fair value. These assets and liabilities are predominantly carried at fair value on a recurring basis (i.e., assets and liabilities that are measured and reported at fair value on the Firm's Consolidated balance sheets). Certain assets, liabilities and unfunded lending-related commitments are measured at fair value on a nonrecurring basis; that is, they are not measured at fair value on an ongoing basis but are subject to fair value adjustments only in certain circumstances (for example, when there is evidence of impairment).

Fair value is defined as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. Fair value is based on quoted market prices or inputs, where available. If prices or quotes are not available, fair value is based on valuation models and other valuation techniques that consider relevant transaction characteristics (such as maturity) and use, as inputs, observable or unobservable market parameters, including yield curves, interest rates, volatilities, prices (such as commodity, equity or debt prices), correlations, foreign exchange rates and credit curves. Fair value may also incorporate valuation adjustments.

The level of precision in estimating unobservable market inputs or other factors can affect the amount of gain or loss recorded for a particular position. Furthermore, while the Firm believes its valuation methods are appropriate and consistent with those of other market participants, the methods and assumptions used reflect management judgment and may vary across the Firm's businesses and portfolios.

The Firm uses various methodologies and assumptions in the determination of fair value. The use of different methodologies or assumptions by other market participants compared with those used by the Firm could result in the Firm deriving a different estimate of fair value at the reporting date.

Valuation process

Risk-taking functions are responsible for providing fair value estimates for assets and liabilities carried on the Consolidated balance sheets at fair value. The Firm's Valuation Control Group ("VCG"), which is part of the Firm's Finance function and independent of the risk-taking functions, is responsible for verifying these estimates and determining any fair value adjustments that may be required to ensure that the Firm's positions are recorded at fair value. In addition, the Firm's Valuation Governance Forum ("VGF"), which is composed of senior finance and risk executives, is responsible for overseeing the management of risks arising from valuation activities conducted across the Firm. The Firmwide VGF is chaired by the Firmwide head of the VCG (under the direction of the Firm's Controller), and includes sub-forums covering the CIB, CCB, AWM and certain corporate functions including Treasury and CIO.

Price verification process

The VCG verifies fair value estimates provided by the risk-taking functions by leveraging independently derived prices, valuation inputs and other market data, where available. Where independent prices or inputs are not available, the VCG performs additional review to ensure the reasonableness of the estimates. The additional review may include evaluating the limited market activity including client unwinds, benchmarking valuation inputs to those used for similar instruments, decomposing the valuation of structured instruments into individual components, comparing expected to actual cash flows, reviewing profit and loss trends, and reviewing trends in collateral valuation. There are also additional levels of management review for more significant or complex positions.

The VCG determines any valuation adjustments that may be required to the estimates provided by the risk-taking functions. No adjustments to quoted prices are applied for instruments classified within level 1 of the fair value hierarchy (refer to the discussion of the fair value hierarchy on page 182 for further information). For other positions, judgment is required to assess the need for valuation adjustments to appropriately reflect liquidity considerations, unobservable parameters, and, for certain portfolios that meet specified criteria, the size of the net open risk position. The determination of such adjustments follows a consistent framework across the Firm:

- Liquidity valuation adjustments are considered where an observable external price or valuation parameter exists but is of lower reliability, potentially due to lower market activity. Liquidity valuation adjustments are made based on current market conditions. Factors that may be considered in determining the liquidity adjustment include analysis of: (1) the estimated bid-offer spread for the instrument being traded; (2) alternative pricing points for similar instruments in active markets; and (3) the range of reasonable values that the price or parameter could take.
- The Firm manages certain portfolios of financial instruments on the basis of net open risk exposure and, as permitted by U.S. GAAP, has elected to estimate the fair value of such portfolios on the basis of a transfer of the entire net open risk position in an orderly transaction. Where this is the case, valuation adjustments may be necessary to reflect the cost of exiting a larger-than-normal market-size net open risk position. Where applied, such adjustments are

based on factors that a relevant market participant would consider in the transfer of the net open risk position, including the size of the adverse market move that is likely to occur during the period required to sufficiently reduce the net open risk position.

- Uncertainty adjustments related to unobservable parameters may be made when positions are valued using prices or input parameters to valuation models that are unobservable due to a lack of market activity or because they cannot be implied from observable market data. Such prices or parameters must be estimated and are, therefore, subject to management judgment. Adjustments are made to reflect the uncertainty inherent in the resulting valuation estimate.
- Where appropriate, the Firm also applies adjustments to its estimates of fair value in order to appropriately reflect counterparty credit quality (CVA), the Firm's own creditworthiness (DVA) and the impact of funding (FVA), using a consistent framework across the Firm. Refer to Credit and funding adjustments on page 198 of this Note for more information on such adjustments.

Valuation model review and approval

If prices or quotes are not available for an instrument or a similar instrument, fair value is generally determined using valuation models that consider relevant transaction terms such as maturity and use as inputs market-based or independently sourced parameters. Where this is the case the price verification process described above is applied to the inputs in those models.

Under the Firm's Estimations and Model Risk Management Policy, MRGR reviews and approves new models, as well as material changes to existing models, prior to implementation in the operating environment. In certain circumstances exceptions may be granted to the Firm's policy to allow a model to be used prior to review or approval. MRGR may also require the user to take appropriate actions to mitigate the model risk if it is to be used in the interim. These actions will depend on the model and may include, for example, limitation of trading activity.

Fair value hierarchy

A three-level fair value hierarchy has been established under U.S. GAAP for disclosure of fair value measurements. The fair value hierarchy is based on the observability of inputs to the valuation of an asset or liability as of the measurement date. The three levels are defined as follows.

- Level 1 – inputs to the valuation methodology are quoted prices (unadjusted) for identical assets or liabilities in active markets.
- Level 2 – inputs to the valuation methodology include quoted prices for similar assets and liabilities in active markets, and inputs that are observable for the asset or liability, either directly or indirectly, for substantially the full term of the financial instrument.
- Level 3 – one or more inputs to the valuation methodology are unobservable and significant to the fair value measurement.

A financial instrument's categorization within the fair value hierarchy is based on the lowest level of input that is significant to the fair value measurement.

The following table describes the valuation methodologies generally used by the Firm to measure its significant products/instruments at fair value, including the general classification of such instruments pursuant to the fair value hierarchy.

Product/instrument	Valuation methodology	Classifications in the fair value hierarchy
Securities financing agreements	Valuations are based on discounted cash flows, which consider: • Derivative features: refer to the discussion of derivatives below for further information • Market rates for the respective maturity • Collateral characteristics	Predominantly level 2
Loans and lending-related commitments — wholesale Loans carried at fair value (trading loans and non-trading loans) and associated lending-related commitments	Where observable market data is available, valuations are based on: • Observed market prices (circumstances are infrequent) • Relevant broker quotes • Observed market prices for similar instruments Where observable market data is unavailable or limited, valuations are based on discounted cash flows, which consider the following: • Credit spreads derived from the cost of CDS; or benchmark credit curves developed by the Firm, by industry and credit rating • Prepayment speed • Collateral characteristics	Level 2 or 3
Loans — consumer Loans carried at fair value — residential mortgage loans expected to be sold	Fair value is based on observable market prices for mortgage-backed securities with similar collateral and incorporates adjustments to these prices to account for differences between the securities and the value of the underlying loans, which include credit characteristics, portfolio composition, and liquidity.	Predominantly level 2
Investment and trading securities	Quoted market prices	Level 1
	In the absence of quoted market prices, securities are valued based on: • Observable market prices for similar securities • Relevant broker quotes • Discounted cash flows In addition, the following inputs to discounted cash flows are used for the following products: Mortgage- and asset-backed securities specific inputs: • Collateral characteristics • Deal-specific payment and loss allocations • Current market assumptions related to yield, prepayment speed, conditional default rates and loss severity Collateralized loan obligations ("CLOs") specific inputs: • Collateral characteristics • Deal-specific payment and loss allocations • Expected prepayment speed, conditional default rates, loss severity • Credit spreads • Credit rating data	Level 2 or 3
Physical commodities	Valued using observable market prices or data.	Predominantly Level 1 or 2

Product/instrument	Valuation methodology	Classifications in the fair value hierarchy
Derivatives	Actively traded derivatives, e.g., exchange-traded derivatives, that are valued using quoted prices.	Level 1
	Derivatives that are valued using models such as the Black-Scholes option pricing model, simulation models, or a combination of models that may use observable or unobservable valuation inputs as well as considering the contractual terms. The key valuation inputs used will depend on the type of derivative and the nature of the underlying instruments and may include equity prices, commodity prices, foreign exchange rates, volatilities, correlations, CDS spreads, recovery rates and prepayment speed. In addition, specific inputs used for derivatives that are valued based on models with significant unobservable inputs are as follows: Interest rate and FX exotic derivatives specific inputs include: • Interest rate curve • Interest rate volatility • Interest rate spread volatility • Bermudan switch value • Interest rate correlation • Interest rate-FX correlation • Foreign exchange correlation Credit derivatives specific inputs include: • Credit correlation between the underlying debt instruments Equity derivatives specific inputs include: • Forward equity price • Equity volatility • Equity correlation • Equity-FX correlation • Equity-IR correlation Commodity derivatives specific inputs include: • Forward commodity price • Commodity volatility • Commodity correlation Additionally, adjustments are made to reflect counterparty credit quality (CVA) and the impact of funding (FVA). Refer to page 198 of this Note.	Level 2 or 3
Mortgage servicing rights	Refer to Mortgage servicing rights in Note 15.	Level 3
Private equity direct investments	Fair value is estimated using all available information; the range of potential inputs include: • Transaction prices • Trading multiples of comparable public companies • Operating performance of the underlying portfolio company • Adjustments as required, since comparable public companies are not identical to the company being valued, and for company-specific issues including lack of liquidity • Additional available inputs relevant to the investment	Level 2 or 3

Product/instrument	Valuation methodology	Classification in the fair value hierarchy
Fund investments (e.g., mutual/collective investment funds, private equity funds, hedge funds, and real estate funds)	Net asset value	
	• NAV is supported by the ability to redeem and purchase at the NAV level	Level 1
	• Adjustments to the NAV as required, for restrictions on redemption (e.g., lock-up periods or withdrawal limitations) or where observable activity is limited	Level 2 or 3[(a)]
Beneficial interests issued by consolidated VIEs	Valued using observable market information, where available. In the absence of observable market information, valuations are based on the fair value of the underlying assets held by the VIE.	Level 2 or 3
Structured notes (included in deposits, short-term borrowings and long-term debt)	Valuations are based on discounted cash flow analyses that consider the embedded derivative and the terms and payment structure of the note. The embedded derivative features are considered using models such as the Black-Scholes option pricing model, simulation models, or a combination of models that may use observable or unobservable valuation inputs, depending on the embedded derivative. The specific inputs used vary according to the nature of the embedded derivative features, as described in the discussion above regarding derivatives valuation. Adjustments are then made to this base valuation to reflect the Firm's own credit risk (DVA). Refer to page 198 of this Note.	Level 2 or 3

(a) Excludes certain investments that are measured at fair value using the net asset value per share (or its equivalent) as a practical expedient.

The following table presents the assets and liabilities reported at fair value as of December 31, 2024 and 2023, by major product category and fair value hierarchy.

Assets and liabilities measured at fair value on a recurring basis

	Fair value hierarchy				
December 31, 2024 (in millions)	Level 1	Level 2	Level 3	Derivative netting adjustments[(e)]	Total fair value
Federal funds sold and securities purchased under resale agreements	$ —	$ 286,771	$ —	$ —	$ 286,771
Securities borrowed	—	83,962	—	—	83,962
Trading assets:					
Debt instruments:					
Mortgage-backed securities:					
U.S. GSEs and government agencies[(a)]	—	104,312	488	—	104,800
Residential – nonagency	—	2,282	5	—	2,287
Commercial – nonagency	—	1,283	10	—	1,293
Total mortgage-backed securities	—	107,877	503	—	108,380
U.S. Treasury, GSEs and government agencies[(a)]	150,580	11,702	—	—	162,282
Obligations of U.S. states and municipalities	—	6,100	1	—	6,101
Certificates of deposit, bankers' acceptances and commercial paper	—	3,950	—	—	3,950
Non-U.S. government debt securities	34,108	54,335	152	—	88,595
Corporate debt securities	—	33,591	390	—	33,981
Loans	—	10,228	1,088	—	11,316
Asset-backed securities	—	2,813	10	—	2,823
Total debt instruments	184,688	230,596	2,144	—	417,428
Equity securities	130,307	1,359	62	—	131,728
Physical commodities[(b)]	5,957	1,533	26	—	7,516
Other	—	19,935	210	—	20,145
Total debt and equity instruments[(c)]	320,952	253,423	2,442	—	576,817
Derivative receivables:					
Interest rate	4,934	282,019	3,781	(265,789)	24,945
Credit	—	10,379	708	(10,273)	814
Foreign exchange	196	261,520	1,204	(237,608)	25,312
Equity	—	82,855	2,365	(79,935)	5,285
Commodity	—	15,232	394	(11,015)	4,611
Total derivative receivables	5,130	652,005	8,452	(604,620)	60,967
Total trading assets[(d)]	326,082	905,428	10,894	(604,620)	637,784
Available-for-sale securities:					
Mortgage-backed securities:					
U.S. GSEs and government agencies[(a)]	—	91,893	—	—	91,893
Residential – nonagency	—	4,811	—	—	4,811
Commercial – nonagency	—	4,057	8	—	4,065
Total mortgage-backed securities	—	100,761	8	—	100,769
U.S. Treasury and government agencies	234,491	288	—	—	234,779
Obligations of U.S. states and municipalities	—	17,913	—	—	17,913
Non-U.S. government debt securities	23,973	12,272	—	—	36,245
Corporate debt securities	—	70	—	—	70
Asset-backed securities:					
Collateralized loan obligations	—	14,943	—	—	14,943
Other[(a)]	—	2,133	—	—	2,133
Total available-for-sale securities	258,464	148,380	8	—	406,852
Loans[(e)]	—	38,934	2,416	—	41,350
Mortgage servicing rights	—	—	9,121	—	9,121
Other assets[(d)]	5,732	6,997	1,344	—	14,073
Total assets measured at fair value on a recurring basis	$ 590,278	$ 1,470,472	$ 23,783	$ (604,620)	$ 1,479,913
Deposits	$ —	$ 31,583	$ 2,185	$ —	$ 33,768
Federal funds purchased and securities loaned or sold under repurchase agreements	—	226,329	—	—	226,329
Short-term borrowings	—	23,045	3,476	—	26,521
Trading liabilities:					
Debt and equity instruments[(c)]	120,719	32,457	46	—	153,222
Derivative payables:					
Interest rate	3,981	266,767	3,480	(264,989)	9,239
Credit	—	12,725	1,071	(11,898)	1,898
Foreign exchange	187	253,196	1,184	(238,970)	15,597
Equity	—	90,908	5,231	(87,491)	8,648
Commodity	—	14,021	467	(10,209)	4,279
Total derivative payables	4,168	637,617	11,433	(613,557)	39,661
Total trading liabilities	124,887	670,074	11,479	(613,557)	192,883
Accounts payable and other liabilities	3,100	2,717	76	—	5,893
Beneficial interests issued by consolidated VIEs	—	1	—	—	1
Long-term debt	—	66,216	34,564	—	100,780
Total liabilities measured at fair value on a recurring basis	$ 127,987	$ 1,019,965	$ 51,780	$ (613,557)	$ 586,175

December 31, 2023 (in millions)	Fair value hierarchy Level 1	Level 2	Level 3	Derivative netting adjustments[e]	Total fair value
Federal funds sold and securities purchased under resale agreements	$ —	$ 259,813	$ —	$ —	$ 259,813
Securities borrowed	—	70,086	—	—	70,086
Trading assets:					
Debt instruments:					
Mortgage-backed securities:					
U.S. GSEs and government agencies[a]	—	73,840	758	—	74,598
Residential – nonagency	—	1,921	5	—	1,926
Commercial – nonagency	—	1,362	12	—	1,374
Total mortgage-backed securities	—	77,123	775	—	77,898
U.S. Treasury, GSEs and government agencies[a]	133,997	9,998	—	—	143,995
Obligations of U.S. states and municipalities	—	5,858	10	—	5,868
Certificates of deposit, bankers' acceptances and commercial paper	—	756	—	—	756
Non-U.S. government debt securities	24,846	55,557	179	—	80,582
Corporate debt securities	—	32,854	484	—	33,338
Loans	—	7,872	684	—	8,556
Asset-backed securities	—	2,199	6	—	2,205
Total debt instruments	158,843	192,217	2,138	—	353,198
Equity securities	107,926	679	127	—	108,732
Physical commodities[b]	2,479	3,305	7	—	5,791
Other	—	17,879	101	—	17,980
Total debt and equity instruments[c]	269,248	214,080	2,373	—	485,701
Derivative receivables:					
Interest rate	2,815	243,578	4,298	(224,367)	26,324
Credit	—	8,644	1,010	(9,103)	551
Foreign exchange	149	204,737	889	(187,756)	18,019
Equity	—	55,167	2,522	(52,761)	4,928
Commodity	—	15,234	205	(10,397)	5,042
Total derivative receivables	2,964	527,360	8,924	(484,384)	54,864
Total trading assets[d]	272,212	741,440	11,297	(484,384)	540,565
Available-for-sale securities:					
Mortgage-backed securities:					
U.S. GSEs and government agencies[a]	—	85,170	—	—	85,170
Residential – nonagency	—	3,639	—	—	3,639
Commercial – nonagency	—	2,803	—	—	2,803
Total mortgage-backed securities	—	91,612	—	—	91,612
U.S. Treasury and government agencies	57,683	122	—	—	57,805
Obligations of U.S. states and municipalities	—	21,367	—	—	21,367
Non-U.S. government debt securities	13,095	8,187	—	—	21,282
Corporate debt securities	—	100	—	—	100
Asset-backed securities:					
Collateralized loan obligations	—	6,752	—	—	6,752
Other[a]	—	2,786	—	—	2,786
Total available-for-sale securities	70,778	130,926	—	—	201,704
Loans[e]	—	35,772	3,079	—	38,851
Mortgage servicing rights	—	—	8,522	—	8,522
Other assets[d]	6,635	3,929	758	—	11,322
Total assets measured at fair value on a recurring basis	$ 349,625	$ 1,241,966	$ 23,656	$ (484,384)	$ 1,130,863
Deposits	$ —	$ 76,551	$ 1,833	$ —	$ 78,384
Federal funds purchased and securities loaned or sold under repurchase agreements	—	169,003	—	—	169,003
Short-term borrowings	—	18,284	1,758	—	20,042
Trading liabilities:					
Debt and equity instruments[c]	107,292	32,252	37	—	139,581
Derivative payables:					
Interest rate	4,409	232,277	3,796	(228,586)	11,896
Credit	—	11,293	745	(10,949)	1,089
Foreign exchange	147	211,289	827	(199,643)	12,620
Equity	—	60,887	4,924	(56,443)	9,368
Commodity	—	15,894	484	(10,504)	5,874
Total derivative payables	4,556	531,640	10,776	(506,125)	40,847
Total trading liabilities	111,848	563,892	10,813	(506,125)	180,428
Accounts payable and other liabilities	3,968	1,617	52	—	5,637
Beneficial interests issued by consolidated VIEs	—	1	—	—	1
Long-term debt	—	60,198	27,726	—	87,924
Total liabilities measured at fair value on a recurring basis	$ 115,816	$ 889,546	$ 42,182	$ (506,125)	$ 541,419

(a) At December 31, 2024 and 2023, included total U.S. GSE obligations of $120.1 billion and $78.5 billion, respectively, which were mortgage-related.

(b) Physical commodities inventories are generally accounted for at the lower of cost or net realizable value. "Net realizable value" is a term defined in U.S. GAAP as not exceeding fair value less costs to sell ("transaction costs"). Transaction costs for the Firm's physical commodities inventories are either not applicable or immaterial to the value of the inventory. Therefore, net realizable value approximates fair value for the Firm's physical commodities inventories. When fair value hedging has been applied (or when net realizable value is below cost), the carrying value of physical commodities approximates fair value, because under fair value hedge accounting, the cost basis is adjusted for changes in fair value. Refer to Note 5 for a further discussion of the Firm's hedge accounting relationships. To provide consistent fair value disclosure information, all physical commodities inventories have been included in each period presented.

(c) Balances reflect the reduction of securities owned (long positions) by the amount of identical securities sold but not yet purchased (short positions).

(d) Certain investments that are measured at fair value using the net asset value per share (or its equivalent) as a practical expedient are not required to be classified in the fair value hierarchy. At both December 31, 2024 and 2023, the fair values of these investments, which include certain hedge funds, private equity funds, real estate and other funds, were $1.0 billion, primarily reported in other assets.

(e) As permitted under U.S. GAAP, the Firm has elected to net derivative receivables and derivative payables and the related cash collateral received and paid when a legally enforceable master netting agreement exists. The level 3 balances would be reduced if netting were applied, including the netting benefit associated with cash collateral.

Level 3 valuations

The Firm has established well-structured processes for determining fair value, including for instruments where fair value is estimated using significant unobservable inputs (level 3). Refer to pages 181–185 of this Note for further information on the Firm's valuation process and a detailed discussion of the determination of fair value for individual financial instruments.

Estimating fair value requires the application of judgment. The type and level of judgment required is largely dependent on the amount of observable market information available to the Firm. For instruments valued using internally developed valuation models and other valuation techniques that use significant unobservable inputs and are therefore classified within level 3 of the fair value hierarchy, judgments used to estimate fair value are more significant than those required when estimating the fair value of instruments classified within levels 1 and 2.

In arriving at an estimate of fair value for an instrument within level 3, management must first determine the appropriate valuation model or other valuation technique to use. Second, due to the lack of observability of significant inputs, management must assess relevant empirical data in deriving valuation inputs including transaction details, yield curves, interest rates, prepayment speed, default rates, volatilities, correlations, prices (such as commodity, equity or debt prices), valuations of comparable instruments, foreign exchange rates and credit curves.

The following table presents the Firm's primary level 3 financial instruments, the valuation techniques used to measure the fair value of those financial instruments, the significant unobservable inputs, the range of values for those inputs and the weighted or arithmetic averages of such inputs. While the determination to classify an instrument within level 3 is based on the significance of the unobservable inputs to the overall fair value measurement, level 3 financial instruments typically include observable components (that is, components that are actively quoted and can be validated to external sources) in addition to the unobservable components. The level 1 and/or level 2 inputs are not included in the table. In addition, the Firm manages the risk of the observable components of level 3 financial instruments using securities and derivative positions that are classified within levels 1 or 2 of the fair value hierarchy.

The range of values presented in the table is representative of the highest and lowest level input used to value the significant groups of instruments within a product/instrument classification. Where provided, the weighted averages of the input values presented in the table are calculated based on the fair value of the instruments that the input is being used to value.

In the Firm's view, the input range, weighted and arithmetic average values do not reflect the degree of input uncertainty or an assessment of the reasonableness of the Firm's estimates and assumptions. Rather, they reflect the characteristics of the various instruments held by the Firm and the relative distribution of instruments within the range of characteristics. For example, two option contracts may have similar levels of market risk exposure and valuation uncertainty, but may have significantly different implied volatility levels because the option contracts have different underlyings, tenors, or strike prices. The input range and weighted and arithmetic average values will therefore vary from period-to-period and parameter-to-parameter based on the characteristics of the instruments held by the Firm at each balance sheet date.

Level 3 inputs[(a)]

December 31, 2024						
Product/Instrument	**Fair value** (in millions)	**Principal valuation technique**	**Unobservable inputs[(g)]**	**Range of input values**		**Average[(i)]**
Residential mortgage-backed securities and loans[(b)]	$ 861	Discounted cash flows	Yield	0%	103%	8%
			Prepayment speed	3%	13%	8%
			Conditional default rate	0%	7%	0%
			Loss severity	0%	110%	5%
Commercial mortgage-backed securities and loans[(c)]	1,424	Market comparables	Price	$0	$90	$81
Corporate debt securities	390	Market comparables	Price	$0	$148	$95
Loans[(d)]	1,730	Market comparables	Price	$0	$107	$79
Non-U.S. government debt securities	152	Market comparables	Price	$0	$103	$95
Net interest rate derivatives	293	Option pricing	Interest rate volatility	9bps	1,097bps	115bps
			Interest rate spread volatility	37bps	77bps	64bps
			Bermudan switch value	0%	45%	17%
			Interest rate correlation	(82)%	97%	64%
			IR-FX correlation	(35)%	60%	8%
	8	Discounted cash flows	Prepayment speed	0%	21%	7%
Net credit derivatives	(393)	Discounted cash flows	Credit correlation	31%	79%	47%
			Credit spread	0bps	2,717bps	331bps
			Recovery rate	10%	90%	61%
	30	Market comparables	Price	$0	$115	$74
Net foreign exchange derivatives	62	Option pricing	IR-FX correlation	(40)%	60%	22%
	(42)	Discounted cash flows	Prepayment speed	11%		11%
			Interest rate curve	1%	27%	7%
Net equity derivatives	(2,866)	Option pricing	Forward equity price[(h)]	76%	153%	100%
			Equity volatility	5%	135%	32%
			Equity correlation	17%	100%	56%
			Equity-FX correlation	(80)%	65%	(32)%
			Equity-IR correlation	5%	25%	14%
Net commodity derivatives	(73)	Option pricing	Oil commodity forward	$87 / BBL	$291 / BBL	$160 / BBL
			Natural gas commodity forward	$2 / MMBTU	$7 / MMBTU	$4 / MMBTU
			Commodity volatility	2%	43%	5%
			Commodity correlation	(35)%	98%	(9)%
MSRs	9,121	Discounted cash flows	Refer to Note 15			
Long-term debt, short-term borrowings, and deposits[(e)]	38,901	Option pricing	Interest rate volatility	9bps	1,097bps	115bps
			Bermudan switch value	0%	45%	17%
			Interest rate correlation	(82)%	97%	64%
			IR-FX correlation	(35)%	60%	8%
			Equity volatility	2%	137%	28%
			Equity correlation	17%	100%	56%
			Equity-FX correlation	(80)%	65%	(32)%
			Equity-IR correlation	5%	25%	14%
	1,324	Discounted cash flows	Credit correlation	30%	57%	46%
			Credit spread	2bps	270bps	83bps
			Recovery rate	20%	40%	35%
			Yield	5%	20%	10%
			Loss severity	0%	100%	50%
Other level 3 assets and liabilities, net[(f)]	1,531					

(a) The categories presented in the table have been aggregated based upon the product type, which may differ from their classification on the Consolidated balance sheets. Furthermore, the inputs presented for each valuation technique in the table are, in some cases, not applicable to every instrument valued using the technique as the characteristics of the instruments can differ.
(b) Comprises U.S. GSE and government agency securities of $488 million, nonagency securities of $5 million and non-trading loans of $368 million.
(c) Comprises nonagency securities of $18 million, trading loans of $65 million and non-trading loans of $1.3 billion.
(d) Comprises trading loans of $1.0 billion and non-trading loans of $707 million.
(e) Long-term debt, short-term borrowings and deposits include structured notes issued by the Firm that are financial instruments that typically contain embedded derivatives. The estimation of the fair value of structured notes includes the derivative features embedded within the instrument. The significant unobservable inputs are broadly consistent with those presented for derivative receivables.
(f) Includes equity securities of $734 million including $672 million in Other assets, for which quoted prices are not readily available and the fair value is generally based on internal valuation techniques such as EBITDA multiples and comparable analysis. All other level 3 assets and liabilities are insignificant both individually and in aggregate.
(g) Price is a significant unobservable input for certain instruments. When quoted market prices are not readily available, reliance is generally placed on price-based internal valuation techniques. The price input is expressed assuming a par value of $100.
(h) Forward equity price is expressed as a percentage of the current equity price.
(i) Amounts represent weighted averages except for derivative related inputs where arithmetic averages are used.

Changes in and ranges of unobservable inputs

The following discussion provides a description of the impact on a fair value measurement of a change in each unobservable input in isolation, and the interrelationship between unobservable inputs, where relevant and significant. The impact of changes in inputs may not be independent, as a change in one unobservable input may give rise to a change in another unobservable input. Where relationships do exist between two unobservable inputs, those relationships are discussed below. Relationships may also exist between observable and unobservable inputs (for example, as observable interest rates rise, unobservable prepayment rates decline); such relationships have not been included in the discussion below. In addition, for each of the individual relationships described below, the inverse relationship would also generally apply.

The following discussion also provides a description of attributes of the underlying instruments and external market factors that affect the range of inputs used in the valuation of the Firm's positions.

Yield – The yield of an asset is the interest rate used to discount future cash flows in a discounted cash flow calculation. An increase in the yield, in isolation, would result in a decrease in a fair value measurement.

Credit spread – The credit spread is the amount of additional annualized return over the market interest rate that a market participant would demand for taking exposure to the credit risk of an instrument. The credit spread for an instrument forms part of the discount rate used in a discounted cash flow calculation. Generally, an increase in the credit spread would result in a decrease in a fair value measurement.

The yield and the credit spread of a particular mortgage-backed security primarily reflect the risk inherent in the instrument. The yield is also impacted by the absolute level of the coupon paid by the instrument (which may not correspond directly to the level of inherent risk). Therefore, the range of yield and credit spreads reflects the range of risk inherent in various instruments owned by the Firm. The risk inherent in mortgage-backed securities is driven by the subordination of the security being valued and the characteristics of the underlying mortgages within the collateralized pool, including borrower FICO scores, LTV ratios for residential mortgages and the nature of the property and/or any tenants for commercial mortgages. For corporate debt securities, obligations of U.S. states and municipalities and other similar instruments, credit spreads reflect the credit quality of the obligor and the tenor of the obligation.

Prepayment speed – The prepayment speed is a measure of the voluntary unscheduled principal repayments of a prepayable obligation in a collateralized pool. Prepayment speeds generally decline as borrower delinquencies rise. An increase in prepayment speeds, in isolation, would result in a decrease in a fair value measurement of assets valued at a premium to par and an increase in a fair value measurement of assets valued at a discount to par.

Prepayment speeds may vary from collateral pool to collateral pool, and are driven by the type and location of the underlying borrower, and the remaining tenor of the obligation as well as the level and type (e.g., fixed or floating) of interest rate being paid by the borrower. Typically collateral pools with higher borrower credit quality have a higher prepayment rate than those with lower borrower credit quality, all other factors being equal.

Conditional default rate – The conditional default rate is a measure of the reduction in the outstanding collateral balance underlying a collateralized obligation as a result of defaults. While there is typically no direct relationship between conditional default rates and prepayment speeds, collateralized obligations for which the underlying collateral has high prepayment speeds will tend to have lower conditional default rates. An increase in conditional default rates would generally be accompanied by an increase in loss severity and an increase in credit spreads. An increase in the conditional default rate, in isolation, would result in a decrease in a fair value measurement. Conditional default rates reflect the quality of the collateral underlying a securitization and the structure of the securitization itself. Based on the types of securities owned in the Firm's market-making portfolios, conditional default rates are most typically at the lower end of the range presented.

Loss severity – The loss severity (the inverse concept is the recovery rate) is the expected amount of future realized losses resulting from the ultimate liquidation of a particular loan, expressed as the net amount of loss relative to the outstanding loan balance. An increase in loss severity is generally accompanied by an increase in conditional default rates. An increase in the loss severity, in isolation, would result in a decrease in a fair value measurement.

The loss severity applied in valuing a mortgage-backed security depends on factors relating to the underlying mortgages, including the LTV ratio, the nature of the lender's lien on the property and other instrument-specific factors.

Correlation – Correlation is a measure of the relationship between the movements of two variables. Correlation is a pricing input for a derivative product where the payoff is driven by one or more underlying risks. Correlation inputs are related to the type of derivative (e.g., interest rate, credit, equity, foreign exchange and commodity) due to the nature of the underlying risks. When parameters are positively correlated, an increase in one parameter will result in an increase in the other parameter. When parameters are negatively correlated, an increase in one parameter will result in a decrease in the other parameter. An increase in correlation can result in an increase or a decrease in a fair value measurement. Given a short correlation position, an increase in correlation, in isolation, would generally result in a decrease in a fair value measurement.

The level of correlation used in the valuation of derivatives with multiple underlying risks depends on a number of factors including the nature of those risks. For example, the correlation between two credit risk exposures would be different than that between two interest rate risk exposures. Similarly, the tenor of the transaction may also impact the correlation input, as the relationship between the underlying risks may be different over different time periods. Furthermore, correlation levels are dependent on market conditions and could have a relatively wide range of levels within or across asset classes over time, particularly in volatile market conditions.

Volatility – Volatility is a measure of the variability in possible returns for an instrument, parameter or market index given how much the particular instrument, parameter or index changes in value over time. Volatility is a pricing input for options, including equity options, commodity options, and interest rate options. Given a long position in an option, an increase in volatility, in isolation, would generally result in an increase in a fair value measurement.

The level of volatility used in the valuation of a particular option-based derivative depends on a number of factors, including the nature of the risk underlying the option (e.g., the volatility of a particular equity security may be significantly different from that of a particular commodity index), the tenor of the derivative as well as the strike price of the option.

Bermudan switch value – The switch value is the difference between the overall value of a Bermudan swaption, which can be exercised at multiple points in time, and its most expensive European swaption and reflects the additional value that the multiple exercise dates provide the holder. Switch values are dependent on market conditions and can vary greatly depending on a number of factors, such as the tenor of the underlying swap as well as the strike price of the option. An increase in switch value, in isolation, would generally result in an increase in a fair value measurement.

Interest rate curve – The interest rate curve represents the relationship of interest rates over differing tenors. The interest rate curve is used to set interest rate and foreign exchange derivative cash flows and is also a pricing input used in the discounting of any derivative cash flow.

Forward price – The forward price is the price at which the buyer agrees to purchase the asset underlying a forward contract on the predetermined future delivery date, and is such that the value of the contract is zero at inception.

The forward price is used as an input in the valuation of certain derivatives and depends on a number of factors including interest rates, the current price of the underlying asset, and the expected income to be received and costs to be incurred by the seller as a result of holding that asset until the delivery date. An increase in the forward can result in an increase or a decrease in a fair value measurement.

Changes in level 3 recurring fair value measurements

The following tables include a rollforward of the Consolidated balance sheets amounts (including changes in fair value) for financial instruments classified by the Firm within level 3 of the fair value hierarchy for the years ended December 31, 2024, 2023 and 2022. When a determination is made to classify a financial instrument within level 3, the determination is based on the significance of the unobservable inputs to the overall fair value measurement. However, level 3 financial instruments typically include, in addition to the unobservable or level 3 components, observable components (that is, components that are actively quoted and can be validated to external sources); accordingly, the gains and losses in the table below include changes in fair value due in part to observable factors that are part of the valuation methodology. The Firm risk-manages the observable components of level 3 financial instruments using securities and derivative positions that are classified within level 1 or 2 of the fair value hierarchy; as these level 1 and level 2 risk management instruments are not included below, the gains or losses in the following tables do not reflect the effect of the Firm's risk management activities related to such level 3 instruments.

Year ended December 31, 2024 (in millions)	Fair value measurements using significant unobservable inputs: Fair value at Jan. 1, 2024	Total realized/unrealized gains/(losses)	Purchases[g]	Sales	Settlements[h]	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2024	Change in unrealized gains/(losses) related to financial instruments held at Dec. 31, 2024
Assets:[a]									
Federal funds sold and securities purchased under resale agreements	$ —	$ —	$ —	$ —	$ —	$ —	$ —	$ —	$ —
Trading assets:									
Debt instruments:									
Mortgage-backed securities:									
U.S. GSEs and government agencies	758	18	46	(260)	(81)	7	—	488	(3)
Residential – nonagency	5	7	—	(5)	(2)	4	(4)	5	—
Commercial – nonagency	12	(2)	—	—	—	—	—	10	(1)
Total mortgage-backed securities	775	23	46	(265)	(83)	11	(4)	503	(4)
Obligations of U.S. states and municipalities	10	—	—	—	(3)	—	(6)	1	—
Non-U.S. government debt securities	179	(6)	175	(183)	—	17	(30)	152	(10)
Corporate debt securities	484	36	459	(354)	(181)	13	(67)	390	45
Loans	684	63	800	(642)	(74)	839	(582)	1,088	29
Asset-backed securities	6	—	9	(5)	(8)	8	—	10	—
Total debt instruments	2,138	116	1,489	(1,449)	(349)	888	(689)	2,144	60
Equity securities	127	(21)	138	(123)	(1)	85	(143)	62	(308)
Physical commodities	7	17	3	—	(1)	—	—	26	16
Other	101	144	53	—	(68)	28	(48)	210	108
Total trading assets – debt and equity instruments	2,373	256 (c)	1,683	(1,572)	(419)	1,001	(880)	2,442	(124) (c)
Net derivative receivables:[b]									
Interest rate	502	745	387	(197)	(608)	(172)	(356)	301	(362)
Credit	265	(208)	(2)	(17)	(333)	(61)	(7)	(363)	(265)
Foreign exchange	62	248	178	(538)	(30)	128	(28)	20	353
Equity	(2,402)	(321)	904	(2,488)	953	(91)	579	(2,866)	783
Commodity	(279)	64	32	(215)	310	15	—	(73)	102
Total net derivative receivables	(1,852)	528 (c)	1,499	(3,455)	292	(181)	188	(2,981)	611 (c)
Available-for-sale securities:									
Mortgage-backed securities:									
Commercial – nonagency	—	—	—	—	—	8	—	8	—
Corporate debt securities	—	—	—	—	—	—	—	—	—
Total available-for-sale securities	—	— (d)	—	—	—	8	—	8	— (d)
Loans	3,079	266 (c)	431	(756)	(993)	816	(427)	2,416	251 (c)
Mortgage servicing rights	8,522	762 (e)	926	(21)	(1,068)	—	—	9,121	762 (e)
Other assets	758	105 (c)	623	(62)	(58)	5	(27)	1,344	88 (c)

Year ended December 31, 2024 (in millions)	Fair value measurements using significant unobservable inputs: Fair value at Jan. 1, 2024	Total realized/unrealized (gains)/losses	Purchases	Sales	Issuances	Settlements[h]	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2024	Change in unrealized (gains)/losses related to financial instruments held at Dec. 31, 2024
Liabilities:[a]										
Deposits	$ 1,833	$ (14) (c)(f)	$ —	$ —	$ 2,006	$ (1,522)	$ 34	$ (152)	$ 2,185	$ (44) (c)(f)
Short-term borrowings	1,758	180 (c)(f)	—	—	7,752	(6,230)	23	(7)	3,476	58 (c)(f)
Trading liabilities – debt and equity instruments	37	(47) (c)	(45)	70	—	—	48	(17)	46	18 (c)
Accounts payable and other liabilities	52	(6) (c)	(35)	63	—	—	5	(3)	76	(6) (c)
Long-term debt	27,726	1,475 (c)(f)	—	—	23,920	(18,432)	738	(863)	34,564	1,212 (c)(f)

Year ended December 31, 2023 (in millions)	Fair value at Jan. 1, 2023	Total realized/ unrealized gains/ (losses)	Purchases[g]	Sales	Settlements[h]	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2023	Change in unrealized gains/(losses) related to financial instruments held at Dec. 31, 2023
	Fair value measurements using significant unobservable inputs								
Assets:[a]									
Federal funds sold and securities purchased under resale agreements	$ —	$ —	$ —	$ —	$ —	$ —	$ —	$ —	$ —
Trading assets:									
Debt instruments:									
Mortgage-backed securities:									
U.S. GSEs and government agencies	759	4	249	(133)	(107)	—	(14)	758	1
Residential – nonagency	5	6	—	(6)	(1)	1	—	5	1
Commercial – nonagency	7	6	—	—	(1)	8	(8)	12	7
Total mortgage-backed securities	771	16	249	(139)	(109)	9	(22)	775	9
Obligations of U.S. states and municipalities	7	—	1	—	(1)	3	—	10	—
Non-U.S. government debt securities	155	74	217	(254)	—	22	(35)	179	74
Corporate debt securities	463	36	322	(172)	(41)	114	(238)	484	35
Loans	759	(15)	1,027	(499)	(441)	382	(529)	684	30
Asset-backed securities	23	—	7	(12)	(1)	5	(16)	6	—
Total debt instruments	2,178	111	1,823	(1,076)	(593)	535	(840)	2,138	148
Equity securities	665	(53)	164	(239)	(384)	192	(218)	127	(422)
Physical commodities	2	—	7	—	(2)	—	—	7	—
Other	64	(58)	141	—	(5)	1	(42)	101	(28)
Total trading assets – debt and equity instruments	2,909	—	2,135	(1,315)	(984)	728	(1,100)	2,373	(302) [c]
Net derivative receivables:[b]									
Interest rate	701	556	251	(255)	654	(1,117)	(288)	502	419
Credit	13	304	(60)	(25)	47	15	(29)	265	230
Foreign exchange	489	31	151	(144)	(187)	144	(422)	62	(80)
Equity	(384)	191	928	(1,931)	(1,306)	700	(600)	(2,402)	(646)
Commodity	(146)	(59)	59	(290)	(51)	(11)	219	(279)	(144)
Total net derivative receivables	673	1,023 [c]	1,329	(2,645)	(843)	(269)	(1,120)	(1,852)	(221) [c]
Available-for-sale securities:									
Mortgage-backed securities:									
Commercial – nonagency	—	—	—	—	—	—	—	—	—
Corporate debt securities	239	24	—	(225)	—	—	(38)	—	—
Total available-for-sale securities	239	24 [d]	—	(225)	—	—	(38)	—	— [d]
Loans	1,418	289 [c]	2,398	(120)	(1,147)	1,306	(1,065)	3,079	293 [c]
Mortgage servicing rights	7,973	467 [e]	1,281	(188)	(1,011)	—	—	8,522	467 [e]
Other assets	405	(36) [c]	525	(20)	(147)	45	(14)	758	(82) [c]

Year ended December 31, 2023 (in millions)	Fair value at Jan. 1, 2023	Total realized/ unrealized (gains)/ losses	Purchases	Sales	Issuances	Settlements[h]	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2023	Change in unrealized (gains)/losses related to financial instruments held at Dec. 31, 2023
	Fair value measurements using significant unobservable inputs									
Liabilities:[a]										
Deposits	$ 2,162	$ 95 [c][f]	$ —	$ —	$ 940	$ (1,043)	$ —	$ (321)	$ 1,833	$ 73 [c][f]
Short-term borrowings	1,401	201 [c][f]	—	—	4,522	(4,345)	3	(24)	1,758	14 [c][f]
Trading liabilities – debt and equity instruments	84	(21) [c]	(32)	9	—	(2)	19	(20)	37	—
Accounts payable and other liabilities	53	(4) [c]	(16)	24	—	—	8	(13)	52	(4) [c]
Long-term debt	24,092	3,010 [c][f]	—	—	12,679	(11,555)	229	(729)	27,726	2,870 [c][f]

	Fair value measurements using significant unobservable inputs								
Year ended December 31, 2022 (in millions)	Fair value at Jan. 1, 2022	Total realized/ unrealized gains/ (losses)	Purchases[g]	Sales	Settlements[h]	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2022	Change in unrealized gains/ (losses) related to financial instruments held at Dec. 31, 2022
Assets:[a]									
Federal funds sold and securities purchased under resale agreements	$ —	$ —	$ 1	$ (1)	$ (1)	$ 1	$ —	$ —	$ —
Trading assets:									
Debt instruments:									
Mortgage-backed securities:									
U.S. GSEs and government agencies	265	31	673	(125)	(84)	4	(5)	759	29
Residential – nonagency	28	(1)	7	(5)	(12)	—	(12)	5	—
Commercial – nonagency	10	—	—	(1)	—	3	(5)	7	—
Total mortgage-backed securities	303	30	680	(131)	(96)	7	(22)	771	29
Obligations of U.S. states and municipalities	7	—	—	—	—	—	—	7	—
Non-U.S. government debt securities	81	(92)	494	(338)	(4)	84	(70)	155	(153)
Corporate debt securities	332	(30)	404	(178)	(100)	357	(322)	463	(48)
Loans	708	(51)	652	(605)	(230)	925	(640)	759	(26)
Asset-backed securities	26	5	19	(24)	(1)	5	(7)	23	1
Total debt instruments	1,457	(138)	2,249	(1,276)	(431)	1,378	(1,061)	2,178	(197)
Equity securities	662	(1,036)	473	(377)	(2)	1,066	(121)	665	(840)
Physical commodities	—	(1)	3	—	—	—	—	2	(1)
Other	160	93	37	—	(221)	1	(6)	64	46
Total trading assets – debt and equity instruments	2,279	(1,082) [c]	2,762	(1,653)	(654)	2,445	(1,188)	2,909	(992) [c]
Net derivative receivables:[b]									
Interest rate	(16)	187	325	(483)	329	732	(373)	701	332
Credit	74	226	17	(9)	(271)	5	(29)	13	170
Foreign exchange	(419)	726	215	(114)	83	3	(5)	489	459
Equity	(3,626)	5,016	1,226	(2,530)	96	(656)	90	(384)	3,435
Commodity	(907)	571	110	(331)	350	5	56	(146)	369
Total net derivative receivables	(4,894)	6,726 [c]	1,893	(3,467)	587	89	(261)	673	4,765 [c]
Available-for-sale securities:									
Mortgage-backed securities:									
Commercial – nonagency	—	—	—	—	—	—	—	—	—
Corporate debt securities	161	5	88	—	(15)	—	—	239	5
Total available-for-sale securities	161	5 [d]	88	—	(15)	—	—	239	5 [d]
Loans	1,933	(158) [c]	568	(261)	(886)	1,053	(831)	1,418	(76) [c]
Mortgage servicing rights	5,494	2,039 [e]	2,198	(822)	(936)	—	—	7,973	2,039 [e]
Other assets	306	194 [c]	50	(38)	(103)	2	(6)	405	191 [c]

	Fair value measurements using significant unobservable inputs									
Year ended December 31, 2022 (in millions)	Fair value at Jan. 1, 2022	Total realized/ unrealized (gains)/ losses	Purchases	Sales	Issuances	Settlements[h]	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2022	Change in unrealized (gains)/ losses related to financial instruments held at Dec. 31, 2022
Liabilities:[a]										
Deposits	$ 2,317	$ (292) [c][f]	$ —	$ —	$ 531	$ (114)	$ —	$ (280)	$ 2,162	$ (76) [c][f]
Short-term borrowings	2,481	(358) [c][f]	—	—	3,963	(4,685)	15	(15)	1,401	90 [c][f]
Trading liabilities – debt and equity instruments	30	(31) [c]	(41)	77	—	—	57	(8)	84	101 [c]
Accounts payable and other liabilities	69	(16) [c]	(37)	42	—	—	1	(6)	53	(16) [c]
Long-term debt	24,374	(3,869) [c][f]	—	—	12,714	(8,876)	793	(1,044)	24,092	(3,447) [c][f]

(a) Level 3 assets at fair value as a percentage of total Firm assets at fair value (including assets measured at fair value on a nonrecurring basis) were 2% at December 31, 2024, 2023 and 2022. Level 3 liabilities at fair value as a percentage of total Firm liabilities at fair value (including liabilities measured at fair value on a nonrecurring basis) were 9% at December 31, 2024, and 8% at both December 31, 2023 and 2022.
(b) All level 3 derivatives are presented on a net basis, irrespective of the underlying counterparty.
(c) Primarily reported in principal transactions revenue, except for changes in fair value for CCB mortgage loans and lending-related commitments originated with the intent to sell, and mortgage loan purchase commitments, which are reported in mortgage fees and related income.
(d) Realized gains/(losses) on AFS securities are reported in investment securities gains/(losses). Unrealized gains/(losses) are reported in OCI. Realized and unrealized gains/(losses) recorded on level 3 AFS securities were not material for the years ended December 31, 2024, 2023 and 2022.
(e) Changes in fair value for MSRs are reported in mortgage fees and related income.
(f) Realized (gains)/losses due to DVA for fair value option elected liabilities are reported in principal transactions revenue, and were not material for the years ended December 31, 2024, 2023 and 2022. Unrealized (gains)/losses are reported in OCI, and were $(50) million, $(158) million and $(529) million for the years ended December 31, 2024, 2023 and 2022, respectively.
(g) Loan originations are included in purchases.
(h) Includes financial assets and liabilities that have matured, been partially or fully repaid, impacts of modifications, deconsolidations associated with beneficial interests in VIEs and other items.

Level 3 analysis

Consolidated balance sheets changes

The following describes significant changes to level 3 assets since December 31, 2023, for those items measured at fair value on a recurring basis. Refer to Assets and liabilities measured at fair value on a nonrecurring basis on page 199 for further information on changes impacting items measured at fair value on a nonrecurring basis.

For the year ended December 31, 2024

Level 3 assets were $23.8 billion at December 31, 2024, reflecting an increase of $127 million from December 31, 2023.

The increase for the year ended December 31, 2024 was driven by:

- $599 million increase in MSRs.
- $586 million increase in other assets primarily due to purchases,

offset by:

- $472 million decrease in gross derivative receivables due to sales and settlements predominantly offset by gains, purchases and net transfers.
- $663 million decrease in non-trading loans due to sales and settlements largely offset by gains, purchases and net transfers.

Refer to Note 15 for information on MSRs.

Refer to the sections below for additional information.

Transfers between levels for instruments carried at fair value on a recurring basis

During the year ended December 31, 2024, significant transfers from level 2 into level 3 included the following:

- $1.0 billion of total debt and equity instruments, predominantly trading loans, driven by a decrease in observability.
- $959 million of gross interest rate derivative receivables and $1.1 billion of gross interest rate derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- $1.6 billion of gross equity derivative receivables and $1.7 billion of gross equity derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- $816 million of non-trading loans driven by a decrease in observability.

During the year ended December 31, 2024, significant transfers from level 3 into level 2 included the following:

- $880 million of total debt and equity instruments, predominantly trading loans and equity securities, driven by an increase in observability.
- $1.4 billion of gross equity derivative receivables and $2.0 billion of gross equity derivative payables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- $863 million of long-term debt as a result of an increase in observability and a decrease in the significance of unobservable inputs.

During the year ended December 31, 2023, significant transfers from level 2 into level 3 included the following:

- $951 million of gross interest rate derivative receivables as a result of a decrease in observability and an increase in the significance of unobservable inputs and $2.1 billion of gross interest rate derivative payables as a result of transition to term SOFR for certain interest rate options.
- $1.5 billion of gross equity derivative receivables and $829 million of gross equity derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- $1.3 billion of non-trading loans driven by a decrease in observability.

During the year ended December 31, 2023, significant transfers from level 3 into level 2 included the following:

- $1.1 billion of total debt and equity instruments, partially due to trading loans, driven by an increase in observability.
- $921 million of gross interest rate derivative receivables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- $2.3 billion of gross equity derivative receivables and $1.7 billion of gross equity derivative payables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- $1.1 billion of non-trading loans as a result of an increase in observability and a decrease in the significance of unobservable inputs.

During the year ended December 31, 2022, significant transfers from level 2 into level 3 included the following:

- $2.4 billion of total debt and equity instruments, predominantly due to equity securities of $1.1 billion driven by a decrease in observability predominantly as a result of restricted access to certain markets and trading loans of $925 million driven by a decrease in observability.
- $1.6 billion of gross interest rate derivative receivables and $878 million of gross interest rate derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- $1.6 billion of gross equity derivative receivables and $2.3 billion of gross equity derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- $1.1 billion of non-trading loans driven by a decrease in observability.
- $793 million of long-term debt driven by a decrease in observability and an increase in the significance of unobservable inputs for structured notes.

During the year ended December 31, 2022, significant transfers from level 3 into level 2 included the following:

- $1.2 billion of total debt and equity instruments, largely due to trading loans, driven by an increase in observability.
- $1.2 billion of gross interest rate derivative receivables and $807 million of gross interest rate derivative payables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- $2.2 billion of gross equity derivative receivables and $2.3 billion of gross equity derivative payables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- $831 million of non-trading loans driven by an increase in observability.
- $1.0 billion of long-term debt driven by an increase in observability and a decrease in the significance of unobservable inputs for structured notes.

All transfers are based on changes in the observability and/or significance of the valuation inputs and are assumed to occur at the beginning of the quarterly reporting period in which they occur.

Gains and losses

The following describes significant components of total realized/unrealized gains/(losses) for instruments measured at fair value on a recurring basis for the years ended December 31, 2024, 2023 and 2022. These amounts exclude any effects of the Firm's risk management activities where the financial instruments are classified as level 1 and 2 of the fair value hierarchy. Refer to Changes in level 3 recurring fair value measurements rollforward tables on pages 192–196 for further information on these instruments.

2024

- $1.9 billion of net gains on assets, predominantly driven by gains in net interest rate derivative receivables due to market movements and gains in MSRs reflecting lower prepayment speeds on higher rates.
- $1.6 billion of net losses on liabilities, predominantly driven by losses in long-term debt due to market movements.

2023

- $1.8 billion of net gains on assets, largely driven by gains in net interest rate derivative receivables due to market movements and gains in MSRs reflecting lower prepayment speeds on higher rates.
- $3.3 billion of net losses on liabilities, predominantly driven by losses in long-term debt due to market movements.

2022

- $7.7 billion of net gains on assets, predominantly driven by gains in net equity derivative receivables due to market movements and gains in MSRs reflecting lower prepayment speeds on higher rates.
- $4.6 billion of net gains on liabilities, predominantly driven by a decline in the fair value of long-term debt due to market movements.

Refer to Note 15 for information on MSRs.

Credit and funding adjustments – derivatives

Derivatives are generally valued using models that use as their basis observable market parameters. These market parameters generally do not consider factors such as counterparty nonperformance risk, the Firm's own credit quality, and funding costs. Therefore, it is generally necessary to make adjustments to the base estimate of fair value to reflect these factors.

CVA represents the adjustment, relative to the relevant benchmark interest rate, necessary to reflect counterparty nonperformance risk. The Firm estimates CVA using a scenario analysis to estimate the expected positive credit exposure across all of the Firm's existing positions with each counterparty, and then estimates losses based on the probability of default and estimated recovery rate as a result of a counterparty credit event considering contractual factors designed to mitigate the Firm's credit exposure, such as collateral and legal rights of offset. The key inputs to this methodology are (i) the probability of a default event occurring for each counterparty, as derived from observed or estimated CDS spreads; and (ii) estimated recovery rates implied by CDS spreads, adjusted to consider the differences in recovery rates as a derivative creditor relative to those reflected in CDS spreads, which generally reflect senior unsecured creditor risk.

FVA represents the adjustment to reflect the impact of funding and is recognized where there is evidence that a market participant in the principal market would incorporate it in a transfer of the instrument. The Firm's FVA framework, applied to uncollateralized (including partially collateralized) over-the-counter ("OTC") derivatives incorporates key inputs such as: (i) the expected funding requirements arising from the Firm's positions with each counterparty and collateral arrangements; and (ii) the estimated market funding cost in the principal market which, for derivative liabilities, considers the Firm's credit risk (DVA). For collateralized derivatives, the fair value is estimated by discounting expected future cash flows at the relevant overnight indexed swap rate given the underlying collateral agreement with the counterparty, and therefore a separate FVA is not necessary.

The following table provides the impact of credit and funding adjustments on principal transactions revenue in the respective periods, excluding the effect of any associated hedging activities. The FVA presented below includes the impact of the Firm's own credit quality on the inception value of liabilities as well as the impact of changes in the Firm's own credit quality over time.

Year ended December 31, (in millions)	**2024**	2023	2022
Credit and funding adjustments:			
Derivatives CVA	**$ 29**	$ 221	$ 22
Derivatives FVA	**99**	114	42

Valuation adjustments on fair value option elected liabilities

The valuation of the Firm's liabilities for which the fair value option has been elected requires consideration of the Firm's own credit risk. DVA on fair value option elected liabilities reflects changes (subsequent to the issuance of the liability) in the Firm's probability of default and LGD, which are estimated based on changes in the Firm's credit spread observed in the bond market. Realized (gains)/losses due to DVA for fair value option elected liabilities are reported in principal transactions revenue. Unrealized (gains)/ losses are reported in OCI. Refer to page 196 in this Note and Note 24 for further information.

Assets and liabilities measured at fair value on a nonrecurring basis

The following tables present the assets held as of December 31, 2024 and 2023, for which nonrecurring fair value adjustments were recorded during the years ended December 31, 2024 and 2023, by major product category and fair value hierarchy. There were no liabilities measured at fair value on a nonrecurring basis at both December 31, 2024 and 2023.

December 31, 2024 (in millions)	Fair value hierarchy Level 1	Level 2	Level 3	Total fair value
Loans	**$ —**	**$ 738**	**$ 694**	**$ 1,432**
Other assets[a]	**—**	**9**	**1,048**	**1,057**
Total assets measured at fair value on a nonrecurring basis	**$ —**	**$ 747**	**$ 1,742**	**$ 2,489**

December 31, 2023 (in millions)	Fair value hierarchy Level 1	Level 2	Level 3	Total fair value
Loans	$ —	$ 599	$ 1,156	$ 1,755
Other assets	—	52	1,334	1,386
Total assets measured at fair value on a nonrecurring basis	$ —	$ 651	$ 2,490	$ 3,141

(a) Included equity securities without readily determinable fair values that were adjusted based on observable price changes in orderly transactions from an identical or similar investment of the same issuer (measurement alternative). Of the $1.0 billion in level 3 assets measured at fair value on a nonrecurring basis as of December 31, 2024, $668 million related to equity securities adjusted based on the measurement alternative. These equity securities are classified as level 3 due to the infrequency of the observable prices and/or the restrictions on the shares. Also, included impairments on certain equity method investments.

Nonrecurring fair value changes

The following table presents the total change in value of assets and liabilities for which fair value adjustments have been recognized for the years ended December 31, 2024, 2023 and 2022, related to assets and liabilities held at those dates.

December 31, (in millions)	**2024**	2023	2022
Loans	**$ (302)**	$ (276)	$ (55)
Other assets[a]	**(610)**	(789)	(409)
Accounts payable and other liabilities	**—**	—	(83)
Total nonrecurring fair value gains/(losses)	**$ (912)**	$ (1,065)	$ (547)

(a) Included $(197) million, $(232) million and $(338) million for the years ended December 31, 2024, 2023 and 2022, respectively, of net gains/(losses) as a result of the measurement alternative. The current period also included impairments on certain equity method investments.

Equity securities without readily determinable fair values

The Firm measures certain equity securities without readily determinable fair values at cost less impairment (if any), plus or minus observable price changes from an identical or similar investment of the same issuer (i.e., measurement alternative), with such changes recognized in other income.

In its determination of the new carrying values upon observable price changes, the Firm may adjust the prices if deemed necessary to arrive at the Firm's estimated fair values. Such adjustments may include adjustments to reflect the different rights and obligations of similar securities, and other adjustments that are consistent with the Firm's valuation techniques for private equity direct investments.

The following table presents the carrying value of equity securities without readily determinable fair values held as of December 31, 2024 and 2023, that are measured under the measurement alternative and the related adjustments recorded during the periods presented for those securities with observable price changes. These securities are included in the nonrecurring fair value tables when applicable price changes are observable.

As of or for the year ended December 31, (in millions)	2024	2023
Other assets		
Carrying value[(a)]	**$ 3,737**	$ 4,457
Upward carrying value changes[(b)]	**89**	93
Downward carrying value changes/impairment[(c)]	**(286)**	(325)

(a) The period-end carrying values reflect cumulative purchases and sales in addition to upward and downward carrying value changes.
(b) The cumulative upward carrying value changes between January 1, 2018 and December 31, 2024 were $1.1 billion.
(c) The cumulative downward carrying value changes/impairment between January 1, 2018 and December 31, 2024 were $(1.5) billion.

Included in other assets above is the Firm's interest in approximately 18.6 million Visa Class B-2 common shares ("Visa B-2 shares") and 37.2 million Visa Class B common shares ("Visa B shares") reflected in the Firm's principal investment portfolio as of December 31, 2024 and 2023, respectively.

The Visa B shares were redenominated to Visa Class B-1 common shares ("Visa B-1 shares") on January 24, 2024. On April 8, 2024, Visa commenced an initial exchange offer for any and all outstanding Visa B-1 shares. On May 6, 2024, the Firm announced that Visa had accepted the Firm's tender of its 37.2 million Visa B-1 shares in exchange for a combination of Visa B-2 shares and Visa Class C common shares ("Visa C shares"), resulting in an initial gain of $8.0 billion based on the fair value of the Visa C shares. In addition, the second quarter of 2024 also reflected other Visa-related activity, including the fair value changes of the Visa C shares and derivative instruments, as well as dividends, resulting in the $7.9 billion net gain on Visa shares in the quarter. As of September 30, 2024, the Firm had disposed of all of its Visa C shares through sales in the second and third quarters of 2024 and through a $1.0 billion contribution to the Firm's Foundation in the second quarter of 2024.

The Visa B-2 shares are subject to certain transfer restrictions and are convertible into Visa Class A common shares ("Visa A shares") at a specified conversion rate upon final resolution of certain litigation matters involving Visa. The conversion rate of Visa B-2 shares to Visa A shares was 1.5430 at December 31, 2024 and may be adjusted by Visa depending on developments related to the litigation matters. The outcome of those litigation matters, and the effect that the resolution of those matters may have on the conversion rate, is unknown. Accordingly, as of December 31, 2024, there is significant uncertainty regarding when the transfer restrictions on Visa B-2 shares may be terminated and what the final conversion rate for the Visa B-2 shares will be. As a result of these considerations, as well as differences in voting rights, Visa B-2 shares are not considered to be similar to Visa A shares, and are held at their nominal carryover basis.

Separately, in connection with sales of Visa B shares prior to 2024, the Firm has entered into derivative instruments with the purchasers of the shares under which the Firm retains the risk associated with changes in the conversion rate. Under the terms of the derivative instruments, the Firm will (a) make or receive payments based on subsequent changes in the conversion rate and (b) make periodic interest payments to the purchasers of the Visa B shares. The payments under the derivative instruments will continue as long as the Visa B-2 shares associated with the previously sold Visa B shares remain subject to transfer restrictions. The derivative instruments are accounted for at fair value using a discounted cash flow methodology based upon the Firm's estimate of the timing and magnitude of final resolution of the litigation matters. The derivative instruments are recorded in trading liabilities, and changes in fair value are recognized in other income. The notional amount of shares associated with those derivative instruments has been adjusted as a result of the Visa exchange offer. As of December 31, 2024, the Firm held derivative instruments associated with 11.6 million Visa B-2 shares related to Visa B share sales prior to 2024, which are all subject to similar terms and conditions.

Additional disclosures about the fair value of financial instruments that are not carried on the Consolidated balance sheets at fair value

U.S. GAAP requires disclosure of the estimated fair value of certain financial instruments, which are included in the following table. However, this table does not include other items, such as nonfinancial assets, intangible assets, certain financial instruments, and customer relationships. In the opinion of management, these items, in the aggregate, add significant value to JPMorganChase.

Financial instruments for which carrying value approximates fair value

Certain financial instruments that are not carried at fair value on the Consolidated balance sheets are carried at amounts that approximate fair value, due to their short-term nature and generally negligible credit risk. These instruments include cash and due from banks, deposits with banks, federal funds sold, securities purchased under resale agreements and securities borrowed, short-term receivables and accrued interest receivable, short-term borrowings, federal funds purchased, securities loaned and sold under repurchase agreements, accounts payable, and accrued liabilities. In addition, U.S. GAAP requires that the fair value of deposit liabilities with no stated maturity (i.e., demand, savings and certain money market deposits) be equal to their carrying value; recognition of the inherent funding value of these instruments is not permitted.

The following table presents, by fair value hierarchy classification, the carrying values and estimated fair values at December 31, 2024 and 2023, of financial assets and liabilities, excluding financial instruments that are carried at fair value on a recurring basis, and their classification within the fair value hierarchy.

	December 31, 2024					December 31, 2023				
		Estimated fair value hierarchy					Estimated fair value hierarchy			
(in billions)	Carrying value	Level 1	Level 2	Level 3	Total estimated fair value	Carrying value	Level 1	Level 2	Level 3	Total estimated fair value
Financial assets										
Cash and due from banks	**$ 23.4**	**$ 23.4**	**$ —**	**$ —**	**$ 23.4**	$ 29.1	$ 29.1	$ —	$ —	$ 29.1
Deposits with banks	**445.9**	**445.8**	**0.1**	**—**	**445.9**	595.1	594.6	0.5	—	595.1
Accrued interest and accounts receivable	**101.1**	**—**	**101.0**	**0.1**	**101.1**	107.1	—	107.0	0.1	107.1
Federal funds sold and securities purchased under resale agreements	**8.2**	**—**	**8.2**	**—**	**8.2**	16.3	—	16.3	—	16.3
Securities borrowed	**135.6**	**—**	**135.6**	**—**	**135.6**	130.3	—	130.3	—	130.3
Investment securities, held-to-maturity	**274.5**	**97.4**	**150.5**	**—**	**247.9**	369.8	160.6	182.2	—	342.8
Loans, net of allowance for loan losses(a)	**1,282.3**	**—**	**268.7**	**1,007.8**	**1,276.5**	1,262.5	—	285.6	964.6	1,250.2
Other	**82.7**	**—**	**81.3**	**1.6**	**82.9**	76.1	—	74.9	1.4	76.3
Financial liabilities										
Deposits	**$2,372.3**	**$ —**	**$2,372.5**	**$ —**	**$ 2,372.5**	$ 2,322.3	$ —	$ 2,322.6	$ —	$ 2,322.6
Federal funds purchased and securities loaned or sold under repurchase agreements	**70.5**	**—**	**70.5**	**—**	**70.5**	47.5	—	47.5	—	47.5
Short-term borrowings	**26.4**	**—**	**26.3**	**—**	**26.3**	24.7	—	24.7	—	24.7
Accounts payable and other liabilities(b)	**232.8**	**—**	**219.6**	**12.6**	**232.2**	241.8	—	233.3	8.1	241.4
Beneficial interests issued by consolidated VIEs	**27.3**	**—**	**27.4**	**—**	**27.4**	23.0	—	23.0	—	23.0
Long-term debt	**300.6**	**—**	**251.2**	**50.7**	**301.9**	303.9	—	252.2	51.3	303.5

(a) Fair value is typically estimated using a discounted cash flow model that incorporates the characteristics of the underlying loans (including principal, contractual interest rate and contractual fees) and other key inputs, including expected lifetime credit losses, interest rates, prepayment rates, and primary origination or secondary market spreads. For certain loans, the fair value is measured based on the value of the underlying collateral. Carrying value of the loan takes into account the loan's allowance for loan losses, which represents the loan's expected credit losses over its remaining expected life. The difference between the estimated fair value and carrying value of a loan is generally attributable to changes in market interest rates, including credit spreads, market liquidity premiums and other factors that affect the fair value of a loan but do not affect its carrying value.

(b) Excludes lending-related commitments disclosed in the table below.

The majority of the Firm's lending-related commitments are not carried at fair value on a recurring basis on the Consolidated balance sheets. The carrying value and the estimated fair value of these wholesale lending-related commitments were as follows for the periods indicated.

	December 31, 2024					December 31, 2023				
		Estimated fair value hierarchy					Estimated fair value hierarchy			
(in billions)	Carrying value(a)(b)(c)	Level 1	Level 2	Level 3	Total estimated fair value	Carrying value(a)(b)(c)	Level 1	Level 2	Level 3	Total estimated fair value
Wholesale lending-related commitments	**$ 2.7**	**$ —**	**$ —**	**$ 4.4**	**$ 4.4**	$ 3.0	$ —	$ —	$ 4.8	$ 4.8

(a) Excludes the current carrying values of the guarantee liability and the offsetting asset, each of which is recognized at fair value at the inception of the guarantees.
(b) Includes the wholesale allowance for lending-related commitments.
(c) As of December 31, 2024 and 2023, included fair value adjustments associated with First Republic for other unfunded commitments to extend credit totaling $699 million and $1.1 billion, respectively, recorded in accounts payable and other liabilities on the Consolidated balance sheets. Refer to Notes 28 and 34 for additional information.

The Firm does not estimate the fair value of consumer off-balance sheet lending-related commitments. In many cases, the Firm can reduce or cancel these commitments with or without notice to the borrower, as permitted by law, or in accordance with the contract. Refer to page 183 of this Note for a further discussion of the valuation of lending-related commitments.

Note 3 – Fair value option

The fair value option provides an option to elect fair value for selected financial assets, financial liabilities, unrecognized firm commitments, and written loan commitments.

The Firm has elected to measure certain instruments at fair value for several reasons including to mitigate income statement volatility caused by the differences between the measurement basis of elected instruments (e.g., certain instruments that otherwise would be accounted for on an accrual basis) and the associated risk management arrangements that are accounted for on a fair value basis, as well as to better reflect those instruments that are managed on a fair value basis.

The Firm's election of fair value includes the following instruments:

- Loans purchased or originated as part of securitization warehousing activity, subject to bifurcation accounting, or managed on a fair value basis, including lending-related commitments
- Certain securities financing agreements
- Owned beneficial interests in securitized financial assets that contain embedded credit derivatives, which would otherwise be required to be separately accounted for as a derivative instrument
- Structured notes and other hybrid instruments, which are predominantly financial instruments that contain embedded derivatives, that are issued or transacted as part of client-driven activities
- Certain long-term beneficial interests issued by CIB's consolidated securitization trusts where the underlying assets are carried at fair value

Changes in fair value under the fair value option election

The following table presents the changes in fair value included in the Consolidated statements of income for the years ended December 31, 2024, 2023 and 2022, for items for which the fair value option was elected. The profit and loss information presented below only includes the financial instruments that were elected to be measured at fair value; related risk management instruments, which are required to be measured at fair value, are not included in the table.

December 31, (in millions)	2024			2023			2022		
	Principal transactions	All other income	Total changes in fair value recorded[e]	Principal transactions	All other income	Total changes in fair value recorded[e]	Principal transactions	All other income	Total changes in fair value recorded[e]
Federal funds sold and securities purchased under resale agreements	$ 144	$ —	$ 144	$ 300	$ —	$ 300	$ (384)	$ —	$ (384)
Securities borrowed	347	—	347	164	—	164	(499)	—	(499)
Trading assets:									
Debt and equity instruments, excluding loans	7,205	—	7,205	3,656	—	3,656	(1,703)	—	(1,703)
Loans reported as trading assets:									
Changes in instrument-specific credit risk	346	—	346	248	—	248	(136)	—	(136)
Other changes in fair value	9	10 [c]	19	3	5 [c]	8	(59)	—	(59)
Loans:									
Changes in instrument-specific credit risk	517	(6) [c]	511	322	(4) [c]	318	(242)	21 [c]	(221)
Other changes in fair value	75	371 [c]	446	427	216 [c]	643	(1,421)	(794) [c]	(2,215)
Other assets	63	—	63	282	(4) [d]	278	39	(6) [d]	33
Deposits[a]	(3,398)	—	(3,398)	(2,582)	—	(2,582)	901	—	901
Federal funds purchased and securities loaned or sold under repurchase agreements	(12)	—	(12)	(121)	—	(121)	181	—	181
Short-term borrowings[a]	(922)	—	(922)	(567)	—	(567)	473	—	473
Trading liabilities	(1)	—	(1)	(24)	—	(24)	43	—	43
Beneficial interests issued by consolidated VIEs	—	—	—	—	—	—	(1)	—	(1)
Other liabilities	(11)	—	(11)	(16)	—	(16)	(11)	—	(11)
Long-term debt[a][b]	(2,711)	(6) [c][d]	(2,717)	(5,875)	(78) [c][d]	(5,953)	8,990	98 [c][d]	9,088

(a) Unrealized gains/(losses) due to instrument-specific credit risk (DVA) for liabilities for which the fair value option has been elected are recorded in OCI, while realized gains/(losses) are recorded in principal transactions revenue. Realized gains/(losses) due to instrument-specific credit risk recorded in principal transactions revenue were not material for the years ended December 31, 2024, 2023 and 2022.
(b) Long-term debt measured at fair value predominantly relates to structured notes. Although the risk associated with the structured notes is actively managed, the gains/(losses) reported in this table do not include the income statement impact of the risk management instruments used to manage such risk.
(c) Reported in mortgage fees and related income.
(d) Reported in other income.
(e) Changes in fair value exclude contractual interest, which is included in interest income and interest expense for all instruments other than certain hybrid financial instruments in CIB. Refer to Note 7 for further information regarding interest income and interest expense.

Determination of instrument-specific credit risk for items for which the fair value option was elected

The following describes how the gains and losses that are attributable to changes in instrument-specific credit risk, were determined.

- Loans and lending-related commitments: For floating-rate instruments, all changes in value are attributed to instrument-specific credit risk. For fixed-rate instruments, an allocation of the changes in value for the period is made between those changes in value that are interest rate-related and changes in value that are credit-related. Allocations are generally based on an analysis of borrower-specific credit spread and recovery information, where available, or benchmarking to similar entities or industries.
- Long-term debt: Changes in value attributable to instrument-specific credit risk were derived principally from observable changes in the Firm's credit spread as observed in the bond market.
- Securities financing agreements: Generally, for these types of agreements, there is a requirement that collateral be maintained with a market value equal to or in excess of the principal amount loaned; as a result, there would be no adjustment or an immaterial adjustment for instrument-specific credit risk related to these agreements.

Difference between aggregate fair value and aggregate remaining contractual principal balance outstanding

The following table reflects the difference between the aggregate fair value and the aggregate remaining contractual principal balance outstanding as of December 31, 2024 and 2023, for loans, long-term debt and long-term beneficial interests for which the fair value option has been elected.

	2024			2023		
December 31, (in millions)	Contractual principal outstanding	Fair value	Fair value over/(under) contractual principal outstanding	Contractual principal outstanding	Fair value	Fair value over/(under) contractual principal outstanding
Loans						
Nonaccrual loans						
Loans reported as trading assets	**$ 3,429**	**$ 464**	**$ (2,965)**	$ 2,987	$ 588	$ (2,399)
Loans	**1,711**	**1,492**	**(219)**	838	732	(106)
Subtotal	**5,140**	**1,956**	**(3,184)**	3,825	1,320	(2,505)
90 or more days past due and government guaranteed						
Loans[(a)]	**50**	**45**	**(5)**	65	59	(6)
All other performing loans[(b)]						
Loans reported as trading assets	**12,171**	**10,852**	**(1,319)**	9,547	7,968	(1,579)
Loans[(c)]	**40,342**	**39,813**	**(529)**	38,948	38,060	(888)
Subtotal	**52,513**	**50,665**	**(1,848)**	48,495	46,028	(2,467)
Total loans	**$ 57,703**	**$ 52,666**	**$ (5,037)**	$ 52,385	$ 47,407	$ (4,978)
Long-term debt						
Principal-protected debt	**$ 57,414** [(e)]	**$ 47,780**	**$ (9,634)**	$ 47,768 [(e)]	$ 38,882	$ (8,886)
Nonprincipal-protected debt[(d)]	**NA**	**53,000**	**NA**	NA	49,042	NA
Total long-term debt	**NA**	**$ 100,780**	**NA**	NA	$ 87,924	NA
Long-term beneficial interests						
Nonprincipal-protected debt[(d)]	**NA**	**$ 1**	**NA**	NA	$ 1	NA
Total long-term beneficial interests	**NA**	**$ 1**	**NA**	NA	$ 1	NA

(a) These balances are excluded from nonaccrual loans as the loans are insured and/or guaranteed by U.S. government agencies.
(b) There were no performing loans that were ninety days or more past due as of December 31, 2024 and 2023.
(c) Includes loans insured and/or guaranteed by U.S. government agencies less than 90 days past due.
(d) Remaining contractual principal is not applicable to nonprincipal-protected structured notes and long-term beneficial interests. Unlike principal-protected structured notes and long-term beneficial interests, for which the Firm is obligated to return a stated amount of principal at maturity, nonprincipal-protected structured notes and long-term beneficial interests do not obligate the Firm to return a stated amount of principal at maturity, but for structured notes to return an amount based on the performance of an underlying variable or derivative feature embedded in the note. However, investors are exposed to the credit risk of the Firm as issuer for both nonprincipal-protected and principal-protected notes.
(e) Where the Firm issues principal-protected zero-coupon or discount notes, the balance reflects the contractual principal payment at maturity or, if applicable, the contractual principal payment at the Firm's next call date.

At December 31, 2024 and 2023, the contractual amount of lending-related commitments for which the fair value option was elected was $12.2 billion and $9.7 billion, respectively, with a corresponding fair value of $50 million and $97 million, respectively. Refer to Note 28 for further information regarding off-balance sheet lending-related financial instruments.

Structured note products by balance sheet classification and risk component

The following table presents the fair value of structured notes, by balance sheet classification and the primary risk type.

	December 31, 2024				December 31, 2023			
(in millions)	Long-term debt	Short-term borrowings	Deposits	Total	Long-term debt	Short-term borrowings	Deposits	Total
Risk exposure								
Interest rate	**$ 46,220**	**$ 1,065**	**$ 28,871**	**$ 76,156**	$ 38,604	$ 654	$ 74,526	$ 113,784
Credit	**6,213**	**1,242**	**—**	**7,455**	5,444	350	—	5,794
Foreign exchange	**2,309**	**1,058**	**416**	**3,783**	2,605	941	187	3,733
Equity	**44,149**	**7,881**	**2,986**	**55,016**	38,685	5,483	2,905	47,073
Commodity	**1,331**	**62**	**1** (a)	**1,394**	1,862	11	1 (a)	1,874
Total structured notes	**$ 100,222**	**$ 11,308**	**$ 32,274**	**$ 143,804**	$ 87,200	$ 7,439	$ 77,619	$ 172,258

(a) Excludes deposits linked to precious metals for which the fair value option has not been elected of $869 million and $627 million for the years ended December 31, 2024 and 2023, respectively.

Note 4 – Credit risk concentrations

Concentrations of credit risk arise when a number of clients, counterparties or customers are engaged in similar business activities or activities in the same geographic region, or when they have similar economic features that would cause their ability to meet contractual obligations to be similarly affected by changes in economic conditions.

JPMorganChase regularly monitors various segments of its credit portfolios to assess potential credit risk concentrations and to obtain additional collateral when deemed necessary and permitted under the Firm's agreements. Senior management is significantly involved in the credit approval and review process, and risk levels are adjusted as needed to reflect the Firm's risk appetite.

In the Firm's consumer portfolio, concentrations are managed primarily by product and by U.S. geographic region, with a key focus on trends and concentrations at the portfolio level, where potential credit risk concentrations can be remedied through changes in underwriting policies and portfolio guidelines. Refer to Note 12 for additional information on the geographic composition of the Firm's consumer loan portfolios. In the wholesale portfolio, credit risk concentrations are evaluated primarily by industry and monitored regularly on both an aggregate portfolio level and on an individual client or counterparty basis.

The Firm's wholesale exposure is managed through loan syndications and participations, loan sales, securitizations, credit derivatives, master netting agreements, collateral and other risk-reduction techniques. Refer to Note 12 for additional information on loans.

The Firm does not believe that its exposure to any particular loan product or industry segment results in a significant concentration of credit risk.

Terms of loan products and collateral coverage are included in the Firm's assessment when extending credit and establishing its allowance for credit losses. Refer to Note 13 for additional information on the allowance for credit losses.

The table below presents both on–balance sheet and off–balance sheet consumer and wholesale credit exposure by the Firm's three credit portfolio segments as of December 31, 2024 and 2023. The wholesale industry of risk category is generally based on the client or counterparty's primary business activity.

	2024				2023			
		On-balance sheet				On-balance sheet		
December 31, (in millions)	Credit exposure[h]	Loans	Derivatives	Off-balance sheet[i]	Credit exposure[h]	Loans	Derivatives	Off-balance sheet[i]
Consumer, excluding credit card	**$ 437,654**	**$ 392,810**	**$ —**	**$ 44,844**	$ 455,496	$ 410,093	$ —	$ 45,403
Credit card[a]	**1,234,171**	**232,860**	**—**	**1,001,311**	1,126,781	211,123	—	915,658
Total consumer[a]	**1,671,825**	**625,670**	**—**	**1,046,155**	1,582,277	621,216	—	961,061
Wholesale[b]								
Real Estate	**207,050**	**169,506**	**310**	**37,234**	208,261	166,372	420	41,469
Individuals and Individual Entities[c]	**144,145**	**130,317**	**1,259**	**12,569**	145,849	126,339	725	18,785
Asset Managers	**135,541**	**58,720**	**15,695**	**61,126**	129,574	52,178	9,925	67,471
Consumer & Retail	**129,815**	**46,509**	**1,608**	**81,698**	127,086	46,274	2,013	78,799
Technology, Media & Telecommunications	**84,716**	**21,449**	**2,448**	**60,819**	77,296	22,450	2,451	52,395
Industrials	**72,530**	**24,011**	**2,035**	**46,484**	75,092	26,548	1,335	47,209
Healthcare	**64,224**	**23,243**	**616**	**40,365**	65,025	23,169	1,577	40,279
Banks & Finance Companies	**61,287**	**40,239**	**3,890**	**17,158**	57,177	33,941	2,898	20,338
Utilities	**35,871**	**6,172**	**2,631**	**27,068**	36,061	7,067	3,396	25,598
State & Municipal Govt[d]	**35,039**	**19,279**	**372**	**15,388**	35,986	20,019	442	15,525
Automotive	**34,336**	**17,696**	**794**	**15,846**	33,977	17,459	428	16,090
Oil & Gas	**31,724**	**7,226**	**1,153**	**23,345**	34,475	8,480	705	25,290
Insurance	**24,267**	**2,533**	**9,703**	**12,031**	20,501	2,535	7,138	10,828
Chemicals & Plastics	**20,782**	**6,176**	**267**	**14,339**	20,773	6,458	441	13,874
Transportation	**17,019**	**5,380**	**769**	**10,870**	16,060	5,080	555	10,425
Metals & Mining	**15,860**	**4,425**	**564**	**10,871**	15,508	4,655	274	10,579
Central Govt	**13,862**	**4,715**	**6,285**	**2,862**	17,704	5,463	10,669	1,572
Securities Firms	**9,443**	**1,878**	**3,197**	**4,368**	8,689	865	3,285	4,539
Financial Markets Infrastructure	**4,446**	**16**	**2,410**	**2,020**	4,251	86	2,155	2,010
All other[e]	**140,873**	**100,906**	**4,961**	**35,006**	134,777	97,034	4,032	33,711
Subtotal	**1,282,830**	**690,396**	**60,967**	**531,467**	1,264,122	672,472	54,864	536,786
Loans held-for-sale and loans at fair value	**31,922**	**31,922**	**—**	**—**	30,018	30,018	—	—
Receivables from customers[f]	**51,929**	**—**	**—**	**—**	47,625	—	—	—
Total wholesale	**1,366,681**	**722,318**	**60,967**	**531,467**	1,341,765	702,490	54,864	536,786
Total exposure[g]	**$3,038,506**	**$1,347,988**	**$ 60,967**	**$1,577,622**	$2,924,042	$1,323,706	$ 54,864	$1,497,847

(a) Also includes commercial card lending-related commitments primarily in CIB.
(b) The industry rankings presented in the table as of December 31, 2023, are based on the industry rankings of the corresponding exposures at December 31, 2024, not actual rankings of such exposures at December 31, 2023.
(c) Individuals and Individual Entities predominantly consists of Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB, and includes exposure to personal investment companies and personal and testamentary trusts.
(d) In addition to the credit risk exposure to states and municipal governments (both U.S. and non-U.S.) at December 31, 2024 and 2023, noted above, the Firm held: $6.1 billion and $5.9 billion, respectively, of trading assets; $17.9 billion and $21.4 billion, respectively, of AFS securities; and $9.3 billion and $9.9 billion, respectively, of HTM securities, issued by U.S. state and municipal governments. Refer to Note 2 and Note 10 for further information.
(e) All other includes: SPEs and Private education and civic organizations, representing approximately 94% and 6%, respectively, at both December 31, 2024 and 2023. Refer to Note 14 for more information on exposures to SPEs.
(f) Receivables from customers reflect held-for-investment margin loans to brokerage clients in CIB, CCB and AWM that are collateralized by assets maintained in the clients' brokerage accounts (including cash on deposit, and primarily liquid and readily marketable debt or equity securities).
(g) Excludes cash placed with banks of $459.2 billion and $614.1 billion, at December 31, 2024 and 2023, respectively, which is predominantly placed with various central banks, primarily Federal Reserve Banks.
(h) Credit exposure is net of risk participations and excludes the benefit of credit derivatives used in credit portfolio management activities held against derivative receivables or loans and liquid securities and other cash collateral held against derivative receivables.
(i) Represents lending-related financial instruments.

Note 5 – Derivative instruments

Derivative contracts derive their value from underlying asset prices, indices, reference rates, other inputs or a combination of these factors and may expose counterparties to risks and rewards of an underlying asset or liability without having to initially invest in, own or exchange the asset or liability. JPMorganChase makes markets in derivatives for clients and also uses derivatives to hedge or manage its own risk exposures. Predominantly all of the Firm's derivatives are entered into for market-making or risk management purposes.

Market-making derivatives
The majority of the Firm's derivatives are entered into for market-making purposes. Clients use derivatives to mitigate or modify interest rate, credit, foreign exchange, equity and commodity risks. The Firm actively manages the risks from its exposure to these derivatives by entering into other derivative contracts or by purchasing or selling other financial instruments that partially or fully offset the exposure from client derivatives.

Risk management derivatives
The Firm manages certain market and credit risk exposures using derivative instruments, including derivatives in hedge accounting relationships and other derivatives that are used to manage risks associated with specified assets and liabilities.

The Firm generally uses interest rate derivatives to manage the risk associated with changes in interest rates. Fixed-rate assets and liabilities appreciate or depreciate in market value as interest rates change. Similarly, interest income and expense increase or decrease as a result of variable-rate assets and liabilities resetting to current market rates, and as a result of the repayment and subsequent origination or issuance of fixed-rate assets and liabilities at current market rates. Gains and losses on the derivative instruments related to these assets and liabilities are expected to substantially offset this variability.

Foreign currency derivatives are used to manage the foreign exchange risk associated with certain foreign currency–denominated (i.e., non-U.S. dollar) assets and liabilities and forecasted transactions, as well as the Firm's net investments in certain non-U.S. subsidiaries or branches whose functional currencies are not the U.S. dollar. As a result of fluctuations in foreign currencies, the U.S. dollar–equivalent values of the foreign currency–denominated assets and liabilities or the forecasted revenues or expenses increase or decrease. Gains or losses on the derivative instruments related to these foreign currency–denominated assets or liabilities, or forecasted transactions, are expected to substantially offset this variability.

Commodities derivatives are used to manage the price risk of certain commodities inventories. Gains or losses on these derivative instruments are expected to substantially offset the depreciation or appreciation of the related inventory.

Credit derivatives are used to manage the counterparty credit risk associated with loans and lending-related commitments. Credit derivatives compensate the purchaser when the entity referenced in the contract experiences a credit event, such as bankruptcy or a failure to pay an obligation when due. Credit derivatives primarily consist of CDS. Refer to the Credit derivatives section on pages 222–224 of this Note for a further discussion of credit derivatives.

Refer to the risk management derivatives gains and losses table on page 221 and the hedge accounting gains and losses tables on pages 218–221 of this Note for more information about risk management derivatives.

Derivative counterparties and settlement types
The Firm enters into OTC derivatives, which are negotiated and settled bilaterally with the derivative counterparty. The Firm also enters into, as principal, certain ETD such as futures and options, and OTC-cleared derivative contracts with CCPs. ETD contracts are generally standardized contracts traded on an exchange and cleared by the CCP, which is the Firm's counterparty from the inception of the transactions. OTC-cleared derivatives are traded on a bilateral basis and then novated to the CCP for clearing.

Derivative clearing services
The Firm provides clearing services for clients in which the Firm acts as a clearing member at certain exchanges and clearing houses. The Firm does not reflect the clients' derivative contracts in its Consolidated Financial Statements. Refer to Note 28 for further information on the Firm's clearing services.

Accounting for derivatives
All free-standing derivatives that the Firm executes for its own account are required to be recorded on the Consolidated balance sheets at fair value.

As permitted under U.S. GAAP, the Firm nets derivative assets and liabilities, and the related cash collateral receivables and payables, when a legally enforceable master netting agreement exists between the Firm and the derivative counterparty. Refer to Note 1 for further discussion of the offsetting of assets and liabilities. The accounting for changes in value of a derivative depends on whether or not the transaction has been designated and qualifies for hedge accounting. Derivatives that are not designated as hedges are reported and measured at fair value through earnings. The tabular disclosures on pages 213–221 of this Note provide additional information on the amount of, and reporting for, derivative assets, liabilities, gains and losses. Refer to Notes 2 and 3 for a further discussion of derivatives embedded in structured notes.

Derivatives designated as hedges

The Firm applies hedge accounting to certain derivatives executed for risk management purposes – generally interest rate, foreign exchange and commodity derivatives. However, JPMorganChase does not seek to apply hedge accounting to all of the derivatives associated with the Firm's risk management activities. For example, the Firm does not apply hedge accounting to purchased CDS used to manage the credit risk of loans and lending-related commitments, because of the difficulties in qualifying such contracts as hedges. For the same reason, the Firm does not apply hedge accounting to certain interest rate, foreign exchange, and commodity derivatives used for risk management purposes.

To qualify for hedge accounting, a derivative must be highly effective at reducing the risk associated with the exposure being hedged. In addition, for a derivative to be designated as a hedge, the risk management objective and strategy must be documented. Hedge documentation must identify the derivative hedging instrument, the asset or liability or forecasted transaction and type of risk to be hedged, and how the effectiveness of the derivative is assessed prospectively and retrospectively. To assess effectiveness, the Firm uses statistical methods such as regression analysis, nonstatistical methods such as dollar-value comparisons of the change in the fair value of the derivative to the change in the fair value or cash flows of the hedged item, and qualitative comparisons of critical terms and the evaluation of any changes in those terms. The extent to which a derivative has been, and is expected to continue to be, highly effective at offsetting changes in the fair value or cash flows of the hedged item must be assessed and documented at least quarterly. If it is determined that a derivative is not highly effective at hedging the designated exposure, hedge accounting is discontinued.

There are three types of hedge accounting designations: fair value hedges, cash flow hedges and net investment hedges. JPMorganChase uses fair value hedges primarily to hedge fixed-rate long-term debt, AFS securities and certain commodities inventories. For qualifying fair value hedges, the changes in the fair value of the derivative, and in the value of the hedged item for the risk being hedged, are recognized in earnings. Certain amounts excluded from the assessment of effectiveness are recorded in OCI and recognized in earnings over the life of the derivative. If the hedge relationship is terminated, then the adjustment to the hedged item continues to be reported as part of the basis of the hedged item and, for interest-bearing financial instruments, is amortized to earnings as a yield adjustment. Derivative amounts affecting earnings are recognized consistent with the classification of the hedged item – primarily net interest income and principal transactions revenue.

The Firm employs the portfolio layer method to manage the interest rate risk of portfolios of fixed-rate assets. Throughout the life of the open hedge, basis adjustments are maintained at the portfolio level and are only allocated to individual assets under certain circumstances. These include instances where the portfolio amount falls below the hedged layer amounts, or in cases of voluntary de-designation.

JPMorganChase uses cash flow hedges primarily to hedge the exposure to variability in forecasted cash flows from floating-rate assets and liabilities and foreign currency–denominated revenue and expense. For qualifying cash flow hedges, changes in the fair value of the derivative are recorded in OCI and recognized in earnings as the hedged item affects earnings. Derivative amounts affecting earnings are recognized consistent with the classification of the hedged item – primarily noninterest revenue, net interest income and compensation expense. If the hedge relationship is terminated, then the change in value of the derivative recorded in AOCI is recognized in earnings when the cash flows that were hedged affect earnings. For hedge relationships that are discontinued because a forecasted transaction is expected to not occur according to the original hedge forecast, any related derivative values recorded in AOCI are immediately recognized in earnings.

JPMorganChase uses net investment hedges to protect the value of the Firm's net investments in certain non-U.S. subsidiaries or branches whose functional currencies are not the U.S. dollar. For qualifying net investment hedges, changes in the fair value of the derivatives due to changes in spot foreign exchange rates are recorded in OCI as translation adjustments. Amounts excluded from the assessment of effectiveness are recorded directly in earnings.

The following table outlines the Firm's primary uses of derivatives and the related hedge accounting designation or disclosure category.

Type of Derivative	Use of Derivative	Designation and disclosure	Affected segment or unit	Page reference
Manage specifically identified risk exposures in qualifying hedge accounting relationships:				
• Interest rate	Hedge fixed rate assets and liabilities	Fair value hedge	Corporate	218-219
• Interest rate	Hedge floating-rate assets and liabilities	Cash flow hedge	Corporate	220
• Foreign exchange	Hedge foreign currency-denominated assets and liabilities	Fair value hedge	Corporate	218-219
• Foreign exchange	Hedge foreign currency-denominated forecasted revenue and expense	Cash flow hedge	Corporate	220
• Foreign exchange	Hedge the value of the Firm's investments in non-U.S. dollar functional currency entities	Net investment hedge	Corporate	221
• Commodity	Hedge commodity inventory	Fair value hedge	CIB, AWM	218-219
Manage specifically identified risk exposures not designated in qualifying hedge accounting relationships:				
• Interest rate	Manage the risk associated with mortgage commitments, warehouse loans and MSRs	Specified risk management	CCB	221
• Credit	Manage the credit risk associated with wholesale lending exposures	Specified risk management	CIB, AWM	221
• Interest rate and foreign exchange	Manage the risk associated with certain other specified assets and liabilities	Specified risk management	Corporate, CIB	221
Market-making derivatives and other activities:				
• Various	Market-making and related risk management	Market-making and other	CIB	221
• Various	Other derivatives	Market-making and other	CIB, AWM, Corporate	221

Notional amount of derivative contracts

The following table summarizes the notional amount of free-standing derivative contracts outstanding as of December 31, 2024 and 2023.

	Notional amounts[(b)]	
December 31, (in billions)	**2024**	2023
Interest rate contracts		
Swaps	**$ 20,437**	$ 23,251
Futures and forwards	**3,067**	2,690
Written options	**3,067**	3,370
Purchased options	**3,089**	3,362
Total interest rate contracts	**29,660**	32,673
Credit derivatives[(a)]	**1,191**	1,045
Foreign exchange contracts		
Cross-currency swaps	**4,509**	4,721
Spot, futures and forwards	**7,005**	6,957
Written options	**1,015**	830
Purchased options	**984**	798
Total foreign exchange contracts	**13,513**	13,306
Equity contracts		
Swaps	**850**	639
Futures and forwards	**206**	157
Written options	**914**	778
Purchased options	**788**	698
Total equity contracts	**2,758**	2,272
Commodity contracts		
Swaps	**148**	115
Spot, futures and forwards	**191**	157
Written options	**137**	130
Purchased options	**125**	115
Total commodity contracts	**601**	517
Total derivative notional amounts	**$ 47,723**	$ 49,813

(a) Refer to the Credit derivatives discussion on pages 222–224 for more information on volumes and types of credit derivative contracts.
(b) Represents the sum of gross long and gross short third-party notional derivative contracts.

While the notional amounts disclosed above give an indication of the volume of the Firm's derivatives activity, the notional amounts significantly exceed, in the Firm's view, the possible losses that could arise from such transactions. For most derivative contracts, the notional amount is not exchanged; it is simply a reference amount used to calculate payments.

Impact of derivatives on the Consolidated balance sheets

The following table summarizes information on derivative receivables and payables (before and after netting adjustments) that are reflected on the Firm's Consolidated balance sheets as of December 31, 2024 and 2023, by accounting designation (e.g., whether the derivatives were designated in qualifying hedge accounting relationships or not) and contract type.

Free-standing derivative receivables and payables[(a)]

	Gross derivative receivables				Gross derivative payables			
December 31, 2024 (in millions)	Not designated as hedges	Designated as hedges	Total derivative receivables	Net derivative receivables[(b)]	Not designated as hedges	Designated as hedges	Total derivative payables	Net derivative payables[(b)]
Trading assets and liabilities								
Interest rate	**$ 290,734**	**$ —**	**$ 290,734**	**$ 24,945**	**$ 274,226**	**$ 2**	**$ 274,228**	**$ 9,239**
Credit	**11,087**	**—**	**11,087**	**814**	**13,796**	**—**	**13,796**	**1,898**
Foreign exchange	**261,035**	**1,885**	**262,920**	**25,312**	**253,289**	**1,278**	**254,567**	**15,597**
Equity	**85,220**	**—**	**85,220**	**5,285**	**96,139**	**—**	**96,139**	**8,648**
Commodity	**15,490**	**136**	**15,626**	**4,611**	**14,415**	**73**	**14,488**	**4,279**
Total fair value of trading assets and liabilities	**$ 663,566**	**$ 2,021**	**$ 665,587**	**$ 60,967**	**$ 651,865**	**$ 1,353**	**$ 653,218**	**$ 39,661**

	Gross derivative receivables				Gross derivative payables			
December 31, 2023 (in millions)	Not designated as hedges	Designated as hedges	Total derivative receivables	Net derivative receivables[(b)]	Not designated as hedges	Designated as hedges	Total derivative payables	Net derivative payables[(b)]
Trading assets and liabilities								
Interest rate	$ 250,689	$ 2	$ 250,691	$ 26,324	$ 240,482	$ —	$ 240,482	$ 11,896
Credit	9,654	—	9,654	551	12,038	—	12,038	1,089
Foreign exchange	205,010	765	205,775	18,019	210,623	1,640	212,263	12,620
Equity	57,689	—	57,689	4,928	65,811	—	65,811	9,368
Commodity	15,228	211	15,439	5,042	16,286	92	16,378	5,874
Total fair value of trading assets and liabilities	$ 538,270	$ 978	$ 539,248	$ 54,864	$ 545,240	$ 1,732	$ 546,972	$ 40,847

(a) Balances exclude structured notes for which the fair value option has been elected. Refer to Note 3 for further information.
(b) As permitted under U.S. GAAP, the Firm has elected to net derivative receivables and derivative payables and the related cash collateral receivables and payables when a legally enforceable master netting agreement exists.

Derivatives netting

The following tables present, as of December 31, 2024 and 2023, gross and net derivative receivables and payables by contract and settlement type. Derivative receivables and payables, as well as the related cash collateral from the same counterparty, have been netted on the Consolidated balance sheets where the Firm has obtained an appropriate legal opinion with respect to the master netting agreement. Where such a legal opinion has not been either sought or obtained, amounts are not eligible for netting on the Consolidated balance sheets, and those derivative receivables and payables are shown separately in the tables.

In addition to the cash collateral received and transferred that is presented on a net basis with derivative receivables and payables, the Firm receives and transfers additional collateral (financial instruments and cash). These amounts mitigate counterparty credit risk associated with the Firm's derivative instruments, but are not eligible for net presentation:

- collateral that consists of liquid securities and other cash collateral held at third-party custodians, which are shown separately as "Collateral not nettable on the Consolidated balance sheets" in the tables, up to the fair value exposure amount. For the purpose of this disclosure, the definition of liquid securities is consistent with the definition of high quality liquid assets as defined in the LCR rule;
- the amount of collateral held or transferred that exceeds the fair value exposure at the individual counterparty level, as of the date presented, which is excluded from the tables; and
- collateral held or transferred that relates to derivative receivables or payables where an appropriate legal opinion has not been either sought or obtained with respect to the master netting agreement, which is excluded from the tables.

December 31, (in millions)	2024 Gross derivative receivables	2024 Amounts netted on the Consolidated balance sheets	2024 Net derivative receivables	2023 Gross derivative receivables	2023 Amounts netted on the Consolidated balance sheets	2023 Net derivative receivables
U.S. GAAP nettable derivative receivables						
Interest rate contracts:						
OTC	**$ 158,202**	**$ (134,791)**	**$ 23,411**	$ 176,901	$ (152,703)	$ 24,198
OTC–cleared	**130,989**	**(130,810)**	**179**	71,419	(71,275)	144
Exchange-traded[(a)]	**190**	**(188)**	**2**	402	(389)	13
Total interest rate contracts	**289,381**	**(265,789)**	**23,592**	248,722	(224,367)	24,355
Credit contracts:						
OTC	**8,680**	**(8,030)**	**650**	7,637	(7,226)	411
OTC–cleared	**2,267**	**(2,243)**	**24**	1,904	(1,877)	27
Total credit contracts	**10,947**	**(10,273)**	**674**	9,541	(9,103)	438
Foreign exchange contracts:						
OTC	**259,608**	**(236,931)**	**22,677**	203,624	(187,295)	16,329
OTC–cleared	**685**	**(677)**	**8**	469	(459)	10
Exchange-traded[(a)]	**34**	**—**	**34**	6	(2)	4
Total foreign exchange contracts	**260,327**	**(237,608)**	**22,719**	204,099	(187,756)	16,343
Equity contracts:						
OTC	**33,269**	**(30,742)**	**2,527**	25,001	(23,677)	1,324
Exchange-traded[(a)]	**51,040**	**(49,193)**	**1,847**	30,462	(29,084)	1,378
Total equity contracts	**84,309**	**(79,935)**	**4,374**	55,463	(52,761)	2,702
Commodity contracts:						
OTC	**8,340**	**(5,848)**	**2,492**	8,049	(5,084)	2,965
OTC–cleared	**126**	**(84)**	**42**	133	(123)	10
Exchange-traded[(a)]	**5,179**	**(5,083)**	**96**	5,214	(5,190)	24
Total commodity contracts	**13,645**	**(11,015)**	**2,630**	13,396	(10,397)	2,999
Derivative receivables with appropriate legal opinion	**658,609**	**(604,620)**	**53,989** [(d)]	531,221	(484,384)	46,837 [(d)]
Derivative receivables where an appropriate legal opinion has not been either sought or obtained	**6,978**		**6,978**	8,027		8,027
Total derivative receivables recognized on the Consolidated balance sheets	**$ 665,587**		**$ 60,967**	$ 539,248		$ 54,864
Collateral not nettable on the Consolidated balance sheets[(b)(c)]			**(28,160)**			(22,461)
Net amounts			**$ 32,807**			$ 32,403

December 31, (in millions)	2024 Gross derivative payables	2024 Amounts netted on the Consolidated balance sheets	2024 Net derivative payables	2023 Gross derivative payables	2023 Amounts netted on the Consolidated balance sheets	2023 Net derivative payables
U.S. GAAP nettable derivative payables						
Interest rate contracts:						
OTC	**$ 138,215**	**$ (130,375)**	**$ 7,840**	$ 161,901	$ (152,467)	$ 9,434
OTC–cleared	**134,555**	**(134,262)**	**293**	76,007	(75,729)	278
Exchange-traded[(a)]	**363**	**(352)**	**11**	436	(390)	46
Total interest rate contracts	**273,133**	**(264,989)**	**8,144**	238,344	(228,586)	9,758
Credit contracts:						
OTC	**11,381**	**(10,133)**	**1,248**	10,332	(9,313)	1,019
OTC–cleared	**1,779**	**(1,765)**	**14**	1,639	(1,636)	3
Total credit contracts	**13,160**	**(11,898)**	**1,262**	11,971	(10,949)	1,022
Foreign exchange contracts:						
OTC	**251,860**	**(238,292)**	**13,568**	209,386	(199,173)	10,213
OTC–cleared	**772**	**(678)**	**94**	552	(470)	82
Exchange-traded[(a)]	**14**	**—**	**14**	6	—	6
Total foreign exchange contracts	**252,646**	**(238,970)**	**13,676**	209,944	(199,643)	10,301
Equity contracts:						
OTC	**44,394**	**(38,298)**	**6,096**	29,999	(27,360)	2,639
Exchange-traded[(a)]	**49,578**	**(49,193)**	**385**	33,137	(29,083)	4,054
Total equity contracts	**93,972**	**(87,491)**	**6,481**	63,136	(56,443)	6,693
Commodity contracts:						
OTC	**6,918**	**(5,206)**	**1,712**	8,788	(5,192)	3,596
OTC–cleared	**84**	**(84)**	**—**	120	(120)	—
Exchange-traded[(a)]	**5,182**	**(4,919)**	**263**	5,376	(5,192)	184
Total commodity contracts	**12,184**	**(10,209)**	**1,975**	14,284	(10,504)	3,780
Derivative payables with appropriate legal opinion	**645,095**	**(613,557)**	**31,538** [(d)]	537,679	(506,125)	31,554 [(d)]
Derivative payables where an appropriate legal opinion has not been either sought or obtained	**8,123**		**8,123**	9,293		9,293
Total derivative payables recognized on the Consolidated balance sheets	**$ 653,218**		**$ 39,661**	$ 546,972		$ 40,847
Collateral not nettable on the Consolidated balance sheets[(b)(c)]			**(10,163)**			(4,547)
Net amounts			**$ 29,498**			$ 36,300

(a) Exchange-traded derivative balances that relate to futures contracts are settled daily.
(b) Includes liquid securities and other cash collateral held at third-party custodians related to derivative instruments where an appropriate legal opinion has been obtained. For some counterparties, the collateral amounts of financial instruments may exceed the derivative receivables and derivative payables balances. Where this is the case, the total amount reported is limited to the net derivative receivables and net derivative payables balances with that counterparty.
(c) Derivative collateral relates only to OTC and OTC-cleared derivative instruments.
(d) Net derivatives receivable included cash collateral netted of $51.9 billion and $48.3 billion at December 31, 2024 and 2023, respectively. Net derivatives payable included cash collateral netted of $60.8 billion and $70.0 billion at December 31, 2024 and 2023, respectively. Derivative cash collateral relates to OTC and OTC-cleared derivative instruments.

Liquidity risk and credit-related contingent features

In addition to the specific market risks introduced by each derivative contract type, derivatives expose JPMorganChase to credit risk — the risk that derivative counterparties may fail to meet their payment obligations under the derivative contracts and the collateral, if any, held by the Firm proves to be of insufficient value to cover the payment obligation. It is the policy of JPMorganChase to actively pursue, where possible, the use of legally enforceable master netting arrangements and collateral agreements to mitigate derivative counterparty credit risk inherent in derivative receivables.

While derivative receivables expose the Firm to credit risk, derivative payables expose the Firm to liquidity risk, as the derivative contracts typically require the Firm to post cash or securities collateral with counterparties as the fair value of the contracts moves in the counterparties' favor or upon specified downgrades in the Firm's and its subsidiaries' respective credit ratings. Certain derivative contracts also provide for termination of the contract, generally upon a downgrade of either the Firm or the counterparty, at the fair value of the derivative contracts. The following table shows the aggregate fair value of net derivative payables related to OTC and OTC-cleared derivatives that contain contingent collateral or termination features that may be triggered upon a ratings downgrade, and the associated collateral the Firm has posted in the normal course of business, at December 31, 2024 and 2023.

OTC and OTC-cleared derivative payables containing downgrade triggers

(in millions)	**December 31, 2024**	December 31, 2023
Aggregate fair value of net derivative payables	**$ 15,371**	$ 14,655
Collateral posted	**15,204**	14,673

The following table shows the impact of a single-notch and two-notch downgrade of the long-term issuer ratings of JPMorgan Chase & Co. and its subsidiaries, predominantly JPMorgan Chase Bank, N.A., at December 31, 2024 and 2023, related to OTC and OTC-cleared derivative contracts with contingent collateral or termination features that may be triggered upon a ratings downgrade. Derivatives contracts generally require additional collateral to be posted or terminations to be triggered when the predefined rating threshold is breached. A downgrade by a single rating agency that does not result in a rating lower than a preexisting corresponding rating provided by another major rating agency will generally not result in additional collateral (except in certain instances in which additional initial margin may be required upon a ratings downgrade), nor in termination payment requirements. The liquidity impact in the table is calculated based upon a downgrade below the lowest current rating of the rating agencies referred to in the derivative contract.

Liquidity impact of downgrade triggers on OTC and OTC-cleared derivatives

	December 31, 2024		December 31, 2023	
(in millions)	Single-notch downgrade	Two-notch downgrade	Single-notch downgrade	Two-notch downgrade
Amount of additional collateral to be posted upon downgrade[(a)]	**$ 119**	**$ 1,205**	$ 75	$ 1,153
Amount required to settle contracts with termination triggers upon downgrade[(b)]	**78**	**458**	93	592

(a) Includes the additional collateral to be posted for initial margin.
(b) Amounts represent fair values of derivative payables, and do not reflect collateral posted.

Derivatives executed in contemplation of a sale of the underlying financial asset

In certain instances the Firm enters into transactions in which it transfers financial assets but maintains the economic exposure to the transferred assets by entering into a derivative with the same counterparty in contemplation of the initial transfer. The Firm generally accounts for such transfers as collateralized financing transactions as described in Note 11, but in limited circumstances they may qualify to be accounted for as a sale and a derivative under U.S. GAAP. The amount of such transfers accounted for as a sale where the associated derivative was outstanding was not material at December 31, 2024 and 2023.

Impact of derivatives on the Consolidated statements of income

The following tables provide information related to gains and losses recorded on derivatives based on their hedge accounting designation or purpose.

Fair value hedge gains and losses

The following tables present derivative instruments, by contract type, used in fair value hedge accounting relationships, as well as pre-tax gains/(losses) recorded on such derivatives and the related hedged items for the years ended December 31, 2024, 2023 and 2022, respectively. The Firm includes gains/(losses) on the hedging derivative in the same line item in the Consolidated statements of income as the related hedged item.

	Gains/(losses) recorded in income			Income statement impact of excluded components[e]		OCI impact
Year ended December 31, 2024 (in millions)	Derivatives	Hedged items	Income statement impact	Amortization approach	Changes in fair value	Derivatives - Gains/(losses) recorded in OCI[f]
Contract type						
Interest rate[(a)(b)]	**$ 711**	**$ (65)**	**$ 646**	**$ —**	**$ 699**	**$ —**
Foreign exchange[(c)]	**(177)**	**402**	**225**	**(532)**	**225**	**(115)**
Commodity[(d)]	**293**	**(160)**	**133**	**—**	**122**	**—**
Total	**$ 827**	**$ 177**	**$ 1,004**	**$ (532)**	**$ 1,046**	**$ (115)**

	Gains/(losses) recorded in income			Income statement impact of excluded components[e]		OCI impact
Year ended December 31, 2023 (in millions)	Derivatives	Hedged items	Income statement impact	Amortization approach	Changes in fair value	Derivatives - Gains/(losses) recorded in OCI[f]
Contract type						
Interest rate[(a)(b)]	$ 1,554	$ (1,248)	$ 306	$ —	$ 157	$ —
Foreign exchange[(c)]	722	(483)	239	(601)	239	(134)
Commodity[(d)]	1,227	(706)	521	—	525	—
Total	$ 3,503	$ (2,437)	$ 1,066	$ (601)	$ 921	$ (134)

	Gains/(losses) recorded in income			Income statement impact of excluded components[e]		OCI impact
Year ended December 31, 2022 (in millions)	Derivatives	Hedged items	Income statement impact	Amortization approach	Changes in fair value	Derivatives - Gains/(losses) recorded in OCI[f]
Contract type						
Interest rate[(a)(b)]	$ (14,352)	$ 14,047	$ (305)	$ —	$ (262)	$ —
Foreign exchange[(c)]	(1,317)	1,423	106	(528)	106	130
Commodity[(d)]	106	(70)	36	—	48	—
Total	$ (15,563)	$ 15,400	$ (163)	$ (528)	$ (108)	$ 130

(a) Primarily consists of hedges of the benchmark (e.g., Secured Overnight Financing Rate ("SOFR")) interest rate risk of fixed-rate long-term debt and AFS securities. Gains and losses were recorded in net interest income.
(b) Includes the amortization of income/expense associated with the inception hedge accounting adjustment applied to the hedged item. Excludes the accrual of interest on interest rate swaps and the related hedged items.
(c) Primarily consists of hedges of the foreign currency risk of long-term debt and AFS securities for changes in spot foreign currency rates. Gains and losses related to the derivatives and the hedged items due to changes in foreign currency rates and the income statement impact of excluded components were recorded primarily in principal transactions revenue and net interest income.
(d) Consists of overall fair value hedges of physical commodities inventories that are generally carried at the lower of cost or net realizable value (net realizable value approximates fair value). Gains and losses were recorded in principal transactions revenue.
(e) The assessment of hedge effectiveness excludes certain components of the changes in fair values of the derivatives and hedged items such as forward points on foreign exchange forward contracts, time values and cross-currency basis spreads. Excluded components may impact earnings either through amortization of the initial amount over the life of the derivative or through fair value changes recognized in the current period.
(f) Represents the change in value of amounts excluded from the assessment of effectiveness under the amortization approach, predominantly cross-currency basis spreads. The amount excluded at inception of the hedge is recognized in earnings over the life of the derivative.

As of December 31, 2024 and 2023, the following amounts were recorded on the Consolidated balance sheets related to certain cumulative fair value hedge basis adjustments that are expected to reverse through the income statement in future periods as an adjustment to yield.

December 31, 2024 (in millions)	Carrying amount of the hedged items[a][b]		Cumulative amount of fair value hedging adjustments included in the carrying amount of hedged items:[d] Active hedging relationships	Discontinued hedging relationships[e]	Total
Assets					
Investment securities - AFS	**$ 203,141**	(c)	**$ (1,675)**	**$ (1,959)**	**$ (3,634)**
Liabilities					
Long-term debt	**$ 211,288**		**$ (3,711)**	**$ (9,332)**	**$ (13,043)**
Beneficial interests issued by consolidated VIEs	**$ 5,312**		**$ (30)**	**$ (5)**	**$ (35)**

December 31, 2023 (in millions)	Carrying amount of the hedged items[a][b]		Cumulative amount of fair value hedging adjustments included in the carrying amount of hedged items:[d] Active hedging relationships	Discontinued hedging relationships[e]	Total
Assets					
Investment securities - AFS	$ 151,752	(c)	$ 549	$ (2,010)	$ (1,461)
Liabilities					
Long-term debt	$ 195,455		$ (2,042)	$ (9,727)	$ (11,769)
Beneficial interests issued by consolidated VIEs	$ —		$ —	$ —	—

(a) Excludes physical commodities with a carrying value of $6.2 billion and $5.6 billion at December 31, 2024 and 2023, respectively, to which the Firm applies fair value hedge accounting. As a result of the application of hedge accounting, these inventories are carried at fair value, thus recognizing unrealized gains and losses in current periods. Since the Firm exits these positions at fair value, there is no incremental impact to net income in future periods.

(b) Excludes hedged items where only foreign currency risk is the designated hedged risk, as basis adjustments related to foreign currency hedges will not reverse through the income statement in future periods. At December 31, 2024 and 2023, the carrying amount excluded for AFS securities was $28.7 billion and $19.3 billion, respectively. At December 31, 2024 and 2023, the carrying amount excluded for long-term debt was $518 million and zero, respectively.

(c) Carrying amount represents the amortized cost, net of allowance if applicable. At December 31, 2024 and December 31, 2023, the amortized cost of the portfolio layer method closed portfolios was $72.8 billion and $83.9 billion, of which $41.2 billion and $68.0 billion was designated as hedged, respectively. The amount designated as hedged is the sum of the notional amounts of all outstanding layers in each portfolio, which includes both spot starting and forward starting layers. At December 31, 2024 and December 31, 2023, the cumulative amount of basis adjustments was $(1.7) billion and $(165) million, which is comprised of $(1.2) billion and $73 million for active hedging relationships, and $(566) million and $(238) million for discontinued hedging relationships, respectively. Refer to Note 10 for additional information.

(d) Positive (negative) amounts related to assets represent cumulative fair value hedge basis adjustments that will reduce (increase) net interest income in future periods. Positive (negative) amounts related to liabilities represent cumulative fair value hedge basis adjustments that will increase (reduce) net interest income in future periods.

(e) Represents basis adjustments existing on the balance sheet date associated with hedged items that have been de-designated from qualifying fair value hedging relationships.

Cash flow hedge gains and losses

The following tables present derivative instruments, by contract type, used in cash flow hedge accounting relationships, and the pre-tax gains/(losses) recorded on such derivatives, for the years ended December 31, 2024, 2023 and 2022, respectively. The Firm includes the gains/(losses) on the hedging derivative in the same line item in the Consolidated statements of income as the change in cash flows on the related hedged item.

Year ended December 31, 2024 (in millions)	Derivatives gains/(losses) recorded in income and other comprehensive income/(loss)		
	Amounts reclassified from AOCI to income	Amounts recorded in OCI	Total change in OCI for period
Contract type			
Interest rate[(a)]	**$ (2,668)**	**$ (3,603)**	**$ (935)**
Foreign exchange[(b)]	**89**	**(139)**	**(228)**
Total	**$ (2,579)**	**$ (3,742)**	**$ (1,163)**

Year ended December 31, 2023 (in millions)	Derivatives gains/(losses) recorded in income and other comprehensive income/(loss)		
	Amounts reclassified from AOCI to income	Amounts recorded in OCI	Total change in OCI for period
Contract type			
Interest rate[(a)]	$ (1,839)	$ 274	$ 2,113
Foreign exchange[(b)]	64	209	145
Total	$ (1,775)	$ 483	$ 2,258

Year ended December 31, 2022 (in millions)	Derivatives gains/(losses) recorded in income and other comprehensive income/(loss)		
	Amounts reclassified from AOCI to income	Amounts recorded in OCI	Total change in OCI for period
Contract type			
Interest rate[(a)]	$ (153)	$ (7,131)	$ (6,978)
Foreign exchange[(b)]	(267)	(342)	(75)
Total	$ (420)	$ (7,473)	$ (7,053)

(a) Primarily consists of hedges of SOFR-indexed floating-rate assets. Gains and losses were recorded in net interest income.

(b) Primarily consists of hedges of the foreign currency risk of non-U.S. dollar-denominated revenue and expense. The income statement classification of gains and losses follows the hedged item – primarily noninterest revenue and compensation expense.

The Firm did not experience any forecasted transactions that failed to occur for the years ended 2024, 2023 and 2022.

Over the next 12 months, the Firm expects that approximately $(1.6) billion (after-tax) of net losses recorded in AOCI at December 31, 2024, related to cash flow hedges will be recognized in income. For cash flow hedges that have been terminated, the maximum length of time over which the derivative results recorded in AOCI will be recognized in earnings is approximately seven years, corresponding to the timing of the originally hedged forecasted cash flows. For open cash flow hedges, the maximum length of time over which forecasted transactions are hedged is approximately seven years. The Firm's longer-dated forecasted transactions relate to core lending and borrowing activities.

Net investment hedge gains and losses

The following table presents hedging instruments, by contract type, that were used in net investment hedge accounting relationships, and the pre-tax gains/(losses) recorded on such instruments for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions)	**2024**		2023		2022	
	Amounts recorded in income[(a)(b)]	**Amounts recorded in OCI**	Amounts recorded in income[(a)(b)]	Amounts recorded in OCI	Amounts recorded in income[(a)(b)]	Amounts recorded in OCI
Foreign exchange derivatives	**$467**	**$4,411**	$384	$(1,732)	$(123)	$3,591

(a) Certain components of hedging derivatives are permitted to be excluded from the assessment of hedge effectiveness, such as forward points on foreign exchange forward contracts. The Firm elects to record changes in fair value of these amounts directly in other income.

(b) Excludes amounts reclassified from AOCI to income on the sale or liquidation of hedged entities. During the year ended December 31, 2024, the Firm reclassified a net pre-tax gain of $89 million to other income/expense. During the year ended December 31, 2023, the Firm reclassified a net pre-tax loss of $(35) million to other revenue including the impact of the acquisition of CIFM. During the year ended December 31, 2022, the Firm reclassified net pre-tax gains of $38 million to other income/expense related to the liquidation of certain legal entities. Refer to Note 24 for further information.

Gains and losses on derivatives used for specified risk management purposes

The following table presents pre-tax gains/(losses) recorded on a limited number of derivatives, not designated in hedge accounting relationships, that are used to manage risks associated with certain specified assets and liabilities, including certain risks arising from mortgage commitments, warehouse loans, MSRs, wholesale lending exposures, and foreign currency denominated assets and liabilities.

Year ended December 31, (in millions)	Derivatives gains/(losses) recorded in income		
	2024	2023	2022
Contract type			
Interest rate[(a)]	**$ (425)**	$ (135)	$ (827)
Credit[(b)]	**(604)**	(441)	51
Foreign exchange[(c)]	**(10)**	(2)	(48)
Total	**$ (1,039)**	$ (578)	$ (824)

(a) Primarily represents interest rate derivatives used to hedge the interest rate risk inherent in mortgage commitments, warehouse loans and MSRs, as well as written commitments to originate warehouse loans. Gains and losses were recorded predominantly in mortgage fees and related income.

(b) Relates to credit derivatives used to mitigate credit risk associated with lending exposures in the Firm's wholesale businesses. These derivatives do not include credit derivatives used to mitigate counterparty credit risk arising from derivative receivables, which is included in gains and losses on derivatives related to market-making activities and other derivatives. Gains and losses were recorded in principal transactions revenue.

(c) Primarily relates to derivatives used to mitigate foreign exchange risk of specified foreign currency-denominated assets and liabilities. Gains and losses were recorded in principal transactions revenue.

Gains and losses on derivatives related to market-making activities and other derivatives

The Firm makes markets in derivatives in order to meet the needs of customers and uses derivatives to manage certain risks associated with net open risk positions from its market-making activities, including the counterparty credit risk arising from derivative receivables. All derivatives not included in the hedge accounting or specified risk management categories above are included in this category. Gains and losses on these derivatives are primarily recorded in principal transactions revenue. Refer to Note 6 for information on principal transactions revenue.

Credit derivatives

Credit derivatives are financial instruments whose value is derived from the credit risk associated with the debt of a third-party issuer (the reference entity) and which allow one party (the protection purchaser) to transfer that risk to another party (the protection seller). Credit derivatives expose the protection purchaser to the creditworthiness of the protection seller, as the protection seller is required to make payments under the contract when the reference entity experiences a credit event, such as a bankruptcy, a failure to pay its obligation or a restructuring. The seller of credit protection receives a premium for providing protection but has the risk that the underlying instrument referenced in the contract will be subject to a credit event.

The Firm is both a purchaser and seller of protection in the credit derivatives market and uses these derivatives for two primary purposes. First, in its capacity as a market-maker, the Firm actively manages a portfolio of credit derivatives by purchasing and selling credit protection, predominantly on corporate debt obligations, to meet the needs of customers. Second, as an end-user, the Firm uses credit derivatives to manage credit risk associated with lending exposures (loans and unfunded commitments) in its wholesale and consumer businesses and derivatives counterparty exposures in its wholesale businesses, and to manage the credit risk arising from certain financial instruments in the Firm's market-making businesses. Following is a summary of various types of credit derivatives.

Credit default swaps

Credit derivatives may reference the credit of either a single reference entity ("single-name"), broad-based index or portfolio. The Firm purchases and sells protection on both single- name and index-reference obligations. Single-name CDS and index CDS contracts are either OTC or OTC-cleared derivative contracts. Single-name CDS are used to manage the default risk of a single reference entity, while index CDS contracts are used to manage the credit risk associated with the broader credit markets or credit market segments. Like the S&P 500 and other market indices, a CDS index consists of a portfolio of CDS across many reference entities. New series of CDS indices are periodically established with a new underlying portfolio of reference entities to reflect changes in the credit markets. If one of the reference entities in the index experiences a credit event, then the reference entity that defaulted is removed from the index. CDS can also be referenced against specific portfolios of reference names or against customized exposure levels: for example, to provide protection against the first $1 million of realized credit losses in a $10 million portfolio of exposure. Such structures are commonly known as tranche CDS.

For both single-name CDS contracts and index CDS contracts, upon the occurrence of a credit event, under the terms of a CDS contract neither party to the CDS contract has recourse to the reference entity. The protection purchaser has recourse to the protection seller for the difference between the face value of the CDS contract and the fair value of the reference obligation at settlement of the credit derivative contract, also known as the recovery value. The protection purchaser does not need to hold the debt instrument of the underlying reference entity in order to receive amounts due under the CDS contract when a credit event occurs.

Credit-related notes

A credit-related note is a funded derivative with a credit risk component where the issuer of the credit-related note purchases from the note investor credit protection on a reference entity or an index. Under the contract, the investor pays the issuer the par value of the note at the inception of the transaction, and in return, the issuer makes periodic payments to the investor, based on the credit risk of the referenced entity. The issuer also repays the investor the par value of the note at maturity unless the reference entity (or one of the entities that makes up a reference index) experiences a specified credit event. If a credit event occurs, the issuer is not obligated to repay the par value of the note, but rather, the issuer pays the investor the difference between the par value of the note and the fair value of the defaulted reference obligation at the time of settlement. Neither party to the credit-related note has recourse to the defaulting reference entity.

The following tables present a summary of the notional amounts of credit derivatives and credit-related notes the Firm sold and purchased as of December 31, 2024 and 2023. Upon a credit event, the Firm as a seller of protection would typically pay out a percentage of the full notional amount of net protection sold, as the amount actually required to be paid on the contracts takes into account the recovery value of the reference obligation at the time of settlement. The Firm manages the credit risk on contracts to sell protection by purchasing protection with identical or similar underlying reference entities. Other purchased protection referenced in the following tables includes credit derivatives bought on related, but not identical, reference positions (including indices, portfolio coverage and other reference points) as well as protection purchased by CIB through credit-related notes. Other purchased protection also includes credit protection against certain loans in the retained lending portfolio through the issuance of credit derivatives and credit-related notes.

The Firm does not use notional amounts of credit derivatives as the primary measure of risk management for such derivatives, because the notional amount does not take into account the probability of the occurrence of a credit event, the recovery value of the reference obligation, or related cash instruments and economic hedges, each of which reduces, in the Firm's view, the risks associated with such derivatives.

Total credit derivatives and credit-related notes

December 31, 2024 (in millions)	Maximum payout/Notional amount			
	Protection sold	Protection purchased with identical underlyings[c]	Net protection (sold)/ purchased[d]	Other protection purchased[e]
Credit derivatives				
Credit default swaps	**$ (450,184)**	**$ 474,554**	**$ 24,370**	**$ 6,858**
Other credit derivatives[a]	**(110,913)**	**137,927**	**27,014**	**10,169**
Total credit derivatives	**(561,097)**	**612,481**	**51,384**	**17,027**
Credit-related notes[b]	**—**	**—**	**—**	**10,471**
Total	**$ (561,097)**	**$ 612,481**	**$ 51,384**	**$ 27,498**

December 31, 2023 (in millions)	Maximum payout/Notional amount			
	Protection sold	Protection purchased with identical underlyings[c]	Net protection (sold)/ purchased[d]	Other protection purchased[e]
Credit derivatives				
Credit default swaps	$ (450,172)	$ 473,823	$ 23,651	$ 7,517
Other credit derivatives[a]	(38,846)	45,416	6,570	29,206
Total credit derivatives	(489,018)	519,239	30,221	36,723
Credit-related notes[b]	—	—	—	9,788
Total	$ (489,018)	$ 519,239	$ 30,221	$ 46,511

(a) Other credit derivatives predominantly consist of credit swap options and total return swaps.
(b) Predominantly represents Other protection purchased by CIB.
(c) Represents the total notional amount of protection purchased where the underlying reference instrument is identical to the reference instrument on protection sold; the notional amount of protection purchased for each individual identical underlying reference instrument may be greater or lower than the notional amount of protection sold.
(d) Does not take into account the fair value of the reference obligation at the time of settlement, which would generally reduce the amount the seller of protection pays to the buyer of protection in determining settlement value.
(e) Represents protection purchased by the Firm on referenced instruments (single-name, portfolio or index) where the Firm has not sold any protection on the identical reference instrument. Also includes credit protection against certain loans and lending-related commitments in the retained lending portfolio through the issuance of credit derivatives and credit-related notes.

The following tables summarize the notional amounts by the ratings, maturity profile, and total fair value, of credit derivatives as of December 31, 2024 and 2023, where JPMorganChase is the seller of protection. The maturity profile is based on the remaining contractual maturity of the credit derivative contracts. The ratings profile is based on the rating of the reference entity on which the credit derivative contract is based. The ratings and maturity profile of credit derivatives where JPMorganChase is the purchaser of protection are comparable to the profile reflected below.

Protection sold – credit derivatives ratings[(a)]/maturity profile

December 31, 2024 (in millions)	<1 year	1–5 years	>5 years	Total notional amount	Fair value of receivables[(b)]	Fair value of payables[(b)]	Net fair value
Risk rating of reference entity							
Investment-grade	**$ (135,950)**	**$ (277,052)**	**$ (33,379)**	**$ (446,381)**	**$ 4,593**	**$ (904)**	**$ 3,689**
Noninvestment-grade	**(42,149)**	**(70,525)**	**(2,042)**	**(114,716)**	**1,889**	**(1,738)**	**151**
Total	**$ (178,099)**	**$ (347,577)**	**$ (35,421)**	**$ (561,097)**	**$ 6,482**	**$ (2,642)**	**$ 3,840**

December 31, 2023 (in millions)	<1 year	1–5 years	>5 years	Total notional amount	Fair value of receivables[(b)]	Fair value of payables[(b)]	Net fair value
Risk rating of reference entity							
Investment-grade	$ (89,981)	$ (263,834)	$ (29,470)	$ (383,285)	$ 3,659	$ (1,144)	$ 2,515
Noninvestment-grade	(31,419)	(69,515)	(4,799)	(105,733)	2,466	(1,583)	883
Total	$ (121,400)	$ (333,349)	$ (34,269)	$ (489,018)	$ 6,125	$ (2,727)	$ 3,398

(a) The ratings scale is primarily based on external credit ratings defined by S&P and Moody's.
(b) Amounts are shown on a gross basis, before the benefit of legally enforceable master netting agreements including cash collateral netting.

Note 6 – Noninterest revenue and noninterest expense

Noninterest revenue

The Firm records noninterest revenue from certain contracts with customers in investment banking fees, deposit-related fees, asset management fees, commissions and other fees, and components of card income. The related contracts are often terminable on demand and the Firm has no remaining obligation to deliver future services. For arrangements with a fixed term, the Firm may commit to deliver services in the future. Revenue associated with these remaining performance obligations typically depends on the occurrence of future events or underlying asset values, and is not recognized until the outcome of those events or values are known.

Investment banking fees

This revenue category includes debt and equity underwriting and advisory fees. As an underwriter, the Firm helps clients raise capital via public offering and private placement of various types of debt and equity instruments. Underwriting fees are primarily based on the issuance price and quantity of the underlying instruments, and are recognized as revenue typically upon execution of the client's transaction. The Firm also manages and syndicates loan arrangements. Credit arrangement and syndication fees, included within debt underwriting fees, are recorded as revenue after satisfying certain retention, timing and yield criteria.

The Firm also provides advisory services by assisting its clients with mergers and acquisitions, divestitures, restructuring and other complex transactions. Advisory fees are recognized as revenue typically upon execution of the client's transaction.

The following table presents the components of investment banking fees.

Year ended December 31, (in millions)	**2024**	2023	2022
Underwriting			
Equity	**$ 1,687**	$ 1,149	$ 975
Debt	**3,945**	2,610	2,732
Total underwriting	**5,632**	3,759	3,707
Advisory	**3,278**	2,760	2,979
Total investment banking fees	**$ 8,910**	$ 6,519	$ 6,686

Investment banking fees are earned primarily by CIB.

Principal transactions

Principal transactions revenue is driven by many factors, including:

- the bid-offer spread, which is the difference between the price at which a market participant is willing and able to sell an instrument to the Firm and the price at which another market participant is willing and able to buy it from the Firm, and vice versa; and
- realized and unrealized gains and losses on financial instruments and commodities transactions, including those accounted for under the fair value option, primarily used in client-driven market-making activities.
  - Realized gains and losses result from the sale of instruments, closing out or termination of transactions, or interim cash payments.
  - Unrealized gains and losses result from changes in valuation.

In connection with its client-driven market-making activities, the Firm transacts in debt and equity instruments, derivatives and commodities, including physical commodities inventories and financial instruments that reference commodities.

Principal transactions revenue also includes realized and unrealized gains and losses related to:

- derivatives designated in qualifying hedge accounting relationships, primarily fair value hedges of commodity and foreign exchange risk;
- derivatives used for specific risk management purposes, primarily to mitigate credit, foreign exchange and interest rate risks.

Refer to Note 5 for further information on the income statement classification of gains and losses from derivatives activities.

In the financial commodity markets, the Firm transacts in OTC derivatives (e.g., swaps, forwards, options) and ETD that reference a wide range of underlying commodities. In the physical commodity markets, the Firm primarily purchases and sells precious and base metals, natural gas, and may hold other commodities inventories under financing and other arrangements with clients.

The following table presents all realized and unrealized gains and losses recorded in principal transactions revenue. This table excludes interest income and interest expense on trading assets and liabilities, which are an integral part of the overall performance of the Firm's client-driven market-making activities in CIB and fund deployment activities in Treasury and CIO. Refer to Note 7 for further information on interest income and interest expense.

Trading revenue is presented primarily by instrument type. The Firm's client-driven market-making businesses generally utilize a variety of instrument types in connection with their market-making and related risk-management activities; accordingly, the trading revenue presented in the table below is not representative of the total revenue of any individual LOB.

Year ended December 31, (in millions)	2024	2023	2022
Trading revenue by instrument type			
Interest rate[(a)]	**$ 3,631**	$ 5,607	$ 3,010
Credit[(b)]	**1,545**	1,434	1,412 [(c)]
Foreign exchange	**4,874**	5,082	5,119
Equity	**13,476**	10,229	8,068
Commodity	**1,194**	2,202	2,348
Total trading revenue	**24,720**	24,554	19,957
Private equity gains/(losses)	**67**	(94)	(45)
Principal transactions	**$24,787**	$ 24,460	$ 19,912

(a) Includes the impact of changes in funding valuation adjustments on derivatives.
(b) Includes the impact of changes in credit valuation adjustments on derivatives, net of the associated hedging activities.
(c) Includes net markdowns on held-for-sale positions, primarily unfunded commitments, in the bridge financing portfolio.

Principal transactions revenue is earned primarily by CIB.

Lending- and deposit-related fees

Lending-related fees include fees earned from loan commitments, standby letters of credit, financial guarantees, and other loan-servicing activities. Deposit-related fees include fees earned from performing cash management activities, and providing overdraft and other deposit account services. Deposit-related fees also include the impact of credits earned by clients that reduce such fees. Lending- and deposit-related fees are recognized over the period in which the related service is provided. Refer to Note 28 for further information on lending-related commitments.

The following table presents the components of lending- and deposit-related fees.

Year ended December 31, (in millions)	2024	2023	2022
Lending-related fees	**$ 2,192** [(a)]	$ 2,365 [(a)]	$ 1,468
Deposit-related fees	**5,414**	5,048	5,630
Total lending- and deposit-related fees	**$7,606**	$ 7,413	$ 7,098

(a) Includes the amortization of the fair value discount on certain acquired lending-related commitments associated with First Republic, predominantly in AWM and CIB. The discount is deferred in other liabilities and recognized on a straight-line basis over the commitment period and was largely recognized in 2023 as the commitments were generally short term. Refer to Note 34 for additional information.

Lending- and deposit-related fees are earned by CCB, CIB and AWM.

Asset management fees

Investment management fees include fees associated with assets the Firm manages on behalf of its clients, including investors in Firm-sponsored funds and owners of separately managed investment accounts. Management fees are typically based on the value of assets under management and are collected and recognized at the end of each period over which the management services are provided and the value of the managed assets is known. The Firm also receives performance-based management fees, which are earned based on exceeding certain benchmarks or other performance targets and are accrued and recognized when the probability of reversal is remote, typically at the end of the related billing period.

All other asset management fees include commissions earned on the sales or distribution of mutual funds to clients. These fees are recorded as revenue at the time the service is rendered or, in the case of certain distribution fees, based on the underlying fund's asset value or investor redemption activity.

The following table presents the components of asset management fees.

Year ended December 31, (in millions)	2024	2023	2022
Asset management fees			
Investment management fees	**$ 17,425**	$ 14,908	$ 13,765
All other asset management fees	**376**	312	331
Total asset management fees	**$ 17,801**	$ 15,220	$ 14,096

Asset management fees earned primarily by AWM and CCB.

Commissions and other fees

This revenue category includes commissions and fees from brokerage and custody services, and other products.

Brokerage commissions represents commissions earned when the Firm acts as a broker, by facilitating its clients' purchases and sales of securities and other financial instruments. Brokerage commissions are collected and recognized as revenue upon occurrence of the client transaction. The Firm reports certain costs paid to third-party clearing houses and exchanges net against commission revenue.

Administration fees predominantly include fees for custody, funds services, securities lending and securities clearance. These fees are recorded as revenue over the period in which the related service is provided.

The following table presents the components of commissions and other fees.

Year ended December 31, (in millions)	2024	2023	2022
Commissions and other fees			
Brokerage commissions and fees	**$ 3,119**	$ 2,820	$ 2,831
Administration fees	**2,526**	2,310	2,348
All other commissions and fees[(a)]	**1,885**	1,706	1,402
Total commissions and other fees	**$ 7,530**	$ 6,836	$ 6,581

(a) Includes annuity sales commissions, depositary receipt-related service fees and travel-related sales commissions, as well as other service fees, which are recognized as revenue when the services are rendered.

Commissions and other fees are earned primarily by CIB, CCB and AWM.

Mortgage fees and related income

This revenue category reflects CCB's Home Lending production and net mortgage servicing revenue.

Production revenue includes fees and income recognized as earned on mortgage loans originated with the intent to sell, and the impact of risk management activities associated with the mortgage pipeline and warehouse loans. Production revenue also includes gains and losses on sales and lower of cost or fair value adjustments on mortgage loans held-for-sale (excluding certain repurchased loans insured by U.S. government agencies), and changes in the fair value of financial instruments measured under the fair value option. Net mortgage servicing revenue includes operating revenue earned from servicing third-party mortgage loans, which is recognized over the period in which the service is provided; changes in the fair value of MSRs; the impact of risk management activities associated with MSRs; and gains and losses on securitization of excess mortgage servicing. Net mortgage servicing revenue also includes gains and losses on sales and lower of cost or fair value adjustments of certain repurchased loans insured by U.S. government agencies.

Refer to Note 15 for further information on risk management activities and MSRs.

Net interest income from mortgage loans is recorded in interest income.

Card income

This revenue category includes interchange and other income from credit and debit card transactions; and fees earned from processing card transactions for merchants, both of which are recognized when purchases are made by a cardholder and presented net of certain transaction-related costs. Card income also includes account origination costs and annual fees, which are deferred and recognized on a straight-line basis over a 12-month period.

Certain credit card products offer the cardholder the ability to earn points based on account activity, which the cardholder can choose to redeem for cash and non-cash rewards. The cost to the Firm related to these proprietary rewards programs varies based on multiple factors including the terms and conditions of the rewards programs, cardholder activity, cardholder reward redemption rates and cardholder reward selections. The Firm maintains a liability for its obligations under its rewards programs and reports the current-period cost as a reduction of card income.

Credit card revenue sharing agreements

The Firm has contractual agreements with numerous co-brand partners that grant the Firm exclusive rights to issue co-branded credit card products and market them to the customers of such partners. These partners endorse the co-brand credit card programs and provide their customer or member lists to the Firm. The partners may also conduct marketing activities and provide rewards redeemable under their own loyalty programs that the Firm will grant to co-brand credit cardholders based on account activity. The terms of these agreements generally range from five to ten years.

The Firm typically makes payments to the co-brand credit card partners based on the cost of partners' marketing activities and loyalty program rewards provided to credit cardholders, new account originations and sales volumes. Payments to partners based on marketing efforts undertaken by the partners are expensed by the Firm as incurred and reported as marketing expense. Payments for partner loyalty program rewards are reported as a reduction of card income when incurred. Payments to partners based on new credit card account originations are accounted for as direct loan origination costs and are deferred and recognized as a reduction of card income on a straight-line basis over a 12-month period. Payments to partners based on sales volumes are reported as a reduction of card income when the related interchange income is earned.

The following table presents the components of card income:

Year ended December 31, (in millions)	2024	2023	2022
Interchange and merchant processing income	**$ 33,847**	$ 31,021	$ 28,085
Reward costs and partner payments	**(26,784)**	(24,601)	(22,162)
All other[(a)]	**(1,566)**	(1,636)	(1,503)
Total card income	**$ 5,497**	$ 4,784	$ 4,420

(a) Predominantly represents the amortization of account origination costs and annual fees, which are deferred and recognized on a straight-line basis over a 12-month period.

Card income is earned primarily by CCB and CIB.

Other income

This revenue category includes operating lease income, as well as losses associated with the Firm's tax-oriented investments, predominantly alternative energy equity-method investments in CIB. The losses associated with these tax-oriented investments are more than offset by lower income tax expense from the associated tax credits.

The following table presents certain components of other income:

Year ended December 31, (in millions)	2024	2023	2022
Operating lease income	**$ 2,795**	$ 2,843	$ 3,654
Losses on tax-oriented investments	**(97)**	(1,538)	(1,491)
Gain on Visa shares	**7,990** (b)	—	914 (c)
Estimated bargain purchase gain associated with the First Republic acquisition	**103**	2,775	—
Gain related to the acquisition of CIFM (a)	**—**	339	—

(a) Gain on the original minority interest in CIFM upon the Firm's acquisition of the remaining 51% of the entity.
(b) Relates to the initial gain recognized on May 6, 2024 on the Visa C shares. Refer to Note 2 for additional information.
(c) Relates to the sale of Visa B shares.

Refer to Note 18 for additional information on operating leases.

Proportional Amortization Method: Effective January 1, 2024, as a result of adopting updates to the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance, the amortization of certain of the Firm's alternative energy tax-oriented investments that was previously recognized in other income is now recognized in income tax expense, which aligns with the associated tax credits and other tax benefits. Refer to Notes 1, 14 and 25 for additional information.

Noninterest expense

Other expense

Other expense on the Firm's Consolidated statements of income included:

Year ended December 31, (in millions)	2024	2023	2022
Legal expense	**$ 740**	$1,436	$ 266
FDIC-related expense	**1,893** (c)	4,203 (c)	860
Operating losses	**1,417**	1,228	1,101
Contribution of Visa shares(a)	**1,000**	—	—
First Republic-related expense(b)	**777**	1,060	—

(a) Represents the contribution of a portion of Visa C shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Note 2 for additional information.
(b) Reflects the expenses classified within other expense, including $488 million and $360 million of integration and restructuring costs associated with First Republic for the full years ended December 31, 2024 and 2023, respectively. Additionally, the second quarter of 2023 included payments to the FDIC for the First Republic individuals who were not employees of the Firm until July 2, 2023. Refer to Note 34 for additional information on the First Republic acquisition.
(c) The first quarter of 2024 included an increase of $725 million to the FDIC special assessment reflecting the FDIC's revised estimate of Deposit Insurance Fund losses. The fourth quarter of 2023 included the $2.9 billion FDIC special assessment.

Refer to Note 32 for additional information on noninterest revenue and expense by segment.

Note 7 – Interest income and interest expense

Interest income and interest expense are recorded in the Consolidated statements of income and classified based on the nature of the underlying asset or liability.

Interest income and interest expense includes the current-period interest accruals for financial instruments measured at fair value, except for derivatives and certain financial instruments containing embedded derivatives; for those instruments, all changes in fair value including any interest elements, are primarily reported in principal transactions revenue. For financial instruments that are not measured at fair value, the related interest is included within interest income or interest expense, as applicable. Interest income and interest expense also includes the effect of derivatives that qualify for hedge accounting where applicable.

Interest income on loans and securities include the amortization and accretion of purchase premiums and discounts, as well as net deferred fees and costs on loans. These amounts are deferred in loans and investment securities, respectively, and recognized on a level-yield basis.

Refer to Notes 5, 10, 11, 12, and 20 for further information on accounting for interest income and interest expense related to hedge accounting, investment securities, securities financing activities (i.e., securities purchased or sold under resale or repurchase agreements; securities borrowed; and securities loaned), loans and long-term debt, respectively.

The following table presents the components of interest income and interest expense:

Year ended December 31, (in millions)	2024	2023	2022
Interest income			
Loans	**$ 92,353** [d]	$ 83,384 [d]	$ 52,736
Taxable securities	**21,947**	17,390	10,372
Non-taxable securities[a]	**1,197**	1,336	975
Total investment securities	**23,144** [d]	18,726 [d]	11,347
Trading assets - debt instruments	**20,327**	15,950	9,053
Federal funds sold and securities purchased under resale agreements	**18,299**	15,079	4,632
Securities borrowed	**9,208**	7,983	2,237
Deposits with banks	**22,297**	21,797	9,039
All other interest-earning assets[b]	**8,305**	7,669	3,763
Total interest income	**$193,933**	$170,588	$ 92,807
Interest expense			
Interest bearing deposits	**$ 49,559**	$ 40,016	$ 10,082
Federal funds purchased and securities loaned or sold under repurchase agreements	**19,149**	13,259	3,721
Short-term borrowings	**2,101**	1,894	747
Trading liabilities - debt and all other interest-bearing liabilities[c]	**10,238**	9,396	3,246
Long-term debt	**18,920**	15,803	8,075
Beneficial interest issued by consolidated VIEs	**1,383**	953	226
Total interest expense	**$101,350**	$ 81,321	$ 26,097
Net interest income	**$ 92,583**	$ 89,267	$ 66,710
Provision for credit losses	**10,678**	9,320	6,389
Net interest income after provision for credit losses	**$ 81,905**	$ 79,947	$ 60,321

(a) Represents securities that are tax-exempt for U.S. federal income tax purposes.
(b) Includes interest earned on brokerage-related held-for-investment customer receivables, which are classified in accrued interest and accounts receivable, and all other interest-earning assets, which are classified in other assets on the Consolidated balance sheets.
(c) All other interest-bearing liabilities includes interest expense on brokerage-related customer payables.
(d) Includes the accretion of the purchase discount on certain acquired loans and investment securities associated with First Republic. Refer to Note 34 for additional information.

Note 8 – Pension and other postretirement employee benefit plans

The Firm has various defined benefit pension plans and OPEB plans that provide benefits to its employees in the U.S. and certain non-U.S. locations. Substantially all the defined benefit pension plans are closed to new participants. The principal defined benefit pension plan in the U.S., which covered substantially all U.S. employees, was closed to new participants and frozen for existing participants on January 1, 2020, (and January 1, 2019 for new hires on or after December 2, 2017). Interest credits continue to accrue to participants' accounts based on their accumulated balances.

The Firm maintains funded and unfunded postretirement benefit plans that provide medical and life insurance for certain eligible employees and retirees as well as their dependents covered under these programs. None of these plans have a material impact on the Firm's Consolidated Financial Statements.

The Firm also provides a qualified defined contribution plan in the U.S. and maintains other similar arrangements in certain non-U.S. locations. The most significant of these plans is the JPMorgan Chase 401(k) Savings Plan ("the 401(k) Savings Plan"), which covers substantially all U.S. employees. Employees can contribute to the 401(k) Savings Plan on a pretax and/or Roth 401(k) after-tax basis. The Firm makes annual matching and pay credit contributions to the 401(k) Savings Plan on behalf of eligible participants.

The following table presents the pretax benefit obligations, plan assets, the net funded status, and the amounts recorded in AOCI on the Consolidated balance sheets for the Firm's significant defined benefit pension and OPEB plans.

As of or for the year ended December 31, (in millions)		**2024**		2023
Projected benefit obligations	**$**	**(14,459)**	$	(14,740)
Fair value of plan assets		**22,201**		22,013
Net funded status		**7,742**		7,273
Accumulated other comprehensive income/(loss)		**(1,649)**		(1,517)

The weighted-average discount rate used to value the benefit obligations as of December 31, 2024 and 2023, was 5.49% and 5.16%, respectively.

Gains and losses

Gains or losses resulting from changes in the benefit obligation and the fair value of plan assets are recorded in OCI. Amortization of net gains or losses are recognized as part of the net periodic benefit cost over subsequent periods, if, as of the beginning of the year, the net gain or loss exceeds 10% of the greater of the projected benefit obligation or the fair value of the plan assets. Amortization is generally over the average expected remaining lifetime of plan participants, given the frozen status of most plans. For the year ended December 31, 2024, the net loss was attributable to lower than expected returns on plan assets, partially offset by projected benefit obligation net gains primarily related to changes in the discount rate. For the year ended December 31, 2023, the net gain was attributable to market-driven increases in the fair value of plan assets, partially offset by changes in the discount rate and interest crediting rate.

The following table presents the net periodic benefit costs reported in the Consolidated statements of income for the Firm's defined benefit pension, defined contribution and OPEB plans, and in other comprehensive income for the defined benefit pension and OPEB plans.

Year ended December 31, (in millions)		**2024**		2023		2022
Total net periodic defined benefit plan cost/(credit)[(a)]	**$**	**(462)**	$	(393)	$	(192) [(b)]
Total defined contribution plans		**1,733**		1,609		1,408
Total pension and OPEB cost included in noninterest expense	**$**	**1,271**	$	1,216	$	1,216
Total recognized in other comprehensive (income)/loss	**$**	**131**	$	(421)	$	1,459

(a) The service cost component of net periodic defined benefit cost is reported in compensation expense; all other components of net periodic defined benefit costs are reported in other expense in the Consolidated statements of income.
(b) Includes pension settlement losses of $92 million for the year ended December 31, 2022.

The following table presents the weighted-average actuarial assumptions used to determine the net periodic benefit costs for the defined benefit pension and OPEB plans.

Year ended December 31,	**2024**	2023	2022
Discount rate	**5.16 %**	5.14 %	2.54 %
Expected long-term rate of return on plan assets	**6.15 %**	5.74 %	3.68 %

Plan assumptions

The Firm's expected long-term rate of return is a blended weighted average, by asset allocation of the projected long-term returns for the various asset classes, taking into consideration local market conditions and the specific allocation of plan assets. Returns on asset classes are developed using a forward-looking approach and are not strictly based on historical returns, with consideration given to current market conditions and the portfolio mix of each plan.

The discount rates used in determining the benefit obligations are generally provided by the Firm's actuaries, with the Firm's principal defined benefit pension plan using a rate that was selected by reference to the yields on portfolios of bonds with maturity dates and coupons that closely match the plan's projected annual cash flows.

Investment strategy and asset allocation

The assets of the Firm's defined benefit pension plans are held in various trusts and are invested in well-diversified portfolios of equity and fixed income securities, cash and cash equivalents, and alternative investments. The Firm regularly reviews the asset allocations and asset managers, as well as other factors that could impact the portfolios, which are rebalanced when deemed necessary. As of December 31, 2024, the approved asset allocation ranges by asset class for the Firm's principal defined benefit plan are 41-100% debt securities, 0-40% equity securities, 0-1% real estate, and 0-8% alternatives.

Assets held by the Firm's defined benefit pension and OPEB plans do not include securities issued by JPMorganChase or its affiliates, except through indirect exposures through investments in exchange traded funds, mutual funds and collective investment funds managed by third-parties. The defined benefit pension and OPEB plans hold investments that are sponsored or managed by affiliates of JPMorganChase in the amount of $1.8 billion as of both December 31, 2024 and 2023.

Fair value measurement of the plans' assets and liabilities

Refer to Note 2 for information on fair value measurements, including descriptions of level 1, 2, and 3 of the fair value hierarchy and the valuation methods employed by the Firm.

Defined benefit pension and OPEB plans assets and liabilities measured at fair value

	2024				2023			
December 31, (in millions)	Level 1[(a)]	Level 2[(b)]	Level 3[(c)]	Total fair value	Level 1[(a)]	Level 2[(b)]	Level 3[(c)]	Total fair value
Assets measured at fair value classified in the fair value hierarchy	**$ 6,910**	**$ 9,693**	**$ 3,956**	**$ 20,559**	$ 6,521	$ 10,713	$ 3,124	$ 20,358
Assets measured at fair value using NAV as a practical expedient				**2,101**				2,097
Net defined benefit pension plan payables				**(459)**				(442)
Total fair value of plan assets				**$ 22,201**				$ 22,013

(a) Consists predominantly of equity securities, U.S. federal, state, and local and non-U.S. government debt securities, and cash equivalents.
(b) Consists of corporate debt securities, fund investments, mortgage-backed securities, and U.S. federal, state, and local and non-U.S. government debt securities.
(c) Consists predominantly of corporate-owned life insurance policies.

Changes in level 3 fair value measurements using significant unobservable inputs

Investments classified in level 3 of the fair value hierarchy increased in 2024 to $4.0 billion, due to $536 million of transfers in and $415 million in unrealized gains, partially offset by $123 million in settlements. The net increase in 2023 was due to $400 million in unrealized gains and $173 million of transfers in, partially offset by $59 million in settlements.

Estimated future benefit payments

The following table presents benefit payments expected to be paid for the defined benefit pension and OPEB plans for the years indicated.

Year ended December 31, (in millions)		
2025	$	1,186
2026		1,155
2027		1,134
2028		1,095
2029		1,093
Years 2030–2034		5,229

Note 9 – Employee share-based incentives

Employee share-based awards

In 2024, 2023 and 2022, JPMorganChase granted long-term share-based awards to certain employees under its LTIP, as amended and restated effective May 18, 2021, and subsequently amended effective May 21, 2024. Under the terms of the LTIP, as of December 31, 2024, 81 million shares of common stock were available for issuance through May 2028. The LTIP is the only active plan under which the Firm is currently granting share-based incentive awards. In the following discussion, the LTIP constitutes the Firm's share-based incentive plans.

RSUs are awarded at no cost to the recipient upon their grant. Generally, RSUs are granted annually and vest at a rate of 50% after two years and 50% after three years and are converted into shares of common stock as of the vesting date. In addition, RSUs typically include full-career eligibility provisions, which allow employees to continue to vest upon voluntary termination based on age and/or service-related requirements, subject to post-employment and other restrictions. All RSU awards are subject to forfeiture until vested and contain clawback provisions that may result in cancellation under certain specified circumstances. Predominantly all RSUs entitle the recipient to receive cash payments equivalent to any dividends paid on the underlying common stock during the period the RSUs are outstanding.

Performance share units ("PSUs") are granted annually, and approved by the Firm's Board of Directors, to members of the Firm's Operating Committee under the variable compensation program. PSUs are subject to the Firm's achievement of specified performance criteria over a three-year period. The number of awards that vest can range from zero to 150% of the grant amount. In addition, dividends that accrue during the vesting period are reinvested in dividend equivalent share units. PSUs and the related dividend equivalent share units are converted into shares of common stock after vesting.

Once the PSUs and dividend equivalent share units have vested, the shares of common stock that are delivered, after applicable tax withholding, must be retained for an additional holding period, for a total combined vesting and holding period of approximately five to eight years from the grant date depending on regulations in certain countries.

Under the LTI Plans, stock appreciation rights ("SARs") were granted with an exercise price equal to the fair value of JPMorganChase's common stock on the grant date. SARs expire ten years after the grant date. There were no grants of SARs in 2024, 2023 or 2022.

The Firm separately recognizes compensation expense for each tranche of each award, net of estimated forfeitures, as if it were a separate award with its own vesting date. Generally, for each tranche granted, compensation expense is recognized on a straight-line basis from the grant date until the vesting date of the respective tranche, provided that the employees will not become full-career eligible during the vesting period. For awards with full-career eligibility provisions and awards granted with no future substantive service requirement, the Firm accrues the estimated value of awards expected to be awarded to employees as of the grant date without giving consideration to the impact of post-employment restrictions. For each tranche granted to employees who will become full-career eligible during the vesting period, compensation expense is recognized on a straight-line basis from the grant date until the earlier of the employee's full-career eligibility date or the vesting date of the respective tranche.

The Firm's policy for issuing shares upon settlement of employee share-based incentive awards is to issue either new shares of common stock or treasury shares. During 2024, 2023 and 2022, the Firm settled all of its employee share-based awards by issuing treasury shares.

Refer to Note 23 for further information on the classification of share-based awards for purposes of calculating earnings per share.

RSUs, PSUs and SARs activity

Generally, compensation expense for RSUs and PSUs is measured based on the number of units granted multiplied by the stock price at the grant date, and for SARs, is measured at the grant date using the Black-Scholes valuation model. Compensation expense for these awards is recognized in net income as described previously. The following table summarizes JPMorganChase's RSUs, PSUs and SARs activity for 2024.

Year ended December 31, 2024 (in thousands, except weighted-average data, and where otherwise stated)	RSUs/PSUs Number of units	RSUs/PSUs Weighted-average grant date fair value	SARs Number of awards	SARs Weighted-average exercise price	SARs Weighted-average remaining contractual life (in years)	SARs Aggregate intrinsic value
Outstanding, January 1	**52,243**	**$ 141.31**	**2,250**	**$ 152.19**		
Granted	**20,020**	**166.74**	**—**	**—**		
Exercised or vested	**(19,542)**	**143.02**	**—**	**—**		
Forfeited	**(2,112)**	**147.41**	**—**	**—**		
Canceled	**NA**	**NA**	**—**	**—**		
Outstanding, December 31	**50,609**	**$ 150.41**	**2,250**	**$ 152.19**	**6.7**	**$ 198,113**
Exercisable, December 31	**NA**	**NA**	**—**	**—**	**—**	**—**

The total fair value of RSUs and PSUs that vested during the years ended December 31, 2024, 2023 and 2022, was $3.5 billion, $2.5 billion and $3.2 billion, respectively. There were no SARs exercised in 2024. The total intrinsic value of SARs exercised during the years ended December 31, 2023 and 2022, was $24 million and $75 million, respectively.

Compensation expense

The Firm recognized the following noncash compensation expense related to its various employee share-based incentive plans in its Consolidated statements of income.

Year ended December 31, (in millions)	**2024**	2023	2022
Cost of prior grants of RSUs, PSUs and SARs that are amortized over their applicable vesting periods	**$ 1,622**	$ 1,510	$ 1,253
Accrual of estimated costs of share-based awards to be granted in future periods, predominantly those to full-career eligible employees	**1,882**	1,607	1,541
Total noncash compensation expense related to employee share-based incentive plans	**$3,504**	$ 3,117	$ 2,794

At December 31, 2024, approximately $963 million (pretax) of compensation expense related to unvested awards had not yet been charged to net income. That cost is expected to be amortized into compensation expense over a weighted-average period of 1.6 years. The Firm does not capitalize any compensation expense related to share-based compensation awards to employees.

Tax benefits

Income tax benefits (including tax benefits from dividends or dividend equivalents) related to share-based incentive arrangements recognized in the Firm's Consolidated statements of income for the years ended December 31, 2024, 2023 and 2022, were $1.0 billion, $836 million and $901 million, respectively.

Note 10 – Investment securities

Investment securities consist of debt securities that are classified as AFS or HTM. Debt securities classified as trading assets are discussed in Note 2. Predominantly all of the Firm's AFS and HTM securities are held by Treasury and CIO in connection with its asset-liability management activities.

AFS securities are carried at fair value on the Consolidated balance sheets. Unrealized gains and losses, after any applicable hedge accounting adjustments or allowance for credit losses, are reported in AOCI. The specific identification method is used to determine realized gains and losses on AFS securities, which are included in investment securities gains/(losses) on the Consolidated statements of income. HTM securities, which the Firm has the intent and ability to hold until maturity, are carried at amortized cost, net of allowance for credit losses, on the Consolidated balance sheets.

For both AFS and HTM securities, purchase discounts or premiums are generally amortized into interest income on a level-yield basis over the contractual life of the security. However, premiums on certain callable debt securities are amortized to the earliest call date.

The amortized costs and estimated fair values of the investment securities portfolio were as follows for the dates indicated.

December 31, (in millions)	2024 Amortized cost[(d)(e)]	2024 Gross unrealized gains	2024 Gross unrealized losses	2024 Fair value	2023 Amortized cost[(d)(e)]	2023 Gross unrealized gains	2023 Gross unrealized losses	2023 Fair value
Available-for-sale securities								
Mortgage-backed securities:								
U.S. GSEs and government agencies	**$ 95,671**	**$ 251**	**$ 4,029**	**$ 91,893**	$ 88,377	$ 870	$ 4,077	$ 85,170
Residential:								
U.S.	**4,242**	**16**	**50**	**4,208**	2,086	10	68	2,028
Non-U.S.	**600**	**3**	**—**	**603**	1,608	4	1	1,611
Commercial	**4,115**	**20**	**70**	**4,065**	2,930	12	139	2,803
Total mortgage-backed securities	**104,628**	**290**	**4,149**	**100,769**	95,001	896	4,285	91,612
U.S. Treasury and government agencies	**235,495**	**545**	**1,261**	**234,779**	58,051	276	522	57,805
Obligations of U.S. states and municipalities	**18,337**	**110**	**534**	**17,913**	21,243	390	266	21,367
Non-U.S. government debt securities	**36,655**	**94**	**504**	**36,245**	21,387	254	359	21,282
Corporate debt securities	**71**	**—**	**1**	**70**	128	—	28	100
Asset-backed securities:								
Collateralized loan obligations	**14,887**	**59**	**3**	**14,943**	6,769	11	28	6,752
Other	**2,125**	**17**	**9**	**2,133**	2,804	8	26	2,786
Unallocated portfolio layer fair value basis adjustments[(a)]	**(1,153)**	**—**	**(1,153)**	**NA**	73	(73)	—	NA
Total available-for-sale securities	**411,045**	**1,115**	**5,308**	**406,852**	205,456	1,762	5,514	201,704
Held-to-maturity securities[(b)]								
Mortgage-backed securities:								
U.S. GSEs and government agencies	**97,177**	**6**	**13,531**	**83,652**	105,614	39	11,643	94,010
U.S. Residential	**8,605**	**4**	**904**	**7,705**	9,709	4	970	8,743
Commercial	**8,817**	**24**	**389**	**8,452**	10,534	13	581	9,966
Total mortgage-backed securities	**114,599**	**34**	**14,824**	**99,809**	125,857	56	13,194	112,719
U.S. Treasury and government agencies	**108,632**	**—**	**11,212**	**97,420**	173,666	—	13,074	160,592
Obligations of U.S. states and municipalities	**9,310**	**32**	**631**	**8,711**	9,945	74	591	9,428
Asset-backed securities:								
Collateralized loan obligations	**40,573**	**84**	**14**	**40,643**	58,565	47	352	58,260
Other	**1,354**	**2**	**39**	**1,317**	1,815	1	61	1,755
Total held-to-maturity securities[(c)]	**274,468**	**152**	**26,720**	**247,900**	369,848	178	27,272	342,754
Total investment securities, net of allowance for credit losses	**$ 685,513**	**$ 1,267**	**$ 32,028**	**$ 654,752**	$ 575,304	$ 1,940	$ 32,786	$ 544,458

(a) Represents the amount of portfolio layer method basis adjustments related to AFS securities hedged in a closed portfolio. Under U.S. GAAP portfolio layer method basis adjustments are not allocated to individual securities, however the amounts impact the unrealized gains or losses in the table for the types of securities being hedged. Refer to Note 1 and Note 5 for additional information.

(b) The Firm purchased $4.7 billion, $4.1 billion and $33.7 billion of HTM securities for the years ended December 31, 2024, 2023 and 2022, respectively.

(c) Effective January 1, 2023, the Firm adopted the portfolio layer method hedge accounting guidance which permitted a transfer of HTM securities to AFS upon adoption. The Firm transferred obligations of U.S. states and municipalities with a carrying value of $7.1 billion resulting in the recognition of $38 million net pre-tax unrealized losses in AOCI. This transfer was a non-cash transaction. Refer to Note 24 for additional information.

(d) The amortized cost of investment securities is reported net of allowance for credit losses of $152 million, $128 million and $96 million at December 31, 2024, 2023 and 2022, respectively.

(e) Excludes $3.7 billion and $2.8 billion of accrued interest receivable at December 31, 2024 and 2023, respectively, included in accrued interest and accounts receivable on the Consolidated balance sheets. The Firm generally does not recognize an allowance for credit losses on accrued interest receivable, consistent with its policy to write them off no later than 90 days past due by reversing interest income. The Firm did not reverse through interest income any accrued interest receivable for the years ended December 31, 2024 and 2023.

At December 31, 2024, the investment securities portfolio consisted of debt securities with an average credit rating of AA+ (based upon external ratings where available, and where not available, based primarily upon internal risk ratings). Risk ratings are used to identify the credit quality of securities and differentiate risk within the portfolio. The Firm's internal risk ratings generally align with the qualitative characteristics (e.g., borrower capacity to meet financial commitments and vulnerability to changes in the economic environment) defined by S&P and Moody's, however the quantitative characteristics (e.g., probability of default ("PD") and loss given default ("LGD")) may differ as they reflect internal historical experiences and assumptions. Risk ratings are assigned at acquisition, reviewed on a regular and ongoing basis by Credit Risk Management and adjusted as necessary over the life of the investment for updated information affecting the issuer's ability to fulfill its obligations.

AFS securities impairment

The following tables present the fair value and gross unrealized losses by aging category for AFS securities at December 31, 2024 and 2023. The tables exclude U.S. Treasury and government agency securities and U.S. GSE and government agency MBS with unrealized losses of $5.3 billion and $4.6 billion, at December 31, 2024 and 2023, respectively; changes in the value of these securities are generally driven by changes in interest rates rather than changes in their credit profile given the explicit or implicit guarantees provided by the U.S. government.

	Available-for-sale securities with gross unrealized losses					
	Less than 12 months		12 months or more			
Year ended December 31, 2024 (in millions)	Fair value	Gross unrealized losses	Fair value	Gross unrealized losses	Total fair value	Total gross unrealized losses
Available-for-sale securities						
Mortgage-backed securities:						
Residential:						
U.S.	**$ 1,505**	**$ 6**	**$ 925**	**$ 44**	**$ 2,430**	**$ 50**
Non-U.S.	**—**	**—**	**30**	**—**	**30**	**—**
Commercial	**763**	**8**	**1,184**	**62**	**1,947**	**70**
Total mortgage-backed securities	**2,268**	**14**	**2,139**	**106**	**4,407**	**120**
Obligations of U.S. states and municipalities	**10,037**	**233**	**2,412**	**301**	**12,449**	**534**
Non-U.S. government debt securities	**14,234**	**234**	**4,184**	**270**	**18,418**	**504**
Corporate debt securities	**9**	**—**	**30**	**1**	**39**	**1**
Asset-backed securities:						
Collateralized loan obligations	**2**	**—**	**375**	**3**	**377**	**3**
Other	**214**	**1**	**200**	**8**	**414**	**9**
Total available-for-sale securities with gross unrealized losses	**$ 26,764**	**$ 482**	**$ 9,340**	**$ 689**	**$ 36,104**	**$ 1,171**

	Available-for-sale securities with gross unrealized losses					
	Less than 12 months		12 months or more			
Year ended December 31, 2023 (in millions)	Fair value	Gross unrealized losses	Fair value	Gross unrealized losses	Total fair value	Total gross unrealized losses
Available-for-sale securities						
Mortgage-backed securities:						
Residential:						
U.S.	$ 81	$ —	$ 1,160	$ 68	$ 1,241	$ 68
Non-U.S.	—	—	722	1	722	1
Commercial	228	3	1,775	136	2,003	139
Total mortgage-backed securities	309	3	3,657	205	3,966	208
Obligations of U.S. states and municipalities	2,134	20	2,278	246	4,412	266
Non-U.S. government debt securities	7,145	23	4,987	336	12,132	359
Corporate debt securities	9	—	79	28	88	28
Asset-backed securities:						
Collateralized loan obligations	932	2	3,744	26	4,676	28
Other	208	1	1,288	25	1,496	26
Total available-for-sale securities with gross unrealized losses	$ 10,737	$ 49	$ 16,033	$ 866	$ 26,770	$ 915

AFS securities are considered impaired if the fair value is less than the amortized cost.

The Firm recognizes impairment losses in earnings if the Firm has the intent to sell the debt security, or if it is more likely than not that the Firm will be required to sell the debt security before recovery of its amortized cost. In these circumstances the impairment loss is recognized in investment securities gains/(losses) in the Consolidated Statements of Income and is equal to the full difference between the amortized cost (net of allowance if applicable) and the fair value of the security.

For impaired debt securities that the Firm has the intent and ability to hold, the securities are evaluated to determine if a credit loss exists. If it is determined that a credit loss exists, that loss is recognized as an allowance for credit losses through the provision for credit losses in the Consolidated Statements of Income, limited by the amount of impairment. Any impairment on debt securities that the Firm has the intent and ability to hold not due to credit losses is recorded in OCI.

Factors considered in evaluating credit losses include adverse conditions specifically related to the industry, geographic area or financial condition of the issuer or underlying collateral of a security; and payment structure of the security.

When assessing securities issued in a securitization for credit losses, the Firm estimates cash flows considering relevant market and economic data, underlying loan-level data, and structural features of the securitization, such as subordination, excess spread, overcollateralization or other forms of credit enhancement, and compares the losses projected for the underlying collateral ("pool losses") against the level of credit enhancement in the securitization structure to determine whether these features are sufficient to absorb the pool losses, or whether a credit loss exists.

For beneficial interests in securitizations that are rated below "AA" at their acquisition, or that can be contractually prepaid or otherwise settled in such a way that the Firm would not recover substantially all of its recorded investment, the Firm evaluates impairment for credit losses when there is an adverse change in expected cash flows.

HTM securities – credit risk

Allowance for credit losses

The allowance for credit losses on HTM securities represents expected credit losses over the remaining expected life of the securities.

The allowance for credit losses on HTM obligations of U.S. states and municipalities and commercial mortgage-backed securities is calculated by applying statistical credit loss factors (estimated PD and LGD) to the amortized cost. The credit loss factors are derived using a weighted average of five internally developed eight-quarter macroeconomic scenarios, followed by a single year straight-line interpolation to revert to long run historical information for periods beyond the forecast period. Refer to Note 13 for further information on the eight-quarter macroeconomic forecast.

The allowance for credit losses on HTM collateralized loan obligations and U.S. residential mortgage-backed securities is calculated as the difference between the amortized cost and the present value of the cash flows expected to be collected, discounted at the security's effective interest rate. These cash flow estimates are developed based on expectations of underlying collateral performance derived using the eight-quarter macroeconomic forecast and the single year straight-line interpolation, as well as considering the structural features of the security.

The application of different inputs and assumptions into the calculation of the allowance for credit losses is subject to significant management judgment, and emphasizing one input or assumption over another, or considering other inputs or assumptions, could affect the estimate of the allowance for credit losses on HTM securities.

Credit quality indicator

The primary credit quality indicator for HTM securities is the risk rating assigned to each security. At both December 31, 2024 and 2023, all HTM securities were rated investment grade and were current and accruing, with approximately 99% rated at least AA+ (based upon external ratings where available, and where not available, based primarily upon internal risk ratings).

Allowance for credit losses on investment securities

The allowance for credit losses on investment securities was $152 million, $128 million and $96 million as of December 31, 2024, 2023 and 2022, respectively, which included a cumulative-effect adjustment to retained earnings related to the transfer of HTM securities to AFS for the year ended December 31, 2023.

Selected impacts of investment securities on the Consolidated statements of income

Year ended December 31, (in millions)	2024	2023	2022
Realized gains	**$ 593**	$ 622	$ 198
Realized losses	**(1,614)**	(3,802)	(2,578)
Investment securities losses	**$(1,021)**	$(3,180)	$(2,380)
Provision for credit losses	**$ 24**	$ 38	$ 54

Contractual maturities and yields

The following table presents the amortized cost and estimated fair value at December 31, 2024, of JPMorganChase's investment securities portfolio by contractual maturity.

By remaining maturity December 31, 2024 (in millions)	Due in one year or less	Due after one year through five years	Due after five years through 10 years	Due after 10 years[c]	Total
Available-for-sale securities					
Mortgage-backed securities					
Amortized cost	$ 132	$ 9,410	$ 4,059	$ 91,027	$ 104,628
Fair value	130	9,345	4,041	87,253	100,769
Average yield[a]	3.32 %	4.58 %	5.25 %	4.75 %	4.75 %
U.S. Treasury and government agencies					
Amortized cost	$ 20,685	$ 169,162	$ 38,667	$ 6,981	$ 235,495
Fair value	20,730	169,145	38,619	6,285	234,779
Average yield[a]	4.50 %	4.57 %	4.96 %	5.53 %	4.65 %
Obligations of U.S. states and municipalities					
Amortized cost	$ 4	$ 16	$ 92	$ 18,225	$ 18,337
Fair value	4	16	90	17,803	17,913
Average yield[a]	1.59 %	3.95 %	4.46 %	5.32 %	5.32 %
Non-U.S. government debt securities					
Amortized cost	$ 13,331	$ 11,769	$ 7,609	$ 3,946	$ 36,655
Fair value	13,327	11,701	7,403	3,814	36,245
Average yield[a]	4.24 %	4.32 %	2.82 %	4.15 %	3.96 %
Corporate debt securities					
Amortized cost	$ 106	$ 9	$ 5	$ —	$ 120
Fair value	57	9	4	—	70
Average yield[a]	14.09 %	4.06 %	4.19 %	— %	12.92 %
Asset-backed securities					
Amortized cost	$ —	$ 368	$ 1,609	$ 15,035	$ 17,012
Fair value	—	370	1,619	15,087	17,076
Average yield[a]	— %	6.14 %	6.04 %	5.96 %	5.97 %
Total available-for-sale securities					
Amortized cost[b]	$ 34,258	$ 190,734	$ 52,041	$ 135,214	$ 412,247
Fair value	34,248	190,586	51,776	130,242	406,852
Average yield[a]	4.42 %	4.56 %	4.70 %	4.98 %	4.70 %
Held-to-maturity securities					
Mortgage-backed securities					
Amortized cost	$ 104	$ 7,994	$ 6,077	$ 100,495	$ 114,670
Fair value	101	7,453	5,352	86,903	99,809
Average yield[a]	0.97 %	2.63 %	2.67 %	2.95 %	2.92 %
U.S. Treasury and government agencies					
Amortized cost	$ 20,083	$ 40,497	$ 48,052	$ —	$ 108,632
Fair value	19,500	37,715	40,205	—	97,420
Average yield[a]	0.43 %	1.23 %	1.25 %	— %	1.09 %
Obligations of U.S. states and municipalities					
Amortized cost	$ —	$ 9	$ 307	$ 9,026	$ 9,342
Fair value	—	9	276	8,426	8,711
Average yield[a]	— %	4.76 %	3.25 %	3.97 %	3.95 %
Asset-backed securities					
Amortized cost	$ —	$ 134	$ 22,433	$ 19,360	$ 41,927
Fair value	—	134	22,466	19,360	41,960
Average yield[a]	— %	5.94 %	5.57 %	5.73 %	5.65 %
Total held-to-maturity securities					
Amortized cost[b]	$ 20,187	$ 48,634	$ 76,869	$ 128,881	$ 274,571
Fair value	19,601	45,311	68,299	114,689	247,900
Average yield[a]	0.43 %	1.47 %	2.63 %	3.44 %	2.64 %

(a) Average yield is computed using the effective yield of each security owned at the end of the period, weighted based on the amortized cost of each security. The effective yield considers the contractual coupon, amortization of premiums and accretion of discounts, and the effect of related hedging derivatives, including closed portfolio hedges. Taxable-equivalent amounts are used where applicable. The effective yield excludes unscheduled principal prepayments; and accordingly, actual maturities of securities may differ from their contractual or expected maturities as certain securities may be prepaid. However, for certain callable debt securities, the average yield is calculated to the earliest call date.

(b) For purposes of this table, the amortized cost of available-for-sale securities excludes the allowance for credit losses of $49 million and the portfolio layer fair value hedge basis adjustments of $(1.2) billion at December 31, 2024. The amortized cost of held-to-maturity securities also excludes the allowance for credit losses of $103 million at December 31, 2024.

(c) Substantially all of the Firm's U.S. residential MBS and collateralized mortgage obligations are due in 10 years or more, based on contractual maturity. The estimated weighted-average life, which reflects anticipated future prepayments, is approximately eight years for agency residential MBS, six years for agency residential collateralized mortgage obligations, and five years for nonagency residential collateralized mortgage obligations.

Note 11 – Securities financing activities

JPMorganChase enters into resale, repurchase, securities borrowed and securities loaned agreements (collectively, "securities financing agreements") primarily to finance the Firm's inventory positions, acquire securities to cover short sales, accommodate customers' financing needs, settle other securities obligations and to deploy the Firm's excess cash.

Securities financing agreements are treated as collateralized financings on the Firm's Consolidated balance sheets. Where appropriate under applicable accounting guidance, securities financing agreements with the same counterparty are reported on a net basis. Refer to Note 1 for further discussion of the offsetting of assets and liabilities. Fees received and paid in connection with securities financing agreements are recorded over the life of the agreement in interest income and interest expense on the Consolidated statements of income.

The Firm has elected the fair value option for certain securities financing agreements. Refer to Note 3 for further information regarding the fair value option. The securities financing agreements for which the fair value option has been elected are reported within securities purchased under resale agreements, securities loaned or sold under repurchase agreements, and securities borrowed on the Consolidated balance sheets. Generally, for agreements carried at fair value, current-period interest accruals are recorded within interest income and interest expense, with changes in fair value reported in principal transactions revenue. However, for financial instruments containing embedded derivatives that would be separately accounted for in accordance with accounting guidance for hybrid instruments, all changes in fair value, including any interest elements, are reported in principal transactions revenue.

Securities financing agreements not elected under the fair value option are measured at amortized cost. As a result of the Firm's credit risk mitigation practices described below, the Firm did not hold any allowance for credit losses with respect to resale and securities borrowed arrangements as of December 31, 2024 and 2023.

Credit risk mitigation practices

Securities financing agreements expose the Firm primarily to credit and liquidity risk. To manage these risks, the Firm monitors the value of the underlying securities (predominantly high-quality securities collateral, including government-issued debt and U.S. GSEs and government agencies MBS) that it has received from or provided to its counterparties compared to the value of cash proceeds and exchanged collateral, and either requests additional collateral or returns securities or collateral when appropriate. Margin levels are initially established based upon the counterparty, the type of underlying securities, and the permissible collateral, and are monitored on an ongoing basis.

In resale and securities borrowed agreements, the Firm is exposed to credit risk to the extent that the value of the securities received is less than initial cash principal advanced and any collateral amounts exchanged. In repurchase and securities loaned agreements, credit risk exposure arises to the extent that the value of underlying securities advanced exceeds the value of the initial cash principal received, and any collateral amounts exchanged.

Additionally, the Firm typically enters into master netting agreements and other similar arrangements with its counterparties, which provide for the right to liquidate the underlying securities and any collateral amounts exchanged in the event of a counterparty default. It is also the Firm's policy to take possession, where possible, of the securities underlying resale and securities borrowed agreements. Refer to Note 29 for further information regarding assets pledged and collateral received in securities financing agreements.

The table below summarizes the gross and net amounts of the Firm's securities financing agreements, as of December 31, 2024 and 2023. When the Firm has obtained an appropriate legal opinion with respect to a master netting agreement with a counterparty and where other relevant netting criteria under U.S. GAAP are met, the Firm nets, on the Consolidated balance sheets, the balances outstanding under its securities financing agreements with the same counterparty. In addition, the Firm exchanges securities and/or cash collateral with its counterparty to reduce the economic exposure with the counterparty, but such collateral is not eligible for net Consolidated balance sheet presentation. Where the Firm has obtained an appropriate legal opinion with respect to the counterparty master netting agreement, such collateral, along with securities financing balances that do not meet all these relevant netting criteria under U.S. GAAP, is presented in the table below as "Amounts not nettable on the Consolidated balance sheets," and reduces the "Net amounts" presented. Where a legal opinion has not been either sought or obtained, the securities financing balances are presented gross in the "Net amounts" below. In transactions where the Firm is acting as the lender in a securities-for-securities lending agreement and receives securities that can be pledged or sold as collateral, the Firm recognizes the securities received at fair value within other assets and the obligation to return those securities within accounts payable and other liabilities on the Consolidated balance sheets.

	December 31, 2024				
(in millions)	Gross amounts	Amounts netted on the Consolidated balance sheets	Amounts presented on the Consolidated balance sheets	Amounts not nettable on the Consolidated balance sheets[(b)]	Net amounts[(c)]
Assets					
Securities purchased under resale agreements	**$ 607,154**	**$ (312,183)**	**$ 294,971**	**$ (282,220)**	**$ 12,751**
Securities borrowed	**267,917**	**(48,371)**	**219,546**	**(170,702)**	**48,844**
Liabilities					
Securities sold under repurchase agreements	**$ 603,683**	**$ (312,183)**	**$ 291,500**	**$ (249,763)**	**$ 41,737**
Securities loaned and other[(a)]	**58,989**	**(48,371)**	**10,618**	**(10,557)**	**61**

	December 31, 2023				
(in millions)	Gross amounts	Amounts netted on the Consolidated balance sheets	Amounts presented on the Consolidated balance sheets	Amounts not nettable on the Consolidated balance sheets[(b)]	Net amounts[(c)]
Assets					
Securities purchased under resale agreements	$ 523,308	$ (247,181)	$ 276,127	$ (267,582)	$ 8,545
Securities borrowed	244,046	(43,610)	200,436	(144,543)	55,893
Liabilities					
Securities sold under repurchase agreements	$ 459,985	$ (247,181)	$ 212,804	$ (182,011)	$ 30,793
Securities loaned and other[(a)]	52,142	(43,610)	8,532	(8,501)	31

(a) Includes securities-for-securities lending agreements of $5.9 billion and $5.6 billion at December 31, 2024 and 2023, respectively, accounted for at fair value, where the Firm is acting as lender.
(b) In some cases, collateral exchanged with a counterparty exceeds the net asset or liability balance with that counterparty. In such cases, the amounts reported in this column are limited to the related net asset or liability with that counterparty.
(c) Includes securities financing agreements that provide collateral rights, but where an appropriate legal opinion with respect to the master netting agreement has not been either sought or obtained. At December 31, 2024 and 2023, included $8.7 billion and $7.1 billion, respectively, of securities purchased under resale agreements; $42.9 billion and $50.7 billion, respectively, of securities borrowed; $40.9 billion and $30.0 billion, respectively, of securities sold under repurchase agreements; and securities loaned and other which were not material.

The tables below present as of December 31, 2024 and 2023 the types of financial assets pledged in securities financing agreements and the remaining contractual maturity of the securities financing agreements.

	Gross liability balance			
	2024		2023	
December 31, (in millions)	Securities sold under repurchase agreements	Securities loaned and other	Securities sold under repurchase agreements	Securities loaned and other
Mortgage-backed securities:				
U.S. GSEs and government agencies	**$ 82,645**	**$ —**	$ 71,064	$ —
Residential - nonagency	**2,610**	**—**	2,292	—
Commercial - nonagency	**2,344**	**—**	2,669	—
U.S. Treasury, GSEs and government agencies	**300,022**	**759**	216,467	1,034
Obligations of U.S. states and municipalities	**1,872**	**—**	2,323	—
Non-U.S. government debt	**117,614**	**1,852**	97,400	1,455
Corporate debt securities	**44,495**	**4,033**	39,247	2,025
Asset-backed securities	**4,619**	**—**	2,703	—
Equity securities	**47,462**	**52,345**	25,820	47,628
Total	**$ 603,683**	**$ 58,989**	$ 459,985	$ 52,142

	Remaining contractual maturity of the agreements				
December 31, 2024 (in millions)	Overnight and continuous	Up to 30 days	30 – 90 days	Greater than 90 days	Total
Total securities sold under repurchase agreements	**$ 308,392**	**$ 171,346**	**$ 19,932**	**$ 104,013**	**$ 603,683**
Total securities loaned and other	**54,066**	**1,463**	**1**	**3,459**	**58,989**

	Remaining contractual maturity of the agreements				
December 31, 2023 (in millions)	Overnight and continuous	Up to 30 days	30 – 90 days	Greater than 90 days	Total
Total securities sold under repurchase agreements	$ 259,048	$ 102,941	$ 20,960	$ 77,036	$ 459,985
Total securities loaned and other	49,610	1,544	—	988	52,142

Transfers not qualifying for sale accounting

At December 31, 2024 and 2023, the Firm held $805 million and $505 million, respectively, of financial assets for which the rights have been transferred to third parties; however, the transfers did not qualify as a sale in accordance with U.S. GAAP. These transfers have been recognized as collateralized financing transactions. The transferred assets are recorded in trading assets and loans, and the corresponding liabilities are recorded primarily in short-term borrowings and long-term debt on the Consolidated balance sheets.

Note 12 – Loans

Loan accounting framework

The accounting for a loan depends on management's strategy for the loan. The Firm accounts for loans based on the following categories:

- Originated or purchased loans held-for-investment (i.e., "retained")
- Loans held-for-sale
- Loans at fair value

The following provides a detailed accounting discussion of the Firm's loans by category:

Loans held-for-investment

Originated or purchased loans held-for-investment, including PCD, are recorded at amortized cost, reflecting the principal amount outstanding, net of the following: unamortized deferred loan fees, costs, premiums or discounts; charge-offs; collection of cash; and foreign exchange. Credit card loans also include billed finance charges and fees.

Interest income

Interest income on performing loans held-for-investment is accrued and recognized as interest income at the contractual rate of interest. Purchase price discounts or premiums, as well as net deferred loan fees or costs, are recognized in interest income over the contractual life of the loan as an adjustment of yield.

The Firm classifies accrued interest on loans, including accrued but unbilled interest on credit card loans, in accrued interest and accounts receivables on the Consolidated balance sheets. For credit card loans, accrued interest once billed is then recognized in the loan balances, with the related allowance recorded in the allowance for credit losses. Changes in the allowance for credit losses on accrued interest on credit card loans are recognized in the provision for credit losses and charge-offs are recognized by reversing interest income. For other loans, the Firm generally does not recognize an allowance for credit losses on accrued interest receivables, consistent with its policy to write them off no later than 90 days past due by reversing interest income.

Nonaccrual loans

Nonaccrual loans are those on which the accrual of interest has been suspended. Loans (other than credit card loans and certain consumer loans insured by U.S. government agencies) are placed on nonaccrual status and considered nonperforming when full payment of principal and interest is not expected, regardless of delinquency status, or when principal and interest has been in default for a period of 90 days or more, unless the loan is both well-secured and in the process of collection. A loan is determined to be past due when the minimum payment is not received from the borrower by the contractually specified due date or for certain loans (e.g., residential real estate loans), when a monthly payment is due and unpaid for 30 days or more. Finally, collateral-dependent loans are typically maintained on nonaccrual status.

On the date a loan is placed on nonaccrual status, all interest accrued but not collected is reversed against interest income. In addition, the amortization of deferred amounts is suspended. Interest income on nonaccrual loans may be recognized as cash interest payments are received (i.e., on a cash basis) if the recorded loan balance is deemed fully collectible; however, if there is doubt regarding the ultimate collectibility of the recorded loan balance, all interest cash receipts are applied to reduce the carrying value of the loan (the cost recovery method). For consumer loans, application of this policy typically results in the Firm recognizing interest income on nonaccrual consumer loans on a cash basis.

A loan may be returned to accrual status when repayment is reasonably assured and there has been demonstrated performance under the terms of the loan or, if applicable, the terms of the restructured loan.

As permitted by regulatory guidance, credit card loans are generally exempt from being placed on nonaccrual status; accordingly, interest and fees related to credit card loans continue to accrue until the loan is charged off or paid in full.

Allowance for loan losses

The allowance for loan losses represents the estimated expected credit losses in the held-for-investment loan portfolio at the balance sheet date and is recognized on the balance sheet as a contra asset, which brings the amortized cost to the net carrying value. Changes in the allowance for loan losses resulting from lending-related activity, macroeconomic variables, changes in credit and other inputs are recorded in the provision for credit losses on the Firm's Consolidated statements of income. Refer to Note 13 for further information on the Firm's accounting policies for the allowance for loan losses.

Charge-offs

Consumer loans are generally charged off or charged down to the lower of the amortized cost or the net realizable value of the underlying collateral (i.e., fair value less estimated costs to sell), with an offset to the allowance for loan losses, upon reaching specified stages of delinquency in accordance with standards established by the FFIEC. Residential real estate loans, unmodified credit card loans and scored business banking loans are generally charged off no later than 180 days past due. Scored auto and closed-end consumer loans, including modified credit card accounts placed on a fixed payment plan, are charged off no later than 120 days past due.

Certain consumer loans are charged off or charged down to their net realizable value earlier than the FFIEC charge-off standards in the following circumstances:

- Loans modified to borrowers experiencing financial difficulty that are determined to be collateral-dependent.
- Loans to borrowers who have experienced an event that suggests a loss is either known or highly certain are subject to accelerated charge-off standards (e.g., residential real estate and auto loans are charged off or charged down within 60 days of receiving notification of a bankruptcy filing).
- Auto loans upon repossession of the automobile.

Other than in certain limited circumstances, the Firm typically does not recognize charge-offs on the government-guaranteed portion of loans.

Wholesale loans are charged off when it is highly certain that a loss has been realized. The determination of whether to recognize a charge-off includes many factors, including the prioritization of the Firm's claim in bankruptcy, expectations of the workout/restructuring of the loan and valuation of the borrower's equity or the loan collateral.

When a loan is charged down to the lower of its amortized cost or the estimated net realizable value of the underlying collateral, the determination of the fair value of the collateral depends on the type of collateral (e.g., securities, real estate). In cases where the collateral is in the form of liquid securities, the fair value is based on quoted market prices or broker quotes. For illiquid securities or other financial assets, the fair value of the collateral is generally estimated using a discounted cash flow model.

For residential real estate loans, collateral values are based upon external valuation sources. When it becomes likely that a borrower is either unable or unwilling to pay, the Firm utilizes a broker's price opinion, appraisal and/or an automated valuation model of the home based on an exterior-only valuation ("exterior opinions"), which is then updated at least every 12 months, or more frequently depending on various market factors. As soon as practicable after the Firm receives the property in satisfaction of a debt (e.g., by taking legal title or physical possession), the Firm generally obtains an appraisal based on an inspection that includes the interior of the home ("interior appraisals"). Exterior opinions and interior appraisals are discounted based upon the Firm's experience with actual liquidation values as compared with the estimated values provided by exterior opinions and interior appraisals, considering state-specific factors.

For commercial real estate loans, collateral values are generally based on appraisals from internal and external valuation sources. Collateral values are typically updated every six to twelve months, either by obtaining a new appraisal or by performing an internal analysis, in accordance with the Firm's policies. The Firm also considers both borrower- and market-specific factors, which may result in obtaining appraisal updates or broker price opinions at more frequent intervals.

Loans held-for-sale

Loans held-for-sale are measured at the lower of cost or fair value, with valuation changes recorded in noninterest revenue. For consumer loans, the valuation is performed on a portfolio basis. For wholesale loans, the valuation is performed on an individual loan basis.

Interest income on loans held-for-sale is accrued and recognized based on the contractual rate of interest.

Loan origination fees or costs and purchase price discounts or premiums are deferred in a contra loan account until the related loan is sold. The deferred fees or costs and discounts or premiums are an adjustment to the basis of the loan and therefore are included in the periodic determination of the lower of cost or fair value adjustments and/or the gain or loss recognized at the time of sale.

Because these loans are recognized at the lower of cost or fair value, the Firm's allowance for loan losses and charge-off policies do not apply to these loans. However, loans held-for-sale are subject to the Firm's nonaccrual policies.

Loans at fair value

Loans for which the fair value option has been elected are measured at fair value, with changes in fair value recorded in noninterest revenue.

Interest income on these loans is accrued and recognized based on the contractual rate of interest. Changes in fair value are recognized in noninterest revenue. Loan origination fees are recognized upfront in noninterest revenue. Loan origination costs are recognized in the associated expense category as incurred.

Because these loans are recognized at fair value, the Firm's allowance for loan losses and charge-off policies do not apply to these loans. However, loans at fair value are subject to the Firm's nonaccrual policies.

Refer to Note 3 for further information on the Firm's elections of fair value accounting under the fair value option. Refer to Note 2 and Note 3 for further information on loans carried at fair value and classified as trading assets.

Loan classification changes

Loans in the held-for-investment portfolio that management decides to sell are transferred to the held-for-sale portfolio at the lower of cost or fair value on the date of transfer. Credit-related losses are charged against the allowance for loan losses; non-credit related losses such as those due to changes in interest rates or foreign currency exchange rates are recognized in noninterest revenue.

In the event that management decides to retain a loan in the held-for-sale portfolio, the loan is transferred to the held-for-investment portfolio at amortized cost on the date of transfer. These loans are subsequently assessed for impairment based on the Firm's allowance methodology. Refer to Note 13 for a further discussion of the methodologies used in establishing the Firm's allowance for loan losses.

Loan modifications

The Firm seeks to modify certain loans in conjunction with its loss mitigation activities. Through the modification, JPMorganChase grants one or more concessions to a borrower who is experiencing financial difficulty in order to minimize the Firm's economic loss and avoid foreclosure or repossession of the collateral, and to ultimately maximize payments received by the Firm from the borrower. The concessions granted vary by program and by borrower-specific characteristics, and may include interest rate reductions, term extensions, other-than-insignificant payment delays or principal forgiveness.

Loans, except for credit card loans, reported as FDMs are generally placed on nonaccrual status, although in many cases such loans were already on nonaccrual status prior to modification. These loans may be returned to performing status (the accrual of interest is resumed) if the following criteria are met: (i) the borrower has performed under the modified terms for a minimum of six months and/or six payments, and (ii) the Firm has an expectation that repayment of the modified loan is reasonably assured based on, for example, the borrower's debt capacity and level of future earnings, collateral values, LTV ratios, and other current market considerations. In certain limited and well-defined circumstances in which the loan is current at the modification date, such loans are not placed on nonaccrual status at the time of modification.

The allowance for credit losses associated with FDMs is measured using the Firm's established allowance methodology, which considers the expected default rates for the modified loans. Refer to Note 13 for further discussion.

Foreclosed property

The Firm acquires property from borrowers through loan restructurings, workouts, and foreclosures. Property acquired may include real property (e.g., residential real estate, land, and buildings) and other commercial and personal property (e.g., automobiles, aircraft, railcars, and ships).

The Firm recognizes foreclosed property upon receiving assets in satisfaction of a loan (e.g., by taking legal title or physical possession). For loans collateralized by real property, the Firm generally recognizes the asset received at foreclosure sale or upon the execution of a deed in lieu of foreclosure transaction with the borrower. Foreclosed assets are reported in other assets on the Consolidated balance sheets and initially recognized at fair value less estimated costs to sell. Each quarter the fair value of the acquired property is reviewed and adjusted, if necessary, to the lower of cost or fair value. Subsequent adjustments to fair value are charged/credited to noninterest revenue. Operating expense, such as real estate taxes and maintenance, are charged to other expense.

Loan portfolio

The Firm's loan portfolio is divided into three portfolio segments, which are the same segments used by the Firm to determine the allowance for loan losses: Consumer, excluding credit card; Credit card; and Wholesale. Within each portfolio segment the Firm monitors and assesses the credit risk in the following classes of loans, based on the risk characteristics of each loan class.

Consumer, excluding credit card	Credit card	Wholesale[c][d]
• Residential real estate[a] • Auto and other[b]	• Credit card loans	• Secured by real estate • Commercial and industrial • Other[e]

(a) Includes scored mortgage and home equity loans held in CCB and AWM, and scored mortgage loans held in CIB.
(b) Includes scored auto, business banking and consumer unsecured loans as well as overdrafts, primarily in CCB.
(c) Includes loans held in CIB, AWM, Corporate, and risk-rated exposure held in CCB, for which the wholesale methodology is applied when determining the allowance for loan losses.
(d) The wholesale portfolio segment's classes align with loan classifications as defined by the bank regulatory agencies, based on the loan's collateral, purpose, and type of borrower.
(e) Includes loans to financial institutions, SPEs, personal investment companies and trusts, individuals and individual entities (predominantly Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB), states and political subdivisions, as well as loans to nonprofits. Refer to Note 14 for more information on SPEs.

The following tables summarize the Firm's loan balances by portfolio segment.

December 31, 2024 (in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total[a][b]
Retained	**$ 376,334**	**$ 232,860**	**$ 690,396**	**$ 1,299,590**
Held-for-sale	**945**	**—**	**6,103**	**7,048**
At fair value	**15,531**	**—**	**25,819**	**41,350**
Total	**$ 392,810**	**$ 232,860**	**$ 722,318**	**$ 1,347,988**

December 31, 2023 (in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total[a][b]
Retained	$ 397,275	$ 211,123	$ 672,472	$ 1,280,870
Held-for-sale	487	—	3,498	3,985
At fair value	12,331	—	26,520	38,851
Total	$ 410,093	$ 211,123	$ 702,490	$ 1,323,706

(a) Excludes $6.6 billion and $6.8 billion of accrued interest receivable at December 31, 2024 and 2023, respectively. The Firm wrote off accrued interest receivable of $84 million and $49 million for the years ended December 31, 2024 and 2023, respectively.
(b) Loans (other than those for which the fair value option has been elected) are presented net of unamortized discounts and premiums and net deferred loan fees or costs. These amounts were not material as of December 31, 2024 and 2023. For the discount associated with First Republic loans, refer to Note 34 on pages 319–321.

The following tables provide information about the carrying value of retained loans purchased, sold and reclassified to held-for-sale during the periods indicated. Loans that were reclassified to held-for-sale and sold in a subsequent period are excluded from the sales line of these tables.

	2024			
Year ended December 31, (in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total
Purchases	**$ 647** [b][c]	**$ —**	**$ 1,432**	**$ 2,079**
Sales	**10,440**	**—**	**45,147**	**55,587**
Retained loans reclassified to held-for-sale[a]	**1,656**	**—**	**749**	**2,405**

	2023			
Year ended December 31, (in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total
Purchases	$ 92,205 [b][c][d]	$ —	$ 60,300 [d]	$ 152,505
Sales	2,202	—	43,949	46,151
Retained loans reclassified to held-for-sale[a]	274	—	1,486	1,760

Year ended December 31, (in millions)	2022			
	Consumer, excluding credit card	Credit card	Wholesale	Total
Purchases	$ 1,625 [b][c]	$ —	$ 1,088	$ 2,713
Sales	2,884	—	41,934	44,818
Retained loans reclassified to held-for-sale[a]	229	—	1,055	1,284

(a) Reclassifications of loans to held-for-sale are non-cash transactions.
(b) Includes purchases of residential real estate loans, including the Firm's voluntary repurchases of certain delinquent loans from loan pools as permitted by Government National Mortgage Association ("Ginnie Mae") guidelines for the years ended December 31, 2024, 2023 and 2022. The Firm typically elects to repurchase these delinquent loans as it continues to service them and/or manage the foreclosure process in accordance with applicable requirements of Ginnie Mae, FHA, RHS, and/or VA.
(c) Excludes purchases of retained loans of $902 million, $5.1 billion and $12.4 billion for the years ended December 31, 2024, 2023 and 2022, respectively, which are predominantly sourced through the correspondent origination channel and underwritten in accordance with the Firm's standards.
(d) Includes loans acquired in the First Republic acquisition consisting of $91.9 billion in Consumer, excluding credit card and $59.2 billion in Wholesale. Refer to Note 34 for additional information.

Gains and losses on sales of loans

Net gains/(losses) on sales of loans and lending-related commitments (including adjustments to record loans and lending-related commitments held-for-sale at the lower of cost or fair value) recognized in noninterest revenue were $154 million for the year ended December 31, 2024, of which $113 million were related to loans. Net gains/(losses) on sales of loans and lending-related commitments were $56 million for the year ended December 31, 2023, of which $62 million were related to loans. Net gains/(losses) on sales of loans and lending-related commitments were $(186) million for the year ended December 31, 2022, of which $(48) million were related to loans. In addition, the sale of loans may also result in write downs, recoveries or changes in the allowance recognized in the provision for credit losses.

Consumer, excluding credit card loan portfolio

Consumer loans, excluding credit card loans, consist primarily of scored residential mortgages, home equity loans and lines of credit, auto and business banking loans, with a focus on serving the prime consumer credit market. These loans include home equity loans secured by junior liens, prime mortgage loans with an interest-only payment period, and certain payment-option loans that may result in negative amortization.

The following table provides information about retained consumer loans, excluding credit card, by class.

December 31, (in millions)	2024	2023
Residential real estate	**$ 309,513**	$ 326,409
Auto and other	**66,821**	70,866
Total retained loans	**$ 376,334**	$ 397,275

Delinquency rates are the primary credit quality indicator for consumer loans. Loans that are more than 30 days past due provide an early warning of borrowers who may be experiencing financial difficulties and/or who may be unable or unwilling to repay the loan. As the loan continues to age, it becomes more clear whether the borrower is likely to be unable or unwilling to pay. In the case of residential real estate loans, late-stage delinquencies (greater than 150 days past due) are a strong indicator of loans that will ultimately result in a foreclosure or similar liquidation transaction. In addition to delinquency rates, other credit quality indicators for consumer loans vary based on the class of loan, as follows:

- For residential real estate loans, the current estimated LTV ratio, or the combined LTV ratio in the case of junior lien loans, is an indicator of the potential loss severity in the event of default. Additionally, LTV or combined LTV ratios can provide insight into a borrower's continued willingness to pay, as the delinquency rate of high-LTV loans tends to be greater than that for loans where the borrower has equity in the collateral. The geographic distribution of the loan collateral also provides insight as to the credit quality of the portfolio, as factors such as the regional economy, home price changes and specific events such as natural disasters, will affect credit quality. The borrower's current or "refreshed" FICO score is a secondary credit quality indicator for certain loans, as FICO scores are an indication of the borrower's credit payment history. Thus, a loan to a borrower with a low FICO score (less than 660) is considered to be of higher risk than a loan to a borrower with a higher FICO score. Further, a loan to a borrower with a high LTV ratio and a low FICO score is at greater risk of default than a loan to a borrower that has both a high LTV ratio and a high FICO score.
- For scored auto and business banking loans, geographic distribution is an indicator of the credit performance of the portfolio. Similar to residential real estate loans, geographic distribution provides insights into the portfolio performance based on regional economic activity and events.

Residential real estate

Delinquency is the primary credit quality indicator for retained residential real estate loans. The following tables provide information on delinquency and gross charge-offs.

	December 31, 2024								
	Term loans by origination year[c]						Revolving loans		
(in millions, except ratios)	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans	Total
Loan delinquency[a]									
Current	**$ 12,301**	**$ 17,280**	**$ 61,337**	**$ 79,760**	**$ 52,289**	**$70,270**	**$ 6,974**	**$ 7,088**	**$307,299**
30–149 days past due	**13**	**54**	**139**	**110**	**59**	**747**	**53**	**204**	**1,379**
150 or more days past due	**—**	**11**	**71**	**68**	**49**	**501**	**8**	**127**	**835**
Total retained loans	**$ 12,314**	**$ 17,345**	**$ 61,547**	**$ 79,938**	**$ 52,397**	**$71,518**	**$ 7,035**	**$ 7,419**	**$309,513**
% of 30+ days past due to total retained loans[b]	**0.11 %**	**0.37 %**	**0.34 %**	**0.22 %**	**0.21 %**	**1.72 %**	**0.87 %**	**4.46 %**	**0.71 %**
Gross charge-offs	**$ —**	**$ —**	**$ 1**	**$ 1**	**$ —**	**$ 176**	**$ 21**	**$ 7**	**$ 206**

	December 31, 2023								
	Term loans by origination year[c]						Revolving loans		
(in millions, except ratios)	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans	Total
Loan delinquency[a]									
Current	$ 23,216	$ 64,366	$ 84,496	$ 55,546	$ 21,530	$59,563	$ 7,479	$ 8,151	$324,347
30–149 days past due	33	74	89	70	41	801	49	223	1,380
150 or more days past due	1	10	17	8	21	456	5	164	682
Total retained loans	$ 23,250	$ 64,450	$ 84,602	$ 55,624	$ 21,592	$60,820	$ 7,533	$ 8,538	$326,409
% of 30+ days past due to total retained loans[b]	0.15 %	0.13 %	0.13 %	0.14 %	0.29 %	2.04 %	0.72 %	4.53 %	0.63 %
Gross charge-offs	$ —	$ —	$ —	$ —	$ 4	$ 167	$ 26	$ 7	$ 204

(a) Individual delinquency classifications include mortgage loans insured by U.S. government agencies which were not material at December 31, 2024 and 2023.
(b) Excludes mortgage loans that are 30 or more days past due insured by U.S. government agencies which were not material at December 31, 2024 and 2023. These amounts have been excluded based upon the government guarantee.
(c) Purchased loans are included in the year in which they were originated.

Approximately 38% of the total revolving loans are senior lien loans; the remaining balance are junior lien loans. The lien position the Firm holds is considered in the Firm's allowance for credit losses. Revolving loans that have been converted to term loans have higher delinquency rates than those that are still within the revolving period. That is primarily because the fully-amortizing payment that is generally required for those products is higher than the minimum payment options available for revolving loans within the revolving period.

Nonaccrual loans and other credit quality indicators

The following table provides information on nonaccrual and other credit quality indicators for retained residential real estate loans.

(in millions, except weighted-average data)	**December 31, 2024**	December 31, 2023
Nonaccrual loans[(a)(b)(c)(d)]	**$ 2,984**	$ 3,466
Current estimated LTV ratios[(e)(f)(g)]		
Greater than 125% and refreshed FICO scores:		
Equal to or greater than 660	**$ 72**	$ 72
Less than 660	**3**	—
101% to 125% and refreshed FICO scores:		
Equal to or greater than 660	**161**	223
Less than 660	**5**	4
80% to 100% and refreshed FICO scores:		
Equal to or greater than 660	**4,962**	6,491
Less than 660	**73**	102
Less than 80% and refreshed FICO scores:		
Equal to or greater than 660	**294,797**	309,251
Less than 660	**8,534**	9,277
No FICO/LTV available[(h)]	**906**	989
Total retained loans	**$ 309,513**	$ 326,409
Weighted-average LTV ratio[(e)(i)]	**47 %**	49 %
Weighted-average FICO[(f)(i)]	**774**	770
Geographic region[(h)(j)]		
California	**$ 120,944**	$ 127,072
New York	**46,854**	48,815
Florida	**21,820**	22,778
Texas	**14,531**	15,506
Massachusetts	**13,511**	14,213
Colorado	**10,465**	10,800
Illinois	**9,835**	10,856
Washington	**9,372**	9,923
New Jersey	**7,554**	8,050
Connecticut	**6,854**	7,163
All other	**47,773**	51,233
Total retained loans	**$ 309,513**	$ 326,409

(a) Includes collateral-dependent residential real estate loans that are charged down to the fair value of the underlying collateral less costs to sell. The Firm reports, in accordance with regulatory guidance, residential real estate loans that have been discharged under Chapter 7 bankruptcy and not reaffirmed by the borrower ("Chapter 7 loans") as collateral-dependent nonaccrual loans, regardless of their delinquency status. At December 31, 2024, approximately 8% of Chapter 7 residential real estate loans were 30 days or more past due.
(b) Mortgage loans insured by U.S. government agencies excluded from nonaccrual loans were not material at December 31, 2024 and 2023.
(c) Generally, all consumer nonaccrual loans have an allowance. In accordance with regulatory guidance, certain nonaccrual loans that are considered collateral-dependent have been charged down to the lower of amortized cost or the fair value of their underlying collateral less costs to sell. If the value of the underlying collateral improves subsequent to charge down, the related allowance may be negative.
(d) Interest income on nonaccrual loans recognized on a cash basis was $160 million and $180 million for the years ended December 31, 2024 and 2023, respectively.
(e) Represents the aggregate unpaid principal balance of loans divided by the estimated current property value. Current property values are estimated, at a minimum, quarterly, based on home valuation models using nationally recognized home price index valuation estimates incorporating actual data to the extent available and forecasted data where actual data is not available. Current estimated combined LTV for junior lien home equity loans considers all available lien positions, as well as unused lines, related to the property.
(f) Refreshed FICO scores represent each borrower's most recent credit score, which is obtained by the Firm on at least a quarterly basis.
(g) Includes residential real estate loans, primarily held in LLCs in AWM that did not have a refreshed FICO score. These loans have been included in a FICO band based on management's estimation of the borrower's credit quality.
(h) Included U.S. government-guaranteed loans as of December 31, 2024 and 2023.
(i) Excludes loans with no FICO and/or LTV available.
(j) The geographic regions presented in the table are ordered based on the magnitude of the corresponding loan balances at December 31, 2024.

Loan modifications

The Firm grants certain modifications of residential real estate loans to borrowers experiencing financial difficulty. The Firm's proprietary modification programs as well as government programs, including U.S. GSE programs, that generally provide various modifications to borrowers experiencing financial difficulty including, but not limited to, interest rate reductions, term extensions, other-than-insignificant payment deferral and principal forgiveness that would otherwise have been required under the terms of the original agreement, are considered FDMs.

In addition, the Firm offers trial modifications of residential real estate loans, which generally include a three-month trial payment period during which the borrower makes monthly payments under the proposed modified loan terms. Loans in a trial payment period continue to age and accrue interest in accordance with the original contractual terms. At the completion of a trial period, the loan modification is considered permanent.

Financial effects of FDMs

For the year ended December 31, 2024, retained residential real estate FDMs were $206 million. The financial effects of the FDMs, which were predominantly in the form of term extensions and interest rate reductions, included extending the weighted-average life of the loans by 15 years, and reducing the weighted-average contractual interest rate from 7.53% to 5.44% for the year ended December 31, 2024.

For the year ended December 31, 2023, retained residential real estate FDMs were $136 million. The financial effects of the FDMs, which were predominantly in the form of term extensions and interest rate reductions, included extending the weighted-average life of the loans by 20 years, and reducing the weighted-average contractual interest rate from 7.21% to 4.44% for the year ended December 31, 2023.

As of December 31, 2024 and 2023, there were no additional commitments to lend to borrowers experiencing financial difficulty whose loans have been modified as FDMs.

For the years ended December 31, 2024 and 2023, loans subject to a trial modification, where the terms of the loans have not been permanently modified, and Chapter 7 loans were not material.

Payment status of FDMs

The following table provides information on the payment status of retained residential real estate FDMs during the years ended December 31, 2024 and 2023.

Year ended December 31, (in millions)	Amortized cost basis	
	2024	2023
Current	**$ 139**	$ 107
30-149 days past due	**47**	13
150 or more days past due	**20**	16
Total	**$ 206**	$ 136

Defaults of FDMs

Retained residential real estate FDMs that defaulted during the year ended December 31, 2024 and that were reported as FDMs in the twelve months prior to the default were $93 million. Retained residential real estate FDMs that defaulted during the year ended December 31, 2023 and that were reported as FDMs on or after January 1, 2023, the date that the Firm adopted the changes to the TDR accounting guidance were not material. Refer to Note 1 for further information.

Nature and extent of TDRs
For periods ending prior to January 1, 2023, modifications of residential real estate loans where the Firm granted concessions to borrowers who were experiencing financial difficulty were generally accounted for and reported as TDRs. Loans with short-term or other insignificant modifications that were not considered concessions were not TDRs. For the year ended December 31, 2022, new TDRs were $362 million, and there were no additional commitments to lend to borrowers whose residential real estate loans were modified in TDRs.

The Firm's proprietary modification programs as well as government programs, including U.S. GSE programs, generally provide various concessions to financially troubled borrowers including, but not limited to, interest rate reductions, term or payment extensions and delays of principal and/or interest payments that would otherwise have been required under the terms of the original agreement.

The following table provides information about how residential real estate loans were modified in TDRs during the period presented. This table excludes loans with short-term or other insignificant modifications that are not considered concessions.

Year ended December 31,	2022
Number of loans approved for a trial modification	3,902
Number of loans permanently modified	4,182
Concession granted:(a)	
Interest rate reduction	54 %
Term or payment extension	67
Principal and/or interest deferred	10
Principal forgiveness	1
Other(b)	37

(a) Represents concessions granted in permanent modifications as a percentage of the number of loans permanently modified. The sum of the percentages exceeds 100% because predominantly all of the modifications include more than one type of concession. Concessions offered on trial modifications are generally consistent with those granted on permanent modifications.
(b) Includes variable interest rate to fixed interest rate modifications and payment delays that meet the definition of a TDR.

Financial effects of TDRs and defaults
The following table provides information about the financial effects of the various concessions granted in modifications of residential real estate loans and about defaults of certain loans modified in TDRs for the period presented. The following table presents only the financial effects of permanent modifications and does not include temporary concessions offered through trial modifications. This table also excludes loans with short-term or other insignificant modifications that were not considered concessions.

Year ended December 31, (in millions, except weighted - average data)		2022
Weighted-average interest rate of loans with interest rate reductions – before TDR		4.75 %
Weighted-average interest rate of loans with interest rate reductions – after TDR		3.35
Weighted-average remaining contractual term (in years) of loans with term or payment extensions – before TDR		22
Weighted-average remaining contractual term (in years) of loans with term or payment extensions – after TDR		38
Charge-offs recognized upon permanent modification	$	1
Principal deferred		16
Principal forgiven		2
Balance of loans that defaulted within one year of permanent modification(a)	$	147

(a) Represents loans permanently modified in TDRs that experienced a payment default in the periods presented, and for which the payment default occurred within one year of the modification. The dollar amounts presented represent the balance of such loans at the end of the reporting period in which such loans defaulted.

Active and suspended foreclosure

At December 31, 2024 and 2023, the Firm had retained residential real estate loans, excluding those insured by U.S. government agencies, with a carrying value of $576 million and $566 million, respectively, that were not included in REO, but were in the process of active or suspended foreclosure.

Auto and other

Delinquency is the primary credit quality indicator for retained auto and other loans. The following tables provide information on delinquency and gross charge-offs.

	December 31, 2024								
	Term loans by origination year						Revolving loans		
(in millions, except ratios)	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans	Total
Loan delinquency									
Current	**$26,165**	**$15,953**	**$ 9,201**	**$ 7,014**	**$ 2,895**	**$ 624**	**$ 3,714**	**$ 148**	**$65,714**
30–119 days past due	**190**	**283**	**259**	**179**	**53**	**23**	**40**	**34**	**1,061**
120 or more days past due	**1**	**1**	**—**	**5**	**6**	**—**	**3**	**30**	**46**
Total retained loans	**$26,356**	**$16,237**	**$ 9,460**	**$ 7,198**	**$ 2,954**	**$ 647**	**$ 3,757**	**$ 212**	**$66,821**
% of 30+ days past due to total retained loans	**0.72 %**	**1.75 %**	**2.74**	**2.50 %**	**1.76 %**	**3.55 %**	**1.14 %**	**30.19 %**	**1.64 %**
Gross charge-offs	**$ 269**	**$ 348**	**$ 224**	**$ 126**	**$ 37**	**$ 82**	**$ 1**	**$ 6**	**$ 1,093**

	December 31, 2023								
	Term loans by origination year						Revolving loans		
(in millions, except ratios)	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans	Total
Loan delinquency									
Current	$30,328	$ 14,797	$ 12,825	$ 6,538	$ 1,777	$ 511	$ 2,984	$ 102	$69,862
30–119 days past due	276	279	231	78	43	17	19	24	967
120 or more days past due	1	1	7	8	—	—	3	17	37
Total retained loans	$30,605	$ 15,077	$ 13,063	$ 6,624	$ 1,820	$ 528	$ 3,006	$ 143	$70,866
% of 30+ days past due to total retained loans	0.91 %	1.86 %	1.75 %	1.15 %	2.36 %	3.22 %	0.73 %	28.67 %	1.39 %
Gross charge-offs	$ 333	$ 297	$ 161	$ 53	$ 35	$ 64	$ —	$ 4	$ 947

Nonaccrual loans and other credit quality indicators
The following table provides information on nonaccrual and geographic region as a credit quality indicator for retained auto and other consumer loans.

	Total Auto and other	
(in millions)	**December 31, 2024**	December 31, 2023
Nonaccrual loans[(a)(b)]	**$ 249**	$ 177
Geographic region[(c)]		
California	**$ 10,321**	$ 10,959
Texas	**7,772**	8,502
Florida	**5,428**	5,684
New York	**4,905**	4,938
Illinois	**2,890**	3,147
New Jersey	**2,468**	2,609
Pennsylvania	**2,012**	1,900
Georgia	**1,716**	1,912
Arizona	**1,643**	1,779
North Carolina	**1,597**	1,714
All other	**26,069**	27,722
Total retained loans	**$ 66,821**	$ 70,866

(a) Generally, all consumer nonaccrual loans have an allowance. In accordance with regulatory guidance, certain nonaccrual loans that are considered collateral-dependent have been charged down to the lower of amortized cost or the fair value of their underlying collateral less costs to sell. If the value of the underlying collateral improves subsequent to charge down, the related allowance may be negative.
(b) Interest income on nonaccrual loans recognized on a cash basis was not material for the years ended December 31, 2024 and 2023.
(c) The geographic regions presented in this table are ordered based on the magnitude of the corresponding loan balances at December 31, 2024.

Loan modifications

The Firm grants certain modifications of auto and other loans to borrowers experiencing financial difficulty.

For the years ended December 31, 2024 and 2023, retained auto and other FDMs were not material.

As of December 31, 2024 and 2023, there were no additional commitments to lend to borrowers modified as FDMs.

For periods ending prior to January 1, 2023, modifications of auto and other loans where the Firm granted concessions to borrowers who were experiencing financial difficulty were generally accounted for and reported as TDRs. Loans with short-term or other insignificant modifications that were not considered concessions were not TDRs. For the year ended December 31, 2022, auto and other TDRs were not material.

Credit card loan portfolio

The credit card portfolio segment includes credit card loans originated and purchased by the Firm. Delinquency rates are the primary credit quality indicator for credit card loans as they provide an early warning that borrowers may be experiencing difficulties (30 days past due); information on those borrowers that have been delinquent for a longer period of time (90 days past due) is also considered. In addition to delinquency rates, the geographic distribution of the loans provides insight as to the credit quality of the portfolio based on the regional economy.

While the borrower's credit score is another general indicator of credit quality, the Firm does not view credit scores as a primary indicator of credit quality because the borrower's credit score tends to be a lagging indicator. The distribution of such scores provides a general indicator of credit quality trends within the portfolio; however, the score does not capture all factors that would be predictive of future credit performance. Refreshed FICO score information, which is obtained at least quarterly, for a statistically significant random sample of the credit card portfolio is indicated in other credit quality indicators. FICO is considered to be the industry benchmark for credit scores.

The Firm generally originates new credit card accounts to prime consumer borrowers. However, certain cardholders' FICO scores may decrease over time, depending on the performance of the cardholder and changes in the credit score calculation.

The following tables provide information on delinquency and gross charge-offs.

	December 31, 2024		
(in millions, except ratios)	Within the revolving period	Converted to term loans	Total
Loan delinquency			
Current and less than 30 days past due and still accruing	$ **226,532**	$ **1,284**	$ **227,816**
30–89 days past due and still accruing	**2,291**	**109**	**2,400**
90 or more days past due and still accruing	**2,591**	**53**	**2,644**
Total retained loans	$ **231,414**	$ **1,446**	$ **232,860**
Loan delinquency ratios			
% of 30+ days past due to total retained loans	**2.11 %**	**11.20 %**	**2.17 %**
% of 90+ days past due to total retained loans	**1.12**	**3.67**	**1.14**
Gross charge-offs	$ **7,951**	$ **247**	$ **8,198**

	December 31, 2023		
(in millions, except ratios)	Within the revolving period	Converted to term loans	Total
Loan delinquency			
Current and less than 30 days past due and still accruing	$ 205,731	$ 882	$ 206,613
30–89 days past due and still accruing	2,217	84	2,301
90 or more days past due and still accruing	2,169	40	2,209
Total retained loans	$ 210,117	$ 1,006	$ 211,123
Loan delinquency ratios			
% of 30+ days past due to total retained loans	2.09 %	12.33 %	2.14 %
% of 90+ days past due to total retained loans	1.03	3.98	1.05
Gross charge-offs	$ 5,325	$ 166	$ 5,491

Other credit quality indicators

The following table provides information on other credit quality indicators for retained credit card loans.

(in millions, except ratios)	**December 31, 2024**	December 31, 2023
Geographic region[(a)]		
California	**$ 36,385**	$ 32,652
Texas	**24,423**	22,086
New York	**18,525**	16,915
Florida	**17,236**	15,103
Illinois	**12,442**	11,364
New Jersey	**9,644**	8,688
Ohio	**6,976**	6,424
Colorado	**6,962**	6,307
Pennsylvania	**6,558**	6,088
Arizona	**5,796**	5,209
All other	**87,913**	80,287
Total retained loans	**$ 232,860**	$ 211,123
Percentage of portfolio based on carrying value with estimated refreshed FICO scores		
Equal to or greater than 660	**85.5 %**	85.8 %
Less than 660	**14.3**	14.0
No FICO available	**0.2**	0.2

(a) The geographic regions presented in the table are ordered based on the magnitude of the corresponding loan balances at December 31, 2024.

Loan modifications

The Firm grants certain modifications of credit card loans to borrowers experiencing financial difficulty. These modifications may involve placing the customer's credit card account on a fixed payment plan, generally for 60 months, which typically includes reducing the interest rate on the credit card account. If the borrower does not make the contractual payments when due under the modified payment terms, the credit card loan continues to age and will be charged-off in accordance with the Firm's standard charge-off policy. In most cases, the Firm does not reinstate the borrower's line of credit.

Financial effects of FDMs

The following table provides information on retained credit card FDMs.

Year ended December 31, (in millions)	Loan modifications	
	2024	2023
Term extension and interest rate reduction[(a)(b)]		
Amortized cost basis	**$ 926**	$ 648
% of total modifications to total retained credit card loans	**0.40 %**	0.31 %
Financial effect of loan modifications	**Term extension with a reduction in the weighted average contractual interest rate from 23.64% to 3.20%**	Term extension with a reduction in the weighted average contractual interest rate from 23.19% to 3.64%

(a) Term extension includes credit card loans whose terms have been modified under long-term programs by placing the customer's credit card account on a fixed payment plan.

(b) The interest rates represent weighted average at the time of modification.

Payment status of FDMs

The following table provides information on the payment status of retained credit card FDMs during the years ended December 31, 2024 and 2023.

Year ended December 31, (in millions)	Amortized cost basis 2024	2023
Current and less than 30 days past due and still accruing	**$ 811**	$ 558
30-89 days past due and still accruing	**74**	59
90 or more days past due and still accruing	**41**	31
Total	**$ 926**	$ 648

Defaults of FDMs

Retained credit card FDMs that defaulted during the year ended December 31, 2024 and that were reported as FDMs in the twelve months prior to the default were not material. Retained credit card FDMs that defaulted during the year ended December 31, 2023 and that were reported as FDMs on or after January 1, 2023, the date that the Firm adopted the changes to the TDR accounting guidance were not material. Refer to Note 1 for further information.

For credit card loans modified as FDMs, payment default is deemed to have occurred when the borrower misses two consecutive contractual payments. Defaulted modified credit card loans remain in the modification program and continue to be charged off in accordance with the Firm's standard charge-off policy.

Financial effects of TDRs and defaults

For periods ending prior to January 1, 2023, modifications of credit card loans where the Firm granted concessions to borrowers who were experiencing financial difficulty were generally accounted for and reported as TDRs. The Firm granted concessions for most of the credit card loans under long-term programs. These concessions involved placing the customer's credit card account on a fixed payment plan, generally for 60 months, and typically included reducing the interest rate on the credit card account. Substantially all modifications under the Firm's long-term programs were considered to be TDRs. Loans with short-term or other insignificant modifications that were not considered concessions were not reported as TDRs.

The following table provides information about the financial effects of the concessions granted on credit card loans modified in TDRs and defaults for the periods presented. For the period disclosed, new enrollments were less than 1% of total retained credit card loans.

Year ended December 31, (in millions, except weighted-average data)	2022
Balance of new TDRs[(a)]	$ 418
Weighted-average interest rate of loans – before TDR	19.86 %
Weighted-average interest rate of loans – after TDR	4.13
Balance of loans that defaulted within one year of modification[(b)]	$ 34

(a) Represents the outstanding balance prior to modification.
(b) Represents loans modified in TDRs that experienced a payment default in the periods presented, and for which the payment default occurred within one year of the modification. The amounts presented represent the balance of such loans as of the end of the quarter in which they defaulted.

For credit card loans modified in TDRs, payment default was deemed to have occurred when the borrower missed two consecutive contractual payments. Defaulted modified credit card loans remained in the modification program and continued to be charged of in accordance with the Firm's standard charge-off policy.

Wholesale loan portfolio

Wholesale loans include loans made to a variety of clients, ranging from large corporate and institutional clients to small businesses and high-net-worth individuals.

The primary credit quality indicator for wholesale loans is the internal risk rating assigned to each loan. Risk ratings are used to identify the credit quality of loans and differentiate risk within the portfolio. Risk ratings on loans consider the PD and the LGD. The PD is the likelihood that a loan will default. The LGD is the estimated loss on the loan that would be realized upon the default of the borrower and takes into consideration collateral and structural support for each credit facility.

Management considers several factors to determine an appropriate internal risk rating, including the obligor's debt capacity and financial flexibility, the level of the obligor's earnings, the amount and sources for repayment, the level and nature of contingencies, management strength, and the industry and geography in which the obligor operates. The Firm's internal risk ratings generally align with the qualitative characteristics (e.g., borrower capacity to meet financial commitments and vulnerability to changes in the economic environment) defined by S&P and Moody's, however the quantitative characteristics (e.g., PD and LGD) may differ as they reflect internal historical experiences and assumptions. The Firm generally considers internal ratings with qualitative characteristics equivalent to BBB-/Baa3 or higher as investment grade, and these ratings have a lower PD and/or lower LGD than non-investment grade ratings.

Noninvestment-grade ratings are further classified as noncriticized and criticized, and the criticized portion is further subdivided into performing and nonaccrual loans, representing management's assessment of the collectibility of principal and interest. Criticized loans have a higher PD than noncriticized loans. The Firm's definition of criticized aligns with the U.S. banking regulatory definition of criticized exposures, which consist of special mention, substandard and doubtful categories. Refer to Note 1 for additional information.

Risk ratings are reviewed on a regular and ongoing basis by Credit Risk Management and are adjusted as necessary for updated information affecting the obligor's ability to fulfill its obligations.

As noted above, the risk rating of a loan considers the industry in which the obligor conducts its operations. As part of the overall credit risk management framework, the Firm focuses on the management and diversification of its industry and client exposures, with particular attention paid to industries with an actual or potential credit concern. Refer to Note 4 for further detail on industry concentrations.

Internal risk rating is the primary credit quality indicator for retained wholesale loans. The following tables provide information on internal risk rating and gross charge-offs for the year ended December 31, 2024.

December 31, (in millions, except ratios)	Secured by real estate		Commercial and industrial		Other[(a)]		Total retained loans	
	2024	2023	**2024**	2023	**2024**	2023	**2024**	2023
Loans by risk ratings								
Investment-grade	**$ 114,280**	$ 120,405	**$ 70,862**	$ 72,624	**$286,528**	$ 265,809	**$ 471,670**	$ 458,838
Noninvestment-grade:								
Noncriticized	**37,422**	34,241	**83,191**	80,637	**72,743**	75,178	**193,356**	190,056
Criticized performing	**9,291**	7,291	**10,977**	12,684	**1,160**	1,257	**21,428**	21,232
Criticized nonaccrual	**1,439**	401	**1,760**	1,221	**743**	724	**3,942**	2,346
Total noninvestment-grade	**48,152**	41,933	**95,928**	94,542	**74,646**	77,159	**218,726**	213,634
Total retained loans	**$ 162,432**	$ 162,338	**$ 166,790**	$ 167,166	**$ 361,174**	$ 342,968	**$690,396**	$ 672,472
% of investment-grade to total retained loans	**70.36 %**	74.17 %	**42.49 %**	43.44 %	**79.33 %**	77.50 %	**68.32 %**	68.23 %
% of total criticized to total retained loans	**6.61**	4.74	**7.64**	8.32	**0.53**	0.58	**3.67**	3.51
% of criticized nonaccrual to total retained loans	**0.89**	0.25	**1.06**	0.73	**0.21**	0.21	**0.57**	0.35

(a) Includes loans to financial institutions, SPEs, personal investment companies and trusts, individuals and individual entities (predominantly Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB), states and political subdivisions, as well as loans to nonprofits. As of December 31, 2024, predominantly consisted of $114.8 billion to individuals and individual entities, $94.0 billion to financial institutions, and $92.5 billion to SPEs. Refer to Note 14 for more information on SPEs.

	Secured by real estate **December 31, 2024**								
	Term loans by origination year						Revolving loans		
(in millions)	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans	Total
Loans by risk ratings									
Investment-grade	**$ 10,002**	**$ 9,834**	**$ 25,284**	**$ 22,796**	**$ 15,548**	**$ 29,488**	**$ 1,328**	**$ —**	**$ 114,280**
Noninvestment-grade	**4,238**	**5,366**	**14,717**	**8,567**	**3,462**	**10,392**	**1,317**	**93**	**48,152**
Total retained loans	**$ 14,240**	**$ 15,200**	**$ 40,001**	**$ 31,363**	**$ 19,010**	**$ 39,880**	**$ 2,645**	**$ 93**	**$ 162,432**
Gross charge-offs	**$ 72**	**$ 18**	**$ 43**	**$ 2**	**$ 109**	**$ 80**	**$ —**	**$ —**	**$ 324**

	Secured by real estate December 31, 2023								
	Term loans by origination year						Revolving loans		
(in millions)	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans	Total
Loans by risk ratings									
Investment-grade	$ 10,687	$ 28,874	$ 25,784	$ 16,820	$ 15,677	$ 21,108	$ 1,455	$ —	$ 120,405
Noninvestment-grade	4,477	12,579	7,839	3,840	3,987	7,918	1,291	2	41,933
Total retained loans	$ 15,164	$ 41,453	$ 33,623	$ 20,660	$ 19,664	$ 29,026	$ 2,746	$ 2	$ 162,338
Gross charge-offs	$ 20	$ 48	$ 22	$ —	$ 23	$ 78	$ —	$ 1	$ 192

	Commercial and industrial								
	December 31, 2024								
	Term loans by origination year						Revolving loans		
(in millions)	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans	Total
Loans by risk ratings									
Investment-grade	**$ 11,564**	**$ 6,285**	**$ 6,588**	**$ 3,119**	**$ 1,067**	**$ 1,139**	**$ 41,099**	**$ 1**	**$ 70,862**
Noninvestment-grade	**21,251**	**11,350**	**10,942**	**5,322**	**783**	**975**	**45,181**	**124**	**95,928**
Total retained loans	**$ 32,815**	**$ 17,635**	**$ 17,530**	**$ 8,441**	**$ 1,850**	**$ 2,114**	**$ 86,280**	**$ 125**	**$ 166,790**
Gross charge-offs	**$ 25**	**$ 22**	**$ 128**	**$ 24**	**$ 1**	**$ 50**	**$ 270**	**$ 5**	**$ 525**

	Commercial and industrial								
	December 31, 2023								
	Term loans by origination year						Revolving loans		
(in millions)	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans	Total
Loans by risk ratings									
Investment-grade	$ 14,875	$ 10,642	$ 4,276	$ 2,291	$ 1,030	$ 1,115	$ 38,394	$ 1	$ 72,624
Noninvestment-grade	18,890	16,444	9,299	1,989	1,144	1,006	45,696	74	94,542
Total retained loans	$ 33,765	$ 27,086	$ 13,575	$ 4,280	$ 2,174	$ 2,121	$ 84,090	$ 75	$ 167,166
Gross charge-offs	$ 25	$ 8	$ 110	$ 55	$ 2	$ 12	$ 259	$ 8	$ 479

	Other(a)								
	December 31, 2024								
	Term loans by origination year						Revolving loans		
(in millions)	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans	Total
Loans by risk ratings									
Investment-grade	**$ 30,484**	**$ 17,039**	**$ 13,272**	**$ 6,288**	**$ 8,632**	**$ 7,382**	**$ 201,949**	**$ 1,482**	**$ 286,528**
Noninvestment-grade	**11,784**	**7,248**	**5,918**	**3,296**	**1,366**	**1,886**	**42,954**	**194**	**74,646**
Total retained loans	**$ 42,268**	**$ 24,287**	**$ 19,190**	**$ 9,584**	**$ 9,998**	**$ 9,268**	**$ 244,903**	**$ 1,676**	**$ 361,174**
Gross charge-offs	**$ —**	**$ 38**	**$ 3**	**$ 36**	**$ 40**	**$ 50**	**$ 6**	**$ —**	**$ 173**

	Other(a)								
	December 31, 2023								
	Term loans by origination year						Revolving loans		
(in millions)	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans	Total
Loans by risk ratings									
Investment-grade	$ 38,338	$ 18,034	$ 10,033	$ 10,099	$ 3,721	$ 6,662	$ 176,728	$ 2,194	$ 265,809
Noninvestment-grade	14,054	8,092	6,169	2,172	811	2,001	43,801	59	77,159
Total retained loans	$ 52,392	$ 26,126	$ 16,202	$ 12,271	$ 4,532	$ 8,663	$ 220,529	$ 2,253	$ 342,968
Gross charge-offs	$ 5	$ 298	$ 8	$ 8	$ —	$ 8	$ 13	$ —	$ 340

(a) Includes loans to financial institutions, SPEs, personal investment companies and trusts, individuals and individual entities (predominantly Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB), states and political subdivisions, as well as loans to nonprofits. Refer to Note 14 for more information on SPEs.

The following table presents additional information on retained loans secured by real estate, which consists of loans secured wholly or substantially by a lien or liens on real property at origination. Multifamily lending includes financing for acquisition, leasing and construction of apartment buildings. Other commercial lending largely includes financing for acquisition, leasing and construction, largely for office, retail and industrial real estate. Included in secured by real estate loans were $12.2 billion and $10.2 billion as of December 31, 2024 and 2023, respectively, of construction and development loans made to finance land development and on-site construction of commercial, industrial, residential, or farm buildings.

December 31, (in millions, except ratios)	Multifamily		Other Commercial		Total retained loans secured by real estate	
	2024	2023	**2024**	2023	**2024**	2023
Retained loans secured by real estate	**$ 101,114**	$100,725	**$ 61,318**	$ 61,613	**$162,432**	$162,338
Criticized	**4,700**	3,596	**6,030**	4,096	**10,730**	7,692
% of criticized to total retained loans secured by real estate	**4.65 %**	3.57 %	**9.83 %**	6.65 %	**6.61 %**	4.74 %
Criticized nonaccrual	**$ 337**	$ 76	**$ 1,102**	$ 325	**$ 1,439**	$ 401
% of criticized nonaccrual loans to total retained loans secured by real estate	**0.33 %**	0.08 %	**1.80 %**	0.53 %	**0.89 %**	0.25 %

Geographic distribution and delinquency

The following table provides information on the geographic distribution and delinquency for retained wholesale loans.

December 31, (in millions)	Secured by real estate		Commercial and industrial		Other		Total retained loans	
	2024	2023	**2024**	2023	**2024**	2023	**2024**	2023
Loans by geographic distribution[(a)]								
Total U.S.	**$ 159,209**	$ 159,499	**$ 127,626**	$ 127,638	**$278,077**	$ 262,499	**$564,912**	$ 549,636
Total non-U.S.	**3,223**	2,839	**39,164**	39,528	**83,097**	80,469	**125,484**	122,836
Total retained loans	**$ 162,432**	$ 162,338	**$166,790**	$ 167,166	**$ 361,174**	$ 342,968	**$690,396**	$ 672,472
Loan delinquency								
Current and less than 30 days past due and still accruing	**$ 159,949**	$ 161,314	**$ 164,104**	$ 164,899	**$ 359,191**	$ 341,128	**$683,244**	$ 667,341
30–89 days past due and still accruing	**918**	473	**868**	884	**1,152**	1,090	**2,938**	2,447
90 or more days past due and still accruing[(b)]	**126**	150	**58**	162	**88**	26	**272**	338
Criticized nonaccrual	**1,439**	401	**1,760**	1,221	**743**	724	**3,942**	2,346
Total retained loans	**$ 162,432**	$ 162,338	**$166,790**	$ 167,166	**$ 361,174**	$ 342,968	**$690,396**	$ 672,472

(a) The U.S. and non-U.S. distribution is determined based predominantly on the domicile of the borrower.
(b) Represents loans that are considered well-collateralized and therefore still accruing interest.

Nonaccrual loans

The following table provides information on retained wholesale nonaccrual loans.

December 31, (in millions)	Secured by real estate		Commercial and industrial		Other		Total retained loans	
	2024	2023	**2024**	2023	**2024**	2023	**2024**	2023
Nonaccrual loans								
With an allowance	**$ 366**	$ 129	**$ 1,362**	$ 776	**$ 555**	$ 492	**$ 2,283**	$ 1,397
Without an allowance[(a)]	**1,073**	272	**398**	445	**188**	232	**1,659**	949
Total nonaccrual loans[(b)]	**$ 1,439**	$ 401	**$ 1,760**	$ 1,221	**$ 743**	$ 724	**$ 3,942**	$ 2,346

(a) When the discounted cash flows or collateral value equals or exceeds the amortized cost of the loan, the loan does not require an allowance. This typically occurs when the loans have been partially charged off and/or there have been interest payments received and applied to the loan balance.
(b) Interest income on nonaccrual loans recognized on a cash basis was $51 million and $19 million for the years ended December 31, 2024 and 2023, respectively.

Loan modifications

The Firm grants certain modifications of wholesale loans to borrowers experiencing financial difficulty.

Financial effects of FDMs

The following tables provide information on retained wholesale loan modifications considered FDMs.

	Secured by real estate		
	Year ended December 31, 2024		
(in millions, except ratios)	Amortized cost basis	% of loan modifications to total retained Secured by real estate loans	Financial effect of loan modifications
Single modifications			
Term extension	**$ 271**	**0.17 %**	Extended loans by a weighted-average of 21 months
Other-than-insignificant payment deferral	**37**	**0.02**	Provided payment deferrals with delayed amounts re-amortized over the remaining tenor
Multiple modifications			
Other-than-insignificant payment deferral and interest rate reduction	**46**	**0.03**	Provided payment deferrals with delayed amounts recaptured at maturity and reduced weighted-average contractual interest by 162 bps
Total	**$ 354**		

	Secured by real estate		
	Year ended December 31, 2023		
(in millions, except ratios)	Amortized cost basis	% of loan modifications to total retained Secured by real estate loans	Financial effect of loan modifications
Single modifications			
Term extension	$ 149	0.09 %	Extended loans by a weighted-average of 14 months
Other-than-insignificant payment deferral	3	—	Provided payment deferrals with delayed amounts primarily re-amortized over the remaining life of the loan
Multiple modifications			
Other-than-insignificant payment deferral and interest rate reduction	5	—	Provided payment deferrals with delayed amounts primarily recaptured at maturity and reduced weighted-average contractual interest by 184 bps
Other[(a)]	3	—	NM
Total	$ 160		

(a) Includes a loan with multiple modifications.

(in millions, except ratios)	Commercial and industrial **Year ended December 31, 2024** Amortized cost basis		% of loan modifications to total retained Commercial and industrial loans	Financial effect of loan modifications
Single modifications				
Term extension	**$**	**1,180**	**0.71 %**	Extended loans by a weighted-average of 20 months
Other-than-insignificant payment deferral		**464**	**0.28**	Provided payment deferrals with delayed amounts primarily re-amortized over the remaining tenor
Multiple modifications				
Other-than-insignificant payment deferral and term extension		**175**	**0.10**	Provided payment deferrals with delayed amounts recaptured at maturity and extended loans by a weighted-average of 18 months
Interest rate reduction and term extension		**51**	**0.03**	Reduced weighted-average contractual interest by 434 bps and extended loans by a weighted-average of 36 months
Other[(a)]		**30**	**0.02**	NM
Total	**$**	**1,900**		

(a) Includes loans with single and multiple modifications.

(in millions, except ratios)	Commercial and industrial Year ended December 31, 2023 Amortized cost basis		% of loan modifications to total retained Commercial and industrial loans	Financial effect of loan modifications
Single modifications				
Term extension	$	916	0.55 %	Extended loans by a weighted-average of 17 months
Other-than-insignificant payment deferral		402	0.24	Provided payment deferrals with delayed amounts primarily recaptured at the end of the deferral period
Multiple modifications				
Other-than-insignificant payment deferral and term extension		35	0.02	Provided payment deferrals with delayed amounts primarily re-amortized over the remaining life of the loan and extended loans by a weighted-average of 7 months
Interest rate reduction and term extension		1	—	Reduced weighted average contractual interest rate over the life of the loan as a result of converting from variable to fixed rate and extended loans by a weighted-average of 16 months
Other[(a)]		9	—	NM
Total	$	1,363		

(a) Include loans with multiple modifications.

(in millions, except ratios)	Other Year ended December 31, 2024 Amortized cost basis	% of loan modifications to total retained Other loans	Financial effect of loan modification
Single modifications			
Term extension	**$ 268**	0.07 %	Extended loans by a weighted-average of 28 months
Multiple modifications			
Other-than-insignificant payment deferral and term extension	**2**	—	Provided payment deferrals with delayed amounts recaptured at maturity and extended loans by a weighted-average of 6 months
Other[(a)]	**5**	—	NM
Total	**$ 275**		

(a) Includes loans with a single modification.

(in millions, except ratios)	Other Year ended December 31, 2023 Amortized cost basis	% of loan modifications to total retained Other loans	Financial effect of loan modifications
Single modifications			
Term extension	$ 355	0.10 %	Extended loans by a weighted-average of 23 months
Multiple modifications			
Other-than-insignificant payment deferral and term extension	245	0.07	Provided payment deferrals with delayed amounts primarily recaptured at the end of the deferral period and extended loans by a weighted-average of 137 months
Other[(a)]	$ 9	—	NM
Total	$ 609		

(a) Includes a loan with a single modification.

Payment status of FDMs

The following table provides information on the payment status of retained wholesale FDMs during the year ended December 31, 2024 and 2023.

(in millions)	Amortized cost basis					
	Secured by real estate	Commercial and industrial	Other	Secured by real estate	Commercial and industrial	Other
	Year ended December 31, 2024			Year ended December 31, 2023		
Current and less than 30 days past due and still accruing	**$ 264**	**$ 1,215**	**$ 240**	$ 118	$ 947	$ 400
30-89 days past due and still accruing	**3**	**13**	**9**	2	42	—
Criticized nonaccrual	**87**	**672**	**26**	40	374	209
Total	**$ 354**	**$ 1,900**	**$ 275**	$ 160	$ 1,363	$ 609

Defaults of FDMs

The following table provides information on retained wholesale FDMs that defaulted during the year ended December 31, 2024 that were reported as FDMs in the twelve months prior to the default, and FDMs that defaulted during the year ended December 31, 2023 that were reported as FDMs on or after January 1, 2023, the date that the Firm adopted the changes to the TDR accounting guidance.

(in millions)	Amortized cost basis					
	Secured by real estate	Commercial and industrial	Other	Secured by real estate	Commercial and industrial	Other
	Year ended December 31, 2024			Year ended December 31, 2023		
Term extension	**$ 3**	**$ 92**	**$ 22**	$ 1	$ 49	$ 31
Other-than-insignificant payment deferral	**—**	**118**	**—**	2	—	—
Interest rate reduction and term extension	**—**	**—**	**—**	3	1	—
Total[(a)]	**$ 3**	**$ 210**	**$ 22**	$ 6	$ 50	$ 31

(a) Represents FDMs that were 30 days or more past due.

As of December 31, 2024 and 2023, additional unfunded commitments on modified loans to borrowers experiencing financial difficulty were $1.8 billion at each period in Commercial and industrial, and $69 million and $4 million, respectively, in Other loan class. Additional commitments on modified loans to borrowers experiencing financial difficulty whose loans have been modified as FDMs in Secured by real estate were not material at each period.

Nature and extent of TDRs

Prior to January 1, 2023, certain loan modifications were considered TDRs. These loan modifications provided various concessions to borrower who were experiencing financial difficulty. Loans with short-term or other insignificant modifications that were not considered concessions were not TDRs nor were loans for which the Firm elected to suspend TDR accounting guidance under the option provided by the CARES Act.

For the year ended December 31, 2022, new TDRs were $801 million. New TDRs for the year ended December 31, 2022 reflected extended maturity dates and covenant waivers primarily in the Commercial and Industrial loan class. For the year ended December 31, 2022, the impact of these modifications resulting in new TDRs was not material to the Firm.

As a result of the elimination of the requirement to assess whether a modification is reasonably expected or involves a concession, the population of loans considered FDMs is greater than the population previously considered TDRs.

Note 13 – Allowance for credit losses

The Firm's allowance for credit losses represents management's estimate of expected credit losses over the remaining expected life of the Firm's financial assets measured at amortized cost and certain off-balance sheet lending-related commitments. The allowance for credit losses generally comprises:

- the allowance for loan losses, which covers the Firm's retained loan portfolios (scored and risk-rated),
- the allowance for lending-related commitments, which is presented on the Consolidated balance sheets in accounts payable and other liabilities, and
- the allowance for credit losses on investment securities, which is reflected in investment securities on the Consolidated balance sheets.

The income statement effect of all changes in the allowance for credit losses is recognized in the provision for credit losses.

Determining the appropriateness of the allowance for credit losses is complex and requires significant judgment by management about the effect of matters that are inherently uncertain. At least quarterly, the allowance for credit losses is reviewed by the CRO, the CFO and the Controller of the Firm. Subsequent evaluations of credit exposures, considering the macroeconomic conditions, forecasts and other factors then prevailing, may result in significant changes in the allowance for credit losses in future periods.

The Firm's policies used to determine its allowance for loan losses and its allowance for lending-related commitments are described in the following paragraphs. Refer to Note 10 for a description of the policies used to determine the allowance for credit losses on investment securities.

Methodology for allowances for loan losses and lending-related commitments

The allowance for loan losses and allowance for lending-related commitments represents expected credit losses over the remaining expected life of retained loans and lending-related commitments that are not unconditionally cancellable. The Firm does not record an allowance for future draws on unconditionally cancellable lending-related commitments (e.g., credit cards). Expected losses related to accrued interest on credit card loans are considered in the Firm's allowance for loan losses. However, the Firm does not record an allowance on other accrued interest receivables, due to its policy to write these receivables off no later than 90 days past due by reversing interest income.

The expected life of each instrument is determined by considering its contractual term, expected prepayments, cancellation features, and certain extension and call options. The expected life of funded credit card loans is generally estimated by considering expected future payments on the credit card account, and determining how much of those amounts should be allocated to repayments of the funded loan balance (as of the balance sheet date) versus other account activity. This allocation is made using an approach that incorporates the payment application requirements of the Credit Card Accountability Responsibility and Disclosure Act of 2009, generally paying down the highest interest rate balances first.

The estimate of expected credit losses includes expected recoveries of amounts previously charged off or expected to be charged off, even if such recoveries result in a negative allowance.

Collective and Individual Assessments

When calculating the allowance for loan losses and the allowance for lending-related commitments, the Firm assesses whether exposures share similar risk characteristics. If similar risk characteristics exist, the Firm estimates expected credit losses collectively, considering the risk associated with a particular pool and the probability that the exposures within the pool will deteriorate or default. The assessment of risk characteristics is subject to significant management judgment. Emphasizing one characteristic over another or considering additional characteristics could affect the allowance.

- Relevant risk characteristics for the consumer portfolio include product type, delinquency status, current FICO scores, geographic distribution, and, for collateralized loans, current LTV ratios.
- Relevant risk characteristics for the wholesale portfolio include risk rating, delinquency status, tenor, level and type of collateral, LOB, geography, industry, credit enhancement, product type, facility purpose, and payment terms.

The majority of the Firm's credit exposures share risk characteristics with other similar exposures, and as a result are collectively assessed for impairment ("portfolio-based component"). The portfolio-based component covers consumer loans, performing risk-rated loans and certain lending-related commitments.

If an exposure does not share risk characteristics with other exposures, the Firm generally estimates expected credit losses on an individual basis, considering expected repayment and conditions impacting that individual exposure ("asset-specific component"). The asset-specific component covers collateral-dependent loans and risk-rated loans that have been placed on nonaccrual status.

Portfolio-based component

The portfolio-based component begins with a quantitative calculation that considers the likelihood of the borrower changing delinquency status or moving from one risk rating to another. The quantitative calculation covers expected credit losses over an instrument's expected life and is estimated by applying credit loss factors to the Firm's estimated exposure at default. The credit loss factors incorporate the probability of borrower default as well as loss severity in the event of default. They are derived using a weighted average of five internally developed macroeconomic scenarios over an eight-quarter forecast period, followed by a single year straight-line interpolation to revert to long run historical information for periods beyond the eight-quarter forecast period. The five macroeconomic scenarios consist of a central, relative adverse, extreme adverse, relative upside and extreme upside scenario, and are updated by the Firm's central forecasting team. The scenarios take into consideration the Firm's macroeconomic outlook, internal perspectives from subject matter experts across the Firm, and market consensus and involve a governed process that incorporates feedback from senior management across LOBs, Corporate Finance and Risk Management.

The quantitative calculation is adjusted to take into consideration model imprecision, emerging risk assessments, trends and other subjective factors that are not yet reflected in the calculation. These adjustments are accomplished in part by analyzing the historical loss experience, including during stressed periods, for each major product or model. Management applies judgment in making this adjustment, including taking into account uncertainties associated with the economic and political conditions, quality of underwriting standards, borrower behavior, credit concentrations or deterioration within an industry, product or portfolio, as well as other relevant internal and external factors affecting the credit quality of the portfolio. In certain instances, the interrelationships between these factors create further uncertainties.

The application of different inputs into the quantitative calculation, and the assumptions used by management to adjust the quantitative calculation, are subject to significant management judgment, and emphasizing one input or assumption over another, or considering other inputs or assumptions, could affect the estimate of the allowance for loan losses and the allowance for lending-related commitments.

Asset-specific component
To determine the asset-specific component of the allowance, collateral-dependent loans (including those loans for which foreclosure is probable) and nonaccrual risk-rated loans in the wholesale portfolio segment are generally evaluated individually.

For collateral-dependent loans, the fair value of collateral less estimated costs to sell, as applicable, is used to determine the charge-off amount for declines in value (to reduce the amortized cost of the loan to the fair value of collateral) or the amount of negative allowance that should be recognized (for recoveries of prior charge-offs associated with improvements in the fair value of the collateral).

For non-collateral dependent loans, the Firm generally measures the asset-specific allowance as the difference between the amortized cost of the loan and the present value of the cash flows expected to be collected, discounted at the loan's effective interest rate. Subsequent changes in impairment are generally recognized as an adjustment to the allowance for loan losses. The asset-specific component of the allowance for non-collateral dependent loans incorporates the effect of the modification on the loan's expected cash flows including changes in interest rates, principal forgiveness, and other concessions, as well as management's expectation of the borrower's ability to repay under the modified terms.

Estimating the timing and amounts of future cash flows is highly judgmental as these cash flow projections rely upon estimates such as loss severities, asset valuations, the amounts and timing of interest or principal payments (including any expected prepayments) or other factors that are reflective of current and expected market conditions. These estimates are, in turn, dependent on factors such as the duration of current overall economic conditions, industry, portfolio, or borrower-specific factors, the expected outcome of insolvency proceedings as well as, in certain circumstances, other economic factors. All of these estimates and assumptions require significant management judgment and certain assumptions are highly subjective.

Other financial assets
In addition to loans and investment securities, the Firm holds other financial assets that are measured at amortized cost on the Consolidated balance sheets, including credit exposures arising from lending activities subject to collateral maintenance requirements. Management estimates the allowance for other financial assets using various techniques considering historical losses and current economic conditions.

Credit risk arising from lending activities subject to collateral maintenance requirements is generally mitigated by factors such as the short-term nature of the activity, the fair value of collateral held and the Firm's right to call for, and the borrower's obligation to provide additional margin when the fair value of the collateral declines. Because of these mitigating factors, these exposures generally do not require an allowance for credit losses. However, management may also consider other factors such as the borrower's ongoing ability to provide collateral to satisfy margin requirements, or whether collateral is significantly concentrated in an individual issuer or in securities with similar risk characteristics. If in management's judgment, an allowance for credit losses for these exposures is required, the Firm estimates expected credit losses based on the value of the collateral and probability of borrower default.

Allowance for credit losses and related information

The table below summarizes information about the allowances for credit losses and includes a breakdown of loans and lending-related commitments by impairment methodology. Refer to Note 10 for further information on the allowance for credit losses on investment securities.

(Table continued on next page)

	2024			
Year ended December 31, (in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total
Allowance for loan losses				
Beginning balance at January 1,	$ 1,856	$ 12,450	$ 8,114	$ 22,420
Cumulative effect of a change in accounting principle[(a)]	NA	NA	NA	NA
Gross charge-offs	1,299	8,198	1,022	10,519
Gross recoveries collected	(625)	(1,056)	(200)	(1,881)
Net charge-offs	674	7,142	822	8,638
Provision for loan losses	624	9,292	578	10,494
Other	1	—	68	69
Ending balance at December 31,	$ 1,807	$ 14,600	$ 7,938	$ 24,345
Allowance for lending-related commitments				
Beginning balance at January 1,	$ 75	$ —	$ 1,899	$ 1,974
Provision for lending-related commitments	7	—	121	128
Other	—	—	(1)	(1)
Ending balance at December 31,	$ 82	$ —	$ 2,019	$ 2,101
Total allowance for investment securities	NA	NA	NA	$ 152
Total allowance for credit losses[(b)]	$ 1,889	$ 14,600	$ 9,957	$ 26,598
Allowance for loan losses by impairment methodology				
Asset-specific[(c)]	$ (728)	$ —	$ 526	$ (202)
Portfolio-based	2,535	14,600	7,412	24,547
Total allowance for loan losses	$ 1,807	$ 14,600	$ 7,938	$ 24,345
Loans by impairment methodology				
Asset-specific[(c)]	$ 2,805	$ —	$ 3,912	$ 6,717
Portfolio-based	373,529	232,860	686,484	1,292,873
Total retained loans	$ 376,334	$ 232,860	$ 690,396	$ 1,299,590
Collateral-dependent loans				
Net charge-offs	$ 1	$ —	$ 324	$ 325
Loans measured at fair value of collateral less cost to sell	2,696	—	1,834	4,530
Allowance for lending-related commitments by impairment methodology				
Asset-specific	$ —	$ —	$ 109	$ 109
Portfolio-based	82	—	1,910	1,992
Total allowance for lending-related commitments[(d)]	$ 82	$ —	$ 2,019	$ 2,101
Lending-related commitments by impairment methodology				
Asset-specific	$ —	$ —	$ 737	$ 737
Portfolio-based[(e)]	25,608	19	510,254	535,881
Total lending-related commitments	$ 25,608	$ 19	$ 510,991	$ 536,618

(a) Represents the impact to the allowance for loan losses upon the adoption of the Financial Instruments - Credit Losses: Troubled Debt Restructurings accounting guidance. Refer to Note 1 for further information.
(b) At December 31, 2024 and 2023 and 2022, in addition to the allowance for credit losses in the table above, the Firm also had an allowance for credit losses of $268 million, $243 million and $21 million, respectively, associated with certain accounts receivable in CIB.
(c) Includes collateral-dependent loans, including those for which foreclosure is deemed probable, and nonaccrual risk-rated loans.
(d) The allowance for lending-related commitments is reported in accounts payable and other liabilities on the Consolidated balance sheets.
(e) At December 31, 2024, 2023 and 2022, lending-related commitments excluded $19.2 billion, $17.2 billion and $13.1 billion, respectively, for the consumer, excluding credit card portfolio segment; $1.0 trillion, $915.7 billion and $821.3 billion, respectively, for the credit card portfolio segment; and $20.5 billion, $19.7 billion and $9.8 billion, respectively, for the wholesale portfolio segment, which were not subject to the allowance for lending-related commitments.

(table continued from previous page)

2023				2022			
Consumer, excluding credit card	Credit card	Wholesale	Total	Consumer, excluding credit card	Credit card	Wholesale	Total
$ 2,040	$ 11,200	$ 6,486	$ 19,726	$ 1,765	$ 10,250	$ 4,371	$ 16,386
(489)	(100)	2	(587)	NA	NA	NA	NA
1,151	5,491	1,011	7,653	812	3,192	322	4,326
(519)	(793)	(132)	(1,444)	(543)	(789)	(141)	(1,473)
632	4,698	879	6,209	269	2,403	181	2,853
936	6,048	2,484	9,468	543	3,353	2,293	6,189
1	—	21	22	1	—	3	4
$ 1,856	$ 12,450	$ 8,114	$ 22,420	$ 2,040	$ 11,200	$ 6,486	$ 19,726
$ 76	$ —	$ 2,306	$ 2,382	$ 113	$ —	$ 2,148	$ 2,261
(1)	—	(407)	(408)	(37)	—	157	120
—	—	—	—	—	—	1	1
$ 75	$ —	$ 1,899	$ 1,974	$ 76	$ —	$ 2,306	$ 2,382
NA	NA	NA	$ 128	NA	NA	NA	$ 96
$ 1,931	$ 12,450	$ 10,013	$ 24,522	$ 2,116	$ 11,200	$ 8,792	$ 22,204
$ (876)	$ —	$ 392	$ (484)	$ (624)	$ 223	$ 467	$ 66
2,732	12,450	7,722	22,904	2,664	10,977	6,019	19,660
$ 1,856	$ 12,450	$ 8,114	$ 22,420	$ 2,040	$ 11,200	$ 6,486	$ 19,726
$ 3,287	$ —	$ 2,338	$ 5,625	$ 11,978	$ 796	$ 2,189	$ 14,963
393,988	211,123	670,134	1,275,245	288,775	184,379	601,481	1,074,635
$ 397,275	$ 211,123	$ 672,472	$ 1,280,870	$ 300,753	$ 185,175	$ 603,670	$ 1,089,598
$ 6	$ —	$ 180	$ 186	$ (33)	$ —	$ 16	$ (17)
3,216	—	1,012	4,228	3,585	—	464	4,049
$ —	$ —	$ 89	$ 89	$ —	$ —	$ 90	$ 90
75	—	1,810	1,885	76	—	2,216	2,292
$ 75	$ —	$ 1,899	$ 1,974	$ 76	$ —	$ 2,306	$ 2,382
$ —	$ —	$ 464	$ 464	$ —	$ —	$ 455	$ 455
28,248	—	516,577	544,825	20,423	—	461,688	482,111
$ 28,248	$ —	$ 517,041	$ 545,289	$ 20,423	$ —	$ 462,143	$ 482,566

Discussion of changes in the allowance

The allowance for credit losses as of December 31, 2024 was $26.9 billion, reflecting a net addition of $2.1 billion from December 31, 2023.

The net addition to the allowance for credit losses included:

- $2.1 billion in **consumer**, reflecting:
  - a $2.2 billion net addition in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years,

  partially offset by

  - a $125 million net reduction in Home Lending in the first quarter of 2024, and
- a net reduction of $30 million in **wholesale**, reflecting:
  - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in Markets, and a reduction due to charge-offs largely from collateral-dependent loans,

  predominantly offset by

  - net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024.

The Firm's qualitative adjustments continued to include additional weight placed on the adverse scenarios to reflect ongoing uncertainties and downside risks related to the geopolitical and macroeconomic environment.

The Firm's allowance for credit losses is estimated using a weighted average of five internally developed macroeconomic scenarios. The adverse scenarios incorporate more punitive macroeconomic factors than the central case assumptions provided in the table below, resulting in:

- a weighted average U.S. unemployment rate peaking at 5.5% in the fourth quarter of 2025, and
- a weighted average U.S. real GDP level that is 1.9% lower than the central case at the end of the second quarter of 2026.

The following table presents the Firm's central case assumptions for the periods presented:

	Central case assumptions at December 31, 2024		
	2Q25	**4Q25**	**2Q26**
U.S. unemployment rate[(a)]	**4.5 %**	**4.3 %**	**4.3 %**
YoY growth in U.S. real GDP[(b)]	**2.0 %**	**1.9 %**	**1.8 %**

	Central case assumptions at December 31, 2023		
	2Q24	4Q24	2Q25
U.S. unemployment rate[(a)]	4.1 %	4.4 %	4.1 %
YoY growth in U.S. real GDP[(b)]	1.8 %	0.7 %	1.0 %

(a) Reflects quarterly average of forecasted U.S. unemployment rate.
(b) The year over year growth in U.S. real GDP in the forecast horizon of the central scenario is calculated as the percentage change in U.S. real GDP levels from the prior year.

Subsequent changes to this forecast and related estimates will be reflected in the provision for credit losses in future periods.

Refer to Note 12 for additional information on the consumer and wholesale credit portfolios.

Refer to Critical Accounting Estimates Used by the Firm on pages 161–164 for further information on the allowance for credit losses and related management judgments.

Note 14 – Variable interest entities

Refer to Note 1 on page 177 for a further description of the Firm's accounting policies regarding consolidation of and involvement with VIEs.

The following table summarizes the most significant types of Firm-sponsored VIEs by business segment. The Firm considers a "Firm-sponsored" VIE to include any entity where: (1) JPMorganChase is the primary beneficiary of the structure; (2) the VIE is used by JPMorganChase to securitize Firm assets; (3) the VIE issues financial instruments with the JPMorganChase name; or (4) the entity is a JPMorganChase–administered asset-backed commercial paper conduit.

Line of Business	*Transaction Type*	*Activity*	*2024 Form 10-K page references*
CCB	Credit card securitization trusts	Securitization of originated credit card receivables	pages 271–272
	Mortgage securitization trusts	Servicing and securitization of both originated and purchased residential mortgages	pages 272–274
CIB	Mortgage and other securitization trusts	Securitization of both originated and purchased residential and commercial mortgages, and other consumer loans	pages 272–274
	Multi-seller conduits	Assisting clients in accessing the financial markets in a cost-efficient manner and structuring transactions to meet investor needs	page 274
	Municipal bond vehicles	Financing of municipal bond investments	pages 274–275

The Firm's other business segments and Corporate are also involved with VIEs (both third-party and Firm-sponsored), but to a lesser extent, as follows:

- Asset & Wealth Management: AWM sponsors and manages certain funds that are deemed VIEs. As asset manager of the funds, AWM earns a fee based on assets managed; the fee varies with each fund's investment objective and is competitively priced. For fund entities that qualify as VIEs, AWM's interests are, in certain cases, considered to be significant variable interests that result in consolidation of the financial results of these entities.
- Corporate: Corporate is involved with entities that may meet the definition of VIEs; however these entities are generally subject to specialized investment company accounting, which does not require the consolidation of investments, including VIEs. In addition, Treasury and CIO invest in securities generally issued by third parties which may meet the definition of VIEs (e.g., issuers of asset-backed securities). In general, the Firm does not have the power to direct the significant activities of these entities and therefore does not consolidate these entities. Refer to Note 10 for further information on the Firm's investment securities portfolio.

In addition, CIB also invests in and provides financing, lending-related services and other services to VIEs sponsored by third parties. Refer to page 276 of this Note for more information on the VIEs sponsored by third parties.

Significant Firm-sponsored VIEs

Credit card securitizations

CCB's Card Services business may securitize originated credit card loans, primarily through the Chase Issuance Trust (the "Trust"). The Firm's continuing involvement in credit card securitizations includes servicing the receivables, retaining an undivided seller's interest in the receivables, retaining certain senior and subordinated securities and maintaining escrow accounts.

The Firm consolidates the assets and liabilities of its sponsored credit card trusts as it is considered to be the primary beneficiary of these securitization trusts based on the Firm's ability to direct the activities of these VIEs through its servicing responsibilities and other duties, including making decisions as to the receivables that are transferred into those trusts and as to any related modifications and workouts. Additionally, the nature and extent of the Firm's other continuing involvement with the trusts, as indicated above, obligates the Firm to absorb losses and gives the Firm the right to receive certain benefits from these VIEs that could potentially be significant.

The underlying securitized credit card receivables and other assets of the securitization trusts are available only for payment of the beneficial interests issued by the securitization trusts; they are not available to pay the Firm's other obligations or the claims of the Firm's creditors.

The agreements with the credit card securitization trusts require the Firm to maintain a minimum undivided interest in the credit card trusts (generally 5%). As of December 31, 2024 and 2023, the Firm held undivided interests in Firm-sponsored credit card securitization trusts of $6.6 billion and $4.9 billion, respectively. The Firm maintained an average undivided interest in principal receivables owned by those trusts of approximately 45% and 65% for the years ended December 31, 2024 and 2023, respectively. The Firm did not retain any senior securities and retained $1.5 billion of subordinated securities in certain of its credit card securitization trusts at both December 31, 2024 and 2023. The Firm's undivided interests in the credit card trusts and securities retained are eliminated in consolidation.

Firm-sponsored mortgage and other securitization trusts

The Firm securitizes (or has securitized) originated and purchased residential mortgages, commercial mortgages and other consumer loans primarily in its CCB and CIB businesses. Depending on the particular transaction, as well as the respective business involved, the Firm may act as the servicer of the loans and/or retain certain beneficial interests in the securitization trusts.

The following tables present the total unpaid principal amount of assets held in Firm-sponsored private-label securitization entities, including those in which the Firm has continuing involvement, and those that are consolidated by the Firm. Continuing involvement includes servicing the loans, holding senior interests or subordinated interests (including amounts required to be held pursuant to credit risk retention rules), recourse or guarantee arrangements, and derivative contracts. In certain instances, the Firm's only continuing involvement is servicing the loans. The Firm's maximum loss exposure from retained and purchased interests is the carrying value of these interests. Refer to page 279 of this Note for information on the securitization-related loan delinquencies and liquidation losses.

	Principal amount outstanding			JPMorganChase interest in securitized assets in nonconsolidated VIEs[c][d][e]			
December 31, 2024 (in millions)	Total assets held by securitization VIEs	Assets held in consolidated securitization VIEs	Assets held in nonconsolidated securitization VIEs with continuing involvement	Trading assets	Investment securities	Other financial assets	Total interests held by JPMorgan Chase
Securitization-related[a]							
Residential mortgage:							
Prime/Alt-A and option ARMs	**$ 71,085**	**$ 615**	**$ 50,846**	**$ 613**	**$ 1,850**	**$ 614**	**$ 3,077**
Subprime	**8,824**	**—**	**1,847**	**44**	**19**	**—**	**63**
Commercial and other[b]	**186,293**	**243**	**125,510**	**530**	**5,768**	**1,074**	**7,372**
Total	**$ 266,202**	**$ 858**	**$ 178,203**	**$ 1,187**	**$ 7,637**	**$ 1,688**	**$ 10,512**

	Principal amount outstanding			JPMorganChase interest in securitized assets in nonconsolidated VIEs[c][d][e]			
December 31, 2023 (in millions)	Total assets held by securitization VIEs	Assets held in consolidated securitization VIEs	Assets held in nonconsolidated securitization VIEs with continuing involvement	Trading assets	Investment securities	Other financial assets	Total interests held by JPMorgan Chase
Securitization-related[a]							
Residential mortgage:							
Prime/Alt-A and option ARMs	$ 58,570	$ 675	$ 39,319	$ 595	$ 1,981	$ 60	$ 2,636
Subprime	8,881	—	1,312	3	—	—	3
Commercial and other[b]	168,042	—	120,262	831	5,638	1,354	7,823
Total	$ 235,493	$ 675	$ 160,893	$ 1,429	$ 7,619	$ 1,414	$ 10,462

(a) Excludes U.S. GSEs and government agency securitizations and re-securitizations, which are not Firm-sponsored.
(b) Consists of securities backed by commercial real estate loans and non-mortgage-related consumer receivables.
(c) Excludes the following: retained servicing; securities retained from loan sales and securitization activity related to U.S. GSEs and government agencies; interest rate and foreign exchange derivatives primarily used to manage interest rate and foreign exchange risks of securitization entities; senior securities of $256 million and $52 million at December 31, 2024 and 2023, respectively, and subordinated securities which were not material at December 31, 2024 and 2023, which the Firm purchased in connection with CIB's secondary market-making activities.
(d) Includes interests held in re-securitization transactions.
(e) At both December 31, 2024 and 2023, 77% of the Firm's retained securitization interests, which are predominantly carried at fair value and include amounts required to be held pursuant to credit risk retention rules, were risk-rated "A" or better, on an S&P-equivalent basis. The retained interests in prime residential mortgages consisted of $2.9 billion and $2.5 billion of investment-grade retained interests at December 31, 2024 and 2023, respectively, and $216 million and $88 million of noninvestment-grade retained interests at December 31, 2024 and 2023, respectively. The retained interests in commercial and other securitization trusts consisted of $6.0 billion and $6.1 billion of investment-grade retained interests, and $1.4 billion and $1.7 billion of noninvestment-grade retained interests at December 31, 2024 and 2023, respectively.

Residential mortgage
The Firm securitizes residential mortgage loans originated by CCB, as well as residential mortgage loans purchased from third parties by either CCB or CIB. CCB generally retains servicing for all residential mortgage loans it originated or purchased, and for certain mortgage loans purchased by CIB. For securitizations of loans serviced by CCB, the Firm has the power to direct the significant activities of the VIE because it is responsible for decisions related to loan modifications and workouts. CCB may also retain an interest upon securitization.

In addition, CIB engages in underwriting and trading activities involving securities issued by Firm-sponsored securitization trusts. As a result, CIB at times retains senior and/or subordinated interests (including residual interests and amounts required to be held pursuant to credit risk retention rules) in residential mortgage securitizations at the time of securitization, and/or reacquires positions in the secondary market in the normal course of business. In certain instances, as a result of the positions retained or reacquired by CIB or held by Treasury and CIO or CCB, when considered together with the servicing arrangements entered into by CCB, the Firm is deemed to be the primary beneficiary of certain securitization trusts.

The Firm does not consolidate residential mortgage securitizations (Firm-sponsored or third-party-sponsored) when it is not the servicer (and therefore does not have the power to direct the most significant activities of the trust) or does not hold a beneficial interest in the trust that could potentially be significant to the trust.

Commercial mortgages and other consumer securitizations
CIB originates and securitizes commercial mortgage loans, and engages in underwriting and trading activities involving the securities issued by securitization trusts. CIB may retain unsold senior and/or subordinated interests (including amounts required to be held pursuant to credit risk retention rules) in commercial mortgage securitizations at the time of securitization but, generally, the Firm does not service commercial loan securitizations. Treasury and CIO may choose to invest in these securitizations as well. For commercial mortgage securitizations the power to direct the significant activities of the VIE generally is held by the servicer or investors in a specified class of securities ("controlling class"). The Firm generally does not retain an interest in the controlling class in its sponsored commercial mortgage securitization transactions.

Re-securitizations
The Firm engages in certain re-securitization transactions in which debt securities are transferred to a VIE in exchange for new beneficial interests. These transfers occur in connection with both U.S. GSEs and government agency sponsored VIEs, which are backed by residential mortgages. The Firm's consolidation analysis is largely dependent on the Firm's role and interest in the re-securitization trusts.

The following table presents the principal amount of securities transferred to re-securitization VIEs.

Year ended December 31, (in millions)	**2024**	2023	2022
Transfers of securities to VIEs			
U.S. GSEs and government agencies	**$ 44,456**	$ 18,864	$ 16,128

Most re-securitizations with which the Firm is involved are client-driven transactions in which a specific client or group of clients is seeking a specific return or risk profile. For these transactions, the Firm has concluded that the decision-making power of the entity is shared between the Firm and its clients, considering the joint effort and decisions in establishing the re-securitization trust and its assets, as well as the significant economic interest the client holds in the re-securitization trust; therefore the Firm does not consolidate the re-securitization VIE.

The Firm did not transfer any private label securities to re-securitization VIEs during 2024, 2023 and 2022, and retained interests in any such Firm-sponsored VIEs as of December 31, 2024 and 2023 were not material.

Additionally, the Firm may invest in beneficial interests of third-party-sponsored re-securitizations and generally purchases these interests in the secondary market. In these circumstances, the Firm does not have the unilateral ability to direct the most significant activities of the re-securitization trust, either because it was not involved in the initial design of the trust, or the Firm was involved with an independent third-party sponsor and demonstrated shared power over the creation of the trust; therefore, the Firm does not consolidate the re-securitization VIE.

The following table presents information on the Firm's interests in nonconsolidated re-securitization VIEs.

December 31, (in millions)	Nonconsolidated re-securitization VIEs 2024	2023
U.S. GSEs and government agencies		
Interest in VIEs	**$ 3,219**	$ 3,371

As of December 31, 2024 and 2023, the Firm did not consolidate any U.S. GSE and government agency re-securitization VIEs or any Firm-sponsored private-label re-securitization VIEs.

Multi-seller conduits

Multi-seller conduit entities are separate bankruptcy remote entities that provide secured financing, collateralized by pools of receivables and other financial assets, to customers of the Firm. The conduits fund their financing facilities through the issuance of highly rated commercial paper. The primary source of repayment of the commercial paper is the cash flows from the pools of assets. In most instances, the assets are structured with deal-specific credit enhancements provided to the conduits by the customers (i.e., sellers) or other third parties. Deal-specific credit enhancements are generally structured to cover a multiple of historical losses expected on the pool of assets, and are typically in the form of overcollateralization provided by the seller. The deal-specific credit enhancements mitigate the Firm's potential losses on its agreements with the conduits.

To ensure timely repayment of the commercial paper, and to provide the conduits with funding to provide financing to customers in the event that the conduits do not obtain funding in the commercial paper market, each asset pool financed by the conduits has a minimum 100% deal-specific liquidity facility associated with it provided by JPMorgan Chase Bank, N.A. JPMorgan Chase Bank, N.A. also provides the multi-seller conduit vehicles with uncommitted program-wide liquidity facilities and program-wide credit enhancement in the form of standby letters of credit. The amount of program-wide credit enhancement required is based upon commercial paper issuance and approximates 10% of the outstanding balance of commercial paper.

The Firm consolidates its Firm-administered multi-seller conduits, as the Firm has both the power to direct the significant activities of the conduits and a potentially significant economic interest in the conduits. As administrative agent and in its role in structuring transactions, the Firm makes decisions regarding asset types and credit quality, and manages the commercial paper funding needs of the conduits. The Firm's interests that could potentially be significant to the VIEs include the fees received as administrative agent and liquidity and program-wide credit enhancement provider, as well as the potential exposure created by the liquidity and credit enhancement facilities provided to the conduits.

In the normal course of business, JPMorganChase makes markets in and invests in commercial paper issued by the Firm-administered multi-seller conduits. The Firm held $2.9 billion and $9.8 billion of the commercial paper issued by the Firm-administered multi-seller conduits at December 31, 2024 and 2023, respectively, which have been eliminated in consolidation. The Firm's investments reflect the Firm's funding needs and capacity and were not driven by market illiquidity. Other than the amounts required to be held pursuant to credit risk retention rules, the Firm is not obligated under any agreement to purchase the commercial paper issued by the Firm-administered multi-seller conduits.

Deal-specific liquidity facilities, program-wide liquidity and credit enhancement provided by the Firm have been eliminated in consolidation. The Firm or the Firm-administered multi-seller conduits provide lending-related commitments to certain clients of the Firm-administered multi-seller conduits. The unfunded commitments were $10.3 billion and $10.8 billion at December 31, 2024 and 2023, respectively, and are reported as off-balance sheet lending-related commitments in other unfunded commitments to extend credit. Refer to Note 28 for more information on off-balance sheet lending-related commitments.

Municipal bond vehicles

Municipal bond vehicles or tender option bond ("TOB") trusts allow institutions to finance their municipal bond investments at short-term rates. In a typical TOB transaction, the trust purchases highly rated municipal bond(s) of a single issuer and funds the purchase by issuing two types of securities: (1) puttable floating-rate certificates ("floaters") and (2) inverse floating-rate residual interests ("residuals"). The floaters are typically purchased by money market funds or other short-term investors and may be tendered, with requisite notice, to the TOB trust. The residuals are retained by the investor seeking to finance its municipal bond investment. TOB transactions where the residual is held by a third-party investor are typically known as customer TOB trusts, and non-customer TOB trusts are transactions where the Residual is retained by the Firm. Customer TOB trusts are sponsored by a third party. The Firm serves as sponsor for all non-customer TOB transactions. The Firm may provide various services to a TOB trust, including remarketing agent, liquidity or tender option provider, and/or sponsor.

J.P. Morgan Securities LLC may serve as a remarketing agent on the floaters for TOB trusts. The remarketing agent is responsible for establishing the periodic variable rate on the floaters, conducting the initial placement and remarketing tendered floaters. The remarketing agent may, but is not obligated to, make markets in floaters. Floaters held by the Firm were not material during 2024 and 2023.

JPMorgan Chase Bank, N.A. or J.P. Morgan Securities LLC often serves as the sole liquidity or tender option provider for the TOB trusts. The liquidity provider's obligation to perform is conditional and is limited by certain events ("Termination Events"), which include bankruptcy or failure to pay by the municipal bond issuer or credit enhancement provider, an event of taxability on the municipal bonds or the immediate downgrade of the municipal bond to below investment grade. In addition, the liquidity provider's exposure is typically further limited by the high credit quality of the underlying municipal bonds, the excess collateralization in the vehicle, or, in certain transactions, the reimbursement agreements with the Residual holders.

Holders of the floaters may "put," or tender, their floaters to the TOB trust. If the remarketing agent cannot successfully remarket the floaters to another investor, the liquidity provider either provides a loan to the TOB trust for the TOB trust's purchase of the floaters, or it directly purchases the tendered floaters.

TOB trusts are considered to be variable interest entities. The Firm consolidates non-customer TOB trusts because as the Residual holder, the Firm has the right to make decisions that significantly impact the economic performance of the municipal bond vehicle, and it has the right to receive benefits and bear losses that could potentially be significant to the municipal bond vehicle.

Consolidated VIE assets and liabilities

The following table presents information on assets and liabilities related to VIEs consolidated by the Firm as of December 31, 2024 and 2023.

December 31, 2024 (in millions)	Assets				Liabilities		
	Trading assets	Loans	Other[c]	Total assets[d]	Beneficial interests in VIE assets[e]	Other[f]	Total liabilities
VIE program type							
Firm-sponsored credit card trusts	**$ —**	**$ 13,531**	**$ 168**	**$ 13,699**	**$ 5,312**	**$ 10**	**$ 5,322**
Firm-administered multi-seller conduits	**1**	**20,383**	**133**	**20,517**	**18,228**	**26**	**18,254**
Municipal bond vehicles	**3,388**	**—**	**22**	**3,410**	**3,617**	**15**	**3,632**
Mortgage securitization entities[a]	**—**	**630**	**8**	**638**	**115**	**48**	**163**
Other	**496**	**1,966** [b]	**350**	**2,812**	**51**	**355**	**406**
Total	**$ 3,885**	**$ 36,510**	**$ 681**	**$ 41,076**	**$ 27,323**	**$ 454**	**$ 27,777**

December 31, 2023 (in millions)	Assets				Liabilities		
	Trading assets	Loans	Other[c]	Total assets[d]	Beneficial interests in VIE assets[e]	Other[f]	Total liabilities
VIE program type							
Firm-sponsored credit card trusts	$ —	$ 9,460	$ 117	$ 9,577	$ 2,998	$ 6	$ 3,004
Firm-administered multi-seller conduits	1	27,372	194	27,567	17,781	30	17,811
Municipal bond vehicles	2,056	—	22	2,078	2,116	11	2,127
Mortgage securitization entities[a]	—	693	8	701	125	57	182
Other	113	86	250	449	—	159	159
Total	$ 2,170	$ 37,611	$ 591	$ 40,372	$ 23,020	$ 263	$ 23,283

(a) Includes residential mortgage securitizations.
(b) Primarily includes consumer loans in CIB.
(c) Includes assets classified as cash and other assets on the Consolidated balance sheets.
(d) The assets of the consolidated VIEs included in the program types above are used to settle the liabilities of those entities. The assets and liabilities include third-party assets and liabilities of consolidated VIEs and exclude intercompany balances that eliminate in consolidation.
(e) The interest-bearing beneficial interest liabilities issued by consolidated VIEs are classified on the Consolidated balance sheets as "Beneficial interests issued by consolidated VIEs". The holders of these beneficial interests generally do not have recourse to the general credit of JPMorganChase. Included in beneficial interests in VIE assets are long-term beneficial interests of $5.5 billion and $3.1 billion at December 31, 2024 and 2023, respectively.
(f) Includes liabilities classified as accounts payable and other liabilities on the Consolidated balance sheets.

VIEs sponsored by third parties

The Firm enters into transactions with VIEs structured by other parties. These include, for example, acting as a derivative counterparty, liquidity provider, investor, underwriter, placement agent, remarketing agent, trustee or custodian. These transactions are conducted at arm's-length, and individual credit decisions are based on the analysis of the specific VIE, taking into consideration the quality of the underlying assets. Where the Firm does not have the power to direct the activities of the VIE that most significantly impact the VIE's economic performance, or a variable interest that could potentially be significant, the Firm generally does not consolidate the VIE, but it records and reports these positions on its Consolidated balance sheets in the same manner it would record and report positions in respect of any other third-party transaction.

Tax credit vehicles

The Firm holds investments in unconsolidated tax credit vehicles, which are limited partnerships and similar entities that own and operate affordable housing, alternative energy, and other projects. These entities are primarily considered VIEs. A third party is typically the general partner or managing member and has control over the significant activities of the tax credit vehicles, and accordingly the Firm does not consolidate tax credit vehicles. The Firm generally invests in these partnerships as a limited partner and earns a return primarily through the receipt of tax credits allocated to the projects. The maximum loss exposure, represented by equity investments and funding commitments, was $35.2 billion and $35.1 billion, of which $15.0 billion and $14.7 billion was unfunded at December 31, 2024 and 2023, respectively. The Firm assesses each project and to reduce the risk of loss, may withhold varying amounts of its capital investment until the project qualifies for tax credits. Refer to Note 28 for more information on off-balance sheet lending-related commitments.

Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance which expanded the types of tax-oriented investments, beyond affordable housing tax credit investments, that the Firm can elect on a program by program basis, to be accounted for using the proportional amortization method. Refer to Notes 1, 6 and 25 for further information.

The proportional amortization method requires the cost of eligible investments, within an elected program, be amortized in proportion to the tax benefits received with the resulting amortization reported directly in income tax expense, which aligns with the associated tax credits and other tax benefits. Investments must meet certain criteria to be eligible, including that substantially all of the return is from income tax credits and other income tax benefits.

In addition, under this method deferred taxes are generally not recorded as the investment is now amortized in proportion to the income tax credits and other income tax benefits received. Delayed equity contributions that are unconditional and legally binding or conditional and probable of occurring are recorded in other liabilities with a corresponding increase in the carrying value of the investment. The guidance also requires a reevaluation of eligible investments when significant modifications or events occur that result in a change in the nature of the investment or a change in the Firm's relationship with the underlying project. During the period, there were no significant modifications or events that resulted in a change in the nature of an eligible investment or a change in the Firm's relationship with the underlying project.

The following table provides information on tax-oriented investments for which the Firm elected to apply the proportional amortization method.

Year ended December 31, (in millions)	Alternative energy and affordable housing programs[(d)]		
	2024	2023	2022
Programs for which the Firm elected proportional amortization:			
Carrying value[(a)]	**$ 31,978**	$ 14,644	$ 12,052
Tax credits and other tax benefits[(b)]	**6,379**	2,044	1,786
Investments that qualify to be accounted for using proportional amortization:			
Amortization losses recognized as a component of income tax expense	**(5,018)**	(1,561)	(1,353)
Non-income-tax-related gains/(losses) and other returns received that are recognized outside of income tax expense[(c)]	**142**	(1)	(1)

(a) Recorded in Other assets on the Consolidated balance sheets. Excludes programs to which the Firm does not apply the proportional amortization method, such as historic tax credit and new market tax credit programs.
(b) Reflected in Income tax expense on the Consolidated statements of income and Operating activities on the Consolidated statements of cash flows.
(c) Recorded in Other income on the Consolidated statements of income and Operating activities on the Consolidated statements of cash flows.
(d) As of December 31, 2023 and 2022 represents eligible affordable housing investments.

Customer municipal bond vehicles (TOB trusts)
The Firm may provide various services to customer TOB trusts, including remarketing agent, liquidity or tender option provider. In certain customer TOB transactions, the Firm, as liquidity provider, has entered into a reimbursement agreement with the Residual holder. In those transactions, upon the termination of the vehicle, the Firm has recourse to the third-party Residual holders for any shortfall. The Firm does not have any intent to protect Residual holders from potential losses on any of the underlying municipal bonds. The Firm does not consolidate customer TOB trusts, since the Firm does not have the power to make decisions that significantly impact the economic performance of the municipal bond vehicle.

The Firm's maximum exposure as a liquidity provider to customer TOB trusts at December 31, 2024 and 2023, was $5.8 billion and $5.1 billion, respectively. The fair value of assets held by such VIEs at December 31, 2024 and 2023 was $8.1 billion and $7.3 billion, respectively.

Loan securitizations

The Firm has securitized and sold a variety of loans, including residential mortgages, credit card receivables, commercial mortgages and other consumer loans. The purposes of these securitization transactions were to satisfy investor demand and to generate liquidity for the Firm.

For loan securitizations in which the Firm is not required to consolidate the trust, the Firm records the transfer of the loan receivable to the trust as a sale when all of the following accounting criteria for a sale are met: (1) the transferred financial assets are legally isolated from the Firm's creditors; (2) the transferee or beneficial interest holder can pledge or exchange the transferred financial assets; and (3) the Firm does not maintain effective control over the transferred financial assets (e.g., the Firm cannot repurchase the transferred assets before their maturity and it does not have the ability to unilaterally cause the holder to return the transferred assets).

For loan securitizations accounted for as a sale, the Firm recognizes a gain or loss based on the difference between the value of proceeds received (including cash, beneficial interests, or servicing assets received) and the carrying value of the assets sold. Gains and losses on securitizations are reported in noninterest revenue.

Securitization activity

The following table provides information related to the Firm's securitization activities for the years ended December 31, 2024, 2023 and 2022, related to assets held in Firm-sponsored securitization entities that were not consolidated by the Firm, and where sale accounting was achieved at the time of the securitization.

	2024		2023		2022	
Year ended December 31, (in millions)	Residential mortgage[d]	Commercial and other[e]	Residential mortgage[d]	Commercial and other[e]	Residential mortgage[d]	Commercial and other[e]
Principal securitized	**$ 19,988**	**$ 17,683**	$ 7,678	$ 3,901	$ 10,218	$ 9,036
All cash flows during the period:[a]						
Proceeds received from loan sales as financial instruments[b][c]	**$ 19,870**	**$ 17,346**	$ 7,251	$ 3,896	$ 9,783	$ 8,921
Servicing fees collected	**35**	**35**	24	5	62	2
Cash flows received on interests	**405**	**1,303**	325	425	489	285

(a) Excludes re-securitization transactions.
(b) Primarily includes Level 2 assets.
(c) The carrying value of the loans accounted for at fair value approximated the proceeds received upon loan sale.
(d) Represents prime mortgages. Excludes loan securitization activity related to U.S. GSEs and government agencies.
(e) Includes commercial mortgages and auto loans.

Key assumptions used to value retained interests originated during the year are shown in the table below.

Year ended December 31,	2024	2023	2022
Residential mortgage retained interest:			
Weighted-average life (in years)	**4.3**	9.6	10.8
Weighted-average discount rate	**7.1 %**	4.8 %	4.0 %
Commercial and other retained interest:			
Weighted-average life (in years)	**4.5**	3.0	5.9
Weighted-average discount rate	**6.2 %**	4.6 %	2.9 %

Loans and excess MSRs sold to U.S. government-sponsored enterprises and loans in securitization transactions pursuant to Ginnie Mae guidelines

In addition to the amounts reported in the securitization activity tables above, the Firm, in the normal course of business, sells originated and purchased mortgage loans and certain originated excess MSRs on a nonrecourse basis, predominantly to U.S. GSEs. These loans and excess MSRs are sold primarily for the purpose of securitization by the U.S. GSEs, who provide certain guarantee provisions (e.g., credit enhancement of the loans). The Firm also sells loans into securitization transactions pursuant to Ginnie Mae guidelines; these loans are typically insured or guaranteed by another U.S. government agency. The Firm does not consolidate the securitization vehicles underlying these transactions as it is not the primary beneficiary. For a limited number of loan sales, the Firm is obligated to share a portion of the credit risk associated with the sold loans with the purchaser. Refer to Note 28 for additional information about the Firm's loan sales- and securitization-related indemnifications and Note 15 for additional information about the impact of the Firm's sale of certain excess MSRs.

The following table summarizes the activities related to loans sold to the U.S. GSEs, and loans in securitization transactions pursuant to Ginnie Mae guidelines.

Year ended December 31, (in millions)	2024	2023	2022
Carrying value of loans sold	**$ 25,765**	$ 19,906	$ 48,891
Proceeds received from loan sales as cash	**$ 2,380**	$ 300	$ 22
Proceeds from loan sales as securities[a][b]	**23,178**	19,389	48,096
Total proceeds received from loan sales[c]	**$ 25,558**	$ 19,689	$ 48,118
Gains/(losses) on loan sales[d][e]	**$ —**	$ —	$ (25)

(a) Includes securities from U.S. GSEs and Ginnie Mae that are generally sold shortly after receipt or retained as part of the Firm's investment securities portfolio.
(b) Included in level 2 assets.
(c) Excludes the value of MSRs retained upon the sale of loans.
(d) Gains/(losses) on loan sales include the value of MSRs.
(e) The carrying value of the loans accounted for at fair value approximated the proceeds received upon loan sale.

Options to repurchase delinquent loans

In addition to the Firm's obligation to repurchase certain loans due to material breaches of representations and warranties as discussed in Note 28, the Firm also has the option to repurchase delinquent loans that it services for Ginnie Mae loan pools, as well as for other U.S. government agencies under certain arrangements. The Firm typically elects to repurchase delinquent loans from Ginnie Mae loan pools as it continues to service them and/or manage the foreclosure process in accordance with the applicable requirements, and such loans continue to be insured or guaranteed. When the Firm's repurchase option becomes exercisable, such loans must be reported on the Consolidated balance sheets as a loan with a corresponding liability. Refer to Note 12 for additional information.

The following table presents loans the Firm repurchased or had an option to repurchase, real estate owned, and foreclosed government-guaranteed residential mortgage loans recognized on the Firm's Consolidated balance sheets as of December 31, 2024 and 2023. Substantially all of these loans and real estate are insured or guaranteed by U.S. government agencies.

December 31, (in millions)	**2024**	2023
Loans repurchased or option to repurchase[(a)]	**$ 577**	$ 597
Real estate owned	**6**	8
Foreclosed government-guaranteed residential mortgage loans[(b)]	**10**	22

(a) Primarily all of these amounts relate to loans that have been repurchased from Ginnie Mae loan pools.
(b) Relates to voluntary repurchases of loans, which are included in accrued interest and accounts receivable.

Loan delinquencies and liquidation losses

The table below includes information about components of and delinquencies related to nonconsolidated securitized financial assets held in Firm-sponsored private-label securitization entities, in which the Firm has continuing involvement as of December 31, 2024 and 2023. For loans sold or securitized where servicing is the Firm's only form of continuing involvement, the Firm generally experiences a loss only if the Firm was required to repurchase a delinquent loan or foreclosed asset due to a breach in representations and warranties associated with its loan sale or servicing contracts.

As of or for the year ended December 31, (in millions)	Securitized assets		90 days past due		Net liquidation losses / (recoveries)	
	2024	2023	**2024**	2023	**2024**	2023
Securitized loans						
Residential mortgage:						
Prime/ Alt-A & option ARMs	**$ 50,846**	$ 39,319	**$ 501**	$ 440	**$ 10**	$ 14
Subprime	**1,847**	1,312	**113**	131	**2**	5
Commercial and other	**125,510**	120,262	**1,715**	2,874	**77**	60
Total loans securitized	**$ 178,203**	$ 160,893	**$ 2,329**	$ 3,445	**$ 89**	$ 79

Note 15 – Goodwill, mortgage servicing rights, and other intangible assets

Goodwill

Goodwill is recorded upon completion of a business combination as the difference between the purchase price and the fair value of the net assets acquired, and can be adjusted up to one year from the acquisition date as additional information pertaining to facts and circumstances that existed as of the acquisition date is obtained about the fair value of assets acquired and liabilities assumed. Subsequent to initial recognition, goodwill is not amortized but is tested for impairment during the fourth quarter of each fiscal year, or more often if events or circumstances, such as adverse changes in the business climate, indicate that there may be an impairment.

The goodwill associated with each business combination is allocated to the related reporting units, which are generally determined based on how the Firm's businesses are managed and how they are reviewed. The following table presents goodwill attributed to the reportable business segments and Corporate.

December 31, (in millions)	**2024**	2023	2022
Consumer & Community Banking	**$ 32,116**	$ 32,116	$ 32,121
Commercial & Investment Bank	**11,236**	11,251	10,993
Asset & Wealth Management	**8,521**	8,582	7,902
Corporate	**692**	685	646
Total goodwill	**$ 52,565**	$ 52,634	$ 51,662

The following table presents changes in the carrying amount of goodwill.

(in millions)	**2024**	2023	2022
Balance at beginning of period	**$ 52,634**	$ 51,662	$ 50,315
Changes during the period from:			
Business combinations[(a)]	**29**	917	1,426
Other[(b)]	**(98)**	55	(79)
Balance at December 31,	**$ 52,565**	$ 52,634	$ 51,662

(a) For 2024, includes estimated goodwill associated with the acquisition of LayerOne Financial in CIB. For 2023, predominantly represents estimated goodwill associated with the acquisition of the remaining 51% interest in CIFM in AWM and the acquisition of Aumni Inc., predominantly in CIB. For 2022, represents estimated goodwill associated with the acquisitions of Global Shares PLC in AWM, Frosch Travel Group, LLC and Figg, Inc. in CCB, and Renovite Technologies, Inc. and Volkswagen Payments S.A. in CIB.
(b) Primarily foreign currency adjustments.

Goodwill impairment testing

The Firm's goodwill was not impaired at December 31, 2024, 2023 and 2022.

The goodwill impairment test is generally performed by comparing the current fair value of each reporting unit with its carrying value. If the fair value is in excess of the carrying value, then the reporting unit's goodwill is considered not to be impaired. If the fair value is less than the carrying value, then an impairment is recognized for the amount by which the reporting unit's carrying value exceeds its fair value, up to the amount of goodwill allocated to that reporting unit.

The Firm uses the reporting units' allocated capital plus goodwill and other intangible assets as a proxy for the carrying values of equity for the reporting units in the goodwill impairment testing. Reporting unit equity is determined on a similar basis as the allocation of capital to the LOBs which takes into consideration a variety of factors including capital levels of similarly rated peers and applicable regulatory capital requirements. LOB's allocated capital levels are incorporated into the Firm's annual budget process, which is reviewed by the Firm's Board of Directors and Operating Committee.

The primary method the Firm uses to estimate the fair value of its reporting units is the income approach. This approach projects cash flows for the forecast period and uses the perpetuity growth method to calculate terminal values. These cash flows and terminal values, which are based on the reporting units' annual budgets and forecasts are then discounted using an appropriate discount rate. The discount rate used for each reporting unit represents an estimate of the cost of equity for that reporting unit and is determined considering the Firm's overall estimated cost of equity (estimated using the Capital Asset Pricing Model), as adjusted for the risk characteristics specific to each reporting unit (for example, for higher levels of risk or uncertainty associated with the business or management's forecasts and assumptions). To assess the reasonableness of the discount rates used for each reporting unit, management compares the discount rate to the estimated cost of equity for publicly traded institutions with similar businesses and risk characteristics. In addition, the weighted average cost of equity (aggregating the various reporting units) is compared with the Firm's overall estimated cost of equity for reasonableness. The valuations derived from the discounted cash flow analysis are then compared with market-based trading and transaction multiples for relevant competitors. Trading and transaction comparables are used as general indicators to assess the overall reasonableness of the estimated fair values, although precise conclusions generally cannot be drawn due to the differences that naturally exist between the Firm's businesses and competitor institutions.

The Firm also takes into consideration a comparison between the aggregate fair values of the Firm's reporting units and JPMorganChase's market capitalization. In evaluating this comparison, the Firm considers several factors, including (i) a control premium that would exist in a market transaction, (ii) factors related to the level of execution risk that would exist at the Firmwide level that do not exist at the reporting unit level and (iii) short-term market volatility and other factors that do not directly affect the value of individual reporting units.

Unanticipated declines in business performance, increases in credit losses, increases in capital requirements, as well as deterioration in economic or market conditions, adverse regulatory or legislative changes or increases in the estimated market cost of equity, could cause the estimated fair values of the Firm's reporting units to decline in the future, which could result in a material impairment loss to earnings in a future period related to some portion of the associated goodwill.

Mortgage servicing rights

MSRs represent the fair value of expected future cash flows for performing servicing activities for others. The fair value considers estimated future servicing fees and ancillary revenue, offset by estimated costs to service the loans, and generally declines over time as net servicing cash flows are received, effectively amortizing the MSR asset against contractual servicing and ancillary fee income. MSRs are either purchased from third parties or recognized upon sale or securitization of mortgage loans if servicing is retained.

As permitted by U.S. GAAP, the Firm has elected to account for its MSRs at fair value. The Firm treats its MSRs as a single class of servicing assets based on the availability of market inputs used to measure the fair value of its MSR asset and its treatment of MSRs as one aggregate pool for risk management purposes. The Firm estimates the fair value of MSRs using an option-adjusted spread ("OAS") model, which projects MSR cash flows over multiple interest rate scenarios in conjunction with the Firm's prepayment model, and then discounts these cash flows at risk-adjusted rates. The model considers portfolio characteristics, contractually specified servicing fees, prepayment assumptions, delinquency rates, costs to service, late charges and other ancillary revenue, and other economic factors. The Firm compares fair value estimates and assumptions to observable market data where available, and also considers recent market activity and actual portfolio experience.

The fair value of MSRs is sensitive to changes in interest rates, including their effect on prepayment speeds. MSRs typically decrease in value when interest rates decline because declining interest rates tend to increase prepayments and therefore reduce the expected life of the net servicing cash flows that comprise the MSR asset. Conversely, securities (e.g., mortgage-backed securities), and certain derivatives (e.g., those for which the Firm receives fixed-rate interest payments) increase in value when interest rates decline. JPMorganChase uses combinations of derivatives and securities to manage the risk of changes in the fair value of MSRs. The intent is to offset any interest-rate related changes in the fair value of MSRs with changes in the fair value of the related risk management instruments.

The following table summarizes MSR activity for the years ended December 31, 2024, 2023 and 2022.

As of or for the year ended December 31, (in millions, except where otherwise noted)	**2024**	2023	2022
Fair value at beginning of period	**$ 8,522**	$ 7,973	$ 5,494
MSR activity:			
Originations of MSRs	**325**	253	798
Purchase of MSRs[(a)]	**601**	1,028	1,400
Disposition of MSRs[(b)]	**(21)**	(188)	(822)
Net additions/(dispositions)	**905**	1,093	1,376
Changes due to collection/realization of expected cash flows	**(1,068)**	(1,011)	(936)
Changes in valuation due to inputs and assumptions:			
Changes due to market interest rates and other[(c)]	**670**	424	2,022
Changes in valuation due to other inputs and assumptions:			
Projected cash flows (e.g., cost to service)	**102**	(22)	14
Discount rates	**14**	14	—
Prepayment model changes and other[(d)]	**(24)**	51	3
Total changes in valuation due to other inputs and assumptions	**92**	43	17
Total changes in valuation due to inputs and assumptions	**762**	467	2,039
Fair value at December 31,	**$ 9,121**	$ 8,522	$ 7,973
Change in unrealized gains/(losses) included in income related to MSRs held at December 31,	**$ 762**	$ 467	$ 2,039
Contractual service fees, late fees and other ancillary fees included in income	**1,606**	1,590	1,535
Third-party mortgage loans serviced at December 31, (in billions)	**652**	632	584
Servicer advances, net of an allowance for uncollectible amounts, at December 31[(e)]	**577**	659	758

(a) Includes purchase price adjustments associated with MSRs purchased, primarily as a result of loans that prepaid within 90 days of settlement, allowing the Firm to recover the purchase price.
(b) Includes excess MSRs transferred to agency-sponsored trusts in exchange for stripped mortgage-backed securities ("SMBS"). In each transaction, a portion of the SMBS was acquired by third parties at the transaction date; the Firm acquired the remaining balance of those SMBS as trading securities.
(c) Represents both the impact of changes in estimated future prepayments due to changes in market interest rates, and the difference between actual and expected prepayments.
(d) Represents changes in prepayments other than those attributable to changes in market interest rates.
(e) Represents amounts the Firm pays as the servicer (e.g., scheduled principal and interest, taxes and insurance), which will generally be reimbursed within a short period of time after the advance from future cash flows from the trust or the underlying loans. The Firm's credit risk associated with these servicer advances is minimal because reimbursement of the advances is typically senior to all cash payments to investors. In addition, the Firm maintains the right to stop payment to investors if the collateral is insufficient to cover the advance. However, certain of these servicer advances may not be recoverable if they were not made in accordance with applicable rules and agreements.

The following table presents the components of mortgage fees and related income (including the impact of MSR risk management activities) for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions)	2024	2023	2022
CCB mortgage fees and related income			
Production revenue	**$ 627**	$ 421	$ 497
Net mortgage servicing revenue:			
Operating revenue:			
Loan servicing revenue	**1,659**	1,634	1,582
Changes in MSR asset fair value due to collection/ realization of expected cash flows	**(1,067)**	(1,011)	(936)
Total operating revenue	**592**	623	646
Risk management:			
Changes in MSR asset fair value due to market interest rates and other[(a)]	**670**	424	2,022
Other changes in MSR asset fair value due to other inputs and assumptions in model[(b)]	**92**	43	17
Change in derivative fair value and other	**(603)**	(336)	(1,946)
Total risk management	**159**	131	93
Total net mortgage servicing revenue	**751**	754	739
Total CCB mortgage fees and related income	**1,378**	1,175	1,236
All other	**23**	1	14
Mortgage fees and related income	**$ 1,401**	$ 1,176	$ 1,250

(a) Represents both the impact of changes in estimated future prepayments due to changes in market interest rates, and the difference between actual and expected prepayments.
(b) Represents the aggregate impact of changes in model inputs and assumptions such as projected cash flows (e.g., cost to service), discount rates and changes in prepayments other than those attributable to changes in market interest rates (e.g., changes in prepayments due to changes in home prices).

Changes in fair value based on variations in assumptions generally cannot be easily extrapolated, because the relationship of the change in the assumptions to the change in fair value are often highly interrelated and may not be linear. In the following table, the effect that a change in a particular assumption may have on the fair value is calculated without changing any other assumption. In reality, changes in one factor may result in changes in another, which would either magnify or counteract the impact of the initial change.

The table below outlines the key economic assumptions used to determine the fair value of the Firm’s MSRs at December 31, 2024 and 2023, and outlines the sensitivities of those fair values to immediate adverse changes in those assumptions, as defined below.

December 31, (in millions, except rates)	2024	2023
Weighted-average prepayment speed assumption (constant prepayment rate)	**6.19 %**	6.29 %
Impact on fair value of 10% adverse change	**$ (209)**	$ (206)
Impact on fair value of 20% adverse change	**(406)**	(401)
Weighted-average option adjusted spread[(a)]	**5.97 %**	6.10 %
Impact on fair value of 100 basis points adverse change	**$ (391)**	$ (369)
Impact on fair value of 200 basis points adverse change	**(751)**	(709)

(a) Includes the impact of operational risk and regulatory capital.

Other intangible assets

The Firm’s finite-lived and indefinite-lived other intangible assets are initially recorded at their fair value primarily upon completion of a business combination. Subsequently, the Firm’s finite-lived intangible assets, including core deposit intangibles, customer relationship intangibles, and certain other intangible assets, are amortized over their useful lives, estimated based on the expected future economic benefits to the Firm of the intangible asset. The Firm’s intangible assets with indefinite lives, such as asset management contracts, are not subject to amortization and are assessed periodically for impairment.

As of December 31, 2024 and 2023, the gross carrying values of other intangible assets were $3.8 billion and $4.2 billion, respectively, and the accumulated amortization was $879 million and $994 million, respectively.

As of December 31, 2024 and 2023, the net carrying values consist of finite-lived intangible assets of $1.7 billion and $2.0 billion, respectively, as well as indefinite-lived intangible assets, which are not subject to amortization, of $1.2 billion at both periods.

As of December 31, 2024, other intangible assets reflected core deposit and certain wealth management customer relationship intangibles related to the First Republic acquisition, and asset management contracts related to the Firm’s acquisition of the remaining 51% interest in CIFM. Refer to Note 34 for additional information on the First Republic acquisition.

For the years ended December 31, 2024 and 2023, amortization expense was $339 million and $315 million, respectively.

The following table presents estimated future amortization expense.

December 31, (in millions)	Finite-lived intangible assets
2025	$ 288
2026	285
2027	284
2028	267
2029	261

Impairment testing

The Firm’s finite-lived and indefinite-lived other intangible assets are assessed for impairment annually or more often if events or changes in circumstances indicate that the asset might be impaired. Once the Firm determines that an impairment exists for an intangible asset, the impairment is recognized in other expense.

Note 16 – Premises and equipment

Premises and equipment includes land carried at cost, as well as buildings, leasehold improvements, internal-use software and furniture and equipment carried at cost less accumulated depreciation and amortization. The Firm's operating lease right-of-use assets are also included in Premises and equipment. Refer to Note 18 for a further discussion of the Firm's right-of-use assets.

The following table presents certain components of Premises and equipment.

December 31, (in millions)	2024	2023
Land, buildings and leasehold improvements	**$ 16,874**	$ 14,862
Right-of-use assets[(a)]	**7,930**	7,917
Other premises and equipment[(b)]	**7,419**	7,378
Total premises and equipment	**$ 32,223**	$ 30,157

(a) Excluded $564 million and $514 million of right-of-use assets that were recorded in Other assets at December 31, 2024 and 2023, respectively.
(b) Other premises and equipment is comprised of internal-use software and furniture and equipment.

JPMorganChase computes depreciation using the straight-line method over the estimated useful life for buildings and furniture and equipment. The Firm depreciates leasehold improvements over the lesser of the remainder of the lease term or the estimated useful life. The Firm also capitalizes certain costs associated with the acquisition or development of internal-use software. Once the software is ready for its intended use, these costs are amortized on a straight-line basis over the software's expected useful life. The estimated useful lives range from 10 to 50 years for buildings and leasehold improvements, and 3 to 10 years for internal-use software and furniture and equipment.

Impairment is assessed when events or changes in circumstances indicate that the carrying value of an asset may not be fully recoverable.

Note 17 – Deposits

As of December 31, 2024 and 2023, noninterest-bearing and interest-bearing deposits were as follows:

December 31, (in millions)	2024	2023
U.S. offices		
Noninterest-bearing (included **$28,904** and $75,393 at fair value)[(a)]	**$ 592,500**	$ 643,748
Interest-bearing (included **$1,101** and $573 at fair value)[(a)]	**1,345,914**	1,303,100
Total deposits in U.S. offices	**1,938,414**	1,946,848
Non-U.S. offices		
Noninterest-bearing (included **$2,255** and $1,737 at fair value)[(a)]	**26,806**	23,097
Interest-bearing (included **$1,508** and $681 at fair value)[(a)]	**440,812**	430,743
Total deposits in non-U.S. offices	**467,618**	453,840
Total deposits	**$2,406,032**	$ 2,400,688

(a) Includes structured notes classified as deposits for which the fair value option has been elected. Refer to Note 3 for further discussion.

As of December 31, 2024 and 2023, time deposits in denominations that met or exceeded the insured limit were as follows:

December 31, (in millions)	2024	2023
U.S. offices	**$ 149,239**	$ 132,654
Non-U.S. offices[(a)]	**92,639**	90,187
Total	**$ 241,878**	$ 222,841

(a) Represents all time deposits in non-U.S. offices as these deposits typically exceed the insured limit.

As of December 31, 2024, the remaining maturities of interest-bearing time deposits were as follows:

December 31, (in millions)	U.S.	Non-U.S.	Total
2025	$222,676	$ 89,427	$ 312,103
2026	749	87	836
2027	482	2	484
2028	149	18	167
2029	314	721	1,035
After 5 years	162	129	291
Total	$224,532	$ 90,384	$ 314,916

Note 18 - Leases

Firm as lessee

At December 31, 2024, JPMorganChase and its subsidiaries were obligated under a number of noncancellable leases, predominantly operating leases for premises and equipment used primarily for business purposes. These leases generally have terms of 20 years or less, determined based on the contractual maturity of the lease, and include periods covered by options to extend or terminate the lease when the Firm is reasonably certain that it will exercise those options. All leases with lease terms greater than twelve months are reported as a lease liability with a corresponding right-of-use ("ROU") asset. None of these lease agreements impose restrictions on the Firm's ability to pay dividends, engage in debt or equity financing transactions or enter into further lease agreements. Certain of these leases contain escalation clauses that will increase rental payments based on maintenance, utility and tax increases, which are non-lease components. The Firm elected not to separate lease and non-lease components of a contract for its real estate leases. As such, real estate lease payments represent payments on both lease and non-lease components.

Operating lease liabilities and ROU assets are recognized at the lease commencement date based on the present value of the future minimum lease payments over the lease term. The future lease payments are discounted at a rate that estimates the Firm's collateralized borrowing rate for financing instruments of a similar term and are included in accounts payable and other liabilities. The operating lease ROU assets, predominantly included in premises and equipment, also include any lease prepayments made, plus initial direct costs incurred, less any lease incentives received. Rental expense associated with operating leases is recognized on a straight-line basis over the lease term, and generally included in occupancy expense in the Consolidated statements of income.

The carrying values of the Firm's operating leases were as follows:

December 31, (in millions, except where otherwise noted)	2024	2023
Right-of-use assets	$ 8,494	$ 8,431
Lease liabilities	8,900	8,833
Weighted average remaining lease term (in years)	8.3	8.4
Weighted average discount rate	4.24 %	4.01 %
Supplemental cash flow information		
Cash paid for amounts included in the measurement of lease liabilities - operating cash flows	$ 1,734	$ 1,662
Supplemental non-cash information		
Right-of-use assets obtained in exchange for operating lease obligations	$ 1,565	$ 2,094

Year ended December 31, (in millions)	2024	2023
Rental expense		
Gross rental expense	$ 2,231	$ 2,079
Sublease rental income	(41)	(72)
Net rental expense	$ 2,190	$ 2,007

The following table presents future payments under operating leases as of December 31, 2024.

Year ended December 31, (in millions)	
2025	1,709
2026	1,553
2027	1,412
2028	1,248
2029	1,048
After 2029	3,721
Total future minimum lease payments	10,691
Less: Imputed interest	(1,791)
Total	$ 8,900

In addition to the table above, as of December 31, 2024, the Firm had additional future operating lease commitments of $887 million that were signed but had not yet commenced. These operating leases will commence between 2025 and 2027 with lease terms up to 21 years.

Firm as lessor

The Firm provides auto and equipment lease financing to its customers through lease arrangements with lease terms that may contain renewal, termination and/or purchase options. The Firm's lease financings are predominantly auto operating leases. These assets subject to operating leases are recognized in other assets on the Firm's Consolidated balance sheets and are depreciated on a straight-line basis over the lease term to reduce the asset to its estimated residual value. Depreciation expense is included in technology, communications and equipment expense in the Consolidated statements of income. The Firm's lease income is generally recognized on a straight-line basis over the lease term and is included in other income in the Consolidated statements of income.

On a periodic basis, the Firm assesses leased assets for impairment, and if the carrying amount of the leased asset exceeds the undiscounted cash flows from the lease payments and the estimated residual value upon disposition of the leased asset, an impairment is recognized.

The risk of loss on auto and equipment leased assets relating to the residual value of the leased assets is monitored through projections of the asset residual values at lease origination and periodic review of residual values, and is mitigated through arrangements with certain manufacturers or lessees.

The following table presents the carrying value of assets subject to leases reported on the Consolidated balance sheets.

December 31, (in millions)	**2024**	2023
Carrying value of assets subject to operating leases, net of accumulated depreciation	**$ 12,988**	$ 10,663
Accumulated depreciation	**2,509**	3,288

The following table presents the Firm's operating lease income and the related depreciation expense on the Consolidated statements of income.

Year ended December 31, (in millions)	**2024**	2023	2022
Operating lease income	**$ 2,795**	$ 2,843	$ 3,654
Depreciation expense	**1,685**	1,778	2,475

The following table presents future receipts under operating leases as of December 31, 2024.

Year ended December 31, (in millions)	
2025	$ 2,381
2026	1,707
2027	704
2028	47
2029	5
After 2029	4
Total future minimum lease receipts	**$ 4,848**

Note 19 – Accounts payable and other liabilities

Accounts payable and other liabilities consist of brokerage payables, which include payables to customers and payables related to security purchases that did not settle, as well as other accrued expenses, such as compensation accruals, credit card rewards liability, merchant servicing payables, operating lease liabilities, accrued interest payables, income tax payables and litigation reserves.

The following table presents the components of accounts payable and other liabilities.

December 31, (in millions)	**2024**	2023
Brokerage payables	**$ 153,153**	$ 161,960
Other payables and liabilities[(a)]	**127,519**	128,347
Total accounts payable and other liabilities	**$ 280,672**	$ 290,307

(a) Includes credit card rewards liability of $14.4 billion and $13.2 billion at December 31, 2024 and 2023, respectively.

The credit card rewards liability represents the estimated cost of rewards points earned and expected to be redeemed by cardholders. The liability is accrued as the cardholder earns the benefit and is reduced when the cardholder redeems points. The redemption rate and cost per point assumptions are key assumptions to estimate the liability and the current period impact is recognized in Card Income.

Refer to Notes 7, 18, 25 and 30 for additional information on accrued interest, operating lease liabilities, income taxes and litigation reserves, respectively.

Note 20 – Long-term debt

JPMorganChase issues long-term debt denominated in various currencies, predominantly U.S. dollars, with both fixed and variable interest rates. Included in senior and subordinated debt below are various equity-linked or other indexed instruments, which the Firm has elected to measure at fair value. Changes in fair value are recorded in principal transactions revenue in the Consolidated statements of income, except for unrealized gains/(losses) due to DVA which are recorded in OCI. The following table is a summary of long-term debt carrying values (including unamortized premiums and discounts, issuance costs, valuation adjustments and fair value adjustments, where applicable) by remaining contractual maturity as of December 31, 2024.

By remaining maturity at December 31, (in millions, except rates)		2024				2023
		Under 1 year	1-5 years	After 5 years	**Total**	Total
Parent company						
Senior debt:	Fixed rate	**$ 7,112**	**$ 90,132**	**$ 117,667**	**$ 214,911**	$ 200,984
	Variable rate	**255**	**6,838**	**1,562**	**8,655**	8,105
	Interest rates[f]	**3.04 %**	**3.40 %**	**4.02 %**	**3.71 %**	3.32 %
Subordinated debt:	Fixed rate	**$ 302**	**$ 5,582**	**$ 8,573**	**$ 14,457**	$ 17,725
	Variable rate	**—**	**—**	**—**	**—**	—
	Interest rates[f]	**7.75 %**	**4.72 %**	**4.69 %**	**4.76 %**	4.62 %
	Subtotal	**$ 7,669**	**$ 102,552**	**$ 127,802**	**$ 238,023**	$ 226,814
Subsidiaries						
Federal Home Loan Banks advances:	Fixed rate	**$ 7,582**	**$ 1,651**	**$ 24**	**$ 9,257**	$ 23,246
	Variable rate	**4,000**	**16,000**	**—**	**20,000**	18,000
	Interest rates[f]	**4.42 %**	**4.84 %**	**5.91 %**	**4.67 %**	4.89 %
Purchase Money Note[a]:	Fixed rate	**$ —**	**$ 49,208**	**$ —**	**$ 49,208**	$ 48,989
	Interest rates[f]	**— %**	**3.40 %**	**— %**	**3.40 %**	3.40 %
Senior debt:	Fixed rate	**$ 2,361**	**$ 16,695**	**$ 7,489**	**$ 26,545**	$ 20,745
	Variable rate	**19,350**	**30,981**	**6,451**	**56,782**	52,048
	Interest rates[f]	**5.39 %**	**5.18%**	**1.44 %**	**3.81 %**	3.91 %
Subordinated debt:	Fixed rate	**$ —**	**$ —**	**$ —**	**$ —**	$ 255
	Variable rate	**—**	**—**	**—**	**—**	—
	Interest rates[f]	**— %**	**— %**	**— %**	**— %**	8.25 %
	Subtotal	**$ 33,293**	**$ 114,535**	**$ 13,964**	**$ 161,792**	$ 163,283
Junior subordinated debt:	Fixed rate	**$ —**	**$ —**	**$ 488**	**$ 488**	$ 518
	Variable rate	**—**	**421**	**694**	**1,115**	1,210
	Interest rates[f]	**— %**	**5.35 %**	**7.01 %**	**6.58 %**	7.14 %
	Subtotal	**$ —**	**$ 421**	**$ 1,182**	**$ 1,603**	$ 1,728
Total long-term debt[b][c][d]		**$ 40,962**	**$ 217,508**	**$ 142,948**	**$ 401,418** [g][h]	$ 391,825
Long-term beneficial interests:	Fixed rate	**$ 999**	**$ 4,313**	**$ —**	**$ 5,312**	$ 2,998
	Variable rate	**—**	**27**	**139**	**166**	125
	Interest rates[f]	**3.97 %**	**4.82 %**	**2.92 %**	**4.62 %**	4.69 %
Total long-term beneficial interests[e]		**$ 999**	**$ 4,340**	**$ 139**	**$ 5,478**	$ 3,123

(a) Reflects the Purchase Money Note associated with First Republic. Refer to Note 34 for additional information.
(b) Included long-term debt of $80.9 billion and $93.0 billion secured by assets totaling $185.5 billion and $218.5 billion at December 31, 2024 and 2023, respectively. The amount of long-term debt secured by assets does not include amounts related to hybrid instruments.
(c) Included $100.8 billion and $87.9 billion of long-term debt accounted for at fair value at December 31, 2024 and 2023, respectively.
(d) Included $13.5 billion and $12.5 billion of outstanding zero-coupon notes at December 31, 2024 and 2023, respectively. The aggregate principal amount of these notes at their respective maturities is $50.2 billion and $47.9 billion, respectively. The aggregate principal amount reflects the contractual principal payment at maturity, which may exceed the contractual principal payment at the Firm's next call date, if applicable.
(e) Included on the Consolidated balance sheets in beneficial interests issued by consolidated VIEs. Also included amounts accounted for at fair value which were not material as of December 31, 2024 and 2023. Excluded short-term commercial paper and other short-term beneficial interests of $21.8 billion and $19.9 billion at December 31, 2024 and 2023, respectively.
(f) The interest rates shown are the weighted average of contractual rates in effect at December 31, 2024 and 2023, respectively, including non-U.S. dollar fixed- and variable-rate issuances, which excludes the effects of the associated derivative instruments used in hedge accounting relationships, if applicable. The interest rates shown exclude structured notes accounted for at fair value.
(g) As of December 31, 2024, long-term debt in the aggregate of $297.1 billion was redeemable at the option of JPMorganChase, in whole or in part, prior to maturity, based on the terms specified in the respective instruments.
(h) The aggregate carrying values of debt that matures in each of the five years subsequent to 2024 is $41.0 billion in 2025, $64.5 billion in 2026, $32.7 billion in 2027, $93.4 billion in 2028 and $26.9 billion in 2029.

The weighted-average contractual interest rates for total long-term debt excluding structured notes accounted for at fair value were 3.82% and 3.65% as of December 31, 2024 and 2023, respectively. In order to modify exposure to interest rate and currency exchange rate movements, JPMorganChase utilizes derivative instruments, primarily interest rate and cross-currency interest rate swaps, in conjunction with some of its debt issuances. The use of these instruments modifies the Firm's interest expense on the associated debt. The modified weighted-average interest rates for total long-term debt, including the effects of related derivative instruments, were 5.15% and 5.20% as of December 31, 2024 and 2023, respectively.

JPMorgan Chase & Co. has guaranteed certain long-term debt of its subsidiaries, including structured notes. These guarantees rank pari passu with the Firm's other unsecured and unsubordinated indebtedness. The amount of such guaranteed long-term debt and structured notes was $41.2 billion and $41.1 billion at December 31, 2024 and 2023, respectively.

The Firm's unsecured debt does not contain requirements that would call for an acceleration of payments, maturities or changes in the structure of the existing debt, provide any limitations on future borrowings or require additional collateral, based on unfavorable changes in the Firm's credit ratings, financial ratios, earnings or stock price.

Note 21 – Preferred stock

At December 31, 2024 and 2023, JPMorganChase was authorized to issue 200 million shares of preferred stock, in one or more series, with a par value of $1 per share. In the event of a liquidation or dissolution of the Firm, JPMorganChase's preferred stock then outstanding takes precedence over the Firm's common stock with respect to the payment of dividends and the distribution of assets.

The following is a summary of JPMorganChase's non-cumulative preferred stock outstanding as of December 31, 2024 and 2023, and the quarterly dividend declarations for the years ended December 31, 2024, 2023 and 2022.

	Shares[a]		Carrying value (in millions)			Contractual rate in effect at December 31, 2024	Earliest redemption date[b]	Floating annualized rate[c]	Dividend declared per share[d]		
	December 31,		December 31,						Year ended December 31,		
	2024	2023	2024	2023	Issue date				2024	2023	2022
Fixed-rate:											
Series DD	**169,625**	169,625	**1,696**	1,696	9/21/2018	**5.750**	12/1/2023	NA	**575.00**	575.00	575.00
Series EE	**185,000**	185,000	**1,850**	1,850	1/24/2019	**6.000**	3/1/2024	NA	**600.00**	600.00	600.00
Series GG	**90,000**	90,000	**900**	900	11/7/2019	**4.750**	12/1/2024	NA	**475.00**	475.00	475.00
Series JJ	**150,000**	150,000	**1,500**	1,500	3/17/2021	**4.550**	6/1/2026	NA	**455.00**	455.00	455.00
Series LL	**185,000**	185,000	**1,850**	1,850	5/20/2021	**4.625**	6/1/2026	NA	**462.52**	462.52	462.52
Series MM	**200,000**	200,000	**2,000**	2,000	7/29/2021	**4.200**	9/1/2026	NA	**420.00**	420.00	420.00
Fixed-to-floating rate:											
Series I	**—**	—	**$ —**	$ —	4/23/2008	**— %**	4/30/2018	— %	**$ —**	$ —	$ 375.03
Series Q	**—**	150,000	**—**	1,500	4/23/2013	**—**	5/1/2023	SOFR + 3.25	**220.45**	801.41 (g)	515.00
Series R	**—**	150,000	**—**	1,500	7/29/2013	**—**	8/1/2023	SOFR + 3.30	**221.70**	756.73 (h)	600.00
Series S	**—**	200,000	**—**	2,000	1/22/2014	**—**	2/1/2024	SOFR + 3.78	**233.70** (e)	675.00	675.00
Series U	**—**	100,000	**—**	1,000	3/10/2014	**—**	4/30/2024	SOFR + 3.33	**153.13**	612.50	612.50
Series V	**—**	—	**—**	—	6/9/2014	**—**	7/1/2019	—	**—**	—	340.91
Series X	**—**	160,000	**—**	1,600	9/23/2014	**—**	10/1/2024	SOFR + 3.33	**457.50**	610.00	610.00
Series CC	**125,750**	125,750	**1,258**	1,258	10/20/2017	**SOFR + 2.58**	11/1/2022	SOFR + 2.58	**812.73**	804.08	526.27 (i)
Series FF	**—**	225,000	**—**	2,250	7/31/2019	**—**	8/1/2024	SOFR + 3.38	**250.00**	500.00	500.00
Series HH	**300,000**	300,000	**3,000**	3,000	1/23/2020	**4.600**	2/1/2025	SOFR + 3.125	**460.00**	460.00	460.00
Series II	**150,000**	150,000	**1,500**	1,500	2/24/2020	**4.000**	4/1/2025	SOFR + 2.745	**400.00**	400.00	400.00
Series KK	**200,000**	200,000	**2,000**	2,000	5/12/2021	**3.650**	6/1/2026	CMT + 2.85	**365.00**	365.00	365.00
Series NN	**250,000**	NA	**2,496**	NA	3/12/2024	**6.875**	6/1/2029	CMT + 2.737	**494.63** (f)	NA	NA
Total preferred stock	**2,005,375**	2,740,375	**$20,050**	$ 27,404							

(a) Represented by depositary shares.
(b) Each series of fixed-to-floating rate preferred stock converts to a floating rate at the earliest redemption date.
(c) Effective June 30, 2023, CME Term SOFR became the replacement reference rate for fixed-to-floating rate preferred stock issued by the Firm that formerly referenced U.S. dollar LIBOR. References in the table to "SOFR" mean a floating annualized rate equal to three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spreads noted. References to "CMT" mean a floating annualized rate equal to the five-year Constant Maturity Treasury ("CMT") rate plus the spreads noted.
(d) Dividends on preferred stock are discretionary and non-cumulative. When declared, dividends are declared quarterly. Dividends are payable quarterly on fixed-rate preferred stock. Dividends are payable semiannually on fixed-to-floating rate preferred stock while at a fixed rate, and payable quarterly after converting to a floating rate.
(e) The dividend rate for Series S preferred stock became floating and payable quarterly starting on February 1, 2024; prior to which the dividend rate was fixed at 6.75% or $337.50 per share payable semiannually. The dividend rate for each quarterly dividend period commencing on February 1, 2024 was three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spread of 3.78%.
(f) The initial dividend declared was prorated based on the number of days outstanding for the period. Dividends were declared quarterly thereafter at the contractual rate.
(g) The dividend rate for Series Q preferred stock became floating and payable quarterly starting on May 1, 2023; prior to which the dividend rate was fixed at 5.15% or $257.50 per share payable semiannually. The dividend rate for each quarterly dividend period commencing on August 1, 2023 was three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spread of 3.25%.
(h) The dividend rate for Series R preferred stock became floating and payable quarterly starting on August 1, 2023; prior to which the dividend rate was fixed at 6.00% or $300.00 per share payable semiannually. The dividend rate for each quarterly dividend period commencing on August 1, 2023 was three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spread of 3.30%.
(i) The dividend rate for Series CC preferred stock became floating and payable quarterly starting on November 1, 2022; prior to which the dividend rate was fixed at 4.625% or $231.25 per share payable semiannually. The dividend rate for each quarterly dividend period commencing on August 1, 2023 was three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spread of 2.58%.

Each series of preferred stock has a liquidation value and redemption price per share of $10,000, plus accrued but unpaid dividends. The aggregate liquidation value was $20.2 billion at December 31, 2024.

Issuances

On February 4, 2025, the Firm issued $3.0 billion of fixed-rate reset non-cumulative preferred stock, Series OO.

On March 12, 2024, the Firm issued $2.5 billion of fixed-rate reset non-cumulative preferred stock, Series NN.

Redemptions

On February 1, 2025, the Firm redeemed all $3.0 billion of its fixed-to-floating rate non-cumulative preferred stock, Series HH.

On October 1, 2024, the Firm redeemed all $1.6 billion of its fixed-to-floating rate non-cumulative preferred stock, Series X.

On August 1, 2024, the Firm redeemed all $2.3 billion of its fixed-to-floating rate non-cumulative preferred stock, Series FF.

On May 1, 2024, the Firm redeemed all $5.0 billion of its fixed-to-floating rate non-cumulative preferred stock, Series Q, Series R and Series S.

On April 30, 2024, the Firm redeemed all $1.0 billion of its fixed-to-floating rate non-cumulative preferred stock, Series U.

Redemption rights

Each series of the Firm’s preferred stock may be redeemed on any dividend payment date on or after the earliest redemption date for that series. All outstanding preferred stock series may also be redeemed following a “capital treatment event,” as described in the terms of each series. Any redemption of the Firm’s preferred stock is subject to non-objection from the Board of Governors of the Federal Reserve System (the “Federal Reserve”).

Note 22 – Common stock

At December 31, 2024 and 2023, JPMorganChase was authorized to issue 9.0 billion shares of common stock with a par value of $1 per share.

Common shares issued which were reissued from treasury by the Firm during the years ended December 31, 2024, 2023 and 2022 were as follows.

Year ended December 31, (in millions)	2024	2023	2022
Total issued – balance at January 1	**4,104.9**	4,104.9	4,104.9
Treasury – balance at January 1	**(1,228.3)**	(1,170.7)	(1,160.8)
Repurchase	**(91.7)**	(69.5)	(23.1)
Reissuance:			
Employee benefits and compensation plans	**11.9**	10.9	12.0
Employee stock purchase plans	**0.8**	1.0	1.2
Total reissuance	**12.7**	11.9	13.2
Total treasury – balance at December 31	**(1,307.3)**	(1,228.3)	(1,170.7)
Outstanding at December 31	**2,797.6**	2,876.6	2,934.2

On June 28, 2024, the Firm announced that its Board of Directors had authorized a new $30 billion common share repurchase program, effective July 1, 2024. Through June 30, 2024, the Firm was authorized to purchase up to $30 billion of common shares under its previously-approved common share repurchase program that was announced on April 13, 2022.

The following table sets forth the Firm's repurchases of common stock for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions)	2024	2023	2022[(b)]
Total number of shares of common stock repurchased	**91.7**	69.5	23.1
Aggregate purchase price of common stock repurchases[(a)]	**$18,841**	$ 9,898	$ 3,122

(a) Excludes excise tax and commissions. As part of the Inflation Reduction Act of 2022, a 1% excise tax is imposed on net share repurchases commencing January 1, 2023.
(b) In the second half of 2022, the Firm temporarily suspended share repurchases, which it resumed in the first quarter of 2023 under its common share repurchase program.

The Board of Directors' authorization to repurchase common shares is utilized at management's discretion. The $30 billion common share repurchase program approved by the Board of Directors does not establish specific price targets or timetables. Management determines the amount and timing of common share repurchases based on various factors, including market conditions; legal and regulatory considerations affecting the amount and timing of repurchase activity; the Firm's capital position (taking into account goodwill and intangibles); organic capital generation; current and proposed future capital requirements; and other investment opportunities. The amount of common shares that the Firm repurchases in any period may be substantially more or less than the amounts estimated or actually repurchased in prior periods, reflecting the dynamic nature of the decision-making process. The Firm's common share repurchases may be suspended by management at any time; and may be executed through open market purchases or privately negotiated transactions, or utilizing Rule 10b5-1 plans, which are written trading plans that the Firm may enter into from time to time under Rule 10b5-1 of the Securities Exchange Act of 1934 and which allow the Firm to repurchase its common shares during periods when it may otherwise not be repurchasing common shares — for example, during internal trading blackout periods.

As of December 31, 2024, approximately 58.8 million shares of common stock were reserved for issuance under various employee incentive, compensation, option and stock purchase plans, and directors' compensation plans.

Note 23 – Earnings per share

Basic earnings per share ("EPS") is calculated using the two-class method. Under the two-class method, all earnings (distributed and undistributed) are allocated to common stock and participating securities. JPMorganChase grants RSUs under its share-based compensation programs, predominantly all of which entitle recipients to receive nonforfeitable dividends during the vesting period on a basis equivalent to dividends paid to holders of the Firm's common stock. These unvested RSUs meet the definition of participating securities based on their respective rights to receive nonforfeitable dividends, and they are treated as a separate class of securities in computing basic EPS. Participating securities are not included as incremental shares in computing diluted EPS; refer to Note 9 for additional information.

Diluted EPS incorporates the potential impact of contingently issuable shares, including awards which require future service as a condition of delivery of the underlying common stock. Diluted EPS is calculated under both the two-class and treasury stock methods, and the more dilutive amount is reported. For each of the periods presented in the table below, diluted EPS calculated under the two-class method was more dilutive.

The following table presents the calculation of net income applicable to common stockholders and basic and diluted EPS for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions, except per share amounts)	**2024**	2023	2022
Basic earnings per share			
Net income	**$ 58,471**	$ 49,552	$ 37,676
Less: Preferred stock dividends	**1,259**	1,501	1,595
Net income applicable to common equity	**57,212**	48,051	36,081
Less: Dividends and undistributed earnings allocated to participating securities	**344**	291	189
Net income applicable to common stockholders	**$ 56,868**	$ 47,760	$ 35,892
Total weighted-average basic shares outstanding	**2,873.9**	2,938.6	2,965.8
Net income per share	**$ 19.79**	$ 16.25	$ 12.10
Diluted earnings per share			
Net income applicable to common stockholders	**$ 56,868**	$ 47,760	$ 35,892
Total weighted-average basic shares outstanding	**2,873.9**	2,938.6	2,965.8
Add: Dilutive impact of unvested PSUs, nondividend-earning RSUs and SARs	**5.1**	4.5	4.2
Total weighted-average diluted shares outstanding	**2,879.0**	2,943.1	2,970.0
Net income per share	**$ 19.75**	$ 16.23	$ 12.09

Note 24 – Accumulated other comprehensive income/(loss)

AOCI includes the after-tax change in unrealized gains and losses on investment securities, foreign currency translation adjustments (including the impact of related derivatives), fair value changes of excluded components on fair value hedges, cash flow hedging activities, net gain/(loss) related to the Firm's defined benefit pension and OPEB plans, and fair value option-elected liabilities arising from changes in the Firm's own credit risk (DVA).

Year ended December 31, (in millions)	Unrealized gains/(losses) on investment securities	Translation adjustments, net of hedges	Fair value hedges	Cash flow hedges	Defined benefit pension and OPEB plans	DVA on fair value option elected liabilities	Accumulated other comprehensive income/(loss)
Balance at December 31, 2021	$ 2,640	$ (934)	$ (131)	$ (296)	$ (210)	$ (1,153)	$ (84)
Net change	(11,764)	(611)	98	(5,360)	(1,241)	1,621	(17,257)
Balance at December 31, 2022	$ (9,124) [(a)]	$ (1,545)	$ (33)	$ (5,656)	$ (1,451)	$ 468	$ (17,341)
Net change	5,381	329	(101)	1,724	373	(808)	6,898
Balance at December 31, 2023	$ (3,743) [(a)]	$ (1,216)	$ (134)	$ (3,932)	$ (1,078)	$ (340)	$ (10,443)
Net change	(87)	(858)	(87)	(882)	(63)	(36)	(2,013)
Balance at December 31, 2024	**$ (3,830)** [(a)]	**$ (2,074)**	**$ (221)**	**$ (4,814)**	**$ (1,141)**	**$ (376)**	**$ (12,456)**

(a) Included after-tax net unamortized unrealized gains/(losses) of $(651) million, $(895) million, and $(1.3) billion for the years ended 2024, 2023 and 2022, respectively, related to AFS securities that have been transferred to HTM. As of December 31, 2023 included after-tax net unamortized unrealized gains/(losses) of $(29) million related to HTM securities that have been transferred to AFS as permitted by the new hedge accounting guidance adopted on January 1, 2023. Refer to Note 10 for further information.

The following table presents the pre-tax and after-tax changes in the components of OCI.

	2024			2023			2022		
Year ended December 31, (in millions)	Pre-tax	Tax effect	After-tax	Pre-tax	Tax effect	After-tax	Pre-tax	Tax effect	After-tax
Unrealized gains/(losses) on investment securities:									
Net unrealized gains/(losses) arising during the period	**$ (1,135)**	**$ 274**	**$ (861)**	$ 3,891	$ (922)	$ 2,969	$ (17,862)	$ 4,290	$ (13,572)
Reclassification adjustment for realized (gains)/losses included in net income[(a)]	**1,021**	**(247)**	**774**	3,180	(768)	2,412	2,380	(572)	1,808
Net change	**(114)**	**27**	**(87)**	7,071	(1,690)	5,381	(15,482)	3,718	(11,764)
Translation adjustments[(b)]:									
Translation	**(4,385)**	**250**	**(4,135)**	1,714	(95)	1,619	(3,574)	265	(3,309)
Hedges	**4,322**	**(1,045)**	**3,277**	(1,697)	407	(1,290)	3,553	(855)	2,698
Net change	**(63)**	**(795)**	**(858)**	17	312	329	(21)	(590)	(611)
Fair value hedges, net change[(c)]	**(115)**	**28**	**(87)**	(134)	33	(101)	130	(32)	98
Cash flow hedges:									
Net unrealized gains/(losses) arising during the period	**(3,742)**	**904**	**(2,838)**	483	(114)	369	(7,473)	1,794	(5,679)
Reclassification adjustment for realized (gains)/losses included in net income[(d)]	**2,579**	**(623)**	**1,956**	1,775	(420)	1,355	420	(101)	319
Net change	**(1,163)**	**281**	**(882)**	2,258	(534)	1,724	(7,053)	1,693	(5,360)
Defined benefit pension and OPEB plans, net change[(e)]	**(131)**	**68**	**(63)**	421	(48)	373	(1,459)	218	(1,241)
DVA on fair value option elected liabilities, net change	**(45)**	**9**	**(36)**	(1,066)	258	(808)	2,141	(520)	1,621
Total other comprehensive income/(loss)	**$ (1,631)**	**$ (382)**	**$ (2,013)**	$ 8,567	$ (1,669)	$ 6,898	$ (21,744)	$ 4,487	$ (17,257)

(a) The pre-tax amount is reported in Investment securities gains/(losses) in the Consolidated statements of income.
(b) Reclassifications of pre-tax realized gains/(losses) on translation adjustments and related hedges are reported in other income/expense in the Consolidated statements of income. During the year ended December 31, 2024, the Firm reclassified a net pre-tax gain of $7 million to other income/expense, of which $89 million gain related to net investment hedges and $(82) million loss related to cumulative translation adjustments. During the year ended December 31, 2023, the Firm reclassified a net pre-tax loss of $(3) million. During the year ended December 31, 2022, the Firm reclassified a net pre-tax loss of $(8) million.
(c) Represents changes in fair value of cross-currency swaps attributable to changes in cross-currency basis spreads, which are excluded from the assessment of hedge effectiveness and recorded in other comprehensive income. The initial cost of cross-currency basis spreads is recognized in earnings as part of the accrual of interest on the cross-currency swaps.
(d) The pre-tax amounts are primarily recorded in noninterest revenue, net interest income and compensation expense in the Consolidated statements of income.
(e) During the year ended December 31, 2022, a remeasurement of the Firm's U.S. principal defined benefit plan in the third quarter, was required as a result of a pension settlement. The remeasurement resulted in a net decrease of $1.4 billion in pre-tax AOCI.

Note 25 – Income taxes

JPMorganChase and its eligible subsidiaries file a consolidated U.S. federal income tax return. JPMorganChase uses the asset and liability method to provide for income taxes on all transactions recorded in the Consolidated Financial Statements. This method requires that income taxes reflect the expected future tax consequences of temporary differences between the carrying amounts of assets or liabilities for book and tax purposes. Accordingly, a deferred tax asset or liability for each temporary difference is determined based on the tax rates that the Firm expects to be in effect when the underlying items of income and expense are realized. JPMorganChase's expense for income taxes includes the current and deferred portions of that expense. A valuation allowance is established to reduce deferred tax assets to the amount the Firm expects to realize.

Due to the inherent complexities arising from the nature of the Firm's businesses, and from conducting business and being taxed in a substantial number of jurisdictions, significant judgments and estimates are required to be made. Agreement of tax liabilities between JPMorganChase and the many tax jurisdictions in which the Firm files tax returns may not be finalized for several years. Thus, the Firm's final tax-related assets and liabilities may ultimately be different from those currently reported.

Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance, under the modified retrospective method. Refer to Notes 1, 6 and 14 for additional information.

Effective tax rate and expense

The following table presents a reconciliation of the applicable statutory U.S. federal income tax rate to the effective tax rate.

Effective tax rate

Year ended December 31,	**2024**	2023	2022
Statutory U.S. federal tax rate	**21.0 %**	21.0 %	21.0 %
Increase/(decrease) in tax rate resulting from:			
U.S. state and local income taxes, net of U.S. federal income tax benefit	**3.1**	2.8	3.5
Tax-exempt income	**(0.7)**	(0.9)	(0.9)
Non-U.S. earnings	**1.4**	1.5	0.4
Business tax credits	**(2.4)**	(4.4)	(5.4)
Other, net	**(0.3)**	(0.4)	(0.2)
Effective tax rate	**22.1 %**	19.6 % [(a)]	18.4 %

(a) Income tax expense associated with the First Republic acquisition was reflected in the estimated bargain purchase gain, which resulted in a reduction in the Firm's effective tax rate.

The following table reflects the components of income tax expense/(benefit) included in the Consolidated statements of income.

Income tax expense/(benefit)

Year ended December 31, (in millions)	**2024**	2023	2022
Current income tax expense/(benefit)			
U.S. federal	**$ 7,091**	$ 8,973	$ 5,606
Non-U.S.	**4,753**	4,355	2,992
U.S. state and local	**2,762**	3,266	2,630
Total current income tax expense/(benefit)	**14,606**	16,594	11,228
Deferred income tax expense/(benefit)			
U.S. federal	**1,771**	(3,475)	(2,004)
Non-U.S.	**72**	35	(154)
U.S. state and local	**161**	(1,094)	(580)
Total deferred income tax expense/(benefit)	**2,004**	(4,534)	(2,738)
Total income tax expense	**$ 16,610**	$ 12,060	$ 8,490

Total income tax expense includes $314 million, $68 million and $331 million of tax benefits in 2024, 2023, and 2022, respectively, resulting from the resolution of tax audits.

Tax effect of items recorded in stockholders' equity

The preceding table does not reflect the tax effect of certain items that are recorded each period directly in stockholders' equity, which are predominantly reflected in OCI as disclosed in Note 24. For the year ended December 31, 2024, stockholders' equity reflected the tax effect associated with the Firm's adoption of the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance. For the year ended December 31, 2023, stockholders' equity reflected the tax effect associated with the Firm's adoption of the TDR accounting guidance. Both of the respective adoptions were recognized in retained earnings. Refer to Note 1, 6 and 14 for further information.

Results from U.S. and non-U.S. earnings

The following table presents the U.S. and non-U.S. components of income before income tax expense.

Year ended December 31, (in millions)	**2024**	2023	2022
U.S.	**$ 59,472**	$46,868	$34,626
Non-U.S.[(a)]	**15,609**	14,744	11,540
Income before income tax expense	**$ 75,081**	$ 61,612	$46,166

(a) For purposes of this table, non-U.S. income is defined as income generated from operations located outside the U.S.

The Firm will recognize any U.S. income tax expense it may incur on global intangible low tax income as income tax expense in the period in which the tax is incurred.

Deferred taxes

Deferred income tax expense/(benefit) reflects the differences between assets and liabilities measured for financial reporting purposes versus income tax return purposes. Deferred tax assets are recognized if, in management's judgment, their realizability is determined to be more likely than not. If a deferred tax asset is determined to be unrealizable, a valuation allowance is established. The significant components of deferred tax assets and liabilities are reflected in the following table, the net deferred tax assets are reflected in other assets on the Firm's Consolidated balance sheets.

December 31, (in millions)	2024	2023
Deferred tax assets		
Allowance for loan losses	**$ 6,117**	$ 5,809
Employee benefits	**1,165**	1,247
Accrued expenses and other	**8,881**	9,887 (a)
Depreciation and amortization	**386**	—
Non-U.S. operations	**948**	860
Tax attribute carryforwards	**352**	290
Gross deferred tax assets	**17,849**	18,093
Valuation allowance	**(249)**	(183)
Deferred tax assets, net of valuation allowance	**$ 17,600**	$ 17,910
Deferred tax liabilities		
Depreciation and amortization	**$ —**	$ 779
Mortgage servicing rights, net of hedges	**1,912**	1,794
Leasing transactions	**2,249**	2,254
Other, net	**1,264**	2,935
Gross deferred tax liabilities	**5,425**	7,762
Net deferred tax assets	**$ 12,175**	$ 10,148

(a) Includes the estimated net deferred tax asset associated with the First Republic acquisition.

JPMorganChase has recorded deferred tax assets of $352 million at December 31, 2024 in connection with tax attribute carryforwards. State and local capital loss carryforwards were $914 million, U.S. federal NOL carryforwards were $496 million, non-U.S. NOL carryforwards were $958 million, and other U.S. federal tax attributes were $111 million. If not utilized, a portion of the U.S. federal NOL carryforwards and other U.S. federal tax attributes will expire between 2026 and 2036 whereas others have an unlimited carryforward period. Similarly, certain non-U.S. NOL carryforwards will expire between 2026 and 2041 whereas others have an unlimited carryforward period. The state and local capital loss carryforwards will expire in 2026.

The valuation allowance at December 31, 2024, was predominantly driven by certain non-U.S. deferred tax assets, including NOL carryforwards.

Unrecognized tax benefits

At December 31, 2024, 2023 and 2022, JPMorganChase's unrecognized tax benefits, excluding related interest expense and penalties, were $6.2 billion, $5.4 billion and $5.0 billion, respectively, of which $4.4 billion, $3.9 billion and $3.8 billion, respectively, if recognized, would reduce the annual effective tax rate. Included in the amount of unrecognized tax benefits are certain items that would not affect the effective tax rate if they were recognized in the Consolidated statements of income. These unrecognized items include the tax effect of certain temporary differences, the portion of gross state and local unrecognized tax benefits that would be offset by the benefit from associated U.S. federal income tax deductions, and the portion of gross non-U.S. unrecognized tax benefits that would have offsets in other jurisdictions. JPMorganChase evaluates the need for changes in unrecognized tax benefits based on its anticipated tax return filing positions as part of its U.S. federal and state and local tax returns. In addition, the Firm is presently under audit by a number of taxing authorities, most notably by the Internal Revenue Service, as summarized in the Tax examination status table below. The evaluation of unrecognized tax benefits as well as the potential for audit settlements make it reasonably possible that over the next 12 months the gross balance of unrecognized tax benefits may increase or decrease by as much as approximately $2.4 billion. The change in the unrecognized tax benefit would result in a payment or income statement recognition.

The following table presents a reconciliation of the beginning and ending amount of unrecognized tax benefits.

(in millions)	**2024**	2023	2022
Balance at January 1,	**$ 5,401**	$ 5,043	$ 4,636
Increases based on tax positions related to the current period	**1,721**	1,440	1,234
Increases based on tax positions related to prior periods	**92**	37	123
Decreases based on tax positions related to prior periods	**(907)**	(1,110)	(824)
Decreases related to cash settlements with taxing authorities	**(148)**	(9)	(126)
Balance at December 31,	**$ 6,159**	$ 5,401	$ 5,043

After-tax interest expense/(benefit) and penalties related to income tax liabilities recognized in income tax expense were $288 million, $229 million and $141 million in 2024, 2023 and 2022, respectively.

At December 31, 2024 and 2023, in addition to the liability for unrecognized tax benefits, the Firm had accrued $1.7 billion and $1.6 billion, respectively, for income tax-related interest and penalties.

Tax examination status

JPMorganChase is continually under examination by the Internal Revenue Service, by taxing authorities throughout the world, and by many state and local jurisdictions throughout the U.S. The following table summarizes the status of tax years that remain subject to income tax examination of JPMorganChase and its consolidated subsidiaries by significant jurisdictions as of December 31, 2024.

	Periods under examination	Status
JPMorganChase – U.S.	2011 – 2013	Field examination of amended returns; certain matters at Appellate level
JPMorganChase – U.S.	2014 - 2020	Field examination of original and amended returns; certain matters at Appellate level
JPMorganChase – New York City	2015 - 2018	Field Examination
JPMorganChase – U.K.	2017 – 2022	Field examination of certain select entities

Note 26 – Restricted cash, other restricted assets and intercompany funds transfers

Restricted cash and other restricted assets

Certain of the Firm's cash and other assets are restricted as to withdrawal or usage. These restrictions are imposed by various regulatory authorities based on the particular activities of the Firm's subsidiaries.

The business of JPMorgan Chase Bank, N.A. is subject to examination and regulation by the OCC. The Bank is a member of the U.S. Federal Reserve System, and its deposits in the U.S. are insured by the FDIC, subject to applicable limits.

The Firm is required to maintain cash reserves at certain non-US central banks.

The Firm is also subject to rules and regulations established by other U.S. and non U.S. regulators. As part of its compliance with the respective regulatory requirements, the Firm's broker-dealer activities are subject to certain restrictions on cash and other assets.

The following table presents the components of the Firm's restricted cash:

December 31, (in billions)		2024		2023
Segregated for the benefit of securities and cleared derivative customers		**18.7**		10.3
Cash reserves at non-U.S. central banks and held for other general purposes		**8.8**		9.3
Total restricted cash[(a)]	**$**	**27.5**	$	19.6

(a) Comprises $26.1 billion and $18.2 billion in deposits with banks, and $1.4 billion and $1.4 billion in cash and due from banks on the Consolidated balance sheets as of December 31, 2024 and 2023, respectively.

Also, as of December 31, 2024 and 2023, the Firm had the following other restricted assets:

- Cash and securities pledged with clearing organizations for the benefit of customers of $40.7 billion and $40.5 billion, respectively.
- Securities with a fair value of $26.8 billion and $20.5 billion, respectively, were also restricted in relation to customer activity.

Intercompany funds transfers

Restrictions imposed by U.S. federal law prohibit JPMorgan Chase Bank, N.A., and its subsidiaries, from lending to JPMorgan Chase & Co. ("Parent Company") and certain of its affiliates unless the loans are secured in specified amounts. Such secured loans provided by any banking subsidiary to the Parent Company or to any particular affiliate, together with certain other transactions with such affiliate (collectively referred to as "covered transactions"), must be made on terms and conditions that are consistent with safe and sound banking practices. In addition, unless collateralized with cash or US Government debt obligations, covered transactions are generally limited to 10% of the banking subsidiary's total capital, as determined by the risk-based capital guidelines; the aggregate amount of covered transactions between any banking subsidiary and all of its affiliates is limited to 20% of the banking subsidiary's total capital.

The Parent Company's two principal subsidiaries are JPMorgan Chase Bank, N.A. and JPMorgan Chase Holdings LLC, an intermediate holding company (the "IHC"). The IHC generally holds the stock of JPMorganChase's subsidiaries other than JPMorgan Chase Bank, N.A. and its subsidiaries. The IHC also owns other assets and provides intercompany loans to the Parent Company. The Parent Company is obligated to contribute to the IHC substantially all the net proceeds received from securities issuances (including issuances of senior and subordinated debt securities and of preferred and common stock).

The principal sources of income and funding for the Parent Company are dividends from JPMorgan Chase Bank, N.A. and dividends and extensions of credit from the IHC. In addition to dividend restrictions set forth in statutes and regulations, the Federal Reserve, the OCC and the FDIC have authority under the Financial Institutions Supervisory Act to prohibit or to limit the payment of dividends by the banking organizations they supervise, including the Parent Company and its subsidiaries that are banks or bank holding companies, if, in the banking regulator's opinion, payment of a dividend would constitute an unsafe or unsound practice in light of the financial condition of the banking organization. The IHC is prohibited from paying dividends or extending credit to the Parent Company if certain capital or liquidity "thresholds" are breached or if limits are otherwise imposed by the Parent Company's management or Board of Directors.

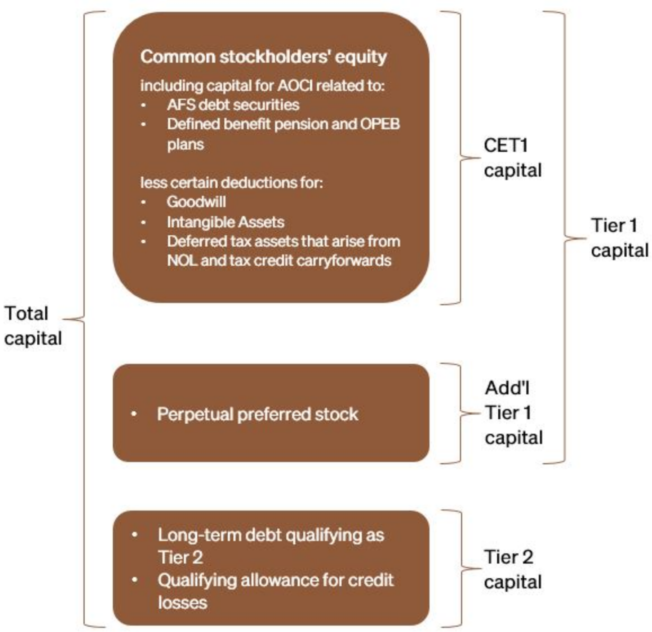
At January 1, 2025, the Parent Company's banking subsidiaries could pay, in the aggregate, approximately $15.5 billion in dividends to their respective bank holding companies without the prior approval of their relevant banking regulators. The capacity to pay dividends in 2025 will be supplemented by the banking subsidiaries' earnings during the year.

Note 27 – Regulatory capital

The Federal Reserve establishes capital requirements, including well-capitalized standards, for the Firm as a consolidated financial holding company. The OCC establishes similar minimum capital requirements and standards for the Firm's principal IDI subsidiary, JPMorgan Chase Bank, N.A.

The capital rules under Basel III establish minimum capital ratios and overall capital adequacy standards for large and internationally active U.S. bank holding companies and banks, including the Firm and JPMorgan Chase Bank, N.A. Under the rules currently in effect, two comprehensive approaches are prescribed for calculating RWA: a standardized approach ("Basel III Standardized"), and an advanced approach ("Basel III Advanced"). For each of these risk-based capital ratios, the capital adequacy of the Firm and JPMorgan Chase Bank, N.A. is evaluated against the lower of the Standardized or Advanced approaches compared to their respective regulatory capital ratio requirements.

The three components of regulatory capital under the Basel III rules and their primary drivers are as illustrated below:



Under the risk-based capital and leverage-based guidelines of the Federal Reserve, JPMorgan Chase & Co. is required to maintain minimum ratios for CET1 capital, Tier 1 capital, Total capital, Tier 1 leverage and the SLR. Failure to meet these minimum requirements could cause the Federal Reserve to take action. JPMorgan Chase Bank, N.A. is also subject to these capital requirements established by its primary regulators.

The following table presents the risk-based regulatory capital ratio requirements and well-capitalized ratios to which the Firm and JPMorgan Chase Bank, N.A. were subject as of December 31, 2024 and 2023.

	Standardized capital ratio requirements		Advanced capital ratio requirements		Well-capitalized ratios	
	BHC[(a)(b)]	IDI[(c)]	BHC[(a)(b)]	IDI[(c)]	BHC[(d)]	IDI[(e)]
Risk-based capital ratios						
CET1 capital	12.3 %	7.0 %	11.5 %	7.0 %	NA	6.5 %
Tier 1 capital	13.8	8.5	13.0	8.5	6.0 %	8.0
Total capital	15.8	10.5	15.0	10.5	10.0	10.0

Note: The table above is as defined by the regulations issued by the Federal Reserve, OCC and FDIC and to which the Firm and JPMorgan Chase Bank, N.A. are subject.

(a) Represents the regulatory capital ratio requirements applicable to the Firm. The CET1, Tier 1 and Total capital ratio requirements each include a respective minimum requirement plus a GSIB surcharge of 4.5% as calculated under Method 2; plus a 3.3% SCB for Basel III Standardized ratios and a fixed 2.5% capital conservation buffer for Basel III Advanced ratios. The countercyclical buffer is currently set to 0% by the federal banking agencies.

(b) For the period ended December 31, 2023, the CET1, Tier 1, and Total capital ratio requirements under Basel III Standardized applicable to the Firm were 11.4%, 12.9%, and 14.9%, respectively; the Basel III Advanced CET1, Tier 1, and Total capital ratio requirements applicable to the Firm were 11.0%, 12.5%, and 14.5%, respectively.

(c) Represents requirements for JPMorgan Chase Bank, N.A. The CET1, Tier 1 and Total capital ratio requirements include a fixed capital conservation buffer requirement of 2.5% that is applicable to JPMorgan Chase Bank, N.A. JPMorgan Chase Bank, N.A. is not subject to the GSIB surcharge.

(d) Represents requirements for bank holding companies pursuant to regulations issued by the Federal Reserve.

(e) Represents requirements for JPMorgan Chase Bank, N.A. pursuant to regulations issued under the FDIC Improvement Act.

The following table presents the leverage-based regulatory capital ratio requirements and well-capitalized ratios to which the Firm and JPMorgan Chase Bank, N.A. were subject as of December 31, 2024 and 2023.

	Capital ratio requirements[(a)]		Well-capitalized ratios	
	BHC	IDI	BHC[(b)]	IDI
Leverage-based capital ratios				
Tier 1 leverage	4.0 %	4.0 %	NA	5.0 %
SLR	5.0	6.0	NA	6.0

Note: The table above is as defined by the regulations issued by the Federal Reserve, OCC and FDIC and to which the Firm and JPMorgan Chase Bank, N.A. are subject.

(a) Represents minimum SLR requirement of 3.0%, as well as supplementary leverage buffer requirements of 2.0% and 3.0% for BHC and JPMorgan Chase Bank, N.A., respectively.

(b) The Federal Reserve's regulations do not establish well-capitalized thresholds for these measures for BHCs.

CECL Regulatory Capital Transition

Beginning January 1, 2022, the $2.9 billion CECL capital benefit, provided by the Federal Reserve in response to the COVID-19 pandemic, is being phased out at 25% per year over a three-year period. As of December 31, 2024 and 2023, the Firm's CET1 capital reflected the remaining benefit of $720 million and $1.4 billion, respectively, associated with the CECL capital transition provisions.

Similarly, as of January 1, 2024, the Firm has phased out 75% of the other CECL capital transition provisions which impacted Tier 2 capital, adjusted average assets, total leverage exposure and RWA, as applicable.

The following tables present risk-based capital metrics under both the Basel III Standardized and Basel III Advanced approaches and leverage-based capital metrics for JPMorgan Chase & Co. and JPMorgan Chase Bank, N.A. As of December 31, 2024 and 2023, JPMorgan Chase & Co. and JPMorgan Chase Bank, N.A. were well-capitalized and met all capital requirements to which each was subject.

December 31, 2024 (in millions, except ratios)	**Basel III Standardized** JPMorgan Chase & Co.	**Basel III Standardized** JPMorgan Chase Bank, N.A.	**Basel III Advanced** JPMorgan Chase & Co.	**Basel III Advanced** JPMorgan Chase Bank, N.A.
Risk-based capital metrics:[(a)]				
CET1 capital	**$ 275,513**	**$ 275,732**	**$ 275,513**	**$ 275,732**
Tier 1 capital	**294,881**	**275,737**	**294,881**	**275,737**
Total capital	**325,589**	**296,041**	**311,898** (b)	**282,328** (b)
Risk-weighted assets	**1,757,460**	**1,718,777**	**1,740,429** (b)	**1,594,072** (b)
CET1 capital ratio	**15.7 %**	**16.0 %**	**15.8 %**	**17.3 %**
Tier 1 capital ratio	**16.8**	**16.0**	**16.9**	**17.3**
Total capital ratio	**18.5**	**17.2**	**17.9**	**17.7**

December 31, 2023 (in millions, except ratios)	Basel III Standardized JPMorgan Chase & Co.	Basel III Standardized JPMorgan Chase Bank, N.A.	Basel III Advanced JPMorgan Chase & Co.	Basel III Advanced JPMorgan Chase Bank, N.A.
Risk-based capital metrics: [(a)]				
CET1 capital	$ 250,585	$ 262,030	$ 250,585	$ 262,030
Tier 1 capital	277,306	262,032	277,306	262,032
Total capital	308,497	281,308	295,417 (b)	268,392 (b)
Risk-weighted assets	1,671,995	1,621,789	1,669,156 (b)	1,526,952 (b)
CET1 capital ratio	15.0 %	16.2 %	15.0 %	17.2 %
Tier 1 capital ratio	16.6	16.2	16.6	17.2
Total capital ratio	18.5	17.3	17.7	17.6

(a) The capital metrics reflect the CECL capital transition provisions.

(b) Includes the impacts of certain assets associated with First Republic to which the Standardized approach has been applied as permitted by the transition provisions in the U.S. capital rules.

Three months ended (in millions, except ratios)	**December 31, 2024** JPMorgan Chase & Co.	**December 31, 2024** JPMorgan Chase Bank, N.A.	December 31, 2023 JPMorgan Chase & Co.	December 31, 2023 JPMorgan Chase Bank, N.A.
Leverage-based capital metrics: [(a)]				
Adjusted average assets[(b)]	**$ 4,070,499**	**$ 3,491,283**	$ 3,831,200	$ 3,337,842
Tier 1 leverage ratio	**7.2 %**	**7.9 %**	7.2 %	7.9 %
Total leverage exposure	**$ 4,837,568**	**$ 4,246,516**	$ 4,540,465	$ 4,038,739
SLR	**6.1 %**	**6.5 %**	6.1 %	6.5 %

(a) The capital metrics reflect the CECL capital transition provisions.

(b) Adjusted average assets, for purposes of calculating the leverage ratios, includes quarterly average assets adjusted for on-balance sheet assets that are subject to deduction from Tier 1 capital, predominantly goodwill, inclusive of estimated equity method goodwill, and other intangible assets.

Note 28 – Off–balance sheet lending-related financial instruments, guarantees, and other commitments

JPMorganChase provides lending-related financial instruments (e.g., commitments and guarantees) to address the financing needs of its customers and clients. The contractual amount of these financial instruments represents the maximum possible credit risk to the Firm should the customer or client draw upon the commitment or the Firm be required to fulfill its obligation under the guarantee, and should the customer or client subsequently fail to perform according to the terms of the contract. Most of these commitments and guarantees have historically been refinanced, extended, cancelled, or expired without being drawn or a default occurring. As a result, the total contractual amount of these instruments is not, in the Firm’s view, representative of its expected future credit exposure or funding requirements.

To provide for expected credit losses in wholesale and certain consumer lending-related commitments, an allowance for credit losses on lending-related commitments is maintained. Refer to Note 13 for further information regarding the allowance for credit losses on lending-related commitments.

The following table summarizes the contractual amounts and carrying values of off-balance sheet lending-related financial instruments, guarantees and other commitments at December 31, 2024 and 2023. The amounts in the table below for credit card, home equity and certain scored business banking lending-related commitments represent the total available credit for these products. The Firm has not experienced, and does not anticipate, that all available lines of credit for these commitments will be utilized at the same time. The Firm can reduce or cancel these commitments, in accordance with the contract, or to the extent otherwise permitted by law, including when there has been a demonstrable decline in the creditworthiness of the borrower or significant decrease in the value of underlying property.

Off–balance sheet lending-related financial instruments, guarantees and other commitments

By remaining maturity as of December 31, (in millions)	Contractual amount 2024: Expires in 1 year or less	Contractual amount 2024: Expires after 1 year through 3 years	Contractual amount 2024: Expires after 3 years through 5 years	Contractual amount 2024: Expires after 5 years	Contractual amount 2024: Total	Contractual amount 2023: Total	Carrying value[h][i] 2024	Carrying value[h][i] 2023
Lending-related								
Consumer, excluding credit card:								
Residential Real Estate[a]	**$ 10,838**	**$ 7,240**	**$ 4,601**	**$ 7,670**	**$ 30,349**	$ 30,125	**$ 534** [j]	$ 678 [j]
Auto and other	**10,833**	**14**	**5**	**3,643**	**14,495**	15,278	**37** [j]	148 [j]
Total consumer, excluding credit card	**21,671**	**7,254**	**4,606**	**11,313**	**44,844**	45,403	**571**	826
Credit card[b]	**1,001,311**	**—**	**—**	**—**	**1,001,311**	915,658	**—**	—
Total consumer[c]	**1,022,982**	**7,254**	**4,606**	**11,313**	**1,046,155**	961,061	**571**	826
Wholesale:								
Other unfunded commitments to extend credit[d]	**101,500**	**199,878**	**172,066**	**24,993**	**498,437**	503,526	**2,608** [j]	2,797 [j]
Standby letters of credit and other financial guarantees[d]	**15,825**	**8,506**	**3,780**	**565**	**28,676**	28,872	**473**	479
Other letters of credit[d]	**3,958**	**278**	**21**	**97**	**4,354**	4,388	**37**	37
Total wholesale[c]	**121,283**	**208,662**	**175,867**	**25,655**	**531,467**	536,786	**3,118**	3,313
Total lending-related	**$1,144,265**	**$ 215,916**	**$ 180,473**	**$ 36,968**	**$1,577,622**	$1,497,847	**$ 3,689**	$ 4,139
Other guarantees and commitments								
Securities lending indemnification agreements and guarantees[e]	**$ 310,046**	**$ —**	**$ —**	**$ —**	**$ 310,046**	$ 283,664	**$ —**	$ —
Derivatives qualifying as guarantees	**902**	**343**	**9,890**	**38,493**	**49,628**	54,562	**113**	89
Unsettled resale and securities borrowed agreements	**115,939**	**—**	**—**	**—**	**115,939**	95,106	**2**	—
Unsettled repurchase and securities loaned agreements	**66,986**	**—**	**—**	**—**	**66,986**	60,724	**(2)**	—
Loan sale and securitization-related indemnifications:								
Mortgage repurchase liability	**NA**	**NA**	**NA**	**NA**	**NA**	NA	**45**	76
Loans sold with recourse	**NA**	**NA**	**NA**	**NA**	**1,189**	803	**23**	24
Exchange & clearing house guarantees and commitments[f]	**401,486**	**—**	**—**	**—**	**401,486**	265,887	**—**	—
Other guarantees and commitments[g]	**10,652**	**425**	**435**	**884**	**12,396**	15,074	**28**	38

(a) Includes certain commitments to purchase loans from correspondents.
(b) Also includes commercial card lending-related commitments primarily in CIB.
(c) Predominantly all consumer and wholesale lending-related commitments are in the U.S.
(d) As of December 31, 2024 and 2023, reflected the contractual amount net of risk participations totaling $85 million and $88 million, respectively, for other unfunded commitments to extend credit; $9.5 billion and $8.2 billion, respectively, for standby letters of credit and other financial guarantees; and $556 million and $589 million, respectively, for other letters of credit. In regulatory filings with the Federal Reserve these commitments are shown gross of risk participations.
(e) As of December 31, 2024 and 2023, collateral held by the Firm in support of securities lending indemnification agreements was $328.7 billion and $300.3 billion, respectively. Securities lending collateral primarily consists of cash, G7 government securities, and securities issued by U.S. GSEs and government agencies.
(f) As of December 31, 2024 and 2023, includes guarantees to the Fixed Income Clearing Corporation under the sponsored member repo program and commitments and guarantees associated with the Firm's membership in certain clearing houses.
(g) As of December 31, 2024 and 2023, primarily includes unfunded commitments to purchase secondary market loans, other equity investment commitments, and unfunded commitments related to certain tax-oriented equity investments, and reflects the impact of adopting updates to the Accounting for Investments in Tax Credit Structures guidance effective January 1, 2024.
(h) For lending-related products, the carrying value includes the allowance for lending-related commitments and the guarantee liability; for derivative-related products, and lending-related commitments for which the fair value option was elected, the carrying value represents the fair value.
(i) For lending-related commitments, the carrying value also includes fees and any purchase discounts or premiums that are deferred and recognized in accounts payable and other liabilities on the Consolidated balance sheets. Deferred amounts for revolving commitments and commitments not expected to fund, are amortized to lending- and deposit-related fees on a straight line basis over the commitment period. For all other commitments the deferred amounts remain deferred until the commitment funds or is sold.
(j) As of December 31, 2024 and 2023, includes fair value adjustments associated with First Republic for residential real estate lending-related commitments totaling $459 million and $630 million, respectively; for auto and other lending-related commitments totaling $37 million and $148 million, respectively; and for other unfunded commitments to extend credit totaling $699 million and $1.1 billion, respectively. Refer to Note 34 for additional information.

Other unfunded commitments to extend credit

Other unfunded commitments to extend credit generally consist of commitments for working capital and general corporate purposes, extensions of credit to support commercial paper facilities and bond financings in the event that those obligations cannot be remarketed to new investors, as well as committed liquidity facilities to clearing organizations. The Firm also issues commitments under multipurpose facilities which could be drawn upon in several forms, including the issuance of a standby letter of credit.

Guarantees

U.S. GAAP requires that a guarantor recognize, at the inception of a guarantee, a liability in an amount equal to the fair value of the obligation undertaken in issuing the guarantee. U.S. GAAP defines a guarantee as a contract that contingently requires the guarantor to pay the guaranteed party based upon: (a) changes in an underlying asset, liability or equity security of the guaranteed party; or (b) a third party's failure to perform under a specified agreement. The Firm considers the following off–balance sheet arrangements to be guarantees under U.S. GAAP: standby letters of credit and other financial guarantees, securities lending indemnifications, certain indemnification agreements included within third-party contractual arrangements, certain derivative contracts and the guarantees under the sponsored member repo program.

As required by U.S. GAAP, the Firm initially records guarantees at the inception date fair value of the non-contingent obligation assumed (e.g., the amount of consideration received or the net present value of the premium receivable). For these obligations, the Firm records this fair value amount in other liabilities with an offsetting entry recorded in cash (for premiums received), or other assets (for premiums receivable). Any premium receivable recorded in other assets is reduced as cash is received under the contract, and the fair value of the liability recorded at inception is amortized into income as lending and deposit-related fees over the life of the guarantee contract. The lending-related contingent obligation is recognized based on expected credit losses in addition to, and separate from, any non-contingent obligation.

Non-lending-related contingent obligations are recognized when the liability becomes probable and reasonably estimable. These obligations are not recognized if the estimated amount is less than the carrying amount of any non-contingent liability recognized at inception (adjusted for any amortization). Examples of non-lending-related contingent obligations include indemnifications provided in sales agreements, where a portion of the sale proceeds is allocated to the guarantee, which adjusts the gain or loss that would otherwise result from the transaction. For these indemnifications, the initial liability is amortized to income as the Firm's risk is reduced (i.e., over time or when the indemnification expires).

The contractual amount and carrying value of guarantees and indemnifications are included in the table on page 303.

For additional information on the guarantees, see below.

Standby letters of credit and other financial guarantees

Standby letters of credit and other financial guarantees are conditional lending commitments issued by the Firm to guarantee the performance of a client or customer to a third party under certain arrangements, such as commercial paper facilities, bond financings, acquisition financings, trade financings and similar transactions.

The following table summarizes the contractual amount and carrying value of standby letters of credit and other financial guarantees and other letters of credit arrangements as of December 31, 2024 and 2023.

Standby letters of credit, other financial guarantees and other letters of credit

December 31, (in millions)	2024 Standby letters of credit and other financial guarantees	2024 Other letters of credit	2023 Standby letters of credit and other financial guarantees	2023 Other letters of credit
Investment-grade[a]	**$ 20,443**	**$ 3,380**	$ 19,694	$ 3,552
Noninvestment-grade[a]	**8,233**	**974**	9,178	836
Total contractual amount	**$ 28,676**	**$ 4,354**	$ 28,872	$ 4,388
Allowance for lending-related commitments	**$ 94**	**$ 37**	$ 110	$ 37
Guarantee liability	**379**	**—**	369	—
Total carrying value	**$ 473**	**$ 37**	$ 479	$ 37
Commitments with collateral	**$ 16,805**	**$ 357**	$ 16,861	$ 539

(a) The ratings scale is based on the Firm's internal risk ratings. Refer to Note 12 for further information on internal risk ratings.

Securities lending indemnifications
Through the Firm's securities lending program, counterparties' securities, via custodial and non-custodial arrangements, may be lent to third parties. As part of this program, the Firm provides an indemnification in the lending agreements which protects the lender against the failure of the borrower to return the lent securities. To minimize its liability under these indemnification agreements, the Firm obtains cash or other highly liquid collateral with a market value exceeding 100% of the value of the securities on loan from the borrower. Collateral is marked to market daily to help assure that collateralization is adequate. Additional collateral is called from the borrower if a shortfall exists, or collateral may be released to the borrower in the event of overcollateralization. If a borrower defaults, the Firm would use the collateral held to purchase replacement securities in the market or to credit the lending client or counterparty with the cash equivalent thereof.

The cash collateral held by the Firm may be invested on behalf of the client in indemnified resale agreements, whereby the Firm indemnifies the client against the loss of principal invested. To minimize its liability under these agreements, the Firm obtains collateral with a market value exceeding 100% of the principal invested.

Derivatives qualifying as guarantees
The Firm transacts in certain derivative contracts that have the characteristics of a guarantee under U.S. GAAP. These contracts include written put options that require the Firm to purchase assets upon exercise by the option holder at a specified price by a specified date in the future. The Firm may enter into written put option contracts in order to meet client needs, or for other trading purposes. The terms of written put options are typically five years or less.

Derivatives deemed to be guarantees also includes stable value contracts, commonly referred to as "stable value products", that require the Firm to make a payment of the difference between the market value and the book value of a counterparty's reference portfolio of assets in the event that market value is less than book value and certain other conditions have been met. Stable value products are transacted in order to allow investors to realize investment returns with less volatility than an unprotected portfolio. These contracts are typically longer-term or may have no stated maturity, but allow the Firm to elect to terminate the contract under certain conditions.

The notional value of derivative guarantees generally represents the Firm's maximum exposure. However, exposure to certain stable value products is contractually limited to a substantially lower percentage of the notional amount.

The fair value of derivative guarantees reflects the probability, in the Firm's view, of whether the Firm will be required to perform under the contract. The Firm reduces exposures to these contracts by entering into offsetting transactions, or by entering into contracts that hedge the market risk related to the derivative guarantees.

The following table summarizes the derivatives qualifying as guarantees as of December 31, 2024 and 2023.

(in millions)	**December 31, 2024**	December 31, 2023
Notional amounts		
Derivative guarantees	**$ 49,628**	$ 54,562
Stable value contracts with contractually limited exposure	**32,939**	32,488
Maximum exposure of stable value contracts with contractually limited exposure	**1,740**	1,652
Fair value		
Derivative payables	**113**	89

In addition to derivative contracts that meet the characteristics of a guarantee, the Firm is both a purchaser and seller of credit protection in the credit derivatives market. Refer to Note 5 for a further discussion of credit derivatives.

Unsettled securities financing agreements
In the normal course of business, the Firm enters into resale and securities borrowed agreements. At settlement, these commitments result in the Firm advancing cash to and receiving securities collateral from the counterparty. The Firm also enters into repurchase and securities loaned agreements. At settlement, these commitments result in the Firm receiving cash from and providing securities collateral to the counterparty. Such agreements settle at a future date. These agreements generally do not meet the definition of a derivative, and therefore, are not recorded on the Consolidated balance sheets until settlement date. These agreements predominantly have regular-way settlement terms. Refer to Note 11 for a further discussion of securities financing agreements.

Loan sales- and securitization-related indemnifications
Mortgage repurchase liability
In connection with the Firm's mortgage loan sale and securitization activities with U.S. GSEs the Firm has made representations and warranties that the loans sold meet certain requirements, and that may require the Firm to repurchase mortgage loans and/or indemnify the loan purchaser if such representations and warranties are breached by the Firm.

Private label securitizations
The liability related to repurchase demands associated with private label securitizations is separately evaluated by the Firm in establishing its litigation reserves.

Refer to Note 30 for additional information regarding litigation.

Loans sold with recourse
The Firm provides servicing for mortgages and certain commercial lending products on both a recourse and nonrecourse basis. In nonrecourse servicing, the principal credit risk to the Firm is the cost of temporary servicing advances of funds (i.e., normal servicing advances). In recourse servicing, the servicer agrees to share credit risk with the owner of the mortgage loans, such as Fannie Mae or Freddie Mac or a private investor, insurer or guarantor. Losses on recourse servicing predominantly occur when foreclosure sales proceeds of the property underlying a defaulted loan are less than the sum of the outstanding principal balance, plus accrued interest on the loan and the cost of holding and disposing of the underlying property. The Firm's securitizations are predominantly nonrecourse, thereby effectively transferring the risk of future credit losses to the purchaser of the mortgage-backed securities issued by the trust. The unpaid principal balance of loans sold with recourse as well as the carrying value of the related liability that the Firm has recorded in accounts payable and other liabilities on the Consolidated balance sheets, which is representative of the Firm's view of the likelihood it will have to perform under its recourse obligations, are disclosed in the table on page 303.

Other off-balance sheet arrangements

Indemnification agreements – general
In connection with issuing securities to investors outside the U.S., the Firm may agree to pay additional amounts to the holders of the securities in the event that, due to a change in tax law, certain types of withholding taxes are imposed on payments on the securities. The terms of the securities may also give the Firm the right to redeem the securities if such additional amounts are payable. The Firm may also enter into indemnification clauses such as in connection with the licensing of software to clients ("software licensees") or when it sells a business or assets to a third party ("third-party purchasers"), pursuant to which it indemnifies software licensees for claims of liability or damages that may occur subsequent to the licensing of the software, or third-party purchasers for losses they may incur due to actions taken by the Firm prior to the sale of the business or assets. It is difficult to estimate the Firm's maximum exposure under these indemnification arrangements, since this would require an assessment of future changes in tax law and future claims that may be made against the Firm that have not yet occurred. However, based on historical experience, management expects the risk of loss to be remote.

Merchant charge-backs
Under the rules of payment networks, in its role as a merchant acquirer, the Firm's Merchant Services business in CIB Payments, retains a contingent liability for disputed processed credit and debit card transactions that result in a charge-back to the merchant. If a dispute is resolved in the cardholder's favor, the Firm will (through the cardholder's issuing bank) credit or refund the amount to the cardholder and will charge back the transaction to the merchant. If the Firm is unable to collect the amount from the merchant, the Firm will bear the loss for the amount credited or refunded to the cardholder. The Firm mitigates this risk by withholding future settlements, retaining cash reserve accounts or obtaining other collateral. In addition, the Firm recognizes a valuation allowance that covers the payment or performance risk related to charge-backs.

Clearing Services – Client Credit Risk
The Firm provides clearing services for clients by entering into securities purchases and sales and derivative contracts with CCPs, including ETDs such as futures and options, as well as OTC-cleared derivative contracts. As a clearing member, the Firm stands behind the performance of its clients, collects cash and securities collateral (margin) as well as any settlement amounts due from or to clients, and remits them to the relevant CCP or client in whole or part. There are two types of margin: variation margin is posted on a daily basis based on the value of clients' derivative contracts and initial margin is posted at inception of a derivative contract, generally on the basis of the potential changes in the variation margin requirement for the contract.

As a clearing member, the Firm is exposed to the risk of nonperformance by its clients, but is not liable to clients for the performance of the CCPs. Where possible, the Firm seeks to mitigate its risk to the client through the collection of appropriate amounts of margin at inception and throughout the life of the transactions. The Firm can also cease providing clearing services if clients do not adhere to their obligations under the clearing agreement. In the event of nonperformance by a client, the Firm would close out the client's positions and access available margin. The CCP would utilize any margin it holds to make itself whole, with any remaining shortfalls required to be paid by the Firm as a clearing member.

The Firm reflects its exposure to nonperformance risk of the client through the recognition of margin receivables from clients and margin payables to CCPs; the clients' underlying securities or derivative contracts are not reflected in the Firm's Consolidated Financial Statements.

It is difficult to estimate the Firm's maximum possible exposure through its role as a clearing member, as this would require an assessment of transactions that clients may execute in the future. However, based upon historical experience, and the credit risk mitigants available to the Firm, management believes it is unlikely that the Firm will have to make any material payments under these arrangements and the risk of loss is expected to be remote.

Refer to Note 5 for information on the derivatives that the Firm executes for its own account and records in its Consolidated Financial Statements.

Exchange & Clearing House Memberships
The Firm is a member of several securities and derivative exchanges and clearing houses, both in the U.S. and other countries, and it provides clearing services to its clients. Membership in some of these organizations requires the Firm to pay a pro rata share of the losses incurred by the organization as a result of the default of another member. Such obligations vary with different organizations. These obligations may be limited to the amount (or a multiple of the amount) of the Firm's contribution to the guarantee fund maintained by a clearing house or exchange as part of the resources available to cover any losses in the event of a member default. Alternatively, these obligations may also include a pro rata share of the residual losses after applying the guarantee fund. Additionally, certain clearing houses require the Firm as a member to pay a pro rata share of losses that may result from the clearing house's investment of guarantee fund contributions and initial margin, unrelated to and independent of the default of another member. Generally a payment would only be required should such losses exceed the resources of the clearing house or exchange that are contractually required to absorb the losses in the first instance. In certain cases, it is difficult to estimate the Firm's maximum possible exposure under these membership agreements, since this would require an assessment of future claims that may be made against the Firm that have not yet occurred. However, based on historical experience, management expects the risk of loss to the Firm to be remote. Where the Firm's maximum possible exposure can be estimated, the amount is disclosed in the table on page 303, in the Exchange & clearing house guarantees and commitments line.

Sponsored member repo program
The Firm acts as a sponsoring member to clear eligible overnight and term resale and repurchase agreements through the Government Securities Division of the Fixed Income Clearing Corporation ("FICC") on behalf of clients that become sponsored members under the FICC's rules. The Firm also guarantees to the FICC the prompt and full payment and performance of its sponsored member clients' respective obligations under the FICC's rules. The Firm minimizes its liability under these guarantees by obtaining a security interest in the cash or high-quality securities collateral that the clients place with the clearing house; therefore, the Firm expects the risk of loss to be remote. The Firm's maximum possible exposure, without taking into consideration the associated collateral, is included in the Exchange & clearing house guarantees and commitments line on page 303. Refer to Note 11 for additional information on credit risk mitigation practices on resale agreements and the types of collateral pledged under repurchase agreements.

Guarantees of subsidiaries
In the normal course of business, the Parent Company may provide counterparties with guarantees of certain of the trading and other obligations of its subsidiaries on a contract-by-contract basis, as negotiated with the Firm's counterparties. The obligations of the subsidiaries are included on the Firm's Consolidated balance sheets or are reflected as off-balance sheet commitments; therefore, the Parent Company has not recognized a separate liability for these guarantees. The Firm believes that the occurrence of any event that would trigger payments by the Parent Company under these guarantees is remote.

The Parent Company has guaranteed certain long-term debt and structured notes of its subsidiaries, including JPMorgan Chase Financial Company LLC ("JPMFC"), a 100%-owned finance subsidiary. All securities issued by JPMFC are fully and unconditionally guaranteed by the Parent Company and no other subsidiary of the Parent Company guarantees these securities. These guarantees, which rank pari passu with the Firm's unsecured and unsubordinated indebtedness, are not included in the table on page 303 of this Note. Refer to Note 20 for additional information.

Note 29 – Pledged assets and collateral

Pledged assets

The Firm pledges financial assets that it owns to maintain potential borrowing capacity at discount windows with Federal Reserve banks, various other central banks and FHLBs. Additionally, the Firm pledges assets for other purposes, including to collateralize repurchase and other securities financing agreements, to cover short sales and to collateralize derivative contracts and deposits. Certain of these pledged assets may be sold or repledged or otherwise used by the secured parties and are parenthetically identified on the Consolidated balance sheets as assets pledged.

The following table presents the carrying value of the Firm's pledged assets.

December 31, (in billions)	2024	2023
Assets that may be sold or repledged or otherwise used by secured parties	**$ 152.5**	$ 145.0
Assets that may not be sold or repledged or otherwise used by secured parties	**297.9**	244.2
Assets pledged at Federal Reserve banks and FHLBs	**724.0**	675.6
Total pledged assets	**$ 1,174.4**	$ 1,064.8

Total pledged assets do not include assets of consolidated VIEs; these assets are used to settle the liabilities of those entities. Refer to Note 14 for additional information on assets and liabilities of consolidated VIEs. Refer to Note 11 for additional information on the Firm's securities financing activities. Refer to Note 20 for additional information on the Firm's long-term debt. The significant components of the Firm's pledged assets were as follows.

December 31, (in billions)	2024	2023
Investment securities	**$ 89.6**	$ 108.6
Loans	**740.9**	681.7
Trading assets and other	**343.9**	274.5
Total pledged assets	**$ 1,174.4**	$ 1,064.8

Collateral

The Firm accepts financial assets as collateral that it is permitted to sell or repledge, deliver or otherwise use. This collateral is generally obtained under resale and other securities financing agreements, prime brokerage-related held-for-investment customer receivables and derivative contracts. Collateral is generally used under repurchase and other securities financing agreements, to cover short sales, and to collateralize derivative contracts and deposits.

The following table presents the fair value of collateral accepted.

December 31, (in billions)	2024	2023
Collateral permitted to be sold or repledged, delivered, or otherwise used	**$ 1,544.0**	$ 1,303.9
Collateral sold, repledged, delivered or otherwise used	**1,210.7**	982.8

Note 30 – Litigation

Contingencies

As of December 31, 2024, the Firm and its subsidiaries and affiliates are defendants or respondents in numerous evolving legal proceedings, including private proceedings, public proceedings, government investigations, regulatory enforcement matters, and the matters described below. The litigations range from individual actions involving a single plaintiff to class action lawsuits with potentially millions of class members. Investigations and regulatory enforcement matters involve both formal and informal proceedings, by both governmental agencies and self-regulatory organizations. These legal proceedings are at varying stages of adjudication, arbitration or investigation, and involve each of the Firm's lines of business and several geographies and a wide variety of claims (including common law tort and contract claims and statutory antitrust, securities and consumer protection claims), some of which present novel legal theories.

The Firm believes the estimate of the aggregate range of reasonably possible losses, in excess of reserves established, for its legal proceedings is from $0 to approximately $1.4 billion at December 31, 2024. This estimated aggregate range of reasonably possible losses was based upon information available as of that date for those proceedings in which the Firm believes that an estimate of reasonably possible loss can be made. For certain matters, the Firm does not believe that such an estimate can be made, as of that date. The Firm's estimate of the aggregate range of reasonably possible losses involves significant judgment, given:

- the number, variety and varying stages of the proceedings, including the fact that many are in preliminary stages,
- the existence in many such proceedings of multiple defendants, including the Firm, whose share of liability (if any) has yet to be determined,
- the numerous yet-unresolved issues in many of the proceedings, including issues regarding class certification and the scope of many of the claims, and
- the uncertainty of the various potential outcomes of such proceedings, including where the Firm has made assumptions concerning future rulings by the court or other adjudicator, or about the behavior or incentives of adverse parties or regulatory authorities, and those assumptions prove to be incorrect.

In addition, the outcome of a particular proceeding may be a result which the Firm did not take into account in its estimate because the Firm had deemed the likelihood of that outcome to be remote. Accordingly, the Firm's estimate of the aggregate range of reasonably possible losses will change from time to time, and actual losses may vary significantly.

Set forth below are descriptions of the Firm's material legal proceedings.

1MDB Litigation. J.P. Morgan (Suisse) SA was named as a defendant in a civil litigation filed in May 2021 in Malaysia by 1Malaysia Development Berhad ("1MDB"), a Malaysian state-owned and controlled investment fund. The claim alleges "dishonest assistance" against J.P. Morgan (Suisse) SA in relation to payments of $300 million and $500 million, from 2009 and 2010, respectively, received from 1MDB and paid into an account at J.P. Morgan (Suisse) SA held by 1MDB PetroSaudi Limited, a joint venture company between 1MDB and PetroSaudi Holdings (Cayman) Limited. In March 2024, the Court upheld the Firm's challenge to the validity of service and the Malaysian Court's jurisdiction to hear the claim. That decision has been appealed by 1MDB. In August 2023, the Court denied an application by 1MDB to discontinue its claim with permission to re-file a new claim in the future. That decision was appealed by both 1MDB and the Firm, and an appeals court is scheduled to hear both appeals in November 2025. In its appeal, the Firm seeks to prevent any claim from continuing.

In addition, in November 2023, the Federal Office of the Attorney General (OAG) in Switzerland notified J.P. Morgan (Suisse) SA that it is conducting an investigation into possible criminal liability in connection with transactions arising from J.P. Morgan (Suisse) SA's relationship with the 1MDB PetroSaudi joint venture and its related persons for the period September 2009 through August 2015. The OAG investigation is ongoing.

Amrapali. India's Enforcement Directorate ("ED") is investigating J.P. Morgan India Private Limited in connection with investments made in 2010 and 2012 by two offshore funds formerly managed by JPMorganChase entities into residential housing projects developed by the Amrapali Group ("Amrapali") relating to delays in delivering or failure to deliver residential units. In August 2021, the ED issued an order fining J.P. Morgan India Private Limited approximately $31.5 million, and the Firm is appealing that order. Relatedly, in July 2019, the Supreme Court of India issued an order making preliminary findings that Amrapali and other parties, including unspecified JPMorganChase entities and the offshore funds that had invested in the projects, violated certain criminal currency control and money laundering provisions, and ordered the ED to conduct a further inquiry. The Firm is responding to and cooperating with the inquiry.

Foreign Exchange Investigations and Litigation. The Firm previously reported settlements with certain government authorities relating to its foreign exchange ("FX") sales and trading activities and controls related to those activities. Among those resolutions, in May 2015, the Firm pleaded guilty to a single violation of federal antitrust law. The Department of Labor ("DOL") granted the Firm exemptions that permit the Firm and its affiliates to continue to rely on the Qualified Professional Asset Manager exemption under the Employee Retirement Income Security Act ("ERISA") through the ten-year disqualification period following the antitrust plea. The only remaining FX-related governmental inquiry is a South Africa Competition Commission matter which is currently pending before the South Africa Competition Tribunal.

With respect to civil litigation matters, some FX-related individual and putative class actions filed outside the U.S., including in the U.K., Israel, the Netherlands, Brazil and Australia, remain. In July 2023, the U.K. Court of Appeal overturned the Competition Appeal Tribunal's earlier denial of a request for class certification on an opt-out basis. The defendants have appealed this decision to the U.K. Supreme Court. In Israel, a settlement in principle has been reached on the putative class action, which remains subject to court approval.

Interchange Litigation. Groups of merchants and retail associations filed a series of class action complaints alleging that Visa and Mastercard, as well as certain banks, conspired to set the price of credit and debit card interchange fees and enacted related rules in violation of antitrust laws.

In September 2018, the parties settled the class action seeking monetary relief, with the defendants collectively contributing approximately $6.2 billion. The settlement has been approved by the United States District Court for the Eastern District of New York and affirmed on appeal. Based on the percentage of merchants that opted out of the settlement, $700 million has been returned to the defendants from the settlement escrow. A separate class action seeking injunctive relief continues, and in September 2021, the District Court granted plaintiffs' motion for class certification in part, and denied the motion in part. In June 2024, the District Court denied preliminary approval of a settlement of the injunctive class action in which Visa and Mastercard agreed to certain changes to their respective network rules and system-wide reductions in interchange rates for U.S.-based merchants. The parties are considering next steps.

Of the merchants who opted out of the damages class settlement, certain merchants filed individual actions raising similar allegations against Visa and Mastercard, as well as against the Firm and other banks. While some of those actions remain pending, the defendants have reached settlements with the merchants who opted out representing over 70% of the combined Mastercard-branded and Visa-branded payment card sales volume.

LIBOR and Other Benchmark Rate Investigations and Litigation. JPMorganChase has responded to inquiries from various governmental agencies and entities around the world relating primarily to the British Bankers Association's ("BBA") London Interbank Offered Rate ("LIBOR") for various currencies and the European Banking Federation's Euro Interbank Offered Rate ("EURIBOR"). The Swiss Competition Commission's investigation relating to EURIBOR, to which the Firm and one other bank remain subject, continues. The Firm appealed a December 2016 decision by the European Commission against the Firm and other banks finding an infringement of European antitrust rules relating to EURIBOR. In December 2023, the European General Court annulled the fine imposed by the European Commission, but exercised its discretion to re-impose a fine in an identical amount. In March 2024, the Firm filed an appeal of this decision with the Court of Justice of the European Union.

In addition, the Firm has been named as a defendant along with other banks in various individual and putative class actions related to benchmark rates, including U.S. dollar LIBOR. In actions related to U.S. dollar LIBOR during the period that it was administered by the BBA, the United States District Court for the Southern District of New York granted class certification of antitrust claims related to bonds and interest rate swaps sold directly by the defendants, including the Firm. The Firm has obtained dismissal of certain actions and resolved certain other actions, and as to all remaining actions has moved for summary judgment. In addition, a lawsuit filed by a group of individual plaintiffs asserting antitrust claims, alleging that the Firm and other defendants were engaged in an unlawful agreement to set U.S. dollar LIBOR and conspired to monopolize the market for LIBOR-based consumer loans and credit cards was dismissed in October 2023 and affirmed on appeal by the United States Court of Appeals for the Ninth Circuit in December 2024. The Firm has resolved all non-U.S. dollar LIBOR actions.

Russian Litigation. The Firm is obligated to comply with international sanctions laws, which mandate the blocking of certain assets. These laws apply when assets associated with individuals, companies, products or services are within the scope of the sanctions. The Firm has faced actual and threatened litigation in Russia seeking payments that the Firm cannot make under, and is contractually excused from paying as a result of, relevant sanctions laws. In claims involving the Firm and claims filed against other financial institutions, Russian courts have disregarded the parties' contractual agreements concerning forum selection and did not recognize foreign sanctions laws as a basis for not making payment. Russian courts have entered judgment against the Firm in a number of claims, including one for $439 million, and a judgment has been executed against assets held onshore by the Firm in Russia. The total amount of the judgments exceeds the total amount of available assets that the Firm holds in Russia. The Firm continues to appeal the Russian courts' decisions, and judgments may not be executed while on appeal. Russian courts have also ordered interim freezes of Firm assets in Russia (including, among other things, funds in bank accounts, securities, shares in authorized capital, and certain trademarks, of the named defendants) pending a determination of certain underlying claims against the Firm. The Firm has challenged claims being pursued in the Russian courts and related freeze orders in other jurisdictions provided for by the parties' contractual forum selections. If further claims are enforced despite the actions taken by the Firm to challenge the claims and orders and to seek the proper application of law, the Firm's assets in Russia could be seized in full, and certain client assets could also be seized, or the Firm could be prevented from complying with its obligations.

SEC Inquiries. In October 2024, the Firm entered into settlements with the SEC to resolve inquiries related to, among other things, conflict disclosures concerning the selection of portfolio managers and the timing of the Firm's liquidation of shares distributed in-kind to certain investment vehicles that invest in third-party managed private funds and certain other matters. The resolutions required the Firm to pay a combined $151 million in civil penalties and voluntary payments to customers. The Firm continues to cooperate in connection with SEC inquiries concerning the aggregation of accounts for fee billing and various other matters.

Shareholder Litigation. Several shareholder putative class actions, as well as shareholder derivative actions purporting to act on behalf of the Firm, have been filed against the Firm, its Board of Directors and certain of its current and former officers.

Certain of these shareholder suits relate to historical trading practices by former employees in the precious metals and U.S. treasuries markets and related conduct which were the subject of the Firm's resolutions with the DOJ, CFTC and SEC in September 2020, and fiduciary activities that were separately the subject of a resolution between JPMorgan Chase Bank, N.A. and the OCC in November 2020. One of these shareholder derivative suits was filed in the Supreme Court of the State of New York in May 2022, asserting breach of fiduciary duty and unjust enrichment claims relating to the historical trading practices and related conduct and fiduciary activities which were the subject of the resolutions described above. In December 2022, the court granted defendants' motion to dismiss this action in full, and in July 2023, the plaintiff filed an appeal, which remains pending.

A second shareholder derivative action relating to the historical trading practices and related conduct was filed in the United States District Court for the Eastern District of New York in December 2022. Defendants have moved to dismiss the complaint.

Zelle Network Litigation. In December 2024, the Consumer Financial Protection Bureau ("CFPB") filed a complaint against Early Warning Services, LLC ("EWS"), Bank of America, N.A., Wells Fargo Bank, N.A. and JPMorgan Chase Bank, N.A. in the United States District Court for the District of Arizona. The CFPB alleges that EWS and the defendant banks have failed to take sufficient efforts to prevent fraud on the Zelle network. The defendants will file a response to the complaint.

* * *

In addition to the various legal proceedings discussed above, JPMorganChase and its subsidiaries are named as defendants or are otherwise involved in a substantial number of other legal proceedings. The Firm believes it has meritorious defenses to the claims asserted against it in its currently outstanding legal proceedings and it intends to defend itself vigorously. Additional legal proceedings may be initiated from time to time in the future.

The Firm has established reserves for several hundred of its currently outstanding legal proceedings. In accordance with the provisions of U.S. GAAP for contingencies, the Firm accrues for a litigation-related liability when it is probable that such a liability has been incurred and the amount of the loss can be reasonably estimated. The Firm evaluates its outstanding legal proceedings each quarter to assess its litigation reserves, and makes adjustments in such reserves, upward or downward, as appropriate, based on management's best judgment after consultation with counsel. The Firm's legal expense was $740 million, $1.4 billion and $266 million for the years ended December 31, 2024, 2023 and 2022, respectively. There is no assurance that the Firm's litigation reserves will not need to be adjusted in the future.

In view of the inherent difficulty of predicting the outcome of legal proceedings, particularly where the claimants seek very large or indeterminate damages, or where the matters present novel legal theories, involve a large number of parties or are in early stages of discovery, the Firm cannot state with confidence what will be the eventual outcomes of the currently pending matters, the timing of their ultimate resolution or the eventual losses, fines, penalties or consequences related to those matters. JPMorganChase believes, based upon its current knowledge and after consultation with counsel, consideration of the material legal proceedings described above and after taking into account its current litigation reserves and its estimated aggregate range of possible losses, that the other legal proceedings currently pending against it should not have a material adverse effect on the Firm's consolidated financial condition. The Firm notes, however, that in light of the uncertainties involved in such proceedings, there is no assurance that the ultimate resolution of these matters will not significantly exceed the reserves it has currently accrued or that a matter will not have material reputational consequences. As a result, the outcome of a particular matter may be material to JPMorganChase's operating results for a particular period, depending on, among other factors, the size of the loss or liability imposed and the level of JPMorganChase's income for that period.

Note 31 – International operations

The following table presents income statement and balance sheet-related information for JPMorganChase by major international geographic area. The Firm defines international activities for purposes of this footnote presentation as business transactions that involve clients residing outside of the U.S., and the information presented below is based predominantly on the domicile of the client, the location from which the client relationship is managed, booking location or the location of the trading desk. However, many of the Firm's U.S. operations serve international businesses.

As the Firm's operations are highly integrated, estimates and subjective assumptions have been made to apportion revenue and expense between U.S. and international operations. These estimates and assumptions are consistent with the allocations used for the Firm's segment reporting as set forth in Note 32.

The Firm's long-lived assets for the periods presented are not considered by management to be significant in relation to total assets. The majority of the Firm's long-lived assets are located in the U.S.

As of or for the year ended December 31, (in millions)	Revenue[b]	Expense[c]	Income before income tax expense	Net income	Total assets	
2024						
Europe/Middle East/Africa	**$ 22,353**	**$ 12,843**	**$ 9,510**	**$ 6,713**	**$ 552,407**	[d]
Asia-Pacific	**11,995**	**6,922**	**5,073**	**3,615**	**296,430**	
Latin America/Caribbean	**3,885**	**1,895**	**1,990**	**1,512**	**73,631**	
Total international	**38,233**	**21,660**	**16,573**	**11,840**	**922,468**	
North America[a]	**139,323**	**80,815**	**58,508**	**46,631**	**3,080,346**	
Total	**$ 177,556**	**$ 102,475**	**$ 75,081**	**$ 58,471**	**$ 4,002,814**	
2023						
Europe/Middle East/Africa	$ 20,974	$ 11,947	$ 9,027	$ 6,402	$ 529,335	[d]
Asia-Pacific	10,605	6,550	4,055	2,709	251,588	
Latin America/Caribbean	3,294	1,971	1,323	994	83,003	
Total international	34,873	20,468	14,405	10,105	863,926	
North America[a]	123,231	76,024	47,207	39,447	3,011,467	
Total	$ 158,104	$ 96,492	$ 61,612	$ 49,552	$ 3,875,393	
2022						
Europe/Middle East/Africa	$ 18,765	$ 11,754	$ 7,011	$ 5,158	$ 558,430	[d]
Asia-Pacific	10,025	6,763	3,262	2,119	281,479	
Latin America/Caribbean	3,178	1,697	1,481	1,156	78,673	
Total international	31,968	20,214	11,754	8,433	918,582	
North America[a]	96,727	62,315	34,412	29,243	2,747,161	
Total	$ 128,695	$ 82,529	$ 46,166	$ 37,676	$ 3,665,743	

(a) Substantially reflects the U.S.
(b) Revenue is composed of net interest income and noninterest revenue.
(c) Expense is composed of noninterest expense and the provision for credit losses.
(d) Total assets for the U.K. were approximately $369 billion, $352 billion and $357 billion at December 31, 2024, 2023 and 2022, respectively.

Note 32 – Business segments & Corporate

Business segment reorganization: Effective in the second quarter of 2024, the Firm reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank. As a result of the reorganization, the Firm has three reportable business segments – Consumer & Community Banking, Commercial & Investment Bank, and Asset & Wealth Management – with the remaining activities in Corporate.

Adoption of accounting standard — Segment Reporting — Improvements to Reportable Segment Disclosures: This guidance was adopted retrospectively for the Firm's annual Consolidated Financial Statements for the year ended December 31, 2024. The adoption of this guidance requires additional reportable segment disclosures, primarily relating to significant segment expenses and the chief operating decision maker ("CODM"). Adoption of this guidance did not result in changes to the identification of the Firm's reportable business segments, or of its CODM.

The Firm is managed on an LOB basis. The business segments are determined based on the products and services provided, or the type of customer served, and they reflect the manner in which financial information is evaluated by the Firm's Operating Committee, whose members act collectively as the Firm's CODM. Segment results are presented on a managed basis. Refer to Segment & Corporate results in this footnote for a further discussion of JPMorganChase's reportable business segments and Corporate.

The following is a description of each of the Firm's reportable business segments, and the products and services that they provide to their respective client bases, as well as a description of Corporate activities.

Consumer & Community Banking

Consumer & Community Banking offers products and services to consumers and small businesses through bank branches, ATMs, digital (including mobile and online) and telephone banking. CCB is organized into Banking & Wealth Management (including Consumer Banking, Business Banking and J.P. Morgan Wealth Management), Home Lending (including Home Lending Production, Home Lending Servicing and Real Estate Portfolios) and Card Services & Auto. Banking & Wealth Management offers deposit, investment and lending products, cash management, payments and services. Home Lending includes mortgage origination and servicing activities, as well as portfolios consisting of residential mortgages and home equity loans. Card Services issues credit cards and offers travel services. Auto originates and services auto loans and leases.

Commercial & Investment Bank

The Commercial & Investment Bank is comprised of the Banking & Payments and Markets & Securities Services businesses. These businesses offer investment banking, lending, payments, market-making, financing, custody and securities products and services to a global base of corporate and institutional clients. Banking & Payments offers products and services in all major capital markets, including advising on corporate strategy and structure, capital-raising in equity and debt markets, and loan origination and syndication. Banking & Payments also provides services that enable clients to manage payments globally across liquidity and account solutions, commerce solutions, clearing, trade, and working capital. Markets & Securities Services includes Markets, which is a global market-maker across products, including cash and derivative instruments, and also offers sophisticated risk management solutions, lending, prime brokerage, clearing and research. Markets & Securities Services also includes Securities Services, a leading global custodian that provides custody, fund services, liquidity and trading services, and data solutions products.

Asset & Wealth Management

Asset & Wealth Management, with client assets of $5.9 trillion, is a global leader in investment and wealth management.

Asset Management

Offers multi-asset investment management solutions across equities, fixed income, alternatives and money market funds to institutional and retail investors providing for a broad range of clients' investment needs.

Global Private Bank

Provides retirement products and services, brokerage, custody, estate planning, lending, deposits and investment management to high net worth clients.

The majority of AWM's client assets are in actively managed portfolios.

Corporate

Corporate consists of Treasury and Chief Investment Office ("CIO") and Other Corporate. Treasury and CIO is predominantly responsible for measuring, monitoring, reporting and managing the Firm's liquidity, funding, capital, structural interest rate and foreign exchange risks.

Other Corporate includes staff functions and expense that is centrally managed as well as certain Firm initiatives and activities not solely aligned to a specific LOB. The major Other Corporate functions include Real Estate, Technology, Legal, Corporate Finance, Human Resources, Internal Audit, Risk Management, Compliance, Control Management, Corporate Responsibility and various Other Corporate groups.

Description of business segment reporting methodology

Results of the reportable business segments are intended to present each segment as if it were a stand-alone business. The management reporting process that derives business segment results includes the allocation of certain income and expense items. The Firm periodically assesses the assumptions, methodologies and reporting classifications used for segment reporting, and therefore further refinements may be implemented in future periods. The Firm also assesses the level of capital required for each LOB on at least an annual basis. The Firm's LOBs also provide various business metrics which are utilized by the Firm and its investors and analysts in assessing performance.

Revenue sharing

When business segments or businesses within each segment join efforts to sell products and services to the Firm's clients and customers, the participating businesses may agree to share revenue from those transactions. Revenue is generally recognized in the segment responsible for the related product or service, with allocations to the other segments or businesses involved in the transaction. The segment and business results reflect these revenue-sharing agreements.

Expense allocation

Where business segments use services provided by Corporate support units, or another business segment, the costs of those services are allocated to the respective business segments. The expense is generally allocated based on the actual cost and use of services provided. In contrast, certain costs and investments related to Corporate that are not currently utilized by any LOB are not allocated to the business segments and are retained in Corporate. Expense retained in Corporate generally includes costs that would not be incurred if the segments were stand-alone businesses, and other items not solely aligned with a particular reportable business segment.

Funds transfer pricing

Funds transfer pricing ("FTP") is the process by which the Firm allocates interest income and expense to the LOBs and Other Corporate and transfers the primary interest rate risk and liquidity risk to Treasury and CIO.

The funds transfer pricing process considers the interest rate and liquidity risk characteristics of assets and liabilities and off-balance sheet products.

Periodically, the methodology and assumptions utilized in the FTP process are adjusted to reflect economic conditions and other factors, which may impact the allocation of net interest income to the segments. Effective in the fourth quarter of 2024, the Firm updated its FTP with respect to consumer deposits, which resulted in an increase in the funding benefit reflected within CCB's net interest income that is fully offset in Corporate, with no effect on the Firm's net interest income.

Foreign exchange risk

Foreign exchange risk is transferred from the LOBs and Other Corporate to Treasury and CIO for certain revenues and expenses. Treasury and CIO manages these risks centrally and reports the impact of foreign exchange rate movements related to the transferred risk in its results.

Debt expense and preferred stock dividend allocation

As part of the FTP process, almost all of the cost of the credit spread component of outstanding unsecured long-term debt and preferred stock dividends is allocated to the reportable business segments, while the balance of the cost is retained in Corporate. The methodology to allocate the cost of unsecured long-term debt and preferred stock dividends to the business segments is aligned with the relevant regulatory capital requirements and funding needs of the LOBs, as applicable. The allocated cost of unsecured long-term debt is included in a business segment's net interest income, and net income is reduced by preferred stock dividends, to arrive at a business segment's net income applicable to common equity.

Capital allocation

Each LOB and Corporate is allocated capital by taking into consideration a variety of factors including capital levels of similarly rated peers and applicable regulatory capital requirements. ROE is measured and internal targets for expected returns are established as key measures of an LOB's performance.

The Firm's current equity allocation methodology incorporates Basel III Standardized RWA and the GSIB surcharge, both under rules currently in effect, as well as a simulation of capital depletion in a severe stress environment. At least annually, the assumptions, judgments and methodologies used to allocate capital are reassessed and, as a result, the capital allocated to the LOBs and Corporate may change.

Segment & Corporate results

The following table provides a summary of results for the Firm's reportable business segments and Corporate activities as of or for the years ended December 31, 2024, 2023 and 2022, on a managed basis. The Firm's definition of managed basis starts with the reported U.S. GAAP results and includes certain reclassifications to present total net revenue for the Firm as a whole (and for each of the reportable business segments and Corporate) on an FTE basis. Accordingly, revenue from investments that receive tax credits and tax-exempt securities is presented in the managed results on a basis comparable to taxable investments and securities. This allows management to assess the comparability of revenue from year-to-year arising from both taxable and tax-exempt sources. The corresponding income tax impact related to tax-exempt items is recorded within income tax expense/(benefit). These adjustments have no impact on net income as reported by the Firm as a whole or by the each of the LOBs and Corporate.

The Operating Committee reviews segment results including net interest income, noninterest revenue, noninterest expense, provision for credit losses and net income on a managed basis. The Operating Committee uses these measures to evaluate segment performance and to make key operating decisions, including resource and capital allocations.

Segment & Corporate results and reconciliation[a]

(Table continued on next page)

As of or for the year ended December 31, (in millions, except ratios)	Consumer & Community Banking			Commercial & Investment Bank			Asset & Wealth Management		
	2024	2023	2022	**2024**	2023	2022	**2024**	2023	2022
Noninterest revenue	**$ 16,649**	$ 15,118	$ 14,886 [e]	**$ 48,253**	$ 43,809	$ 39,538 [b]	**$ 15,023**	$ 13,560	$ 12,507
Net interest income	**54,858**	55,030	39,928	**21,861**	20,544	20,097	**6,555**	6,267	5,241
Total net revenue	**71,507**	70,148	54,814	**70,114**	64,353	59,635	**21,578**	19,827	17,748
Provision for credit losses	**9,974**	6,899	3,813	**762**	2,091	2,426	**(68)**	159	128
Compensation expense[b]	**17,045**	15,171	13,092	**18,191**	17,105	16,214	**7,984**	7,115	6,336
Noncompensation expense[c][d]	**20,991**	19,648	18,116	**17,162**	16,867	15,855	**6,430**	5,665	5,493
Total noninterest expense	**38,036**	34,819	31,208 [e]	**35,353**	33,972	32,069 [b]	**14,414**	12,780	11,829
Income/(loss) before income tax expense/(benefit)	**23,497**	28,430	19,793	**33,999**	28,290	25,140	**7,232**	6,888	5,791
Income tax expense/(benefit)	**5,894**	7,198	4,877 [e]	**9,153**	8,018	6,002 [b]	**1,811**	1,661	1,426
Net income/(loss)	**$ 17,603**	$ 21,232	$ 14,916	**$ 24,846**	$ 20,272	$ 19,138	**$ 5,421**	$ 5,227	$ 4,365
Average equity	**$ 54,500**	$ 54,349	$ 50,000	**$ 132,000**	$ 137,507	$ 128,000	**$ 15,500**	$ 16,671	$ 17,000
Total assets	**650,268**	642,951	514,085	**1,773,194**	1,638,493	1,591,402	**255,385**	245,512	232,037
Return on equity	**32 %**	38 %	29 %	**18 %**	14 %	14 %	**34 %**	31 %	25 %
Overhead ratio	**53**	50	57	**50**	53	54	**67**	64	67

(Table continued from previous page)

As of or for the year ended December 31, (in millions, except ratios)	Corporate			Reconciling Items[a]			Total		
	2024	2023	2022	**2024**	2023	2022	**2024**	2023	2022
Noninterest revenue	**$ 7,608** [f]	$ 132	$ (1,798)	**$ (2,560)**	$ (3,782)	$ (3,148)	**$ 84,973** [f]	$ 68,837	$ 61,985
Net interest income	**9,786**	7,906	1,878	**(477)**	(480)	(434)	**92,583**	89,267	66,710
Total net revenue	**17,394**	8,038	80	**(3,037)**	(4,262)	(3,582)	**177,556**	158,104	128,695
Provision for credit losses	**10**	171	22	**—**	—	—	**10,678**	9,320	6,389
Total noninterest expense[d]	**3,994** [g]	5,601	1,034	**—**	—	—	**91,797** [g]	87,172	76,140
Income/(loss) before income tax expense/(benefit)	**13,390**	2,266	(976)	**(3,037)**	(4,262)	(3,582)	**75,081**	61,612	46,166
Income tax expense/(benefit)	**2,789**	(555)	(233)	**(3,037)**	(4,262)	(3,582)	**16,610**	12,060	8,490
Net income/(loss)	**$ 10,601**	$ 2,821	$ (743)	**$ —**	$ —	$ —	**$ 58,471**	$ 49,552	$ 37,676
Average equity	**$ 110,370**	$ 73,529	$ 58,068	**$ —**	$ —	$ —	**$ 312,370**	$ 282,056	$ 253,068
Total assets	**1,323,967**	1,348,437	1,328,219	**NA**	NA	NA	**4,002,814**	3,875,393	3,665,743
Return on equity	**NM**	NM	NM	**NM**	NM	NM	**18 %**	17 %	14 %
Overhead ratio	**NM**	NM	NM	**NM**	NM	NM	**52**	55	59

(a) Segment results on a managed basis reflect revenue on a FTE basis with the corresponding income tax impact recorded within income tax expense/(benefit). These adjustments are eliminated in reconciling items to arrive at the Firm's reported U.S. GAAP results. In addition, effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures guidance, under the modified retrospective method. Refer to Notes 1, 6, 14 and 25 for additional information.

(b) Excludes expense related to services provided by Corporate support units, which is allocated from Corporate to each respective reportable business segment, as applicable, through noncompensation expense.

(c) Reflects occupancy; technology, communications and equipment; professional and outside services; marketing; and other expense. Refer to Note 6 for additional information on other expense.

(d) Certain services are provided by Corporate and used by each of the reportable business segments. The costs of these services, including compensation-related costs, are allocated from Corporate to the respective reportable business segments, with the allocations recorded in noncompensation expense.

(e) In the first quarter of 2023, the allocations of revenue and expense to CCB associated with a Merchant Services revenue sharing agreement were discontinued and are now retained in Payments in CIB. Prior-period amounts have been revised to conform with the current presentation.

(f) Included a $7.9 billion net gain related to Visa shares recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.

(g) Included a $1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.

Note 33 – Parent Company

The following tables present Parent Company-only financial statements.

Statements of income and comprehensive income

Year ended December 31, (in millions)	2024	2023	2022
Income			
Dividends from subsidiaries and affiliates:			
Bank and bank holding company	**$ 37,000**	$ 61,000	$ 40,500
Non-bank	**—**	—	—
Interest income from subsidiaries	**1,228**	1,166	498
Other income/(expense) from subsidiaries:			
Bank and bank holding company	**555**	1,801	(3,497)
Non-bank	**172**	250	335
Other income/(expense)	**1,252**	(654)	5,271
Total income	**40,207**	63,563	43,107
Expense			
Interest expense/(income) to subsidiaries and affiliates[a]	**7,433**	2,258	22,731
Other interest expense/ (income)[a]	**8,068**	11,714	(14,658)
Noninterest expense	**3,280**	3,431	2,817
Total expense	**18,781**	17,403	10,890
Income before income tax benefit and undistributed net income of subsidiaries	**21,426**	46,160	32,217
Income tax benefit	**1,264**	1,525	1,260
Equity in undistributed net income of subsidiaries	**35,781**	1,867	4,199
Net income	**$ 58,471**	$ 49,552	$ 37,676
Other comprehensive income/ (loss), net	**(2,013)**	6,898	(17,257)
Comprehensive income	**$ 56,458**	$ 56,450	$ 20,419

Balance sheets

December 31, (in millions)	2024	2023
Assets		
Cash and due from banks	**$ 38**	$ 42
Deposits with banking subsidiaries	**9,762**	9,804
Trading assets	**43,214**	3,198
Advances to, and receivables from, subsidiaries:		
Bank and bank holding company	**142**	152
Non-bank	**79**	21
Investments (at equity) in subsidiaries and affiliates:		
Bank and bank holding company	**603,044**	568,472
Non-bank	**1,238**	1,045
Other assets	**12,097**	8,962
Total assets	**$669,614**	$ 591,696
Liabilities and stockholders' equity		
Borrowings from, and payables to, subsidiaries and affiliates	**$ 72,881**	$ 22,777
Short-term borrowings	**—**	999
Other liabilities	**12,349**	11,500
Long-term debt[b][c]	**239,626**	228,542
Total liabilities[c]	**324,856**	263,818
Total stockholders' equity	**344,758**	327,878
Total liabilities and stockholders' equity	**$669,614**	$ 591,696

Statements of cash flows

Year ended December 31, (in millions)	2024	2023	2022
Operating activities			
Net income	**$ 58,471**	$ 49,552	$ 37,676
Less: Net income of subsidiaries and affiliates	**72,781**	62,868	44,699
Parent company net loss	**(14,310)**	(13,316)	(7,023)
Cash dividends from subsidiaries and affiliates	**37,000**	61,000	40,500
Other operating adjustments	**(44,671)**	9,412	(23,747)
Net cash provided by/(used in) operating activities	**(21,981)**	57,096	9,730
Investing activities			
Net change in:			
Advances to and investments in subsidiaries and affiliates, net	**—**	(25,000)	—
All other investing activities, net	**21**	25	31
Net cash provided by/(used in) investing activities	**21**	(24,975)	31
Financing activities			
Net change in:			
Borrowings from subsidiaries and affiliates	**49,902**	(2,249)	(4,491)
Short-term borrowings	**(999)**	—	—
Proceeds from long-term borrowings	**44,997**	19,398	41,389
Payments of long-term borrowings	**(29,753)**	(25,105)	(18,294)
Proceeds from issuance of preferred stock	**2,500**	—	—
Redemption of preferred stock	**(9,850)**	—	(7,434)
Treasury stock repurchased	**(18,830)**	(9,824)	(3,162)
Dividends paid	**(14,783)**	(13,463)	(13,562)
All other financing activities, net	**(1,270)**	(879)	(1,205)
Net cash provided by/(used in) financing activities	**21,914**	(32,122)	(6,759)
Net increase/(decrease) in cash and due from banks and deposits with banking subsidiaries	**(46)**	(1)	3,002
Cash and due from banks and deposits with banking subsidiaries at the beginning of the year	**9,846**	9,847	6,845
Cash and due from banks and deposits with banking subsidiaries at the end of the year	**$ 9,800**	$ 9,846	$ 9,847
Cash interest paid	**$ 14,851**	$ 13,742	$ 7,462
Cash income taxes paid, net[(d)]	**6,252**	10,291	6,941

(a) Includes interest expense for intercompany derivative hedges on the Firm's LTD and related fair value adjustments, which is offset by related amounts in Other interest expense/(income).

(b) At December 31, 2024, long-term debt that contractually matures in 2025 through 2029 totaled $7.7 billion, $29.3 billion, $20.2 billion, $35.0 billion, and $18.5 billion, respectively.

(c) Refer to Notes 20 and 28 for information regarding the Parent Company's guarantees of its subsidiaries' obligations.

(d) Represents payments, net of refunds, made by the Parent Company to various taxing authorities and includes taxes paid on behalf of certain of its subsidiaries that are subsequently reimbursed. The reimbursements were $5.0 billion, $13.2 billion, and $11.3 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

Note 34 – Business combinations

On May 1, 2023, JPMorganChase acquired certain assets and assumed certain liabilities of First Republic Bank (the "First Republic acquisition") from the Federal Deposit Insurance Corporation (“FDIC”), as receiver. The acquisition resulted in a bargain purchase gain, which represents the excess of the estimated fair value of the net assets acquired above the purchase price.

The Firm has determined that this acquisition constitutes a business combination under U.S. GAAP. Accordingly, the initial recognition of the assets acquired and liabilities assumed were generally measured at their estimated fair values as of May 1, 2023. The determination of those fair values required management to make certain market-based assumptions about expected future cash flows, discount rates and other valuation inputs at the time of the acquisition. The Firm believes that the fair value estimates of the assets acquired and liabilities assumed provide a reasonable basis for determining the estimated bargain purchase gain.

The First Republic acquisition resulted in a preliminary estimated bargain purchase gain of $2.7 billion. The final bargain purchase gain of $2.9 billion reflects adjustments of $103 million and $63 million for the years ended December 31, 2024 and 2023, respectively, made during the one-year measurement period, as permitted by U.S. GAAP, to finalize management's fair value estimates for the assets acquired and liabilities assumed. As of December 31, 2024, certain matters related to the final settlement remained outstanding between the Firm and the FDIC.

On January 17, 2025, the Firm reached an agreement with the FDIC with respect to certain outstanding items. As a result of the agreement, the Firm made a payment of $609 million to the FDIC on January 31, 2025 and reduced its additional payable to the FDIC, which will result in a gain of approximately $600 million to be recorded in other income in the first quarter of 2025.

In connection with the First Republic acquisition, the Firm and the FDIC entered into two shared-loss agreements with respect to certain loans and lending-related commitments (the "shared-loss assets"): the Commercial Shared-Loss Agreement ("CSLA") and the Single-Family Shared-Loss Agreement (“SFSLA”). The CSLA covers 80% of credit losses, on a pari passu basis, over 5 years with a subsequent 3-year recovery period for certain acquired commercial loans and other real estate exposure. The SFSLA covers 80% of credit losses, on a pari passu basis, for 7 years for certain acquired loans secured by mortgages on real property or shares in cooperative property constituting a primary residence. The indemnification assets, which represent the fair value of the CSLA and SFSLA on the acquisition date, are reflected in the total assets acquired.

As part of the consideration paid, JPMorganChase issued a five-year, $50 billion secured note to the FDIC (the "Purchase Money Note"). The Purchase Money Note bears interest at a fixed rate of 3.4% and is secured by certain of the acquired loans. The Purchase Money Note is prepayable upon notice to the holder.

The Firm had placed a $5 billion deposit with First Republic Bank on March 16, 2023, as part of $30 billion of deposits provided by a consortium of large U.S. banks. The Firm's $5 billion deposit was effectively settled as part of the acquisition and the associated allowance for credit losses was released upon closing. The Firm subsequently repaid the remaining $25 billion of deposits to the consortium of banks, including accrued interest through the payment date on May 9, 2023.

The computation of the purchase price, the fair values of the assets acquired and liabilities assumed as part of the First Republic acquisition and the related bargain purchase gain are presented below, which reflects adjustments made during the measurement period to the acquisition-date fair value of the net assets acquired. The measurement period ended on April 30, 2024.

(in millions)		Fair value purchase price allocation as of May 1, 2023
Purchase price consideration		
Amounts paid/due to the FDIC, net of cash acquired[(a)]	$	13,555
Purchase Money Note (at fair value)[(b)]		48,848
Settlement of First Republic deposit and other related party transactions[(c)]		5,447
Contingent consideration - Shared-loss agreements		15
Purchase price consideration	$	67,865
Assets		
Securities	$	30,285
Loans		153,242
Core deposit and customer relationship intangibles		1,455
Indemnification assets - Shared-loss agreements		675
Accounts receivable and other assets[(d)]		6,740
Total assets acquired	$	192,397
Liabilities		
Deposits	$	87,572
FHLB advances		27,919
Lending-related commitments		2,614
Accounts payable and other liabilities[(d)]		2,792
Deferred tax liabilities		757
Total liabilities assumed	$	121,654
Fair value of net assets acquired	$	70,743
Gain on acquisition, after income taxes	$	2,878

(a) Net of cash acquired of $680 million, and including disputed amounts with the FDIC as of April 30, 2024.
(b) As part of the consideration paid, JPMorganChase issued a five-year, $50 billion secured note to the FDIC (the "Purchase Money Note").
(c) Includes $447 million of securities financing transactions with First Republic Bank that were effectively settled on the acquisition date.
(d) Other assets include $1.2 billion in tax-oriented investments and $683 million of lease right-of-use assets. Other liabilities include the related tax-oriented investment liabilities of $669 million and lease liabilities of $748 million.

The following describes the accounting policies and fair value methodologies generally used by the Firm for the following assets acquired and liabilities assumed: core deposit and customer relationship intangibles, shared-loss agreements and the related indemnification assets, Purchase Money Note, and FHLB advances.

For further discussion of the Firm's accounting policies and valuation methodologies, refer to Notes 2 and 3 for fair value measurement, Note 10 for investment securities, Note 12 for loans, Note 17 for deposits, and Note 28 for lending-related commitments.

Core deposit and customer relationship intangibles

Core deposit and certain wealth management customer relationship intangibles were acquired as part of the First Republic acquisition. The core deposit intangible of $1.3 billion was valued by discounting estimated after-tax cost savings over the remaining useful life of the deposits using the favorable source of funds method. The after-tax cost savings were estimated based on the difference between the cost of maintaining the core deposit base relative to the cost of next best alternative funding sources available to market participants. The customer relationship intangibles of $180 million were valued by discounting estimated after-tax earnings over their remaining useful lives using the multi-period excess earnings method. Both intangible asset valuations utilized assumptions that the Firm believes a market participant would use to estimate fair values, such as growth and attrition rates, projected fee income as well as related costs to service the relationships, and discount rates. The core deposit and customer relationship intangibles are amortized over a projected period of future cash flows of approximately 7 years. Refer to Note 15 for further discussion on other intangible assets.

Indemnification assets - Shared-loss agreements

The indemnification assets represent forecasted recoveries from the FDIC associated with the shared-loss assets over the respective shared-loss recovery periods. The indemnification assets were recorded at fair value in other assets on the Consolidated balance sheets on the acquisition date. The fair values of the indemnification assets were estimated based on the timing of the forecasted losses underlying the related allowance for credit losses. The subsequent quarterly remeasurement of the indemnification assets is based on changes in the amount and timing of forecasted losses in the allowance for credit losses associated with the shared-loss assets and is recorded in other income. Under certain circumstances, the Firm may be required to make a payment to the FDIC upon termination of the shared-loss agreements based on the level of actual losses and recoveries on the shared-loss assets. The estimated potential future payment is reflected as contingent consideration as part of the purchase price consideration.

Purchase Money Note and FHLB advances

The Purchase Money Note is recorded in long-term debt on the Consolidated balance sheets. The fair value of the Purchase Money Note was estimated based on a discounted cash flow methodology and incorporated estimated market discount rates.

The FHLB advances assumed in the acquisition are recorded in short-term borrowings and in long-term debt. The fair values of the FHLB advances were based on a discounted cash flow methodology and considered the observed FHLB advance issuance rates.

Loans

The following table presents the unpaid principal balance ("UPB") and fair values of the loans acquired as of May 1, 2023, and reflects adjustments made during the measurement period to the acquisition-date fair value of the loans acquired.

	May 1, 2023	
(in millions)	UPB	Fair value
Residential real estate	$ **106,240**	$ **92,053**
Auto and other	**3,093**	**2,030**
Total consumer	**109,333**	**94,083**
Secured by real estate	**37,117**	**33,602**
Commercial & industrial	**4,332**	**3,932**
Other	**23,499**	**21,625**
Total wholesale	**64,948**	**59,159**
Total loans	$ **174,281**	$ **153,242**

Unaudited pro forma condensed combined financial information

The following table presents certain unaudited pro forma financial information for the year ended December 31, 2023 and 2022 as if the First Republic acquisition had occurred on January 1, 2022, including recognition of the estimated bargain purchase gain of $2.8 billion and the provision for credit losses of $1.2 billion. Additional adjustments include the interest on the Purchase Money Note and the impact of amortizing and accreting certain estimated fair value adjustments related to intangible assets, loans and lending-related commitments.

The Firm expects to achieve operating cost savings and other business synergies resulting from the acquisition that are not reflected in the pro forma amounts. The pro forma information is not necessarily indicative of the historical results of operations had the acquisition occurred on January 1, 2022, nor is it indicative of the results of operations in future periods.

	Year ended December 31,	
(in millions)	2023	2022
Noninterest revenue	$ **65,816**	$ 66,510
Net interest income	**90,856**	71,005
Net income	**48,665**	41,089

Supplementary Information: Distribution of assets, liabilities and stockholders' equity; interest rates and interest differentials

Consolidated average balance sheets, interest and rates

Provided below is a summary of JPMorganChase's consolidated average balances, interest and rates on a taxable-equivalent basis for the years 2022 through 2024. Income computed on a taxable-equivalent basis is the income reported in the Consolidated statements of income, adjusted to present interest income and rates earned on assets exempt from income taxes (i.e., federal taxes) on a basis comparable with other taxable investments. The incremental tax rate used for calculating the taxable-equivalent adjustment was approximately 24% in 2024, 2023 and 2022.

(Table continued on next page)

(Unaudited)	**2024**		
Year ended December 31, (Taxable-equivalent interest and rates; in millions, except rates)	Average balance[f]	Interest[f]	Rate
Assets			
Deposits with banks	$ 490,205	$ 22,297	4.55 %
Federal funds sold and securities purchased under resale agreements	359,197	18,299	5.09
Securities borrowed	209,744	9,208	4.39
Trading assets – debt instruments	456,029	20,373	4.47
Taxable securities	583,329	21,947	3.76
Non-taxable securities[a]	27,912	1,393	4.99
Total investment securities	611,241	23,340	3.82 [i]
Loans	1,322,425	92,588 [h]	7.00
All other interest-earning assets[b][c]	88,726	8,305	9.36
Total interest-earning assets	3,537,567	194,410	5.50
Allowance for loan losses	(22,877)		
Cash and due from banks	22,591		
Trading assets – equity and other instruments	208,534		
Trading assets – derivative receivables	57,005		
Goodwill, MSRs and other intangible assets	64,393		
All other noninterest-earning assets	218,709		
Total assets	$ 4,085,922		
Liabilities			
Interest-bearing deposits	$ 1,748,050	$ 49,559	2.84 %
Federal funds purchased and securities loaned or sold under repurchase agreements	363,820	19,149	5.26
Short-term borrowings	39,593	2,101	5.31
Trading liabilities – debt and all other interest-bearing liabilities[d][e]	314,054	10,238	3.26
Beneficial interests issued by consolidated VIEs	26,515	1,383	5.22
Long-term debt	344,346	18,920	5.49
Total interest-bearing liabilities	2,836,378	101,350	3.57
Noninterest-bearing deposits	638,592		
Trading liabilities – equity and other instruments[e]	32,025		
Trading liabilities – derivative payables	39,497		
All other liabilities, including the allowance for lending-related commitments	203,006		
Total liabilities	3,749,498		
Stockholders' equity			
Preferred stock	24,054		
Common stockholders' equity	312,370		
Total stockholders' equity	336,424 [g]		
Total liabilities and stockholders' equity	$ 4,085,922		
Interest rate spread			1.93 %
Net interest income and net yield on interest-earning assets		$ 93,060	2.63

(a) Represents securities that are tax-exempt for U.S. federal income tax purposes.
(b) Includes brokerage-related held-for-investment customer receivables, which are classified in accrued interest and accounts receivable, and all other interest-earning assets, which are classified in other assets on the Consolidated Balance Sheets.
(c) The rates reflect the impact of interest earned on cash collateral where the cash collateral has been netted against certain derivative payables.
(d) All other interest-bearing liabilities include brokerage-related customer payables.
(e) The combined balance of trading liabilities – debt and equity instruments was $185.4 billion, $153.3 billion and $138.1 billion for the years ended December 31, 2024, 2023 and 2022, respectively.
(f) Includes the effect of derivatives that qualify for hedge accounting where applicable. Taxable-equivalent amounts are used, also where applicable. Refer to Note 5 for additional information on hedge accounting.

(Table continued from previous page)

2023			2022		
Average balance[f]	Interest[f]	Rate	Average balance[f]	Interest[f]	Rate
$ 499,396	$ 21,797	4.36 %	$ 670,773	$ 9,039	1.35 %
317,159	15,079	4.75	307,150	4,632	1.51
193,228	7,983	4.13	205,516	2,237	1.09
376,928	16,001	4.25	283,108	9,097	3.21
573,914	17,390	3.03	626,122	10,372	1.66
30,886	1,560	5.05	27,863	1,224	4.39
604,800	18,950	3.13 [i]	653,985	11,596	1.77 [i]
1,248,076	83,589 [h]	6.70	1,100,318	52,877 [h]	4.81
86,121	7,669	8.90	128,229	3,763	2.93
3,325,708	171,068	5.14	3,349,079	93,241	2.78
(20,762)			(17,399)		
24,853			27,601		
160,087			140,778		
64,227			78,606		
63,212			59,467		
204,899			215,408		
$ 3,822,224			$ 3,853,540		
$ 1,698,529	$ 40,016	2.36 %	$ 1,748,666	$ 10,082	0.58 %
256,086	13,259	5.18	242,762	3,721	1.53
37,468	1,894	5.05	46,063	747	1.62
286,605	9,396	3.28	268,019	3,246	1.21
18,648	953	5.11	11,208	226	2.02
296,433	15,803	5.33	250,080	8,075	3.23
2,593,769	81,321	3.14	2,566,798	26,097	1.02
660,538			719,249		
30,501			39,155		
46,355			57,388		
181,601			185,989		
3,512,764			3,568,579		
27,404			31,893		
282,056			253,068		
309,460 [g]			284,961 [g]		
$ 3,822,224			$ 3,853,540		
		2.00 %			1.76 %
	$ 89,747	2.70		$ 67,144	2.00

(g) The ratio of average stockholders' equity to average assets was 8.2%, 8.1% and 7.4% for the years ended December 31, 2024, 2023 and 2022, respectively. The return on average stockholders' equity, based on net income, was 17.4%, 16.0% and 13.2% for the years ended December 31, 2024, 2023 and 2022, respectively.

(h) Included fees and commissions on loans of $3.6 billion, $2.2 billion and $1.8 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

(i) The annualized rate for securities based on amortized cost was 3.79%, 3.09% and 1.75% for the years ended December 31, 2024, 2023 and 2022, respectively, and does not give effect to changes in fair value that are reflected in AOCI.

Within the Consolidated average balance sheets, interest and rates summary, the principal amounts of nonaccrual loans have been included in the average loan balances used to determine the average interest rate earned on loans. Refer to Note 12 for additional information on nonaccrual loans, including interest accrued.

Interest rates and interest differential analysis of net interest income – U.S. and non-U.S.

Presented below is a summary of interest and rates segregated between U.S. and non-U.S. operations for the years 2022 through 2024. The segregation of U.S. and non-U.S. components is based on the location of the office recording the transaction.

(Table continued on next page)

(Unaudited) Year ended December 31, (Taxable-equivalent interest and rates; in millions, except rates)	2024 Average balance	2024 Interest	2024 Rate
Interest-earning assets			
Deposits with banks:			
U.S.	$ **284,913**	$ **15,157**	**5.32** %
Non-U.S.	**205,292**	**7,140**	**3.48**
Federal funds sold and securities purchased under resale agreements:			
U.S.	**193,210**	**10,686**	**5.53**
Non-U.S.	**165,987**	**7,613**	**4.59**
Securities borrowed:			
U.S.	**150,251**	**7,330**	**4.88**
Non-U.S.	**59,493**	**1,878**	**3.16**
Trading assets – debt instruments:			
U.S.	**309,568**	**13,579**	**4.39**
Non-U.S.	**146,461**	**6,794**	**4.64**
Investment securities:			
U.S.	**567,784**	**21,458**	**3.78**
Non-U.S.	**43,457**	**1,882**	**4.33**
Loans:			
U.S.	**1,211,978**	**85,621**	**7.06**
Non-U.S.	**110,447**	**6,967**	**6.31**
All other interest-earning assets, predominantly U.S.[(a)]	**88,726**	**8,305**	**9.36**
Total interest-earning assets	**3,537,567**	**194,410**	**5.50**
Interest-bearing liabilities			
Interest-bearing deposits:			
U.S.	**1,307,000**	**33,173**	**2.54**
Non-U.S.	**441,050**	**16,386**	**3.72**
Federal funds purchased and securities loaned or sold under repurchase agreements:			
U.S.	**294,476**	**15,949**	**5.42**
Non-U.S.	**69,344**	**3,200**	**4.61**
Trading liabilities – debt, short-term and all other interest-bearing liabilities:			
U.S.	**222,710**	**8,289**	**3.72**
Non-U.S.	**130,937**	**4,050**	**3.09**
Beneficial interests issued by consolidated VIEs, predominantly U.S.	**26,515**	**1,383**	**5.22**
Long-term debt:			
U.S.	**338,166**	**18,760**	**5.55**
Non-U.S.	**6,180**	**160**	**2.59**
Total interest-bearing liabilities	**2,836,378**	**101,350**	**3.57**
Noninterest-bearing liabilities[(b)]	**701,189**		
Total investable funds	$ **3,537,567**	$ **101,350**	**2.86** %
Net interest income and net yield:		$ **93,060**	**2.63** %
U.S.		**80,913**	**2.92**
Non-U.S.		**12,147**	**1.58**
Percentage of total assets and liabilities attributable to non-U.S. operations:			
Assets			**24.3**
Liabilities			**20.5**

(a) The rates reflect the impact of interest earned on cash collateral where that cash collateral has been netted against certain derivative payables.
(b) Represents the amount of noninterest-bearing liabilities funding interest-earning assets.

Refer to the "Net interest income" discussion in Consolidated Results of Operations on pages 59–62 for further information.

(Table continued from previous page)

2023			2022		
Average balance	Interest	Rate	Average balance	Interest	Rate
$ 296,784	$ 15,348	5.17 %	$ 456,366	$ 7,418	1.63 %
202,612	6,449	3.18	214,407	1,621	0.76
155,304	8,330	5.36	130,213	2,191	1.68
161,855	6,749	4.17	176,937	2,441	1.38
133,805	6,239	4.66	142,736	1,811	1.27
59,423	1,744	2.93	62,780	426	0.68
248,541	10,721	4.31	170,975	5,414	3.17
128,387	5,280	4.11	112,133	3,683	3.28
568,505	17,469	3.07	623,285	10,994	1.76
36,295	1,481	4.08	30,700	602	1.96
1,137,162	76,884	6.76	985,187	48,953	4.97
110,914	6,705	6.05	115,131	3,924	3.41
86,121	7,669	8.90	128,229	3,763	2.93
3,325,708	171,068	5.14	3,349,079	93,241	2.78
1,290,110	26,253	2.03	1,358,322	7,026	0.52
408,419	13,763	3.37	390,344	3,056	0.78
197,049	10,639	5.40	173,016	3,083	1.78
59,037	2,620	4.44	69,746	638	0.91
205,388	7,774	3.79	194,570	2,384	1.23
118,685	3,516	2.96	119,512	1,609	1.35
18,648	953	5.11	11,208	226	2.02
293,218	15,749	5.37	246,670	8,026	3.25
3,215	54	1.68	3,410	49	1.44
2,593,769	81,321	3.14	2,566,798	26,097	1.02
731,939			782,281		
$ 3,325,708	$ 81,321	2.45 %	$ 3,349,079	$ 26,097	0.78 %
	$ 89,747	2.70 %		$ 67,144	2.00 %
	77,923	3.01		58,950	2.27
	11,824	1.61		8,194	1.09
		24.7			24.9
		20.2			20.6

Changes in net interest income, volume and rate analysis

The table below presents an attribution of net interest income between volume and rate. The attribution between volume and rate is calculated using annual average balances for each category of assets and liabilities shown in the table and the corresponding annual rates (refer to pages 322-325 for more information on average balances and rates). In this analysis, when the change cannot be isolated to either volume or rate, it has been allocated to volume. The annual rates include the impact of changes in market rates, as well as the impact of any change in composition of the various products within each category of asset or liability. This analysis is calculated separately for each category without consideration of the relationship between categories (for example, the net spread between the rates earned on assets and the rates paid on liabilities that fund those assets). As a result, changes in the granularity or groupings considered in this analysis would produce a different attribution result, and due to the complexities involved, precise allocation of changes in interest rates between volume and rates is inherently complex and judgmental.

	2024 versus 2023			2023 versus 2022		
(Unaudited)	Increase/(decrease) due to change in:			Increase/(decrease) due to change in:		
Year ended December 31, (On a taxable-equivalent basis; in millions)	Volume	Rate	Net change	Volume	Rate	Net change
Interest-earning assets						
Deposits with banks:						
U.S.	**$ (636)**	**$ 445**	**$ (191)**	$ (8,225)	$ 16,155	$ 7,930
Non-U.S.	**83**	**608**	**691**	(361)	5,189	4,828
Federal funds sold and securities purchased under resale agreements:						
U.S.	**2,092**	**264**	**2,356**	1,347	4,792	6,139
Non-U.S.	**184**	**680**	**864**	(629)	4,937	4,308
Securities borrowed:						
U.S.	**797**	**294**	**1,091**	(411)	4,839	4,428
Non-U.S.	**(3)**	**137**	**134**	(95)	1,413	1,318
Trading assets – debt instruments:						
U.S.	**2,659**	**199**	**2,858**	3,358	1,949	5,307
Non-U.S.	**834**	**680**	**1,514**	666	931	1,597
Investment securities:						
U.S.	**(47)**	**4,036**	**3,989**	(1,690)	8,165	6,475
Non-U.S.	**310**	**91**	**401**	228	651	879
Loans:						
U.S.	**5,326**	**3,411**	**8,737**	10,296	17,635	27,931
Non-U.S.	**(26)**	**288**	**262**	(258)	3,039	2,781
All other interest-earning assets, predominantly U.S.	**240**	**396**	**636**	(3,749)	7,655	3,906
Change in interest income	**11,813**	**11,529**	**23,342**	477	77,350	77,827
Interest-bearing liabilities						
Interest-bearing deposits:						
U.S.	**340**	**6,580**	**6,920**	(1,284)	20,511	19,227
Non-U.S.	**1,194**	**1,429**	**2,623**	597	10,110	10,707
Federal funds purchased and securities loaned or sold under repurchase agreements:						
U.S.	**5,271**	**39**	**5,310**	1,293	6,263	7,556
Non-U.S.	**480**	**100**	**580**	(480)	2,462	1,982
Trading liabilities – debt, short-term and all other interest-bearing liabilities:						
U.S.	**659**	**(144)**	**515**	409	4,981	5,390
Non-U.S.	**380**	**154**	**534**	(17)	1,924	1,907
Beneficial interests issued by consolidated VIEs, predominantly U.S.	**409**	**21**	**430**	381	346	727
Long-term debt:						
U.S.	**2,483**	**528**	**3,011**	2,494	5,229	7,723
Non-U.S.	**77**	**29**	**106**	(3)	8	5
Change in interest expense	**11,293**	**8,736**	**20,029**	3,390	51,834	55,224
Change in net interest income	**$ 520**	**$ 2,793**	**$ 3,313**	$ (2,913)	$ 25,516	$ 22,603

Glossary of Terms and Acronyms

2023 Form 10-K: Annual report on Form 10-K for the year ended December 31, 2023, filed with the U.S. Securities and Exchange Commission.

ABS: Asset-backed securities

Active foreclosures: Loans referred to foreclosure where formal foreclosure proceedings are ongoing. Includes both judicial and non-judicial states.

AFS: Available-for-sale

ALCO: Asset Liability Committee

Alternative assets "Alternatives": The following types of assets constitute alternative investments - hedge funds, currency, real estate, private equity and other investment funds designed to focus on nontraditional strategies

Amortized cost: Amount at which a financing receivable or investment is originated or acquired, adjusted for accretion or amortization of premium, discount, and net deferred fees or costs, collection of cash, charge-offs, foreign exchange, and fair value hedge accounting adjustments. For AFS securities, amortized cost is also reduced by any impairment losses recognized in earnings. Amortized cost is not reduced by the allowance for credit losses, except where explicitly presented net.

AOCI: Accumulated other comprehensive income/(loss)

ARM: Adjustable rate mortgage(s)

AUC: "Assets under custody": Represents assets held directly or indirectly on behalf of clients under safekeeping, custody and servicing arrangements.

AUM: "Assets under management": Represent assets managed by AWM on behalf of its Private Banking, Institutional and Retail clients. Includes "Committed capital not Called."

Auto loan and lease origination volume: Dollar amount of auto loans and leases originated.

AWM: Asset & Wealth Management

Beneficial interests issued by consolidated VIEs: Represents the interest of third-party holders of debt, equity securities, or other obligations, issued by VIEs that JPMorganChase consolidates.

Benefit obligation: Refers to the projected benefit obligation for pension plans and the accumulated postretirement benefit obligation for OPEB plans.

BHC: Bank holding company

BWM: Banking & Wealth Management

Bridge Financing Portfolio: A portfolio of held-for-sale unfunded loan commitments and funded loans. The unfunded commitments include both short-term bridge loan commitments that will ultimately be replaced by longer term financing as well as term loan commitments. The funded loans include term loans and funded revolver facilities.

CB: Commercial Banking

CCAR: Comprehensive Capital Analysis and Review

CCB: Consumer & Community Banking

CCB Consumer customer: A unique individual that has financial ownership or decision-making power with respect to accounts; excludes customers under the age of 18. Where a customer uses the same identifier as both a Consumer and a Small business, the customer is included in both metrics.

CCB Small business customer: A unique business or legal entity that has financial ownership or decision-making power with respect to accounts. Where a customer uses the same identifier as both a Consumer and a Small business, the customer is included in both metrics.

CCO: Chief Compliance Officer

CCP: "Central counterparty" is a clearing house that interposes itself between counterparties to contracts traded in one or more financial markets, becoming the buyer to every seller and the seller to every buyer and thereby ensuring the future performance of open contracts. A CCP becomes a counterparty to trades with market participants through novation, an open offer system, or another legally binding arrangement.

CDS: Credit default swaps

CECL: Current Expected Credit Losses

CEO: Chief Executive Officer

CET1 Capital: Common equity Tier 1 capital

CFO: Chief Financial Officer

CFP: Contingency funding plan

CFTC: Commodity Futures Trading Commission

CIB: Commercial & Investment Bank

CIO: Chief Investment Office

Client assets: Represent assets under management as well as custody, brokerage, administration and deposit accounts.

Client deposits and other third-party liabilities: Deposits, as well as deposits that are swept to on-balance sheet liabilities (e.g., commercial paper, federal funds purchased and securities loaned or sold under repurchase agreements) as part of client cash management programs.

Client investment assets: Represent assets under management as well as custody, brokerage and annuity accounts, and deposits held in investment accounts.

CLO: Collateralized loan obligations

CLTV: Combined loan-to-value

CMT: Constant Maturity Treasury

Collateral-dependent: A loan is considered to be collateral-dependent when repayment of the loan is expected to be provided substantially through the operation or sale of the collateral when the borrower is experiencing financial difficulty, including when foreclosure is deemed probable based on borrower delinquency.

Commercial Card: provides a wide range of payment services to corporate and public sector clients worldwide through the commercial card products. Services include procurement, corporate travel and entertainment, expense management services, and business-to-business payment solutions.

Credit derivatives: Financial instruments whose value is derived from the credit risk associated with the debt of a third-party issuer (the reference entity) which allow one party (the protection purchaser) to transfer that risk to another party (the protection seller). Upon the occurrence of a credit event by the reference entity, which may include, among other events, the bankruptcy or failure to pay its obligations, or certain restructurings of the debt of the reference entity, neither party has recourse to the reference entity. The protection purchaser has recourse to the protection seller for the difference between the face value of the CDS contract and the fair value at the time of settling the credit derivative contract. The determination as to whether a credit event has occurred is generally made by the relevant International Swaps and Derivatives Association ("ISDA") Determinations Committee.

Criticized: Criticized loans, lending-related commitments and derivative receivables that are classified as special mention, substandard and doubtful categories for regulatory purposes and are generally consistent with a rating of CCC+/Caa1 and below, as defined by S&P and Moody's.

CRO: Chief Risk Officer

CRR: Capital Requirements Regulation

CTC: CIO, Treasury and Corporate

Custom lending: Loans to AWM's Global Private Bank clients, including loans to private investment funds and loans that are collateralized by nontraditional asset types, such as art work, aircraft, etc.

CVA: Credit valuation adjustment

Debit and credit card sales volume: Dollar amount of card member purchases, net of returns.

Deposit margin: Represents net interest income expressed as a percentage of average deposits.

Distributed denial-of-service attack: The use of a large number of remote computer systems to electronically send a high volume of traffic to a target website to create a service outage at the target. This is a form of cyberattack.

Dodd-Frank Act: Wall Street Reform and Consumer Protection Act

DVA: Debit valuation adjustment

EC: European Commission

Eligible HQLA: Eligible high-quality liquid assets, for purposes of calculating the LCR, is the amount of unencumbered HQLA that satisfy certain operational considerations as defined in the LCR rule.

Eligible LTD: Long-term debt satisfying certain eligibility criteria

Embedded derivatives: are implicit or explicit terms or features of a financial instrument that affect some or all of the cash flows or the value of the instrument in a manner similar to a derivative. An instrument containing such terms or features is referred to as a "hybrid." The component of the hybrid that is the non-derivative instrument is referred to as the "host." For example, callable debt is a hybrid instrument that contains a plain vanilla debt instrument (i.e., the host) and an embedded option that allows the issuer to redeem the debt issue at a specified date for a specified amount (i.e., the embedded derivative). However, a floating rate instrument is not a hybrid composed of a fixed-rate instrument and an interest rate swap.

EPS: Earnings per share

ERISA: Employee Retirement Income Security Act of 1974

ETD: "Exchange-traded derivatives": Derivative contracts that are executed on an exchange and settled via a central clearing house.

EU: European Union

Expense categories:

- Volume- and/or revenue-related expenses generally correlate with changes in the related business/ transaction volume or revenue. Examples include commissions and incentive compensation within the LOBs, depreciation expense related to operating lease assets, and brokerage expense related to trading transaction volume.
- Investments in the business include expenses associated with supporting medium- to longer-term strategic plans of the Firm. Examples include front office growth, market expansion, initiatives in technology (including related compensation), marketing, and acquisitions.
- Structural expenses are those associated with the day-to-day cost of running the Firm and are expenses not included in the above two categories. Examples include employee salaries and benefits, certain other incentive compensation, and costs related to real estate.

Fannie Mae: Federal National Mortgage Association

FASB: Financial Accounting Standards Board

FCA: Financial Conduct Authority

FCC: Firmwide Control Committee

FDIC: Federal Deposit Insurance Corporation

FDM: "Financial difficulty modification" applies to loan modifications effective January 1, 2023, and is deemed to occur when the Firm modifies specific terms of the original loan agreement. The following types of modifications are considered FDMs: principal forgiveness, interest rate reduction, other-than-insignificant payment delay, term extension or a combination of these modifications.

Federal Reserve: The Board of the Governors of the Federal Reserve System

FFIEC: Federal Financial Institutions Examination Council

FHA: Federal Housing Administration

FHLB: Federal Home Loan Bank

FICC: The Fixed Income Clearing Corporation

FICO score: A measure of consumer credit risk provided by credit bureaus, typically produced from statistical models by Fair Isaac Corporation utilizing data collected by the credit bureaus.

FINRA: Financial Industry Regulatory Authority

Firm: JPMorgan Chase & Co.

Forward points: Represents the interest rate differential between two currencies, which is either added to or subtracted from the current exchange rate (i.e., "spot rate") to determine the forward exchange rate.

FRC: Firmwide Risk Committee

Freddie Mac: Federal Home Loan Mortgage Corporation

Free standing derivatives: a derivative contract entered into either separate and apart from any of the Firm's other financial instruments or equity transactions. Or, in conjunction with some other transaction and is legally detachable and separately exercisable.

FSB: Financial Stability Board

FTE: Fully taxable equivalent

FVA: Funding valuation adjustment

FX: Foreign exchange

G7: Group of Seven nations: Countries in the G7 are Canada, France, Germany, Italy, Japan, the U.K. and the U.S.

G7 government securities: Securities issued by the government of one of the G7 nations.

Ginnie Mae: Government National Mortgage Association

GSIB: Global systemically important banks

HELOC: Home equity line of credit

Home equity – senior lien: Represents loans and commitments where JPMorganChase holds the first security interest on the property.

Home equity – junior lien: Represents loans and commitments where JPMorganChase holds a security interest that is subordinate in rank to other liens.

HQLA: "High-quality liquid assets" consist of cash and certain high-quality liquid securities as defined in the LCR rule.

HTM: Held-to-maturity

IBOR: Interbank Offered Rate

ICAAP: Internal capital adequacy assessment process

IDI: Insured depository institutions

IHC: JPMorgan Chase Holdings LLC, an intermediate holding company

Indirect tax expense: Refers to taxes that are imposed on goods and services rather than on income. Examples of indirect taxes include value-added tax ("VAT") and sales tax, among others.

Investment-grade: An indication of credit quality based on JPMorganChase's internal risk assessment. The Firm considers ratings of BBB-/Baa3 or higher as investment-grade.

IPO: Initial public offering

ISDA: International Swaps and Derivatives Association

JPMorganChase: JPMorgan Chase & Co.

JPMorgan Chase Bank, N.A.: JPMorgan Chase Bank, National Association

JPMorgan Chase Foundation or the Firm's Foundation: A not-for-profit organization that makes contributions for charitable and educational purposes.

J.P. Morgan Securities: J.P. Morgan Securities LLC

JPMSE: J.P. Morgan SE

LCR: Liquidity coverage ratio

LDA: Loss Distribution Approach

LGD: Loss given default

LIBOR: London Interbank Offered Rate

LLC: Limited Liability Company

LOB: Line of business

LOB CROs: Line of Business and CTC Chief Risk Officers

LTIP: Long-term incentive plan

LTV: "Loan-to-value": For residential real estate loans, the relationship, expressed as a percentage, between the principal amount of a loan and the appraised value of the collateral (i.e., residential real estate) securing the loan.

Origination date LTV ratio
The LTV ratio at the origination date of the loan. Origination date LTV ratios are calculated based on the actual appraised values of collateral (i.e., loan-level data) at the origination date.

Current estimated LTV ratio
An estimate of the LTV as of a certain date. The current estimated LTV ratios are calculated using estimated collateral values derived from a nationally recognized

home price index measured at the metropolitan statistical area ("MSA") level. These MSA-level home price indices consist of actual data to the extent available and forecasted data where actual data is not available. As a result, the estimated collateral values used to calculate these ratios do not represent actual appraised loan-level collateral values; as such, the resulting LTV ratios are necessarily imprecise and should therefore be viewed as estimates.

Combined LTV ratio
The LTV ratio considering all available lien positions, as well as unused lines, related to the property. Combined LTV ratios are used for junior lien home equity products.

Macro businesses: the macro businesses include Rates, Currencies and Emerging Markets, Fixed Income Financing and Commodities in CIB's Fixed Income Markets.

Managed basis: A non-GAAP presentation of Firmwide financial results that includes reclassifications to present revenue on a fully taxable-equivalent basis. Management also uses this financial measure at the segment level, because it believes this provides information to enable investors to understand the underlying operational performance and trends of the particular business segment and facilitates a comparison of the business segment with the performance of competitors.

Markets: consists of CIB's Fixed Income Markets and Equity Markets businesses.

Master netting agreement: A single agreement with a counterparty that permits multiple transactions governed by that agreement to be terminated or accelerated and settled through a single payment in a single currency in the event of a default (e.g., bankruptcy, failure to make a required payment or securities transfer or deliver collateral or margin when due).

MBS: Mortgage-backed securities

MD&A: Management's discussion and analysis

Measurement alternative: Measures equity securities without readily determinable fair values at cost less impairment (if any), plus or minus observable price changes from an identical or similar investment of the same issuer.

Merchant Services: offers merchants payment processing capabilities, fraud and risk management, data and analytics, and other payments services. Through Merchant Services, merchants of all sizes can accept payments via credit and debit cards and payments in multiple currencies.

MEV: Macroeconomic variable

Moody's: Moody's Investor Services

Mortgage origination channels:

Retail – Borrowers who buy or refinance a home through direct contact with a mortgage banker employed by the Firm using a branch office, the Internet or by phone. Borrowers are frequently referred to a mortgage banker by a banker in a Chase branch, real estate brokers, home builders or other third parties.

Correspondent – Banks, thrifts, other mortgage banks and other financial institutions that sell closed loans to the Firm.

Mortgage product types:

Alt-A
Alt-A loans are generally higher in credit quality than subprime loans but have characteristics that would disqualify the borrower from a traditional prime loan. Alt-A lending characteristics may include one or more of the following: (i) limited documentation; (ii) a high CLTV ratio; (iii) loans secured by non-owner occupied properties; or (iv) a debt-to-income ratio above normal limits. A substantial proportion of the Firm's Alt-A loans are those where a borrower does not provide complete documentation of his or her assets or the amount or source of his or her income.

Option ARMs
The option ARM real estate loan product is an adjustable-rate mortgage loan that provides the borrower with the option each month to make a fully amortizing, interest-only or minimum payment. The minimum payment on an option ARM loan is based on the interest rate charged during the introductory period. This introductory rate is usually significantly below the fully indexed rate. The fully indexed rate is calculated using an index rate plus a margin. Once the introductory period ends, the contractual interest rate charged on the loan increases to the fully indexed rate and adjusts monthly to reflect movements in the index. The minimum payment is typically insufficient to cover interest accrued in the prior month, and any unpaid interest is deferred and added to the principal balance of the loan. Option ARM loans are subject to payment recast, which converts the loan to a variable-rate fully amortizing loan upon meeting specified loan balance and anniversary date triggers.

Prime
Prime mortgage loans are made to borrowers with good credit records who meet specific underwriting requirements, including prescriptive requirements related to income and overall debt levels. New prime mortgage borrowers provide full documentation and generally have reliable payment histories.

Subprime
Subprime loans are loans that, prior to mid-2008, were offered to certain customers with one or more high risk characteristics, including but not limited to: (i) unreliable or poor payment histories; (ii) a high LTV ratio of greater than 80% (without borrower-paid mortgage insurance); (iii) a high debt-to-income ratio; (iv) an occupancy type for the loan is other than the borrower's

primary residence; or (v) a history of delinquencies or late payments on the loan.

MREL: Minimum requirements for own funds and eligible liabilities

MSR: Mortgage servicing rights

Multi-asset: Any fund or account that allocates assets under management to more than one asset class.

NA: Data is not applicable or available for the period presented.

NAV: Net Asset Value

Net Capital Rule: Rule 15c3-1 under the Securities Exchange Act of 1934.

Net charge-off/(recovery) rate: Represents net charge-offs/(recoveries) (annualized) divided by average retained loans for the reporting period.

Net interchange income includes the following components:

- **Interchange income**: Fees earned by credit and debit card issuers on sales transactions.
- **Rewards costs**: The cost to the Firm for points earned by cardholders enrolled in credit card rewards programs generally tied to sales transactions.
- **Partner payments**: Payments to co-brand credit card partners based on the cost of loyalty program rewards earned by cardholders on credit card transactions.

Net mortgage servicing revenue: Includes operating revenue earned from servicing third-party mortgage loans, which is recognized over the period in which the service is provided; changes in the fair value of MSRs; the impact of risk management activities associated with MSRs; and gains and losses on securitization of excess mortgage servicing. Net mortgage servicing revenue also includes gains and losses on sales and lower of cost or fair value adjustments of certain repurchased loans insured by U.S. government agencies.

Net revenue rate: Represents Card Services net revenue (annualized) expressed as a percentage of average loans for the period.

Net yield on interest-earning assets: The average rate for interest-earning assets less the average rate paid for all sources of funds.

NFA: National Futures Association

NM: Not meaningful

NOL: Net operating loss

Nonaccrual loans: Loans for which interest income is not recognized on an accrual basis. Loans (other than credit card loans and certain consumer loans insured by U.S. government agencies) are placed on nonaccrual status when full payment of principal and interest is not expected, regardless of delinquency status, or when principal and interest have been in default for a period of 90 days or more unless the loan is both well-secured and in the process of collection. Collateral-dependent loans are typically maintained on nonaccrual status.

Nonperforming assets: Nonperforming assets include nonaccrual loans, nonperforming derivatives and certain assets acquired in loan satisfactions, predominantly real estate owned and other commercial and personal property.

NSFR: Net Stable Funding Ratio

OAS: Option-adjusted spread

OCC: Office of the Comptroller of the Currency

OCI: Other comprehensive income/(loss)

OPEB: Other postretirement employee benefit

Operating losses: Primarily refer to fraud losses associated with customer deposit accounts, credit and debit cards; exclude legal expense.

Over-the-counter (“OTC”) derivatives: Derivative contracts that are negotiated, executed and settled bilaterally between two derivative counterparties, where one or both counterparties is a derivatives dealer.

Over-the-counter cleared (“OTC-cleared”) derivatives: Derivative contracts that are negotiated and executed bilaterally, but subsequently settled via a central clearing house, such that each derivative counterparty is only exposed to the default of that clearing house.

Overhead ratio: Noninterest expense as a percentage of total net revenue.

Parent Company: JPMorgan Chase & Co.

Participating securities: Represents unvested share-based compensation awards containing nonforfeitable rights to dividends or dividend equivalents (collectively, “dividends”), which are included in the earnings per share calculation using the two-class method. JPMorganChase grants RSUs to certain employees under its share-based compensation programs, which entitle the recipients to receive nonforfeitable dividends during the vesting period on a basis equivalent to the dividends paid to holders of common stock. These unvested awards meet the definition of participating securities. Under the two-class method, all earnings (distributed and undistributed) are allocated to each class of common stock and participating securities, based on their respective rights to receive dividends.

PCAOB: Public Company Accounting Oversight Board

PCD: “Purchased credit deteriorated” assets represent acquired financial assets that as of the date of acquisition have experienced a more-than-insignificant deterioration in credit quality since origination, as determined by the Firm.

PD: Probability of default

Pillar 1: The Basel framework consists of a three “Pillar” approach. Pillar 1 establishes minimum capital requirements, defines eligible capital instruments, and prescribes rules for calculating RWA.

Pillar 3: The Basel framework consists of a three "Pillar" approach. Pillar 3 encourages market discipline through disclosure requirements which allow market participants to assess the risk and capital profiles of banks.

PRA: Prudential Regulation Authority

Preferred stock dividends: reflects dividends declared and deemed dividends upon redemption of preferred stock

Pre-provision profit/(loss): Represents total net revenue less noninterest expense. The Firm believes that this financial measure is useful in assessing the ability of a lending institution to generate income in excess of its provision for credit losses.

Pre-tax margin: Represents income before income tax expense divided by total net revenue, which is, in management's view, a comprehensive measure of pretax performance derived by measuring earnings after all costs are taken into consideration. It is one basis upon which management evaluates the performance of AWM against the performance of their respective competitors.

Principal transactions revenue: Principal transactions revenue is driven by many factors, including:

- the bid-offer spread, which is the difference between the price at which a market participant is willing and able to sell an instrument to the Firm and the price at which another market participant is willing and able to buy it from the Firm, and vice versa; and
- realized and unrealized gains and losses on financial instruments and commodities transactions, including those accounted for under the fair value option, primarily used in client-driven market-making activities.
  - Realized gains and losses result from the sale of instruments, closing out or termination of transactions, or interim cash payments.
  - Unrealized gains and losses result from changes in valuation.

In connection with its client-driven market-making activities, the Firm transacts in debt and equity instruments, derivatives and commodities, including physical commodities inventories and financial instruments that reference commodities.

Principal transactions revenue also includes realized and unrealized gains and losses related to:

- derivatives designated in qualifying hedge accounting relationships, primarily fair value hedges of commodity and foreign exchange risk;
- derivatives used for specific risk management purposes, primarily to mitigate credit, foreign exchange and interest rate risks.

Production revenue: Includes fees and income recognized as earned on mortgage loans originated with the intent to sell, and the impact of risk management activities associated with the mortgage pipeline and warehouse loans. Production revenue also includes gains and losses on sales and lower of cost or fair value adjustments on mortgage loans held-for-sale (excluding certain repurchased loans insured by U.S. government agencies), and changes in the fair value of financial instruments measured under the fair value option.

PSU(s): Performance share units

Regulatory VaR: Daily aggregated VaR calculated in accordance with regulatory rules.

REO: Real estate owned

Reported basis: Financial statements prepared under U.S. GAAP, which excludes the impact of taxable-equivalent adjustments.

Retained loans: Loans that are held-for-investment (i.e., excludes loans held-for-sale and loans at fair value).

Revenue wallet: Proportion of fee revenue based on estimates of investment banking fees generated across the industry (i.e., the revenue wallet) from investment banking transactions in M&A, equity and debt underwriting, and loan syndications. Source: Dealogic, a third-party provider of investment banking competitive analysis and volume-based league tables for the above noted industry products.

RHS: Rural Housing Service of the U.S. Department of Agriculture

ROA: Return on assets

ROE: Return on equity

ROTCE: Return on tangible common equity

ROU assets: Right-of-use assets

RSU(s): Restricted stock units

RWA "Risk-weighted assets": Basel III establishes two comprehensive approaches for calculating RWA (a Standardized approach and an Advanced approach) which include capital requirements for credit risk, market risk, and in the case of Basel III Advanced, also operational risk. Key differences in the calculation of credit risk RWA between the Standardized and Advanced approaches are that for Basel III Advanced, credit risk RWA is based on risk-sensitive approaches which largely rely on the use of internal credit models and parameters, whereas for Basel III Standardized, credit risk RWA is generally based on supervisory risk-weightings which vary primarily by counterparty type and asset class. Market risk RWA is calculated on a generally consistent basis between Basel III Standardized and Basel III Advanced.

S&P: Standard and Poor's

SAR as it pertains to Hong Kong: Special Administrative Region

SAR(s) as it pertains to employee stock awards: Stock appreciation rights

SCB: Stress capital buffer

Scored portfolios: Consumer loan portfolios that predominantly include residential real estate loans, credit card loans, auto loans to individuals and certain small business loans.

SEC: U.S. Securities and Exchange Commission

Securities financing agreements: Include resale, repurchase, securities borrowed and securities loaned agreements

Securitized Products Group: Comprised of Securitized Products and tax-oriented investments.

Seed capital: Initial JPMorgan capital invested in products, such as mutual funds, with the intention of ensuring the fund is of sufficient size to represent a viable offering to clients, enabling pricing of its shares, and allowing the manager to develop a track record. After these goals are achieved, the intent is to remove the Firm's capital from the investment.

Shelf securities: Securities registered with the SEC under a shelf registration statement that have not been issued, offered or sold. These securities are not included in league tables until they have actually been issued.

Single-name: Single reference-entities

SLR: Supplementary leverage ratio

SMBS: Stripped mortgage-backed securities

SOFR: Secured Overnight Financing Rate

SPEs: Special purpose entities

Structural interest rate risk: Represents interest rate risk of the non-trading assets and liabilities of the Firm.

Structured notes: Structured notes are financial instruments whose cash flows are linked to the movement in one or more indexes, interest rates, foreign exchange rates, commodities prices, prepayment rates, underlying reference pool of loans or other market variables. The notes typically contain embedded (but not separable or detachable) derivatives. Contractual cash flows for principal, interest, or both can vary in amount and timing throughout the life of the note based on non-traditional indexes or non-traditional uses of traditional interest rates or indexes.

Suspended foreclosures: Loans referred to foreclosure where formal foreclosure proceedings have started but are currently on hold, which could be due to bankruptcy or loss mitigation. Includes both judicial and non-judicial states.

Taxable-equivalent basis: In presenting results on a managed basis, the total net revenue for each of the reportable business segments and Corporate, and the Firm as a whole, is presented on a tax-equivalent basis. Accordingly, revenue from investments that receive tax credits and tax-exempt securities is presented in managed basis results on a level comparable to taxable investments and securities; the corresponding income tax impact related to tax-exempt items is recorded within income tax expense.

TBVPS: Tangible book value per share

TCE: Tangible common equity

TDR: "Troubled debt restructuring" applies to loan modifications granted prior to January 1, 2023 and is deemed to occur when the Firm modifies the original terms of a loan agreement by granting a concession to a borrower that is experiencing financial difficulty. Loans with short-term and other insignificant modifications that are not considered concessions are not TDRs.

TLAC: Total Loss Absorbing Capacity

U.K.: United Kingdom

Unaudited: Financial statements and/or information that have not been subject to auditing procedures by an independent registered public accounting firm.

U.S.: United States of America

U.S. GAAP: Accounting principles generally accepted in the U.S.

U.S. government agencies: U.S. government agencies include, but are not limited to, agencies such as Ginnie Mae and FHA, and do not include Fannie Mae and Freddie Mac which are U.S. government-sponsored enterprises ("U.S. GSEs"). In general, obligations of U.S. government agencies are fully and explicitly guaranteed as to the timely payment of principal and interest by the full faith and credit of the U.S. government in the event of a default.

U.S. GSE(s): "U.S. government-sponsored enterprises" are quasi-governmental, privately-held entities established or chartered by the U.S. government to serve public purposes as specified by the U.S. Congress to improve the flow of credit to specific sectors of the economy and provide certain essential services to the public. U.S. GSEs include Fannie Mae and Freddie Mac, but do not include Ginnie Mae or FHA. U.S. GSE obligations are not explicitly guaranteed as to the timely payment of principal and interest by the full faith and credit of the U.S. government.

U.S. Treasury: U.S. Department of the Treasury

VA: U.S. Department of Veterans Affairs

VaR: "Value-at-risk" is a measure of the dollar amount of potential loss from adverse market moves in an ordinary market environment.

VCG: Valuation Control Group

VGF: Valuation Governance Forum

VIEs: Variable interest entities

Warehouse loans: Consist of prime mortgages originated with the intent to sell that are accounted for at fair value and classified as loans.

Signatures

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on behalf of the undersigned, thereunto duly authorized.

JPMorgan Chase & Co.
(Registrant)

By: /s/ JAMES DIMON
(James Dimon
Chairman and Chief Executive Officer)

February 14, 2025

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacity and on the date indicated. JPMorgan Chase & Co. does not exercise the power of attorney to sign on behalf of any Director.

	Capacity	Date
/s/ JAMES DIMON (James Dimon)	Director, Chairman and Chief Executive Officer (Principal Executive Officer)	
/s/ LINDA B. BAMMANN (Linda B. Bammann)	Director	
/s/ STEPHEN B. BURKE (Stephen B. Burke)	Director	
/s/ TODD A. COMBS (Todd A. Combs)	Director	
/s/ ALICIA BOLER DAVIS (Alicia Boler Davis)	Director	
/s/ ALEX GORSKY (Alex Gorsky)	Director	February 14, 2025
/s/ MELLODY HOBSON (Mellody Hobson)	Director	
/s/ PHEBE N. NOVAKOVIC (Phebe N. Novakovic)	Director	
/s/ VIRGINIA M. ROMETTY (Virginia M. Rometty)	Director	
/s/ BRAD D. SMITH (Brad D. Smith)	Director	
/s/ MARK A. WEINBERGER (Mark A. Weinberger)	Director	
/s/ JEREMY BARNUM (Jeremy Barnum)	Executive Vice President and Chief Financial Officer (Principal Financial Officer)	
/s/ ELENA KORABLINA (Elena Korablina)	Managing Director and Firmwide Controller (Principal Accounting Officer)	

Exhibit 4.6

DESCRIPTION OF SECURITIES OF JPMORGAN CHASE & CO. REGISTERED PURSUANT TO SECTION 12 OF THE SECURITIES EXCHANGE ACT OF 1934

As of the filing date of the Annual Report on Form 10-K to which this Exhibit is attached (the “Form 10-K”), the following outstanding securities issued by JPMorgan Chase & Co. are registered pursuant to Section 12 of the Securities Exchange Act of 1934: (i) common stock; (ii) six series of preferred stock represented by depositary shares; (iii) JPMorgan Chase & Co.’s guarantee of the Alerian MLP Index ETNs due January 28, 2044 issued by JPMorgan Chase Financial Company LLC; and (iv) JPMorgan Chase & Co.’s guarantee of the Callable Fixed Rate Notes due June 10, 2032 issued by JPMorgan Chase Financial Company LLC. All references herein to “JPMorganChase,” “we” or “us” are to JPMorgan Chase & Co.

DESCRIPTION OF COMMON STOCK

The following summary is not complete. You should refer to the applicable provisions of our Restated Certificate of Incorporation and our By-laws, each of which are incorporated by reference as Exhibits to the Form 10-K, and to the Delaware General Corporation Law (“DGCL”), for a complete statement of the terms and rights of our common stock, par value $1.00 per share, which we refer to herein as common stock. We encourage you to read our Restated Certificate of Incorporation, which we refer to herein as our certificate of incorporation, By-laws and the relevant provisions of the DGCL for additional information.

Authorized Shares

We are authorized to issue up to 9,000,000,000 shares of common stock.

Dividends

Holders of common stock are entitled to receive dividends if, as and when declared by our board of directors out of funds legally available for payment, subject to the rights of holders of our preferred stock.

Voting Rights

Each holder of common stock is entitled to one vote per share. Subject to the rights, if any, of the holders of any series of preferred stock under its applicable certificate of designations and applicable law, all voting rights are vested in the holders of shares of our common stock. Holders of shares of our common stock have noncumulative voting rights, which means that the holders of more than 50% of the shares voting for the election of directors can elect 100% of the directors and the holders of the remaining shares will not be able to elect any directors.

Rights Upon Liquidation

In the event of our voluntary or involuntary liquidation, dissolution or winding-up, the holders of our common stock will be entitled to share equally in any of our assets available for distribution after we have paid in full all of our debts and after the holders of all series of our outstanding preferred stock have received their liquidation preferences in full.

Miscellaneous

The issued and outstanding shares of common stock are fully paid and nonassessable. Holders of shares of our common stock are not entitled to preemptive rights or to the benefit of any sinking funds. Our common stock is not convertible into shares of any other class of our capital stock. Computershare Inc. is the transfer agent, registrar and dividend disbursement agent for our common stock.

Listing

Our common stock is listed on the New York Stock Exchange (“NYSE”) under the trading symbol “JPM”.

DESCRIPTION OF LISTED PREFERRED STOCK

The following summary is not complete. You should refer to our certificate of incorporation and to the Certificate of Designations, Powers, Preferences and Rights relating to each series of Listed Preferred Stock (as defined below), which we refer to herein as a certificate of designations, for the complete terms of that series of preferred stock. Copies of our certificate of incorporation and the certificate of designations for each series of Listed Preferred Stock are incorporated by reference as Exhibits to the Form 10-K. We encourage you to read our certificate of incorporation and the relevant certificates of designations for additional information.

Authorized Shares

Under our certificate of incorporation, our board of directors is authorized, without further stockholder action, to issue up to 200,000,000 shares of preferred stock, $1 par value per share, which we refer herein to as preferred stock, in one or more series, and to determine the voting powers and the designations, preferences and relative, participating, optional or other special rights, and qualifications, limitations or restrictions of each series. We may amend our certificate of incorporation to increase or decrease the number of authorized shares of preferred stock in a manner permitted by our certificate of incorporation and the DGCL.

Outstanding Preferred Stock

As of the filing date of the Form 10-K, we have 11 series of preferred stock issued and outstanding. The shares of each series of our preferred stock are represented by depositary shares, with each depositary share representing a fractional interest in a share of preferred stock of the relevant series. Of the 11 series of our issued and outstanding preferred stock, depositary shares representing the following six series of preferred stock are registered pursuant to Section 12 of the Securities Exchange Act of 1934, with each depositary share representing a 1/400th interest in a share of preferred stock of the relevant series:

a. 5.75% Non-Cumulative Preferred Stock, Series DD;
b. 6.00% Non-Cumulative Preferred Stock, Series EE;
c. 4.75 Non-Cumulative Preferred Stock, Series GG;
d. 4.55% Non-Cumulative Preferred Stock, Series JJ;
e. 4.625% Non-Cumulative Preferred Stock, Series LL; and
f. 4.20% Non-Cumulative Preferred Stock, Series MM.

We refer to the above six series of preferred stock herein collectively as the "Listed Preferred Stock".

The Listed Preferred Stock is fully paid and nonassessable.

The terms of the depositary shares are summarized below under "Description of Depositary Shares".

Ranking

The Listed Preferred Stock ranks, as to payment of dividends and distribution of assets upon our liquidation, dissolution or winding-up, on a parity with any series of preferred stock ranking on a parity with the Listed Preferred Stock and senior to our common stock and to any series of preferred stock ranking junior to the Listed Preferred Stock. The Listed Preferred Stock is subordinate to our existing and future indebtedness.

Dividend Rights

Holders of the Listed Preferred Stock are entitled to receive, when, as and if declared by our board of directors or any duly authorized committee of our board, cash dividends at the rates and on the dates described below under "Specific Terms of Listed Preferred Stock". We will pay each dividend to the holders of record as they appear on our stock register on record dates determined by our board of directors or a duly authorized committee of our board. Dividends on the Listed Preferred Stock are noncumulative. If a dividend is not declared on any series of Listed Preferred Stock, because the dividends are noncumulative, then the right of holders of that series to receive that dividend will be lost, and we will have no obligation to pay the dividend for that dividend period, whether or not dividends are declared for any future dividend period.

We may not declare or pay or set aside for payment full dividends on any series of preferred stock ranking, as to dividends, equally with or junior to a series of Listed Preferred Stock unless we have previously declared and paid or set aside for payment, or we contemporaneously declare and pay or set aside for payment, full dividends on that series of Listed Preferred Stock for the most recently completed dividend period. When dividends are not paid in full on a particular series of Listed Preferred Stock and any other series of preferred stock ranking on a parity as to dividends with that series, all dividends declared and paid upon the shares of that series of Listed Preferred Stock and any other series of preferred stock ranking on a parity as to dividends with the that series will be declared and paid pro rata. For purposes of calculating the pro rata allocation of partial dividend payments, we will allocate dividend payments based on the ratio between the then-current dividends due on shares of that Listed Preferred Stock and (i) in the case of any series of non-cumulative preferred stock ranking on a parity as to dividends with that Listed Preferred Stock, the aggregate of the current and unpaid dividends due on such series of preferred stock and (ii) in the case of any series of cumulative preferred stock ranking on a parity as to dividends with that Listed Preferred Stock, the aggregate of the current and accumulated and unpaid dividends due on such series of preferred stock.

In addition, unless full dividends on all outstanding shares of the Listed Preferred Stock have been declared and paid or a sum sufficient for the payment thereof set aside for such payment in respect of the applicable most recently completed dividend period, with respect to a particular series of Listed Preferred Stock:

a. no dividend (other than a dividend in common stock or in any other capital stock ranking junior to that Listed Preferred Stock as to dividends and upon liquidation, dissolution or winding-up) will be declared or paid or a sum sufficient for the payment thereof set aside for such payment or other distribution declared or made upon our common stock or upon any other capital stock ranking junior to that Listed Preferred Stock as to dividends or upon liquidation, dissolution or winding-up, and
b. no common stock or other capital stock ranking junior or equally with that Listed Preferred Stock as to dividends or upon liquidation, dissolution or winding-up will be redeemed, purchased or otherwise acquired for any consideration (or any moneys be paid to or made available for a sinking fund for the redemption of any shares of any such capital stock) by us, except:

    i. by conversion into or exchange for capital stock ranking junior to that Listed Preferred Stock;
    ii. as a result of reclassification into capital stock ranking junior to that Listed Preferred Stock;
    iii. through the use of the proceeds of a substantially contemporaneous sale of shares of capital stock ranking junior to that Listed Preferred Stock or, in the case of capital stock ranking on a parity with that Listed Preferred Stock, through the use of the proceeds of a substantially contemporaneous sale of other shares of capital stock ranking on a parity with that Listed Preferred Stock;
    iv. in the case of capital stock ranking on a parity with that Listed Preferred Stock, pursuant to pro rata offers to purchase all or a pro rata portion of the shares of that Listed Preferred Stock and such capital stock ranking on a parity with that Listed Preferred Stock;
    v. in connection with the satisfaction of our obligations pursuant to any contract entered into in the ordinary course prior to the beginning of the most recently completed dividend period; or
    vi. any purchase, redemption or other acquisition of capital stock ranking junior to that Listed Preferred Stock pursuant to any of our or our subsidiaries' employee, consultant or director incentive or benefit plans or arrangements (including any employment, severance or consulting arrangements) adopted before or after the issuance of that Listed Preferred Stock).

However, the foregoing will not restrict our ability or the ability of any of our affiliates to engage in underwriting, stabilization, market-making or similar transactions in our capital stock in the ordinary course of business. Subject to the conditions described above, and not otherwise, dividends (payable in cash, capital stock, or otherwise), as may be determined by our board of directors or a duly authorized committee of our board, may be declared and paid on our common stock and any other capital stock ranking junior to or on a parity with the Listed Preferred Stock from time to time out of any assets legally available for such payment, and the holders of the Listed Preferred Stock will not be entitled to participate in those dividends.

As used herein, "junior to a series of Listed Preferred Stock" and like terms refer to our common stock and any other class or series of our capital stock over which the Listed Preferred Stock has preference or priority, either as to dividends or upon liquidation, dissolution or winding-up, or both, as the context may require; "parity preferred stock" and "on a parity with a series of Listed Preferred Stock" and like terms refer to any class or series of our capital stock that ranks on a parity with the shares of a particular series of Listed Preferred Stock, either as to dividends or upon liquidation, dissolution or winding-up, or both, as the context may require; and "senior to a series of Listed Preferred Stock" and like terms refer to any class or series of our capital stock that ranks senior to a particular series of Listed Preferred Stock, either as to dividends or upon liquidation, dissolution or winding-up, or both, as the context may require.

We will compute the amount of dividends payable by annualizing the applicable dividend rate and dividing by the number of dividend periods in a year, except that the amount of dividends payable for any period greater or less than a full dividend period, other than the initial dividend period, will be computed on the basis of a 360-day year consisting of twelve 30-day months and, for any period less than a full month, the actual number of days elapsed in the period. Dollar amounts resulting from that calculation will be rounded to the nearest cent, with one-half cent being rounded upward.

Rights Upon Liquidation

In the event of our voluntary or involuntary liquidation, dissolution or winding-up, holders of each series of Listed Preferred Stock will be entitled to receive and to be paid out of our assets legally available for distribution to our stockholders the amount of $10,000 per share, plus any declared and unpaid dividends, without accumulation of undeclared dividends, before we make any distribution of assets to the holders of our common stock or any other class or series of shares ranking junior to the Listed Preferred Stock of such series. After the payment to such holders of the full preferential amounts to which they are entitled, such holders will have no right or claim to any of our remaining assets.

If, upon our voluntary or involuntary liquidation, dissolution or winding-up, we fail to pay in full the amounts payable with respect to a particular series of Listed Preferred Stock, and any stock having the same rank as that series, the holders of that series and of that other stock will share ratably in any such distribution of our assets in proportion to the full respective distributions to which they are entitled. For any series of Listed Preferred Stock, neither the sale of all or substantially all of our property or business, nor our merger or consolidation into or with any other entity will be considered a liquidation, dissolution or winding-up, voluntary of or involuntary, of us.

Because we are a holding company, our rights and the rights of our creditors and our stockholders, including the holders of the Listed Preferred Stock, to participate in the assets of any of our subsidiaries upon that subsidiary's liquidation, dissolution, winding-up or recapitalization may be subject to the prior claims of that subsidiary's creditors, except to the extent that we are a creditor with recognized claims against the subsidiary.

Holders of the Listed Preferred Stock are subordinate to all of our indebtedness and to other non-equity claims on us and our assets, including in the event that we enter into a receivership, insolvency, liquidation or similar proceeding. In addition, holders of the Listed Preferred Stock may be fully subordinated to interests held by the U.S. government in the event that we enter into a receivership, insolvency, liquidation or similar proceeding.

Redemption

We may redeem each series of Listed Preferred Stock on the dates and at the redemption prices set forth below under "Specific Terms of Listed Preferred Stock". In addition, we may redeem each series of Listed Preferred Stock in whole, but not in part, at a redemption price equal to $10,000 per share (equivalent to $25 per depositary share), plus any declared and unpaid dividends, following the occurrence of a capital treatment event. For these purposes, "capital treatment event" means the good faith determination by JPMorganChase that, as a result of any:

a. amendment to, or change or any announced prospective change in, the laws or regulations of the United States or any political subdivision of or in the United States that is enacted or becomes effective after the initial issuance of any shares of such series of Listed Preferred Stock;
b. proposed change in those laws or regulations that is announced or becomes effective after the initial issuance of any shares of such series of Listed Preferred Stock; or
c. official administrative decision or judicial decision or administrative action or other official pronouncement interpreting or applying those laws or regulations that is announced or becomes effective after the initial issuance of any shares of such series of Listed Preferred Stock,

there is more than an insubstantial risk that JPMorganChase will not be entitled to treat an amount equal to the full liquidation amount of all shares of such series of Listed Preferred Stock then outstanding as "additional Tier 1 capital" (or its equivalent) for purposes of the capital adequacy guidelines or regulations of the appropriate federal banking agency, as then in effect and applicable, for as long as any share of such series of Listed Preferred Stock is outstanding. Redemption of any Listed Preferred Stock may be subject to our receipt of any required approvals from the Federal Reserve Board or any other regulatory authority.

If we elect to redeem shares of a series of Listed Preferred Stock, we will provide notice by first class mail, postage prepaid, addressed to the holders of record of such shares to be redeemed. Such mailing will be at least 30 days and not more than 60 days before the date fixed for redemption. Any notice so mailed will be conclusively presumed to have been duly given, whether or not the holder receives such notice, but failure to duly give such notice by mail, or any defect in such notice or in the mailing thereof, to any holder of shares of the series designated for redemption will not affect the validity of the proceedings for the redemption of any other shares of that series. Each notice of redemption will state:

a. the redemption date;
b. the number of shares of the series of Listed Preferred Stock to be redeemed and, if fewer than all the shares held by such holder are to be redeemed, the number of such shares to be redeemed from such holder;
c. the redemption price;
d. the place or places where the certificates representing such shares are to be surrendered for payment of the redemption price; and
e. that dividends on the shares to be redeemed will cease to accrue on the redemption date.

Notwithstanding the foregoing, if the series of Listed Preferred Stock is held in book-entry form through The Depository Trust Company, or "DTC", we may give such notice in any manner permitted or required by DTC. For each series of Listed Preferred Stock, neither the holders of a series nor the holders of the related depositary shares have the right to require redemption of such series of Listed Preferred Stock.

In the case of any redemption of only part of the shares a series of Listed Preferred Stock at the time outstanding, the shares of the series to be redeemed will be selected either pro rata from the holders of record of that series in proportion to the number of shares held by such holders or by lot. From and after the redemption date, dividends will cease to accumulate on the shares of Listed Preferred Stock called for redemption up to the redemption date and all rights of the holders of those shares, except the right to receive the redemption price, will cease.

In the event that we fail to pay full dividends, including accumulated but unpaid dividends, if any, on any series of Listed Preferred Stock, we may not redeem that series in part and we may not purchase or acquire any shares of that series, except by a purchase or exchange offer made on the same terms to all holders of that series.

Preemptive and Conversion Rights

The Listed Preferred Stock is not subject to any preemptive rights and is not convertible into property or shares of any other class or series of our capital stock.

Depositary, Transfer Agent, and Registrar

Computershare Inc. is the depositary, transfer agent and registrar for each series of the Listed Preferred Stock and the related depositary shares.

Voting Rights

Except as indicated below or except as expressly required by applicable law, the holders of the Listed Preferred Stock are not entitled to vote. Each share of Listed Preferred Stock a series is entitled to one vote on matters on which holders of that series are entitled to vote. The voting power of each series of Listed Preferred Stock depends on the number of shares in that series, and not on the aggregate liquidation preference or initial offering price of the shares of that series.

If, at any time or times, the equivalent of an aggregate of six quarterly dividends, whether or not consecutive, for any series of Listed Preferred Stock has not been paid, the number of directors constituting our board of directors will be automatically increased by two and the holders of each outstanding series of Listed Preferred Stock with such voting rights, together with holders of such other shares of any other class or series of parity preferred stock outstanding at the time upon which like voting rights have been conferred and are exercisable, which we refer to as “voting parity stock,” voting together as a class, will be entitled to elect those additional two directors, which we refer to as “preferred directors,” at that annual meeting and at each subsequent annual meeting of stockholders until full dividends have been paid for at least four quarterly consecutive dividend periods. At that time such right will terminate, except as expressly provided in the applicable certificate of designations or by law, subject to revesting. Upon any termination of the right of the holders of the Listed Preferred Stock and voting parity stock as a class to vote for directors as provided above, the preferred directors will cease to be qualified as directors, the term of office of all preferred directors then in office will terminate immediately and the authorized number of directors will be reduced by the number of preferred directors elected. Any preferred director may be removed and replaced at any time, with cause as provided by law or without cause by the affirmative vote of the holders of shares of the Listed Preferred Stock, voting together as a class with the holders of shares of voting parity stock, to the extent the voting rights of such holders described above are then exercisable. Any vacancy created by removal with or without cause may be filled only as described in the preceding sentence. If the office of any preferred director becomes vacant for any reason other than removal, the remaining preferred director may choose a successor who will hold office for the unexpired term in respect of which such vacancy occurred.

So long as any shares of a particular series of Listed Preferred Stock remains outstanding, we will not, without the affirmative vote of the holders of at least 66 2/3% in voting power of that series and any voting parity stock, voting together as a class, authorize, create or issue any capital stock ranking senior to that series as to dividends or upon liquidation, dissolution or winding-up, or reclassify any authorized capital stock into any such shares of such capital stock or issue any obligation or security convertible into or evidencing the right to purchase any such shares of capital stock. So long as any shares of a particular series of Listed Preferred Stock remain outstanding, we will not, without the affirmative vote of the holders of at least 66 2/3% in voting power of that series, amend, alter or repeal any provision of the applicable certificate of designations or our certificate of incorporation, including by merger, consolidation or otherwise, so as to adversely affect the powers, preferences or special rights of that series.

Notwithstanding the foregoing, none of the following will be deemed to adversely affect the powers, preferences or special rights of any series of Listed Preferred Stock:

a. any increase in the amount of authorized common stock or authorized preferred stock, or any increase or decrease in the number of shares of any series of preferred stock, or the authorization, creation and issuance of other classes or series of capital stock, in each case ranking on a parity with or junior to that series of Listed Preferred Stock as to dividends or upon liquidation, dissolution or winding-up;
b. a merger or consolidation of JPMorganChase with or into another entity in which the shares of that series remain outstanding; and
c. a merger or consolidation of JPMorganChase with or into another entity in which the shares of the that series are converted into or exchanged for preference securities of the surviving entity or any entity, directly or indirectly,

controlling such surviving entity and such new preference securities have powers, preferences and special rights that are not materially less favorable than that series;

provided that if the amendment would adversely affect such series but not any other series of preferred stock outstanding, then the amendment will only need to be approved by holders of at least two-thirds of the shares of the series of Listed Preferred Stock adversely affected.

In exercising the voting rights described above or when otherwise granted voting rights by operation of law or by us, each share of Listed Preferred Stock with respect to a series will be entitled to one vote (equivalent to $1/400^{th}$ of a vote per relevant depositary share).

If we redeem or call for redemption all outstanding shares of a series of Listed Preferred Stock and irrevocably deposit in trust sufficient funds to effect such redemption, at or prior to the time when the act with respect to which such vote would otherwise be required or upon which the holders of such series will be entitled to vote will be effected, the voting provisions described above will not apply.

Our board of directors may also from time to time, without notice to or consent of holders of a series of Listed Preferred Stock, issue additional shares of such series. Delaware law provides that the holders of preferred stock will have the right to vote separately as a class on any amendment to our certificate of incorporation (including any certificate of designations) that would increase or decrease the aggregate number of authorized shares of such class, increase or decrease the par value of such class or adversely affect the powers, preferences and special rights of the shares of preferred stock. Notwithstanding the foregoing, as permitted by law, our certificate of incorporation provides that any increase or decrease in our authorized capital stock may be adopted by the affirmative vote of holders of capital stock representing not less than a majority of the voting power represented by the outstanding shares of our capital stock entitled to vote. If any proposed amendment would alter or change the powers, preferences or special rights of one or more series of preferred stock so as to affect them adversely, but would not so affect the entire class of preferred stock, only the shares of the series so affected will be considered a separate class for purposes of this vote on the amendment. This right is in addition to any voting rights that may be provided for in our certificate of incorporation (including any certificate of designations).

Under regulations adopted by the Federal Reserve Board, if the holders of any series of our preferred stock become entitled to vote for the election of directors because dividends on that series are in arrears, that series may then be deemed a “class of voting securities.” In such a case, a holder of 25% or more of the series, or a holder of 5% or more if that holder would also be considered to exercise a “controlling influence” over JPMorganChase, may then be subject to regulation as a bank holding company in accordance with the Bank Holding Company Act. In addition, (1) any other bank holding company may be required to obtain the prior approval of the Federal Reserve Board to acquire or retain 5% or more of that series, and (2) any person other than a bank holding company may be required to provide notice to the Federal Reserve Board prior to acquiring or retaining 10% or more of that series.

Description of Depositary Shares

The following summary of the terms of the depositary shares representing each series of the Listed Preferred Stock is not complete. You should refer to each of the deposit agreements among us, the depositary, and the holders from time to time of the depositary receipts evidencing the depositary shares relating to each series of the Listed Preferred Stock for the complete terms of those depositary shares. Each of those deposit agreements has been filed as an exhibit to a Current Report on Form 8-K filed in connection with the issuance of the depositary shares representing each series of the Listed Preferred Stock.

General. Each depositary share represents a $1/400^{th}$ interest in a share of the relevant series of Listed Preferred Stock, and is evidenced by depositary receipts. In connection with the issuance of each series of Listed Preferred Stock, we deposited shares of that series of Listed Preferred Stock with Computershare Inc., as depositary under the deposit agreement relating to that series of Listed Preferred Stock. Subject to the terms of each deposit agreement, the depositary shares are entitled to all the powers, preferences and special rights of the relevant series of Listed Preferred Stock, as applicable, in proportion to the applicable fraction of a share of Listed Preferred Stock those depositary shares represent.

Dividends and Other Distributions. Each dividend payable on a depositary share will be in an amount equal to $1/400^{th}$ of the dividend declared and payable on the related share of the Listed Preferred Stock.

The depositary will distribute all dividends and other cash distributions received on the relevant series of Listed Preferred Stock to the holders of record of the related depositary receipts in proportion to the number of depositary shares held by each holder. In the event of a distribution other than in cash, the depositary will distribute property received by it to the holders of record of the depositary receipts as nearly as practicable in proportion to the number of depositary shares held by each holder, unless the depositary determines that this distribution is not feasible, in which case the depositary may, with our approval, adopt a method of distribution that it deems practicable, including the sale of the property and distribution of the net proceeds of that sale to the holders of the depositary receipts.

Record dates for the payment of dividends and other matters relating to the depositary shares will be the same as the corresponding record dates for the related shares of Listed Preferred Stock.

The amount paid as dividends or otherwise distributable by the depositary with respect to the depositary shares or the underlying Listed Preferred Stock will be reduced by any amounts required to be withheld by us or the depositary on account of taxes or other governmental charges.

Redemption of Depositary Shares. If we redeem a series of Listed Preferred Stock, in whole or from time to time in part, the corresponding depositary shares also will be redeemed with the proceeds received by the depositary from the redemption of the Listed Preferred Stock held by the depositary. The redemption price per depositary share will be $1/400^{th}$ of the redemption price per share payable with respect to the Listed Preferred Stock, plus any declared and unpaid dividends, without accumulation of undeclared dividends.

If we redeem shares of a series of Listed Preferred Stock held by the depositary, the depositary will redeem, as of the same redemption date, the number of depositary shares representing those shares of the Listed Preferred Stock so redeemed. If we redeem less than all of the outstanding depositary shares, the depositary will select pro rata or by lot those depositary shares to be redeemed. The depositary will mail notice of redemption to record holders of the depositary receipts not less than 30 and not more than 60 days prior to the date fixed for redemption of the Listed Preferred Stock and the related depositary shares.

The redemption of depositary shares that are held in book-entry form through DTC will be effected in accordance with the applicable procedures of DTC.

Voting the Listed Preferred Stock. Because each depositary share represents a $1/400^{th}$ interest in a share of Listed Preferred Stock, holders of depositary receipts will be entitled to $1/400^{th}$ of a vote per depositary share under those limited circumstances in which holders of the Listed Preferred Stock are entitled to a vote.

When the depositary receives notice of any meeting at which the holders of a series of Listed Preferred Stock are entitled to vote, the depositary will mail the information contained in the notice to the record holders of the depositary shares relating to that Listed Preferred Stock. Each record holder of the depositary shares on the record date, which will be the same date as the record date for the applicable Listed Preferred Stock, may instruct the depositary to vote the amount of the Listed Preferred Stock represented by the holder’s depositary shares. To the extent practicable, the depositary will vote the amount of the Listed Preferred Stock represented by depositary shares in accordance with the instructions it receives. We will agree to take all actions that the depositary determines are necessary to enable the depositary to vote as instructed. If the depositary does not receive specific instructions from the holders of any depositary shares representing the Listed Preferred Stock, it will abstain from voting with respect to such shares.

Withdrawal of Listed Preferred Stock. Underlying shares of Listed Preferred Stock may be withdrawn from the depositary arrangement upon surrender of depositary receipts at the depositary’s office and upon payment of the taxes, charges and fees provided for in the deposit agreement. Subject to the terms of the relevant deposit agreement, the holder of depositary receipts will receive the appropriate number of shares of Listed Preferred Stock represented by such depositary shares. Only whole shares of Listed Preferred Stock may be withdrawn; if a holder holds an amount other than a whole multiple of 400 depositary shares, the depositary will deliver along with the withdrawn shares of Listed Preferred Stock a new depositary receipt evidencing the excess number of depositary shares. Holders of withdrawn shares of Listed Preferred Stock will not be entitled to redeposit such shares or to receive depositary shares.

Form and Notices. Each series of Listed Preferred Stock was issued in registered form to the depositary, and the depositary shares representing that Listed Preferred Stock were issued in book-entry only form through DTC. The depositary will forward to the holders of depositary shares all reports, notices, and communications from us that are delivered to the depositary and that we are required to furnish to the holders of the Listed Preferred Stock.

Amendment and Termination of the Deposit Agreement. We and the depositary may amend any form of depositary receipt evidencing depositary shares and any provision of any deposit agreement at any time regarding any depositary shares. However, any amendment that materially and adversely alters the rights of the holders of depositary shares representing a particular series of Listed Preferred Stock or would be materially and adversely inconsistent with the rights granted to holders of that underlying Listed Preferred Stock pursuant to our certificate of incorporation will not be effective unless the amendment has been approved by the holders of at least a majority of the related depositary shares then outstanding. The deposit agreement relating to the depositary shares representing a particular series of Listed Preferred Stock may be terminated by us or by the depositary only if:

a. all such outstanding depositary shares have been redeemed; or

b. there has been a final distribution of the relevant underlying Listed Preferred Stock in connection with our liquidation, dissolution or winding up and the preferred stock has been distributed to the holders of depositary receipts.

Charges of Depositary. We will pay all transfer and other taxes and governmental charges arising solely from the existence of the depositary arrangements regarding any depositary shares. We also pay charges of the depositary in connection with the initial deposit of each series of Listed Preferred Stock and any redemption of the Listed Preferred Stock. Holders of depositary receipts will pay transfer and other taxes and governmental charges and other charges with respect to their depositary receipts as expressly provided in the deposit agreement.

Resignation and Removal of Depositary. With respect to the depositary shares representing each series of Listed Preferred Stock, the depositary may resign at any time by delivering a notice to us of its election to do so. We may remove the depositary at any time. Any such resignation or removal will take effect upon the appointment of a successor depositary and its acceptance of its appointment. We must appoint a successor depositary within 60 days after delivery of the notice of resignation or removal.

Miscellaneous. The depositary will forward to holders of applicable depositary receipts all reports and communications from us that we deliver to the depositary and that we are required to furnish to the holders of the relevant Listed Preferred Stock.

Neither we nor the depositary will be liable if either of us is prevented or delayed by law or any circumstance beyond our control in performing our respective obligations under any deposit agreement. Our obligations and those of the depositary will be limited to performing in good faith our respective duties under any deposit agreement. Neither we nor the depositary will be obligated to prosecute or defend any legal proceeding relating to any depositary shares or Listed Preferred Stock unless satisfactory indemnity is furnished. We and the depositary may rely upon written advice of counsel or accountants, or upon information provided by persons presenting preferred stock for deposit, holders of depositary receipts or other persons we believe to be competent, and on documents we believe to be genuine.

Specific Terms of Listed Preferred Stock

5.75% Non-Cumulative Preferred Stock, Series DD

On September 21, 2018, we issued an aggregate of 169,625 shares of 5.75% Non-Cumulative Preferred Stock, Series DD, $1 par value, with a liquidation preference of $10,000 per share (the “Series DD Preferred Stock”). Shares of the Series DD Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series DD Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 5.75% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on December 1, 2018. Dividends on the Series DD Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series DD Preferred Stock may be redeemed on any dividend payment date on or after December 1, 2023, in whole or from time to time in part, at a redemption price equal to $10,000 per share (equivalent to $25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series DD Preferred Stock following the occurrence of a “capital treatment event,” as described above.

Listing. The depositary shares representing the Series DD Preferred Stock are listed on the NYSE under the trading symbol “JPM PR D”.

6.00% Non-Cumulative Preferred Stock, Series EE

On January 24, 2019, we issued an aggregate of 185,000 shares of 6.00% Non-Cumulative Preferred Stock, Series EE, $1 par value, with a liquidation preference of $10,000 per share (the “Series EE Preferred Stock”). Shares of the Series EE Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series EE Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 6.00% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on June 1, 2019. Dividends on the Series EE Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series EE Preferred Stock may be redeemed on any dividend payment date on or after March 1, 2024, in whole or from time to time in part, at a redemption price equal to $10,000 per share (equivalent to $25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series EE Preferred Stock following the occurrence of a “capital treatment event,” as described above.

Listing. The depositary shares representing the Series EE Preferred Stock are listed on the NYSE under the trading symbol "JPM PR C".

4.75% Non-Cumulative Preferred Stock, Series GG

On November 7, 2019, we issued an aggregate of 90,000 shares of 4.75% Non-Cumulative Preferred Stock, Series GG, $1 par value, with a liquidation preference of $10,000 per share (the "Series GG Preferred Stock"). Shares of the Series GG Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series GG Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 4.75% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on March 1, 2020. Dividends on the Series GG Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series GG Preferred Stock may be redeemed on any dividend payment date on or after December 1, 2024, in whole or from time to time in part, at a redemption price equal to $10,000 per share (equivalent to $25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series GG Preferred Stock following the occurrence of a "capital treatment event," as described above.

Listing. The depositary shares representing the Series GG Preferred Stock are listed on the NYSE under the trading symbol "JPM PR J".

4.55% Non-Cumulative Preferred Stock, Series JJ

On March 17, 2021, we issued an aggregate of 150,000 shares of 4.55% Non-Cumulative Preferred Stock, Series JJ, $1 par value, with a liquidation preference of $10,000 per share (the "Series JJ Preferred Stock"). Shares of the Series JJ Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series JJ Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 4.55% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on June 1, 2021. Dividends on the Series JJ Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series JJ Preferred Stock may be redeemed on any dividend payment date on or after June 1, 2026, in whole at any time or from time to time in part, at a redemption price equal to $10,000 per share (equivalent to $25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series JJ Preferred Stock following the occurrence of a "capital treatment event," as described above.

Listing. The depositary shares representing the Series JJ Preferred Stock are listed on the NYSE under the trading symbol "JPM PR K".

4.625% Non-Cumulative Preferred Stock, Series LL

On May 20, 2021, we issued an aggregate of 185,000 shares of 4.625% Non-Cumulative Preferred Stock, Series LL, $1 par value, with a liquidation preference of $10,000 per share (the "Series LL Preferred Stock"). Shares of the Series LL Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series LL Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 4.625% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on September 1, 2021. Dividends on the Series LL Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series LL Preferred Stock may be redeemed on any dividend payment date on or after June 1, 2026, in whole at any time or from time to time in part, at a redemption price equal to $10,000 per share (equivalent to $25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series LL Preferred Stock following the occurrence of a "capital treatment event," as described above.

Listing. The depositary shares representing the Series LL Preferred Stock are listed on the NYSE under the trading symbol "JPM PR L".

4.20% Non-Cumulative Preferred Stock, Series MM

On July 29, 2021, we issued an aggregate of 200,000 shares of 4.20% Non-Cumulative Preferred Stock, Series MM, $1 par value, with a liquidation preference of $10,000 per share (the "Series MM Preferred Stock"). Shares of the Series MM

Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series MM Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 4.20% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on December 1, 2021. Dividends on the Series MM Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series MM Preferred Stock may be redeemed on any dividend payment date on or after September 1, 2026, in whole at any time or from time to time in part, at a redemption price equal to $10,000 per share (equivalent to $25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series MM Preferred Stock following the occurrence of a "capital treatment event," as described above.

Listing. The depositary shares representing the Series MM Preferred Stock are listed on the NYSE under the trading symbol "JPM PR M".

DESCRIPTION OF JPMORGAN CHASE FINANCIAL COMPANY LLC'S THE ALERIAN MLP INDEX ETNS DUE JANUARY 28, 2044, FULLY AND UNCONDITIONALLY GUARANTEED BY JPMORGAN CHASE & CO.

The following description of Alerian MLP Index ETNs due January 28, 2044 (the "Alerian ETNs") is a summary and does not purport to be complete. It is subject to and qualified in its entirety by reference to the indenture dated February, 2016 (as may be amended or supplemented from time to time, the "2016 Indenture"), among JPMorgan Chase Financial Company LLC, as issuer ("JPMorgan Financial" or the "Issuer"), JPMorganChase, as guarantor (the "Guarantor"), and Deutsche Bank Trust Company Americas, as trustee (the "Trustee"), which is incorporated by reference as an Exhibit to the Form 10-K. We encourage you to read the 2016 Indenture for additional information.

General

The Alerian ETNs were originally issued on January 30, 2024 and are part of a series of JPMorgan Financial's debt securities entitled "Global Medium-Term Notes, Series A" (the "Series A Notes") that JPMorgan Financial may issue under the 2016 Indenture from time to time. For more information about the Series A Notes, please see the section titled "— Description of JPMorgan Chase Financial Company LLC's Callable Fixed Rate Notes Due June 10, 2032, Fully and Unconditionally Guaranteed by JPMorgan Chase & Co. — General Terms of the Series A Notes" below.

The Alerian ETNs are unsecured and unsubordinated obligations of JPMorgan Financial, the payment of which is fully and unconditionally guaranteed by JPMorganChase, the Guarantor. The Alerian ETNs will rank *pari passu* with all of JPMorgan Financial's other unsecured and unsubordinated obligations. JPMorgan Chase & Co.'s guarantee of the Alerian ETNs will rank *pari passu* with all of JPMorgan Chase & Co.'s other unsecured and unsubordinated obligations. The Alerian ETNs do not guarantee any return of principal at, or prior to, maturity or upon early redemption or repurchase. **Any payment on the Alerian ETNs is subject to the credit risk of JPMorgan Financial, as issuer of the Alerian ETNs, and JPMorgan Chase & Co., as guarantor of the Alerian ETNs.**

The Alerian ETNs are not bank deposits and are not insured by the Federal Deposit Insurance Corporation or by any other governmental agency, nor are they obligations of, or guaranteed by, a bank.

Unless otherwise specified, references herein to "holders" mean those in whose names the Alerian ETNs are registered on the books that JPMorgan Financial or the Trustee, or any successor trustee, as applicable, maintains for this purpose, and not those who own beneficial interests in the Alerian ETNs (registered in street name or otherwise).

Key Terms of the Alerian ETNs

General

Principal Amount: $26 per Alerian ETN (equal to the Initial VWAP Level *divided* by ten, rounded to the nearest cent).

Index: The return on the Alerian ETNs is linked to the performance of the Alerian MLP Index® (Bloomberg ticker: AMZ), which we refer to as the Index, as measured by its VWAP level, and to cash distributions on its components. See "— Terms Relating to Closing Intrinsic Note Value — VWAP Level" below.

The Index measures the composite performance of energy-oriented Master Limited Partnerships, or MLPs, that earn the majority of their cash flows from qualified activities involving energy commodities (the "Index Components") using a capped, float-adjusted, capitalization-weighted methodology. MLPs are limited partnerships primarily engaged in the exploration, marketing, mining, processing, production, refining, storage, or transportation of any mineral or natural resource. The Index is calculated and maintained by VettaFi LLC ("VettaFi").

Coupon Payments: For each Alerian ETN a holder holds on a Coupon Record Date, the holder will receive on the immediately following Coupon Payment Date an amount in cash equal to the Coupon Amount, if any, as of the immediately preceding Coupon Valuation Date.

Coupon Amount: The Coupon Amount as of any Coupon Valuation Date will equal:

a. the Reference Distribution Amount with respect to that Coupon Valuation Date, *minus*
b. the Accrued Investor Fee with respect to that Coupon Valuation Date,

provided that the Coupon Amount will not be less than $0.

If the Reference Distribution Amount on a Coupon Valuation Date is less than the Accrued Investor Fee on that Coupon Valuation Date, an amount equal to the excess of the Accrued Investor Fee over the Reference Distribution Amount (the "Investor Fee Shortfall") will be included in the Accrued Investor Fee with respect to the next Coupon Valuation Date.

The Accrued Investor Fee will reduce each Coupon Amount. In addition, no Coupon Payment will be payable with respect to a Coupon Valuation Date if the Reference Distribution Amount is less than the Accrued Investor Fee, even if that Reference Distribution Amount is positive. The holders are not guaranteed any Coupon Payments.

The Coupon Amount is subject to adjustment in the event of a split or reverse split of the Alerian ETNs as described under "— Split or Reverse Split of the Alerian ETNs" below.

Payment at Maturity: For each Alerian ETN, unless earlier repurchased or redeemed, holders will receive at maturity a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Final Valuation Date. If that amount is less than or equal to zero, the payment at maturity will be $0.

Issuer Redemption: On any Business Day on or after July 26, 2024, the Issuer may, in its sole discretion, redeem the Alerian ETNs, in whole or in part. If the Issuer exercises its right to redeem the Alerian ETNs prior to maturity, for each Alerian ETN that is redeemed, holders will receive on the Redemption Settlement Date a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Redemption Valuation Date. If that amount is less than or equal to zero, the payment upon early redemption will be $0.

Weekly Repurchase: On a weekly basis, holders may request that we repurchase a minimum of 50,000 Alerian ETNs (subject to adjustment in the event of a split or reverse split of the Alerian ETNs) if holders comply with the required procedures, subject to a repurchase fee of 0.125%. For each Alerian ETN that is repurchased, holders will receive on the relevant Repurchase Date a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Repurchase Valuation Date *minus* the Repurchase Fee Amount with respect to the Repurchase Valuation Date.

If the amount calculated above is less than or equal to zero, the payment upon early repurchase will be $0.

Terms Relating to Closing Intrinsic Note Value

Closing Intrinsic Note Value: As of any date of determination, an amount per Alerian ETN equal to:

a. the Principal Amount, multiplied by the Index Ratio as of that date, *plus*
b. the Coupon Amount as of that date, calculated as if that date were a Coupon Valuation Date (the "Stub Coupon Amount"), *minus*
c. any Investor Fee Shortfall determined in calculating that Stub Coupon Amount.

In addition, if the Coupon Ex-Date with respect to the Coupon Amount as of the immediately preceding Coupon Valuation Date has not yet occurred, the Closing Intrinsic Note Value will also include that Coupon Amount (an "Unpaid Coupon Amount").

For purposes of determining the Closing Intrinsic Note Value over any Measurement Period, the Index Ratio is determined based on the arithmetic average of the VWAP Levels over that Measurement Period, and the date of determination for purposes of determining the Coupon Amount, any Investor Fee Shortfall and any Unpaid Coupon Amount is the final day of that Measurement Period.

The Closing Intrinsic Note Value is not the closing price or any other trading price of the Alerian ETNs in the secondary market and is not intended as a price or quotation, or as an offer or solicitation for the purchase or sale of the Alerian ETNs or as a recommendation to transact in the Alerian ETNs at the stated price. The trading price of the Alerian ETNs at any time may vary significantly from the Closing Intrinsic Note Value due to, among other things, imbalances of supply and demand (including as a result of any decision of ours to issue, stop issuing or resume issuing additional Alerian ETNs), lack of liquidity, transaction costs, credit considerations and bid-offer spreads.

If the Reference Distribution Amount used to calculate the Stub Coupon Amount on any Index Business Day is less than the Accrued Investor Fee used to calculate that Stub Coupon Amount, the resulting Investor Fee Shortfall will be deducted in calculating the Closing Intrinsic Note Value on that Index Business Day. Accordingly, the payment at maturity or upon early repurchase or redemption will be reduced by the amount of any Investor Fee Shortfall reflected in the relevant Closing Intrinsic Note Value.

Index Ratio: As of any date of determination, the Index Ratio is equal to:

$$\frac{\text{VWAP Level}}{\text{Initial VWAP Level}}$$

provided that, solely for purposes of determining the Closing Intrinsic Note Value in connection with any payment at maturity or upon early repurchase or redemption, the Index Ratio determined over the relevant Measurement Period is equal to:

$$\frac{\text{Final VWAP Level}}{\text{Initial VWAP Level}}$$

Initial VWAP Level: 260.0267 which is the arithmetic average of the VWAP Levels determined over a period of five Index Business Days ending on the Inception Date.

Final VWAP Level: With respect to a Measurement Period, the arithmetic average of the VWAP Levels on the five Index Business Days in that Measurement Period, as calculated by the Note Calculation Agent.

Measurement Period: With respect to the Final Valuation Date or any Repurchase Valuation Date or Redemption Valuation Date, the five Index Business Days starting from and including the Final Valuation Date or that Repurchase Valuation Date or Redemption Valuation Date, as applicable (or, if that day is not an Index Business Day, the five Index Business Days immediately following that day).

The Measurement Period is subject to postponement in the event of a market disruption event and as described under "— Postponement of an Averaging Date."

VWAP Level: On any Index Business Day, as calculated by the VWAP Calculation Agent, the sum of the products for each Index Component of:

a. the VWAP of that Index Component as of that day; and

b. the Index Units of that Index Component as of that day, *divided* by the Index Divisor as of that day.

The calculation of the VWAP Level may be modified in circumstances described under "— Postponement of an Averaging Date" and "— Discontinuation of an Index; Alternation of Method of Calculation" below. The official closing level of the Index may vary significantly from the VWAP Level.

VWAP: With respect to each Index Component, as of any date of determination, the volume-weighted average price of one share of that Index Component as determined by the VWAP Calculation Agent based on the Primary Exchange for that Index Component.

Index Units: With respect to each Index Component, as of any date of determination, the number of units of that Index Component included in the Index for purposes of the calculation of the official level of the Index by the Index Calculation Agent.

Index Divisor: As of any date of determination, the divisor used in the calculation of the official level of the Index by the Index Calculation Agent.

Inception Date: January 26, 2024

Initial Issue Date: January 30, 2024

Final Valuation Date: January 20, 2044

Maturity Date: January 28, 2044. The Maturity Date is subject to postponement in the event of a market disruption event and as described under "— Postponement of a Payment Date" below.

Terms Relating to Coupon Payments

Accrued Investor Fee: The Accrued Investor Fee accrues at a rate of 0.85% per annum each day. In addition, the Accrued Investor Fee carries forward any shortfall if the Reference Distribution Amount determined in connection with any Coupon Payment is less than the Accrued Investor Fee at that time.

With respect to each Coupon Valuation Date, the Accrued Investor Fee is an amount equal to:

a. the Periodic Investor Fee with respect to that Coupon Valuation Date, *plus*
b. the Investor Fee Shortfall, if any, as of the immediately preceding Coupon Valuation Date, if any.

Periodic Investor Fee: With respect to each Coupon Valuation Date, an amount equal to the product of

a. the investor fee of 0.85% per annum;
b. the Principal Amount multiplied by the Index Ratio as of the immediately preceding Index Business Day that is not a Disrupted Day for any Index Component; and
c. the day count fraction, calculated using a 30/360 day count convention as described under "— Additional Terms — Day Count Fraction" below.

Reference Distribution Amount: With respect to each Coupon Valuation Date, an amount equal to the sum of the gross cash distributions that a Reference Holder would have been entitled to receive in respect of each Index Component held by that Reference Holder on the "record date" with respect to that Index Component, for those cash distributions whose "ex-dividend date" occurs during the Coupon Accrual Period for that Coupon Valuation Date.

Notwithstanding the foregoing, with respect to cash distributions for an Index Component that are scheduled to be paid prior to the applicable Coupon Ex-Date, if the issuer of that Index Component fails to pay the distribution to holders of that Index Component by the scheduled payment date for that distribution, that distribution will be assumed to be zero for the purposes of calculating the applicable Reference Distribution Amount.

Reference Holder: As of any date of determination, a hypothetical holder of a number of shares of each Index Component equal to:

a. the Index Units of that Index Component as of that date, *divided* by
b. the Index Divisor as of that date *multiplied* by 10,

provided that solely for purposes of determining the Reference Distribution Amount included in any Stub Coupon Amount payable at maturity or upon early repurchase or redemption, the Reference Holder will be deemed to hold four-fifths, three-fifths, two-fifths and one-fifth of the shares of each Index Component it would otherwise hold on the second, third, fourth and fifth Index Business Days, respectively, in the relevant Measurement Period.

The Accrued Investor Fee, the Periodic Investor Fee, the Reference Distribution Amount and the Reference Holder are each subject to adjustment in the event of a split or reverse split of the Alerian ETNs as described under "— Split or Reverse Split of the Alerian ETNs" below.

Coupon Accrual Period: With respect to each Coupon Valuation Date, the period from but excluding the immediately preceding Coupon Valuation Date (or, in the case of the first Coupon Valuation Date, from but excluding November 15, 2023) to and including that Coupon Valuation Date.

Coupon Valuation Date: The first Index Business Day occurring on or after the 15th of February, May, August and November of each calendar year during the term of the Alerian ETNs, beginning on February 15, 2024.

Coupon Ex-Date: With respect to a Coupon Amount, the first Exchange Business Day on which the Alerian ETNs trade without the right to receive that Coupon Amount. Under current NYSE Arca practice, the Coupon Ex-Date will generally be the first Exchange Business Day immediately preceding the applicable Coupon Record Date; however, beginning May 28, 2024, under NYSE Arca practice, the Coupon Ex-Date is expected to generally be the applicable Coupon Record Date. For purposes of this paragraph, "Exchange Business Day" means any day on which the primary exchange or market for trading of the Alerian ETNs is scheduled to be open for trading.

Coupon Record Date: The 9th Index Business Day following each Coupon Valuation Date.

Coupon Payment Date: The 15th Index Business Day following each Coupon Valuation Date, subject to postponement in the event of a market disruption event as described under "— Postponement of a Payment Date" below.

Terms Relating to Issuer Redemption

Early Redemption: On any Business Day on or after July 26, 2024, the Issuer may, in its sole discretion, redeem the Alerian ETNs, in whole or in part. If the Issuer exercises its right to redeem the Alerian ETNs, the Issuer will deliver an irrevocable redemption notice (the "Redemption Notice") to DTC (the holder of the master note evidencing the Alerian ETNs) at least five Business Days prior to the Redemption Valuation Date specified in the Redemption Notice. If fewer than all the Alerian ETNs are to be redeemed, the Issuer will specify in the Redemption Notice the principal amount of the Alerian ETNs to be redeemed, and the Trustee will select the Alerian ETNs to be redeemed pro rata, by lot or in such manner as it deems appropriate and fair.

Payment upon Early Redemption: If the Issuer exercises its right to redeem any Alerian ETNs prior to maturity, for each Alerian ETNs selected for redemption by the Trustee, holders will receive on the Redemption Settlement Date a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Redemption Valuation Date. If that amount is less than or equal to zero, the payment upon early redemption will be $0.

Redemption Valuation Date: The date specified as the Redemption Valuation Date in the Redemption Notice.

Redemption Settlement Date: Unless otherwise specified in the Redemption Notice, the day that follows the final day in the Measurement Period with respect to the Redemption Valuation Date by a number of Business Days corresponding to the standard settlement cycle, which is currently two Business Days and which is expected to be one Business Day beginning May 28, 2024. In no event will the Redemption Notice specify a Redemption Settlement Date that follows the final day in the Measurement Period by more than five Business Days.

Terms Relating to Weekly Repurchase Right

Early Repurchase: On a weekly basis, holders may request that we repurchase a minimum of 50,000 Alerian ETNs (subject to adjustment in the event of a split or reverse split of the Alerian ETNs) if holders comply with the procedures described under "— Repurchase Procedures" below and unless holders have delivered a Redemption Notice to DTC to redeem all of the outstanding notes. The Issuer may from time to time, in its sole discretion, reduce the minimum number of the Alerian ETNs required for an early repurchase on a consistent basis for all holders of the Alerian ETNs, but the Issuer is under no obligation to do so.

Payment upon Early Repurchase: Subject to holders' compliance with the required procedures, for each Alerian ETN that is repurchased, holders will receive on the relevant Repurchase Date a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Repurchase Valuation Date *minus* the Repurchase Fee Amount with respect to the Repurchase Valuation Date. If that amount is less than or equal to zero, the payment upon early redemption will be $0.

Repurchase Fee Amount: With respect to any Repurchase Valuation Date, an amount per Alerian ETN in cash equal to 0.125% of the Closing Intrinsic Note Value with respect to that Repurchase Valuation Date (but excluding any Unpaid Coupon Amount included in that Closing Intrinsic Note Value).

Repurchase Valuation Date: The last Index Business Day of each week, generally Friday.

Repurchase Date: Unless otherwise specified in the Issuer's acknowledgement, the day that follows the final day in the Measurement Period with respect to the Repurchase Valuation Date by a number of Business Days corresponding to the standard settlement cycle, which is currently two Business Days and which is expected to be one Business Day beginning May 28, 2024. In no event will the Issuer's acknowledgement specify a Repurchase Date that follows the final day in the Measurement Period by more than five Business Days.

Repurchase Notice: A repurchase notice in the form specified by the Issuer.

Repurchase Procedures: In order to request that the Issuer repurchase a holder's Alerian ETN, holders must instruct their broker or other person through which they hold Alerian ETNs to take the following steps:

a. send a completed Repurchase Notice to the Issuer via email at ETN_Repurchase@jpmorgan.com by no later than 4:00 p.m., New York City time, on the Business Day immediately preceding the applicable Repurchase Valuation Date;
b. instruct the holder's DTC custodian to book a delivery versus payment trade with respect to that holder's Alerian ETNs on the final day in the Measurement Period with respect to the relevant Repurchase Valuation Date at a price equal to the amount payable upon early repurchase of the Alerian ETNs; and
c. cause the holder's DTC custodian to deliver the trade as booked for settlement via DTC at or prior to 10:00 a.m., New York City time, on the relevant Repurchase Date.

Different brokerage firms may have different deadlines for accepting instructions from their customers. Accordingly, holders should consult the brokerage firm through which they own their interest in the Alerian ETNs in respect of those deadlines.

Once delivered, a Repurchase Notice may not be revoked. If the Issuer does not receive a holder's Repurchase Notice by the deadline, such holder's Repurchase Notice will not be effective. The Issuer or its affiliate must acknowledge receipt of the Repurchase Notice on the same Business Day for it to be effective, such acknowledgment will be deemed to evidence its acceptance of a holder's repurchase request. The Note Calculation Agent will, in its sole discretion, resolve any questions that may arise as to the validity of a Repurchase Notice and the timing of receipt of a Repurchase Notice.

Additional Terms

Business Day: Any day other than a day on which the banking institutions in the City of New York are authorized or required by law, regulation or executive order to close or a day on which transactions in dollars are not conducted.

Index Business Day: Any day on which the Primary Exchange and the Related Exchange with respect to each Index Component are scheduled to be open for trading.

Primary Exchange: With respect to each Index Component, the primary exchange or market of trading of that Index Component.

Related Exchange: With respect to each Index Component, each exchange or quotation system where trading has a material effect (as determined by the Note Calculation Agent) on the overall market for futures or options contracts relating to that Index Component.

Disrupted Day: With respect to an Index Component, a day on which the Primary Exchange or any Related Exchange with respect to that Index Component fails to open for trading during its regular trading session or on which a market disruption event (as described under "— Market Disruption Events") with respect to that Index Component has occurred or is continuing, and, in each case, the occurrence of which is determined by the Note Calculation Agent to have a material effect on the VWAP Level.

Index Sponsor and Index Calculation Agent: VettaFi

Note Calculation Agent, VWAP Calculation Agent and Published ETN Value Calculation Agent: J.P. Morgan Securities LLC ("JPMS"), one of our and JPMorgan Financial's affiliates, will act as the Note Calculation Agent. Solactive AG will act as the VWAP Calculation Agent and the Published ETN Value Calculation Agent. We and JPMorgan Financial may appoint a different Note Calculation Agent, VWAP Calculation Agent or Published ETN Value Calculation Agent from time to time without holders' consent and without notifying holders.

The Note Calculation Agent will make all necessary calculations and determinations in connection with the Alerian ETNs, including calculations and determinations relating to any payments on the Alerian ETNs and the assumptions used to determine the pricing of the Alerian ETNs, other than determinations to be made by the VWAP Calculation Agent. The VWAP Calculation Agent will determine the VWAP Level and the VWAP of each Index Component on each Index Business Day, except as described below under "— Postponement of an Averaging Date." The Published ETN Value Calculation Agent is responsible for calculating the Daily Closing Intrinsic Note Value, the Intraday Intrinsic Note Value and the Interim Coupon or Shortfall Amount for purposes of publication. All determinations made by the Note Calculation Agent, the VWAP Calculation Agent or the Published ETN Value Calculation Agent will be at the sole discretion of the Note Calculation Agent, the VWAP Calculation Agent or the Published ETN Value Calculation Agent, as applicable, and will, in the absence of manifest error, be conclusive for all purposes and binding on holders, us and JPMorgan Financial.

The Note Calculation Agent will provide written notice to the Trustee at its New York office, on which notice the Trustee may conclusively rely, of any amount payable on the Alerian ETNs at or prior to 11:00 a.m., New York City time, on the date on which payment is to be made.

All values with respect to calculations in connection with the Alerian ETNs will be rounded to the nearest one hundred-thousandth, with five one-millionths rounded upward (e.g., 0.876545 would be rounded to 0.87655). Notwithstanding the foregoing, all dollar amounts related to determination of any payment on the Alerian ETNs per note will be rounded to the nearest ten-thousandth, with five one hundred-thousandths rounded upward (e.g., 0.76545 would be rounded up to 0.7655), and all dollar amounts payable, if any, on the aggregate principal amount of Alerian ETNs per holder will be rounded to the nearest cent, with one-half cent rounded upward.

Trustee: Deutsche Bank Trust Company Americas

Day Count Fraction: With respect to each Coupon Valuation Date, the day count fraction is calculated as follows:

$$\frac{[360 \times (Y2 - Y1) + [30 \times (M2 - M1)]] + (D2 - D1)}{360}$$

where:

a. "Y1" is the year, expressed as a number, in which the first day of the Coupon Accrual Period with respect to that Coupon Valuation Date falls;
b. "Y2" is the year, expressed as a number, in which the day immediately following the last day included in the Coupon Accrual Period with respect to that Coupon Valuation Date falls;

c. "M1" is the calendar month, expressed as a number, in which the first day of the Coupon Accrual Period with respect to that Coupon Valuation Date falls;
d. "M2" is the calendar month, expressed as a number, in which the day immediately following the last day included in the Coupon Accrual Period with respect to that Coupon Valuation Date falls;
e. "D1" is the first calendar day, expressed as a number, of the Coupon Accrual Period with respect to that Coupon Valuation Date, unless that number would be 31, in which case D1 will be 30; and
f. "D2" is the calendar day, expressed as a number, immediately following the last day included in the Coupon Accrual Period with respect to that Coupon Valuation Date, unless that number would be 31 and D1 is greater than 29, in which case D2 will be 30.

Split or Reverse Split of the Alerian ETNs

The Issuer may initiate a split or reverse split of the Alerian ETNs at any time. The Issuer will determine the ratio of such split or reverse split, as applicable, using relevant market indicia in its sole discretion. If the Issuer decides to initiate a split or reverse split, as applicable, the Issuer will issue a notice to holders of the Alerian ETNs and a press release announcing the split or reverse split and the ratio and specifying the effective date of the split or reverse split, which will be at least three Business Days after the date on which the split or reverse split, as applicable, is announced, which we refer to as the "announcement date." The record date for a share split or reverse split will be the Business Day immediately preceding the effective date.

If the Alerian ETNs undergo a split or reverse split, the Issuer will adjust the terms of the Alerian ETNs as may be necessary or desirable to effectuate that split or reverse split, as applicable, including, without limitation, the Principal Amount, any Coupon Amount, the Closing Intrinsic Note Value, the Accrued Investor Fee, the Periodic Investor Fee, the Reference Distribution Amount, the number of shares held by a Reference Holder and the minimum number of Alerian ETNs holders may request that the Issuer repurchase. For example, if the Alerian ETNs undergo a 4-for-1 split, each Alerian ETN holder who holds one Alerian ETN via DTC prior to the split will, after the split, hold four Alerian ETNs, and the Principal Amount, any Coupon Amount, the Closing Intrinsic Note Value, the Accrued Investor Fee, the Periodic Investor Fee, the Reference Distribution Amount and the number of shares held by a Reference Holder will be adjusted as may be necessary or desirable to equal 1/4 of their respective values that would have prevailed in the absence of the 4-for-1 split. In addition, the minimum number of Alerian ETNs holders may request that the Issuer repurchase will be increased to equal 4 times its prior value. These adjustments may be applied retroactively for purposes of adjusting any Coupon Amount that has been determined but not yet paid.

In the case of a reverse split, holders who hold a number of Alerian ETNs that is not evenly divisible by the relevant ratio will receive the same treatment as all other holders for the maximum number of Alerian ETNs they hold that is evenly divisible by the relevant ratio, and the Issuer will have the right to compensate holders for their remaining or "partial" Alerian ETNs in a manner determined by the Note Calculation Agent in its sole discretion. The Issuer's current intention is to provide holders with a cash payment for their partial Alerian ETNs in an amount equal to the appropriate percentage of the Closing Intrinsic Note Value (which may be calculated using an Index Ratio that reflects the average VWAP Level over a measurement period of five Index Business Days) on a specified Index Business Day no later than 20 Index Business Days following the announcement date, with payment to be made on a specified Business Day no later than five Business Days following the date on which the amount of the payment is determined.

Postponement of a Payment Date

In this section, we refer to each Coupon Payment Date and the Maturity Date as "Payment Dates." If any scheduled Payment Date is not a Business Day, then that Payment Date will be the next succeeding Business Day following the scheduled Payment Date. If, due to a market disruption event or otherwise, any Averaging Date referenced in the determination of a payment on the Alerian ETNs that will or may be payable on any Payment Date is postponed so that it falls less than three Business Days prior to that scheduled Payment Date, that Payment Date will be the third Business Day following the latest such Averaging Date, as postponed. If any Payment Date is adjusted as the result of a non-Business Day, a market disruption event or otherwise, any amount payable on that Payment Date will be made on that Payment Date as postponed, with the same force and effect as if that Payment Date had not been postponed, but no interest will accrue or be payable as a result of the delayed payment.

Postponement of an Averaging Date

The Final VWAP Level used to determine the Closing Intrinsic Note Value in connection with any payment at maturity or upon early repurchase or redemption will reflect the arithmetic average of the VWAP Levels on the five Index Business Days in the relevant Measurement Period, and the VWAP Level on each Index Business Day is calculated by reference to the VWAP and the Index Units of each Index Component, as well as the Index Divisor. In this section, we refer to each of the five Index Business Days in a Measurement Period as an "Averaging Date." As set forth below, if any Index Component is disrupted on any Averaging Date, determinations with respect to that Index Component for purposes of calculating the

VWAP Level applicable to that Averaging Date will be postponed. In addition, determinations with respect to that disrupted Index Component for purposes of calculating the VWAP Level applicable to each following Averaging Date will also be postponed so as to avoid using the same VWAP of an Index Component on a given Index Business Day to calculate the VWAP Levels for more than one Averaging Date.

If any Averaging Date is an Unavailable Day for any Index Component, the VWAP Level on that Averaging Date will be deemed to be the Adjusted VWAP Level with respect to that Averaging Date, and that Averaging Date will be postponed to the earliest date on which that Adjusted VWAP Level can be calculated as set forth below. For the avoidance of doubt, no Averaging Date will be postponed to a date that is after the applicable Final Disrupted Averaging Date.

The "Adjusted VWAP Level" with respect to an Averaging Date that is an Unavailable Day for any Index Component will be determined by the VWAP Calculation Agent and will be calculated in accordance with the formula for calculating the VWAP Level, using:

a. with respect to each Index Component for which the originally scheduled Averaging Date is not an Unavailable Day, the VWAP and the Index Units of that Index Component, and the Index Divisor, in each case as of the originally scheduled Averaging Date; and
b. with respect to each Index Component for which the originally scheduled Averaging Date is an Unavailable Day, the VWAP and the Index Units of that Index Component, and the Index Divisor, in each case, as of the first Index Business Day immediately following the originally scheduled Averaging Date that is not an Unavailable Day with respect to that Index Component, provided that, if each day from and including the originally scheduled Averaging Date to and including the applicable Final Disrupted Averaging Date is an Unavailable Day with respect to that Index Component, the VWAP of that Index Component will be determined by the VWAP Calculation Agent in good faith based on its assessment of the market value of that Index Component on that Final Disrupted Averaging Date and the Index Units of that Index Component and the Index Divisor will each be determined by the VWAP Calculation Agent based on its good faith estimate.

An Index Business Day will be an "Unavailable Day" with respect to an Index Component if it is a Disrupted Day for that Index Component or a day that is a Prior Valuation Date with respect to that Index Component and the relevant Averaging Date.

A "Disrupted Day" means, with respect to an Index Component, a day on which the Primary Exchange or any Related Exchange with respect to that Index Component fails to open for trading during its regular trading session or on which a Market Disruption Event with respect to that Index Component has occurred or is continuing, and, in each case, the occurrence of which is determined by the Note Calculation Agent to have a material effect on the VWAP Level.

An Index Business Day will be a "Prior Valuation Date" with respect to an Index Component and an Averaging Date if the VWAP and the Index Units of that Index Component, and the Index Divisor, in each case as of that Index Business Day, were used in calculating the Adjusted VWAP Level with respect to an earlier occurring originally scheduled Averaging Date within the same Measurement Period as that Averaging Date.

With respect to an Averaging Date, the "Final Disrupted Averaging Date" means the third Index Business Day after that Averaging Date, as originally scheduled.

Market Disruption Events

With respect to an Index Component, a "Market Disruption Event" means, the occurrence or existence of a condition specified below:

a. any suspension, absence or limitation of trading in that Index Component on the Primary Exchange with respect to that Index Component, whether by reason of movements in price exceeding limits permitted by the Primary Exchange or otherwise;
b. any suspension, absence or limitation of trading in futures or options contracts related to that Index Component, if available, on any Related Exchange with respect to that Index Component, whether by reason of movements in price exceeding limits permitted by that Related Exchange or otherwise;
c. any event that disrupts or impairs the ability of market participants in general to effect transactions in, or obtain market values for, that Index Component or futures or options contracts relating to that Index Component; or
d. the closure on any Index Business Day of the Primary Exchange or any Related Exchange with respect to that Index Component prior to its Scheduled Closing Time unless the earlier closing time is announced by the Primary Exchange or that Related Exchange, as applicable, at least one hour prior to the earlier of (a) the actual closing time for the regular trading session on the Primary Exchange or that Related Exchange, as applicable, on that Index Business Day and (b) the submission deadline for orders to be entered into the system for execution at the close of trading on that Index Business Day for the Primary Exchange or that Related Exchange, as applicable;

in each case determined by the Note Calculation Agent in its sole discretion; and

a. a determination by the Note Calculation Agent in its sole discretion that the applicable event described above materially interfered with the Issuer's ability or the ability of any of its affiliates to adjust or unwind all or a material portion of any hedge with respect to the Alerian ETNs.

For purposes of determining whether a Market Disruption Event with respect to an Index Component has occurred:

a. a limitation on the hours or number of days of trading will not constitute a Market Disruption Event if it results from an announced change in the regular business hours of the Primary Exchange or any Related Exchange; and
b. limitations pursuant to the rules of the Primary Exchange or any Relevant Exchange similar to NYSE Rule 80B (or any applicable rule or regulation enacted or promulgated by any other self-regulatory organization or any government agency of scope similar to NYSE Rule 80B as determined by the Note Calculation Agent) on trading during significant market fluctuations will constitute a suspension, absence or material limitation of trading.

"Scheduled Closing Time" means, with respect to the Primary Exchange or the Related Exchange, on any Index Business Day, the scheduled weekday closing time of the Primary Exchange or such Related Exchange on such Index Business Day, without regard to after hours or any other trading outside of the regular trading session hours.

Discontinuation of the Index; Alteration of Method of Calculation

If the Index Sponsor discontinues publication of, or otherwise fails to publish, the Index, or if the Index Sponsor does not make the Index Components, their Index Units or the Index Divisor available to the VWAP Calculation Agent, and the Index Sponsor or another entity publishes a successor or substitute index that the Note Calculation Agent determines, in its sole discretion, to be comparable to the discontinued Index and for which the index components, their Index Units and the index divisor, or equivalent information, is available to the VWAP Calculation Agent (such index being referred to in this section as a "successor index"), then the VWAP Level on any subsequent relevant date will be determined by reference to the VWAPs of the index components of that successor index, their Index Units and the index divisor of that successor index, or equivalent information, published with respect to that day and a fixed adjustment factor determined by the Note Calculation Agent upon the selection of that successor index to account for the difference in the levels of the Index and that successor index at that time for purposes of maintaining comparability between prior VWAP Levels and succeeding VWAP Levels.

Upon any selection by the Note Calculation Agent of a successor index, the Note Calculation Agent will cause written notice thereof to be promptly furnished to the Trustee, us, the Issuer and DTC, as holder of the Alerian ETNs.

Payment upon an Event of Default

In case an event of default with respect to the Alerian ETNs shall have occurred and be continuing, the amount declared due and payable per Alerian ETN upon any acceleration of the Alerian ETNs will be determined by the Note Calculation Agent and will be an amount in cash equal to the amount payable at maturity per Alerian ETN calculated as if the date of acceleration were (a) the final day of the Measurement Period with respect to the Final Valuation Date and (b) the Final Disrupted Averaging Date for the final day of the Measurement Period with respect to the Final Valuation Date (if the date of acceleration is a Disrupted Day). Under these circumstances, the Measurement Period with respect to the Final Valuation Date will consist of the date of acceleration and the four immediately preceding Index Business Days.

The amount determined as described above will constitute the final payment on the Alerian ETNs, and no additional amounts will accrue with respect to the Alerian ETNs following the date of acceleration.

If the maturity of the Alerian ETNs is accelerated because of an event of default as described above, the Issuer will, or will cause the Note Calculation Agent to, provide written notice to the Trustee at its New York office, on which notice the Trustee may conclusively rely, and to DTC, as holder of the Alerian ETNs, of the cash amount due with respect to the Alerian ETNs as promptly as possible and in no event later than two Business Days after the date of acceleration.

Listing

The Alerian ETNs have been listed on the NYSE Arca under the ticker symbol "AMJB." No assurance can be given as to the continued listing for the term of the Alerian ETNs, or the liquidity or trading market for the Alerian ETNs. The Issuer is not required to maintain a listing on NYSE Arca or any other exchange.

Book-Entry Only Issuance — The Depository Trust Company

DTC will act as securities depositary for the Alerian ETNs. The Issuer will issue Alerian ETNs only in fully registered form as book-entry notes. The Alerian ETNs will be represented by one or more permanent global notes deposited with, or on behalf of, DTC and registered in the name of DTC or its nominee. These certificates name DTC or its nominee as the owner of the Alerian ETNs. DTC maintains a computerized system that will reflect the interests held by its participants in the global notes. An investor's beneficial interest will be reflected in the records of DTC's direct or indirect participants through an account maintained by the investor with its broker/dealer, bank, trust company or other representative.

The Alerian ETNs will initially be represented by a type of global note referred to as a master note. A master note represents multiple securities that may be issued at different times and that may have different terms. Unless otherwise specified, in connection with each issuance of the Alerian ETNs, the Trustee and/or paying agent will, in accordance with instructions from the Issuer, make appropriate entries or notations in its records relating to the master note representing the Alerian ETNs to indicate that the master note evidences the Alerian ETNs of that issuance.

References to "holders" mean those who own Alerian ETNs registered in their own names, on the books that the Issuer or the paying agent maintain for this purpose, and not those who own beneficial interests in Alerian ETNs registered in street name or in Alerian ETNs issued in book-entry form through DTC.

Denominations

Book-entry interests in Alerian ETNs will be issued in minimum denominations equal to the Principal Amount and in integral multiples thereof.

Registrar, Transfer Agent and Paying Agent

Payment of amounts due at maturity on the Alerian ETNs will be payable and the transfer of the Alerian ETNs will be registrable at the principal corporate trust office of The Bank of New York Mellon in the City of New York.

The Bank of New York Mellon or one of its affiliates will act as registrar and transfer agent for the Alerian ETNs. The Bank of New York Mellon will also act as paying agent for the Alerian ETNs and may designate additional paying agents.

Registration of transfers of the Alerian ETNs will be effected without charge by or on behalf of The Bank of New York Mellon but upon payment (with the giving of such indemnity as The Bank of New York Mellon may require) in respect of any tax or other governmental charges that may be imposed in relation to it.

Reopening Issuances

In Issuer's sole discretion, and without providing holders notice or obtaining their consent, the Issuer may decide to issue and sell additional Alerian ETNs from time to time. These further issuances, if any, will be consolidated to form a single class with the originally issued Alerian ETNs, will have the same CUSIP number and will trade interchangeably with the originally issued Alerian ETNs immediately upon settlement.

However, the Issuer is under no obligation to issue or sell additional Alerian ETNs at any time, and, if the Issuer does sell additional Alerian ETNs, the Issuer may limit or restrict those sales, and the Issuer may stop and subsequently resume selling additional Alerian ETNs at any time. If the Issuer limits, restricts or stops sales of such additional Alerian ETNs, or if the Issuer subsequently resumes sales of such additional Alerian ETNs, the liquidity and trading price of the Alerian ETNs in the secondary market could be materially and adversely affected. Unless the Issuer indicates otherwise, if the Issuer suspends selling additional Alerian ETNs, the Issuer reserves the right to resume selling additional Alerian ETNs at any time, which might result in the reduction or elimination of any premium in the trading price. If holders pay a premium for the Alerian ETNs above the Closing Intrinsic Note Value and the Intraday Intrinsic Note Value, holders could incur significant losses if holders sell their Alerian ETNs at a time when the premium is no longer present in the market.

A suspension of additional issuances of the Alerian ETNs could result in a significant reduction in the number of outstanding Alerian ETNs if noteholders subsequently exercise their right to have the Alerian ETNs repurchased by the Issuer. Accordingly, the number of outstanding Alerian ETNs, and their liquidity, could vary substantially over the term of the Alerian ETNs.

The Alerian ETNs do not limit the Issuer's ability to incur other indebtedness or to issue other securities. Also, the Issuer is not subject to financial or similar restrictions by the terms of the Alerian ETNs.

The Issuer has no obligation to take holders' interests into account when deciding whether to issue additional Alerian ETNs. In addition, the Issuer is under no obligation to reopen any series of notes or to issue any additional Alerian ETNs.

General Terms of the Series A Notes

For a summary of the material provisions of the 2016 Indenture and the Series A Notes and guarantees issued thereunder, including provisions on events of defaults, modification of the 2016 Indenture and governing law, please see "— Description of JPMorgan Chase Financial Company LLC's Callable Fixed Rate Notes Due June 10, 2032, Fully and Unconditionally Guaranteed by JPMorgan Chase & Co. — General Terms of the Series A Notes" below.

DESCRIPTION OF JPMORGAN CHASE FINANCIAL COMPANY LLC'S CALLABLE FIXED RATE NOTES DUE JUNE 10, 2032, FULLY AND UNCONDITIONALLY GUARANTEED BY JPMORGAN CHASE & CO.

The following description of the Callable Fixed Rate Notes due June 10, 2032 (the "Callable Notes") is a summary and does not purport to be complete. It is subject to and qualified in its entirety by reference to the indenture dated February 19, 2016 (as may be amended or supplemented from time to time, the "2016 Indenture"), among JPMorgan Chase Financial Company LLC, as issuer ("JPMorgan Financial" or the "Issuer"), JPMorganChase, as guarantor (the "Guarantor"), and Deutsche Bank Trust Company Americas, as trustee (the "Trustee"), which is incorporated by reference as an Exhibit to the Form 10-K. We encourage you to read the 2016 Indenture for additional information.

General

As of December 31, 2024, $5,000,000 aggregate principal amount of the Callable Notes were outstanding.

The Callable Notes are unsecured and unsubordinated obligations of JPMorgan Financial, the payment of which is fully and unconditionally guaranteed by JPMorgan Chase & Co. The Callable Notes will rank *pari passu* with all of the Issuer's other unsecured and unsubordinated obligations. JPMorgan Chase & Co.'s guarantee of the Callable Notes will rank *pari passu* with all of JPMorgan Chase & Co.'s other unsecured and unsubordinated obligations. **Any payment on the Callable Notes issued by JPMorgan Financial is subject to the credit risk of JPMorgan Finance Callable Notes, and the credit risk of JPMorgan Chase & Co., as guarantor of the Callable Notes.**

The Callable Notes are part of a series of the Issuer's debt securities entitled "Global Medium-Term Notes, Series A" (the "Series A Notes") that the Issuer may issue under the 2016 Indenture from time to time. For more information about the Series A Notes, please see the section titled "— General Terms of the Series A Notes" below.

The Callable Notes are not bank deposits and are not insured by the Federal Deposit Insurance Corporation or by any other governmental agency, nor are they obligations of, or guaranteed by, a bank.

Unless otherwise specified, references herein to "holders" mean those in whose names the Callable Notes are registered on the books that the Issuer or the Trustee, or any successor trustee, as applicable, maintain for this purpose, and not those who own beneficial interests in the Callable Notes (registered in street name or otherwise).

Key Terms of the Callable Notes

We issued the Callable Notes on June 10, 2021, in minimum denominations of $1,000 and in integral multiples of $1,000 thereafter. The Maturity Date of the Callable Notes is June 10, 2032. Interest on the Callable Notes is payable semiannually on June 10th and December 10th of each year, beginning on December 10, 2021 to and including the Maturity Date (each, an "Interest Payment Date"), subject to any earlier redemption, at the Interest Rate of 2.60% per annum.

On June 10, 2031 (the "Redemption Date"), the Issuer may redeem the holders' Callable Notes, in whole but not in part, at a price equal to the principal amount being redeemed *plus* any accrued and unpaid interest. Any accrued and unpaid interest on the Callable Notes redeemed will be paid to the person who is the holder of record on such Callable Notes at the close of business one (1) business day prior to the Redemption Date. To redeem the Callable Notes, the Issuer will deliver notice to DTC, as holder of the Callable Notes, at least 5 business days and not more than 15 business days prior to the Redemption Date.

Calculation Agent

JPMS (the "Calculation Agent") will make all necessary calculations and determinations in connection with the Callable Notes, including calculations and determinations relating to any payments on the Callable Notes.

Payment upon an Event of Default

In case an event of default with respect to the Callable Notes shall have occurred and be continuing, the amount declared due and payable per $1,000 principal amount note upon any acceleration of the Callable Notes will be determined by the Calculation Agent and will be an amount in cash equal to $1,000 per $1,000 principal amount note *plus* accrued and unpaid interest, calculated as if the date of acceleration were the Maturity Date. In such case, interest will be calculated on the basis of a 360-day year and the actual number of days in such adjusted Interest Period and will be based on the Interest Rate on the applicable date immediately preceding such adjusted Interest Period.

If the maturity of the Callable Notes is accelerated because of an event of default as described above, the Issuer will, or will cause the Calculation Agent to, provide written notice to the Trustee at its New York office, on which notice the Trustee may conclusively rely, and to DTC of the cash amount due with respect to the Callable Notes as promptly as possible and in no event later than two business days after the date of acceleration.

Listing

The Callable Notes are listed and admitted to trading on the NYSE under the trading symbol "JPM/32." No assurance can be given as to the continued listing for the term of the Callable Notes, or the liquidity or trading market for the Callable Notes.

Book-Entry Only Issuance — The Depository Trust Company

DTC will act as securities depositary for the Callable Notes. The Callable Notes have been issued only as fully registered securities registered in the name of Cede & Co. (DTC's nominee). One or more fully registered global note certificates, representing the total aggregate principal amount of the Callable Notes, have been issued and have been deposited with DTC. We will not issue definitive notes in exchange for the global notes except in limited circumstances.

Registrar, Transfer Agent and Paying Agent

The Bank of New York Mellon or one of its affiliates will act as registrar and transfer agent for the Callable Notes. The Bank of New York Mellon will also act as paying agent for the Callable Notes and may designate additional paying agents.

Reopening Issuances

The Issuer may, in its sole discretion, "reopen" the Callable Notes based upon market conditions at that time. These further issuances, if any, will be consolidated with, have the same CUSIP number as and trade interchangeably with the respective originally issued Callable Notes immediately upon settlement and, consequently, will increase the aggregate principal amount of such outstanding Callable Notes. The price of any additional offering will be determined at the time of pricing of that offering.

General Terms of the Series A Notes

In this "General Terms of the Series A Notes" section, all references to the "debt securities" refer to Series A Notes issued by JPMorgan Chase Financial Company LLC.

The following description of the terms of the debt securities contains certain general terms that may apply to the debt securities, including the Callable Notes and the Alerian ETNs.

We have summarized below the material provisions of the 2016 Indenture and the debt securities and guarantees issued under the 2016 Indenture.

These descriptions are only summaries, and each investor should refer to the 2016 Indenture, which describes completely the terms and definitions summarized below and contains additional information regarding the debt securities issued under it. Where appropriate, we use parentheses to refer you to the particular sections of the 2016 Indenture. Any reference to particular sections or defined terms of the 2016 Indenture in any statement under this heading qualifies the entire statement and incorporates by reference the applicable section or definition into that statement.

The debt securities will be the Issuer's direct, unsecured general obligations, the payment on which is fully and unconditionally guaranteed by the Guarantor, and will have the same rank in liquidation as all of the Issuer's other unsecured and unsubordinated debt.

The Guarantor is a holding company that holds the stock of JPMorgan Chase & Co. Bank, National Association (the "Bank"), and JPMorgan Chase & Co. Holdings LLC (the "IHC"), its "intermediate holding company." The Guarantor conducts substantially all of its operations through subsidiaries, including the Bank and the IHC. As a result, claims of the holders of the debt securities against the Guarantor under the guarantee will generally have a junior position to claims of creditors of the Guarantor's subsidiaries. Claims of the Guarantor's subsidiaries' creditors other than the Guarantor include substantial amounts of deposit liabilities, long-term debt and other liabilities. In addition, the Guarantor is obligated to contribute to the IHC substantially all the net proceeds that the Guarantor receives from the issuance of securities (including the Alerian ETNs), and the ability of the Bank and the IHC to make payments to the Guarantor is limited. As a result of these arrangements, the Guarantor's ability to make various payments is dependent on its receiving dividends from the Bank and dividends and extensions of credit from the IHC. These limitations could affect the Guarantor's ability to pay interest on its debt securities, redeem or repurchase outstanding securities and fulfill its other payment obligations, including payment obligations under the guarantees of our debt securities.

Events of Default and Waivers

An "Event of Default" with respect to a series of debt securities issued under the 2016 Indenture is defined in the 2016 Indenture as:

a. default in the payment of interest on any debt securities of that series and continuance of such default for 30 days;

b. default in the payment of principal or other amounts payable on any debt securities of that series when due, at maturity, upon redemption, by declaration, or otherwise;
c. default in the performance, or breach, of any other covenants or warranties applicable to the Issuer contained in the 2016 Indenture applicable to that series, and continuation of such default or breach for 90 days after written notice has been given by the Trustee to the Issuer and the Guarantor or given by holders of at least 25% in aggregate principal amount of the outstanding securities of all series affected thereby to the Issuer, the Guarantor and the Trustee;
d. certain events of the Issuer's bankruptcy, insolvency, receivership, winding up or liquidation, whether voluntary or involuntary;
e. the guarantee ceases to be in full force and effect, other than in accordance with the 2016 Indenture, or the Guarantor denies or disaffirms its obligations under the guarantee, *provided* that no Event of Default with respect to the guarantee will occur as a result of, or because it is related directly or indirectly to, the insolvency of the Guarantor or the commencement of proceedings under Title 11 of the United States Code, or the appointment of a receiver for the Guarantor under Title II of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010, or the Federal Deposit Insurance Corporation having separately repudiated the Guarantee in receivership, or the commencement of or certain other events of the Guarantor's bankruptcy, insolvency, resolution, receivership, winding up or liquidation; or
f. any other event of default provided in the applicable supplemental indentures to the 2016 Indenture or form of security. (Section 5.01)

If an Event of Default occurs and is continuing because of a default in the payment of principal, interest or other amounts payable on the debt securities, a failure in the performance, or breach, of any covenant or agreement applicable to the Issuer, the guarantee ceasing to be in full force and effect, or any other event of default provided in the applicable supplemental indentures to the 2016 Indenture or form of security, either the Trustee or the holders of not less than 25% in aggregate principal amount of the debt securities of such series then outstanding, treated as one class, by written notice to the Issuer and the Guarantor, may declare the principal of all outstanding debt securities of such series and any interest accrued thereon, to be due and payable immediately. If a default due to specified events of the Issuer's bankruptcy, insolvency, receivership, winding up or liquidation, occurs and is continuing, the principal of all outstanding debt securities and any interest accrued thereon will automatically, and without any declaration or other action on the part of the Trustee or any holder, become immediately due and payable. Subject to certain conditions such declarations may be annulled and past defaults may be waived by the holders of a majority in aggregate principal amount of the outstanding debt securities of the series affected. (Sections 5.01 and 5.10)

Events of bankruptcy, insolvency, resolution, receivership, winding up or liquidation relating to the Guarantor will not constitute an Event of Default with respect to any series of debt securities. In addition, failure by the Guarantor to perform any of its covenants or warranties (other than a payment default) will not constitute an Event of Default with respect to any series of debt securities. Therefore, events of bankruptcy, resolution, receivership, insolvency, winding up or liquidation relating to the Guarantor (in the absence of any such event occurring with respect to the Issuer) will not permit any of the debt securities to be declared due and payable and the Trustee is not authorized to exercise any remedy against the Issuer or the Guarantor upon the occurrence or continuation of these events with respect to the Guarantor. Instead, even if an event of bankruptcy, insolvency, resolution, receivership, winding up or liquidation relating to the Guarantor has occurred, the Trustee and the holders of debt securities of a series will not be able to declare the relevant debt securities to be immediately due and payable unless there is an Event of Default with respect to that series as described above, such as the Issuer's bankruptcy, insolvency, receivership, winding up or liquidation or a payment default by the Issuer or the Guarantor on the relevant debt securities. **The value holders receive on any series of debt securities may be significantly less than what holders would have otherwise received had the Issuer's debt securities been declared due and payable immediately or the Trustee been authorized to exercise any remedy against the Issuer or the Guarantor upon the occurrence or continuation of these events with respect to the Guarantor.**

An Event of Default with respect to one series of debt securities does not necessarily constitute an Event of Default with respect to any other series of debt securities. The 2016 Indenture requires the Trustee to provide notice of default with respect to the debt securities within 90 days, unless the default is cured, but provides that the Trustee may withhold notice to the holders of the debt securities of any default if the board of directors, the executive committee, or a trust committee of directors or Trustees and/or responsible officers of the Trustee determines in good faith that it is in the interest of the holders of the debt securities of the applicable series to do so. The Trustee may not withhold notice of a default in the payment of principal of, interest on or any other amounts due under, such debt securities. (Section 5.11)

The 2016 Indenture provides that the holders of a majority in aggregate principal amount of outstanding debt securities of each series affected, with all such series voting as a single class, may direct the time, method, and place of conducting any proceeding for any remedy available to the Trustee, or exercising any trust or power conferred on the Trustee. The Trustee may decline to act if the direction is contrary to law and in certain other circumstances set forth in the 2016

Indenture. (Section 5.09) The Trustee is not obligated to exercise any of its rights or powers under the 2016 Indenture at the request or direction of the holders of debt securities unless the holders offer the Trustee security or indemnity satisfactory to it against the costs, expenses and liabilities incurred therein or thereby. (Section 6.02(d))

No holder of any debt security of any affected series has the right to institute any action for remedy unless such holder has previously given to the Trustee written notice of default, the Trustee has failed to take action for 60 days after the holders of not less than 25% in aggregate principal amount of the debt securities of each affected series make written request upon the Trustee to institute such action and have offered reasonable indemnity in connection with the same and the holders of a majority in aggregate principal amount of the debt securities of each affected series (voting as a single class) have not given direction to the Trustee that is inconsistent with the written request referred to above. (Section 5.06)

However, the right of any holder of a debt security or coupon to receive payment of the principal of and interest on that debt security or coupon on or after its due date, or to institute suit for the enforcement of any such payment, may not be impaired or affected without the consent of that holder. (Section 5.07)

The 2016 Indenture requires the Issuer and the Guarantor to file annually with the Trustee a written statement as to whether or not the Issuer or the Guarantor, as the case may be, has knowledge of a default. (Section 3.05)

The Issuer, the Guarantor and certain of their affiliates have a wide range of banking relationships with Deutsche Bank Trust Company Americas and The Bank of New York Mellon. If an actual or potential event of default occurs with respect to any debt securities, the Trustee may be considered to have a conflicting interest for purposes of the Trust Indenture Act of 1939, as amended. In that case, the Trustee may be required to resign under the 2016 Indenture, and the Issuer would be required to appoint a successor trustee. For this purpose, a “potential” event of default means an event that would be an event of default if the requirements for giving the Issuer default notice or for the default having to exist for a specific period of time were disregarded.

Modification of the 2016 Indenture

The 2016 Indenture contains provisions permitting the Issuer, the Guarantor and the Trustee to modify the 2016 Indenture or the rights of the holders of debt securities with the consent of the holders of not less than a majority in aggregate principal amount of each outstanding series of debt securities affected by the modification. Each holder of an affected debt security must consent to a modification that would:

a. extend the final maturity date of the principal of, or of any interest on, or other amounts payable under any debt security;
b. reduce the principal amount of, rate of interest on, or any other amounts due under any debt security;
c. change the currency or currency unit of payment of any debt security or certain provisions of the 2016 Indenture applicable to debt securities in foreign currencies;
d. change the method in which amounts of payments of principal, interest or other amounts are determined on any debt security;
e. reduce any amount payable upon redemption of any debt security;
f. impair the right of a holder to institute suit for the payment of a debt security or, if the debt securities provide, any right of repurchase at the option of the holder of a debt security;
g. reduce the percentage of debt securities of any series, the consent of the holders of which is required for any modification; or
h. make any change in the guarantee that would adversely affect the holders of the debt securities of such series or release the Guarantor from the guarantee other than pursuant to the terms of the 2016 Indenture. (Section 8.02)

The 2016 Indenture also permits the Issuer, the Guarantor and the Trustee to amend the 2016 Indenture in certain circumstances without the consent of the holders of debt securities to evidence the Issuer’s or the Guarantor’s merger or the replacement of the Trustee, to cure any ambiguity or to correct or supplement any defective or inconsistent provision, to make any change to the 2016 Indenture or the Issuer’s debt securities that the Issuer deems necessary or desirable and that does not materially and adversely affect the interests of holders of the debt securities and for certain other purposes. (Section 8.01)

Consolidations, Mergers, Sales and Transfers of Assets

Neither the Issuer nor the Guarantor may merge or consolidate with any other entity or sell, convey or transfer all or substantially all of their respective assets to any other entity, unless:

a. with respect to the Issuer:
    i. either the Issuer is the continuing company in the case of a merger or consolidation or the successor entity in the case of a merger or consolidation (including an affiliate of the Guarantor) or the entity to whom those assets are sold, conveyed or transferred in the case of a sale, conveyance or transfer is a

United States corporation or limited liability company that expressly assumes the due and punctual payment of the principal of, any interest on, or any other amounts due under the debt securities and the due and punctual performance and observance of all the covenants and conditions of the 2016 Indenture binding upon the Issuer, and

ii. no Event of Default and no event which, with notice or lapse of time or both, would become an Event of Default has occurred or would be continuing, immediately after the merger or consolidation, or the sale, conveyance or transfer, and

a. with respect to the Guarantor:

   i. either the Guarantor is the continuing corporation in the case of a merger or consolidation or the successor corporation in the case of a merger or consolidation or the entity to whom those assets are sold, conveyed or transferred in the case of a sale, conveyance or transfer is a United States corporation that expressly assumes the full and unconditional guarantee of the full and punctual payment of the principal of, any interest on, or any other amounts due under the debt securities and the due and punctual performance and observance of all the covenants and conditions of the 2016 Indenture binding upon the Guarantor, and
   ii. no Event of Default and no event which, with notice or lapse of time or both, would become an Event of Default has occurred or would be continuing, immediately after the merger or consolidation, or the sale, conveyance or transfer. (Sections 9.01 and 9.02)

Any transfer of material assets of the Guarantor to any other entity that occurs as a result of, or because it is related directly or indirectly to, any proceedings relative to the Guarantor under Title 11 of the United States Code or under a receivership under Title II of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 or under any other applicable federal or state bankruptcy, insolvency, resolution or other similar law will be deemed to be a sale, conveyance or transfer of all or substantially all of the Guarantor's assets.

There are no covenants or other provisions in the 2016 Indenture that would afford holders of debt securities additional protection in the event of a recapitalization transaction involving the Issuer or the Guarantor, a change of control of the Issuer or the Guarantor or a highly leveraged transaction involving the Issuer or the Guarantor. The merger covenant described above would apply only if the recapitalization transaction, change of control or highly leveraged transaction were structured to include a merger or consolidation of the Issuer or the Guarantor or a sale or conveyance of all or substantially all of the Issuer's or the Guarantor's assets. However, the Issuer may provide specific protections, such as a put right or increased interest, for particular debt securities, which the Issuer would describe in the applicable prospectus supplement.

JPMorganChase Guarantee

The Guarantor will fully and unconditionally guarantee the full and punctual payment of the principal of, interest on, and all other amounts payable under the debt securities when the same becomes due and payable, whether at maturity, upon redemption, repurchase at the option of the holders of the applicable debt securities or upon acceleration. If for any reason the Issuer does not make any required payment in respect of the Issuer's debt securities when due, the Guarantor will on demand pay the unpaid amount at the same place and in the same manner that applies to payments made by the Issuer under the 2016 Indenture. The guarantee is of payment and not of collection. (Section 14.01)

The Guarantor's obligations under the guarantee are unconditional and absolute. However, (1) the Guarantor will not be liable for any amount of payment that the Issuer is excused from making or any amount in excess of the amount actually due and owing by the Issuer, and (2) any defense or counterclaims available to the Issuer (except those resulting solely from, or on account of, the Issuer's insolvency or the Issuer's status as debtor or subject of a bankruptcy or insolvency proceeding) will also be available to the Guarantor to the same extent as these defense or counterclaims are available to the Issuer, whether or not asserted by the Issuer. (Section 14.02)

Concerning the Trustee, Paying Agent, Registrar and Transfer Agent

We, the Guarantor and certain of their affiliates have a wide range of banking relationships with Deutsche Bank Trust Company Americas, The Bank of New York Mellon and The Bank of New York Mellon, London Branch. The Bank of New York Mellon and, for notes settled through Euroclear Bank SA/NV or Clearstream Banking, S.A., Luxembourg, The Bank of New York Mellon, London Branch, will be the paying agents, authenticating agents, registrars and transfer agents for debt securities issued under the 2016 Indenture.

Deutsche Bank Trust Company Americas is initially serving as the Trustee for the debt securities issued under our 2016 Indenture, to which JPMorganChase acts as a guarantor, the warrants issued under our warrant indenture, to which JPMorganChase acts as a guarantor, and the debt securities issued under JPMorganChase's indenture.

Governing Law and Judgments

The debt securities and the 2016 Indenture, including the guarantee, will be governed by, and construed in accordance with, the laws of the State of New York. (Section 11.08)

Exhibit 10.19

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF JANUARY 21, 2025
RESTRICTED STOCK UNIT AWARD
OPERATING COMMITTEE

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of January 21, 2025 ("Grant Date") awarding Restricted Stock Units ("RSUs") pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan ("Plan"). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in "Definitions" below or elsewhere in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as "JPMorgan Chase" and together with its subsidiaries as the "Firm".

Form and Purpose of Award

Each RSU represents a non-transferable right to receive one share of Common Stock as of the applicable vesting date as set forth in your Award Agreement.

The purpose of this award is to motivate your future performance for services to be provided during the vesting period and to align your interests with those of the Firm and its shareholders.

Dividend Equivalents

If dividends are paid on Common Stock while RSUs under this award are outstanding, you will be paid an amount equal to the dividend paid on one share of Common Stock, multiplied by the number of RSUs outstanding under this award as of the dividend record date.

Vesting Period

The period from the Grant Date to the last vesting date is the "vesting period" (see subsections captioned "--Amendment" pursuant to which the Firm may extend the vesting period and "--No Ownership Rights/Other Limitations" pursuant to which the Firm may place restrictions on delivered shares of Common Stock following a vesting date).

Protection-Based Vesting

This award is intended and expected to vest on the vesting date(s), provided that you are continuously employed by the Firm through such vesting date, or you meet the requirements for continued vesting described under the subsections "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability". However, vesting and the number of RSUs in which you vest are subject to these terms and conditions (including, but not limited to, the section captioned "Remedies" and the following protection-based vesting provision).

Up to a total of fifty percent of your award that would otherwise be distributable to you during the vesting period ("At Risk RSUs") may be cancelled if the Chief Executive Officer of JPMorgan Chase ("CEO") determines in his or her sole discretion that cancellation of all or portion of the At Risk RSUs is appropriate in light of any one or a combination of the following factors:

- Your performance in relation to the priorities for your position, or the Firm's performance in relation to the priorities for which you share responsibility as a member of the Operating Committee, have been unsatisfactory for a sustained period of time. Among the factors the CEO may consider in assessing performance are net income, total net revenue, return on equity, earnings per share and capital ratios of the Firm, both on an absolute basis and, as appropriate, relative to peer firms.

- For any calendar year ending during the vesting period, JPMorgan Chase's annual pre-tax pre-provision income at the Firm level is negative.
- Awards granted to participants in a Line of Business for which you exercise, or during the vesting period exercised, direct or indirect responsibility, were in whole or in part cancelled because the Line of Business did not meet its annual Line of Business Financial Threshold.
- The Firm does not meet the Firmwide Financial Threshold.

In the event that your employment terminates due to "Job Elimination", "Full Career Eligibility", "Government Office" or "Disability" thereby entitling you to continued vesting in your award (or potentially acceleration due to satisfaction of the Government Office Requirements), the cancellation circumstances described above will continue to apply to your At Risk RSUs pursuant to the subsection captioned "Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity".

Any determination above with respect to protection-based vesting provisions is subject to ratification by the Compensation and Management Development Committee of the Board of Directors of JPMorgan Chase ("Committee"). In the case of an award to the CEO, all such determinations shall be made by the Committee and ratified by the Board.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled "Other Matters":

https://about.jpmorganchase.com/about/governance/corporate-governance-principles

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

Termination of Employment

Except as explicitly set forth below under the subsections captioned "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability" or under the section captioned "Death", any RSUs outstanding under this award will be cancelled effective on the date your employment with the Firm terminates for any reason.

Subject to these terms and conditions (including, but not limited to, sections captioned "Protection-Based Vesting", "Bonus Recoupment", "Your Obligations" and "Remedies"), you will be eligible to continue to vest (as you otherwise would vest if you were still employed by JPMorgan Chase) with respect to your award in accordance with its terms and conditions following the termination of your employment if one of the following circumstances applies to you:

- **Job Elimination**
  In the event that the Director of Human Resources or nominee in his or her sole discretion determines that
  - the Firm terminated your employment because your job was eliminated, and
  - after you are notified that your job will be eliminated, you provided such services as requested by the Firm in a cooperative and professional manner, and
  - you satisfied the Release/Certification Requirements set forth below.

- **Full Career Eligibility**
  In the event that the Director of Human Resources or nominee in his or her sole discretion determines that
  - (i) you voluntarily terminated your employment with the Firm, and (ii) had completed at least five years of Recognized Service (as defined below) on your date of termination with at least three years of continuous service with the Firm immediately preceding your termination date or if you are over 60 years of age with at least three years of continuous service with the Firm immediately preceding your termination date, and
  - the sum of your age and Recognized Service (as defined below) on your date of termination equaled or exceeded 60 and
  - you provided at least 180 days advance written notice to the Firm of your intention to voluntarily terminate your employment under this provision, during which notice period you provided such services as requested by the Firm in a cooperative and professional manner and you did not perform any services for any other employer, and
  - continued vesting shall be appropriate, which determination shall be made prior to your termination and will be based on your performance and conduct (before and after providing notice), and

- you satisfied the Release/Certification Requirements set forth below, and
- for 36 months from the date of grant of this award you do not either perform services in any capacity (such as an employee, contractor, consultant, advisor, or self-employed individual, whether paid or unpaid) for a Financial Services Company (as defined below) or work in your profession (whether or not for a Financial Services Company); provided that you may do one or more of the following:
  - work in a government, education or Not-for-Profit Organization (as defined below),
  - subject to disclosure and confirmation of the following, serve on the board of directors of a public or non-public company that is not a Financial Services Company.

After receipt of such advance written notice, the Firm may choose to have you continue to provide services during such 180-day period as a condition to continued vesting or shorten the length of the 180-day period at the Firm's sole discretion, but to a date no earlier than the date you would otherwise meet the age and service requirements.

Additional advance notice requirements may apply for employees subject to notice period policies (see "Notice Period" below).

- **Government Office**
  In the event that you voluntarily terminate your employment with the Firm to accept a Government Office or become a candidate for an elective Government Office, as described at the end of these terms and conditions under the section captioned "Government Office Requirements". See also definition of Government Office in the section captioned "Definitions".

- **Disability**
  In the event that
  - your employment with the Firm terminates because (i) you are unable to return to work while you are receiving benefits under the JPMorgan Chase Long Term Disability Plan, or for non-U.S. employees, under the equivalent JPMorgan Chase sponsored local country plan (in either case, "LTD Plan"), or (ii) if you are not covered by a LTD Plan, you are unable to return to work due to a long-term disability that would qualify for benefits under the applicable LTD Plan, as determined by the Firm or a third-party designated by the Firm; provided that you (x) request in writing continued vesting due to such disability within 30 days of the date your employment terminates, and (y) provide any requested supporting documentation and (z) receive the Firm's written consent to such treatment, and
  - you satisfied the Release/Certification Requirements set forth below.

Release/Certification

To qualify for continued vesting after termination of your employment under any of the foregoing circumstances:

- you must timely execute and deliver a release of claims in favor of the Firm, having such form and terms as the Firm shall specify,
- with respect to "Full Career Eligibility", prior to the termination of your employment, you must confirm with management that you meet the eligibility criteria (including providing at least 180 days advance written notification), advise that you are seeking to be treated as an individual eligible for "Full Career Eligibility", and receive written consent to such continued vesting,
- with respect to "Full Career Eligibility" and "Government Office", it is your responsibility to (i) notify the Firm within 15 days after the date you are no longer in compliance with the employment restrictions (as described herein) or (ii) take the appropriate steps to certify to the Firm prior to each vesting date while the employment restrictions are outstanding, on the authorized form of, and by the due date set by, the Firm, that you have complied with the employment restrictions applicable to you (as described herein) from your date of termination of employment through the applicable vesting date,
- with respect to "Disability", you must satisfy the notice and documentation described above and receive written consent to such continued vesting, and
- in all cases, complied with all other terms of the Award Agreement. (See section captioned "Your Obligations".)

Death

If you die while you are eligible to vest in RSUs under this award, the RSUs will immediately vest and will be distributed in shares of Common Stock (after applicable tax withholding) to your designated beneficiary on file with the Firm's Stock Administration Department, or if no beneficiary has been designated or survives you or if beneficiary designation is not recognized locally, then to your estate unless otherwise required by local legislation. Any shares will be distributed no later than the end of the calendar year immediately following the calendar year which contains your date of death; however, our administrative practice is to register such shares in the name of your beneficiary or estate within 60 days of the Firm's receipt of any required documentation.

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned "--Confidentiality & Non-Solicitation", "--False Statements", "--Cooperation", "--Compliance with Award Agreement" and "--Notice Period."

- **Confidentiality & Non-Solicitation**

  During your employment by the Firm and for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm's then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual's employment terminated because his or her job was eliminated, or the individual's employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm's confidential or proprietary information.

  These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm's policies.

  You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm's business or its customers except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. "Confidential Information" includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm's private and internal information through telephonic, intranet or internet means.

  For Employees with a Work Location in California, you agree that the restrictions in the first paragraph of this "Confidentiality and Non-Solicitation" provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use "Confidential Information" in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

  Nothing in this award precludes you from reporting to the Firm's management or directors, the government, a regulator, a self-regulatory agency, your attorneys or a court, conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

  If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**

You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of the law or from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**
  You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**
  You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third party who might be affected thereby, including your prospective employer.

- **Notice Period**
  If you are subject to a notice period or become subject to a notice period after the Grant Date, whether by contract or by policy, that requires you to provide advance written notice of your intention to terminate your employment ("Notice Period"), then as consideration for this award and continued employment, you will provide the Firm with the necessary advance written notice that applies to you, as specified by such contract or policy.

  After receipt of your notice, the Firm may choose to have you continue to provide services during the applicable Notice Period or may place you on a paid leave for all or part of the applicable Notice Period. During the Notice Period, you shall continue to devote your full time and loyalty to the Firm by providing services in a cooperative and professional manner and not perform any services for any other employer and shall receive your base salary and certain benefits until your employment terminates. You and the Firm may mutually agree to waive or modify the length of the Notice Period.

  Regardless of whether a Notice Period applies to you, you must comply with the 180-day advance notice period described under the subsection captioned "--Full Career Eligibility" in the event you wish to terminate employment under that same subsection.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**
  In addition to the cancellation provisions described under the sections captioned "Protection-Based Vesting", "Bonus Recoupment", and "Termination of Employment", up to 100% of your outstanding RSUs under this award may be cancelled if the Firm in its sole discretion determines that:
  - Any of the following detrimental and risk-related conduct has occurred:
    - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
    - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
    - this award was based on a material misrepresentation by you, or
    - you improperly or with gross negligence failed to identify, raise or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
    - your employment was terminated for Cause (see section captioned "Definitions" below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
  - you have failed to comply with any of the advance notice/cooperation requirements or employment restrictions applicable to your termination of employment, or
  - you have failed to return the required forms specified under the section captioned "Release/Certification" by the specified deadline, or
  - you have violated any of the provisions as set forth above in the section captioned "Your Obligations".

To the extent provided under the subsection captioned "--Amendment" below, JPMorgan Chase reserves the right to suspend vesting of this award and/or distribution of shares under this award, including, without limitation, during any period that JPMorgan Chase is evaluating whether this award is subject to cancellation and/or recovery and/or whether the conditions for distributions of shares under this award are satisfied. JPMorgan Chase is not responsible for any price fluctuations during any period of suspension and, if applicable, suspended units will be reinstated consistent with Plan administration procedures. See also subsection captioned "--No Ownership Rights/Other Limitations".

- **Recovery**
  In addition, you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares of Common Stock previously distributed under this award as follows:
  - Payment may be required with respect to any shares distributed within the three year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
    - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment, or
    - you violated any of the provisions as set forth above in the section captioned "Your Obligations", or
    - you violated the employment restrictions set forth in the subsection "--Full Career Eligibility" or "--Government Office" following the termination of your employment.
  - In addition, payment may be required with respect to any shares distributed within the one year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above "Detrimental Conduct, Risk Related and Other Cancellation/Recapture" subsection.

  Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm's employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase's rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

  For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm's intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

  Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

  Nothing in the section in any way limits your obligations under "Bonus Recoupment".

- **Right to an Injunction**
  You acknowledge that a violation or attempted violation of any of the provisions set forth in "Your Obligations" herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in "Your Obligations"; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares, dividend equivalents and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award, shares that vest under this award, and/or dividend equivalents, and (iii) retaining shares that vest under this award or dividend equivalents until you pay any taxes associated with the award, vested shares and/or the dividend equivalents directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt owed by you under this award without your consent. This restriction on the Firm does not apply to accounts described and authorized in "No Ownership Rights/Other Limitations" described below.

No Ownership Rights/Other Limitations: RSUs do not convey the rights of ownership of Common Stock and do not carry voting rights. No shares of Common Stock will be issued to you until after the RSUs have vested. Shares will be issued in accordance with JPMorgan Chase's procedures for issuing stock. By accepting this award, you authorize the Firm, in its sole discretion, to establish on your behalf a brokerage account in your name with the Firm or book-entry account with our stock plan administrator and/or transfer agent and deliver to that account any vested shares derived from the award. You also acknowledge that should there be a determination that the cancellation provisions of this award apply during the period when the vesting of any outstanding RSUs has been suspended, then you agree that such RSUs may be cancelled in whole or part. (See Sections captioned "Protection-Based Vesting", "Bonus Recoupment", "Termination of Employment" and "Remedies", as well as the subsection captioned "--Amendment" permitting suspension of vesting.)

With respect to any applicable vesting date, JPMorgan Chase may impose for any reason, as of such vesting date for such period as it may specify in its sole discretion, such restrictions on the Common Stock to be issued to you as it may deem appropriate, including, but not limited to, restricting the sale, transfer, pledging, assignment, hedging or encumbrance of such shares of Common Stock. Such restrictions described in the last sentence shall not impact your right to vote or receive dividends with respect to the Common Stock. By accepting this award, you acknowledge that during such specified period should there be a determination that the recovery provisions of this award apply, then you agree that you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares subject to such restrictions (notwithstanding the limitation set forth in the "Right to Set Off" subsection above). (See Sections captioned "Bonus Recoupment" and "Remedies".)

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is "at-will" and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares and cash hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Notwithstanding anything herein to the contrary, if you (i) are subject to taxation under the Code, (ii) are a specified employee as defined in the JPMorgan Chase 2005 Deferred Compensation Plan and (iii) have incurred a separation from service (as defined in that Plan with the exception of death) and if any units/shares under this award represent deferred compensation as defined in Section 409A and such shares are distributable (under the terms of this award) within six months following, and as a result of your separation from service, then those shares will be delivered to you during the first calendar month after the expiration of six full months from date of your separation from service. Further, if your award is not subject to a substantial risk of forfeiture as defined by regulations issued under Section 409A of the Code, then the remainder of each calendar year immediately following (i) each applicable vesting date set forth in your Award Agreement shall be a payment date for purposes of distributing the vested portion of the award and (ii) each date that JPMorgan

Chase specifies for payment of dividends declared on its Common Stock, shall be the payment date(s) for purposes of distributing dividend equivalent payments.

Change in Outstanding Shares: In the event of any change in the outstanding shares of Common Stock by reason of any stock dividend or split, recapitalization, issuance of a new class of common stock, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to stockholders of Common Stock other than regular cash dividends, the Committee will make an equitable substitution or proportionate adjustment, in the number or kind of shares of Common Stock or other securities issued or reserved for issuance pursuant to the Plan and to any RSUs outstanding under this award for such corporate events.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under "No Ownership Rights/Other Limitations", in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity: Upon receipt of satisfactory evidence that applicable United States federal, state, local, foreign or supranational ethics or conflict of interest laws or regulations require you to divest your interest in JPMorgan Chase RSUs, the Firm may accelerate the distribution of all or part of your outstanding award effective on or before the required divestiture date; provided that no accelerated distribution shall occur if the Firm determines that such acceleration will violate Section 409A of the Code. Accelerated distribution under this paragraph does not impact the dates as set forth in the "Remedies" section above. The time period for recovery shall be determined by the originally scheduled vesting date or distribution date prior to any acceleration event.

If you have voluntarily terminated your employment and have satisfied the requirements of the section captioned "Government Office Requirements", acceleration shall apply (to extent required) to the percentage of your outstanding award that would continue to vest under that section. In the case of a termination of employment where the award is outstanding as a result of the subsections entitled "--Job Elimination" or "--Full Career Eligibility", then acceleration shall apply, to the extent required, to the full outstanding award.

Notwithstanding accelerated distribution pursuant to the foregoing, you will remain subject to the applicable terms of your Award Agreement as if your award had remained outstanding for the duration of the original vesting period and shares had been distributed as scheduled as of each applicable vesting date, including, but not limited to, repayment obligations set forth in the section captioned "Remedies" and the employment restrictions in the sections captioned "Protection-Based Vesting" and "Government Office Requirements" and the subsection "--Full Career Eligibility".

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes).   Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm's legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm's By-Laws in the event of such a suit by the Firm.

Non-transferability: Neither this award or any other outstanding awards of RSUs, nor your interests or rights in any such awards, shall be assigned, pledged, transferred, hedged, hypothecated or subject to any lien. An award may be transferred following your death by will, the laws of descent or by a beneficiary designation on file with the Firm.

Outstanding Awards: The Administrative provisions set forth above shall apply to any award of RSUs outstanding as of the date hereof, and such awards are hereby amended.

Definitions

"**Cause**" means a determination by the Firm that your employment terminated as a result of your (i) violation of any law, rule or regulation (including rules of self-regulatory bodies) related to the Firm's business, (ii) indictment or conviction of a felony, (iii) commission of a fraudulent act, (iv) violation of the JPMorgan Chase Code of Conduct or other Firm policies or misconduct related to your duties to the Firm (other than immaterial and inadvertent violations or misconduct), (v) grossly inadequate performance of the duties associated with your position or job function or failure to follow reasonable directives of your manager, or (vi) any act or failure to act that is injurious to the interests of the Firm or its relationship with a customer, client or an employee.

"**Financial Services Company**" means a business enterprise that engages in any of the following services (itself or through an affiliate or subsidiary), regardless of whether such services are the principle strategy or revenue-generating activity:

- commercial or retail banking, including, but not limited to, commercial, institutional and personal trust, custody and/or lending and processing services, internet banking, originating and servicing mortgages, issuing and servicing credit cards, payment servicing or processing or merchant services,
- insurance, including but not limited to, guaranteeing against loss, harm, damage, illness, disability or death, providing and issuing annuities, acting as principal, agent or broker for purpose of the forgoing,
- financial, investment or economic advisory services, including but not limited to, investment banking services (such as advising on mergers or dispositions, underwriting, dealing in, or making a market in securities or other

similar activities), brokerage services, investment management services, asset management services, foreign exchange services, interbank networks and hedge funds,

- issuing, trading or selling instruments representing interests in pools of assets or in derivatives instruments,
- financial technology companies, such as those selling blockchain services, or offering or selling financial products/services,
- advising on, or investing in, private equity or real estate funds or ventures, or
- any similar activities that the Director of Human Resources or nominee determines in his or her sole discretion constitute financial services,

- however, subject to disclosure and confirmation of the following, it shall not be considered a "Financial Services Company" for the purpose of this Award Agreement if such family office is solely funded by you and/or your family's own personal funds without external investors directly or indirectly.

"Firmwide Financial Threshold" means a cumulative return on tangible common equity for calendar years 2025, 2026 and 2027 of not less than 15%. Cumulative return on tangible common equity means (i) the sum of the Firm's reported net income for all three calendar years, divided by (ii) reported year-end tangible equity averaged over the three years.

"Government Office" means (i) a full-time position in an elected or appointed office in local, state, or federal government (including equivalent positions outside the U.S. or in a supranational organization), not reasonably anticipated to be a full-career position, or (ii) conducting a bona fide full-time campaign for such an elective public office after formally filing for candidacy, where it is customary and reasonably necessary to campaign full-time for the office.

"Line of Business" means a business unit of the Firm (or one or more business units designated below under the definition "Line of Business Financial Threshold" of the Commercial & Investment Bank). All Corporate Functions (including the functions of the Chief Investment Office) are considered a single Line of Business.

"Line of Business Financial Threshold" means the financial threshold set forth below for the following Lines of Business based on the Firm's management reporting system:

Asset & Wealth Management	Annual negative pre-tax pre-provision income[1]
Card and Auto	Annual negative pre-tax pre-loan loss reserve income[2]
Commercial & Investment Bank	Annual negative pre-tax pre-provision income[1] for CIB overall or annual negative allocated product revenues (excluding XVA) for: • Fixed Income • Equities • Securities Services • Global Investment Banking • Payments
Consumer Banking, J.P. Morgan Wealth Management and Business Banking	Annual negative pre-tax pre-loan loss reserve income[2]
Corporate Functions (including Chief Investment Office)	Annual negative pre-tax pre-provision income[1] at the Firm level
Home Lending	Annual negative pre-tax pre-loan loss reserve income[2]
[1]Pre-tax pre-provision income means Revenue less Expenses [2]Pre-tax pre-loan loss reserve income means Revenue less (Expenses plus Net Charge-offs)	

"Not-for-Profit Organization" means an entity exempt from tax under state law and under Section 501(c)(3) of the Code. Section 501(c)(3) only includes entities organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals. Not-for-Profit Organization shall also mean entities outside the United States exempt from local and national tax laws because they are organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals.

"Recognized Service" means the period of service as an employee set forth in the Firm's applicable service-related policies.

Government Office Requirements

You may be eligible to continue vesting in all or part of your award if you voluntarily resign to accept a Government Office (as defined above) or to become a candidate for an elective Government Office.

Full Career Eligibility:
"Government Office Requirements" does not apply to you if you satisfy the subsection captioned "--Full Career Eligibility" as of the date that you voluntarily terminate your employment with the Firm.

Eligibility:
Eligibility for continued vesting is conditioned on your providing the Firm:

- At least 60 days' advance written notice of your intention to resign to accept or pursue a Government Office (see section captioned "Definitions"), during which period you must perform in a cooperative and professional manner services requested by the Firm and not provide services for any other employer. The Firm may elect to shorten this notice period at the Firm's sole discretion.
- Confirmation, in a form of satisfactory to the firm, that vesting in this award pursuant to this provision would not violate any applicable law, regulation or rule.
- Documentation in a form satisfactory to the Firm that your resignation is for the purpose of accepting a Government Office or becoming a candidate for a Government Office. (See Section captioned "Definitions".)

Portion of Your Awards Subject to Continued Vesting:
Subject to the conditions below, the percentage of your outstanding awards that will continue to vest in accordance with this award's original schedule will be based on your years of continuous service completed with the Firm immediately preceding your termination date, as follows:

- 50% if you have at least 3 but less than 4 years of continuous service,
- 75% if you have at least 4 but less than 5 years of continuous service, or
- 100% if you have 5 or more years of continuous service.

The portion of each award subject to continued vesting above is referred to as the "CV Award" and the portion not subject to continued vesting will be cancelled on the date your employment terminates.

Conditions for Continued Vesting of Awards:

- You must remain in a non-elective Government Office for two or more years after your employment with the Firm terminates to receive in full your CV Award; provided that if your non-elective Government Office is for a period less than two years, you will be entitled to retain any portion of the CV Award with a vesting date during your period of Government Service; or
- In the case of resignation from the Firm to campaign for an elective Government Office, your name must be on the primary or final public ballot for the election. (If you are not elected, see below for employment restrictions.)

Satisfaction of Conditions:
If your service in a Government Office ends two years or more after your employment with the Firm terminates, or in the case of resignation from the Firm to campaign for a Government Office, your name is on the primary or final public ballot for the election and you are not elected, any CV Awards then outstanding and any such awards that would have then been outstanding but for an accelerated distribution of shares (as described in the subsection captioned "Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity") will be subject for the remainder of the applicable vesting period to the same terms and conditions of this Award Agreement, including employment restrictions during the vesting period, as if you had resigned from the Firm having met the requirements for Full Career Eligibility.

Failure to Satisfy Conditions:
If you do not satisfy the above "Conditions for Continued Vesting of Awards", any outstanding RSUs under each CV Award will be cancelled. You also will be required to repay the Fair Market Value of the number of shares (before tax and other withholdings) of Common Stock distributed to you that would have been outstanding as RSUs on the date you failed to satisfy the "Condition for Continued Vesting of Awards" but for their accelerated distribution (as described in the subsection captioned "Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity"). Fair Market Value for this purpose will be determined as the date that the shares were distributed.

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF JANUARY 21, 2025
RESTRICTED STOCK UNIT AWARD
OPERATING COMMITTEE

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of January 21, 2025 ("Grant Date") awarding Restricted Stock Units ("RSUs") pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan ("Plan"). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in "Definitions" below or elsewhere in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as "JPMorgan Chase" and together with its subsidiaries as the "Firm".

Form and Purpose of Award

Each RSU represents a non-transferable right to receive one share of Common Stock as of the applicable vesting date as set forth in your Award Agreement.

The purpose of this award is to motivate your future performance for services to be provided during the vesting period and to align your interests with those of the Firm and its shareholders.

Dividend Equivalents

This award is not eligible for dividend equivalent payments.

Vesting Period

The period from the Grant Date to the last vesting date is the "vesting period" (see subsections captioned "--Amendment" pursuant to which the Firm may extend the vesting period and "--No Ownership Rights/Other Limitations" pursuant to which the Firm may place restrictions on delivered shares of Common Stock following a vesting date).

Protection-Based Vesting

This award is intended and expected to vest on the vesting date(s), provided that you are continuously employed by the Firm through such vesting date, or you meet the requirements for continued vesting described under the subsections "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability". However, vesting and the number of RSUs in which you vest are subject to these terms and conditions (including, but not limited to, the section captioned "Remedies" and the following protection-based vesting provision).

Up to a total of fifty percent of your award that would otherwise be distributable to you during the vesting period ("At Risk RSUs") may be cancelled if the Chief Executive Officer of JPMorgan Chase ("CEO") determines in his or her sole discretion that cancellation of all or portion of the At Risk RSUs is appropriate in light of any one or a combination of the following factors:

- Your performance in relation to the priorities for your position, or the Firm's performance in relation to the priorities for which you share responsibility as a member of the Operating Committee, have been unsatisfactory for a sustained period of time. Among the factors the CEO may consider in assessing performance are net income, total net revenue, return on equity, earnings per share and capital ratios of the Firm, both on an absolute basis and, as appropriate, relative to peer firms.
- For any calendar year ending during the vesting period, JPMorgan Chase's annual pre-tax pre-provision income at the Firm level is negative.
- Awards granted to participants in a Line of Business for which you exercise, or during the vesting period exercised, direct or indirect responsibility, were in whole or in part cancelled because the Line of Business did not meet its annual Line of Business Financial Threshold.
- The Firm does not meet the Firmwide Financial Threshold.

In the event that your employment terminates due to “Job Elimination”, ”Full Career Eligibility”, “Government Office” or “Disability” thereby entitling you to continued vesting in your award (or potentially acceleration due to satisfaction of the Government Office Requirements), the cancellation circumstances described above will continue to apply to your At Risk RSUs pursuant to the subsection captioned “Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”.

Any determination above with respect to protection-based vesting provisions is subject to ratification by the Compensation and Management Development Committee of the Board of Directors of JPMorgan Chase (“Committee”). In the case of an award to the CEO, all such determinations shall be made by the Committee and ratified by the Board.

Holding Requirement

As of each vesting date, you shall be entitled to a distribution equal to the Fair Market Value of the number of RSUs vesting on such date, less the number being withheld to satisfy tax withholding obligations. You agree that the distribution made to you will be held in an account in your name with restrictions preventing you from transferring, assigning, hedging, selling, pledging, or otherwise encumbering such distribution for a twelve-month period commencing with the vesting date. Such restrictions shall lapse in event of your death.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled “Other Matters”:

https://about.jpmorganchase.com/about/governance/corporate-governance-principles

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

EMEA Malus and Clawback Policy – Identified Staff

In consideration of grant of this award, and without prejudice to any other provision of this Award Agreement, you agree that you are subject to the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff (and any applicable supplement(s) to that policy) or successor policy as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award. You can access this policy as currently in effect in My Rewards through the following link: https://myrewards.jpmorganchase.com

The provisions of the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff set out the terms and conditions applying to the grant of this award which ensure that the Firm is able to meet its regulatory obligations to operate malus (reduce) and/or clawback (recover) to awards in certain circumstances. These include, but are not limited to, where (i) there is a material downturn in the Firm’s financial performance or (ii) where the Firm is required to hold more capital. The circumstances in which the events at (i) and (ii) would occur are analogous to some of the circumstances considered under the existing Firmwide terms and conditions, in particular the Bonus Recoupment Policy and the Protection Based Vesting provisions.

Termination of Employment

Except as explicitly set forth below under the subsections captioned “--Job Elimination”, “--Full Career Eligibility”, “--Government Office” or “--Disability” or under the section captioned “Death”, any RSUs outstanding under this award will be cancelled effective on the date your employment with the Firm terminates for any reason.

Subject to these terms and conditions (including, but not limited to, sections captioned “Protection-Based Vesting”, “Bonus Recoupment”, “EMEA Malus and Clawback Policy – Identified Staff”, “Your Obligations” and “Remedies”), you will be eligible to continue to vest (as you otherwise would vest if you were still employed by JPMorgan Chase) with respect to your award in accordance with its terms and conditions following the termination of your employment if one of the following circumstances applies to you:

- **Job Elimination**

  In the event that the Director of Human Resources or nominee in his or her sole discretion determines that
  - the Firm terminated your employment because your job was eliminated, and
  - after you are notified that your job will be eliminated, you provided such services as requested by the Firm in a cooperative and professional manner, and

  - you satisfied the Release/Certification Requirements set forth below.

- **Full Career Eligibility**
  In the event that the Director of Human Resources or nominee in his or her sole discretion determines that
  - (i) you voluntarily terminated your employment with the Firm, and (ii) had completed at least five years of Recognized Service (as defined below) on your date of termination with at least three years of continuous service with the Firm immediately preceding your termination date or if you are over 60 years of age with at least three years of continuous service with the Firm immediately preceding your termination date, and
  - your Recognized Service (as defined below) on your date of termination equaled or exceeded 15 years, or your combined Recognized Service with the Firm and external professional experience (as attested by you to the Firm) equaled or exceeded 30 years, and
  - you provided at least 180 days advance written notice to the Firm of your intention to voluntarily terminate your employment under this provision, during which notice period you provided such services as requested by the Firm in a cooperative and professional manner and you did not perform any services for any other employer, and
  - continued vesting shall be appropriate, which determination shall be made prior to your termination and will be based on your performance and conduct (before and after providing notice), and
  - you satisfied the Release/Certification Requirements set forth below, and
  - for 36 months from the date of grant of this award you do not either perform services in any capacity (such as an employee, contractor, consultant, advisor, or self-employed individual, whether paid or unpaid) for a Financial Services Company (as defined below) or work in your profession (whether or not for a Financial Services Company); provided that you may do one or more of the following:
    - work in a government, education or Not-for-Profit Organization (as defined below),
    - subject to disclosure and confirmation of the following, serve on the board of directors of a public or non-public company that is not a Financial Services Company.

  After receipt of such advance written notice, the Firm may choose to have you continue to provide services during such 180-day period as a condition to continued vesting or shorten the length of the 180-day period at the Firm's sole discretion, but to a date no earlier than the date you would otherwise meet the service requirement.

  Additional advance notice requirements may apply for employees subject to notice period policies (see "Notice Period" below).

- **Government Office**
  In the event that you voluntarily terminate your employment with the Firm to accept a Government Office or become a candidate for an elective Government Office, as described at the end of these terms and conditions under the section captioned "Government Office Requirements". See also definition of Government Office in the section captioned "Definitions".

- **Disability**
  In the event that
  - your employment with the Firm terminates because (i) you are unable to return to work while you are receiving benefits under the JPMorgan Chase Long Term Disability Plan, or for non-U.S. employees, under the equivalent JPMorgan Chase sponsored local country plan (in either case, "LTD Plan"), or (ii) if you are not covered by a LTD Plan, you are unable to return to work due to a long-term disability that would qualify for benefits under the applicable LTD Plan, as determined by the Firm or a third party designated by the Firm; provided that you (x) request in writing continued vesting due to such disability within 30 days of the date your employment terminates, and (y) provide any requested supporting documentation and (z) receive the Firm's written consent to such treatment, and
  - you satisfied the Release/Certification Requirements set forth below.

Release/Certification

To qualify for continued vesting after termination of your employment under any of the foregoing circumstances:
- you must timely execute and deliver a release of claims in favor of the Firm, having such form and terms as the Firm shall specify,
- with respect to "Full Career Eligibility", prior to the termination of your employment, you must confirm with management that you meet the eligibility criteria (including providing at least 180 days advance written

notification), advise that you are seeking to be treated as an individual eligible for "Full Career Eligibility", and receive written consent to such continued vesting,
- with respect to "Full Career Eligibility" and "Government Office", it is your responsibility to (i) notify the Firm within 15 days after the date you are no longer in compliance with the employment restrictions (as described herein) or (ii) take the appropriate steps to certify to the Firm prior to each vesting date while the employment restrictions are outstanding, on the authorized form of, and by the due date set by, the Firm, that you have complied with the employment restrictions applicable to you (as described herein) from your date of termination of employment through the applicable vesting date,
- with respect to "Disability", you must satisfy the notice and documentation described above and receive written consent to such continued vesting, and
- in all cases, complied with all other terms of the Award Agreement. (See section captioned "Your Obligations".)

Death
If you die while you are eligible to vest in RSUs under this award, the RSUs will immediately vest and will be distributed in shares of Common Stock (after applicable tax withholding) to your designated beneficiary on file with the Firm's Stock Administration Department, or if no beneficiary has been designated or survives you or if beneficiary designation is not recognized locally, then to your estate unless otherwise required by local legislation. Any shares will be distributed no later than the end of the calendar year immediately following the calendar year which contains your date of death; however, our administrative practice is to register such shares in the name of your beneficiary or estate within 60 days of the Firm's receipt of any required documentation.

Your Obligations
In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned "—Confidentiality & Non-Solicitation", "—False Statements", "--Cooperation", "--Compliance with Award Agreement" and "--Notice Period."

- **Confidentiality & Non-Solicitation**

  During your employment by the Firm and for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm's then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual's employment terminated because his or her job was eliminated, or the individual's employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm's confidential or proprietary information.

  These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm's policies.

  You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm's business or its customers except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. "Confidential Information" includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm's private and internal information through telephonic, intranet or internet means.

  For Employees with a Work Location in California, you agree that the restrictions in the first paragraph of this "Confidentiality and Non-Solicitation" provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use "Confidential Information" in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

Nothing in this award precludes you from reporting to the Firm's management or directors, the government, a regulator, a self-regulatory agency, your attorneys, or a court conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**
  You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of the law or from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**
  You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate, and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**
  You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third party who might be affected thereby, including your prospective employer.

- **Notice Period**
  If you are subject to a notice period or become subject to a notice period after the Grant Date, whether by contract or by policy, that requires you to provide advance written notice of your intention to terminate your employment ("Notice Period"), then as consideration for this award and continued employment, you will provide the Firm with the necessary advance written notice that applies to you, as specified by such contract or policy.

  After receipt of your notice, the Firm may choose to have you continue to provide services during the applicable Notice Period or may place you on a paid leave for all or part of the applicable Notice Period. During the Notice Period, you shall continue to devote your full time and loyalty to the Firm by providing services in a cooperative and professional manner and not perform any services for any other employer and shall receive your base salary and certain benefits until your employment terminates. You and the Firm may mutually agree to waive or modify the length of the Notice Period.

  Regardless of whether a Notice Period applies to you, you must comply with the 180-day advance notice period described under the subsection captioned "--Full Career Eligibility" in the event you wish to terminate employment under that same subsection.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**
  In addition to the cancellation provisions described under the sections captioned "Protection-Based Vesting", "Bonus Recoupment", "EMEA Malus and Clawback Policy - Identified Staff", and "Termination of Employment", up to 100% of your outstanding RSUs under this award may be cancelled if the Firm in its sole discretion determines that:
  - Any of the following detrimental and risk-related conduct has occurred:

    - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
    - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
    - this award was based on a material misrepresentation by you, or
    - you improperly or with gross negligence failed to identify, raise or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
    - your employment was terminated for Cause (see section captioned "Definitions" below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
- you have failed to comply with any of the advance notice/cooperation requirements or employment restrictions applicable to your termination of employment, or
- you have failed to return the required forms specified under the section captioned "Release/Certification" by the specified deadline, or
- you have violated any of the provisions as set forth above in the section captioned "Your Obligations".

To the extent provided under the subsection captioned "--Amendment" below, JPMorgan Chase reserves the right to suspend vesting of this award and/or distribution of shares under this award, including, without limitation, during any period that JPMorgan Chase is evaluating whether this award is subject to cancellation and/or recovery and/or whether the conditions for distributions of shares under this award are satisfied. JPMorgan Chase is not responsible for any price fluctuations during any period of suspension and, if applicable, suspended units will be reinstated consistent with Plan administration procedures. See also subsection captioned "--No Ownership Rights/Other Limitations".

- **Recovery**

In addition, you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares of Common Stock previously distributed under this award as follows:

- Payment may be required with respect to any shares distributed within the three-year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
    - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment, or
    - you violated any of the provisions as set forth above in the section captioned "Your Obligations", or
    - you violated the employment restrictions set forth in the subsection "--Full Career Eligibility" or "--Government Office" following the termination of your employment.
- In addition, payment may be required with respect to any shares distributed within the one-year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above "Detrimental Conduct, Risk Related and Other Cancellation/Recapture" subsection.

Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm's employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase's rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm's intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under "Bonus Recoupment" and "EMEA Malus and Clawback Policy - Identified Staff".

- **Right to an Injunction**
  You acknowledge that a violation or attempted violation of any of the provisions set forth in "Your Obligations" herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in "Your Obligations"; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award and shares that vest under this award, and (iii) retaining shares that vest under this award until you pay any taxes associated with the award and/or vested shares directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt owed by you under this award without your consent. This restriction on the Firm does not apply to accounts described and authorized in "No Ownership Rights/Other Limitations" described below.

No Ownership Rights/Other Limitations: RSUs do not convey the rights of ownership of Common Stock and do not carry voting rights. No shares of Common Stock will be issued to you until after the RSUs have vested. Shares will be issued in accordance with JPMorgan Chase's procedures for issuing stock. By accepting this award, you authorize the Firm, in its sole discretion, to establish on your behalf a brokerage account in your name with the Firm or book-entry account with our stock plan administrator and/or transfer agent and deliver to that account any vested shares derived from the award. You also acknowledge that should there be a determination that the cancellation provisions of this award apply during the period when the vesting of any outstanding RSUs has been suspended, then you agree that such RSUs may be cancelled in whole or part. (See Sections captioned "Protection-Based Vesting", "Bonus Recoupment", "EMEA Malus and Clawback Policy - Identified Staff", "Termination of Employment" and "Remedies", as well as the subsection captioned "--Amendment" permitting suspension of vesting.)

With respect to any applicable vesting date, JPMorgan Chase may impose for any reason, as of such vesting date for such period as it may specify in its sole discretion, such restrictions on the Common Stock to be issued to you as it may deem appropriate, including, but not limited to, restricting the sale, transfer, pledging, assignment, hedging or encumbrance of such shares of Common Stock. Such restrictions described in the last sentence shall not impact your right to vote or receive dividends with respect to the Common Stock. By accepting this award, you acknowledge that during such specified period should there be a determination that the recovery provisions of this award apply, then you agree that you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares subject to such restrictions (notwithstanding the limitation set forth in the "Right to Set Off" subsection above). (See Sections captioned "Bonus Recoupment" and "Remedies".)

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is "at-will" and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Notwithstanding anything herein to the contrary, if you (i) are subject to taxation under the Code, (ii) are a specified employee as defined in the JPMorgan Chase 2005 Deferred Compensation Plan and (iii) have incurred a separation from service (as defined in that Plan with the exception of death) and if any units/shares under this award represent deferred compensation as defined in Section 409A and such shares are distributable (under the terms of this award) within six months following, and as a result of your separation from service, then those shares will be delivered to you during the first calendar month after the expiration of six full months from date of your separation from service. Further, if your award is not subject to a substantial risk of forfeiture as defined by regulations issued under Section 409A of the Code, then the remainder of each calendar year immediately following each applicable vesting date set forth in your Award Agreement shall be a payment date for purposes of distributing the vested portion of the award.

Change in Outstanding Shares: In the event of any change in the outstanding shares of Common Stock by reason of any stock dividend or split, recapitalization, issuance of a new class of common stock, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to stockholders of Common Stock other than regular cash dividends, the Committee will make an equitable substitution or proportionate adjustment, in the number or kind of shares of Common Stock or other securities issued or reserved for issuance pursuant to the Plan and to any RSUs outstanding under this award for such corporate events.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under "No Ownership Rights/Other Limitations", in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area, or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity: Upon receipt of satisfactory evidence that applicable United States federal, state, local, foreign or supranational ethics or conflict of interest laws or regulations require you to divest your interest in JPMorgan Chase RSUs, the Firm may accelerate the distribution of all or part of your outstanding award effective on or before the required divestiture date; provided that no accelerated distribution shall occur if the Firm determines that such acceleration will violate Section 409A of the Code. Accelerated distribution under this paragraph does not impact the dates as set forth in the "Remedies" section above. The time period for recovery shall be determined by the originally scheduled vesting date or distribution date prior to any acceleration event.

If you have voluntarily terminated your employment and have satisfied the requirements of the section captioned "Government Office Requirements", acceleration shall apply (to extent required) to the percentage of your outstanding award that would continue to vest under that section. In the case of a termination of employment where the award is outstanding as a result of the subsections entitled "--Job Elimination" or "--Full Career Eligibility", then acceleration shall apply, to the extent required, to the full outstanding award.

Notwithstanding accelerated distribution pursuant to the foregoing, you will remain subject to the applicable terms of your Award Agreement as if your award had remained outstanding for the duration of the original vesting period and shares had been distributed as scheduled as of each applicable vesting date, including, but not limited to, repayment obligations set forth in the section captioned "Remedies" and the employment restrictions in the sections captioned "Protection-Based Vesting" and "Government Office Requirements" and the subsection "--Full Career Eligibility".

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm's legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm's By-Laws in the event of such a suit by the Firm.

Non-transferability: Neither this award or any other outstanding awards of RSUs, nor your interests or rights in any such awards, shall be assigned, pledged, transferred, hedged, hypothecated or subject to any lien. An award may be transferred following your death by will, the laws of descent or by a beneficiary designation on file with the Firm.

Outstanding Awards: The Administrative provisions set forth above shall apply to any award of RSUs outstanding as of the date hereof, and such awards are hereby amended.

Definitions

"**Cause**" means a determination by the Firm that your employment terminated as a result of your (i) violation of any law, rule or regulation (including rules of self-regulatory bodies) related to the Firm's business, (ii) indictment or conviction of a felony, (iii) commission of a fraudulent act, (iv) violation of the JPMorgan Chase Code of Conduct or other Firm policies or misconduct related to your duties to the Firm (other than immaterial and inadvertent violations or misconduct), (v) grossly inadequate performance of the duties associated with your position or job function or failure to follow reasonable directives of your manager, or (vi) any act or failure to act that is injurious to the interests of the Firm or its relationship with a customer, client or an employee.

“**Financial Services Company**” means a business enterprise that engages in any of the following services (itself or through an affiliate or subsidiary), regardless of whether such services are the principle strategy or revenue-generating activity:

- commercial or retail banking, including, but not limited to, commercial, institutional and personal trust, custody and/or lending and processing services, internet banking, originating and servicing mortgages, issuing and servicing credit cards, payment servicing or processing or merchant services,
- insurance, including but not limited to, guaranteeing against loss, harm, damage, illness, disability, or death, providing and issuing annuities, acting as principal, agent or broker for purpose of the forgoing,
- financial, investment or economic advisory services, including but not limited to, investment banking services (such as advising on mergers or dispositions, underwriting, dealing in, or making a market in securities or other similar activities), brokerage services, investment management services, asset management services, foreign exchange services, interbank networks, and hedge funds,
- issuing, trading, or selling instruments representing interests in pools of assets or in derivatives instruments,
- financial technology companies, such as those selling blockchain services, or offering or selling financial products/services,
- advising on, or investing in, private equity or real estate funds or ventures, or
- any similar activities that the Director of Human Resources or nominee determines in his or her sole discretion constitute financial services.

- however, subject to disclosure and confirmation of the following, it shall not be considered a “Financial Services Company” for the purpose of this Award Agreement if such family office is solely funded by you and/or your family’s own personal funds without external investors directly or indirectly.

“Firmwide Financial Threshold” means a cumulative return on tangible common equity for calendar years 2025, 2026 and 2027 of not less than 15%. Cumulative return on tangible common equity means (i) the sum of the Firm’s reported net income for all three calendar years, divided by (ii) reported year-end tangible equity averaged over the three years.

“Government Office” means (i) a full-time position in an elected or appointed office in local, state, or federal government (including equivalent positions outside the U.S. or in a supranational organization), not reasonably anticipated to be a full-career position, or (ii) conducting a bona fide full-time campaign for such an elective public office after formally filing for candidacy, where it is customary and reasonably necessary to campaign full-time for the office.

“Line of Business” means a business unit of the Firm (or one or more business units designated below under the definition “Line of Business Financial Threshold” of the Commercial & Investment Bank). All Corporate Functions (including the functions of the Chief Investment Office) are considered a single Line of Business.

“Line of Business Financial Threshold” means the financial threshold set forth below for the following Lines of Business based on the Firm’s management reporting system:

Asset & Wealth Management	Annual negative pre-tax pre-provision income[1]
Card and Auto	Annual negative pre-tax pre-loan loss reserve income[2]
Commercial & Investment Bank	Annual negative pre-tax pre-provision income[1] for CIB overall or annual negative allocated product revenues (excluding XVA) for: • Fixed Income • Equities • Securities Services • Global Investment Banking • Payments
Consumer Banking, J.P. Morgan Wealth Management and Business Banking	Annual negative pre-tax pre-loan loss reserve income[2]
Corporate Functions (including Chief Investment Office)	Annual negative pre-tax pre-provision income[1] at the Firm level
Home Lending	Annual negative pre-tax pre-loan loss reserve income[2]
[1]Pre-tax pre-provision income means Revenue less Expenses [2]Pre-tax pre-loan loss reserve income means Revenue less (Expenses plus Net Charge-offs)	

“Not-for-Profit Organization” means an entity exempt from tax under state law and under Section 501(c)(3) of the Code. Section 501(c)(3) only includes entities organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals. Not-for-Profit Organization shall also mean entities outside the United States exempt from local and national tax laws because they are organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals.

"Recognized Service" means the period of service as an employee set forth in the Firm's applicable service-related policies.

Government Office Requirements

You may be eligible to continue vesting in all or part of your award if you voluntarily resign to accept a Government Office (as defined above) or to become a candidate for an elective Government Office.

Full Career Eligibility:
"Government Office Requirements" does not apply to you if you satisfy the subsection captioned "--Full Career Eligibility" as of the date that you voluntarily terminate your employment with the Firm.

Eligibility:
Eligibility for continued vesting is conditioned on your providing the Firm:

- At least 60 days' advance written notice of your intention to resign to accept or pursue a Government Office (see section captioned "Definitions"), during which period you must perform in a cooperative and professional manner services requested by the Firm and not provide services for any other employer. The Firm may elect to shorten this notice period at the Firm's sole discretion.
- Confirmation, in a form of satisfactory to the firm, that vesting in this award pursuant to this provision would not violate any applicable law, regulation or rule.
- Documentation in a form satisfactory to the Firm that your resignation is for the purpose of accepting a Government Office or becoming a candidate for a Government Office. (See Section captioned "Definitions".)

Portion of Your Awards Subject to Continued Vesting:
Subject to the conditions below, the percentage of your outstanding awards that will continue to vest in accordance with this award's original schedule will be based on your years of continuous service completed with the Firm immediately preceding your termination date, as follows:

- 50% if you have at least 3 but less than 4 years of continuous service,
- 75% if you have at least 4 but less than 5 years of continuous service, or
- 100% if you have 5 or more years of continuous service.

The portion of each award subject to continued vesting above is referred to as the "CV Award" and the portion not subject to continued vesting will be cancelled on the date your employment terminates.

Conditions for Continued Vesting of Awards:

- You must remain in a non-elective Government Office for two or more years after your employment with the Firm terminates to receive in full your CV Award; provided that if your non-elective Government Office is for a period less than two years, you will be entitled to retain any portion of the CV Award with a vesting date during your period of Government Service; or
- In the case of resignation from the Firm to campaign for an elective Government Office, your name must be on the primary or final public ballot for the election. (If you are not elected, see below for employment restrictions.)

Satisfaction of Conditions:
If your service in a Government Office ends two years or more after your employment with the Firm terminates, or in the case of resignation from the Firm to campaign for a Government Office, your name is on the primary or final public ballot for the election and you are not elected, any CV Awards then outstanding and any such awards that would have then been outstanding but for an accelerated distribution of shares (as described in the subsection captioned "Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity") will be subject for the remainder of the applicable vesting period to the same terms and conditions of this Award Agreement, including employment restrictions during the vesting period, as if you had resigned from the Firm having met the requirements for Full Career Eligibility.

Failure to Satisfy Conditions:

If you do not satisfy the above “Conditions for Continued Vesting of Awards”, any outstanding RSUs under each CV Award will be cancelled. You also will be required to repay the Fair Market Value of the number of shares (before tax and other withholdings) of Common Stock distributed to you that would have been outstanding as RSUs on the date you failed to satisfy the “Condition for Continued Vesting of Awards” but for their accelerated distribution (as described in the subsection captioned “Accelerated Distribution for Ethics or Conflict Reasons Resulting from Employment by a Government Entity”). Fair Market Value for this purpose will be determined as the date that the shares were distributed.

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF ________________, 20___

PERFORMANCE SHARE UNIT AWARD

OPERATING COMMITTEE

(Protection-Based Vesting Provisions)

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of ___________________ ("Grant Date") awarding performance share units ("PSUs") pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan ("Plan"). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement and Plan. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in "Definitions" below or elsewhere in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as "JPMorgan Chase", and together with its subsidiaries as the "Firm".

Form and Purpose of Award

Each PSU represents a non-transferable right to receive one share of Common Stock as of the vesting date as set forth in your Award Agreement.
The purpose of this award is to further emphasize sustained long-term performance and to align your interests with those of the Firm and its shareholders.

Number to Vest on the Vesting Date

Subject to any cancellation in whole or part of your award pursuant to these terms and conditions:

Performance calculation: On the vesting date set forth in your Award Agreement, you will vest in a number of PSUs derived by multiplying the Target Award Number by the Award Payout Percentage determined using the Performance Table. See sections captioned "Calculation of Performance Ranking" and "Definitions".

You will also vest in additional shares of Common Stock as calculated under the section captioned, "Reinvested Dividend Equivalent Share Units". Delivery of vested shares to your account will be made not later than the date specified in the last sentence of the subsection captioned "Section 409A Compliance".

Reinvested Dividend Equivalent Share Units

If dividends are paid on Common Stock during the Vesting Period while the award is outstanding, you will receive on the vesting date additional units representing shares of Common Stock as calculated in this section. The number, if any, will be based on the dividends that would have been paid during the Vesting Period as of each dividend payment date on the actual number of shares of Common Stock distributable to you resulting from the vesting of the PSUs, if any, and treated as reinvested in additional shares of Common Stock on each dividend payment based on the Fair Market Value of one share of Common Stock on each dividend payment date ("Reinvested Dividend Equivalent Share Units").

Holding Requirement

As of the vesting date set forth in your Award Agreement, you shall be entitled to be issued a number of shares of the Common Stock of JPMorgan Chase equal to the number of PSUs, if any, plus any additional Reinvested Dividend Equivalent Share Units, vesting on such date, less the number withheld to satisfy tax withholding obligations. The net number of shares issued to you will be held in an account in your name with restrictions preventing you from transferring, assigning, hedging, selling, pledging, or otherwise encumbering such shares for a two-year period commencing as of the vesting date and ending as of the second anniversary of the vesting date. Such restrictions shall only lapse, prior to the expiration of the two-year holding period, in the event of your death or for an accelerated distribution for ethics or conflict reasons. See section captioned, "Death" and subsection captioned, "Accelerated Distribution for Ethics or Conflict Reasons Resulting from Employment by a Government Entity".

Calculation of Performance Ranking

For purposes of the Performance Ranking, the ranking of the Firm and of each Performance Company for the Performance Period shall be determined and calculated by the Calculation Agent, using the definitions of "Average Tangible Common Equity" (if otherwise applicable), "Calculated PSUs", "Firm Reported ROTCE", "Performance Table" and "ROTCE" as set forth in the "Definitions" section of these terms and conditions. See section captioned "Definitions". Except for Firm's externally published Reported ROTCE, calculations will be expressed as a decimal to the second place (i.e. xx.yy%), rounded to the nearest hundredth. See section captioned, "Definitions--Performance Table" in the event of a tie. All performance-based calculations as set forth herein are binding and conclusive on you and your successors.

Capital Ratio Performance Threshold

Unvested PSUs are subject to reduction if the Firm's Common Equity Tier 1 (CET1) capital ratio at any year end falls below a predetermined threshold of _______%.

If the Firm's CET1 capital ratio at any year end during the Performance Period is below this predetermined threshold, up to one-third of the Target Award Number of PSUs will be subject to downward adjustment by the Compensation & Management Development Committee of the Board of Directors of JPMorgan Chase ("Committee") for each such year.

Vesting Period

The period from the Grant Date to the vesting date is the "Vesting Period". (See "Administrative Provision--Amendment" pursuant to which the Firm may extend the vesting period and "No Ownership Rights/Other Limitations" pursuant to which the Firm may place restrictions on delivered shares of Common Stock following the vesting date and section captioned, "Holding Period" above.)

Protection-Based Vesting

This award is intended and expected to vest on the vesting date, provided that you are continuously employed by the Firm through such vesting date, or you meet the requirements for continued vesting described under the subsections "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability". However, vesting and the number of PSUs that will vest are subject to these terms and conditions (including, but not limited to, sections captioned "Number to Vest on the Vesting Date", "Capital Ratio Performance Threshold", "Remedies", and the following protection-based vesting provision).

Up to a total of fifty percent of your award (including any associated Reinvested Dividend Equivalent Share Units) that would otherwise be distributable to you on the vesting date ("At Risk PSUs") may be cancelled if the Chief Executive Officer of JPMorgan Chase ("CEO") determines in his or her sole discretion that cancellation of all or portion of the At Risk PSUs is appropriate in light of any one or a combination of the following factors:

- Your performance in relation to the priorities for your position, or the Firm's performance in relation to the priorities for which you share responsibility as a member of the Operating Committee, have been unsatisfactory for a sustained period of time. Among the factors the CEO may consider in assessing performance are: net income, total net revenue, earnings per share and capital ratios of the Firm, both on an absolute basis and, as appropriate, relative to peer firms.
- For any calendar year ending during the vesting period, JPMorgan Chase's annual pre-tax pre-provision income at the Firm level is negative.
- RSU awards granted to participants in a Line of Business for which you exercise, or during the vesting period exercised, direct or indirect responsibility, were in whole or in part cancelled because the Line of Business did not meet its annual Line of Business Financial Threshold.
- The Firm does not meet the Firmwide Financial Threshold.

For avoidance of doubt, cancellation of the At Risk PSUs, in whole or part, for one or more of the above factors may occur prior to the end of the Performance Period and the maximum number of At Risk PSUs subject to cancellation prior to the end of the Performance Period will be up to fifty percent of the Target Award Number.

In the event that your employment terminates due to "Job Elimination", "Full Career Eligibility", "Government Office" or "Disability" thereby entitling you to continued vesting in your award, (or potentially acceleration due to satisfaction of the Government Office Requirements), the cancellation circumstances described above will continue to apply.

Any determination above with respect to protection-based vesting provisions is subject to ratification by the Committee. In the case of an award to any current or former CEO, all such determinations shall be made by the Committee and ratified by the Board.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 20XX and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled "Other Matters":

https://about.jpmorganchase.com/about/governance/corporate-governance-principles

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

Recovery of Erroneously Awarded Incentive-Based Compensation

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation (or successor policy) as in effect from time to time as it applies to this award. You can access this policy as currently in effect by clicking the following link to the Firmwide Policy & Standard Portal, or alternatively typing go/dogma then searching for the Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation.

Firmwide Policy & Standard Portal > Recovery of Erroneously Awarded Incentive-Based Compensation Policy

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation (or successor policy).

Termination of Employment

Except as explicitly set forth below under the subsections captioned "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability" or under the section captioned "Death", this award (for avoidance of doubt, including any associated Reinvested Dividend Equivalent Share Units) will be cancelled in full effective on the date your employment with the Firm terminates for any reason.

Subject to these terms and conditions (including, but not limited to, sections captioned "Protection-Based Vesting", "Number to Vest on the Vesting Date", "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation", "Your Obligations" and "Remedies") you will be eligible to continue to vest (as you otherwise would vest if you were still employed by JPMorgan Chase) with respect to your award in accordance with its terms and conditions following the termination of your employment if one of the following circumstances applies to you:

- **Job Elimination**

  In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

  - the Firm terminated your employment because your job was eliminated, and
  - after you are notified that your job will be eliminated, you provided such services as requested by the Firm in a cooperative and professional manner, and
  - you satisfied the Release/Certification Requirements set forth below.

- **Full Career Eligibility**

  In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

  - (i) you voluntarily terminated your employment with the Firm, and (ii) had completed at least five years of Recognized Service (as defined below) on your date of termination with at least three years of continuous service with the Firm immediately preceding your termination date or if you are over 60 years of age with at least three years of continuous service with the Firm immediately preceding your termination date, and
  - the sum of your age and Recognized Service (as defined below) on your date of termination equaled or exceeded 60 and
  - you provided at least 180 days advance written notice to the Firm of your intention to voluntarily terminate your employment under this provision, during which notice period you provided such services as requested by the Firm in a cooperative and professional manner and you did not perform any services for any other employer, and
  - continued vesting shall be appropriate, which determination shall be made prior to your termination and will be based on your performance and conduct (before and after providing notice), and
  - you satisfied the Release/Certification Requirements set forth below, and

- for 36 months from the date of grant of this award you do not either perform services in any capacity (such as an employee, contractor, consultant, advisor, or self-employed individual, whether paid or unpaid) for a Financial Services Company (as defined below) or work in your profession (whether or not for a Financial Services Company); provided that you may do one or more of the following:
    - work in a government, education or Not-for-Profit Organization (as defined below),
    - subject to disclosure and confirmation of the following, serve on the board of directors of a public or non-public company that is not a Financial Services Company.

After receipt of such advance written notice, the Firm may choose to have you continue to provide services during such 180-day period as a condition to continued vesting or shorten the length of the 180-day period at the Firm's sole discretion, but to a date no earlier than the date you would otherwise meet the age and service requirements.

Additional advance notice requirements may apply for employees subject to notice period policies (see "Notice Period" below.)

- **Government Office**

  In the event that you voluntarily terminate your employment with the Firm to accept a Government Office or become a candidate for an elective Government Office, as described at the end of these terms and conditions under the section captioned "Government Office Requirements". See also definition of Government Office in the section captioned "Definitions".

- **Disability**

  In the event that

  - your employment with the Firm terminates because (i) you are unable to return to work while you are receiving benefits under the JPMorgan Chase Long Term Disability Plan, or for non-U.S. employees, under the equivalent JPMorgan Chase-sponsored local country plan (in either case, "LTD Plan"), or (ii) if you are not covered by a LTD Plan, you are unable to return to work due to a long-term disability that would qualify for benefits under the applicable LTD Plan, as determined by the Firm or a third party designated by the Firm; provided that you (x) request in writing continued vesting due to such disability within 30 days of the date your employment terminates, and (y) provide any requested supporting documentation and (z) receive the Firm's written consent to such treatment, and
  - you satisfied the Release/Certification Requirements set forth below.

Release/Certification

To qualify for continued vesting after termination of your employment under any of the foregoing circumstances:

- you must timely execute and deliver a release of claims in favor of the Firm, having such form and terms as the Firm shall specify,
- with respect to "Full Career Eligibility", prior to the termination of your employment, you must confirm with management that you meet the eligibility criteria (including providing at least 180 days advance written notification), advise that you are seeking to be treated as an individual eligible for Full Career Eligibility, and receive written consent to such continued vesting,
- with respect to "Full Career Eligibility" and "Government Office", it is your responsibility to (i) notify the Firm within 15 days after the date you are no longer in compliance with the employment restrictions (as described herein) or (ii) take the appropriate steps to certify to the Firm prior to the vesting date while the employment restrictions are outstanding, on the authorized form of, and by the due date set by, the Firm that you have complied with the employment restrictions applicable to you (as described herein) from your date of termination of employment through the applicable vesting date,
- with respect to "Disability", you must satisfy the notice and documentation described above and receive written consent to such continued vesting, and
- in all cases, otherwise complied with all other terms of the Award Agreement. (See section captioned "Your Obligations" below.)

Death

If you die while you are eligible to vest in this award, your designated beneficiary on file with the Firm's Stock Administration Department, or if no beneficiary has been designated or survives you or if beneficiary designation is not recognized locally, then to your estate unless otherwise required by local legislation, may be entitled to receive a distribution of a number of shares of Common Stock associated with your award. The Award Payout Percentage in the case of death is based on the "Number to Vest on the Vesting Date" calculation described above using the average performance of all completed calendar years, multiplied by one-third of the Target Award Number of PSUs for each completed calendar year in the Performance Period, and using the Award Payout Percentage equal to 100 percent for any remaining calendar years in the Performance Period.

In addition, your beneficiary or your estate shall receive additional shares of Common Stock, i.e., Reinvested Dividend Equivalent Share Units, as set forth in the section captioned, "Reinvested Dividend Equivalent Share Units" but based on dividend equivalents up to the date of your death.

Any shares will be distributed no later than the end of the calendar year immediately following the calendar year which contains your date of death; however, our administrative practice is to register such shares in the name of your beneficiary or estate within 60 days of the Firm's receipt of any required documentation.

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned "--Confidentiality & Non-Solicitation", "--False Statements", "--Cooperation", "--Compliance with Award Agreement", and "--Notice Period".

- **Confidentiality & Non-Solicitation**
  During your employment by the Firm and for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm's then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual's employment terminated because his or her job was eliminated, or the individual's employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm's confidential or proprietary information.

  These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm's policies.

  You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm's business or its customers except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. "Confidential Information" includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm's private and internal information through telephonic, intranet or internet means.

  For Employees with a Work location in California, you agree that the restrictions in the first paragraph of the "Confidentiality and Non-Solicitation" provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use "Confidential Information" in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

  Nothing in this award precludes you from reporting to the Firm's management or directors, the government, a regulator, a self-regulatory agency, your attorneys, or a court, conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**
  You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of the law or from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**
  You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate, and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**
  You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third party who might be affected thereby, including your prospective employer.

- **Notice Period**
  If you are subject to a notice period or become subject to a notice period after the Grant Date, whether by contract or by policy, that requires you to provide advance written notice of your intention to terminate your employment ("Notice Period"), then as consideration for this award and continued employment, you will provide the Firm with the necessary advance written notice that applies to you, as specified by such contract or policy.

  After receipt of your notice, the Firm may choose to have you continue to provide services during the applicable Notice Period or may place you on a paid leave for all or part of the applicable Notice Period. During the Notice Period, you shall continue to devote your full time and loyalty to the Firm by providing services in a cooperative and professional manner and not perform any services for any other employer and shall receive your base salary and certain benefits until your employment terminates. You and the Firm may mutually agree to waive or modify the length of the Notice Period.

  Regardless of whether a Notice Period applies to you, you must comply with the 180-day advance notice period described under the subsection captioned "-- Full Career Eligibility" in the event you wish to terminate employment under that same subsection.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**

  In addition to the cancellation provisions described under the sections captioned "Protection-Based Vesting", "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation" and "Termination of Employment", up to 100% of your outstanding PSUs under this award (for the avoidance of doubt, including any associated Reinvested Dividend Equivalent Share Units as well as the Calculated PSUs) may be cancelled if the Firm in its sole discretion determines that:

  - Any of the following detrimental and risk-related conduct has occurred:
    - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
    - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
    - this award was based on a material misrepresentation by you, or

    - you improperly or with gross negligence failed to identify, raise or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
    - your employment was terminated for Cause (see section captioned "Definitions" below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
- you have failed to comply with any of the advance notice/cooperation requirements or employment restrictions applicable to your termination of employment, or
- you have failed to return the required forms specified under the section captioned "Release/Certification" by the specified deadline, or
- you have violated any of the provisions as set forth above in the section captioned "Your Obligations".

To the extent provided under the subsection captioned "--Amendment" below, JPMorgan Chase reserves the right to suspend vesting of this award and/or distribution of shares under this award, including, without limitation, during any period that JPMorgan Chase is evaluating whether this award is subject to cancellation and/or recovery and/or whether the conditions for distributions of shares under this award are satisfied. The Firm is not responsible for any price fluctuations during any period of suspension and, if applicable, suspended units will be reinstated consistent with Plan administration procedures. See also "Administrative Provisions—No Ownership Rights/Other Limitations".

In addition, you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date or acceleration date) of the gross number of shares previously distributed, including vested shares subject to the Holding Requirements, under this award as follows:
- Payment may be required with respect to any shares of Common Stock distributed within the three-year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
  - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment;
  - you violated any of the provisions as set forth above in the section captioned "Your Obligations;" or
  - you violated the employment restrictions set forth in the subsection "Full Career Eligibility" or "Government Office" following the termination of your employment.
- In addition, payment may be required with respect to any shares distributed within the one year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above "Detrimental Conduct, Risk Related and Other Cancellation/Recapture" subsection.

Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm's employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase's rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm's intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under "Bonus Recoupment" or "Recovery of Erroneously Awarded Incentive-Based Compensation".

- **Right to an Injunction**

You acknowledge that a violation or attempted violation of any of the provisions set forth in "Your Obligations" herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in "Your Obligations"; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award and shares that vest under this award, and (iii) retaining shares that vest under this award until you pay any taxes associated with the award and vested shares directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt or obligation owed by you under this award without your consent. This restriction on the Firm does not apply to accounts described and authorized in "No Ownership Rights/Other Limitations" described below.

No Ownership Rights/Other Limitations: PSUs do not convey the rights of ownership of Common Stock and do not carry voting rights. No shares of Common Stock will be issued to you until after the number of PSUs have been determined, if any, and have vested. Shares will be issued in accordance with JPMorgan Chase's procedures for issuing stock. By accepting this award, you authorize the Firm, in its sole discretion, to establish on your behalf a brokerage account in your name with the Firm or book-entry account with our stock plan administrator and/or transfer agent and deliver to that account any vested shares derived from the award. You also acknowledge that should there be a determination that the cancellation provisions of this award apply during the period when the vesting of any outstanding PSUs has been suspended, then you agree that such PSUs may be cancelled in whole or part. (See Sections captioned "Protection-Based Vesting", "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation", "Termination of Employment" and "Remedies", as well as the subsection captioned "—Amendment" permitting suspension of vesting.)

With respect to any applicable vesting date, JPMorgan Chase may impose for any reason, as of such vesting date for such period as it may specify in its sole discretion, such restrictions on the Common Stock to be issued to you as it may deem appropriate, including, but not limited to, restricting the sale, transfer, pledging, assignment, hedging or encumbrance of such shares of Common Stock. Such restrictions described in the last sentence shall not impact your right to vote or receive dividends with respect to the Common Stock. By accepting this award, you acknowledge that during such specified period should there be a determination that the recovery provisions of this award apply, then you agree that you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares subject to such restrictions (notwithstanding the limitation set forth in the "Right to Set Off" subsection above). (See sections captioned "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation" and "Remedies".)

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is "at-will" and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Notwithstanding anything herein to the contrary, if you (i) are subject to taxation under the Code, (ii) are a specified employee as defined in the JPMorgan Chase 2005 Deferred Compensation Plan and (iii) have incurred a separation from service (as defined in that Plan with the exception of death) and if any units/shares under this award represent deferred compensation as defined in Section 409A and such shares are distributable (under the terms of this award) within six months following, and as a result of your separation from service, then those shares will be delivered during the first calendar month after the expiration of six full months from date of your separation from service. Further, if your award is not subject to a substantial risk of forfeiture as defined by regulations issued under Section 409A of the Code, then the remainder of each calendar year immediately following the vesting date set forth in your Award Agreement shall be a payment date for purposes of distributing the vested portion of the award.

Change in Outstanding Shares: In the event of any change in the outstanding shares of Common Stock by reason of any stock dividend or split, recapitalization, issuance of a new class of common stock, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to stockholders of Common Stock other than regular cash dividends, the Committee will make an equitable substitution or proportionate adjustment, in the number or kind of shares of Common Stock or other securities issued or reserved for issuance pursuant to the Plan and to any PSUs outstanding under this award for such corporate events.

Other Equitable Adjustments: The Committee may make adjustments (up or down) to the award as it deems to be equitable, to maintain the intended economics of the award in light of changed circumstances, which may include unusual or non-recurring events affecting the Firm (or the Performance Companies) or its financial statements in each case resulting from changes in accounting methods, practices or policies, changes in capital structure by reason of legal or regulatory requirements and such other changed circumstances, as the Committee may deem appropriate.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under "No Ownership Rights/Other Limitations", in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area, or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity: Upon receipt of satisfactory evidence that applicable United States federal, state, local, foreign or supranational ethics or conflict of interest laws or regulations require you to divest your interest in JPMorgan Chase PSUs, the Firm may accelerate the distribution of all or part of your outstanding award, including Reinvested Dividend Equivalent Share Units, effective on or before the required divestiture date and waive the Holding Requirement; provided that no accelerated distribution shall occur if the Firm determines that such acceleration will violate Section 409A of the Code. Accelerated distribution under this paragraph does not impact the dates as set forth in the "Remedies" section above. The time period for recovery shall be determined by the originally scheduled vesting date or distribution date prior to any acceleration event.

If you have voluntarily terminated your employment and have satisfied the requirements of the section captioned "Government Office Requirements", acceleration shall apply (to extent required) to the percentage of your outstanding award that would continue to vest under that section. In the case of a termination of employment where the award is outstanding as a result of the subsections entitled "Job Elimination" or "Full Career Eligibility", then acceleration shall apply, to the extent required, to the full outstanding award. Subject to the two foregoing sections, the number of shares of

Common Stock to be received on acceleration shall be determined using the methodology set forth under the section captioned "Death".

To the extent you have vested shares under this award subject to the Holding Requirement and become subject to divestiture requirement as forth herein, the Firm may waive the holding period to the extent required.

Notwithstanding an accelerated distribution or waiver of the Holding Requirement pursuant to the foregoing, you will remain subject to the applicable terms of your Award Agreement as if your award had remained outstanding for the duration of the original vesting period and shares had been distributed as scheduled as of the vesting date, including, but not limited to, repayment obligations set forth in the section captioned "Remedies" and the employment restrictions in the sections captioned "Protection-Based Vesting" and "Government Office Requirements" and the subsection "Full Career Eligibility".

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm's legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm's By-Laws in the event of such a suit by the Firm.

Non-transferability: Neither this award or any other outstanding awards of restricted stock units or of performance-based share units, nor your interests or rights in any such awards, shall be assigned, pledged, transferred, hedged, hypothecated or subject to any lien. An award may be transferred following your death by will, the laws of descent or by a beneficiary designation on file with the Firm.

Definitions

"Average Tangible Common Equity" means annual average common stockholders' equity less annual average goodwill and annual average identifiable intangible assets. Annual averages of the components of Average Tangible Common Equity will be calculated using quarterly balances as reported in publicly available financial disclosures. In the event that quarterly balances are not available, annual year end balances will be used. This calculation is used solely for purposes of the Performance Ranking.

"Award Payout Percentage" means the applicable percentage specified in the Performance Table.

"Calculated PSUs" means the number of PSUs determined by multiplying the Target Award Number (after giving effect to any cancellation thereof, in whole or in part) by the Award Payout Percentage corresponding to the Firm's Performance Ranking based on the three-year average performance for the Performance Period (both percentage and ranking, as set forth in the Performance Table); provided that if the average of the Firm's Reported ROTCE, expressed as a decimal rounded to the nearest hundredth, for the Performance Period either equals or exceeds ______% or is less than ______%, ______ percent or _______, respectively as the case may be, shall be substituted for the Performance Period's Award Payout Percentage in calculating the number of PSUs to distribute. For avoidance of doubt, any cancellation of this award (in whole or in part) during the Performance Period will reduce the Target Award Number.

"Calculation Agent" means a third party entity not owned or controlled by the Firm, such as an accounting or consulting firm, retained from time to time by the Director of Human Resources or his/her delegate.

"Cause" means a determination by the Firm that your employment terminated as a result of your (i) violation of any law, rule or regulation (including rules of self-regulatory bodies) related to the Firm's business, (ii) indictment or conviction of a felony, (iii) commission of a fraudulent act, (iv) violation of the JPMorgan Chase Code of Conduct or other Firm policies or misconduct related to your duties to the Firm (other than immaterial and inadvertent violations or misconduct), (v) grossly inadequate performance of the duties associated with your position or job function or failure to follow reasonable directives of your manager, or (vi) any act or failure to act that is injurious to the interests of the Firm or its relationship with a customer, client or an employee.

"Financial Services Company" means a business enterprise that engages in any of the following services (itself or through an affiliate or subsidiary), regardless of whether such services are the principle strategy or revenue-generating activity:

- commercial or retail banking, including, but not limited to, commercial, institutional and personal trust, custody and/or lending and processing services, internet banking, originating and servicing mortgages, issuing and servicing credit cards, payment servicing or processing or merchant services,
- insurance, including but not limited to, guaranteeing against loss, harm, damage, illness, disability, or death, providing and issuing annuities, acting as principal, agent or broker for purpose of the forgoing,
- financial, investment or economic advisory services, including but not limited to, investment banking services (such as advising on mergers or dispositions, underwriting, dealing in, or making a market in securities or other similar activities), brokerage services, investment management services, asset management services, foreign exchange services, interbank networks and hedge funds,
- issuing, trading or selling instruments representing interests in pools of assets or in derivatives instruments,
- financial technology companies, such as those selling blockchain services, or offering or selling financial products/ services,
- advising on, or investing in, private equity or real estate funds or ventures, or
- any similar activities that the Director of Human Resources or nominee determines in his or her sole discretion constitute financial services,
- however, subject to disclosure and confirmation of the following, it shall not be considered a "Financial Services Company" for the purpose of this Award Agreement if such family office is solely funded by you and/or your family's own personal funds without external investors directly or indirectly.

"Firm Reported ROTCE" means the Firm's percentage return on tangible common equity for each year in the Performance Period.

"Firmwide Financial Threshold" means a cumulative return on tangible common equity for calendar years 20XX, 20XX and 20XX of not less than _______%. Cumulative return on tangible common equity means (i) the sum of the Firm's reported net income for all three calendar years, divided by (ii) reported year-end tangible equity averaged over the three years.

"Government Office" means (i) a full-time position in an elected or appointed office in local, state, or federal government (including equivalent positions outside the U.S. or in a supranational organization), not reasonably anticipated to be a full-career position, or (ii) conducting a bona fide full-time campaign for such an elective public office after formally filing for candidacy, where it is customary and reasonably necessary to campaign full-time for the office.

"Line of Business" means a business unit of the Firm (or one or more business units designated below under the definition "Line of Business Financial Threshold" of the Commercial & Investment Bank). All Corporate Functions (including the functions of the Chief Investment Office) are considered a single Line of Business.

"Line of Business Financial Threshold" means the financial threshold set forth below: for the following Lines of Business based on the Firm's management reporting system:

Asset & Wealth Management	Annual negative pre-tax pre-provision income[1]
Card and Auto	Annual negative pre-tax pre-loan loss reserve income[2]
Commercial & Investment Bank	Annual negative pre-tax pre-provision income[1] for CIB overall or annual negative allocated product revenues (excluding XVA) for: • Fixed Income • Equities • Securities Services • Global Investment Banking • Payments
Consumer Banking, J.P. Morgan Wealth Management and Business Banking	Annual negative pre-tax pre-loan loss reserve income[2]
Corporate Functions (including Chief Investment Office)	Annual negative pre-tax pre-provision income[1] at the Firm level
Home Lending	Annual negative pre-tax pre-loan loss reserve income[2]
[1]Pre-tax pre-provision income means Revenue less Expenses [2]Pre-tax pre-loan loss reserve income means Revenue less (Expenses plus Net Charge-offs)	

"Not-for-Profit Organization" means an entity exempt from tax under state law and under Section 501(c)(3) of the Code. Section 501(c)(3) only includes entities organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals. Not-for-Profit Organization shall also mean entities outside the United States exempt from local and national tax laws because they are organized and operated exclusively for purposes identical to those applicable to Section 501(c)(3) organization.

"Performance Companies" mean the following institutions which have business activities that overlap with a significant portion of the Firm's revenue mix: ____, and ________________.

If, during the Performance Period, one or more Performance Companies shall merge, engage in a spin-off or otherwise experience a material change in its revenue mix or business activities or its existence or its primary businesses shall terminate or cease due to receivership, bankruptcy, sale, or otherwise, then the Committee may eliminate such institution from the list of Performance Companies or make such other equitable adjustments, such as adding an acquirer or a new company to the list of Performance Companies, as it deems appropriate, with any such changes having effect for purposes of all calculations hereunder on a prospective basis from the date the applicable change is made.

"Performance Period" means calendar years 20XX, 20XX and 20XX.

"Performance Ranking" means the ranking of the average ROTCE of the Firm as compared to the ranking of the average ROTCE of the Performance Companies as specified in the Performance Table for the Performance Period.

"Performance Table" means the table used in the calculation of PSUs for the Performance Period as set forth below:

Firm Reported ROTCE (average performance)	Award Payout Percentage
>=________%	______%
______% to ______%	Pay by relative ROTCE scale
<______%	_____%

Firm Performance Ranking (average performance)	Award Payout Percentage
#1	_______%
#2	_______%
#3	_______%
#4	_______%
#5	_______%
#6	_______%
#7	_______%
#8	_______%
#9	_______%
#10 +	_______%

If, after the calculation of the Performance Ranking, there is a tie, the tie shall be disregarded for purposes of determining the Award Payout Percentage. For example, in the case of a tie for the fourth ranking between the Firm and a Performance Company, the Firm shall be treated as having satisfied that ranking. In the case of that same tie among Performance Companies, the fourth and fifth rankings will be deemed to have been satisfied.

"Recognized Service" means the period of service as an employee set forth in the Firm's applicable service-related policies.

"ROTCE" means for the Firm and each of the Performance Companies a percentage derived by, for each year in the Performance Period, dividing (i) annual earnings from continuing operations less dividends on preferred stock as set forth in published financial disclosures by (ii) the Average Tangible Common Equity for the year**.** If, prior to the end of the vesting period, the Firm or any Performance Company restates its published financial statements for any year in the Performance Period, ROTCE for that year shall be recalculated for the Firm or Performance Company with the Performance Ranking adjusted, if necessary. This calculation is used solely for purposes of the Performance Ranking.

"Target Award Number" means the number of PSUs designated as such in the Award Agreement.

Government Office Requirements

You may be eligible to continue vesting in all or part of your award if you voluntarily resign to accept a Government Office (as defined above) or to become a candidate for an elective Government Office.

Full Career Eligibility:
"Government Office Requirements" does not apply to you if you satisfy the subsection captioned "--Full Career Eligibility" as of the date that you voluntarily terminate your employment with the Firm.

Eligibility:
Eligibility for continued vesting is conditioned on your providing the Firm:

- At least 60 days' advance written notice of your intention to resign to accept or pursue a Government Office (see section captioned "Definitions"), during which period you must perform in a cooperative and professional manner services requested by the Firm and not provide services for any other employer. The Firm may elect to shorten this notice period at the Firm's sole discretion.
- Confirmation, in a form satisfactory to the Firm, that vesting in this award pursuant to this provision would not violate any applicable law, regulation or rule.
- Documentation in a form satisfactory to the Firm that your resignation is for the purpose of accepting a Government Office or becoming a candidate for a Government Office. (See section captioned "Definitions".)

Portion of Your Award Subject to Continued Vesting:
Subject to the conditions below, the percentage of this award that will continue to vest in accordance with this award's original schedule will be based on your years of continuous service completed with the Firm immediately preceding your termination date, as follows:

- 50% if you have at least 3 but less than 4 years of continuous service,
- 75% if you have at least 4 but less than 5 years of continuous service, or
- 100% if you have 5 or more years of continuous service.

The portion of this award subject to continued vesting above is referred to as the “CV Award” and the portion not subject to continued vesting will be cancelled as of the date your employment terminates.

Conditions for Continued Vesting of Award:

- You must remain in a non-elective Government Office for two or more years after your employment with the Firm terminates to be eligible to receive the CV Award; provided that if your non-elective Government Office is for a period less than two years, you will be eligible to receive the CV Award if it has a vesting date during your period of Government Service; or
- In the case of resignation from the Firm to campaign for an elective Government Office, your name must be on the primary or final public ballot for the election. (If you are not elected, see below for employment restrictions.)

For avoidance of doubt, the performance criteria and protection-based vesting set forth in these terms and conditions continue to apply to a CV Award.

Satisfaction of Conditions:

If your service in a Government Office ends two years or more after your employment with the Firm terminates, or in the case of resignation from the Firm to campaign for a Government Office, your name is on the primary or final public ballot for the election and you are not elected, any CV Awards then outstanding and any such awards that would have then been outstanding but for an accelerated distribution of shares (as described in the subsection captioned “--Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”) will be subject for the remainder of the applicable vesting period to the same terms and conditions of this Award Agreement, including employment restrictions during the vesting period, as if you had resigned from the Firm having met the requirements for Full Career Eligibility.

Failure to Satisfy Conditions:

If you do not satisfy the above “Conditions for Continued Vesting of Awards”, any outstanding PSUs under the CV Award will be cancelled. You also will be required to repay the Fair Market Value of the number of shares (before tax and other withholdings) of Common Stock distributed to you that would have been outstanding as PSUs on the date you failed to satisfy the “Conditions for Continued Vesting of Award” but for their accelerated distribution (as described in subsection captioned, “Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”). Fair Market Value for this purpose will be determined as the date that the shares were distributed.

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF ________________, 20___
PERFORMANCE SHARE UNIT AWARD
OPERATING COMMITTEE
(Protection-Based Vesting Provisions)

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of ___________________ ("Grant Date") awarding performance share units ("PSUs") pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan ("Plan"). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement and Plan. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in "Definitions" below or elsewhere in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as "JPMorgan Chase", and together with its subsidiaries as the "Firm".

Form and Purpose of Award

Each PSU represents a non-transferable right to receive one share of Common Stock following each vesting date as set forth in your Award Agreement.

The purpose of this award is to further emphasize sustained long-term performance and to align your interests with those of the Firm and its shareholders.

Number of Performance Share Units at End of Performance Period

Subject to any cancellation in whole or part of your award pursuant to these terms and conditions:

Performance calculation: The number of PSUs at the end of the Performance Period will be derived by multiplying the Target Award Number by the Award Payout Percentage determined using the Performance Table.

The number of PSUs determined above will be subject to the Qualitative Performance Factor (as detailed below), which if the Committee determines that such an adjustment is appropriate, will be applied following the end of each year during the Performance Period, to adjust downward one-third of the Target Award Number of PSUs for each calendar year in the Performance Period. Additionally, the Committee, in its discretion, may make a qualitative performance assessment based on the entire three-year Performance Period and apply the Qualitative Performance Factor to the entire number of PSUs determined above.

See sections captioned "Calculation of Performance Ranking" and "Definitions".

Delivery of vested shares of common stock to your account will be made not later than the date specified in the last sentence of the subsection captioned "Section 409A Compliance".

Reinvested Dividend Equivalent Share Units

This award is not eligible for reinvested dividend equivalent share units.

Holding Requirement

The net number of shares of Common Stock (after tax and all other lawful withholdings) in which you have vested, if any, as of the vesting date set forth in your Award Agreement will be held in an account in your name with restrictions preventing you from transferring, assigning, hedging, selling, pledging or otherwise encumbering such shares for (i) a twelve month period measured from each vesting date; and (ii) a two-year period for such shares vesting on ______________, with the holding periods running concurrently. Such restrictions shall only lapse, prior to the expiration of the two-year holding period, in the event of your death or for an accelerated distribution for ethics or conflict reasons. See section captioned, "Death" and subsection captioned, "Accelerated Distribution for Ethics or Conflict Reasons Resulting from Employment by a Government Entity".

Calculation of Performance Ranking

For purposes of the Performance Ranking, the ranking of the Firm and of each Performance Company for the Performance Period shall be determined and calculated by the Calculation Agent, using the definitions of "Average Tangible Common Equity" (if otherwise applicable), "Calculated PSUs", "Firm Reported ROTCE", "Performance Table" and "ROTCE" as set forth in the "Definitions" section of these terms and conditions. See section captioned "Definitions". Except for Firm's externally published Reported ROTCE, calculations will be expressed as a decimal to the second place (i.e. xx.yy%), rounded to the nearest hundredth. See section captioned, "Definitions--Performance Table" in the event of a tie. All performance-based calculations as set forth herein are binding and conclusive on you and your successors.

Capital Ratio Performance Threshold

Unvested PSUs are subject to reduction if the Firm's Common Equity Tier 1 (CET1) capital ratio at any year end falls below a predetermined threshold of ______%.

If the Firm's CET1 capital ratio at any year end during the Performance Period is below this predetermined threshold, up to one-third of the Target Award Number of PSUs will be subject to downward adjustment by the Compensation & Management Development Committee of the Board of Directors of JPMorgan Chase ("Committee") for each such year.

Qualitative Performance

Determination of Qualitative Performance Factor. Annually during the Performance Period, the Committee will formally assess your qualitative performance based on four broad categories: (1) Client/Customer Focus; (2) Risk, Controls & Conduct; (3) Teamwork & Leadership; and (4) Business Results. If the Committee determines that your performance "Meets" expectations, no downward adjustment to one-third of the Target Award Number of PSUs for that year shall take place (and the Qualitative Performance Factor shall be 100%). If the Committee determines that your performance did "Not Meet" expectations, the Committee shall determine whether a downward adjustment is appropriate, and if so, to what extent. A downward adjustment could result in a Qualitative Performance Factor of between 0% and 99%, depending on the circumstances. During the Performance Period, a 0% Performance Factor for each year in the Performance Period would reduce your Target Award Number of PSUs to zero, resulting in the cancellation of award with no shares vesting.

Additionally, the Committee may, in its sole discretion, make such assessment of your qualitative performance based on your performance during the entire three-year Performance Period and apply the Qualitative Performance Factor to the entire number of PSUs determined under section captioned "Number of Performance Share Units at the end of the Performance Period". In the case of a Qualitative Performance Factor of 0%, the award would be cancelled.

The assessment will have regard to feedback solicited from the Chair of the UK Remuneration Committee to incorporate qualitative performance against local regulatory responsibilities as a "Senior Manager" of the relevant UK-regulated entities.

The Qualitative Performance Factor shall only be applied, if applicable, in respect of a period of your employment with the Firm, or as soon as administratively practical.

Protection-Based Vesting

This award is intended and expected to vest on each vesting date set forth in your Award Agreement, provided that you are continuously employed by the Firm through such vesting date, or you meet the requirements for continued vesting described under the subsections "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability". However, vesting and the number of PSUs that will vest are subject to these terms and conditions (including, but not limited to, sections captioned "Number of Performance Share Units at End of Performance Period", "Capital Ratio Performance Threshold", "Remedies", and the following protection-based vesting provision).

Up to a total of fifty percent of your award that would otherwise be distributable to you as of any vesting date ("At Risk PSUs") may be cancelled if the Chief Executive Officer of JPMorgan Chase ("CEO") determines in his or her sole discretion that cancellation of all or portion of the At Risk PSUs is appropriate in light of any one or a combination of the following factors:

- Your performance in relation to the priorities for your position, or the Firm's performance in relation to the priorities for which you share responsibility as a member of the Operating Committee, have been unsatisfactory for a sustained period of time. Among the factors the CEO may consider in assessing performance are: net income, total net revenue, earnings per share and capital ratios of the Firm, both on an absolute basis and, as appropriate, relative to peer firms.
- For any calendar year ending during the vesting period, JPMorgan Chase's annual pre-tax pre-provision income at the Firm level is negative.

- RSU awards granted to participants in a Line of Business for which you exercise, or during the vesting period exercised, direct or indirect responsibility, were in whole or in part cancelled because the Line of Business did not meet its annual Line of Business Financial Threshold.
- The Firm does not meet the Firmwide Financial Threshold.

For avoidance of doubt, cancellation of the At Risk PSUs, in whole or part, for one or more of the above factors may occur prior to the end of the Performance Period and the maximum number of At Risk PSUs subject to cancellation prior to the end of the Performance Period will be up to fifty percent of the Target Award Number.

In the event that your employment terminates due to "Job Elimination", "Full Career Eligibility", "Government Office" or "Disability" thereby entitling you to continued vesting in your award, (or potentially acceleration due to satisfaction of the Government Office Requirements), the cancellation circumstances described above will continue to apply.

Any determination above with respect to protection-based vesting provisions is subject to ratification by the Committee. In the case of an award to any current or former CEO, all such determinations shall be made by the Committee and ratified by the Board.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 20______ and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled "Other Matters":

https://about.jpmorganchase.com/about/governance/corporate-governance-principles

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

Recovery of Erroneously Awarded Incentive-Based Compensation

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Recovery of Erroneously Awarded Incentive-Based Compensation Policy (or successor policy) as in effect from time to time as it applies to this award. You can access this policy as currently in effect by clicking the following link to the Firmwide Policy & Standard Portal, or alternatively typing go/dogma then searching for the Recovery of Erroneously Awarded Incentive-Based Compensation Policy.
Firmwide Policy & Standard Portal > Recovery of Erroneously Awarded Incentive-Based Compensation Policy

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Recovery of Erroneously Awarded Incentive-Based Compensation Policy (or successor policy).

EMEA Malus and Clawback Policy - Identified Staff

In consideration of grant of this award, and without prejudice to any other provision of this Award Agreement, you agree that you are subject to the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff (and any applicable supplement(s) to that policy) or successor policy as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year ______ and to this award.

The provisions of the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff set out the terms and conditions applying to the grant of this award which ensure that the Firm is able to meet its regulatory obligations to operate malus (reduce) and/or clawback (recover) to awards in certain circumstances. These include, but are not limited to, where (i) there is a material downturn in the Firm's financial performance or (ii) where the Firm is required to hold more capital. The circumstances in which the events at (i) and (ii) would occur are analogous to some of the circumstances considered under the existing Firmwide terms and conditions, in particular the Bonus Recoupment Policy and the Protection Based Vesting provisions. You can access this policy as currently in effect in My Rewards through the following link: https://myrewards.jpmorganchase.com

Termination of Employment

Except as explicitly set forth below under the subsections captioned "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability" or under the section captioned "Death", this award will be cancelled in full effective on the date your employment with the Firm terminates for any reason.

Subject to these terms and conditions (including, but not limited to, sections captioned "Protection-Based Vesting", "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation", "EMEA Malus and Clawback Policy – Identified Staff", "Your Obligations" and "Remedies") you will be eligible to continue to vest (as you otherwise would vest if you were still employed by JPMorgan Chase) with respect to your award in accordance with its terms and conditions following the termination of your employment if one of the following circumstances applies to you:

- **Job Elimination**

  In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

  - the Firm terminated your employment because your job was eliminated, and
  - after you are notified that your job will be eliminated, you provided such services as requested by the Firm in a cooperative and professional manner, and
  - you satisfied the Release/Certification Requirements set forth below.

- **Full Career Eligibility**

  In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

  - (i) you voluntarily terminated your employment with the Firm, and (ii) had completed at least five years of Recognized Service (as defined below) on your date of termination with at least three years of continuous service with the Firm immediately preceding your termination date or if you are over 60 years of age with at least three years of continuous service with the Firm immediately preceding your termination date, and
  - the sum of your age and Recognized Service (as defined below) on your date of termination equaled or exceeded 60 and
  - you provided at least 180 days advance written notice to the Firm of your intention to voluntarily terminate your employment under this provision, during which notice period you provided such services as requested by the Firm in a cooperative and professional manner and you did not perform any services for any other employer, and
  - continued vesting shall be appropriate, which determination shall be made prior to your termination and will be based on your performance and conduct (before and after providing notice), and
  - you satisfied the Release/Certification Requirements set forth below, and
  - for 36 months from the date of grant of this award you do not either perform services in any capacity (such as an employee, contractor, consultant, advisor, or self-employed individual, whether paid or unpaid) for a Financial Services Company (as defined below) or work in your profession (whether or not for a Financial Services Company); provided that you may do one or more of the following:
    - work in a government, education or Not-for-Profit Organization (as defined below),
    - subject to disclosure and confirmation of the following, serve on the board of directors of a public or non-public company that is not a Financial Services Company.

  After receipt of such advance written notice, the Firm may choose to have you continue to provide services during such 180-day period as a condition to continued vesting or shorten the length of the 180-day period at the Firm's sole discretion, but to a date no earlier than the date you would otherwise meet the age and service requirements.

  Additional advance notice requirements may apply for employees subject to notice period policies (see "Notice Period" below.)

- **Government Office**

  In the event that you voluntarily terminate your employment with the Firm to accept a Government Office or become a candidate for an elective Government Office, as described at the end of these terms and conditions under the section captioned "Government Office Requirements". See also definition of Government Office in the section captioned "Definitions".

- **Disability**

  In the event that

- your employment with the Firm terminates because (i) you are unable to return to work while you are receiving benefits under the JPMorgan Chase Long Term Disability Plan, or for non-U.S. employees, under the equivalent JPMorgan Chase-sponsored local country plan (in either case, "LTD Plan"), or (ii) if you are not covered by a LTD Plan, you are unable to return to work due to a long-term disability that would qualify for benefits under the applicable LTD Plan, as determined by the Firm or a third party designated by the Firm; provided that you (x) request in writing continued vesting due to such disability within 30 days of the date your employment terminates, and (y) provide any requested supporting documentation and (z) receive the Firm's written consent to such treatment, and
- you satisfied the Release/Certification Requirements set forth below.

Release/Certification

To qualify for continued vesting after termination of your employment under any of the foregoing circumstances:

- you must timely execute and deliver a release of claims in favor of the Firm, having such form and terms as the Firm shall specify,
- with respect to "Full Career Eligibility", prior to the termination of your employment, you must confirm with management that you meet the eligibility criteria (including providing at least 180 days advance written notification), advise that you are seeking to be treated as an individual eligible for Full Career Eligibility, and receive written consent to such continued vesting,
- with respect to "Full Career Eligibility" and "Government Office", it is your responsibility to (i) notify the Firm within 15 days after the date you are no longer in compliance with the employment restrictions (as described herein) or (ii) take the appropriate steps to certify to the Firm prior to each vesting date while the employment restrictions are outstanding, on the authorized form of, and by the due date set by, the Firm that you have complied with the employment restrictions applicable to you (as described herein) from your date of termination of employment through the applicable vesting date,
- with respect to "Disability", you must satisfy the notice and documentation described above and receive written consent to such continued vesting, and
- in all cases, otherwise complied with all other terms of the Award Agreement. (See section captioned "Your Obligations" below.)

Death

If you die while you are eligible to vest in this award, your designated beneficiary on file with the Firm's Stock Administration Department (or your estate or if no beneficiary has been designated or survives you or if beneficiary designation is not recognized locally) may be entitled to receive a distribution of a number of shares of Common Stock associated with your award.

Should you die after the end of the Performance Period, your beneficiary will receive shares of Common Stock equal to any outstanding PSUs.

Should you die during the Performance Period, your Beneficiary will receive shares of Common Stock based on the average performance of all completed calendar years, multiplied by one-third of the Target Award Number of PSUs for each completed calendar year in the Performance Period, and using the Award Payout Percentage equal to 100 percent for any remaining calendar years in the Performance Period.

Any shares will be distributed no later than the end of the calendar year immediately following the calendar year which contains your date of death; however, our administrative practice is to register such shares in the name of your beneficiary or estate within 60 days of the Firm's receipt of any required documentation.

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned "--Confidentiality & Non-Solicitation", "--False Statements", "--Cooperation", "--Compliance with Award Agreement", and "--Notice Period".

- **Confidentiality & Non-Solicitation**

  During your employment by the Firm and for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm's then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual's employment terminated because his or her job was eliminated, or the individual's employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm's confidential or proprietary information.

These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm's policies.

You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm's business or its customers except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. "Confidential Information" includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm's private and internal information through telephonic, intranet or internet means. .

For Employees with a Work location in California, you agree that the restrictions in the first paragraph of the "Confidentiality and Non-Solicitation" provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use "Confidential Information" in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

Nothing in this award precludes you from reporting to the Firm's management or directors, the government, a regulator, a self-regulatory agency, your attorneys, or a court, conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**

You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of the law or from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**

You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate, and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**

  You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third party who might be affected thereby, including your prospective employer.

- **Notice Period**

  If you are subject to a notice period or become subject to a notice period after the Grant Date, whether by contract or by policy, that requires you to provide advance written notice of your intention to terminate your employment ("Notice Period"), then as consideration for this award and continued employment, you will provide the Firm with the necessary advance written notice that applies to you, as specified by such contract or policy.

  After receipt of your notice, the Firm may choose to have you continue to provide services during the applicable Notice Period or may place you on a paid leave for all or part of the applicable Notice Period. During the Notice Period, you shall continue to devote your full time and loyalty to the Firm by providing services in a cooperative and professional manner and not perform any services for any other employer and shall receive your base salary and certain benefits until your employment terminates. You and the Firm may mutually agree to waive or modify the length of the Notice Period.

  Regardless of whether a Notice Period applies to you, you must comply with the 180-day advance notice period described under the subsection captioned "-- Full Career Eligibility" in the event you wish to terminate employment under that same subsection.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**

  In addition to the cancellation provisions described under the sections captioned "Protection-Based Vesting", "Qualitative Performance Factor", "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation", "EMEA Malus and Clawback Policy – Identified Staff" and "Termination of Employment", up to 100% of your outstanding PSUs under this award (for the avoidance of doubt, including any associated Reinvested Dividend Equivalent Share Units as well as the Calculated PSUs) may be cancelled if the Firm in its sole discretion determines that:

  - Any of the following detrimental and risk-related conduct has occurred:
    - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
    - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
    - this award was based on a material misrepresentation by you, or
    - you improperly or with gross negligence failed to identify, raise or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
    - your employment was terminated for Cause (see section captioned "Definitions" below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
  - you have failed to comply with any of the advance notice/cooperation requirements or employment restrictions applicable to your termination of employment, or
  - you have failed to return the required forms specified under the section captioned "Release/Certification" by the specified deadline, or
  - you have violated any of the provisions as set forth above in the section captioned "Your Obligations".

  To the extent provided under the subsection captioned "--Amendment" below, JPMorgan Chase reserves the right to suspend vesting of this award and/or distribution of shares under this award, including, without limitation, during any period that JPMorgan Chase is evaluating whether this award is subject to cancellation and/or recovery and/or whether the conditions for distributions of shares under this award are satisfied. The Firm is not responsible for any price fluctuations during any period of suspension and, if applicable, suspended units will be reinstated consistent with Plan administration procedures. See also "Administrative Provisions—No Ownership Rights/Other Limitations".

In addition, you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date or acceleration date) of the gross number of shares of Common Stock previously distributed, including vested shares subject to the Holding Requirements, under this award as follows:

- Payment may be required with respect to any shares of Common Stock distributed within the three-year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
  - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment;
  - you violated any of the provisions as set forth above in the section captioned "Your Obligations"; or
  - you violated the employment restrictions set forth in the subsection "Full Career Eligibility" or "Government Office" following the termination of your employment.
- In addition, payment may be required with respect to any shares distributed within the one-year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above "Detrimental Conduct, Risk Related and Other Cancellation/Recapture" subsection.

Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm's employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase's rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm's intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation" or "EMEA Malus and Clawback Policy – Identified Staff".

- **Right to an Injunction**
  You acknowledge that a violation or attempted violation of any of the provisions set forth in "Your Obligations" herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in "Your Obligations"; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award and shares that vest under this award, and (iii) retaining shares that vest under this award until you pay any taxes associated with the award and vested shares directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt or obligation owed by you under this award without your consent. This restriction on the Firm does not apply to accounts described and authorized in "No Ownership Rights/Other Limitations" described below.

No Ownership Rights/Other Limitations: PSUs do not convey the rights of ownership of Common Stock and do not carry voting rights. No shares of Common Stock will be issued to you until after the number of PSUs have been determined, if any, and have vested. Shares will be issued in accordance with JPMorgan Chase's procedures for issuing stock. By accepting this award, you authorize the Firm, in its sole discretion, to establish on your behalf a brokerage account in your name with the Firm or book-entry account with our stock plan administrator and/or transfer agent and deliver to that account any vested shares derived from the award. You also acknowledge that should there be a determination that the cancellation provisions of this award apply during the period when the vesting of any outstanding PSUs has been suspended, then you agree that such PSUs may be cancelled in whole or part. (See Sections captioned "Protection-Based Vesting", "Qualitative Performance Factor", "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation", "EMEA Malus and Clawback Policy – Identified Staff", "Termination of Employment" and "Remedies", as well as the subsection captioned "—Amendment" permitting suspension of vesting.)

With respect to any applicable vesting date, JPMorgan Chase may impose for any reason, as of such vesting date for such period as it may specify in its sole discretion, such restrictions on the Common Stock to be issued to you as it may deem appropriate, including, but not limited to, restricting the sale, transfer, pledging, assignment, hedging or encumbrance of such shares of Common Stock. Such restrictions described in the last sentence shall not impact your right to vote or receive dividends with respect to the Common Stock. By accepting this award, you acknowledge that during such specified period should there be a determination that the recovery provisions of this award apply, then you agree that you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares subject to such restrictions (notwithstanding the limitation set forth in the "Right to Set Off" subsection above). (See sections captioned "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation" and "Remedies".)

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is "at-will" and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Notwithstanding anything herein to the contrary, if you (i) are subject to taxation under the Code, (ii) are a specified employee as defined in the JPMorgan Chase 2005 Deferred Compensation Plan and (iii) have incurred a separation from service (as defined in that Plan with the exception of death) and if any units/shares under this award represent deferred compensation as defined in Section 409A and such shares are distributable (under the terms of this award) within six months following, and as a result of your separation from service, then those shares will be delivered during the first calendar month after the expiration of six full months from date of your separation from service. Further, if your award is not subject to a substantial risk of forfeiture as defined by regulations issued under Section 409A of the Code, then the remainder of each calendar year immediately following each vesting date set forth in your Award Agreement shall be a payment date for purposes of distributing the vested portion of the award.

Change in Outstanding Shares: In the event of any change in the outstanding shares of Common Stock by reason of any stock dividend or split, recapitalization, issuance of a new class of common stock, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to stockholders of Common Stock other than regular cash dividends, the Committee will make an equitable substitution or proportionate adjustment, in the number or kind of shares of Common Stock or other securities issued or reserved for issuance pursuant to the Plan and to any PSUs outstanding under this award for such corporate events.

Other Equitable Adjustments: Except for the "Qualitative Performance Factor", the Committee may make adjustments (up or down) to the award as it deems to be equitable, to maintain the intended economics of the award in light of changed circumstances, which may include unusual or non-recurring events affecting the Firm (or the Performance Companies) or its financial statements in each case resulting from changes in accounting methods, practices or policies, changes in capital structure by reason of legal or regulatory requirements and such other changed circumstances, as the Committee may deem appropriate.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under "No Ownership Rights/Other Limitations", in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area, or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity: Upon receipt of satisfactory evidence that applicable United States federal, state, local, foreign or supranational ethics or conflict of interest laws or regulations require you to divest your interest in JPMorgan Chase PSUs, the Firm may accelerate the distribution of all or part of your outstanding award, effective on or before the required divestiture date and waive the Holding Requirement; provided that no accelerated distribution shall occur if the Firm determines that such acceleration will violate Section 409A of the Code. Accelerated distribution under this paragraph does not impact the dates as set forth in the "Remedies" section above. The time period for recovery shall be determined by the originally scheduled vesting date set forth in your Award Agreement or distribution date prior to any acceleration event.

If you have voluntarily terminated your employment and have satisfied the requirements of the section captioned "Government Office Requirements", acceleration shall apply (to extent required) to the percentage of your outstanding award that would continue to vest under that section. In the case of a termination of employment where the award is outstanding as a result of the subsections entitled "Job Elimination" or "Full Career Eligibility", then acceleration shall apply, to the extent required, to the full outstanding award. Subject to the two foregoing sections, the number of shares of Common Stock to be received on acceleration shall be determined using the methodology set forth under the section captioned "Death".

To the extent you have vested shares under this award subject to the Holding Requirement and become subject to divestiture requirement as forth herein, the Firm may waive the holding period to the extent required.

Notwithstanding an accelerated distribution or waiver of the Holding Requirement pursuant to the foregoing, you will remain subject to the applicable terms of your Award Agreement as if your award had remained outstanding for the duration of the vesting period and shares had been distributed as scheduled as of each vesting date, including, but not limited to, repayment obligations set forth in the section captioned "Remedies" and the employment restrictions in the sections captioned "Protection-Based Vesting" and "Government Office Requirements" and the subsection "Full Career Eligibility".

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm's legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm's By-Laws in the event of such a suit by the Firm.

Non-transferability: Neither this award or any other outstanding awards of restricted stock units or of performance-based share units, nor your interests or rights in any such awards, shall be assigned, pledged, transferred, hedged, hypothecated or subject to any lien. An award may be transferred following your death by will, the laws of descent or by a beneficiary designation on file with the Firm.

Definitions

"Average Tangible Common Equity" means annual average common stockholders' equity less annual average goodwill and annual average identifiable intangible assets. Annual averages of the components of Average Tangible Common Equity will be calculated using quarterly balances as reported in publicly available financial disclosures. In the event that quarterly balances are not available, annual year end balances will be used. This calculation is used solely for purposes of the Performance Ranking.

"Award Payout Percentage" means the applicable percentage specified in the Performance Table.

"Calculated PSUs" means the number of PSUs determined by multiplying the Target Award Number (after giving effect to any cancellation thereof, in whole or in part) by the Award Payout Percentage corresponding to the Firm's Performance Ranking based on the three-year average performance for the Performance Period (both percentage and ranking, as set forth in the Performance Table); provided that if the average of the Firm's Reported ROTCE, expressed as a decimal rounded to the nearest hundredth, for the Performance Period either equals or exceeds ______% or is less than ______%, _______ percent or ________, respectively as the case may be, shall be substituted for the Performance Period's Award Payout Percentage in calculating the number of PSUs to distribute. For avoidance of doubt, any cancellation of this award (in whole or in part) during the Performance Period will reduce the Target Award Number.

"Calculation Agent" means a third-party entity not owned or controlled by the Firm, such as an accounting or consulting firm, retained from time to time by the Director of Human Resources or his/her delegate.

"Cause" means a determination by the Firm that your employment terminated as a result of your (i) violation of any law, rule or regulation (including rules of self-regulatory bodies) related to the Firm's business, (ii) indictment or conviction of a felony, (iii) commission of a fraudulent act, (iv) violation of the JPMorgan Chase Code of Conduct or other Firm policies or misconduct related to your duties to the Firm (other than immaterial and inadvertent violations or misconduct), (v) grossly inadequate performance of the duties associated with your position or job function or failure to follow reasonable directives of your manager, or (vi) any act or failure to act that is injurious to the interests of the Firm or its relationship with a customer, client or an employee.

"Financial Services Company" means a business enterprise that engages in any of the following services (itself or through an affiliate or subsidiary), regardless of whether such services are the principle strategy or revenue-generating activity:

- commercial or retail banking, including, but not limited to, commercial, institutional and personal trust, custody and/or lending and processing services, internet banking, originating and servicing mortgages, issuing and servicing credit cards, payment servicing or processing or merchant services,
- insurance, including but not limited to, guaranteeing against loss, harm, damage, illness, disability, or death, providing and issuing annuities, acting as principal, agent or broker for purpose of the forgoing,
- financial, investment or economic advisory services, including but not limited to, investment banking services (such as advising on mergers or dispositions, underwriting, dealing in, or making a market in securities or other similar activities), brokerage services, investment management services, asset management services, foreign exchange services, interbank networks and hedge funds,
- issuing, trading or selling instruments representing interests in pools of assets or in derivatives instruments,
- financial technology companies, such as those selling blockchain services, or offering or selling financial products/ services,
- advising on, or investing in, private equity or real estate funds or ventures, or
- any similar activities that the Director of Human Resources or nominee determines in his or her sole discretion constitute financial services,
- however, subject to disclosure and confirmation of the following, it shall not be considered a "Financial Services Company" for the purpose of this Award Agreement if such family office is solely funded by you and/or your family's own personal funds without external investors directly or indirectly.

"Firm Reported ROTCE" means the Firm's percentage return on tangible common equity for each year in the Performance Period.

"Firmwide Financial Threshold" means a cumulative return on tangible common equity for calendar years 20XX, 20XX and 20XX of not less than ______%. Cumulative return on tangible common equity means (i) the sum of the Firm's reported net income for all three calendar years, divided by (ii) reported year-end tangible equity averaged over the three years.

"Government Office" means (i) a full-time position in an elected or appointed office in local, state, or federal government (including equivalent positions outside the U.S. or in a supranational organization), not reasonably anticipated to be a full-career position, or (ii) conducting a bona fide full-time campaign for such an elective public office after formally filing for candidacy, where it is customary and reasonably necessary to campaign full-time for the office.

"Line of Business" means a business unit of the Firm (or one or more business units designated below under the definition "Line of Business Financial Threshold" of the Commercial & Investment Bank). All Corporate Functions (including the functions of the Chief Investment Office) are considered a single Line of Business.

"Line of Business Financial Threshold" means the financial threshold set forth below: for the following Lines of Business based on the Firm's management reporting system:

Asset & Wealth Management	Annual negative pre-tax pre-provision income[1]
Card and Auto	Annual negative pre-tax pre-loan loss reserve income[2]
Commercial & Investment Bank	Annual negative pre-tax pre-provision income[1] for CIB overall or annual negative allocated product revenues (excluding XVA) for: • Fixed Income • Equities • Securities Services • Global Investment Banking • Payments
Consumer Banking, J.P. Morgan Wealth Management and Business Banking	Annual negative pre-tax pre-loan loss reserve income[2]
Corporate Functions (including Chief Investment Office)	Annual negative pre-tax pre-provision income[1] at the Firm level
Home Lending	Annual negative pre-tax pre-loan loss reserve income[2]
[1]Pre-tax pre-provision income means Revenue less Expenses [2]Pre-tax pre-loan loss reserve income means Revenue less (Expenses plus Net Charge-offs)	

"Not-for-Profit Organization" means an entity exempt from tax under state law and under Section 501(c)(3) of the Code. Section 501(c)(3) only includes entities organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals. Not-for-Profit Organization shall also mean entities outside the United States exempt from local and national tax laws because they are organized and operated exclusively for purposes identical to those applicable to Section 501(c)(3) organization.

"Performance Companies" mean the following institutions which have business activities that overlap with a significant portion of the Firm's revenue mix: ____, and ________________.

If, during the Performance Period, one or more Performance Companies shall merge, engage in a spin-off or otherwise experience a material change in its revenue mix or business activities or its existence or its primary businesses shall terminate or cease due to receivership, bankruptcy, sale, or otherwise, then the Committee may eliminate such institution from the list of Performance Companies or make such other equitable adjustments, such as adding an acquirer or a new company to the list of Performance Companies, as it deems appropriate, with any such changes having effect for purposes of all calculations hereunder on a prospective basis from the date the applicable change is made.

"Performance Period" means calendar years 20XX, 20XX and 20XX.

"Performance Ranking" means the ranking of the average ROTCE of the Firm as compared to the ranking of the average ROTCE of the Performance Companies as specified in the Performance Table for the Performance Period.

"Performance Table" means the table used in the calculation of PSUs for the Performance Period as set forth below:

Firm Reported ROTCE (average performance)	Award Payout Percentage
>=________%	______%
______% to ______%	Pay by relative ROTCE scale
<______%	_____%

Firm Performance Ranking (average performance)	Award Payout Percentage
#1	_______%
#2	_______%
#3	_______%
#4	_______%
#5	_______%
#6	_______%
#7	_______%
#8	_______%
#9	_______%
#10 +	_______%

If, after the calculation of the Performance Ranking, there is a tie, the tie shall be disregarded for purposes of determining the Award Payout Percentage. For example, in the case of a tie for the fourth ranking between the Firm and a Performance Company, the Firm shall be treated as having satisfied that ranking. In the case of that same tie among Performance Companies, the fourth and fifth rankings will be deemed to have been satisfied.

"Recognized Service" means the period of service as an employee set forth in the Firm's applicable service-related policies.

"ROTCE" means for the Firm and each of the Performance Companies a percentage derived by, for each year in the Performance Period, dividing (i) annual earnings from continuing operations less dividends on preferred stock as set forth in published financial disclosures by (ii) the Average Tangible Common Equity for the year. If, prior to the end of the vesting period, the Firm or any Performance Company restates its published financial statements for any year in the Performance Period, ROTCE for that year shall be recalculated for the Firm or Performance Company with the Performance Ranking adjusted, if necessary. This calculation is used solely for purposes of the Performance Ranking.

"Target Award Number" means the number of PSUs designated as such in the Award Agreement.

Government Office Requirements

You may be eligible to continue vesting in all or part of your award if you voluntarily resign to accept a Government Office (as defined above) or to become a candidate for an elective Government Office.

Full Career Eligibility:
"Government Office Requirements" does not apply to you if you satisfy the subsection captioned "--Full Career Eligibility" as of the date that you voluntarily terminate your employment with the Firm.

Eligibility:
Eligibility for continued vesting is conditioned on your providing the Firm:

- At least 60 days' advance written notice of your intention to resign to accept or pursue a Government Office (see section captioned "Definitions"), during which period you must perform in a cooperative and professional manner services requested by the Firm and not provide services for any other employer. The Firm may elect to shorten this notice period at the Firm's sole discretion.
- Confirmation, in a form satisfactory to the Firm, that vesting in this award pursuant to this provision would not violate any applicable law, regulation or rule.
- Documentation in a form satisfactory to the Firm that your resignation is for the purpose of accepting a Government Office or becoming a candidate for a Government Office. (See section captioned "Definitions".)

Portion of Your Award Subject to Continued Vesting:
Subject to the conditions below, the percentage of this award that will continue to vest in accordance with this award's original schedule will be based on your years of continuous service completed with the Firm immediately preceding your termination date, as follows:

- 50% if you have at least 3 but less than 4 years of continuous service,
- 75% if you have at least 4 but less than 5 years of continuous service, or
- 100% if you have 5 or more years of continuous service.

The portion of this award subject to continued vesting above is referred to as the "CV Award" and the portion not subject to continued vesting will be cancelled as of the date your employment terminates.

Conditions for Continued Vesting of Award:

- You must remain in a non-elective Government Office for two or more years after your employment with the Firm terminates to be eligible to receive the CV Award; provided that if your non-elective Government Office is for a period less than two years, you will be eligible to receive the CV Award if it has a vesting date during your period of Government Service; or
- In the case of resignation from the Firm to campaign for an elective Government Office, your name must be on the primary or final public ballot for the election. (If you are not elected, see below for employment restrictions.)

For avoidance of doubt, the performance criteria and protection-based vesting set forth in these terms and conditions continue to apply to a CV Award.

Satisfaction of Conditions:
If your service in a Government Office ends two years or more after your employment with the Firm terminates, or in the case of resignation from the Firm to campaign for a Government Office, your name is on the primary or final public ballot for the election and you are not elected, any CV Awards then outstanding and any such awards that would have then been outstanding but for an accelerated distribution of shares (as described in the subsection captioned "--Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity") will be subject for the remainder of the applicable vesting period to the same terms and conditions of this Award Agreement, including employment restrictions during the vesting period, as if you had resigned from the Firm having met the requirements for Full Career Eligibility.

Failure to Satisfy Conditions:
If you do not satisfy the above "Conditions for Continued Vesting of Awards", any outstanding PSUs under the CV Award will be cancelled. You also will be required to repay the Fair Market Value of the number of shares (before tax and other withholdings) of Common Stock distributed to you that would have been outstanding as PSUs on the date you failed to satisfy the "Conditions for Continued Vesting of Award" but for their accelerated distribution (as described in subsection

captioned, “Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”). Fair Market Value for this purpose will be determined as the date that the shares were distributed.

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF JANUARY 21, 2025
RESTRICTED STOCK UNIT AWARD
Identified Staff – Retained Award

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of January 21, 2025 ("Grant Date") awarding Restricted Stock Units ("RSUs") pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan ("Plan"). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as "JPMorgan Chase" and together with its subsidiaries as the "Firm".

Form of Award

Each RSU represents a non-transferable right to receive one share of Common Stock as of the applicable vesting date as set forth in your Award Agreement.

Holding Requirement

As of the vesting date, you shall be entitled to a distribution equal to the Fair Market Value of the number of RSUs vesting on such date, less the number being withheld to satisfy tax withholding obligations. You agree that the distribution made to you will be held in an account in your name with restrictions preventing you from transferring, assigning, hedging, selling, pledging, or otherwise encumbering such distribution for a twelve-month period commencing with the vesting date. Such restrictions shall lapse in event of your death.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled "Other Matters":

https://about.jpmorganchase.com/about/governance/corporate-governance-principles

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

EMEA Malus and Clawback Policy – Identified Staff

In consideration of grant of this award, and without prejudice to any other provision of this Award Agreement, you agree that you are subject to the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff (and any applicable supplement(s) to that policy) or successor policy as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award.

The provisions of the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff set out the terms and conditions applying to the grant of this award which ensure that the Firm is able to meet its regulatory obligations to operate malus (reduce) and/or clawback (recover) to awards in certain circumstances. These include, but are not limited to, where (i) there is a material downturn in the Firm's financial performance or (ii) where the Firm is required to hold more capital. The circumstances in which the events at (i) and (ii) would occur are analogous to some of the circumstances considered under the existing Firmwide terms and conditions, in particular the Bonus Recoupment Policy and the Protection Based Vesting provisions. You can access this policy as currently in effect in My Rewards through the following link: https://myrewards.jpmorganchase.com

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned "--Confidentiality & Non-Solicitation", "--False Statements", "--Cooperation", "--Compliance with Award Agreement" and "--Notice Period."

- **Confidentiality & Non-Solicitation**

  During your employment by the Firm and for the one year period following the termination of your employment, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm's then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual's employment terminated because his or her job was eliminated, or the individual's employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm's confidential or proprietary information.

  These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm's policies.

  You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm's business or its customers, except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. "Confidential Information" includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm's private and internal information through telephonic, intranet or internet means.

  For Employees with a Work Location in California, you agree that the restrictions in the first paragraph of this "Confidentiality and Non-Solicitation" provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use "Confidential Information" in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

  Nothing in this award precludes you from reporting to the Firm's management or directors, the government, a regulator, a self-regulatory agency, your attorneys or a court, conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

  If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**

  You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of the law or preclude you from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**
  You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**
  You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third-party who might be affected thereby, including your prospective employer.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**
  Notwithstanding any terms of this Award Agreement to the contrary, JPMorgan Chase reserves the right to recover from you up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares of Common Stock previously distributed (including shares withheld for tax purposes) under this award if the Firm in its sole discretion determines that:
  - Any of the following detrimental and risk-related conduct has occurred:
    - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
    - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
    - this award was based on a material misrepresentation by you, or
    - you improperly or with gross negligence failed to identify, raise, or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
    - your employment was terminated for Cause (see section captioned "Definitions" below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
- **Recovery**
  You may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares of Common Stock previously distributed under this award as follows:
  - Payment may be required with respect to any shares distributed within the three-year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
    - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment, or
    - you violated any of the provisions as set forth above in the section captioned "Your Obligations", or
  - In addition, payment may be required with respect to any shares distributed within the one-year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above "Detrimental Conduct, Risk Related and Other Cancellation/Recapture" subsection.

  Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm's employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase's rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

  For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm's intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under "Bonus Recoupment" and "EMEA Malus and Clawback Policy - Identified Staff".

- **Right to an Injunction**
  You acknowledge that a violation or attempted violation of any of the provisions set forth in "Your Obligations" herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in "Your Obligations"; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award and shares that vest under this award, and (iii) retaining shares that vest under this award until you pay any taxes associated with the award and/or vested shares directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt owed by you under this award without your consent.

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is "at-will" and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under "No Ownership Rights/Other Limitations", in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area, or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm's legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm's By-Laws in the event of such a suit by the Firm.

Exhibit 19

Insider Trading Policy – Firmwide

Effective Date: January 31, 2025

1. Summary or Rationale

This Insider Trading Policy – Firmwide (“Policy”) provides guidelines to promote compliance by Firm Personnel and their Connected Persons with the provisions of the U.S. federal securities laws that prohibit persons who are in possession of Firm Material Nonpublic Information from: (i) trading in Firm Securities; or (ii) impermissibly providing Firm Material Nonpublic Information to other persons who may trade on the basis of that information (i.e., “tipping”).

This Policy applies to Directors, Employees and Non-Employees (each, a “Firm Person” and, collectively, "Firm Personnel"), as well as their Connected Persons, and extends to the securities accounts of such persons and any other securities account that such persons could be expected to influence or control.

This Policy applies to all Firm Personnel and their Connected Persons in all jurisdictions in accordance with and subject to Firm policies, even if the activities prohibited in this Policy are not illegal in the jurisdiction where any particular Firm Person or Connected Person is located.

In addition, neither JPMC nor any Firm subsidiary may trade in Firm Securities while in possession of Firm Material Nonpublic Information, consistent with applicable law.

2. Defined Terms

CEO	The Chief Executive Officer of JPMC
CFO	The Chief Financial Officer of JPMC
Closed Window Period	Any period outside of an Open Window Period
Connected Persons	The following individuals and entities who, based on their relationship with a Firm Person, are subject to provisions of this Policy: • The spouse, domestic partner or minor children (even if financially independent) of a Firm Person • Anyone to whom a Firm Person provides significant financial support or for whom a Firm Person, or anyone listed above, has or shares the power, directly or indirectly, to make investment decisions • Any corporations, partnerships, trusts or other entities that are controlled by a Firm Person and/or anyone listed above

Corporate Officers	Members of the JPMC Operating Committee and the Firmwide Controller
ESPP	JPMorgan Chase Employee Stock Purchase Plan
Exchange Act	The Securities Exchange Act of 1934, as amended
Firm	JPMC and its direct and indirect subsidiaries
Firm MNPI	Material Nonpublic Information relating to the Firm or Firm Securities
Firm Personnel	The following persons are Firm Personnel (each a Firm Person) within the meaning of this Policy: • Directors: Any member of the Board of Directors of JPMC • Employees: Any person directly employed by the Firm • Non-Employees: Contingent workers and other types of non-employee personnel (including any director of a Firm subsidiary) who are not directly employed by the Firm but have access to JPMC facilities and/or internal JPMC systems
Firm Securities	Firm Securities means: • Securities and other financial instruments (such as common stock, preferred stock and bonds) issued by JPMC or any Firm subsidiary • Options, put or call options, forward contracts, warrants, swaps (including credit default swaps) and other derivative securities tied to any of the above Baskets, indices and exchange traded funds that (i) consist of 30 or fewer components and (ii) are comprised of a considerable portion (greater than 3%) of the above are also deemed to be Firm Securities for purposes of this Policy
GC	The General Counsel of JPMC
GRAT	Grantor Retained Annuity Trust
JPMC	JPMorgan Chase & Co.
Material Nonpublic Information or MNPI	Information about an issuer of financial instruments (including private companies that issue publicly traded financial instruments) that is not known by the public, but if it were, is reasonably likely to affect the market price of the financial instruments to which the information relates or is likely to influence a reasonable investor to buy or sell those financial instruments, or would be likely to be used as part their investment decision regarding those financial instruments, as applicable

Open Window Period	Generally, each period beginning the day after public release of JPMC's quarterly earnings and ending on the 15th day of the following month, provided that an Open Window Period may be closed or suspended at any time if circumstances require such action with respect to some or all Window List Personnel
Policy	This Insider Trading Policy – Firmwide
SAR	Stock Appreciation Right
Trading or Transactions	For purposes of this Policy, the terms "trading" or "transactions" (and variations thereof) include any self-directed (i) purchases or sales of, or any other transactions in, Firm Securities; (ii) making or modifying any instructions or investment options relating to any transaction in Firm Securities; and (iii) gifts of Firm Securities
Window List Personnel	Firm Personnel whose position, the nature of their work or their access to information at the Firm indicates that they have or may have access to Firm MNPI Firm Personnel are notified when they become Window List Personnel.
10b5-1 Plan	A plan that meets the requirements set forth in Rule 10b5-1(c)(1) under the Exchange Act

3. Prohibition on Insider Trading

Insider trading is illegal and strictly prohibited by the Firm. Firm Personnel and their Connected Persons must never trade or engage in any transaction in Firm Securities while in possession of Firm MNPI. In addition, neither JPMC nor any Firm subsidiary may trade in Firm Securities when in possession of Firm MNPI, consistent with applicable law. When JPMC possesses or may be deemed to possess Firm MNPI, it will generally observe a "blackout period" during which it will not trade in Firm Securities. However, the Firm may repurchase its common stock under a 10b5-1 Plan during the blackout period preceding each quarterly earnings announcement.

4. Prohibition on Unauthorized Disclosure

Firm Personnel and their Connected Persons must not impermissibly provide Firm MNPI to any person who may trade on the basis of that information (i.e., "tipping"). In addition, Firm MNPI must not be shared with any person unless the disclosure complies with the Firm's policies concerning the safeguarding of confidential information, including the Code of Conduct.

5. Policies Applicable to Window List Personnel

5.1 Window List Personnel

Firm Personnel who have or may have access to Firm MNPI are designated as Window List Personnel. Firm Personnel are notified via email when they become Window List Personnel.

Window List Personnel and their Connected Persons are prohibited from trading in Firm Securities during any Closed Window Period. During any Open Window Period, Window List Personnel and their Connected Persons are permitted to trade in Firm Securities after obtaining preapproval from the Compliance Department. Any such approval is valid for the length indicated on the approval notification which is generally the end of the next business day, unless otherwise stipulated. An Open Window Period may be closed or suspended at any time if circumstances require such action with respect to some or all persons designated as Window List Personnel. If preapproval for a transaction is requested during a Closed Window Period, the Window List Person making the request will receive a denial notification from Compliance and will not be authorized to execute the trade. In extremely rare circumstances, an exception may be granted in consultation with the Window List Person's manager, Compliance and Legal.

Non-Window List Personnel are not subject to the preapproval requirements described in this Section 5.1 and in Section 6.2 or the minimum holding period requirements described in Section 7.3, but regional or line of business-specific restrictions may apply.

For additional requirements for Corporate Officers and Directors, see Section 6.

5.2 Window Covered Transactions

Window List Personnel and their Connected Persons may only engage in the following types of transactions in Firm Securities, subject to compliance with preapproval requirements and the requirement that the transaction be conducted during an Open Window Period:

- Direct purchases, sales and gifts of JPMC common stock or other Firm Securities, including self-directed sales to meet a margin call;
- Liquidating holdings of Firm Securities under the ESPP or the Firm's deferred compensation plan;
- Placing, cancelling or amending limit orders (i.e., an order to buy or sell a stock at a specific price or better) with respect to Firm Securities;
- Entering into, amending or terminating 10b5-1 Plans with respect to Firm Securities; and

- Use of previously-owned JPMC shares or share withholding for option exercises, unless as part of a 10b5-1 Plan entered into during an Open Window Period.

Additional restrictions apply to JPMC Employee Stock Options and JPMC Stock Appreciation Rights.

The following transactions are not considered to be restricted transactions for Window List Personnel and their Connected Persons and are not required to be preapproved:

- Dividend reinvestments and automatic purchases of JPMC common stock in accordance with previously made employee benefit elections (e.g., the ESPP) entered into (or amended) and preapproved during an Open Window Period;
- Transactions under a 10b5-1 Plan entered into (or amended) and preapproved during an Open Window Period; and
- Limit orders and derivatives that are executed during a Closed Window Period that were placed during an Open Window Period.

6. Policies Applicable to Corporate Officers and Directors

6.1 Transactions that are Required to be Pursuant to a 10b5-1 Plan

When a Corporate Officer or Director wishes to engage in an open market sale of Firm Securities, the transaction must be made pursuant to a 10b5-1 Plan.

When a Corporate Officer or Director wishes to (i) exercise and hold a SAR or option, (ii) make a charitable gift of Firm Securities, (iii) create a GRAT relating to Firm Securities or (iv) transfer Firm Securities to or from a GRAT or another trust, the transaction requires pre-approval under this Policy but is not required to be pursuant to a 10b5-1 Plan.

A Corporate Officer or Director may enter into a 10b5-1 Plan subject to compliance with the Firm's other share retention policies. To enter into a 10b5-1 Plan, a Corporate Officer or Director must comply with the preapproval requirements described in Section 6.2 of this Policy.

All 10b5-1 Plans must comply with all of the requirements set forth in Rule 10b5-1(c) and may be entered into, amended or terminated only during an Open Window Period.

6.2 Pre-approval for Firm Securities Transactions and Entry into 10b5-1 Plans

During an Open Window Period, Corporate Officers and Directors are permitted to trade in Firm Securities after obtaining the preapprovals described below.

- Corporate Officers and Directors must receive preapproval from the CEO and the GC for any transactions in Firm Securities that require preapproval under Section 5 and for entry into, amendment of or termination of a 10b5-1 Plan under Section 6.1.

- In the case of transactions or 10b5-1 Plans entered into by the GC, preapproval is required from the CEO,
- in the case of transactions or 10b5-1 Plans entered into by the Firmwide Controller, approval is required from the GC and the CFO, and
- in the case of transactions or 10b5-1 Plans entered into by the CEO, approval is required from the GC.

The preapprovals described above for Corporate Officers are submitted to the Compliance Department for its preapproval, which is valid for the length indicated on the approval notification which is generally the end of the next business day, unless otherwise stipulated.

7. Other Prohibitions

Other restrictions that are applicable to transactions in Firm Securities are described below.

7.1 Prohibited Trading Activity

Trading with respect to Firm Securities is prohibited for Window List Personnel and all other Firm Personnel who are subject to personal account dealing restrictions under Firm policies in the following:

- Forwards and futures (all asset classes)
- Swaps (all asset classes)
- Uncovered option strategies except Long-Term Equity Anticipation Securities (LEAPS)
- Short sales

Except as set forth in Section 7.2, direct hedging strategies (e.g., covered calls) with respect to Firm Securities are permitted but must be proportionate to the risk (i.e., fully covered with underlying shares).

All Employees are prohibited from hedging or pledging unvested shares of JPMC common stock received as compensation.

7.2 Anti-Hedging/Anti-Pledging

Shares of JPMC common stock owned outright or through deferred compensation by a member of the JPMC Operating Committee or Director may not be held in margin accounts or otherwise pledged as collateral, nor may the economic risk of such shares be hedged.

7.3 Minimum Holding Period

Window List Personnel and their Connected Persons are required to hold Firm Securities for a minimum holding period (specific to each region and/or line of business) prior to transacting in those Firm Securities.

Vested shares of JPMC common stock or shares of JPMC common stock received in connection with purchases under the ESPP are deemed to have satisfied the minimum holding period requirement.

8. Post-termination Transactions

If a Firm Person is aware of Firm MNPI when their employment, directorship or other service relationship with the Firm terminates, that Firm Person and their Connected Persons may not trade in Firm Securities or disclose any Firm Material Nonpublic Information until such information has become public or is no longer material.

9. Personal Responsibility for Compliance with this Policy and Potential Disciplinary Action

This Policy is designed to reduce the risk of violations of the insider trading provisions of the U.S. federal securities laws, but compliance with this Policy is not an assurance that an insider trading violation will not be found to have occurred. The ultimate responsibility for adhering to this Policy and avoiding improper trading rests exclusively with Firm Personnel and their Connected Persons. Any action on the part of Firm Personnel and their Connected Persons pursuant to this Policy (or otherwise) does not insulate an individual from liability under applicable securities laws and this Policy does not constitute legal advice in any way. The existence of a personal financial emergency does not excuse anyone subject to this Policy from compliance with this Policy and the laws prohibiting insider trading.

In addition to potential criminal and civil liability, failure to comply with the requirements of this Policy may result in the receipt of Policy reminders or violations which may lead to disciplinary action, up to and including termination of employment (subject to local laws and regulations). The Firm also reserves the right to instruct an individual to cancel any trade at his or her expense.

For further information or questions regarding this Policy, Firm Personnel may reach out to the Compliance Department or Office of the Secretary. Firm Personnel should also refer to the Code of Conduct and other related Firm policies.

Exhibit 21

JPMorgan Chase & Co.

List of subsidiaries

While there are a number of subsidiaries that are required to be reported for various purposes to bank regulators, the following is a list of JPMorgan Chase & Co.'s significant legal entity subsidiaries as of December 31, 2024, as defined by SEC rules. The list includes the parent company of significant subsidiaries even if the parent company did not meet the definition of a significant subsidiary. Excluded from the list are subsidiaries that, if considered in the aggregate, would not constitute a significant subsidiary under SEC rules as of December 31, 2024.

Also included in the list are certain subsidiaries that have been designated as material legal entities for resolution planning purposes under the Dodd-Frank Act that did not meet the definition of a significant subsidiary under SEC rules.

December 31, 2024 Name	Organized Under The Laws Of
JPMorgan Chase Bank, National Association	United States
Paymentech, LLC	United States
J.P. Morgan International Finance Limited	United States
JPMorgan Securities Japan Co., Ltd.	Japan
J.P. Morgan Capital Holdings Limited	United Kingdom
J.P. Morgan Securities plc	United Kingdom
J.P. Morgan SE	Germany
JPMorgan Chase Holdings LLC	United States
J.P. Morgan Services India Private Limited	India
JPMorgan Asset Management Holdings Inc.	United States
JPMorgan Distribution Services, Inc.	United States
JPMorgan Asset Management International Limited	United Kingdom
JPMorgan Asset Management (UK) Limited	United Kingdom
JPMorgan Asset Management Holdings (Luxembourg) S.à r.l.	Luxembourg
JPMorgan Asset Management (Europe) S.à r.l.	Luxembourg
J.P. Morgan Investment Management Inc.	United States
J.P. Morgan Broker-Dealer Holdings Inc.	United States
J.P. Morgan Securities LLC	United States

Exhibit 22.2

JPMorgan Chase & Co.

JPMorgan Chase & Co. guarantee of subsidiary issuances

Securities	Guarantor
JPMorgan Chase Financial Company LLC issued, from time to time, its Global Medium-Term Notes, Series A, under the Indenture dated February 19, 2016 ("Series A Notes"), that are each fully and unconditionally guaranteed by JPMorgan Chase & Co. In addition, JPMorgan Chase Financial Company LLC may issue, from time to time, debt securities (including its Series A Notes) and warrants that are each fully and unconditionally guaranteed by JPMorgan Chase & Co. under the Registration Statement on Form S-3 (Registration Statement Nos. 333-270004 and 333-270004-01), which was declared effective on April 13, 2023.	JPMorgan Chase & Co.

Exhibit 23

CONSENT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

We hereby consent to the incorporation by reference in the Registration Statements on:

Form S-3
(No. 333-270004)
(No. 333-270004-01)
(No. 333-263304)

Form S-8
(No. 333-272306)
(No. 333-272303)
(No. 333-272302)
(No. 333-272299)
(No. 333-219702)
(No. 333-219701)
(No. 333-219699)
(No. 333-185584)
(No. 333-185582)
(No. 333-185581)
(No. 333-175681)
(No. 333-158325)
(No. 333-142109)
(No. 333-125827)
(No. 333-112967)

of JPMorgan Chase & Co. or JPMorgan Chase Financial Company LLC of our report dated February 14, 2025 relating to the financial statements and the effectiveness of internal control over financial reporting, which appears in this Form 10-K.

/s/ PricewaterhouseCoopers LLP
New York, New York
February 14, 2025

Exhibit 31.1

JPMorgan Chase & Co.

CERTIFICATION

I, James Dimon, certify that:

1. I have reviewed this Annual Report on Form 10-K of JPMorgan Chase & Co.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
   (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
   (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent function):
   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: February 14, 2025

/s/ James Dimon

James Dimon
Chairman and Chief Executive Officer

Exhibit 31.2

JPMorgan Chase & Co.

CERTIFICATION

I, Jeremy Barnum, certify that:

1. I have reviewed this Annual Report on Form 10-K of JPMorgan Chase & Co.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

   (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent function):
   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: February 14, 2025

/s/ Jeremy Barnum

Jeremy Barnum
Executive Vice President and Chief Financial Officer

Exhibit 32

JPMorgan Chase & Co.

CERTIFICATION PURSUANT TO 18 U.S.C. SECTION 1350, AS ADOPTED PURSUANT TO SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002

In connection with the Annual Report of JPMorgan Chase & Co. on Form 10-K for the period ended December 31, 2024 as filed with the Securities and Exchange Commission on the date hereof (the "Report"), each of the undersigned officers of JPMorgan Chase & Co., certify, pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that:

1. The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and
2. The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of JPMorgan Chase & Co.

Date: February 14, 2025

By: /s/ James Dimon

James Dimon
Chairman and Chief Executive Officer

Date: February 14, 2025

By: /s/ Jeremy Barnum

Jeremy Barnum
Executive Vice President and Chief Financial Officer

This certification accompanies this Annual Report and shall not be deemed "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, or otherwise subject to the liability of that Section.

A signed original of this written statement required by Section 906 has been provided to, and will be retained by, JPMorgan Chase & Co. and furnished to the Securities and Exchange Commission or its staff upon request.

Exhibit 97

Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation

Effective Date: October 2, 2023

1. Summary or Rationale

In accordance with the applicable rules of The New York Stock Exchange (the "NYSE") Listed Company Manual, Section 10D of the Securities Exchange Act of 1934, as amended ("Exchange Act") and Rule 10D-1 thereunder ("Rule 10D-1"), JPMorganChase (the "Firm") has adopted this policy (this "Policy") to provide for the recovery of Erroneously Awarded Incentive-Based Compensation from Executive Officers in the event of a Restatement. All capitalized terms used and not otherwise defined herein shall have the meanings set forth in Section 5, below.

2. Scope

	Subject to	Role to Play
Lines of Business	• All	
Corporate Functions	• All	• Human Resources
Locations	• All	
Legal Entities	• All	

2.1. Included in Scope

This Policy applies to Incentive-Based Compensation received by a person:

- after beginning service as an Executive Officer,
- who served as an Executive Officer at any time during the performance period for that Incentive-Based Compensation,
- while the Firm has a class of securities listed on a national securities exchange, and
- during the Recovery Period.

Notwithstanding the look-back requirement, the Firm is only required to apply this Policy to Incentive-Based Compensation received on or after October 2, 2023. This Policy will continue to apply after an employee has terminated.

For purposes of this Policy, Incentive-Based Compensation shall be deemed "received" in the Firm's fiscal period during which the Financial Reporting Measure specified in the Incentive-Based Compensation award is attained, even if the payment or grant of the Incentive-Based Compensation occurs after the end of that period.

3. Changes from Previous Version

This is the first version of this Policy.

4. Policy Statement

The Firm shall recover reasonably promptly the Erroneously Awarded Incentive-Based Compensation in the event that the Firm is required to prepare a Restatement.

For clarity, the Firm's obligation to recover Erroneously Awarded Incentive-Based Compensation under this Policy is not dependent on if or when a Restatement is filed.

The Firm shall recover Erroneously Awarded Incentive-Based Compensation in compliance with this Policy except to the extent provided under Section 4.1.2 below.

4.1 Recovery of Erroneously Awarded Incentive-Based Compensation

4.1.1 In the event of a Restatement, the Firm will reasonably promptly recover the Erroneously Awarded Incentive-Based Compensation received in accordance with the NYSE rules and Rule 10D-1 as follows:

- For purposes of determining the relevant Recovery Period, the date that the Firm is required to prepare the Restatement ("Restatement Date") is the earlier to occur of:
  - the date the Board of Directors of the Firm (the "Board"), a committee of the Board, or the officer or officers of the Firm authorized to take such action if Board action is not required, concludes, or reasonably should have concluded, that the Firm is required to prepare a Restatement, or
  - the date a court, regulator, or other legally authorized body directs the Firm to prepare a Restatement.
- The Compensation & Management Development Committee of the Board (the "Committee") composed of independent directors responsible for executive compensation, or in the absence of such committee, a majority of the independent directors serving on the Board, shall determine the Erroneously Awarded Incentive-Based Compensation received by each Executive Officer during the relevant Recovery Period.
- The amount of Incentive-Based Compensation subject to recovery under this Policy is the amount of Incentive-Based Compensation received that exceeds the amount of Incentive-Based Compensation that otherwise would have been received had it been determined based on the restated amounts, computed without regard to any taxes paid (the "Erroneously Awarded Incentive-Based Compensation").
- For Incentive-Based Compensation based on (or derived from) the Firm's stock price or total shareholder return, where the Erroneously Awarded Incentive-Based Compensation is not subject to mathematical recalculation directly from the information in the applicable Restatement:
  - The amount to be recovered shall be based on a reasonable estimate of the effect of the Restatement on the Firm's stock price or total shareholder return upon which the Incentive-Based Compensation was received; and
  - The Firm shall maintain documentation of the determination of such reasonable estimate and provide such documentation to the NYSE.
- The Committee shall have discretion to determine the appropriate means of recovering Erroneously Awarded Incentive-Based Compensation based on the

particular facts and circumstances. Notwithstanding the foregoing, except as set forth in 4.1.2 below, in no event may the Firm settle for an amount that is less than the Erroneously Awarded Incentive-Based Compensation that is subject to recovery under this Policy.

- To the extent that the Executive Officer has already reimbursed the Firm for any Erroneously Awarded Incentive-Based Compensation received under any duplicative recovery obligations established by the Firm or applicable law, it shall be appropriate for any such reimbursed amount to be credited to the Erroneously Awarded Incentive-Based Compensation that is subject to recovery under this Policy.

1. **Notwithstanding anything herein to the contrary, the Firm shall not be required to take the actions contemplated by Section 4.1.1 above if it is determined by the Committee that recovery would be impracticable and any of the following conditions are met:**

- The direct expenses paid to a third party to assist in enforcing this Policy would exceed the amount to be recovered. Before making this determination, the Firm shall make a reasonable attempt to recover the Erroneously Awarded Incentive-Based Compensation, document such attempt(s) and provide such documentation to the NYSE;
- Recovery would violate home country law where that law was adopted prior to November 28, 2022, provided that, before determining that it would be impracticable to recover any Erroneously Awarded Incentive-Based Compensation based on violation of home country law, the Firm shall obtain an opinion of home country counsel, acceptable to the NYSE, that recovery would result in such a violation and a copy of the opinion shall be provided to the NYSE; or
- Recovery would likely cause an otherwise tax-qualified retirement plan, under which benefits are broadly available to employees of the Firm, to fail to meet the requirements of Section 401(a)(13) or Section 411(a) of the Internal Revenue Code of 1986, as amended, and regulations thereunder.

4.2 Disclosure Requirements

The Firm shall file all disclosures with respect to this Policy and recoveries under this Policy in accordance with the requirements of the U.S. Federal securities laws, including the disclosure required by the applicable Securities and Exchange Commission ("SEC") filings.

4.3 Prohibition of Indemnification

Notwithstanding the terms of any indemnification arrangement or insurance policy with any individual covered by this Policy, the Firm shall not indemnify any Executive Officer or former Executive Officer against the loss of Erroneously Awarded Incentive-Based Compensation, including any payment or reimbursement for the cost of insurance obtained by any such covered individual to fund amounts recoverable under this Policy.

4.4 Administration; Amendment; Termination

All determinations under this Policy shall be made by the Committee, including determinations regarding how any recovery under this Policy is effected. Any determinations of the Committee shall be final, binding and conclusive and need not be uniform with respect to each individual covered by this Policy.

The Committee may amend this Policy from time to time and may terminate this Policy at any time, in its discretion.

4.5 Other Recovery Rights

Any right of recovery under this Policy is in addition to, and not in lieu of, any other remedies or rights of recovery that may be available to the Firm under applicable law, regulation or rule or pursuant to the terms of any policy of the Firm or any provision in any employment agreement, equity award agreement, compensation plan, or any other agreement or arrangement.

5. Defined Terms

Executive Officer	An Executive Officer is the Firm's president, principal financial officer, principal accounting officer (or if there is no such accounting officer, the controller), any vice-president of the Firm in charge of a principal business unit, division, or function (such as sales, administration, or finance), any other officer who performs a policy-making function, or any other person who performs similar policy-making functions for the Firm. Policy-making function is not intended to include policy-making functions that are not significant. Identification of an Executive Officer for purposes of this Policy will include at a minimum executive officers identified pursuant to 17 CFR 229.401(b).
Financial Reporting Measure	Any measure that is determined and presented in accordance with the accounting principles used in preparing the Firm's financial statements, and any measures that are derived wholly or in part from such measure. Stock price and total shareholder return are also Financial Reporting Measures. A Financial Reporting Measure need not be presented within the Firm's financial statements or included in a filing with the SEC.
Incentive-Based Compensation	Any compensation that is granted, earned or vested based wholly or in part upon the attainment of a Financial Reporting Measure.
Recovery Period	With respect to any accounting restatement, the three completed fiscal years immediately preceding the Restatement Date, and if the Firm changes its fiscal year, any transition period of less than nine months within or immediately following those three completed fiscal years.
Restatement	An accounting restatement due to the material noncompliance of the Firm with any financial reporting requirement under the securities laws, including any required accounting restatement to correct an error in previously issued financial statements that is material to the previously issued financial statements (a "Big R" restatement), or that would result in a material misstatement if the error were corrected in the current period or left uncorrected in the current period (a "little r" restatement).

UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2024

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to

Commission File Number: 001-35727

Netflix, Inc.
(Exact name of registrant as specified in its charter)

Delaware	**77-0467272**
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)

121 Albright Way, Los Gatos, California 95032
(Address and zip code of principal executive offices)
(408) 540-3700
(Registrant’s telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock, par value $0.001 per share	**NFLX**	**NASDAQ Global Select Market**

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant: (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically, every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “smaller reporting company,” and “emerging growth company” in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant’s executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☒

As of June 30, 2024 the aggregate market value of voting stock held by non-affiliates of the registrant, based upon the closing sales price for the registrant’s common stock, as reported in the NASDAQ Global Select Market System, was $287,180,927,160. Shares of common stock beneficially owned by each executive officer and director of the registrant and by each person known by the registrant to beneficially own 10% or more of the outstanding common stock have been excluded in that such persons may be deemed to be affiliates. This determination of affiliate status is not necessarily a conclusive determination for any other purpose.

As of December 31, 2024, there were 427,757,100 shares of the registrant’s common stock, par value $0.001, outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Parts of the registrant’s Proxy Statement for the registrant’s 2025 Annual Meeting of Stockholders are incorporated by reference into Part III of this Annual Report on Form 10-K.

NETFLIX, INC.

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PART I

Forward-Looking Statements

This Annual Report on Form 10-K contains forward-looking statements within the meaning of the federal securities laws. These forward-looking statements include, but are not limited to, statements regarding: our core strategy; our ability to improve our content offerings and service; our future financial performance, including expectations regarding revenues, deferred revenue, operating income and margin, net income, expenses, and profitability; liquidity, including the sufficiency of our capital resources, net cash provided by (used in) operating activities, access to financing sources, and free cash flows; capital allocation strategies, including any stock repurchases or repurchase programs; seasonality; stock price volatility; impact of foreign exchange rate fluctuations; expectations regarding hedging activity; impact of interest rate fluctuations; adequacy of existing facilities; future regulatory changes and their impact on our business; intellectual property; cybersecurity; price changes and testing; accounting treatment for changes related to content assets; acquisitions; actions by competitors; membership growth, including impact of content and pricing changes on membership growth; partnerships; advertising; multi-household usage; member viewing patterns; dividends; future contractual obligations, including unknown content obligations and timing of payments; our global content and marketing investments, including investments in original programming, consumer products and live experiences; impact of work stoppages; content amortization; resolutions of tax examinations; tax expense; unrecognized tax benefits; deferred tax assets; resolutions of disputes and other proceedings; our ability to effectively manage change and growth; our company culture; and our ability to attract and retain qualified employees and key personnel. These forward-looking statements are subject to risks and uncertainties that could cause actual results and events to differ. A detailed discussion of these and other risks and uncertainties that could cause actual results and events to differ materially from such forward-looking statements is included throughout this filing and particularly in Item 1A: "Risk Factors" section set forth in this Annual Report on Form 10-K. All forward-looking statements included in this document are based on information available to us on the date hereof, and we assume no obligation to revise or publicly release any revision to any such forward-looking statement, except as may otherwise be required by law.

Item 1. Business

ABOUT US

Netflix, Inc. (“Netflix”, “the Company”, “registrant”, “we”, or “us”) is one of the world’s leading entertainment services with approximately 302 million paid memberships in over 190 countries enjoying TV series, films and games across a wide variety of genres and languages. Members can play, pause and resume watching as much as they want, anytime, anywhere, and can change their plans at any time.

Our core strategy is to grow our business globally within the parameters of our operating margin target. We strive to continuously improve our members' experience by offering compelling content that delights them and attracts new members. We aim to offer a range of pricing plans, including our ad-supported subscription plan, to meet a variety of consumer needs. We seek to drive conversation around our content to further enhance member joy, and we are continuously enhancing our user interface to help our members more easily choose content that they will find enjoyable.

BUSINESS SEGMENTS

We operate as one operating segment. Our revenues are primarily derived from monthly membership fees for services related to streaming content to our members. See Note 12, *Segment and Geographic Information*, in the accompanying notes to our consolidated financial statements for further detail.

COMPETITION

The market for entertainment video is intensely competitive and subject to rapid change. We compete with a broad set of activities for consumers’ leisure time, including other entertainment video providers, such as linear television, streaming entertainment providers (including those that provide pirated content), video gaming providers, as well as user-generated content, some of which are by professional content creators, and more broadly against other sources of entertainment, such as social media, that our members could choose in their moments of free time. We also compete against entertainment video providers and content producers in obtaining content for our service, both for licensed content and for original content projects.

While consumers may maintain simultaneous relationships with multiple entertainment sources, we strive for consumers to choose us in their moments of free time. We have often referred to this choice as our objective of "winning moments of truth." In attempting to win these moments of truth with our members, we seek to continually improve our service, including both our technology and our content offerings.

SEASONALITY

Our membership growth exhibits a seasonal pattern that reflects variations when consumers buy internet-connected screens and when they tend to increase their viewing. Historically, the fourth quarter represents our greatest streaming membership growth. In addition, our membership growth can be impacted by our content release schedule and changes to pricing and plans.

INTELLECTUAL PROPERTY

We regard our trademarks, service marks, copyrights, patents, domain names, trade dress, trade secrets, proprietary technologies and similar intellectual property as important to our success. We use a combination of patent, trademark, copyright and trade secret laws and confidentiality agreements to protect our proprietary intellectual property. Our intellectual property rights extend to our technology, business processes, the content we produce and distribute through our service, and the consumer products and experiences based thereon. We use the intellectual property of third parties in creating some of our content, merchandising our products and marketing our service. Our ability to provide our members with content they can watch depends on studios, content providers and other rights holders licensing rights, including distribution rights, to such content and certain related elements thereof, such as the public performance of music contained within the content we distribute. The license periods and the terms and conditions of such licenses vary. Our ability to protect and enforce our intellectual property rights is subject to certain risks and from time to time we encounter disputes over rights and obligations concerning intellectual property. We cannot provide assurance that we will prevail in any intellectual property disputes.

REGULATION

The media landscape and the internet delivery of content have seen growing regulatory action. Historically, media has been highly regulated in many countries. We are seeing some of these legacy regulatory frameworks be updated and expanded to address services like ours. In particular, we are seeing some countries update their cultural support legislation to include services like Netflix. This includes content quotas, levies and investment obligations. Some even restrict the extent of ownership rights we can have both in our service and in our content. In certain countries, regulators are also looking at restrictions that could require formal reviews of and/or adjustments to content that appears on our service in their country. In general these regulations impact all services and may make operating in certain jurisdictions more expensive or restrictive as to the content offerings we may provide.

HUMAN CAPITAL

We view our employees and our culture as key to our success. As of December 31, 2024, we had approximately 14,000 full-time employees. Of these, approximately 9,600 (69%) were located in the United States and Canada, 2,200 (16%) in Europe, Middle East, and Africa, 600 (4%) in Latin America and 1,600 (11%) in Asia-Pacific. We also have a number of employees engaged in content production, some of whom are part-time or temporary, and whose numbers fluctuate throughout the year and may be covered by collective bargaining agreements. We believe an important component of our success is our company culture. This culture, which is detailed in a "Culture Memo" located on our website, is often described as providing a unique environment for our employees to perform the best work of their lives in pursuit of excellence. We aim to attract and retain great people - representing a diverse array of perspectives and skills - to work together as a dream team. As our business grows and changes, we seek to improve our culture to meet the new needs of our business. We believe our dynamic culture helps us create a better experience for our members, employees, creators and partners.

As we have expanded our offices globally, our company culture remains an important aspect of our operations. We are mindful of cultural differences across and within regions. Fostering a work environment that is culturally diverse, inclusive and equitable is a major focus for us. We work to build diversity, inclusion and equity into all aspects of our operations globally, with the goal of having diversity and inclusion function as a critical lens through which each Netflix employee carries out their role. We want more people and cultures to see themselves reflected on screen - so it’s important that our employee base is diverse and represents the communities we serve. We look to help increase representation by educating our people leaders and recruiters on how to hire more inclusively, and to help the company and senior leaders diversify their networks. We also support numerous employee resource groups (ERGs), representing employees and allies from a broad array of historically underrepresented and/or marginalized communities. We publish annually an update on our inclusion initiatives and progress, which further highlights our approach to diversity and inclusion, and publish our EEO-1 reports on our website.

We believe in fostering great leaders. We offer various experiences and training to inform our employees about our culture and other context that we believe is important for success at Netflix. We aim to equip our people leaders (officers, VPs, directors, and manager-level employees) to lead the business and our teams in alignment with our expectations and strategic objectives.

We aim generally to pay our employees at their personal top of market, and they generally are able to choose the form of their compensation between cash and stock options. This permits employee compensation to be highly personalized and reflective of each employee's individual needs and preferences. We conduct pay equity analyses at least annually, and have adopted practices to help ensure that employees from underrepresented groups are not being underpaid based on gender identity (globally) and race or ethnicity (U.S.) relative to others doing the same or similar work under comparable circumstances. We aim to rectify any pay gaps that we find through this analysis.

We care about the health and well-being of our employees and their families and provide a variety of benefit programs based on region, including health benefits. In the U.S., employees generally receive an annual cash health benefit allowance that they may allocate to medical, dental and vision premiums in a way that makes sense for them. Employees have access to a host of other benefits, including mental health, childcare, family planning and a company match for charitable donations.

We believe that our approach to human capital resources has been instrumental in our growth, and has made Netflix a desirable destination for employees.

OTHER INFORMATION

We maintain a website at *www.netflix.com.* The contents of our website are not incorporated in, or otherwise to be regarded as part of, this Annual Report on Form 10-K. We make available, free of charge on our website, access to our Annual Report on Form 10-K, our Quarterly Reports on Form 10-Q, our Current Reports on Form 8-K and amendments to those reports filed or furnished pursuant to Section 13(a) or 15(d) of the Securities Exchange Act of 1934, as amended (the "Exchange Act"), as soon as reasonably practicable after we file or furnish them electronically with the Securities and Exchange Commission ("SEC").

Investors and others should note that we announce material financial and other information to our investors using our investor relations website (*ir.netflix.net*), SEC filings, press releases, public conference calls and webcasts. We use these channels as well as social media and blogs to communicate with our members and the public about our company, our services and other issues. It is possible that the information we post on social media and blogs could be deemed to be material information. Therefore, we encourage investors, the media, and others interested in our company to review the information we post on the social media channels and blogs listed on our investor relations website.

Item 1A. Risk Factors

If any of the following risks actually occur, our business, financial condition and results of operations could be harmed. In that case, the trading price of our common stock could decline, and you could lose all or part of your investment.

Risks Related to Our Business

If our efforts to attract and retain members are not successful, our business will be adversely affected.

We must continually add new members both to replace canceled memberships and to grow our business beyond our current membership base. Our ability to continue to attract and retain members will depend in part on our ability to consistently provide our members in countries around the globe with compelling content choices that keep our members engaged with our service, effectively drive conversation around our content and service, as well as provide a quality experience for choosing and enjoying TV series, films and games. Our penetration and growth rates have fluctuated and vary across the jurisdictions where we provide our service. In countries where we have been operating for many years or where we are highly penetrated, our membership growth is slower than in newer or less penetrated countries. Furthermore, the relative service levels, content offerings, pricing and related features of competitors to our service may adversely impact our ability to attract and retain members. Competitors include other entertainment video providers, such as linear television, and streaming entertainment providers (including those that provide pirated content), video gaming providers, as well as user-generated content, some of which are by professional content creators, and more broadly other sources of entertainment, such as social media, that our members could choose in their moments of free time.

Members cancel our service for many reasons, including a perception that they do not use the service sufficiently, that they need to cut household expenses, dissatisfaction with content, a preference for competitive services and customer service issues that they believe are not satisfactorily resolved. Membership growth is also impacted by seasonality, with the fourth quarter historically representing our greatest growth, as well as the timing of our content release schedules. Adverse macroeconomic conditions, including inflation, may also adversely impact our ability to attract and retain members. If we do not grow as expected, given, in particular, that our content costs are largely fixed in nature, we may not be able to adjust our expenditures or increase our (per membership) revenues, including by adjusting membership pricing, commensurate with the lowered growth rate such that our margins, liquidity and results of operations may be adversely impacted. If we are unable to successfully compete with current and new competitors in providing compelling content, retaining our existing members and attracting new members, our business will be adversely affected.

If we do not continuously provide value to our members, including making improvements to our service in a manner that is favorably received by them, our revenue, results of operations and business will be adversely affected.

If consumers do not perceive our service offering to be of value, including if we introduce new or adjust existing features, adjust pricing or service offerings, or change the mix of content in a manner that is not favorably received by them, we may not be able to attract and retain members, and accordingly, our revenue and results of operations may be adversely affected. We expanded our entertainment video offering to include games and, more recently, live programming. If our efforts to sustain and improve our existing TV and film offering, as well as develop and expand our video entertainment options, are not done in a manner valued by our current and future members, our ability to attract and retain members may be negatively impacted. We may also seek to extend our business into new products and services to help drive growth. For example, we continue to expand our offering of consumer products and live experiences. To the extent we cannot successfully find and develop new products, experiences and services to help drive growth, or we do not realize the expected benefits or generate additional revenue from such new products, experiences, and services, our future results of operations and growth may be adversely impacted.

We have and may, from time to time, adjust our membership pricing, our membership plans, or our pricing model itself, including for example, the lower-priced ad-supported subscription plan. Similarly, we have increased enforcement of and will continue to enforce our terms of use to limit multi-household usage and shared viewing outside of a household. These and other adjustments we make may not be well-received by consumers, and could negatively impact our ability to attract and retain members, revenue and our results of operations. In addition, many of our members rejoin our service or originate from word-of-mouth advertising from existing members. If our efforts to satisfy our existing members or adjustments to our service are not successful, we may not be able to attract or retain members, and as a result, our ability to maintain and/or grow our business will be adversely affected.

Changes in competitive offerings for entertainment video could adversely impact our business.

The market for entertainment is intensely competitive and subject to rapid change. Through new and existing distribution channels, consumers have increasing options to access entertainment video. The various economic models underlying these channels include subscription, transactional, ad-supported and piracy-based models. All of these have the potential to capture meaningful segments of the entertainment video market. We face competition from traditional providers of entertainment video, including broadcasters and cable network operators, as well as internet based e-commerce or entertainment video providers and platforms. Several of these competitors have long operating histories, large customer bases, strong brand recognition, exclusive rights to certain content, large content libraries, and significant financial, marketing and other resources. They may offer more compelling content or secure better terms from suppliers, adopt more aggressive pricing and devote more resources to product development, technology, infrastructure, content acquisitions and marketing. New entrants may enter the market or existing providers may adjust their services with unique offerings or approaches to providing entertainment video. In addition, new technological developments, including the development and use of generative artificial intelligence, are rapidly evolving. If our competitors gain an advantage by using such technologies more effectively to satisfy consumer demand, our ability to compete successfully and our results of operations could be adversely impacted. Companies also may enter into business combinations or alliances that strengthen their competitive positions. Piracy also threatens to damage our business, as its fundamental proposition to consumers is so compelling and difficult to compete against: virtually all content for free. In light of the compelling consumer proposition, piracy services are subject to rapid global growth, and our efforts to prevent that growth may be insufficient. If we are unable to successfully or profitably compete with current and new competitors and win moments of truth, our business will be adversely affected, and we may not be able to increase or maintain market share, revenues or profitability.

We face risks, such as unforeseen costs and potential liability in connection with content we acquire, produce, license and/or distribute through our service.

As a producer and distributor of content, we face potential liability for negligence, copyright and trademark infringement, or other claims based on the nature and content of materials that we acquire, produce, license and/or distribute. We also may face potential liability for content used in promoting our service, including marketing materials. We devote significant resources toward the development, production, marketing and distribution of original programming. We believe that original and exclusive programming can help differentiate our service from other offerings, enhance our brand and otherwise attract and retain members. To the extent our programming does not meet our expectations, in particular, in terms of costs, viewing and popularity, our business, including our brand and results of operations may be adversely impacted. As a content producer, we are responsible for production costs and other expenses, such as ongoing guild payments, and take on risks associated with production, such as completion and key talent risk. Further, negotiations or renewals related to entertainment industry collective bargaining agreements have, and in the future, could negatively impact timing and costs associated with our productions. We contract with third parties related to the development, production, marketing and distribution of our original programming. We may face potential liability or may suffer significant losses in connection with these arrangements, including but not limited to if such third parties violate applicable law, become insolvent or engage in fraudulent behavior. To the extent we create and sell physical or digital merchandise relating to our programming, and/or license such rights to third parties, we could become subject to product liability, intellectual property or other claims related to such merchandise. We may decide to remove content from our service, not to place licensed or produced content on our service or discontinue or alter production of original content if we believe such content might not be well received by our members, is prohibited by law, or could be damaging to our brand or business.

To the extent we do not accurately anticipate costs or mitigate risks, including for content that we obtain but ultimately does not appear on or is removed from our service, or if we become liable for content we acquire, produce, license and/or distribute, our business may suffer. Litigation to defend these claims could be costly and the expenses and damages arising from any liability or unforeseen production risks could harm our results of operations. We may not be indemnified against claims or costs of these types and we may not have insurance coverage for these types of claims.

If we are not able to manage change and growth, our business could be adversely affected.

We are expanding our operations, scaling our streaming service to effectively and reliably handle anticipated growth in both members and features related to our services, such as introducing games and advertising on our service, as well as offering live programming and expanding our consumer products and experiences. We are also scaling our own studio operations to produce original content. As our international offering evolves, we are managing and adjusting our business to address varied content offerings, consumer customs and practices, in particular those dealing with e-commerce and streaming video, as well as differing legal and regulatory environments. As we scale our streaming service and introduce new features such as our ad-supported subscription plan and live programming, we are developing technology and utilizing third-party “cloud” computing, technology and other services. As we scale our content production, including games, and introduce new features and expand our consumer products and experience offerings, we are building out expertise in a number of disciplines, including creative, marketing, legal, finance, licensing, merchandising and other resources, which requires significant resources and management attention. Further, we may expand our content and service offerings in a manner that is not well received by consumers. As we grow our operations, we may face integration and operational challenges as well as potential unknown liabilities and reputational concerns in connection with partners we work with or companies we may acquire or control. If we are not able to manage the growing complexity of our business, including improving, refining or evolving our corporate culture, as well as our systems and operational practices related to our streaming operations and original content, our business may be adversely affected.

If we fail to maintain a positive reputation concerning our service and the content we offer, we may not be able to attract or retain members, we may face regulatory scrutiny and our operating results may be adversely affected.

We believe that a positive reputation concerning our service is important in attracting and retaining members. To the extent our content is perceived as low quality, offensive or otherwise not compelling to consumers, our ability to establish and maintain a positive reputation may be adversely impacted. To the extent our content is deemed controversial or offensive by government regulators, we may face direct or indirect retaliatory action or behavior, including being required to remove such content from our service, our entire service could be banned and/or become subject to heightened regulatory scrutiny across our business and operations. We could also face boycotts which could adversely affect our business. Furthermore, to the extent our response to government action or our marketing, customer service and public relations efforts are not effective or result in negative reaction, our ability to establish and maintain a positive reputation may likewise be adversely impacted. There is an increasing focus from regulators, investors, members and other stakeholders on environmental, social, and governance (“ESG”) matters, both in the United States and internationally, including the adoption of new disclosure and regulatory frameworks. To the extent we are unable to meet regulatory or industry standards or investor expectations on ESG issues or the content we distribute and the manner in which we produce content creates ESG related concerns, our reputation may be harmed.

Our business could be adversely impacted by costs and challenges associated with strategic acquisitions and investments.

From time to time, we acquire or invest in businesses, content, and technologies that support our business. The risks associated with such acquisitions or investments (some of which may be unforeseen) include the difficulty of integrating solutions, operations, and personnel; inheriting liabilities and exposure to litigation; failure to realize anticipated benefits and expected synergies; and diversion of management’s time and attention, among other acquisition-related risks.

We may not be successful in overcoming such risks, and such acquisitions and investments may negatively impact our business. In addition, if we do not complete an announced acquisition transaction or integrate an acquired business successfully and in a timely manner, we may not realize the benefits of the acquisition to the extent anticipated. Acquisitions and investments may contribute to fluctuations in our quarterly financial results. These fluctuations could arise from transaction-related costs and charges associated with eliminating redundant expenses or write-offs of impaired assets recorded in connection with acquisitions and investments, and could negatively impact our financial results.

We rely upon a number of partners to make our service available on their devices.

We currently offer members the ability to receive streaming content through a host of internet-connected devices, including TVs, digital video players, TV set-top boxes and mobile devices. We have agreements with various cable, satellite and telecommunications operators to make our service available through the TV set-top boxes of these service providers, some of which compete directly with us or have investments in competing streaming content providers. In many instances, our agreements also include provisions by which the partner bills consumers directly for the Netflix service or otherwise offers services or products in connection with offering our service. If partners or other providers do a better job of connecting consumers with content they want to watch, for example through multi-service discovery interfaces, our service may be adversely impacted. We intend to continue to broaden our relationships with existing partners and to increase our capability to stream content and offer games to other platforms and partners over time. If we are not successful in maintaining existing and creating new relationships, or if we encounter technological, content licensing, regulatory, business or other impediments to delivering our streaming content to our members via these devices, our ability to retain members and grow our business could be adversely impacted.

Our agreements with our partners are typically between one and three years in duration and our business could be adversely affected if, upon expiration, a number of our partners do not continue to provide access to our service or are unwilling to do so on terms acceptable to us, which terms may include the degree of accessibility and prominence of our service. Furthermore, devices are manufactured and sold by entities other than Netflix and while these entities should be responsible for the devices’ performance, the connection between these devices and our service may nonetheless result in consumer dissatisfaction toward us and such dissatisfaction could result in claims against us or otherwise adversely impact our business. In addition, technology changes to our streaming functionality may require that partners update their devices, and from time to time, lead to us to stop supporting the delivery of our service on certain legacy devices. If partners do not update or otherwise modify their devices, or if we discontinue support for certain devices, our service and our members' use and enjoyment could be negatively impacted.

We are subject to payment processing risk.

Our members pay for our service using a variety of different payment methods, including credit and debit cards, gift cards, prepaid cards, direct debit, online wallets and direct carrier and partner billing. We rely on internal systems and those of third parties to process payment. Acceptance and processing of these payment methods are subject to certain rules, regulations, and industry standards, including data storage requirements, additional authentication requirements for certain payment methods, and require payment of interchange and other fees. To the extent there are increases in payment processing fees, material changes in the payment ecosystem, such as large re-issuances of payment cards, delays in receiving payments from payment processors, changes to rules, regulations or industry standards concerning payments, loss of payment partners and/or disruptions or failures in our payment processing systems, partner systems or payment products, including products we use to update payment information, our revenue, operating expenses and results of operations could be adversely impacted. In certain instances, we leverage third parties such as our cable and other partners to bill members on our behalf. If these third parties become unwilling or unable to continue processing payments on our behalf, we would have to transition members or otherwise find alternative methods of collecting payments, which could adversely impact member acquisition and retention. In addition, from time to time, we encounter fraudulent use of payment methods, which could impact our results of operations and if not adequately controlled and managed could create negative consumer perceptions of our service. If we are unable to maintain our fraud and chargeback rate at acceptable levels, card networks may impose fines, our card approval rate may be impacted and we may be subject to additional card authentication requirements. The termination of our ability to process payments on any major payment method would significantly impair our ability to operate our business.

If government regulations relating to the internet or other areas of our business change, we may need to alter the manner in which we conduct our business, or incur greater operating expenses.

The adoption or modification of laws or regulations relating to the internet or other areas of our business could limit or otherwise adversely affect the manner in which we currently conduct our business. As our service and others like us gain traction in international markets, governments are increasingly looking to introduce new or extend legacy regulations to these services, in particular those related to broadcast media and tax. For example, European law enables individual member states to impose levies and other financial obligations on media operators that provide services in their jurisdiction. Several European Union (EU) Member States have and others may, over time, impose financial and regulatory obligations on us. In addition, the continued growth and development of the market for online commerce may lead to more stringent consumer protection laws, which may impose additional burdens on us. Rules governing new technological developments, including generative artificial intelligence, are nascent and rapidly evolving such that the impact on areas related to our business remains uncertain. For example, in Europe, the Digital Markets Act remains subject to non-compliance investigations, the result of which could change how we interact with digital gatekeepers like Apple and Google. If we are required to comply with new regulations or legislation or new interpretations of existing regulations or legislation, this compliance could cause us to incur additional expenses or alter our business model.

Changes in laws or regulations that adversely affect the growth, popularity or use of the internet, including laws impacting net neutrality, requiring payment of network access fees or payment of network support taxes, could decrease the demand for our service and increase our cost of doing business. Certain laws intended to prevent network operators from discriminating against the legal traffic that traverse their networks have been implemented in many countries, including across the EU and several U.S. states. In others, the laws may be nascent, evolving or non-existent. For example, a U.S. federal appeals court recently overturned the Federal Communications Commission's net neutrality rules. Given uncertainty around these rules, including changing interpretations, amendments or repeal, coupled with potentially significant political and economic power of local network operators, we could experience discriminatory or anti-competitive practices that could impede our growth, cause us to incur additional expense or otherwise negatively affect our business.

We are engaged in legal proceedings that could cause us to incur unforeseen expenses and could occupy a significant amount of our management's time and attention.

From time to time, we are subject to litigation or claims that could negatively affect our business operations and financial position. As we have grown, we have seen a rise in the number of litigation matters against us. These matters have included copyright and other claims related to our content, products, patent infringement claims, tax litigation, employment related litigation, as well as consumer and securities class actions, each of which are typically expensive to defend. Litigation disputes could cause us to incur unforeseen expenses, result in content unavailability, service disruptions, and otherwise occupy a significant amount of our management's time and attention, any of which could negatively affect our business operations and financial position. We also receive inquiries and subpoenas and other types of information requests from government authorities, and we may become subject to related claims and other actions related to our business activities. While the ultimate outcome of investigations, inquiries, information requests and related legal proceedings is difficult to predict, such matters can be expensive, time-consuming and distracting, and adverse resolutions or settlements of those matters may result in, among other things, modification of our business practices, reputational harm or costs and significant payments, any of which could negatively affect our business operations and financial position.

Our advertising offering is new and subject to various risks and uncertainties, which may adversely affect our business.

We have limited experience and operating history offering advertising on our service, and our advertising revenue may not grow as we expect. Our ability to generate advertising revenue is subject to various risks and will depend on a number of factors, including:

- our ability to attract and retain advertisers;
- fluctuations in memberships, including those selecting the ad-supported subscription plan, and member engagement;
- the quantity or quality of ads shown to our members;
- our ability to compete effectively for advertising spend;
- the impact of seasonal, cyclical or other shifts in advertising spend, including the impact of macroeconomic conditions;
- the availability, accuracy, utility, and security of analytics and measurement solutions offered by us or third parties that demonstrate the value of our ads to marketers, or our ability to further improve such tools;
- changes in the way advertising on devices, connected TVs or on personal computers is measured or priced;
- adverse legal developments relating to advertising or measurement tools;
- changes in third-party policies, which may negatively impact the ability to measure, deliver and select ads to be served;
- regulatory, legislative and industry developments relating to the collection and use of information and other privacy considerations, including regulations related to ad targeting and measurement tools;
- any liability or reputational harm from advertisements shown on our service;

- our relationship with third-party service providers for the management, operation, sale and technology to support advertisements on our service;
- our ability to develop and expand an advertising sales and advertising technology organization team;
- our ability to develop the technology and related infrastructure to support advertising and drive value to advertisers;
- the impact of our content and reputation on advertisers' willingness to spend with us; and
- any member dissatisfaction due to advertisements.

Risks Related to Intellectual Property

If studios, content providers or other rights holders refuse to license streaming content or other rights upon terms acceptable to us, our business could be adversely affected.

Our ability to provide our members with content they can enjoy depends on obtaining various rights from third parties upon terms acceptable to us, including necessary distribution rights, to such content and certain related elements thereof, such as the public performance of music contained within the content we distribute. The license periods and the terms and conditions of such rights vary. As content providers develop their own streaming services, they may be unwilling to provide us with access to certain content, including popular series or movies. If the studios, content providers and other rights holders are not or are no longer willing or able to license us content upon terms acceptable to us, our ability to stream content to our members may be adversely affected and/or our costs could increase. Certain licenses for content provide for the studios or other content providers to withdraw content from our service relatively quickly. Because of these provisions and other actions we may take, content available through our service can be withdrawn on short notice. As competition increases, we see the cost of certain programming increase. As we seek to differentiate our service, we are often focused on securing certain exclusive rights when obtaining content, including original content. We are also focused on programming an overall mix of content that delights our members in a cost efficient manner. Within this context, we are selective about the titles we add and renew to our service. If we do not maintain a compelling mix of content, our member acquisition and retention may be adversely affected.

Music and certain authors' performances contained within content we distribute may require us to obtain licenses for such distribution. In this regard, we engage in negotiations with collection management organizations ("CMOs") and similar entities that hold certain rights to music and/or other interests in intellectual property (e.g., remuneration rights) in connection with streaming content into various territories. If we are unable to reach mutually acceptable terms with these organizations, we could become involved in litigation and/or could be enjoined from distributing certain content, which could adversely impact our business. Additionally, pending and ongoing litigation and negotiations between certain CMOs and other third parties in various territories could adversely impact our negotiations with CMOs, or result in music publishers represented by certain CMOs unilaterally withdrawing rights, and thereby adversely impact our ability to reach licensing agreements reasonably acceptable to us. Failure to reach such licensing agreements could expose us to potential liability for copyright infringement or otherwise increase our costs. Additionally, as the market for the digital distribution of content grows, a broader role for CMOs in the remuneration of authors, performers and other beneficiaries of neighboring rights is likely to expose us to greater distribution expenses in certain markets.

If our trademarks and other proprietary rights are not adequately protected to prevent use or appropriation by third parties, the value of our brand and other intangible assets may be diminished, and our business may be adversely affected.

We rely and expect to continue to rely on a combination of confidentiality and license agreements with our employees, consultants and third parties with whom we have relationships, as well as trademark, copyright, patent and trade secret protection laws, to protect our proprietary rights. We may also seek to enforce our proprietary rights through court proceedings or other legal actions. We have filed and we expect to file from time to time for trademark, copyright, and patent applications. Nevertheless, these applications may not be approved, third parties may challenge any copyrights, patents or trademarks issued to or held by us, third parties may knowingly or unknowingly infringe our intellectual property rights, and we may not be able to prevent infringement or misappropriation without substantial expense to us. If the protection of our intellectual property rights is inadequate to prevent use or misappropriation by third parties, the value of our brand, content, and other intangible assets may be diminished, competitors may be able to more effectively mimic our service and methods of operations, the perception of our business and service to members and potential members may become confused in the marketplace, and our ability to attract members may be adversely affected.

We currently hold various domain names relating to our brand, including Netflix.com. Failure to protect our domain names could adversely affect our reputation and brand and make it more difficult for users to find our website and our service. We may be unable, without significant cost or at all, to prevent third parties from acquiring domain names that are similar to, infringe upon or otherwise decrease the value of our trademarks and other proprietary rights.

Intellectual property claims against us could be costly and result in the loss of significant rights related to, among other things, our technology, business processes, and content.

Trademark, copyright, patent and other intellectual property rights are important to us and other companies. Our intellectual property rights extend to our technology, business processes, the content we produce and distribute through our service, and consumer products, experiences, and marketing assets based thereon. We use the intellectual property of third parties in creating some of our content, merchandising our products and experiences, and marketing our service. From time to time, third parties allege that we have infringed or otherwise violated their intellectual property rights. If we are unable to obtain sufficient rights, successfully defend our use, or develop non-infringing technology or otherwise alter our business practices on a timely basis in response to claims against us for infringement, misappropriation, misuse or other violation of third-party intellectual property rights, our business and competitive position may be adversely affected. In addition, the use or adoption of new and emerging technologies may increase our exposure to intellectual property claims, and the availability of copyright and other intellectual property protection for AI-generated material is uncertain. Many companies are devoting significant resources to developing patents that could potentially affect many aspects of our business. There are numerous patents that broadly claim means and methods of conducting business on the internet. We have not searched for patents relative to our technology. Defending ourselves against intellectual property claims, whether they are with or without merit or are determined in our favor, results in costly litigation and diversion of technical and management personnel. It also may result in our inability to use our current technology and products, our recommendation and merchandising technology or inability to market our service or merchandise our products. We may also have to remove content from our service, or remove consumer products or marketing materials from the marketplace. As a result of a dispute, we may have to develop non-infringing technology, enter into royalty or licensing agreements, adjust our content, merchandising or marketing activities or take other actions to resolve the claims. These actions, if required, may be costly or unavailable on terms acceptable to us.

Risks Related to Information Technology

Any significant disruption in or unauthorized access to our computer systems or those of third parties that we utilize in our operations, including those relating to cybersecurity or arising from cyber-attacks, could result in a loss or degradation of service, unauthorized access, disclosure or destruction of data, including member and corporate information, or theft of intellectual property, including digital content assets, which could adversely impact our business.

Our reputation and ability to attract, retain and serve our members is dependent upon the reliable performance and security of our computer systems and those of third parties that we utilize in our operations. These systems may be subject to damage or interruption from, among other things, earthquakes, adverse weather conditions, other natural disasters, public health issues such as pandemics or epidemics, terrorist attacks, rogue employees, power loss, telecommunications failures, cybersecurity risks and incidents, and other interruptions beyond our control. Interruptions in these systems, or with the internet in general, could make our service unavailable or degraded or otherwise hinder our ability to deliver our service. Service interruptions, errors in our software or the unavailability of computer systems or data used in our operations could diminish the overall attractiveness of our service to existing and potential members.

Our computer systems and those of third parties we use in our operations are subject to constantly evolving cybersecurity threats, including cyber-attacks such as computer viruses, malware, ransomware, denial of service attacks, physical or electronic break-ins, or insider threats, as well as misconfigurations in information systems, networks, software or hardware, and similar disruptions or errors. These systems and those of third parties with which we do business have experienced and may continue to experience directed attacks intended to lead to interruptions and delays in our service and operations as well as loss, misuse or theft of personal information (of third parties, employees, and our members) and other data, confidential information or intellectual property. We and many of the third parties we work with rely on open source software and libraries that are integrated into a variety of applications, tools and systems, which may increase our exposure to vulnerabilities. The addition of new features or upgrades also increases our exposure to vulnerabilities, and generative artificial intelligence could intensify these cybersecurity risks. Additionally, outside parties may attempt to induce employees, vendors, partners, or users to disclose sensitive or confidential information in order to gain access to data. Any attempt by hackers to obtain our data (including member and corporate information) or intellectual property (including digital content assets), disrupt our service, or otherwise access our systems, or those of third parties we use, if successful, could harm our business, be expensive to remedy and damage our reputation. We have implemented certain systems and processes to thwart hackers and protect our data and systems. However, the techniques used to gain unauthorized access to data and software are constantly evolving, and we may be unable to anticipate, detect or prevent unauthorized access or address all cybersecurity incidents that occur. Because of our prominence, we (and/or third parties we use) have been and may continue to be a particularly attractive target for such attacks, and from time to time, we have experienced unauthorized releases of certain digital content assets and unintended disclosure of personal information due to incidents related to third parties. However, to date these unauthorized releases have not had a material impact on our service, systems or business. There is no assurance that hackers may not have a material impact on our service or systems in the future. We do not carry insurance to cover expenses related to such disruptions or unauthorized access. Efforts to prevent hackers from disrupting our service or otherwise accessing our systems are expensive to develop, implement and maintain. These efforts require ongoing monitoring and updating as technologies change and efforts to overcome security measures become more sophisticated, and may limit the functionality of or otherwise negatively impact our service offering and systems. Any significant disruption to our service or access to our systems could result in a loss of members, damage our reputation, and adversely affect our business and results of operations. Further, a penetration of our systems or a third-party's systems or other misappropriation or misuse of personal information could subject us to business, regulatory, litigation and reputation risk, which could have a negative effect on our business, financial condition and results of operations.

We utilize our own communications and computer hardware systems located either in our facilities or in that of a third-party provider. In addition, we utilize third-party "cloud" computing services in connection with our business operations. We also utilize our own and third-party content delivery networks ("CDN") to help us stream content and offer games in high volume to Netflix members over the internet. Problems faced by us or our third-party “cloud” computing or other network providers, including technological or business-related disruptions, as well as cybersecurity threats and regulatory interference, could adversely impact the experience of our members.

We rely upon Amazon Web Services to operate certain aspects of our service and any disruption of or interference with our use of the Amazon Web Services operation would impact our operations and our business would be adversely impacted.

Amazon Web Services (“AWS”) provides a distributed computing infrastructure platform for business operations, or what is commonly referred to as a "cloud" computing service. We have architected our software and computer systems so as to utilize data processing, storage capabilities and other services provided by AWS. Currently, we run the vast majority of our computing on AWS. Given this, along with the fact that we cannot easily switch our AWS operations to another cloud provider, any commercial disputes related to, disruption of or interference with our use of AWS would impact our operations and our business would be adversely impacted. While the retail side of Amazon competes with us, we do not believe that Amazon will use the AWS operation in such a manner as to gain competitive advantage against our service, although if it were to do so it could harm our business.

If the technology we use in operating our business fails, is unavailable, or does not operate to expectations, our business and results of operations could be adversely impacted.

We utilize a combination of proprietary and third-party technology to operate our business. This includes the technology that we have developed to recommend and merchandise content to our consumers as well as enable fast and efficient delivery of content to our members and their various consumer electronic devices. For example, we have built and deployed our own CDN. To the extent Internet Service Providers (“ISPs”) do not interconnect with our CDN or charge us to access their networks, or if we experience difficulties in our CDN’s operation, our ability to efficiently and effectively deliver our streaming content to our members could be adversely impacted and our business and results of operations could be adversely affected. Likewise, if our recommendation and merchandising technology does not enable us to predict and recommend titles that our members will enjoy, our ability to attract and retain members may be adversely affected. We also utilize third-party technology to help market our service, process payments, and otherwise manage the daily operations of our business. If our technology or that of third-parties we utilize in our operations fails or otherwise operates improperly, including as a result of “bugs” or other errors in our development and deployment of software, our ability to operate our service, retain existing members and add new members may be impaired. Any harm to our members’ devices caused by software used in our operations could have an adverse effect on our business, results of operations and financial condition.

Changes in how network operators handle and charge for access to data that travel across their networks could adversely impact our business.

We rely upon the ability of consumers to access our service through the internet. If network operators block, restrict or otherwise impair access to our service over their networks, our service and business could be negatively affected. To the extent that network operators implement usage based pricing, including meaningful bandwidth caps, or otherwise try to monetize access to their networks by data providers, we could incur greater operating expenses and our member acquisition and retention could be negatively impacted. Furthermore, to the extent network operators create tiers of internet access service and either charge us for or prohibit us from being available through these tiers, our business could be negatively impacted.

Most network operators that provide consumers with access to the internet also provide these consumers with multichannel video programming. As such, many network operators have an incentive to use their network infrastructure in a manner adverse to our continued growth and success. While we believe that consumer demand, regulatory oversight and competition will help counterbalance these incentives, to the extent that network operators are able to provide preferential treatment to their data as opposed to ours or otherwise implement discriminatory network management practices, our business could be negatively impacted. The extent to which these incentives limit operator behavior differs across markets.

Risks Related to Privacy

Privacy concerns could limit our ability to collect and leverage member personal information and disclosure of member personal information could adversely impact our business and reputation.

In the ordinary course of business and in particular in connection with content acquisition, merchandising our service to our members and our advertising offering, we collect and utilize information supplied by our members and third parties, which may include personal information and other data. We are subject to laws, rules and regulations relating to privacy and the collection, use and security of personal information, including but not limited to Regulation (EU) 2016/679 (also known as the General Data Protection Regulation or “GDPR”) and the California Privacy Rights Act ("CPRA"). Any actual or perceived failure to comply with the GDPR, the California Consumer Privacy Act/CPRA, other data privacy laws or regulations, or related contractual or other obligations, or any perceived privacy rights violation, have and could in the future lead to investigations, claims, and proceedings by governmental entities and private parties, which to date have not been material but may result in significant damages for contract breach, and other significant costs, penalties, and other liabilities, as well as harm to our reputation and market position.

Other businesses have been criticized by privacy groups and governmental bodies for attempts to link personal identities and other information to data collected on the internet regarding users' browsing and other habits. Increased regulation of data utilization practices, including self-regulation or findings under existing laws that limit our ability to collect, transfer and use personal information, could have an adverse effect on our business. In addition, if we were to disclose personal information about our members in a manner that was objectionable to them, our business reputation could be adversely affected, and we could face potential legal claims that could impact our operating results. Internationally, we may become subject to additional and/or more stringent legal obligations concerning our treatment of member and other personal information, such as laws regarding data localization, data transfer and/or restrictions on data export. Failure to comply with these obligations could subject us to liability, and to the extent that we need to alter our business model or practices to adapt to these obligations, we could incur additional expenses.

Our reputation and relationships with members would be harmed if member personal information, particularly billing data, were to be accessed by unauthorized persons.

We maintain personal information regarding our members, including names, age, gender and billing information. This personal information is maintained on our own systems as well as that of third parties we use in our operations. With respect to billing information, such as credit card numbers, we rely on encryption and authentication technology to secure such information. We take measures to protect against unauthorized intrusion into our members' information. Despite these measures and technologies we, our payment processing services or other third-party services we use such as AWS, could experience an unauthorized intrusion into our members' information. In the event of such a breach, current and potential members may become unwilling to provide the information to us necessary for them to remain or become members. We have notified and may in the future be required to notify regulators about any actual or perceived data breach (including various state Attorneys General, one or more EU data protection authorities, or other data protection authorities) as well as the individuals who are affected by the incident within strict time periods. Additionally, we could face legal claims or regulatory fines or penalties for such a breach. The costs relating to any data breach could be material, and we currently do not carry insurance against the risk of a data breach. We also maintain personal information concerning our employees, as well as personal information of others working on our productions. Should an unauthorized intrusion into our members' or employees' personal information and/or production personal information occur, our business could be adversely affected and our larger reputation with respect to data protection could be negatively impacted.

Risks Related to Liquidity

The long-term and largely fixed cost nature of our content commitments may limit our operating flexibility and could adversely affect our liquidity and results of operations.

In connection with licensing streaming content, we typically enter into multi-year commitments with studios and other content providers. We also enter into multi-year commitments for content that we produce, either directly or through third parties, including elements associated with these productions such as non-cancelable commitments under talent agreements. The payment terms of these agreements are not tied to member usage or the size of our membership base ("fixed cost") but may be determined by costs of production or tied to such factors as titles licensed and/or theatrical exhibition receipts. Such commitments, to the extent estimable under accounting standards, are included in the Contractual Obligations section of Part II, Item 7, "Management's Discussion and Analysis of Financial Condition and Results of Operations" and Note 8, Commitments and Contingencies in the accompanying notes to our consolidated financial statements included in Part II, Item 8, "Financial Statements and Supplementary Data" of this Annual Report on Form 10-K. Given the multiple-year duration and largely fixed cost nature of content commitments, if business performance does not meet our expectations, our margins may be adversely impacted. Further, we may be unable to react to any reduction in our cash flows from operations, including those caused by a downturn in the economy, by reducing our streaming content obligations in the near-term. Payment terms for certain content commitments, such as content we directly produce, will typically require more up-front cash payments than other content licenses or arrangements whereby we do not fund the production of such content. To the extent revenue growth does not meet our expectations, our liquidity and results of operations could be adversely affected as a result of content commitments and accelerated payment requirements of certain agreements. In addition, the long-term and largely fixed cost nature of our content commitments may limit our flexibility in planning for, or reacting to changes in our business and the market segments in which we operate. If we license and/or produce content that is not favorably received by consumers in a territory, or is unable to be shown in a territory, acquisition and retention may be adversely impacted and given the long-term and fixed cost nature of our content commitments, we may not be able to adjust our content offerings quickly and our results of operations may be adversely impacted.

We may seek additional capital that may result in stockholder dilution or that may have rights senior to those of our common stockholders.

From time to time, we may seek to obtain additional capital, either through equity, equity-linked or debt securities. The decision to obtain additional capital will depend on, among other things, our business plans, operating performance and condition of the capital markets. Rising interest rates or any disruption in the capital markets could make it more difficult and expensive for us to raise additional capital or refinance our existing indebtedness. If we raise additional funds through the issuance of equity, equity-linked or debt securities, those securities may have rights, preferences or privileges senior to the rights of our common stock, and our stockholders may experience dilution. Any large equity or equity-linked offering could also negatively impact our stock price.

We have a substantial amount of indebtedness and other obligations, including streaming content obligations, which could adversely affect our financial position, and we may not be able to generate sufficient cash to service our debt and other obligations.

We have a substantial amount of indebtedness and other obligations, including streaming content obligations. Moreover, we may incur additional indebtedness in the future and incur other obligations, including additional streaming content obligations. Our ability to make payments on our debt and other obligations will depend on our financial and operating performance, which is subject to prevailing economic and competitive conditions and to certain financial, business and other factors beyond our control. If we are unable to service our debt and other obligations from cash flows, we may need to refinance or restructure all or a portion of such obligations prior to maturity. If the financial markets become difficult or costly to access, including due to rising interest rates, fluctuations in foreign currency exchange rates or other changes in economic conditions, our ability to raise additional capital may be negatively impacted, and any refinancing or restructuring could be at higher interest rates and may require us to comply with more onerous covenants, which could further restrict our business operations.

As of December 31, 2024, we had the equivalent of $15.7 billion aggregate principal amount of senior notes outstanding (“Notes”), some of which is denominated in currencies other than the U.S. dollar. In addition, we have entered into a revolving credit agreement that provides for a $3 billion unsecured revolving credit facility. As of December 31, 2024, we have not borrowed any amount under this revolving credit facility. As of December 31, 2024, we had approximately $6.2 billion of total content liabilities as reflected on our consolidated balance sheet, some of which is denominated in currencies other than the U.S. dollar. Such amount does not include streaming content commitments that do not meet the criteria for liability recognition, the amounts of which are significant. Our substantial indebtedness and other obligations, including streaming content obligations, may require us to use a substantial portion of our cash flow from operations to make debt service payments and pay our other obligations when due, limit our ability to borrow additional funds, limit our flexibility to plan for, or react to, changes in our business and industry and place us at a competitive disadvantage compared to our less leveraged competitors. For more information on our streaming content obligations, including those not on our consolidated balance sheet, see Note 8, *Commitments and Contingencies,* in the accompanying notes to our consolidated financial statements included in Part II, Item 8, "Financial Statements and Supplementary Data" of this Annual Report on Form 10-K.

Risks Related to International Operations

We could be subject to economic, political, regulatory and other risks arising from our international operations.

Operating in international markets requires significant resources and management attention and will subject us to economic, political, regulatory and other risks that may be different from or incremental to those in the U.S. In addition to the risks that we face in the U.S., our international operations involve risks that could adversely affect our business, including:

- the need to adapt our content and user interfaces for specific cultural and language differences;
- difficulties and costs associated with staffing and managing foreign operations;
- political or social unrest, global hostilities, and economic instability;
- compliance with laws such as the Foreign Corrupt Practices Act, UK Bribery Act and other anti-corruption laws, export controls and economic sanctions, and local laws prohibiting corrupt payments to government officials;
- difficulties in understanding and complying with local laws, regulations and customs in foreign jurisdictions;
- regulatory requirements or government action, whether in response to enforcement of actual or purported legal and regulatory requirements or otherwise, that results in disruption or non-availability of our service or particular content, administrative enforcement actions, fines, and civil and criminal liability, or increased operating costs in the applicable jurisdiction;
- application of foreign intellectual property laws, which requires the business to adapt to bespoke compliance rules, or changes to such laws, among other issues, may impact the economics of creating or distributing content, anti-piracy efforts, or our ability to protect or exploit intellectual property rights;
- adverse tax consequences such as those related to changes in tax laws or tax rates or their interpretations, and the related application of judgment in determining our global provision for income taxes, deferred tax assets or liabilities or other tax liabilities given the ultimate tax determination is uncertain;
- fluctuations in currency exchange rates, which have and may continue to impact revenues and expenses of our international operations and expose us to foreign currency exchange rate risk, and while we use derivative and non-derivative instruments to hedge certain exposures to fluctuations in exchange rates, the use of such hedging activities may not be effective in offsetting an adverse financial impact and may introduce or heighten counterparty risk and we may choose not to hedge certain exposures;
- rates of inflation;
- profit repatriation, currency control regulations and other restrictions on the transfer of funds;
- differing payment processing systems as well as consumer use and acceptance of electronic payment methods, such as payment cards;
- new and different sources of competition;

- censorship requirements that cause us to remove or edit popular content, leading to consumer disappointment, brand tarnishment or dissatisfaction with our service;
- low usage and/or penetration of internet-connected consumer electronic devices;
- different and more stringent user protection, data protection, privacy and other laws, including data localization and/or restrictions on data export, and local ownership requirements;
- availability of reliable broadband connectivity and wide area networks in targeted areas for expansion;
- differing laws and consumer understanding/attitudes regarding the illegality of piracy;
- negative impacts from trade disputes; and
- implementation of regulations designed to stimulate the local production of film and TV series in order to promote and preserve local culture and economic activity, including local content quotas, investment obligations, and levies to support local film funds. For example, the EU revised its Audio Visual Media Services Directive in 2018 to require that European works comprise at least thirty percent (30%) of media service providers' catalogs, and to require prominence of those works, and certain EU Member States have imposed additional investment obligations and levies.

These and other factors may cause us to adjust our business plans, including expanding or ceasing certain operations in certain countries, and the execution of our strategies. Our failure to manage any of these risks successfully could harm our international operations and our overall business, and results of our operations.

We are subject to taxation related risks in multiple jurisdictions.

We are a U.S.-based multinational company subject to tax in multiple U.S. and foreign tax jurisdictions. Significant judgment is required in determining our global provision for income taxes, deferred taxes and other tax liabilities and receivables, and in evaluating our tax positions and other tax attributes on a worldwide basis. We are subject to the periodic examination of our domestic and foreign tax returns by the Internal Revenue Service, state, local, and foreign tax authorities, some of whom are challenging our tax positions. We regularly assess the likelihood of adverse outcomes from these examinations in determining the adequacy of our provision for income taxes and other tax liabilities. We consider many factors when evaluating and estimating our tax positions and tax benefits, which may require periodic adjustments and may not accurately forecast actual tax audit outcomes. If the ultimate determination of income and other tax liabilities differ from the amounts recorded or accrued, our business, financial condition or results of operations may be adversely impacted.

Tax laws are regularly being re-examined and evaluated globally. New laws and interpretations of the law are taken into account for financial statement purposes in the quarter or year that they become applicable. Tax authorities are increasingly scrutinizing the tax positions of companies and we have tax audits pending in several jurisdictions. The U.S. federal and state governments, countries in the EU, as well as a number of other countries and organizations such as the Organization for Economic Cooperation and Development, are actively considering changes to existing tax laws that, if enacted, could increase our tax obligations in jurisdictions where we do business. If U.S. or other tax authorities change applicable tax laws or successfully challenge how or where our profits are currently recognized, our overall taxes could increase, and our business, financial condition or results of operations may be adversely impacted.

Risks Related to Human Resources

We may lose key employees or may be unable to hire qualified employees, and the failure to maintain and improve our company culture may adversely affect our business.

We rely on the continued service of our senior management, including our Co-Chief Executive Officers, Ted Sarandos and Greg Peters, members of our executive team and other key employees. In our industry, there is substantial and continuous competition for highly-skilled business, product development, technical, creative and other personnel. If we experience high executive turnover, fail to adapt our business practices to industry expectations, fail to maintain a positive reputation, fail to implement succession plans for key employees, encounter difficulties associated with the transition of members of our management team, are not successful in recruiting new personnel or in retaining and motivating existing personnel, in instilling our culture in new employees, or maintaining and improving our culture as we grow, our operations may be disrupted, which could adversely affect our results of operations.

Labor disputes may have an adverse effect on the Company's business.

We and our partners, suppliers, and vendors engage writers, directors, actors, other talent, trade employees and others who are subject to collective bargaining agreements in the motion picture industry, both in the U.S. and internationally. Expiring collective bargaining agreements may be renewed on terms that are unfavorable to us. If expiring collective bargaining agreements cannot be renewed, affected unions have, and could in the future, take action in the form of strikes or work stoppages. Such work stoppages have resulted, and may in the future result, in halted productions and delays in our ability to provide new content to our members. Such actions, as well as higher costs or operating complexities in connection with these collective bargaining agreements or a significant labor dispute, could have an adverse effect on our business by causing delays in production, added costs or by reducing profit margins, and our ability to provide new content to our members could likewise be delayed or dropped.

Risks Related to Our Stock Ownership

Provisions in our charter documents and under Delaware law could discourage a takeover that stockholders may consider favorable, although we have announced plans to modify some of these provisions over time.

Our charter documents in their current form may discourage, delay or prevent a merger or acquisition that a stockholder may consider favorable because they:

- authorize our board of directors, without stockholder approval, to issue up to 10,000,000 shares of undesignated preferred stock;
- provide for a classified board of directors until our annual meeting of stockholders to be held in 2025;
- prohibit our stockholders from acting by written consent; and
- establish advance notice requirements for proposing matters to be approved by stockholders at stockholder meetings.

As a Delaware corporation, we are also subject to certain Delaware anti-takeover provisions. Under Delaware law, a corporation may not engage in a business combination with any holder of 15% or more of its capital stock unless the holder has held the stock for three years or, among other things, the board of directors has approved the transaction. Our board of directors could rely on Delaware law to prevent or delay an acquisition of us.

In addition, a merger or acquisition may trigger retention payments to certain executive employees under the terms of our Amended and Restated Executive Severance and Retention Incentive Plan, thereby increasing the cost of such a transaction.

Our stock price is volatile.

The price at which our common stock has traded has fluctuated significantly. The price may continue to be volatile due to a number of factors including the following, some of which are beyond our control:

- variations in our operating results, including our membership acquisition and retention, revenues, operating income, net income, net cash provided by operating activities and free cash flow;
- variations between our actual operating results and the expectations of securities analysts, investors and the financial community;
- announcements of developments affecting our business, systems or expansion plans by us or others;
- competition, including the introduction of new competitors, their pricing strategies and services;
- market volatility in general;
- the level of demand for our stock, including the amount of short interest in our stock;
- the impact of our current stock repurchase program and any future stock repurchase program we may adopt;
- the operating results of our competitors; and
- other risks and uncertainties described in these risk factors.

As a result of these and other factors, investors in our common stock may not be able to resell their shares at or above their original purchase price.

Following certain periods of volatility in the market price of our securities, we became the subject of securities litigation. We may experience more such litigation following future periods of volatility. This type of litigation may result in substantial costs and a diversion of management's attention and resources.

Preparing and forecasting our financial results requires us to make judgments and estimates which may differ materially from actual results.

Given the dynamic nature of our business, and the inherent limitations in predicting the future, forecasts of our revenues, operating margins, net income and other financial and operating data may differ materially from actual results. Also, predicting consumer adoption of various pricing strategies, such as the ad-supported subscription plan or efforts to limit multi-household usage, and new revenue streams, such as advertising revenue, is inherently difficult given the lack of operating history with respect to such offerings, and actual results may differ significantly from the expectations of our management, securities analysts or investors. Such discrepancies could cause a decline in the trading price of our common stock. In addition, the preparation of consolidated financial statements in conformity with accounting principles generally accepted in the United States of America also requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities, disclosures of contingent assets and liabilities at the date of the financial statements, and the reported amounts of revenues and expenses during the reported periods. We base such estimates on historical experience and on various other assumptions that we believe to be reasonable under the circumstances, but actual results may differ from these estimates. For example, we estimate the content amortization pattern, beginning with the month of first availability, of any particular licensed or produced television series, documentary or feature film based upon various factors including historical and estimated viewing patterns. If actual viewing patterns differ from these estimates, the pattern and/or period of amortization would be changed and could affect the timing or recognition of content amortization. If we revise such estimates it could result in greater in-period expenses, which could cause us to miss our earnings guidance or negatively impact the results we report which could negatively impact our stock price. Further, events outside of our control may cause actual results to differ from our forecast.

Item 1B. Unresolved Staff Comments

None.

Item 1C. Cybersecurity

We have an enterprise-wide information security program designed to identify, protect, detect and respond to and manage reasonably foreseeable cybersecurity risks and threats. To protect our information systems from cybersecurity threats, we use various security tools that help prevent, identify, escalate, investigate, resolve and recover from identified vulnerabilities and security incidents in a timely manner. These include, but are not limited to, internal reporting, monitoring and detection tools, and a bug bounty program to allow security researchers to assist us in identifying vulnerabilities in our products before they are exploited by malicious threat actors. We also maintain a third party security program to identify, prioritize, assess, mitigate and remediate third party risks; however, we rely on the third parties we use to implement security programs commensurate with their risk, and we cannot ensure in all circumstances that their efforts will be successful.

We regularly assess risks from cybersecurity and technology threats and monitor our information systems for potential vulnerabilities. We use a widely-adopted risk quantification model to identify, measure and prioritize cybersecurity and technology risks and develop related security controls and safeguards. We conduct regular reviews and tests of our information security program and also leverage audits by our internal audit team, tabletop exercises, penetration and vulnerability testing, red team exercises, simulations, and other exercises to evaluate the effectiveness of our information security program and improve our security measures and planning. We also engage an external auditor to conduct an annual payment card industry data security standard review of our security controls protecting payment information, as well as third-party penetration testing of our cardholder environment and related systems. The results of these assessments are reported to the Audit Committee.

Our systems and those of third parties with which we do business have experienced and may continue to experience directed attacks intended to lead to interruptions and delays in our service and operations as well as loss, misuse or theft of personal information (of third parties, employees, and our members) and other data, confidential information or intellectual property, and we have experienced unauthorized releases of certain digital content assets and unintended disclosure of personal information due to incidents related to third parties. However, to date these incidents have not had a material impact on our service, systems or business. Any significant disruption to our service or access to our systems could result in a loss of members and adversely affect our business and results of operation. Further, a penetration of our systems or a third-party's systems or other misappropriation or misuse of personal information could subject us to business, regulatory, litigation and reputation risk, which could have a negative effect on our business, financial condition and results of operations. See "Risk Factors - Any significant disruption in or unauthorized access to our computer systems or those of third parties that we utilize in our operations, including those relating to cybersecurity or arising from cyber-attacks, could result in a loss or degradation of service, unauthorized access, disclosure or destruction of data, including member and corporate information, or theft of intellectual property, including digital content assets, which could adversely impact our business."

The Vice President of Security and Privacy Engineering leads our global information security organization responsible for overseeing the Netflix information security program. Our VP of Security and Privacy Engineering has over 30 years of industry experience, including serving in similar roles leading and overseeing cybersecurity programs at other public companies. Team members who support our information security program have relevant educational and industry experience, including holding similar positions at large technology companies. The teams provide regular reports to senior management and other relevant teams on various cybersecurity threats, assessments and findings.

The Board oversees our annual enterprise risk assessment, where we assess key risks within the company, including security and technology risks and cybersecurity threats. The Audit Committee of the Board oversees our cybersecurity risk and receives regular reports from our VP of Security and Privacy Engineering on various cybersecurity matters, including risk assessments, mitigation strategies, areas of emerging risks, incidents and industry trends, and other areas of importance.

Item 2. Properties

We have leased principal properties in both Los Gatos, California, which is the location of our corporate headquarters, and in Los Angeles, California. In addition, we own and lease various office and production space throughout the world.

We believe that our existing facilities are adequate to meet current requirements, and that suitable additional or substitute space will be available as needed to accommodate any further physical expansion of operations and for any additional offices.

Item 3. Legal Proceedings

Information with respect to this item may be found in Note 8 *Commitments and Contingencies* in the accompanying notes to our consolidated financial statements included in Part II, Item 8, "Financial Statements and Supplementary Data" of this Annual Report on Form 10-K, under the caption "Legal Proceedings" which information is incorporated herein by reference.

Item 4. Mine Safety Disclosures

Not applicable.

PART II

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

Market Information

Our common stock is traded on the NASDAQ Global Select Market under the symbol "NFLX".

Holders

As of December 31, 2024, there were approximately 2,752 stockholders of record of our common stock, although there is a significantly larger number of beneficial owners of our common stock.

Dividend Policy

We have never declared or paid any cash dividends on our capital stock, and we do not currently anticipate paying any cash dividends in the foreseeable future.

Company Purchases of Equity Securities

Stock repurchases during the three months ended December 31, 2024 were as follows:

Period	Total Number of Shares Purchased (1)	Average Price Paid per Share (2)	Total Number of Shares Purchased as Part of Publicly Announced Programs (1)	Approximate Dollar Value of Shares that May Yet Be Purchased Under the Program (1)
				(in thousands)
October 1 - 31, 2024	519,883	$ 724.15	519,883	$ 2,678,384
November 1 - 30, 2024	457,732	$ 792.49	457,732	$ 2,315,637
December 1 - 31, 2024	188,212	$ 913.13	188,212	$ 17,143,775
Total	1,165,827		1,165,827	

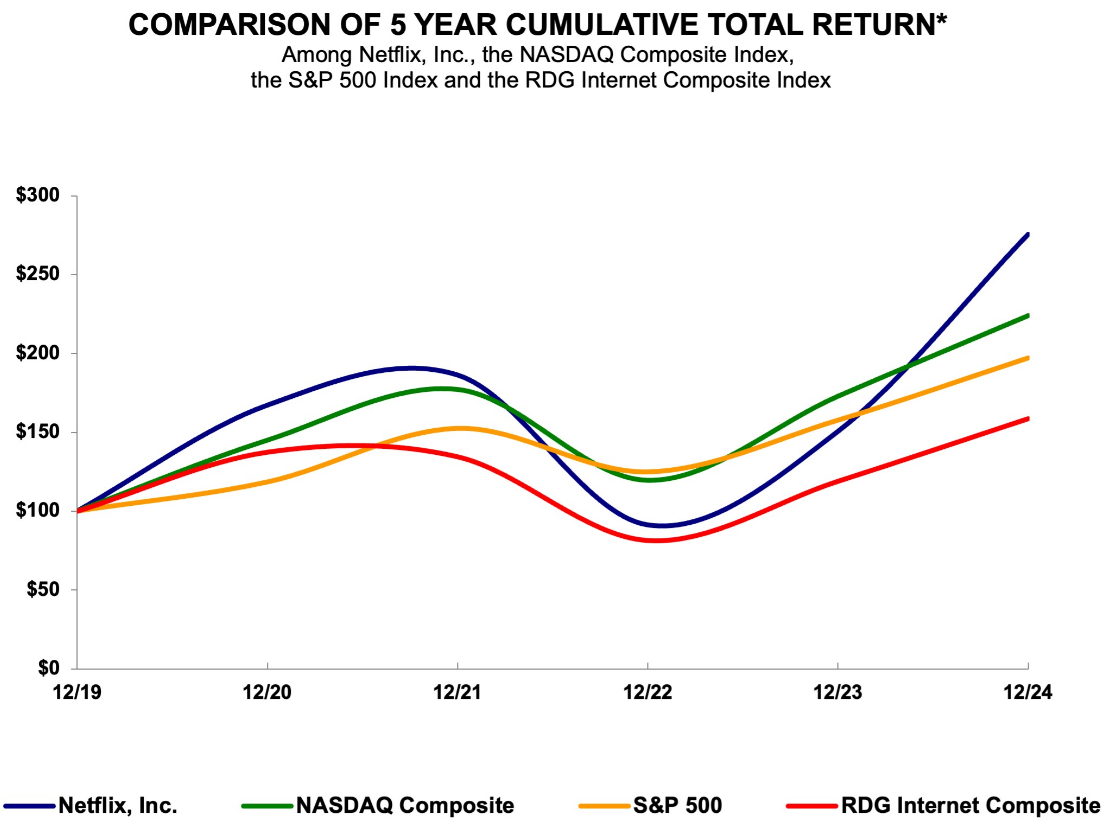
(1) In September 2023, the Board of Directors authorized the repurchase of up to $10 billion of its common stock, with no expiration date, and in December 2024, the Board of Directors increased the share repurchase authorization by an additional $15 billion, also with no expiration date. For further information regarding stock repurchase activity, see Note 9 *Stockholders' Equity* to the consolidated financial statements in this Annual Report.

(2) Average price paid per share includes costs associated with the repurchases but excludes the 1% excise tax on stock repurchases imposed by the Inflation Reduction Act of 2022.

Stock Performance Graph

Notwithstanding any statement to the contrary in any of our previous or future filings with the Securities and Exchange Commission, the following information relating to the price performance of our common stock shall not be deemed "filed" with the Commission or "soliciting material" under the Exchange Act and shall not be incorporated by reference into any such filings.

The following graph compares, for the five year period ended December 31, 2024, the total cumulative stockholder return on the Company's common stock with the total cumulative return of the NASDAQ Composite Index, the S&P 500 Index and the RDG Internet Composite Index. Measurement points are the last trading day of each of the Company's fiscal years ended December 31, 2019, December 31, 2020, December 31, 2021, December 31, 2022, December 31, 2023 and December 31, 2024. Total cumulative stockholder return assumes $100 invested at the beginning of the period in the Company's common stock, the stocks represented in the NASDAQ Composite Index, the stocks represented in the S&P 500 Index and the stocks represented in the RDG Internet Composite Index, respectively, and reinvestment of any dividends. Historical stock price performance should not be relied upon as an indication of future stock price performance.



*$100 invested on 12/31/19 in stock or index, including reinvestment of dividends.
Fiscal year ending December 31.

Item 6. [Reserved]

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

This section of this Form 10-K generally discusses 2024 and 2023 items and year-to-year comparisons between 2024 and 2023. Discussions of 2022 items and year-to-year comparisons between 2023 and 2022 that are not included in this Form 10-K can be found in "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Part II, Item 7 of the Company's Annual Report on Form 10-K for the fiscal year ended December 31, 2023.

Results of Operations

The following represents our consolidated performance highlights:

	As of/Year Ended December 31,			Change
	2024	2023	2022	2024 vs. 2023
	(in thousands, except revenue per membership and percentages)			
Financial Results:				
Streaming revenues	$ 39,000,966	$ 33,640,458	$ 31,469,852	16 %
DVD revenues (1)	—	82,839	145,698	(100)%
Total revenues	**$ 39,000,966**	**$ 33,723,297**	**$ 31,615,550**	16 %
Operating income	$ 10,417,614	$ 6,954,003	$ 5,632,831	50 %
Operating margin	27 %	21 %	18 %	
Global Streaming Memberships:				
Paid net membership additions	41,350	29,529	8,903	40 %
Paid memberships at end of period (2)	301,626	260,276	230,747	16 %
Average paying memberships	277,730	240,889	222,924	15 %
Average monthly revenue per paying membership	$ 11.70	$ 11.64	$ 11.76	1 %
Constant currency change (3)				4 %

(1) We discontinued our DVD-by-mail service in the year ended December 31, 2023. The discontinuance of our DVD business had an immaterial impact on our operations and financial results.

(2) A paid membership (also referred to as a paid subscription) is defined as a membership that has the right to receive Netflix service following sign-up and a method of payment being provided, and that is not part of a free trial or certain other promotions that may be offered by the Company to new or rejoining members. Certain members have the option to add extra member sub accounts. These extra member sub accounts are not included in paid memberships. A membership is canceled and ceases to be reflected in the above metrics as of the effective cancellation date. Voluntary cancellations generally become effective at the end of the prepaid membership period. Involuntary cancellations, as a result of a failed method of payment, become effective immediately. Memberships are assigned to territories based on the geographic location used at time of sign-up as determined by the Company's internal systems, which utilize industry standard geo-location technology.

(3) We believe the non-GAAP financial measure of constant currency revenue is useful in analyzing the underlying trends in average monthly revenue per paying membership ("ARM") absent foreign currency fluctuations. However, this non-GAAP financial measure should be considered in addition to, not as a substitute for, or superior to other financial measures prepared in accordance with GAAP.

In order to exclude the effect of foreign currency rate fluctuations on ARM, we calculate current period revenue assuming foreign exchange rates had remained constant with foreign exchange rates from each of the corresponding months of the prior-year period and exclude the impact of hedging gains or losses realized as revenues. Constant currency percentage change in ARM is calculated as the percentage change between current period constant currency ARM and the prior comparative period ARM. The impact of hedging gains or losses is excluded from both the current and prior periods. For the year ended December 31, 2024, our revenues would have been approximately $1,424 million higher, excluding the impact of hedging and had foreign currency exchange rates remained constant with those for the year ended December 31, 2023. The unfavorable foreign exchange rate impacts in the year ended December 31, 2024 were primarily driven by the devaluation of the Argentine peso relative to the U.S. dollar coupled with significant price increases in the local currency in this jurisdiction.

Operating margin for the year ended December 31, 2024 increased six percentage points as compared to the prior comparative period, primarily due to revenues growing at a faster rate as compared to the growth in cost of revenues, sales and marketing, and technology and development expenses, coupled with lower general and administrative expenses.

Streaming Revenues

We primarily derive revenues from monthly membership fees for services related to streaming content to our members. We offer a variety of streaming membership plans, the price of which varies by country and the features of the plan. As of December 31, 2024, pricing on our paid plans ranged from the U.S. dollar equivalent of $1 to $32 per month, and pricing on our extra member sub accounts ranged from the U.S. dollar equivalent of $2 to $8 per month. We expect that from time to time the prices of our membership plans in each country may change and we may test other plan and price variations.

We also earn revenue from advertisements presented on our streaming service, consumer products, live events and various other sources. Revenues earned from sources other than monthly membership fees were not a material component of streaming revenues for the years ended December 31, 2024, 2023, and 2022.

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except percentages)				
Streaming revenues	$ 39,000,966	$ 33,640,458	$ 31,469,852	$ 5,360,508	16 %

Streaming revenues for the year ended December 31, 2024 increased 16% as compared to the year ended December 31, 2023, primarily due to the growth in average paying memberships and price increases, partially offset by unfavorable changes in foreign exchange rates.

The following tables summarize streaming revenues and other streaming membership information by region for the years ended December 31, 2024, 2023 and 2022. Hedging gains of $124 million are included in "Streaming revenues" for the year ended December 31, 2024. No hedging gains and losses were recognized as "Streaming revenues" in the comparative prior year periods. See Note 7 *Derivative Financial Instruments and Hedging Activities* to the consolidated financial statements for further information regarding the Company's derivative and non-derivative financial instruments.

United States and Canada (UCAN)

	As of/Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except revenue per membership and percentages)				
Streaming revenues	$ 17,359,369	$ 14,873,783	$ 14,084,643	$ 2,485,586	17 %
Paid net membership additions (losses)	9,497	5,832	(919)	3,665	63 %
Paid memberships at end of period	89,625	80,128	74,296	9,497	12 %
Average paying memberships	84,112	76,126	74,001	7,986	10 %
Average monthly revenue per paying membership	$ 17.20	$ 16.28	$ 15.86	$ 0.92	6 %
Constant currency change					6 %

Europe, Middle East, and Africa (EMEA)

	As of/Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except revenue per membership and percentages)				
Streaming revenues	$ 12,387,035	$ 10,556,487	$ 9,745,015	$ 1,830,548	17 %
Paid net membership additions	12,320	12,084	2,693	236	2 %
Paid memberships at end of period	101,133	88,813	76,729	12,320	14 %
Average paying memberships	94,200	80,928	73,904	13,272	16 %
Average monthly revenue per paying membership	$ 10.96	$ 10.87	$ 10.99	$ 0.09	1 %
Constant currency change					1 %

Latin America (LATAM)

	As of/Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except revenue per membership and percentages)				
Streaming revenues	$ 4,839,816	$ 4,446,461	$ 4,069,973	$ 393,355	9 %
Paid net membership additions	7,330	4,298	1,738	3,032	71 %
Paid memberships at end of period	53,327	45,997	41,699	7,330	16 %
Average paying memberships	48,954	42,802	40,000	6,152	14 %
Average monthly revenue per paying membership	$ 8.24	$ 8.66	$ 8.48	$ (0.42)	(5)%
Constant currency change					21 %

Asia-Pacific (APAC)

	As of/Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except revenue per membership and percentages)				
Streaming revenues	$ 4,414,746	$ 3,763,727	$ 3,570,221	$ 651,019	17 %
Paid net membership additions	12,203	7,315	5,391	4,888	67 %
Paid memberships at end of period	57,541	45,338	38,023	12,203	27 %
Average paying memberships	50,466	41,033	35,019	9,433	23 %
Average monthly revenue per paying membership	$ 7.29	$ 7.64	$ 8.50	$ (0.35)	(5)%
Constant currency change					(3)%

Cost of Revenues

Cost of revenues primarily consists of the amortization of content assets. Other costs of revenues include expenses associated with the acquisition, licensing and production of content, streaming delivery costs, and other operating costs.

Expenses related to the acquisition, licensing and production of content not included in content amortization may include payroll, stock-based compensation, facilities, and other personnel-related expenses, costs associated with obtaining rights to music included in our content, overall deals with talent, miscellaneous production-related costs and participations and residuals. Streaming delivery costs are primarily related to our global content delivery network ("Open Connect"). We have built our own Open Connect network to help us efficiently stream a high volume of content to our members over the internet. Delivery expenses, therefore, include equipment costs related to Open Connect, payroll and related personnel expenses and all third-party costs, such as cloud computing costs, associated with delivering content over the internet. Other operating costs include customer service and payment processing fees, including those we pay to our integrated payment partners, as well as other costs directly incurred in making our content available to members.

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except percentages)				
Cost of revenues	$ 21,038,464	$ 19,715,368	$ 19,168,285	$ 1,323,096	7 %
As a percentage of revenues	54 %	58 %	61 %		

The increase in cost of revenues for the year ended December 31, 2024 as compared to the year ended December 31, 2023 was due to a $1,104 million increase in content amortization relating to our existing and new content.

Sales and Marketing

Sales and marketing expenses consist primarily of advertising expenses and certain payments made to marketing and advertising sales partners, including consumer electronics ("CE") manufacturers, multichannel video programming distributors ("MVPDs"), mobile operators, and ISPs. Marketing expenses include promotional activities such as digital and television advertising. Sales and marketing expenses also include payroll, stock-based compensation, facilities, and other related expenses for personnel that support advertising sales and marketing activities.

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except percentages)				
Sales and marketing	$ 2,917,554	$ 2,657,883	$ 2,530,502	$ 259,671	10 %
As a percentage of revenues	7 %	8 %	8 %		

The increase in sales and marketing expenses for the year ended December 31, 2024 as compared to the year ended December 31, 2023 was primarily driven by a $131 million increase in personnel-related costs due to the growth in advertising sales headcount. Other sales and marketing expenses increased $129 million primarily due to a $54 million increase in marketing expenses due to the timing of marketing spend on our content slate, coupled with an increase in expenses incurred in connection with our advertising offering, including increased payments to advertising sales partners and other advertising distribution expenses.

Technology and Development

Technology and development expenses consist primarily of payroll, stock-based compensation, facilities, and other related expenses for technology personnel responsible for making improvements to our service offerings, including testing, maintaining and modifying our user interface, our recommendations, merchandising and infrastructure. Technology and development expenses also include costs associated with general use computer hardware and software.

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except percentages)				
Technology and development	$ 2,925,295	$ 2,675,758	$ 2,711,041	$ 249,537	9 %
As a percentage of revenues	8 %	8 %	9 %		

The increase in technology and development expenses for the year ended December 31, 2024 as compared to the year ended December 31, 2023 was primarily due to a $224 million increase in personnel-related costs.

General and Administrative

General and administrative expenses consist of payroll, stock-based compensation, facilities, and other related expenses for corporate personnel. General and administrative expenses also include professional fees and other general corporate expenses.

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except percentages)				
General and administrative	$ 1,702,039	$ 1,720,285	$ 1,572,891	$ (18,246)	(1)%
As a percentage of revenues	4 %	5 %	5 %		

General and administrative expenses for the year ended December 31, 2024 as compared to the year ended December 31, 2023 remained relatively flat.

Interest Expense

Interest expense consists primarily of the interest associated with our outstanding debt obligations, including the amortization of debt issuance costs. See Note 6 *Debt* in the accompanying notes to our consolidated financial statements for further detail on our debt obligations.

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except percentages)				
Interest expense	$ 718,733	$ 699,826	$ 706,212	$ 18,907	3 %
As a percentage of revenues	2 %	2 %	2 %		

Interest expense primarily consists of interest on our Notes of $718 million for the year ended December 31, 2024. The increase in interest expense for the year ended December 31, 2024 as compared to the year ended December 31, 2023 was due to the increase in debt.

Interest and Other Income (Expense)

Interest and other income (expense) consists primarily of foreign exchange gains and losses on foreign currency denominated balances, gains and losses on certain derivative instruments, and interest earned on cash, cash equivalents and short-term investments.

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except percentages)				
Interest and other income (expense)	$ 266,776	$ (48,772)	$ 337,310	$ 315,548	647 %
As a percentage of revenues	1 %	— %	1 %		

Interest and other income (expense) increased for the year ended December 31, 2024 primarily due to foreign exchange losses of $18 million, net of the impacts of derivatives and hedging, compared to the losses of $293 million for the corresponding period in 2023. In the year ended December 31, 2024, the foreign exchange losses were primarily driven by the remeasurement of cash and content liability positions in currencies other than the functional currencies, partially offset by a non-cash gain of $122 million, net of hedging impacts, from the remeasurement of our €5,170 million Senior Notes. The foreign exchange loss in the year ended December 31, 2023 was primarily driven by a non-cash loss of $176 million from the remeasurement of our Senior Notes denominated in euros, coupled with the remeasurement of cash and content liability positions in currencies other than the functional currencies.

Provision for Income Taxes

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands, except percentages)				
Provision for income taxes	$ 1,254,026	$ 797,415	$ 772,005	$ 456,611	57 %
Effective tax rate	13 %	13 %	15 %		

The effective tax rate for the year ended December 31, 2024 remained relatively flat as compared to the year ended December 31, 2023. See Note 10 *Income Taxes* to the consolidated financial statements for further information regarding income taxes.

Liquidity and Capital Resources

	As of December 31,		Change	
	2024	2023	2024 vs. 2023	
	(in thousands, except percentages)			
Cash, cash equivalents, restricted cash and short-term investments	$ 9,586,343	$ 7,139,488	$ 2,446,855	34 %
Short-term and long-term debt	15,582,804	14,543,261	1,039,543	7 %

Cash, cash equivalents, restricted cash and short-term investments increased $2,447 million in the year ended December 31, 2024 primarily due to cash provided by operations, issuance of debt, and proceeds from issuance of common stock, partially offset by the repurchase of stock and repayment of debt.

Debt, net of debt issuance costs and discounts, increased $1,040 million primarily due to the issuance of $1,800 million in additional Senior Notes, partially offset by the repayment upon maturity of the $400 million aggregate principal amount of our 5.750% Senior Notes and the remeasurement of our euro-denominated notes in the year ended December 31, 2024. The amount of principal and interest due in the next twelve months is $2,487 million. As of December 31, 2024, no amounts had been borrowed under our $3 billion Revolving Credit Agreement. See Note 6 *Debt* in the accompanying notes to our consolidated financial statements.

We anticipate that we may periodically raise additional debt capital. Our ability to obtain this or any additional financing that we may choose or need, including for the refinancing of upcoming maturities or potential strategic acquisitions and investments, will depend on, among other things, our development efforts, business plans, operating performance, and the condition of the capital markets at the time we seek financing. We may not be able to obtain such financing on terms acceptable to us or at all. If we raise additional funds through the issuance of equity or debt securities, those securities may have rights, preferences or privileges senior to the rights of our common stock, and our stockholders may experience dilution.

In September 2023, the Board of Directors authorized the repurchase of up to $10 billion of our common stock, with no expiration date, and in December 2024, the Board of Directors increased the share repurchase authorization by an additional $15 billion, also with no expiration date. Stock repurchases may be effected through open market repurchases in compliance with Rule 10b-18 under the Exchange Act, including through the use of trading plans intended to qualify under Rule 10b5-1 under the Exchange Act, privately-negotiated transactions, accelerated stock repurchase plans, block purchases, or other similar purchase techniques and in such amounts as management deems appropriate. We are not obligated to repurchase any specific number of shares, and the timing and actual number of shares repurchased will depend on a variety of factors, including our stock price, general economic, business and market conditions, and alternative investment opportunities. We may discontinue any repurchases of our common stock at any time without prior notice. In the fiscal year ended December 31, 2024, the Company repurchased 9,861,935 shares of common stock for an aggregate amount of $6,211 million (excluding the 1% excise tax on stock repurchases as a result of the Inflation Reduction Act of 2022). As of December 31, 2024, $17.1 billion remains available for repurchases.

Our primary uses of cash include the acquisition, licensing and production of content, marketing programs, streaming delivery and personnel-related costs, as well as strategic acquisitions and investments. Cash payment terms for non-original content have historically been in line with the amortization period. Investments in original content, and in particular content that we produce and own, require more cash upfront relative to licensed content. For example, production costs are paid as the content is created, well in advance of when the content is available on the service and amortized. We expect to continue to significantly invest in global content, particularly in original content, which will impact our liquidity. We currently anticipate that cash flows from operations, available funds and access to financing sources, including our revolving credit facility, will continue to be sufficient to meet our cash needs for the next twelve months and beyond.

Our material cash requirements from known contractual and other obligations primarily relate to our content, debt and lease obligations. As of December 31, 2024, the expected timing of those payments are as follows:

Contractual obligations (in thousands):	Total	Next 12 Months	Beyond 12 Months
Content obligations (1)	$ 23,248,931	$ 11,424,696	$ 11,824,235
Debt (2)	19,841,462	2,486,945	17,354,517
Operating lease obligations (3)	2,761,120	514,625	2,246,495
Total	$ 45,851,513	$ 14,426,266	$ 31,425,247

(1) As of December 31, 2024, content obligations were comprised of $4.4 billion included in "Current content liabilities" and $1.8 billion of "Non-current content liabilities" on the Consolidated Balance Sheets and $17.0 billion of obligations that are not reflected on the Consolidated Balance Sheets as they did not then meet the criteria for recognition.

Content obligations include amounts related to the acquisition, licensing and production of content. An obligation for the production of content includes non-cancelable commitments under creative talent and employment agreements and other production related commitments. An obligation for the acquisition and licensing of content is incurred at the time we enter into an agreement to obtain future titles. Once a title becomes available, a content liability is recorded on the Consolidated Balance Sheets. Certain agreements include the obligation to license rights for unknown future titles, the ultimate quantity and/or fees for which are not yet determinable as of the reporting date. Traditional film output deals, or certain TV series license agreements where the number of seasons to be aired is unknown, are examples of these types of agreements. The contractual obligations table above does not include any estimated obligation for the unknown future titles, payment for which could range from less than one year to more than five years. However, these unknown obligations are expected to be significant and we believe could include approximately $1 billion to $4 billion over the next three years, with the payments for the vast majority of such amounts expected to occur after the next twelve months. The foregoing range is based on considerable management judgments and the actual amounts may differ. Once we know the title that we will receive and the license fees, we include the amount in the contractual obligations table above.

(2) Debt obligations include our Notes consisting of principal and interest payments. See Note 6 *Debt* in the accompanying notes to our consolidated financial statements for further details.

(3) Operating lease obligations are comprised of operating lease liabilities included in "Accrued expenses and other liabilities" and "Other non-current liabilities" on the Consolidated Balance Sheets, inclusive of imputed interest. Operating lease obligations also include additional obligations that are not reflected on the Consolidated Balance Sheets as they did not meet the criteria for recognition. As of December 31, 2024, the Company has additional operating leases for real estate that have not yet commenced of $38 million which has been included above. See Note 5 *Balance Sheet Components* in the accompanying notes to our consolidated financial statements for further details regarding leases.

As of December 31, 2024, we had gross unrecognized tax benefits of $432 million, of which $302 million was classified in “Other non-current liabilities" in the Consolidated Balance Sheets. At this time, an estimate of the range of reasonably possible adjustments to the balance of unrecognized tax benefits cannot be made. In addition, we may be required to pay deposits of approximately $800 million related to certain direct and indirect taxes in the next twelve months, which are in excess of our typical annual obligations.

Cash Flows

The following table summarizes our cash flows:

	Year Ended December 31,			Change	
	2024	2023	2022	2024 vs. 2023	
	(in thousands)				
Net cash provided by operating activities	$ 7,361,364	$ 7,274,301	$ 2,026,257	$ 87,063	1 %
Net cash provided by (used in) investing activities	(2,181,784)	541,751	(2,076,392)	(2,723,535)	(503)%
Net cash used in financing activities	(4,074,427)	(5,950,803)	(664,254)	(1,876,376)	(32)%

Net cash provided by operating activities for the year ended December 31, 2024 increased $87 million as compared to the year ended December 31, 2023, primarily driven by a $3,304 million or 61% increase in net income, an increase in adjustments for non-cash expenses, and favorable changes in working capital, partially offset by an increase in payments for content assets. The payments for content assets increased $3,862 million, from $13,140 million to $17,003 million, or 29%.

Net cash provided by (used in) investing activities for the year ended December 31, 2024 decreased $2,724 million as compared to the year ended December 31, 2023, primarily due to there being no maturities of investments in the year ended December 31, 2024, as compared to maturities of investments of $1,395 million in the year ended December 31, 2023, coupled with an increase in purchases of investments of $1,237 million and an increase in purchases of property and equipment of $91 million.

Net cash used in financing activities for the year ended December 31, 2024 decreased $1,876 million as compared to the year ended December 31, 2023, primarily due to proceeds from the issuance of debt of $1,794 million in the year ended December 31, 2024 and a $663 million increase in the proceeds from the issuance of common stock. These cash inflows were partially offset by the repayment upon maturity of the $400 million aggregate principal amount of our 5.750% Senior Notes in the year ended December 31, 2024 as compared to no repayments of debt in the corresponding period in 2023, coupled with a $218 million increase in the repurchases of common stock.

Indemnifications

The information set forth under Note 8 *Commitments and Contingencies* in the accompanying notes to our consolidated financial statements included in Part II, Item 8, "Financial Statements and Supplementary Data" of this Annual Report on Form 10-K is incorporated herein by reference.

Critical Accounting Estimates

The preparation of consolidated financial statements in conformity with accounting principles generally accepted in the United States of America requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities, disclosures of contingent assets and liabilities at the date of the financial statements, and the reported amounts of revenues and expenses during the reported periods. The Securities and Exchange Commission ("SEC") has defined a company’s critical accounting policies as the ones that are most important to the portrayal of a company’s financial condition and results of operations, and which require a company to make its most difficult and subjective judgments. Based on this definition, we have identified the critical accounting policies and judgments addressed below. We base our estimates on historical experience and on various other assumptions that we believe to be reasonable under the circumstances. Actual results may differ from these estimates.

Content

We acquire, license and produce content, including original programming, in order to offer our members unlimited viewing of video entertainment. The content licenses are for a fixed fee and specific windows of availability. Payment terms for certain content licenses and the production of content require more upfront cash payments relative to the amortization expense. Payments for content, including additions to content assets and the changes in related liabilities, are classified within "Net cash provided by operating activities" on the Consolidated Statements of Cash Flows.

We recognize content assets (licensed and produced) as "Content assets, net" on the Consolidated Balance Sheets. For licensed content, we capitalize the fee per title and record a corresponding liability at the gross amount of the liability when the license period begins, the cost of the title is known and the title is accepted and available for streaming. For produced content, we capitalize costs associated with the production, including development costs, direct costs and production overhead.

Based on factors including historical and estimated viewing patterns, we amortize the content assets (licensed and produced) in “Cost of revenues” on the Consolidated Statements of Operations over the shorter of each title's contractual window of availability or estimated period of use or ten years, beginning with the month of first availability. The amortization is on an accelerated basis, as we typically expect more upfront viewing, and film amortization is more accelerated than TV series amortization. On average, over 90% of a licensed or produced content asset is expected to be amortized within four years after its month of first availability. We review factors that impact the amortization of the content assets on a regular basis. Our estimates related to these factors require considerable management judgment.

In the normal course of business, we, or a third-party producing content on our behalf, may qualify for tax incentives through eligible spend on productions. The accounting for tax incentives is dependent on the particular type of incentive, including the nature of the benefit and the location the incentive is earned. In general, tax incentives are realized as cash receipts and may be received prior to or after a title launches on our service. Upon a title’s launch, any amounts we are eligible for through qualified production spend but have not received, are recognized in “Other current assets” or “Other non-current assets” on the Consolidated Balance Sheets as receivables. Tax incentives are generally accounted for as a reduction to the cost basis of content assets (presented in “Content assets, net”) and reduce content amortization over the life of the title (as presented in “Cost of revenues”) on the Consolidated Statements of Operations.

Our business model is subscription based as opposed to a model generating revenues at a specific title level. Content assets (licensed and produced) are predominantly monetized as a group and therefore are reviewed at a group level when an event or change in circumstances indicates a change in the expected usefulness of the content or that the fair value may be less than unamortized cost. To date, we have not identified any such event or changes in circumstances. If such changes are identified in the future, these aggregated content assets will be stated at the lower of unamortized cost or fair value. In addition, unamortized costs for assets that have been, or are expected to be, abandoned are written off.

Income Taxes

We record a provision for income taxes for the anticipated tax consequences of our reported results of operations using the asset and liability method. Deferred income taxes are recognized by applying enacted statutory tax rates applicable to future years to differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases as well as net operating loss and tax credit carryforwards. The effect on deferred tax assets and liabilities of a change in tax rates is recognized in income in the period that includes the enactment date. The measurement of deferred tax assets is reduced, if necessary, by a valuation allowance when it is more likely than not they will not be realized.

Although we believe our assumptions, judgments and estimates are reasonable, changes in tax laws or our interpretation of tax laws and the resolution of any tax audits could significantly impact the amounts provided for income taxes in our consolidated financial statements.

In evaluating our ability to recover our deferred tax assets, in full or in part, we consider all available positive and negative evidence, including our past operating results, our forecast of future earnings and future taxable income, and prudent and feasible tax planning strategies. The assumptions utilized in determining future taxable income require significant judgment and are consistent with the plans and estimates we are using to manage the underlying business. Actual operating results in future years could differ from our current assumptions, judgments and estimates.

We do not recognize certain tax benefits from uncertain tax positions within the provision for income taxes. We may recognize a tax benefit only if it is more likely than not the tax position will be sustained on examination by the taxing authorities, based on the technical merits of the position. The tax benefits recognized in the financial statements from such positions are then measured based on the largest benefit that has a greater than 50% likelihood of being realized upon settlement. Due to uncertainties in any tax audit outcome, our estimates of the ultimate settlement of our unrecognized tax positions may change and the actual tax benefits may differ significantly from the estimates.

See Note 10 *Income Taxes* to the consolidated financial statements for further information regarding income taxes.

Recent Accounting Pronouncements

The information set forth under Note 1 to the consolidated financial statements under the caption “Basis of Presentation and Summary of Significant Accounting Policies” is incorporated herein by reference.

Item 7A. Quantitative and Qualitative Disclosures about Market Risk

We are exposed to market risks related to interest rate changes and the corresponding changes in the market values of our debt and foreign currency fluctuations.

Interest Rate Risk

At December 31, 2024, our cash equivalents were generally invested in money market funds and time deposits. Interest paid on such funds fluctuates with the prevailing interest rate. Our short-term investments are primarily comprised of investments in government securities. These securities are classified as available-for-sale and are recorded at fair value with unrealized gains and losses, net of tax, included in accumulated other comprehensive income (“AOCI”) within Stockholders' equity in the Consolidated Balance Sheets. Changes in interest rates could adversely affect the market value of these securities.

As of December 31, 2024, we had $15.7 billion of debt, consisting of fixed rate unsecured debt in fifteen tranches due between 2025 and 2054. Refer to Note 6 *Debt* to the consolidated financial statements for details about all issuances. The fair value of our debt will fluctuate with movements of interest rates, increasing in periods of declining rates of interest and declining in periods of increasing rates of interest. The fair value of our debt will also fluctuate based on changes in foreign currency rates, as discussed below.

Foreign Currency Risk

We operate our business globally and transact in multiple currencies. Currencies denominated in other than the U.S. dollar accounted for 56% of revenue and 29% of operating expenses for the year ended December 31, 2024. We therefore have foreign currency risk related to these currencies, which are primarily the euro, the British pound, the Brazilian real, Argentine peso, and the Mexican peso.

Accordingly, volatility in exchange rates, and in particular a weakening of foreign currencies relative to the U.S. dollar may negatively affect our revenue and operating income as expressed in U.S. dollars. Excluding the impact of hedging gains or losses realized as revenues, our revenues for the year ended December 31, 2024 would have been approximately $1,424 million higher had foreign currency exchange rates remained constant with those in the same period of 2023. The unfavorable foreign exchange rate impact in the year ended December 31, 2024 was primarily driven by the devaluation of the Argentine peso relative to the U.S. dollar coupled with significant price increases in the local currency in this jurisdiction. See Part II, Item 7, "Management’s Discussion and Analysis of Financial Condition and Results of Operations" for further information regarding our non-GAAP financial measure of constant currency.

We enter into foreign exchange forward contracts to mitigate fluctuations in forecasted U.S. dollar-equivalent revenues from changes in foreign currency exchange rates. These contracts may reduce, but do not entirely eliminate, the effect of foreign currency exchange fluctuations, and we may choose not to hedge certain exposures. We designate these contracts as cash flow hedges of forecasted foreign currency revenue and initially record the gains or losses on these derivative instruments as a component of AOCI and reclassify the amounts into “Revenues” on the Consolidated Statements of Operations in the same period the forecasted transaction affects earnings. If the U.S dollar weakened by 10% as of December 31, 2024 and December 31, 2023, the amount recorded in AOCI related to our foreign exchange contracts, before taxes, would have been approximately $1.9 billion and $958 million lower, respectively. This adverse change in AOCI would be expected to offset a corresponding favorable foreign currency change in the underlying forecasted revenues when recognized in earnings.

We enter into foreign exchange forward contracts to mitigate fluctuations in forecasted and firmly committed U.S. dollar-equivalent transactions related to the licensing and production of content assets from changes in foreign currency exchange rates. These contracts may reduce, but do not entirely eliminate, the effect of foreign currency exchange fluctuations, and we may choose not to hedge certain exposures. We designate these contracts as cash flow hedges and initially record the gains or losses on these derivative instruments as a component of AOCI and reclassify the amounts into “Cost of Revenues” to offset the hedged exposures as they affect earnings, which occurs as the underlying hedged content assets are amortized. If the U.S dollar strengthened by 10% as of December 31, 2024 and December 31, 2023, the amount recorded in AOCI related to our foreign exchange contracts, before taxes, would have been approximately $187 million and $71 million lower, respectively. This adverse change in AOCI would be expected to offset a corresponding favorable foreign currency change in the underlying exposures when recognized in earnings.

We use non-derivative instruments to mitigate foreign exchange risk related to our net investments in certain foreign subsidiaries. These non-derivative instruments may reduce, but do not entirely eliminate, the effect of foreign currency exchange fluctuations, and we may choose not to hedge certain exposures. We designate a portion of our foreign currency-denominated Senior Notes in euros as net investment hedges and the gains or losses on these non-derivative instruments are reported as a component of AOCI and remain in AOCI until the hedged net investment is sold or liquidated, at which point the amounts recognized in AOCI are reclassified into earnings.

We have also experienced and will continue to experience fluctuations in our net income as a result of gains (losses) on the settlement and the remeasurement of monetary assets and liabilities denominated in currencies that are not the functional currency. In the year ended December 31, 2024, we began entering into foreign exchange forward contracts to mitigate the foreign exchange risk on intercompany transactions and monetary assets and liabilities that are not denominated in the functional currencies of the Company and its subsidiaries. These contracts may reduce, but do not entirely eliminate, the effect of foreign currency exchange fluctuations, and we may choose not to hedge certain exposures. Certain contracts are not designated as hedging instruments and the gains or losses on these derivative instruments are recorded in “Interest and other income (expense)” in the Consolidated Statements of Operations. We also designate certain contracts as fair value hedges to mitigate the foreign exchange risk on the remeasurement of our foreign-currency denominated debt. The gains or losses on these derivative instruments included in the assessment of hedge effectiveness are recorded in “Interest and other income (expense),” net with the offsetting foreign currency remeasurement gains and losses on the hedged items. If an adverse change in exchange rates of 10% was applied to our monetary assets and liabilities denominated in currencies other than the functional currencies as of December 31, 2024 and December 31, 2023, income before income taxes would have been approximately $38 million and $516 million lower, respectively, after considering the offsetting impact of the foreign currency exchange contracts and our net investment hedges. The decrease in the hypothetical adverse change in income before taxes from $516 million as of December 31, 2023 to $38 million as of December 31, 2024 is primarily driven by our use of non-derivative and derivative instruments to mitigate foreign exchange risk related to the remeasurement of foreign-currency denominated balances during the year ended December 31, 2024.

Item 8. Financial Statements and Supplementary Data

The consolidated financial statements and accompanying notes listed in Part IV, Item 15(a)(1) of this Annual Report on Form 10-K are included immediately following Part IV hereof and incorporated by reference herein.

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

None.

Item 9A. Controls and Procedures

(a) Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our co-Chief Executive Officers and Chief Financial Officer, evaluated the effectiveness of our disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act) as of the end of the period covered by this Annual Report on Form 10-K. Based on that evaluation, our co-Chief Executive Officers and Chief Financial Officer concluded that our disclosure controls and procedures as of the end of the period covered by this Annual Report on Form 10-K were effective in providing reasonable assurance that information required to be disclosed by us in reports that we file or submit under the Exchange Act, is recorded, processed, summarized and reported within the time periods specified in the Securities and Exchange Commission's rules and forms, and that such information is accumulated and communicated to our management, including our co-Chief Executive Officers and Chief Financial Officer, as appropriate, to allow timely decisions regarding required disclosures.

Our management, including our co-Chief Executive Officers and Chief Financial Officer, does not expect that our disclosure controls and procedures or our internal controls will prevent all error and all fraud. A control system, no matter how well conceived and operated, can provide only reasonable, not absolute, assurance that the objectives of the control system are met. Further, the design of a control system must reflect the fact that there are resource constraints, and the benefits of controls must be considered relative to their costs. Because of the inherent limitations in all control systems, no evaluation of controls can provide absolute assurance that all control issues and instances of fraud, if any, within Netflix have been detected.

(b) Management's Annual Report on Internal Control Over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting (as defined in Rule 13a-15(f) of the Exchange Act). Our management assessed the effectiveness of our internal control over financial reporting as of December 31, 2024. In making this assessment, our management used the criteria set forth by the Committee of Sponsoring Organizations of the Treadway Commission ("COSO") in *Internal Control—Integrated Framework* (2013 framework). Based on our assessment under the framework in *Internal Control—Integrated Framework* (2013 framework), our management concluded that our internal control over financial reporting was effective as of December 31, 2024. The effectiveness of our internal control over financial reporting as of December 31, 2024 has been audited by Ernst & Young LLP, an independent registered public accounting firm, as stated in their report that is included herein.

(c) Changes in Internal Control Over Financial Reporting

There was no change in our internal control over financial reporting that occurred during the quarter ended December 31, 2024 that has materially affected, or is reasonably likely to materially affect, our internal control over financial reporting.

Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors of Netflix, Inc.

Opinion on Internal Control Over Financial Reporting

We have audited Netflix, Inc.'s internal control over financial reporting as of December 31, 2024, based on criteria established in Internal Control-Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, Netflix, Inc. (the Company) maintained, in all material respects, effective internal control over financial reporting as of December 31, 2024, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2024 and 2023, the related consolidated statements of operations, comprehensive income, stockholders' equity and cash flows for each of the three years in the period ended December 31, 2024, and the related notes and our report dated January 27, 2025 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Annual Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP

San Jose, California
January 27, 2025

Item 9B. Other Information

Rule 10b5-1 Trading Plans

The adoption or termination of contracts, instructions or written plans for the purchase or sale of our securities by our Section 16 officers and directors for the three months ended December 31, 2024, each of which is intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) under the Exchange Act ("Rule 10b5-1 Plan"), were as follows:

Name	Title	Action	Date Adopted	Expiration Date	Aggregate # of Securities to be Purchased/Sold
Greg Peters (1)	Co-CEO and Director	Termination	10/30/2024	N/A	187,913
Greg Peters (2)	Co-CEO and Director	Adoption	10/30/2024	11/1/2027	158,583
David Hyman (3)	Chief Legal Officer	Adoption	10/29/2024	1/30/2026	104,154
Ted Sarandos (4)	Co-CEO and Director	Adoption	10/25/2024	2/27/2026	199,063
Spencer Neumann (5)	Chief Financial Officer	Adoption	10/29/2024	12/31/2025	33,406
Ann Mather (6)	Director	Adoption	10/31/2024	12/31/2025	2,682
Strive Masiyiwa (7)	Director	Adoption	11/7/2024	12/31/2025	2,813
Jeffrey Karbowski (8)	Chief Accounting Officer	Adoption	10/29/2024	1/31/2026	3,820

(1) On October 30, 2024, Greg Peters, Co-CEO and a member of the Board of Directors, terminated a pre-arranged stock trading plan pursuant to Rule 10b5-1, which was adopted on July 26, 2023 and provided for the potential exercise of vested stock options and the associated sale of up to 187,913 shares of Netflix common stock until November 1, 2027 or the earlier completion of all authorized transactions under the plan.

(2) Upon termination of Mr. Peters' prior plan (described in footnote 1), on October 30, 2024, Mr. Peters entered into a pre-arranged stock trading plan pursuant to Rule 10b5-1 that provides for the potential exercise of vested stock options and the associated sale of up to 158,583 shares of Netflix common stock. This figure includes a grant of 31,112 Performance Share Units (PSUs) that are expected to vest during the term of the 10b5-1 plan, which are assumed to vest at 100% of the target award amount. The actual number of PSUs that may vest can vary between 0% - 200% of the target award of PSUs, subject to the achievement of certain performance conditions as set forth in the PSU award agreement, less shares to be withheld for tax withholding obligations. The plan expires on November 1, 2027, or upon the earlier completion of all authorized transactions under the plan.

(3) David Hyman, Chief Legal Officer, entered into a pre-arranged stock trading plan pursuant to Rule 10b5-1 on October 29, 2024. Mr. Hyman's plan provides for the potential exercise of vested stock options and the associated sale of up to 104,154 shares of Netflix common stock. This figure includes 2,174 PSUs that are expected to vest during the term of the 10b5-1 plan, which are assumed to vest at 100% of the target award amount. The actual number of PSUs that may vest can vary between 0% - 200% of the target award of PSUs, subject to the achievement of certain performance conditions as set forth in the PSU award agreement, less shares to be withheld for tax withholding obligations. The plan expires on January 30, 2026, or upon the earlier completion of all authorized transactions under the plan.

(4) Ted Sarandos, co-CEO and a member of the Board of Directors, entered into a pre-arranged stock trading plan pursuant to Rule 10b5-1 on October 25, 2024. Mr. Sarandos' plan provides for the potential exercise of vested stock options and the associated sale of up to 199,063 shares of Netflix common stock. The plan expires on February 27, 2026, or upon the earlier completion of all authorized transactions under the plan.

(5) Spencer Neumann, Chief Financial Officer, entered into a pre-arranged stock trading plan pursuant to Rule 10b5-1 on October 29, 2024. Mr. Neumann's plan provides for the potential exercise of vested stock options and the associated sale of up to 33,406 shares of Netflix common stock. The plan expires on December 31, 2025, or upon the earlier completion of all authorized transactions under the plan.

(6) Ann Mather, a member of the Board of Directors, entered into a pre-arranged stock trading plan pursuant to Rule 10b5-1 on October 31, 2024. Ms. Mather's plan provides for the potential exercise of vested stock options and the associated sale of up to 2,682 shares of Netflix common stock. The plan expires on December 31, 2025, or upon the earlier completion of all authorized transactions under the plan.

(7) Strive Masiyiwa, a member of the Board of Directors, entered into a pre-arranged stock trading plan pursuant to Rule 10b5-1 on November 7, 2024. Mr. Masiyiwa's plan provides for the potential exercise of vested stock options and the associated sale of up to 2,813 shares of Netflix common stock. The plan expires on December 31, 2025, or upon the earlier completion of all authorized transactions under the plan.

(8) Jeffrey Karbowski, Chief Accounting Officer, entered into a pre-arranged stock trading plan pursuant to Rule 10b5-1 on October 29, 2024. Mr. Karbowski's plan provides for the potential exercise of vested stock options and the associated sale of up to 3,820 shares of Netflix common stock. The plan expires on January 31, 2026, or upon the earlier completion of all authorized transactions under the plan.

Other than those disclosed above, none of our directors or officers adopted or terminated a "non-Rule 10b5-1 trading arrangement" as defined in Item 408 of Regulation S-K.

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections

Not applicable.

PART III

Item 10. Directors, Executive Officers and Corporate Governance

Information regarding our directors and executive officers is incorporated by reference from the information contained under the sections “Proposal One: Election of Directors,” and “Code of Ethics” in our Proxy Statement for the Annual Meeting of Stockholders.

The Company has adopted an insider trading policy which governs transactions in our securities by the Company and its directors, officers, employees, consultants, and contractors and is designed to promote compliance with insider trading laws, rules and regulations applicable to the Company. A copy of our insider trading policy is filed with this Annual Report on Form 10-K as Exhibit 19.1.

Item 11. Executive Compensation

Information required by this item is incorporated by reference from information contained under the sections “Compensation Discussion and Analysis” and “Compensation of Named Executive Officers and Other Matters” in our Proxy Statement for the Annual Meeting of Stockholders.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters

Information required by this item is incorporated by reference from information contained under the sections “Security Ownership of Certain Beneficial Owners and Management” and “Equity Compensation Plan Information” in our Proxy Statement for the Annual Meeting of Stockholders.

Item 13. Certain Relationships and Related Transactions, and Director Independence

Information required by this item is incorporated by reference from information contained under the section “Certain Relationships and Related Transactions” and “Director Independence” in our Proxy Statement for the Annual Meeting of Stockholders.

Item 14. Principal Accountant Fees and Services

Information with respect to principal independent registered public accounting firm fees and services is incorporated by reference from the information under the caption “Proposal Two: Ratification of Appointment of Independent Registered Public Accounting Firm” in our Proxy Statement for the Annual Meeting of Stockholders.

PART IV

Item 15. Exhibits, Financial Statement Schedules

(a) The following documents are filed as part of this Annual Report on Form 10-K:

(1) Financial Statements:

The financial statements are filed as part of this Annual Report on Form 10-K under “Item 8. Financial Statements and Supplementary Data.”

(2) Financial Statement Schedules:

The financial statement schedules are omitted as they are either not applicable or the information required is presented in the financial statements and notes thereto under “Item 8. Financial Statements and Supplementary Data.”

(3) Exhibits:

See Exhibit Index immediately following the signature page of this Annual Report on Form 10-K.

Item 16. Form 10–K Summary

None.

NETFLIX, INC.

INDEX TO FINANCIAL STATEMENTS

Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors of Netflix, Inc.

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Netflix, Inc. (the Company) as of December 31, 2024 and 2023, the related consolidated statements of operations, comprehensive income, stockholders' equity and cash flows for each of the three years in the period ended December 31, 2024, and the related notes (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2024 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2024, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2024, based on criteria established in Internal Control-Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework), and our report dated January 27, 2025 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

	Content Amortization
Description of the Matter	As disclosed in Note 1 to the consolidated financial statements “Organization and Summary of Significant Accounting Policies”, the Company acquires, licenses and produces content, including original programming (“Content”). The Company amortizes Content based on factors including historical and estimated viewing patterns. Auditing the amortization of the Company’s Content is complex and subjective due to the judgmental nature of amortization which is based on an estimate of future viewing patterns. Estimated viewing patterns are based on historical and forecasted viewing. If actual viewing patterns differ from these estimates, the pattern and/or period of amortization would be changed and could affect the timing of recognition of content amortization.
How We Addressed the Matter in Our Audit	We obtained an understanding, evaluated the design and tested the operating effectiveness of controls over the content amortization process. For example, we tested controls over management’s review of the content amortization method and the significant assumptions, including the historical and forecasted viewing hour consumption, used to develop estimated viewing patterns. We also tested management’s controls to determine that the data used in the model was complete and accurate. To test content amortization, our audit procedures included, among others, evaluating the content amortization method, testing the significant assumptions used to develop the estimated viewing patterns and testing the completeness and accuracy of the underlying data. For example, we assessed management’s assumptions by comparing them to current viewing trends and current operating information including comparing previous estimates of viewing patterns to actual results. We also performed sensitivity analyses to evaluate the potential changes in the content amortization recorded that could result from changes in the assumptions.

/s/ Ernst & Young LLP

We have served as the Company's auditor since 2012.

San Jose, California
January 27, 2025

NETFLIX, INC.

CONSOLIDATED STATEMENTS OF OPERATIONS
(in thousands, except per share data)

	Year ended December 31,		
	2024	2023	2022
Revenues	$ 39,000,966	$ 33,723,297	$ 31,615,550
Cost of revenues	21,038,464	19,715,368	19,168,285
Sales and marketing	2,917,554	2,657,883	2,530,502
Technology and development	2,925,295	2,675,758	2,711,041
General and administrative	1,702,039	1,720,285	1,572,891
Operating income	10,417,614	6,954,003	5,632,831
Other income (expense):			
Interest expense	(718,733)	(699,826)	(706,212)
Interest and other income (expense)	266,776	(48,772)	337,310
Income before income taxes	9,965,657	6,205,405	5,263,929
Provision for income taxes	(1,254,026)	(797,415)	(772,005)
Net income	$ 8,711,631	$ 5,407,990	$ 4,491,924
Earnings per share:			
Basic	$ 20.28	$ 12.25	$ 10.10
Diluted	$ 19.83	$ 12.03	$ 9.95
Weighted-average shares of common stock outstanding:			
Basic	429,519	441,571	444,698
Diluted	439,261	449,498	451,290

See accompanying notes to consolidated financial statements.

NETFLIX, INC.

CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME
(in thousands)

	Year ended December 31,		
	2024	2023	2022
Net income	$ 8,711,631	$ 5,407,990	$ 4,491,924
Other comprehensive income (loss):			
Foreign currency translation adjustments, net of income tax (expense) benefit of $(7) million, $0, and $0, respectively	(247,949)	113,384	(176,811)
Change in unrealized gains on available-for-sale securities, net of income tax (expense) benefit of $(1) million, $0, and $0, respectively	2,511	—	—
Cash flow hedges:			
Net unrealized gains (losses)	921,227	(120,023)	—
Reclassification of net gains included in net income	(96,795)	—	—
Net change, net of income tax (expense) benefit of $(246) million, $36 million, and $0, respectively	824,432	(120,023)	—
Fair value hedges:			
Net unrealized gains excluded from the assessment of effectiveness, net of income tax (expense) benefit of $(2) million, $0, and $0, respectively	7,113	—	—
Total other comprehensive income (loss)	586,107	(6,639)	(176,811)
Comprehensive income	$ 9,297,738	$ 5,401,351	$ 4,315,113

See accompanying notes to consolidated financial statements.

NETFLIX, INC.

CONSOLIDATED STATEMENTS OF CASH FLOWS
(in thousands)

	Year Ended December 31, 2024	2023	2022
Cash flows from operating activities:			
Net income	$ 8,711,631	$ 5,407,990	$ 4,491,924
Adjustments to reconcile net income to net cash provided by operating activities:			
Additions to content assets	(16,223,617)	(12,554,703)	(16,839,038)
Change in content liabilities	(779,135)	(585,602)	179,310
Amortization of content assets	15,301,517	14,197,437	14,026,132
Depreciation and amortization of property, equipment and intangibles	328,914	356,947	336,682
Stock-based compensation expense	272,588	339,368	575,452
Foreign currency remeasurement loss (gain) on debt	(121,539)	176,296	(353,111)
Other non-cash items	494,778	512,075	533,543
Deferred income taxes	(590,698)	(459,359)	(166,550)
Changes in operating assets and liabilities:			
Other current assets	22,180	(181,003)	(353,834)
Accounts payable	121,353	93,502	(158,543)
Accrued expenses and other liabilities	191,899	103,565	(55,513)
Deferred revenue	77,844	178,708	27,356
Other non-current assets and liabilities	(446,351)	(310,920)	(217,553)
Net cash provided by operating activities	7,361,364	7,274,301	2,026,257
Cash flows from investing activities:			
Purchases of property and equipment	(439,538)	(348,552)	(407,729)
Acquisitions	—	—	(757,387)
Purchases of investments	(1,742,246)	(504,862)	(911,276)
Proceeds from maturities of investments	—	1,395,165	—
Net cash provided by (used in) investing activities	(2,181,784)	541,751	(2,076,392)
Cash flows from financing activities:			
Proceeds from issuance of debt	1,794,460	—	—
Repayments of debt	(400,000)	—	(700,000)
Proceeds from issuance of common stock	832,887	169,990	35,746
Repurchases of common stock	(6,263,746)	(6,045,347)	—
Taxes paid related to net share settlement of equity awards	(8,285)	—	—
Other financing activities	(29,743)	(75,446)	—
Net cash used in financing activities	(4,074,427)	(5,950,803)	(664,254)
Effect of exchange rate changes on cash, cash equivalents and restricted cash	(416,331)	82,684	(170,140)
Net increase (decrease) in cash, cash equivalents and restricted cash	688,822	1,947,933	(884,529)
Cash, cash equivalents and restricted cash, beginning of year	7,118,515	5,170,582	6,055,111
Cash, cash equivalents and restricted cash, end of year	$ 7,807,337	$ 7,118,515	$ 5,170,582
Supplemental disclosure:			
Income taxes paid	$ 1,641,530	$ 1,154,973	$ 811,720
Interest paid	674,502	684,504	701,693

See accompanying notes to consolidated financial statements.

NETFLIX, INC.

CONSOLIDATED BALANCE SHEETS
(in thousands, except share and per share data)

	As of December 31,	
	2024	**2023**
Assets		
Current assets:		
Cash and cash equivalents	$ 7,804,733	$ 7,116,913
Short-term investments	1,779,006	20,973
Other current assets	3,516,640	2,780,247
Total current assets	13,100,379	9,918,133
Content assets, net	32,452,462	31,658,056
Property and equipment, net	1,593,756	1,491,444
Other non-current assets	6,483,777	5,664,359
Total assets	$ 53,630,374	$ 48,731,992
Liabilities and Stockholders' Equity		
Current liabilities:		
Current content liabilities	$ 4,393,681	$ 4,466,470
Accounts payable	899,909	747,412
Accrued expenses and other liabilities	2,156,544	1,803,960
Deferred revenue	1,520,813	1,442,969
Short-term debt	1,784,453	399,844
Total current liabilities	10,755,400	8,860,655
Non-current content liabilities	1,780,806	2,578,173
Long-term debt	13,798,351	14,143,417
Other non-current liabilities	2,552,250	2,561,434
Total liabilities	28,886,807	28,143,679
Commitments and contingencies (Note 8)		
Stockholders' equity:		
Preferred stock, $0.001 par value; 10,000,000 shares authorized at December 31, 2024 and December 31, 2023; no shares issued and outstanding at December 31, 2024 and December 31, 2023	—	—
Common stock, $0.001 par value; 4,990,000,000 shares authorized at December 31, 2024 and December 31, 2023; 427,757,100 and 432,759,584 issued and outstanding at December 31, 2024 and December 31, 2023, respectively	6,252,126	5,145,172
Treasury stock at cost (25,953,460 and 16,078,268 shares at December 31, 2024 and December 31, 2023)	(13,171,638)	(6,922,200)
Accumulated other comprehensive income (loss)	362,162	(223,945)
Retained earnings	31,300,917	22,589,286
Total stockholders' equity	24,743,567	20,588,313
Total liabilities and stockholders' equity	$ 53,630,374	$ 48,731,992

See accompanying notes to consolidated financial statements.

NETFLIX, INC.

CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY
(in thousands, except share data)

	Common Stock and Additional Paid-in Capital		Treasury Stock	Accumulated Other Comprehensive Income (Loss)	Retained Earnings	Total Stockholders' Equity
	Shares	Amount				
Balances as of December 31, 2021	443,963,107	$ 4,024,561	$ (824,190)	$ (40,495)	$ 12,689,372	$ 15,849,248
Net income	—	—	—	—	4,491,924	4,491,924
Other comprehensive loss	—	—	—	(176,811)	—	(176,811)
Issuance of common stock	1,383,669	37,588	—	—	—	37,588
Stock-based compensation expense	—	575,452	—	—	—	575,452
Balances as of December 31, 2022	445,346,776	$ 4,637,601	$ (824,190)	$ (217,306)	$ 17,181,296	$ 20,777,401
Net income	—	—	—	—	5,407,990	5,407,990
Other comprehensive loss	—	—	—	(6,639)	—	(6,639)
Issuance of common stock	1,926,598	168,203	—	—	—	168,203
Repurchases of common stock	(14,513,790)	—	(6,098,010)	—	—	(6,098,010)
Stock-based compensation expense	—	339,368	—	—	—	339,368
Balances as of December 31, 2023	432,759,584	$ 5,145,172	$ (6,922,200)	$ (223,945)	$ 22,589,286	$ 20,588,313
Net income	—	—	—	—	8,711,631	8,711,631
Other comprehensive income	—	—	—	586,107	—	586,107
Issuance of common stock	4,872,708	834,366	—	—	—	834,366
Repurchases of common stock	(9,861,935)	—	(6,241,153)	—	—	(6,241,153)
Shares withheld related to net share settlement of equity awards	(13,257)	—	(8,285)	—	—	(8,285)
Stock-based compensation expense	—	272,588	—	—	—	272,588
Balances as of December 31, 2024	427,757,100	$ 6,252,126	$(13,171,638)	$ 362,162	$ 31,300,917	$ 24,743,567

See accompanying notes to consolidated financial statements.

NETFLIX, INC.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

1. Organization and Summary of Significant Accounting Policies

Description of Business

Netflix, Inc. (the “Company”) was incorporated on August 29, 1997 and began operations on April 14, 1998. The Company is one of the world’s leading entertainment services with approximately 302 million paid memberships in over 190 countries enjoying TV series, films and games across a wide variety of genres and languages. Members can play, pause and resume watching as much as they want, anytime, anywhere, and can change their plans at any time.

Basis of Presentation

The consolidated financial statements include the accounts of the Company and its wholly-owned subsidiaries. Intercompany balances and transactions have been eliminated.

Use of Estimates

The preparation of consolidated financial statements in conformity with generally accepted accounting principles ("GAAP") in the United States of America requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities, disclosure of contingent assets and liabilities at the date of the financial statements, and the reported amounts of revenues and expenses during the reporting periods. Significant items subject to such estimates and assumptions include the content asset amortization policy and the recognition and measurement of income tax assets and liabilities. The Company bases its estimates on historical experience and on various other assumptions that the Company believes to be reasonable under the circumstances. On an ongoing basis, the Company evaluates these assumptions, judgments and estimates. Actual results may differ from these estimates.

Recently issued accounting pronouncements not yet adopted

In December 2023, the Financial Accounting Standards Board (“FASB”) issued Accounting Standards Update (“ASU”) 2023-09, *Income Taxes (Topic 740): Improvements to Income Tax Disclosures*, which requires public entities, on an annual basis, to provide disclosure of specific categories in the rate reconciliation, as well as disclosure of income taxes paid disaggregated by jurisdiction. ASU 2023-09 is effective for fiscal years beginning after December 15, 2024, with early adoption permitted. The Company is currently evaluating the impact of adopting ASU 2023-09.

In November 2024, the FASB issued ASU 2024-03, *Income Statement-Reporting Comprehensive Income-Expense Disaggregation Disclosures (Subtopic 220-40): Disaggregation of Income Statement Expenses*, requiring public entities to disclose additional information about specific expense categories in the notes to the financial statements on an interim and annual basis. ASU 2024-03 is effective for fiscal years beginning after December 15, 2026, and for interim periods beginning after December 15, 2027, with early adoption permitted. The Company is currently evaluating the impact of adopting ASU 2024-03.

Recently adopted accounting pronouncements

In November 2023, the FASB issued ASU 2023-07, *Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures*, requiring public entities to disclose information about their reportable segments’ significant expenses and other segment items on an interim and annual basis. Public entities with a single reportable segment are required to apply the disclosure requirements in ASU 2023-07, as well as all existing segment disclosures and reconciliation requirements in ASC 280 on an interim and annual basis. The Company adopted ASU 2023-07 during the year ended December 31, 2024. See Note 12 *Segment and Geographic Information* in the accompanying notes to the consolidated financial statements for further detail.

Cash Equivalents and Short-term Investments

The Company considers investments in instruments purchased with an original maturity of 90 days or less to be cash equivalents. The Company also classifies amounts in transit from payment processors for customer credit card and debit card transactions that it expects to settle within several days as cash equivalents.

The Company classifies short-term investments, which consist of marketable securities with original maturities in excess of 90 days as available-for-sale. Short-term investments are reported at fair value, with allowances for credit losses included in “Interest and other income (expense)” in the Consolidated Statements of Operations and unrealized gains and losses included in “Accumulated other comprehensive income (loss)” within Stockholders’ equity in the Consolidated Balance Sheets. The Company uses the specific identification method to determine cost in calculating realized gains and losses upon the sale of short-term investments.

Short-term investments are reviewed periodically for allowances for credit losses and impairment. When evaluating the investments, the Company reviews factors such as the extent to which the fair value of the security is less than the amortized cost basis, adverse conditions specifically related to the security, the financial condition of the issuer, the Company’s intent to sell, and whether it would be more likely than not that the Company would be required to sell the investments before the recovery of their amortized cost basis.

Content

The Company acquires, licenses and produces content, including original programming, in order to offer members unlimited viewing of video entertainment. The content licenses are for a fixed fee and specific windows of availability. Payment terms for certain content licenses and the production of content require more upfront cash payments relative to the amortization expense. Payments for content, including additions to content assets and the changes in related liabilities, are classified within "Net cash provided by operating activities" on the Consolidated Statements of Cash Flows.

The Company recognizes content assets (licensed and produced) as “Content assets, net” on the Consolidated Balance Sheets. For licensed content, the Company capitalizes the fee per title and records a corresponding liability at the gross amount of the liability when the license period begins, the cost of the title is known and the title is accepted and available for streaming. For produced content, the Company capitalizes costs associated with the production, including development costs, direct costs and production overhead.

Based on factors including historical and estimated viewing patterns, the Company amortizes the content assets (licensed and produced) in “Cost of revenues” on the Consolidated Statements of Operations over the shorter of each title's contractual window of availability or estimated period of use over ten years, beginning with the month of first availability. The amortization is on an accelerated basis, as the Company typically expects more upfront viewing, and film amortization is more accelerated than TV series amortization. On average, over 90% of a licensed or produced content asset is expected to be amortized within four years after its month of first availability. The Company reviews factors impacting the amortization of the content assets on a regular basis. The Company's estimates related to these factors require considerable management judgment.

In the normal course of business, the Company, or a third-party producing content on the Company's behalf, may qualify for tax incentives through eligible spend on productions. The accounting for tax incentives is dependent on the particular type of incentive, including the nature of the benefit and the location the incentive is earned. In general, tax incentives are realized as cash receipts and may be received prior to or after a title launches on the Company’s service. Upon a title’s launch, any amounts the Company is eligible for through qualified production spend but has not received, are recognized in “Other current assets” or “Other non-current assets” on the Company’s Consolidated Balance Sheets as receivables. Tax incentives are generally accounted for as a reduction to the cost basis of the Company’s content assets (presented in “Content assets, net”) and reduce content amortization over the life of the title (as presented in “Cost of revenues”) on the Consolidated Statements of Operations.

The Company's business model is subscription based as opposed to a model generating revenues at a specific title level. Content assets (licensed and produced) are predominantly monetized as a group and therefore are reviewed in aggregate at a group level when an event or change in circumstances indicates a change in the expected usefulness of the content or that the fair value may be less than unamortized cost. To date, the Company has not identified any such event or changes in circumstances. If such changes are identified in the future, these aggregated content assets will be stated at the lower of unamortized cost or fair value. In addition, unamortized costs for assets that have been, or are expected to be, abandoned are written off.

Acquisitions

The Company uses its best estimates and assumptions to assign fair value to the tangible and intangible assets acquired and liabilities assumed at the acquisition date. In addition, uncertain tax positions, tax-related valuation allowances and pre-acquisition contingencies are initially recorded in connection with a business combination as of the acquisition date.

Property and Equipment

Property and equipment are carried at cost less accumulated depreciation. Depreciation is calculated using the straight-line method over the shorter of the estimated useful lives of the respective assets, generally up to 30 years, or the expected lease term for leasehold improvements, if applicable.

Trade Receivables

Trade receivables primarily consist of membership and advertising fees due to the Company. The Company evaluates the need for an allowance for credit losses based on historical collection trends, the financial condition of its payment partners, and external market factors.

Revenue Recognition

The Company's primary source of revenues is from monthly membership fees. Members are billed in advance of the start of their monthly membership and revenues are recognized ratably over each monthly membership period. Revenues are presented net of the taxes that are collected from members and remitted to governmental authorities. The Company is the principal in all its relationships where partners, including consumer electronics ("CE") manufacturers, multichannel video programming distributors ("MVPDs"), mobile operators and internet service providers ("ISPs"), provide access to the service as the Company retains control over service delivery to its members. In circumstances in which the price that the member pays is established by a partner and there is no standalone price for the Netflix service (for instance, in a bundle), the net amount collected from the partner is recognized as revenue.

The Company also earns revenue from advertisements presented on its streaming service, consumer products, live events and various other sources. Revenues earned from sources other than monthly membership fees were not a material component of revenues for the years ended December 31, 2024, 2023, and 2022. See Note 2 *Revenue Recognition* to the consolidated financial statements for further information regarding revenues.

Sales and Marketing

Sales and marketing expenses consist primarily of advertising expenses and certain payments made to marketing and advertising sales partners, including CE manufacturers, MVPDs, mobile operators and ISPs. Sales and marketing expenses also include payroll, stock-based compensation, facilities, and other related expenses for personnel that support the Company's advertising sales and marketing activities. Marketing expenses are expensed as incurred and include promotional activities such as digital and television advertising. Advertising expenses were $1,779 million, $1,732 million and $1,586 million for the years ended December 31, 2024, 2023 and 2022, respectively.

Income Taxes

The Company records a provision for income taxes for the anticipated tax consequences of the reported results of operations using the asset and liability method. Deferred income taxes are recognized by applying enacted statutory tax rates applicable to future years to differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases as well as net operating loss and tax credit carryforwards. The effect on deferred tax assets and liabilities of a change in tax rates is recognized in income in the period that includes the enactment date. The measurement of deferred tax assets is reduced, if necessary, by a valuation allowance for any tax benefits for which future realization is uncertain. The Company accounts for the tax effects of global intangible low tax income as a current period expense.

The Company does not recognize certain tax benefits from uncertain tax positions within the provision for income taxes. The Company may recognize a tax benefit only if it is more likely than not the tax position will be sustained on examination by the taxing authorities, based on the technical merits of the position. The tax benefits recognized in the financial statements from such positions are then measured based on the largest benefit that has a greater than 50% likelihood of being realized upon settlement. The Company recognizes interest and penalties related to uncertain tax positions in income tax expense. See Note 10 *Income Taxes* to the consolidated financial statements for further information regarding income taxes.

Foreign Currency

The functional currency for the Company's subsidiaries is determined based on the primary economic environment in which the subsidiary operates. The Company translates the assets and liabilities of its non-U.S. dollar functional currency subsidiaries into U.S. dollars using exchange rates in effect at the end of each period. Revenues and expenses for these subsidiaries are translated using rates that approximate those in effect during the period. Gains and losses from these translations are recognized in cumulative translation adjustment included in "Accumulated other comprehensive income" in Stockholders' equity on the Consolidated Balance Sheets.

The Company remeasures monetary assets and liabilities that are not denominated in the functional currency at exchange rates in effect at the end of each period. Gains and losses from these remeasurements are recognized in "Interest and other income (expense)" in the Consolidated Statements of Operations. Foreign exchange gains (losses) were $(18) million, $(293) million, and $282 million for the years ended December 31, 2024, 2023, and 2022, respectively. These gains and losses were primarily due to the non-cash remeasurement of our Senior Notes denominated in euros and the remeasurement of cash and content liability positions denominated in currencies other than functional currencies. Foreign exchange gains (losses) for the year ended December 31, 2024 were net of hedging impacts. No hedging gains or losses were recognized in the Consolidated Statements of Operations in the years ended December 31, 2023 and 2022. See Note 7 *Derivative Financial Instruments and Hedging Activities* for further information.

Derivative Financial Instruments and Hedging Activities

The Company uses derivative and non-derivative instruments to manage foreign exchange risk related to its ongoing business operations with the primary objective of reducing earnings and cash flow volatility associated with fluctuations in foreign exchange rates.

The Company recognizes derivative instruments at fair value as either assets (presented in "Other current assets" and "Other non-current assets") or liabilities (presented in "Accrued expenses and other liabilities" and "Other non-current liabilities") on the Company's Consolidated Balance Sheets. The Company classifies derivative instruments in the Level 2 category within the fair value hierarchy.

Cash flow hedges

The Company enters into forward contracts to manage the foreign exchange risk on forecasted revenue transactions denominated in currencies other than the U.S. dollar, as well as the foreign exchange risk on forecasted transactions and firm commitments related to the licensing and production of foreign currency-denominated content assets. These forward contracts are designated as cash flow hedges of foreign currency firm commitments and forecasted transactions and generally have maturities of 36 months or less. The hedging contracts may reduce, but do not entirely eliminate, the effect of foreign currency exchange movements, and the Company may choose not to hedge certain exposures.

The gain or loss on derivative instruments designated as cash flow hedges of forecasted foreign currency revenue is initially reported as a component of accumulated other comprehensive income ("AOCI") and reclassified into "Revenues" on the Consolidated Statements of Operations in the same period the forecasted transaction affects earnings. The gain or loss on derivative instruments designated as cash flow hedges of firmly committed or forecasted transactions related to the licensing and production of content assets is initially reported as a component of AOCI and reclassified into "Cost of Revenues" on the Consolidated Statements of Operations in the same period the hedged transaction affects earnings, which occurs as the underlying hedged content assets are amortized. Cash flows from hedging activities are classified in the same category as the cash flows for the underlying item being hedged within "Net cash provided by operating activities" on the Consolidated Statements of Cash Flows.

In the event that the likelihood of occurrence of the underlying forecasted transactions is determined to be probable not to occur, the gains or losses on the related cash flow hedges are reclassified from AOCI to "Interest and other income (expense)" in the Consolidated Statements of Operations in the period of dedesignation.

Fair value hedges

The Company designates forward contracts as fair value hedges to manage the foreign exchange risk on its foreign-currency denominated debt. These hedges may reduce, but do not entirely eliminate, the effect of foreign currency exchange movements, and the Company may choose not to hedge the full amount of its exposure. The gain or loss on derivative instruments designated as fair value hedges included in the assessment of hedge effectiveness is recognized in "Interest and other income (expense)," net with the offsetting foreign currency remeasurement gains and losses on the hedged items. The Company excludes forward points from the assessment of hedge effectiveness and recognizes the initial value of the excluded component over the life of the hedging instrument in "Interest and other income (expense)" on the Consolidated Statements of Operations. The difference between changes in fair value of the excluded component and the amount recognized in earnings is recognized as a component in AOCI. Cash flows from hedging activities are classified in the same category as the cash flows for the underlying item being hedged within "Net cash provided by (used in) financing activities" on the Consolidated Statements of Cash Flows.

Net investment hedges

The Company designates a portion of its foreign currency-denominated debt as net investment hedges to manage the foreign exchange risk on its investment in certain foreign subsidiaries. These hedges may reduce, but do not entirely eliminate, the effect of foreign currency exchange movements, and the Company may choose not to hedge certain exposures. The gains or losses on these non-derivative instruments are reported as a component of AOCI as part of the cumulative translation adjustment on the Company's Consolidated Balance Sheets. The accumulated gains and losses remain in AOCI until the hedged net investment is sold or liquidated, at which point the amounts recognized in AOCI are reclassified into earnings.

Derivative instruments not designated as hedging instruments

The Company enters into forward contracts to manage the foreign exchange risk on intercompany transactions and monetary assets and liabilities that are not denominated in the functional currencies of the Company and its subsidiaries. These derivative instruments are not designated as hedging instruments and may reduce, but do not entirely eliminate, the effect of foreign currency exchange movements. The gains or losses on derivative instruments not designated as hedging instruments are recorded in "Interest and other income (expense)" in the Consolidated Statements of Operations. Cash flows related to these derivative instruments are classified within "Net cash provided by operating activities" on the Consolidated Statements of Cash Flows.

See Note 7 *Derivative Financial Instruments and Hedging Activities* to the consolidated financial statements for further information regarding the Company's derivative and non-derivative financial instruments.

Stock-Based Compensation

The Company grants non-qualified stock options to its employees on a monthly basis. For certain executive officers, the Company grants restricted stock units ("RSUs") and performance-based restricted stock units ("PSUs"). Stock-based compensation expense is based on the fair value of the stock awards at the grant date and is recognized, net of forfeitures, over the requisite service period. See Note 9 *Stockholders' Equity* to the consolidated financial statements for further information regarding stock-based compensation.

2. Revenue Recognition

The following tables summarize streaming revenues, paid net membership additions (losses), and ending paid memberships by region for the years ended December 31, 2024, 2023 and 2022, respectively. Hedging gains of $124 million are included in "Streaming revenues" for the year ended December 31, 2024. No hedging gains and losses were recognized as "Streaming revenues" in the comparative prior year periods. See Note 7 *Derivative Financial Instruments and Hedging Activities* for further information.

United States and Canada (UCAN)

	As of/Year Ended December 31,		
	2024	2023	2022
		(in thousands)	
Streaming revenues	$ 17,359,369	$ 14,873,783	$ 14,084,643
Paid net membership additions (losses)	9,497	5,832	(919)
Paid memberships at end of period (1)	89,625	80,128	74,296

Europe, Middle East, and Africa (EMEA)

	As of/Year Ended December 31,		
	2024	2023	2022
		(in thousands)	
Streaming revenues	$ 12,387,035	$ 10,556,487	$ 9,745,015
Paid net membership additions	12,320	12,084	2,693
Paid memberships at end of period (1)	101,133	88,813	76,729

Latin America (LATAM)

	As of/Year Ended December 31,		
	2024	2023	2022
		(in thousands)	
Streaming revenues	$ 4,839,816	$ 4,446,461	$ 4,069,973
Paid net membership additions	7,330	4,298	1,738
Paid memberships at end of period (1)	53,327	45,997	41,699

Asia-Pacific (APAC)

	As of/Year Ended December 31,		
	2024	2023	2022
		(in thousands)	
Streaming revenues	$ 4,414,746	$ 3,763,727	$ 3,570,221
Paid net membership additions	12,203	7,315	5,391
Paid memberships at end of period (1)	57,541	45,338	38,023

(1) A paid membership (also referred to as a paid subscription) is defined as a membership that has the right to receive Netflix service following sign-up and a method of payment being provided, and that is not part of a free trial or certain other promotions that may be offered by the Company to new or rejoining members. Certain members have the option to add extra member sub accounts. These extra member sub accounts are not included in paid memberships. A membership is canceled and ceases to be reflected in the above metrics as of the effective cancellation date. Voluntary cancellations generally become effective at the end of the prepaid membership period. Involuntary cancellations, as a result of a failed method of payment, become effective immediately. Memberships are assigned to territories based on the geographic location used at time of sign-up as determined by the Company's internal systems, which utilize industry standard geo-location technology.

Deferred revenue primarily consists of membership fees billed that have not been recognized, as well as gift and other prepaid memberships that have not been fully redeemed. As of December 31, 2024, total deferred revenue was $1,521 million, the vast majority of which was related to membership fees billed that are expected to be recognized as revenue within the next month. The remaining deferred revenue balance, which is related to gift cards and other prepaid memberships, will be recognized as revenue over the period of service after redemption, which is expected to occur over the next 12 months. The $78 million increase in deferred revenue as compared to the balance of $1,443 million for the year ended December 31, 2023, is a result of the increase in membership fees billed due to increased memberships and price increases.

3. Earnings per Share

Basic earnings per share is computed using the weighted-average number of outstanding shares of common stock during the period. Diluted earnings per share is computed using the weighted-average number of outstanding shares of common stock and, when dilutive, potential outstanding shares of common stock during the period. Potential outstanding shares of common stock are calculated using the treasury-stock method and consist of incremental shares issuable upon the assumed exercise of stock options and vesting of time-based and performance-based restricted stock units. The computation of earnings per share is as follows:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands, except per share data)		
Basic earnings per share:			
Net income	$ 8,711,631	$ 5,407,990	$ 4,491,924
Shares used in computation:			
Weighted-average shares of common stock outstanding	429,519	441,571	444,698
Basic earnings per share	$ 20.28	$ 12.25	$ 10.10
Diluted earnings per share:			
Net income	$ 8,711,631	$ 5,407,990	$ 4,491,924
Shares used in computation:			
Weighted-average shares of common stock outstanding	429,519	441,571	444,698
Effect of dilutive stock-based awards	9,742	7,927	6,592
Weighted-average number of shares	439,261	449,498	451,290
Diluted earnings per share	$ 19.83	$ 12.03	$ 9.95

The following table summarizes the potential shares of common stock excluded from the diluted calculation as their inclusion would have been anti-dilutive:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands)		
Stock-based awards	243	4,109	6,790

4. Cash, Cash Equivalents, Restricted Cash, and Short-term Investments

The Company classifies short-term investments, which consist of marketable securities with original maturities in excess of 90 days as available-for-sale ("AFS"). The Company does not buy and hold securities principally for the purpose of selling them in the near future. The Company's policy is focused on the preservation of capital, liquidity and return. From time to time, the Company may sell certain securities but the objectives are generally not to generate profits on short-term differences in price.

The following tables summarize the Company's cash, cash equivalents, restricted cash and short-term investments as of December 31, 2024 and 2023:

	As of December 31, 2024							
	Amortized Cost	Gross Unrealized Gains	Gross Unrealized Losses	Estimated Fair Value	Cash and cash equivalents	Short-term investments	Other Current Assets	Non-current Assets
	(in thousands)							
Cash	$ 4,866,753	$ —	$ —	$ 4,866,753	$ 4,864,207	$ —	$ 2,472	$ 74
Level 1 securities:								
Money market funds	2,676,314	—	—	2,676,314	2,676,256	—	—	58
Level 2 securities:								
Time Deposits (1)	301,374	—	—	301,374	264,270	37,104	—	—
Government securities (2)	1,738,642	3,260	—	1,741,902	—	1,741,902	—	—
	$ 9,583,083	$ 3,260	$ —	$ 9,586,343	$ 7,804,733	$ 1,779,006	$ 2,472	$ 132

	As of December 31, 2023							
	Amortized Cost	Gross Unrealized Gains	Gross Unrealized Losses	Estimated Fair Value	Cash and cash equivalents	Short-term investments	Other Current Assets	Non-current Assets
	(in thousands)							
Cash	$ 5,988,176	$ —	$ —	$ 5,988,176	$ 5,986,629	$ —	$ 1,466	$ 81
Level 1 securities:								
Money market funds	925,707	—	—	925,707	925,652	—	—	55
Level 2 securities:								
Time Deposits (1)	225,605	—	—	225,605	204,632	20,973	—	—
	$ 7,139,488	$ —	$ —	$ 7,139,488	$ 7,116,913	$ 20,973	$ 1,466	$ 136

(1) The majority of the Company's time deposits are international deposits, which mature within one year.

(2) The Company's government securities mature within one year.

Other current assets include restricted cash for deposits related to self-insurance. Non-current assets include restricted cash related to letter of credit agreements. The fair value of AFS securities, cash equivalents and short-term investments included in the Level 2 category is based on observable inputs, such as quoted prices for similar assets at the measurement date; quoted prices in markets that are not active; or other inputs that are observable, either directly or indirectly.

See Note 6 *Debt* and Note 7 *Derivative Financial Instruments and Hedging Activities* to the consolidated financial statements for further information regarding the fair value of the Company's senior notes and derivative financial instruments.

5. Balance Sheet Components

Content Assets, Net

Content assets consisted of the following:

	As of December 31,	
	2024	2023
	(in thousands)	
Licensed content, net	$ 12,422,309	$ 12,722,701
Produced content, net		
Released, less amortization	10,151,543	9,843,150
In production	9,317,367	8,247,578
In development and pre-production	561,243	844,627
	20,030,153	18,935,355
Content assets, net	$ 32,452,462	$ 31,658,056

As of December 31, 2024, approximately $6,409 million, $2,465 million, and $1,431 million of the $12,422 million unamortized cost of the licensed content is expected to be amortized in each of the next three years. As of December 31, 2024, approximately $3,954 million, $2,767 million, and $1,718 million of the $10,152 million unamortized cost of the produced content that has been released is expected to be amortized in each of the next three years.

The following table summarizes the amortization of content assets:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands)		
Licensed content	$ 7,689,014	$ 7,145,446	$ 7,681,978
Produced content (1)	7,612,503	7,051,991	6,344,154
Total	$ 15,301,517	$ 14,197,437	$ 14,026,132

(1) Tax incentives earned on qualified production spend generally reduce the cost-basis of content assets and result in lower content amortization over the life of the title. For the years ended December 31, 2024, 2023 and 2022, tax incentives resulted in lower content amortization on produced content of approximately $899 million, $835 million and $719 million, respectively.

Property and Equipment, Net

Property and equipment and accumulated depreciation consisted of the following:

	As of December 31,		Estimated Useful Lives (in Years)
	2024	2023	
	(in thousands)		
Land	$ 85,000	$ 85,000	
Buildings and improvements	475,684	154,165	30 years
Leasehold improvements	1,026,593	1,032,492	Over life of lease
Furniture and fixtures	134,987	144,737	3 years
Information technology	446,419	414,092	3 years
Corporate aircraft	99,175	99,175	8-10 years
Machinery and equipment	15,135	10,334	3-5 years
Capital work-in-progress	228,300	406,492	
Property and equipment, gross	2,511,293	2,346,487	
Less: Accumulated depreciation	(917,537)	(855,043)	
Property and equipment, net	$ 1,593,756	$ 1,491,444	

Leases

The Company has entered into operating leases primarily for real estate. These leases generally have terms which range from 1 year to 15 years, and often include one or more options to renew. These renewal terms can extend the lease term from 1 year to 20 years, and are included in the lease term when it is reasonably certain that the Company will exercise the option. These operating leases are included in "Other non-current assets" on the Company's Consolidated Balance Sheets, and represent the Company's right to use the underlying asset for the lease term. The Company's obligations to make lease payments are included in "Accrued expenses and other liabilities" and "Other non-current liabilities" on the Company's Consolidated Balance Sheets. Operating lease right-of-use assets and liabilities are recognized at the commencement date based on the present value of lease payments over the lease term. The Company has entered into various short-term operating leases with an initial term of twelve months or less. These leases are not recorded on the Company's Consolidated Balance Sheets. All operating lease expense is recognized on a straight-line basis over the lease term. Because the rate implicit in each lease is not readily determinable, the Company uses its incremental borrowing rate to determine the present value of the lease payments. The Company has certain contracts for real estate which may contain lease and non-lease components which it has elected to treat as a single lease component.

The components of lease costs for the years ended December 31, 2024, 2023 and 2022 were as follows:

	Year ended December 31,		
	2024	2023	2022
	(in thousands)		
Operating lease cost	$ 468,282	$ 430,856	$ 413,664
Short-term lease cost	197,691	207,822	194,764
Total lease cost	$ 665,973	$ 638,678	$ 608,428

Information related to the Company's operating right-of-use assets and related operating lease liabilities were as follows:

	Year ended December 31,		
	2024	2023	2022
	(in thousands)		
Cash paid for operating lease liabilities	$ 509,296	$ 451,525	$ 413,034
Right-of-use assets obtained in exchange for new operating lease obligations	442,391	196,639	252,393

	As of December 31,	
	2024	2023
	(in thousands, except lease term and discount rate)	
Operating lease right-of-use assets, net	$ 2,102,310	$ 2,076,899
Current operating lease liabilities	$ 428,482	$ 383,312
Non-current operating lease liabilities	1,983,688	2,046,801
Total operating lease liabilities	$ 2,412,170	$ 2,430,113
Weighted-average remaining lease term	6.9 years	7.5 years
Weighted-average discount rate	3.5 %	3.3 %

Maturities of operating lease liabilities as of December 31, 2024 were as follows (in thousands):

Due in 12 month period ended December 31,		
2025	$	505,201
2026		477,256
2027		415,983
2028		345,912
2029		276,109
Thereafter		696,367
		2,716,828
Less imputed interest		(304,658)
Total operating lease liabilities	$	2,412,170

The Company has additional operating leases for real estate of $38 million which have not commenced as of December 31, 2024, and as such, have not been recognized on the Company's Consolidated Balance Sheets. These operating leases are expected to commence in 2025 with lease terms between 3 and 7 years.

Other Current Assets

Other current assets consisted of the following:

	As of	
	December 31, 2024	**December 31, 2023**
	(in thousands)	
Trade receivables	$ 1,335,304	$ 1,287,054
Prepaid expenses	431,924	408,936
Other (1)	1,749,412	1,084,257
Total other current assets	$ 3,516,640	$ 2,780,247

(1) $653 million and $555 million of receivables related to tax incentives earned on production spend are included in Other as of December 31, 2024 and 2023, respectively.

6. Debt

As of December 31, 2024, the Company had aggregate outstanding notes of $15,583 million, net of $70 million of issuance costs and discounts, with varying maturities (the "Notes"). Of the outstanding balance, $1,784 million, net of issuance costs, is classified as short-term debt on the Consolidated Balance Sheets. As of December 31, 2023, the Company had aggregate outstanding notes of $14,543 million, net of $65 million of issuance costs. Each of the Notes are senior unsecured obligations of the Company. Interest is payable semi-annually at fixed rates.

A portion of the outstanding Notes is denominated in foreign currency (comprised of €5,170 million) and is remeasured into U.S. dollars at each balance sheet date (with remeasurement gain, net of hedging impacts, totaling $122 million for the year ended December 31, 2024). See Note 7 *Derivative Financial Instruments and Hedging Activities* to the consolidated financial statements for further information regarding the Company’s derivative and non-derivative financial instruments.

The following table provides a summary of the Company's outstanding debt and the fair values based on quoted market prices in less active markets as of December 31, 2024 and December 31, 2023:

	Principal Amount at Par				Level 2 Fair Value as of	
	December 31, 2024	December 31, 2023	Issuance Date	Maturity	December 31, 2024	December 31, 2023
	(in millions)				(in millions)	
5.750% Senior Notes	$ —	$ 400	February 2014	March 2024	$ —	$ 400
5.875% Senior Notes	800	800	February 2015	February 2025	801	807
3.000% Senior Notes (1)	487	519	April 2020	June 2025	487	516
3.625% Senior Notes	500	500	April 2020	June 2025	497	491
4.375% Senior Notes	1,000	1,000	October 2016	November 2026	998	996
3.625% Senior Notes (1)	1,346	1,434	May 2017	May 2027	1,375	1,454
4.875% Senior Notes	1,600	1,600	October 2017	April 2028	1,607	1,621
5.875% Senior Notes	1,900	1,900	April 2018	November 2028	1,970	2,009
4.625% Senior Notes (1)	1,139	1,215	October 2018	May 2029	1,220	1,300
6.375% Senior Notes	800	800	October 2018	May 2029	848	872
3.875% Senior Notes (1)	1,242	1,325	April 2019	November 2029	1,293	1,372
5.375% Senior Notes	900	900	April 2019	November 2029	918	931
3.625% Senior Notes (1)	1,139	1,215	October 2019	June 2030	1,174	1,237
4.875% Senior Notes	1,000	1,000	October 2019	June 2030	996	1,012
4.900% Senior Notes	1,000	—	August 2024	August 2034	982	—
5.400% Senior Notes	800	—	August 2024	August 2054	782	—
	$ 15,653	$ 14,608			$ 15,948	$ 15,018

(1) The following Senior Notes have a principal amount denominated in euros: 3.000% Senior Notes for €470 million, 3.625% Senior Notes for €1,300 million, 4.625% Senior Notes for €1,100 million, 3.875% Senior Notes for €1,200 million, and 3.625% Senior Notes for €1,100 million.

In the year ended December 31, 2024, the Company repaid upon maturity the $400 million aggregate principal amount of its 5.750% Senior Notes.

Each of the Notes are repayable in whole or in part upon the occurrence of a change of control, at the option of the holders, at a purchase price in cash equal to 101% of the principal plus accrued interest. The Company may redeem the Notes prior to maturity in whole or in part at an amount equal to the principal amount thereof plus accrued and unpaid interest and an applicable premium. The Notes include, among other terms and conditions, limitations on the Company's ability to create, incur or allow certain liens, and consolidate or merge with, or convey, transfer or lease all or substantially all of the Company's and its subsidiaries assets, to another person. Certain of the Notes additionally limit the ability to enter into sale and lease-back transactions and create, assume, incur or guarantee additional indebtedness of certain of the Company's subsidiaries. As of December 31, 2024 and December 31, 2023, the Company was in compliance with all related covenants.

Revolving Credit Facility

On April 12, 2024, the Company entered into a five-year, $3 billion unsecured revolving credit facility that matures on April 12, 2029 (the "Revolving Credit Agreement"), to replace its previous $1 billion unsecured revolving credit facility. As of December 31, 2024, no amounts have been borrowed under the Revolving Credit Agreement.

The borrowings under the Revolving Credit Agreement bear interest, at the Company's option, of either (i) a floating rate per annum equal to a base rate (the "Alternate Base Rate") plus an applicable margin or (ii) a per annum rate equal to an adjusted term SOFR rate (the "Adjusted Term SOFR Rate") plus an applicable margin. The applicable margin for Alternate Base Rate loans will range from 0.00% to 0.25%, and the applicable margin for Adjusted Term SOFR Rate loans will range from 0.75% to 1.25%, each based on the Company's credit ratings.

The Revolving Credit Agreement contains customary affirmative covenants and negative covenants (and customary baskets and exceptions with respect thereto) for a credit facility of this size and type and requires the Company to maintain a minimum ratio of consolidated EBITDA to consolidated interest expense of 3.0 to 1.0 as of the last day of each fiscal quarter. As of December 31, 2024 and December 31, 2023, the Company was in compliance with all related covenants and ratios.

7. Derivative Financial Instruments and Hedging Activities

The Company uses derivative and non-derivative instruments to manage foreign exchange risk related to its ongoing business operations with the primary objective of reducing earnings and cash flow volatility associated with fluctuations in foreign exchange rates. The Company did not use any derivative instruments prior to the fiscal year ended December 31, 2023.

Notional Amount of Derivative Contracts

The net notional amounts of the Company's outstanding derivative instruments were as follows:

	As of December 31,	
	2024	2023
	(in thousands)	
Derivatives designated as hedging instruments:		
Foreign exchange contracts		
Cash flow hedges	$ 18,508,390	$ 8,783,273
Fair value hedges	3,819,817	—
Derivatives not designated as hedging instruments:		
Foreign exchange contracts	1,432,136	—
Total	$ 23,760,343	$ 8,783,273

As of December 31, 2024, approximately $1.0 billion of the Company's euro–denominated Senior Notes was designated as a hedge of the foreign exchange risk of the Company's net investment in certain foreign subsidiaries. No amounts were designated as net investment hedges as of December 31, 2023.

As of December 31, 2024, the carrying amount of the Company's euro-denominated Senior Notes (included in "Long-term debt" on the Company's Consolidated Balance Sheets) which were designated as the hedged items in fair value hedges was approximately $3.6 billion. No amounts were designated as fair value hedges as of December 31, 2023.

See Note 6 *Debt* for further information on the Company's debt obligations.

Fair Value of Derivative Contracts

The fair value of the Company's outstanding derivative instruments was as follows:

	As of December 31, 2024			
	Derivative Assets		Derivative Liabilities	
	Other current assets	Other non-current assets	Accrued expenses and other liabilities	Other non-current liabilities
	(in thousands)			
Derivatives designated as hedging instruments:				
Foreign exchange contracts	$ 580,065	$ 406,677	$ 303,425	$ 83
Derivatives not designated as hedging instruments:				
Foreign exchange contracts	16,211	—	14,492	—
Total	$ 596,276	$ 406,677	$ 317,917	$ 83

	As of December 31, 2023			
	Derivative Assets		Derivative Liabilities	
	Other current assets	Other non-current assets	Accrued expenses and other liabilities	Other non-current liabilities
	(in thousands)			
Derivatives designated as hedging instruments:				
Foreign exchange contracts	$ 26,416	$ 4,518	$ 140,089	$ 46,575
Derivatives not designated as hedging instruments:				
Foreign exchange contracts	—	—	—	—
Total	$ 26,416	$ 4,518	$ 140,089	$ 46,575

The Company classifies derivative instruments in the Level 2 category within the fair value hierarchy. These instruments are valued using industry standard valuation models that use observable inputs such as interest rate yield curves, and forward and spot prices for currencies.

As of December 31, 2024, the pre-tax net accumulated gain on our foreign currency cash flow hedges included in AOCI on the Consolidated Balance Sheets expected to be recognized in earnings within the next 12 months is $564 million.

Master Netting Agreements

In order to mitigate counterparty credit risk, the Company enters into master netting agreements with its counterparties for its foreign currency exchange contracts which permit the parties to settle amounts on a net basis under certain conditions. The Company has elected to present its derivative assets and liabilities on a gross basis on its Consolidated Balance Sheets.

The Company also enters into collateral security arrangements with its counterparties that require the parties to post cash collateral when certain contractual thresholds are met. Cash collateral received is presented in “Accrued expenses and other liabilities” representing the Company’s obligation to return counterparty cash collateral. Cash collateral posted is presented in “Other current assets” representing the Company’s right to reclaim the cash collateral. The Company does not offset the fair value of its derivative instruments against the fair value of cash collateral posted or received.

The potential offsetting effect to the Company’s derivative assets and liabilities under its master netting agreements and collateral security agreements were as follows:

	As of December 31, 2024					
				Gross Amount Not Offset in the Consolidated Balance Sheets		
	Gross Amount Recognized in the Consolidated Balance Sheets	Gross Amount Offset in the Consolidated Balance Sheets	Net Amount Presented in the Consolidated Balance Sheets	Financial Instruments	Collateral Received and Posted	Net Amount
	(in thousands)					
Derivative assets	$ 1,002,953	$ —	$ 1,002,953	$ (316,320)	$ (1,800)	$ 684,833
Derivative liabilities	318,000	—	318,000	(316,320)	—	$ 1,680

	As of December 31, 2023					
				Gross Amount Not Offset in the Consolidated Balance Sheets		
	Gross Amount Recognized in the Consolidated Balance Sheets	Gross Amount Offset in the Consolidated Balance Sheets	Net Amount Presented in the Consolidated Balance Sheets	Financial Instruments	Collateral Received and Posted	Net Amount
	(in thousands)					
Derivative assets	$ 30,934	$ —	$ 30,934	$ (27,246)	$ —	$ 3,688
Derivative liabilities	186,664	—	186,664	(27,246)	—	159,418

Effect of Derivative and Non-Derivative Instruments on Consolidated Financial Statements

The pre-tax gains (losses) on the Company's cash flow hedges, fair value hedges, and net investment hedges recognized in AOCI were as follows:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands)		
Cash flow hedges:			
Foreign exchange contracts			
Amount included in the assessment of effectiveness	$ 1,195,738	$ (155,730)	$ —
Fair value hedges:			
Foreign exchange contracts			
Amount excluded from the assessment of effectiveness	(14,334)	—	—
Net investment hedges:			
Foreign currency-denominated debt			
Amount included in the assessment of effectiveness	32,400	—	—
Total	$ 1,213,804	$ (155,730)	$ —

The gains (losses) on hedged items and derivative instruments recognized in the Consolidated Statement of Operations were as follows:

	Year Ended December 31,		
	2024		
	Revenues	Cost of Revenues	Interest and other income (expense)
	(in thousands)		
Total amounts presented in the Consolidated Statements of Operations	$ 39,000,966	$ 21,038,464	$ 266,776
Gains on derivatives in cash flow hedging relationship			
Foreign exchange contracts			
Amount of gains reclassified from AOCI	124,010	1,629	—
Gains (losses) on derivatives in fair value hedging relationship			
Foreign exchange contracts			
Hedged items	—	—	196,660
Derivatives designated as hedging instruments	—	—	(201,239)
Amount excluded from assessment of effectiveness and recognized in earnings based on amortization approach	—	—	(23,567)
Gains on derivatives not designated as hedging instruments			
Foreign exchange contracts	—	—	63,291

No gains or losses on derivative instruments were reclassified from AOCI into the Consolidated Statements of Operations in the years ended December 31, 2023 and December 31, 2022.

8. Commitments and Contingencies

Content

At December 31, 2024, the Company had $23.2 billion of obligations comprised of $4.4 billion included in "Current content liabilities" and $1.8 billion of "Non-current content liabilities" on the Consolidated Balance Sheets and $17.0 billion of obligations that are not reflected on the Consolidated Balance Sheets as they did not yet meet the criteria for recognition.

At December 31, 2023, the Company had $21.7 billion of obligations comprised of $4.5 billion included in "Current content liabilities" and $2.6 billion of "Non-current content liabilities" on the Consolidated Balance Sheets and $14.6 billion of obligations that are not reflected on the Consolidated Balance Sheets as they did not yet meet the criteria for recognition.

The expected timing of payments for these content obligations is as follows:

	As of December 31,	
	2024	2023
	(in thousands)	
Less than one year	$ 11,424,696	$ 10,328,923
Due after one year and through three years	8,113,910	8,784,302
Due after three years and through five years	2,809,834	2,016,358
Due after five years	900,491	583,766
Total content obligations	$ 23,248,931	$ 21,713,349

Content obligations include amounts related to the acquisition, licensing and production of content. Obligations that are in non-U.S. dollar currencies are translated to the U.S. dollar at period end rates. An obligation for the production of content includes non-cancelable commitments under creative talent and employment agreements as well as other production related commitments. An obligation for the acquisition and licensing of content is incurred at the time the Company enters into an agreement to obtain future titles. Once a title becomes available, a content liability is recorded on the Consolidated Balance Sheets. Certain agreements include the obligation to license rights for unknown future titles, the ultimate quantity and/or fees for which are not yet determinable as of the reporting date. Traditional film output deals, or certain TV series license agreements where the number of seasons to be aired is unknown, are examples of such license agreements. The Company does not include any estimated obligation for these future titles beyond the known minimum amount. However, the unknown obligations are expected to be significant.

Legal Proceedings

From time to time, in the normal course of its operations, the Company is subject to litigation matters and claims, including claims relating to employee relations, business practices and patent infringement. Litigation can be expensive and disruptive to normal business operations. Moreover, the results of complex legal proceedings are difficult to predict and the Company's view of these matters may change in the future as the litigation and events related thereto unfold. The Company expenses legal fees as incurred. The Company records a provision for contingent losses when it is both probable that a liability has been incurred and the amount of the loss can be reasonably estimated. An unfavorable outcome to any legal matter, if material, could have an adverse effect on the Company's operations or its financial position, liquidity or results of operations.

The Company is involved in litigation matters not listed herein but does not consider the matters to be material either individually or in the aggregate at this time. The Company's view of the matters not listed may change in the future as the litigation and events related thereto unfold.

Non-Income Taxes

The Company is routinely under audit by various tax authorities with regard to non-income tax matters. The subject matter of non-income tax audits primarily arises from disputes on the tax treatment and tax rate applied to our revenue in certain jurisdictions. We accrue non-income taxes that may result from examinations by, or any negotiated agreements with, these tax authorities when a loss is probable and reasonably estimable.

Similar to other U.S. companies doing business in Brazil, the Company is involved in a number of matters with Brazilian tax authorities regarding non-income tax assessments. Although the Company believes it has meritorious defenses to these matters, there is inherent complexity and uncertainty with respect to these matters, and the final outcome may be materially different from our expectations. The current potential exposure with respect to the various issues with Brazilian tax authorities regarding non-income tax assessments is estimated to be approximately $400 million, which is expected to increase over time.

Guarantees—Indemnification Obligations

In the ordinary course of business, the Company has entered into contractual arrangements under which it has agreed to provide indemnification of varying scope and terms to business partners and other parties with respect to certain matters, including, but not limited to, losses arising out of the Company's breach of such agreements and out of intellectual property infringement claims made by third parties. In these circumstances, payment may be conditional on the other party making a claim pursuant to the procedures specified in the particular contract.

The Company's obligations under these agreements may be limited in terms of time or amount, and in some instances, the Company may have recourse against third parties for certain payments. In addition, the Company has entered into indemnification agreements with its directors and certain of its officers that will require it, among other things, to indemnify them against certain liabilities that may arise by reason of their status or service as directors or officers. The terms of such obligations vary.

It is not possible to make a reasonable estimate of the maximum potential amount of future payments under these or similar agreements due to the conditional nature of the Company's obligations and the unique facts and circumstances involved in each particular agreement. No amount has been accrued in the accompanying consolidated financial statements with respect to these indemnification guarantees.

9. Stockholders' Equity

Voting Rights

The holders of each share of common stock shall be entitled to one vote per share on all matters to be voted upon by the Company's stockholders.

Stock Option Plan

The Netflix, Inc. 2020 Stock Plan is a stockholder-approved plan that provides for the grant of incentive stock options to employees and for the grant of non-statutory stock options, stock appreciation rights, restricted stock and restricted stock units to employees, directors and consultants.

Restricted Stock Unit Awards

The Company grants time-based restricted stock unit ("RSU") awards and performance-based restricted stock unit ("PSU") awards to certain executive officers. RSU awards vest quarterly over a three-year period subject to the executive's continued employment or service with the Company through the vesting date. PSU awards have performance periods ranging from one to three years and vest depending on the Company's achievement of predetermined market-based performance targets.

Stock Option Activity

The following table summarizes the activities related to the Company's stock options:

	Options Outstanding			
	Number of Shares	Weighted- Average Exercise Price (per share)	Weighted- Average Remaining Contractual Term (in years)	Aggregate Intrinsic Value (in thousands)
Balances as of December 31, 2021	17,595,851	$ 219.83		
Granted	3,691,257	267.94		
Exercised	(1,383,669)	27.19		
Expired	(6,578)	11.10		
Balances as of December 31, 2022	19,896,861	$ 242.22		
Granted	1,729,218	372.49		
Exercised	(1,926,598)	87.30		
Expired	(4,372)	36.39		
Balances as of December 31, 2023	19,695,109	$ 268.86		
Granted	575,856	620.92		
Exercised	(4,846,048)	172.17		
Expired	(5,915)	56.05		
Balances as of December 31, 2024	15,419,002	$ 312.48	5.16	$ 8,925,315
Vested and expected to vest as of December 31, 2024	15,419,002	$ 312.48	5.16	$ 8,925,315
Exercisable as of December 31, 2024	15,401,292	$ 312.30	5.16	$ 8,917,827

The aggregate intrinsic value in the table above represents the total pretax intrinsic value (the difference between the Company's closing stock price on the last trading day of 2024 and the exercise price, multiplied by the number of in-the-money options) that would have been received by the option holders had all option holders exercised their options on the last trading day of 2024. This amount changes based on the fair market value of the Company's common stock.

A summary of the amounts related to option exercises, is as follows:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands)		
Total intrinsic value of options exercised	$ 2,352,829	$ 610,594	$ 345,839
Cash received from options exercised	832,887	169,990	35,746

The total fair value of stock options that vested during the years ended December 31, 2024 and 2023 was $242 million and $311 million, respectively. The Company did not grant any stock options subject to vesting conditions in the year ended December 31, 2022.

Restricted Stock Unit Activity

The following table summarizes the activities related to the Company's unvested RSUs and PSUs:

	Unvested Restricted Stock Units	
	Number of Shares	Weighted-Average Grant-Date Fair Value (per share)
Balances as of December 31, 2023	—	$ —
Granted (1)	159,978	686.36
Vested (1)	(26,660)	562.00
Forfeited	—	—
Balances as of December 31, 2024	133,318	$ 711.23

(1) Amounts exclude 26,660 incremental PSU awards that will be granted and 53,320 incremental PSU awards that will vest based on the achievement of market-based performance targets during the period presented, but have not been settled as of December 31, 2024.

The total fair value of RSUs that vested during the year ended December 31, 2024 was $15 million. No RSUs or PSUs were granted in the years ended December 31, 2023 and December 31, 2022.

Stock-Based Compensation

The following table summarizes total stock-based compensation expense and the related income tax impact:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands)		
Total stock-based compensation expense	$ 272,588	$ 339,368	$ 575,452
Total income tax impact on provision	43,876	61,588	127,289

As of December 31, 2024, $46 million of total unrecognized compensation cost related to unvested stock options and RSUs is expected to be recognized over a weighted-average period of 1.4 years.

Stock Repurchases

In September 2023, the Board of Directors authorized the repurchase of up to $10 billion, with no expiration date, and in December 2024, the Board of Directors increased the share repurchase authorization by an additional $15 billion, also with no expiration date. Stock repurchases may be effected through open market repurchases in compliance with Rule 10b-18 under the Exchange Act, including through the use of trading plans intended to qualify under Rule 10b5-1 under the Exchange Act, privately-negotiated transactions, accelerated stock repurchase plans, block purchases, or other similar purchase techniques and in such amounts as management deems appropriate. The Company is not obligated to repurchase any specific number of shares, and the timing and actual number of shares repurchased will depend on a variety of factors, including the Company's stock price, general economic, business and market conditions, and alternative investment opportunities. The Company may discontinue any repurchases of its common stock at any time without prior notice. During the year ended December 31, 2024, the Company repurchased 9,861,935 shares for an aggregate amount of $6,211 million (excluding the 1% excise tax on stock repurchases as a result of the Inflation Reduction Act of 2022). As of December 31, 2024, $17.1 billion remains available for repurchases. Shares repurchased by the Company are accounted for when the transaction is settled. As of December 31, 2024, there were no unsettled share repurchases. Direct costs incurred to acquire the shares are included in the total cost of the shares.

Accumulated Other Comprehensive Income (Loss)

The following table summarizes the changes in accumulated balances of other comprehensive income (loss):

	Foreign Currency Translation Adjustments	Net Investment Hedge Gains (Losses)	Change in Unrealized Gains (Losses) on Cash Flow Hedges	Change in Unrealized Gains (Losses) on Excluded Component of Fair Value Hedges	Change in Unrealized Gains (Losses) on AFS Securities	Tax (Expense) Benefit	Total
				(in thousands)			
Balances as of December 31, 2021	$ (40,495)	$ —	$ —	$ —	$ —	$ —	$ (40,495)
Other comprehensive income (loss) before reclassifications	(176,811)	—	—	—	—	—	(176,811)
Amounts reclassified from accumulated other comprehensive income (loss)	—	—	—	—	—	—	—
Net change in accumulated other comprehensive income (loss)	(176,811)	—	—	—	—	—	(176,811)
Balances as of December 31, 2022	(217,306)	—	—	—	—	—	(217,306)
Other comprehensive income (loss) before reclassifications	113,384	—	(155,730)	—	—	35,707	(6,639)
Amounts reclassified from accumulated other comprehensive income (loss)	—	—	—	—	—	—	—
Net change in accumulated other comprehensive income (loss)	113,384	—	(155,730)	—	—	35,707	(6,639)
Balances as of December 31, 2023	(103,922)	—	(155,730)	—	—	35,707	(223,945)
Other comprehensive income (loss) before reclassifications	(272,911)	32,400	1,195,738	(14,334)	3,260	(279,408)	664,745
Amounts reclassified from accumulated other comprehensive income (loss)	—	—	(125,639)	23,567	—	23,434	(78,638)
Net change in accumulated other comprehensive income (loss)	(272,911)	32,400	1,070,099	9,233	3,260	(255,974)	586,107
Balances as of December 31, 2024	$ (376,833)	$ 32,400	$ 914,369	$ 9,233	$ 3,260	$ (220,267)	$ 362,162

The following table summarizes the amounts reclassified from AOCI to the Consolidated Statement of Operations:

	Year Ended December 31, 2024				
	Revenues	Cost of Revenues	Interest and other income (expense)	Provision for Income Taxes	Total Reclassifications
			(in thousands)		
Gains (losses) on derivatives in cash flow hedging relationship					
Foreign exchange contracts					
Amount of gains (losses) reclassified from AOCI	$ 124,010	$ 1,629	$ —	$ (28,844)	$ 96,795
Gains (losses) on derivatives in fair value hedging relationship					
Foreign exchange contracts					
Amount excluded from assessment of effectiveness and recognized in earnings based on amortization approach	—	—	(23,567)	5,410	(18,157)
Total	$ 124,010	$ 1,629	$ (23,567)	$ (23,434)	$ 78,638

No amounts were reclassified from AOCI into the Consolidated Statements of Operations in the years ended December 31, 2023 and 2022.

10. Income Taxes

Income before provision for income taxes was as follows:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands)		
United States	$ 9,101,391	$ 5,602,762	$ 4,623,218
Foreign	864,266	602,643	640,711
Income before income taxes	$ 9,965,657	$ 6,205,405	$ 5,263,929

The components of provision for income taxes for all periods presented were as follows:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands)		
Current tax provision:			
Federal	$ 1,093,667	$ 854,170	$ 109,910
State	214,814	181,684	119,795
Foreign	536,915	304,539	676,827
Total current	1,845,396	1,340,393	906,532
Deferred tax provision:			
Federal	(520,510)	(412,760)	(52,434)
State	(41,700)	(55,475)	(30,691)
Foreign	(29,160)	(74,743)	(51,402)
Total deferred	(591,370)	(542,978)	(134,527)
Provision for income taxes	$ 1,254,026	$ 797,415	$ 772,005

A reconciliation of the provision for income taxes to the amount computed by applying the 21% statutory Federal income tax rate to income before income taxes is as follows:

	Year Ended December 31,		
	2024	2023	2022
	(in thousands)		
Expected tax expense at U.S. Federal statutory tax rate	$ 2,092,710	$ 1,303,123	$ 1,105,428
State income taxes, net of Federal income tax effect	166,311	104,717	92,084
Foreign earnings at other than U.S. rates	13,909	(32,292)	104,665
Research and development tax credit	(185,312)	(87,036)	(146,615)
Excess tax benefits on stock-based compensation	(435,909)	(119,043)	(75,211)
Foreign-derived intangible income deduction	(502,968)	(426,597)	(361,013)
Nontaxable and nondeductible items	70,386	41,782	44,046
Other	34,899	12,761	8,621
Provision for income taxes	$ 1,254,026	$ 797,415	$ 772,005
Effective Tax Rate	13 %	13 %	15 %

The components of deferred tax assets and liabilities were as follows:

	As of December 31, 2024	2023
	(in thousands)	
Deferred tax assets:		
Stock-based compensation	$ 440,889	$ 486,876
Tax credits and net operating loss carryforwards	834,402	544,431
Capitalized research expenses	1,075,474	593,439
Accruals and reserves	152,142	137,251
Operating lease liabilities	522,489	516,574
OCI hedging losses	—	35,707
Unrealized losses	12,157	26,506
Other	18,197	11,615
Total deferred tax assets	3,055,750	2,352,399
Valuation allowance	(540,272)	(442,293)
Net deferred tax assets	2,515,478	1,910,106
Deferred tax liabilities:		
Depreciation & amortization	(370,709)	(357,477)
Operating right-of-use lease assets	(449,661)	(435,216)
OCI hedging gains	(220,009)	—
Acquired intangibles	(282,187)	(233,433)
Other	(15,354)	(9,430)
Total deferred tax liabilities	(1,337,920)	(1,035,556)
Net deferred tax assets	$ 1,177,558	$ 874,550

The following table shows the deferred tax assets and liabilities within our Consolidated Balance Sheets:

	As of December 31, 2024	2023
	(in thousands)	
Total deferred tax assets:		
Other non-current assets	$ 1,290,160	$ 1,000,760
Total deferred tax liabilities:		
Other non-current liabilities	(112,602)	(126,210)
Net deferred tax assets	$ 1,177,558	$ 874,550

As of December 31, 2024, for tax return purposes, the Company had $694 million of California R&D tax credit carryforwards which can be carried forward indefinitely, $943 million of state net operating loss carryforwards which will begin to expire in 2026, $41 million of foreign tax credit carryforwards which will begin to expire in 2033, and $420 million of foreign net operating loss carryforwards which will begin to expire in 2025.

In evaluating its ability to realize the net deferred tax assets, the Company considered all available positive and negative evidence, including its past operating results and the forecast of future market growth, forecasted earnings, future taxable income, and prudent and feasible tax planning strategies. As of December 31, 2024, the valuation allowance of $540 million was primarily related to California R&D tax credits, state net operating loss carryforwards, and foreign tax credits that the Company does not expect to realize.

At December 31, 2024, we have not provided for applicable U.S. income and foreign withholding taxes on an immaterial amount of undistributed foreign earnings that we intend to indefinitely reinvest. For the balance of undistributed earnings for which we are not indefinitely reinvested, we have provided the appropriate taxes.

The unrecognized tax benefits that are not expected to result in payment or receipt of cash within one year are classified as “Other non-current liabilities” and a reduction of deferred tax assets which is classified as "Other non-current assets" in the Consolidated Balance Sheets. As of December 31, 2024 and 2023, the total amount of gross unrecognized tax benefits was $432 million and $327 million, respectively, of which $251 million and $188 million, respectively, if recognized, would favorably impact the Company's effective tax rate. The aggregate changes in the Company's total gross amount of unrecognized tax benefits are summarized as follows:

	As of December 31,		
	2024	2023	2022
	(in thousands)		
Balance at the beginning of the year	$ 327,105	$ 226,977	$ 202,557
Increases related to tax positions taken during the current period	93,325	65,630	26,865
Increases related to tax positions taken during prior periods	15,751	76,794	—
Decreases related to tax positions taken during prior periods	(3,901)	(10,117)	(2,445)
Decreases related to settlements with taxing authorities	—	(32,179)	—
Decreases related to expiration of statute of limitations	—	—	—
Balance at the end of the year	$ 432,280	$ 327,105	$ 226,977

The Company includes interest and penalties related to unrecognized tax benefits within the provision for income taxes and in "Other non-current liabilities" in the Consolidated Balance Sheets. During the years ended December 31, 2024, 2023 and 2022, the Company recorded $16 million, $25 million, and $2 million, respectively, net of interest and penalties in the provision for income taxes. The amount of interest and penalties accrued at December 31, 2024 and 2023 was $44 million and $28 million, respectively.

The Company files U.S. Federal, state and foreign tax returns. The Company is currently under examination by the IRS for years 2016 through 2018 and is subject to examination for 2019 through 2023. The Company is also generally subject to examination by various state and foreign jurisdictions for years 2017 through 2023. While the Company is in various stages of inquiry and examination with certain taxing authorities and believes that its tax positions will more likely than not be sustained, it is nonetheless possible that future obligations related to these matters could arise. The Company believes that adequate amounts have been reserved for any adjustments that may ultimately result from an examination.

Given the potential outcome of current examinations, it is reasonably possible that the balance of unrecognized tax benefits could significantly change within the next twelve months. However, an estimate of the range of reasonably possible adjustments cannot be made at this time.

11. Employee Benefit Plan

The Company maintains a 401(k) savings plan covering substantially all of its employees. Eligible employees may contribute up to 80% of their annual salary through payroll deductions, but not more than the statutory limits set by the Internal Revenue Service. The Company matches employee contributions at the discretion of the Board. During the years ended December 31, 2024, 2023 and 2022, the Company's matching contributions totaled $128 million, $114 million and $102 million, respectively.

Multiemployer Benefit Plans

The Company contributes to various multiemployer defined pension plans under the terms of collective bargaining agreements that cover our union-represented employees. The risks of participating in multiemployer pension plans are different from single-employer plans such that (i) contributions made by the Company to the multiemployer pension plans may be used to provide benefits to employees of other participating employers; (ii) if the Company chooses to stop participating in the multiemployer pension plans, it may be required to pay those plans an amount based on the underfunded status of the plan; and (iii) if a company stops contributing to the multiemployer pension plan, the unfunded obligations of the plan may become the obligation of the remaining participating employers. The Company also contributes to various other multiemployer benefit plans that provide health and welfare benefits to both active and retired participants. The Company does not participate in any multiemployer benefit plans that are individually significant to the Company.

The following table summarizes the Company's contributions to multiemployer pension and health plans for the years ended December 31, 2024, 2023 and 2022, respectively:

	Year Ended December 31,		
	2024	2023	2022
		(in thousands)	
Pension benefits	$ 89,707	$ 57,285	$ 127,885
Health benefits	134,079	85,157	96,285
Total contributions	$ 223,786	$ 142,442	$ 224,170

12. Segment and Geographic Information

The Company operates as one operating segment. The Company's chief operating decision maker ("CODM") is its co-chief executive officers, who review financial information presented on a consolidated basis. The CODM uses consolidated operating margin and net income to assess financial performance and allocate resources. These financial metrics are used by the CODM to make key operating decisions, such as the determination of the rate at which the Company seeks to grow global operating margin and the allocation of budget between cost of revenues, sales and marketing, technology and development, and general and administrative expenses.

The following table presents selected financial information with respect to the Company’s single operating segment for the years ended December 31, 2024, 2023 and 2022:

	Year Ended December 31,		
	2024	2023	2022
		(in thousands)	
Revenues	$ 39,000,966	$ 33,723,297	$ 31,615,550
Less:			
Content amortization	15,301,517	14,197,437	14,026,132
Other cost of revenues	5,736,947	5,517,931	5,142,153
Sales and marketing	2,917,554	2,657,883	2,530,502
Technology and development	2,925,295	2,675,758	2,711,041
General and administrative	1,702,039	1,720,285	1,572,891
Operating income	10,417,614	6,954,003	5,632,831
Operating margin	27 %	21 %	18 %
Other income (expense)			
Interest expense	(718,733)	(699,826)	(706,212)
Interest and other income (expense) (1)	266,776	(48,772)	337,310
Income before income taxes	9,965,657	6,205,405	5,263,929
Provision for income taxes	(1,254,026)	(797,415)	(772,005)
Net income	$ 8,711,631	$ 5,407,990	$ 4,491,924

(1) Includes interest income of $294 million, $281 million and $60 million for the years ended December 31, 2024, 2023 and 2022, respectively.

See the consolidated financial statements for other financial information regarding the Company’s operating segment.

Total U.S. revenues were $16.1 billion, $13.8 billion and $13.0 billion for the years ended December 31, 2024, 2023 and 2022, respectively. See Note 2 *Revenue Recognition* for additional information about streaming revenue by region.

The Company's long-lived tangible assets, as well as the Company's operating lease right-of-use assets recognized on the Consolidated Balance Sheets were located as follows:

	As of December 31,	
	2024	2023
	(in thousands)	
United States	$ 2,769,828	$ 2,724,710
International	926,238	843,633

EXHIBIT INDEX

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
3.1	Restated Certificate of Incorporation	8-K	001-35727	3.1	June 8, 2022	
3.2	Amended and Restated Bylaws	8-K	001-35727	3.2	February 24, 2023	
4.1	Form of Common Stock Certificate	S-1/A	333-83878	4.1	April 16, 2002	
4.2	Indenture, dated as of February 5, 2015, by and between the Company and Wells Fargo Bank, National Association, as Trustee.	8-K	001-35727	4.2	February 5, 2015	
4.3	Indenture, dated as of October 27, 2016, by and between the Company and Wells Fargo Bank, National Association, as Trustee.	8-K	001-35727	4.1	October 27, 2016	
4.4	Indenture, dated as of May 2, 2017, by and between the Company and Wells Fargo Bank, National Association, as Trustee.	8-K	001-35727	4.1	May 3, 2017	
4.5	Indenture, dated as of October 26, 2017, by and between the Company and Wells Fargo Bank National Association, as Trustee	8-K	001-35727	4.1	October 26, 2017	
4.6	Indenture, dated as of April 26, 2018, by and between the Company and Wells Fargo Bank National Association, as Trustee	8-K	001-35727	4.1	April 26, 2018	
4.7	Indenture, dated as of October 26, 2018, by and between the Company and Wells Fargo Bank National Association, as Trustee (6.375% Senior Notes due 2029)	8-K	001-35727	4.1	October 26, 2018	
4.8	Indenture, dated as of October 26, 2018, by and between the Company and Wells Fargo Bank National Association, as Trustee (4.625% Senior Notes due 2029)	8-K	001-35727	4.3	October 26, 2018	
4.9	Indenture, dated as of April 29, 2019, by and between the Company and Wells Fargo Bank National Association, as Trustee (5.375% Senior Notes due 2029)	8-K	001-35727	4.1	April 29, 2019	
4.10	Indenture, dated as of April 29, 2019, by and between the Company and Wells Fargo Bank National Association, as Trustee (3.875% Senior Notes due 2029)	8-K	001-35727	4.3	April 29, 2019	
4.11	Indenture, dated as of October 25, 2019, by and between the Company and Wells Fargo Bank National Association, as Trustee (4.875% Senior Notes due 2030)	8-K	001-35727	4.1	October 25, 2019	
4.12	Indenture, dated as of October 25, 2019, by and between the Company and Wells Fargo Bank National Association, as Trustee (3.625% Senior Notes due 2030)	8-K	001-35727	4.3	October 25, 2019	
4.13	Indenture, dated as of April 28, 2020, by and between the Company and Wells Fargo Bank National Association, as Trustee (3.625% Senior Notes due 2025)	8-K	001-35727	4.1	April 28, 2020	
4.14	Indenture, dated as of April 28, 2020, by and between the Company and Wells Fargo Bank National Association, as Trustee (3.000% Senior Notes due 2025)	8-K	001-35727	4.3	April 28, 2020	
4.15	Indenture, dated as of July 29, 2024, by and between the Company and Computershare Trust Company, National Association, as Trustee.	S-3	333-281071	4.1	July 29, 2024	
4.16	Supplemental Indenture, dated as of August 1, 2024, by and between the Company and Computershare Trust Company, National Association, as Trustee.	8-K	001-35727	4.2	August 1, 2024	
4.17	Description of Registrant's Securities Registered Pursuant to Section 12 of the Securities Exchange Act of 1934	10-K	001-35727	4.18	January 26, 2023	
10.1†	Form of Indemnification Agreement entered into by the registrant with each of its executive officers and directors	S-1/A	333-83878	10.1	March 20, 2002	
10.2†	2011 Stock Plan	Def 14A	000-49802	A	April 20, 2011	
10.3†	2020 Stock Plan	Def 14A	001-35727	A	April 22, 2020	
10.4†	Description of Director Equity Compensation Plan	8-K	001-35727	Item 5.02	January 24, 2018	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.5†	Amended and Restated Performance Bonus Plan	8-K	001-35727	10.1	December 9, 2022	
10.6†	Form of Stock Option Agreement under the 2011 Stock Plan	10-K	001-35727	10.11	January 27, 2022	
10.7†	Form of Stock Option Agreement under the 2020 Stock Plan	10-K	001-35727	10.11	January 26, 2023	
10.8†	Form of Stock Option Agreement under the 2020 Stock Plan (Options Subject to Vesting)	8-K	001-35727	10.1	December 23, 2022	
10.9†	Netflix, Inc. 2020 Stock Plan Form of Restricted Stock Unit Award Agreement	8-K	001-35727	10.1	December 8, 2023	
10.10†	Netflix, Inc. 2020 Stock Plan Form of Performance-Based Restricted Stock Unit Award Agreement	8-K	001-35727	10.2	December 8, 2023	
10.11†	Netflix, Inc. Executive Officer Severance Plan	8-K	001-35727	10.3	December 8, 2023	
19.1	Netflix, Inc. Insider Trading Policy					X
21.1	List of Significant Subsidiaries					X
23.1	Consent of Ernst & Young LLP					X
24	Power of Attorney (see signature page)					
31.1	Certification of Co-Chief Executive Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002					X
31.2	Certification of Co-Chief Executive Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002					X
31.3	Certification of Chief Financial Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002					X
32.1*	Certifications of Co-Chief Executive Officers and Chief Financial Officer Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002					X
97.1	Netflix, Inc. Clawback Policy	10-K	001-35727	97.1	January 26, 2024	
101	The following financial statements from the Company's Annual Report on Form 10-K for the year ended December 31, 2024, formatted in Inline XBRL: (i) Consolidated Statements of Operations, (ii) Consolidated Statements of Comprehensive Income, (iii) Consolidated Statements of Cash Flows, (iv) Consolidated Balance Sheets, (v) Consolidated Statements of Stockholders' Equity and (vi) Notes to Consolidated Financial Statements, tagged as blocks of text and including detailed tags					X
104	The cover page from the Company's Annual Report on Form 10-K for the year ended December 31, 2024, formatted in Inline XBRL					X

* These certifications are not deemed filed by the SEC and are not to be incorporated by reference in any filing we make under the Securities Act of 1933 or the Securities Exchange Act of 1934, irrespective of any general incorporation language in any filings.

† Indicates a management contract or compensatory plan

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

			Netflix, Inc.
Dated:	January 27, 2025	By:	/S/ TED SARANDOS
			Ted Sarandos Co-Chief Executive Officer *(principal executive officer)*
Dated:	January 27, 2025	By:	/S/ GREG PETERS
			Greg Peters Co-Chief Executive Officer *(principal executive officer)*

POWER OF ATTORNEY

KNOWN ALL PERSONS BY THESE PRESENTS, that each person whose signature appears below constitutes and appoints Ted Sarandos, Greg Peters, and Spencer Neumann, and each of them, as his true and lawful attorneys-in-fact and agents, with full power of substitution and resubstitution, for him and in his name, place, and stead, in any and all capacities, to sign any and all amendments to this Report, and to file the same, with all exhibits thereto, and other documents in connection therewith, with the Securities and Exchange Commission, granting unto said attorneys-in-fact and agents, and each of them, full power and authority to do and perform each and every act and thing requisite and necessary to be done in connection therewith, as fully to all intents and purposes as he might or could do in person, hereby ratifying and confirming that all said attorneys-in-fact and agents, or any of them or their or his substitute or substituted, may lawfully do or cause to be done by virtue thereof.

Pursuant to the requirements of the Securities and Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Ted Sarandos Ted Sarandos	Co-Chief Executive Officer and Director (principal executive officer)	January 27, 2025
/s/ Greg Peters Greg Peters	Co-Chief Executive Officer and Director (principal executive officer)	January 27, 2025
/s/ Spencer Neumann Spencer Neumann	Chief Financial Officer (principal financial officer)	January 27, 2025
/s/ Jeffrey Karbowski Jeffrey Karbowski	Chief Accounting Officer (principal accounting officer)	January 27, 2025
/s/ Reed Hastings Reed Hastings	Executive Chairman and Director	January 27, 2025
/s/ Richard Barton Richard Barton	Director	January 27, 2025
/s/ Mathias Döpfner Mathias Döpfner	Director	January 27, 2025
/s/ Timothy M. Haley Timothy M. Haley	Director	January 27, 2025
/s/ Jay C. Hoag Jay C. Hoag	Director	January 27, 2025
/s/ Leslie J. Kilgore Leslie J. Kilgore	Director	January 27, 2025
/s/ Strive Masiyiwa Strive Masiyiwa	Director	January 27, 2025
/s/ Ann Mather Ann Mather	Director	January 27, 2025

/S/ SUSAN RICE Susan Rice	Director	January 27, 2025
/S/ BRAD SMITH Brad Smith	Director	January 27, 2025
/S/ ANNE SWEENEY Anne Sweeney	Director	January 27, 2025

Insider Trading Policy

In order to take an active role in the prevention of insider trading violations by executive officers, directors, employees and other related individuals of Netflix, Inc. (the "Company") and its subsidiaries, the Company has adopted this Insider Trading Policy (the "Policy").

Statement of Intent

The Company opposes the misuse of material nonpublic information in the trading of securities and it is the intent of this Policy to implement procedures designed to prevent trading based on material nonpublic information regarding the Company, including any of its subsidiaries. The Company also wishes to discourage certain trading in its securities by its executive officers, directors, employees and other related individuals that may be contrary to the interests of our shareholders. The term "executive officer" herein shall have the same meaning as the term "officer" as defined under Rule 16a-1(f) of the Securities Exchange Act of 1934, as amended (the "Exchange Act").

It is also Company policy to comply with applicable securities laws concerning trading in Company securities on the Company's behalf.

Definition of Material Nonpublic Information

It is not possible to define all categories of material information. However, information should be regarded as material if there is a substantial likelihood that it would be considered important to a reasonable investor in making an investment decision regarding transactions of the Company's securities. Put another way, there must be a substantial likelihood that the information would be viewed by the reasonable investor as having significantly altered the total mix of information available in the market concerning the Company. Either positive or negative information may be material.

Questions concerning whether nonpublic information is material can be directed to the Compliance Officer.

Covered Parties

The Policy covers executive officers, directors and all employees of, or consultants or contractors to, the Company or its subsidiaries, as well as their immediate family members who reside with them, anyone else who lives in their household, and any other individuals who do not live in their household but whose transactions in Company securities are directed by them or are subject to their influence or control (such as parents or children who consult with them before trading in Company securities) ("Insiders"). This Policy applies to trades of Company securities in which an Insider has any "beneficial" or other interest, or over which they exercise investment control.

Covered Transactions

This Policy applies to all transactions in the Company's securities, including common stock, options for common stock and any other securities the Company may issue from time to time, such as preferred stock, warrants and convertible debentures, as well as to derivative securities relating to the Company's securities, whether or not issued by the Company, such as publicly-traded options. Transactions subject to this Policy include purchases, sales and bona fide gifts of the Company's securities (which may include gifts to trusts for estate planning purposes, as well as donations to a charitable organization).

Prohibited Transactions

No Insider shall engage in any transaction involving the Company's securities, including any offer to purchase or offer to sell, during any period commencing with the date that the Insider possesses material nonpublic information concerning the Company or its subsidiaries, and ending at the beginning of the trading day following the date of public disclosure of that information, or at such time as such nonpublic information is no longer material.

No Insider shall disclose ("tip") material nonpublic information about the Company or its subsidiaries to any other person where such information may be used by such person to their profit by trading in the securities of companies to which such information relates, nor shall such Insider or related person make recommendations or express opinions on the basis of material nonpublic information as to trading in the Company's securities.

No Insider shall engage in any transaction involving another company's securities while in possession of material nonpublic information about such company when that information is obtained in the course of employment with, or the performance of services on behalf of, the Company and the other company is a business partner of the Company such as one with which the Company is discussing a proposed transaction, collaborator, customer, distributor, vendor or supplier.

Except for participation in the Company's stock option program, no executive officer or director shall, at any time, engage in any transactions involving any hedging or derivatives of Company securities, including trading in futures and derivative securities and engaging in hedging activities relating to our securities, including exchange traded options, puts, calls, collars, forward sale contracts, equity swaps, exchange funds or other arrangements or instruments designed to hedge or offset decreases in the market value of our securities, provided that it shall not be a violation of this Policy for an executive officer or director to engage in any such transaction initiated prior to March 4, 2020.

Except as may otherwise be authorized under this Policy, executive officers and directors shall not initiate any transactions subsequent to March 4, 2020 that involve pledging any Company securities as collateral for a loan or holding them as security in a margin account.

Problematic Transactions

While employees (other than executive officers, except as otherwise provided) are not prohibited by this Policy from using Company securities as collateral for loans or in margin accounts with respect to the Company's securities or engaging in transactions involving hedging or derivatives of Company securities, including by means of exchange traded options, puts, calls, collars, forward sale contracts, equity swaps, exchange funds or other arrangements or instruments designed to hedge or offset decreases in the market value of our securities, the Company discourages employees from such activity because, among other problems, these types of transactions may result in transactions in Company securities occurring outside the Open Window (defined below). Limit orders with brokers should not extend beyond any Open Window and be cancellable upon an imposition of a black-out period. Employees interested in trading outside of the Open Window should look into adopting a 10b5-1 trading plan, as described below. Exercising stock options issued pursuant to the Company's stock option plan, as otherwise permitted under this Policy, is not considered problematic.

The Company's Trading Window

The Company has determined that all executive officers, directors, and those other persons identified on Attachment 1 (as may be amended from time to time by the Compliance Officer), shall be prohibited from buying, selling or otherwise effecting transactions in any stock or other securities of the Company or derivative securities thereof EXCEPT during the following trading window:

Beginning at the open of market on the trading day following the date of public disclosure of the Company's financial results for a preceding calendar quarter or year and ending at the close of market on the 10th day of the second calendar month of the current calendar quarter (the "Open Window").

In addition, the Company, through the Compliance Officer, may authorize longer or additional trading windows in which buying, selling, gifting or otherwise effecting transactions in the Company's securities would be permitted pursuant to this Policy as if it were the "Open Window." Similarly, the Company, through the Compliance Officer, may impose special black-out periods during which certain persons will be prohibited from buying, selling or otherwise effecting transactions in the Company's securities, even though the trading window would otherwise be open. If a special black-out period is imposed, the Company will notify affected individuals, who should thereafter not engage in any transaction involving the Company's securities and should not disclose to others the fact of such suspension of trading.

It should be noted that even during the Open Window, any person possessing material nonpublic information should not engage in any transactions in the Company's securities until the beginning of the trading day following the date of public disclosure of such information, whether or not the Company has recommended a suspension of trading to that person.

Pre-Clearance of Trades and Other Transactions

All executive officers and directors of the Company and other persons designated by the Compliance Officer as being subject to pre-clearance requirements must refrain from trading in or gifting the Company's securities, even during the Open Window, without first contacting the Company's Compliance Officer and obtaining pre-clearance to commence trading in the Company's securities.

In addition, all executive officers and directors are required to comply with Section 16 of the Exchange Act and related rules and regulations, which set forth reporting obligations as well as limitations on "short swing" transactions. The Company is available to assist in filing Section 16 reporting; however, the obligation to comply with Section 16 is personal. Please direct any inquiries concerning compliance to the Compliance Officer. Executive officers and directors who wish to pledge the Company's stock as collateral after March 4, 2020 must first contact the Company's Compliance Officer to request an exemption and pre-clearance to enter into the transaction.

Adoption and Effect of 10b5-1 Trading Plans

The Company permits all directors, executive officers and employees to adopt trading plans in accordance with Rule 10b5-1(c) under the Exchange Act ("10b5-1 trading plans"). A 10b5-1 trading plan must comply with the requirements of Rule 10b5-1 and may only be adopted or amended in an Open Window and when the person adopting the plan is not otherwise in possession of material nonpublic information about the Company, its subsidiaries or its securities. All 10b5-1 trading plans must be pre-approved by the Compliance Officer or his delegate. The restrictions on trading set forth in this Policy shall not apply to trades made pursuant to a 10b5-1 trading plan. More information concerning trading plans is available from the Compliance Officer.

Exemptions from this Policy

The exercise of stock options under the Company's stock option plan with a cash payment of the exercise price is exempt from this Policy, since the other party to these transactions is the Company itself and the price does not vary with the market, but is fixed by the terms of the option agreement, or the Company withholding a portion of the shares underlying the options to satisfy exercise price or tax withholding requirements. This exemption does not apply to the sale of any shares issued upon such exercise and it does not apply to a cashless exercise of options, which is accomplished by a sale of a portion of the shares issued upon exercise of an option.

Post-Termination Transactions

If a person is in possession of material nonpublic information when their service terminates, that individual should not trade in Company securities until that information has become public or is no longer material.

Consequences for Violation

Directors and employees who violate this Policy shall also be subject to disciplinary action by the Company, which may include ineligibility for future participation in the Company's equity stock option and other incentive plans or termination of employment or service.

Pursuant to U.S. federal and state securities laws, Insiders may be subject to criminal and civil fines and penalties as well as imprisonment for engaging in transactions in the Company's securities at a time when they are in possession of material nonpublic information regarding the Company or its subsidiaries. In addition, Insiders may be liable for improper transactions by any person (commonly referred to as a "tippee") to whom they have disclosed material nonpublic information regarding the Company or its subsidiaries or to whom they have made recommendations or expressed opinions on the basis of such information as to trading in the Company's securities.

Individual Responsibility

Every executive officer, director and employee, consultant and contractor has the individual responsibility to comply with this Policy, and the applicable laws of their jurisdiction. An Insider may, from time to time, have to forego a proposed transaction in the Company's securities even if he or she planned to make the transaction before learning of the material nonpublic information and even though the Insider believes he or she may suffer an economic loss or forego anticipated profit by waiting. **Trading in the Company's securities during the trading window (including with pre-clearance) should not be considered a "safe harbor," and all directors, executive officers and other persons should use good judgment at all times.**

Compliance Officer

- The Company's Chief Legal Officer shall serve as the Insider Trading Compliance Officer (the "Compliance Officer"). The duties of the Compliance Officer shall include, but not be limited to, the following:
- Pre-clearing transactions as required under this Policy.
- Assisting, as requested, in the preparation and filing of Section 16 reports (Forms 3, 4 and 5) for Section 16 reporting persons.
- Serving as the designated recipient at the Company of copies of reports filed with the Securities and Exchange Commission by Section 16 reporting persons under Section 16 of the Exchange Act.
- Circulating the Policy (and/or a summary thereof) to all employees, including Section 16 reporting persons, on an annual basis.
- Assisting the Company in implementation of the Policy.
- Coordinating with Company counsel regarding compliance activities with respect to Rule 144 requirements and regarding changing requirements and recommendations for compliance with Section 16 of the Exchange Act and insider trading laws to ensure that the Policy is amended as necessary to comply with such requirements.

The duties may be delegated by the Compliance Officer to such other individuals as the Compliance Officer deems appropriate.

* * * * *

Attachment 1

Persons subject to trading window restrictions

Insiders, as defined in the Company’s Insider Trading Policy, except for the following:

- Employees, contractors and consultants who are employed or retained through a direct or indirect subsidiary of the Company that is specifically and solely engaged in the production of original content and not otherwise included as a matter of business practice in the distribution of corporate information about the Company and its financial performance, unless such person has been notified in writing by the Compliance Officer that the trading window applies to them.
- Employees of subsidiaries that are not integrated into the Company's email systems and/or who do not have access to consolidated financial data.

EXHIBIT 21.1

NETFLIX, INC.

LIST OF SIGNIFICANT SUBSIDIARIES*

Legal Name	Jurisdiction	Percent Owned
Netflix Entretenimento Brasil LTDA	Brazil	100 %
Netflix International B.V.	The Netherlands	100 %
Netflix Studios, LLC	United States	100 %
Netflix Global, LLC	United States	100 %
Netflix México S. de R.L. de C.V.	Mexico	100 %
Netflix Pte. Ltd.	Singapore	100 %
Netflix Services France S.A.S.	France	100 %
Netflix Services UK Limited	United Kingdom	100 %
Netflix Services Germany GmbH	Germany	100 %
Netflix Worldwide Entertainment, LLC	United States	100 %
Netflix Services Canada ULC	Canada	100 %

* Pursuant to Item 601(b)(21)(ii) of Regulation S-K, the names of other subsidiaries of Netflix Inc. are omitted because, considered in the aggregate, they would not constitute a significant subsidiary as of the end of the year covered by this report.

EXHIBIT 23.1

Consent of Independent Registered Public Accounting Firm

We consent to the incorporation by reference in the following Registration Statements:

(1) Registration Statement (Form S-8 No. 333-89024) pertaining to Netflix.com, Inc. 1997 Stock Plan, as amended, Netflix, Inc. 2002 Stock Plan, Netflix, Inc. 2002 Employee Stock Purchase Plan, and Netflix.com, Inc. Stand-Alone Stock Option Agreements,

(2) Registration Statement (Form S-8 No. 333-104250) pertaining to Netflix, Inc. 2002 Stock Plan and Netflix, Inc. 2002 Employee Stock Purchase Plan,

(3) Registration Statement (Form S-8 No. 333-113198) pertaining to Netflix, Inc. 2002 Stock Plan and Netflix, Inc. 2002 Employee Stock Purchase Plan,

(4) Registration Statement (Form S-8 No. 333-123501) pertaining to Netflix, Inc. 2002 Stock Plan and Netflix, Inc. 2002 Employee Stock Purchase Plan,

(5) Registration Statement (Form S-8 No. 333-136403) pertaining to Netflix, Inc. Amended and Restated 2002 Stock Plan and Netflix, Inc. 2002 Employee Stock Purchase Plan,

(6) Registration Statement (Form S-8 No. 333-145147) pertaining to Netflix, Inc. 2002 Employee Stock Purchase Plan,

(7) Registration Statement (Form S-8 No. 333-160946) pertaining to Netflix, Inc. 2002 Employee Stock Purchase Plan,

(8) Registration Statement (Form S-8 No. 333-177561) pertaining to Netflix, Inc. 2011 Stock Plan,

(9) Registration Statement (Form S-8 No. 333-239465) pertaining to Netflix, Inc. 2020 Stock Plan,

(10) Registration Statement (Form S-8 No. 333-239468) pertaining to Netflix, Inc. Post-Effective Amendment No.1 for the 2011 Stock Plan and the 2020 Stock Plan, and

(11) Registration Statement (Form S-3ASR No. 333-281071) pertaining to Netflix, Inc.

of our reports dated January 27, 2025, with respect to the consolidated financial statements of Netflix, Inc. and the effectiveness of internal control over financial reporting of Netflix, Inc. included in this Annual Report (Form 10-K) of Netflix, Inc. for the year ended December 31, 2024.

/s/ Ernst & Young LLP

San Jose, California
January 27, 2025

EXHIBIT 31.1

CERTIFICATION OF CO-CHIEF EXECUTIVE OFFICER
PURSUANT TO SECTION 302 OF THE SARBANES-OXLEY ACT OF 2002

I, Ted Sarandos, certify that:

1. I have reviewed this Annual Report on Form 10-K of Netflix, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officers and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
    a. designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
    b. designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
    c. evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
    d. disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officers and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of registrant's board of directors (or persons performing the equivalent function):
    a. all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
    b. any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Dated: January 27, 2025	By:	/S/ TED SARANDOS
		Ted Sarandos
		Co-Chief Executive Officer

EXHIBIT 31.2

CERTIFICATION OF CO-CHIEF EXECUTIVE OFFICER
PURSUANT TO SECTION 302 OF THE SARBANES-OXLEY ACT OF 2002

I, Greg Peters, certify that:

1. I have reviewed this Annual Report on Form 10-K of Netflix, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officers and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
    a. designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
    b. designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
    c. evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
    d. disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officers and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of registrant's board of directors (or persons performing the equivalent function):
    a. all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
    b. any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Dated: January 27, 2025 By: /S/ GREG PETERS

Greg Peters
Co-Chief Executive Officer

EXHIBIT 31.3

CERTIFICATION OF CHIEF FINANCIAL OFFICER
PURSUANT TO SECTION 302 OF THE SARBANES-OXLEY ACT OF 2002

I, Spencer Neumann, certify that:

1. I have reviewed this Annual Report on Form 10-K of Netflix, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant’s other certifying officers and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
    a. designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
    b. designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
    c. evaluated the effectiveness of the registrant’s disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
    d. disclosed in this report any change in the registrant’s internal control over financial reporting that occurred during the registrant’s most recent fiscal quarter (the registrant’s fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant’s internal control over financial reporting; and
5. The registrant’s other certifying officers and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant’s auditors and the audit committee of registrant’s board of directors (or persons performing the equivalent function):
    a. all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant’s ability to record, process, summarize and report financial information; and
    b. any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant’s internal control over financial reporting.

Dated: January 27, 2025	By:	/S/ SPENCER NEUMANN
		Spencer Neumann
		Chief Financial Officer

EXHIBIT 32.1

CERTIFICATIONS OF CO-CHIEF EXECUTIVE OFFICERS AND CHIEF FINANCIAL OFFICER PURSUANT TO SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002

I, Ted Sarandos, certify, pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that the Annual Report on Form 10-K of Netflix, Inc. for the year ended December 31, 2024 fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and that information contained in such report fairly presents, in all material respects, the financial condition and results of operations of Netflix, Inc.

Dated: January 27, 2025	By:	/S/ TED SARANDOS
		Ted Sarandos
		Co-Chief Executive Officer

I, Greg Peters, certify, pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that the Annual Report on Form 10-K of Netflix, Inc. for the year ended December 31, 2024 fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and that information contained in such report fairly presents, in all material respects, the financial condition and results of operations of Netflix, Inc.

Dated: January 27, 2025	By:	/S/ GREG PETERS
		Greg Peters
		Co-Chief Executive Officer

I, Spencer Neumann, certify, pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that the Annual Report on Form 10-K of Netflix, Inc. for the year ended December 31, 2024 fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and that information contained in such report fairly presents, in all material respects, the financial condition and results of operations of Netflix, Inc.

Dated: January 27, 2025	By:	/S/ SPENCER NEUMANN
		Spencer Neumann
		Chief Financial Officer

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2024

or

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from to .

Commission File No. 001-42297

BlackRock, Inc.
(Exact name of registrant as specified in its charter)

Delaware	**99-1116001**
(State or Other Jurisdiction of Incorporation or Organization)	**(I.R.S. Employer Identification No.)**

50 Hudson Yards, New York, NY 10001
(Address of Principal Executive Offices)

(212) 810-5800
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock, $.01 par value	**BLK**	**New York Stock Exchange**

Securities registered pursuant to Section 12(g) of the Act:
None

Indicate by check mark if the registrant is a well-known, seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☒	Accelerated filer ☐
Non-accelerated filer ☐	Smaller reporting company ☐
	Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

The aggregate market value of the voting common stock held by nonaffiliates of the registrant as of June 30, 2024 was approximately $118 billion.

As of January 31, 2025, there were 155,253,443 shares of the registrant's common stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

The following documents are incorporated by reference herein:

Portions of the definitive Proxy Statement of BlackRock, Inc. to be filed pursuant to Regulation 14A of the general rules and regulations under the Securities Exchange Act of 1934, as amended, for the 2025 annual meeting of stockholders ("Proxy Statement") are incorporated by reference into Part III of this Form 10-K.

BlackRock, Inc.
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PART I

Item 1. Business

OVERVIEW

BlackRock, Inc. (together, with its subsidiaries, unless the context otherwise indicates, “BlackRock” or the “Company”) is a leading publicly traded investment management firm with $11.6 trillion of assets under management (“AUM”) at December 31, 2024. With approximately 21,100 employees in more than 30 countries who serve clients in over 100 countries across the globe, BlackRock provides a broad range of investment management and technology services to institutional and retail clients worldwide. On October 1, 2024, BlackRock completed the acquisition of 100% of the issued and outstanding limited liability company interests of Global Infrastructure Management, LLC ("GIP" or the "GIP Transaction"). As a result of the closing of the GIP Transaction, BlackRock, Inc. (formerly known as BlackRock Funding, Inc. ("BlackRock Funding")) (“New BlackRock”) became the ultimate parent company of BlackRock Finance, Inc. (formerly known as BlackRock, Inc.) (“Old BlackRock”), GIP and their respective subsidiaries. See Note 1, *Business Overview* and Note 3, *Acquisitions,* in the notes to the consolidated financial statements contained in Part II, Item 8 for additional information.

BlackRock’s diverse platform of alpha-seeking active, private markets, index and cash management investment strategies across asset classes enables the Company to offer choice and tailor investment and asset allocation solutions for clients. Product offerings include single- and multi-asset portfolios investing in equities, fixed income, private markets, liquid alternatives and money market instruments. Products are offered directly and through intermediaries in a variety of vehicles, including open-end and closed-end mutual funds, iShares® exchange-traded funds (“ETFs”), separate accounts, collective trust funds and other pooled investment vehicles. BlackRock also offers technology services, including the investment and risk management technology platform, Aladdin®, Aladdin Wealth™, eFront®, and Cachematrix®, as well as advisory services and solutions to a broad base of institutional and wealth management clients. The Company is highly regulated and manages its clients’ assets as a fiduciary. The Company does not engage in proprietary trading activities that could conflict with the interests of its clients.

BlackRock serves a diverse mix of institutional and retail clients across the globe. Clients include tax-exempt institutions, such as defined benefit and defined contribution pension plans, charities, foundations and endowments; official institutions, such as central banks, sovereign wealth funds, supranationals and other government entities; taxable institutions, including insurance companies, financial institutions, corporations and third-party fund sponsors, and retail intermediaries.

BlackRock maintains a significant global sales and marketing presence that is focused on establishing and maintaining retail and institutional investment management and technology service relationships by marketing its services to investors directly and through third-party distribution relationships, including financial professionals and pension consultants.

BlackRock is an independent, publicly traded company, with no single majority shareholder and over 80% of its Board of Directors consisting of independent directors.

Management seeks to deliver value for stockholders over time by, among other things, capitalizing on BlackRock’s differentiated competitive position, including:

- the Company’s longstanding model of client choice, through which it offers a wide range of index, active, private markets, and whole portfolio solutions across broad markets, themes, regions, and investment styles;
- the Company's focus on strong investment performance, seeking the best risk-adjusted returns for client portfolios, within the mandates given by clients, to help them meet their investment objectives;
- the Company's research, data and analytics, which are at the center of BlackRock's investment approach and processes. They inform BlackRock's pursuit of the best risk-adjusted returns, and underpin product creation and innovation;
- the Company’s global reach and commitment to best practices around the world, with approximately 59% of employees outside the United States (“US”) serving clients locally and supporting local investment capabilities. Approximately 35% of total AUM is managed for clients domiciled outside the US;
- the Company’s differentiated client relationships and fiduciary focus, which enable effective positioning toward changing client needs and industry trends including the secular shift to ETFs; growing allocations to private markets, such as infrastructure and private credit; increasing demand for outsourcing and whole portfolio solutions using index, active and private markets products; anticipated re-allocations to fixed income; demand for high-performing active strategies; interest in sustainable investment strategies; and a continued focus on income and retirement; and
- the Company’s longstanding commitment to innovation, technology services and the continued development of, and increased interest in, BlackRock technology products and solutions, including Aladdin, Aladdin Wealth, eFront, and Cachematrix. This commitment is further extended by minority investments in financial technology and digital distribution providers, data and whole portfolio capabilities including Securitize, Upvest, Avaloq, Human Interest, Circle, Clarity AI, Envestnet, Acorns, Scalable Capital and iCapital.

BlackRock operates in a global marketplace impacted by changing market dynamics and economic uncertainty, factors that can significantly affect earnings and stockholder returns in any given period.

The Company’s ability to increase revenue, earnings and stockholder value over time is predicated on its ability to generate new business, including business in Aladdin and other technology products and services. New business efforts depend on BlackRock’s ability to achieve clients’ investment objectives, in a manner consistent with their risk preferences, to deliver excellent client service and to innovate in technology to serve clients’ evolving needs. All of these efforts require the commitment and contributions of BlackRock employees. Accordingly, the ability to attract, develop and retain qualified professionals is critical to the Company’s long-term success.

FINANCIAL HIGHLIGHTS

(in millions, except per share data) GAAP:	2024	2023	2022	2021	2020
Total revenue	**$ 20,407**	$ 17,859	$ 17,873	$ 19,374	$ 16,205
Operating income	**$ 7,574**	$ 6,275	$ 6,385	$ 7,450	$ 5,695
Operating margin	**37.1 %**	35.1 %	35.7 %	38.5 %	35.1 %
Nonoperating income (expense)(1)	**$ 578**	$ 706	$ 89	$ 419	$ 475
Net income attributable to BlackRock, Inc.	**$ 6,369**	$ 5,502	$ 5,178	$ 5,901	$ 4,932
Diluted earnings per common share	**$ 42.01**	$ 36.51	$ 33.97	$ 38.22	$ 31.85

(in millions, except per share data) As adjusted(2):	2024	2023	2022	2021	2020
Operating income	**$ 8,110**	$ 6,593	$ 6,711	$ 7,747	$ 6,433
Operating margin	**44.5 %**	41.7 %	42.8 %	46.8 %	46.0 %
Nonoperating income (expense)(1)	**$ 533**	$ 648	$ 89	$ 419	$ 353
Net income attributable to BlackRock, Inc.	**$ 6,612**	$ 5,692	$ 5,391	$ 6,254	$ 5,352
Diluted earnings per common share	**$ 43.61**	$ 37.77	$ 35.36	$ 40.51	$ 34.57

(1) Net of net income (loss) attributable to noncontrolling interests (redeemable and nonredeemable).

(2) BlackRock reports its financial results in accordance with accounting principles generally accepted in the United States ("GAAP"); however, management believes evaluating the Company's ongoing operating results may be enhanced if investors have additional non-GAAP financial measures.

Beginning in the first quarter of 2022, BlackRock updated the definitions of operating income, as adjusted, operating margin, as adjusted, and net income attributable to BlackRock, Inc., as adjusted, to include new adjustments. Such measures have been recast for all prior periods to reflect the inclusion of such new adjustments. In addition, beginning in the first quarter of 2023, BlackRock updated the definitions of its non-GAAP financial measures to exclude the impact of market valuation changes on certain deferred cash compensation plans which the Company began economically hedging in 2023. For further information on non-GAAP financial measures and for as adjusted items for 2024 and 2023, see Item 7. *Management's Discussion and Analysis of Financial Condition and Results of Operations — Non-GAAP Financial Measures*. For further information on non-GAAP financial measures and for as adjusted items for 2022, 2021 and 2020, see Item 7. *Management's Discussion and Analysis of Financial Condition and Results of Operations — Non-GAAP Financial Measures*, of the Company's Annual Report on Form 10-K for the year ended December 31, 2022.

ASSETS UNDER MANAGEMENT

The Company's AUM by product type for the years 2020 through 2024 is presented below.

	December 31,					
(in millions)	2024	2023	2022	2021	2020	5-Year CAGR(1)
Equity	**$ 6,310,191**	$ 5,293,344	$ 4,435,354	$ 5,342,360	$ 4,419,806	11%
Fixed income	**2,905,669**	2,804,026	2,536,823	2,822,041	2,674,488	5%
Multi-asset	**992,921**	870,804	684,904	816,494	658,733	12%
Alternatives	**421,807**	275,984	266,210	264,881	235,042	19%
Long-term	**10,630,588**	**9,244,158**	**7,923,291**	**9,245,776**	**7,988,069**	**9%**
Cash management	**920,663**	764,837	671,194	755,057	666,252	11%
Advisory	**—**	—	—	9,310	22,359	—
Total	**$ 11,551,251**	**$ 10,008,995**	**$ 8,594,485**	**$ 10,010,143**	**$ 8,676,680**	**9%**

(1) Percentage represents compound annual growth rate ("CAGR") over a five-year period (December 31, 2019 – December 31, 2024).

Component changes in AUM by product type for the five years ended December 31, 2024 are presented below.

(in millions)	December 31, 2019	Net inflows (outflows)	Acquisitions(1)	Market change	FX impact	December 31, 2024
Equity	$ 3,820,329	$ 469,857	$ 45,398	$ 2,134,992	$ (160,385)	**$ 6,310,191**
Fixed income	2,315,392	944,834	—	(220,628)	(133,929)	**2,905,669**
Multi-asset	568,121	276,730	—	180,410	(32,340)	**992,921**
Alternatives	178,072	116,114	72,051	60,258	(4,688)	**421,807**
Long-term	**6,881,914**	**1,807,535**	**117,449**	**2,155,032**	**(331,342)**	**10,630,588**
Cash management	545,949	361,995	—	19,205	(6,486)	**920,663**
Advisory	1,770	(2,423)	—	636	17	**—**
Total	**$ 7,429,633**	**$ 2,167,107**	**$ 117,449**	**$ 2,174,873**	**$ (337,811)**	**$ 11,551,251**

(1) Amounts include the following: (a) net AUM from the acquisition of Aperio Group, LLC ("Aperio Transaction") in February 2021, (b) net AUM from the acquisition of Kreos Capital (the "Kreos Transaction") in August 2023, (c) net AUM from the acquisition of SpiderRock Advisors (the "SpiderRock Transaction") in May 2024 and (d) net AUM from the GIP Transaction in October 2024.

AUM represents the broad range of financial assets managed for clients on a discretionary basis pursuant to investment management and trust agreements that are expected to continue for at least 12 months. In general, reported AUM reflects the valuation methodology that corresponds to the basis used for determining revenue (for example, net asset value). Reported AUM does not include assets for which BlackRock provides risk management or other forms of nondiscretionary advice, or assets that the Company is retained to manage on a short-term, temporary basis.

Investment management fees are typically earned as a percentage of AUM. BlackRock also earns performance fees on certain portfolios relative to an agreed-upon benchmark or return hurdle. On some products, the Company also may earn securities lending revenue. In addition, BlackRock offers its proprietary Aladdin investment system as well as risk management, outsourcing, advisory and other technology services, to institutional investors and wealth management intermediaries. Revenue for these services may be based on several criteria including value of positions, number of users, implementation go-lives and software solution delivery and support.

At December 31, 2024, total AUM was $11.6 trillion, representing a CAGR of 9% over the last five years. AUM growth during the period was achieved through the combination of net market valuation gains, net inflows and acquisitions, including the net AUM impact from the Aperio Transaction, which added $41 billion of AUM in February 2021, the Kreos Transaction, which added $2 billion of AUM in August 2023, the SpiderRock Transaction, which added $4 billion of AUM in May 2024, and the GIP Transaction, which added $70 billion of AUM in October 2024. BlackRock's AUM mix encompasses a broadly diversified product range, as described below.

The Company considers the categorization of its AUM by client type, product type, investment style, and client region useful to understanding its business. The following discussion of the Company's AUM will be organized as follows:

Client Type	Product Type	Investment Style	Client Region
• Retail	• Equity	• Active	• Americas
• ETFs	• Fixed Income	• Index and ETFs	• Europe, the Middle East and Africa ("EMEA")
• Institutional	• Multi-asset		• Asia-Pacific
	• Alternatives		
	• Cash Management		

CLIENT TYPE

BlackRock serves a diverse mix of institutional and retail clients across the globe, with a regionally focused business model. BlackRock leverages the benefits of scale across global investment, risk and technology platforms while at the same time using local distribution presence to deliver solutions for clients. Furthermore, the Company's structure facilitates strong teamwork globally across both functions and regions in order to enhance BlackRock's ability to leverage best practices to serve our clients and continue to develop our talent.

Clients include tax-exempt institutions, such as defined benefit and defined contribution pension plans, charities, foundations and endowments; official institutions, such as central banks, sovereign wealth funds, supranationals and other government entities; taxable institutions, including insurance companies, financial institutions, corporations and third-party fund sponsors, and retail intermediaries.

ETFs are a growing component of both institutional and retail client portfolios. However, as ETFs are traded on exchanges, complete transparency on the ultimate end-investor is unavailable. Therefore, ETFs are presented as a separate client type below, with investments in ETFs by institutions and retail clients excluded from figures and discussions in their respective sections.

AUM by investment style and client type at December 31, 2024 is presented below.

(in millions)	Retail	ETFs	Institutional	Total
Active	$ 733,907	$ —	$ 2,136,749	$ 2,870,656
Non-ETF index	281,920	—	3,247,637	3,529,557
ETFs	—	4,230,375	—	4,230,375
Long-term	**1,015,827**	**4,230,375**	**5,384,386**	**10,630,588**
Cash management	12,247	—	908,416	920,663
Total	**$ 1,028,074**	**$ 4,230,375**	**$ 6,292,802**	**$ 11,551,251**

Retail

BlackRock serves retail investors globally through a wide array of products across the investment spectrum, including separate accounts, open-end and closed-end funds, unit trusts and private investment funds. Retail investors are served principally through intermediaries, including broker-dealers, banks, trust companies, insurance companies and independent financial advisors. Technology solutions, digital distribution tools and a shift toward portfolio construction are increasing the number of financial advisors and end-retail investors using BlackRock products.

Retail represented 10% of long-term AUM at December 31, 2024 and 28% of long-term investment advisory and administration fees (collectively "base fees") and securities lending revenue for 2024.

ETFs have a significant retail component but are shown separately below. With the exclusion of ETFs, the majority of retail AUM is comprised of active mutual funds. In the aggregate, active and index mutual funds totaled $738 billion, or approximately 75%, of retail long-term AUM at year-end, with the remainder invested in private investment funds and separately managed accounts. Approximately 70% of retail long-term AUM is invested in active products.

Component changes in retail long-term AUM for 2024 are presented below.

(in millions)	December 31, 2023	Net inflows (outflows)	Acquisitions(1)	Market change	FX impact	December 31, 2024
Equity	$ 435,734	$ 15,285	$ 4,074	$ 54,257	$ (4,232)	**$ 505,118**
Fixed income	312,799	11,671	—	1,483	(7,312)	**318,641**
Multi-asset	139,537	(2,328)	—	14,420	(651)	**150,978**
Alternatives	41,627	(261)	—	69	(345)	**41,090**
Total	**$ 929,697**	**$ 24,367**	**$ 4,074**	**$ 70,229**	**$ (12,540)**	**$ 1,015,827**

(1) Amounts include AUM attributable to the SpiderRock Transaction.

The retail client base is diversified geographically, with 70% of long-term AUM managed for investors based in the Americas, 25% in EMEA and 5% in Asia-Pacific at year-end 2024.

- *US retail* long-term net inflows of $19 billion were driven by $9 billion of net inflows into both fixed income and equity. Fixed income net inflows were primarily into active mutual funds, while equity net inflows reflected demand for Aperio, BlackRock's customized index equity solution.
- *International retail* long-term net inflows of $6 billion reflected equity and fixed income net inflows of $6 billion and $2 billion, respectively. Equity net inflows were driven by wealth outsourcing solutions, and fixed income inflows were led by active mutual funds.

ETFs

BlackRock is the leading ETF provider in the world with $4.2 trillion of iShares ETF AUM as of December 31, 2024. BlackRock generated ETF net inflows of $390 billion in 2024. The Company offers both index-tracking ETFs, which seek to replicate the performance of a specific index, and active ETFs. Active ETFs are managed by professional portfolio managers who actively select and adjust the fund's holdings in an effort to outperform the market, deliver a specific outcome or gain exposure to hard-to-index markets.

Equity ETF net inflows of $236 billion were driven by flows into core ETFs, as well as continued client use of BlackRock's broad-based precision exposure ETFs to express risk preferences and make tactical allocation changes. Fixed income ETF net inflows of $112 billion were diversified across exposures, led by flows into treasury, core and corporate credit ETFs. Alternative ETFs had net inflows of $41 billion, driven by the launch of cryptocurrency exchange-traded products ("ETPs") in the year.

ETFs represented 40% of long-term AUM at December 31, 2024 and 45% of long-term base fees and securities lending revenue for 2024.

Component changes in ETFs AUM for 2024 are presented below.

(in millions)	December 31, 2023	Net inflows (outflows)	Market change	FX impact	December 31, 2024
Equity	$ 2,532,631	$ 236,357	$ 359,322	$ (21,912)	**$ 3,106,398**
Fixed income	898,403	112,341	(16,291)	(8,801)	**985,652**
Multi-asset	9,140	1,025	841	(272)	**10,734**
Alternatives(1)	59,125	40,710	27,919	(163)	**127,591**
Total	**$ 3,499,299**	**$ 390,433**	**$ 371,791**	**$ (31,148)**	**$ 4,230,375**

(1) Amounts include cryptocurrency and commodity ETFs and ETPs.

BlackRock's ETF product range offers investors a precise, transparent and efficient way to gain exposure to a full range of asset classes and global markets that have been difficult for many investors to access, as well as the liquidity required to make adjustments to their exposures quickly and cost-efficiently.

- *US ETF** AUM ended 2024 at $3.1 trillion with $280 billion of net inflows, led by net inflows into core equity, fixed income, cryptocurrency and other precision exposure ETFs.
- *International ETF** AUM ended 2024 at $1.1 trillion with $110 billion of net inflows, diversified across product categories, and led by net inflows into core equity, fixed income and precision exposure ETFs.

* Regional ETF amounts based on jurisdiction of product, not underlying client.

Institutional

BlackRock serves institutional investors on six continents in sub-categories including: pensions, endowments and foundations, official institutions, and financial institutions; institutional AUM is diversified across product and region.

Component changes in institutional long-term AUM for 2024 are presented below.

(in millions)	December 31, 2023	Net inflows (outflows)	Acquisition(1)	Market change	FX impact	December 31, 2024
Active:						
Equity	$ 186,688	$ 5,380	$ —	$ 30,876	$ (4,096)	**$ 218,848**
Fixed income	836,823	(2,843)	—	16,885	(10,537)	**840,328**
Multi-asset	717,182	54,887	—	72,798	(16,828)	**828,039**
Alternatives	171,980	7,023	69,875	3,618	(2,962)	**249,534**
Active subtotal	1,912,673	64,447	69,875	124,177	(34,423)	**2,136,749**
Index:						
Equity	2,138,291	(31,454)	—	420,860	(47,870)	**2,479,827**
Fixed income	756,001	42,500	—	(5,068)	(32,385)	**761,048**
Multi-asset	4,945	(1,906)	—	204	(73)	**3,170**
Alternatives	3,252	234	—	165	(59)	**3,592**
Index subtotal	2,902,489	9,374	—	416,161	(80,387)	**3,247,637**
Total	**$ 4,815,162**	**$ 73,821**	**$ 69,875**	**$ 540,338**	**$ (114,810)**	**$ 5,384,386**

(1) Amounts include AUM attributable to the GIP Transaction.

Institutional active AUM ended 2024 at $2.1 trillion, reflecting $64 billion of net inflows, driven by the funding of several significant outsourcing mandates and continued growth in BlackRock's *LifePath®* target-date and private markets platforms.

Multi-asset net inflows of $55 billion reflected continued growth from significant pension outsourcing mandates and *LifePath* target-date offerings. Fixed income net outflows of $3 billion were impacted by an approximately $20 billion active fixed income redemption in the second quarter from a large insurance client linked to M&A activity. Equity net inflows of $5 billion were primarily into quantitative equity strategies.

Alternatives net inflows of $7 billion were led by infrastructure, private credit and private equity. Excluding return of capital and investment of $13 billion, alternatives net inflows were $20 billion. At year-end, BlackRock had approximately $45 billion of non-fee paying, unfunded, uninvested commitments to deploy for institutional clients, which is not included in AUM.

Institutional active represented 20% of long-term AUM and 21% of long-term base fees and securities lending revenue for 2024.

Institutional index AUM totaled $3.2 trillion at December 31, 2024, reflecting $9 billion of net inflows, driven by fixed income.

Institutional index represented 30% of long-term AUM and 6% of long-term base fees and securities lending revenue for 2024.

The Company’s institutional clients consist of the following:

- Pensions, Foundations and Endowments BlackRock is among the world’s largest managers of pension plan assets with $3.4 trillion, or 63%, of long-term institutional AUM managed for defined benefit, defined contribution and other pension plans for corporations, governments and unions at December 31, 2024. The market landscape continues to shift from defined benefit to defined contribution, and BlackRock's defined contribution channel represented $1.7 trillion of total pension AUM. BlackRock remains well positioned for the ongoing evolution of the defined contribution market and demand for outcome-oriented investments. An additional $90 billion, or 2%, of long-term institutional AUM was managed for other tax-exempt investors, including charities, foundations and endowments.
- Official Institutions BlackRock managed $294 billion, or 5%, of long-term institutional AUM for official institutions, including central banks, sovereign wealth funds, supranationals, multilateral entities and government ministries and agencies at year-end 2024.
- Financial and Other Institutions BlackRock is a top independent manager of assets for insurance companies, which accounted for $695 billion, or 13%, of long-term institutional AUM at year-end 2024. Assets managed for other taxable institutions, including corporations, banks and third-party fund sponsors for which the Company provides sub-advisory services, totaled $890 billion, or 17%, of long-term institutional AUM at year-end.

CLIENT TYPE AND PRODUCT TYPE

Component changes in AUM by client type and product type for 2024 are presented below.

(in millions)	December 31, 2023	Net inflows (outflows)	Acquisition[(1)]	Market change	FX impact	December 31, 2024
Retail:						
Equity	$ 435,734	$ 15,285	$ 4,074	$ 54,257	$ (4,232)	**$ 505,118**
Fixed income	312,799	11,671	—	1,483	(7,312)	**318,641**
Multi-asset	139,537	(2,328)	—	14,420	(651)	**150,978**
Alternatives	41,627	(261)	—	69	(345)	**41,090**
Retail subtotal	929,697	24,367	4,074	70,229	(12,540)	**1,015,827**
ETFs:						
Equity	2,532,631	236,357	—	359,322	(21,912)	**3,106,398**
Fixed income	898,403	112,341	—	(16,291)	(8,801)	**985,652**
Multi-asset	9,140	1,025	—	841	(272)	**10,734**
Alternatives	59,125	40,710	—	27,919	(163)	**127,591**
ETFs subtotal	3,499,299	390,433	—	371,791	(31,148)	**4,230,375**
Institutional:						
Active:						
Equity	186,688	5,380	—	30,876	(4,096)	**218,848**
Fixed income	836,823	(2,843)	—	16,885	(10,537)	**840,328**
Multi-asset	717,182	54,887	—	72,798	(16,828)	**828,039**
Alternatives	171,980	7,023	69,875	3,618	(2,962)	**249,534**
Active subtotal	1,912,673	64,447	69,875	124,177	(34,423)	**2,136,749**
Index:						
Equity	2,138,291	(31,454)	—	420,860	(47,870)	**2,479,827**
Fixed income	756,001	42,500	—	(5,068)	(32,385)	**761,048**
Multi-asset	4,945	(1,906)	—	204	(73)	**3,170**
Alternatives	3,252	234	—	165	(59)	**3,592**
Index subtotal	2,902,489	9,374	—	416,161	(80,387)	**3,247,637**
Institutional subtotal	4,815,162	73,821	69,875	540,338	(114,810)	**5,384,386**
Long-term	**9,244,158**	**488,621**	**73,949**	**982,358**	**(158,498)**	**10,630,588**
Cash management	764,837	152,730	—	10,606	(7,510)	**920,663**
Total	**$ 10,008,995**	**$ 641,351**	**$ 73,949**	**$ 992,964**	**$ (166,008)**	**$ 11,551,251**

(1) Amounts include AUM attributable to the GIP Transaction and the SpiderRock Transaction.

Long-term product offerings include active and index strategies. BlackRock's active strategies seek to earn attractive returns in excess of a market benchmark or performance hurdle while maintaining an appropriate risk profile and leverage fundamental research and quantitative models to drive portfolio construction. In contrast, index strategies seek to closely track the returns of a corresponding index, generally by investing in substantially the same underlying securities within the index or in a subset of those securities selected to approximate a similar risk and return profile of the index. Index products include both BlackRock's non-ETF index products and iShares ETFs.

Although many clients use both active and index strategies, the application of these strategies may differ. For example, clients may use index products to gain exposure to a market or asset class or may use a combination of index strategies to target active returns. In addition, institutional non-ETF index assignments tend to be very large (multi-billion dollars) and typically reflect low fee rates. Net flows in institutional index products generally have a small impact on BlackRock’s revenues and earnings.

Equity

Year-end 2024 equity AUM totaled $6.3 trillion, reflecting net inflows of $226 billion. Net inflows included ETF net inflows of $236 billion, partially offset by net outflows of $6 billion and $4 billion from active and non-ETF index, respectively.

BlackRock’s effective fee rates fluctuate due to changes in AUM mix. Approximately half of BlackRock’s equity AUM is tied to international market strategies, including emerging markets, which tend to have higher fee rates than US equity strategies. Accordingly, fluctuations in international equity markets, which may not consistently move in tandem with US markets, have a greater impact on BlackRock’s equity revenues and effective fee rate.

Equity represented 60% of long-term AUM and 54% of long-term base fees and securities lending revenue for 2024.

Fixed Income

Fixed income AUM ended 2024 at $2.9 trillion, reflecting net inflows of $164 billion. Net inflows included $112 billion, $42 billion and $9 billion into ETFs, non-ETF index and active, respectively. Fixed income ETF net inflows of $112 billion reflected the benefit of BlackRock's diverse product offering and included strong flows into treasury, core and corporate credit ETFs.

Fixed income represented 27% of long-term AUM and 25% of long-term base fees and securities lending revenue for 2024.

Multi-Asset

BlackRock manages a variety of multi-asset funds and bespoke mandates for a diversified client base that leverages the Company's broad investment expertise in global equities, bonds, and private markets, and the Company's extensive risk management capabilities. Investment solutions may include a combination of long-only portfolios and private markets investments as well as tactical asset allocation overlays.

Multi-asset represented 9% of long-term AUM and 8% of long-term base fees and securities lending revenue for 2024.

Component changes in multi-asset AUM for 2024 are presented below.

(in millions)	December 31, 2023	Net inflows (outflows)	Market change	FX impact	December 31, 2024
Target date/risk	$ 489,136	$ 30,388	$ 66,679	$ (4,661)	$ 581,542
Asset allocation and balanced	246,127	21,494	16,428	(6,920)	277,129
Fiduciary	135,541	(204)	5,156	(6,243)	134,250
Total	**$ 870,804**	**$ 51,678**	**$ 88,263**	**$ (17,824)**	**$ 992,921**

Multi-asset net inflows reflected ongoing institutional demand for BlackRock's solutions-based advice with $53 billion of net inflows coming from institutional clients, including the funding of several significant outsourcing mandates. Defined contribution plans remained a significant driver of flows and contributed $24 billion to institutional multi-asset net inflows in 2024, primarily into target date and target risk product offerings.

The Company’s multi-asset strategies include the following:

- *Target date and target risk strategies* generated net inflows of $30 billion. Institutional investors represented 90% of target date and target risk AUM, with defined contribution plans representing 81% of AUM. Flows were driven by defined contribution investments across BlackRock's *LifePath* franchise, including BlackRock's *LifePath Paycheck* offering launched in 2024. *LifePath* products utilize a proprietary active asset allocation model that seeks to balance risk and return over an investment horizon based on the investor’s expected retirement timing. Underlying investments are primarily index products.
- *Asset allocation and balanced strategies* generated $21 billion of net inflows. These strategies combine equity, fixed income and private markets components for investors seeking a tailored solution relative to a specific benchmark and within a risk budget. In certain cases, these strategies seek to minimize downside risk through diversification, derivatives strategies and tactical asset allocation decisions. Flows in this category included pension outsourcing mandates that funded during the year. Flagship products also include BlackRock's Global Allocation and Multi-Asset Income fund families.
- *Fiduciary management services* are complex mandates in which pension plan sponsors or endowments and foundations retain BlackRock to assume responsibility for some or all aspects of investment management, often with BlackRock acting as outsourced chief investment officer. These customized services require strong partnership with the clients’ investment staff and trustees in order to tailor investment strategies to meet client-specific risk budgets and return objectives.

Alternatives

BlackRock alternatives focus on sourcing and managing high-alpha investments with lower correlation to public markets and developing a holistic approach to address client needs in alternatives investing. The Company's alternatives products fall into three main categories — (1) private markets, (2) liquid alternatives, and (3) currency (including cryptocurrency) and commodities. Private markets include offerings in infrastructure, private credit, private equity, real estate and multi-alternative solutions. Liquid alternatives include offerings in direct hedge funds and hedge fund solutions (funds of funds).

In 2024, private markets generated $9 billion of net inflows, or $22 billion excluding return of capital / return on investment of $13 billion. The largest contributors to return of capital / return on investment were private equity, private credit and infrastructure. Net inflows were driven by infrastructure, private credit and private equity. At year-end, BlackRock had approximately $45 billion of non-fee paying, unfunded, uninvested commitments, which are expected to be deployed in future years; these commitments are not included in AUM or flows until they are fee-paying. Liquid alternatives saw net outflows of $3 billion driven by hedge fund solutions. Currency and commodities saw $41 billion of net inflows, primarily into cryptocurrency ETPs.

Private markets continue to see increasing allocations from institutional and wealth investors, and BlackRock’s highly diversified franchise is well positioned to continue to meet growing demand. The closing of the GIP transaction added $116 billion of client AUM and $70 billion of fee-paying AUM in October. In addition to GIP, the Company’s planned acquisitions of Preqin and HPS Investment Partners ("HPS"), each position its platform ahead of evolving client needs and structural industry trends. The private markets capabilities the Company expects to add through these transactions will allow it to serve clients even more comprehensively and position the Company to raise significant private capital.

Alternatives represented 4% of long-term AUM and 13% of long-term base fees and securities lending revenue for 2024.

Component changes in alternatives AUM for 2024 are presented in the table below.

(in millions)	December 31, 2023	Net inflows (outflows)	Acquisition[1]	Market change	FX impact	December 31, 2024	Memo: return of capital/ investment[2]	Memo: committed capital[3]
Private markets:								
Infrastructure	$ 35,701	$ 5,633	$ 69,875	$ (586)	$ (1,017)	**$ 109,606**	$ (3,089)	$ 28,821
Private equity	35,208	771	—	490	(142)	**36,327**	(5,492)	6,986
Private credit	31,128	2,629	—	(620)	(712)	**32,425**	(3,209)	5,561
Real estate	27,558	314	—	(1,190)	(535)	**26,147**	(370)	630
Multi-alternatives	7,314	110	—	103	(58)	**7,469**	(713)	3,167
Total private markets	136,909	9,457	69,875	(1,803)	(2,464)	**211,974**	(12,873)	45,165
Liquid alternatives:								
Direct hedge fund strategies	46,318	(28)	—	2,769	(285)	**48,774**	—	—
Hedge fund solutions	27,915	(2,581)	—	2,713	(431)	**27,616**	—	—
Total liquid alternatives	74,233	(2,609)	—	5,482	(716)	**76,390**	—	—
Currency and commodities[4]	64,842	40,858	—	28,092	(349)	**133,443**	—	—
Total	**$ 275,984**	**$ 47,706**	**$ 69,875**	**$ 31,771**	**$ (3,529)**	**$ 421,807**	**$ (12,873)**	**$ 45,165**

(1) Amounts include AUM attributable to the GIP Transaction.
(2) Return of capital/investment is included in outflows.
(3) Amount represents client assets that are uninvested commitments, which are currently non-fee paying and are not included in AUM. These commitments are expected to generate fees and will be counted in AUM and flows as the capital is deployed over time.
(4) Amounts include cryptocurrency and commodity ETFs and ETPs.

Private Markets

The Company's private markets strategies include the following:

- Infrastructure includes offerings across equity, debt and solutions, which totaled $110 billion in AUM. Net inflows were $6 billion, or $9 billion excluding return of capital / return on investment. In 2024, the GIP transaction added $70 billion to AUM.
- Private Equity AUM of $36 billion included AUM of $28 billion in private equity solutions, and $8 billion in a direct private equity strategy. Private equity net inflows were $1 billion, or $6 billion excluding return of capital / return on investment.
- Private Credit primarily represents direct lending, opportunistic and venture debt strategies, with AUM of $32 billion at December 31, 2024. Net inflows were $3 billion, or $6 billion excluding return of capital / return on investment. Private credit does not include private credit assets included in infrastructure and real estate debt, as well as assets in private placements and multi-strategy credit funds, which are reported within fixed income and multi-asset.
- Real Estate had $26 billion in AUM at December 31, 2024.
- Multi-Alternatives represents highly customized portfolios of alternative investments, with $7 billion in AUM at December 31, 2024.

Liquid Alternatives

The Company's liquid alternatives products' net outflows of $3 billion reflected redemptions from hedge fund solutions. Direct hedge fund strategies includes a variety of single- and multi-strategy offerings.

In addition, the Company manages $90 billion in liquid credit strategies which is included in active fixed income.

Currency and Commodities

The Company's currency and commodities products include a range of active and index products, including cryptocurrency ETPs.

Currency and commodities products had $41 billion of net inflows, primarily from ETPs. Cryptocurrency and commodities ETPs represented $55 billion and $73 billion of AUM, respectively, and are not eligible for performance fees.

Cash Management

Cash management AUM totaled a record $921 billion at December 31, 2024, reflecting $153 billion of net inflows. Cash management products include taxable and tax-exempt money market funds, short-term investment funds and customized separate accounts. Portfolios are denominated in US dollars, Canadian dollars, Australian dollars, euros, Swiss francs, New Zealand dollars or British pounds.

CLIENT REGION

BlackRock's footprints in the Americas, EMEA and Asia-Pacific regions reflect strong relationships with intermediaries and an established ability to deliver the Company's global investment expertise in funds and other products tailored to local regulations and requirements.

AUM by product type and client region at December 31, 2024 is presented below.

(in millions)	Americas	EMEA	Asia-Pacific	Total
Equity	$ 4,377,383	$ 1,490,961	$ 441,847	$ 6,310,191
Fixed income	1,835,652	756,127	313,890	2,905,669
Multi-asset	715,826	212,174	64,921	992,921
Alternatives	287,347	99,946	34,514	421,807
Long-term	**7,216,208**	**2,559,208**	**855,172**	**10,630,588**
Cash management	645,638	259,850	15,175	920,663
Total	**$ 7,861,846**	**$ 2,819,058**	**$ 870,347**	**$ 11,551,251**

Component changes in AUM by client region for 2024 are presented below.

(in millions)	December 31, 2023	Net inflows (outflows)	Acquisition[1]	Market change	FX impact	December 31, 2024
Americas	$ 6,728,310	$ 406,091	$ 71,308	$ 685,291	$ (29,154)	**$ 7,861,846**
EMEA	2,478,810	207,459	112	203,851	(71,174)	**2,819,058**
Asia-Pacific	801,875	27,801	2,529	103,822	(65,680)	**870,347**
Total	**$ 10,008,995**	**$ 641,351**	**$ 73,949**	**$ 992,964**	**$ (166,008)**	**$ 11,551,251**

(1) Amounts include AUM attributable to the GIP Transaction and the SpiderRock Transaction.

Americas

Americas net inflows of $406 billion were driven by net inflows into equity, fixed income, cash, multi-asset, and cryptocurrency products of $139 billion, $108 billion, $90 billion, $23 billion, and $41 billion, respectively. During the year, BlackRock served clients through offices across the US as well as in Mexico, Canada, Brazil, Colombia, Chile, the Dominican Republic, and Peru.

The Americas represented 68% of total AUM and 65% of total base fees and securities lending revenue for 2024.

EMEA

EMEA net inflows of $207 billion were driven by equity, cash, fixed income, and multi-asset net inflows of $102 billion, $59 billion, $37 billion, and $9 billion, respectively, while alternatives added modest inflows. Offerings include fund families in the United Kingdom ("UK"), the Netherlands, Luxembourg and Dublin and ETFs listed on stock exchanges throughout Europe, as well as separate accounts and pooled investment products.

EMEA represented 24% of total AUM and 29% of total base fees and securities lending revenue for 2024.

Asia-Pacific

Asia-Pacific net inflows of $28 billion were driven by fixed income and multi-asset net inflows of $20 billion and $19 billion, respectively. These were partially offset by equity net outflows of $16 billion. Clients in the Asia-Pacific region are served through offices in India, Singapore, Hong Kong, Japan, Australia, China, Taiwan, Korea and New Zealand.

Asia-Pacific represented 8% of total AUM and 6% of total base fees and securities lending revenue for 2024.

INVESTMENT PERFORMANCE

Investment performance across active and index products as of December 31, 2024 was as follows:

	One-year period	Three-year period	Five-year period
Fixed income:			
Actively managed AUM above benchmark or peer median			
Taxable	69%	79%	82%
Tax-exempt	69%	42%	45%
Index AUM within or above applicable tolerance	97%	99%	98%
Equity:			
Actively managed AUM above benchmark or peer median			
Fundamental	47%	44%	64%
Systematic	93%	89%	93%
Index AUM within or above applicable tolerance	94%	99%	100%

Performance Notes

Past performance is not indicative of future results. Except as specified, the performance information shown is as of December 31, 2024 and is based on preliminary data available at that time. The performance data shown reflects information for all actively and passively managed equity and fixed income accounts, including US registered investment companies, European-domiciled retail funds and separate accounts for which performance data is available, including performance data for high net worth accounts available as of November 30, 2024. The performance data does not include accounts terminated prior to December 31, 2024 and accounts for which data has not yet been verified. If such accounts had been included, the performance data provided may have substantially differed from that shown.

Performance comparisons shown are gross-of-fees for institutional and high net worth separate accounts, and net-of-fees for retail funds. The performance tracking shown for index accounts is based on gross-of-fees performance and includes all institutional accounts and all iShares funds globally using an index strategy. AUM information is based on AUM available as of December 31, 2024 for each account or fund in the asset class shown without adjustment for overlapping management of the same account or fund. Fund performance reflects the reinvestment of dividends and distributions.

Performance shown is derived from applicable benchmarks or peer median information, as selected by BlackRock. Peer medians are based in part on data either from Lipper, Inc. or Morningstar, Inc. for each included product.

TECHNOLOGY SERVICES

BlackRock offers investment management technology systems, risk management services, and wealth management and digital distribution tools on a fee basis. Aladdin is the Company's proprietary technology platform, providing an end-to-end, SaaS solution for investment and risk management for both BlackRock and a growing number of institutional and retail investors around the world. BlackRock offers risk reporting capabilities via Aladdin Risk, as well as investment accounting capabilities. Aladdin Provider is a tool used by asset servicers, connecting them to the platform used by asset managers and owners to add operational efficiency. In 2019, BlackRock acquired eFront, a leading end-to-end alternative investment management software and solutions provider to enable clients to manage portfolios and risk across public and private asset classes on a single platform. eFront is offered to clients both as a standalone offering and as part of an integrated “Whole Portfolio” solution that provides transparency across clients’ public and private assets. Through the Company's Cachematrix platform, BlackRock is also a leading provider of financial technology which simplifies the cash management process for banks and their corporate clients in a streamlined, open-architecture platform.

BlackRock offers a number of wealth management technology tools offering personalized digital advice, portfolio construction capabilities and risk analytics for retail distributors. These tools include Aladdin Wealth, which provides wealth management firms and their financial professionals with institutional-quality business management, portfolio construction, modeling and risk analytics capabilities.

At year-end, BlackRock technology services clients included banks, insurance companies, official institutions, pension funds, asset managers, asset servicers, retail distributors and other investors across North America, South America, Europe, the Middle East, Asia, Africa and Australia.

Technology services revenue of $1.6 billion was up 8% year-over-year, and annual contract value (“ACV”) increased 12% year-over-year. ACV growth was driven by strong net sales of Aladdin in 2024, with over half of new client mandates spanning multiple Aladdin products. Aladdin assignments are typically long-term contracts that provide recurring revenue. At the end of any period, BlackRock generally has recurring revenue contracts in place for a large portion of total annual revenue. BlackRock measures the fees related to these agreements and refers to this as ACV. For further information on ACV, see Item 7. *Management’s Discussion and Analysis of Financial Condition and Results of Operations — Non-GAAP Financial Measures*.

Aladdin, which represented the majority of technology services revenue for the year, continues to benefit from trends favoring global platform and multi-asset risk solutions across public and private markets. Approximately 25% of Aladdin’s revenue was denominated in non-US currencies. In addition, while Aladdin is a multi-asset system, the majority of positions managed on the platform are fixed income. Revenue growth in 2024 reflected the successful onboarding of a number of new clients and expanding relationships with existing clients. 2024 revenue growth also reflects the prior year revenue impact of several clients’ renewals of eFront “on-premises” licenses.

BlackRock is focused on enhancing Aladdin, with continued investment into areas such as whole portfolio, private markets, wealth and sustainable investing solutions. BlackRock continues to evolve and enable clients to further simplify their operating infrastructure with Aladdin. Clients increasingly want to tailor how they use Aladdin to meet their specific needs, and BlackRock is providing them with choice and flexibility. BlackRock is empowering clients with data and opening Aladdin by creating connectivity with ecosystem providers and third-party technology solutions, which include asset servicers, cloud providers, digital asset platforms, trading systems and others. This connectivity helps clients work in their Aladdin environments with a more customized and seamless end-to-end experience.

In addition, BlackRock has made minority investments in financial technology and digital distribution providers, data and whole portfolio capabilities including Securitize, Upvest, Avaloq, Human Interest, Circle, Clarity AI, Envestnet, Acorns, Scalable Capital and iCapital. BlackRock records its share of income related to certain minority investments accounted for under the equity method within nonoperating income (expense) beginning in the first quarter of 2024 and within other revenue in 2023. In addition, BlackRock records gains and losses related to changes in value of other minority investments in nonoperating income (expense).

In June 2024, BlackRock announced that it had entered into a definitive agreement to acquire Preqin, a leading independent provider of private markets data, for £2.55 billion (or approximately $3.2 billion based on the GBP/USD foreign exchange rate at December 31, 2024) in cash. The Company believes bringing together Preqin's data and research tools with the complementary workflows of Aladdin and eFront in a unified platform will create a preeminent private markets technology and data provider. The Preqin Transaction is anticipated to close in the first quarter of 2025, subject to customary closing conditions.

SECURITIES LENDING

Securities lending is managed by a dedicated team, supported by quantitative analysis, proprietary technology and disciplined risk management. BlackRock receives both cash (primarily for US domiciled portfolios) and noncash collateral under securities lending arrangements. The cash management team invests the cash received as collateral for securities on loan in other portfolios. Fees for securities lending for US domiciled portfolios can be structured as a share of earnings, or as a management fee based on a percentage of the value of the cash collateral or both. The value of the securities on loan and the revenue earned are captured in the corresponding asset class being managed. The value of the collateral is not included in AUM.

Outstanding loan balances ended the year at approximately $412 billion, up from $359 billion at year-end 2023. More demand for general collateral securities resulted in higher balances year-over-year. Intrinsic lending spreads decreased, and cash reinvestment spreads increased year-over-year.

BlackRock employs a conservative investment style for cash and securities lending collateral that emphasizes quality, liquidity, and interest rate risk management. Disciplined risk management, including a rigorous credit surveillance process, is an integral part of the investment process. BlackRock’s Cash Management Credit Committee has established risk limits, such as aggregate issuer exposure limits and maturity limits, across many of the products BlackRock manages, including over all of its cash management products. In the ordinary course of BlackRock's business, there may be instances when a portfolio may exceed an internal risk limit or when an internal risk limit may be changed. No such instances, individually or in the aggregate, have been material to the Company. To the extent that daily evaluation and reporting of the profile of the portfolios identify that a limit has been exceeded, the relevant portfolio will be adjusted. To the extent a portfolio manager would like to obtain a temporary waiver of a risk limit, the portfolio manager must obtain approval from the credit research team, which is independent from the cash management portfolio managers. While a risk limit may be waived temporarily, such waivers are infrequent.

RISK AND QUANTITATIVE ANALYSIS

Across all asset classes, in addition to the efforts of the portfolio management teams, the Risk and Quantitative Analysis ("RQA") group at BlackRock draws on extensive analytical systems and proprietary and third-party data to identify, measure and manage a wide range of risks. RQA provides risk management advice and independent risk oversight of the investment management processes, identifies and helps manage counterparty and enterprise risks, coordinates standards for firm wide investment performance measurement and determines risk management-related analytical and information requirements. Where appropriate, RQA will work with portfolio managers and developers to facilitate the development or improvement of risk models and analytics.

COMPETITION

BlackRock competes with investment management firms, mutual fund complexes, insurance companies, banks, brokerage firms, financial technology providers and other financial institutions that offer products that are similar to, or alternatives to, those offered by BlackRock. In order to grow its business, BlackRock must be able to compete effectively for AUM. Key competitive factors include investment performance track records, the efficient delivery of beta for index products, investment style and discipline, price, client service and brand name recognition. Historically, the Company has competed principally on the basis of its long-term investment performance track record, its investment process, its risk management and analytic capabilities and the quality of its client service.

HUMAN CAPITAL

With approximately 21,100 employees in more than 30 countries as of December 31, 2024, BlackRock provides a broad range of investment management and technology services to institutional and retail clients in more than 100 countries across the globe. As an asset manager, BlackRock's long-term success depends on its people and how it manages its workforce.

Culture and Principles

BlackRock believes that maintaining a strong corporate culture is an important component of its human capital management practices and critical to the firm's long-term success. BlackRock's culture is underpinned by five core principles that unify its workforce and guide how it interacts with its employees, its clients, the communities in which it operates and its other stakeholders: (1) We are a fiduciary to our clients; (2) We are One BlackRock (1BLK); (3) We are passionate about performance; (4) We take emotional ownership; and (5) We are committed to a better future.

Connectivity and Inclusivity

BlackRock's approach to building a connected and inclusive culture is aligned with the firm's business priorities and long-term objectives. Delivering for the firm's clients requires attracting the best people from across the world. BlackRock is committed to creating an environment that supports top talent and fosters diverse perspectives to avoid groupthink. Creating a connected and inclusive culture where people with new ideas and fresh perspectives can thrive is core to our 1BLK principle. These values have been fundamental to BlackRock since its founding 37 years ago.

In 2024, BlackRock continued to focus on (1) supporting its talent and culture across the globe, (2) expanding investment choices and business partnership opportunities for interested clients, and (3) helping more and more people experience financial wellbeing through BlackRock philanthropy and employee-led volunteer efforts. Last year also marked the start of a significant period at BlackRock as the firm prepared to welcome thousands of new colleagues through acquisitions. These employees will play a crucial role in driving the firm's success forward, and a connected and inclusive culture is imperative to enabling that objective.

Since 2020, BlackRock has published annual SASB-aligned disclosure and EEO-1 reports, which provide information about the firm's workforce, including workforce composition. Of the Company's approximately 21,100 employees as of December 31, 2024, 43% were based in the Americas, 31% were based in EMEA, and 26% were based in the Asia-Pacific region.

Board Oversight of Human Capital Management

BlackRock's Board of Directors (the "Board") plays an important role in the oversight of human capital management and devotes one Board meeting annually to an in-depth review of BlackRock's culture, talent development, retention and recruiting initiatives, human capital management strategy, leadership and succession planning and employee feedback. Moreover, the Board's Management Development and Compensation Committee periodically reviews efforts and developments related to the firm's human capital management strategy.

Succession planning for BlackRock's Chief Executive Officer and other senior executives is a key part of the Board's annual review of human capital management issues. As part of this review, the Board focuses on whether BlackRock has the right people in place to execute the Company's long-term strategic plans, and on BlackRock's ability to identify, attract, develop, promote and retain future senior executives. An important element of the succession planning across the organization is a commitment to building leadership from within.

Employee Engagement

BlackRock values continuous dialogue with its employees to better understand their experiences at the firm and assess the efficacy of its human capital management practices. The Company uses several employee engagement mechanisms, including: (1) global employee opinion surveys; (2) interactive events and communications; (3) the sponsorship of employee networks; and (4) local community involvement. The employee opinion pulse surveys, which BlackRock conducts throughout the year, provide the Company with actionable feedback for its teams and for the Company as a whole. Additionally, BlackRock uses ongoing lifecycle surveys to collect feedback at various points along the employee journey. BlackRock works to keep employees informed and engaged through a regular cadence of communications and events, including newsletters, global and local townhalls and messages from leaders with timely business and organizational updates and culture-building opportunities.

BlackRock believes that employees value opportunities to give back to their communities. Through local, employee-led BlackRock Gives committees, the Company supports nonprofit organizations nominated by employees in the communities where it operates. In addition, the Company has a matching gifts program that matches an employee's donations to IRS qualified charitable organizations for up to $10,000 a year. BlackRock also matches volunteer time with eligible charities and full-time employees may take up to two paid volunteer days.

Compensation, Wellness and Benefits

BlackRock is committed to responsible business practices and believes that investing in the physical, emotional, mental and financial well-being of its employees is a critical component of the firm's human capital management strategy. To that end, the Company designs its compensation and benefits practices to: (1) attract, motivate and retain talented employees; (2) align employee incentives and risk-taking with that of the firm and the interests of its clients; and (3) support employees and their families across many aspects of their lives. The Company has a strong pay-for-performance culture and an annual compensation process that takes into consideration firmwide results, individual business results and employee performance, as well as market benchmarks. BlackRock also offers a wide range of benefits that it regularly reviews in accordance with market practices and the local requirements of its offices, including, where applicable, retirement savings plans, a Flexible Time Off ("FTO") policy and flexible working arrangements, parental leave and family forming benefits, such as fertility benefits, adoption and surrogacy assistance, and backup elder and childcare benefits. The Company provides comprehensive healthcare and mental-health benefits to eligible employees, including medical, dental and vision coverage, health savings and spending accounts, counseling services, an employee assistance program and access to telemedicine services, where available. The Company also offers a Mental Health Ambassador program that is comprised of global volunteers across office locations who are trained in empathetic listening skills and direct interested colleagues to benefits, tools and resources to support mental health.

BlackRock prioritizes protecting the rights of its workforce. The Company has implemented policies related to harassment prevention and compliance with applicable equal employment opportunity and overtime regulations. BlackRock is also committed to providing a safe and healthy work environment for its workforce. To do this, it designs global programs, including environmental and occupational health and safety programs, to meet or exceed local requirements. Moreover, BlackRock encourages all of its employees to raise issues of concern and assures employees that they may do so without fear of retaliation.

Recruiting, Training and Development

BlackRock recognizes that, like all companies, it is operating in an increasingly competitive environment. As such, the Company engages in efforts to reach top talent. In the spirit of attracting talent from broad backgrounds, BlackRock also provides formal recruiting programs for Veterans (former service members transitioning to civilian careers) and Returners (individuals who have taken a career break of 18 months or more).

BlackRock is also committed to innovation, learning and reinvention in all areas of its business and believes that developing the capabilities of its employees is integral to delivering long-term value. To that end, the Company's human capital management practices are designed to provide opportunities for employees to learn, innovate and enhance their skillsets at every stage of their career. One example is the BlackRock Academies, the firm's online suite of interactive resources and courses, which enable employees to build skills in specific facets of BlackRock's business and purpose. The Company believes these opportunities play an important role in engaging BlackRock's employees.

In addition, BlackRock believes that a critical driver of its future success is its ability to grow strong leaders and people managers. The Company invests in leadership development programs designed to foster career growth. People managers have access to coaching and the Managing at BlackRock program to assist in building foundational skills.

REGULATION

Virtually all aspects of BlackRock's business are subject to various laws and regulations around the world, some of which are summarized below. These laws and regulations are primarily intended to protect investment advisory clients, investors in registered and unregistered investment companies, and trust and other fiduciary clients of BlackRock Institutional Trust Company, N.A. ("BTC"). Under these laws and regulations, agencies that regulate investment advisers, investment funds and trust banks and other individuals and entities have broad administrative powers, including the power to limit, restrict or prohibit the regulated entity or person from carrying on business if it fails to comply with such laws and regulations. Possible sanctions for significant compliance failures include the suspension of individual employees, limitations on engaging in certain lines of business for specified periods of time, revocation of investment adviser and other registrations or bank charters, censures and fines both for individuals and BlackRock. The rules governing the regulation of financial institutions and their holding companies and subsidiaries are very detailed and technical. Accordingly, the discussion below is general in nature, does not purport to be complete and is current only as of the date of this report.

BlackRock's business may be impacted by numerous regulatory reform initiatives occurring around the world. Any such initiative, or any new laws or regulations or changes to, or in the enforcement of, existing laws or regulations, could materially and adversely impact the scope or profitability of BlackRock's business activities, lead to business disruptions, require BlackRock to alter its business or operating activities and expose BlackRock to additional costs (including compliance and legal costs) as well as reputational harm. BlackRock's profitability also could be materially and adversely affected by modification of the rules and regulations that impact the business and financial communities in general, including changes to the laws governing banking, securities, taxation, antitrust regulation and electronic commerce.

GLOBAL REGULATORY REFORM

Policymaking workstreams focused on the financial services sector led by global standard setters, such as the Financial Stability Board ("FSB") and International Organization of Securities Commissions ("IOSCO"), may lead to or inform new regulations in multiple jurisdictions in which BlackRock operates. Such workstreams have focused on areas such as money market funds ("MMFs"), open-ended funds ("OEFs") and sustainability regulations.

Macroprudential Policies for Asset Managers

Concerns about liquidity and leverage risks in the asset management industry and wider market-based finance sector have prompted a broad review of existing regulations globally, including an assessment of the adequacy of certain structural market components in mitigating risks, by the FSB, IOSCO, the US Securities and Exchange Commission (the “SEC”) and the Financial Stability Oversight Council (“FSOC”). In November 2022, the SEC proposed amendments to rules governing OEF liquidity risk management. The EU launched a consultation on macroprudential policies in 2024, including enhanced requirements for liquidity management tools, which may lead to new restrictions on management of OEFs. The UK proposed introducing liquidity facilities to certain asset owners, which could result in regulatory burdens on asset managers. If any of these regulatory or policy actions result in broad application of macroprudential tools to OEFs or require changes to structural features of certain OEFs, it could limit BlackRock’s ability to offer products to certain clients and/or result in clients altering their investment strategies or allocations in a manner that is adverse to BlackRock.

Global MMF Reforms

Following the market events of March 2020, US, UK and EU authorities initiated a review of existing regulatory frameworks with the aim of improving the resilience of MMFs in market downturns. In the US, the SEC adopted changes to Rule 2a-7, the primary rule under the Investment Company Act of 1940 governing MMFs, including changes to required liquidity levels and requiring mandatory liquidity fees under certain circumstances. The UK released a consultation in December 2023 indicating their intent to change regulatory requirements for MMFs domiciled in the UK, including material increases in required liquidity levels. The EU consultation on macroprudential policies mentioned above may also result in changes to the regulations of EU-domiciled MMFs. Depending on the terms of the final UK and EU reforms, certain of BlackRock’s MMF products could be adversely impacted.

Sustainability

Sustainability has been the subject of regulatory focus across jurisdictions. Disclosure standards aligned with the International Sustainability Standards Board's (“ISSB”) inaugural disclosure standards have been adopted by several national regulators, including in Hong Kong, Singapore and Australia, while others are expected to propose ISSB-aligned standards, such as the UK, Canada and Japan. However, in the US, final rules issued by the SEC requiring corporate issuers to make climate-related disclosures in their periodic reports are pending litigation, and as of February 2025, the SEC was revisiting its litigation position. The SEC has previously proposed rules requiring enhanced ESG disclosures by investment companies and investment advisers in fund and adviser filings, including disclosures on ESG strategies and how ESG factors are considered and GHG emissions disclosure by certain environmentally focused funds. It also increased scrutiny of disclosure and compliance issues relating to investment advisers’ and funds’ ESG strategies, policies and procedures. In addition, the US Department of Labor (“DOL”) issued final rules clarifying that Employee Retirement Income Security Act of 1974, as amended (“ERISA”) plan fiduciaries can consider the economic effects of ESG factors for purposes of investing ERISA plan assets and exercising voting rights with respect to plan investments. In 2023, California passed several laws requiring certain companies doing business in California to publish certain types of climate-related disclosures, and other states may adopt similar laws.

The EU has enacted numerous sustainability regulations, including (1) the Sustainable Finance Disclosure Regulation, requiring sustainability-related disclosures by financial market participants; (2) the EU Taxonomy Regulation, requiring asset managers to report against an EU-wide taxonomy of environmentally sustainable activities and make detailed disclosures relating to ESG characteristics of funds and portfolios; (3) the Corporate Sustainability Reporting Directive ("CSRD"), requiring enhanced sustainability reporting for EU-based and EU-listed companies, and from 2028, for a wider group of global companies; and (4) the Corporate Sustainability Due Diligence Directive ("CSDDD"), requiring in-scope EU companies and certain non-EU companies to manage actual or potential adverse impacts of their activities and their supply chains on human rights and environmental matters. The European Commission (“EC”) is reviewing and may amend aspects of the CSRD, CSDDD and EU Taxonomy Regulation. Meanwhile, the UK continues to work on implementation of its Sustainability Disclosure Requirements.

The EU and the UK Financial Conduct Authority (“FCA”) have issued rules and guidelines on the use of ESG or sustainability related terms in fund names. In addition, the EU adopted regulations on ESG rating providers applicable in mid-2026 while the UK is expected to propose new legislation on ESG rating providers. Japan and Singapore have published codes of conduct for ESG data and rating providers, with Hong Kong considering a similar approach, while India introduced a regulatory framework for ESG rating providers in July 2023.

As jurisdictions continue to develop and implement sustainability regulations and litigation challenging such regulations increases, BlackRock faces greater fragmentation risk related to local application of regulations, resulting in complex and conflicting compliance obligations and legal and regulatory uncertainty.

Taxation

BlackRock’s businesses may be directly or indirectly affected by tax legislation and regulation, or the modification of existing tax laws, by US or non-US tax authorities. Legislation at both the US federal and state level has been previously proposed to enact a financial transaction tax (“FTT”) on stocks, bonds and a broad range of financial instruments and derivative transactions. In the EU, certain Member States have also enacted similar FTTs and the European Commission (“EC”) has proposed legislation to harmonize these taxes and provide for the adoption of EU-level legislation applicable to some (but not all) EU Member States. If enacted as proposed, FTTs could have an adverse effect on BlackRock’s financial results and clients’ performance results.

The Organisation for Economic Cooperation and Development (“OECD”) has proposed certain international tax reforms, which, among other things, would (1) shift taxing rights to the jurisdiction of the consumer (“Pillar One”) and (2) establish a global minimum tax for multinational companies of 15% (“Pillar Two”). In response, EU member states and several other countries, including the UK, have since adopted laws implementing the OECD’s minimum tax rules under Pillar Two, effective starting in 2024. As a result of these developments, the tax laws of certain countries in which BlackRock does business have changed and may continue to change, and any such changes could increase its tax liabilities. The Company is continuing to monitor legislative developments and evaluate the potential impact of the Pillar Two Framework on future periods.

The application of tax regulations involves numerous uncertainties and, in the normal course of business, US and non-US tax authorities may review and challenge tax positions adopted by BlackRock. These challenges may result in adjustments to, or impact the timing or amount of, taxable income, deductions or other tax allocations, which may adversely affect BlackRock’s effective tax rate and overall financial condition. Similarly, the Company manages assets in products and accounts that have investment objectives which may conform to tax positions adopted by BlackRock or to specific tax rules. To the extent there are changes in tax law or policy, or regulatory challenges to tax positions adopted by BlackRock, the value or attractiveness of such investments may be diminished and BlackRock may suffer financial or reputational harm.

US REGULATORY REFORM

Antitrust Rules and Guidance

In October 2024, the Federal Trade Commission ("FTC"), with concurrence from the Antitrust Division of the Department of Justice (the "DOJ") approved amendments to rules enacted under the Hart-Scott-Rodino Antitrust Improvements Act of 1976 ("HSR") that require parties in certain transactions to provide the FTC and DOJ prior notice and observe a waiting period before consummation of such transactions. The final amendments significantly expanded the information required to be reported and documents to be submitted in connection with an HSR filing, which will likely substantially increase any pre-merger notification expenses and may delay transactions. In December 2023, the FTC and DOJ also jointly issued new merger guidelines, which could impact (1) the ability of the Company to expand its services through strategic investments or acquisitions and (2) funds that engage in transactions reportable under HSR.

Designation as a Systemically Important Financial Institution ("SIFI")

The FSOC has the authority to designate nonbank financial institutions as SIFIs in the US under the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 (the "Dodd-Frank Act"). In November 2023, the FSOC finalized amendments to its existing interpretive guidance to remove the prioritization of an activities-based approach over an entity-specific approach to designation in connection with addressing potential risks to financial stability, although the amendment clarified that the FSOC retained the ability to use an activities-based approach when appropriate. If BlackRock is designated as a SIFI, it could become subject to enhanced regulatory and capital requirements and direct supervision by the Federal Reserve.

SEC Rules Governing Security-Based Swaps

In 2021, the SEC proposed rules in connection with security-based swaps ("SBS") transactions to require public reporting of large SBS positions, which, if adopted as proposed, may affect the types of transactions BlackRock may choose to execute in SBS or other SBS-related assets, introduce or increase costs relating to such transactions, and impact the liquidity in the SBS markets in which BlackRock transacts.

SEC Rules on Form PF

In 2023 and 2024, the SEC adopted new rules and amendments to Form PF for registered investment advisers requiring new disclosures, filing obligations and enhanced reporting. Implementing these rules and amendments may significantly increase BlackRock's reporting, disclosure and compliance obligations and create operational complexity for BlackRock's alternatives products.

US DOL Fiduciary Rule

The US DOL adopted new regulation redefining the meaning of "investment advice fiduciary" under ERISA as well as amendments to several prohibited transaction exemptions applicable to investment advice fiduciaries, which would substantially expand when a person would be considered a fiduciary subject to ERISA and could require BlackRock to revise a number of its distribution relationships, create compliance and operational challenges for BlackRock and its distribution partners, and limit BlackRock's ability to provide certain services to applicable clients. In July 2024, federal courts issued stays on the regulation and implementation has been postponed pending further notice.

SEC US Treasury Clearing Mandate

In December 2023, the SEC adopted rules mandating central clearing of US Treasury repurchases and certain other Treasury transactions. The rules require many market participants, including a large number of BlackRock funds and accounts, to clear Treasury repurchase transactions and potentially certain cash Treasury securities transactions through a clearing agency registered with the SEC, which could increase transaction costs for BlackRock's clients.

Proposed Rules on Equity Market Structure

In 2023, the SEC proposed equity market structure reforms that would significantly change how national market system ("NMS") stock orders are priced, executed and reported. The reforms include: (1) a requirement for certain retail orders to be subject to order-by-order competition, (2) a best execution rule and (3) an adjustment to the tick sizes at which NMS stocks can be quoted or traded. In 2024, the SEC adopted the rule adjusting NMS tick sizes. If the other proposed rules are enacted as proposed, their collective impact may adversely affect market efficiency and execution costs, which would result in negative effects for BlackRock's business and clients.

SEC Rules on Short Sales and Reporting of Securities Loans

In 2023, the SEC adopted a new rule requiring certain institutional managers to report short positions and activity to the SEC for publication on an aggregate basis, which could impact investment strategies and result in greater operational burdens and cost for BlackRock. The SEC also adopted a new rule requiring certain persons to report information on securities loan transactions to a registered national securities association which will then publish certain information. The rule may increase BlackRock's operational burdens and costs.

SEC Predictive Data Analytics Rules

The SEC proposed new rules in 2023 that would require broker-dealers and investment advisers, when engaging or communicating with investors using predictive data analytics ("PDA") and PDA-like technologies, to evaluate such technologies for conflicts of interest and, where identified, eliminate or neutralize the conflict of interest. If adopted as proposed, the rules could encompass a wide range of forward-looking uses of technology applications and impose significant operational burdens and costs.

Financial Crimes Enforcement Network Rule for Registered Investment Advisers

In August 2024, the Financial Crime Enforcement Network ("FinCEN") issued a final rule which will require registered investment advisers to adopt new anti-money laundering requirements beginning in 2026. Under the rule, registered investment advisers will be required to establish written risk-based anti-money laundering programs and report suspicious activity to FinCEN under the Bank Secrecy Act of 1970 (the "Bank Secrecy Act"), as well as comply with Bank Secrecy Act reporting and recordkeeping requirements, which may increase BlackRock's compliance burdens and costs.

SEC Rulemakings for US Registered Funds and Investment Advisers

The SEC has engaged in various initiatives and reviews impacting regulatory structure governing the asset management industry and registered investment companies. For example, the SEC adopted rules requiring certain funds to provide tailored fund shareholder reports, adopted final amendments to the rule governing fund names, expanding the scope of the rule to fund names including growth, value, ESG or similar terms, and proposed rules governing outsourcing of certain functions by investment advisers to service providers.

INTERNATIONAL REGULATORY REFORM

Enhanced Regulatory Scrutiny of Technology Service Providers to Financial Services Firms

The EU's Digital Operational Resilience Act ("DORA"), which focuses on direct regulation of providers and users of technology and data services, became applicable beginning in January 2025. DORA, among other things: (1) introduces additional governance, risk management, incident reporting, resilience testing and information sharing requirements to several of BlackRock's European entities and certain Aladdin clients; and (2) may potentially subject Aladdin to additional oversight. The European Supervisory Authorities will use data collected under DORA to assess which third party suppliers should be designated as critical to the EU financial system and become subject to further regulatory oversight. In 2024, the UK issued final policies regulating services provided by certain third parties designated by HMT as "critical" to the financial sector, which became effective in January 2025. Entities designated as "critical" will be required to provide additional information to financial regulators, engage in resilience testing and report major incidents like cyber-attacks, natural disasters and power outages.

Retail Investment Strategy

The EU continues to consider a proposed Retail Investment Strategy package of amendments intended to enhance protections for retail investors. When enacted, these changes could impact clients' product preferences and may increase costs for BlackRock in European markets due to additional requirements on distributors and product providers.

EMIR 3.0

The EU legislative package known as "EMIR 3.0" introduces key changes to clearing, margining and reporting requirements in the European Market Infrastructure Regulation (EMIR), including: (1) a requirement to hold an "active account" with an EU central counterparty for clearing certain euro-denominated instruments; (2) new reporting requirements for cleared trades; (3) revised clearing thresholds for financial and non-financial counterparties; and (4) amendments related to clearing to the UCITS directive. EMIR 3.0 is expected to impact EU counterparties as well as UK and non-EU entities trading with EU firms, and the collective impact of the package may increase operational complexity, necessitate a reassessment of clearing and trading strategies, and lead to higher transaction costs for BlackRock and its clients.

FSMA 2023

The Financial Services and Markets Act 2023 ("FSMA") reflects significant changes to the UK framework for financial services regulation, including changes that: (1) revoke, amend or retain EU law on financial services regulation, (2) amend the UK Markets in Financial Instruments Directive and Markets in Financial Instruments Regulation frameworks, (3) establish a new designated activities regime and (4) reform the financial promotion regime for unauthorized firms. Other reforms building upon the FSMA and potentially impacting the asset management sector include: (1) replacement of the packaged retail and insurance based investment products ("PRIIPs") Regulation; (2) review of the UK's green finance strategy, including regulation of ESG data providers, UK taxonomy and disclosure requirements; (3) review of governance through the Senior Managers and Certification Regime; (4) repeal of EU legislation on the European Long-Term Investment Fund; (5) market infrastructure reforms, including transition to T+1 settlement; (6) reassessment of the boundary between investment advice and financial guidance; (7) a new UK cryptoasset regime; and (8) continuing reforms to the UK listing regime.

Overseas Fund Regime ("OFR")

OFR, the simplified regime through which non-UK funds can register with the FCA to be marketed to UK retail investors, was enacted in February 2022 and continues to be implemented through 2025. For certain types of funds, OFR requires consumer protection regimes in EU countries where such BlackRock funds are domiciled to be found equivalent to the UK's regime in order to market the funds in the UK.

Conduct Regulation

The FCA continues to focus on conduct regulation, including the implementation of the Consumer Duty by all asset management firms, including BlackRock's UK subsidiaries. The Consumer Duty rules require firms to act to deliver good outcomes for retail customers in their manufacture and distribution of products and services, in respect of price and value, consumer understanding and consumer support. Any failure to meet the FCA's regulatory expectations could expose BlackRock to regulatory sanctions and increased reputational risk.

UK Stewardship Code Review

In 2024, the UK Financial Reporting Council released a consultation on reforms to the UK Stewardship Code, including tailored reporting requirements for proxy advisers and investment consultants, which may impact BlackRock's activities on behalf of its clients.

Regulatory Environment in China

The Company's operations in China are subject to a number of regulatory risks, including an evolving regulatory environment and complex data security and data transfer regulations. These factors may increase compliance risk and costs, limit the Company's ability to source and execute new investment opportunities and lead to impairment losses on its investments. Restrictions on transfers of certain types of onshore data of the Company's Chinese entities to offshore entities also may limit BlackRock's ability to aggregate, report and monitor such data on its global platform. In addition, a number of regulators in China have jurisdiction over BlackRock's business operations, increasing operational and regulatory engagement complexity. These risks may be further heightened by additional scrutiny by Chinese regulators of certain sectors, such as technology and other industries that might be deemed to be of national importance.

EXISTING US REGULATION — OVERVIEW

BlackRock and certain of its US subsidiaries are currently subject to extensive regulation, primarily at the federal level, by the SEC, the DOL, the Federal Reserve, the Office of the Comptroller of the Currency ("OCC"), the Financial Industry Regulatory Authority ("FINRA"), the National Futures Association ("NFA"), the FTC, the Department of Justice, the CFTC and other federal government agencies and regulatory bodies. In addition, BlackRock's interactions with government entities, public officials, and candidates for public office are subject to federal, state, and local laws and rules.

Certain of BlackRock's US subsidiaries are also subject to various anti-terrorist financing, privacy, anti-money laundering and economic sanctions laws and regulations established by various agencies. In addition, the Investment Advisers Act of 1940 (the "Advisers Act") imposes numerous obligations on registered investment advisers such as certain BlackRock subsidiaries, including record-keeping, operational and marketing requirements, disclosure obligations and prohibitions on fraudulent activities. State level oversight and regulation through attorneys general, insurance commissioners and other state level agencies also applies to certain BlackRock activities.

The Investment Company Act of 1940 (the "Investment Company Act") imposes stringent governance, compliance, operational, disclosure and related obligations on registered investment companies and their investment advisers and distributors, such as certain BlackRock subsidiaries and affiliates. The SEC is authorized to institute proceedings and impose sanctions for violations of the Advisers Act and the Investment Company Act, ranging from fines and censure to termination of an investment adviser's registration. Investment advisers also are subject to certain state securities laws and regulations. Non-compliance with the Advisers Act, the Investment Company Act or other federal or state securities laws and regulations could result in investigations, sanctions, disgorgement, fines and reputational damage.

BlackRock's trading and investment activities for client accounts are regulated under the Securities Exchange Act of 1934, as amended (the "Exchange Act"), the Securities Act of 1933, as amended and the Commodity Exchange Act, as well as the rules of various securities exchanges and self-regulatory organizations, including laws governing trading on inside information, market manipulation and a broad number of technical requirements (e.g., short sale limits, volume limitations and reporting obligations) and market regulation policies. Violation of any of these laws and regulations could result in fines or sanctions, as well as restrictions on BlackRock's activities and damage to its reputation. Furthermore, the Dodd-Frank Act requires one of BlackRock's subsidiaries, BTC, to register as a municipal advisor (as that term is defined in the Exchange Act) with the SEC and Municipal Securities Rulemaking Board ("MSRB"). BTC's registration as a municipal advisor subjects BTC to additional regulation by the SEC and MSRB.

BlackRock and its subsidiaries are more broadly subject to comprehensive regulation of their derivatives businesses, including regulations promulgated by US regulators such as the CFTC, the NFA and the SEC that, among other things, impose margin requirements, public and regulatory reporting, central clearing and mandatory trading on regulated exchanges or execution facilities for certain types of swaps and security-based swaps.

BlackRock manages a variety of private pools of capital, including hedge funds, funds of hedge funds, private equity funds, collateralized debt obligations, collateralized loan obligations, real estate funds, collective trust funds, managed futures funds and hybrid funds. Congress, regulators, tax authorities and others continue to explore, on their own and in response to demands from the investment community and the public, increased regulation related to private pools of capital, including changes with respect to investor eligibility, certain limitations on trading activities, record-keeping and reporting, the scope of anti-fraud protections, safekeeping of client assets and a variety of other matters. BlackRock may be materially and adversely affected by new legislation, rulemaking or changes in the interpretation or enforcement of existing rules and regulations imposed by various regulators in this area.

The activities of certain BlackRock subsidiaries are subject to ERISA, and to regulations promulgated thereunder by the DOL, insofar as they act as a "fiduciary" under ERISA with respect to benefit plan clients that are subject to ERISA. ERISA and applicable provisions of the Internal Revenue Code, including applicable related regulations, impose certain duties on persons who are fiduciaries under ERISA, prohibit certain transactions involving ERISA plan clients (clients subject to the prohibited transaction rules under Section 975 of the Code), and, among other things, mandate certain required periodic reporting and disclosures and require certain BlackRock entities to carry bonds insuring against losses caused by fraud or dishonesty. ERISA and other applicable regulations also impose additional compliance, reporting and operational requirements on BlackRock that otherwise are not applicable to clients that are not subject to ERISA. Excise taxes and other potential penalties could apply as a result of violations of the above-described prohibitions and requirements.

BlackRock has eight subsidiaries that are registered as commodity pool operators and/or commodity trading advisors with the CFTC and are members of the NFA. The CFTC and NFA each administer a comparable regulatory system covering futures contracts and various other financial instruments, including swaps as a result of the Dodd-Frank Act, in which certain BlackRock clients may invest. In addition, two of BlackRock's subsidiaries are registered with the SEC as broker-dealers and are member-firms of FINRA. Each broker-dealer has a membership agreement with FINRA that limits the scope of such broker-dealer's permitted activities. The broker-dealers are also members of the MSRB and are subject to MSRB rules.

BlackRock's business activity in California that involves the processing of personal information is subject to the California Consumer Privacy Act ("CCPA") and the California Privacy Rights Act ("CPRA"), which provide for enhanced consumer protections for California residents. The CCPA and CPRA impose obligations on BlackRock for the handling, disclosure and deletion of personal information for California residents. In addition, several other US states have proposed or adopted similar privacy laws. Any failure by BlackRock to comply with the CCPA, CPRA or similar state privacy laws may result in fines, heightened regulatory scrutiny, litigation and/or reputational harm.

US Banking Regulation

One of BlackRock's subsidiaries, BTC, is organized as a nationally-chartered limited purpose trust company that does not accept deposits or make commercial loans. Accordingly, BTC is examined and supervised by the OCC and is subject to various banking laws and regulations enforced by the OCC, such as laws and regulations governing capital adequacy, fiduciary activities, conflicts of interest, self-dealing, and the prevention of financial crime, including money laundering. BTC is also a member of the Federal Reserve System and is subject to various Federal Reserve regulations applicable to member institutions, such as regulations restricting transactions with affiliates. Many of these laws and regulations are meant for the protection of BTC and/or BTC's customers rather than BlackRock, its affiliates or stockholders.

EXISTING INTERNATIONAL REGULATION — OVERVIEW

BlackRock's international operations are subject to the laws and regulations of a number of international jurisdictions, as well as oversight by numerous regulatory agencies and bodies in those jurisdictions. In some instances, these operations are also affected by US laws and regulations that have extra-territorial application.

Below is a summary of certain international regulatory standards to which BlackRock is subject. It is not meant to be comprehensive as there are parallel legal and regulatory arrangements in force in many jurisdictions where BlackRock's subsidiaries conduct business.

Of note among the various other international regulations to which BlackRock is subject, are the extensive and complex regulatory reporting requirements that necessitate the monitoring and reporting of issuer exposure levels (thresholds) across the holdings of managed funds and accounts and those of the Company.

Regulation in EMEA

The FCA currently regulates certain BlackRock subsidiaries in the UK. It is also responsible for the conduct of business regulation of the UK branch of one of BlackRock's US subsidiaries. In addition, the Prudential Regulation Authority ("PRA") regulates one BlackRock UK insurance subsidiary. Authorization by the FCA and (where relevant) the PRA is required to conduct certain financial services-related business in the UK under the Financial Services and Markets Act 2000 (the "FSMA"). The FCA's rules adopted under the FSMA govern the majority of a firm's capital and liquidity resources requirements, senior management arrangements, conduct of business and client assets requirements, interaction with clients, and systems and controls, whereas the rules of the PRA focus solely on the prudential requirements that apply to BlackRock's UK-based insurance subsidiary. The FCA supervises BlackRock's UK-regulated subsidiaries through a combination of proactive engagement, event-driven and reactive supervision and thematic reviews in order to monitor BlackRock's compliance with regulatory requirements. Breaches of the FCA's rules may result in a wide range of disciplinary actions against BlackRock's UK-regulated subsidiaries and/or its employees. The FCA's Consumer Duty requires BlackRock's UK authorized firms to act to deliver good outcomes for retail customers taking into account products and services, price and value and consumer understanding and support.

In addition, BlackRock has regulated entities in France, Germany, Ireland, Jersey, Luxembourg, the Netherlands and Switzerland. Each of these entities is required to comply with regulatory rules in the country in which it has been established, including the branches of the Netherlands entity which operate across the EU.

BlackRock's EU subsidiaries and branches must comply with the EU regulatory regime set out in MiFID II. BlackRock's UK-regulated subsidiaries and branches must comply with the UK version of MiFID II, which regulates the provision of investment services and activities in the UK. MiFID II, and the UK equivalent of MiFID II, set out detailed requirements governing the organization and conduct of business of investment firms and regulated markets. The legislation also includes pre- and post-trade transparency requirements for equity and non-equity markets and extensive transaction reporting requirements. BlackRock's UK insurance subsidiary must also comply with the UK regulation which implemented Solvency II and the Insurance Distribution Directive. In addition, relevant entities must comply with revised obligations on capital resources for certain investment firms arising out of the Investment Firms Prudential Review. These include requirements to ensure capital adequacy, as well as matters of liquidity, governance and remuneration. Relevant BlackRock entities must also comply with the requirements of the UCITS Directive and the AIFMD, as implemented in the relevant EU Member States and in the UK, which impose obligations on the authorization and capital, conduct of business, organization, transparency and marketing of retail and alternative investment funds respectively that are sold in, or marketed to, the EU. The obligations introduced through these regulations and directives (and the UK implementation and onshoring of the same) will affect certain of BlackRock's European and UK operations. Compliance with the UCITS Directives and the AIFMD (and the UK implementation and onshoring of the same) may subject BlackRock to additional expenses associated with depositary oversight and other organizational requirements. The UK Government and the FCA have also enacted the new OFR providing a fast-track framework for non-UK funds to be recognized and registered for marketing to retail investors in the UK after Brexit.

The EU has seen an increase in Common Supervisory Actions by ESMA to coordinate supervisory action by national EU regulators, most notably in areas such as sustainability-related product features and disclosures, product governance, liquidity management and fund costs and charges. BlackRock's EU operations may be affected to the extent this initiative results in formal legislation or action.

EU Member States, the UK and many other non-US jurisdictions have adopted statutes and/or regulations concerning privacy and data protection and requiring notification of personal data security breaches if certain thresholds are met. For example, the EU adopted the General Data Protection Regulation ("GDPR"), which became effective in 2018. Following Brexit, the UK onshored the GDPR into national law ("UK GDPR"), which became effective in 2021. In June 2021, the EC published a new set of standard contractual clauses, which only apply to the transfer of personal data outside of the EU to a country not approved by the EU as providing an adequate level of protection for the processing of personal data. The EU's adequacy decision with respect to the UK, which allows the continued flow of personal data from the EU to the UK, will be regularly reviewed and may be revoked if the UK diverges from its current adequate data protection laws. The UK has developed its own international data transfer agreement, which was implemented in March 2022. In June 2023, the EU-US Data Protection Framework came into force, which allows organizations to self-certify their compliance under the framework for data transfers from the EU, UK and Switzerland to the US. Saudi Arabia and Dubai have also adopted privacy and data protection regulations aligned with GDPR. GDPR and UK GDPR, as well as other statutes and/or regulations concerning privacy and data protection, increase compliance obligations, affect BlackRock's collection, processing, retention and transfer of personal data and reporting of personal data security breaches, and provide for increased penalties for non-compliance.

BlackRock also maintains two offices in the Middle East, one in Dubai, which is regulated by the Dubai Financial Services Authority, and one in Riyadh, Saudi Arabia, which is regulated by the Saudi Capital Markets Authority. Both offices are authorized to provide certain investment services and support BlackRock's provision of investment products and services in their countries of domicile. Other countries across the Middle Eastern region are serviced on a cross-border basis.

Regulation in the Asia-Pacific Region

In Japan, a BlackRock subsidiary is subject to the Financial Instruments and Exchange Act ("FIEA") and the Act on Investment Trusts and Investment Corporations. These laws are administered and enforced by the Japanese Financial Services Agency ("JFSA"), which establishes standards for compliance, including capital adequacy and financial soundness requirements, customer protection requirements and conduct of business rules. The JFSA is empowered to conduct administrative proceedings that can result in censure, fines, cease and desist orders or the suspension or revocation of registrations and licenses granted under the FIEA. This Japanese subsidiary also holds a license for real estate brokerage activities which subjects it to the regulations set forth in the Real Estate Brokerage Act.

In Australia, BlackRock's main operating entities are principally regulated under the Corporations Act 2001 (Cth) by the Australian Securities and Investments Commission ("ASIC"), which includes holding an Australian financial services license and operating registered managed investment schemes. ASIC is Australia's integrated corporate, markets, financial services and consumer credit regulator.

In New Zealand, certain BlackRock subsidiaries are primarily regulated by the Financial Markets Authority ("FMA"). The FMA is responsible for overseeing and enforcing financial markets legislation including the licensing of firms to provide certain financial products and services in New Zealand and administering anti-money laundering and terrorism financing legislation, amongst other functions.

The activities of certain BlackRock subsidiaries in Hong Kong are subject to the Securities and Futures Ordinance ("SFO"), which governs the securities and futures markets, and regulates, among others, offers of investments to the public, and provides for the licensing of intermediaries. The SFO is administered by the Securities & Futures Commission ("SFC"). The SFC is also empowered to establish standards for compliance as well as codes and guidelines. The relevant BlackRock subsidiaries and the employees conducting any of the regulated activities specified in the SFO are required to be licensed with the SFC, and are subject to the rules, codes and guidelines issued by the SFC.

BlackRock's operations in Taiwan are subject to the Securities Investment Trust and Consulting Act and other regulations, rules or guidelines thereunder (collectively, "SITE and SICE Requirements"). BlackRock's subsidiary in Taiwan is governed and regulated by the Securities & Futures Bureau ("SFB") under the Taiwan Financial Supervisory Commission, which is responsible for regulating securities markets (including the Taiwan Stock Exchange, the Taipei Exchange and the Taiwan Futures Exchange), the asset management industry, the broker and futures commission merchant sector, the banking industry and the insurance sector. The relevant BlackRock subsidiary and employees conducting regulated activities are required to be licensed with the SFB and subject to the SITE and SICE Requirements.

BlackRock's Fund Management Company in China ("BlackRock FMC") is regulated by the China Securities Regulatory Commission and is subject to the Securities Investment Fund Law and Measures for the Supervision and Administration of Mutual Fund Managers for the overall oversight from incorporation to the corporate governance and operations of fund managers and funds. BlackRock FMC is also subject to the China Securities Law and various other financial laws and regulations. BlackRock CCB Wealth Management Limited, which is BlackRock's wealth management joint venture company with CCB Wealth Management Co., Ltd. and Fullerton Management Pte Ltd. in China, is regulated by the National Financial Regulatory Administration ("NFRA", formerly known as the China Banking and Insurance Regulatory Commission). They have enacted Bank Wealth Management Supervision and Management Measures and Management Measures of Bank Wealth Management Subsidiaries and other relevant rules to regulate the setup, conduct of business and risk management of bank wealth management companies.

In Singapore, a BlackRock subsidiary is regulated by the Monetary Authority of Singapore ("MAS") and its business activities are subject to the Securities and Futures Act 2001 ("SFA"). The SFA governs the regulation of activities and institutions in the securities and derivatives industry, including fund management, dealing in capital markets products and leveraged foreign exchange trading. The MAS is Singapore's central bank and integrated financial regulator, which regulates the financial services sector in Singapore and conducts integrated supervision of financial services and financial stability surveillance. This BlackRock subsidiary and the employees conducting any of the regulated activities specified in the SFA are required to be licensed with the MAS, and are subject to the SFA and the regulations, rules, codes, notices and guidelines issued by the MAS.

In India, the Jio BlackRock joint venture entities, Jio BlackRock Asset Management Private Limited, Jio BlackRock Investment Advisers Private Limited and Jio BlackRock Trustee Private Limited, are governed by the Companies Act, 2013 and regulated by the Ministry of Corporate Affairs in India. Following approval to act as sponsors, Jio Financial Services and BlackRock through their Jio BlackRock joint venture entities have made applications to the Securities and Exchange Board of India seeking to manage and operate a licensed asset management company.

Other financial regulators oversee BlackRock subsidiaries, branches and representative offices across the Asia-Pacific region, including in South Korea. Regulators in all of these jurisdictions have authority with respect to financial services including, among other things, the authority to grant, suspend or cancel required licenses or registrations. In addition, these regulators may subject certain BlackRock subsidiaries to net capital requirements.

AVAILABLE INFORMATION

BlackRock files annual, quarterly and current reports, proxy statements and all amendments to these reports and other information with the SEC. BlackRock makes available free-of-charge, on or through its website at https://www.blackrock.com, the Company's Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, proxy statements and all amendments to those filings, as soon as reasonably practicable after such material is electronically filed with or furnished to the SEC. The Company also makes available on its website the charters for the Audit Committee, Management Development and Compensation Committee, Nominating, Governance and Sustainability Committee and Risk Committee of the Board of Directors, its Code of Business Conduct and Ethics, its Code of Ethics for Chief Executive and Senior Financial Officers and its Corporate Governance Guidelines. Further, BlackRock will provide, without charge, upon written request, a copy of the Company's Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, proxy statements and all amendments to those filings as well as the committee charters, its Code of Business Conduct and Ethics, its Code of Ethics for Chief Executive and Senior Financial Officers and its Corporate Governance Guidelines. Requests for copies should be addressed to Investor Relations, BlackRock, Inc., 50 Hudson Yards, New York, New York 10001. Reports, proxy statements and other information regarding issuers that file electronically with the SEC, including BlackRock's filings, are also available to the public from the SEC's website at https://www.sec.gov.

Item 1A. Risk Factors

As a global investment management firm, risk is an inherent part of BlackRock's business. Global markets, by their nature, are prone to uncertainty and subject participants to a variety of risks. While BlackRock devotes significant resources across all of its operations to identify, measure, monitor, manage and analyze market, operating, legal, compliance, reputational, fiduciary and investment risks, BlackRock's business, financial condition, operating results and nonoperating results could be materially adversely affected and the Company's stock price could decline as a result of any of these risks and uncertainties, including the ones discussed below.

RISKS RELATED TO MARKET AND COMPETITION

Changes in the value levels of equity, debt, real assets, commodities, foreign exchange or other asset markets, including from the impact of global trade policies and tariffs, may cause assets under management ("AUM"), revenue and earnings to decline.

BlackRock's investment management revenue is primarily comprised of fees based on a percentage of the value of AUM and, in some cases, performance fees which are normally expressed as a percentage of returns to the client. Numerous factors, including price movements in the equity, debt or currency markets, or movements in the price of real assets, commodities, digital assets or other alternative investments in which BlackRock invests on behalf of its clients, including from the impact of global fiscal, monetary and trade policies, could cause:

- the value of AUM, or BlackRock's returns on AUM, to decrease;
- client redemptions from BlackRock's products;
- client rebalancing or reallocating of assets into BlackRock products that yield lower fees;
- an impairment to the value of intangible assets and goodwill; or
- a decrease in the value of seed or co-investment capital.

These risks may also be heightened by market volatility, illiquid market conditions or other market disruptions. The occurrence of any of the above events may cause the Company's AUM, revenue and earnings to decline.

Changes in interest or foreign exchange rates and/or divergent beta may cause BlackRock's AUM and base fees to fluctuate and introduce volatility to the Company's net income and operating cash flows.

BlackRock's business is directly and indirectly affected by changes in global interest rates, as well as changes in global markets, which have experienced substantial volatility in recent years. Similarly, due to the global nature of BlackRock's operations, a portion of its business is conducted in currencies other than the United States ("US") dollar. Fluctuations in BlackRock's AUM related to its exposure to foreign exchange rates relative to the US dollar and interest rates may introduce volatility to the Company's base fees, net income and operating cash flows.

In addition, beta divergence between equity markets, where certain markets perform differently than others, may lead to an increase in the proportion of BlackRock AUM weighted toward lower fee equity products, resulting in a decline in BlackRock's effective fee rate. Divergent market factors may also erode the correlation between the growth rates of AUM and base fees.

BlackRock's investment advisory contracts may be terminated or may not be renewed by clients and fund boards on favorable terms and the liquidation of certain funds may be accelerated at the option of investors.

BlackRock derives a substantial portion of its revenue from providing investment advisory services. The advisory or management contracts BlackRock has entered into with its clients, including the agreements that govern many of BlackRock's investment funds, provide investors or, in some cases, the independent directors of applicable investment funds, with significant latitude to terminate such contracts, withdraw funds or liquidate funds, or to remove BlackRock as a fund's investment advisor (or equivalent). BlackRock also manages its US mutual funds, closed-end and exchange-traded funds under management contracts that must be renewed and approved annually by the funds' respective boards of directors, a majority of whom are independent from the Company. BlackRock's fee arrangements under any of its advisory or management contracts may be reduced (including at the behest of a fund's board of directors). In addition, shareholder activism involving closed-end funds has increased, including public campaigns to demand that a fund consider significant transactions such as a tender offer, merger or liquidation or seek other actions such as the termination of the fund's management contract. If a number of BlackRock's clients terminate their contracts, or otherwise remove BlackRock from its advisory roles, liquidate funds or fail to renew management contracts on similar terms, the fees or carried interest BlackRock earns could be reduced, which may cause BlackRock's AUM, revenue and earnings to decline.

The failure or negative performance of products offered by competitors may cause AUM in similar BlackRock products to decline irrespective of BlackRock's performance.

Many competitors offer similar products to those offered by BlackRock and the failure or negative performance of competitors' products could lead to a loss of confidence in similar BlackRock products, irrespective of the performance of such BlackRock products. Any loss of confidence in a product type could lead to withdrawals, redemptions and liquidity issues in such products, which may cause the Company's AUM, revenue and earnings to decline.

Increased competition may cause BlackRock's AUM, revenue and earnings to decline.

The investment management industry is highly competitive, and BlackRock competes based on a number of factors including: investment performance, liquidity, its technology and portfolio construction offerings, the level of fees charged, the quality and breadth of services and products provided, name recognition and reputation, and its ability to develop new investment strategies and products to meet the changing needs of investors. In addition, over the past several years, the asset management industry has continued to evolve as investors increasingly seek out firms that have the capacity to deliver broad multi-asset investment capabilities and technological expertise, including in a manner that is responsive to ever more localized needs. This evolution, together with the introduction of new technologies, as well as regulatory changes, continues to alter the competitive landscape for investment managers, which may lead to additional fee compression or require BlackRock to invest more to modify or adapt its product offerings to attract and retain customers and remain competitive with the products, services and geographic diversity offered by other financial institutions, technology companies, advisory or asset management firms. Increased competition on the basis of any of these factors, including competition leading to fee reductions on existing or new business, may cause the Company's AUM, revenue and earnings to decline.

Failure to maintain Aladdin's competitive position in a dynamic market could lead to a loss of clients and could impede BlackRock's productivity and growth.

The sophisticated risk analytics, portfolio management, trade execution and investment operations that BlackRock provides via its technology platform to support investment advisory and Aladdin clients are important elements of BlackRock's competitive success. Aladdin's competitive position is based in part on its ability to combine risk analytics with portfolio management, trading and operations tools on a single platform. Increased competition from risk analytics and investment management technology providers, including from competitors with increasingly sophisticated and comprehensive product offerings, or a shift in client demand toward standalone or internally developed solutions, whether due to price competition, perceived client market share, platform offerings or flexibility, or market-based or regulatory factors, may weaken Aladdin's competitive position and may cause the Company's revenue and earnings to decline. In addition, to the extent that Aladdin competitors are able to innovate more effectively than BlackRock or leverage delivery models that provide clients faster time to market, lower costs or the ability to more seamlessly combine or bundle with other service offerings, BlackRock may lose existing clients or fail to capture future market share, which may impede its productivity and growth. Moreover, although BlackRock takes steps to safeguard against infringements of its intellectual property ("IP"), there can be no assurance that the Company will be able to effectively protect and enforce its IP rights in Aladdin.

BlackRock may be unable to develop new products and services and the development of new products and services may expose BlackRock to reputational harm, additional costs or operational risk.

BlackRock's financial performance depends, in part, on its ability to react to changes in the asset management industry, respond to evolving client demands and technological advances, and develop, market and manage new investment products and services. The development and introduction of new products and services, including the creation of increasingly customizable products, requires continued innovative effort on the part of BlackRock and may require significant time and resources as well as ongoing support and investment. Substantial risk and uncertainties are associated with the introduction of new products and services, including the implementation of new and appropriate operational controls and procedures, shifting client and market preferences, the introduction of competing products or services, constraints on BlackRock's ability to manage growth within client mandates, compliance with regulatory and disclosure requirements and IP-related lawsuits or claims, which may not be fully evident or identified prior to the introduction of any such product or service. A growing number of BlackRock's products and services also depend on data provided by third parties as analytical inputs and are subject to additional risks, including with respect to data quality, cost, availability and provider relationships. Data sets for certain developing analytics, such as those in the sustainability space, continue to evolve and difficulties approximating gaps in the data, sourcing data from reliable sources, or validating the data could adversely impact the accuracy and effectiveness of such analytics. There can be no assurance that BlackRock will be able to innovate effectively in order to develop new products or services that address the needs of its clients on the timeline they require. Any failure to successfully develop and support new products and services, or effectively manage associated operational risks, could have an adverse impact on BlackRock's growth, harm BlackRock's reputation and expose the Company to additional costs, which may cause its AUM, revenue and earnings to decline.

Changes in the value of seed and co-investments that BlackRock owns could affect its income and could increase the volatility of its earnings.

At December 31, 2024, BlackRock's net economic investment exposure of approximately $3.9 billion in its investments (see Item 7, *Management's Discussion and Analysis of Financial Condition and Results of Operations – Investments*) primarily resulted from co-investments and seed investments in its sponsored investment funds. Movements in the equity, debt or currency markets, or in the price of real assets, commodities or other alternative investments, could lower the value of these investments, increase the volatility of BlackRock's earnings and cause earnings to decline.

BlackRock indemnifies certain securities lending clients for specified losses as a result of a borrower default.

BlackRock provides borrower default indemnification to certain of its securities lending clients. In the event of a borrower default, BlackRock may use the collateral provided by the defaulting borrower to repurchase securities out on loan to such borrower in order to replace them in a client's account. Borrower default indemnification is limited to the shortfall that occurs in the event the collateral available at the time of the borrower's default is insufficient to repurchase those securities out on loan. BlackRock requires all borrowers to mark to market their posted collateral daily to levels in excess of the value of the securities out on loan which mitigates the likelihood of the indemnity being triggered. Where the collateral is in the form of cash, the borrower default indemnification BlackRock provides does not guarantee, assume or otherwise insure the investment performance or return of any cash collateral vehicle into which that cash collateral is invested. The amount of securities on loan as of December 31, 2024 and subject to this type of indemnification was approximately $305 billion. In the Company's capacity as lending agent, cash and securities totaling approximately $324 billion was held as collateral for indemnified securities on loan at December 31, 2024. Significant borrower defaults occurring simultaneously with rapid declines in the value of collateral and/or increases in the value of the securities loaned may create collateral shortfalls, which could result in material liabilities under these indemnities and may cause the Company's revenue and earnings to decline.

BlackRock's decision on whether to provide support to particular investment products from time to time, or the inability to provide support, may cause AUM, revenue and earnings to decline.

While not legally mandated, BlackRock, at its option and from time to time, has and may in the future choose to seed, warehouse or otherwise support investment products through capital or credit support for commercial or other reasons. Any decision by BlackRock on whether to support products may utilize capital and liquidity that would otherwise be available for other corporate purposes. BlackRock's ability to seed, warehouse or otherwise support certain products may be restricted by regulation or by the Company's failure to have or make available sufficient capital or liquidity. Moreover, inherent constraints arising from the business models of certain asset managers, including BlackRock, may during periods of market volatility result in BlackRock having fewer options for accessing liquidity than asset managers with alternate business models, which may adversely impact its ability to support certain products. Any decision by BlackRock to support particular investment products, or its inability or unwillingness to provide such support, may result in losses or affect BlackRock's capital or liquidity, which may cause AUM, revenue and earnings to decline.

Geopolitical unrest and other events outside of BlackRock's control could adversely affect the global economy or specific international, regional and domestic markets, which may cause BlackRock's AUM, revenue and earnings to decline.

Geopolitical risks, including those arising from trade tension and/or the imposition of trade tariffs, terrorist activity or acts of civil or international hostility, could have an adverse impact on BlackRock. For instance, the Ukraine-Russia and Middle East conflicts have and may continue to result in geopolitical instability and adversely affect the global economy, supply chains, specific markets and operations. Strategic competition between the US and China and resulting tensions and heightened levels of political polarization have also contributed to uncertainty in the geopolitical and regulatory landscapes. Similarly, other events outside of BlackRock's control, including the impact of natural disasters, climate-related events, pandemics or health crises may arise from time to time and be accompanied by governmental actions that may increase international tension or impact the US or global economy in ways that are uncertain. Any such events and responses, including regulatory developments, may cause significant volatility and declines in the global markets, disproportionate impacts to certain industries or sectors, disruptions to commerce (including to economic activity, travel and supply chains), loss of life and property damage, and may adversely affect the global economy or capital markets, as well as the Company's products, operations, clients, vendors and employees, which may cause BlackRock's AUM, revenue and earnings to decline. BlackRock's exposure to geopolitical risks may be heightened to the extent such risks arise in countries in which BlackRock currently operates or seeks to expand its presence.

Climate-related risks could adversely affect BlackRock's business, products, operations and clients, which may cause BlackRock's AUM, revenue and earnings to decline.

BlackRock's business and those of its clients could be impacted by climate-related risks. Climate-related risks may impact BlackRock through changes in the physical climate or from the process of transitioning to a low-carbon economy. Climate-related physical risks arise from the direct impacts of a changing climate in the short- and long-term. Such risks may include the risks of extreme weather events and changes in temperature, which may damage infrastructure and facilities, including BlackRock's physical assets, as well as disrupt connectivity or supply chains. Climate-related transition risks arise from exposure to the transition to a low-carbon economy through policy, regulatory, technology and market changes. For instance, divergent existing and future climate regulations or guidance, as well as differing perspectives of stakeholders regarding climate impacts, have affected and may continue to affect BlackRock's business activities and reputation, increase scrutiny and complicate compliance requirements.

Climate-related physical and transition risks could also impact BlackRock's business both directly and indirectly through adverse impacts to its clients' investments, including as a result of declines in asset values, changes in client preferences, increased regulatory and compliance costs and significant business disruptions. Any of these risks may cause the Company's AUM, revenue and earnings to decline.

RISKS RELATED TO INVESTMENT PERFORMANCE

Poor investment performance could lead to the loss of clients and may cause AUM, revenue and earnings to decline.

The Company's management believes that investment performance, including the efficient delivery of beta, is one of the most important factors for the growth and retention of AUM. Poor investment performance relative to applicable portfolio benchmarks, aggregate fee levels or competitors may cause AUM, revenue and earnings to decline as a result of:

- client withdrawals in favor of better performing products offered by competitors;
- client shifts to products that charge lower fees;
- the diminishing ability to attract additional funds from existing and new clients;
- reduced, minimal or no performance fees;
- an impairment to the value of intangible assets and goodwill; or
- a decrease in the valuations of seed and co-investment capital.

Performance fees may increase volatility of both revenue and earnings.

A portion of BlackRock's revenue is derived from performance fees on investment advisory assignments. Performance fees represented $1.2 billion, or 6%, of total revenue for the year ended December 31, 2024. Generally, the Company is entitled to a performance fee only if the agreement under which it is managing the assets provides for one and if returns on the related portfolio exceed agreed-upon periodic or cumulative return targets. If these targets are not exceeded, a performance fee for that period will not be earned and, if targets are based on cumulative returns, the Company may not earn performance fees in future periods. The volatility of the Company's future revenue and earnings may also be affected due to private markets becoming an increasing component of the overall composition of the Company's performance fee generating assets, including from the Company's acquisition of GIP (the "GIP Acquisition") and its proposed acquisition of HPS (the "HPS Acquisition"). In particular, the Company expects that as it manages more private markets products, its performance fees will generally be recognized over substantially longer multi-year periods than those associated with more liquid products.

Failure to identify errors in the quantitative models BlackRock utilizes to manage its business could adversely affect product performance and client relationships.

BlackRock employs various quantitative models to support its investment processes, including those related to risk assessment, portfolio management, trading and hedging activities and product valuations. Any errors or limitations in the underlying models, model inputs or assumptions, including those from third-party sources, as well as any failure of BlackRock's governance, approval, testing, validation and monitoring standards in respect of such models, model inputs or assumptions, the failure to timely update such models, model inputs or assumptions or errors in how such models are used, could have adverse effects on BlackRock's business and reputation. These risks may be heightened by the rapid growth and complexity of new models, evolving data sets and standards, and market volatility.

TECHNOLOGY AND OPERATIONAL RISKS

A failure in, or disruption to, BlackRock's operations, systems or infrastructure, including business continuity plans, could adversely affect operations, damage the Company's reputation and cause BlackRock's AUM, revenue and earnings to decline.

BlackRock's infrastructure, including its technological capacity, data centers and office space, is vital to the competitiveness of its business. Moreover, a significant portion of BlackRock's critical business operations is concentrated in a limited number of geographic areas, including San Francisco, New York, London, Edinburgh, Budapest, Atlanta, Gurgaon, Mumbai and Belgrade. The failure to maintain an infrastructure commensurate with the size and scope of BlackRock's business, or the occurrence of a business outage or event outside BlackRock's control, including a major earthquake, hurricane, fire, terrorist act, pandemic, health crisis or other catastrophic event, or the actions of individuals or groups seeking to disrupt BlackRock's operations in any location at which BlackRock maintains a major presence, could materially impact operations, result in business disruption or impede the Company's growth.

Despite BlackRock's efforts to ensure business continuity, if it fails to keep business continuity plans up-to-date or if such plans, including secure back-up facilities and systems and the availability of back-up employees, are improperly implemented or deployed during a disruption, the Company's ability to operate could be adversely impacted which may cause AUM, revenue and earnings to decline or impact the Company's ability to comply with regulatory obligations or contractual obligations leading to reputational harm, legal liability, regulatory fines and/or sanctions.

A cyber-attack or a failure to implement effective information and cybersecurity policies, procedures and capabilities could disrupt operations and lead to financial losses and reputational harm, which may cause BlackRock's AUM, revenue and earnings to decline.

BlackRock is dependent on the effectiveness of the information and cybersecurity policies, procedures and capabilities it maintains to protect its computer and telecommunications systems and the data that resides on or is transmitted through them, including data provided by third parties that is significant to portions of BlackRock's business and products. An information security incident or disruption, such as a cyber-attack including social engineering, deepfakes, phishing scams, business email compromise, malware, denial-of-service or ransomware attacks, or failures to control access to sensitive systems, could materially interrupt business operations or cause disclosure or modification of sensitive or confidential client or competitive information. Moreover, developments in BlackRock's use of process automation and artificial intelligence ("AI"), as well as the use of remote access by employees and mobile and cloud technologies, heightens these and other operational risks, as certain aspects of the security of such technologies may be complex, unpredictable or beyond BlackRock's control. BlackRock's growing exposure to the public Internet, as well as reliance on mobile or cloud technology or any failure by mobile technology and cloud service providers to adequately safeguard their systems and prevent cyber-attacks, could disrupt BlackRock's operations and result in misappropriation, corruption or loss of personal, confidential or proprietary information or third-party data. In addition, there is a risk that encryption and other protective measures may be circumvented, particularly to the extent that new computing technologies including quantum computing increase the speed and computing power available.

The financial services industry has been the subject of cyber-attacks involving the dissemination, theft and destruction of corporate information or other assets, as a result of failure to follow procedures by employees or contractors or as a result of actions by third parties, including nation state actors, terrorist organizations, cyber criminals and hacktivists. BlackRock has been and continues to be the target of cyber-attacks, as well as the co-opting of its brand, and continues to monitor and develop its systems to protect its technology infrastructure and data from misappropriation or corruption, as the failure to do so could disrupt BlackRock's operations and cause financial losses. Advances in technology, including generative AI, and use of such technology by malicious actors heightens these risks. Although BlackRock has implemented policies and controls, and takes protective measures involving significant expense, to help prevent and address potential data breaches, inadvertent disclosures, increasingly sophisticated cyber-attacks and cyber-related fraud, there can be no assurance that any of these measures proves fully effective. In addition, given the evolving nature of cyber threat actors and the increasing sophistication of cyber-attack methodology, a successful cyber-attack may persist for an extended period of time before being detected, and it may take a considerable amount of time for an investigation to be completed and the severity and potential impact to be known. Moreover, due to the complexity and interconnectedness of BlackRock's systems, the process of upgrading or patching the Company's protective measures could itself create a risk of security issues or system disruptions for the Company, as well as for clients who rely upon, or have exposure to, BlackRock's systems.

In addition, due to BlackRock's interconnectivity with third-party vendors, advisors, central agents, exchanges, clearing houses and other financial institutions, BlackRock or any such third-party may be adversely affected if any of them (or their service providers) is subject to a successful cyber-attack or other information security event, including those arising due to the use of mobile technology or a third-party cloud environment. BlackRock also routinely transmits and receives personal, confidential or proprietary information by email and other electronic means. The Company collaborates with clients, vendors and other third parties to develop secure transmission capabilities and protect against cyber-attacks. However, BlackRock or such third parties may not have all appropriate controls in place to protect the confidentiality of such information.

Any information security incident or cyber-attack against BlackRock or third parties with whom it is connected, including any interception, mishandling or misuse of personal, confidential or proprietary information or failure to disclose or communicate a cybersecurity incident appropriately, could result in material financial loss, loss of competitive position, regulatory fines and/or sanctions, breach of client contracts, reputational harm or legal liability, which, in turn, may cause BlackRock's AUM, revenue and earnings to decline. In addition, BlackRock's cybersecurity insurance may not cover all losses and damages from such events and BlackRock's ability to maintain or obtain sufficient insurance coverage in the future may be limited.

Failure or unavailability of third-party dependencies may adversely affect Aladdin operations, which could cause reputational harm, lead to a loss of clients and impede BlackRock's productivity and growth.

BlackRock must maintain effective infrastructure, including a robust and secure technological framework, in order to maximize the benefit of the Aladdin platform. In so doing, it relies in part on certain third-party service providers, including for cloud hosting and technologies supporting cloud-based operations. For example, Aladdin's data architecture depends on third-party providers of technology solutions, including the ability of such parties to scale and perform in response to Aladdin's growth. In addition, the analytical capabilities of Aladdin depend on the ability of a number of third parties to provide data and other information as inputs into Aladdin's analytical calculations. Although BlackRock has implemented internal controls and procedures and maintains a robust vendor management program designed to perform diligence and monitor third parties that support the Aladdin platform, there can be no assurance that these measures will prove effective. Any failure by third parties to maintain infrastructure that is commensurate with Aladdin's size and growth, or provide the data or information required to support its varying capabilities, could compromise Aladdin's resilience, result in operational difficulties, cause reputational harm and adversely impact BlackRock's ability to provide services to its investment advisory and Aladdin clients.

Continuing enhancements to Aladdin's capabilities, as well as the expansion of the Aladdin platform into new markets and geographies, have led to significant growth in Aladdin's processing scale, which may expose BlackRock to reputational harm, increased regulatory scrutiny and heightened operational, data management, cyber- and information-security risks.

The operation of BlackRock's Aladdin platform routinely involves updating existing capabilities, configuration change management, developing, testing and rolling out new functionalities and expanding coverage into new markets and geographies, including in connection with inorganic transactions or to address client or regulatory requirements. These updates and expansion initiatives, which have led to significant growth in Aladdin*'s* processing scale, frequently occur on accelerated time frames and may expose BlackRock to additional cyber- and information-security risks, as well as increased execution, operational and data management risks. If BlackRock is unable to manage the pace of, or provide the operational resiliency and stability for, the expansion of Aladdin and associated growth of its processing scale, BlackRock may experience client attrition, reduced business, increased costs, reputational harm or regulatory fines and/or sanctions, which may cause BlackRock's AUM, revenue and earnings to decline.

In addition, the highly regulated business activities of many Aladdin clients may expose BlackRock to heightened regulatory scrutiny. For example, the changing political and regulatory environment in certain jurisdictions in which Aladdin clients are based has required BlackRock to open new data centers in those jurisdictions in order to host client data in the client's home location. Operating new data centers in foreign jurisdictions may expose BlackRock to increased operational complexity, as well as additional regulatory risks associated with the compliance requirements of such jurisdictions. In addition, there has been increased regulatory scrutiny globally on technology and information providers, which may impact Aladdin and certain functionalities and tools.

A failure to effectively manage the development and use of AI, combined with an evolving regulatory environment, could have an adverse effect on BlackRock's growth, reputation or business.

BlackRock uses machine learning and AI in its business and expects to continue to expand its AI capabilities, including through generative AI. AI methods are complex and rapidly evolving, and the introduction of AI into new or existing processes may result in new or enhanced governmental or regulatory scrutiny, IP or other litigation, data protection, confidentiality or information security risks, social or ethical concerns, competitive harm or other complications. For example, the use of datasets to develop and test AI models, the content generated by AI systems, or the application of AI systems may be found to be insufficient, biased or harmful, or lead to adverse business decisions or operating errors. AI technologies, including generative AI, may create content that appears correct but is factually inaccurate or flawed. In addition, IP ownership and license rights, including copyright, surrounding AI technologies are still being developed and have not been fully addressed by US courts or federal, state or non-US laws or regulation. Furthermore, regulatory scrutiny of AI technologies and controls continues to evolve globally with new and forthcoming laws and regulations. Efforts around use of these technologies require additional investment in operational controls and procedures, development and implementation of appropriate protections and safeguards for handling the use of data with AI, including with respect to data leakage, fraud prevention and regulatory compliance costs. AI technologies may also disrupt the competitive landscape for investment management and technology services, including in commercial and operational areas such as data aggregation and quantitative models. Any failure to successfully integrate AI technologies, respond to client or market demands, accurately communicate AI initiatives, comply with AI-related regulations, identify or address any legal or regulatory issues associated with AI or effectively manage the related risks could harm BlackRock's growth and reputation, adversely impact product offerings, client interactions or business initiatives, and expose the Company to legal and regulatory liabilities and additional costs, including regulatory fines or sanctions, which may cause its AUM, revenue and earnings to decline.

Failure to maintain adequate corporate and contingent liquidity may cause BlackRock's AUM, liquidity and earnings to decline, as well as harm its prospects for growth.

BlackRock's ability to meet anticipated cash needs depends upon a number of factors, including its creditworthiness and ability to generate operating cash flows. In addition, while BlackRock, Inc. is not subject to regulatory capital or liquidity requirements, certain of its subsidiaries are subject to regulatory capital and liquidity frameworks as well as certain other prudential requirements and standards, which require them to maintain certain levels of capital and liquidity. Failure to maintain adequate liquidity could lead to unanticipated costs and force BlackRock to revise existing or future strategic and business initiatives. BlackRock's access to equity and debt markets and its ability to issue public or private debt, or obtain lines of credit or commercial paper back-up lines, on reasonable terms may be limited by adverse market conditions, a reduction in its long- or short-term credit ratings, or changes in government regulations, including tax and interest rates. Failure to obtain funds and/or financing, or any adverse change to the cost of obtaining such funds and/or financing, may cause BlackRock's AUM, liquidity and earnings to decline, curtail its operations and limit or impede its prospects for growth.

Operating risks associated with BlackRock's securities lending program may result in client losses.

BlackRock lends securities to banks and broker-dealers as agent on behalf of certain of its clients. In these securities lending transactions, the borrower is required to provide and maintain collateral at or above regulatory minimums. Securities on loan are marked to market daily to determine if the borrower is required to provide additional collateral. BlackRock must manage this process and is charged with mitigating the associated operational risks. The failure of BlackRock's controls to mitigate such operational risks could result in financial losses for the Company's clients that participate in its securities lending programs (separate from any losses related to the risks of collateral investments), and BlackRock may be held liable for any failure to manage such risks.

Inorganic transactions may harm the Company's competitive or financial position if they are not successful.

BlackRock employs a variety of organic and inorganic strategies intended to enhance earnings, increase product offerings, deliver whole-portfolio solutions, access new clients, leverage advances in technology and expand into new geographies. Inorganic strategies have included hiring smaller-sized investment teams, making minority investments in early- to mid-stage technological and other ventures, entering into strategic joint ventures and acquiring investment management and technology businesses, analytics, models and other IP. Inorganic transactions involve a number of financial, accounting, tax, regulatory, geographical and operational challenges and uncertainties, including in some cases, the assumption of pre-existing liabilities, which must be managed in order for BlackRock to realize the benefit of such transactions, and such transactions may be the subject of unanticipated liabilities arising from commercial, client or other disputes, information security vulnerabilities or breaches and IP or other legal claims.

The success of BlackRock's inorganic strategy also depends in large part on its ability to integrate the workforce, operations, strategies, technologies and other components of a target business following the completion of an acquisition. BlackRock may be required to commit significant management time, as well as create new, or grow existing, operational and support functions, to facilitate the integration of acquired businesses, manage combined future growth and maintain a cohesive corporate culture. There can be no assurance that BlackRock will be able to successfully integrate acquired businesses, retain associated talent, scale support functions, effectively manage growth or realize other intended benefits of its inorganic strategy in the timeframe BlackRock expects, or at all. Moreover, the challenges associated with BlackRock's inorganic strategy may be heightened when inorganic transactions are in new geographic locations, involve new markets, products, business lines or early stage investments or are delivered via technology and systems that differ from those employed by BlackRock or that overlap with existing BlackRock businesses. In addition, in the case of minority investments and joint ventures, including BlackRock's joint venture to provide investment solutions in India, BlackRock may be subject to risks due to reputational harm, liability or loss resulting from, or relating to operating systems, risk management controls, and employees that are outside of BlackRock's control, risks related to the jurisdictions or markets in which such investees or joint ventures operate and risks related to the joint venture partners and investees. Any failure to identify and mitigate the risks associated with acquisitions, joint ventures or minority investments through due diligence, governance or oversight rights, indemnification provisions and/or operational expertise, or to manage the integration of acquisitions effectively, could result in losses or impairments related to such transactions and have an adverse effect on BlackRock's reputation or cause its AUM, revenue and earnings to decline, which may harm the Company's competitive position in the investment management industry.

BlackRock is subject to risks associated with its recent and proposed acquisitions, including completion of proposed acquisitions in the anticipated timeframes or at all, and any failure to realize anticipated benefits of such acquisitions.

In October 2024, BlackRock completed the GIP Acquisition. BlackRock also previously announced (1) its proposed acquisition of Preqin (the "Preqin Acquisition") which is currently expected to close in the first quarter of 2025, subject to customary closing conditions and (2) the HPS Acquisition (together with the Preqin Acquisition, the "Proposed Acquisitions") which is currently expected to close in mid-2025, subject to regulatory approvals and customary closing conditions. BlackRock is subject to risks and uncertainties associated with the Proposed Acquisitions, including the risk that a condition to closing may not be satisfied or waived, the possibility of failure to obtain any outstanding necessary regulatory approvals, which may be outside the control of BlackRock or the acquired company, or the possibility that a Proposed Acquisition does not close in the anticipated timeframe or at all. BlackRock may not be able to realize the anticipated benefits of GIP Acquisition or the Proposed Acquisitions, including synergies, value creation or other benefits of such acquisition, fully or at all, or on the timeline BlackRock expects. At times, the resources of BlackRock and the acquired companies or the attention of certain members of their management may be focused on completion and integration of the acquisition and diverted from day-to-day business operations, which may disrupt ongoing business. In addition, the process of integrating each acquired company may have an adverse impact on the Company, including from risks related to significant transaction and integration costs, unknown liabilities, employee turnover, divergence of management attention, litigation and/or regulatory actions related to the acquisition or if the acquired business does not perform as expected, which may cause BlackRock's AUM, revenue and earnings to decline.

BlackRock's alternatives products include investments in early-stage companies, private equity portfolio companies and real assets, such as real estate, infrastructure and energy assets, which expose BlackRock and its funds and accounts to new or increased risks and liabilities, as well as reputational harm.

BlackRock's alternatives products include investments in early-stage companies, private equity portfolio companies and real assets, including real estate, infrastructure and energy assets, which expose BlackRock and its funds and accounts to increased risks and liabilities that are inherent in the ownership and management of such investments and portfolio companies. These include:

- risks related to the potential illiquidity, valuation and disposition of such investments;
- risks related to emerging and less established companies that have, among other things, short operating histories, new technologies and products, nascent control functions, quickly evolving markets and limited financial resources;
- construction risks, including as a result of force majeure, labor disputes or work stoppages, shortages of material or interruptions to the availability of necessary equipment;
- accidents, pandemics, health crises or catastrophic events, such as explosions, fires or terrorist activity beyond BlackRock's control;
- climate-related risks, including greater frequency or intensity of adverse weather and natural disasters;
- personal injury or property damage;
- failures on the part of third-party servicers and operators, including managers and contractors, appointed in connection with investments or projects to adequately perform their contractual duties or operate in accordance with applicable laws;
- risks related to investments in emerging markets, including economic and political risks and differences in legal or regulatory environments, which may make enforcement of legal obligations more difficult;
- exposure to stringent and complex non-US, federal, state and local laws, ordinances and regulations, including those related to financial crime, permits, government contracting, conservation, exploration and production, tenancy, occupational health and safety, foreign investment and environmental protection;
- environmental hazards, such as natural gas leaks, product and waste spills, pipeline and tank ruptures, and unauthorized discharges of products, wastes and other pollutants;
- changes to the supply and demand for properties and/or tenancies or fluctuations in the price of commodities;
- risks related to the availability, cost, coverage and other limitations on insurance;
- risks related to governance and oversight, including board oversight, of portfolio companies;
- the financial resources of tenants; and
- contingent liabilities on disposition of investments.

The above risks may expose BlackRock's funds and accounts to additional expenses and liabilities, including costs associated with delays or remediation, and increased legal or regulatory costs, all of which could impact the returns earned by BlackRock's clients. These risks could also result in direct liability for BlackRock by exposing BlackRock to losses, regulatory sanctions or litigation, including claims for compensatory or punitive damages. Similarly, market conditions may change during the course of developments or projects in which BlackRock invests and those changes may make such developments or projects less attractive than at the time they were commenced and potentially harm the investment returns of BlackRock's clients. These risks may be heightened as the Company expands its alternative products, including through the GIP Acquisition. The occurrence of any such events may expose BlackRock to reputational harm, divert management's attention away from BlackRock's other business activities or cause its AUM, revenue and earnings to decline.

Operating in international markets increases BlackRock's operational, political, regulatory and other risks.

As a result of BlackRock's extensive international operations, the Company faces associated operational, regulatory, reputational, political and foreign exchange rate risks, many of which are outside of the Company's control. Operating outside the US may also expose BlackRock to increased compliance risks, as well as higher costs to comply with US and non-US anti-corruption, anti-money laundering and sanctions laws and regulations. Similarly, certain jurisdictions in which BlackRock operates may not have comparable levels of protection for corporate assets, such as IP, and client information and records, to the US. As a result, there may also be heightened information security or privacy risks in those jurisdictions. Any theft or unauthorized use of data, technology or IP may negatively impact BlackRock's business operations and reputation. In addition, changes to the political or regulatory environment in a jurisdiction in which BlackRock operates, including increased restrictions or scrutiny, may adversely impact BlackRock's business or operating activities. The failure of the Company's systems of internal control to mitigate such risks, or of its operating infrastructure to support its global activities, could result in operational failures and regulatory fines and/or sanctions and impede the Company's growth, which may cause the Company's AUM, revenue and earnings to decline.

RISKS RELATED TO HUMAN CAPITAL

The potential for human error in connection with BlackRock's operational systems could disrupt operations, cause losses, lead to regulatory fines or damage the Company's reputation and may cause BlackRock's AUM, revenue and earnings to decline.

Many of BlackRock's operations are highly complex and are dependent on the Company's ability to process and monitor a large number of transactions, many of which occur across numerous markets and currencies at high volumes and frequencies. Although BlackRock expends considerable resources on systemic controls, supervision, technology and training in an effort to ensure that such transactions do not violate client guidelines, applicable rules and regulations or information barriers, or adversely affect clients, counterparties or the Company, BlackRock's operations are dependent on its employees. From time-to-time, employees make mistakes that are not always immediately detected by systems, controls, policies and procedures intended to prevent and detect such errors. These can include calculation errors, errors in software implementation or development, failure to ensure data security, follow processes, patch systems or timely report issues, or errors in judgment. Such risks may be exacerbated in times of increased market volatility, high trading volumes, significant changes in operation or business offerings, and workforce turnover (including turnover related to acquisitions). Human errors, even if promptly discovered and remediated, may disrupt operations or result in regulatory fines and/or sanctions, breach of client contracts, reputational harm or legal liability, which, in turn, may cause BlackRock's AUM, revenue and earnings to decline.

Fraud, the circumvention of controls or the violation of risk management and workplace policies could have an adverse effect on BlackRock's reputation, which may cause the Company's AUM, revenue and earnings to decline.

BlackRock seeks to foster a positive workplace culture, has adopted a comprehensive risk management framework and continues to enhance various controls, procedures, policies and systems to monitor and manage risks. Notwithstanding these measures, BlackRock cannot ensure that its workplace culture or such controls, procedures, policies and systems will successfully identify and manage internal and external risks, and BlackRock employees have in the past engaged in improper conduct. In addition, BlackRock is subject to the risk that its employees, contractors or other third parties may in the future deliberately or recklessly seek to circumvent established controls to commit fraud, pay or solicit bribes or otherwise act in ways that are inconsistent with the Company's controls, policies, procedures, workplace culture or principles. This risk may be heightened as BlackRock expands into new markets, increases the breadth of its business offerings and integrates acquisitions, all of which introduce additional complexity to its risk management program. The changing nature of the office environment, such as return to office arrangements and remote and alternative work models, could cause employees to become disconnected with corporate culture and policies, which may increase operational issues. Persistent attempts to circumvent policies and controls or repeated incidents involving fraud, conflicts of interests or transgressions of policies and controls could have an adverse effect on BlackRock's reputation, cause adverse publicity, and result in litigation, regulatory inquiries, fines and/or sanctions, which may cause the Company's AUM, revenue and earnings to decline.

The failure to recruit, train and retain employees and develop and implement effective executive succession could lead to the loss of clients and may cause AUM, revenue and earnings to decline.

BlackRock's success is largely dependent on the talents and efforts of its highly skilled workforce and the Company's ability to plan for the future long-term growth of the business by identifying and developing those employees who can ultimately transition into key roles within BlackRock. The global market for qualified fund managers, investment analysts, technology and risk specialists and other professionals is highly competitive. Factors that affect BlackRock's ability to attract, train and retain highly qualified employees include the Company's reputation and workplace culture, the immigration and public health policies in the jurisdictions in which BlackRock has offices, its approach to remote and alternative work models, the compensation and benefits it provides, the impact of acquisitions and its commitment to effectively managing executive succession, including the development and training of qualified individuals.

In addition, BlackRock pays certain of its employees in deferred compensation that is tied to the Company's share price or through incentive fees and carried interest related to certain investment funds. As such, decreases in BlackRock's share price or poor performance of the investment funds related to the incentive fees and carried interest could impair the retention value of such deferred compensation. There can be no assurance that the Company will continue to be successful in its efforts to recruit and retain employees and effectively manage executive succession. If BlackRock is unable to offer competitive compensation or otherwise attract, develop and retain talented individuals, or if it fails to effectively manage executive succession, the Company's ability to compete effectively and retain its existing clients may be materially impacted.

RISKS RELATED TO KEY THIRD-PARTY RELATIONSHIPS

The impairment or failure of third parties may negatively impact the performance of products and accounts that BlackRock manages, which may cause BlackRock's AUM, revenue and earnings to decline.

BlackRock's investment management activities expose the products and accounts it manages for its clients to many different industries and counterparties, including distributors, brokers and dealers, commercial and investment banks, clearing organizations, mutual and hedge funds, and other institutional clients. Transactions with counterparties expose BlackRock's clients to credit risk in the event the applicable counterparty defaults. Although BlackRock regularly assesses risks posed by its counterparties, such counterparties may be subject to sudden swings in the financial and credit markets that may impair their ability to perform or they may fail to meet their obligations. Counterparties may also experience lapses in their internal controls or risk management systems or expose BlackRock and/or its clients to losses resulting from employee malfeasance, negligence or human error. In addition, the concentration of certain financial institutions that BlackRock uses to facilitate securities and derivatives transactions for its clients, including clearing organizations, exchanges and central agents, increases the risk that a technical or operational issue at, or default by, one such institution could introduce operational issues or delays impacting multiple BlackRock clients. Any such operational issue, impairment or failure could negatively impact the performance of products that BlackRock manages for its clients, which may lead to client attrition and, in turn, cause BlackRock's AUM, revenue and earnings to decline.

The failure of key third-party providers to BlackRock to fulfill their obligations or a failure by BlackRock to maintain its relationships with key third-party providers could have a material adverse effect on BlackRock's growth, reputation or business, which may cause the Company's AUM, revenue and earnings to decline.

BlackRock depends on a number of key third-party providers for various fund administration, accounting, custody, market and environmental, social and governance ("ESG") data, market indices, insurance, technology and AI, cloud hosting and transfer agent roles and other distribution and operational needs. Further, BlackRock relies upon a relatively concentrated group of third-party index providers to deliver services that are integral to its clients' investment decisions. The index provider industry is characterized by large vendors and the use of long-term contracts remains the market standard. This industry structure may limit BlackRock's ability to renegotiate its index provider contracts on favorable terms or at all. While BlackRock performs focused diligence on its vendors in an effort to ensure they operate in accordance with expectations and required obligations, to the extent any significant deficiencies are uncovered, there may be few, or no, alternative vendors available. In addition, BlackRock's operations and processes rely on commercially available data provided by third parties as well as providers of services, including technology services, and operating errors, process delays and failures or failures to comply with data usage requirements with respect to these service providers may adversely impact BlackRock. Data providers commonly disclaim the accuracy and completeness of data and BlackRock does not have the ability to validate or verify the accuracy and completeness of commercially sourced datasets. Moreover, in situations where BlackRock has limited access to alternative vendors, or where the nature of BlackRock's arrangement with a vendor requires a long term-commitment, BlackRock may be dependent on such vendor for continuous operational reliability and may incur additional costs if such vendor introduces required upgrades to its services.

BlackRock may from time to time transfer key contracts from one third-party provider to another. Key contract transfers may be costly and complex and expose BlackRock to heightened operational and legal risks. Any failure to mitigate such risks could result in reputational harm as well as financial losses to BlackRock and its clients. The failure or inability of BlackRock to diversify its sources for key services or the failure of any key third-party provider to fulfill its obligations could result in activities inconsistent with clients' investment management or other agreements, have an adverse financial impact on BlackRock products or lead to operational, legal and regulatory issues for the Company, which could result in reputational harm or legal liability, fines and/or sanctions and may cause BlackRock's AUM, revenue and earnings to decline.

Any disruption to the Company's distribution channels may cause BlackRock's AUM, revenue and earnings to decline.

BlackRock relies on a number of third parties to provide distribution, portfolio administration and servicing for certain BlackRock investment management products and services through their various distribution channels. BlackRock's ability to maintain strong relationships with its distributors may impact the Company's future performance, and its relationships with distributors are subject to periodic renegotiation that may result in increased distribution costs and/or reductions in the amount of BlackRock products and services being marketed or distributed. Moreover, new fiduciary regulations could lead to significant shifts in distributors' business models and more limited product offerings, potentially resulting in reduced distribution and/or marketing of certain of the Company's products and services and fee compression. If BlackRock is unable to distribute its products and services successfully or if it is unable to replace or renew existing distribution arrangements, BlackRock's AUM, revenue and earnings may decline. In addition, improper activities, as well as inadequate anti-money laundering and sanctions diligence conducted by third-party distributors, could create reputational and regulatory harm to BlackRock.

Key technology partnerships may expose BlackRock to increased regulatory oversight, as well as migration, execution, technology and operational risks.

BlackRock has a number of key strategic partnerships, including with Microsoft. For instance, the Aladdin infrastructure and environment for BlackRock and its external Aladdin clients are hosted on Microsoft Azure. BlackRock has also migrated certain systems that support its corporate functions to cloud-based platforms. The benefits of cloud-based platforms are significant and BlackRock has adopted a robust risk-based approach to its migration strategies. However, these partnerships also introduce new risks, including: (1) risks associated with relying on third-parties for aspects of infrastructure reliability and stability; (2) software and information security risks arising from the use of cloud technology; (3) operational and execution risks; and (4) risks related to increased regulatory oversight and new compliance obligations, which risks may be further exacerbated as BlackRock and the Aladdin platform continue to grow. A prolonged global failure of cloud services could also impact BlackRock's other systems. Failures by BlackRock to manage these risks, and/or risks associated with future technology partnerships, may result in escalating costs, financial loss, client dissatisfaction or attrition, regulatory fines and/or sanctions, reputational harm or legal liability, which, in turn, may cause BlackRock's AUM, revenue and earnings to decline.

Disruption to the operations of third parties whose functions are integral to BlackRock's exchange-traded fund ("ETF") platform may adversely affect the prices at which ETFs trade, particularly during periods of market volatility.

BlackRock is the largest provider of ETFs globally. Shares of ETFs trade on stock exchanges at prices at, above or below the ETF's most recent net asset value ("NAV"). The NAV of an ETF is calculated at least once daily, generally at the end of each business day, and fluctuates with changes in the market value of the ETF's holdings. The trading price of the ETF's shares fluctuates continuously throughout trading hours. The creation/redemption feature and arbitrage mechanism of an ETF are designed to make it more likely that the ETF's shares normally will trade at prices close to the NAV. Notwithstanding these features, exchange prices have in the past deviated measurably from the NAV of certain ETFs and may under certain circumstances do so in the future. ETF market prices are subject to numerous potential risks, including trading halts invoked by a stock exchange, and the inability or unwillingness of market makers, authorized participants, settlement systems or other market participants to perform functions necessary for an ETF's arbitrage mechanism to function effectively. These risks may be heightened as a result of significant market volatility, the growth of the ETF industry combined with increased market activity, as well as the complexity associated with certain products or asset classes, such as digital assets. Moreover, if market events lead to incidents where ETFs trade at prices that deviate meaningfully from an ETF's NAV, or trading halts are invoked by the relevant stock exchange or market, investors may lose confidence in ETF products and redeem their holdings, which may cause BlackRock's AUM, revenue and earnings to decline.

LEGAL, REGULATORY AND REPUTATIONAL RISKS

BlackRock is subject to extensive regulation around the world, which increases its cost of doing business.

BlackRock's business is subject to extensive regulation around the world. These regulations subject BlackRock's business activities to an array of increasingly detailed operational requirements, compliance with which is costly and complex.

In addition, many of BlackRock's legal entities are subject to laws and regulations aimed at preventing corruption, money laundering, inappropriate employment practices, illegal payments and engaging in business activities with certain individuals, countries or groups, including but not limited to the US Foreign Corrupt Practices Act, the USA PATRIOT Act, the Bank Secrecy Act, the EU Anti-Money Laundering Directives, the Money Laundering, Terrorist Financing and Transfer of Funds Regulations 2017, the UK Bribery Act, sanctions imposed by the US Treasury's Office of Foreign Assets Control, the United Nations and the EU and its member states, as well as those imposed by other countries in which BlackRock operates, such as His Majesty's Treasury's ("HMT") Office of Financial Sanctions Implementation.

BlackRock is also subject to certain risk retention rules and regulation, as well as regulatory capital requirements, which require the Company to maintain capital to support certain of its businesses. Furthermore, many jurisdictions in which BlackRock operates have laws and regulations relating to data privacy, cybersecurity and protection of personal information, including the GDPR and UK GDPR, which impose stringent data protection rules for individuals within the European Economic Area ("EEA") and UK, respectively, and for personal data exported outside the EEA and UK.

BlackRock is additionally subject to scrutiny from various government agencies that focus on antitrust and competition laws and regulations within the US and internationally, including in connection with merger control proceedings and proposed investments. Any determination of a failure to comply with any such laws or regulations could result in fines and/or sanctions against the Company, as well as reputational harm. Moreover, to the extent that these laws and regulations become more stringent, or if BlackRock is required to hold increased levels of capital to support its businesses, the Company's financial performance or plans for growth may be adversely impacted.

BlackRock may also be adversely affected by a failure to comply with existing laws and regulations or by changes in the interpretation or enforcement of such laws and regulations, including those discussed above. Challenges associated with interpreting regulations issued in numerous countries in a globally consistent manner may add to such risks if regulators in different jurisdictions have inconsistent views or provide only limited regulatory guidance. In particular, violation of applicable laws or regulations could result in fines and/or sanctions, temporary or permanent prohibition of certain activities, reputational harm and related client terminations, suspensions of employees or revocation of their licenses, suspension or termination of investment adviser, broker-dealer or other registrations, or suspension or termination of BTC's bank charter or other sanctions, which could have a material adverse effect on BlackRock's reputation or business and may cause the Company's AUM, revenue and earnings to decline. For a more extensive discussion of the laws, regulations and regulators to which BlackRock is subject and regulated by, see Item 1, *Business – Regulation*.

New regulations informed by global standard setters and/or developed by various national authorities may expose BlackRock to increasing regulatory scrutiny and compliance costs in the jurisdictions in which it operates.

Policymaking workstreams focused on the financial services sector led by global standard setters, such as the Financial Stability Board ("FSB") and International Organization of Securities Commissions ("IOSCO"), may lead to or inform new regulations in multiple jurisdictions in which BlackRock operates. Such workstreams have focused on areas such as money market funds ("MMFs"), open-ended funds ("OEFs") and sustainability regulations. BlackRock is, and may become, subject to increasing regulation in these areas, see Item 1, *Business – Regulation*, including:

- *Macroprudential Policies for Asset Managers:* Concerns about liquidity and leverage risks in the asset management industry and wider market-based finance sector have prompted a broad review of existing regulations globally, including an assessment of the adequacy of certain structural market components in mitigating risks, by the FSB, IOSCO, the US Securities and Exchange Commission (the "SEC") and the Financial Stability Oversight Council ("FSOC"). In November 2022, the SEC proposed amendments to rules governing OEF liquidity risk management. The EU launched a consultation on macroprudential policies in 2024, including enhanced requirements for liquidity management tools, which may lead to new restrictions on management of OEFs. The UK proposed introducing liquidity facilities to certain asset owners, which could result in regulatory burdens on asset managers. If any of these regulatory or policy actions result in broad application of macroprudential tools to OEFs or require changes to structural features of certain OEFs, it could limit BlackRock's ability to offer products to certain clients and/or result in clients altering their investment strategies or allocations in a manner that is adverse to BlackRock.

- *Global MMF Reforms:* Following the market events of March 2020, US, UK and EU authorities initiated a review of existing regulatory frameworks with the aim of improving the resilience of MMFs in market downturns. In the US, the SEC adopted changes to Rule 2a-7, the primary rule under the Investment Company Act of 1940 governing MMFs, including changes to required liquidity levels and requiring mandatory liquidity fees under certain circumstances. The UK released a consultation in December 2023 indicating their intent to change regulatory requirements for MMFs domiciled in the UK, including material increases in required liquidity levels. The EU consultation on macroprudential policies mentioned above may also result in changes to the regulations of EU-domiciled MMFs. Depending on the terms of the final UK and EU reforms, certain of BlackRock's MMF products could be adversely impacted.
- *Sustainability:* Sustainability has been the subject of regulatory focus across jurisdictions. Disclosure standards aligned with the International Sustainability Standards Board's ("ISSB") inaugural disclosure standards have been adopted by several national regulators, including in Hong Kong, Singapore and Australia, while others are expected to propose ISSB-aligned standards, such as the UK, Canada and Japan. However in the US, final rules issued by the SEC requiring corporate issuers to make climate-related disclosures in their periodic reports are pending litigation, and as of February 2025, the SEC was revisiting its litigation position. The SEC has previously proposed rules requiring enhanced ESG disclosures by investment companies and investment advisers in fund and adviser filings, including disclosures on ESG strategies and how ESG factors are considered and GHG emissions disclosure by certain environmentally focused funds. It also increased scrutiny of disclosure and compliance issues relating to investment advisers' and funds' ESG strategies, policies and procedures. In addition, the US Department of Labor ("DOL") issued final rules clarifying that Employee Retirement Income Security Act of 1974, as amended ("ERISA") plan fiduciaries can consider the economic effects of ESG factors for purposes of investing ERISA plan assets and exercising voting rights with respect to plan investments. In 2023, California passed several laws requiring certain companies doing business in California to publish certain types of climate-related disclosures, and other states may adopt similar laws.

  The EU has enacted numerous sustainability regulations, including (1) the Sustainable Finance Disclosure Regulation, requiring sustainability-related disclosures by financial market participants; (2) the EU Taxonomy Regulation, requiring asset managers to report against an EU-wide taxonomy of environmentally sustainable activities and make detailed disclosures relating to ESG characteristics of funds and portfolios; (3) the Corporate Sustainability Reporting Directive ("CSRD"), requiring enhanced sustainability reporting for EU-based and EU-listed companies, and from 2028, for a wider group of global companies; and (4) the Corporate Sustainability Due Diligence Directive ("CSDDD"), requiring in-scope EU companies and certain non-EU companies to manage actual or potential adverse impacts of their activities and their supply chains on human rights and environmental matters. The European Commission ("EC") is reviewing and may amend aspects of the CSRD, CSDDD and EU Taxonomy Regulation. Meanwhile, the UK continues to work on implementation of its Sustainability Disclosure Requirements.

  The EU and the UK Financial Conduct Authority ("FCA") have issued rules and guidelines on the use of ESG or sustainability related terms in fund names. In addition, the EU adopted regulations on ESG rating providers applicable in mid-2026 while the UK is expected to propose new legislation on ESG rating providers. Japan and Singapore have published codes of conduct for ESG data and rating providers, with Hong Kong considering a similar approach, while India introduced a regulatory framework for ESG rating providers in July 2023.

  As jurisdictions continue to develop and implement sustainability regulations and litigation challenging such regulations increases, BlackRock faces greater fragmentation risk related to local application of regulations, resulting in complex and conflicting compliance obligations and legal and regulatory uncertainty.

Global regulatory reforms could require BlackRock to alter its future business or operating activities, which could be time-consuming and increase costs, including costs related to regulatory compliance, result in litigation, impede the Company's growth and cause its AUM, revenue and earnings to decline. Regulatory reform may also impact BlackRock's clients, which could cause them to change their investment strategies or allocations in manners that may be adverse to BlackRock.

Regulatory reforms in the US expose BlackRock to increasing regulatory scrutiny, as well as regulatory uncertainty.

In recent years, a number of regulatory reforms have been proposed or fully or partially implemented in the US, and the level of regulatory scrutiny to which BlackRock is subject has increased. BlackRock, as well as its clients, vendors and distributors, have expended resources and altered certain of their business or operating activities to prepare for, address and meet the requirements that such regulatory reforms impose. New or proposed changes to laws, regulations, policies, initiatives and other government actions may be difficult to anticipate, which provides additional uncertainty and may heighten the Company's risks related to such actions. While BlackRock is, and may become, subject to numerous reform initiatives in the US, see Item 1, *Business – Regulation*, key regulatory reforms that may impact the Company include:

- *Antitrust Rules and Guidance:* In October 2024, the Federal Trade Commission ("FTC"), with concurrence from the Antitrust Division of the Department of Justice (the "DOJ") approved amendments to rules enacted under the Hart-Scott-Rodino Antitrust Improvements Act of 1976 ("HSR") that require parties in certain transactions to provide the FTC and DOJ prior notice and observe a waiting period before consummation of such transactions. The final amendments significantly expanded the information required to be reported and documents to be submitted in connection with an HSR filing, which will likely substantially increase any pre-merger notification expenses and may delay transactions. In December 2023, the FTC and DOJ also jointly issued new merger guidelines, which could impact (1) the ability of the Company to expand its services through strategic investments or acquisitions and (2) funds that engage in transactions reportable under HSR.
- *Designation as a Systemically Important Financial Institution ("SIFI"):* The FSOC has the authority to designate nonbank financial institutions as SIFIs in the US under the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010. In November 2023, the FSOC finalized amendments to its existing interpretive guidance to remove the prioritization of an activities-based approach over an entity-specific approach to designation in connection with addressing potential risks to financial stability, although the amendment clarified that the FSOC retained the ability to use an activities-based approach when appropriate. If BlackRock is designated as a SIFI, it could become subject to enhanced regulatory and capital requirements and direct supervision by the Federal Reserve.
- *SEC Rules Governing Security-Based Swaps:* In 2021, the SEC proposed rules in connection with security-based swaps ("SBS") transactions to require public reporting of large SBS positions, which, if adopted as proposed, may affect the types of transactions BlackRock may choose to execute in SBS or other SBS-related assets, introduce or increase costs relating to such transactions, and impact the liquidity in the SBS markets in which BlackRock transacts.

- *SEC Rules on Form PF*: In 2023 and 2024, the SEC adopted new rules and amendments to Form PF for registered investment advisers requiring new disclosures, filing obligations and enhanced reporting. Implementing these rules and amendments may significantly increase BlackRock's reporting, disclosure and compliance obligations and create operational complexity for BlackRock's alternatives products.
- *US DOL Fiduciary Rule:* The US DOL adopted new regulation redefining the meaning of "investment advice fiduciary" under ERISA as well as amendments to several prohibited transaction exemptions applicable to investment advice fiduciaries, which would substantially expand when a person would be considered a fiduciary subject to ERISA and could require BlackRock to revise a number of its distribution relationships, create compliance and operational challenges for BlackRock and its distribution partners, and limit BlackRock's ability to provide certain services to applicable clients. In July 2024, federal courts issued stays on the regulation and implementation has been postponed pending further notice.
- *SEC US Treasury Clearing Mandate:* In December 2023, the SEC adopted rules mandating central clearing of US Treasury repurchases and certain other Treasury transactions. The rules require many market participants, including a large number of BlackRock funds and accounts, to clear Treasury repurchase transactions and potentially certain cash Treasury securities transactions through a clearing agency registered with the SEC, which could increase transaction costs for BlackRock's clients.
- *Proposed Rules on Equity Market Structure:* In 2023, the SEC proposed equity market structure reforms that would significantly change how national market system ("NMS") stock orders are priced, executed and reported. The reforms include: (1) a requirement for certain retail orders to be subject to order-by-order competition, (2) a best execution rule and (3) an adjustment to the tick sizes at which NMS stocks can be quoted or traded. In 2024, the SEC adopted the rule adjusting NMS tick sizes. If the other proposed rules are enacted as proposed, their collective impact may adversely affect market efficiency and execution costs, which would result in negative effects for BlackRock's business and clients.
- *SEC Rules on Short Sales and Reporting of Securities Loans:* In 2023, the SEC adopted a new rule requiring certain institutional managers to report short positions and activity to the SEC for publication on an aggregate basis, which could impact investment strategies and result in greater operational burdens and cost for BlackRock. The SEC also adopted a new rule requiring certain persons to report information on securities loan transactions to a registered national securities association which will then publish certain information. The rule may increase BlackRock's operational burdens and costs.
- *SEC Predictive Data Analytics Rules:* The SEC proposed new rules in 2023 that would require broker-dealers and investment advisers, when engaging or communicating with investors using predictive data analytics ("PDA") and PDA-like technologies, to evaluate such technologies for conflicts of interest and, where identified, eliminate or neutralize the conflict of interest. If adopted as proposed, the rules could encompass a wide range of forward-looking uses of technology applications and impose significant operational burdens and costs.
- *Financial Crimes Enforcement Network Rule for Registered Investment Advisers:* In August 2024, the Financial Crime Enforcement Network ("FinCEN") issued a final rule which will require registered investment advisers to adopt new anti-money laundering requirements beginning in 2026. Under the rule, registered investment advisers will be required to establish written risk-based anti-money laundering programs and report suspicious activity to FinCEN under the Bank Secrecy Act of 1970 (the "Bank Secrecy Act"), as well as comply with Bank Secrecy Act reporting and recordkeeping requirements, which may increase BlackRock's compliance burdens and costs.
- *SEC Rulemakings for US Registered Funds and Investment Advisers:* The SEC has engaged in various initiatives and reviews impacting regulatory structure governing the asset management industry and registered investment companies. For example, the SEC adopted rules requiring certain funds to provide tailored fund shareholder reports, adopted final amendments to the rule governing fund names, expanding the scope of the rule to fund names including growth, value, ESG or similar terms, and proposed rules governing outsourcing of certain functions by investment advisers to service providers.

Regulatory reforms in the US could require BlackRock to alter its future business or operating activities, which could be time-consuming and costly, increase regulatory compliance costs, result in litigation, impede the Company's growth and cause its AUM, revenue and earnings to decline. Regulatory reform may also impact BlackRock's clients, which could cause them to change their investment strategies or allocations in manners that may be adverse to BlackRock.

International regulatory reforms expose BlackRock to increasing regulatory scrutiny, as well as regulatory uncertainty.

BlackRock's business and operating activities are subject to increasing regulatory oversight outside of the US and the Company may be affected by several proposed or implemented reform initiatives in the EMEA and the Asia-Pacific regions, as well as volatility associated with international regulatory uncertainty. While BlackRock is, and may become, subject to numerous reform initiatives internationally, see Item 1, *Business – Regulation*, key reforms in these regions include:

European Union

- *Enhanced Regulatory Scrutiny of Technology Service Providers to Financial Services Firms:* The EU's Digital Operational Resilience Act ("DORA"), which focuses on direct regulation of providers and users of technology and data services, became applicable beginning in January 2025. DORA, among other things: (1) introduces additional governance, risk management, incident reporting, resilience testing and information sharing requirements to several of BlackRock's European entities and certain Aladdin clients; and (2) may potentially subject Aladdin to additional oversight. The European Supervisory Authorities will use data collected under DORA to assess which third party suppliers should be designated as critical to the EU financial system and become subject to further regulatory oversight. In 2024, the UK issued final policies regulating services provided by certain third parties designated by HMT as "critical" to the financial sector, which became effective in January 2025. Entities designated as "critical" will be required to provide additional information to financial regulators, engage in resilience testing and report major incidents like cyber-attacks, natural disasters and power outages.
- *Retail Investment Strategy:* The EU continues to consider a proposed Retail Investment Strategy package of amendments intended to enhance protections for retail investors. When enacted, these changes could impact clients' product preferences and may increase costs for BlackRock in European markets due to additional requirements on distributors and product providers.

- *EMIR 3.0:* The EU legislative package known as "EMIR 3.0" introduces key changes to clearing, margining and reporting requirements in the European Market Infrastructure Regulation (EMIR), including: (1) a requirement to hold an "active account" with an EU central counterparty for clearing certain euro-denominated instruments; (2) new reporting requirements for cleared trades; (3) revised clearing thresholds for financial and non-financial counterparties; and (4) amendments related to clearing to the UCITS directive. EMIR 3.0 is expected to impact EU counterparties as well as UK and non-EU entities trading with EU firms, and the collective impact of the package may increase operational complexity, necessitate a reassessment of clearing and trading strategies, and lead to higher transaction costs for BlackRock and its clients.

United Kingdom

- *FSMA 2023:* The Financial Services and Markets Act 2023 ("FSMA") reflects significant changes to the UK framework for financial services regulation, including changes that: (1) revoke, amend or retain EU law on financial services regulation, (2) amend the UK Markets in Financial Instruments Directive and Markets in Financial Instruments Regulation frameworks, (3) establish a new designated activities regime and (4) reform the financial promotion regime for unauthorized firms. Other reforms building upon the FSMA and potentially impacting the asset management sector include: (1) replacement of the packaged retail and insurance based investment products ("PRIIPs") Regulation; (2) review of the UK's green finance strategy, including regulation of ESG data providers, UK taxonomy and disclosure requirements; (3) review of governance through the Senior Managers and Certification Regime; (4) repeal of EU legislation on the European Long-Term Investment Fund; (5) market infrastructure reforms, including transition to T+1 settlement; (6) reassessment of the boundary between investment advice and financial guidance; (7) a new UK cryptoasset regime; and (8) continuing reforms to the UK listing regime.
- *Overseas Fund Regime ("OFR"):* OFR, the simplified regime through which non-UK funds can register with the FCA to be marketed to UK retail investors, was enacted in February 2022 and continues to be implemented through 2025. For certain types of funds, OFR requires consumer protection regimes in EU countries where such BlackRock funds are domiciled to be found equivalent to the UK's regime in order to market the funds in the UK.
- *Conduct Regulation:* The FCA continues to focus on conduct regulation, including the implementation of the Consumer Duty by all asset management firms, including BlackRock's UK subsidiaries. The Consumer Duty rules require firms to act to deliver good outcomes for retail customers in their manufacture and distribution of products and services, in respect of price and value, consumer understanding and consumer support. Any failure to meet the FCA's regulatory expectations could expose BlackRock to regulatory sanctions and increased reputational risk.
- *UK Stewardship Code Review:* In 2024, the UK Financial Reporting Council released a consultation on reforms to the UK Stewardship Code, including tailored reporting requirements for proxy advisers and investment consultants, which may impact BlackRock's activities on behalf of its clients.

Asia-Pacific

- *Regulatory Environment in China:* The Company's operations in China are subject to a number of regulatory risks, including an evolving regulatory environment and complex data security and data transfer regulations. These factors may increase compliance risk and costs, limit the Company's ability to source and execute new investment opportunities and lead to impairment losses on its investments. Restrictions on transfers of certain types of onshore data of the Company's Chinese entities to offshore entities also may limit BlackRock's ability to aggregate, report and monitor such data on its global platform. In addition, a number of regulators in China have jurisdiction over BlackRock's business operations, increasing operational and regulatory engagement complexity. These risks may be further heightened by additional scrutiny by Chinese regulators of certain sectors, such as technology and other industries that might be deemed to be of national importance.

International regulatory reforms could require BlackRock to alter its future business or operating activities, which could be time-consuming and costly, increase regulatory compliance costs, result in litigation, impede the Company's growth and cause its AUM, revenue and earnings to decline. Regulatory reform may also impact BlackRock's clients, which could cause them to change their investment strategies or allocations in manners that may be adverse to BlackRock.

Legal proceedings may cause the Company's AUM, revenue and earnings to decline.

BlackRock is subject to a number of sources of potential legal liability and the Company, certain of the investment funds it manages and certain of its subsidiaries and employees have been named as defendants in various legal actions, including arbitrations, class actions and other litigation arising in connection with BlackRock's activities. Certain of BlackRock's subsidiaries and employees are also subject to periodic examination, special inquiries and potential proceedings by regulatory authorities, including the SEC, Office of the Comptroller of the Currency ("OCC"), DOL, Commodity Futures Trading Commission, the FCA, Commission de Surveillance du Secteur Financial and Federal Reserve. Similarly, from time to time, BlackRock receives subpoenas or other requests for information from various US state and federal as well as non-US governmental and regulatory authorities in connection with certain industry-wide, company-specific or other investigations, proceedings or litigations. Governmental or regulatory authorities have and could in the future institute proceedings and/or seek to impose sanctions for violations. Any such action may also result in litigation by investors in BlackRock's funds, other BlackRock clients or BlackRock's shareholders. Such legal proceedings could harm the Company's reputation and may cause its AUM, revenue and earnings to decline, potentially harm the investment returns of the applicable fund, or result in the Company being liable for damages.

In addition, when clients retain BlackRock to manage their assets or provide them with products or services, they typically specify contractual requirements or guidelines that BlackRock must observe in the provision of its services. A failure to comply with these guidelines or requirements could expose BlackRock to lawsuits, harm its reputation or cause clients to withdraw assets or terminate contracts.

BlackRock faces increasing focus from regulators, officials, clients and other stakeholders regarding environmental and social-related matters, which may adversely impact its reputation and business.

BlackRock faces increasing focus from regulators, officials, clients and other stakeholders regarding environmental and social-related matters. BlackRock offers choice to its clients who have a variety of goals and preferences, including those who want to increase their exposure to the low-carbon transition and those who choose not to invest in products or strategies with sustainable investment objectives. BlackRock is subject to competing demands from different stakeholder groups with divergent views on environmental and social-related matters, including in countries in which BlackRock operates and invests, as well as in countries, states and localities where BlackRock serves public sector clients. This divergence has and continues to increase the risk that any perceived or actual action or lack thereof by BlackRock on such matters on behalf of its clients will be viewed differently by various stakeholders and adversely impact BlackRock's reputation and business, including through withdrawals, redemptions, terminations or decisions not to commit or invest new capital by clients, as well as legal and governmental action and scrutiny. Some US states and state officials have adopted or proposed legislation or otherwise raised concerns about BlackRock's business practices. In certain instances, this has led them to take official positions restricting or prohibiting state government entities from doing certain business with entities identified by the state as "boycotting" or "discriminating" against particular industries or considering environmental and social factors in their investment processes and proxy voting. Other states and localities may adopt similar legislation or other environmental and social-related laws and positions that adversely impact BlackRock's business. BlackRock has previously communicated and may communicate certain initiatives and goals for its corporate activities related to the environment, human capital management, and other environmental and social-related matters. BlackRock faces criticism for the scope or nature of certain initiatives or goals and may face additional criticism for revisions thereto. If BlackRock is not able to successfully manage environmental and social-related expectations across varied stakeholder interests, it may adversely affect BlackRock's reputation, ability to attract and retain clients, employees, shareholders and business partners or result in litigation, legal or governmental action, which may cause its AUM, revenue and earnings to decline.

Damage to BlackRock's reputation may harm its business.

BlackRock's reputation is critical to its relationships with its clients, employees, shareholders and business partners. BlackRock's reputation may be harmed by, among other factors, regulatory, enforcement or other governmental actions, technology or operational failures, poor investment performance, ineffective management or monitoring of key third-party relationships, ransomware or other cybersecurity incidents, privacy incidents, employee errors or misconduct, failures to manage risks or conflicts of interest, or legal actions related to BlackRock or its products and services. In addition, BlackRock's business, scale and investments subject it to significant media coverage and increasing attention from a broad range of stakeholders. This heightened scrutiny has resulted in negative publicity and adverse actions for BlackRock and may continue to do so in the future. Any perceived or actual action or lack thereof, or perceived lack of transparency, by BlackRock on matters subject to scrutiny, such as environmental and social matters, may be viewed differently by various stakeholders and adversely impact BlackRock's reputation and business, including through redemptions or terminations by clients, and legal and governmental action and scrutiny. BlackRock's global presence and investments on behalf of its clients around the world could also lead to heightened scrutiny and criticism in an increasingly fragmented geopolitical landscape. For example, BlackRock has received criticism from some stakeholders because of its operations and investments in certain countries on behalf of clients, including China. These criticisms could adversely impact BlackRock's reputation and business. In addition, the increasing popularity of social media and non-mainstream Internet news sources may lead to faster and wider dissemination of adverse publicity, inaccurate information or disinformation campaigns about BlackRock, making effective remediation more difficult. Damage to BlackRock's reputation may impact BlackRock's ability to attract and retain clients, employees, shareholders and business partners, which may cause its AUM, revenue and earnings to decline.

A failure to effectively manage potential conflicts of interest could result in litigation or enforcement actions and/or adversely affect BlackRock's business and reputation, which may cause BlackRock's AUM, revenue and earnings to decline.

As a global investment management firm that provides investment and technology services to a diverse range of clients, the Company must routinely address and manage conflicts of interest, as well as the perception of conflicts of interest, between itself and its clients, employees or vendors. While BlackRock has policies, controls and disclosure protocols in place to manage and address potential conflicts of interest, identifying and mitigating conflicts of interest can be complex and is the subject of increasing regulatory and media scrutiny. It is possible that actual, potential or perceived conflicts could give rise to investor or client dissatisfaction, adverse publicity, litigation or enforcement actions. In particular, BlackRock's broad range of investment, advisory and technology offerings, and its focus on providing clients with whole portfolio solutions, may result in clients working with multiple BlackRock businesses and/or BlackRock being engaged by institutions that have a nexus to industries or jurisdictions in which BlackRock operates, which may increase the potential for actual or perceived conflicts of interest and improper information sharing. To the extent that BlackRock fails, or appears to fail, to deal appropriately with any conflict of interest, it may face adverse publicity, reputational damage, litigation, regulatory proceedings, client attrition, penalties, fines and/or sanctions, any of which may cause BlackRock's AUM, revenue and earnings to decline.

A subsidiary of BlackRock is subject to US banking regulations that may limit its business activities.

BlackRock's trust bank subsidiary, which is a national banking association chartered by the OCC, is subject to OCC regulation and capital requirements that may limit its business activities. The OCC has broad supervisory and enforcement authority over BlackRock's trust bank. Having a subsidiary subject to banking regulation may put BlackRock at a competitive disadvantage because certain of its competitors are not subject to the limitations imposed by such regulation.

The implications of complying with threshold limits and/or any failure to comply with ownership reporting requirements could result in harm to BlackRock's reputation, impact the performance of certain BlackRock funds and may cause its AUM, revenue and earnings to decline.

Of note among the various regulations to which BlackRock is subject are the extensive and increasingly stringent regulatory reporting requirements that necessitate the monitoring and reporting of issuer exposure levels (thresholds) across the holdings of managed funds and accounts and those of the Company. The specific triggers and the reporting methods that these threshold filings entail vary significantly by regulator and across jurisdictions. BlackRock continues to invest in technology, training and its employees to further enhance its monitoring and reporting functions. Despite these investments, the complexity of the various threshold reporting requirements combined with the breadth of the assets managed by the Company and high volume of securities trading have caused errors and omissions to occur in the past and pose a risk that errors or omissions may occur in the future. Any such errors may expose BlackRock to monetary penalties or other sanctions, which could have an adverse effect on BlackRock's reputation and may cause its AUM, revenue and earnings to decline.

Moreover, as BlackRock's business grows it is becoming subject to a greater number of regulatory, industry-level or issuer-specific threshold limits and scrutiny that may prevent BlackRock from holding positions in certain equity securities, securities convertible into equity securities or futures contracts in excess of certain thresholds. In addition, regulators are reviewing and considering changes to their regulatory frameworks on threshold limits and ownership reporting requirements. These changes may result in significant operational impacts and costs to BlackRock and its products depending on their scope. Although BlackRock is actively engaged in regulatory, issuer-specific and structural initiatives to create additional investment capacity, threshold limits may nonetheless prevent the purchase of certain securities which may, in turn, impact the performance of certain BlackRock index funds by increasing tracking error relative to the funds' benchmarks, impact the performance of certain BlackRock actively managed funds by preventing them from taking advantage of alpha generating opportunities, and impede the Company's growth.

BlackRock has been the subject of commentary citing concerns about the scale of its index investing business, as well as purported competition issues relating to the common ownership theory.

As a leader in the index investing and asset management industry, BlackRock has been the subject of commentary citing concerns about the growth of index investing and concentrated proxy voting power. Some commentators have argued that continued growth of index funds has the potential to impact stock market competitiveness by exacerbating stock price moves and market volatility. Some commentators, regulators and lawmakers have also argued that index managers have accumulated outsized influence through the proxy voting power their clients have assigned them. Some have proposed limitations on the ability of index fund managers to vote or engage on behalf of their clients, or indicated that voting and engagement on certain topics should trigger changes in regulatory status. Additional commentary focuses on the common ownership theory, an academic theory stating that minority ownership of multiple companies within a single industry by the same investor leads to anticompetitive effects. This theory purports to link aggregated equity positions in certain industries with higher consumer prices and executive compensation and lower wages and employment rates, among other things. In 2021, the FTC identified common ownership as a key enforcement area and passed a resolution empowering individual commissioners to investigate shareholder conduct in connection with common ownership. In 2023, the FTC and DOJ released new merger guidelines recognizing that common ownership may reduce competitive incentives and in 2024, the FTC and DOJ submitted a joint comment letter to the Federal Energy Regulatory Commission ("FERC") encouraging FERC to consider common ownership as a relevant factor in updating regulatory relief available to asset managers. Common ownership may be given greater consideration in regulatory investigations, studies, rule proposals, policy decisions and/or the scrutiny of mergers and acquisitions. The debate on common ownership has been on the agenda of lawmakers, policymakers and competition regulators globally, and common ownership may continue to be a consideration for the EC and European Parliament, among others. There is substantial literature casting doubt on the assumptions, data, methodology and conclusions associated with the common ownership theory, including research conducted by staff of regulatory agencies. Competition regulators, including at the FTC and UK Competition & Markets Authority ("CMA"), have acknowledged that the debate around the theory remains unsettled. Nevertheless, some commentators have proposed remedies, including limits on the ownership stakes of common owners that, if enacted into policy, could have a negative impact on the capital markets, as well as increase costs and limit the availability of products for investors. Such policy solutions could, in turn, adversely affect BlackRock.

New tax legislation or changes to existing US and non-US tax laws, treaties and regulations or challenges to BlackRock's historical taxation practices may adversely affect BlackRock's effective tax rate, business and overall financial condition.

BlackRock's businesses may be directly or indirectly affected by tax legislation and regulation, or the modification of existing tax laws, by US or non-US tax authorities. Legislation at both the US federal and state level has been previously proposed to enact a financial transaction tax ("FTT") on stocks, bonds and a broad range of financial instruments and derivative transactions. In the EU, certain Member States have also enacted similar FTTs and the EC has proposed legislation to harmonize these taxes and provide for the adoption of EU-level legislation applicable to some (but not all) EU Member States. If enacted as proposed, FTTs could have an adverse effect on BlackRock's financial results and clients' performance results.

The Organisation for Economic Cooperation and Development ("OECD") has proposed certain international tax reforms, which, among other things, would (1) shift taxing rights to the jurisdiction of the consumer ("Pillar One") and (2) establish a global minimum tax for multinational companies of 15% ("Pillar Two"). In response, EU member states and several other countries, including the UK, have since adopted laws implementing the OECD's minimum tax rules under Pillar Two, effective starting in 2024. As a result of these developments, the tax laws of certain countries in which BlackRock does business have changed and may continue to change, and any such changes could increase its tax liabilities. The Company is continuing to monitor legislative developments and evaluate the potential impact of the Pillar Two Framework on future periods.

The application of tax regulations involves numerous uncertainties, and in the normal course of business US and non-US tax authorities may review and challenge tax positions adopted by BlackRock. These challenges may result in adjustments to, or impact the timing or amount of, taxable income, deductions or other tax allocations, which may adversely affect BlackRock's effective tax rate and overall financial condition. Similarly, the Company manages assets in products and accounts that have investment objectives which may conform to tax positions adopted by BlackRock or to specific tax rules. To the extent there are changes in tax law or policy, or regulatory challenges to tax positions adopted by BlackRock, the value or attractiveness of such investments may be diminished and BlackRock may suffer financial or reputational harm.

Item 1B. Unresolved Staff Comments

The Company has no unresolved comments from the Securities and Exchange Commission ("SEC") staff relating to BlackRock's periodic or current reports filed with the SEC pursuant to the Exchange Act.

Item 1C. Cybersecurity

CYBERSECURITY RISK MANAGEMENT AND STRATEGY

BlackRock recognizes the importance of identifying, assessing, and managing material risks associated with cybersecurity threats. Cybersecurity represents an important component of the Company's approach to enterprise risk management ("ERM"). The Company leverages a multi-lines-of-defense model with cybersecurity operational processes executed by global information security and other teams across the firm and dedicated internal audit technology and technology risk management ("TRM") teams that independently review technology risks. The Company's cybersecurity program is fully integrated into its ERM framework and is aligned with recognized frameworks, such as NIST CSF, FFIEC CAT, FedRAMP, SOC 1/2, ISO 27001/2 and others. BlackRock aims to inform and continuously improve its cybersecurity program through engagement with regulatory, client, insurer, vendor, partner, peer, government and industry organizations and associations, as well as external audit, technology risk, information security and other assessments.

BlackRock seeks to address cybersecurity risks through a global, multilayered strategy of control programs that are designed to preserve the confidentiality, integrity and availability of the information that BlackRock collects and stores by identifying, preventing and mitigating cybersecurity threats and incidents. As one of the critical elements of the Company's overall ERM framework, BlackRock's cybersecurity program is focused on the following key areas:

- **Governance:** As discussed in more detail under the heading "Cybersecurity Governance" below, the Board's oversight of cybersecurity risk management is supported by the Risk Committee, which regularly interacts with the Company's risk management function, the Company's Chief Risk Officer ("CRO") and Chief Information Security Officer ("CISO"), along with other members of management. In addition, technology and cybersecurity risks are formally overseen by a dedicated management risk governance committee, the Technology Risk and Cybersecurity Committee ("TRCC"), which is a sub-committee of the firmwide Enterprise Risk Committee ("ERC").
- **Cross-Functional Approach:** The Company has implemented a global, cross-functional approach to identifying, preventing, and mitigating cybersecurity threats and incidents, while also implementing layered preventative, detective, reactive and recovery controls to identify and manage cybersecurity risks.
- **Safeguards:** The Company deploys a range of people, process and technical controls that are designed to protect the Company's information systems from cybersecurity threats, which may include, among others: physical security controls, perimeter controls, technical assessments, firewalls, network segregation, intrusion detection and prevention, tabletop exercises, ongoing vulnerability and patch management, vendor due diligence, multi-factor authentication, device encryption, application security, code testing and penetration testing; endpoint security, including anti-malware protection, threat intel and response, managed detection and response, security configuration management, portable storage device lockdown, and restricted administrative privileges, employee awareness, training, and phishing testing, data loss prevention program and monitoring, information security incident reporting and monitoring; and layered and comprehensive access controls.
- **Incident Response and Recovery Planning:** The Company maintains incident response and recovery plans that address the Company's response to a cybersecurity incident, including processes designed to assess, escalate, contain, investigate and remediate an incident, as well as to comply with applicable legal obligations and mitigate potential reputational damage. These plans are evaluated on a periodic basis.
- **Third-Party Risk Management:** The Company maintains a risk-based approach to identifying and overseeing cybersecurity risks presented by third parties, including vendors, service providers, counterparties and clients, as well as the systems of third parties that could significantly and adversely impact the Company's business in the event of a cybersecurity incident affecting those third-party systems. Operational incidents can arise as a result of failures by third parties with which the Company does business, such as failures by internet, communication technology and cloud service providers or other vendors to adequately follow processes and procedures, safeguard their systems, or prevent system disruptions or cyber-attacks. Third-party risks are included within BlackRock's ERM framework, and risk identification and mitigation are supported by the Company's cybersecurity program. BlackRock also performs due diligence on certain third parties and monitors cybersecurity threats and risks identified through such diligence.
- **Education and Awareness:** The Company's employees and contractors are required to complete annual information security training to equip them with effective tools to address cybersecurity threats, and receive communications on the Company's evolving information security policies and procedures.

The Company's global information security team, in collaboration with the technology risk and internal audit teams, engages in the periodic assessment and testing of the Company's cyber risks and cybersecurity program. These efforts may include a wide range of activities, including audits, assessments, wargames and "tabletop" exercises, threat modeling, vulnerability testing and other exercises focused on evaluating the effectiveness of the Company's cybersecurity measures and planning. BlackRock also participates in financial services industry and government forums in an effort to improve both internal and sector cybersecurity defense. The Company regularly engages third parties and advisors to assess its cybersecurity control environment. The results of certain program and control assessments are reported to the Risk Committee, and BlackRock adjusts its cybersecurity program as appropriate based on the information provided by these assessments.

As of December 31, 2024, BlackRock is not aware of any cybersecurity risks that have materially affected or are reasonably likely to materially affect BlackRock's business strategy, results of operations, or financial condition. For additional information on whether and how risks from cybersecurity threats are reasonably likely to materially affect BlackRock, see "A cyber-attack or a failure to implement effective information and cybersecurity policies, procedures and capabilities could disrupt operations and lead to financial losses and reputational harm, which may cause BlackRock's AUM, revenue and earnings to decline." under Part I, Item 1A, *Risk Factors* herein.

CYBERSECURITY GOVERNANCE

BlackRock's Board of Directors is actively engaged in the oversight of BlackRock's risk management program. The Risk Committee assists the Board with its oversight of the Company's levels of risk, risk assessment, risk management and related policies and processes, including risks arising from cybersecurity threats. The Risk Committee receives regular reports on the Company's cybersecurity program, technology resilience risk management and related developments from members of the Company's information security team, including the CISO. The Board and the Risk Committee also receive information regarding cybersecurity incidents that meet certain reporting thresholds. On an annual basis, senior members of BlackRock's technology, risk and information security teams provide a comprehensive overview of BlackRock's cyber risk and related programs to a joint session of the Board's Risk and Audit Committees.

Technology and cybersecurity risks at BlackRock are also overseen by the TRCC, a dedicated management risk governance committee and sub-committee of the firmwide ERC. The chair of the TRCC is appointed by the head of Enterprise Risk Management at the Company and its members include the CISO as well as a broad range of senior business stakeholders across BlackRock. The TRCC is responsible for oversight of BlackRock's technology and cybersecurity risk management practices and helps ensure that technology and cybersecurity risks remain within firmwide risk tolerances and technology and cybersecurity risk issues are escalated as appropriate to the ERC and other committees.

BlackRock's cybersecurity risk management and strategy processes, which are discussed in greater detail above, are led by the Company's CISO. As of December 31, 2024, the CISO had over 30 years of experience in information technology with a 26-year concentration in information security, including previously serving as the CISO at several global financial institutions. He also holds the Certified Information Systems Security Professional certification. The CISO works closely with the leadership team and other subject matter experts in the global cybersecurity group, who collectively have extensive prior work experience in various roles involving managing information security, developing cybersecurity strategy, implementing effective information and cybersecurity programs and overseeing cybersecurity controls in technology risk and audit functions, as well as having relevant degrees and industry-leading certifications.

The CISO and members of the TRCC monitor the prevention, detection, mitigation and remediation of cybersecurity incidents through their management of, and participation in, the cybersecurity risk management processes described above, including the operation of BlackRock's incident response plan.

Item 2. Properties

BlackRock's principal office, which is leased, is located at 50 Hudson Yards, New York, New York. BlackRock leases additional office space throughout the world, including Atlanta, Belgrade (Serbia), Budapest, Edinburgh, Gurgaon (India), Hong Kong, London, Mumbai (India), Princeton (New Jersey), San Francisco and Singapore. The Company also owns an 84,500 square foot office building in Wilmington, Delaware and a 43,000 square foot data center in Amherst, New York.

Item 3. Legal Proceedings

For a discussion of the Company's legal proceedings, see Note 16, *Commitments and Contingencies*, in the notes to the consolidated financial statements contained in Part II, Item 8.

Item 4. Mine Safety Disclosures

Not applicable.

PART II

Item 5. Market for Registrant’s Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

BlackRock’s common stock is listed on the NYSE and is traded under the symbol “BLK”. At the close of business on January 31, 2025, there were 203 common stockholders of record. Common stockholders include institutional or omnibus accounts that hold common stock for many underlying investors.

The following table sets forth for the periods indicated the dividends declared per share for the common stock as reported on the NYSE:

	Cash Dividend Declared
2024	
First Quarter	**$ 5.10**
Second Quarter	**$ 5.10**
Third Quarter	**$ 5.10**
Fourth Quarter	**$ 5.10**
2023	
First Quarter	**$ 5.00**
Second Quarter	**$ 5.00**
Third Quarter	**$ 5.00**
Fourth Quarter	**$ 5.00**

The closing price of BlackRock’s common stock as of February 24, 2025 was $952.80.

DIVIDENDS

On January 29, 2025, the Board of Directors approved BlackRock’s quarterly dividend of $5.21 per share to be paid on March 24, 2025 to stockholders of record at the close of business on March 7, 2025.

ISSUER PURCHASES OF EQUITY SECURITIES

During the three months ended December 31, 2024, the Company made the following purchases of its common stock, which is registered pursuant to Section 12(b) of the Exchange Act.

	Total Number of Shares Purchased(1)	Average Price Paid per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs	Maximum Number of Shares That May Yet Be Purchased Under the Plans or Programs(1)
October 1, 2024 through October 31, 2024	**5,489**	**$ 942.62**	**—**	**4,180,939**
November 1, 2024 through November 30, 2024	**2,246**	**$ 979.87**	**—**	**4,180,939**
December 1, 2024 through December 31, 2024	**362,622**	**$ 1,046.61**	**358,288**	**3,822,651**
Total	**370,357**	**$ 1,044.67**	**358,288**	

(1) Consists of purchases made by the Company primarily to satisfy income tax withholding obligations of employees and members of the Company’s Board of Directors related to the vesting of certain restricted stock or restricted stock unit awards and purchases made by the Company as part of the share repurchase program that the Company announced in July 2010, which initially authorized the repurchase of 5.1 million shares with no stated expiration. In January 2023, the Company announced that the Board of Directors authorized the repurchase of an additional seven million shares under the Company’s existing share repurchase program, for a total of up to approximately 7.9 million shares of BlackRock common stock.

Item 6. [Reserved]

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

FORWARD-LOOKING STATEMENTS

This report, and other statements that BlackRock may make, may contain forward-looking statements within the meaning of the Private Securities Litigation Reform Act, with respect to BlackRock's future financial or business performance, strategies or expectations. Forward-looking statements are typically identified by words or phrases such as "trend," "potential," "opportunity," "pipeline," "believe," "comfortable," "expect," "anticipate," "current," "intention," "estimate," "position," "assume," "outlook," "continue," "remain," "maintain," "sustain," "seek," "achieve," and similar expressions, or future or conditional verbs such as "will," "would," "should," "could," "may" and similar expressions.

BlackRock cautions that forward-looking statements are subject to numerous assumptions, risks and uncertainties, which change over time and may contain information that is not purely historical in nature. Such information may include, among other things, projections and forecasts. There is no guarantee that any forecasts made will come to pass. Forward-looking statements speak only as of the date they are made, and BlackRock assumes no duty to and does not undertake to update forward-looking statements. Actual results could differ materially from those anticipated in forward-looking statements and future results could differ materially from historical performance.

BlackRock has previously disclosed risk factors in its Securities and Exchange Commission ("SEC") reports. These risk factors and those identified elsewhere in this report, among others, could cause actual results to differ materially from forward-looking statements or historical performance and include: (1) the introduction, withdrawal, success and timing of business initiatives and strategies; (2) changes and volatility in political, economic or industry conditions, the interest rate environment, foreign exchange rates or financial and capital markets, which could result in changes in demand for products or services or in the value of assets under management ("AUM"); (3) the relative and absolute investment performance of BlackRock's investment products; (4) BlackRock's ability to develop new products and services that address client preferences; (5) the impact of increased competition; (6) the impact of recent or future acquisitions or divestitures, including the planned acquisitions of HPS Investment Partners ("HPS" or the "HPS Transaction") and Preqin Holdings Limited ("Preqin" or the "Preqin Transaction"), and the acquisition of Global Infrastructure Management, LLC ("GIP" or the "GIP Transaction" and together with the HPS Transaction and the Preqin Transaction, the "Transactions"); (7) BlackRock's ability to integrate acquired businesses successfully, including the Transactions; (8) risks related to the HPS Transaction and the Preqin Transaction, including delays in the expected closing date of the HPS Transaction or the Preqin Transaction, the possibility that either or both of the HPS Transaction or the Preqin Transaction do not close, including, but not limited to, due to the failure to satisfy the closing conditions; the possibility that expected synergies and value creation from the HPS Transaction or the Preqin Transaction will not be realized, or will not be realized within the expected time period; and the risk of impacts to business and operational relationships related to disruptions from the HPS Transaction or the Preqin Transaction; (9) the unfavorable resolution of legal proceedings; (10) the extent and timing of any share repurchases; (11) the impact, extent and timing of technological changes and the adequacy of intellectual property, data, information and cybersecurity protection; (12) the failure to effectively manage the development and use of artificial intelligence; (13) attempts to circumvent BlackRock's operational control environment or the potential for human error in connection with BlackRock's operational systems; (14) the impact of legislative and regulatory actions and reforms, regulatory, supervisory or enforcement actions of government agencies and governmental scrutiny relating to BlackRock; (15) changes in law and policy and uncertainty pending any such changes; (16) any failure to effectively manage conflicts of interest; (17) damage to BlackRock's reputation; (18) increasing focus from stakeholders regarding environmental and social matters; (19) geopolitical unrest, terrorist activities, civil or international hostilities, and other events outside BlackRock's control, including wars, natural disasters and health crises, which may adversely affect the general economy, domestic and local financial and capital markets, specific industries or BlackRock; (20) climate-related risks to BlackRock's business, products, operations and clients; (21) the ability to attract, train and retain highly qualified professionals; (22) fluctuations in the carrying value of BlackRock's economic investments; (23) the impact of changes to tax legislation, including income, payroll and transaction taxes, and taxation on products, which could affect the value proposition to clients and, generally, the tax position of BlackRock; (24) BlackRock's success in negotiating distribution arrangements and maintaining distribution channels for its products; (25) the failure by key third-party providers to fulfill their obligations to BlackRock; (26) operational, technological and regulatory risks associated with BlackRock's major technology partnerships; (27) any disruption to the operations of third parties whose functions are integral to BlackRock's exchange-traded funds ("ETFs") platform; (28) the impact of BlackRock electing to provide support to its products from time to time and any potential liabilities related to securities lending or other indemnification obligations; and (29) the impact of problems, instability or failure of other financial institutions or the failure or negative performance of products offered by other financial institutions.

OVERVIEW

BlackRock, Inc. (together, with its subsidiaries, unless the context otherwise indicates, "BlackRock" or the "Company") is a leading publicly traded investment management firm with $11.6 trillion of AUM at December 31, 2024. With approximately 21,100 employees in more than 30 countries, BlackRock provides a broad range of investment management and technology services to institutional and retail clients in more than 100 countries across the globe. For further information see Note 1, *Business Overview*, and Note 27, *Segment Information*, in the notes to the consolidated financial statements contained in Part II, Item 8.

On October 1, 2024, BlackRock completed the acquisition of 100% of the issued and outstanding limited liability company interests of GIP for a total consideration, at close, of approximately $3 billion in cash and 6.9 million shares, valued at $5.9 billion. The remaining deferred consideration, all in stock, initially valued at $4.2 billion, is subject to the satisfaction of certain post-closing events. As a result of the closing of the GIP Transaction, BlackRock, Inc. (formerly known as BlackRock Funding, Inc.) ("New BlackRock") became the ultimate parent company of BlackRock Finance, Inc. (formerly known as BlackRock, Inc.) ("Old BlackRock"), GIP and their respective subsidiaries. In addition, New BlackRock became the publicly listed company and retained the ticker symbol "BLK". References herein to BlackRock or the Company for any period (1) prior to the closing of the GIP Transaction on October 1, 2024 refer to Old BlackRock and (2) thereafter refer to New BlackRock. For additional information related to this reorganization, see Note 1, *Business Overview* and Note 3, *Acquisitions* in the notes to the consolidated financial statements contained in Part II, Item 8.

In May 2024, BlackRock completed the acquisition of the remaining equity interest in SpiderRock Advisors ("SRA"), a leading provider of customized option overlay strategies in the United States ("US") wealth market (the "SpiderRock Transaction"). This transaction expands on BlackRock's minority investment in SRA made in 2021 and reinforces BlackRock's commitment to personalized separately managed accounts.

In June 2024, BlackRock announced that it had entered into a definitive agreement to acquire Preqin, a leading independent provider of private markets data, for £2.55 billion (or approximately $3.2 billion based on the GBP/USD foreign exchange rate at December 31, 2024) in cash. The Company believes bringing together Preqin's data and research tools with the complementary workflows of Aladdin and eFront in a unified platform will create a preeminent private markets technology and data provider. The Preqin Transaction is anticipated to close in the first quarter of 2025, subject to customary closing conditions.

In December 2024, BlackRock announced that it had entered into a definitive agreement to acquire 100% of the business and assets of HPS, a leading global credit investment manager with 100% of the consideration paid in BlackRock equity. The equity will generally be delivered in units of a wholly-owned subsidiary of BlackRock ("SubCo Units") which will be exchangeable on a one-for-one basis (subject to certain adjustments) into BlackRock common stock (accordingly, the value of each unit delivered will be based on the price of a share of BlackRock's common stock and the specific terms of the SubCo Units). Approximately 9.2 million SubCo Units and restricted stock units ("RSUs") will be paid at closing. Approximately 2.9 million SubCo Units, will be paid in approximately five years, subject to the satisfaction of certain post-closing conditions. In addition, there is potential for additional consideration to be earned of up to 1.6 million SubCo Units that is based on financial performance milestones measured and paid in approximately five years. Of the total deal consideration, up to 0.7 million units will be used to fund an equity retention pool for HPS employees. In aggregate, inclusive of all SubCo Units paid at closing, eligible to be paid in approximately five years, and potentially earned through achievement of financial performance milestones as well as BlackRock RSUs to be issued in the transaction, the maximum amount of common stock issuable upon exchange of such SubCo Units would be approximately 13.7 million shares. The Company expects the addition of HPS will create an integrated private credit platform to provide both public and private income solutions for clients across their whole portfolios. The HPS Transaction is anticipated to close in mid-2025 subject to regulatory approvals and customary closing conditions.

The following discussion includes a comparison of BlackRock's results for 2024 and 2023. For a discussion of BlackRock's results for 2022 and a comparison of results for 2023 and 2022, see Item 7, *Management's Discussion and Analysis of Financial Condition and Results of Operations,* of the Company's Annual Report on Form 10-K for the year ended December 31, 2023, which was filed with the SEC on February 23, 2024.

Business Outlook

BlackRock's strategy continues to be guided by the Company's clients' needs and focus on the long-term, which the Company believes better enables it to deliver durable returns for shareholders and create value for all of its stakeholders. BlackRock's highly diversified multi-product platform was created to meet client needs in all market environments and provide clients with choice in how they seek to achieve their unique financial goals. BlackRock is positioned to provide alpha-seeking active, private markets, index and cash management investment strategies across asset classes and geographies. In addition, BlackRock leverages its world-class risk management, analytics and technology capabilities, including the Aladdin platform, on behalf of clients. BlackRock serves a diverse mix of institutional and retail clients across the globe, as well as investors in ETFs, maintaining differentiated client relationships and a fiduciary focus. The diversity of BlackRock's platform facilitates the generation of organic growth in various market environments, and as client preferences evolve. BlackRock's long-term strategy remains to keep alpha at the heart of BlackRock; drive growth in ETFs, private markets, and technology; be the global leader in sustainable investing; and lead as a whole portfolio advisor.

BlackRock's framework for long-term shareholder value creation is predicated on generating differentiated organic growth, leveraging scale to increase operating margins over time, and returning capital to shareholders on a consistent basis. BlackRock's diversified platform, in terms of style, product, client and geography, enables it to generate more stable cash flows through market cycles, positioning BlackRock to invest for the long-term by striking an appropriate balance between investing for future growth and prudent discretionary expense management.

BlackRock has invested to serve the full breadth of client needs. Clients increasingly want to build portfolios that are seamlessly integrated across public and private markets, and that are underpinned by data, risk management and technology. The Company is differentiated in being able to deliver across public and private markets, equity and debt, and in the way that best serves each client – from broad-based ETFs to customized whole portfolio solutions. The Company also offers its Aladdin technology to support integrated public-private portfolios.

2024 was a milestone year of organic and inorganic programmatic actions grounded in client needs, investment capability expansion, technology and scale. Clients entrusted BlackRock with a record $641 billion of net inflows in 2024, led by two consecutive record flows quarters in the second half of the year. The Company's closing of the GIP transaction and planned acquisitions of HPS and Preqin are expected to expand and enhance private markets investment and data capabilities.

BlackRock expects 2025 to be a dynamic investing environment. Mega forces like artificial intelligence ("AI") and an ongoing evolution in debt financing are transforming economies and their long-term growth trajectories. Capital markets will play a key role in these transformations. Private markets assets are an increasingly vital part of capital markets, and blending both public and private markets will be critical in fully capturing growth opportunities.

BlackRock is well-positioned to capitalize on structural growth opportunities against a backdrop of economic and capital markets evolution. The Company has made coordinated investments to build the premier long-term capital partner and technology provider across public and private markets. The acquisition of GIP, and the planned acquisitions of Preqin and HPS, each position BlackRock's platform ahead of evolving client needs and structural industry trends.

BlackRock has built a broad private markets platform with $212 billion of AUM across infrastructure, private credit, real estate, private equity and multi-alternatives. As of December 31, 2024, BlackRock had approximately $45 billion of committed capital to deploy for institutional clients in a variety of private markets strategies, and remains confident in its ability to accelerate growth as a leader in private markets. BlackRock also manages $76 billion in liquid alternatives, as well as $90 billion in liquid credit strategies, included within fixed income AUM. With the close of the GIP transaction, and the planned acquisition of HPS, BlackRock's alternatives platform is expected to represent approximately $600 billion in client assets making it a top five alternatives provider.

The private markets – and clients' allocations to them – are expected to continue to grow. Standardized, transparent private markets data and analytics will be increasingly important. As with Aladdin, BlackRock believes it can add even more value to Preqin as both a user and provider of private markets data and risk analytics. Aladdin expanded into new asset classes and markets as BlackRock and its clients evolved, and it expects the same for Preqin.

In addition to private markets, BlackRock is executing on a strong opportunity set across multiple growth channels. These include ETFs, whole portfolio solutions like outsourced mandates and models, and technology.

The ETF industry has been growing rapidly, driven by structural tailwinds including the use of ETFs as active tools, the migration from commission-based to fee-based wealth management, growth in model portfolios, expansion of digital wealth platforms, and the modernization of the bond market. BlackRock's ETF growth strategy is centered on increasing scale and pursuing global growth themes in client and product segments, including Core, Strategic, which includes Fixed Income, Factors, Sustainable and Thematic ETFs, and Precision Exposures. BlackRock views ETFs as a technology that facilitates investing, and ETFs have become core to asset management. The Company believes that ETFs will continue to be a structural growth area as clients turn to ETFs as the preferred vehicle for investing strategies of all types. BlackRock has been a leader in expanding the market for ETFs by making them more accessible and by delivering new asset classes like bonds or cryptocurrency and investment strategies like active. Approximately a quarter of 2024 ETF net inflows of $390 billion were into products launched in the last five years. Active ETFs delivered $22 billion in net inflows in 2024, while BlackRock's Bitcoin exchange-traded product ("ETP") was the largest ETF launch in history, growing to over $50 billion of AUM in less than a year. BlackRock will continue to innovate at the product and portfolio level and accelerate distribution capabilities to deliver differentiated investment solutions.

As the asset management landscape shifts globally from individual product selection to a whole-portfolio approach, BlackRock's strategy is focused on creating outcome-oriented client solutions for both retail investors and institutions. This includes having a diverse platform of alpha-seeking active, index and private markets products, as well as enhanced distribution and portfolio construction technology offerings. Digital wealth tools are an important component of BlackRock's retail strategy, as BlackRock scales and customizes model portfolios, extends Aladdin Wealth and digital wealth partnerships globally, and helps advisors build better portfolios through portfolio construction and risk management, powered by Aladdin. BlackRock has also seen strong momentum in outsourcing solutions among institutional clients, including the funding of several significant mandates in 2024, and anticipates continued outsourcing opportunities in the future.

BlackRock continues to invest in technology services offerings, which enhance the ability to manage portfolios and risk, effectively serve clients and operate efficiently. Market volatility, growing cost pressures, and complexity in optimizing whole portfolios underscore the need for enterprise operating and risk management technology, and should continue to drive demand for holistic and flexible technology solutions. BlackRock continues to evolve and enable clients to further simplify their operating infrastructure with Aladdin. Clients increasingly want to tailor how they use Aladdin to meet their specific needs, and BlackRock is providing them with choice and flexibility. Through the integration of Aladdin and eFront, clients are able to better manage and analyze risk across their whole portfolio spanning public and private markets. BlackRock is empowering clients with data and opening Aladdin by creating connectivity with ecosystem providers and third-party technology solutions, which include asset servicers, cloud providers, digital asset platforms, trading systems and others. This connectivity helps clients work in their Aladdin environments with a more customized and seamless end-to-end experience. Investments in Aladdin AI copilots, enhancements in openness supporting ecosystem partnerships, and advancing whole portfolio solutions including private markets and digital assets are expected to further augment the value of using Aladdin. BlackRock's planned acquisition of Preqin is expected to expand capabilities beyond private markets investment management and technology to data.

Across BlackRock, many clients are focusing on the impact of sustainability factors on their portfolios. This shift has been driven by an increased understanding of how sustainability-related factors can affect economic growth, asset values, and financial markets as a whole. As a fiduciary, BlackRock is committed to providing clients with choice and then executing in accordance with their chosen objectives – for some clients, this includes investing in sustainable strategies. The Company aims to deliver the best risk-adjusted returns within the mandates clients choose, underpinned by research, data, and analytics.

BlackRock's investment management revenue is primarily comprised of fees earned as a percentage of AUM and, in some cases, performance fees, which are normally expressed as a percentage of fund returns to the client. Numerous factors, including price movements in the equity, debt or currency markets, or in the price of real assets, commodities or alternative investments in which BlackRock invests on behalf of clients, and BlackRock's ability to maintain strong investment performance, could impact BlackRock's AUM, revenue and earnings.

Central banks globally have taken actions to reduce or maintain interest rates, after a rapid rate hiking regime in 2022 and much of 2023, in an effort to moderate inflation. BlackRock's business is directly and indirectly affected by changes in global interest rates. Changes in global interest rates may cause BlackRock's AUM to fluctuate and introduce volatility to the Company's base fees, net income and operating cash flows. BlackRock's business may also be impacted by governmental changes, as well as potential regulations, foreign and trade policies and fiscal spending that may arise as a result of such changes. See Part I, Item 1A, *Risk Factors* herein for information on the possible future effects of changes in global interest rates and governmental changes on the Company's results.

BlackRock manages $2.9 trillion in fixed income assets, nearly two-thirds of which are owned by institutions for strategic or liability-matching purposes. BlackRock believes it is well positioned for a stabilizing rate environment due to the breadth, diversification and investment performance of its fixed income platform which encompasses active, ETFs and non-ETF index fixed income products, and a range of strategies, including unconstrained, high yield, total return and short-duration.

BlackRock manages $6.3 trillion of equity assets across markets globally. Beta divergence between equity markets, where certain markets perform differently than others, may lead to an increase in the proportion of BlackRock AUM weighted toward lower fee equity products, resulting in a decline in BlackRock's effective fee rate. Divergent market factors may also erode the correlation between the growth rates of AUM and investment advisory and administration fees (collectively "base fees") and securities lending revenue.

BlackRock believes its strategy aligns with expected future client demand and structural growth opportunities in areas including private markets, such as infrastructure and private credit; ETFs; whole portfolio solutions including outsourcing and models; and integrated public-private markets enterprise technology through Aladdin, eFront and Preqin upon the transaction's closing.

BlackRock enters 2025 with strong momentum across its franchise, including its newly enhanced private markets platform. The Company is positioned ahead of market opportunities that it believes will drive outsized growth for BlackRock in the years to come.

EXECUTIVE SUMMARY

(in millions, except per share data)	2024	2023
GAAP basis[1]:		
Total revenue	**$ 20,407**	$ 17,859
Total expense	**12,833**	11,584
Operating income	**$ 7,574**	$ 6,275
Operating margin	***37.1 %***	*35.1 %*
Nonoperating income (expense), less net income (loss) attributable to noncontrolling interests	**578**	706
Income tax expense	**1,783**	1,479
Net income attributable to BlackRock	**$ 6,369**	$ 5,502
Diluted earnings per common share	**$ 42.01**	$ 36.51
Effective tax rate	**21.9 %**	21.2 %
As adjusted[2]:		
Operating income	**$ 8,110**	$ 6,593
Operating margin	***44.5 %***	*41.7 %*
Nonoperating income (expense), less net income (loss) attributable to noncontrolling interests	**$ 533**	$ 648
Net income attributable to BlackRock	**$ 6,612**	$ 5,692
Diluted earnings per common share	**$ 43.61**	$ 37.77
Effective tax rate	**23.5 %**	21.4 %
Other:		
Assets under management (end of period)	**$ 11,551,251**	$ 10,008,995
Diluted weighted-average common shares outstanding	**151.6**	150.7
Shares outstanding (end of period)	**154.9**	148.5
Book value per share[3]	**$ 306.52**	$ 264.96
Cash dividends declared and paid per share	**$ 20.40**	$ 20.00

(1) Accounting principles generally accepted in the United States ("GAAP").
(2) As adjusted items are described in more detail in *Non-GAAP Financial Measures*.
(3) Total BlackRock stockholders' equity, divided by total shares outstanding at December 31 of the respective year-end.

2024 Compared With 2023

GAAP. Operating income of $7.6 billion increased $1.3 billion and operating margin of 37.1% increased 200 bps from 2023. Increases in operating income and operating margin reflected higher base fees, driven by the positive impact of markets on average AUM, organic base fee growth and fees on AUM acquired in the GIP Transaction, as well as higher performance fees and technology services revenue, partially offset by higher employee compensation and benefits expense, sales, asset and account expense, and general and administration expense. Expense for 2024 was impacted by the GIP Transaction, including nonrecurring retention-related deferred compensation expense, acquisition-related costs and amortization of intangible assets acquired in the GIP Transaction. In addition, expense for 2024 included a $50 million noncash impairment charge related to certain indefinite-lived open-end management contracts. Expense for 2023 included a restructuring charge of $61 million in connection with initiatives to reorganize specific platforms, primarily Aladdin and private markets, to stay ahead of client needs.

Nonoperating income (expense) less net income (loss) attributable to noncontrolling interests ("NCI") decreased $128 million from 2023, driven primarily by lower mark-to-market revaluation of private equity co-investments and higher interest expense, partially offset by higher interest and dividend income, a pre-tax gain of approximately $66 million in connection with a transaction related to a minority investment in EquiLend Holdings, LLC (the "EquiLend Transaction"), and higher mark-to-market gains on unhedged seed capital investments and certain minority investments.

Income tax expense for 2024 included discrete tax benefits of $137 million recognized in connection with the reorganization and establishment of a more efficient global intellectual property and technology platform and corporate structure, $63 million related to the realization of capital losses from changes in the Company's organizational tax structure, $37 million related to vested stock-based compensation awards, and a net noncash discrete tax expense of $14 million related to the revaluation of deferred income tax liabilities. Income tax expense for 2023 included $242 million discrete tax net benefits related to the resolution of certain outstanding tax matters and stock-based compensation awards that vested in 2023.

Earnings per diluted common share increased $5.50, or 15%, from 2023, primarily reflecting higher operating income, partially offset by a higher effective tax rate and lower nonoperating income.

As Adjusted. Operating income of $8.1 billion increased $1.5 billion and operating margin of 44.5% increased 280 bps from 2023. Earnings per diluted common share increased $5.84, or 15%, from 2023, reflecting higher operating income, partially offset by a higher effective tax rate and lower nonoperating income. The acquisition related expenses and the noncash impairment charge of $50 million described above have been excluded from as adjusted results for 2024. In addition, income tax expense for 2024 excluded the $137 million of benefit and the $14 million net noncash tax expense described above. The pre-tax restructuring charge of $61 million described above has been excluded from as adjusted results for 2023.

See *Non-GAAP Financial Measures* for further information on as adjusted items and the reconciliation to GAAP.

For further discussion of BlackRock's revenue, expense, nonoperating results and income tax expense, see *Discussion of Financial Results* herein.

NON-GAAP FINANCIAL MEASURES

BlackRock reports its financial results in accordance with GAAP; however, management believes evaluating the Company's ongoing operating results may be enhanced if investors have additional non-GAAP financial measures. Adjustments to GAAP financial measures ("non-GAAP adjustments") include certain items management deems nonrecurring or that occur infrequently, transactions that ultimately will not impact BlackRock's book value or certain tax items that do not impact cash flow. Management reviews non-GAAP financial measures, in addition to GAAP financial measures, to assess ongoing operations and considers them to be helpful, for both management and investors, in evaluating BlackRock's financial performance over time. Management also uses non-GAAP financial measures as a benchmark to compare its performance with other companies and to enhance comparability for the reporting periods presented. Non-GAAP financial measures may pose limitations because they do not include all of BlackRock's revenue and expense. BlackRock's management does not advocate that investors consider such non-GAAP financial measures in isolation from, or as a substitute for, financial information prepared in accordance with GAAP. Non-GAAP financial measures may not be comparable to other similarly titled measures of other companies.

Computations and reconciliations for all periods are derived from the consolidated statements of income as follows:

(1) Operating income, as adjusted, and operating margin, as adjusted:

(in millions)	2024	2023
Operating income, GAAP basis	**$ 7,574**	$ 6,275
Non-GAAP expense adjustments:		
Compensation expense related to appreciation (depreciation) on deferred cash compensation plans (a)	**43**	57
Amortization and impairment of intangible assets (b)	**291**	151
Acquisition-related compensation costs (b)	**148**	17
Acquisition-related transaction costs (b)[(1)]	**90**	7
Contingent consideration fair value adjustments (b)	**(36)**	3
Lease costs - New York (c)	**—**	14
Restructuring charge (d)	**—**	61
Reduction of indemnification asset (e)[(1)]	**—**	8
Operating income, as adjusted	**$ 8,110**	$ 6,593
Revenue, GAAP basis	**$ 20,407**	$ 17,859
Non-GAAP adjustments:		
Distribution fees	**(1,273)**	(1,262)
Investment advisory fees	**(898)**	(789)
Revenue used for operating margin measurement	**$ 18,236**	$ 15,808
Operating margin, GAAP basis	**37.1 %**	35.1 %
Operating margin, as adjusted	**44.5 %**	41.7 %

(1) Amount included within general and administration expense.

(2) Nonoperating income (expense), less net income (loss) attributable to NCI, as adjusted:

(in millions)	2024	2023
Nonoperating income (expense), GAAP basis	**$ 721**	$ 880
Less: Net income (loss) attributable to NCI	**143**	174
Nonoperating income (expense), net of NCI	**578**	706
Less: Hedge gain (loss) on deferred cash compensation plans (a)	**45**	58
Nonoperating income (expense), less net income (loss) attributable to NCI, as adjusted	**$ 533**	$ 648

(3) Net income attributable to BlackRock, Inc., as adjusted:

(in millions, except per share data)	2024	2023
Net income attributable to BlackRock, Inc., GAAP basis	**$ 6,369**	$ 5,502
Non-GAAP adjustments[(1)]:		
Net impact of hedged deferred cash compensation plans (a)	**(1)**	(1)
Amortization and impairment of intangible assets (b)	**218**	114
Acquisition-related compensation costs (b)	**110**	12
Acquisition-related transaction costs (b)	**66**	5
Contingent consideration fair value adjustments (b)	**(27)**	3
Lease costs - New York (c)	**—**	11
Restructuring charge (d)	**—**	46
Income tax matters	**(123)**	—
Net income attributable to BlackRock, Inc., as adjusted	**$ 6,612**	$ 5,692
Diluted weighted-average common shares outstanding	**151.6**	150.7
Diluted earnings per common share, GAAP basis	**$ 42.01**	$ 36.51
Diluted earnings per common share, as adjusted	**$ 43.61**	$ 37.77

(1) Non-GAAP adjustments, excluding income tax matters, are net of tax.

(1) Operating income, as adjusted, and operating margin, as adjusted: Management believes operating income, as adjusted, and operating margin, as adjusted, are effective indicators of BlackRock's financial performance over time, and, therefore, provide useful disclosure to investors. Management believes that operating margin, as adjusted, reflects the Company's long-term ability to manage ongoing costs in relation to its revenues. The Company uses operating margin, as adjusted, to assess the Company's financial performance, to determine the long-term and annual compensation of the Company's senior-level employees and to evaluate the Company's relative performance against industry peers. Furthermore, this metric eliminates margin variability arising from the accounting of revenues and expenses related to distributing different product structures in multiple distribution channels utilized by asset managers.

- Operating income, as adjusted, includes the following non-GAAP expense adjustments:

  (a) *Compensation expense related to appreciation (depreciation) on deferred cash compensation plans*. The Company excludes compensation expense related to the market valuation changes on certain deferred cash compensation plans, which the Company hedges economically. For these deferred cash compensation plans, the final value of the deferred amount to be distributed to employees in cash upon vesting is determined based on the returns on specified investment funds. The Company recognizes compensation expense for the appreciation (depreciation) of the deferred cash compensation liability in proportion to the vested amount of the award during a respective period, while the net gain (loss) to economically hedge these plans is immediately recognized in nonoperating income (expense), which creates a timing difference impacting net income. This timing difference will reverse and offset to zero over the life of the award at the end of the multi-year vesting period. Management believes excluding market valuation changes related to the deferred cash compensation plans in the calculation of operating income, as adjusted, provides useful disclosure to both management and investors of the Company's financial performance over time as these amounts are economically hedged, while also increasing comparability with other companies.

  (b) *Acquisition-related costs*. Acquisition-related costs include adjustments related to amortization and noncash impairment of intangible assets, other acquisition-related costs, including professional services expense and compensation costs for nonrecurring retention-related deferred compensation, and contingent consideration fair value adjustments incurred in connection with certain acquisitions. Management believes excluding the impact of these expenses when calculating operating income, as adjusted, provides a helpful indication of the Company's financial performance over time, thereby providing helpful information for both management and investors while also increasing comparability with other companies.

  (c) *Lease costs – New York*. In 2023, the Company continued to recognize lease expense within general and administration expense for both its current headquarters located at 50 Hudson Yards in New York and prior headquarters until the Company's lease on its prior headquarters expired in April 2023. The Company began lease payments related to its current headquarters in May 2023, but began recording lease expense in August 2021 when it obtained access to the building to begin its tenant improvements. Prior to the Company's move to its current headquarters in February 2023, the impact of lease costs related to 50 Hudson Yards was excluded from operating income, as adjusted. In February 2023, the Company completed the majority of its move to 50 Hudson Yards and no longer excluded the impact of these lease costs. Subsequently, from February 2023 through April 2023, the Company excluded the impact of lease costs related to the Company's prior headquarters. Management believes excluding the impact of these respective New York lease costs ("Lease costs – New York") when calculating operating income, as adjusted, is useful to assess the Company's financial performance and ongoing operations, and enhances comparability among periods presented.

  (d) *Restructuring charge*. In the fourth quarter of 2023, the Company recorded a restructuring charge, comprised of severance and compensation expense for accelerated vesting of previously granted deferred compensation awards, in connection with initiatives to reorganize specific platforms, primarily Aladdin and private markets. Management believes excluding the impact of these restructuring charges when calculating operating income, as adjusted, is useful to assess the Company's financial performance and ongoing operations, and enhances comparability among periods presented.

  (e) *Reduction of indemnification asset.* In connection with a previous acquisition, BlackRock recorded an $8 million indemnification asset. Due to the resolution of certain tax matters in the third quarter of 2023, BlackRock recorded $8 million of general and administration expense to reflect the reduction of the indemnification asset and an offsetting $8 million tax benefit. The $8 million general and administrative expense and $8 million tax benefit have been excluded from as adjusted results as there was no impact on BlackRock's book value.

- Revenue used for calculating operating margin, as adjusted, is reduced to exclude all of the Company's distribution fees, which are recorded as a separate line item on the consolidated statements of income, as well as a portion of investment advisory fees received that is used to pay distribution and servicing costs. For certain products, based on distinct arrangements, distribution fees are collected by the Company and then passed-through to third-party client intermediaries. For other products, investment advisory fees are collected by the Company and a portion is passed-through to third-party client intermediaries. However, in both structures, the third-party client intermediary similarly owns the relationship with the retail client and is responsible for distributing the product and servicing the client. The amount of distribution and investment advisory fees fluctuates each period primarily based on a predetermined percentage of the value of AUM during the period. These fees also vary based on the type of investment product sold and the geographic location where it is sold. In addition, the Company may waive fees on certain products that could result in the reduction of payments to the third-party intermediaries.

(2) Nonoperating income (expense), less net income (loss) attributable to NCI, as adjusted: Management believes nonoperating income (expense), less net income (loss) attributable to NCI, as adjusted, is an effective measure for reviewing BlackRock’s nonoperating contribution to its results and provides comparability of this information among reporting periods. Nonoperating income (expense), less net income (loss) attributable to NCI, as adjusted, excludes the gain (loss) on the economic hedge of certain deferred cash compensation plans. As the gain (loss) on investments and derivatives used to hedge these compensation plans over time substantially offsets the compensation expense related to the market valuation changes on these deferred cash compensation plans, which is included in operating income, GAAP basis, management believes excluding the gain (loss) on the economic hedge of the deferred cash compensation plans when calculating nonoperating income (expense), less net income (loss) attributable to NCI, as adjusted, provides a useful measure for both management and investors of BlackRock’s nonoperating results that impact book value.

(3) Net income attributable to BlackRock, Inc., as adjusted: Management believes net income attributable to BlackRock, Inc., as adjusted, and diluted earnings per common share, as adjusted, are useful measures of BlackRock’s profitability and financial performance. Net income attributable to BlackRock, Inc., as adjusted, equals net income attributable to BlackRock, Inc., GAAP basis, adjusted for certain items management deems nonrecurring or that occur infrequently, transactions that ultimately will not impact BlackRock’s book value or certain tax items that do not impact cash flow.

For each period presented, the non-GAAP adjustments were tax effected at the respective blended rates applicable to the adjustments. Amounts for income tax matters in 2024 include a discrete tax benefit of $137 million recognized in connection with the reorganization and establishment of a more efficient global intellectual property and technology platform and corporate structure. This discrete tax benefit has been excluded from as adjusted results due to the nonrecurring nature of the intellectual property reorganization. In addition, amounts for 2024 include a net noncash expense of $14 million associated with the revaluation of deferred tax liabilities related to intangible assets and goodwill as a result of tax rate changes. This discrete tax expense has been excluded from the as adjusted results as it does not have a cash flow impact as well as to ensure comparability among periods presented.

Per share amounts reflect net income attributable to BlackRock, Inc., as adjusted, divided by diluted weighted-average common shares outstanding.

(4) Annual Contract Value ("ACV"): Management believes ACV is an effective metric for reviewing BlackRock’s technology services’ ongoing contribution to its operating results and provides comparability of this information among reporting periods while also providing a useful supplemental metric for both management and investors of BlackRock’s growth in technology services revenue over time, as it is linked to the net new business in technology services. ACV represents forward-looking, annualized estimated value of the recurring subscription fees under client contracts, assuming all client contracts that come up for renewal are renewed, unless we have received a notice of termination, even though such notice may not be effective until a later date. ACV also includes the annualized estimated value of new sales, for existing and new clients, when we execute client contracts, even though the recurring fees may not be effective until a later date and excludes nonrecurring fees such as implementation and consulting fees.

ASSETS UNDER MANAGEMENT

AUM for reporting purposes generally is based upon how investment advisory and administration fees are calculated for each portfolio. Net asset values, total assets, committed assets or other measures may be used to determine portfolio AUM.

AUM and Net Inflows (Outflows) by Client Type and Product Type

	AUM		Net inflows (outflows)	
(in millions)	**2024**	**2023**	**2024**	**2023**
Retail	**$ 1,015,827**	$ 929,697	**$ 24,367**	$ (8,473)
ETFs	**4,230,375**	3,499,299	**390,433**	185,942
Institutional:				
Active	**2,136,749**	1,912,673	**64,447**	87,106
Index	**3,247,637**	2,902,489	**9,374**	(55,125)
Institutional subtotal	**5,384,386**	4,815,162	**73,821**	31,981
Long-term	**10,630,588**	**9,244,158**	**488,621**	**209,450**
Cash management	**920,663**	764,837	**152,730**	79,245
Total	**$ 11,551,251**	**$ 10,008,995**	**$ 641,351**	**$ 288,695**

AUM and Net Inflows (Outflows) by Investment Style and Product Type

	AUM		Net inflows (outflows)	
(in millions)	**2024**	**2023**	**2024**	**2023**
Active	**$ 2,870,656**	$ 2,621,178	**$ 62,164**	$ 59,221
Index and ETFs	**7,759,932**	6,622,980	**426,457**	150,229
Long-term	**10,630,588**	**9,244,158**	**488,621**	**209,450**
Cash management	**920,663**	764,837	**152,730**	79,245
Total	**$ 11,551,251**	**$ 10,008,995**	**$ 641,351**	**$ 288,695**

AUM and Net Inflows (Outflows) by Product Type

	AUM		Net inflows (outflows)	
(in millions)	**2024**	**2023**	**2024**	**2023**
Equity	**$ 6,310,191**	$ 5,293,344	**$ 225,568**	$ (11,490)
Fixed income	**2,905,669**	2,804,026	**163,669**	143,087
Multi-asset	**992,921**	870,804	**51,678**	82,787
Alternatives:				
Private markets	**211,974**	136,909	**9,457**	13,665
Liquid alternatives	**76,390**	74,233	**(2,609)**	(11,370)
Currency and commodities[(1)]	**133,443**	64,842	**40,858**	(7,229)
Alternatives subtotal	**421,807**	275,984	**47,706**	(4,934)
Long-term	**10,630,588**	**9,244,158**	**488,621**	**209,450**
Cash management	**920,663**	764,837	**152,730**	79,245
Total	**$ 11,551,251**	**$ 10,008,995**	**$ 641,351**	**$ 288,695**

(1) Amounts include cryptocurrency and commodity ETFs and ETPs.

The following table presents the component changes in BlackRock's AUM for 2024 and 2023.

(in millions)	2024	2023
Beginning AUM	$ 10,008,995	$ 8,594,485
Net inflows (outflows):		
Long-term	488,621	209,450
Cash management	152,730	79,245
Total net inflows (outflows)	641,351	288,695
Acquisitions(1)	73,949	2,177
Market change	992,964	1,073,550
FX impact(2)	(166,008)	50,088
Total change	1,542,256	1,414,510
Ending AUM	$ 11,551,251	$ 10,008,995

(1) Amount for 2024 includes AUM attributable to the GIP Transaction and the SpiderRock Transaction. Amount for 2023 includes AUM attributable to the acquisition of Kreos Capital in August 2023 (the "Kreos Transaction").
(2) Foreign exchange reflects the impact of translating non-US dollar denominated AUM into US dollars for reporting purposes.

BlackRock has historically grown AUM through organic growth and acquisitions. Management believes that the Company will be able to continue to grow AUM organically by focusing on strong investment performance, efficient delivery of beta for index products, client service, developing new products and optimizing distribution capabilities.

Component Changes in AUM for 2024

The following table presents the component changes in AUM by client type and product type for 2024.

(in millions)	December 31, 2023	Net inflows (outflows)	Acquisition(1)	Market change	FX impact(2)	December 31, 2024	Full year average AUM(3)
Retail:							
Equity	$ 435,734	$ 15,285	$ 4,074	$ 54,257	$ (4,232)	$ 505,118	$ 485,161
Fixed income	312,799	11,671	—	1,483	(7,312)	318,641	316,520
Multi-asset	139,537	(2,328)	—	14,420	(651)	150,978	147,169
Alternatives	41,627	(261)	—	69	(345)	41,090	41,087
Retail subtotal	929,697	24,367	4,074	70,229	(12,540)	1,015,827	989,937
ETFs:							
Equity	2,532,631	236,357	—	359,322	(21,912)	3,106,398	2,845,456
Fixed income	898,403	112,341	—	(16,291)	(8,801)	985,652	948,250
Multi-asset	9,140	1,025	—	841	(272)	10,734	9,451
Alternatives	59,125	40,710	—	27,919	(163)	127,591	89,331
ETFs subtotal	3,499,299	390,433	—	371,791	(31,148)	4,230,375	3,892,488
Institutional:							
Active:							
Equity	186,688	5,380	—	30,876	(4,096)	218,848	207,929
Fixed income	836,823	(2,843)	—	16,885	(10,537)	840,328	841,830
Multi-asset	717,182	54,887	—	72,798	(16,828)	828,039	774,210
Alternatives	171,980	7,023	69,875	3,618	(2,962)	249,534	191,190
Active subtotal	1,912,673	64,447	69,875	124,177	(34,423)	2,136,749	2,015,159
Index:							
Equity	2,138,291	(31,454)	—	420,860	(47,870)	2,479,827	2,333,824
Fixed income	756,001	42,500	—	(5,068)	(32,385)	761,048	759,871
Multi-asset	4,945	(1,906)	—	204	(73)	3,170	3,693
Alternatives	3,252	234	—	165	(59)	3,592	2,912
Index subtotal	2,902,489	9,374	—	416,161	(80,387)	3,247,637	3,100,300
Institutional subtotal	4,815,162	73,821	69,875	540,338	(114,810)	5,384,386	5,115,459
Long-term	**9,244,158**	**488,621**	**73,949**	**982,358**	**(158,498)**	**10,630,588**	**9,997,884**
Cash management	764,837	152,730	—	10,606	(7,510)	920,663	806,123
Total	**$ 10,008,995**	**$ 641,351**	**$ 73,949**	**$ 992,964**	**$ (166,008)**	**$ 11,551,251**	**$ 10,804,007**

(1) Amounts include AUM attributable to the GIP Transaction and the SpiderRock Transaction.
(2) Foreign exchange reflects the impact of translating non-US dollar denominated AUM into US dollars for reporting purposes.
(3) Average AUM is calculated as the average of the month-end spot AUM amounts for the trailing thirteen months.

The following table presents the component changes in AUM by investment style and product type for 2024.

(in millions)	December 31, 2023	Net inflows (outflows)	Acquisition(1)	Market change	FX impact(2)	December 31, 2024	Full year average AUM(3)
Active:							
Equity	$ 427,448	$ (6,333)	$ 4,074	$ 48,479	$ (6,505)	**$ 467,163**	$ 461,583
Fixed income	1,123,422	9,184	—	18,516	(17,248)	**1,133,874**	1,133,152
Multi-asset	856,705	52,553	—	87,221	(17,478)	**979,001**	921,364
Alternatives	213,603	6,760	69,875	3,687	(3,307)	**290,618**	232,274
Active subtotal	2,621,178	62,164	73,949	157,903	(44,538)	**2,870,656**	2,748,373
Index and ETFs:							
ETFs:							
Equity	2,532,631	236,357	—	359,322	(21,912)	**3,106,398**	2,845,456
Fixed income	898,403	112,341	—	(16,291)	(8,801)	**985,652**	948,250
Multi-asset	9,140	1,025	—	841	(272)	**10,734**	9,451
Alternatives	59,125	40,710	—	27,919	(163)	**127,591**	89,331
ETFs subtotal	3,499,299	390,433	—	371,791	(31,148)	**4,230,375**	3,892,488
Non-ETF index:							
Equity	2,333,265	(4,456)	—	457,514	(49,693)	**2,736,630**	2,565,331
Fixed income	782,201	42,144	—	(5,216)	(32,986)	**786,143**	785,069
Multi-asset	4,959	(1,900)	—	201	(74)	**3,186**	3,708
Alternatives	3,256	236	—	165	(59)	**3,598**	2,915
Non-ETF index subtotal	3,123,681	36,024	—	452,664	(82,812)	**3,529,557**	3,357,023
Index & ETFs subtotal	6,622,980	426,457	—	824,455	(113,960)	**7,759,932**	7,249,511
Long-term	**9,244,158**	**488,621**	**73,949**	**982,358**	**(158,498)**	**10,630,588**	**9,997,884**
Cash management	764,837	152,730	—	10,606	(7,510)	**920,663**	806,123
Total	**$ 10,008,995**	**$ 641,351**	**$ 73,949**	**$ 992,964**	**$ (166,008)**	**$ 11,551,251**	**$ 10,804,007**

The following table presents the component changes in AUM by product type for 2024.

(in millions)	December 31, 2023	Net inflows (outflows)	Acquisition(1)	Market change	FX impact(2)	December 31, 2024	Full year average AUM(3)
Equity	$ 5,293,344	$ 225,568	$ 4,074	$ 865,315	$ (78,110)	**$ 6,310,191**	$ 5,872,370
Fixed income	2,804,026	163,669	—	(2,991)	(59,035)	**2,905,669**	2,866,471
Multi-asset	870,804	51,678	—	88,263	(17,824)	**992,921**	934,523
Alternatives:							
Private markets	136,909	9,457	69,875	(1,803)	(2,464)	**211,974**	154,597
Liquid alternatives	74,233	(2,609)	—	5,482	(716)	**76,390**	75,402
Currency and commodities(4)	64,842	40,858	—	28,092	(349)	**133,443**	94,521
Alternatives subtotal	275,984	47,706	69,875	31,771	(3,529)	**421,807**	324,520
Long-term	**9,244,158**	**488,621**	**73,949**	**982,358**	**(158,498)**	**10,630,588**	**9,997,884**
Cash management	764,837	152,730	—	10,606	(7,510)	**920,663**	806,123
Total	**$ 10,008,995**	**$ 641,351**	**$ 73,949**	**$ 992,964**	**$ (166,008)**	**$ 11,551,251**	**$ 10,804,007**

(1) Amounts include AUM attributable to the GIP Transaction and the SpiderRock Transaction.
(2) Foreign exchange reflects the impact of translating non-US dollar denominated AUM into US dollars for reporting purposes.
(3) Average AUM is calculated as the average of the month-end spot AUM amounts for the trailing thirteen months.
(4) Amounts include cryptocurrency and commodity ETFs and ETPs.

AUM increased $1.5 trillion to $11.6 trillion at December 31, 2024 from $10.0 trillion at December 31, 2023, driven primarily by net market appreciation, net inflows, led by flows into equity, bond and cryptocurrency products, cash management, and significant outsourcing mandates, and AUM added from the GIP Transaction, partially offset by the negative impact of foreign exchange movements.

Net market appreciation of $993 billion was primarily driven by global equity market appreciation.

AUM decreased $166 billion due to the impact of foreign exchange movements, primarily due to the strengthening of the US dollar, largely against the euro, the Japanese yen, the Canadian dollar and the British pound.

For further discussion on AUM, see Part I, Item 1 – *Business – Assets Under Management*.

Component Changes in AUM for 2023

The following table presents the component changes in AUM by client type and product type for 2023.

(in millions)	December 31, 2022	Net inflows (outflows)	Acquisition[1]	Market change	FX impact[2]	December 31, 2023	Full year average AUM[3]
Retail:							
Equity	$ 370,612	$ 2,810	$ —	$ 58,248	$ 4,064	$ 435,734	$ 403,530
Fixed income	299,114	(2,471)	—	11,821	4,335	312,799	306,232
Multi-asset	125,168	(236)	—	14,022	583	139,537	131,236
Alternatives	48,581	(8,576)	—	1,286	336	41,627	45,319
Retail subtotal	843,475	(8,473)	—	85,377	9,318	929,697	886,317
ETFs:							
Equity	2,081,742	81,223	—	362,885	6,781	2,532,631	2,262,361
Fixed income	758,093	111,956	—	24,544	3,810	898,403	824,832
Multi-asset	8,875	(746)	—	949	62	9,140	8,024
Alternatives	60,900	(6,491)	—	4,626	90	59,125	61,439
ETFs subtotal	2,909,610	185,942	—	393,004	10,743	3,499,299	3,156,656
Institutional:							
Active:							
Equity	168,734	(13,301)	—	29,088	2,167	186,688	174,967
Fixed income	774,955	4,714	—	53,538	3,616	836,823	798,832
Multi-asset	544,469	85,665	—	79,644	7,404	717,182	642,051
Alternatives	153,433	10,028	2,177	4,925	1,417	171,980	162,871
Active subtotal	1,641,591	87,106	2,177	167,195	14,604	1,912,673	1,778,721
Index:							
Equity	1,814,266	(82,222)	—	401,047	5,200	2,138,291	1,979,704
Fixed income	704,661	28,888	—	17,774	4,678	756,001	713,802
Multi-asset	6,392	(1,896)	—	559	(110)	4,945	5,882
Alternatives	3,296	105	—	(138)	(11)	3,252	3,263
Index subtotal	2,528,615	(55,125)	—	419,242	9,757	2,902,489	2,702,651
Institutional subtotal	4,170,206	31,981	2,177	586,437	24,361	4,815,162	4,481,372
Long-term	**7,923,291**	**209,450**	**2,177**	**1,064,818**	**44,422**	**9,244,158**	**8,524,345**
Cash management	671,194	79,245	—	8,732	5,666	764,837	696,355
Total	**$ 8,594,485**	**$ 288,695**	**$ 2,177**	**$ 1,073,550**	**$ 50,088**	**$ 10,008,995**	**$ 9,220,700**

(1) Amounts include AUM attributable to the Kreos Transaction.
(2) Foreign exchange reflects the impact of translating non-US dollar denominated AUM into US dollars for reporting purposes.
(3) Average AUM is calculated as the average of the month-end spot AUM amounts for the trailing thirteen months.

The following table presents the component changes in AUM by investment style and product type for 2023.

(in millions)	December 31, 2022	Net inflows (outflows)	Acquisition(1)	Market change	FX impact(2)	December 31, 2023	Full year average AUM(3)
Active:							
Equity	$ 392,836	$ (26,772)	$ —	$ 57,431	$ 3,953	$ 427,448	$ 409,687
Fixed income	1,053,083	(882)	—	64,203	7,018	1,123,422	1,080,917
Multi-asset	669,629	85,424	—	93,665	7,987	856,705	773,278
Alternatives	202,012	1,451	2,177	6,210	1,753	213,603	208,189
Active subtotal	2,317,560	59,221	2,177	221,509	20,711	2,621,178	2,472,071
Index and ETFs:							
ETFs:							
Equity	2,081,742	81,223	—	362,885	6,781	2,532,631	2,262,361
Fixed income	758,093	111,956	—	24,544	3,810	898,403	824,832
Multi-asset	8,875	(746)	—	949	62	9,140	8,024
Alternatives	60,900	(6,491)	—	4,626	90	59,125	61,439
ETFs subtotal	2,909,610	185,942	—	393,004	10,743	3,499,299	3,156,656
Non-ETF index:							
Equity	1,960,776	(65,941)	—	430,952	7,478	2,333,265	2,148,514
Fixed income	725,647	32,013	—	18,930	5,611	782,201	737,949
Multi-asset	6,400	(1,891)	—	560	(110)	4,959	5,891
Alternatives	3,298	106	—	(137)	(11)	3,256	3,264
Non-ETF index subtotal	2,696,121	(35,713)	—	450,305	12,968	3,123,681	2,895,618
Index & ETFs subtotal	5,605,731	150,229	—	843,309	23,711	6,622,980	6,052,274
Long-term	**7,923,291**	**209,450**	**2,177**	**1,064,818**	**44,422**	**9,244,158**	**8,524,345**
Cash management	671,194	79,245	—	8,732	5,666	764,837	696,355
Total	**$ 8,594,485**	**$ 288,695**	**$ 2,177**	**$ 1,073,550**	**$ 50,088**	**$ 10,008,995**	**$ 9,220,700**

The following table presents the component changes in AUM by product type for 2023.

(in millions)	December 31, 2022	Net inflows (outflows)	Acquisition(1)	Market change	FX impact(2)	December 31, 2023	Full year average AUM(3)
Equity	$ 4,435,354	$ (11,490)	$ —	$ 851,268	$ 18,212	$ 5,293,344	$ 4,820,562
Fixed income	2,536,823	143,087	—	107,677	16,439	2,804,026	2,643,698
Multi-asset	684,904	82,787	—	95,174	7,939	870,804	787,193
Alternatives:							
Private markets	117,751	13,665	2,177	1,885	1,431	136,909	127,655
Liquid alternatives	80,654	(11,370)	—	4,548	401	74,233	77,595
Currency and commodities(4)	67,805	(7,229)	—	4,266	—	64,842	67,642
Alternatives subtotal	266,210	(4,934)	2,177	10,699	1,832	275,984	272,892
Long-term	**7,923,291**	**209,450**	**2,177**	**1,064,818**	**44,422**	**9,244,158**	**8,524,345**
Cash management	671,194	79,245	—	8,732	5,666	764,837	696,355
Total	**$ 8,594,485**	**$ 288,695**	**$ 2,177**	**$ 1,073,550**	**$ 50,088**	**$ 10,008,995**	**$ 9,220,700**

(1) Amounts include AUM attributable to the Kreos Transaction.
(2) Foreign exchange reflects the impact of translating non-US dollar denominated AUM into US dollars for reporting purposes.
(3) Average AUM is calculated as the average of the month-end spot AUM amounts for the trailing thirteen months.
(4) Amounts include commodity ETFs and ETPs.

AUM increased $1.4 trillion to $10.0 trillion at December 31, 2023 from $8.6 trillion at December 31, 2022, driven primarily by net market appreciation, net inflows, led by flows into bond and equity ETFs, cash management, significant outsourcing mandates and growth in private markets.

Net market appreciation of $1.1 trillion was primarily driven by global equity market appreciation.

AUM increased $50 billion due to the impact of foreign exchange movements, primarily due to the weakening of the US dollar largely against the British pound and the euro, partially offset by the strengthening of the US dollar against the Japanese yen.

DISCUSSION OF FINANCIAL RESULTS

Introduction

The Company derives a substantial portion of its revenue from investment advisory and administration fees, which are recognized as the services are performed over time because the customer is receiving and consuming the benefits as they are provided by the Company. Fees are primarily based on agreed-upon percentages of AUM and recognized for services provided during the period, which are distinct from services provided in other periods. Such fees are affected by changes in AUM, including market appreciation or depreciation, foreign exchange translation and net inflows or outflows. Net inflows or outflows represent the sum of new client assets, additional fundings from existing clients (including dividend reinvestment), withdrawals of assets from, and termination of, client accounts and distributions to investors representing return of capital and return on investments. Market appreciation or depreciation includes current income earned on, and changes in the fair value of, securities held in client accounts. Foreign exchange translation reflects the impact of translating non-US dollar denominated AUM into US dollars for reporting purposes.

The Company also earns revenue by lending securities on behalf of clients, primarily to highly rated banks and broker-dealers. The securities loaned are secured by collateral in the form of cash or securities, with minimum collateral generally ranging from approximately 102% to 112% of the value of the loaned securities. Generally, the revenue earned is shared between the Company and the funds or accounts managed by the Company from which the securities are borrowed.

Investment advisory agreements for certain separate accounts and investment funds provide for performance fees based upon relative and/or absolute investment performance, in addition to base fees based on AUM. Investment advisory performance fees generally are earned after a given period of time when investment performance exceeds a contractual threshold, and when it is determined that the fees are no longer probable of significant reversal. As such, the timing of recognition of performance fees may increase the volatility of the Company's revenue and earnings. The magnitude of performance fees can fluctuate quarterly due to the timing of carried interest recognition on private market products and a greater number and size of liquid products with performance measurement periods that end in the third and fourth quarters.

The Company offers investment management technology systems, risk management services, wealth management and digital distribution tools, all on a fee basis. Clients include banks, insurance companies, official institutions, pension funds, asset managers, retail distributors and other investors. Fees earned for technology services are primarily recorded as services are performed over time and are generally determined using the value of positions on the Aladdin platform, or on a fixed-rate basis. Revenue derived from the sale of software licenses is recognized upon the granting of access rights.

The Company earns distribution and service fees for distributing investment products and providing support services to investment portfolios. The fees are primarily based on AUM and are recognized when the amount of fees is known.

The Company advises global financial institutions, regulators, and government entities across a range of risk, regulatory, capital markets and strategic services. Fees earned for advisory services, which are included in advisory and other revenue, are determined using fixed-rate fees and are recognized over time as the related services are completed.

The Company earns fees for transition management services primarily comprised of commissions recognized in connection with buying and selling securities on behalf of its customers. Commissions related to transition management services, which are included in advisory and other revenue, are recorded on a trade-date basis as transactions occur.

Beginning in the first quarter of 2024, BlackRock updated the presentation of the Company's expense line items within the consolidated statements of income by including a new "sales, asset and account expense" income statement caption. Such expense line items have been recast for 2023 to conform to this new presentation. For a recast of 2023 expense line items, see page 12 of Exhibit 99.1 to the Current Report on Form 8-K furnished on April 12, 2024. Operating expense reflects employee compensation and benefits, sales, asset and account expense, general and administration expense, and amortization and impairment of intangible assets.

- Employee compensation and benefits expense includes salaries, commissions, temporary help, incentive compensation, employer payroll taxes, severance and related benefit costs.
- Sales, asset and account expense includes distribution and servicing costs, direct fund expense, and sub-advisory and other expense. These expenses primarily vary over time with fluctuations in AUM, number of shareholder accounts, or other attributes directly related to volume of business.
  - Distribution and servicing costs, which are primarily AUM driven, include payments to third parties, primarily associated with distribution and servicing of client investments in certain Company products.
  - Direct fund expense primarily consists of third-party non-advisory expenses incurred by the Company related to certain funds for the use of index trademarks, reference data for indices, custodial services, fund administration, fund accounting, transfer agent services, shareholder reporting services, legal expense, audit and tax services as well as other fund-related expenses directly attributable to the non-advisory operations of the fund.
  - Sub-advisory and other expense is primarily related to the contracts where third-party advisors provide investment advisory services on the Company's behalf.
- General and administration expense includes marketing and promotional (including travel and entertainment expense), occupancy and office-related, portfolio services (including market data costs), technology, professional services, communications, contingent consideration fair value adjustments, product launch costs, the net impact of foreign currency remeasurement, and other general and administration expense.

Approximately 80% of the Company's revenue is generated in US dollars. The Company's revenue and expense generated in foreign currencies (primarily the euro and British pound) are impacted by foreign exchange rates. Any effect of foreign exchange rate change on revenue is partially offset by a change in expense driven by the Company's considerable non-dollar expense base related to its operations outside the US.

Nonoperating income (expense) includes the effect of changes in the valuations on investments and earnings on equity method investments as well as interest and dividend income and interest expense. The Company primarily holds seed and co-investments in sponsored investment products that invest in a variety of asset classes, including private equity, private credit, hedge funds and real assets. Investments generally are made for co-investment purposes, to establish a performance track record or for regulatory purposes, including Federal Reserve Bank stock. The Company does not engage in proprietary trading activities that could conflict with the interests of its clients.

In addition, nonoperating income (expense) includes the impact of changes in the valuations of consolidated sponsored investment products ("CIPs"). The portion of nonoperating income (expense) not attributable to the Company is allocated to NCI on the consolidated statements of income.

Revenue

The table below presents detail of revenue for 2024 and 2023 and includes the product type mix of base fees and securities lending revenue and performance fees.

(in millions)	2024	2023
Revenue:		
Investment advisory, administration fees and securities lending revenue:		
Equity:		
Active	**$ 2,166**	$ 2,000
ETFs	**5,124**	4,418
Non-ETF index	**784**	743
Equity subtotal	**8,074**	7,161
Fixed income:		
Active	**1,952**	1,897
ETFs	**1,367**	1,230
Non-ETF index	**369**	353
Fixed income subtotal	**3,688**	3,480
Multi-asset	**1,278**	1,203
Alternatives:		
Private markets	**1,196**	889
Liquid alternatives	**568**	572
Currency and commodities(1)	**247**	185
Alternatives subtotal	**2,011**	1,646
Long-term	**15,051**	13,490
Cash management	**1,049**	909
Total investment advisory, administration fees and securities lending revenue(2)	**16,100**	14,399
Investment advisory performance fees:		
Equity	**161**	99
Fixed income	**34**	4
Multi-asset	**24**	28
Alternatives:		
Private markets	**308**	273
Liquid alternatives	**680**	150
Alternatives subtotal	**988**	423
Total investment advisory performance fees	**1,207**	554
Technology services revenue	**1,603**	1,485
Distribution fees	**1,273**	1,262
Advisory and other revenue:		
Advisory	**49**	81
Other	**175**	78
Total advisory and other revenue	**224**	159
Total revenue	**$ 20,407**	$ 17,859

(1) Amounts include cryptocurrency and commodity ETFs and ETPs.
(2) Amounts include $615 million and $675 million of securities lending revenue for 2024 and 2023, respectively.

The table below lists a percentage breakdown of base fees and securities lending revenue and average AUM by product type:

	Percentage of Base Fees and Securities Lending Revenue		Percentage of Average AUM by Product Type(1)	
	2024	**2023**	**2024**	**2023**
Equity:				
Active	**13 %**	14 %	**4 %**	4 %
ETFs	**32 %**	31 %	**27 %**	24 %
Non-ETF index	**5 %**	5 %	**24 %**	23 %
Equity subtotal	**50 %**	50 %	**55 %**	51 %
Fixed income:				
Active	**12 %**	13 %	**10 %**	12 %
ETFs	**8 %**	9 %	**9 %**	9 %
Non-ETF index	**2 %**	2 %	**7 %**	8 %
Fixed income subtotal	**22 %**	24 %	**26 %**	29 %
Multi-asset	**8 %**	8 %	**9 %**	9 %
Alternatives:				
Private markets	**7 %**	7 %	**1 %**	1 %
Liquid alternatives	**4 %**	4 %	**1 %**	1 %
Currency and commodities(2)	**2 %**	1 %	**1 %**	1 %
Alternatives subtotal	**13 %**	12 %	**3 %**	3 %
Long-term	**93 %**	**94 %**	**93 %**	**92 %**
Cash management	**7 %**	6 %	**7 %**	8 %
Total AUM	**100 %**	**100 %**	**100 %**	**100 %**

(1) Average AUM is calculated as the average of the month-end spot AUM amounts for the trailing thirteen months.
(2) Amounts include cryptocurrency and commodity ETFs and ETPs.

Revenue increased $2.5 billion, or 14%, from 2023, primarily driven by the positive impact of market beta on average AUM, organic base fee growth, and fees on AUM acquired in the GIP Transaction, as well as higher performance fees and technology services revenue.

Investment advisory, administration fees and securities lending revenue of $16.1 billion in 2024 increased $1.7 billion from $14.4 billion in 2023, primarily driven by the impact of market beta on average AUM, organic base fee growth and approximately $230 million of fees related to AUM acquired in the GIP Transaction, partially offset by lower securities lending revenue. Securities lending revenue of $615 million decreased $60 million from $675 million in 2023, primarily reflecting lower spreads, partially offset by higher average balances of securities on loan.

Investment advisory performance fees of $1.2 billion in 2024 increased $653 million from $554 million in 2023, primarily driven by higher revenue from liquid alternative products, including strong performance from a single hedge fund in 2024, and higher revenue from long-only and private market products.

Technology services revenue of $1.6 billion in 2024 increased $118 million from $1.5 billion in 2023, reflecting sustained demand for Aladdin technology offerings and the successful onboarding of a number of new clients.

Advisory and other revenue of $224 million in 2024 increased $65 million from $159 million in 2023, reflecting the impact of presenting earnings (losses) from certain equity method minority investments within nonoperating income (expense) beginning in the first quarter of 2024 and higher transition management assignments, partially offset by lower revenue from advisory assignments.

Expense

The following table presents expense for 2024 and 2023.

(in millions)	2024	2023
Expense:		
Employee compensation and benefits	**$ 6,546**	$ 5,779
Sales, asset and account expense(1):		
Distribution and servicing costs	**2,171**	2,051
Direct fund expense	**1,464**	1,331
Sub-advisory and other	**140**	116
Total sales, asset and account expense	**3,775**	3,498
General and administration expense:		
Marketing and promotional	**314**	309
Occupancy and office related	**421**	418
Portfolio services	**262**	270
Technology	**674**	607
Professional services	**277**	195
Communications	**39**	47
Foreign exchange remeasurement	**—**	(6)
Contingent consideration fair value adjustments	**(36)**	3
Other general and administration	**270**	252
Total general and administration expense	**2,221**	2,095
Restructuring charge	**—**	61
Amortization and impairment of intangible assets	**291**	151
Total expense	**$ 12,833**	**$ 11,584**

(1) Beginning in the first quarter of 2024, BlackRock updated the presentation of the Company's expense line items within the consolidated statements of income by including a new "sales, asset and account expense" income statement caption. Such expense line items have been recast for 2023 to conform to this new presentation. For a recast of 2023 expense line items, see page 12 of Exhibit 99.1 to the Current Report on Form 8-K furnished on April 12, 2024.

Expense increased $1.2 billion, or 11%, from 2023, reflecting higher employee compensation and benefits expense, sales, asset and account expense, amortization and impairment of intangible assets and general and administration expense. Expense for 2024 was impacted by the previously described expenses incurred in connection with the GIP Transaction and the $50 million noncash impairment charge(1). Expense for 2023 included a previously mentioned restructuring charge of $61 million(1).

Employee compensation and benefits expense of $6.5 billion in 2024 increased $767 million from $5.8 billion in 2023, reflecting higher incentive compensation, primarily as a result of higher operating income and performance fees. 2024 employee compensation and benefits expense was also impacted by the GIP Transaction, including nonrecurring retention-related deferred compensation expense(1).

Sales, asset and account expense of $3.8 billion in 2024 increased $277 million from $3.5 billion in 2023, driven by higher distribution and servicing costs and direct fund expense, primarily reflecting higher average AUM.

General and administration expense of $2.2 billion in 2024 increased $126 million from $2.1 billion in 2023, primarily associated with acquisition-related costs(1) in connection with the GIP Transaction, including transaction costs recorded in professional services expense, and higher technology expense, partially offset by lower contingent consideration fair value adjustments.

Restructuring charge(1) of $61 million, comprised of severance and compensation expense for accelerated vesting of previously granted deferred compensation awards, was recorded in 2023, as previously described.

Amortization and impairment of intangible assets(1) of $291 million in 2024 increased $140 million from $151 million in 2023, primarily due to the amortization of intangible assets acquired in the GIP Transaction and the previously described $50 million noncash impairment charge.

(1) These expenses have been excluded from the Company's as adjusted financial results under the expense adjustments for acquisition-related costs and restructuring charge, as applicable. See *Non-GAAP Financial Measures* for further information on as adjusted items. See Note 12, *Intangible Assets*, in the notes to the consolidated financial statements and *Critical Accounting Policies and Estimates* for further information on the impairment charge.

Nonoperating Results

The summary of nonoperating income (expense), less net income (loss) attributable to NCI for 2024 and 2023 was as follows:

(in millions)	2024	2023
Nonoperating income (expense), GAAP basis	**$ 721**	$ 880
Less: Net income (loss) attributable to NCI	**143**	174
Nonoperating income (expense), net of NCI	**578**	706
Less: Hedge gain (loss) on deferred cash compensation plans(1)	**45**	58
Nonoperating income (expense), net of NCI, as adjusted(2)	**$ 533**	$ 648

(in millions)	2024	2023
Net gain (loss) on investments, net of NCI		
Private equity	**$ (10)**	$ 349
Real assets	**14**	13
Other alternatives(3)	**41**	49
Other investments(4)	**127**	66
Hedge gain (loss) on deferred cash compensation plans(1)	**45**	58
Subtotal	**217**	535
Other income/gain (expense/loss)(5)	**132**	(10)
Total net gain (loss) on investments, net of NCI	**349**	525
Interest and dividend income	**767**	473
Interest expense	**(538)**	(292)
Net interest income (expense)	**229**	181
Nonoperating income (expense), net of NCI	**578**	706
Less: Hedge gain (loss) on deferred cash compensation plans(1)	**45**	58
Nonoperating income (expense), net of NCI, as adjusted(2)	**$ 533**	$ 648

(1) Amount relates to the gain (loss) from economically hedging BlackRock's deferred cash compensation plans.
(2) Management believes nonoperating income (expense), net of NCI, as adjusted, is an effective measure for reviewing BlackRock's nonoperating results, which ultimately impacts BlackRock's book value. See *Non-GAAP Financial Measures* for further information on other non-GAAP financial measures.
(3) Amounts primarily include net gains (losses) related to credit funds, direct hedge fund strategies and hedge fund solutions.
(4) Amounts primarily include net gains (losses) related to BlackRock's seed investment portfolio, net of the impact of certain hedges.
(5) Amounts for 2024 include earnings (losses) from certain equity method minority investments, which the Company recorded within nonoperating income (expense) beginning in the first quarter of 2024 and noncash pre-tax gains (losses) related to the revaluation of certain minority investments. In addition, amount for 2024 includes a pre-tax gain of approximately $66 million in connection with a transaction related to a minority investment in the EquiLend Transaction and a noncash pre-tax gain in connection with the SpiderRock Transaction of approximately $19 million.

Income Tax Expense

	GAAP		As Adjusted	
(in millions)	2024	2023	2024	2023
Operating income(1)	**$ 7,574**	$ 6,275	**$ 8,110**	$ 6,593
Total nonoperating income (expense)(1)(2)	**$ 578**	$ 706	**$ 533**	$ 648
Income before income taxes(2)	**$ 8,152**	$ 6,981	**$ 8,643**	$ 7,241
Income tax expense	**$ 1,783**	$ 1,479	**$ 2,031**	$ 1,549
Effective tax rate	**21.9 %**	21.2 %	**23.5 %**	21.4 %

(1) As adjusted items are described in more detail in *Non-GAAP Financial Measures.*
(2) Net of net income (loss) attributable to NCI.

The Company's tax rate is affected by tax rates in foreign jurisdictions and the relative amount of income earned in those jurisdictions, which the Company expects to be fairly consistent in the near term. The significant foreign jurisdictions that have different statutory tax rates than the US federal statutory rate of 21% include the UK, Germany, Canada and Ireland.

2024 Income tax expense (GAAP) reflected:

- a discrete tax benefit of $137 million recognized in connection with the reorganization and establishment of a more efficient global intellectual property and technology platform and corporate structure;
- a discrete tax benefit of $63 million related to the realization of capital losses from changes in the Company's organizational tax structure;
- a net discrete tax benefit of $37 million related to stock-based compensation awards that vested in 2024; and
- a $14 million net noncash tax expense related to the revaluation of certain deferred income tax liabilities.

The as adjusted effective tax rate of 23.5% for 2024 excluded the $137 million discrete tax benefit due to the nonrecurring nature of the intellectual property reorganization and the $14 million net noncash tax expense, as it does not have a cash flow impact as well as to ensure comparability among periods presented.

The Organisation for Economic Cooperation and Development ("OECD") has proposed certain international tax reforms, which, among other things, would (1) shift taxing rights to the jurisdiction of the consumer ("Pillar One") and (2) establish a global minimum tax for multinational companies of 15% ("Pillar Two"). In response, European Union member states and several other countries, including the United Kingdom, have since adopted laws implementing the OECD's minimum tax rules under Pillar Two, effective starting in 2024. As a result of these developments, the tax laws of certain countries in which BlackRock does business have changed and may continue to change, and any such changes could increase its tax liabilities. The Pillar Two Framework did not have a material impact on BlackRock's consolidated financial statements for 2024 and the Company is continuing to monitor legislative developments and evaluate the potential impact of the Pillar Two Framework on future periods.

2023 Income tax expense (GAAP) reflected:

- a discrete tax benefit of $201 million, related to the resolution of certain outstanding tax matters; and
- a discrete tax benefit of $41 million, related to stock-based compensation awards that vested in 2023.

STATEMENT OF FINANCIAL CONDITION OVERVIEW

As Adjusted Statement of Financial Condition

The following table presents a reconciliation of the consolidated statement of financial condition presented on a GAAP basis to the consolidated statement of financial condition, excluding the impact of separate account assets and separate account collateral held under securities lending agreements (directly related to lending separate account securities) and separate account liabilities and separate account collateral liabilities under securities lending agreements and CIPs.

The Company presents the as adjusted statement of financial condition as additional information to enable investors to exclude certain assets that have equal and offsetting liabilities or NCI that ultimately do not have an impact on stockholders' equity or cash flows. Management views the as adjusted statement of financial condition, which contains non-GAAP financial measures, as an economic presentation of the Company's total assets and liabilities; however, it does not advocate that investors consider such non-GAAP financial measures in isolation from, or as a substitute for, financial information prepared in accordance with GAAP.

Separate Account Assets and Liabilities and Separate Account Collateral Held under Securities Lending Agreements

Separate account assets are maintained by BlackRock Life Limited, a wholly owned subsidiary of the Company that is a registered life insurance company in the UK, and represent segregated assets held for purposes of funding individual and group pension contracts. The Company records equal and offsetting separate account liabilities. The separate account assets are not available to creditors of the Company and the holders of the pension contracts have no recourse to the Company's assets. The net investment income attributable to separate account assets accrues directly to the contract owners and is not reported on the consolidated statements of income. While BlackRock has no economic interest in these assets or liabilities, BlackRock earns an investment advisory fee for the service of managing these assets on behalf of its clients.

In addition, the Company records on its consolidated statements of financial condition the separate account collateral obtained under BlackRock Life Limited securities lending arrangements for which it has legal title as its own asset in addition to an equal and offsetting separate account collateral liability for the obligation to return the collateral. The collateral is not available to creditors of the Company, and the borrowers under the securities lending arrangements have no recourse to the Company's assets.

Consolidated Sponsored Investment Products

The Company consolidates certain sponsored investment products accounted for as variable interest entities ("VIEs") and voting rights entities ("VREs"). See Note 2, *Significant Accounting Policies*, in the notes to the consolidated financial statements contained in Part II, Item 8 of this filing for more information on the Company's consolidation policy.

The Company cannot readily access cash and cash equivalents or other assets held by CIPs to use in its operating activities. In addition, the Company cannot readily sell investments held by CIPs in order to obtain cash for use in the Company's operations.

(in millions)	December 31, 2024 GAAP Basis	Separate Account Assets/ Collateral(1)	CIPs(2)	As Adjusted
Assets				
Cash and cash equivalents	$ 12,762	$ —	$ 169	$ 12,593
Accounts receivable	4,304	—	—	4,304
Investments	9,769	—	1,875	7,894
Separate account assets and collateral held under securities lending agreements	58,870	58,870	—	—
Operating lease right-of-use assets	1,519	—	—	1,519
Other assets(3)	4,699	—	76	4,623
Subtotal	91,923	58,870	2,120	30,933
Goodwill and intangible assets, net	46,692	—	—	46,692
Total assets	$ 138,615	$ 58,870	$ 2,120	$ 77,625
Liabilities				
Accrued compensation and benefits	$ 2,964	$ —	$ —	$ 2,964
Accounts payable and accrued liabilities	1,536	—	—	1,536
Borrowings	12,314	—	—	12,314
Separate account liabilities and collateral liabilities under securities lending agreements	58,870	58,870	—	—
Contingent consideration liabilities	4,302	—	—	4,302
Deferred income tax liabilities(4)	3,334	—	—	3,334
Operating lease liabilities	1,908	—	—	1,908
Other liabilities	4,032	—	318	3,714
Total liabilities	89,260	58,870	318	30,072
Equity				
Total BlackRock, Inc. stockholders' equity	47,495	—	—	47,495
Noncontrolling interests	1,860	—	1,802	58
Total equity	49,355	—	1,802	47,553
Total liabilities and equity	$ 138,615	$ 58,870	$ 2,120	$ 77,625

(1) Amounts represent segregated client assets and related liabilities, in which BlackRock has no economic interest. BlackRock earns an investment advisory fee for the service of managing these assets on behalf of its clients.
(2) Amounts represent the impact of consolidating CIPs.
(3) Amount includes property and equipment and other assets.
(4) Amount includes approximately $4.2 billion of deferred income tax liabilities related to goodwill and intangibles. See Note 25, *Income Taxes*, in the notes to the consolidated financial statements contained in Part II, Item 8 of this filing for more information.

The following discussion summarizes the significant changes in assets and liabilities on a GAAP basis. Please see the consolidated statements of financial condition as of December 31, 2024 and 2023 contained in Part II, Item 8 of this filing. The discussion does not include changes related to assets and liabilities that are equal and offsetting and have no impact on BlackRock's stockholders' equity.

Assets. Cash and cash equivalents at December 31, 2024 included $169 million of cash held by CIPs (see *Liquidity and Capital Resources* for details on the change in cash and cash equivalents during 2024). Accounts receivable at December 31, 2024 increased $388 million from December 31, 2023, primarily due to higher base and performance fee receivables. Investments at December 31, 2024 increased $29 million from December 31, 2023 (for more information see *Investments* herein). Goodwill and intangible assets, net at December 31, 2024 increased $12.9 billion from December 31, 2023, primarily due to the GIP Transaction and the SpiderRock Transaction, partially offset by amortization of intangible assets and a $50 million noncash impairment charge related to certain indefinite-lived intangible assets. Other assets at December 31, 2024 decreased $261 million from December 31, 2023, primarily related to a decrease in unit trust receivables (substantially offset by a decrease in unit trust payables recorded within other liabilities), partially offset by an increase in certain minority investments.

Liabilities. Accrued compensation and benefits at December 31, 2024 increased $571 million from December 31, 2023, primarily due to higher 2024 incentive compensation accruals. Accounts payable and accrued liabilities at December 31, 2024 increased $296 million from December 31, 2023, primarily due to higher interest on borrowings and increased accruals, including accruals related to direct fund expense. Contingent consideration liabilities at December 31, 2024 increased $4.2 billion from December 31, 2023, primarily due to the contingent consideration liability in connection with the GIP Transaction, which will be settled all in stock, subject to the achievement of specified performance targets (See Note 3, *Acquisitions*, in the notes to the consolidated financial statements contained in Part II, Item 8 of this filing for more information). Other liabilities at December 31, 2024 decreased $343 million from December 31, 2023, primarily due to lower unit trust payables (substantially offset by a decrease in unit trust receivables recorded within other assets). Deferred income tax liabilities at December 31, 2024 decreased $172 million from December 31, 2023, primarily due to the effects of temporary differences associated with the intellectual property reorganization and the GIP Transaction.

Investments

The Company's investments were $9.8 billion and $9.7 billion at December 31, 2024 and 2023, respectively. Investments include CIPs accounted for as VIEs and VREs. Management reviews BlackRock's investments on an "economic" basis, which eliminates the portion of investments that does not impact BlackRock's book value or net income attributable to BlackRock. BlackRock's management does not advocate that investors consider such non-GAAP financial measures in isolation from, or as a substitute for, financial information prepared in accordance with GAAP.

The Company presents investments, as adjusted, to enable investors to understand the economic portion of investments that is owned by the Company as a gauge to measure the impact of changes in net nonoperating income (expense) on investments to net income (loss) attributable to BlackRock.

The Company further presents net "economic" investment exposure, net of deferred cash compensation investments and hedged exposures, to reflect another helpful measure for investors. The economic impact of investments held pursuant to deferred cash compensation plans is substantially offset by a change in associated compensation expense, and the impact of the portfolio of seed investments is mitigated by futures entered into as part of the Company's macro hedging strategy. Carried interest capital allocations are excluded as there is no impact to BlackRock's stockholders' equity until such amounts are realized as performance fees. Finally, the Company's regulatory investment in Federal Reserve Bank stock, which is not subject to market or interest rate risk, is excluded from the Company's net economic investment exposure.

(in millions)		December 31, 2024		December 31, 2023
Investments, GAAP	**$**	**9,769**	$	9,740
Investments held by CIPs		**(5,752)**		(5,977)
Net interest in CIPs[1]		**3,877**		4,111
Investments, as adjusted		**7,894**		7,874
Investments related to deferred cash compensation plans		**(185)**		(264)
Hedged exposures		**(1,757)**		(1,771)
Federal Reserve Bank stock		**(93)**		(92)
Carried interest		**(1,983)**		(1,975)
Total "economic" investment exposure[2]	**$**	**3,876**	$	3,772

(1) Amounts included $1.9 billion of carried interest (VIEs) at both December 31, 2024 and 2023, which has no impact on the Company's "economic" investment exposure.
(2) Amounts do not include investments in corporate minority investments included in other assets on the consolidated statements of financial condition.

The following table represents the carrying value of the Company's economic investment exposure, by asset type, at December 31, 2024 and 2023:

(in millions)		December 31, 2024		December 31, 2023
Equity/Fixed income/Multi-asset[1]	**$**	**3,025**	$	2,786
Alternatives:				
Private equity		**1,199**		1,491
Real assets		**629**		509
Other alternatives[2]		**780**		757
Alternatives subtotal		**2,608**		2,757
Hedged exposures		**(1,757)**		(1,771)
Total "economic" investment exposure	**$**	**3,876**	$	3,772

(1) Amounts include seed investments in equity, fixed income, and multi-asset mutual funds/strategies.
(2) Other alternatives primarily include co-investments in credit funds, direct hedge fund strategies, and hedge fund solutions.

As adjusted investment activity for 2024 and 2023 was as follows:

(in millions)		2024		2023
Investments, as adjusted, beginning balance	**$**	**7,874**	$	6,419
Purchases/capital contributions		**2,214**		1,403
Sales/maturities		**(1,888)**		(914)
Distributions[1]		**(466)**		(111)
Market appreciation(depreciation)/earnings from equity method investments		**220**		607
Carried interest capital allocations/(distributions)		**8**		425
Other[2]		**(68)**		45
Investments, as adjusted, ending balance	**$**	**7,894**	$	7,874

(1) Amount includes distributions representing return of capital and return on investments.
(2) Amount includes the impact of foreign exchange movements.

LIQUIDITY AND CAPITAL RESOURCES

BlackRock Cash Flows Excluding the Impact of CIPs

The consolidated statements of cash flows include the cash flows of the CIPs. The Company uses an adjusted cash flow statement, which excludes the impact of CIPs, as a supplemental non-GAAP measure to assess liquidity and capital requirements. The Company believes that its cash flows, excluding the impact of the CIPs, provide investors with useful information on the cash flows of BlackRock relating to its ability to fund additional operating, investing and financing activities. BlackRock's management does not advocate that investors consider such non-GAAP measures in isolation from, or as a substitute for, its cash flows presented in accordance with GAAP.

The following table presents a reconciliation of the consolidated statements of cash flows presented on a GAAP basis to the consolidated statements of cash flows, excluding the impact of the cash flows of CIPs:

(in millions)	GAAP Basis	Impact on Cash Flows of CIPs	Cash Flows Excluding Impact of CIPs
Cash, cash equivalents and restricted cash, December 31, 2022	$ 7,433	$ 265	$ 7,168
Net cash provided by/(used in) operating activities	4,165	(1,519)	5,684
Net cash provided by/(used in) investing activities	(959)	(26)	(933)
Net cash provided by/(used in) financing activities	(1,992)	1,568	(3,560)
Effect of exchange rate changes on cash, cash equivalents and restricted cash	106	—	106
Net increase/(decrease) in cash, cash equivalents and restricted cash	1,320	23	1,297
Cash, cash equivalents and restricted cash, December 31, 2023	$ 8,753	$ 288	$ 8,465
Net cash provided by/(used in) operating activities	**4,956**	**(2,311)**	**7,267**
Net cash provided by/(used in) investing activities	**(3,004)**	**(127)**	**(2,877)**
Net cash provided by/(used in) financing activities	**2,236**	**2,319**	**(83)**
Effect of exchange rate changes on cash, cash equivalents and restricted cash	**(162)**	**—**	**(162)**
Net increase/(decrease) in cash, cash equivalents and restricted cash	**4,026**	**(119)**	**4,145**
Cash, cash equivalents and restricted cash, December 31, 2024	**$ 12,779**	**$ 169**	**$ 12,610**

Sources of BlackRock's operating cash primarily include base fees and securities lending revenue, performance fees, technology services revenue, advisory and other revenue and distribution fees. BlackRock uses its cash to pay all operating expenses, interest and principal on borrowings, income taxes, dividends and repurchases of the Company's stock, acquisitions, capital expenditures and purchases of co-investments and seed investments.

For details of the Company's GAAP cash flows from operating, investing and financing activities, see the consolidated statements of cash flows contained in Part II, Item 8 of this filing.

Cash flows provided by/(used in) operating activities, excluding the impact of CIPs, primarily include the receipt of base fees, securities lending revenue, performance fees and technology services revenue, offset by the payment of operating expenses incurred in the normal course of business, including year-end incentive and deferred cash compensation accrued during prior years, and income tax payments.

Cash flows used in investing activities, excluding the impact of CIPs, for 2024 were $2.9 billion, primarily reflecting $2.9 billion related to the GIP Transaction, $255 million of purchases of property and equipment, $74 million related to the SpiderRock Transaction, and $52 million of net investment purchases, partially offset by $366 million of distributions of capital from equity method investees.

Cash flows used in financing activities, excluding the impact of CIPs, for 2024 were $83 million, primarily resulting from $3.1 billion of cash dividend payments, $1.9 billion of share repurchases, including $1.6 billion in open market transactions and $0.3 billion of employee tax withholdings related to employee stock transactions, and $1.0 billion of repayment of long-term borrowings, partially offset by $5.5 billion of proceeds from long-term borrowings related to the issuances of senior notes to fund a portion of the cash consideration for the GIP Transaction and the Preqin Transaction, which is anticipated to close in the first quarter of 2025, and $0.5 billion in proceeds from stock options exercised.

The Company manages its financial condition and funding to maintain appropriate liquidity for the business. Management believes that the Company's liquid assets, continuing cash flows from operations, borrowing capacity under the Company's existing revolving credit facility and uncommitted commercial paper private placement program, provide sufficient resources to meet the Company's short-term and long-term cash needs, including operating, debt and other obligations as they come due and anticipated future capital requirements. Liquidity resources at December 31, 2024 and 2023 were as follows:

(in millions)	December 31, 2024	December 31, 2023
Cash and cash equivalents(1)	**$ 12,762**	$ 8,736
Cash and cash equivalents held by CIPs(2)	**(169)**	(288)
Subtotal(3)	**12,593**	8,448
Credit facility — undrawn	**5,400**	5,000
Total liquidity resources	**$ 17,993**	$ 13,448

(1) Amounts exclude restricted cash.
(2) The Company cannot readily access such cash and cash equivalents to use in its operating activities.
(3) The percentage of cash and cash equivalents held by the Company's US subsidiaries was approximately 65% and 50% at December 31, 2024 and 2023, respectively. See *Net Capital Requirements* herein for more information on net capital requirements in certain regulated subsidiaries.

Total liquidity resources increased $4.5 billion during 2024, primarily reflecting $4.5 billion of net proceeds from long-term borrowings, a $400 million increase in the aggregate commitment amount under the credit facility, and positive cash flows from operating activities, partially offset by cash dividend payments of $3.1 billion, $2.9 billion related to the GIP Transaction, share repurchases of $1.9 billion and $74 million related to the SpiderRock Transaction.

A significant portion of the Company's $7.9 billion of investments, as adjusted, is illiquid in nature and, as such, cannot be readily convertible to cash.

Share Repurchases. During 2024, the Company repurchased 1.9 million common shares under the Company's existing share repurchase program for approximately $1.6 billion. At December 31, 2024, there were approximately 3.8 million shares still authorized to be repurchased under the program. The timing and actual number of shares repurchased will depend on a variety of factors, including legal limitations, price and market conditions.

Net Capital Requirements. The Company is required to maintain net capital in certain regulated subsidiaries within a number of jurisdictions, which is partially maintained by retaining cash and cash equivalent investments in those subsidiaries or jurisdictions. As a result, such subsidiaries of the Company may be restricted in their ability to transfer cash between different jurisdictions and to their parents. Additionally, transfers of cash between international jurisdictions may have adverse tax consequences that could discourage such transfers.

BlackRock Institutional Trust Company, N.A. ("BTC") is chartered as a national bank that does not accept deposits or make commercial loans and whose powers are limited to trust and other fiduciary activities. BTC provides investment management and other fiduciary services, including investment advisory and securities lending agency services, to institutional clients. BTC is subject to regulatory capital and liquid asset requirements administered by the US Office of the Comptroller of the Currency.

At both December 31, 2024 and 2023, the Company was required to maintain approximately $1.8 billion in net capital in certain regulated subsidiaries, including BTC, entities regulated by the Financial Conduct Authority and Prudential Regulation Authority in the UK, and the Company's broker-dealers. The Company was in compliance with all applicable regulatory net capital requirements.

Undistributed Earnings of Foreign Subsidiaries. As a result of the 2017 Tax Cuts and Jobs Act and the one-time mandatory deemed repatriation tax on untaxed accumulated foreign earnings, US income taxes were provided on the Company's undistributed foreign earnings. The financial statement basis in excess of tax basis of its foreign subsidiaries remains indefinitely reinvested in foreign operations. The Company will continue to evaluate its capital management plans.

Short-Term Borrowings

2024 Revolving Credit Facility. The Company maintains an unsecured revolving credit facility, which is available for working capital and general corporate purposes (the "2024 Credit Facility"). In March 2024, the 2024 Credit Facility was amended to, among other things, (1) permit the GIP Transaction and the transactions contemplated in connection with the GIP Transaction, (2) add New BlackRock as a borrower under the existing credit agreement, (3) add New BlackRock as a guarantor of the payment and performance of the obligations, liabilities and indebtedness of Old BlackRock and certain of its other subsidiaries and (4) update the sustainability-linked pricing mechanics to remove existing metrics and allow new metrics, if any, to be set following the consummation of the GIP Transaction. In May 2024, the 2024 Credit Facility was further amended to, among other things, (1) increase the aggregate commitment amount by $400 million to $5.4 billion and (2) extend the maturity date to March 2029 for lenders (other than one non-extending lender) pursuant to the Company's option to request extensions of the maturity date available under the 2024 Credit Facility (with the commitment of the non-extending lender maturing in March 2028). The 2024 Credit Facility permits the Company to request up to an additional $1.0 billion of borrowing capacity, subject to lender credit approval, which could increase the overall size of the 2024 Credit Facility to an aggregate principal amount of up to $6.4 billion. Interest on outstanding borrowings accrues at an applicable benchmark rate for the denominated currency of the loan, plus a spread. The 2024 Credit Facility requires the Company not to exceed a maximum leverage ratio (ratio of net debt to earnings before interest, taxes, depreciation and amortization, where net debt equals total debt less unrestricted cash) of 3 to 1, which was satisfied with a ratio of less than 1 to 1 at December 31, 2024. At December 31, 2024, the Company had no amount outstanding under the 2024 Credit Facility.

Commercial Paper Program. On November 7, 2024, BlackRock established a commercial paper program (the "CP Program") under which the Company may issue short-term, unsecured commercial paper notes (the "CP Notes") on a private-placement basis up to a maximum aggregate amount outstanding at any time of $5 billion. The payments of the CP Notes have been unconditionally guaranteed by Old BlackRock (the "CP Notes Guarantee"). The CP Notes will rank equal in right of payment with all of BlackRock's other unsubordinated indebtedness, and the obligations of Old BlackRock under the CP Notes Guarantee will rank equal in right of payment with all of Old BlackRock's other unsubordinated indebtedness. Net proceeds of issuances of the CP Notes are expected to be used for general corporate purposes. The CP Program is currently supported by the 2024 Credit Facility. The CP Program replaced the Company's prior $4 billion commercial paper program, which was terminated concurrently with the establishment of this CP Program, and has been put into place in connection with the Company's organizational structure following the closing of the GIP Transaction. At December 31, 2024, BlackRock had no CP Notes outstanding.

Subsidiary Credit Facility. In January 2024, BlackRock Investment Management (UK) Limited ("BIM UK"), a wholly owned subsidiary of the Company, entered into a revolving credit facility (the "Subsidiary Credit Facility") in the amount of £25 million (or approximately $31 million based on the GBP/USD foreign exchange rate at December 31, 2024) with a rolling 364-day term structure. The Subsidiary Credit Facility is available for BIM UK's general corporate and working capital purposes. At December 31, 2024, there was no amount outstanding under the Subsidiary Credit Facility.

Long-Term Borrowings

The carrying value of long-term borrowings at December 31, 2024 included the following:

(in millions)	Maturity Amount	Carrying Value	Maturity
1.25% Notes[1][2]	$ 725	$ 725	May 2025
3.20% Notes[2]	700	698	March 2027
4.60% Notes	800	797	July 2027
3.25% Notes[2]	1,000	993	April 2029
4.70% Notes	500	497	March 2029
2.40% Notes[2]	1,000	997	April 2030
1.90% Notes[2]	1,250	1,243	January 2031
2.10% Notes[2]	1,000	989	February 2032
4.75% Notes[2]	1,250	1,233	May 2033
5.00% Notes	1,000	993	March 2034
4.90% Notes	500	495	January 2035
5.25% Notes	1,500	1,468	March 2054
5.35% Notes	1,200	1,186	January 2055
Total long-term borrowings	$ 12,425	$ 12,314	

(1) The carrying value of the 1.25% Notes is calculated using the EUR/USD foreign exchange rate as of December 31, 2024.
(2) Issued by Old BlackRock and guaranteed by New BlackRock.

In March 2024, New BlackRock issued $3.0 billion in aggregate principal amount of senior unsecured and unsubordinated notes. These notes were issued as three separate series of senior debt securities including $500 million of 4.70% notes maturing on March 14, 2029 (the "2029 Notes"), $1.0 billion of 5.00% notes maturing on March 14, 2034 (the "2034 Notes") and $1.5 billion of 5.25% notes maturing on March 14, 2054 (the "2054 Notes") (collectively, the "March 2024 Notes"). Net proceeds were used to fund a portion of the cash consideration for the GIP Transaction, which closed in October 2024. Interest on the March 2024 Notes of approximately $152 million per year is payable semi-annually on March 14 and September 14 of each year, which commenced on September 14, 2024. The March 2024 Notes are fully and unconditionally guaranteed (the “March 2024 Notes Guarantee”) on a senior unsecured basis by Old BlackRock. The March 2024 Notes and the March 2024 Notes Guarantee rank equally in right of payment with all of BlackRock and Old BlackRock's other unsubordinated indebtedness, respectively. The March 2024 Notes may be redeemed prior to maturity at any time in whole or in part at the option of BlackRock at the redemption prices set forth in the applicable series of March 2024 Notes.

In March 2024, the Company fully repaid $1.0 billion of 3.50% Notes at maturity.

In July 2024, New BlackRock issued $2.5 billion in aggregate principal amount of senior unsecured and unsubordinated notes. These notes were issued as three separate series of senior debt securities including $800 million of 4.60% notes maturing on July 26, 2027 (the "2027 Notes"), $500 million of 4.90% notes maturing on January 8, 2035 (the "2035 Notes") and $1.2 billion of 5.35% notes maturing on January 8, 2055 (the "2055 Notes") (collectively, the "July 2024 Notes"). Net proceeds are intended to be used to fund a portion of the cash consideration for the Preqin Transaction, which is anticipated to close in the first quarter of 2025, subject to customary closing conditions. The July 2024 Notes are fully and unconditionally guaranteed (the “July 2024 Notes Guarantee”, and together with the March 2024 Notes Guarantee "Notes Guarantees") on a senior unsecured basis by Old BlackRock. The July 2024 Notes and the July 2024 Notes Guarantee rank equally in right of payment with all of BlackRock and Old BlackRock’s other unsubordinated indebtedness, respectively. Interest on the 2027 Notes of approximately $37 million per year is payable semi-annually on January 26 and July 26 of each year, beginning January 26, 2025. Interest on the 2035 Notes and 2055 Notes of approximately $25 million and $64 million per year, respectively, is payable semi-annually on January 8 and July 8 of each year, beginning January 8, 2025. The July 2024 Notes may be redeemed prior to maturity at any time in whole or in part at the option of BlackRock at the redemption prices set forth in the applicable series of July 2024 Notes. In addition, if the Preqin Transaction is not consummated, the Company will be required to redeem all outstanding 2027 Notes (the “Special Mandatory Redemption”) at a Special Mandatory Redemption price equal to 101% of the aggregate principal amount of the applicable series of 2027 Notes, plus accrued and unpaid interest, if any, to, but excluding, the Special Mandatory Redemption date.

For more information on the Company’s borrowings, see Note 15, *Borrowings*, in the notes to the consolidated financial statements contained in Part II, Item 8 of this filing.

Supplemental Guarantor Information

On October 1, 2024, in connection with the closing of the GIP Transaction, BlackRock, Inc. (formerly known as BlackRock Funding, Inc.) ("New BlackRock") became the ultimate parent company of BlackRock Finance, Inc. (formerly known as BlackRock, Inc.) ("Old BlackRock"), GIP and their respective subsidiaries. See *Overview* for additional information.

New BlackRock is the issuer of the previously described March 2024 Notes and July 2024 Notes (collectively the "2024 Notes"), which are fully and unconditionally guaranteed on a senior unsecured basis by Old BlackRock. The 2024 Notes and the Notes Guarantees rank equally in right of payment with all of New BlackRock's and Old BlackRock's other unsubordinated indebtedness, respectively. No other subsidiary of Old BlackRock or New BlackRock guarantees the 2024 Notes. The Notes Guarantees will be automatically and unconditionally released and discharged, and Old BlackRock will be released from all obligations under the indenture in its capacity as guarantor, in certain circumstances as described in the indenture governing the 2024 Notes. See Note 15, *Borrowings*, in the notes to the consolidated financial statements for further information on the 2024 Notes, Old BlackRock's senior unsecured notes and the New BlackRock Guarantee.

On October 1, 2024, in connection with the closing of the GIP Transaction, New BlackRock also entered into a guarantee (the “New BlackRock Guarantee”) pursuant to which New BlackRock fully and unconditionally guaranteed, on a senior unsecured basis, the obligations of Old BlackRock with respect to its previously issued senior unsecured notes. The New BlackRock Guarantee ranks equally in right of payment with all of New BlackRock's other unsubordinated indebtedness. In certain circumstances as described in the New BlackRock Guarantee, the New BlackRock Guarantee will be automatically and unconditionally released and discharged, and New BlackRock will be released from all obligations under the New BlackRock Guarantee.

The following presents unaudited summarized financial information of New BlackRock and Old BlackRock (together with the New BlackRock, the "Obligor Group") on a combined basis as of December 31, 2024 and for the period from October 1, 2024 to December 31, 2024. Intercompany balances and transactions between New BlackRock and Old BlackRock have been eliminated, and balances and transactions with subsidiaries, which are not part of the Obligor Group, have been separately presented, and investments in and equity in earnings related to subsidiaries of New BlackRock and Old BlackRock, which are not members of the Obligor Group, have been excluded.

Summarized Balance Sheet (unaudited)

(in millions)	**December 31, 2024**
Assets	
Receivables from non-guarantor subsidiaries	**$ 7,681**
Goodwill and intangible assets	**27,273**
Other assets	**362**
Total assets	**$ 35,316**
Liabilities	
Borrowings	**$ 12,314**
Payables to non-guarantor subsidiaries	**10,206**
Other liabilities	**3,278**
Total liabilities	**$ 25,798**

Summarized Income Statement (unaudited)

For 2024, net loss of the Obligor Group was $767 million and primarily comprised of $87 million amortization expense, a gain of $31 million related to a contingent consideration fair value adjustment, a $35 million impairment charge, $391 million of interest expense, and related taxes. Revenue during this period was not material.

Contractual Obligations, Commitments and Contingencies

The Company’s material contractual obligations, commitments and contingencies at December 31, 2024 include borrowings, operating leases, investment commitments, compensation and benefits obligations, purchase obligations, and contingent consideration liabilities.

Borrowings. At December 31, 2024, the Company had outstanding borrowings with varying maturities for an aggregate principal amount of $12.4 billion, of which $725 million is payable within 12 months. Future interest payments associated with these borrowings total $6.4 billion, of which $470 million is payable within 12 months. See Note 15, *Borrowings*, in the notes to the consolidated financial statements contained in Part II, Item 8 of this filing.

Operating Leases. The Company leases its primary office locations under agreements that expire on varying dates through 2043. At December 31, 2024, the Company had operating lease payment obligations of approximately $2.3 billion, of which $198 million is payable within 12 months. See Note 13, *Leases*, in the notes to the consolidated financial statements contained in Part II, Item 8 of this filing.

Investment Commitments. At December 31, 2024, the Company had $1.2 billion of various capital commitments to fund sponsored investment products, including CIPs. These products include various private market products, including private equity funds, real assets funds, and opportunistic funds. This amount excludes additional commitments made by consolidated funds of funds to underlying third-party funds as third-party noncontrolling interest holders have the legal obligation to fund the respective commitments of such funds of funds. Generally, the timing of the funding of these commitments is unknown and the commitments are callable on demand at any time prior to the expiration of the commitment. These unfunded commitments are not recorded on the consolidated statements of financial condition. These commitments do not include potential future commitments approved by the Company that are not yet legally binding. The Company intends to make additional capital commitments from time to time to fund additional investment products for, and with, its clients.

Compensation and Benefit Obligations. The Company has various compensation and benefit obligations, including bonuses, commissions and incentive payments payable, defined contribution plan matching contribution obligations, and deferred compensation arrangements. Accrued compensation and benefits at December 31, 2024 totaled $3.0 billion and included annual incentive compensation of $2.0 billion, deferred compensation of $0.4 billion and other compensation and benefits related obligations of $0.5 billion. Substantially all of the incentive compensation liability was paid in the first quarter of 2025, while the deferred compensation obligations are payable over various periods, with the majority payable over periods of up to three years.

Purchase Obligations. In the ordinary course of business, BlackRock enters into contracts or purchase obligations with third parties whereby the third parties provide services to or on behalf of BlackRock. Purchase obligations represent executory contracts, which are either noncancelable or cancelable with a penalty. At December 31, 2024, the Company's obligations primarily reflected standard service contracts for market data, technology, office-related services, marketing and promotional services, and obligations for equipment. Purchase obligations are recorded on the consolidated financial statements when services are provided and, as such, obligations for services and equipment not received are not included in the consolidated statement of financial condition at December 31, 2024. At December 31, 2024, the Company had purchase obligations of approximately $615 million, of which $260 million is payable within 12 months.

Contingent Consideration Liabilities. In connection with certain acquisitions, BlackRock is required to make contingent payments, subject to the achievement of specified performance targets or satisfaction of certain post-closing events. The fair value of this contingent consideration is estimated at the time of acquisition closing and is included in contingent consideration liabilities on the consolidated statements of financial condition. The fair value of the remaining aggregate contingent payments at December 31, 2024 totaled $4.3 billion, including $4.2 billion related to the GIP Transaction, which, if any, will be settled all in stock, for a number of shares ranging from 4.0 million to 5.2 million shares, subject to achieving certain performance targets. See Note 3, *Acquisitions*, in the notes to the consolidated financial statements contained in Part II, Item 8 of this filing for more information.

CRITICAL ACCOUNTING POLICIES AND ESTIMATES

The preparation of consolidated financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities at the date of the consolidated financial statements and the reported amounts of revenue and expense during the reporting periods. Actual results could differ significantly from those estimates. These estimates, judgments and assumptions are affected by the Company's application of accounting policies. Management considers the following accounting policies and estimates critical to understanding the consolidated financial statements. These policies and estimates are considered critical because they had a material impact, or are reasonably likely to have a material impact on the Company's consolidated financial statements and because they require management to make significant judgments, assumptions or estimates. For a summary of these and additional accounting policies see Note 2, *Significant Accounting Policies*, in the notes to the consolidated financial statements included in Part II, Item 8 of this filing.

Consolidation

The Company consolidates entities in which the Company has a controlling financial interest. The Company has a controlling financial interest when it owns a majority of the VRE or is a primary beneficiary ("PB") of a VIE. Assessing whether an entity is a VIE or a VRE involves judgment and analysis on a structure-by-structure basis. Factors considered in this assessment include the entity's legal organization, the entity's capital structure, the rights of equity investment holders, the Company's contractual involvement with and economic interest in the entity and any related party or de facto agent implications of the Company's involvement with the entity. Entities that are determined to be VREs are consolidated if the Company can exert absolute control over the financial and operating policies of the investee, which generally exists if there is greater than 50% voting interest. Entities that are determined to be VIEs are consolidated if the Company is the PB of the entity. BlackRock is deemed to be the PB of a VIE if it (1) has the power to direct the activities that most significantly impact the entities' economic performance and (2) has the obligation to absorb losses or the right to receive benefits that potentially could be significant to the VIE. There is judgment involved in assessing whether the Company is the PB of a VIE. In addition, the Company's ownership interest in VIEs is subject to variability and is impacted by actions of other investors such as ongoing redemptions and contributions. The Company generally consolidates VIEs in which it holds an economic interest of 10% or greater and deconsolidates such VIEs once its economic interest falls below 10%. As of December 31, 2024, the Company was deemed to be the PB of approximately 110 VIEs. See Note 6, *Consolidated Sponsored Investment Products*, in the notes to the consolidated financial statements contained in Part II, Item 8 of this filing for more information.

Fair Value Measurements

The Company's assessment of the significance of a particular input to the fair value measurement according to the fair value hierarchy (i.e., Level 1, 2 and 3 inputs, as defined) in its entirety requires judgment and considers factors specific to the financial instrument. See Note 2, *Significant Accounting Policies*, and Note 8, *Fair Value Disclosures*, in the consolidated financial statements contained in Part II, Item 8 of this filing for more information on fair value measurements.

Changes in Valuation. Changes in value on $7.1 billion of investments will impact the Company's nonoperating income (expense), $640 million are held at cost or amortized cost and the remaining $2.0 billion relates to carried interest, which will not impact nonoperating income (expense). At December 31, 2024, changes in fair value of $3.8 billion of CIPs will impact BlackRock's net income (loss) attributable to NCI on the consolidated statements of income. BlackRock's net exposure to changes in fair value of CIPs was $2.0 billion.

Goodwill and Intangible Assets

The Company accounts for business combinations using the acquisition method of accounting, where the purchase price is allocated to the assets acquired and liabilities assumed based on their fair values at the date of the transaction. Any excess purchase consideration over the fair value of net assets acquired is recorded as goodwill. The Company determines fair value of the tangible and identifiable intangible assets acquired and liabilities assumed, using the best available information which incorporates various estimates and assumptions, including, but not limited to, future expected cash flows, fundraising assumptions, useful lives, and discount rates. These estimates are based on historical data, internal estimates, and external sources. Unanticipated events may affect the validity of these assumptions.

During 2024, BlackRock recorded approximately $2.7 billion of finite-lived management contracts and investor relationships in connection with the GIP Transaction. The acquisition date fair values were determined using an income approach, which applied certain significant assumptions, which are inherently uncertain and unpredictable. These assumptions primarily included discounts rates ranging from 7%-11%, as well as estimated revenue projections based on AUM, AUM growth rates, revenue basis points, operating margins, and tax rates. While the Company believes these assumptions to be reasonable and appropriate, changes in these estimates could produce different fair value amounts.

Goodwill. The Company assesses its goodwill for impairment at least annually, considering qualitative factors such as entity-specific and macroeconomic factors as potential impairment indicators as well as quantitative factors such as the book value and the market capitalization of the Company. The impairment assessment performed as of July 31, 2024 indicated no impairment charge was required. The Company continues to monitor various impairment indicators as well as its book value per share compared with closing prices of its common stock for potential indicators of impairment. At December 31, 2024, the Company had $25.9 billion of goodwill, including $10.3 billion in connection with the GIP Transaction and the Company's common stock closed at $1,025, which exceeded its book value of $307 per share.

Indefinite-lived and finite-lived intangibles. Indefinite-lived intangible assets represent the value of advisory contracts acquired in business acquisitions to manage AUM in proprietary open-end investment funds, collective trust funds and certain other commingled products without a specified termination date. The assignment of indefinite lives to such contracts primarily is based upon the following: (1) the assumption that there is no foreseeable limit on the contract period to manage these products; (2) the Company expects to, and has the ability to, continue to operate these products indefinitely; (3) the products have multiple investors and are not reliant on a single investor or small group of investors for their continued operation; (4) current competitive factors and economic conditions do not indicate a finite life; and (5) there is a high likelihood of continued renewal based on historical experience. In addition, trade names/trademarks are considered indefinite-lived intangibles if they are expected to generate cash flows indefinitely. Indefinite-lived intangible assets are not amortized.

Finite-lived intangible assets represent finite-lived investor/customer relationships, technology related assets, and management contracts, which relate to acquired separate accounts and funds, that are expected to contribute to the future cash flows of the Company for a specified period of time. Finite-lived intangible assets are amortized over their remaining expected useful lives, which, at December 31, 2024 ranged from approximately 1 to 14 years with a weighted-average remaining estimated useful life of approximately 9 years.

The Company performs assessments to determine if any intangible assets are impaired at least annually, as of July 31, or more frequently if events or changes in circumstances indicate that it is more likely than not that the intangible assets might be impaired.

In evaluating whether it is more likely than not that the fair value of indefinite-lived intangibles is less than its carrying value, BlackRock performs certain quantitative assessments and assessed various significant quantitative factors including AUM, revenue basis points, projected AUM growth rates, operating margins, tax rates and discount rates. In addition, the Company considered other qualitative factors including: (1) macroeconomic conditions such as a deterioration in general economic conditions, limitations on accessing capital, fluctuations in foreign exchange rates, or other developments in equity and credit markets; (2) industry and market considerations such as a deterioration in the environment in which the Company operates, an increased competitive environment, a decline in market-dependent multiples or metrics, a change in the market for an entity's services, or regulatory, legal or political developments; and (3) Company-specific events, such as a change in management or key personnel, overall financial performance and litigation that could affect significant inputs used to determine the fair value of the indefinite-lived intangible asset. If an indefinite-lived intangible is determined to be more likely than not impaired, then the fair value of the asset, which is generally determined using an income approach, is compared with its carrying value and any excess of the carrying value over the fair value would be recognized as an expense in the period in which the impairment occurs.

For finite-lived intangible assets, if potential impairment circumstances are considered to exist, the Company will perform a recoverability test, using an undiscounted cash flow analysis. Factors included in evaluating finite-lived customer relationships, technology related assets and trade names include technology services revenue trends, customer attrition rates, obsolescence rates, and royalty rates. For finite-lived management contracts, evaluation is based on changes in assumptions including AUM, revenue basis points, projected AUM growth rates, operating margins, tax rates and discount rates. Actual results could differ from these cash flow estimates, which could materially impact the impairment conclusion. If the carrying value of the asset is determined not to be recoverable based on the undiscounted cash flow test, the difference between the book value of the asset and its current estimated fair value would be recognized as an expense in the period in which the impairment occurs.

In addition, management judgment is required to estimate the period over which finite-lived intangible assets will contribute to the Company's cash flows and the pattern in which these assets will be consumed and whether the indefinite-life and finite-life classifications are still appropriate. A change in the remaining useful life of any of these assets, or the reclassification of an indefinite-lived intangible asset to a finite-lived intangible asset, could have a significant impact on the Company's amortization expense, which was $241 million, $151 million and $151 million for 2024, 2023 and 2022, respectively.

In 2024, 2023 and 2022, the Company performed its annual impairment assessment, including evaluating various qualitative factors and performing certain quantitative assessments. In 2024, based on this assessment, the Company determined that the indefinite-lived intangible assets related to certain acquired open-end management contracts were impaired, and as a result, recorded a noncash impairment charge of $50 million, included within amortization and impairment of intangible assets expense on the consolidated statements of income. The impairment was primarily the result of a decrease in certain quantitative factors, including reduced growth expectation, lower revenue basis points and net client outflows, which caused the fair value to decline below its carrying value. While the Company believes all assumptions utilized in the analysis are reasonable and appropriate, changes in these estimates could produce different fair value amounts, which could drive additional impairment in future periods. In addition, the Company determined, that no impairment charges were required for any other intangible assets, and that the classification of indefinite-lived versus finite-lived intangibles was still appropriate and no changes were required to the expected lives of the finite-lived intangibles. In 2023 and 2022, the Company determined that no impairment charges were required and that the classification of indefinite-lived versus finite-lived intangibles was still appropriate and no changes were required to the expected lives of the finite-lived intangibles. The Company continuously monitors various factors, including AUM, for potential indicators of impairment.

Contingent Consideration Liabilities

In connection with certain acquisitions, BlackRock is required to make contingent payments, subject to the achievement of specified performance targets or satisfaction of certain post-closing events. The fair value of this contingent consideration is estimated at the time of acquisition closing and is included in contingent consideration liabilities on the consolidated statements of financial condition. The fair value of the remaining aggregate contingent payments at December 31, 2024 totaled $4.3 billion, including $4.2 billion related to the GIP Transaction, which, if any, will be settled all in stock, ranging from 4.0 million to 5.2 million shares, subject to achieving certain performance targets. The fair value of the GIP Transaction contingent consideration was estimated using the income approach, which included certain significant inputs such as a risk-free discount rate of approximately 4.3% as well as current estimates of the timing and amounts of fundraising forecasts, stock and AUM volatility, and correlation between stock price and AUM (Level 3 inputs). As the estimated fair value of the contingent consideration subsequently changes, contingent consideration liabilities are adjusted, resulting in contingent consideration fair value adjustments recorded within general and administration expense of the consolidated statements of income until the contingency is resolved. Accordingly, changes in the key inputs and assumptions described can materially impact the amount of contingent consideration expense recorded in a reporting period.

Revenue Recognition

The Company recognizes revenues when its obligations related to the services are satisfied and it is probable that a significant reversal of the revenue amount would not occur in future periods. The Company enters into contracts that can include multiple services, which are accounted for separately if they are determined to be distinct. Management judgment is required in assessing the probability of significant revenue reversal and in identification of distinct services.

The Company derives a substantial portion of its revenue from investment advisory and administration fees which are recognized as the services are performed over time because the customer is receiving and consuming the benefits as they are provided by the Company. Fees are primarily based on agreed-upon percentages of AUM and recognized for services provided during the period, which are distinct from services provided in other periods. Such fees are affected by changes in AUM, including market appreciation or depreciation, foreign exchange translation and net inflows or outflows. AUM represents the broad range of financial assets the Company manages for clients on a discretionary basis pursuant to investment management and trust agreements that are expected to continue for at least 12 months. In general, reported AUM reflects the valuation methodology that corresponds to the basis used for determining revenue (for example, net asset values).

The Company receives investment advisory performance fees, including incentive allocations (carried interest) from certain actively managed investment funds and certain separately managed accounts ("SMAs"). These performance fees are dependent upon exceeding specified relative or absolute investment return thresholds, which vary by product or account, and include monthly, quarterly, annual or longer measurement periods.

Performance fees, including carried interest, are generated on certain management contracts when performance hurdles are achieved. Such performance fees are recognized when the contractual performance criteria have been met and when it is determined that they are no longer probable of significant reversal. Given the unique nature of each fee arrangement, contracts with customers are evaluated on an individual basis to determine the timing of revenue recognition. Significant judgment is involved in making such determination. Performance fees typically arise from investment management services that began in prior reporting periods. Consequently, a portion of the fees the Company recognizes may be partially related to the services performed in prior periods that meet the recognition criteria in the current period. At each reporting date, the Company considers various factors in estimating performance fees to be recognized, including carried interest. These factors include but are not limited to whether: (1) the amounts are dependent on the financial markets and, thus, are highly susceptible to factors outside the Company's influence; (2) the ultimate payments have a large number and a broad range of possible amounts; and (3) the funds or SMAs have the ability to (a) invest or reinvest their sales proceeds or (b) distribute their sales proceeds, and determine the timing of such distributions.

The Company is allocated/distributed carried interest from certain alternative investment products upon exceeding performance thresholds. The Company may be required to reverse/return all, or part, of such carried interest allocations/distributions depending upon future performance of these products. Carried interest subject to such clawback provisions is recorded in investments or cash and cash equivalents to the extent that it is distributed, on the Company's consolidated statements of financial condition.

The Company records a liability for deferred carried interest to the extent it receives cash or capital allocations related to carried interest prior to meeting the revenue recognition criteria. At December 31, 2024 and 2023, the Company had $1.9 billion and $1.8 billion, respectively, of deferred carried interest recorded in other liabilities on the consolidated statements of financial condition. A portion of the deferred carried interest may also be paid to certain employees and other third parties, which may be subject to clawback. The ultimate timing of the recognition of performance fee revenue and related compensation expense, if any, is unknown. See Note 17, *Revenue*, in the notes to the consolidated financial statements for detailed changes in the deferred carried interest liability balance for 2024 and 2023.

The Company earns revenue for providing technology services. Determining the amount of revenue to recognize requires judgment and estimates. Complex arrangements with nonstandard terms and conditions may require contract interpretation to determine the appropriate accounting, including whether promised goods and services specified in an arrangement, are distinct performance obligations, and should be accounted for separately. Other judgments include determining whether performance obligations are satisfied over time or at a point in time. Fees earned for technology services are primarily recorded as services are performed over time and are generally determined using the value of positions on the Aladdin platform or on a fixed-rate basis. Revenue derived from the sale of software licenses is recognized upon the granting of access rights.

Adjustments to revenue arising from initial estimates recorded historically have been immaterial since the majority of BlackRock's investment advisory and administration revenue is calculated based on AUM, recognized when known, and given the Company does not record performance fee revenue until: (1) performance thresholds have been exceeded and (2) management determines the fees are no longer probable of significant reversal. See Note 2, *Significant Accounting Policies*, in the consolidated financial statements contained in Part II, Item 8 of this filing for more information on revenue recognition, including other revenue streams.

Income Taxes

The Company records income taxes based upon its estimated income tax liability or benefit. The Company's actual tax liability or benefit may differ from the estimated income tax liability or benefit.

Deferred income tax assets and liabilities are recognized for future tax consequences attributable to temporary differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases using currently enacted tax rates in effect for the year in which the differences are expected to reverse. The effect of a change in tax rates on deferred tax assets and liabilities is recognized in income in the period that includes the enactment date.

Significant management judgment is required in estimating the ranges of possible outcomes and determining the probability of favorable or unfavorable tax outcomes and potential interest and penalties related to such unfavorable outcomes. Actual future tax consequences relating to uncertain tax positions may be materially different than the Company's current estimates. At December 31, 2024, BlackRock had $517 million of gross unrecognized tax benefits, of which $431 million, if recognized, would affect the effective tax rate.

Management is required to estimate the timing of the recognition of deferred tax assets and liabilities, make assumptions about the future deductibility of deferred income tax assets and assess deferred income tax liabilities based on enacted tax rates for the appropriate tax jurisdictions to determine the amount of such deferred income tax assets and liabilities. At December 31, 2024, the Company had deferred income tax assets of $181 million and deferred income tax liabilities of $3.3 billion on the consolidated statement of financial condition. Changes in deferred tax assets and liabilities may occur in certain circumstances, including statutory income tax rate changes, statutory tax law changes, changes in the anticipated timing of recognition of deferred tax assets and liabilities or changes in the structure or tax status of the Company.

The Company assesses whether a valuation allowance should be established against its deferred income tax assets based on consideration of all available evidence, both positive and negative, using a more likely than not standard. The assessment considers, among other matters, the nature, frequency and severity of recent losses, forecast of future profitability, the duration of statutory carry back and carry forward periods, the Company's experience with tax attributes expiring unused, and tax planning alternatives.

Accounting Developments

For accounting pronouncements that the Company adopted during 2024 and for accounting pronouncements not yet adopted by the Company, see Note 2, *Significant Accounting Policies*, in the consolidated financial statements contained in Part II, Item 8 of this filing.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

AUM Market Price Risk. BlackRock's investment advisory and administration fees are primarily comprised of fees based on a percentage of the value of AUM and, in some cases, performance fees expressed as a percentage of the returns realized on AUM. At December 31, 2024, the majority of the Company's investment advisory and administration fees were based on average or period end AUM of the applicable investment funds or separate accounts. Movements in equity market prices, interest rates/credit spreads, foreign exchange rates or all three could cause the value of AUM to decline, which would result in lower investment advisory and administration fees.

Corporate Investments Portfolio Risks. As a leading investment management firm, BlackRock devotes significant resources across all of its operations to identifying, measuring, monitoring, managing and analyzing market and operating risks, including the management and oversight of its own investment portfolio. The Board of Directors of the Company has adopted guidelines for the review of investments (or commitments to invest) to be made by the Company, requiring, among other things, that certain investments be referred to the Board of Directors, depending on the circumstances, for notification or approval.

In the normal course of its business, BlackRock is exposed to equity market price risk, interest rate/credit spread risk and foreign exchange rate risk associated with its corporate investments.

BlackRock has investments primarily in sponsored investment products that invest in a variety of asset classes, including real assets, private equity and hedge funds. Investments generally are made for co-investment purposes, to establish a performance track record, to hedge exposure to certain deferred cash compensation plans or for regulatory purposes. The Company has a seed capital hedging program in which it enters into futures to hedge market and interest rate exposure with respect to its total portfolio of seed investments in sponsored investment products. The Company had outstanding futures related to its seed capital hedging program with an aggregate notional value of approximately $1.8 billion at both December 31, 2024 and 2023.

At December 31, 2024 and 2023, approximately $5.8 billion and $6.0 billion, respectively, of BlackRock's investments were held in consolidated sponsored investment products accounted for as variable interest entities or voting rights entities. Excluding the impact of the Federal Reserve Bank stock, carried interest, investments made to hedge exposure to certain deferred cash compensation plans and certain investments that are hedged via the seed capital hedging program, the Company's economic exposure to its investment portfolio at December 31, 2024 and 2023 were $3.9 billion and $3.8 billion, respectively. See Item 7, *Management's Discussion and Analysis of Financial Condition and Results of Operations – Statement of Financial Condition Overview – Investments* for further information on the Company's investments.

Equity Market Price Risk. Investments subject to market price risk include public and private equity and real assets investments, hedge funds and funds of funds as well as mutual funds. The following table provides the Company's net exposure to equity market price risk and hypothetical exposure to a 10% adverse change in market prices:

	As of December 31,			
	2024		**2023**	
(in millions)	**Net Exposure**	**Effect of -10% Change**	**Net Exposure**	**Effect of -10% Change**
Equity Market Price Risk				
Investments	**$ 1,899**	**$ 190**	$ 1,684	$ 168

Interest Rate/Credit Spread Risk. Investments subject to interest rate and credit spread risk include debt securities and sponsored investment products that invest primarily in debt securities. The following table provides the Company's exposure to interest rate risk and credit spread risk and hypothetical exposure to an adverse 100 basis point fluctuation in interest rates or credit spreads:

	As of December 31,			
	2024		**2023**	
(in millions)	**Exposure**	**Effect of -100 Basis Point Change**	**Exposure**	**Effect of -100 Basis Point Change**
Interest Rate/Credit Spread Risk				
Investments	**$ 1,977**	**$ 54**	$ 2,088	$ 53

Foreign Exchange Rate Risk. As discussed above, the Company invests in sponsored investment products that invest in a variety of asset classes. The carrying value of the total economic investment exposure denominated in foreign currencies are primarily based in the British pound and euro. The following table provides the Company's exposure to foreign currencies and hypothetical exposure to a 10% adverse change in the applicable foreign exchange rates:

	As of December 31,			
	2024		**2023**	
(in millions)	**Exposure**	**Effect of -10% Change**	**Exposure**	**Effect of -10% Change**
Foreign Exchange Rate Risk				
Investments	**$ 1,111**	**$ 111**	$ 1,125	$ 112

Other Market Risks. The Company executes forward foreign currency exchange contracts to mitigate the risk of certain foreign exchange risk movements. At December 31, 2024 and 2023, the Company had outstanding forward foreign currency exchange contracts with an aggregate notional value of approximately $3.6 billion and $3.1 billion, with expiration dates in January 2025 and 2024, respectively. In addition, the Company entered into futures to hedge economically the exposure to market movements on certain deferred cash compensation plans. At December 31, 2024 and 2023, the Company had outstanding exchange traded futures with aggregate notional values related to its deferred cash compensation hedging program of approximately $197 million and $204 million, with expiration dates during the first quarter of 2025 and 2024, respectively.

Item 8. Financial Statements and Supplemental Data

The report of the independent registered public accounting firm and financial statements listed in the accompanying index are included in Item 15 of this report. See Index to the consolidated financial statements on page F-1 of this Form 10-K.

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

There have been no disagreements on accounting and financial disclosure matters. BlackRock has not changed accountants in the two most recent fiscal years.

Item 9A. Controls and Procedures

Disclosure Controls and Procedures. Under the direction of BlackRock's Chief Executive Officer and Chief Financial Officer, BlackRock evaluated the effectiveness of its disclosure controls and procedures (as such term is defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act) as of the end of the period covered by this annual report on Form 10-K. Based on this evaluation, BlackRock's Chief Executive Officer and Chief Financial Officer have concluded that BlackRock's disclosure controls and procedures were effective.

Internal Control over Financial Reporting. There were no changes in our internal control over financial reporting that occurred during the fourth quarter of the fiscal year ending December 31, 2024 that have materially affected or are reasonably likely to materially affect our internal control over financial reporting.

Management’s Report on Internal Control Over Financial Reporting

Management of BlackRock, Inc. (the “Company”) is responsible for establishing and maintaining effective internal control over financial reporting. Internal control over financial reporting is defined in Rules 13a-15(f) and 15d-15(f) under the Securities Exchange Act of 1934, as amended, as a process designed by, or under the supervision of, the Company’s principal executive and principal financial officers, or persons performing similar functions, and effected by the Company’s board of directors, management and other personnel, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with accounting principles generally accepted in the United States of America and includes those policies and procedures that:

- pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the Company;
- provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with accounting principles generally accepted in the United States of America, and that receipts and expenditures of the Company are being made only in accordance with the authorizations of management and directors of the Company; and
- provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of the Company’s assets that could have a material effect on the financial statements.

Because of the inherent limitations of internal control over financial reporting, including the possibility of collusion or improper management override of controls, material misstatements due to error or fraud may not be prevented or detected on a timely basis. Also, projections of any evaluation of effectiveness of the internal control over financial reporting to future periods are subject to the risks that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Management assessed the effectiveness of the Company’s internal control over financial reporting as of December 31, 2024 based on the criteria established in *Internal Control — Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission. Based on this assessment, management concluded that, as of December 31, 2024, the Company’s internal control over financial reporting is effective.

The Company’s independent registered public accounting firm has issued an attestation report on the effectiveness of the Company’s internal control over financial reporting.

February 25, 2025

Report of Independent Registered Public Accounting Firm

To the Board of Directors and Stockholders of BlackRock, Inc.:

Opinion on Internal Control over Financial Reporting

We have audited the internal control over financial reporting of BlackRock, Inc. and subsidiaries (the "Company") as of December 31, 2024, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO). In our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2024, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by COSO.

We have also audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated financial statements as of and for the year ended December 31, 2024, of the Company and our report dated February 25, 2025, expressed an unqualified opinion on those financial statements.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects. Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Deloitte & Touche LLP
New York, New York
February 25, 2025

Item 9B. Other Information

The Company is furnishing no other information in this Form 10-K.

Item 9C. Disclosure Regarding Foreign Jurisdictions That Prevent Inspections

Not applicable.

PART III

Item 10. Directors, Executive Officers and Corporate Governance

The information regarding directors and executive officers set forth under the captions "Item 1: Election of Directors – Director Nominee Biographies" and "Corporate Governance – Other Executive Officers" of the Proxy Statement is incorporated herein by reference.

Information regarding compliance with Section 16(a) of the Exchange Act required by Item 10, if any, is set forth under the caption "Delinquent Section 16(a) Reports" of the Proxy Statement and incorporated herein by reference.

The information regarding BlackRock's Code of Ethics for Chief Executive and Senior Financial Officers under the caption "Corporate Governance – Our Corporate Governance Framework" of the Proxy Statement is incorporated herein by reference.

The information regarding BlackRock's Audit Committee under the caption "Corporate Governance – Board Committees" of the Proxy Statement is incorporated herein by reference.

Item 11. Executive Compensation

The information contained in the sections captioned "Management Development & Compensation Committee Interlocks and Insider Participation," "Executive Compensation – Compensation Discussion and Analysis" and "Corporate Governance – 2024 Director Compensation" of the Proxy Statement is incorporated herein by reference.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters

The information contained in the sections captioned "Ownership of BlackRock Common Stock" and "Executive Compensation – Compensation Discussion and Analysis – 6. Executive Compensation Tables – Equity Compensation Plan Information" of the Proxy Statement is incorporated herein by reference.

Item 13. Certain Relationships and Related Transactions, and Director Independence

The information contained in the sections captioned "Certain Relationships and Related Transactions" and "Item 1: Election of Directors – Criteria for Board Membership – Director Independence" of the Proxy Statement is incorporated herein by reference.

Item 14. Principal Accountant Fees and Services

The information regarding BlackRock's independent auditor fees and services in the section captioned "Item 3: Ratification of the Appointment of the Independent Registered Public Accounting Firm" of the Proxy Statement is incorporated herein by reference.

PART IV

Item 15. Exhibits and Financial Statement Schedules

1. Financial Statements

The Company's consolidated financial statements are included beginning on page F-1.

2. Financial Statement Schedules

Financial statement schedules have been omitted because they are not applicable, not required or the information required is included in the Company's consolidated financial statements or notes thereto.

3. Exhibit Index

Please note that the agreements included as exhibits to this Form 10-K are included to provide information regarding their terms and are not intended to provide any other factual or disclosure information about BlackRock or the other parties to the agreements. The agreements contain representations and warranties by each of the parties to the applicable agreement that have been made solely for the benefit of the other parties to the applicable agreement and may not describe the actual state of affairs as of the date they were made or at any other time.

Exhibit No.	Description
3.1	Restated Certificate of Incorporation of BlackRock, Inc.
3.2(1)	Amended and Restated Bylaws of BlackRock, Inc.
4.1(2)	Indenture, dated September 17, 2007, between BlackRock Finance, Inc. and The Bank of New York, as trustee
4.2(3)	Form of 1.250% Notes due 2025
4.3(4)	Form of 3.200% Notes due 2027
4.4(5)	Form of 3.250% Notes due 2029
4.5(6)	Form of 2.400% Notes due 2030
4.6(7)	Form of 1.900% Notes due 2031
4.7(8)	Form of 2.10% Notes due 2032
4.8(9)	Form of 4.750% Notes due 2033
4.9(3)	Officers' Certificate, dated May 6, 2015, for the 1.250% Notes due 2025 issued pursuant to the Indenture
4.10(10)	Indenture, dated March 14, 2024, among BlackRock, Inc., BlackRock Finance, Inc. and The Bank of New York Mellon, as trustee.
4.11(10)	First Supplemental Indenture, dated March 14, 2024, among BlackRock, Inc., BlackRock Finance, Inc. and The Bank of New York Mellon, as trustee.
4.12(11)	Second Supplemental Indenture, dated July 26, 2024, among BlackRock, Inc., BlackRock Finance, Inc. and The Bank of New York Mellon, as trustee
4.13(10)	Form of Note for the 4.700% Notes due 2029
4.14(10)	Form of Note for the 5.000% Notes due 2034
4.15(10)	Form of Note for the 5.250% Notes due 2054
4.16(11)	Form of Note for the 4.600% Notes due 2027
4.17(11)	Form of Note for the 4.900% Notes due 2035
4.18(11)	Form of Note for the 5.350% Notes due 2055
4.19(1)	Guarantee of BlackRock Finance, Inc. Indebtedness, dated October 1, 2024
4.20	Description of Securities
10.1(12)	BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan+
10.2(13)	Amended and Restated BlackRock, Inc. 1999 Annual Incentive Performance Plan+
10.3(14)	Amendment No. 1 to the BlackRock, Inc. Amended and Restated 1999 Annual Incentive Performance Plan+
10.4(15)	Form of Restricted Stock Unit Agreement under the BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan+
10.5(15)	Form of Performance-Based Restricted Stock Unit Agreement (BPIP) under the BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan+
10.6(16)	Form of Performance-Based Stock Option Agreement under the BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan+
10.7(17)	Form of Stock Option Agreement expected to be used in connection with future grants of Stock Options under the BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan+
10.8(17)	Form of Restricted Stock Agreement expected to be used in connection with future grants of Restricted Stock under the BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan+
10.9(17)	Form of Directors' Restricted Stock Unit Agreement expected to be used in connection with future grants of Restricted Stock Units under the BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan+
10.10(18)	BlackRock, Inc. Amended and Restated Voluntary Deferred Compensation Plan, as amended and restated as of November 16, 2015+
10.11(19)	Five-Year Revolving Credit Agreement, dated as of March 10, 2011, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender and L/C agent, Sumitomo Mitsui Banking Corporation, as Japanese Yen lender, a group of lenders, Wells Fargo Securities, LLC, Citigroup Global Markets Inc., Merrill Lynch, Pierce, Fenner & Smith Incorporated, Barclays Capital, J.P. Morgan Securities LLC and Morgan Stanley Senior Funding, Inc., as joint lead arrangers and joint bookrunners, Citibank, N.A., as syndication agent and Bank of America, N.A., Barclays Bank PLC, JPMorgan Chase Bank, N.A. and Morgan Stanley Senior Funding, Inc., as documentation agents
10.12(20)	Amendment No. 1, dated as of March 30, 2012, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein

Exhibit No.	Description
10.13(21)	Amendment No. 2, dated as of March 28, 2013, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.14(22)	Amendment No. 3, dated as of March 28, 2014, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.15(23)	Amendment No. 4, dated as of April 2, 2015, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.16(24)	Amendment No. 5, dated as of April 8, 2016, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.17(25)	Amendment No. 6, dated as of April 6, 2017, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.18(26)	Amendment No. 7, dated as of April 3, 2018, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.19(27)	Amendment No. 8, dated as of March 29, 2019, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.20(28)	Amendment No. 9, dated as of March 31, 2020, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, swingline lender, issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.21(29)	Amendment No. 10, dated as of March 31, 2021, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, a swingline lender, an issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.22(30)	Amendment No. 11, dated as of December 13, 2021, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, a swingline lender, an issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.23(31)	Amendment No. 12, dated as of March 31, 2022, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, a swingline lender, an issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.24(32)	Amendment No. 13, dated as of March 31, 2023, by and among BlackRock Finance, Inc., certain of its subsidiaries, Wells Fargo Bank, National Association, as administrative agent, a swingline lender, an issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.25(33)	Amendment No. 14, dated as of March 12, 2024, by and among BlackRock, Inc., BlackRock Finance, Inc., certain subsidiaries of BlackRock, Inc., Wells Fargo Bank, National Association, as administrative agent, a swingline lender, an issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein**
10.26(34)	Amendment No. 15, dated as of May 31, 2024, by and among BlackRock, Inc., BlackRock Finance, Inc., certain subsidiaries of BlackRock, Inc., Wells Fargo Bank, National Association, as administrative agent, a swingline lender, an issuing lender, L/C agent and a lender, and the banks and other financial institutions referred to therein
10.27(35)	Form of Commercial Paper Dealer Agreement among BlackRock, Inc., as issuer, BlackRock Finance, Inc., as guarantor, and the applicable Dealer party thereto
10.28(36)	Lease Agreement, dated as of February 17, 2010, among BlackRock Investment Management (UK) Limited and Mourant & Co Trustees Limited and Mourant Property Trustees Limited as Trustees of the Drapers Gardens Unit Trust for the lease of Drapers Gardens, 12 Throgmorton Avenue, London, EC2, United Kingdom
10.29(37)	Lease, by and between BlackRock Finance, Inc. and 50 HYMC Holdings LLC*
10.30(38)	BlackRock, Inc. Leadership Retention Carry Plan+
10.31(39)	Form of Percentage Points Award Agreement pursuant to the BlackRock, Inc. Leadership Retention Carry Plan+
10.32(40)	Form of Performance-Based Stock Option Agreement under the BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan+
10.33	Adebayo O. Ogunlesi Offer Letter+
10.34	Form of Carry Points Award Letter for Adebayo O. Ogunlesi+
19.1	Global Insider Trading Policy
21.1	Subsidiaries of Registrant
22.1	Subsidiary Guarantor and Issuer of Guaranteed Securities
23.1	Deloitte & Touche LLP Consent
31.1	Section 302 Certification of Chief Executive Officer
31.2	Section 302 Certification of Chief Financial Officer
32.1	Section 906 Certification of Chief Executive Officer and Chief Financial Officer
97.1(41)	Policy Relating to Recovery of Erroneously Awarded Compensation
101.INS	Inline XBRL Instance Document – the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document.
101.SCH	Inline XBRL Taxonomy Extension Schema with Embedded Linkbases Document.
104	Cover Page Interactive Data File (formatted as Inline XBRL and contained in Exhibit 101).

(1) Incorporated by reference to BlackRock, Inc.'s Current Report on Form 8-K12B filed on October 1, 2024.

(2) Incorporated by reference to BlackRock Finance, Inc.'s Annual Report on Form 10-K for the year ended December 31, 2007.

(3) Incorporated by reference to BlackRock Finance, Inc.'s Current Report on Form 8-K filed on May 6, 2015.

(4) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on March 28, 2017.
(5) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 29, 2019.
(6) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on January 27, 2020.
(7) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on May 6, 2020.
(8) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on December 10, 2021.
(9) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on May 25, 2023.
(10) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on March 14, 2024.
(11) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on July 26, 2024.
(12) Incorporated by reference to BlackRock’ Finance’s Definitive Proxy Statement on Form DEF 14A filed on April 4, 2024.
(13) Incorporated by reference to BlackRock Holdco 2, Inc.’s Annual Report on Form 10-K for the year ended December 31, 2002.
(14) Incorporated by reference to BlackRock Holdco 2, Inc.’s Current Report on Form 8-K filed on May 24, 2006.
(15) Incorporated by reference to BlackRock Finance, Inc.’s Quarterly Report on Form 10-Q for the quarter ended June 30, 2015.
(16) Incorporated by reference to BlackRock Finance, Inc.’s Quarterly Report on Form 10-Q for the quarter ended March 31, 2018.
(17) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on October 5, 2006.
(18) Incorporated by reference to BlackRock Finance, Inc.’s Annual Report on Form 10-K for the year ended December 31, 2015.
(19) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K/A filed on August 24, 2012.
(20) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 4, 2012.
(21) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 3, 2013.
(22) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on March 28, 2014.
(23) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 3, 2015.
(24) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 14, 2016.
(25) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 11, 2017.
(26) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 6, 2018.
(27) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on March 29, 2019.
(28) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 1, 2020.
(29) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 6, 2021.
(30) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on December 13, 2021.
(31) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 1, 2022.
(32) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on April 3, 2023.
(33) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on March 15, 2024.
(34) Incorporated by reference to BlackRock Finance, Inc.’s Current Report on Form 8-K filed on May 31, 2024.
(35) Incorporated by reference to BlackRock, Inc.’s Current Report on Form 8-K filed on November 8, 2024.
(36) Incorporated by reference to BlackRock Finance, Inc.’s Annual Report on Form 10-K for the year ended December 31, 2009.
(37) Incorporated by reference to BlackRock Finance, Inc.’s Quarterly Report on Form 10-Q for the quarter ended June 30, 2017.
(38) Incorporated by reference to BlackRock Finance, Inc.’s Annual Report on Form 10-K for the year ended December 31, 2019.
(39) Incorporated by reference to BlackRock Finance, Inc.’s Quarterly Report on Form 10-Q for the quarter ended June 30, 2019.
(40) Incorporated by reference to BlackRock Finance, Inc.’s Quarterly Report on Form 10-Q for the quarter ended June 30, 2023.
(41) Incorporated by reference to BlackRock Finance, Inc.’s Annual Report on Form 10-K for the year ended December 31, 2023.

+ Denotes compensatory plans or arrangements.

* Portions of this exhibit have been omitted pursuant to a confidential treatment order from the SEC.

** Portions of this exhibit have been omitted pursuant to Item 601(b)(10)(iv) of Regulation S-K.

Item 16. Form 10-K Summary

Not applicable.

Signatures

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

BLACKROCK, INC.

By: /s/ Laurence D. Fink

Laurence D. Fink
Chairman, Chief Executive Officer and Director

February 25, 2025

Each of the officers and directors of BlackRock, Inc. whose signature appears below, in so signing, also makes, constitutes and appoints Laurence D. Fink, Martin S. Small, Christopher J. Meade, Laura Hildner and R. Andrew Dickson III, his or her true and lawful attorneys-in-fact, with full power and substitution, for him or her in any and all capacities, to execute and cause to be filed with the Securities and Exchange Commission any and all amendments to the Annual Report on Form 10-K, with exhibits thereto and other documents connected therewith and to perform any acts necessary to be done in order to file such documents, and hereby ratifies and confirms all that said attorney-in-fact or his or her substitute or substitutes may do or cause to be done by virtue hereof.

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Laurence D. Fink **Laurence D. Fink**	Chairman, Chief Executive Officer and Director (Principal Executive Officer)	February 25, 2025
/s/ Martin S. Small **Martin S. Small**	Senior Managing Director and Chief Financial Officer (Principal Financial Officer)	February 25, 2025
/s/ Marc D. Comerchero **Marc D. Comerchero**	Managing Director and Chief Accounting Officer (Principal Accounting Officer)	February 25, 2025
/s/ Pamela Daley **Pamela Daley**	Director	February 25, 2025
/s/ William E. Ford **William E. Ford**	Director	February 25, 2025
/s/ Fabrizio Freda **Fabrizio Freda**	Director	February 25, 2025
/s/ Murry S. Gerber **Murry S. Gerber**	Director	February 25, 2025
/s/ Margaret L. Johnson **Margaret L. Johnson**	Director	February 25, 2025
/s/ Robert S. Kapito **Robert S. Kapito**	Director	February 25, 2025
/s/ Cheryl D. Mills **Cheryl D. Mills**	Director	February 25, 2025
/s/ Amin H. Nasser **Amin H. Nasser**	Director	February 25, 2025
/s/ Gordon M. Nixon **Gordon M. Nixon**	Director	February 25, 2025
/s/ Adebayo Ogunlesi **Adebayo Ogunlesi**	Director	February 25, 2025
/s/ Kristin Peck **Kristin Peck**	Director	February 25, 2025
/s/ Charles H. Robbins **Charles H. Robbins**	Director	February 25, 2025
/s/ Marco Antonio Slim Domit **Marco Antonio Slim Domit**	Director	February 25, 2025
/s/ Hans E. Vestberg **Hans E. Vestberg**	Director	February 25, 2025
/s/ Susan L. Wagner **Susan L. Wagner**	Director	February 25, 2025
/s/ Mark Wilson **Mark Wilson**	Director	February 25, 2025

Index to Financial Statements

Report of Independent Registered Public Accounting Firm

To the Board of Directors and Stockholders of BlackRock, Inc.:

Opinion on the Financial Statements

We have audited the accompanying consolidated statements of financial condition of BlackRock, Inc. and subsidiaries (the "Company") as of December 31, 2024 and 2023, the related consolidated statements of income, comprehensive income, changes in equity, and cash flows, for each of the three years in the period ended December 31, 2024, and the related notes (collectively referred to as the "financial statements"). In our opinion, the financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2024 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2024, in conformity with accounting principles generally accepted in the United States of America.

We have also audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2024, based on criteria established in *Internal Control — Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission and our report dated February 25, 2025, expressed an unqualified opinion on the Company's internal control over financial reporting.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current-period audit of the financial statements that were communicated or required to be communicated to the audit committee and that (1) relate to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Acquisition of Global Infrastructure Management, LLC (“GIP”) – Fair Value of intangible assets related to management contracts and contingent consideration — Refer to Notes 2 and 3 to the financial statements

Critical Audit Matter Description

On October 1, 2024, the Company completed the acquisition of GIP for consideration at close of approximately $3 billion in cash and 6.9 million shares of BlackRock common stock. In addition, as part of the purchase consideration, a contingent consideration payment, all in stock, may be due subject to achieving certain performance targets. The Company accounts for the acquisition under the acquisition method of accounting for business combinations. Accordingly, the Company allocated the purchase price to the assets acquired and liabilities assumed, based on their estimated fair values at the date of the transaction close. The Company recorded approximately $1.8 billion of intangible assets related to management contracts as of the acquisition date (the “acquired management contracts”). Additionally, the Company recorded an estimated contingent consideration liability of approximately $4.2 billion.

Acquired management contracts are valued using discounted cash flow methods based on future cash flows specific to the type of intangible asset acquired. The determination of fair value requires management to make estimates and assumptions related to forecasted revenue and cash flows, and the determination of the discount rates.

The fair value of the contingent consideration was determined using an income approach where management develops projections during the contingent consideration period utilizing various potential pay-out scenarios. The determination of fair value requires management to make estimates and assumptions related to projected assets under management (“AUM”), discount rate and common stock volatility.

Given the fair value determination of acquired management contracts and contingent consideration requires management to make significant estimates and assumptions, performing audit procedures to evaluate the reasonableness of these estimates and assumptions required a high degree of auditor judgment and an increased extent of effort, including the need to involve our fair value specialists.

How the Critical Audit Matter Was Addressed in the Audit

Our audit procedures related to the valuation of acquired management contracts and contingent consideration included the following, among others:

- We tested the design and operating effectiveness of controls over the Company’s valuation of acquired management contracts and contingent consideration, including management’s controls over forecasts of future revenue and cash flows, projected AUM, and determination of discount rates.
- We evaluated the reasonableness of management’s forecasted revenue and cash flows and projected AUM, by comparing management’s projections to historical results, contractual revenue agreements in place, and relevant industry reports and evaluated whether the forecasted revenue and cash flows and projected AUM were consistent with evidence obtained in other areas of the audit.

- With the assistance of our fair value specialists, we evaluated the reasonableness of the Company’s valuation methodologies and valuation assumptions used in the fair value analysis by:
  - Testing the source information underlying the determination of the discount rates and the mathematical accuracy of the valuation.
  - For acquired management contracts, evaluating the reasonableness of the methodology used, and developing a range of independent estimates and comparing those to the discount rates selected by management.
  - For contingent consideration, evaluating the reasonableness of the methodology used, the acceptability of the valuation assumptions used, as well as evaluating the probability of contingent future events to calculate an independent estimate of fair value.

Impairment of indefinite-lived intangible assets related to certain management contracts — Refer to Notes 2 and 12 to the financial statements

Critical Audit Matter Description

The Company’s indefinite-lived intangible assets are comprised of management contracts, trade names/trademarks and licenses acquired in business acquisitions. The Company performs its impairment assessment of its indefinite-lived intangible assets at least annually, as of July 31st. In 2024, the annual assessment excluded indefinite-lived intangible assets related to GIP as the acquisition was finalized October 1, 2024; refer to Acquisition of Global Infrastructure Management, LLC (GIP) – fair value of intangible assets related to management contracts and contingent consideration critical audit matter for discussion of the acquisition of GIP. In evaluating whether it is more likely than not that the fair value of indefinite-lived intangibles is less than carrying value, the Company performs certain quantitative assessments and assesses various significant qualitative factors. If an indefinite-lived intangible asset is determined to be more likely than not impaired, the fair value of the asset is then compared with its carrying value. Any excess of the carrying value over the fair value would be recognized as an expense in the period in which the impairment occurs. The determination of fair value requires management to make estimates and assumptions related to revenue basis points, projected AUM growth rates, operating margins, tax rates, and discount rates.

Given the significant judgments made by management to estimate the fair value of indefinite-lived intangible assets related to certain management contracts, performing audit procedures to evaluate the reasonableness of management’s estimates and assumptions related to projected AUM growth rates, revenue basis points, operating margins, tax rates, and discount rates, required a high degree of auditor judgment and an increased extent of effort, including the need to involve our fair value specialists.

How the Critical Audit Matter Was Addressed in the Audit

Our audit procedures related to the determination of fair value of indefinite-lived intangible assets related to certain management contracts included the following, among others:

- We tested the design and operating effectiveness of controls over the Company’s indefinite-lived intangible asset impairment analysis, including those related to management’s assessment of the factors that impact the fair value of the Company’s indefinite-lived intangible assets. This includes controls related to management’s revenue basis points, projected AUM growth rates, operating margins, tax rates, and the determination of the discount rates.
- We evaluated the reasonableness of management’s AUM, revenue basis points, projected AUM growth rates, operating margins, tax rates and discount rates by comparing management’s projections to historical amounts, internal communications to management and the Board of Directors, and forecasted information included in analyst and industry reports for the Company and certain of its peer companies.
- We evaluated management’s ability to accurately project revenue basis points, AUM growth rates, operating margins and tax rates, by comparing actual results to management’s historical forecasts.
- With the assistance of our fair value specialists, we evaluated the reasonableness of the Company’s valuation methodology and assumptions, including the determination of the discount rates by: (1) testing the source information underlying the determination of the discount rate and the mathematical accuracy of the evaluation and (2) developing a range of independent estimates and comparing those to the discount rate selected by management.
- We evaluated the impact of changes in management’s forecasts from July 31, 2024, the annual impairment assessment date, to December 31, 2024.

/s/ Deloitte & Touche LLP
New York, New York
February 25, 2025

We have served as the Company's auditor since 2002.

BlackRock, Inc.
Consolidated Statements of Financial Condition

(in millions, except shares and per share data)		December 31, 2024		December 31, 2023
Assets				
Cash and cash equivalents[1]	**$**	**12,762**	$	8,736
Accounts receivable		**4,304**		3,916
Investments[1]		**9,769**		9,740
Separate account assets		**52,811**		56,098
Separate account collateral held under securities lending agreements		**6,059**		4,558
Property and equipment (net of accumulated depreciation and amortization of $1,553 and $1,439 at December 31, 2024 and 2023, respectively)		**1,103**		1,112
Intangible assets (net of accumulated amortization of $782 and $618 at December 31, 2024 and 2023, respectively)		**20,743**		18,258
Goodwill		**25,949**		15,524
Operating lease right-of-use assets		**1,519**		1,421
Other assets[1]		**3,596**		3,848
Total assets	**$**	**138,615**	$	123,211
Liabilities				
Accrued compensation and benefits	**$**	**2,964**	$	2,393
Accounts payable and accrued liabilities		**1,536**		1,240
Borrowings		**12,314**		7,918
Separate account liabilities		**52,811**		56,098
Separate account collateral liabilities under securities lending agreements		**6,059**		4,558
Contingent consideration liabilities		**4,302**		99
Deferred income tax liabilities		**3,334**		3,506
Operating lease liabilities		**1,908**		1,784
Other liabilities[1]		**4,032**		4,375
Total liabilities		**89,260**		81,971
Commitments and contingencies (Note 16)				
Temporary equity				
Redeemable noncontrolling interests		**1,691**		1,740
Permanent equity				
BlackRock, Inc. stockholders' equity				
Common stock, $0.01 par value;		**2**		2
Shares authorized: 500,000,000 at December 31, 2024 and 2023;				
Shares issued: 155,318,170 and 172,075,373 at December 31, 2024 and 2023, respectively;				
Shares outstanding: 154,947,813 and 148,500,074 at December 31, 2024 and 2023, respectively				
Additional paid-in capital		**13,446**		19,833
Retained earnings		**35,611**		32,343
Accumulated other comprehensive loss		**(1,178)**		(840)
Treasury stock, common, at cost (370,357 and 23,575,299 shares held at December 31, 2024 and 2023, respectively)		**(386)**		(11,991)
Total BlackRock, Inc. stockholders' equity		**47,495**		39,347
Nonredeemable noncontrolling interests		**169**		153
Total permanent equity		**47,664**		39,500
Total liabilities, temporary equity and permanent equity	**$**	**138,615**	$	123,211

(1) At December 31, 2024, cash and cash equivalents, investments, other assets and other liabilities include $125 million, $5.1 billion, $45 million and $2.1 billion, respectively, related to consolidated variable interest entities ("VIEs"). At December 31, 2023, cash and cash equivalents, investments, other assets and other liabilities include $234 million, $5.0 billion, $83 million and $2.2 billion, respectively, related to consolidated VIEs.

See accompanying notes to consolidated financial statements.

BlackRock, Inc.
Consolidated Statements of Income

(in millions, except per share data)	2024	2023	2022
Revenue			
Investment advisory, administration fees and securities lending revenue:			
Related parties	**$ 12,050**	$ 10,757	$ 10,848
Other third parties	**4,050**	3,642	3,603
Total investment advisory, administration fees and securities lending revenue	**16,100**	14,399	14,451
Investment advisory performance fees	**1,207**	554	514
Technology services revenue	**1,603**	1,485	1,364
Distribution fees	**1,273**	1,262	1,381
Advisory and other revenue	**224**	159	163
Total revenue	**20,407**	17,859	17,873
Expense			
Employee compensation and benefits	**6,546**	5,779	5,681
Sales, asset and account expense:			
Distribution and servicing costs	**2,171**	2,051	2,179
Direct fund expense	**1,464**	1,331	1,226
Sub-advisory and other	**140**	116	103
Total sales, asset and account expense	**3,775**	3,498	3,508
General and administration expense	**2,221**	2,095	2,057
Restructuring charge	**—**	61	91
Amortization and impairment of intangible assets	**291**	151	151
Total expense	**12,833**	11,584	11,488
Operating income	**7,574**	6,275	6,385
Nonoperating income (expense)			
Net gain (loss) on investments	**492**	699	(35)
Interest and dividend income	**767**	473	152
Interest expense	**(538)**	(292)	(212)
Total nonoperating income (expense)	**721**	880	(95)
Income before income taxes	**8,295**	7,155	6,290
Income tax expense	**1,783**	1,479	1,296
Net income	**6,512**	5,676	4,994
Less:			
Net income (loss) attributable to noncontrolling interests	**143**	174	(184)
Net income attributable to BlackRock, Inc.	**$ 6,369**	$ 5,502	$ 5,178
Earnings per share attributable to BlackRock, Inc. common stockholders:			
Basic	**$ 42.45**	$ 36.85	$ 34.31
Diluted	**$ 42.01**	$ 36.51	$ 33.97
Weighted-average common shares outstanding:			
Basic	**150.0**	149.3	150.9
Diluted	**151.6**	150.7	152.4

See accompanying notes to consolidated financial statements.

BlackRock, Inc.
Consolidated Statements of Comprehensive Income

(in millions)	2024	2023	2022
Net income	**$ 6,512**	$ 5,676	$ 4,994
Other comprehensive income (loss):			
Foreign currency translation adjustments[1]	**(338)**	261	(551)
Comprehensive income (loss)	**6,174**	5,937	4,443
Less: Comprehensive income (loss) attributable to noncontrolling interests	**143**	174	(184)
Comprehensive income attributable to BlackRock, Inc.	**$ 6,031**	$ 5,763	$ 4,627

(1) Amount for 2024 includes a gain from a net investment hedge of $37 million (net of tax expense of $12 million). Amount for 2023 includes a loss from a net investment hedge of $20 million (net of tax benefit of $6 million). Amount for 2022 includes a gain from a net investment hedge of $37 million (net of tax expense of $12 million).

See accompanying notes to consolidated financial statements.

BlackRock, Inc.
Consolidated Statements of Changes in Equity

(in millions)	Additional Paid-in Capital[(1)]	Retained Earnings	Accumulated Other Comprehensive Income (Loss)	Treasury Stock Common	Total BlackRock Stockholders' Equity	Nonredeemable Noncontrolling Interests	Total Permanent Equity	Redeemable Noncontrolling Interests / Temporary Equity
December 31, 2021	$ 19,642	$ 27,688	$ (550)	$ (9,087)	$ 37,693	$ 113	$ 37,806	$ 1,087
Net income	—	5,178	—	—	5,178	6	5,184	(190)
Dividends declared ($19.52 per share)	—	(2,990)	—	—	(2,990)	—	(2,990)	—
Stock-based compensation	708	—	—	—	708	—	708	—
Issuance of common shares related to employee stock transactions	(576)	—	—	614	38	—	38	—
Employee tax withholdings related to employee stock transactions	—	—	—	(457)	(457)	—	(457)	—
Shares repurchased	—	—	—	(1,875)	(1,875)	—	(1,875)	—
Subscriptions (redemptions/distributions) — noncontrolling interest holders	—	—	—	—	—	4	4	614
Net consolidations (deconsolidations) of sponsored investment funds	—	—	—	—	—	9	9	(602)
Other comprehensive income (loss)	—	—	(551)	—	(551)	—	(551)	—
December 31, 2022	$ 19,774	$ 29,876	$ (1,101)	$ (10,805)	$ 37,744	$ 132	$ 37,876	$ 909
Net income	—	5,502	—	—	5,502	16	5,518	158
Dividends declared ($20.00 per share)	—	(3,035)	—	—	(3,035)	—	(3,035)	—
Stock-based compensation	630	—	—	—	630	—	630	—
Issuance of common shares related to employee stock transactions	(569)	—	—	698	129	—	129	—
Employee tax withholdings related to employee stock transactions	—	—	—	(375)	(375)	—	(375)	—
Shares repurchased	—	—	—	(1,509)	(1,509)	—	(1,509)	—
Subscriptions (redemptions/distributions) — noncontrolling interest holders	—	—	—	—	—	(16)	(16)	1,643
Net consolidations (deconsolidations) of sponsored investment funds	—	—	—	—	—	21	21	(970)
Other comprehensive income (loss)	—	—	261	—	261	—	261	—
December 31, 2023	$ 19,835	$ 32,343	$ (840)	$ (11,991)	$ 39,347	$ 153	$ 39,500	$ 1,740
Net income	—	6,369	—	—	6,369	—	6,369	143
Dividends declared ($20.40 per share)	—	(3,101)	—	—	(3,101)	—	(3,101)	—
Stock-based compensation	753	—	—	—	753	—	753	—
Issuance of common shares related to employee stock transactions	(215)	—	—	706	491	—	491	—
Issuance of common shares in connection with the GIP Transaction	5,904	—	—	—	5,904	—	5,904	—
Cancellation of treasury stock, common in connection with the GIP Transaction	(12,829)	—	—	12,829	—	—	—	—
Employee tax withholdings related to employee stock transactions	—	—	—	(305)	(305)	—	(305)	—
Shares repurchased	—	—	—	(1,625)	(1,625)	—	(1,625)	—
Subscriptions (redemptions/distributions) — noncontrolling interest holders	—	—	—	—	—	16	16	2,389
Net consolidations (deconsolidations) of sponsored investment funds	—	—	—	—	—	—	—	(2,581)
Other comprehensive income (loss)	—	—	(338)	—	(338)	—	(338)	—
December 31, 2024	**$ 13,448**	**$ 35,611**	**$ (1,178)**	**$ (386)**	**$ 47,495**	**$ 169**	**$ 47,664**	**$ 1,691**

(1) Amounts include $2 million of common stock at December 31, 2024, 2023, 2022 and 2021.

See accompanying notes to consolidated financial statements.

BlackRock, Inc.
Consolidated Statements of Cash Flows

(in millions)	2024	2023	2022
Operating activities			
Net income	**$ 6,512**	$ 5,676	$ 4,994
Adjustments to reconcile net income to net cash provided by/(used in) operating activities:			
Depreciation and amortization	**529**	427	418
Impairment of intangible assets	**50**	—	—
Noncash lease expense	**129**	140	165
Stock-based compensation	**753**	630	708
Deferred income tax expense (benefit)	**(106)**	124	602
Contingent consideration fair value adjustments	**(36)**	3	3
Other investment gains	**(126)**	—	(268)
Net (gains) losses within CIPs	**(269)**	(380)	400
Net (purchases) proceeds within CIPs	**(2,672)**	(1,780)	(1,190)
(Earnings) losses from equity method investees	**(41)**	(378)	(29)
Distributions of earnings from equity method investees	**57**	49	50
Changes in operating assets and liabilities:			
Accounts receivable	**(443)**	(586)	416
Investments, trading	**58**	72	196
Other assets	**317**	(326)	(166)
Accrued compensation and benefits	**367**	145	(711)
Accounts payable and accrued liabilities	**259**	(26)	(151)
Other liabilities	**(382)**	375	(481)
Net cash provided by/(used in) operating activities	**4,956**	4,165	4,956
Investing activities			
Purchases of investments	**(818)**	(846)	(824)
Proceeds from sales and maturities of investments	**766**	400	242
Distributions of capital from equity method investees	**366**	46	70
Net consolidations (deconsolidations) of sponsored investment funds	**(127)**	(26)	(85)
Acquisitions, net of cash acquired	**(2,936)**	(189)	—
Purchases of property and equipment	**(255)**	(344)	(533)
Net cash provided by/(used in) investing activities	**(3,004)**	(959)	(1,130)
Financing activities			
Repayments of long-term borrowings	**(1,000)**	—	(750)
Proceeds from long-term borrowings	**5,474**	1,238	—
Cash dividends paid	**(3,101)**	(3,035)	(2,990)
Proceeds from stock options exercised	**464**	95	11
Repurchases of common stock	**(1,930)**	(1,884)	(2,332)
Net proceeds from (repayments of) borrowings by CIPs	**(58)**	(59)	(26)
Net (redemptions/distributions paid)/subscriptions received from noncontrolling interest holders	**2,405**	1,627	618
Other financing activities	**(18)**	26	27
Net cash provided by/(used in) financing activities	**2,236**	(1,992)	(5,442)
Effect of exchange rate changes on cash, cash equivalents and restricted cash	**(162)**	106	(291)
Net increase/(decrease) in cash, cash equivalents and restricted cash	**4,026**	1,320	(1,907)
Cash, cash equivalents and restricted cash, beginning of year	**8,753**	7,433	9,340
Cash, cash equivalents and restricted cash, end of year	**$ 12,779**	$ 8,753	$ 7,433
Supplemental disclosure of cash flow information:			
Cash paid for:			
Interest	**$ 289**	$ 200	$ 177
Income taxes (net of refunds)	**$ 1,699**	$ 1,392	$ 1,067
Supplemental schedule of noncash investing and financing transactions:			
Issuance of common shares related to employee stock transactions	**$ 215**	$ 569	$ 576
Issuance of common shares in connection with the GIP Transaction	**$ 5,904**	$ —	$ —
Cancellation of treasury stock, common in connection with the GIP Transaction	**$ (12,829)**	$ —	$ —
Increase/(decrease) in noncontrolling interests due to net consolidation (deconsolidation) of sponsored investment funds	**$ (2,581)**	$ (949)	$ (593)
Established contingent consideration liabilities in connection with acquisitions	**$ 4,246**	$ —	$ —

See accompanying notes to consolidated financial statements.

BlackRock, Inc.
Notes to the Consolidated Financial Statements

1. Business Overview

BlackRock, Inc. (together, with its subsidiaries, unless the context otherwise indicates, "BlackRock" or the "Company") is a leading publicly traded investment management firm providing a broad range of investment management and technology services to institutional and retail clients worldwide. On October 1, 2024, BlackRock completed the acquisition of 100% of the issued and outstanding limited liability company interests of Global Infrastructure Management, LLC ("GIP" or the "GIP Transaction"). See Note 3, *Acquisitions,* for information on the GIP Transaction. As a result of the closing of the GIP Transaction, (1) BlackRock, Inc. (formerly known as BlackRock Funding, Inc. ("BlackRock Funding")) ("New BlackRock") became the ultimate parent company of BlackRock Finance, Inc. (formerly known as BlackRock, Inc.) ("Old BlackRock"), GIP and their respective subsidiaries and (2) each share of common stock, $0.01 par value, of Old BlackRock issued and outstanding immediately prior to the closing of the GIP Transaction was converted automatically into one share of common stock, $0.01 par value, of New BlackRock. New BlackRock also changed its name from "BlackRock Funding, Inc." to "BlackRock, Inc." and Old BlackRock changed its name from "BlackRock, Inc." to "BlackRock Finance, Inc." In addition, New BlackRock became the publicly listed company and retained the ticker symbol "BLK". References herein to BlackRock or the Company for any period (1) prior to the closing of the GIP Transaction on October 1, 2024 refer to Old BlackRock and (2) thereafter refer to New BlackRock.

BlackRock's diverse platform of alpha-seeking active, private markets, index and cash management investment strategies across asset classes enables the Company to offer choice and tailor investment and asset allocation solutions for clients. Product offerings include single- and multi-asset portfolios investing in equities, fixed income, private markets, liquid alternatives and money market instruments. Products are offered directly and through intermediaries in a variety of vehicles, including open-end and closed-end mutual funds, iShares® exchange-traded funds ("ETFs"), separate accounts, collective trust funds and other pooled investment vehicles. BlackRock also offers technology services, including the investment and risk management technology platform, Aladdin®, Aladdin Wealth™, eFront®, and Cachematrix®, as well as advisory services and solutions to a broad base of institutional and wealth management clients.

2. Significant Accounting Policies

Basis of Presentation

These consolidated financial statements have been prepared in accordance with accounting principles generally accepted in the United States ("GAAP") and include the accounts of the Company and its controlled subsidiaries. Noncontrolling interests ("NCI") on the consolidated statements of financial condition represent the portion of consolidated sponsored investment products ("CIPs") and a consolidated affiliate in which the Company does not have direct equity ownership. Intercompany balances and transactions have been eliminated upon consolidation.

The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenue and expense during the reporting periods. Actual results could differ from those estimates.

Certain prior period presentations were reclassified to ensure comparability with current period classifications.

Accounting Pronouncements Adopted in 2024

Segment Reporting. In November 2023, the Financial Accounting Standards Board ("FASB") issued Accounting Standards Update ("ASU") 2023-07, *Improvements to Reportable Segment Disclosures* ("ASU 2023-07"), which requires incremental disclosures about reportable segments but does not change the definition of a segment or the guidance for determining reportable segments. The new guidance requires disclosure of significant segment expenses that are (1) regularly provided to (or easily computed from information regularly provided to) the chief operating decision maker ("CODM") and (2) included in the reported measure of segment profit or loss. The new standard also requires companies to disclose the title and position of the individual (or the name of the committee) identified as the CODM, allows companies to disclose multiple measures of segment profit or loss if those measures are used to assess performance and allocate resources, and is applicable to companies with a single reportable segment. The Company adopted disclosure requirements of ASU 2023-07 during the year ended December 31, 2024. See Note 27, *Segment Information,* for further detail.

Recent Accounting Pronouncements Not Yet Adopted

Income Tax Disclosure Requirements. In December 2023, the FASB issued ASU 2023-09, *Improvements to Income Tax Disclosures* ("ASU 2023-09"), which enhances interim and annual income tax disclosures. The two primary enhancements disaggregate existing income tax disclosures related to the effective tax rate reconciliation and income taxes paid. The additional disclosure requirements under ASU 2023-09 are required to be applied prospectively and are effective for the Company on January 1, 2025. The Company does not expect the additional disclosure requirements under ASU 2023-09 to have a material impact on the consolidated financial statements.

Disaggregation of Income Statement Expenses. In November 2024, the FASB issued ASU 2024-03, *Disaggregation of Income Statement Expenses* ("ASU 2024-03"), which requires entities to disaggregate in a tabular presentation disclosures about specific types of expenses included in the expense captions presented on the face of the income statement, as well as disclosures about selling expenses. Specifically, ASU 2024-03 requires disaggregation of expense captions that include any of the following natural expenses: (1) purchases of inventory, (2) employee compensation, (3) depreciation, (4) intangible asset amortization, and (5) depreciation, depletion, and amortization recognized as part of oil- and gas-producing activities or other types of depletion expenses. The requirements are effective for fiscal years beginning after December 15, 2026, and interim periods within fiscal years beginning after December 15, 2027 and are required to be applied prospectively with the option for retrospective application. Early adoption is permitted. The Company does not expect the additional disclosure requirements under ASU 2024-03 to have a material impact on the consolidated financial statements.

Cash and Cash Equivalents. Cash and cash equivalents primarily consists of cash, money market funds and short-term, highly liquid investments with original maturities of three months or less. Cash and cash equivalent balances that are legally restricted from use by the Company are recorded in other assets on the consolidated statements of financial condition. Cash balances maintained by consolidated VIEs and voting rights entities ("VREs") are not considered legally restricted and are included in cash and cash equivalents on the consolidated statements of financial condition.

Investments

Investments in Debt Securities. The Company classifies debt investments as held-to-maturity or trading based on the Company's intent and ability to hold the debt security to maturity or its intent to sell the security.

Held-to-maturity securities are purchased with the positive intent and ability to be held to maturity and are recorded at amortized cost on the consolidated statements of financial condition.

Trading securities are those investments that are purchased principally for the purpose of selling them in the near term. Trading securities are carried at fair value on the consolidated statements of financial condition with changes in the fair value recorded through net income ("FVTNI") within nonoperating income (expense). Trading securities include certain investments in collateralized loan obligations ("CLOs") for which the fair value option is elected in order to reduce operational complexity of bifurcating embedded derivatives.

Investments in Equity Securities. Equity securities are generally carried at fair value on the consolidated statements of financial condition with changes in the FVTNI within nonoperating income (expense). For nonmarketable equity securities, the Company generally elects to apply the practicality exception to fair value measurement, under which such securities will be measured at cost, less impairment, plus or minus observable price changes for identical or similar securities of the same issuer with such changes recorded through net income within nonoperating income (expense). Dividends received are recorded as dividend income within nonoperating income (expense).

Equity Method. The Company applies the equity method of accounting for equity investments where the Company does not consolidate the investee, but can exert significant influence over the financial and operating policies of the investee. The evaluation of whether the Company exerts control or significant influence over the financial and operational policies of its investees is based on the facts and circumstances surrounding each individual investment and is generally considered to exist when the Company's ownership interest in the investee is between 20% and 50%, or lower for co-investments in certain sponsored investment funds generally structured as partnerships or similar vehicles. Factors considered in these evaluations may include the type of investment, the legal structure of the investee, the terms of BlackRock's contractual agreements, including investor voting or other rights, any influence BlackRock may have on the governing board of the investee, the legal rights of other investors in the entity pursuant to the entity's operating documents and the relationship between BlackRock and other investors in the entity. The Company's share of the investee's underlying net income or loss is recorded as net gain (loss) on investments within nonoperating income (expense) and as other revenue for certain strategic investments since such investees are considered to be an extension of the Company's core business. The Company's share of net income of the investee is recorded based upon the most current information available at the time, which may precede the date of the consolidated statement of financial condition. Distributions received reduce the Company's carrying value of the investment and the cost basis if deemed to be a return of capital. The Company classifies distributions in the consolidated statements of cash flows as either distributions of earnings (operating) or distributions of capital (investing) based on the nature of the distribution.

Impairments of Investments. Management periodically assesses equity method, nonmarketable investments, and held-to-maturity investments for impairment. If impairment exists, an impairment charge would be recorded for the excess of the carrying amount of the investment over its estimated fair value in the consolidated statements of income.

For equity method investments and nonmarketable investments, impairment evaluation considers qualitative factors, including the financial conditions and specific events related to an investee, that may indicate the fair value of the investment is less than its carrying value. For held-to-maturity investments, impairment is evaluated using market values, where available, or the expected future cash flows of the investment.

For the Company's investments in CLOs, the Company reviews cash flow estimates over the life of each CLO investment. On a quarterly basis, if the present value of the estimated future cash flows is lower than the carrying value of the investment and there is an adverse change in estimated cash flows, an impairment is considered to be other-than-temporary.

Consolidation. The Company performs an analysis for investment products to determine if the product is a VIE or a VRE. Factors considered in this analysis include the entity's legal organization, the entity's capital structure, the rights of equity investment holders and the Company's contractual involvement with, and economic interest in, the entity and any related party or de facto agent implications of the Company's involvement with the entity. Entities that are determined to be VIEs are consolidated if the Company is the primary beneficiary ("PB") of the entity. VREs are typically consolidated if the Company holds the majority voting interest. Upon the occurrence of certain events (such as contributions and redemptions, either by the Company, or third parties, or amendments to an entity's governing documents), management reviews and reconsiders its previous conclusion regarding the status of an entity as a VIE or a VRE.

Consolidation of Variable Interest Entities. Certain investment products for which a controlling financial interest is achieved through arrangements that do not involve or are not directly linked to voting interests are deemed consolidated VIEs. BlackRock reviews factors, including whether or not (1) the entity has equity at risk that is sufficient to permit the entity to finance its activities without additional subordinated support from other parties and (2) the equity holders at risk have the obligation to absorb losses, the right to receive residual returns, and the right to direct the activities of the entity that most significantly impact the entity's economic performance, to determine if the investment product is a VIE.

The PB of a VIE is defined as the variable interest holder that has a controlling financial interest in the VIE. A controlling financial interest is defined as (1) the power to direct the activities of the VIE that most significantly impact its economic performance and (2) the obligation to absorb losses of the entity or the right to receive benefits from the entity that potentially could be significant to the VIE. The Company generally consolidates VIEs in which it holds an economic interest of 10% or greater and deconsolidates such VIEs once economic interest falls below 10%. Management continually reconsiders whether the Company is deemed to be a VIE's PB.

Consolidation of Voting Rights Entities. BlackRock is required to consolidate an investee to the extent that BlackRock can exert absolute control over the financial and operating policies of the investee, which generally exists if there is a greater than 50% voting equity interest.

Retention of Specialized Investment Company Accounting Principles. Upon consolidation of sponsored investment products, the Company retains the specialized investment company accounting principles of the underlying funds. All of the underlying investments held by such CIPs are carried at fair value with corresponding changes in the investments' fair values reflected in net income within nonoperating income (expense). When the Company no longer controls these funds due to reduced ownership percentage or other reasons, the funds are deconsolidated and accounted for as an equity method investment or equity securities FVTNI.

Separate Account Assets and Liabilities. Separate account assets are maintained by BlackRock Life Limited, a wholly owned subsidiary of the Company, which is a registered life insurance company in the United Kingdom ("UK"), and represent segregated assets held for purposes of funding individual and group pension contracts. The life insurance company does not underwrite any insurance contracts that involve any insurance risk transfer from the insured to the life insurance company. The separate account assets primarily include equity securities, debt securities, money market funds and derivatives. The separate account assets are not subject to general claims of the creditors of BlackRock. These separate account assets and the related equal and offsetting liabilities are recorded as separate account assets and separate account liabilities on the consolidated statements of financial condition.

The net investment income attributable to separate account assets supporting individual and group pension contracts accrues directly to the contract owner and is not reported on the consolidated statements of income. While BlackRock has no economic interest in these separate account assets and liabilities, BlackRock earns policy administration and management fees associated with these products, which are included in investment advisory, administration fees and securities lending revenue on the consolidated statements of income.

Separate Account Collateral Assets Held and Liabilities Under Securities Lending Agreements. The Company facilitates securities lending arrangements whereby securities held by separate accounts maintained by BlackRock Life Limited are lent to third parties under global master securities lending agreements. In exchange, the Company receives collateral by obtaining either (1) legal title or (2) first ranking priority security interest. The minimum collateral values generally range from approximately 102% to 112% of the value of the securities lent in order to reduce counterparty risk. The required collateral value is calculated on a daily basis. The global master securities lending agreements provide the Company the right to request additional collateral or, in the event of borrower default, the right to liquidate collateral. The securities lending transactions entered into by the Company are accompanied by an agreement that entitles the Company to request the borrower to return the securities at any time; therefore, these transactions are not reported as sales.

In situations where the Company receives the legal title to collateral under these securities lending arrangements, the Company records an asset on the consolidated statements of financial condition and an equal collateral liability for the obligation to return the collateral. Additionally, in situations where the Company obtains a first ranking priority security interest in the collateral, the Company does not have the ability to pledge or resell the collateral and therefore does not record the collateral on the consolidated statements of financial condition. At December 31, 2024 and 2023, the fair value of loaned securities held by separate accounts was approximately $9.9 billion and $9.3 billion, respectively, and the fair value of the collateral under these securities lending agreements was approximately $10.6 billion and $10.1 billion, respectively, of which approximately $6.1 billion as of 2024 and $4.6 billion as of 2023 was recognized on the consolidated statements of financial condition. During 2024 and 2023, the Company had not resold or repledged any of the collateral received under these arrangements. The securities lending revenue earned from lending securities held by the separate accounts is included in investment advisory, administration fees and securities lending revenue on the consolidated statements of income.

Property and Equipment. Property and equipment are recorded at cost less accumulated depreciation. Depreciation is generally determined by cost less any estimated residual value using the straight-line method over the estimated useful lives of the various classes of property and equipment. Leasehold improvements are amortized using the straight-line method over the shorter of the estimated useful life or the remaining lease term.

The Company capitalizes certain costs incurred in connection with developing or obtaining software within property and equipment. Capitalized software costs are amortized, beginning when the software product is ready for its intended use, over the estimated useful life of the software of approximately three years.

Goodwill and Intangible Assets. Goodwill represents the cost of a business acquisition in excess of the fair value of the net assets acquired. The Company has determined that it has one reporting unit for goodwill impairment testing purposes, the consolidated BlackRock single operating segment, which is consistent with internal management reporting and management's oversight of operations. The Company performs an impairment assessment of its goodwill at least annually, as of July 31. In its assessment of goodwill for impairment, the Company considers such factors as the book value and market capitalization of the Company as well as other qualitative factors. See Note 11, *Goodwill*, for further information on the Company's goodwill.

Intangible assets are comprised of indefinite-lived intangible assets and finite-lived intangible assets acquired in a business acquisition. The value of contracts to manage assets in proprietary open-end funds and collective trust funds and certain other commingled products without a specified termination date is generally classified as indefinite-lived intangible assets. In addition, trade names/trademarks are considered indefinite-lived intangible assets when they are expected to generate cash flows indefinitely.

Indefinite-lived intangible assets and goodwill are not amortized. Finite-lived investor/customer relationships, technology-related assets, and management contracts, which relate to acquired separate accounts and funds, that are expected to contribute to the future cash flows of the Company for a specified period of time, are amortized over their estimated useful lives. On a quarterly basis, the Company considers whether the indefinite-lived and finite-lived classifications are still appropriate.

The Company performs assessments to determine if any intangible assets are potentially impaired at least annually, as of July 31. The carrying value of finite-lived assets and their remaining useful lives are reviewed to determine if circumstances exist which may indicate a potential impairment or revisions to the amortization period.

In evaluating whether it is more likely than not that the fair value of indefinite-lived intangibles is less than its carrying value, BlackRock assesses various significant quantitative factors, including assets under management (“AUM”), revenue basis points, projected AUM growth rates, operating margins, tax rates and discount rates. If an indefinite-lived intangible is determined to be more likely than not impaired, then the fair value of the asset is compared with its carrying value and any excess of the carrying value over the fair value would be recognized as an expense in the period in which the impairment occurs. See Note 12, *Intangible Assets*, for further information on the Company’s intangible assets.

For finite-lived intangible assets, if potential impairment circumstances are considered to exist, the Company will perform a recoverability test using an undiscounted cash flow analysis. If the carrying value of the asset is determined not to be recoverable based on the undiscounted cash flow test, the difference between the carrying value of the asset and its current fair value would be recognized as an expense in the period in which the impairment occurs.

Consolidated Affiliate. The Company owns 50.1% of an asset management company in China - BlackRock CCB Wealth Management Company Ltd. ("WMC"). The Company consolidates WMC, which it deems to be a VRE, because it exerts control over the financial and operating policies of the entity, based on the Company's 50.1% ownership and voting rights.

Noncontrolling Interests. NCI consist of third-party ownership interests in the Company's CIPs ("NCI – CIPs") and the WMC. The Company reports NCI in stockholders' equity, separate from the parent's equity, on the consolidated statements of financial condition. NCI that are redeemable at the option of the holders are classified as temporary equity at estimated redemption value and nonredeemable NCI are classified as a component of permanent equity in the consolidated statements of financial condition. In addition, the Company reports net income (loss) attributable to redeemable and nonredeemable NCI holders in net income (loss) attributable to NCI in the consolidated statements of income.

Treasury Stock. The Company records common stock purchased for treasury at cost. At the date of subsequent reissuance, the treasury stock account is reduced by the cost of such stock using the average cost method.

Revenue Recognition. Revenue is recognized upon transfer of control of promised services to customers in an amount to which the Company expects to be entitled in exchange for those services. The Company enters into contracts that can include multiple services, which are accounted for separately if they are determined to be distinct. Consideration for the Company's services is generally in the form of variable consideration because the amount of fees is subject to market conditions that are outside of the Company's influence. The Company includes variable consideration in revenue when it is no longer probable of significant reversal, i.e. when the associated uncertainty is resolved. For some contracts with customers, the Company has discretion to involve a third-party in providing services to the customer. Generally, the Company is deemed to be the principal in these arrangements because the Company controls the promised services before they are transferred to customers, and accordingly presents the revenue gross of related costs.

Investment Advisory, Administration Fees and Securities Lending Revenue. Investment advisory and administration fees are recognized as the services are performed over time because the customer is receiving and consuming the benefits as they are provided by the Company. Fees are primarily based on agreed-upon percentages of AUM and recognized for services provided during the period, which are distinct from services provided in other periods. Such fees are affected by changes in AUM, including market appreciation or depreciation, foreign exchange translation and net inflows or outflows. Investment advisory and administration fees for investment funds are shown net of fee waivers. In addition, the Company may contract with third parties to provide sub-advisory services on its behalf. The Company presents the investment advisory fees and associated costs to such third-party advisors on a gross basis where it is deemed to be the principal and on a net basis where it is deemed to be the agent. Management judgment involved in making these assessments is focused on ascertaining whether the Company is primarily responsible for fulfilling the promised service.

The Company also earns revenue by lending securities on behalf of clients, primarily to highly rated banks and broker-dealers. The securities loaned are collateralized by either cash or securities, generally ranging from 102% to 112% of the value of the loaned securities. Securities lending fees are based on (1) a percentage of the notional value of the loaned securities and (2) a spread between the interest earned on the reinvested cash collateral and the amount rebated to the borrower. Revenue is recognized over time as services are performed. Generally, the securities lending fees are shared between the Company and the funds or other third-party accounts managed by the Company from which the securities are borrowed. For 2024, 2023 and 2022, securities lending revenue earned by the Company totaled $615 million, $675 million and $599 million, respectively, and is recorded in investment advisory, administration and securities lending revenue on the consolidated statements of income. Investment advisory, administration fees and securities lending revenue are reported together as the fees for these services often are agreed upon with clients as a bundled fee.

Money Market Fee Waivers. The Company may voluntarily waive a portion of its management fees on certain money market funds to ensure that they maintain a targeted level of daily net investment income (the "Yield Support Waivers"). There were no Yield Support Waivers during 2024 and 2023. During 2022, these waivers resulted in a reduction of management fees of approximately $72 million, which was partially offset by a reduction of BlackRock's distribution and servicing costs paid to financial intermediaries. The Company may increase or decrease the level of Yield Support Waivers in future periods.

Investment Advisory Performance Fees / Carried Interest. The Company receives investment advisory performance fees, including incentive allocations (carried interest) from certain actively managed investment funds and certain separately managed accounts. These performance fees are dependent upon exceeding specified relative or absolute investment return thresholds, which vary by product or account, and include monthly, quarterly, annual or longer measurement periods.

Performance fees, including carried interest, are generated on certain management contracts when performance hurdles are achieved. Such performance fees are recognized when the contractual performance criteria have been met and when it is determined that they are no longer probable of significant reversal. Given the unique nature of each fee arrangement, contracts with customers are evaluated on an individual basis to determine the timing of revenue recognition. Significant judgment is involved in making such determination. Performance fees typically arise from investment management services that began in prior reporting periods. Consequently, a portion of the fees the Company recognizes may be partially related to the services performed in prior periods that meet the recognition criteria in the current period. At each reporting date, the Company considers various factors in estimating performance fees to be recognized, including carried interest.

The Company is allocated carried interest from certain alternative investment products upon exceeding performance thresholds. The Company may be required to reverse/return all, or part, of such carried interest allocations/distributions depending upon future performance of these funds. Carried interest subject to such clawback provisions is recorded in investments or cash and cash equivalents to the extent that it is distributed, on the Company's consolidated statements of financial condition.

The Company records a liability for deferred carried interest to the extent it receives cash or capital allocations related to carried interest prior to meeting the revenue recognition criteria. A portion of the deferred carried interest may also be paid to certain employees. The ultimate timing of the recognition of performance fee revenue and related compensation expense, if any, is unknown.

Technology services revenue. The Company offers investment management technology systems, risk management services, wealth management and digital distribution tools, all on a fee basis. Clients include banks, insurance companies, official institutions, pension funds, asset managers, retail distributors and other investors. Fees earned for technology services are primarily recorded as services are performed over time and are generally determined using the value of positions on the Aladdin platform, or on a fixed-rate basis. Revenue derived from the sale of software licenses is recognized upon the granting of access rights.

Distribution Fees. The Company earns distribution and service fees related to distributing investment products and shareholder support services for investment portfolios. Distribution fees are passed-through to third-party distributors, which perform various fund distribution services and shareholder servicing of certain funds on the Company's behalf, and are recognized as distribution and servicing costs. The Company presents distribution fees and related distribution and servicing costs incurred on a gross basis.

Distribution fees primarily consist of ongoing distribution fees, shareholder servicing fees and upfront sales commissions for serving as the principal underwriter and/or distributor for certain managed mutual funds. The service of distribution is satisfied at the point in time when an investor makes an investment in a share class of the managed mutual funds. Fees are generally considered variable consideration because they are based on the value of AUM and are uncertain on trade date. Accordingly, the Company recognizes distribution fees when the amounts become known and the portion recognized in the current period may relate to distribution services performed in prior periods. Upfront sales commissions are recognized on a trade date basis. Shareholder servicing fees are based on AUM and recognized in revenue as the services are performed.

Advisory and other revenue. Advisory and other revenue primarily includes fees earned for advisory services, fees earned for transition management services primarily comprised of commissions recognized in connection with buying and selling securities on behalf of customers, and equity method investment earnings related to certain strategic investments.

Advisory services fees are determined using fixed-rate fees and are recognized over time as the related services are completed.

Commissions related to transition management services are recorded on a trade-date basis as transactions occur.

Stock-based Compensation. The Company recognizes compensation cost for equity classified awards based on the grant-date fair value of the award. The compensation cost is recognized over the period during which an employee is required to provide service (usually the vesting period) in exchange for the stock-based award.

The Company generally measures the grant-date fair value of restricted stock units ("RSUs") using the Company's stock price on the date of grant. The grant-date fair value related to the October 2024 incentive retention RSUs granted in connection with the GIP Transaction, are reduced by the present value of the dividends expected to be paid on the shares during the vesting period discounted at the appropriate risk-free interest rate, given that they are not entitled to participate in dividends until they vest (See Note 3, *Acquisitions* and Note 18, *Stock-Based Compensation* for further information on the GIP Transaction). Stock-based awards may have performance, market and/or service conditions. For employee stock options and awards with market conditions, the Company uses pricing models. Compensation cost for awards containing performance conditions is recognized if it is probable that the conditions will be achieved. The probability of achievement is assessed on a quarterly basis. If a stock-based award is modified after the grant-date, incremental compensation cost is recognized for an amount equal to the excess of the fair value of the modified award over the fair value of the original award immediately before the modification. Awards under the Company's stock-based compensation plans vest over various periods. Compensation cost is recorded by the Company on a straight-line basis over the requisite service period for each separate vesting portion of the award as if the award is, in-substance, multiple awards and is adjusted for actual forfeitures as they occur.

The Company amortizes the grant-date fair value of stock-based compensation awards made to retirement-eligible employees over the requisite service period. Upon notification of retirement, the Company accelerates the unamortized portion of the award over the contractually required retirement notification period.

The Company recognizes all excess tax benefits and deficiencies in income tax expense on the consolidated statements of income, which results in volatility of income tax expense as a result of fluctuations in the Company's stock price. Accordingly, the Company recorded a discrete income tax benefit of $37 million, $41 million and $87 million during 2024, 2023 and 2022, respectively, for vested RSUs where the grant date stock price was lower than the vesting date stock price.

Distribution and Servicing Costs. Distribution and servicing costs include payments to third parties, primarily associated with distribution and servicing of client investments in certain BlackRock products. Distribution and servicing costs are expensed as incurred.

Direct Fund Expense. Direct fund expense, which is expensed as incurred, primarily consists of third-party non-advisory expense incurred by BlackRock related to certain investment products for the use of certain index trademarks, reference data for certain indices, custodial services, fund administration, fund accounting, transfer agent services, shareholder reporting services, audit and tax services as well as other fund-related expense directly attributable to the non-advisory operations of the fund.

Leases. The Company determines if a contract is a lease or contains a lease at inception. The Company accounts for its office facility leases as operating leases, which may include escalation clauses that are based on an index or market rate. The Company accounts for lease and non-lease components, including common areas maintenance charges, as a single component for its leases. The Company elected the short-term lease exception for leases with an initial term of 12 months or less. Consequently, such leases are not recorded on the consolidated statements of financial condition. The Company's lease terms include options to extend or terminate the lease when it is reasonably certain they will be exercised or not.

The Company recognizes operating right-of-use ("ROU") assets and operating lease liabilities on the consolidated statements of financial condition based on the present value of future lease payments over the lease term at the commencement date discounted using an incremental borrowing rate ("IBR"). The IBR for individual leases is estimated considering the Company's or a subsidiary's credit rating using various financial metrics, such as revenue, operating margin and revenue growth, and, as appropriate, performing market analysis of yields on publicly traded bonds (secured or unsecured) with similar terms of comparable companies in a similar economic environment. ROU assets are tested for impairment when there is an indication that the carrying value of an asset may not be recoverable. Fixed lease payments made over the lease term are recorded as lease expense on a straight-line basis. Variable lease payments based on usage, changes in an index or market rate are expensed as incurred.

Foreign Exchange. Foreign currency transactions are recorded at the exchange rates prevailing on the dates of the transactions. Monetary assets and liabilities that are denominated in foreign currencies are subsequently remeasured into the functional currencies of the Company's subsidiaries at the rates prevailing at each statement of financial condition date. Gains and losses arising on remeasurement are included in general and administration expense on the consolidated statements of income. Revenue and expenses are translated at average exchange rates during the period. Gains or losses resulting from translating foreign currency financial statements into United States ("US") dollars are included in accumulated other comprehensive income (loss) ("AOCI"), a separate component of stockholders' equity, on the consolidated statements of financial condition.

Income Taxes. Deferred income tax assets and liabilities are recognized for the future tax consequences attributable to temporary differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases using currently enacted tax rates in effect for the year in which the differences are expected to reverse. The effect of a change in tax rates on deferred income tax assets and liabilities is recognized on the consolidated statements of income in the period that includes the enactment date.

Management periodically assesses the recoverability of its deferred income tax assets based upon expected future earnings, taxable income in prior carryback years, future deductibility of the asset, changes in applicable tax laws and other factors. If management determines that it is not more likely than not that the deferred tax asset will be fully recoverable in the future, a valuation allowance will be established for the difference between the asset balance and the amount expected to be recoverable in the future. This allowance will result in additional income tax expense. Further, the Company records its income taxes receivable and payable based upon its estimated income tax position.

Earnings per Share ("EPS"). Basic EPS is calculated by dividing net income applicable to common shareholders by the weighted-average number of shares outstanding during the period. Diluted EPS includes the determinants of basic EPS and common stock equivalents outstanding during the period. Diluted EPS is computed using the treasury stock method.

Business Segments. The Company's management directs BlackRock's operations as one business, the asset management business. The Company utilizes a consolidated approach to assess performance and allocate resources. As such, the Company operates in one business segment.

Fair Value Measurements

Hierarchy of Fair Value Inputs. The Company uses a fair value hierarchy that prioritizes inputs to valuation approaches used to measure fair value. The fair value hierarchy gives the highest priority to quoted prices (unadjusted) in active markets for identical assets or liabilities and the lowest priority to unobservable inputs. Assets and liabilities measured and reported at fair value are classified and disclosed in one of the following categories:

<u>Level 1 Inputs:</u>

Quoted prices (unadjusted) in active markets for identical assets or liabilities at the reporting date.

- Level 1 assets may include listed mutual funds, ETFs, listed equities, commodities and certain exchange-traded derivatives.

<u>Level 2 Inputs:</u>

Quoted prices for similar assets or liabilities in active markets; quoted prices for identical or similar assets or liabilities that are not active; quotes from pricing services or brokers for which the Company can determine that orderly transactions took place at the quoted price or that the inputs used to arrive at the price are observable; and inputs other than quoted prices that are observable, such as models or other valuation methodologies.

- Level 2 assets may include debt securities, loans held within consolidated CLOs, short-term floating-rate notes, asset-backed securities, as well as over-the-counter derivatives, including interest rate swaps and foreign currency exchange contracts that have inputs to the valuations that generally can be corroborated by observable market data.

<u>Level 3 Inputs:</u>

Unobservable inputs for the valuation of the asset or liability, which may include nonbinding broker quotes. Level 3 assets include investments for which there is little, if any, market activity. These inputs require significant management judgment or estimation.

- Level 3 assets may include direct private equity investments, including those held within CIPs, investments in CLOs and loans held within consolidated CLOs and CIPs.
- Level 3 liabilities may include borrowings of consolidated CLOs and contingent liabilities related to acquisitions valued using the income approach based on unobservable market data, or other valuation techniques.

Significance of Inputs. The Company's assessment of the significance of a particular input to the fair value measurement in its entirety requires judgment and considers factors specific to the financial instrument.

Valuation Approaches. The fair values of certain Level 3 assets and liabilities were determined using various valuation approaches as appropriate, including third-party pricing vendors, broker quotes and market and income approaches.

A significant number of inputs used to value equity, debt securities, and loans held within CLOs and CIPs are sourced from third-party pricing vendors. Generally, prices obtained from pricing vendors are categorized as Level 1 inputs for identical securities traded in active markets and as Level 2 for other similar securities if the vendor uses observable inputs in determining the price.

In addition, quotes obtained from brokers generally are nonbinding and categorized as Level 3 inputs. However, if the Company is able to determine that market participants have transacted for the asset in an orderly manner near the quoted price or if the Company can determine that the inputs used by the broker are observable, the quote is classified as a Level 2 input.

Investments Measured at Net Asset Value. As a practical expedient, the Company uses net asset value (“NAV”) as the fair value for certain investments. The inputs to value these investments may include the Company’s capital accounts for its partnership interests in various alternative investments, including hedge funds, real assets and private equity funds. The various partnerships are investment companies, which record their underlying investments at fair value based on fair value policies established by management of the underlying fund. Fair value policies at the underlying fund generally require the fund to utilize pricing/valuation information from third-party sources, including independent appraisals. However, in some instances, current valuation information for illiquid securities or securities in markets that are not active may not be available from any third-party source or fund management may conclude that the valuations that are available from third-party sources are not reliable. In these instances, fund management may perform model-based analytical valuations that could be used as an input to value these investments.

Fair Value Assets and Liabilities of Consolidated CLO. The Company applies the fair value option provisions for eligible assets, including loans, held by a consolidated CLO. As the fair value of the financial assets of the consolidated CLO is more observable than the fair value of the borrowings of the consolidated CLO, the Company measures the fair value of the borrowings of the consolidated CLO equal to the fair value of the assets of the consolidated CLO less the fair value of the Company’s economic interest in the CLO.

Derivatives and Hedging Activities. The Company does not use derivative financial instruments for trading or speculative purposes. The Company uses derivative financial instruments primarily for purposes of hedging exposures to fluctuations in foreign currency exchange rates of certain assets and liabilities, and market price and interest rate exposures with respect to its total portfolio of seed investments in sponsored investment products. Certain CIPs also utilize derivatives as a part of their investment strategies.

In addition, the Company uses derivatives and makes investments to economically hedge market valuation changes on certain deferred cash compensation plans, for which the final value of the deferred amount distributed to employees in cash upon vesting is determined based on the returns of specified investment funds. The Company recognizes compensation expense for the appreciation (depreciation) of the deferred cash compensation liability in proportion to the vested amount of the award during a respective period, while the gain (loss) to economically hedge these plans is immediately recognized in nonoperating income (expense). See Note 5, *Investments*, and Note 9, *Derivatives and Hedging*, for further information on the Company’s investments and derivatives, respectively, used to economically hedge these deferred cash compensation plans.

The Company records all derivative financial instruments as either assets or liabilities at fair value on a gross basis in the consolidated statements of financial condition. Credit risks are managed through master netting and collateral support agreements. The amounts related to the right to reclaim or the obligation to return cash collateral may not be used to offset amounts due under the derivative instruments in the normal course of settlement. Therefore, such amounts are not offset against fair value amounts recognized for derivative instruments with the same counterparty and are included in other assets and other liabilities. Changes in the fair value of the Company’s derivative financial instruments are recognized in earnings and, where applicable, are offset by the corresponding gain or loss on the related foreign-denominated or hedged assets or liabilities, on the consolidated statements of income.

The Company may also use financial instruments designated as net investment hedges for accounting purposes to hedge net investments in international subsidiaries, the functional currency of which is not US dollars. The gain or loss from revaluing net investment hedges at the spot rate is deferred and reported within AOCI on the consolidated statements of financial condition. The Company reassesses the effectiveness of its net investment hedge at least quarterly.

3. Acquisitions

GIP

As discussed in Note 1, *Business Overview,* on October 1, 2024, BlackRock completed the acquisition of GIP, a leading independent infrastructure fund manager. BlackRock expects the combination of GIP with BlackRock’s complementary infrastructure offerings will create a broad global infrastructure franchise with differentiated origination and asset management capabilities.

Consideration at close included approximately $3 billion in cash, funded through the issuance of long-term notes in March 2024 (See Note 15, *Borrowings* for further information), and 6.9 million of unregistered shares of BlackRock common stock. The shares were valued at $5.9 billion at close, based on the price of BlackRock's common stock on September 30, 2024 of approximately $950, discounted for security-specific registration restrictions for two years after closing, resulting in a value of approximately $855 per share. In addition, as part of the purchase consideration, a contingent consideration payment, all in stock, may be due subject to achieving certain performance targets. The contingent consideration payment, if any, ranges from 4.0 million to 5.2 million shares, and will be valued based on the price of BlackRock's common stock at the time the contingency is resolved. The payment is expected to be payable no later than December 31, 2028 or based on the achievement of the agreed upon performance targets. The fair value of the contingent consideration payment, which was determined by using the income approach with the assistance of a third-party fair value specialist, was $4.2 billion at close, and is recorded within contingent consideration liabilities in the consolidated statements of financial condition. Certain significant inputs were used to determine the fair value, including assumptions on discount rates as well as estimates of the timing and amounts of fundraising forecasts, stock and AUM volatility, and correlation between stock price and AUM (Level 3 inputs). The contingent consideration payment is classified as a liability as the monetary value of the consideration to be delivered in shares is predominately based on achieving certain performance targets.

In addition, on October 1, 2024, in connection with the GIP Transaction, the Company granted incentive retention awards of approximately $415 million of cliff-vesting RSUs and approximately $178 million of performance-based RSUs recognized as compensation expense over a period of approximately five years. See Note 18, *Stock-Based Compensation*, for additional information on the incentive retention awards issued in connection with the GIP Transaction. The Company also granted approximately $100 million of deferred cash awards in connection with the GIP Transaction, which are recognized as compensation expense over a period of approximately 50% over one year and the remaining 50% over five years.

In connection with the GIP Transaction, the Company entered into several stockholder agreements with the former owners of GIP and their respective affiliates. Each agreement, among other things, (1) sets forth various restrictions, limitations and other terms concerning the transfer of equity securities of BlackRock, which includes an initial lock-up period ending on the two-year anniversary of the closing, and (2) contains customary standstill provisions such as limiting the purchase of additional shares.

The GIP Transaction was accounted for as a business combination under the acquisition method of accounting. Accordingly, the purchase price was allocated to the assets acquired and liabilities assumed based upon their estimated fair values at the date of the transaction. The goodwill recognized in connection with the acquisition includes future benefits for BlackRock as a result of scale and anticipated synergies from a combined global infrastructure franchise. The amount of goodwill expected to be deductible for tax purposes is approximately $200 million.

The following table summarizes the consideration paid for GIP and the fair values of the assets acquired and liabilities assumed recognized at the acquisition date in this acquisition:

(in millions)	Fair Value Estimate
Finite-lived intangible assets:	
Management contracts[(1)]	**$ 1,840**
Investor relationships[(1)]	**820**
Trade name[(2)]	**80**
Goodwill	**10,297**
Operating lease ROU assets[(3)]	**75**
Other assets[(3)]	**114**
Accrued compensation and benefits[(3)]	**(154)**
Operating lease liabilities[(3)]	**(96)**
Other liabilities assumed[(3)]	**(10)**
Total consideration, net of cash acquired	**$ 12,966**
Summary of consideration, net of cash acquired:	
Cash paid	**$ 2,930**
Cash acquired	**(68)**
Closing stock consideration at fair value	**5,904**
Deferred stock consideration at fair value	**4,200**
Total cash and stock consideration	**$ 12,966**

(1) The fair value for management contracts and investor relationships was determined based on a discounted cash flow analysis (Level 3 inputs), have weighted-average estimated useful lives of approximately 8 years and 14 years, respectively, and are amortized based on their pattern of economic benefit.
(2) The fair value was determined based upon a relief from royalty method (Level 3 inputs), has a weighted-average estimated useful life of approximately 10 years and is amortized based on its pattern of economic benefit.
(3) Acquired operating lease ROU assets and operating lease liabilities were determined based on current lease guidance. The acquired book values of the remaining assets and liabilities approximated their fair values.

Transaction and integration costs incurred in connection with the GIP Transaction were approximately $195 million in 2024. These costs were primarily comprised of $130 million of compensation expense, mostly for nonrecurring retention-related deferred compensation and $65 million of other acquisition-related transaction costs, largely related to advisory fees, legal fees and consulting expenses, recorded in general and administration expense.

Finite-lived intangible assets are amortized over their useful lives, which range from 3 to 14 years. Amortization expense related to the finite-lived intangible assets was $86 million for 2024. The finite-lived intangible assets had a weighted-average remaining useful life of approximately 10 years with remaining amortization expense as follows:

(in millions) **Year**	Amount
2025	**$ 281**
2026	**323**
2027	**277**
2028	**196**
2029	**228**
2030	**269**
Thereafter	**1,079**
Total	**$ 2,653**

The following unaudited pro forma information presents combined results of operations of the Company as if the GIP Transaction and related $3.0 billion in aggregate notes issuance (see Note 15, *Borrowings*, for more information) had occurred on January 1, 2023 and are not indicative of the actual results of operations that would have been achieved nor are they indicative of future results of operations of the combined Company. The pro forma combined provision for income taxes may not represent the amount that would have resulted had BlackRock and GIP filed consolidated tax returns during the years presented.

(Unaudited) (in millions)	2024[(1)]	2023
Total revenue	**$ 21,087**	$ 18,664
Net income attributable to BlackRock, Inc.	**$ 6,318**	$ 5,025

(1) Subsequent to the closing of the GIP Transaction on October 1, 2024, GIP contributed approximately $240 million of revenue and $56 million of net income.

For purposes of the pro forma financial information above, the following pro forma adjustments and related tax effects were included as if the GIP Transaction had occurred on January 1, 2023 (unaudited):

- Compensation expense included retention-related deferred compensation awards in connection with the GIP Transaction of $285 million for the year ended December 31, 2023, and $95 million for the nine months ended September 30, 2024 (see Note 18, *Stock-Based Compensation*, for further information on retention related deferred compensation issued in connection with the GIP Transaction);
- Acquisition-related transaction costs of $65 million, which were recorded in 2024 were included in 2023 results and removed from 2024 results;
- Amortization of finite-lived intangible assets of $300 million for the year ended December 31, 2023 and $240 million for the nine months ended September 30, 2024;
- Interest expense for the $3.0 billion notes, which were issued in March 2024 in connection with the GIP Transaction, of $155 million for the year ended December 31, 2023 and $115 million for the nine months ended September 30, 2024 were included as if the issuance occurred on January 1, 2023; and
- Adjustments to reflect the tax effects of the GIP Transaction, as if GIP had been included in the Company's results as of January 1, 2023.

SpiderRock Advisors

In May 2024, BlackRock completed the acquisition of the remaining equity interest in SpiderRock Advisors ("SRA"), a leading provider of customized option overlay strategies in the US wealth market (the "SpiderRock Transaction"). This transaction expands on BlackRock's minority investment in SRA made in 2021 and reinforces BlackRock's commitment to personalized separately managed accounts. Pro forma financial information for SRA has not been presented in the table above, as the effects were not material to the Company's historical consolidated financial statements.

Kreos Capital

In August 2023, BlackRock completed the acquisition of Kreos Capital, a provider of growth and venture debt financing to companies in the technology and healthcare industries (the "Kreos Transaction"). The acquisition adds to BlackRock's position as a leading global credit asset manager and advances its ambitions to provide clients with a diverse range of private market investment products and solutions. Total consideration for the transaction was approximately $250 million, which included contingent consideration. Pro forma financial information for Kreos has not been presented in the table above, as the effects were not material to the Company's historical consolidated financial statements.

4. Cash, Cash Equivalents, and Restricted Cash

The following table provides a reconciliation of cash and cash equivalents reported within the consolidated statements of financial condition to the cash, cash equivalents, and restricted cash reported within the consolidated statements of cash flows.

(in millions)	**December 31, 2024**	December 31, 2023
Cash and cash equivalents	**$ 12,762**	$ 8,736
Restricted cash included in other assets	**17**	17
Total cash, cash equivalents and restricted cash	**$ 12,779**	$ 8,753

5. Investments

A summary of the carrying value of total investments is as follows:

(in millions)	**December 31, 2024**	**December 31, 2023**
Debt securities:		
Trading securities (including $1,743 and $1,829 held by CIPs at December 31, 2024 and December 31, 2023, respectively)	**$ 1,823**	$ 1,871
Held-to-maturity investments	**547**	617
Total debt securities	**2,370**	2,488
Equity securities at FVTNI (including $1,556 and $1,429 held by CIPs at December 31, 2024 and December 31, 2023, respectively)(1)	**1,950**	1,585
Equity method investments:		
Equity method investments(2)	**2,610**	2,515
Investments related to deferred cash compensation plans(1)	**173**	241
Total equity method investments	**2,783**	2,756
Loans held by CIPs	**145**	205
Federal Reserve Bank stock(3)	**93**	92
Carried interest(4)	**1,983**	1,975
Other investments(5)	**445**	639
Total investments	**$ 9,769**	$ 9,740

(1) Amounts include investments held to economically hedge the impact of market valuation changes on certain deferred cash compensation plans comprised of equity method investments of $173 million and $241 million at December 31, 2024 and 2023, respectively. In addition, amounts related to deferred cash compensation plans include equity securities held at FVTNI of $12 million and $14 million at December 31, 2024 and 2023, respectively.
(2) Equity method investments include BlackRock's direct investments in certain BlackRock sponsored investment funds.
(3) Federal Reserve Bank stock is held for regulatory purposes and is restricted from sale.
(4) Carried interest represents allocations to BlackRock's general partner capital accounts from certain sponsored investment funds. These balances are subject to change upon cash distributions, additional allocations or reallocations back to limited partners within the respective funds.
(5) Other investments include BlackRock's investments in nonmarketable equity securities, which are measured at cost, adjusted for observable price changes, and private equity, real asset, and commodity investments held by CIPs, which are measured at fair value.

Held-to-Maturity Investments

Held-to-maturity investments included certain investments in BlackRock sponsored CLOs. The amortized cost (carrying value) of these investments approximated fair value (primarily a Level 2 input). At December 31, 2024, \$9 million of these investments mature in less than one year, \$24 million of these investments mature between one and five years, \$326 million of these investments mature between five and ten years and \$188 million of these investments mature after ten years.

Trading Debt Securities and Equity Securities at FVTNI

A summary of the cost and carrying value of trading debt securities and equity securities at FVTNI is as follows:

	December 31, 2024		December 31, 2023	
(in millions)	**Cost**	**Fair Value**	**Cost**	**Fair Value**
Trading debt securities:				
Corporate debt	**$ 1,047**	**$ 1,061**	$ 1,225	$ 1,218
Government debt	**578**	**557**	501	489
Asset/mortgage-backed debt	**222**	**205**	185	164
Total trading debt securities	**$ 1,847**	**$ 1,823**	$ 1,911	$ 1,871
Equity securities at FVTNI:				
Equity securities/mutual funds	**$ 1,843**	**$ 1,950**	$ 1,520	$ 1,585

6. Consolidated Sponsored Investment Products

In the normal course of business, the Company is the manager of various types of sponsored investment products, which may be considered VIEs or VREs. The Company consolidates certain sponsored investment funds accounted for as VREs because it is deemed to control such funds. In addition, the Company may from time to time own equity or debt securities issued by vehicles or enter into derivatives or loan arrangements with the vehicles, each of which are considered variable interests. The Company's involvement in financing the operations of the VIEs is generally limited to its economic interest in the entity. The Company's consolidated VIEs include certain sponsored investment products in which BlackRock has an economic interest and as the investment manager, is deemed to have both the power to direct the most significant activities of the products and the right to receive benefits (or the obligation to absorb losses) that could potentially be significant to these sponsored investment products. The assets of these VIEs are not available to creditors of the Company. In addition, the investors in these VIEs have no recourse to the credit of the Company.

The following table presents the balances related to these CIPs accounted for as VIEs and VREs that were recorded on the consolidated statements of financial condition, including BlackRock's net interest in these products:

	December 31, 2024			December 31, 2023		
(in millions)	**VIEs**	**VREs**	**Total**	**VIEs**	**VREs**	**Total**
Cash and cash equivalents[1]	**$ 125**	**$ 44**	**$ 169**	$ 234	$ 54	$ 288
Investments:						
Trading debt securities	**1,497**	**246**	**1,743**	1,423	406	1,829
Equity securities at FVTNI	**1,179**	**377**	**1,556**	1,059	370	1,429
Loans	**141**	**4**	**145**	195	10	205
Other investments	**370**	**33**	**403**	427	171	598
Carried interest	**1,905**	**—**	**1,905**	1,916	—	1,916
Total investments	**5,092**	**660**	**5,752**	5,020	957	5,977
Other assets	**45**	**31**	**76**	83	39	122
Other liabilities[2]	**(2,130)**	**(93)**	**(2,223)**	(2,233)	(108)	(2,341)
Noncontrolling interest - CIPs	**(1,672)**	**(130)**	**(1,802)**	(1,625)	(226)	(1,851)
BlackRock's net interest in CIPs	**$ 1,460**	**$ 512**	**$ 1,972**	$ 1,479	$ 716	$ 2,195

(1) The Company generally cannot readily access cash and cash equivalents held by CIPs to use in its operating activities.
(2) At both December 31, 2024 and 2023, other liabilities of VIEs primarily include deferred carried interest liabilities and borrowings of a consolidated CLO.

BlackRock's total exposure to CIPs represents the value of its economic interest in these CIPs. Valuation changes associated with financial instruments held at fair value by these CIPs are reflected in nonoperating income (expense) and partially offset in net income (loss) attributable to NCI for the portion not attributable to BlackRock.

Net gain (loss) related to consolidated VIEs is presented in the following table:

(in millions)	**2024**	**2023**	**2022**
Nonoperating net gain (loss) on consolidated VIEs	**$ 234**	$ 310	$ (311)
Net income (loss) attributable to NCI on consolidated VIEs	**$ 132**	$ 174	$ (161)

7. Variable Interest Entities

Nonconsolidated VIEs. At December 31, 2024 and 2023, the Company's carrying value of assets and liabilities included on the consolidated statements of financial condition pertaining to nonconsolidated VIEs and its maximum risk of loss related to VIEs for which it held a variable interest, but for which it was not the PB, was as follows:

(in millions)	Investments	Advisory Fee Receivables	Other Net Assets (Liabilities)	Maximum Risk of Loss[1]
December 31, 2024				
Sponsored investment products	**$ 2,330**	**$ 158**	**$ (11)**	**$ 2,505**
December 31, 2023				
Sponsored investment products	$ 2,377	$ 116	$ (11)	$ 2,510

(1) At both December 31, 2024 and 2023, BlackRock's maximum risk of loss associated with these VIEs primarily related to BlackRock's investments and the collection of receivables.

The net assets of sponsored investment products that are nonconsolidated VIEs approximated $46 billion and $39 billion at December 31, 2024 and 2023, respectively.

8. Fair Value Disclosures

Fair Value Hierarchy

Assets and liabilities measured at fair value on a recurring basis

December 31, 2024 *(in millions)*	Quoted Prices in Active Markets for Identical Assets (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)	Investments Measured at NAV[1]	Other[2]	December 31, 2024
Assets:						
Investments						
Debt securities:						
Trading securities	$ —	$ 1,744	$ 79	$ —	$ —	**$ 1,823**
Held-to-maturity investments	—	—	—	—	547	**547**
Total debt securities	—	1,744	79	—	547	**2,370**
Equity securities at FVTNI:						
Equity securities/mutual funds	1,950	—	—	—	—	**1,950**
Equity method:						
Equity, fixed income, and multi-asset mutual funds	347	131	—	—	—	**478**
Hedge funds/funds of hedge funds/other	—	—	—	552	—	**552**
Private equity funds	—	—	—	1,060	—	**1,060**
Real assets funds	—	—	—	520	—	**520**
Investments related to deferred cash compensation plans	—	—	—	173	—	**173**
Total equity method	347	131	—	2,305	—	**2,783**
Loans held by CIPs	—	10	135	—	—	**145**
Federal Reserve Bank Stock	—	—	—	—	93	**93**
Carried interest	—	—	—	—	1,983	**1,983**
Other investments	18	—	—	274	153	**445**
Total investments	2,315	1,885	214	2,579	2,776	**9,769**
Other assets[3]	—	7	149	—	—	**156**
Separate account assets	32,933	19,346	—	—	532	**52,811**
Separate account collateral held under securities lending agreements:						
Equity securities	2,719	—	—	—	—	**2,719**
Debt securities	—	3,340	—	—	—	**3,340**
Total separate account collateral held under securities lending agreements	2,719	3,340	—	—	—	**6,059**
Total	$ 37,967	$ 24,578	$ 363	$ 2,579	$ 3,308	**$ 68,795**
Liabilities:						
Separate account collateral liabilities under securities lending agreements	$ 2,719	$ 3,340	$ —	$ —	$ —	**$ 6,059**
Contingent consideration liabilities	—	—	4,302	—	—	**4,302**
Other liabilities[4]	—	46	129	—	—	**175**
Total	$ 2,719	$ 3,386	$ 4,431	$ —	$ —	**$ 10,536**

(1) Amounts are comprised of certain investments measured at fair value using NAV (or its equivalent) as a practical expedient.
(2) Amounts are comprised of investments held at amortized cost and cost, adjusted for observable price changes, and carried interest.
(3) Level 3 amount includes corporate minority private debt investments with changes in fair value recorded in AOCI, net of tax.
(4) Level 2 amount primarily includes fair value of derivatives (See Note 9, *Derivatives and Hedging*, for more information). Level 3 amount primarily includes borrowings of a consolidated CLO classified based on the significance of unobservable inputs used for calculating the fair value of consolidated CLO assets.

December 31, 2023 (in millions)	Quoted Prices in Active Markets for Identical Assets (Level 1)	Significant Other Observable Inputs (Level 2)	Significant Unobservable Inputs (Level 3)	Investments Measured at NAV[(1)]	Other[(2)]	December 31, 2023
Assets:						
Investments						
Debt securities:						
Trading securities	$ —	$ 1,829	$ 42	$ —	$ —	$ 1,871
Held-to-maturity investments	—	—	—	—	617	617
Total debt securities	—	1,829	42	—	617	2,488
Equity securities at FVTNI:						
Equity securities/mutual funds	1,585	—	—	—	—	1,585
Equity method:						
Equity, fixed income, and multi-asset mutual funds	246	—	—	—	—	246
Hedge funds/funds of hedge funds/other	—	—	—	588	—	588
Private equity funds	—	—	—	1,264	—	1,264
Real assets funds	—	—	—	417	—	417
Investments related to deferred cash compensation plans	—	—	—	241	—	241
Total equity method	246	—	—	2,510	—	2,756
Loans held by CIPs	—	30	175	—	—	205
Federal Reserve Bank Stock	—	—	—	—	92	92
Carried interest	—	—	—	—	1,975	1,975
Other investments	15	—	—	467	157	639
Total investments	1,846	1,859	217	2,977	2,841	9,740
Other assets[(3)]	117	19	120	—	—	256
Separate account assets	34,621	20,810	—	—	667	56,098
Separate account collateral held under securities lending agreements:						
Equity securities	1,686	—	—	—	—	1,686
Debt securities	—	2,872	—	—	—	2,872
Total separate account collateral held under securities lending agreements	1,686	2,872	—	—	—	4,558
Total	$ 38,270	$ 25,560	$ 337	$ 2,977	$ 3,508	$ 70,652
Liabilities:						
Separate account collateral liabilities under securities lending agreements	$ 1,686	$ 2,872	$ —	$ —	$ —	$ 4,558
Other[(4)]	—	17	279	—	—	296
Total	$ 1,686	$ 2,889	$ 279	$ —	$ —	$ 4,854

(1) Amounts are comprised of certain investments measured at fair value using NAV (or its equivalent) as a practical expedient.
(2) Amounts are comprised of investments held at amortized cost and cost, adjusted for observable price changes, and carried interest.
(3) Level 1 amount includes a minority investment in a publicly traded company. Level 3 amount includes a corporate minority private debt investment with changes in fair value recorded in AOCI, net of tax.
(4) Level 2 amount primarily includes fair value of derivatives (See Note 9, *Derivatives and Hedging*, for more information). Level 3 amount primarily includes borrowings of a consolidated CLO classified based on the significance of unobservable inputs used for calculating the fair value of consolidated CLO assets, and contingent liabilities related to certain acquisitions.

Level 3 Assets. Level 3 assets predominantly include investments in nonconsolidated CLOs, loans of consolidated CIPs, and corporate minority private debt investments. Investments in CLOs and loans were valued based on single-broker nonbinding quotes or quotes from pricing services which use significant unobservable inputs. BlackRock's corporate minority private debt investments were primarily valued using the income approach by discounting the expected cash flows to a single present value. For investments utilizing a discounted cashflow valuation technique, an increase (decrease) in the discount rate or risk premium in isolation could have resulted in a significantly lower (higher) fair value measurement as of December 31, 2024 and 2023.

Level 3 Liabilities. Level 3 liabilities primarily include borrowings of a consolidated CLO, which were valued based on the fair value of the assets of the consolidated CLO less the fair value of the Company's economic interest in the CLO, as well as contingent consideration liabilities related to certain acquisitions, which were valued based upon discounted cash flow analyses using unobservable market data inputs or other valuation techniques.

At December 31, 2024, the contingent consideration liability related to the GIP Transaction was estimated using the income approach, with certain significant inputs including a risk-free discount rate of approximately 4.3% as well as current estimates of the timing and amounts of fundraising forecasts, stock and AUM volatility, and correlation between stock price and AUM (Level 3 inputs). Accordingly, changes in key inputs and assumptions described can materially impact the amount of contingent consideration expense recorded in a reporting period until the contingency is resolved. Changes in fair value are recorded within general and administration expense of the consolidated statements of income.

Nonrecurring Fair Value Measurements. The Company assessed its intangible assets for impairment during the annual impairment assessment as of July 31, 2024 and concluded that an impairment charge was required for indefinite-lived intangible assets related to certain open-end management contracts, which reduced the carrying value of these management contracts to $87 million. See Note 12, *Intangible Assets*, for more information. The fair value of these contracts was determined using a discounted cash flow analysis. The most sensitive assumptions used to determine present value were growth expectations, revenue / operating margin forecast, and the discount rate applied to the cash flow forecast, which are considered Level 3 inputs in the valuation hierarchy.

Changes in Level 3 Assets and Liabilities Measured at Fair Value on a Recurring Basis for 2024

(in millions)	December 31, 2023	Realized and Unrealized Gains (Losses)	Purchases	Sales and Maturities	Issuances and Other Settlements(1)	Transfers into Level 3	Transfers out of Level 3	December 31, 2024	Total Net Unrealized Gains (Losses) Included in Earnings(2)
Assets:									
Investments:									
Trading debt securities	$ 42	$ 3	$ 35	$ (1)	$ —	$ —	$ —	**$ 79**	$ 3
Loans	175	7	402	(455)	—	12	(6)	**135**	7
Total investments	217	10	437	(456)	—	12	(6)	**214**	10
Other assets	120	(8)	37	—	—	—	—	**149**	(8)
Total assets	$ 337	$ 2	$ 474	$ (456)	$ —	$ 12	$ (6)	**$ 363**	$ 2
Liabilities:									
Contingent consideration liabilities	$ 99	$ 42	$ —	$ —	$ 4,245	$ —	$ —	**$ 4,302**	$ 42
Other liabilities	180	(7)	—	—	(58)	—	—	**129**	(7)
Total liabilities	$ 279	$ 35	$ —	$ —	$ 4,187	$ —	$ —	**$ 4,431**	$ 35

(1) Issuances and other settlements amounts include contingent consideration liabilities related to the SpiderRock and GIP Transactions and repayments of borrowings of a consolidated CLO.
(2) Earnings attributable to the change in unrealized gains (losses) relating to assets and liabilities still held at the reporting date.

Changes in Level 3 Assets and Liabilities Measured at Fair Value on a Recurring Basis for 2023

(in millions)	December 31, 2022	Realized and Unrealized Gains (Losses)	Purchases	Sales and Maturities	Issuances and Other Settlements(1)	Transfers into Level 3	Transfers out of Level 3	December 31, 2023	Total Net Unrealized Gains (Losses) Included in Earnings(2)
Assets:									
Investments:									
Trading debt securities	$ 52	$ —	$ 8	$ (18)	$ —	$ —	$ —	**$ 42**	$ —
Loans	248	13	76	(58)	(122)	38	(20)	**175**	(1)
Total investments	300	13	84	(76)	(122)	38	(20)	**217**	(1)
Other assets	—	7	113	—	—	—	—	**120**	7
Total assets	$ 300	$ 20	$ 197	$ (76)	$ (122)	$ 38	$ (20)	**$ 337**	$ 6
Liabilities:									
Other	$ 280	$ 1	$ —	$ —	$ —	$ —	$ —	**$ 279**	$ 1

(1) Issuances and other settlements amounts include a deconsolidation related to a previously consolidated VRE. In addition, issuances and other settlements include a contingent liability in connection with the Kreos Transaction, offset by repayments of borrowings of a consolidated CLO.
(2) Earnings attributable to the change in unrealized gains (losses) relating to assets and liabilities still held at the reporting date.

Realized and Unrealized Gains (Losses) for Level 3 Assets and Liabilities. Realized and unrealized gains (losses) recorded for Level 3 assets and liabilities are primarily reported in nonoperating income (expense) on the consolidated statements of income. A portion of net income (loss) related to securities held by CIPs is allocated to NCI to reflect net income (loss) not attributable to the Company.

Transfers in and/or out of Levels. Transfers in and/or out of levels are reflected when significant inputs, including market inputs or performance attributes, used for the fair value measurement become observable/unobservable.

Disclosures of Fair Value for Financial Instruments Not Held at Fair Value. At December 31, 2024 and 2023, the fair value of the Company's financial instruments not held at fair value are categorized in the table below.

	December 31, 2024		December 31, 2023		
(in millions)	**Carrying Amount**	**Estimated Fair Value**	**Carrying Amount**	**Estimated Fair Value**	**Fair Value Hierarchy**
Financial Assets(1):					
Cash and cash equivalents	**$ 12,762**	**$ 12,762**	$ 8,736	$ 8,736	Level 1 (2)(3)
Other assets	**86**	**86**	80	80	Level 1 (2)(4)
Financial Liabilities:					
Long-term borrowings	**$ 12,314**	**$ 11,680**	$ 7,918	$ 7,413	Level 2 (5)

(1) See Note 5*, Investments*, for further information on investments not held at fair value.
(2) Cash and cash equivalents, other than money market funds, are carried at either cost or amortized cost, which approximates fair value due to their short-term maturities.
(3) At December 31, 2024 and 2023, approximately $6.2 billion and $3.4 billion, respectively, of money market funds were recorded within cash and cash equivalents on the consolidated statements of financial condition. Money market funds are valued based on quoted market prices, or $1.00 per share, which generally is the NAV of the fund.
(4) At December 31, 2024 and 2023, other assets included cash collateral of approximately $69 million and $63 million, respectively. See Note 9, *Derivatives and Hedging* for further information on derivatives held by the Company. In addition, other assets included $17 million of restricted cash at both December 31, 2024 and 2023.
(5) Long-term borrowings are recorded at amortized cost, net of debt issuance costs. The fair value of the long-term borrowings, including the current portion of long-term borrowings, is determined using market prices and the EUR/USD foreign exchange rate at the end of December 2024 and 2023, respectively. See Note 15, *Borrowings*, for the fair value of each of the Company's long-term borrowings.

Investments in Certain Entities that Calculate NAV Per Share

As a practical expedient to value certain investments that do not have a readily determinable fair value and have attributes of an investment company, the Company uses NAV as the fair value. The following tables list information regarding all investments that use a fair value measurement to account for both their financial assets and financial liabilities in their calculation of a NAV per share (or equivalent).

December 31, 2024

(in millions)	Ref	Fair Value	Total Unfunded Commitments	Redemption Frequency	Redemption Notice Period
Equity method[(1)]:					
Hedge funds/funds of hedge funds/other	**(a)**	**$ 552**	**$ 138**	**Daily/Monthly (2%) Quarterly (10%) N/R (88%)**	**1 – 90 days**
Private equity funds	**(b)**	**1,060**	**227**	**N/R**	**N/R**
Real assets funds	**(c)**	**520**	**710**	**Quarterly (7%) N/R (93%)**	**60 days**
Investments related to deferred cash compensation plan	**(d)**	**173**	**—**	**Monthly**	**1 – 90 days**
Consolidated sponsored investment products:					
Real assets funds	**(c)**	**175**	**40**	**N/R**	**N/R**
Private equity funds	**(e)**	**7**	**42**	**N/R**	**N/R**
Hedge funds/other	**(a)**	**92**	**58**	**Quarterly (64%) N/R (36%)**	**90 days**
Total		**$ 2,579**	**$ 1,215**		

December 31, 2023

(in millions)	Ref	Fair Value	Total Unfunded Commitments	Redemption Frequency	Redemption Notice Period
Equity method[(1)]:					
Hedge funds/funds of hedge funds/other	(a)	$ 588	$ 134	Daily/Monthly (4%) Quarterly (8%) N/R (88%)	1 – 90 days
Private equity funds	(b)	1,264	218	N/R	N/R
Real assets funds	(c)	417	210	Quarterly (10%) N/R (90%)	60 days
Investments related to deferred cash compensation plan	(d)	241	—	Monthly	1 – 90 days
Consolidated sponsored investment products:					
Real assets funds	(c)	154	62	N/R	N/R
Private equity funds	(e)	145	37	N/R	N/R
Hedge funds/other	(a)	168	64	Quarterly (83%) N/R (17%)	90 days
Total		$ 2,977	$ 725		

N/R – Not Redeemable

(1) Comprised of equity method investments, which include investment companies that account for their financial assets and most financial liabilities under fair value measures; therefore, the Company's investment in such equity method investees approximates fair value.

(a) This category includes hedge funds, funds of hedge funds, and other funds that invest primarily in equities, fixed income securities, private credit, opportunistic and mortgage instruments and other third-party hedge funds. The fair values of the investments have been estimated using the NAV of the Company's ownership interest in partners' capital. The liquidation period for the investments in the funds that are not subject to redemption is unknown at both December 31, 2024 and 2023.

(b) This category includes private equity funds that initially invest in nonmarketable securities of private companies, which ultimately may become public in the future. The fair values of these investments have been estimated using capital accounts representing the Company's ownership interest in the funds and may also include other performance inputs. The Company's investment in each fund is not subject to redemption and is normally returned through distributions as a result of the liquidation of the underlying assets of the private equity funds. The liquidation period for the investments in these funds is unknown at both December 31, 2024 and 2023.

(c) This category includes several real assets funds that invest directly and indirectly in real estate or infrastructure. The fair values of the investments have been estimated using capital accounts representing the Company's ownership interest in the funds. The Company's investments that are not subject to redemption or are not currently redeemable are normally returned through distributions and realizations of the underlying assets of the funds. The liquidation period for the investments in the funds that are not subject to redemptions is unknown at both December 31, 2024 and 2023. The total remaining unfunded commitments were $750 million and $272 million at December 31, 2024 and 2023, respectively. The Company's portion of the total remaining unfunded commitments was $736 million and $248 million at December 31, 2024 and 2023, respectively.

(d) This category includes hedge funds and funds of hedge funds that invest primarily in equities, fixed income securities, mortgage instruments and other third-party hedge funds. The fair values of the investments have been estimated using the NAV of the Company's ownership interest in partners' capital. The investments in hedge funds will be redeemed upon settlement of certain deferred cash compensation liabilities.

(e) This category includes the underlying third-party private equity funds within consolidated BlackRock sponsored private equity funds of funds. These investments are not subject to redemption or are not currently redeemable; however, for certain funds, the Company may sell or transfer its interest, which may need approval by the general partner of the underlying funds. Due to the nature of the investments in this category, the Company reduces its investment by distributions that are received through the realization of the underlying assets of the funds. The liquidation period for the underlying assets of these funds is unknown.

Fair Value Option

At December 31, 2024 and 2023, the Company elected the fair value option for certain investments in CLOs of approximately $72 million and $42 million, respectively, reported within investments.

In addition, the Company elected the fair value option for bank loans and borrowings of a consolidated CLO, recorded within investments and other liabilities, respectively. The following table summarizes the information related to these bank loans and borrowings at December 31, 2024 and 2023:

(in millions)	December 31, 2024	December 31, 2023
CLO loans:		
Aggregate principal amounts outstanding	**$ 156**	$ 203
Fair value	**141**	194
Aggregate principal balance in excess of (less than) fair value	**$ 15**	$ 9
CLO borrowings:		
Aggregate principal amounts outstanding	**$ 146**	$ 190
Fair value	**129**	180
Aggregate principal balance in excess of (less than) fair value	**$ 17**	$ 10

At December 31, 2024, the principal amounts outstanding of the borrowings issued by the consolidated CLO mature in 2030, and may be repaid prior to maturity at any time.

During the year ended December 31, 2024 and 2023, the net gains (losses) from the change in fair value of the bank loans and borrowings held by the consolidated CLO were not material and were recorded in net gain (loss) on the consolidated statements of income. The change in fair value of the assets and liabilities included interest income and expense, respectively.

9. Derivatives and Hedging

The Company maintains a program to enter into exchange traded futures as a macro hedging strategy to hedge market price and interest rate exposures with respect to its total portfolio of seed investments in sponsored investment products. The Company had outstanding exchange traded futures related to this macro hedging strategy with aggregate notional values of approximately $1.8 billion at both December 31, 2024 and 2023, with expiration dates during the first quarter of 2025 and 2024, respectively.

In addition, the Company enters into exchange traded futures to economically hedge the exposure to market movements on certain deferred cash compensation plans. The Company had outstanding exchange traded futures with aggregate notional values related to its deferred cash compensation hedging program of approximately $197 million and $204 million at December 31, 2024 and 2023, with expiration dates during the first quarter of 2025 and 2024, respectively.

Changes in the value of the futures contracts are recognized as gains or losses within nonoperating income (expense). Variation margin payments, which represent settlements of profit/loss, are generally received or made daily, and are reflected in other assets and other liabilities on the consolidated statements of financial condition. These amounts were not material as of December 31, 2024 and 2023.

The Company executes forward foreign currency exchange contracts to mitigate the risk of certain foreign exchange movements. At December 31, 2024 and 2023, the Company had outstanding forward foreign currency exchange contracts with aggregate notional values of approximately $3.6 billion, with expiration dates in January 2025, and $3.1 billion, with expiration dates in January 2024, respectively.

At both December 31, 2024 and 2023, the Company had a derivative providing credit protection with a notional amount of approximately $17 million to a counterparty, representing the Company's maximum risk of loss with respect to the derivative. The Company carries the derivative at fair value based on the expected discounted future cash outflows under the arrangement.

The following table presents the fair values of derivative instruments recognized in the consolidated statements of financial condition at December 31, 2024:

	Assets			Liabilities		
(in millions)	Statement of Financial Condition Classification	December 31, 2024	December 31, 2023	Statement of Financial Condition Classification	December 31, 2024	December 31, 2023
Derivative instruments						
Forward foreign currency exchange contracts	Other assets	**$ 7**	$ 19	Other liabilities	**$ 35**	$ 6

The following table presents realized and unrealized gains (losses) recognized in the consolidated statements of income on derivative instruments:

		Gains (Losses)		
(in millions)	Statement of Income Classification	2024	2023	2022
Derivative Instruments				
Exchange traded futures[(1)]	Net gain (loss) on investments	**$ (30)**	$ (88)	$ 36
Forward foreign currency exchange contracts	General and administration expense	**5**	98	(222)
Total return swaps	Net gain (loss) on investments	**—**	—	83
Total gain (loss) from derivative instruments		**$ (25)**	$ 10	$ (103)

(1) Amounts for 2024, 2023 and 2022 include $48 million of losses, $112 million of losses and $36 million of gains on futures used as a macro hedging strategy of seed investments, respectively. In addition, amounts for 2024 and 2023 include $18 million and $24 million of gains on futures used to economically hedge certain deferred cash compensation plans, respectively.

The Company's CIPs may utilize derivative instruments as a part of the funds' investment strategies. The change in fair value of such derivatives, which is recorded in nonoperating income (expense), was not material for 2024, 2023 and 2022.

See Note 15, *Borrowings*, for more information on the Company's net investment hedge.

10. Property and Equipment

Property and equipment consists of the following:

(in millions)	Estimated Useful Life-In Years	December 31, 2024		December 31, 2023	
Property and equipment:					
Land	N/A	**$**	**6**	$	6
Building	39		**33**		33
Building improvements	15		**32**		31
Leasehold improvements	1-15		**1,048**		1,036
Equipment and computer software	3		**1,136**		1,088
Other transportation equipment	8-10		**198**		192
Furniture and fixtures	7		**101**		99
Construction in progress	N/A		**102**		66
Total			**2,656**		2,551
Less: Accumulated depreciation and amortization			**1,553**		1,439
Property and equipment, net		**$**	**1,103**	$	1,112

N/A – Not Applicable

Qualifying software costs of approximately $105 million, $103 million and $91 million have been capitalized within equipment and computer software during 2024, 2023 and 2022, respectively, and are being amortized over an estimated useful life of three years.

Depreciation and amortization expense was $270 million, $263 million and $251 million for 2024, 2023 and 2022, respectively.

11. Goodwill

Goodwill activity during 2024 and 2023 was as follows:

(in millions)	2024		2023	
Beginning of year balance	**$**	**15,524**	$	15,341
Acquisitions[(1)]		**10,428**		184
Other		**(3)**		(1)
End of year balance	**$**	**25,949**	$	15,524

(1) 2024 amount represents goodwill of $10.3 billion related to the GIP Transaction and $131 million related to the SpiderRock Transaction. 2023 amount represents goodwill in connection with the Kreos Transaction. See Note 3, *Acquisitions*, for further information.

BlackRock assessed its goodwill for impairment as of July 31, 2024, 2023 and 2022 and considered such factors as the book value and the market capitalization of the Company. The impairment assessment indicated no impairment charges were required. The Company continues to monitor its book value per share compared with closing prices of its common stock as well as qualitative factors for potential indicators of impairment. At December 31, 2024, the Company's common stock closed at a market price of $1,025, which exceeded its book value of $307 per share.

12. Intangible Assets

Intangible assets at December 31, 2024 and 2023 consisted of the following:

(in millions)	Remaining Weighted-Average Estimated Useful Life (years)	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount
At December 31, 2024				
Indefinite-lived intangible assets:				
Management contracts	**N/A**	**$ 16,119**	**$ —**	**$ 16,119**
Trade names/trademarks	**N/A**	**1,403**	**—**	**1,403**
License	**N/A**	**6**	**—**	**6**
Total indefinite-lived intangible assets		**17,528**	**—**	**17,528**
Finite-lived intangible assets[(1)]:				
Management contracts	**7.6**	**2,028**	**215**	**1,813**
Investor/customer relationships	**11.2**	**1,623**	**414**	**1,209**
Technology-related	**3.5**	**257**	**144**	**113**
Trade names/trademarks	**9.6**	**89**	**9**	**80**
Total finite-lived intangible assets	**8.9**	**3,997**	**782**	**3,215**
Total intangible assets		**$ 21,525**	**$ 782**	**$ 20,743**
At December 31, 2023				
Indefinite-lived intangible assets:				
Management contracts	N/A	$ 16,169	$ —	$ 16,169
Trade names/trademarks	N/A	1,403	—	1,403
License	N/A	6	—	6
Total indefinite-lived intangible assets		17,578	—	17,578
Finite-lived intangible assets[(2)]:				
Management contracts	3.7	244	156	88
Investor/customer relationships	6.0	785	338	447
Technology-related	4.6	260	118	142
Trade names/trademarks	1.8	9	6	3
Total finite-lived intangible assets	5.4	1,298	618	680
Total intangible assets		$ 18,876	$ 618	$ 18,258

N/A – Not Applicable

(1) In connection with the GIP Transaction, the Company acquired approximately $1.8 billion of finite-lived management contracts, $820 million of finite-lived investor relationships and $80 million of a finite-lived trade name (see Note 3, *Acquisitions*, for further information). In connection with the SpiderRock Transaction, the Company acquired approximately $29 million of finite-lived customer relationships and $8 million of finite-lived technology-related intangible assets with weighted-average estimated lives of approximately eleven and five years, respectively.

(2) In connection with the Kreos Transaction, the Company acquired approximately $67 million of finite-lived management contracts and $39 million of finite-lived investor relationships with weighted-average estimated lives of approximately five and ten years, respectively.

The Company assessed its intangible assets for impairment as of July 31, 2024 and concluded that an impairment charge was required for indefinite-lived intangible assets related to certain acquired open-end management contracts, primarily driven by quantitative factors, such as reduced growth expectations, a decrease in revenue basis points and net client outflows. As a result, the Company recorded a noncash impairment charge of $50 million, which is included within amortization and impairment of intangible assets expense on the consolidated statements of income for the year ended December 31, 2024. No impairment charges were required for any other intangible assets.

The impairment tests performed for intangible assets as of July 31, 2023 and 2022 indicated no impairment charges were required.

Estimated amortization expense for finite-lived intangible assets for each of the five succeeding years is as follows:

(in millions) **Year**	**Amount**
2025	$ 429
2026	457
2027	384
2028	283
2029	271

13. Leases

The following table presents components of lease cost included in general and administration expense on the consolidated statements of income:

(in millions)	2024	2023	2022
Lease cost:			
Operating lease cost[1]	**$ 183**	$ 189	$ 216
Variable lease cost[2]	**60**	49	39
Total lease cost	**$ 243**	$ 238	$ 255

(1) Amounts include short-term leases, which are immaterial for 2024, 2023 and 2022.
(2) Amounts include operating lease payments, which may be adjusted based on usage, changes in an index or market rate, as well as common area maintenance charges and other variable costs not included in the measurement of ROU assets and operating lease liabilities.

Supplemental information related to operating leases is summarized below:

(in millions)	2024	2023	2022
Supplemental cash flow information:			
Operating cash flows from operating leases included in the measurement of operating lease liabilities	**$ 183**	$ 142	$ 162
Supplemental noncash information:			
ROU assets in exchange for operating lease liabilities[1]	**$ 235**	$ 32	$ 115

(1) Amount for 2024 includes $75 million of ROU assets obtained in connection with the GIP Transaction. See Note 3, *Acquisitions,* for further information.

	December 31, 2024	December 31, 2023
Lease term and discount rate:		
Weighted-average remaining lease term	**14 years**	15 years
Weighted-average discount rate	**3 %**	3 %

(in millions) **Maturity of operating lease liabilities at December 31, 2024**	Amount
2025	**$ 198**
2026	**185**
2027	**179**
2028	**170**
2029	**156**
Thereafter	**1,439**
Total lease payments	**2,327**
Less: Imputed interest	**(419)**
Present value of lease liabilities	**$ 1,908**

14. Other Assets

The Company records certain corporate investments, which exclude seed and co-investments in the Company's sponsored investment products, within other assets on the consolidated statements of financial condition.

At December 31, 2024 and 2023, the Company had $888 million and $773 million, respectively, of corporate equity method investments, recorded within other assets. At December 31, 2024 and 2023, the Company's ownership interest in its minority investment in iCapital Network Inc. ("iCapital") was approximately 24% and 25%, respectively, and the carrying value of the Company's interest was $652 million and $641 million, respectively. In accordance with GAAP, certain equity method investees, including iCapital, do not account for both their financial assets and liabilities under fair value measures; therefore, the Company’s investment in such equity method investees may not represent fair value.

At December 31, 2024 and 2023, the Company had $438 million and $484 million, respectively, of other nonequity method corporate minority investments recorded within other assets. These investments include equity securities, generally measured at fair value or under the measurement alternative to fair value for nonmarketable securities, and corporate minority private debt investments measured at fair value. Changes in value of the equity securities are recorded in nonoperating income (expense) and changes in value of the debt securities are recorded in AOCI, net of tax. See Note 2, *Significant Accounting Policies*, for further information.

15. Borrowings

Short-Term Borrowings

2024 Revolving Credit Facility. The Company maintains an unsecured revolving credit facility, which is available for working capital and general corporate purposes (the “2024 Credit Facility”). In March 2024, the 2024 Credit Facility was amended to, among other things, (1) permit the GIP Transaction and the transactions contemplated in connection with the GIP Transaction, (2) add New BlackRock as a borrower under the existing credit agreement, (3) add New BlackRock as a guarantor of the payment and performance of the obligations, liabilities and indebtedness of Old BlackRock and certain of its other subsidiaries and (4) update the sustainability-linked pricing mechanics to remove existing metrics and allow new metrics, if any, to be set following the consummation of the GIP Transaction. In May 2024, the 2024 Credit Facility was further amended to, among other things, (1) increase the aggregate commitment amount by $400 million to $5.4 billion and (2) extend the maturity date to March 2029 for lenders (other than one non-extending lender) pursuant to the Company’s option to request extensions of the maturity date available under the 2024 Credit Facility (with the commitment of the non-extending lender maturing in March 2028). The 2024 Credit Facility permits the Company to request up to an additional $1.0 billion of borrowing capacity, subject to lender credit approval, which could increase the overall size of the 2024 Credit Facility to an aggregate principal amount of up to $6.4 billion. Interest on outstanding borrowings accrues at an applicable benchmark rate for the denominated currency of the loan, plus a spread. The 2024 Credit Facility requires the Company not to exceed a maximum leverage ratio (ratio of net debt to earnings before interest, taxes, depreciation and amortization, where net debt equals total debt less unrestricted cash) of 3 to 1, which was satisfied with a ratio of less than 1 to 1 at December 31, 2024. At December 31, 2024, the Company had no amount outstanding under the 2024 Credit Facility.

Commercial Paper Program. On November 7, 2024, BlackRock established a commercial paper program (the "CP Program") under which the Company may issue short-term, unsecured commercial paper notes (the "CP Notes") on a private-placement basis up to a maximum aggregate amount outstanding at any time of $5 billion. The payments of the CP Notes have been unconditionally guaranteed by Old BlackRock (the "CP Notes Guarantee"). In addition, the CP Notes will rank at least equal in right of payment with all of New BlackRock's other unsubordinated indebtedness, and the obligations of Old BlackRock under the guarantee will rank at least equal in right of payment with all of Old BlackRock's other unsubordinated indebtedness. Net proceeds of issuances of the CP Notes are expected to be used for general corporate purposes. The CP Program is currently supported by the 2024 Credit Facility. The CP Program replaced the Company's prior $4 billion commercial paper program, which was terminated concurrently with the establishment of this CP Program, and has been put into place in connection with the Company's organizational structure following the closing of the GIP Transaction. At December 31, 2024, BlackRock had no CP Notes outstanding.

Subsidiary Credit Facility. In January 2024, BlackRock Investment Management (UK) Limited ("BIM UK"), a wholly owned subsidiary of the Company, entered into a revolving credit facility (the “Subsidiary Credit Facility”) in the amount of £25 million (or approximately $31 million based on the GBP/USD foreign exchange rate at December 31, 2024) with a rolling 364-day term structure. The Subsidiary Credit Facility is available for BIM UK's general corporate and working capital purposes. At December 31, 2024, there was no amount outstanding under the Subsidiary Credit Facility.

Long-Term Borrowings

The carrying value and fair value of long-term borrowings determined using market prices and EUR/USD foreign exchange rate at December 31, 2024 included the following:

(in millions)	Maturity Amount	Unamortized Discount and Debt Issuance Costs(1)	Carrying Value	Fair Value
1.25% Notes due 2025(2)	$ 725	$ —	$ 725	$ 720
3.20% Notes due 2027(2)	700	(2)	698	682
4.60% Notes due 2027	800	(3)	797	801
3.25% Notes due 2029(2)	1,000	(7)	993	944
4.70% Notes due 2029	500	(3)	497	500
2.40% Notes due 2030(2)	1,000	(3)	997	891
1.90% Notes due 2031(2)	1,250	(7)	1,243	1,055
2.10% Notes due 2032(2)	1,000	(11)	989	823
4.75% Notes due 2033(2)	1,250	(17)	1,233	1,224
5.00% Notes due 2034	1,000	(7)	993	986
4.90% Notes due 2035	500	(5)	495	489
5.25% Notes due 2054	1,500	(32)	1,468	1,418
5.35% Notes due 2055	1,200	(14)	1,186	1,147
Total long-term borrowings	$ 12,425	$ (111)	$ 12,314	$ 11,680

(1) The unamortized discount and debt issuance costs are being amortized over the term of the notes.
(2) Issued by Old BlackRock and guaranteed by New BlackRock.

Long-term borrowings at December 31, 2023 had a carrying value of $7.9 billion and a fair value of $7.4 billion determined using market prices at the end of December 2023.

March 2024 Notes. In March 2024, the Company issued $3.0 billion in aggregate principal amount of senior unsecured and unsubordinated notes. These notes were issued as three separate series of senior debt securities including $500 million of 4.70% notes maturing on March 14, 2029 (the "March 2029 Notes"), $1.0 billion of 5.00% notes maturing on March 14, 2034 (the "2034 Notes") and $1.5 billion of 5.25% notes maturing on March 14, 2054 (the "2054 Notes") (collectively, the "March 2024 Notes"). Net proceeds were used to fund a portion of the cash consideration for the GIP Transaction, which closed in October 2024. Interest on the March 2024 Notes of approximately $152 million per year is payable semi-annually on March 14 and September 14 of each year, which commenced on September 14, 2024. The March 2024 Notes are fully and unconditionally guaranteed (the “March 2024 Notes Guarantee”) on a senior unsecured basis by Old BlackRock. The March 2024 Notes and the March 2024 Notes Guarantee rank equally in right of payment with all of BlackRock and Old BlackRock’s other unsubordinated indebtedness, respectively. The March 2024 Notes may be redeemed prior to maturity at any time in whole or in part at the option of BlackRock at the redemption prices set forth in the applicable series of March 2024 Notes.

July 2024 Notes. In July 2024, the Company issued $2.5 billion in aggregate principal amount of senior unsecured and unsubordinated notes. These notes were issued as three separate series of senior debt securities including $800 million of 4.60% notes maturing on July 26, 2027 (the "July 2027 Notes"), $500 million of 4.90% notes maturing on January 8, 2035 (the "2035 Notes") and $1.2 billion of 5.35% notes maturing on January 8, 2055 (the "2055 Notes") (collectively, the "July 2024 Notes"). Net proceeds are intended to be used to fund a portion of the cash consideration for the acquisition of Preqin Holding Limited ("Preqin" or the "Preqin Transaction"), which is anticipated to close in the first quarter of 2025, subject to customary closing conditions. The July 2024 Notes are fully and unconditionally guaranteed (the "July 2024 Notes Guarantee") on a senior unsecured basis by Old BlackRock. The July 2024 Notes and the July 2024 Notes Guarantee rank equally in right of payment with all of BlackRock's and Old BlackRock's other unsubordinated indebtedness, respectively. Interest on the July 2027 Notes of approximately $37 million per year is payable semi-annually on January 26 and July 26 of each year, beginning January 26, 2025. Interest on the 2035 Notes and 2055 Notes of approximately $25 million and $64 million per year, respectively, is payable semi-annually on January 8 and July 8 of each year, beginning January 8, 2025. The July 2024 Notes may be redeemed prior to maturity at any time in whole or in part at the option of BlackRock at the redemption prices set forth in the applicable series of July 2024 Notes. In addition, if the Preqin Transaction is not consummated, the Company will be required to redeem all outstanding July 2027 Notes (the "Special Mandatory Redemption") at a Special Mandatory Redemption price equal to 101% of the aggregate principal amount of the applicable series of July 2027 Notes, plus accrued and unpaid interest, if any, to, but excluding, the Special Mandatory Redemption date.

2033 Notes. In May 2023, the Company issued $1.25 billion in aggregate principal amount of 4.75% senior unsecured notes maturing on May 25, 2033 (the "2033 Notes"). The net proceeds of the 2033 Notes are being used for general corporate purposes. Interest of approximately $59 million per year is payable semi-annually on May 25 and November 25 of each year, commencing on November 25, 2023. The 2033 Notes may be redeemed at the option of the Company, in whole or in part, at any time prior to February 25, 2033 at a "make-whole" redemption price, or thereafter at 100% of the principal amount of the 2033 Notes, in each case plus accrued but unpaid interest.

2032 Notes. In December 2021, the Company issued $1 billion in aggregate principal amount of 2.10% senior unsecured and unsubordinated notes maturing on February 25, 2032 (the "2032 Notes"). The net proceeds of the 2032 Notes were used for general corporate purposes, which included the repayment of the $750 million 3.375% Notes in June 2022. Interest of approximately $21 million per year is payable semi-annually on February 25 and August 25 of each year, which commenced on February 25, 2022. The 2032 Notes may be redeemed prior to November 25, 2031 in whole or in part at any time, at the option of the Company, at a "make-whole" redemption price or at 100% of the principal amount of the 2032 Notes thereafter.

2031 Notes. In April 2020, the Company issued $1.25 billion in aggregate principal amount of 1.90% senior unsecured and unsubordinated notes maturing on January 28, 2031 (the "2031 Notes"). The net proceeds of the 2031 Notes were used for general corporate purposes. Interest of approximately $24 million per year is payable semi-annually on January 28 and July 28 of each year, which commenced on July 28, 2020. The 2031 Notes may be redeemed prior to October 28, 2030 in whole or in part at any time, at the option of the Company, at a "make-whole" redemption price or at 100% of the principal amount of the 2031 Notes thereafter.

2030 Notes. In January 2020, the Company issued $1 billion in aggregate principal amount of 2.40% senior unsecured and unsubordinated notes maturing on April 30, 2030 (the "2030 Notes"). The net proceeds of the 2030 Notes were used for general corporate purposes. Interest of approximately $24 million per year is payable semi-annually on April 30 and October 30 of each year, which commenced on April 30, 2020. The 2030 Notes may be redeemed prior to January 30, 2030 in whole or in part at any time, at the option of the Company, at a "make-whole" redemption price or at 100% of the principal amount of the 2030 Notes thereafter.

2029 Notes. In April 2019, the Company issued $1 billion in aggregate principal amount of 3.25% senior unsecured and unsubordinated notes maturing on April 30, 2029 (the "2029 Notes"). The net proceeds of the 2029 Notes were used for general corporate purposes, which included a portion of the purchase price for the acquisition of eFront Holdings SAS, repayment of a portion of the $1 billion 5.00% notes in December 2019 and repayment of borrowings under its commercial paper program. Interest is payable semi-annually on April 30 and October 30 of each year, which commenced on October 30, 2019, and is approximately $33 million per year. The 2029 Notes may be redeemed prior to January 30, 2029 in whole or in part at any time, at the option of the Company, at a "make-whole" redemption price or at par thereafter.

2027 Notes. In March 2017, the Company issued $700 million in aggregate principal amount of 3.20% senior unsecured and unsubordinated notes maturing on March 15, 2027 (the "2027 Notes"). The net proceeds of the 2027 Notes were used to fully repay $700 million in aggregate principal amount outstanding of 6.25% notes in April 2017 prior to their maturity in September 2017. Interest is payable semi-annually on March 15 and September 15 of each year, and is approximately $22 million per year. The 2027 Notes may be redeemed prior to maturity at any time in whole or in part at the option of the Company at a "make-whole" redemption price.

2025 Notes. In May 2015, the Company issued €700 million of 1.25% senior unsecured notes maturing on May 6, 2025 (the "2025 Notes" and, together with the 2027 Notes, the 2029 Notes, the 2030 Notes, the 2031 Notes, the 2032 Notes and the 2033 Notes, the "Old BlackRock Notes"). The notes are listed on the Official List of The International Stock Exchange. The net proceeds of the 2025 Notes were used for general corporate purposes, including refinancing of outstanding indebtedness. Interest of approximately $11 million per year based on current exchange rates is payable annually on May 6 of each year. The 2025 Notes may be redeemed in whole or in part prior to maturity at any time at the option of the Company at a "make-whole" redemption price.

Upon conversion to US dollars the Company designated the €700 million debt offering as a net investment hedge to offset its currency exposure relating to its net investment in certain euro functional currency operations. A gain of $37 million (net of tax expense of $12 million), loss of $20 million (net of tax benefit of $6 million), and a gain of $37 million (net of tax expense of $12 million) were recognized in other comprehensive income for 2024, 2023 and 2022, respectively. No hedge ineffectiveness was recognized during 2024, 2023 and 2022.

2024 Notes. In March 2014, the Company issued $1 billion in aggregate principal amount of 3.50% senior unsecured and unsubordinated notes, which were repaid in March 2024 at maturity (the "2024 Notes"). The net proceeds of the 2024 Notes were used to refinance certain indebtedness which matured in the fourth quarter of 2014. Interest was payable semi-annually in arrears on March 18 and September 18 of each year, or approximately $35 million per year.

New BlackRock Guarantee. On October 1, 2024, in connection with the closing of the GIP Transaction, New BlackRock also entered into a guarantee (the "New BlackRock Guarantee") pursuant to which New BlackRock fully and unconditionally guaranteed, on a senior unsecured basis, the obligations of Old BlackRock with respect to the Old BlackRock Notes. The New BlackRock Guarantee ranks equally in right of payment with all of New BlackRock's other unsubordinated indebtedness. The New BlackRock Guarantee will be automatically and unconditionally released and discharged, and New BlackRock will be released from all obligations under the New BlackRock Guarantee, in certain circumstances as described in the New BlackRock Guarantee.

16. Commitments and Contingencies

Investment Commitments. At December 31, 2024, the Company had $1.2 billion of various capital commitments to fund sponsored investment products, including CIPs. These products include various private markets products, including private equity funds, real assets funds, and opportunistic funds. This amount excludes additional commitments made by consolidated funds of funds to underlying third-party funds as third-party noncontrolling interest holders have the legal obligation to fund the respective commitments of such funds of funds. Generally, the timing of the funding of these commitments is unknown and the commitments are callable on demand at any time prior to the expiration of the commitment. These unfunded commitments are not recorded on the consolidated statements of financial condition. These commitments do not include potential future commitments approved by the Company that are not yet legally binding. The Company intends to make additional capital commitments from time to time to fund additional investment products for, and with, its clients.

Contingencies

Contingent Consideration Liabilities. In connection with certain acquisitions, BlackRock is required to make contingent payments, subject to the achievement of specified performance targets or satisfaction of certain post-closing events. The fair value of this contingent consideration is estimated at the time of acquisition closing and is included in contingent consideration liabilities on the consolidated statements of financial condition. The fair value of the remaining aggregate contingent payments at December 31, 2024 totaled $4.3 billion, including $4.2 billion related to the GIP Transaction, which, if any, will be settled, all in stock, ranging from 4.0 million to 5.2 million shares, subject to achieving certain performance targets. See Note 3, *Acquisitions*, for more information.

Legal Proceedings. From time to time, BlackRock receives subpoenas or other requests for information from various US federal and state governmental and regulatory authorities and international governmental and regulatory authorities in connection with industry-wide or other investigations or proceedings. It is BlackRock's policy to cooperate fully with such matters. In 2023, BlackRock responded to requests from the SEC in connection with a publicly reported, industry-wide investigation of investment advisers' compliance with record retention requirements relating to certain types of electronic communications.

The Company, certain of its subsidiaries and employees have been named as defendants in various legal actions, including arbitrations and other litigation arising in connection with BlackRock's activities. Additionally, BlackRock-advised investment portfolios may be subject to lawsuits, any of which potentially could harm the investment returns of the applicable portfolio or result in the Company being liable to the portfolios for any resulting damages.

Management, after consultation with legal counsel, currently does not anticipate that the aggregate liability arising out of regulatory matters or lawsuits will have a material effect on BlackRock's results of operations, financial position, or cash flows. However, there is no assurance as to whether any such pending or threatened matters will have a material effect on BlackRock's results of operations, financial position or cash flows in any future reporting period. Due to uncertainties surrounding the outcome of these matters, management cannot reasonably estimate the possible loss or range of loss that may arise from these matters.

Indemnifications. In the ordinary course of business or in connection with certain acquisition agreements, BlackRock enters into contracts pursuant to which it may agree to indemnify third parties in certain circumstances. The terms of these indemnities vary from contract to contract and the amount of indemnification liability, if any, cannot be determined or the likelihood of any liability is considered remote. Consequently, no liability has been recorded on the consolidated statements of financial condition.

In connection with securities lending transactions, BlackRock has agreed to indemnify certain securities lending clients against potential loss resulting from a borrower's failure to fulfill its obligations under the securities lending agreement should the value of the collateral pledged by the borrower at the time of default be insufficient to cover the borrower's obligation under the securities lending agreement. The amount of securities on loan as of December 31, 2024 and subject to this type of indemnification was approximately $305 billion. In the Company's capacity as lending agent, cash and securities totaling approximately $324 billion were held as collateral for indemnified securities on loan at December 31, 2024. The fair value of these indemnifications was not material at December 31, 2024.

17. Revenue

The table below presents detail of revenue for 2024, 2023 and 2022 and includes the product mix of investment advisory, administration fees and securities lending revenue and performance fees.

(in millions)	2024	2023	2022
Revenue:			
Investment advisory, administration fees and securities lending revenue:			
Equity:			
Active	**$ 2,166**	$ 2,000	$ 2,147
ETFs	**5,124**	4,418	4,345
Non-ETF index	**784**	743	711
Equity subtotal	**8,074**	7,161	7,203
Fixed income:			
Active	**1,952**	1,897	1,977
ETFs	**1,367**	1,230	1,122
Non-ETF index	**369**	353	396
Fixed income subtotal	**3,688**	3,480	3,495
Multi-asset	**1,278**	1,203	1,299
Alternatives:			
Private markets	**1,196**	889	741
Liquid alternatives	**568**	572	633
Currency and commodities(1)	**247**	185	216
Alternatives subtotal	**2,011**	1,646	1,590
Long-term	**15,051**	13,490	13,587
Cash management	**1,049**	909	864
Total investment advisory, administration fees and securities lending revenue(2)	**16,100**	14,399	14,451
Investment advisory performance fees:			
Equity	**161**	99	49
Fixed income	**34**	4	25
Multi-asset	**24**	28	25
Alternatives:			
Private markets	**308**	273	296
Liquid alternatives	**680**	150	119
Alternatives subtotal	**988**	423	415
Total investment advisory performance fees	**1,207**	554	514
Technology services revenue	**1,603**	1,485	1,364
Distribution fees	**1,273**	1,262	1,381
Advisory and other revenue:			
Advisory	**49**	81	56
Other	**175**	78	107
Total advisory and other revenue	**224**	159	163
Total revenue	**$ 20,407**	$ 17,859	$ 17,873

(1) Amounts include cryptocurrency and commodity ETFs and exchange-traded products.
(2) Amounts include $615 million, $675 million and $599 million of securities lending revenue for 2024, 2023 and 2022, respectively.

The tables below present the investment advisory, administration fees and securities lending revenue by client type and investment style:

(in millions)	2024	2023	2022
By client type:			
Retail	**$ 4,284**	$ 4,115	$ 4,442
ETFs	**6,743**	5,834	5,671
Institutional:			
Active	**3,089**	2,623	2,535
Index	**935**	918	939
Total institutional	**4,024**	3,541	3,474
Long-term	**15,051**	13,490	13,587
Cash management	**1,049**	909	864
Total	**$ 16,100**	$ 14,399	$ 14,451
By investment style:			
Active	**$ 7,117**	$ 6,534	$ 6,789
Index and ETFs	**7,934**	6,956	6,798
Long-term	**15,051**	13,490	13,587
Cash management	**1,049**	909	864
Total	**$ 16,100**	$ 14,399	$ 14,451

Investment Advisory and Administration Fees – Remaining Performance Obligation

The tables below present estimated investment advisory and administration fees expected to be recognized in the future related to the unsatisfied portion of the performance obligations at December 31, 2024 and 2023:

December 31, 2024

(in millions)	2025	2026	2027	Thereafter	Total
Investment advisory and administration fees:					
Alternatives(1)(2)	**$ 427**	**$ 381**	**$ 353**	**$ 142**	**$ 1,303**

December 31, 2023

(in millions)	2024	2025	2026	Thereafter	Total
Investment advisory and administration fees:					
Alternatives(1)(2)	$ 204	$ 174	$ 152	$ 164	$ 694

(1) Investment advisory and administration fees include management fees related to certain private markets products, which are determined based on known contractual committed capital outstanding at December 31, 2024 and 2023. Revenue attributed to future periods could be subject to change due to a change in business activities and actual amounts could differ from amounts disclosed in the table above.

(2) The Company elected practical expedients to exclude amounts related to (a) performance obligations with an original duration of one year or less, and (b) variable consideration related to future service periods.

Change in Deferred Carried Interest Liability

The table below presents changes in the deferred carried interest liability, which is included in other liabilities on the consolidated statements of financial condition, for the year ended December 31, 2024 and 2023:

(in millions)	2024	2023
Beginning balance	**$ 1,783**	$ 1,420
Net increase (decrease) in unrealized allocations	**364**	577
Performance fee revenue recognized	**(287)**	(214)
Ending balance	**$ 1,860**	$ 1,783

Technology Services Revenue – Remaining Performance Obligation

The tables below present estimated technology services revenue expected to be recognized in the future related to the unsatisfied portion of the performance obligations at December 31, 2024 and 2023:

December 31, 2024

(in millions)	2025	2026	2027	Thereafter	Total
Technology services revenue(1)(2)	**$ 134**	**$ 81**	**$ 50**	**$ 69**	**$ 334**

December 31, 2023

(in millions)	2024	2025	2026	Thereafter	Total
Technology services revenue(1)(2)	$ 131	$ 73	$ 56	$ 59	$ 319

(1) Technology services revenue primarily includes upfront payments from customers, which the Company generally recognizes as services are performed. Revenue attributed to future periods could be subject to change due to a change in business activities and actual amounts could differ from amounts disclosed in the table above.

(2) The Company elected practical expedients to exclude amounts related to (a) performance obligations with an original duration of one year or less, and (b) variable consideration related to future service periods.

In addition to amounts disclosed in the tables above, certain technology services contracts require fixed minimum fees, which are billed on a monthly or quarterly basis in arrears. The Company recognizes such revenue as services are performed. As of December 31, 2024, the estimated annual fixed minimum fees for 2025 for outstanding contracts approximated $1.2 billion. The term for these contracts, which are either in their initial or renewal period, ranges from one to five years.

The table below presents changes in the technology services deferred revenue liability for the year ended December 31, 2024 and 2023, which is included in other liabilities on the consolidated statements of financial condition:

(in millions)	2024	2023
Beginning balance	**$ 133**	$ 125
Additions(1)	**84**	92
Revenue recognized that was included in the beginning balance	**(93)**	(84)
Ending balance	**$ 124**	$ 133

(1) Amounts are net of revenue recognized.

18. Stock-Based Compensation

Prior to May 15, 2024, the Company maintained the BlackRock, Inc. Second Amended and Restated 1999 Stock Award and Incentive Plan. On May 15, 2024, the Company adopted, pursuant to shareholder approval, the BlackRock, Inc. Third Amended and Restated 1999 Stock Award and Incentive Plan (the "Award Plan"). Any awards granted on or after May 15, 2024 are granted pursuant to such plan.

The components of stock-based compensation expense are as follows:

(in millions)	2024	2023	2022
Stock-based compensation:			
RSUs(1)	**$ 718**	$ 596	$ 686
Stock options	**35**	34	22
Total stock-based compensation(2)	**$ 753**	$ 630	$ 708

(1) Amount for 2024 includes $71 million of incentive retention awards granted in connection with the GIP Transaction.

(2) Amount for 2023 and 2022 includes $14 million and $33 million of compensation expense for accelerated vesting of previously granted stock-based compensation awards, respectively, recognized as part of the restructuring charge disclosed in Note 24, *Restructuring Charge.*

Stock Award and Incentive Plan. Pursuant to the Award Plan, options to purchase shares of the Company's common stock at an exercise price not less than the market value of BlackRock's common stock on the date of grant in the form of stock options, restricted stock or RSUs may be granted to employees and nonemployee directors. A maximum of 48,500,000 shares of common stock were authorized for issuance under the Award Plan. Of this amount, 7,708,389 shares remain available for future awards at December 31, 2024. Upon exercise of employee stock options, the issuance of restricted stock or the vesting of RSUs, the Company generally issues shares out of treasury to the extent available.

RSUs

Time-Based RSUs

Pursuant to the Award Plan, RSUs may be granted to certain employees. Substantially all RSUs vest over periods ranging from one to five years and are expensed using the straight-line method over the requisite service period for each separately vesting portion of the award as if the award was, in-substance, multiple awards. RSUs are not considered participating securities for purposes of calculating EPS as the dividend equivalents are subject to forfeiture prior to vesting of the award.

RSU activity for 2024 is summarized below.

Outstanding at	RSUs	Weighted-Average Grant Date Fair Value
December 31, 2023	1,772,639	$ 757.49
Granted(1)	1,395,311	$ 817.10
Converted	(779,146)	$ 760.38
Forfeited	(91,139)	$ 748.23
December 31, 2024	**2,297,665**	**$ 793.08**

(1) During the first quarter of 2024, in connection with the GIP Transaction, the Company granted incentive retention awards of approximately 106,000 RSUs to certain employees that vest between two to five years from the grant date. Additionally, in October 2024, in connection with the GIP Transaction, the Company granted incentive retention awards of approximately 500,000 RSUs to certain employees that cliff vest 100% on October 1, 2029. The weighted-average grant date fair value of these October 2024 awards was $830.90.

The Company values RSUs at their grant-date fair value as measured by BlackRock's common stock price. The incentive retention RSUs granted in October 2024, in connection with the GIP Transaction, are not entitled to participate in dividends until they vest, hence the grant-date fair value of the awards are reduced by the present value of the dividends expected to be paid on the common shares during the vesting period, discounted at a risk-free interest rate. The total fair market value of RSUs granted to employees during 2024, 2023 and 2022 was $1.1 billion, $565 million and $662 million, respectively. The total grant-date fair market value of RSUs converted to common stock during 2024, 2023 and 2022 was $592 million, $592 million and $461 million, respectively.

RSUs granted under the Award Plan in connection with annual incentive compensation and incentive retention awards in connection with the GIP Transaction primarily related to the following:

	2024	2023	2022
Awards granted that vest ratably over three years from the date of grant	**346,831**	342,706	498,633
Awards granted that vest with varying vesting periods	**204,622**	169,764	117,169
Awards granted that cliff vest 100% on:			
January 31, 2025	**—**	—	197,817
January 31, 2026	**—**	259,465	—
January 31, 2027	**343,418**	—	—
October 1, 2029(1)	**500,440**	—	—
Total awards granted	**1,395,311**	771,935	813,619

(1) Incentive retention awards granted in October 2024, in connection with the GIP Transaction.

At December 31, 2024, the intrinsic value of outstanding RSUs was $2.4 billion, reflecting a closing stock price of $1,025.

At December 31, 2024, total unrecognized stock-based compensation expense related to unvested RSUs was $913 million. The unrecognized compensation cost is expected to be recognized over the remaining weighted-average period of 2.0 years.

In January 2025, pursuant to the Award Plan, the Company granted approximately:

- 332,000 RSUs to employees as part of annual incentive compensation that vest ratably over three years from the date of grant;
- 216,000 RSUs to employees that cliff vest 100% on January 31, 2028; and
- 27,000 RSUs to employees with various vesting schedules.

Performance-Based RSUs

Pursuant to the Award Plan, performance-based RSUs may be granted to certain employees. Each performance-based award consists of a "base" number of RSUs granted to the employee. The number of shares that an employee ultimately receives at vesting will be equal to the base number of performance-based RSUs granted, multiplied by a predetermined percentage determined in accordance with the level of attainment of Company performance measures during the performance period and could be higher or lower than the original RSU grant. Performance-based RSUs are not considered participating securities as the dividend equivalents are subject to forfeiture prior to vesting of the award.

In the first quarter of 2024, 2023 and 2022, the Company granted 165,631, 169,938 and 143,846, respectively, performance-based RSUs to certain employees that cliff vest 100% on January 31, 2027, 2026 and 2025, respectively. These awards are amortized over a service period of three years. In 2024, the Company reduced the number of original shares granted in 2021 by 42,579 RSUs based on the level of attainment of Company performance measures during the performance period.

In October 2024, in connection with the GIP Transaction, the Company awarded a target amount of approximately 210,000 incentive retention performance-based RSUs to certain employees subject to achievement of certain performance targets. The number of shares that an employee ultimately receives at vesting could be higher or lower than the original target amount, based on the achievement of certain performance targets.

Performance-based RSU activity for 2024 is summarized below.

Outstanding at	Performance-Based RSUs	Weighted-Average Grant Date Fair Value	Performance-Based RSUs in Connection with the GIP Transaction	Weighted-Average Grant Date Fair Value	Total Performance-Based RSUs	Weighted-Average Grant Date Fair Value
December 31, 2023	456,384	$ 767.69	—	$ —	456,384	$ 767.69
Granted	165,631	$ 798.83	210,505	$ 845.48	376,136	$ 824.94
Reduction of shares due to performance measures	(42,579)	$ 739.64	—	$ —	(42,579)	$ 739.64
Converted	(116,029)	$ 739.32	—	$ —	(116,029)	$ 739.32
Forfeited	(12,365)	$ 784.44	—	$ —	(12,365)	$ 784.44
December 31, 2024	**451,042**	**$ 788.61**	**210,505**	**$ 845.48**	**661,547**	**$ 806.71**

The Company values performance-based RSUs at their grant-date fair value as measured by BlackRock's common stock price. The incentive retention performance-based RSUs granted in October 2024 in connection with GIP Transaction mentioned above, are not entitled to participate in dividends until they vest, hence the grant-date fair value of the awards are reduced by the present value of the dividends expected to be paid on the common shares during the vesting period, discounted at a risk-free interest rate.

The total grant-date fair market value of performance-based RSUs granted (including impact due to performance measures) to employees during 2024, 2023 and 2022 was $279 million, $142 million and $164 million, respectively. At December 31, 2024, the intrinsic value of outstanding performance-based RSUs was $678 million reflecting a closing stock price of $1,025. At December 31, 2024, total unrecognized stock-based compensation expense related to unvested performance-based awards was $311 million. The unrecognized compensation cost is expected to be recognized over the remaining weighted-average period of 2.1 years.

In January 2025, the Company granted approximately 136,000 performance-based RSUs to certain employees that cliff vest 100% on January 31, 2028. These awards are amortized over a service period of three years. The number of shares distributed at vesting could be higher or lower than the original grant based on the level of attainment of predetermined Company performance measures.

Stock Options

Stock option activity and ending balance for year-end December 31, 2024 is summarized below.

	2017 Performance-based Options		2023 Performance-based Options		2023 Time-based Options	
	Shares Under Option	Weighted Average Exercise Price	Shares Under Option	Weighted Average Exercise Price	Shares Under Option	Weighted Average Exercise Price
Outstanding at December 31, 2023	1,549,080	$ 513.50	807,695	$ 673.58	326,391	$ 673.58
Exercised	(905,219)	$ 513.50	—	$ —	—	$ —
Forfeited	(18,036)	$ 513.50	(40,725)	$ 673.58	(26,705)	$ 673.58
Outstanding at December 31, 2024	**625,825**	**$ 513.50**	**766,970**	**$ 673.58**	**299,686**	**$ 673.58**

	Options Outstanding				Options Exercisable			
Option Type	Exercise Prices	Options Outstanding	Weighted Average Remaining Life (years)	Aggregate Intrinsic Value (in millions)	Exercise Prices	Options Exercisable	Weighted Average Remaining Life (years)	Aggregate Intrinsic Value (in millions)
2017 Performance-based	**$ 513.50**	**625,825**	**1.9**	**$ 320**	**$ 513.50**	**625,825**	**1.9**	**$ 320**
2023 Performance-based	**$ 673.58**	**766,970**	**7.4**	**270**	**$ 673.58**	**—**	**—**	**—**
2023 Time-based	**$ 673.58**	**299,686**	**7.4**	**105**	**$ 673.58**	**—**	**—**	**—**
Total		**1,692,481**	**5.4**	**$ 695**		**625,825**	**1.9**	**$ 320**

At December 31, 2024, total unrecognized stock-based compensation expense related to unvested performance-based and time-based stock options was $113 million. The unrecognized compensation cost is expected to be recognized over the remaining weighted-average period of 3.7 years.

Performance-Based Stock Options

In 2017, pursuant to the Award Plan, the Company awarded performance-based stock option grants to certain employees ("2017 Performance-based Options"). Vesting of 2017 Performance-based Options was contingent upon the achievement of obtaining 125% of BlackRock's grant-date stock price within five years from the grant date and the attainment of Company performance measures during the four-year performance period. Both hurdles have been achieved, and each of the three tranches of the awards vested in equal installments at the end of 2022, 2023 and 2024, respectively. Vested 2017 Performance-based Options are exercisable for up to nine years following the grant date. The expense for each tranche is amortized over the respective requisite service period. The total fair value of 2017 Performance-based Options vested during 2024 was $52 million. The aggregate intrinsic value of 2017 Performance-based Options exercised during 2024 was $373 million.

The options have a strike price of $513.50, which was the closing price of the shares on the grant date. The grant-date fair value of the awards issued in 2017 was $208 million and was estimated using a Monte Carlo simulation with an embedded lattice model using the assumptions included in the following table:

Grant Year	Expected Term (Years)(1)	Expected Stock Volatility(2)	Expected Dividend Yield(3)	Risk-Free Interest Rate(4)
2017	6.56	22.23%	2.16%	2.33%

(1) The expected term was derived using a Monte Carlo simulation with the embedded lattice model and represents the period of time that options granted are expected to be outstanding.
(2) The expected stock volatility was based upon an average of historical stock price fluctuations of BlackRock's common stock and an implied volatility at the grant date.
(3) The expected dividend yield was calculated as the most recent quarterly dividend divided by the average three-month stock price as of the grant date.
(4) The risk-free interest rate is based on the US Treasury Constant Maturities yield curve at grant date.

On May 30, 2023, pursuant to the Award Plan, the Company awarded performance-based options to purchase 814,482 shares of BlackRock common stock to certain employees as long-term incentive compensation ("2023 Performance-based Options"). Vesting of 2023 Performance-based Options is contingent upon the achievement of obtaining 130% of grant-date stock price over 60 calendar days within four years from the grant date and attainment of a predetermined Company performance measure during the three-year performance period. As of December 31, 2024, the price hurdle was achieved and the Company assumes that the performance measure will be achieved. Accordingly, the awards are expected to vest in three tranches of 25%, 25% and 50% in May of 2027, 2028 and 2029, respectively. Vested 2023 Performance-based Options are exercisable for up to nine years following the grant date, and the awards are forfeited if the employee resigns before the respective vesting date. The expense for each tranche is amortized over the respective requisite service period.

The 2023 Performance-based Options have a strike price of $673.58 which was the closing price of the shares on the grant date. The grant-date fair value of the 2023 Performance-based Options was $120 million and was estimated using a Monte Carlo simulation with an embedded lattice model using the assumptions included in the following table:

Grant Year	Expected Term (Years)(1)	Expected Stock Volatility(2)	Expected Dividend Yield(3)	Risk-Free Interest Rate(4)
2023	6.02	27.73%	3.02%	3.61%

(1) The expected term was derived using a Monte Carlo simulation with the embedded lattice model and represents the period of time that options granted are expected to be outstanding.
(2) The expected stock volatility was based upon an average of historical stock price fluctuations of BlackRock's common stock and an implied volatility at the grant date.
(3) The expected dividend yield was calculated as the most recent quarterly dividend divided by the average three-month stock price as of the grant date.
(4) The risk-free interest rate is based on the US Treasury Constant Maturities yield curve at grant date.

Time-Based Stock Options

On May 30, 2023, pursuant to the Award Plan, the Company awarded time-based stock options to purchase 326,391 shares of BlackRock common stock to certain employees as long-term incentive compensation ("2023 Time-based Options"). These awards will vest in three tranches of 25%, 25% and 50% in May 2027, 2028 and 2029, respectively. Vested 2023 Time-based Options can be exercised up to nine years following the grant date, and the awards are forfeited if the employee resigns before the respective vesting date.

The 2023 Time-based Options have a strike price of $673.58 which was the closing price of the shares on the grant date. The grant-date fair value of the 2023 Time-based Options was $55 million and was estimated using a Black-Scholes-Merton model using the assumptions included in the following table:

Grant Year	Expected Term (Years)(1)	Expected Stock Volatility(2)	Expected Dividend Yield(3)	Risk-Free Interest Rate(4)
2023	7.13	28.29%	3.02%	3.65%

(1) The expected term represents the period of time that options granted are expected to be outstanding, and was calculated as the midpoint between the weighted average time to vest and expiration.
(2) The expected stock volatility was based upon an average of historical stock price fluctuations of BlackRock's common stock and an implied volatility at the grant date.
(3) The expected dividend yield was calculated as the most recent quarterly dividend divided by the average three-month stock price as of the grant date.
(4) The risk-free interest rate is based on the US Treasury Constant Maturities yield curve at grant date.

Employee Stock Purchase Plan ("ESPP"). The ESPP allows eligible employees to purchase the Company's common stock at 95% of the fair market value on the last day of each three-month offering period; therefore, the Company does not record compensation expense related to employees purchasing shares under the ESPP.

19. Deferred Cash Compensation and Employee Benefit Plans

Deferred Cash Compensation Plans

The components of deferred cash compensation expense are as follows:

(in millions)	**2024**	2023	2022
Deferred cash compensation expense:			
IPDCP	**$ 155**	$ 195	$ 228
VDCP	**26**	17	(18)
Other[1]	**31**	14	14
Total deferred cash compensation expense	**$ 212**	$ 226	$ 224

(1) Amounts primarily relate to deferred cash compensation in connection with certain acquisitions.

Investment Professional Deferred Compensation Program ("IPDCP"). The Company adopted IPDCP for the purpose of providing deferred compensation and retention incentives to certain employees. For this plan, the final value of the deferred amount to be distributed in cash upon vesting is associated with investment returns of certain investment funds. In January 2024, 2023 and 2022, the Company granted approximately $114 million, $90 million, and $257 million of deferred compensation that will fluctuate with investment returns and will vest ratably over three years from the date of grant. The liabilities for this plan were $220 million and $313 million at December 31, 2024 and 2023, respectively, and are reflected in the consolidated statements of financial condition as accrued compensation and benefits. In January 2025, the Company granted approximately $264 million of additional deferred compensation that will fluctuate with investment returns and will vest ratably over three years from the date of grant.

Voluntary Deferred Compensation Plan. The Company adopted a Voluntary Deferred Compensation Plan ("VDCP") that allows eligible employees in the US to elect to defer between 1% and 100% of their annual cash incentive compensation. The participants must specify a deferral period of up to 10 years from the year of deferral and additionally elect to receive distributions in the form of a lump sum or in up to 10 annual installments. VDCP deferred cash compensation expense includes the mark-to-market impact of investment returns. The liability balance of $170 million and $144 million at December 31, 2024 and 2023, respectively, is reflected on the consolidated statements of financial condition as accrued compensation and benefits.

Other Deferred Cash Plans. The liabilities related to other deferred cash plans were $34 million and $82 million at December 31, 2024 and 2023, respectively, primarily related to deferred cash plans granted in connection with certain acquisitions.

In 2019, the Company adopted a carried interest retention incentive program referred to as the BlackRock Leadership Retention Carry Plan, pursuant to which senior-level employees (but not including the Chief Executive Officer), as may be determined by the Company from time to time, will be eligible to receive a portion of the cash payments, based on their percentage points, in the total carried interest distributions paid to the Company from participating carry funds. Cash payments, if any, with respect to these percentage points will be made over time following the recipient's termination of employment due to qualified retirement, death or disability, subject to his or her execution of a release of claims and continued compliance with his or her restrictive covenant obligations following termination. There was no material impact to the consolidated financial statements.

Defined Contribution Plans

The Company has several defined contribution plans primarily in the US and UK.

Certain of the Company's US employees participate in a defined contribution plan. Employee contributions of up to 8% of eligible compensation, as defined by the plan and subject to Internal Revenue Code limitations, are matched by the Company at 50% up to a maximum of $5,000 annually. In addition, the Company makes an annual retirement contribution to eligible participants generally equal to 3-5% of eligible compensation. The Company's contribution expense related to this plan was $149 million in 2024, $86 million in 2023, and $83 million in 2022.

Certain UK wholly owned subsidiaries of the Company contribute to defined contribution plans for their employees. The contributions range between 9% and 15% of each employee's eligible compensation as of December 31, 2024. The Company's contribution expense related to these plans was $66 million in 2024, $64 million in 2023, and $60 million in 2022.

In addition, the contribution expense related to defined contribution plans in other regions was $50 million in 2024, $42 million in 2023 and $41 million in 2022.

Defined Benefit Plans. The Company has several defined benefit pension plans with plan assets of approximately $27 million and $28 million at December 31, 2024 and 2023, respectively. The underfunded obligations at December 31, 2024 and 2023 were not material. Benefit payments for the next five years and in aggregate for the five years thereafter are not expected to be material.

20. Related Party Transactions

Determination of Related Parties

Registered Investment Companies and Equity Method Investments. The Company considers the registered investment companies that it manages, which include mutual funds and ETFs, to be related parties as a result of the Company's advisory relationship. In addition, equity method investments are considered related parties, due to the Company's influence over the financial and operating policies of the investee.

Revenue from Related Parties

Revenue for services provided by the Company to these and other related parties are as follows:

(in millions)	2024	2023	2022
Investment advisory, administration fees and securities lending revenue(1)	**$ 12,050**	$ 10,757	$ 10,848
Investment advisory performance fees(1)	**728**	286	244
Other	**16**	(31)	(31)
Total revenue from related parties	**$ 12,794**	$ 11,012	$ 11,061

(1) Amounts primarily include revenue from registered investment companies and equity method investees.

The Company provides investment advisory and administration services to its open- and closed-end funds and other commingled or pooled funds and separate accounts in which related parties invest.

Receivables and Payables with Related Parties. Due from related parties, which is included within other assets on the consolidated statements of financial condition, was $245 million and $203 million at December 31, 2024 and 2023, respectively, and primarily represented receivables from certain investment products managed by BlackRock. Accounts receivable at December 31, 2024 and 2023 included $1.3 billion and $1.1 billion, respectively, related to receivables from BlackRock mutual funds and ETFs, for investment advisory and administration services.

Due to related parties, which is included within other liabilities on the consolidated statements of financial condition, was $11 million and $21 million at December 31, 2024 and 2023, respectively, and primarily represented payables to certain investment products managed by BlackRock.

21. Net Capital Requirements

The Company is required to maintain net capital in certain regulated subsidiaries within a number of jurisdictions, which is partially maintained by retaining cash and cash equivalent investments in those subsidiaries or jurisdictions. As a result, such subsidiaries of the Company may be restricted in their ability to transfer cash between different jurisdictions and to their parents. Additionally, transfers of cash between international jurisdictions may have adverse tax consequences that could discourage such transfers.

Banking Regulatory Requirements. BlackRock Institutional Trust Company, N.A. ("BTC"), a wholly owned subsidiary of the Company, which is chartered as a national bank whose powers are limited to trust and other fiduciary activities and which is subject to regulatory capital requirements administered by the US Office of the Comptroller of the Currency. Federal banking regulators would be required to take certain actions and permitted to take other actions in the event of BTC's failure to meet minimum capital requirements that, if undertaken, could have a direct material effect on the Company's consolidated financial statements.

Quantitative measures established by regulators to ensure capital adequacy require BTC to maintain a minimum Common Equity Tier 1 capital and Tier 1 leverage ratio, as well as Tier 1 and total risk-based capital ratios. Based on BTC's calculations as of December 31, 2024 and 2023, it exceeded the applicable capital adequacy requirements.

	Actual		For Capital Adequacy Purposes		To Be Well Capitalized Under Prompt Corrective Action Provisions	
(in millions)	Amount	Ratio	Amount	Ratio	Amount	Ratio
December 31, 2024						
Total capital (to risk weighted assets)	**$ 961**	**153.2 %**	**$ 50**	**8.0 %**	**$ 63**	**10.0 %**
Common Equity Tier 1 capital (to risk weighted assets)	**$ 953**	**151.9 %**	**$ 28**	**4.5 %**	**$ 41**	**6.5 %**
Tier 1 capital (to risk weighted assets)	**$ 953**	**151.9 %**	**$ 38**	**6.0 %**	**$ 50**	**8.0 %**
Tier 1 capital (to average assets)	**$ 953**	**71.3 %**	**$ 53**	**4.0 %**	**$ 67**	**5.0 %**
December 31, 2023						
Total capital (to risk weighted assets)	$ 775	145.8 %	$ 43	8.0 %	$ 53	10.0 %
Common Equity Tier 1 capital (to risk weighted assets)	$ 771	145.1 %	$ 24	4.5 %	$ 35	6.5 %
Tier 1 capital (to risk weighted assets)	$ 771	145.1 %	$ 32	6.0 %	$ 43	8.0 %
Tier 1 capital (to average assets)	$ 771	65.9 %	$ 47	4.0 %	$ 59	5.0 %

Broker-dealers. BlackRock Investments, LLC and BlackRock Execution Services are registered broker-dealers and wholly owned subsidiaries of BlackRock that are subject to the Uniform Net Capital requirements under the Securities Exchange Act of 1934, which requires maintenance of certain minimum net capital levels.

Capital Requirements. At both December 31, 2024 and 2023, the Company was required to maintain approximately $1.8 billion in net capital in certain regulated subsidiaries, including BTC, entities regulated by the Financial Conduct Authority and Prudential Regulation Authority in the UK, and the Company's broker-dealers. The Company was in compliance with all applicable regulatory net capital requirements.

22. Accumulated Other Comprehensive Income (Loss)

The following table presents changes in AOCI for 2024, 2023 and 2022:

(in millions)	2024	2023	2022
Beginning balance	**$ (840)**	$ (1,101)	$ (550)
Foreign currency translation adjustments(1)	**(338)**	261	(551)
Ending balance	**$ (1,178)**	$ (840)	$ (1,101)

(1) Amount for 2024 includes a gain from a net investment hedge of $37 million (net of tax expense of $12 million). Amount for 2023 includes a loss from a net investment hedge of $20 million (net of tax benefit of $6 million). Amount for 2022 includes a gain from a net investment hedge of $37 million (net of tax expense of $12 million).

23. Capital Stock

Cash Dividends for Common Shares / RSUs. During 2024, 2023 and 2022, the Company paid cash dividends of $20.40 per share (or $3.1 billion), $20.00 per share (or $3.0 billion) and $19.52 per share (or $3.0 billion), respectively.

The GIP Transaction. On October 1, 2024, as part of the closing of the GIP Transaction, (1) BlackRock issued 6,908,416 shares of common stock as part of the total consideration paid, (2) each share of common stock, $0.01 par value, of Old BlackRock issued and outstanding, other than common stock held in treasury, immediately prior to the closing of the GIP Transaction was converted automatically into one share of common stock, $0.01 par value, of New BlackRock and (3) shares of Old BlackRock common stock held in treasury were cancelled. See Note 3, *Acquisitions*, and Note 1*, Business Overview*, for additional information.

Share Repurchases. During 2024, the Company repurchased 1.9 million common shares under the Company's existing share repurchase program for approximately $1.6 billion. At December 31, 2024, there were approximately 3.8 million shares still authorized to be repurchased under the program. The timing and actual number of shares repurchased will depend on a variety of factors, including legal limitations, price and market conditions.

The Company's common shares issued and outstanding and related activity consist of the following:

	Shares Issued		Shares Outstanding
	Common Shares	**Treasury Common Shares**	**Common Shares**
December 31, 2021	172,075,373	(20,390,882)	151,684,491
Shares repurchased	—	(2,710,821)	(2,710,821)
Net issuance of common shares related to employee stock transactions	—	782,822	782,822
December 31, 2022	172,075,373	(22,318,881)	149,756,492
Shares repurchased	—	(2,176,538)	(2,176,538)
Net issuance of common shares related to employee stock transactions	—	920,120	920,120
December 31, 2023	172,075,373	(23,575,299)	148,500,074
Shares repurchased	—	(1,909,964)	(1,909,964)
Net issuance of common shares related to employee stock transactions	456,182	993,105	1,449,287
Issuance of common shares in connection with the GIP Transaction	6,908,416	—	6,908,416
Cancellation of treasury stock, common in connection with the GIP Transaction	(24,121,801)	24,121,801	—
December 31, 2024	**155,318,170**	**(370,357)**	**154,947,813**

24. Restructuring Charge

In the fourth quarter of 2023, a restructuring charge of $61 million ($46 million after-tax), comprised of $47 million of severance and $14 million of compensation expense for accelerated vesting of previously granted deferred compensation awards, was recorded in connection with initiatives to reorganize specific platforms, primarily Aladdin and private markets.

The table below presents a rollforward of the Company's restructuring liability for 2024 and 2023, which is included in other liabilities on the consolidated statements of financial condition:

(in millions)		
Liability as of December 31, 2022	$	58
Cash payments		(58)
Additions		61
Accelerated vesting expense of deferred compensation awards		(14)
Liability as of December 31, 2023	$	47
Cash payments		(47)
Liability as of December 31, 2024	**$**	**—**

25. Income Taxes

The components of income tax expense for 2024, 2023 and 2022, are as follows:

(in millions)	2024	2023	2022
Current income tax expense:			
Federal	**$ 960**	$ 641	$ 255
State and local	**142**	176	(9)
Foreign	**787**	538	448
Total net current income tax expense	**1,889**	1,355	694
Deferred income tax expense (benefit):			
Federal	**(105)**	101	562
State and local	**—**	11	64
Foreign	**(1)**	12	(24)
Total net deferred income tax expense (benefit)	**(106)**	124	602
Total income tax expense	**$ 1,783**	$ 1,479	$ 1,296

Income tax expense has been based on the following components of income before taxes, less net income (loss) attributable to NCI:

(in millions)	2024	2023	2022
Domestic	**$ 5,139**	$ 4,565	$ 4,604
Foreign	**3,013**	2,416	1,870
Total	**$ 8,152**	$ 6,981	$ 6,474

The foreign income before taxes includes countries that have statutory tax rates that are different than the US federal statutory tax rate of 21%, such as the UK, Germany, Canada and Ireland.

A reconciliation of income tax expense with expected federal income tax expense computed at the applicable federal income tax rate of 21% for 2024, 2023 and 2022 is as follows:

(in millions)	2024		2023		2022	
Statutory income tax expense	**$ 1,712**	**21%**	$ 1,466	21%	$ 1,360	21%
Increase (decrease) in income taxes resulting from:						
State and local taxes (net of federal benefit)	**130**	**2**	110	2	115	2
Impact of federal, foreign, state, and local tax rate changes on deferred taxes	**12**	**—**	—	—	(25)	—
Stock-based compensation awards	**(37)**	**—**	(41)	(1)	(87)	(1)
Resolution of outstanding tax matters	**—**	**—**	(204)	(3)	(143)	(2)
Intellectual property reorganization	**(137)**	**(2)**	—	—	—	—
Effect of foreign tax rates	**84**	**1**	112	2	23	—
Other	**19**	**—**	36	—	53	—
Income tax expense	**$ 1,783**	**22%**	$ 1,479	21%	$ 1,296	20%

Deferred income taxes are provided for the effects of temporary differences between the tax basis of an asset or liability and its reported amount in the consolidated financial statements. These temporary differences result in taxable or deductible amounts in future years.

The components of deferred income tax assets and liabilities are shown below:

	December 31,	
(in millions)	2024	2023
Deferred income tax assets:		
Compensation and benefits	**$ 354**	$ 375
Loss carryforwards	**103**	95
Foreign tax credit carryforward	**39**	—
Capitalized costs	**276**	216
Other	**795**	825
Gross deferred tax assets	**1,567**	1,511
Less: Deferred tax valuation allowances	**(69)**	(59)
Deferred tax assets net of valuation allowances	**1,498**	1,452
Deferred income tax liabilities:		
Goodwill and acquired indefinite-lived intangibles	**4,199**	4,299
Acquired finite-lived intangibles	**53**	86
Unrealized investment gains	**58**	25
Other	**341**	340
Gross deferred tax liabilities	**4,651**	4,750
Net deferred tax (liabilities)	**$ (3,153)**	$ (3,298)

Deferred income tax assets and liabilities are recorded net when related to the same tax jurisdiction. At December 31, 2024, the Company recorded on the consolidated statement of financial condition deferred income tax assets, within other assets, and deferred income tax liabilities of $181 million and $3.3 billion, respectively. At December 31, 2023, the Company recorded on the consolidated statement of financial condition deferred income tax assets, within other assets, and deferred income tax liabilities of $208 million and $3.5 billion, respectively.

Income tax expense for 2024 included discrete tax benefits of $137 million, recognized in connection with the reorganization and establishment of a more efficient global intellectual property and technology platform and corporate structure, $63 million related to the realization of capital losses from changes in the Company's organizational tax structure, $37 million related to vested stock-based compensation awards, and a net noncash discrete tax expense of $14 million related to the revaluation of deferred income tax liabilities. Income tax expense for 2023 included $242 million of net discrete tax net benefits related to the resolution of certain outstanding tax matters and stock-based compensation awards that vested in 2023.

At December 31, 2024 and 2023, the Company had available state net operating loss carryforwards of $2.9 billion and $2.7 billion, respectively, which will begin to expire in 2032. At December 31, 2024 and 2023, the Company had foreign net operating loss carryforwards of $193 million and $164 million, respectively, of which $10 million will begin to expire in 2025. At December 31, 2024, the Company had foreign tax credit carryforwards for income tax purposes of $39 million which will begin to expire in 2034.

At December 31, 2024 and 2023, the Company had $69 million and $59 million of valuation allowances for deferred income tax assets, respectively, recorded on the consolidated statements of financial condition.

Current income taxes are recorded net on the consolidated statements of financial condition when related to the same tax jurisdiction. At December 31, 2024, the Company had current income taxes receivable and payable of $215 million and $134 million, respectively, recorded in other assets and accounts payable and accrued liabilities, respectively. At December 31, 2023, the Company had current income taxes receivable and payable of $252 million and $85 million, respectively, recorded in other assets and accounts payable and accrued liabilities, respectively.

The following tabular reconciliation presents the total amounts of gross unrecognized tax benefits:

(in millions)	**2024**	2023	2022
Balance at January 1	**$ 749**	$ 912	$ 1,022
Additions for tax positions of prior years	**30**	25	13
Reductions for tax positions of prior years	**(10)**	(22)	(75)
Additions based on tax positions related to current year	**51**	49	55
Additions related to business combinations	**—**	16	—
Settlements	**(303)**	(231)	(103)
Balance at December 31	**$ 517**	$ 749	$ 912

Included in the balance of unrecognized tax benefits at December 31, 2024, 2023 and 2022, respectively, are $431 million, $505 million and $497 million of tax benefits that, if recognized, would affect the effective tax rate.

The Company recognizes interest and penalties related to income tax matters as a component of income tax expense. Related to the unrecognized tax benefits noted above, the Company accrued interest and penalties of $63 million during 2024 and in total, as of December 31, 2024, had recognized a liability for interest and penalties of $203 million. The Company accrued interest and penalties of $(20) million during 2023 and in total, as of December 31, 2023, had recognized a liability for interest and penalties of $140 million. The Company accrued interest and penalties of $(40) million during 2022 and in total, as of December 31, 2022, had recognized a liability for interest and penalties of $160 million.

BlackRock is subject to US federal income tax, state and local income tax, and foreign income tax in multiple jurisdictions. Tax years after 2015 remain open to US federal income tax examination.

During 2020 and 2021, the Internal Revenue Service commenced its examination of BlackRock's 2017 through 2018 tax years and 2019 tax year, respectively. During 2023, the Internal Revenue Service commenced its examination of BlackRock's 2016 tax year.

The Company is currently under audit in several state and local jurisdictions. The significant state and local income tax examinations are in New York State for tax years 2015 through 2020, and New York City for tax years 2012 through 2014. No open state and local tax examinations cover years earlier than 2012.

Upon conclusion of its examination, Her Majesty's Revenue and Customs ("HMRC") issued a closure notice during 2017 for various UK BlackRock subsidiaries for tax years 2009 and years after. At that time, the Company decided to pursue litigation for the tax matters included on such notice. During 2020, the Company received a favorable decision from the First Tier Tribunal ("FTT"), however, HMRC appealed to the Upper Tribunal ("UT") and the UT ruled in HMRC's favor, overturning the FTT's decision in July 2022. BlackRock appealed the UT's decision to the UK Court of Appeal, who dismissed the Company's appeal. The Company applied for permission to appeal to the UK Supreme Court, who refused permission to appeal in October 2024. The resolution did not result in a material impact to the consolidated financial statements.

From time to time, BlackRock may receive or be subject to tax authorities' assessments and challenges related to income taxes. BlackRock does not currently expect the ultimate resolution of any other existing matters to be material to the consolidated financial statements.

At December 31, 2024, it is reasonably possible the total amounts of unrecognized tax benefits will change within the next twelve months due to completion of tax authorities' exams or the expiration of statues of limitations. Management estimates that the existing liability for uncertain tax positions could decrease by approximately $160 million to $220 million within the next twelve months.

26. Earnings Per Share

The following table sets forth the computation of basic and diluted EPS for 2024, 2023 and 2022 under the treasury stock method:

(in millions, except shares and per share data)	**2024**	2023	2022
Net income attributable to BlackRock, Inc.	**$ 6,369**	$ 5,502	$ 5,178
Basic weighted-average shares outstanding	**150,042,269**	149,327,558	150,921,161
Dilutive effect of:			
Nonparticipating RSUs	**1,034,323**	969,089	1,119,829
Stock options	**538,493**	409,804	399,481
Total diluted weighted-average shares outstanding	**151,615,085**	150,706,451	152,440,471
Basic earnings per share	**$ 42.45**	$ 36.85	$ 34.31
Diluted earnings per share	**$ 42.01**	$ 36.51	$ 33.97

The amount of anti-dilutive RSUs and stock options were immaterial for 2024 and 2022. For 2023, 194,240 shares primarily related to stock options were excluded from the calculation of diluted EPS because to include them would have an anti-dilutive effect. Certain performance-based RSUs for 2024, 2023, and 2022, and certain performance-based stock options for 2023 were excluded from the diluted EPS calculation because the designated contingencies were not met.

27. Segment Information

The Company's management directs BlackRock's operations as one business, the asset management business. As such, the Company operates in one asset management operating segment. The Company's CODM is its Chairman and Chief Executive Officer, who reviews financial information presented, including significant expenses on a consolidated basis, as presented in the consolidated statements of income. The CODM utilizes a consolidated approach to assess performance and allocates resources using key financial metrics including total revenue, operating income and net income attributable to BlackRock, Inc. These financial metrics are used by the CODM to make key operating decisions, including capital allocation, determining annual and long-term compensation and managing costs in relation to revenue. Furthermore, these financial metrics are used to evaluate financial performance based on consolidated specific business objectives, contributions to the total firm operating margin and to evaluate the Company's relative performance against industry peers. See the consolidated financial statements for key financial metrics used by CODM and for more financial information regarding the Company's operating segment. The measure of segment assets is reported on the balance sheet as total consolidated assets.

The following table illustrates total revenue for 2024, 2023 and 2022 by geographic region. These amounts are aggregated on a legal entity basis and do not necessarily reflect where the customer resides, or affiliated services are provided.

(in millions)	2024	2023	2022
Revenue			
Americas	**$ 13,411**	$ 11,899	$ 11,931
Europe	**6,137**	5,209	5,164
Asia-Pacific	**859**	751	778
Total revenue	**$ 20,407**	$ 17,859	$ 17,873

See Note 17, *Revenue*, for further information on the Company's sources of revenue.

The following table illustrates long-lived assets that consist of goodwill and property and equipment at December 31, 2024 and 2023 by geographic region. These amounts are aggregated on a legal entity basis and do not necessarily reflect where the asset is physically located.

(in millions)	2024	2023
Long-lived Assets		
Americas	**$ 25,515**	$ 15,017
Europe	**1,437**	1,521
Asia-Pacific	**100**	98
Total long-lived assets	**$ 27,052**	$ 16,636

Americas is primarily comprised of the US, and also includes Latin America and Canada. Europe is primarily comprised of the UK, Luxembourg and the Netherlands, and also includes Switzerland, Ireland and France. Asia-Pacific is primarily comprised of Hong Kong, Japan, Singapore and Australia.

28. Subsequent Events

In June 2024, BlackRock announced that it had entered into a definitive agreement to acquire Preqin, a leading independent provider of private markets data, for £2.55 billion (or approximately $3.2 billion based on the GBP/USD foreign exchange rate at December 31, 2024) in cash. The Company believes bringing together Preqin's data and research tools with the complementary workflows of Aladdin and eFront in a unified platform will create a preeminent private markets technology and data provider. The Preqin Transaction is anticipated to close in the first quarter of 2025, subject to customary closing conditions.

In December 2024, BlackRock announced that it had entered into a definitive agreement to acquire 100% of the business and assets of HPS Investment Partners ("HPS"), a leading global credit investment manager with 100% of the consideration paid in BlackRock equity (the "HPS Transaction"). The equity will generally be delivered in units of a wholly-owned subsidiary of BlackRock ("SubCo Units") which will be exchangeable on a one-for-one basis (subject to certain adjustments) into BlackRock common stock (accordingly, the value of each unit delivered will be based on the price of a share of BlackRock's common stock and the specific terms of the SubCo Units). Approximately 9.2 million SubCo Units and RSUs will be paid at closing. Approximately 2.9 million SubCo Units, will be paid in approximately five years, subject to the satisfaction of certain post-closing conditions. In addition, there is potential for additional consideration to be earned of up to 1.6 million SubCo Units that is based on financial performance milestones measured and paid in approximately five years. Of the total deal consideration, up to 0.7 million units will be used to fund an equity retention pool for HPS employees. In aggregate, inclusive of all SubCo Units paid at closing, eligible to be paid in approximately five years, and potentially earned through achievement of financial performance milestones as well as BlackRock RSUs to be issued in the transaction, the maximum amount of common stock issuable upon exchange of such SubCo Units would be approximately 13.7 million shares. The Company expects the addition of HPS will create an integrated private credit platform to provide both public and private income solutions for clients across their whole portfolios. The HPS Transaction is anticipated to close in mid-2025 subject to regulatory approvals and customary closing conditions.

On January 29, 2025, the Company announced that the Board of Directors approved BlackRock's quarterly dividend of $5.21 per share to be paid on March 24, 2025 to stockholders of record at the close of business on March 7, 2025.

The Company conducted a review for additional subsequent events and determined that no subsequent events had occurred that would require accrual or additional disclosures.

Exhibit 3.1

RESTATED CERTIFICATE OF INCORPORATION

OF

BLACKROCK, INC.

November 20, 2024

BlackRock, Inc., a corporation organized and existing under the laws of the State of Delaware (the "Corporation"), does hereby certify as follows:

1. The name of the Corporation is BlackRock, Inc. The Corporation was originally incorporated under the name "New Banana, Inc." and the original Certificate of Incorporation of the Corporation was filed with the Secretary of State of the State of Delaware on January 8, 2024 and subsequently amended via the filing of a Certificate of Amendment to the Certificate of Incorporation of the Corporation with the Secretary of State of the State of Delaware on January 24, 2024 pursuant to which the Corporation changed its name to BlackRock Funding, Inc. (collectively, the "Original Certificate of Incorporation").

2. The Original Certificate of Incorporation was amended and restated on October 1, 2024 and subsequently amended via the filing of a Certificate of Amendment on October 1, 2024, pursuant to which the Corporation changed its name from BlackRock Funding, Inc. to BlackRock, Inc. (collectively, the "Amended and Restated Certificate of Incorporation").

3. This Restated Certificate of Incorporation was adopted in accordance with Section 245 of the General Corporation Law of the State of Delaware on November 20, 2024. This Restated Certificate of Incorporation only restates and integrates and does not further amend the provisions of the Amended and Restated Certificate of Incorporation, as heretofore amended or supplemented, and there is no discrepancy between the provisions of the Amended and Restated Certificate of Incorporation, as heretofore amended or supplemented, and the provisions of this Restated Certificate of Incorporation.

4. The text of the Amended and Restated Certificate of Incorporation of the Corporation, as heretofore amended or supplemented, is hereby integrated and restated in its entirety as follows:

FIRST: The name of the corporation is BlackRock, Inc. (hereinafter the "Corporation").

SECOND: The registered office of the Corporation within the State of Delaware shall be at 251 Little Falls Drive, City of Wilmington, County of New Castle, State of Delaware 19808, and the name of the registered agent of the Corporation at such address shall be Corporation Service Company.

THIRD: The purpose of the Corporation is to engage in any lawful act or activity for which corporations may be organized under the General Corporation Law of the State of Delaware (the "DGCL").

FOURTH: A. Authorized Shares. The Corporation shall be authorized to issue 1,000,000,000 shares of stock, of which (i) 500,000,000 shares shall be shares of Common Stock, par value $0.01 per share (the "Common Stock") and (ii) 500,000,000 shares shall be shares of Preferred Stock, par value $0.01 per share (the "Preferred Stock").

B. Preferred Stock. The Preferred Stock may be issued from time to time in one or more classes or series. The Board of Directors is hereby authorized to provide for the issuance of shares of Preferred Stock in one or more classes or series and, by filing a certificate pursuant to the DGCL (hereinafter referred to as a "Preferred Stock Designation"), to establish from time to time the number of shares to be included in each such class or series, and to fix the designations, voting powers (if any), privileges, preferences and relative, participating, optional or other special rights of the shares of each such class or series and the qualifications, limitations and restrictions thereon. The authority of the Board of Directors with respect to each class or series shall include, but not be limited to, determination of the following:

(1) the designation of the class or series, which may be by distinguishing number, letter or title;

(2) the number of shares of the class or series, which number the Board of Directors may thereafter (except where otherwise provided in the Preferred Stock Designation) increase or decrease (but not below the number of shares thereof then outstanding) in the manner permitted by law;

(3) the rate of any dividends (or method of determining the dividends) payable to the holders of the shares of such class or series, any conditions upon which such dividends shall be paid, the form of payment thereof (whether cash, securities of the Corporation, securities of another person or other assets) and the date or dates or the method for determining the date or dates upon which such dividends shall be payable;

(4) whether dividends, if any, shall be cumulative or noncumulative and, in the case of shares of any class or series having cumulative dividend rights, the date or dates or method of determining the date or dates from which dividends on the shares of such class or series shall cumulate;

(5) if the shares of such class or series may be redeemed by the Corporation, the price or prices (or method of determining such price or prices) at which, the form of payment of such price or prices (which may be cash, property or rights, including securities of the Corporation or of another corporation or other entity) for which, the period or periods within which and the other terms and conditions upon which the shares of such class or series may be redeemed, in whole or in part, at the option of the Corporation or at the option of the holder or holders thereof or upon the happening of a specified event or events, if any, including the obligation, if any, of the Corporation to purchase or redeem shares of such class or series pursuant to a sinking fund or otherwise;

(6) the amount payable out of the assets of the Corporation to the holders of shares of the class or series in the event of any voluntary or involuntary liquidation, dissolution or winding up of the affairs of the Corporation;

(7) provisions, if any, for the conversion or exchange of the shares of such class or series, at any time or times, at the option of the holder or holder thereof or at the option of the Corporation or upon the happening of a specified event or events, into shares of any other class or classes or any other series of the same class of capital stock of the Corporation or into any other security of the Corporation, or into the stock or other securities of any other corporation or other entity, and the price or

prices or rate or rates of conversion or exchange and any adjustments applicable thereto, and all other terms and conditions upon which each conversion or exchange may be made;

(8) restrictions on the issuance of shares of the same class or series or of any other class or series of capital stock of the Corporation, if any; and

(9) the voting rights and powers, if any, of the holders of shares of the class or series.

C. Common Stock. No stockholder shall be entitled to exercise any right of cumulative voting.

FIFTH: A. Stockholder Meetings.

(1) Meetings of stockholders may be held within or without the State of Delaware, as the Bylaws may provide. An annual meeting of the stockholders of the Corporation for the election of directors and for the transaction of such other business as may come before the meeting shall be held at such time and place as shall be determined in accordance with the Bylaws. Elections of directors need not be by written ballot unless otherwise provided in the Bylaws.

(2) Except as otherwise required by law and subject to the rights of the holders of any class or series of stock having a preference over the Common Stock as to dividends or distributions upon liquidation, special meetings of stockholders of the Corporation of any class or series for any purpose or purposes may be called only by:

(a) the Chairman of the Board of Directors;

(b) the President of the Corporation;

(c) a majority of the Board of Directors;

(d) any committee of the Board of Directors the powers and authority of which include the power and authority to call such meetings; or

(e) the Chairman of the Board of Directors, the President or the Corporate Secretary of the Corporation upon a written request of one or more record holders of shares of stock of the Corporation representing ownership of not less than 15 percent of the voting power of all outstanding shares of stock in the Corporation, which request complies with the procedures for calling a special meeting of stockholders as set forth in the Bylaws, as may be amended from time to time.

B. Action by Written Consent. Any action required or permitted to be taken at any annual or special meeting of the stockholders may be effected by written consent of such stockholders pursuant to Section 228 of the DGCL if such action has been approved in advance by the requisite vote of the Board of Directors.

SIXTH: A. Powers of the Board of Directors. The business and affairs of the Corporation shall be managed by or under the direction of the Board of Directors, which shall be constituted as provided in this Article and as provided by law.

B. Number of Directors. Subject to the rights of holders of Preferred Stock, the number of directors shall be fixed from time to time by resolution adopted by the affirmative vote of a majority of the entire Board of Directors then in office.

C. Election and Term. The directors of the Corporation shall be elected annually for terms expiring at the next annual meeting of stockholders and until such directors' successors shall have been elected and qualified. Any vacancies created in the Board of Directors through and increase in the number of directors or otherwise may be filled in accordance with the Bylaws of the Corporation and the applicable laws of the State of Delaware.

D. Removal of Directors. Except as may be provided in a resolution or resolutions providing for any class or series of Preferred Stock with respect to any directors elected by the holders of such class or series, any director, or the entire Board of Directors, may be removed, with or without cause, by the holders of a majority of the votes of capital stock then entitled to vote in the election of directors at a meeting of stockholders called for that purpose.

E. Meetings of the Board of Directors. Meetings of the Board of Directors may be held within or without the State of Delaware, as the Bylaws may provide.

F. Quorum; Required Vote. Except as otherwise provided by law:

(1) at all meetings of the Board of Directors, a majority of the entire Board of Directors shall constitute a quorum for the transaction of business, and

(2) the act of a majority of the directors present at any meeting at which there is a quorum shall be the act of the Board of Directors.

SEVENTH: A. Liability. A director of the Corporation shall, to the maximum extent permitted by the laws of the State of Delaware, as now or hereafter in effect, have no personal liability to the Corporation or its stockholders for monetary damages for breach of fiduciary duty as a director.

B. Indemnification.

(1) The Corporation shall indemnify its directors and officers to the fullest extent authorized or permitted by law, as now or hereafter in effect and such right to indemnification shall continue as to a person who has ceased to be a director or officer of the Corporation and shall inure to the benefit of his or her heirs, executors and personal and legal representatives; *provided*, *however*, that, except for proceedings to enforce rights to indemnification, the Corporation shall not be obligated to indemnify any director or officer (or his or her heirs, executors or personal or legal representatives) in connection with a proceeding (or part thereof) initiated by such person unless such proceeding (or part thereof) was authorized or consented to by the Board of Directors. The right to indemnification conferred by this Article Seventh shall include the right to be paid by the Corporation the expenses incurred in defending or otherwise participating in any proceeding in advance of its final disposition.

(2) The Corporation may, to the extent authorized from time to time by the Board of Directors, provide rights to indemnification and to the advancement of expenses to employees and agents of the Corporation similar to those conferred in this Article Seventh to directors and officers of the Corporation.

(3) The rights to indemnification and to the advance of expenses conferred in this Article Seventh shall not be exclusive of any other right which any person may have or hereafter

acquire under this Certificate of Incorporation, the Bylaws of the Corporation, any statute, agreement, vote of stockholders or disinterested directors or otherwise.

C. Modification.

(1) Any repeal or modification of this Article Seventh by the stockholders of the Corporation shall not adversely affect any exclusion of liability, rights to indemnification and to the advancement of expenses or other protection of a director, officer, employee or agent of the Corporation existing at the time of such repeal or modification with respect to any acts or omissions occurring prior to such repeal or modification.

(2) Any repeal or modification of the laws of the State of Delaware, as are now or hereafter in effect, shall not adversely affect any rights to indemnification and to the advancement of expenses or other protection of a director, officer, employee or agent of the Corporation existing at the time of such repeal or modification with respect to any acts or omissions occurring prior to such repeal or modification.

EIGHTH: The Corporation shall be subject to Section 203 of the DGCL.

NINTH: The books of the Corporation may be kept (subject to any provision contained in the DGCL or other applicable statutes) outside the State of Delaware at such place or places as may be designated from time to time by the Board of Directors or in the Bylaws of the Corporation.

TENTH: Whenever a compromise or arrangement is proposed between the Corporation and its creditors or any class of them and/or between the Corporation and its stockholders or any class of them, any court of equitable jurisdiction within the State of Delaware may, in a summary way, on the application of the Corporation or of any creditor or stockholder thereof or on the application of any receiver or receivers appointed for the Corporation under the provisions of Section 291 of the DGCL or on the application of trustees in dissolution or of any receiver or receivers appointed for the Corporation under the provisions of Section 279 of the DGCL, order a meeting of the creditors or class of creditors, and/or of the stockholders or class of stockholders of the Corporation, as the case maybe, to be summoned in such manner as said court directs. If a majority in number representing three-fourths in value of the creditors or class of creditors, and/or of the stockholders or class of stockholders of the Corporation, as the case may be, agree to any compromise or arrangement and to any reorganization of the Corporation as a consequence of such compromise or arrangement, the said compromise or arrangement and the said reorganization shall, if sanctioned by the court to which the said application has been made, be binding on all the creditors or class of creditors, and/or on all the stockholders or class of stockholders, of the Corporation, as the case may be, and also on the Corporation.

ELEVENTH: Bylaw Amendments. The Bylaws of the Corporation may be adopted, consistent with law and the provisions of this Certificate of Incorporation (including any Preferred Stock Designation), and once adopted, any Bylaw may be altered or repealed by: (1) the affirmative vote of at least a majority of the members of the Board of Directors then in office, or (2) the affirmative vote of at least a majority of the voting power of the Voting Stock.

TWELFTH: General Right to Amend Certificate of Incorporation.

(1) The Corporation hereby reserves the right at any time and from time to time to amend, alter, change or repeal any provision contained in this Certificate of Incorporation, and to add thereto any other provision authorized by the laws of the state of Delaware at the time in force, and except as may otherwise be explicitly provided by any provision of this Certificate of Incorporation, all

rights, preferences and privileges of whatsoever nature conferred upon stockholders, directors or officers of the Corporation or any other person whomsoever by and pursuant to this Certificate of Incorporation in its present form, or as hereafter amended, are granted subject to the right reserved in this paragraph (1).

(2) Subject to the rights of the holders of Preferred Stock, the provisions of this Certificate of Incorporation may only be altered, amended or repealed, and any inconsistent provision adopted, with such action (if any) of the Board of Directors as is provided by law, and in addition to any other vote of stockholders (if any) required by law, and notwithstanding that a lower vote (or a no vote) of stockholders otherwise would be required, by the approval of at least a majority of the voting power of all Voting Stock.

THIRTEENTH: The Corporation shall have perpetual existence.

FOURTEENTH: For purposes of this Certificate of Incorporation, the following definitions shall apply:

(1) "Capital Stock" means any and all shares (however designated, whether voting or non-voting) of capital stock issued by the Corporation.

(2) each reference to a "person" shall be deemed to include not only a natural person, but also a corporation, partnership, joint venture, association or legal entity of any kind; each reference to a "natural person" (or to a "record holder" of shares, if a natural person) shall be deemed to include, in his, her or its representative capacity, a guardian, committee, executor, administrator or other legal representative of such natural person or record holder.

(3) "Voting Stock" shall mean the then outstanding shares of Capital Stock of the Corporation entitled to vote generally on the election of directors and shall exclude any class or series of capital stock of the Corporation only entitled to vote in the event of dividend arrearages or any default under any provision of such series thereon, whether or not at the time of determination there are any such dividend arrearages or defaults.

IN WITNESS WHEREOF, BlackRock, Inc. has caused this Restated Certificate of Incorporation to be executed by a duly authorized officer of the Corporation on this 20th day of November, 2024.

BLACKROCK, INC.

By: /s/ Andrew Dickson
Name: Andrew Dickson
Title: Managing Director and Secretary

Exhibit 4.20

DESCRIPTION OF THE REGISTRANT'S SECURITIES REGISTERED PURSUANT TO SECTION 12 OF THE SECURITIES EXCHANGE ACT OF 1934

As of December 31, 2024, our common stock is the only class of securities that we have registered under Section 12 of the Securities Exchange Act, as amended.

The description of our common stock below does not purport to be complete and is qualified in its entirety by reference to our restated certificate of incorporation, as restated on November 20, 2024 (as amended and restated, the "Restated Certificate of Incorporation") and our Amended and Restated Bylaws, as in effect since October 1, 2024 (the "Amended and Restated Bylaws"), each of which is incorporated herein by reference as an exhibit to the Annual Report on Form 10-K filed with the Securities and Exchange Commission, of which this Exhibit 4.20 is a part. When used in this exhibit, the terms "BlackRock," "we," "our" and "us" refer solely to BlackRock, Inc. (formerly known as "BlackRock Funding, Inc") and not to its subsidiaries. We urge you to read our Restated Certificate of Incorporation, our Amended and Restated Bylaws and the applicable provision of the Delaware General Corporation Law.

DESCRIPTION OF COMMON STOCK

Authorized Shares of Capital Stock

Our Restated Certificate of Incorporation provides that we are authorized to issue one billion shares of capital stock, consisting of 500,000,000 shares of common stock, par value $0.01 per share, and 500,000,000 shares of preferred stock, par value $0.01 per share.

As of December 31, 2024, we had 154,947,813 shares of common stock issued and outstanding and no shares of preferred stock issued and outstanding.

Voting Rights

The holders of common stock are entitled to one vote for each share on all matters submitted to a vote of stockholders.

Dividends and Liquidation Rights

Subject to the preferential rights of any outstanding series of preferred stock created by our board of directors from time to time, the holders of common stock are entitled to such dividends as may be declared from time to time by our board of directors from funds available therefor, and, upon liquidation, holders of common stock are entitled to share pro rata in any distribution of our assets after payment, or providing for the payment of, our liabilities.

Miscellaneous

The outstanding shares of our common stock, are fully paid and nonassessable. Our common stock has no preemptive or conversion rights and there are no redemption or sinking fund provisions applicable thereto.

Listing

Our common stock is listed on the New York Stock Exchange under the ticker symbol "BLK."

Transfer Agent and Registrar

The transfer agent and registrar for our common stock is Computershare Investor Services, 480 Washington Boulevard, Jersey City, New Jersey 07310-1900, telephone (800) 903-8567.

Anti-Takeover Considerations

The Delaware General Corporation Law, our Restated Certificate of Incorporation and our Amended and Restated Bylaws contain provisions which could serve to discourage or to make more difficult a change in control of us without the support of our board of directors or without meeting various other conditions.

Extraordinary Corporate Transactions

Delaware law provides that the holders of a majority of the shares entitled to vote must approve any fundamental corporate transactions such as mergers, sales of all or substantially all of a corporation's assets, dissolutions, etc.

State Takeover Legislation

Section 203 of the Delaware General Corporation Law, in general, prohibits a business combination between a corporation and an interested stockholder within three years of the time such stockholder became an interested stockholder, unless (a) prior to such time, the board of directors of the corporation approved either the business combination or the transaction that resulted in the stockholder becoming an interested stockholder, (b) upon consummation of the transaction that resulted in the stockholder becoming an interested stockholder, the interested stockholder owned at least 85% of the voting stock of the corporation outstanding at the time the transaction commenced, exclusive of shares owned by directors who are also officers and by certain employee stock plans or (c) at or subsequent to such time, the business combination is approved by the board of directors and authorized by the affirmative vote at a stockholders' meeting, and not by written consent, of at least 66 2/3% of the outstanding voting stock which is not owned by the interested stockholder. The restrictions of Section 203 of the Delaware General Corporation Law do not apply to certain business combinations or to corporations that have elected, in the manner provided therein, not to be subject to Section 203 of the Delaware General Corporation Law or, with certain exceptions, which do not have a class of voting stock that is listed on a national securities exchange or held of record by more than 2,000 stockholders. We have elected to be governed by Section 203 of the Delaware General Corporation Law.

Rights of Dissenting Stockholders

Delaware law does not afford appraisal rights in a merger transaction to holders of shares that are either listed on a national securities exchange or held of record by more than 2,000 stockholders, *provided* that such shares are converted into stock of the surviving corporation or another corporation, which corporation in either case must also be listed on a national securities exchange or held of record by more than 2,000 stockholders. In addition, Delaware law denies appraisal rights to stockholders of the surviving corporation in a merger if the surviving corporation's stockholders were not required to approve the merger.

Stockholder Action

Delaware law provides that, unless otherwise stated in our Restated Certificate of Incorporation, any action which may be taken at an annual meeting or special meeting of stockholders may be taken without a meeting, if a consent in writing is signed by the holders of the outstanding stock having the minimum number of votes necessary to authorize the action at a meeting of stockholders. Our Restated Certificate of Incorporation provides that stockholders may take action by written consent if such action has been approved in advance by the majority vote of our board of directors.

Meetings of Stockholders

Our Restated Certificate of Incorporation provides that special meetings of the stockholders may be called at any time by the chairman of the board of directors, the president, a majority of the board of directors, any committee of the board of directors that has the power to call such meetings, or the chairman of the board of directors, the president or the corporate secretary upon a written request of one or more record holders of shares of our stock representing ownership of not less than 15 percent of the voting power of all outstanding shares of our stock, which request complies with the procedures for calling a special meeting of stockholders as set forth in our Amended and Restated Bylaws, as they may be further amended from time to time.

Cumulative Voting

Delaware law permits stockholders to cumulate their votes and either cast them for one candidate or distribute them among two or more candidates in the election of directors only if expressly authorized in a corporation's certificate of incorporation. Our Restated Certificate of Incorporation does not authorize cumulative voting.

Removal of Directors

Delaware law provides that, except in the case of a classified board of directors or where cumulative voting applies, a director, or the entire board of directors, of a corporation may be removed, with or without cause, by the affirmative vote of a majority of the shares of the corporation entitled to vote at an election of directors.

Our Restated Certificate of Incorporation provides that any or all of the directors may be removed, with or without cause, by the holders of a majority of the votes of capital stock then entitled to vote in the election of directors at a meeting of stockholders called for that purpose.

Vacancies

Delaware law provides that vacancies and newly created directorships resulting from a resignation or any increase in the authorized number of directors elected by all of the stockholders having the right to vote as a single class may be filled by a majority of the directors then in office, unless the governing documents of a corporation provide otherwise.

Our Amended and Restated Bylaws provide that newly created directorships resulting from an increase in the number of directors and vacancies occurring in the board of directors for any reason, may be filled by vote of a majority of the directors then in office, although less than a quorum, or by a sole remaining director or by the stockholders if the vacancy resulted from the action of stockholders.

No Preemptive Rights

Holders of common stock do not have any preemptive rights to subscribe for any additional shares of capital stock or other obligations convertible into or exercisable for shares of capital stock that we may issue in the future.

Board Term

Our directors are elected annually for terms of one year.

Exhibit 10.33

BlackRock

September 30, 2024

Dear Mr. Adebayo O. Ogunlesi,

Welcome! On behalf of BlackRock, we are excited about the prospect of having you as part of the BlackRock team. We look forward to all we can achieve as we bring our teams together to serve clients.

As you are aware, BlackRock, Inc. ("**BlackRock**") and Global Infrastructure Management, LLC ("**GIP**"), together with certain other parties, have entered into the Transaction Agreement, dated as of January 12, 2024 (the "**Transaction Agreement**") and the Governance Agreement, dated as of the date hereof (the "**Governance Agreement**"). If anything contained in this letter is found to conflict with or otherwise be inconsistent with any provision of the Governance Agreement, the provisions of the Governance Agreement shall control any such conflict or inconsistency.

We are providing this letter to briefly describe the compensation and benefits that you will receive or will be eligible for, as applicable, following the close of the transaction contemplated by the Transaction Agreement (the "**Closing**", with the date of such Closing, the "**Closing Date**"), and to provide other important information relative to your employment with BlackRock. Please note that your employment with BlackRock is subject to and contingent upon the Closing occurring and this letter will be void if the Closing fails to occur for any reason.

This confirms that upon the Closing, (a) your title will be "Senior Managing Director, Chairman and Chief Executive Officer" of Global Infrastructure Partners, a part of BlackRock, reporting to Laurence D. Fink, BlackRock's Chairman and Chief Executive Officer, and (b) you will become a member of BlackRock's Global Executive Committee. Following the Closing, you will continue to be employed on a full-time basis by your current employing entity, which will become part of the BlackRock group of companies. You will also become a member of the GIP Office of the Chairman (as defined in the Governance Agreement) and your day-to-day executive duties, responsibilities and authorities will be determined in accordance with the governance and operating principles set forth in the Governance Agreement. You will continue to be based in New York.

Section 1. Compensation and Benefits.

(a) Base Compensation. Your base salary will be at an annualized rate of $250,000, payable semi-monthly and in accordance with BlackRock's standard payroll practices. Beginning on January 1, 2026 (or such earlier date as may be mutually agreed upon by the GIP Office of the Chairman and BlackRock), your annualized base salary may be changed to harmonize such amount with BlackRock's compensation standards and align your compensation with similarly-situated BlackRock employees, but shall not affect any compensation payable in respect of any period prior to January 1, 2026 that is paid following such date. All compensation and benefits paid to you by BlackRock shall be subject to reduction, as may be applicable, to reflect all required tax withholdings and other payroll deductions.

(b) Bonus Compensation and Total Compensation.

(i) 2024 Annual Bonus. In recognition of your performance in the 2024 calendar year, you will be eligible for an annual discretionary cash bonus, payable in a manner consistent with GIP pre-Closing practice, targeted at no less than $3,750,000.

(ii) 2025 Annual Bonus. In recognition of your performance in the 2025 calendar year, you will be eligible for an annual discretionary cash bonus, payable in a manner consistent with GIP pre-Closing practice.

(iii) Total Compensation (after 2025). Annual discretionary bonuses reflect the financial performance of BlackRock, the performance of your business unit and your individual performance for the relevant performance year, in each case, as determined by the Management Development & Compensation Committee of the Board of Directors of BlackRock (the "**MDCC**"). The amount and form of annual discretionary bonus compensation is not guaranteed. A bonus awarded to you in one year does not mean that any bonus will be awarded to you in a subsequent year.

Notwithstanding any provision to the contrary above, starting on January 1, 2026 (or such earlier date as may be mutually agreed upon by the GIP Office of the Chairman and BlackRock), the form of delivery of your total compensation (inclusive of base salary, annual bonus and other incentive compensation but excluding any distributions in respect of (A) carried interest, performance fees, or similar entitlements and (B) any commitments made to investment vehicles or accounts sponsored or managed by the Company Business (as defined in the Governance Agreement) (such amounts under the foregoing clauses (A) and (B), together, the "**PRE**")) in respect of 2026 and subsequent calendar years may be changed to harmonize with BlackRock's total compensation approach for similarly-situated individuals, which provides that compensation may be delivered in a mix of cash and deferred equity under the terms of BlackRock's standard guidelines applicable to similarly-situated BlackRock employees (consistent with the terms of the Governance Agreement).

BlackRock has a strong pay for performance culture. The firm maintains a "total compensation" philosophy and strives to create a competitive total package consisting of base salary, annual bonus, and long-term awards that is intended to align individual performance with the financial success of the firm.

To be eligible to receive any bonus award discussed above, you must be employed with BlackRock, not have given notice of resignation or been informed that your employment has been terminated, in each case, through the date of payment, which is generally no later than February 1st of the year following the performance year.

(c) Employee Benefits. For the period commencing on the Closing Date through December 31, 2025 (or such earlier date as may be mutually agreed upon by the GIP Office of the Chairman and BlackRock) (such period, the "**GIP Benefit Period**"), you will continue to participate or be eligible to participate in all legacy GIP employee health, welfare, voluntary insurance, retirement savings and other benefit plans that were provided to you by the Company Group (as defined in the Transaction Agreement) immediately prior to the Closing Date subject to the terms and conditions of the applicable plan as may be in effect during the GIP Benefit Period or accrued during the GIP Benefit Period (excluding retention, transaction or change in control payments, deferred compensation, severance, retiree health and welfare benefits, defined benefit pension benefits, PRE, profits interests and allocations related thereto, and profit sharing). During the GIP Benefit Period, you will be eligible for leave and paid time off in accordance with GIP policy as in effect immediately prior to the Closing Date.

Effective as of January 1, 2026 (or such earlier date as may be mutually agreed upon by the GIP Office of the Chairman and BlackRock) and subsequent calendar years, you will be eligible to participate in BlackRock's employee benefit plans and programs, as in effect from time to time. You will be given full credit for all periods of service you had with the Company Group on or before the Closing Date to the extent it was recognized as service by the applicable member of the Company Group for purposes of vesting, eligibility to participate and level of benefits under BlackRock's employee benefits plans and programs in which you are eligible to participate following the Closing Date for service accrued or deemed accrued on or prior to the Closing Date with the Company Group (or any predecessor thereto) to the same extent that such credit was recognized by the Company Group under an analogous employee benefit plan or program of the Company Group immediately prior to the Closing, except to the extent that such credit would result in any duplication of benefits for the same period of service.

In addition, and notwithstanding anything to the contrary in this letter, for so long as you remain an employee of BlackRock (even after the GIP Benefit Period), you shall retain your benefits with respect to (i) the use of private and first-class air travel for business purposes (including personally owned private aircraft); (ii) your participation in the enhanced medical benefits under ArmadaCare at your level of participation as of the Closing Date throughout your BlackRock employment; and (iii) lifetime retiree medical benefits provided by GIP to retired founders, subject to the continuity of your coverage under GIP's and BlackRock's medical benefits plans, and your not being eligible for medical coverage with another employer; in each case of the foregoing clauses (i)-(iii) (collectively, the "**Protected Benefits**"), in a manner consistent with GIP's past practice prior to the Closing Date.

For the avoidance of doubt, you will not be eligible to receive any awards pursuant to the Retention Plan (as defined in the Transaction Agreement).

Except as expressly provided above with respect to the Protected Benefits, nothing in this letter should be construed as a guarantee of any particular level of benefits or of your participation in any benefit plan. Subject to the specific terms of this letter, the Transaction Agreement, and the Governance Agreement, BlackRock and GIP reserve the right to amend, modify or terminate any or all benefits and compensation plans and programs at any time.

(d) PRE. During your employment with GIP prior to the Closing, you have participated in allocations and distributions with respect to PRE. Your current allocations and distributions with respect to such PRE in Existing Funds (as defined in the Governance Agreement) will remain fully intact and shall be governed by the associated award letters, compensation statements, Organizational Documents (as defined in the Governance Agreement), the Governance Agreement and the Transaction Agreement, as applicable (and nothing in this letter is intended to create any additional rights or entitlements in PRE beyond those contained in such documentation). You will remain responsible for the co- investment requirement associated with such grants of PRE consistent with past practices.

Section 2. Pre-Employment Requirements; Conditions of Employment Continuation.

Your employment and continuation of employment with BlackRock upon the Closing is contingent upon your (i) signing and returning of this letter, (ii) successful completion of BlackRock's standard background checks, (iii) your provision to BlackRock of appropriate documentation authorizing you to work in the United States, and (iv) your completion of BlackRock's Initial Political Contributions Disclosure form, if applicable, and the absence of restrictions on your employment due to prior political contributions.

(a) Business Protection Agreement (Restrictive Covenants). In connection with the Transaction Agreement, you have entered into that certain Business Protection Agreement, dated as of January 12, 2024, by and among you, GIP and New Banana, Inc. (the "**BPA"**), setting forth certain restrictive covenants and other agreements among the parties thereto. This letter shall only become effective upon the Closing if the BPA becomes effective upon the Closing without repudiation by you. Notwithstanding the foregoing, nothing herein shall be construed to limit any of your rights or entitlements to consideration pursuant to the Transaction Agreement and such rights and entitlements to consideration shall not be contingent on your employment with BlackRock or any of its affiliates.

(b) Screening Process. BlackRock's background screening process consists of the following: (i) a background check (which dependent upon job location or employing entity may include processing of fingerprints for purposes of a criminal records check); and (ii) your provision to BlackRock of appropriate documentation authorizing you to work in the United States.

(c) BlackRock Policies. There are a number of BlackRock policies relating to Compliance and other areas that will also require your attention. Information on these policies will be made available to you either before or shortly after the Closing Date.

(d) Confidentiality and Employment Policy. You will be required to sign a confidentiality and solicitation agreement (the "**Confidentiality and Employment Policy**") as a condition of employment upon the Closing, a copy of which is enclosed with this letter.

(e) Notice Period. Notwithstanding your "at-will" status, written notice of the intention to leave BlackRock is required of any Employee as detailed in the Confidentiality and Employment Policy. For clarity, the notice period required for Senior Managing Directors is 90 days.

(f) Outside Activities. BlackRock maintains an Outside Activity Policy, which describes certain outside activities for which employees must seek prior approval. A copy of the policy is provided separately for your review. The Outside Activity Policy is subject to change at any time in the sole discretion of BlackRock. Upon your commencement of employment, you will be required to complete a form (which will be provided to you) disclosing any activities covered by the policy or confirming that there are none. While BlackRock encourages its employees to be involved in activities outside of the workplace, activities covered by the policy must be approved by BlackRock's Legal & Compliance Department or the Outside Activity Review Committee and any current activities in which you are engaged will be reviewed in the same manner as applicable to new and current BlackRock employees.

Section 3. Acknowledgments and Representations.

(a) You hereby acknowledge and agree that (i) you are not permitted to make any unauthorized use of documents or other information in your employment with BlackRock which could properly be considered or construed to be confidential or proprietary information of another individual or company (for clarity, this excludes the Company Group); (ii) you may not bring any confidential documents or other form of tangible information onto the premises of BlackRock relating to your prior employer(s)' business, (for clarity, this excludes the Company Group); and (iii) you are not subject to any contractual or other restriction or obligation that would prohibit you from becoming employed by BlackRock or performing your duties and responsibilities with BlackRock.

Notwithstanding the foregoing, this clause shall not be construed to restrict you from retaining and using in connection with your BlackRock employment information pertaining to the business being acquired pursuant to the terms of the Transaction Agreement.

(b) Entire Agreement. This letter, together with the Confidentiality and Employment Policy and the Governance Agreement, reflects the complete agreement between you and BlackRock regarding the terms of your employment with BlackRock and supersedes any other statements or agreements regarding your employment with BlackRock on or before the date of your signing of this letter between you and BlackRock and any of its affiliates (including, without limitation, following the Closing, GIP, and including any employment agreement or arrangement or other restrictive covenant between you and GIP which, as of the Closing, shall become null and void). Notwithstanding the foregoing, this letter does not supersede the BPA or any other restrictive covenant agreement or arrangement to which you are a party or bound, including but not limited to those set forth in any Protective Covenants Agreement or Organizational Documents, or any amendment thereto. This letter may not be modified, except by a writing signed by an authorized representative of BlackRock and you.

(c) Employment Status. At all times during your employment with BlackRock, you will be considered an employee at-will. This means that both you and BlackRock have the right, subject to Section 2(e) hereof, to terminate your employment at any time, with or without cause.

Please signify your acceptance of the terms and conditions of this letter by signing and dating where indicated below and returning the executed letter.

If you have any questions, please contact a member of the GIP HR team at Human.Resources@global-infra.com, who will refer matters to BlackRock HR as appropriate.

We are excited about the future and look forward to the Closing and welcoming you to the team. Together, we will build on the work you have already accomplished to create a world leading infrastructure private markets investment platform. We look forward to all that we will achieve together.

Sincerely,

/s/ Martin Small

Martin Small
Senior Managing Director

Agreed and Accepted this 30th day of September, 2024

Adebayo O. Ogunlesi

Exhibit 10.34

FORM OF CARRY POINTS AWARD LETTER

Date: [_____]

To: [_____]

From: [_____]

CC: [_____]

Subject: Global Infrastructure GP V, L.P. – Tranche [____] Allocation Award

As you have been informed, you have been awarded a portion of the carried interest of the Global Infrastructure Partners Carry Plan as implemented in Global Infrastructure GP V, L.P. (the "Partnership"). The Partnership divides investments into Tranches. [Describe Tranche] The award is subject to the terms of the Partnership as set forth in the Amended and Restated Agreement of Exempted Limited Partnership of the Partnership. The Tranche Allocation Date for Tranche [____] is [_____]. Your award is as set forth below:

	Awarded Points
Tranche [______]	[Points]

Documentation:

To document your carried interest allocation, please:

1) Sign the enclosed Award Acceptance Form;

2) Complete, sign and return two copies of the 83(b) Election Form. This election is required to be made within 30 days of a transfer of property. We will ensure that these elections are filed within the required period of time.

Please return this form to [______], [______] by [______]. If you have any questions, please call [______].

AWARD ACCEPTANCE FORM – TRANCHE [____]

	Awarded Points
Tranche [______]	[Points]

I agree and accept the Point allocation set forth above.

______
Print name

Signature

Date

GLOBAL INFRASTRUCTURE GP V, L.P.

By: Global Infrastructure Investors V, LLC, *its general partner*

By: ____________
Name:
Title:

ANNEX A

**Excerpt of Certain Provisions from the
Amended and Restated Agreement of Exempted Limited Partnership
of Global Infrastructure GP V, L.P.**

Certain information has been excluded from this Annex A because it is both not material and it is the type that the Registrant treats as private or confidential.

"**Additional Partners**" means any Person that becomes a Limited Partner following the execution and delivery of a counterpart of this Agreement at the discretion of the General Partner.

"**Administrative Expenses**" means the Partnership's share of "Offering and Organizational Expenses" (as such term is defined in the relevant Fund Agreement) of a Fund.

"**Affiliate**" means, with respect to any Person, any Person that directly or indirectly through one or more intermediaries, controls, is controlled by or is under common control with such Person, and the term "Affiliated" shall have a correlative meaning. The term "control" means the possession, directly or indirectly, of the power to direct or cause the direction of the management and policies of a Person, whether through the ownership of voting securities, by contract or otherwise.

"**Agreement**" means this Amended and Restated Agreement of Exempted Limited Partnership, as amended from time to time.

"**Available Capital Commitment**" means, with respect to any Partner at any time, such Partner's Capital Commitment, reduced by the sum of all of such Partner's prior Capital Contributions, and increased by distributions or other amounts which would reduce the Partnership's "Funded Commitment" (as such term is defined in the relevant Fund Agreement), and subject to adjustment as provided in this Agreement.

"**Award Letter**" means the letter allocating Tranche Points to a Limited Partner to which this Annex A is attached.

"**Capital Account**" means a capital account established for each Partner on the books and records of the Partnership.

"**Capital Commitment**" means the commitment of each Limited Partner to make Capital Contributions in the amounts set forth on the books and records of the Partnership, as adjusted from time to time pursuant to the terms of this Agreement.

"**Capital Contribution**" means, with respect to any Partner, the capital contributed by such Partner to the Partnership pursuant to this Agreement, unless such capital is not treated as a Capital Contribution by the express terms of this Agreement.

“**Capital Interest**” means the portion of a Partner’s Interest other than any portion of such Interest attributable to Points and Carry Proceeds, including the portion of a Partner’s Interest relating to Special Profits Distributions.

“**Carry Proceeds**” means, with respect to Proceeds received by the Partnership from a Fund with respect to any Fund Investment, Proceeds representing distributions on account of the Fund Agreement for the Primary Fund or similar provisions in any other Fund Agreement, excluding any such amounts received by the Partnership on account of the Capital Contribution of the Partnership or the general partner of any Fund with respect to such Fund Investment. “Carry Proceeds” also means, with respect to any Fund Investment, any amount contributed by the Partners pursuant to Section 6.04 hereof with respect to such Fund Investment at the time of the disposition of such Fund Investment or at the time of a Fund Clawback, which contributed amount is available for distribution to the Partners.

“**Cause**” means, as to any GIM Limited Partner (or a GIM Limited Partner’s Designated Individual), that such Person:

(a) has been convicted or pleaded guilty or no contest to an act constituting a felony under the laws of the United States or any state or political subdivision thereof, a crime of moral turpitude or any crime involving the Partnership or its Affiliates;

(b) has violated any laws, rules or regulations applicable to the private equity industry when such violation is materially injurious to the business, reputation, character or standing of the Partnership or any of its Affiliates, after notice and, if capable of cure, a thirty (30) day-opportunity to cure;

(c) commits an act or acts constituting (i) a breach of fiduciary duty, Gross Negligence or willful misconduct or actual fraud or (ii) which constitutes a material breach of any such Person’s material obligations under this Agreement (including, without limitation, breach of the Restrictive Covenants), the governing documents of any Partnership Affiliate to which such Person is a party, or such Person’s Services Agreement, in each case which conduct is detrimental to the business, reputation, character or standing of GIM, after notice and, if capable of cure, a thirty (30) day-opportunity to cure;

(d) engages in insubordination or willful failure to perform his duties under his Services Agreement, after notice and, if capable of cure, a thirty (30) day-opportunity to cure; or

(e) defaults, directly or indirectly, under any Fund Agreement with respect to his direct or indirect obligation to fund his Capital Commitments and other payments to the Fund or defaults under this Agreement with respect to his obligations to make payments thereunder; *provided* that it shall not be a default hereunder, if in certain hardship circumstances, the General Partner has waived the requirements under this clause (e) as determined by the General Partner in its sole discretion and where remaining Limited Partners have agreed to fund any shortfall,

provided, *further* that, with regard to any GIM Limited Partner who is a Designated Portfolio Company Employee, then the foregoing definition will be deemed to include and incorporate a reference to the GIP Portfolio Company that employs them wherever the "Partnership" is referenced.

"**Clawback Contribution**" has the meaning set forth in Section 6.04(a).

"**Clawback Pool**" has the meaning set forth in Section 6.04(d).

"**Competitive Business**" has the meaning set forth in Section 3.15.

"**Continuing GIM Employee**" means, at any time, any Person who at such time is a director, officer or employee of GIM or that is a Designated Portfolio Company Employee or Designated Professional.

"**Designated Individual**" means, with respect to any Related Vehicle, the Continuing GIM Employee designated as such by the General Partner at the time of admission of the Related Vehicle as a Limited Partner, which designation shall be made by and shall reflect the individual who formed the trust, partnership, limited partnership, limited liability company or other similar entity.

"**Designated Portfolio Company Employee**" means any individual who is a director, officer or employee of a GIP Portfolio Company and who is not a Partner-Level Service Provider nor an officer or employee (but who may previously have been an officer or employee) of GIM, in each case as such individual is so designated by the General Partner to be a "Designated Portfolio Company Employee" entitled to participate in this Partnership as a Limited Partner.

"**Designated Professional"** means (*a*) any individual who is seconded by GIM, or (*b*) any individual who has previously been an officer or employee of GIM (but is not presently a Partner-Level Service Provider nor an officer or employee of GIM), in each case as such individual is so designated by the General Partner to be a "Designated Professional" entitled to participation in this Partnership as a Limited Partner.

"**Disability**" means, with respect to any GIM Limited Partner (or a Limited Partner's Designated Individual), a termination of such Limited Partner's (or such Designated Individual's) employment as a consequence of being deemed disabled under his Services Agreement or the long-term employee benefit plans of GIM (or of a GIP Portfolio Company with regard to a Designated Portfolio Company Employee) applicable to such Limited Partner (or such Designated Individual).

"**Distributable Share**" means, as to any Partner with respect to any Fund Investment at any time, the percentage of the total Carry Proceeds distributable to all Partners pursuant to Section 6.02(b)(i) that would be distributable to such Partner pursuant to such Section if Carry Proceeds were received by the Partnership with respect to such Fund Investment.

"**Fair Value**" means the value of any property held by the Partnership.

"**Former GIM Employees**" has the meaning set forth in Section 3.15(c).

"**Founding Limited Partner**" means each of Jonathan D. Bram, Matthew C. Harris, Michael McGhee, Adebayo O. Ogunlesi, Rajaram Rao and William Woodburn, to the extent that they are GIM Limited Partners, including any Related Vehicles of the foregoing.

"**Fund**" means (*a*) the Primary Fund; (*b*) Global Infrastructure Partners V-B Feeder Fund, L.P., a Cayman Islands exempted limited partnership; (*c*) Global Infrastructure Partners V-C, L.P., a Cayman Islands exempted limited partnership; (*d*) Global Infrastructure Partners V-C Intermediate, L.P., a Cayman Islands exempted limited partnership; (*e*) Global Infrastructure Partners V-C2 Intermediate, L.P., a Luxembourg special limited partnership; (*f*) Global Infrastructure Partners V Friends & Family Fund, L.P., a Delaware limited partnership; and (*g*) any other entity for which the Partnership may serve as a general partner or (as applicable) managing member or from which the Partnership may be entitled to receive direct or indirect distributions of carried interest from time to time.

"**Fund Agreement**" means any agreement of limited partnership (or similar constituent document) of any Fund, as the same may be amended and/or restated from time to time.

"**Fund Clawback**" has the meaning set forth in Section 6.04(a).

"**Fund Expenses**" means the Partnership's share of "Partnership Expenses" (as such term is defined in the relevant Fund Agreement) of a Fund. For the avoidance of doubt, Fund Expenses shall not include any Administrative Expenses.

"**Fund Investment**" means a "Portfolio Investment" (as such term is defined in the relevant Fund Agreement) of a Fund.

"**General Partner**" means Global Infrastructure Investors V, LLC, a Delaware limited liability company and any successor general partner of the Partnership appointed in accordance with its terms.

"**GIM**" means Global Infrastructure Management, LLC, a Delaware limited liability company, its successors, assigns and/or Affiliates, in each case, as the context so requires.

"**GIM Employees**" has the meaning set forth in Section 3.15(c).

"**GIM Limited Partner**" means each Limited Partner who is, or whose Designated Individual is, an employee or partner of GIM or who is a Designated Portfolio Company Employee or a Designated Professional.

"**GIM Related Person**" means GIM and all other Persons that directly or indirectly control, are controlled by or are under common control with GIM.

"**GIP Funds**" means each of the Funds and any other investment vehicle managed or advised by GIM.

"**GIP Portfolio Company**" means a "Portfolio Company" of a GIP Fund (as such term is defined in the applicable Fund Agreement).

"**Gross Negligence**" has the meaning given under, and shall be interpreted in accordance with, the laws of the State of Delaware, and "**Grossly Negligent**" shall be construed accordingly.

"**Incentive Capital Contribution**" has the meaning set forth in the Fund Agreement for the Primary Fund (or any other applicable Fund Agreement).

"**Interest**" means the partnership interest of a Partner in the Partnership at any particular time, including the right of such Partner to any and all benefits to which such Partner may be entitled to under this Agreement, together with the obligations of such Partner to comply with all the terms and provisions of this Agreement with respect to which such Partner is required to comply.

"**Investment Percentage**" of any Partner means, with respect to any Fund Investment, the percentage derived by (*a*) dividing the sum of such Partner's Capital Contributions in respect of such Fund Investment by the aggregate of such sums for all Partners (except as otherwise provided herein) in respect of such Fund Investment and (*b*) multiplying such quotient by 100.

"**Limited Partner**" means, at any time, any Person who is at such time a limited partner of the Partnership and shown as such on the books and records of the Partnership, in such Person's capacity as a limited partner of the Partnership.

"**Memorandum Account**" means a memorandum account established for each Partner on the books and records of the Partnership.

"**Non-Competition Period**" means, with respect to a GIM Limited Partner: (*a*) the period during which such GIM Limited Partner is a Continuing GIM Employee *plus* the following 12 months; or (*b*) such other period agreed to by the General Partner and such GIM Limited Partner.

"**Participating Partner**" means, with respect to any Fund Investment, any Partner who has made a Capital Contribution in respect of such Fund Investment.

"**Partner**" means the General Partner or any Limited Partner and "**Partners**" means, collectively, the General Partner and the Limited Partners.

"**Partner-Level Service Provider**" means a person who has been granted the title of "Partner" by GIM.

"**Partnership**" means Global Infrastructure GP V, L.P., a Cayman Islands exempted limited partnership.

"**Person**" means any individual, partnership, corporation, limited liability company, trust or other entity.

"**Points**" means the points allocated to Partners pursuant to Section 4.03. For the avoidance of doubt, all Tranche Points are Points.

"**Point Share**" means, as to any Partner with respect to any Fund Investment at any time, the amount (expressed as a percentage) of such Partner's Points at such time with respect to such

Fund Investment (as adjusted as provided in Section 4.03, if applicable) divided by 100. For the avoidance of doubt, the sum of the Point Shares of all Partners shall at all times equal 100%.

"**Primary Fund**" means Global Infrastructure Partners V-A/B, L.P., a Delaware limited partnership.

"**Prime Rate**" means, the prime rate announced, from time to time, by Citibank N.A. or if not so announced, the rate of interest publicly announced from time to time by any money center bank as its prime rate in effect at its principal office, as identified in writing by the General Partner to the Limited Partners.

"**Proceeds**" means, with respect to any Fund Investment, (*a*) the sum of the cash and noncash proceeds received by the Partnership from any sale or distribution of a Fund Investment, exchange of a Fund Investment, receipt of any dividends, interest or other distributions in connection with a Fund Investment, less (*b*) any expenses incurred by the Partnership in connection with such receipt. The value of any noncash Proceeds shall be the Fair Value thereof, as determined in good faith by the General Partner in its discretion.

"**Related Partner**" means (*a*) with respect to a Limited Partner who is an individual, any Related Vehicle with respect to which such Limited Partner has been designated a Designated Individual and (*b*) with respect to a Limited Partner that is a Related Vehicle, such Limited Partner's Designated Individual and any other Related Vehicle of such Limited Partner's Designated Individual.

"**Related Vehicle**" means any trust, partnership, limited partnership, limited liability company or other similar entity that is (*a*) formed by an individual who is or was a Continuing GIM Employee at the time of such entity's admission as a Limited Partner and is, in the determination of the General Partner, for the benefit of such individual and/or his family members and (*b*) designated by the General Partner as a Related Vehicle.

"**Release Condition**" means the condition that is satisfied at any date and as to any amount of Carry Proceeds that have been retained by the Partnership pursuant to Section 6.02(b)(iii), if: (*a*) such release would not breach the Fund Agreements or any Side Letters with Fund investors; and (*b*) the General Partner determines in its sole discretion that such amounts may be released.

"**Repurchase Period**" for any Limited Partner means the period from the date of the Limited Partner's ceasing to be a Continuing GIM Employee until the later of (*a*) December 31 of the year following the year in which the Limited Partner's employment is terminated and (*b*) the last day of the eighteenth (18th) calendar month following the month in which the Limited Partner's employment is terminated.

"**Repurchase Value**" means, with respect to the Capital Interest of any Limited Partner at any time, an amount determined by the General Partner equal to the lesser of (*a*) the implied cost of such Interest based on the cost of the Fund Investments attributable to such Interest as originally recorded on the books of the relevant Fund plus interest equal to the Prime Rate compounded annually and (*b*) the Fair Value of such Interest at such time, as determined by the General Partner in its reasonable discretion, which amount will be based on the Fair Value of the Fund Investments attributable to such Interest as carried on the books of the relevant Fund at such time.

"**Restrictive Covenants**" has the meaning set forth in Section 3.15(f).

"**Retirement**" means, with respect to a Founding Limited Partner, the retirement of such Founding Limited Partner at such time that the sum of (*a*) such Person's years of employment with GIM or its Affiliates (including as a member of GIM UK) and (*b*) such Person's age equals at least sixty (60) years.

"**Return Proceeds**" means, with respect to Proceeds received by the Partnership from a Fund with respect to any Fund Investment, all such Proceeds other than Carry Proceeds. For the avoidance of doubt, Return Proceeds include any Special Profits Distributions.

"**Services Agreement**" means:

> with respect to any GIM Limited Partner (or GIM Limited Partner's Designated Individual), (*i*) the employment agreement or equivalent between GIM (or its Affiliate) and such GIM Limited Partner (or such GIM Limited Partner's Designated Individual) and (*ii*) any Side Letter between GIM and such GIM Limited Partner (or such GIM Limited Partner's Designated Individual);
>
> with respect to a GIM Limited Partner (or GIM Limited Partner's Designated Individual) who is also a Designated Portfolio Company Employee, the employment agreement (or any other agreement relating to the performance of services) between such Designated Portfolio Company Employee and the applicable GIP Portfolio Company; and
>
> with respect to a GIM Limited Partner (or GIM Limited Partner's Designated Individual) who is also a Designated Professional, the agreement with respect to such Designated Professional between (i) GIM and such Designated Professional or the Person that employs or is associated with such Designated Professional or (*ii*) the Designated Professional and the Person that employs such Designated Professional.

"**Services Nonrenewal Notice**" means the notice from a GIM Limited Partner to GIM indicating his or her decision to not renew such GIM Limited Partner's Services Agreement.

"**Side Letter**" means a side letter or similar agreement entered into by and between the General Partner (on its own behalf or on behalf of the Partnership, and without the approval of any Limited Partner or other Person) and one or more Limited Partners, which has the effect of establishing rights under, or altering or supplementing the terms hereof or of any subscription agreement with respect to such individual Limited Partner.

"**Special Profits Distributions**" means distributions in respect of any profits interest in any Fund (and not, for the avoidance of doubt, distributions attributable to the Partnership's capital contributions to such Fund) that are received by the Partnership from the Fund pursuant to the Fund Agreement for the Primary Fund (or the equivalent section of any other Fund Agreement) and that were apportioned to the Partnership based on its Partner Interest for a Fund Investment or other investment (including distributions received from the Fund pursuant to the Fund Agreement for the Primary Fund (or any equivalent section of any other Fund Agreement)). For purposes of this definition, "Partner Interest" has the meaning set forth in the Fund Agreement for the Primary Fund (or any other applicable Fund Agreement).

“**Special Profits Percentage**” has the meaning set forth in Section 4.05.

“**Successor Fund**” has the meaning set forth in the Fund Agreement of the Primary Fund.

“**Tax Amount**” means, with respect to the income associated with any distribution of Proceeds hereunder to a Partner, the portion thereof which the General Partner determines in its reasonable discretion to be necessary to enable the Partners to discharge their U.S. federal, state and local (and, as the General Partner shall determine, non-U.S.) income tax liabilities attributable to such distribution. The amounts distributable shall be determined by the General Partner based on such assumptions as the General Partner determines to be appropriate.

“**Terminated Partner**” means any GIM Limited Partner (and, if applicable, such GIM Limited Partner’s Related Vehicle) who is who is not (*a*) providing services to the Partnership or a GIP Portfolio Company, (*b*) a GIM Related Person or (*c*) a Continuing GIM Employee.

“**Termination Event**” means, with respect to (*a*) any GIM Limited Partner, such Limited Partner’s ceasing to be a Continuing GIM Employee, and (*b*) any Limited Partner that is a Related Vehicle, such Related Vehicle’s Designated Individual’s ceasing to be a Continuing GIM Employee.

“**Tranche Allocation Date**” means as to any Tranche Points, the date of the first closing of the Fund and each subsequent date specified by the General Partner as the allocation date for such Tranche Points.

“**Tranche Period**” means each of the successive periods beginning on the date of the first closing of the Fund and ending the day immediately preceding the next Tranche Allocation Date.

“**Tranche Points**” means, as to any Limited Partner at any time and with respect to any Tranche Period, the number of Points allocated by the General Partner to such Limited Partner with respect to such Tranche Period under Section 4.03, which number shall be set forth from time to time on the books and records of the Partnership.

Section 3.15 *Restrictive Covenants*.

(a) Non-Competition. Each GIM Limited Partner agrees that during the Non-Competition Period (as defined below), (*i*) such Limited Partner shall not engage in any other business (whether as an employee, consultant, director, partner, member, shareholder or otherwise) that invests or operates primarily in the infrastructure area (each a “**Competitive Business**”) and (*ii*) such Limited Partner shall not knowingly adversely affect or interfere with the ability of GIM to manage a Fund (or any other GIP Fund). Notwithstanding anything in this Section 3.15 to the contrary, (*A*) direct ownership of less than five percent of the outstanding common stock of a publicly traded corporation; (*B*) participation as a passive investor in investment partnerships; (*C*) other personal investments specifically approved in writing by the General Partner; and (*D*) participation in a Successor Fund shall not constitute engaging in a Competitive Business.

(b) Non-Solicitation of Investors. During the Non-Competition Period (including any period after notice of termination has been given and prior to the date of termination), each GIM Limited Partner shall not, on behalf of any entity that competes with a Fund, solicit or otherwise receive funding from any prospective investor or client with whom GIM previously had material contact, or any investor or limited partner of GIM or a Fund (including any former investor of a Fund and any person that becomes an investor or limited partner of a Fund) or any other person who has a material business relationship with GIM or a Fund.

(c) Non-Solicitation of Employees. During the Non-Competition Period, each GIM Limited Partner shall not, without the Partnership's prior written consent, directly or indirectly, (*i*) solicit or induce, or cause others to solicit or induce, any employees or members of GIM or GIP Portfolio Companies ("**GIM Employees**") to leave or in any way modify their relationship with GIM or GIP Portfolio Companies, (*ii*) hire, or cause others to hire, any GIM Employees or any employee, member or independent contractor who has left the employment or other service of GIM or GIP Portfolio Companies within the one-year period which follows the termination of such employee's or member's or independent contractor's employment or other service with GIM or GIP Portfolio Companies ("**Former GIM Employees**", who shall not include any persons whose employment or other service with GIM was terminated for unrelated reasons and who were only approached after such termination), (*iii*) encourage or assist in the (*A*) hiring process of any GIM Employees or Former GIM Employees on behalf of any third party or (*B*) modification of any such employee's or member's relationship with GIM or GIP Portfolio Companies or (*iv*) cause others to participate, encourage or assist in the hiring process of any GIM Employee or Former GIM Employee.

(d) Non-Disparagement; Non-Interference. Each GIM Limited Partner shall not, and shall not cause or induce others to: (*i*) make any false, defamatory or disparaging statements about GIM its directors, employees and officers, a GIP Fund or a GIP Portfolio Company; or (*ii*) cause, induce or encourage any actual or prospective client, investor or limited partner of GIM or a GIP Fund or GIP Portfolio Company (including any former investor of a GIP Fund and any person that becomes an investor or limited partner of a GIP Fund) or any other person who has a material business relationship with GIM or a GIP Fund or GIP Portfolio Company, to terminate or modify any such actual or prospective relationship. This Section 3.15(d) shall apply to each Limited Partner while they are a Limited Partner and after they have ceased to be a Limited Partner, and shall survive the termination of this Agreement.

Section 4.03 *Allocation of Points*.

(a) For each Fund Investment there shall be established a pool of 100 Points which shall be allocated by the General Partner to Limited Partners as provided in this Section 4.03. Points not allocated under this Section 4.03 shall be allocated in the sole discretion of the General Partner.

(i) The General Partner shall allocate Tranche Points for each Tranche Period to each Limited Partner (OMITTED) as of each Tranche Allocation Date. Tranche Points may be allocated by way of an Award Letter. Except as provided elsewhere in this Agreement, a Limited Partner will be allocated Points in respect of each Fund Investment made during a Tranche Period equal to such Limited Partner's Tranche Points for such Tranche Period.

(ii) OMITTED.

(iii) The General Partner may allocate Points to Additional Partners as provided for in Section 5.12.

If a Fund Investment is committed to be made during one Tranche Period but the Fund Investment closes in another Tranche Period, the General Partner shall deem such Fund Investment to have been made in the Tranche Period in effect on the earlier to occur of (*A*) the date on which a binding agreement has been entered into with respect to such Fund Investment and (*B*) the date on which information about the Fund Investment has been disseminated to the Limited Partners for purposes of allocating Points to such Fund Investment pursuant to this Section 4.03. The Points allocated to any Limited Partner may be reduced, including pursuant to Section 4.04, and such Points shall revert to the General Partner, which shall reallocate such Points in its sole discretion.

Section 4.04 *Termination of Services*.

(a) The Points allocated to a GIM Limited Partner will be subject to reduction upon a Termination Event with respect to such Limited Partner as follows:

Date of Termination Event	Reduction
Prior to July 1 of the first calendar year next succeeding the applicable Tranche Allocation Date	100%
On or after July 1 of the first calendar year next succeeding the applicable Tranche Allocation Date, through and including June 30 of the second calendar year following the applicable Tranche Allocation Date	80%
On or after July 1 of the second calendar year following the applicable Tranche Allocation Date, through and including June 30 of the third calendar year following the applicable Tranche Allocation Date	60%
On or after July 1 of the third calendar year following the applicable Tranche Allocation Date, through and including June 30 of the fourth calendar year following the applicable Tranche Allocation Date	40%
On or after July 1 of the fourth calendar year following the applicable Tranche Allocation Date, through and including June 30 of the fifth calendar year following the applicable Tranche Allocation Date	20%

For the avoidance of doubt, such reductions shall not affect distributions made to a GIM Limited Partner before a Termination Event occurs with respect to such GIM Limited Partner.

(b) Notwithstanding the foregoing provisions of this Section 4.04:

(i) Cause. Upon a Termination Event with respect to a GIM Limited Partner for Cause (*A*) all of such GIM Limited Partner's vested and unvested Points shall be reduced to zero and (for the avoidance of doubt) such GIM Limited Partner shall not have the right to participate in any Carry Proceeds realized from any Fund Investments made after the date of such Termination Event and (*B*) the General Partner shall have the right exercisable by it or by one or more other Limited Partners designated by the General Partner in its discretion, at any time during the Repurchase Period, to repurchase the Capital Interest of such GIM Limited Partner at the Repurchase Value at the time of repurchase. The General Partner shall reallocate all Points forfeited pursuant to this 4.04(b)(i) in its sole discretion.

(ii) Death or Disability. Upon a Termination Event with respect to a GIM Limited Partner by reason of the death or Disability of such GIM Limited Partner, (*A*) all of such GIM Limited Partner's unvested Points will become fully vested, (*B*) the General Partner shall have no repurchase rights in respect of any portion (Points or Capital Interest) of the GIM Limited Partner's Interest and (*C*) such Limited Partner shall not have the right to participate in any Carry Proceeds realized from any Fund Investment made after the date of such Termination Event; *provided* that such GIM Limited Partner shall have the right to participate in any Carry Proceeds realized from any Fund Investment made after the date of such Termination Event (*1*) that is the subject of a binding agreement entered into by the relevant Fund prior to the date of such Termination Event or (*2*) about which the relevant Fund has disseminated information to its limited partners prior to the date of such Termination Event.

(iii) Voluntary Resignation. Upon a voluntary Termination Event with respect to a GIM Limited Partner for any reason, other than, with respect to a Founding Limited Partner, such Founding Limited Partner's Retirement, including but not limited to such GIM Limited Partner's decision not to renew his Services Agreement, (*A*) such GIM Limited Partner's unvested Points will be reduced to zero, (*B*) such GIM Limited Partner shall not have the right to participate in any Carry Proceeds realized from any Fund Investment made after the date of such Termination Event and (C) the General Partner shall have no repurchase rights in respect of any portion (Points or Capital Interest) of the GIM Limited Partner's Interests; *provided* that such GIM Limited Partner shall have the right to participate in any Carry Proceeds realized from any Fund Investment made after the date of such Termination Event (*1*) that is the subject of a binding agreement entered into by the relevant Fund prior to the date of such Termination Event or (*3*) about which the relevant Fund has disseminated information to its limited partners prior to the date of such Services Nonrenewal Notice or notice of a voluntary Termination Event. The General Partner shall reallocate all Points forfeited pursuant to this Section 4.04(b)(iii) in its sole discretion.

(iv) Retirement. Upon the Retirement of a Founding Limited Partner, then upon the effectiveness of such Retirement (*A*) 50% of the unvested Points of such Founding Limited Partner (or such greater percentage as the General Partner determines in its discretion) will become fully vested, (*B*) the remainder of the unvested Points of such

Founding Limited Partner will continue to vest as determined by the General Partner in its discretion, (*C*) the General Partner shall have no repurchase rights in respect of any portion (Points or Capital Interest) of such Founding Limited Partner's Interest and (*D*) such Founding Limited Partner shall not have the right to participate in any Carry Proceeds realized from any Fund Investment made after the date of such Retirement; *provided* that such Founding Limited Partner shall have the right to participate in any Carry Proceeds realized from any Fund Investment made after the date of such Retirement (*1*) that is the subject of a binding agreement entered into by the relevant Fund prior to the date of such Retirement or (*2*) about which the relevant Fund has disseminated information to its limited partners prior to the date of such Retirement.

(v) Termination without Cause. Upon an involuntary Termination Event with respect to a GIM Limited Partner other than pursuant to Section 4.04(b)(i) or 4.04(b)(ii), including the decision of GIM not to renew such GIM Limited Partner's Services Agreement, then upon the effectiveness of such involuntary Termination Event (*A*) 50% of the unvested Points of such GIM Limited Partner (or such greater percentage as the General Partner determines in its discretion) will become fully vested, (*B*) the remainder of the unvested Points of such GIM Limited Partner will continue to vest as determined by the General Partner in its discretion, (*C*) the General Partner shall have no repurchase rights in respect of any portion (Points or Capital Interest) of such GIM Limited Partner's Interest and (*D*) such GIM Limited Partner shall not have the right to participate in any Carry Proceeds realized from any Fund Investment made after the date of such involuntary Termination Event; *provided* that such GIM Limited Partner shall have the right to participate in any Carry Proceeds realized from any Fund Investment made after the date of such involuntary Termination Event (*1*) that is the subject of a binding agreement entered into by the relevant Fund prior to the date of such involuntary Termination Event or (*2*) about which the relevant Fund has disseminated information to its limited partners prior to the date of such involuntary Termination Event.

(vi) Competition Event. If a GIM Limited Partner breaches Sections 3.15(a), (b), (c) or (d) within 12 months of, or the confidentiality provisions of this Agreement at any time after, in each case, the Termination Event relating to such GIM Limited Partner, then all of such GIM Limited Partner's vested and unvested Points shall be reduced to zero. The General Partner shall reallocate all Points forfeited pursuant to this Section 4.04(b)(vi) in its sole discretion.

(c) Notwithstanding the other provisions of this Section 4.04, the General Partner may, in its sole discretion, reduce the Point reduction described in Section 4.04 or otherwise upon the occurrence of a Termination Event.

Section 4.05 *Special Profits Percentage*. For each Tranche Period and for each Limited Partner there shall be established a "Special Profits Percentage", which shall be determined by the General Partner, and which shall apply to each Fund Investment that closes in such Tranche Period. The sum of the Special Profits Percentages for each Tranche Period shall be 100%. A Partner's Special Profits Percentage may be decreased to in connection with the default by any Limited Partner or the admission of an Additional Partner. The books and records of the Partnership shall be revised to reflect the foregoing. Notwithstanding the foregoing, the General Partner may its sole discretion:

(a) require any Partner to make loans or capital contributions (which shall not reduce such Partner's Available Capital Commitment) to the Partnership or an Affiliate thereof in proportion to its share of the Incentive Capital Contributions (determined based on the Partners' Special Profits Percentages for each Fund Investment) and make any appropriate adjustments in connection therewith; or

(b) reduce the amount distributable to each Partner that is a Terminated Partner in proportion to its share of the Incentive Capital Contributions (determined based on the Partners' Special Profits Percentages for each Fund Investment), in which case the Partnership's use of any such undistributed amounts shall be determined by the General Partner.

Section 6.01 *Distributions Generally*. Distributions shall be made in accordance with this Article 6. Except as expressly set forth below, all calculations with respect to distributions shall be made on a Fund Investment by Fund Investment and Participating Partner by Participating Partner basis.

Section 6.02 *Dispositions of Fund Investments*.

(a) Return Proceeds.

(i) The Return Proceeds (other than Special Profits Distributions) realized on any date from any Fund Investment shall, subject to the provisions of Section 6.05, be distributed to the Participating Partners *pro rata* in accordance with each Participating Partner's Investment Percentage, if any, with respect to the Fund Investment giving rise to such Return Proceeds.

(ii) Special Profits Distributions realized on any date from any Fund Investment shall, subject to the provisions of Section 6.05, be distributed to the Participating Partners *pro rata* in accordance with each Participating Partner's Special Profits Percentage, if any, with respect to the Fund Investment giving rise to such Special Profits Distributions.

(b) Carry Proceeds.

(i) The amount distributed to each Participating Partner in respect of receipts by the Partnership representing Carry Proceeds from any Fund Investment shall reflect the General Partner's good-faith determination of an equitable allocation of Fund Expenses and Administrative Expenses across all Fund Investments and Tranche Periods. Any Carry Proceeds shall be distributed to the Partners *pro rata* in accordance with the positive balances of their Memorandum Accounts immediately prior to such distribution; *provided* that in no event shall any Partner receive a distribution of Carry Proceeds pursuant to this Section 6.02(b)(i) that results in a negative balance in such Partner's Memorandum Account.

(ii) For the avoidance of doubt, the amount of Carry Proceeds to be distributed to any Limited Partner pursuant to this Section 6.02(b) shall not be affected by any reduction of Points after the realization by the Partnership of such Carry Proceeds. Each Participating Partner's Point Share of any Carry Proceeds realized on any date by the Partnership shall be distributed by the Partnership to such Participating Partner in the manner hereinafter set forth, subject to Sections 6.04 and 6.05.

(iii) Amounts representing Carry Proceeds to be distributed to any Limited Partner pursuant to this Section 6.02(b) at any time shall be distributed (*A*) in the case of any Founding Limited Partner who is no longer a Continuing GIM Employee because of Retirement at such time, (*1*) with respect to Carry Proceeds consisting of cash, 80% in cash and 20% shall be retained by the Partnership and distributed upon satisfaction of the Release Condition and (*2*) with respect to Carry Proceeds consisting of securities or other property, 80% in the form of such securities or other property and 20% shall be retained by the Partnership and the Partnership shall in turn distribute to such Partner an interest in such retained securities or other property upon satisfaction of the Release Condition, and (*B*) in the case of (*y*) any Limited Partners or their Designated Individuals (other than any Founding Limited Partners) who are no longer Continuing GIM Employees because of any Termination Event or (*x*) any Founding Limited Partner who is no longer a Continuing GIM Employee because of any Termination Event other than Retirement, and except to the extent the General Partner determines in its sole discretion to distribute a greater percentage of cash or securities, (*1*) with respect to Carry Proceeds consisting of cash, 50% in cash and 50% shall be retained by the Partnership and distributed upon satisfaction of the Release Condition and (*2*) with respect to Carry Proceeds consisting of securities or other property, 50% in the form of such securities or other property and 50% shall be retained by the Partnership and the Partnership shall in turn distribute to such Partner an interest in such retained securities or other property upon satisfaction of the Release Condition.

(iv) With respect to any securities or other property retained by the Partnership in respect of any Limited Partner in connection with any in-kind distribution, such Limited Partner may at any time request the Partnership to sell all or any portion of such securities or other property. In such event, the Partnership shall use its reasonable best efforts to sell such securities or other property for cash and the amount retained by the Partnership for subsequent distribution pursuant to Section 6.02(b)(iii) shall be increased by an aggregate amount equal to the amount of the net cash proceeds from such sale, *provided*, in each case, that to the extent permitted by law, any taxable gain or loss recognized by the Partnership upon the disposition of such securities shall be allocated equitably pursuant to the tax allocation provisions of this Agreement among those Limited Partners with respect to which such cash was retained as a result of such sale or exchange among those Limited Partners who requested the sale.

(v) Any amounts retained by the Partnership pursuant to Section 6.02(b)(iii) may be, in the sole discretion of the General Partner, (A) invested in short-term investments pending application to the repayment of amounts retained by the Partnership pursuant to Section 6.02(b)(iii), (B) held in a segregated account with a perfected security interest in favor of the Fund or released to the Limited Partner pursuant to Section 6.05(c), or (C) subject to such other conditions or requirements as the General Partner may determine.

Section 6.04 *Clawback.*

(a) The Partners shall be required to make a contribution to the Partnership in the manner hereinafter set forth (a "**Clawback Contribution**") at such time as (*i*) the Partnership is required to make a contribution to a Fund pursuant to any clawback or similar provision in any Fund Agreement, (a "**Fund Clawback**") or (*ii*), if determined necessary or advisable by the General Partner.

(b) Contributions by the Partners on account of any Clawback Contribution will be made from the Clawback Pool at the direction of the General Partner *pro rata* in accordance with the negative balances of the Partners' Memorandum Accounts immediately prior to such contribution.

(c) If, at any time, the Partnership makes a contribution to a Fund pursuant to a Fund Clawback and, in the reasonable discretion of the General Partner such contribution is attributable to a particular Fund Investment, then each Partner's Memorandum Account shall be debited by such Partner's *pro rata* share of the contribution, based on such Partner's original Distributable Share (as adjusted for a subsequent reallocation of Points among Partners but without giving effect to reductions pursuant to Section 4.04) with respect to such Fund Investment. If, in the reasonable discretion of the General Partner, such contribution is not attributable to a particular Fund Investment, the General Partner shall determine in good faith the amount, if any, by which each Partner's Memorandum Account shall be debited.

(d) Contributions by a Limited Partner on account of a Clawback Contribution initially shall be made from any or all of the following sources, at the General Partner's discretion, (collectively, the "**Clawback Pool**"): (*i*) by reduction of any amount retained by the Partnership pursuant to Section 6.02(b)(iii); (*ii*) by reduction of such Limited Partner's interest in any securities or other property retained by the Partnership in respect of such Limited Partner in connection with any in kind distribution (it being understood that the Partnership shall use its reasonable best efforts to sell such securities or other property for cash and shall reduce such Limited Partner's Clawback Contribution by an amount equal to the amount of the net cash proceeds from such sale); and (*iii*) by deduction from any future distributions on account of Carry Proceeds and Return Proceeds (but excluding distributions that are a return of a Capital Contribution) from the Partnership to such Limited Partner.

It is understood that Clawback Contributions shall be made from time to time from the above sources as and to the extent such sources become available to satisfy the obligation to make such Clawback Contributions.

(e) Notwithstanding anything in this Section 6.04 to the contrary, to the extent the Clawback Pool is insufficient to satisfy Clawback Contributions, each Partner shall return distributions to the Partnership in satisfaction of the remainder of such Clawback Contributions *pro rata* in accordance with the negative balances of the Partners' Memorandum Accounts immediately prior to such return of distributions *provided, however*, that a Partner shall not be required to return more to the Partnership pursuant to this Section 6.04(f) than the excess of (*i*) the aggregate distributions to such Partner, as the case may be, pursuant to Section 6.02(b) over (*ii*) the aggregate distributions that were made, or that could have been made, to such Partner pursuant to Section 6.03(a). The Partners shall execute a separate guarantee of their obligations under this Section 6.04(f) as contemplated by the Fund Agreement before receiving any Carry Proceeds under this Agreement. Any payments made by a Partner to a Fund pursuant to such guarantee in respect of a Fund Clawback shall be deemed to be returns of distributions to the Partnership pursuant to this Section 6.04(f).

(f) The General Partner shall maintain records of (*i*) each Limited Partner's required Clawback Contributions, (*ii*) the outstanding amounts retained pursuant to Section 6.02(b)(iii) and owing to such Limited Partner, (*iii*) such Limited Partner's interest in any securities or other property retained by the Partnership in connection with any in-kind distribution and (*iv*) any amounts that may be deducted from future distributions by the Partnership to such Limited Partner on account of any Clawback Contribution, which records shall be made available to such Limited Partner upon the reasonable request of such Limited Partner.

(g) Clawback Contributions by a Limited Partner shall not be deemed Capital Contributions and shall not affect the obligations of a Partner to make Capital Contributions hereunder, and the obligation on the part of each Partner to make such Clawback Contributions shall be in addition to any liability such Partner may otherwise have under this Agreement.

Section 6.05 *Other General Principles of Distributions*.

(a) Distributions of Cash. Subject to Sections 6.04 and distributions upon dissolutions of the Partnership and the remaining provisions of this Section 6.05, distributions shall be made promptly after the receipt of cash or noncash proceeds by the Partnership.

(b) Distributions in Kind. Any securities or other property constituting all or any portion of a Fund Investment may be distributed in kind to the Partnership at such time and in such amounts as the General Partner shall in its discretion determine, and any such distribution shall be made to the Participating Partners in respect of such Fund Investment in accordance with the amounts that would have been distributed in respect thereof had such property been disposed of by the Partnership immediately prior to such distribution. In any distribution of property in kind, the General Partner shall not, subject to the provisions of Sections 6.04, 6.05(c) and 6.05(e), discriminate among the Limited Partners but shall in any such distribution, (*i*) distribute to the applicable Limited Partners property of the same type and (*ii*) if cash and property in kind are to be distributed simultaneously in respect of any Fund Investment, distribute cash and property in kind in the same proportion to each such Partner. For purposes of allocations to Capital Accounts, property to be distributed in kind shall be valued at the Fair Value thereof by the General Partner in its discretion on a date as near as reasonably practicable to the date of such distribution.

(c) Withholding of Certain Amounts. Notwithstanding anything else contained in this Agreement, the General Partner may, in its discretion, withhold from any distribution to any Limited Partner pursuant to this Agreement, any amounts due from such Partner to the Partnership or the General Partner pursuant to this Agreement to the extent not otherwise paid. In addition, if the General Partner shall have determined in good faith, that there will likely be a requirement for a Clawback Contribution, then the General Partner may withhold all or any portion of distributions of Carry Proceeds to any Limited Partner under this Agreement as determined by the General Partner; *provided* that the General Partner shall in any such distribution distribute an amount in cash to each Limited Partner at least equal to the Tax Amount with respect to such distributions and *provided*, *further*, that the General Partner, in its sole discretion, may permit any Limited Partner to substitute such amount withheld with a letter of credit issued by a financial institution that has a long-term credit rating (and maintains a long-term credit rating) of BBB+ by S&P, Baa1 by Moody's or higher. If any amount withheld in accordance with the immediately preceding sentence is cash, such amount shall be added from time to time to the amounts retained and owing to such Limited Partner pursuant to Section 6.02(b)(iii), and, if any amount withheld in accordance with the immediately preceding sentence is securities or other property, such amount shall increase such Limited Partner's interest in such securities or other property retained by the Partnership.

(d) Amounts Held in Reserve. In addition to the rights set forth in Section 6.05(b) and Section 6.05(c), the General Partner shall have the right, in its reasonable discretion, to withhold amounts otherwise distributable by the Partnership to the Partners in order to maintain the Partnership in a sound financial and cash position and to make such provision as the General Partner in its reasonable discretion deems necessary or advisable for any and all liabilities and obligations, contingent or otherwise, of the Partnership (including, but not limited to, any obligation to repay any loan owed by the Partnership to any Fund pursuant to section 8.3(d) of the Fund Agreement of the Primary Fund or the equivalent section of any other Fund Agreement, in which case the General Partner shall make appropriate adjustments to distributions pursuant to this Agreement to reflect such repayment) or if, following such distribution, the aggregate of the balances on any Partner's Capital Account would be less than zero.

(e) Equal Treatment. Subject to the Related Partners provisions of this Agreement, nothing in this Agreement shall obligate the General Partner to treat all Limited Partners alike, and the exercise of any power or discretion by the General Partner in the case of any one such Limited Partner shall not create any obligation on the part of the General Partner to take any similar action in the case of any other such Limited Partner, it being understood that any power or discretion conferred upon the General Partner shall be treated as having been so conferred upon the General Partner as to each such Limited Partner separately.

(f) Fund Agreement. Subject to Section 6.02(a) and Section 6.04(d), Return Proceeds realized from any Fund Investment shall be distributed to the Participating Partners to the extent a corresponding distribution was made to limited partners of the Primary Fund pursuant to the Fund Agreement for the Primary Fund (or to partners of any other Fund pursuant to similar provisions of such other Fund Agreement).

Exhibit 19.1

BlackRock

Global Insider Trading Policy

Objective and Scope

This policy governs trading in the securities of BlackRock, Inc. by directors and officers of BlackRock, Inc. (the "directors and officers") and by employees[1] (collectively with the directors and officers, "Covered Persons") of BlackRock, Inc. and its wholly-owned subsidiaries (collectively, "BlackRock") and the handling of material, nonpublic information ("MNPI", as further defined below) under applicable securities laws in the U.S. or the equivalent in the other jurisdictions in which BlackRock operates. Covered Persons play a critical role in maintaining the integrity of BlackRock's reputation and must handle MNPI and proprietary or confidential information properly. BlackRock has adopted a number of policies that deal with the handling of MNPI, including but not limited to a Code of Business Conduct and Ethics that obligates Covered Persons to maintain the confidentiality of information entrusted to them, a policy that establishes controls on the use and sharing of MNPI, and confidentiality and employment policies that obligate Covered Persons to hold information relating to the business of BlackRock in strict confidence.

Applicable laws and regulations prohibit any behaviors that lead to market abuse. Covered Persons are prohibited by law from buying or selling BlackRock securities or any other company's securities (including, but not limited to, common stock, options to purchase common stock, preferred stock, debt, convertible debentures, and warrants, as well as derivative securities, such as exchange-traded put or call options or swaps) while in possession of MNPI with respect to those securities or companies, whether for their own account, a family[2] member's account, organization or firm account, or for a client's account. In addition, if a Covered Person has MNPI, they are prohibited from "tipping" or disclosing such information to others or donating shares with the expectation of receiving a tax benefit.

BlackRock employees may acquire MNPI through BlackRock's customers, suppliers, affiliates, and companies in which BlackRock, its products, funds, or accounts, may invest. In certain circumstances, it may be necessary to establish an information barrier in order to wall off the employee in possession of MNPI.

Policy / Document Requirements and Statements

1. Material, Nonpublic Information

- **Material Information:** Information is "material" if there is a substantial likelihood that a reasonable person would consider the information important when making an investment decision or the information, if made public, would likely affect the market price of a company's securities. Information may be material to transactions in the securities of more than one company. For example, in some circumstances, the same information may be material to transactions in securities of the company from which the information originated as well as suppliers, customers, or competitors of that company.
  Material information concerning BlackRock (including information relating to its subsidiaries or affiliates) or other such companies may include the following:
  - Sales and earnings results or estimates (or changes thereto, if previously published);
  - Changes in product offerings;
  - Significant additions or losses of client accounts;
  - Proposed mergers, acquisitions, divestitures, or joint ventures;

1 For purposes of this policy, the term "employee" includes all contingent workers and individual services providers, unless their agreement with BlackRock contains express conditions to the contrary.

2 Family members include Family Relationships as further described in the Global Relationships at Work Policy.

- Stock repurchase plans and stock splits;
- Securities offerings;
- Litigation and investigations;
- Changes in control or extraordinary management developments;
- Extraordinary borrowings or other liquidity problems;
- Cybersecurity risks or incidents; and
- Other similar items.

- **Nonpublic Information:** Information is considered to be "nonpublic" unless it has been disclosed to the public adequately. Examples of adequate disclosure include public filings with securities regulatory authorities, the issuance of press releases, and may also include meetings that are generally open to members of the press and the public. By contrast, information would likely not be considered public if it is available only to BlackRock employees, or if it is available to a select group of analysts, brokers, and/or investors.

2. Restriction on Tipping

Covered Persons are prohibited from disclosing MNPI to another person even if such person does not purchase or sell securities on the basis of such information, or pass on the information to another person (also known as "tipping").

Covered Persons may not disclose MNPI to:

- Any person outside of the firm (not employed by BlackRock), unless any such disclosure is made in accordance with BlackRock's policies regarding the protection, and authorized external disclosure, of information; or
- Any BlackRock employee unless such employee needs to know the information for a valid business reason. MNPI may not be shared with anyone in a different information barrier group without obtaining prior approval from Legal & Compliance.

3. Trading in BlackRock Securities

- **No Trading When in Possession of MNPI:** No Covered Person may purchase or sell BlackRock securities when in possession of MNPI regarding BlackRock, even if the transaction has been pre-approved and the trading window is open.
- **Pre-Clearance Required:** All transactions in BlackRock's securities by a Covered Person must be pre-cleared by Legal & Compliance and the transaction must be within the prescribed trading window. This includes purchases, sales, stock option exercises, estate planning transactions and gifts.
  - Employees, including officers, must submit a pre-clearance request in the Personal Trading Assistant and receive an approval before undertaking any transactions permitted under this policy.
  - Pre-clearance approvals, whether for market orders[3] or limit orders[4], are valid **only** on the day the approval is received. The order must be executed on the same day by the closing time of the market on which the security is traded.
  - Employees can request exemption from the pre-clearance requirement for trades in a spousal account in which the employee has no investment discretion. Spousal accounts require disclosure, regardless of pre-clearance exemption status, and are subject to periodic monitoring. Employees may be required to supply a quarterly statement for such accounts. When such requests are made, employees must provide the statements to the Legal & Compliance within 30 days of the request.
  - **Pre-Clearance Not Required**: Pre-clearance approval is not required to transact in the following:
    - Purchases of common stock under an Employee Stock Purchase Plan, vesting of Restricted Share Units, or acquisitions of common stock in connection with director compensation (however, sales of the same must be pre-cleared);

[3] Buy or sell transactions placed at current market price.

[4] Buy or sell transactions placed at a pre-determined price (detailed within the pre-clearance request).

        - 529 Plans, Direct Stock Purchase Plans, and any securities purchased pursuant to a dividend reinvestment plan;
        - Securities acquired by an exercise of rights to the holders of a class of securities (however, sales of the same must be pre-cleared);
        - Stock dividend, stock split, or similar corporate distribution;
        - Conversion of employee stock options (however, sales of the same must be pre-cleared); and
        - Transfer of securities with no change in beneficial ownership (e.g., transfer from one account in your name to another account in your name).
    - **Trading Plan Exception:** Sales of BlackRock securities may be effected for Covered Persons without seeking pre-clearance, regardless of their awareness of MNPI and outside of the prescribed trading windows (see below) solely if the transaction is made under a pre-arranged written trading plan that is:
        - In compliance with the requirements of Rule 10b5-1(c) under the Securities Exchange Act of 1934, as amended;
        - Pre-approved by Legal & Compliance prior to adoption and execution; and
        - Entered into (or amended) when the Covered Person is not in possession of MNPI and is adopted (or amended) during a prescribed trading window period.
- **Trading Windows:** If not in possession of MNPI, Covered Persons are only permitted to transfer (resulting in change to beneficial ownership), gift, or trade (purchase, sell, or exercise employee stock options) BlackRock securities (upon pre-clearance approval) during trading windows as determined and announced by Legal & Compliance. The opening and closing dates of each trading window are announced by email to all Covered Persons by Legal & Compliance. Typically, the trading window opens at the beginning of the second full day of trading following the public release of BlackRock's quarterly financial information and closes at the end of the second trading day of the last month of the quarter in which such quarterly financial information was released. The trading window may be opened and closed by Legal & Compliance at other times.
- **Prohibition on Hedging and Pledging BlackRock Securities:** A Covered Person may not:
    - Enter into any transactions that have the effect of hedging the economic risks and rewards of BlackRock securities held by such Covered Person, other than pursuant to a contractual right negotiated in connection with a merger or acquisition.
    - Hold BlackRock securities in margin accounts or pledge BlackRock securities as collateral for a loan.
- **Restrictions on Trading in BlackRock Securities:** In addition, employees may not:
    - Trade in options or warrants on BlackRock securities;
    - Engage in day trading of BlackRock securities;
    - Engage in any short selling of BlackRock securities;
    - Purchase any single-stock futures contracts on BlackRock securities; and
    - Trade in BlackRock securities in managed accounts.
- **BlackRock's Trading in Its Own Securities:** It is BlackRock's policy to comply with applicable securities laws concerning trading in BlackRock securities on BlackRock's behalf.

4. Post-Termination Transactions

Although the pre-clearance procedures specified in this policy will cease to apply to a Covered Person upon the conclusion of their service with BlackRock, the Covered Person will remain subject to applicable securities laws pertaining to trading while in possession of MNPI.

5. Other Trading Restrictions

This policy should be read in conjunction with other BlackRock policies, including the Global Personal Trading Policy which contains other restrictions on trading the securities of other companies for BlackRock employees.

6. Other Barriers

BlackRock implements information barriers for a variety of reasons in addition to the prevention of insider trading. Barriers established for another explicit purpose, such as avoidance of potential coordinated trading, will be outlined in those specific policies and/or procedures.

7. Consequences of Violations

The penalties for insider trading violations are severe and could include significant fines and imprisonment. In addition, an employee's failure to comply with this Policy may subject the employee to disciplinary action, including dismissal for cause, whether or not the employee's failure to comply results in a violation of applicable law.

8. Questions

Please contact Legal & Compliance in your respective region with questions regarding this policy.

Exhibit 21.1

SUBSIDIARIES OF REGISTRANT

The following table lists the direct and indirect subsidiaries of BlackRock, Inc. as of December 31, 2024*.

Name of Subsidiary	Jurisdiction/State of Incorporation
Acero Holdings I LLC	Delaware
Acero LLC	Delaware
Amethyst Intermediate, LLC	Delaware
AnalytX Hosting LLC	Florida
AnalytX LLC	Delaware
AnalytX Software LLC	Virginia
Aperio Group, LLC	California
Aperio Holdings, LLC	Delaware
Asia Property AU 1 Pty Ltd	Australia
Asia Property AU 2 Pty Ltd	Australia
Asia-Pacific Private Credit Opportunities Fund I (GenPar) (GP) Ltd.	Cayman Islands
AU Infra 1 Pty Ltd	Australia
AU Infra 2 Pty Ltd	Australia
BAA Holdings, LLC	Delaware
Beijing eFront Software Company Limited	China
BFM Holdco, LLC	Delaware
Blackhawk Investment Holding, LLC	Delaware
BlackRock (Barbados) Finco 1 SRL	Barbados
BlackRock (BVI) Finco 1, Ltd.	British Virgin Islands
BlackRock (BVI) Holdco 1, Ltd.	British Virgin Islands
BlackRock (BVI) Holdco 2, Ltd.	British Virgin Islands
BlackRock (Channel Islands) Limited	Jersey
BlackRock (Luxembourg) S.A.	Luxembourg
BlackRock (Netherlands) B.V.	Netherlands
BlackRock (Singapore) Holdco Pte. Limited	Singapore
BlackRock (Singapore) Limited	Singapore
BlackRock Advisors (UK) Limited	United Kingdom
BlackRock Advisors Singapore Pte. Limited	Singapore
BlackRock Advisors, LLC	Delaware
BlackRock Aladdin (eFront)	France
BlackRock Aladdin Luxembourg S.à.r.l.	Luxembourg
BlackRock Alternative Advisors GP Holdings, LLC	Delaware
BlackRock Alternatives Management, LLC	Delaware
BlackRock AP Investment Holdco, LLC	Delaware
BlackRock Asset Management Canada Limited/Gestion d’actifs BlackRock Canada Limitée	Canada
BlackRock Asset Management Deutschland AG	Germany
BlackRock Asset Management International Inc.	Delaware
BlackRock Asset Management Investor Services Limited	United Kingdom
BlackRock Asset Management Ireland Limited	Ireland
BlackRock Asset Management North Asia Limited	Hong Kong
BlackRock Asset Management Schweiz AG	Switzerland
BlackRock Asset Management UK Limited	United Kingdom
BlackRock Australia Holdco Pty. Ltd.	Australia
BlackRock Brasil Gestora de Investimentos Ltda.	Brazil
BlackRock Cal 1 Investor, LLC	Delaware
BlackRock Canada Holdings ULC	Canada
BlackRock Capital Holdings, Inc.	Delaware
BlackRock Capital Investment Advisors, LLC	Delaware
BlackRock Capital Management, Inc.	Delaware
BlackRock Capital Markets EMEA S.à r.l	Luxembourg
BlackRock Capital Markets, LLC	Delaware
BlackRock Cayco Limited	Cayman Islands
BlackRock Cayman 1 LP	Cayman Islands
BlackRock Cayman 2 LP	Cayman Islands
BlackRock Cayman Capital Holdings Limited	Cayman Islands
BlackRock Cayman Finco 2 Limited	Cayman Islands
BlackRock Cayman Finco 3 Limited	Cayman Islands
BlackRock Cayman Finco Limited	Cayman Islands
BlackRock Cayman West Bay Finco Limited	Cayman Islands
BlackRock Cayman West Bay IV Limited	Cayman Islands
BlackRock Cayman Z Limited	Cayman Islands
BlackRock CCB Wealth Management Limited	China
BlackRock Channel Islands Holdco Limited	Jersey
BlackRock Chile Asesorias Limitada	Chile
BlackRock Colombia Holdco, LLC	Delaware
BlackRock Colombia Infraestructura S.A.S.	Colombia
BlackRock Colombia SAS	Colombia
BlackRock Company Secretarial Services (UK) Limited	United Kingdom
BlackRock Corporation US Inc.	California
BlackRock Crane Limited	United Kingdom
BlackRock Delaware Holdings Inc.	Delaware
BlackRock Enterprise Management (Shanghai) Co. Ltd.	China
BlackRock Execution Services	California

Name of Subsidiary	Jurisdiction/State of Incorporation
BlackRock Finance Europe Limited	United Kingdom
BlackRock Finance, Inc.	Delaware
BlackRock Financial Management, Inc.	Delaware
BlackRock Finco UK Ltd.	United Kingdom
BlackRock Finco, LLC	Delaware
BlackRock First Partner Limited	Jersey
BlackRock France SAS	France
BlackRock Fund Advisors	California
BlackRock Fund Management Co., Ltd.	China
BlackRock Fund Management Company S.A.	Luxembourg
BlackRock Fund Managers Limited	United Kingdom
BlackRock Funding International, Ltd.	Cayman Islands
BlackRock Funds Services Group, LLC	Delaware
BlackRock GPCo, (Scotland) Ltd.	Scotland
BlackRock Group Limited	United Kingdom
BlackRock HK Holdco Limited	Hong Kong
BlackRock Holdco 10, Ltd	Cayman Islands
BlackRock Holdco 11, Ltd	Cayman Islands
BlackRock Holdco 12, Ltd	Cayman Islands
BlackRock Holdco 2, Inc.	Delaware
BlackRock Holdco 3, LLC	Delaware
BlackRock Holdco 4, LLC	Delaware
BlackRock Holdco 5, LLC	Delaware
BlackRock Holdco 6, LLC	Delaware
BlackRock Holdco 7, LLC	Delaware
BlackRock Holdco 8, Ltd	Cayman Islands
BlackRock Holdco 9, Ltd	Cayman Islands
BlackRock Hungary Kft.	Hungary
BlackRock Index Services, LLC	Delaware
BlackRock Infrastructure Management I, LLC	Cayman Islands
BlackRock Institutional Services, Inc.	Delaware
BlackRock Institutional Trust Company, National Association	United States
BlackRock International Holdings, Inc.	Delaware
BlackRock International Limited	Scotland
BlackRock Investment Management (Australia) Limited	Australia
BlackRock Investment Management (Korea) Limited	Korea
BlackRock Investment Management (New Zealand) Limited	New Zealand
BlackRock Investment Management (Shanghai) Co., Ltd.	China
BlackRock Investment Management (Taiwan) Limited	Taiwan
BlackRock Investment Management (UK) Limited	United Kingdom
BlackRock Investment Management Ireland Holdings Unlimited Company	Ireland
BlackRock Investment Management, LLC	Delaware
BlackRock Investments, LLC	Delaware
BlackRock Japan Co., Ltd.	Japan
BlackRock Japan Holdings GK	Japan
BlackRock Jersey Finco 2 Limited	Jersey
BlackRock Latin America Holdco, LLC	Delaware
BlackRock Latin America B.V.	Netherlands
BlackRock Life Limited	United Kingdom
BlackRock Lux Finco S.à r.l.	Luxembourg
BlackRock Luxembourg Holdco S.à r.l.	Luxembourg
BlackRock México Infraestructura I, S. de R.L. de C.V.	Mexico
BlackRock México Infraestructura II, S. de R.L. de C.V.	Mexico
BlackRock México Infraestructura III, S. de R.L. de C.V.	Mexico
BlackRock México Manager II, S. de R.L. de C.V.	Mexico
BlackRock México Manager III, S. de R.L. de C.V.	Mexico
BlackRock México Manager, S. de R.L. de C.V.	Mexico
BlackRock México Operadora, S.A. de C.V., Sociedad Operadora de Fondos de Inversion	Mexico
BlackRock Niagara LLC	Delaware
BlackRock Operations (Luxembourg) S.à r.l.	Luxembourg
BlackRock Overseas Investment Fund Management (Shanghai) Co., Ltd.	China
BlackRock PC Holdings, LLC	Delaware
BlackRock Pensions Limited	United Kingdom
BlackRock Peru Asesorías S.A.	Peru
BlackRock Portfolio Management LLC	Delaware
BlackRock Property Consulting (Beijing) Co., Ltd.	China
BlackRock Property France President SAS	France
BlackRock Property France S.a.r.l.	France
BlackRock Property Lux S.à.r.l.	Luxembourg
BlackRock Realty Advisors, Inc.	Delaware
BlackRock Saturn Subco, LLC	Delaware
BlackRock Saudi Arabia	Saudi Arabia
BlackRock Scale Holdings, LLC	Delaware
BlackRock Services India Private Limited	India
BlackRock Strategic Investors GP, LLC	Delaware
BlackRock Strategic Investors, LP	Delaware
BlackRock Technology Holdings, LLC	Delaware
BlackRock Titan I, LLC	Delaware
BlackRock Titan II, LP	Delaware

Name of Subsidiary	Jurisdiction/State of Incorporation
BlackRock Titan Partner I, LLC	Delaware
BlackRock Titan Partner II, LLC	Delaware
BlackRock UK (Alpha) Limited	United Kingdom
BlackRock UK (Beta) Limited	United Kingdom
BlackRock UK (Delta) LP	United Kingdom
BlackRock UK (Gamma) Limited	United Kingdom
BlackRock UK (Sigma) Limited	United Kingdom
BlackRock UK 2 LLP	United Kingdom
BlackRock UK 3 LLP	United Kingdom
BlackRock UK 4 LLP	United Kingdom
BlackRock UK A LLP	United Kingdom
BlackRock UK Holdco 2 Limited	United Kingdom
BlackRock UK Holdco Limited	United Kingdom
BlackRock US Finco 2 LLC	Delaware
BlackRock US Finco 3 LLC	Delaware
BLK (Gallatin) Holdings, LLC	Delaware
BLK SMI, LLC	Delaware
BR Acquisition Mexico S.A. de C.V.	Mexico
BR Jersey International Holdings L.P.	Jersey
Cachematrix Holdings LLC	Colorado
Cachematrix Integrations Private Limited	India
Cachematrix Software Solutions LLC	Colorado
DMLT East, LLC	Florida
DMLT, LLC	New York
eFront (Jersey) Limited	Jersey
eFront Cayman 2 LP	Cayman Islands
eFront Cayman Ltd	Cayman Islands
eFront d.o.o. Beograd	Serbia
eFront DMLT Holdings, LLC	Delaware
eFront DMLT Holdings, S.R.L	Dominican Republic
eFront Do Brasil Soluções Informáticas Para Sistemas Financeiros Ltda.	Brazil
eFront DR, S.R.L	Dominican Republic
eFront Financial Solutions Inc.	Delaware
eFront FZ-LLC	United Arab Emirates
eFront GmbH	Germany
eFront Holding SAS	France
eFront Hong Kong Limited	Hong Kong
eFront Kabushiki Kaisha	Japan
eFront Ltd	United Kingdom
eFront Singapore Pte. Ltd.	Singapore
eFront Solutions Financières Inc.	Canada
eFront U.S., LLC	Delaware
EG Holdings Blocker, LLC	Delaware
GIM Advisory Services, LLC	Delaware
GIM EM Manager, LLC	Delaware
GIM India Fund Holdings, LLC	Delaware
GIM India Holdings, LP	Delaware
GIM LLC de Mexico S. de. R.L.	Mexico
GIM UK Holdings, LLC	Delaware
GIMAS Finance Pty Limited	Australia
GIP Australia Guernsey Co Limited	Guernsey
Global Energy & Power Infrastructure Advisors, LLC	Delaware
Global Energy & Power Infrastructure II Advisors, LLC	Delaware
Global Infrastructure Management Australia Pty Limited	Australia
Global Infrastructure Management EM HK, Limited	Hong Kong
Global Infrastructure Management HK, LLC	Delaware
Global Infrastructure Management Holdings LP	Delaware
Global Infrastructure Management LLP	United Kingdom
Global Infrastructure Management Participation LLC	Delaware
Global Infrastructure Management, LLC	Delaware
Global Infrastructure Partners India Private Limited	India
Global Infrastructure Partners Singapore Manager Pte. Ltd.	Singapore
Global Infrastructure Scots GP Limited	Scotland
Grosvenor Alternate Partner Limited	United Kingdom
Grosvenor Ventures Limited	United Kingdom
HLX Financial Holdings, LLC	Delaware
iShares (DE) I Investmentaktiengesellschaft mit Teilgesellschaftsvermögen	Germany
iShares Delaware Trust Sponsor LLC	Delaware
Kreos Capital Israel Ltd.	Israel
Kreos Capital Management (UK) Limited	United Kingdom
Kreos Capital Management Limited	Jersey
Mercury Private Equity MUST 3 (Jersey) Limited	Jersey
MyLifePath Digital Services, Inc.	Delaware
New Zealand Climate Infrastructure Fund GP Limited	New Zealand
Object Capital Technology, Inc.	Delaware
Otto Holdco, LLC	Delaware
Phoenix Acquisition LLC	Netherlands
Phoenix Acquisitions Holdings, LLC	Delaware
Portfolio Administration & Management Ltd.	Cayman Islands

Name of Subsidiary	Jurisdiction/State of Incorporation
Prestadora de Servicios Integrales BlackRock Mexico, S.A. de C.V.	Mexico
Retirement Product Insurance Agency Inc.	Delaware
SpiderRock Advisors, LLC	Delaware
St. Albans House Nominees (Jersey) Ltd.	Jersey
SVOF/MM, LLC	Delaware
Tennenbaum Capital Partners, LLC	Delaware
Tlali Acero, S.A. de C.V.	Mexico
Trident Merger, LLC	Delaware
Web Holdings, LLC	Delaware

* Certain subsidiaries that are not significant have been omitted.

Exhibit 22.1

Subsidiary Guarantor and Issuer of Guaranteed Securities

BlackRock Finance, Inc.(formerly known as BlackRock, Inc.), a Delaware corporation (“Old BlackRock”), is a direct wholly-owned subsidiary of BlackRock, Inc. (formerly known as BlackRock Funding, Inc.), a Delaware corporation (“New BlackRock”). Old BlackRock is a guarantor of New BlackRock’s registered debt securities. Old BlackRock is also an issuer of registered debt securities, which are guaranteed by New BlackRock. As of December 31, 2024, the registered debt securities were as follows:

New BlackRock registered debt securities guaranteed by Old BlackRock:

Name of Issuer	State of Formation of Issuer	Description of Registered Debt Securities
BlackRock, Inc.	Delaware	4.700% Notes due 2029
BlackRock, Inc.	Delaware	5.000% Notes due 2034
BlackRock, Inc.	Delaware	5.250% Notes due 2054
BlackRock, Inc.	Delaware	4.600% Notes due 2027
BlackRock, Inc.	Delaware	4.900% Notes due 2035
BlackRock, Inc.	Delaware	5.350% Notes due 2055

Old BlackRock registered debt securities guaranteed by New BlackRock:

Name of Issuer	State of Formation of Issuer	Description of Registered Debt Securities
BlackRock Finance, Inc.	Delaware	1.250% Notes due 2025
BlackRock Finance, Inc.	Delaware	3.200% Notes due 2027
BlackRock Finance, Inc.	Delaware	3.250% Notes due 2029
BlackRock Finance, Inc.	Delaware	2.400% Notes due 2030
BlackRock Finance, Inc.	Delaware	1.900% Notes due 2031
BlackRock Finance, Inc.	Delaware	2.10% Notes due 2032
BlackRock Finance, Inc.	Delaware	4.750% Notes due 2033

Exhibit 23.1

CONSENT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

We consent to the incorporation by reference in Registration Statement No. 333-278583 on Form S-3ASR and Registration Statement Nos. 333-137708, 333-169329, 333-197764, 333-225372, and 333-279595 on Form S-8 of our reports dated February 25, 2025, relating to the financial statements of BlackRock, Inc., and the effectiveness of BlackRock, Inc.’s internal control over financial reporting, appearing in this Annual Report on Form 10-K of BlackRock, Inc. for the year ended December 31, 2024.

/s/ Deloitte & Touche LLP

New York, New York
February 25, 2025

Exhibit 31.1

CEO CERTIFICATION

I, Laurence D. Fink, certify that:

1. I have reviewed this Annual Report on Form 10-K, for the fiscal year ended December 31, 2024 of BlackRock, Inc.;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant’s other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)), for the registrant and have:

   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

   (c) Evaluated the effectiveness of the registrant’s disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   (d) Disclosed in this report any change in the registrant’s internal control over financial reporting that occurred during the registrant’s most recent fiscal quarter (the registrant’s fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant’s internal control over financial reporting; and

5. The registrant’s other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant’s auditors and the audit committee of the registrant’s board of directors (or persons performing the equivalent functions):

   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant’s ability to record, process, summarize and report financial information; and

   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant’s internal control over financial reporting.

Date: February 25, 2025

By: /s/ Laurence D. Fink
Laurence D. Fink
Chairman & Chief Executive Officer

Exhibit 31.2

CFO CERTIFICATION

I, Martin S. Small, certify that:

1. I have reviewed this Annual Report on Form 10-K, for the fiscal year ended December 31, 2024 of BlackRock, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)), for the registrant and have:
   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
   (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
   (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: February 25, 2025

By: /s/ Martin S. Small
Martin S. Small
Senior Managing Director & Chief Financial Officer

Exhibit 32.1

Certification of CEO and CFO Pursuant to
18 U.S.C. Section 1350,
as Adopted Pursuant to
Section 906 of the Sarbanes-Oxley Act of 2002

In connection with the Annual Report on Form 10-K of BlackRock, Inc. (the “Company”) for the annual period ending December 31, 2024 as filed with the Securities and Exchange Commission on the date hereof (the “Report”), Laurence D. Fink, as Chief Executive Officer of the Company, and Martin S. Small, as Chief Financial Officer of the Company, each hereby certifies, pursuant to 18 U.S.C. § 1350, as adopted pursuant to § 906 of the Sarbanes-Oxley Act of 2002, that:

(1) The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and

(2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

/s/ Laurence D. Fink
Name: Laurence D. Fink
Title: Chairman & Chief Executive Officer
Date: February 25, 2025

/s/ Martin S. Small
Name: Martin S. Small
Title: Senior Managing Director &
Chief Financial Officer
Date: February 25, 2025

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549
FORM 10-K

(Mark One)

x **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023
OR

o **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from _________ to _________

Commission File Number: 001-34756

Tesla, Inc.
(Exact name of registrant as specified in its charter)

Delaware	**91-2197729**
(State or other jurisdiction of incorporation or organization)	**(I.R.S. Employer Identification No.)**
1 Tesla Road Austin, Texas	**78725**
(Address of principal executive offices)	**(Zip Code)**

(512) 516-8177
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock	TSLA	The Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act:
None

Indicate by check mark whether the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes x No o

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or 15(d) of the Act. Yes o No x

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 ("Exchange Act") during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes x No o

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes x No o

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act:

Large accelerated filer	x	Accelerated filer	o
Non-accelerated filer	o	Smaller reporting company	o
Emerging growth company	o		

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. o

Indicate by check mark whether the Registrant has filed a report on and attestation to its management’s assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. x

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. o

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant’s executive officers during the relevant recovery period pursuant to §240.10D-1(b). o

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes o No x

The aggregate market value of voting stock held by non-affiliates of the registrant, as of June 30, 2023, the last day of the registrant’s most recently completed second fiscal quarter, was $722.52 billion (based on the closing price for shares of the registrant’s Common Stock as reported by the NASDAQ Global Select Market on June 30, 2023). Shares of Common Stock held by each executive officer and director have been excluded in that such persons may be deemed to be affiliates. This determination of affiliate status is not necessarily a conclusive determination for other purposes.

As of January 22, 2024, there were 3,184,790,415 shares of the registrant’s common stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant’s Proxy Statement for the 2024 Annual Meeting of Stockholders are incorporated herein by reference in Part III of this Annual Report on Form 10-K to the extent stated herein. Such proxy statement will be filed with the Securities and Exchange Commission within 120 days of the registrant’s fiscal year ended December 31, 2023.

TESLA, INC.

ANNUAL REPORT ON FORM 10-K FOR THE YEAR ENDED DECEMBER 31, 2023

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Forward-Looking Statements

The discussions in this Annual Report on Form 10-K contain forward-looking statements reflecting our current expectations that involve risks and uncertainties. These forward-looking statements include, but are not limited to, statements concerning supply chain constraints, our strategy, competition, future operations and production capacity, future financial position, future revenues, projected costs, profitability, expected cost reductions, capital adequacy, expectations regarding demand and acceptance for our technologies, growth opportunities and trends in the markets in which we operate, prospects and plans and objectives of management. The words "anticipates," "believes," "could," "estimates," "expects," "intends," "may," "plans," "projects," "will," "would" and similar expressions are intended to identify forward-looking statements, although not all forward-looking statements contain these identifying words. We may not actually achieve the plans, intentions or expectations disclosed in our forward-looking statements and you should not place undue reliance on our forward-looking statements. Actual results or events could differ materially from the plans, intentions and expectations disclosed in the forward-looking statements that we make. These forward-looking statements involve risks and uncertainties that could cause our actual results to differ materially from those in the forward-looking statements, including, without limitation, the risks set forth in Part I, Item 1A, "Risk Factors" of the Annual Report on Form 10-K for the fiscal year ended December 31, 2023 and that are otherwise described or updated from time to time in our other filings with the Securities and Exchange Commission (the "SEC"). The discussion of such risks is not an indication that any such risks have occurred at the time of this filing. We do not assume any obligation to update any forward-looking statements.

PART I

ITEM 1. BUSINESS

Overview

We design, develop, manufacture, sell and lease high-performance fully electric vehicles and energy generation and storage systems, and offer services related to our products. We generally sell our products directly to customers, and continue to grow our customer-facing infrastructure through a global network of vehicle showrooms and service centers, Mobile Service, body shops, Supercharger stations and Destination Chargers to accelerate the widespread adoption of our products. We emphasize performance, attractive styling and the safety of our users and workforce in the design and manufacture of our products and are continuing to develop full self-driving technology for improved safety. We also strive to lower the cost of ownership for our customers through continuous efforts to reduce manufacturing costs and by offering financial and other services tailored to our products.

Our mission is to accelerate the world's transition to sustainable energy. We believe that this mission, along with our engineering expertise, vertically integrated business model and focus on user experience differentiate us from other companies.

Segment Information

We operate as two reportable segments: (i) automotive and (ii) energy generation and storage.

The automotive segment includes the design, development, manufacturing, sales and leasing of high-performance fully electric vehicles as well as sales of automotive regulatory credits. Additionally, the automotive segment also includes services and other, which includes sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise. The energy generation and storage segment includes the design, manufacture, installation, sales and leasing of solar energy generation and energy storage products and related services and sales of solar energy systems incentives.

Our Products and Services

Automotive

We currently manufacture five different consumer vehicles – the Model 3, Y, S, X and Cybertruck. Model 3 is a four-door mid-size sedan that we designed for manufacturability with a base price for mass-market appeal. Model Y is a compact sport utility vehicle ("SUV") built on the Model 3 platform with seating for up to seven adults. Model S is a four-door full-size sedan and Model X is a mid-size SUV with seating for up to seven adults. Model S and Model X feature the highest performance characteristics and longest ranges that we offer in a sedan and SUV, respectively. In November 2023, we entered the consumer pickup truck market with first deliveries of the Cybertruck, a full-size electric pickup truck with a stainless steel exterior that has the utility and strength of a truck while featuring the speed of a sports car.

In 2022, we also began early production and deliveries of a commercial electric vehicle, the Tesla Semi. We have planned electric vehicles to address additional vehicle markets, and to continue leveraging developments in our proprietary Full Self-Driving ("FSD") Capability features, battery cell and other technologies.

Energy Generation and Storage

Energy Storage Products

Powerwall and Megapack are our lithium-ion battery energy storage products. Powerwall, which we sell directly to customers, as well as through channel partners, is designed to store energy at a home or small commercial facility. Megapack is an energy storage solution for commercial, industrial, utility and energy generation customers, multiple of which may be grouped together to form larger installations of gigawatt hours ("GWh") or greater capacity.

We also continue to develop software capabilities for remotely controlling and dispatching our energy storage systems across a wide range of markets and applications, including through our real-time energy control and optimization platforms.

Solar Energy Offerings

We sell retrofit solar energy systems to customers and channel partners and also make them available through power purchase agreement ("PPA") arrangements. We purchase most of the components for our retrofit solar energy systems from multiple sources to ensure competitive pricing and adequate supply. We also design and manufacture certain components for our solar energy products.

We sell our Solar Roof, which combines premium glass roof tiles with energy generation, directly to customers, as well as through channel customers. We continue to improve our installation capability and efficiency, including through collaboration with real estate developers and builders on new homes.

Technology

Automotive

Battery and Powertrain

Our core vehicle technology competencies include powertrain engineering and manufacturing and our ability to design vehicles that utilize the unique advantages of an electric powertrain. We have designed our proprietary powertrain systems to be adaptable, efficient, reliable and cost-effective while withstanding the rigors of an automotive environment. We offer dual motor powertrain vehicles, which use two electric motors to maximize traction and performance in an all-wheel drive configuration, as well as vehicle powertrain technology featuring three electric motors for further increased performance in certain versions of Model S and Model X, Cybertruck and the Tesla Semi.

We maintain extensive testing and R&D capabilities for battery cells, packs and systems, and have built an expansive body of knowledge on lithium-ion cell chemistry types and performance characteristics. In order to enable a greater supply of cells for our products with higher energy density at lower costs, we have developed a new proprietary lithium-ion battery cell and improved manufacturing processes.

Vehicle Control and Infotainment Software

The performance and safety systems of our vehicles and their battery packs utilize sophisticated control software. Control systems in our vehicles optimize performance, customize vehicle behavior, manage charging and control all infotainment functions. We develop almost all of this software, including most of the user interfaces, internally and update our vehicles' software regularly through over-the-air updates.

Self-Driving Development and Artificial Intelligence

We have expertise in developing technologies, systems and software to enable self-driving vehicles using primarily vision-based technologies. Our FSD Computer runs our neural networks in our vehicles, and we are also developing additional computer hardware to better enable the massive amounts of field data captured by our vehicles to continually train and improve these neural networks for real-world performance.

Currently, we offer in our vehicles certain advanced driver assist systems under our Autopilot and FSD Capability options. Although at present the driver is ultimately responsible for controlling the vehicle, our systems provide safety and convenience functionality that relieves drivers of the most tedious and potentially dangerous aspects of road travel much like the system that airplane pilots use, when conditions permit. As with other vehicle systems, we improve these functions in our vehicles over time through over-the-air updates.

We intend to establish in the future an autonomous Tesla ride-hailing network, which we expect would also allow us to access a new customer base even as modes of transportation evolve.

We are also applying our artificial intelligence learnings from self-driving technology to the field of robotics, such as through Optimus, a robotic humanoid in development, which is controlled by the same AI system.

Energy Generation and Storage

Energy Storage Products

We leverage many of the component-level technologies from our vehicles in our energy storage products. By taking a modular approach to the design of battery systems, we can optimize manufacturing capacity of our energy storage products. Additionally, our expertise in power electronics enables our battery systems to interconnect with electricity grids while providing fast-acting systems for power injection and absorption. We have also developed software to remotely control and dispatch our energy storage systems.

Solar Energy Systems

We have engineered Solar Roof over numerous iterations to combine aesthetic appeal and durability with power generation. The efficiency of our solar energy products is aided by our own solar inverter, which incorporates our power electronics technologies. We designed both products to integrate with Powerwall.

Design and Engineering

Automotive

We have established significant in-house capabilities in the design and test engineering of electric vehicles and their components and systems. Our team has significant experience in computer-aided design as well as durability, strength and crash test simulations, which reduces the product development time of new models. We have also achieved complex engineering feats in stamping, casting and thermal systems, and developed a method to integrate batteries directly with vehicle body structures without separate battery packs to optimize manufacturability, weight, range and cost characteristics.

We are also expanding our manufacturing operations globally while taking action to localize our vehicle designs and production for particular markets, including country-specific market demands and factory optimizations for local workforces. As we increase our capabilities, particularly in the areas of automation, die-making and line-building, we are also making strides in the simulations modeling these capabilities prior to construction.

Energy Generation and Storage

Our expertise in electrical, mechanical, civil and software engineering allows us to design, engineer, manufacture and install energy generating and storage products and components, including at the residential through utility scale. For example, the modular design of our Megapack utility-scale battery line is intended to significantly reduce the amount of assembly required in the field. We also customize solutions including our energy storage products, solar energy systems and/or Solar Roof for customers to meet their specific needs.

Sales and Marketing

Historically, we have been able to achieve sales without traditional advertising and at relatively low marketing costs. We continue to monitor our public narrative and brand, and tailor our marketing efforts accordingly, including through investments in customer education and advertising as necessary.

Automotive

Direct Sales

Our vehicle sales channels currently include our website and an international network of company-owned stores. In some jurisdictions, we also have galleries to educate and inform customers about our products, but such locations do not transact in the sale of vehicles. We believe this infrastructure enables us to better control costs of inventory, manage warranty service and pricing, educate consumers about electric vehicles, make our vehicles more affordable, maintain and strengthen the Tesla brand and obtain rapid customer feedback.

We reevaluate our sales strategy both globally and at a location-by-location level from time to time to optimize our sales channels. However, sales of vehicles in the automobile industry tend to be cyclical in many markets, which may expose us to volatility from time to time.

Used Vehicle Sales

Our used vehicle business supports new vehicle sales by integrating the trade-in of a customer’s existing Tesla or non-Tesla vehicle with the sale of a new or used Tesla vehicle. The Tesla and non-Tesla vehicles we acquire as trade-ins are subsequently remarketed, either directly by us or through third parties. We also remarket used Tesla vehicles acquired from other sources including lease returns.

Public Charging

We have a growing global network of Tesla Superchargers, which are our industrial-grade, high-speed vehicle chargers. Where possible, we co-locate Superchargers with our solar and energy storage systems to reduce costs and promote renewable power. Supercharger stations are typically placed along well-traveled routes and in and around dense city centers to allow vehicle owners the ability to enjoy quick, reliable charging along an extensive network with convenient stops. Use of the Supercharger network either requires payment of a fee or is free under certain sales programs. In November 2021, we began to offer Supercharger access to non-Tesla vehicles in certain locations in support of our mission to accelerate the world’s transition to sustainable energy, and in November 2022, we opened up our previously proprietary charging connector as the North American Charging Standard (NACS). This enables all electric vehicles and charging stations to interoperate — which makes charging easier and more efficient for everyone and advances our mission to accelerate the world’s transition to sustainable energy. Following this, a number of major automotive companies announced their adoption of NACS, with their access to the Supercharger network beginning in phases in 2024 and their production of NACS vehicles beginning no later than 2025. We also engaged SAE International to govern NACS as an industry standard, now named J3400. We continue to monitor and increase our network of Tesla Superchargers in anticipation of future demand.

We also work with a wide variety of hospitality, retail and public destinations, as well as businesses with commuting employees, to offer additional charging options for our customers, as well as single-family homeowners and multi-family residential entities, to deploy home charging solutions.

In-App Upgrades

As our vehicles are capable of being updated remotely over-the-air, our customers may purchase additional paid options and features through the Tesla app or through the in-vehicle user interface. We expect that this functionality will also allow us to offer certain options and features on a subscription basis in the future.

Energy Generation and Storage

We market and sell our solar and energy storage products to residential, commercial and industrial customers and utilities through a variety of channels, including through our website, stores and galleries, as well as through our network of channel partners, and in the case of some commercial customers, through PPA transactions. We emphasize simplicity, standardization and accessibility to make it easy and cost-effective for customers to adopt clean energy, while reducing our customer acquisition costs.

Service and Warranty

Automotive

Service

We provide service for our electric vehicles at our company-owned service locations and through Tesla Mobile Service technicians who perform work remotely at customers’ homes or other locations. Servicing the vehicles ourselves allows us to identify problems and implement solutions and improvements faster than traditional automobile manufacturers and their dealer networks. The connectivity of our vehicles also allows us to diagnose and remedy many problems remotely and proactively.

Vehicle Limited Warranties and Extended Service Plans

We provide a manufacturer’s limited warranty on all new and used Tesla vehicles we sell directly to consumers, which may include limited warranties on certain components, specific types of damage or battery capacity retention. We also currently offer optional extended service plans that provide coverage beyond the new vehicle limited warranties for certain models in specified regions.

Energy Generation and Storage

We provide service and repairs to our energy product customers, including under warranty where applicable. We generally provide manufacturer's limited warranties with our energy storage products and offer certain extended limited warranties that are available at the time of purchase of the system. If we install a system, we also provide certain limited warranties on our installation workmanship.

For retrofit solar energy systems, we provide separate limited warranties for workmanship and against roof leaks, and for Solar Roof, we also provide limited warranties for defects and weatherization. For components not manufactured by us, we generally pass-through the applicable manufacturers' warranties.

As part of our solar energy system and energy storage contracts, we may provide the customer with performance guarantees that commit that the underlying system will meet or exceed the minimum energy generation or performance requirements specified in the contract.

Financial Services

Automotive

Purchase Financing and Leases

We offer leasing and/or loan financing arrangements for our vehicles in certain jurisdictions in North America, Europe and Asia ourselves and through various financial institutions. Under certain of such programs, we have provided resale value guarantees or buyback guarantees that may obligate us to cover a resale loss up to a certain limit or repurchase the subject vehicles at pre-determined values.

Insurance

In 2021, we launched our insurance product using real-time driving behavior in select states, which offers rates that are often better than other alternatives and promotes safer driving. Our insurance products are currently available in 12 states and we plan to expand the markets in which we offer insurance products, as part of our ongoing effort to decrease the total cost of ownership for our customers.

Energy Generation and Storage

We offer certain financing options to our solar customers, which enable the customer to purchase and own a solar energy system, Solar Roof or integrated solar and Powerwall system. Our solar PPAs, offered primarily to commercial customers, charge a fee per kilowatt-hour based on the amount of electricity produced by our solar energy systems.

Manufacturing

We currently have manufacturing facilities in the U.S. in Northern California, in Buffalo, New York, Gigafactory New York; in Austin, Texas, Gigafactory Texas and near Reno, Nevada, Gigafactory Nevada. At these facilities, we manufacture and assemble, among other things, vehicles, certain vehicle parts and components, such as our battery packs and battery cells, energy storage components and solar products and components.

Internationally, we also have manufacturing facilities in China (Gigafactory Shanghai) and Germany (Gigafactory Berlin-Brandenburg), which allows us to increase the affordability of our vehicles for customers in local markets by reducing transportation and manufacturing costs and eliminating the impact of unfavorable tariffs. In March 2023, we announced the location of our next Gigafactory in Monterrey, Mexico. Generally, we continue to expand production capacity at our existing facilities. We also intend to further increase cost-competitiveness in our significant markets by strategically adding local manufacturing.

Supply Chain

Our products use thousands of parts that are sourced from hundreds of suppliers across the world. We have developed close relationships with vendors of key parts such as battery cells, electronics and complex vehicle assemblies. Certain components purchased from these suppliers are shared or are similar across many product lines, allowing us to take advantage of pricing efficiencies from economies of scale.

As is the case for some automotive companies, some of our procured components and systems are sourced from single suppliers. Where multiple sources are available for certain key components, we work to qualify multiple suppliers for them where it is sensible to do so in order to minimize potential production risks due to disruptions in their supply. We also mitigate risk by maintaining safety stock for key parts and assemblies and die banks for components with lengthy procurement lead times.

Our products use various raw materials including aluminum, steel, cobalt, lithium, nickel and copper. Pricing for these materials is governed by market conditions and may fluctuate due to various factors outside of our control, such as supply and demand and market speculation. We strive to execute long-term supply contracts for such materials at competitive pricing when feasible, and we currently believe that we have adequate access to raw materials supplies to meet the needs of our operations.

Governmental Programs, Incentives and Regulations

Globally, the ownership of our products by our customers is impacted by various government credits, incentives, and policies. Our business and products are also subject to numerous governmental regulations that vary among jurisdictions.

The operation of our business is also impacted by various government programs, incentives, and other arrangements. See Note 2, *Summary of Significant Accounting Policies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Programs and Incentives

Inflation Reduction Act

On August 16, 2022, the Inflation Reduction Act of 2022 (“IRA”) was enacted into law and is effective for taxable years beginning after December 31, 2022, and remains subject to future guidance releases. The IRA includes multiple incentives to promote clean energy, electric vehicles, battery and energy storage manufacture or purchase, including through providing tax credits to consumers. For example, qualifying Tesla customers may receive up to $7,500 in federal tax credits for the purchase of qualified electric vehicles in the U.S. through 2032.

Automotive Regulatory Credits

We earn tradable credits in the operation of our business under various regulations related to zero-emission vehicles (“ZEVs”), greenhouse gas, fuel economy and clean fuel. We sell these credits to other regulated entities who can use the credits to comply with emission standards and other regulatory requirements. Sales of these credits are recognized within automotive regulatory credits revenue in our consolidated statements of operations included elsewhere in this Annual Report on Form 10-K.

Energy Storage System Incentives and Policies

While the regulatory regime for energy storage projects is still under development, there are various policies, incentives and financial mechanisms at the federal, state and local levels that support the adoption of energy storage.

For example, energy storage systems that are charged using solar energy may be eligible for the solar energy-related U.S. federal tax credits described below. The Federal Energy Regulatory Commission (“FERC”) has also taken steps to enable the participation of energy storage in wholesale energy markets. In addition, California and a number of other states have adopted procurement targets for energy storage, and behind-the-meter energy storage systems qualify for funding under the California Self Generation Incentive Program. Our customers primarily benefit directly under these programs. In certain instances our customers may transfer such credits to us as contract consideration. In such transactions, they are included as a component of energy generation and storage revenues in our consolidated statements of operations included elsewhere in this Annual Report on Form 10-K.

Pursuant to the IRA, under Sections 48, 48E and 25D of the Internal Revenue Code (”IRC”), standalone energy storage technology is eligible for a tax credit between 6% and 50% of qualified expenditures, regardless of the source of energy, which may be claimed by our customers for storage systems they purchase or by us for arrangements where we own the systems. These tax credits are primarily for the benefit of our customers and are currently scheduled to phase-out starting in 2032 or later.

Solar Energy System Incentives and Policies

U.S. federal, state and local governments have established various policies, incentives and financial mechanisms to reduce the cost of solar energy and to accelerate the adoption of solar energy. These incentives include tax credits, cash grants, tax abatements and rebates.

In particular, pursuant to the IRA, Sections 48, 48E and 25D of the IRC provides a tax credit between 6% and 70% of qualified commercial or residential expenditures for solar energy systems, which may be claimed by our customers for systems they purchase, or by us for arrangements where we own the systems for properties that meet statutory requirements. These tax credits are primarily for the direct benefit of our customers and are currently scheduled to phase-out starting in 2032 or later.

Regulations

Vehicle Safety and Testing

In the U.S., our vehicles are subject to regulation by the National Highway Traffic Safety Administration (“NHTSA”), including all applicable Federal Motor Vehicle Safety Standards (“FMVSS”) and the NHTSA bumper standard. Numerous FMVSS apply to our vehicles, such as crash-worthiness and occupant protection requirements. Our current vehicles fully comply and we expect that our vehicles in the future will fully comply with all applicable FMVSS with limited or no exemptions, however, FMVSS are subject to change from time to time. As a manufacturer, we must self-certify that our vehicles meet all applicable FMVSS and the NHTSA bumper standard, or otherwise are exempt, before the vehicles may be imported or sold in the U.S.

We are also required to comply with other federal laws administered by NHTSA, including the Corporate Average Fuel Economy standards, Theft Prevention Act requirements, labeling requirements and other information provided to customers in writing, Early Warning Reporting requirements regarding warranty claims, field reports, death and injury reports and foreign recalls, a Standing General Order requiring reports regarding crashes involving vehicles equipped with advanced driver assistance systems, and additional requirements for cooperating with compliance and safety investigations and recall reporting. The U.S. Automobile Information and Disclosure Act also requires manufacturers of motor vehicles to disclose certain information regarding the manufacturer’s suggested retail price, optional equipment and pricing. In addition, federal law requires inclusion of fuel economy ratings, as determined by the U.S. Department of Transportation and the Environmental Protection Agency (the “EPA”), and New Car Assessment Program ratings as determined by NHTSA, if available.

Our vehicles sold outside of the U.S. are subject to similar foreign compliance, safety, environmental and other regulations. Many of those regulations are different from those applicable in the U.S. and may require redesign and/or retesting. Some of those regulations impact or prevent the rollout of new vehicle features.

Self-Driving Vehicles

Generally, laws pertaining to self-driving vehicles are evolving globally, and in some cases may create restrictions on features or vehicle designs that we develop. While there are currently no federal U.S. regulations pertaining specifically to self-driving vehicles or self-driving equipment, NHTSA has published recommended guidelines on self-driving vehicles, apart from the FMVSS and manufacturer reporting obligations, and retains the authority to investigate and/or take action on the safety or compliance of any vehicle, equipment or features operating on public roads. Certain U.S. states also have legal restrictions on the operation, registration or licensure of self-driving vehicles, and many other states are considering them. This regulatory patchwork increases the legal complexity with respect to self-driving vehicles in the U.S.

In markets that follow the regulations of the United Nations Economic Commission for Europe (“ECE markets”), some requirements restrict the design of advanced driver-assistance or self-driving features, which can compromise or prevent their use entirely. Other applicable laws, both current and proposed, may hinder the path and timeline to introducing self-driving vehicles for sale and use in the markets where they apply.

Other key markets, including China, continue to consider self-driving regulation. Any implemented regulations may differ materially from the U.S. and ECE markets, which may further increase the legal complexity of self-driving vehicles and limit or prevent certain features.

Automobile Manufacturer and Dealer Regulation

In the U.S., state laws regulate the manufacture, distribution, sale and service of automobiles, and generally require motor vehicle manufacturers and dealers to be licensed in order to sell vehicles directly to residents. Certain states have asserted that the laws in such states do not permit automobile manufacturers to be licensed as dealers or to act in the capacity of a dealer, or that they otherwise restrict a manufacturer's ability to deliver or perform warranty repairs on vehicles. To sell vehicles to residents of states where we are not licensed as a dealer, we generally conduct the sale out of the state. In certain such states, we have opened "galleries" that serve an educational purpose and where sales may not occur.

Some automobile dealer trade associations have both challenged the legality of our operations in court and used administrative and legislative processes to attempt to prohibit or limit our ability to operate existing stores or expand to new locations. Certain dealer associations have also actively lobbied state licensing agencies and legislators to interpret existing laws or enact new laws in ways not favorable to our ownership and operation of our own retail and service locations. We expect such challenges to continue, and we intend to actively fight any such efforts.

Battery Safety and Testing

Our battery packs are subject to various U.S. and international regulations that govern transport of "dangerous goods," defined to include lithium-ion batteries, which may present a risk in transportation. We conduct testing to demonstrate our compliance with such regulations.

We use lithium-ion cells in our high voltage battery packs in our vehicles and energy storage products. The use, storage and disposal of our battery packs are regulated under existing laws and are the subject of ongoing regulatory changes that may add additional requirements in the future. We have agreements with third party battery recycling companies to recycle our battery packs, and we are also piloting our own recycling technology.

Solar Energy—General

We are subject to certain state and federal regulations applicable to solar and battery storage providers and sellers of electricity. To operate our systems, we enter into standard interconnection agreements with applicable utilities. Sales of electricity and non-sale equipment leases by third parties, such as our leases and PPAs, have faced regulatory challenges in some states and jurisdictions.

Solar Energy—Net Metering

Most states in the U.S. make net energy metering, or net metering, available to solar customers. Net metering typically allows solar customers to interconnect their solar energy systems to the utility grid and offset their utility electricity purchases by receiving a bill credit for excess energy generated by their solar energy system that is exported to the grid. In certain jurisdictions, regulators or utilities have reduced or eliminated the benefit available under net metering or have proposed to do so.

Competition

Automotive

The worldwide automotive market is highly competitive and we expect it will become even more competitive in the future as a significant and growing number of established and new automobile manufacturers, as well as other companies, have entered, or are reported to have plans to enter the electric vehicle market.

We believe that our vehicles compete in the market based on both their traditional segment classification as well as their propulsion technology. For example, Cybertruck competes with other pickup trucks, Model S and Model X compete primarily with premium sedans and premium SUVs and Model 3 and Model Y compete with small to medium-sized sedans and compact SUVs, which are extremely competitive markets. Competing products typically include internal combustion vehicles from more established automobile manufacturers; however, many established and new automobile manufacturers have entered or have announced plans to enter the market for electric and other alternative fuel vehicles. Overall, we believe these announcements and vehicle introductions, including the introduction of electric vehicles into rental car company fleets, promote the development of the electric vehicle market by highlighting the attractiveness of electric vehicles relative to the internal combustion vehicle. Many major automobile manufacturers have electric vehicles available today in major markets including the U.S., China and Europe, and other current and prospective automobile manufacturers are also developing electric vehicles. In addition, several manufacturers offer hybrid vehicles, including plug-in versions.

We believe that there is also increasing competition for our vehicle offerings as a platform for delivering self-driving technologies, charging solutions and other features and services, and we expect to compete in this developing market through continued progress on our Autopilot, FSD and neural network capabilities, Supercharger network and our infotainment offerings.

Energy Generation and Storage

Energy Storage Systems

The market for energy storage products is also highly competitive, and both established and emerging companies have introduced products that are similar to our product portfolio or that are alternatives to the elements of our systems. We compete with these companies based on price, energy density and efficiency. We believe that the specifications and features of our products, our strong brand and the modular, scalable nature of our energy storage products give us a competitive advantage in our markets.

Solar Energy Systems

The primary competitors to our solar energy business are the traditional local utility companies that supply energy to our potential customers. We compete with these traditional utility companies primarily based on price and the ease by which customers can switch to electricity generated by our solar energy systems. We also compete with solar energy companies that provide products and services similar to ours. Many solar energy companies only install solar energy systems, while others only provide financing for these installations. We believe we have a significant expansion opportunity with our offerings and that the regulatory environment is increasingly conducive to the adoption of renewable energy systems.

Intellectual Property

We place a strong emphasis on our innovative approach and proprietary designs which bring intrinsic value and uniqueness to our product portfolio. As part of our business, we seek to protect the underlying intellectual property rights of these innovations and designs such as with respect to patents, trademarks, copyrights, trade secrets, confidential information and other measures, including through employee and third-party nondisclosure agreements and other contractual arrangements. For example, we place a high priority on obtaining patents to provide the broadest and strongest possible protection to enable our freedom to operate our innovations and designs across all of our products and technologies as well as to protect and defend our product portfolio. We have also adopted a patent policy in which we irrevocably pledged that we will not initiate a lawsuit against any party for infringing our patents through activity relating to electric vehicles or related equipment for so long as such party is acting in good faith. We made this pledge in order to encourage the advancement of a common, rapidly-evolving platform for electric vehicles, thereby benefiting ourselves, other companies making electric vehicles and the world.

Environmental, Social and Governance (ESG) and Human Capital Resources

ESG

The very purpose of Tesla's existence is to accelerate the world's transition to sustainable energy. We believe the world cannot reduce carbon emissions without addressing both energy generation and consumption, and we are designing and manufacturing a complete energy and transportation ecosystem to achieve this goal. As we expand, we are building each new factory to be more efficient and sustainably designed than the previous one, including with respect to per-unit waste reduction and resource consumption, including water and energy usage. We are focused on further enhancing sustainability of operations outside of our direct control, including reducing the carbon footprint of our supply chain.

We are committed to sourcing only responsibly produced materials, and our suppliers are required to provide evidence of management systems that ensure social, environmental and sustainability best practices in their own operations, as well as to demonstrate a commitment to responsible sourcing into their supply chains. We have a zero-tolerance policy when it comes to child or forced labor and human trafficking by our suppliers and we look to the Organization for Economic Co-operation and Development Due Diligence Guidelines to inform our process and use feedback from our internal and external stakeholders to find ways to continually improve. We are also driving safety in our own factories by focusing on worker engagement. Our incidents per vehicle continue to drop even as our production volumes increase. We also strive to be an employer of choice by offering compelling, impactful jobs with best in-industry benefits.

We believe that sound corporate governance is critical to helping us achieve our goals, including with respect to ESG. We continue to evolve a governance framework that exercises appropriate oversight of responsibilities at all levels throughout the company and manages its affairs consistent with high principles of business ethics. Our ESG Sustainability Council is made up of leaders from across our company, and regularly presents to our Board of Directors, which oversees our ESG impacts, initiatives and priorities.

Human Capital Resources

A competitive edge for Tesla is its ability to attract and retain high quality employees. During the past year, Tesla made substantial investments in its workforce, further strengthening its standing as one of the most desirable and innovative companies to work for. As of December 31, 2023, our employee headcount worldwide was 140,473.

We have created an environment that fosters growth opportunities, and as of this report, nearly two-thirds (65%) of our managers were promoted from an internal, non-manager position, and 43% of our management employees have been with Tesla for more than five years. Tesla's growth of 35% over the past two years has offered internal career development to our employees as well as the ability to make a meaningful contribution to a sustainable future.

We are able to retain our employees, in part, not only because employees can enjoy ownership in Tesla through stock (of which 89% have been given the opportunity to), but because we also provide them with excellent health benefits such as free counseling, paid parental leave, paid time off and zero-premium medical plan options that are made available on the first day of employment.

We recognize the positive impact that leaders can have on their teams and offer fundamental skills training and continuous development to all leaders through various programs globally.

We don't stop there. Tesla has several other programs strategically designed to increase paths for greater career opportunity such as:

- **Technician Trainee (Service)** – The Tesla Technician Trainee Program provides on-the-job automotive maintenance training at Tesla, resulting in an industry certification. Targeted at individuals with limited experience, whether in industry or vocational schools, the program prepares trainees for employment as technicians. In 2023, we hired over 1,900 Technician Trainees across the U.S., Germany and China.

- **START (Manufacturing and Service)** – Tesla START is an intensive training program that complements the Technician Trainee program and equips individuals with the skills needed for a successful technician role at Tesla. We have partnered with colleges and technical academies to launch Tesla START in the U.S., United Kingdom and Germany. In 2023, we hired over 350 trainees for manufacturing and service roles through this program, providing an opportunity to transition into full-time employment.

- **Internships** – Annually, Tesla hires over 6,000 university and college students from around the world. We recruit from diverse student organizations and campuses, seeking top talent passionate about our mission. Our interns engage in meaningful work from day one, and we often offer them full-time positions post-internship.

- **Military Fellowship and Transition Programs –** The Military Fellowship and Transition Programs are designed to offer exiting military service members in the U.S. and Europe with career guidance on transitioning into the civil workforce. We partner with the career transition services of European Defence Ministries across five countries, as well as the U.S. Chamber of Commerce's Hire our Heroes. These programs aim to convert high-performing individuals to full-time roles and create a veteran talent pipeline.

- **Apprenticeships** – Tesla Apprenticeships are offered globally, providing academic and on-the-job training to prepare specialists in skilled trades. Apprentices will complete between one to four years of on-the-job training. Apprentice programs have seen skilled trade hires across the U.S., Australia, Hong Kong, Korea and Germany.

- **Manufacturing Development Program** – Tesla's manufacturing pathway program is designed to provide graduating high school seniors with the financial resources, coursework and experience they need to start a successful manufacturing career at Tesla. We hired 373 graduates through this program in 2023, and our goal in 2024 is grow this program to over 600 students annually across our Fremont Factory, Gigafactory Nevada, Gigafactory Texas and Gigafactory New York.

- **Engineering Development Program** – Launched in January 2024, this program targets recent college and university graduates for specialized engineering fields. In collaboration with Austin Community College, the program educates early-career engineers in controls engineering, enhancing their knowledge of high-demand technologies for U.S. manufacturing.

We will continue to expand the opportunities for our employees to add skills and develop professionally with a new Employee Educational Assistance Program launching in the U.S. in the spring of 2024 to help employees pursue select certificates or degrees. With virtual, self-paced education options available, employees can pursue a new path or expand their knowledge while continuing to grow their career.

At Tesla, our employees show up passionate about making a difference in the world and for each other. We remain unwavering in our demand that our factories, offices, stores and service centers are places where our employees feel respected and appreciated. Our policies are designed to promote fairness and respect for everyone. We hire, evaluate and promote employees based on their skills and performance. Everyone is expected to be trustworthy, demonstrate excellence in their performance and collaborate with others. With this in mind, we will not tolerate certain behaviors. These include harassment, retaliation, violence, intimidation and discrimination of any kind on the basis of race, color, religion, national origin, gender, sexual orientation, gender identity, gender expression, age, disability or veteran status.

Anti-harassment training is conducted on day one of new hire orientation for all employees and reoccurring for leaders. In addition, we run various leadership development programs throughout the year aimed at enhancing leaders' skills, and in particular, helping them to understand how to appropriately respond to and address employee concerns.

Employees are encouraged to speak up both in regard to misconduct and safety concerns and can do so by contacting the integrity line, submitting concerns through our Take Charge process, or notifying their Human Resource Partner or any member of management. Concerns are reviewed in accordance with established protocols by investigators with expertise, who also review for trends and outcomes for remediation and appropriate controls. Responding to questions timely is key so Human Resource Partners for each functional area are visible throughout facilities and are actively involved in driving culture and engagement alongside business leaders.

Available Information

We file or furnish periodic reports and amendments thereto, including our Annual Reports on Form 10-K, our Quarterly Reports on Form 10-Q and Current Reports on Form 8-K, proxy statements and other information with the SEC. In addition, the SEC maintains a website (www.sec.gov) that contains reports, proxy and information statements, and other information regarding issuers that file electronically. Our website is located at www.tesla.com, and our reports, amendments thereto, proxy statements and other information are also made available, free of charge, on our investor relations website at ir.tesla.com as soon as reasonably practicable after we electronically file or furnish such information with the SEC. The information posted on our website is not incorporated by reference into this Annual Report on Form 10-K.

ITEM 1A. RISK FACTORS

You should carefully consider the risks described below together with the other information set forth in this report, which could materially affect our business, financial condition and future results. The risks described below are not the only risks facing our company. Risks and uncertainties not currently known to us or that we currently deem to be immaterial also may materially adversely affect our business, financial condition and operating results.

Risks Related to Our Ability to Grow Our Business

We may experience delays in launching and ramping the production of our products and features, or we may be unable to control our manufacturing costs.

We have previously experienced and may in the future experience launch and production ramp delays for new products and features. For example, we encountered unanticipated supplier issues that led to delays during the initial ramp of our first Model X and experienced challenges with a supplier and with ramping full automation for certain of our initial Model 3 manufacturing processes. In addition, we may introduce in the future new or unique manufacturing processes and design features for our products. As we expand our vehicle offerings and global footprint, there is no guarantee that we will be able to successfully and timely introduce and scale such processes or features.

In particular, our future business depends in large part on increasing the production of mass-market vehicles. In order to be successful, we will need to implement, maintain and ramp efficient and cost-effective manufacturing capabilities, processes and supply chains and achieve the design tolerances, high quality and output rates we have planned at our manufacturing facilities in California, Nevada, Texas, China, Germany and any future sites such as Mexico. We will also need to hire, train and compensate skilled employees to operate these facilities. Bottlenecks and other unexpected challenges such as those we experienced in the past may arise during our production ramps, and we must address them promptly while continuing to improve manufacturing processes and reducing costs. If we are not successful in achieving these goals, we could face delays in establishing and/or sustaining our product ramps or be unable to meet our related cost and profitability targets.

We have experienced, and may also experience similar future delays in launching and/or ramping production of our energy storage products and Solar Roof; new product versions or variants; new vehicles; and future features and services based on artificial intelligence. Likewise, we may encounter delays with the design, construction and regulatory or other approvals necessary to build and bring online future manufacturing facilities and products.

Any delay or other complication in ramping the production of our current products or the development, manufacture, launch and production ramp of our future products, features and services, or in doing so cost-effectively and with high quality, may harm our brand, business, prospects, financial condition and operating results.

Our suppliers may fail to deliver components according to schedules, prices, quality and volumes that are acceptable to us, or we may be unable to manage these components effectively.

Our products contain thousands of parts purchased globally from hundreds of suppliers, including single-source direct suppliers, which exposes us to multiple potential sources of component shortages. Unexpected changes in business conditions, materials pricing, including inflation of raw material costs, labor issues, wars, trade policies, natural disasters, health epidemics such as the global COVID-19 pandemic, trade and shipping disruptions, port congestions, cyberattacks and other factors beyond our or our suppliers' control could also affect these suppliers' ability to deliver components to us or to remain solvent and operational. For example, a global shortage of semiconductors beginning in early 2021 has caused challenges in the manufacturing industry and impacted our supply chain and production. Additionally, if our suppliers do not accurately forecast and effectively allocate production or if they are not willing to allocate sufficient production to us, or face other challenges such as insolvency, it may reduce our access to components and require us to search for new suppliers. The unavailability of any component or supplier could result in production delays, idle manufacturing facilities, product design changes and loss of access to important technology and tools for producing and supporting our products, as well as impact our capacity expansion and our ability to fulfill our obligations under customer contracts. Moreover, significant increases in our production or product design changes by us have required and may in the future require us to procure additional components in a short amount of time. We have faced in the past, and may face suppliers who are unwilling or unable to sustainably meet our timelines or our cost, quality and volume needs, which may increase our costs or require us to replace them with other sources. Finally, as we construct new manufacturing facilities and add production lines to existing facilities, we may experience issues in correspondingly increasing the level of localized procurement at those facilities. While we believe that we will be able to secure additional or alternate sources or develop our own replacements for most of our components, there is no assurance that we will be able to do so quickly or at all. Additionally, we may be unsuccessful in our continuous efforts to negotiate with existing suppliers to obtain cost reductions and avoid unfavorable changes to terms, source less expensive suppliers for certain parts and redesign certain parts to make them less expensive to produce, especially in the case of increases in materials pricing. Any of these occurrences may harm our business, prospects, financial condition and operating results.

As the scale of our vehicle production increases, we will also need to accurately forecast, purchase, warehouse and transport components at high volumes to our manufacturing facilities and servicing locations internationally. If we are unable to accurately match the timing and quantities of component purchases to our actual needs or successfully implement automation, inventory management and other systems to accommodate the increased complexity in our supply chain and parts management, we may incur unexpected production disruption, storage, transportation and write-off costs, which may harm our business and operating results.

We may be unable to meet our projected construction timelines, costs and production ramps at new factories, or we may experience difficulties in generating and maintaining demand for products manufactured there.

Our ability to increase production of our vehicles on a sustained basis, make them affordable globally by accessing local supply chains and workforces and streamline delivery logistics is dependent on the construction and ramp of our current and future factories. The construction of and commencement and ramp of production at these factories are subject to a number of uncertainties inherent in all new manufacturing operations, including ongoing compliance with regulatory requirements, procurement and maintenance of construction, environmental and operational licenses and approvals for additional expansion, supply chain constraints, hiring, training and retention of qualified employees and the pace of bringing production equipment and processes online with the capability to manufacture high-quality units at scale. Moreover, we will have to establish and ramp production of our proprietary battery cells and packs at our new factories, and we additionally intend to incorporate sequential design and manufacturing changes into vehicles manufactured at each new factory. If we experience any issues or delays in meeting our projected timelines, costs, capital efficiency and production capacity for our new factories, expanding and managing teams to implement iterative design and production changes there, maintaining and complying with the terms of any debt financing that we obtain to fund them or generating and maintaining demand for the vehicles we manufacture there, our business, prospects, operating results and financial condition may be harmed.

We may be unable to grow our global product sales, delivery and installation capabilities and our servicing and vehicle charging networks, or we may be unable to accurately project and effectively manage our growth.

Our success will depend on our ability to continue to expand our sales capabilities. We are targeting a global mass demographic with a broad range of potential customers, in which we have relatively limited experience projecting demand and pricing our products. We currently produce numerous international variants at a limited number of factories, and if our specific demand expectations for these variants prove inaccurate, we may not be able to timely generate deliveries matched to the vehicles that we produce in the same timeframe or that are commensurate with the size of our operations in a given region. Likewise, as we develop and grow our energy products and services worldwide, our success will depend on our ability to correctly forecast demand in various markets.

Because we do not have independent dealer networks, we are responsible for delivering all of our vehicles to our customers. As our production volumes continue to grow, we have faced in the past, and may face challenges with deliveries at increasing volumes, particularly in international markets requiring significant transit times. We have also deployed a number of delivery models, such as deliveries to customers' homes and workplaces and touchless deliveries, but there is no guarantee that such models will be scalable or be accepted globally. Likewise, as we ramp our energy products, we are working to substantially increase our production and installation capabilities. If we experience production delays or inaccurately forecast demand, our business, financial condition and operating results may be harmed.

Moreover, because of our unique expertise with our vehicles, we recommend that our vehicles be serviced by us or by certain authorized professionals. If we experience delays in adding servicing capacity or servicing our vehicles efficiently, or experience unforeseen issues with the reliability of our vehicles, particularly higher-volume additions to our fleet such as Model 3 and Model Y, it could overburden our servicing capabilities and parts inventory. Similarly, the increasing number of Tesla vehicles also requires us to continue to rapidly increase the number of our Supercharger stations and connectors throughout the world.

There is no assurance that we will be able to ramp our business to meet our sales, delivery, installation, servicing and vehicle charging targets globally, that our projections on which such targets are based will prove accurate or that the pace of growth or coverage of our customer infrastructure network will meet customer expectations. These plans require significant cash investments and management resources and there is no guarantee that they will generate additional sales or installations of our products, or that we will be able to avoid cost overruns or be able to hire additional personnel to support them. As we expand, we will also need to ensure our compliance with regulatory requirements in various jurisdictions applicable to the sale, installation and servicing of our products, the sale or dispatch of electricity related to our energy products and the operation of Superchargers. If we fail to manage our growth effectively, it may harm our brand, business, prospects, financial condition and operating results.

We will need to maintain and significantly grow our access to battery cells, including through the development and manufacture of our own cells, and control our related costs.

We are dependent on the continued supply of lithium-ion battery cells for our vehicles and energy storage products, and we will require substantially more cells to grow our business according to our plans. Currently, we rely on suppliers such as Panasonic and Contemporary Amperex Technology Co. Limited (CATL) for these cells. We have to date fully qualified only a very limited number of such suppliers and have limited flexibility in changing suppliers. Any disruption in the supply of battery cells from our suppliers could limit production of our vehicles and energy storage products. In the long term, we intend to supplement cells from our suppliers with cells manufactured by us, which we believe will be more efficient, manufacturable at greater volumes and more cost-effective than currently available cells. However, our efforts to develop and manufacture such battery cells have required, and may continue to require, significant investments, and there can be no assurance that we will be able to achieve these targets in the timeframes that we have planned or at all. If we are unable to do so, we may have to curtail our planned vehicle and energy storage product production or procure additional cells from suppliers at potentially greater costs, either of which may harm our business and operating results.

In addition, the cost and mass production of battery cells, whether manufactured by our suppliers or by us, depends in part upon the prices and availability of raw materials such as lithium, nickel, cobalt and/or other metals. The prices for these materials fluctuate and their available supply may be unstable, depending on market conditions and global demand for these materials. For example, as a result of increased global production of electric vehicles and energy storage products, suppliers of these raw materials may be unable to meet our volume needs. Additionally, our suppliers may not be willing or able to reliably meet our timelines or our cost and quality needs, which may require us to replace them with other sources. Any reduced availability of these materials may impact our access to cells and our growth, and any increases in their prices may reduce our profitability if we cannot recoup such costs through increased prices. Moreover, our inability to meet demand and any product price increases may harm our brand, growth, prospects and operating results.

Our future growth and success are dependent upon consumers' demand for electric vehicles and specifically our vehicles in an automotive industry that is generally competitive, cyclical and volatile.

Though we continue to see increased interest and adoption of electric vehicles, if the market for electric vehicles in general and Tesla vehicles in particular does not develop as we expect, develops more slowly than we expect, or if demand for our vehicles decreases in our markets or our vehicles compete with each other, our business, prospects, financial condition and operating results may be harmed.

In addition, electric vehicles still constitute a small percentage of overall vehicle sales. As a result, the market for our vehicles could be negatively affected by numerous factors, such as:

- perceptions about electric vehicle features, quality, safety, performance and cost;
- perceptions about the limited range over which electric vehicles may be driven on a single battery charge, and access to charging facilities;
- competition, including from other types of alternative fuel vehicles, plug-in hybrid electric vehicles and high fuel-economy internal combustion engine vehicles;
- volatility in the cost of oil, gasoline and energy;
- government regulations and economic incentives and conditions; and
- concerns about our future viability.

The target demographics for our vehicles are highly competitive. Sales of vehicles in the automotive industry tend to be cyclical in many markets, which may expose us to further volatility. We also cannot predict the duration or direction of current global trends or their sustained impact on consumer demand. Ultimately, we continue to monitor macroeconomic conditions to remain flexible and to optimize and evolve our business as appropriate, and attempt to accurately project demand and infrastructure requirements globally and deploy our production, workforce and other resources accordingly. Rising interest rates may lead to consumers to increasingly pull back spending, including on our products, which may harm our demand, business and operating results. If we experience unfavorable global market conditions, or if we cannot or do not maintain operations at a scope that is commensurate with such conditions or are later required to or choose to suspend such operations again, our business, prospects, financial condition and operating results may be harmed.

We face strong competition for our products and services from a growing list of established and new competitors.

The worldwide automotive market is highly competitive today and we expect it will become even more so in the future. A significant and growing number of established and new automobile manufacturers, as well as other companies, have entered, or are reported to have plans to enter, the market for electric and other alternative fuel vehicles, including hybrid, plug-in hybrid and fully electric vehicles, as well as the market for self-driving technology and other vehicle applications and software platforms. In some cases, our competitors offer or will offer electric vehicles in important markets such as China and Europe, and/or have announced an intention to produce electric vehicles exclusively at some point in the future. In addition, certain government and economic incentives which provide benefits to manufacturers who assemble domestically or have local suppliers, may provide a greater benefit to our competitors, which could negatively impact our profitability. Many of our competitors have significantly more or better-established resources than we do to devote to the design, development, manufacturing, distribution, promotion, sale and support of their products. Increased competition could result in our lower vehicle unit sales, price reductions, revenue shortfalls, loss of customers and loss of market share, which may harm our business, financial condition and operating results.

We also face competition in our energy generation and storage business from other manufacturers, developers, installers and service providers of competing energy technologies, as well as from large utilities. Decreases in the retail or wholesale prices of electricity from utilities or other renewable energy sources could make our products less attractive to customers and lead to an increased rate of customer defaults.

Risks Related to Our Operations

We may experience issues with lithium-ion cells or other components manufactured at our Gigafactories, which may harm the production and profitability of our vehicle and energy storage products.

Our plan to grow the volume and profitability of our vehicles and energy storage products depends on significant lithium-ion battery cell production, including by our partner Panasonic at Gigafactory Nevada. We also produce several vehicle components at our Gigafactories, such as battery modules and packs and drive units, and manufacture energy storage products. If we are unable to or otherwise do not maintain and grow our respective operations, or if we are unable to do so cost-effectively or hire and retain highly-skilled personnel there, our ability to manufacture our products profitably would be limited, which may harm our business and operating results.

Finally, the high volumes of lithium-ion cells and battery modules and packs manufactured by us and by our suppliers are stored and recycled at our various facilities. Any mishandling of these products may cause disruption to the operation of such facilities. While we have implemented safety procedures related to the handling of the cells, there can be no assurance that a safety issue or fire related to the cells would not disrupt our operations. Any such disruptions or issues may harm our brand and business.

We face risks associated with maintaining and expanding our international operations, including unfavorable and uncertain regulatory, political, economic, tax and labor conditions.

We are subject to legal and regulatory requirements, political uncertainty and social, environmental and economic conditions in numerous jurisdictions, including markets in which we generate significant sales, over which we have little control and which are inherently unpredictable. Our operations in such jurisdictions, particularly as a company based in the U.S., create risks relating to conforming our products to regulatory and safety requirements and charging and other electric infrastructures; organizing local operating entities; establishing, staffing and managing foreign business locations; attracting local customers; navigating foreign government taxes, regulations and permit requirements; enforceability of our contractual rights; trade restrictions, customs regulations, tariffs and price or exchange controls; and preferences in foreign nations for domestically manufactured products. For example, we monitor tax legislation changes on a global basis, including changes arising as a result of the Organization for Economic Cooperation and Development’s multi-jurisdictional plan of action to address base erosion and profit shifting. Such conditions may increase our costs, impact our ability to sell our products and require significant management attention, and may harm our business if we are unable to manage them effectively.

Our business may suffer if our products or features contain defects, fail to perform as expected or take longer than expected to become fully functional.

If our products contain design or manufacturing defects, whether relating to our software or hardware, that cause them not to perform as designed or intended or that require repair, or certain features of our vehicles such as new Autopilot or FSD Capability features take longer than expected to become enabled, are legally restricted or become subject to onerous regulation, our ability to develop, market and sell our products and services may be harmed, and we may experience delivery delays, product recalls, allegations of product liability, breach of warranty and related consumer protection claims and significant warranty and other expenses. While we are continuously working to develop and improve our products’ capability and performance, there is no guarantee that any incremental changes in the specific software or equipment we deploy in our vehicles over time will not result in initial functional disparities from prior iterations or will perform as forecast in the timeframe we anticipate, or at all. Although we attempt to remedy any issues we observe in our products as effectively and rapidly as possible, such efforts may not be timely, may hamper production or may not completely satisfy our customers. We have performed, and continue to perform, extensive internal testing on our products and features, though, like the rest of the industry, we currently have a limited frame of reference by which to evaluate certain aspects of their long-term quality, reliability, durability and performance characteristics, including exposure to or consequence of external attacks. While we attempt to identify and address or remedy defects we identify pre-production and sale, there may be latent defects that we may be unable to detect or control for in our products, and thereby address, prior to their sale to or installation for customers.

We may be required to defend or insure against product liability claims.

The automobile industry generally experiences significant product liability claims, and as such we face the risk of such claims in the event our vehicles do not perform or are claimed to not have performed as expected. As is true for other automakers, our vehicles have been involved and we expect in the future will be involved in accidents resulting in death or personal injury, and such accidents where Autopilot, Enhanced Autopilot or FSD Capability features are engaged are the subject of significant public attention, especially in light of NHTSA’s Standing General Order requiring reports regarding crashes involving vehicles with advanced driver assistance systems. We have experienced, and we expect to continue to face, claims and regulatory scrutiny arising from or related to misuse or claimed failures or alleged misrepresentations of such new technologies that we are pioneering. In addition, the battery packs that we produce make use of lithium-ion cells. On rare occasions, lithium-ion cells can rapidly release the energy they contain by venting smoke and flames in a manner that can ignite nearby materials as well as other lithium-ion cells. While we have designed our battery packs to passively contain any single cell’s release of energy without spreading to neighboring cells, there can be no assurance that a field or testing failure of our vehicles or other battery packs that we produce will not occur, in particular due to a high-speed crash. Likewise, as our solar energy systems and energy storage products generate and store electricity, they have the potential to fail or cause injury to people or property. Any product liability claim may subject us to lawsuits and substantial monetary damages, product recalls or redesign efforts, and even a meritless claim may require us to defend it, all of which may generate negative publicity and be expensive and time-consuming. In most jurisdictions, we generally self-insure against the risk of product liability claims for vehicle exposure, meaning that any product liability claims will likely have to be paid from company funds and not by insurance.

We will need to maintain public credibility and confidence in our long-term business prospects in order to succeed.

In order to maintain and grow our business, we must maintain credibility and confidence among customers, suppliers, analysts, investors, ratings agencies and other parties in our long-term financial viability and business prospects. Maintaining such confidence may be challenging due to our limited operating history relative to established competitors; customer unfamiliarity with our products; any delays we may experience in scaling manufacturing, delivery and service operations to meet demand; competition and uncertainty regarding the future of electric vehicles or our other products and services; our quarterly production and sales performance compared with market expectations; and other factors including those over which we have no control. In particular, Tesla’s products, business, results of operations, and statements and actions of Tesla and its management are subject to significant amounts of commentary by a range of third parties. Such attention can include criticism, which may be exaggerated or unfounded, such as speculation regarding the sufficiency or stability of our management team. Any such negative perceptions, whether caused by us or not, may harm our business and make it more difficult to raise additional funds if needed.

We may be unable to effectively grow, or manage the compliance, residual value, financing and credit risks related to, our various financing programs.

We offer financing arrangements for our vehicles in North America, Europe and Asia primarily ourselves and through various financial institutions. We also currently offer vehicle financing arrangements directly through our local subsidiaries in certain markets. Depending on the country, such arrangements are available for specified models and may include operating leases directly with us under which we typically receive only a very small portion of the total vehicle purchase price at the time of lease, followed by a stream of payments over the term of the lease. We have also offered various arrangements for customers of our solar energy systems whereby they pay us a fixed payment to lease or finance the purchase of such systems or purchase electricity generated by them. If we do not successfully monitor and comply with applicable national, state and/or local financial regulations and consumer protection laws governing these transactions, we may become subject to enforcement actions or penalties.

The profitability of any directly-leased vehicles returned to us at the end of their leases depends on our ability to accurately project our vehicles’ residual values at the outset of the leases, and such values may fluctuate prior to the end of their terms depending on various factors such as supply and demand of our used vehicles, economic cycles and the pricing of new vehicles. We have made in the past and may make in the future certain adjustments to our prices from time to time in the ordinary course of business, which may impact the residual values of our vehicles and reduce the profitability of our vehicle leasing program. The funding and growth of this program also rely on our ability to secure adequate financing and/or business partners. If we are unable to adequately fund our leasing program through internal funds, partners or other financing sources, and compelling alternative financing programs are not available for our customers who may expect or need such options, we may be unable to grow our vehicle deliveries. Furthermore, if our vehicle leasing business grows substantially, our business may suffer if we cannot effectively manage the resulting greater levels of residual risk.

Similarly, we have provided resale value guarantees to vehicle customers and partners for certain financing programs, under which such counterparties may sell their vehicles back to us at certain points in time at pre-determined amounts. However, actual resale values are subject to fluctuations over the term of the financing arrangements, such as from the vehicle pricing changes discussed above. If the actual resale values of any vehicles resold or returned to us pursuant to these programs are materially lower than the pre-determined amounts we have offered, our financial condition and operating results may be harmed.

Finally, our vehicle and solar energy system financing programs and our energy storage sales programs also expose us to customer credit risk. In the event of a widespread economic downturn or other catastrophic event, our customers may be unable or unwilling to satisfy their payment obligations to us on a timely basis or at all. If a significant number of our customers default, we may incur substantial credit losses and/or impairment charges with respect to the underlying assets.

We must manage ongoing obligations under our agreement with the Research Foundation for the State University of New York relating to our Gigafactory New York.

We are party to an operating lease and a research and development agreement through the State University of New York (the “SUNY Foundation”). These agreements provide for the construction and use of our Gigafactory New York, which we have primarily used for the development and production of our Solar Roof and other solar products and components, energy storage components and Supercharger components, and for other lessor-approved functions. Under this agreement, we are obligated to, among other things, meet employment targets as well as specified minimum numbers of personnel in the State of New York and in Buffalo, New York and spend or incur $5.00 billion in combined capital, operational expenses, costs of goods sold and other costs in the State of New York during a period that was initially 10 years beginning April 30, 2018. As of December 31, 2023, we are currently in excess of such targets relating to investments and personnel in the State of New York and Buffalo. While we expect to have and grow significant operations at Gigafactory New York and the surrounding Buffalo area, any failure by us in any year over the course of the term of the agreement to meet all applicable future obligations may result in our obligation to pay a “program payment” of $41 million to the SUNY Foundation for such year, the termination of our lease at Gigafactory New York which may require us to pay additional penalties, and/or the need to adjust certain of our operations. Any of the foregoing events may harm our business, financial condition and operating results.

If we are unable to attract, hire and retain key employees and qualified personnel, our ability to compete may be harmed.

The loss of the services of any of our key employees or any significant portion of our workforce could disrupt our operations or delay the development, introduction and ramp of our products and services. In particular, we are highly dependent on the services of Elon Musk, Technoking of Tesla and our Chief Executive Officer. None of our key employees is bound by an employment agreement for any specific term and we may not be able to successfully attract and retain senior leadership necessary to grow our business. Our future success also depends upon our ability to attract, hire and retain a large number of engineering, manufacturing, marketing, sales and delivery, service, installation, technology and support personnel, especially to support our planned high-volume product sales, market and geographical expansion and technological innovations. If we are not successful in managing these risks, our business, financial condition and operating results may be harmed.

Employees may leave Tesla or choose other employers over Tesla due to various factors, such as a very competitive labor market for talented individuals with automotive or technology experience, or any negative publicity related to us. In regions where we have or will have operations, particularly significant engineering and manufacturing centers, there is strong competition for individuals with skillsets needed for our business, including specialized knowledge of electric vehicles, engineering and electrical and building construction expertise. We also compete with both mature and prosperous companies that have far greater financial resources than we do and start-ups and emerging companies that promise short-term growth opportunities.

Finally, our compensation philosophy for all of our personnel reflects our startup origins, with an emphasis on equity-based awards and benefits in order to closely align their incentives with the long-term interests of our stockholders. We periodically seek and obtain approval from our stockholders for future increases to the number of awards available under our equity incentive and employee stock purchase plans. If we are unable to obtain the requisite stockholder approvals for such future increases, we may have to expend additional cash to compensate our employees and our ability to retain and hire qualified personnel may be harmed.

We are highly dependent on the services of Elon Musk, Technoking of Tesla and our Chief Executive Officer.

We are highly dependent on the services of Elon Musk, Technoking of Tesla and our Chief Executive Officer. Although Mr. Musk spends significant time with Tesla and is highly active in our management, he does not devote his full time and attention to Tesla. Mr. Musk also currently serves as Chief Executive Officer and Chief Technical Officer of Space Exploration Technologies Corp., a developer and manufacturer of space launch vehicles, Chairman and Chief Technical Officer of X Corp., a social media company, and is involved in other emerging technology ventures.

Our information technology systems or data, or those of our service providers or customers or users could be subject to cyber-attacks or other security incidents, which could result in data breaches, intellectual property theft, claims, litigation, regulatory investigations, significant liability, reputational damage and other adverse consequences.

We continue to expand our information technology systems as our operations grow, such as product data management, procurement, inventory management, production planning and execution, sales, service and logistics, dealer management, financial, tax and regulatory compliance systems. This includes the implementation of new internally developed systems and the deployment of such systems in the U.S. and abroad. While, we maintain information technology measures designed to protect us against intellectual property theft, data breaches, sabotage and other external or internal cyber-attacks or misappropriation, our systems and those of our service providers are potentially vulnerable to malware, ransomware, viruses, denial-of-service attacks, phishing attacks, social engineering, computer hacking, unauthorized access, exploitation of bugs, defects and vulnerabilities, breakdowns, damage, interruptions, system malfunctions, power outages, terrorism, acts of vandalism, security breaches, security incidents, inadvertent or intentional actions by employees or other third parties, and other cyber-attacks.

To the extent any security incident results in unauthorized access or damage to or acquisition, use, corruption, loss, destruction, alteration or dissemination of our data, including intellectual property and personal information, or our products or vehicles, or for it to be believed or reported that any of these occurred, it could disrupt our business, harm our reputation, compel us to comply with applicable data breach notification laws, subject us to time consuming, distracting and expensive litigation, regulatory investigation and oversight, mandatory corrective action, require us to verify the correctness of database contents, or otherwise subject us to liability under laws, regulations and contractual obligations, including those that protect the privacy and security of personal information. This could result in increased costs to us and result in significant legal and financial exposure and/or reputational harm.

We also rely on service providers, and similar incidents relating to their information technology systems could also have a material adverse effect on our business. There have been and may continue to be significant supply chain attacks. Our service providers, including our workforce management software provider, have been subject to ransomware and other security incidents, and we cannot guarantee that our or our service providers' systems have not been breached or that they do not contain exploitable defects, bugs, or vulnerabilities that could result in a security incident, or other disruption to, our or our service providers' systems. Our ability to monitor our service providers' security measures is limited, and, in any event, malicious third parties may be able to circumvent those security measures.

Further, the implementation, maintenance, segregation and improvement of these systems require significant management time, support and cost, and there are inherent risks associated with developing, improving and expanding our core systems as well as implementing new systems and updating current systems, including disruptions to the related areas of business operation. These risks may affect our ability to manage our data and inventory, procure parts or supplies or manufacture, sell, deliver and service products, adequately protect our intellectual property or achieve and maintain compliance with, or realize available benefits under, tax laws and other applicable regulations.

Moreover, if we do not successfully implement, maintain or expand these systems as planned, our operations may be disrupted, our ability to accurately and/or timely report our financial results could be impaired and deficiencies may arise in our internal control over financial reporting, which may impact our ability to certify our financial results. Moreover, our proprietary information, including intellectual property and personal information, could be compromised or misappropriated and our reputation may be adversely affected. If these systems or their functionality do not operate as we expect them to, we may be required to expend significant resources to make corrections or find alternative sources for performing these functions.

Any unauthorized control or manipulation of our products' systems could result in loss of confidence in us and our products.

Our products contain complex information technology systems. For example, our vehicles and energy storage products are designed with built-in data connectivity to accept and install periodic remote updates from us to improve or update their functionality. While we have implemented security measures intended to prevent unauthorized access to our information technology networks, our products and their systems, malicious entities have reportedly attempted, and may attempt in the future, to gain unauthorized access to modify, alter and use such networks, products and systems to gain control of, or to change, our products' functionality, user interface and performance characteristics or to gain access to data stored in or generated by our products. We encourage reporting of potential vulnerabilities in the security of our products through our security vulnerability reporting policy, and we aim to remedy any reported and verified vulnerability. However, there can be no assurance that any vulnerabilities will not be exploited before they can be identified, or that our remediation efforts are or will be successful.

Any unauthorized access to or control of our products or their systems or any loss of data could result in legal claims or government investigations. In addition, regardless of their veracity, reports of unauthorized access to our products, their systems or data, as well as other factors that may result in the perception that our products, their systems or data are capable of being hacked, may harm our brand, prospects and operating results. We have been the subject of such reports in the past.

Our business may be adversely affected by any disruptions caused by union activities.

It is not uncommon for employees of certain trades at companies such as ours to belong to a union, which can result in higher employee costs and increased risk of work stoppages. Moreover, regulations in some jurisdictions outside of the U.S. mandate employee participation in industrial collective bargaining agreements and work councils with certain consultation rights with respect to the relevant companies' operations. Although we work diligently to provide the best possible work environment for our employees, they may still decide to join or seek recognition to form a labor union, or we may be required to become a union signatory. From time to time, labor unions have engaged in campaigns to organize certain of our operations, as part of which such unions have filed unfair labor practice charges against us with the National Labor Relations Board (the "NLRB"), and they may do so in the future. Any unfavorable ultimate outcome for Tesla may have a negative impact on the perception of Tesla's treatment of our employees. Furthermore, we are directly or indirectly dependent upon companies with unionized work forces, such as suppliers and trucking and freight companies. Any work stoppages or strikes organized by such unions could delay the manufacture and sale of our products and may harm our business and operating results.

We may choose to or be compelled to undertake product recalls or take other similar actions.

As a manufacturing company, we must manage the risk of product recalls with respect to our products. Recalls for our vehicles have resulted from various hardware and software-related safety concerns or non-compliance determinations. In addition to recalls initiated by us for various causes, testing of or investigations into our products by government regulators or industry groups may compel us to initiate product recalls or may result in negative public perceptions about the safety of our products, even if we disagree with the defect determination or have data that contradicts it. In the future, we may voluntarily or involuntarily initiate recalls if any of our products are determined by us or a regulator to contain a safety defect or be noncompliant with applicable laws and regulations, such as U.S. Federal Motor Vehicle Safety Standards. Such recalls, whether voluntary or involuntary or caused by systems or components engineered or manufactured by us or our suppliers, could result in significant expense, supply chain complications and service burdens, and may harm our brand, business, prospects, financial condition and operating results.

Our current and future warranty reserves may be insufficient to cover future warranty claims.

We provide a manufacturer's warranty on all new and used Tesla vehicles we sell directly to customers. We also provide certain warranties with respect to the energy generation and storage systems we sell, including on their installation and maintenance. For components not manufactured by us, we generally pass through to our customers the applicable manufacturers' warranties, but may retain some warranty responsibilities for some or all of the life of such components. As part of our energy generation and storage system contracts, we may provide the customer with performance guarantees that guarantee that the underlying system will meet or exceed the minimum energy generation or other energy performance requirements specified in the contract. Under these performance guarantees, we generally bear the risk of electricity production or other performance shortfalls, including in some cases shortfalls caused by failures in components from third party manufacturers. These risks are exacerbated in the event such manufacturers cease operations or fail to honor their warranties.

If our warranty reserves are inadequate to cover future warranty claims on our products, our financial condition and operating results may be harmed. Warranty reserves include our management’s best estimates of the projected costs to repair or to replace items under warranty, which are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims. Such estimates are inherently uncertain and changes to our historical or projected experience, especially with respect to products that we have introduced relatively recently and/or that we expect to produce at significantly greater volumes than our past products, may cause material changes to our warranty reserves in the future.

Our insurance coverage strategy may not be adequate to protect us from all business risks.

We may be subject, in the ordinary course of business, to losses resulting from products liability, accidents, acts of God and other claims against us, for which we may have no insurance coverage. As a general matter, we do not maintain as much insurance coverage as many other companies do, and in some cases, we do not maintain any at all. Additionally, the policies that we do have may include significant deductibles or self-insured retentions, policy limitations and exclusions, and we cannot be certain that our insurance coverage will be sufficient to cover all future losses or claims against us. A loss that is uninsured or which exceeds policy limits may require us to pay substantial amounts, which may harm our financial condition and operating results.

Our debt agreements contain covenant restrictions that may limit our ability to operate our business.

The terms of certain of our debt facilities contain, and any of our other future debt agreements may contain, covenant restrictions that may limit our ability to operate our business, including restrictions on our and/or our subsidiaries’ ability to, among other things, incur additional debt or create liens. In addition, under certain circumstances we are required to maintain a certain amount of liquidity. As a result of these covenants, our ability to respond to changes in business and economic conditions and engage in beneficial transactions, including to obtain additional financing as needed, may be restricted. Furthermore, our failure to comply with our debt covenants could result in a default under our debt agreements, which could permit the holders to accelerate our obligation to repay the debt. If any of our debt is accelerated, we may not have sufficient funds available to repay it.

Additional funds may not be available to us when we need or want them.

Our business and our future plans for expansion are capital-intensive, and the specific timing of cash inflows and outflows may fluctuate substantially from period to period. We may need or want to raise additional funds through the issuance of equity, equity-related or debt securities or through obtaining credit from financial institutions to fund, together with our principal sources of liquidity, the costs of developing and manufacturing our current or future products, to pay any significant unplanned or accelerated expenses or for new significant strategic investments, or to refinance our significant consolidated indebtedness, even if not required to do so by the terms of such indebtedness. We cannot be certain that additional funds will be available to us on favorable terms when required, or at all. If we cannot raise additional funds when we need them, our financial condition, results of operations, business and prospects could be materially and adversely affected.

We may be negatively impacted by any early obsolescence of our manufacturing equipment.

We depreciate the cost of our manufacturing equipment over their expected useful lives. However, product cycles or manufacturing technology may change periodically, and we may decide to update our products or manufacturing processes more quickly than expected. Moreover, improvements in engineering and manufacturing expertise and efficiency may result in our ability to manufacture our products using less of our currently installed equipment. Alternatively, as we ramp and mature the production of our products to higher levels, we may discontinue the use of already installed equipment in favor of different or additional equipment. The useful life of any equipment that would be retired early as a result would be shortened, causing the depreciation on such equipment to be accelerated, and our results of operations may be harmed.

There is no guarantee that we will have sufficient cash flow from our business to pay our indebtedness or that we will not incur additional indebtedness.

As of December 31, 2023, we and our subsidiaries had outstanding $4.68 billion in aggregate principal amount of indebtedness (see Note 11, *Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K). Our consolidated indebtedness may increase our vulnerability to any generally adverse economic and industry conditions. We and our subsidiaries may, subject to the limitations in the terms of our existing and future indebtedness, incur additional debt, secure existing or future debt or recapitalize our debt.

Our ability to make scheduled payments of the principal and interest on our indebtedness when due, to make payments upon conversion or repurchase demands with respect to our convertible senior notes or to refinance our indebtedness as we may need or desire, depends on our future performance, which is subject to economic, financial, competitive and other factors beyond our control. Our business may not continue to generate cash flow from operations in the future sufficient to satisfy our obligations under our existing indebtedness and any future indebtedness we may incur, and to make necessary capital expenditures. If we are unable to generate such cash flow, we may be required to adopt one or more alternatives, such as reducing or delaying investments or capital expenditures, selling assets, refinancing or obtaining additional equity capital on terms that may be onerous or highly dilutive. Our ability to refinance existing or future indebtedness will depend on the capital markets and our financial condition at such time. In addition, our ability to make payments may be limited by law, by regulatory authority or by agreements governing our future indebtedness. We may not be able to engage in these activities on desirable terms or at all, which may result in a default on our existing or future indebtedness and harm our financial condition and operating results.

We are exposed to fluctuations in currency exchange rates.

We transact business globally in multiple currencies and have foreign currency risks related to our revenue, costs of revenue, operating expenses and localized subsidiary debt denominated in currencies other than the U.S. dollar. To the extent we have significant revenues denominated in such foreign currencies, any strengthening of the U.S. dollar would tend to reduce our revenues as measured in U.S. dollars, as we have historically experienced, and are currently experiencing. In addition, a portion of our costs and expenses have been, and we anticipate will continue to be, denominated in foreign currencies. If we do not have fully offsetting revenues in these currencies and if the value of the U.S. dollar depreciates significantly against these currencies, our costs as measured in U.S. dollars as a percent of our revenues will correspondingly increase and our margins will suffer. As a result, our operating results may be harmed.

We may not be able to adequately protect or defend ourselves against intellectual property infringement claims, which may be time-consuming and expensive, or affect the freedom to operate our business.

Our competitors or other third parties may hold or obtain patents, copyrights, trademarks or other proprietary rights that could prevent, limit or interfere with our ability to make, use, develop, sell or market our products and services, which could make it more difficult for us to operate our business. From time to time, the holders of such intellectual property rights may assert their rights and urge us to take licenses and/or may bring suits alleging infringement or misappropriation of such rights, which could result in substantial costs, negative publicity and management attention, regardless of merit.

In addition, the effective protection for our brands, technologies, and proprietary information may be limited or unavailable in certain countries, making it difficult to protect our intellectual property from misappropriation or infringement. Although we make reasonable efforts to maintain the confidentiality of our proprietary information, we cannot guarantee that these actions will deter or prevent misappropriation of our intellectual property. The theft or unauthorized use or publication of our trade secrets and confidential information could affect our competitive position.

While we endeavor to obtain and protect the intellectual property rights that we expect will allow us to retain or advance our strategic initiatives in these circumstances, there can be no assurance that we will be able to adequately identify and protect the portions of intellectual property that are strategic to our business, or mitigate the risk of potential suits or other legal demands by third parties. Accordingly, we may consider the entering into licensing agreements with respect to such rights, although no assurance can be given that such licenses can be obtained on acceptable terms or that litigation will not occur, and such licenses and associated litigation could significantly increase our operating expenses. Further, if we are determined to have or believe there is a high likelihood that we have infringed upon a third party’s intellectual property rights, we may be required to cease making, selling or incorporating certain components or intellectual property into the goods and services we offer, to pay substantial damages and/or license royalties, to redesign our products and services and/or to establish and maintain alternative branding for our products and services. In the event that we are required to take one or more such actions, our brand, business, financial condition and operating results may be harmed.

Increased scrutiny and changing expectations from stakeholders with respect to the Company’s ESG practices may result in additional costs or risks.

Companies across many industries are facing increasing scrutiny related to their environmental, social and governance (ESG) practices. Investor advocacy groups, certain institutional investors, investment funds and other influential investors are also increasingly focused on ESG practices and in recent years have placed increasing importance on the non-financial impacts of their investments. While our mission is to accelerate the world’s transition to sustainable energy, if our ESG practices do not meet investor or other industry stakeholder expectations, which continue to evolve, we may incur additional costs and our brand, ability to attract and retain qualified employees and business may be harmed.

Our operations could be adversely affected by events outside of our control, such as natural disasters, wars or health epidemics.

We may be impacted by natural disasters, wars, health epidemics, weather conditions, the long-term effects of climate change, power outages or other events outside of our control. For example, our Fremont Factory and Gigafactory Nevada are located in seismically active regions in Northern California and Nevada, and our Gigafactory Shanghai is located in a flood-prone area. Moreover, the area in which our Gigafactory Texas is located experienced severe winter storms in the first quarter of 2021 that had a widespread impact on utilities and transportation. If major disasters such as earthquakes, floods or other climate-related events occur, or our information system or communication breaks down or operates improperly, our headquarters and production facilities may be seriously damaged, or we may have to stop or delay production and shipment of our products. In addition, the global COVID-19 pandemic has impacted economic markets, manufacturing operations, supply chains, employment and consumer behavior in nearly every geographic region and industry across the world, and we have been, and may in the future be, adversely affected as a result. Also, the broader consequences in the current conflict between Russia and Ukraine, which may include further embargoes, regional instability and geopolitical shifts; airspace bans relating to certain routes, or strategic decisions to alter certain routes; and potential retaliatory action by the Russian government against companies, and the extent of the conflict on our business and operating results cannot be predicted. We may incur expenses or delays relating to such events outside of our control, which could have a material adverse impact on our business, operating results and financial condition.

Risks Related to Government Laws and Regulations

Demand for our products and services may be impacted by the status of government and economic incentives supporting the development and adoption of such products.

Government and economic incentives that support the development and adoption of electric vehicles in the U.S. and abroad, including certain tax exemptions, tax credits and rebates, may be reduced, eliminated, amended or exhausted from time to time. For example, previously available incentives favoring electric vehicles in certain areas have expired or were cancelled or temporarily unavailable, and in some cases were not eventually replaced or reinstituted, which may have negatively impacted sales. In addition, certain government and economic incentives may also be implemented or amended to provide benefits to manufacturers who assemble domestically, have local suppliers or have other characteristics that may not apply to Tesla. Such developments could negatively impact demand for our vehicles, and we and our customers may have to adjust to them, including through pricing modifications.

In addition, certain governmental rebates, tax credits and other financial incentives that are currently available with respect to our solar and energy storage product businesses allow us to lower our costs and encourage customers to buy our products and investors to invest in our solar financing funds. However, these incentives may expire when the allocated funding is exhausted, reduced or terminated as renewable energy adoption rates increase, sometimes without warning. Likewise, in jurisdictions where net metering is currently available, our customers receive bill credits from utilities for energy that their solar energy systems generate and export to the grid in excess of the electric load they use. The benefit available under net metering has been or has been proposed to be reduced, altered or eliminated in several jurisdictions, and has also been contested and may continue to be contested before the Federal Energy Regulatory Commission. Any reductions or terminations of such incentives may harm our business, prospects, financial condition and operating results by making our products less competitive for customers, increasing our cost of capital and adversely impacting our ability to attract investment partners and to form new financing funds for our solar and energy storage assets.

Finally, we and our fund investors claim these U.S. federal tax credits and certain state incentives in amounts based on independently appraised fair market values of our solar and energy storage systems. Some governmental authorities have audited such values and in certain cases have determined that these values should be lower, and they may do so again in the future. Such determinations may result in adverse tax consequences and/or our obligation to make indemnification or other payments to our funds or fund investors.

We are subject to evolving laws and regulations that could impose substantial costs, legal prohibitions or unfavorable changes upon our operations or products.

As we grow our manufacturing operations in additional regions, we are or will be subject to complex environmental, manufacturing, health and safety laws and regulations at numerous jurisdictional levels in the U.S., China, Germany and other locations abroad, including laws relating to the use, handling, storage, recycling, disposal and/or human exposure to hazardous materials, product material inputs and post-consumer products and with respect to constructing, expanding and maintaining our facilities. New, or changes in, environmental and climate change laws, regulations or rules could also lead to increased costs of compliance, including remediations of any discovered issues, and changes to our operations, which may be significant, and any failures to comply could result in significant expenses, delays or fines. In addition, as we have increased our employee headcount and operations, we are and may continue to be subject to increased scrutiny, including litigation and government investigations, that we will need to defend against. If we are unable to successfully defend ourselves in such litigation or government investigations, it may harm our brand, ability to attract and retain qualified employees, business and financial condition. We are also subject to laws and regulations applicable to the supply, manufacture, import, sale, service and performance of our products both domestically and abroad. For example, in countries outside of the U.S., we are required to meet standards relating to vehicle safety, fuel economy and emissions that are often materially different from equivalent requirements in the U.S., thus resulting in additional investment into the vehicles and systems to ensure regulatory compliance in all countries. This process may include official review and certification of our vehicles by foreign regulatory agencies prior to market entry, as well as compliance with foreign reporting and recall management systems requirements.

In particular, we offer in our vehicles in certain markets Autopilot and FSD Capability features that today assist drivers with certain tedious and potentially dangerous aspects of road travel, but which currently require drivers to remain fully engaged in the driving operation. We are continuing to develop our Autopilot and FSD Capability technology. There are a variety of international, federal and state regulations that may apply to, and may adversely affect, the design and performance, sale, marketing, registration and operation of Autopilot and FSD Capability, and future capability, including full self-driving vehicles that may not be operated by a human driver. This includes many existing vehicle standards that were not originally intended to apply to vehicles that may not be operated by a human driver. Such regulations continue to rapidly change, which increases the likelihood of a patchwork of complex or conflicting regulations, or may delay, restrict or prohibit the availability of certain functionalities and vehicle designs, which could adversely affect our business.

Finally, as a manufacturer, installer and service provider with respect to solar generation and energy storage systems, a supplier of electricity generated and stored by certain of the solar energy and energy storage systems we install for customers, and a provider of grid services through virtual power plant models, we are impacted by federal, state and local regulations and policies concerning the import or export of components, electricity pricing, the interconnection of electricity generation and storage equipment with the electrical grid and the sale of electricity generated by third party-owned systems. If regulations and policies are introduced that adversely impact the import or export of components, or the interconnection, maintenance or use of our solar and energy storage systems, they could deter potential customers from purchasing our solar and energy storage products and services, threaten the economics of our existing contracts and cause us to cease solar and energy storage system sales and services in the relevant jurisdictions, which may harm our business, financial condition and operating results.

Any failure by us to comply with a variety of U.S. and international privacy and consumer protection laws may harm us.

Any failure by us or our vendors or other business partners to comply with our public privacy notice or with federal, state or international privacy, data protection or security laws or regulations relating to the processing, collection, use, retention, security and transfer of personally identifiable information could result in regulatory or litigation-related actions against us, legal liability, fines, damages, ongoing audit requirements and other significant costs. Substantial expenses and operational changes may be required in connection with maintaining compliance with such laws, and even an unsuccessful challenge by customers or regulatory authorities of our activities could result in adverse publicity and could require a costly response from and defense by us. In addition, certain privacy laws are still subject to a high degree of uncertainty as to their interpretation, application and impact, and may require extensive system and operational changes, be difficult to implement, increase our operating costs, adversely impact the cost or attractiveness of the products or services we offer, or result in adverse publicity and harm our reputation. For example, the General Data Protection Regulation applies to the processing of personal information collected from individuals located in the European Union requiring certain data protection measures when handling, with a significant risk of fines for noncompliance. Similarly, our North American operations are subject to complex and changing federal and US state-specific data privacy laws and regulations, such as the California Consumer Privacy Act which imposes certain legal obligations on our use and processing of personal information related to California residents. Finally, additional privacy and cybersecurity laws have come into effect in China.

These laws continue to develop and may be inconsistent from jurisdiction to jurisdiction. Complying with emerging and changing requirements may cause us to incur substantial costs and make enhancements to relevant data practices. Noncompliance could result in significant penalties or legal liability.

In addition to the risks related to general privacy regulation, we may also be subject to specific vehicle manufacturer obligations relating to cybersecurity, data privacy and data localization requirements which place additional risks to our international operations. Risks and penalties could include ongoing audit requirements, data protection authority investigations, legal proceedings by international governmental entities or others resulting in mandated disclosure of sensitive data or other commercially unfavorable terms. Notwithstanding our efforts to protect the security and integrity of our customers' personal information, we may be required to expend significant resources to comply with data breach requirements if, for example, third parties improperly obtain and use the personal information of our customers or we otherwise experience a data loss with respect to the personal information we process and handle. A major breach of our network security and systems may occur despite defensive measures, and may result in fines, penalties and damages and harm our brand, prospects and operating results.

We could be subject to liability, penalties and other restrictive sanctions and adverse consequences arising out of certain governmental investigations and proceedings.

We are cooperating with certain government investigations as discussed in Note 15, *Commitments and Contingencies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K. To our knowledge, no government agency in any such ongoing investigation has concluded that any wrongdoing occurred. However, we cannot predict the outcome or impact of any such ongoing matters, and there exists the possibility that we could be subject to liability, penalties and other restrictive sanctions and adverse consequences if the SEC, the U.S. Department of Justice or any other government agency were to pursue legal action in the future. Moreover, we expect to incur costs in responding to related requests for information and subpoenas, and if instituted, in defending against any governmental proceedings.

We may face regulatory challenges to or limitations on our ability to sell vehicles directly.

While we intend to continue to leverage our most effective sales strategies, including sales through our website, we may not be able to sell our vehicles through our own stores in certain states in the U.S. with laws that may be interpreted to impose limitations on this direct-to-consumer sales model. It has also been asserted that the laws in some states limit our ability to obtain dealer licenses from state motor vehicle regulators, and such assertions persist. In certain locations, decisions by regulators permitting us to sell vehicles have been, and may be, challenged by dealer associations and others as to whether such decisions comply with applicable state motor vehicle industry laws. We have prevailed in many of these lawsuits and such results have reinforced our continuing belief that state franchise laws were not intended to apply to a manufacturer that does not have franchise dealers anywhere in the world. In some states, there have also been regulatory and legislative efforts by dealer associations to propose laws that, if enacted, would prevent us from obtaining dealer licenses in their states given our current sales model. A few states have passed legislation that clarifies our ability to operate, but at the same time limits the number of dealer licenses we can obtain or stores that we can operate. The application of state laws applicable to our operations continues to be difficult to predict.

Internationally, there may be laws in jurisdictions we have not yet entered or laws we are unaware of in jurisdictions we have entered that may restrict our sales or other business practices. Even for those jurisdictions we have analyzed, the laws in this area can be complex, difficult to interpret and may change over time. Continued regulatory limitations and other obstacles interfering with our ability to sell vehicles directly to consumers may harm our financial condition and operating results.

Risks Related to the Ownership of Our Common Stock

The trading price of our common stock is likely to continue to be volatile.

The trading price of our common stock has been highly volatile and could continue to be subject to wide fluctuations in response to various factors, some of which are beyond our control. Our common stock has experienced over the last 52 weeks an intra-day trading high of $299.29 per share and a low of $152.37 per share. The stock market in general, and the market for technology companies in particular, has experienced extreme price and volume fluctuations that have often been unrelated or disproportionate to the operating performance of those companies. In particular, a large proportion of our common stock has been historically and may in the future be traded by short sellers which may put pressure on the supply and demand for our common stock, further influencing volatility in its market price. Public perception of our company or management and other factors outside of our control may additionally impact the stock price of companies like us that garner a disproportionate degree of public attention, regardless of actual operating performance. In addition, in the past, following periods of volatility in the overall market or the market price of our shares, securities class action litigation has been filed against us. While we defend such actions vigorously, any judgment against us or any future stockholder litigation could result in substantial costs and a diversion of our management's attention and resources.

Our financial results may vary significantly from period to period due to fluctuations in our operating costs and other factors.

We expect our period-to-period financial results to vary based on our operating costs, which we anticipate will fluctuate as the pace at which we continue to design, develop and manufacture new products and increase production capacity by expanding our current manufacturing facilities and adding future facilities, may not be consistent or linear between periods. Additionally, our revenues from period to period may fluctuate as we introduce existing products to new markets for the first time and as we develop and introduce new products. As a result of these factors, we believe that quarter-to-quarter comparisons of our financial results, especially in the short term, are not necessarily meaningful and that these comparisons cannot be relied upon as indicators of future performance. Moreover, our financial results may not meet expectations of equity research analysts, ratings agencies or investors, who may be focused only on short-term quarterly financial results. If any of this occurs, the trading price of our stock could fall substantially, either suddenly or over time.

We may fail to meet our publicly announced guidance or other expectations about our business, which could cause our stock price to decline.

We provide from time to time guidance regarding our expected financial and business performance. Correctly identifying key factors affecting business conditions and predicting future events is inherently an uncertain process, and our guidance may not ultimately be accurate and has in the past been inaccurate in certain respects, such as the timing of new product manufacturing ramps. Our guidance is based on certain assumptions such as those relating to anticipated production and sales volumes (which generally are not linear throughout a given period), average sales prices, supplier and commodity costs and planned cost reductions. If our guidance varies from actual results, such as due to our assumptions not being met or the impact on our financial performance that could occur as a result of various risks and uncertainties, the market value of our common stock could decline significantly.

If Elon Musk were forced to sell shares of our common stock, either that he has pledged to secure certain personal loan obligations, or in satisfaction of other obligations, such sales could cause our stock price to decline.

Certain banking institutions have made extensions of credit to Elon Musk, our Chief Executive Officer, a portion of which was used to purchase shares of common stock in certain of our public offerings and private placements at the same prices offered to third-party participants in such offerings and placements. We are not a party to these loans, which are partially secured by pledges of a portion of the Tesla common stock currently owned by Mr. Musk. If the price of our common stock were to decline substantially, Mr. Musk may be forced by one or more of the banking institutions to sell shares of Tesla common stock to satisfy his loan obligations if he could not do so through other means. Any such sales could cause the price of our common stock to decline further. Further, Mr. Musk from time to time may commit to investing in significant business or other ventures, and as a result, be required to sell shares of our common stock in satisfaction of such commitments.

Anti-takeover provisions contained in our governing documents, applicable laws and our convertible senior notes could impair a takeover attempt.

Our certificate of incorporation and bylaws afford certain rights and powers to our board of directors that may facilitate the delay or prevention of an acquisition that it deems undesirable. We are also subject to Section 203 of the Delaware General Corporation Law and other provisions of Delaware law that limit the ability of stockholders in certain situations to effect certain business combinations. In addition, the terms of our convertible senior notes may require us to repurchase such notes in the event of a fundamental change, including a takeover of our company. Any of the foregoing provisions and terms that has the effect of delaying or deterring a change in control could limit the opportunity for our stockholders to receive a premium for their shares of our common stock, and could also affect the price that some investors are willing to pay for our common stock.

ITEM 1B. UNRESOLVED STAFF COMMENTS

None.

ITEM 1C. CYBERSECURITY

Cybersecurity Risk Management and Strategy

We recognize the importance of assessing, identifying, and managing material risks associated with cybersecurity threats, as such term is defined in Item 106(a) of Regulation S-K. These risks include, among other things: operational risks, intellectual property theft, fraud, extortion, harm to employees or customers and violation of data privacy or security laws.

Identifying and assessing cybersecurity risk is integrated into our overall risk management systems and processes. Cybersecurity risks related to our business, technical operations, privacy and compliance issues are identified and addressed through a multi-faceted approach including third party assessments, internal IT Audit, IT security, governance, risk and compliance reviews. To defend, detect and respond to cybersecurity incidents, we, among other things: conduct proactive privacy and cybersecurity reviews of systems and applications, audit applicable data policies, perform penetration testing using external third-party tools and techniques to test security controls, operate a bug bounty program to encourage proactive vulnerability reporting, conduct employee training, monitor emerging laws and regulations related to data protection and information security (including our consumer products) and implement appropriate changes.

We have implemented incident response and breach management processes which have four overarching and interconnected stages: 1) preparation for a cybersecurity incident, 2) detection and analysis of a security incident, 3) containment, eradication and recovery, and 4) post-incident analysis. Such incident responses are overseen by leaders from our Information Security, Product Security, Compliance and Legal teams regarding matters of cybersecurity.

Security events and data incidents are evaluated, ranked by severity and prioritized for response and remediation. Incidents are evaluated to determine materiality as well as operational and business impact, and reviewed for privacy impact.

We also conduct tabletop exercises to simulate responses to cybersecurity incidents. Our team of cybersecurity professionals then collaborate with technical and business stakeholders across our business units to further analyze the risk to the company, and form detection, mitigation and remediation strategies.

As part of the above processes, we regularly engage external auditors and consultants to assess our internal cybersecurity programs and compliance with applicable practices and standards. As of 2023, our Information Security Management System has been certified to conform to the requirements of ISO/IEC 27001:2013.

Our risk management program also assesses third party risks, and we perform third-party risk management to identify and mitigate risks from third parties such as vendors, suppliers, and other business partners associated with our use of third-party service providers. Cybersecurity risks are evaluated when determining the selection and oversight of applicable third-party service providers and potential fourth-party risks when handling and/or processing our employee, business or customer data. In addition to new vendor onboarding, we perform risk management during third-party cybersecurity compromise incidents to identify and mitigate risks to us from third-party incidents.

We describe whether and how risks from identified cybersecurity threats, including as a result of any previous cybersecurity incidents, have materially affected or are reasonably likely to materially affect us, including our business strategy, results of operations, or financial condition, under the heading “Our information technology systems or data, or those of our service providers or customers or users could be subject to cyber-attacks or other security incidents, which could result in data breaches, intellectual property theft, claims, litigation, regulatory investigations, significant liability, reputational damage and other adverse consequences” included as part of our risk factor disclosures at Item 1A of this Annual Report on Form 10-K.

Cybersecurity Governance

Cybersecurity is an important part of our risk management processes and an area of focus for our Board and management. Our Audit Committee is responsible for the oversight of risks from cybersecurity threats. Members of the Audit Committee receive updates on a quarterly basis from senior management, including leaders from our Information Security, Product Security, Compliance and Legal teams regarding matters of cybersecurity. This includes existing and new cybersecurity risks, status on how management is addressing and/or mitigating those risks, cybersecurity and data privacy incidents (if any) and status on key information security initiatives. Our Board members also engage in ad hoc conversations with management on cybersecurity-related news events and discuss any updates to our cybersecurity risk management and strategy programs.

Our cybersecurity risk management and strategy processes are overseen by leaders from our Information Security, Product Security, Compliance and Legal teams. Such individuals have an average of over 15 years of prior work experience in various roles involving information technology, including security, auditing, compliance, systems and programming. These individuals are informed about, and monitor the prevention, mitigation, detection and remediation of cybersecurity incidents through their management of, and participation in, the cybersecurity risk management and strategy processes described above, including the operation of our incident response plan, and report to the Audit Committee on any appropriate items.

ITEM 2. PROPERTIES

We are headquartered in Austin, Texas. Our principal facilities include a large number of properties in North America, Europe and Asia utilized for manufacturing and assembly, warehousing, engineering, retail and service locations, Supercharger sites and administrative and sales offices. Our facilities are used to support both of our reporting segments, and are suitable and adequate for the conduct of our business. We generally lease such facilities with the primary exception of some manufacturing facilities. The following table sets forth the location of our primary owned and leased manufacturing facilities.

Primary Manufacturing Facilities	Location	Owned or Leased
Gigafactory Texas	Austin, Texas	Owned
Fremont Factory	Fremont, California	Owned
Gigafactory Nevada	Sparks, Nevada	Owned
Gigafactory Berlin-Brandenburg	Grunheide, Germany	Owned
Gigafactory Shanghai	Shanghai, China	*
Gigafactory New York	Buffalo, New York	Leased
Megafactory	Lathrop, California	Leased

* We own the building and the land use rights with an initial term of 50 years. The land use rights are treated as operating lease right-of-use assets.

ITEM 3. LEGAL PROCEEDINGS

For a description of our material pending legal proceedings, please see Note 15, *Commitments and Contingencies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

In addition, each of the matters below is being disclosed pursuant to Item 103 of Regulation S-K because it relates to environmental regulations and aggregate civil penalties that we currently believe could potentially exceed $1 million. We believe that any proceeding that is material to our business or financial condition is likely to have potential penalties far in excess of such amount.

District attorneys in certain California counties conducted an investigation into Tesla’s waste segregation practices pursuant to Cal. Health & Saf. Code § 25100 et seq. and Cal. Civil Code § 1798.80. Tesla has implemented various remedial measures, including conducting training and audits, and enhancements to its site waste management programs, and settlement discussions are ongoing. While the outcome of this matter cannot be determined at this time, it is not currently expected to have a material adverse impact on our business.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Market Information

Our common stock has traded on The NASDAQ Global Select Market under the symbol "TSLA" since it began trading on June 29, 2010. Our initial public offering was priced at approximately $1.13 per share on June 28, 2010 as adjusted to give effect to the three-for-one stock split effected in the form of a stock dividend in August 2022 (the "2022 Stock Split") and the five-for-one stock split effected in the form of a stock dividend in August 2020 (the "2020 Stock Split").

Holders

As of January 22, 2024, there were 9,300 holders of record of our common stock. A substantially greater number of holders of our common stock are "street name" or beneficial holders, whose shares are held by banks, brokers and other financial institutions.

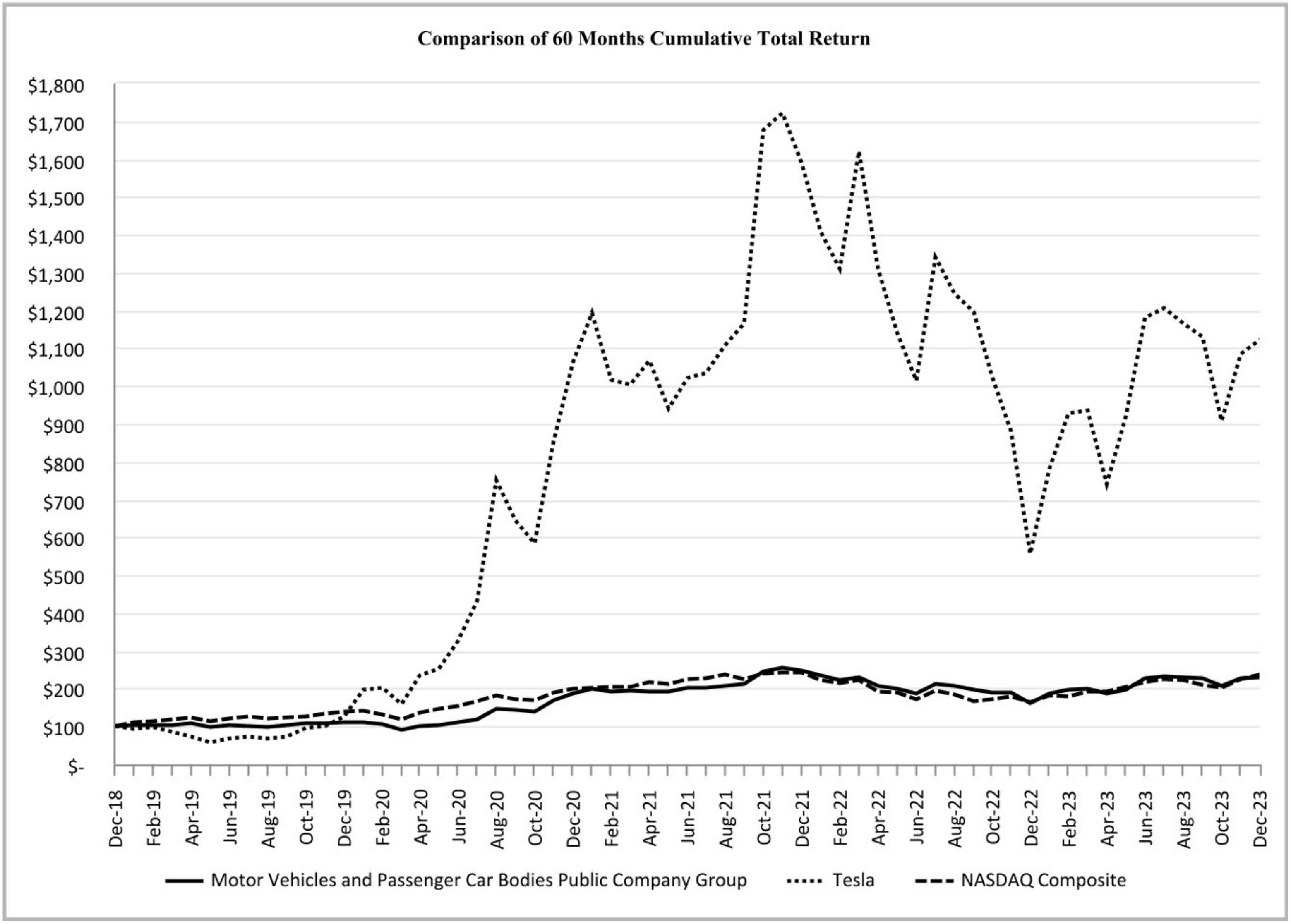
Dividend Policy

We have never declared or paid cash dividends on our common stock. We currently do not anticipate paying any cash dividends in the foreseeable future. Any future determination to declare cash dividends will be made at the discretion of our board of directors, subject to applicable laws, and will depend on our financial condition, results of operations, capital requirements, general business conditions and other factors that our board of directors may deem relevant.

Stock Performance Graph

This performance graph shall not be deemed "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, as amended (the "Exchange Act"), or incorporated by reference into any filing of Tesla, Inc. under the Securities Act of 1933, as amended (the "Securities Act"), or the Exchange Act, except as shall be expressly set forth by specific reference in such filing.

The following graph shows a comparison, from January 1, 2019 through December 31, 2023, of the cumulative total return on our common stock, The NASDAQ Composite Index and a group of all public companies sharing the same SIC code as us, which is SIC code 3711, "Motor Vehicles and Passenger Car Bodies" (Motor Vehicles and Passenger Car Bodies Public Company Group). Such returns are based on historical results and are not intended to suggest future performance. Data for The NASDAQ Composite Index and the Motor Vehicles and Passenger Car Bodies Public Company Group assumes an investment of $100 on January 1, 2019 and reinvestment of dividends. We have never declared or paid cash dividends on our common stock nor do we anticipate paying any such cash dividends in the foreseeable future.



Unregistered Sales of Equity Securities and Use of Proceeds

None.

Purchases of Equity Securities by the Issuer and Affiliated Purchasers

None.

ITEM 6. [RESERVED]

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

The following discussion and analysis should be read in conjunction with the consolidated financial statements and the related notes included elsewhere in this Annual Report on Form 10-K. For further discussion of our products and services, technology and competitive strengths, refer to Item 1- Business. For discussion related to changes in financial condition and the results of operations for fiscal year 2022-related items, refer to Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations in our Annual Report on Form 10-K for fiscal year 2022, which was filed with the Securities and Exchange Commission on January 31, 2023.

Overview and 2023 Highlights

Our mission is to accelerate the world's transition to sustainable energy. We design, develop, manufacture, lease and sell high-performance fully electric vehicles, solar energy generation systems and energy storage products. We also offer maintenance, installation, operation, charging, insurance, financial and other services related to our products. Additionally, we are increasingly focused on products and services based on artificial intelligence, robotics and automation.

In 2023, we produced 1,845,985 consumer vehicles and delivered 1,808,581 consumer vehicles. We are currently focused on increasing vehicle production, capacity and delivery capabilities, reducing costs, improving and developing our vehicles and battery technologies, vertically integrating and localizing our supply chain, improving and further deploying our FSD capabilities, increasing the affordability and efficiency of our vehicles, bringing new products to market and expanding our global infrastructure, including our service and charging infrastructure.

In 2023, we deployed 14.72 GWh of energy storage products and 223 megawatts of solar energy systems. We are currently focused on ramping production of energy storage products, improving our Solar Roof installation capability and efficiency, and increasing market share of retrofit solar energy systems.

In 2023, we recognized total revenues of $96.77 billion, representing an increase of $15.31 billion, compared to the prior year. We continue to ramp production, build new manufacturing capacity and expand our operations to enable increased deliveries and deployments of our products, and invest in research and development to accelerate our AI, software and fleet-based profits for further revenue growth.

In 2023, our net income attributable to common stockholders was $15.00 billion, representing a favorable change of $2.44 billion, compared to the prior year. This included a one-time non-cash tax benefit of $5.93 billion for the release of valuation allowance on certain deferred tax assets. We continue to focus on further cost reductions and operational efficiencies while maximizing delivery volumes.

We ended 2023 with $29.09 billion in cash and cash equivalents and investments, representing an increase of $6.91 billion from the end of 2022. Our cash flows provided by operating activities in 2023 and 2022 were $13.26 billion and $14.72 billion, respectively, representing a decrease of $1.47 billion. Capital expenditures amounted to $8.90 billion in 2023, compared to $7.16 billion in 2022, representing an increase of $1.74 billion. Sustained growth has allowed our business to generally fund itself, and we will continue investing in a number of capital-intensive projects and research and development in upcoming periods.

Management Opportunities, Challenges and Uncertainties and 2024 Outlook

Automotive—Production

The following is a summary of the status of production of each of our announced vehicle models in production and under development, as of the date of this Annual Report on Form 10-K:

Production Location	Vehicle Model(s)	Production Status
Fremont Factory	Model S / Model X	Active
	Model 3 / Model Y	Active
Gigafactory Shanghai	Model 3 / Model Y	Active
Gigafactory Berlin-Brandenburg	Model Y	Active
Gigafactory Texas	Model Y	Active
	Cybertruck	Active
Gigafactory Nevada	Tesla Semi	Pilot production
Various	Next Generation Platform	In development
TBD	Tesla Roadster	In development

We are focused on growing our manufacturing capacity, which includes capacity for manufacturing new vehicle models such as our Cybertruck and next generation platform, and ramping all of our production vehicles to their installed production capacities as well as increasing production rate and efficiency at our current factories. The next phase of production growth will depend on the continued ramp at our factories and the introduction of our next generation platform, as well as our ability to add to our available sources of battery cell supply by manufacturing our own cells that we are developing to have high-volume output, lower capital and production costs and longer range. Our goals are to improve vehicle performance, decrease production costs and increase affordability and customer awareness.

These plans are subject to uncertainties inherent in establishing and ramping manufacturing operations, which may be exacerbated by new product and manufacturing technologies we introduce, the number of concurrent international projects, any industry-wide component constraints, labor shortages and any future impact from events outside of our control. For example, during the third quarter of 2023, we experienced a sequential decline in production volumes due to pre-planned shutdowns for upgrades at various factories. Moreover, we have set ambitious technological targets with our plans for battery cells as well as for iterative manufacturing and design improvements for our vehicles with each new factory.

Automotive—Demand, Sales, Deliveries and Infrastructure

Our cost reduction efforts, cost innovation strategies, and additional localized procurement and manufacturing are key to our vehicles' affordability and have allowed us to competitively price our vehicles. We will also continue to generate demand and brand awareness by improving our vehicles' performance and functionality, including through products based on artificial intelligence such as Autopilot, FSD Capability, and other software features and delivering new vehicles, such as our Cybertruck. Moreover, we expect to continue to benefit from ongoing electrification of the automotive sector and increasing environmental regulations and initiatives.

However, we operate in a cyclical industry that is sensitive to political and regulatory uncertainty, including with respect to trade and the environment, all of which can be compounded by inflationary pressures, rising energy prices, interest rate fluctuations and the liquidity of enterprise customers. For example, inflationary pressures have increased across the markets in which we operate. In an effort to curb this trend, central banks in developed countries raised interest rates rapidly and substantially, impacting the affordability of vehicle lease and finance arrangements. Further, sales of vehicles in the automotive industry also tend to be cyclical in many markets, which may expose us to increased volatility as we expand and adjust our operations. Moreover, as additional competitors enter the marketplace and help bring the world closer to sustainable transportation, we will have to adjust and continue to execute well to maintain our momentum. Additionally, our suppliers' liquidity and allocation plans may be affected by current challenges in the North American automotive industry, which could reduce our access to components or result in unfavorable changes to cost. These macroeconomic and industry trends have had, and will likely continue to have, an impact on the pricing of, and order rate for our vehicles, and in turn our operating margin. Changes in government and economic incentives in relation to electric vehicles may also impact our sales. We will continue to adjust accordingly to such developments, and we believe our ongoing cost reduction, including improved production innovation and efficiency at our newest factories and lower logistics costs, and focus on operating leverage will continue to benefit us in relation to our competitors, while our new products will help enable future growth.

As our production increases, we must work constantly to similarly increase vehicle delivery capability so that it does not become a bottleneck on our total deliveries. We are also committed to reducing the percentage of vehicles delivered in the third month of each quarter, which will help to reduce the cost per vehicle. As we expand our manufacturing operations globally, we will also have to continue to increase and staff our delivery, servicing and charging infrastructure accordingly, maintain our vehicle reliability and optimize our Supercharger locations to ensure cost effectiveness and customer satisfaction. In particular, as other automotive manufacturers have announced their adoption of the North American Charging Standard (“NACS”) and agreements with us to utilize our Superchargers, we must correspondingly expand our network in order to ensure adequate availability to meet customer demands. We also remain focused on continued enhancements of the capability and efficiency of our servicing operations.

Energy Generation and Storage Demand, Production and Deployment

The long-term success of this business is dependent upon increasing margins through greater volumes. We continue to increase the production of our energy storage products to meet high levels of demand, including the construction of a new Megafactory in Shanghai and the ongoing ramp at our Megafactory in Lathrop, California. For Megapack, energy storage deployments can vary meaningfully quarter to quarter depending on the timing of specific project milestones. We remain committed to growing our retrofit solar energy business by offering a low-cost and simplified online ordering experience. In addition, we continue to seek to improve our installation capabilities and price efficiencies for Solar Roof. As these product lines grow, we will have to maintain adequate battery cell supply for our energy storage products and ensure the availability of qualified personnel, particularly skilled electricians, to support the ramp of Solar Roof.

Cash Flow and Capital Expenditure Trends

Our capital expenditures are typically difficult to project beyond the short-term given the number and breadth of our core projects at any given time, and may further be impacted by uncertainties in future global market conditions. We are simultaneously ramping new products, building or ramping manufacturing facilities on three continents, piloting the development and manufacture of new battery cell technologies, expanding our Supercharger network and investing in autonomy and other artificial intelligence enabled training and products, and the pace of our capital spend may vary depending on overall priority among projects, the pace at which we meet milestones, production adjustments to and among our various products, increased capital efficiencies and the addition of new projects. Owing and subject to the foregoing as well as the pipeline of announced projects under development, all other continuing infrastructure growth and varying levels of inflation, we currently expect our capital expenditures to exceed $10.00 billion in 2024 and be between $8.00 to $10.00 billion in each of the following two fiscal years.

Our business has been consistently generating cash flow from operations in excess of our level of capital spend, and with better working capital management resulting in shorter days sales outstanding than days payable outstanding, our sales growth is also generally facilitating positive cash generation. We have and will continue to utilize such cash flows, among other things, to do more vertical integration, expand our product roadmap and provide financing options to our customers. At the same time, we are likely to see heightened levels of capital expenditures during certain periods depending on the specific pace of our capital-intensive projects and other potential variables such as rising material prices and increases in supply chain and labor expenses resulting from changes in global trade conditions and labor availability. Overall, we expect our ability to be self-funding to continue as long as macroeconomic factors support current trends in our sales.

Critical Accounting Policies and Estimates

The consolidated financial statements are prepared in accordance with accounting principles generally accepted in the U.S. (“GAAP”). The preparation of the consolidated financial statements requires us to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenues, costs and expenses and related disclosures. We base our estimates on historical experience, as appropriate, and on various other assumptions that we believe to be reasonable under the circumstances. Changes in the accounting estimates are reasonably likely to occur from period to period. Accordingly, actual results could differ significantly from the estimates made by our management. We evaluate our estimates and assumptions on an ongoing basis. To the extent that there are material differences between these estimates and actual results, our future financial statement presentation, financial condition, results of operations and cash flows may be affected.

The estimates used for, but not limited to, determining significant economic incentive for resale value guarantee arrangements, sales return reserves, the collectability of accounts and financing receivables, inventory valuation, warranties, fair value of long-lived assets, goodwill, fair value of financial instruments, fair value and residual value of operating lease vehicles and solar energy systems subject to leases could be impacted. We have assessed the impact and are not aware of any specific events or circumstances that required an update to our estimates and assumptions or materially affected the carrying value of our assets or liabilities as of the date of issuance of this Annual Report on Form 10-K. These estimates may change as new events occur and additional information is obtained. Actual results could differ materially from these estimates under different assumptions or conditions.

Revenue Recognition

Automotive Sales

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, and specific other features and services that meet the definition of a performance obligation under Accounting Standards Codification ("ASC") 606, *Revenue from Contracts with Customers* ("ASC 606"), including access to our FSD Capability features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. We recognize revenue on automotive sales upon delivery to the customer, which is when the control of a vehicle transfers. Payments are typically received at the point control transfers or in accordance with payment terms customary to the business, except sales we finance for which payments are collected over the contractual loan term. We also recognize a sales return reserve based on historical experience plus consideration for expected future market values, when we offer resale value guarantees or similar buyback terms. Other features and services such as access to our internet connectivity, unlimited free Supercharging and over-the-air software updates are provisioned upon control transfer of a vehicle and recognized over time on a straight-line basis as we have a stand-ready obligation to deliver such services to the customer. Other limited free Supercharging incentives are recognized based on actual usage or expiration, whichever is earlier. We recognize revenue related to these other features and services over the performance period, which is generally the expected ownership life of the vehicle. Revenue related to FSD Capability features is recognized when functionality is delivered to the customer and their ongoing maintenance is recognized over time. For our obligations related to automotive sales, we estimate standalone selling price by considering costs used to develop and deliver the service, third-party pricing of similar options and other information that may be available.

Inventory Valuation

Inventories are stated at the lower of cost or net realizable value. Cost is computed using standard cost for vehicles and energy products, which approximates actual cost on a first-in, first-out basis. We record inventory write-downs for excess or obsolete inventories based upon assumptions about current and future demand forecasts. If our inventory on-hand is in excess of our future demand forecast, the excess amounts are written-off.

We also review our inventory to determine whether its carrying value exceeds the net amount realizable upon the ultimate sale of the inventory. This requires us to determine the estimated selling price of our vehicles less the estimated cost to convert the inventory on-hand into a finished product. Once inventory is written-down, a new, lower cost basis for that inventory is established and subsequent changes in facts and circumstances do not result in the restoration or increase in that newly established cost basis.

Should our estimates of future selling prices or production costs change, additional and potentially material write-downs may be required. A small change in our estimates may result in a material charge to our reported financial results.

Warranties

We provide a manufacturer's warranty on all new and used vehicles and a warranty on the installation and components of the energy generation and storage systems we sell for periods typically between 10 to 25 years. We accrue a warranty reserve for the products sold by us, which includes our best estimate of the projected costs to repair or replace items under warranties and recalls if identified. These estimates are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims. These estimates are inherently uncertain and changes to our historical or projected warranty experience may cause material changes to the warranty reserve in the future. The warranty reserve does not include projected warranty costs associated with our vehicles subject to operating lease accounting and our solar energy systems under lease contracts or PPAs, as the costs to repair these warranty claims are expensed as incurred. The portion of the warranty reserve expected to be incurred within the next 12 months is included within Accrued liabilities and other, while the remaining balance is included within Other long-term liabilities on the consolidated balance sheets. For liabilities that we are entitled to receive indemnification from our suppliers, we record receivables for the contractually obligated amounts on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Warranty expense is recorded as a component of Cost of revenues in the consolidated statements of operations. Due to the magnitude of our automotive business, our accrued warranty balance is primarily related to our automotive segment.

Stock-Based Compensation

We use the fair value method of accounting for our stock options and restricted stock units ("RSUs") granted to employees and for our employee stock purchase plan (the "ESPP") to measure the cost of employee services received in exchange for the stock-based awards. The fair value of stock option awards with only service and/or performance conditions is estimated on the grant or offering date using the Black-Scholes option-pricing model. The Black-Scholes option-pricing model requires inputs such as the risk-free interest rate, expected term and expected volatility. These inputs are subjective and generally require significant judgment. The fair value of RSUs is measured on the grant date based on the closing fair market value of our common stock. The resulting cost is recognized over the period during which an employee is required to provide service in exchange for the awards, usually the vesting period, which is generally four years for stock options and RSUs and six months for the ESPP. Stock-based compensation expense is recognized on a straight-line basis, net of actual forfeitures in the period.

For performance-based awards, stock-based compensation expense is recognized over the expected performance achievement period of individual performance milestones when the achievement of each individual performance milestone becomes probable.

As we accumulate additional employee stock-based awards data over time and as we incorporate market data related to our common stock, we may calculate significantly different volatilities and expected lives, which could materially impact the valuation of our stock-based awards and the stock-based compensation expense that we will recognize in future periods. Stock-based compensation expense is recorded in Cost of revenues, Research and development expense and Selling, general and administrative expense in the consolidated statements of operations.

Income Taxes

We are subject to income taxes in the U.S. and in many foreign jurisdictions. Significant judgment is required in determining our provision for income taxes, our deferred tax assets and liabilities and any valuation allowance recorded against our net deferred tax assets that are not more likely than not to be realized. We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. In completing our assessment of realizability of our deferred tax assets, we consider our history of income (loss) measured at pre-tax income (loss) adjusted for permanent book-tax differences on a jurisdictional basis, volatility in actual earnings, excess tax benefits related to stock-based compensation in recent prior years, and impacts of the timing of reversal of existing temporary differences. We also rely on our assessment of the Company's projected future results of business operations, including uncertainty in future operating results relative to historical results, volatility in the market price of our common stock and its performance over time, variable macroeconomic conditions impacting our ability to forecast future taxable income, and changes in business that may affect the existence and magnitude of future taxable income. Our valuation allowance assessment is based on our best estimate of future results considering all available information.

Furthermore, significant judgment is required in evaluating our tax positions. In the ordinary course of business, there are many transactions and calculations for which the ultimate tax settlement is uncertain. As a result, we recognize the effect of this uncertainty on our tax attributes or taxes payable based on our estimates of the eventual outcome. These effects are recognized when, despite our belief that our tax return positions are supportable, we believe that it is more likely than not that some of those positions may not be fully sustained upon review by tax authorities. We are required to file income tax returns in the U.S. and various foreign jurisdictions, which requires us to interpret the applicable tax laws and regulations in effect in such jurisdictions. Such returns are subject to audit by the various federal, state and foreign taxing authorities, who may disagree with respect to our tax positions. We believe that our consideration is adequate for all open audit years based on our assessment of many factors, including past experience and interpretations of tax law. We review and update our estimates in light of changing facts and circumstances, such as the closing of a tax audit, the lapse of a statute of limitations or a change in estimate. To the extent that the final tax outcome of these matters differs from our expectations, such differences may impact income tax expense in the period in which such determination is made.

Results of Operations

Revenues

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Automotive sales	$ 78,509	$ 67,210	$ 44,125	$ 11,299	17 %	$ 23,085	52 %
Automotive regulatory credits	1,790	1,776	1,465	14	1 %	311	21 %
Automotive leasing	2,120	2,476	1,642	(356)	(14)%	834	51 %
Total automotive revenues	82,419	71,462	47,232	10,957	15 %	24,230	51 %
Services and other	8,319	6,091	3,802	2,228	37 %	2,289	60 %
Total automotive & services and other segment revenue	90,738	77,553	51,034	13,185	17 %	26,519	52 %
Energy generation and storage segment revenue	6,035	3,909	2,789	2,126	54 %	1,120	40 %
Total revenues	$ 96,773	$ 81,462	$ 53,823	$ 15,311	19 %	$ 27,639	51 %

Automotive & Services and Other Segment

Automotive sales revenue includes revenues related to cash and financing deliveries of new Model S, Model X, Semi, Model 3, Model Y, and Cybertruck vehicles, including access to our FSD Capability features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. These deliveries are vehicles that are not subject to lease accounting.

Automotive regulatory credits includes sales of regulatory credits to other automotive manufacturers. Our revenue from automotive regulatory credits is directly related to our new vehicle production, sales and pricing negotiated with our customers. We monetize them proactively as new vehicles are sold based on standing arrangements with buyers of such credits, typically as close as possible to the production and delivery of the vehicle or changes in regulation impacting the credits.

Automotive leasing revenue includes the amortization of revenue for vehicles under direct operating lease agreements. Additionally, automotive leasing revenue includes direct sales-type leasing programs where we recognize all revenue associated with the sales-type lease upon delivery to the customer.

Services and other revenue consists of sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise.

2023 compared to 2022

Automotive sales revenue increased $11.30 billion, or 17%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022, primarily due to an increase of 473,382 combined Model 3 and Model Y cash deliveries from production ramping of Model Y globally. The increase was partially offset by a lower average selling price on our vehicles driven by overall price reductions year over year, sales mix, and a negative impact from the United States dollar strengthening against other foreign currencies in the year ended December 31, 2023 compared to the prior year.

Automotive regulatory credits revenue increased $14 million, or 1%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022.

Automotive leasing revenue decreased $356 million, or 14%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The decrease was primarily due to a decrease in direct sales-type leasing revenue driven by lower deliveries year over year, partially offset by an increase from our growing direct operating lease portfolio.

Services and other revenue increased $2.23 billion, or 37%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The increase was primarily due to higher used vehicle revenue driven by increases in volume, body shop and part sales revenue, non-warranty maintenance services revenue, paid Supercharging revenue and insurance services revenue, all of which are primarily attributable to our growing fleet. The increases were partially offset by a decrease in the average selling price of used vehicles.

Energy Generation and Storage Segment

Energy generation and storage revenue includes sales and leasing of solar energy generation and energy storage products, financing of solar energy generation products, services related to such products and sales of solar energy systems incentives.

2023 compared to 2022

Energy generation and storage revenue increased $2.13 billion, or 54%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The increase was primarily due to an increase in deployments of Megapack.

Cost of Revenues and Gross Margin

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Cost of revenues							
Automotive sales	$ 65,121	$ 49,599	$ 32,415	$ 15,522	31 %	$ 17,184	53 %
Automotive leasing	1,268	1,509	978	(241)	(16)%	531	54 %
Total automotive cost of revenues	66,389	51,108	33,393	15,281	30 %	17,715	53 %
Services and other	7,830	5,880	3,906	1,950	33 %	1,974	51 %
Total automotive & services and other segment cost of revenues	74,219	56,988	37,299	17,231	30 %	19,689	53 %
Energy generation and storage segment	4,894	3,621	2,918	1,273	35 %	703	24 %
Total cost of revenues	$ 79,113	$ 60,609	$ 40,217	$ 18,504	31 %	$ 20,392	51 %
Gross profit total automotive	$ 16,030	$ 20,354	$ 13,839				
Gross margin total automotive	19.4 %	28.5 %	29.3 %				
Gross profit total automotive & services and other segment	$ 16,519	$ 20,565	$ 13,735				
Gross margin total automotive & services and other segment	18.2 %	26.5 %	26.9 %				
Gross profit energy generation and storage segment	$ 1,141	$ 288	$ (129)				
Gross margin energy generation and storage segment	18.9 %	7.4 %	(4.6)%				
Total gross profit	$ 17,660	$ 20,853	$ 13,606				
Total gross margin	18.2 %	25.6 %	25.3 %				

Automotive & Services and Other Segment

Cost of automotive sales revenue includes direct and indirect materials, labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, shipping and logistic costs, vehicle connectivity costs, FSD ongoing maintenance costs, allocations of electricity and infrastructure costs related to our Supercharger network and reserves for estimated warranty expenses. Cost of automotive sales revenues also includes adjustments to warranty expense and charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of automotive sales revenue benefits from manufacturing credits earned.

Cost of automotive leasing revenue includes the depreciation of operating lease vehicles, cost of goods sold associated with direct sales-type leases and warranty expense related to leased vehicles.

Costs of services and other revenue includes cost of used vehicles including refurbishment costs, costs associated with providing non-warranty after-sales services, costs associated with our body shops and part sales, costs of paid Supercharging, costs to provide vehicle insurance and costs for retail merchandise.

2023 compared to 2022

Cost of automotive sales revenue increased $15.52 billion, or 31%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. Cost of automotive sales revenue increased in line with the change in deliveries year over year, as discussed above. The increase was partially offset by a decrease in the average combined cost per unit of our vehicles primarily due to sales mix, lower inbound freight, a decrease in material costs and lower manufacturing costs from better fixed cost absorption. Our costs of revenue were also positively impacted by the United States dollar strengthening against our foreign currencies as compared to the prior periods and by the IRA manufacturing credits earned during the current year.

Cost of automotive leasing revenue decreased $241 million, or 16%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The decrease was primarily due to a decrease in direct sales-type leasing cost of revenue driven by lower deliveries year over year.

Cost of services and other revenue increased $1.95 billion, or 33%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The increase was generally in line with the changes in services and other revenue as discussed above.

Gross margin for total automotive decreased from 28.5% to 19.4% in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The decrease was primarily due to a lower average selling price on our vehicles partially offset by the favorable change in our average combined cost per unit of our vehicles and IRA manufacturing credits earned as discussed above.

Gross margin for total automotive & services and other segment decreased from 26.5% to 18.2% in the year ended December 31, 2023 as compared to the year ended December 31, 2022, primarily due to the automotive gross margin decrease discussed above.

Energy Generation and Storage Segment

Cost of energy generation and storage revenue includes direct and indirect material and labor costs, warehouse rent, freight, warranty expense, other overhead costs and amortization of certain acquired intangible assets. Cost of energy generation and storage revenue also includes charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of energy generation and storage revenue benefits from manufacturing credits earned. In agreements for solar energy systems and PPAs where we are the lessor, the cost of revenue is primarily comprised of depreciation of the cost of leased solar energy systems, maintenance costs associated with those systems and amortization of any initial direct costs.

2023 compared to 2022

Cost of energy generation and storage revenue increased $1.27 billion, or 35%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022, in line with the increase in Megapack deployments year over year, as discussed above. This increase was partially offset by an improvement in production ramping that drove down the average cost per MWh of Megapack as well as IRA manufacturing credits earned during the current year.

Gross margin for energy generation and storage increased from 7.4% to 18.9% in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The increase was driven by an improvement in our Megapack gross margin from lower average cost per MWh and a higher proportion of Megapack, which operated at a higher gross margin, within the segment as compared to the prior year periods. Additionally, there was a margin benefit from IRA manufacturing credits earned.

Research and Development Expense

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Research and development	$ 3,969	$ 3,075	$ 2,593	$ 894	29 %	$ 482	19 %
As a percentage of revenues	4 %	4 %	5 %				

Research and development ("R&D") expenses consist primarily of personnel costs for our teams in engineering and research, manufacturing engineering and manufacturing test organizations, prototyping expense, contract and professional services and amortized equipment expense.

R&D expenses increased $894 million, or 29%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The overall increase was primarily driven by additional costs in the current year related to the pre-production phase for Cybertruck, AI and other programs.

R&D expenses as a percentage of revenue stayed consistent at 4% in the year ended December 31, 2023 as compared to the year ended December 31, 2022. Our R&D expenses have increased proportionately with total revenues as we continue to expand our product roadmap and technologies.

Selling, General and Administrative Expense

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Selling, general and administrative	$ 4,800	$ 3,946	$ 4,517	$ 854	22 %	$ (571)	(13)%
As a percentage of revenues	5 %	5 %	8 %				

Selling, general and administrative ("SG&A") expenses generally consist of personnel and facilities costs related to our stores, marketing, sales, executive, finance, human resources, information technology and legal organizations, as well as fees for professional and contract services and litigation settlements.

SG&A expenses increased $854 million, or 22%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. This was driven by a $447 million increase in employee and labor costs primarily from increased headcount, including professional services and a $363 million increase in facilities related expenses.

Restructuring and Other

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Restructuring and other	$ —	$ 176	$ (27)	$ (176)	(100)%	$ 203	Not meaningful

During the year ended December 31, 2022, we recorded an impairment loss of $204 million as well as realized gains of $64 million in connection with converting our holdings of digital assets into fiat currency. We also recorded other expenses of $36 million during the second quarter of the year ended December 31, 2022, related to employee terminations.

Interest Income

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Interest income	$ 1,066	$ 297	$ 56	$ 769	259 %	$ 241	430 %

Interest income increased $769 million, or 259%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. This increase was primarily due to higher interest earned on our cash and cash equivalents and short-term investments in the year ended December 31, 2023 as compared to the prior year due to rising interest rates and our increasing portfolio balance.

Other Income (Expense), Net

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Other income (expense), net	$ 172	$ (43)	$ 135	$ 215	Not meaningful	$ (178)	Not meaningful

Other income (expense), net, consists primarily of foreign exchange gains and losses related to our foreign currency-denominated monetary assets and liabilities. We expect our foreign exchange gains and losses will vary depending upon movements in the underlying exchange rates.

Other income, net, changed favorably by $215 million in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The favorable change was primarily due to fluctuations in foreign currency exchange rates on our intercompany balances.

(Benefit from) Provision for Income Taxes

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
(Benefit from) provision for income taxes	$ (5,001)	$ 1,132	$ 699	$ (6,133)	Not meaningful	$ 433	62 %
Effective tax rate	(50)%	8 %	11 %				

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. As of December 31, 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which is objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that our U.S. federal and certain state deferred tax assets are realizable. As such, we released $6.54 billion of our valuation allowance associated with the U.S. federal and state deferred tax assets, with the exception of our California deferred tax assets. Approximately $5.93 billion of the total valuation allowance release was related to deferred tax assets to be realized in the future years and the remainder benefited us during the year ended December 31, 2023. We continue to maintain a full valuation allowance against our California deferred tax assets as of December 31, 2023, because we concluded they are not more likely than not to be realized as we expect our California deferred tax assets generation in future years to exceed our ability to use these deferred tax assets.

Our (benefit from) provision for income taxes changed by $6.13 billion in the year ended December 31, 2023 as compared to the year ended December 31, 2022, primarily due to the release of $6.54 billion of our valuation allowance associated with the U.S. federal and certain state deferred tax assets.

Our effective tax rate changed from an expense of 8% to a benefit of 50% in the year ended December 31, 2023 as compared to the year ended December 31, 2022, primarily due to the release of the valuation allowance regarding our U.S. federal and certain state deferred tax assets.

See Note 14, *Income Taxes*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Liquidity and Capital Resources

We expect to continue to generate net positive operating cash flow as we have done in the last five fiscal years. The cash we generate from our core operations enables us to fund ongoing operations and production, our research and development projects for new products and technologies including our proprietary battery cells, additional manufacturing ramps at existing manufacturing facilities, the construction of future factories, and the continued expansion of our retail and service locations, body shops, Mobile Service fleet, Supercharger, including to support NACS, energy product installation capabilities and autonomy and other artificial intelligence enabled products.

In addition, because a large portion of our future expenditures will be to fund our growth, we expect that if needed we will be able to adjust our capital and operating expenditures by operating segment. For example, if our near-term manufacturing operations decrease in scale or ramp more slowly than expected, including due to global economic or business conditions, we may choose to correspondingly slow the pace of our capital expenditures. Finally, we continually evaluate our cash needs and may decide it is best to raise additional capital or seek alternative financing sources to fund the rapid growth of our business, including through drawdowns on existing or new debt facilities or financing funds. Conversely, we may also from time to time determine that it is in our best interests to voluntarily repay certain indebtedness early.

Accordingly, we believe that our current sources of funds will provide us with adequate liquidity during the 12-month period following December 31, 2023, as well as in the long-term.

See the sections below for more details regarding the material requirements for cash in our business and our sources of liquidity to meet such needs.

Material Cash Requirements

From time to time in the ordinary course of business, we enter into agreements with vendors for the purchase of components and raw materials to be used in the manufacture of our products. However, due to contractual terms, variability in the precise growth curves of our development and production ramps, and opportunities to renegotiate pricing, we generally do not have binding and enforceable purchase orders under such contracts beyond the short-term, and the timing and magnitude of purchase orders beyond such period is difficult to accurately project.

As discussed in and subject to the considerations referenced in Part II, Item 7, Management's Discussion and Analysis of Financial Condition and Results of Operations—Management Opportunities, Challenges and Uncertainties and 2023 Outlook—Cash Flow and Capital Expenditure Trends in this Annual Report on Form 10-K, we currently expect our capital expenditures to support our projects globally to exceed $10.00 billion in 2024 and be between $8.00 to $10.00 billion in each of the following two fiscal years. In connection with our operations at Gigafactory New York, we have an agreement to spend or incur $5.00 billion in combined capital, operational expenses, costs of goods sold and other costs in the State of New York through December 31, 2029 (pursuant to a deferral of our required timelines to meet such obligations that was granted in April 2021, and which was memorialized in an amendment to our agreement with the SUNY Foundation in August 2021). For details regarding these obligations, refer to Note 15, *Commitments and Contingencies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

As of December 31, 2023, we and our subsidiaries had outstanding $4.68 billion in aggregate principal amount of indebtedness, of which $1.98 billion is scheduled to become due in the succeeding 12 months. As of December 31, 2023, our total minimum lease payments was $5.96 billion, of which $1.31 billion is due in the succeeding 12 months. For details regarding our indebtedness and lease obligations, refer to Note 11, *Debt*, and Note 12, *Leases*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

Sources and Conditions of Liquidity

Our sources to fund our material cash requirements are predominantly from our deliveries and servicing of new and used vehicles, sales and installations of our energy storage products and solar energy systems, proceeds from debt facilities and proceeds from equity offerings, when applicable.

As of December 31, 2023, we had $16.40 billion and $12.70 billion of cash and cash equivalents and short-term investments, respectively. Balances held in foreign currencies had a U.S. dollar equivalent of $4.43 billion and consisted primarily of Chinese yuan and euros. We had $5.03 billion of unused committed credit amounts as of December 31, 2023. For details regarding our indebtedness, refer to Note 11, *Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

We continue adapting our strategy to meet our liquidity and risk objectives, such as investing in U.S. government securities and other investments, to do more vertical integration, expand our product roadmap and provide financing options to our customers.

Summary of Cash Flows

(Dollars in millions)	Year Ended December 31, 2023	2022	2021
Net cash provided by operating activities	$ 13,256	$ 14,724	$ 11,497
Net cash used in investing activities	$ (15,584)	$ (11,973)	$ (7,868)
Net cash provided by (used in) financing activities	$ 2,589	$ (3,527)	$ (5,203)

Cash Flows from Operating Activities

Our cash flows from operating activities are significantly affected by our cash investments to support the growth of our business in areas such as research and development and selling, general and administrative and working capital. Our operating cash inflows include cash from vehicle sales and related servicing, customer lease and financing payments, customer deposits, cash from sales of regulatory credits and energy generation and storage products, and interest income on our cash and investments portfolio. These cash inflows are offset by our payments to suppliers for production materials and parts used in our manufacturing process, operating expenses, operating lease payments and interest payments on our financings.

Net cash provided by operating activities decreased by $1.47 billion to $13.26 billion during the year ended December 31, 2023 from $14.72 billion during the year ended December 31, 2022. This decrease was primarily due to the decrease in net income excluding non-cash expenses, gains and losses of $2.93 billion, partially offset by favorable changes in net operating assets and liabilities of $1.46 billion.

Cash Flows from Investing Activities

Cash flows from investing activities and their variability across each period related primarily to capital expenditures, which were $8.90 billion for the year ended December 31, 2023 and $7.16 billion for the year ended December 31, 2022, mainly for global factory expansion and machinery and equipment as we expand our product roadmap. We also purchased $6.62 billion and $5.81 billion of investments, net of proceeds from maturities and sales, for the year ended December 31, 2023 and 2022, respectively. Additionally, proceeds from sales of digital assets was $936 million in the year ended December 31, 2022.

Cash Flows from Financing Activities

Net cash from financing activities changed by $6.12 billion to $2.59 billion net cash provided by financing activities during the year ended December 31, 2023 from $3.53 billion net cash used in financing activities during the year ended December 31, 2022. The change was primarily due to a $3.93 billion increase in proceeds from issuances of debt and a $2.01 billion decrease in repayments of debt. See Note 11, *Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details regarding our debt obligations.

Recent Accounting Pronouncements

See Note 2, *Summary of Significant Accounting Policies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

Foreign Currency Risk

We transact business globally in multiple currencies and hence have foreign currency risks related to our revenue, costs of revenue and operating expenses denominated in currencies other than the U.S. dollar (primarily the Chinese yuan and euro in relation to our current year operations). In general, we are a net receiver of currencies other than the U.S. dollar for our foreign subsidiaries. Accordingly, changes in exchange rates affect our operating results as expressed in U.S. dollars as we do not typically hedge foreign currency risk.

We have also experienced, and will continue to experience, fluctuations in our net income as a result of gains (losses) on the settlement and the re-measurement of monetary assets and liabilities denominated in currencies that are not the local currency (primarily consisting of our intercompany and cash and cash equivalents balances).

We considered the historical trends in foreign currency exchange rates and determined that it is reasonably possible that adverse changes in foreign currency exchange rates of 10% for all currencies could be experienced in the near-term. These changes were applied to our total monetary assets and liabilities denominated in currencies other than our local currencies at the balance sheet date to compute the impact these changes would have had on our net income before income taxes. These changes would have resulted in a gain or loss of $1.01 billion at December 31, 2023 and $473 million at December 31, 2022, assuming no foreign currency hedging.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

Index to Consolidated Financial Statements

Report of Independent Registered Public Accounting Firm

To the Board of Directors and Stockholders of Tesla, Inc.

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of Tesla, Inc. and its subsidiaries (the "Company") as of December 31, 2023 and 2022, and the related consolidated statements of operations, of comprehensive income, of redeemable noncontrolling interests and equity and of cash flows for each of the three years in the period ended December 31, 2023, including the related notes (collectively referred to as the "consolidated financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control - Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control - Integrated Framework (2013) issued by the COSO.

Changes in Accounting Principles

As discussed in Note 2 to the consolidated financial statements, the Company changed the manner in which it accounts for convertible debt in 2021.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in Management's Report on Internal Control over Financial Reporting appearing under Item 9A. Our responsibility is to express opinions on the Company's consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company’s internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company’s internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company’s assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that (i) relates to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Automotive Warranty Reserve

As described in Note 2 to the consolidated financial statements, total accrued warranty, which primarily relates to the automotive segment, was $5,152 million as of December 31, 2023. The Company provides a manufacturer’s warranty on all new and used Tesla vehicles. A warranty reserve is accrued for these products sold, which includes management’s best estimate of the projected costs to repair or replace items under warranty and recalls if identified. These estimates are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims.

The principal considerations for our determination that performing procedures relating to the automotive warranty reserve is a critical audit matter are the significant judgment by management in determining the automotive warranty reserve for certain Tesla vehicle models; this in turn led to significant auditor judgment, subjectivity, and effort in performing procedures to evaluate management’s significant assumptions related to the nature, frequency and costs of future claims for certain Tesla vehicle models, and the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to management’s estimate of the automotive warranty reserve for certain Tesla vehicle models, including controls over management’s significant assumptions related to the nature, frequency and costs of future claims as well as the completeness and accuracy of actual claims incurred to date. These procedures also included, among others, performing one of the following: (i) testing management’s process for determining the automotive warranty reserve for certain Tesla vehicle models or (ii) developing an independent estimate of the automotive warranty reserve for certain Tesla vehicle models and comparing the independent estimate to management’s estimate to evaluate the reasonableness of the estimate. Testing management’s process involved evaluating the reasonableness of significant assumptions related to the nature and frequency of future claims and the related costs to repair or replace items under warranty. Evaluating the assumptions related to the nature and frequency of future claims and the related costs to repair or replace items under warranty involved evaluating whether the assumptions used were reasonable by performing a lookback analysis comparing prior period forecasted claims to actual claims incurred. Developing the independent estimate involved testing the completeness and accuracy of historical vehicle claims processed and testing that such claims were appropriately used by management in the estimation of future claims. Professionals with specialized skill and knowledge were used to assist in developing an independent estimate of the automotive warranty reserve for certain Tesla vehicle models and in evaluating the appropriateness of certain aspects of management’s significant assumptions related to the nature and frequency of future claims.

/s/ PricewaterhouseCoopers LLP

San Jose, California
January 26, 2024

We have served as the Company’s auditor since 2005.

Tesla, Inc.

Consolidated Balance Sheets
(in millions, except per share data)

	December 31, 2023	December 31, 2022
Assets		
Current assets		
Cash and cash equivalents	$ 16,398	$ 16,253
Short-term investments	12,696	5,932
Accounts receivable, net	3,508	2,952
Inventory	13,626	12,839
Prepaid expenses and other current assets	3,388	2,941
Total current assets	49,616	40,917
Operating lease vehicles, net	5,989	5,035
Solar energy systems, net	5,229	5,489
Property, plant and equipment, net	29,725	23,548
Operating lease right-of-use assets	4,180	2,563
Digital assets, net	184	184
Intangible assets, net	178	215
Goodwill	253	194
Deferred tax assets	6,733	328
Other non-current assets	4,531	3,865
Total assets	$ 106,618	$ 82,338
Liabilities		
Current liabilities		
Accounts payable	$ 14,431	$ 15,255
Accrued liabilities and other	9,080	8,205
Deferred revenue	2,864	1,747
Current portion of debt and finance leases	2,373	1,502
Total current liabilities	28,748	26,709
Debt and finance leases, net of current portion	2,857	1,597
Deferred revenue, net of current portion	3,251	2,804
Other long-term liabilities	8,153	5,330
Total liabilities	43,009	36,440
Commitments and contingencies (Note 15)		
Redeemable noncontrolling interests in subsidiaries	242	409
Equity		
Stockholders' equity		
Preferred stock; $0.001 par value; 100 shares authorized; no shares issued and outstanding	—	—
Common stock; $0.001 par value; 6,000 shares authorized; 3,185 and 3,164 shares issued and outstanding as of December 31, 2023 and 2022, respectively	3	3
Additional paid-in capital	34,892	32,177
Accumulated other comprehensive loss	(143)	(361)
Retained earnings	27,882	12,885
Total stockholders' equity	62,634	44,704
Noncontrolling interests in subsidiaries	733	785
Total liabilities and equity	$ 106,618	$ 82,338

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Operations
(in millions, except per share data)

	Year Ended December 31,		
	2023	**2022**	**2021**
Revenues			
Automotive sales	$ 78,509	$ 67,210	$ 44,125
Automotive regulatory credits	1,790	1,776	1,465
Automotive leasing	2,120	2,476	1,642
Total automotive revenues	82,419	71,462	47,232
Energy generation and storage	6,035	3,909	2,789
Services and other	8,319	6,091	3,802
Total revenues	96,773	81,462	53,823
Cost of revenues			
Automotive sales	65,121	49,599	32,415
Automotive leasing	1,268	1,509	978
Total automotive cost of revenues	66,389	51,108	33,393
Energy generation and storage	4,894	3,621	2,918
Services and other	7,830	5,880	3,906
Total cost of revenues	79,113	60,609	40,217
Gross profit	17,660	20,853	13,606
Operating expenses			
Research and development	3,969	3,075	2,593
Selling, general and administrative	4,800	3,946	4,517
Restructuring and other	—	176	(27)
Total operating expenses	8,769	7,197	7,083
Income from operations	8,891	13,656	6,523
Interest income	1,066	297	56
Interest expense	(156)	(191)	(371)
Other income (expense), net	172	(43)	135
Income before income taxes	9,973	13,719	6,343
(Benefit from) provision for income taxes	(5,001)	1,132	699
Net income	14,974	12,587	5,644
Net (loss) income attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	(23)	31	125
Net income attributable to common stockholders	$ 14,997	$ 12,556	$ 5,519
Net income per share of common stock attributable to common stockholders			
Basic	$ 4.73	$ 4.02	$ 1.87
Diluted	$ 4.30	$ 3.62	$ 1.63
Weighted average shares used in computing net income per share of common stock			
Basic	3,174	3,130	2,959
Diluted	3,485	3,475	3,386

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Comprehensive Income
(in millions)

	Year Ended December 31,		
	2023	2022	2021
Net income	$ 14,974	$ 12,587	$ 5,644
Other comprehensive income (loss):			
Foreign currency translation adjustment	198	(392)	(308)
Unrealized net gain (loss) on investments	16	(23)	(1)
Adjustment for net loss realized and included in net income	4	—	—
Comprehensive income	15,192	12,172	5,335
Less: Comprehensive (loss) income attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	(23)	31	125
Comprehensive income attributable to common stockholders	$ 15,215	$ 12,141	$ 5,210

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Redeemable Noncontrolling Interests and Equity
(in millions)

	Redeemable Noncontrolling Interests	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive Income (Loss)	(Accumulated Deficit) Retained Earnings	Total Stockholders' Equity	Noncontrolling Interests in Subsidiaries	Total Equity
		Shares	Amount						
Balance as of December 31, 2020	$ 604	2,879	$ 3	$ 27,260	$ 363	$ (5,401)	$ 22,225	$ 850	$ 23,075
Adjustments for prior periods from adopting ASU 2020-06	—	—	—	(474)	—	211	(263)	—	(263)
Exercises of conversion feature of convertible senior notes	—	2	—	6	—	—	6	—	6
Settlements of warrants	—	112	—	—	—	—	—	—	—
Issuance of common stock for equity incentive awards	—	107	—	707	—	—	707	—	707
Stock-based compensation	—	—	—	2,299	—	—	2,299	—	2,299
Contributions from noncontrolling interests	2	—	—	—	—	—	—	—	—
Distributions to noncontrolling interests	(66)	—	—	—	—	—	—	(106)	(106)
Buy-outs of noncontrolling interests	(15)	—	—	5	—	—	5	—	5
Net income	43	—	—	—	—	5,519	5,519	82	5,601
Other comprehensive loss	—	—	—	—	(309)	—	(309)	—	(309)
Balance as of December 31, 2021	$ 568	3,100	$ 3	$ 29,803	$ 54	$ 329	$ 30,189	$ 826	$ 31,015
Settlements of warrants	—	37	—	—	—	—	—	—	—
Issuance of common stock for equity incentive awards	—	27	—	541	—	—	541	—	541
Stock-based compensation	—	—	—	1,806	—	—	1,806	—	1,806
Distributions to noncontrolling interests	(46)	—	—	—	—	—	—	(113)	(113)
Buy-outs of noncontrolling interests	(11)	—	—	27	—	—	27	(61)	(34)
Net (loss) income	(102)	—	—	—	—	12,556	12,556	133	12,689
Other comprehensive loss	—	—	—	—	(415)	—	(415)	—	(415)
Balance as of December 31, 2022	$ 409	3,164	$ 3	$ 32,177	$ (361)	$ 12,885	$ 44,704	$ 785	$ 45,489
Issuance of common stock for equity incentive awards	—	21	—	700	—	—	700	—	700
Stock-based compensation	—	—	—	2,013	—	—	2,013	—	2,013
Distributions to noncontrolling interests	(32)	—	—	—	—	—	—	(108)	(108)
Buy-outs of noncontrolling interests	(39)	—	—	2	—	—	2	(17)	(15)
Net (loss) income	(96)	—	—	—	—	14,997	14,997	73	15,070
Other comprehensive income	—	—	—	—	218	—	218	—	218
Balance as of December 31, 2023	$ 242	3,185	$ 3	$ 34,892	$ (143)	$ 27,882	$ 62,634	$ 733	$ 63,367

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Cash Flows
(in millions)

	Year Ended December 31,		
	2023	2022	2021
Cash Flows from Operating Activities			
Net income	$ 14,974	$ 12,587	$ 5,644
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation, amortization and impairment	4,667	3,747	2,911
Stock-based compensation	1,812	1,560	2,121
Inventory and purchase commitments write-downs	463	177	140
Foreign currency transaction net unrealized (gain) loss	(144)	81	(55)
Deferred income taxes	(6,349)	(196)	(149)
Non-cash interest and other operating activities	81	340	245
Digital assets loss (gain), net	—	140	(27)
Changes in operating assets and liabilities:			
Accounts receivable	(586)	(1,124)	(130)
Inventory	(1,195)	(6,465)	(1,709)
Operating lease vehicles	(1,952)	(1,570)	(2,114)
Prepaid expenses and other assets	(2,652)	(3,713)	(1,540)
Accounts payable, accrued and other liabilities	2,605	8,029	5,367
Deferred revenue	1,532	1,131	793
Net cash provided by operating activities	13,256	14,724	11,497
Cash Flows from Investing Activities			
Purchases of property and equipment excluding finance leases, net of sales	(8,898)	(7,158)	(6,482)
Purchases of solar energy systems, net of sales	(1)	(5)	(32)
Purchases of digital assets	—	—	(1,500)
Proceeds from sales of digital assets	—	936	272
Purchase of intangible assets	—	(9)	—
Purchases of investments	(19,112)	(5,835)	(132)
Proceeds from maturities of investments	12,353	22	—
Proceeds from sales of investments	138	—	—
Receipt of government grants	—	76	6
Business combinations, net of cash acquired	(64)	—	—
Net cash used in investing activities	(15,584)	(11,973)	(7,868)
Cash Flows from Financing Activities			
Proceeds from issuances of debt	3,931	—	8,883
Repayments of debt	(1,351)	(3,364)	(14,167)
Collateralized lease repayments	—	—	(9)
Proceeds from exercises of stock options and other stock issuances	700	541	707
Principal payments on finance leases	(464)	(502)	(439)
Debt issuance costs	(29)	—	(9)
Proceeds from investments by noncontrolling interests in subsidiaries	—	—	2
Distributions paid to noncontrolling interests in subsidiaries	(144)	(157)	(161)
Payments for buy-outs of noncontrolling interests in subsidiaries	(54)	(45)	(10)
Net cash provided by (used in) financing activities	2,589	(3,527)	(5,203)
Effect of exchange rate changes on cash and cash equivalents and restricted cash	4	(444)	(183)
Net increase (decrease) in cash and cash equivalents and restricted cash	265	(1,220)	(1,757)
Cash and cash equivalents and restricted cash, beginning of period	16,924	18,144	19,901
Cash and cash equivalents and restricted cash, end of period	$ 17,189	$ 16,924	$ 18,144
Supplemental Non-Cash Investing and Financing Activities			
Acquisitions of property and equipment included in liabilities	$ 2,272	$ 2,148	$ 2,251
Supplemental Disclosures			
Cash paid during the period for interest, net of amounts capitalized	$ 126	$ 152	$ 266
Cash paid during the period for income taxes, net of refunds	$ 1,119	$ 1,203	$ 561

The accompanying notes are an integral part of these consolidated financial statements.

Note 1 – Overview

Tesla, Inc. (“Tesla”, the “Company”, “we”, “us” or “our”) was incorporated in the State of Delaware on July 1, 2003. We design, develop, manufacture, sell and lease high-performance fully electric vehicles and energy generation and storage systems, and offer services related to our products. Our Chief Executive Officer, as the chief operating decision maker (“CODM”), organizes our company, manages resource allocations and measures performance among two operating and reportable segments: (i) automotive and (ii) energy generation and storage.

Note 2 – Summary of Significant Accounting Policies

Principles of Consolidation

The accompanying consolidated financial statements have been prepared in conformity with GAAP and reflect our accounts and operations and those of our subsidiaries in which we have a controlling financial interest. In accordance with the provisions of ASC 810, *Consolidation* (“ASC 810”), we consolidate any variable interest entity (“VIE”) of which we are the primary beneficiary. We have formed VIEs with financing fund investors in the ordinary course of business in order to facilitate the funding and monetization of certain attributes associated with solar energy systems and leases under our direct vehicle leasing programs. The typical condition for a controlling financial interest ownership is holding a majority of the voting interests of an entity; however, a controlling financial interest may also exist in entities, such as VIEs, through arrangements that do not involve controlling voting interests. ASC 810 requires a variable interest holder to consolidate a VIE if that party has the power to direct the activities of the VIE that most significantly impact the VIE’s economic performance and the obligation to absorb losses of the VIE that could potentially be significant to the VIE or the right to receive benefits from the VIE that could potentially be significant to the VIE. We do not consolidate a VIE in which we have a majority ownership interest when we are not considered the primary beneficiary. We have determined that we are the primary beneficiary of all the VIEs (see Note 16*, Variable Interest Entity Arrangements*). We evaluate our relationships with all the VIEs on an ongoing basis to ensure that we continue to be the primary beneficiary. All intercompany transactions and balances have been eliminated upon consolidation.

Use of Estimates

The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenues, costs and expenses and related disclosures in the accompanying notes. The estimates used for, but not limited to, determining significant economic incentive for resale value guarantee arrangements, sales return reserves, income taxes, the collectability of accounts and finance receivables, inventory valuation, warranties, fair value of long-lived assets, goodwill, fair value of financial instruments, fair value and residual value of operating lease vehicles and solar energy systems subject to leases could be impacted. We have assessed the impact and are not aware of any specific events or circumstances that required an update to our estimates and assumptions or materially affected the carrying value of our assets or liabilities as of the date of issuance of this Annual Report on Form 10-K. These estimates may change as new events occur and additional information is obtained. Actual results could differ materially from these estimates under different assumptions or conditions.

Reclassifications

Certain prior period balances have been reclassified to conform to the current period presentation in the consolidated financial statements and the accompanying notes.

Revenue Recognition

Revenue by source

The following table disaggregates our revenue by major source (in millions):

	Year Ended December 31,		
	2023	2022	2021
Automotive sales	$ 78,509	$ 67,210	$ 44,125
Automotive regulatory credits	1,790	1,776	1,465
Energy generation and storage sales	5,515	3,376	2,279
Services and other	8,319	6,091	3,802
Total revenues from sales and services	94,133	78,453	51,671
Automotive leasing	2,120	2,476	1,642
Energy generation and storage leasing	520	533	510
Total revenues	$ 96,773	$ 81,462	$ 53,823

Automotive Segment

Automotive Sales

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, and specific other features and services that meet the definition of a performance obligation under ASC 606, including access to our FSD Capability features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. We recognize revenue on automotive sales upon delivery to the customer, which is when the control of a vehicle transfers. Payments are typically received at the point control transfers or in accordance with payment terms customary to the business, except sales we finance for which payments are collected over the contractual loan term. We also recognize a sales return reserve based on historical experience plus consideration for expected future market values, when we offer resale value guarantees or similar buyback terms. Other features and services such as access to our internet connectivity, unlimited free Supercharging and over-the-air software updates are provisioned upon control transfer of a vehicle and recognized over time on a straight-line basis as we have a stand-ready obligation to deliver such services to the customer. Other limited free Supercharging incentives are recognized based on actual usage or expiration, whichever is earlier. We recognize revenue related to these other features and services over the performance period, which is generally the expected ownership life of the vehicle. Revenue related to FSD Capability features is recognized when functionality is delivered to the customer and their ongoing maintenance is recognized over time. For our obligations related to automotive sales, we estimate standalone selling price by considering costs used to develop and deliver the service, third-party pricing of similar options and other information that may be available.

Any fees that are paid or payable by us to a customer’s lender when we arrange the financing are recognized as an offset against automotive sales revenue. Costs to obtain a contract mainly relate to commissions paid to our sales personnel for the sale of vehicles. As our contract costs related to automotive sales are typically fulfilled within one year, the costs to obtain a contract are expensed as incurred. Amounts billed to customers related to shipping and handling are classified as automotive sales revenue, and we have elected to recognize the cost for freight and shipping when control over vehicles, parts or accessories have transferred to the customer as an expense in cost of automotive sales revenue. Our policy is to exclude taxes collected from a customer from the transaction price of automotive contracts.

We offer resale value guarantees to our commercial banking partners in connection with certain vehicle leasing programs. Under these programs, we originate the lease with our end customer and immediately transfer the lease and the underlying vehicle to our commercial banking partner, with the transaction being accounted for as a sale under ASC 606. We receive upfront payment for the vehicle, do not bear casualty and credit risks during the lease term, and we provide a guarantee capped to a limit if they are unable to sell the vehicle at or above the vehicle's contract residual value at the end of the lease term. We estimate a guarantee liability in accordance with ASC 460, *Guarantees* and record it within other liabilities on our consolidated balance sheet. On a quarterly basis, we assess the estimated market value of vehicles sold under this program to determine whether there have been changes to the amount of expected resale value guarantee payments. As we accumulate more data related to the resale values of our vehicles or as market conditions change, there may be material changes to their estimated values. The total guarantee liability on vehicles sold under this program was immaterial as of December 31, 2023.

Deferred revenue related to the access to our FSD Capability features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates primarily on automotive sales consisted of the following (in millions):

	Year Ended December 31,	
	2023	**2022**
Deferred revenue— beginning of period	$ 2,913	$ 2,382
Additions	1,201	1,178
Net changes in liability for pre-existing contracts	17	(67)
Revenue recognized	(595)	(580)
Deferred revenue— end of period	$ 3,536	$ 2,913

Deferred revenue is equivalent to the total transaction price allocated to the performance obligations that are unsatisfied, or partially unsatisfied, as of the balance sheet date. Revenue recognized from the deferred revenue balance as of December 31, 2022 was $469 million for the year ended December 31, 2023. We had recognized revenue of $472 million from the deferred revenue balance as of December 31, 2021, for the year ended December 31, 2022, primarily related to the general FSD Capability feature release in North America in the fourth quarter of 2022. Of the total deferred revenue balance as of December 31, 2023, we expect to recognize $926 million of revenue in the next 12 months. The remaining balance will be recognized at the time of transfer of control of the product or over the performance period as discussed above in Automotive Sales.

We have been providing loans for financing our automotive deliveries in volume since fiscal year 2022. As of December 31, 2023 and 2022, we have recorded net financing receivables on the consolidated balance sheets, of which $242 million and $128 million, respectively, is recorded within Accounts receivable, net, for the current portion and $1.04 billion and $665 million, respectively, is recorded within Other non-current assets for the long-term portion.

Automotive Regulatory Credits

We earn tradable credits in the operation of our automotive business under various regulations related to ZEVs, greenhouse gas, fuel economy and clean fuel. We sell these credits to other regulated entities who can use the credits to comply with emission standards and other regulatory requirements.

Payments for automotive regulatory credits are typically received at the point control transfers to the customer, or in accordance with payment terms customary to the business. We recognize revenue on the sale of automotive regulatory credits, which have negligible incremental costs associated with them, at the time control of the regulatory credits is transferred to the purchasing party. Deferred revenue related to sales of automotive regulatory credits was immaterial as of December 31, 2023 and 2022. Revenue recognized from the deferred revenue balance as of December 31, 2022 and 2021 was immaterial for the years ended December 31, 2023 and 2022. During the year ended December 31, 2022, we had also recognized $288 million in revenue due to changes in regulation which entitled us to additional consideration for credits sold previously.

Automotive Leasing Revenue

Direct Vehicle Operating Leasing Program

We have outstanding leases under our direct vehicle operating leasing programs in the U.S., Canada and in certain countries in Europe. Qualifying customers are permitted to lease a vehicle directly from Tesla for up to 48 months. At the end of the lease term, customers are generally required to return the vehicles to us. We account for these leasing transactions as operating leases. We record leasing revenues to automotive leasing revenue on a straight-line basis over the contractual term, and we record the depreciation of these vehicles to cost of automotive leasing revenue. For the years ended December 31, 2023, 2022 and 2021, we recognized $1.86 billion, $1.75 billion and $1.25 billion of direct vehicle leasing revenue, respectively. As of December 31, 2023 and 2022, we had deferred $458 million and $407 million, respectively, of lease-related upfront payments, which will be recognized on a straight-line basis over the contractual terms of the individual leases.

Our policy is to exclude taxes collected from a customer from the transaction price of automotive contracts.

Direct Sales-Type Leasing Program

We have outstanding direct leases and vehicles financed by us under loan arrangements accounted for as sales-type leases under ASC 842, *Leases* ("ASC 842"), in certain countries in Asia and Europe. Depending on the specific program, customers may or may not have a right to return the vehicle to us during or at the end of the lease term. If the customer does not have a right to return, the customer will take title to the vehicle at the end of the lease term after making all contractual payments. Under the programs for which there is a right to return, the purchase option is reasonably certain to be exercised by the lessee and we therefore expect the customer to take title to the vehicle at the end of the lease term after making all contractual payments. Our arrangements under these programs can have terms for up to 72 months. We recognize all revenue and costs associated with the sales-type lease as automotive leasing revenue and automotive leasing cost of revenue, respectively, upon delivery of the vehicle to the customer. Interest income based on the implicit rate in the lease is recorded to automotive leasing revenue over time as customers are invoiced on a monthly basis. For the years ended December 31, 2023, 2022 and 2021, we recognized $215 million, $683 million and $369 million, respectively, of sales-type leasing revenue and $164 million, $427 million and $234 million, respectively, of sales-type leasing cost of revenue.

Services and Other Revenue

Services and other revenue consists of sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise.

Revenues related to repair, maintenance and vehicle insurance services are recognized over time as services are provided and extended service plans are recognized over the performance period of the service contract as the obligation represents a stand-ready obligation to the customer. We sell used vehicles, services, service plans, vehicle components and merchandise separately and thus use standalone selling prices as the basis for revenue allocation to the extent that these items are sold in transactions with other performance obligations. Payment for used vehicles, services, vehicle components, and merchandise are typically received at the point when control transfers to the customer or in accordance with payment terms customary to the business. Payments received for prepaid plans are refundable upon customer cancellation of the related contracts and are included within Customer deposits on the consolidated balance sheets. We record in Deferred revenue any non-refundable prepayment amounts that are collected from customers and unearned insurance premiums, which is recognized as revenue ratably over the respective customer contract term. Deferred revenue excluding unearned insurance premiums was immaterial as of December 31, 2023 and 2022.

Energy Generation and Storage Segment

<u>Energy Generation and Storage Sales</u>

Energy generation and storage sales revenue consists of the sale of solar energy systems and energy storage systems to residential, small commercial, large commercial and utility grade customers. Sales of solar energy systems to residential and small scale commercial customers consist of the engineering, design and installation of the system. Residential and small scale commercial customers pay the full purchase price of the solar energy system upfront. Revenue for the design and installation obligation is recognized when control transfers, which is when we install a solar energy system and the system passes inspection by the utility or the authority having jurisdiction. Sales of energy storage systems to residential and small scale commercial customers consist of the installation of the energy storage system and revenue is recognized when control transfers, which is when the product has been delivered or, if we are performing installation, when installed and commissioned. Payment for such storage systems is made upon invoice or in accordance with payment terms customary to the business.

For large commercial and utility grade energy storage system sales which consist of the engineering, design and installation of the system, customers make milestone payments that are consistent with contract-specific phases of a project. Revenue from such contracts is recognized over time using the percentage of completion method based on cost incurred as a percentage of total estimated contract costs for energy storage system sales.

In instances where there are multiple performance obligations in a single contract, we allocate the consideration to the various obligations in the contract based on the relative standalone selling price method. Standalone selling prices are estimated based on estimated costs plus margin or by using market data for comparable products. Costs to obtain a contract relate mainly to commissions paid to our sales personnel related to the sale of energy storage systems. As our contract costs related to energy storage system sales are typically fulfilled within one year, the costs to obtain a contract are expensed as incurred.

As part of our energy storage system contracts, we may provide the customer with performance guarantees that warrant that the underlying system will meet or exceed the minimum energy performance requirements specified in the contract. If an energy storage system does not meet the performance guarantee requirements, we may be required to pay liquidated damages. Other forms of variable consideration related to our large commercial and utility grade energy storage system contracts include variable customer payments that will be made based on our energy market participation activities. Such guarantees and variable customer payments represent a form of variable consideration and are estimated at contract inception at their most likely amount and updated at the end of each reporting period as additional performance data becomes available. Such estimates are included in the transaction price only to the extent that it is probable a significant reversal of revenue will not occur.

We record as deferred revenue any non-refundable amounts that are collected from customers related to fees charged for prepayments, which is recognized as revenue ratably over the respective customer contract term. As of December 31, 2023 and 2022, deferred revenue related to such customer payments amounted to $1.60 billion and $863 million, respectively, mainly due to contractual payment terms. Revenue recognized from the deferred revenue balance as of December 31, 2022 and 2021 was $571 million and $171 million for the years ended December 31, 2023 and 2022, respectively. We have elected the practical expedient to omit disclosure of the amount of the transaction price allocated to remaining performance obligations for energy generation and storage sales with an original expected contract length of one year or less and the amount that we have the right to invoice when that amount corresponds directly with the value of the performance to date. As of December 31, 2023, total transaction price allocated to performance obligations that were unsatisfied or partially unsatisfied for contracts with an original expected length of more than one year was $3.43 billion. Of this amount, we expect to recognize $1.05 billion in the next 12 months and the rest over the remaining performance obligation period.

We have been providing loans for financing our energy generation products in volume since fiscal year 2022. As of December 31, 2023 and 2022, we have recorded net financing receivables on the consolidated balance sheets, of which $31 million and $24 million, respectively, is recorded within Accounts receivable, net, for the current portion and $578 million and $387 million, respectively, is recorded within Other non-current assets for the long-term portion.

Energy Generation and Storage Leasing

For revenue arrangements where we are the lessor under operating lease agreements for energy generation and storage products, we record lease revenue from minimum lease payments, including upfront rebates and incentives earned from such systems, on a straight-line basis over the life of the lease term, assuming all other revenue recognition criteria have been met. The difference between the payments received and the revenue recognized is recorded as deferred revenue or deferred asset on the consolidated balance sheet.

For solar energy systems where customers purchase electricity from us under PPAs prior to January 1, 2019, we have determined that these agreements should be accounted for as operating leases pursuant to ASC 840, *Leases*. Revenue is recognized based on the amount of electricity delivered at rates specified under the contracts, assuming all other revenue recognition criteria are met.

We record as deferred revenue any amounts that are collected from customers, including lease prepayments, in excess of revenue recognized, which is recognized as revenue ratably over the respective customer contract term. As of December 31, 2023 and 2022, deferred revenue related to such customer payments amounted to $181 million and $191 million, respectively. Deferred revenue also includes the portion of rebates and incentives received from utility companies and various local and state government agencies, which is recognized as revenue over the lease term. As of December 31, 2023 and 2022, deferred revenue from rebates and incentives was immaterial.

We capitalize initial direct costs from the execution of agreements for solar energy systems and PPAs, which include the referral fees and sales commissions, as an element of solar energy systems, net, and subsequently amortize these costs over the term of the related agreements.

Cost of Revenues

Automotive Segment

Automotive Sales

Cost of automotive sales revenue includes direct and indirect materials, labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, shipping and logistic costs, vehicle connectivity costs, FSD Capability ongoing maintenance costs, allocations of electricity and infrastructure costs related to our Supercharger network and reserves for estimated warranty expenses. Cost of automotive sales revenues also includes adjustments to warranty expense and charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of automotive sales revenue benefits from manufacturing credits earned.

Automotive Leasing

Cost of automotive leasing revenue includes the depreciation of operating lease vehicles, cost of goods sold associated with direct sales-type leases and warranty expense related to leased vehicles.

Services and Other

Costs of services and other revenue includes cost of used vehicles including refurbishment costs, costs associated with providing non-warranty after-sales services, costs associated with our body shops and part sales, costs of paid Supercharging, costs to provide vehicle insurance and costs for retail merchandise.

Energy Generation and Storage Segment

Energy Generation and Storage

Cost of energy generation and storage revenue includes direct and indirect material and labor costs, overhead costs, freight, warranty expense, and amortization of certain acquired intangible assets. Cost of energy generation and storage revenue also includes charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of energy generation and storage revenue benefits from manufacturing credits earned. In agreements for solar energy systems and PPAs where we are the lessor, the cost of revenue is primarily comprised of depreciation of the cost of leased solar energy systems, maintenance costs associated with those systems and amortization of any initial direct costs.

Research and Development Costs

Research and development costs are expensed as incurred.

Income Taxes

We are subject to income taxes in the U.S. and in many foreign jurisdictions. Income taxes are computed using the asset and liability method, under which deferred tax assets and liabilities are determined based on the difference between the financial statement and tax bases of assets and liabilities using enacted tax rates in effect for the year in which the differences are expected to affect taxable income. Valuation allowances are established when necessary to reduce deferred tax assets to the amount expected to be realized.

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. Significant judgment is required in determining our provision for income taxes, our deferred tax assets and liabilities and any valuation allowance recorded against our net deferred tax assets that are not more likely than not to be realized. In completing our assessment of realizability of our deferred tax assets, we consider our history of income (loss) measured at pre-tax income (loss) adjusted for permanent book-tax differences on a jurisdictional basis, volatility in actual earnings, excess tax benefits related to stock-based compensation in recent prior years, and impacts of the timing of reversal of existing temporary differences. We also rely on our assessment of the Company’s projected future results of business operations, including uncertainty in future operating results relative to historical results, volatility in the market price of our common stock and its performance over time, variable macroeconomic conditions impacting our ability to forecast future taxable income, and changes in business that may affect the existence and magnitude of future taxable income. Our valuation allowance assessment is based on our best estimate of future results considering all available information.

We record liabilities related to uncertain tax positions when, despite our belief that our tax return positions are supportable, we believe that it is more likely than not that those positions may not be fully sustained upon review by tax authorities. Accrued interest and penalties related to unrecognized tax benefits are classified as income tax expense.

The Tax Cuts and Jobs Act subjects a U.S. shareholder to tax on global intangible low-taxed income (“GILTI”) earned by certain foreign subsidiaries. Under GAAP, we can make an accounting policy election to either treat taxes due on the GILTI inclusion as a current period expense or factor such amounts into our measurement of deferred taxes. We elected the deferred method, under which we recorded the corresponding deferred tax assets and liabilities in our consolidated balance sheets.

Comprehensive Income

Comprehensive income is comprised of net income and other comprehensive income (loss). Other comprehensive income (loss) consists of foreign currency translation adjustments and unrealized net gains and losses on investments that have been excluded from the determination of net income.

Stock-Based Compensation

We use the fair value method of accounting for our stock options and RSUs granted to employees and for our ESPP to measure the cost of employee services received in exchange for the stock-based awards. The fair value of stock option awards with only service and/or performance conditions is estimated on the grant or offering date using the Black-Scholes option-pricing model. The Black-Scholes option-pricing model requires inputs such as the risk-free interest rate, expected term and expected volatility. These inputs are subjective and generally require significant judgment. The fair value of RSUs is measured on the grant date based on the closing fair market value of our common stock. The resulting cost is recognized over the period during which an employee is required to provide service in exchange for the awards, usually the vesting period, which is generally four years for stock options and RSUs and six months for the ESPP. Stock-based compensation expense is recognized on a straight-line basis, net of actual forfeitures in the period.

For performance-based awards, stock-based compensation expense is recognized over the expected performance achievement period of individual performance milestones when the achievement of each individual performance milestone becomes probable.

As we accumulate additional employee stock-based awards data over time and as we incorporate market data related to our common stock, we may calculate significantly different volatilities and expected lives, which could materially impact the valuation of our stock-based awards and the stock-based compensation expense that we will recognize in future periods. Stock-based compensation expense is recorded in Cost of revenues, Research and development expense and Selling, general and administrative expense in the consolidated statements of operations.

Noncontrolling Interests and Redeemable Noncontrolling Interests

Noncontrolling interests and redeemable noncontrolling interests represent third-party interests in the net assets under certain funding arrangements, or funds, that we have entered into to finance the costs of solar energy systems and vehicles under operating leases. We have determined that the contractual provisions of the funds represent substantive profit-sharing arrangements. We have further determined that the methodology for calculating the noncontrolling interest and redeemable noncontrolling interest balances that reflects the substantive profit-sharing arrangements is a balance sheet approach using the hypothetical liquidation at book value ("HLBV") method. We, therefore, determine the amount of the noncontrolling interests and redeemable noncontrolling interests in the net assets of the funds at each balance sheet date using the HLBV method, which is presented on the consolidated balance sheet as noncontrolling interests in subsidiaries and redeemable noncontrolling interests in subsidiaries. Under the HLBV method, the amounts reported as noncontrolling interests and redeemable noncontrolling interests in the consolidated balance sheet represent the amounts the third parties would hypothetically receive at each balance sheet date under the liquidation provisions of the funds, assuming the net assets of the funds were liquidated at their recorded amounts determined in accordance with GAAP and with tax laws effective at the balance sheet date and distributed to the third parties. The third parties' interests in the results of operations of the funds are determined as the difference in the noncontrolling interest and redeemable noncontrolling interest balances in the consolidated balance sheets between the start and end of each reporting period, after taking into account any capital transactions between the funds and the third parties. However, the redeemable noncontrolling interest balance is at least equal to the redemption amount. The redeemable noncontrolling interest balance is presented as temporary equity in the mezzanine section of the consolidated balance sheet since these third parties have the right to redeem their interests in the funds for cash or other assets. For certain funds, there have been significant fluctuations in net (loss) income attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries due to changes in the liquidation provisions as time-based milestones have been reached.

Net Income per Share of Common Stock Attributable to Common Stockholders

Basic net income per share of common stock attributable to common stockholders is calculated by dividing net income attributable to common stockholders by the weighted-average shares of common stock outstanding for the period. Potentially dilutive shares, which are based on the weighted-average shares of common stock underlying outstanding stock-based awards, warrants and convertible senior notes using the treasury stock method or the if-converted method, as applicable, are included when calculating diluted net income per share of common stock attributable to common stockholders when their effect is dilutive.

Furthermore, in connection with the offerings of our convertible senior notes, we entered into convertible note hedges and warrants (see Note 11, *Debt*). However, our convertible note hedges are not included when calculating potentially dilutive shares since their effect is always anti-dilutive. The strike price on the warrants were below our average share price during the period and were included in the tables below. Warrants are included in the weighted-average shares used in computing basic net income per share of common stock in the period(s) they are settled.

The following table presents the reconciliation of net income attributable to common stockholders to net income used in computing basic and diluted net income per share of common stock (in millions):

	Year Ended December 31,		
	2023	**2022**	**2021**
Net income attributable to common stockholders	$ 14,997	$ 12,556	$ 5,519
Less: Buy-out of noncontrolling interest	(2)	(27)	(5)
Net income used in computing basic net income per share of common stock	14,999	12,583	5,524
Less: Dilutive convertible debt	—	(1)	(9)
Net income used in computing diluted net income per share of common stock	$ 14,999	$ 12,584	$ 5,533

The following table presents the reconciliation of basic to diluted weighted average shares used in computing net income per share of common stock attributable to common stockholders (in millions):

	Year Ended December 31,		
	2023	2022	2021
Weighted average shares used in computing net income per share of common stock, basic	3,174	3,130	2,959
Add:			
Stock-based awards	298	310	292
Convertible senior notes	2	3	29
Warrants	11	32	106
Weighted average shares used in computing net income per share of common stock, diluted	3,485	3,475	3,386

The following table presents the potentially dilutive shares that were excluded from the computation of diluted net income per share of common stock attributable to common stockholders, because their effect was anti-dilutive (in millions):

	Year Ended December 31,		
	2023	2022	2021
Stock-based awards	12	4	1

Business Combinations

We account for business acquisitions under ASC 805, *Business Combinations*. The total purchase consideration for an acquisition is measured as the fair value of the assets given, equity instruments issued and liabilities assumed at the acquisition date. Costs that are directly attributable to the acquisition are expensed as incurred. Identifiable assets (including intangible assets), liabilities assumed (including contingent liabilities) and noncontrolling interests in an acquisition are measured initially at their fair values at the acquisition date. We recognize goodwill if the fair value of the total purchase consideration and any noncontrolling interests is in excess of the net fair value of the identifiable assets acquired and the liabilities assumed. We recognize a bargain purchase gain within Other income (expense), net, in the consolidated statement of operations if the net fair value of the identifiable assets acquired and the liabilities assumed is in excess of the fair value of the total purchase consideration and any noncontrolling interests. We include the results of operations of the acquired business in the consolidated financial statements beginning on the acquisition date.

Cash and Cash Equivalents

All highly liquid investments with an original maturity of three months or less at the date of purchase are considered cash equivalents. Our cash equivalents are primarily comprised of U.S. government securities, money market funds and commercial paper.

Restricted Cash

We maintain certain cash balances restricted as to withdrawal or use. Our restricted cash is comprised primarily of cash held to service certain payments under various secured debt facilities. In addition, restricted cash includes cash held as collateral for sales to lease partners with a resale value guarantee, letters of credit, real estate leases and deposits held for our insurance services. We record restricted cash as other assets in the consolidated balance sheets and determine current or non-current classification based on the expected duration of the restriction.

Our total cash and cash equivalents and restricted cash, as presented in the consolidated statements of cash flows, was as follows (in millions):

	December 31, 2023	December 31, 2022	December 31, 2021
Cash and cash equivalents	$ 16,398	$ 16,253	$ 17,576
Restricted cash included in prepaid expenses and other current assets	543	294	345
Restricted cash included in other non-current assets	248	377	223
Total as presented in the consolidated statements of cash flows	$ 17,189	$ 16,924	$ 18,144

Investments

Investments may be comprised of a combination of marketable securities, including U.S. government securities, corporate debt securities, commercial paper, time deposits, and certain certificates of deposit, which are all designated as available-for-sale and reported at estimated fair value, with unrealized gains and losses recorded in accumulated other comprehensive income which is included within stockholders’ equity. Available-for-sale marketable securities with maturities greater than three months at the date of purchase are included in short-term investments in our consolidated balance sheets. Interest, dividends, amortization and accretion of purchase premiums and discounts on these investments are included within Interest income in our consolidated statements of operations.

The cost of available-for-sale investments sold is based on the specific identification method. Realized gains and losses on the sale of available-for-sale investments are recorded in Other income (expense), net.

We regularly review all of our investments for declines in fair value. The review includes but is not limited to (i) the consideration of the cause of the decline, (ii) any currently recorded expected credit losses and (iii) the creditworthiness of the respective security issuers. The amortized cost basis of our investments approximates its fair value.

Accounts Receivable and Allowance for Doubtful Accounts

Accounts receivable primarily include amounts related to receivables from financial institutions and leasing companies offering various financing products to our customers, sales of energy generation and storage products, sales of regulatory credits to other automotive manufacturers and government rebates already passed through to customers. We provide an allowance against accounts receivable for the amount we expect to be uncollectible. We write-off accounts receivable against the allowance when they are deemed uncollectible.

Depending on the day of the week on which the end of a fiscal quarter falls, our accounts receivable balance may fluctuate as we are waiting for certain customer payments to clear through our banking institutions and receipts of payments from our financing partners, which can take up to approximately two weeks based on the contractual payment terms with such partners. Our accounts receivable balances associated with our sales of regulatory credits are dependent on contractual payment terms. Additionally, government rebates can take up to a year or more to be collected depending on the customary processing timelines of the specific jurisdictions issuing them. These various factors may have a significant impact on our accounts receivable balance from period to period. As of December 31, 2023 and 2022, we had $207 million and $753 million, respectively, of long-term government rebates receivable in Other non-current assets in our consolidated balance sheets.

Financing Receivables

We provide financing options to our customers for our automotive and energy products. Financing receivables are carried at amortized cost, net of allowance for loan losses. Provisions for loan losses are charged to operations in amounts sufficient to maintain the allowance for loan losses at levels considered adequate to cover expected credit losses on the financing receivables. In determining expected credit losses, we consider our historical level of credit losses, current economic trends, and reasonable and supportable forecasts that affect the collectability of the future cash flows.

When originating consumer receivables, we review the credit application, the proposed contract terms, credit bureau information (e.g., FICO score) and other information. Our evaluation emphasizes the applicant’s ability to pay and creditworthiness focusing on payment, affordability, and applicant credit history as key considerations. Generally, all customers in this portfolio have strong creditworthiness at loan origination.

After origination, we review the credit quality of retail financing based on customer payment activity and aging analysis. For all financing receivables, we define “past due” as any payment, including principal and interest, which is at least 31 days past the contractual due date. As of December 31, 2023 and 2022, the vast majority of our financing receivables were at current status with only an immaterial balance being past due. As of December 31, 2023, the majority of our financing receivables, excluding MyPower notes receivable, were originated in 2023 and 2022, and as of December 31, 2022, the majority of our financing receivables, excluding MyPower notes receivable, were originated in 2022.

We have customer notes receivable under the legacy MyPower loan program, which provided residential customers with the option to finance the purchase of a solar energy system through a 30-year loan and were all originated prior to year 2018. The outstanding balances, net of any allowance for expected credit losses, are presented on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. As of December 31, 2023 and 2022, the total outstanding balance of MyPower customer notes receivable, net of allowance for expected credit losses, was $266 million and $280 million, respectively, of which $5 million and $7 million were due in the next 12 months as of December 31, 2023 and 2022, respectively. As of December 31, 2023 and 2022, the allowance for expected credit losses was $36 million and $37 million, respectively.

Concentration of Risk

Credit Risk

Financial instruments that potentially subject us to a concentration of credit risk consist of cash, cash equivalents, investments, restricted cash, accounts receivable and other finance receivables. Our cash and investments balances are primarily on deposit at high credit quality financial institutions or invested in money market funds. These deposits are typically in excess of insured limits. As of December 31, 2023 and 2022, no entity represented 10% or more of our total receivables balance.

Supply Risk

We are dependent on our suppliers, including single source suppliers, and the inability of these suppliers to deliver necessary components of our products in a timely manner at prices, quality levels and volumes acceptable to us, or our inability to efficiently manage these components from these suppliers, could have a material adverse effect on our business, prospects, financial condition and operating results.

Inventory Valuation

Inventories are stated at the lower of cost or net realizable value. Cost is computed using standard cost for vehicles and energy products, which approximates actual cost on a first-in, first-out basis. We record inventory write-downs for excess or obsolete inventories based upon assumptions about current and future demand forecasts. If our inventory on-hand is in excess of our future demand forecast, the excess amounts are written-off.

We also review our inventory to determine whether its carrying value exceeds the net amount realizable upon the ultimate sale of the inventory. This requires us to determine the estimated selling price of our vehicles less the estimated cost to convert the inventory on-hand into a finished product. Once inventory is written-down, a new, lower cost basis for that inventory is established and subsequent changes in facts and circumstances do not result in the restoration or increase in that newly established cost basis.

Should our estimates of future selling prices or production costs change, additional and potentially material write-downs may be required. A small change in our estimates may result in a material charge to our reported financial results.

Operating Lease Vehicles

Vehicles that are leased as part of our direct vehicle leasing program are classified as operating lease vehicles at cost less accumulated depreciation. We generally depreciate their cost, less residual value, using the straight-line-method to cost of automotive leasing revenue over the contractual period. The gross cost of operating lease vehicles as of December 31, 2023 and 2022 was $7.36 billion and $6.08 billion, respectively. Operating lease vehicles on the consolidated balance sheets are presented net of accumulated depreciation of $1.38 billion and $1.04 billion as of December 31, 2023 and 2022, respectively.

Digital Assets, Net

We currently account for all digital assets held as indefinite-lived intangible assets in accordance with ASC 350, *Intangibles—Goodwill and Other*. We have ownership of and control over our digital assets and we may use third-party custodial services to secure it. The digital assets are initially recorded at cost and are subsequently remeasured on the consolidated balance sheet at cost, net of any impairment losses incurred since acquisition.

We determine the fair value of our digital assets on a nonrecurring basis in accordance with ASC 820, *Fair Value Measurement* ("ASC 820"), based on quoted prices on the active exchange(s) that we have determined is the principal market for such assets (Level I inputs). We perform an analysis each quarter to identify whether events or changes in circumstances, principally decreases in the quoted prices on active exchanges, indicate that it is more likely than not that our digital assets are impaired. In determining if an impairment has occurred, we consider the lowest market price of one unit of digital asset quoted on the active exchange since acquiring the digital asset. When the then current carrying value of a digital asset exceeds the fair value determined each quarter, an impairment loss has occurred with respect to those digital assets in the amount equal to the difference between their carrying values and the prices determined.

Impairment losses are recognized within Restructuring and other in the consolidated statements of operations in the period in which the impairment is identified. Gains are not recorded until realized upon sale(s), at which point they are presented net of any impairment losses for the same digital assets held within Restructuring and other. In determining the gain to be recognized upon sale, we calculate the difference between the sales price and carrying value of the digital assets sold immediately prior to sale.

See Note 3, *Digital Assets, Net*, for further information regarding digital assets.

Solar Energy Systems, Net

We are the lessor of solar energy systems. Solar energy systems are stated at cost less accumulated depreciation.

Depreciation and amortization is calculated using the straight-line method over the estimated useful lives of the respective assets, as follows:

Solar energy systems in service	30 to 35 years
Initial direct costs related to customer solar energy system lease acquisition costs	Lease term (up to 25 years)

Solar energy systems pending interconnection will be depreciated as solar energy systems in service when they have been interconnected and placed in-service. Solar energy systems under construction represents systems that are under installation, which will be depreciated as solar energy systems in service when they are completed, interconnected and placed in service. Initial direct costs related to customer solar energy system agreement acquisition costs are capitalized and amortized over the term of the related customer agreements.

Property, Plant and Equipment, Net

Property, plant and equipment, net, including leasehold improvements, are recognized at cost less accumulated depreciation. Depreciation is generally computed using the straight-line method over the estimated useful lives of the respective assets, as follows:

Machinery, equipment, vehicles and office furniture	3 to 15 years
Tooling	4 to 7 years
Building and building improvements	15 to 30 years
Computer equipment and software	3 to 10 years

Leasehold improvements are depreciated on a straight-line basis over the shorter of their estimated useful lives or the terms of the related leases.

Upon the retirement or sale of our property, plant and equipment, the cost and associated accumulated depreciation are removed from the consolidated balance sheet, and the resulting gain or loss is reflected on the consolidated statement of operations. Maintenance and repair expenditures are expensed as incurred while major improvements that increase the functionality, output or expected life of an asset are capitalized and depreciated ratably over the identified useful life.

Interest expense on outstanding debt is capitalized during the period of significant capital asset construction. Capitalized interest on construction in progress is included within Property, plant and equipment, net and is amortized over the life of the related assets.

Long-Lived Assets Including Acquired Intangible Assets

We review our property, plant and equipment, solar energy systems, long-term prepayments and intangible assets for impairment whenever events or changes in circumstances indicate that the carrying amount of an asset (or asset group) may not be recoverable. We measure recoverability by comparing the carrying amount to the future undiscounted cash flows that the asset is expected to generate. If the asset is not recoverable, its carrying amount would be adjusted down to its fair value. For the years ended December 31, 2023, 2022 and 2021, we have recognized no material impairments of our long-lived assets.

Intangible assets with definite lives are amortized on a straight-line basis over their estimated useful lives, which range from seven to thirty years.

Goodwill

We assess goodwill for impairment annually in the fourth quarter, or more frequently if events or changes in circumstances indicate that it might be impaired, by comparing its carrying value to the reporting unit's fair value. For the years ended December 31, 2023, 2022, and 2021, we did not recognize any impairment of goodwill.

Capitalization of Software Costs

We capitalize costs incurred in the development of internal use software, during the application development stage to Property, plant and equipment, net on the consolidated balance sheets. Costs related to preliminary project activities and post-implementation activities are expensed as incurred. Such costs are amortized on a straight-line basis over their estimated useful life of three to five years.

Software development costs incurred in development of software to be sold, leased, or otherwise marketed, incurred subsequent to the establishment of technological feasibility and prior to the general availability of the software are capitalized when they are expected to become significant. Such costs are amortized over the estimated useful life of the applicable software once it is made generally available to our customers.

We evaluate the useful lives of these assets on an annual basis, and we test for impairment whenever events or changes in circumstances occur that could impact the recoverability of these assets. For the years ended December 31, 2023, 2022, and 2021, we have recognized no impairments of capitalized software costs.

Foreign Currency

We determine the functional and reporting currency of each of our international subsidiaries and their operating divisions based on the primary currency in which they operate. In cases where the functional currency is not the U.S. dollar, we recognize a cumulative translation adjustment created by the different rates we apply to current period income or loss and the balance sheet. For each subsidiary, we apply the monthly average functional exchange rate to its monthly income or loss and the month-end functional currency rate to translate the balance sheet.

Foreign currency transaction gains and losses are a result of the effect of exchange rate changes on transactions denominated in currencies other than the functional currency of the respective subsidiary. Transaction gains and losses are recognized in Other income (expense), net, in the consolidated statements of operations. For the years ended December 31, 2023, 2022 and 2021, we recorded a net foreign currency transaction gain of $122 million, loss of $89 million and gain of $97 million, respectively.

Warranties

We provide a manufacturer's warranty on all new and used vehicles and a warranty on the installation and components of the energy generation and storage systems we sell for periods typically between 10 to 25 years. We accrue a warranty reserve for the products sold by us, which includes our best estimate of the projected costs to repair or replace items under warranties and recalls if identified. These estimates are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims. These estimates are inherently uncertain and changes to our historical or projected warranty experience may cause material changes to the warranty reserve in the future. The warranty reserve does not include projected warranty costs associated with our vehicles subject to operating lease accounting and our solar energy systems under lease contracts or PPAs, as the costs to repair these warranty claims are expensed as incurred. The portion of the warranty reserve expected to be incurred within the next 12 months is included within Accrued liabilities and other, while the remaining balance is included within Other long-term liabilities on the consolidated balance sheets. For liabilities that we are entitled to receive indemnification from our suppliers, we record receivables for the contractually obligated amounts on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Warranty expense is recorded as a component of Cost of revenues in the consolidated statements of operations. Due to the magnitude of our automotive business, our accrued warranty balance is primarily related to our automotive segment. Accrued warranty activity consisted of the following (in millions):

	Year Ended December 31,		
	2023	**2022**	**2021**
Accrued warranty—beginning of period	$ 3,505	$ 2,101	$ 1,468
Warranty costs incurred	(1,225)	(803)	(525)
Net changes in liability for pre-existing warranties, including expirations and foreign exchange impact	539	522	102
Provision for warranty	2,333	1,685	1,056
Accrued warranty—end of period	$ 5,152	$ 3,505	$ 2,101

Customer Deposits

Customer deposits primarily consist of refundable cash payments from customers at the time they place an order or reservation for a vehicle or an energy product and any additional payments up to the point of delivery or the completion of installation. Customer deposits also include prepayments on contracts that can be cancelled without significant penalties, such as vehicle maintenance plans. Customer deposits are included in Accrued liabilities and other on the consolidated balance sheets until refunded, forfeited or applied towards the customer's purchase balance.

Government Assistance Programs and Incentives

Globally, the operation of our business is impacted by various government programs, incentives, and other arrangements. Government incentives are recorded in our consolidated financial statements in accordance with their purpose as a reduction of expense, or an offset to the related capital asset. The benefit is generally recorded when all conditions attached to the incentive have been met or are expected to be met and there is reasonable assurance of their receipt.

The IRA Incentives

On August 16, 2022, the IRA was enacted into law and is effective for taxable years beginning after December 31, 2022. The IRA includes multiple incentives to promote clean energy, electric vehicles, battery and energy storage manufacture or purchase, in addition to a new corporate alternative minimum tax of 15% on adjusted financial statement income of corporations with profits greater than $1 billion. Some of these measures are expected to materially affect our consolidated financial statements. For the year ended December 31, 2023, the impact from our IRA incentive was primarily a reduction of our material costs in our consolidated statement of operations. We will continue to evaluate the effects of the IRA as more guidance is issued and the relevant implications to our consolidated financial statements.

Gigafactory New York—New York State Investment and Lease

We have a lease through the Research Foundation for the SUNY Foundation with respect to Gigafactory New York. Under the lease and a related research and development agreement, we are continuing to designate further buildouts at the facility. We are required to comply with certain covenants, including hiring and cumulative investment targets. Under the terms of the arrangement, the SUNY Foundation paid for a majority of the construction costs related to the manufacturing facility and the acquisition and commissioning of certain manufacturing equipment; and we are responsible for any construction or equipment costs in excess of such amount (refer to Note 15, *Commitments and Contingencies*). This incentive reduces the related lease costs of the facility within the Energy generation and storage cost of revenues and operating expense line items in our consolidated statements of operations.

Gigafactory Shanghai—Land Use Rights and Economic Benefits

We have an agreement with the local government of Shanghai for land use rights at Gigafactory Shanghai. Under the terms of the arrangement, we are required to meet a cumulative capital expenditure target and an annual tax revenue target starting at the end of 2023. In addition, the Shanghai government has granted to our Gigafactory Shanghai subsidiary certain incentives to be used in connection with eligible capital investments at Gigafactory Shanghai (refer to Note 15, *Commitments and Contingencies*). For the year ended December 31, 2022, we received grant funding of $76 million. These incentives offset the related costs of our facilities and are recorded as a reduction of the cost of the capital investment within the Property, plant and equipment, net line item in our consolidated balance sheets. The incentive therefore reduces the depreciation expense over the useful lives of the related equipment.

Nevada Tax Incentives

In connection with the construction of Gigafactory Nevada, we entered into agreements with the State of Nevada and Storey County in Nevada that provide abatements for specified taxes, discounts to the base tariff energy rates and transferable tax credits of up to $195 million in consideration of capital investment and hiring targets that were met at Gigafactory Nevada.

Gigafactory Texas Tax Incentives

In connection with the construction of Gigafactory Texas, we entered into a 20-year agreement with Travis County in Texas pursuant to which we would receive grant funding equal to 70-80% of property taxes paid by us to Travis County and a separate 10-year agreement with the Del Valle Independent School District in Texas pursuant to which a portion of the taxable value of our property would be capped at a specified amount, in each case subject to our meeting certain minimum economic development metrics through our construction and operations at Gigafactory Texas. This incentive is recorded as a reduction of the related expenses within the Cost of automotive revenues and operating expense line items of our consolidated statements of operations. As of December 31, 2023, the grant funding related to property taxes paid were immaterial.

Defined Contribution Plan

We have a 401(k) savings plan in the U.S. that is intended to qualify as a deferred salary arrangement under Section 401(k) of the Internal Revenue Code and a number of savings plans internationally. Under the 401(k) savings plan, participating employees may elect to contribute up to 90% of their eligible compensation, subject to certain limitations. Beginning in January 2022, we began to match 50% of each employee’s contributions up to a maximum of 6% (capped at $3,000) of the employee’s eligible compensation, vested upon one year of service. During the years ended December 31, 2023 and 2022, we recognized $99 million and $91 million, respectively, of expenses related to employer contributions for the 401(k) savings plan.

Recent Accounting Pronouncements

Recently issued accounting pronouncements not yet adopted

In November 2023, the Financial Accounting Standards Board ("FASB") issued ASU No. 2023-07, Improvements to Reportable Segment Disclosures (Topic 280). This ASU updates reportable segment disclosure requirements by requiring disclosures of significant reportable segment expenses that are regularly provided to the Chief Operating Decision Maker ("CODM") and included within each reported measure of a segment's profit or loss. This ASU also requires disclosure of the title and position of the individual identified as the CODM and an explanation of how the CODM uses the reported measures of a segment's profit or loss in assessing segment performance and deciding how to allocate resources. The ASU is effective for annual periods beginning after December 15, 2023, and interim periods within fiscal years beginning after December 15, 2024. Adoption of the ASU should be applied retrospectively to all prior periods presented in the financial statements. Early adoption is also permitted. This ASU will likely result in us including the additional required disclosures when adopted. We are currently evaluating the provisions of this ASU and expect to adopt them for the year ending December 31, 2024.

In December 2023, the FASB issued ASU No. 2023-08, Accounting for and Disclosure of Crypto Assets (Subtopic 350-60). This ASU requires certain crypto assets to be measured at fair value separately in the balance sheet and income statement each reporting period. This ASU also enhances the other intangible asset disclosure requirements by requiring the name, cost basis, fair value, and number of units for each significant crypto holding. The ASU is effective for annual periods beginning after December 15, 2024, including interim periods within those fiscal years. Adoption of the ASU requires a cumulative-effect adjustment to the opening balance of retained earnings as of the beginning of the annual reporting period in which an entity adopts the amendments. Early adoption is also permitted, including adoption in an interim period. However, if the ASU is early adopted in an interim period, an entity must adopt the ASU as of the beginning of the fiscal year that includes the interim period. This ASU will result in gains and losses recorded in the consolidated financial statements of operations and additional disclosures when adopted. We are currently evaluating the adoption of this ASU and it will affect the carrying value of our crypto assets held and the gains and losses relating thereto, once adopted.

In December 2023, the FASB issued ASU No. 2023-09, Improvements to Income Tax Disclosures (Topic 740). The ASU requires disaggregated information about a reporting entity's effective tax rate reconciliation as well as additional information on income taxes paid. The ASU is effective on a prospective basis for annual periods beginning after December 15, 2024. Early adoption is also permitted for annual financial statements that have not yet been issued or made available for issuance. This ASU will result in the required additional disclosures being included in our consolidated financial statements, once adopted.

Recently adopted accounting pronouncements

In October 2021, the FASB issued ASU No. 2021-08, Accounting for Contract Assets and Contract Liabilities from Contracts with Customers (Topic 805). This ASU requires an acquirer in a business combination to recognize and measure contract assets and contract liabilities (deferred revenue) from acquired contracts using the revenue recognition guidance in Topic 606. At the acquisition date, the acquirer applies the revenue model as if it had originated the acquired contracts. The ASU is effective for annual periods beginning after December 15, 2022, including interim periods within those fiscal years. We adopted this ASU prospectively on January 1, 2023. This ASU has not and is currently not expected to have a material impact on our consolidated financial statements.

In March 2022, the FASB issued ASU 2022-02, Troubled Debt Restructurings and Vintage Disclosures. This ASU eliminates the accounting guidance for troubled debt restructurings by creditors that have adopted ASU 2016-13, Measurement of Credit Losses on Financial Instruments, which we adopted on January 1, 2020. This ASU also enhances the disclosure requirements for certain loan refinancing and restructurings by creditors when a borrower is experiencing financial difficulty. In addition, the ASU amends the guidance on vintage disclosures to require entities to disclose current period gross write-offs by year of origination for financing receivables and net investments in leases within the scope of ASC 326-20. The ASU is effective for annual periods beginning after December 15, 2022, including interim periods within those fiscal years. We adopted the ASU prospectively on January 1, 2023. This ASU has not and is currently not expected to have a material impact on our consolidated financial statements.

ASU 2020-06

In August 2020, the FASB issued ASU 2020-06, Accounting for Convertible Instruments and Contracts in an Entity's Own Equity. The ASU simplifies the accounting for convertible instruments by removing certain separation models in ASC 470-20, Debt—Debt with Conversion and Other Options, for convertible instruments. The ASU updates the guidance on certain embedded conversion features that are not required to be accounted for as derivatives under Topic 815, Derivatives and Hedging, or that do not result in substantial premiums accounted for as paid-in capital, such that those features are no longer required to be separated from the host contract. The convertible debt instruments will be accounted for as a single liability measured at amortized cost. This will also result in the interest expense recognized for convertible debt instruments to be typically closer to the coupon interest rate when applying the guidance in Topic 835, Interest. Further, the ASU made amendments to the EPS guidance in Topic 260 for convertible debt instruments, the most significant impact of which is requiring the use of the if-converted method for diluted EPS calculation, and no longer allowing the net share settlement method. The ASU also made revisions to Topic 815-40, which provides guidance on how an entity must determine whether a contract qualifies for a scope exception from derivative accounting. The amendments to Topic 815-40 change the scope of contracts that are recognized as assets or liabilities.

On January 1, 2021, we adopted the ASU using the modified retrospective method. We recognized a favorable $211 million cumulative effect of initially applying the ASU as an adjustment to the January 1, 2021 opening balance of accumulated deficit. Due to the recombination of the equity conversion component of our convertible debt remaining outstanding, additional paid in capital was reduced by $474 million and convertible senior notes (mezzanine equity) was reduced by $51 million. The removal of the remaining debt discounts recorded for this previous separation had the effect of increasing our net debt balance by $269 million and we reduced property, plant and equipment by $45 million related to previously capitalized interest. The prior period consolidated financial statements have not been retrospectively adjusted and continue to be reported under the accounting standards in effect for those periods.

Note 3 – Digital Assets, Net

During the years ended December 31, 2023 and 2022, we purchased and/or received immaterial amounts of digital assets. During the year ended December 31, 2023, we recorded an immaterial amount of impairment losses on digital assets. During the year ended December 31, 2022, we recorded $204 million of impairment losses on digital assets and realized gains of $64 million in connection with converting our holdings of digital assets into fiat currency. The gains are presented net of impairment losses in Restructuring and other in the consolidated statements of operations. As of December 31, 2023 and 2022, the carrying value of our digital assets held reflects cumulative impairment of $204 million.

Note 4 – Goodwill and Intangible Assets

Goodwill increased $59 million within the automotive segment from $194 million as of December 31, 2022 to $253 million as of December 31, 2023 primarily from a business combination, net of the impact of a divestiture. There were no accumulated impairment losses as of December 31, 2023 and 2022.

The net carrying value of our intangible assets decreased from $215 million as of December 31, 2022 to $178 million as of December 31, 2023 mainly from amortization.

Note 5 – Fair Value of Financial Instruments

ASC 820, *Fair Value Measurements* ("ASC 820") states that fair value is an exit price, representing the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants. As such, fair value is a market-based measurement that should be determined based on assumptions that market participants would use in pricing an asset or a liability. The three-tiered fair value hierarchy, which prioritizes which inputs should be used in measuring fair value, is comprised of: (Level I) observable inputs such as quoted prices in active markets; (Level II) inputs other than quoted prices in active markets that are observable either directly or indirectly and (Level III) unobservable inputs for which there is little or no market data. The fair value hierarchy requires the use of observable market data when available in determining fair value. Our assets and liabilities that were measured at fair value on a recurring basis were as follows (in millions):

	December 31, 2023				December 31, 2022			
	Fair Value	Level I	Level II	Level III	Fair Value	Level I	Level II	Level III
Money market funds	$ 109	$ 109	$ —	$ —	$ 2,188	$ 2,188	$ —	$ —
U.S. government securities	5,136	—	5,136	—	894	—	894	—
Corporate debt securities	480	—	480	—	885	—	885	—
Certificates of deposit and time deposits	6,996	—	6,996	—	4,253	—	4,253	—
Commercial paper	470	—	470	—	—	—	—	—
Total	$ 13,191	$ 109	$ 13,082	$ —	$ 8,220	$ 2,188	$ 6,032	$ —

All of our money market funds were classified within Level I of the fair value hierarchy because they were valued using quoted prices in active markets. Our U.S. government securities, certificates of deposit, commercial paper, time deposits and corporate debt securities are classified within Level II of the fair value hierarchy and the market approach was used to determine fair value of these investments.

Our cash, cash equivalents and investments classified by security type as of December 31, 2023 and 2022 consisted of the following (in millions):

	December 31, 2023					
	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Short-Term Investments
Cash	$ 15,903	$ —	$ —	$ 15,903	$ 15,903	$ —
Money market funds	109	—	—	109	109	—
U.S. government securities	5,136	1	(1)	5,136	277	4,859
Corporate debt securities	485	1	(6)	480	—	480
Certificates of deposit and time deposits	6,995	1	—	6,996	—	6,996
Commercial paper	470	—	—	470	109	361
Total cash, cash equivalents and short-term investments	$ 29,098	$ 3	$ (7)	$ 29,094	$ 16,398	$ 12,696

	December 31, 2022					
	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Short-Term Investments
Cash	$ 13,965	$ —	$ —	$ 13,965	$ 13,965	$ —
Money market funds	2,188	—	—	2,188	2,188	—
U.S. government securities	897	—	(3)	894	—	894
Corporate debt securities	907	—	(22)	885	—	885
Certificates of deposit and time deposits	4,252	1	—	4,253	100	4,153
Total cash, cash equivalents and short-term investments	$ 22,209	$ 1	$ (25)	$ 22,185	$ 16,253	$ 5,932

We record gross realized gains, losses and credit losses as a component of Other income (expense), net in the consolidated statements of operations. For the years ended December 31, 2023 and 2022, we did not recognize any material gross realized gains, losses or credit losses. The ending allowance balances for credit losses were immaterial as of December 31, 2023 and 2022. We have determined that the gross unrealized losses on our investments as of December 31, 2023 and 2022 were temporary in nature.

The following table summarizes the fair value of our investments by stated contractual maturities as of December 31, 2023 (in millions):

Due in 1 year or less	$ 12,374
Due in 1 year through 5 years	297
Due in 5 years through 10 years	25
Total	$ 12,696

Disclosure of Fair Values

Our financial instruments that are not re-measured at fair value include accounts receivable, financing receivables, other receivables, digital assets, accounts payable, accrued liabilities, customer deposits and debt. The carrying values of these financial instruments materially approximate their fair values, other than our 2.00% Convertible Senior Notes due in 2024 ("2024 Notes") and digital assets.

We estimate the fair value of the 2024 Notes using commonly accepted valuation methodologies and market-based risk measurements that are indirectly observable, such as credit risk (Level II). In addition, we estimate the fair values of our digital assets based on quoted prices in active markets (Level I). The following table presents the estimated fair values and the carrying values (in millions):

	December 31, 2023		December 31, 2022	
	Carrying Value	**Fair Value**	**Carrying Value**	**Fair Value**
2024 Notes	$ 37	$ 443	$ 37	$ 223
Digital assets, net	$ 184	$ 487	$ 184	$ 191

Note 6 – Inventory

Our inventory consisted of the following (in millions):

	December 31, 2023	**December 31, 2022**
Raw materials	$ 5,390	$ 6,137
Work in process	2,016	2,385
Finished goods (1)	5,049	3,475
Service parts	1,171	842
Total	$ 13,626	$ 12,839

(1) Finished goods inventory includes products in transit to fulfill customer orders, new vehicles available for sale, used vehicles and energy products available for sale.

We write-down inventory for any excess or obsolete inventories or when we believe that the net realizable value of inventories is less than the carrying value. During the years ended December 31, 2023, 2022 and 2021 we recorded write-downs of $233 million, $144 million and $106 million, respectively, in Cost of revenues in the consolidated statements of operations.

Note 7 – Solar Energy Systems, Net

Our solar energy systems, net, consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Solar energy systems in service	$ 6,755	$ 6,785
Initial direct costs related to customer solar energy system lease acquisition costs	104	104
	6,859	6,889
Less: accumulated depreciation and amortization (1)	(1,643)	(1,418)
	5,216	5,471
Solar energy systems under construction	1	2
Solar energy systems pending interconnection	12	16
Solar energy systems, net (2)	$ 5,229	$ 5,489

(1) Depreciation and amortization expense during the years ended December 31, 2023, 2022 and 2021 was $235 million, $235 million and $236 million, respectively.

(2) As of December 31, 2023 and 2022, there were $740 million and $802 million, respectively, of gross solar energy systems under lease pass-through fund arrangements with accumulated depreciation of $157 million and $148 million, respectively.

Note 8 – Property, Plant and Equipment, Net

Our property, plant and equipment, net, consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Machinery, equipment, vehicles and office furniture	$ 16,372	$ 13,558
Tooling	3,147	2,579
Leasehold improvements	3,168	2,366
Land and buildings	9,505	7,751
Computer equipment, hardware and software	3,799	2,072
Construction in progress	5,791	4,263
	41,782	32,589
Less: Accumulated depreciation	(12,057)	(9,041)
Total	$ 29,725	$ 23,548

Construction in progress is primarily comprised of ongoing construction and expansion of our facilities, and equipment and tooling related to the manufacturing of our products. Completed assets are transferred to their respective asset classes and depreciation begins when an asset is ready for its intended use.

Depreciation expense during the years ended December 31, 2023, 2022 and 2021 was $3.33 billion, $2.42 billion and $1.91 billion, respectively.

Panasonic has partnered with us on Gigafactory Nevada with investments in the production equipment that it uses to manufacture and supply us with battery cells. Under our arrangement with Panasonic, we plan to purchase the full output from their production equipment at negotiated prices. As the terms of the arrangement convey a finance lease under ASC 842, we account for their production equipment as leased assets when production commences. We account for each lease and any non-lease components associated with that lease as a single lease component for all asset classes, except production equipment classes embedded in supply agreements. This results in us recording the cost of their production equipment within Property, plant and equipment, net, on the consolidated balance sheets with a corresponding liability recorded to debt and finance leases. Depreciation on Panasonic production equipment is computed using the units-of-production method whereby capitalized costs are amortized over the total estimated productive life of the respective assets. As of December 31, 2023 and 2022, we had cumulatively capitalized gross costs of $2.02 billion and $2.01 billion, respectively, on the consolidated balance sheets in relation to the production equipment under our Panasonic arrangement.

Note 9 – Accrued Liabilities and Other

Our accrued liabilities and other current liabilities consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Accrued purchases (1)	$ 2,721	$ 2,747
Accrued warranty reserve, current portion	1,546	1,025
Payroll and related costs	1,325	1,026
Taxes payable (2)	1,204	1,235
Customer deposits	876	1,063
Operating lease liabilities, current portion	672	485
Sales return reserve, current portion	219	270
Other current liabilities	517	354
Total	$ 9,080	$ 8,205

(1) Accrued purchases primarily reflects receipts of goods and services for which we had not yet been invoiced. As we are invoiced for these goods and services, this balance will reduce and accounts payable will increase.

(2) Taxes payable includes value added tax, income tax, sales tax, property tax and use tax payables.

Note 10 – Other Long-Term Liabilities

Our other long-term liabilities consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Operating lease liabilities	$ 3,671	$ 2,164
Accrued warranty reserve	3,606	2,480
Other non-current liabilities	876	686
Total other long-term liabilities	$ 8,153	$ 5,330

Note 11 – Debt

The following is a summary of our debt and finance leases as of December 31, 2023 (in millions):

	Net Carrying Value		Unpaid Principal Balance	Unused Committed Amount (1)	Contractual Interest Rates	Contractual Maturity Date
	Current	Long-Term				
Recourse debt:						
2024 Notes	$ 37	$ —	$ 37	$ —	2.00 %	May 2024
RCF Credit Agreement	—	—	—	5,000	Not applicable	January 2028
Solar Bonds	—	7	7	—	4.70-5.75%	March 2025 - January 2031
Other	—	—	—	28	Not applicable	December 2026
Total recourse debt	37	7	44	5,028		
Non-recourse debt:						
Automotive Asset-backed Notes	1,906	2,337	4,259	—	0.60-6.57%	July 2024-May 2031
Solar Asset-backed Notes	4	8	13	—	4.80 %	December 2026
Cash Equity Debt	28	330	367	—	5.25-5.81%	July 2033-January 2035
Total non-recourse debt	1,938	2,675	4,639	—		
Total debt	1,975	2,682	$ 4,683	$ 5,028		
Finance leases	398	175				
Total debt and finance leases	$ 2,373	$ 2,857				

The following is a summary of our debt and finance leases as of December 31, 2022 (in millions):

	Net Carrying Value		Unpaid Principal Balance	Unused Committed Amount (2)	Contractual Interest Rates	Contractual Maturity Date
	Current	Long-Term				
Recourse debt:						
2024 Notes	$ —	$ 37	$ 37	$ —	2.00 %	May 2024
Credit Agreement	—	—	—	2,266	Not applicable	July 2023
Solar Bonds	—	7	7	—	4.70-5.75%	March 2025 - January 2031
Total recourse debt	—	44	44	2,266		
Non-recourse debt:						
Automotive Asset-backed Notes	984	613	1,603	—	0.36-4.64%	December 2023-September 2025
Solar Asset-backed Notes	4	13	17	—	4.80 %	December 2026
Cash Equity Debt	28	359	397	—	5.25-5.81%	July 2033-January 2035
Automotive Lease-backed Credit Facilities	—	—	—	151	Not applicable	September 2024
Total non-recourse debt	1,016	985	2,017	151		
Total debt	1,016	1,029	$ 2,061	$ 2,417		
Finance leases	486	568				
Total debt and finance leases	$ 1,502	$ 1,597				

(1) There are no restrictions on draw-down or use for general corporate purposes with respect to any available committed funds under our RCF Credit Agreement, except certain specified conditions prior to draw-down. Refer to the section below for the terms of the facility.

(2) There were no restrictions on draw-down or use for general corporate purposes with respect to any available committed funds under our credit facilities, except certain specified conditions prior to draw-down, including pledging to our lenders sufficient amounts of qualified receivables, inventories, leased vehicles and our interests in those leases or various other assets as described below.

Recourse debt refers to debt that is recourse to our general assets. Non-recourse debt refers to debt that is recourse to only assets of our subsidiaries. The differences between the unpaid principal balances and the net carrying values are due to debt discounts or deferred issuance costs. As of December 31, 2023, we were in material compliance with all financial debt covenants.

2024 Notes

The closing price of our common stock continued to exceed 130% of the applicable conversion price of our 2024 Notes on at least 20 of the last 30 consecutive trading days of each quarter in 2023, causing the 2024 Notes to be convertible by their holders in the subsequent quarter. As of December 31, 2023, the if-converted value of the notes exceeds the outstanding principal amount by $406 million. Upon conversion, the 2024 Notes will be settled in cash, shares of our common stock or a combination thereof, at our election.

Credit Agreement

In June 2015, we entered into a senior asset-based revolving credit agreement (as amended from time to time, the “Credit Agreement”) with a syndicate of banks. Borrowed funds bear interest, at our option, at an annual rate of (a) 1% plus LIBOR or (b) the highest of (i) the federal funds rate plus 0.50%, (ii) the lenders’ “prime rate” or (iii) 1% plus LIBOR. The fee for undrawn amounts is 0.25% per annum. The Credit Agreement is secured by certain of our accounts receivable, inventory and equipment. Availability under the Credit Agreement is based on the value of such assets, as reduced by certain reserves.

In January 2023, we entered into a 5-year senior unsecured revolving credit facility (the “RCF Credit Agreement”) with a syndicate of banks to replace the existing Credit Agreement, which was terminated. The RCF Credit Agreement contains two optional one-year extensions and has a total commitment of up to $5.00 billion, which could be increased up to $7.00 billion under certain circumstances. The underlying borrowings may be used for general corporate purposes. Borrowed funds accrue interest at a variable rate equal to: (i) for dollar-denominated loans, at our election, (a) Term SOFR (the forward-looking secured overnight financing rate) plus 0.10%, or (b) an alternate base rate; (ii) for loans denominated in pounds sterling, SONIA (the sterling overnight index average reference rate); or (iii) for loans denominated in euros, an adjusted EURIBOR (euro interbank offered rate); in each case, plus an applicable margin. The applicable margin will be based on the rating assigned to our senior, unsecured long-term indebtedness (the “Credit Rating”) from time to time. The fee for undrawn amounts is variable based on the Credit Rating and is currently 0.125% per annum.

Automotive Asset-backed Notes

From time to time, we transfer receivables and/or beneficial interests related to certain vehicles (either leased or financed) into special purpose entities ("SPEs") and issue Automotive Asset-backed Notes, backed by these automotive assets to investors. The SPEs are consolidated in the financial statements. The cash flows generated by these automotive assets are used to service the principal and interest payments on the Automotive Asset-backed Notes and satisfy the SPEs' expenses, and any remaining cash is distributed to the owners of the SPEs. We recognize revenue earned from the associated customer lease or financing contracts in accordance with our revenue recognition policy. The SPEs' assets and cash flows are not available to our other creditors, and the creditors of the SPEs, including the Automotive Asset-backed Note holders, have no recourse to our other assets.

In 2023, we transferred beneficial interests related to certain leased vehicles and financing receivables into SPEs and issued $3.93 billion in aggregate principal amount of Automotive Asset-backed Notes, with terms similar to our other previously issued Automotive Asset-backed Notes. The proceeds from the issuance, net of debt issuance costs, were $3.92 billion.

Cash Equity Debt

In connection with the cash equity financing deals closed in 2016, our subsidiaries issued $502 million in aggregate principal amount of debt that bears interest at fixed rates. This debt is secured by, among other things, our interests in certain financing funds and is non-recourse to our other assets.

Automotive Lease-backed Credit Facilities

In the third quarter of 2023, we terminated our Automotive Lease-backed Credit Facilities and the previously committed funds are no longer available for future borrowings.

Pledged Assets

As of December 31, 2023 and 2022, we had pledged or restricted $4.64 billion and $2.02 billion of our assets (consisting principally of operating lease vehicles, financing receivables, restricted cash, and equity interests in certain SPEs) as collateral for our outstanding debt.

Schedule of Principal Maturities of Debt

The future scheduled principal maturities of debt as of December 31, 2023 were as follows (in millions):

	Recourse debt	Non-recourse debt	Total
2024	$ 37	$ 1,941	$ 1,978
2025	4	1,663	1,667
2026	—	494	494
2027	—	276	276
2028	—	44	44
Thereafter	3	221	224
Total	$ 44	$ 4,639	$ 4,683

Note 12 – Leases

We have entered into various operating and finance lease agreements for certain of our offices, manufacturing and warehouse facilities, retail and service locations, data centers, equipment, vehicles, and solar energy systems, worldwide. We determine if an arrangement is a lease, or contains a lease, at inception and record the leases in our financial statements upon lease commencement, which is the date when the underlying asset is made available for use by the lessor.

We have lease agreements with lease and non-lease components, and have elected to utilize the practical expedient to account for lease and non-lease components together as a single combined lease component, from both a lessee and lessor perspective with the exception of direct sales-type leases and production equipment classes embedded in supply agreements. From a lessor perspective, the timing and pattern of transfer are the same for the non-lease components and associated lease component and, the lease component, if accounted for separately, would be classified as an operating lease.

We have elected not to present short-term leases on the consolidated balance sheet as these leases have a lease term of 12 months or less at lease inception and do not contain purchase options or renewal terms that we are reasonably certain to exercise. All other lease assets and lease liabilities are recognized based on the present value of lease payments over the lease term at commencement date. Because most of our leases do not provide an implicit rate of return, we used our incremental borrowing rate based on the information available at lease commencement date in determining the present value of lease payments.

Our leases, where we are the lessee, often include options to extend the lease term for up to 10 years. Some of our leases also include options to terminate the lease prior to the end of the agreed upon lease term. For purposes of calculating lease liabilities, lease terms include options to extend or terminate the lease when it is reasonably certain that we will exercise such options.

Lease expense for operating leases is recognized on a straight-line basis over the lease term as cost of revenues or operating expenses depending on the nature of the leased asset. Certain operating leases provide for annual increases to lease payments based on an index or rate. We calculate the present value of future lease payments based on the index or rate at the lease commencement date for new leases. Differences between the calculated lease payment and actual payment are expensed as incurred. Amortization of finance lease assets is recognized over the lease term as cost of revenues or operating expenses depending on the nature of the leased asset. Interest expense on finance lease liabilities is recognized over the lease term within Interest expense in the consolidated statements of operations.

The balances for the operating and finance leases where we are the lessee are presented as follows (in millions) within our consolidated balance sheets:

	December 31, 2023	December 31, 2022
Operating leases:		
Operating lease right-of-use assets	$ 4,180	$ 2,563
Accrued liabilities and other	$ 672	$ 485
Other long-term liabilities	3,671	2,164
Total operating lease liabilities	$ 4,343	$ 2,649
Finance leases:		
Solar energy systems, net	$ 23	$ 25
Property, plant and equipment, net	601	1,094
Total finance lease assets	$ 624	$ 1,119
Current portion of long-term debt and finance leases	$ 398	$ 486
Long-term debt and finance leases, net of current portion	175	568
Total finance lease liabilities	$ 573	$ 1,054

The components of lease expense are as follows (in millions) within our consolidated statements of operations:

	Year Ended December 31,		
	2023	2022	2021
Operating lease expense:			
Operating lease expense (1)	$ 1,153	$ 798	$ 627
Finance lease expense:			
Amortization of leased assets	$ 506	$ 493	$ 415
Interest on lease liabilities	45	72	89
Total finance lease expense	$ 551	$ 565	$ 504
Total lease expense	$ 1,704	$ 1,363	$ 1,131

(1) Includes short-term leases and variable lease costs, which are immaterial.

Other information related to leases where we are the lessee is as follows:

	December 31, 2023	December 31, 2022
Weighted-average remaining lease term:		
Operating leases	7.4 years	6.4 years
Finance leases	2.3 years	3.1 years
Weighted-average discount rate:		
Operating leases	5.6 %	5.3 %
Finance leases	5.5 %	5.7 %

Supplemental cash flow information related to leases where we are the lessee is as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash outflows from operating leases	$ 1,084	$ 754	$ 616
Operating cash outflows from finance leases (interest payments)	$ 47	$ 75	$ 89
Leased assets obtained in exchange for finance lease liabilities	$ 10	$ 58	$ 486
Leased assets obtained in exchange for operating lease liabilities	$ 2,170	$ 1,059	$ 818

As of December 31, 2023, the maturities of our operating and finance lease liabilities (excluding short-term leases) are as follows (in millions):

	Operating Leases	Finance Leases
2024	$ 892	$ 418
2025	831	81
2026	706	57
2027	603	38
2028	508	2
Thereafter	1,820	4
Total minimum lease payments	5,360	600
Less: Interest	1,017	27
Present value of lease obligations	4,343	573
Less: Current portion	672	398
Long-term portion of lease obligations	$ 3,671	$ 175

As of December 31, 2023, we have excluded from the table above additional operating leases that have not yet commenced with aggregate rent payments of $1.53 billion. These operating leases will commence between fiscal year 2024 and 2025 with lease terms of 2 years to 20 years.

Operating Lease and Sales-type Lease Receivables

We are the lessor of certain vehicle and solar energy system arrangements as described in Note 2, *Summary of Significant Accounting Policies*. As of December 31, 2023, maturities of our operating lease and sales-type lease receivables from customers for each of the next five years and thereafter were as follows (in millions):

	Operating Leases	Sales-type Leases
2024	$ 1,405	$ 227
2025	960	214
2026	461	210
2027	227	102
2028	197	25
Thereafter	1,492	2
Gross lease receivables	$ 4,742	$ 780

The above table does not include vehicle sales to customers or leasing partners with a resale value guarantee as the cash payments were received upfront. For our solar PPA arrangements, customers are charged solely based on actual power produced by the installed solar energy system at a predefined rate per kilowatt-hour of power produced. The future payments from such arrangements are not included in the above table as they are a function of the power generated by the related solar energy systems in the future.

Net Investment in Sales-type Leases

Net investment in sales-type leases, which is the sum of the present value of the future contractual lease payments, is presented on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Lease receivables relating to sales-type leases are presented on the consolidated balance sheets as follows (in millions):

	December 31, 2023	December 31, 2022
Gross lease receivables	$ 780	$ 837
Unearned interest income	(78)	(95)
Allowance for expected credit losses	(6)	(4)
Net investment in sales-type leases	$ 696	$ 738
Reported as:		
Prepaid expenses and other current assets	$ 189	$ 164
Other non-current assets	507	574
Net investment in sales-type leases	$ 696	$ 738

Lease Pass-Through Financing Obligation

As of December 31, 2023, we have five transactions referred to as “lease pass-through fund arrangements.” Under these arrangements, our wholly owned subsidiaries finance the cost of solar energy systems with investors through arrangements contractually structured as master leases for an initial term ranging between 10 and 25 years. These solar energy systems are subject to lease or PPAs with customers with an initial term not exceeding 25 years.

Under a lease pass-through fund arrangement, the investor makes a large upfront payment to the lessor, which is one of our subsidiaries, and in some cases, subsequent periodic payments. As of December 31, 2023, the future minimum master lease payments to be received from investors, for each of the next five years and thereafter, were as follows (in millions):

2024	$ 18
2025	27
2026	28
2027	29
2028	29
Thereafter	337
Total	$ 468

Note 13 – Equity Incentive Plans

In June 2019, we adopted the 2019 Equity Incentive Plan (the “2019 Plan”). The 2019 Plan provides for the grant of stock options, restricted stock, RSUs, stock appreciation rights, performance units and performance shares to our employees, directors and consultants. Stock options granted under the 2019 Plan may be either incentive stock options or nonstatutory stock options. Incentive stock options may only be granted to our employees. Nonstatutory stock options may be granted to our employees, directors and consultants. Generally, our stock options and RSUs vest over four years and our stock options are exercisable over a maximum period of 10 years from their grant dates. Vesting typically terminates when the employment or consulting relationship ends.

As of December 31, 2023, 131.1 million shares were reserved and available for issuance under the 2019 Plan.

The following table summarizes our stock option and RSU activity for the year ended December 31, 2023:

	Stock Options				RSUs	
	Number of Options (in thousands)	Weighted-Average Exercise Price	Weighted-Average Remaining Contractual Life (years)	Aggregate Intrinsic Value (in billions)	Number of RSUs (in thousands)	Weighted-Average Grant Date Fair Value
Beginning of period	343,564	$ 30.65			21,333	$ 162.32
Granted	9,521	$ 226.50			11,743	$ 228.33
Exercised or released	(7,626)	$ 43.07			(11,085)	$ 116.47
Cancelled	(1,438)	$ 194.23			(2,903)	$ 192.22
End of period	344,021	$ 35.11	4.31	$ 73.57	19,088	$ 225.01
Vested and expected to vest, December 31, 2023	340,884	$ 33.38	4.27	$ 73.45	18,446	$ 225.76
Exercisable and vested, December 31, 2023	329,124	$ 27.07	4.11	$ 72.90		

The weighted-average grant date fair value of RSUs granted in the years ended December 31, 2023, 2022 and 2021 was $228.33, $239.85 and $261.33, respectively. The aggregate release date fair value of RSUs in the years ended December 31, 2023, 2022 and 2021 was $2.50 billion, $4.32 billion and $5.70 billion, respectively.

The aggregate intrinsic value of options exercised in the years ended December 31, 2023, 2022, and 2021 was $1.33 billion, $1.90 billion and $26.88 billion, respectively. During the year ended December 31, 2021, our CEO exercised all of the remaining vested options from the 2012 CEO Performance Award, which amounted to an intrinsic value of $23.45 billion.

ESPP

Our employees are eligible to purchase our common stock through payroll deductions of up to 15% of their eligible compensation, subject to any plan limitations. The purchase price would be 85% of the lower of the fair market value on the first and last trading days of each six-month offering period. During the years ended December 31, 2023, 2022 and 2021, under the ESPP we issued 2.1 million, 1.4 million and 1.5 million shares, respectively. As of December 31, 2023, there were 97.8 million shares available for issuance under the ESPP.

Fair Value Assumptions

We use the fair value method in recognizing stock-based compensation expense. Under the fair value method, we estimate the fair value of each stock option award with service or service and performance conditions and the ESPP on the grant date generally using the Black-Scholes option pricing model. The weighted-average assumptions used in the Black-Scholes model for stock options are as follows:

	Year Ended December 31,		
	2023	2022	2021
Risk-free interest rate	3.90 %	3.11 %	0.66 %
Expected term (in years)	4.5	4.1	4.3
Expected volatility	63 %	63 %	59 %
Dividend yield	0.0 %	0.0 %	0.0 %
Grant date fair value per share	$ 121.62	$ 114.51	$ 128.02

The fair value of RSUs with service or service and performance conditions is measured on the grant date based on the closing fair market value of our common stock. The risk-free interest rate is based on the U.S. Treasury yield for zero-coupon U.S. Treasury notes with maturities approximating each grant's expected life. We use our historical data in estimating the expected term of our employee grants. The expected volatility is based on the average of the implied volatility of publicly traded options for our common stock and the historical volatility of our common stock.

2018 CEO Performance Award

In March 2018, our stockholders approved the Board of Directors' grant of 304.0 million stock option awards, as adjusted to give effect to the 2020 Stock Split and the 2022 Stock Split, to our CEO (the "2018 CEO Performance Award"). The 2018 CEO Performance Award consisted of 12 vesting tranches with a vesting schedule based entirely on the attainment of both operational milestones (performance conditions) and market conditions, assuming continued employment either as the CEO or as both Executive Chairman and Chief Product Officer and service through each vesting date. Each of the 12 vesting tranches of the 2018 CEO Performance Award vested upon certification by the Board of Directors that both (i) the market capitalization milestone for such tranche, which began at $100.0 billion for the first tranche and increases by increments of $50.0 billion thereafter (based on both a six calendar month trailing average and a 30 calendar day trailing average, counting only trading days), had been achieved, and (ii) any one of the following eight operational milestones focused on total revenue or any one of the eight operational milestones focused on Adjusted EBITDA had been achieved for the four consecutive fiscal quarters on an annualized basis and subsequently reported by us in our consolidated financial statements filed with our Forms 10-Q and/or 10-K. Adjusted EBITDA was defined as net income (loss) attributable to common stockholders before interest expense, provision (benefit) for income taxes, depreciation and amortization and stock-based compensation. Upon vesting and exercise, including the payment of the exercise price of $23.34 per share as adjusted to give effect to the 2020 Stock Split and the 2022 Stock Split, our CEO must hold shares that he acquires for five years post-exercise, other than a cashless exercise where shares are simultaneously sold to pay for the exercise price and any required tax withholding.

The achievement status of the operational milestones as of December 31, 2023 is provided below.

Total Annualized Revenue			Annualized Adjusted EBITDA		
Milestone (in billions)		Achievement Status	Milestone (in billions)		Achievement Status
$	20.0	Achieved	$	1.5	Achieved
$	35.0	Achieved	$	3.0	Achieved
$	55.0	Achieved	$	4.5	Achieved
$	75.0	Achieved	$	6.0	Achieved
$	100.0	-	$	8.0	Achieved
$	125.0	-	$	10.0	Achieved
$	150.0	-	$	12.0	Achieved
$	175.0	-	$	14.0	Achieved

Stock-based compensation under the 2018 CEO Performance Award represented a non-cash expense and was recorded as a Selling, general, and administrative operating expense in our consolidated statements of operations. In each quarter since the grant of the 2018 CEO Performance Award, we had recognized expense, generally on a pro-rated basis, for only the number of tranches (up to the maximum of 12 tranches) that corresponded to the number of operational milestones that had been achieved or had been determined probable of being achieved in the future, in accordance with the following principles.

On the grant date, a Monte Carlo simulation was used to determine for each tranche (i) a fixed amount of expense for such tranche and (ii) the future time when the market capitalization milestone for such tranche was expected to be achieved, or its "expected market capitalization milestone achievement time." Separately, based on a subjective assessment of our future financial performance each quarter, we determined whether it was probable that we would achieve each operational milestone that had not previously been achieved or deemed probable of achievement and if so, the future time when we expected to achieve that operational milestone, or its "expected operational milestone achievement time."

As of December 31, 2022, all remaining unrecognized stock-based compensation expense under the 2018 CEO Performance Award had been recognized. For the years ended December 31, 2022 and 2021, we recorded stock-based compensation expense of $66 million and $910 million, respectively, related to the 2018 CEO Performance Award.

Other Performance-Based Grants

From time to time, the Compensation Committee of our Board of Directors grants certain employees performance-based RSUs and stock options.

As of December 31, 2023, we had unrecognized stock-based compensation expense of $655 million under these grants to purchase or receive an aggregate 5.3 million shares of our common stock. For awards probable of achievement, we estimate the unrecognized stock-based compensation expense of $110 million will be recognized over a weighted-average period of 4.0 years.

For the years ended December 31, 2023 and 2022, we recorded $57 million and $159 million, respectively, of stock-based compensation expense related to these grants, net of forfeitures.

Summary Stock-Based Compensation Information

The following table summarizes our stock-based compensation expense by line item in the consolidated statements of operations (in millions):

	Year Ended December 31,		
	2023	2022	2021
Cost of revenues	$ 741	$ 594	$ 421
Research and development	689	536	448
Selling, general and administrative	382	430	1,252
Total	$ 1,812	$ 1,560	$ 2,121

Our income tax benefits recognized from stock-based compensation arrangements were immaterial while we were under full valuation allowances on our U.S. deferred tax assets during the years ended December 31, 2022 and 2021. With the release of the valuation allowance associated with our federal and certain state deferred tax assets in 2023, income tax benefits recognized from stock-based compensation expense were $326 million during the year ended December 31, 2023. During the years ended December 31, 2023, 2022 and 2021, stock-based compensation expense capitalized to our consolidated balance sheets was $199 million, $245 million and $182 million, respectively. As of December 31, 2023, we had $4.82 billion of total unrecognized stock-based compensation expense related to non-performance awards, which will be recognized over a weighted-average period of 2.8 years.

Note 14 – Income Taxes

Our income before (benefit from) provision for income taxes for the years ended December 31, 2023, 2022 and 2021 was as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Domestic	$ 3,196	$ 5,524	$ (130)
Noncontrolling interest and redeemable noncontrolling interest	(23)	31	125
Foreign	6,800	8,164	6,348
Income before income taxes	$ 9,973	$ 13,719	$ 6,343

A (benefit from) provision for income taxes of $(5.00) billion, $1.13 billion and $699 million has been recognized for the years ended December 31, 2023, 2022 and 2021, respectively. The components of the (benefit from) provision for income taxes for the years ended December 31, 2023, 2022 and 2021 consisted of the following (in millions):

	Year Ended December 31,		
	2023	**2022**	**2021**
Current:			
Federal	$ 48	$ —	$ —
State	57	62	9
Foreign	1,243	1,266	839
Total current	1,348	1,328	848
Deferred:			
Federal	(5,246)	26	—
State	(653)	1	—
Foreign	(450)	(223)	(149)
Total deferred	(6,349)	(196)	(149)
Total (Benefit from) provision for income taxes	$ (5,001)	$ 1,132	$ 699

The reconciliation of taxes at the federal statutory rate to our (benefit from) provision for income taxes for the years ended December 31, 2023, 2022 and 2021 was as follows (in millions):

	Year Ended December 31,		
	2023	**2022**	**2021**
Tax at statutory federal rate	$ 2,094	$ 2,881	$ 1,332
State tax, net of federal benefit	(372)	51	6
Nondeductible executive compensation	23	14	201
Excess tax benefits related to stock-based compensation	(288)	(745)	(7,123)
Nontaxable manufacturing credit	(101)	—	—
Foreign income rate differential	(816)	(923)	(668)
U.S. tax credits	(593)	(276)	(328)
GILTI inclusion	670	1,279	1,008
Unrecognized tax benefits	183	252	28
Change in valuation allowance	(5,962)	(1,532)	6,165
Other	161	131	78
(Benefit from) provision for income taxes	$ (5,001)	$ 1,132	$ 699

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. As of December 31, 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which is objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that our U.S. federal and certain state deferred tax assets are realizable. As such, we released $6.54 billion of our valuation allowance associated with the U.S. federal and state deferred tax assets, with the exception of our California deferred tax assets. We continue to maintain a full valuation allowance against our California deferred tax assets as of December 31, 2023, because we concluded they are not more likely than not to be realized as we expect our California deferred tax assets generation in future years to exceed our ability to use these deferred tax assets.

Deferred tax assets (liabilities) as of December 31, 2023 and 2022 consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Deferred tax assets:		
Net operating loss carry-forwards	$ 2,826	$ 4,486
Research and development credits	1,358	1,184
Other tax credits and attributes	827	217
Deferred revenue	1,035	751
Inventory and warranty reserves	1,258	819
Stock-based compensation	230	185
Operating lease right-of-use liabilities	930	554
Capitalized research and development costs	1,344	693
Deferred GILTI tax assets	760	466
Accruals and others	206	178
Total deferred tax assets	10,774	9,533
Valuation allowance	(892)	(7,349)
Deferred tax assets, net of valuation allowance	9,882	2,184
Deferred tax liabilities:		
Depreciation and amortization	(2,122)	(1,178)
Investment in certain financing funds	(133)	(238)
Operating lease right-of-use assets	(859)	(506)
Other	(116)	(15)
Total deferred tax liabilities	(3,230)	(1,937)
Deferred tax assets (liabilities), net of valuation allowance	$ 6,652	$ 247

As of December 31, 2023, we maintained valuation allowances of $892 million for deferred tax assets that are not more likely than not to be realized, which primarily included deferred tax assets in the state of California and certain foreign operating losses. The valuation allowance on our net deferred tax assets decreased by $6.46 billion and $1.73 billion during the years ended December 31, 2023 and 2022, respectively, and increased by $6.14 billion during the year ended December 31, 2021. The valuation allowance decrease during the year ended December 31, 2023 was primarily due to the release of our valuation allowance with respect to our U.S. federal and certain state deferred tax assets. The changes in valuation allowances during the years ended December 31, 2022 and 2021 were primarily due to changes in our U.S. deferred tax assets and liabilities in the respective year. Among our deferred tax assets in foreign jurisdictions, we recorded a valuation allowance on certain foreign net operating losses that are not more likely than not to be realized. The remainder of our foreign deferred tax assets are more likely than not to be realized given the expectation of future earnings in these jurisdictions.

As of December 31, 2023, we had $10.31 billion of federal and $10.36 billion of state net operating loss carry-forwards available to offset future taxable income, some of which, if not utilized, will begin to expire in 2024 for federal and state purposes. Federal and state laws can impose substantial restrictions on the utilization of net operating loss and tax credit carry-forwards in the event of an "ownership change," as defined in Section 382 of the Internal Revenue Code. We have determined that no significant limitation would be placed on the utilization of our net operating loss and tax credit carry-forwards due to prior ownership changes or expirations.

As of December 31, 2023, we had federal research and development tax credits of $1.10 billion, federal renewable energy tax credits of $605 million, and state research and development tax credits of $923 million. Most of our state research and development tax credits were in the state of California. If not utilized, some of the federal tax credits may expire in various amounts beginning in 2036. However, California research and development tax credits can be carried forward indefinitely.

The local government of Shanghai granted a beneficial corporate income tax rate of 15% to certain eligible enterprises, compared to the 25% statutory corporate income tax rate in China. Our Gigafactory Shanghai subsidiary was granted this beneficial income tax rate of 15% for 2019 through 2023. Starting in 2024, Gigafactory Shanghai is subject to 25% statutory corporate income tax rate in China.

As of December 31, 2023, we intend to indefinitely reinvest our foreign earnings and cash unless such repatriation results in no or minimal tax costs. We have recorded the taxes associated with the foreign earnings we intend to repatriate in the future. For the earnings we intend to indefinitely reinvest, no deferred tax liabilities for foreign withholding or other taxes have been recorded. The estimated amount of such unrecognized withholding tax liability associated with the indefinitely reinvested earnings is approximately $245 million.

Uncertain Tax Positions

The changes to our gross unrecognized tax benefits were as follows (in millions):

December 31, 2020	$	380
Increases in balances related to prior year tax positions		117
Decreases in balances related to prior year tax positions		(90)
Increases in balances related to current year tax positions		124
December 31, 2021		531
Increases in balances related to prior year tax positions		136
Decreases in balances related to prior year tax positions		(12)
Increases in balances related to current year tax positions		222
Decreases in balances related to expiration of the statute of limitations		(7)
December 31, 2022		870
Increases in balances related to prior year tax positions		59
Decreases related to settlement with tax authorities		(6)
Increases in balances related to current year tax positions		255
Decreases in balances related to expiration of the statute of limitations		(4)
December 31, 2023	$	1,174

We include interest and penalties related to unrecognized tax benefits in income tax expense. We recognized net interest and penalties related to unrecognized tax benefits in provision for income taxes line of our consolidated statements of operations of $17 million, $27 million and $4 million for the years ended December 31, 2023, 2022 and 2021, respectively. As of December 31, 2023, and 2022, we have accrued $47 million and $31 million, respectively, related to interest and penalties on our unrecognized tax benefits. Unrecognized tax benefits of $901 million, if recognized, would affect our effective tax rate.

We file income tax returns in the U.S. and various state and foreign jurisdictions. We are currently under examination by the Internal Revenue Service ("IRS") for the years 2015 to 2018. Additional tax years within the periods 2004 to 2014 and 2019 to 2022 remain subject to examination for federal income tax purposes. All net operating losses and tax credits generated to date are subject to adjustment for U.S. federal and state income tax purposes. Our returns for 2004 and subsequent tax years remain subject to examination in U.S. state and foreign jurisdictions.

Given the uncertainty in timing and outcome of our tax examinations, an estimate of the range of the reasonably possible change in gross unrecognized tax benefits within twelve months cannot be made at this time.

Note 15 – Commitments and Contingencies

Operating Lease Arrangement in Buffalo, New York

We have an operating lease arrangement through the Research Foundation for the SUNY Foundation with respect to Gigafactory New York. Under the lease and a related research and development agreement, we are continuing to further develop the facility.

Under this agreement, we are obligated to, among other things, meet employment targets as well as specified minimum numbers of personnel in the State of New York and in Buffalo, New York and spend or incur $5.00 billion in combined capital, operational expenses, costs of goods sold and other costs in the State of New York during the 10-year period beginning April 30, 2018. On an annual basis during the initial lease term, as measured on each anniversary of such date, if we fail to meet these specified investment and job creation requirements, then we would be obligated to pay a $41 million "program payment" to the SUNY Foundation for each year that we fail to meet these requirements. Furthermore, if the arrangement is terminated due to a material breach by us, then additional amounts may become payable by us.

In 2021, an amendment was executed to extend our overall agreement to spend or incur $5.00 billion in combined capital, operational expenses, costs of goods sold and other costs in the State of New York through December 31, 2029. On February 1, 2022, we reported to the State of New York that we had met and exceeded our annual requirements for jobs and investment in Buffalo and New York State. As of December 31, 2023, we have met and expect to meet the requirements under this arrangement based on our current and anticipated level of operations. However, if our expectations as to the costs and timelines of our investment and operations at Buffalo prove incorrect, we may incur additional expenses or be required to make substantial payments to the SUNY Foundation.

Operating Lease Arrangement in Shanghai, China

We have an operating lease arrangement for an initial term of 50 years with the local government of Shanghai for land use rights where we have been constructing Gigafactory Shanghai. Under the terms of the arrangement, we are required to spend RMB 14.08 billion in capital expenditures by the end of 2023, which has been achieved in 2023, and to generate RMB 2.23 billion of annual tax revenues starting at the end of 2023. As of December 31, 2023, we have met and expect to meet the tax revenue requirements based on our current level of spend and sales.

Legal Proceedings

Litigation Relating to 2018 CEO Performance Award

On June 4, 2018, a purported Tesla stockholder filed a putative class and derivative action in the Delaware Court of Chancery against Elon Musk and the members of Tesla's board of directors as then constituted, alleging corporate waste, unjust enrichment and that such board members breached their fiduciary duties by approving the stock-based compensation plan awarded to Elon Musk in 2018. Trial was held November 14-18, 2022. Post-trial briefing and argument are now complete.

Litigation Related to Directors' Compensation

On June 17, 2020, a purported Tesla stockholder filed a derivative action in the Delaware Court of Chancery, purportedly on behalf of Tesla, against certain of Tesla's current and former directors regarding compensation awards granted to Tesla's directors, other than Elon Musk, between 2017 and 2020. The suit asserts claims for breach of fiduciary duty and unjust enrichment and seeks declaratory and injunctive relief, unspecified damages and other relief. Defendants filed their answer on September 17, 2020.

On July 14, 2023, the parties filed a Stipulation and Agreement of Compromise and Settlement, which does not involve an admission of any wrongdoing by any party. If the settlement is approved by the Court, this action will be fully settled and dismissed with prejudice. Pursuant to the terms of the agreement, Tesla provided notice of the proposed settlement to stockholders of record as of July 14, 2023. The Court held a hearing regarding the settlement on October 13, 2023, after which it took the settlement and plaintiff counsels' fee request under advisement. The settlement is not expected to have an adverse impact on our results of operations, cash flows or financial position.

Litigation Relating to Potential Going Private Transaction

Between August 10, 2018 and September 6, 2018, nine purported stockholder class actions were filed against Tesla and Elon Musk in connection with Mr. Musk's August 7, 2018 Twitter post that he was considering taking Tesla private. On January 16, 2019, Plaintiffs filed their consolidated complaint in the United States District Court for the Northern District of California and added as defendants the members of Tesla's board of directors. The consolidated complaint asserts claims for violations of the federal securities laws and seeks unspecified damages and other relief. The parties stipulated to certification of a class of stockholders, which the court granted on November 25, 2020. Trial started on January 17, 2023, and on February 3, 2023, a jury rendered a verdict in favor of the defendants on all counts. After trial, plaintiffs filed a motion for judgment as a matter of law and a motion for new trial, which the Court denied and judgement was entered in favor of defendants on July 11, 2023. On July 14, 2023, plaintiffs filed a notice of appeal.

Between October 17, 2018 and March 8, 2021, seven derivative lawsuits were filed in the Delaware Court of Chancery, purportedly on behalf of Tesla, against Mr. Musk and the members of Tesla's board of directors, as constituted at relevant times, in relation to statements made and actions connected to a potential going private transaction, with certain of the lawsuits challenging additional Twitter posts by Mr. Musk, among other things. Five of those actions were consolidated, and all seven actions have been stayed pending resolution of the appeal in the above-referenced consolidated purported stockholder class action. In addition to these cases, two derivative lawsuits were filed on October 25, 2018 and February 11, 2019 in the U.S. District Court for the District of Delaware, purportedly on behalf of Tesla, against Mr. Musk and the members of the Tesla board of directors as then constituted. Those cases have also been consolidated and stayed pending resolution of the appeal in the above-referenced consolidated purported stockholder class action.

On October 21, 2022, a lawsuit was filed in the Delaware Court of Chancery by a purported shareholder of Tesla alleging, among other things, that board members breached their fiduciary duties in connection with their oversight of the Company's 2018 settlement with the SEC, as amended. Among other things, the plaintiff seeks reforms to the Company's corporate governance and internal procedures, unspecified damages, and attorneys' fees. The parties reached an agreement to stay the case until March 5, 2024.

On November 15, 2021, JPMorgan Chase Bank ("JP Morgan") filed a lawsuit against Tesla in the Southern District of New York alleging breach of a stock warrant agreement that was entered into as part of a convertible notes offering in 2014. In 2018, JP Morgan informed Tesla that it had adjusted the strike price based upon Mr. Musk's August 7, 2018 Twitter post that he was considering taking Tesla private. Tesla disputed JP Morgan's adjustment as a violation of the parties' agreement. In 2021, Tesla delivered shares to JP Morgan per the agreement, which they duly accepted. JP Morgan now alleges that it is owed approximately $162 million as the value of additional shares that it claims should have been delivered as a result of the adjustment to the strike price in 2018. On January 24, 2022, Tesla filed multiple counterclaims as part of its answer to the underlying lawsuit, asserting among other points that JP Morgan should have terminated the stock warrant agreement in 2018 rather than make an adjustment to the strike price that it should have known would lead to a commercially unreasonable result. Tesla believes that the adjustments made by JP Morgan were neither proper nor commercially reasonable, as required under the stock warrant agreements. JP Morgan filed a motion for judgment on the pleadings, which Tesla opposed, and that motion is currently pending before the Court.

Litigation and Investigations Relating to Alleged Discrimination and Harassment

On October 4, 2021, in a case captioned Diaz v. Tesla, a jury in the Northern District of California returned a verdict against Tesla on claims by a former contingent worker that he was subjected to race discrimination while assigned to work at Tesla's Fremont Factory from 2015-2016. A retrial was held starting on March 27, 2023, after which a jury returned a verdict of $3,175,000. As a result, the damages awarded against Tesla were reduced from an initial $136.9 million (October 4, 2021) down to $15 million (April 13, 2022), and then further down to $3.175 million (April 3, 2023). On November 2, 2023, the plaintiff filed a notice of appeal, and on November 16, 2023, Tesla filed a notice of cross appeal.

On February 9, 2022, shortly after the first Diaz jury verdict, the California Civil Rights Department ("CRD," formerly "DFEH") filed a civil complaint against Tesla in Alameda County, California Superior Court, alleging systemic race discrimination, hostile work environment and pay equity claims, among others. CRD's amended complaint seeks monetary damages and injunctive relief. On September 22, 2022, Tesla filed a cross complaint against CRD, alleging that it violated the Administrative Procedures Act by failing to follow statutory pre-requisites prior to filing suit and that cross complaint was subject to a sustained demurrer, which Tesla later amended and refiled. The case is currently in discovery.

Additionally, on June 1, 2022 the Equal Employment Opportunity Commission ("EEOC") issued a cause finding against Tesla that closely parallels the CRD's allegations. On September 28, 2023, the EEOC filed a civil complaint against Tesla in the United States District Court for the Northern District of California asserting claims for race harassment and retaliation and seeking, among other things, monetary and injunctive relief. On December 18, 2023, Tesla filed a motion to stay the case. Separately, on December 26, 2023, Tesla filed a motion to dismiss the case.

On June 16, 2022, two Tesla stockholders filed separate derivative actions in the U.S. District Court for the Western District of Texas, purportedly on behalf of Tesla, against certain of Tesla's current and former directors. Both suits assert claims for breach of fiduciary duty, unjust enrichment, and violation of the federal securities laws in connection with alleged race and gender discrimination and sexual harassment. Among other things, plaintiffs seek declaratory and injunctive relief, unspecified damages payable to Tesla, and attorneys' fees. On July 22, 2022, the Court consolidated the two cases and on September 6, 2022, plaintiffs filed a consolidated complaint. On November 7, 2022, the defendants filed a motion to dismiss the case and on September 15, 2023, the Court dismissed the action but granted plaintiffs leave to file an amended complaint. On November 2, 2023, plaintiff filed an amended complaint purportedly on behalf of Tesla, against Elon Musk. On December 19, 2023, the defendants moved to dismiss the amended complaint.

Other Litigation Related to Our Products and Services

We are also subject to various lawsuits that seek monetary and other injunctive relief. These lawsuits include proposed class actions and other consumer claims that allege, among other things, purported defects and misrepresentations related to our products and services. For example, on September 14, 2022, a proposed class action was filed against Tesla, Inc. and related entities in the U.S. District Court for the Northern District of California, alleging various claims about the Company’s driver assistance technology systems under state and federal law. This case was later consolidated with several other proposed class actions, and a Consolidated Amended Complaint was filed on October 28, 2022, which seeks damages and other relief on behalf of all persons who purchased or leased from Tesla between January 1, 2016 to the present. On October 5, 2022 a proposed class action complaint was filed in the U.S. District Court for the Eastern District of New York asserting similar state and federal law claims against the same defendants. On September 30, 2023, the Court dismissed this action with leave to amend the complaint. On November 20, 2023, the plaintiff moved to amend the complaint, which Tesla opposed. On March 22, 2023, the plaintiffs in the Northern District of California consolidated action filed a motion for a preliminary injunction to order Tesla to (1) cease using the term “Full Self-Driving Capability” (FSD Capability), (2) cease the sale and activation of FSD Capability and deactivate FSD Capability on Tesla vehicles, and (3) provide certain notices to consumers about proposed court-findings about the accuracy of the use of the terms Autopilot and FSD Capability. Tesla opposed the motion. On September 30, 2023, the Court denied the request for a preliminary injunction, compelled four of five plaintiffs to arbitration, and dismissed the claims of the fifth plaintiff with leave to amend the complaint. On October 31, 2023, the remaining plaintiff in the Northern District of California action filed an amended complaint, which Tesla has moved to dismiss. On October 2, 2023, a similar proposed class action was filed in San Diego County Superior Court in California. Tesla subsequently removed the San Diego County case to federal court and on January 8, 2024, the federal court granted Tesla’s motion to transfer the case to the U.S. District Court for the Northern District of California.

On February 27, 2023, a proposed class action was filed in the U.S. District Court for the Northern District of California against Tesla, Inc., Elon Musk and certain current and former Company executives. The complaint alleges that the defendants made material misrepresentations and omissions about the Company’s Autopilot and FSD Capability technologies and seeks money damages and other relief on behalf of persons who purchased Tesla stock between February 19, 2019 and February 17, 2023. An amended complaint was filed on September 5, 2023, naming only Tesla, Inc. and Elon Musk as defendants. On November 6, 2023, Tesla moved to dismiss the amended complaint.

On March 14, 2023, a proposed class action was filed against Tesla, Inc. in the U.S. District Court for the Northern District of California. Several similar complaints have also been filed in the same court and these cases have now all been consolidated. These complaints allege that Tesla violates federal antitrust and warranty laws through its repair, service, and maintenance practices and seeks, among other relief, damages for persons who paid Tesla for repairs services or Tesla compatible replacement parts from March 2019 to March 2023. On July 17, 2023, these plaintiffs filed a consolidated amended complaint. On September 27, 2023, the court granted Tesla’s motion to compel arbitration as to three of the plaintiffs, and on November 17, 2023, the court granted Tesla’s motion to dismiss without prejudice. The plaintiffs filed a Consolidated Second Amended Complaint on December 12, 2023, which Tesla has moved to dismiss. Plaintiffs have also appealed the court’s arbitration order. Trial is currently set for July 7, 2025.

The Company intends to vigorously defend itself in these matters; however, we cannot predict the outcome or impact. We are unable to reasonably estimate the possible loss or range of loss, if any, associated with these claims, unless noted.

Certain Investigations and Other Matters

We regularly receive requests for information, including subpoenas, from regulators and governmental authorities such as the National Highway Traffic Safety Administration, the National Transportation Safety Board, the Securities and Exchange Commission (“SEC”), the Department of Justice (“DOJ”), and various local, state, federal, and international agencies. The ongoing requests for information include topics such as operations, technology (e.g., vehicle functionality, Autopilot and FSD Capability), compliance, finance, data privacy, and other matters related to Tesla’s business, its personnel, and related parties. We routinely cooperate with such formal and informal requests for information, investigations, and other inquiries. To our knowledge no government agency in any ongoing investigation has concluded that any wrongdoing occurred. We cannot predict the outcome or impact of any ongoing matters. Should the government decide to pursue an enforcement action, there exists the possibility of a material adverse impact on our business, results of operation, prospects, cash flows, financial position or brand.

We are also subject to various other legal proceedings, risks and claims that arise from the normal course of business activities. For example, during the second quarter of 2023, a foreign news outlet reported that it obtained certain misappropriated data including, purportedly non-public Tesla business and personal information. Tesla has made notifications to potentially affected individuals (current and former employees) and regulatory authorities and we are working with certain law enforcement and other authorities. On August 5, 2023, a putative class action was filed in the United States District Court for the Northern District of California, purportedly on behalf of all U.S. individuals impacted by the data incident, followed by several additional lawsuits, that each assert claims under various state laws and seeks monetary damages and other relief. If an unfavorable ruling or development were to occur in these or other possible legal proceedings, risks and claims, there exists the possibility of a material adverse impact on our business, results of operations, prospects, cash flows, financial position or brand.

Letters of Credit

As of December 31, 2023, we had $525 million of unused letters of credit outstanding.

Note 16 – Variable Interest Entity Arrangements

We have entered into various arrangements with investors to facilitate the funding and monetization of our solar energy systems and vehicles. In particular, our wholly owned subsidiaries and fund investors have formed and contributed cash and assets into various financing funds and entered into related agreements. We have determined that the funds are VIEs and we are the primary beneficiary of these VIEs by reference to the power and benefits criterion under ASC 810. We have considered the provisions within the agreements, which grant us the power to manage and make decisions that affect the operation of these VIEs, including determining the solar energy systems and the associated customer contracts to be sold or contributed to these VIEs, redeploying solar energy systems and managing customer receivables. We consider that the rights granted to the fund investors under the agreements are more protective in nature rather than participating.

As the primary beneficiary of these VIEs, we consolidate in the financial statements the financial position, results of operations and cash flows of these VIEs, and all intercompany balances and transactions between us and these VIEs are eliminated in the consolidated financial statements. Cash distributions of income and other receipts by a fund, net of agreed upon expenses, estimated expenses, tax benefits and detriments of income and loss and tax credits, are allocated to the fund investor and our subsidiary as specified in the agreements.

Generally, our subsidiary has the option to acquire the fund investor’s interest in the fund for an amount based on the market value of the fund or the formula specified in the agreements.

Upon the sale or liquidation of a fund, distributions would occur in the order and priority specified in the agreements.

Pursuant to management services, maintenance and warranty arrangements, we have been contracted to provide services to the funds, such as operations and maintenance support, accounting, lease servicing and performance reporting. In some instances, we have guaranteed payments to the fund investors as specified in the agreements. A fund’s creditors have no recourse to our general credit or to that of other funds. Certain assets of the funds have been pledged as collateral for their obligations.

The aggregate carrying values of the VIEs' assets and liabilities, after elimination of any intercompany transactions and balances, in the consolidated balance sheets were as follows (in millions):

	December 31, 2023	December 31, 2022
Assets		
Current assets		
Cash and cash equivalents	$ 66	$ 68
Accounts receivable, net	13	22
Prepaid expenses and other current assets	361	274
Total current assets	440	364
Solar energy systems, net	3,278	4,060
Other non-current assets	369	404
Total assets	$ 4,087	$ 4,828
Liabilities		
Current liabilities		
Accrued liabilities and other	$ 67	$ 69
Deferred revenue	6	10
Current portion of debt and finance leases	1,564	1,013
Total current liabilities	1,637	1,092
Deferred revenue, net of current portion	99	149
Debt and finance leases, net of current portion	2,041	971
Other long-term liabilities	—	3
Total liabilities	$ 3,777	$ 2,215

Note 17 – Related Party Transactions

In relation to our CEO's exercise of stock options and sale of common stock from the 2012 CEO Performance Award, Tesla withheld the appropriate amount of taxes. However, given the significant amounts involved, our CEO entered into an indemnification agreement with us in November 2021 for additional taxes owed, if any.

Tesla periodically does business with certain entities with which its CEO and directors are affiliated, such as SpaceX and X Corp., in accordance with our Related Person Transactions Policy. Such transactions have not had to date, and are not currently expected to have, a material impact on our consolidated financial statements.

Note 18 – Segment Reporting and Information about Geographic Areas

We have two operating and reportable segments: (i) automotive and (ii) energy generation and storage. The automotive segment includes the design, development, manufacturing, sales and leasing of electric vehicles as well as sales of automotive regulatory credits. Additionally, the automotive segment is also comprised of services and other, which includes sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise. The energy generation and storage segment includes the design, manufacture, installation, sales and leasing of solar energy generation and energy storage products and related services and sales of solar energy systems incentives. Our CODM does not evaluate operating segments using asset or liability information. The following table presents revenues and gross profit by reportable segment (in millions):

	Year Ended December 31,		
	2023	2022	2021
Automotive segment			
Revenues	$ 90,738	$ 77,553	$ 51,034
Gross profit	$ 16,519	$ 20,565	$ 13,735
Energy generation and storage segment			
Revenues	$ 6,035	$ 3,909	$ 2,789
Gross profit	$ 1,141	$ 288	$ (129)

The following table presents revenues by geographic area based on the sales location of our products (in millions):

	Year Ended December 31,		
	2023	2022	2021
United States	$ 45,235	$ 40,553	$ 23,973
China	21,745	18,145	13,844
Other international	29,793	22,764	16,006
Total	$ 96,773	$ 81,462	$ 53,823

The following table presents long-lived assets by geographic area (in millions):

	December 31, 2023	December 31, 2022
United States	$ 26,629	$ 21,667
Germany	4,258	3,547
China	2,820	2,978
Other international	1,247	845
Total	$ 34,954	$ 29,037

The following table presents inventory by reportable segment (in millions):

	December 31, 2023	December 31, 2022
Automotive	$ 11,139	$ 10,996
Energy generation and storage	2,487	1,843
Total	$ 13,626	$ 12,839

Note 19 – Restructuring and Other

During the years ended December 31, 2022 and 2021, we recorded $204 million and $101 million, respectively, of impairment losses on digital assets. During the years ended December 31, 2022 and 2021 we also realized gains of $64 million and $128 million, respectively, in connection with converting our holdings of digital assets into fiat currency. We also recorded other expenses of $36 million during the second quarter of the year ended December 31, 2022, related to employee terminations.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our Chief Executive Officer and our Chief Financial Officer, evaluated the effectiveness of our disclosure controls and procedures pursuant to Rule 13a-15 under the Securities Exchange Act of 1934, as amended (the “Exchange Act”). In designing and evaluating the disclosure controls and procedures, our management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving the desired control objectives. In addition, the design of disclosure controls and procedures must reflect the fact that there are resource constraints and that our management is required to apply its judgment in evaluating the benefits of possible controls and procedures relative to their costs.

Based on this evaluation, our Chief Executive Officer and our Chief Financial Officer concluded that, as of December 31, 2023, our disclosure controls and procedures were designed at a reasonable assurance level and were effective to provide reasonable assurance that the information we are required to disclose in reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC rules and forms, and that such information is accumulated and communicated to our management, including our Chief Executive Officer and our Chief Financial Officer, as appropriate, to allow timely decisions regarding required disclosures.

Management’s Report on Internal Control over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is a process designed by, or under the supervision of, our Chief Executive Officer and Chief Financial Officer to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles and includes those policies and procedures that (1) pertain to the maintenance of records that in reasonable detail accurately and fairly reflect the transactions and dispositions of our assets; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that our receipts and expenditures are being made only in accordance with authorizations of our management and directors and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of our assets that could have a material effect on the financial statements.

Under the supervision and with the participation of our management, including our Chief Executive Officer and Chief Financial Officer, we conducted an evaluation of the effectiveness of our internal control over financial reporting based on criteria established in Internal Control – Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (“COSO”). Our management concluded that our internal control over financial reporting was effective as of December 31, 2023.

Our independent registered public accounting firm, PricewaterhouseCoopers LLP, has audited the effectiveness of our internal control over financial reporting as of December 31, 2023, as stated in their report which is included herein.

Limitations on the Effectiveness of Controls

Because of inherent limitations, internal control over financial reporting may not prevent or detect misstatements and projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Changes in Internal Control over Financial Reporting

There was no change in our internal control over financial reporting that occurred during the quarter ended December 31, 2023, which has materially affected, or is reasonably likely to materially affect, our internal control over financial reporting.

ITEM 9B. OTHER INFORMATION

None of the Company’s directors or officers adopted, modified or terminated a Rule 10b5-1 trading arrangement or a non-Rule 10b5-1 trading arrangement during the Company’s fiscal quarter ended December 31, 2023, as such terms are defined under Item 408(a) of Regulation S-K, except as follows:

On October 23, 2023, Robyn Denholm, one of our directors, adopted a Rule 10b5-1 trading arrangement for the potential sale of up to 281,116 shares of our common stock, subject to certain conditions. The trading arrangement covers stock options that expire in August 2024. The arrangement's expiration date is August 16, 2024.

On November 13, 2023, Andrew Baglino, Senior Vice President, Powertrain and Energy Engineering, adopted a Rule 10b5-1 trading arrangement for the potential sale of up to 115,500 shares of our common stock, subject to certain conditions. The arrangement's expiration date is December 31, 2024.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

The information required by this Item 10 of Form 10-K will be included in our 2024 Proxy Statement to be filed with the Securities and Exchange Commission in connection with the solicitation of proxies for our 2024 Annual Meeting of Stockholders and is incorporated herein by reference. The 2024 Proxy Statement will be filed with the Securities and Exchange Commission within 120 days after the end of the fiscal year to which this report relates.

ITEM 11. EXECUTIVE COMPENSATION

The information required by this Item 11 of Form 10-K will be included in our 2024 Proxy Statement and is incorporated herein by reference.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information required by this Item 12 of Form 10-K will be included in our 2024 Proxy Statement and is incorporated herein by reference.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS AND DIRECTOR INDEPENDENCE

The information required by this Item 13 of Form 10-K will be included in our 2024 Proxy Statement and is incorporated herein by reference.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

The information required by this Item 14 of Form 10-K will be included in our 2024 Proxy Statement and is incorporated herein by reference.

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

1. Financial statements (see *Index to Consolidated Financial Statements* in Part II, Item 8 of this report)
2. All financial statement schedules have been omitted since the required information was not applicable or was not present in amounts sufficient to require submission of the schedules, or because the information required is included in the consolidated financial statements or the accompanying notes
3. The exhibits listed in the following *Index to Exhibits* are filed or incorporated by reference as part of this report

INDEX TO EXHIBITS

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
3.1	Amended and Restated Certificate of Incorporation of the Registrant.	10-K	001-34756	3.1	March 1, 2017	
3.2	Certificate of Amendment to the Amended and Restated Certificate of Incorporation of the Registrant.	10-K	001-34756	3.2	March 1, 2017	
3.3	Amended and Restated Bylaws of the Registrant.	8-K	001-34756	3.1	April 5, 2023	
4.1	Specimen common stock certificate of the Registrant.	10-K	001-34756	4.1	March 1, 2017	
4.2	Fifth Amended and Restated Investors’ Rights Agreement, dated as of August 31, 2009, between Registrant and certain holders of the Registrant’s capital stock named therein.	S-1	333-164593	4.2	January 29, 2010	
4.3	Amendment to Fifth Amended and Restated Investors’ Rights Agreement, dated as of May 20, 2010, between Registrant and certain holders of the Registrant’s capital stock named therein.	S-1/A	333-164593	4.2A	May 27, 2010	
4.4	Amendment to Fifth Amended and Restated Investors’ Rights Agreement between Registrant, Toyota Motor Corporation and certain holders of the Registrant’s capital stock named therein.	S-1/A	333-164593	4.2B	May 27, 2010	
4.5	Amendment to Fifth Amended and Restated Investor’s Rights Agreement, dated as of June 14, 2010, between Registrant and certain holders of the Registrant’s capital stock named therein.	S-1/A	333-164593	4.2C	June 15, 2010	
4.6	Amendment to Fifth Amended and Restated Investor’s Rights Agreement, dated as of November 2, 2010, between Registrant and certain holders of the Registrant’s capital stock named therein.	8-K	001-34756	4.1	November 4, 2010	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.7	Waiver to Fifth Amended and Restated Investor’s Rights Agreement, dated as of May 22, 2011, between Registrant and certain holders of the Registrant’s capital stock named therein.	S-1/A	333-174466	4.2E	June 2, 2011	
4.8	Amendment to Fifth Amended and Restated Investor’s Rights Agreement, dated as of May 30, 2011, between Registrant and certain holders of the Registrant’s capital stock named therein.	8-K	001-34756	4.1	June 1, 2011	
4.9	Sixth Amendment to Fifth Amended and Restated Investors’ Rights Agreement, dated as of May 15, 2013 among the Registrant, the Elon Musk Revocable Trust dated July 22, 2003 and certain other holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	May 20, 2013	
4.10	Waiver to Fifth Amended and Restated Investor’s Rights Agreement, dated as of May 14, 2013, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.2	May 20, 2013	
4.11	Waiver to Fifth Amended and Restated Investor’s Rights Agreement, dated as of August 13, 2015, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	August 19, 2015	
4.12	Waiver to Fifth Amended and Restated Investors’ Rights Agreement, dated as of May 18, 2016, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	May 24, 2016	
4.13	Waiver to Fifth Amended and Restated Investors’ Rights Agreement, dated as of March 15, 2017, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	March 17, 2017	
4.14	Waiver to Fifth Amended and Restated Investors’ Rights Agreement, dated as of May 1, 2019, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	May 3, 2019	
4.15	Indenture, dated as of May 22, 2013, by and between the Registrant and U.S. Bank National Association.	8-K	001-34756	4.1	May 22, 2013	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.16	Fifth Supplemental Indenture, dated as of May 7, 2019, by and between Registrant and U.S. Bank National Association, related to 2.00% Convertible Senior Notes due May 15, 2024.	8-K	001-34756	4.2	May 8, 2019	
4.17	Form of 2.00% Convertible Senior Notes due May 15, 2024 (included in Exhibit 4.16).	8-K	001-34756	4.2	May 8, 2019	
4.18	Indenture, dated as of October 15, 2014, between SolarCity and U.S. Bank National Association, as trustee.	S-3ASR(1)	333-199321	4.1	October 15, 2014	
4.19	Tenth Supplemental Indenture, dated as of March 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/6-10.	8-K(1)	001-35758	4.3	March 9, 2015	
4.20	Eleventh Supplemental Indenture, dated as of March 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/7-15.	8-K(1)	001-35758	4.4	March 9, 2015	
4.21	Fifteenth Supplemental Indenture, dated as of March 19, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C4-10.	8-K(1)	001-35758	4.5	March 19, 2015	
4.22	Sixteenth Supplemental Indenture, dated as of March 19, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C5-15.	8-K(1)	001-35758	4.6	March 19, 2015	
4.23	Twentieth Supplemental Indenture, dated as of March 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C9-10.	8-K(1)	001-35758	4.5	March 26, 2015	
4.24	Twenty-First Supplemental Indenture, dated as of March 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C10-15.	8-K(1)	001-35758	4.6	March 26, 2015	
4.25	Twenty-Sixth Supplemental Indenture, dated as of April 2, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C14-10.	8-K(1)	001-35758	4.5	April 2, 2015	
4.26	Thirtieth Supplemental Indenture, dated as of April 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C19-10.	8-K(1)	001-35758	4.5	April 9, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.27	Thirty-First Supplemental Indenture, dated as of April 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C20-15.	8-K(1)	001-35758	4.6	April 9, 2015	
4.28	Thirty-Fifth Supplemental Indenture, dated as of April 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C24-10.	8-K(1)	001-35758	4.5	April 14, 2015	
4.29	Thirty-Sixth Supplemental Indenture, dated as of April 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C25-15.	8-K(1)	001-35758	4.6	April 14, 2015	
4.30	Thirty-Eighth Supplemental Indenture, dated as of April 21, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C27-10.	8-K(1)	001-35758	4.3	April 21, 2015	
4.31	Thirty-Ninth Supplemental Indenture, dated as of April 21, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C28-15.	8-K(1)	001-35758	4.4	April 21, 2015	
4.32	Forty-Third Supplemental Indenture, dated as of April 27, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C32-10.	8-K(1)	001-35758	4.5	April 27, 2015	
4.33	Forty-Fourth Supplemental Indenture, dated as of April 27, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C33-15.	8-K(1)	001-35758	4.6	April 27, 2015	
4.34	Forty-Eighth Supplemental Indenture, dated as of May 1, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/12-10.	8-K(1)	001-35758	4.5	May 1, 2015	
4.35	Forty-Ninth Supplemental Indenture, dated as of May 1, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/13-15.	8-K(1)	001-35758	4.6	May 1, 2015	
4.36	Fifty-Second Supplemental Indenture, dated as of May 11, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C36-10.	8-K(1)	001-35758	4.4	May 11, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.37	Fifty-Third Supplemental Indenture, dated as of May 11, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C37-15.	8-K(1)	001-35758	4.5	May 11, 2015	
4.38	Fifty-Seventh Supplemental Indenture, dated as of May 18, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C40-10.	8-K(1)	001-35758	4.4	May 18, 2015	
4.39	Fifty-Eighth Supplemental Indenture, dated as of May 18, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C41-15.	8-K(1)	001-35758	4.5	May 18, 2015	
4.40	Sixty-First Supplemental Indenture, dated as of May 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C44-10.	8-K(1)	001-35758	4.4	May 26, 2015	
4.41	Sixty-Second Supplemental Indenture, dated as of May 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C45-15.	8-K(1)	001-35758	4.5	May 26, 2015	
4.42	Seventieth Supplemental Indenture, dated as of June 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C52-10.	8-K(1)	001-35758	4.4	June 16, 2015	
4.43	Seventy-First Supplemental Indenture, dated as of June 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C53-15.	8-K(1)	001-35758	4.5	June 16, 2015	
4.44	Seventy-Fourth Supplemental Indenture, dated as of June 22, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C56-10.	8-K(1)	001-35758	4.4	June 23, 2015	
4.45	Seventy-Fifth Supplemental Indenture, dated as of June 22, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C57-15.	8-K(1)	001-35758	4.5	June 23, 2015	
4.46	Eightieth Supplemental Indenture, dated as of June 29, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C61-10.	8-K(1)	001-35758	4.5	June 29, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.47	Eighty-First Supplemental Indenture, dated as of June 29, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C62-15.	8-K(1)	001-35758	4.6	June 29, 2015	
4.48	Ninetieth Supplemental Indenture, dated as of July 20, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C71-10.	8-K(1)	001-35758	4.5	July 21, 2015	
4.49	Ninety-First Supplemental Indenture, dated as of July 20, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C72-15.	8-K(1)	001-35758	4.6	July 21, 2015	
4.50	Ninety-Fifth Supplemental Indenture, dated as of July 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/20-10.	8-K(1)	001-35758	4.5	July 31, 2015	
4.51	Ninety-Sixth Supplemental Indenture, dated as of July 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/21-15.	8-K(1)	001-35758	4.6	July 31, 2015	
4.52	One Hundred-and-Fifth Supplemental Indenture, dated as of August 10, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C81-10.	8-K(1)	001-35758	4.5	August 10, 2015	
4.53	One Hundred-and-Eleventh Supplemental Indenture, dated as of August 17, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C87-15.	8-K(1)	001-35758	4.6	August 17, 2015	
4.54	One Hundred-and-Sixteenth Supplemental Indenture, dated as of August 24, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C92-15.	8-K(1)	001-35758	4.6	August 24, 2015	
4.55	One Hundred-and-Twenty-First Supplemental Indenture, dated as of August 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C97-15.	8-K(1)	001-35758	4.6	August 31, 2015	
4.56	One Hundred-and-Twenty-Eighth Supplemental Indenture, dated as of September 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C101-10.	8-K(1)	001-35758	4.5	September 15, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.57	One Hundred-and-Twenty-Ninth Supplemental Indenture, dated as of September 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C102-15.	8-K(1)	001-35758	4.6	September 15, 2015	
4.58	One Hundred-and-Thirty-Third Supplemental Indenture, dated as of September 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C106-10.	8-K(1)	001-35758	4.5	September 29, 2015	
4.59	One Hundred-and-Thirty-Fourth Supplemental Indenture, dated as of September 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C107-15.	8-K(1)	001-35758	4.6	September 29, 2015	
4.60	One Hundred-and-Thirty-Eighth Supplemental Indenture, dated as of October 13, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C111-10.	8-K(1)	001-35758	4.5	October 13, 2015	
4.61	One Hundred-and-Forty-Third Supplemental Indenture, dated as of October 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/25-10.	8-K(1)	001-35758	4.5	October 30, 2015	
4.62	One Hundred-and-Forty-Fourth Supplemental Indenture, dated as of October 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/26-15.	8-K(1)	001-35758	4.6	October 30, 2015	
4.63	One Hundred-and-Forty-Eighth Supplemental Indenture, dated as of November 4, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C116-10.	8-K(1)	001-35758	4.5	November 4, 2015	
4.64	One Hundred-and-Fifty-Third Supplemental Indenture, dated as of November 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C121-10.	8-K(1)	001-35758	4.5	November 17, 2015	
4.65	One Hundred-and-Fifty-Fourth Supplemental Indenture, dated as of November 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C122-15.	8-K(1)	001-35758	4.6	November 17, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.66	One Hundred-and-Fifty-Eighth Supplemental Indenture, dated as of November 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C126-10.	8-K(1)	001-35758	4.5	November 30, 2015	
4.67	One Hundred-and-Fifty-Ninth Supplemental Indenture, dated as of November 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C127-15.	8-K(1)	001-35758	4.6	November 30, 2015	
4.68	One Hundred-and-Sixty-Third Supplemental Indenture, dated as of December 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C131-10.	8-K(1)	001-35758	4.5	December 14, 2015	
4.69	One Hundred-and-Sixty-Fourth Supplemental Indenture, dated as of December 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C132-15.	8-K(1)	001-35758	4.6	December 14, 2015	
4.70	One Hundred-and-Sixty-Eighth Supplemental Indenture, dated as of December 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C136-10.	8-K(1)	001-35758	4.5	December 28, 2015	
4.71	One Hundred-and-Sixty-Ninth Supplemental Indenture, dated as of December 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C137-15.	8-K(1)	001-35758	4.6	December 28, 2015	
4.72	One Hundred-and-Seventy-Third Supplemental Indenture, dated as of January 29, 2016, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2016/4-10.	8-K(1)	001-35758	4.5	January 29, 2016	
4.73	One Hundred-and-Seventy-Fourth Supplemental Indenture, dated as of January 29, 2016, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2016/5-15.	8-K(1)	001-35758	4.6	January 29, 2016	
4.74	Description of Registrant's Securities	10-K	001-34756	4.119	February 13, 2020	
10.1**	Form of Indemnification Agreement between the Registrant and its directors and officers.	S-1/A	333-164593	10.1	June 15, 2010	
10.2**	2003 Equity Incentive Plan.	S-1/A	333-164593	10.2	May 27, 2010	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.3**	Form of Stock Option Agreement under 2003 Equity Incentive Plan.	S-1	333-164593	10.3	January 29, 2010	
10.4**	Amended and Restated 2010 Equity Incentive Plan.	10-K	001-34756	10.4	February 23, 2018	
10.5**	Form of Stock Option Agreement under 2010 Equity Incentive Plan.	10-K	001-34756	10.6	March 1, 2017	
10.6**	Form of Restricted Stock Unit Award Agreement under 2010 Equity Incentive Plan.	10-K	001-34756	10.7	March 1, 2017	
10.7**	Amended and Restated 2010 Employee Stock Purchase Plan, effective as of February 1, 2017.	10-K	001-34756	10.8	March 1, 2017	
10.8**	2019 Equity Incentive Plan.	S-8	333-232079	4.2	June 12, 2019	
10.9**	Form of Stock Option Agreement under 2019 Equity Incentive Plan.	S-8	333-232079	4.3	June 12, 2019	
10.10**	Form of Restricted Stock Unit Award Agreement under 2019 Equity Incentive Plan.	S-8	333-232079	4.4	June 12, 2019	
10.11**	Employee Stock Purchase Plan, effective as of June 12, 2019.	S-8	333-232079	4.5	June 12, 2019	
10.12**	2007 SolarCity Stock Plan and form of agreements used thereunder.	S-1(1)	333-184317	10.2	October 5, 2012	
10.13**	2012 SolarCity Equity Incentive Plan and form of agreements used thereunder.	S-1(1)	333-184317	10.3	October 5, 2012	
10.14**	2010 Zep Solar, Inc. Equity Incentive Plan and form of agreements used thereunder.	S-8(1)	333-192996	4.5	December 20, 2013	
10.15**	Offer Letter between the Registrant and Elon Musk dated October 13, 2008.	S-1	333-164593	10.9	January 29, 2010	
10.16**	Performance Stock Option Agreement between the Registrant and Elon Musk dated January 21, 2018.	DEF 14A	001-34756	Appendix A	February 8, 2018	
10.17**	Maxwell Technologies, Inc. 2005 Omnibus Equity Incentive Plan, as amended through May 6, 2010	8-K(2)	001-15477	10.1	May 10, 2010	
10.18**	Maxwell Technologies, Inc. 2013 Omnibus Equity Incentive Plan	DEF 14A(2)	001-15477	Appendix A	June 2, 2017	
10.19	Indemnification Agreement, effective as of June 23, 2020, between Registrant and Elon R. Musk.	10-Q	001-34756	10.4	July 28, 2020	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.20	Indemnification Agreement, dated as of February 27, 2014, by and between the Registrant and J.P. Morgan Securities LLC.	8-K	001-34756	10.1	March 5, 2014	
10.21	Form of Call Option Confirmation relating to 1.25% Convertible Senior Notes Due March 1, 2021.	8-K	001-34756	10.3	March 5, 2014	
10.22	Form of Warrant Confirmation relating to 1.25% Convertible Senior Notes Due March 1, 2021.	8-K	001-34756	10.5	March 5, 2014	
10.23	Form of Call Option Confirmation relating to 2.00% Convertible Senior Notes due May 15, 2024.	8-K	001-34756	10.1	May 3, 2019	
10.24	Form of Warrant Confirmation relating to 2.00% Convertible Senior Notes due May 15, 2024.	8-K	001-34756	10.2	May 3, 2019	
10.25†	Supply Agreement between Panasonic Corporation and the Registrant dated October 5, 2011.	10-K	001-34756	10.50	February 27, 2012	
10.26†	Amendment No. 1 to Supply Agreement between Panasonic Corporation and the Registrant dated October 29, 2013.	10-K	001-34756	10.35A	February 26, 2014	
10.27	Agreement between Panasonic Corporation and the Registrant dated July 31, 2014.	10-Q	001-34756	10.1	November 7, 2014	
10.28†	General Terms and Conditions between Panasonic Corporation and the Registrant dated October 1, 2014.	8-K	001-34756	10.2	October 11, 2016	
10.29	Letter Agreement, dated as of February 24, 2015, regarding addition of co-party to General Terms and Conditions, Production Pricing Agreement and Investment Letter Agreement between Panasonic Corporation and the Registrant.	10-K	001-34756	10.25A	February 24, 2016	
10.30†	Amendment to Gigafactory General Terms, dated March 1, 2016, by and among the Registrant, Panasonic Corporation and Panasonic Energy Corporation of North America.	8-K	001-34756	10.1	October 11, 2016	
10.31††	Amended and Restated General Terms and Conditions for Gigafactory, entered into on June 10, 2020, by and among Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation and Panasonic Corporation of North America.	10-Q	001-34756	10.2	July 28, 2020	
10.32†	Production Pricing Agreement between Panasonic Corporation and the Registrant dated October 1, 2014.	10-Q	001-34756	10.3	November 7, 2014	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.33†	Investment Letter Agreement between Panasonic Corporation and the Registrant dated October 1, 2014.	10-Q	001-34756	10.4	November 7, 2014	
10.34	Amendment to Gigafactory Documents, dated April 5, 2016, by and among the Registrant, Panasonic Corporation, Panasonic Corporation of North America and Panasonic Energy Corporation of North America.	10-Q	001-34756	10.2	May 10, 2016	
10.35††	2019 Pricing Agreement (Japan Cells) with respect to 2011 Supply Agreement, executed September 20, 2019, by and among the Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation and SANYO Electric Co., Ltd.	10-Q	001-34756	10.6	October 29, 2019	
10.36††	2020 Pricing Agreement (Gigafactory 2170 Cells), entered into on June 9, 2020, by and among Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation and Panasonic Corporation of North America.	10-Q	001-34756	10.3	July 28, 2020	
10.37††	2021 Pricing Agreement (Japan Cells) with respect to 2011 Supply Agreement, executed December 29, 2020, by and among the Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation of North America and SANYO Electric Co., Ltd.	10-K	001-34756	10.39	February 8, 2021	
10.38††	Amended and Restated Factory Lease, executed as of March 26, 2019, by and between the Registrant and Panasonic Energy North America, a division of Panasonic Corporation of North America, as tenant.	10-Q	001-34756	10.3	July 29, 2019	
10.39††	Lease Amendment, executed September 20, 2019, by and among the Registrant, Panasonic Corporation of North America, on behalf of its division Panasonic Energy of North America, with respect to the Amended and Restated Factory Lease, executed as of March 26, 2019.	10-Q	001-34756	10.7	October 29, 2019	
10.40††	Second Lease Amendment, entered into on June 9, 2020, by and between the Registrant and Panasonic Energy of North America, a division of Panasonic Corporation of North America, with respect to the Amended and Restated Factory Lease dated January 1, 2017.	10-Q	001-34756	10.1	July 28, 2020	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.41	Amendment and Restatement in respect of ABL Credit Agreement, dated as of March 6, 2019, by and among certain of the Registrant's and Tesla Motors Netherlands B.V.'s direct or indirect subsidiaries from time to time party thereto, as borrowers, Wells Fargo Bank, National Association, as documentation agent, JPMorgan Chase Bank, N.A., Goldman Sachs Bank USA, Morgan Stanley Senior Funding Inc. and Bank of America, N.A., as syndication agents, the lenders from time to time party thereto, and Deutsche Bank AG New York Branch, as administrative agent and collateral agent.	S-4/A	333-229749	10.68	April 3, 2019	
10.42	First Amendment to Amended and Restated ABL Credit Agreement, dated as of December 23, 2020, in respect of the Amended and Restated ABL Credit Agreement, dated as of March 6, 2019, by and among certain of the Registrant's and Tesla Motors Netherlands B.V.'s direct or indirect subsidiaries from time to time party thereto, as borrowers, Wells Fargo Bank, National Association, as documentation agent, JPMorgan Chase Bank, N.A., Goldman Sachs Bank USA, Morgan Stanley Senior Funding Inc. and Bank of America, N.A., as syndication agents, the lenders from time to time party thereto, and Deutsche Bank AG New York Branch, as administrative agent and collateral agent.	10-K	001-34756	10.44	February 8, 2021	
10.43†	Agreement for Tax Abatement and Incentives, dated as of May 7, 2015, by and between Tesla Motors, Inc. and the State of Nevada, acting by and through the Nevada Governor's Office of Economic Development.	10-Q	001-34756	10.1	August 7, 2015	
10.44	Purchase Agreement, dated as of August 11, 2017, by and among the Registrant, SolarCity and Goldman Sachs & Co. LLC and Morgan Stanley & Co. LLC as representatives of the several initial purchasers named therein.	8-K	001-34756	10.1	August 23, 2017	
10.45	Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of September 2, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-Q(1)	001-35758	10.16	November 6, 2014	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.46	First Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 31, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-K(1)	001-35758	10.16a	February 24, 2015	
10.47	Second Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of December 15, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-K(1)	001-35758	10.16b	February 24, 2015	
10.48	Third Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of February 12, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-Q(1)	001-35758	10.16c	May 6, 2015	
10.49	Fourth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of March 30, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-Q(1)	001-35758	10.16d	May 6, 2015	
10.50	Fifth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of June 30, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q(1)	001-35758	10.16e	July 30, 2015	
10.51	Sixth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of September 1, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q(1)	001-35758	10.16f	October 30, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.52	Seventh Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 9, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q(1)	001-35758	10.16g	October 30, 2015	
10.53	Eighth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 26, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q(1)	001-35758	10.16h	October 30, 2015	
10.54	Ninth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of December 9, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-K(1)	001-35758	10.16i	February 10, 2016	
10.55	Tenth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of March 31, 2017, by and between The Research Foundation For The State University of New York, on behalf of the Colleges of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q	001-34756	10.8	May 10, 2017	
10.56	Eleventh Amendment to Amended and Restated Agreement for Research & Development Alliance on Triex Module Technology, effective as of July 22, 2020, among the Research Foundation for the State University of New York, Silevo, LLC and Tesla Energy Operations, Inc.	10-Q	001-34756	10.6	July 28, 2020	
10.57	Twelfth Amendment to Amended and Restated Agreement for Research & Development Alliance on Triex Module Technology, effective as of May 1, 2021, among the Research Foundation for the State University of New York, Silevo, LLC and Tesla Energy Operations, Inc.	10-Q	001-34756	10.1	October 25, 2021	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.58††	Grant Contract for State-Owned Construction Land Use Right, dated as of October 17, 2018, by and between Shanghai Planning and Land Resource Administration Bureau, as grantor, and Tesla (Shanghai) Co., Ltd., as grantee (English translation).	10-Q	001-34756	10.2	July 29, 2019	
10.59	Credit Agreement, dated as of January 20, 2023, among Tesla, Inc., the Lenders and Issuing Banks from time to time party thereto, Citibank, N.A., as Administrative Agent and Deutsche Bank Securities, Inc., as Syndication Agent	10-K	001-34756	10.59	January 31, 2023	
21.1	List of Subsidiaries of the Registrant	—	—	—	—	X
23.1	Consent of PricewaterhouseCoopers LLP, Independent Registered Public Accounting Firm	—	—	—	—	X
31.1	Rule 13a-14(a) / 15(d)-14(a) Certification of Principal Executive Officer	—	—	—	—	X
31.2	Rule 13a-14(a) / 15(d)-14(a) Certification of Principal Financial Officer	—	—	—	—	X
32.1*	Section 1350 Certifications	—	—	—	—	X
97	Tesla, Inc. Clawback Policy	—	—	—	—	X
101.INS	Inline XBRL Instance Document	—	—	—	—	X
101.SCH	Inline XBRL Taxonomy Extension Schema Document	—	—	—	—	X
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document.	—	—	—	—	X
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document	—	—	—	—	X
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document	—	—	—	—	X
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document	—	—	—	—	X
104	Cover Page Interactive Data File (formatted as inline XBRL with applicable taxonomy extension information contained in Exhibits 101)					

* Furnished herewith
** Indicates a management contract or compensatory plan or arrangement
† Confidential treatment has been requested for portions of this exhibit
†† Portions of this exhibit have been redacted in compliance with Regulation S-K Item 601(b)(10).

(1) Indicates a filing of SolarCity
(2) Indicates a filing of Maxwell Technologies, Inc.

ITEM 16. SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

Tesla, Inc.

Date: January 26, 2024

/s/ Elon Musk

Elon Musk

Chief Executive Officer

(Principal Executive Officer)

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Elon Musk Elon Musk	Chief Executive Officer and Director (Principal Executive Officer)	January 26, 2024
/s/ Vaibhav Taneja Vaibhav Taneja	Chief Financial Officer (Principal Financial Officer and Principal Accounting Officer)	January 26, 2024
/s/ Robyn Denholm Robyn Denholm	Director	January 26, 2024
/s/ Ira Ehrenpreis Ira Ehrenpreis	Director	January 26, 2024
/s/ Joseph Gebbia Joseph Gebbia	Director	January 26, 2024
/s/ James Murdoch James Murdoch	Director	January 26, 2024
/s/ Kimbal Musk Kimbal Musk	Director	January 26, 2024
/s/ JB Straubel JB Straubel	Director	January 26, 2024
/s/ Kathleen Wilson-Thompson Kathleen Wilson-Thompson	Director	January 26, 2024

Exhibit 21.1

SUBSIDIARIES OF TESLA, INC.

Name of Subsidiary	Jurisdiction of Incorporation or Organization
Alabama Service LLC	Delaware
All EV Holdings, LLC	Delaware
Allegheny Solar 1, LLC	Delaware
Allegheny Solar Manager 1, LLC	Delaware
Alset Transport GmbH	Germany
Alset Warehouse GmbH	Germany
Ancon Holdings II, LLC	Delaware
Ancon Holdings III, LLC	Delaware
Ancon Holdings, LLC	Delaware
Ancon Solar Corporation	Delaware
Ancon Solar I, LLC	Delaware
Ancon Solar II Lessee Manager, LLC	Delaware
Ancon Solar II Lessee, LLC	Delaware
Ancon Solar II Lessor, LLC	Delaware
Ancon Solar III Lessee Manager, LLC	Delaware
Ancon Solar III Lessee, LLC	Delaware
Ancon Solar III Lessor, LLC	Delaware
Ancon Solar Managing Member I, LLC	Delaware
Arpad Solar Borrower, LLC	Delaware
Arpad Solar I, LLC	Delaware
Arpad Solar Manager I, LLC	Delaware
AU Solar 1, LLC	Delaware
AU Solar 2, LLC	Delaware
Banyan SolarCity Manager 2010, LLC	Delaware
Banyan SolarCity Owner 2010, LLC	Delaware
Basking Solar I, LLC	Delaware
Basking Solar II, LLC	Delaware
Basking Solar Manager II, LLC	Delaware
Beatrix Solar I, LLC	Delaware
Bernese Solar Manager I, LLC	Delaware
Blue Skies Solar I, LLC	Delaware
Blue Skies Solar II, LLC	Delaware
BT Connolly Storage, LLC	Texas
Caballero Solar Managing Member I, LLC	Delaware
Caballero Solar Managing Member II, LLC	Delaware
Caballero Solar Managing Member III, LLC	Delaware
Cardinal Blue Solar, LLC	Delaware
Castello Solar I, LLC	Delaware
Castello Solar II, LLC	Delaware
Castello Solar III, LLC	Delaware
Chaparral SREC Borrower, LLC	Delaware
Chaparral SREC Holdings, LLC	Delaware
Chompie Solar I, LLC	Delaware
Chompie Solar II, LLC	Delaware
Chompie Solar Manager I, LLC	Delaware
Chompie Solar Manager II, LLC	Delaware

Clydesdale SC Solar I, LLC	Delaware
Colorado River Project, LLC	Delaware
Community Solar Partners, LLC	Delaware
Connecticut Auto Repair and Service LLC	Delaware
Compass Automation Incorporated	Illinois
Dom Solar General Partner I, LLC	Delaware
Dom Solar Lessor I, LP	Cayman Islands
Domino Solar Ltd.	Cayman Islands
Dom Solar Limited Partner I, LLC	Delaware
El Rey EV, LLC	Delaware
Falconer Solar Manager I, LLC	Delaware
Firehorn Solar I, LLC	Cayman Islands
Firehorn Solar Manager I, LLC	Delaware
FocalPoint Solar Borrower, LLC	Delaware
FocalPoint Solar I, LLC	Delaware
FocalPoint Solar Manager I, LLC	Delaware
Fontane Solar I, LLC	Delaware
Fotovoltaica GI 4, S. de R.L. de C.V.	Mexico
Fotovoltaica GI 5, S. de R.L. de C.V.	Mexico
FP System Owner, LLC	Delaware
Giga Insurance Texas, Inc.	Texas
Giga Texas Energy, LLC	Delaware
Grohmann Engineering Trading (Shanghai) Co. Ltd.	China
Grohmann USA, Inc.	Delaware
Guilder Solar, LLC	Delaware
Hamilton Solar, LLC	Delaware
Harborfields LLC	Delaware
Hangzhou Silevo Electric Power Co., Ltd.	China
Harpoon Solar I, LLC	Delaware
Harpoon Solar Manager I, LLC	Delaware
Haymarket Holdings, LLC	Delaware
Haymarket Manager 1, LLC	Delaware
Haymarket Solar 1, LLC	Delaware
Hibar Systems Europe GmbH	Germany
Hive Battery Inc.	Delaware
Ikehu Manager I, LLC	Delaware
IL Buono Solar I, LLC	Delaware
Iliosson, S.A. de C.V.	Mexico
Industrial Maintenance Technologies, Inc.	California
Kansas Repair LLC	Delaware
Klamath Falls Solar 1, LLC	Delaware
Knight Solar Managing Member I, LLC	Delaware
Knight Solar Managing Member II, LLC	Delaware
Knight Solar Managing Member III, LLC	Delaware
Landlord 2008-A, LLC	Delaware
Lincoln Auto Repair and Service LLC	Delaware
Louis Solar II, LLC	Delaware
Louis Solar III, LLC	Delaware
Louis Solar Manager II, LLC	Delaware
Louis Solar Manager III, LLC	Delaware

Louis Solar Master Tenant I, LLC	Delaware
Louis Solar MT Manager I, LLC	Delaware
Louis Solar Owner I, LLC	Delaware
Louis Solar Owner Manager I, LLC	Delaware
Master Tenant 2008-A, LLC	Delaware
Matterhorn Solar I, LLC	Delaware
Maxwell Technologies, Inc.	Delaware
Megalodon Solar, LLC	Delaware
Monte Rosa Solar I, LLC	Delaware
Mound Solar Manager V, LLC	Delaware
Mound Solar Manager VI, LLC	Delaware
Mound Solar Manager X, LLC	Delaware
Mound Solar Manager XI, LLC	Delaware
Mound Solar Manager XII, LLC	Delaware
Mound Solar Master Tenant IX, LLC	Delaware
Mound Solar Master Tenant V, LLC	California
Mound Solar Master Tenant VI, LLC	Delaware
Mound Solar Master Tenant VII, LLC	Delaware
Mound Solar Master Tenant VIII, LLC	Delaware
Mound Solar MT Manager IX, LLC	Delaware
Mound Solar MT Manager VII, LLC	Delaware
Mound Solar MT Manager VIII, LLC	Delaware
Mound Solar Owner IX, LLC	Delaware
Mound Solar Owner Manager IX, LLC	Delaware
Mound Solar Owner Manager VII, LLC	Delaware
Mound Solar Owner Manager VIII, LLC	Delaware
Mound Solar Owner V, LLC	California
Mound Solar Owner VI, LLC	Delaware
Mound Solar Owner VII, LLC	Delaware
Mound Solar Owner VIII, LLC	Delaware
Mound Solar Partnership X, LLC	Delaware
Mound Solar Partnership XI, LLC	Delaware
Mound Solar Partnership XII, LLC	Delaware
MS SolarCity 2008, LLC	Delaware
MS SolarCity Commercial 2008, LLC	Delaware
MS SolarCity Residential 2008, LLC	Delaware
New Mexico Sales and Vehicle Service LLC	Delaware
NBA SolarCity AFB, LLC	California
NBA SolarCity Commercial I, LLC	California
NBA SolarCity Solar Phoenix, LLC	California
Northern Nevada Research Co., LLC	Nevada
Oranje Solar I, LLC	Delaware
Oranje Solar Manager I, LLC	Delaware
Palmetto Auto Repair and Service LLC	Delaware
Paramount Energy Fund I Lessee, LLC	Delaware
Paramount Energy Fund I Lessor, LLC	Delaware
PEF I MM, LLC	Delaware
Perbix Machine Company, Inc.	Minnesota
Presidio Solar I, LLC	Delaware
Presidio Solar II, LLC	Delaware

Presidio Solar III, LLC	Delaware
Pukana La Solar I, LLC	Delaware
R9 Solar 1, LLC	Delaware
Roadster Automobile Sales and Service (Beijing) Co., Ltd.	China
Roadster Finland Oy	Finland
SA VPP Holding Trust	Australia
SA VPP Project Trust	Australia
Sequoia Pacific Holdings, LLC	Delaware
Sequoia Pacific Manager I, LLC	Delaware
Sequoia Pacific Solar I, LLC	Delaware
Sequoia SolarCity Owner I, LLC	Delaware
Sierra Solar Power (Hong Kong) Limited	Hong Kong
SiiLion, Inc.	Delaware
Silevo, LLC	Delaware
Solar Aquarium Holdings, LLC	Delaware
Solar Energy of America 1, LLC	Delaware
Solar Energy of America Manager 1, LLC	Delaware
Solar Explorer, LLC	Delaware
Solar Gezellig Holdings, LLC	Delaware
Solar House I, LLC	Delaware
Solar House II, LLC	Delaware
Solar House III, LLC	Delaware
Solar House IV, LLC	Delaware
Solar Integrated Fund I, LLC	Delaware
Solar Integrated Fund II, LLC	Delaware
Solar Integrated Fund III, LLC	Delaware
Solar Integrated Fund IV-A, LLC	Delaware
Solar Integrated Fund V, LLC	Delaware
Solar Integrated Fund VI, LLC	Delaware
Solar Integrated Manager I, LLC	Delaware
Solar Integrated Manager II, LLC	Delaware
Solar Integrated Manager III, LLC	Delaware
Solar Integrated Manager IV-A, LLC	Delaware
Solar Integrated Manager V, LLC	Delaware
Solar Integrated Manager VI, LLC	Delaware
Solar Services Company, LLC	Delaware
Solar Ulysses Manager I, LLC	Delaware
Solar Ulysses Manager II, LLC	Delaware
Solar Voyager, LLC	Delaware
Solar Warehouse Manager I, LLC	Delaware
Solar Warehouse Manager II, LLC	Delaware
Solar Warehouse Manager III, LLC	Delaware
Solar Warehouse Manager IV, LLC	Delaware
SolarCity Alpine Holdings, LLC	Delaware
SolarCity Amphitheatre Holdings, LLC	Delaware
SolarCity Arbor Holdings, LLC	Delaware
SolarCity Arches Holdings, LLC	Delaware
SolarCity AU Holdings, LLC	Delaware
SolarCity Cruyff Holdings, LLC	Delaware
SolarCity Electrical, LLC	Delaware

SolarCity Electrical New York Corporation	Delaware
SolarCity Finance Company, LLC	Delaware
SolarCity Finance Holdings, LLC	Delaware
SolarCity Foxborough Holdings, LLC	Delaware
SolarCity FTE Series 1, LLC	Delaware
SolarCity FTE Series 2, LLC	Delaware
SolarCity Fund Holdings, LLC	Delaware
SolarCity Grand Canyon Holdings, LLC	Delaware
SolarCity Holdings 2008, LLC	Delaware
SolarCity International, Inc.	Delaware
SolarCity Leviathan Holdings, LLC	Delaware
SolarCity LMC Series I, LLC	Delaware
SolarCity LMC Series II, LLC	Delaware
SolarCity LMC Series III, LLC	Delaware
SolarCity LMC Series IV, LLC	Delaware
SolarCity LMC Series V, LLC	Delaware
SolarCity Mid-Atlantic Holdings, LLC	Delaware
SolarCity Nitro Holdings, LLC	Delaware
SolarCity Orange Holdings, LLC	Delaware
SolarCity Series Holdings I, LLC	Delaware
SolarCity Series Holdings II, LLC	Delaware
SolarCity Series Holdings IV, LLC	Delaware
SolarCity Steep Holdings, LLC	Delaware
SolarCity Ulu Holdings, LLC	Delaware
SolarCity Village Holdings, LLC	Delaware
SolarRock, LLC	Delaware
SolarStrong, LLC	Delaware
Sparrowhawk Solar I, LLC	Delaware
SREC Holdings, LLC	Delaware
TALT Holdings, LLC	Delaware
TALT TBM Holdings, LLC	Delaware
TBM Partnership II, LLC	Delaware
TEO Engineering, Inc.	California
TES 2017-1, LLC	Delaware
TES 2017-2, LLC	Delaware
TES Holdings 2017-1, LLC	Delaware
Tesla 2014 Warehouse SPV LLC	Delaware
Tesla Auto Lease Trust 2021-A	Delaware
Tesla Auto Lease Trust 2021-B	Delaware
Tesla Auto Lease Trust 2022-A	Delaware
Tesla Auto Lease Trust 2023-A	Delaware
Tesla Auto Lease Trust 2023-B	Delaware
Tesla Electric Vehicle Trust 2023-1	Delaware
Tesla Autobidder International B.V.	Netherlands
Tesla Automation GmbH	Germany
Tesla Automobile Information Service (Dalian) Co., Ltd.	China
Tesla Automobile Management and Service (Haikou) Co., Ltd.	China
Tesla Automobile Sales and Service (Beijing) Co., Ltd.	China
Tesla Automobile Sales and Service (Changchun) Co., Ltd.	China
Tesla Automobile Sales and Service (Changsha) Co., Ltd.	China

Tesla Automobile Sales and Service (Chengdu) Co., Ltd.	China
Tesla Automobile Sales and Service (Chongqing) Co., Ltd.	China
Tesla Automobile Sales and Service (Dalian) Co., Ltd.	China
Tesla Automobile Sales and Service (Fuzhou) Co., Ltd.	China
Tesla Automobile Sales and Service (Guangzhou) Co., Ltd.	China
Tesla Automobile Sales and Service (Guiyang) Co., Ltd.	China
Tesla Automobile Sales and Service (Haerbin) Co., Ltd.	China
Tesla Automobile Sales and Service (Hangzhou) Co., Ltd.	China
Tesla Automobile Sales and Service (Hefei) Co., Ltd.	China
Tesla Automobile Sales and Service (Hohhot) Co., Ltd.	China
Tesla Automobile Sales and Service (Jinan) Co., Ltd.	China
Tesla Automobile Sales and Service (Kunming) Co., Ltd.	China
Tesla Automobile Sales and Service (Lanzhou) Co., Ltd.	China
Tesla Automobile Sales and Service (Nanchang) Co., Ltd.	China
Tesla Automobile Sales and Service (Nanjing) Co., Ltd.	China
Tesla Automobile Sales and Service (Nanning) Co., Ltd.	China
Tesla Automobile Sales and Service (Ningbo) Co., Ltd.	China
Tesla Automobile Sales and Service (Qingdao) Co., Ltd.	China
Tesla Automobile Sales and Service (Shanghai) Co., Ltd.	China
Tesla Automobile Sales and Service (Shenyang) Co., Ltd.	China
Tesla Automobile Sales and Service (Shijiazhuang) Co., Ltd.	China
Tesla Automobile Sales and Service (Suzhou) Co. Ltd.	China
Tesla Automobile Sales and Service (Taiyuan) Co., Ltd.	China
Tesla Automobile Sales and Service (Tianjin) Co. Ltd.	China
Tesla Automobile Sales and Service (Urumqi) Co. Ltd.	China
Tesla Automobile Sales and Service (Wenzhou) Co., Ltd.	China
Tesla Automobile Sales and Service (Wuhan) Co., Ltd.	China
Tesla Automobile Sales and Service (Wuxi) Co., Ltd.	China
Tesla Automobile Sales and Service (Xi'an) Co., Ltd.	China
Tesla Automobile Sales and Service (Xiamen) Co., Ltd.	China
Tesla Automobile Sales and Service (Xining) Co., Ltd.	China
Tesla Automobile Sales and Service (Yinchuan) Co., Ltd.	China
Tesla Automobile Sales and Service (Zhengzhou) Co. Ltd.	China
Tesla Automobiles Sales and Service Mexico, S. de R.L. de C.V.	Mexico
Tesla (Beijing) New Energy R&D Co., Ltd.	China
Tesla Belgium BV	Belgium
Tesla Canada Finance ULC	Canada
Tesla Canada GP Inc.	Canada
Tesla Canada LP	Canada
Tesla Charging, LLC	Delaware
Tesla Chile SpA	Chile
Tesla Construction (Shanghai) Co., Ltd.	China
Tesla Czech Republic s.r.o.	Czech Republic
Tesla Energia Macau Limitada	Macau
Tesla Engineering Germany GmbH	Germany
Tesla Energy d.o.o.	Slovenia
Tesla Energy Management LLC	Delaware
Tesla Energy Operations, Inc.	Delaware
Tesla Energy Ventures Australia Pty Ltd	Australia
Tesla Energy Ventures Limited	United Kingdom

Tesla Energy Ventures Holdings B.V.	Netherlands
Tesla Finance LLC	Delaware
Tesla Financial Leasing (China) Co., Ltd.	China
Tesla Financial Services GmbH	Germany
Tesla Financial Services Holdings B.V.	Netherlands
Tesla Financial Services Limited	United Kingdom
Tesla France S.à r.l.	France
Tesla Germany GmbH	Germany
Tesla General Insurance, Inc.	Arizona
Tesla Greece Single Member P.C.	Greece
Tesla Hrvatska d.o.o.	Croatia
Tesla Hungary Kft.	Hungary
Tesla India Motors and Energy Private Limited	India
Tesla Insurance Brokers Co., Ltd.	China
Tesla Insurance Holdings, LLC	Delaware
Tesla Insurance, Inc.	Delaware
Tesla Insurance Ltd.	Malta
Tesla Insurance Company	California
Tesla Insurance Services, Inc.	California
Tesla Insurance Services of Texas, Inc.	Texas
Tesla International B.V.	Netherlands
Tesla Investments LLC	Delaware
Tesla Italy S.r.l.	Italy
Tesla Jordan Car Trading LLC	Jordan
Tesla Korea Limited	Republic of Korea
Tesla Lease Trust	Delaware
Tesla LLC	Delaware
Tesla Manufacturing Brandenburg SE	Germany
Tesla Manufacturing Mexico, S. de R.L. de C.V.	Mexico
Tesla Manufacturing Mexico Holding, S. de R.L. de C.V.	Mexico
Tesla Michigan, Inc.	Michigan
Tesla Mississippi LLC	Delaware
Tesla Motors Australia, Pty Ltd	Australia
Tesla Motors Austria GmbH	Austria
Tesla Motors (Beijing) Co., Ltd.	China
Tesla Motors Canada ULC	Canada
Tesla Motors Colombia S.A.S	Colombia
Tesla Motors Holding B.V.	Netherlands
Tesla Motors Denmark ApS	Denmark
Tesla Motors FL, Inc.	Florida
Tesla Motors HK Limited	Hong Kong
Tesla Motors Iceland ehf.	Iceland
Tesla Motors Ireland Limited	Ireland
Tesla Motors Israel Ltd.	Israel
Tesla Motors Japan GK	Japan
Tesla Motors Limited	United Kingdom
Tesla Motors Luxembourg S.à r.l.	Luxembourg
Tesla Motors MA, Inc.	Massachusetts
Tesla Motors Netherlands B.V.	Netherlands
Tesla Motors New York LLC	New York

Tesla Motors NL LLC	Delaware
Tesla Motors NV, Inc.	Nevada
Tesla Motors PA, Inc.	Pennsylvania
Tesla Motors Romania S.R.L.	Romania
Tesla Motors Sales and Service LLC	Turkey
Tesla Motors Singapore Holdings Pte. Ltd.	Singapore
Tesla Motors Singapore Private Limited	Singapore
Tesla Motors Stichting	Netherlands
Tesla Motors Taiwan Limited	Taiwan
Tesla Motors TN, Inc.	Tennessee
Tesla Motors TX, Inc.	Texas
Tesla Motors UT, Inc.	Utah
Tesla Nambe LLC	Delaware
Tesla New Zealand ULC	New Zealand
Tesla Norway AS	Norway
Tesla Poland sp. z o.o.	Poland
Tesla Property &Casualty, Inc.	California
Tesla Portugal, Sociedade Unipessoal LDA	Portugal
Tesla Puerto Rico LLC	Puerto Rico
Tesla Qatar LLC	Qatar
Tesla Sales, Inc.	Delaware
Tesla Sdn. Bhd.	Malaysia
Tesla Shanghai Co., Ltd	China
Tesla (Shanghai) New Energy Co., LTD.	China
Tesla Spain, S.L. Unipersonal	Spain
Tesla Switzerland GmbH	Switzerland
Tesla (Thailand) Ltd.	Thailand
Tesla TH1 LLC	Delaware
Tesla TH2 LLC	Delaware
Telsa Toronto Automation ULC	Canada
Tesla Toronto International Holdings ULC	Canada
Tesla Transport B.V.	Netherlands
The Big Green Solar I, LLC	Delaware
The Big Green Solar Manager I, LLC	Delaware
Three Rivers Solar 1, LLC	Delaware
Three Rivers Solar 2, LLC	Delaware
Three Rivers Solar 3, LLC	Delaware
Three Rivers Solar Manager 1, LLC	Delaware
Three Rivers Solar Manager 2, LLC	Delaware
Three Rivers Solar Manager 3, LLC	Delaware
TM International C.V.	Netherlands
TM Sweden AB	Sweden
USB SolarCity Manager IV, LLC	Delaware
USB SolarCity Master Tenant IV, LLC	California
USB SolarCity Owner IV, LLC	California
Visigoth Solar 1, LLC	Delaware
Visigoth Solar Holdings, LLC	Delaware
Visigoth Solar Managing Member 1, LLC	Delaware
VPP Project 1 (SA) Pty Ltd.	Australia
Weisshorn Solar I, LLC	Cayman Islands

Weisshorn Solar Manager I, LLC	Delaware
Zep Solar LLC	California

Exhibit 23.1

CONSENT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

We hereby consent to the incorporation by reference in the Registration Statements on Form S-8 (Nos. 333-232079, 333-223169, 333-216376, 333-209696, 333-198002, 333-187113, 333-183033, and 333-167874) of Tesla, Inc. of our report dated January 26, 2024 relating to the financial statements and the effectiveness of internal control over financial reporting, which appears in this Form 10-K.

/s/ PricewaterhouseCoopers LLP

San Jose, California
January 26, 2024

Exhibit 31.1

CERTIFICATIONS

I, Elon Musk, certify that:

1. I have reviewed this Annual Report on Form 10-K of Tesla, Inc.;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant’s other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:

   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

   (c) Evaluated the effectiveness of the registrant’s disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   (d) Disclosed in this report any change in the registrant’s internal control over financial reporting that occurred during the registrant’s most recent fiscal quarter (the registrant’s fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant’s internal control over financial reporting; and

5. The registrant’s other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant’s auditors and the audit committee of the registrant’s Board of Directors (or persons performing the equivalent functions):

   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant’s ability to record, process, summarize and report financial information; and

   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant’s internal control over financial reporting.

Date: January 26, 2024

/s/ Elon Musk

Elon Musk
Chief Executive Officer
(Principal Executive Officer)

Exhibit 31.2

CERTIFICATIONS

I, Vaibhav Taneja, certify that:

1. I have reviewed this Annual Report on Form 10-K of Tesla, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
   - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
   - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
   - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
   - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's Board of Directors (or persons performing the equivalent functions):
   - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
   - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: January 26, 2024

/s/ Vaibhav Taneja

Vaibhav Taneja
Chief Financial Officer
(Principal Financial Officer and
Principal Accounting Officer)

Exhibit 32.1

SECTION 1350 CERTIFICATIONS

I, Elon Musk, certify, pursuant to 18 U.S.C. Section 1350, that, to my knowledge, the Annual Report of Tesla, Inc. on Form 10-K for the annual period ended December 31, 2023, (i) fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and (ii) that the information contained in such Form 10-K fairly presents, in all material respects, the financial condition and results of operations of Tesla, Inc.

Date: January 26, 2024

/s/ Elon Musk
Elon Musk
Chief Executive Officer
(Principal Executive Officer)

I, Vaibhav Taneja, certify, pursuant to 18 U.S.C. Section 1350, that, to my knowledge, the Annual Report of Tesla, Inc. on Form 10-K for the annual period ended December 31, 2023, (i) fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934 and (ii) that the information contained in such Form 10-K fairly presents, in all material respects, the financial condition and results of operations of Tesla, Inc.

Date: January 26, 2024

/s/ Vaibhav Taneja
Vaibhav Taneja
Chief Financial Officer
(Principal Financial Officer and
Principal Accounting Officer)

Exhibit 97

Tesla, Inc.

CLAWBACK POLICY

The Compensation Committee (the "Committee") of the Board of Directors (the "Board") of Tesla, Inc. (the "Company") believes that it is appropriate for the Company to adopt this Clawback Policy (this "Policy") to be applied to the certain executive officers of the Company and adopts this Policy to be effective as of the Effective Date.

1. Definitions

For purposes of this Policy, the following definitions shall apply:

a) "Company Group" means the Company and each of its Subsidiaries, as applicable.

b) "Covered Compensation" means any Incentive-Based Compensation Received by a person who served as an Executive Officer at any time during the performance period for such Incentive-Based Compensation and that was Received (i) on or after October 2, 2023 and (ii) after the person became an Executive Officer.

c) "Effective Date" means November 15, 2023.

d) "Erroneously Awarded Compensation" means the amount of Covered Compensation Received by a person during the fiscal period when the applicable Financial Reporting Measure relating to such Covered Compensation was attained that exceeds the amount of Covered Compensation that otherwise would have been Received by such person had such amount been determined based on the applicable Restatement, computed without regard to any taxes paid (i.e., on a pre-tax basis). For Covered Compensation based on stock price or total shareholder return, where the amount of Erroneously Awarded Compensation is not subject to mathematical recalculation directly from the information in a Restatement, the Committee will determine the amount of such Covered Compensation that constitutes Erroneously Awarded Compensation, if any, based on a reasonable estimate of the effect of the Restatement on the stock price or total shareholder return upon which the Covered Compensation was granted, vested or paid and the Committee shall maintain documentation of such determination and provide such documentation to Nasdaq.

e) "Exchange Act" means the U.S. Securities Exchange Act of 1934, as amended.

f) "Executive Officer" means each "executive officer" of the Company (as defined in Rule 10D-1(d) under the Exchange Act), which shall be deemed to include any individuals identified by the Company as executive officers pursuant to Item 401(b) of Regulation S-K under the Exchange Act. Both current and former Executive Officers are subject to the Policy in accordance with its terms.

g) "Financial Reporting Measure" means (i) any measure that is determined and presented in accordance with the accounting principles used in preparing the Company's financial statements, and any measures derived wholly or in part from such measures and may consist of GAAP or non-GAAP financial measures (as defined under Regulation G of the Exchange Act and Item 10 of Regulation S-K under the Exchange Act), (ii) stock price or (iii) total shareholder return. Financial Reporting Measures may or may not be filed with the SEC and may be presented outside the Company's financial statements, such as in Managements' Discussion and Analysis of Financial Conditions and Result of Operations or in the performance graph required under Item 201(e) of Regulation S-K under the Exchange Act.

h) "Incentive-Based Compensation" means any compensation that is granted, earned or vested based wholly or in part upon the attainment of a Financial Reporting Measure.

i) "Lookback Period" means the three completed fiscal years (plus any transition period of less than nine months that is within or immediately following the three completed fiscal years and that results from a change in the Company's fiscal year) immediately preceding the date on which the Company is required to prepare a Restatement for a given reporting period, with such date being the earlier of: (i) the date the Board, a committee of the Board, or the officer or officers of the Company authorized to take such action if Board action is not required, concludes, or reasonably should have concluded, that the Company is required to prepare a Restatement, or (ii) the date a court, regulator or other legally authorized body directs the Company to prepare a Restatement. Recovery of any Erroneously Awarded Compensation under the Policy is not dependent on if or when the Restatement is actually filed.

j) "Nasdaq" means the Nasdaq Stock Market.

k) "Received": Incentive-Based Compensation is deemed "Received" in the Company's fiscal period during which the Financial Reporting Measure specified in or otherwise relating to the Incentive-Based Compensation award is attained, even if the grant, certification of achievement, vesting or payment of the Incentive-Based Compensation occurs after the end of that period.

l) "Restatement" means an accounting restatement required due to the material noncompliance by the Company with any financial reporting requirement under the securities laws, including (i) to correct an error in previously issued financial statements that is material to the previously issued financial statements (commonly referred to as a "Big R" restatement) or (ii) to correct an error in previously issued financial statements that is not material to the previously issued financial statements but that would result in a material misstatement if the error were corrected in the current period or left uncorrected in the current period (commonly referred to as a "little r" restatement). Changes to the Company's financial statements that do not represent error corrections under the then-current relevant accounting standards will not constitute Restatements.

m) "SEC" means the U.S. Securities and Exchange Commission.

n) "Subsidiary" means any domestic or foreign corporation, partnership, association, joint stock company, joint venture, trust or unincorporated organization "affiliated" with the Company, that is, directly or indirectly, through one or more intermediaries, "controlling", "controlled by" or "under common control with", the Company. "Control" for this purpose means the possession, direct or indirect, of the power to direct or cause the direction of the management and policies of such person, whether through the ownership of voting securities, contract or otherwise.

2. Recoupment of Erroneously Awarded Compensation

In the event of a Restatement, any Erroneously Awarded Compensation Received during the Lookback Period that (a) is then-outstanding but has not yet been paid shall be automatically and immediately forfeited or (b) has been paid to any person shall be subject to reasonably prompt repayment to the Company Group in accordance with Section 3 of this Policy. The Committee must pursue (and shall not have the discretion to waive) the forfeiture and/or repayment of such Erroneously Awarded Compensation in accordance with Section 3 of this Policy, except as provided below. Recovery of any Erroneously Awarded Compensation under the Policy is not dependent on fraud or misconduct by any person in connection with the Restatement.

Notwithstanding the foregoing, the Committee (or, if the Committee is not a committee of the Board responsible for the Company's executive compensation decisions and composed entirely of independent directors, a majority of the independent directors serving on the Board) may determine not to pursue the forfeiture and/or recovery of Erroneously Awarded Compensation from any person if the Committee determines that such forfeiture and/or recovery would be impracticable due to any of the following circumstances: (i) the direct expense paid to a third party (for example, reasonable legal expenses and consulting fees) to assist in enforcing the Policy would exceed the amount to be recovered (following reasonable attempts by the Company Group to recover such Erroneously Awarded Compensation, the documentation of such attempts, and the provision of such documentation to the Nasdaq) or (ii) recovery would likely cause any otherwise tax-qualified retirement plan, under which benefits are broadly available to employees of Company Group, to fail to meet the requirements of 26 U.S.C.
401(a)(13) or 26 U.S.C. 411(a) and regulations thereunder.

3. Means of Repayment

In the event that the Committee determines that any person shall repay any Erroneously Awarded Compensation, the Committee shall provide written notice to such person by email or certified mail to the physical address on file with the Company Group for such person, and the person shall satisfy such repayment in a manner and on such terms as required by the Committee, and the Company Group shall be entitled to set off the repayment amount against any amount owed to the person by the Company Group, to require the forfeiture of any award granted by the Company Group to the person, or to take any and all necessary actions to reasonably promptly recoup the repayment amount from the person, in each case, to the fullest extent permitted under applicable law, including without limitation, Section 409A of the U.S.

Internal Revenue Code and the regulations and guidance thereunder. If the Committee does not specify a repayment timing in the written notice described above, the applicable person shall be required to repay the Erroneously Awarded Compensation to the Company Group by wire, cash or cashier's check no later than thirty (30) days after receipt of such notice.

4. No Indemnification

No person shall be indemnified, insured or reimbursed by the Company Group in respect of any loss of compensation by such person in accordance with this Policy, nor shall any person receive any advancement of expenses for disputes related to any loss of compensation by such person in accordance with this Policy, and no person shall be paid or reimbursed by the Company Group for any premiums paid by such person for any third-party insurance policy covering potential recovery obligations under this Policy.

5. Miscellaneous

This Policy generally will be administered and interpreted by the Committee, provided that the Board may, from time to time, exercise discretion to administer and interpret this Policy, in which case, all references herein to "Committee" shall be deemed to refer to the Board. Any determination by the Committee with respect to this Policy shall be final, conclusive and binding on all interested parties. Any discretionary determinations of the Committee under this Policy, if any, need not be uniform with respect to all persons, and may be made selectively amongst persons, whether or not such persons are similarly situated.

This Policy is intended to satisfy the requirements of Section 954 of the Dodd-Frank Wall Street Reform and Consumer Protection Act, as it may be amended from time to time, Rule 10D- 1 under the Exchange Act, and any related rules or regulations promulgated by the SEC or the Nasdaq, including any additional or new requirements that become effective after the Effective Date which upon effectiveness shall be deemed to automatically amend this Policy to the extent necessary to comply with such additional or new requirements.

The provisions in this Policy are intended to be applied to the fullest extent of the law. To the extent that any provision of this Policy is found to be unenforceable or invalid under any applicable law, such provision will be applied to the maximum extent permitted and shall automatically be deemed amended in a manner consistent with its objectives to the extent necessary to conform to applicable law. The invalidity or unenforceability of any provision of this Policy shall not affect the validity or enforceability of any other provision of this Policy. Recoupment of Erroneously Awarded Compensation under this Policy is not dependent upon the Company Group satisfying any conditions in this Policy, including any requirements to provide applicable documentation to the Nasdaq.

The rights of the Company Group under this Policy to seek forfeiture or reimbursement are in addition to, and not in lieu of, any rights of recoupment, or remedies or rights other than recoupment, that may be available to the Company Group pursuant to the terms of any law, government regulation or stock exchange listing requirement or any other policy, code of conduct, employee handbook, employment agreement, equity award agreement, or other plan or agreement of the Company Group.

6. Amendment and Termination

To the extent permitted by, and in a manner consistent with applicable law, including SEC and Nasdaq rules, the Committee may terminate, suspend or amend this Policy at any time in its discretion.

7. Successors

This Policy shall be binding and enforceable against all persons and their respective beneficiaries, heirs, executors, administrators or other legal representatives with respect to any Covered Compensation granted, vested or paid to or administered by such persons or entities.

8. Agreement to Arbitrate; Governing Law; Venue

This Policy and all rights and obligations hereunder will be governed by and interpreted in accordance with Texas law. Any dispute arising from or relating to this Policy shall be subject to binding arbitration in accordance with the then-current Streamlined Arbitration Rules of the Judicial Arbitration and Mediation Services ("JAMS"). The existence, content and result of the arbitration shall be held in confidence by the parties, their representatives, any other participants and the arbitrator. The arbitration will be conducted by a single arbitrator selected by agreement of the parties or, failing such agreement, appointed in accordance with the JAMS rules. The arbitration shall be conducted in English and in Austin, Texas. Each party will bear its own expenses in the arbitration and will share equally the costs of the arbitration; *provided*, however, that the arbitrator may, in its discretion, award reasonable costs and fees to the prevailing party. Judgment upon the award rendered in the arbitration may be entered in any court of competent jurisdiction. If any dispute in arbitration under this Policy is substantially the same or involves common issues of law or fact, either party shall be entitled to require that any such dispute be consolidated with the relevant arbitration pursuant hereto, and the other party shall permit, and co-operate in, such consolidation.

Tesla, Inc.

CLAWBACK POLICY

ACKNOWLEDGMENT, CONSENT AND AGREEMENT

I acknowledge that I have received and reviewed a copy of the Tesla, Inc. Clawback Policy (as may be amended from time to time, the "Policy") and I have been given an opportunity to ask questions about the Policy and review it with my counsel. I knowingly, voluntarily and irrevocably consent to and agree to be bound by and subject to the Policy's terms and conditions, including that I will return any Erroneously Awarded Compensation that is required to be repaid in accordance with the Policy. I further acknowledge, understand and agree that (i) the compensation that I receive, have received or may become entitled to receive from the Company Group is subject to the Policy, and the Policy may affect such compensation and (ii) I have no right to indemnification, insurance payments or other reimbursement by or from the Company Group for any compensation that is subject to recoupment and/or forfeiture under the Policy.

Capitalized terms used but not defined herein have the meanings set forth in the Policy.

Signed: ______________________________

Print Name: ______________________________

Date: ______________________________

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☒ **Annual report pursuant to section 13 or 15(d) of the Securities Exchange Act of 1934**

For the fiscal year ended January 31, 2024, or

☐ **Transition report pursuant to section 13 or 15(d) of the Securities Exchange Act of 1934**

Commission file number 001-06991.

WALMART INC.
(Exact name of registrant as specified in its charter)

Delaware	**71-0415188**
(State or other jurisdiction of incorporation or organization)	(IRS Employer Identification No.)
702 S.W. 8th Street **Bentonville, AR**	**72716**
(Address of principal executive offices)	(Zip Code)

Registrant's telephone number, including area code: (479) 273-4000

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock, par value $0.10 per share	WMT	New York Stock Exchange
2.550% Notes due 2026	WMT26	New York Stock Exchange
1.050% Notes due 2026	WMT26A	New York Stock Exchange
1.500% Notes due 2028	WMT28C	New York Stock Exchange
4.875% Notes due 2029	WMT29B	New York Stock Exchange
5.750% Notes due 2030	WMT30B	New York Stock Exchange
1.800% Notes due 2031	WMT31A	New York Stock Exchange
5.625% Notes due 2034	WMT34	New York Stock Exchange
5.250% Notes due 2035	WMT35A	New York Stock Exchange
4.875% Notes due 2039	WMT39	New York Stock Exchange

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act.
Yes ý No ¨

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Exchange Act.
Yes ¨ No ý

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for at least the past 90 days.

Yes ý No ¨

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files).

Yes ý No ¨

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large Accelerated Filer	☒	Accelerated Filer	☐
Non-Accelerated Filer	☐	Smaller Reporting Company	☐
		Emerging Growth Company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ¨

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ¨

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act).

Yes ☐ No ☒

As of July 31, 2023, the aggregate market value of the voting common stock of the registrant held by non-affiliates of the registrant, based on the closing sale price of those shares on the New York Stock Exchange reported on July 31, 2023, was $228,694,206,501. For the purposes of this disclosure only, the registrant has assumed that its directors, executive officers (as defined in Rule 3b-7 under the Exchange Act) and the beneficial owners of 5% or more of the registrant's outstanding common stock are the affiliates of the registrant.

The registrant had 8,058,048,674 shares of common stock outstanding as of March 13, 2024.

DOCUMENTS INCORPORATED BY REFERENCE

Document	Parts Into Which Incorporated
Portions of the registrant's Proxy Statement for the Annual Meeting of Shareholders to be held June 5, 2024 (the "Proxy Statement")	Part III

Walmart Inc.
Form 10-K
For the Fiscal Year Ended January 31, 2024

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WALMART INC.

ANNUAL REPORT ON FORM 10-K
FOR THE FISCAL YEAR ENDED JANUARY 31, 2024

All references in this Annual Report on Form 10-K, the information incorporated into this Annual Report on Form 10-K by reference to information in the Proxy Statement of Walmart Inc. for its Annual Shareholders' Meeting to be held on June 5, 2024 and in the exhibits to this Annual Report on Form 10-K to "Walmart Inc.," "Walmart," "the Company," "our Company," "we," "us" and "our" are to the Delaware corporation named "Walmart Inc." and, except where expressly noted otherwise or the context otherwise requires, that corporation's consolidated subsidiaries.

On February 23, 2024, the Company effected a 3-for-1 forward split of its common stock and a proportionate increase in the number of authorized shares. All share and per share information, including share based compensation, throughout this Annual Report on Form 10-K has been retroactively adjusted to reflect the stock split.

PART I

Cautionary Statement Regarding Forward-Looking Statements

This Annual Report on Form 10-K and other reports, statements and information that Walmart Inc. (which individually or together with its subsidiaries, as the context otherwise requires, is referred to as "we," "Walmart" or the "Company") has filed with or furnished to the Securities and Exchange Commission ("SEC") or may file with or furnish to the SEC in the future, and prior or future public announcements and presentations that we or our management have made or may make, include or may include, or incorporate or may incorporate by reference, statements that may be deemed to be "forward-looking statements" within the meaning of Section 21E of the Securities Exchange Act of 1934, as amended (the "Exchange Act"), that are intended to enjoy the protection of the safe harbor for forward-looking statements provided by the Exchange Act as well as protections afforded by other federal securities laws.

Nature of Forward-Looking Statements

Such forward-looking statements are not statements of historical facts, but instead express our estimates or expectations for our consolidated, or one of our segment's, economic performance or results of operations for future periods or as of future dates or events or developments that may occur in the future or discuss our plans, objectives or goals. These forward-looking statements may relate to:

- macroeconomic, geopolitical, and business conditions, trends and events around the world and in the markets in which we operate, including inflation or deflation, generally, and in certain product categories, the impact of supply chain challenges, and recessionary pressures;
- the growth of our business or change in our competitive position in the future, or in or over particular periods, both generally, and with respect to particular markets, segments or lines of business, including, but not limited to, advertising, fulfillment, healthcare and financial services;
- the amount, number, growth, increase, reduction or decrease in or over certain periods, of or in certain financial items or measures or operating measures, including our earnings per share, net sales, growth rates, comparable store and club sales, our eCommerce sales, liabilities, expenses of certain categories, expense leverage, operating income, returns, capital and operating investments or expenditures of particular types and new store and club openings, inventory levels and associated costs, product mix and demand for certain merchandise, consumer confidence, disposable income, credit availability, spending levels, shopping patterns and debt levels;
- our increasing investments in eCommerce, technology, automation, supply chain, new stores and clubs as well as remodels and other omni-channel customer initiatives, such as same day pickup and delivery;
- investments and capital expenditures we will make and how certain of those investments and capital expenditures are expected to be financed;
- our workforce strategy, including the availability of necessary personnel to staff our stores, clubs and other facilities and the potential impact of changes to the costs of labor;
- volatility in currency exchange rates affecting our consolidated, or one or more of our segments' results of operations;
- the Company continuing to provide returns to shareholders through share repurchases and dividends, the use of share repurchase authorization over a certain period or the source of funding of a certain portion of our share repurchases;
- our sources of liquidity, including our cash, continuing to be adequate or sufficient to fund our operations, finance our global investment and expansion activities, pay dividends and fund share repurchases;
- cash flows from operations, our current cash position and access to capital markets or credit will continue to be sufficient to meet our anticipated operating cash needs;
- the reclassification of amounts related to our derivatives;
- our effective tax rate for certain periods and the realization of certain net deferred tax assets and the effects of resolutions of tax-related matters;

- the adoption or creation of new, and modification of existing, governmental policies, programs, initiatives and actions in the markets in which we operate and elsewhere and actions with respect to such policies, programs and initiatives (including, but not limited to, changes in the enforcement priorities of regulatory authorities);
- the effect of adverse decisions in, or settlement of, litigation or other proceedings or investigations to which we are subject;
- the effect on our results of operations or financial position of our adoption of certain new, or amendments to existing, accounting standards; or
- our commitments, intentions, plans or goals related to environmental, social, and governance ("ESG") priorities, including, but not limited to, the sustainability of our environment and supply chains, the promotion of economic opportunity or other societal initiatives.

Our forward-looking statements may also include statements of our strategies, plans and objectives for our operations, including areas of future focus in our operations, and the assumptions underlying any of the forward-looking statements we make. The forward-looking statements we make can typically be identified by the use therein of words and phrases such as "aim," "anticipate," "believe," "continue," "could be," "could increase," "could occur," "could result," "estimate," "expansion," "expect," "expectation," "expected to be," "focus," "forecast," "goal," "grow," "guidance," "intend," "invest," "is expected," "may continue," "may fluctuate," "may grow," "may impact," "may result," "objective," "plan," "priority," "project," "should," "strategy," "to be," "we'll," "we will," "will add," "will allow," "will be," "will benefit," "will change," "will come in at," "will continue," "will decrease," "will grow," "will have," "will impact," "will include," "will increase," "will open," "will remain," "will result," "will stay," "will strengthen," "would be," "would decrease" and "would increase," variations of such words or phrases, other phrases commencing with the word "will" or similar words and phrases denoting anticipated or expected occurrences or results.

The forward-looking statements that we make or that are made by others on our behalf are based on our knowledge of our business and our operating environment and assumptions that we believe to be or will believe to be reasonable when such forward-looking statements were or are made. As a consequence of the factors described above, the other risks, uncertainties and factors we disclose below and in the other reports as mentioned above, other risks not known to us at this time, changes in facts, assumptions not being realized or other circumstances, our actual results may differ materially from those discussed in or implied or contemplated by our forward-looking statements. Consequently, this cautionary statement qualifies all forward-looking statements we make or that are made on our behalf, including those made herein and incorporated by reference herein. We cannot assure you that the results or developments expected or anticipated by us will be realized or, even if substantially realized, that those results or developments will result in the expected consequences for us or affect us, our business, our operations or our operating results in the manner or to the extent we expect. We caution readers not to place undue reliance on such forward-looking statements, which speak only as of their dates. We undertake no obligation to revise or update any of the forward-looking statements to reflect subsequent events or circumstances except to the extent required by applicable law.

ITEM 1. BUSINESS

General

Walmart Inc. ("Walmart," the "Company" or "we") is a people-led, technology-powered omni-channel retailer dedicated to helping people around the world save money and live better – anytime and anywhere – by providing the opportunity to shop in both retail stores and through eCommerce, and to access our other service offerings. Through innovation, we strive to continuously improve a customer-centric experience that seamlessly integrates our eCommerce and retail stores in an omni-channel offering that saves time for our customers. Each week, we serve approximately 255 million customers who visit more than 10,500 stores and numerous eCommerce websites in 19 countries.

Our strategy is to make every day easier for busy families, operate with discipline, sharpen our culture and become more digital, and make trust a competitive advantage. Making life easier for busy families includes our commitment to price leadership, which has been and will remain a cornerstone of our business, as well as increasing convenience to save our customers time. By leading on price, we earn the trust of our customers every day by providing a broad assortment of quality merchandise and services at everyday low prices ("EDLP"). EDLP is our pricing philosophy under which we price items at a low price every day so our customers trust that our prices will not change under frequent promotional activity. Everyday low cost ("EDLC") is our commitment to control expenses so our cost savings can be passed along to our customers.

Our operations comprise three reportable segments: Walmart U.S., Walmart International and Sam's Club. Our fiscal year ends on January 31 for our United States ("U.S.") and Canadian operations. We consolidate all other operations generally using a one-month lag and on a calendar year basis. Our discussion is as of, and for the fiscal years ended, January 31, 2024 ("fiscal 2024"), January 31, 2023 ("fiscal 2023") and January 31, 2022 ("fiscal 2022"). During fiscal 2024, we generated total revenues of $648.1 billion, which was comprised primarily of net sales of $642.6 billion.

We maintain our principal offices in Bentonville, Arkansas. Our common stock trades on the New York Stock Exchange under the symbol "WMT."

The Development of Our Company

The businesses conducted by our founders began in 1945 when Sam M. Walton opened a franchise Ben Franklin variety store in Newport, Arkansas. In 1946, his brother, James L. Walton, opened a similar store in Versailles, Missouri. Until 1962, our founders' business was devoted entirely to the operation of variety stores, at which time we began to open discount stores. We completed our initial public offering in 1970. In 1983, we opened our first Sam's Club, and in 1988, we opened our first supercenter. In 1998, we opened our first Walmart Neighborhood Market. In 1991, we began our first international initiative when we entered into a joint venture in Mexico and, as of January 31, 2024, our Walmart International segment conducted business in 18 countries.

In 2000, we began our first eCommerce initiative by creating both walmart.com and samsclub.com. Since then, our eCommerce presence has continued to grow. In 2007, leveraging our physical stores, walmart.com launched its Site-to-Store service, enabling customers to make a purchase online and pick up merchandise in stores. To date, we now have over 8,000 pickup and over 7,800 delivery locations globally. In recent years, we expanded our eCommerce and digital presence through acquisitions with our majority stakes in Flipkart and PhonePe in India. We continue to heavily invest in omni-channel and eCommerce innovation, which enables us to leverage technology, talent and expertise, and expand our assortment and service offerings.

We are enhancing our omni-channel capabilities through a combination of stores, eCommerce websites and service offerings, as well as our supply chain, combined with approximately 2.1 million associates as of January 31, 2024, to better serve our customers. Together, these elements produce a global retail ecosystem that we believe allows customers to view Walmart as their primary retail destination. In the U.S., our Walmart+ membership incorporates several service offerings which provide enhanced omni-channel shopping experiences and benefits for members. As we execute on our strategy globally, our business is expanding through offerings such as advertising, marketplace and fulfillment services, healthcare and financial services. These offerings represent mutually reinforcing pieces of our omni-channel model centered on our customers around the world who are increasingly seeking convenience.

Information About Our Segments

We are engaged in global operations of retail, wholesale and other units, as well as eCommerce, located throughout the U.S., Africa, Canada, Central America, Chile, China, India and Mexico. We also previously operated in the United Kingdom and Japan prior to the sale of those operations in the first quarter of fiscal 2022. Refer to Note 12 to our Consolidated Financial Statements for information on these divestitures. Our operations are conducted in three reportable segments: Walmart U.S., Walmart International and Sam's Club, which are further described below. Each segment contributes to the Company's operating results differently. However, each has generally maintained a consistent contribution rate to the Company's net sales in recent years other than minor changes to the contribution rate for the Walmart International segment due to the exit of certain markets and fluctuations in currency exchange rates. Additional information on our operating segments and geographic information is contained in Note 13 to our Consolidated Financial Statements.

Walmart U.S. Segment

Walmart U.S. is our largest segment and operates in the U.S., including in all 50 states, Washington D.C. and Puerto Rico. Walmart U.S. is a mass merchandiser of consumer products, operating under the "Walmart" and "Walmart Neighborhood Market" brands, including walmart.com. Walmart U.S. had net sales of $441.8 billion for fiscal 2024, representing 69% of our fiscal 2024 consolidated net sales, and had net sales of $420.6 billion and $393.2 billion for fiscal 2023 and 2022, respectively. Of our three segments, Walmart U.S. has historically had the highest gross profit as a percentage of net sales ("gross profit rate"). In addition, Walmart U.S. has historically contributed the greatest amount to the Company's net sales and operating income.

Omni-channel. Walmart U.S. provides an omni-channel experience to customers, integrating retail stores and eCommerce, through services such as pickup and delivery, in-home delivery, ship-from-store and digital pharmacy fulfillment options. As of January 31, 2024, the vast majority of our stores have pickup locations and more than 4,300 locations offer same-day delivery. Our Walmart+ membership offering provides enhanced omni-channel shopping benefits including unlimited free shipping on eligible items with no order minimum, unlimited delivery from store, fuel discounts, mobile Scan & Go and access to additional member benefits. We define eCommerce sales as sales initiated by customers digitally and fulfilled by a number of methods including our dedicated eCommerce fulfillment centers and leveraging our stores, as well as certain other business offerings that are part of our ecosystem, such as our Walmart Connect advertising business. The following table provides the approximate size of our retail stores as of January 31, 2024:

	Minimum Square Feet	Maximum Square Feet	Average Square Feet
Supercenters (general merchandise and grocery)	69,000	260,000	178,000
Discount stores (general merchandise and limited grocery)	30,000	206,000	105,000
Neighborhood markets[1] (grocery)	28,000	65,000	42,000

(1) Excludes other small formats.

Merchandise. Walmart U.S. does business primarily in three strategic merchandise units, listed below:

- Grocery consists of a full line of grocery items, including dry grocery, snacks, dairy, meat, produce, deli & bakery, frozen foods, alcoholic and nonalcoholic beverages, as well as consumables such as health and beauty aids, pet supplies, household chemicals, paper goods and baby products;
- General merchandise includes:
  - Entertainment (e.g., electronics, toys, seasonal merchandise, wireless, video games, movies, music and books);
  - Hardlines (e.g., automotive, hardware and paint, sporting goods, outdoor living and stationery);
  - Fashion (e.g., apparel for adults and children, as well as shoes, jewelry and accessories); and
  - Home (e.g., housewares and small appliances, bed & bath, furniture and home organization, home furnishings, home decor, fabrics and crafts).
- Health and wellness includes pharmacy, over-the-counter drugs and other medical products, optical services and other clinical services.

Other categories in the Walmart U.S. business include fuel and various service offerings such as in-house advertising via Walmart Connect, supply chain and fulfillment capabilities to online marketplace sellers via Walmart Fulfillment Services, initiatives such as business-to-business last mile delivery services via Walmart GoLocal, and a suite of data products for merchants and suppliers via Walmart Luminate. Additional service offerings include financial services and related products (including through our digital channels, stores and our fintech venture, ONE), such as money orders, prepaid access, money transfers, check cashing, bill payment and certain types of installment lending.

Brand name merchandise represents a significant portion of the merchandise sold in Walmart U.S. We also market lines of merchandise under our private brands, including brands such as: "Allswell," "Athletic Works," "Equate," "Free Assembly," "Freshness Guaranteed," "George," "Great Value," "Holiday Time," "Hyper Tough," "Mainstays," "Marketside," "No Boundaries," "onn.," "Ozark Trail," "Parent's Choice," "Sam's Choice," "Scoop," "Spring Valley," "Time and Tru," "Way to Celebrate" and "Wonder Nation." The Company also markets lines of merchandise under licensed brands, some of which include: "Avia," "Better Homes & Gardens," "Love & Sports," "Sofia Jeans by Sofia Vergara," and "The Pioneer Woman."

Periodically, revisions are made to the categorization of the components comprising our strategic merchandise units. When revisions are made, the previous periods' presentation is adjusted to maintain comparability.

Operations. Walmart U.S. is available to customers through supercenters, discount stores and neighborhood markets, as well as online or through the mobile application 24 hours a day. Consistent with its strategy, Walmart U.S. continues to develop technology tools and services to better serve customers and help stores operate more efficiently, such as pickup and delivery, Walmart+, ship-from-store and other initiatives which provide convenient and seamless omni-channel shopping experiences.

Seasonal Aspects of Operations. Walmart U.S.'s business is seasonal to a certain extent due to calendar events and national and religious holidays, as well as different weather patterns. Historically, its highest sales volume has occurred in the fiscal quarter ending January 31.

Competition. Walmart U.S. competes with brick and mortar, eCommerce and omni-channel retailers operating discount, department, retail and wholesale grocers, drug, dollar, variety and specialty stores, supermarkets, hypermarkets and supercenter-type stores, social commerce platforms, as well as companies that offer services in digital advertising, fulfillment and delivery services, health and wellness and financial services. Each of these landscapes is highly competitive and rapidly evolving, and new business models and the entry of new, well-funded competitors continue to intensify this competition. Some of our competitors have longer histories in these lines of business, more customers and greater brand recognition. They may be able to obtain more favorable terms from suppliers and business partners and to devote greater resources to the development of these businesses. In addition, for eCommerce and other internet-based businesses, newer or smaller businesses may be better able to innovate and compete with us.

Our ability to develop and operate units at the right locations and to deliver a customer-centric omni-channel experience largely determines our competitive position within the retail industry. We compete in a variety of ways, including the prices at which we sell our merchandise, merchandise and selection availability, services offered to customers, the quality of the products and services we offer, location, store hours, in-store amenities, the shopping convenience and overall shopping experience we offer, the attractiveness and ease of use of our digital platforms, cost and speed of and options for delivery to customers of merchandise purchased through our digital platforms or through our omni-channel integration of our physical and digital operations. We employ many strategies and programs designed to meet competitive pressures within our industry. These strategies include the following:

- EDLP: our pricing philosophy under which we price items at everyday low prices so our customers trust that our prices will not change under frequent promotional activity;
- EDLC: everyday low cost is our commitment to control expenses so our cost savings can be passed along to our customers;
- Omni-channel offerings such as pickup and delivery and our Walmart+ membership offering, all of which enhance convenience and seek to serve customers in the ways they want to be served; and
- Expanding our ecosystem and the products and services we offer in areas such as digital advertising, fulfillment services, health and wellness, and financial services to provide our customers a broader set of offerings to meet expanding needs.

Distribution. We continue to invest in supply chain automation and utilize a total of 162 distribution facilities which are located strategically throughout the U.S. For fiscal 2024, the majority of Walmart U.S.'s purchases of store merchandise were shipped through these facilities, while most of the remaining store merchandise we purchased was shipped directly from suppliers. General merchandise and dry grocery merchandise is transported primarily through the segment's private truck fleet; however, we contract with common carriers to transport the majority of our perishable grocery merchandise. We ship merchandise purchased by customers on our eCommerce platforms by a number of methods from multiple locations including from our 30 dedicated eCommerce fulfillment centers, as well as leveraging our ability to ship or deliver directly from more than 4,300 stores, some of which include market fulfillment centers, which are positioned inside or attached to our stores to fill online orders more efficiently.

Walmart International Segment

Walmart International is our second largest segment and operates in 18 countries outside of the U.S. Walmart International operates through our wholly-owned subsidiaries in Canada, Chile, China, and Africa (which includes Botswana, Lesotho, Malawi, Mozambique, Namibia, South Africa, Eswatini, and Zambia), and our majority-owned subsidiaries in India, as well as Mexico and Central America (which includes Costa Rica, El Salvador, Guatemala, Honduras and Nicaragua). Walmart International previously operated in the United Kingdom and Japan prior to the sale of those operations in the first quarter of fiscal 2022. Refer to Note 12 to our Consolidated Financial Statements for discussion of recent divestitures.

Walmart International includes numerous formats divided into two major categories: retail and wholesale. These categories consist of many formats, including: supercenters, supermarkets, hypermarkets, warehouse clubs (including Sam's Clubs) and cash & carry, as well as eCommerce through websites and mobile applications, including walmart.com.mx, walmart.ca, flipkart.com, PhonePe and other sites. Walmart International had net sales of $114.6 billion for fiscal 2024, representing 18% of our fiscal 2024 consolidated net sales, and had net sales of $101.0 billion for both fiscal 2023 and 2022. The gross profit rate is slightly lower than that of Walmart U.S. primarily because of its format and channel mix.

Walmart International's strategy is to create strong local businesses powered by Walmart, which means being locally relevant and customer-focused in each of the markets it operates. We are being deliberate about where and how we choose to operate and continue to re-shape the portfolio to best enable long-term, sustainable and profitable growth. As such, we have taken certain strategic actions to strengthen our Walmart International portfolio for the long-term, which include the following highlights over the last three years:

- Divested of Asda Group Limited ("Asda"), our retail operations in the U.K., in February 2021.
- Divested of a majority stake in Seiyu, our retail operations in Japan, in March 2021.
- Bought out the noncontrolling interest shareholders of our Massmart subsidiary in November 2022 and exited operations in certain countries in Africa.
- Increased our ownership in PhonePe, our digital payments platform in India, as part of the separation from Flipkart in December 2022.

Omni-channel. Walmart International provides an omni-channel experience to customers, integrating retail stores and eCommerce, such as through pickup and delivery services in most of our markets and our marketplaces such as Flipkart in India. Our financial services offerings continue to grow with our digital payment platform at PhonePe in India. We continue to expand our marketplace offerings, which also unlock fulfillment and advertising services.

Generally, retail units' selling areas range in size from 1,400 square feet to 186,000 square feet. Our wholesale stores' selling areas generally range in size from 25,000 square feet to 158,000 square feet. As of January 31, 2024, Walmart International had over 2,800 pickup and over 2,900 delivery locations.

Merchandise. The merchandising strategy for Walmart International is similar to that of our operations in the U.S. in terms of the breadth and scope of merchandise offered for sale. While brand name merchandise accounts for a majority of our sales, we have both leveraged U.S. private brands and developed market specific private brands to serve our customers with high quality, low priced items. Along with the private brands we market globally, such as "Equate," "George," "Great Value," "Holiday Time," "Mainstays," "Marketside" and "Parent's Choice," our international markets have developed market specific brands including "Aurrera" and "Lider." In addition, we have developed and continue to grow our relationships with regional and local suppliers in each market to ensure reliable sources of quality merchandise that is equal to national brands at low prices.

Consistent with its strategy, Walmart International continues to build mutually reinforcing businesses in areas such as advertising, marketplace and fulfillment services, healthcare and financial services. Our businesses in Mexico and Canada, for example, offer prepaid cards and money transfers, and our PhonePe business in India continues to grow, providing a platform that offers mobile and bill payment, person-to-person (P2P) payment, investment and insurance solutions, financial services and advertising. In Mexico, we also offer a value-based internet and telephone service allowing customers to enjoy digital connectivity. Combined, these offerings did not represent a significant portion of annual segment revenues.

Operations. The hours of operation for operating units in Walmart International vary by country and by individual markets within countries, depending upon local and national ordinances governing hours of operation. Customers can also access online and mobile applications 24 hours a day. Consistent with its strategy, Walmart International continues to develop technology tools and services to better serve customers and help its various formats operate more efficiently, as well as to provide convenient and seamless omni-channel shopping experiences.

Seasonal Aspects of Operations. Walmart International's business is seasonal to a certain extent. Historically, its highest sales volume has occurred in the fourth quarter of our fiscal year. The seasonality of the business varies by country due to different national and religious holidays, festivals and customs, as well as different weather patterns.

Competition. Walmart International competes with brick and mortar, eCommerce and omni-channel retailers who operate department, drug, discount, variety and specialty stores, supermarkets, hypermarkets and supercenter-type stores, wholesale clubs, home-improvement stores, specialty electronics stores, cash & carry operations and convenience stores, and direct to consumer offerings, as well as companies that offer services in digital advertising, fulfillment services, health and wellness and financial services. Our ability to develop and operate units at the right locations and to deliver a customer-centric omni-channel experience largely determines our competitive position within the retail industry. We believe price leadership is a critical part of our business model and we continue to progress our markets towards an EDLP approach. Additionally, our ability to operate food departments effectively has a significant impact on our competitive position in many of the markets where we operate. Each of these landscapes is highly competitive and rapidly evolving, and new business models and the entry of new or well-funded competitors continue to intensify this competition. Some of our competitors have longer histories in these lines of business, more customers and greater brand recognition and therefore they may be able to obtain more favorable terms from suppliers and business partners and to devote greater resources to the development of these businesses. In addition, for eCommerce and other internet-based businesses, newer or smaller businesses may be better able to innovate and compete with us.

Distribution. We utilize a total of 176 distribution facilities located in Canada, Central America, Chile, China, India, Mexico and Africa. Through these facilities, we process and distribute both imported and domestic products to the operating units of the Walmart International segment. During fiscal 2024, the majority of Walmart International's purchases passed through these distribution facilities. Suppliers ship the remainder of Walmart International's purchases directly to our stores in the various markets in which we operate. Across the segment, we have efficient networks connecting physical stores and distribution and fulfillment centers, which facilitate the movement of goods to where our customers live. We ship merchandise purchased by customers on our eCommerce platforms by a number of methods from multiple locations, such as in India where we utilize a combination of more than 3,500 eCommerce fulfillment centers, sort centers and last-mile delivery facilities, as well as our physical retail stores.

Sam's Club Segment

Sam's Club operates in 44 states in the U.S. and in Puerto Rico. Sam's Club is a membership-only warehouse club that also operates samsclub.com. Sam's Club had net sales of $86.2 billion for fiscal 2024, representing 13% of our consolidated fiscal 2024 net sales, and had net sales of $84.3 billion and $73.6 billion for fiscal 2023 and 2022, respectively. As a membership-only warehouse club, membership income is a significant component of the segment's operating income. Sam's Club operates with a lower gross profit rate and lower operating expenses as a percentage of net sales than our other segments.

Membership. The following two options are available to members:

	Plus Membership	Club Membership
Annual Membership Fee	$110	$50
Number of Add-on Memberships ($45 each)	Up to 16	Up to 8

All memberships include a spouse/household card at no additional cost. Plus Members are also eligible for free curbside pickup and free shipping on the majority of merchandise, with no minimum order size, and receive discounts on prescriptions and glasses. Beginning in fiscal 2023, Sam's Club launched a single loyalty rewards currency called Sam's Cash which merges and replaces existing Cash Rewards for Plus members and Cash Back for Sam's Club Mastercard holders. Members may redeem Sam's Cash on purchases in the club and online, to pay for membership fees or for cash in clubs. Sam's Cash does not expire and is available for monthly redemption.

Omni-channel. Sam's Club provides an omni-channel experience to members, integrating warehouse clubs and eCommerce through such services as Curbside Pickup, mobile Scan & Go, ship-from-club, and delivery-from-club. Members have access to a broad assortment of merchandise and services, including those not found in our clubs, online at samsclub.com and through our mobile commerce applications. The warehouse facility sizes generally range between 32,000 and 168,000 square feet, with an average size of approximately 134,000 square feet.

Merchandise. Sam's Club offers merchandise in the following five merchandise categories:

- Grocery and consumables includes dairy, meat, bakery, deli, produce, dry, chilled or frozen packaged foods, alcoholic and nonalcoholic beverages, floral, snack foods, candy, other grocery items, health and beauty aids, paper goods, laundry and home care, baby care, pet supplies and other consumable items;
- Fuel, tobacco and other categories;
- Home and apparel includes home improvement, outdoor living, gardening, furniture, apparel, jewelry, tools and power equipment, housewares, toys, seasonal items, mattresses and tire and battery centers;
- Health and wellness includes pharmacy, optical and hearing services and over-the-counter drugs; and

- Technology, office and entertainment includes consumer electronics and accessories, software, video games, office supplies, appliances and third-party gift cards.

Within the categories above, the Member's Mark private label brand continues to expand its assortment and deliver member value.

Operations. Sam's Club is available to members through warehouse club locations, as well as online or through the mobile application 24 hours a day. Club locations offer Plus Members the ability to shop before regular operating hours. Consistent with its strategy, Sam's Club continues to develop technology tools to drive a great member experience. Curbside Pickup is available at clubs to help provide fast, easy and contact-free shopping for members. Sam's Club also offers Scan & Go, a mobile checkout and payment solution, which allows members to bypass the checkout line.

Seasonal Aspects of Operations. Sam's Club's business is seasonal to a certain extent due to calendar events and national and religious holidays, as well as different weather patterns. Historically, its highest sales volume has occurred in the fiscal quarter ending January 31.

Competition. Sam's Club competes with other membership-only warehouse clubs, the largest of which is Costco, as well as with discount retailers, retail and wholesale grocers, general merchandise wholesalers and distributors, gasoline stations as well as omni-channel and eCommerce retailers and catalog businesses. At Sam's Club, we provide value at members-only prices, a quality merchandise assortment, and bulk sizing to serve both our Plus and Club members. Our eCommerce website and mobile commerce applications have increasingly become important factors in our ability to compete.

Distribution. We utilize 30 dedicated distribution facilities located strategically throughout the U.S., as well as some of the Walmart U.S. segment's distribution facilities which service the Sam's Club segment for certain items. During fiscal 2024, the majority of Sam's Club's non-fuel club purchases were shipped from these facilities, while the remainder of our purchases were shipped directly to Sam's Club locations by suppliers. Sam's Club ships merchandise purchased on samsclub.com and through its mobile commerce applications by a number of methods including shipments made directly from clubs, 15 dedicated eCommerce fulfillment centers and other distribution centers.

Sam's Club uses a combination of our private truck fleet, as well as common carriers, to transport perishable and non-perishable merchandise from distribution facilities to clubs.

Intellectual Property

We regard our trademarks, service marks, copyrights, patents, domain names, trade dress, trade secrets, proprietary technologies and similar intellectual property as important to our success, and with respect to our associates, customers and others, we rely on trademark, copyright, and patent laws, trade-secret protection, and confidentiality and/or license agreements to protect our proprietary rights. We have registered, or applied for the registration of, a number of U.S. and international domain names, trademarks, service marks and copyrights. Additionally, we have filed U.S. and international patent applications covering certain of our proprietary technology. We have licensed in the past, and expect that we may license in the future, certain of our proprietary rights to third parties.

Suppliers and Supply Chain

As a retailer and warehouse club operator, we utilize a global supply chain that includes both U.S. and international suppliers from whom we purchase the merchandise that we sell in our stores, clubs and online. In many instances, we purchase merchandise from producers located near the stores and clubs in which such merchandise will be sold, particularly products in the "fresh" category. Consistent with applicable laws, we offer our suppliers the opportunity to efficiently sell significant quantities of their products to us. These relationships enable us to obtain pricing that reflects the volume, certainty and cost-effectiveness these arrangements provide to such suppliers, which in turn enables us to provide low prices to our customers. Our suppliers are subject to standards of conduct, including requirements that they comply with local labor laws, local worker safety laws and other applicable laws. Our ability to acquire from our suppliers the assortment and volume of products we wish to offer to our customers, to receive those products within the required time through our supply chain and to distribute those products to our stores and clubs, determines, along with other supply chain logistics matters (such as containers or port access for example), in part, our in-stock levels in our stores and clubs and the attractiveness of our merchandise assortment we offer to our customers and members.

Government Regulation

As a company with global operations, we are subject to the laws of the United States and multiple foreign jurisdictions in which we operate and the rules and regulations of various governing bodies, which may differ among jurisdictions. For additional information, see the risk factors herein in "Item 1A. Risk Factors" under the sub-caption "Legal, Tax, Regulatory, Compliance, Reputational and Other Risks."

Environmental, Social and Governance ("ESG") Priorities

Our ESG strategy is centered on the concept of creating shared value: we believe we maximize long-term value and create competitive advantage for the Company by serving our stakeholders, including our customers, associates, shareholders, suppliers, business partners and communities. We believe that addressing such societal needs builds the value of our business, including by enhancing customer and associate trust, creating new business opportunities, managing cost and risk, building capabilities for future advantage and strengthening the underlying systems on which Walmart and our stakeholders rely.

We prioritize the ESG issues that offer the greatest potential for Walmart to create shared value: issues that rank high in relevance to our business and stakeholders and which Walmart is positioned to make a positive impact. Our current ESG priorities are categorized into four broad themes: opportunity, sustainability, community and ethics and integrity.

- **Opportunity**. Retail can be a powerful engine for inclusive economic opportunity. We aim to advance belonging, diversity, equity and inclusion, and create opportunity for Walmart associates (as further described in the Human Capital Management section below), our suppliers and workers in supply chains, and the communities in which we operate. Doing so helps us fulfill our customer mission, strengthens our business and helps people build a better life for themselves and their families.
- **Sustainability**. Walmart's sustainability efforts focus on our ability to create and preserve long-term value for both people and planet. With respect to people, our sustainability efforts include sourcing responsibly, helping prevent forced labor, empowering women, creating inclusive economic opportunity and selling safer, healthier products. With respect to the planet, our efforts aim to enhance the sustainability of product supply chains by reducing emissions, protecting and restoring nature and reducing waste. To help address the effects of climate change, Walmart has set science-based targets for emissions reduction, including our goal to achieve zero emissions in our operations by 2040—without offsets—and to reduce, avoid or sequester one billion metric tons of emissions in our value chain by 2030 under our Project Gigaton initiative.
- **Community**. Walmart aims to serve and strengthen communities by operating our business in a way that meets the needs of our customer and community stakeholder groups, including by providing safer, healthier and more affordable food and other products, disaster support, associate volunteerism, local grant programs and community cohesion initiatives.
- **Ethics and Integrity**. At every level of our Company, we work to create a culture that inspires trust among our associates, with our customers and in the communities we serve.

We periodically publish information on our ESG priorities, strategies and progress on our corporate website and may update those disclosures from time to time. Nothing on our website, including our ESG reporting, documents or sections thereof, shall be deemed incorporated by reference into this Annual Report on Form 10-K or incorporated by reference into any of our other filings with the SEC.

Human Capital Management

Our associates – powered by technology – play a critical role in delivering on our purpose to help people save money and live better. Our business is focused on serving people and this is delivered by our approximately 2.1 million associates around the world with approximately 1.6 million associates in the U.S. and approximately 0.5 million associates internationally. In the U.S., approximately 92% of our associates are hourly and approximately 69% of our associates are full-time.

We believe our people make the difference, and we are focused on investing in the growth and well-being of our associates, investing in digital experiences to improve their quality of work and creating a culture of belonging. An important part of our focus is to provide opportunities for associates to grow and learn. For some, we are a foundational entry point to develop critical skills that are relevant for a variety of careers. We are focused on developing, rewarding and retaining associates in an ever-changing environment. Our people ultimately make Walmart a better place to work and a better place to shop. Our workforce strategy includes the following strategic priorities: belonging, well-being, growth and digital.

Belonging - *Focus on creating a workplace where all associates feel seen, supported and connected through a culture of belonging.* We publish our diversity representation twice yearly and hold ourselves accountable to providing recurring belonging, diversity, equity and inclusion updates to senior leadership – including our President and CEO – and members of the Board of Directors. Of the approximately 2.1 million associates employed worldwide, 52% identify as women. In the U.S., 51% of the approximately 1.6 million associates identify as people of color.

Our Belonging programs aim to create equitable opportunities for everyone, regardless of background, so that every eligible and qualified individual can thrive and perform. We regularly review our processes around fair-pay practices and are committed to creating a performance culture where associates are rewarded based on meaningful factors such as qualifications, experience, performance and the work they do.

To build a company where associates feel seen, supported and connected, we gather and respond to associate feedback in a variety of ways, including in-person dialogue; real-time digital insights such as pulse surveys; formal surveys like our associate engagement survey; and always-on confidential channels, including our Open Door process and ethics channels.

Well-being - *Prioritize the emotional, physical and financial well-being of associates.* We invest in our associates by offering competitive wages, as well as a broad range of benefits to meet the diverse needs of our global associate population and their eligible dependents. In the U.S., this includes company-paid benefits such as 401(k) match, family building support, maternity leave, fertility benefits, a paid parental leave program to all full-time associates, paid time off, Associate Stock Purchase Plan match, life insurance, behavioral and mental health services, a Walmart discount card or Sam's Club membership and predictable scheduling that helps associates plan for life outside of work and know what to expect in their paychecks.

Additional information about how we invest in our associates' well-being, including wage structure and pay, can be found in our Human Capital brief in our most recent ESG reporting, which is available on our corporate website. Nothing on our website, including our ESG reporting documents, or sections thereof, shall be deemed incorporated by reference into this Annual Report on Form 10-K or incorporated by reference into any of our other filings with the SEC. Certain information relating to retirement-related benefits we provide to our associates is included in Note 11 to our Consolidated Financial Statements.

Growth - *Provide ongoing growth, development and learning opportunities for associates and continue to attract talent with new skills.* We are invested in the growth of our associates – in support of our business and their success – by offering good jobs that lead to great careers and better lives.

Approximately 75% of our U.S. salaried store, club and supply chain management started their careers in hourly positions. Our focus on providing a path of opportunity for our associates through robust training, competitive wages and benefits and career advancement creates a strong associate value proposition and strengthens our workforce. In the U.S., we seek to enable these pathways through programs like Walmart Academy and Live Better U (LBU). Walmart Academy offers training for on-the-job retail skills, leadership and well-being, serving our associates through a combination of digital and in-person offerings. Additionally, our LBU program provides access to educational opportunities for our part-time and full-time frontline eligible associates in the U.S. through high school diplomas, short-form certificates and credentials or college degrees.

Digital - *Drive a digital transformation that improves the associate experience and powers the business*. To deliver a seamless customer and associate experience, we continue to invest in consumer-grade digital tools designed to improve associate productivity and efficiency, engagement and performance, allowing associates to spend more time generating new ideas, developing strategy and building relationships. This capability has been expanded to certain international markets.

Technology is also used to improve associate experience, including an app developed to capture real-time associate feedback. Walmart also supports associates on the U.S. Medical Plan with free virtual visits for medical doctor urgent care and mental health care with psychiatrists and psychologists.

Information About Our Executive Officers

The following chart names the executive officers of the Company as of the date of the filing of this Annual Report on Form 10-K with the SEC, each of whom is elected by, and serves at the pleasure of, the Board of Directors. The business experience shown for each officer has been his or her principal occupation for at least the past five years, unless otherwise noted.

Name	Business Experience	Current Position Held Since	Age
Daniel J. Bartlett	Executive Vice President, Corporate Affairs, effective June 2013. From November 2007 to June 2013, he served as the Chief Executive Officer and President of U.S. Operations at Hill & Knowlton, Inc., a public relations company.	2013	52
Rachel Brand	Executive Vice President, Global Governance, Chief Legal Officer and Corporate Secretary, effective April 2018. From May 2017 to February 2018, she served as Associate Attorney General in the United States Department of Justice.	2018	50
David M. Chojnowski	Senior Vice President and Controller effective January 2017. From October 2014 to January 2017, he served as Vice President and Controller, Walmart U.S.	2017	54
John Furner	Executive Vice President, President and Chief Executive Officer, Walmart U.S. effective November 2019. From February 2017 until November 2019, he served as President and Chief Executive Officer, Sam's Club.	2019	49
Suresh Kumar	Executive Vice President, Global Chief Technology Officer and Chief Development Officer effective July 2019. From February 2018 until June 2019, Mr. Kumar was Vice President and General Manager at Google LLC.	2019	59
Kathryn McLay	Executive Vice President, President and Chief Executive Officer, Walmart International, effective August 2023. From 2019 to 2023, she served as Executive Vice President, President and Chief Executive Officer, Sam's Club. From February 2019 to November 2019, she served as Executive Vice President, Walmart U.S. Neighborhood Markets. From December 2015 until February 2019, she served as Senior Vice President, U.S. Supply Chain.	2023	50
C. Douglas McMillon	President and Chief Executive Officer, effective February 2014. From February 2009 to January 2014, he served as Executive Vice President, President and Chief Executive Officer, Walmart International.	2014	57
Donna Morris	Executive Vice President, Global People, and Chief People Officer, effective February 2020. From April 2002 to January 2020, she worked at Adobe Inc. in various roles, including most recently, Chief Human Resources Officer and Executive Vice President, Employee Experience.	2020	56
Christopher Nicholas	Executive Vice President, President and Chief Executive Officer, Sam's Club effective September 2023. From October 2021 to September 2023, he served as Executive Vice President, Chief Operating Officer, Walmart U.S. From February 2021 until October 2021, he served as Executive Vice President, Chief Financial Officer Walmart U.S. From January 2020 until February 2021, he served as Executive Vice President, Chief Financial Officer Walmart International. He joined the Company in August 2018 as Senior Vice President and Deputy Chief Financial Officer, Walmart International.	2023	47
John David Rainey	Executive Vice President and Chief Financial Officer, effective June 2022. From September 2015 to June 2022, he served as Chief Financial Officer and Executive Vice President, Global Customer Operations for PayPal Holdings, Inc.	2022	53

Our Website and Availability of SEC Reports and Other Information

Our corporate website is located at www.stock.walmart.com. We file with, or furnish to, the SEC Annual Reports on Form 10-K, Quarterly Reports on Form 10-Q, Current Reports on Form 8-K, amendments to those reports, proxy statements and annual reports to shareholders, and, from time to time, other documents. The reports and other documents filed with, or furnished to, the SEC are available to investors on or through our corporate website free of charge as soon as reasonably practicable after we electronically file them with or furnish them to the SEC. The SEC maintains a website that contains reports, proxy and information statements and other information regarding issuers, such as the Company, that file electronically with the SEC. The address of that website is www.sec.gov. Our SEC filings, our Reporting Protocols for Senior Financial Officers and our Code of Conduct can be found on our website at www.stock.walmart.com. These documents are available in print to any shareholder who requests a copy by writing or calling our Investor Relations Department, which is located at our principal offices.

A description of any substantive amendment or waiver of Walmart's Reporting Protocols for Senior Financial Officers or our Code of Conduct for our chief executive officer, our chief financial officer and our controller, who is our principal accounting officer, will be disclosed on our website at www.stock.walmart.com under the Corporate Governance section. Any such description will be located on our website for a period of 12 months following the amendment or waiver.

ITEM 1A. RISK FACTORS

The risks described below could, in ways we may or may not be able to accurately predict, materially and adversely affect our business, results of operations, financial position and liquidity. Our business operations could also be affected by additional factors that apply to all companies operating in the U.S. and globally. The following risk factors do not identify all risks that we may face.

Strategic Risks

Failure to successfully execute our omni-channel strategy and the cost of our investments in eCommerce and technology may materially adversely affect our market position, net sales and financial performance.

The retail business continues to rapidly evolve and consumers increasingly embrace digital shopping. As a result, the portion of total consumer expenditures with retailers and wholesale clubs occurring through digital platforms is increasing and the pace of this increase could continue to accelerate.

Our strategy, which includes investments in eCommerce, technology, including the use of artificial intelligence technology, talent, supply chain automation, acquisitions, joint ventures, store remodels and other customer initiatives may not adequately or effectively allow us to continue to grow our eCommerce business, increase comparable sales, maintain or grow our overall market position or otherwise offset the impact on the growth of our business of a moderated pace of new store and club openings and sustain the current pace of remodels. The success of this strategy will depend in large measure on our ability to continue building and delivering a seamless omni-channel shopping experience and interconnected ecosystem for our customers that deepens and maintains our relationships with our customers across our various businesses and partnerships and reinforces our overall enterprise strategy. The success of this strategy is further subject to the related risks discussed in this Item 1A. With the interconnected components of this enterprise strategy and an increasing allocation of capital expenditures focused on these initiatives, changes in customer or member perceptions about our reputation or our failure to successfully execute on individual components of this strategy may adversely affect our market position, net sales and financial performance, which could also result in impairment charges to intangible assets or other long-lived assets. In addition, a greater concentration of eCommerce sales, including increasing online grocery sales, could result in a reduction in the amount of traffic in our stores and clubs, which would, in turn, reduce the opportunities for cross-store or cross-club sales of merchandise that such traffic creates and could reduce our sales within our stores and clubs and materially adversely affect our financial performance.

Furthermore, the cost of certain investments in eCommerce, technology, talent and automation, including any operating losses incurred for those initiatives, will adversely impact our financial performance in the short-term and failure to realize the benefits of these investments may adversely impact our financial performance over the longer term.

If we do not timely identify or effectively respond to consumer trends or preferences, it could negatively affect our relationship with our customers, demand for the products and services we sell, our market share and the growth of our business.

It is difficult to predict consistently and successfully the products and services our customers will demand and changes in their shopping patterns, tastes and preferences. The success of our business depends in part on how accurately we predict consumer demand, availability of merchandise, the related impact on the demand for existing products and services and the competitive environment. Our business is dependent on our ability to make critical decisions and predictions with respect to merchandise categories that quickly respond to changing consumer spending patterns, tastes and preferences, any incorrect calculations by us may result in lower sales, spoilage and inventory markdowns, which could adversely impact our results of operations. Our ability to predict and adapt to changing tastes and preferences depends on many factors, including obtaining accurate and relevant data on customer preferences, emphasizing relevant merchandise categories, effectively managing our inventory levels, and implementing competitive and effective pricing and promotion strategies. Price transparency, assortment of products, customer experience, convenience, ease and the speed and cost of shipping are of primary importance to customers and continue to increase in importance, particularly as a result of digital tools and social media available to consumers and the choices available to consumers for purchasing products. We must continue to preserve our reputation, which is impacted based on public perceptions. It may be difficult to address negative publicity across media channels, regardless of whether it is accurate. Negative incidents involving us, our workforce (including the loss of merchandise as a result of shrink or theft) or others with whom we do business could quickly erode trust and confidence in our business and could result in consumer boycotts, workforce unrest and government investigations. Negative reputational incidents or negative perceptions of us could adversely impact our business and results of operations, including through lower sales, the termination of business relationships and associate retention and recruiting efforts. Moreover, failure to adequately predict customer demand and consumer spending patterns or otherwise optimize and operate our distribution and fulfillment centers could result in excess or insufficient inventory, service interruptions and increased costs, any of which could significantly harm our business. As we continue to add new fulfillment centers, our fulfillment and technology networks become increasingly complex and operating them becomes more challenging. There can be no assurance that we will be able to operate our networks effectively.

We face strong competition from other retailers, wholesale club operators, omni-channel retailers and other businesses which could materially adversely affect our financial performance.

Each of our segments competes for customers, employees, digital prominence, products and services and in other important aspects of its business with many other local, regional, national and global physical, eCommerce and omni-channel retailers, social commerce platforms, wholesale club operators and retail intermediaries, as well as companies that offer services in digital advertising, fulfillment and delivery services, health and wellness and financial services. The omni-channel retail landscape is highly competitive and rapidly evolving, and the entry of new, well-funded competitors may increase competitive pressures. In addition, for eCommerce and other internet-based businesses, newer or smaller businesses may be better able to innovate and compete with us.

We compete in a variety of ways, including the prices at which we sell our merchandise, merchandise selection and availability, services offered to customers, location, store hours, in-store amenities, the shopping convenience and overall shopping experience we offer, the attractiveness and ease of use of our digital platforms, cost and speed of and options for delivery to customers of merchandise purchased through our digital platforms or through our omni-channel integration of our physical and digital operations.

A failure to respond effectively to competitive pressures and changes in the retail and other markets in which we operate, omni-channel innovations and omni-channel ecosystems developed by our competitors or delays or failure in execution of our strategy could materially adversely affect our financial performance. See "Item 1. Business" above for additional discussion of the competitive situation of each of our reportable segments.

Certain segments of the retail industry are undergoing consolidation or substantially reducing operations, whether due to bankruptcy, consolidation or other factors. Such consolidation, or other business combinations or alliances, competitive omni-channel ecosystems or reductions in operations may result in competitors with greatly improved financial resources, improved access to merchandise, greater market penetration and other improvements in their competitive positions. Such business combinations or alliances could allow these companies to provide a wider variety of products and services at competitive prices, which could adversely affect our financial performance.

General or macro-economic factors, both domestically and internationally, may materially adversely affect our financial performance.

General economic conditions and other economic factors, globally or in one or more of the markets we serve, may adversely affect our financial performance. Higher interest rates, higher prices of petroleum products, including crude oil, natural gas, gasoline and diesel fuel, higher costs for electricity and other energy, weakness in the housing market, inflation, deflation, increased costs of essential services, such as medical care and utilities, higher levels of unemployment, decreases in consumer disposable income, unavailability of consumer credit, higher consumer debt levels, changes in consumer spending and shopping patterns, fluctuations in currency exchange rates, higher tax rates, imposition of new taxes or other changes in tax laws, changes in healthcare laws, other regulatory changes, the imposition of tariffs or other measures that create barriers to or increase the costs associated with international trade, overall economic slowdown or recession and other economic factors in the U.S., or in any of the other markets in which we operate, could adversely affect consumer demand for the products and services we sell in the U.S. or such other markets, change the mix of products we sell to one with a lower average gross margin, cause a slowdown in discretionary purchases of goods, adversely affect our net sales, growth rates, operating income and result in slower inventory turnover and greater markdowns of inventory, or otherwise materially adversely affect our operations and operating results and could result in impairment charges to intangible assets, goodwill or other long-lived assets.

In addition, the economic factors listed above, any other economic factors or circumstances resulting in higher transportation, labor, insurance or healthcare costs or commodity prices, including energy prices, and other economic factors in the U.S. and other countries in which we operate can increase our cost of sales and operating, selling, general and administrative expenses and otherwise materially adversely affect our operations and operating results.

The economic factors that affect our operations may also adversely affect the operations of our suppliers, which can result in an increase in the cost to us of the goods we sell to our customers or, in more extreme cases, in certain suppliers not producing goods in the volume typically available to us for sale, or adversely impact product margins due to higher labor and material costs of our suppliers that we are unable, or choose not, to pass on to our customers.

The performance of strategic alliances and other business relationships to support the expansion of our business could materially adversely affect our financial performance.

We may enter into strategic alliances and other business relationships in the countries in which we have existing operations or in other markets to expand our business. These arrangements (such as ONE, our fintech venture, and our healthcare initiative with UnitedHealth Group) may not generate the level of sales or profitability we anticipate when entering into the arrangement or may otherwise adversely impact our business and competitive position relative to the results we could have achieved in the absence of such alliance. In addition, any investment we make in connection with a strategic alliance, business relationship or in certain of our divested markets, could materially adversely affect our financial performance.

Operational Risks

Global or regional health pandemics or epidemics, such as COVID-19, could negatively impact our business, financial position and results of operations.

The emergence, severity, magnitude and duration of global or regional pandemics, epidemics or other health crises are uncertain and difficult to predict. A pandemic, such as COVID-19, or other epidemic or contagious disease outbreak could impact our business operations, demand for our products and services, in-stock positions, costs of doing business, access to inventory, supply chain operations, the extent and duration of measures to try to contain the spread of a virus or other disease (such as travel bans and restrictions, quarantines, shelter-in-place orders, limitations on large gatherings, business and government shutdowns and other restrictions on retailers), our ability to predict future performance, exposure to litigation and our financial performance, among other things. In the event of any global or regional health crisis, customer demand for certain products may fluctuate, customer behaviors may change and consumer disposable income could be negatively impacted, which may challenge our ability to anticipate and/or adjust inventory levels to meet that demand. Other factors and uncertainties may include, but are not limited to: the severity and duration of pandemics, epidemics or other health crises, including disease outbreaks in areas in which we and our suppliers operate; increased operational costs; evolving macroeconomic factors, including general economic uncertainty, unemployment rates and recessionary pressures; unknown consequences on our business performance and initiatives stemming from the substantial investment of time, capital and other resources to a pandemic or other health crisis response; the effectiveness and extent of administration of vaccinations and medical treatments; the pace of recovery when any such pandemic or other health crisis subsides; and the long-term impact of a pandemic or other health crisis on our business, including consumer behaviors. These risks and their impacts are difficult to predict and could otherwise disrupt and adversely affect our operations and our financial performance.

To the extent that a future pandemic or epidemic occurs, such events may also heighten other risks described in this section, including but not limited to those related to consumer behavior and expectations, competition, our reputation, implementation of strategic initiatives, cybersecurity threats, payment-related risks, technology systems disruption, supply chain disruptions, labor availability and cost, litigation and regulatory requirements.

Natural disasters, climate change, geopolitical events, catastrophic and other events could materially adversely affect our financial performance.

The occurrence of one or more natural disasters, such as hurricanes, tropical storms, floods, fires, earthquakes, tsunamis, cyclones, typhoons; weather conditions such as major or extended winter storms, droughts and tornadoes, whether as a result of climate change or otherwise; geopolitical tensions or events; and catastrophic and other events, such as war, civil unrest (including theft, looting or vandalism), terrorist attacks or other acts of violence, including active shooter situations (such as those that have occurred in our U.S. stores), or the loss of merchandise as a result of shrink or theft in countries in which we operate, in which our suppliers are located or regions goods are transported from or through, or in other areas of the world (such as in Ukraine and Israel where wars currently exist, and armed hostilities in the Red Sea and surrounding areas through which ocean carrier vessels travel to the Suez Canal) could adversely affect our operations and financial performance.

Such events could result in physical damage to, or the complete loss of, one or more of our properties, the closure of one or more stores, clubs and distribution or fulfillment centers, limitations on store or club operating hours, the lack of an adequate work force in a market, the inability of customers and associates to reach or have transportation to our stores and clubs affected by such events, the evacuation of the populace from areas in which our stores, clubs and distribution and fulfillment centers are located, the unavailability of our digital platforms to our customers, changes in the purchasing patterns of consumers (including the frequency of visits by consumers to physical retail locations, whether as a result of limitations on large gatherings, travel and movement limitations or otherwise), such as Hurricane Otis that impacted our stores and operations around Acapulco, Mexico in fiscal 2024, although not material to our consolidated financial performance. Moreover, these disasters and events can negatively impact consumers' disposable income, the temporary or long-term disruption in the supply of products from some suppliers, the disruption in the transport of goods from overseas, the disruption or delay in the delivery of goods to our distribution and fulfillment centers or stores within a country in which we are operating, the reduction in the availability of products in our stores, increases in the costs of procuring products as a result of either reduced availability or economic sanctions, increased transportation costs (whether due to fuel prices, fuel supply or otherwise), the disruption (whether directly or indirectly) of critical infrastructure systems, banking systems, utility services or energy availability to our stores, clubs and our facilities and the disruption in our communications with our stores, clubs and our other facilities.

Furthermore, the long-term impacts of climate change, whether involving physical risks (such as extreme weather conditions, drought or rising sea levels) or transition risks (such as regulatory or technology changes) are expected to be widespread and unpredictable. Certain impacts of physical risk may include: temperature changes that increase the heating and cooling costs at stores, clubs and distribution or fulfillment centers; extreme weather patterns that affect the production or sourcing of certain commodities; flooding and extreme storms that damage or destroy our buildings and inventory; and heat and extreme weather events that cause long-term disruption or threats to the habitability of the communities in which we operate. Relative to transition risk, certain impacts may include: changes in energy and commodity prices driven by climate-related weather events; prolonged climate-related events affecting macroeconomic conditions with related effects on consumer spending and confidence; stakeholder perception of our engagement in climate-related policies; and new regulatory requirements resulting in higher compliance risk and operational costs.

We bear the risk of losses incurred as a result of physical damage to, or destruction of, any stores, clubs and distribution or fulfillment centers; theft, loss or spoilage of inventory; and business interruption caused by such events. These events and their impacts could otherwise disrupt and adversely affect our operations and could materially adversely affect our financial performance. Moreover, our operations in the U.S. comprise a significant portion of our financial and operational performance. Therefore, any of the above matters that uniquely impact or are specifically concentrated in the U.S. could materially adversely affect our financial and operational performance.

Risks associated with our suppliers could materially adversely affect our financial performance.

The products we sell are sourced from a wide variety of domestic and international suppliers. Global sourcing of many of the products we sell is an important factor in our financial performance. We expect our suppliers to comply with applicable laws, including labor, safety, anti-corruption and environmental laws, and to otherwise meet our required supplier standards of conduct. Our ability to find qualified suppliers who uphold our standards, and to access products in a timely and efficient manner and in the large volumes we may demand, is a significant challenge, especially with respect to suppliers located and goods sourced outside the U.S.

Political and economic instability, as well as other impactful events and circumstances (such as pandemic recovery related challenges, including supply chain disruption and production, labor shortages and increases in labor costs) in the countries in which our suppliers and their manufacturers are located or regions goods are transported from or through, the financial instability of suppliers, suppliers not having the financial ability or capacity to fulfill their indemnification obligations to us if called upon, thereby exposing us to the full cost of risks and claims, suppliers' failure to meet our terms and conditions or our supplier standards (including our responsible sourcing standards), labor problems experienced by our suppliers and their manufacturers, the availability of raw materials to suppliers, merchandise safety and quality issues, disruption or delay in the transportation of merchandise from the suppliers and manufacturers to our stores, clubs and other facilities, including as a result of labor slowdowns at any port at which a material amount of merchandise we purchase enters into the markets in which we operate, currency exchange rates, transport availability and cost, transport security, inflation and other factors relating to the suppliers and the countries in which they are located are beyond our control.

In addition, U.S. and international trade policies, tariffs and other restrictions on the exportation and importation of goods, trade sanctions imposed between certain countries and entities, the limitation on the exportation or importation of certain types of goods or of goods containing certain materials from other countries and other factors relating to foreign trade are beyond our control. These and other factors affecting our suppliers and our access to products could adversely affect our operations and financial performance.

If the quality or safety of products we sell in stores or online fails to meet our customers' expectations or regulatory standards, we could lose customers, incur liability for any injuries caused by a product we sell or otherwise experience a material impact to our brand, reputation and financial performance.

Our customers count on us to provide them with quality products at an affordable price. Occasionally, the quality of products that we source from our suppliers fails to meet customer expectations. In many cases, these products are subject to regulatory action or recall. For general merchandise, this could be because the product fails to meet safety standards. For food products, it could be because the product is a source of foodborne illness. For health and wellness products, it could be because the product does not produce the expected result for the customer or harms the customer. Any of these factors could cause customers to avoid purchasing certain products from us, or to choose to buy products from a different retailer, even if the quality issue is outside of our control. Any lost confidence on the part of our customers would be difficult and costly to reestablish. When a product we sell does not meet quality or safety standards, there is an increased risk of liability for harm the product may cause our customers. While we rely on our suppliers to meet our safety and quality expectations, and to indemnify us if their products do not, certain suppliers may not have the financial capacity or ability to fulfill their indemnification obligations. In that case, we are exposed to the full cost of liability claims. Any issue regarding the quality or safety of products we sell, regardless of the cause, could adversely affect our brand, reputation and financial performance.

If the quality or safety of products offered for sale on our third-party marketplace fails to meet our customers' expectations or regulatory standards, we could be held directly liable, lose customers, become subject to regulatory enforcement or otherwise experience reputational harm.

Some of the products customers buy from our website are sold by third parties, which we refer to as marketplace transactions. While that transaction ultimately occurs between the third-party seller and the customer, some regulators and courts have taken a view that the retailer is responsible for marketplace transactions that occur on a retailer's digital platform. Unsettled law on whether a retailer is responsible for intellectual property or product liability claims related to marketplace transactions creates additional risk. Any unfavorable changes or legal interpretations could further expose us to liability. In addition, poor quality or safety of third-party products offered for sale on our platforms could erode customer trust, leading to loss of sales, reduction in transactions and deterioration of our competitive position. In addition, we may face reputational, financial and other risks, including liability, for third-party products offered for sale on our platform that are controversial, counterfeit, pirated or stolen or that infringe the intellectual property rights of others. We may not be able to collect sufficient damages for these types of breaches from third-party sellers. Furthermore, a regulator may view us as having responsibility for regulatory compliance of the third-party products offered for sale on our platform. Although we have marketplace compliance controls in place and impose contractual terms on sellers to prohibit sales of non-compliant products, we may not be able prevent sellers from offering prohibited items for sale, enforce such terms, or fully protect against regulatory risk. Any of these events could have a material adverse impact on our business and results of operations and impede the execution of our eCommerce growth and enterprise strategy.

We rely extensively on information and financial systems to process transactions, summarize results and manage our business. Disruptions in our systems could harm our ability to conduct our operations.

Given the number of individual transactions we have each year, it is crucial that we maintain uninterrupted operation of our business-critical information systems. Our information systems are subject to damage or interruption from power outages, computer and telecommunications failures, computer viruses, worms, other malicious computer programs, denial-of-service attacks, security incidents and breaches from a variety of threat actors, including both cybercriminals and nation state-sponsored actors, catastrophic events such as fires, major or extended winter storms, tornadoes, earthquakes and hurricanes, usage errors by our associates or contractors, civil or political unrest or armed hostilities. The availability of our information systems and the integrity of data are essential to our business operations, including the processing of transactions, management of our associates, facilities, logistics, inventories, physical stores and clubs and our online operations. Our information systems are not fully redundant and our disaster recovery planning cannot account for all eventualities. If our systems are damaged, breached, attacked, interrupted or otherwise cease to function properly, we may have to make a significant investment to repair or replace them, and may experience loss or corruption of data as well as suffer interruptions in our business operations in the interim. Any interruption to the availability of our information systems or corruption of our data may have a material adverse effect on our business or results of operations. In addition, the cost of securing our systems against failure or attack is considerable, and increases in these costs, particularly in the wake of a breach or failure, could be significant.

In addition, we frequently update our information technology hardware, software, processes and systems. The risk of system disruption is increased when significant system changes are undertaken. If we fail to timely or successfully integrate and update our information systems and processes, we may fail to realize the cost savings or operational benefits anticipated to be derived from these initiatives. For example, during the first quarter of the fiscal year ended January 31, 2024, we initiated an upgrade to our existing financial system, including our general ledger and other applications. If we are unable to implement this upgrade as planned, the effectiveness of our internal control over financial reporting could be adversely affected; our ability to assess those controls adequately could be delayed; and our reputation, business, results of operations, financial condition and cash flows could be negatively impacted.

If the technology-based systems that give our customers the ability to shop with us online and enable us to deliver products and services do not function effectively, our operating results, as well as our ability to grow our omni-channel business globally, could be materially adversely affected.

Increasingly, customers are using computers, tablets and smart phones to shop with us and with our competitors and to do comparison shopping. We use social media, online advertising and email to interact with our customers and as a means to enhance their shopping experience. As a part of our omni-channel sales strategy, we offer various pickup, delivery and shipping programs including options where many products available for purchase online can be picked up by the customer or member at a local Walmart store or Sam's Club, which provides additional customer traffic at such stores and clubs. Omni-channel retailing is a rapidly evolving part of the retail industry and of our operations around the world, and we continue to make investments in supply chain automation to support our omni-channel strategy. We must anticipate and meet our customers' changing expectations while adjusting for technology investments and developments in our competitors' operations through focusing on the building and delivery of a seamless shopping experience across all channels by each operating segment. Moreover, some of the various technology systems and services on which we rely are provided and managed by third-party service providers. To the extent either our or such other third-party systems and services do not perform or function as anticipated, whether because of an inherent flaw in the technology, a faulty implementation or a cybersecurity incident, such failure can significantly interfere with our ability to meet our customers' changing expectations. Any disruption or failure on our part to provide attractive, user-friendly and secure digital platforms that offer a wide assortment of merchandise and services at competitive prices and with low cost and rapid delivery options and that continually meet the changing expectations of online shoppers and developments in online and digital platform merchandising and related technology in a cost-efficient manner could place us at a competitive disadvantage, result in the loss of eCommerce and other sales, harm our reputation with customers, have a material adverse impact on the growth of our eCommerce business globally and have a material adverse impact on our business and results of operations.

Any failure to maintain the privacy or security of the information relating to our company, customers, members, associates, business partners and vendors, whether as a result of cyberattacks on our information systems or otherwise, could damage our reputation, result in litigation or other legal actions against us, result in fines, penalties, and liability, cause us to incur substantial additional costs and materially adversely affect our business and operating results.

Like most retailers, we process in our information systems personal information and/or payment information about our customers and members, and we also process information concerning our associates and vendors. In addition, our health and wellness business operations, the Walmart Health locations, and third-party service providers who handle information on our behalf, store and maintain personal health information. Some of this information is stored digitally in connection with the digital platforms and technologies that we use to conduct and facilitate our various businesses. We utilize third-party service providers for a variety of reasons, including, without limitation, for digital storage technology, compute capacity, content delivery to customers and members, back-office support and other functions. Such providers may have access to information we hold about our customers, members, associates, business partners or vendors. In addition, our eCommerce operations depend upon the secure transmission of confidential information over public networks, including information permitting cashless payments.

Cyber threats are rapidly evolving and those threats and the means for disrupting or obtaining access to information systems or information stored in digital and other storage media are becoming increasingly sophisticated and frequent, and in some cases, they may lead to successful attacks. Unauthorized activities directed against information systems and devices, whether our own or those of our third-party service providers and vendors, have resulted in cybersecurity incidents, including malware, ransomware, denial of service attacks or phishing incidents. We expect that our information systems and those of our third-party service providers, vendors and suppliers will continue to experience such attempted attacks in the future, which could include disruptions to our supply chain system. Cyberattacks and threat actors can be sponsored by particular nation-states, or be the work of sophisticated criminal organizations, insiders (including our associates or contractors) or third parties, each with a wide-range of motives and expertise. We and the businesses with which we interact have experienced and continue to experience incidents and threats to data and information systems. These incidents and threats have included and are likely to continue to include both random and targeted cyberattacks, computer viruses, phishing incidents, worms, bot attacks, ransomware or other destructive or disruptive software and attempts to misappropriate customer information, including credit card and payment information, and cause system failures and disruptions. The increased use of remote work infrastructure in recent years has also increased the possible attack surfaces to be exploited. Our logging capabilities, or the logging capabilities of third parties, are also not always complete or sufficiently detailed, affecting our ability to fully investigate and understand the scope of security events. As noted above, some of our information systems and those of our third-party service providers have experienced cybersecurity incidents or breaches and, although to date they have not had a material adverse effect on our operating results, there can be no assurance of a similar result in the future.

Our digital platforms, which are increasingly important to our business and continue to grow in complexity and scope, and the systems on which they run, including those applications and systems used in legacy operations and acquired eCommerce, technology or other businesses, are regularly subject to cyberattacks. Those attacks involve attempts to impede the operations of our system or gain unauthorized access to our eCommerce websites (including marketplace platforms) or mobile commerce applications to obtain and misuse customers' or members' information including personal information and/or payment information and related risks discussed in this Item 1A. Such attacks, if successful, may result in potential data misuse and/or loss and may create denials of service or otherwise disable, degrade or sabotage the information systems that enable or support one or more of our digital platforms or otherwise significantly disrupt our customers' and members' shopping experience, our supply chain integrity and continuity and our ability to efficiently operate our business. If we are unable to maintain the security of the information systems that enable or support our digital platforms and keep them operating within acceptable parameters, we could suffer loss of sales, reductions in transactions, reputational damage and deterioration of our competitive position and incur liability for any damage to customers, members or others whose personal or confidential information is unlawfully obtained and misused, any of which events could have a material adverse impact on our business and results of operations and impede the execution of our strategy for the growth of our business.

Associate error or malfeasance, faulty password management, social engineering or other vulnerabilities and irregularities may also result in a defeat of our security measures or those of our third-party service providers and a compromise or breach of our or their information systems. Moreover, the hardware, software or applications that comprise our information system and networked environment may have vulnerabilities or defects of design, coding, manufacture or operations that could be intentionally exploited or inadvertently used in a manner that could compromise information security. Given the age, size and complexity of these information systems and our networked environment, patches for certain vulnerabilities may not exist and, even where patches or other risk-mitigating activities are available, the deployment of patches or execution of risk-mitigating actions may not occur before an underlying vulnerability is exploited by threat actors or inadvertently results in the compromise of our information systems or data.

Any compromise of our information systems or of those of businesses with which we interact, which results in regulated data or confidential information being accessed, obtained, damaged, disclosed, destroyed, modified, lost or used by unauthorized persons could harm our reputation and expose us to regulatory actions (including, with respect to health information, liability under the Health Insurance Portability and Accountability Act of 1996, or "HIPAA"), customer attrition, remediation expenses and claims from customers, members, associates, vendors, financial institutions, payment card networks and other persons, any of which could materially and adversely affect our business operations, financial position and results of operations.

Because the techniques used to obtain unauthorized access, disable or degrade service, or sabotage systems or data change frequently and may not immediately produce signs of a compromise, we may be unable to anticipate these techniques or to implement adequate preventative measures, or detect the activities of a threat actor. Even if we detect a cybersecurity incident, the nature and extent of that cybersecurity incident may not be immediately clear. Based on the sophistication of the threat actors and the size and complexity of our information systems and networked environment, among other factors, an investigation into a cybersecurity incident could take a significant amount of time to complete. We may not understand or appreciate that what is detected and treated as multiple individual cybersecurity incidents or events may be associated with the coordinated actions of a single threat actor. In addition, while our investigation of a cybersecurity incident is ongoing, we may not know the full extent of the harm caused by a threat actor, and such harm may spread both internally and to certain customers, vendors, or other third parties. These factors may inhibit our ability to provide rapid, complete and reliable information about the cybersecurity incident to customers, counterparties and regulators, as well as the public. It may also not be clear how best to contain and remediate any harm caused by the cybersecurity incident, and certain errors or actions could be repeated or compounded before they are discovered and remediated. Any or all of these factors could further increase the costs and consequences of a cybersecurity incident on our business operations, financial position and results of operations.

To the extent that any cyberattack, ransomware or incursion in our or one of our third-party service provider's information systems results in the loss, damage, misappropriation or other compromise of information, we may be materially adversely affected by claims from customers, members, financial institutions, regulatory authorities, payment card networks and others.

Our compliance programs, information technology and enterprise risk management efforts cannot eliminate all systemic risk. Disruptions in our systems caused by associate error or malfeasance, security incidents, breaches or cyberattacks – including attacks on those parties we do business with (such as strategic partners, suppliers, banks or utility companies) – could harm our ability to conduct our operations, which may have a material effect on us, may result in losses that could have a material adverse effect on our financial position or results of operations, or may have a cascading effect that adversely impacts our partners, third-party service providers, customers, members, financial services firms and other third parties that we interact with on a regular basis.

Our reputation with our customers and members is important to the success of our enterprise strategy, which combines traditional retail, membership models, marketplaces, financial services, healthcare, and other customer and business services into a series of interconnected assets to make it seamless for customers to interact with us. Security-related events could be widely publicized and could materially adversely affect our reputation with our customers, members, associates, vendors and shareholders, could harm our competitive position particularly with respect to our eCommerce operations, and could result in a material reduction in our net sales in our eCommerce operations, as well as in our stores thereby materially adversely affecting our operations, net sales, growth rates, operating income, results of operations, financial position, cash flows and liquidity. Such events could also result in the release to the public of confidential information about our operations and financial position and performance and could result in litigation or other legal actions against us or the imposition of penalties, fines, fees or liabilities, which may not be covered by our insurance policies. Moreover, a security compromise or operationally impactful malware event, such as ransomware, could require us to devote significant management resources to address the problems created by the issue and to expend significant additional resources to upgrade further the security measures we employ to guard personal and confidential information against cyberattacks and other attempts to access or otherwise compromise such information and could result in a disruption of our operations, particularly our digital operations.

We accept payments using a variety of methods, including cash, checks, credit and debit cards, electronic benefits transfer (EBT) cards, mobile payments and our private label credit cards and gift cards, and we may offer new payment options over time, which may have information security risk implications. As a retailer accepting debit and credit cards for payment, we are subject to various industry data protection standards and protocols, such as payment network security operating guidelines and the Payment Card Industry Data Security Standard. We cannot be certain that the security measures we or our third-party suppliers maintain are able to detect, prevent or contain cyberattacks, cyberterrorism, security incidents, breaches or other compromises from malware, ransomware or other threats that are known or may be developed in the future. In certain circumstances, our contracts with payment card processors and payment card networks (such as Visa, Mastercard, American Express and Discover) generally require us to adhere to payment card network rules which could make us liable to payment card issuers and others if information in connection with payment cards and payment card transactions that we process is compromised, which liabilities could be substantial.

Additionally, through various financial service partners and our ONE fintech venture, we offer various services such as money transfers, digital payment platforms, bill payment, money orders, check cashing, prepaid access, co-branded credits cards, installment lending and earned wage access. These products and services require us to comply with legal and regulatory requirements, including those related to privacy, information security, anti-money laundering and sanctions regimes and consumer protection, under both U.S. state and federal laws and regulations, as well as those of certain other countries. Failure to comply with these laws and regulations could result in fines, sanctions, penalties and harm to our reputation.

We also have compliance obligations associated with privacy laws enacted to protect and regulate the collection, use, retention, disclosure and transfer of personal information, which include liability for security and privacy breaches. Among other obligations, breaches may trigger obligations under U.S. federal and state laws and those in certain other countries to notify affected individuals, government agencies and the media. Consequently, cybersecurity incidents that result in a data breach could subject us to fines, sanctions and other legal liability and harm our reputation.

Changes in third-party reimbursements and contracts, type or scope of offerings of our health and wellness business or the Walmart Health business could adversely affect our overall results of operations, cash flows and liquidity.

We have retail pharmacy operations in our Walmart U.S. and Sam's Club segments across the U.S. and in various of our international markets such as Canada and Mexico. We also provide management services to Walmart Health centers that offer medical, dental, behavioral health and other health services in a number of states, as well as a national telehealth service provider. In addition, our 10-year collaboration with UnitedHealth Group includes agreements for Walmart Health to provide value-based care to patients in certain areas of the U.S., among other initiatives.

A large majority of our retail pharmacy net sales are generated by filling prescriptions for which we receive payment through established contractual relationships with third-party payers and payment administrators, such as private insurers, governmental agencies and pharmacy benefit managers ("PBMs"). Our retail pharmacy operations are subject to numerous risks, including: reductions in the third-party reimbursement rates for drugs; changes in our payer mix (i.e., shifts in the relative distribution of our pharmacy customers across drug insurance plans and programs toward plans and programs with less favorable reimbursement terms); changes in third-party payer drug formularies (i.e., the schedule of prescription drugs approved for reimbursement or which otherwise receive preferential coverage treatment); growth in, and our participation in or exclusion from, pharmacy payer network arrangements, including exclusive and preferred pharmacy network arrangements operated by PBMs and/or any insurance plan or program; increases in the prices we pay for brand name and generic prescription drugs we sell; increases in the administrative burdens associated with seeking third-party reimbursement; changes in the frequency with which new brand name pharmaceuticals become available to consumers; introduction of lower cost generic drugs as substitutes for existing brand name drugs for which there was no prior generic drug competition; changes in drug mix (i.e., the relative distribution of drugs customers purchase at our pharmacies between brands and generics); changes in the health insurance market generally; changes in the scope of or the elimination of Medicare Part D or Medicaid drug programs; increased competition from other retail pharmacy operations including competitors offering online retail pharmacy options and/or home delivery options; further consolidation and strategic alliances among third-party payers, PBMs or purchasers of drugs; overall economic conditions and the ability of our pharmacy customers to pay for drugs prescribed for them to the extent the costs are not reimbursed by a third-party; failure to meet any performance or incentive thresholds to which our level of third-party reimbursement may be subject; changes in laws or regulations or the practices of third-party payers and PBMs related to the use of third-party financial assistance to assist our pharmacy customers with paying for drugs prescribed for them; and any additional changes in the state or federal regulatory environment for the retail pharmacy industry and the pharmaceutical industry, including as a result of health reform efforts, and other changes to or novel interpretations of existing state or federal laws, rules and regulations that affect our retail pharmacy business.

If the supply of certain pharmaceuticals provided by one or more of our vendors were to be disrupted for any reason, our pharmacy operations could be severely affected until at least such time as we could obtain a new supplier for such pharmaceuticals. Any such disruption could cause reputational damage and result in a significant number of our pharmacy customers transferring their prescriptions to other pharmacies.

Walmart Health clinical operations are also subject to numerous risks, including but not limited to: reductions in the third-party reimbursement rates for services; changes in our payer mix; changes in the health insurance market generally; our inability to retain and negotiate favorable contracts with private third-party payers, including managed care plans; competition for patients from other healthcare providers, including those that offer telehealth services; changes to healthcare provider utilization practices and treatment methodologies; trends toward value-based purchasing and price transparency; overall economic conditions and the ability of patients to pay for services; staffing challenges, including retention of a sufficient number and quality of healthcare professionals; compliance with the complex and extensive laws and regulations governing the healthcare industry; changes in laws and regulations, including as a result of health reform efforts; and healthcare technology initiatives, including those related to patient data and interoperability; and public health conditions.

One or a combination of the factors above may adversely affect the volumes of brand name and generic pharmaceuticals we sell, our cost of sales associated with our retail pharmacy operations, and the net sales and gross margin of those operations or result in the loss of cross-store or cross-club selling opportunities. In addition, these and other factors may adversely affect the type, volume and mix of services we provide, the reimbursement we receive for health and wellness services rendered, and the scope and pace of expansion of Walmart Health and related offerings. Any of these developments could, in turn, adversely affect our overall net sales, other results of operations, cash flows and liquidity.

Our failure to attract and retain qualified associates, increases in wage and benefit costs, changes in laws and other labor issues could materially adversely affect our financial performance.

Our ability to continue to conduct and expand our operations depends on our ability to attract and retain a large and growing number of qualified associates globally. Our ability to meet our labor needs, including our ability to find qualified personnel to fill positions that become vacant at our existing stores, clubs, distribution and fulfillment centers and corporate offices, while controlling our associate wage and related labor costs, is generally subject to numerous external factors, including the availability of a sufficient number of qualified persons in the work force of the markets in which we operate, unemployment levels within those markets, prevailing wage rates, changing demographics, health and other insurance costs and adoption of new or revised employment and labor laws and regulations. Additionally, our ability to successfully execute organizational changes, including our enterprise strategy and management transitions within our senior leadership, and to effectively motivate and retain associates are critical to our business success. We compete for talent with other retail and non-retail businesses, including, for example, technology, health and wellness and fintech businesses, and invest significant resources in training and motivating our associates. Increased competition among potential employers at all levels, including senior management and executive levels, could result in increased associate costs or make it more difficult to recruit and retain associates. If we are unable to locate, attract or retain qualified personnel, or manage leadership transition successfully, the quality of service we provide to our customers may decrease and our financial performance may be adversely affected.

In addition, if our costs of labor or related costs increase for other reasons or if new, revised or novel interpretations of existing labor laws, rules or regulations or healthcare laws, including those related to worker classification, are adopted or implemented that further increase our labor costs, our financial performance could be materially adversely affected.

Financial Risks

Failure to meet market expectations for our financial performance could adversely affect the market price and volatility of our stock.

We believe that the price of our stock generally reflects high market expectations for our future operating results. Any failure to meet or delay in meeting these expectations, including our consolidated net sales, consolidated operating income, growth rates, eCommerce growth rates, advertising and other higher-margin initiatives (which is expected to help drive our operating income growth at a rate faster than net sales over the long term), capital expenditures, comparable store and club sales growth rates or earnings and adjusted earnings per share could cause the market price of our stock to decline, as could changes in our dividend or stock repurchase programs or policies, changes in our effective tax rates, changes in our financial estimates and recommendations by securities analysts or, failure of our performance to compare favorably to that of other retailers may have a negative effect on the price of our stock.

Fluctuations in foreign exchange rates may materially adversely affect our financial performance and our reported results of operations.

Our operations in countries other than the U.S. are conducted primarily in the local currencies of those countries. Our Consolidated Financial Statements are denominated in U.S. dollars, and to prepare those financial statements we must translate the amounts of the assets, liabilities, net sales, other revenues and expenses of our operations outside of the U.S. from local currencies into U.S. dollars using exchange rates for the current period. In recent years, fluctuations in currency exchange rates that were unfavorable have had adverse effects on our reported results of operations.

As a result of such translations, fluctuations in currency exchange rates from period-to-period that are unfavorable to us may also result in our Consolidated Financial Statements reflecting significant adverse period-over-period changes in our financial performance or reflecting a period-over-period improvement in our financial performance that is not as robust as it would be without such fluctuations in the currency exchange rates. Such unfavorable currency exchange rate fluctuations will adversely affect the reported performance of our Walmart International operating segment and have a corresponding adverse effect on our reported consolidated results of operations.

We may pay for products we purchase for sale in our stores, clubs and eCommerce platforms around the world with a currency other than the local currency of the country in which the goods will be sold. When we must acquire the currency to pay for such products and the exchange rates for the payment currency fluctuate in a manner unfavorable to us, our cost of sales may increase and we may be unable or unwilling to change the prices at which we sell those goods to address that increase in our costs, with a corresponding adverse effect on our gross profit. Consequently, unfavorable fluctuations in currency exchange rates have adversely affected, and may continue to adversely affect, our results of operations.

Legal, Tax, Regulatory, Compliance, Reputational and Other Risks

Our international operations subject us to legislative, judicial, accounting, legal, regulatory, tax, political and economic risks and conditions specific to the countries or regions in which we operate, which could materially adversely affect our business or financial performance.

In addition to our U.S. operations, we operate retail and eCommerce businesses in Africa, Canada, Central America, Chile, China, India and Mexico.

During fiscal 2024, our Walmart International operations generated approximately 18% of our consolidated net sales. Walmart International's operations in various countries also source goods and services from other countries. Our future operating results in these countries could be negatively affected by a variety of factors, most of which are beyond our control. These factors include political conditions, including political instability, local and global economic conditions, legal and regulatory constraints, such as regulation of product and service offerings including regulatory restrictions (such as foreign ownership restrictions) on eCommerce and retail operations in international markets, such as India, restrictive governmental actions (such as trade protection measures or nationalization), antitrust and competition law regulatory matters (such as those underway in Canada, Mexico and India (relating to our Flipkart subsidiary)), local product safety and environmental laws, tax regulations, local labor laws, anti-money laundering laws and regulations, trade policies, foreign exchange or currency regulations, laws and regulations regarding consumer and data protection, and other matters in any of the countries or regions in which we operate, now or in the future.

Changing our operations in accordance with new or changed restrictions on international trade or newly imposed sanctions can be expensive, time-consuming and disruptive to our operations, and such restrictions can be announced with little or no advance notice and we may not be able to effectively mitigate all adverse impacts from such measures. In addition, tensions between nation-state governments and conflicts of laws may lead to challenges for our operations. If disputes and conflicts further escalate in the future, actions by governments in response could be significantly more severe and restrictive and could adversely affect our business or financial performance and our reputation. Political uncertainty surrounding trade and other international disputes could also have a negative effect on consumer confidence and spending, which could also adversely affect our business or financial performance and our reputation. The economies of some of the countries in which we have operations have in the past suffered from high rates of inflation and currency devaluations, which, if they recur, could adversely affect our financial performance. Other factors which may impact our international operations include foreign trade, monetary and fiscal policies of the U.S. and of other countries, laws, regulations and other activities of foreign governments, agencies and similar organizations, and risks associated with having numerous facilities located in countries that have historically been less stable than the U.S. Additional risks inherent in our international operations generally include, among others, the costs and difficulties of managing international operations, adverse tax consequences and greater difficulty in enforcing intellectual property rights in countries other than the U.S. The various risks inherent in doing business in the U.S. generally also exist when doing business outside of the U.S., and may be exaggerated by the difficulty of doing business in numerous sovereign jurisdictions due to differences in culture, geopolitical tensions or events, laws and regulations.

In foreign countries in which we have operations, a risk exists that our associates, contractors or agents could, in contravention of our policies, engage in business practices prohibited by U.S. laws and regulations applicable to us, such as the Foreign Corrupt Practices Act or U.S. sanctions laws and regulations or the laws and regulations of other countries. We maintain global policies which appropriately regulate such business practices and have in place global compliance programs designed to ensure compliance with these laws and regulations. Nevertheless, we remain subject to the risk that one or more of our associates, contractors or agents, including those based in or from countries where practices that violate such U.S. laws and regulations or the laws and regulations of other countries may be customary, will engage in business practices that are appropriately regulated by our policies, circumvent our compliance programs and, by doing so, violate such laws and regulations. Any such violations, even if prohibited by our internal policies, could subject us to fines and penalties and adversely affect our business or financial performance and our reputation.

Changes in tax and trade laws, regulations and interpretations could materially adversely affect our financial performance.

In fiscal 2024, our Walmart U.S. and Sam's Club operating segments generated approximately 82% of our consolidated net sales. Significant changes in tax and trade policies, including tariffs and government regulations affecting trade between the U.S. and other countries where we source many of the products we sell in our stores and clubs could have an adverse effect on our business and financial performance. A significant portion of the general merchandise we sell in our U.S. stores and clubs is manufactured in other countries. Any such actions including the imposition of further tariffs on imports could increase the cost to us of such merchandise (whether imported directly or indirectly) and cause increases in the prices at which we sell such merchandise to our customers, which could materially adversely affect the financial performance of our U.S. and international operations as well as our business.

We are subject to income taxes, other taxes and tax collection and reporting obligations in both the U.S. and the foreign jurisdictions in which we currently operate or have historically operated. The determination of our worldwide provision for income taxes and current and deferred tax assets and liabilities requires judgment and estimation. Our taxes could be materially adversely affected by earnings being lower than anticipated in jurisdictions that have lower statutory tax rates and higher than anticipated in jurisdictions that have higher statutory tax rates, by changes in the valuation of our deferred tax assets and liabilities, or by changes in worldwide tax laws, tax rates, regulations or accounting principles and the interpretations of those rules. In addition, we also may not have sufficient notice to enable us to build systems and adopt processes to properly comply with new reporting or collection obligations by the effective date of those obligations.

We are also exposed to future tax legislation, as well as the issuance of future regulations and changes in administrative interpretations of existing tax laws, and changes in transfer pricing arrangements with our subsidiaries, any of which can impact our or our subsidiaries current and future years' tax provision. The effect of such changes in tax law, changes in administrative interpretations of existing tax laws or changes in transfer pricing arrangements could also have a material effect on our business, financial position and results of operations. In the U.S., the Tax Cuts and Jobs Act of 2017 (the "Tax Act") significantly changed federal income tax laws that affect U.S. corporations. As further guidance is issued by the U.S. Treasury Department, the Internal Revenue Service, and other standard-setting bodies, any resulting changes in our estimates will be treated in accordance with the relevant accounting guidance. Compliance with the Tax Act and any other new tax rules, regulations, guidance and interpretations, including collecting information not regularly produced by us or unexpected changes in our estimates, may require us to incur additional costs and could affect our results of operations.

In addition, legislatures and taxing authorities in many jurisdictions in which we operate may enact changes to or seek to enforce novel interpretations of their tax rules. These changes could include modifications that have temporary effect and more permanent changes. For example, the Organization for Economic Cooperation and Development (the "OECD"), the European Union and other countries (including countries in which we operate) have committed to enacting substantial changes to numerous long-standing tax principles impacting how large multinational enterprises are taxed. In particular, the OECD's Pillar Two initiative introduces a 15% global minimum tax applied on a country-by-country basis, which became effective in many jurisdictions in which we operate starting January 1, 2024. The impact of these potential new rules as well as any other changes in domestic and international tax rules and regulations could have a material effect on our effective tax rate.

Furthermore, we are subject to regular review and audit by both domestic and foreign tax authorities as well as subject to the prospective and retrospective effects of changing tax regulations, legislation and interpretations. Although we believe our tax estimates are reasonable, the ultimate tax outcome may materially differ from the tax amounts recorded in our Consolidated Financial Statements and may materially affect our income tax provision, net income or cash flows in the period or periods for which such determination and settlement is made.

Changes in and/or failure to comply with other laws, regulations and interpretations of such laws and regulations specific to the businesses and jurisdictions in which we operate could materially adversely affect our reputation, market position or our business and financial performance.

We operate in complex regulated environments in the U.S. and in other countries in which we operate and could be materially adversely affected by changes to existing legal requirements including the related interpretations and enforcement practices, new legal requirements and/or any failure to comply with applicable regulations. In addition, the degree of regulatory, political, and media scrutiny we face increases the likelihood that our efforts to adhere our practices and procedures to comply with these laws and legal requirements may be subject to frequent or increasing challenges.

Our health and wellness operations in the U.S. and the operations of the Walmart Health locations are subject to numerous federal, state and local laws and regulations including, but not limited to, those related to: licensing, reimbursement arrangements and other requirements and restrictions; registration and regulation of pharmacies; dispensing and sale of controlled substances and products containing pseudoephedrine; governmental (including Medicare and Medicaid) and commercial reimbursement; data privacy and security and the sharing and interoperability of data, including obligations and restrictions related to health information (such as those imposed under HIPAA); billing and coding for healthcare services and properly handling overpayments; debt collection; necessity and adequacy of healthcare services; relationships with referral sources and referral recipients and other fraud and abuse issues, such as those addressed by anti-kickback and false claims laws and patient inducement regulations; qualification of healthcare practitioners; quality and standards of medical services and equipment; and the practice of the professions of pharmacy, medical, dental and behavioral healthcare services, including limitations on the corporate practice of medicine in certain states.

Health-related legislation at the federal and state level may have an adverse effect on our business or require us to modify certain aspects of our operations. For example, in the U.S., the Drug Enforcement Administration ("DEA") and various other regulatory authorities regulate the purchase, distribution, maintenance and dispensing of pharmaceuticals and controlled substances. We are required to hold valid DEA and state-level licenses, meet various security and operating standards and comply with the federal and various state-controlled substance acts and related regulations governing the sale, dispensing, disposal and holding of controlled substances. The DEA, the U.S. Food and Drug Administration and state regulatory authorities have broad enforcement powers, including the ability to seize or recall products and impose significant criminal, civil and administrative sanctions for violations of these laws and regulations. In addition, there has been recent heightened governmental and public scrutiny of pharmaceutical product pricing, which has resulted in federal and state legislation and regulations, executive orders and other initiatives and proposals designed to increase transparency in pharmaceutical product pricing and reform government program reimbursement methodologies (for example, the Inflation Reduction Act, which includes, among other matters, policies designed to impact drug prices and reduce drug spending by the federal government). Other health reform efforts at the federal and state levels may also impact our business or require us to modify certain aspects of our operations. We may not be able to predict the nature or success of reform initiatives, and the resulting uncertainties may have an adverse effect on our business.

We are also governed by foreign, national and state laws and regulations of general applicability, including laws and regulations related to competition and antitrust matters; protection of the environment and health and safety matters, including exposure to, and the management and disposal of, hazardous substances; food and drug safety, including drug supply chain security requirements; trade, consumer protection, and safety, including the availability, sale, price label accuracy, advertisement and promotion of products we sell and the financial services we offer (including through our digital channels, stores and clubs, as well as our ONE fintech venture); anti-money laundering prohibitions; consumer financial protection laws; economic, trade and other sanctions matters; licensure, certification and enrollment with government programs; data privacy and cybersecurity and the sharing and interoperability of data; working conditions, workplace health and safety, equal employment opportunity, worker classification, employee benefit and other labor and employment matters; and health and wellness related regulations for our pharmacy operations outside of the U.S. In addition, certain financial services we offer or make available are subject to U.S. and international legal and regulatory requirements, including those intended to help detect and prevent money laundering, fraud and other illicit activity, as well as consumer financial protections and sanctions laws. Failure to meet these requirements could affect the profitability of our business activities; limit our ability to pursue business opportunities or conduct business in certain jurisdictions; require changes to business practices or governance or alter our relationships with our customers, partners and other third parties, including our ability to continue certain relationships in Mexico, India or other international jurisdictions; result in increased costs related to regulatory oversight and compliance, litigation-related settlements, judgments or expenses, restitution to customers or the imposition of fines or monetary penalties. Increased U.S. regulation of non-bank financial institutions may also result in additional requirements and scrutiny of certain financial services we offer.

Increasing governmental and societal attention to ESG matters, including expanding mandatory and voluntary reporting diligence, and disclosure topics such as climate change, sustainability (including with respect to our supply chain), natural resources, waste reduction, energy, human capital and risk oversight could expand the nature, scope and complexity of matters that we are required to control, assess and report.

Data privacy and protection laws or customer expectations relating to the collection, use, retention, disclosure, transfer and processing of personal information continue to undergo a rapid transformation in the U.S. and in non-U.S. jurisdictions. Recently enacted state laws, such as the California Consumer Privacy Act ("CCPA"), in a number of states that have become effective, or will soon be effective, have created a substantially more complex regulatory regime associated with data-handling practices. Moreover, other laws and regulations related to data-handling and privacy that apply to our business, such as the Illinois Biometric Information Privacy Act, the European Union's General Data Protection Regulation ("GDPR"), the United Kingdom's General Data Protection Regulation (which implements the GDPR into U.K. law), China's Personal Information Protection Act ("PIPL"), and similar legislation in Quebec Canada further increases the compliance obligations of our business. Certain of these laws have required us to modify our data processing practices and policies and to incur substantial costs and expenses to comply, which we anticipate will continue in the future. These and other privacy and cybersecurity laws may carry significant potential damages and civil penalties for noncompliance. For example, in the case of non-compliance with a material provision of the GDPR (such as non-adherence to the core principles of processing personal data), regulators have the authority to levy a fine in an amount that is up to the greater of €20 million or 4% of global annual turnover (i.e., revenue) in the prior year. These administrative fines are discretionary and based, in each case, on a multi-factored approach. Further, PIPL took effect in China in November 2021. PIPL raises the protection requirements for processing personal information and requires government approval to either allow the access of personal information in China by someone outside of China or conduct personal data transfers outside of China. Fines for PIPL violations range from approximately RMB 50 million to up to 5% of the infringing company's previous year's revenues generated from within China. We have made changes, and we may in the future make additional adjustments to our business practices, to comply with the personal information protection laws and regulations in China as they evolve. Residents in jurisdictions with comprehensive privacy laws have rights to access, correct and require deletion of their personal information, opt out of certain personal information sharing and receive detailed information about how their personal information is used and may have a private right of action for data breaches. Furthermore, our marketing and customer engagement activities are subject to communications privacy laws such as the Telephone Consumer Protection Act. We may be subjected to penalties and other consequences for noncompliance, including changing some portions of our business. Even an unsuccessful challenge by customer or regulatory authorities of our activities could result in adverse publicity, impact our reputation and could require a costly response from and defense by us.

The impact of new laws, regulations and policies and the related interpretations, as well as changes in enforcement practices or regulatory scrutiny as to existing laws and regulations (including, but not limited to, in the U.S., shifting enforcement priorities for existing antitrust, competition and pricing laws, as well as proposed new rules and regulations) generally cannot be predicted, and changes in applicable laws, regulations and policies and the related interpretations and enforcement practices of existing laws and regulations may require extensive system and operational changes, be difficult to implement, increase our operating costs, require significant capital expenditures, or adversely impact the cost or attractiveness of the products or services we offer, or result in adverse publicity and harm our reputation. If we fail to predict or respond adequately to changes, including by implementing strategic and operational initiatives, or do not respond as effectively as our competitors, our business, operations and financial performance may be adversely affected.

In addition, we may face audits or investigations by one or more government agencies relating to our compliance with applicable laws and regulations. The regulatory, political and media scrutiny we face, which may continue, amplifies these risks. To the extent a regulator or court disagrees with our interpretation of these laws and determines that our practices are not in compliance with applicable laws and regulations, we could be subject to civil and criminal penalties that could adversely affect the continued operation of our businesses, including: suspension of payments from government programs; loss of required licenses and certifications; loss of authorizations to participate in or exclusion from government programs, including the Medicare and Medicaid programs in the U.S.; termination from contractual relationships, including those with our drug suppliers and third-party payers; and significant fines or monetary damages. Failure to comply with applicable legal or regulatory requirements in the U.S. or in any of the countries in which we operate could result in significant legal and financial exposure, damage to our reputation and have a material adverse effect on our business operations, financial position and results of operations.

We are subject to risks related to litigation and other legal proceedings that may materially adversely affect our results of operations, financial position and liquidity.

We operate in a highly regulated and litigious environment. We are involved in legal proceedings, including litigation, arbitration and other claims, and investigations, inspections, audits, claims, inquiries and similar actions by pharmacy, healthcare, tax, environmental and other governmental authorities. We may also have indemnification obligations for legal commitments of certain businesses we have divested. Legal proceedings, in general, and securities, derivative action and class action and multi-district litigation, in particular, can be expensive and disruptive. Some of these suits may purport or may be determined to be class actions and/or involve parties seeking large and/or indeterminate amounts, including punitive or exemplary damages, and may remain unresolved for several years. For example, we are currently a defendant in a number of cases containing class or collective-action allegations, or both, in which the plaintiffs have brought claims under federal and state wage and hour laws, as well as a number of cases containing class-action allegations in which the plaintiffs have brought claims under federal and state consumer laws.

We have been responding to subpoenas, information requests and investigations from governmental entities related to nationwide controlled substance dispensing and distribution practices involving opioids. We are also a defendant in numerous litigation proceedings related to opioids, including the consolidated multidistrict litigation entitled In re National Prescription Opiate Litigation (MDL No. 2804) currently pending in the U.S. District Court for the Northern District of Ohio and a lawsuit filed against us by the United States Department of Justice in the District of Delaware in 2020. Similar cases that name us also have been filed in state courts by state, local and tribal governments, healthcare providers and other plaintiffs. Plaintiffs in these cases are seeking compensatory and punitive damages, as well as injunctive relief including abatement. We have entered into a settlement framework to resolve certain of these matters and accrued a liability for approximately $3.3 billion, almost all of which was paid in fiscal 2024. We cannot predict the ultimate number of opioids-related claims that may be filed or their outcomes and cannot reasonably estimate any loss or range of loss that may arise from opioids-related matters.

In addition, in July 2021, the Directorate of Enforcement in India issued a show cause notice to Flipkart and other parties requesting the recipients show cause as to why further proceedings under India's Foreign Direct Investment rules and regulations should not be initiated against them based on alleged violations that related to a period prior to our acquisition of a majority stake in Flipkart in 2018. Also, in October 2023, the main Mexican operating subsidiary of Wal-Mart de México was notified of the initiation of a quasi-judicial administrative process against it for alleged relative monopolistic practices in connection with the supply and wholesale distribution of certain consumer goods, retail marketing practices of such consumer goods and related services. Because this process is at an early stage, we cannot provide any assurance as to the scope and outcome of this matter, and cannot reasonably estimate any loss or range of loss that may arise from this matter.

We can provide no assurance as to the scope or outcome of any proceeding that might result from the notice, the amount of proceeds we may receive in indemnification, and can provide no assurance as to whether there will be a material adverse effect to our business or Consolidated Financial Statements. We are also a defendant in litigation with the Federal Trade Commission regarding our money transfer agent services and is also cooperating with and responding to subpoenas issued by the U.S. Attorney's Office for the Middle District of Pennsylvania on behalf of the U.S. Department of Justice regarding our consumer fraud prevention program and anti-money laundering compliance related to our money transfer services, where we are an agent. We are unable to predict the outcome of the litigation or investigations or any other related actions by governmental entities regarding these matters and can provide no assurance as to the scope and outcome of these matters and whether our business, financial position, results of operations or cash flows will not be materially adversely affected. We discuss in more detail these cases and other litigation to which we are party below under the caption "Item 3. Legal Proceedings" and in Note 10 in the "Notes to our Consolidated Financial Statements," which are part of this Annual Report on Form 10-K.

Our amended and restated bylaws designate the Court of Chancery of the State of Delaware as the sole and exclusive forum for certain types of actions and proceedings that may be initiated by our shareholders, which could increase the costs for our shareholders to bring claims, discourage our shareholders from bringing claims, or limit our shareholders' ability to obtain a favorable judicial forum for disputes with us or our directors, officers, associates or shareholders in such capacity.

Our bylaws provide that, unless we consent in writing to the selection of an alternative forum, the Court of Chancery of the State of Delaware will, to the fullest extent permitted by law, be the sole and exclusive forum for claims, including derivative claims that are based upon a violation of a duty by a current or former director, officer, associate or shareholder in such capacity or as to which the Delaware General Corporation Law confers jurisdiction upon the Court of Chancery. The exclusive forum provision may increase the costs for a shareholder to bring a claim or limit a shareholder's ability to bring a claim in a judicial forum that the shareholder finds favorable for disputes with us or our directors, officers, associates or shareholders in such capacity, which may discourage such lawsuits against us and such persons. Alternatively, if a court were to find these provisions of our bylaws inapplicable to, or unenforceable in respect of, the claims as to which they are intended to apply, then we may incur additional costs associated with resolving such matters in other jurisdictions, which could adversely affect our business, financial position or results of operations. While the exclusive forum provision applies to state and federal law claims, our shareholders will not be deemed to have waived our compliance with, and the exclusive forum provision will not preclude or contract the scope of exclusive federal or concurrent jurisdiction for actions brought under, the federal securities laws, including the Exchange Act, as amended, or the Securities Act of 1933, as amended, and the rules and regulations promulgated thereunder.

Our reputation may be adversely affected if we are not able to satisfy varied stakeholder expectations with respect to our ESG goals.

We strive to deliver shared value through our business and our diverse stakeholders expect us to make significant progress in certain ESG priority issue areas. Stakeholder expectations regarding ESG matters continue to evolve and are not uniform. We have established, and may continue to establish, various goals and initiatives on these matters, including with respect to climate change initiatives. We cannot guarantee that we will achieve these goals and initiatives. Any failure, or perceived failure, by us to achieve these goals and initiatives or to otherwise meet evolving and varied stakeholder expectations could adversely affect our reputation. We periodically publish information about our ESG priorities, strategies and progress on our corporate website and update our ESG reporting from time to time. Achievement of these aspirations and goals is subject to risks and uncertainties, many of which are outside of our control, and it is possible that we may fail, or be perceived to have failed, in the achievement of our ESG goals or that certain of our customers, associates, shareholders, investors, suppliers, business partners, government agencies and non-governmental organizations might not be satisfied with our goals or our efforts toward achieving those goals. Certain challenges we face in the achievement of our ESG objectives are also captured within our ESG reporting, which is not incorporated by reference into and does not form any part of this Annual Report on Form 10-K. A failure or perceived failure to meet our goals could adversely affect public perception of our business, associate morale or customer or shareholder support.

ITEM 1B. UNRESOLVED STAFF COMMENTS

None.

ITEM 1C. CYBERSECURITY

Walmart seeks to build and maintain the trust of customers, associates, shareholders and other stakeholders with respect to our use of technology and data. Our digital trust commitments, in line with our Company's values of service, excellence, integrity and respect for the individual, provide a foundation for our approach to cybersecurity.

The Board of Directors, committees of the Board of Directors and management coordinate risk oversight and management responsibilities, and cybersecurity represents an important component of our overall approach to enterprise risk management. In general, we seek to address cybersecurity risks through a cross-functional approach focused on protecting business operations and preserving the confidentiality, integrity and availability of information by identifying, preventing and mitigating cybersecurity threats and effectively responding to cybersecurity incidents when they occur.

Board of Directors' oversight of risks from cybersecurity threats

Our Board of Directors, which has primary responsibility for overseeing risk management, has delegated risk management oversight responsibility for information systems, information security, data privacy and cybersecurity to the Audit Committee. Several of our Board members, including certain members of our Audit Committee, have backgrounds or professional experience in risk management, digital platforms, information technology or cybersecurity.

The Audit Committee receives periodic updates from our Chief Information Security Officer ("CISO"), Chief Technology Officer ("CTO") and other members of management on risks related to information systems, information security, data privacy and cybersecurity. Specific topics may include updates to our company's approach to cybersecurity risk management; recent developments; key initiatives; the threat landscape; trends; and the results of certain assessments and testing. The Board of Directors receives regular reports from the Audit Committee chair on these and other risk-related matters as deemed necessary. Our CISO or other members of management provide information to the Audit Committee pursuant to risk-based escalation protocols for cybersecurity incidents that exceed established reporting thresholds.

Management's role in assessing and managing material risks from cybersecurity threats

Our CISO leads Walmart's Information Security organization and has responsibility for overseeing our Company's cybersecurity program. To operationalize our program, we deploy multidisciplinary teams, including cybersecurity personnel and professionals, to address cybersecurity threats and respond to cybersecurity incidents, including for those non-wholly owned subsidiaries whose systems have not been fully integrated into Walmart's networks. Through ongoing engagement with these teams and certain third-party service providers, our CISO monitors the prevention, detection, mitigation and remediation of cybersecurity threats and incidents and reports cybersecurity incidents that reach established thresholds to senior management and the Audit Committee, which are also analyzed for external reporting requirements.

Our CISO has been a Walmart associate for over 30 years, has served in various roles in information technology and information security at Walmart for almost 20 years, and has received industry-recognized information security certifications. Our CTO, to whom the CISO reports, has served as Walmart's CTO since 2019 and prior to that had experience managing technology and other risks at several other large public companies.

Risk Management and Strategy

Our cybersecurity program is informed by various industry frameworks including the National Institute of Standards and Technology Cybersecurity Framework for Improving Critical Infrastructure Cybersecurity (NIST-CSF Version 1.1), which are reflected in our related policies, standards, processes and practices. We may implement changes to our cybersecurity program when deemed necessary based on updates to industry standards among other things. We have multiple layers of security designed to detect and block cybersecurity events, as well as dedicated teams of cybersecurity personnel and professionals, which assist our CISO in helping to assess, identify, monitor, detect and manage cybersecurity risks, threats, vulnerabilities and incidents. We collaborate with public and private entities and industry groups and engage third-party service providers to expand the capabilities and capacity of our cybersecurity program when deemed necessary. Certain key components of our cybersecurity program include the following:

Protecting our technology and information systems: When we implement significant changes to our technologies or information systems, we conduct risk-based security and privacy impact assessments and deploy technical safeguards that are designed to reasonably protect our technology and information systems from cybersecurity threats. We actively monitor and proactively research potential cybersecurity threats to our technologies and information systems. We use what we learn to evolve our security controls over time to mitigate risks posed by such threats.

Incident response and recovery planning: We maintain incident response and recovery plans that address our response to cybersecurity incidents, including incidents that we become aware of at third parties that support our operations. These plans guide how we evaluate and assign incident severity levels and reporting thresholds; escalate and engage incident response teams; and manage and mitigate the related risks.

Third-party risk management: We maintain a risk-based approach to identifying and managing cybersecurity threats presented to Walmart by third-party systems that support our operations, as well as third-party users of our data and systems, including vendors, service providers and subcontractors.

Training and awareness: We provide recurring information security training (which includes cybersecurity training) to our associates and certain third parties based on access, risk, roles, policies, standards and behaviors.

Assessments and testing: We engage in periodic assessment and testing of our policies, standards, processes and practices that are designed to address cybersecurity threats. These efforts include tabletop exercises, threat modeling, vulnerability testing and other exercises focused on evaluating the effectiveness of our cybersecurity measures and planning. We regularly engage third parties to assist with our assessments and testing. Where appropriate we adjust our cybersecurity policies, standards, processes and practices accordingly based on internal and external assessment and testing results.

Certain of Walmart's systems and those of our third-party service providers have experienced cybersecurity incidents and threats. Based on the information available as of the date of this Annual Report on Form 10-K, we are not aware of any risks from cybersecurity threats, including as a result of any cybersecurity incidents, which have materially affected us or are reasonably likely to materially affect us, including our business strategy, results of operations, or financial condition. Despite our security measures, however, there can be no assurance that we, or the third parties with which we interact, will not experience a cybersecurity incident in the future that will materially affect us. Additional information about cybersecurity risks we face is discussed in "Item 1A. Risk Factors," which should be read in conjunction with the information above.

ITEM 2. PROPERTIES

As of January 31, 2024[1], retail unit counts for Walmart U.S., Sam's Club and International are summarized as follows:

	Total	Square feet[2]
Walmart U.S.		
Supercenters	3,560	632,771
Discount Stores	360	37,816
Neighborhood Markets and other small formats	695	28,414
Walmart U.S. Total	4,615	699,001
Sam's Club	599	80,199
Domestic Total	5,214	779,200
International		
Retail	5,075	236,180
Wholesale	327	37,685
International Total	5,402	273,865
Total Company	10,616	1,053,065

(1) Walmart International unit counts, with the exception of Canada, are as of December 31, 2023, to correspond with the balance sheet date of the related geographic market. Canada unit counts are as of January 31, 2024.
(2) Square feet reported in thousands.

Owned and Leased Properties

The following table provides further details of our retail units and distribution facilities, including return facilities and dedicated eCommerce fulfillment centers, as of January 31, 2024[1]:

	Owned	Leased[2]	Total
Retail Units			
Walmart U.S. retail units	4,041	574	4,615
Sam's Club retail units	512	87	599
International retail units	1,469	3,933	5,402
Total retail units	6,022	4,594	10,616
Distribution Facilities			
Walmart U.S. distribution facilities	112	50	162
Sam's Club distribution facilities	10	20	30
International distribution facilities	22	154	176
Total distribution facilities	144	224	368

(1) Walmart International properties, with the exception of Canada, are as of December 31, 2023, to correspond with the balance sheet date of the related geographic market. Canada unit counts are as of January 31, 2024.
(2) Also includes U.S. and international distribution facilities which are third-party owned and operated.

We own office facilities in Bentonville, Arkansas, that serve as our principal office and own and lease office facilities throughout the U.S. and internationally for operations as well as for field and market management. The land on which our stores are located is either owned or leased by the Company. We use independent contractors to construct our buildings. All store leases provide for annual rentals, some of which escalate during the original lease or provide for additional rent based on sales volume. Substantially all of the Company's store and club leases have renewal options, some of which include rent escalation clauses. For further information on our distribution centers, see the caption "Distribution" provided for each of our segments under "Item 1. Business."

ITEM 3. LEGAL PROCEEDINGS

I. SUPPLEMENTAL INFORMATION: The Company is involved in legal proceedings arising in the normal course of its business, including litigation, arbitration and other claims, and investigations, inspections, subpoenas, audits, claims, inquiries and similar actions by governmental authorities. We discuss certain legal proceedings in Note 10 to our Consolidated Financial Statements included in "Item 8. Financial Statements and Supplementary Data," which is captioned "Contingencies," under the sub-caption "Legal Proceedings." We refer you to that discussion for important information concerning those legal proceedings, including the basis for such actions and, where known, the relief sought. We provide the following additional information concerning those legal proceedings, including the name of the lawsuit, the court in which the lawsuit is pending, and the date on which the petition commencing the lawsuit was filed.

Prescription Opiate Litigation: *In re National Prescription Opiate Litigation* (MDL No. 2804) (the "MDL"). The MDL is pending in the U.S. District Court for the Northern District of Ohio and includes over 340 cases as of March 4, 2024. The liability phase of a single, two-county trial in one of the MDL cases against a number of parties, including the Company, regarding opioid dispensing claims resulted in a jury verdict on November 23, 2021, finding in favor of the plaintiffs as to the liability of all defendants, including the Company. The abatement phase of the single, two-county trial resulted in a judgment on August 17, 2022, that ordered all three defendants, including the Company, to pay an aggregate amount of approximately $0.7 billion over fifteen years, on a joint and several liability basis, and granted the plaintiffs injunctive relief. The Company has filed an appeal with the Sixth Circuit Court of Appeals. The monetary aspect of the judgment is stayed pending appeal, and the injunctive portion of the judgment went into effect on February 20, 2023. On September 11, 2023, the Sixth Circuit Court of Appeals issued an order certifying certain questions in the appeal for review by the Supreme Court of Ohio. On November 29, 2023, the Supreme Court of Ohio accepted the request for certification, and the matter remains pending with that court. On October 25, 2023, the MDL designated four cases brought by third-party payers as bellwether cases to proceed through discovery. Additional bellwethers of cases brought by hospitals and other healthcare providers may be designated in the future. In addition, there are over 50 other cases pending in state and federal courts throughout the country against the Company as of March 4, 2024, as well as other cases in Canada against Wal-Mart Canada Corp. and certain other subsidiaries of the Company. The case citations and currently scheduled trial dates, where applicable, are listed on Exhibit 99.1 to this Annual Report on Form 10-K.

Opioid Settlement Framework: On November 15, 2022, the Company announced that it had agreed to a Settlement Framework to resolve substantially all opioids-related lawsuits filed against the Company by states, political subdivisions, and Native American tribes (other than the single, two-county trial on appeal to the Sixth Circuit Court of Appeals as described above), as described in more detail in Note 10 to the Consolidated Financial Statements. The Company now has settlement agreements with all 50 states, including four states that previously settled with the Company, as well as the District of Columbia, Puerto Rico and three other U.S. territories, that are intended to resolve substantially all opioids-related lawsuits brought by state and local governments against the Company. As described in more detail in Note 10 to the Consolidated Financial Statements, the Settlement Framework became effective on September 6, 2023, and as of January 31, 2024 substantially all of the original approximately $3.3 billion accrued liability for the Settlement Framework and other settlements have been paid.

DOJ Opioid Civil Litigation: A civil complaint pending in the U.S. District Court for the District of Delaware has been filed by the U.S. Department of Justice (the "DOJ") against the Company, in which the DOJ alleges violations of the Controlled Substances Act related to nationwide distribution and dispensing of opioids. U.S. v. Walmart Inc., et al., USDC, Dist. of DE, 12/22/20. The Company filed a motion to dismiss the DOJ complaint on February 22, 2021. After the parties had fully briefed the Company's motion to dismiss, the DOJ filed an amended complaint on October 7, 2022. On November 7, 2022, the Company filed a partial motion to dismiss the amended complaint. The Court held a hearing on the partial motion to dismiss on January 18, 2024, and ordered the DOJ to file an amended complaint. The DOJ filed that amended complaint on February 1, 2024, and Walmart filed a partial motion to dismiss that complaint on February 6, 2024. On March 11, 2024, the Court granted in-part Walmart's motion by dismissing the entirety of the DOJ's claims related to distribution and dismissing the DOJ's claims arising under one of the DOJ's two dispensing liability theories. The DOJ's claims arising under its other dispensing liability theory remain pending.

Opioids Related Securities Class Actions and Derivative Litigation: Three derivative complaints and two securities class actions drawing heavily on the allegations of the DOJ complaint have been filed in Delaware naming the Company and various current and former directors and certain current and former officers as defendants. The plaintiffs in the derivative suits (in which the Company is a nominal defendant) allege, among other things, that the defendants breached their fiduciary duties in connection with oversight of opioids dispensing and distribution and that the defendants violated Section 14(a) of the Exchange Act, and are liable for contribution under Section 10(b) of the Exchange Act in connection with the Company's disclosures about opioids. Two of the derivative suits have been filed in the U.S. District Court in Delaware and those suits have been stayed pending further developments in other opioids litigation matters. The other derivative suit has been filed in the Delaware Court of Chancery. The defendants in the derivative suit pending in the Delaware Court of Chancery moved to dismiss and/or to stay that case on December 21, 2021; the plaintiffs responded by filing an amended complaint on February 22, 2022. On April 20, 2022, the defendants moved to dismiss and/or stay proceedings on the amended complaint. In two orders issued on April 12 and 26, 2023, the Court of Chancery granted the defendants' motion to dismiss with respect to claims involving the Company's distribution practices and denied the remainder of the motion, including the Company's request to stay the litigation. On May 5, 2023, the Company's Board of Directors appointed an independent Special Litigation Committee (the "SLC") to investigate the allegations regarding certain current and former officers and directors named in the various proceedings regarding oversight with respect to opioids. The Board has authorized the SLC to retain independent legal counsel and such other advisors as the SLC deems appropriate in carrying out its duties. The derivative matter pending in the Delaware Court of Chancery is stayed until the SLC completes its investigation.

The securities class actions, alleging violations of Sections 10(b) and 20(a) of the Exchange Act regarding the Company's disclosures with respect to opioids, purport to be filed on behalf of a class of investors who acquired Walmart stock from March 30, 2016, through December 22, 2020. On May 11, 2021, the U.S. District Court in Delaware consolidated the class actions and appointed a lead plaintiff and lead counsel. The defendants moved to dismiss the consolidated securities class action on October 8, 2021. On October 14, 2022, plaintiffs filed an amended complaint, which revised the applicable putative class of investors to those who acquired Walmart stock from March 31, 2017, through December 22, 2020. On November 16, 2022, the Company moved to dismiss the amended complaint. That motion remains pending.

Derivative Lawsuits: *Abt v. Alvarez et al.*, USDC, Dist. of DE, 2/9/21; *Nguyen v. McMillon et al.,* USDC, Dist. of DE, 4/16/21: *Ontario Provincial Council of Carpenters' Pension Trust Fund et al. v. Walton et al.,* DE Court of Chancery, 9/27/21.

Securities Class Actions: *Stanton v. Walmart Inc. et al.*, USDC, Dist. of DE, 1/20/21 and *Martin v. Walmart Inc. et al.,* USDC, Dist. of DE, 3/5/21, consolidated into *In re Walmart Inc. Securities Litigation*, USDC, Dist. of DE, 5/11/21.

ASDA Equal Value Claims: Ms S Brierley & Others v. ASDA Stores Ltd (2406372/2008 & Others – Manchester Employment Tribunal); Abbas & Others v Asda Stores limited (KB-2022-003243); and Abusubih & Others v Asda Stores limited (KB-2022-003240).

Money Transfer Agent Services Litigation: Federal Trade Commission v. Walmart Inc. (CV-3372), USDC, N. Dist. Of Ill, 6/28/22.

Mexico Antitrust Matter: Comisión Federal de Competencia Económica of México, Investigative Authority v. Nueva Wal-Mart de México, S.de R.L. de C.V. (Docket IO-002-2020, consolidated with Docket DE-026-2020), Mexico, 10/6/23.

II. CERTAIN OTHER MATTERS:

Foreign Direct Investment Matters: In July 2021, the Directorate of Enforcement in India issued a show cause notice to Flipkart Private Limited and one of its subsidiaries ("Flipkart"), and to unrelated companies and individuals, including certain current and former shareholders and directors of Flipkart. The notice requests the recipients to show cause as to why further proceedings under India's Foreign Direct Investment rules and regulations (the "Rules") should not be initiated against them based on alleged violations during the period from 2009 to 2015, prior to the Company's acquisition of a majority stake in Flipkart in 2018. The notice is an initial stage of proceedings under the Rules which could, depending upon the conclusions at the end of the initial stage, lead to a hearing to consider the merits of the allegations described in the notice. If a hearing is initiated and if it is determined that violations of the Rules occurred, the regulatory authority has the authority to impose monetary and/or non-monetary relief. Flipkart has begun the process of responding to the notice and, if the matter progresses to a consideration of the merits of the allegations described in the notice is initiated, Flipkart intends to defend against the allegations vigorously. Due to the fact that this process is in an early stage, the Company is unable to predict whether the notice will lead to a hearing on the merits or, if it does, the final outcome of the resulting proceedings. While the Company does not currently believe that this matter will have a material adverse effect on its business, financial condition, results of operations or cash flows, the Company can provide no assurance as to the scope or outcome of any proceeding that might result from the notice, the amount of the proceeds the Company may receive in indemnification from individuals and entities that sold shares to the Company under the 2018 agreement pursuant to which the Company acquired its majority stake in Flipkart, and can provide no assurance as to whether there will be a material adverse effect to its business or its Consolidated Financial Statements.

III. ENVIRONMENTAL MATTERS: Item 103 of SEC Regulation S-K requires disclosure of certain environmental matters when a governmental authority is a party to the proceedings and such proceedings involve potential monetary sanctions that the Company reasonably believes will exceed an applied threshold not to exceed $1 million.

In December 2021, the Office of the Attorney General of the State of California filed suit against the Company, bringing enforcement claims regarding Walmart's management of waste consumer products at its California facilities that are alleged to be hazardous. The suit was filed in Superior Court of Alameda County, California, Case No. 21CV004367, People v. Walmart Inc. Trial is currently set for September 30, 2024. The Company believes it has strong defenses and is vigorously defending this litigation matter. While the Company cannot predict the ultimate outcome of this matter, the potential for penalties or settlement costs could exceed $1 million. Although the Company does not believe that this matter will have a material adverse effect on its business, financial position, results of operations or cash flows, the Company can provide no assurance as to the scope and outcome of this matter and no assurance as to whether there will be a material adverse effect to its business or its Consolidated Financial Statements.

In October 2023, the Company received a Finding of Violation from the U.S. Environmental Protection Agency (the "EPA") alleging violations of the Clean Air Act in connection with the Company's refrigeration leak detection and repair program at certain of its facilities. The Company is evaluating the findings and cooperating with the EPA in its investigation. The EPA may seek to impose monetary and non-monetary penalties for the alleged violations of the Clean Air Act. Due to the fact that this process is in an early stage, the Company is unable to predict the final outcome of this matter. Although the Company does not believe this matter will have a material adverse effect on its business, financial position, results of operations or cash flows, the Company can provide no assurance as to the scope or outcome of this matter and no assurance as to whether there will be a material adverse effect to its business or its Consolidated Financial Statements.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Market for Common Stock

The principal market on which Walmart's common stock is listed for trading is the New York Stock Exchange. The common stock trades under the symbol "WMT."

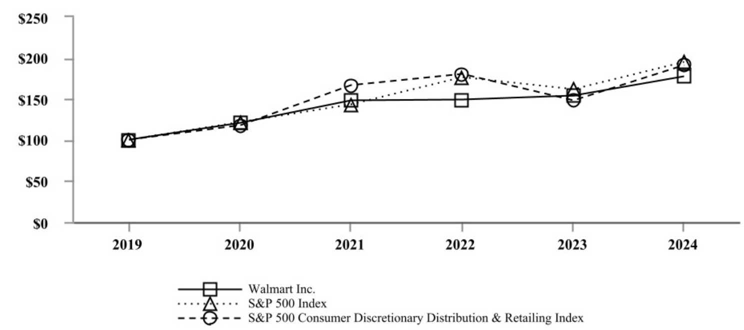
Holders of Record of Common Stock

As of March 13, 2024, there were 200,344 holders of record of Walmart's common stock, although there is a much larger number of beneficial owners.

Stock Performance Chart

This graph compares the cumulative total shareholder return on Walmart's common stock during the five fiscal years ended through fiscal 2024 to the cumulative total returns on the S&P 500 Consumer Discretionary Distribution & Retailing Index (formerly named the S&P 500 Retailing Index) and the S&P 500 Index. The comparison assumes $100 was invested on February 1, 2019 in shares of our common stock and in each of the indices shown and assumes that all of the dividends were reinvested.

COMPARISON OF 5 YEAR CUMULATIVE TOTAL RETURN*
Among Walmart Inc., the S&P 500 Index
and S&P 500 Consumer Discretionary Distribution & Retailing Index
(Fiscal Years Ended January 31)



*Assumes $100 Invested on February 1, 2019
Assumes Dividends Reinvested
Fiscal Year ended January 31, 2024

	Fiscal Years Ended January 31,					
	2019	2020	2021	2022	2023	2024
Walmart Inc.	$ 100.00	$ 120.27	$ 148.41	$ 148.47	$ 153.58	$ 177.30
S&P 500 Index	100.00	121.68	142.67	175.90	161.45	195.06
S&P 500 Consumer Discretionary Distribution & Retailing Index	100.00	117.54	166.19	180.56	147.66	190.67

Issuer Repurchases of Equity Securities

From time to time, the Company repurchases shares of our common stock under share repurchase programs authorized by the Company's Board of Directors. All repurchases made during fiscal 2024 were made under the current $20.0 billion share repurchase program approved in November 2022, which has no expiration date or other restrictions limiting the period over which the Company can make repurchases. As of January 31, 2024, authorization for $16.5 billion of share repurchases remained under the share repurchase program. Any repurchased shares are constructively retired and returned to an unissued status.

Share repurchase activity under our share repurchase programs, on a trade date basis, for each month in the quarter ended January 31, 2024, was as follows:

Fiscal Period	Total Number of Shares Repurchased[1]	Average Price Paid per Share (in dollars)[1]	Total Number of Shares Repurchased as Part of Publicly Announced Plans or Programs[1]	Approximate Dollar Value of Shares that May Yet Be Repurchased Under the Plans or Programs[2] (in billions)
November 1-30, 2023	5,340,951	$ 51.99	5,340,951	$ 17.8
December 1-31, 2023	13,913,403	51.41	13,913,403	17.1
January 1-31, 2024	10,211,025	53.62	10,211,025	16.5
Total	29,465,379		29,465,379	

[1] Share and per share information in this table has been adjusted to reflect the 3-for-1 common stock split effected on February 23, 2024. Refer to Note 1.
[2] Represents the approximate dollar value of shares that could have been repurchased under the current plan at the end of the month.

ITEM 6. RESERVED

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

Overview

This discussion, which presents our results for the fiscal years ended January 31, 2024 ("fiscal 2024"), January 31, 2023 ("fiscal 2023") and January 31, 2022 ("fiscal 2022"), should be read in conjunction with our Consolidated Financial Statements and the accompanying notes. We intend for this discussion to provide the reader with information that will assist in understanding our financial statements, the changes in certain key items in those financial statements from period to period and the primary factors that accounted for those changes. We also discuss certain performance metrics that management uses to assess the Company's performance. Additionally, the discussion provides information about the financial results of each of the three segments to provide a better understanding of how each of those segments and its results of operations affect the financial position and results of operations of the Company as a whole.

Throughout this Item 7, we discuss segment operating income, comparable store and club sales and other measures. Management measures the results of the Company's segments using each segment's operating income, including certain corporate overhead allocations, as well as other measures. From time to time, we revise the measurement of each segment's operating income and other measures as determined by the information regularly reviewed by our chief operating decision maker.

Management also measures the results of comparable store and club sales, or comparable sales, a metric that indicates the performance of our existing stores and clubs by measuring the change in sales for such stores and clubs, for a particular period from the corresponding period in the previous year. Walmart's definition of comparable sales includes sales from stores and clubs open for the previous 12 months, including remodels, relocations, expansions and conversions, as well as eCommerce sales. We measure the eCommerce sales impact by including all sales initiated digitally, including omni-channel transactions which are fulfilled through our stores and clubs as well as certain other business offerings that are part of our ecosystem, such as our Walmart Connect advertising business. Sales at a store that has changed in format are excluded from comparable sales when the conversion of that store is accompanied by a relocation or expansion that results in a change in the store's retail square feet of more than five percent. Sales related to divested businesses are excluded from comparable sales, and sales related to acquisitions are excluded until such acquisitions have been owned for 12 months. Comparable sales are also referred to as "same-store" sales by others within the retail industry. The method of calculating comparable sales varies across the retail industry. As a result, our calculation of comparable sales is not necessarily comparable to similarly titled measures reported by other companies.

In discussing our operating results, the term currency exchange rates refers to the currency exchange rates we use to convert the operating results for countries where the functional currency is not the U.S. dollar into U.S. dollars. We calculate the effect of changes in currency exchange rates as the difference between current period activity translated using the current period's currency exchange rates and the comparable prior year period's currency exchange rates. Additionally, no currency exchange rate fluctuations are calculated for non-USD acquisitions until owned for 12 months. Throughout our discussion, we refer to the results of this calculation as the impact of currency exchange rate fluctuations. Volatility in currency exchange rates may impact the results, including net sales and operating income, of the Company and the Walmart International segment in the future.

On February 23, 2024, the Company effected a 3-for-1 forward split of its common stock and a proportionate increase in the number of authorized shares. All share and per share information, including share based compensation, throughout this Annual Report on Form 10-K has been retroactively adjusted to reflect the stock split.

We have taken certain strategic actions across our segments, including an increased emphasis on investments in automation and supply chain as well as diversifying our earnings streams through category and business mix. Additionally, in the Walmart International segment, we have taken strategic actions to reshape our portfolio including the following highlights over the last three years:

- In February 2021, we completed the sale of Asda for net consideration of $9.6 billion. Refer to Note 12.
- In March 2021, we completed the sale of Seiyu for net consideration of $1.2 billion. Refer to Note 12.
- In November 2022, we completed the buyout of the noncontrolling interest shareholders of our Massmart subsidiary (Refer to Note 3) and in December 2022, we exited operations in certain countries in Africa.
- In December 2022, we increased our ownership in PhonePe as part of the separation from our majority-owned Flipkart subsidiary. Refer to Note 3.

We operate in a highly competitive omni-channel retail industry in all of the markets we serve. We face strong sales competition from other discount, department, drug, dollar, variety and specialty stores, warehouse clubs and supermarkets, as well as eCommerce, health and wellness, financial services, advertising and data service businesses. Many of these competitors are national, regional or international chains or have a national or international omni-channel or eCommerce presence. We compete with a number of companies for attracting and retaining quality associates. We, along with other retail companies, are influenced by a number of factors including, but not limited to: catastrophic events, weather and other risks related to climate change, global health epidemics and pandemics, competitive pressures, consumer disposable income, consumer debt levels and buying patterns, consumer credit availability, disruptions in supply chain, inventory management, cost and availability of goods, currency exchange rate fluctuations, customer preferences, inflation, deflation, fuel and energy prices, general economic conditions, insurance costs, interest rates, labor availability and costs, tax rates, the imposition of tariffs, cybersecurity attacks and unemployment. Further information on the factors that can affect our operating results and on certain risks to our Company and an investment in its securities can be found herein under "Item 1A. Risk Factors."

We are committed to helping customers save money and live better through everyday low prices, supported by everyday low costs. Merchandise costs for fiscal 2024 continued to be impacted by inflation, however at a lower rate than we experienced in fiscal 2023. The impact to our net sales and gross profit margin is influenced in part by our pricing and merchandising strategies in response to cost increases. Those pricing strategies include but are not limited to: absorbing cost increases instead of passing those cost increases on to our customers and members; reducing prices in certain merchandise categories; focusing on opening price points for certain food categories; and when necessary, passing cost increases on to our customers and members. Merchandising strategies include, but are not limited to: working with our suppliers to reduce product costs and share in absorbing cost increases; focusing on private label brands and smaller pack sizes; earlier-than-usual purchasing and in greater volumes or moderating purchasing in certain categories; and securing ocean carrier and container capacity. These strategies have and may continue to impact gross profit as a percentage of net sales.

We expect continued uncertainty in our business and the global economy due to inflationary trends; swings in macroeconomic conditions and their effect on consumer confidence; volatility in employment trends; supply chain pressures; and ongoing uncertainties related to global health epidemics or pandemics, any of which may impact our results. For a detailed discussion on results of operations by reportable segment, refer to "Results of Operations" below.

Company Performance Metrics

We are committed to helping customers save money and live better through everyday low prices, supported by everyday low costs. At times, we adjust our business strategies to maintain and strengthen our competitive positions in the countries in which we operate. We define our financial priorities as follows:

- Growth - serve customers through a seamless omni-channel experience;
- Margin - improve our operating income margin through productivity initiatives as well as category and business mix; and
- Returns - improve our Return on Investment ("ROI") through margin improvement and disciplined capital spend.

Growth

Our objective of prioritizing growth means we will focus on serving customers and members however they want to shop through our omni-channel business model. This includes increasing comparable store and club sales through increasing membership at Sam's Club and through Walmart+, accelerating eCommerce sales growth and expansion of omni-channel initiatives that complement our strategy.

Comparable sales is a metric that indicates the performance of our existing stores and clubs by measuring the change in sales for such stores and clubs, including eCommerce sales, for a particular period over the corresponding period in the previous year. The retail industry generally reports comparable sales using the retail calendar (also known as the 4-5-4 calendar). To be consistent with the retail industry, we provide comparable sales using the retail calendar in our quarterly earnings releases. However, when we discuss our comparable sales below, we are referring to our calendar comparable sales calculated using our fiscal calendar, which may result in differences when compared to comparable sales using the retail calendar. We focus on comparable sales in the U.S. as we believe it is a meaningful metric within the context of the U.S. retail market where there is a single currency, one inflationary market and generally consistent store and club formats from year to year.

Calendar comparable sales, as well as the impact of fuel, for fiscal 2024 and 2023, were as follows:

	Fiscal Years Ended January 31,			
	2024	2023	2024	2023
	With Fuel		Fuel Impact	
Walmart U.S.	5.5%	7.0%	(0.1)%	0.4%
Sam's Club	2.3%	14.6%	(2.6)%	4.2%
Total U.S.	4.9%	8.2%	(0.6)%	1.0%

Comparable sales in the U.S., including fuel, increased 4.9% and 8.2% in fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. Walmart U.S. comparable sales increased 5.5% and 7.0% in fiscal 2024 and 2023, respectively. For fiscal 2024, comparable sales growth was driven by growth in transactions combined with growth in average ticket, including strong sales in grocery and health and wellness. For fiscal 2023, comparable sales growth was driven by growth in average ticket, including strong food sales and higher inflation impacts in certain merchandise categories, as well as growth in transactions. Walmart U.S. eCommerce sales positively contributed approximately 2.6% and 0.7% to comparable sales for fiscal 2024 and 2023, respectively, which was primarily driven by store pickup and delivery.

Comparable sales at Sam's Club increased 2.3% and 14.6% in fiscal 2024 and 2023, respectively. For fiscal 2024, Sam's Club comparable sales benefited from growth in transactions and average ticket, including strong sales in grocery and health and wellness. Sam's Club comparable sales for fiscal 2023 benefited from growth in transactions and average ticket and included higher inflation impacts in certain merchandise categories. Sam's Club eCommerce sales positively contributed approximately 1.7% and 0.8% to comparable sales for fiscal 2024 and 2023, respectively.

Margin

Our objective of prioritizing margin focuses on growth with a focus on incremental margin accretion through a combination of productivity improvements as well as category and business mix. We invest in technology and process improvements to increase productivity, manage inventory and reduce costs and we operate with discipline by managing expenses and optimizing the efficiency of how we work. Additionally, we focus on our mix of businesses, including the expansion of connected value streams with higher margins, such as advertising and membership income. Our objective is to achieve operating income leverage, which we define as growing operating income at a faster rate than net sales.

	Fiscal Years Ended January 31,	
(Amounts in millions, except unit counts)	**2024**	**2023**
Net sales	$ 642,637	$ 605,881
Percentage change from comparable period	6.1 %	6.7 %
Gross profit as a percentage of net sales	23.7 %	23.5 %
Operating, selling, general and administrative expenses as a percentage of net sales	20.4 %	21.0 %
Operating income	$ 27,012	$ 20,428
Operating income as a percentage of net sales	4.2 %	3.4 %

Gross profit as a percentage of net sales ("gross profit rate") increased 27 and decreased 98 basis points for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. For fiscal 2024, the increase was primarily driven by the Walmart U.S. segment, due to managing prices aligned to our competitive historic price gaps and lapping higher markdowns incurred in the prior year, partially offset by product mix shifts into lower margin categories. For fiscal 2023, the decrease was primarily due to markdowns and merchandise mix in the U.S., higher supply chain costs and inflation related LIFO charges in the Sam's Club segment.

For fiscal 2024, operating expenses as a percentage of net sales decreased 60 basis points when compared to the previous fiscal year. Operating expenses as a percentage of net sales were positively impacted by lapping charges of $3.3 billion related to opioid-related legal settlements and $0.8 billion related to the reorganization and restructuring of certain businesses in the Walmart International segment in the prior year.

For fiscal 2023, operating expenses as a percentage of net sales increased 23 basis points when compared to the previous fiscal year. Operating expenses as a percentage of net sales were negatively impacted by the charges related to opioid-related legal settlements and the reorganization and restructuring of certain businesses in the Walmart International segment discussed above. These charges were partially offset by growth in net sales and lower incremental COVID-19 costs.

Operating income as a percentage of net sales increased 83 basis points and decreased 120 basis points for fiscal 2024 and 2023, respectively, due to the factors described above.

Returns

As we execute our financial framework, we believe our return on capital will improve over time. We measure return on capital with our return on assets, return on investment and free cash flow metrics. We also provide returns in the form of share repurchases and dividends, which are discussed in the Liquidity and Capital Resources section.

Return on Assets and Return on Investment

We include Return on Assets ("ROA"), the most directly comparable measure based on our financial statements presented in accordance with generally accepted accounting principles in the U.S. ("GAAP"), and Return on Investment ("ROI") as metrics to assess returns on assets. While ROI is considered a non-GAAP financial measure, management believes ROI is a meaningful metric to share with investors because it helps investors assess how effectively Walmart is deploying its assets. Trends in ROI can fluctuate over time as management balances long-term strategic initiatives with possible short-term impacts. ROA was 6.6% and 4.6% for fiscal 2024 and 2023, respectively. The increase in ROA was primarily due to the increase in consolidated net income, which was driven by higher operating income. ROI was 15.0% and 12.7% for fiscal 2024 and 2023, respectively. The increase in ROI was the result of an increase in operating income, primarily due to lapping charges associated with opioid-related legal settlements as well as reorganization and restructuring expenses, all recorded in fiscal 2023, as well as improvements in business performance, partially offset by an increase in average invested capital, primarily due to higher purchases of property and equipment.

We define ROI as adjusted operating income (operating income plus interest income, depreciation and amortization, and rent expense) for the trailing twelve months divided by average invested capital during that period. We consider average invested capital to be the average of our beginning and ending total assets, plus average accumulated depreciation and average amortization, less average accounts payable and average accrued liabilities for that period.

Our calculation of ROI is considered a non-GAAP financial measure because we calculate ROI using financial measures that exclude and include amounts that are included and excluded in the most directly comparable GAAP financial measure. For example, we exclude the impact of depreciation and amortization from our reported operating income in calculating the numerator of our calculation of ROI. As mentioned above, we consider ROA to be the financial measure computed in accordance with GAAP most directly comparable to our calculation of ROI. ROI differs from ROA (which is consolidated net income for the period divided by average total assets for the period) because ROI: adjusts operating income to exclude certain expense items and adds interest income; and adjusts total assets for the impact of accumulated depreciation and amortization, accounts payable and accrued liabilities to arrive at total invested capital. Because of the adjustments mentioned above, we believe ROI more accurately measures how we are deploying our key assets and is more meaningful to investors than ROA. Although ROI is a standard financial measure, numerous methods exist for calculating a company's ROI. As a result, the method used by management to calculate our ROI may differ from the methods used by other companies to calculate their ROI.

The calculation of ROA and ROI, along with a reconciliation of ROI to the calculation of ROA, the most comparable GAAP financial measure, is as follows:

	Fiscal Years Ended January 31,	
(Amounts in millions)	**2024**	**2023**
CALCULATION OF RETURN ON ASSETS		
Numerator		
Consolidated net income	$ 16,270	$ 11,292
Denominator		
Average total assets[(1)]	$ 247,798	$ 244,029
Return on assets (ROA)	6.6 %	4.6 %
CALCULATION OF RETURN ON INVESTMENT		
Numerator		
Operating income	$ 27,012	$ 20,428
+ Interest income	546	254
+ Depreciation and amortization	11,853	10,945
+ Rent	2,277	2,306
ROI operating income	$ 41,688	$ 33,933
Denominator		
Average total assets[(1)]	$ 247,798	$ 244,029
+ Average accumulated depreciation and amortization[(1)]	114,944	106,249
- Average accounts payable[(1)]	55,277	54,502
- Average accrued liabilities[(1)]	29,943	28,593
Average invested capital	$ 277,522	$ 267,183
Return on investment (ROI)	15.0 %	12.7 %

[(1)] The average is based on the addition of the account balance at the end of the current period to the account balance at the end of the prior period and dividing by 2.

	As of January 31,		
	2024	**2023**	**2022**
Certain Balance Sheet Data			
Total assets	$ 252,399	$ 243,197	$ 244,860
Accumulated depreciation and amortization	119,602	110,286	102,211
Accounts payable	56,812	53,742	55,261
Accrued liabilities	28,759	31,126	26,060

Strategic Capital Allocation

Our strategy includes allocating the majority of our capital to higher-return areas focused on automation such as eCommerce, supply chain and store and club investments. The following table provides additional detail regarding our capital expenditures:

(Amounts in millions)	Fiscal Years Ended January 31,	
Allocation of Capital Expenditures	**2024**	**2023**
Supply chain, customer-facing initiatives and technology	$ 11,828	$ 9,209
Store and club remodels	5,792	4,990
New stores and clubs, including expansions and relocations	75	33
Total U.S.	$ 17,695	$ 14,232
Walmart International	2,911	2,625
Total capital expenditures	$ 20,606	$ 16,857

Free Cash Flow

Free cash flow is considered a non-GAAP financial measure. Management believes, however, that free cash flow, which measures our ability to generate additional cash from our business operations, is an important financial measure for use in evaluating the Company's financial performance. Free cash flow should be considered in addition to, rather than as a substitute for, consolidated net income as a measure of our performance and net cash provided by operating activities as a measure of our liquidity. See "Liquidity and Capital Resources" for discussions of GAAP metrics including net cash provided by operating activities, net cash used in investing activities and net cash used in financing activities.

We define free cash flow as net cash provided by operating activities in a period minus payments for property and equipment made in that period. We had net cash provided by operating activities of $35.7 billion, $28.8 billion and $24.2 billion for fiscal 2024, 2023 and 2022, respectively. We generated free cash flow of $15.1 billion, $12.0 billion and $11.1 billion for fiscal 2024, 2023 and 2022, respectively. The increase in net cash provided by operating activities in fiscal 2024 is primarily due to higher cash provided by operating income, as well as timing of certain payments and strategic inventory management as part of working capital initiatives, partially offset by payment of the remaining accrued opioid legal charges. Free cash flow for fiscal 2024 increased when compared to fiscal 2023 due to the increase in operating cash flows described above, partially offset by an increase of $3.7 billion in capital expenditures to support our investment strategy. Net cash provided by operating activities for fiscal 2023 increased when compared to fiscal 2022 primarily due to moderated levels of inventory purchases, partially offset by a decline in operating income and the timing of certain payments. Free cash flow for fiscal 2023 increased when compared to fiscal 2022 due to the increase in net cash provided by operating activities described above, partially offset by an increase of $3.8 billion in capital expenditures to support our investment strategy.

Walmart's definition of free cash flow is limited in that it does not represent residual cash flows available for discretionary expenditures due to the fact that the measure does not deduct the payments required for debt service and other contractual obligations or payments made for business acquisitions. Therefore, we believe it is important to view free cash flow as a measure that provides supplemental information to our Consolidated Statements of Cash Flows.

Although other companies report their free cash flow, numerous methods may exist for calculating a company's free cash flow. As a result, the method used by management to calculate our free cash flow may differ from the methods used by other companies to calculate their free cash flow.

The following table sets forth a reconciliation of free cash flow, a non-GAAP financial measure, to net cash provided by operating activities, which we believe to be the GAAP financial measure most directly comparable to free cash flow, as well as information regarding net cash used in investing activities and net cash used in financing activities.

	Fiscal Years Ended January 31,		
(Amounts in millions)	**2024**	**2023**	**2022**
Net cash provided by operating activities	$ 35,726	$ 28,841	$ 24,181
Payments for property and equipment	(20,606)	(16,857)	(13,106)
Free cash flow	$ 15,120	$ 11,984	$ 11,075
Net cash used in investing activities[1]	$ (21,287)	$ (17,722)	$ (6,015)
Net cash used in financing activities	(13,414)	(17,039)	(22,828)

[1] "Net cash used in investing activities" includes payments for property and equipment, which is also included in our computation of free cash flow.

Results of Operations

Consolidated Results of Operations

(Amounts in millions, except unit counts)	Fiscal Years Ended January 31, 2024	2023	2022
Total revenues	$ 648,125	$ 611,289	$ 572,754
Percentage change from comparable period	6.0 %	6.7 %	2.4 %
Net sales	$ 642,637	$ 605,881	$ 567,762
Percentage change from comparable period	6.1 %	6.7 %	2.3 %
Total U.S. calendar comparable sales increase	4.9 %	8.2 %	7.7 %
Gross profit rate	23.7 %	23.5 %	24.4 %
Operating income	$ 27,012	$ 20,428	$ 25,942
Operating income as a percentage of net sales	4.2 %	3.4 %	4.6 %
Loss on extinguishment of debt	$ —	$ —	$ 2,410
Other (gains) and losses	$ 3,027	$ 1,538	$ 3,000
Consolidated net income	$ 16,270	$ 11,292	$ 13,940
Unit counts at period end[(1)]	10,616	10,623	10,593
Retail square feet at period end[(1)]	1,053	1,056	1,060

(1) Unit counts and associated retail square feet are presented for stores and clubs generally open as of period end, and reflects the removal of stores in the U.K. and Japan subsequent to closing the divestitures in fiscal 2022. Permanently closed locations are not included in these metrics.

Our total revenues, which includes net sales and membership and other income, increased $36.8 billion or 6.0% and $38.5 billion or 6.7% for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. These increases in revenues were primarily due to increases in net sales, which increased $36.8 billion or 6.1% and $38.1 billion or 6.7% for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. For fiscal 2024, the increase was primarily due to positive comparable sales for the Walmart U.S. and Sam's Club segments, which were driven by growth in transactions, combined with growth in average ticket, including strong sales in grocery and health and wellness, along with positive comparable sales across our international markets. Net sales were positively impacted by $3.0 billion of fluctuations in currency exchange rates during fiscal 2024. For fiscal 2023, the increase was primarily due to strong positive comparable sales for the Walmart U.S. and Sam's Club segments which was driven by growth in average ticket, including strong food sales and higher inflation impacts in certain merchandise categories, as well as growth in transactions, along with positive comparable sales in all of our international markets. Additionally, net sales were negatively impacted by a decrease of $5.0 billion related to the divestiture of our operations in the U.K. and Japan, which closed in the first quarter of fiscal 2022 and $3.7 billion of fluctuations in currency exchange rates during fiscal 2023.

Our gross profit rate increased 27 basis points and decreased 98 basis points for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. For fiscal 2024, the increase was primarily driven by the Walmart U.S. segment, due to managing prices aligned to our competitive historic price gaps and lapping higher markdowns incurred in the prior year, partially offset by product mix shifts into lower margin categories. For fiscal 2023, the decrease was primarily due to markdowns and merchandise mix in the U.S., higher supply chain costs and inflation related LIFO charges in the Sam's Club segment.

For fiscal 2024, operating expenses as a percentage of net sales decreased 60 basis points when compared to the previous fiscal year. Operating expenses as a percentage of net sales were positively impacted by lapping charges of $3.3 billion related to opioid-related legal settlements and $0.8 billion related to the reorganization and restructuring of certain businesses in the Walmart International segment in the prior year. For fiscal 2023, operating expenses as a percentage of net sales increased 23 basis points when compared to the previous fiscal year. Operating expenses as a percentage of net sales were negatively impacted by the charges related to opioid-related legal settlements and the reorganization and restructuring of certain businesses in the Walmart International segment discussed above. These charges were partially offset by growth in net sales and lower incremental COVID-19 costs.

Loss on extinguishment of debt was $2.4 billion in fiscal 2022 due to the early retirement of certain higher rate long-term debt to reduce interest expense in future periods. There were no such early retirements of debt in fiscal 2024 and fiscal 2023.

Other gains and losses consist of certain non-operating items, such as the change in the fair value of our investments and gains or losses on business dispositions, which by their nature can fluctuate from period to period. Other gains and losses consisted of a net loss of $3.0 billion and $1.5 billion for fiscal 2024 and 2023, respectively. The net loss in fiscal 2024 primarily consists of net losses associated with the fair value changes of our equity and other investments. The net loss in fiscal 2023 primarily consists of: net losses associated with the fair value changes of our equity and other investments; a gain of $0.4 billion recognized on the sale of our remaining equity method investment in Brazil; and a $0.2 billion dividend from one of our investments.

Our effective income tax rate was 25.5%, 33.6%, and 25.4% for fiscal 2024, 2023 and 2022, respectively. The higher effective tax rate in fiscal 2023 as compared to both fiscal 2024 and fiscal 2022 is primarily due to the tax impact of the business reorganization resulting in the full separation of PhonePe from Flipkart in fiscal 2023. Our effective income tax rate may also fluctuate as a result of various factors, including changes in our assessment of unrecognized tax benefits, valuation allowances, changes in tax law, outcomes of administrative audits, the impact of discrete items and the mix and size of earnings among our U.S. operations and international operations, which are subject to statutory rates that are generally higher than the U.S. statutory rate. The reconciliation from the U.S. statutory rate to the effective income tax rates for fiscal 2024, 2023 and 2022 is presented in Note 9.

As a result of the factors discussed above, we reported $16.3 billion and $11.3 billion of consolidated net income for fiscal 2024 and 2023, respectively, which represents an increase of $5.0 billion and a decrease of $2.6 billion for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. Diluted net income per common share attributable to Walmart ("EPS") was $1.91, $1.42 and $1.62 for fiscal 2024, 2023 and 2022, respectively.

Walmart U.S. Segment

	Fiscal Years Ended January 31,		
(Amounts in millions, except unit counts)	2024	2023	2022
Net sales	$ 441,817	$ 420,553	$ 393,247
Percentage change from comparable period	5.1 %	6.9 %	6.3 %
Calendar comparable sales increase	5.5 %	7.0 %	6.4 %
Operating income	$ 22,154	$ 20,620	$ 21,587
Operating income as a percentage of net sales	5.0 %	4.9 %	5.5 %
Unit counts at period end	4,615	4,717	4,742
Retail square feet at period end	699	702	703

Net sales for the Walmart U.S. segment increased $21.3 billion or 5.1% and $27.3 billion or 6.9% for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. The increases in net sales were primarily due to increases in comparable sales of 5.5% and 7.0% for fiscal 2024 and 2023, respectively. Comparable sales in fiscal 2024 were driven by growth in transactions combined with growth in average ticket, including strong sales in grocery and health and wellness. Comparable sales in fiscal 2023 were driven by growth in average ticket, including strong food sales and higher inflation impacts in certain merchandise categories, as well as growth in transactions. Walmart U.S. eCommerce sales positively contributed approximately 2.6% and 0.7% to comparable sales for fiscal 2024 and 2023, respectively, which was primarily driven by store pickup and delivery.

Gross profit rate increased 20 basis points for fiscal 2024 and decreased 85 basis points for fiscal 2023, when compared to the respective previous fiscal year. The increase in fiscal 2024 gross profit rate was primarily due to managing prices aligned to our competitive historic price gaps and lapping higher net markdowns incurred in the prior year, partially offset by product mix shifts into lower margin categories. The decrease in fiscal 2023 gross profit rate was primarily due to net markdowns and product mix shifts into lower margin categories and increased supply chain costs, partially offset by price management impacts driven by cost inflation.

Operating expenses as a percentage of segment net sales increased 9 basis points for fiscal 2024 when compared to the previous fiscal year primarily driven by higher variable pay relative to last year as a result of exceeding our planned performance. For fiscal 2023, operating expenses as a percentage of segment net sales decreased 25 basis points primarily driven by strong sales growth and lower incremental COVID-19 related costs, partially offset by increased wage costs.

As a result of the factors discussed above, segment operating income increased $1.5 billion and decreased $1.0 billion for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year.

Walmart International Segment

(Amounts in millions, except unit counts)	Fiscal Years Ended January 31, 2024	2023	2022
Net sales	$ 114,641	$ 100,983	$ 100,959
Percentage change from comparable period	13.5 %	— %	(16.8)%
Operating income	$ 4,909	$ 2,965	$ 3,758
Operating income as a percentage of net sales	4.3 %	2.9 %	3.7 %
Unit counts at period end	5,402	5,306	5,251
Retail square feet at period end	274	273	277

Net sales for the Walmart International segment increased $13.7 billion or 13.5% for fiscal 2024 and were flat for 2023, when compared to the previous fiscal year. For fiscal 2024, the increase was primarily due to positive comparable sales across our international markets and positive fluctuations in currency exchange rates of $3.0 billion during fiscal 2024. For fiscal 2023, net sales benefited from positive comparable sales across all of our international markets, offset by the impacts of a decrease of $5.0 billion related to the divestiture of our operations in the U.K. and Japan, which closed in the first quarter of fiscal 2022, as well as $3.7 billion of fluctuations in currency exchange rates during fiscal 2023.

Gross profit rate increased 20 basis points and decreased 50 basis points for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. For fiscal 2024, the increase was primarily driven by supply chain efficiencies partially offset by ongoing format and channel shifts. For fiscal 2023, the decrease was primarily driven by continued growth in lower margin formats and channels in China and category mix shifts into lower margin categories.

Operating expenses as a percentage of segment net sales decreased 152 basis points and increased 41 basis points for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. The decrease in operating expenses as a percentage of segment net sales for fiscal 2024 was primarily due to the lapping of business reorganization and restructuring charges incurred related to Flipkart and Massmart in fiscal 2023 and an increase in sales in the current year. The increase in operating expenses as a percentage of segment net sales for fiscal 2023, was primarily due to incurring these business reorganization and restructuring charges.

As a result of the factors discussed above, segment operating income increased $1.9 billion and decreased $0.8 billion for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year.

Sam's Club Segment

(Amounts in millions, except unit counts)	Fiscal Years Ended January 31, 2024	2023	2022
Including Fuel			
Net sales	$ 86,179	$ 84,345	$ 73,556
Percentage change from comparable period	2.2 %	14.7 %	15.1 %
Calendar comparable sales increase	2.3 %	14.6 %	15.0 %
Operating income	$ 2,192	$ 1,964	$ 2,259
Operating income as a percentage of net sales	2.5 %	2.3 %	3.1 %
Unit counts at period end	599	600	600
Retail square feet at period end	80	80	80
Excluding Fuel [(1)]			
Net sales	$ 75,057	$ 71,665	$ 64,860
Percentage change from comparable period	4.7 %	10.5 %	9.6 %
Operating income	$ 1,659	$ 1,352	$ 1,923
Operating income as a percentage of net sales	2.2 %	1.9 %	3.0 %

(1) We believe the "Excluding Fuel" information is useful to investors because it permits investors to understand the effect of the Sam's Club segment's fuel sales on its results of operations, which are impacted by the volatility of fuel prices. Volatility in fuel prices may continue to impact the operating results of the Sam's Club segment in the future. Management uses such information to better measure underlying operating results in the segment.

Net sales for the Sam's Club segment increased $1.8 billion or 2.2% and $10.8 billion or 14.7% for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. The increases in net sales were primarily due to increases in comparable sales, including fuel, of 2.3% and 14.6% for fiscal 2024 and 2023, respectively. Comparable sales benefited from growth in transactions and average ticket, including strong sales in grocery and health and wellness. Additionally, fiscal 2024 growth was partially offset by lower fuel sales due to deflation in this category. Sam's Club eCommerce sales positively contributed approximately 1.7% and 0.8% to comparable sales for fiscal 2024 and 2023, respectively, which was primarily driven by curbside pickup and ship to home.

Gross profit rate increased 55 basis points and decreased 155 basis points for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. For fiscal 2024, the increase in gross profit rate was primarily due to the lapping of elevated markdowns in the prior year, partially offset by product mix shifts into lower margin categories. For fiscal 2023, the decrease in gross profit rate was primarily due to inventory markdowns, elevated supply chain and eCommerce fulfillment costs and inflation related LIFO charges.

Membership and other income increased 7.5% and 7.0% for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. For fiscal 2024 and 2023, the increases were primarily due to growth in membership base and Plus penetration. Fiscal 2024 was also positively impacted by higher Plus renewals, as well as the expiration of a promotional offering offsetting membership fee increases during the fourth quarter of fiscal 2024.

Operating expenses as a percentage of segment net sales increased 46 basis points and decreased 97 basis points for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year. Fiscal 2024 operating expenses as a percentage of net sales increased primarily due to lower fuel sales and elevated technology spend. Fiscal 2023 operating expenses as a percentage of net sales decreased primarily due to higher sales.

As a result of the factors discussed above, segment operating income increased $0.2 billion and decreased $0.3 billion for fiscal 2024 and 2023, respectively, when compared to the previous fiscal year.

Liquidity and Capital Resources

Liquidity

The strength and stability of our operations have historically supplied us with a significant source of liquidity. Our cash flows provided by operating activities, supplemented with our long-term debt and short-term borrowings, have been sufficient to fund our operations while allowing us to invest in activities that support the long-term growth of our operations. Generally, some or all of the remaining available cash flow has been used to fund dividends on our common stock and share repurchases. We believe our sources of liquidity will continue to be sufficient to fund operations, finance our global investment activities, pay dividends and fund our share repurchases for at least the next 12 months and for the foreseeable future.

Net Cash Provided by Operating Activities

	Fiscal Years Ended January 31,		
(Amounts in millions)	**2024**	**2023**	**2022**
Net cash provided by operating activities	$ 35,726	$ 28,841	$ 24,181

Net cash provided by operating activities was $35.7 billion, $28.8 billion and $24.2 billion for fiscal 2024, 2023 and 2022, respectively. The increase in net cash provided by operating activities in fiscal 2024 is primarily due to higher cash provided by operating income, as well as timing of certain payments and strategic inventory management as part of working capital initiatives, partially offset by payment of the remaining accrued opioid legal charges. The increase in net cash provided by operating activities for fiscal 2023, when compared to the previous fiscal year, was primarily due to moderated levels of inventory purchases, partially offset by a decline in operating income and the timing of certain payments.

Cash Equivalents and Working Capital Deficit

Cash and cash equivalents were $9.9 billion and $8.6 billion as of January 31, 2024 and 2023, respectively. Our working capital deficit, defined as total current assets less total current liabilities, was $15.5 billion and $16.5 billion as of January 31, 2024 and 2023, respectively. The decrease in our working capital deficit is primarily driven by a decrease in accrued liabilities primarily due to the payment of the remaining accrued opioid legal charges and an increase in cash, partially offset by an increase in accounts payable and a decrease in inventories as part of working capital initiatives. We generally operate with a working capital deficit due to our efficient use of cash in funding operations, consistent access to the capital markets and returns provided to our shareholders in the form of payments of cash dividends and share repurchases.

We use intercompany financing arrangements in an effort to ensure cash can be made available in the country in which it is needed with the minimum cost possible. Additionally, from time-to-time, we repatriate earnings and related cash from jurisdictions outside of the U.S. Under current law, repatriations of foreign earnings will generally be free of U.S. federal tax, but may incur other taxes such as withholding or state taxes. We do not expect current local laws, or other existing limitations on anticipated future repatriations of cash amounts held outside the U.S. to have a material effect on our overall liquidity, financial position or results of operations.

As of January 31, 2024 and 2023, cash and cash equivalents of $3.5 billion and $2.9 billion, respectively, may not be freely transferable to the U.S. due to local laws or other restrictions or are subject to the approval of the noncontrolling interest shareholders.

Net Cash Used in Investing Activities

	Fiscal Years Ended January 31,		
(Amounts in millions)	2024	2023	2022
Net cash used in investing activities	$ (21,287)	$ (17,722)	$ (6,015)

Net cash used in investing activities was $21.3 billion, $17.7 billion and $6.0 billion for fiscal 2024, 2023 and 2022, respectively, and generally consisted of capital expenditures. Net cash used in investing activities increased $3.6 billion for fiscal 2024 when compared to the previous fiscal year primarily due to increased payments for property and equipment. Net cash used in investing activities increased $11.7 billion for fiscal 2023 when compared to the previous fiscal year, primarily due to the result of lapping the net proceeds received from the divestitures of our operations in the U.K. and Japan and an increase in capital expenditures to support our investment strategy.

Capital expenditures

Refer to the "Strategic Capital Allocation" section in our Company Performance Metrics for capital expenditure detail for fiscal 2024 and 2023. For the fiscal year ending January 31, 2025 ("fiscal 2025"), we project capital expenditures will be approximately $20 billion to $24 billion, with a focus on technology, supply chain, and customer-facing initiatives.

Net Cash Used in Financing Activities

	Fiscal Years Ended January 31,		
(Amounts in millions)	2024	2023	2022
Net cash used in financing activities	$ (13,414)	$ (17,039)	$ (22,828)

Net cash from financing activities generally consists of debt transactions, dividends paid, repurchases of Company stock and transactions with noncontrolling interest shareholders. Fiscal 2024 net cash used in financing activities decreased $3.6 billion when compared to the previous fiscal year. The decrease is primarily due to fewer repurchases of Company stock, partially offset by the purchase of certain noncontrolling interests. Fiscal 2023 net cash used in financing activities decreased $5.8 billion when compared to the previous fiscal year. The decrease was primarily due to repayments of long-term debt and related payment of premiums for the early extinguishment of certain notes in the prior fiscal year, partially offset by the equity funding from the sale of subsidiary stock in the prior fiscal year.

*Purchase and Sale of Subsidiary Stoc*k

During fiscal 2024, we paid $3.5 billion to acquire shares from certain Flipkart noncontrolling interest holders and settle the liability to former noncontrolling interest holders of PhonePe. Additionally, we received $0.7 billion related to new rounds of equity funding for the Company's majority owned PhonePe subsidiary.

During fiscal 2023, we completed a $0.4 billion buyout of the noncontrolling interest shareholders of our Massmart subsidiary and completed a $0.4 billion acquisition of Alert Innovation, bringing our ownership to approximately 100% of both Massmart and Alert Innovation.

During fiscal 2022, we received $3.2 billion primarily related to a new equity funding for our majority-owned Flipkart subsidiary.

Short-term Borrowings

We generally utilize the liquidity provided by short-term borrowings to provide funding for our operations, dividend payments, share repurchases, capital expenditures and other cash requirements. The following table includes additional information related to the our short-term borrowings for fiscal 2024, 2023 and 2022:

	Fiscal Years Ended January 31,		
(Amounts in millions)	2024	2023	2022
Maximum amount outstanding at any month-end	$ 9,942	$ 11,432	$ 716
Average daily short-term borrowings	4,295	7,250	626
Annual weighted-average interest rate	5.1 %	2.4 %	3.7 %

Short-term borrowings as of January 31, 2024 and 2023 were $0.9 billion and $0.4 billion, respectively, with weighted-average interest rates of 7.7% and 6.6%, respectively. We also have $15.0 billion of various undrawn committed lines of credit in the U.S. as of January 31, 2024 that provide additional liquidity, if needed. Additionally, we maintain access to various credit facilities outside of the U.S. to further support our Walmart International segment operations, as needed.

As of January 31, 2024, we have $2.1 billion of syndicated and fronted letters of credit available, of which $1.7 billion was drawn and represents an unrecorded current obligation.

Long-term Debt

The following table provides the changes in our long-term debt for fiscal 2024:

(Amounts in millions)	Long-term debt due within one year	Long-term debt	Total
Balances as of February 1, 2023	$ 4,191	$ 34,649	$ 38,840
Proceeds from issuance of long-term debt	—	4,967	4,967
Repayments of long-term debt	(4,213)	(4)	(4,217)
Reclassifications of long-term debt	3,486	(3,486)	—
Currency and other adjustments	(17)	6	(11)
Balances as of January 31, 2024	$ 3,447	$ 36,132	$ 39,579

Our total outstanding long-term debt increased $0.7 billion during fiscal 2024, primarily due to the issuance of new long-term debt in April 2023, partially offset by the maturities of certain long-term debt. Refer to Note 6 to our Consolidated Financial Statements for details on the issuances of long-term debt.

Estimated contractual interest payments associated with our long-term debt amount to $20.2 billion, with approximately $1.8 billion expected to be paid in fiscal 2025. Estimated interest payments are based on our principal amounts and expected maturities of all debt outstanding as of January 31, 2024, and assumes interest rates remain at current levels for our variable rate instruments.

Dividends

Our total dividend payments were $6.1 billion, $6.1 billion and $6.2 billion for fiscal 2024, 2023 and 2022, respectively. Effective February 20, 2024, the Company approved the fiscal 2025 annual dividend of $0.83 per share, an increase over the fiscal 2024 annual dividend of $0.76 per share. For fiscal 2025, the annual dividend will be paid in four quarterly installments of $0.2075 per share, according to the following record and payable dates:

Record Date	Payable Date
March 15, 2024	April 1, 2024
May 10, 2024	May 28, 2024
August 16, 2024	September 3, 2024
December 13, 2024	January 6, 2025

Company Share Repurchase Program

From time to time, the Company repurchases shares of its common stock under share repurchase programs authorized by the Company's Board of Directors. All repurchases made during fiscal 2024 were made under the current $20.0 billion share repurchase program approved in November 2022, which has no expiration date or other restrictions limiting the period over which the Company can make repurchases. As of January 31, 2024, authorization for $16.5 billion of share repurchases remained under the share repurchase program. Any repurchased shares are constructively retired and returned to an unissued status.

We regularly review share repurchase activity and consider several factors in determining when to execute share repurchases, including, among other things, current cash needs, capacity for leverage, cost of borrowings, our results of operations and the market price of our common stock. We anticipate that a majority of the ongoing share repurchase program will be funded through the Company's free cash flow.

The following table provides, on a settlement date basis, the number of shares repurchased, average price paid per share and total amount paid for share repurchases for fiscal 2024, 2023 and 2022:

	Fiscal Years Ended January 31,		
(Amounts in millions, except per share data)	2024	2023	2022
Total number of shares repurchased	54.6	221.8	209.1
Average price paid per share	$ 50.87	$ 44.72	$ 46.82
Total amount paid for share repurchases	$ 2,779	$ 9,920	$ 9,787

Material Cash Requirements

Material cash requirements from operating activities primarily consist of inventory purchases, employee related costs, taxes, interest and other general operating expenses, which we expect to be primarily satisfied by our cash from operations. Other material cash requirements from known contractual and other obligations include short-term borrowings, long-term debt and related interest payments, leases and purchase obligations. See Note 6 and Note 7 to our Consolidated Financial Statements for information regarding outstanding short-term borrowings and long-term debt, and leases, respectively.

As of January 31, 2024, the Company has $34.3 billion of unrecorded purchase obligations outstanding, of which $14.6 billion is due within one year. Purchase obligations include legally binding contracts, such as firm commitments for inventory and utility purchases, as well as commitments to make capital expenditures, software acquisition and license commitments and legally binding service contracts. Contractual obligations for the purchase of goods or services are defined as agreements that are enforceable and legally binding and that specify all significant terms, including: fixed or minimum quantities to be purchased; fixed, minimum or variable price provisions; and the approximate timing of the transaction. Contracts that specify the Company will purchase all or a portion of its requirements of a specific product or service from a supplier, but do not include a fixed or minimum quantity, are excluded from the obligations quantified above. Accordingly, purchase orders for inventory are also excluded as purchase orders represent authorizations to purchase rather than binding agreements. Our purchase orders are based on our current inventory needs and are fulfilled by our suppliers within short time periods. We also enter into contracts for outsourced services; however, the obligations under these contracts are not significant and the contracts generally contain clauses allowing for cancellation without significant penalty. Timing of payments and actual amounts paid may be different depending on the timing of receipt of goods or services or changes to agreed-upon amounts for some obligations.

Capital Resources

We believe our cash flows from operations, current cash position, short-term borrowings and access to capital markets will continue to be sufficient to meet our anticipated cash requirements and contractual obligations, which includes funding seasonal buildups in merchandise inventories and funding our capital expenditures, acquisitions, dividend payments and share repurchases.

We have strong commercial paper and long-term debt ratings that have enabled and should continue to enable us to refinance our debt as it becomes due at favorable rates in capital markets. As of January 31, 2024, the ratings assigned to our commercial paper and rated series of our outstanding long-term debt were as follows:

Rating agency	Commercial paper	Long-term debt
Standard & Poor's	A-1+	AA
Moody's Investors Service	P-1	Aa2
Fitch Ratings	F1+	AA

Credit rating agencies review their ratings periodically and, therefore, the credit ratings assigned to us by each agency may be subject to revision at any time. Accordingly, we are not able to predict whether our current credit ratings will remain consistent over time. Factors that could affect our credit ratings include changes in our operating performance, the general economic environment, conditions in the retail industry, our financial position, including our total debt and capitalization, and changes in our business strategy. Any downgrade of our credit ratings by a credit rating agency could increase our future borrowing costs or impair our ability to access capital and credit markets on terms commercially acceptable to us. In addition, any downgrade of our current short-term credit ratings could impair our ability to access the commercial paper markets with the same flexibility that we have experienced historically, potentially requiring us to rely more heavily on more expensive types of debt financing. The credit rating agency ratings are not recommendations to buy, sell or hold our commercial paper or debt securities. Each rating may be subject to revision or withdrawal at any time by the assigning rating organization and should be evaluated independently of any other rating. Moreover, each credit rating is specific to the security to which it applies.

Other Matters

In Note 10 to our Consolidated Financial Statements, which is captioned "Contingencies" and appears in Part II of this Annual Report on Form 10-K under the caption "Item 8. Financial Statements and Supplementary Data," we discuss, under the sub-captions "*Settlement Framework Regarding Multidistrict and State or Local Opioid Related Litigation*," and "*Other Opioid Related Litigation*" the Prescription Opiate Litigation, the Settlement Framework, and other matters, including certain risks arising therefrom. In that Note 10, we also discuss under the sub-caption "*Asda Equal Value Claims"* the Company's indemnification obligation for the Asda Equal Value Claims matter, under the sub-caption "*Money Transfer Agent Services Matters,*" a United States Federal Trade Commission complaint related to money transfers and the Company's anti-fraud program and a government investigation by the U.S. Attorney's Office for the Middle District of Pennsylvania into the Company's consumer fraud prevention and anti-money laundering compliance related to the Company's money transfer agent services as well as under the sub-caption "*Mexico Antitrust Matter*," we disclose the main Mexican operating subsidiary of Wal-Mart de México was notified of the initiation of a quasi-judicial administrative process against it for alleged relative monopolistic practices in connection with the supply and wholesale distribution of certain consumer goods, retail marketing practices of such consumer goods and related services. We discuss various legal proceedings related to the Federal and State Prescription Opiate Litigation, the Settlement Framework, DOJ Opioid Civil Litigation and Opioids Related Securities Class Actions and Derivative Litigation in Part I of this Annual Report on Form 10-K under the caption "Item 3. Legal Proceedings," under the sub-caption "I. Supplemental Information." We also discuss the Foreign Direct Investment Matters in Part I of this Annual Report on Form 10-K under the caption "Item 3. Legal Proceedings," under the sub-caption "II. Certain Other Matters." We also discuss an environmental matter with the State of California in Part I of this Annual Report on Form 10-K under the caption "Item 3. Legal Proceedings," under the sub-caption "III. Environmental Matters." The foregoing matters and other matters described elsewhere in this Annual Report on Form 10-K represent contingent liabilities of the Company that may or may not result in the incurrence of a material liability by the Company upon their final resolution.

Summary of Critical Accounting Estimates

Management strives to report our financial results in a clear and understandable manner, although in some cases accounting and disclosure rules are complex and require us to use technical terminology. In preparing the Company's Consolidated Financial Statements, we follow accounting principles generally accepted in the U.S. These principles require us to make certain estimates and apply judgments that affect our financial position and results of operations as reflected in our financial statements. These judgments and estimates are based on past events and expectations of future outcomes. Actual results may differ from our estimates.

Management continually reviews our accounting policies including how they are applied and how they are reported and disclosed in our financial statements. Following is a summary of our critical accounting estimates and how they are applied in preparation of the financial statements.

Contingencies

We are involved in a number of legal proceedings and certain regulatory matters. We record a liability when it is probable that a loss has been incurred and the amount is reasonably estimable. We also perform an assessment of the materiality of loss contingencies where a loss is either reasonably possible or it is reasonably possible that a loss could be incurred in excess of amounts accrued. If a loss or an additional loss has at least a reasonable possibility of occurring and the impact on the financial statements would be material, we provide disclosure of the loss contingency in the footnotes to our financial statements. We review all contingencies at least quarterly to determine whether the likelihood of loss has changed and to assess whether a reasonable estimate of the loss or the range of the loss can be made. Although we are not able to predict the outcome or reasonably estimate a range of possible losses in certain matters described in Note 10 to our Consolidated Financial Statements and have not recorded an associated accrual related to these matters, an adverse judgment or negotiated resolution in any of these matters could have a material adverse effect on our business, reputation, financial position, results of operations or cash flows.

Uncertain Tax Positions

We are subject to income taxes in the U.S. and numerous foreign jurisdictions. Our tax returns are routinely audited and settlements of issues raised in these audits sometimes affect our tax provisions. The benefits of uncertain tax positions are recorded in our financial statements only after determining a more likely than not probability that the uncertain tax positions will withstand challenge, if any, from taxing authorities. When facts and circumstances change, we reassess these probabilities and record any changes in the financial statements as appropriate. We account for uncertain tax positions by determining the minimum recognition threshold that a tax position is required to meet before being recognized in the financial statements. Accordingly, the determination of our uncertain tax positions requires judgment, the use of estimates in certain cases and the interpretation and application of complex tax laws.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

Market Risk

In addition to the risks inherent in our operations, we are exposed to certain market risks, including changes in interest rates, currency exchange rates and the fair values of certain equity and equity method investments measured on a recurring basis.

The analysis presented below for each of our market risk sensitive instruments is based on a hypothetical scenario used to calibrate potential risk and does not represent our view of future market changes. The effect of a change in a particular assumption is calculated without adjusting any other assumption. In reality, however, a change in one factor could cause a change in another, which may magnify or negate other sensitivities.

Interest Rate Risk

We are exposed to changes in interest rates as a result of our short-term borrowings and long-term debt. We hedge a portion of our interest rate risk by managing the mix of fixed and variable rate debt and by entering into interest rate swaps. For fiscal 2024, the net fair value of our interest rate swaps increased $35 million primarily due to fluctuations in market interest rates.

The table below provides information about our financial instruments that are sensitive to changes in interest rates. For long-term debt, the table represents the principal cash flows and related weighted-average interest rates by expected maturity dates. For interest rate swaps, the table represents the contractual cash flows and weighted-average interest rates by the contractual maturity date, unless otherwise noted. The notional amounts are used to calculate contractual cash flows to be exchanged under the contracts. The weighted-average variable rates are based upon prevailing market rates as of January 31, 2024.

	Expected Maturity Date						
(Amounts in millions)	**Fiscal 2025**	**Fiscal 2026**	**Fiscal 2027**	**Fiscal 2028**	**Fiscal 2029**	**Thereafter**	**Total**
Liabilities							
Short-term borrowings:							
Variable rate	$ 878	$ —	$ —	$ —	$ —	$ —	$ 878
Weighted-average interest rate	7.7 %	— %	— %	— %	— %	— %	7.7 %
Long-term debt[(1)]:							
Fixed rate	$ 3,447	$ 2,600	$ 3,483	$ 1,760	$ 3,458	$ 24,831	$ 39,579
Weighted-average interest rate	3.0 %	3.8 %	2.5 %	3.6 %	3.0 %	4.5 %	3.9 %
Interest rate derivatives							
Interest rate swaps:							
Fixed to variable	$ 1,500	$ —	$ —	$ —	$ 1,250	$ 3,521	$ 6,271
Weighted-average pay rate	6.7 %	— %	— %	— %	5.7 %	6.9 %	6.6 %
Weighted-average receive rate	3.3 %	— %	— %	— %	1.5 %	2.9 %	2.7 %

(1) Includes deferred loan costs, discounts, fair value hedges, foreign-held debt and secured debt.

As of January 31, 2024, our variable rate borrowings, including the effect of our commercial paper and interest rate swaps, represented 18% of our total short-term and long-term debt. Based on January 31, 2024 debt levels, a 100 basis point change in prevailing market rates would cause our annual interest costs to change by approximately $0.1 billion.

Foreign Currency Risk

We are exposed to fluctuations in currency exchange rates as a result of our investments and operations in countries other than the U.S., as well as our foreign-currency-denominated long-term debt. For fiscal 2024, movements in currency exchange rates and the related impact on the translation of the balance sheets resulted in the $0.3 billion net gain in the currency translation and other category of accumulated other comprehensive loss.

We hedge a portion of our foreign currency risk by entering into currency swaps. The aggregate fair value of these swaps was in a liability position of $1.3 billion and $1.4 billion as of January 31, 2024 and January 31, 2023, respectively. The change in the fair value of these swaps was due to fluctuations in currency exchange rates, primarily due to the strengthening of certain currencies relative to the U.S. dollar in fiscal 2024. The hypothetical result of a uniform 10% weakening in the value of the U.S. dollar relative to other currencies underlying these swaps would have resulted in a change in the value of the swaps of $0.7 billion. A hypothetical 10% change in interest rates underlying these swaps from the market rates in effect as of January 31, 2024 would have resulted in a change in the value of the swaps of $0.1 billion.

In certain countries, we also enter into immaterial foreign currency forward contracts to hedge the purchase and payment of purchase commitments denominated in non-functional currencies.

Investment Risk

We are exposed to investment risk primarily related to changes in the fair value of certain equity investments, including certain immaterial equity method investments where we have elected the fair value option, measured on a recurring basis. The amounts of gains and losses included in earnings from fair value changes for these investments are recorded within other gains and losses and resulted in a net loss of $3.8 billion in fiscal 2024 primarily due to net decreases in the underlying stock prices of these investments. As of January 31, 2024, the fair value of our equity investments, including certain equity method investments, measured on a recurring basis was $7.2 billion. As of January 31, 2024, a hypothetical 10% change in the stock price of such investments would have changed the fair value of such investments by approximately $0.7 billion.

<u>ITEM 8.</u> **<u>FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA</u>**

Consolidated Financial Statements of Walmart Inc.
For the Fiscal Year Ended January 31, 2024

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Report of Independent Registered Public Accounting Firm

To the Shareholders and the Board of Directors of Walmart Inc.

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Walmart Inc. (the Company) as of January 31, 2024 and 2023, the related consolidated statements of income, comprehensive income, shareholders' equity and cash flows for each of the three years in the period ended January 31, 2024, and the related notes (collectively referred to as the "Consolidated Financial Statements"). In our opinion, the Consolidated Financial Statements present fairly, in all material respects, the financial position of the Company at January 31, 2024 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended January 31, 2024, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of January 31, 2024, based on criteria established in Internal Control-Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) and our report dated March 15, 2024 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the Consolidated Financial Statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the account or disclosures to which it relates.

	Contingencies
Description of the Matter	As described in Note 10 to the Consolidated Financial Statements, at January 31, 2024, the Company is involved in a number of legal proceedings and certain regulatory matters. The Company records a liability for those legal proceedings and regulatory matters when management determines it is probable that a loss has been incurred and the amount of the loss can be reasonably estimated. The Company also discloses when it is reasonably possible that a material loss may be incurred. In assessing the probability of occurrence and whether an estimate of loss can be reasonably estimated for a particular legal proceeding, management exercises judgment on matters relevant to each proceeding. Auditing management's accounting for, and disclosure of, loss contingencies was complex and highly judgmental as it involved our assessment of the significant judgments made by management when assessing the probability of loss for contingencies or when determining whether an estimate of the loss or range of loss could be made.

How We Addressed the Matter in Our Audit	We obtained an understanding, evaluated the design and tested the operating effectiveness of controls over the identification and evaluation of contingencies. For example, we tested controls over the Company's assessment of the likelihood of loss and the Company's determinations regarding the measurement of loss. To test the Company's assessment of the probability of loss or determination of an estimate of loss, or range of loss, among other procedures, we read the minutes of the meetings of the board of directors and committees of the board of directors, reviewed documents provided to the Company by certain outside legal counsel, read letters received directly by us from internal and outside legal counsel, evaluated the current status of contingencies based on discussions with internal and outside legal counsel, and obtained representations from management. We also assessed the adequacy of the related disclosures.

/s/ Ernst & Young LLP

We have served as the Company's auditor since 1969.

Rogers, Arkansas
March 15, 2024

Report of Independent Registered Public Accounting Firm

To the Shareholders and the Board of Directors of Walmart Inc.

Opinion on Internal Control Over Financial Reporting

We have audited Walmart Inc.'s internal control over financial reporting as of January 31, 2024, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, Walmart Inc. (the Company) maintained, in all material respects, effective internal control over financial reporting as of January 31, 2024, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of January 31, 2024 and 2023, the related consolidated statements of income, comprehensive income, shareholders' equity and cash flows for each of the three years in the period ended January 31, 2024, and the related notes and our report dated March 15, 2024 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP

Rogers, Arkansas
March 15, 2024

Walmart Inc.
Consolidated Statements of Income

(Amounts in millions, except per share data)	Fiscal Years Ended January 31, 2024	2023	2022
Revenues:			
Net sales	$ 642,637	$ 605,881	$ 567,762
Membership and other income	5,488	5,408	4,992
Total revenues	648,125	611,289	572,754
Costs and expenses:			
Cost of sales	490,142	463,721	429,000
Operating, selling, general and administrative expenses	130,971	127,140	117,812
Operating income	27,012	20,428	25,942
Interest:			
Debt	2,259	1,787	1,674
Finance lease	424	341	320
Interest income	(546)	(254)	(158)
Interest, net	2,137	1,874	1,836
Loss on extinguishment of debt	—	—	2,410
Other (gains) and losses	3,027	1,538	3,000
Income before income taxes	21,848	17,016	18,696
Provision for income taxes	5,578	5,724	4,756
Consolidated net income	16,270	11,292	13,940
Consolidated net (income) loss attributable to noncontrolling interest	(759)	388	(267)
Consolidated net income attributable to Walmart	$ 15,511	$ 11,680	$ 13,673
Net income per common share:			
Basic net income per common share attributable to Walmart	$ 1.92	$ 1.43	$ 1.63
Diluted net income per common share attributable to Walmart	1.91	1.42	1.62
Weighted-average common shares outstanding:			
Basic	8,077	8,171	8,376
Diluted	8,108	8,202	8,415
Dividends declared per common share	$ 0.7600	$ 0.7467	$ 0.7333

See accompanying notes.

Walmart Inc.

Consolidated Statements of Comprehensive Income

(Amounts in millions)	Fiscal Years Ended January 31, 2024	2023	2022
Consolidated net income	$ 16,270	$ 11,292	$ 13,940
Consolidated net (income) loss attributable to noncontrolling interest	(759)	388	(267)
Consolidated net income attributable to Walmart	15,511	11,680	13,673
Other comprehensive income (loss), net of income taxes			
Currency translation and other	899	(1,858)	2,442
Net investment hedges	—	—	(1,202)
Cash flow hedges	56	(203)	(444)
Minimum pension liability	(11)	5	1,974
Other comprehensive income (loss), net of income taxes	944	(2,056)	2,770
Other comprehensive (income) loss attributable to noncontrolling interest	(566)	404	230
Other comprehensive income (loss) attributable to Walmart	378	(1,652)	3,000
Comprehensive income, net of income taxes	17,214	9,236	16,710
Comprehensive (income) loss attributable to noncontrolling interest	(1,325)	792	(37)
Comprehensive income attributable to Walmart	$ 15,889	$ 10,028	$ 16,673

See accompanying notes.

Walmart Inc.
Consolidated Balance Sheets

(Amounts in millions)	As of January 31, 2024	2023
ASSETS		
Current assets:		
Cash and cash equivalents	$ 9,867	$ 8,625
Receivables, net	8,796	7,933
Inventories	54,892	56,576
Prepaid expenses and other	3,322	2,521
Total current assets	76,877	75,655
Property and equipment, net	110,810	100,760
Operating lease right-of-use assets	13,673	13,555
Finance lease right-of-use assets, net	5,855	4,919
Goodwill	28,113	28,174
Other long-term assets	17,071	20,134
Total assets	$ 252,399	$ 243,197
LIABILITIES, REDEEMABLE NONCONTROLLING INTEREST, AND EQUITY		
Current liabilities:		
Short-term borrowings	$ 878	$ 372
Accounts payable	56,812	53,742
Accrued liabilities	28,759	31,126
Accrued income taxes	307	727
Long-term debt due within one year	3,447	4,191
Operating lease obligations due within one year	1,487	1,473
Finance lease obligations due within one year	725	567
Total current liabilities	92,415	92,198
Long-term debt	36,132	34,649
Long-term operating lease obligations	12,943	12,828
Long-term finance lease obligations	5,709	4,843
Deferred income taxes and other	14,629	14,688
Commitments and contingencies		
Redeemable noncontrolling interest	222	237
Equity:		
Common stock	805	808
Capital in excess of par value	4,544	4,430
Retained earnings	89,814	83,135
Accumulated other comprehensive loss	(11,302)	(11,680)
Total Walmart shareholders' equity	83,861	76,693
Noncontrolling interest	6,488	7,061
Total equity	90,349	83,754
Total liabilities, redeemable noncontrolling interest, and equity	$ 252,399	$ 243,197

See accompanying notes.

Walmart Inc.

Consolidated Statements of Shareholders' Equity

(Amounts in millions)	Common Stock Shares	Common Stock Amount	Capital in Excess of Par Value	Retained Earnings	Accumulated Other Comprehensive Income (Loss)	Total Walmart Shareholders' Equity	Noncontrolling Interest	Total Equity
Balances as of February 1, 2021	8,464	$ 846	$ 3,082	$ 88,763	$ (11,766)	$ 80,925	$ 6,606	$ 87,531
Consolidated net income	—	—	—	13,673	—	13,673	267	13,940
Other comprehensive income (loss), net of income taxes	—	—	—	—	3,000	3,000	(230)	2,770
Cash dividends declared ($0.7333 per share)	—	—	—	(6,152)	—	(6,152)	—	(6,152)
Purchase of Company stock	(210)	(21)	(412)	(9,375)	—	(9,808)	—	(9,808)
Cash dividend declared to noncontrolling interest	—	—	—	—	—	—	(416)	(416)
Sale of subsidiary stock	—	—	952	—	—	952	2,287	3,239
Other	30	3	665	(5)	—	663	124	787
Balances as of January 31, 2022	8,284	828	4,287	86,904	(8,766)	83,253	8,638	91,891
Consolidated net income	—	—	—	11,680	—	11,680	(388)	11,292
Other comprehensive (loss), net of income taxes	—	—	—	—	(1,652)	(1,652)	(404)	(2,056)
Cash dividends declared ($0.7467 per share)	—	—	—	(6,114)	—	(6,114)	—	(6,114)
Purchase of Company stock	(221)	(22)	(518)	(9,326)	—	(9,866)	—	(9,866)
Cash dividend declared to noncontrolling interest	—	—	—	—	—	—	(449)	(449)
Purchase of noncontrolling interest	—	—	(18)	—	(1,262)	(1,280)	(493)	(1,773)
Sale of subsidiary stock	—	—	48	—	—	48	18	66
Other	17	2	631	(9)	—	624	139	763
Balances as of January 31, 2023	8,080	808	4,430	83,135	(11,680)	76,693	7,061	83,754
Consolidated net income	—	—	—	15,511	—	15,511	774	16,285
Other comprehensive income, net of income taxes	—	—	—	—	378	378	566	944
Cash dividends declared ($0.7600 per share)	—	—	—	(6,140)	—	(6,140)	—	(6,140)
Purchase of Company stock	(55)	(6)	(150)	(2,635)	—	(2,791)	—	(2,791)
Cash dividend declared to noncontrolling interest	—	—	—	—	—	—	(776)	(776)
Purchase of noncontrolling interest	—	—	(1,076)	—	—	(1,076)	(1,367)	(2,443)
Sale of subsidiary stock	—	—	562	—	—	562	154	716
Other	29	3	778	(57)	—	724	76	800
Balances as of January 31, 2024	8,054	805	4,544	89,814	(11,302)	83,861	6,488	90,349

See accompanying notes.

Walmart Inc.
Consolidated Statements of Cash Flows

(Amounts in millions)	Fiscal Years Ended January 31, 2024	2023	2022
Cash flows from operating activities:			
Consolidated net income	$ 16,270	$ 11,292	$ 13,940
Adjustments to reconcile consolidated net income to net cash provided by operating activities:			
Depreciation and amortization	11,853	10,945	10,658
Net unrealized and realized (gains) and losses	3,193	1,683	2,440
Losses on disposal of business operations	—	—	433
Deferred income taxes	(175)	449	(755)
Loss on extinguishment of debt	—	—	2,410
Other operating activities	2,642	1,919	1,652
Changes in certain assets and liabilities, net of effects of acquisitions and dispositions:			
Receivables, net	(797)	240	(1,796)
Inventories	2,017	(528)	(11,764)
Accounts payable	2,515	(1,425)	5,520
Accrued liabilities	(1,324)	4,393	1,404
Accrued income taxes	(468)	(127)	39
Net cash provided by operating activities	35,726	28,841	24,181
Cash flows from investing activities:			
Payments for property and equipment	(20,606)	(16,857)	(13,106)
Proceeds from the disposal of property and equipment	250	170	394
Proceeds from disposal of certain operations, net of divested cash	135	—	7,935
Payments for business acquisitions, net of cash acquired	(9)	(740)	(359)
Other investing activities	(1,057)	(295)	(879)
Net cash used in investing activities	(21,287)	(17,722)	(6,015)
Cash flows from financing activities:			
Net change in short-term borrowings	512	(34)	193
Proceeds from issuance of long-term debt	4,967	5,041	6,945
Repayments of long-term debt	(4,217)	(2,689)	(13,010)
Premiums paid to extinguish debt	—	—	(2,317)
Dividends paid	(6,140)	(6,114)	(6,152)
Purchase of Company stock	(2,779)	(9,920)	(9,787)
Dividends paid to noncontrolling interest	(763)	(444)	(424)
Purchase of noncontrolling interest	(3,462)	(827)	—
Sale of subsidiary stock	716	66	3,239
Other financing activities	(2,248)	(2,118)	(1,515)
Net cash used in financing activities	(13,414)	(17,039)	(22,828)
Effect of exchange rates on cash, cash equivalents and restricted cash	69	(73)	(140)
Net increase (decrease) in cash, cash equivalents and restricted cash	1,094	(5,993)	(4,802)
Change in cash and cash equivalents reclassified from assets held for sale	—	—	1,848
Cash, cash equivalents and restricted cash at beginning of year	8,841	14,834	17,788
Cash, cash equivalents and restricted cash at end of year	$ 9,935	$ 8,841	$ 14,834
Supplemental disclosure of cash flow information:			
Income taxes paid	$ 5,879	$ 3,310	$ 5,918
Interest paid	2,519	2,051	2,237

See accompanying notes.

Walmart Inc.

Notes to Consolidated Financial Statements

Note 1. Summary of Significant Accounting Policies

General

Walmart Inc. ("Walmart" or the "Company") is a people-led, technology-powered omni-channel retailer dedicated to helping people around the world save money and live better – anytime and anywhere – by providing the opportunity to shop in both retail stores and through eCommerce. Through innovation, the Company is striving to continuously improve a customer-centric experience that seamlessly integrates eCommerce and retail stores in an omni-channel offering that saves time for its customers.

The Company's operations comprise three reportable segments: Walmart U.S., Walmart International and Sam's Club.

Principles of Consolidation

The Consolidated Financial Statements include the accounts of Walmart and its subsidiaries as of and for the fiscal years ended January 31, 2024 ("fiscal 2024"), January 31, 2023 ("fiscal 2023") and January 31, 2022 ("fiscal 2022"). Intercompany accounts and transactions have been eliminated in consolidation. The Company consolidates variable interest entities where it has been determined that the Company is the primary beneficiary of those entities' operations. Investments in common stock or in-substance common stock for which the Company exercises significant influence but does not have control are accounted for under the equity method. These variable interest entities and equity method investments are immaterial to the Company's Consolidated Financial Statements.

The Company's Consolidated Financial Statements are based on a fiscal year ending on January 31 for the United States ("U.S.") and Canadian operations. The Company consolidates all other operations generally using a one-month lag and based on a calendar year. There were no significant intervening events during the month of January 2024 related to the operations consolidated using a lag that materially affected the Consolidated Financial Statements.

Use of Estimates

The Consolidated Financial Statements have been prepared in conformity with U.S. generally accepted accounting principles ("GAAP"). Those principles require management to make estimates and assumptions that affect the reported amounts of assets and liabilities. Management's estimates and assumptions also affect the disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenues and expenses during the reporting period. Actual results may differ from those estimates.

Common Stock Split

On February 23, 2024, the Company effected a 3-for-1 forward split of its common stock and a proportionate increase in the number of authorized shares. All share and per share information, including share based compensation, throughout this Annual Report on Form 10-K has been retroactively adjusted to reflect the stock split. The shares of common stock retain a par value of $0.10 per share. Accordingly, an amount equal to the par value of the increased shares resulting from the stock split was reclassified from capital in excess of par value to common stock.

Cash and Cash Equivalents

The Company considers investments with a maturity when purchased of three months or less to be cash equivalents. All credit card, debit card and electronic transfer transactions that process in less than seven days are classified as cash and cash equivalents. The amounts due from banks for these transactions classified as cash and cash equivalents totaled $2.1 billion and $2.0 billion as of January 31, 2024 and 2023, respectively.

The Company's cash balances are held in various locations around the world. Of the Company's $9.9 billion and $8.6 billion in cash and cash equivalents as of January 31, 2024 and January 31, 2023, approximately 60% and 62% were held outside of the U.S., respectively. Cash and cash equivalents held outside of the U.S. are generally utilized to support liquidity needs in the Company's non-U.S. operations.

The Company uses intercompany financing arrangements in an effort to ensure cash can be made available in the country in which it is needed with the minimum cost possible.

As of January 31, 2024 and 2023, cash and cash equivalents of approximately $3.5 billion and $2.9 billion, respectively, may not be freely transferable to the U.S. due to local laws, other restrictions or are subject to the approval of the noncontrolling interest shareholders.

Receivables

Receivables are stated at their carrying values, net of a reserve for doubtful accounts, and are primarily due from the following: customers, which includes pharmacy insurance companies as well as advertisers, and banks for customer credit, debit cards and electronic transfer transactions that take in excess of seven days to process; suppliers for marketing or incentive programs; governments for income taxes; and real estate transactions. Net receivables from transactions with customers were $3.7 billion as of January 31, 2024 and January 31, 2023.

Inventories

The Company utilizes various inventory methods to account for and value its inventories depending upon the nature of the store formats and businesses in each of its segments, resulting in inventories that are recorded at the lower of cost or market or net realizable value, as appropriate.

- Walmart U.S. Segment - Inventories are primarily accounted for under the retail inventory method of accounting ("RIM") to determine inventory cost, using the last-in, first-out ("LIFO") valuation method. RIM generally results in inventory being valued at the lower of cost or market as permanent markdowns are immediately recorded as a reduction of the retail value of inventory.
- Walmart International Segment – Depending on the store format in each market, inventories are generally accounted for using either the RIM or weighted-average cost method, using the first-in, first-out valuation method.
- Sam's Club Segment - The majority of this segment's inventory is accounted for and valued using the weighted-average cost LIFO method.

For those segments that utilize the LIFO method, the Company records an adjustment each quarter, if necessary, for the projected annual effect of inflation or deflation. These estimates are adjusted to actual results determined at year end for inflation or deflation and inventory levels.

Property and Equipment

Property and equipment are initially recorded at cost. Gains or losses on disposition are recognized as earned or incurred. Costs of major improvements are capitalized, while costs of normal repairs and maintenance are expensed as incurred. The following table summarizes the Company's property and equipment balances and includes the estimated useful lives that are generally used to depreciate the assets on a straight-line basis:

(Dollars in millions)	**Estimated Useful Lives (in Years)**	**As of January 31, 2024**	**2023**
Land	N/A	$ 19,562	$ 19,317
Buildings and improvements	3 - 40	111,767	104,554
Fixtures and equipment	2 - 30	72,161	65,235
Transportation equipment	3 - 15	2,979	2,462
Construction in progress	N/A	13,390	10,802
Property and equipment		219,859	202,370
Accumulated depreciation		(109,049)	(101,610)
Property and equipment, net		$ 110,810	$ 100,760

Leasehold improvements are depreciated or amortized over the shorter of the estimated useful life of the asset or the remaining expected lease term. Total depreciation and amortization expense for property and equipment, property under finance leases and intangible assets for fiscal 2024, 2023 and 2022 was $11.9 billion, $10.9 billion and $10.7 billion, respectively.

Leases

For any new or modified lease, the Company, at the inception of the contract, determines whether a contract is or contains a lease. The Company records right-of-use ("ROU") assets and lease obligations for its finance and operating leases, which are initially recognized based on the discounted future lease payments over the term of the lease. If the rate implicit in the Company's leases is not readily determinable, the Company's applicable incremental borrowing rate is used in calculating the present value of the sum of the lease payments.

Lease term is defined as the non-cancelable period of the lease plus any options to extend or terminate the lease when it is reasonably certain that the Company will exercise the option. The Company has elected not to recognize ROU asset and lease obligations for its short-term leases, which are defined as leases with an initial term of 12 months or less.

For a majority of all classes of underlying assets, the Company has elected to not separate lease from non-lease components. For leases in which the lease and non-lease components have been combined, the variable lease expense includes expenses such as common area maintenance, utilities and repairs and maintenance.

Impairment of Long-Lived Assets

Management reviews long-lived assets for indicators of impairment whenever events or changes in circumstances indicate that the carrying amount may not be recoverable. The evaluation is performed at the lowest level of identifiable cash flows, which is at the individual store or club level. Undiscounted cash flows expected to be generated by the related assets are estimated over the assets' useful lives based on updated projections. If the evaluation indicates that the carrying amount of the assets may not be recoverable, any potential impairment is measured based upon the fair value of the related asset or asset group as determined by an appropriate market appraisal or other valuation technique.

Goodwill and Other Acquired Intangible Assets

Goodwill represents the excess of the purchase price over the fair value of net assets acquired in business combinations and is allocated to the appropriate reporting unit when acquired. Other acquired intangible assets are stated at the fair value acquired as determined by a valuation technique commensurate with the intended use of the related asset. Goodwill and indefinite-lived intangible assets are not amortized; rather, they are evaluated for impairment annually and whenever events or changes in circumstances indicate that the value of the asset may be impaired. Definite-lived intangible assets are considered long-lived assets and are amortized on a straight-line basis over the periods that expected economic benefits will be provided.

Goodwill is typically assigned to the reporting unit which consolidates the acquisition. Components within the same reportable segment are aggregated and deemed a single reporting unit if the components have similar economic characteristics. As of January 31, 2024, the Company's reporting units consisted of Walmart U.S., Walmart International and Sam's Club. Goodwill and other indefinite-lived acquired intangible assets are evaluated for impairment using either a qualitative or quantitative approach for each of the Company's reporting units. Generally, a qualitative assessment is first performed to determine whether a quantitative goodwill impairment test is necessary. If management determines, after performing an assessment based on the qualitative factors, that the fair value of the reporting unit is more likely than not less than the carrying amount, or that a fair value of the reporting unit substantially in excess of the carrying amount cannot be assured, then a quantitative goodwill impairment test would be required. The quantitative test for goodwill impairment is performed by determining the fair value of the related reporting units. Fair value is measured based on the discounted cash flow method and relative market-based approaches. Management has performed its evaluation and determined the fair value of each reporting unit is significantly greater than the carrying amount and, accordingly, the Company has not recorded any impairment charges related to goodwill during fiscal 2024, fiscal 2023 or fiscal 2022.

The following table reflects goodwill activity, by reportable segment, for fiscal 2024 and 2023:

(Amounts in millions)	Walmart U.S.	Walmart International	Sam's Club	Total
Balances as of February 1, 2022	$ 2,941	$ 25,752	$ 321	$ 29,014
Changes in currency translation and other	—	(1,475)	—	(1,475)
Acquisitions	433	202	—	635
Balances as of January 31, 2023	3,374	24,479	321	28,174
Changes in currency translation and other	(10)	(58)	—	(68)
Acquisitions	—	7	—	7
Balances as of January 31, 2024	$ 3,364	$ 24,428	$ 321	$ 28,113

Intangible assets are recorded in other long-term assets in the Company's Consolidated Balance Sheets. As of January 31, 2024 and 2023, the Company had $4.1 billion and $4.3 billion, respectively, in indefinite-lived intangible assets which primarily consists of acquired trade names. There were no significant impairment charges related to intangible assets for fiscal 2024, 2023 or 2022.

Fair Value Measurement

The Company records and discloses certain financial and non-financial assets and liabilities at fair value. The fair value of an asset is the price at which the asset could be sold in an orderly transaction between unrelated, knowledgeable and willing parties able to engage in the transaction. The fair value of a liability is the amount that would be paid to transfer the liability to a new obligor in a transaction between such parties, not the amount that would be paid to settle the liability with the creditor. Refer to Note 8 for more information.

Investments

Investments in equity securities are recorded in other long-term assets in the Consolidated Balance Sheets. Changes in the fair value of certain equity securities, as well as certain immaterial equity method investments where the Company has elected the fair value option, are measured on a recurring basis and recognized within other gains and losses in the Consolidated Statements of Income. These fair value changes, along with certain other immaterial investment activity, resulted in net losses of $3.8 billion, $1.7 billion and $2.4 billion for fiscal 2024, 2023 and 2022, respectively, primarily due to net changes in the underlying stock prices of those investments. Refer to Note 8 for details. Equity investments without readily determinable fair values are carried at cost and adjusted for any observable price changes or impairments within other gains and losses in the Consolidated Statements of Income.

Investments in debt securities classified as trading are reported at fair value and adjustments in fair value are recorded within other gains and losses in the Consolidated Statements of Income. As of January 31, 2024 and January 31, 2023, the Company had $1.2 billion and $0.5 billion, respectively, in debt securities classified as trading.

Indemnification Liabilities

The Company has provided certain indemnifications in connection with its divestitures and has recorded indemnification liabilities equal to the estimated fair value of the obligations upon inception. As of January 31, 2024 and January 31, 2023, the Company had $0.7 billion and $0.6 billion, respectively, of certain legal indemnification liabilities recorded within deferred income taxes and other in the Consolidated Balance Sheets. The maximum of potential future payments under these indemnities was $3.2 billion, based on exchange rates as of January 31, 2024.

Supplier Financing Program Obligations

In September 2022, the FASB issued *ASU 2022-04, Liabilities - Supplier Finance Programs (Subtopic 405-50): Disclosure of Supplier Finance Program Obligations*, which enhances the transparency about the use of supplier finance programs for investors and other allocators of capital. The Company adopted this ASU as of February 1, 2023, other than the roll-forward disclosure requirement, which the Company will adopt in fiscal 2025.

The Company has supplier financing programs with financial institutions, in which the Company agrees to pay the financial institution the stated amount of confirmed invoices on the invoice due date for participating suppliers. Participation in these programs is optional and solely up to the supplier, who negotiates the terms of the arrangement directly with the financial institution and may allow early payment. Supplier participation in these programs has no bearing on the Company's amounts due. The payment terms that the Company has with participating suppliers under these programs generally range between 30 and 90 days. The Company does not have an economic interest in a supplier's participation in the program or a direct financial relationship with the financial institution funding the program. The Company is responsible for ensuring that participating financial institutions are paid according to the terms negotiated with the supplier, regardless of whether the supplier elects to receive early payment from the financial institution. The outstanding payment obligations to financial institutions under these programs were $5.3 billion and $5.2 billion, as of January 31, 2024 and January 31, 2023, respectively. These obligations are generally classified as accounts payable within the Consolidated Balance Sheets. The activity related to these programs is classified as an operating activity within the Consolidated Statements of Cash Flows.

Self Insurance Reserves

The Company self-insures a number of risks, including, but not limited to, workers' compensation, general liability, auto liability, product liability and certain employee-related healthcare benefits. Standard actuarial procedures and data analysis are used to estimate the liabilities associated with these risks on an undiscounted basis. The recorded liabilities reflect the ultimate cost for claims incurred but not paid and any estimable administrative run-out expenses related to the processing of these outstanding claim payments. On a regular basis, the liabilities are evaluated for appropriateness with claims reserve valuations. To limit exposure to some risks, the Company maintains insurance coverage with varying limits and retentions, including stop-loss insurance coverage for workers' compensation, general liability and auto liability.

Derivatives

The Company uses derivatives for hedging purposes to manage its exposure to changes in interest and currency exchange rates, as well as to maintain an appropriate mix of fixed- and variable-rate debt. Use of derivatives in hedging programs subjects the Company to certain risks, such as market and credit risks. The Company may be exposed to credit-related losses in the event of nonperformance by its counterparties to derivatives. Credit risk is monitored through established approval procedures, including setting concentration limits by counterparty, reviewing credit ratings and requiring collateral from the counterparty. The Company enters into derivatives with counterparties rated generally "A-" or better by nationally recognized credit rating agencies. The Company is subject to master netting arrangements which provides set-off and close-out netting of exposures with counterparties, but the Company does not offset derivative assets and liabilities in its Consolidated Balance Sheets. The Company's collateral arrangements require the counterparty in a net liability position in excess of pre-determined thresholds, after considering the effects of netting arrangements, to pledge cash collateral. Cash collateral received from counterparties and cash collateral provided to counterparties under these arrangements was not significant as of January 31, 2024 and 2023.

In order to qualify for hedge accounting, at the inception of the hedging relationship, the Company formally documents its risk management objective and strategy for undertaking the hedging transaction, as well as its designation of the hedge. If a derivative is recorded using hedge accounting, depending on the nature of the hedge, derivative gains and losses are recorded through the same financial statement line item in earnings or are recognized in accumulated other comprehensive loss until the hedged item is recognized in earnings. Derivatives that do not meet the criteria for hedge accounting, or contracts for which the Company has not elected hedge accounting, are recorded at fair value with unrealized gains or losses reported in earnings. Derivatives with an unrealized gain are recorded in the Company's Consolidated Balance Sheets as either current or non-current assets, based on maturity date, and derivatives with an unrealized loss are recorded as either current or non-current liabilities, based on maturity date. Refer to Note 8 for the presentation of the Company's derivative assets and liabilities.

Fair Value Hedges

The Company is a party to receive fixed-rate, pay variable-rate interest rate swaps that the Company uses to hedge the fair value of fixed-rate debt. All interest rate swaps designated as fair value hedges of the related long-term debt meet the shortcut method requirements under U.S. GAAP. Accordingly, changes in the fair values of these interest rate swaps are considered to exactly offset changes in the fair value of the underlying long-term debt. These derivatives will mature on dates ranging from April 2024 to September 2031.

Cash Flow Hedges

The Company is a party to receive fixed-rate, pay fixed-rate cross currency interest rate swaps used to hedge the currency exposure associated with the forecasted payments of principal and interest of certain non-U.S. denominated debt. The Company records changes in the fair value of these swaps in accumulated other comprehensive loss which is subsequently reclassified into earnings in the period that the hedged forecasted transaction affects earnings. These derivatives will mature on dates ranging from July 2024 to January 2039.

Net Investment Hedges

Prior to the divestiture of the Company's operations in the United Kingdom and Japan as discussed in Note 12, the Company was a party to receive fixed-rate, pay fixed-rate cross currency interest rate swaps used to hedge the currency exposure associated with net investments of these foreign operations. Changes in fair value attributable to the hedged risk were recorded in accumulated other comprehensive loss. The Company also previously designated certain foreign currency denominated long-term debt as a hedge of currency exposure associated with the net investment of these divested operations and recorded foreign currency gain or loss associated with designated long-term debt in accumulated other comprehensive loss. Upon closing of the sale of the Company's operations in the U.K. and Japan during the first quarter of fiscal 2022, these amounts were released from accumulated other comprehensive loss as discussed in Note 4.

Income Taxes

Income taxes are accounted for under the balance sheet method. Deferred tax assets and liabilities are recognized for the estimated future tax consequences attributable to differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases ("temporary differences"). Deferred tax assets and liabilities are measured using enacted tax rates in effect for the year in which those temporary differences are expected to be recovered or settled. The effect on deferred tax assets and liabilities of a change in tax rate is recognized in income in the period that includes the enactment date.

Deferred tax assets are evaluated for future realization and reduced by a valuation allowance to the extent that a portion is not more likely than not to be realized. Many factors are considered when assessing whether it is more likely than not that the deferred tax assets will be realized, including recent cumulative earnings, expectations of future taxable income, carryforward periods, and other relevant quantitative and qualitative factors. The recoverability of the deferred tax assets is evaluated by assessing the adequacy of future expected taxable income from all sources, including reversal of taxable temporary differences, forecasted operating earnings and available tax planning strategies. These sources of income rely on estimates.

In determining the provision for income taxes, an annual effective income tax rate is used based on annual income, permanent differences between book and tax income, and statutory income tax rates. Discrete events such as audit settlements or changes in tax laws are recognized in the period in which they occur.

The Company records a liability for unrecognized tax benefits resulting from uncertain tax positions taken or expected to be taken in a tax return. The Company records interest and penalties related to unrecognized tax benefits in interest expense and operating, selling, general and administrative expenses, respectively, in the Company's Consolidated Statements of Income. Refer to Note 9 for additional income tax disclosures.

Redeemable Noncontrolling Interest

Noncontrolling interests that are redeemable outside the Company's control at fixed or determinable prices and dates are presented as temporary equity in the Consolidated Balance Sheets. Redeemable noncontrolling interests are recorded at the greater of the redemption fair value or the carrying value of the noncontrolling interest and adjusted each reporting period for income, loss and any distributions made. Remeasurements to the redemption value of the redeemable noncontrolling interest are recognized in capital in excess of par. As of January 31, 2024, the Company has a redeemable noncontrolling interest related to an acquisition in the Walmart U.S. segment as the minority interest owner holds a put option which may require the Company to purchase its interest beginning in December 2027, with annual options thereafter.

Revenue Recognition

Net Sales

The Company recognizes sales revenue, net of sales taxes and estimated sales returns, at the time it sells merchandise or services to the customer. eCommerce sales include shipping revenue and are recorded upon delivery to the customer. Estimated sales returns are calculated based on expected returns.

Membership Fee Revenue

The Company recognizes membership fee revenue over the term of the membership, which is typically 12 months. Membership fee revenue was $3.1 billion for fiscal 2024, $2.6 billion for fiscal 2023 and $2.2 billion for fiscal 2022. Membership fee revenue is included in membership and other income in the Company's Consolidated Statements of Income. Deferred membership fee revenue is included in accrued liabilities in the Company's Consolidated Balance Sheets.

Gift Cards

Customer purchases of gift cards are not recognized as sales until the card is redeemed and the customer purchases merchandise using the gift card. Gift cards in the U.S. and some countries do not carry an expiration date; therefore, customers and members can redeem their gift cards for merchandise and services indefinitely. Gift cards in some countries where the Company does business have expiration dates. While gift cards are generally redeemed within 12 months, a certain number of gift cards, both with and without expiration dates, will not be fully redeemed. Management estimates unredeemed balances and recognizes revenue for these amounts in membership and other income in the Company's Consolidated Statements of Income over the expected redemption period.

Financial, Advertising and Other Services

The Company recognizes revenue from service transactions at the time the service is performed. Generally, revenue from services is classified as a component of net sales in the Company's Consolidated Statements of Income.

Cost of Sales

Cost of sales includes actual product cost, the cost of transportation to the Company's distribution facilities, stores and clubs from suppliers, the cost of transportation from the Company's distribution facilities to the stores, clubs and customers and the cost of warehousing for the Sam's Club segment and import distribution centers. Cost of sales is reduced by supplier payments that are not a reimbursement of specific, incremental and identifiable costs.

Payments from Suppliers

The Company receives consideration from suppliers for various programs, primarily volume incentives, warehouse allowances and reimbursements for specific programs such as markdowns, margin protection, certain advertising arrangements and supplier-specific fixtures. Payments from suppliers are accounted for as a reduction of cost of sales and recognized in the Company's Consolidated Statements of Income when the related inventory is sold, except in certain limited situations when the payment is a reimbursement of specific, incremental and identifiable costs.

Operating, Selling, General and Administrative Expenses

Operating, selling, general and administrative expenses include all operating costs of the Company, except cost of sales, as described above. As a result, the majority of the cost of warehousing and occupancy for the Walmart U.S. and Walmart International segments' distribution facilities is included in operating, selling, general and administrative expenses. Because the Company only includes a portion of the cost of its Walmart U.S. and Walmart International segments' distribution facilities in cost of sales, its gross profit and gross profit as a percentage of net sales may not be comparable to those of other retailers that may include all costs related to their distribution facilities in cost of sales and in the calculation of gross profit.

Advertising Costs

Advertising costs are expensed as incurred, and consist primarily of digital, television and print advertisements that are recorded in operating, selling, general and administrative expenses in the Company's Consolidated Statements of Income. Advertising costs were $4.4 billion, $4.1 billion and $3.9 billion for fiscal 2024, 2023 and 2022, respectively.

Currency Translation

The assets and liabilities of all international subsidiaries are translated from the respective local currency to the U.S. dollar using exchange rates at the balance sheet date. Related translation adjustments are recorded as a component of accumulated other comprehensive loss. The Company's Consolidated Statements of Income of all international subsidiaries are translated from the respective local currencies to the U.S. dollar using average exchange rates for the period covered by the income statements.

Recent Accounting Pronouncements

In November 2023, the FASB issued ASU 2023-07, *Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures*, which updates reportable segment disclosure requirements primarily through enhanced disclosures about significant segment expenses. The amendments are effective for fiscal years beginning after December 15, 2023, and for interim periods within fiscal years beginning after December 15, 2024. Early adoption is permitted. The amendments should be applied retrospectively to all prior periods presented in the financial statements. Management is currently evaluating this ASU to determine its impact on the Company's disclosures.

In December 2023, the FASB issued ASU 2023-09, *Income Taxes (Topic 740): Improvements to Income Tax Disclosures*, which expands the requirements for income tax disclosures in order to provide greater transparency. The amendments are effective for fiscal years beginning after December 15, 2024. Early adoption is permitted. The amendments should be applied prospectively. Management is currently evaluating this ASU to determine its impact on the Company's disclosures.

Note 2. Net Income Per Common Share

Basic net income per common share attributable to Walmart is based on the weighted-average common shares outstanding during the relevant period. Diluted net income per common share attributable to Walmart is based on the weighted-average common shares outstanding during the relevant period adjusted for the dilutive effect of share-based awards. The Company did not have significant share-based awards outstanding that were antidilutive and not included in the calculation of diluted net income per common share attributable to Walmart for fiscal 2024, 2023 and 2022.

The following table provides a reconciliation of the numerators and denominators used to determine basic and diluted net income per common share attributable to Walmart:

	Fiscal Years Ended January 31,		
(Amounts in millions, except per share data)	**2024**	**2023**	**2022**
Numerator			
Consolidated net income	$ 16,270	$ 11,292	$ 13,940
Consolidated net (income) loss attributable to noncontrolling interest	(759)	388	(267)
Consolidated net income attributable to Walmart	$ 15,511	$ 11,680	$ 13,673
Denominator			
Weighted-average common shares outstanding, basic	8,077	8,171	8,376
Dilutive impact of stock options and other share-based awards	31	31	39
Weighted-average common shares outstanding, diluted	8,108	8,202	8,415
Net income per common share attributable to Walmart			
Basic	$ 1.92	$ 1.43	$ 1.63
Diluted	1.91	1.42	1.62

Note 3. Shareholders' Equity

The total authorized shares of $0.10 par value common stock is 33.0 billion, of which 8.1 billion were issued and outstanding as of January 31, 2024 and 2023. The total authorized shares of $0.10 par value preferred stock is 0.1 billion; none of which were issued or outstanding for any period presented.

Purchases and Sales of Subsidiary Stock

During fiscal 2024, the Company paid $3.5 billion to acquire shares from certain Flipkart noncontrolling interest holders and settle the liability to former noncontrolling interest holders of PhonePe. The Company's ownership of Flipkart increased from approximately 75% as of January 31, 2023 to approximately 85% as of January 31, 2024.

Also during fiscal 2024, the Company received $0.7 billion related to new rounds of equity funding for the Company's majority owned PhonePe subsidiary, which decreased the Company's ownership from approximately 89% as of January 31, 2023 to approximately 84% as of January 31, 2024.

During fiscal 2023, the Company completed a $0.4 billion buyout of the noncontrolling interest shareholders of the Company's Massmart subsidiary. This transaction increased the Company's ownership in Massmart from approximately 53% to 100%. Additionally, the Company completed a $0.4 billion acquisition of Alert Innovation, which was previously consolidated as a variable interest entity, and resulted in the Company becoming a 100% owner.

Also during fiscal 2023, the Company increased its ownership in PhonePe from approximately 76% to approximately 89% as part of the separation from the Company's majority-owned Flipkart subsidiary. In consideration for the transaction, the Company initially recorded a liability to noncontrolling interest holders of $0.9 billion within accrued liabilities in the Company's Consolidated Balance Sheet as of January 31, 2023, which was paid during fiscal 2024.

During fiscal 2022, the Company received $3.2 billion primarily related to a new equity funding for the Company's majority-owned Flipkart subsidiary, which reduced the Company's ownership from approximately 83% as of January 31, 2021 to approximately 75% as of January 31, 2022.

Share-Based Compensation

The Company has awarded share-based compensation to associates and nonemployee directors of the Company. The compensation expense recognized for all stock incentive plans, including expense associated with plans of the Company's consolidated subsidiaries granted in the subsidiaries' respective stock, was $2.1 billion, $1.6 billion and $1.2 billion for fiscal 2024, 2023 and 2022, respectively. Share-based compensation expense is generally included in operating, selling, general and administrative expenses in the Company's Consolidated Statements of Income. The total income tax benefit recognized for share-based compensation was $0.5 billion, $0.4 billion and $0.3 billion for fiscal 2024, 2023 and 2022, respectively. The following table summarizes the Company's share-based compensation expense by award type for all plans:

	Fiscal Years Ended January 31,		
(Amounts in millions)	**2024**	**2023**	**2022**
Restricted stock units	$ 1,258	$ 927	$ 659
Restricted stock and performance-based restricted stock units	609	444	321
Other	226	207	183
Share-based compensation expense	$ 2,093	$ 1,578	$ 1,163

The Walmart Inc. Stock Incentive Plan of 2015 (the "Plan"), as subsequently amended and restated, was established to grant stock options, restricted (non-vested) stock, restricted stock units, performance share units and other equity compensation awards for which 780 million shares of Walmart common stock issued or to be issued under the Plan have been registered under the Securities Act of 1933. The Company believes that such awards serve to align the interests of its associates with those of its shareholders.

The Plan's award types are summarized as follows:

- *Restricted Stock Units.* Restricted stock units provide rights to Company stock after a specified service period. Beginning in fiscal 2023, restricted stock units generally vest at a rate of approximately 8% each quarter over a three year period from the date of grant. For grants made from fiscal 2020 through fiscal 2022, restricted stock units generally vest at a rate of 25% each year over a four year period from the date of the grant. Prior to fiscal 2020, 50% of restricted stock units generally vested three years from the grant date and the remaining 50% were vested five years from the grant date. The fair value of each restricted stock unit is determined on the date of grant using the stock price discounted for the expected dividend yield through the vesting period and is recognized ratably over the vesting period. The expected dividend yield is based on the anticipated dividends over the vesting period. The weighted-average discount for the dividend yield used to determine the fair value of restricted stock units granted in fiscal 2024, 2023 and 2022 was 2.2%, 2.3% and 3.8%, respectively.

- *Restricted Stock and Performance-based Restricted Stock Units.* Restricted stock awards are for shares that vest based on the passage of time and include restrictions related to employment. Performance-based restricted stock units vest based on the passage of time and achievement of performance criteria and generally range from 0% to 150% of the original award amount. Vesting periods for restricted stock are generally between one month and three years. Vesting periods for performance-based restricted stock units are generally between one and three years. Restricted stock and performance-based restricted stock units may be settled or deferred in stock and are accounted for as equity in the Company's Consolidated Balance Sheets. The fair value of restricted stock awards is determined on the date of grant and is expensed ratably over the vesting period. The fair value of performance-based restricted stock units is determined on the date of grant using the Company's stock price discounted for the expected dividend yield through the vesting period and is recognized over the vesting period. The weighted-average discount for the dividend yield used to determine the fair value of performance-based restricted stock units in fiscal 2024, 2023 and 2022 was 3.3%, 3.3% and 4.2%, respectively.

In addition to the Plan, Flipkart and PhonePe have share-based compensation plans for associates under which options to acquire their own common shares may be issued. These plans may be subject to performance or other conditions, including vesting upon an initial public offering. Share-based compensation expense associated with certain of these plans is included in the Other line in the table above.

The following table shows the activity for restricted stock units and restricted stock and performance-based restricted stock units during fiscal 2024:

	Restricted Stock Units		**Restricted Stock and Performance-based Restricted Stock Units**	
(Shares in thousands)	**Shares**	**Weighted-Average Grant-Date Fair Value Per Share**	**Shares**	**Weighted-Average Grant-Date Fair Value Per Share**
Outstanding as of February 1, 2023	48,660	$ 42.67	21,480	$ 46.29
Granted	35,751	48.37	12,999	49.07
Adjustment for performance achievement[1]	—	—	1,713	47.27
Vested/exercised	(31,794)	42.29	(10,383)	45.85
Forfeited	(3,426)	46.47	(2,706)	46.51
Outstanding as of January 31, 2024	49,191	$ 46.79	23,103	$ 48.09

[1] Represents the adjustment to previously granted performance share units for performance achievement.

The following table includes additional information related to restricted stock units and restricted stock and performance-based restricted stock units:

	Fiscal Years Ended January 31,		
(Amounts in millions, except years)	**2024**	**2023**	**2022**
Fair value of restricted stock units vested	$ 1,345	$ 931	$ 703
Fair value of restricted stock and performance-based restricted stock units vested	477	390	264
Unrecognized compensation cost for restricted stock units	1,686	1,323	1,102
Unrecognized compensation cost for restricted stock and performance-based restricted stock units	656	548	417
Weighted average remaining period to expense for restricted stock units (years)	0.9	1.0	1.2
Weighted average remaining period to expense for restricted stock and performance-based restricted stock units (years)	1.3	1.4	1.5

Share Repurchase Program

From time to time, the Company repurchases shares of its common stock under share repurchase programs authorized by the Company's Board of Directors. All repurchases made during fiscal 2024 were made under the current $20.0 billion share repurchase program approved in November 2022, which has no expiration date or other restrictions limiting the period over which the Company can make repurchases. As of January 31, 2024 authorization for $16.5 billion of share repurchases remained under the share repurchase program. Any repurchased shares are constructively retired and returned to an unissued status.

The Company regularly reviews share repurchase activity and considers several factors in determining when to execute share repurchases, including, among other things, current cash needs, capacity for leverage, cost of borrowings, results of operations and the market price of the Company's common stock. The following table provides, on a settlement date basis, the number of shares repurchased, average price paid per share and total amount paid for share repurchases for fiscal 2024, 2023 and 2022:

	Fiscal Years Ended January 31,		
(Amounts in millions, except per share data)	2024	2023	2022
Total number of shares repurchased	54.6	221.8	209.1
Average price paid per share	$ 50.87	$ 44.72	$ 46.82
Total cash paid for share repurchases	$ 2,779	$ 9,920	$ 9,787

Note 4. Accumulated Other Comprehensive Loss

The following table provides the changes in the composition of total accumulated other comprehensive loss for fiscal 2024, 2023 and 2022:

(Amounts in millions and net of immaterial income taxes)	Currency Translation and Other	Net Investment Hedges	Cash Flow Hedges	Minimum Pension Liability	Total
Balances as of February 1, 2021	$ (10,772)	$ 1,296	$ (304)	$ (1,986)	$ (11,766)
Other comprehensive loss before reclassifications, net	(586)	(7)	(540)	—	(1,133)
Reclassifications related to business dispositions, net[1]	3,258	(1,195)	30	1,966	4,059
Reclassifications to income, net	—	—	66	8	74
Balances as of January 31, 2022	(8,100)	94	(748)	(12)	(8,766)
Other comprehensive income (loss) before reclassifications, net	(1,145)	—	(571)	5	(1,711)
Return of currency translation to parent[2]	(1,262)	—	—	—	(1,262)
Reclassifications to income, net	(309)	—	368	—	59
Balances as of January 31, 2023	(10,816)	94	(951)	(7)	(11,680)
Other comprehensive income (loss) before reclassifications, net	333	—	(8)	(11)	314
Reclassifications to income, net	—	—	64	—	64
Balances as of January 31, 2024	$ (10,483)	$ 94	$ (895)	$ (18)	$ (11,302)

[1] Upon closing of the sale of the Company's operations in the U.K. and Japan during the first quarter of fiscal 2022, these amounts were released from accumulated other comprehensive loss, the majority of which was considered in the impairment evaluation when the individual disposal groups met the held for sale classification in fiscal 2021.
[2] Upon closing of the noncontrolling interest shareholder buyout of the Company's Massmart subsidiary during the fourth quarter of fiscal 2023, the cumulative amount of currency translation was reallocated from the Company's noncontrolling interest back to the Company. Refer to Note 3.

Amounts reclassified from accumulated other comprehensive loss for derivative instruments are generally recorded in interest, net, in the Company's Consolidated Statements of Income. The amounts for the minimum pension liability, as well as the cumulative translation resulting from the disposition of a business, are recorded in other gains and losses in the Company's Consolidated Statements of Income. Amounts related to the Company's derivatives expected to be reclassified from accumulated other comprehensive loss to net income during the next 12 months are not significant.

Note 5. Accrued Liabilities

The Company's accrued liabilities consist of the following as of January 31, 2024 and 2023:

	January 31,	
(Amounts in millions)	2024	2023
Accrued wages and benefits[1]	8,590	8,287
Self-insurance[2]	4,916	4,724
Accrued non-income taxes[3]	3,459	3,425
Opioid litigation settlement[4]	—	2,949
Deferred gift card revenue	2,664	2,488
Other[5]	9,130	9,253
Total accrued liabilities	$ 28,759	$ 31,126

(1) Accrued wages and benefits include accrued wages, salaries, vacation, bonuses and other incentive plans.
(2) Self-insurance consists of insurance-related liabilities, such as workers' compensation, general liability, auto liability, product liability and certain employee-related healthcare benefits.
(3) Accrued non-income taxes include accrued payroll, property, value-added, sales and miscellaneous other taxes.
(4) Represents the remaining balance for the opioids litigation settlement (substantially all of the balance outstanding at the end of fiscal 2023 was paid in fiscal 2024, see Note 10.)
(5) Other accrued liabilities includes items such as deferred membership revenue, interest, the purchase of PhonePe stock (see Note 3), supply chain, advertising, and maintenance & utilities.

Note 6. Short-term Borrowings and Long-term Debt

Short-term borrowings consist of commercial paper and lines of credit. Short-term borrowings as of January 31, 2024 and 2023 were $0.9 billion and $0.4 billion, respectively, with weighted-average interest rates of 7.7% and 6.6%, respectively.

The Company has various committed lines of credit in the U.S. to support its commercial paper program and are summarized in the following table:

	January 31, 2024			January 31, 2023		
(Amounts in millions)	Available	Drawn	Undrawn	Available	Drawn	Undrawn
Five-year credit facility[1]	$ 5,000	$ —	$ 5,000	$ 5,000	$ —	$ 5,000
364-day revolving credit facility[1]	10,000	—	10,000	10,000	—	10,000
Total	$ 15,000	$ —	$ 15,000	$ 15,000	$ —	$ 15,000

(1) In April 2023, the Company renewed and extended its existing 364-day revolving credit facility as well as its five year credit facility.

The committed lines of credit in the table above mature in April 2024 and April 2028, carry interest rates of the Secured Overnight Financing Rate plus 55 basis points, and incur commitment fees ranging between 1.5 and 4.0 basis points. In conjunction with the committed lines of credit listed in the table above, the Company has agreed to observe certain covenants, the most restrictive of which relates to the maximum amount of secured debt. Additionally, the Company has syndicated and fronted letters of credit available which totaled $2.1 billion as of January 31, 2024 and 2023, of which $1.7 billion and $1.8 billion was drawn as of January 31, 2024 and 2023, respectively.

The Company's long-term debt, which includes the fair value instruments further discussed in Note 8, consists of the following as of January 31, 2024 and 2023:

	January 31, 2024			January 31, 2023	
(Amounts in millions)	Maturity Dates By Fiscal Year	Amount	Average Rate[1]	Amount	Average Rate[1]
Unsecured debt					
Fixed	2025 - 2054	$ 34,527	3.7%	$ 33,707	3.6%
Total U.S. dollar denominated		34,527		33,707	
Fixed	2027 - 2030	1,789	4.0%	1,790	4.0%
Total Euro denominated		1,789		1,790	
Fixed	2031 - 2039	3,412	5.4%	3,318	5.4%
Total Sterling denominated		3,412		3,318	
Fixed	2025 - 2028	677	0.4%	767	0.4%
Total Yen denominated		677		767	
Total unsecured debt		40,405		39,582	
Total other[2]		(826)		(742)	
Total debt		39,579		38,840	
Less amounts due within one year		(3,447)		(4,191)	
Long-term debt		$ 36,132		$ 34,649	

(1) The average rate represents the weighted-average stated rate for each corresponding debt category, based on year-end balances and year-end interest rates.
(2) Includes deferred loan costs, discounts, fair value hedges, foreign-held debt and secured debt.

Annual maturities of long-term debt during the next five years and thereafter are as follows:

(Amounts in millions) Fiscal Year	Annual Maturities
2025	$ 3,447
2026	2,600
2027	3,483
2028	1,760
2029	3,458
Thereafter	24,831
Total	$ 39,579

Debt Issuances

Information on significant long-term debt issued during fiscal 2024 and 2023, for general corporate purposes, is as follows:

(Amounts in millions)

Issue Date	Principal Amount	Maturity Date	Fixed vs. Floating	Interest Rate	Net Proceeds
April 18, 2023	$750	April 15, 2026	Fixed	4.000%	$ 748
April 18, 2023	$750	April 15, 2028	Fixed	3.900%	746
April 18, 2023	$500	April 15, 2030	Fixed	4.000%	497
April 18, 2023	$1,500	April 15, 2033	Fixed	4.100%	1,491
April 18, 2023	$1,500	April 15, 2053	Fixed	4.500%	1,485
Total					$ 4,967

(Amounts in millions)

Issue Date	Principal Amount	Maturity Date	Fixed vs. Floating	Interest Rate	Net Proceeds
September 9, 2022	$1,750	September 9, 2025	Fixed	3.900%	$ 1,744
September 9, 2022	$1,000	September 9, 2027	Fixed	3.950%	994
September 9, 2022	$1,250	September 9, 2032	Fixed	4.150%	1,239
September 9, 2022	$1,000	September 9, 2052	Fixed	4.500%	992
Total					$ 4,969

These issuances are senior, unsecured notes which rank equally with all other senior, unsecured debt obligations of the Company, and are not convertible or exchangeable. These issuances do not contain any financial covenants which restrict the Company's ability to pay dividends or repurchase Company stock. Additionally, the Company received immaterial proceeds from debt issuances by certain international markets during fiscal 2023.

Maturities and Extinguishments

The following tables provide details of significant long-term debt repayments during fiscal 2024 and 2023:

(Amounts in millions)

Maturity Date	Principal Amount	Fixed vs. Floating	Interest Rate	Repayment
April 11, 2023	$1,750	Fixed	2.550%	$ 1,750
June 26, 2023	$2,280	Fixed	3.400%	2,280
Total repayment of matured debt				$ 4,030

(Amounts in millions)

Maturity Date	Principal Amount	Fixed vs. Floating	Interest Rate	Repayment
April 8, 2022	€850	Fixed	1.900%	$ 927
July 15, 2022	¥70,000	Fixed	0.183%	512
December 15, 2022	$1,250	Fixed	2.350%	1,250
Total repayment of matured debt				$ 2,689

Note 7. Leases

The Company leases certain retail locations, distribution and fulfillment centers, warehouses, office spaces, land and equipment throughout the U.S. and internationally. The Company's lease costs recognized in the Consolidated Statements of Income consist of the following:

	Fiscal years ended January 31,		
(Amounts in millions)	2024	2023	2022
Operating lease cost	$ 2,277	$ 2,306	$ 2,274
Finance lease cost:			
Amortization of right-of-use assets	755	596	565
Interest on lease obligations	326	256	232
Variable lease cost	1,082	899	823

Other lease information is as follows:

(Amounts in millions)	Fiscal years ended January 31, 2024	2023	2022
Cash paid for amounts included in measurement of lease obligations:			
Operating cash flows from operating leases	$ 2,273	2,280	2,234
Operating cash flows from finance leases	315	248	225
Financing cash flows from finance leases	1,055	563	538
Assets obtained in exchange for operating lease obligations	1,514	1,714	1,816
Assets obtained in exchange for finance lease obligations	1,572	1,226	1,044

	As of January 31, 2024	2023
Weighted-average remaining lease term - operating leases	11.7 years	12.0 years
Weighted-average remaining lease term - finance leases	12.4 years	13.3 years
Weighted-average discount rate - operating leases	6.4%	6.0%
Weighted-average discount rate - finance leases	6.8%	6.5%

The aggregate annual lease obligations at January 31, 2024, are as follows:

(Amounts in millions) Fiscal Year	Operating Leases	Finance Leases
2025	$ 2,181	$ 1,071
2026	2,107	1,000
2027	1,970	926
2028	1,815	823
2029	1,645	636
Thereafter	11,295	5,850
Total undiscounted lease obligations	21,013	10,306
Less imputed interest	(6,583)	(3,872)
Net lease obligations	$ 14,430	$ 6,434

Note 8. Fair Value Measurements

Assets and liabilities recorded at fair value are measured using the fair value hierarchy, which prioritizes the inputs used in measuring fair value. The levels of the fair value hierarchy are:

- Level 1: observable inputs such as quoted prices in active markets;
- Level 2: inputs other than quoted prices in active markets that are either directly or indirectly observable; and
- Level 3: unobservable inputs for which little or no market data exists, therefore requiring the Company to develop its own assumptions.

As described in Note 1, the Company measures the fair value of certain equity investments, including certain immaterial equity method investments where the Company has elected the fair value option, on a recurring basis within other long-term assets in the accompanying Consolidated Balance Sheets. The amounts of gains and losses included in earnings from fair value changes for these investments are recognized within other gains and losses in the Consolidated Statements of Income. The fair value of these investments is as follows:

(Amounts in millions)	Fair Value as of January 31, 2024	Fair Value as of January 31, 2023
Equity investments measured using Level 1 inputs	$ 2,835	$ 5,099
Equity investments measured using Level 2 inputs	4,414	5,570
Total	$ 7,249	$ 10,669

Changes in the fair value of these investments were primarily due to gains and losses resulting from net changes in the underlying stock prices, along with certain other immaterial investment activity. The fair value of these investments decreased $3.4 billion and $1.2 billion during fiscal 2024 and 2023, respectively. Equity investments without readily determinable fair values are carried at cost and adjusted for any observable price changes or impairments within other gains and losses in the Consolidated Statements of Income.

Derivatives

The Company also has derivatives recorded at fair value. Derivative fair values are the estimated amounts the Company would receive or pay upon termination of the related derivative agreements as of the reporting dates. The fair values have been measured using the income approach and Level 2 inputs, which include the relevant interest rate and foreign currency forward curves. As of January 31, 2024 and January 31, 2023, the notional amounts and fair values of these derivatives were as follows:

	January 31, 2024		January 31, 2023	
(Amounts in millions)	Notional Amount	Fair Value	Notional Amount	Fair Value
Receive fixed-rate, pay variable-rate interest rate swaps designated as fair value hedges	$ 6,271	$ (654) [1]	$ 8,021	$ (689) [1]
Receive fixed-rate, pay fixed-rate cross-currency swaps designated as cash flow hedges	5,879	(1,302) [1]	5,900	(1,423) [1]
Total	$ 12,150	$ (1,956)	$ 13,921	$ (2,112)

(1) Primarily classified in deferred income taxes and other in the Company's Consolidated Balance Sheets.

Nonrecurring Fair Value Measurements

In addition to assets and liabilities that are recorded at fair value on a recurring basis, the Company's assets and liabilities are also subject to nonrecurring fair value measurements. Generally, assets are recorded at fair value on a nonrecurring basis as a result of impairment charges.

Upon completing the sales of the Company's operations in the U.K. in February 2021 and Japan in March 2021, the Company recorded incremental non-recurring impairment charges of $0.4 billion in the first quarter of fiscal 2022 within other gains and losses in the Consolidated Statements of Income. Refer to Note 12. The Company did not have any material assets or liabilities resulting in nonrecurring fair value measurements as of January 31, 2024 and January 31, 2023.

Other Fair Value Disclosures

The Company records cash and cash equivalents, restricted cash and short-term borrowings at cost. The carrying values of these instruments approximate their fair value due to their short-term maturities.

The Company's long-term debt is also recorded at cost. The fair value is estimated using Level 2 inputs based on the Company's current incremental borrowing rate for similar types of borrowing arrangements. The carrying value and fair value of the Company's long-term debt as of January 31, 2024 and 2023, are as follows:

	January 31, 2024		January 31, 2023	
(Amounts in millions)	Carrying Value	Fair Value	Carrying Value	Fair Value
Long-term debt, including amounts due within one year	$ 39,579	$ 38,431	$ 38,840	$ 38,169

Note 9. Taxes

The components of income before income taxes are as follows:

	Fiscal Years Ended January 31,		
(Amounts in millions)	2024	2023	2022
U.S.	$ 20,092	$ 15,089	$ 15,536
Non-U.S.	1,756	1,927	3,160
Total income before income taxes	$ 21,848	$ 17,016	$ 18,696

A summary of the provision for income taxes is as follows:

	Fiscal Years Ended January 31,		
(Amounts in millions)	2024	2023	2022
Current:			
U.S. federal	$ 3,215	$ 2,030	$ 3,313
U.S. state and local	762	610	649
International	1,772	2,654	1,553
Total current tax provision	5,749	5,294	5,515
Deferred:			
U.S. federal	(438)	608	(671)
U.S. state and local	141	119	41
International	126	(297)	(129)
Total deferred tax expense (benefit)	(171)	430	(759)
Total provision for income taxes	$ 5,578	$ 5,724	$ 4,756

Effective Income Tax Rate Reconciliation

A reconciliation of the significant differences between the U.S. statutory tax rate and the effective income tax rate on pre-tax income from continuing operations is as follows:

	Fiscal Years Ended January 31,		
	2024	2023	2022
U.S. statutory tax rate	21.0 %	21.0 %	21.0 %
U.S. state income taxes, net of federal income tax benefit	3.0 %	3.1 %	2.8 %
Income taxed outside the U.S.	0.1 %	1.1 %	(1.5)%
Separation, disposal and wind-down of certain business operations	— %	6.3 %	0.5 %
Valuation allowance	1.2 %	1.7 %	4.4 %
Net impact of repatriated international earnings	(0.4)%	(0.4)%	(0.3)%
Federal tax credits	(1.5)%	(1.3)%	(1.1)%
Change in unrecognized tax benefits	0.6 %	0.3 %	0.2 %
Other, net	1.5 %	1.8 %	(0.6)%
Effective income tax rate	25.5 %	33.6 %	25.4 %

The following sections regarding deferred taxes, unremitted earnings, net operating losses, tax credit carryforwards, valuation allowances and uncertain tax positions exclude amounts related to operations classified as held for sale.

Deferred Taxes

The significant components of the Company's deferred tax account balances are as follows:

	January 31,	
(Amounts in millions)	2024	2023
Deferred tax assets:		
Loss and tax credit carryforwards	$ 7,136	$ 7,690
Accrued liabilities	3,066	3,312
Share-based compensation	238	237
Lease obligations	4,831	4,653
Other	1,124	839
Total deferred tax assets	16,395	16,731
Valuation allowances	(7,485)	(7,815)
Deferred tax assets, net of valuation allowances	8,910	8,916
Deferred tax liabilities:		
Property and equipment	4,813	4,352
Acquired intangibles	898	932
Inventory	3,035	3,032
Lease right of use assets	4,941	4,727
Mark-to-market investments	322	1,390
Other	486	249
Total deferred tax liabilities	14,495	14,682
Net deferred tax liabilities	$ 5,585	$ 5,766

The deferred taxes noted above are classified as follows in the Company's Consolidated Balance Sheets:

(Amounts in millions)	January 31, 2024	January 31, 2023
Balance Sheet classification		
Assets:		
Other long-term assets	$ 1,663	$ 1,503
Liabilities:		
Deferred income taxes and other	7,248	7,269
Net deferred tax liabilities	$ 5,585	$ 5,766

Unremitted Earnings

Prior to the Tax Cuts and Jobs Act of 2017 (the "Tax Act"), the Company asserted that all unremitted earnings of its foreign subsidiaries were considered indefinitely reinvested. As a result of the Tax Act, the Company reported and paid U.S. tax on the majority of its previously unremitted foreign earnings, and repatriations of foreign earnings will generally be free of U.S. federal tax, but may incur other taxes such as withholding or state taxes. As of January 31, 2024, the Company has not recorded approximately $1 billion of deferred tax liabilities associated with remaining unremitted foreign earnings considered indefinitely reinvested, for which U.S. and foreign income and withholding taxes would be due upon repatriation.

Net Operating Losses, Tax Credit Carryforwards and Valuation Allowances

As of January 31, 2024, the Company's net operating loss and capital loss carryforwards totaled approximately $30.3 billion. Of these carryforwards, approximately $16.5 billion will expire, if not utilized, in various years through 2044. The remaining carryforwards have no expiration.

The realizability of these future tax deductions and credits is evaluated by assessing the adequacy of future expected taxable income from all sources, including taxable income in prior carryback years, reversal of taxable temporary differences, forecasted operating earnings and available tax planning strategies. To the extent the Company does not consider it more likely than not that a deferred tax asset will be recovered, a valuation allowance is generally established. To the extent that a valuation allowance was established and it is subsequently determined that it is more likely than not that the deferred tax assets will be recovered, the change in the valuation allowance is recognized in the Consolidated Statements of Income.

The Company had valuation allowances of approximately $7.5 billion and $7.8 billion as of January 31, 2024 and 2023, respectively, on deferred tax assets associated primarily with the net operating loss carryforwards. Activity in the valuation allowance during fiscal 2024 related to valuation allowance builds in multiple markets, as well as releases due to the expiration of unrealized deferred tax assets.

Uncertain Tax Positions

The benefits of uncertain tax positions are recorded in the Company's Consolidated Financial Statements only after determining a more-likely-than-not probability that the uncertain tax positions will withstand challenge, if any, from taxing authorities.

As of January 31, 2024 and 2023, the amount of gross unrecognized tax benefits related to continuing operations was $3.5 billion and $3.3 billion, respectively. The amount of unrecognized tax benefits that would affect the Company's effective income tax rate was $1.7 billion and $1.5 billion as of January 31, 2024 and 2023, respectively.

A reconciliation of gross unrecognized tax benefits from continuing operations is as follows:

	Fiscal Years Ended January 31,		
(Amounts in millions)	2024	2023	2022
Gross unrecognized tax benefits, beginning of year	$ 3,307	$ 3,245	$ 3,135
Increases related to prior year tax positions	336	79	170
Decreases related to prior year tax positions	(74)	(248)	(97)
Increases related to current year tax positions	102	357	75
Settlements during the period	(102)	(89)	(5)
Lapse in statutes of limitations	(29)	(37)	(33)
Gross unrecognized tax benefits, end of year	$ 3,540	$ 3,307	$ 3,245

The Company classifies interest and penalties related to uncertain tax benefits as interest expense and as operating, selling, general and administrative expenses, respectively. Interest expense and penalties related to these positions were immaterial for fiscal 2024, 2023 and 2022. During the next twelve months, it is reasonably possible that tax audit resolutions could reduce unrecognized tax benefits by an immaterial amount, either because the tax positions are sustained on audit or because the Company agrees to their disallowance. The Company does not expect any change to have a material impact to its Consolidated Financial Statements.

The Company remains subject to income tax examinations for its U.S. federal income taxes generally for fiscal 2018 through 2023. The Company also remains subject to income tax examinations for international income taxes for fiscal 2013 through 2023, and for U.S. state and local income taxes generally for the fiscal years ended 2017 through 2023. With few exceptions, the Company is no longer subject to U.S. federal, state, local or foreign examinations by tax authorities for years before fiscal 2013.

Other Taxes

The Company is subject to tax examinations for value added, sales-based, payroll and other non-income taxes. A number of these examinations are ongoing in various jurisdictions. In certain cases, the Company has received assessments and judgments from the respective taxing authorities in connection with these examinations. Unless otherwise indicated, the possible losses or range of possible losses associated with these matters are individually immaterial, but a group of related matters, if decided adversely to the Company, could result in a liability material to the Company's Consolidated Financial Statements.

Note 10. Contingencies

Legal Proceedings

The Company is involved in a number of legal proceedings and certain regulatory matters. The Company records a liability for those legal proceedings and regulatory matters when it determines it is probable that a loss has been incurred and the amount of the loss can be reasonably estimated. The Company also discloses when it is reasonably possible that a material loss may be incurred. From time to time, the Company may enter into discussions regarding settlement of these matters, and may enter into settlement agreements, if it believes settlement is in the best interest of the Company and its shareholders.

Unless stated otherwise, the matters discussed below, if decided adversely to or settled by the Company, individually or in the aggregate, may result in a liability material to the Company's financial position, results of operations or cash flows.

Settlement Framework Regarding Multidistrict and State or Local Opioid-Related Litigation

During fiscal 2023, the Company accrued a liability for approximately $3.3 billion for the Settlement Framework (described below) and other previously agreed upon state and tribal settlements. The Settlement Framework includes no admission of wrongdoing or liability by the Company, and the Company continues to believe it has substantial factual and legal defenses to opioids-related litigation. As of January 31, 2024, substantially all of the original approximately $3.3 billion accrued liability for the Settlement Framework and other settlements have been paid.

In December 2017, the United States Judicial Panel on Multidistrict Litigation consolidated numerous lawsuits filed against a wide array of defendants by various plaintiffs, including counties, cities, healthcare providers, Native American tribes, individuals and third-party payers, asserting claims generally concerning the impacts of widespread opioid abuse. The consolidated multidistrict litigation is entitled In re National Prescription Opiate Litigation (MDL No. 2804) (the "MDL") and is pending in the U.S. District Court for the Northern District of Ohio. The Company is named as a defendant in some of the cases included in the MDL.

On November 15, 2022, the Company announced it had agreed to financial amounts and payment terms to resolve substantially all opioids-related lawsuits filed against the Company by states, political subdivisions, and Native American tribes whether as part of the MDL (excluding, however, a single, two-county trial described further below) or in state court, as well as all potential claims that could be made against the Company by states, political subdivisions, and Native American tribes for up to approximately $3.1 billion (the "Settlement Amount"). The Settlement Amount includes amounts for remediation of alleged harms as well as attorneys' fees and costs and also includes some, but not all, amounts from previously agreed recent settlements by the Company. One settlement framework with corresponding conditions and participation thresholds applies for the states and political subdivisions, and another settlement framework with corresponding conditions and participation thresholds applies for the Native American tribes. Both settlement frameworks are referred to collectively as the "Settlement Framework."

The Settlement Framework, among other applicable conditions, provides that payments to states and political subdivisions are contingent upon the number of states and political subdivisions, including those states and political subdivisions who have not yet sued the Company, that agree to participate in the Settlement Framework or otherwise have their claims foreclosed within a prescribed deadline. On December 20, 2022, the Company announced that it had settlement agreements with all 50 states, including four states that previously settled with the Company, as well as the District of Columbia, Puerto Rico and three other U.S. territories (the "Settling States"), thus satisfying the initial threshold of required participation by Settling States. On August 22, 2023, the settlement administrator determined that a sufficient number of political subdivisions had agreed to participate in the Settlement Framework, which was a necessary condition for the Settlement Framework to become effective. The Settlement Framework became effective 15 days later, on September 6, 2023. The Company deposited the full portion of the Settlement Amount attributable to the Settling States on October 11, 2023. Although the settlement administrator has determined that sufficient number of political subdivisions have agreed to participate in the Settlement Framework, and thus the

Settlement Framework was effective, eligible political subdivisions still have until July 15, 2025, to join the Settlement Framework.

Other Opioid-Related Litigation

The Company will continue to vigorously defend against any opioid-related litigation not covered or otherwise resolved by the Settlement Framework, including, but not limited to, each of the matters described below; any other actions filed by healthcare providers, individuals, and third-party payers; and any action filed by a political subdivision or Native American tribe that is not resolved by the Settlement Framework. Accordingly, the Company has not accrued a liability for these opioid-related litigation matters nor can the Company reasonably estimate any loss or range of loss that may arise from these matters. The Company can provide no assurance as to the scope and outcome of any of these matters and no assurance that its business, financial position, results of operations or cash flows will not be materially adversely affected.

Two-County Trial and MDL Bellwethers; Canada; and Other Litigation. The liability phase of a single, two-county trial in one of the MDL cases resulted in a jury verdict on November 23, 2021, finding in favor of the plaintiffs as to the liability of all defendants, including the Company. The abatement phase of the single, two-county trial resulted in a judgment on August 17, 2022, that ordered all three defendants, including the Company, to pay an aggregate amount of approximately $0.7 billion over fifteen years, on a joint and several liability basis, and granted the plaintiffs injunctive relief. On September 7, 2022, the Company filed an appeal with the Sixth Circuit Court of Appeals. The monetary aspect of the judgment is stayed pending appeal, and the injunctive aspect of the judgment went into effect on February 20, 2023. On September 11, 2023, the Sixth Circuit Court of Appeals issued an order of certifying certain questions in the appeal for review by the Supreme Court of Ohio. On November 29, 2023, the Supreme Court of Ohio accepted the request for certification, and the matter remains pending with the court.

The MDL designated five additional single-county cases as bellwethers to proceed through discovery; however, these five counties have elected to participate in the Settlement Framework and receive a portion of the Settlement Amount rather than go to trial. On October 25, 2023, the MDL designated four cases brought by third-party payers as bellwether cases to proceed through discovery. Additional bellwethers of cases brought by hospitals and other healthcare providers may be designated in the future.

Wal-Mart Canada Corp. and certain other subsidiaries of the Company have been named as defendants in two putative class action complaints filed in Canada related to dispensing and distribution practices involving opioids. These matters remain pending.

Similar cases that name the Company also have been filed in state and federal courts by state, local, and tribal governments, healthcare providers, and other plaintiffs. Plaintiffs in these cases and in the MDL are seeking compensatory and punitive damages, as well as injunctive relief including abatement. The Company has also been responding to subpoenas, information requests, and investigations from governmental entities related to nationwide controlled substance dispensing and distribution practices involving opioids.

DOJ Opioid Civil Litigation. On December 22, 2020, the U.S. Department of Justice (the "DOJ") filed a civil complaint in the U.S. District Court for the District of Delaware alleging that the Company unlawfully dispensed controlled substances from its pharmacies and unlawfully distributed controlled substances to those pharmacies. The complaint alleges that this conduct resulted in violations of the Controlled Substances Act. The DOJ is seeking civil penalties and injunctive relief. The Company initially moved to dismiss the DOJ complaint on February 22, 2021. After that motion was fully briefed, the DOJ filed an amended complaint on October 7, 2022. On November 7, 2022, the Company filed a partial motion to dismiss the amended complaint. The Court held a hearing on the partial motion to dismiss on January 18, 2024, and ordered the DOJ to file an amended complaint. The DOJ filed that amended complaint on February 1, 2024, and Walmart filed a partial motion to dismiss that complaint on February 6, 2024. On March 11, 2024, the Court granted in-part Walmart's motion by dismissing the entirety of the DOJ's claims related to distribution and dismissing the DOJ's claims arising under one of the DOJ's two dispensing liability theories. The DOJ's claims arising under its other dispensing liability theory remain pending.

Opioid-Related Securities Class Actions and Derivative Litigation. In addition, the Company is the subject of two securities class actions alleging violations of the federal securities laws regarding the Company's disclosures with respect to opioids, filed in the U.S. District Court for the District of Delaware on January 20, 2021 and March 5, 2021, purportedly on behalf of a class of investors who acquired Walmart stock from March 30, 2016 through December 22, 2020. Those cases have been consolidated. On October 8, 2021, the defendants filed a motion to dismiss the consolidated securities action. After the parties had fully briefed the motion to dismiss, on September 9, 2022, the Court entered an order permitting the plaintiffs to file an amended complaint, which was filed on October 14, 2022, and which revised the applicable putative class of investors to those who acquired Walmart stock from March 31, 2017, through December 22, 2020. On November 16, 2022, the defendants filed a motion to dismiss the amended complaint. That motion remains pending.

Derivative actions were also filed by two of the Company's shareholders in the U.S. District Court for the District of Delaware on February 9, 2021 and April 16, 2021, alleging breach of fiduciary duties against certain of its current and former directors with respect to oversight of the Company's distribution and dispensing of opioids and also alleging violations of the federal securities laws and other breaches of duty by certain current and former directors and officers in connection with the Company's opioids disclosures. Those cases have been stayed pending developments in other opioids litigation matters. On September 27, 2021, three shareholders filed a derivative action in the Delaware Court of Chancery alleging that certain members of the Board of Directors and certain former officers breached their fiduciary duties in failing to adequately oversee the Company's prescription opioids business. The defendants moved to dismiss and/or to stay proceedings on December 21, 2021, and the plaintiffs responded by filing an amended complaint on February 22, 2022. On April 20, 2022, the defendants moved to dismiss and/or to stay proceedings with respect to the amended complaint. In two orders issued on April 12 and 26, 2023, the Court of Chancery granted the defendants' motion to dismiss with respect to claims involving the Company's distribution practices and denied the remainder of the motion, including the Company's request to stay the litigation. On May 5, 2023, the Company's Board of Directors (the "Board") appointed an independent Special Litigation Committee (the "SLC") to investigate the allegations regarding certain current and former officers and directors named in the various derivative proceedings regarding oversight with respect to opioids. The Board has authorized the SLC to retain independent legal counsel and such other advisors as the SLC deems appropriate in carrying out its duties. The derivative matter pending in the Delaware Court of Chancery is stayed until the SLC completes its investigation.

Other Legal Proceedings

Asda Equal Value Claims. Asda, formerly a subsidiary of the Company, was and still is a defendant in certain equal value claims that began in 2008 and are proceeding before an Employment Tribunal in Manchester in the United Kingdom on behalf of current and former Asda store employees, as well as additional claims in the High Court of the United Kingdom (the "Asda Equal Value Claims"). Further claims may be asserted in the future. Subsequent to the divestiture of Asda in February 2021, the Company continues to oversee the conduct of the defense of these claims. While potential liability for these claims remains with Asda, the Company has agreed to provide indemnification with respect to certain of these claims up to a contractually determined amount. The Company cannot predict the number of such claims that may be filed, and cannot reasonably estimate any loss or range of loss that may arise related to these proceedings. Accordingly, the Company can provide no assurance as to the scope and outcome of these matters.

Money Transfer Agent Services Matters. The Company has responded to grand jury subpoenas issued by the United States Attorney's Office for the Middle District of Pennsylvania on behalf of the DOJ seeking documents regarding the Company's consumer fraud prevention program and anti-money laundering compliance related to the Company's money transfer services, where Walmart is an agent. The most recent subpoena was issued in August 2020. Walmart's responses to DOJ's subpoenas have been complete since 2021. The Company continues to cooperate with and provide information and documents voluntarily in response to supplemental requests from the DOJ. The Company has also responded to civil investigative demands from the United States Federal Trade Commission (the "FTC") in connection with the FTC's investigation related to money transfers and the Company's anti-fraud program in its capacity as an agent. On June 28, 2022, the FTC filed a complaint against the Company in the U.S. District Court for the Northern District of Illinois alleging that Walmart violated the Federal Trade Commission Act and the Telemarketing Sales Rule regarding its money transfer agent services and is requesting non-monetary relief and civil penalties. On August 29, 2022, the Company filed a motion to dismiss the complaint. On March 27, 2023, the Court issued an opinion dismissing the FTC's claim under the Telemarketing Sales Rule and denying Walmart's motion to dismiss the claim under Section 5 of the Federal Trade Commission Act. On April 12, 2023, Walmart filed a motion to certify the Court's March 27, 2023, order for interlocutory appeal. On June 30, 2023, the FTC filed an amended complaint against Walmart again asserting claims under the Federal Trade Commission Act and Telemarketing Sales Rule. On July 20, 2023, the Court denied Walmart's motion to certify the Court's March 27, 2023, order for interlocutory appeal, finding that it would be more orderly to consider a request for interlocutory appeal after a ruling on Walmart's motion to dismiss the amended complaint. Walmart's motion to dismiss the amended complaint was filed on August 11, 2023. The motion remains pending. No other deadlines have yet been set, and discovery is stayed.

The Company intends to vigorously defend these matters. However, the Company can provide no assurance as to the scope and outcome of these matters and cannot reasonably estimate any loss or range of loss that may arise. Accordingly, the Company can provide no assurance that its business, financial position, results of operations or cash flows will not be materially adversely affected.

Mexico Antitrust Matter. On October 6, 2023, the Comisión Federal de Competencia Económica of México ("COFECE") notified the main Mexican operating subsidiary of Wal-Mart de México, S.A.B. de C.V. ("Walmex"), a majority owned subsidiary of the Company, that COFECE's Investigatory Authority ("IA") had requested COFECE to initiate a quasi-judicial administrative process against Walmex's subsidiary for alleged relative monopolistic practices in connection with the supply and wholesale distribution of certain consumer goods, retail marketing practices of such consumer goods and related services. The quasi-judicial administrative process is the first opportunity for Walmex's subsidiary to respond to and defend against the IA's allegations before COFECE. While COFECE has the authority to impose monetary relief and/or non-structural conduct measures, such relief and conduct measures would be subject to appeal by Walmex's subsidiary. On December 14, 2023, Walmex's subsidiary submitted its defense arguments and will continue to defend against the allegations vigorously, both at the quasi-judicial administrative process and, if required, before any courts. Because this process is at an early stage, the Company can provide no assurance as to the scope and outcome of these matters, cannot reasonably estimate any loss or range of loss that may arise and can provide no assurance that its business, financial position, results of operations or cash flows will not be materially adversely affected.

Note 11. Retirement-Related Benefits

The Company offers a 401(k) plan for associates in the U.S. under which eligible associates can begin contributing to the plan immediately upon hire. The Company also offers a 401(k) type plan for associates in Puerto Rico under which associates can begin to contribute generally after one year of employment. Under these plans, after one year of employment, the Company matches 100% of participant contributions up to 6% of annual eligible earnings. The matching contributions immediately vest at 100% for each associate. Participants can contribute up to 50% of their pre-tax earnings, but not more than the statutory limits.

Associates in international countries who are not U.S. citizens are covered by various defined contribution post-employment benefit arrangements. These plans are administered based upon the legislative and tax requirements in the countries in which they are established.

The following table summarizes the contribution expense related to the Company's defined contribution plans for fiscal 2024, 2023 and 2022:

	Fiscal Years Ended January 31,		
(Amounts in millions)	2024	2023	2022
Defined contribution plans:			
U.S.	$ 1,528	$ 1,491	$ 1,441
International	85	74	39
Total contribution expense for defined contribution plans	$ 1,613	$ 1,565	$ 1,480

Note 12. Disposals, Acquisitions and Related Items

The following dispositions impact the Company's Walmart International segment. Other immaterial transactions have also occurred.

Asda

In February 2021, the Company completed the divestiture of Asda, the Company's retail operations in the U.K., for net consideration of $9.6 billion. Upon closing of the transaction, the Company recorded an incremental pre-tax loss of $0.2 billion in other gains and losses in its Consolidated Statements of Income in the first quarter of fiscal 2022, primarily related to changes in the net assets of the disposal group, currency exchange rate fluctuations and customary purchase price adjustments upon closing. During the first quarter of fiscal 2022, the Company deconsolidated the financial statements of Asda and recognized its retained investment in Asda as a debt security within other long-term assets and also recognized certain legal and tax indemnity liabilities within deferred income taxes and other in the Consolidated Balance Sheet.

Seiyu

In March 2021, the Company completed the divestiture of Seiyu, the Company's retail operations in Japan, for net consideration of $1.2 billion. Upon closing of the transaction, the Company recorded an incremental pre-tax loss of $0.2 billion in other gains and losses in its Consolidated Statements of Income in the first quarter of fiscal 2022, primarily related to changes in the net assets of the disposal group, currency exchange rate fluctuations and customary purchase price adjustments upon closing. During the first quarter of fiscal 2022, the Company deconsolidated the financial statements of Seiyu and recognized its retained 15 percent ownership interest in Seiyu as an equity investment within other long-term assets in the Consolidated Balance Sheet.

Note 13. Segments and Disaggregated Revenue

Segments

The Company is engaged in the operation of retail and wholesale stores and clubs, as well as eCommerce websites and mobile applications, located throughout the U.S., Africa, Canada, Central America, Chile, China, India and Mexico. The Company previously operated in the United Kingdom and Japan prior to the sale of those operations in the first quarter of fiscal 2022. Refer to Note 12 for discussion of recent divestitures. The Company's operations are conducted in three reportable segments: Walmart U.S., Walmart International and Sam's Club. The Company defines its segments as those operations whose results the chief operating decision maker ("CODM") regularly reviews to analyze performance and allocate resources. The Company sells similar individual products and services in each of its segments. It is impracticable to segregate and identify revenues for each of these individual products and services.

The Walmart U.S. segment includes the Company's mass merchant concept in the U.S., as well as eCommerce, which includes omni-channel initiatives and certain other business offerings such as advertising services through Walmart Connect. The

Walmart International segment consists of the Company's operations outside of the U.S., as well as eCommerce and omni-channel initiatives. The Sam's Club segment includes the warehouse membership clubs in the U.S., as well as eCommerce and omni-channel initiatives. Corporate and support consists of corporate overhead and other items not allocated to any of the Company's segments.

The Company measures the results of its segments using, among other measures, each segment's net sales and operating income, which includes certain corporate overhead allocations. From time to time, the Company revises the measurement of each segment's operating income, including any corporate overhead allocations, as determined by the information regularly reviewed by its CODM. Information for the Company's segments, as well as for Corporate and support, including the reconciliation to income before income taxes, is provided in the following table:

(Amounts in millions)	Walmart U.S.	Walmart International	Sam's Club	Corporate and support	Consolidated
Fiscal Year Ended January 31, 2024					
Net sales	$ 441,817	$ 114,641	$ 86,179	$ —	$ 642,637
Operating income (loss)	22,154	4,909	2,192	(2,243)	27,012
Interest, net					(2,137)
Other gains and (losses)					(3,027)
Income before income taxes					$ 21,848
Total assets	$ 137,782	$ 86,136	$ 15,682	$ 12,799	$ 252,399
Depreciation and amortization	7,671	2,159	642	1,381	11,853
Capital expenditures	$ 13,877	$ 2,911	$ 1,041	$ 2,777	$ 20,606
Fiscal Year Ended January 31, 2023					
Net sales	$ 420,553	$ 100,983	$ 84,345	$ —	$ 605,881
Operating income (loss)	20,620	2,965	1,964	(5,121)	20,428
Interest, net					(1,874)
Other gains and (losses)					(1,538)
Income before income taxes					$ 17,016
Total assets	$ 130,659	$ 86,766	$ 15,490	$ 10,282	$ 243,197
Depreciation and amortization	$ 7,054	$ 1,964	$ 609	$ 1,318	10,945
Capital expenditures	$ 11,425	$ 2,625	$ 727	$ 2,080	16,857
Fiscal Year Ended January 31, 2022					
Net sales	$ 393,247	$ 100,959	$ 73,556	$ —	$ 567,762
Operating income (loss)	21,587	3,758	2,259	(1,662)	25,942
Interest, net					(1,836)
Loss on extinguishment of debt					(2,410)
Other gains and (losses)					(3,000)
Income before income taxes					$ 18,696
Total assets	$ 125,044	$ 91,403	$ 14,678	$ 13,735	$ 244,860
Depreciation and amortization	$ 6,773	$ 1,963	$ 601	$ 1,321	10,658
Capital expenditures	$ 8,475	$ 2,497	$ 622	$ 1,512	13,106

Total revenues, consisting of net sales and membership and other income, and long-lived assets, consisting primarily of net property and equipment and lease right-of-use assets, aggregated by the Company's U.S. and non-U.S. operations for fiscal 2024, 2023 and 2022, are as follows:

	Fiscal Years Ended January 31,		
(Amounts in millions)	2024	2023	2022
Revenues			
U.S. operations	$ 532,076	$ 508,685	$ 470,295
Non-U.S. operations	116,049	102,604	102,459
Total revenues	$ 648,125	$ 611,289	$ 572,754
Long-lived assets			
U.S. operations	$ 104,480	$ 95,567	$ 89,795
Non-U.S. operations	25,858	23,667	22,829
Total long-lived assets	$ 130,338	$ 119,234	$ 112,624

No individual country outside of the U.S. had total revenues or long-lived assets that were material to the consolidated totals. Long-lived assets related to operations classified as held for sale are excluded from the table above. Additionally, the Company did not generate material revenues from any single customer.

Disaggregated Revenues

In the following tables, segment net sales are disaggregated by either merchandise category or market. In addition, net sales related to eCommerce, which include omni-channel sales where a customer initiates an order digitally and the order is fulfilled through a store or club, are provided for each segment.

(Amounts in millions)	Fiscal Years Ended January 31,		
Walmart U.S. net sales by merchandise category	**2024**	**2023**	**2022**
Grocery	$ 264,210	$ 247,299	$ 218,944
General merchandise	113,985	118,597	125,876
Health and wellness	54,898	46,591	42,839
Other categories	8,724	8,066	5,588
Total	$ 441,817	$ 420,553	$ 393,247

Of Walmart U.S.'s total net sales, approximately $65.4 billion, $53.4 billion and $47.8 billion related to eCommerce for fiscal 2024, 2023 and 2022, respectively.

(Amounts in millions)	Fiscal Years Ended January 31,		
Walmart International net sales by market	**2024**	**2023**	**2022**
Mexico and Central America	$ 49,726	$ 40,496	$ 35,964
Canada	22,639	22,300	21,773
China	17,011	14,711	13,852
United Kingdom	—	—	3,811
Other	25,265	23,476	25,559
Total	$ 114,641	$ 100,983	$ 100,959

Of Walmart International's total net sales, approximately $24.8 billion, $20.3 billion and $18.5 billion related to eCommerce for fiscal 2024, 2023 and 2022, respectively.

(Amounts in millions)	Fiscal Years Ended January 31,		
Sam's Club net sales by merchandise category	**2024**	**2023**	**2022**
Grocery and consumables	$ 56,449	$ 53,027	$ 46,822
Fuel, tobacco and other categories	12,854	14,636	10,751
Home and apparel	9,263	9,579	9,037
Health and wellness	5,005	4,248	3,956
Technology, office and entertainment	2,608	2,855	2,990
Total	$ 86,179	$ 84,345	$ 73,556

Of Sam's Club's total net sales, approximately $9.9 billion, $8.4 billion and $6.9 billion related to eCommerce for fiscal 2024, 2023 and 2022, respectively.

Note 14. Subsequent Event

Dividends Declared

The Company approved, effective February 20, 2024, the fiscal 2025 annual dividend of $0.83 per share, an increase over the fiscal 2024 dividend of $0.76 per share. For fiscal 2025, the annual dividend will be paid in four quarterly installments of $0.2075 per share, according to the following record and payable dates:

Record Date	**Payable Date**
March 15, 2024	April 1, 2024
May 10, 2024	May 28, 2024
August 16, 2024	September 3, 2024
December 13, 2024	January 6, 2025

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

We maintain disclosure controls and procedures that are designed to provide reasonable assurance that information, which is required to be timely disclosed, is accumulated and communicated to management in a timely fashion. In designing and evaluating such controls and procedures, we recognize that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving the desired control objectives. Our management is necessarily required to use judgment in evaluating controls and procedures. Also, we have investments in unconsolidated entities. Since we do not control or manage those entities, our controls and procedures with respect to those entities are substantially more limited than those we maintain with respect to our consolidated subsidiaries.

In the ordinary course of business, we review our internal control over financial reporting and make changes to our systems and processes to improve such controls and increase efficiency, while ensuring that we maintain an effective internal control environment. Changes may include such activities as implementing new, more efficient systems, updating existing systems, automating manual processes, standardizing controls globally, migrating certain processes to our shared services organizations and increasing monitoring controls. We are currently upgrading our financial system in stages, beginning in our U.S. and Canadian markets, including our general ledger which was upgraded for these markets during fiscal 2024. Our financial system is a significant component of our internal control over financial reporting. We will continue to implement other components of our new financial system in stages, and each implementation will impact our internal control over financial reporting.

An evaluation of the effectiveness of the design and operation of our disclosure controls and procedures as of January 31, 2024 was performed under the supervision and with the participation of management, including our Chief Executive Officer and Chief Financial Officer. Based upon that evaluation, our Chief Executive Officer and Chief Financial Officer concluded that our disclosure controls and procedures are effective to provide reasonable assurance that information required to be disclosed by the Company in the reports that it files or submits under the Securities Exchange Act of 1934, as amended, is accumulated and communicated to management, including our Chief Executive Officer and Chief Financial Officer, as appropriate, to allow timely decisions regarding required disclosure and are effective to provide reasonable assurance that such information is recorded, processed, summarized and reported within the time periods specified by the SEC's rules and forms.

Report on Internal Control Over Financial Reporting

Management has responsibility for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external reporting purposes in accordance with accounting principles generally accepted in the United States. Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Management has assessed the effectiveness of the Company's internal control over financial reporting as of January 31, 2024. In making its assessment, management has utilized the criteria set forth by the Committee of Sponsoring Organizations ("COSO") of the Treadway Commission in Internal Control-Integrated Framework (2013). Management concluded that based on its assessment, Walmart's internal control over financial reporting was effective as of January 31, 2024. The Company's internal control over financial reporting as of January 31, 2024, has been audited by Ernst & Young LLP as stated in their report which appears herein.

Changes in Internal Control Over Financial Reporting

There have been no changes in the Company's internal control over financial reporting as of January 31, 2024, that have materially affected, or are reasonably likely to materially affect, the Company's internal control over financial reporting.

ITEM 9B. OTHER INFORMATION

Security Trading Plans of Directors and Executive Officers

None of the Company's directors or executive officers adopted or terminated a Rule 10b5-1 trading arrangement or a non-Rule 10b5-1 trading arrangement during the Company's fiscal quarter ended January 31, 2024, as such terms are defined under Item 408(a) or Regulation S-K.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

Please see the information concerning our executive officers contained in "Item 1. Business" herein under the caption "Information About Our Executive Officers," which is included in accordance with the Instruction to Item 401 of the SEC's Regulation S-K.

Information required by this Item 10 with respect to the Company's directors and certain family relationships is incorporated by reference to such information under the caption "Proposal No. 1 – Election of Directors" included in our Proxy Statement relating to our 2024 Annual Meeting of Shareholders (our "Proxy Statement").

No material changes have been made to the procedures by which shareholders of the Company may recommend nominees to our Board of Directors since those procedures were disclosed in our proxy statement relating to our 2023 Annual Shareholders' Meeting as previously filed with the SEC.

The information regarding our Audit Committee, including our audit committee financial experts, our Reporting Protocols for Senior Financial Officers and our Code of Conduct applicable to all of our associates, including our Chief Executive Officer, Chief Financial Officer and our Controller, who is our principal accounting officer, required by this Item 10 is incorporated herein by reference to the information under the captions "Corporate Governance" and "Proposal No. 4: Ratification of Independent Accountants" included in our Proxy Statement. "Item 1. Business" above contains information relating to the availability of a copy of our Reporting Protocols for Senior Financial Officers and our Code of Conduct and the posting of amendments to and any waivers of the Reporting Protocols for Senior Financial Officers and our Code of Conduct on our website.

ITEM 11. EXECUTIVE COMPENSATION

The information required by this Item 11 is incorporated herein by reference to the information under the captions "Corporate Governance – Director Compensation" and "Executive Compensation" included in our Proxy Statement.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information required by this Item 12 is incorporated herein by reference to the information that appears under the caption "Stock Ownership" included in our Proxy Statement.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

The information required by this Item 13 is incorporated herein by reference to the information under the caption "Corporate Governance – Board Processes and Practices" included in our Proxy Statement.

ITEM 14. PRINCIPAL ACCOUNTING FEES AND SERVICES

The information required by this Item 14 is incorporated herein by reference to the information under the caption "Proposal No. 4 – Ratification of Independent Accountants" included in our Proxy Statement.

PART IV

ITEM 15. EXHIBITS, FINANCIAL STATEMENT SCHEDULES

(a) Documents filed as part of this report are as follows:

1. Financial Statements: See the Financial Statements in "Item 8. Financial Statements and Supplementary Data."

2. Financial Statement Schedules:

   Certain schedules have been omitted because the required information is not present or is not present in amounts sufficient to require submission of the schedule, or because the information required is included in the Consolidated Financial Statements, including the notes thereto.

3. Exhibits:

   See exhibits listed under part (b) below.

(b) The required exhibits are filed as part of this Form 10-K or are incorporated by reference herein.[(1)]

3.1(a)	Restated Certificate of Incorporation of the Company dated February 1, 2018 is incorporated herein by reference to Exhibit 3.1 to the Report on Form 8-K filed by the Company on February 1, 2018
3.1(b)	Certificate of Amendment to the Restated Certificate of Incorporation of the Company, effective February 23, 2024 is incorporated herein by reference to Exhibit 3.1 to the Report on Form 8-K filed by the Company on February 23, 2024
3.2	Amended and Restated Bylaws of the Company dated November 10, 2022 are incorporated herein by reference to Exhibit 3.1 to the Report on Form 8-K filed by the Company on November 16, 2022
4.1	Indenture dated as of April 1, 1991, between the Company and J.P. Morgan Trust Company, National Association, as successor trustee to Bank One Trust Company, NA, as successor trustee to The First National Bank of Chicago, Trustee, is incorporated herein by reference to Exhibit 4(a) to Registration Statement on Form S-3 (File Number 33-51344) [(P)]
4.2	First Supplemental Indenture dated as of September 9, 1992, to the Indenture dated as of April 1, 1991, between the Company and J.P. Morgan Trust Company, National Association, as successor trustee to Bank One Trust Company, NA, as successor trustee to The First National Bank of Chicago, Trustee, is incorporated herein by reference to Exhibit 4(b) to Registration Statement on Form S-3 (File Number 33-51344) [(P)]
4.3	Indenture dated as of December 11, 2002, between the Company and J.P. Morgan Trust Company, National Association, as successor trustee to Bank One Trust Company, NA, is incorporated by reference to Exhibit 4.5 to Registration Statement on Form S-3 (File Number 333-101847)
4.4	Indenture dated as of July 19, 2005, between the Company and J.P. Morgan Trust Company, National Association is incorporated by reference to Exhibit 4.5 to Registration Statement on Form S-3 (File Number 333-126512)
4.5	First Supplemental Indenture, dated December 1, 2006, between the Company and The Bank of New York Trust Company, N.A., as successor-in-interest to J.P. Morgan Trust Company, National Association, as Trustee, under the Indenture, dated as of July 19, 2005, between the Company and J.P. Morgan Trust Company, National Association, as Trustee, is incorporated herein by reference to Exhibit 4.6 to Post-Effective Amendment No. 1 to Registration Statement on Form S-3 (File Number 333-130569)
4.6	Second Supplemental Indenture, dated December 19, 2014, between the Company and The Bank of New York Trust Company, N.A., as successor-in-interest to J.P. Morgan Trust Company, National Association, as Trustee, under the Indenture, dated as of July 19, 2005, between the Company and J.P. Morgan Trust Company, National Association, as Trustee, is incorporated herein by reference to Exhibit 4.3 to Registration Statement on Form S-3 (File Number 333-201074)
4.7	Third Supplemental Indenture, dated June 26, 2018, between the Company and The Bank of New York Trust Company, N.A., as successor-in-interest to J.P. Morgan Trust Company, National Association, as Trustee, under the Indenture, dated as of July 19, 2005, between the Company and J.P. Morgan Trust Company, National Association, as Trustee, is incorporated herein by reference to Exhibit 4(S) to Current Report on Form 8-K filed on June 26, 2018
4.8*	Description of Registrant's Securities

10.1*	Walmart Inc. Deferred Compensation Matching Plan, as amended and restated effective November 8, 2023 (C)
10.2	Walmart Inc. Management Incentive Plan, as amended effective February 1, 2018 is incorporated by reference to Exhibit 10(b) to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2018, filed on March 30, 2018 (C)
10.3*	Walmart Inc. 2016 Associate Stock Purchase Plan, as amended effective February 1, 2024 (C)
10.4	Walmart Inc. Stock Incentive Plan of 2015, as amended effective February 1, 2018 is incorporated by reference to Exhibit 10(d) to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2018, filed on March 30, 2018 (C)
10.5	Walmart Inc. Supplemental Executive Retirement Plan, as amended and restated effective February 1, 2023 is incorporated by reference to Exhibit 10.5 to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2023 filed on March 17, 2023 (C)
10.6	Walmart Inc. Director Compensation Deferral Plan, as amended effective February 1, 2018 is incorporated by reference to Exhibit 10(f) to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2018, filed on March 30, 2018 (C)
10.7	Form of Post-Termination Agreement and Covenant Not to Compete with attached Schedule of Executive Officers who have executed a Post-Termination Agreement and Covenant Not to Compete is incorporated by reference to Exhibit 10(p) to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2011, filed on March 30, 2011 (C)
10.7(a)*	Amended Schedule of Executive Officers who have executed a Post-Termination Agreement and Covenant Not to Compete in the form filed as Exhibit 10(p) to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2011 (C)
10.8	Form of Walmart Inc. Stock Incentive Plan of 2015 Restricted Stock Notification of Award and Terms and Conditions of Award is incorporated by reference to Exhibit 10.8 to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2022, filed March 18, 2022 (C)
10.9	Form of Walmart Inc. Stock Incentive Plan of 2015 Global Share-Settled Performance-Based Restricted Stock Unit Notification and Terms and Conditions is incorporated by reference to Exhibit 10.9 to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2022, filed on March 18, 2022 (C)
10.10	Walmart Inc. Officer Deferred Compensation Plan, as amended and restated effective February 1, 2023 is incorporated by reference to Exhibit 10.10 to the Annual Report on Form 10-K of the Company for the fiscal year ended January 31, 2023 filed on March 17, 2023 (C)
10.11	Post Termination Agreement and Covenant Not to Compete between the Company and Suresh Kumar dated June 6, 2019 is incorporated herein by reference to Exhibit 10.16 to the Annual Report on Form 10-K for the fiscal year ended January 31, 2020 filed on March 20, 2020 (C)
10.12	Retirement Agreement between the Company and M. Brett Biggs dated November 29, 2021 is incorporated herein by reference to Exhibit 10.1 to the Current Report on Form 8-K filed on November 29, 2021 (C)
10.13	Share Issuance and Acquisition Agreement by and Between Flipkart Private Limited and Walmart Inc. dated as of May 9, 2018 is incorporated herein by reference to Exhibit 10.1 to the Quarterly Report on Form 10-Q of the Company for the fiscal quarter ended July 31, 2018 filed on September 6, 2018 (portions of this exhibit have been omitted and filed separately with the SEC pursuant to a request for confidential treatment.)
10.14	Counterpart Form of Share Purchase Agreement by and Among Wal-Mart International Holdings, Inc., the shareholders of Flipkart Private Limited identified on Schedule I thereto, Fortis Advisors LLC and Walmart Inc. dated as of May 9, 2018 is incorporated herein by reference to Exhibit 10.2 to the Quarterly Report on Form 10-Q of the Company for the fiscal quarter ended July 31, 2018 filed on September 6, 2018 (portions of this exhibit have been omitted and filed separately with the SEC pursuant to a request for confidential treatment.)
10.15	Retirement Agreement between the Company and Judith McKenna dated August 16, 2023 is incorporated by reference to Exhibit 10.1 to the Quarterly Report on Form 10-Q of the Company for the fiscal quarter ended July 31, 2023 filed on September 1, 2023 (C)
21*	List of the Company's Significant Subsidiaries
23*	Consent of Independent Registered Public Accounting Firm

31.1*	Chief Executive Officer Section 302 Certification
31.2*	Chief Financial Officer Section 302 Certification
32.1**	Chief Executive Officer Section 906 Certification
32.2**	Chief Financial Officer Section 906 Certification
97.1*	Walmart Executive Compensation Recoupment Policy
99.1*	Certain Federal and State Court Opioids Litigation Case Citations and Currently Scheduled Trial Dates
101.INS*	Inline XBRL Instance Document
101.SCH*	Inline XBRL Taxonomy Extension Schema Document
101.CAL*	Inline XBRL Taxonomy Extension Calculation Linkbase Document
101.DEF*	Inline XBRL Taxonomy Extension Definition Linkbase Document
101.LAB*	Inline XBRL Taxonomy Extension Label Linkbase Document
101.PRE*	Inline XBRL Taxonomy Extension Presentation Linkbase Document
104	Cover Page Interactive Data File (formatted as Inline XBRL and contained in Exhibit 101)

* Filed herewith as an Exhibit.

** Furnished herewith as an Exhibit.

(C) This Exhibit is a management contract or compensatory plan or arrangement

(P) This Exhibit was originally filed in paper format. Accordingly, a hyperlink has not been provided.

(1) Certain instruments defining the rights of holders of long-term debt securities of the Registrant are omitted pursuant to Item 601(b)(4)(iii) of Regulation S-K. The Company hereby undertakes to furnish to the SEC, upon request, copies of any such instruments.

(c) Financial Statement Schedules: None.

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

Walmart Inc.

Date: March 15, 2024	By	/s/ C. Douglas McMillon
		C. Douglas McMillon President and Chief Executive Officer

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated:

Date: March 15, 2024	By	/s/ C. Douglas McMillon
		C. Douglas McMillon President and Chief Executive Officer and Director (Principal Executive Officer)
Date: March 15, 2024	By	/s/ Gregory B. Penner
		Gregory B. Penner Chairman of the Board and Director
Date: March 15, 2024	By	/s/ John David Rainey
		John David Rainey Executive Vice President and Chief Financial Officer (Principal Financial Officer)
Date: March 15, 2024	By	/s/ David M. Chojnowski
		David M. Chojnowski Senior Vice President and Controller (Principal Accounting Officer)

Signature Page to Walmart Inc.
Form 10-K for the Fiscal Year Ended January 31, 2024

Date: March 15, 2024	By	/s/ Cesar Conde Cesar Conde Director
Date: March 15, 2024	By	/s/ Timothy P. Flynn Timothy P. Flynn Director
Date: March 15, 2024	By	/s/ Sarah Friar Sarah Friar Director
Date: March 15, 2024	By	/s/ Carla A. Harris Carla A. Harris Director
Date: March 15, 2024	By	/s/ Thomas W. Horton Thomas W. Horton Director
Date: March 15, 2024	By	/s/ Marissa A. Mayer Marissa A. Mayer Director
Date: March 15, 2024	By	/s/ Randall L. Stephenson Randall L. Stephenson Director
Date: March 15, 2024	By	/s/ S. Robson Walton S. Robson Walton Director
Date: March 15, 2024	By	/s/ Steuart L. Walton Steuart L. Walton Director

Signature Page to Walmart Inc.
Form 10-K for the Fiscal Year Ended January 31, 2024

Exhibit 4.8

**DESCRIPTION OF THE REGISTRANT'S SECURITIES
REGISTERED PURSUANT TO SECTION 12 OF
THE SECURITIES EXCHANGE ACT OF 1934**

Walmart Inc. has ten classes of securities registered under Section 12 of the Securities Exchange Act of 1934, as amended (the "***Exchange Act***"): (i) Common Stock, $0.10 par value per share ("***Common Stock***"), (ii) 2.550% Notes due 2026 (the "***2.550% 2026 Notes***"), (iii) 1.050% Notes due 2026 (the "***2026 Notes***"), (iv) 1.500% Notes due 2028 (the "***2028 Notes***"), (v) 5.750% Notes due 2030 (the "***2030 Notes***"), (vi) 1.800% Notes due 2031 (the "***2031 Notes***," and, collectively with the 2026 Notes, 2028 Notes and 2030 Notes, the "***Luxembourg Notes***"), (vii) 4.875% Notes due 2029 (the "***2029 Notes***"), (viii) 5.625% Notes due 2034 (the "***2034 Notes***"), (ix) 5.250% Notes due 2035 (the "***2035 Notes***") and (x) 4.875% Notes due 2039 (the "***2039 Notes***," and, collectively with the 2029 Notes, 2034 Notes, and 2035 Notes, the "***Irish Notes***" and, together with the 2.550% 2026 Notes and the Luxembourg Notes, the "***Notes***"). Each of the Company's securities registered under Section 12 of the Exchange Act is listed on The New York Stock Exchange (the "***NYSE***"). Except as the context otherwise requires, or as otherwise specified in this Exhibit 4.8 or the information incorporated by reference into this Exhibit 4.8, the terms "***Walmart Inc.***," "***Walmart***," the "***Company***," "***we***," "***us***," "***our***" and "***our company***" refer to Walmart Inc.

DESCRIPTION OF COMMON STOCK

The following is a description of the rights of holders of Common Stock and related provisions of the Company's Restated Certificate of Incorporation, as amended (the "***Certificate***"), and Amended and Restated Bylaws (the "***Bylaws***"), which are incorporated by reference as Exhibit 3.1(a), Exhibit 3.1(b) and Exhibit 3.2, respectively, to the Annual Report on Form 10-K of which this Exhibit 4.8 is a part, and applicable Delaware law, including the General Corporation Law of Delaware, as amended (the "***DGCL***"). This description is qualified in its entirety by, and should be read in conjunction with, the Certificate, Bylaws and applicable Delaware law.

Authorized Capital Stock

Pursuant to our Certificate, our authorized capital stock consists of 33,100,000,000 shares, with a par value of $0.10 per share, of which 33,000,000,000 shares are designated as Common Stock and 100,000,000 shares are designated as preferred stock. The number of shares of Common Stock issued and outstanding varies from time to time.

Common Stock

Fully Paid and Non-Assessable Shares; No Liability for Corporate Obligations

All of the outstanding shares of Common Stock are fully paid and non-assessable. A share of Common Stock is fully paid and non-assessable if such share has been issued for consideration legally permissible under the DGCL with a value at least equal to the par value per share of Common Stock. Holders of fully paid and non-assessable shares of the Common Stock will not be liable for any obligations or liabilities of the Company that the Company may fail to discharge.

Voting Rights

Each holder of shares of Common Stock is entitled to one vote for each share owned of record on all matters submitted to a vote of shareholders. Except as noted below or as otherwise required by the DGCL, the vote of shareholders required to decide any matter brought before a shareholder meeting at which a quorum is present is a majority of the outstanding shares present in person or represented by proxy at that meeting and entitled to vote on the question subject to the shareholder vote. In a contested election of directors, which is an election in which there are more nominees for election than board positions to be filled, directors are elected by the vote of a plurality of the outstanding shares present in person or represented by proxy at that meeting and entitled to vote on the election of directors. The holders of a majority of the outstanding shares of our stock must approve any amendments to our Certificate, any merger or consolidation to which we are a party (other than parent-subsidiary mergers), any sale of all or substantially all of our assets or our dissolution as a corporation. In addition, the DGCL requires the holders of a majority of the outstanding shares of our stock to approve any conversion of our corporation to another type of entity, such as a limited liability company. Our shareholders do not have cumulative voting rights as to the election of directors.

Dividends

Subject to the preferential rights of any holders of any series of our preferred stock that may be issued in the future, the holders of shares of Common Stock are entitled to such dividends and distributions, whether payable in cash or otherwise, as may be declared from time to time by our board of directors from legally available funds.

Liquidation Distributions

Subject to the preferential rights of any holders of any series of our preferred stock that may be issued in the future, upon our liquidation, dissolution or winding-up and after payment of all prior claims against our assets and our outstanding obligations, the holders shares of Common Stock will be entitled to receive, pro rata, all of our remaining assets.

Preemptive, Conversion, Redemption or Similar Rights

The holders of shares of Common Stock are not entitled to any preemptive or other similar rights to subscribe for or acquire additional shares of Common Stock or any other securities of the Company. The shares of Common Stock are not subject to conversion or redemption by the Company and the holders of shares of Common Stock do not have any right or option to convert such shares into any other security or property of the Company or to cause the Company to redeem such shares of Common Stock. There are no sinking fund provisions applicable to the Common Stock.

Certificate, Bylaws and DGCL

Provisions of the Certificate and Bylaws may delay or discourage transactions involving an actual or potential change in control of the Company or change in the Company's management, including transactions in which shareholders might otherwise receive a premium for their shares, or transactions that shareholders might otherwise deem to be in their best interests. Among other things, the Certificate and Bylaws include the following provisions:

i. vacancies on our board of directors, and any new director positions created by the expansion of our board of directors, may be filled only by a majority of the directors then in office, subject to certain exceptions;

ii. our Bylaws establish an advance notice procedure for shareholders to submit proposed nominations of persons for election to our board of directors and other proposals for business to be brought before an annual meeting of our shareholders;

iii. our Bylaws include enhanced procedural and disclosure requirements with which shareholders must comply in connection proposed nominations of persons for election to our board of directors, including related to the U.S. Securities and Exchange Commission's ("***SEC***") rules regarding universal proxy cards;

iv. our board of directors may issue up to 100,000,000 shares of preferred stock, with designations, rights and preferences as may be determined from time to time by our board of directors; and

v. our Bylaws may be amended by our shareholders or our board of directors.

Anti-Takeover Provisions of Delaware Law, Our Certificate and Our Bylaws

In addition, as a Delaware corporation, the Company is subject to the provisions of Section 203 of the DGCL, which prohibits the Company, subject to certain exceptions described below, from engaging in a "business combination" with:

- a stockholder who owns 15% or more of the Company's outstanding voting stock (otherwise known as an "***interested stockholder***");
- an affiliate of an interested stockholder; or
- an associate of an interested stockholder,

in each case, for three years following the date that the stockholder became an interested stockholder.

A "***business combination***" includes a merger or sale of more than 10% of the Company's assets. However, the above provisions of Section 203 do not apply if:

- the board of directors approves the transaction that made the stockholder an "interested stockholder," prior to the date of the transaction;

- after the completion of the transaction that resulted in the stockholder becoming an interested stockholder, that stockholder owned at least 85% of the Company's voting stock outstanding at the time the transaction commenced, other than statutorily excluded shares of Common Stock; or
- on or subsequent to the date of the transaction, the business combination is approved by the board of directors and authorized at a meeting of the Company's stockholders, and not by written consent, by an affirmative vote of at least two-thirds of the outstanding voting stock not owned by the interested stockholder.

Exclusive Forum Provision

Our Certificate provides that, unless we select or consent in writing to the selection of an alternative forum, the Court of Chancery of the State of Delaware shall, to the fullest extent permitted by law, and subject to applicable jurisdictional requirements, be the sole and exclusive forum for any current or former stockholder (including any current or former beneficial owner) to bring claim, including claims in the right of the corporation (a) that are based upon a violation of a duty by a current or former director, officer, employee or stockholder in such capacity; or (b) as to which the DGCL confers jurisdiction upon the Court of Chancery.

Listing

Shares of the Common Stock are listed for trading on the NYSE under the symbol "WMT."

Transfer Agent and Registrar

The transfer agent and registrar for the Common Stock is Computershare Trust Company, N.A.

DESCRIPTION OF DEBT SECURITIES

The Indentures

The following description of the Notes is qualified in its entirety by reference to:

i. the Indenture, dated as of July 19, 2005, between the Company and The Bank of New York Mellon Trust Company, N.A. ("***BNYM***"), as successor-in-interest to J.P. Morgan Trust Company, National Association ("***JPMTC***"), as trustee, as amended by the First Supplemental Indenture, dated as of December 1, 2006, the Second Supplemental Indenture, dated as of December 19, 2014, and the Third Supplemental Indenture, dated as of June 26, 2018 (the "***2005 Indenture***"), incorporated by reference to Exhibit 4.4, Exhibit 4.5, Exhibit 4.6 and Exhibit 4.7, respectively, to the Annual Report on Form 10-K of which this Exhibit 4.8 is a part;
ii. the Indenture, dated as of December 11, 2002, between the Company and BNYM, as successor trustee to JPMTC , as successor trustee to Bank One Trust Company, NA ("***Bank One***"), as trustee (the "***2002 Indenture***"), incorporated by reference to Exhibit 4.3 to the Annual Report on Form 10-K of which this Exhibit 4.8 is a part; and
iii. the Indenture, dated as of April 1, 1991, between the Company and BNYM, as successor trustee to JPMTC, as successor trustee to Bank One, as successor trustee to the First National Bank of Chicago, as trustee, as amended by the First Supplemental Indenture, dated as of September 9, 1992 (the "***1991 Indenture***"), incorporated by reference to Exhibit 4.1 and Exhibit 4.2, respectively, to the Annual Report on Form 10-K of which this Exhibit 4.8 is a part.

Notes issued under the 2005 Indenture are referred to herein as the "***2005 Indenture Notes***," Notes issued under the 2002 Indenture are referred to herein as the "***2002 Indenture Notes***" and Notes issued under the 1991 Indenture are referred to herein as the "***1991 Indenture Notes***".

Trustee under the Indentures

BNYM is the trustee under each of the Indentures. We have commercial deposits and custodial arrangements with BNYM and its affiliates. We may enter into similar or other banking relationships with BNYM in the future in the normal course of business. In addition, BNYM acts as trustee and as paying agent with respect to other debt securities issued by us, and may do so for future issuances of debt securities by us as well.

The 2005 Indenture

The 2005 Indenture Notes were issued under the 2005 Indenture, which provides that debt securities may be issued under the 2005 Indenture from time to time in one or more series. The 2005 Indenture and the 2005 Indenture Notes are governed by, and construed in accordance with, the laws of the State of New York. The 2005 Indenture does not limit the amount of debt securities that we may issue under the 2005 Indenture. We may, without the consent of the holders of the debt securities of any series, issue additional debt securities ranking equally with, and otherwise similar in all respects to, the debt securities of the series (except for the public offering price and the issue date) so that those additional debt securities will be consolidated and form a single series with the debt securities of the series previously offered and sold.

Covenants

The 2005 Indenture sets forth limited covenants that apply to the 2005 Indenture Notes. However, these covenants do not, among other things:

- limit the amount of indebtedness or lease obligations that may be incurred by us and our subsidiaries;
- limit our ability or that of our subsidiaries to issue, assume or guarantee debt secured by liens; or
- restrict us from paying dividends or making distributions on our capital stock or purchasing or redeeming our capital stock.

Consolidation, Merger and Sale of Assets

The 2005 Indenture provides that we may amalgamate, consolidate with, merge into or transfer our assets substantially as an entirety to any person; provided that the following conditions are satisfied:

- any successor to us (a "***Successor***") assumes our obligations on the 2005 Indenture Notes and under the 2005 Indenture;
- any Successor be an entity incorporated or organized under the laws of the United States;
- after giving effect to such transaction, no event of default, as described below under "—Events of Default," has occurred and is continuing; and
- certain other conditions under the 2005 Indenture are met.

Upon any amalgamation, consolidation, merger, reorganization or arrangement or any conveyance or transfer of the properties and assets of the Company substantially as an entirety, the Successor will succeed to, and be substituted for, and may exercise every right and power of, the Company, as the case may be, under the 2005 Indenture with the same effect as if such Successor had been named as the Company therein.

Any such amalgamation, consolidation, merger or transfer of assets substantially as an entirety that meets the conditions described above would not constitute a default or event of default that would entitle holders of the 2005 Indenture Notes or the trustee, on their behalf, to take any of the actions described below under "—Events of Default."

Events of Default

Each of the following events are defined in the 2005 Indenture as an "***event of default***" with respect to the debt securities of any series, including the 2005 Indenture Notes:

i. we fail to pay interest on any outstanding debt securities of that series when that interest is due and payable and that failure continues for 30 days;
ii. we fail to pay principal of or premium, if any, on any outstanding debt securities of that series when that principal or premium, if any, is due and payable;
iii. we fail to perform or we breach any covenant or warranty in the 2005 Indenture with respect to any outstanding debt securities of that series and that failure continues for 90 days after we receive written notice of that default;
iv. certain events of bankruptcy, insolvency or reorganization occur with respect to us; or
v. any other event occurs that is designated as an event of default with respect to the particular series of debt securities when that particular series of debt security is established.

An event of default with respect to the debt securities of any series issued under the 2005 Indenture does not necessarily constitute an event of default with respect to the debt securities of any other series issued under the 2005 Indenture.

If an event of default with respect to any series of outstanding debt securities occurs and is continuing (other than an event of default relating to certain events of bankruptcy, insolvency or reorganization with respect to us), the trustee or the holders of not less than 25% in aggregate principal amount of the outstanding debt securities of that series may declare the principal amount of the outstanding debt securities of that series to be immediately due and payable. If an event of default relating to certain events of bankruptcy, insolvency or reorganization with respect to us occurs and is continuing, the principal of and accrued and unpaid interest on the then outstanding debt securities of all series issued under the 2005 Indenture will become and be immediately due and payable without any declaration or other act on the part of the trustee or any holders.

The holders of not less than a majority in aggregate principal amount of the outstanding debt securities of any series may rescind a declaration of acceleration and its consequences, if we have deposited certain sums with the trustee and all events of default with respect to the debt securities of such series, other than the nonpayment of the principal which have become due solely by such acceleration, have been cured or waived, as provided in the 2005 Indenture.

An event of default for a particular series of debt securities does not necessarily constitute an event of default for any other series of debt securities issued under the 2005 Indenture.

We are required to file annually with the trustee a written statement as to the existence or non-existence of defaults under the 2005 Indenture or any series of debt securities issued thereunder.

No holder of any debt securities of any series, including the 2005 Indenture Notes, will have any right to institute any proceeding, judicial or otherwise, with respect to the 2005 Indenture, or for the appointment of a receiver or trustee, or for any other remedy, unless:

i. such holder has previously given written notice to the trustee of a continuing event of default with respect to the debt securities of such series;

ii. the holders of not less than 25% in principal amount of the outstanding debt securities of such series have made a written request to the trustee to institute proceedings in respect of such event of default in its own name as trustee;

iii. such holder or holders have offered to the trustee reasonable indemnity against the costs, expenses and liabilities to be incurred in compliance with such request;

iv. the trustee for 60 days after its receipt of such notice, request and offer of indemnity has failed to institute any such proceeding; and

v. no direction inconsistent with such written request has been given to the trustee during such 60-day period by the holders of a majority in principal amount of the outstanding debt securities of such series;

it being understood and intended that no one or more holders of the debt securities of such series has any right in any manner whatever by virtue of, or by availing of, any provision of the 2005 Indenture to affect, disturb or prejudice the rights of any other holders of the debt securities of such series or to obtain or to seek to obtain priority or preference over any other such holders or to enforce any right under the 2005 Indenture, except in the manner herein provided and for the equal and ratable benefit of all the holders of the debt securities of such series.

The holders of a majority in aggregate principal amount of outstanding debt securities of a series will have the right, subject to certain limitations, to direct the time, method and place of conducting any proceeding for any remedy available to the trustee with respect to the debt securities of that series or exercising any trust or power conferred on the trustee, and to waive certain defaults. The 2005 Indenture provides that if an event of default occurs and is continuing, the trustee will exercise such of its rights and powers under the 2005 Indenture, and use the same degree of care and skill in their exercise, as a prudent person would exercise under the circumstances in the conduct of such person's own affairs.

Notwithstanding the foregoing, the holder of any debt security will have the right to institute suit for the enforcement of any payment of principal of and premium, if any, and interest on that debt security or any redemption price or repurchase price when due and that that right will not be impaired without the consent of that holder.

Modification and Waivers

We and the trustee may execute a supplemental indenture to add provisions to or to eliminate or change provisions of the 2005 Indenture or to modify otherwise the rights of the holders of the 2005 Indenture Notes if we have the consent of the holders of not less than a majority in aggregate principal amount of the 2005 Indenture Notes affected by that supplemental indenture. However, we and the trustee may not execute a supplemental indenture without the consent of each holder of the 2005 Indenture Notes affected by that supplemental indenture if that supplement indenture would, among other things:

- change the maturity of the principal of, or the stated maturity of any installment of interest or premium, if any, on, any Note, reduce the principal amount of or the premium, if any, or rate of interest on any Note, change any method for determining the rate of interest on any Note, change the obligation to pay any additional amounts with respect to any Note, reduce the amount due and payable on a Note upon the acceleration of its maturity or upon its repurchase or redemption if the amount payable upon acceleration, repurchase or redemption is otherwise less than the stated principal amount of that Note, change the method of calculating interest on a Note, change the currency in which the principal of or the premium, if any, or interest on a Note is payable, reduce the minimum rate of interest on any Note or impair the right to institute suit for the enforcement of any such payment on or with respect to any such holder's 2005 Indenture Notes;
- reduce the percentage in principal amount of outstanding 2005 Indenture Notes described above as being required to consent to entry into a particular supplemental indenture or for the waiver of certain defaults under the 2005 Indenture and their consequences; or
- modify the provisions of the 2005 Indenture relating to modification of the 2005 Indenture, except in certain specified respects.

The trustee and we, without the consent of any holders, may execute a supplemental indenture to, among other things:

- evidence the succession of another corporation to us and the Successor's assumption to our covenants with respect to the 2005 Indenture Notes and the 2005 Indenture;
- add to our covenants further restrictions or conditions for the benefit of holders of the 2005 Indenture Notes;
- cure ambiguities or correct or supplement any provision contained in the 2005 Indenture or any supplemental indenture that may be inconsistent with another provision;
- add additional events of default with respect to the 2005 Indenture Notes;
- add to, change or eliminate any provision of the Indenture, provided that the addition, change or elimination will not affect any outstanding 2005 Indenture Notes;
- establish new series of debt securities and the form or terms of such series of debt securities and to provide for the issuance of securities of any series so established;
- evidence and provide for the acceptance of appointment of a successor trustee with respect to the 2005 Indenture Notes and to add or change any provision to or of the 2005 Indenture as necessary to have more than one trustee under the 2005 Indenture; and
- comply with the requirements of the SEC in order to maintain the qualification of the 2005 Indenture under the Trust Indenture Act of 1939.

The holders of a majority in aggregate principal amount of the outstanding 2005 Indenture Notes may waive an event of default resulting in acceleration of the 2005 Indenture Notes and rescind and annul that acceleration, but only if all other events of default with respect to the 2005 Indenture Notes have been remedied or waived and all payments due with respect to the 2005 Indenture Notes, other than those becoming due as a result of acceleration, have been made. If an event of default occurs and is continuing with respect to the 2005 Indenture Notes, the trustee may, in its discretion, and will, at the written request of holders of not less than a majority in aggregate principal amount of the outstanding 2005 Indenture Notes and upon reasonable indemnity against the costs, expenses and liabilities to be incurred in compliance with such request and subject to certain other conditions set forth in the 2005 Indenture, proceed to protect the rights of the holders of the 2005 Indenture Notes.

The holders of a majority in aggregate principal amount of the 2005 Indenture Notes may waive any past default under the 2005 Indenture and its consequences except an uncured default in the payment of principal of and premium, if any, or interest on the 2005 Indenture Notes or with respect to any covenant or provision of the 2005 Indenture that the 2005 Indenture or the 2005 Indenture Notes specifically provide cannot be waived without the consent of each holder of the 2005 Indenture Notes. Upon such a waiver, the default and any event of default arising out of the default will be deemed cured for all purposes of the 2005 Indenture Notes.

The 2002 Indenture

The 2002 Indenture Notes were issued under the 2002 Indenture, which provides that debt securities may be issued under the 2002 Indenture from time to time in one or more series. The 2002 Indenture and the 2002 Indenture Notes are governed by, and construed in accordance with, the laws of the State of New York. The 2002 Indenture does not limit the amount of debt securities that we may issue under the 2002 Indenture. We may, without the consent of the holders of the debt securities of any series, issue additional debt securities ranking equally with, and otherwise similar in all respects to, the debt securities of the series (except for the public offering price and the issue date) so that those additional debt securities will be consolidated and form a single series with the debt securities of the series previously offered and sold.

Conversion or Exchange Rights

Debt securities offered under the 2002 Indenture may be convertible into or exchangeable for other securities, including, for example, shares of our equity securities. The terms and conditions may include, among others, the following:

- the conversion or exchange price or prices or the ratio or ratios or method of determining the conversion or exchange prices or ratios;
- the conversion or exchange period;
- provisions regarding our ability or the ability of the holder to convert or exchange the debt securities;
- events requiring adjustment to the conversion or exchange price; and
- provisions affecting conversion or exchange in the event of our redemption of the debt securities.

Events of Default and Waiver

An event of default with respect to debt securities of a series issued under the 2002 Indenture will occur if:

- we fail to pay interest on any outstanding debt securities of that series when it is due and payable and that failure continues for 30 days;
- we fail to pay principal of, or premium, if any, on any outstanding debt securities of that series when it is due and payable;
- we fail to perform or we breach any covenant or warranty in the 2002 Indenture with respect to any outstanding debt securities of that series and that failure continues for 90 days after we receive written notice of that default;
- certain events of bankruptcy, insolvency or reorganization occur with respect to us; or
- any other event occurs that is designated as an event of default with respect to the particular series of debt securities when that particular series of debt securities is established.

An event of default with respect to a particular series of debt securities issued under the 2002 Indenture does not necessarily constitute an event of default with respect to any other series of debt securities issued under the 2002 Indenture. If an event of default with respect to any series of outstanding debt securities occurs and is continuing, the trustee or the holders of not less than 25% in aggregate principal amount of the outstanding debt securities of that series may declare the principal amount of the outstanding debt securities of that series to be immediately due and payable. The holders of a majority in aggregate principal amount of the outstanding debt securities of a series may waive an event of default resulting in acceleration of the debt securities of that series and rescind and annul that acceleration, but only if all other events of default with respect to the debt securities of that series have been remedied or waived and all payments due with respect to the debt securities of that series, other than those due as a result of acceleration, have been made. If an event of default occurs and is continuing with respect to the debt securities of a series, the trustee may, in its discretion, and will, at the written request of holders of not less than a majority in aggregate principal amount of the outstanding debt securities of that series and upon reasonable indemnity against the costs, expenses and liabilities to be incurred in compliance with such request and subject to certain other conditions set forth in the indenture, proceed to protect the rights of the holders of the debt securities of that series. The holders of a majority in aggregate principal amount of the debt securities of that series may waive any past default under the 2002 Indenture and its consequences except a default in the payment of principal of, premium, if any, or interest on, those debt securities and any covenant or provision of the 2002 Indenture that cannot be waived without the consent of each holder of debt securities of that series. Upon such a waiver, the default and any event of default arising out of the default will be deemed cured for all purposes of the debt securities of that series.

The 2002 Indenture provides that upon the occurrence of an event of default described in the first two bullet points in the first paragraph under "—Events of Default and Waiver" with respect to debt securities of a series, we will, upon the trustee's demand, pay to the trustee for the benefit of the holders of the outstanding debt securities of that series, the whole amount then due and payable on the debt securities of that series for principal, premium, if any, and interest. The 2002 Indenture also provides that if we fail to pay such amount forthwith upon such demand, the trustee may, among other things, institute a judicial proceeding for the collection of those amounts.

The 2002 Indenture also provides that, notwithstanding any other provision of the 2002 Indenture, the holder of any debt securities of a series will have the right to institute suit for the enforcement of any payment of principal of, and interest on, the debt securities of that series or any redemption price or repurchase price when due and that that right will not be impaired without the consent of that holder.

The trustee is required, within 90 days after the occurrence of a default with respect to the debt securities of a series, to give to the holders of the debt securities of that series notice of all uncured defaults known to it. However, except in the case of default in the payment of principal or interest on any of the debt securities of that series, the trustee will be protected in withholding that notice if the trustee in good faith determines that the withholding of that notice is in the interest of the holders of the debt securities of that series. The term "***default***," for the purpose of this provision only, means the occurrence of any event that is or would become, after notice or the passage of time or both, an event of default with respect to that series.

We are required to file annually with the trustee a written statement as to the existence or non-existence of defaults under the 2002 Indenture or any series of debt securities.

Legal Defeasance and Covenant Defeasance

We may, at our option and at any time, elect to have all of the obligations discharged with respect to the outstanding debt securities issued under the 2002 Indenture or as to any series thereof, except for:

- the rights of holders of debt securities to receive payments of principal and interest from the trust referred to below when those payments are due;
- our obligations respecting the debt securities concerning issuing temporary notes, registration of transfers of debt securities, mutilated, destroyed, lost or stolen debt securities, the maintenance of an office or agency for payment and money for debt security payments being held in trust;
- the rights, powers, trusts, duties and immunities of the trustee and our obligations in connection therewith; and
- the provisions of the 2002 Indenture relating to such a discharge of obligations.

We refer to a discharge of this type as "***defeasance***."

In addition, other than our covenant to pay the amounts due and owing with respect to a series of debt securities, we may elect to have our obligations as the issuer of a series of debt securities released with respect to covenants relating to that series of debt securities. Thereafter, any failure to comply with those obligations will not constitute a default or event of default with respect to the debt securities of that series. If such a release of our covenants occurs, our failure to perform or our breach of the covenants or warranties defeased will no longer constitute an event of default with respect to those debt securities.

To exercise either of the rights we describe above, certain conditions must be met, including:

- we must irrevocably deposit with the trustee, in trust for the debt security holders' benefit, moneys in the currency in which the securities are denominated, securities issued by a government, governmental agency or central bank of the country in whose currency the securities are denominated, or a combination of cash and such securities, in amounts sufficient to pay the principal of and interest on all of the then outstanding debt securities to be affected by the defeasance at their stated maturity;

- the trustee must receive an opinion of counsel confirming that the holders of the outstanding debt securities will not recognize income, gain or loss for U.S. federal income tax purposes as a result of that defeasance and will be subject to U.S. federal income tax on the same amounts, in the same manner and at the same times as would have been the case if that defeasance had not occurred, which opinion, only in the case of the type of defeasance described first above, will be based on a ruling of the Internal Revenue Service or a change in federal income tax law to that effect occurring after the date of the indenture;
- no default or event of default exists on the date of such deposit, subject to certain exceptions; and
- the trustee must receive an opinion of counsel to the effect that, after the 91st day following the deposit, the trust funds will not be part of any "estate" formed by the bankruptcy or reorganization of the party depositing those funds with the trustee or subject to the "automatic stay" under the United States Bankruptcy Code or, in the case of covenant defeasance, will be subject to a first priority lien in favor of the trustee for the benefit of the holders.

Satisfaction and Discharge

If we so request, the 2002 Indenture will cease to be of further effect, other than as to certain rights of registration of transfer or exchange of the notes, as provided for in the 2002 Indenture, and the trustee, at our expense, will execute proper instruments acknowledging satisfaction and discharge of the 2002 Indenture and the debt securities when:

- either all the debt securities previously authenticated and delivered under the 2002 Indenture, other than destroyed, lost or stolen securities that have been replaced or paid and notes that have been subject to defeasance, have been delivered to the trustee for cancellation; or
- all of the securities issued under the 2002 Indenture not previously delivered to the trustee for cancellation have become due and payable, will become due and payable at their stated maturity within 60 days or will become due and payable at redemption within 60 days under arrangements satisfactory to the trustee for the giving of notice of redemption by the trustee in our name and expense; and
- in each of the foregoing cases, we have irrevocably deposited or caused to be deposited with the trustee cash in U.S. dollars, certain United States government securities or a combination thereof, in trust for the purpose and in an amount sufficient to pay and discharge the entire indebtedness arising under the debt securities issued pursuant to the 2002 Indenture not previously delivered to the trustee for cancellation, for principal and premium, if any, on and interest on these securities to the date of such deposit (in the case of notes that have become due and payable) or to the stated maturity of these securities or redemption date, as the case may be; and
- we have paid or caused to be paid all sums payable under the 2002 Indenture by us; and
- no default or event of default then exists; and
- we have delivered to the trustee an officers' certificate and an opinion of counsel, each stating that all conditions precedent provided in the 2002 Indenture relating to the satisfaction and discharge of the 2002 Indenture and the securities issued under the indenture have been complied with.

Modification of the 2002 Indenture

The 2002 Indenture provides that, with the consent of the holders of not less than a majority in aggregate principal amount of the outstanding debt securities of each affected series, modifications and alterations of the 2002 Indenture may be made which affect the rights of the holders of such debt securities. However, no such modification or alteration may be made without the consent of the holder of each debt security affected if the modification or alteration would, among other things:

- change the maturity of the principal of, or of any installment of interest on, any such debt security, or reduce the principal amount of any such debt security, or change the method of calculation of interest or the currency of payment of principal or interest on, or reduce the minimum rate of interest thereon, or impair the right to institute suit for the enforcement of any such payment on or with respect to any such debt security, or

- reduce the above-stated percentage in principal amount of outstanding debt securities required to modify or alter the 2002 Indenture.

The trustee and we, without the consent of the holders of the debt securities, may execute a supplemental indenture to the 2002 Indenture to, among other things:

- evidence the succession of another corporation to us and the successor's assumption to our respective covenants with respect to the debt securities and the 2002 Indenture;
- add to our covenants further restrictions or conditions that our board of directors and the trustee consider to be for the protection of holders of all or any series of the debt securities and to make the occurrence of a default in any of those additional covenants, restrictions or conditions a default or an event of default under the 2002 Indenture subject to certain limitations;
- cure ambiguities or correct or supplement any provision contained in the 2002 Indenture or any supplemental indenture that may be defective or inconsistent with another provision;
- add additional events of default with respect to all or any series of the debt securities;
- add to, change or eliminate any provision of the 2002 Indenture provided that the addition, change or elimination will not affect any outstanding debt securities;
- provide for the issuance of debt securities whether or not then outstanding under the 2002 Indenture in coupon form and to provide for exchangeability of the coupon form securities with other debt securities issued under the 2002 Indenture in fully registered form;
- establish new series of debt securities and the form or terms of such series of debt securities and to provide for the issuance of securities of any series so established; and
- evidence and provide for the acceptance of appointment of a successor trustee and to change the 2002 Indenture as necessary to have more than one trustee under the 2002 Indenture.

Amalgamation, Consolidation, Merger or Sale of Assets

The 2002 Indenture provides that we may, without the consent of the holders of any of the outstanding debt securities of any series, amalgamate, consolidate with, merge into or transfer our assets substantially as an entirety to any person, provided that:

- any successor to us assumes our obligations on the debt securities and under the 2002 Indenture;
- any successor to us must be an entity incorporated or organized under the laws of the United States;
- after giving effect thereto, no event of default, as defined in the 2002 Indenture, shall have occurred and be continuing; and
- certain other conditions under the 2002 Indenture are met.

Any such amalgamation, consolidation, merger or transfer of assets substantially as an entirety that meets the conditions described above would not constitute a default or event of default that would entitle holders of the debt securities or the trustee, on their behalf, to take any of the actions described above under "—Events of Default and Waiver."

No Limitations on Additional Debt and Liens

The 2002 Indenture does not contain any covenants or other provisions that would limit our right to incur additional indebtedness, enter into any sale and leaseback transaction or grant liens on our assets.

The 1991 Indenture

The 1991 Indenture Notes were issued under the 1991 Indenture, which provides that debt securities may be issued under the 1991 Indenture from time to time in one or more series. The 1991 Indenture and the 1991 Indenture Notes are governed by, and construed in accordance with, the laws of the State of New York. The 1991 Indenture does not limit the amount of debt securities that we may issue under the 1991 Indenture. We may, without the consent of the holders of the debt securities of any series, issue additional debt securities ranking equally with, and otherwise similar in all respects to, the debt securities of the series (except for the public offering price and the issue date) so that those additional debt securities will be consolidated and form a single series with the debt securities of the series previously offered and sold.

Covenants

We will not, and will not permit any of our subsidiaries to issue, assume or guarantee any debt for money we borrow if that debt is secured by any mortgage, deed of trust, security interest, pledge, lien or other encumbrance upon any Operating Property (as defined below) belonging to us or of any of our subsidiaries or any shares of stock or indebtedness of any of our subsidiaries, whether owned at the date of the 1991 Indenture or thereafter acquired, without effectively securing the debt securities equally and ratably with that debt. This restriction does not, however, apply to:

- mortgages on any property acquired, constructed or improved by us or any of our subsidiaries after January 31, 1991, created or assumed within 60 months after the acquisition, or construction or improvement is complete, or within six months after completion pursuant to a firm commitment for financing arrangement that we enter into within that 60-month period, to secure or provide for the payment of the purchase price or cost;
- mortgages existing on any property at the time of its acquisition;
- mortgages existing on any property, shares of stock or debt acquired from a corporation merged with or into us or one of our subsidiaries;
- mortgages on property of any corporation existing at the time it becomes our subsidiary;
- mortgages to secure debt of any of our subsidiaries to us or to another of our subsidiaries;
- mortgages in favor of governmental bodies to secure partial progress, advance or other payments pursuant to any contract or statute or to secure indebtedness incurred to finance the purchase price or cost of constructing or improving the property subject to those mortgages; or
- mortgages for extending, renewing or replacing debt secured by any mortgage referred to in the foregoing items or in this item or any mortgages existing on January 31, 1991.

This restriction does not apply to the issuance, assumption or guarantee by us or any of our subsidiaries of debt secured by a mortgage which would otherwise be subject to the restrictions described above up to an aggregate amount which, together with all of our and our subsidiaries' secured debt, not including secured debt permitted under the foregoing exceptions, and the Value (as defined below) of Sale and Lease-back Transactions (as defined below) existing at that time other than those Sale and Lease-back Transactions the proceeds of which have been applied to the retirement of certain long-term debt or to the purchase of other operating property, and other than those Sale and Lease-back Transactions in which the property involved would have been permitted to be mortgaged under the principle described in the first item above, does not exceed the greater of 10% of our Consolidated Net Tangible Assets (as defined below) or 15% of Consolidated Capitalization (as defined below).

We will not, and will not permit any of our subsidiaries to, engage in Sale and Lease-back Transactions relating to any Operating Property, except for temporary leases for a term, including renewals, of not more than 48 months and except for leases between us and one of our subsidiaries or between our subsidiaries. However, we or our subsidiaries can engage in that type of transaction if the net proceeds of the Sale and Lease-back Transaction are at least equal to the sum of all costs incurred by us in connection with the acquisition of, and construction of any improvement on, the Operating Property to be leased and either:

- we or our subsidiary would be entitled to incur debt secured by a mortgage on the property to be leased without securing the debt securities pursuant to the first exception to the prohibition on liens stated above; or
- the Value thereof would be an amount permitted as described above; or
- we apply an amount equal to the sum of all costs incurred by us in connection with the acquisition of, and the construction of any improvements on, that property (1) to the payment or other retirement of certain of our or one of our subsidiary's long-term debt or (2) to the purchase of Operating Property, other than that involved in that Sale and Lease-back Transaction.

We may merge with or consolidate into another corporation or sell or convey all or substantially all of our property to another corporation that is authorized to purchase and operate our property, as long as:

- immediately after the merger, consolidation, sale or conveyance, the surviving or acquiring corporation is not in default under the 1991 Indenture;

- the surviving or acquiring corporation is a U.S. corporation; and
- the surviving or acquiring corporation assumes, by a supplemental indenture satisfactory to the trustee, the obligation to pay the principal of and interest and any premium on all of the debt securities and to perform our covenants under the 1991 Indenture.

In the case of a merger or consolidation or a sale or conveyance of all or substantially all of our assets and the assumption of our liabilities under the 1991 Indenture by a successor corporation, the successor corporation will assume our place in the 1991 Indenture as if it had originally been a party to the 1991 Indenture. The successor corporation may then issue debt securities under the 1991 Indenture.

Events of Default, Notice and Waiver

An event of default with respect to any series of debt securities under the 1991 Indenture is:

- a default in payment of principal or premium, if any, at maturity;
- a default for 30 days in payment of any interest;
- our failure for 60 days after notice to perform any other of the covenants or agreements in the 1991 Indenture;
- our default in the payment of any of our debt or acceleration of any of that debt under the terms of the instrument under which that debt is issued, if that default in payment is not cured or that acceleration is not annulled within 10 days after written notice;
- certain events in the case of our bankruptcy, insolvency or reorganization; or
- any other event of default provided with respect to any series of debt securities.

If an event of default occurs and is continuing with respect to any series of debt securities, either the trustee or the holders of 25% in principal amount then outstanding of the debt securities of that series may declare the principal of all the debt securities to be due and payable immediately, but upon certain conditions that declaration may be annulled. The holders of a majority in principal amount then outstanding of the debt securities of a series may waive defaults, except an uncured default in the payment of principal of or interest or any premium on the debt securities.

We are required to file annually with the trustee a certificate either stating the absence of any default or specifying any default that exists. The trustee is required, within 90 days after the occurrence of a default with respect to the debt securities of any series, to give to the holders of the debt securities notice of all uncured defaults known to it. However, except in the case of default in the payment of principal and premium, if any, or interest on any of the debt securities of that series, the trustee will be protected in withholding that notice if the trustee in good faith determines that the withholding of that notice is in the interest of the holders of the debt securities of that series. The term “default,” for the purpose of this provision only, means the occurrence of any of the events of default specified above excluding any grace periods.

The trustee is entitled, subject to the duty of the trustee during a default to act with the required standard of care, to be indemnified by the holders of the debt securities of any series before proceeding to exercise any right or power under the 1991 Indenture at the request of those holders. The 1991 Indenture provides that the holders of a majority in principal amount of each series of outstanding debt securities may direct, with regard to that series, the time, method and place of conducting any proceeding for any remedy available to the trustee, or exercising any trust or power conferred on the trustee, provided that the trustee may decline to act if that direction is contrary to law or if the trustee determines in good faith that the proceeding so directed would be illegal or would involve it in personal liability.

Modification of the 1991 Indenture

The trustee and we, with the consent of the holders of not less than 66 2/3% in aggregate principal amount of each series of the debt securities at the time outstanding affected thereby, may execute supplemental indentures amending, changing or eliminating the provisions of the 1991 Indenture or of any supplemental indenture or modifying in any manner the rights of the holders of those debt securities. However, no supplemental indenture of that kind may:

- extend the fixed maturity of any debt securities or the time of payment of interest, reduce the interest rate, the principal amount or any premium to be paid upon redemption or the amount of principal of an original issue discount security that would be payable upon acceleration of maturity, or impair or affect the right of any debt security holder to institute suit for payment or the right of repayment, if any, at the option of the holder of debt securities, without the consent of the holder of each debt securities so affected; or

- reduce the above percentage of debt securities, the holders of which are required to consent to any supplemental indenture of that kind, without the consent of the holders of all the affected debt securities then outstanding.

In some circumstances, the holders of a majority in aggregate principal amount of each series of debt securities may waive all defaults and rescind and annul a declaration that the series of debt securities has become due and payable and the consequences of a declaration of that kind.

The trustee and we, without the consent of the holders of the debt securities, may execute an indenture or supplemental indentures to:

- evidence the succession of another corporation to us and our successor's assumption to our agreements and obligations with respect to the debt securities and the indenture;
- add to our covenants further restrictions or conditions that our board of directors and the trustee consider to be for the protection of holders of all or any series of the debt securities and to make the occurrence of a default in any of those additional covenants, restrictions or conditions a default or an event of default permitting enforcement of all or any of the several remedies provided in the indenture with some permissible limitations;
- cure ambiguities or correct or supplement any provision contained in the indenture or any supplemental indenture that may be defective or inconsistent with another provision;
- provide for the issuance of debt securities whether or not then outstanding under the indenture in coupon form and to provide for exchangeability of the coupon form securities with debt securities issued under the indenture in fully registered form;
- establish the form or terms and to provide for the issuance of any series of debt securities under the indenture; and
- evidence and provide for the acceptance of appointment of a successor trustee and to change the indenture as necessary to have more than one trustee under the indenture.

Defeasance of Offered Debt Securities in Certain Circumstances

The 1991 Indenture provides that our board of directors may provide by resolution that we will be discharged from any and all obligations in respect of the debt securities of any series upon the deposit with the trustee, in trust, of money and/or obligations of, or obligations the principal of and interest on which are fully guaranteed by, the United States of America, which through the payment of interest and principal those debt securities in accordance with their terms will provide money in an amount sufficient to pay any installment of principal of and interest on the debt securities of that series on the stated maturity of that payments in accordance with the terms of the 1991 Indenture and those debt securities. A discharge may only occur if we have received from, or there has been published by, the U.S. Internal Revenue Service a ruling to the effect that the discharge will not be deemed, or result in, a taxable event with respect to holders of the debt securities of that series.

Definitions

The 1991 Indenture contains the following defined terms:

"***Consolidated Capitalization***" means the total of all the assets appearing on our and our subsidiaries' consolidated balance sheets less current liabilities and deferred income taxes.

"***Consolidated Net Tangible Assets***" means the total of all the assets appearing on our and our subsidiaries' consolidated balance sheets less:

- current liabilities;
- reserves for depreciation and other asset valuation reserves;
- intangible assets such as goodwill, trademarks, trade names, patents, and unamortized debt discount and expense; and
- appropriate adjustments on account of minority interests of other persons holding stock in any of our majority-owned subsidiaries.

"***Operating Property***" means any manufacturing or processing plant, office facility, retail store, wholesale club, Supercenter, hypermart, warehouse, distribution center or equipment located within the United States of America or its territories or possessions and owned and operated now or hereafter by us or any of our subsidiaries and having a book value on the date as of which the determination is being made of more than 0.60% of Consolidated Net Tangible Assets; provided, however, that separate items of equipment with an aggregate book value in excess of $200,000,000 that are secured pursuant to the same financing transaction will constitute one "Operating Property."

"***Sale and Lease-back Transaction***" means any arrangement with any person providing for the leasing to us or any of our subsidiaries of any Operating Property, except for temporary leases for a term, including any renewal thereof, of not more than 48 months and except for leases between us and one of our subsidiaries or between our subsidiaries, which Operating Property has been or is to be sold or transferred by us or one of our subsidiaries to that person.

"***Value***" means, with respect to a Sale and Lease-back Transaction, as of any particular time, the amount equal to the greater of:

- the net proceeds from the sale or transfer of the property leased pursuant to that Sale and Lease-back Transaction; or
- the sum of all of our costs incurred in connection with the acquisition of that property and the construction of any improvements thereon, as determined in good faith by us at the time of entering into that Sale and Lease-back Transaction,

in either case multiplied by a fraction, the numerator of which shall be equal to the number of full years of the term of the lease remaining at the time of determination and the denominator of which shall be equal to the number of full years of that term, without regard to any renewal or extension options contained in the lease.

The Notes

The 2.550% 2026 Notes

The 2.550% 2026 Notes were issued under and pursuant to the 2005 Indenture. The 2.550% 2026 Notes were issued in an aggregate principal amount of €650,000,000. The maturity date of the 2.550% 2026 Notes is April 8, 2026, and interest at a rate of 2.550% per annum is paid annually on April 8 of each year and on the maturity date.

Ranking

The 2.550% 2026 Notes are our senior unsecured debt obligations and rank equally among themselves and with all of our existing and future unsecured and unsubordinated debt obligations. Consequently, the holders of the 2.550% 2026 Notes have a right to payment equal to that of our other unsecured creditors. None of our subsidiaries have any obligation as to any of the 2.550% 2026 Notes or will guarantee the payment of amounts owing with respect to any of the 2.550% 2026 Notes. Neither the 2.550% 2026 Notes nor the 2005 Indenture restricts the ability of our subsidiaries to incur indebtedness.

Payment on the 2.550% 2026 Notes

All payments of principal, including payments made upon any redemption of the 2.550% 2026 Notes, of, accrued interest on, and any additional amounts payable with respect to, the 2.550% 2026 Notes are payable in euro; provided, however, if the euro is unavailable to us due to the imposition of exchange controls or other circumstances beyond our control or if the euro is no longer being used by the then member states of the European Monetary Union that have adopted the euro as their currency or for the settlement of transactions by public institutions of or within the international banking community, then all payments in respect of the 2.550% 2026 Notes will be made in U.S. dollars until the euro is again available to us or so used. The amount payable on any date in euro will be converted into U.S. dollars at the rate mandated by the Board of Governors of the U.S. Federal Reserve System as of the close of business on the second Business Day prior to the relevant payment date or, in the event the Board of Governors of the U.S. Federal Reserve System has not mandated a rate of conversion on such date, on the basis of the most recent U.S. dollar/euro exchange rate published in *The Wall Street Journal* on or most recently prior to the second Business Day prior to the relevant payment date. Any payment in respect of the 2.550% 2026 Notes so made in U.S. dollars will not constitute an event of default under such 2.550% 2026 Notes or the 2005 Indenture.

"***Business Day***" means any day, other than a Saturday or a Sunday, (1) which is not a day on which banking institutions are authorized or obligated by law or executive order to close in New York City or London and (2) on which the Trans-European Automated Real-time Gross Settlement Express Transfer system (the TARGET2 system), or any successor thereto, is open.

Payment of Additional Amounts

All payments of principal and interest in respect of the 2.550% 2026 Notes will be made free and clear of, and without deduction or withholding for or on account of any present or future taxes, duties, assessments or other governmental charges of whatsoever nature required to be deducted or withheld by the United States or any political subdivision or taxing authority of or in the United States, unless such withholding or deduction is required by law.

In the event any withholding or deduction on payments in respect of the 2.550% 2026 Notes for or on account of any present or future tax, assessment or other governmental charge is required to be deducted or withheld by the United States or any taxing authority thereof or therein, we will pay such additional amounts on the 2.550% 2026 Notes as will result in receipt by each beneficial owner of a Note that is a Non-U.S. Person (as defined below) of such amounts (after all such withholding or deduction, including on any additional amounts) as would have been received by such beneficial owner had no such withholding or deduction been required. We will not be required, however, to make any payment of additional amounts for or on account of:

i. any tax, assessment or other governmental charge that would not have been imposed but for (A) the existence of any present or former connection (other than a connection arising solely from the ownership of those 2.550% 2026 Notes or the receipt of payments in respect of those 2.550% 2026 Notes) between that beneficial owner, or between a fiduciary, settlor, beneficiary of, member or shareholder of, or possessor of a power over, that beneficial owner, if that beneficial owner is an estate, trust, partnership or corporation, and the United States, including that beneficial owner, or that fiduciary, settlor, beneficiary, member, shareholder or possessor, being or having been a citizen or resident or treated as a resident of the United States or being or having been engaged in trade or business or present in the United States or (B) the presentation of a Note for payment on a date more than 30 days after the later of the date on which that payment becomes due and payable and the date on which payment is duly provided for;

ii. any estate, inheritance, gift, sales, transfer, excise, personal property or similar tax, assessment or other governmental charge;

iii. any tax, assessment or other governmental charge imposed by reason of that beneficial owner's past or present status as a passive foreign investment company, a controlled foreign corporation or a personal holding company with respect to the United States or as a corporation that accumulates earnings to avoid U.S. federal income tax;

iv. any tax, assessment or other governmental charge which is payable otherwise than by withholding from payment of principal of or premium, if any, or interest on such holder's 2.550% 2026 Notes;

v. any tax, assessment or other governmental charge required to be withheld by any paying agent from any payment of principal of and premium, if any, or interest on any Note if that payment can be made without withholding by any other paying agent;

vi. any tax, assessment or other governmental charge which would not have been imposed but for the failure of a beneficial owner or any holder of 2.550% 2026 Notes to comply with our request to comply with certification, information, documentation or other reporting requirements concerning the nationality, residence, identity or connections with the United States of the beneficial owner or any holder of the 2.550% 2026 Notes, if such compliance is required by statute or by regulation of the U.S. Treasury Department as a precondition to relief or exemption from such tax, assessment or other governmental charge, including, without limitation, any withholding required pursuant to Sections 1471 through 1474 of the U.S. Internal Revenue Code of 1986, as amended (the "***Code***");

vii. any tax, assessment or other governmental charge imposed on interest received by (A) a 10% shareholder (as defined in Section 871(h)(3)(B) of the Code and the regulations that may be promulgated thereunder) of our Company or (B) a controlled foreign corporation that is related to us within the meaning of Section 864(d)(4) of the Code;

viii. any withholding or deduction that is imposed on a payment to an individual and is required to be made pursuant to that European Union Directive relating to the taxation of savings adopted on June 3, 2003 by the European Union's Economic and Financial Affairs Council, or any law implementing or complying with, or introduced in order to conform to, such Directive; or

ix. any combination of items (i), (ii), (iii), (iv), (v), (vi), (vii) and (viii).

In addition, we will not pay any additional amounts to any beneficial owner or holder of 2.550% 2026 Notes who is a fiduciary or partnership to the extent that a beneficiary or settlor with respect to that fiduciary or a member of that partnership or a beneficial owner thereof would not have been entitled to the payment of those additional amounts had that beneficiary, settlor, member or beneficial owner been the beneficial owner of those 2.550% 2026 Notes.

As used in the preceding section, "***Non-U.S. Person***" means any corporation, partnership, individual or fiduciary that is, as to the United States, a foreign corporation, a non-resident alien individual who has not made a valid election to be treated as a United States resident, a non-resident fiduciary of a foreign estate or trust or a foreign partnership, one or more of the members of which is, as to the United States, a foreign corporation, a non-resident alien individual or a non-resident fiduciary of a foreign estate or trust.

Redemption for Tax Reasons

If, as a result of any change or amendment to the laws, treaties, regulations or rulings of the United States or any political subdivision or taxing authority thereof, any proposed change in such laws, treaties, regulations or rulings, or any change in the official application, enforcement or interpretation of those laws, treaties, regulations or rulings, we become, or based upon a written opinion of independent counsel selected by us, will become obligated to pay additional amounts as described above under the heading "Payments of Additional Amounts" with respect to the 2.550% 2026 Notes, then we may redeem the 2.550% 2026 Notes, in whole, but not in part, at our option, on not more than 60 days' and not less than 30 days' prior notice, at a redemption price equal to 100% of the principal amount of the 2.550% 2026 Notes, plus accrued but unpaid interest, if any, on the 2.550% 2026 Notes to the date fixed for redemption.

Optional Redemption

We may redeem the 2.550% 2026 Notes, in whole or in part, at our option, at any time. If the 2.550% 2026 Notes are redeemed before January 8, 2026, then the 2.550% 2026 Notes will be redeemed at a redemption price equal to the greater of:

- 100% of the principal amount of the 2.550% 2026 Notes to be redeemed; or
- the sum of the present values of the remaining scheduled payments of principal and interest thereon (not including any portion of such payments of interest accrued as of the date of redemption), discounted to the date of redemption on an annual basis (Actual/Actual (ICMA)) at the applicable Comparable Government Bond Rate (as defined below), plus 15 basis points, plus, in each case, accrued and unpaid interest on the principal amount being redeemed to, but excluding, the date fixed for redemption.

Any 2.550% 2026 Notes redeemed on or after January 8, 2026 will be redeemed at a redemption price equal to 100% of the principal amount of the 2.550% 2026 Notes then outstanding to be redeemed, plus accrued and unpaid interest on the principal amount being redeemed to, but excluding, the date fixed for redemption.

Installments of interest on 2.550% 2026 Notes being redeemed that are due and payable on interest payment dates falling on or prior to a redemption date shall be payable on the interest payment date to the holders as of the close of business on the relevant regular record date according to the 2.550% 2026 Notes and the 2005 Indenture.

"***Comparable Government Bond***" means, in relation to any Comparable Government Bond Rate calculation for the 2.550% 2026 Notes, at the discretion of an independent investment bank selected by us, a German government bond whose maturity is closest to the maturity of the 2.550% 2026 Notes, or if such independent investment bank in its discretion determines that such similar bond is not in issue, such other German government bond as such independent investment bank may, with the advice of three brokers of, and/or market makers in, German government bonds selected by us, determine to be appropriate for determining the Comparable Government Bond Rate.

"***Comparable Government Bond Rate***" means the price, expressed as a percentage (rounded to three decimal places, with 0.0005 being rounded upwards), at which the gross redemption yield on the 2.550% 2026 Notes being redeemed, if they were to be purchased at such price on the third Business Day prior to the date fixed for redemption, would be equal to the gross redemption yield on such Business Day of the Comparable Government Bond on the basis of the middle market price of the Comparable Government Bond prevailing at 11:00 a.m. (London time) on such Business Day as determined by an independent investment bank selected by us. Such independent bank will calculate such gross redemption yield on the 2.550% 2026 Notes to be redeemed and the Comparable Government Bond in accordance with generally accepted market practices at the time of such calculations.

Discharge, Legal Defeasance and Covenant Defeasance

We may, at our option, elect to have all of the obligations discharged with respect to the outstanding 2.550% 2026 Notes, except for:

- the rights of holders of the 2.550% 2026 Notes to receive payments of principal, premium, if any, interest and additional amounts, if any, from the trust referred to below when those payments are due;
- our obligations respecting the 2.550% 2026 Notes concerning issuing temporary debt securities, registration of transfers of debt securities, mutilated, destroyed, lost or stolen debt securities, the maintenance of an office or agency for payment and money for payments with respect to the 2.550% 2026 Notes being held in trust;
- the rights, powers, trusts, duties and immunities of the trustee and our obligations in connection therewith; and
- the provisions of the 2005 Indenture relating to such a discharge of obligations.

We refer to a discharge of this type as "***legal defeasance***."

In addition, other than our covenant to pay the amounts due and owing with respect to the 2.550% 2026 Notes, we may elect to have our obligations as the issuer of the 2.550% 2026 Notes released with respect to covenants relating to the 2.550% 2026 Notes. Thereafter, any failure to comply with those obligations will not constitute a default or event of default with respect to the 2.550% 2026 Notes. If such a release of our covenants occurs, our failure to perform or our breach of the covenants or warranties defeased will no longer constitute an event of default with respect to the 2.550% 2026 Notes. We refer to a discharge of this type as "***covenant defeasance***."

To exercise either of the defeasance rights described above as to the outstanding 2.550% 2026 Notes, certain conditions must be met, including:

- we must irrevocably deposit with the trustee, in trust for the benefit of the holders of the outstanding 2.550% 2026 Notes, moneys in the currency in which the 2.550% 2026 Notes are denominated, securities issued by a government, governmental agency or central bank of the country in whose currency the 2.550% 2026 Notes are denominated or a combination of cash and such securities, in amounts sufficient to pay the principal of and premium, if any, and interest on all of the then outstanding 2.550% 2026 Notes to be affected by the defeasance at their stated maturity;
- no default or event of default exists on the date of such deposit, subject to certain exceptions;
- the trustee must receive an opinion of counsel confirming that the holders of the outstanding 2.550% 2026 Notes will not recognize income, gain or loss for U.S. federal income tax purposes as a result of that legal defeasance or covenant defeasance and will be subject to U.S. federal income tax on the same amounts, in the same manner and at the same times as would have been the case if that defeasance had not occurred, which opinion, only in the case of the legal defeasance of the 2.550% 2026 Notes, will be based on a ruling of the Internal Revenue Service or a change in United States federal income tax law to that effect occurring after the date of the 2005 Indenture; and
- the trustee must receive an opinion of counsel to the effect that, after the 91st day following the deposit, the trust funds will not be part of any "estate" formed by the bankruptcy of the party depositing those funds with the trustee or subject to the "automatic stay" under the United States Bankruptcy Code or, in the case of covenant defeasance, will be subject to a first priority lien in favor of the trustee for the benefit of the holders of the outstanding 2.550% 2026 Notes.

We may exercise our legal defeasance option notwithstanding our prior exercise of our covenant defeasance option.

Book-Entry and Settlement

The 2.550% 2026 Notes were issued in book-entry form and are represented by global notes deposited with, or on behalf of, a common depositary on behalf of Clearstream Banking, société anonyme ("***Clearstream***") and Euroclear Bank SA/NV, as the operator of the Euroclear System ("***Euroclear***"), and are registered in the name of the common depositary or its nominee. Beneficial interests in the global notes will be shown on, and transfers of beneficial interests in the global notes will be made only through, records maintained by Euroclear and Clearstream. Except as described below, certificated notes will not be issued in exchange for beneficial interests in the global notes.

Certificated Notes

Subject to certain conditions, the 2.550% 2026 Notes represented by the global notes are exchangeable for certificated debt securities with the same terms in authorized denominations only if:

- The Depository Trust Company, Euroclear or Clearstream, as the case may be, is unwilling or unable to continue as depositary or ceases to be a clearing agency registered under applicable law, and a successor is not appointed by us within 90 days; or
- we decide to discontinue the book-entry system; or
- an event of default has occurred and is continuing with respect to the 2.550% 2026 Notes.

Any Note that is exchangeable as above is exchangeable for certificated notes issuable in authorized denominations and registered in such names as the common depositary shall direct. Subject to the foregoing, a global note is not exchangeable, except for a global note of the same aggregate denomination to be registered in the name of the common depositary or its nominee.

The 2026 Notes and 2028 Notes

The 2026 Notes and 2028 Notes were issued under and pursuant to the 2005 Indenture. The 2026 Notes and the 2028 Notes are each a separate series of notes under the 2005 Indenture. The 2026 Notes and 2028 Notes were issued in registered book-entry form without interest coupons in minimum denominations of $2,000 and integral multiples of $1,000 in excess thereof. The 2026 Notes and 2028 Notes constitute our senior unsecured debt obligations and rank equally among themselves and with all of our other existing and future senior unsecured debt.

The 2026 Notes were issued in an aggregate principal amount of $1,250,000,000, and the 2028 Notes were issued in an aggregate principal amount of $1,250,000,000.

Interest Rates on the 2026 Notes and 2028 Notes

The 2026 Notes bear interest from September 17, 2021 at a rate equal to 1.050%, and the 2028 Notes bear interest from September 22, 2021 at a rate equal to 1.500%. Interest on the notes is paid semi-annually, in the case of 2026 Notes, on March 17 and September 17 of each year, beginning on March 17, 2022, and in the case of the 2028 Notes, on March 22 and September 22 of each year, beginning on March 22, 2022.

Interest on each 2026 Note or 2028 Note, as applicable, is payable to the person in whose name such note is registered at the close of business, in the case of the 2026 Notes, on the March 2 and September 2 immediately preceding the applicable interest payment date, and in the case of the 2028 Notes, on the March 7 and September 7 immediately preceding the applicable interest payment date. Interest on the 2026 Notes and 2028 Notes is computed on the basis of a 360-day year of twelve 30-day months.

Other Terms of the 2026 and 2028 Notes

If any interest payment date for 2026 Notes or 2028 Notes would otherwise be a day that is not a business day, then the interest payment date for such notes of that series is postponed to the following date that is a business day.

Interest will not accrue as a result of any such postponed payment. The term "***business day***" means any day which is not a day on which banking institutions in The City of New York, or the relevant place of payment are authorized or required by law, regulation or executive order to close.

None of the 2026 Notes or 2028 Notes are subject to a sinking fund or are convertible into or exchangeable for any other securities.

The 2026 Notes or 2028 Notes are subject to defeasance.

Optional Redemption

We may redeem the 2026 Notes at any time prior to August 17, 2026 (one month prior to the maturity date of such notes) and the 2028 Notes at any time prior to July 22, 2028 (two months prior to the maturity date of such notes), in each case, at our option and, as to each series of notes, as a whole or in part, at a redemption price equal to the greater of:

- 100% of the principal amount of such notes to be redeemed, plus any accrued and unpaid interest to, but excluding, the redemption date; and
- the sum of the present values of the Remaining Scheduled Payments (as defined below), plus any accrued and unpaid interest to, but excluding, the redemption date.

We refer to the date on which we may first redeem the notes of each series pursuant to the redemption rights described in this paragraph as the "***Par Call Date***" for the notes of such series. In determining the present value of the Remaining Scheduled Payments, we will discount such payments to the redemption date on a semi-annual basis (assuming a 360-day year consisting of twelve 30-day months) using a discount rate equal to the Applicable Treasury Rate plus, in the case of the 2026 Notes, 5 basis points, and, in the case of the 2028 Notes, 10 basis points.

The term "***Applicable Treasury Rate***" means, with respect to any redemption date, the rate per annum equal to the semi-annual equivalent yield to maturity of the Comparable Treasury Issue. In determining this rate, we will assume a price for the Comparable Treasury Issue (expressed as a percentage of its principal amount) equal to the Comparable Treasury Price for such redemption date.

The term "***Comparable Treasury Issue***" means the United States Treasury security selected by the Independent Investment Banker as having a maturity comparable to the remaining term of the notes of a series to be redeemed (assuming the notes of such series matured on the Par Call Date (as defined below) for the notes of such series) that would be utilized, at the time of selection and in accordance with customary financial practice, in pricing a new issue of corporate debt securities of comparable maturity to the remaining term of the notes of such series.

The term "***Independent Investment Banker***" means one of the Reference Treasury Dealers that we appoint to act as the Independent Investment Banker from time to time.

The term "***Comparable Treasury Price***" means, with respect to any redemption date, the arithmetic average, as determined by the Independent Investment Banker, of the Reference Treasury Dealer Quotations for such redemption date.

The term "***Reference Treasury Dealer Quotations***" means, with respect to each Reference Treasury Dealer and any redemption date, the arithmetic average, as determined by the Independent Investment Banker, of the bid and asked prices for the Comparable Treasury Issue (expressed in each case as a percentage of its principal amount) quoted in writing to the Independent Investment Banker by such Reference Treasury Dealer by 5:00 p.m., New York City time, on the third business day preceding such redemption date.

The term "***Reference Treasury Dealer***" means each of Citigroup Global Markets Inc., Mizuho Securities USA LLC and Wells Fargo Securities, LLC or one of their respective affiliates or successors; provided, however, that if any of the foregoing shall cease to be a primary U.S. Government securities dealer (a "***Primary Treasury Dealer***"), we will substitute another Primary Treasury Dealer for such entity.

The term "***Remaining Scheduled Payments***" means, with respect to any 2026 Note or 2028 Note, the remaining scheduled payments of the principal thereof to be redeemed and interest thereon that would be due after the related redemption date but for such redemption (assuming the notes of such series matured on the Par Call Date for the notes of such series); provided, however, that, if such redemption date is not an interest payment date with respect to such note, then the amount of the next scheduled interest payment thereon will be reduced by the amount of interest accrued thereon to such redemption date.

Pursuant to the redemption procedures with respect to any such note, a notice of redemption will be given to each holder of the 2026 Notes or 2028 Notes to be redeemed not less than 10 days nor more than 60 days prior to the date set for the redemption.

The calculation of the redemption price for any notes to be redeemed and the accrued interest payable upon a redemption of such notes will be made by the Company or on behalf of the Company by a person the Company designates. Such calculation will not be the duty or obligation of BNYM unless otherwise expressly agreed.

Notices

Notices to holders of the 2026 Notes or 2028 Notes will be sent to such holders. Any notice shall be deemed to have been given on the date of mailing. So long as the 2026 Notes or 2028 Notes are represented by one or more global securities deposited with DTC or its nominee, notices to holders of such notes may be given by delivery of those notices to DTC, and such notices shall be deemed to be given on the date of delivery to DTC. BNYM will transmit notices to each registered holder's last known address as it appears in the security register that BNYM maintains. BNYM will only transmit these notices to the registered holder of the 2026 Notes or 2028 Notes. Holders of 2026 Notes or 2028 Notes will not receive notices regarding the 2026 Notes or 2028 Notes directly from us unless we reissue the 2026 Notes or 2028 Notes in fully certificated form.

Other Matters

We may acquire the 2026 Notes or 2028 Notes by means other than a redemption, whether by tender offer, open market purchases, negotiated transactions or otherwise, in accordance with applicable securities laws, so long as such acquisition does not otherwise violate the terms of the 2005 Indenture.

Same-Day Settlement and Payment

We will make all payments of principal and interest on the 2026 Notes or 2028 Notes to DTC in immediately available funds. The 2026 Notes and 2028 Notes will trade in the same-day funds settlement system in the United States until maturity. Purchases of 2026 Notes or 2028 Notes in secondary market trading must be in immediately available funds. Secondary market trading in the 2026 Notes or 2028 Notes between participants in Clearstream and Euroclear will occur in accordance with the applicable rules and operating procedures of Clearstream and Euroclear and will be settled using the procedures applicable to eurobonds in immediately available funds.

The 2030 Notes

The 2030 Notes were issued under and pursuant to the 1991 Indenture. The 2030 Notes were issued in registered book-entry form, without coupons, only in denominations of (Pounds) 1,000, (Pounds) 10,000 and (Pounds) 100,000 and integral multiples of (Pounds) 1,000 in excess of those denominations. The 2030 Notes constitute our senior unsecured debt obligations and rank equally among themselves and with all of our existing and future senior, unsecured and unsubordinated debt.

The 2030 Notes will mature on December 19, 2030 and were issued in a total principal amount of (Pounds) 500 million. Unless previously redeemed or purchased and cancelled, we will repay the 2030 Notes at 100% of their principal amount together with accrued and unpaid interest thereon at maturity. We will pay principal of, interest on and any other amounts payable under the 2030 Notes in pounds sterling or, if the United Kingdom adopts the euro, in euro.

We may, without the consent of the holders of the 2030 Notes, create and issue additional notes ranking equally with the 2030 Notes and otherwise similar in all respects to the 2030 Notes so that these additional notes will be consolidated and form a single series with the 2030 Notes. No additional 2030 Notes may be issued if an event of default under the 1991 Indenture has occurred.

The 2030 Notes are not subject to a sinking fund. The 2030 Notes are not convertible or exchangeable.

The 2030 Notes bear interest from December 19, 2000, at the annual interest rate of 5.750%. Interest is payable semi-annually in arrears on June 19 and December 19 of each year, beginning on June 19, 2001, to the person in whose name a 2030 Note is registered at the close of business on the preceding June 10 or December 10, as the case may be.

If interest is required to be calculated for any period other than from one scheduled interest payment date to the next interest payment date, it will be calculated on the basis of a 360-day year of twelve 30-day months.

If any interest payment date for the 2030 Notes would otherwise be a day that is not a business day, then the interest payment date will be postponed to the following date that is a business day. Interest will not accrue as a result of any delayed payment. The term "business day" means any day, other than a Saturday or Sunday, that is neither a legal holiday nor a day on which banking institutions are generally authorized or required by law or regulation to close in New York City or London.

If, prior to the maturity of the 2030 Notes, the United Kingdom adopts the euro as its lawful currency in accordance with the Treaty establishing the European Communities, as amended by the Treaty on European Union, the 2030 Notes will be re-denominated into euro, and the regulations of the European Commission relating to the euro shall apply to the 2030 Notes. The circumstances and consequences described in this paragraph will not entitle us, the trustee under the 1991 Indenture or any holder of the 2030 Notes to redeem early, rescind, or receive notice relating to the 2030 Notes, repudiate the terms of the 2030 Notes or the 1991 Indenture, or raise any defense, request any compensation or make any claim related to the 2030 Notes or the 1991 Indenture, nor will these circumstances and consequences affect any of our other obligations under the 2030 Notes or the 1991 Indenture.

While the 2030 Notes are represented by a global note deposited with the common depositary for Clearstream, and Euroclear, notices to holders may be given by delivery to Clearstream and Euroclear and such notices shall be deemed to be given on the date of delivery to Clearstream and Euroclear. The trustee will mail notices by first-class mail, postage prepaid, to each registered holder's last known address as it appears in the security register that the trustee maintains. The trustee will only mail these notices to the registered holder of the 2030 Notes. Holders of 2030 Notes will not receive notices regarding the 2030 Notes directly from us unless we reissue the 2030 Notes in fully certificated form.

The trustee will also publish notices regarding the 2030 Notes in a daily newspaper of general circulation in The City of New York and in London. In addition, if the notes are listed on the Luxembourg Stock Exchange, and so long as the rules of the Luxembourg Stock Exchange require notice by publication, the trustee will publish notices regarding the 2030 Notes in a daily newspaper of general circulation in Luxembourg. We expect that publication will be made in The City of New York in *The Wall Street Journal*, in London in the *Financial Times*, and in Luxembourg in the *Luxemburger Wort*. If publication in Luxembourg is not practical, the trustee will publish these notices in an English language newspaper of general circulation elsewhere in Europe. Published notices will be deemed to have been given on the date they are published or, if published more than once, on the date of first publication. If publication as described above becomes impossible, the trustee may publish sufficient notice by alternate means that approximate the terms and conditions described in this paragraph.

Optional Redemption

The 2030 Notes will be redeemable as a whole or in part, at our option, at any time after December 19, 2003, at a redemption price equal to the greater of (i) 100% of the principal amount of such 2030 Notes and (ii) as determined by the Calculation Agent, the price at which the yield on the outstanding principal amount of the 2030 Notes on the Reference Date is equal to the yield on the Benchmark Gilt as of that date as determined by reference to the middle-market price on the Benchmark Gilt at 3:00 p.m., London time, on that date, in either case, plus accrued and unpaid interest on the 2030 Notes up to, but excluding, the date specified as the redemption date.

"***Reference Date***" means the date that is the first dealing day in London prior to the publication of the notice of redemption referred to below.

"***Benchmark Gilt***" means the 4.25% Treasury Stock due June 2032 or such other U.K. government stock as the Calculation Agent, with the advice of three brokers and/or U.K. gilt-edged market makers or three other persons operating in the U.K. gilt-edged market that may be chosen by the Calculation Agent, may determine from time to time to be the most appropriate benchmark U.K. government stock for the notes.

"***Calculation Agent***" means the trustee or any successor entity.

We will give notice of any redemption between 30 and 60 days preceding the redemption date to each holder of the 2030 Notes to be redeemed as described above.

In the case of any partial redemption, selection of the 2030 Notes for redemption will be made by the trustee under the 1991 Indenture in compliance with the rules and requirements of the principal securities exchange, if any, on which the 2030 Notes are listed or, if the 2030 Notes are not so listed or that exchange prescribes no method of selection, on a pro rata basis, by lot or by any other method as the trustee in its sole discretion deems to be fair and appropriate, although no 2030 Note of (Pounds) 1,000 in original principal amount or less shall be redeemed in part. If any 2030 Note is to be redeemed in part only, the notice of redemption relating to the 2030 Note will state the portion of the principal amount thereof to be redeemed. A new 2030 Note in principal amount equal to the unredeemed portion thereof will be issued and delivered to the trustee, or its nominee, or, in the case of 2030 Notes in definitive form, issued in the name of the holder thereof, in each case upon cancellation of the original 2030 Note.

Unless we default in payment of the redemption price of the 2030 Notes, on and after the redemption date, interest will cease to accrue on the notes or the portions of the 2030 Notes called for redemption.

Payment of Additional Amounts

We will pay to the holder of any 2030 Note who is a United States Alien, as defined below, additional amounts as may be necessary so that every net payment of the principal of, interest on and any other amounts payable under that 2030 Note, after deduction or withholding for or on account of any present or future tax, assessment or other governmental charge imposed upon that holder by the United States or any taxing authority thereof or therein, will not be less than the amount provided in that 2030 Note to be then due and payable. We will not be required, however, to make any payment of additional amounts for or on account of:

a. any tax, assessment or other governmental charge that would not have been imposed but for (1) the existence of any present or former connection between that holder, or between a fiduciary, settlor, beneficiary of, member or shareholder of, or possessor of a power over, that holder, if that holder is an estate, trust, partnership or corporation, and the United States including, without limitation, that holder, or that fiduciary, settlor, beneficiary, member, shareholder or possessor, being or having been a citizen or resident or treated as a resident of the United States or being or having been engaged in trade or business in, or present in, the United States or (2) the presentation of a note for payment on a date more than 30 days after the later of the date on which that payment becomes due and payable and the date on which payment is duly provided for;

b. any estate, inheritance, gift, sales, transfer, excise, personal property or similar tax, assessment or other governmental charge;

c. any tax, assessment or other governmental charge imposed by reason of that holder's past or present status as a passive foreign investment company, a controlled foreign corporation, a personal holding company or foreign personal holding company with respect to the United States, or as a corporation which accumulates earnings to avoid United States federal income tax;

d. any tax, assessment or other governmental charge which is payable otherwise than by withholding from payment of the principal of, interest on or any other amounts payable under that note;

e. any tax, assessment or other governmental charge required to be withheld by any paying agent from any payment of the principal of, interest on or any other amounts payable under any note if that payment can be made without withholding by any other paying agent;

f. any tax, assessment or other governmental charge which would not have been imposed but for the failure to comply with certification, information, documentation or other reporting requirements concerning the nationality, residence, identity or connections with the United States of the holder or beneficial owner of that note, if such compliance is required by statute or by regulation of the U.S. Treasury Department as a precondition to relief or exemption from such tax, assessment or other governmental charge;

g. any tax, assessment or other governmental charge imposed on interest received by (1) a 10% shareholder (as defined in Section 871(h)(3)(B) of the U.S. Internal Revenue Code of 1986 and the regulations that may be promulgated thereunder) of our company or (2) a controlled foreign corporation with respect to our company within the meaning of the U.S. Internal Revenue Code; or

h. any combination of items (a), (b), (c), (d), (e), (f) and (g);

nor will we pay any additional amounts to any holder who is a fiduciary or partnership other than the sole beneficial owner of that note to the extent that a beneficiary or settlor with respect to that fiduciary, or a member of that partnership or a beneficial owner thereof would not have been entitled to the payment of those additional amounts had that beneficiary, settlor, member or beneficial owner been the holder of that 2030 Note.

"***United States Alien***" means any corporation, partnership, individual or fiduciary that is, as to the United States, a foreign corporation, a non-resident alien individual who has not made a valid election to be treated as a United States resident, a non-resident fiduciary of a foreign estate or trust, or a foreign partnership one or more of the members of which is, as to the United States, a foreign corporation, a non-resident alien individual or a non-resident fiduciary of a foreign estate or trust.

Redemption Upon Tax Event

The 2030 Notes may be redeemed at our option in whole, but not in part, on not more than 60 days' and not less than 30 days' notice, at a redemption price equal to 100% of their principal amount, plus accrued and unpaid interest on the notes, if we determine that as a result of any change in or amendment to the laws, treaties, regulations or rulings of the United States or any political subdivision or taxing authority thereof, or any proposed change in such laws, treaties, regulations or rulings, or any change in the official application, enforcement or interpretation of those laws, treaties, regulations or rulings, including a holding by a court of competent jurisdiction in the United States, or any other action, other than an action predicated on law generally known on or before December 5, 2000 except for proposals before the Congress before that date, taken by any taxing authority or a court of competent jurisdiction in the United States, or the official proposal of any action, whether or not the action or proposal was taken or made with respect to our company, (A) we have or will become obligated to pay additional amounts on any 2030 Note or (B) there is a substantial possibility that we will be required to pay those additional amounts. Prior to the publication of any notice of redemption, we will deliver to the trustee (1) an officers' certificate stating that we are entitled to effect a redemption and setting forth a statement of facts showing that the conditions precedent to the right of our company to so redeem have occurred and (2) an opinion of counsel to that effect based on that statement of facts. If we redeem the 2030 Notes upon a tax event, we will publish a notice of that redemption in Luxembourg in the *Luxemburger Wort* at the time notice is given to the holders of the 2030 Notes as described above.

Prescription

Under New York's statute of limitations, any legal action to enforce our payment obligations evidenced by the 2030 Notes must be commenced within six years after payment is due. Thereafter our payment obligations will generally become unenforceable.

Replacement of 2030 Notes

If any mutilated 2030 Note is surrendered to the trustee, we will execute and the trustee will authenticate and deliver in exchange for such mutilated 2030 Note a new note of the same series and principal amount. If the trustee and we receive evidence to our satisfaction of the destruction, loss or theft of any 2030 Note and any security or indemnity required by them, then we shall execute and the trustee shall authenticate and deliver, in lieu of such destroyed, lost or stolen 2030 Note, a new note of the same series and principal amount. All expenses associated with issuing the new note shall be borne by the owner of the mutilated, destroyed, lost or stolen 2030 Note.

The 2031 Notes

The 2031 Notes were issued under and pursuant to the 2005 Indenture. The 2031 Notes were issued in registered book-entry form without interest coupons in minimum denominations of $2,000 and integral multiples of $1,000 in excess thereof. The 2031 Notes constitute our senior unsecured debt obligations and rank equally among themselves and with all of our other existing and future senior unsecured debt.

The 2031 Notes will mature on September 22, 2031. Unless previously purchased and cancelled or, to the extent permitted by the notes, redeemed prior to maturity, we will repay the 2031 Notes at 100% of their principal amount, together with accrued and unpaid interest thereon, at their maturity.

The 2031 Notes were issued in an aggregate principal amount of $2,000,000,000. We may, without the consent of the holders of the 2031 Notes, create and issue additional notes ranking equally with and otherwise similar in all respects to the 2031 Notes (except for the public offering price, initial interest accrual date, initial interest payment date, and the issue date) so that those additional notes will be consolidated and form a single series with the other outstanding 2031 Notes; provided, however, that any additional notes issued that are not fungible with the outstanding 2031 Notes for U.S. federal income tax purposes will be issued under one or more separate CUSIP and ISIN numbers. No additional notes may be issued if an event of default under the indenture has occurred and is continuing.

Interest Rates on the 2031 Notes

The 2031 Notes bear interest from September 22, 2021 at the annual interest rate 1.800%. Interest on the 2031 Notes is paid semi-annually on March 22 and September 22 of each year, beginning on March 22, 2022.

Interest on each 2031 Note is payable to the person in whose name the 2031 Note is registered at the close of business on the March 7 and September 7 immediately preceding the applicable interest payment date. Interest on the 2031 Notes is computed on the basis of a 360-day year of twelve 30-day months.

Other Terms of the 2031 Notes

If any interest payment date for the 2031 Notes would otherwise be a day that is not a business day, then the interest payment date will be postponed to the following date that is a business day. Interest will not accrue as a result of any such postponed payment. The term "***business day***" means any day which is not a day on which banking institutions in The City of New York, or the relevant place of payment are authorized or required by law, regulation or executive order to close.

The 2031 Notes are not subject to a sinking fund and are not convertible into or exchangeable for any other securities.

The 2031 Notes are subject to defeasance.

Optional Redemption

We may redeem the 2031 Notes at any time prior to June 22, 2031 (three months prior to the maturity date of the 2031 Notes), at our option and, as a whole or in part, at a redemption price equal to the greater of:

- 100% of the principal amount of the 2031 Notes to be redeemed, plus any accrued and unpaid interest to, but excluding, the redemption date; and
- the sum of the present values of the Remaining Scheduled Payments (as defined below), plus any accrued and unpaid interest to, but excluding, the redemption date.

We refer to the date on which we may first redeem the 2031 Notes pursuant to the redemption rights described in this paragraph as the "***Par Call Date***" for the 2031 Notes. In determining the present value of the Remaining Scheduled Payments, we will discount such payments to the redemption date on a semi-annual basis (assuming a 360-day year consisting of twelve 30-day months) using a discount rate equal to the Applicable Treasury Rate plus 10 basis points.

The term "***Applicable Treasury Rate***" means, with respect to any redemption date, the rate per annum equal to the semi-annual equivalent yield to maturity of the Comparable Treasury Issue. In determining this rate, we will assume a price for the Comparable Treasury Issue (expressed as a percentage of its principal amount) equal to the Comparable Treasury Price for such redemption date.

The term "***Comparable Treasury Issue***" means the United States Treasury security selected by the Independent Investment Banker as having a maturity comparable to the remaining term of the 2031 Notes to be redeemed (assuming the 2031 Notes matured on the Par Call Date for the 2031 Notes) that would be utilized, at the time of selection and in accordance with customary financial practice, in pricing a new issue of corporate debt securities of comparable maturity to the remaining term of the 2031 Notes.

The term "***Independent Investment Banker***" means one of the Reference Treasury Dealers that we appoint to act as the Independent Investment Banker from time to time.

The term "***Comparable Treasury Price***" means, with respect to any redemption date, the arithmetic average, as determined by the Independent Investment Banker, of the Reference Treasury Dealer Quotations for such redemption date.

The term "***Reference Treasury Dealer Quotations***" means, with respect to each Reference Treasury Dealer and any redemption date, the arithmetic average, as determined by the Independent Investment Banker, of the bid and asked prices for the Comparable Treasury Issue (expressed in each case as a percentage of its principal amount) quoted in writing to the Independent Investment Banker by such Reference Treasury Dealer by 5:00 p.m., New York City time, on the third business day preceding such redemption date.

The term "***Reference Treasury Dealer***" means each of BofA Securities, Inc., Citigroup Global Markets Inc. and Morgan Stanley & Co. LLC or one of their respective affiliates or successors; *provided*, *however*, that if any of the foregoing shall cease to be a primary U.S. Government securities dealer (a "***Primary Treasury Dealer***"), we will substitute another Primary Treasury Dealer for such entity.

The term "***Remaining Scheduled Payments***" means, with respect to the 2031 Notes, the remaining scheduled payments of the principal thereof to be redeemed and interest thereon that would be due after the related redemption date but for such redemption (assuming the 2031 Notes matured on the Par Call Date); *provided*, *however*, that, if such redemption date is not an interest payment date with respect to the 2031 Notes, then the amount of the next scheduled interest payment thereon will be reduced by the amount of interest accrued thereon to such redemption date.

A notice of redemption will be given to each holder of the 2031 Notes to be redeemed not less than 10 days nor more than 60 days prior to the date set for the redemption.

The calculation of the redemption price for any 2031 Notes to be redeemed and the accrued interest payable upon a redemption of such 2031 Notes will be made by the Company or on behalf of the Company by a person the Company designates. Such calculation will not be the duty or obligation of BNYM unless otherwise expressly agreed.

Notices

Notices to holders of the 2031 Notes will be sent to such holders. Any notice shall be deemed to have been given on the date of mailing. So long as the 2031 Notes are represented by one or more global securities deposited with DTC or its nominee, notices to holders of the 2031 Notes may be given by delivery of those notices to DTC, and such notices shall be deemed to be given on the date of delivery to DTC. BNYM will transmit notices to each registered holder's last known address as it appears in the security register that BNYM maintains. BNYM will only transmit these notices to the registered holder of the 2031 Notes. Holders of 2031 Notes will not receive notices regarding the 2031 Notes directly from us unless we reissue the 2031 Notes in fully certificated form.

Other Matters

We may acquire the 2031 Notes by means other than a redemption, whether by tender offer, open market purchases, negotiated transactions or otherwise, in accordance with applicable securities laws, so long as such acquisition does not otherwise violate the terms of the 2005 Indenture.

Same-Day Settlement and Payment

We will make all payments of principal and interest on the 2031 Notes to DTC in immediately available funds. The 2031 Notes will trade in the same-day funds settlement system in the United States until maturity. Purchases of 2031 Notes in secondary market trading must be in immediately available funds. Secondary market trading in the 2031 Notes between participants in Clearstream and Euroclear will occur in accordance with the applicable rules and operating procedures of Clearstream and Euroclear and will be settled using the procedures applicable to eurobonds in immediately available funds.

The 2029 Notes

The 2029 Notes were issued under and pursuant to 2005 Indenture. The 2029 Notes were issued in registered book-entry form without interest coupons in denominations of €50,000 and integral multiples of €1,000 in excess thereof. The 2029 Notes constitute our senior unsecured debt obligations and rank equally among themselves and with all of our existing and future senior unsecured debt.

The 2029 Notes mature on September 21, 2029. Unless previously redeemed or purchased and cancelled, we will repay the 2029 Notes at 100% of their principal amount, together with accrued and unpaid interest thereon at their maturity. We will pay principal of and interest on the 2029 Notes in euro.

The 2029 Notes were issued in an aggregate principal amount of €1,000,000,000. We may, without the consent of the holders of the 2029 Notes, create and issue additional notes ranking equally with and otherwise similar in all respects to the 2029 Notes (except for the offering price and the issue date) so that those additional notes will be consolidated and form a single series with the 2029 Notes. No additional 2029 Notes may be issued if an event of default under the 2005 Indenture has occurred and is continuing.

The 2029 Notes bear interest from September 21, 2009 at the annual interest rate of 4.875%. Interest on each 2029 Note is payable annually in arrears on September 21 of each year, beginning on September 21, 2010, to the person in whose name the 2029 Note is registered at the close of business on the immediately preceding September 15. Interest is computed on the basis of the actual number of days in the period for which interest is being calculated and the actual number of days from and including the last date on which interest was paid on the 2029 Notes (or September 21, 2009 if no interest has been paid on the 2029 Notes), to but excluding the next scheduled interest payment date. This payment convention is referred to as Actual/Actual (ICMA) as defined in the rulebook of the International Capital Market Association.

We will pay to beneficial owners of the 2029 Notes who are non-U.S. persons additional amounts in the event of deduction or withholding of taxes, assessments or other governmental charges imposed by the United States or any taxing authority thereof or therein. Any additional amounts will be paid in euro.

The 2029 Notes are not subject to a sinking fund and are not convertible or exchangeable. The 2029 Notes are generally not be redeemable prior to maturity.

The 2029 Notes are subject to defeasance.

If any interest payment date for the 2029 Notes would otherwise be a day that is not a business day, then the interest payment date will be postponed to the following date that is a business day. Interest will not accrue as a result of any delayed payment. The term "***business day***" means any day, other than a Saturday or Sunday, that (i) is neither a legal holiday nor a day on which banking institutions are generally authorized or required by law or regulation to close in The City of New York or London and (ii) is a day on which the Trans-European Automated Real-time Gross Settlement Express Transfer system (the TARGET2 system), or any successor thereto, operates.

Notices to holders of the 2029 Notes will be mailed to such holders. Any notice shall be deemed to have been given on the date of mailing and publication or, if published more than once, on the date of first publication. So long as the 2029 Notes are represented by a global security deposited with BNYM, as the common depositary (the "***Common Depositary***") for Clearstream and Euroclear, notices to holders may be given by delivery to Clearstream and Euroclear, and such notices shall be deemed to be given on the date of delivery to Clearstream and Euroclear. BNYM will mail notices by first-class mail, postage prepaid, to each registered holder's last known address as it appears in the security register that BNYM maintains. BNYM will only mail these notices to the registered holder of the 2029 Notes. Holders of the 2029 Notes will not receive notices regarding the 2029 Notes directly from us unless we reissue the 2029 Notes in fully certificated form.

BNYM will publish any notices regarding the 2029 Notes in a daily newspaper of general circulation in The City of New York and in London. If the 2029 Notes are admitted to the Official List and to trading on its regulated market, the trustee will publish notices regarding the notes in a daily newspaper of general circulation in Dublin, Ireland for so long as such publication is required pursuant to the Prospectus Directive or the rules of the applicable stock exchange. We expect that publication will be made in The City of New York in *The Wall Street Journal*, in London in the *Financial Times* and in Dublin, Ireland in the *Irish Times*. If publication in Dublin, Ireland is not practical, BNYM will publish these notices in an English language newspaper of general circulation elsewhere in Europe. Published notices will be deemed to have been given on the date they are published or, if published more than once, on the date of first publication. If publication as described above becomes impossible, BNYM may publish sufficient notice by alternate means that approximate the terms and conditions described in this paragraph.

Redemption upon Tax Event

We may redeem the 2029 Notes if certain tax-related events occur. If we redeem the 2029 Notes as a result of a tax event and the 2029 Notes are admitted to the Official List and to trading on its regulated market, we will publish a notice of the redemption in Dublin, Ireland. The redemption price paid for the 2029 Notes upon any such redemption will be paid in euro.

Prescription

Under New York law, any legal action to enforce our payment obligations evidenced by the 2029 Notes must be commenced within six years after payment is due. Thereafter, our payment obligations will generally become unenforceable.

Replacement of the 2029 Notes

If any mutilated 2029 Note is surrendered to BNYM, we will execute and the trustee will authenticate and deliver a new note of the same series and principal amount in exchange for such mutilated 2029 Note. If BNYM and we receive evidence to our satisfaction of the destruction, loss or theft of any 2029 Note and any security or indemnity required by it and us, then we shall execute, and BNYM shall authenticate and deliver, in lieu of such destroyed, lost or stolen note, a new 2029 Note of the same series and principal amount. All expenses associated with issuing the new note shall be borne by the owner of the mutilated, destroyed, lost or stolen 2029 Note.

The 2034 Notes

The 2034 Notes were issued under and pursuant to the 2005 Indenture. The 2034 Notes were issued in registered book-entry form without interest coupons in denominations of £50,000 and integral multiples of £1,000 in excess thereof. The 2034 Notes constitute our senior unsecured debt obligations and rank equally among themselves and with all of our existing and future senior unsecured debt.

The 2034 Notes will mature on March 27, 2034. Unless previously redeemed or purchased and cancelled, we will repay the 2034 Notes at 100% of their principal amount, together with accrued and unpaid interest thereon at maturity. We will pay principal of and interest on the 2034 Notes in pounds sterling or, solely if the United Kingdom adopts the euro, in euro.

The 2034 Notes were issued in an aggregate principal amount of £1,000,000,000. We may, without the consent of the holders of the 2034 Notes, create and issue additional notes ranking equally with and otherwise similar in all respects to the 2034 Notes (except for the public offering price and the issue date) so that those additional notes will be consolidated and form a single series with the 2034 Notes. No additional 2034 Notes may be issued if an event of default under the 2005 Indenture has occurred and is continuing.

The 2034 Notes bear interest from March 27, 2009 at the annual interest rate of 5.625%. Interest on each 2024 Note is payable semi-annually in arrears on March 27 and September 27 of each year, beginning on September 27, 2009, to the person in whose name the 2034 Note is registered at the close of business on the immediately preceding March 15 or September 15, as the case may be. Interest on the 2034 Notes is computed on the basis of a 360-day year consisting of twelve 30-day months.

We will pay to beneficial owners of notes who are non-U.S. persons additional amounts in the event of deduction or withholding of taxes, assessments or other governmental charges imposed by the United States or any taxing authority thereof or therein. Any additional amounts will be paid in pounds sterling or, if the United Kingdom adopts the euro, in euro.

The 2034 Notes are redeemable at our option, as described below. The 2034 Notes are not subject to a sinking fund and are not convertible or exchangeable.

The 2024 Notes are subject to defeasance.

If any interest payment date for the 2034 Notes would otherwise be a day that is not a business day, then the interest payment date will be postponed to the following date that is a business day. Interest will not accrue as a result of any delayed payment. The term "***business day***" means any day, other than a Saturday or Sunday, that is neither a legal holiday nor a day on which banking institutions are generally authorized or required by law or regulation to close in The City of New York or London.

If, prior to the maturity of the 2034 Notes, the United Kingdom adopts the euro as its lawful currency in accordance with the Treaty Establishing the European Communities, as amended from time to time, the 2034 Notes will be re-denominated into euro, and the regulations of the European Commission relating to the euro shall apply to the 2034 Notes. The circumstances and consequences described in this paragraph will not entitle us, BNYM under the 2005 Indenture or any holder of the 2034 Notes to redeem early, rescind or receive notice relating to the 2034 Notes, repudiate the terms of the 2034 Notes or the 2005 Indenture, or raise any defense, request any compensation or make any claim with respect to the 2034 Notes or the 2005 Indenture, nor will these circumstances and consequences affect any of our other obligations under the 2034 Notes or the 2005 Indenture.

Notices to holders of the 2034 Notes will be mailed to such holders. Any notice shall be deemed to have been given on the date of mailing and publication or, if published more than once, on the date of first publication. So long as the 2034 Notes are represented by a global security deposited with the Common Depositary for Clearstream and Euroclear, notices to holders may be given by delivery to Clearstream and Euroclear, and such notices shall be deemed to be given on the date of delivery to Clearstream and Euroclear. BNYM will mail notices by first-class mail, postage prepaid, to each registered holder's last known address as it appears in the security register that BNYM maintains. BNYM will only mail these notices to the registered holder of the 2034 Notes. Holders of 2034 Notes will not receive notices regarding the 2034 Notes directly from us unless we reissue the 2034 Notes in fully certificated form.

BNYM will publish any notices regarding the 2034 Notes in a daily newspaper of general circulation in The City of New York and in London. If the 2034 Notes are admitted to the Official List and to trading on its regulated market, BNYM will publish notices regarding the 2034 Notes in a daily newspaper of general circulation in Dublin, Ireland for so long as such publication is required pursuant to the Prospectus Directive or the rules of the applicable stock exchange. We expect that publication will be made in The City of New York in *The Wall Street Journal*, in London in the *Financial Times* and in Dublin, Ireland in the *Irish Times*. If publication in Dublin, Ireland is not practical, BNYM will publish these notices in an English language newspaper of general circulation elsewhere in Europe. Published notices will be deemed to have been given on the date they are published or, if published more than once, on the date of first publication. If publication as described above becomes impossible, BNYM may publish sufficient notice by alternate means that approximate the terms and conditions described in this paragraph.

Redemption upon Tax Event

We may redeem the 2034 Notes if certain tax-related events occur. If we redeem the 2034 Notes as a result of a tax event and the 2034 Notes are admitted to the Official List and to trading on its regulated market, we will publish a notice of the redemption in Dublin, Ireland. The redemption price paid for the 2034 Notes upon any such redemption will be paid in pounds sterling or, if the United Kingdom adopts the euro, in euro.

Prescription

Under New York law, any legal action to enforce our payment obligations evidenced by the 2034 Notes must be commenced within six years after payment is due. Thereafter, our payment obligations will generally become unenforceable.

Replacement of the 2034 Notes

If any mutilated 2034 Note is surrendered to BNYM, we will execute and the trustee will authenticate and deliver a new note of the same series and principal amount in exchange for such mutilated 2034 Note. If BNYM and we receive evidence to our satisfaction of the destruction, loss or theft of any 2034 Note and any security or indemnity required by it and us, then we shall execute, and BNYM shall authenticate and deliver, in lieu of such destroyed, lost or stolen note, a new 2034 Note of the same series and principal amount. All expenses associated with issuing the new note shall be borne by the owner of the mutilated, destroyed, lost or stolen 2034 Note.

The 2035 Notes

The 2035 Notes were issued under and pursuant to the 2002 Indenture. The 2035 Notes were issued in registered book-entry form without interest coupons in denominations of £1,000 and integral multiples of £1,000 in excess thereof. The 2035 Notes constitute our senior, unsecured and unsubordinated debt obligations and rank equally among themselves and with all of our existing and future senior, unsecured and unsubordinated debt.

The 2035 Notes will mature on September 28, 2035. Unless previously redeemed or purchased and cancelled, we will repay the 2035 Notes at 100% of their principal amount together with accrued and unpaid interest thereon at maturity. We will pay principal of, interest on and any other amounts payable under the 2035 Notes in pounds sterling or, if the United Kingdom adopts the euro, in euro.

The 2035 Notes were issued in a total principal amount of £1,000,000,000. We may, without the consent of the holders of the 2035 Notes, create and issue additional notes ranking equally with the 2035 Notes and otherwise similar in all respects to the 2035 Notes so that those additional notes will be consolidated and form a single series with the 2035 Notes. No additional 2035 Notes may be issued if an event of default under the 2002 Indenture has occurred.

The 2035 Notes are redeemable at our option, as described below. The 2035 Notes are not subject to a sinking fund. The 2035 Notes are subject to defeasance. The 2035 Notes are not convertible or exchangeable.

The 2035 Notes bear interest from September 29, 2004 at the annual interest rate of 5.250%. Interest is payable semi-annually in arrears on September 28 and March 28 of each year, beginning on March 28, 2005, to the person in whose name the 2035 Note is registered at the close of business on the preceding September 15 or March 15, as the case may be. Interest on the 2035 Notes is computed on the basis of a 360-day year of twelve 30-day months.

The 2035 Notes do not have the covenant restricting the grant of liens and cross-default event of default provisions.

If any interest payment date for the 2035 Notes would otherwise be a day that is not a business day, then the interest payment date will be postponed to the following date that is a business day. Interest will not accrue as a result of any delayed payment. The term "***business day***" means any day, other than a Saturday or Sunday, that is neither a legal holiday nor a day on which banking institutions are generally authorized or required by law or regulation to close in The City of New York or London.

If, prior to the maturity of the 2035 Notes, the United Kingdom adopts the euro as its lawful currency in accordance with the Treaty establishing the European Communities, as amended from time to time, the 2035 Notes will be re-denominated into euro, and the regulations of the European Commission relating to the euro shall apply to the 2035 Notes. The circumstances and consequences described in this paragraph will not entitle us, the trustee under the 2002 Indenture or any holder of the 2035 Notes to redeem early, rescind, or receive notice relating to the 2035 Notes, repudiate the terms of the 2035 Notes or the 2002 Indenture, or raise any defense, request any compensation or make any claim with respect to the 2035 Notes or the 2002 Indenture, nor will these circumstances and consequences affect any of our other obligations under the 2035 Notes or the 2002 Indenture.

While the 2035 Notes are represented by a global note deposited with the trustee, as common depositary for Clearstream and Euroclear with respect to the 2035 Notes, notices to holders may be given by delivery to Clearstream and Euroclear, and such notices shall be deemed to be given on the date of delivery to Clearstream and Euroclear. The trustee will mail notices by first-class mail, postage prepaid, to each registered holder's last known address as it appears in the security register that the trustee maintains. The trustee will only mail these notices to the registered holder of the 2035 Notes. Holders of 2035 Notes will not receive notices regarding the 2035 Notes directly from us unless we reissue the 2035 Notes in fully certificated form.

The trustee will also publish notices regarding the 2035 Notes in a daily newspaper of general circulation in the City of New York and in London. We expect that publication will be made in the City of New York in *The Wall Street Journal*, in London in the *Financial Times* and in Dublin, Ireland in the *Irish Times*. If publication in Dublin, Ireland is not practical, the trustee will publish these notices in an English language newspaper of general circulation elsewhere in Europe. Published notices will be deemed to have been given on the date they are published or, if published more than once, on the date of first publication. If publication as described above becomes impossible, the trustee may publish sufficient notice by alternate means that approximate the terms and conditions described in this paragraph.

Optional Redemption

The 2035 Notes are redeemable as a whole or in part, at our option, at any time at a redemption price equal to the greater of (1) 100% of the principal amount of such 2035 Notes and (2) as determined by the Calculation Agent, the price at which the yield on the outstanding principal amount of the 2035 Notes on the Reference Date is equal to the yield on the Benchmark Gilt as of that date as determined by reference to the middle-market price on the Benchmark Gilt at 3:00 p.m., London time, on that date, in either case, plus accrued and unpaid interest on the 2035 Notes up to, but excluding, the date specified as the redemption date.

"***Reference Date***" means the date that is the first dealing day in London prior to the publication of the notice of redemption referred to below.

"***Benchmark Gilt***" means the 4.25% Treasury Stock due March 7, 2036 or such other U.K. government stock as the Calculation Agent, with the advice of three brokers and/or U.K. gilt-edged market makers or three other persons operating in the U.K. gilt-edged market that may be chosen by the Calculation Agent, may determine from time to time to be the most appropriate benchmark U.K. government stock for the notes.

"***Calculation Agent***" means J.P. Morgan Trust Company, National Association or any successor entity.

We will give notice of any redemption between 30 and 60 days preceding the redemption date to each holder of the 2035 Notes to be redeemed as described above.

In the case of any partial redemption, selection of the 2035 Notes for redemption will be made by the trustee under the 2002 Indenture in compliance with the rules and requirements of the principal securities exchange, if any, on which the 2035 Notes are listed or, if the 2035 Notes are not so listed or that exchange prescribes no method of selection, on a pro rata basis, by lot or by any other method as the trustee in its sole discretion deems to be fair and appropriate, although no 2035 Note of £1,000 in original principal amount or less shall be redeemed in part. If any 2035 Note is to be redeemed in part only, the notice of redemption relating to the 2035 Note will state the portion of the principal amount thereof to be redeemed. A new 2035 Note in principal amount equal to the unredeemed portion thereof will be issued and delivered to the trustee, or its nominee, or, in the case of 2035 Notes in definitive form, issued in the name of the holder thereof, in each case upon cancellation of the original 2035 Note.

Unless we default in payment of the redemption price of the 2035 Notes, on and after the redemption date, interest will cease to accrue on the 2035 Notes or the portions of the 2035 Notes called for redemption.

Redemption upon Tax Event

The 2035 Notes may be redeemed at our option in whole, but not in part, on not more than 60 days' and not less than 30 days' notice, at a redemption price equal to 100% of their principal amount, if we determine that as a result of any change or amendment to the laws, treaties, regulations or rulings of the United States or any political subdivision or taxing authority thereof, or any proposed change in such laws, treaties, regulations or rulings, or any change in the official application, enforcement or interpretation of those laws, treaties, regulations or rulings, including a holding by a court of competent jurisdiction in the United States, or any other action, other than an action predicated on law generally known on or before September 22, 2004 except for proposals before the Congress before that date, taken by any taxing authority or a court of competent jurisdiction in the United States, or the official proposal of any action, whether or not such action or proposal was taken or made with respect to us, (A) we have or will become obligated to pay additional amounts on any 2035 Note or (B) there is a substantial possibility that we will be required to pay those additional amounts. Prior to the publication of any notice of such a redemption, we will deliver to the trustee (1) an officers' certificate stating that we are entitled to effect such a redemption and setting forth a statement of facts showing that the conditions precedent to the right of our company to so redeem have occurred and (2) an opinion of counsel to that effect based on that statement of facts.

Payment of Additional Amounts

We will pay to beneficial owners of 2035 Notes who are non-U.S. persons additional amounts in the event of deduction or withholding of taxes, assessments or other governmental charges imposed by the United States or any taxing authority thereof or therein.

Prescription

Under New York's statute of limitations, any legal action to enforce our payment obligations evidenced by the 2035 Notes must be commenced within six years after payment is due. Thereafter our payment obligations will generally become unenforceable.

Replacement of 2035 Notes

If any mutilated 2035 Note is surrendered to the trustee, we will execute and the trustee will authenticate and deliver a new note of the same series and principal amount in exchange for such mutilated 2035 Note. If the trustee and we receive evidence to our satisfaction of the destruction, loss or theft of any 2035 Note and any security or indemnity required by it and us, then we shall execute, and the trustee shall authenticate and deliver, in lieu of such destroyed, lost or stolen note, a new 2035 Note of the same series and principal amount. All expenses associated with issuing the new note shall be borne by the owner of the mutilated, destroyed, lost or stolen 2035 Note.

The 2039 Notes

The 2039 Notes were issued under and pursuant to the 2005 Indenture. The 2039 Notes were issued in registered book-entry form without interest coupons in denominations of £50,000 and integral multiples of £1,000 in excess thereof. The 2039 Notes constitute our senior unsecured debt obligations and rank equally among themselves and with all of our existing and future senior unsecured debt.

The 2039 Notes will mature on January 19, 2039. Unless previously redeemed or purchased and cancelled, we will repay the 2039 Notes at 100% of their principal amount, together with accrued and unpaid interest thereon at maturity. We will pay principal of and interest on the notes in pounds sterling or, solely if the United Kingdom adopts the euro, in euro.

The 2039 Notes were issued in a total principal amount of £1,000,000,000. We may, without the consent of the holders of the 2039 Notes, create and issue additional notes ranking equally with and otherwise similar in all respects to the 2039 Notes (except for the public offering price and the issue date) so that those additional notes will be consolidated and form a single series with the 2039 Notes. No additional 2039 Notes may be issued if an event of default under the 2005 Indenture has occurred and is continuing.

The 2039 Notes bear interest from December 19, 2006 at the annual interest rate of 4.875%. Interest on each 2039 Note is payable semi-annually in arrears on January 19 and July 19 of each year, beginning on July 19, 2007, to the person in whose name the 2039 Note is registered at the close of business on the immediately preceding January 15 or July 15, as the case may be. Interest on the 2039 Notes is computed on the basis of a 360-day year consisting of twelve 30-day months.

We will pay to beneficial owners of 2039 Notes who are non-U.S. persons additional amounts in the event of deduction or withholding of taxes, assessments or other governmental charges imposed by the United States or any taxing authority thereof or therein.

The 2039 Notes are redeemable at our option, as described below. The 2039 Notes are not subject to a sinking fund. The 2039 Notes re subject to defeasance. The 2039 Notes are not convertible or exchangeable.

The 2039 Notes do not have a covenant restricting the grant of liens and cross-default event of default provisions.

If any interest payment date for the 2039 Notes would otherwise be a day that is not a business day, then the interest payment date will be postponed to the following date that is a business day. Interest will not accrue as a result of any delayed payment. The term "***business day***" means any day, other than a Saturday or Sunday, that is neither a legal holiday nor a day on which banking institutions are generally authorized or required by law or regulation to close in The City of New York or London.

If, prior to the maturity of the 2039 Notes, the United Kingdom adopts the euro as its lawful currency in accordance with the Treaty Establishing the European Communities, as amended from time to time, the 2039 Notes will be re-denominated into euro, and the regulations of the European Commission relating to the euro shall apply to the 2039 Notes. The circumstances and consequences described in this paragraph will not entitle us, BNYM under the 2005 Indenture or any holder of the 2039 Notes to redeem early, rescind or receive notice relating to the 2039 Notes, repudiate the terms of the 2039 Notes or the 2005 Indenture, or raise any defense, request any compensation or make any claim with respect to the 2039 Notes or the 2005 Indenture, nor will these circumstances and consequences affect any of our other obligations under the 2039 Notes or the 2005 Indenture.

Notices to holders of the 2039 Notes will be mailed to such holders. Any notice shall be deemed to have been given on the date of mailing and publication or, if published more than once, on the date of first publication. So long as the 2039 Notes are represented by a global security deposited with the Common Depositary for Clearstream and Euroclear, notices to holders may be given by delivery to Clearstream and Euroclear, and such notices shall be deemed to be given on the date of delivery to Clearstream and Euroclear. BNYM will mail notices by first-class mail, postage prepaid, to each registered holder's last known address as it appears in the security register that BNYM maintains. BNYM will only mail these notices to the registered holder of the 2039 Notes. Holders of 2039 Notes will not receive notices regarding the 2039 Notes directly from us unless we reissue the 2039 Notes in fully certificated form.

BNYM will publish any notices regarding the 2039 Notes in a daily newspaper of general circulation in The City of New York and in London. We expect that publication will be made in The City of New York in *The Wall Street Journal*, in London in the *Financial Times* and in Dublin, Ireland in the *Irish Times*. If publication in Dublin, Ireland is not practical, BNYM will publish these notices in an English language newspaper of general circulation elsewhere in Europe. Published notices will be deemed to have been given on the date they are published or, if published more than once, on the date of first publication. If publication as described above becomes impossible, BNYM may publish sufficient notice by alternate means that approximate the terms and conditions described in this paragraph.

Optional Redemption

The 2039 Notes are redeemable as a whole or in part, at our option, at any time at a redemption price equal to the greater of (1) 100% of the principal amount of such 2039 Notes and (2) as determined by the Calculation Agent, the price at which the yield on the outstanding principal amount of such 2039 Notes on the Reference Date is equal to the yield on the Benchmark Gilt as of that date as determined by reference to the middle-market price on the Benchmark Gilt at 3:00 p.m., London time, on that date, in either case, plus accrued and unpaid interest on such 2039 Notes up to, but excluding, the date specified as the redemption date.

"***Reference Date***" means the date that is the first dealing day in London prior to the publication of the notice of redemption referred to below.

"***Benchmark Gilt***" means the 4.75% Treasury Stock due December 7, 2038, or such other U.K. government stock as the Calculation Agent, with the advice of three brokers and/or U.K. gilt-edged market makers or three other persons operating in the U.K. gilt-edged market that may be chosen by the Calculation Agent, may determine from time to time to be the most appropriate benchmark U.K. government stock for the notes.

"***Calculation Agent***" means The Bank of New York, or any successor entity.

We will give notice of any redemption between 30 and 60 days preceding the redemption date to each holder of the 2039 Notes to be redeemed as described above.

The redemption price paid for the 2039 Notes upon any such redemption will be paid in pounds sterling or, solely if the United Kingdom adopts the euro, in euro.

In the case of any partial redemption, selection of the 2039 Notes for redemption will be made by BNYM under the 2005 Indenture in compliance with the rules and requirements of the principal securities exchange, if any, on which the 2039 Notes are listed at the time of redemption or, if the 2039 Notes are not so listed or that exchange prescribes no method of selection, on a pro rata basis, by lot or by any other method as BNYM in its sole discretion deems to be fair and appropriate, although no 2039 Note of £50,000 in original principal amount or less shall be redeemed in part. If any 2039 Note is to be redeemed in part only, the notice of redemption relating to the 2039 Note will state the position of the principal amount thereof to be redeemed. A new 2039 Note in principal amount equal to the unredeemed portion thereof will be issued and delivered to BNYM, or its nominee, or, in the case of 2039 Notes in definitive form, issued in the name of the holder thereof, in each case upon cancellation of the original 2039 Note.

Unless we default in payment of the redemption price of the 2039 Notes, on and after the redemption date, interest will cease to accrue on the 2039 Notes or the portions of the 2039 Notes called for redemption.

We may redeem the 2039 Notes if certain tax-related events occur. If we redeem the 2039 Notes as a result of a tax event, we will publish a notice of the redemption in Dublin, Ireland. The redemption price paid for the 2039 Notes upon any such redemption will be paid in pounds sterling or, solely if the United Kingdom adopts the euro, in euro.

Prescription

Under New York law, any legal action to enforce our payment obligations evidenced by the 2039 Notes must be commenced within six years after payment is due. Thereafter, our payment obligations will generally become unenforceable.

Replacement of the 2039 Notes

If any mutilated 2039 Note is surrendered to BNYM, we will execute and BNYM will authenticate and deliver a new note of the same series and principal amount in exchange for such mutilated 2039 Note. If BNYM and we receive evidence to our satisfaction of the destruction, loss or theft of any 2039 Note and any security or indemnity required by it and us, then we shall execute, and BNYM shall authenticate and deliver, in lieu of such destroyed, lost or stolen note, a new 2039 Note of the same series and principal amount. All expenses associated with issuing the new note shall be borne by the owner of the mutilated, destroyed, lost or stolen 2039 Note.

Exhibit 10.1

WALMART DEFERRED COMPENSATION MATCHING PLAN

Amended and Restated Effective November 8, 2023

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WALMART DEFERRED COMPENSATION MATCHING PLAN

ARTICLE I.
GENERAL

1.1 Purpose.

The purpose of the Walmart Deferred Compensation Matching Plan is to enable certain individuals to defer compensation and to be credited with matching allocations and earnings. The Plan is intended to reward such individuals for their contributions to the success of Walmart and its Related Affiliates. The Plan is also intended to assist such individuals in saving for retirement by providing benefits that are in excess of benefits permitted by applicable law under the 401(k) Plan.

1.2 Effective Date.

The effective date of the amended and restated Plan is November 8, 2023.

1.3 Nature of Plan.

The Plan is intended to be (and shall be administered as) an unfunded employee pension plan benefiting a select group of management or highly compensated employees under the provisions of ERISA. The Plan shall be "unfunded" for tax purposes and for purposes of Title I of ERISA. Any and all payments under the Plan shall be made solely from the general assets of Walmart. A Participant's interests under the Plan do not represent or create a claim against specific assets of Walmart or any Employer. Nothing herein shall be deemed to create a trust of any kind or create any fiduciary relationship between the Committee, the Plan Administrator, Walmart or any Employer and a Participant, the Participant's beneficiary or any other person. To the extent any person acquires a right to receive payments from Walmart under this Plan, such right is no greater than the right of any other unsecured general creditor of Walmart. The Plan is intended to be in compliance with Code Section 409A and shall be interpreted, applied and administered at all times in accordance with Code Section 409A and guidance issued thereunder.

ARTICLE II.
DEFINITIONS

2.1 Definitions.

Whenever used in this Plan, the following words and phrases have the meaning set forth below unless the context plainly requires a different meaning:

(a) **Account** means the bookkeeping account maintained on behalf of a Participant under the Plan to reflect such Participant's Class Accounts.

(b) **Class Account** means a bookkeeping subaccount maintained under a Participant's Account to reflect such Participant's Deferral Credits and Employer Matching Contribution Credits, and earnings credited in accordance with Section 4.4.

(1) Each Class Account shall be a Post-2020 Class Account or a Pre-2021 Class Account. A Post-2020 Class Account shall be maintained under a Participant's Account to reflect Deferral Credits and Employer Matching Contribution Credits earned for each Plan Year commencing on or after February 1, 2021. A Pre-2021 Class Account shall be maintained under a Participant's Account to reflect Deferral Credits and Employer Matching Contribution Credits attributable to all Plan Years commencing prior to February 1, 2021.

(2) Each Post-2020 Class Account shall consist of a Deferral Account and a Matching Account. Such Deferral Account may be allocated among one or more Scheduled In-Service Accounts and one or more Retirement Accounts, as elected or deemed elected by the Participant in accordance with Section 3.5. Such Matching Account may be allocated to one Retirement Account, as elected or deemed elected by the Participant in accordance with Section 3.5.

(3) Each Pre-2021 Class Account shall consist of a Deferral Account and a Matching Account. Such Deferral Account may be allocated among one or more Scheduled In-Service Accounts and one or more Retirement Accounts to the extent authorized hereunder and as elected or deemed elected by the Participant in accordance with Section 3.5. Such Matching Account may be allocated among one or more Retirement Accounts as elected or deemed elected by the Participant in accordance with Section 3.5.

(c) **Code** means the Internal Revenue Code of 1986, as amended from time to time.

(d) **Committee** means the Compensation and Management Development Committee of the Board of Directors of Walmart.

(e) **Compensation** means a Participant's base compensation for a Plan Year with respect to services rendered for an Employer. Compensation includes, but is not limited to, short-term disability payments made by an Employer. Compensation does not include military differential payments.

(f) **Deferral Account** means a bookkeeping subaccount maintained under a Participant's Class Account to reflect his or her Deferral Credits.

(g) **Deferral Credit** means the amount of Deferred Compensation credited to a Participant's Deferral Account in accordance with Section 3.1, the amount of Deferred MIP Bonus credited to a Participant's Deferral Account in accordance with Section 3.2, and the amount of Deferred Special Bonus credited to a Participant's Deferral Account in accordance with Section 3.3.

(h) **Deferred Compensation** means the Compensation deferred by a Participant in accordance with Section 3.1.

(i) **Deferred MIP Bonus** means the amount deferred by a Participant in accordance with Section 3.2 from bonuses payable to the Participant under the MIP.

(j) **Deferred Special Bonus** means the amount deferred by a Participant in accordance with Section 3.3 from a Special Bonus payable to the Participant.

(k) **Disabled** means the Participant has incurred a Separation from Service because the Participant, as determined by the Plan Administrator or his or her delegate, is unable to engage in any substantial gainful activity by reason of a medically determinable physical or mental impairment which can be expected to result in death or can be expected to last for a continuous period of not less than twelve (12) months.

(l) **Eligible Officer** means an individual who is a corporate officer of an Employer, and who holds the title of Vice President or above, Treasurer, Controller, or an officer title of similar rank or other position as determined by the Plan Administrator. In no event will any individual constitute an Eligible Officer if he or she is not subject to federal income tax withholding in the United States. Notwithstanding anything in the preceding provisions of this Section 2.1(k), Eligible Officer shall exclude any individual who, pursuant to Walmart's Global Assignment Policy, is seconded to an Employer and, under the terms of his or her offer or assignment letter, he or she is intended to remain on the home country's benefit and pension programs. For purposes of this Plan, effective February 1, 2019 an individual shall not become an Eligible Officer prior to the first day of the month immediately following the month in which the individual would otherwise satisfy the requirements of being an Eligible Officer.

(m) **Eligible Participant** means with respect to a Plan Year an individual who either (1) is an Eligible Officer, or (2) is an employee of an Employer and who as of the October 31 immediately preceding the Plan Year is in an eligible position determined by the Plan Administrator and is a member of a select group of management or highly compensated employees under the provisions of ERISA. For purposes of this Plan, effective February 1, 2019, an individual shall not become an Eligible Participant prior to the first day of the month immediately following the month in which the individual would otherwise satisfy the requirements of being an Eligible Participant.

(n) **Employer** means Walmart and any entity, whether or not incorporated, which is a member of a controlled group of corporations, trades or businesses, as defined in Code Sections 414(b) and 414(c), of which Walmart is a member, and which has been designated by the Committee as a participating employer in the Plan.

(o) **Employer Matching Contribution Credits** means the amount credited to a Participant's Matching Account pursuant to Section 3.4.

(p) **ERISA** means the Employee Retirement Income Security Act of 1974, as amended from time to time.

(q) **Excess Compensation** means for a Plan Year the excess, if any, of (1) the sum of (i) the Participant's base compensation for the Plan Year for services rendered for an Employer, and (ii) the Participant's MIP bonus payable with respect to a performance period that coincides with the Plan Year or that ends within the Plan Year, over (2) the annual compensation limit under Code Section 401(a)(17) (or under a comparable provision of the Internal Revenue Code of the Commonwealth of Puerto Rico if the Participant is an eligible participant under the Walmart Puerto Rico 401(k) Plan) in effect for the calendar year in which the Plan Year begins. For purposes of this paragraph, a Participant's base compensation and a Participant's MIP bonus shall include the cash amounts of such base compensation and MIP bonus payable to the Participant regardless of whether the payment of any or all of such amounts to the Participant is deferred or not made on account of (1) a deferral election by the Participant under the 401(k) Plan, (2) a deferral election by the Participant under this Plan, (3) a pre-tax contribution by the Participant under Code Section 125, (4) a pre-tax contribution by the Participant under Code Section 132(f)(4), or (5) withholding for the payment of employment taxes or income taxes with respect to the Participant.

(r) **401(k) Plan** means the Walmart 401(k) Plan and the Walmart Puerto Rico 401(k) Plan, as amended from time to time.

(s) **Gross Misconduct** means conduct engaged in by the Participant which has been deemed by the Plan Administrator or his or her delegate to be detrimental to the best interests of Walmart or any Related Affiliate or any entity in which Walmart has an ownership interest. Examples of such conduct include, without limitation, disclosure of confidential information in violation of Walmart's Statement of Ethics, theft, the commission of a felony or a crime involving moral turpitude, gross misconduct or similar serious offenses.

(t) **Interest Rate** means a daily rate of simple interest based on the yield on United States Treasury securities (not indexed for inflation) with a constant maturity of ten (10) years, as of the first business day of January preceding such Plan Year, plus two hundred seventy (270) basis points. This rate shall be determined on the basis of Federal Reserve Statistical Release H-15 (or any successor statistical release of the Federal Reserve) and, if there is no such statistical release, on the basis of such other generally recognized source of information concerning the market for United States Treasury securities as the Committee selects.

(u) **Investment Options** means the investment options, determined from time to time by the Plan Administrator, used to credit earnings, gains and losses on Account balances.

(v) **Matching Account** means a bookkeeping subaccount maintained under a Participant's Class Account to reflect his or her Employer Matching Contribution Credits.

(w) **MIP** means the Walmart Inc. Management Incentive Plan, as amended from time to time, without regard to any non-U.S. subplans.

(x) **Participant** means any individual for whom an Account is maintained. An individual will cease to be a Participant at such time that the Participant's Account has been fully distributed or forfeited in accordance with the Plan.

(y) **Plan** means the Walmart Deferred Compensation Matching Plan, as set forth herein, and as amended from time to time.

(z) **Plan Administrator** means the Senior Vice President, Global Total Rewards of Walmart or any successor position.

(aa) **Plan Year** means the twelve (12)-month period commencing on February 1 and ending on January 31.

(bb) **Post-2020 Class Account** means a Class Account maintained under a Participant's Account to reflect Deferral Credits and Employer Matching Contribution Credits earned for each Plan Year commencing on or after February 1, 2021.

(cc) **Post-Investments Portion** means the portion of a Participant's Account, if any, attributable to Deferral Credits and Employer Matching Contribution Credits earned in Plan Years commencing on or after February 1, 2023.

(dd) **Pre-2021 Class Account** means a Class Account maintained under a Participant's Account to reflect Deferral Credits and Employer Matching Contribution Credits attributable to all Plan Years commencing prior to February 1, 2021.

(ee) **Pre-Investments Portion** means the portion of a Participant's Account, if any, attributable to Deferral Credits and Employer Matching Contribution Credits earned in Plan Years commencing prior to February 1, 2023.

(ff) **Related Affiliate** means all persons with whom Walmart would be considered a single employer under Code Sections 414(b) and 414(c), except that in applying Code Sections 1563(a)(1), (2) and (3) for purposes of determining a controlled group of corporations under Code Section 414(b), the language "at least 50 percent" shall be used instead of "at least 80 percent" in each place it appears in Code Sections 1563(a)(1), (2) and (3), and in applying Treas. Regs. Sec. 1.414(c)-2 for purposes of determining a controlled group of trades or businesses under Code Section 414(c), the language "at least 50 percent" shall be used instead of "at least 80 percent" in each place it appears in Treas. Regs. Sec. 1.414(c)-2.

(gg) **Retirement Account** means a bookkeeping subaccount maintained under a Participant's Deferral Account or Matching Account to which Deferral Credits and Employer Matching Contribution Credits (as applicable), and earnings credited in accordance with Section 4.4, may be allocated pursuant to the election or deemed election of the Participant in accordance with Section 3.5.

(hh) **Scheduled In-Service Account** means a bookkeeping subaccount maintained under a Participant's Deferral Account to which Deferral Credits and earnings credited in accordance with Section 4.4 may be allocated pursuant to the election of the Participant in accordance with Section 3.5.

(ii) **Scheduled Pay Date** means, with respect to each Scheduled In-Service Account, the first day of a calendar month designated by the Participant in accordance with Section 3.5. In no event shall such date be earlier than the first day of the second Plan Year beginning after the Plan Year for which Deferral Credits are first allocated to such Scheduled In-Service Account. Once selected, the Scheduled Pay Date with respect to any Scheduled In-Service Account is irrevocable. If a Participant fails to designate a Scheduled Pay Date with respect to a Scheduled In-Service Account, then the Participant is deemed to have designated as the Scheduled Pay Date for such Scheduled In-Service Account the first day of the second Plan Year beginning after the Plan Year for which Deferral Credits are first allocated to such Scheduled In-Service Account.

(jj) **Separation from Service** means the Participant has a termination of employment (other than on account of death) with the Company. For purposes of this paragraph, "Company" means the Employer and any Related Affiliate. Whether a termination of employment has occurred shall be determined based on whether the facts and circumstances indicate the Participant and the Company reasonably anticipate that no further services will be performed by the Participant for the Company; provided, however, that a Participant shall be deemed to have a termination of employment if the level of services he or she actually performs for the Company after a certain date permanently decreases to no more than twenty percent (20%) of the average level of bona fide services performed for the Company by the Participant (whether as an employee or independent contractor) over the immediately preceding 36-month period (or the full period of services for the Company if the Participant has been providing services to the Company for less than 36 months). For this purpose, a Participant is not treated as having a Separation from Service while he or she is on a military leave, sick leave, or other bona fide leave of absence, if the period of such leave does not exceed six (6) months, or if longer, so long as the Participant has a right to reemployment with the Company under an applicable statute or by contract. This definition of Separation from Service is intended to be consistent with the separation from service requirements as defined in Code Section 409A.

(kk) **Separation Pay Date** means the last day of the calendar month in which falls the date that is six (6) months after a Participant's Separation from Service.

(ll) **Special Bonus** means a bonus, other than a bonus payable under the MIP, that is payable to an Eligible Officer with respect to services rendered or to be rendered for an Employer and that is eligible for deferral under the Plan either because (1) the bonus is payable pursuant to an offer letter accepted in writing by the Eligible Officer before commencement of employment and that specifically refers to the deferability of the bonus by explicit reference to this Plan or (2) the bonus is eligible for deferral in accordance with guidelines established by the Plan Administrator, or by an officer to whom the Plan Administrator has delegated authority to establish such guidelines, and the bonus requires as a condition of receipt of the bonus and to avoid forfeiture of the bonus that the recipient continue to perform services for the Employer for a period of at least thirteen (13) months after the date he or she obtains the legally binding right to the bonus.

(mm) **Unforeseeable Emergency** means a severe financial hardship to the Participant resulting from an illness or accident of the Participant, the Participant's spouse, the Participant's beneficiary, or the Participant's dependent (as defined in Code Section 152, without regard to subsections (b)(1), (b)(2) and (d)(1)(B)), the loss of the Participant's property due to casualty, or other similar extraordinary and unforeseeable circumstances arising as a result of events beyond the control of the Participant.

(nn) **Valuation Date** means the last business day of each month of the Plan Year.

(oo) **Walmart** means Walmart Inc., a Delaware corporation.

(pp) **Years of Participation** means a period of Plan Years which includes the first Plan Year with respect to which an Eligible Participant makes a deferral election in accordance with any one or more of Sections 3.1, 3.2 and 3.3 and an amount is credited to the Participant's Account with respect to any such deferral election, and each subsequent Plan Year during all or part of which the Participant remains a Participant and an employee of an Employer. In addition to the preceding definition, a Participant's Years of Participation shall include any period commencing February 1 and ending January 31, whether before or after the effective date of the Plan, during which or with respect to which an account is maintained for the Participant under the Walmart Inc. Officer Deferred Compensation Plan, as such plan may be amended from time to time.

ARTICLE III.
DEFERRAL CREDITS AND MATCHING CONTRIBUTION CREDITS AND

ACCOUNT ALLOCATIONS

3.1 Deferred Compensation.

(a) For each Plan Year, each Eligible Officer may elect to defer, as Deferred Compensation, all or a portion of the Eligible Officer's Compensation to be otherwise paid for such Plan Year by the Employer, provided that: (i) for Plan Years commencing on and after February 1, 2022, an Eligible Officer may elect to defer no more than eighty percent (80%) of the Eligible Officer's Compensation for a Plan Year; and (ii) no election shall be effective to reduce amounts paid by the Employer to an Eligible Officer to an amount which is less than the sum of the amount the Employer is required to withhold for a Plan Year for purposes of federal, state, or local taxes (including, but not limited to, income and FICA withholding) or for insurance premiums or other withholdings as allowed by Code Section 409A. The Eligible Officer's Deferred Compensation will be deferred proratably for each payroll period of the Plan Year. If a payroll period begins in one Plan Year and ends in the following Plan Year, the Deferred Compensation with respect to such payroll period shall be determined by the Eligible Officer's deferral election made with respect to the Plan Year in which the payroll period ends. All deferral elections made under this Section 3.1 must be filed with, and on forms (which may be electronic) approved by, the Plan Administrator.

(b) Compensation deferral elections must be filed:

(1) With respect to an individual who is an Eligible Officer as of the December 31 preceding the Plan Year for which the deferral election is to be effective, no later than such December 31; or

(2) With respect to an individual who first becomes an Eligible Officer during the Plan Year, within thirty (30) days following the first date he or she becomes an Eligible Officer. For purposes of this rule, an Eligible Officer will be treated as first becoming an Eligible Officer during the Plan Year only if:

(A) he or she was not eligible to participate in the Plan or any other plan required by Code Section 409A to be aggregated with the Plan at any time during the twenty-four (24)-month period ending on the date during the Plan Year he or she becomes an Eligible Officer; or

(B) he or she was paid all amounts previously due under the Plan and any other plan required by Code Section 409A to be aggregated with the Plan and, on and before the date of the last such payment, was not eligible to continue to participate in the Plan and any other plan required by Code Section 409A to be aggregated with the Plan for periods after such payment.

A deferral election under this Section 3.1(b)(2) will be effective only with respect to Compensation for payroll periods beginning after the payroll period in which the Eligible Officer's election form (which may be electronic) is received by the Plan Administrator and has become irrevocable. For the avoidance of doubt, an Eligible Officer's election form shall become irrevocable at the end of the thirty (30) day period following the first date he or she becomes an Eligible Officer. In addition, a deferral election under this Section 3.1(b)(2) will be effective only if the deferral election meets the requirements set forth in Code Section 409A(a)(4)(B).

(c) The Deferred Compensation of an Eligible Officer who elects to defer all or a portion of the Eligible Officer's Compensation under this Section 3.1 with respect to a Plan Year shall be credited to the Eligible Officer's Deferral Account under his or her Class Account for such Plan Year and shall be allocated to a Retirement Account or to a Scheduled In-Service Account under such Deferral Account in accordance with Section 3.5.

3.2 Deferred MIP Bonuses.

(a) For each Plan Year, each Eligible Participant may elect to defer all or a portion of the Eligible Participant's bonus (if any) to be otherwise paid to the Eligible Participant under the MIP with respect to a performance period under the MIP that coincides with the Plan Year or that ends within the Plan Year; provided, however, an Eligible Participant who is not an Eligible Officer may elect to defer no more than eighty percent (80%) of the Eligible Participant's MIP bonus for a Plan Year. No election under this Section 3.2 shall be effective to reduce amounts paid by the Employer to an Eligible Participant to an amount which is less than the sum of the amount the Employer is required to withhold for a Plan Year for purposes of federal, state, or local taxes (including, but not limited to, income and FICA withholding) or for insurance premiums or other withholdings as allowed by Code Section 409A. All bonus deferral elections made under this Section 3.2 must be filed with, and on forms (which may be electronic) approved by, the Plan Administrator.

(b) MIP bonus deferral elections must be filed:

(1) No later than the December 31 (or such other date as determined by the Plan Administrator or his or her delegate) preceding the first day of the performance period for which the deferral election is to be effective.

(2) If authorized by the Plan Administrator or his or her delegate with respect to an Eligible Participant, and if the MIP bonus constitutes "performance-based compensation" within the meaning of Code Section 409A based on services performed over a performance period of at least twelve (12) months, and if the Eligible Participant has been continuously employed by an Employer or a Related Affiliate since the first day of the performance period, then no later than the earlier of (i) the date that is six months prior to the last day of the performance period, or (ii) the date in the performance period as of which the amount of the MIP bonus has become both substantially certain to be paid and calculable.

(3) Solely with respect to an Eligible Officer who first becomes an Eligible Participant during the Plan Year, within thirty (30) days following the first date he or she becomes an Eligible Participant, as described in Code

Section 409A(a)(4)(B). For purposes of this rule, an Eligible Officer will be treated as first becoming an Eligible Participant during the Plan Year only if:

(A) he or she was not eligible to participate in the Plan or any other plan required by Code Section 409A to be aggregated with the Plan at any time during the twenty-four (24)-month period ending on the date during the Plan Year he or she becomes an Eligible Participant; or

(B) he or she was paid all amounts previously due under the Plan and any other plan required by Code Section 409A to be aggregated with the Plan and, on and before the date of the last such payment, was not eligible to continue to participate in the Plan and any other plan required by Code Section 409A to be aggregated with the Plan for periods after such payment.

An MIP bonus deferral election under this Section 3.2(b)(3) will be effective only with respect to an MIP bonus paid for services performed after such election. For this purpose, the amount of the MIP bonus payable to the Eligible Officer for services rendered subsequent to the Eligible Officer's election will be determined by multiplying the bonus by a fraction, the numerator of which is the number of calendar days remaining in the performance period after the election and the denominator of which is the total number of calendar days in such performance period. For purposes of this Section 3.2(b)(3), the date of an Eligible Officer's election is the date the executed election form (which may be electronic) is received by the Plan Administrator.

(c) The Deferred MIP Bonus of an Eligible Participant who elects to defer all or a portion of the Eligible Participant's MIP bonus under this Section 3.2 with respect to a performance period that coincides with a Plan Year or that ends within a Plan Year shall be credited to the Eligible Participant's Deferral Account under his or her Class Account for such Plan Year and shall be allocated to a Retirement Account or to a Scheduled In-Service Account under such Deferral Account in accordance with Section 3.5.

3.3 Deferred Special Bonuses.

(a) An Eligible Officer may elect to defer all or a portion of the Eligible Officer's Special Bonus to be otherwise paid to the Eligible Officer in a Plan Year. All Special Bonus deferral elections made under this Section 3.3 must be filed with, and on forms (which may be electronic) approved by, the Plan Administrator. No election under this Section 3.3 shall be effective to reduce amounts paid by the Employer to an Eligible Participant to an amount which is less than the sum of the amount the Employer is required to withhold for a Plan Year for purposes of federal, state, or local taxes (including, but not limited to, income and FICA withholding) for insurance premiums or other withholdings as allowed by Code Section 409A. For purposes of this Section 3.3, the date of an Eligible Officer's election is the date the executed election form (which may be electronic) is received by the Plan Administrator. A deferral election is not permitted with respect to a Special Bonus unless the Special Bonus is a type described in, and the deferral election with respect to the Special Bonus satisfies the applicable conditions of, Section 3.3(b) or Section 3.3(c).

(b) A Special Bonus described in this Section 3.3(b) is one that: (1) requires as a condition of receipt of the Special Bonus and to avoid forfeiture of the Special Bonus that the Eligible Officer continue to perform services for a period of at least thirteen (13) months after the date he or she obtains the legally binding right to the Special Bonus; (2) may not have an earlier vesting date for a good reason termination or the Eligible Officer's retirement; and (3) must otherwise meet the qualifications as described in Code Section 409A. The deferral election with respect to a Special Bonus described in this Section 3.3(b) must be filed within thirty (30) days after the Eligible Officer obtains the legally binding right to the Special Bonus.

(c) A Special Bonus described in this Section 3.3(c) is one payable pursuant to an offer letter accepted in writing by an Eligible Officer before commencement of employment and that specifically refers to the deferability of the Special Bonus by explicit reference to the Plan. The deferral election with respect to a Special Bonus described in this Section 3.3(c) must be filed prior to the time the Eligible Officer renders any services to the Employer, regardless of whether the deferral election relates to all of the Special Bonus or a portion of the Special Bonus.

(d) The Deferred Special Bonus of an Eligible Officer who elects to defer all or a portion of the Eligible Officer's Special Bonus under this Section 3.3 otherwise payable in a Plan Year shall be credited to the Eligible Officer's Deferral Account under his or her Class Account for such Plan Year and shall be allocated to a Retirement Account or to a Scheduled In-Service Account under such Deferral Account in accordance with Section 3.5.

3.4 Employer Matching Contribution Credits.

(a) If a Participant is employed by the Employer or any Related Affiliate on the last day of the Plan Year and has not incurred a Separation from Service during that Plan Year, and if Deferral Credits have been made to the Participant's Account with respect to the Plan Year, then to the extent applicable under the following provisions of this Section 3.4 an Employer Matching Contribution Credit will be made to the Participant's Matching Account under his or her Class Account for such Plan Year and shall be allocated to a Retirement Account under such Matching Account in accordance with Section 3.5. The amount of the Employer Matching Contribution Credit, if any, made to such Matching Account for the Plan Year will equal the total amount of Deferred Compensation and Deferred MIP Bonus credited to the Participant's Account for the Plan Year under Section 3.1(c) and Section 3.2(c); provided, however, in no event shall the Employer Matching Contribution Credit made to such Matching Account for a Plan Year exceed 6% of the Participant's Excess Compensation for such Plan Year. Notwithstanding the preceding provisions of this Section 3.4(a), an Employer

Matching Contribution Credit for a Plan Year shall not be made with respect to any Deferral Credits for the Plan Year that have been withdrawn in accordance with Section 5.5.

(b) A Participant shall become vested in his or her Matching Accounts, including earnings thereon, if the Participant has completed at least three (3) Years of Participation. If a Participant is not otherwise vested in the Participant's Matching Accounts under the preceding sentence of this Section 3.4(b), the Participant will become vested in the Participant's Matching Accounts if the Participant dies prior to the Participant's Separation from Service, or if the Participant is Disabled. Notwithstanding any provision hereunder to the contrary, a Participant's Matching Accounts shall be distributed pursuant to Article V only if the Participant has become vested in the Participant's Matching Accounts under this Section 3.4(b) as of the date of the Participant's Separation from Service.

3.5 Account Allocation Elections

(a) At the same time that an Eligible Participant makes an election to defer Compensation, an MIP bonus, or a Special Bonus to a Deferral Account in accordance with the provisions of the Plan, the Eligible Participant shall also make an election to allocate the amount or amounts subject to each such deferral election either to a Retirement Account or to a Scheduled In-Service Account under such Deferral Account. In addition to the preceding requirement, at the same time that an Eligible Participant makes an election to defer Compensation or an MIP bonus in accordance with the provisions of this Plan, the Eligible Participant shall also make an election to allocate the Employer Matching Contribution Credits (if any) with respect to such Deferred Compensation or Deferred MIP Bonus to one Retirement Account under the Matching Account of his or her Class Account for the Plan Year to which such Employer Matching Contribution Credits relate.

(b) At the time of an Eligible Participant's first election to allocate any amount subject to a deferral election (regardless of whether the amount is Deferred Compensation, Deferred MIP Bonus, Deferred Special Bonus or Employer Matching Contribution Credit) to a Retirement Account, the Eligible Participant shall also designate the form of distribution with respect to such Retirement Account. The form of distribution must be a form permitted under Section 5.4(a).

(c) At the time of an Eligible Participant's first election to allocate any amount subject to a deferral election (regardless of whether the amount is Deferred Compensation, Deferred MIP Bonus or Deferred Special Bonus) to a Scheduled In-Service Account, the Eligible Participant shall also designate the Scheduled Pay Date with respect to such Scheduled In-Service Account.

(d) If at the time of an Eligible Participant's deferral election under the Plan the Eligible Participant fails to make an account allocation election under Section 3.5(a), then:

(1) If the amount subject to such deferral election is for services attributable to a Plan Year commencing prior to February 1, 2021, such amount shall be allocated in the same manner as the same category of deferred amounts (meaning either Deferred Compensation, Deferred MIP Bonus, Deferred Special Bonus or Employer Matching Contribution Credits) were allocated for the most recent preceding Plan Year for which the Eligible Participant made an allocation election, but if none then to the Eligible Participant's Retirement Account under the Deferral Account or Matching Account (as applicable) of his Pre-2021 Class Account if there is only one, or equally to the Eligible Participant's Retirement Accounts under such Deferral Account or Matching Account if the Eligible Participant has more than one Retirement Account thereunder, but if the Eligible Participant has no Retirement Account thereunder then the amount subject to such deferral election shall be allocated to a Retirement Account deemed to be elected by the Participant under such Deferral Account or Matching Account with a lump sum form of payment, and such Retirement Account shall be one of the Participant's permitted Retirement Accounts under the Plan.

(2) If the amount subject to such deferral election is for services attributable to a Plan Year commencing on or after February 1, 2021, such amount shall be allocated in the same manner as the same category of deferred amounts (meaning either Deferred Compensation, Deferred MIP Bonus, Deferred Special Bonus or Employer Matching Contribution Credits) were allocated for the most recent preceding Plan Year for which the Eligible Participant made an allocation election, but if none then such deferral election shall be allocated to a Retirement Account deemed to be elected by the Participant under the Deferral Account or Matching Account (as applicable) of his Class Account for such Plan Year with a lump sum form of payment.

3.6 Irrevocability of Deferral Elections and Account Allocation Elections.

(a) Except as otherwise provided herein, once made for a Plan Year, a deferral election or elections under Sections 3.1(b)(1), 3.2(b)(1) and 3.2(b)(2), and the corresponding account allocation election or elections under Section 3.5, may not be revoked, changed or modified after the applicable deferral election filing deadline specified in Sections 3.1(b)(1), 3.2(b)(1), and 3.2(b)(2), and a deferral election or elections under Sections 3.1(b)(2), 3.2(b)(3), 3.3(b) and 3.3(c), and the corresponding account allocation election or elections under Section 3.5, may not be revoked, changed or modified after the date of each such deferral election as provided in Sections 3.1(b)(2), 3.2(b)(3), 3.3(b) and 3.3(c). A deferral election for one Plan Year will not automatically be given effect for a subsequent Plan Year, so that if a deferral is desired for a subsequent Plan Year, a separate election must be made by the Eligible Participant.

(b) In the event an Eligible Officer has a Separation from Service for any reason, then his or her deferral election under Section 3.1 will terminate as of the date of such Separation from Service (but will be effective with respect to the last regular paycheck issued to such Eligible Officer), regardless of whether the Eligible Officer continues to receive Compensation, or other remuneration, from any Employer or Related Affiliate thereafter. If an Eligible Officer has a Separation from Service for any reason and is rehired (whether or not as an Eligible Officer) within the same Plan Year, his or her deferral election, if any, under Section 3.1 shall be automatically reinstated and shall remain in effect for the remainder of such Plan Year.

(c) In the event an Eligible Participant has a Separation from Service for any reason, then his or her deferral elections, if any, under Sections 3.2 and 3.3 will remain in effect with respect to the bonus, if any, subject to any such deferral election. If an Eligible Participant has a Separation from Service for any reason and is rehired (whether or not as an Eligible Participant) within the same Plan Year or the same performance period, his or her deferral elections, if any, under Sections 3.2 and 3.3 will remain in effect with respect to the bonus, if any, subject to any such deferral elections.

(d) In the event an Eligible Participant who is an Eligible Officer ceases to be an Eligible Officer (other than on account of a Separation from Service) during any Plan Year, then his or her Compensation deferral election, if any, under Section 3.1 will terminate as of the next following January 31. In addition, in the event the Compensation of such individual is reduced as a result of the change in status, his or her deferral election following such loss and through the date of termination of such election as provided in the preceding sentence will be pro rated based on his or her new level of Compensation.

(e) In the event an Eligible Officer receives Company-paid short term disability payments and the Compensation of such individual is reduced as a result of the short term disability status, then following such reduction in Compensation his or her Compensation deferral election, if any, under Section 3.1 will be pro rated based on his or her new level of Compensation through the date of termination of such election.

(f) In the event an Eligible Participant ceases to be an Eligible Participant (other than on account of a Separation from Service) during any Plan Year, then his or her bonus deferral election, if any, under Section 3.2 will terminate for any performance period beginning in the calendar year following the year of the loss of Eligible Participant status.

(g) In the event an Eligible Participant who is an Eligible Officer ceases to be an Eligible Officer (other than on account of Separation from Service) during any Plan Year, then his or her bonus deferral election, if any, under Section 3.3 will remain in effect.

(h) Notwithstanding anything herein to the contrary, in the event an Eligible Officer goes on an unpaid leave of absence, his or her Compensation deferral election, if any, under Section 3.1 shall automatically cease when he or she commences the unpaid leave of absence; provided, however, that if he or she returns from the unpaid leave of absence during the same Plan Year, his or her Compensation deferral election under Section 3.1 shall automatically resume immediately upon return from the leave of absence and shall continue in effect for the balance of the Plan Year. An Eligible Officer's Compensation deferral election under Section 3.1, if any, shall remain in effect with respect to any Compensation to which such election applies that is paid while on a leave of absence. An Eligible Participant's deferral election under Sections 3.2 or 3.3, if any, shall not be affected by his or her leave of absence.

3.7 Automatic Suspension of Deferral Elections.

In the event a Participant requests a distribution pursuant to Section 5.5 due to an Unforeseeable Emergency, or the Participant requests a cancellation of deferrals under the Plan in order to alleviate his or her Unforeseeable Emergency, and the Plan Administrator or his or her delegate determines that the Participant's Unforeseeable Emergency may be relieved through the cessation of deferrals under the Plan, some or all the Participant's deferral elections under Sections 3.1, 3.2 and 3.3, if any, for such Plan Year as determined by the Plan Administrator or his or her delegate, shall be cancelled as soon as administratively practicable following such determination by the Plan Administrator or his or her delegate.

ARTICLE IV.
ACCOUNTS AND TIMING OF CREDITS TO ACCOUNTS

4.1 Nature of Accounts.

Each Participant's Account will be used solely as a measuring device to determine the amount to be paid a Participant under this Plan. The Accounts do not constitute, nor will they be treated as, property or a trust fund of any kind. All amounts at any time attributable to a Participant's Account will be, and remain, the sole property of Walmart. A Participant's rights hereunder are limited to the right to receive Plan benefits as provided herein. The Plan represents an unsecured promise by Walmart to pay the benefits provided by the Plan.

4.2 Deferral Credits and Employer Matching Contribution Credits.

Deferral Credits and Employer Matching Contribution Credits for a Plan Year will be credited to each Participant's Class Account for such Plan Year as follows:

(a) Deferred Compensation will be credited to the Deferral Account of such Class Account as soon as practicable after the date such Compensation would have otherwise been paid in cash.

(b) Deferred MIP Bonuses and Deferred Special Bonuses will be credited to the Deferral Account of such Class Account as soon as practicable after the date the bonus could have otherwise been paid in cash.

(c) Employer Matching Contribution Credits earned in Plan Years commencing prior to February 1, 2023 will be credited to the Matching Account of such Class Account as of the last day of the Plan Year. Employer Matching Contribution Credits earned in Plan Years commencing on or after February 1, 2023 will be credited to the Matching Account of such Class Account as soon as practicable following the last day of the Plan Year.

A Participant's Account, including earnings credited thereto, will be maintained by the Plan Administrator until the Participant's Plan benefits have been paid in full.

4.3 Valuation of Accounts.

Each Participant's Account will be valued monthly as of each Valuation Date.

4.4 Credited Earnings.

(a) Every Valuation Date during a Plan Year, the Pre-Investments Portion of a Participant's Account will be credited with either of the following, as elected by the Participant in accordance with this Section 4.4(a):

(1) Interest at the Interest Rate, which will be for the entire month and calculated based on the value of such Pre-Investments Portion as of the immediately preceding Valuation Date, resulting in the monthly compounding of interest on each Valuation Date; or

(2) Earnings, gains and losses based on the results that would have been achieved had such Pre-Investments Portion been invested as soon as practicable after such election into the Investment Options selected by the Participant.

The Plan Administrator, in his or her sole discretion, shall establish one or more windows of time during which a Participant may irrevocably elect for all or a portion of his or her Pre-Investments Portion to be credited with earnings, gains and losses in accordance with Section 4.4(a)(2). If a Participant does not make such an election in accordance with the procedures established by the Plan Administrator in his or her sole discretion, the Participant's entire Pre-Investments Portion shall be credited with interest in accordance with Section 4.4(a)(1). Once earnings are elected under Section 4.4(a)(2) with respect to any portion of a Participant's Pre-Investments Portion, such Participant shall be prohibited from subsequently electing to receive interest with respect to such portion in accordance with Section 4.4(a)(1).

(b) Every Valuation Date during a Plan Year, the Post-Investments Portion of a Participant's Account, if any, will be credited with earnings, gains and losses based on the results that would have been achieved had amounts credited to his or her Account been invested as soon as practicable after crediting into the Investment Options selected by the Participant.

(c) The Plan Administrator shall specify procedures to allow Participants to make elections as to the deemed investment of amounts in Investment Options in accordance with Sections 4.4(a)(2) and 4.4(b). Nothing in this Section 4.4 or otherwise in the Plan, however, will require Walmart to actually invest any amounts in such Investment Options or otherwise.

ARTICLE V.
PAYMENT OF PLAN BENEFITS

5.1 Scheduled In-Service Benefits.

(a) In-Service Benefits. Each of a Participant's Scheduled In-Service Accounts will be distributed in a lump sum within the 90-day period commencing on the Scheduled Pay Date applicable to such Scheduled In-Service Account. The lump sum amount will be the value of the applicable Participant's Scheduled In-Service Account as of the Scheduled Pay Date.

(b) Intervening Separation or Death. Notwithstanding the preceding, should an event occur prior to the Scheduled Pay Date of any Scheduled In-Service Account that would trigger a distribution under Section 5.2 or 5.3 earlier than the Scheduled Pay Date, such Scheduled In-Service Account or Accounts shall be distributed in accordance with Section 5.2 or 5.3, as applicable, and not in accordance with Section 5.1(a).

5.2 Separation Benefits.

(a) Separation Benefits. In the event of a Participant's Separation from Service, the Participant's Scheduled In-Service Accounts will be distributed in a lump sum under Section 5.2(b) and the Participant's Retirement Accounts will be distributed in one of the forms provided in Section 5.2(b) or 5.2(c) below in accordance with the Participant's distribution election given effect under the provisions of Section 5.4 with respect to each such Retirement Account.

(b) Lump Sum Distributions.

(1) Any lump sum to be paid under this Section 5.2(b) shall be paid within the 90-day period commencing on the Participant's Separation Pay Date.

(2) The lump sum amount will be the value of the Participant's Account, or Retirement Account, as applicable, as of the last day of the month preceding the date of the distribution.

(c) Installment Distributions.

(1) If a Participant's Retirement Account is to be distributed in the form of annual installments, the first such installment shall be made within the 90-day period commencing on the first January 31 following the Participant's Separation from Service; provided, however, that if such January 31 is earlier than the Participant's Separation Pay Date, the first such installment shall be made within the 90-day period commencing on the Participant's Separation Pay Date. Subsequent installments shall be made within the 90-day period commencing on each successive January 31, until the Participant's benefits under such Account are distributed in full.

(2) If any portion of a Participant's Retirement Account is being credited with interest at the Interest Rate in accordance with Section 4.4(a)(1) as of the applicable Valuation Date, such portion will be paid in equal annual installments in an amount which would fully amortize a loan equal to the lump sum value of such portion determined in accordance with Section 5.2(b)(2) (using as the distribution date the date of the first installment) over the installment period, with interest calculated at the Interest Rate in effect for the Plan Year in which the Participant's Separation from Service occurs.

(3) If any portion of a Participant's Retirement Account is being credited with earnings, gains and losses in accordance with Sections 4.4(a)(2) or 4.4(b) as of the applicable Valuation Date, such portion of such Retirement Account will be paid in substantially equal annual installments equal to the quotient of (A) the total remaining balance in the applicable portion of the Retirement Account as of the Valuation Date immediately prior to the date on which such installment payment is scheduled to be paid, divided by (B) the number of installment payments remaining in the applicable period of annual installments.

5.3 Death Benefits.

(a) General. In the event of the Participant's death before incurring a Separation from Service or before commencement of benefits, the Participant's Account will be distributed in one of the forms provided in Section 5.3(b) or 5.3(c) below in accordance with the Participant's distribution election given effect under the provisions of Section 5.4 below. A Participant may elect only one form of payment under the Plan for all beneficiaries (at any level). If the Participant fails to make an effective election as provided in Section 5.4 below, the Participant will be deemed to have elected distribution in a lump sum under Section 5.3(b) for all beneficiary levels.

(b) Lump Sum Distributions.

(1) Any lump sum to be paid under this Section 5.3(b) shall be paid within the 90-day period commencing on the last day of the month in which the Participant's death occurs.

(2) The lump sum amount will be the value of the Participant's Account as of the last day of the month preceding the date of distribution.

(c) Installment Distributions.

(1) If the Participant's Account is to be distributed in the form of annual installments, the first such installment shall be made within the 90-day period commencing on the first January 31 coincident with or next following the Participant's death. Subsequent installments will be made during the 90-day period commencing on each successive January 31, until the Participant's benefits are distributed in full.

(2) If a portion of a Participant's Account is being credited with interest at the Interest Rate in accordance with Section 4.4(a)(1) as of the applicable Valuation Date, such portion will be paid in equal annual installments in an amount which would fully amortize a loan equal to the lump sum value of such portion as of the last day of the month preceding the date of distribution (using as the distribution date the date of the first installment) over the installment period, with interest calculated at the Interest Rate in effect for the Plan Year in which the Participant's death occurs.

(3) If a portion of a Participant's Account is being credited with earnings, gains and losses in accordance with Section 4.4(a)(2) or 4.4(b) as of the applicable Valuation Date, such portion will be paid in substantially equal annual installments equal to the quotient of (A) the total remaining balance of such portion as of the Valuation Date immediately prior to the date on which such installment payment is scheduled to be paid, divided by (B) the number of installment payments remaining in the applicable period of annual installments.

(d) Death After Commencement of Installments. Notwithstanding the preceding, in the event of a Participant's death after installment payments to the Participant have commenced, such installment payments shall continue to be made to the Participant's designated beneficiary in the same manner as they were being distributed to the Participant prior to his or her death, provided, however, that if the Participant's distribution election applicable to Section 5.3(a) is a lump sum payment, the Participant's remaining installments will be distributed in lump sum to the Participant's designated beneficiary within the 90-day period commencing on the last day of the month in which the Participant's death occurs.

(e) Designation of Beneficiary. A Participant may, by written or electronic instrument delivered to the Plan Administrator in the form prescribed by the Plan Administrator, designate primary and contingent beneficiaries (which may be a trust or trusts) to receive any benefit payments which may be payable under this Plan following the Participant's death, and may designate the proportions in which such beneficiaries are to receive such payments. A Participant may change such designation from time to time and the last designation filed with the Plan Administrator in accordance with its procedures prior to the Participant's death will control. In the event no beneficiary is designated, or if all designated beneficiaries predecease the Participant, payment shall be payable to the following "default" beneficiaries of the Participant in the following order of priority: (1) the Participant's surviving spouse known to the Plan Administrator, if any; (2) the Participant's living children known to the Plan Administrator in equal shares; (3) the Participant's living parents known to the Plan Administrator in equal shares; (4) the Participant's surviving siblings known to the Plan Administrator in equal shares; or (5) the beneficiary's estate for distribution in accordance with the terms of the beneficiary's last will and testament or as a court of competent jurisdiction shall determine.

(f) Death of Beneficiary. In the event a beneficiary dies before full payment of the Participant's benefits under the Plan, benefits that would have been paid to such beneficiary shall continue in the same form in equal shares to the remaining beneficiaries at the same level (i.e., primary, contingent) and, if none, to the next level of beneficiaries. If there are no beneficiaries at the next level, then any remaining benefits shall be paid to the following "default" beneficiaries of the last living beneficiary in the following order of priority: (1) the beneficiary's surviving spouse known to the Plan Administrator, if any; (2) the beneficiary's living children known to the Plan Administrator in equal shares; (3) the beneficiary's surviving parents known to the Plan Administrator in equal shares; (4) the beneficiary's surviving siblings known to the Plan Administrator in equal shares; or (5) the beneficiary's estate for distribution in accordance with the terms of the beneficiary's last will and testament or as a court of competent jurisdiction shall determine.

5.4 Form of Distribution.

(a) Forms Available. In the event of a Participant's Separation from Service, or in the event of a Participant's death if the Participant dies prior to Separation from Service, distribution of his or her Retirement Account or, in the event of death, his or her Account, may be made, at the Participant's election per this Section 5.4, in one of the following forms:

(1) a lump sum;

(2) subject to the minimum account value restriction below, substantially equal annual installments over a period not to exceed fifteen (15) years; or

(3) solely with respect to distribution of the Participant's Account in the event of death, partially a lump sum and, subject to the minimum account value restriction below, substantially equal annual installments over a period not to exceed fifteen (15) years;

Notwithstanding the foregoing:

(A) With respect to an installment election applicable to a Retirement Account under a Participant's Pre-2021 Class Account, such election will be given effect only if, as of the date on which any lump sum payment would be valued, the value of such Retirement Account, or, in the event of death, the Participant's Account, is at least fifty thousand dollars ($50,000). If such Retirement Account, or in the event of death, the Participant's Account, is valued at less than fifty thousand dollars ($50,000) as of the date on which any lump sum payment would be valued, such Retirement Account shall be defaulted to a lump sum payment.

(B) With respect to an installment election applicable to a Retirement Account under a Participant's Post-2020 Class Account, such election will be given effect only if, as of the date on which any lump sum payment would be valued, the combined value of all of the Participant's Retirement Accounts subject to an installment election under his or her Post-2020 Class Accounts, or, in the event of death, the Participant's Account, is at least fifty thousand dollars ($50,000). If the total of all such Retirement Accounts, or in the event of death, the Participant's Account, is valued at less than fifty thousand dollars ($50,000) as of the date on which any lump sum payment would be valued, such Retirement Account shall be defaulted to a lump sum payment.

(C) For purposes of clarification, to the extent a Retirement Account, or in the event of death, an Account, is defaulted to a lump sum payment pursuant to paragraph (A) or (B) above, such lump sum payment shall be paid at the same time the first installment with respect to such Retirement Account or Account would have been paid but for the application of paragraph (A) or (B) above.

(b) Subsequent Elections. In accordance with the procedures and rules established by the Plan Administrator, a Participant may change his or her distribution election (or deemed distribution election) with respect to his or her Retirement Account, or, in the event of death, his or her Account, per this Section 5.4 at any time by making a new election (referred to in this subsection as a "subsequent election") on a form (which may be electronic) approved by, and filed with, the Plan Administrator; provided, however, that such subsequent election shall be subject to the following restrictions:

(1) A subsequent election may not take effect until at least twelve (12) months after the date on which such subsequent election is made;

(2) Payment or initial payment pursuant to a subsequent election may not be made earlier than five (5) years from the date such payment would have been made absent the subsequent election (but, for this purpose, installment payments shall not commence until the first January 31 after such delay), unless the distribution is made on account of the Participant's death;

(3) A subsequent election related to a payment must be made not less than twelve (12) months before the date the payment is scheduled to be paid;

(4) Payment of a Participant's Retirement Account or, in the event of death, Account, pursuant to a subsequent election must be completed by the last day of the Plan Year which contains the twentieth (20^{th}) anniversary of the Participant's Separation Pay Date or the Participant's death;

(5) For purposes of this Section 5.4(b) and Code Section 409A, the entitlement to annual installment payments is treated as the entitlement to a single payment.

If a Participant's distribution election does not satisfy the requirements of this Section 5.4(b), it will not be recognized or given effect by the Plan Administrator. In that event, distribution of the benefit will be made in accordance with the Participant's most recent distribution election which does satisfy the requirements of this Section 5.4(b).

(c) Filing of Election. A Participant's distribution election applicable to the Participant's Account in the event of the Participant's death prior to Separation from Service, and a Participant's distribution election with respect to the Participant's Retirement Account or Retirement Accounts, and the Participant's Scheduled Pay Date with respect to the Participant's Scheduled In-Service Accounts, must be filed with, and on forms (which may be electronic) prescribed by, the Plan Administrator.

5.5 Distributions for Unforeseeable Emergencies.

(a) In the event of an Unforeseeable Emergency, the Plan Administrator or his or her delegate, in its sole and absolute discretion and upon written application of a Participant or, following the Participant's death, the beneficiary to whom a Participant's benefits are then being paid, or will be paid, pursuant to Section 5.3, may direct immediate distribution of all or a portion of the Participant's Account (excluding the Participant's Matching Account and related earnings if the Participant is not fully vested in his or her Matching Account). The Plan Administrator will permit distribution on account of an Unforeseeable Emergency only to the extent reasonably necessary to satisfy the emergency need, plus amounts necessary to pay federal, state or local income taxes and penalties reasonably anticipated to result from the distribution, after taking into account the extent to which such need is or may be relieved through reimbursement or compensation by insurance, by liquidation of the Participant's or beneficiary's assets (to the extent the liquidation of such assets would not itself cause severe financial hardship), or by cessation of deferrals under the Plan. Any distribution under this Section 5.5 shall first be made from the Participant's Scheduled In-Service Accounts with respect to Deferral Credits made in the same Plan Year as the Distribution under this Section 5.5, and then from the Participant's Retirement Accounts with respect to Deferral Credits made in the same Plan Year as the Distribution under this Section 5.5, and then proratably from the remaining amount of the Participant's Scheduled In-Service Accounts and then proratably from the Participant's Retirement Accounts.

(b) Notwithstanding anything in the Plan to the contrary, if the Plan Administrator reasonably anticipates that its deduction with respect to any distribution under this Section 5.5 that occurs prior to January 1, 2021 would not be permitted due to the application of Code Section 162(m); such payment shall be suspended to the extent a deduction would not be permitted until the earliest date at which it reasonably anticipates that the deduction of such distribution would not be barred by application of Code Section 162(m); provided, however, that the conditions of Section 5.5(a) are still satisfied as of such date.

5.6 Distributions for Payment of Taxes.

The Plan Administrator may accelerate and pay a portion of a Participant's Plan benefits in a lump sum equal to (a) the Federal Insurance Contributions Act tax imposed on Plan benefits and any income tax withholding related to such amounts, as well as (b) any state, local or foreign tax obligations arising from participation in the Plan (and related withholding under Code Section 3401) that apply to the amounts deferred under the Plan before such amount is paid or made available to the Participant.

5.7 Reductions Arising from a Participant's Gross Misconduct.

Notwithstanding anything herein to the contrary, a Participant's Plan benefits are contingent upon the Participant not engaging in Gross Misconduct while employed with any Employer or Related Affiliate or any entity in which Walmart has an ownership interest, or during such additional period as provided in Walmart's Statement of Ethics. In the event the Plan Administrator determines that the Participant has engaged in Gross Misconduct during the prescribed period, then notwithstanding any provisions hereunder to the contrary: (a) the Participant shall forfeit all Employer Matching Contribution Credits and credited Plan earnings thereon; (b) earnings credited to the Participant's Deferral Accounts shall be recalculated for each Plan Year to reflect the amount which would otherwise have been credited if the applicable Interest Rate (or if a portion of such Deferral Accounts is being credited with earnings, gains and losses, the positive earnings rate (if any) for such Plan Year) were fifty percent (50%) of the Interest Rate in effect (or positive earnings rate, as applicable) for such Plan Year; and (c) if the Participant is then receiving installment payments, any remaining installments shall be recalculated to reflect the amount which would otherwise have been paid if the applicable Interest Rate (or positive earnings rate, as applicable) were fifty percent (50%) of the Interest Rate in effect (or positive earnings rate, as applicable) with respect to such installment payments. For the avoidance of doubt, if a portion of a Participant's Deferral Accounts is being credited with earnings, gains and losses and such portion experiences a net loss for a Plan Year, such net loss shall not be adjusted as a result of the Participant's engaging in Gross Misconduct under this Section 5.7. Under no circumstances will a Participant forfeit any portion of the Participant's Deferred Compensation, Deferred MIP Bonus and Deferred Special Bonus as a result of the Participant's engaging in Gross Misconduct under this Section 5.7. Any payments received hereunder by a Participant (or the Participant's beneficiary) are contingent upon the Participant not engaging (or not having engaged) in Gross Misconduct while employed with any Employer or Related Affiliate or any entity in which Walmart has an ownership interest, or during such additional period as provided in Walmart's Statement of Ethics. If the Plan Administrator determines, after payment of amounts hereunder, that the Participant has engaged in Gross Misconduct during the prescribed period, the Participant (or the Participant's beneficiary) shall repay to Walmart any amount in excess of that to which the Participant is entitled under this Section 5.7.

ARTICLE VI.
ADMINISTRATION

6.1 General.

The Plan Administrator is responsible for the administration of the Plan and is granted the following rights and duties:

(a) The Plan Administrator shall have the exclusive duty, authority and discretion to interpret and construe the provisions of the Plan, to determine eligibility for and the amount of any benefit payable under the Plan, and to decide any dispute which may arise regarding the rights of Participants (or their beneficiaries) under this Plan;

(b) The Plan Administrator shall have the authority to adopt, alter, and repeal such administrative rules, regulations, and practices governing the operation of the Plan as it shall from time to time deem advisable;

(c) The Plan Administrator may appoint a person or persons to act on behalf of, or to assist, the Plan Administrator in the administration of the Plan, establishment of forms (including electronic forms) desirable for Plan operation, and such other matters as the Plan Administrator deems necessary or appropriate;

(d) The decision of the Plan Administrator in matters pertaining to this Plan shall be final, binding, and conclusive upon Walmart, any Related Affiliate, the Participant, the Participant's beneficiary, and upon any person affected by such decision, subject to the claims procedure set forth in Article VII; and

(e) In any matter relating solely to a Plan Administrator's individual rights or benefits under this Plan, the Plan Administrator's duties under this Section 6.1 shall be performed by the Committee and the Plan Administrator shall not participate in any Committee proceeding pertaining to, or vote on, such matter.

6.2 Allocation and Delegation of Duties.

The Plan Administrator shall have the authority to delegate, from time to time, by written instrument filed in its records, all or any part of its responsibilities under the Plan to such person or persons as the Plan Administrator may deem advisable (and may authorize such person to delegate such responsibilities to such other person or persons as the Plan Administrator shall authorize) and in the same manner to revoke any such delegation of responsibility. Any action of the delegate in the exercise of such delegated responsibilities shall have the same force and effect for all purposes hereunder as if such action had been taken by the Plan Administrator. The Plan Administrator shall not be liable for any acts or omissions of any such delegate. The delegate shall periodically report to the Plan Administrator concerning the discharge of the delegated responsibilities.

ARTICLE VII.
CLAIMS PROCEDURE

7.1 General.

Any claim for benefits under the Plan must be filed by the Participant or beneficiary ("claimant") in writing with the Plan Administrator or his or her delegate within one (1) year of the Participant's Separation from Service. If the claim is not filed within one (1) year of the Participant's Separation from Service, neither the Plan nor any Employer nor any Related Affiliate shall have any obligation to pay the benefit and the claimant shall have no further rights under the Plan. If a timely claim for a Plan benefit is wholly or partially denied, notice of the decision will be furnished to the claimant by the Plan Administrator or his or her delegate within a reasonable period of time, not to exceed sixty (60) days, after receipt of the claim by the Plan Administrator or his or her delegate, unless special circumstances require an extension of time for processing, in which case a decision will be rendered within a reasonable period of time, but not later than one hundred twenty (120) days after receipt. Any claimant who is denied a claim for benefits will be furnished written notice setting forth:

(a) the specific reason or reasons for the denial;

(b) specific reference to the pertinent Plan provision upon which the denial is based;

(c) a description of any additional material or information necessary for the claimant to perfect the claim and an explanation of why such material or information is necessary; and

(d) an explanation of the Plan's claim review procedure, including the claimant's right to bring a civil action under ERISA Section 502(a) following an adverse determination on review.

7.2 Appeals Procedure.

To appeal a denial of a claim, a claimant or the claimant's duly authorized representative:

(a) may request a review by written application to the Plan Administrator not later than sixty (60) days after receipt by the claimant of the written notification of denial of a claim;

(b) may review pertinent documents; and

(c) may submit issues and comments in writing.

A decision on review of a denied claim will be made by the Plan Administrator not later than sixty (60) days after receipt of a request for review, unless special circumstances require an extension of time for processing, in which case a decision will be rendered within a reasonable period of time, but not later than one hundred twenty (120) days after receipt of a request for review. The decision on review will be in writing and shall include:

(a) the specific reason or reasons for the adverse determination;

(b) specific reference to pertinent Plan provisions on which the adverse determination is based;

(c) a statement that the claimant is entitled to receive, upon request and free of charge, reasonable access to, and copies of, all documents, records, and other information relevant to the claimant's claim for benefits; and

(d) a statement describing any voluntary appeal procedures offered by the Plan and the claimant's right to obtain the information about such procedures, as well as a statement of the claimant's right to bring an action under ERISA section 502(a).

7.3 Disability Claims.

Claims for disability benefits shall be determined under DOL Regulation section 2560.503-1 which is hereby incorporated by reference.

ARTICLE VIII.
MISCELLANEOUS PROVISIONS

8.1 Amendment, Suspension or Termination of Plan.

Walmart, by action of the Committee, reserves the right to amend, suspend or to terminate the Plan in any manner that it deems advisable. Notwithstanding the preceding sentence, the Plan may not be amended, suspended or terminated to cause a Participant to forfeit the Participant's then-existing Account.

Notwithstanding the preceding, Walmart may, by action of the Committee within the thirty (30) days preceding or twelve (12) months following a change in control (within the meaning of Code Section 409A) of a relevant affiliate, partially terminate the Plan and distribute benefits to all Participants involved in such change in control within twelve (12) months after such action, provided that all plans sponsored by the service recipient immediately after the change in control (which are required to be aggregated with this Plan pursuant to Code Section

409A) are also terminated and liquidated with respect to each Participant involved in the change in control. Any action taken in this Section 8.1 will be done in accordance with Code Section 409A.

8.2 Non-Alienability.

No interest or amounts payable under the Plan may be subject in any manner to anticipation, alienation, sale, transfer, assignment, pledge, encumbrance, charge, garnishment, execution or levy of any kind, whether voluntary or involuntary. Notwithstanding the preceding, distribution may be made to the extent necessary to fulfill a domestic relations order as defined in Code Section 414(p)(1)(B) and in accordance with procedures established by the Plan Administrator from time to time; provided, however, that all such distributions shall be made in a single lump sum payment.

8.3 Recovery of Overpayments.

In the event any payments under the Plan are made on account of a mistake of fact or law, the recipient shall return such payment or overpayment to Walmart as requested by Walmart.

8.4 No Employment Rights.

Nothing contained herein shall be construed as conferring upon any Eligible Participant or Participant the right to continue in the employ of any Employer or any Related Affiliate as an officer or in any other capacity.

8.5 No Right to Bonus.

Nothing contained herein shall be construed as conferring upon the Participant the right to receive a bonus from the MIP or any other bonus or award from any Employer or a Related Affiliate. A Participant's entitlement to such a bonus or award is governed solely by the provisions of the MIP or such other plan or arrangement.

8.6 Withholding and Employment Taxes.

To the extent required by law, the Employer or a Related Affiliate will withhold from a Participant's current compensation such taxes as are required to be withheld for employment taxes. To the extent required by law, the Employer or a Related Affiliate will withhold from a Participant's Plan distributions such taxes as are required to be withheld for federal, Puerto Rican, state or local government income tax purposes.

8.7 Income and Excise Taxes.

The Participant (or the Participant's Beneficiaries) is solely responsible for the payment of all federal, Puerto Rican, state and local income and excise taxes resulting from the Participant's participation in this Plan.

8.8 Successors and Assigns.

The provisions of this Plan are binding upon and inure to the benefit of Walmart and each other Employer, their successors and assigns, and the Participant, the Participant's beneficiaries, heirs, and legal representatives.

8.9 Governing Law.

This Plan shall be subject to and construed in accordance with the laws of the State of Delaware to the extent not preempted by federal law.

Exhibit 10.3

Walmart Inc.
2016 Associate Stock Purchase Plan

(As amended effective February 1, 2024)

Table of Contents

Walmart Inc. 2016
Associate Stock Purchase Plan

I. Definitions

1.1 **“Account”** shall mean a Participant’s account which holds his or her shares of Stock pursuant to the Plan.

1.2 **“Account Administrator”** shall mean the third party administrator for the Accounts as may be from time to time appointed by the Committee.

1.3 **“Account Closure”** shall mean the closing of a Participant’s Account by one of the following means:

(a) **“Automatic Account Closure”** shall mean the closure of a Participant’s Account by the Committee (or the Account Administrator if applicable) at the time such Participant’s Account balance contains no shares (or fractional shares) of Stock on or after his or her termination of employment with the Employer.

(b) **“Participant Account Closure”** shall mean the closure of a Participant’s Account pursuant to a request by the Participant to have his or her Account closed and to have all Stock or proceeds from the sale thereof distributed.

1.4 **“Affiliate”** shall mean any entity that is more than 50% owned or controlled, directly or indirectly, by the Company.

1.5 **“Associate”** shall mean any common law employee of an Employer, but shall not include independent contractors. An individual classified by the Employer as either an independent contractor or an individual who provides services to the Employer through another entity shall not be eligible to participate in this Plan during the period that the individual is so classified, even if such individual is later retroactively reclassified as an Associate during all or any part of such period pursuant to applicable law or otherwise.

1.6 **“Award Program”** shall mean a program established by the Company or a Participating Employer that results in its Associates receiving shares of Stock as an award for job performance.

1.7 **“Board”** shall mean the Board of Directors of the Company.

1.8 **“Committee”** shall mean the Compensation and Management Development Committee of the Board, or such other committee as may be appointed by the Board.

1.9 **“Company”** shall mean Walmart Inc., a Delaware corporation.

1.10 **“Contribution”** shall mean any of the types of contributions that may be made to a Participant’s Account under the Plan, either by the Company, a Participating Employer or a Participant as set forth in Section III.

1.11 **“Employer”** shall mean the Company and its Affiliates.

1.12 **“Participant”** shall mean any Associate of the Company or a Participating Employer who satisfies the eligibility requirements in Section II and who has an Account established under the Plan, and Participant shall also include any former Associate of the Company or a Participating Employer who was a Participant in the Plan at the time of his or her termination of employment until such time as an Account Closure occurs.

1.13 **“Participating Employer”** shall mean an Affiliate whose participation in the Plan has been approved by the Committee. The Committee may require the Participating Employer to make corresponding contributions under the Plan in accordance with rules and procedures established by the Committee. The Committee, in its sole discretion, may terminate any such Affiliate’s Participating Employer status at any time and the Accounts of those Participants who are Associates of such Affiliate will be treated as if such Participants had transferred employment to an Affiliate that is not a Participating Employer as described in Section 5.3 of the Plan.

1.14 “Payroll Deduction” shall mean the payroll deduction from a Participant’s biweekly or weekly regular compensation (including from vacation pay and pay from any paid leave of absence) of an amount authorized by the Participant as a Payroll Deduction Contribution.

1.15 “Plan” shall mean the Walmart Inc. 2016 Associate Stock Purchase Plan (formerly known as the Wal-Mart Stores, Inc. 2016 Associate Stock Incentive Plan, and the Wal-Mart Stores, Inc. 2004 Associate Stock Purchase Plan), as amended, restated and renamed herein, or as it may be further amended from time to time.

1.16 “Plan Year” shall mean April 1 of a calendar year to March 31 of the following calendar year, or such other period as set by the Committee.

1.17 “Section 16 Officers” shall mean those officers of the Company who are subject to subsection 16(a) of the Securities Exchange Act of 1934, as amended.

1.18 “Stock” shall mean the common stock, $.10 par value per share, of the Company.

II. Eligibility

2.1 In General. All Associates (including Section 16 Officers) of the Company or a Participating Employer are eligible to participate in the Plan, subject to the following limitations:

(a) Associates who are restricted or prohibited from participating in the Plan under the applicable law of their state or country of residence may not participate in the Plan, except as may be provided in accordance with rules and procedures established by the Committee.

(b) Associates of the Company and its Affiliates who are members of a collective bargaining unit whose benefits were the subject of good faith collective bargaining are excluded from participation in the Plan.

(c) Participation by Associates of non-U.S. Participating Employers shall only be permitted upon approval by the Committee, which approval may be limited to groups or categories of Associates designated by the non-U.S. Participating Employer.

(d) Section 16 Officers may be restricted in their ability to acquire or sell shares of Stock in order to comply with Section 16 of the Securities Exchange Act of 1934, as amended, in accordance with rules and procedures adopted by the Company’s Audit Committee.

2.2 Leaves of Absence. Participants continue to be eligible to participate in the Plan while on a bona fide leave of absence from the Company or a Participating Employer in accordance with applicable policies of the Company or Participating Employer, or under such other circumstances with the approval of the Committee.

III. Plan Contributions

3.1 Shares Available for Contributions. Subject to stockholder approval of this Plan: (i) 10,943,171 shares of Stock shall be available for purchase from the Company under the Plan for credit to Accounts or for purchase in open market transactions over a national securities exchange under the Plan for credit to Accounts; (ii) 20,000,000 shares of Stock shall be available for purchase from the Company under the Plan for credit to Accounts; and (iii) 100,000,000 shares of Stock shall be available for purchase in open market transactions over a national securities exchange under the Plan for credit to Accounts.

3.2 Plan Contributions. The definitions of the types of Contributions which may be made pursuant to the Plan are as follows (subject to the limits provided in Section 3.3 as applicable):

(a) "Award Contribution" means a contribution under the Plan on behalf of a Participant by the Company or a Participating Employer, as applicable, made pursuant to the Award Program in the sole discretion of the Committee.

(b) "Matching Contribution" means a cash contribution to the Plan on behalf of a Participant by the Company or a Participating Employer, as applicable, which is equal to fifteen percent (15%) of the amount of the Participant's Payroll Deduction (up to a maximum dollar limit).

(c) "Payroll Deduction Contribution" means a contribution to the Plan by a Participant pursuant to a valid authorization for a Payroll Deduction.

(d) "Voluntary Contribution" means a contribution, if and to the extent permitted by the Committee from time to time, of shares of Stock or cash by the Participant to the Participant's Account which is not made by Payroll Deduction.

3.3 Maximum Limits on Contributions.

(a) Matching Contributions and "Outstanding Performance" awards under the Award Program are subject to a maximum dollar limit for the Plan Year as set by the Committee from time to time in its discretion.

(b) During any Plan Year, the combination of Payroll Deduction Contributions and Voluntary Contributions made in cash (not Stock) by a Participant shall not exceed $125,000.

3.4 Payroll Deductions.

(a) Subject to the Committee's authority to adjust the following amounts, a Participant's authorization for Payroll Deduction shall be for a minimum amount of $2.00 per biweekly pay period or $1.00 per weekly pay period, as applicable to the Participant, and such Payroll Deduction shall be in even multiples of $.50.

(b) A Participant's request for Payroll Deduction (or a request for a revision thereto) will become effective as soon as practicable after receipt of such request by the Company or the Participating Employer, as applicable.

(c) A Participant's Payroll Deduction authorization may be revised or terminated at any time by the Participant's request to the Company or the Participating Employer, as applicable.

(d) A Participant's authorization for Payroll Deduction shall remain effective until the earlier of the Participant's (1) request to revise or terminate the Payroll Deduction authorization or (2) termination of employment with the Company or a Participating Employer, subject to Section 8 of the Plan.

(e) All requests to initiate, revise or terminate an authorization for Payroll Deduction as described in this Section 3.4 shall be made in writing or in such other form acceptable to the Committee or its delegate from time to time.

(f) The Senior Vice President, Total Rewards, or any successor position, in his or her discretion, may prohibit a Payroll Deduction from the final paycheck of a Participant. This Section applies to the Participant's final paycheck even if the Participant made a valid Payroll Deduction election applicable to prior paychecks. If a Payroll Deduction is prohibited from a Participant's final paycheck, any corresponding Matching Contributions shall also be prohibited.

3.5 Matching Contributions. The Company or Participating Employer, as applicable, shall make Matching Contributions as provided under the Plan and subject to the limits set forth in Section 3.3.

3.6 Award Contributions. Award Contributions shall be made, in the Committee's sole discretion, by either (1) the Company or the Participating Employer, as applicable, remitting to the Account Administrator on behalf of the Participant funds sufficient to purchase any shares or fractional shares of Stock that have been granted to such Participant under the Award Program or (2) the Participant receiving the Award Contribution directly as a certificate for a share or shares (as applicable) of Stock.

3.7 Voluntary Contributions. Participants may make Voluntary Contributions to the Plan subject to the terms and limitations described herein or that may be prescribed by the Committee from time to time.

3.8 Remittance of Contributions.

(a) The Company or a Participating Employer, as applicable, will forward the total of all Payroll Deductions for the applicable payroll period along with the corresponding Matching Contributions, a list of Participants for whom the Contributions are being made and the amount allocable to each such Participant's Account to the Account Administrator as soon as practicable.

(b) Voluntary Contributions, whether made in cash or shares of Stock, shall be remitted to the Account Administrator directly by the Participant.

(c) As soon as practicable following a grant of an Award Contribution, an Award Contribution shall be made in the Committee's sole discretion as described in Section 3.6 of the Plan.

(d) Prior to the time a Participant's Payroll Deduction and corresponding Matching Contribution is distributed to the Account Administrator, such amounts are considered general assets of the Company or Participating Employer (as applicable) and, as such, are subject to the claims of the Company's or Participating Employer's (as applicable) creditors in the event of insolvency or bankruptcy. In addition, no interest shall be paid on such amounts and all Participants assume the risk of fluctuations in the value or market price of Stock.

IV. Account Purchases, Maintenance & Sales

4.1 Account Establishment. The Account Administrator shall establish an Account in accordance with the Plan for any Associate who becomes a Participant. Upon the Committee's (or its delegate's) request, the Account Administrator shall establish an Account for an Associate who is to be awarded shares under an Award Program and who is not then a Participant.

4.2 Share Purchases. No later than five business days after the Account Administrator receives the remittance of funds for Contributions (including Voluntary Contributions made in cash) made to the Plan, the Account Administrator shall purchase shares of Stock from the Company, which may be purchases from the Company of authorized, but unissued, shares of Stock, shares of Stock held as treasury shares of Stock, in open market transactions over a national securities exchange, or in a combination of the foregoing. Notwithstanding the foregoing, the Committee may from time to time provide instructions to the Account Administrator with respect to the purchase of such shares of Stock but, absent such instructions, the Account Administrator shall determine the source of such Stock purchases in its discretion.

(a) In the case of purchases from the Company of authorized but unissued or treasury shares of Stock, the price of such shares is equal to the Volume Weighted Average Price (VWAP) as reported on the New York Stock Exchange - Composite Transactions on the relevant date of purchase; provided, however, that the Committee may, in its discretion, designate some other methodology for determining the fair market value of such shares of Stock purchased from the Company.

(b) The Account Administrator's purchase of shares of Stock in open market transactions over a national securities exchange and the price per share shall be in accordance with rules and procedures established by the Committee from time to time, the rules of the national securities exchange over which the shares of Stock are purchased and the rules of the Financial Industry Regulatory Authority.

(c) As determined in the discretion of the Account Administrator (in accordance with any applicable rules and procedures of the Committee), funds received as Voluntary Contributions may be bundled into a group for the purpose of purchasing shares of Stock and such shares may be purchased over a time period that is greater than one day. If such shares of Stock are purchased as part of a bundled group, a Participant's purchase price for each share of Stock shall be the average price of all shares of Stock purchased within that group as determined by the Account Administrator.

(d) No provision of this Plan shall limit the ability of the Committee to implement a real-time trading (or other) mechanism for the purchase or sale of shares of Stock under the Plan and, to the extent determined by the Committee, shall replace any other methodology for valuing and allocating shares of Stock purchased or sold under the Plan.

4.3 Share Purchases for Non-U.S. Participants. With respect to non-U.S. Participants, the amounts (1) withheld from such a Participant's compensation pursuant to an authorization for Payroll Deduction or (2) contributed as either a Matching Contribution or an Award Contribution made directly to a Participant's Account shall be converted from the applicable foreign currency to U.S. dollars for the purpose of purchasing shares of Stock, and such conversion shall be pursuant to the exchange rate published in The Wall Street Journal (or other similar source) on a date as soon as practicable prior to the effective date of the cash transfer from the Company or the Participating Employer, as applicable, to the Account Administrator. All such Participants assume the risk of fluctuations in the value or market price of shares of Stock and applicable currency exchange rates. With respect to non-U.S. Participants making Voluntary Contributions in cash, such amounts must be tendered to the Account Administrator in U.S. dollars unless otherwise determined by the Committee.

Notwithstanding anything to the contrary contained in the terms of this Plan, at the discretion of the Committee, purchases of shares of Stock for, or the participation in the Plan by, non-U.S. Participants may be suspended or discontinued if the applicable laws of a country or another governmental authority (such as the European Union) governing such purchases of shares of Stock or participation in the Plan would require the Company to register or qualify such shares of Stock, or its deemed offer or sale of shares of Stock under the Plan to, or to otherwise comply with procedures under such laws with respect to any deemed offer and sale of shares of Stock by the Company under the Plan to, such non-U.S. Participants or the Company would otherwise become subject to the laws of such country or other governmental authority as a result of such purchases of shares of Stock by or on behalf of a non-U.S. Participant unless the Company is already subject in all respects to the jurisdiction of such country or governmental authority.

4.4 Allocation to Accounts. The number of shares (whole and fractional shares) of Stock shall depend upon the purchase price as described in Section 4.2 at the time such purchases are made. Purchases of Stock will be allocated by the Account Administrator based upon the applicable purchase price to each applicable Participant's Account in proportion to the respective amount of Contributions received for each Participant's Account. Allocations of Stock will be made in full shares and in fractional interests in shares to the thousandths of a share.

4.5 Share Ownership. At the time shares of Stock are credited to a Participant's Account, he or she will acquire full ownership of all such shares (as well as any fractional interests) of Stock.

(a) All shares of Stock will be registered in the name of the Account Administrator and will remain so registered until delivery is requested by the Participant. The Participant may request from the Account Administrator that a certificate for any or all full shares of Stock be delivered to the Participant or that the Participant be reflected as the record owner of such shares of Stock in the Direct Registration System of the Depository Trust Company, if the Company participates in that system, at no cost to such Participant at any time.

(b) The Account Administrator shall cause to be delivered at no cost to each Participant as promptly as practicable, by mail, electronic mail, or otherwise, all notices of shareholders' meetings, proxy statements and other material distributed by the Company to its stockholders. The full shares of Stock in each Participant's Account shall be voted in accordance with the Participant's signed proxy voting instructions timely delivered to the Account Administrator. In the event that a Participant does not timely provide the Account Administrator with proxy voting instructions, the Account Administrator may direct the voting of such shares of Stock held in an Account to the extent such action or direction would comply with applicable law and any applicable listing standards of a national securities exchange.

(c) A Participant may not assign or hypothecate any interest in the Plan; provided, however, that upon purchase of shares under the Plan, such shares may be sold, assigned, pledged, hypothecated or otherwise dealt with as would be the case with respect to any other shares of Stock the Participant might otherwise own, subject to the Participant's compliance with the Walmart Inc. Insider Trading Policy.

(d) Neither the Company nor any Participating Employer may make any deductions from amounts properly credited to a Participant's Account. Neither the Company nor any Participating Employer shall have any security interest on the shares of Stock held in a Participant's Account. Notwithstanding the foregoing, a lender may have a security interest on the shares of Stock held in a Participant's Account if the Participant has pledged such Stock as collateral in connection with a line of credit that may be obtained by certain Participants (other than Section 16 Officers) through the Account Administrator's Stock Secured Line of Credit Program, if any.

4.6 **Account Statements.** Each Participant will be sent at least an annual statement reflecting all Account activity during the period covered by the statement.

4.7 **Risk of Loss.** There is no guarantee of the value or market price of shares of Stock acquired pursuant to the Plan. In seeking potential benefits of Stock ownership, each Participant bears the risks associated with Plan participation and ownership of Stock, including the risk of any decrease in the value of market price of shares of Stock acquired pursuant to the Plan.

4.8 **Commission & Maintenance Charges.**

(a) No brokerage commissions are charged to Participants for purchases of Stock under the Plan, however, brokerage commissions and other applicable fees shall be charged to the Participant for all sales of Stock from his or her Account. Such commissions and other applicable fees for sales of Stock held in a Participant's Account shall be at the rates posted by the Account Administrator, which may be changed from time to time by the Account Administrator with approval of the Committee (or its delegate).

(b) The Company or Participating Employer, as applicable, shall pay the applicable periodic maintenance fees (if any) for the Participant's Account until the earlier of (1) a Participant Account Closure occurs or (2) the Participant incurs a termination of employment with the Company or Participating Employer, as applicable, subject to Section 5.3. Any services requested of the Account Administrator by the Participant that are not covered by the Company's arrangement with the Account Administrator shall be paid for solely by the Participant.

(c) At such time as the Company or Participating Employer, as applicable, ceases to pay the applicable Account maintenance fees as set forth subsection (b) above, the Participant shall become responsible for any applicable Account maintenance fees. In this case, periodic maintenance fees and other applicable charges to the Account shall be paid from time to time to the Account Administrator automatically from the proceeds of a sale of a sufficient number of shares of Stock held in the Participant's Account to cover such fees and charges until the earlier of a Participant Account Closure or an Automatic Account Closure occurring.

4.9 **Account Sales.** The Participant may instruct the Account Administrator in writing (or any other method acceptable to the Committee or its delegate) at any time to sell any portion or all of his or her full shares of Stock and the fractional interest in any shares of Stock allocable to his or her Account, and the timing for such sale of Stock shall be in accordance with rules and procedures established by the Committee from time to time. Any account sales shall be subject to the Walmart Inc. Insider Trading Policy, including, but not limited to, the Trading Windows and trading restrictions on Section 16 Officers.

(a) The sale price for a share of Stock under the Plan shall be the average price of all shares of Stock sold by the Account Administrator on the date of the Participant's sale transaction; provided, however, that the Committee reserves the right to implement a real-time trading or similar mechanism for Participants' sales of shares of Stock from their respective Accounts under the Plan and the valuation of shares of Stock would be in accordance with any such mechanism.

(b) Upon such sale, the Account Administrator shall mail to the Participant a check (or such method of payment as approved by the Committee or its delegate) for the proceeds, less the brokerage commission, and other normal charges such as sales fees, which are payable by the Participant.

(c) Such instruction to the Account Administrator, or a request for delivery of Stock certificates held in the Participant's Account, will not affect the Participant's status as a Participant under the Plan unless the delivery of such certificates results in an Account Closure.

(d) With respect to non-U.S. Participants, shares of Stock are sold or traded in U.S. dollars and such amounts can be converted for the purpose of remitting the proceeds to the non-U.S. Participant. If the proceeds from the sale of shares of Stock held in the Participant's Account are converted to a currency other than U.S. dollars, such conversion shall be made pursuant to the applicable exchange rate published in The Wall Street Journal (or other similar source) on the date such transaction is executed. All such Participants assume the risk of fluctuations in the value or market price of shares of Stock and applicable currency exchange rates.

V. Account Closure & Termination of Employment

5.1 **Account Closure.** A Participant who elects to discontinue Payroll Deductions under the Plan shall continue to be a Participant until the earlier of a Participant Account Closure or an Automatic Account Closure occurring. In connection with a Participant Account Closure, the Participant must elect to have his or her Account fully distributed in either (1) Stock (except that the value of any fractional shares of Stock will be distributed in cash) less any applicable fees or (2) cash by directing all full shares (and fractional interests) of Stock to be sold with the proceeds, less applicable brokerage commissions and other applicable fees, being distributed in accordance with the terms, provisions, and conditions of the Plan.

5.2 **By Termination of Employment Other Than Due to Death of Participant.** The Account of a Participant who incurs a termination of employment (other than by reason of death) with the Company or a Participating Employer will continue to be maintained with the periodic fees and any other applicable charges being paid by the Participant in accordance with Section 4.8(c) of the Plan.

5.3 **By Transferring Employment from the Company or a Participating Employer to an Affiliate.** A Participant who transfers employment from the Company or a Participating Employer to an Affiliate who does not sponsor or participate in the Plan may continue to have his or her Account maintained at the expense of the Company while still employed with an Affiliate until the earlier of a Participant Account Closure or an Automatic Account Closure occurring (provided that such Automatic Account Closure can only occur following termination of employment with such Affiliate). In connection with a Participant Account Closure, the Participant must elect to have his or her Account fully distributed in either (1) Stock (except that the value of any fractional shares of Stock will be distributed in cash less any applicable fees) or (2) cash by directing all full shares (and fractional interests) of Stock to be sold with the proceeds, less applicable brokerage commissions and other applicable fees, being distributed, in accordance with the terms, provisions, and conditions of the Plan. Unless and until such Participant re-establishes eligibility to participate in the Plan, such Participant shall no longer be eligible to make or receive Contributions to the Plan (including by Payroll Deduction or Voluntary Contribution).

5.4 **Termination Due to Death of Participant.** Following a Participant's death, the Company or Participating Employer, as applicable, shall cease making Payroll Deductions and Matching Contributions to such Participant's Account as soon as practicable. In addition, as soon as practicable following the Participant's death, the Account Administrator will distribute the proceeds of the deceased Participant's Account less applicable brokerage commissions and other applicable fees in accordance with rules and procedures established by the Committee (which may include a designation by a Participant of a beneficiary or a joint tenant with respect to a Participant's Account) and, in the absence of applicable rules and procedures (or such designations), to the Participant's estate.

VI. Award Program

6.1 **Scope of the Award Program.** The Award Program is designed to provide an incentive to Associates of the Company and Participating Employers who provide exceptional customer service and job performance. Awards under the Award Program are not intended to be given to those who satisfy, but do not exceed, expectations. The Award Program includes an "Outstanding Performance" component.

6.2 Outstanding Performance Component. An "Outstanding Performance" award is an award of Stock to an Associate in recognition of the individual's consistently outstanding performance in his or her specific job-related roles over a month, a quarter, or a year.

(a) Associates who receive "Outstanding Performance" awards may either be issued certificates for shares of Stock or, at the discretion of the Committee, the Company (or Participating Employer) may have the Account Administrator purchase shares of Stock to be credited to the Participant's Account as described in Section 3.6 of the Plan.

(b) "Outstanding Performance" awards are either approved directly by the Committee or by its delegate in accordance with rules and procedures established by the Committee, and are subject to individual maximum dollar limitations as set by the Committee from time to time.

6.3 Former Great Job Component. This component of the Plan was discontinued in 2007 and all outstanding Great Job buttons were cancelled on September 1, 2008.

VII. Administration

7.1 Committee.

(a) Subject to Section 7.2, the Plan shall be administered by the Committee.

(b) The Committee may delegate to officers or managers of the Company or any Affiliate the authority, subject to such terms as the Committee shall determine, to perform specified functions under the Plan. The Committee also may revoke any such delegation of authority at any time.

7.2 Powers of the Committee. Subject to and consistent with the provisions of the Plan, the Committee has full and final authority and sole discretion as follows:

(a) to determine when, to whom and in what types and amounts Contributions should be made;

(b) to make Contributions to eligible Associates in any number, and to determine the terms and conditions applicable to each Contribution;

(c) to determine whether any terms and conditions applicable to a Contribution have been satisfied;

(d) to set minimum and maximum dollar, share or other limitations on the various types of Contributions under the Plan;

(e) to determine whether an Affiliate should be designated as a Participating Employer and whether an Affiliate's Participating Employer status should be terminated;

(f) to determine whether Associates of non-U.S. Participating Employers should be eligible to participate in the Plan;

(g) to construe and interpret the Plan and to make all determinations, including factual determinations, necessary or advisable for the administration of the Plan;

(h) to make, amend, suspend, waive and rescind rules and regulations relating to the Plan (including, but not limited to, such rules and regulations that would allow designations for beneficiaries and/or joint tenants to be made by Participants in connection with Accounts under the Plan);

(i) to appoint such agents as the Committee may deem necessary or advisable to administer the Plan;

(j) to correct any defect or supply any omission or reconcile any inconsistency, and to construe and interpret the Plan, the rules and regulations, and award agreements or any other instrument entered into or relating to a Contribution under the Plan; and

(k) to take any other action with respect to any matters relating to the Plan for which it is responsible and to make all other decisions and determinations as may be required under the terms of the Plan or as the Committee may deem necessary or advisable for the administration of the Plan.

Any action of the Committee with respect to the Plan shall be final, conclusive and binding on all persons, including the Company, its Affiliates, any Associate, any person claiming any rights under the Plan from or through any Participant, and stockholders, except to the extent the Committee may subsequently modify, or take further action not consistent with, its prior action. If not specified in the Plan, the time at which the Committee must or may make any determination shall be determined by the Committee, and any such determination may thereafter be modified by the Committee. The express grant of any specific power to the Committee, and the taking of any action by the Committee, shall not be construed as limiting any power or authority of the Committee.

VIII. Amendment & Termination

8.1 **Right to Amend or Terminate.** The Board, or a duly authorized committee thereof, reserves the right to amend, modify, suspend or discontinue the Plan at any time in its sole discretion without the approval of the Company's stockholders, except that (a) any amendment or modification shall be subject to the approval of the Company's stockholders if such stockholder approval is required by any federal or state law or regulation or the rules of any securities exchange or automated quotation system on which the shares of Stock may then be listed or quoted, and (b) the Board may otherwise, in its discretion, determine to submit other such amendments or modifications to stockholders for approval.

8.2 **Limitation on Right to Amend or Terminate.** Any such amendment, modification, suspension or termination will not result in the forfeiture of (1) subject to Section 3.8(d), any funds contributed but not yet invested in the Participant's Account, (2) any shares (or fractional interests) of Stock purchased on behalf of the Participant under the Plan, or (3) subject to Section 3.8(d), any dividends or other distributions in respect of such shares that are declared subsequent to a Participant's Contribution but prior to the effective date of the amendment, modification, suspension or termination of the Plan.

IX. Miscellaneous Provisions

9.1 **Successors.** All obligations of the Company under the Plan with respect to Contributions made hereunder shall be binding on any successor to the Company, whether the existence of such successor is the result of a direct or indirect purchase, merger, consolidation, or otherwise of all or substantially all of the business and/or assets of the Company.

9.2 **Severability.** If any part of the Plan is declared by any court or governmental authority to be unlawful or invalid, such unlawfulness or invalidity shall not invalidate any other part of the Plan. Any Section or part of a Section so declared to be unlawful or invalid shall, if possible, be construed in a manner which will give effect to the terms of such Section or part of a Section to the fullest extent possible while remaining lawful and valid.

9.3 **Requirements of Law.** The granting of awards, the making of Contributions, and the delivery of shares of Stock under the Plan shall be subject to all applicable laws, rules, and regulations, and to such approvals by any governmental agencies or national securities exchanges as may be required. Notwithstanding any provision of the Plan, Participants shall not be entitled to receive benefits under the Plan, and the Company (and any Affiliate) shall not be obligated to deliver any shares of Stock or deliver benefits to a Participant, if such delivery would constitute a violation by the Participant or the Company or any of its Affiliates of any applicable law or regulation.

9.4 **Securities Law Compliance.**

(a) If the Committee deems it necessary to comply with any applicable securities law or the requirements of any securities exchange upon which shares of Stock may be listed, the Committee may impose any restriction on Contributions, shares of Stock acquired pursuant to Contributions or purchases of shares of Stock under the Plan as it may deem advisable. Shares of Stock credited to the Account of a Participant, delivered to a Participant in certificated form or registered to a Participant in the Direct Registration System that may not be reoffered and resold by such Participant under applicable securities laws except pursuant to an effective registration statement covering or with respect to, or a qualification of, such shares of Stock or of such reoffer and resale of such shares of Stock or in accordance with another compliance procedure permitting a public reoffer and resale of such shares of Stock, in each case, in accordance with applicable securities laws, shall be subject to such stop transfer orders and other restrictions as the Committee may deem advisable under applicable securities laws and the listing requirements and rules for listed companies of any securities exchange upon which shares of Stock are then listed and the Committee may cause a legend or legends to be put on any such certificates to make appropriate reference to such restrictions and, if so requested by the Company, such Participant shall make a written representation to the Company that he or she will not reoffer or resell any shares of Stock held by him or her except in compliance with all applicable requirements for the registration or qualification of such shares of Stock or of such reoffer and resale of such shares of Stock or in accordance with such other compliance procedure permitting the public reoffer and resale of such shares of Stock unless he or she shall have furnished to the Company an opinion of an experienced securities attorney licensed in the jurisdiction whose securities laws govern such reoffer and resale that such registration, qualification or other compliance actions regarding such reoffer and resale are not required for such reoffer and resale of such shares of Stock to be effected in compliance with the applicable securities laws, which opinion shall be in form and substance satisfactory to the Company in its sole discretion.

(b) If the Committee determines that the nonforfeitability of, or delivery of benefits pursuant to, any Contribution or any purchase of shares of Stock, either from the Company or in an open market purchase over a national securities exchange, would result in a violation of any provision of any applicable securities laws or the listing requirements or rules for listed companies of any securities exchange on which the Stock is listed for trading, then the Committee may postpone any such nonforfeitability or delivery, as applicable, and may suspend the purchase of shares of Stock under the Plan, but the Company shall use all reasonable efforts to cause such nonforfeitability or delivery to comply with all such provisions at the earliest practicable date and to permit the Account Administrator to make purchases of shares of Stock in accordance with the Plan at the earliest practicable date; provided that the Company may, in its sole discretion, suspend the participation in the Plan of any Participant or any or all Participants to the extent necessary for the Company to comply or remain in compliance with the securities laws of any jurisdiction, including the United States, and may suspend purchases of shares of Stock that would be deemed to be the offer and sale of shares of Stock by the Company in a transaction that is not registered pursuant to, or exempt from registration under, Section 5 of the Securities Act of 1933, as amended, or is not qualified or otherwise permitted to be made under the securities laws of any jurisdiction outside of the United States, for such period as the Company deems appropriate in order for the Company to file a registration statement with the U.S. Securities and Exchange Commission with respect to the offer and sale of shares of Stock under the Plan and to have such registration statement declared effective by the U.S. Securities and Exchange Commission and to effect all necessary qualifications and to comply with any other compliance procedures required to be adhered to in order for such purchases of such shares of Stock on behalf of Participants to comply with all applicable securities laws, rules, and regulations.

9.5 **No Rights as a Stockholder.** No Participant shall have any rights as a stockholder of the Company with respect to the shares of Stock which may be deliverable to the Participant's Account in connection with a Contribution (other than a Voluntary Contribution of previously-owned shares of Stock) under the Plan until such shares of Stock have been credited to his or her Account or have been delivered to him or her.

9.6 **Nature of Payments.** Matching Contributions and Award Contributions shall be special incentive payments to the Participant and shall not be taken into account in computing the amount of salary or compensation of the Participant for purposes of determining any pension, retirement, death or other benefit under (a) any pension, retirement, profit-sharing, bonus, insurance or other employee benefit plan of the Company or any Affiliate, except as such plan shall otherwise expressly provide, or (b) any agreement between (i) the Company or any Affiliate and (ii) the Participant, except as such agreement shall otherwise expressly provide.

9.7 **Non-Exclusivity of the Plan.** Neither the adoption of the Plan by the Board nor its submission to the stockholders of the Company for approval shall be construed as creating any limitations on the power of the Board to adopt such other compensatory arrangements for Associates as it may deem desirable.

9.8 **Military Service.** The Plan shall be administered in accordance with Section 414(u) of the Internal Revenue Code and the Uniformed Services Employment and Reemployment Rights Act of 1994.

9.9 **Construction.** The following rules of construction will apply to the Plan: (a) the word "or" is disjunctive but not necessarily exclusive, and (b) words in the singular include the plural, words in the plural include the singular, and words in the neuter gender include the masculine and feminine genders and words in the masculine or feminine gender include the other neuter genders.

9.10 **Headings.** The headings of articles and sections are included solely for convenience of reference, and if there is any conflict between such headings and the text of this Plan, the text shall control.

9.11 **Stockholder Approval.** All Contributions made on or after the effective date of the amended and restated Plan and prior to the date the Company's stockholders approve the amended and restated Plan are expressly conditioned upon and subject to approval of the amended and restated Plan by the Company's stockholders.

9.12 **Taxes.** All Payroll Deduction Contributions, Matching Contributions and Award Contributions are subject to withholding for applicable federal, state and local income taxes and will be reported as wage income by the Company. When a Participant authorizes a Payroll Deduction of a specific amount, more than that amount will actually be withheld from his or her compensation to cover the withholding taxes due on the Payroll Deduction Contribution and Matching Contribution. Unless set forth otherwise by applicable law, rule, or regulation, the distribution of shares of Stock from a Participant's Account to a Participant, or cash in lieu of fractional shares, will not be a taxable event.

9.13 **Company-Associate Relationships.** Nothing contained in this Plan shall in any way affect the rights of the Company (including any of its Affiliates) in its relationship with any Associate or affect the Company's (including any of its Affiliates') right to discharge any Associate or increase or reduce any Associate's compensation.

9.14 **Governing Law.** This Plan shall be governed by and construed in accordance with the laws of the State of Delaware, except to the extent it is governed by the federal securities laws or the choice of laws provision contained in the Company's agreement with the Account Administrator.

9.15 **Adjustments upon Changes in Capitalization or Merger.** Subject to any required action by the Company's stockholders, the number and type of shares of Stock available for Contributions under the Plan shall be proportionately adjusted to reflect an extraordinary dividend or other distribution, stock split, reverse stock split, merger, reorganization, subdivision, consolidation or reduction of capital, recapitalization, consolidation, split-up, spin-off, combination or reclassification of the Stock, or any other increase or decrease in the number of outstanding Stock effected without receipt of consideration by the Company, issuance or warrants or other rights to purchase Stock or other securities of the Company or other similar corporate transaction or event that affects the Stock such that an adjustment is determined by the Board or the Committee to be appropriate in order to prevent dilution or enlargement of the benefits or potential benefits intended to be made available under the Plan. That adjustment shall be made by the Board or the Committee, whose determination shall be final, binding, and conclusive as to every person interested under the Plan.

Exhibit 10.7(a)

AMENDED SCHEDULE OF EXECUTIVE OFFICERS WHO HAVE EXECUTED A POST-TERMINATION AGREEMENT AND COVENANT NOT TO COMPETE IN THE FORM FILED AS EXHIBIT 10(p) TO THE ANNUAL REPORT ON FORM 10-K OF THE COMPANY FOR THE FISCAL YEAR ENDED JANUARY 31, 2011 (this "Amended Schedule")

This Amended Schedule amends the Schedule of Executive Officers Who Have Executed a Post-Termination Agreement and Covenant Not to Compete that followed the form of Post-Termination Agreement and Covenant Not to Compete originally filed by Walmart Inc. (formerly Wal-Mart Stores, Inc.) as Exhibit 10(p) to its Annual Report on Form 10-K for the year ended January 31, 2011, as filed on March 30, 2011 (the "Form Agreement"). This Amended Schedule is included pursuant to Instruction 2 of Item 601(a) of Regulation S-K for the purpose of setting forth the details in which the specific agreements executed in the form of the Form Agreement differ from the Form Agreement, in particular to set forth the persons who, with Walmart Inc., were parties to Post-Termination Agreements and Covenants Not to Compete in such form as of January 31, 2024.

Executive Officer Who is a Party to such a Post-Termination Agreement and Covenant Not to Compete	Date of Agreement	Value of Restricted Stock Award Granted in Connection with Agreement
Daniel J. Bartlett	May 16, 2013	Not Applicable
Rachel L. Brand	February 21, 2018	Not Applicable
David Chojnowski	November 16, 2016	Not Applicable
John R. Furner	May 7, 2011	Not Applicable
C. Douglas McMillon	January 19, 2010	$2,000,000
Christopher Nicholas	March 10, 2018	Not Applicable
Kathryn McLay	December 24, 2015	Not Applicable
Donna Morris	December 17, 2019	Not Applicable
John David Rainey	May 23, 2022	Not Applicable

Exhibit 21

Significant Subsidiaries of Walmart Inc.

The following list details certain of the subsidiaries of Walmart Inc. Subsidiaries not included in the list are omitted because, in the aggregate, they are not significant as permitted by Item 601(b)(21) of Regulation S-K.

Subsidiary	Organized or Incorporated	Percent of Equity Securities Owned	Name Under Which Doing Business Other Than Subsidiary's
Wal-Mart Stores East, LP	Delaware, U.S.	100%	Walmart
Wal-Mart Stores Texas, LLC	Delaware, U.S.	100%	Walmart
Wal-Mart Property Company	Delaware, U.S.	100%	NA
Wal-Mart Real Estate Business Trust	Delaware, U.S.	100%	NA
Sam's West, Inc.	Arkansas, U.S.	100%	Sam's Club
Sam's East, Inc.	Arkansas, U.S.	100%	Sam's Club
Sam's Property Company	Delaware, U.S.	100%	NA
Sam's Real Estate Business Trust	Delaware, U.S.	100%	NA
Wal-Mart de Mexico, S.A.B. de C.V.	Mexico	71%	Walmex
Wal-Mart Canada Corp.	Canada	100%	Walmart
Flipkart Private Limited	Singapore	85%	Flipkart
Walmart Chile S.A.[(1)]	Chile	100%	Walmart Chile
Massmart Holdings Ltd.	South Africa	100%	Massmart
Qomolangma Holdings Ltd.	Luxembourg	100%	NA

(1) The Company owns substantially all of Walmart Chile.

Exhibit 23

Consent of Independent Registered Public Accounting Firm

We consent to the incorporation by reference in the following Registration Statements:

(1)	Shareholder Investment Plan of Wal-Mart Stores, Inc.	Form S-3 File No. 333-02089
(2)	Wal-Mart Stores, Inc. Director Compensation Plan	Form S-8 File No. 333-24259
(3)	Wal-Mart Stores, Inc. 401(k) Retirement Savings Plan	Form S-8 File No. 333-29847
(4)	Wal-Mart Puerto Rico, Inc., 401(k) Retirement Savings Plan	Form S-8 File No. 333-44659
(5)	Wal-Mart Stores, Inc. Associate Stock Purchase Plan of 1996	Form S-8 File No. 333-62965
(6)	Wal-Mart Stores, Inc. Stock Incentive Plan of 2015, which amended and restated the 2010 plan	Form S-8 File No. 333-60329
(7)	Wal-Mart Profit Sharing and 401(k) Plan	Form S-8 File No. 333-109421
(8)	Wal-Mart Stores, Inc. Associate Stock Purchase Plan of 1996	Form S-8 File No. 333-109417
(9)	Wal-Mart Puerto Rico Profit Sharing and 401(k) Plan	Form S-8 File No. 333-109414
(10)	Wal-Mart Stores, Inc. Stock Incentive Plan of 2015, which amended and restated the 2010 plan	Form S-8 File No. 333-128204
(11)	Walmart Deferred Compensation Matching Plan	Form S-8 File No. 333-178717
(12)	Wal-Mart Stores, Inc. Common Stock	Form S-3 ASR File No. 333-178385
(13)	Walmart 401(k) Plan	Form S-8 File No. 333-187577
(14)	Wal-Mart Stores, Inc. Associate Stock Purchase Plan	Form S-8 File No. 333-214060
(15)	Walmart Inc. 2016 Associate Stock Purchase Plan	Form S-8 File No. 333-228631
(16)	Walmart Inc. Stock Incentive Plan of 2015	Form S-8 File No. 333-228635
(17)	Walmart 401(k) Plan	Form S-8 File No. 333-233682
(18)	Walmart Inc. Stock Incentive Plan of 2015	Form S-8 File No. 333-275879
(19)	Debt Securities of Walmart Inc.	Form S-3 ASR File No. 333-275878

of our reports dated March 15, 2024, with respect to the Consolidated Financial Statements of Walmart Inc. and the effectiveness of internal control over financial reporting of Walmart Inc. included in this Annual Report (Form 10-K) of Walmart Inc. for the year ended January 31, 2024.

/s/ Ernst & Young LLP

Rogers, Arkansas
March 15, 2024

Exhibit 31.1

I, C. Douglas McMillon, certify that:

1. I have reviewed this Annual Report on Form 10-K of Walmart Inc. (the "registrant");
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
   a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
   b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
   c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report, based on such evaluation; and
   d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting.
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the Audit Committee of registrant's Board of Directors:
   a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
   b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: March 15, 2024

/s/ C. Douglas McMillon

C. Douglas McMillon
President and Chief Executive Officer

Exhibit 31.2

I, John David Rainey, certify that:

1. I have reviewed this Annual Report on Form 10-K of Walmart Inc. (the "registrant");
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
    a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
    b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
    c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report, based on such evaluation; and
    d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting.
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the Audit Committee of registrant's Board of Directors:
    a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
    b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: March 15, 2024

/s/ John David Rainey

John David Rainey
Executive Vice President and Chief Financial Officer

Exhibit 32.1

**CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350 (AS ADOPTED
PURSUANT TO SECTION 906 OF THE
SARBANES-OXLEY ACT OF 2002)**

In connection with the Annual Report of Walmart Inc. (the "Company") on Form 10-K for the period ending January 31, 2024 as filed with the Securities and Exchange Commission on the date hereof (the "Report"), I, C. Douglas McMillon, President and Chief Executive Officer of the Company, certify to my knowledge and in my capacity as an officer of the Company, pursuant to 18 U.S.C. 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that:

1. The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934, as amended; and

2. The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company as of the dates and for the periods expressed in the Report.

IN WITNESS WHEREOF, the undersigned has executed this Certificate, effective as of March 15, 2024.

/s/ C. Douglas McMillon

C. Douglas McMillon
President and Chief Executive Officer

Exhibit 32.2

**CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350 (AS ADOPTED
PURSUANT TO SECTION 906 OF THE
SARBANES-OXLEY ACT OF 2002)**

In connection with the Annual Report of Walmart Inc. (the "Company") on Form 10-K for the period ending January 31, 2024 as filed with the Securities and Exchange Commission on the date hereof (the "Report"), I, John David Rainey, Executive Vice President and Chief Financial Officer of the Company, certify to my knowledge and in my capacity as an officer of the Company, pursuant to 18 U.S.C. 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that:

1. The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934, as amended; and

2. The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company as of the dates and for the periods expressed in the Report.

IN WITNESS WHEREOF, the undersigned has executed this Certificate, effective as of March 15, 2024.

/s/ John David Rainey
John David Rainey
Executive Vice President and Chief Financial Officer

Exhibit 97.1

WALMART INC.
EXECUTIVE COMPENSATION RECOUPMENT POLICY

1. **Purpose**. The purpose of this Policy is to describe the circumstances in which Executive Officers will be required to repay or return Erroneously Awarded Compensation to members of the Company. Each Executive Officer shall be required to sign and return to the Company the Acknowledgement Form attached hereto pursuant to which such Executive Officer will agree to be bound by the terms and comply with this Policy.

2. **Administration**. This Policy shall be administered by the Committee. Any determinations made by the Committee shall be final and binding on all affected individuals.

3. **Definitions**. For purposes of this Policy, the following capitalized terms shall have the meanings set forth below.

    a. "***Accounting Restatement***" shall mean an accounting restatement (i) due to the material noncompliance of the Company with any financial reporting requirement under the securities laws, including any required accounting restatement to correct an error in previously issued financial restatements that is material to the previously issued financial statements (a "Big R" restatement), or (ii) that corrects an error that is not material to previously issued financial statements, but would result in a material misstatement if the error were corrected the current period or left uncorrected in the current period (a "little r" restatement).

    b. "***Board***" shall mean the Board of Directors of the Company.

    c. "***Clawback Eligible Incentive Compensation***" shall mean, in connection with an Accounting Restatement and with respect to each individual who served as an Executive Officer at any time during the applicable performance period for any Incentive-based Compensation (whether or not such Executive Officer is serving at the time the Erroneously Awarded Compensation is required to be repaid to the Company), all Incentive-based Compensation Received by such Executive Officer (i) on or after the Effective Date, (ii) after beginning service as an Executive Officer, (iii) while the Company has a class of securities listed on a national securities exchange or a national securities association, and (iv) during the applicable Clawback Period.

    d. "***Clawback Period***" shall mean, with respect to any Accounting Restatement, the three most recently completed fiscal years of the Company immediately preceding the Restatement Date. In the event the Company changes its fiscal year within or immediately following the above-described three fiscal year period, the Clawback Period will also include the resulting transition period if such transition period is less than nine months. For purposes of this Policy, any transition period of nine to twelve months will be deemed a completed fiscal year.

e. "***Committee***" shall mean the Compensation and Management Development Committee of the Board or any successor committee of the Board that satisfies the requirements of Section 303A.05 of the NYSE Listed Company Manual.

f. "***Company***" shall mean Walmart Inc. a Delaware corporation, together with its subsidiaries and affiliates.

g. "***Effective Date***" shall mean October 2, 2023.

h. "***Erroneously Awarded Compensation***" shall mean, with respect to each Executive Officer in connection with an Accounting Restatement, the amount of Clawback Eligible Incentive Compensation that exceeds the amount of Incentive-based Compensation that otherwise would have been Received had it been determined based on the restated amounts, computed without regard to any taxes paid.

i. "***Executive Officer***" shall mean each individual who is or was designated as an "officer" of the Company in accordance with 17 C.F.R. 240.16a-1(f). For purposes of this policy, Executive Officers shall include at a minimum the executive officers identified pursuant to 17 C.F.R. 229.401(b).

j. "***Financial Reporting Measures***" shall mean: (i) measures that are determined and presented in accordance with the accounting principles used in preparing the Company's financial statements; (ii) measures derived wholly or in part from such measures; (iii) the Company's stock price; and (iv) the Company's total shareholder return. For the avoidance of doubt, a Financial Reporting Measure need not be presented in the Company's financial statements or included in a filing with the SEC.

k. "***Incentive-based Compensation***" shall mean any compensation that is granted, earned or vested based wholly or in part upon the attainment of a Financial Reporting Measure.

l. "***NYSE***" shall mean the New York Stock Exchange.

m. "***Policy***" shall mean this Policy, as the same may be amended and/or restated from time to time.

n. "***Received***" shall, with respect to any Incentive-based Compensation, mean actual or deemed receipt, and Incentive-based Compensation shall be deemed received in the Company's fiscal period during which the Financial Reporting Measure specified in the Incentive-based Compensation award is attained, even if payment or grant of the Incentive-based Compensation occurs after the end of that period.

o. "***Restatement Date***" shall mean the earlier to occur of (i) the date the Board, a committee of the Board, or the officers of the Company authorized to take such action if Board action is not required, concludes, or reasonably should have concluded, that the issuer is required to prepare an Accounting Restatement, or (ii) the date of court, regulator or other legally authorized body directs the issuer to prepare an Accounting Restatement.

p. “***SEC***” shall mean the U.S. Securities and Exchange Commission.

4. **Repayment of Erroneously Awarded Compensation**.

a. In the event of an Accounting Restatement, the Committee shall reasonably promptly (and in all events within ninety (90) days after the Restatement Date) determine the amount of any Erroneously Awarded Compensation for each Executive Officer in connection with such Accounting Restatement and shall promptly thereafter provide each Executive Officer with a written notice containing the amount of Erroneously Awarded Compensation and a demand for repayment or return, as applicable. For Incentive-based Compensation based on (or derived from) stock price or total shareholder return where the amount of Erroneously Awarded Compensation is not subject to mathematical recalculation directly from the information in the applicable Accounting Restatement, the amount shall be determined by the Committee based on a reasonable estimate of the effect of the Accounting Restatement on the stock price or total shareholder return upon which the Incentive-based Compensation was Received (in which case, the Company shall maintain documentation of such determination of that reasonable estimate and provide such documentation to the NYSE.

b. The Committee shall have broad discretion to determine the appropriate means of recovery of Erroneously Awarded Compensation based on all applicable facts and circumstances and taking into account the time value of money and the cost to shareholders of delaying recovery. To the extent that the Committee determines that any method of recovery (other than repayment by the Executive Officer in a lump sum in cash or property) is appropriate, the Company shall offer to enter into a repayment agreement (in a form reasonable acceptable to the Committee) with the Executive Officer. If the Executive Officer accepts such offer and signs the repayment agreement within thirty (30) days after such offer is extended, the Company shall countersign such repayment agreement. If the Executive Officer fails to sign the repayment agreement within thirty (30) days after such offer is extended, the Executive Officer will be required to repay the Erroneously Awarded Compensation in a lump sum in cash (or such property as the Committee agrees to accept with a value equal to such Erroneously Awarded Compensation) on or prior to the date that is one hundred twenty (120) days following the Restatement Date. For the avoidance of doubt, except as set forth in Section 4(d) below, in no event may the Company accept an amount that is less than the amount of Erroneously Awarded Compensation in satisfaction of an Executive Officer’s obligations hereunder.

c. To the extent that an Executive Officer fails to repay all Erroneously Awarded Compensation to the Company when due (as determined in accordance with Section 4(b) above), the Company shall take all actions reasonable and appropriate to recover such Erroneously Awarded Compensation from the applicable Executive Officer. The applicable Executive Officer shall be required to reimburse the

Company for any and all expenses reasonably incurred (including legal fees) by the Company in recovering such Erroneously Awarded Compensation in accordance with the immediately preceding sentence.

d. Notwithstanding anything herein to the contrary, the Company shall not be required to take the actions contemplated by Section 4(b) above if the following conditions are met and the Committee determines that recovery would be impracticable:

   i. The direct expenses paid to a third party to assist in enforcing the Policy against an Executive Officer would exceed the amount to be recovered, after the Company has made a reasonable attempt to recover the applicable Erroneously Awarded Compensation, documented such attempts and provided such documentation to the NYSE;

   ii. Recovery would violate home country law where that law was adopted prior to November 28, 2022, provided that, before determining that it would be impracticable to recover any amount of Erroneously Awarded Compensation based on violation of home country law, the Company has obtained an opinion of home country counsel, acceptable to the NYSE, that recovery would result in such a violation and a copy of the opinion is provided to the NYSE; or

   iii. Recovery would likely cause an otherwise tax-qualified retirement plan, under which benefits are broadly available to employees of the Company, to fail to meet the requirements of 26 U.S.C. 401(a)(13) or 26 U.S.C. 411(a)and regulations thereunder.

5. **Reporting and Disclosure**. The Company shall file all disclosures with respect to this Policy in accordance with the requirement of the federal securities laws and regulations thereunder, including the disclosure required by applicable SEC filings.

6. **Indemnification Prohibition**. The Company shall not be permitted to indemnify any Executive Officer against (i) the loss of any Erroneously Awarded Compensation that is repaid, returned or recovered pursuant to the terms of this Policy, or (ii) any claims relating to the Company’s enforcement of its rights under this Policy. Further, the Company shall not enter into any agreement that exempts any Incentive-based Compensation from the application of this Policy or that waives the Company’s right to recovery of any Erroneously Awarded Compensation, and this Policy shall supersede any such agreement (whether entered into before, on or after the Effective Date).

7. **Interpretation**. The Committee is authorized to interpret and construe this Policy and to make all determinations necessary, appropriate, or advisable for the administration of this Policy.

8. **Effective Date**. This Policy shall be effective as of the Effective Date.

9. **Amendment**; **Termination**. The Committee may amend this Policy from time to time in its discretion and shall amend this Policy as it deems necessary, including as and when it determines that it is legally required by any federal securities laws, SEC rule or the rules of any national securities exchange or national securities association on which the Company's securities are listed. The Committee may terminate this Policy at any time. Notwithstanding anything in this Section 9 to the contrary, no amendment or termination of this Policy shall be effective if such amendment or termination would (after taking into account any actions taken by the Company contemporaneously with such amendment or termination) cause the Company to violate any federal securities laws, SEC rule or the rules of any national securities exchange or national securities association on which the Company's securities are listed.

10. **Other Recoupment Rights**; **No Additional Payments**. The Committee may require that any offer letter, employment agreement, equity award agreement, or any other agreement entered into on or after the Effective Date shall, as a condition to the grant of any benefit thereunder, require an Executive Officer to agree to abide by the terms of this Policy. Any right of recoupment under this Policy is in addition to, and not in lieu of, any other remedies or rights of recoupment that may be available to the Company under applicable law, regulation or rule or pursuant to the terms of any compensatory plan, offer letter, employment agreement, equity award agreement, or similar agreement and any other legal remedies available to the Company.

11. **Successors**. This Policy shall be binding and enforceable against all Executive Officers and their beneficiaries, heirs, executors, administrators or other legal representatives.

* * *

WALMART INC.
EXECUTIVE COMPENSATION RECOUPMENT POLICY

ACKNOWLEDGEMENT FORM

By signing below, the undersigned acknowledges and confirms that the undersigned has received and reviewed a copy of the Walmart Inc. Executive Compensation Recoupment Policy (the "***Policy***"). Capitalized terms used but not otherwise defined in this Acknowledgement Form (this "***Acknowledgement Form***") shall have the meanings ascribed to such terms in the Policy.

By signing this Acknowledgement Form, the undersigned acknowledges and agrees that the undersigned is and will continue to be subject to the Policy and that the Policy will apply both during and after the undersigned's employment with the Company. Further, by signing below, the undersigned agrees to abide by the terms of the Policy, including, without limitation, by returning any Erroneously Awarded Compensation (as defined in the Policy) to the Company to the extent required by, and in a manner permitted by, the Policy.

____ Signature

____ Print Name

____ Date

Exhibit 99.1

Opioids Litigation Case Citations and Currently Scheduled Trial Dates

A. Case Citations For Pending State Court Cases as of March 4, 2024

Commonwealth of Massachusetts v. Walmart, Inc., Mass. Super. Ct., Suffolk Cty., 2/22/2024; State of Nebraska, ex rel. Michael T. Hilgers, Att'y Gen. v. CVS Pharm., Inc., et al., Neb. Dist. Ct., Lancaster Cty., 2/1/2024; State of Ohio, ex rel. Ohio Att'y Gen. Dave Yost v. CVS Health Corp., et al., Ohio Ct. Com. Pl., Franklin Cty., 1/17/2024; Commonwealth of Virginia, ex rel. Jason S. Miyares, Att'y Gen. v. CVS Pharm., Inc., et al., Va. Cir. Ct., City of Richmond, 1/4/2024; People of Illinois v. CVS Pharm., Inc., et al., Ill. Cir. Ct., Cook Cty., Ch. Div., 12/6/2023; State of Arizona ex rel. Kristin K. Mayes, Att'y Gen. v. Walmart, Inc., Ariz. Super. Ct., Maricopa Cty., 11/7/2023; State of North Carolina ex rel. Joshua H. Stein, Att'y Gen. v. CVS Pharm., Inc., et al. N.C. Super. Ct., Wake Cty., 10/31/2023; Att'y Gen. for State of Oregon v. Walmart Inc., Or. Cir. Ct., Multnomah Cty., 10/30/2023; Commonwealth of Pennsylvania ex rel. Allegheny Cty. Dist. Att'y Stephen A. Zappala, Jr. v. CVS Ind., LLC, et al., Pa. Ct. Com. Pl., Delaware Cty., 8/8/2022; Paynter ex rel. Minor Child(ren) Z.N.B. v. McKesson Corp., et al., W. Va. Cir. Ct., Kanawha Cty., 3/28/2022; Blankenship ex rel. Minor Child Z.D.B. v. McKesson Corp., et al., W. Va. Cir. Ct., Kanawha Cty., 1/14/2022; E. Me. Med. Ctr., et al. v. Teva Pharm. USA, Inc., et al., Me. Bus. & Consumer Ct., Cumberland Cty., 9/10/2021; Miss. Baptist Med. Ctr. Inc., et al. v. Amneal Pharm., LLC, et al., Miss. 1st Jud. Dist., Hinds Cty. Cir. Ct., 5/15/2020; Dallas Cty. Hosp. Dist. d/b/a Parkland Health & Hosp. Sys., et al., v. Amneal Pharm., LLC, et al., Tex. Dist. Ct., 152nd Jud. Dist., Harris Cty., 11/20/2019; Fla. Health Scis. Ctr., Inc., et al. v. Sackler, et al., Fla. Cir. Ct., 17th Jud. Cir., Broward Cty., 9/16/2019; City of Yonkers v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 5/29/2019; City of Albany v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 5/3/2019; City of Plattsburgh v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 5/3/2019; City of Troy v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 5/3/2019; City of Schenectady v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 5/3/2019; Cty. of Ulster v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 4/10/2019; Cty. of Washington v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 4/1/2019; Cty. of Montgomery v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 3/26/2019; Cty. of Herkimer v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 3/26/2019; Cty. of Lewis v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 2/26/2019; Cty. of St. Lawrence v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Suffolk Cty., 1/30/2019; City of New York v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 1/16/2019; Cty. of Fulton v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 1/8/2019; Cty. of Ontario v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Suffolk Cty., 1/8/2019; Cty. of Columbia v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 12/1/2018; Cty. of Monroe v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 12/1/2018; Cty. of Wyoming v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/28/2018; Cty. of Greene v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/28/2018; Cty. of Oswego v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/27/2018; Cty. of Schenectady v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/15/2018; Cty. of Broome v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/13/2018; Cty. of Erie v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/13/2018; Cty. of Orange v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/13/2018; Cty. of Dutchess v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/13/2018; Cty. of Seneca v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/13/2018; Cty. of Sullivan v. Purdue Pharma L.P., et al., N.Y. Sup. Ct., Westchester Cty., 11/13/2018.

B. Currently Scheduled Trial Dates In Pending State Court Cases as of March 4, 2024

- 9/8/25 – Florida Health Sciences Center, Inc., et al. v. Richard Sackler, et al., Fla. Cir. Ct., 17th Jud. Dist., Broward Cty.

C. Case Citations For Pending Non-MDL Federal Court Cases as of March 4, 2024

Chaney v. CVS Pharm., et al., Inc., U.S. Dist. Ct., E.D. Ky., 12/11/2023; San Miguel Hosp. Corp. d/b/a Alta Vista Reg'l Hosp. v. Johnson & Johnson, et al., U.S. Dist. Ct., D.N.M., 10/16/2023; Baby Doe, et al., ex rel. Their Guardian Ad Litem v. Endo Health Sols., Inc., et al., U.S. Dist. Ct., M.D. Tenn., 8/3/2022; City of Kingston v. Teva Pharm. USA, Inc., et al., U.S. Dist. Ct., S.D.N.Y., 2/27/2020; Cty. of Cortland v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 1/8/2019; City of Ithaca v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Rensselaer v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Saratoga v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Schoharie v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Westchester v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Genesee v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Niagara v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Hamilton v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Franklin v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Schuyler v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Steuben v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Clinton v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018; Cty. of Tompkins v. Purdue Pharma L.P., et al., U.S. Dist. Ct., S.D.N.Y., 10/23/2018.

D. International Cases as of March 4, 2024

City of Grand Prairie v. Apotex, Calgary 2001-07073 (ABKB)
Lac La Ronge Indian Band v. Apotex, Prince Albert QBG-PA-00072-2021 (SKKB)